Mingchen Xia

Foundations of pluripotential theory

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Introduction

A plurisubharmonic function is the several-variable analogue of a subharmonic function: an upper semicontinuous function whose restriction to every complex line is subharmonic. Such functions arise as local potentials of closed positive currents, as weights of singular Hermitian metrics on line bundles and as solutions of complex Monge–Ampère equations. They may take the value $-\infty$, and much of the geometric information they carry is concentrated in their singularities. Pluripotential theory studies these functions, the nonlinear measures they define, and the ways in which their singularities can be measured and compared.

This book develops the foundations of the theory from the beginning. It is written as a companion to Part I of *Singularities in global pluripotential theory* [Xia], and its purpose is to give complete proofs of the analytic results on which that text relies. We start from classical potential theory in $\mathbb{R}^d$ and end with the machinery behind non-pluripolar products, singularity envelopes, relative energy and the ample toric setting. The chapters are arranged in the order in which the proofs depend on one another: weighted holomorphic estimates come before multiplier ideals, regularization comes before the study of global positive classes, and the Monge–Ampère operator on bounded functions comes before any product involving unbounded potentials.

From distributions to pointwise potentials

A locally integrable function has distributional derivatives of every order, but as a distribution it cannot see its values on a set of measure zero. Potential theory recovers these values from mean-value inequalities. If the distributional Laplacian of a function is a positive measure, the function has a canonical subharmonic representative, whose value at each point is the limit of its averages over small balls. Harmonic replacement on balls then controls maxima, boundary values and upper envelopes. The exceptional sets that occur in these constructions are measured by capacity rather than by Lebesgue measure. For this reason measure theory, the Dirichlet problem, energy and the fine topology together form the first part of the book.

In several variables a further difficulty appears. Plurisubharmonicity requires the submean inequality on every complex line, and an upper bound obtained separately on each line need not be uniform on a neighborhood. The Fourier and incidence estimates of Chapter 6 provide this uniformity. With them we show in Theorem 8.1.3 that upper semicontinuity is automatic for a function that is subharmonic on every complex line, and we identify plurisubharmonic functions with the canonical representatives of distributions whose complex Hessian is positive. Compactness and envelope arguments can therefore be carried out for genuine functions, not only for almost-everywhere classes.

Two ways of measuring a singularity

The first way is local and holomorphic. The Lelong number of a plurisubharmonic function measures its logarithmic concentration at a point, while its multiplier ideals record which holomorphic functions remain square integrable against the corresponding exponential weight. The proofs that multiplier ideals are coherent, and that integrability persists after a small increase of the weight, rely on holomorphic extension. Weighted L^2 estimates for the $\bar{\partial}$ operator are the basic tool: they correct the error introduced by a cutoff function, prescribe values along slices and produce the holomorphic functions used in approximation. Potentials with analytic singularities, defined by finitely many holomorphic functions, make the resulting growth explicit; on a resolution of singularities it is described by divisors.

The second way is nonlinear. For locally bounded plurisubharmonic functions, Bedford and Taylor defined the exterior products of their curvature currents and proved that these products are continuous along decreasing sequences. For an unbounded function one first truncates the potential and then discards the part of the product carried by its pole set. When the products obtained in this way have locally bounded mass, their limit is the non-pluripolar product. This operation can lose mass, and the relation between this loss and the singularities of the potentials is one of the central themes of the global theory.

The two measurements need not agree. Using an atom-free probability measure carried by a polar Cantor set (Proposition 4.2.4), we construct in Example 15.3.8 a potential on the projective line whose Lelong numbers all vanish and whose multiplier ideals are all trivial, but whose non-pluripolar mass is zero. This example, which is [Xia, Example 6.1.3], shows that neither Lelong numbers nor multiplier ideals can replace the mass theory developed in the later chapters.

Global potentials and envelopes

On a compact Kähler manifold, a smooth closed real $(1,1)$-form fixes a cohomology class, and a potential changes the current within that class. Local positivity must

now be reconciled with this global constraint. Elliptic theory for the scalar Laplacian provides global potentials, and compactness and approximation describe pseudo-effective and big classes, including their boundary behavior. Singular Hermitian metrics express the same positivity in the language of line bundles.

An envelope is the largest admissible potential lying below a given obstacle. When an additional constraint on the singularities is imposed, the envelope selects a canonical representative of that constraint. Its Monge–Ampère measure is carried by the set where the envelope touches the obstacle, but proving this requires local replacement arguments and careful passages to the limit. Relative capacities and integration by parts then lead to the relative energy, to Monge–Ampère equations with prescribed singularities and to Perron segments joining two potentials. As an application we prove in Theorem 17.5.3 the identity between ceilings and singularity envelopes of [PSWZ26, Theorem 1.2] in every big class, together with the idempotence of the singularity envelope; the case of zero mass is obtained from positive-mass perturbations together with a uniform estimate of their tails. Finally, convex duality and real Monge–Ampère measures provide the corresponding foundations for invariant potentials on the open torus, and the projective toric manifold of a Delzant polytope, with its sections and the Guillemin metric, provides the ample toric setting.

Prerequisites and how to read this book

We assume undergraduate real analysis, including the differential calculus of several real variables (in particular the inverse and implicit function theorems), Lebesgue integration with its convergence theorems and the change of variables formula, linear algebra, point-set topology and the elementary theory of holomorphic functions of one variable. Distribution theory, the functional analysis needed for weighted estimates and elliptic equations, holomorphic functions of several variables, the commutative algebra used for ideals of holomorphic germs, complex differential geometry and convex analysis are developed in the text. Part I establishes the linear theory and the theory of exceptional sets; Part II develops the local complex theory and then passes to singular potentials on compact manifolds. Each part opens with a short introduction describing its chapters.

Appendix A collects the facts about complex spaces, normalization and modifications that are needed for multiplier ideals and resolutions; it can be consulted when these tools first appear. Two existence theorems are used without proof: the resolution of singularities of complex spaces and the principalization of coherent ideals. They are stated precisely as Theorems A.5.4 and A.5.5, and every consequence of them used in the book is proved. The historical and bibliographical notes in Appendix B give the sources of the results and the precise versions of the cited works, with references to the corresponding statements in the text.

Our notation and normalizations follow [Xia]. The global conventions are listed on the following pages; all other notation is introduced where it is first needed. A few definitions and constructions of [Xia] are repeated here when a complete argument

requires them, while the classification of singularities and its applications are left to that book.

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Conventions

In the whole book, we adopt the following conventions:

- A complex space is always assumed to be *reduced*, *paracompact* and *Hausdorff*.
- A *modification* of a complex space X is a proper bimeromorphic morphism $\pi\colon Y \to X$ that is locally obtained from a finite composition of blow-ups with smooth centers.
- A *subnet* of a net refers to a Willard subnet. More precisely, if $f\colon I \to X$ and $f'\colon I' \to X$ are nets, namely, maps from directed sets I and I' to the same set X, we say f' is a subnet of f if there is an *increasing* final function $h\colon I' \to I$ such that $f \circ h = f'$.
- A *domain* in $\mathbb{R}^d$ or $\mathbb{C}^n$ refers to a connected open subset.
- A *complex manifold* is assumed to be paracompact.
- A *submanifold* of a complex manifold means a closed complex submanifold.
- A *neighborhood* is not necessarily open.
- The set $\mathbb{N}$ of natural numbers includes 0.
- *Increasing functions* and *decreasing functions* are not necessarily strictly monotone.
- *Positive forms* are not necessarily strictly positive.
- The function $\mathbb{R} \ni x \mapsto x^2$ is *convex* instead of concave. This is different from the convention in [CLS11].
- We say a positive measure μ on a measurable space X is *carried* by a measurable subset Y if $\mu(X \setminus Y) = 0$.
- $\mathrm{dd^c}$ means $(2\pi)^{-1}\mathrm{i}\partial\overline{\partial}$.

We also use the following notation throughout the book.

- The indicator function of a set E is denoted by $\mathbf{1}_E$.
- For subsets A and B of a topological space, $A \Subset B$ means that the closure of A is a compact subset of B; we then say that A is *relatively compact* in B.
- The closure, interior and boundary of a set E in its ambient space are denoted by $\overline{E}$, $\operatorname{Int} E$ and ∂E. The support of a function, measure, distribution or current is denoted by Supp.

- For an open set Ω in $\mathbb{R}^d$ or in a manifold, $C^k(\Omega)$ is the space of k times continuously differentiable functions, where $0 \leq k \leq \infty$, and $C_c^k(\Omega)$ is its subspace of functions with compact support. We write $L^p_{\mathrm{loc}}(\Omega)$ for the functions whose restrictions to compact subsets of Ω are p-integrable.
- The imaginary unit is i, and $\operatorname{Re} z$ and $\operatorname{Im} z$ are the real and imaginary parts of a complex number z.

Part I
Potential theory

Introduction to Part I

This part develops the potential theory of the Laplacian on open subsets of $\mathbb{R}^d$, as far as it is needed for the complex theory of Part II. A potential can be described in three ways: by its distributional derivatives, by its mean values, and by the measure that records its singularities. Each description is suited to different operations, namely weak limits, pointwise comparison and the analysis of poles. Our main task is to pass between these descriptions while keeping control of the exceptional sets produced by limits and envelopes.

Measures and distributions come first (Chapter 1), because the curvature currents of later chapters have measures as coefficients. We prove that positive linear functionals are given by measures, approximate distributions by smooth functions through convolution, and establish the compactness principles used to extract weak limits. Harmonic functions (Chapter 2) are the comparison functions of the theory. The Poisson kernel solves the Dirichlet problem on a ball; on a general bounded domain, Perron's method produces a harmonic function in the interior, and barriers decide whether it attains the prescribed boundary values.

Subharmonic functions (Chapter 3) are obtained by replacing the mean-value equality by an inequality. Their distributional Laplacians are positive measures, and a subharmonic function is determined at every point by its almost-everywhere class. This correspondence is indispensable for upper envelopes: the pointwise supremum of a family and its upper semicontinuous regularization may differ, and one must know where. The Riesz decomposition splits a subharmonic function into a harmonic part and the potential of its Laplacian, which links comparison arguments with energy.

Capacity (Chapter 4) measures the size of a set by the potentials it can support. A set of Lebesgue measure zero can have positive capacity, whereas a polar set carries no nonzero measure of finite energy. Equilibrium potentials connect this distinction with comparison arguments. The fine topology (Chapter 5) is the coarsest topology that makes all subharmonic functions continuous. The reduction of a nonnegative superharmonic function on a set is the infimum of the nonnegative superharmonic functions that dominate it there, and balayage sweeps a measure onto a compact set while preserving its potential quasi-everywhere on that set. These constructions

explain why the exceptional sets of potential theory can be discarded only when they have capacity zero, and not merely when they have Lebesgue measure zero.

The last chapter of this part (Chapter 6) prepares the passage to several complex variables. Before subharmonicity along complex lines can produce a plurisubharmonic function, upper bounds along individual lines must be made uniform. An energy estimate, the Fourier transform and a projection argument provide this uniformity. The same Fourier methods give L^p estimates for second derivatives. The two results serve different purposes in Part II: the incidence estimate provides the local upper bounds that identify the pointwise representative of a plurisubharmonic function, while the second-derivative estimates control the Monge–Ampère measure of an envelope lying below a nonsmooth obstacle.

Chapter 1
Measures and distributions

The potentials studied in this book are seldom differentiable. On $\mathbb{R}^2$, for instance, the function $\log|x|$ tends to $-\infty$ at the origin, yet its Laplacian has a precise meaning: it is 2π times the Dirac mass at the origin (Proposition 2.2.1). Such statements are made rigorous by differentiating in the weak sense, that is, by moving derivatives onto smooth compactly supported test functions through integration by parts. The derivatives obtained in this way are distributions. For subharmonic functions, and later for plurisubharmonic ones, the relevant second derivatives are positive distributions, and a positive distribution is a measure. The measure attached to a potential records where its mass lies, including the mass concentrated at its singularities; measures and distributions are therefore the basic language of potential theory.

Throughout this chapter, Ω is an open subset of $\mathbb{R}^d$, where $d \geq 1$. The first section introduces Radon measures and signed Radon measures and recalls, with short proofs that fix our conventions, the parts of Lebesgue integration that we use. The core of the chapter consists of two facts: the Riesz representation theorem (Theorem 1.2.2), which identifies positive linear functionals on compactly supported continuous functions with Radon measures, and the regularization of distributions by convolution with a smooth kernel, which commutes with derivatives and preserves positivity (Proposition 1.4.3). Together they show that positive distributions are measures (Theorem 1.4.5). Since limits of potentials lead to limits of measures, we then study weak convergence and compactness of measures, for sequences and, where sequences do not suffice, for nets. The last section constructs bounded densities and proves the Lebesgue differentiation theorem; as an application, a continuous function with locally bounded distributional Hessian has a second-order Taylor expansion almost everywhere (Proposition 1.7.5).

Historical remarks and references for this chapter are collected in Appendix B.1.

1.1 Radon measures and integration

Unless stated otherwise, all measures in this chapter are positive, that is, they take values in $[0, \infty]$. We use the Lebesgue integral, defined first for nonnegative simple functions and then, for a nonnegative measurable function, as the supremum of the integrals of the simple functions below it. Lebesgue measure on $\mathbb{R}^d$ is denoted by $\mathrm{d}x$, and the Lebesgue measure of a measurable set $E \subset \mathbb{R}^d$ by $|E|$. The Borel σ-algebra of Ω is the smallest σ-algebra containing the open subsets of Ω, and a Borel measure on Ω is a measure defined on it. A measure on a measurable space X is σ-finite if X is a countable union of measurable sets of finite measure. We write $C(\Omega) = C^0(\Omega)$ and $C_c(\Omega) = C_c^0(\Omega)$. Their elements may be real- or complex-valued; the real-valued functions in $C_c(\Omega)$ form the space $C_c(\Omega, \mathbb{R})$, which we use whenever positivity is involved.

Definition 1.1.1 A Radon measure on Ω is a Borel measure μ on Ω which is finite on compact sets and satisfies, for every Borel set $E \subset \Omega$,

$$\mu(E) = \inf_{E \subset U,\ U \text{ open}} \mu(U) = \sup_{K \subset E,\ K \text{ compact}} \mu(K).$$

The two equalities are called outer and inner regularity of μ. A measurable function is locally integrable with respect to μ if its absolute value has finite integral over every compact subset of Ω.

The support $\operatorname{Supp}\mu$ of a Radon measure μ is the set of points of Ω all of whose open neighborhoods have positive measure. Its complement $\Omega \setminus \operatorname{Supp}\mu$ is the union of all open subsets of Ω of measure zero, so it is open and $\operatorname{Supp}\mu$ is relatively closed in Ω. Moreover $\mu(\Omega \setminus \operatorname{Supp}\mu) = 0$ by inner regularity, because every compact subset of $\Omega \setminus \operatorname{Supp}\mu$ is covered by finitely many open sets of measure zero.

We shall often use a compact exhaustion of Ω, that is, a sequence of compact sets $K_j \subset \Omega$, $j \geq 1$, with $K_j \subset \operatorname{Int} K_{j+1}$ for every j and $\bigcup_j \operatorname{Int} K_j = \Omega$. For example,

$$K_j = \{x \in \Omega : |x| \leq j,\ \operatorname{dist}(x, \mathbb{R}^d \setminus \Omega) \geq j^{-1}\},$$

where $\operatorname{dist}(x, \varnothing) = +\infty$ if $\Omega = \mathbb{R}^d$. Indeed, K_j is closed and bounded, hence compact. It lies in the open set $\{x : |x| < j+1,\ \operatorname{dist}(x, \mathbb{R}^d \setminus \Omega) > (j+1)^{-1}\}$, which is contained in K_{j+1}, and every point of Ω lies in $\operatorname{Int} K_j$ for all large j.

If $K \subset \Omega$ is compact and $U \subset \Omega$ is an open set containing K, a cutoff function for K in U is a continuous function $\eta \colon \Omega \to [0, 1]$ such that $\eta = 1$ on K and $\operatorname{Supp}\eta$ is a compact subset of U. Such functions exist. If $K = \varnothing$, take $\eta = 0$; otherwise choose $\delta > 0$ with $2\delta < \operatorname{dist}(K, \mathbb{R}^d \setminus U)$, so that $V = \{x : \operatorname{dist}(x, K) < \delta\}$ is an open set whose closure is a compact subset of U, and put

$$\eta(x) = \frac{\operatorname{dist}(x, \mathbb{R}^d \setminus V)}{\operatorname{dist}(x, \mathbb{R}^d \setminus V) + \operatorname{dist}(x, K)}.$$

The denominator never vanishes, since the first term is positive on $V \supset K$ and the second is positive off K; moreover $\eta = 1$ on K and $\eta = 0$ outside V. Smooth cutoff functions are constructed in Lemma 1.4.1.

Proposition 1.1.2 *Let μ be a Borel measure on Ω which is finite on compact subsets. Then μ is a Radon measure. In particular, μ is σ-finite.*

Proof Assume first that $\mu(\Omega) < \infty$, and let $F \subset \Omega$ be relatively closed; we may assume that $F \neq \emptyset$. The open sets $U_j = \{x \in \Omega : \operatorname{dist}(x, F) < j^{-1}\}$ decrease to F, so $\mu(U_j) \to \mu(F)$ because μ is finite. Hence F can be approximated in measure from outside by open sets. Passing to complements, every open set can be approximated from inside by relatively closed sets.

Let $\mathcal{A}$ be the class of Borel sets E such that for every $\varepsilon > 0$ there are an open set U and a relatively closed set F with $F \subset E \subset U$ and $\mu(U \setminus F) < \varepsilon$. We have just seen that $\mathcal{A}$ contains the relatively closed sets, and $\mathcal{A}$ is closed under complements because the condition is symmetric. To see that $\mathcal{A}$ is closed under countable unions, let $E = \bigcup_j E_j$ with $E_j \in \mathcal{A}$, and choose $F_j \subset E_j \subset U_j$ with $\mu(U_j \setminus F_j) < \varepsilon 2^{-j}$. The open set $U = \bigcup_j U_j$ satisfies $\mu(U \setminus E) < \varepsilon$. Since $\mu(\bigcup_{j \le N} E_j) \to \mu(E)$, there is N such that the relatively closed set $F = \bigcup_{j \le N} F_j$ satisfies $\mu(E \setminus F) \le \mu(E \setminus \bigcup_{j \le N} E_j) + \sum_{j \le N} \mu(E_j \setminus F_j) < 2\varepsilon$, and hence $\mu(U \setminus F) < 3\varepsilon$. Thus $\mathcal{A}$ is a σ-algebra containing the open sets, so it contains every Borel set E. If $F \subset E$ is relatively closed and (K_j) is a compact exhaustion of Ω, then the sets $F \cap K_j$ are compact and $\mu(F \setminus K_j) \to 0$. Hence every Borel set can be approximated in measure from outside by open sets and from inside by compact sets.

In general, let (K_j) be a compact exhaustion of Ω, put $K_0 = \emptyset$, and let $V_j = \operatorname{Int} K_{j+1}$. The measures $\mu_j(E) = \mu(E \cap V_j)$ are finite, hence Radon by the first part. For inner regularity, let E be a Borel set and $\varepsilon > 0$. A compact set $K \subset E$ with $\mu_j(E \setminus K) < \varepsilon$ satisfies $\mu(K) \ge \mu_j(K) > \mu(E \cap V_j) - \varepsilon$, and $\mu(E \cap V_j) \to \mu(E)$ as $j \to \infty$. For outer regularity, note that the Borel sets $B_j = K_j \setminus K_{j-1}$ partition Ω and that $E \cap B_j \subset V_j$. By outer regularity of μ_j there is an open set $O_j \subset V_j$ containing $E \cap B_j$ with $\mu(O_j \setminus (E \cap B_j)) < \varepsilon 2^{-j}$; it suffices to intersect an approximating open set with V_j. Then $O = \bigcup_j O_j$ is an open set containing E with $\mu(O \setminus E) < \varepsilon$. Finally, μ is σ-finite because $\Omega = \bigcup_j K_j$ and each $\mu(K_j)$ is finite. □

Proposition 1.1.3 *Let (X, μ) be a measure space and let f_j, $j \ge 1$, be measurable functions on X.*

(1) *If $f_j \ge 0$ and $f_j \uparrow f$ pointwise, then $\int f_j \, d\mu \uparrow \int f \, d\mu$.*
(2) *If $f_j \ge 0$ for every j, then*

$$\int \varliminf_j f_j \, d\mu \le \varliminf_j \int f_j \, d\mu.$$

(3) *If the f_j are real or complex, converge almost everywhere to a measurable function f, and satisfy $|f_j| \le g$ for an integrable function g, then f is integrable and $\int |f_j - f| \, d\mu \to 0$.*

Proof (1) The integrals $\int f_j \,\mathrm{d}\mu$ increase and are bounded by $\int f \,\mathrm{d}\mu$, so we must show that their limit is at least $\int f \,\mathrm{d}\mu$. Let $s \le f$ be a nonnegative simple function and $0 < t < 1$. The measurable sets $E_j = \{f_j \ge ts\}$ increase with j, and their union contains $\{s > 0\}$, since $f(x) \ge s(x) > ts(x)$ there. Applying continuity from below to the sets $E_j \cap \{s = a\}$ for the finitely many values a of s, we obtain

$$\lim_j \int f_j \,\mathrm{d}\mu \ge \lim_j t \int_{E_j} s \,\mathrm{d}\mu = t \int s \,\mathrm{d}\mu.$$

Taking the supremum over s and then letting $t \uparrow 1$ proves (1); infinite integrals are allowed.

(2) The functions $g_j = \inf_{k \ge j} f_k$ are measurable, they increase to $\underline{\lim}_j f_j$, and $\int g_j \,\mathrm{d}\mu \le \int f_k \,\mathrm{d}\mu$ for every $k \ge j$. Applying (1) to (g_j) gives (2).

(3) Letting $j \to \infty$ in $|f_j| \le g$ gives $|f| \le g$ almost everywhere, so f is integrable. The functions $2g - |f_j - f|$ are nonnegative almost everywhere, and (2), applied after a modification on a null set, gives

$$\int 2g \,\mathrm{d}\mu \le \varliminf_j \int \bigl(2g - |f_j - f|\bigr) \,\mathrm{d}\mu = \int 2g \,\mathrm{d}\mu - \varlimsup_j \int |f_j - f| \,\mathrm{d}\mu.$$

Since $\int 2g \,\mathrm{d}\mu < \infty$, it follows that $\int |f_j - f| \,\mathrm{d}\mu \to 0$. □

Assertions (1), (2) and (3) are the monotone convergence theorem, Fatou's lemma and the dominated convergence theorem. They concern sequences and fail for nets. For instance, the indicator functions of the finite subsets of $[0, 1]$, directed by inclusion, form an increasing net of Borel functions with supremum $\mathbf{1}_{[0,1]}$; each of them has Lebesgue integral zero, while the supremum has integral one. The supremum of an increasing net of measurable functions need not even be measurable: if $A \subset \mathbb{R}$ is not Lebesgue measurable, the indicator functions of the finite subsets of A increase to $\mathbf{1}_A$. Where nets occur later, the passage to the limit therefore requires a separate argument.

Sets that are not Lebesgue measurable exist. Choose a set $V \subset [0, 1)$ containing exactly one point of each coset $x + \mathbb{Q}$ of the rational numbers. The translates $V + q$, $q \in \mathbb{Q} \cap (-1, 1)$, are contained in $(-1, 2)$, and they are pairwise disjoint, because two points of V with rational difference coincide. They cover $[0, 1)$, since every $x \in [0, 1)$ differs from the point of V in its coset by a rational number in $(-1, 1)$. If V were measurable, translation invariance of Lebesgue measure would give $1 \le \sum_q |V| \le 3$, which is impossible for a countably infinite sum of equal terms.

Lemma 1.1.4 (Jensen's inequality) *Let (X, P) be a probability space, let $I \subset \mathbb{R}$ be an open interval, let $\varphi\colon I \to \mathbb{R}$ be convex, and let $f\colon X \to I$ be an integrable function. Then $\int f \,\mathrm{d}P \in I$, the function $\varphi \circ f$ is measurable, its negative part is integrable, and*

$$\varphi\Bigl(\int f \,\mathrm{d}P\Bigr) \le \int \varphi \circ f \,\mathrm{d}P.$$

Proof For $u < v$ in I put $s(u,v) = (\varphi(v) - \varphi(u))/(v-u)$. If $u < v < w$ lie in I, then $v = \lambda u + (1-\lambda)w$ with $\lambda = (w-v)/(w-u) \in (0,1)$, and the convexity inequality $\varphi(v) \le \lambda\varphi(u) + (1-\lambda)\varphi(w)$ is equivalent to each of the inequalities $s(u,v) \le s(u,w)$ and $s(u,w) \le s(v,w)$. Hence $s(u,v)$ is nondecreasing in each of its two arguments. Consequently, if $[x,y] \subset I$ and $x' < x$, $y' > y$ are points of I, then $s(x',x) \le s(u,v) \le s(y,y')$ whenever $x \le u < v \le y$, so φ is Lipschitz on $[x,y]$. In particular, φ is continuous, and $\varphi \circ f$ is measurable.

Let $a = \int f \,\mathrm{d}P$. If I has a finite left endpoint α, then $f - \alpha > 0$ on X, so one of the sets $\{f - \alpha > 1/k\}$, which increase to X, has positive measure, and therefore $a - \alpha = \int (f-\alpha)\,\mathrm{d}P > 0$. In the same way, a is smaller than a finite right endpoint of I; thus $a \in I$. For $u < a < w$ in I we have $s(u,a) \le s(u,w) \le s(a,w)$, so $c = \sup_{u<a} s(u,a)$ is finite and $c \le s(a,w)$ for every $w > a$. Hence $\varphi(y) \ge \varphi(a) + c(y-a)$ for every $y \in I$: for $y < a$ this is the inequality $s(y,a) \le c$, and for $y > a$ it is $s(a,y) \ge c$. It follows that $\varphi \circ f \ge \varphi(a) + c(f-a)$, where the right side is integrable. Therefore the negative part of $\varphi \circ f$ is integrable, and since $\int (f-a)\,\mathrm{d}P = 0$, integration gives $\int \varphi \circ f \,\mathrm{d}P \ge \varphi(a)$. □

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces, and let $\Phi\colon X \to Y$ be measurable, that is, $\Phi^{-1}(E) \in \mathcal{M}$ for every $E \in \mathcal{N}$. The image of a measure μ on X under Φ is the measure $\Phi_*\mu(E) = \mu(\Phi^{-1}(E))$ on Y; it is countably additive because preimages preserve disjoint unions. For every $\mathcal{N}$-measurable function $f \ge 0$ on Y,

$$\int_Y f \,\mathrm{d}\Phi_*\mu = \int_X f \circ \Phi \,\mathrm{d}\mu.$$

This holds for indicator functions by definition, hence for simple functions by linearity, and for general f by monotone convergence, applied to simple functions $s_k \uparrow f$ and to $s_k \circ \Phi \uparrow f \circ \Phi$. Consequently, a function f is integrable with respect to $\Phi_*\mu$ if and only if $f \circ \Phi$ is integrable with respect to μ, and then the identity still holds.

Lemma 1.1.5 *Let μ be a σ-finite measure on a measurable space X. Then every family of pairwise disjoint measurable subsets of X of positive measure is countable.*

Proof Write $X = \bigcup_m X_m$ with measurable sets X_m of finite measure. Every member E of the family satisfies $\mu(E \cap X_m) \ge 1/k$ for some m and some integer $k \ge 1$. For fixed m and k, at most $k\mu(X_m)$ members have this property, because the corresponding sets $E \cap X_m$ are pairwise disjoint subsets of X_m. Hence the family is a countable union of finite subfamilies. □

In particular, a σ-finite Borel measure has at most countably many atoms, that is, points of positive measure. If μ is a Radon measure on Ω, which is σ-finite by Proposition 1.1.2, and $a \in \mathbb{R}^d$, then $\mu(\Omega \cap \{x : |x-a| = r\}) = 0$ for all but countably many radii $r > 0$, since these sets are pairwise disjoint.

Uniqueness questions for measures, and the construction of product measures, rest on the following lemma.

Lemma 1.1.6 *Let $\mathcal{P}$ be a nonempty family of subsets of a set Z which is closed under finite intersections, and let $\sigma(\mathcal{P})$ be the σ-algebra generated by $\mathcal{P}$.*

(1) *Every family of subsets of Z which contains $\mathcal{P}$ and is closed under complements and countable disjoint unions contains $\sigma(\mathcal{P})$.*
(2) *Let λ and λ' be finite measures on a σ-algebra containing $\sigma(\mathcal{P})$. If $\lambda(Z) = \lambda'(Z)$ and $\lambda(A) = \lambda'(A)$ for every $A \in \mathcal{P}$, then $\lambda(E) = \lambda'(E)$ for every $E \in \sigma(\mathcal{P})$.*

Proof (1) Let $\mathcal{L}$ be the intersection of all families of subsets of Z which contain $\mathcal{P}$ and are closed under complements and countable disjoint unions; it is the smallest such family. It contains $Z = A \cup (Z \setminus A)$ for any $A \in \mathcal{P}$. If $B \subset A$ belong to $\mathcal{L}$, so does $A \setminus B$, the complement of the disjoint union of $Z \setminus A$ and B. For $A \in \mathcal{P}$, the sets $B \in \mathcal{L}$ with $A \cap B \in \mathcal{L}$ form a family which contains $\mathcal{P}$ and is closed under countable disjoint unions and under complements, because $A \cap (Z \setminus B) = A \setminus (A \cap B)$; hence it equals $\mathcal{L}$. Thus $A \cap B \in \mathcal{L}$ whenever $A \in \mathcal{P}$ and $B \in \mathcal{L}$. Repeating the argument with $A \in \mathcal{L}$ shows that $\mathcal{L}$ is closed under finite intersections. Writing a countable union $\bigcup_j A_j$ of members of $\mathcal{L}$ as the disjoint union of the sets $A_j \cap (Z \setminus A_1) \cap \cdots \cap (Z \setminus A_{j-1})$, we see that $\mathcal{L}$ is a σ-algebra. Hence $\sigma(\mathcal{P}) \subset \mathcal{L}$, and $\mathcal{L}$ is contained in every family as in (1).

(2) The sets $E \in \sigma(\mathcal{P})$ with $\lambda(E) = \lambda'(E)$ form a family which contains $\mathcal{P}$, is closed under countable disjoint unions by countable additivity, and is closed under complements because $\lambda(Z \setminus E) = \lambda(Z) - \lambda(E)$ and $\lambda'(Z \setminus E) = \lambda'(Z) - \lambda'(E)$ are differences of finite numbers. By (1), this family is all of $\sigma(\mathcal{P})$. □

Assertion (1) is Dynkin's π–λ theorem.

Proposition 1.1.7 *Let (X, μ) and (Y, ν) be σ-finite measure spaces. There is a unique measure $\mu \times \nu$ on the product σ-algebra of $X \times Y$ with $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ for all measurable $A \subset X$ and $B \subset Y$, where $0 \cdot \infty = 0$. If $f \geq 0$ is product-measurable, then $x \mapsto \int_Y f(x, y)\, \mathrm{d}\nu(y)$ and $y \mapsto \int_X f(x, y)\, \mathrm{d}\mu(x)$ are measurable and*

$$\int f \, \mathrm{d}(\mu \times \nu) = \int_X \left(\int_Y f(x, y)\, \mathrm{d}\nu(y) \right) \mathrm{d}\mu(x) = \int_Y \left(\int_X f(x, y)\, \mathrm{d}\mu(x) \right) \mathrm{d}\nu(y).$$

If f is complex-valued, product-measurable and $\int |f| \, \mathrm{d}(\mu \times \nu) < \infty$, then its sections are integrable for almost every value of the other variable, both iterated integrals exist, and the same equalities hold.

Proof The product σ-algebra is generated by the measurable rectangles $A \times B$, and the intersection of two rectangles is a rectangle, so Lemma 1.1.6 applies to the family of rectangles.

Step 1: finite measures. Suppose that μ and ν are finite. The sets $E \subset X \times Y$ whose sections $E_x = \{y : (x, y) \in E\}$ are all measurable form a σ-algebra containing the rectangles, so sections of product-measurable sets are measurable. The product-measurable sets E for which $x \mapsto \nu(E_x)$ is measurable include the rectangles, since $\nu((A \times B)_x) = \mathbf{1}_A(x)\nu(B)$. They are closed under complements, as $\nu((X \times Y \setminus E)_x) = \nu(Y) - \nu(E_x)$, and under countable disjoint unions, so by Lemma 1.1.6(1) they are all product-measurable sets.

By monotone convergence (Proposition 1.1.3), $\lambda(E) = \int_X \nu(E_x)\, \mathrm{d}\mu(x)$ is a measure with $\lambda(A \times B) = \mu(A)\nu(B)$. If λ' is another such measure, then $\lambda'(X \times Y) =$

$\mu(X)\nu(Y) = \lambda(X \times Y)$, so $\lambda' = \lambda$ by Lemma 1.1.6(2). In particular, λ coincides with the measure $E \mapsto \int_Y \mu(E^y)\,\mathrm{d}\nu(y)$, where $E^y = \{x : (x, y) \in E\}$, which is obtained by exchanging the roles of X and Y. This proves uniqueness and both integral identities for indicator functions. If $f \geq 0$ is product-measurable, its sections are measurable, because $\{y : f(x, y) > t\}$ is a section of $\{f > t\}$. Linearity extends the identities, together with the measurability of the inner integrals, to nonnegative simple functions, and monotone convergence, applied to an increasing sequence of such functions converging to f, extends them to f.

Step 2: σ-finite measures. Partition X and Y into countably many measurable sets X_m and Y_n of finite measure, and consider the finite measures $\mu_m(A) = \mu(A \cap X_m)$ on X and $\nu_n(B) = \nu(B \cap Y_n)$ on Y. Let λ_{mn} be the product of μ_m and ν_n given by Step 1, and put $\mu \times \nu = \sum_{m,n} \lambda_{mn}$. This is a measure on the product σ-algebra, and $(\mu \times \nu)(A \times B) = \sum_{m,n} \mu(A \cap X_m)\nu(B \cap Y_n) = \mu(A)\nu(B)$. If λ' is another measure with this property, then $\lambda'_{mn}(E) = \lambda'(E \cap (X_m \times Y_n))$ is a finite measure with $\lambda'_{mn}(A \times B) = \mu_m(A)\nu_n(B)$, so $\lambda'_{mn} = \lambda_{mn}$ by Step 1, and $\lambda' = \sum_{m,n} \lambda'_{mn} = \mu \times \nu$.

The integral of a nonnegative measurable function with respect to a countable sum of measures is the sum of the integrals. This holds for indicator functions by definition and for simple functions by linearity; in general it follows by monotone convergence, which also allows us to interchange a limit with a series of nonnegative terms. Applying this to $\mu \times \nu = \sum_{m,n} \lambda_{mn}$, to $\nu = \sum_n \nu_n$ and to $\mu = \sum_m \mu_m$, and using Step 1 for each λ_{mn}, we obtain for every nonnegative product-measurable f

$$\int f\,\mathrm{d}(\mu\times\nu) = \sum_{m,n} \int_X \left(\int_Y f(x, y)\,\mathrm{d}\nu_n(y) \right) \mathrm{d}\mu_m(x) = \int_X \left(\int_Y f(x, y)\,\mathrm{d}\nu(y) \right) \mathrm{d}\mu(x),$$

where $x \mapsto \int_Y f(x, y)\,\mathrm{d}\nu(y) = \sum_n \int_Y f(x, y)\,\mathrm{d}\nu_n(y)$ is measurable as a countable sum of measurable functions. The other iterated integral is treated in the same way, with the roles of X and Y exchanged.

Now let $\int |f|\,\mathrm{d}(\mu\times\nu) < \infty$. By the nonnegative case applied to $|f|$, the measurable function $x \mapsto \int_Y |f(x, y)|\,\mathrm{d}\nu(y)$ is integrable, so it is finite outside a measurable μ-null set N. For $x \notin N$ the section $f(x, \cdot)$ is integrable, and its integral is the corresponding combination of the integrals of the sections of the positive and negative parts of $\operatorname{Re} f$ and $\operatorname{Im} f$. These four functions have finite integrals with respect to $\mu \times \nu$; defining the inner integral to be zero on N and applying the nonnegative case to each of them, we obtain the first equality for f. The second follows by exchanging the roles of X and Y. □

We refer to these assertions as Tonelli's theorem (for nonnegative f) and Fubini's theorem (for integrable f). To apply them to Borel functions on products of topological spaces, we need to compare the Borel σ-algebra of a product with the product of the Borel σ-algebras.

Lemma 1.1.8 *Let Z_1 and Z_2 be separable metric spaces, and give $Z_1 \times Z_2$ the product topology. Then the Borel σ-algebra of $Z_1 \times Z_2$ is the σ-algebra generated by the products $A_1 \times A_2$ of Borel sets $A_1 \subset Z_1$ and $A_2 \subset Z_2$. Consequently, every Borel function on $Z_1 \times Z_2$ is measurable for the product of the Borel σ-algebras of Z_1 and*

Z_2, and Tonelli's and Fubini's theorems apply to it whenever Z_1 and Z_2 carry σ-finite Borel measures.

Proof Let p_1 and p_2 be the projections of $Z_1 \times Z_2$ onto its factors. For $i = 1, 2$, the sets $E \subset Z_i$ with $p_i^{-1}(E)$ Borel form a σ-algebra which contains the open sets, because p_i is continuous; hence it contains all Borel sets. Thus $A_1 \times A_2 = p_1^{-1}(A_1) \cap p_2^{-1}(A_2)$ is Borel whenever A_1 and A_2 are Borel. Conversely, we may assume that Z_1 and Z_2 are nonempty, and we choose countable dense subsets $D_1 \subset Z_1$ and $D_2 \subset Z_2$. Let $W \subset Z_1 \times Z_2$ be open and $(z_1, z_2) \in W$. There is $r > 0$ with $B(z_1, r) \times B(z_2, r) \subset W$, where $B(z_i, r)$ is the open ball in Z_i. Choose $a_i \in D_i$ with $\operatorname{dist}(a_i, z_i) < r/3$ and a rational number $q \in (r/3, 2r/3)$; then $z_i \in B(a_i, q) \subset B(z_i, r)$ by the triangle inequality. Hence W is the union of those of the countably many products $B(a_1, q_1) \times B(a_2, q_2)$, with $a_i \in D_i$ and positive rational q_i, that are contained in W. Each of them is a product of open sets, so every open subset of $Z_1 \times Z_2$, and therefore every Borel subset, belongs to the σ-algebra generated by the products of Borel sets. □

In particular, the Borel σ-algebra of $\mathbb{R}^{d_1+d_2} = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ is the product of the Borel σ-algebras of the factors. On Borel sets, Lebesgue measure on $\mathbb{R}^{d_1+d_2}$ is the product of the Lebesgue measures on the factors. Indeed, both measures assign to a product of half-open boxes the product of their volumes. These products are closed under finite intersections, and they generate the Borel σ-algebra, since every open set is a countable union of half-open boxes. By Lemma 1.1.6(2), the two measures agree on the Borel subsets of every box $[-N, N)^{d_1+d_2}$, hence on all Borel sets.

Recall that the completion of a measure λ on a σ-algebra $\mathcal{A}$ is its extension to the σ-algebra of all sets $E \cup N'$, where $E \in \mathcal{A}$ and N' is contained in a λ-null set of $\mathcal{A}$; Lebesgue measure is the completion of its restriction to the Borel sets. A function f which is measurable for the completion agrees λ-almost everywhere with an $\mathcal{A}$-measurable function g. Indeed, approximate f pointwise by simple functions measurable for the completion. Replace each of their level sets by a set of $\mathcal{A}$ contained in it and differing from it only inside a λ-null set of $\mathcal{A}$, and let g be the pointwise limit of the modified simple functions outside the countable union of these null sets, and 0 on that union.

In particular, every Lebesgue measurable function on Ω agrees almost everywhere with a Borel function, and in integrals with respect to Lebesgue measure we may use Borel representatives; for instance, if v is a Borel function on $\mathbb{R}^d$, then $(x, y) \mapsto v(x - y)$ is a Borel function on $\mathbb{R}^d \times \mathbb{R}^d$. Tonelli's and Fubini's theorems are often applied to functions that are only measurable for the completion of $\mu \times \nu$; statements about almost every section of such a function refer to a product-measurable representative.

Let α and β be Radon measures on Ω. The difference $\sigma = \alpha - \beta$ is defined on the relatively compact Borel subsets of Ω, and on all Borel subsets if α and β are finite. We call σ a real signed Radon measure, and a finite real signed measure if α and β are finite. The total variation of σ is

$$|\sigma|(A) = \sup_{A=\bigsqcup_{i=1}^m A_i} \sum_{i=1}^m |\sigma(A_i)|,$$

where A is a relatively compact Borel set and the supremum is taken over all finite partitions of A into Borel sets. Since $|\sigma(A_i)| \le \alpha(A_i) + \beta(A_i)$, we have $|\sigma| \le \alpha + \beta$.

We show that $|\sigma|$ is countably additive on the relatively compact Borel sets. It is finitely additive: if $A = A' \sqcup A''$, combining partitions of A' and A'' gives $|\sigma|(A) \ge |\sigma|(A') + |\sigma|(A'')$, and intersecting a partition of A with A' and A'' gives the reverse inequality by the triangle inequality. If A is the union of disjoint Borel sets A_j, finite additivity and monotonicity give $|\sigma|(A) \ge \sum_j |\sigma|(A_j)$. Conversely, for every finite partition $A = \bigsqcup_i B_i$, countable additivity of σ gives $\sum_i |\sigma(B_i)| \le \sum_i \sum_j |\sigma(B_i \cap A_j)| \le \sum_j |\sigma|(A_j)$. This argument uses only the countable additivity of σ on relatively compact Borel sets, the bound $|\sigma| \le \alpha + \beta$ and the triangle inequality.

To extend $|\sigma|$ to all Borel sets, let (K_j) be a compact exhaustion of Ω and $K_0 = \varnothing$. The relatively compact Borel sets $B_j = K_j \setminus K_{j-1}$ partition Ω, and we put $|\sigma|(E) = \sum_j |\sigma|(E \cap B_j)$ for Borel sets $E \subset \Omega$. This extension is countably additive, by rearrangement of double series of nonnegative terms, and it agrees with the previous definition on relatively compact sets by countable additivity there. It is bounded by $\alpha + \beta$, hence finite on compact sets, so it is a Radon measure by Proposition 1.1.2. In the same way, the set functions $\sigma^\pm = (|\sigma| \pm \sigma)/2$, which are nonnegative because $|\sigma(A)| \le |\sigma|(A)$, extend to Radon measures, and $\sigma = \sigma^+ - \sigma^-$ and $|\sigma| = \sigma^+ + \sigma^-$ on relatively compact Borel sets.

For a Borel function f which is integrable with respect to $|\sigma|$, for instance a bounded Borel function with compact support, we put $\int f \,\mathrm{d}\sigma = \int f \,\mathrm{d}\sigma^+ - \int f \,\mathrm{d}\sigma^-$. The measures $\sigma^+ + \beta$ and $\alpha + \sigma^-$ agree on relatively compact Borel sets, hence on all Borel sets, so $\int f \,\mathrm{d}\sigma = \int f \,\mathrm{d}\alpha - \int f \,\mathrm{d}\beta$ whenever f is integrable with respect to $\alpha + \beta$. The inequality $|\int f \,\mathrm{d}\sigma| \le \int |f| \,\mathrm{d}|\sigma|$ holds for simple f because $|\sigma^+(A) - \sigma^-(A)| \le |\sigma|(A)$, and in general by approximation with simple functions $s_k \to f$, $|s_k| \le |f|$, and dominated convergence. In this way, convergence theorems and iterated integrals for signed measures reduce to the positive case.

1.2 Positive linear functionals

A linear functional $L\colon C_c(\Omega, \mathbb{R}) \to \mathbb{R}$ is positive if $L(f) \ge 0$ whenever $f \ge 0$. Integration against a Radon measure is the basic example, and the Riesz representation theorem below shows that there are no others. Positivity alone gives a local bound. Let $K \subset \Omega$ be compact and let η be a cutoff function for K in Ω. If $f \in C_c(\Omega, \mathbb{R})$ and $\operatorname{Supp} f \subset K$, then $\|f\|_\infty \eta \pm f \ge 0$, and therefore

$$|L(f)| \le L(\eta)\|f\|_\infty, \qquad \operatorname{Supp} f \subset K, \tag{1.1}$$

where the constant $L(\eta)$ depends on K but not on f.

The measure representing L is obtained from an outer measure. Recall that an outer measure on a set X is a function m^* from the subsets of X to $[0,\infty]$ such that $m^*(\varnothing)=0$, $m^*(A)\le m^*(B)$ whenever $A\subset B$, and $m^*(\bigcup_j A_j)\le\sum_j m^*(A_j)$ for every sequence of subsets. A set $F\subset X$ is m^*-measurable if $m^*(E)=m^*(E\cap F)+m^*(E\setminus F)$ for every $E\subset X$. For subsets A,B of a metric space, $\operatorname{dist}(A,B)$ is the infimum of the distances between points of A and of B, and $+\infty$ if A or B is empty.

Lemma 1.2.1 *Let m^* be an outer measure on a metric space X such that*

$$m^*(A\cup B)=m^*(A)+m^*(B)\quad \textit{whenever } \operatorname{dist}(A,B)>0.$$

Then every Borel subset of X is m^*-measurable, and the restriction of m^* to the Borel sets is a measure.

Proof We first show that the m^*-measurable sets form a σ-algebra on which m^* is countably additive. The definition is symmetric in F and $X\setminus F$, so complements of measurable sets are measurable. If F and G are measurable and $E\subset X$, splitting E by F and then $E\cap F$ by G gives

$$\begin{aligned} m^*(E) &= m^*(E\cap F\cap G)+m^*((E\cap F)\setminus G)+m^*(E\setminus F)\\ &\ge m^*(E\cap F\cap G)+m^*(E\setminus(F\cap G)), \end{aligned}$$

where the inequality follows from subadditivity, since $E\setminus(F\cap G)=((E\cap F)\setminus G)\cup(E\setminus F)$. The reverse inequality is again subadditivity, so $F\cap G$ is measurable. If $F_1,F_2,\dots$ are disjoint measurable sets with union F, splitting E successively by $F_1,\dots,F_N$ gives

$$m^*(E)=\sum_{j=1}^{N} m^*(E\cap F_j)+m^*\Big(E\setminus\bigcup_{j=1}^{N}F_j\Big)\ge\sum_{j=1}^{N} m^*(E\cap F_j)+m^*(E\setminus F).$$

Letting $N\to\infty$ and using subadditivity, we obtain $m^*(E)\ge m^*(E\cap F)+m^*(E\setminus F)$, so F is measurable, and the choice $E=F$ gives $m^*(F)=\sum_j m^*(F_j)$. As every countable union of measurable sets is a disjoint union of measurable sets, this proves the claim.

It remains to show that every nonempty closed set $F\subset X$ is measurable. By subadditivity, we only need $m^*(E)\ge m^*(E\cap F)+m^*(E\setminus F)$, and only for sets $E\subset X$ with $m^*(E)<\infty$. For $N\ge 1$ put $E_N=\{x\in E:\operatorname{dist}(x,F)\ge N^{-1}\}$. Since $\operatorname{dist}(E\cap F,E_N)\ge N^{-1}$, the hypothesis gives $m^*(E)\ge m^*(E\cap F)+m^*(E_N)$, and it suffices to show that $m^*(E_N)\to m^*(E\setminus F)$. Consider the layers

$$A_k=\{x\in E:(k+1)^{-1}\le\operatorname{dist}(x,F)<k^{-1}\},\qquad k\ge 1.$$

Since $\operatorname{dist}(\cdot,F)$ is 1-Lipschitz, $\operatorname{dist}(A_k,A_{k'})\ge(k+1)^{-1}-(k+2)^{-1}>0$ whenever $k'\ge k+2$. Hence finitely many layers whose indices have the same parity are pairwise at positive distance, and repeated use of the hypothesis shows that the sum of their outer measures is at most $m^*(E)$. Consequently $\sum_k m^*(A_k)\le 2m^*(E)<\infty$. Since F is closed, $E\setminus F\subset E_N\cup\bigcup_{k\ge N}A_k$, and therefore

$$m^*(E_N) \le m^*(E \setminus F) \le m^*(E_N) + \sum_{k \ge N} m^*(A_k).$$

The tails of the convergent series tend to zero, so $m^*(E_N) \to m^*(E \setminus F)$. Thus the σ-algebra of m^*-measurable sets contains the closed sets, and hence all Borel sets.□

This is Carathéodory's criterion for metric outer measures.

Theorem 1.2.2 (Riesz representation) *Let L be a positive linear functional on $C_c(\Omega, \mathbb{R})$. Then there is a unique Radon measure μ on Ω such that $L(f) = \int f \, d\mu$ for every $f \in C_c(\Omega, \mathbb{R})$.*

Proof We construct μ from a metric outer measure (Lemma 1.2.1) and identify L with integration against μ by slicing functions into level sets. Call f admissible for an open set $U \subset \Omega$ if $f \in C_c(\Omega, \mathbb{R})$, $0 \le f \le 1$ and $\operatorname{Supp} f \subset U$, and put

$$m(U) = \sup\{L(f) : f \text{ admissible for } U\}.$$

Then $m(\varnothing) = 0$, and m is monotone.

Step 1: m is countably subadditive on open sets and additive on disjoint ones. Let $U_1, U_2, \ldots$ be open and let f be admissible for $U = \bigcup_j U_j$. Covering $\operatorname{Supp} f$ by finitely many open balls whose closures lie in some U_j, we find n and compact, possibly empty, sets $K_j \subset U_j$, $1 \le j \le n$, which cover $\operatorname{Supp} f$. If η_j is a cutoff function for K_j in U_j, then $\eta = \sum_{j=1}^n \eta_j \ge 1$ on $\operatorname{Supp} f$, so $\eta > 0$ on an open neighborhood W of $\operatorname{Supp} f$. The functions f_j equal to $f\eta_j/\eta$ on W and to 0 on $\Omega \setminus \operatorname{Supp} f$ are continuous, since both formulas give 0 where these open sets overlap; moreover $0 \le f_j \le f$, $\operatorname{Supp} f_j \subset U_j$ and $\sum_j f_j = f$. Hence $L(f) = \sum_{j=1}^n L(f_j) \le \sum_j m(U_j)$, and so $m(U) \le \sum_j m(U_j)$. If the U_j are pairwise disjoint and h_j is admissible for U_j, then the h_j have disjoint supports, so $\sum_{j \le N} h_j$ is admissible for U; taking suprema shows that m is countably additive on disjoint open sets.

Step 2: construction of μ. For $A \subset \Omega$ put $m^*(A) = \inf\{m(U) : U \supset A \text{ open}\}$. Then $m^*(U) = m(U)$ for open U by monotonicity of m. If $\sum_j m^*(A_j) < \infty$ (otherwise there is nothing to prove) and $\varepsilon > 0$, open sets $U_j \supset A_j$ with $m(U_j) < m^*(A_j) + \varepsilon 2^{-j}$ give $m^*(\bigcup_j A_j) \le \sum_j m^*(A_j) + \varepsilon$ by Step 1. Thus m^* is an outer measure. Let $A, B \subset \Omega$ be nonempty with $\operatorname{dist}(A, B) = 2r > 0$, and let $W \supset A \cup B$ be open. The sets $W \cap \{\operatorname{dist}(\cdot, A) < r\}$ and $W \cap \{\operatorname{dist}(\cdot, B) < r\}$ are open neighborhoods of A and B in W, which are disjoint by the triangle inequality, so Step 1 gives $m(W) \ge m^*(A) + m^*(B)$. Taking the infimum over W, we obtain the hypothesis of Lemma 1.2.1 (if A or B is empty, it holds because $m^*(\varnothing) = 0$), and that lemma yields a Borel measure μ with $\mu(U) = m(U)$ for open U.

It remains to show that μ is finite on compact sets. Every compact set $K \subset \Omega$ has a relatively compact open neighborhood V in Ω, for instance the set V used in the construction of cutoff functions in Section 1.1. If η is a cutoff function for $\overline{V}$ in Ω, then (1.1) gives $L(f) \le L(\eta)$ for every f admissible for V, so $\mu(K) \le m(V) \le L(\eta) < \infty$. By Proposition 1.1.2, μ is a Radon measure.

Step 3: $L(f) = \int f \, d\mu$. By linearity, it suffices to treat $f \ge 0$ with $M = \|f\|_\infty > 0$. Fix an integer $N \ge 1$, put $\delta = M/N$, and let $U_k = \{f > k\delta\}$ for $k \ge 0$, so

that $\mu(U_k) \le \mu(\operatorname{Supp} f) < \infty$. If h_k is admissible for U_k, $1 \le k \le N$, then $h_k \le \mathbf{1}_{U_k}$, and since at most $f(x)/\delta$ indices $k \ge 1$ satisfy $f(x) > k\delta$, we have $\delta \sum_{k=1}^N h_k \le \delta \sum_{k=1}^N \mathbf{1}_{U_k} \le f$. Positivity of L and the supremum over the h_k give the lower bound

$$\delta \sum_{k=1}^{N} \mu(U_k) \le L(f).$$

For an upper bound, put $g_k = \min\{1, \max\{0, f/\delta - (k-1)\}\}$ for $1 \le k \le N$; these are continuous with values in $[0,1]$, and $f = \delta \sum_{k=1}^N g_k$. If η is a cutoff function for $\operatorname{Supp} f$ in Ω, then $g_1 \le \eta$, so $L(g_1) \le L(\eta)$. For $k \ge 2$, g_k vanishes outside the compact set $\{f \ge (k-1)\delta\} \subset U_{k-2}$, so it is admissible for U_{k-2} and $L(g_k) \le \mu(U_{k-2})$. Consequently

$$L(f) \le \delta L(\eta) + \delta \sum_{k=2}^{N} \mu(U_{k-2}),$$

and the upper bound exceeds the lower bound by

$$\delta L(\eta) + \delta\big(\mu(U_0) - \mu(U_{N-1}) - \mu(U_N)\big) \le \delta\big(L(\eta) + \mu(\operatorname{Supp} f)\big).$$

On the other hand, the simple function $s_N = \delta \sum_{k=1}^N \mathbf{1}_{U_k}$ satisfies $0 \le f - s_N \le \delta$ and vanishes outside $\operatorname{Supp} f$, so $\int f \, \mathrm{d}\mu$ exceeds the lower bound $\int s_N \, \mathrm{d}\mu$ by at most $\delta\mu(\operatorname{Supp} f)$. Hence $|L(f) - \int f \, \mathrm{d}\mu| \le \delta(L(\eta) + \mu(\operatorname{Supp} f))$, and letting $N \to \infty$ gives $L(f) = \int f \, \mathrm{d}\mu$.

Step 4: uniqueness. Let ν be any Radon measure on Ω and $U \subset \Omega$ open. If $K \subset U$ is compact and η is a cutoff function for K in U, then $\nu(K) \le \int \eta \, \mathrm{d}\nu \le \nu(U)$, so inner regularity gives

$$\nu(U) = \sup\left\{ \int f \, \mathrm{d}\nu : 0 \le f \le 1,\ f \in C_c(\Omega),\ \operatorname{Supp} f \subset U \right\}. \tag{1.2}$$

Hence two Radon measures inducing the same functional agree on open sets, and by outer regularity on all Borel sets. □

Proposition 1.2.3 *Let μ be a Radon measure on Ω and let $1 \le p < \infty$. Then $C_c(\Omega)$ is dense in $L^p(\mu)$.*

Proof Let $u \in L^p(\mu)$ and let (K_j) be a compact exhaustion of Ω. The functions $u\mathbf{1}_{K_j \cap \{|u| \le j\}}$ converge to u in $L^p(\mu)$ by dominated convergence (Proposition 1.1.3), so it suffices to approximate a bounded Borel function v vanishing outside a compact set K. A uniform approximation of $\operatorname{Re} v$ and $\operatorname{Im} v$ by simple functions vanishing outside K, with error δ, has $L^p(\mu)$ error at most $2\delta\mu(K)^{1/p}$; we are thus reduced to indicator functions $\mathbf{1}_E$ of Borel sets $E \subset K$. Given $\varepsilon > 0$, regularity of μ gives a compact set $K' \subset E$ and an open set $U \supset E$ with $\mu(U \setminus K') < \varepsilon$. A cutoff function η for K' in U satisfies $|\eta - \mathbf{1}_E| \le \mathbf{1}_{U \setminus K'}$, so $\|\eta - \mathbf{1}_E\|_{L^p(\mu)}^p < \varepsilon$. □

1.3 Test functions and distributions

A locally integrable function need not be differentiable, but integration by parts shows how its derivatives should act: if $u \in C^1(\Omega)$ and χ is smooth with compact support in Ω, then $\int (\partial_j u)\chi \, \mathrm{d}x = -\int u \, \partial_j \chi \, \mathrm{d}x$, and the right side makes sense for every locally integrable u. Distributions are the continuous linear functionals on such functions χ.

A test function on Ω is an element of $C_c^\infty(\Omega)$. For a nonempty compact set $K \subset \Omega$, let $\mathcal{D}_K(\Omega)$ be the space of test functions supported in K. A multi-index is an element $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}^d$; we write $|\alpha| = \alpha_1 + \cdots + \alpha_d$ and $\partial^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}$. We give $\mathcal{D}_K(\Omega)$ the topology defined by the seminorms

$$p_{K,m}(\chi) = \max_{|\alpha| \le m} \sup_{x \in K} |\partial^\alpha \chi(x)|, \qquad m \in \mathbb{N},$$

and $\mathcal{D}(\Omega) = C_c^\infty(\Omega)$ the topology defined by all seminorms on $C_c^\infty(\Omega)$ whose restriction to every $\mathcal{D}_K(\Omega)$ is continuous.

Here a family $\mathcal{S}$ of seminorms on a complex vector space E defines the topology in which a set $G \subset E$ is open if for every $\chi_0 \in G$ there are $q_1, \ldots, q_k \in \mathcal{S}$ and $r > 0$ with $\{\chi : q_l(\chi - \chi_0) < r \text{ for } 1 \le l \le k\} \subset G$. A net (χ_i) in E converges to χ for this topology whenever $q(\chi_i - \chi) \to 0$ for every $q \in \mathcal{S}$, because every neighborhood of χ contains a set of the above form. A seminorm q on E is continuous if and only if there are $q_1, \ldots, q_k \in \mathcal{S}$ and $C \ge 0$ with $q \le C \max_l q_l$. Indeed, if q is continuous, the open set $\{q < 1\}$ contains a set $\{\chi : q_l(\chi) < r \text{ for } l \le k\}$. For $\chi \in E$ with $M = \max_l q_l(\chi) > 0$, the vector $r\chi/(2M)$ belongs to this set, so $q(\chi) < 2M/r$; if $M = 0$, then $t\chi$ belongs to it for every $t > 0$, so $q(\chi) = 0$. Conversely, such a bound gives $|q(\chi) - q(\chi_0)| \le q(\chi - \chi_0) \le C \max_l q_l(\chi - \chi_0)$, and q is continuous. The same argument, applied to $q = |T|$, shows that a linear functional T on E is continuous if and only if $|T| \le C \max_l q_l$ for some $q_1, \ldots, q_k \in \mathcal{S}$ and $C \ge 0$.

Since $p_{K,m}$ increases with m, a seminorm on $\mathcal{D}_K(\Omega)$ is therefore continuous if and only if it is bounded by a constant multiple of a single $p_{K,m}$. In particular, let test functions χ_j be supported in a fixed compact set K, and suppose that $\partial^\alpha \chi_j \to \partial^\alpha \chi$ uniformly for every α. Then χ is supported in K, and every seminorm q defining the topology of $\mathcal{D}(\Omega)$ satisfies $q(\chi_j - \chi) \le C p_{K,m}(\chi_j - \chi) \to 0$ for suitable C and m; hence $\chi_j \to \chi$ in $\mathcal{D}(\Omega)$.

Definition 1.3.1 A distribution on Ω is a complex linear functional $\chi \mapsto \langle T, \chi \rangle$ on $C_c^\infty(\Omega)$ such that for each nonempty compact set $K \subset \Omega$ there are $C > 0$ and $m \in \mathbb{N}$ with

$$|\langle T, \chi \rangle| \le C p_{K,m}(\chi) \qquad \text{for all } \chi \in \mathcal{D}_K(\Omega).$$

A distribution T is real if $\langle T, \chi \rangle$ is real for every real test function χ, and a real distribution is positive if $\langle T, \chi \rangle \ge 0$ for every nonnegative test function χ. The space of distributions on Ω is denoted by $\mathcal{D}'(\Omega)$.

The constants C and m may depend on K. The estimate is equivalent to continuity of T on $\mathcal{D}(\Omega)$. Indeed, the maximum of finitely many seminorms whose restrictions to every $\mathcal{D}_K(\Omega)$ are continuous is again such a seminorm, so by the criterion above, T

is continuous if and only if $|T| \le Cq$ for a single seminorm q defining the topology of $\mathcal{D}(\Omega)$. If T is continuous, the restriction of this q to $\mathcal{D}_K(\Omega)$ is continuous, hence bounded by a multiple of some $p_{K,m}$, and the estimate follows. Conversely, under the estimate, the restriction of the seminorm $\chi \mapsto |\langle T, \chi\rangle|$ to every $\mathcal{D}_K(\Omega)$ is continuous, so this seminorm is one of those defining the topology of $\mathcal{D}(\Omega)$, and T is continuous.

A locally integrable function u defines a distribution by $\langle u, \chi\rangle = \int_\Omega u\chi \,\mathrm{d}x$; on $\mathcal{D}_K(\Omega)$ the estimate holds with $m = 0$ and $C = 1 + \int_K |u| \,\mathrm{d}x$. Likewise, a Radon measure defines a distribution by integration (Theorem 1.4.5). These pairings are bilinear, without complex conjugation; other conventions are compared in Appendix B.1. Changing u on a null set does not change its distribution, and conversely, by Proposition 1.4.3 below, the distribution determines u up to equality almost everywhere. A distribution T vanishes on an open set $U \subset \Omega$ if $\langle T, \chi\rangle = 0$ whenever $\operatorname{Supp}\chi \subset U$, and its support $\operatorname{Supp} T$ is the complement in Ω of the union of all open sets on which T vanishes.

For a multi-index α, the distributional derivative of $T \in \mathcal{D}'(\Omega)$ is defined by

$$\langle \partial^\alpha T, \chi\rangle = (-1)^{|\alpha|}\langle T, \partial^\alpha \chi\rangle.$$

Since $p_{K,m}(\partial^\alpha \chi) \le p_{K,m+|\alpha|}(\chi)$, an estimate of order m for T on $\mathcal{D}_K(\Omega)$ gives one of order $m + |\alpha|$ for $\partial^\alpha T$. By integration by parts, this definition agrees with the classical derivative for smooth functions. If $a \in C^\infty(\Omega)$, the product aT is defined by $\langle aT, \chi\rangle = \langle T, a\chi\rangle$; it is a distribution because Leibniz's formula bounds the derivatives of $a\chi$ on K by those of χ. Unwinding the definitions gives $\partial_j(aT) = a\partial_j T + (\partial_j a)T$.

Example 1.3.2 For $x \in \Omega$, the Dirac measure δ_x assigns 1 to Borel sets containing x and 0 to the others; as a distribution, $\langle \delta_x, \chi\rangle = \chi(x)$. In one real dimension, with $0 \in \Omega$, we have $\langle \delta_0', \chi\rangle = -\chi'(0)$. This distribution is not given by integration against a real signed Radon measure in the sense of Section 1.1. Indeed, let θ be a test function with $\theta(0) = 1$, for instance a multiple of the function ρ_ε of Section 1.4 with ε small. The test functions $\theta(x)\sin(nx)$ are bounded by $\sup|\theta|$ and supported in $\operatorname{Supp}\theta$, while their pairings with δ_0' equal $-\theta(0)n\cos 0 - \theta'(0)\sin 0 = -n$. Integration against a real signed Radon measure σ would bound these pairings by $\sup|\theta| \cdot |\sigma|(\operatorname{Supp}\theta)$, since $|\int f \,\mathrm{d}\sigma| \le \int |f| \,\mathrm{d}|\sigma|$.

1.4 Convolution and mollifiers

Mollification, that is, convolution with a smooth kernel of small support, is our basic tool for approximating functions and distributions by smooth functions. The convolution of two functions f and g on $\mathbb{R}^d$ is

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y)g(x - y) \,\mathrm{d}y,$$

defined at every point x at which the integral exists. As kernel we take

$$\rho(x) = \begin{cases} c\exp(-(1-|x|^2)^{-1}), & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \qquad \int_{\mathbb{R}^d} \rho \, \mathrm{d}x = 1,$$

where the constant $c > 0$ normalizes the integral. Inside the unit ball, each partial derivative of ρ is a polynomial in x and $(1-|x|^2)^{-1}$ times $\exp(-(1-|x|^2)^{-1})$, and hence tends to zero at the unit sphere, because $t^k \mathrm{e}^{-t} \to 0$ as $t \to \infty$. Since a coordinate line meets the sphere in at most two points, the mean value theorem on coordinate segments shows inductively that all partial derivatives of ρ exist on $\mathbb{R}^d$, vanish on the sphere and are continuous. Thus ρ is a nonnegative function in $C_c^\infty(\mathbb{R}^d)$; it is radial, that is, $\rho(x)$ depends only on $|x|$, and in particular it is even.

We write $B(x,r) = \{y \in \mathbb{R}^d : |y-x| < r\}$ for the open ball with center x and radius $r > 0$; its closure $\overline{B(x,r)} = \{y \in \mathbb{R}^d : |y-x| \leq r\}$ is the closed ball. For $\varepsilon > 0$ the function $\rho_\varepsilon(x) = \varepsilon^{-d}\rho(x/\varepsilon)$ is radial, has integral 1 and is supported in $\overline{B(0,\varepsilon)}$. We call ρ and the functions ρ_ε mollifiers. Put

$$\Omega_\varepsilon = \{x \in \Omega : \operatorname{dist}(x, \mathbb{R}^d \setminus \Omega) > \varepsilon\}.$$

For $u \in L^1_{\mathrm{loc}}(\Omega)$, extended by zero outside Ω, the mollification of u is

$$u_\varepsilon(x) = (u * \rho_\varepsilon)(x) = \int_\Omega u(y)\rho_\varepsilon(x-y)\,\mathrm{d}y, \qquad x \in \Omega_\varepsilon.$$

The mollification u_ε is smooth on Ω_ε. To see this, let $\psi \in C_c^\infty(\mathbb{R}^d)$ be supported in $\overline{B(0,\varepsilon)}$, fix $x \in \Omega_\varepsilon$, and let $L \subset \Omega_\varepsilon$ be a closed ball about x. For $x' \in L$, only points y of the compact set $L' = \{y : \operatorname{dist}(y,L) \leq \varepsilon\}$ contribute to $(u*\psi)(x')$. Every point of L' lies within distance ε of a point of L, so $L' \subset \Omega$, and u is integrable on L'. Hence $|(u*\psi)(x') - (u*\psi)(x)| \leq \sup_y |\psi(x'-y) - \psi(x-y)| \int_{L'} |u|\,\mathrm{d}y$ for $x' \in L$, so $u*\psi$ is continuous at x by the uniform continuity of ψ. Moreover, if e_j denotes the jth coordinate unit vector, the mean value theorem and the uniform continuity of $\partial_j\psi$ show that $t^{-1}(\psi(x+te_j-y) - \psi(x-y)) \to \partial_j\psi(x-y)$ uniformly in y as $t \to 0$. Integrating against u over L', we obtain $\partial_j(u*\psi)(x) = (u*\partial_j\psi)(x)$. Applying this to $\psi = \partial^\alpha\rho_\varepsilon$, we find by induction on $|\alpha|$ that $u_\varepsilon \in C^\infty(\Omega_\varepsilon)$ and $\partial^\alpha u_\varepsilon = u * \partial^\alpha\rho_\varepsilon$.

Lemma 1.4.1 *In the following assertions, limits are taken as $\varepsilon \to 0$.*

(1) *If $u \in C(\Omega)$, then $u_\varepsilon \to u$ uniformly on every compact subset of Ω.*
(2) *Let $1 \leq p < \infty$. Translation is continuous in $L^p(\mathbb{R}^d)$: if $v \in L^p(\mathbb{R}^d)$, then $\|v(\cdot - h) - v\|_p \to 0$ as $h \to 0$. If $u \in L^p_{\mathrm{loc}}(\Omega)$, then $u_\varepsilon \to u$ in $L^p(K)$ for every compact set $K \subset \Omega$.*
(3) *If $\chi \in C_c^\infty(\Omega)$ is extended by zero to $\mathbb{R}^d$, then for small ε the functions $\chi * \rho_\varepsilon$ are supported in a fixed compact subset of Ω, and $\partial^\alpha(\chi*\rho_\varepsilon) \to \partial^\alpha\chi$ uniformly for every multi-index α.*
(4) *If $K \subset U \subset \Omega$ with K compact and U open, there is a smooth cutoff function for K in U which equals 1 on a neighborhood of K.*

(5) *If $f \in C_c(\Omega)$ is extended by zero to $\mathbb{R}^d$, then for small ε the functions $f * \rho_\varepsilon$ are test functions supported in a fixed compact subset of Ω, and they converge uniformly to f. Consequently, $C_c^\infty(\Omega)$ is dense in $L^p(\mu)$ for every Radon measure μ on Ω and every $1 \le p < \infty$.*

Proof (1) Since $\int \rho_\varepsilon = 1$, we have for $x \in \Omega_\varepsilon$

$$|u_\varepsilon(x) - u(x)| \le \int |u(x-h) - u(x)| \rho_\varepsilon(h)\, \mathrm{d}h \le \sup_{|h| \le \varepsilon} |u(x-h) - u(x)|.$$

If $K \subset \Omega$ is compact, choose $\delta > 0$ with $K' = \{x : \operatorname{dist}(x, K) \le \delta\} \subset \Omega$. For $\varepsilon < \delta$ we have $K \subset \Omega_\varepsilon$, because the closed ball of radius δ about each point of K lies in Ω, and the bound tends to zero uniformly on K by the uniform continuity of u on the compact set K'.

(2) We first prove the continuity of translation. For $v \in C_c(\mathbb{R}^d)$ it follows from uniform continuity, since the functions $v(\cdot - h)$, $|h| \le 1$, are supported in a fixed compact set; the general case follows by approximating v in L^p by such functions (Proposition 1.2.3 for Lebesgue measure), since translation preserves the L^p norm. Since $\int \rho_\varepsilon = 1$, we have $v * \rho_\varepsilon(x) - v(x) = \int (v(x-h) - v(x)) \rho_\varepsilon(h)\, \mathrm{d}h$. Hence, by the triangle inequality and Jensen's inequality (Lemma 1.1.4) for the convex function $t \mapsto |t|^p$ on $\mathbb{R}$, applied to the function $h \mapsto |v(x-h) - v(x)|$, which is integrable for the probability measure $\rho_\varepsilon(h)\, \mathrm{d}h$ because $|v| \le 1 + |v|^p$, we obtain

$$|v * \rho_\varepsilon(x) - v(x)|^p \le \int |v(x-h) - v(x)|^p \rho_\varepsilon(h)\, \mathrm{d}h \qquad \text{for every } x.$$

We may take v to be a Borel function, so that the integrand is a Borel function of (x, h). Integrating in x and using Tonelli's theorem, we obtain

$$\|v * \rho_\varepsilon - v\|_p^p \le \int \rho_\varepsilon(h) \|v(\cdot - h) - v\|_p^p\, \mathrm{d}h \le \sup_{|h| \le \varepsilon} \|v(\cdot - h) - v\|_p^p \to 0.$$

Now let $u \in L^p_{\mathrm{loc}}(\Omega)$ and $K \subset \Omega$ compact, choose $\delta > 0$ with $\{x : \operatorname{dist}(x, K) \le 2\delta\} \subset \Omega$, and let $v \in L^p(\mathbb{R}^d)$ equal u on $W = \{x : \operatorname{dist}(x, K) < 2\delta\}$ and 0 elsewhere. For $\varepsilon < \delta$, the values of u_ε on K only involve u on W, so $\|u_\varepsilon - u\|_{L^p(K)} \le \|v * \rho_\varepsilon - v\|_p \to 0$.

(3) The function $\chi * \rho_\varepsilon$ vanishes outside the set of points at distance at most ε from $\operatorname{Supp} \chi$, which is a compact subset of Ω for small ε. Differentiation under the integral sign gives $\partial^\alpha(\chi * \rho_\varepsilon) = (\partial^\alpha \chi) * \rho_\varepsilon$, and the bound displayed in (1), applied on $\mathbb{R}^d$ to the uniformly continuous function $\partial^\alpha \chi$ in place of u, proves uniform convergence.

(4) Choose $\delta > 0$ with $3\delta < \operatorname{dist}(K, \mathbb{R}^d \setminus U)$, and let η be a cutoff function for $\{x : \operatorname{dist}(x, K) \le 2\delta\}$ in $\{x : \operatorname{dist}(x, K) < 3\delta\}$. If $\varepsilon < \delta$ is smaller than the distance from $\operatorname{Supp} \eta$ to $\mathbb{R}^d \setminus U$, then $\eta * \rho_\varepsilon$ is smooth with values in $[0, 1]$ and compact support in U, and it equals 1 on $\{x : \operatorname{dist}(x, K) < \delta\}$, where the average only involves points at which $\eta = 1$.

(5) The support statement is proved as in (3), the functions $f * \rho_\varepsilon$ are smooth by differentiation under the integral sign, and uniform convergence follows from (1) applied on $\mathbb{R}^d$, since all these functions vanish outside a fixed compact set. If $g \in L^p(\mu)$, choose $f \in C_c(\Omega)$ close to g in $L^p(\mu)$ by Proposition 1.2.3. If all $f * \rho_\varepsilon$ are supported in the compact set L, then so is their uniform limit f, and $\|f * \rho_\varepsilon - f\|_{L^p(\mu)} \le \sup|f * \rho_\varepsilon - f|\, \mu(L)^{1/p} \to 0$. □

Smooth cutoff functions yield partitions of unity, which localize test functions and distributions.

Lemma 1.4.2 *Let $K \subset \Omega$ be compact, and let $U_1, \dots, U_n \subset \Omega$ be open sets covering K. Then there are smooth functions $\eta_j \colon \Omega \to [0,1]$ with compact support in U_j, $1 \le j \le n$, such that $\sum_j \eta_j \le 1$ on Ω and $\sum_j \eta_j = 1$ on a neighborhood of K. Consequently, every test function χ with $\operatorname{Supp}\chi \subset U_1 \cup \cdots \cup U_n$ is the sum of the test functions $\chi\eta_j$, which are supported in U_j and are nonnegative if χ is nonnegative. In particular, a distribution on Ω that vanishes on each member of a family of open subsets of Ω vanishes on their union.*

Proof For every $x \in K$ choose an index $j(x)$ with $x \in U_{j(x)}$ and a radius $r_x > 0$ with $\overline{B(x, r_x)} \subset U_{j(x)}$. Finitely many of the balls $B(x, r_x)$ cover K; let L_j be the union of the closures of those among them with $j(x) = j$. Then L_j is a compact, possibly empty, subset of U_j, and $K \subset L_1 \cup \cdots \cup L_n$. By Lemma 1.4.1(4), there are smooth cutoff functions ψ_j for L_j in U_j which equal 1 on neighborhoods V_j of L_j. Put $\eta_1 = \psi_1$ and $\eta_j = \psi_j(1 - \psi_1) \cdots (1 - \psi_{j-1})$ for $2 \le j \le n$. These functions are smooth with values in $[0,1]$, and $\operatorname{Supp}\eta_j \subset \operatorname{Supp}\psi_j$ is a compact subset of U_j. By induction on n,

$$\sum_{j=1}^{n} \eta_j = 1 - \prod_{j=1}^{n} (1 - \psi_j),$$

which takes values in $[0,1]$ and equals 1 on the neighborhood $V_1 \cup \cdots \cup V_n$ of K. If χ is a test function with $\operatorname{Supp}\chi \subset U_1 \cup \cdots \cup U_n$, we apply this to $K = \operatorname{Supp}\chi$; then $\chi = \sum_j \chi\eta_j$, and $\operatorname{Supp}(\chi\eta_j) \subset \operatorname{Supp}\eta_j$ is a compact subset of U_j. Finally, let a distribution T vanish on each member of a family of open subsets of Ω, and let χ be a test function supported in their union. The compact set $\operatorname{Supp}\chi$ is covered by finitely many members $U_1, \dots, U_n$ of the family, and the decomposition $\chi = \sum_j \chi\eta_j$ just obtained gives $\langle T, \chi\rangle = \sum_j \langle T, \chi\eta_j\rangle = 0$. □

By the last assertion of Lemma 1.4.2, every distribution T vanishes on $\Omega \setminus \operatorname{Supp} T$.

A distribution is mollified by letting it act on the translated kernel. Let $T \in \mathcal{D}'(\Omega)$, let $s > 0$, and let $\phi \in C_c^\infty(\mathbb{R}^d)$ be supported in $\overline{B(0,s)}$. If $x \in \Omega_s$, then $\phi(x - \cdot)$ is a test function on Ω, supported in $\overline{B(x,s)}$, and we put

$$(T * \phi)(x) = \langle T, \phi(x - \cdot)\rangle, \qquad x \in \Omega_s.$$

For T given by $u \in L^1_{\mathrm{loc}}(\Omega)$, this is the convolution $u * \phi$ of u, extended by zero, with ϕ. The function $T_\varepsilon = T * \rho_\varepsilon$ is called the mollification of T; for T given by u, it is the mollification u_ε of u.

Proposition 1.4.3 *Let $T \in \mathcal{D}'(\Omega)$ and $\varepsilon > 0$, and let $T_\varepsilon(x) = \langle T, \rho_\varepsilon(x - \cdot)\rangle$ for $x \in \Omega_\varepsilon$.*

(1) *If $s > 0$ and $\phi \in C_c^\infty(\mathbb{R}^d)$ is supported in $\overline{B(0,s)}$, then $T * \phi$ is smooth on Ω_s, and $\partial^\alpha(T*\phi) = T*\partial^\alpha\phi = (\partial^\alpha T)*\phi$ for every multi-index α. In particular, T_ε is smooth on Ω_ε, and $\partial^\alpha T_\varepsilon = (\partial^\alpha T)_\varepsilon$.*
(2) *For every test function χ, $\int T_\varepsilon\chi\,\mathrm{d}x \to \langle T,\chi\rangle$ as $\varepsilon \to 0$; the integral is defined once $\operatorname{Supp}\chi \subset \Omega_\varepsilon$.*
(3) *If T is positive, then $T_\varepsilon \geq 0$.*
(4) *A locally integrable function on Ω is determined by its distribution up to equality almost everywhere, and a Radon measure on Ω is determined by its distribution.*

Proof (1) Fix $x \in \Omega_s$ and $\delta > 0$ with $\overline{B(x,\delta)} \subset \Omega_s$. For $|x'-x| \leq \delta$, the test functions $\phi(x'-\cdot)$ are supported in the compact set $K = \overline{B(x,\delta+s)}$, which lies in Ω because $\overline{B(x',s)} \subset \Omega$ for every such x'. Taylor's formula, applied to all derivatives of ϕ, shows that the difference quotients $t^{-1}(\phi(x+te_j-\cdot)-\phi(x-\cdot))$, where e_j is a coordinate unit vector, converge to $(\partial_j\phi)(x-\cdot)$ in every seminorm $p_{K,m}$ as $t \to 0$; similarly, $\phi(x'-\cdot) \to \phi(x-\cdot)$ in every $p_{K,m}$ as $x' \to x$. By the estimate of Definition 1.3.1 on $\mathcal{D}_K(\Omega)$, $T*\phi$ is continuous at x and $\partial_j(T*\phi)(x) = \langle T, (\partial_j\phi)(x-\cdot)\rangle = (T*\partial_j\phi)(x)$. Applying the same argument to the derivatives of ϕ, which are also supported in $\overline{B(0,s)}$, we find by induction that $T*\phi$ is smooth and $\partial^\alpha(T*\phi) = T*\partial^\alpha\phi$.

Since $\partial_y^\alpha[\phi(x-y)] = (-1)^{|\alpha|}(\partial^\alpha\phi)(x-y)$, we have $((\partial^\alpha T)*\phi)(x) = (-1)^{|\alpha|}\langle T, \partial_y^\alpha[\phi(x-y)]\rangle = (T*\partial^\alpha\phi)(x)$. The case $\phi = \rho_\varepsilon$, $s = \varepsilon$, gives the assertions about T_ε.

(2) Let χ be a test function with $\operatorname{Supp}\chi \subset \Omega_\varepsilon$. We first prove that

$$\int T_\varepsilon(x)\chi(x)\,\mathrm{d}x = \langle T, \chi * \rho_\varepsilon\rangle. \tag{1.3}$$

Since ρ is even, $(\chi*\rho_\varepsilon)(y) = \int \chi(x)\rho_\varepsilon(x-y)\,\mathrm{d}x$, and $\chi*\rho_\varepsilon$ is supported in the compact set $K \subset \Omega$ of points at distance at most ε from $\operatorname{Supp}\chi$. Divide a closed cube containing $\operatorname{Supp}\chi$ into N^d congruent subcubes Q_l with centers c_l, and put $S_N(y) = \sum_l \chi(c_l)\rho_\varepsilon(c_l-y)|Q_l|$. Only centers $c_l \in \operatorname{Supp}\chi$ contribute, so $S_N \in \mathcal{D}_K(\Omega)$ and $\langle T, S_N\rangle = \sum_l \chi(c_l)T_\varepsilon(c_l)|Q_l|$, a Riemann sum for the continuous function $T_\varepsilon\chi$, extended by zero; it tends to $\int T_\varepsilon\chi\,\mathrm{d}x$.

On the other hand, each derivative $\partial^\beta S_N(y)$ is a Riemann sum for the integral over the cube of $x \mapsto \chi(x)\partial_y^\beta[\rho_\varepsilon(x-y)]$, and this integral is $\partial^\beta(\chi*\rho_\varepsilon)(y)$ by differentiation under the integral sign. The integrand is uniformly continuous in (x,y), so the difference between the Riemann sum and the integral is at most the volume of the cube times the largest oscillation of the integrand in x over a subcube, which tends to zero uniformly in y. Hence $S_N \to \chi*\rho_\varepsilon$ in every seminorm $p_{K,m}$, and the estimate for T on $\mathcal{D}_K(\Omega)$ gives $\langle T, S_N\rangle \to \langle T, \chi*\rho_\varepsilon\rangle$. This proves (1.3). By Lemma 1.4.1(3), the functions $\chi*\rho_\varepsilon$ are supported in a fixed compact set for small ε and converge to χ with all derivatives uniformly, so $\langle T, \chi*\rho_\varepsilon\rangle \to \langle T,\chi\rangle$.

(3) If T is positive, then $T_\varepsilon(x) \geq 0$ because $\rho_\varepsilon(x-\cdot)$ is a nonnegative test function.

(4) If $u \in L^1_{\mathrm{loc}}(\Omega)$ defines the zero distribution, then $u_\varepsilon = T_\varepsilon = 0$ for every ε, and $u_\varepsilon \to u$ in $L^1(K)$ for every compact K by Lemma 1.4.1(2); hence $u = 0$ almost everywhere. If two Radon measures μ and ν define the same distribution, then $\int f \,\mathrm{d}\mu = \int f \,\mathrm{d}\nu$ for every $f \in C_c(\Omega)$, because f is the uniform limit of test functions supported in a fixed compact set (Lemma 1.4.1(5)). The uniqueness in Theorem 1.2.2 gives $\mu = \nu$. □

Corollary 1.4.4 *Let $T \in \mathcal{D}'(\Omega)$ satisfy $\partial_j T = 0$ for $1 \le j \le d$. Then T is given by a locally constant function on Ω. If Ω is connected, T is a constant function.*

Proof Let B be an open ball with $\overline{B} \subset \Omega$, and fix $\theta \in C^\infty_c(B)$ with $\int \theta \,\mathrm{d}x = 1$, for instance a translate of ρ_s with $s > 0$ small. Since $\overline{B}$ is compact, $\overline{B} \subset \Omega_\varepsilon$ for all small ε. By Proposition 1.4.3(1), the smooth function T_ε has gradient $((\partial_j T)_\varepsilon)_j = 0$ on Ω_ε, so it is equal to a constant c_ε on the convex set B. By Proposition 1.4.3(2), $c_\varepsilon = \int T_\varepsilon \theta \,\mathrm{d}x \to \langle T, \theta\rangle =: c_B$, and therefore, for every $\chi \in C^\infty_c(B)$,

$$\langle T, \chi\rangle = \lim_{\varepsilon\to 0} \int T_\varepsilon \chi \,\mathrm{d}x = \lim_{\varepsilon\to 0} c_\varepsilon \int \chi \,\mathrm{d}x = c_B \int \chi \,\mathrm{d}x.$$

If two such balls B and B' intersect, testing with a nonnegative function $\chi \neq 0$ in $C^\infty_c(B \cap B')$ gives $c_B = c_{B'}$. Any two such balls containing a point $x \in \Omega$ intersect, so $c(x) = c_B$, for any such ball B containing x, is a well-defined function on Ω; it is locally constant, since $c = c_B$ on B. Every test function χ has compact support covered by finitely many such balls B_l, and by Lemma 1.4.2 it is a sum of test functions χ_l supported in B_l. Hence $\langle T, \chi\rangle = \sum_l c_{B_l} \int \chi_l \,\mathrm{d}x = \int c\chi \,\mathrm{d}x$. Finally, if Ω is connected, the locally constant function c is constant, because for each value a the set $\{c = a\}$ is open and so is its complement. □

Theorem 1.4.5 *Let T be a positive distribution on Ω. Then there is a unique Radon measure μ on Ω such that $\langle T, \chi\rangle = \int \chi \,\mathrm{d}\mu$ for every test function χ. Conversely, every Radon measure defines a positive distribution in this way.*

Proof Let $K \subset \Omega$ be compact and let η be a smooth cutoff function for K in Ω (Lemma 1.4.1(4)). For a real test function χ with $\operatorname{Supp}\chi \subset K$, the test functions $\|\chi\|_\infty \eta \pm \chi$ are nonnegative, so positivity gives, as in (1.1),

$$|\langle T, \chi\rangle| \le \langle T, \eta\rangle \|\chi\|_\infty.$$

We use this bound to extend T to $C_c(\Omega, \mathbb{R})$. If $f \in C_c(\Omega, \mathbb{R})$, the real test functions $f * \rho_\varepsilon$ are supported in a fixed compact set for small ε and converge uniformly to f (Lemma 1.4.1(5)). By the displayed estimate, $\langle T, f * \rho_\varepsilon\rangle$ converges as $\varepsilon \to 0$, and its limit $\Lambda(f)$ is also the limit of $\langle T, \chi_k\rangle$ for any real test functions χ_k supported in a common compact set and converging uniformly to f. Approximating two functions of $C_c(\Omega, \mathbb{R})$ by test functions with a common compact support, we see that Λ is linear. If $f \ge 0$, then $f * \rho_\varepsilon \ge 0$, so $\Lambda(f) \ge 0$.

By Theorem 1.2.2, there is a Radon measure μ with $\Lambda(f) = \int f \,\mathrm{d}\mu$. For a real test function χ, the constant sequence $\chi_k = \chi$ shows that $\Lambda(\chi) = \langle T, \chi\rangle$;

applying this to $\operatorname{Re}\chi$ and $\operatorname{Im}\chi$ gives $\langle T,\chi\rangle = \int \chi\,\mathrm{d}\mu$ for all test functions. The measure μ is unique by Proposition 1.4.3(4). Conversely, if μ is a Radon measure, then $|\int \chi\,\mathrm{d}\mu| \le (1+\mu(K))p_{K,0}(\chi)$ for $\chi \in \mathcal{D}_K(\Omega)$, so integration against μ is a distribution, and it is positive. □

1.5 Weak convergence of measures

The families of measures whose limits we shall need are not always sequences, so we work with nets. A directed set is a nonempty set I with a preorder $\le$ in which every two elements have a common upper bound. A net $(x_i)_{i\in I}$ in a topological space X is a map $i \mapsto x_i$ from a directed set I to X; it converges to $x \in X$ if for every neighborhood U of x there is $i_0 \in I$ with $x_i \in U$ for all $i \ge i_0$. For a net (a_i) in $[-\infty,\infty]$, $\underline{\lim}_i a_i = \sup_{i_0}\inf_{i\ge i_0} a_i$ and $\overline{\lim}_i a_i = \inf_{i_0}\sup_{i\ge i_0} a_i$.

The weak topology on $\mathcal{D}'(\Omega)$ is the weakest topology for which all maps $T \mapsto \langle T,\chi\rangle$, $\chi \in C_c^\infty(\Omega)$, are continuous. A set containing T is a neighborhood of T for this topology if and only if it contains a set $\{S : |\langle S-T,\chi_l\rangle| < r \text{ for } 1\le l\le k\}$ with finitely many test functions χ_l and $r>0$. Since the index set of a net is directed, it follows that a net (T_i) converges to T exactly when $\langle T_i,\chi\rangle \to \langle T,\chi\rangle$ for every test function χ, and we then say that $T_i \to T$ in the sense of distributions.

Likewise, the vague topology on the set of Radon measures on Ω is the weakest topology for which all maps $\mu \mapsto \int f\,\mathrm{d}\mu$, $f \in C_c(\Omega)$, are continuous. For the same reason, a net (μ_i) converges to μ in this topology, and we say that $\mu_i \to \mu$ vaguely, if and only if $\int f\,\mathrm{d}\mu_i \to \int f\,\mathrm{d}\mu$ for every $f \in C_c(\Omega)$. A net of finite Radon measures converges narrowly to a finite Radon measure μ if the same holds for every bounded continuous function f on Ω. Narrow convergence implies vague convergence, but the converse fails, because mass can escape to infinity or to the boundary of Ω (Example 1.5.4).

Proposition 1.5.1 *Let (T_i) be a net of positive distributions on Ω converging in the sense of distributions to $T \in \mathcal{D}'(\Omega)$. Then T is positive, and the Radon measures associated with the T_i by Theorem 1.4.5 converge vaguely to the Radon measure associated with T. Conversely, if a net of Radon measures converges vaguely, the associated distributions converge in the sense of distributions.*

Proof For a nonnegative test function χ, the numbers $\langle T_i,\chi\rangle$ are nonnegative, hence so is their limit $\langle T,\chi\rangle$; likewise $\langle T,\chi\rangle$ is real for real χ. Thus T is positive, and T and the T_i are given by Radon measures μ and μ_i. Let $f \in C_c(\Omega)$. By Lemma 1.4.1(5), f is the uniform limit of test functions supported in a fixed compact set L; in particular, f vanishes outside L. If η is a smooth cutoff function for L in Ω, then $\mu_i(L) \le \langle T_i,\eta\rangle \to \langle T,\eta\rangle$. A convergent net of real numbers need not be bounded, but it is bounded from some index on; hence there is i_0 with $\mu_i(L) \le C = \langle T,\eta\rangle + 1$ for $i \ge i_0$, and also $\mu(L) \le \langle T,\eta\rangle < C$. For every test function χ supported in L and every $i \ge i_0$,

$$\left|\int f\,\mathrm{d}\mu_i - \int f\,\mathrm{d}\mu\right| \le 2C\|f-\chi\|_\infty + |\langle T_i,\chi\rangle - \langle T,\chi\rangle|,$$

so $\overline{\lim}_i \left|\int f\,\mathrm{d}\mu_i - \int f\,\mathrm{d}\mu\right| \le 2C\|f-\chi\|_\infty$, which can be made arbitrarily small. Hence $\mu_i \to \mu$ vaguely. The converse holds because test functions belong to $C_c(\Omega)$.□

Thus, by the uniqueness in Theorem 1.4.5, a net of Radon measures converges vaguely to a Radon measure if and only if it converges to it in the sense of distributions. In Part II, where measures appear as currents of top degree, this mode of convergence is called weak convergence.

Proposition 1.5.2 *Let (μ_i) be a net of Radon measures on Ω converging vaguely to a Radon measure μ. Then for every open set $U \subset \Omega$ and every compact set $K \subset \Omega$,*

$$\mu(U) \le \underline{\lim_i}\,\mu_i(U), \qquad \overline{\lim_i}\,\mu_i(K) \le \mu(K).$$

Consequently, if $E \Subset \Omega$ is a Borel set with $\mu(\partial E) = 0$, then $\mu_i(E) \to \mu(E)$.

Proof If $f \in C_c(\Omega)$, $0 \le f \le 1$ and $\operatorname{Supp} f \subset U$, then $\int f\,\mathrm{d}\mu = \lim_i \int f\,\mathrm{d}\mu_i \le \underline{\lim}_i\,\mu_i(U)$; taking the supremum over such f and using (1.2) gives the first inequality. For the second, let $\varepsilon > 0$, choose by outer regularity an open set $V \supset K$ with $\mu(V) < \mu(K) + \varepsilon$, and let η be a cutoff function for K in V. Then $\mu_i(K) \le \int \eta\,\mathrm{d}\mu_i \to \int \eta\,\mathrm{d}\mu \le \mu(V)$, so $\overline{\lim}_i\,\mu_i(K) \le \mu(K) + \varepsilon$. Finally, for E as in the last assertion, the closure $\overline{E}$ is a compact subset of Ω, and $\overline{E} \setminus \operatorname{Int} E = \partial E$ is μ-null, so $\mu(\operatorname{Int} E) = \mu(E) = \mu(\overline{E})$. Since $\mu_i(\operatorname{Int} E) \le \mu_i(E) \le \mu_i(\overline{E})$, the two inequalities, applied to $\operatorname{Int} E$ and $\overline{E}$, give

$$\mu(E) = \mu(\operatorname{Int} E) \le \underline{\lim_i}\,\mu_i(E) \le \overline{\lim_i}\,\mu_i(E) \le \mu(\overline{E}) = \mu(E).$$

Proposition 1.5.3 *Let (μ_i) be a net of finite Radon measures on Ω converging vaguely to μ. Suppose that the net is eventually uniformly tight: for every $\varepsilon > 0$ there are a compact set $K \subset \Omega$ and an index i_0 such that $\mu_i(\Omega \setminus K) < \varepsilon$ for all $i \ge i_0$. Then μ is finite and $\mu_i \to \mu$ narrowly. In particular, $\mu_i(\Omega) \to \mu(\Omega)$.*

Proof Let $\varepsilon > 0$, and choose K and i_0 as above. Since $\Omega \setminus K$ is open, Proposition 1.5.2 gives $\mu(\Omega \setminus K) \le \varepsilon$, and since $\mu(K) < \infty$, the measure μ is finite. Let f be a bounded continuous function on Ω and η a cutoff function for K in Ω. Then $f\eta \in C_c(\Omega)$, so $\int f\eta\,\mathrm{d}\mu_i \to \int f\eta\,\mathrm{d}\mu$, while $1-\eta$ takes values in $[0,1]$ and vanishes on K, so that for $i \ge i_0$

$$\left|\int f(1-\eta)\,\mathrm{d}\mu_i\right| \le \|f\|_\infty\varepsilon, \qquad \left|\int f(1-\eta)\,\mathrm{d}\mu\right| \le \|f\|_\infty\varepsilon.$$

Hence $\overline{\lim}_i \left|\int f\,\mathrm{d}\mu_i - \int f\,\mathrm{d}\mu\right| \le 2\|f\|_\infty\varepsilon$. Letting $\varepsilon \downarrow 0$ proves narrow convergence, and $f = 1$ gives the last assertion. □

Example 1.5.4 On $\mathbb{R}$, the Dirac measures δ_j, $j \geq 1$, converge vaguely to zero, since every $f \in C_c(\mathbb{R})$ vanishes at j for large j; yet each has total mass one. On $\Omega = (0,1)$ the measures $\delta_{1/j}$, $j \geq 2$, behave in the same way, the mass escaping through the boundary point 0, which does not belong to Ω. Neither sequence is eventually uniformly tight: a compact subset of $\mathbb{R}$ does not contain j for large j, and a compact subset of $(0,1)$ does not contain $1/j$ for large j, so the mass outside a fixed compact set is eventually equal to one.

Theorem 1.5.5 *Let (μ_j) be a sequence of Radon measures on Ω such that $\sup_j \mu_j(K) < \infty$ for every compact set $K \subset \Omega$. Then (μ_j) has a vaguely convergent subsequence. If, moreover, the μ_j are finite and the sequence is eventually uniformly tight, then it has a narrowly convergent subsequence with a finite limit.*

Proof *Step 1: a countable dense family.* Let $\varphi(t) = \max\{0, 1-|t|\}$, so that $\sum_{n\in\mathbb{Z}} \varphi(t-n) = 1$ for $t \in \mathbb{R}$. For $k \geq 1$ and $n \in \mathbb{Z}^d$ put $\varphi_{k,n}(x) = \prod_{l=1}^d \varphi(2^k x_l - n_l)$. These are nonnegative continuous functions with $\sum_n \varphi_{k,n} = 1$, and $\varphi_{k,n}(x) \neq 0$ only if $|x - 2^{-k}n|_\infty < 2^{-k}$, where $|\cdot|_\infty$ is the maximum norm. Let $\mathcal{F}$ be the countable family of all finite sums $\sum_n q_n \varphi_{k,n}$, $k \geq 1$, with rational coefficients q_n, whose support is a compact subset of Ω. Given $f \in C_c(\Omega, \mathbb{R})$, extended by zero to $\mathbb{R}^d$, choose rational numbers $q_{k,n}$ with $|q_{k,n} - f(2^{-k}n)| \leq 2^{-k}$ and $q_{k,n} = 0$ if $f(2^{-k}n) = 0$, and put $f_k = \sum_n q_{k,n}\varphi_{k,n}$. Then

$$|f_k(x) - f(x)| \leq \sum_n |q_{k,n} - f(x)|\varphi_{k,n}(x) \leq 2^{-k} + \sup\{|f(y) - f(x)| : |y-x|_\infty < 2^{-k}\},$$

which tends to zero uniformly in x because f is uniformly continuous. Moreover, f_k vanishes at points of $|\cdot|_\infty$-distance more than 2^{-k} from $\operatorname{Supp} f$, so for large k all the f_k belong to $\mathcal{F}$ and are supported in a fixed compact set $L \subset \Omega$.

Step 2: diagonal extraction. Enumerate $\mathcal{F} = \{g_1, g_2, \ldots\}$. For each l, the numbers $\int g_l \, d\mu_j$ are bounded by $\|g_l\|_\infty \sup_j \mu_j(\operatorname{Supp} g_l)$. Choosing successively convergent subsequences for $g_1, g_2, \ldots$ and passing to the diagonal subsequence, we find (μ_{j_k}) such that $\int g_l \, d\mu_{j_k}$ converges for every l. If $f \in C_c(\Omega, \mathbb{R})$ and L, f_m are as in Step 1, then with $C_L = \sup_j \mu_j(L)$,

$$\left| \int f \, d\mu_{j_k} - \int f \, d\mu_{j_{k'}} \right| \leq 2C_L \|f - f_m\|_\infty + \left| \int f_m \, d\mu_{j_k} - \int f_m \, d\mu_{j_{k'}} \right|,$$

so $(\int f \, d\mu_{j_k})_k$ is a Cauchy sequence. Its limit is a positive linear functional of f, hence integration against a Radon measure μ by Theorem 1.2.2, and $\mu_{j_k} \to \mu$ vaguely; complex f are handled by linearity. If the sequence is eventually uniformly tight, so is every subsequence, and Proposition 1.5.3 gives the last assertion. □

1.6 Nets and compactness

In a general topological space, compactness cannot be tested with sequences alone. For instance, the product $[0,1]^{[0,1]}$ of copies of $[0,1]$, whose points are the functions $[0,1] \to [0,1]$, is compact by Tychonoff's theorem (Lemma 1.6.2 below), and convergence of a sequence in it is pointwise convergence. Let $f_n(t) = \lfloor 2^n t \rfloor - 2\lfloor 2^{n-1} t \rfloor$ be the nth binary digit of t. Given $n_1 < n_2 < \cdots$, the number $t = \sum_{k \text{ even}} 2^{-n_k}$ has a binary expansion with infinitely many zeros, whose digit at the place n_k is 1 for even k and 0 for odd k. Hence $f_{n_k}(t)$ alternates between 1 and 0, and no subsequence of (f_n) converges.

For nets, subnets take the place of subsequences. Recall from the Conventions that subnets are Willard subnets: if $(x_i)_{i \in I}$ is a net and $h\colon J \to I$ is an increasing map from a directed set J which is final, meaning that for every $i_0 \in I$ there is $j_0 \in J$ with $h(j) \geq i_0$ for all $j \geq j_0$, then $(x_{h(j)})_{j \in J}$ is a subnet of (x_i). Every subnet of a net converging to x converges to x: given a neighborhood U of x, choose i_0 with $x_i \in U$ for $i \geq i_0$, and then j_0 as in the definition of finality.

Proposition 1.6.1 *A topological space X is compact if and only if every net in X has a convergent Willard subnet.*

Proof Suppose that X is compact, and let $(x_i)_{i \in I}$ be a net in X. The closures T_{i_0} of the tails $\{x_i : i \geq i_0\}$ have the finite intersection property, since finitely many of them contain the tail of a common upper bound of their indices. As X is compact, they have a common point x; otherwise their complements would form an open cover without a finite subcover. Thus every neighborhood of x meets every tail of the net.

We now construct a subnet converging to x. Let J be the set of triples (i, U, k) with $i, k \in I$, $k \geq i$, U an open neighborhood of x and $x_k \in U$, preordered by

$$(i, U, k) \leq (i', U', k') \quad \text{if } i \leq i',\ U' \subset U,\ k \leq k'.$$

It contains (i, X, i) for every i, and it is directed: given two triples (i_1, U_1, k_1) and (i_2, U_2, k_2), choose $i \geq i_1, i_2, k_1, k_2$, put $U = U_1 \cap U_2$, and choose $k \geq i$ with $x_k \in U$, which is possible because U meets the tail of i; then (i, U, k) dominates both triples. The map $h(i, U, k) = k$ is increasing, and it is final because $(i, U, k) \geq (i_0, X, i_0)$ implies $k \geq i_0$. The subnet $(x_{h(j)})_{j \in J}$ converges to x: if V is a neighborhood of x, choose an open neighborhood $U \subset V$ of x and a triple $(i, U, k) \in J$; then $x_{k'} \in U' \subset V$ for every $(i', U', k') \geq (i, U, k)$.

Conversely, suppose that X is not compact, and let $\mathcal{U}$ be an open cover without a finite subcover. The finite subfamilies $F \subset \mathcal{U}$, ordered by inclusion, form a directed set, and we may choose points $x_F \notin \bigcup F$. If a subnet $(x_{h(j)})_{j \in J}$ converged to a point x, let $U \in \mathcal{U}$ contain x. Then $x_{h(j)} \in U$ for all large j, while finality gives $h(j) \supset \{U\}$, and hence $x_{h(j)} \notin U$, for all large j. Since J is directed, some j satisfies both conditions, a contradiction. □

Lemma 1.6.2 *A product of compact topological spaces is compact in the product topology.*

Proof This is Tychonoff's theorem; we prove it with ultrafilters, using the axiom of choice in the form of Zorn's lemma. A filter on a set Z is a family $\mathcal{F}$ of subsets of Z such that $Z \in \mathcal{F}$, $\varnothing \notin \mathcal{F}$, and $\mathcal{F}$ is closed under finite intersections and under passing to supersets. An ultrafilter is a filter which is maximal for inclusion. Since the union of a chain of filters containing a given filter is again a filter, Zorn's lemma shows that every filter is contained in an ultrafilter.

A filter $\mathcal{U}$ on Z is an ultrafilter if and only if, for every $A \subset Z$, either $A \in \mathcal{U}$ or $Z \setminus A \in \mathcal{U}$. Indeed, let $\mathcal{U}$ be an ultrafilter and $A \subset Z$. If A meets every member of $\mathcal{U}$, the sets containing some $U \cap A$ with $U \in \mathcal{U}$ form a filter containing $\mathcal{U}$ and A, so $A \in \mathcal{U}$ by maximality; otherwise some $U \in \mathcal{U}$ is disjoint from A, and $Z \setminus A \supset U$ belongs to $\mathcal{U}$. Conversely, if $\mathcal{U}$ has this property, a strictly larger filter would contain some $A \notin \mathcal{U}$ together with $Z \setminus A$, hence $\varnothing$.

A filter on a topological space Z converges to z if every neighborhood of z belongs to it. We show that Z is compact if and only if every ultrafilter on Z converges. If Z is compact and $\mathcal{U}$ is an ultrafilter, the closures of the members of $\mathcal{U}$ have the finite intersection property and hence a common point z. If an open neighborhood N of z were not in $\mathcal{U}$, the closed set $Z \setminus N$ would belong to $\mathcal{U}$. Since z lies in the closure of every member of $\mathcal{U}$, this would give $z \in Z \setminus N$, which is absurd. Hence $\mathcal{U}$ converges to z. Conversely, suppose that every ultrafilter converges, and let C be a family of closed sets with the finite intersection property. The sets containing a finite intersection of members of C form a filter, which lies in an ultrafilter $\mathcal{U}$ converging to some z. If $z \notin C$ for some $C \in C$, then both C and its complement, a neighborhood of z, would belong to $\mathcal{U}$. Hence z lies in every member of C, and Z is compact.

Now let $Z = \prod_\alpha X_\alpha$ with compact factors and projections p_α; if some X_α is empty, so is Z, and there is nothing to prove. Let $\mathcal{U}$ be an ultrafilter on Z. For each α, the family $\{A \subset X_\alpha : p_\alpha^{-1}(A) \in \mathcal{U}\}$ is a filter with the property characterizing ultrafilters, hence an ultrafilter, and since X_α is compact it converges to some $z_\alpha \in X_\alpha$. Put $z = (z_\alpha)$. A basic neighborhood $\bigcap_{l=1}^n p_{\alpha_l}^{-1}(V_l)$ of z, with V_l a neighborhood of z_{α_l}, is a finite intersection of members of $\mathcal{U}$, so every neighborhood of z belongs to $\mathcal{U}$. Thus every ultrafilter on Z converges, and Z is compact. □

Theorem 1.6.3 *Let $(K_r)_{r\geq 1}$ be a compact exhaustion of Ω. For each r, let $\chi_r \in C_c(\Omega)$ be a nonnegative function equal to 1 on K_r, and let $0 \leq M_r < \infty$. Then*

$$\mathcal{M} = \left\{ \mu \text{ Radon measure on } \Omega : \int \chi_r \, \mathrm{d}\mu \leq M_r \text{ for every } r \right\}$$

is compact in the vague topology. In particular, every net in $\mathcal{M}$ has a Willard subnet converging vaguely to an element of $\mathcal{M}$.

Proof We embed $\mathcal{M}$ homeomorphically onto a closed subset of a product of compact intervals. For $f \in C_c(\Omega, \mathbb{R})$, choose $r(f)$ with $\operatorname{Supp} f \subset K_{r(f)}$; this is possible because $\operatorname{Supp} f$ is compact and the open sets $\operatorname{Int} K_r$ increase to Ω. For $\mu \in \mathcal{M}$ we have $|f| \leq \|f\|_\infty \chi_{r(f)}$, so $\int f \, \mathrm{d}\mu$ lies in $I_f = [-M_{r(f)}\|f\|_\infty, M_{r(f)}\|f\|_\infty]$. The product $P = \prod_f I_f$ over all $f \in C_c(\Omega, \mathbb{R})$ is compact by Lemma 1.6.2. The map $\Phi \colon \mathcal{M} \to P$, $\Phi(\mu) = (\int f \, \mathrm{d}\mu)_f$, is injective by the uniqueness in Theorem 1.2.2, and it is a

homeomorphism onto its image. Indeed, since $\int f\,\mathrm{d}\mu = \int \operatorname{Re} f\,\mathrm{d}\mu + \mathrm{i}\int \operatorname{Im} f\,\mathrm{d}\mu$, the vague topology on $\mathcal{M}$ is the weakest topology for which the maps $\mu \mapsto \int f\,\mathrm{d}\mu$ with real f are continuous. The subspace topology of $\Phi(\mathcal{M})$ is the weakest topology for which the restrictions of the coordinate projections are continuous, and Φ intertwines these two families of maps.

The image consists of the points $(a_f) \in P$ such that $a_{sf+tg} = sa_f + ta_g$ for all f, g and real s, t, $a_f \geq 0$ whenever $f \geq 0$, and $a_{\chi_r} \leq M_r$ for every r: a point satisfying these conditions defines a positive linear functional on $C_c(\Omega, \mathbb{R})$, which by Theorem 1.2.2 is integration against a Radon measure belonging to $\mathcal{M}$. Each condition involves finitely many coordinates and is closed, so $\Phi(\mathcal{M})$ is closed in P and hence compact. Therefore $\mathcal{M}$ is vaguely compact, and the assertion about nets follows from Proposition 1.6.1. □

Proposition 1.6.4 (Arzelà–Ascoli) *Let K be a compact metric space, and let $\mathcal{F} \subset C(K)$ be equicontinuous and pointwise bounded, that is, $\sup_{f\in\mathcal{F}} |f(x)| < \infty$ for every $x \in K$. Then $\mathcal{F}$ is relatively compact for the uniform topology. In particular, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.*

Proof We may assume that $K \neq \varnothing$. Let $\varepsilon > 0$. By equicontinuity, every $x \in K$ is the center of an open ball on which $|f - f(x)| < \varepsilon$ for all $f \in \mathcal{F}$; by compactness, K is covered by finitely many such balls B_l with centers x_l, $1 \leq l \leq N$. Then every $f \in \mathcal{F}$ is bounded on K by $M = \varepsilon + \max_l \sup_{g\in\mathcal{F}} |g(x_l)|$, which is finite by pointwise boundedness. Let G be a finite set of scalars such that every scalar of absolute value at most M lies at distance less than ε from some element of G. For each map $\gamma\colon \{1, \dots, N\} \to G$, the functions $f \in \mathcal{F}$ with $|f(x_l) - \gamma(l)| < \varepsilon$ for all l form a set $\mathcal{F}_\gamma$, and these finitely many sets cover $\mathcal{F}$. If $f, g \in \mathcal{F}_\gamma$ and $y \in B_l$, then

$$|f(y) - g(y)| \leq |f(y) - f(x_l)| + |f(x_l) - g(x_l)| + |g(x_l) - g(y)| < 4\varepsilon.$$

Choosing one function in each nonempty $\mathcal{F}_\gamma$, we obtain a finite set within uniform distance 4ε of $\mathcal{F}$, and hence within 5ε of its closure $\overline{\mathcal{F}}$ in $C(K)$. Thus $\overline{\mathcal{F}}$ is totally bounded. It is also complete, because $C(K)$ is complete for the uniform metric: a uniform Cauchy sequence converges pointwise, and the limit is continuous since the convergence is uniform.

A complete and totally bounded metric space Y is sequentially compact and compact. Given a sequence in Y, cover Y by finitely many sets of diameter less than 1 and keep one containing infinitely many terms. Cover it by finitely many sets of diameter less than $1/2$, keep one containing infinitely many of those terms, and so on. A diagonal choice gives a Cauchy subsequence, which converges because Y is complete.

Suppose now that an open cover of Y had no finite subcover, and put $S_0 = Y$. Covering S_{n-1} by finitely many subsets of diameter less than $1/n$, we see that one of them, S_n, is not covered by finitely many members of the cover, since otherwise S_{n-1} would be. This gives nested nonempty sets $S_1 \supset S_2 \supset \cdots$ with $\operatorname{diam} S_n < 1/n$, none of which is covered by finitely many members of the cover. Points $y_n \in S_n$ form a Cauchy sequence with a limit y. If the open ball of radius s about y in Y lies

in a member U of the cover, then S_n lies in this ball, and hence in U, as soon as $1/n + \operatorname{dist}(y_n, y) < s$, a contradiction. Hence $\overline{\mathcal{F}}$ is compact, and every sequence in $\mathcal{F}$ has a uniformly convergent subsequence. □

1.7 Densities and differentiation of integrals

If a measure μ is dominated by a constant multiple of a measure ν, then μ has a bounded density with respect to ν. We construct it directly from averages over dyadic cubes and deduce the same for real distributions dominated by a multiple of ν. When ν is Lebesgue measure, the Lebesgue differentiation theorem recovers a density at almost every point from its averages over small cubes. Together with a Poincaré inequality on cubes, these results relate a distributional Hessian to a pointwise second-order expansion (Proposition 1.7.5), and they will later describe the coefficients of positive currents. For a Radon measure ν on Ω and a Borel function $g \geq 0$, we write $g\nu$ for the measure $E \mapsto \int_E g \, \mathrm{d}\nu$ on Borel sets, and $g \, \mathrm{d}x$ when ν is Lebesgue measure.

Lemma 1.7.1 *Let ν be a Radon measure and μ a Borel measure on Ω.*

(1) *If there is a constant $C \in [0, \infty)$ with $\mu(E) \leq C\nu(E)$ for every Borel set $E \subset \Omega$, then $\mu = g\nu$ for a Borel function g with values in $[0, C]$.*
(2) *If every relatively compact open set $U \subset \Omega$ admits a constant $C_U \in [0, \infty)$ with $\mu(E) \leq C_U \nu(E)$ for every Borel set $E \subset U$, then $\mu = g\nu$ for a nonnegative Borel function g such that $g \leq C_U$ ν-almost everywhere on U for every such U; in particular, g is locally essentially bounded with respect to ν.*

In both cases the density g is unique up to ν-null sets. In particular, if $\mu(E) \leq C|E|$ for every Borel set E, then $\mu = g \, \mathrm{d}x$ with $0 \leq g \leq C$.

Proof We first construct the density on a half-open cube as an L^2 limit of averages of μ over dyadic subcubes. Let $Q = \prod_{i=1}^{d} [a_i, a_i + s)$ be a half-open cube whose closure lies in Ω, so that $\nu(Q) < \infty$, and suppose that $\mu(E) \leq C\nu(E)$ for Borel sets $E \subset Q$; in particular, $\mu(Q) \leq C\nu(Q) < \infty$. For $k \geq 0$, let $\mathcal{Q}_k$ be the partition of Q into 2^{kd} half-open dyadic subcubes of side $2^{-k}s$. On each $Q' \in \mathcal{Q}_k$, let $g_k = \mu(Q')/\nu(Q')$ if $\nu(Q') > 0$ and $g_k = 0$ otherwise; in the latter case also $\mu(Q') = 0$, by domination. Thus $0 \leq g_k \leq C$, and $\int_{Q'} g_l \, \mathrm{d}\nu = \mu(Q')$ whenever $Q' \in \mathcal{Q}_k$ and $l \geq k$, because Q' is a disjoint union of cubes of $\mathcal{Q}_l$. As g_k is constant on the cubes of $\mathcal{Q}_k$, it follows that $\int_Q g_l g_k \, \mathrm{d}\nu = \int_Q g_k^2 \, \mathrm{d}\nu$, and hence, in $L^2(Q, \nu)$,

$$\|g_l - g_k\|_2^2 = \|g_l\|_2^2 - \|g_k\|_2^2 \qquad (l \geq k).$$

The squared norms increase and are bounded by $C^2\nu(Q)$, so (g_k) is a Cauchy sequence in $L^2(Q, \nu)$. Choose $k_1 < k_2 < \cdots$ with $\|g_{k_{j+1}} - g_{k_j}\|_2 \leq 2^{-j}$. By the Cauchy–Schwarz inequality, $\int_Q |g_{k_{j+1}} - g_{k_j}| \, \mathrm{d}\nu \leq \nu(Q)^{1/2} 2^{-j}$, so by monotone convergence $\sum_j |g_{k_{j+1}} - g_{k_j}|$ has finite integral; hence it is finite, and (g_{k_j}) converges,

ν-almost everywhere. Let g be the limit of (g_{k_j}) on the Borel set where it converges, and 0 elsewhere. Then g is a Borel function with values in $[0, C]$, $g_{k_j} \to g$ in $L^2(Q, \nu)$ by dominated convergence, and, the sequence being Cauchy, $g_k \to g$ in $L^2(Q, \nu)$ and hence in $L^1(Q, \nu)$. Consequently $\int_{Q'} g \, \mathrm{d}\nu = \mu(Q')$ for every dyadic subcube Q' of Q.

The dyadic subcubes of Q, together with $\varnothing$, form a family closed under finite intersections, since two of them are nested or disjoint. If U is open, every point of $U \cap Q$ lies in arbitrarily small dyadic subcubes, hence in one contained in U, so $U \cap Q$ is a countable union of dyadic subcubes. Thus these cubes generate the Borel σ-algebra of Q, and Lemma 1.1.6(2) shows that the finite measures μ and $g\nu$, which agree on all dyadic subcubes including Q, agree on every Borel subset of Q.

For uniqueness, let $g, h \geq 0$ be Borel functions with $g\nu = h\nu = \mu$ on the Borel subsets of such a cube Q. The set $A = Q \cap \{g > h\}$ satisfies $\int_A (g - h) \, \mathrm{d}\nu = \mu(A) - \mu(A) = 0$, because $\mu(A) < \infty$, so $\nu(A) = 0$; by symmetry, $g = h$ ν-almost everywhere on Q. Since Ω is a countable union of such cubes, densities on Ω are unique up to ν-null sets.

To construct the density on Ω in case (1), cover Ω by countably many half-open cubes $Q_1, Q_2, \ldots$ with closures in Ω, for instance all half-open dyadic cubes with this property, let $g^{(k)}$ be the density on Q_k just constructed, and put $g = g^{(k)}$ on $P_k = Q_k \setminus \bigcup_{l<k} Q_l$. For every Borel set $E \subset \Omega$,

$$\mu(E) = \sum_k \mu(E \cap P_k) = \sum_k \int_{E \cap P_k} g^{(k)} \, \mathrm{d}\nu = \int_E g \, \mathrm{d}\nu,$$

so $\mu = g\nu$ with $0 \leq g \leq C$. In case (2), every cube Q with closure in Ω lies in a relatively compact open set $U \subset \Omega$, for instance a set of points at distance less than δ from $\overline{Q}$ with δ small, and $\mu \leq C_U \nu$ on Borel subsets of Q; the same construction gives $\mu = g\nu$ with $g \geq 0$ Borel. If $U \subset \Omega$ is relatively compact and open, the sets $A_j = U \cap \{g > C_U + 1/j\}$ satisfy $(C_U + 1/j)\nu(A_j) \leq \mu(A_j) \leq C_U \nu(A_j)$ and $\nu(A_j) < \infty$, so $\nu(A_j) = 0$; thus $g \leq C_U$ ν-almost everywhere on U. The final assertion is case (1) for Lebesgue measure, which is a Radon measure by Proposition 1.1.2. □

Distributions bounded by a multiple of a measure are themselves given by bounded densities.

Corollary 1.7.2 *Let ν be a Radon measure on Ω, and let T be a real distribution on Ω.*

(1) *If there is a constant $C \geq 0$ with $|\langle T, \chi \rangle| \leq C \int \chi \, \mathrm{d}\nu$ for every nonnegative test function χ, then there is a Borel function g with values in $[-C, C]$ such that $\langle T, \chi \rangle = \int \chi g \, \mathrm{d}\nu$ for every test function χ.*

(2) *Suppose that every point of Ω has an open neighborhood $U \subset \Omega$ with a constant $C_U \geq 0$ such that $|\langle T, \chi \rangle| \leq C_U \int \chi \, \mathrm{d}\nu$ for every nonnegative test function χ supported in U. Then there is a real Borel function g such that $|g| \leq C_U$ ν-almost everywhere on each such U and $\langle T, \chi \rangle = \int \chi g \, \mathrm{d}\nu$ for every test function χ.*

In both cases, g is unique up to ν-null sets among the real Borel functions that are locally integrable with respect to ν.

Proof (1) Denote also by $C\nu$ the positive distribution defined by integration against the Radon measure $C\nu$. For every nonnegative test function χ we have $\langle C\nu \pm T, \chi\rangle = C\int \chi \,\mathrm{d}\nu \pm \langle T, \chi\rangle \geq 0$, so the real distributions $C\nu \pm T$ are positive, and by Theorem 1.4.5 they are given by Radon measures λ_+ and λ_-. The Radon measures $\lambda_+ + \lambda_-$ and $2C\nu$ define the same distribution, so they are equal by Proposition 1.4.3(4). In particular, $\lambda_+(E) \leq 2C\nu(E)$ for every Borel set $E \subset \Omega$, and Lemma 1.7.1(1) gives $\lambda_+ = h\nu$ for a Borel function h with values in $[0, 2C]$. Hence $\langle T, \chi\rangle = \int \chi \,\mathrm{d}\lambda_+ - C\int \chi \,\mathrm{d}\nu = \int \chi(h - C)\,\mathrm{d}\nu$ for every test function χ, and $g = h - C$ has the required properties.

Uniqueness. Let g and g' be real Borel functions, locally integrable with respect to ν, with $\int \chi g \,\mathrm{d}\nu = \int \chi g' \,\mathrm{d}\nu$ for every test function χ, and put $w = g - g'$. The Borel measures $w^+\nu$ and $w^-\nu$, where $w^\pm = \max\{\pm w, 0\}$, are finite on compact sets, hence Radon by Proposition 1.1.2, and they define the same distribution. By Proposition 1.4.3(4) they are equal, and evaluating them on the set $\{w > 0\}$ gives $\int_{\{w>0\}} w \,\mathrm{d}\nu = 0$, so that $\nu(\{w > 0\}) = 0$. Similarly $\nu(\{w < 0\}) = 0$, and $g = g'$ ν-almost everywhere.

(2) Let $\mathcal{U}$ be the family of the open sets U in the hypothesis. Each set of a compact exhaustion of Ω is covered by finitely many members of $\mathcal{U}$, so countably many members $U_1, U_2, \ldots$ of $\mathcal{U}$ cover Ω. Let $U \in \mathcal{U}$. The restriction of ν to the Borel subsets of U is a Radon measure on U by Proposition 1.1.2, and test functions on U, extended by zero, are test functions on Ω. Hence (1), applied on U, gives a Borel function g_U on U with values in $[-C_U, C_U]$ which represents T on U. By the uniqueness assertion, applied on the open sets $U_k \cap U_l$, we have $g_{U_k} = g_{U_l}$ ν-almost everywhere on $U_k \cap U_l$. Define $g = g_{U_k}$ on $P_k = U_k \setminus \bigcup_{l<k} U_l$. Since $P_l \cap U_k = \varnothing$ for $l > k$, we have $g = g_{U_k}$ ν-almost everywhere on U_k for every k. Hence g is locally integrable with respect to ν, and $\langle T, \chi\rangle = \int \chi g \,\mathrm{d}\nu$ for every test function χ supported in some U_k; by Lemma 1.4.2, this holds for every test function. Finally, for every $U \in \mathcal{U}$, the uniqueness assertion on U gives $g = g_U$ ν-almost everywhere on U, so $|g| \leq C_U$ ν-almost everywhere on U. □

Theorem 1.7.3 (Lebesgue differentiation) *Let f be a locally integrable function on Ω. Then almost every $x \in \Omega$ is a Lebesgue point of f, that is, $f(x)$ is finite and*

$$\lim_{r\downarrow 0} \frac{1}{|Q(x,r)|}\int_{Q(x,r)} |f(y) - f(x)|\,\mathrm{d}y = 0, \qquad Q(x,r) = x + (-r,r)^d.$$

At every Lebesgue point x of f, the mollifications satisfy $f_\varepsilon(x) \to f(x)$ as $\varepsilon \to 0$.

Proof We prove a maximal inequality on $\mathbb{R}^d$, deduce the theorem for integrable functions by approximation with continuous ones, and then localize. The set N where f is infinite is a null set. Changing f to 0 on N changes neither the averages in the statement nor the values $f(x)$ for $x \notin N$, so the modified function has the same

Lebesgue points outside N and the same mollifications. We may therefore assume that f is finite everywhere.

Consider first $f \in L^1(\mathbb{R}^d)$, and define its maximal function $Mf(x)$ as the supremum over $r > 0$ of the averages of $|f|$ over $Q(x,r)$. For fixed r, the average of $|f|$ over $Q(x,r)$ depends continuously on x, because $\int_{Q(x,r)} |f|$ and $\int_{Q(x',r)} |f|$ differ by at most $\|f(\cdot + x - x') - f\|_1$, which tends to zero as $x' \to x$ by the continuity of translation in L^1 (Lemma 1.4.1(2)). As a supremum of continuous functions, Mf is lower semicontinuous, so $\{Mf > t\}$ is open. We claim the maximal inequality

$$|\{Mf > t\}| \le \frac{3^d}{t}\|f\|_1, \qquad t > 0.$$

By inner regularity of Lebesgue measure, it suffices to bound $|L|$ for compact $L \subset \{Mf > t\}$. Each point of L is the center of an open cube over which the average of $|f|$ exceeds t, and finitely many of these cubes cover L. Select one of largest side length, discard all cubes meeting it, and repeat with the remaining ones. The selected cubes are disjoint, and each discarded cube meets a selected cube of at least the same side length, so it lies in the concentric cube with three times that side length. These enlarged cubes cover L, and therefore $|L|$ is at most 3^d times the total volume of the selected cubes, which is less than $t^{-1}\|f\|_1$.

Now let $g \in C_c(\mathbb{R}^d)$. For every x and r,

$$\frac{1}{|Q(x,r)|}\int_{Q(x,r)} |f(y) - f(x)|\,dy \le M(f-g)(x) + |g(x) - f(x)| + \frac{1}{|Q(x,r)|}\int_{Q(x,r)} |g(y) - g(x)|\,dy,$$

and the last term tends to zero as $r \downarrow 0$ because g is continuous at x. Hence the upper limit $\Lambda f(x)$ of the left side satisfies $\Lambda f \le M(f-g) + |f-g|$, and for $t > 0$ the set $\{\Lambda f > 2t\}$ is contained in $\{M(f-g) > t\} \cup \{|f-g| > t\}$, whose measure is at most $(3^d+1)t^{-1}\|f-g\|_1$ by the maximal inequality and Chebyshev's inequality. By Proposition 1.2.3, $\|f-g\|_1$ can be made arbitrarily small, so $\{\Lambda f > 2t\}$ is contained in measurable sets of arbitrarily small measure and is therefore a null set. Letting t run through the positive rationals shows that $\Lambda f = 0$ almost everywhere.

Next, let f be locally integrable on Ω, and let (K_j) be a compact exhaustion of Ω. The function equal to f on $\operatorname{Int} K_{j+1}$ and to 0 elsewhere belongs to $L^1(\mathbb{R}^d)$, and for $x \in \operatorname{Int} K_j$ and small r the averages in the statement are the same for f and for this function. Hence almost every point of each $\operatorname{Int} K_j$, and therefore of Ω, is a Lebesgue point of f.

Finally, let x be a Lebesgue point of f and let $\varepsilon > 0$ be so small that $x \in \Omega_\varepsilon$. The function $\rho_\varepsilon(x - \cdot)$ is bounded by $\varepsilon^{-d}\sup\rho = 2^d \sup\rho/|Q(x,\varepsilon)|$ and vanishes outside $B(x,\varepsilon) \subset Q(x,\varepsilon)$. Since $\int \rho_\varepsilon = 1$,

$$|f_\varepsilon(x) - f(x)| = \left| \int (f(y) - f(x))\rho_\varepsilon(x-y)\,\mathrm{d}y \right|$$
$$\leq \frac{2^d \sup \rho}{|Q(x,\varepsilon)|} \int_{Q(x,\varepsilon)} |f(y) - f(x)|\,\mathrm{d}y,$$

which tends to zero as $\varepsilon \to 0$. □

For Proposition 1.7.5 we also need the following Poincaré inequality on cubes.

Lemma 1.7.4 *Let $Q \subset \mathbb{R}^d$ be a closed cube of side length l, and let F be a locally integrable function on an open neighborhood V of Q whose distributional first derivatives $\partial_k F$ on V are locally integrable. Put $F_Q = |Q|^{-1} \int_Q F\,\mathrm{d}x$. Then*

$$\int_Q |F - F_Q|\,\mathrm{d}x \leq l \sum_{k=1}^{d} \int_Q |\partial_k F|\,\mathrm{d}x.$$

The same assertion holds componentwise for vector-valued F.

Proof Suppose first that F is continuously differentiable near Q. For $x, y \in Q$, the points $z^{(k)} = (y_1, \ldots, y_k, x_{k+1}, \ldots, x_d)$, $0 \leq k \leq d$, lead from x to y by changing one coordinate at a time. If I_k is the kth side interval of Q, the fundamental theorem of calculus gives

$$|F(y) - F(x)| \leq \sum_{k=1}^{d} \int_{I_k} |\partial_k F(y_1, \ldots, y_{k-1}, t, x_{k+1}, \ldots, x_d)|\,\mathrm{d}t.$$

Integrate over $(x, y) \in Q \times Q$. The kth term does not depend on the $d+1$ coordinates $x_1, \ldots, x_k, y_k, \ldots, y_d$, which contribute a factor l^{d+1}, and integration over the remaining coordinates and t gives $\int_Q |\partial_k F|\,\mathrm{d}x$. Since $|F(x) - F_Q| \leq |Q|^{-1} \int_Q |F(x) - F(y)|\,\mathrm{d}y$ and $|Q| = l^d$, the inequality follows. In general, the mollifications F_ε are smooth on $V_\varepsilon = \{x \in V : \operatorname{dist}(x, \mathbb{R}^d \setminus V) > \varepsilon\}$, which is a neighborhood of Q for small ε. By Lemma 1.4.1(2) and Proposition 1.4.3(1), $F_\varepsilon \to F$ and $\partial_k F_\varepsilon = (\partial_k F)_\varepsilon \to \partial_k F$ in $L^1(Q)$. Since $|(F_\varepsilon)_Q - F_Q| \leq |Q|^{-1} \|F_\varepsilon - F\|_{L^1(Q)}$, the left sides of the inequality for F_ε and for F differ by at most $2\|F_\varepsilon - F\|_{L^1(Q)}$, and the inequality for F_ε passes to the limit. □

Proposition 1.7.5 *Let $f : \Omega \to \mathbb{R}$ be continuous, and suppose that its real distributional Hessian has locally bounded measurable coefficients, that is, $\partial_a \partial_b f = A_{ab}$ for measurable functions A_{ab} on Ω that are bounded on compact sets, $1 \leq a, b \leq d$. Then $f \in C^{1,1}_{\mathrm{loc}}(\Omega)$: its first derivatives exist and are locally Lipschitz. More precisely, if $V \subset \Omega$ is a convex open set and $|A_{ab}| \leq C$ almost everywhere on V for all a and b, then ∇f is Lipschitz on V with constant dC. Moreover, with $A = (A_{ab})$, almost every $x \in \Omega$ satisfies*

$$f(x+h) = f(x) + \nabla f(x) \cdot h + \tfrac{1}{2} h^t A(x) h + o(|h|^2) \qquad (h \to 0).$$

Conversely, the hypothesis on the Hessian holds if the Hessians of smooth regularizations of f are locally uniformly bounded. More precisely, it suffices that every ball $B \Subset \Omega$ carry smooth functions f_j which converge to f in the sense of distributions on B and whose second derivatives are bounded on B uniformly in j.

Proof Steps 1 and 2 show by mollification that $f \in C^{1,1}_{\mathrm{loc}}(\Omega)$, Step 3 proves the Taylor expansion at the common Lebesgue points of the A_{ab} by means of Lemma 1.7.4, and Step 4 proves the converse.

Step 1: $f \in C^1(\Omega)$. Let $B = B(x_0, R)$ and $B' = B(x_0, R+s)$, where $R, s > 0$ are such that $\overline{B(x_0, R+2s)} \subset \Omega$, and let C bound the functions $|A_{ab}|$ on $B(x_0, R+2s)$. For $\varepsilon < s$, the closed ball $\overline{B'}$ lies in Ω_ε, the mollification f_ε is smooth near $\overline{B'}$, and $\partial_a \partial_b f_\varepsilon = (\partial_a \partial_b f)_\varepsilon = A_{ab} * \rho_\varepsilon$ there by Proposition 1.4.3(1). At points of $\overline{B'}$ this average involves only values of A_{ab} on $B(x_0, R+2s)$, so $|\partial_a \partial_b f_\varepsilon| \le C$ on $\overline{B'}$. By Lemma 1.4.1(1), $f_\varepsilon \to f$ uniformly on $\overline{B'}$, so $|f_\varepsilon| \le M$ on $\overline{B'}$ for some M and all small ε. For $x \in \overline{B}$ and a unit vector e, Taylor's formula

$$f_\varepsilon(x + se) = f_\varepsilon(x) + s\, \partial_e f_\varepsilon(x) + \int_0^s (s-t)\, \partial_e^2 f_\varepsilon(x + te)\, \mathrm{d}t,$$

in which $|\partial_e^2 f_\varepsilon| = |\sum_{a,b} e_a e_b \partial_a \partial_b f_\varepsilon| \le dC$, gives $|\partial_e f_\varepsilon(x)| \le 2M/s + dCs/2$. Thus the gradients ∇f_ε are bounded on $\overline{B}$ uniformly in small ε. They are also Lipschitz on the convex set $\overline{B}$ with constant dC. Indeed,

$$\nabla f_\varepsilon(y) - \nabla f_\varepsilon(x) = \int_0^1 D^2 f_\varepsilon(x + t(y-x))(y-x)\, \mathrm{d}t,$$

where $D^2 f_\varepsilon$ is the Hessian matrix, and a $d \times d$ matrix H with entries bounded by C satisfies $|Hv| \le \sqrt{d}\, C \sum_b |v_b| \le dC|v|$ for $v \in \mathbb{R}^d$, by the Cauchy–Schwarz inequality.

Applying Proposition 1.6.4 successively to the components of the maps $\nabla f_{1/k}$, $k > 1/s$, we find $\varepsilon_k \to 0$ such that ∇f_{ε_k} converges uniformly on $\overline{B}$ to a continuous map $g = (g_1, \ldots, g_d)$. Since $\partial_a f_\varepsilon = (\partial_a f)_\varepsilon \to \partial_a f$ in the sense of distributions by Proposition 1.4.3(1) and (2), the distribution $\partial_a f$ on B is the continuous function g_a. Letting $\varepsilon = \varepsilon_k \to 0$ in the identity

$$f_\varepsilon(y) - f_\varepsilon(x) = \int_0^1 \nabla f_\varepsilon(x + t(y-x)) \cdot (y-x)\, \mathrm{d}t, \qquad x, y \in B,$$

we obtain the same identity for f and g. Since g is continuous, it follows that f is differentiable on B with $\nabla f = g$. Covering Ω by such balls B, we conclude that $f \in C^1(\Omega)$ and that its distributional first derivatives coincide with the continuous functions $\partial_a f$ on each such ball, hence on Ω by the last assertion of Lemma 1.4.2. Consequently, $\partial_a f_\varepsilon = (\partial_a f)_\varepsilon \to \partial_a f$ uniformly on compact subsets of Ω, by Lemma 1.4.1(1).

Step 2: the Lipschitz bound. Let $V \subset \Omega$ be convex and open with $|A_{ab}| \le C$ almost everywhere on V, and let $x, y \in V$. The segment $[x, y]$ is a compact subset of V;

choose $\delta > 0$ smaller than its distance to $\mathbb{R}^d \setminus V$. For $\varepsilon < \delta$ and $z \in [x, y]$, the closed ball $\overline{B(z, \varepsilon)}$ lies in V, so $z \in \Omega_\varepsilon$ and $|\partial_a \partial_b f_\varepsilon(z)| = |(A_{ab} * \rho_\varepsilon)(z)| \le C$. As in Step 1, $|\nabla f_\varepsilon(y) - \nabla f_\varepsilon(x)| \le dC|y-x|$, and letting $\varepsilon \to 0$ gives $|\nabla f(y) - \nabla f(x)| \le dC|y-x|$. Every point of Ω lies in a ball $V \Subset \Omega$, on which the A_{ab} are bounded, so ∇f is locally Lipschitz and $f \in C^{1,1}_{\mathrm{loc}}(\Omega)$.

Step 3: the Taylor expansion. By Theorem 1.7.3, almost every point of Ω is a common Lebesgue point of the functions A_{ab}. Let x be such a point, let $Q = (-1, 1)^d$, and choose $r_0 > 0$ with $x + 3r_0\overline{Q} \subset \Omega$. The functions $|A_{ab}|$ are bounded on the compact set $x + 3r_0\overline{Q}$ by some constant C, so by Step 2, ∇f is Lipschitz with constant $\lambda = dC$ on the convex open set $x + 3r_0 Q$, which contains $x + 2r_0\overline{Q}$. For $0 < r \le r_0$, define on $2Q$

$$G_r(y) = \frac{\nabla f(x + ry) - \nabla f(x)}{r} - A(x)y.$$

We show that $G_r \to 0$ uniformly on $\overline{Q}$ as $r \downarrow 0$; the expansion then follows by integrating ∇f along segments. The maps G_r vanish at 0 and are Lipschitz with the common constant $\lambda + |A(x)|$, where $|A(x)|$ is the operator norm, and hence also with $\Lambda = \lambda + |A(x)| + 1 > 0$.

To compute the derivatives of G_r, let ψ be a test function on $2Q$. The function $z \mapsto \psi((z - x)/r)$ is a test function on Ω, and the distributional derivative ∂_b of the continuous function $\partial_a f$ is $\partial_b \partial_a f = \partial_a \partial_b f = A_{ab}$, since partial derivatives of distributions commute. Hence the change of variables $z = x + ry$ gives

$$\begin{aligned} \int \frac{\partial_a f(x + ry)}{r}\, \partial_b \psi(y)\, \mathrm{d}y &= r^{-d} \int \partial_a f(z)\, \partial_{z_b}\big[\psi((z - x)/r)\big]\, \mathrm{d}z \\ &= -\int A_{ab}(x + ry)\psi(y)\, \mathrm{d}y. \end{aligned}$$

Thus $\partial_{y_b}(G_r)_a(y) = A_{ab}(x + ry) - A_{ab}(x)$ in the sense of distributions on $2Q$, and since x is a Lebesgue point,

$$\int_Q |A_{ab}(x + ry) - A_{ab}(x)|\, \mathrm{d}y = \frac{1}{r^d} \int_{Q(x,r)} |A_{ab}(z) - A_{ab}(x)|\, \mathrm{d}z \to 0 \qquad (r \downarrow 0).$$

Lemma 1.7.4, applied on the cube $\overline{Q}$ of side length 2 to each component of G_r, gives

$$\int_Q |G_r - (G_r)_Q|\, \mathrm{d}y \le 2 \sum_{a,b} \int_Q |A_{ab}(x + ry) - A_{ab}(x)|\, \mathrm{d}y \to 0 \qquad (r \downarrow 0).$$

We deduce that $\sup_{\overline{Q}} |G_r - (G_r)_Q| \to 0$. Otherwise there would be $\kappa > 0$, radii $r_k \to 0$ and points $y_k \in \overline{Q}$ with $|G_{r_k}(y_k) - (G_{r_k})_Q| \ge \kappa$, and by the common Lipschitz bound, $|G_{r_k} - (G_{r_k})_Q| \ge \kappa/2$ on $Q \cap B(y_k, \kappa/(2\Lambda))$. For $y \in \overline{Q}$ and $0 < \sigma \le 1$, the open cube $\prod_l I_l$, with $I_l = (y_l - \sigma, y_l)$ if $y_l > 0$ and $I_l = (y_l, y_l + \sigma)$ if $y_l \le 0$, lies in Q and within distance $\sqrt{d}\sigma$ of y. With $\sigma = \min\{1, \kappa/(2\Lambda\sqrt{d})\}$, this shows that $Q \cap B(y_k, \kappa/(2\Lambda))$ has measure at least σ^d for every k, so that $\int_Q |G_{r_k} - (G_{r_k})_Q|\, \mathrm{d}y \ge \sigma^d \kappa/2$, contradicting the convergence of these integrals

to zero. Since $G_r(0) = 0$, also $|(G_r)_Q| = |G_r(0) - (G_r)_Q| \to 0$, and hence $\sup_{\overline{Q}} |G_r| \to 0$ as $r \downarrow 0$.

Now let $0 < |h| \le r_0$, and put $r = |h|$ and $y = h/|h| \in \overline{Q}$. For $0 < t \le 1$ we have $\nabla f(x + th) - \nabla f(x) = rt\, G_{rt}(y) + tA(x)h$, so, as f is continuously differentiable,

$$f(x + h) - f(x) - \nabla f(x) \cdot h - \tfrac{1}{2} h^t A(x) h = \int_0^1 rt\, G_{rt}(y) \cdot h \, \mathrm{d}t.$$

The absolute value of the right side is at most $|h|^2 \sup_{0<\varrho\le|h|} \sup_{\overline{Q}} |G_\varrho| = o(|h|^2)$.

Step 4: the converse. Let $B \Subset \Omega$ be a ball, and let f_j be as in the statement, with $|\partial_a \partial_b f_j| \le C_B$ on B for all j. For every nonnegative test function χ supported in B, two integrations by parts give

$$\langle \partial_a \partial_b f, \chi \rangle = \lim_j \int f_j\, \partial_a \partial_b \chi \, \mathrm{d}x = \lim_j \int (\partial_a \partial_b f_j) \chi \, \mathrm{d}x,$$

so $|\langle \partial_a \partial_b f, \chi \rangle| \le C_B \int \chi \, \mathrm{d}x$. The distribution $\partial_a \partial_b f$ is real, because f is real, and every point of Ω lies in such a ball B. By Corollary 1.7.2(2), applied with ν equal to Lebesgue measure, $\partial_a \partial_b f$ is given by a Borel function A_{ab} with $|A_{ab}| \le C_B$ almost everywhere on every such ball B. Each set of a compact exhaustion of Ω is covered by finitely many such balls, so countably many such balls B_k cover Ω. Redefine A_{ab} to be 0 on the null set $\bigcup_k \{x \in B_k : |A_{ab}(x)| > C_{B_k}\}$. The new function still represents $\partial_a \partial_b f$, and it is bounded on every compact set, because a compact set is covered by finitely many of the balls B_k. Finally, since $f_\varepsilon \to f$ in the sense of distributions (Proposition 1.4.3(2)), the mollifications $f_j = f_{\varepsilon_j}$ with $\varepsilon_j \to 0$ satisfy the hypothesis of the converse whenever their second derivatives are bounded on each ball $B \Subset \Omega$ uniformly in small ε. □

Chapter 2
Harmonic functions and the Dirichlet problem

Harmonic functions are the solutions of the Laplace equation $\Delta h = 0$. In potential theory they play the role that affine functions play for convex functions: the subharmonic functions of Chapter 3 are studied by comparison with them, and the fundamental solution of the Laplacian, together with the Green functions obtained from it by harmonic corrections, defines the potentials and energies of Chapters 4 and 5. Unless stated otherwise, Ω denotes an open subset of $\mathbb{R}^d$, where $d \geq 1$.

We begin with mean values. After introducing surface measure, polar coordinates and the divergence theorem on balls, we show that a continuous function is harmonic exactly when its value at the center of every closed ball in its domain equals its mean over the boundary sphere (Theorem 2.1.2). Averaging against a mollifier then shows that harmonic functions are smooth, with interior estimates for all their derivatives. A distribution in the sense of Chapter 1 need not have pointwise values; nevertheless, Weyl's lemma (Theorem 2.1.4) shows that a distribution with vanishing Laplacian is a smooth harmonic function. The fundamental solution of the Laplacian (Proposition 2.2.1) fixes the sign conventions for all later potentials.

The Dirichlet problem asks for a harmonic function with prescribed continuous boundary values. On a ball it is solved explicitly by the Poisson integral (Theorem 2.3.1), whose kernel also yields the Harnack inequalities and the compactness of families of nonnegative harmonic functions. On a general bounded open set there is no such formula, and we use Perron's method instead. The candidate solution is the pointwise supremum of all continuous subsolutions whose boundary values do not exceed the data; replacing a subsolution on a small ball by the Poisson integral of its values shows that this envelope is harmonic (Theorem 2.4.2). The envelope also depends linearly on the data (Theorem 2.4.4). A boundary point is regular if the envelope attains the prescribed value there for all data. Regularity is decided by the existence of a barrier, which need only be constructed near the point (Theorem 2.4.6 and Proposition 2.4.7); for $d \geq 2$, regularity fails at the center of a punctured ball. When every boundary point is regular, the solution operator is integration against harmonic measure, and correcting the fundamental solution by the harmonic function with the same boundary values gives the Green function (Theorem 2.5.2).

Historical remarks and references for this chapter are collected in Appendix B.2.

2.1 The Laplacian and mean values

Recall from Section 1.4 that $B(x,r)$ is the open ball with center x and radius $r>0$; its closure $\overline{B(x,r)}$ is the closed ball. We write $S^{d-1}=\{y\in\mathbb{R}^d : |y|=1\}$ for the unit sphere. The Laplacian is $\Delta=\sum_{j=1}^d \partial_{x_j}^2$. A real function $h\in C^2(\Omega)$ is *harmonic* if $\Delta h=0$, and a complex function is harmonic if its real and imaginary parts are harmonic. In dimension one the harmonic functions are the functions that are affine on each component interval of Ω, because $h''=0$ on an interval forces h' to be constant there.

We define surface measure on the unit sphere through the Lebesgue measure of cones: for a Borel set $A\subset S^{d-1}$ put

$$S(A)=d\,\bigl|\{t\omega : 0<t<1,\ \omega\in A\}\bigr|.$$

The cone over A is the inverse image of A under the continuous map $y\mapsto y/|y|$ on $B(0,1)\setminus\{0\}$, so it is a Borel set; moreover, the cone over a countable union is the union of the cones, and cones over disjoint sets are disjoint. Hence S is a finite Borel measure on S^{d-1}, the *surface measure*. It is invariant under every orthogonal transformation Q of $\mathbb{R}^d$: the cone over QA is the image under Q of the cone over A, and Lebesgue measure is invariant under orthogonal transformations. By linearity and monotone convergence (Proposition 1.1.3), it follows that $\int_{S^{d-1}} g\circ Q\,\mathrm{d}S=\int_{S^{d-1}} g\,\mathrm{d}S$ for every nonnegative or integrable Borel function g.

We write $s_{d-1}=S(S^{d-1})$ for the area of the unit sphere, so that the unit ball has volume s_{d-1}/d. For instance, $s_0=2$, because S is counting measure on $S^0=\{\pm1\}$, and $s_1=2\pi$, because the unit disc has area π. On a sphere $\partial B(x,r)$ we use the measure obtained by translation and dilation, again denoted $\mathrm{d}S$:

$$\int_{\partial B(x,r)} g\,\mathrm{d}S=r^{d-1}\int_{S^{d-1}} g(x+r\omega)\,\mathrm{d}S(\omega).$$

Next we express Lebesgue measure in polar coordinates. Let $0\le a<b$ and let $A\subset S^{d-1}$ be a Borel set. By dilation, the cone $\{t\omega : 0<t<b,\ \omega\in A\}$ has Lebesgue measure $b^dS(A)/d$. The set $\{t\omega : 0<t\le a,\ \omega\in A\}$ is the intersection of the decreasing cones $\{t\omega : 0<t<a+1/n,\ \omega\in A\}$, so its measure is $\lim_n (a+1/n)^dS(A)/d=a^dS(A)/d$. Subtracting and translating, we find that for every $x\in\mathbb{R}^d$ the *annular sector* $\{x+t\omega : a<t<b,\ \omega\in A\}$ has Lebesgue measure

$$\frac{(b^d-a^d)S(A)}{d}=S(A)\int_a^b t^{d-1}\,\mathrm{d}t.$$

Now fix $x\in\mathbb{R}^d$ and $R>0$, let λ be the measure $t^{d-1}\,\mathrm{d}t$ on the Borel sets of $(0,R)$, and let $\Phi(t,\omega)=x+t\omega$. The map Φ is a homeomorphism from $(0,R)\times S^{d-1}$ onto $Z=B(x,R)\setminus\{x\}$, with inverse $y\mapsto(|y-x|,(y-x)/|y-x|)$, and the annular sectors with center x contained in Z are the images under Φ of the rectangles $(a,b)\times A$ with $b\le R$. These sectors generate the Borel σ-algebra of Z. Indeed, the sets $(a,b)\times(S^{d-1}\cap B(\omega,q))$, where $0<a<b<R$ are rational, ω runs through a

countable dense subset of S^{d-1} and q through the positive rationals, form a countable base of the topology of $(0,R)\times S^{d-1}$. Hence every open subset of Z is a countable union of their images, which are sectors.

The sets $E\subset Z$ for which $\Phi^{-1}(E)$ belongs to the product σ-algebra form a σ-algebra containing the sectors, and hence all Borel subsets of Z. Consequently $\mu(E)=(\lambda\times S)(\Phi^{-1}(E))$ defines a finite Borel measure μ on Z, and μ agrees with Lebesgue measure on the sectors. The sectors are closed under finite intersections, and Z itself is a sector, so the two finite measures agree on every Borel subset of Z by part (2) of Lemma 1.1.6. Hence $\int_Z f\,\mathrm{d}y=\int f\circ\Phi\,\mathrm{d}(\lambda\times S)$ for indicator functions of Borel sets, by linearity for nonnegative simple Borel functions, and by monotone convergence (Proposition 1.1.3) for all nonnegative Borel functions f. Since the point x is a Lebesgue null set, Tonelli's theorem (Proposition 1.1.7) yields the formula of integration in polar coordinates

$$\int_{B(x,R)} f(y)\,\mathrm{d}y=\int_0^R r^{d-1}\int_{S^{d-1}} f(x+r\omega)\,\mathrm{d}S(\omega)\,\mathrm{d}r \tag{2.1}$$

for every nonnegative Borel function f on $B(x,R)$. Applying it to the positive and negative parts of $\operatorname{Re} f$ and $\operatorname{Im} f$ and using Fubini's theorem, we obtain (2.1) for every Borel function f which is integrable on $B(x,R)$. Letting $R\to\infty$ and using monotone convergence, we see that (2.1) also holds with $R=\infty$, where $B(x,\infty)=\mathbb{R}^d$. In particular, every sphere $\partial B(x,r)$ is a Lebesgue null set, since its indicator function vanishes at $x+t\omega$ unless $t=r$.

Next we prove the divergence theorem on balls. Let $B=B(c,R)$, let $\nu(y)=(y-c)/R$ be the outward unit normal on ∂B, and let $F=(F_1,\dots,F_d)$ be a C^1 vector field on an open neighborhood U of $\overline{B}$, with divergence $\operatorname{div}F=\sum_{j=1}^d\partial_jF_j$. Then

$$\int_B \operatorname{div}F\,\mathrm{d}y=\int_{\partial B}F\cdot\nu\,\mathrm{d}S. \tag{2.2}$$

It suffices to prove, for each j, that $\int_B\partial_jF_j\,\mathrm{d}y=\int_{\partial B}F_j\nu_j\,\mathrm{d}S$. To do so, we approximate $\mathbf{1}_B$ by smooth radial cutoff functions χ_ε and integrate $\partial_j(F_j\chi_\varepsilon)$ over $\mathbb{R}^d$. Let $0<\varepsilon<R$, and let ψ_ε be a smooth nonnegative function on $\mathbb{R}$ with support in $[R-\varepsilon,R]$ and integral one, for instance a translate and dilate of the mollifier ρ of Section 1.4 in dimension one. The function $\theta_\varepsilon(t)=\int_t^\infty\psi_\varepsilon(s)\,\mathrm{d}s$ is smooth with values in $[0,1]$, equals 1 for $t\le R-\varepsilon$ and 0 for $t\ge R$, and satisfies $\theta_\varepsilon'=-\psi_\varepsilon$. Hence $\chi_\varepsilon(y)=\theta_\varepsilon(|y-c|)$ is smooth on $\mathbb{R}^d$, because $|y-c|$ is smooth for $y\ne c$ and $\chi_\varepsilon=1$ on $B(c,R-\varepsilon)$. Moreover, χ_ε vanishes outside $\overline{B}$, and

$$\partial_j\chi_\varepsilon(y)=-\psi_\varepsilon(|y-c|)\,\frac{y_j-c_j}{|y-c|}\qquad(y\ne c).$$

The function $F_j\chi_\varepsilon$, extended by zero outside U, is a C^1 function with compact support on $\mathbb{R}^d$, because it is C^1 on U and vanishes on the open set $\mathbb{R}^d\setminus\overline{B}$. Integrating first in the variable y_j (Fubini's theorem) and using the fundamental theorem of calculus, we get $\int_{\mathbb{R}^d}\partial_j(F_j\chi_\varepsilon)\,\mathrm{d}y=0$. Since $\partial_j\chi_\varepsilon$ vanishes outside B, it follows, by

(2.1) with center c, that

$$\int_{\mathbb{R}^d} \chi_\varepsilon \, \partial_j F_j \, \mathrm{d}y = -\int_B F_j \, \partial_j \chi_\varepsilon \, \mathrm{d}y = \int_B F_j(y) \, \psi_\varepsilon(|y-c|) \, \frac{y_j - c_j}{|y-c|} \, \mathrm{d}y$$
$$= \int_0^R \psi_\varepsilon(t) g(t) \, \mathrm{d}t,$$

where

$$g(t) = t^{d-1} \int_{S^{d-1}} F_j(c + t\omega) \, \omega_j \, \mathrm{d}S(\omega), \qquad 0 < t \le R.$$

Since F_j is uniformly continuous on $\overline{B}$, the function g is continuous on $(0, R]$, and as $\psi_\varepsilon \ge 0$ has integral one and support in $[R-\varepsilon, R]$, we have $|\int_0^R \psi_\varepsilon g \, \mathrm{d}t - g(R)| \le \sup_{[R-\varepsilon,R]} |g - g(R)|$. Hence the right side tends to $g(R)$ as $\varepsilon \to 0$. On the left, $\chi_\varepsilon \to \mathbf{1}_B$ pointwise and $|\chi_\varepsilon \partial_j F_j| \le |\partial_j F_j| \mathbf{1}_{\overline{B}}$, so by dominated convergence the left side tends to $\int_B \partial_j F_j \, \mathrm{d}y$. Finally, $\nu(c + R\omega) = \omega$, so the definition of surface measure on ∂B gives $g(R) = \int_{\partial B} F_j \nu_j \, \mathrm{d}S$. Summing over j proves (2.2).

We also need the corresponding identity on an annulus. Let $0 < r < R$, and let F be a C^1 vector field on an open neighborhood U of the closed annulus $\{r \le |y-c| \le R\}$. Choose a smooth function η with compact support in U which equals one near the annulus (Lemma 1.4.1(4)). Then ηF, extended by zero outside U, is a C^1 vector field on $\mathbb{R}^d$, and it equals F near the closed annulus. We apply (2.2) to ηF on $B(c, R)$ and on $B(c, r)$ and subtract. On the left we obtain the integral of $\operatorname{div}(\eta F)$ over $\{r \le |y-c| < R\}$, which is the integral of $\operatorname{div} F$ over the open annulus, because the sphere $\partial B(c, r)$ is a Lebesgue null set. Hence

$$\int_{r<|y-c|<R} \operatorname{div} F \, \mathrm{d}y = \int_{\partial B(c,R)} F \cdot \nu \, \mathrm{d}S - \int_{\partial B(c,r)} F \cdot \nu \, \mathrm{d}S, \tag{2.3}$$

where $\nu(y) = (y-c)/|y-c|$ on both spheres.

For a continuous function u on Ω and a closed ball $\overline{B(x,r)} \subset \Omega$, we write

$$M_u(x, r) = \frac{1}{s_{d-1}} \int_{S^{d-1}} u(x + r\omega) \, \mathrm{d}S(\omega)$$

for the spherical mean of u over the sphere $\partial B(x, r)$. We say that u has the *sphere mean-value property* if $u(x) = M_u(x, r)$ for every closed ball $\overline{B(x,r)} \subset \Omega$, and the *ball mean-value property* if $u(x) = |B(x,r)|^{-1} \int_{B(x,r)} u \, \mathrm{d}y$ for every such ball.

Proposition 2.1.1 *Let $u \in C^2(\Omega)$ and $\overline{B(x,R)} \subset \Omega$. Then, for $0 < r < R$,*

$$\frac{\mathrm{d}}{\mathrm{d}r} M_u(x, r) = \frac{1}{s_{d-1} r^{d-1}} \int_{B(x,r)} \Delta u(y) \, \mathrm{d}y. \tag{2.4}$$

In particular, harmonic functions have both the sphere and the ball mean-value properties.

Proof Fix $0 < r < R$. For $\omega \in S^{d-1}$ and small $h \neq 0$, the fundamental theorem of calculus gives

$$\frac{u(x+(r+h)\omega) - u(x+r\omega)}{h} = \int_0^1 \nabla u(x+(r+sh)\omega)\cdot\omega \,\mathrm{d}s.$$

Since ∇u is uniformly continuous on the compact set $\overline{B(x,R)}$, these difference quotients converge to $\nabla u(x+r\omega)\cdot\omega$ uniformly in ω as $h \to 0$. Integrating over the finite measure space (S^{d-1}, S) and using the definition of surface measure on $\partial B(x,r)$, where $\nu(x+r\omega) = \omega$, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}r} M_u(x,r) = \frac{1}{s_{d-1}} \int_{S^{d-1}} \nabla u(x+r\omega)\cdot\omega \,\mathrm{d}S(\omega) = \frac{1}{s_{d-1} r^{d-1}} \int_{\partial B(x,r)} \nabla u \cdot \nu \,\mathrm{d}S.$$

Applying (2.2) on $B(x,r)$ to $F = \nabla u$, which is C^1 on Ω with divergence Δu, gives (2.4).

Now let u be harmonic. Then $M_u(x,\cdot)$ has zero derivative, and hence is constant, on $(0,R)$. Since $|M_u(x,r) - u(x)| \le \max_{|y-x|=r} |u(y) - u(x)|$, which tends to zero as $r \downarrow 0$ because u is continuous at x, the constant is $u(x)$. If $\overline{B(x,r)} \subset \Omega$, then the compact set $\overline{B(x,r)}$ has positive distance from the closed set $\mathbb{R}^d \setminus \Omega$, so there is $R > r$ with $\overline{B(x,R)} \subset \Omega$, and therefore $u(x) = M_u(x,r)$; this is the sphere mean-value property. Since $M_u(x,t) = u(x)$ for $0 < t \le r$, (2.1) gives $\int_{B(x,r)} u \,\mathrm{d}y = s_{d-1} r^d u(x)/d = |B(x,r)|\, u(x)$, which is the ball mean-value property. □

Theorem 2.1.2 *A continuous function on Ω is harmonic if and only if it has the sphere mean-value property. Every harmonic function is smooth. If h is harmonic on Ω and $\overline{B(x,r)} \subset \Omega$, then for every multi-index α,*

$$|\partial^\alpha h(x)| \le C_{d,\alpha} r^{-d-|\alpha|} \int_{B(x,r)} |h(y)| \,\mathrm{d}y, \tag{2.5}$$

where the constant $C_{d,\alpha}$ depends only on d and α. Moreover, every harmonic function h on Ω satisfies $h * \rho_\varepsilon = h$ on Ω_ε for every $\varepsilon > 0$, where ρ_ε and Ω_ε are as in Section 1.4.

Proof As in Section 1.4, we write $u_\varepsilon = u * \rho_\varepsilon$. Let u be continuous with the sphere mean-value property, and let $\overline{B(x,\varepsilon)} \subset \Omega$. The compact set $\overline{B(x,\varepsilon)}$ has positive distance from $\mathbb{R}^d \setminus \Omega$, so $x \in \Omega_\varepsilon$, and $M_u(x,r) = u(x)$ for $0 < r \le \varepsilon$. Since ρ is radial, we may write $\rho_\varepsilon(y) = \phi_\varepsilon(|y|)$; as ρ_ε is even and vanishes outside $B(0,\varepsilon)$, (2.1) gives

$$\begin{aligned} u_\varepsilon(x) &= \int_{B(0,\varepsilon)} u(x+y)\rho_\varepsilon(y) \,\mathrm{d}y = s_{d-1} \int_0^\varepsilon r^{d-1}\phi_\varepsilon(r) M_u(x,r) \,\mathrm{d}r \\ &= u(x)\, s_{d-1} \int_0^\varepsilon r^{d-1}\phi_\varepsilon(r) \,\mathrm{d}r = u(x) \int_{\mathbb{R}^d} \rho_\varepsilon \,\mathrm{d}y = u(x). \end{aligned} \tag{2.6}$$

Every point of Ω_ε is the center of such a closed ball, so $u = u_\varepsilon$ on the open set Ω_ε, and u_ε is smooth there (Section 1.4). As Ω is the union of the sets Ω_ε, the function u is smooth.

Next we show that $\Delta u = 0$. Fix $x \in \Omega$. Taylor's formula gives

$$u(x + r\omega) = u(x) + r \sum_j \partial_j u(x)\, \omega_j + \frac{r^2}{2} \sum_{j,k} \partial_j \partial_k u(x)\, \omega_j \omega_k + o(r^2),$$

uniformly in $\omega \in S^{d-1}$. We average over the sphere. The reflections $\omega \mapsto -\omega$ and $\omega_j \mapsto -\omega_j$ preserve S, so $\int \omega_j \, dS = 0$ and $\int \omega_j \omega_k \, dS = 0$ for $j \neq k$. Permutations of the coordinates also preserve S, so the integrals $\int \omega_j^2 \, dS$ are all equal, and their sum is s_{d-1} because $\sum_j \omega_j^2 = 1$. Hence

$$M_u(x, r) = u(x) + \frac{r^2}{2d} \Delta u(x) + o(r^2).$$

Since $M_u(x, r) = u(x)$ for all small r, it follows that $\Delta u(x) = 0$. Thus u is harmonic. Conversely, harmonic functions have the sphere mean-value property by Proposition 2.1.1; in particular they are smooth, and they satisfy $h = h_\varepsilon = h * \rho_\varepsilon$ on Ω_ε by the first part of the proof.

Now let h be harmonic and $\overline{B(x, r)} \subset \Omega$. Then x lies in the open set Ω_r, and $h = h_r$ on Ω_r by the first part of the proof. By the differentiation formula $\partial^\alpha h_r = h * \partial^\alpha \rho_r$ of Section 1.4, and since $\partial^\alpha \rho_r$ vanishes outside $B(0, r)$, we get

$$\partial^\alpha h(x) = \int_{B(x,r)} h(y)\, (\partial^\alpha \rho_r)(x - y)\, dy.$$

Since $\partial^\alpha \rho_r(z) = r^{-d-|\alpha|} (\partial^\alpha \rho)(z/r)$, this proves (2.5) with $C_{d,\alpha} = \sup |\partial^\alpha \rho|$, where ρ is the mollifier on $\mathbb{R}^d$ fixed in Section 1.4. □

Proposition 2.1.3 *Let h be a real harmonic function on an open set $D \subset \mathbb{R}^d$.*

(1) *If D is connected and h attains its maximum or its minimum over D at a point of D, then h is constant.*
(2) *If D is bounded and $\overline{\lim}_{D \ni x \to \zeta} h(x) \leq 0$ for every $\zeta \in \partial D$, then $h \leq 0$ on D.*

Consequently, a harmonic function on a bounded open set which is continuous on its closure is determined by its boundary values.

Proof For the first assertion, suppose that h attains its maximum m at a point of D, and let $E = \{x \in D : h(x) = m\}$. This set is nonempty and closed in D. If $x \in E$ and $\overline{B(x, r)} \subset D$, the ball mean-value property gives $\int_{B(x,r)} (m - h)\, dy = 0$. The integrand is continuous and nonnegative, so it vanishes on $B(x, r)$. Thus E is open, and $E = D$ by connectedness. For the minimum, apply this to $-h$.

For the second assertion, let $\varepsilon > 0$, and suppose that $h > \varepsilon$ somewhere in D; we derive a contradiction. Each $\zeta \in \partial D$ has a ball $B(\zeta, r_\zeta)$ such that $h < \varepsilon$ on $D \cap B(\zeta, r_\zeta)$. Since D is bounded, ∂D is compact, and finitely many of these balls

cover it; let W be their union. Then $K = \overline{D} \setminus W = D \setminus W$ is compact. Since $h < \varepsilon$ on $D \setminus K = D \cap W$, the set K is nonempty. The maximum of h over K is attained at some point x_0, and $h(x_0) > \varepsilon$; hence $h(x_0)$ is also the maximum of h over D. Let D_0 be the component of D containing x_0; it is open, because every point of D lies in a ball contained in D, and balls are connected. By the first assertion, $h = h(x_0)$ on D_0.

Since $\mathbb{R}^d$ is connected and D_0 is a nonempty, bounded, open subset, D_0 is not closed, so it has a boundary point ζ. This point does not lie in D: otherwise the component of D containing ζ would be an open neighborhood of ζ, so it would meet D_0 and hence equal D_0, which does not contain its boundary point ζ. Thus $\zeta \in \partial D$ and $\overline{\lim}_{D \ni x \to \zeta} h(x) \geq h(x_0) > 0$, contradicting the hypothesis. Hence $h \leq \varepsilon$ on D, and letting $\varepsilon \downarrow 0$ gives $h \leq 0$.

Finally, if two harmonic functions on a bounded open set are continuous on its closure and agree on its boundary, apply the second assertion to the real and imaginary parts of their difference and of its negative. □

Theorem 2.1.4 (Weyl's lemma) *Let $T \in \mathcal{D}'(\Omega)$ satisfy $\Delta T = 0$. Then there is a unique smooth harmonic function h on Ω such that $\langle T, \chi \rangle = \int_\Omega h\chi \, \mathrm{d}x$ for every $\chi \in C_c^\infty(\Omega)$. In particular, a locally integrable function whose distributional Laplacian vanishes agrees almost everywhere with a harmonic function.*

Proof Recall from Section 1.4 that $(T * \phi)(y) = \langle T, \phi(y - \cdot) \rangle$ for a test function ϕ on $\mathbb{R}^d$ supported in $\overline{B(0,s)}$ and $y \in \Omega_s$, and that $T_\varepsilon = T * \rho_\varepsilon$. By part (1) of Proposition 1.4.3, $T * \phi$ is smooth on Ω_s and $\partial^\alpha (T * \phi) = T * \partial^\alpha \phi$; moreover $\Delta T_\varepsilon = (\Delta T)_\varepsilon = 0$, so T_ε is a smooth harmonic function on Ω_ε. We show that T_ε converges to the required h, locally uniformly with all derivatives, by writing $T_\varepsilon = T * (\rho_\varepsilon * \rho_r)$ near each point for a fixed $r > 0$.

Fix $x \in \Omega$ and $r > 0$ with $\overline{B(x,3r)} \subset \Omega$, and let $0 < \varepsilon < r$. Since $\operatorname{dist}(x, \mathbb{R}^d \setminus \Omega) > 3r$, every point of $B(x,2r)$ has distance greater than r from $\mathbb{R}^d \setminus \Omega$, and every point of $B(x,r)$ has distance greater than $2r$; thus $B(x,2r) \subset \Omega_\varepsilon$ and $B(x,r) \subset \Omega_{2r}$. Let $y \in B(x,r)$. The closed ball $\overline{B(y,r)}$ lies in $B(x,2r)$, where T_ε is harmonic, so the last assertion of Theorem 2.1.2, applied to T_ε on $B(x,2r)$, gives $T_\varepsilon(y) = \int T_\varepsilon(z)\rho_r(y-z)\,\mathrm{d}z$. The test function $\chi = \rho_r(y - \cdot)$ is supported in $\overline{B(y,r)} \subset \Omega_\varepsilon$, so (1.3) gives $T_\varepsilon(y) = \langle T, \chi * \rho_\varepsilon \rangle$. Substituting $z = w + v$ and using that ρ_ε is even, we find

$$\begin{aligned}(\chi * \rho_\varepsilon)(w) &= \int \rho_r(y-z)\rho_\varepsilon(w-z)\,\mathrm{d}z \\ &= \int \rho_\varepsilon(v)\rho_r(y-w-v)\,\mathrm{d}v = (\rho_\varepsilon * \rho_r)(y-w).\end{aligned}$$

The function $\rho_\varepsilon * \rho_r$ is a test function supported in $\overline{B(0,\varepsilon+r)} \subset \overline{B(0,2r)}$, so we have shown that

$$T_\varepsilon = T * (\rho_\varepsilon * \rho_r) \qquad \text{on } B(x,r).$$

Since $\rho_\varepsilon * \rho_r = \rho_r * \rho_\varepsilon$, part (3) of Lemma 1.4.1, applied on $\mathbb{R}^d$ to ρ_r, shows that $\partial^\alpha(\rho_\varepsilon * \rho_r) \to \partial^\alpha \rho_r$ uniformly for every multi-index α as $\varepsilon \to 0$. Let C and m be

the constants of Definition 1.3.1 for T and the compact set $\overline{B(x,3r)}$, which contains the supports of $(\rho_\varepsilon * \rho_r)(y-\cdot)$ and $\rho_r(y-\cdot)$ for $y \in B(x,r)$. For $y \in B(x,r)$ and a multi-index α, the difference $\partial^\alpha T_\varepsilon(y) - \partial^\alpha (T * \rho_r)(y)$ is the pairing of T with $w \mapsto \psi(y-w)$, where $\psi = \partial^\alpha(\rho_\varepsilon * \rho_r - \rho_r)$, and the derivatives of order β of this test function are $\pm(\partial^\beta \psi)(y-w)$. Hence

$$\sup_{y\in B(x,r)} \bigl|\partial^\alpha T_\varepsilon(y) - \partial^\alpha (T*\rho_r)(y)\bigr| \le C \max_{|\beta|\le m} \sup \bigl|\partial^{\alpha+\beta}(\rho_\varepsilon * \rho_r - \rho_r)\bigr|,$$

which tends to zero. Thus $T_\varepsilon \to T * \rho_r$ on $B(x,r)$, uniformly with all derivatives.

Every point of Ω is the center of such a ball. Therefore $h(y) = \lim_{\varepsilon\to 0} T_\varepsilon(y)$ exists for every $y \in \Omega$, and on each ball $B(x,r)$ as above it equals $T * \rho_r$ and is the limit of T_ε uniformly with all derivatives. Thus h is smooth and $\Delta h = \lim_{\varepsilon\to 0} \Delta T_\varepsilon = 0$. Every compact subset of Ω is covered by finitely many of these balls, so $T_\varepsilon \to h$ uniformly on compact sets, and part (2) of Proposition 1.4.3 gives $\int h\chi \,\mathrm{d}x = \lim_{\varepsilon\to 0} \int T_\varepsilon \chi \,\mathrm{d}x = \langle T, \chi\rangle$ for every test function χ. If two continuous functions define the same distribution, they agree almost everywhere by part (4) of Proposition 1.4.3, and hence everywhere, since nonempty open sets have positive measure. This proves uniqueness. Finally, if T is given by a locally integrable function u, then u and h define the same distribution, so $u = h$ almost everywhere by the same proposition. □

2.2 Fundamental solutions

A fundamental solution of Δ is a distribution E on $\mathbb{R}^d$ with $\Delta E = \delta_0$. We fix the following choice.

Proposition 2.2.1 *The locally integrable functions*

$$E_d(x) = \begin{cases} |x|/2, & d = 1, \\ (2\pi)^{-1} \log|x|, & d = 2, \\ -((d-2)s_{d-1})^{-1}|x|^{2-d}, & d \ge 3 \end{cases} \tag{2.7}$$

are fundamental solutions of Δ. For $d \ge 2$ we set $E_d(0) = -\infty$; this does not affect the associated distribution. Each E_d is harmonic on $\mathbb{R}^d \setminus \{0\}$. For $f \in C_c^\infty(\mathbb{R}^d)$, the convolution $(E_d * f)(x) = \int E_d(x-y) f(y)\,\mathrm{d}y$ is smooth and satisfies $\Delta(E_d * f) = f$. Moreover, if $0 \le a < b \le \infty$ and $F \in C^2(a,b)$, then the radial function $x \mapsto F(|x|)$ is of class C^2 on the annulus $\{a < |x| < b\}$, and its Laplacian is

$$\Delta F(|x|) = F''(|x|) + \frac{d-1}{|x|} F'(|x|) = |x|^{1-d} \frac{\mathrm{d}}{\mathrm{d}t}\bigl(t^{d-1} F'(t)\bigr)\Big|_{t=|x|}. \tag{2.8}$$

Proof We begin with (2.8). Let $F \in C^2(a,b)$. The function $x \mapsto |x|$ is smooth on $\mathbb{R}^d \setminus \{0\}$, so $F(|x|)$ is of class C^2 on the annulus. For $a < |x| < b$ the chain rule gives

$\partial_j F(|x|) = F'(|x|)x_j/|x|$ and

$$\partial_j^2 F(|x|) = F''(|x|)\frac{x_j^2}{|x|^2} + F'(|x|)\Big(\frac{1}{|x|} - \frac{x_j^2}{|x|^3}\Big);$$

summing over j gives the first expression in (2.8), and the product rule gives the second.

Write $E_d(x) = F_d(|x|)$ for $x \neq 0$. Differentiating the three cases of (2.7) and using $s_0 = 2$ and $s_1 = 2\pi$, we find

$$F_d'(t) = \frac{1}{s_{d-1}t^{d-1}}, \qquad t > 0,$$

in every dimension $d \geq 1$. Hence $t^{d-1}F_d'(t)$ is constant, and E_d is harmonic on $\mathbb{R}^d\backslash\{0\}$ by (2.8). The functions E_d are continuous off the origin, and they are locally integrable. For $d \geq 2$ this follows from (2.1): $\int_{B(0,1)} |E_d|\,\mathrm{d}x = s_{d-1}\int_0^1 r^{d-1}|F_d(r)|\,\mathrm{d}r$, which is a constant multiple of $\int_0^1 r|\log r|\,\mathrm{d}r$ when $d = 2$ and of $\int_0^1 r\,\mathrm{d}r$ when $d \geq 3$, hence finite, while E_d is bounded on every compact set disjoint from $B(0,1)$. In dimension one, E_1 is continuous, and for every test function χ on $\mathbb{R}$ an integration by parts on each half-line gives

$$\int_0^\infty \frac{x}{2}\chi''(x)\,\mathrm{d}x = -\frac{1}{2}\int_0^\infty \chi'(x)\,\mathrm{d}x = \frac{\chi(0)}{2},$$
$$\int_{-\infty}^0 \frac{-x}{2}\chi''(x)\,\mathrm{d}x = \frac{1}{2}\int_{-\infty}^0 \chi'(x)\,\mathrm{d}x = \frac{\chi(0)}{2}.$$

Thus $\langle E_1'', \chi\rangle = \int E_1\chi''\,\mathrm{d}x = \chi(0)$, that is, $E_1'' = \delta_0$.

Let $d \geq 2$, let χ be a test function, and choose R with $\operatorname{Supp}\chi \subset B(0,R)$. The vector field $E_d\nabla\chi - \chi\nabla E_d$ is smooth on $\mathbb{R}^d \setminus \{0\}$, which is an open neighborhood of every closed annulus $\{\varepsilon \leq |x| \leq R\}$, and its divergence is

$$\nabla E_d \cdot \nabla\chi + E_d\Delta\chi - \nabla\chi \cdot \nabla E_d - \chi\Delta E_d = E_d\Delta\chi,$$

because $\Delta E_d = 0$ off the origin. The vector field vanishes near the sphere $|x| = R$, since $\operatorname{Supp}\chi$ is a compact subset of $B(0,R)$, and $\Delta\chi = 0$ for $|x| \geq R$. Hence (2.3), with $c = 0$ and $\nu = x/|x|$, gives

$$\int_{|x|>\varepsilon} E_d\Delta\chi\,\mathrm{d}x = \int_{|x|=\varepsilon} (-E_d\partial_r\chi + \chi\partial_r E_d)\,\mathrm{d}S,$$

where ∂_r denotes the derivative in the direction of $x/|x|$. The first boundary term tends to zero as $\varepsilon \to 0$: its absolute value is at most $\sup|\nabla\chi|\,|F_d(\varepsilon)|\,s_{d-1}\varepsilon^{d-1}$, which is $\sup|\nabla\chi|$ times $\varepsilon|\log\varepsilon|$ if $d = 2$, and times $\varepsilon/(d-2)$ if $d \geq 3$. On the sphere $|x| = \varepsilon$ we have $\partial_r E_d = F_d'(\varepsilon) = (s_{d-1}\varepsilon^{d-1})^{-1}$, so the second term equals $s_{d-1}^{-1}\int_{S^{d-1}}\chi(\varepsilon\omega)\,\mathrm{d}S(\omega) = M_\chi(0,\varepsilon)$, which tends to $\chi(0)$ because χ is continuous

at 0. On the left, $E_d\Delta\chi$ is integrable, so dominated convergence gives the limit $\int E_d\Delta\chi\,\mathrm{d}x$. Hence $\langle\Delta E_d,\chi\rangle = \langle E_d,\Delta\chi\rangle = \chi(0)$, that is, $\Delta E_d = \delta_0$.

Finally, let $f \in C_c^\infty(\mathbb{R}^d)$ and write $(E_d * f)(x) = \int E_d(y) f(x-y)\,\mathrm{d}y$. Fix a compact set $K \subset \mathbb{R}^d$, and let L be the compact set of points $x'-z$ with $\operatorname{dist}(x',K) \le 1$ and $z \in \operatorname{Supp} f$. If $g \in C_c(\mathbb{R}^d)$ is supported in $\operatorname{Supp} f$, then for $x \in K$ the function $y \mapsto E_d(y)g(x-y)$ vanishes outside L and is bounded by $\sup|g|\,|E_d|\mathbf{1}_L$, which is integrable; by dominated convergence, $E_d * g$ is continuous on K. Moreover, for $x \in K$, a coordinate vector e_k and $0 < |t| \le 1$, the mean value theorem gives

$$\left| E_d(y)\,\frac{f(x+te_k-y)-f(x-y)}{t}\right| \le \sup|\partial_k f|\,|E_d(y)|\,\mathbf{1}_L(y),$$

so dominated convergence shows that $\partial_k(E_d * f) = E_d * \partial_k f$ on K. Since K is arbitrary and $\partial_k f$ is again a test function supported in $\operatorname{Supp} f$, induction on $|\alpha|$ shows that $E_d * f$ is smooth and $\partial^\alpha(E_d * f) = E_d * \partial^\alpha f$. In particular, since $\Delta_y[f(x-y)] = (\Delta f)(x-y)$,

$$\Delta(E_d * f)(x) = \int E_d(y)\,\Delta_y[f(x-y)]\,\mathrm{d}y = \langle \Delta E_d, f(x-\cdot)\rangle = f(x).$$

We shall also use $\Gamma_d = -E_d$, which satisfies $-\Delta\Gamma_d = \delta_0$; the Green functions of Section 2.5 are normalized by this sign. For $d \ge 3$ the function Γ_d is positive. In dimension two, $\Gamma_2(x) = (2\pi)^{-1}\log(1/|x|)$ changes sign at $|x| = 1$; adding a constant, which changes neither the singularity at the origin nor the equation $-\Delta\Gamma_2 = \delta_0$, makes it nonnegative on any prescribed bounded range of distances, as is done in Section 4.1.

2.3 Poisson kernels and Harnack inequalities

Let $D \subset \mathbb{R}^d$ be a bounded open set. The *Dirichlet problem* on D asks, for a given $f \in C(\partial D)$, for a function h which is harmonic on D and continuous on $\overline{D}$ with $h = f$ on ∂D. By Proposition 2.1.3 it has at most one solution. On a ball $B = B(c,R)$ the solution is given by an explicit integral formula: for $x \in B$ and $\zeta \in \partial B$ put

$$P_B(x,\zeta) = \frac{R^2 - |x-c|^2}{s_{d-1}R|x-\zeta|^d}. \tag{2.9}$$

This is the Poisson kernel of B. In dimension one, $B = (a,b)$, the boundary $\partial B = \{a,b\}$ carries counting measure and $s_0 = 2$, so that $P_B(x,a) = (b-x)/(b-a)$ and $P_B(x,b) = (x-a)/(b-a)$; the Poisson integral below is then the affine interpolation of the two endpoint values.

Theorem 2.3.1 *Let B be a ball and $f \in C(\partial B)$. The Poisson integral*

$$H_B f(x) = \int_{\partial B} P_B(x,\zeta) f(\zeta)\, dS(\zeta), \qquad x \in B,$$

is the unique harmonic function on B which extends continuously to $\overline{B}$ with boundary values f. The kernel is positive, and $\int_{\partial B} P_B(x,\zeta)\, dS(\zeta) = 1$ for every $x \in B$. The operator H_B is linear, preserves order, and satisfies $\sup_B |H_B f| \le \max_{\partial B} |f|$. Every function harmonic on a neighborhood of $\overline{B}$ is the Poisson integral of its restriction to ∂B.

Proof If $d = 1$ and $B = (a,b)$, then $H_B f$ is the affine function with the values $f(a)$ and $f(b)$ at the endpoints, as noted above. It is harmonic on B and extends continuously to $\overline{B}$ with boundary values f, and the kernel is positive with $P_B(x,a) + P_B(x,b) = 1$. The remaining assertions are proved below in all dimensions.

Now let $d \ge 2$; after a translation we may assume that $c = 0$. Fix $\zeta \in \partial B$ and put $A(x) = R^2 - |x|^2$ and $t(x) = |x - \zeta|$ for $x \ne \zeta$. We have $\nabla A = -2x$, $\Delta A = -2d$ and $\nabla t^{-d} = -dt^{-d-2}(x - \zeta)$, and (2.8), applied to $F(t) = t^{-d}$ after translating ζ to the origin, gives $\Delta t^{-d} = \big(d(d+1) - d(d-1)\big)t^{-d-2} = 2dt^{-d-2}$. By the product rule $\Delta(At^{-d}) = (\Delta A)t^{-d} + 2\nabla A \cdot \nabla t^{-d} + A\Delta t^{-d}$, and since $t^2 = |x|^2 - 2x\cdot\zeta + |\zeta|^2$,

$$\Delta(At^{-d}) = 2dt^{-d-2}\big(-t^2 + 2x\cdot(x-\zeta) + R^2 - |x|^2\big) = 2dt^{-d-2}(R^2 - |\zeta|^2) = 0.$$

Thus $P_B(\cdot,\zeta)$ is harmonic on B for every $\zeta \in \partial B$. Every partial derivative $\partial_x^\alpha P_B(x,\zeta)$ is continuous on $B \times \partial B$, hence bounded on $K \times \partial B$ for each compact set $K \subset B$, where $|x - \zeta| \ge \operatorname{dist}(K, \partial B) > 0$. By the mean value theorem and dominated convergence on the finite measure space $(\partial B, S)$, we may therefore differentiate under the integral sign any number of times, so $H_B f$ is smooth on B and $\Delta H_B f(x) = \int_{\partial B} \Delta_x P_B(x,\zeta) f(\zeta)\, dS(\zeta) = 0$.

Next we show that the kernel has integral one. Taking $f = 1$, we see that $I(x) = \int_{\partial B} P_B(x,\zeta)\, dS(\zeta)$ is harmonic on B. It is also radial. Indeed, $P_B(Qx, Q\zeta) = P_B(x,\zeta)$ for every orthogonal transformation Q, and surface measure on ∂B is invariant under Q, so $I(Qx) = I(x)$. Moreover, if $|x| = |x'|$ and $x \ne x'$, the orthogonal reflection $y \mapsto y - 2(y\cdot e)e$ with $e = (x - x')/|x - x'|$ maps x to x'. Hence $I(x) = F(|x|)$, where $F(r) = I(re_1)$ is smooth on $(-R, R)$ and e_1 is the first coordinate vector. By (2.8), $(r^{d-1}F'(r))' = 0$ on $(0,R)$, so $r^{d-1}F'(r)$ is constant there. Since F' is bounded near 0 and $d \ge 2$, this constant is $\lim_{r\to 0} r^{d-1}F'(r) = 0$. Hence F is constant on $(0,R)$, and since I is continuous at the origin, I is constant on B, with value $I(0) = R^2 \cdot s_{d-1}R^{d-1}/(s_{d-1}R \cdot R^d) = 1$.

For the boundary behavior, fix $\zeta_0 \in \partial B$ and $\varepsilon > 0$, and choose $\delta > 0$ such that $|f(\zeta) - f(\zeta_0)| < \varepsilon$ for $\zeta \in \partial B$ with $|\zeta - \zeta_0| < \delta$. Since the kernel is positive with integral one,

$$\begin{aligned} |H_B f(x) - f(\zeta_0)| &\le \int_{\partial B} P_B(x,\zeta)|f(\zeta) - f(\zeta_0)|\, dS(\zeta) \\ &\le \varepsilon + 2\max_{\partial B}|f| \int_{\{\zeta\in\partial B:\,|\zeta-\zeta_0|\ge\delta\}} P_B(x,\zeta)\, dS(\zeta). \end{aligned}$$

If $|x-\zeta_0|<\delta/2$, then $|x-\zeta|\ge\delta/2$ in the last integral, so $P_B(x,\zeta)\le(R^2-|x-c|^2)/(s_{d-1}R(\delta/2)^d)$ there, and the integral is at most $R^{d-2}(2/\delta)^d(R^2-|x-c|^2)$. This tends to zero as $x\to\zeta_0$, because $|\zeta_0-c|=R$. Hence $\overline{\lim}_{x\to\zeta_0}|H_Bf(x)-f(\zeta_0)|\le\varepsilon$ for every $\varepsilon>0$. Since f is continuous on ∂B, it follows that the function equal to H_Bf on B and to f on ∂B is continuous on $\overline{B}$.

Uniqueness follows from the last assertion of Proposition 2.1.3. The operator H_B is linear by definition, and since the kernel is positive with integral one, H_B preserves order and $|H_Bf(x)|\le\int_{\partial B}P_B(x,\zeta)|f(\zeta)|\,dS(\zeta)\le\max_{\partial B}|f|$. Finally, if h is harmonic on a neighborhood of $\overline{B}$, then h and $H_B(h|_{\partial B})$ are harmonic on B, continuous on $\overline{B}$ and equal on ∂B, so they coincide. □

We often regard H_Bf as the continuous function on $\overline{B}$ which equals f on ∂B.

Proposition 2.3.2 (1) *Let $h\ge0$ be harmonic on $B(0,R)$, let $x\in B(0,R)$ and put $r=|x|$. Then*

$$\frac{R^{d-2}(R-r)}{(R+r)^{d-1}}h(0)\le h(x)\le\frac{R^{d-2}(R+r)}{(R-r)^{d-1}}h(0). \tag{2.10}$$

(2) *A nonnegative harmonic function on a domain is either positive everywhere or identically zero.*

(3) *Let Ω be a domain, $K\subset\Omega$ compact and $x_0\in\Omega$. Then there is a constant $C=C(K,x_0,\Omega)>0$ such that*

$$\sup_K h\le Ch(x_0)$$

for every nonnegative harmonic function h on Ω.

Proof (1) Let $r<R'<R$. By Theorem 2.3.1 on $B(0,R')$, both $h(x)$ and $h(0)$ are integrals of the nonnegative boundary values of h against the Poisson kernel. For $|\zeta|=R'$ we have $R'-r\le|x-\zeta|\le R'+r$, so the ratio

$$\frac{P_{B(0,R')}(x,\zeta)}{P_{B(0,R')}(0,\zeta)}=\frac{R'^{d-2}(R'^2-r^2)}{|x-\zeta|^d}$$

lies between the two coefficients in (2.10) with R' in place of R. Integrating and letting $R'\uparrow R$ gives (2.10).

(2) If a nonnegative harmonic function on a domain vanishes at a point, it attains its minimum there and is identically zero by Proposition 2.1.3.

(3) We first apply (1), after a translation, to a ball $B(a,2t)\subset\Omega$: for $y\in B(a,t)$ we have $r=|y-a|<t$, and the right inequality in (2.10) on $B(a,2t)$ gives

$$h(y)\le\frac{(2t)^{d-2}(2t+r)}{(2t-r)^{d-1}}h(a)\le\frac{(2t)^{d-2}\cdot3t}{t^{d-1}}h(a)=C_0h(a),\qquad C_0=3\cdot2^{d-2}.$$

Now let $y\in\Omega$. Since Ω is connected, y can be joined to x_0 by a polygonal path in Ω: the set of points that can be so joined is open, and so is its complement in Ω,

because balls are convex. The path is compact, so there is $t > 0$ with $B(z, 2t) \subset \Omega$ for every point z of the path. Choose points $x_0 = p_0, p_1, \dots, p_N = y$ along the path with $|p_{i+1} - p_i| < t$. Applying the ball estimate on $B(p_i, 2t)$ for $i = 0, \dots, N-1$ gives $h(y) \le C_0^N h(x_0)$, and once more on $B(y, 2t)$ gives $h \le C_0^{N+1} h(x_0)$ on $B(y, t)$. Here t and N depend on y, x_0 and Ω, but not on h. Finitely many of the balls $B(y, t)$, $y \in K$, cover K, and the largest of the corresponding constants C_0^{N+1} is the required C. □

A sequence of smooth functions on an open set $D \subset \mathbb{R}^d$ converges to g *in* C^∞ *on compact subsets* if, for every multi-index α, the derivatives ∂^α of its terms converge to $\partial^\alpha g$ uniformly on compact subsets of D.

Corollary 2.3.3 *Let $D \subset \mathbb{R}^d$ be a domain.*

(1) *Every sequence of nonnegative harmonic functions on D which is bounded at one point of D has a subsequence converging in C^∞ on compact subsets; the limit is harmonic.*
(2) *An increasing sequence of harmonic functions on D either converges in C^∞ on compact subsets to a finite harmonic function, or tends to $+\infty$ uniformly on compact subsets.*
(3) *A decreasing sequence of harmonic functions on D either converges in C^∞ on compact subsets to a finite harmonic function, or tends to $-\infty$ uniformly on compact subsets.*

Proof Let $h_j \ge 0$ be harmonic on D with $h_j(x_0) \le M$ for some $x_0 \in D$. Let $K \subset D$ be compact and choose $\delta > 0$ such that $K_\delta = \{y : \operatorname{dist}(y, K) \le \delta\}$ is contained in D. By Proposition 2.3.2(3), $h_j \le C(K_\delta, x_0, D)M$ on K_δ for all j, and (2.5) with radius δ then bounds each derivative $\partial^\alpha h_j$ on K uniformly in j. Taking for K a closed ball in D and applying this bound to the first derivatives of $\partial^\alpha h_j$, we see that the functions $\partial^\alpha h_j$ are uniformly Lipschitz on every closed ball in D, and hence equicontinuous on compact subsets of D.

Let K_m be the sets of the compact exhaustion of D constructed in Section 1.1; every compact subset of D lies in some K_m. Applying Proposition 1.6.4 on each K_m and making a diagonal extraction over all m and all multi-indices α, we obtain a subsequence along which every $\partial^\alpha h_j$ converges uniformly on compact subsets of D, say to g_α. To identify the limits, let e_k be the kth standard basis vector. Passing to the uniform limit in

$$\partial^\alpha h_j(y + te_k) - \partial^\alpha h_j(y) = \int_0^t \partial^{\alpha + e_k} h_j(y + se_k)\, \mathrm{d}s,$$

which holds whenever the segment from y to $y + te_k$ lies in D, we find that $\partial_k g_\alpha = g_{\alpha + e_k}$. By induction, $g = g_0$ is smooth with $\partial^\alpha g = g_\alpha$, so the subsequence converges to g in C^∞ on compact subsets, and $\Delta g = \lim_j \Delta h_j = 0$. This proves the first assertion.

We also need a lower Harnack bound: for compact $K \subset D$ and $x_0 \in D$ there is $c > 0$ such that $h \ge c\, h(x_0)$ on K for every nonnegative harmonic h on D. Indeed, if

$B(a,2t) \subset D$ and $y \in B(a,t)$, then $r = |y-a| < t$, and the left inequality in (2.10) on $B(a,2t)$ gives $h(y) \geq (2t)^{d-2}(2t-r)(2t+r)^{1-d}h(a) \geq c_0 h(a)$ with $c_0 = 2^{d-2}3^{1-d}$. Cover K by finitely many balls $B(a_k,t_k)$ with $B(a_k,2t_k) \subset D$. Proposition 2.3.2(3), applied to the compact set $\{x_0\}$ and the base point a_k, gives $h(x_0) \leq C_k h(a_k)$. Hence $h \geq (c_0/\max_k C_k)h(x_0)$ on K.

Now let (h_j) be increasing and fix $x_0 \in D$. The functions $h_j - h_1$ are nonnegative and harmonic. If $h_j(x_0)$ is bounded, the first assertion applies to every subsequence of $(h_j - h_1)$. Since the sequence increases and, by Proposition 2.3.2(3), is bounded on compact sets, it has a finite pointwise limit, and every subsequence has a further subsequence converging in C^∞ on compact subsets to this limit. Hence the whole sequence converges in C^∞ on compact subsets, and the limit is a finite harmonic function. If $h_j(x_0) \to +\infty$, the lower Harnack bound gives $h_j - h_1 \geq c(h_j(x_0) - h_1(x_0))$ on each compact K, and since h_1 is bounded on K, $h_j \to +\infty$ uniformly on K. For a decreasing sequence, apply this to $(-h_j)$. □

Corollary 2.3.4 *A real harmonic function on $\mathbb{R}^d$ which is bounded above or bounded below is constant.*

Proof After changing the sign and adding a constant, we may assume that the function h is nonnegative. Fix $x \in \mathbb{R}^d$ and put $r = |x|$. For $R > r$, (2.10) on $B(0,R)$ gives $c_R h(0) \leq h(x) \leq C_R h(0)$, where the coefficients c_R and C_R of (2.10) tend to one as $R \to \infty$. Letting $R \to \infty$ gives $h(x) = h(0)$. Hence h is constant. □

2.4 Perron envelopes and boundary barriers

We now solve the Dirichlet problem on a general bounded open set by Perron's method, and we characterize by barriers the boundary points at which the solution attains the prescribed values. Throughout this section Ω is a nonempty bounded open set, and all functions and boundary data are real-valued. The boundary $\partial\Omega$ is compact, and it is nonempty: otherwise Ω would be a nonempty proper subset of the connected space $\mathbb{R}^d$ which is both open and closed.

Let D be an open set. A *continuous subsolution* on D is a continuous function $v\colon D \to \mathbb{R}$ such that $v \leq H_B(v|_{\partial B})$ on B for every ball B with $\overline{B} \subset D$. A function w is a *continuous supersolution* if $-w$ is a continuous subsolution. Harmonic functions are both, by the Poisson representation in Theorem 2.3.1. At the center c of $B = B(c,r)$ the Poisson kernel equals $(s_{d-1}r^{d-1})^{-1}$, so a continuous subsolution satisfies the sphere submean inequality $v(c) \leq M_v(c,r)$ whenever $\overline{B(c,r)} \subset D$.

Conversely, a continuous function v on D such that every $c \in D$ satisfies $v(c) \leq M_v(c,r)$ for all sufficiently small $r > 0$ is a continuous subsolution. The proof rests on the identity $M_{|\cdot|^2}(c,r) = |c|^2 + r^2$ for the mean of $|y|^2$ over the sphere $\partial B(c,r)$, which holds because $|c + r\omega|^2 = |c|^2 + 2r\, c\cdot\omega + r^2$ and $\int_{S^{d-1}} \omega \, dS(\omega) = 0$, by the invariance of S under $\omega \mapsto -\omega$.

To prove the claim, let B be a ball with $\overline{B} \subset D$, let H be the Poisson integral of $v|_{\partial B}$, regarded as a continuous function on $\overline{B}$, and for $\varepsilon > 0$ put $w(y) = v(y) - H(y) + \varepsilon|y|^2$.

Since $M_H(c,r) = H(c)$ whenever $\overline{B(c,r)} \subset B$ (Proposition 2.1.1), we obtain $w(c) \le M_w(c,r) - \varepsilon r^2$ for every $c \in B$ and all small $r > 0$. Hence the continuous function w cannot attain its maximum over the compact set $\overline{B}$ at a point $c \in B$, for we would then have $M_w(c,r) \le w(c)$ for all small r. The maximum is therefore attained on ∂B, where $H = v$ and hence $w(y) = \varepsilon|y|^2$. Thus $v - H \le \varepsilon \max_{\overline{B}} |y|^2$ on B, and letting $\varepsilon \downarrow 0$ gives $v \le H$.

In particular, being a continuous subsolution is a local property: if every point of D has an open neighborhood on which v is a continuous subsolution, then v is a continuous subsolution on D. Moreover, a real-valued continuous function is subharmonic in the sense of Definition 3.1.1 exactly when every point c satisfies $v(c) \le M_v(c,r)$ for all small $r > 0$, so the continuous subsolutions are exactly the real-valued continuous subharmonic functions.

These functions are stable under the usual operations. If v_1 and v_2 are continuous subsolutions, then so is $\max\{v_1, v_2\}$, because H_B preserves order, so that $v_i \le H_B(v_i|_{\partial B}) \le H_B(\max\{v_1, v_2\}|_{\partial B})$. Sums and multiples by nonnegative constants of continuous subsolutions are continuous subsolutions, by linearity of H_B, and so is $v - h$ for every harmonic h, because $h = H_B(h|_{\partial B})$ on B. Correspondingly, the minimum of two continuous supersolutions is a continuous supersolution. Finally, a real function $v \in C^2(D)$ with $\Delta v \ge 0$ is a continuous subsolution. Indeed, if $\overline{B(c,R)} \subset D$, then by (2.4) the mean $M_v(c,r)$ is nondecreasing in $r \in (0,R)$, and it tends to $v(c)$ as $r \downarrow 0$ because v is continuous at c; hence $v(c) \le M_v(c,r)$ for $0 < r < R$, and the converse statement above applies.

Lemma 2.4.1 *Let D be a bounded open set and v a continuous subsolution on D. If $\overline{\lim}_{D \ni x \to \zeta} v(x) \le 0$ for every $\zeta \in \partial D$, then $v \le 0$ on D. Moreover, if B is a ball with $\overline{B} \subset D$, then the function v^B which equals $H_B(v|_{\partial B})$ on B and v on $D \setminus B$ is a continuous subsolution on D with $v^B \ge v$.*

Proof For the first assertion we may assume that $D \ne \varnothing$. Let $\varepsilon > 0$, and put $w(y) = v(y) + \varepsilon|y|^2$ for $y \in D$. If $\overline{B(c,r)} \subset D$, then $v(c) \le M_v(c,r)$, and the identity $M_{|\cdot|^2}(c,r) = |c|^2 + r^2$ gives $w(c) \le M_w(c,r) - \varepsilon r^2$. Consequently w does not attain its supremum over D at any point $c \in D$, for we would then have $M_w(c,r) \le w(c)$ for small r. Choose $x_k \in D$ with $w(x_k) \to \sup_D w$. Since D is bounded, we may assume, after passing to a subsequence, that x_k converges to a point $p \in \overline{D}$. If p were in D, the continuity of w would give $w(p) = \sup_D w$, which we have excluded. Hence $p \in \partial D$. Since $\sup_D w = \lim_k \big(v(x_k) + \varepsilon|x_k|^2\big)$, the hypothesis gives $\sup_D w \le \overline{\lim}_{D \ni x \to p} v(x) + \varepsilon|p|^2 \le \varepsilon \max_{\overline{D}} |y|^2$. Thus $v \le w \le \varepsilon \max_{\overline{D}} |y|^2$ on D, and letting $\varepsilon \downarrow 0$ gives $v \le 0$.

For the second assertion, v^B is harmonic on B and equals v on $D \setminus B$; it is continuous at the points of ∂B because $H_B(v|_{\partial B})$ extends continuously to v on ∂B (Theorem 2.3.1). Moreover $v^B \ge v$, because $v \le H_B(v|_{\partial B})$ on B. To verify the subsolution inequality, let A be a ball with $\overline{A} \subset D$, and let $H = H_A(v^B|_{\partial A})$, regarded as a continuous function on $\overline{A}$. Since $v \le v^B$ and H_A preserves order, $v \le H_A(v|_{\partial A}) \le H$ on A; the same inequality holds on ∂A, where $H = v^B \ge v$. On the open set $A \cap B$, the function $v^B - H$ is harmonic, and it is continuous on the closure of $A \cap B$. A boundary point ζ of $A \cap B$ lies either on ∂A, where $v^B(\zeta) = H(\zeta)$,

or in $A \cap \partial B$, where $v^B(\zeta) = v(\zeta) \le H(\zeta)$. By part (2) of Proposition 2.1.3, applied to $v^B - H$ on the bounded open set $A \cap B$, we get $v^B \le H$ on $A \cap B$. On $A \setminus B$ we have $v^B = v \le H$. Hence $v^B \le H_A(v^B|_{\partial A})$ on A, which is the subsolution inequality for v^B on A. □

We call v^B the *harmonic lifting* of v on B.

For $f \in C(\partial\Omega)$, let $\mathcal{S}_f$ be the set of continuous subsolutions v on Ω such that $\overline{\lim}_{\Omega \ni x \to \zeta} v(x) \le f(\zeta)$ for every $\zeta \in \partial\Omega$. The *lower Perron envelope* of f is

$$U_f(x) = \sup_{v \in \mathcal{S}_f} v(x), \qquad x \in \Omega.$$

The constant $\min_{\partial\Omega} f$ belongs to $\mathcal{S}_f$, and every $v \in \mathcal{S}_f$ satisfies $v \le \max_{\partial\Omega} f$, by Lemma 2.4.1 applied to $v - \max_{\partial\Omega} f$.

Theorem 2.4.2 *Let $f \in C(\partial\Omega)$. Then U_f is harmonic on Ω and $\min_{\partial\Omega} f \le U_f \le \max_{\partial\Omega} f$. If a harmonic function on Ω extends continuously to $\overline{\Omega}$ with boundary values f, then it equals U_f.*

Proof The bounds for U_f were proved above. To prove harmonicity, fix a ball B with $\overline{B} \subset \Omega$ and a point $x_0 \in B$; we show that U_f agrees on B with a harmonic function. If $v \in \mathcal{S}_f$, its harmonic lifting v^B on B is a continuous subsolution by Lemma 2.4.1, and it coincides with v near $\partial\Omega$ because $\overline{B}$ is a compact subset of Ω; hence $v^B \in \mathcal{S}_f$.

Choose $v_j \in \mathcal{S}_f$ with $v_j(x_0) \to U_f(x_0)$, and put $w_j = \max\{v_1, \dots, v_j, \min_{\partial\Omega} f\}$. Then (w_j) is an increasing sequence in $\mathcal{S}_f$ with values in $[\min_{\partial\Omega} f, \max_{\partial\Omega} f]$, and $w_j(x_0) \to U_f(x_0)$ because $v_j \le w_j \le U_f$. The liftings w_j^B belong to $\mathcal{S}_f$ and satisfy $w_j \le w_j^B \le U_f$. On B they are harmonic, increase with j because H_B preserves order, and are bounded by $\max_{\partial\Omega} f$. By part (2) of Corollary 2.3.3, applied on the domain B to this sequence, which is bounded above, they converge on B to a harmonic function h, and $h \le U_f$ on B, while $h(x_0) = U_f(x_0)$.

Now let $v \in \mathcal{S}_f$ be arbitrary. The same argument applied to the increasing sequence $\max\{w_j, v\} \in \mathcal{S}_f$ shows that the liftings $(\max\{w_j, v\})^B$ increase on B to a harmonic function k. Since these liftings dominate w_j^B and v and lie in $\mathcal{S}_f$, we have $h \le k \le U_f$ and $k \ge v$ on B. In particular $k(x_0) \le U_f(x_0) = h(x_0)$, so the nonnegative harmonic function $k - h$ on B vanishes at x_0 and therefore vanishes identically, by Proposition 2.3.2(2). Hence $v \le h$ on B. Taking the supremum over $v \in \mathcal{S}_f$ gives $U_f \le h$, so $U_f = h$ on B. Since B was arbitrary, U_f is harmonic on Ω.

Finally, let H be harmonic on Ω and continuous on $\overline{\Omega}$ with $H = f$ on $\partial\Omega$. Then $H \in \mathcal{S}_f$, so $H \le U_f$. Conversely, for $v \in \mathcal{S}_f$ the function $v - H$ is a continuous subsolution with $\overline{\lim}_{x \to \zeta}(v - H)(x) \le 0$ at every $\zeta \in \partial\Omega$, so $v \le H$ by Lemma 2.4.1. Hence $U_f = H$. □

The *upper Perron envelope* is $V_f = -U_{-f}$. Equivalently, V_f is the pointwise infimum of the set $-\mathcal{S}_{-f}$, which consists of the continuous supersolutions w on Ω with $\underline{\lim}_{\Omega \ni x \to \zeta} w(x) \ge f(\zeta)$ for every $\zeta \in \partial\Omega$. If $v \in \mathcal{S}_f$ and $w \in -\mathcal{S}_{-f}$, then $v - w$ is a continuous subsolution with $\overline{\lim}_{x \to \zeta}(v - w)(x) \le f(\zeta) - f(\zeta) = 0$ at every

$\zeta \in \partial\Omega$, so $v \le w$ by Lemma 2.4.1. Hence $U_f \le V_f$. To prove equality, we first treat data that extend smoothly to a neighborhood of $\overline{\Omega}$ and then approximate, using the following lemma.

Lemma 2.4.3 *Let $K \subset \mathbb{R}^d$ be compact and $f \in C(K)$. Then f is the uniform limit on K of restrictions to K of functions in $C_c^\infty(\mathbb{R}^d)$, which can be taken real-valued if f is real-valued.*

Proof We may assume that $K \neq \varnothing$. Let $\delta > 0$. By compactness there are points $\zeta_1, \dots, \zeta_N \in K$ with $K \subset \bigcup_i B(\zeta_i, \delta)$. Put $\chi_i(x) = \rho((x - \zeta_i)/(2\delta))$, where ρ is the mollifier of Section 1.4; this function is smooth, positive on $B(\zeta_i, 2\delta)$ and zero outside it. The function $\sigma = \sum_i \chi_i$ is positive on the open set $W = \bigcup_i B(\zeta_i, 2\delta) \supset K$. Choose a smooth function η with compact support in W which equals one near K (Lemma 1.4.1(4)), and put

$$F = \eta \, \frac{\sum_i f(\zeta_i)\chi_i}{\sigma},$$

extended by zero outside W. This is a smooth function with compact support on $\mathbb{R}^d$, because it is smooth on W, where $\sigma > 0$, and vanishes on the open set $\mathbb{R}^d \setminus \operatorname{Supp}\eta$; it is real-valued if f is. For $\zeta \in K$ we have $\eta(\zeta) = 1$, so $F(\zeta)$ is a convex combination of values $f(\zeta_i)$ with $|\zeta_i - \zeta| < 2\delta$, and therefore $|F(\zeta) - f(\zeta)| \le \sup\{|f(y) - f(y')| : y, y' \in K,\ |y - y'| < 2\delta\}$. By the uniform continuity of f on the compact set K, this bound tends to zero as $\delta \to 0$. □

Theorem 2.4.4 *For every $f \in C(\partial\Omega)$ we have $U_f = V_f$. The resulting map $f \mapsto U_f$ from $C(\partial\Omega)$ to harmonic functions on Ω is linear and positive, and it is a contraction for the supremum norm:* $\sup_\Omega |U_f| \le \max_{\partial\Omega} |f|$.

Proof Write $\|f\| = \max_{\partial\Omega} |f|$, and let $\mathcal{R}$ be the set of $f \in C(\partial\Omega)$ with $U_f = V_f$. We show that $\mathcal{R}$ is a closed linear subspace of $C(\partial\Omega)$ containing a dense subset.

Step 1: $\mathcal{R}$ is a closed linear subspace, and $f \mapsto U_f$ is linear on it. If $v \in \mathcal{S}_f$ and $w \in \mathcal{S}_g$, then $v + w \in \mathcal{S}_{f+g}$; taking suprema over v and w at a fixed point gives $U_f + U_g \le U_{f+g}$. Applying this to $-f$ and $-g$ and changing signs gives $V_{f+g} \le V_f + V_g$. Together with the inequality $U_{f+g} \le V_{f+g}$ proved above, we obtain

$$U_f + U_g \le U_{f+g} \le V_{f+g} \le V_f + V_g.$$

Moreover $\mathcal{S}_{tf} = t\,\mathcal{S}_f$ for $t > 0$, so $U_{tf} = tU_f$ and $V_{tf} = tV_f$, while $U_{-f} = -V_f$ and $V_{-f} = -U_f$ by definition. Also $U_0 = 0$, because $0 \in \mathcal{S}_0$ and every member of $\mathcal{S}_0$ is nonpositive by Lemma 2.4.1; hence $V_0 = -U_0 = 0$ and $0 \in \mathcal{R}$. Consequently, if $f, g \in \mathcal{R}$, the display forces $U_{f+g} = V_{f+g} = U_f + U_g$, and $tf \in \mathcal{R}$ for every real t. Thus $\mathcal{R}$ is a linear subspace, and $f \mapsto U_f$ is linear on it.

To see that $\mathcal{R}$ is closed, we first show that U_f and V_f depend 1-Lipschitz continuously on f for the supremum norm. If $f \le g$, then $\mathcal{S}_f \subset \mathcal{S}_g$ and hence $U_f \le U_g$; and $\mathcal{S}_{f+t} = \mathcal{S}_f + t$ for constants t, so $U_{f+t} = U_f + t$. Since $f \le g + \|f - g\|$, we get $U_f \le U_g + \|f - g\|$, and by symmetry $|U_f - U_g| \le \|f - g\|$; as $V_f = -U_{-f}$ and $V_g = -U_{-g}$, also $|V_f - V_g| \le \|f - g\|$. If $f_n \in \mathcal{R}$ converge uniformly to f, then

$0 \le V_f - U_f \le |V_f - V_{f_n}| + |U_{f_n} - U_f| \le 2\|f - f_n\| \to 0$, so $f \in \mathcal{R}$. Hence $\mathcal{R}$ is closed.

Step 2: $\mathcal{R}$ contains the restriction to $\partial\Omega$ of every real smooth function on a neighborhood of $\overline{\Omega}$. Suppose first that F is a real smooth function on a neighborhood of $\overline{\Omega}$ with $\Delta F \ge 0$ there. Then F is a continuous subsolution on Ω, and $F(x) \to F(\zeta)$ as $\Omega \ni x \to \zeta \in \partial\Omega$, so $F \in \mathcal{S}_{F|_{\partial\Omega}}$ and $U_{F|_{\partial\Omega}} \ge F$. Consequently the harmonic function $U_{F|_{\partial\Omega}}$ satisfies $\underline{\lim}_{x\to\zeta} U_{F|_{\partial\Omega}}(x) \ge F(\zeta)$ at every $\zeta \in \partial\Omega$, so it belongs to $-\mathcal{S}_{-F|_{\partial\Omega}}$ and $V_{F|_{\partial\Omega}} \le U_{F|_{\partial\Omega}}$. Thus $F|_{\partial\Omega} \in \mathcal{R}$.

Now let F be an arbitrary real smooth function on an open neighborhood W of $\overline{\Omega}$, and choose $\delta > 0$ such that the compact set $L = \{x : \operatorname{dist}(x, \overline{\Omega}) \le \delta\}$ lies in W. Since ΔF is continuous, hence bounded below, on L, there is a constant $A \ge 0$ with $\Delta F + 2dA \ge 0$ on L. As $\Delta(A|x|^2) = 2dA$, both $F + A|x|^2$ and $A|x|^2$ are smooth with nonnegative Laplacian on the open neighborhood $\{x : \operatorname{dist}(x, \overline{\Omega}) < \delta\}$ of $\overline{\Omega}$. By the first case their restrictions to $\partial\Omega$ lie in $\mathcal{R}$, and by Step 1 so does the restriction of their difference F.

Step 3: these restrictions are dense in $C(\partial\Omega)$. This is Lemma 2.4.3, applied to real-valued functions on the compact set $\partial\Omega$.

By Steps 1–3, $\mathcal{R} = C(\partial\Omega)$, and $f \mapsto U_f$ is linear. By Step 1 it preserves order and satisfies $|U_f - U_g| \le \|f - g\|$. Since $U_0 = 0$, it follows that $U_f \ge 0$ whenever $f \ge 0$, and that $|U_f| \le \|f\|$. □

Definition 2.4.5 A boundary point $\xi \in \partial\Omega$ is *regular* if $U_f(x) \to f(\xi)$ as $\Omega \ni x \to \xi$, for every $f \in C(\partial\Omega)$. A *barrier* at ξ is a positive continuous supersolution b on Ω such that $b(x) \to 0$ as $x \to \xi$ and, for every neighborhood N of ξ with $\Omega \setminus N \ne \emptyset$,

$$\inf_{\Omega\setminus N} b > 0.$$

Theorem 2.4.6 *A boundary point of Ω is regular if and only if it has a barrier. If every boundary point of Ω is regular, then for every $f \in C(\partial\Omega)$ the Perron envelope U_f is the unique harmonic function on Ω which extends continuously to $\overline{\Omega}$ with boundary values f. If $\xi \in \partial\Omega$ and there is a ball $B(a,r)$ with $\overline{B(a,r)} \cap \overline{\Omega} = \{\xi\}$, then ξ has a barrier.*

Proof Suppose that b is a barrier at ξ, and let $f \in C(\partial\Omega)$ and $\varepsilon > 0$. We show that $|U_f - f(\xi)| \le \varepsilon + Cb$ on Ω for a suitable constant C. Choose $\delta > 0$ such that $|f(\zeta) - f(\xi)| < \varepsilon$ for all $\zeta \in \partial\Omega$ with $|\zeta - \xi| < 2\delta$, and put $c = \inf_{\Omega\setminus B(\xi,\delta)} b > 0$; if $\Omega \subset B(\xi,\delta)$, then $|\zeta - \xi| \le \delta$ for every $\zeta \in \partial\Omega$, and we put $c = 1$. If $\zeta \in \partial\Omega$ and $|\zeta - \xi| \ge 2\delta$, then $\Omega \cap B(\zeta,\delta)$ lies outside $B(\xi,\delta)$, so $\underline{\lim}_{x\to\zeta} b(x) \ge c$. Let $M = \max_{\partial\Omega} |f|$ and fix $C \ge 2M/c$, which depends on f, ε and b. We claim that, for every $\zeta \in \partial\Omega$,

$$\overline{\lim_{x\to\zeta}}\big(f(\xi) - \varepsilon - Cb(x)\big) \le f(\zeta) \le \underline{\lim_{x\to\zeta}}\big(f(\xi) + \varepsilon + Cb(x)\big).$$

If $|\zeta - \xi| < 2\delta$, this holds because $b > 0$ and $f(\xi) - \varepsilon < f(\zeta) < f(\xi) + \varepsilon$. If $|\zeta - \xi| \ge 2\delta$, it holds because $f(\xi) - Cc \le M - 2M \le f(\zeta)$ and $f(\xi) + Cc \ge -M + 2M \ge f(\zeta)$.

Hence the continuous subsolution $f(\xi) - \varepsilon - Cb$ belongs to $\mathcal{S}_f$, and the continuous supersolution $f(\xi) + \varepsilon + Cb$ belongs to $-\mathcal{S}_{-f}$, so that it dominates every member of $\mathcal{S}_f$, as shown after Theorem 2.4.2. Therefore $|U_f - f(\xi)| \le \varepsilon + Cb$ on Ω. Letting $x \to \xi$ and then $\varepsilon \downarrow 0$ shows that $U_f(x) \to f(\xi)$. Thus ξ is regular.

Conversely, let ξ be regular, and take the boundary function $f(\zeta) = |\zeta - \xi|$. The smooth function $v(x) = |x - \xi|^2/\operatorname{diam}\Omega$ satisfies $\Delta v \ge 0$, and $v \le f$ on $\partial\Omega$ because $|\zeta - \xi| \le \operatorname{diam}\overline{\Omega} = \operatorname{diam}\Omega$ for $\zeta \in \partial\Omega$; hence $v \in \mathcal{S}_f$. Thus $b = U_f$ is harmonic by Theorem 2.4.2, and hence a continuous supersolution, with $b \ge |x - \xi|^2/\operatorname{diam}\Omega$. In particular b is positive on Ω, and if N is a neighborhood of ξ containing a ball $B(\xi, t)$, then $\inf_{\Omega\setminus N} b \ge t^2/\operatorname{diam}\Omega > 0$. By regularity, $b(x) \to f(\xi) = 0$ as $x \to \xi$. Hence b is a barrier.

Now suppose that every boundary point is regular, and let $f \in C(\partial\Omega)$. Then U_f is harmonic on Ω by Theorem 2.4.2. By regularity and the continuity of f on $\partial\Omega$, the function equal to U_f on Ω and to f on $\partial\Omega$ is continuous on $\overline{\Omega}$. Uniqueness follows from Proposition 2.1.3.

Finally, suppose that $\overline{B(a,r)} \cap \overline{\Omega} = \{\xi\}$. Then $|\xi - a| = r$, since otherwise the open ball $B(a,r)$ would be a neighborhood of ξ and would therefore contain points of Ω, which differ from $\xi \notin \Omega$. In particular $a \notin \overline{\Omega}$, and every point of $\overline{\Omega} \setminus \{\xi\}$ satisfies $|x - a| > r$. Put

$$b(x) = \begin{cases} |x-a| - r, & d = 1, \\ \log(|x-a|/r), & d = 2, \\ r^{2-d} - |x-a|^{2-d}, & d \ge 3. \end{cases}$$

For $d \ge 2$ this is harmonic on $\mathbb{R}^d \setminus \{a\}$ by Proposition 2.2.1, being a positive multiple of $E_d(x-a)$ plus a constant. In dimension one, each component of Ω is an open interval disjoint from $[a-r, a+r]$, so it lies on one side of this interval; hence $|x-a|$ is affine on each component, and b is harmonic on Ω as well. In all dimensions b is an increasing function of $|x-a|$ which vanishes when $|x-a| = r$, so it is continuous on the compact set $\overline{\Omega}$, positive on $\overline{\Omega} \setminus \{\xi\}$ and zero at ξ. Hence $b(x) \to 0$ as $x \to \xi$, and if N is a neighborhood of ξ with $\Omega \setminus N \ne \varnothing$, the infimum of b over the nonempty compact set $\overline{\Omega} \setminus \operatorname{Int} N$ is attained and positive. Thus b is a barrier at ξ. □

It is often easier to construct a barrier only near ξ; the next proposition shows that this suffices.

Proposition 2.4.7 *Let $\xi \in \partial\Omega$ and let W be an open neighborhood of ξ. Suppose that b is a positive continuous supersolution on $\Omega \cap W$ which extends continuously to $\overline{\Omega} \cap W$, and that the extension vanishes at ξ but nowhere else on $\overline{\Omega} \cap W$. Then ξ is regular.*

Proof Choose $r > 0$ with $\overline{B(\xi, 2r)} \subset W$ and $0 < t < r$. The set $L = \overline{\Omega} \cap \{r - t \le |x - \xi| \le r + t\}$ is compact, and b is positive and continuous on it, so there is $c > 0$ with $b \ge 2c$ on L (if L is empty, any $c > 0$ will do). Define $\tilde{b} = \min\{b, c\}$ on $\Omega \cap B(\xi, r)$ and $\tilde{b} = c$ on $\Omega \setminus B(\xi, r)$. Since $b \ge 2c$ on L, we have $\tilde{b} = \min\{b, c\}$ on $\Omega \cap B(\xi, r + t)$ and $\tilde{b} = c$ on $\Omega \setminus \overline{B(\xi, r)}$. These two open sets cover Ω, and on each of them $\tilde{b}$ is a continuous supersolution, being a minimum of two continuous

supersolutions or a constant. By the locality of the subsolution property, noted after the definition of continuous subsolutions, $\tilde b$ is a continuous supersolution on Ω.

The function $\tilde b$ is positive, and it tends to zero at ξ, because $0 < \tilde b \le b$ on $\Omega \cap B(\xi, r)$ and the extension of b vanishes at ξ. If N is a neighborhood of ξ, then $\tilde b \ge c$ on $\Omega \setminus B(\xi, r)$, while on $(\overline{\Omega \cap B(\xi, r)}) \setminus N$ it is at least the minimum of c and of b over the compact set $(\overline{\Omega} \cap \overline{B(\xi, r)}) \setminus \operatorname{Int} N$, on which b is positive. Hence $\tilde b$ is a barrier at ξ, and Theorem 2.4.6 shows that ξ is regular. □

Example 2.4.8 Let $d \ge 2$. The punctured ball $\Omega = B(0,1) \setminus \{0\}$ is a bounded domain with boundary $S^{d-1} \cup \{0\}$, and the origin is an irregular boundary point, that is, a boundary point that is not regular. The punctured ball is connected, because it is the image of $(0,1) \times S^{d-1}$ under the continuous map $(t, \omega) \mapsto t\omega$ and the sphere S^{d-1} is path-connected when $d \ge 2$. Indeed, two points ω, ω' with $\omega' \ne -\omega$ are joined by the path $s \mapsto ((1-s)\omega + s\omega')/|(1-s)\omega + s\omega'|$, $0 \le s \le 1$, whose denominator does not vanish, and two antipodal points can be joined through a unit vector orthogonal to them, which exists because $d \ge 2$.

To see that the origin is irregular, let f equal one at the origin and zero on S^{d-1}; it is continuous on $\partial\Omega$ because the origin is an isolated point of $\partial\Omega$. Let $g(x) = \Gamma_d(x) - \Gamma_d(e_1)$, that is, $g(x) = (2\pi)^{-1} \log(1/|x|)$ if $d = 2$ and $g(x) = ((d-2)s_{d-1})^{-1}(|x|^{2-d} - 1)$ if $d \ge 3$. This function is harmonic on $\mathbb{R}^d \setminus \{0\}$ by Proposition 2.2.1, it vanishes on S^{d-1}, and it tends to $+\infty$ at the origin. Let $v \in \mathcal{S}_f$ and $\varepsilon > 0$; recall that $v \le \max_{\partial\Omega} f = 1$. For every sufficiently small $\delta \in (0,1)$ we have $\varepsilon g \ge 1$ on the sphere $|x| = \delta$. On the annulus $D = \{\delta < |x| < 1\}$, the function $v - \varepsilon g$ is a continuous subsolution; it is continuous and nonpositive on the inner sphere $|x| = \delta$, and its upper limit at every point of S^{d-1} is at most $f = 0$, because g is continuous with value 0 there. By Lemma 2.4.1, $v \le \varepsilon g$ on D. Letting $\delta \downarrow 0$ and then $\varepsilon \downarrow 0$ gives $v \le 0$ on Ω. Since $0 \in \mathcal{S}_f$, the Perron envelope U_f vanishes identically, so it does not tend to $f(0) = 1$ at the origin.

In dimension one the situation is different: the bounded open set $(-1,0) \cup (0,1)$ has boundary $\{-1, 0, 1\}$. If f is a function on this boundary, the function which is affine on each of the two intervals with the boundary values given by f is harmonic and extends continuously to $[-1,1]$ with boundary values f, so it equals U_f by Theorem 2.4.2. Hence the origin is a regular boundary point of $(-1,0) \cup (0,1)$.

2.5 Harmonic measure and Green functions

A nonempty bounded open set is called *regular* if all its boundary points are regular in the sense of Definition 2.4.5. Balls are regular, by the first and the last assertion of Theorem 2.4.6: for $\zeta \in \partial B(c,R)$, the closed ball $\overline{B(2\zeta - c, R)}$ meets $\overline{B(c,R)}$ only at ζ. Indeed, ζ is the midpoint of c and $2\zeta - c$, so every $y \in \mathbb{R}^d$ satisfies $|y-c|^2 + |y-(2\zeta-c)|^2 = 2|y-\zeta|^2 + 2R^2$, and if y lies in both closed balls, the left side is at most $2R^2$, whence $y = \zeta$.

In this section Ω is a regular bounded open set. For real $f \in C(\partial\Omega)$ we write $H_\Omega f$ for the unique harmonic function on Ω which extends continuously to $\overline{\Omega}$ with boundary

values f (Theorem 2.4.6). By uniqueness, H_Ω is linear, and by Proposition 2.1.3 it preserves order and satisfies $\sup_\Omega |H_\Omega f| \le \max_{\partial\Omega} |f|$. For complex boundary data we put $H_\Omega(f + ig) = H_\Omega f + iH_\Omega g$, where f and g are real and continuous. When Ω is a ball, H_Ω is the Poisson integral of Theorem 2.3.1, again by uniqueness.

Proposition 2.5.1 *For every $x \in \Omega$ there is a unique Borel probability measure $\omega_x = \omega_x^\Omega$ on $\partial\Omega$ such that*

$$H_\Omega f(x) = \int_{\partial\Omega} f \, d\omega_x, \qquad f \in C(\partial\Omega).$$

For each $f \in C(\partial\Omega)$, the function $x \mapsto \int f \, d\omega_x$ is harmonic on Ω. If Ω is a ball, then $d\omega_x = P_\Omega(x, \cdot)\, dS$.

Proof The map $g \mapsto H_\Omega(g|_{\partial\Omega})(x)$ is a positive linear functional on $C_c(\mathbb{R}^d, \mathbb{R})$, so by Theorem 1.2.2 it is integration against a Radon measure μ on $\mathbb{R}^d$. If $g \in C_c(\mathbb{R}^d)$ is supported in the open set $\mathbb{R}^d \setminus \partial\Omega$, then $g|_{\partial\Omega} = 0$ and $\int g \, d\mu = 0$; by (1.2), $\mu(\mathbb{R}^d \setminus \partial\Omega) = 0$. If $\eta \in C_c(\mathbb{R}^d)$ equals one on the compact set $\partial\Omega$, then $\mu(\mathbb{R}^d) = \mu(\partial\Omega) = \int \eta \, d\mu = H_\Omega 1(x) = 1$. Let ω_x be the restriction of μ to the Borel subsets of $\partial\Omega$; it is a probability measure.

Every $f \in C(\partial\Omega)$ is a uniform limit of restrictions of functions in $C_c(\mathbb{R}^d)$, by Lemma 2.4.3. Both sides of the asserted identity are continuous for uniform convergence on $\partial\Omega$, the right side because ω_x is finite and the left side because $|H_\Omega f| \le \max_{\partial\Omega} |f|$. Hence the identity holds for real f, and by linearity for complex f. If ω' is another Borel probability measure on $\partial\Omega$ with the same property, then ω_x and ω', extended by zero to $\mathbb{R}^d$, are finite Borel measures, hence Radon measures by Proposition 1.1.2, with the same integrals against every $g \in C_c(\mathbb{R}^d)$. The uniqueness assertion of Theorem 1.2.2 gives $\omega' = \omega_x$. The function $x \mapsto \int f \, d\omega_x = H_\Omega f(x)$ is harmonic by the definition of H_Ω. If Ω is a ball, $P_\Omega(x, \cdot)\, dS$ is a Borel probability measure on $\partial\Omega$ with the required property, by Theorem 2.3.1, so it equals ω_x. □

The probability measure ω_x is called the *harmonic measure* of Ω at x. When Ω is a ball centered at x, it is normalized surface measure on $\partial\Omega$; in general it depends on x.

Theorem 2.5.2 *Let Ω be a bounded regular domain. For $x \in \Omega$, define the* Green function *of Ω by*

$$G_\Omega(x, y) = \Gamma_d(y - x) - H_\Omega(\Gamma_d(\cdot - x)|_{\partial\Omega})(y), \qquad y \in \Omega,$$

so that $G_\Omega(x, x) = \Gamma_d(0) = +\infty$ when $d \ge 2$.

(1) *For every $x \in \Omega$, the function $G_\Omega(x, \cdot)$ is positive on Ω, harmonic on $\Omega \setminus \{x\}$ and locally integrable on Ω, it satisfies $-\Delta_y G_\Omega(x, \cdot) = \delta_x$ in the sense of distributions on Ω, and $G_\Omega(x, y) \to 0$ as y tends to any point of $\partial\Omega$. These properties determine $G_\Omega(x, y)$ uniquely for $y \ne x$.*

(2) *If $d = 1$ and $\Omega = (a, b)$, then*

$$G_{(a,b)}(x,y) = \frac{(\min\{x,y\}-a)(b-\max\{x,y\})}{b-a}, \qquad x,y \in (a,b).$$

(3) *Green functions are monotone in the domain: if $D' \subset D''$ are bounded regular domains, then $G_{D'}(x,y) \le G_{D''}(x,y)$ for all $x, y \in D'$.*

(4) *Let $D_1 \subset D_2 \subset \cdots$ be bounded regular domains whose union D is a bounded regular domain. Then for $x \ne y$ in D, the values $G_{D_j}(x,y)$, defined once $x, y \in D_j$, increase to $G_D(x,y)$, and for fixed x the convergence holds in C^∞ on compact subsets of $D \setminus \{x\}$, as functions of y.*

Proof (1) Fix $x \in \Omega$ and let $\gamma = \Gamma_d(\cdot - x)|_{\partial\Omega}$, which is continuous because $\Gamma_d(\cdot - x)$ is continuous on $\mathbb{R}^d \setminus \{x\}$. The function $H_\Omega\gamma$ is harmonic on Ω, and $\Gamma_d(\cdot - x)$ is harmonic on $\mathbb{R}^d \setminus \{x\}$ by Proposition 2.2.1, so $G_\Omega(x,\cdot)$ is harmonic on $\Omega \setminus \{x\}$. By the same proposition, $\Gamma_d(\cdot - x)$ is locally integrable, and for every test function χ on Ω, which we extend by zero to $\mathbb{R}^d$,

$$\int \Gamma_d(y-x)\Delta\chi(y)\,\mathrm{d}y = -\int E_d(z)(\Delta\chi)(x+z)\,\mathrm{d}z = -\langle \Delta E_d, \chi(x+\cdot)\rangle = -\chi(x).$$

Since $H_\Omega\gamma$ is smooth on Ω with $\Delta H_\Omega\gamma = 0$, integration by parts gives $\int H_\Omega\gamma\,\Delta\chi\,\mathrm{d}y = 0$. Hence $G_\Omega(x,\cdot)$ is locally integrable on Ω and $\langle -\Delta_y G_\Omega(x,\cdot), \chi\rangle = \chi(x)$, that is, $-\Delta_y G_\Omega(x,\cdot) = \delta_x$. As $y \to \zeta \in \partial\Omega$, both $\Gamma_d(y-x)$ and $H_\Omega\gamma(y)$ tend to $\gamma(\zeta)$, so $G_\Omega(x,y) \to 0$.

We prove positivity when $d \ge 2$; for $d = 1$ it follows from (2) below. First we show that $G_\Omega(x,\cdot) \ge 0$. Let $v \in \mathcal{S}_\gamma$; we claim that $v \le \Gamma_d(\cdot - x)$ on $\Omega \setminus \{x\}$. Since $v \le \max_{\partial\Omega}\gamma$ and $\Gamma_d(y-x) \to +\infty$ as $y \to x$, there is $\varepsilon_0 > 0$ such that $\overline{B(x,\varepsilon_0)} \subset \Omega$ and $v \le \Gamma_d(\cdot - x)$ on $\partial B(x,\varepsilon)$ for $0 < \varepsilon < \varepsilon_0$. On the bounded open set $\Omega \setminus \overline{B(x,\varepsilon)}$, whose boundary lies in $\partial\Omega \cup \partial B(x,\varepsilon)$, the function $v - \Gamma_d(\cdot - x)$ is a continuous subsolution. At a point $\zeta \in \partial\Omega$ its upper limit is at most $\gamma(\zeta) - \gamma(\zeta) = 0$, and on $\partial B(x,\varepsilon)$ it is continuous and nonpositive. By Lemma 2.4.1, $v \le \Gamma_d(\cdot - x)$ on $\Omega \setminus \overline{B(x,\varepsilon)}$, and letting $\varepsilon \downarrow 0$ gives the claim. Taking the supremum over v and using $H_\Omega\gamma = U_\gamma$ (Theorem 2.4.6), we obtain $H_\Omega\gamma \le \Gamma_d(\cdot - x)$ on $\Omega \setminus \{x\}$, that is, $G_\Omega(x,\cdot) \ge 0$.

Strict positivity follows from the minimum principle, because $\Omega \setminus \{x\}$ is connected when $d \ge 2$. To see this, let $y_0, y_1 \in \Omega \setminus \{x\}$. They can be joined by a polygonal path $p\colon [0,1] \to \Omega$, because the points of the domain Ω that can be joined to y_0 by a polygonal path in Ω form an open set whose complement in Ω is also open, balls being convex. Suppose that p passes through x, and choose $r > 0$ with $\overline{B(x,r)} \subset \Omega$ and $r < \min\{|y_0 - x|, |y_1 - x|\}$. By the intermediate value theorem, the closed set of times t with $|p(t) - x| = r$ is nonempty; let t_1 and t_2 be its smallest and largest elements. Then $|p(t) - x| > r$ for $t < t_1$ and for $t > t_2$, and replacing p on $[t_1, t_2]$ by a path from $p(t_1)$ to $p(t_2)$ in the sphere $\partial B(x,r)$, which is path-connected because $d \ge 2$ (Example 2.4.8), we obtain a path from y_0 to y_1 in $\Omega \setminus \{x\}$. Now if $G_\Omega(x,\cdot)$ vanished at some point of the domain $\Omega \setminus \{x\}$, it would attain its minimum there, and part (1) of Proposition 2.1.3 would make it identically zero on $\Omega \setminus \{x\}$, contradicting $G_\Omega(x,y) \to +\infty$ as $y \to x$. Hence $G_\Omega(x,\cdot) > 0$.

For uniqueness, let G' be another function with the stated properties. The locally integrable function $G_\Omega(x,\cdot) - G'$ has vanishing distributional Laplacian on Ω, so by Theorem 2.1.4 it agrees almost everywhere with a harmonic function h on Ω. Away from x, the functions $G_\Omega(x,\cdot)$ and G' are harmonic, so their difference is continuous there and therefore equals h everywhere on $\Omega \setminus \{x\}$. Then h is real and has zero boundary limits, because $G_\Omega(x,\cdot)$ and G' do, and $h = 0$ by Proposition 2.1.3, applied to h and to $-h$. This argument does not use positivity, and it applies in every dimension.

(2) Let $d = 1$, $\Omega = (a,b)$ and $x \in (a,b)$, and let $g(y)$ denote the right side of the formula in (2). The function g is continuous, positive on (a,b), and tends to zero at a and b. It is affine on each side of x, with slope $g'_- = (b-x)/(b-a)$ on (a,x) and $g'_+ = -(x-a)/(b-a)$ on (x,b), so $g'_+ - g'_- = -1$. For a test function χ on (a,b), an integration by parts on each of the intervals (a,x) and (x,b), on which g' is constant, gives

$$\int_a^b g\chi''\,\mathrm{d}y = \big(g(x)\chi'(x) - g'_-\chi(x)\big) + \big(-g(x)\chi'(x) + g'_+\chi(x)\big) = -\chi(x),$$

so $-g'' = \delta_x$ in the sense of distributions. Moreover g is harmonic on $(a,b) \setminus \{x\}$. By the uniqueness argument just given, g coincides with $G_{(a,b)}(x,\cdot)$ on $(a,b) \setminus \{x\}$, and also at x, since both functions are continuous there. This proves (2), and with it the positivity in (1) when $d = 1$.

(3) Let $D' \subset D''$ be bounded regular domains and $x \in D'$. On $D' \setminus \{x\}$ the fundamental solutions cancel, and

$$G_{D''}(x,\cdot) - G_{D'}(x,\cdot) = H_{D'}(\Gamma_d(\cdot - x)|_{\partial D'}) - H_{D''}(\Gamma_d(\cdot - x)|_{\partial D''}),$$

which is harmonic on all of D'. At every $\zeta \in \partial D'$ its lower limit is nonnegative, because $G_{D'}(x,y) \to 0$ as $y \to \zeta$ and $G_{D''}(x,y) \geq 0$. By Proposition 2.1.3, applied to its negative, this harmonic function is nonnegative on D', so $G_{D'}(x,y) \leq G_{D''}(x,y)$ for $y \in D' \setminus \{x\}$. For $y = x$, both sides are $+\infty$ if $d \geq 2$; if $d = 1$, the display also holds at $y = x$, so the inequality holds there as well.

(4) Let D_j and D be as in the statement, fix $x \in D$, and consider only j large enough that $x \in D_j$. Put $h_j = H_{D_j}(\Gamma_d(\cdot - x)|_{\partial D_j})$ and $h = H_D(\Gamma_d(\cdot - x)|_{\partial D})$; we show that h_j decreases to h in C^∞ on compact subsets of D. By (3), $G_{D_j}(x,\cdot) \leq G_{D_{j+1}}(x,\cdot) \leq G_D(x,\cdot)$ on $D_j \setminus \{x\}$, that is, $h_j \geq h_{j+1} \geq h$ there, and also at x, since these functions are continuous at x. Let B be a ball with $\overline{B} \subset D$. The compact set $\overline{B}$ is covered by the increasing open sets D_j, so $\overline{B} \subset D_j$ for all large j. On B the functions h_j then form a decreasing sequence of harmonic functions bounded below by h, and part (3) of Corollary 2.3.3 shows that they converge in C^∞ on compact subsets of B to a finite harmonic function. These local limits are pointwise limits, so they define a harmonic function h_∞ on D. Every compact subset of D is covered by finitely many balls $B(a,s)$ with $\overline{B(a,2s)} \subset D$, and $\overline{B(a,s)}$ is a compact subset of $B(a,2s)$; hence $h_j \to h_\infty$ in C^∞ on compact subsets of D.

It remains to show that $h_\infty = h$. The function $\Gamma_d(\cdot - x) - h_\infty$ is the pointwise limit of $G_{D_j}(x, \cdot)$, so it lies between 0 and $G_D(x, \cdot)$. Therefore the harmonic function $h_\infty - h = G_D(x, \cdot) - (\Gamma_d(\cdot - x) - h_\infty)$ satisfies $0 \le h_\infty - h \le G_D(x, \cdot)$ on $D \setminus \{x\}$, so it has zero boundary limits and vanishes by Proposition 2.1.3. Thus $h_j \to h$, which means that $G_{D_j}(x, y)$ increases to $G_D(x, y)$ for every $y \neq x$. Moreover $h_j - h \to 0$ in C^∞ on compact subsets of D, and $G_{D_j}(x, \cdot) - G_D(x, \cdot) = h - h_j$ on $D_j \setminus \{x\}$; since every compact subset of $D \setminus \{x\}$ lies in D_j for large j, $G_{D_j}(x, \cdot) \to G_D(x, \cdot)$ in C^∞ on compact subsets of $D \setminus \{x\}$. □

Chapter 3
Subharmonic functions

Subharmonic functions are, roughly speaking, the functions u with $\Delta u \geq 0$. They are the real-variable model for the plurisubharmonic functions of Part II, and they include the potentials $E_d * \nu$ of positive measures ν, which are studied further in Chapter 4. The condition $\Delta u \geq 0$ cannot be imposed in the classical sense, because the most important examples are singular: for $d \geq 2$ the fundamental solution E_d of Proposition 2.2.1 tends to $-\infty$ at the origin. We therefore define subharmonicity pointwise, by upper semicontinuity and a submean inequality on small spheres, and we allow the value $-\infty$. Throughout the chapter, Ω is an open subset of $\mathbb{R}^d$ and $d \geq 1$.

From the submean inequality we first derive the comparison principle and the monotonicity of spherical means. We then show that locally integrable subharmonic functions correspond exactly to real distributions with positive Laplacian (Theorem 3.2.2). No pointwise information is lost in passing to the distribution: the value of a subharmonic function at each point is the limit of its ball averages, and every real distribution with positive Laplacian has exactly one subharmonic representative, its canonical representative. This allows us to pass between pointwise and almost-everywhere statements, and it leads to the Riesz decomposition of a subharmonic function into a potential and a harmonic function.

The last three sections concern limits and their exceptional sets. A sequence of subharmonic functions on a domain that is uniformly bounded above on compact sets either tends to $-\infty$ locally uniformly or has a subsequence converging in L^1_{loc}, and by Hartogs' lemma the limit controls the sequence from above, uniformly on compact sets. The pointwise supremum of a family of subharmonic functions need not be upper semicontinuous; with Choquet's countable selection lemma we show that its upper semicontinuous regularization is subharmonic and differs from it only on a Lebesgue null set. Finally, we introduce polar sets, which lie locally in the pole set $\{\psi = -\infty\}$ of a subharmonic function ψ that is not identically $-\infty$, and prove that relatively closed polar sets are removable for subharmonic functions that are bounded above near them.

Historical remarks and references for this chapter are collected in Appendix B.3.

3.1 Mean inequalities and comparison

Recall that a function $u\colon \Omega \to [-\infty, \infty)$ is upper semicontinuous if the set $\{u < a\}$ is open for every $a \in \mathbb{R}$; equivalently, $\overline{\lim}_{y\to x} u(y) \le u(x)$ for every $x \in \Omega$. Such a function is Borel measurable, because the sets $\{u < a\}$ are open. It is bounded above on every compact set $K \subset \Omega$, because the increasing open sets $\{u < n\}$, $n \in \mathbb{N}$, cover K, so that one of them contains K. A similar covering argument shows that u attains its maximum on every nonempty compact set K: otherwise the open sets $\{u < a\}$ with $a < \sup_K u$ would cover K, and a finite subcover would give $\sup_K u < \sup_K u$. The same arguments apply to upper semicontinuous functions on any compact metric space.

We shall often integrate upper semicontinuous functions over products. Subsets X and Y of Euclidean spaces are separable metric spaces, so by Lemma 1.1.8 every Borel function on $X \times Y$ is measurable for the product of the Borel σ-algebras of X and Y. Hence Tonelli's and Fubini's theorems (Proposition 1.1.7) apply to Borel functions on such products. As in Section 2.1, $\mathrm{d}S$ denotes surface measure on the unit sphere S^{d-1}, and s_{d-1} is its total mass.

Definition 3.1.1 A function $u\colon \Omega \to [-\infty, \infty)$ is subharmonic if it is upper semicontinuous and every $x \in \Omega$ admits an $r_x > 0$ with $\overline{B(x, r_x)} \subset \Omega$ such that

$$u(x) \le \frac{1}{s_{d-1}} \int_{S^{d-1}} u(x + r\omega)\, \mathrm{d}S(\omega) \qquad \text{for } 0 < r < r_x. \tag{3.1}$$

The function identically equal to $-\infty$ is subharmonic. We write $\mathrm{SH}(\Omega)$ for the set of subharmonic functions on Ω. A function v is superharmonic if $-v$ is subharmonic.

As for continuous functions in Section 2.1, we denote the right-hand side of (3.1) by $M_u(x, r)$, the spherical mean of u over $\partial B(x, r)$, for every upper semicontinuous u and every $r > 0$ with $\overline{B(x, r)} \subset \Omega$. It is well defined in $[-\infty, \infty)$, since u is Borel measurable and bounded above on the compact sphere $x + rS^{d-1}$.

Subharmonicity is a local property: a function is subharmonic on Ω if every point of Ω has an open neighborhood on which it is subharmonic. Indeed, upper semicontinuity is a local condition, and if u is subharmonic on an open neighborhood $W \subset \Omega$ of x, then a radius r_x as in Definition 3.1.1 for $u|_W$, with $\overline{B(x, r_x)} \subset W$, also serves for u on Ω. By Proposition 2.1.1, every real harmonic function on Ω is subharmonic. To compare subharmonic functions with harmonic functions on balls, we shall approximate upper semicontinuous boundary values from above by continuous functions.

Lemma 3.1.2 *Let X be a metric space and let $f\colon X \to [-\infty, \infty)$ be upper semicontinuous and bounded above; the latter holds automatically if X is compact. Then f is the pointwise limit of a decreasing sequence of bounded, real-valued continuous functions f_n on X. For every finite Borel measure μ on X, the integrals $\int_X f_n \,\mathrm{d}\mu$ decrease to $\int_X f \,\mathrm{d}\mu \in [-\infty, \infty)$.*

Proof Since f is bounded above, we can choose a positive integer $n_0 > \sup_X f$. For $n \geq n_0$ and $x \in X$ put

$$f_n(x) = \sup_{y \in X} \bigl(\max\{f(y), -n\} - n \operatorname{dist}(x, y)\bigr).$$

Taking $y = x$ gives $f_n(x) \geq \max\{f(x), -n\} \geq f(x)$, and $f_n \leq \max\{\sup_X f, -n\} < n_0$. Thus $-n \leq f_n < n_0$, so f_n is real-valued and bounded, and as a supremum of n-Lipschitz functions of x it is n-Lipschitz. Both $\max\{f(y), -n\}$ and $-n \operatorname{dist}(x, y)$ decrease as n increases, so the sequence (f_n) is decreasing.

We show that $f_n(x) \to f(x)$ for every $x \in X$. Otherwise there is a real number c with $\lim_n f_n(x) > c > f(x)$. For each n choose $y_n \in X$ with $\max\{f(y_n), -n\} - n \operatorname{dist}(x, y_n) > c$. Then $n \operatorname{dist}(x, y_n) < n_0 - c$, so $y_n \to x$. Moreover $\max\{f(y_n), -n\} > c$, which forces $f(y_n) > c$ as soon as $n > -c$. On the other hand, $\{f < c\}$ is an open neighborhood of x by upper semicontinuity, so it contains y_n for all large n. This contradiction proves that $f_n \downarrow f$.

Finally, the nonnegative functions $n_0 - f_n$ increase to $n_0 - f$, so the monotone convergence theorem gives $\int (n_0 - f_n)\, \mathrm{d}\mu \uparrow \int (n_0 - f)\, \mathrm{d}\mu$. Subtracting these integrals from the finite number $n_0 \mu(X)$ proves the assertion about the integrals of f_n. □

Theorem 3.1.3 *Let $u \in \mathrm{SH}(\Omega)$.*

(1) *Let $D \subset \Omega$ be a bounded open set, and let h be a real function which is harmonic on D and continuous on $\overline{D}$. If*

$$\overline{\lim_{D \ni x \to \zeta}} u(x) \leq h(\zeta) \qquad \text{for every } \zeta \in \partial D,$$

then $u \leq h$ on D.

(2) *Let $x \in \Omega$ and $\overline{B(x, R)} \subset \Omega$. Then the spherical mean $M_u(x, r)$ increases with $r \in (0, R]$ and decreases to $u(x)$ as $r \downarrow 0$. Moreover,*

$$u(x) \leq \frac{1}{|B(x, R)|} \int_{B(x,R)} u(y)\, \mathrm{d}y.$$

Conversely, an upper semicontinuous function $u \colon \Omega \to [-\infty, \infty)$ is subharmonic if assertion (1) *holds for u whenever D is a ball with $\overline{D} \subset \Omega$.*

Proof We begin with (1). Fix $\varepsilon > 0$ and consider the upper semicontinuous function $w(y) = u(y) - h(y) + \varepsilon |y|^2$ on D. For $x \in D$ and small $r > 0$ we have $M_h(x, r) = h(x)$ by Proposition 2.1.1, and $M_{|\cdot|^2}(x, r) = |x|^2 + r^2$ by the identity for spherical means of $|y|^2$ proved in Section 2.4. Together with (3.1) this gives

$$M_w(x, r) \geq w(x) + \varepsilon r^2$$

for every $x \in D$ and every sufficiently small $r > 0$. Consequently, w cannot attain a finite maximum over D at a point $x \in D$, since we would then have $M_w(x, r) \leq w(x)$ for small r.

Now let $\delta > 0$. Because h is continuous on $\overline{D}$, the boundary hypothesis gives $\overline{\lim}_{D \ni x \to \zeta} (u - h)(x) \le 0$, so every $\zeta \in \partial D$ has an open neighborhood N_ζ with $u - h < \delta$ on $N_\zeta \cap D$. Finitely many of these neighborhoods cover the compact set ∂D. If N denotes their union, then $K = \overline{D} \setminus N$ is a compact subset of D. Put $c = \delta + \varepsilon \sup_{y \in \overline{D}} |y|^2$, so that $w < c$ on $D \setminus K$. If w took a value larger than c, then $\sup_D w = \sup_K w$ would be a finite maximum attained at a point of K, which we have just excluded. Hence $u - h \le w \le c$ on D, and letting $\varepsilon \downarrow 0$ and $\delta \downarrow 0$ gives $u \le h$.

For (2), let $B = B(x, R)$. By Lemma 3.1.2 there are continuous functions f_n on ∂B that decrease to $u|_{\partial B}$. Let $H_n = H_B f_n$ be their Poisson integrals; by Theorem 2.3.1, H_n is harmonic on B and continuous on $\overline{B}$ with boundary values f_n. For $\zeta \in \partial B$, upper semicontinuity gives $\overline{\lim}_{B \ni y \to \zeta} u(y) \le u(\zeta) \le f_n(\zeta)$, so (1) yields $u \le H_n$ on B. The value of a Poisson integral at the center of the ball is the mean of its boundary values, because $P_B(x, \cdot)$ is the constant $(s_{d-1} R^{d-1})^{-1}$ by (2.9). Using also the mean-value property of H_n, we obtain for $0 < r < R$

$$u(x) \le H_n(x) = M_{f_n}(x, R), \qquad M_u(x, r) \le M_{H_n}(x, r) = H_n(x) = M_{f_n}(x, R).$$

By Lemma 3.1.2, applied to surface measure on ∂B, $M_{f_n}(x, R)$ decreases to $M_u(x, R)$. Therefore $u(x) \le M_u(x, R)$, and $M_u(x, r) \le M_u(x, R)$ for $0 < r < R$. As R may be replaced by any smaller radius, $M_u(x, \cdot)$ is increasing on $(0, R]$. Moreover, $u(x) \le M_u(x, r) \le \sup_{\partial B(x,r)} u$, and the upper limit of the right-hand side as $r \downarrow 0$ is at most $u(x)$ by upper semicontinuity; hence $M_u(x, r)$ decreases to $u(x)$ as $r \downarrow 0$.

Finally, let M be a real upper bound for u on $\overline{B}$. Integration in polar coordinates (2.1), applied to the nonnegative function $M - u$, and the substitution $r = Rt$ give

$$\frac{1}{|B|} \int_B u(y)\, \mathrm{d}y = \frac{d}{R^d} \int_0^R r^{d-1} M_u(x, r)\, \mathrm{d}r = d \int_0^1 t^{d-1} M_u(x, Rt)\, \mathrm{d}t, \tag{3.2}$$

where all integrals lie in $[-\infty, \infty)$; here we used $|B| = s_{d-1} R^d / d$. Since $M_u(x, r) \ge u(x)$ for $0 < r \le R$ and $d \int_0^1 t^{d-1}\, \mathrm{d}t = 1$, the right-hand side is at least $u(x)$.

The proof of (2) used the subharmonicity of u only through (1) for the ball B. Hence, if u is upper semicontinuous and satisfies (1) for all balls B with $\overline{B} \subset \Omega$, then $u(x) \le M_u(x, R)$ whenever $\overline{B(x, R)} \subset \Omega$, so u is subharmonic. □

We call the first assertion of Theorem 3.1.3 the comparison principle. Applied with $\Omega = D$, it holds for every bounded open set D and every $u \in \mathrm{SH}(D)$, even when u is not defined near ∂D.

Elementary operations preserve subharmonicity. If $u, v \in \mathrm{SH}(\Omega)$ and $c \ge 0$, then $u + v$, cu and $\max\{u, v\}$ are subharmonic; here we use the convention $0 \cdot (-\infty) = 0$, so that $0 \cdot u$ is the zero function. Indeed, these functions are upper semicontinuous: $\{cu < a\} = \{u < a/c\}$ if $c > 0$, $\{\max\{u, v\} < a\} = \{u < a\} \cap \{v < a\}$, and if $u(x) + v(x) < a$, we can choose real numbers $s > u(x)$ and $t > v(x)$ with $s + t \le a$, so that $u + v < a$ on the open neighborhood $\{u < s\} \cap \{v < t\}$ of x. The identities $M_{u+v} = M_u + M_v$ and $M_{cu} = c M_u$, together with the inequality $M_{\max\{u,v\}} \ge \max\{M_u, M_v\}$, give the submean inequality at x for all radii smaller

than the radii r_x for u and for v. In particular, adding a real harmonic function to a subharmonic function produces a subharmonic function.

Proposition 3.1.4 *Let $u \in \mathrm{SH}(\Omega)$. If Ω is connected and u attains its maximum over Ω, then u is constant. If Ω is bounded and $\overline{\lim}_{\Omega \ni x \to \zeta} u(x) \le 0$ for every $\zeta \in \partial\Omega$, then $u \le 0$ on Ω.*

Proof Let $m = \max_\Omega u$. If $m = -\infty$, then u is identically $-\infty$, so assume that m is finite, and let $E = \{u = m\} = \{u \ge m\}$. This set is nonempty, and it is closed in Ω because u is upper semicontinuous. Let $x \in E$. For every sufficiently small $r > 0$, (3.1) gives $m = u(x) \le M_u(x, r) \le m$, so

$$\int_{S^{d-1}} \bigl(m - u(x + r\omega)\bigr)\, dS(\omega) = 0.$$

Suppose that $u(x + r\omega_0) < a < m$ for some $\omega_0 \in S^{d-1}$ and some real number a. By upper semicontinuity, the set $A = \{\omega \in S^{d-1} : u(x + r\omega) < a\}$ is relatively open in S^{d-1}, and it contains ω_0. The cone $\{t\omega : 0 < t < 1,\ \omega \in A\}$ is the inverse image of A under the continuous map $y \mapsto y/|y|$ on $B(0, 1) \setminus \{0\}$, so it is a nonempty open set, and therefore $S(A) > 0$ by the definition of surface measure in Section 2.1. Since $m - u(x + r\omega) > m - a > 0$ for $\omega \in A$, the integral above would be at least $(m - a)S(A) > 0$, a contradiction. Hence $u = m$ on every sufficiently small sphere centered at x, and E contains a ball around x. Thus E is open and closed in Ω, and $E = \Omega$ because Ω is connected.

The second assertion is the comparison principle on the bounded open set $D = \Omega$ with $h = 0$, in the form noted after Theorem 3.1.3. □

Proposition 3.1.5 *Let $u \in \mathrm{SH}(\Omega)$. On each connected component of Ω, either u is identically $-\infty$, or u is locally integrable and hence finite almost everywhere.*

Proof Let $x \in \Omega$ with $u(x) > -\infty$, and let $\overline{B(x, r)} \subset \Omega$. If M is a real upper bound for u on $B(x, r)$, the ball submean inequality of Theorem 3.1.3 gives

$$\int_{B(x,r)} (M - u)\, dy \le |B(x, r)|\, (M - u(x)) < \infty.$$

Since $M - u \ge 0$, it follows that u is integrable on $B(x, r)$.

Now let Ω' be a connected component of Ω on which u is not identically $-\infty$, and let Y be the set of points of Ω' that have a neighborhood on which u is integrable. Then Y is open, and it is nonempty by the first paragraph. Let $y \in \Omega'$ lie in the closure of Y, and choose $r > 0$ with $\overline{B(y, 3r)} \subset \Omega$. The set $Y \cap B(y, r)$ contains a point z, and u is integrable on an open neighborhood U of z; hence u is finite almost everywhere on the nonempty open set $U \cap B(y, r)$, which has positive measure, and there is a point $x \in B(y, r)$ with $u(x) > -\infty$. Since $|x - y| < r$, the ball $B(x, 2r)$ contains y and its closure lies in $\overline{B(y, 3r)} \subset \Omega$. By the first paragraph, u is integrable on $B(x, 2r)$, hence $y \in Y$. Thus Y is also closed in Ω', and $Y = \Omega'$ because Ω' is connected. Finally, a locally integrable function is finite almost everywhere. □

A subharmonic function on a domain is called nontrivial if it is not identically $-\infty$; by Proposition 3.1.5, the nontrivial subharmonic functions on a domain are exactly the locally integrable ones. The set $\{u = -\infty\}$ is called the pole set of u. In particular, the pole set of a nontrivial subharmonic function on a domain has Lebesgue measure zero.

Theorem 3.1.6 *Let (u_j) be a decreasing sequence in $\mathrm{SH}(\Omega)$ with pointwise limit u. Then $u \in \mathrm{SH}(\Omega)$. On each connected component Ω' of Ω, either u is identically $-\infty$ and $u_j \to -\infty$ uniformly on compact subsets of Ω', or u is locally integrable on Ω' and $u_j \to u$ in L^1 on compact subsets of Ω'.*

Proof The limit u is upper semicontinuous, since $\{u < a\} = \bigcup_j \{u_j < a\}$ is open for every real a. Let $x \in \Omega$ and $\overline{B(x,r)} \subset \Omega$. By Theorem 3.1.3, $u_j(x) \le M_{u_j}(x,r)$ for every j. If M is a real upper bound for u_1 on $\partial B(x,r)$, the nonnegative functions $M - u_j$ increase to $M - u$ on this sphere, so the monotone convergence theorem gives $M_{u_j}(x,r) \downarrow M_u(x,r)$. Hence $u(x) \le u_j(x) \le M_{u_j}(x,r)$ for all j, and in the limit $u(x) \le M_u(x,r)$. Thus u is subharmonic.

Let Ω' be a connected component of Ω. By Proposition 3.1.5, u is either identically $-\infty$ or locally integrable on Ω'. In the first case, fix a compact set $K \subset \Omega'$ and a real number A. The open sets $\{u_j < -A\}$ increase with j and cover Ω', because $u_j(x) \downarrow -\infty$ for every $x \in \Omega'$. Finitely many of them, and hence one of them, cover K; thus $u_j < -A$ on K for all large j. In the second case, $u_1 \ge u$ is not identically $-\infty$ on Ω' and is therefore locally integrable there as well. Then $0 \le u_j - u \le u_1 - u$ almost everywhere on Ω', and $u_j - u \to 0$ wherever u is finite, so dominated convergence gives $\int_K (u_j - u)\,\mathrm{d}x \to 0$ for every compact $K \subset \Omega'$. □

3.2 Riesz measures and pointwise representatives

We now compare subharmonic functions with distributions. Recall from Proposition 2.2.1 that the fundamental solutions E_d of (2.7) satisfy $\Delta E_d = \delta_0$, and that $E_d(0) = -\infty$ for $d \ge 2$, while $E_1(0) = 0$. Superpositions of translates of E_d produce subharmonic functions whose Laplacian is a prescribed positive measure. Conversely, by the Riesz decomposition (Theorem 3.2.3), every locally integrable subharmonic function is locally such a superposition plus a harmonic function.

Proposition 3.2.1 *Let ν be a finite positive Borel measure on $\mathbb{R}^d$ with compact support. Then its fundamental potential*

$$p(x) = E_d * \nu(x) = \int E_d(x - y)\,\mathrm{d}\nu(y)$$

is subharmonic and locally integrable on $\mathbb{R}^d$, and $\Delta p = \nu$ in the sense of distributions.

Proof By Proposition 1.1.2, ν is a Radon measure; let L be its compact support, so that $\nu(\mathbb{R}^d \setminus L) = 0$ (Section 1.1). For each x, the function $E_d(x - \cdot)$ is bounded

above on L, so $p(x)$ is well defined in $[-\infty, \infty)$. Next, E_d is subharmonic on $\mathbb{R}^d$. For $d \geq 2$ it is harmonic on $\mathbb{R}^d \setminus \{0\}$; it is upper semicontinuous at the origin because $E_d(x) \to -\infty$ as $x \to 0$, and the submean inequality at the origin holds because $E_d(0) = -\infty$. Since subharmonicity is local, $E_d \in \mathrm{SH}(\mathbb{R}^d)$. For $d = 1$, the triangle inequality gives $|x| \leq \frac{1}{2}(|x + r| + |x - r|)$, which is the submean inequality for $E_1 = |x|/2$.

To show that p is subharmonic, we truncate the kernel. The functions $E_d^{(m)} = \max\{E_d, -m\}$, $m \in \mathbb{N}$, are continuous and subharmonic, and they decrease to E_d. Put $p_m = E_d^{(m)} * \nu$. For x in a compact set K, the differences $x - y$ with $y \in L$ lie in the compact set $K - L$, on which $E_d^{(m)}$ is uniformly continuous; hence p_m is continuous. By Theorem 3.1.3, each translate $E_d^{(m)}(\cdot - y)$ satisfies the submean inequality on every sphere. Integrating in y and using Fubini's theorem for the bounded continuous integrand gives $p_m(x) \leq M_{p_m}(x, r)$ for all x and $r > 0$, so $p_m \in \mathrm{SH}(\mathbb{R}^d)$. For fixed x, the functions $E_d^{(m)}(x - y)$ decrease to $E_d(x - y)$ and are bounded above, uniformly in m and $y \in L$, by the finite constant $\max\{0, \sup_{y \in L} E_d(x-y)\}$. Monotone convergence, applied to the differences from this constant, gives $p_m(x) \downarrow p(x)$. Hence p is subharmonic by Theorem 3.1.6.

To prove local integrability, let K be compact and choose $R > 0$ with $K - L \subset B(0, R)$. Since E_d is locally integrable, Tonelli's theorem gives

$$\int_K \int |E_d(x - y)| \, d\nu(y) \, dx \leq \nu(L) \int_{B(0,R)} |E_d(z)| \, dz < \infty.$$

Hence p is locally integrable. Finally, let χ be a test function. The preceding estimate with $K = \operatorname{Supp} \chi$ justifies the use of Fubini's theorem, and the identity $\Delta E_d = \delta_0$, translated to the point y, gives

$$\int p \Delta\chi \, dx = \int \left(\int E_d(x - y) \Delta\chi(x) \, dx \right) d\nu(y) = \int \chi(y) \, d\nu(y).$$

This is the distributional equation $\Delta p = \nu$. □

Theorem 3.2.2 *Let $\Omega \subset \mathbb{R}^d$ be open.*

(1) *Let $u \in \mathrm{SH}(\Omega)$ be not identically $-\infty$ on any connected component of Ω. Then u is locally integrable, and Δu is a positive distribution, hence a positive Radon measure.*

(2) *Let $u \in \mathrm{SH}(\Omega)$ and $x \in \Omega$. For $0 < \varepsilon < \operatorname{dist}(x, \mathbb{R}^d \setminus \Omega)$ put $u * \rho_\varepsilon(x) = \int u(x - y)\rho_\varepsilon(y) \, dy$, with the mollifier ρ_ε of Section 1.4. This integral and the integral of u over a ball $B(x, r)$ with $\overline{B(x, r)} \subset \Omega$ are defined in $[-\infty, \infty)$, because u is bounded above on compact sets; if u is locally integrable, $u * \rho_\varepsilon$ is the mollification of Section 1.4. Then*

$$u * \rho_\varepsilon(x) \downarrow u(x) \quad (\varepsilon \downarrow 0), \qquad u(x) = \lim_{r \downarrow 0} \frac{1}{|B(x, r)|} \int_{B(x,r)} u(y) \, dy. \tag{3.3}$$

In particular, two subharmonic functions on Ω that are equal almost everywhere are equal everywhere.

(3) *Let T be a real distribution on Ω with $\Delta T \geq 0$. Then there is exactly one locally integrable $u \in \mathrm{SH}(\Omega)$ such that $\langle T, \chi\rangle = \int_\Omega u\chi \,\mathrm{d}x$ for every test function χ.*

Proof (1) By Proposition 3.1.5, u is locally integrable. To show that $\Delta u \geq 0$, fix $\varepsilon > 0$ and $x \in \Omega_\varepsilon$, with Ω_ε as in Section 1.4. For sufficiently small $r > 0$, all the closed balls $\overline{B(x-z,r)}$ with $|z| \leq \varepsilon$ lie in Ω, so $u(x-z) \leq M_u(x-z,r)$ for these z by Theorem 3.1.3. We multiply by $\rho_\varepsilon(z)$ and integrate over z. Since u is bounded above on the compact set $\overline{B(x,\varepsilon+r)} \subset \Omega$, Fubini's theorem applies to the difference between an upper bound and u; as $\overline{B(x,r)} \subset \Omega_\varepsilon$, we obtain

$$u * \rho_\varepsilon(x) \leq M_{u*\rho_\varepsilon}(x,r).$$

The function $v = u * \rho_\varepsilon$ is smooth and real on Ω_ε by Proposition 1.4.3, and we have just shown that $v(x) \leq M_v(x,r)$ for every $x \in \Omega_\varepsilon$ and every sufficiently small $r > 0$. This forces $\Delta v \geq 0$ on Ω_ε. Indeed, suppose that $\Delta v(x) < 0$. Since Δv is continuous, there is $R > 0$ such that $\overline{B(x,R)} \subset \Omega_\varepsilon$, $\Delta v < 0$ on $B(x,R)$ and $v(x) \leq M_v(x,r)$ for $0 < r < R$. By (2.4), the function $r \mapsto M_v(x,r)$ has negative derivative on $(0,R)$, so it is strictly decreasing there. Moreover, $M_v(x,s) \to v(x)$ as $s \downarrow 0$, because $|M_v(x,s) - v(x)| \leq \max_{|y-x|=s} |v(y) - v(x)|$ and v is continuous at x. Hence $M_v(x,r) < M_v(x,r/2) \leq \lim_{s\downarrow 0} M_v(x,s) = v(x)$ for $0 < r < R$, a contradiction.

Now let χ be a nonnegative test function. Then $\operatorname{Supp}\chi \subset \Omega_\varepsilon$ for small ε, and $u * \rho_\varepsilon \to u$ in L^1 on $\operatorname{Supp}\chi$ by Lemma 1.4.1. Hence

$$\langle \Delta u, \chi\rangle = \int u\Delta\chi \,\mathrm{d}x = \lim_{\varepsilon\downarrow 0} \int (u*\rho_\varepsilon)\Delta\chi \,\mathrm{d}x = \lim_{\varepsilon\downarrow 0} \int \Delta(u*\rho_\varepsilon)\chi \,\mathrm{d}x \geq 0.$$

By Theorem 1.4.5, the positive distribution Δu is a positive Radon measure.

(2) Fix $x \in \Omega$ and $0 < \varepsilon < \operatorname{dist}(x, \mathbb{R}^d \setminus \Omega)$, let C be a real upper bound for u on $\overline{B(x,\varepsilon)}$, and let e_1 be the first coordinate unit vector. Since ρ is radial, $\rho_\varepsilon(r\omega) = \varepsilon^{-d}\rho((r/\varepsilon)e_1)$ for $\omega \in S^{d-1}$, and the spherical mean of $\omega \mapsto u(x - r\omega)$ equals $M_u(x,r)$ by the symmetry $\omega \mapsto -\omega$. Integration in polar coordinates (2.1), applied to the nonnegative function $(C - u(x-y))\rho_\varepsilon(y)$ on $B(0,\varepsilon)$, and the substitution $r = \varepsilon t$ therefore give, with $w(t) = s_{d-1}\rho(te_1)t^{d-1}$,

$$\int_{B(0,\varepsilon)} \big(C - u(x-y)\big)\rho_\varepsilon(y) \,\mathrm{d}y = \int_0^1 \big(C - M_u(x,\varepsilon t)\big)\, w(t) \,\mathrm{d}t.$$

The same computation, applied to ρ_ε alone, shows that $\int_0^1 w \,\mathrm{d}t = \int \rho_\varepsilon \,\mathrm{d}y = 1$, so $w(t)\,\mathrm{d}t$ is a probability measure on $(0,1)$, and we obtain

$$u * \rho_\varepsilon(x) = \int_0^1 M_u(x,\varepsilon t)\, w(t) \,\mathrm{d}t \in [-\infty, \infty).$$

By Theorem 3.1.3, $M_u(x,\varepsilon t)$ increases with ε and is at least $u(x)$. Hence $u*\rho_\varepsilon(x)$ decreases as $\varepsilon\downarrow 0$ and is at least $u(x)$. On the other hand, $u*\rho_\varepsilon(x)\le \sup_{B(x,\varepsilon)} u$, and the upper limit of the right-hand side as $\varepsilon\downarrow 0$ is at most $u(x)$ by upper semicontinuity, also when $u(x)=-\infty$. Thus $u*\rho_\varepsilon(x)\downarrow u(x)$. The ball averages are treated in the same way: by (3.2), the average of u over $B(x,r)$ is the average of $M_u(x,rt)$, $0<t<1$, with respect to the probability measure $d\,t^{d-1}\,\mathrm{d}t$, so it lies between $u(x)$ and $\sup_{B(x,r)} u$ and tends to $u(x)$. This proves (3.3). The ball averages in (3.3) depend only on the almost-everywhere class of u; hence two subharmonic functions that agree almost everywhere agree everywhere.

(3) Since a distribution determines a locally integrable function representing it up to null sets (Proposition 1.4.3), (2) shows that T has at most one locally integrable subharmonic representative. It remains to construct one; we do this on balls and then glue.

By Theorem 1.4.5, $\mu=\Delta T$ is a positive Radon measure. Let B be a ball with $\overline{B}\subset\Omega$, and choose a nonnegative cutoff $\eta\in C_c^\infty(\Omega)$ equal to one on an open set $U\supset\overline{B}$ (Lemma 1.4.1). The measure $\eta\mu$, extended by zero to $\mathbb{R}^d$, is finite and has compact support, so by Proposition 3.2.1 its fundamental potential $p=E_d*(\eta\mu)$ is subharmonic and locally integrable, and $\Delta p=\eta\mu=\mu$ on U. Thus $\Delta(T-p)=0$ on U, and by Weyl's lemma (Theorem 2.1.4) the real distribution $T-p$ is represented on U by a smooth harmonic function h; since $\operatorname{Re} h$ also represents $T-p$, the uniqueness in Theorem 2.1.4 shows that $h=\operatorname{Re} h$ is real. Then $u_B=p+h$ is a locally integrable subharmonic function on B representing T there.

If B' is another such ball, u_B and $u_{B'}$ both represent T on $B\cap B'$, so they agree almost everywhere there (Proposition 1.4.3), and hence everywhere on $B\cap B'$ by (2). Hence there is a function u on Ω with $u=u_B$ on every ball B with $\overline{B}\subset\Omega$. It is subharmonic, because subharmonicity is local, and it is locally integrable.

To see that u represents T on all of Ω, let $T_\varepsilon(x)=\langle T,\rho_\varepsilon(x-\cdot)\rangle$ be the mollification of Proposition 1.4.3. If $x\in\Omega_\varepsilon$, then $\overline{B(x,\varepsilon')}\subset\Omega$ for some $\varepsilon'>\varepsilon$, and the test function $\rho_\varepsilon(x-\cdot)$ is supported in the ball $B=B(x,\varepsilon')$, on which $u=u_B$ represents T; hence $T_\varepsilon(x)=u*\rho_\varepsilon(x)$. For every test function χ, Proposition 1.4.3 and Lemma 1.4.1 now give

$$\langle T,\chi\rangle=\lim_{\varepsilon\downarrow0}\int T_\varepsilon\chi\,\mathrm{d}x=\lim_{\varepsilon\downarrow0}\int (u*\rho_\varepsilon)\chi\,\mathrm{d}x=\int u\chi\,\mathrm{d}x.$$

The function u of Theorem 3.2.2(3) is called the canonical representative of T, and every locally integrable subharmonic function is the canonical representative of the distribution it defines. Subharmonicity is a property of a pointwise function, not of its almost-everywhere class. For instance, if we raise the value of a subharmonic function u at a single point x_0, the spherical means $M_u(x_0,r)$ do not change and still tend to $u(x_0)$ as $r\downarrow0$ (Theorem 3.1.3), so the submean inequality fails at x_0. On the other hand, by Theorem 3.2.2(2), the almost-everywhere class of a subharmonic function determines it at every point, even without integrability assumptions.

Suppose that locally integrable subharmonic functions u_j on Ω converge in the sense of distributions to $T\in\mathcal{D}'(\Omega)$. Then T is real as a limit of real distributions. Since

each u_j is locally integrable, it is not identically $-\infty$ on any connected component of Ω, so $\Delta u_j \geq 0$ by Theorem 3.2.2(1), and hence $\langle \Delta T, \chi \rangle = \lim_j \langle \Delta u_j, \chi \rangle \geq 0$ for every nonnegative test function χ. Therefore T has a canonical representative by Theorem 3.2.2(3).

For a locally integrable $u \in \mathrm{SH}(\Omega)$, the positive Radon measure $\mu_u = \Delta u$ is called the Riesz measure of u. Some texts on potential theory in the plane define the Riesz measure as $(2\pi)^{-1}\Delta u$; with our normalization, $\mu_{\log|z|} = 2\pi\delta_0$ on $\mathbb{C} = \mathbb{R}^2$, since $\log|z| = 2\pi E_2(z)$.

Theorem 3.2.3 (Riesz decomposition) *Let $u \in \mathrm{SH}(\Omega)$ be locally integrable. Let η be a bounded nonnegative Borel function on Ω which vanishes outside a compact subset of Ω and equals one on an open set $V \subset \Omega$; for instance, $\eta = \mathbf{1}_V$ for an open set $V \Subset \Omega$, or $\eta \in C_c^\infty(\Omega)$ with values in $[0,1]$ and $\eta = 1$ on V. Let $\eta\mu_u$ denote the measure $E \mapsto \int_{E\cap\Omega} \eta \, \mathrm{d}\mu_u$ on $\mathbb{R}^d$. Then there is a smooth real harmonic function h on V such that*

$$u = E_d * (\eta\mu_u) + h \qquad \text{on } V.$$

This equality holds at every point of V, including the points where $u = -\infty$.

Proof If η vanishes outside the compact set $C \subset \Omega$, then $\eta\mu_u$ is a positive Borel measure on $\mathbb{R}^d$ with support in C and total mass at most $\sup\eta \cdot \mu_u(C) < \infty$. Let $p = E_d * (\eta\mu_u)$. By Proposition 3.2.1, p is subharmonic and locally integrable on $\mathbb{R}^d$, and $\Delta p = \eta\mu_u$. If χ is a test function with $\operatorname{Supp}\chi \subset V$, then $\int \chi\eta \, \mathrm{d}\mu_u = \int \chi \, \mathrm{d}\mu_u$ because $\eta = 1$ on V; hence $\Delta p = \Delta u$ on V. By Weyl's lemma (Theorem 2.1.4), the real distribution $u - p$ is represented on V by a smooth harmonic function h; since $\operatorname{Re} h$ also represents $u - p$, the uniqueness in Theorem 2.1.4 shows that $h = \operatorname{Re} h$ is real. Then $p + h$ is a locally integrable subharmonic function on V that agrees with u almost everywhere on V (Proposition 1.4.3), and Theorem 3.2.2(2), applied on V, gives $u = p + h$ at every point of V. □

Example 3.2.4 Let $I \subset \mathbb{R}$ be an open interval. Every $u \in \mathrm{SH}(I)$ other than the constant $-\infty$ is a finite, continuous, convex function. Indeed, u is locally integrable by Proposition 3.1.5. Given $x \in I$, choose an open interval $V \ni x$ with $\overline{V} \subset I$ and apply Theorem 3.2.3 with $\eta = \mathbf{1}_V$. On V we have $u = E_1 * (\eta\mu_u) + h$ with h affine, and the potential $E_1 * (\eta\mu_u)(\xi) = \frac{1}{2}\int |\xi - y| \, \eta(y) \, \mathrm{d}\mu_u(y)$ is finite and Lipschitz in ξ, since $\big||\xi - y| - |\xi' - y|\big| \leq |\xi - \xi'|$ and $\eta\mu_u$ is a finite measure. Hence u is finite and continuous.

In dimension one the submean inequality reads $u(x) \leq \frac{1}{2}(u(x-r) + u(x+r))$, and by Theorem 3.1.3 it holds whenever $[x-r, x+r] \subset I$. Thus u satisfies the midpoint inequality $u\big((a+b)/2\big) \leq \big(u(a) + u(b)\big)/2$ for all $a, b \in I$. By induction on n, it follows that $u((1-\lambda)a + \lambda b) \leq (1-\lambda)u(a) + \lambda u(b)$ for every dyadic number $\lambda = k2^{-n} \in [0,1]$, because a dyadic number of order $n+1$ is the midpoint of two adjacent dyadic numbers of order n. Since u is continuous and the dyadic numbers are dense in $[0,1]$, the same inequality holds for every $\lambda \in [0,1]$.

Conversely, a finite convex function f on I is continuous. For $p < q$ in I let $\sigma(p,q) = (f(q) - f(p))/(q-p)$ be the slope of the chord of f over $[p,q]$. If $p < q < r$ lie in I, then $q = (1-\lambda)p + \lambda r$ with $\lambda = (q-p)/(r-p) \in (0,1)$, and

convexity gives $f(q) \le (1-\lambda)f(p)+\lambda f(r)$, that is, $f(q)-f(p) \le \lambda(f(r)-f(p))$ and $f(r)-f(q) \ge (1-\lambda)(f(r)-f(p))$. Dividing by $q-p=\lambda(r-p)$ and by $r-q=(1-\lambda)(r-p)$, respectively, we obtain

$$\sigma(p,q) \le \sigma(p,r) \le \sigma(q,r).$$

Hence $\sigma(p,q)$ increases with q for fixed p, and with p for fixed q. Now let $[a,b]$ be a compact subinterval of I, choose $a',b' \in I$ with $a' < a$ and $b < b'$, and let $a \le s < t \le b$. Applying these monotonicity properties and the displayed inequality to the triples (a',s,t) and (s,t,b'), we find

$$\sigma(a',a) \le \sigma(a',s) \le \sigma(s,t) \le \sigma(t,b') \le \sigma(b,b'),$$

so $|f(t)-f(s)| \le \max\{|\sigma(a',a)|,|\sigma(b,b')|\}\,(t-s)$. Thus f is Lipschitz on every compact subinterval of I, and in particular continuous. Since f satisfies the midpoint inequality, which is the submean inequality in dimension one, it is subharmonic. Hence the subharmonic functions on I other than $-\infty$ are exactly the finite convex functions.

Next let $D \subset \mathbb{C} = \mathbb{R}^2$ be a domain, and let f be holomorphic on D and not identically zero. Then $\log|f|$ is subharmonic, and

$$\Delta \log|f| = 2\pi \sum_{f(a)=0} \operatorname{ord}_a(f)\,\delta_a,$$

where $\operatorname{ord}_a(f)$ is the order of the zero of f at a. Indeed, the zeros of f form a locally finite subset of D, and away from them $\log|f|$ is locally the real part of a holomorphic logarithm of f, hence harmonic. Near a zero a of order m we can write $f(z) = (z-a)^m g(z)$ with g holomorphic and nonvanishing, so that $\log|f(z)| = 2\pi m E_2(z-a) + \log|g(z)|$, where $\log|g|$ is harmonic and $E_2(\cdot - a)$, the fundamental potential of δ_a, is subharmonic with $\Delta E_2(\cdot - a) = \delta_a$ (Proposition 3.2.1). Hence $\log|f|$ is subharmonic near every point, and therefore on D. Moreover, the right-hand side of the displayed formula is a positive Radon measure, as is $\Delta\log|f|$ (Theorem 3.2.2(1)). The two measures agree on a neighborhood of every point of D (Proposition 1.4.3), and hence on D: cover D by countably many such neighborhoods V_k, and decompose a Borel set $E \subset D$ into the disjoint sets $E \cap V_k \setminus (V_1 \cup \cdots \cup V_{k-1})$. For $\log|f|^2 = 2\log|f|$ the coefficient 2π becomes 4π.

3.3 Compactness and Hartogs' lemma

We now study sequences of subharmonic functions that are uniformly bounded above on compact sets. The ball submean inequality turns a lower bound at a single point into L^1 bounds on compact sets, and these bounds give compactness in L^1_{loc}.

Lemma 3.3.1 *Let Ω be a domain and let (u_j) be a sequence in $\mathrm{SH}(\Omega)$ that is uniformly bounded above on every compact subset of Ω. Suppose that there are a*

compact set $K \subset \Omega$, points $x_j \in K$ and a constant C_0 such that $u_j(x_j) \geq -C_0$ for all j. Then there is a subsequence (u_{j_k}) such that $\sup_k \int_L |u_{j_k}|\,dx < \infty$ for every compact set $L \subset \Omega$.

Proof We first record a local estimate. Let $a \in \Omega$ and $r > 0$ with $\overline{B(a,4r)} \subset \Omega$, and let $M \geq 0$ be a common upper bound of the functions u_j on $B(a,4r)$. Suppose that $u_j(z) \geq -C$ for some index j, some point $z \in B(a,r)$ and some constant C. Then $B(a,r) \subset B(z,2r)$ and $\overline{B(z,2r)} \subset B(a,4r)$. Since $M - u_j \geq 0$ on $B(z,2r)$, the ball submean inequality of Theorem 3.1.3 gives

$$\int_{B(a,r)} (M - u_j)\,dx \leq \int_{B(z,2r)} (M - u_j)\,dx \leq |B(0,2r)|\,(M + C),$$

and since $|u_j| \leq (M - u_j) + M$, we obtain

$$\int_{B(a,r)} |u_j|\,dx \leq |B(0,2r)|\,(M + C) + |B(0,r)|\,M. \tag{3.4}$$

Passing to a subsequence, we may assume that $x_j \to x_0 \in K$. Choose $r > 0$ with $\overline{B(x_0,4r)} \subset \Omega$. Discarding finitely many terms, we may assume that $|x_j - x_0| < r$ for all j. Then (3.4), with $a = x_0$, $z = x_j$ and $C = C_0$, bounds $\int_{B(x_0,r)} |u_j|\,dx$ uniformly in j.

Let Y be the set of points $y \in \Omega$ that have a neighborhood U with $\sup_j \int_U |u_j|\,dx < \infty$. Then Y is open and contains x_0. Let $y \in \Omega$ lie in the closure of Y. Choose $r > 0$ with $\overline{B(y,4r)} \subset \Omega$ and an open ball A with $\overline{A} \subset Y \cap B(y,r)$. Covering the compact set $\overline{A}$ by finitely many of the neighborhoods in the definition of Y, we find $C_A = \sup_j \int_A |u_j|\,dx < \infty$. Put $C = C_A/|A|$. For each j there is a point $z_j \in A$ with $u_j(z_j) \geq -C$, since otherwise $\int_A |u_j|\,dx \geq \int_A (-u_j)\,dx > C|A| = C_A$. By (3.4), with $a = y$ and $z = z_j$, the integrals $\int_{B(y,r)} |u_j|\,dx$ are bounded uniformly in j, so $y \in Y$. Thus Y is closed in Ω, and $Y = \Omega$ because Ω is connected. Finally, every compact subset of Ω is covered by finitely many of the neighborhoods in the definition of Y.□

Theorem 3.3.2 *Let $\Omega \subset \mathbb{R}^d$ be open, and let (u_j) be a sequence in $\mathrm{SH}(\Omega)$ that is uniformly bounded above on every compact subset of Ω.*

(1) *If Ω is connected, then either $u_j \to -\infty$ uniformly on compact subsets of Ω, or a subsequence of (u_j) converges in $L^1_{\mathrm{loc}}(\Omega)$ to a locally integrable subharmonic function.*

(2) *If every u_j is locally integrable and (u_j) converges in the sense of distributions to some $T \in \mathcal{D}'(\Omega)$, then T is real, $\Delta T \geq 0$, and $u_j \to u$ in $L^1_{\mathrm{loc}}(\Omega)$, where u is the canonical representative of T.*

Proof We prove (2) first, because (1) reduces to it. As observed after Theorem 3.2.2, T is real, $\Delta T \geq 0$, and T has a canonical representative u. We begin by showing that (u_j) is bounded in $L^1(L)$ for every compact set $L \subset \Omega$. Fix such a set L, let χ be

a nonnegative test function equal to one near L, and let $M \geq 0$ bound all u_j from above on $\operatorname{Supp}\chi$. Then $|u_j| \leq 2M - u_j$ on $\operatorname{Supp}\chi$, so

$$\int_L |u_j|\,\mathrm{d}x \leq 2M \int \chi\,\mathrm{d}x - \int u_j\chi\,\mathrm{d}x,$$

which is bounded because $\int u_j\chi\,\mathrm{d}x$ converges.

Next we show that the mollifications $u_j * \rho_\varepsilon$ converge locally uniformly. Fix $\varepsilon > 0$. For $x \in \Omega_\varepsilon$, the function $\rho_\varepsilon(x-\cdot)$ is a test function, so $u_j * \rho_\varepsilon(x) \to \langle T, \rho_\varepsilon(x-\cdot)\rangle = u * \rho_\varepsilon(x)$. Let $L \subset \Omega_\varepsilon$ be compact; then $L' = \{y : \operatorname{dist}(y, L) \leq \varepsilon\}$ is a compact subset of Ω, and the mean value theorem, applied to ρ_ε, gives

$$|u_j * \rho_\varepsilon(x) - u_j * \rho_\varepsilon(x')| \leq \sup|\nabla\rho_\varepsilon| \cdot |x - x'| \int_{L'} |u_j|\,\mathrm{d}y, \qquad x, x' \in L.$$

Thus the functions $u_j * \rho_\varepsilon$ are A-Lipschitz on L, where $A = \sup|\nabla\rho_\varepsilon| \cdot \sup_j \int_{L'} |u_j|\,\mathrm{d}y$, and so is their pointwise limit $u * \rho_\varepsilon$. Given $\tau > 0$, cover L by finitely many balls $B(x_i, \tau/(3A+1))$ with $x_i \in L$, and choose j_0 such that $|u_j * \rho_\varepsilon(x_i) - u * \rho_\varepsilon(x_i)| < \tau/3$ for all i and all $j \geq j_0$. Then $|u_j * \rho_\varepsilon - u * \rho_\varepsilon| < \tau$ on L for $j \geq j_0$, so the convergence is uniform on L.

Now let χ be a nonnegative test function, choose ε so small that $\operatorname{Supp}\chi \subset \Omega_\varepsilon$, let $\delta > 0$, and put $g = u * \rho_\varepsilon + \delta$. By (3.3), $u_j \leq u_j * \rho_\varepsilon$ and $u \leq u * \rho_\varepsilon$ on Ω_ε. By the uniform convergence just proved, $u_j \leq g$ on $\operatorname{Supp}\chi$ for all large j; also $u \leq g$. Hence $|u_j - u| \leq (g - u_j) + (g - u)$ almost everywhere on $\operatorname{Supp}\chi$, and since $\int u_j\chi\,\mathrm{d}x \to \int u\chi\,\mathrm{d}x$,

$$\begin{aligned}\overline{\lim_j} \int |u_j - u|\chi\,\mathrm{d}x &\leq \lim_j \int (g - u_j)\chi\,\mathrm{d}x + \int (g - u)\chi\,\mathrm{d}x \\ &= 2\int (u * \rho_\varepsilon + \delta - u)\chi\,\mathrm{d}x.\end{aligned}$$

Let $\delta \downarrow 0$, and then $\varepsilon \downarrow 0$; since $u * \rho_\varepsilon \to u$ in L^1 on $\operatorname{Supp}\chi$ (Lemma 1.4.1), the right-hand side tends to zero. Choosing χ equal to one on a prescribed compact set proves L^1 convergence on that set. This proves (2).

(1) Suppose that u_j does not tend to $-\infty$ uniformly on compact subsets. Then there are a compact set $K \subset \Omega$, a constant C_0 and infinitely many indices j with $\sup_K u_j > -C_0$, and for these j we can choose $x_j \in K$ with $u_j(x_j) > -C_0$. By Lemma 3.3.1, a subsequence, which we again denote by (u_j), is bounded in $L^1(L)$ for every compact set $L \subset \Omega$; in particular, every u_j is locally integrable. By (2), it suffices to show that a further subsequence converges in the sense of distributions.

Choose open sets $U_m \Subset \Omega$ increasing to Ω, for instance the interiors of the sets of a compact exhaustion (Section 1.1), and constants $M_m \geq 0$ with $u_j \leq M_m$ on $\overline{U_m}$ for all j. On U_m, the Borel measures $(M_m - u_j)\,\mathrm{d}x$ are positive, and their masses on a compact set $L \subset U_m$ are at most $M_m|L| + \sup_j \int_L |u_j|\,\mathrm{d}x$, uniformly in j; in particular, they are Radon measures by Proposition 1.1.2. By Theorem 1.5.5 and a diagonal extraction over m, we may assume that for each m these measures converge

vaguely on U_m. Every $\chi \in C_c(\Omega)$ has its support in some U_m, so

$$\int u_j \chi \, \mathrm{d}x = M_m \int \chi \, \mathrm{d}x - \int (M_m - u_j)\chi \, \mathrm{d}x$$

converges. Let $T(\chi)$ denote the limit when χ is a test function. If $\operatorname{Supp}\chi \subset L$ with L compact, then $|T(\chi)| \le \sup_j \int_L |u_j| \, \mathrm{d}x \cdot \sup|\chi|$. Hence T is a distribution of order zero, and $u_j \to T$ in the sense of distributions. □

Theorem 3.3.3 (Hartogs' lemma) *Let (u_j) be a sequence of locally integrable subharmonic functions on Ω converging in $L^1_{\mathrm{loc}}(\Omega)$ to a function $v \in L^1_{\mathrm{loc}}(\Omega)$. Then $\Delta v \ge 0$ in the sense of distributions. Let u be the canonical representative of v, that is, the locally integrable subharmonic function equal to v almost everywhere. Then, for every $x \in \Omega$ and every sequence $x_j \to x$ in Ω,*

$$\overline{\lim_j} \, u_j(x_j) \le u(x).$$

For every compact set $K \subset \Omega$ and every lower semicontinuous function $f : K \to \mathbb{R}$, for instance every real continuous function on K,

$$\overline{\lim_j} \sup_K (u_j - f) \le \sup_K (u - f).$$

In particular, if $f : K \to \mathbb{R}$ is lower semicontinuous and $u < f$ on K, then $u_j < f$ on K for all sufficiently large j.

Proof Convergence in L^1_{loc} implies convergence in the sense of distributions, because $|\int (u_j - v)\chi \, \mathrm{d}x| \le \sup|\chi| \int_{\operatorname{Supp}\chi} |u_j - v| \, \mathrm{d}x$. Hence, as observed after Theorem 3.2.2, the distribution defined by v is real, has positive Laplacian and has a canonical representative u.

Let $x_j \to x$, and choose $\varepsilon > 0$ with $\overline{B(x, 2\varepsilon)} \subset \Omega$. For $y \in B(x, \varepsilon)$ we have

$$|u_j * \rho_\varepsilon(y) - u * \rho_\varepsilon(y)| \le \sup \rho_\varepsilon \int_{B(x,2\varepsilon)} |u_j - v| \, \mathrm{d}z,$$

so $u_j * \rho_\varepsilon \to u * \rho_\varepsilon$ uniformly on $B(x, \varepsilon)$. For large j we have $x_j \in B(x, \varepsilon)$. Since $u_j \le u_j * \rho_\varepsilon$ by (3.3) and $u * \rho_\varepsilon$ is continuous, we obtain

$$\overline{\lim_j} \, u_j(x_j) \le \overline{\lim_j} \, u_j * \rho_\varepsilon(x_j) = u * \rho_\varepsilon(x).$$

Letting $\varepsilon \downarrow 0$ and using (3.3) again proves the first assertion.

For the second assertion we may assume $K \ne \varnothing$. Since u and $-f$ are upper semicontinuous on the compact set K, they are bounded above there, so $s = \sup_K (u - f)$ lies in $[-\infty, \infty)$. Suppose that $\overline{\lim}_j \sup_K (u_j - f) > c > s$ for some real c. Then $\sup_K (u_j - f) > c$ for infinitely many j, so along a subsequence there are points $x_j \in K$ with $u_j(x_j) - f(x_j) > c$, and after passing to a further subsequence we may assume $x_j \to x \in K$. Since $\underline{\lim}_j f(x_j) \ge f(x)$ by lower semicontinuity, the

first assertion, applied to this subsequence, gives

$$c \le \overline{\lim_j}\big(u_j(x_j) - f(x_j)\big) \le \overline{\lim_j} u_j(x_j) - \underline{\lim_j} f(x_j) \le u(x) - f(x) \le s < c,$$

a contradiction. Finally, if $u < f$ on K, then the upper semicontinuous function $u - f$ attains a negative maximum on K, so $\sup_K(u - f) < 0$, and the second assertion gives $\sup_K(u_j - f) < 0$ for all large j. □

3.4 Upper envelopes and countable selection

The pointwise supremum of a family of subharmonic functions need not be upper semicontinuous (Example 3.4.3), so we regularize it. Let $F\colon \Omega \to [-\infty, \infty)$ be locally bounded above, that is, bounded above on a neighborhood of each point. Its upper semicontinuous regularization is

$$F^*(x) = \inf_{r>0} \sup_{B(x,r)\cap\Omega} F.$$

Since $x \in B(x, r)$, we have $F^* \ge F$. The function F^* is upper semicontinuous: if $F^*(x) < c$, then $\sup_{B(x,r)\cap\Omega} F < c$ for some $r > 0$, and hence $F^* < c$ on $B(x, r) \cap \Omega$. It is the smallest upper semicontinuous function that dominates F: if G is such a function and $c > G(x)$, then $G < c$ on some ball $B(x, r)$, so $\sup_{B(x,r)} F \le c$ and $F^*(x) \le c$. We take the supremum of an empty family of functions to be the constant $-\infty$. A family of functions on Ω is locally uniformly bounded above if every point of Ω has a neighborhood on which all members of the family are bounded above by one constant. The following lemma reduces such families to countable subfamilies.

Lemma 3.4.1 (Choquet's lemma) *Let $\mathcal{F}$ be a nonempty family of functions $\Omega \to [-\infty, \infty)$ that is locally uniformly bounded above, and let $F = \sup_{w\in\mathcal{F}} w$. Then there is a countable subfamily $\{v_j\}_{j\ge1} \subset \mathcal{F}$ such that $v = \sup_j v_j$ satisfies $v^* = F^*$.*

Proof Call $B(q, s)$ a rational ball if $q \in \mathbb{Q}^d$ and s is a positive rational number. Enumerate the rational balls B with $\overline{B} \subset \Omega$ as $B_1, B_2, \ldots$, in such a way that each of them occurs infinitely often. Since F is bounded above on the compact set $\overline{B_j}$, we have $\sup_{B_j} F < \infty$. If $\sup_{B_j} F > -\infty$, choose $v_j \in \mathcal{F}$ with $\sup_{B_j} v_j > \sup_{B_j} F - 1/j$; otherwise let v_j be any member of $\mathcal{F}$. Put $v = \sup_j v_j \le F$. If B is a rational ball with $\overline{B} \subset \Omega$, then $B = B_j$ for infinitely many j, so $\sup_B v = \sup_B F$. Every ball $B(x, r)$ contains a rational ball that contains x: choose $q \in \mathbb{Q}^d$ with $|q - x| < r/3$ and a rational number $s \in (r/3, 2r/3)$; then $x \in B(q, s) \subset B(x, r)$. Conversely, a rational ball $B(q, s)$ containing x contains $B(x, s - |q - x|)$. Hence $F^*(x)$ is the infimum of $\sup_B F$ over the rational balls $B \ni x$ with $\overline{B} \subset \Omega$, and similarly for $v^*(x)$. Therefore $v^* = F^*$. □

Theorem 3.4.2 *Let $\{u_t\}_{t\in I}$ be a family in $\mathrm{SH}(\Omega)$ that is uniformly bounded above on every compact subset of Ω, and let $F = \sup_t u_t$. Then F^* is subharmonic. Moreover,*

F is Lebesgue measurable and $F = F^*$ almost everywhere. The functions F and F^* need not coincide everywhere (Example 3.4.3).

Proof If the family is empty, then $F = F^* = -\infty$. Otherwise, we replace the family by an increasing sequence, using Choquet's lemma, and identify the regularized limit of this sequence with a canonical representative. The connected components of Ω are open, because balls are connected; hence they are pairwise disjoint sets of positive Lebesgue measure, and there are at most countably many of them by Lemma 1.1.5. As a countable union of null sets is null, it suffices to prove the assertions on each component Ω'. On Ω' the regularization F^* coincides with the regularization of $F|_{\Omega'}$ in Ω', because small balls about a point of Ω' lie in Ω'. If every u_t is identically $-\infty$ on Ω', then $F = F^* = -\infty$ there.

Otherwise fix t_0 such that u_{t_0} is nontrivial, hence locally integrable, on Ω'. By Lemma 3.4.1, applied on Ω', there are indices $t_1, t_2, \ldots$ with $(\sup_{k\geq 1} u_{t_k})^* = F^*$ on Ω'. Put $v_j = \max\{u_{t_0}, u_{t_1}, \ldots, u_{t_j}\}$ and $v = \sup_j v_j$. Since $\sup_{k\geq 1} u_{t_k} \leq v \leq F$, we still have $v^* = F^*$ on Ω'. The functions v_j are subharmonic, increase with j, are uniformly bounded above on compact sets and satisfy $v_j \geq u_{t_0}$; in particular, they are locally integrable on Ω'. If $K \subset \Omega'$ is compact and M bounds all v_j from above on K, then $0 \leq M - v_j \leq M - u_{t_0}$ on K with $M - u_{t_0}$ integrable, and $M - v_j$ decreases to $M - v$. Dominated convergence shows that v is locally integrable and $v_j \to v$ in $L^1_{\mathrm{loc}}(\Omega')$, hence in the sense of distributions. As observed after Theorem 3.2.2, v therefore has a canonical representative w: a subharmonic function on Ω' with $w = v$ almost everywhere.

We claim that $w = v^*$. For each j we have $v_j \leq v$, so $v_j * \rho_\varepsilon \leq v * \rho_\varepsilon = w * \rho_\varepsilon$, and letting $\varepsilon \downarrow 0$ in (3.3) gives $v_j \leq w$ everywhere. Hence $v \leq w$, and $v^* \leq w$ because w is upper semicontinuous. Conversely, by (3.3), $w(x)$ is the limit of the averages of w over $B(x, r)$; these coincide with the averages of v, which are at most $\sup_{B(x,r)} v$. Letting $r \downarrow 0$ gives $w(x) \leq v^*(x)$. Thus $F^* = v^* = w$ is subharmonic on Ω'.

Finally, $v \leq F \leq F^* = w$ on Ω', and $v = w$ almost everywhere. The functions v and w are Borel measurable, so the set where $F \neq F^*$ is contained in the Borel null set $\{v \neq w\}$. Since Lebesgue measure is complete, F is Lebesgue measurable and $F = F^*$ almost everywhere. □

Example 3.4.3 Let $d \geq 2$. The fundamental solution E_d is negative on $B(0, 1)$ and equals $-\infty$ at the origin. The functions $u_j = \max\{-1, E_d/j\}$, $j \geq 1$, are subharmonic on $B(0, 1)$, take values in $[-1, 0)$ and increase with j. Their limit is 0 on $B(0, 1) \setminus \{0\}$ and -1 at the origin, whereas its upper semicontinuous regularization is identically zero. Thus even an increasing sequence of bounded subharmonic functions can have a limit that is not upper semicontinuous.

3.5 Polar sets and removable singularities

The exceptional sets of potential theory are the sets that lie locally in the pole set of a nontrivial subharmonic function. They are Lebesgue null, and relatively closed ones are removable for subharmonic functions that are bounded above near them.

Definition 3.5.1 A set $A \subset \Omega$ is polar if every point of A has a connected open neighborhood $V \subset \Omega$ on which there is a nontrivial $\psi \in \mathrm{SH}(V)$ with $A \cap V \subset \{\psi = -\infty\}$.

Every polar set A has Lebesgue measure zero. Indeed, since $\mathbb{R}^d$ has a countable base, A is covered by countably many of the neighborhoods V in the definition, and each $A \cap V$ lies in a pole set $\{\psi = -\infty\}$, which is null by Proposition 3.1.5. Subsets of polar sets are polar. If $d \geq 2$, every one-point set $\{a\} \subset \Omega$ is polar, since $E_d(\cdot - a)$ is a nontrivial subharmonic function with a pole at a. If $d = 1$, the only polar set is the empty set, because nontrivial subharmonic functions on intervals are finite (Example 3.2.4). In particular, a Lebesgue null set need not be polar. Polar sets are the natural exceptional sets of the limit theorems above: for $d \geq 2$, Theorem 4.4.4 in the next chapter shows that the set where $F \neq F^*$ in Theorem 3.4.2 is not only Lebesgue null but polar.

Theorem 3.5.2 *Let $A \subset \Omega$ be a relatively closed polar set, and let $u \in \mathrm{SH}(\Omega \setminus A)$ be bounded above on $K \setminus A$ for every compact set $K \subset \Omega$. Then u has a unique subharmonic extension $\widetilde{u}$ to Ω; it is given on A by*

$$\widetilde{u}(x) = \overline{\lim_{\Omega\setminus A \ni y \to x}} u(y).$$

If Ω is a domain, then $\Omega \setminus A$ is connected.

Proof Put $\widetilde{u} = u$ on $\Omega \setminus A$, and define $\widetilde{u}$ on A by the displayed formula. Since A has Lebesgue measure zero, it has no interior points, so every point of A is a limit of points of $\Omega \setminus A$; together with the local upper bound for u, this shows that $\widetilde{u}(x) \in [-\infty, \infty)$ is well defined for $x \in A$. We first show that $\widetilde{u}$ is upper semicontinuous and locally bounded above. On the open set $\Omega \setminus A$ both properties hold because $\widetilde{u} = u$ there. Let $x \in A$ and $\overline{B(x,r)} \subset \Omega$. For $y \in A \cap B(x,r)$, the definition gives $\widetilde{u}(y) \leq \sup_{B(x,r)\setminus A} u$, hence

$$\sup_{B(x,r)} \widetilde{u} = \sup_{B(x,r)\setminus A} u.$$

The right-hand side is bounded above by hypothesis and tends to $\widetilde{u}(x)$ as $r \downarrow 0$, which proves both claims at x.

Next we show that $\widetilde{u}$ is subharmonic near each point $x \in A$; near points of $\Omega \setminus A$ this holds because $\widetilde{u} = u$. The idea is to approximate u from below by functions $u + \varepsilon\psi$ that tend to $-\infty$ at A. Let V and ψ be as in Definition 3.5.1. By Proposition 3.1.5, ψ is locally integrable on the connected set V, so it is not identically $-\infty$ on any ball in V, and its pole set $\{\psi = -\infty\}$ is a null set. Replacing V by a ball centered at x whose

closure lies in V, and ψ by ψ minus an upper bound on that ball, we may therefore assume that V is a ball, that $\psi \le 0$ on V, and that ψ is nontrivial on V.

For $\varepsilon > 0$ define $v_\varepsilon = u + \varepsilon\psi$ on $V \setminus A$ and $v_\varepsilon = -\infty$ on $A \cap V$. On the open set $V \setminus A$, v_ε is subharmonic. At a point $y \in A \cap V$, u is bounded above near y, and $\psi(z) \to -\infty$ as $z \to y$ because ψ is upper semicontinuous with $\psi(y) = -\infty$; hence $v_\varepsilon(z) \to -\infty$ as $z \to y$. So v_ε is upper semicontinuous at y, and the submean inequality at y holds because $v_\varepsilon(y) = -\infty$. Thus $v_\varepsilon \in \mathrm{SH}(V)$.

As $\varepsilon \downarrow 0$, the functions v_ε increase because $\psi \le 0$, and they are bounded above by u on $V \setminus A$, hence uniformly bounded above on compact subsets of V. By Theorem 3.4.2, the regularized supremum $w = (\sup_{\varepsilon>0} v_\varepsilon)^*$ is subharmonic on V and equals $\sup_{\varepsilon>0} v_\varepsilon$ almost everywhere. This supremum equals u on $V \setminus (A \cup \{\psi = -\infty\})$, so $w = u$ almost everywhere on $V \setminus A$. Since w and u are subharmonic on the open set $V \setminus A$, they agree everywhere there by Theorem 3.2.2(2).

It remains to show that $w = \widetilde{u}$ on $A \cap V$. Let $y \in A \cap V$. Since w is upper semicontinuous and $w = u$ on $V \setminus A$, we have $w(y) \ge \overline{\lim}_{V\setminus A \ni z \to y} u(z) = \widetilde{u}(y)$. Conversely, by (3.3), $w(y)$ is the limit of the averages of w over $B(y, r)$. Since A is null and $w = u$ off A, these averages are at most $\sup_{B(y,r)\setminus A} u$, which tends to $\widetilde{u}(y)$ as $r \downarrow 0$. Hence $\widetilde{u} = w$ on V, and $\widetilde{u}$ is subharmonic on V. This proves that $\widetilde{u} \in \mathrm{SH}(\Omega)$.

For uniqueness, let $u' \in \mathrm{SH}(\Omega)$ agree with u on $\Omega \setminus A$. Since A is null, $u' = \widetilde{u}$ almost everywhere on Ω, and hence everywhere by Theorem 3.2.2(2).

Finally, let Ω be a domain, and suppose that $\Omega \setminus A$ is the union of two disjoint nonempty open sets W_0 and W_1. The function g equal to 0 on W_0 and to 1 on W_1 is harmonic, hence subharmonic, and bounded on $\Omega \setminus A$. Its subharmonic extension $\widetilde{g}$ to Ω takes values in $[0, 1]$ and equals 1 on W_1, so it attains the finite maximum 1. By Proposition 3.1.4, $\widetilde{g}$ is identically 1 on Ω, contradicting $\widetilde{g} = 0$ on W_0. □

Chapter 4
Potentials, energy and capacity

Polar sets were introduced in Section 3.5 as the sets that lie locally in the pole set of a nontrivial subharmonic function. Every polar set has Lebesgue measure zero, but the converse fails: a sphere is a Lebesgue null set, yet its normalized surface measure has a bounded potential (Example 4.3.6), and we shall see that this prevents the sphere from being polar. In this chapter we introduce two quantities adapted to the Laplacian, the energy of a measure and the capacity of a set. They characterize the polar sets, and they show that the exceptional sets of the limit theorems of Chapter 3 are polar.

The energy of a positive measure is the integral of its potential against the measure itself. The Newtonian kernel, and the logarithmic kernel on measures of total mass zero, define positive definite energies, and minimizing the energy over the probability measures on a compact set gives its equilibrium measure. Measures with finite continuous potentials give no mass to polar sets, and by the continuity principle (Theorem 4.2.2) every compactly supported measure whose potential is finite almost everywhere with respect to the measure itself is a countable sum of such measures. We also construct a polar Cantor set carrying an atom-free probability measure (Proposition 4.2.4); this measure reappears in Example 15.3.8.

Capacity is defined by an extremal problem for subharmonic functions. For a compact set K in a ball B, let u_K be the upper semicontinuous regularization of the supremum of all nonpositive subharmonic functions on B that are at most -1 on K. The relative capacity $C_B(K)$ is the total mass of the Riesz measure of u_K. A compact set is nonpolar if and only if it carries a nonzero measure of finite energy, if and only if its relative capacity is positive (Theorem 4.3.3). With Choquet's capacitability theorem this extends to arbitrary sets with compact closure in B: such a set is polar if and only if its outer capacity vanishes (Corollary 4.4.2). The main application, Theorem 4.4.4, has two parts. If subharmonic functions v_j converge to v in L^1_{loc}, then the inequality $\overline{\lim}_j v_j \le v$ of Hartogs' lemma (Theorem 3.3.3) is an equality outside a polar set. Moreover, the upper envelope of a family of subharmonic functions that is locally uniformly bounded above agrees with its upper semicontinuous regularization outside a polar set, which sharpens Theorem 3.4.2. The chapter closes with the Dirichlet energy of Green potentials and the quasi-continuity of subharmonic functions.

Throughout this chapter $d \geq 2$; in dimension one the empty set is the only polar set (Section 3.5).

Historical remarks and references for this chapter are collected in Appendix B.4.

4.1 Kernel energies and equilibrium measures

Recall the fundamental solution E_d of (2.7), the kernel $\Gamma_d = -E_d$ of $-\Delta$, and the area s_{d-1} of the unit sphere. For energy estimates we need kernels that are nonnegative on the sets under consideration, so we fix a compact set $K \subset \mathbb{R}^d$. For $d \geq 3$ we put

$$k(x,y) = \Gamma_d(x-y) = \frac{|x-y|^{2-d}}{(d-2)s_{d-1}}.$$

In dimension two we fix a scale $D > \operatorname{diam} K$ and put

$$k_D(x,y) = \frac{1}{2\pi}\log\frac{D}{|x-y|} = \Gamma_2(x-y) + \frac{\log D}{2\pi}.$$

Both kernels equal $+\infty$ on the diagonal and are nonnegative on $K \times K$.

By a measure on K we mean a finite positive Borel measure on $\mathbb{R}^d$ supported on K; it is a Radon measure by Proposition 1.1.2. The potential and the energy of such a measure μ are

$$U^\mu(x) = \int k(x,y)\,\mathrm{d}\mu(y), \qquad I(\mu) = \iint k(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y),$$

with k_D in place of k when $d = 2$; when the scale must be specified, we write I_D. We call U^μ and $I(\mu)$ the Newtonian potential and the Newtonian energy of μ if $d \geq 3$, and its logarithmic potential and logarithmic energy if $d = 2$. For two measures μ and ν on K, their mutual energy is $I(\mu,\nu) = \iint k(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y)$. The same integrals define the energy and the mutual energy of finite signed measures on K whenever the kernel is integrable for the product of the total variations.

The potential U^μ is lower semicontinuous on $\mathbb{R}^d$: the kernel is a lower semicontinuous function of x with values in $(-\infty,\infty]$, and it is bounded below for x in a compact set and $y \in K$, so Fatou's lemma applies after adding a constant. Moreover, U^μ is related to the fundamental potential $E_d * \mu$ of Proposition 3.2.1 by $U^\mu = -E_d * \mu$ if $d \geq 3$ and $U^\mu = -E_2 * \mu + (2\pi)^{-1}\mu(K)\log D$ if $d = 2$. Replacing D by another scale $D' > \operatorname{diam} K$ adds $(2\pi)^{-1}\mu(K)^2\log(D'/D)$ to the energy; in particular, whether μ has finite energy does not depend on the scale.

Fundamental potentials satisfy a reciprocity law. Let μ and ν be finite positive measures with compact supports, and let M be an upper bound of E_d on the compact set $\operatorname{Supp}\mu - \operatorname{Supp}\nu$. Applying Tonelli's theorem to the function $(x,y) \mapsto M - E_d(x-y)$, which is nonnegative on $\operatorname{Supp}\mu \times \operatorname{Supp}\nu$, using $E_d(-z) = E_d(z)$ and subtracting the finite number $M\mu(\mathbb{R}^d)\nu(\mathbb{R}^d)$, we obtain

$$\int (E_d * \nu)\,\mathrm{d}\mu = \int (E_d * \mu)\,\mathrm{d}\nu, \tag{4.1}$$

where both sides lie in $[-\infty, \infty)$.

The positivity of the energy rests on the Gaussian kernels $\mathrm{e}^{-t|x-y|^2}$, $t > 0$: each of them is an L^2 inner product of two Gaussians, and both k and k_D are obtained from them by integration in t.

Lemma 4.1.1 *Let σ and τ be finite real signed measures on $\mathbb{R}^d$, and for $t > 0$ put*

$$Q_t(\sigma,\tau) = \iint \mathrm{e}^{-t|x-y|^2}\,\mathrm{d}\sigma(x)\,\mathrm{d}\tau(y), \qquad h_t^\sigma(z) = \int \mathrm{e}^{-2t|z-x|^2}\,\mathrm{d}\sigma(x),$$

and $Q_t(\sigma) = Q_t(\sigma,\sigma)$. Then h_t^σ is a bounded continuous function, and

$$Q_t(\sigma,\tau) = \left(\frac{4t}{\pi}\right)^{d/2} \int_{\mathbb{R}^d} h_t^\sigma(z)\, h_t^\tau(z)\,\mathrm{d}z. \tag{4.2}$$

In particular $Q_t(\sigma) \geq 0$. If $Q_t(\sigma) = 0$ for arbitrarily large t, then $\sigma = 0$.

Proof The function h_t^σ is bounded by $|\sigma|(\mathbb{R}^d)$, and it is continuous by dominated convergence. Completing the square gives $|z-x|^2+|z-y|^2 = 2|z-\frac{1}{2}(x+y)|^2+\frac{1}{2}|x-y|^2$. Since $\int_{\mathbb{R}^d} \mathrm{e}^{-4t|w|^2}\,\mathrm{d}w = (\pi/4t)^{d/2}$, it follows that

$$\mathrm{e}^{-t|x-y|^2} = \left(\frac{4t}{\pi}\right)^{d/2} \int_{\mathbb{R}^d} \mathrm{e}^{-2t|z-x|^2}\,\mathrm{e}^{-2t|z-y|^2}\,\mathrm{d}z.$$

We integrate this identity against $\sigma\otimes\tau$. Fubini's theorem applies, because the integral of the absolute value of the integrand with respect to $\mathrm{d}z\otimes|\sigma|\otimes|\tau|$ is at most $(\pi/4t)^{d/2}|\sigma|(\mathbb{R}^d)|\tau|(\mathbb{R}^d)$, and integrating first in x and in y gives (4.2). For $\tau = \sigma$ it shows that $Q_t(\sigma)$ is a positive multiple of $\int (h_t^\sigma)^2\,\mathrm{d}z$. Hence $Q_t(\sigma) \geq 0$, and if $Q_t(\sigma) = 0$, then $h_t^\sigma = 0$ almost everywhere, hence everywhere, because h_t^σ is continuous.

Now suppose that $Q_{t_j}(\sigma) = 0$ for a sequence $t_j \to \infty$. The functions $g_t(z) = (2t/\pi)^{d/2}\mathrm{e}^{-2t|z|^2}$ are nonnegative and have integral one, and $\int_{|z|\geq\delta} g_t\,\mathrm{d}z \to 0$ as $t \to \infty$ for every $\delta > 0$. Hence $g_t * f \to f$ uniformly on $\mathbb{R}^d$ for every $f \in C_c(\mathbb{R}^d)$: in the integral defining $g_t * f - f$ we use the uniform continuity of f on the region $|z| < \delta$ and the smallness of the Gaussian tail on the region $|z| \geq \delta$. Since $g_{t_j} * \sigma = (2t_j/\pi)^{d/2} h_{t_j}^\sigma = 0$, Fubini's theorem and the symmetry of g_t give

$$\int (g_{t_j} * f)\,\mathrm{d}\sigma = \int f\,(g_{t_j} * \sigma)\,\mathrm{d}z = 0.$$

Letting $j \to \infty$, we get $\int f\,\mathrm{d}\sigma = 0$ for every $f \in C_c(\mathbb{R}^d)$. Thus the positive and negative parts of σ define the same positive linear functional on $C_c(\mathbb{R}^d)$, and the uniqueness assertion of Theorem 1.2.2 shows that they are equal, that is, $\sigma = 0$. □

Proposition 4.1.2 *Let μ and ν be measures on K with finite energy.*

(1) *If $d \geq 3$, then $I(\mu,\nu)$ is finite and $I(\mu,\nu)^2 \leq I(\mu)I(\nu)$. The energy is a positive definite quadratic form on the real linear span of the measures on K with finite energy.*
(2) *If $d = 2$, then $I_D(\mu,\nu)$ is finite. The energy I_D is a positive definite quadratic form on the space of finite signed measures on K of total mass zero whose positive and negative parts have finite energy, and it satisfies the Cauchy–Schwarz inequality on this space.*

Proof By (4.2), the expression $Q_t(\sigma,\tau)$ of Lemma 4.1.1 is $(4t/\pi)^{d/2}$ times the L^2 inner product of h_t^σ and h_t^τ. Hence, by the Cauchy–Schwarz inequality in z,

$$Q_t(\mu,\nu) \leq Q_t(\mu)^{1/2} Q_t(\nu)^{1/2} \tag{4.3}$$

for positive measures μ and ν.

The case $d \geq 3$. Put $a = (d-2)/2 > 0$ and $c_a = \int_0^\infty s^{a-1}\mathrm{e}^{-s}\,\mathrm{d}s \in (0,\infty)$. The substitution $s = tr^2$ gives

$$r^{2-d} = \frac{1}{c_a}\int_0^\infty t^{a-1}\mathrm{e}^{-tr^2}\,\mathrm{d}t, \qquad r > 0,$$

and both sides are $+\infty$ for $r = 0$. By Tonelli's theorem,

$$I(\mu,\nu) = \kappa\int_0^\infty t^{a-1}Q_t(\mu,\nu)\,\mathrm{d}t, \qquad \kappa = \frac{1}{(d-2)s_{d-1}c_a}.$$

Applying (4.3) and then the Cauchy–Schwarz inequality for the measure $t^{a-1}\,\mathrm{d}t$, we obtain $I(\mu,\nu) \leq I(\mu)^{1/2}I(\nu)^{1/2}$. Now let σ be a real linear combination of measures on K with finite energy. Its total variation is dominated by a combination of the same measures with nonnegative coefficients, so the inequality just proved shows that $\iint k\,\mathrm{d}|\sigma|\,\mathrm{d}|\sigma| < \infty$. Consequently Fubini's theorem applies to the integrand $t^{a-1}\mathrm{e}^{-t|x-y|^2}$, and

$$I(\sigma) = \kappa\int_0^\infty t^{a-1}Q_t(\sigma)\,\mathrm{d}t \geq 0$$

by Lemma 4.1.1. If $I(\sigma) = 0$, then $Q_t(\sigma) = 0$ for almost every $t > 0$, in particular for arbitrarily large t, and Lemma 4.1.1 gives $\sigma = 0$.

The case $d = 2$. For $0 < r \leq D$ we have

$$\log\frac{D}{r} = \frac{1}{2}\int_0^\infty \frac{\mathrm{e}^{-tr^2} - \mathrm{e}^{-tD^2}}{t}\,\mathrm{d}t.$$

Indeed, the right side vanishes for $r = D$, and differentiation under the integral sign, which is legitimate for r in a compact subinterval of $(0,\infty)$, shows that its derivative in r is $-r\int_0^\infty \mathrm{e}^{-tr^2}\,\mathrm{d}t = -1/r$. The integrand is nonnegative, and since $|\mathrm{e}^{-tr^2} - \mathrm{e}^{-tD^2}| \leq tD^2$, its integral over $0 < t < 1$ is at most D^2. Since $|x-y| \leq \operatorname{diam} K < D$ for $x, y \in K$, Tonelli's theorem gives

$$4\pi I_D(\mu) = \int_0^\infty \bigl(Q_t(\mu) - \mu(K)^2 \mathrm{e}^{-tD^2}\bigr)\,\frac{\mathrm{d}t}{t},$$

with a nonnegative integrand. As $\int_1^\infty \mathrm{e}^{-tD^2}\,\mathrm{d}t/t < \infty$, it follows that $\int_1^\infty Q_t(\mu)\,\mathrm{d}t/t < \infty$, and likewise for ν. By (4.3) and the Cauchy–Schwarz inequality for the measure $\mathrm{d}t/t$ on $(1,\infty)$, also $\int_1^\infty Q_t(\mu,\nu)\,\mathrm{d}t/t < \infty$. Since the part of the integral over $0 < t < 1$ contributes at most $D^2\mu(K)\nu(K)$ to $4\pi I_D(\mu,\nu)$, the mutual energy $I_D(\mu,\nu)$ is finite.

Now let σ be a signed measure on K of total mass zero whose positive and negative parts $\sigma^\pm$ have finite energy. Then

$$\iint k_D\,\mathrm{d}|\sigma|\,\mathrm{d}|\sigma| = I_D(\sigma^+) + 2I_D(\sigma^+,\sigma^-) + I_D(\sigma^-) < \infty,$$

so Fubini's theorem applies to the integrand $(\mathrm{e}^{-t|x-y|^2} - \mathrm{e}^{-tD^2})/t$ with respect to $\mathrm{d}t \otimes \sigma \otimes \sigma$. Since $\sigma(K) = 0$, the term e^{-tD^2} integrates to zero, and

$$I_D(\sigma) = \frac{1}{4\pi}\int_0^\infty Q_t(\sigma)\,\frac{\mathrm{d}t}{t} \ge 0.$$

In particular $I_D(\sigma)$ does not depend on D. As in the case $d \ge 3$, $I_D(\sigma) = 0$ forces $\sigma = 0$ by Lemma 4.1.1.

It remains to prove the Cauchy–Schwarz inequality. Let $\mathcal{E}$ be the real linear span of the measures on K with finite energy if $d \ge 3$, and the set of signed measures on K of total mass zero whose positive and negative parts have finite energy if $d = 2$. In the second case $\mathcal{E}$ is also a vector space. Indeed, let $\sigma, \tau \in \mathcal{E}$ and $c \in \mathbb{R}$. Since $|\sigma+\tau| \le |\sigma| + |\tau|$, we have $(\sigma+\tau)^\pm = \frac{1}{2}\bigl(|\sigma+\tau| \pm (\sigma+\tau)\bigr) \le \sigma^\pm + \tau^\pm$, while $(c\sigma)^\pm = c\sigma^\pm$ if $c \ge 0$ and $(c\sigma)^\pm = |c|\sigma^\mp$ if $c < 0$. A positive measure on K dominated by the sum of two measures with finite energy has finite energy, because the kernel is nonnegative on $K \times K$ and the mutual energy of two measures with finite energy is finite. Hence $\sigma + \tau$ and $c\sigma$ belong to $\mathcal{E}$.

In both cases, the total variation of an element of $\mathcal{E}$ is dominated by a sum of measures on K with finite energy, so for $\sigma, \tau \in \mathcal{E}$ the kernel is integrable with respect to $|\sigma| \otimes |\tau|$. Consequently $I(\sigma,\tau) = \iint k\,\mathrm{d}\sigma\,\mathrm{d}\tau$ (with k_D in place of k when $d = 2$) is a symmetric bilinear form on $\mathcal{E}$, and $I(\sigma) = I(\sigma,\sigma) \ge 0$ by what we have proved. Hence, for $\sigma, \tau \in \mathcal{E}$, the quadratic polynomial $s \mapsto I(\sigma + s\tau) = I(\sigma) + 2sI(\sigma,\tau) + s^2 I(\tau)$ is nonnegative on $\mathbb{R}$. Its discriminant is therefore nonpositive, which is the Cauchy–Schwarz inequality $I(\sigma,\tau)^2 \le I(\sigma)I(\tau)$. □

In dimension two, the restriction to measures of total mass zero in part (2) is what makes the logarithmic energy independent of the scale. For a positive measure, the sign of I_D depends on how the scale compares with the support. Identify $\mathbb{R}^2$ with $\mathbb{C}$, and let σ_r be the normalized arc length measure on the circle $|x| = r$. For $|x| = r$, the rotation invariance of σ_r gives

$$\int \log|x-y|\,\mathrm{d}\sigma_r(y) = \log r + \frac{1}{2\pi}\int_{-\pi}^{\pi} \log|1 - \mathrm{e}^{\mathrm{i}\theta}|\,\mathrm{d}\theta,$$

and the last integral vanishes. Indeed, for $0 < a < 1$ the function $z \mapsto \log|1 - az|$ is harmonic on a neighborhood of the closed unit disc, so its mean over the unit circle equals its value 0 at the origin. We may let $a \uparrow 1$ by dominated convergence, since $|\theta|/\pi \le |1 - ae^{i\theta}| \le 2$ for $1/2 \le a < 1$ and $|\theta| \le \pi$. Hence, for every $D > 0$, the function $\int k_D(\cdot, y)\, d\sigma_r(y)$ equals $(2\pi)^{-1}\log(D/r)$ on the circle, and $\iint k_D\, d\sigma_r\, d\sigma_r = (2\pi)^{-1}\log(D/r)$. For a fixed scale D, this energy is negative as soon as $r > D$. Thus no single scale makes the logarithmic energy nonnegative on all positive measures with compact support.

Theorem 4.1.3 *Let $K \subset \mathbb{R}^d$ be a compact set that carries a probability measure of finite energy; when $d = 2$, the scale $D > \operatorname{diam} K$ is fixed. Then there is a unique probability measure μ_K on K of minimal energy, the equilibrium measure of K. Put $V(K) = I(\mu_K)$, and $V_D(K) = I_D(\mu_K)$ when $d = 2$. Every probability measure ν on K with finite energy satisfies*

$$I(\mu_K, \nu) \ge V(K)$$

(with I_D and $V_D(K)$ when $d = 2$). If $d \ge 3$, then $V(K) > 0$, and the Newtonian capacity of K is defined to be $1/V(K)$. If $d = 2$, the logarithmic capacity of K is defined to be $De^{-2\pi V_D(K)}$; neither μ_K nor this number depends on D. If K carries no probability measure of finite energy, in particular if $K = \varnothing$, its Newtonian or logarithmic capacity is defined to be zero.

Proof *Existence.* Let μ_j be probability measures on K whose energies converge to the infimum m, which is finite by hypothesis. By Theorem 1.5.5, after passing to a subsequence, μ_j converges vaguely to a Radon measure μ on $\mathbb{R}^d$. By Proposition 1.5.2, applied to the open set $\mathbb{R}^d \setminus K$, we have $\mu(\mathbb{R}^d \setminus K) \le \underline{\lim}_j\, \mu_j(\mathbb{R}^d \setminus K) = 0$, so μ is carried by K; testing against a function in $C_c(\mathbb{R}^d)$ equal to one on K then shows that $\mu(K) = 1$. To prove that $I(\mu) \le m$, we first show that

$$\iint F\, d\mu_j\, d\mu_j \longrightarrow \iint F\, d\mu\, d\mu$$

for every continuous function F on $\mathbb{R}^d \times \mathbb{R}^d$.

Given $\delta > 0$, cover K by finitely many balls $B(x_i, \delta)$ with $x_i \in K$. By Lemma 1.4.2, there are functions $0 \le \varphi_i \in C_c^\infty(B(x_i, \delta))$ with $\sum_i \varphi_i = 1$ on K. For $x, y \in K$, a term $\varphi_i(x)\varphi_l(y)$ is nonzero only if $|x - x_i| < \delta$ and $|y - x_l| < \delta$, so

$$\Bigl|F(x, y) - \sum_{i,l} F(x_i, x_l)\varphi_i(x)\varphi_l(y)\Bigr| \le \omega(2\delta),$$

where ω is the modulus of continuity of F on $K \times K$. The double integrals of the finite sums converge, since $\int \varphi_i\, d\mu_j \to \int \varphi_i\, d\mu$. As all the measures are probability measures carried by K and $\omega(2\delta) \to 0$ as $\delta \downarrow 0$, this proves the asserted convergence.

For $n > 0$, the truncated kernel $\min\{k, n\}$ is continuous on $\mathbb{R}^d \times \mathbb{R}^d$, including the diagonal, and it increases to k as $n \to \infty$. Applying the convergence just proved to $F = \min\{k, n\}$, we get

$$\iint \min\{k,n\}\,\mathrm{d}\mu\,\mathrm{d}\mu = \lim_j \iint \min\{k,n\}\,\mathrm{d}\mu_j\,\mathrm{d}\mu_j \le \varliminf_j I(\mu_j) = m.$$

Since $k \ge 0$ on $K \times K$, monotone convergence in n gives $I(\mu) \le m$. Hence μ is a minimizer.

Uniqueness. Let μ and μ' be minimizers. The signed measure $\sigma = \mu - \mu'$ has total mass zero. Since $|\sigma| \le \mu + \mu'$, its positive and negative parts satisfy $\sigma^+ = \frac{1}{2}(|\sigma| + \sigma) \le \mu$ and $\sigma^- \le \mu'$, so they have finite energy, the kernel being nonnegative on $K \times K$. All mutual energies are finite by Proposition 4.1.2, so we may expand and obtain

$$I\big(\tfrac{1}{2}(\mu+\mu')\big) = \tfrac{1}{2}I(\mu) + \tfrac{1}{2}I(\mu') - \tfrac{1}{4}I(\sigma) = m - \tfrac{1}{4}I(\sigma).$$

Since $\frac{1}{2}(\mu+\mu')$ is a probability measure on K, this energy is at least m, so $I(\sigma) \le 0$. By Proposition 4.1.2, $\sigma = 0$.

The inequality. Let ν be a probability measure on K with finite energy. For $0 < t \le 1$ the probability measure $(1-t)\mu_K + t\nu$ competes in the minimization, so

$$V(K) \le I\big((1-t)\mu_K + t\nu\big) = (1-t)^2 V(K) + 2t(1-t)I(\mu_K,\nu) + t^2 I(\nu),$$

all terms being finite. Subtracting $V(K)$, dividing by t and letting $t \downarrow 0$ gives $0 \le -2V(K) + 2I(\mu_K,\nu)$.

The capacities. Let $d \ge 3$. A point mass has infinite energy, so K is not a single point and $\operatorname{diam} K > 0$. The kernel is at least $((d-2)s_{d-1})^{-1}(\operatorname{diam} K)^{2-d} > 0$ on $K \times K$, hence $V(K) > 0$. Let $d = 2$. Replacing D by $D' > \operatorname{diam} K$ changes the energy of every probability measure on K by the same constant $(2\pi)^{-1}\log(D'/D)$. Hence μ_K does not depend on the scale, $V_{D'}(K) = V_D(K) + (2\pi)^{-1}\log(D'/D)$, and $D'\mathrm{e}^{-2\pi V_{D'}(K)} = D\mathrm{e}^{-2\pi V_D(K)}$. □

4.2 Continuous potentials and polar sets

To relate energy to polarity we need measures whose fundamental potentials $E_d * \mu$, or equivalently whose potentials U^μ, are finite and continuous. We show that such measures are plentiful: every compactly supported measure whose potential is finite almost everywhere with respect to the measure itself is a countable sum of them. We begin with a version of Lusin's theorem.

Lemma 4.2.1 *Let μ be a finite Radon measure on $\mathbb{R}^d$, and let f be a Borel function with values in $[-\infty,\infty]$ that is finite μ-almost everywhere. Then for every $\varepsilon > 0$ there is a compact set L with $\mu(\mathbb{R}^d \setminus L) < \varepsilon$ such that f is finite on L and the restriction $f|_L$ is continuous.*

Proof The sets $\{|f| > N\}$ decrease to the μ-null set $\{|f| = \infty\}$ as $N \to \infty$, and μ is finite, so we can choose N with $\mu(\{|f| > N\}) < \varepsilon/3$. The truncation $f_N = \max\{-N, \min\{f, N\}\}$ is bounded, hence belongs to $L^1(\mu)$. By Proposition 1.2.3

there are continuous functions g_j, $j \geq 1$, with $\int |g_j - f_N| \, d\mu < \varepsilon 2^{-2j}/3$. By Chebyshev's inequality, the set $E_j = \{|g_j - f_N| > 2^{-j}\}$ satisfies $\mu(E_j) < \varepsilon 2^{-j}/3$, so $\mu(\bigcup_j E_j) < \varepsilon/3$. On the Borel set $F = \mathbb{R}^d \setminus (\{|f| > N\} \cup \bigcup_j E_j)$ we have $f = f_N$ and $|g_j - f| \leq 2^{-j}$ for every j, so $g_j \to f$ uniformly on F. Since $\mu(\mathbb{R}^d \setminus F) < 2\varepsilon/3$, inner regularity gives a compact set $L \subset F$ with $\mu(F \setminus L) < \varepsilon/3$. Then $\mu(\mathbb{R}^d \setminus L) < \varepsilon$, and $f|_L$ is finite and continuous as a uniform limit of continuous functions. □

Theorem 4.2.2 *Let $K \subset \mathbb{R}^d$ be compact, let μ be a finite positive measure on $\mathbb{R}^d$ with* $\operatorname{Supp}\mu \subset K$, *and put $p = E_d * \mu$.*

(1) *If $x \in K$ and the restriction $p|_K$ is continuous at x, then p is continuous at x as a function on $\mathbb{R}^d$ with values in $[-\infty, \infty)$.*
(2) *If p is finite μ-almost everywhere, then $\mu = \sum_j \mu_j$, where the μ_j are the restrictions of μ to pairwise disjoint compact sets and every $E_d * \mu_j$ is finite and continuous on $\mathbb{R}^d$.*

Proof (1) Since p is upper semicontinuous (Proposition 3.2.1), $\overline{\lim}_{z\to x} p(z) \leq p(x)$; in particular p is continuous at x if $p(x) = -\infty$. Assume that $p(x)$ is finite; we must show that $\underline{\lim}_{z\to x} p(z) \geq p(x)$. For $z \in \mathbb{R}^d$ choose a point $z' \in K$ closest to z. Then $|z' - x| \leq |z' - z| + |z - x| \leq 2|z - x|$, so $z' \to x$ as $z \to x$, and $|z' - y| \leq |z' - z| + |z - y| \leq 2|z - y|$ for every $y \in K$. Consequently, for $y \in K$,

$$E_d(z-y) \geq 2^{d-2} E_d(z'-y) \quad (d \geq 3), \qquad E_2(z-y) \geq E_2(z'-y) - \frac{\log 2}{2\pi} \quad (d = 2).$$

Fix $\delta > 0$ and split $p = p'_\delta + p''_\delta$, where $p'_\delta(w) = \int_{B(x,\delta)} E_d(w - y) \, d\mu(y)$ is the near part and p''_δ the far part. The far part is continuous at x, since $E_d(w - y) \to E_d(x - y)$ uniformly for $y \in K$ with $|y - x| \geq \delta$ as $w \to x$, by the uniform continuity of E_d on compact subsets of $\mathbb{R}^d \setminus \{0\}$. The near part, restricted to K, is continuous at x, being the difference of $p|_K$ and $p''_\delta|_K$. Since μ is carried by K, we may integrate the inequalities above with respect to μ over $B(x, \delta)$; letting $z \to x$, we obtain for $d \geq 3$

$$\underline{\lim_{z\to x}} \, p(z) \geq p''_\delta(x) + 2^{d-2} \underline{\lim_{z\to x}} \, p'_\delta(z') = p(x) + (2^{d-2} - 1) p'_\delta(x),$$

and for $d = 2$ the bound $\underline{\lim}_{z\to x} p(z) \geq p(x) - (2\pi)^{-1} \mu(B(x, \delta)) \log 2$. Since $p(x)$ is finite, the function $E_d(x - \cdot)$ is μ-integrable and $\mu(\{x\}) = 0$. By dominated convergence and continuity of μ from above, $p'_\delta(x) \to 0$ and $\mu(B(x, \delta)) \to 0$ as $\delta \downarrow 0$, which proves $\underline{\lim}_{z\to x} p(z) \geq p(x)$.

(2) Let $M \geq 0$ be an upper bound of E_d on the compact set $K - K$. If ν is the restriction of μ to a Borel set, then $p = E_d * \nu + E_d * (\mu - \nu)$ and $E_d * (\mu - \nu) \leq M\mu(K)$ on K, so $E_d * \nu \geq p - M\mu(K)$ on K; hence $E_d * \nu$ is finite ν-almost everywhere.

We construct inductively pairwise disjoint compact sets $K_1, K_2, \ldots \subset K$ such that the residual measures $\nu_j = \mu|_{K \setminus (K_1 \cup \cdots \cup K_j)}$ satisfy $\nu_j(\mathbb{R}^d) < 2^{-j}$, and such that each $E_d * (\mu|_{K_j})$ is finite and continuous on $\mathbb{R}^d$. Put $\nu_0 = \mu$, and suppose $K_1, \ldots, K_{j-1}$ have been chosen. Apply Lemma 4.2.1 to $\nu = \nu_{j-1}$ and $E_d * \nu$: there is a compact

set L with $\nu(\mathbb{R}^d \setminus L) < 2^{-j-1}$ on which $E_d * \nu$ is finite and continuous. Since ν is carried by K and does not charge $K_1 \cup \cdots \cup K_{j-1}$, inner regularity gives a compact set $K_j \subset (L \cap K) \setminus (K_1 \cup \cdots \cup K_{j-1})$ with $\nu(\mathbb{R}^d \setminus K_j) < 2^{-j}$. Put $\mu_j = \nu|_{K_j} = \mu|_{K_j}$; then $\nu_j = \nu - \mu_j$ has mass less than 2^{-j}.

It remains to show that $E_d * \mu_j$ is finite and continuous. On K_j the two functions $E_d * \mu_j$ and $E_d * (\nu - \mu_j)$ are upper semicontinuous and less than $+\infty$, and their sum $E_d * \nu$ is finite and continuous; hence both are finite on K_j. Moreover, for $x \in K_j$ and $K_j \ni w \to x$,

$$\varliminf_{w \to x} E_d * \mu_j(w) \geq \lim_{w \to x} E_d * \nu(w) - \varlimsup_{w \to x} E_d * (\nu - \mu_j)(w) \geq E_d * \mu_j(x),$$

so the restriction of $E_d * \mu_j$ to K_j is continuous. The support of μ_j is contained in K_j, so part (1), applied to μ_j and the compact set K_j, shows that $E_d * \mu_j$ is continuous at every point of K_j. At a point $x \notin K_j$, the function $E_d * \mu_j$ is finite and continuous, because $E_d(w - y) \to E_d(x - y)$ uniformly in $y \in K_j$ as $w \to x$, as for the far part in (1). Thus $E_d * \mu_j$ is finite and continuous on $\mathbb{R}^d$. Finally, $\mu - \sum_{i \leq j} \mu_i = \nu_j$ has mass less than 2^{-j}, so $\mu = \sum_j \mu_j$. □

Since the definition of polarity is local, a subset of an open set Ω is polar in Ω if and only if it is polar in $\mathbb{R}^d$. Indeed, if ψ is a nontrivial subharmonic function on a connected open set V, as in Definition 3.5.1, then its restriction to every ball contained in V is nontrivial, because ψ is locally integrable on V (Proposition 3.1.5). Hence the neighborhoods in Definition 3.5.1 may be shrunk to balls contained in Ω. We therefore speak simply of polar sets. The next result shows that a bounded polar set lies in the pole set of a single potential.

Proposition 4.2.3 *If $A \subset \mathbb{R}^d$ is polar and has compact closure, there is a finite positive measure ν with compact support such that $E_d * \nu = -\infty$ on A. Countable unions of polar sets are polar.*

Proof Fix a bounded open set $W \supset \overline{A}$; we construct ν with support in $\overline{W}$. Let $a \in A$. By Definition 3.5.1, after shrinking the neighborhood as explained above, there are an open ball $V \subset W$ centered at a and a nontrivial function $\psi \in \mathrm{SH}(V)$ with $A \cap V \subset \{\psi = -\infty\}$; by Proposition 3.1.5, ψ is locally integrable. Choose an open ball B_a centered at a with $\overline{B_a} \subset V$, and a cutoff $\eta \in C_c^\infty(V)$ with $0 \leq \eta \leq 1$ and $\eta = 1$ near $\overline{B_a}$. By the Riesz decomposition (Theorem 3.2.3), $\psi = E_d * (\eta \mu_\psi) + h$ on B_a, where μ_ψ is the Riesz measure of ψ and h is harmonic, hence finite; therefore $E_d * (\eta \mu_\psi) = -\infty$ on $A \cap B_a$. The measure $\eta \mu_\psi$ is finite, positive and supported in $\overline{W}$. Since $\mathbb{R}^d$ has a countable base, countably many of the balls B_a cover A. Let ν_j be the corresponding measures.

Choose $c_j > 0$ with $\sum_j c_j \nu_j(\mathbb{R}^d) < \infty$, and put $\nu = \sum_j c_j \nu_j$, a finite positive measure supported in $\overline{W}$. Let $S \subset \mathbb{R}^d$ be compact and let M be an upper bound of E_d on the compact set $S - \overline{W}$. For $x \in S$, Tonelli's theorem applied to the nonnegative functions $M - E_d(x - \cdot)$ gives

$$M\nu(\mathbb{R}^d) - E_d * \nu(x) = \sum_j c_j \big(M\nu_j(\mathbb{R}^d) - E_d * \nu_j(x)\big).$$

Thus $E_d*\nu(x) = -\infty$ whenever $E_d*\nu_j(x) = -\infty$ for some j; in particular $E_d*\nu = -\infty$ on A. By Proposition 3.2.1, $E_d * \nu$ is subharmonic and locally integrable, so it is not identically $-\infty$.

Now let $A = \bigcup_k A_k$ with polar sets A_k, let $x \in \mathbb{R}^d$, and let B' be an open ball centered at x. Apply the construction above to the bounded polar sets $A_k \cap B'$, with one and the same bounded open set $W \supset \overline{B'}$. It gives measures $\nu^{(k)}$ supported in $\overline{W}$ with $E_d * \nu^{(k)} = -\infty$ on $A_k \cap B'$. Choosing $c_k > 0$ with $\sum_k c_k \nu^{(k)}(\mathbb{R}^d) < \infty$, the Tonelli argument above shows that $\nu = \sum_k c_k \nu^{(k)}$ satisfies $E_d * \nu = -\infty$ on $A \cap B'$. Since $E_d * \nu$ is subharmonic on B' and not identically $-\infty$ there, A is polar. □

Although polar sets are Lebesgue null sets, they can carry measures without atoms. The next proposition constructs a polar Cantor set carrying an atom-free probability measure whose fundamental potential equals $-\infty$ at every point of the set.

Proposition 4.2.4 *There are a compact Cantor set $K \subset [0, 1/10] \subset \mathbb{C}$ and an atom-free probability measure μ supported on K such that*

$$p(z) = \int_K \log|z - a|^2 \, \mathrm{d}\mu(a)$$

is locally integrable and subharmonic on $\mathbb{C}$, finite and harmonic on $\mathbb{C} \setminus K$, and equal to $-\infty$ at every point of K. In particular, K is polar and μ has infinite logarithmic energy.

Proof *The set.* Put $a_j = \mathrm{e}^{-4^j}$ for $j \geq 1$, and let K be the set of all sums $\sum_{j\geq 1} \varepsilon_j a_j$ with $\varepsilon_j \in \{0, 1\}$. Since $a_{j+1}/a_j = \mathrm{e}^{-3\cdot 4^j} \leq \mathrm{e}^{-12} < 1/2$, the tails $R_k = \sum_{j>k} a_j$ satisfy

$$R_k \leq a_{k+1} \sum_{i\geq 0} 2^{-i} = 2a_{k+1} \quad (k \geq 0), \qquad 2a_{k+1} < a_k \quad (k \geq 1).$$

In particular $K \subset [0, R_0] \subset [0, 2\mathrm{e}^{-4}] \subset [0, 1/10]$. Let $\mathcal{S} = \prod_{j\geq 1}\{0, 1\}$ carry the product topology, and let $c\colon \mathcal{S} \to \mathbb{R}$ be the coding map $c(\varepsilon) = \sum_j \varepsilon_j a_j$. If ε and ε' agree in their first k digits, then $|c(\varepsilon) - c(\varepsilon')| \leq R_k$; hence c is continuous, and $K = c(\mathcal{S})$ is compact by Lemma 1.6.2. If ε and ε' first differ at the k-th digit, say $\varepsilon_k = 1$ and $\varepsilon'_k = 0$, then $c(\varepsilon) - c(\varepsilon') \geq a_k - R_k > 0$; hence c is injective.

For a word $\sigma \in \{0, 1\}^k$, let K_σ be the image of the sequences beginning with σ. The 2^k sets K_σ are compact, pairwise disjoint by injectivity, cover K, and have diameter at most R_k. Consequently each K_σ is open and closed in K, and two distinct points of K lie in different sets K_σ once k is large; thus K is totally disconnected. Changing the k-th digit of ε produces a point of K different from $c(\varepsilon)$ at distance a_k, so K has no isolated points. Hence K is a Cantor set: a nonempty compact metric space that is totally disconnected and has no isolated points.

The measure. For $t \in [0, 1)$ let $\varepsilon_j(t)$ be its binary digits, using the terminating expansion at dyadic rationals, and put $\Phi(t) = \sum_j \varepsilon_j(t) a_j$. Each ε_j is a Borel step

function, so Φ is Borel as a uniform limit of Borel functions. Let μ be the image of Lebesgue measure on $[0,1)$ under Φ. It is a probability measure carried by the compact set K, and it is a Radon measure by Proposition 1.1.2. For $\sigma \in \{0,1\}^k$, injectivity of c shows that $\Phi(t) \in K_\sigma$ if and only if the first k binary digits of t form the word σ, and this happens exactly on a dyadic interval of length 2^{-k}. Hence $\mu(K_\sigma) = 2^{-k}$. Every $x \in K$ lies in one of the sets K_σ with σ of length k, so $\mu(\{x\}) \le 2^{-k}$ for every k; thus μ has no atoms.

The potential. Since $\log|z-a|^2 = 4\pi E_2(z-a)$, we have $p = 4\pi E_2 * \mu$, which is subharmonic and locally integrable on $\mathbb{C}$ by Proposition 3.2.1. Let $z_0 \notin K$ and $\delta = \frac{1}{2}\operatorname{dist}(z_0, K)$. For $|z - z_0| < \delta$ and $a \in K$, the kernel and all its derivatives in z are bounded uniformly, so differentiation under the integral sign shows that p is smooth on $B(z_0,\delta)$ with $\Delta p(z) = \int \Delta_z \log|z-a|^2 \, d\mu(a) = 0$. Thus p is finite and harmonic on $\mathbb{C} \setminus K$.

Polarity. Let $x \in K$ and $k \ge 1$, and let K_σ, with σ of length k, be the piece containing x. Since $\operatorname{diam} K \le R_0 < 1$, every term $\log|x-a|^2$ with $a \in K$ is nonpositive, and for $a \in K_\sigma$ we have $|x-a| \le R_k \le 2a_{k+1}$. Consequently

$$p(x) \le \int_{K_\sigma} \log|x-a|^2 \, d\mu(a) \le \mu(K_\sigma)\log\bigl(4a_{k+1}^2\bigr) = 2^{-k}\bigl(\log 4 - 2\cdot 4^{k+1}\bigr).$$

The right side tends to $-\infty$ as $k \to \infty$, so $p(x) = -\infty$. Since p is subharmonic on $\mathbb{C}$ and not identically $-\infty$, the set K is polar. For $D > \operatorname{diam} K$, the logarithmic potential of the probability measure μ is $U^\mu = (2\pi)^{-1}\log D - p/(4\pi)$, which equals $+\infty$ on K. Since μ is carried by K, Tonelli's theorem gives $I_D(\mu) = \int U^\mu \, d\mu = +\infty$. □

4.3 Relative extremal functions and capacity

The Newtonian and logarithmic capacities of Section 4.1, which we call kernel capacities, are defined by minimizing energy over measures. We now define a second capacity by maximizing over subharmonic functions. Its values depend on a surrounding ball, but it is increasing, continuous along decreasing sequences and strongly subadditive (Proposition 4.3.4). These properties allow us to extend it to arbitrary sets (Theorem 4.3.5), and we shall show that its null sets are exactly the polar sets.

From now on, $B = B(0,R)$ is an open ball centered at the origin; the results hold for every radius $R > 0$, and in some proofs we take R large. For a set $A \Subset B$, we say that a function v is admissible for A if $v \in \mathrm{SH}(B)$, $v \le 0$ on B and $v \le -1$ on A. The relative extremal function of A and its regularization are

$$f_A = \sup\{v : v \text{ is admissible for } A\}, \qquad u_A = f_A^*.$$

The constant -1 is admissible, so $-1 \le f_A \le u_A \le 0$, and u_A is subharmonic by Theorem 3.4.2. The condition $v \le -1$ is imposed at every point of A, but the

regularization u_A may exceed -1 at some points of A. For instance, let $A = \{0\}$. Since E_d is a radial function that increases with $|x|$, the function $\varepsilon(E_d - c_R)$, where c_R is the value of E_d on ∂B, is nonpositive on B for every $\varepsilon > 0$. It is subharmonic by Proposition 3.2.1 and equals $-\infty$ at the origin, so it is admissible for A. Letting $\varepsilon \downarrow 0$, we obtain $f_A = 0$ on $B \setminus \{0\}$, and therefore $u_A(0) = 0$.

Extremal functions will be represented as Green potentials. On a ball, the Green function G_B of Theorem 2.5.2 is given by an explicit formula, from which we read off its symmetry and its smoothness near ∂B.

Lemma 4.3.1 *The Green function of B is symmetric: $G_B(x,y) = G_B(y,x)$ for $x, y \in B$. For $x, y \in B$ with $y \neq 0$, put $y^* = R^2 y/|y|^2$. Then for $d \geq 3$,*

$$G_B(x,y) = \Gamma_d(x-y) - \left(\frac{R}{|y|}\right)^{d-2} \Gamma_d(x-y^*),$$

and for $d = 2$,

$$G_B(x,y) = \frac{1}{2\pi} \log \frac{|y|\,|x-y^*|}{R|x-y|}.$$

For $y = 0$, $G_B(x,0) = \Gamma_d(x) - \Gamma_d(Re_1)$ if $d \geq 3$ and $G_B(x,0) = (2\pi)^{-1} \log(R/|x|)$ if $d = 2$; here e_1 is the first coordinate unit vector. Moreover, for $0 < r < R$, these formulas extend the difference $G_B(x,y) - \Gamma_d(x-y)$ to a smooth function of (x,y) on the set where $|y| \leq r$ and $|x| < R^2/r$; this range of x contains a neighborhood of $\overline{B}$.

Proof Fix $y \in B \setminus \{0\}$. Since $|y^*| = R^2/|y| > R$, the function $x \mapsto \Gamma_d(x-y^*)$ is harmonic on a neighborhood of $\overline{B}$. Expanding the square gives

$$|y|^2|x-y^*|^2 = R^4 - 2R^2 x \cdot y + |x|^2|y|^2, \tag{4.4}$$

and therefore $|y|^2|x-y^*|^2 - R^2|x-y|^2 = (R^2 - |x|^2)(R^2 - |y|^2)$. Each displayed expression, as a function of x, is the sum of $\Gamma_d(x-y)$ and a function harmonic near $\overline{B}$. By Proposition 2.2.1, it is therefore harmonic on $B \setminus \{y\}$ and locally integrable on B, with distributional Laplacian $-\delta_y$ on B. Moreover, the last identity shows that $|y|\,|x-y^*| > R|x-y|$ on B, with equality on ∂B. Hence the expression is positive on $B \setminus \{y\}$ and tends to zero at every point of ∂B. The ball is a regular domain by Theorem 2.4.6, and the uniqueness assertion of Theorem 2.5.2 identifies the displayed expression with $G_B(y,x)$ for $x \neq y$. The same argument applies to $y = 0$, where the stated expressions differ from $\Gamma_d(x)$ by constants.

Let $Q(x,y)$ denote the right side of (4.4); it is symmetric in x and y, and $Q(x,0) = R^4$. The formulas just proved can be written, for all $x, y \in B$ including $y = 0$, as

$$G_B(y,x) = \Gamma_d(x-y) - \frac{R^{d-2} Q(x,y)^{(2-d)/2}}{(d-2)s_{d-1}} \qquad (d \geq 3),$$

$$G_B(y,x) = \Gamma_2(x-y) + \frac{1}{4\pi} \log \frac{Q(x,y)}{R^2} \qquad (d = 2).$$

Both right sides are symmetric in x and y, so $G_B(y,x) = G_B(x,y)$; this also proves the formulas in the order stated. Finally, $Q(x,y) \geq (R^2 - |x||y|)^2 > 0$ when $|y| \leq r$ and $|x| < R^2/r$, so the correction terms $G_B(x,y) - \Gamma_d(x-y)$ in the last display are smooth functions of (x,y) there. □

For a finite positive measure λ with compact support in B, its Green potential is

$$G_B\lambda(x) = \int G_B(x,y)\,\mathrm{d}\lambda(y), \qquad x \in B.$$

Proposition 4.3.2 *Let $K \subset B$ be compact.*

(1) *The function f_K is the supremum of the continuous admissible functions for K; it is lower semicontinuous, equals -1 on K, and equals u_K on $B \setminus K$. The function u_K is harmonic on $B \setminus K$ and tends to zero at ∂B.*

(2) *Let λ be a finite positive measure with compact support in B, and put $u = -G_B\lambda$. Then $u = E_d * \lambda - H_B\big((E_d * \lambda)|_{\partial B}\big)$ on B, where H_B is the Poisson integral of Theorem 2.3.1; in particular, u is subharmonic on B with Riesz measure λ. If $\operatorname{Supp}\lambda \subset B(0,r)$ with $0 < r < R$, then the restriction of u to the annulus $r < |x| < R$ extends to a harmonic function on the annulus $r < |x| < R^2/r$ that vanishes on ∂B. Moreover,*

$$\lambda(B) = \int_{\partial B} \partial_n u\,\mathrm{d}S,$$

where ∂_n is the derivative along the outward unit normal of ∂B. If λ' is another such measure and $-G_B\lambda \leq -G_B\lambda'$ on B, then $\lambda(B) \geq \lambda'(B)$. Conversely, if v is a locally integrable subharmonic function on B whose Riesz measure λ has compact support in B, and if $v(x) \to 0$ as x tends to any point of ∂B, then $v = -G_B\lambda$.

(3) *The Riesz measure $\lambda_K = \Delta u_K$ is a finite positive measure supported on K, and $u_K = -G_B\lambda_K$.*

The number $C_B(K) = \lambda_K(K)$ is called the relative capacity of K with respect to B; by (2) *and* (3)*, $C_B(K) = \int_{\partial B} \partial_n u_K\,\mathrm{d}S$.*

Proof (1) Since K is a compact subset of B, there is $c > 0$ such that $q(x) = c(|x|^2 - R^2) \leq -1$ on K. The function q is smooth and subharmonic, since $\Delta q = 2dc > 0$, and it is nonpositive on B; hence it is admissible, and $q \leq f_K \leq u_K \leq 0$. In particular f_K and u_K tend to zero at ∂B.

We show that f_K is the supremum of the continuous admissible functions. Let v be admissible. We first extend $\max\{v,q\}$ to a subharmonic function on $\mathbb{R}^d$: put $w = \max\{v,q\}$ on B and $w = q$ on $\mathbb{R}^d \setminus B$. At $\zeta \in \partial B$ we have $w(\zeta) = 0$ and $\varliminf_{x\to\zeta} w(x) \leq 0$, because $v \leq 0$ on B and q is continuous with $q(\zeta) = 0$; so w is upper semicontinuous. We verify the comparison principle of Theorem 3.1.3 for w on an arbitrary ball D. Let h be harmonic on D and continuous on $\overline{D}$, with $\varliminf_{D\ni x\to\zeta} w(x) \leq h(\zeta)$ for $\zeta \in \partial D$. Since $w \geq q$ and q is continuous, $q \leq h$ on ∂D, hence on D by the comparison principle for q; in particular $w = q \leq h$ on

$D \setminus B$. On the bounded open set $D \cap B$, the function $v - h$ has nonpositive boundary upper limits: at points of $\partial D \cap B$ because $v \le w$, and at points of ∂B because $\overline{\lim}\, v \le 0 = q \le h$ there. By the comparison principle on bounded open sets (the remark after Theorem 3.1.3), $v \le h$ on $D \cap B$. Thus $w \le h$ on D, and the converse part of Theorem 3.1.3 shows that w is subharmonic on $\mathbb{R}^d$.

Since $w \ge q$, it is locally integrable. Its mollifications $w * \rho_\varepsilon$ are smooth and satisfy $\Delta(w * \rho_\varepsilon) = (\Delta w) * \rho_\varepsilon \ge 0$, so they are subharmonic, and they decrease to w as $\varepsilon \downarrow 0$ by (3.3). Fix $\delta > 0$. On B we have $w * \rho_\varepsilon \le \sup_{B(0,R+\varepsilon)} w \le c((R+\varepsilon)^2 - R^2)$, because $w \le 0$ on B and $w = q$ outside B; hence $w * \rho_\varepsilon - \delta \le 0$ on B for small ε. On K we have $w \le -1$. Since the continuous functions $w * \rho_\varepsilon$ decrease to w, the open sets $\{w * \rho_\varepsilon < -1 + \delta\}$ increase and cover the compact set K, so $w * \rho_\varepsilon < -1 + \delta$ on K for small ε. For such ε the function $w * \rho_\varepsilon - \delta$ is continuous and admissible, and it is at least $w - \delta \ge v - \delta$. Since v and δ are arbitrary, f_K is the supremum of the continuous admissible functions; in particular f_K is lower semicontinuous. Every admissible function is at most -1 on K, so $f_K = -1$ on K.

Let D be an open ball with $\overline{D} \subset B \setminus K$; we show that $f_K = u_K$ on D and that u_K is harmonic there. For a continuous admissible function v, let v^D be its harmonic lifting on D, which equals the Poisson integral of $v|_{\partial D}$ on D and v elsewhere. As noted in Section 2.4, the continuous subsolutions defined there are exactly the real-valued continuous subharmonic functions. Hence Lemma 2.4.1 shows that v^D is continuous and subharmonic and that $v^D \ge v$. It is admissible: $v^D \le 0$ because the Poisson integral preserves order, and $v^D = v$ on K.

Apply Choquet's lemma (Lemma 3.4.1) to the family of continuous admissible functions, whose supremum is f_K: there are continuous admissible functions $v_1, v_2, \ldots$ with $(\sup_j v_j)^* = u_K$. The functions $w_j = \max\{-1, v_1, \ldots, v_j\}$ are continuous, admissible and increasing, and so are the functions w_j^D. On D, the harmonic functions w_j^D take values in $[-1, 0]$ and increase, so by Corollary 2.3.3 they converge locally uniformly to a harmonic function H. On the one hand, each w_j^D is admissible, so $H \le f_K \le u_K$ on D. On the other hand, $H \ge \sup_j w_j$. The upper semicontinuous regularization of $\sup_j w_j = \max\{-1, \sup_j v_j\}$ is $\max\{-1, u_K\} = u_K$, so $\sup_j w_j = u_K$ almost everywhere by Theorem 3.4.2. Hence $H = u_K$ almost everywhere on D; since both functions are subharmonic and locally integrable, $H = u_K$ on D by Theorem 3.2.2. Therefore $f_K = u_K = H$ on D. As D is arbitrary, $f_K = u_K$ on $B \setminus K$, and u_K is harmonic there.

(2) Fix $0 < r < R$ with $\operatorname{Supp} \lambda \subset B(0, r)$. By the definition of G_B in Theorem 2.5.2 and the symmetry in Lemma 4.3.1,

$$u(x) = -\int G_B(y, x)\, \mathrm{d}\lambda(y) = E_d * \lambda(x) - \int H_B\big(E_d(\cdot - y)|_{\partial B}\big)(x)\, \mathrm{d}\lambda(y).$$

By Fubini's theorem the last integral is the Poisson integral of the continuous function $\zeta \mapsto E_d * \lambda(\zeta)$ on ∂B, so it is harmonic on B. Hence u is subharmonic with Riesz measure λ, by Proposition 3.2.1. By Lemma 4.3.1, for $y \in \operatorname{Supp} \lambda$ the formula given there for $G_B(x, y)$ defines a harmonic function of x on the annulus $r < |x| < R^2/r$, which vanishes for $|x| = R$. Since this function is jointly smooth in

(x, y), its derivatives in x are bounded uniformly in $y \in \operatorname{Supp}\lambda$ on compact subsets of the annulus. Differentiating under the integral sign, we see that $-\int G_B(x,y)\,\mathrm{d}\lambda(y)$ defines a harmonic function on this annulus that extends u across ∂B and vanishes on ∂B; we denote the extension again by u.

Choose $r < r' < R$ and $\chi \in C_c^\infty(B)$ with $\chi = 1$ on a neighborhood of $\overline{B(0,r')}$. Then $\lambda(B) = \int \chi\,\mathrm{d}\lambda = \int_B u\Delta\chi\,\mathrm{d}x$. The integrand vanishes outside the annulus $r' < |x| < R$, near whose closure u is smooth and harmonic. The divergence identity (2.3) on this annulus, applied to the field $u\nabla\chi - \chi\nabla u$, whose divergence there is $u\Delta\chi$, gives

$$\int_B u\Delta\chi\,\mathrm{d}x = \int_{|x|=r'} \partial_r u\,\mathrm{d}S,$$

because χ and its derivatives vanish near $|x| = R$, while $\chi = 1$ near $|x| = r'$; here ∂_r is the radial derivative. Applying the divergence identity on the same annulus to ∇u, whose divergence vanishes there, we see that this integral equals $\int_{\partial B} \partial_n u\,\mathrm{d}S$. This proves the flux identity.

For the comparison of masses, let $u' = -G_B\lambda'$ be a second function as above with $u \le u'$ on B. Then $u - u' \le 0$ on B and $u - u' = 0$ on ∂B, and since $u - u'$ is smooth near ∂B, its derivative at a point of ∂B along the outward normal is the limit of difference quotients $((u-u')(\zeta - sn) - 0)/(-s) \ge 0$ as $s \downarrow 0$. Hence $\partial_n u \ge \partial_n u'$ on ∂B, and integration gives $\lambda(B) \ge \lambda'(B)$.

For the converse, let v and λ be as stated, extend λ by zero to $\mathbb{R}^d$, and put $p = E_d * \lambda$. By Proposition 3.2.1, $\Delta(v - p) = 0$ on B in the sense of distributions, so by Theorem 2.1.4 the locally integrable function $v - p$ agrees almost everywhere on B with a harmonic function h. Then $p + h$ is a locally integrable subharmonic function on B that agrees with v almost everywhere, hence everywhere by Theorem 3.2.2. The function p is finite and continuous on the neighborhood $\mathbb{R}^d \setminus \operatorname{Supp}\lambda$ of ∂B: if $x \notin \operatorname{Supp}\lambda$, then $E_d(w - y) \to E_d(x - y)$ uniformly in $y \in \operatorname{Supp}\lambda$ as $w \to x$, because E_d is uniformly continuous on compact subsets of $\mathbb{R}^d \setminus \{0\}$. As v tends to zero at ∂B, the harmonic function $h = v - p$ extends continuously to $\overline{B}$ with boundary values $-p|_{\partial B}$, and the uniqueness in Theorem 2.3.1 gives $h = -H_B(p|_{\partial B})$. By the formula for $-G_B\lambda$ proved at the beginning of (2), $v = p - H_B(p|_{\partial B}) = -G_B\lambda$.

(3) By (1), u_K is harmonic on $B \setminus K$, so its Riesz measure λ_K, a positive Radon measure on B by Theorem 3.2.2, is supported on the compact set K; in particular it is finite. Moreover $-1 \le u_K \le 0$, so u_K is locally integrable, and u_K tends to zero at ∂B by (1). The converse in (2) therefore gives $u_K = -G_B\lambda_K$. The flux formula for $C_B(K)$ is now (2) with $\lambda = \lambda_K$. □

Theorem 4.3.3 *For a compact set $K \subset \mathbb{R}^d$, the following conditions are equivalent:*

(1) *K is not polar;*
(2) *K carries a nonzero positive measure with finite energy;*
(3) *K carries a nonzero positive measure μ such that $E_d * \mu$ is finite and continuous on $\mathbb{R}^d$.*

If $K \subset B$, they are also equivalent to $C_B(K) > 0$. Moreover, every finite positive measure with compact support and finite energy gives zero mass to polar sets: every polar set is contained in a Borel set of measure zero.

Proof Conditions (1)–(3) do not involve B, so for their equivalence we may enlarge R and assume that $K \subset B$. We first show, for every ball $B = B(0, R)$ containing K, that K is polar if and only if $C_B(K) = 0$. Since $u_K = -G_B\lambda_K$ and $C_B(K) = \lambda_K(K)$ (Proposition 4.3.2), we have $C_B(K) = 0$ if and only if $u_K = 0$. Suppose $u_K = 0$. Then $f_K = 0$ almost everywhere by Theorem 3.4.2. Choose a point $x \in B$ with $f_K(x) = 0$ and admissible functions v_j with $v_j(x) > -2^{-j}$. The partial sums $\sum_{j\le N} v_j$ are subharmonic and decrease with N, and their values at x stay above -1. By Theorem 3.1.6, the limit $\sum_j v_j$ is subharmonic on B and not identically $-\infty$. It equals $-\infty$ on K, since every v_j is at most -1 there. Thus K is polar.

Conversely, if K is polar, Proposition 4.2.3 gives a function $E_d * \nu$, not identically $-\infty$, that equals $-\infty$ on K; subtracting a constant, we may assume it is negative on B. All its positive multiples are then admissible, so $f_K = 0$ wherever $E_d * \nu > -\infty$, that is, almost everywhere. Hence $u_K = 0$ almost everywhere, and therefore everywhere by Theorem 3.2.2.

Suppose that K is not polar, so that $\lambda_K \neq 0$. By parts (2) and (3) of Proposition 4.3.2, $u_K = E_d * \lambda_K + h$ with the harmonic function $h = -H_B\big((E_d * \lambda_K)|_{\partial B}\big)$ on B. Since $u_K \ge -1$, it follows that $E_d * \lambda_K \ge -1 - h > -\infty$ on K. Theorem 4.2.2 writes λ_K as a sum of measures with finite continuous potentials, and one of them, μ, is nonzero; so (3) holds. A measure μ as in (3) has finite energy, because its potential U^μ is bounded on the compact support of μ; so (3) implies (2).

Next, let μ be a finite positive measure with compact support such that $E_d * \mu$ is finite and continuous, and let ν be any finite positive measure with compact support. We show that μ gives measure zero to the Borel set $\{E_d * \nu = -\infty\}$; by Proposition 4.2.3, every bounded polar set lies in a set of this form. By the reciprocity law (4.1), $\int (E_d * \nu)\, d\mu = \int (E_d * \mu)\, d\nu$. The right side is finite because $E_d * \mu$ is continuous, hence so is the left side, and therefore $E_d * \nu > -\infty$ μ-almost everywhere.

Now let μ be a finite positive measure with compact support and finite energy. Then $U^\mu < \infty$ μ-almost everywhere, that is, $E_d * \mu$ is finite μ-almost everywhere, and Theorem 4.2.2 writes μ as a sum of measures with finite continuous potentials. If A is a bounded polar set and ν is chosen for A as in Proposition 4.2.3, the previous paragraph shows that each of these measures, and hence μ, gives measure zero to the Borel set $\{E_d * \nu = -\infty\}$, which contains A. An arbitrary polar set is a countable union of bounded polar sets, so it is contained in a countable union of Borel sets of μ-measure zero; this proves the final assertion. In particular, a nonzero measure with finite energy cannot be carried by a polar compact set, so (2) implies (1). □

Proposition 4.3.4 *The relative capacity of compact subsets of B has the following properties. It is increasing: $C_B(K) \le C_B(L)$ for compact sets $K \subset L \subset B$. It is continuous along decreasing sequences: if compact sets $K_j \subset B$ decrease to K, then $C_B(K_j) \to C_B(K)$. It is strongly subadditive: for compact sets $K, L \subset B$,*

$$C_B(K \cup L) + C_B(K \cap L) \le C_B(K) + C_B(L).$$

Proof If $K \subset L$, every function admissible for L is admissible for K, so $u_L \le u_K$. By Proposition 4.3.2, $u_L = -G_B\lambda_L$ and $u_K = -G_B\lambda_K$, and part (2) of that proposition gives $C_B(L) \ge C_B(K)$.

Let K_j be compact sets decreasing to K. Their extremal functions increase with j and are bounded above by f_K. Let v be admissible for K and $\varepsilon > 0$. The function $v + \varepsilon(|x|^2 - R^2)$ is less than -1 on K, so by upper semicontinuity it is less than -1 on an open neighborhood U of K. The compact sets $K_j \setminus U$ decrease and have empty intersection, so one of them is empty; thus $K_j \subset U$ for all large j. Hence $v + \varepsilon(|x|^2 - R^2)$ is admissible for K_j, and therefore at most f_{K_j}, when j is large. Letting $\varepsilon \downarrow 0$ and taking the supremum over v, we obtain $f_{K_j} \to f_K$ pointwise. Since $f_{K_j} = u_{K_j}$ and $f_K = u_K$ almost everywhere (Theorem 3.4.2) and all these functions take values in $[-1, 0]$, bounded convergence gives $u_{K_j} \to u_K$ in $L^1(B)$. Choose $\chi \in C_c^\infty(B)$ equal to one near K_1. Since all the Riesz measures are supported in K_1,

$$C_B(K_j) = \int \chi \, \mathrm{d}\lambda_{K_j} = \int u_{K_j} \Delta\chi \, \mathrm{d}x \longrightarrow \int u_K \Delta\chi \, \mathrm{d}x = C_B(K).$$

For strong subadditivity we first prove that $F = f_{K\cup L} + f_{K\cap L}$ satisfies $F \ge f_K + f_L$. At a point of K we have $f_{K\cup L} = -1 = f_K$, and $f_{K\cap L} \ge f_L$ because $K \cap L \subset L$; the case of a point of L is symmetric. So the inequality holds on $K \cup L$. By Proposition 4.3.2, F is harmonic on the open set $U = B \setminus (K \cup L)$, lower semicontinuous on B, and tends to zero at ∂B. Let v and w be continuous admissible functions for K and L. At a boundary point ζ of U in B, which lies in $K \cup L$, we have

$$\begin{aligned} \overline{\lim_{U \ni x \to \zeta}} (v + w)(x) = v(\zeta) + w(\zeta) &\le f_K(\zeta) + f_L(\zeta) \\ &\le F(\zeta) \le \overline{\lim_{U \ni x \to \zeta}} F(x), \end{aligned}$$

and at points of ∂B the upper limits of $v + w$ are nonpositive while F tends to zero. By Proposition 3.1.4, applied to the subharmonic function $v + w - F$ on the bounded open set U, we get $v + w \le F$ on U. Taking the supremum over v and w, which by Proposition 4.3.2 gives f_K and f_L, proves the inequality on U, hence on B.

Since each f equals the corresponding u almost everywhere, we obtain $u_{K\cup L} + u_{K\cap L} \ge u_K + u_L$ almost everywhere; taking ball averages and letting the radius tend to zero, (3.3) gives this inequality everywhere. The two sides are the negatives of the Green potentials of $\lambda_{K\cup L} + \lambda_{K\cap L}$ and of $\lambda_K + \lambda_L$, so part (2) of Proposition 4.3.2 gives the asserted inequality of masses. □

We extend the relative capacity to arbitrary sets with compact closure in B. For an open set $O \Subset B$ and a set $A \Subset B$ put

$$\begin{aligned} C_B(O) &= \sup\{C_B(K) : K \subset O \text{ compact}\}, \\ C_B^*(A) &= \inf\{C_B(O) : O \text{ open}, \ A \subset O \Subset B\}. \end{aligned}$$

We call $C_B^*(A)$ the outer relative capacity of A. For an open set $O \Subset B$ we have $C_B^*(O) = C_B(O)$: the set O itself is allowed in the infimum, and every open $O' \supset O$ satisfies $C_B(O') \geq C_B(O)$, because compact subsets of O are compact subsets of O'. The outer capacity C_B^* is increasing by definition, and $C_B^*(\varnothing) = 0$.

Theorem 4.3.5 *The outer relative capacity has the following properties.*

(1) $C_B^*(K) = C_B(K)$ *for every compact set* $K \subset B$.
(2) *It is increasing and countably subadditive:* $C_B^*(\bigcup_j A_j) \leq \sum_j C_B^*(A_j)$ *whenever* $\bigcup_j A_j \Subset B$.
(3) *If* $A_1 \subset A_2 \subset \cdots$ *and* $A = \bigcup_j A_j \Subset B$, *then* $C_B^*(A_j) \to C_B^*(A)$.
(4) *If* $C_B^*(A) = 0$, *then* A *is polar.*

Proof (1) Let $K \subset B$ be compact; we may assume that $K \neq \varnothing$, since $C_B^*(\varnothing) = 0 = C_B(\varnothing)$. For $0 < \delta < \operatorname{dist}(K, \mathbb{R}^d \setminus B)$, the compact sets $K_j = \{x : \operatorname{dist}(x, K) \leq \delta/j\}$ are neighborhoods of K in B that decrease to K; let O_j be their interiors. Every open set containing K has capacity at least $C_B(K)$, so by Proposition 4.3.4

$$C_B(K) \leq C_B^*(K) \leq C_B(O_j) \leq C_B(K_j) \longrightarrow C_B(K).$$

Strong subadditivity. Parts (2) and (3) rest on two inequalities for C_B^*, the first of which is $C_B^*(A \cup A') + C_B^*(A \cap A') \leq C_B^*(A) + C_B^*(A')$ for $A, A' \Subset B$. Consider first open sets $O, O' \Subset B$, and let $K \subset O \cup O'$ and $L \subset O \cap O'$ be compact. Cover K by finitely many closed balls, each contained in O or in O', and let K' and K'' be the intersections of K with the union of those contained in O and in O', respectively. Then $K' \cup L \subset O$ and $K'' \cup L \subset O'$ are compact, their union contains K and their intersection contains L. By monotonicity and Proposition 4.3.4,

$$\begin{aligned} C_B(K) + C_B(L) &\leq C_B\big((K' \cup L) \cup (K'' \cup L)\big) + C_B\big((K' \cup L) \cap (K'' \cup L)\big) \\ &\leq C_B(K' \cup L) + C_B(K'' \cup L) \leq C_B(O) + C_B(O'). \end{aligned}$$

Taking the supremum over K and L proves the inequality for open sets. For arbitrary $A, A' \Subset B$, apply it to open sets $O \supset A$ and $O' \supset A'$ with compact closures in B, and take the infimum.

An excess inequality. Let $A_j \subset E_j$ be sets with $E_1 \cup E_2 \Subset B$. All the capacities involved are finite. Strong subadditivity applied to E_1 and $A_1 \cup E_2$, and to $A_1 \cup A_2$ and E_2, gives

$$\begin{aligned} C_B^*(E_1 \cup E_2) + C_B^*(A_1) &\leq C_B^*(E_1) + C_B^*(A_1 \cup E_2), \\ C_B^*(A_1 \cup E_2) + C_B^*(A_2) &\leq C_B^*(A_1 \cup A_2) + C_B^*(E_2), \end{aligned}$$

where we have used monotonicity to bound the capacities of the intersections from below. Adding these inequalities and cancelling $C_B^*(A_1 \cup E_2)$ gives the case $N = 2$ of

$$C_B^*\left(\bigcup_{j=1}^N E_j\right) - C_B^*\left(\bigcup_{j=1}^N A_j\right) \leq \sum_{j=1}^N \big(C_B^*(E_j) - C_B^*(A_j)\big), \tag{4.5}$$

valid whenever $A_j \subset E_j$ and $\bigcup_j E_j \Subset B$. Induction on N, applied to the pairs $\bigcup_{j<N} A_j \subset \bigcup_{j<N} E_j$ and $A_N \subset E_N$, proves (4.5) in general.

(2) and (3). If O_j are open sets increasing to an open set $O \Subset B$, every compact subset of O lies in some O_j, so $C_B(O_j) \to C_B(O)$. Now let A_j be arbitrary sets with $A = \bigcup_j A_j \Subset B$, fix an open set W with $\overline{A} \subset W \Subset B$, and let $\varepsilon > 0$. Intersecting almost optimal open sets in the definition of $C_B^*(A_j)$ with W, we find open sets E_j with $A_j \subset E_j \subset W$ and $C_B(E_j) < C_B^*(A_j) + \varepsilon 2^{-j}$. By (4.5) and the result for open sets,

$$C_B^*\left(\bigcup_j A_j\right) \le C_B\left(\bigcup_j E_j\right) = \lim_{N\to\infty} C_B\left(\bigcup_{j=1}^N E_j\right) \le \lim_{N\to\infty} C_B^*\left(\bigcup_{j=1}^N A_j\right) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. If the A_j increase, the right side is $\lim_N C_B^*(A_N) \le C_B^*(A)$, which proves (3). In general, strong subadditivity gives finite subadditivity, and the displayed inequality extends it to countable subadditivity, which proves (2).

(4) Suppose $C_B^*(A) = 0$. We construct subharmonic functions v_j on B that equal -1 on A and whose Riesz measures have mass less than 2^{-j}. Their sum will be a nontrivial subharmonic function equal to $-\infty$ on A. Fix an open set W with $\overline{A} \subset W \Subset B$, and choose open sets O_j with $A \subset O_j \subset W$ and $C_B(O_j) < 2^{-j}$. Exhaust O_j by compact sets $L_{j,m}$, increasing in m, whose interiors cover O_j. Choose $c > 0$ with $q(x) = c(|x|^2 - R^2) \le -1$ on $\overline{W}$; then q is admissible for every $L_{j,m}$, so $u_{L_{j,m}} \ge q$. The functions $u_{L_{j,m}}$ decrease as m increases, and their limit v_j is subharmonic by Theorem 3.1.6, lies between q and 0, and hence tends to zero at ∂B. Moreover, $v_j = -1$ on O_j, since $f_{L_{j,m}}$ and hence $u_{L_{j,m}}$ equal -1 on the interior of $L_{j,m}$.

The functions $u_{L_{j,m}}$ converge to v_j in $L^1(B)$ by bounded convergence, so their Riesz measures, which are supported in $\overline{W}$, converge in the sense of distributions on B, and hence vaguely by Proposition 1.5.1, to the Riesz measure λ_j of v_j. By Proposition 1.5.2, applied to the open set $B \setminus \overline{W}$, the measure λ_j is supported in $\overline{W}$. Testing against a function in $C_c^\infty(B)$ equal to one near $\overline{W}$ gives $\lambda_j(B) = \lim_m C_B(L_{j,m}) = C_B(O_j)$, since every compact subset of O_j lies in some $L_{j,m}$. The function v_j is bounded, hence locally integrable, tends to zero at ∂B, and has a Riesz measure with compact support in B; by the converse in part (2) of Proposition 4.3.2, $v_j = -G_B\lambda_j$.

The measure $\nu = \sum_j \lambda_j$ is finite, since $\lambda_j(B) = C_B(O_j) < 2^{-j}$, and it is supported in $\overline{W}$. By Tonelli's theorem for the nonnegative kernel G_B, we have $\sum_j v_j = -G_B\nu$. By part (2) of Proposition 4.3.2, $-G_B\nu$ is subharmonic on B with Riesz measure ν, so it is locally integrable and not identically $-\infty$. Since every v_j equals -1 on A, the sum $\sum_j v_j$ equals $-\infty$ on A. Hence A is polar. □

Example 4.3.6 Let $0 < r < R$ and $K = \overline{B(0,r)}$. The function $x \mapsto G_B(x,0)$ is radial and decreasing in $|x|$ by Lemma 4.3.1. We claim that $u_K = f_K = w$, where

$$w = \max\left\{-1, -\frac{G_B(\cdot,0)}{G_B(re_1,0)}\right\}.$$

The function w equals -1 on K and $-G_B(x,0)/G_B(re_1,0)$ on the annulus $r<|x|<R$. It is subharmonic, because $-G_B(\cdot,0)/G_B(re_1,0)$ is a positive multiple of E_d plus a constant on B, and it is nonpositive; hence it is admissible and $w \le f_K$. Conversely, let v be admissible. On the annulus, w is harmonic and continuous up to the boundary, with boundary values -1 on $|x|=r$ and 0 on $|x|=R$. Since $v \le -1$ on the inner sphere and $v \le 0$ on B, upper semicontinuity shows that the boundary upper limits of v are at most these values. The comparison principle (Theorem 3.1.3) on the annulus gives $v \le w$ there, and $v \le -1 = w$ on K. Hence $f_K = w$, and since w is continuous, $u_K = w$. Near ∂B, u_K is $-1/G_B(re_1,0)$ times $G_B(\cdot,0)$, whose outward normal derivative on ∂B is $-(s_{d-1}R^{d-1})^{-1}$. Therefore the flux formula of Proposition 4.3.2 gives

$$C_B(\overline{B(0,r)}) = \frac{1}{G_B(re_1,0)} = \begin{cases} 2\pi/\log(R/r), & d=2, \\ (d-2)s_{d-1}/(r^{2-d}-R^{2-d}), & d\ge 3. \end{cases}$$

These are relative capacities. To compute the kernel capacities of the same ball, let σ_r be the normalized surface measure on the sphere $|x|=r$ and $p = E_d * \sigma_r$. If $|x|>r$, then $p(x)$ is the spherical mean over $\partial B(0,r)$ of the function $y \mapsto E_d(x-y)$, which is harmonic on a neighborhood of $\overline{B(0,r)}$; hence $p(x) = E_d(x)$ by the mean-value property (Theorem 2.1.2). Surface measure is invariant under orthogonal transformations (Section 2.1), which act transitively on spheres centered at the origin, so p is radial. On the ball $|x|<r$, the function p is harmonic: by Proposition 3.2.1 and Theorem 2.1.4 it agrees almost everywhere with a harmonic function there, hence everywhere by Theorem 3.2.2. Since p is radial, for $0<|x|<r$ the value $p(x)$ is the spherical mean of p over $\partial B(0,|x|)$, which equals $p(0)$ by the mean-value property; moreover, $p(0) = E_d(re_1)$ because E_d is radial. Finally, p is constant on the sphere $|x|=r$, which is the support of σ_r, so part (1) of Theorem 4.2.2 shows that p is continuous at the points of the sphere. Hence $p = E_d(re_1)$ on K.

Consequently the Newtonian potential of σ_r equals $\Gamma_d(re_1)$ on K and $\Gamma_d(x)$ for $|x|>r$, and the logarithmic potential with a scale $D>2r$ equals $(2\pi)^{-1}\log(D/r)$ on K and $(2\pi)^{-1}\log(D/|x|)$ for $|x|>r$. In particular σ_r has finite energy, and its potential U^{σ_r} has a constant value V on K. Since σ_r is a nonzero measure with finite energy carried by the sphere $|x|=r$, Theorem 4.3.3 shows that this sphere is not polar, although it has Lebesgue measure zero: for $0<\varepsilon<r$ it is contained in $B(0,r+\varepsilon)\setminus B(0,r-\varepsilon)$, whose measure $s_{d-1}\big((r+\varepsilon)^d-(r-\varepsilon)^d\big)/d$ tends to zero as $\varepsilon \downarrow 0$.

Let ν be a probability measure on K with finite energy. Then $\nu-\sigma_r$ has total mass zero, and its positive and negative parts are dominated by ν and σ_r, so they have finite energy because the kernel is nonnegative on $K\times K$. Moreover, $\int U^{\sigma_r}\,\mathrm{d}(\nu-\sigma_r) = V-V = 0$. Hence Proposition 4.1.2 gives

$$I(\nu) = I(\sigma_r) + 2\int U^{\sigma_r}\,\mathrm{d}(\nu-\sigma_r) + I(\nu-\sigma_r) \ge I(\sigma_r).$$

Thus σ_r is the equilibrium measure of K, the Newtonian capacity of K is $(d-2)s_{d-1}r^{d-2}$ if $d \geq 3$, and its logarithmic capacity is r if $d = 2$. A probability measure on a compact set is also one on every larger compact set, so the kernel capacities are increasing; when $d = 2$, we compute both logarithmic capacities with a scale exceeding the diameter of the larger set, as Theorem 4.1.3 allows. Letting $r \downarrow 0$, we recover the fact that a point has kernel capacity zero.

4.4 Capacitability and exceptional sets

Theorem 4.3.3 characterizes the polar compact sets. To extend this characterization to Borel sets, in particular to the exceptional sets of limits of subharmonic functions, we approximate sets from inside by compact sets. A set $A \Subset B$ is capacitable if

$$C_B^*(A) = \sup\{C_B(K) : K \subset A \text{ compact}\}.$$

Open sets are capacitable by definition, and compact sets by Theorem 4.3.5. So is every set of the form $A = \bigcup_j K_j \Subset B$ with compact sets K_j: the compact sets $K_1 \cup \cdots \cup K_j$ increase to A, so part (3) of Theorem 4.3.5 gives $C_B^*(A) = \lim_j C_B(K_1 \cup \cdots \cup K_j)$. The exceptional sets $\{\overline{\lim}_j v_j < v\}$ that we meet below are Borel sets, but a Borel set need not be a countable union of closed sets.

For instance, let $0 < s < R$ and consider the Borel set $A = \{te_1 : 0 \leq t \leq s,\ t \notin \mathbb{Q}\}$, a segment with countably many points removed. Suppose that $A = \bigcup_n F_n$ with closed sets F_n. The closed sets $\{t \in [0,s] : te_1 \in F_n\}$ and the one-point sets $\{q\}$ with $q \in \mathbb{Q} \cap [0,s]$ form a sequence $D_1, D_2, \ldots$ of closed sets covering $[0,s]$, none of which contains an interval of positive length, since the former consist of irrational numbers. Put $J_0 = [0,s]$, and choose closed intervals $J_k \subset J_{k-1}$ of positive length with $J_k \cap D_k = \varnothing$: the interior of J_{k-1} contains a point outside D_k, and since D_k is closed, a short closed interval around this point lies in J_{k-1} and misses D_k. The nested compact intervals J_k have a common point, which lies in no D_k; this is a contradiction. Choquet's theorem shows that, nevertheless, every Borel set is capacitable.

Theorem 4.4.1 *Let $L \subset B$ be compact. Every Borel set $A \subset L$ is capacitable. More generally, let X be a compact Hausdorff space, $f : X \to L$ a continuous map, and $\widetilde{A} = \bigcap_{i\geq 1}\bigcup_{k\geq 1} K_{ik}$ with compact sets $K_{ik} \subset X$; then $f(\widetilde{A})$ is capacitable. In particular, every Borel set $A \subset L$ contains an increasing sequence of compact sets whose capacities increase to $C_B^*(A)$.*

Proof *Capacitability of $f(\widetilde{A})$.* Replacing K_{ik} by $K_{i1} \cup \cdots \cup K_{ik}$, we may assume that K_{ik} increases with k. Put $A = f(\widetilde{A})$ and fix $\varepsilon > 0$. We choose integers $h_1, h_2, \ldots$ inductively. Put $\widetilde{A}_0 = \widetilde{A}$, and suppose $\widetilde{A}_{p-1}$ has been defined. Since $\widetilde{A}_{p-1} \subset \bigcup_k K_{pk}$, the sets $\widetilde{A}_{p-1} \cap K_{ph}$ increase with h and have union $\widetilde{A}_{p-1}$, so their images under f increase to $f(\widetilde{A}_{p-1}) \subset L$. By part (3) of Theorem 4.3.5, we can choose h_p such that $\widetilde{A}_p = \widetilde{A}_{p-1} \cap K_{p,h_p}$ satisfies

$$C_B^*\big(f(\widetilde{A}_p)\big) > C_B^*\big(f(\widetilde{A}_{p-1})\big) - \varepsilon 2^{-p}.$$

Hence $C_B^*(f(\widetilde{A}_p)) > C_B^*(A) - \varepsilon$ for every p.

The sets $L_p = K_{1,h_1} \cap \cdots \cap K_{p,h_p}$ are compact, because compact subsets of the Hausdorff space X are closed; they decrease and contain $\widetilde{A}_p$, so the compact sets $f(L_p) \subset L$ decrease and, by part (1) of Theorem 4.3.5 and the monotonicity of C_B^*, satisfy $C_B(f(L_p)) = C_B^*(f(L_p)) \geq C_B^*(f(\widetilde{A}_p)) > C_B^*(A) - \varepsilon$. Moreover $\bigcap_p f(L_p) = f(\bigcap_p L_p)$: if $y \in f(L_p)$ for every p, the sets $f^{-1}(y) \cap L_p$ form a decreasing sequence of nonempty compact sets, since $f^{-1}(y)$ is closed, and hence have a common point. Since $\bigcap_p L_p = \bigcap_i K_{i,h_i} \subset \widetilde{A}$, the compact set $\bigcap_p f(L_p)$ is contained in A, and by Proposition 4.3.4 its capacity is $\lim_p C_B(f(L_p)) \geq C_B^*(A) - \varepsilon$. As ε is arbitrary, A is capacitable.

Borel sets. If $L = \varnothing$ there is nothing to prove, so let $L \neq \varnothing$. Call a subset of L analytic if it has the form $f(\widetilde{A})$ considered above; this is the notion of descriptive set theory and is unrelated to complex analytic sets. In such a representation we may assume that X is nonempty, since the empty set is the image of the empty subset of a one-point space under a constant map. We may also assume that $X = Y \times L$ for a nonempty compact Hausdorff space Y and that f is the projection onto L. Indeed, put $Y = X$. The map $g\colon X \to Y \times L$, $g(x) = (x, f(x))$, is continuous and injective, so $g(\widetilde{A}) = \bigcap_i \bigcup_k g(K_{ik})$ with compact sets $g(K_{ik})$, and the projection of $g(\widetilde{A})$ onto L is $f(\widetilde{A})$.

Analytic sets are closed under countable intersections and countable unions. To see this, let $A_j = \pi(\widetilde{A}_j)$, $j \geq 1$, be analytic, with $\widetilde{A}_j \subset Y_j \times L$ as above and π denoting projections onto L. The space $X = (\prod_j Y_j) \times L$ is compact by Lemma 1.6.2 and Hausdorff, as a product of Hausdorff spaces. Let $\widehat{A}_j \subset X$ be the inverse image of $\widetilde{A}_j$ under the continuous map $\pi_j\colon (y, z) \mapsto (y_j, z)$. If $\widetilde{A}_j = \bigcap_i \bigcup_k K_{ik}^{(j)}$ with compact sets $K_{ik}^{(j)} \subset Y_j \times L$, then $\widehat{A}_j = \bigcap_i \bigcup_k \pi_j^{-1}(K_{ik}^{(j)})$. Each $K_{ik}^{(j)}$ is closed, being a compact subset of a Hausdorff space, so $\pi_j^{-1}(K_{ik}^{(j)})$ is a closed and hence compact subset of X. Thus each $\widehat{A}_j$, and therefore also $\bigcap_j \widehat{A}_j$, is a countable intersection of countable unions of compact subsets of X.

A point $z \in L$ lies in $\bigcap_j A_j$ if and only if for every j there is $y_j \in Y_j$ with $(y_j, z) \in \widetilde{A}_j$, that is, if and only if $(y, z) \in \bigcap_j \widehat{A}_j$ for some $y = (y_j)_j \in \prod_j Y_j$. Hence the projection of $\bigcap_j \widehat{A}_j$ onto L is $\bigcap_j A_j$, and countable intersections of analytic sets are analytic. Since every Y_i is nonempty, the same reasoning shows that the projection of each $\widehat{A}_j$ onto L is A_j.

For unions, write $\widehat{A}_j = \bigcap_n \bigcup_k H_{jnk}$ with compact $H_{jnk} \subset X$, and consider, in the product of X with the one-point compactification $\{1, 2, \ldots\} \cup \{\infty\}$ of the positive integers (a compact Hausdorff space, homeomorphic to $\{1/j : j \geq 1\} \cup \{0\} \subset \mathbb{R}$), the set

$$\bigcap_n \bigcup_{j,k} (H_{jnk} \times \{j\}).$$

A point (x, j) belongs to it if and only if $j \neq \infty$ and $x \in \widehat{A}_j$, so its projection onto L is $\bigcup_j A_j$. Each $H_{jnk} \times \{j\}$ is compact, so countable unions of analytic sets are analytic.

Closed subsets of the compact metric space L are compact, hence analytic, and an open set $U \subset L$ is the union of the compact sets $\{z \in L : \operatorname{dist}(z, L \setminus U) \geq 1/n\}$, hence analytic. The family of sets $E \subset L$ such that both E and $L \setminus E$ are analytic is closed under complements, and under countable unions, because the complement of a countable union is the countable intersection of the complements. It is therefore a σ-algebra containing all open subsets of L, so every Borel subset of L is analytic and hence capacitable.

Increasing compact sets. Let $A \subset L$ be Borel. Since A is capacitable, there are compact sets $K_j \subset A$ with $C_B(K_j) \to C_B^*(A)$. The compact sets $K_1 \cup \cdots \cup K_j \subset A$ increase, and by monotonicity their capacities increase to $C_B^*(A)$. □

Corollary 4.4.2 *Let $A \Subset B$. Then A is polar if and only if $C_B^*(A) = 0$. A compact set has Newtonian or logarithmic capacity zero if and only if it is polar.*

Proof By Theorem 4.3.5, it remains to show that a polar set has outer capacity zero. By Proposition 4.2.3, A is contained in the pole set of a function $E_d * \nu$ with ν finite, positive and compactly supported. Let N be a compact neighborhood of $\overline{A}$ in B. The set $P = N \cap \{E_d * \nu = -\infty\}$ is Borel, being the intersection of N with the open sets $\{E_d * \nu < -n\}$, $n \geq 1$. Every compact subset of P is polar and therefore has capacity zero by Theorem 4.3.3. By Theorem 4.4.1, $C_B^*(P) = 0$, and monotonicity gives $C_B^*(A) = 0$. For the kernel capacities, a compact set has capacity zero if and only if it carries no probability measure of finite energy (Theorem 4.1.3), that is, no nonzero positive measure of finite energy; by Theorem 4.3.3 this happens if and only if it is polar. □

The main step towards the polarity of exceptional sets is the following lemma, which tests L^1_{loc} convergence of subharmonic functions against measures with continuous potentials.

Lemma 4.4.3 *Let $\Omega \subset \mathbb{R}^d$ be open, and let v_j and v be locally integrable subharmonic functions on Ω with $v_j \to v$ in $L^1_{\mathrm{loc}}(\Omega)$. If σ is a finite positive measure with compact support in Ω such that $E_d * \sigma$ is finite and continuous, then*

$$\int v_j \, \mathrm{d}\sigma \longrightarrow \int v \, \mathrm{d}\sigma > -\infty, \qquad \overline{\lim_j} \, v_j = v \quad \sigma\text{-almost everywhere}.$$

Proof Choose a cutoff $\chi \in C_c^\infty(\Omega)$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on an open neighborhood V of $\operatorname{Supp}\sigma$, and put $\nu_j = \chi \Delta v_j$ and $\nu = \chi \Delta v$. Since $\Delta v_j \to \Delta v$ distributionally, Proposition 1.5.1 shows that $\nu_j \to \nu$ vaguely, and the total masses $\nu_j(\Omega) = \int \chi \, \mathrm{d}\Delta v_j$ are bounded. By the Riesz decomposition (Theorem 3.2.3), $v_j = E_d * \nu_j + h_j$ and $v = E_d * \nu + h$ on V, with h_j and h harmonic.

We show that $h_j \to h$ uniformly on compact subsets of V. For a test function φ, Fubini's theorem gives $\int (E_d * \nu_j)\varphi \, \mathrm{d}x = \int (E_d * \varphi) \, \mathrm{d}\nu_j$, and $E_d * \varphi$ is continuous by Proposition 2.2.1; since the ν_j are supported in the fixed compact set $\operatorname{Supp}\chi$, it

follows that $E_d * \nu_j \to E_d * \nu$ distributionally, and hence $h_j \to h$ distributionally on V. The functions h_j are also bounded in L^1 on compact subsets of V. This holds for the ν_j, which converge in L^1_{loc}, and for the $E_d * \nu_j$: if $Q \subset V$ is compact and $s > 0$ is such that $Q - \operatorname{Supp}\chi \subset B(0,s)$, then Tonelli's theorem gives

$$\int_Q |E_d * \nu_j| \, dx \le \int \left(\int_Q |E_d(x-y)| \, dx \right) d\nu_j(y) \le \nu_j(\Omega) \int_{B(0,s)} |E_d| \, dz,$$

and the masses $\nu_j(\Omega)$ are bounded.

By the interior estimates (2.5), the derivatives of the h_j are therefore bounded on compact subsets of V, so the h_j are locally equicontinuous. Moreover, $h_j \to h$ pointwise on V. Indeed, if $x \in V$ and $\varepsilon > 0$ is less than the distance from x to $\mathbb{R}^d \setminus V$, then $h_j(x) = (h_j * \rho_\varepsilon)(x)$ and $h(x) = (h * \rho_\varepsilon)(x)$ by the last assertion of Theorem 2.1.2, and $(h_j * \rho_\varepsilon)(x) \to (h * \rho_\varepsilon)(x)$ by distributional convergence. To upgrade this to uniform convergence, let $Q \subset V$ be compact and $\theta > 0$. By equicontinuity, every point of Q has a neighborhood on which each h_j, and also the continuous function h, varies by less than θ. Finitely many of these neighborhoods, centered at points $x_1, \dots, x_N$, cover Q, and for all large j we have $|h_j(x_i) - h(x_i)| < \theta$ for every i; hence $|h_j - h| < 3\theta$ on Q.

Next, the reciprocity law (4.1), applied to σ and ν_j and to σ and ν, gives

$$\int (E_d * \nu_j) \, d\sigma = \int (E_d * \sigma) \, d\nu_j \longrightarrow \int (E_d * \sigma) \, d\nu = \int (E_d * \nu) \, d\sigma,$$

where the convergence holds because $E_d * \sigma$ is continuous and the ν_j converge vaguely with supports in $\operatorname{Supp}\chi$. All these integrals are finite, since $E_d * \sigma$ is bounded on $\operatorname{Supp}\chi$. Adding the integrals of h_j, which converge to $\int h \, d\sigma$, proves that $\int \nu_j \, d\sigma \to \int \nu \, d\sigma > -\infty$.

By Hartogs' lemma (Theorem 3.3.3), $\overline{\lim}_j \nu_j \le \nu$ everywhere, and the ν_j are bounded above on $\operatorname{Supp}\sigma$ by a common constant for large j. Reverse Fatou's lemma therefore gives

$$\int \overline{\lim_j} \nu_j \, d\sigma \ge \lim_j \int \nu_j \, d\sigma = \int \nu \, d\sigma.$$

Since $\nu - \overline{\lim}_j \nu_j \ge 0$ and $\int \nu \, d\sigma$ is finite, it follows that $\overline{\lim}_j \nu_j = \nu$ σ-almost everywhere. □

Theorem 4.4.4 *Let $\Omega \subset \mathbb{R}^d$ be open.*

(1) *If ν_j and ν are locally integrable subharmonic functions on Ω with $\nu_j \to \nu$ in $L^1_{\mathrm{loc}}(\Omega)$, then the set $\{\overline{\lim}_j \nu_j < \nu\}$ is polar.*
(2) *Let $\{u_t\}$ be a family of subharmonic functions on Ω with a common upper bound on every compact subset, and let $F = \sup_t u_t$. Then the set $\{F < F^*\}$, where the supremum differs from its upper semicontinuous regularization, is polar.*

Proof (1) Let $N \subset \Omega$ be compact, and take R so large that $N \subset B$. The set $E = \{x \in N : \overline{\lim}_j \nu_j(x) < \nu(x)\}$ is the union, over rational numbers $a < b$ and integers $J \ge 1$, of the Borel sets

$$N \cap \{v \geq b\} \cap \bigcap_{j \geq J} \{v_j \leq a\},$$

so it is Borel. Suppose some compact subset S of E were not polar. By Theorem 4.3.3, S would carry a nonzero positive measure σ with $E_d * \sigma$ finite and continuous, and Lemma 4.4.3 would give $\overline{\lim}_j v_j = v$ σ-almost everywhere on S, which is impossible. Hence every compact subset of E is polar and has capacity zero. By Theorem 4.4.1 and Corollary 4.4.2, E is polar. Since countably many compact sets N cover Ω, Proposition 4.2.3 completes the proof.

(2) If the family is empty, or if all its members are identically $-\infty$ on a component of Ω, then $F = F^* = -\infty$ there. Let Ω' be a component on which some member u_{t_0} is not identically $-\infty$; by Proposition 3.1.5, u_{t_0} is locally integrable on Ω'. By Lemma 3.4.1, applied on Ω', there are members $u_{t_1}, u_{t_2}, \ldots$ of the family with $(\sup_{k \geq 1} u_{t_k})^* = F^*$ on Ω'. Put $w_j = \max\{u_{t_0}, \ldots, u_{t_j}\}$ and $w = \sup_j w_j$. Since $\sup_{k \geq 1} u_{t_k} \leq w \leq F$, we have $w^* = F^*$ on Ω', and Theorem 3.4.2, applied to the family (w_j), shows that F^* is subharmonic on Ω' and that $w = F^*$ almost everywhere on Ω'.

We show next that $w_j \to F^*$ in $L^1_{\mathrm{loc}}(\Omega')$. As $u_{t_0} \leq w_j \leq F^*$, the functions w_j and F^* are locally integrable on Ω'. If $Q \subset \Omega'$ is compact and M is a common upper bound of the w_j on Q, then $0 \leq M - w_j \leq M - u_{t_0}$ on Q and $M - w_j$ decreases to $M - w$, so dominated convergence gives $w_j \to w = F^*$ in $L^1(Q)$. Since $\overline{\lim}_j w_j = w$, part (1) shows that $\{w < F^*\} \cap \Omega'$ is polar, and $\{F < F^*\} \cap \Omega'$ is contained in it because $w \leq F$. The components of Ω are pairwise disjoint nonempty open sets, hence of positive Lebesgue measure, so there are at most countably many of them (Lemma 1.1.5), and the union over them is polar by Proposition 4.2.3. □

Capacitability also gives Frostman's description of the potential of the equilibrium measure.

Proposition 4.4.5 *Let K be a compact set carrying a probability measure of finite energy, and let μ_K be its equilibrium measure (Theorem 4.1.3). When $d = 2$, U^{μ_K} and $V(K)$ stand for the potential and the minimal energy with respect to a fixed scale $D > \operatorname{diam} K$. Then $U^{\mu_K} \geq V(K)$ on K outside a polar set, $U^{\mu_K} = V(K)$ almost everywhere with respect to μ_K, and $U^{\mu_K} \leq V(K)$ on all of $\mathbb{R}^d$.*

Proof Take R so large that $K \subset B$. Let $\varepsilon > 0$, and consider the set $E_\varepsilon = \{x \in K : U^{\mu_K}(x) < V(K) - \varepsilon\}$, which is Borel since U^{μ_K} is lower semicontinuous. If $C^*_B(E_\varepsilon) > 0$, then by Theorem 4.4.1 E_ε contains a compact set of positive capacity, which is not polar by Theorem 4.3.3; that theorem provides a probability measure ν on this compact set with finite continuous potential, hence with finite energy. Then $I(\mu_K, \nu) = \int U^{\mu_K} \, \mathrm{d}\nu \leq V(K) - \varepsilon$, contradicting Theorem 4.1.3. Thus $C^*_B(E_\varepsilon) = 0$ for every $\varepsilon > 0$, and countable subadditivity (Theorem 4.3.5) and Corollary 4.4.2 show that $\{x \in K : U^{\mu_K}(x) < V(K)\}$ is polar. The finite-energy measure μ_K gives zero mass to this set (Theorem 4.3.3), so $U^{\mu_K} \geq V(K)$ μ_K-almost everywhere. Since $\int U^{\mu_K} \, \mathrm{d}\mu_K = V(K)$, it follows that $U^{\mu_K} = V(K)$ μ_K-almost everywhere.

Let $x \in \operatorname{Supp} \mu_K$. Every neighborhood of x has positive μ_K-measure and therefore contains points where $U^{\mu_K} = V(K)$; by lower semicontinuity, $U^{\mu_K}(x) \leq V(K)$.

To extend this bound to $\mathbb{R}^d$, decompose $\mu_K = \sum_j \mu_j$ by Theorem 4.2.2, which applies because $E_d * \mu_K$ is finite μ_K-almost everywhere, and let $\pi_N = \sum_{j\le N} \mu_j$. The potential U^{π_N} is continuous on $\mathbb{R}^d$ and, by differentiation under the integral sign, harmonic off the compact set $S_N = \operatorname{Supp} \pi_N \subset \operatorname{Supp} \mu_K$. Since the kernel is nonnegative on $K \times K$, we have $U^{\pi_N} \le U^{\mu_K} \le V(K)$ on S_N.

We claim that $U^{\pi_N} \le V(K)$ on $\mathbb{R}^d$. If $\pi_N = 0$, this holds because $V(K) \ge 0$, so let $\pi_N \ne 0$. As $|x| \to \infty$, the Newtonian potential $U^{\pi_N}(x)$ tends to $0 < V(K)$, and the logarithmic potential tends to $-\infty$. Applying the maximum principle (Proposition 2.1.3) to U^{π_N} on the bounded open sets $B(0,\rho) \setminus S_N$ with ρ large, we obtain the claim. Finally, fix $x \in \mathbb{R}^d$. The kernel $k(x,\cdot)$ is bounded below on K by a constant $-c_x$ (with $c_x = 0$ in the Newtonian case), and monotone convergence applied to $k(x,\cdot) + c_x$ gives $U^{\pi_N}(x) \to U^{\mu_K}(x)$. Hence $U^{\mu_K}(x) \le V(K)$. □

4.5 Dirichlet energy and quasi-continuity

The Green energy of a finite positive measure μ with compact support in B is the integral $\int G_B \mu \, d\mu$ of its Green potential against the measure itself. The mutual Green energy of two such measures μ and ν is $\iint G_B(x,y)\, d\mu(x)\, d\nu(y)$. Both lie in $[0,\infty]$ because $G_B > 0$. When the Green potentials are bounded, these energies are Dirichlet integrals of the potentials. This identity allows us to bound capacities by L^2 norms of gradients, and it leads to the quasi-continuity of subharmonic functions.

Proposition 4.5.1 *Let u and v be bounded subharmonic functions on B that tend to zero at ∂B and whose Riesz measures μ and ν are supported in compact subsets of B. Then their distributional gradients belong to $L^2(B)$, and*

$$\int_B |\nabla u|^2 \, dx = -\int_B u \, d\mu,$$
$$\int_B \nabla u \cdot \nabla v \, dx = -\int_B u \, d\nu = \iint G_B(x,y)\, d\mu(x)\, d\nu(y).$$

In particular, the mutual Green energy on the right satisfies the Cauchy–Schwarz inequality, and $\|\nabla u_K\|_2^2 \le C_B(K)$ for every compact set $K \subset B$. Moreover, u admits smooth regularizations with the same boundary behavior: there are a compact set $Q \subset B$ and functions u_ε on B, defined for all sufficiently small $\varepsilon > 0$, with the following properties. Each u_ε is the restriction of a smooth function on a neighborhood of $\overline{B}$, satisfies the hypotheses imposed on u with Riesz measure supported in Q, and equals u on $B \setminus Q$. As $\varepsilon \downarrow 0$, the functions u_ε decrease to u pointwise on B, and $\nabla u_\varepsilon \to \nabla u$ in $L^2(B)$.

Proof We extend u harmonically across ∂B, regularize the extension by convolution, and derive the identities from the divergence theorem for the regularizations. Since u is bounded, hence locally integrable, the converse in part (2) of Proposition 4.3.2 shows that $u = -G_B\mu$; likewise $v = -G_B\nu$. Choose $0 < r < R$ with $\operatorname{Supp}\mu \subset B(0,r)$, and

put $R' = R^2/r$. By part (2) of that proposition, u extends to a function $\widetilde{u}$ on $B(0,R')$ that is harmonic on the annulus $r < |x| < R'$ and vanishes on ∂B. This extension is subharmonic on $B(0,R')$, because subharmonicity is a local property and the open sets B and $\{r < |x| < R'\}$ cover $B(0,R')$. Moreover, the distribution $\Delta\widetilde{u}$ on $B(0,R')$ equals μ, extended by zero: the two agree on B, and both vanish on the annulus, so they agree on $B(0,R')$ by the last assertion of Lemma 1.4.2.

Fix r' with $r < r' < R$, put $Q = \overline{B(0,r')}$, and let $u_\varepsilon = \widetilde{u} * \rho_\varepsilon$ for $0 < \varepsilon < \min\{r'-r, R'-R\}$. Since $\widetilde{u}$ is harmonic on the annulus $r < |x| < R'$, the last assertion of Theorem 2.1.2 gives $u_\varepsilon = \widetilde{u}$ on $r+\varepsilon < |x| < R'-\varepsilon$. Hence u_ε is smooth on a neighborhood of $\overline{B}$, equals u on $B \setminus Q$, vanishes on ∂B, and satisfies $\Delta u_\varepsilon = (\Delta\widetilde{u}) * \rho_\varepsilon = \mu * \rho_\varepsilon \geq 0$ on B by Proposition 1.4.3(1). In particular, u_ε is a bounded subharmonic function on B (a smooth function with nonnegative Laplacian is subharmonic, Section 2.4), it tends to zero at ∂B, and its Riesz measure $(\mu * \rho_\varepsilon)\,\mathrm{d}x$ is supported in $\overline{B(0,r+\varepsilon)} \subset Q$. By (3.3), applied to $\widetilde{u}$, the functions u_ε decrease to u on B as $\varepsilon \downarrow 0$.

For small $\varepsilon, \delta > 0$, the divergence identity (2.2) on B, applied to the field $u_\varepsilon \nabla u_\delta$, together with Fubini's theorem and the symmetry of ρ_δ, gives

$$\int_B \nabla u_\varepsilon \cdot \nabla u_\delta \,\mathrm{d}x = -\int_B u_\varepsilon (\mu * \rho_\delta)\,\mathrm{d}x = -\int (\widetilde{u} * \rho_\varepsilon * \rho_\delta)\,\mathrm{d}\mu.$$

We let ε and δ tend to zero. On $\operatorname{Supp}\mu$, the functions $\varphi_{\varepsilon,\delta} = \widetilde{u} * \rho_\varepsilon * \rho_\delta$ are bounded by $\sup|\widetilde{u}|$ near $\overline{B}$, which is finite. By (3.3), $\widetilde{u} * \rho_\varepsilon$ decreases to $\widetilde{u}$ as $\varepsilon \downarrow 0$; convolving with $\rho_\delta \geq 0$ preserves order, so $\varphi_{\varepsilon,\delta}$ decreases as either radius decreases, and $\varphi_{\varepsilon,\delta} \geq \widetilde{u} * \rho_\delta \geq \widetilde{u}$. If $\varepsilon, \delta \leq \eta \leq \eta'$, then $\varphi_{\varepsilon,\delta} \leq \varphi_{\eta,\eta} \leq \varphi_{\eta',\eta}$, and $\varphi_{\eta',\eta} \to \widetilde{u} * \rho_{\eta'}$ as $\eta \downarrow 0$ because $\widetilde{u} * \rho_{\eta'}$ is continuous. Letting $\eta \downarrow 0$ and then $\eta' \downarrow 0$, we see that $\varphi_{\varepsilon,\delta} \to u$ pointwise on $\operatorname{Supp}\mu$ as $\max\{\varepsilon,\delta\} \to 0$.

By dominated convergence, $\int \nabla u_\varepsilon \cdot \nabla u_\delta \,\mathrm{d}x \to -\int u\,\mathrm{d}\mu$. Expanding $\|\nabla u_\varepsilon - \nabla u_\delta\|_2^2$, we find that the gradients ∇u_ε form a Cauchy family in $L^2(B)$. Their limit is the distributional gradient of u, as one sees by integrating against test functions and using $u_\varepsilon \to u$ in $L^1_{\mathrm{loc}}(B)$. Hence $\nabla u \in L^2(B)$ and $\|\nabla u\|_2^2 = \lim_\varepsilon \|\nabla u_\varepsilon\|_2^2 = -\int u\,\mathrm{d}\mu$.

The function $u+v$ satisfies the same hypotheses, with Riesz measure $\mu+\nu$. Applying the first identity to u, v and $u+v$ gives $2\int \nabla u \cdot \nabla v\,\mathrm{d}x = -\int u\,\mathrm{d}\nu - \int v\,\mathrm{d}\mu$. Since $u = -G_B\mu$ and $v = -G_B\nu$, Tonelli's theorem and the symmetry of G_B (Lemma 4.3.1) give

$$-\int u\,\mathrm{d}\nu = \iint G_B(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y) = -\int v\,\mathrm{d}\mu,$$

which proves the second line of the display. All these integrals are finite because u and v are bounded. The Cauchy–Schwarz inequality in $L^2(B)$ now gives the Cauchy–Schwarz inequality for the Green energy. Finally, u_K satisfies the hypotheses by Proposition 4.3.2, and since $u_K \geq -1$ and λ_K has total mass $C_B(K)$, we get $\|\nabla u_K\|_2^2 = -\int u_K\,\mathrm{d}\lambda_K \leq C_B(K)$. □

Theorem 4.5.2 *Let $\Omega \subset \mathbb{R}^d$ be an open set with $\overline{B} \subset \Omega$, and let $u \in \mathrm{SH}(\Omega)$ be not identically $-\infty$ on B. Then u is quasi-continuous: for every compact set $L \subset B$ and*

every $\eta > 0$ there is an open set $O \Subset B$ with $C_B(O) < \eta$ such that the restriction of u to $L \setminus O$ is finite and continuous.

Proof *The case of a bounded Green potential.* Suppose first that u is a bounded subharmonic function on B that tends to zero at ∂B and whose Riesz measure μ has compact support in B. Let $Q \subset B$ and u_ε be the compact set and the regularizations given by the last assertion of Proposition 4.5.1. These regularizations are continuous on $\overline{B}$, decrease to u, equal u on $B \setminus Q$, and $\nabla u_\varepsilon \to \nabla u$ in $L^2(B)$. For $\delta > 0$, the set $O_{\delta,\varepsilon} = \{u_\varepsilon - u > \delta\}$ is open, because $u_\varepsilon - u$ is lower semicontinuous, and it is contained in Q, so it has compact closure in B. Let $K \subset O_{\delta,\varepsilon}$ be compact. The function $u_\varepsilon - u$ is a difference of two functions satisfying the hypotheses of Proposition 4.5.1, and λ_K is supported on K, where $u_\varepsilon - u > \delta$. By the bilinear identity of that proposition, the Cauchy–Schwarz inequality and the bound $\|\nabla u_K\|_2^2 \le C_B(K)$,

$$\delta C_B(K) \le \int (u_\varepsilon - u)\,\mathrm{d}\lambda_K = -\int \nabla(u_\varepsilon - u)\cdot\nabla u_K\,\mathrm{d}x \le \|\nabla(u_\varepsilon - u)\|_2\, C_B(K)^{1/2}.$$

Hence $C_B(K) \le \delta^{-2}\|\nabla(u_\varepsilon - u)\|_2^2$, and taking the supremum over K gives $C_B(O_{\delta,\varepsilon}) \le \delta^{-2}\|\nabla(u_\varepsilon - u)\|_2^2$.

Choose $\delta_j \downarrow 0$ and then radii $\varepsilon_j \downarrow 0$ such that $C_B(O_{\delta_j,\varepsilon_j}) < \eta 2^{-j}$, and let $O = \bigcup_j O_{\delta_j,\varepsilon_j}$, which is open with compact closure in B. By countable subadditivity (Theorem 4.3.5), $C_B(O) < \eta$. Outside O we have $0 \le u_{\varepsilon_j} - u \le \delta_j$ for all j, so the continuous functions u_{ε_j} converge uniformly to u on $B \setminus O$. Hence u is finite and continuous on $B \setminus O$, in particular on $L \setminus O$.

The case of a function bounded near $\overline{B}$. Suppose next that u is bounded on a neighborhood of $\overline{B}$; subtracting a constant, we may assume that $u \le -1$ there. Choose $r < R$ with $L \subset B(0,r)$, and $M > 0$ with $-M < \inf_B u$. Let $q = M u_{\overline{B(0,r)}}$, where the extremal function $u_{\overline{B(0,r)}}$ was computed in Example 4.3.6. Then q is subharmonic on B, equals $-M$ on $\overline{B(0,r)}$, is harmonic on $r < |x| < R$, and tends to zero at ∂B. The function $w = \max\{u, q\}$ is subharmonic and bounded on B; it equals u on $B(0,r)$, and it equals q near ∂B, where $q > -1 \ge u$. Hence w tends to zero at ∂B, and its Riesz measure has compact support in B. The first case, applied to w, gives the required open set, since $w = u$ near L.

The general case. The function u is bounded above near $\overline{B}$, so the functions $u_N = \max\{u, -N\}$, $N \ge 1$, are subharmonic and bounded near $\overline{B}$. Fix an open set W with $L \subset W \Subset B$. By the second case, there are open sets $O_N \subset W$ with $C_B(O_N) < \eta 2^{-N-1}$ such that u_N is continuous on $L \setminus O_N$ (intersecting with W does not affect this). On $L \setminus \bigcup_N O_N$, every u_N is continuous, and e^{u_N} converges uniformly to e^u, because $0 \le \mathrm{e}^{u_N} - \mathrm{e}^u \le \mathrm{e}^{-N}$. Hence u is continuous there as a function with values in $[-\infty, \infty)$. Let Ω_0 be the connected component of Ω that contains the connected set $\overline{B}$. Since u is not identically $-\infty$ on Ω_0, the set $\{u = -\infty\} \cap \Omega_0$ is polar by the definition of polarity. By Corollary 4.4.2, $L \cap \{u = -\infty\}$ is contained in an open set $O' \Subset B$ with $C_B(O') < \eta/2$. The open set $O = O' \cup \bigcup_N O_N$ has compact closure in B and capacity less than η by countable subadditivity, and the restriction of u to $L \setminus O$ is finite and continuous. □

Chapter 5
Fine topology and balayage

Throughout this chapter $\Omega \subset \mathbb{R}^d$ is open and $d \geq 2$. Subharmonic functions are upper semicontinuous but in general not continuous. For instance, let e_1 be the first coordinate unit vector, and let $u = E_d * \nu$ be the fundamental potential of the probability measure $\nu = \sum_{j\geq 1} 2^{-j}\delta_{e_1/j}$, which is supported on the compact set $\{0\} \cup \{e_1/j : j \geq 1\}$. By Proposition 3.2.1, u is subharmonic on $\mathbb{R}^d$. At each atom we have $u(e_1/j) = -\infty$, because the integrand $E_d(e_1/j - \cdot)$ is bounded above on the support of ν and equals $-\infty$ at e_1/j. On the other hand, $u(0) = \sum_j 2^{-j}E_d(e_1/j)$ is finite, since $|E_d(e_1/j)|$ equals $(2\pi)^{-1}\log j$ if $d = 2$ and a constant multiple of j^{d-2} if $d \geq 3$. Hence u is not continuous at 0.

By Theorem 4.5.2, the discontinuities of subharmonic functions are nevertheless controlled by capacity: on a compact subset of a ball, a nontrivial subharmonic function becomes continuous once an open set of arbitrarily small capacity has been removed. In this chapter we study two further constructions, the fine topology and balayage, in which the polar sets of Chapters 3 and 4 appear as exceptional sets. Both have counterparts in the pluripotential theory of Part II.

The fine topology is the coarsest topology that contains the Euclidean topology and makes every subharmonic function continuous. In this topology, thinness of a set at a point becomes an ordinary neighborhood condition (Proposition 5.1.3). Polar sets are finely closed, and the fine topology induces the discrete topology on them (Proposition 5.1.5). Since there are uncountable compact polar sets, the fine topology is neither Lindelöf nor second countable (Example 5.1.7). Up to polar sets, however, it has the Lindelöf property: by Choquet's lemma and Theorem 4.4.4, every union of finely open sets agrees with a countable subunion outside a polar set (Theorem 5.4.1). Consequently finely open sets are measurable, and together with quasi-continuity this shows that they are open up to sets of small capacity (Proposition 5.4.2).

Balayage is based on reduction. For a nonnegative superharmonic function s and a set A, the reduction R_s^A of s on A is the infimum of all nonnegative superharmonic functions that majorize s on A: it keeps the values of s on A and removes as much of s elsewhere as superharmonicity permits. Its lower semicontinuous regularization $\widehat{R}_s^A$ is superharmonic and agrees with R_s^A outside a polar set. On a ball, when A is compact and s is the Green potential of a measure μ whose potential is bounded on

A or whose Green energy is finite, $\widehat{R}_s^A$ is the Green potential of a measure μ^A carried by A, the balayage of μ onto A. We characterize μ^A by an orthogonality relation for the Green energy, which is a positive definite quadratic form on signed measures of finite energy (Proposition 5.2.2 and Theorems 5.3.1 and 5.3.2), and express the balayage of a Dirac mass through harmonic measure (Example 5.3.3).

Finally, we compare the fine topology of $\mathbb{C}^n = \mathbb{R}^{2n}$ with the topology generated by the functions that are subharmonic on complex lines, which becomes the plurifine topology of Chapter 10; for $n \geq 2$ the two topologies differ (Proposition 5.5.1).

Historical remarks and references for this chapter are collected in Appendix B.5.

5.1 Fine neighborhoods and thinness

A subharmonic function u is upper semicontinuous, so its sublevel sets $\{u < a\}$ are open. It is continuous exactly when its superlevel sets $\{u > a\}$ are open as well. The fine topology is obtained by declaring all these superlevel sets open.

Definition 5.1.1 The *classical fine topology* on Ω is the coarsest topology on Ω that contains the Euclidean topology and makes every subharmonic function $u \in \mathrm{SH}(V)$, on every Euclidean open set $V \subset \Omega$, continuous as a map from V with the induced topology to $[-\infty, \infty)$. A set is *finely open* if it is open in this topology, and a *fine neighborhood* of a point is a neighborhood of it in this topology.

Since the sublevel sets of upper semicontinuous functions are Euclidean open, a subbasis of the fine topology consists of the Euclidean open sets and the sets $\{x \in V : u(x) > a\}$, where $V \subset \Omega$ is Euclidean open, $u \in \mathrm{SH}(V)$ and $a \in \mathbb{R}$. Restricting finitely many such functions to the intersection of their domains, we see that the sets

$$V \cap \{u_1 > a_1\} \cap \cdots \cap \{u_m > a_m\}, \qquad m \geq 0, \quad u_i \in \mathrm{SH}(V), \quad a_i \in \mathbb{R},$$

with $V \subset \Omega$ Euclidean open, form a basis of the fine topology. Superharmonic functions, being negatives of subharmonic functions, are finely continuous as well.

The fine topology depends on the pointwise values of subharmonic functions. These values are determined by the almost-everywhere class: by Theorem 3.2.2, a locally integrable subharmonic function is the canonical representative (3.3) of its distribution class. A function that merely agrees almost everywhere with a subharmonic function need not be finely continuous. For instance, the indicator function $\mathbf{1}_{\{x_0\}}$ of a point $x_0 \in \Omega$ agrees almost everywhere with the zero function, but $\{\mathbf{1}_{\{x_0\}} > 1/2\} = \{x_0\}$ is not finely open by Proposition 5.1.4 below.

In dimension one every subharmonic function is continuous, since on each component interval of its domain it is either identically $-\infty$ or finite, continuous and convex (Example 3.2.4). The fine topology is then the Euclidean topology, and this is why we assume $d \geq 2$.

Definition 5.1.2 A set $A \subset \Omega$ is *thin* at a point $x \in \Omega$ if either x does not belong to the Euclidean closure of $A \setminus \{x\}$, or there is a subharmonic function u on a Euclidean open neighborhood of x such that

$$u(x) > \overline{\lim_{\substack{y \to x \\ y \in A \setminus \{x\}}}} u(y).$$

Proposition 5.1.3 *Let $A \subset \Omega$ and $x \in \Omega$. Then A is thin at x if and only if some fine neighborhood of x does not meet $A \setminus \{x\}$. Moreover, every fine neighborhood of x contains a set $\{y \in V : u(y) > a\}$, where V is a Euclidean open neighborhood of x, $u \in \mathrm{SH}(V)$, $a \in \mathbb{R}$ and $u(x) > a$.*

Proof Suppose first that A is thin at x. If x does not belong to the Euclidean closure of $A \setminus \{x\}$, some Euclidean neighborhood of x does not meet $A \setminus \{x\}$. Otherwise let u be a subharmonic function on a Euclidean open neighborhood V of x as in Definition 5.1.2, and choose $a \in \mathbb{R}$ strictly between the upper limit in that definition and $u(x)$. By the definition of the upper limit, after shrinking V we have $u < a$ on $(A \setminus \{x\}) \cap V$. Hence $\{y \in V : u(y) > a\}$ is a fine neighborhood of x that does not meet $A \setminus \{x\}$.

Conversely, let N be a fine neighborhood of x. By the description of the basis above, N contains a set

$$N_0 = V \cap \bigcap_{i=1}^{m} \{u_i > a_i\}, \qquad u_i \in \mathrm{SH}(V), \quad u_i(x) > a_i,$$

where V is a Euclidean open neighborhood of x; adding the function 0 with the constant -1 if necessary, we may assume that $m \geq 1$. Each $u_i(x)$ is finite, since $u_i(x) > a_i$. Put $b_i = u_i(x) - a_i > 0$ and

$$v = \sum_{i=1}^{m} \frac{u_i - u_i(x)}{b_i} \in \mathrm{SH}(V).$$

Then $v(x) = 0$. We claim that

$$\overline{\lim_{\substack{y \to x \\ y \in V \setminus N_0}}} v(y) \leq -1.$$

There is nothing to prove if x is not an accumulation point of $V \setminus N_0$. Otherwise let $y_k \to x$ be a sequence in $V \setminus N_0$ along which $v(y_k)$ converges to the upper limit. For each k some index i satisfies $u_i(y_k) \leq a_i$, that is, $(u_i(y_k) - u_i(x))/b_i \leq -1$; passing to a subsequence, we may assume that this index i is the same for all k. For every other index j, upper semicontinuity of u_j at x gives $\overline{\lim}_k (u_j(y_k) - u_j(x))/b_j \leq 0$. Hence $\lim_k v(y_k) \leq -1$, which proves the claim.

If N does not meet $A \setminus \{x\}$, then $(A \setminus \{x\}) \cap V \subset V \setminus N_0$, and the claim shows that either x is not in the Euclidean closure of $A \setminus \{x\}$ or the upper limit of v along $A \setminus \{x\}$ is at most $-1 < v(x)$. In both cases A is thin at x. For the last assertion, the

claim provides a Euclidean open neighborhood $V' \subset V$ of x such that $v < -1/2$ on $V' \setminus N_0$. Thus $\{y \in V' : v(y) > -1/2\} \subset N_0 \subset N$, and $v(x) = 0 > -1/2$. □

Proposition 5.1.4 *Let N be a fine neighborhood of a point $x \in \Omega$. Then every Euclidean ball centered at x contains a Borel subset of N of positive Lebesgue measure. In particular, the fine topology has no isolated points.*

Proof By Proposition 5.1.3, N contains a set $\{y \in V : u(y) > a\}$ with $u \in \mathrm{SH}(V)$ and $u(x) > a$. It suffices to consider balls $B(x,r)$ with $\overline{B(x,r)} \subset V$. The set $E = \{y \in B(x,r) : u(y) > a\}$ is a Borel subset of N, because u is Borel measurable. Let M be a finite upper bound for u on $\overline{B(x,r)}$; then $M \geq u(x) > a$. Since $u \leq M$ on E and $u \leq a$ on $B(x,r) \setminus E$, the ball submean inequality of Theorem 3.1.3 gives

$$u(x) \leq \frac{1}{|B(x,r)|} \int_{B(x,r)} u(y)\,\mathrm{d}y \leq a + (M-a)\frac{|E|}{|B(x,r)|}.$$

Hence $|E| \geq |B(x,r)|\,(u(x)-a)/(M-a) > 0$. As a singleton has Lebesgue measure zero, it is not a fine neighborhood of its point. □

Proposition 5.1.5 *Let $A \subset \Omega$ be polar. For every $x \in \Omega$ there is a fine neighborhood N of x such that $N \cap A \subset \{x\}$. Consequently, A is finely closed, and the topology induced on A by the fine topology is discrete.*

Proof We construct a subharmonic function on $\mathbb{R}^d$ that is finite at x and equals $-\infty$ on $A \setminus \{x\}$ near x. Fix $x \in \Omega$ and $\rho > 0$ with $\overline{B(x,\rho)} \subset \Omega$. For $k \geq 0$ put $r_k = 2^{-k}\rho$, let $S_k = \{y : r_{k+1} \leq |y-x| \leq r_k\}$ be the closed annulus, and let $A_k = A \cap S_k$; these sets cover $A \cap \overline{B(x,\rho)} \setminus \{x\}$. Each A_k is polar and has compact closure, so Proposition 4.2.3 provides a finite positive measure ν'_k of compact support with $E_d * \nu'_k = -\infty$ on A_k. We restrict it to the open set $U_k = \{y : r_{k+2} < |y-x| < 2\rho\}$, which contains S_k, and put $\nu_k = \nu'_k|_{U_k}$. For $y \in A_k$ and $z \in \operatorname{Supp}\nu'_k \setminus U_k$ we have $|y-z| \geq \operatorname{dist}(S_k, \mathbb{R}^d \setminus U_k) > 0$, so $E_d(y-z)$ is bounded on this set of pairs and

$$E_d * \nu_k(y) = E_d * \nu'_k(y) - \int_{\mathbb{R}^d \setminus U_k} E_d(y-z)\,\mathrm{d}\nu'_k(z) = -\infty, \qquad y \in A_k.$$

Moreover, ν_k is supported in $\overline{U_k}$, which does not meet $B(x, r_{k+2})$, so $p_k = E_d * \nu_k$ is finite at x.

Choose $c_k > 0$ with

$$\sum_k c_k\big(\nu_k(\mathbb{R}^d) + |p_k(x)|\big) < \infty,$$

and put $\nu = \sum_k c_k \nu_k$, a finite measure supported in $\overline{B(x,2\rho)}$, and $p = E_d * \nu$. Let C be an upper bound for $E_d(y-z)$ when $|y-z| \leq 3\rho$. For $y \in \overline{B(x,\rho)}$, monotone convergence applied to the nonnegative functions $C - E_d(y-\cdot)$ gives

$$p(y) - C\nu(\mathbb{R}^d) = \sum_k c_k\big(p_k(y) - C\nu_k(\mathbb{R}^d)\big),$$

where every term on the right is nonpositive. Hence $p(x)$ is finite, and $p = -\infty$ on $A \cap \overline{B(x,\rho)} \setminus \{x\}$, because there one term equals $-\infty$. Since p is subharmonic on $\mathbb{R}^d$ (Proposition 3.2.1), the set

$$N = \{y \in B(x,\rho) : p(y) > p(x) - 1\}$$

is a fine neighborhood of x, and $N \cap A \subset \{x\}$. If $x \notin A$, this shows that $\Omega \setminus A$ is a fine neighborhood of each of its points, so A is finely closed. If $x \in A$, it shows that $\{x\}$ is relatively open in A for the fine topology. □

Singletons are polar since $d \geq 2$, and countable unions of polar sets are polar by Proposition 4.2.3; hence every countable set is polar. In particular, if $\Omega \neq \emptyset$, the set $\Omega \cap \mathbb{Q}^d$ is finely closed but Euclidean dense in Ω, so the fine topology is strictly finer than the Euclidean topology. As we show in Example 5.1.7 below, the fine topology is not even Lindelöf; this is why the countability theorem of Section 5.4 must allow a polar exceptional set. To produce uncountable compact polar sets in every dimension, we use the following observation: a subharmonic function of some of the variables is subharmonic in all of them.

Lemma 5.1.6 *Let $1 \leq k < d$, write $x = (z,y) \in \mathbb{R}^k \times \mathbb{R}^{d-k}$, let $V \subset \mathbb{R}^k$ be open, and let $g \in \mathrm{SH}(V)$. Then the function $G(z,y) = g(z)$ is subharmonic on $V \times \mathbb{R}^{d-k}$.*

Proof The function G is upper semicontinuous, because $\{G < a\} = \{g < a\} \times \mathbb{R}^{d-k}$ for every real a. Since subharmonicity is a local property, it suffices to show that G is subharmonic on $D \times \mathbb{R}^{d-k}$ for every open ball $D = B(z_0,s) \subset \mathbb{R}^k$ with $\overline{B(z_0,2s)} \subset V$. If g is identically $-\infty$ on $B(z_0,2s)$, then so is G on $B(z_0,2s) \times \mathbb{R}^{d-k}$, and there is nothing to prove. Otherwise g is locally integrable on the ball $B(z_0,2s)$ by Proposition 3.1.5.

For $0 < \varepsilon < s$, the k-dimensional mollification $g_\varepsilon = g * \rho_\varepsilon$ is smooth on D and satisfies $\Delta_z g_\varepsilon = (\Delta g) * \rho_\varepsilon \geq 0$ there, by part (1) of Theorem 3.2.2 and parts (1) and (3) of Proposition 1.4.3; moreover, g_ε decreases to g on D as $\varepsilon \downarrow 0$, by (3.3). The function $G_\varepsilon(z,y) = g_\varepsilon(z)$ is smooth on $D \times \mathbb{R}^{d-k}$, with $\Delta_x G_\varepsilon(z,y) = \Delta_z g_\varepsilon(z) \geq 0$. By (2.4), its sphere means in $\mathbb{R}^d$ increase with the radius, and they tend to its value at the center as the radius tends to zero, because G_ε is continuous. Hence G_ε satisfies the submean inequality and is subharmonic on $D \times \mathbb{R}^{d-k}$. Along the sequence $\varepsilon_j = s/(j+1)$, the functions G_{ε_j} decrease to G, which is therefore subharmonic on $D \times \mathbb{R}^{d-k}$ by Theorem 3.1.6. □

Example 5.1.7 Every Euclidean ball in $\mathbb{R}^d$ contains an uncountable compact polar set. Consequently, if Ω is nonempty, the fine topology on Ω is neither Lindelöf nor second countable.

Proof A similarity $x \mapsto a + tx$ with $t > 0$ preserves upper semicontinuity and sphere means, and hence maps polar sets to polar sets. It therefore suffices to find one uncountable compact polar subset of $\mathbb{R}^d$. For $d = 2$ we take the Cantor set K of Proposition 4.2.4. It is compact and polar, and it is uncountable because it carries an atom-free probability measure.

For $d > 2$, write $x = (z, y) \in \mathbb{R}^2 \times \mathbb{R}^{d-2}$. By Lemma 5.1.6, applied with $k = 2$ to the subharmonic function p of Proposition 4.2.4, the function $P(x) = p(z)$ is subharmonic on $\mathbb{R}^d$. It is finite when $z \notin K$ and equals $-\infty$ on $K \times \mathbb{R}^{d-2}$. Hence $K \times \{0\}$ is an uncountable compact polar subset of $\mathbb{R}^d$.

Now let $\Omega \neq \varnothing$, and let $K' \subset \Omega$ be an uncountable compact polar set. By Proposition 5.1.5, K' is finely closed, and its induced fine topology is discrete. A closed subspace F of a Lindelöf space X is Lindelöf: extend the members of an open cover of F to open subsets of X, add $X \setminus F$, and pass to a countable subcover. On the other hand, the cover of an uncountable discrete space by its singletons has no countable subcover. Hence the fine topology on Ω is not Lindelöf. Nor is it second countable, because a second countable space is Lindelöf: given an open cover and a countable basis, choose, for each basis element contained in some member of the cover, one such member; these countably many members cover the space. □

5.2 Green energy and compact reduction

This section prepares balayage: the Green energy is a positive definite quadratic form on signed measures of finite energy (Proposition 5.2.2), and a regularized reduction on a compact set is the Green potential of a measure carried by that set (Theorem 5.2.5).

Fix an open ball $B \subset \mathbb{R}^d$ and let G_B be its Green function (Theorem 2.5.2); it is nonnegative and symmetric (Lemma 4.3.1). The results of Chapter 4 about balls are stated for balls centered at the origin; we apply them to B after a translation. In this section and the next, unless signed measures are explicitly mentioned, a *measure* means a finite positive Borel measure with compact support in B. Recall from Section 4.3 that the Green potential of a measure μ is $G_B\mu(x) = \int G_B(x, y)\, \mathrm{d}\mu(y)$. We denote the mutual Green energy of measures μ and ν and the Green energy of μ (Section 4.5) by

$$I_B(\mu, \nu) = \iint G_B(x, y)\, \mathrm{d}\mu(x)\, \mathrm{d}\nu(y), \qquad I_B(\mu) = I_B(\mu, \mu).$$

By Lemma 4.3.1, $G_B(x, y) - \Gamma_d(x - y)$ is continuous on $B \times B$, so G_B is a continuous function on $B \times B$ with values in $[0, \infty]$, and these integrals are defined in $[0, \infty]$ by Tonelli's theorem. For finite signed measures we use the same formulas whenever G_B is integrable for the product of the total variations.

Let μ be supported in a compact set $K \subset B$. By the definition of G_B in Theorem 2.5.2 and by Theorem 2.3.1, the harmonic correction $G_B(x, y) - \Gamma_d(x - y) = -H_B(\Gamma_d(\cdot - x)|_{\partial B})(y)$ has absolute value at most $\max_{\zeta \in \partial B} |\Gamma_d(\zeta - x)|$, which is bounded for $x \in K$ because K has positive distance from ∂B. Hence, on $K \times K$, the Green function differs by a bounded function from the kernel k of Section 4.1 if $d \geq 3$, and from the kernel k_D with $D > \operatorname{diam} K$ if $d = 2$, because $k(x, y) = \Gamma_d(x - y)$ and $k_D(x, y) - \Gamma_2(x - y)$ is constant. Consequently μ has finite Green energy if and only if it has finite energy in the sense of Section 4.1. In particular, by Theorem 4.3.3,

a measure μ of finite Green energy does not charge polar sets: every polar set is contained in a Borel set of μ-measure zero.

Definition 5.2.1 A property holds *quasi-everywhere* on a set $E \subset \mathbb{R}^d$ if it holds at every point of E outside a polar set.

Thus a property that holds quasi-everywhere on a compact set $K \subset B$ holds μ-almost everywhere for every measure μ on K with finite Green energy. In Part II the exceptional sets are pluripolar, and *quasi-everywhere* will mean outside a pluripolar set (Definition 10.1.1).

Proposition 5.2.2 *Let μ and ν be measures with finite Green energy. Then the distributional gradients $\nabla G_B\mu$ and $\nabla G_B\nu$ belong to $L^2(B)$, the mutual energy $I_B(\mu,\nu)$ is finite, and*

$$I_B(\mu,\nu) = \int_B \nabla G_B\mu \cdot \nabla G_B\nu \, \mathrm{d}x.$$

The same identity holds for finite signed measures whose positive and negative parts have finite Green energy. On such signed measures, I_B is a positive definite quadratic form, that is, $I_B(\sigma) \geq 0$ with equality only for $\sigma = 0$, and it satisfies the Cauchy–Schwarz inequality $I_B(\sigma,\tau)^2 \leq I_B(\sigma)I_B(\tau)$. Moreover, every measure μ with finite Green energy is the sum of its restrictions $\mu^{(1)}, \mu^{(2)}, \ldots$ to countably many pairwise disjoint compact sets, and for every n the Green potential of $\mu^{(1)} + \cdots + \mu^{(n)}$ is finite, continuous and bounded on B.

Proof We first prove the gradient identity for the partial sums of a decomposition of μ and ν into pieces with bounded Green potentials, to which Proposition 4.5.1 applies, and then pass to the limit. Translating, we may assume that $B = B(0,R)$ for some $R > 0$, and we choose $r < R$ such that the supports of μ and ν lie in $B(0,r)$. By Lemma 4.3.1, the function $\kappa(x,y) = G_B(x,y) - \Gamma_d(x-y)$ extends to a continuous function on the compact set $\overline{B} \times \overline{B(0,r)}$; let C be a bound for $|\kappa|$ there. Since $\Gamma_d = -E_d$, every finite positive measure λ supported in $B(0,r)$ satisfies

$$G_B\lambda(x) = -E_d * \lambda(x) + \int \kappa(x,y)\, \mathrm{d}\lambda(y), \qquad x \in B,$$

and the last integral is a continuous function of $x \in \overline{B}$ bounded by $C\lambda(B)$.

Step 1: decomposition. Since $\int G_B\mu \, \mathrm{d}\mu = I_B(\mu) < \infty$, the Green potential $G_B\mu$ is finite μ-almost everywhere, and by the display so is $E_d * \mu$. By part (2) of Theorem 4.2.2, μ is therefore the sum of its restrictions to countably many disjoint compact sets, each of which has a finite and continuous fundamental potential. Let μ_n be the sum of the first n of these restrictions. Since $E_d * \mu_n$ is finite and continuous, the display shows that $G_B\mu_n$ is finite, continuous and bounded on B; this proves the last assertion of the proposition. By part (2) of Proposition 4.3.2, $-G_B\mu_n$ is a subharmonic function on B with Riesz measure μ_n, which extends harmonically across ∂B and vanishes on ∂B; in particular, it tends to zero at ∂B. Thus Proposition 4.5.1 applies to the functions $-G_B\mu_n$, $n \geq 1$. Hence $\nabla G_B\mu_n \in L^2(B)$ and

$$\int_B \nabla G_B\mu_n \cdot \nabla G_B\mu_m \,\mathrm{d}x = I_B(\mu_n, \mu_m).$$

Step 2: the gradient of $G_B\mu$. For $m > n$, expanding the square with the last display gives

$$\|\nabla G_B\mu_m - \nabla G_B\mu_n\|^2_{L^2(B)} = I_B(\mu_m - \mu_n) \le I_B(\mu - \mu_n),$$

where the inequality holds because $0 \le \mu_m - \mu_n \le \mu - \mu_n$ and $G_B \ge 0$. The measure $\mu - \mu_n$ is the restriction of μ to a decreasing sequence of sets whose intersection is μ-null. Since G_B is nonnegative and integrable for $\mu \otimes \mu$, dominated convergence gives $I_B(\mu - \mu_n) \to 0$. Thus $(\nabla G_B\mu_n)$ is a Cauchy sequence in $L^2(B)$; let g be its limit.

On the other hand, $G_B\mu_n \uparrow G_B\mu$ pointwise by monotone convergence, and $G_B\mu \in L^1(B)$. Indeed, for $y \in \operatorname{Supp}\mu$, the bound $|\kappa| \le C$ and the local integrability of Γ_d (Proposition 2.2.1) give

$$\int_B G_B(x,y)\,\mathrm{d}x \le \int_B \big(|\Gamma_d(x-y)| + C\big)\,\mathrm{d}x \le \int_{B(0,2R)} |\Gamma_d(z)|\,\mathrm{d}z + C|B| < \infty,$$

and integrating against μ, by Tonelli's theorem, shows that $\int_B G_B\mu\,\mathrm{d}x$ is at most $\mu(B)$ times the right side. Hence $G_B\mu_n \to G_B\mu$ in $L^1(B)$ by monotone convergence. For a smooth compactly supported vector field φ on B we obtain

$$\int_B g\cdot\varphi\,\mathrm{d}x = \lim_n \int_B \nabla G_B\mu_n\cdot\varphi\,\mathrm{d}x = -\lim_n \int_B G_B\mu_n \operatorname{div}\varphi\,\mathrm{d}x = -\int_B G_B\mu \operatorname{div}\varphi\,\mathrm{d}x,$$

so $g = \nabla G_B\mu$. Since $\|\nabla G_B\mu_n\|_2^2 = I_B(\mu_n) \uparrow I_B(\mu)$ by monotone convergence, we get $\|\nabla G_B\mu\|_2^2 = I_B(\mu)$.

Step 3: the mutual energy. Construct the partial sums ν_n of ν in the same way. By Proposition 4.5.1 and the Cauchy–Schwarz inequality in $L^2(B)$,

$$I_B(\mu_n,\nu_n) = \int_B \nabla G_B\mu_n\cdot\nabla G_B\nu_n\,\mathrm{d}x \le I_B(\mu_n)^{1/2} I_B(\nu_n)^{1/2} \le I_B(\mu)^{1/2} I_B(\nu)^{1/2}.$$

As $n \to \infty$, the left side increases to $I_B(\mu,\nu)$ by monotone convergence, so the mutual energy is finite, while the integral converges to $\int_B \nabla G_B\mu\cdot\nabla G_B\nu\,\mathrm{d}x$ by convergence in $L^2(B)$. This proves the identity for positive measures. For signed measures, we expand both sides bilinearly in the positive and negative parts.

Step 4: positive definiteness. Let σ be a signed measure as in the statement, and put $G_B\sigma = G_B\sigma^+ - G_B\sigma^-$, a function in $L^1(B)$ by Step 2. Then $I_B(\sigma) = \|\nabla G_B\sigma\|_2^2 \ge 0$. Suppose that $I_B(\sigma) = 0$, so that $\nabla G_B\sigma = 0$ as a distribution on B. Since B is connected, Corollary 1.4.4 shows that the distribution defined by $G_B\sigma$ is a constant. On the other hand, $-\Delta G_B\sigma = \sigma$ in the sense of distributions, by part (2) of Proposition 4.3.2. Hence σ^+ and σ^-, which are Radon measures by Proposition 1.1.2, define the same distribution, so $\sigma^+ = \sigma^-$ by part (4) of Proposition 1.4.3, that is, $\sigma = 0$. Finally, the Cauchy–Schwarz inequality for I_B follows from the gradient identity and the Cauchy–Schwarz inequality in $L^2(B)$. □

Recall from Section 4.3 the relative extremal function u_K of a compact set $K \subset B$, its Riesz measure $\lambda_K = \Delta u_K$, the relative capacity $C_B(K) = \lambda_K(K)$, and the outer capacity C_B^*.

Proposition 5.2.3 *Let $K \subset B$ be compact. Then*

$$C_B(K) = \left(\inf_{\substack{\tau \geq 0,\ \operatorname{Supp} \tau \subset K \\ \tau(K)=1}} I_B(\tau) \right)^{-1},$$

with the conventions that the infimum of an empty family is $+\infty$ and that $1/(+\infty) = 0$. If $C_B(K) > 0$, then $\lambda_K / C_B(K)$ is the unique probability measure on K minimizing I_B. Moreover, if $D \subset B$ are open balls and $A \Subset D$, then $C_B^*(A) \leq C_D^*(A)$.

Proof By part (3) of Proposition 4.3.2, the function $w_K = -u_K$ is the Green potential $G_B \lambda_K$, and $0 \leq w_K \leq 1$ because $-1 \leq u_K \leq 0$. Since u_K is the upper semicontinuous regularization of the envelope f_K, which equals -1 on K, part (2) of Theorem 4.4.4 shows that $w_K = 1$ quasi-everywhere on K. In particular $I_B(\lambda_K) = \int w_K \, \mathrm{d}\lambda_K \leq \lambda_K(K)$, so λ_K has finite Green energy. As measures of finite Green energy do not charge polar sets, we obtain, with $m = C_B(K)$,

$$I_B(\lambda_K) = \int_K w_K \, \mathrm{d}\lambda_K = m, \qquad I_B(\lambda_K, \tau) = \int_K w_K \, \mathrm{d}\tau = 1$$

for every probability measure τ on K with finite Green energy; in the second identity we used the symmetry of G_B.

Suppose that $m > 0$, and let τ be a probability measure on K. If $I_B(\tau) < \infty$, the Cauchy–Schwarz inequality of Proposition 5.2.2 gives $1 = I_B(\lambda_K, \tau)^2 \leq m I_B(\tau)$; thus $I_B(\tau) \geq 1/m$ in all cases. Equality holds for $\tau = \lambda_K / m$, whose energy is $m/m^2 = 1/m$. Hence the infimum equals $1/m$, which proves the formula. If $\tau_1 \neq \tau_2$ were two minimizers, their midpoint would again be a probability measure on K, and by positive definiteness

$$I_B\Big(\frac{\tau_1 + \tau_2}{2}\Big) = \frac{I_B(\tau_1) + I_B(\tau_2)}{2} - \frac{I_B(\tau_1 - \tau_2)}{4} < \frac{1}{m},$$

a contradiction. Suppose next that $m = 0$. Then K is polar by Theorem 4.3.3, so it carries no probability measure of finite energy, and the infimum is $+\infty$ (if $K = \varnothing$, the family is empty).

For the last assertion, recall that balls are regular domains (Theorem 2.4.6). The monotonicity of Green functions in the domain (Theorem 2.5.2) gives $G_D \leq G_B$ on $D \times D$. Hence $I_B(\tau) \geq I_D(\tau)$ for every measure τ supported on a compact set $K \subset D$, and the variational formula, applied in B and in D, gives $C_B(K) \leq C_D(K)$. Taking suprema over compact subsets, we obtain $C_B(O) \leq C_D(O)$ for every open $O \Subset D$. Every such O containing A is also allowed in the infimum defining $C_B^*(A)$, so $C_B^*(A) \leq C_D^*(A)$. □

We now turn to reduction. Let U be an open set, s a nonnegative superharmonic function on U, and $A \subset U$ an arbitrary subset. The *reduction* of s on A, also called the réduite, and the *regularized reduction* are

$$R_s^A = \inf\{h : h \geq 0 \text{ is superharmonic on } U,\ h \geq s \text{ on } A\}, \qquad \widehat{R}_s^A = (R_s^A)_*.$$

Here $F_* = -(-F)^*$, with $(-F)^*$ the upper semicontinuous regularization of Section 3.4, is the lower semicontinuous regularization of a function F that is locally bounded below; explicitly, $F_*(x) = \sup_{r>0} \inf_{B(x,r)\cap U} F$. We call the functions h in this infimum *admissible* for R_s^A; the inequality $h \geq s$ is required at every point of A. If $U = B$, $s = 1$ and $A \Subset B$, the admissible functions are exactly the negatives of the functions admissible for A in the sense of Section 4.3, so that $R_1^A = -f_A$ and $\widehat{R}_1^A = -u_A$.

Since s itself is admissible, $0 \leq R_s^A \leq s$, and $R_s^A = s$ on all of A. The negatives of the admissible functions form a family of subharmonic functions bounded above by zero, whose supremum is $-R_s^A$. Since $(-R_s^A)^* = -\widehat{R}_s^A$, Theorem 3.4.2 shows that $\widehat{R}_s^A$ is superharmonic, and part (2) of Theorem 4.4.4 shows that $R_s^A = \widehat{R}_s^A$ outside a polar set. Thus $\widehat{R}_s^A = s$ quasi-everywhere on A, though in general not everywhere on A: if U is bounded, $s = 1$ and $A \subset U$ is a nonempty polar set, then $\widehat{R}_s^A = 0$ by Proposition 5.2.4 below.

The ball-average formula (3.3) holds for every subharmonic function, integrable or not. Applied to $-w$, it shows that every superharmonic function w on an open set satisfies

$$w(x) = \lim_{r\downarrow 0} \frac{1}{|B(x,r)|} \int_{B(x,r)} w(y)\,\mathrm{d}y$$

at every point x of that set. Consequently, two superharmonic functions satisfying an inequality almost everywhere satisfy it everywhere.

Proposition 5.2.4 *Let s and t be nonnegative superharmonic functions on an open set U, let $A \subset A' \subset U$, and let $c \geq 0$. Then*

$$R_s^A \leq R_s^{A'}, \qquad R_{cs}^A = cR_s^A, \qquad R_{s+t}^A \leq R_s^A + R_t^A,$$

and $R_s^A \leq R_t^A$ if $s \leq t$. The same relations hold for the regularized reductions; in particular,

$$\widehat{R}_{s+t}^A \leq \widehat{R}_s^A + \widehat{R}_t^A.$$

If U is bounded and A is polar, then there is a locally integrable nonnegative superharmonic function p on U with $p = +\infty$ on A, and $\widehat{R}_s^A = 0$.

Proof Enlarging A or s shrinks the class of admissible functions, which proves the two monotonicity assertions. For $c > 0$, multiplication by c is a bijection between the admissible functions for s and those for cs; for $c = 0$, both sides vanish (with $0 \cdot \infty = 0$), since the zero function is admissible. If h_1 and h_2 are admissible for s and t respectively, then $h_1 + h_2$ is admissible for $s+t$, so $R_{s+t}^A \leq R_s^A + R_t^A$. Monotonicity and homogeneity are preserved by lower semicontinuous regularization. For subadditivity,

recall that each reduction agrees with its regularization outside a polar set, hence almost everywhere. Therefore $\widehat{R}^A_{s+t} \le \widehat{R}^A_s + \widehat{R}^A_t$ almost everywhere. Both sides are superharmonic, so the inequality holds everywhere by the ball-average formula above.

Now let U be bounded and A polar. Then A has compact closure, so Proposition 4.2.3 provides a finite positive measure ν of compact support with $E_d * \nu = -\infty$ on A. The kernel E_d is bounded above on the bounded set of differences $x - y$ with $x \in U$ and $y \in \operatorname{Supp}\nu$, so adding a constant to $-E_d * \nu$ gives a nonnegative superharmonic function p on U with $p = +\infty$ on A; by Proposition 3.2.1, p is locally integrable. For every $\varepsilon > 0$, the function εp is admissible, even at points of A where $s = +\infty$. Hence $R^A_s \le \varepsilon p$ for every $\varepsilon > 0$, and so $R^A_s = 0$ on the set $\{p < \infty\}$. This set has full Lebesgue measure, because p is locally integrable, and is therefore dense in U. Consequently $\widehat{R}^A_s = 0$. □

Theorem 5.2.5 *Let $A \subset B$ be compact, and let s be a nonnegative superharmonic function on B which is bounded on A. Then $r = \widehat{R}^A_s$ is bounded, harmonic on $B \setminus A$, and tends to zero at ∂B. Moreover, there is a finite positive measure ν supported on A, with finite Green energy, such that*

$$r = G_B\nu \quad \text{on } B, \qquad r = s \quad \text{quasi-everywhere on } A.$$

Proof We show that r is harmonic off A by replacing admissible functions with their Poisson integrals on balls in $B \setminus A$ (Steps 1–3), and then identify r with the Green potential of its Riesz measure (Step 4).

Step 1: capped admissible functions. Choose a closed ball $L \subset B$ whose interior contains A, and a constant $M \ge 0$ with $s \le M$ on A. The relative extremal function u_L takes values in $[-1, 0]$ and tends to zero at ∂B by part (1) of Proposition 4.3.2; it equals -1 on $\operatorname{Int} L$, because it is the upper semicontinuous regularization of the envelope f_L, which equals -1 on L. Hence $h_0 = -Mu_L$ is a nonnegative superharmonic function on B, bounded by M, which majorizes s on A and tends to zero at ∂B. If h is admissible, so is $\min\{h, h_0\}$. Therefore R^A_s is also the infimum of the admissible functions h with $h \le h_0$; we call these *capped*. In particular $0 \le r \le R^A_s \le h_0 \le M$, so r is bounded and tends to zero at ∂B.

Step 2: Poisson modification. Let D be an open ball with $\overline{D} \subset B \setminus A$, and let h be a capped admissible function. Its restriction to ∂D is bounded and lower semicontinuous, so by Lemma 3.1.2, applied to $-h$, it is the increasing limit of continuous functions g_k on ∂D. By monotone convergence, the Poisson integrals $H_D g_k$ of Theorem 2.3.1 increase to

$$H(y) = \int_{\partial D} P_D(y, \zeta) h(\zeta)\, \mathrm{d}S(\zeta), \qquad y \in D.$$

Since $0 \le h \le M$ and the Poisson kernel has integral one, $0 \le H \le M$, and H is harmonic by part (2) of Corollary 2.3.3. For $\zeta \in \partial D$, lower semicontinuity gives $\varliminf_{y \to \zeta} h(y) \ge h(\zeta) \ge g_k(\zeta)$, so the comparison principle for the superharmonic function h (Theorem 3.1.3) gives $H_D g_k \le h$ on D; hence $H \le h$ on D. On the other hand, $H \ge H_D g_k$, and $H_D g_k$ extends continuously to $\overline{D}$ with boundary values g_k; letting $k \to \infty$, we get

$$\varliminf_{D \ni y \to \zeta} H(y) \geq h(\zeta), \qquad \zeta \in \partial D.$$

Define $h^D = H$ on D and $h^D = h$ on $B \setminus D$. The last inequality and the lower semicontinuity of h show that h^D is lower semicontinuous, and $0 \leq h^D \leq h$.

We check that h^D is superharmonic, using the converse part of Theorem 3.1.3. Let E be a ball with $\overline{E} \subset B$, and let q be harmonic on E and continuous on $\overline{E}$ with $\varliminf_{E \ni y \to \zeta} h^D(y) \geq q(\zeta)$ for $\zeta \in \partial E$. Since $h \geq h^D$, the function h satisfies the same boundary inequality, so $h \geq q$ on E. In particular $h^D = h \geq q$ on $E \setminus D$. On the open set $E \cap D$ consider the harmonic function $H - q$. At boundary points of $E \cap D$ lying on ∂E, its lower limits are nonnegative by the hypothesis on q. At boundary points $\zeta \in E \cap \partial D$, they are at least $h(\zeta) - q(\zeta) \geq 0$ by the inequality displayed above. The minimum principle (Proposition 2.1.3) gives $H \geq q$ on $E \cap D$. Hence $h^D \geq q$ on E, and h^D is superharmonic. Since $D \cap A = \varnothing$, the function h^D is again a capped admissible function. Finally, h^D depends monotonically on h, by the integral formula for H.

Step 3: harmonicity off A. By Choquet's lemma (Lemma 3.4.1), applied to the negatives of the capped admissible functions, there is a sequence of capped admissible functions whose infimum has the same lower semicontinuous regularization r as R_s^A. By Theorem 3.4.2, applied to the negatives of this sequence, the infimum equals r almost everywhere. Replacing the functions of the sequence by their successive minima, we obtain a decreasing sequence (h_j) of capped admissible functions whose limit equals r almost everywhere.

Now fix a ball D as in Step 2. On D, the modified functions h_j^D are harmonic, decrease, and lie between 0 and M; by part (3) of Corollary 2.3.3, they converge on D to a harmonic function Φ. Each h_j^D is admissible, so $h_j^D \geq R_s^A \geq r$; moreover $h_j^D \leq h_j$. Thus $r \leq \Phi \leq \lim_j h_j = r$ almost everywhere on D. The bounded superharmonic functions r and Φ therefore agree almost everywhere, hence everywhere on D by Theorem 3.2.2. Since such balls cover $B \setminus A$, the function r is harmonic on $B \setminus A$.

Step 4: Green representation. The function $-r$ is bounded and subharmonic on B, so $\nu = -\Delta r$ is a positive measure by part (1) of Theorem 3.2.2. As r is harmonic on $B \setminus A$, the measure ν is supported on the compact set A and is therefore finite. The function $-r$ is bounded, hence locally integrable, its Riesz measure ν has compact support in B, and it tends to zero at ∂B; so the converse in part (2) of Proposition 4.3.2 gives $-r = -G_B\nu$, that is, $r = G_B\nu$ on B. The reduction R_s^A equals s on A, and $r = R_s^A$ outside a polar set, so $r = s$ quasi-everywhere on A. Finally $I_B(\nu) = \int r \, d\nu \leq M\nu(A) < \infty$. □

5.3 Sweeping measures and harmonic measure

We keep the ball B and the conventions of Section 5.2. Applied to the Green potential of a measure, Theorem 5.2.5 sweeps the measure onto a compact set without changing its potential there, apart from a polar set.

The measure obtained in this way is characterized by an energy identity, and the uniqueness part of this characterization rests on the following consequence of Proposition 5.2.2. Let ν_1 and ν_2 be measures with finite Green energy such that $I_B(\nu_1, \tau) = I_B(\nu_2, \tau)$ for $\tau = \nu_1$ and for $\tau = \nu_2$. Then $\nu_1 = \nu_2$. Indeed, the mutual energies $I_B(\nu_i, \nu_k)$ are finite by Proposition 5.2.2, so the hypothesis gives $I_B(\nu_1 - \nu_2) = I_B(\nu_1 - \nu_2, \nu_1) - I_B(\nu_1 - \nu_2, \nu_2) = 0$. The positive and negative parts of $\nu_1 - \nu_2$, being dominated by ν_1 and ν_2 respectively, have finite Green energy because $G_B \geq 0$, and $\nu_1 = \nu_2$ by positive definiteness.

Theorem 5.3.1 *Let $A \subset B$ be compact, and let μ be a measure such that $G_B\mu$ is bounded on A. Then there is a unique positive measure μ^A supported on A whose Green potential is $\widehat{R}^A_{G_B\mu}$. It has finite Green energy and satisfies*

$$G_B\mu^A \leq G_B\mu, \qquad G_B\mu^A = G_B\mu \text{ quasi-everywhere on } A, \qquad \mu^A(A) \leq \mu(B).$$

For every positive measure τ supported on A with finite Green energy,

$$I_B(\mu, \tau) = I_B(\mu^A, \tau). \tag{5.1}$$

This identity characterizes μ^A among the positive measures on A with finite Green energy. Consequently, sweeping is additive and positively homogeneous: if μ_1 and μ_2 satisfy the boundedness hypothesis, then $(\mu_1 + \mu_2)^A = \mu_1^A + \mu_2^A$ and $(c\mu_1)^A = c\mu_1^A$ for $c \geq 0$.

Proof The function $G_B\mu$ is nonnegative, superharmonic by part (2) of Proposition 4.3.2, and bounded on A. Applying Theorem 5.2.5 to $s = G_B\mu$, we see that $\widehat{R}^A_{G_B\mu}$ is the Green potential of a positive measure μ^A on A with finite Green energy, and that $G_B\mu^A = G_B\mu$ quasi-everywhere on A. Since $G_B\mu$ is itself admissible, $G_B\mu^A = \widehat{R}^A_{G_B\mu} \leq G_B\mu$. The measure μ^A is unique because $-\Delta G_B\nu = \nu$ for every measure ν, by part (2) of Proposition 4.3.2, so that a measure is determined by its Green potential. Since $-G_B\mu \leq -G_B\mu^A$ on B, the comparison of boundary fluxes in part (2) of Proposition 4.3.2, applied with $\lambda = \mu$ and $\lambda' = \mu^A$, gives the mass bound $\mu^A(A) \leq \mu(B)$.

Let τ be a positive measure on A with finite Green energy. It does not charge polar sets, so we may integrate the equality $G_B\mu = G_B\mu^A$, valid quasi-everywhere on A, against τ. Both integrals are finite, since $0 \leq G_B\mu^A \leq G_B\mu$ and $G_B\mu$ is bounded on A. By Tonelli's theorem and the symmetry of G_B, the two integrals are $I_B(\mu, \tau)$ and $I_B(\mu^A, \tau)$, which proves (5.1).

Suppose that ν_1 and ν_2 are positive measures on A with finite Green energy which both satisfy (5.1) in place of μ^A. Then $I_B(\nu_1, \tau) = I_B(\mu, \tau) = I_B(\nu_2, \tau)$ for $\tau = \nu_1$ and $\tau = \nu_2$, so $\nu_1 = \nu_2$ by the uniqueness criterion at the beginning of this section. If μ_1 and μ_2 satisfy the boundedness hypothesis, so does $\mu_1 + \mu_2$, and $\mu_1^A + \mu_2^A$ is a positive measure on A with finite Green energy satisfying (5.1) for the source $\mu_1 + \mu_2$. By the characterization, $(\mu_1 + \mu_2)^A = \mu_1^A + \mu_2^A$. The same argument gives $(c\mu_1)^A = c\mu_1^A$. □

The measure μ^A is called the *balayage*, or *swept measure*, of μ onto A, relative to B. We now replace the boundedness of $G_B\mu$ on A by the assumption that μ has finite Green energy.

Theorem 5.3.2 *Let $A \subset B$ be compact, and let μ be a measure with finite Green energy. Then the regularized reduction $\widehat{R}^A_{G_B\mu}$ is the Green potential of a unique positive measure μ^A supported on A, and μ^A has finite Green energy. All the conclusions of Theorem 5.3.1 remain valid, additivity and homogeneity now holding for sources of finite Green energy. Moreover,*

$$I_B(\mu, \mu^A) = I_B(\mu^A), \qquad I_B(\mu - \mu^A) = I_B(\mu) - I_B(\mu^A) \geq 0.$$

If μ is supported on A, then $\mu^A = \mu$.

Proof We sweep the partial sums of a decomposition of μ into pieces with bounded Green potentials, show that the swept measures increase to a measure ν satisfying (5.1), and then identify $G_B\nu$ with the regularized reduction. By the last assertion of Proposition 5.2.2, μ is the sum of its restrictions to countably many pairwise disjoint compact sets, and the partial sums μ_n of this series, which satisfy $\mu_n \uparrow \mu$, have bounded Green potentials. Hence Theorem 5.3.1 applies to the measures μ_n and to the differences $\mu_m - \mu_n$ with $m > n$, whose Green potentials are bounded by $G_B\mu_m$.

By additivity, $\mu_m^A = \mu_n^A + (\mu_m - \mu_n)^A \geq \mu_n^A$. Hence the measures μ_n^A increase setwise, and $\nu(E) = \lim_n \mu_n^A(E)$ defines a measure ν on A: if a Borel set E is the disjoint union of Borel sets E_k, the monotone convergence theorem for series gives $\nu(E) = \lim_n \sum_k \mu_n^A(E_k) = \sum_k \nu(E_k)$. Moreover $\nu(A) = \lim_n \mu_n^A(A) \leq \mu(B)$. For every nonnegative Borel function f we have

$$\int f \, \mathrm{d}\mu_n^A \uparrow \int f \, \mathrm{d}\nu \qquad (n \to \infty). \tag{5.2}$$

This is clear for simple functions, and in general it follows by approximating f from below by an increasing sequence of simple functions f_k and interchanging the suprema over k and n of $\int f_k \, \mathrm{d}\mu_n^A$.

Testing (5.1) for μ_n against $\tau = \mu_n^A$ and using the Cauchy–Schwarz inequality, we get

$$I_B(\mu_n^A) = I_B(\mu_n, \mu_n^A) \leq I_B(\mu_n)^{1/2} I_B(\mu_n^A)^{1/2},$$

so that $I_B(\mu_n^A) \leq I_B(\mu_n) \leq I_B(\mu)$. We claim that $I_B(\mu_n^A) \uparrow I_B(\nu)$. The sequence increases and is bounded by $I_B(\nu)$, because $\mu_n^A \leq \mu_{n+1}^A \leq \nu$ and $G_B \geq 0$. Conversely, applying (5.2) to $f = G_B(x, \cdot)$, we get $G_B\mu_m^A \uparrow G_B\nu$ pointwise as $m \to \infty$. For $m \leq n$ we have $I_B(\mu_n^A) \geq \int G_B\mu_m^A \, \mathrm{d}\mu_n^A$, and by (5.2) the right side tends to $\int G_B\mu_m^A \, \mathrm{d}\nu$ as $n \to \infty$; letting $m \to \infty$ and using monotone convergence for the fixed measure ν, we obtain $\lim_n I_B(\mu_n^A) \geq \int G_B\nu \, \mathrm{d}\nu = I_B(\nu)$. This proves the claim, and hence $I_B(\nu) \leq I_B(\mu) < \infty$.

Now let τ be a positive measure on A with finite Green energy. We have $I_B(\mu_n, \tau) = I_B(\mu_n^A, \tau)$ for every n. As $n \to \infty$, the left side $\int G_B\tau \, \mathrm{d}\mu_n$ increases to $\int G_B\tau \, \mathrm{d}\mu = I_B(\mu, \tau)$ by monotone convergence, since μ_n is the restriction of μ to an increasing

sequence of sets whose union carries μ, and the right side $\int G_B\tau \, d\mu_n^A$ increases to $I_B(\nu,\tau)$ by (5.2). Hence $I_B(\mu,\tau) = I_B(\nu,\tau)$, which is (5.1) for ν.

By monotone convergence, $G_B\mu_n \uparrow G_B\mu$ pointwise on B, and we have seen that $G_B\mu_n^A \uparrow G_B\nu$. Since $G_B\mu_n^A \le G_B\mu_n$, it follows that $G_B\nu \le G_B\mu$ everywhere. Moreover, $G_B\mu_n^A = G_B\mu_n$ on A outside a polar set E_n, so $G_B\nu = G_B\mu$ on $A \setminus E$, where $E = \bigcup_n E_n$ is polar by Proposition 4.2.3.

We now identify $\widehat{R}^A_{G_B\mu}$ with $G_B\nu$. Since B is bounded and E is polar, Proposition 5.2.4, applied with $U = B$ and with E in place of A, provides a nonnegative superharmonic function p on B, locally integrable and hence finite almost everywhere, with $p = +\infty$ on E. For every $\varepsilon > 0$, the function $G_B\nu + \varepsilon p$ is nonnegative and superharmonic, and it is at least $G_B\mu$ at every point of A: on $A \setminus E$ because $G_B\nu = G_B\mu$ there, and on E because $p = +\infty$. Hence $R^A_{G_B\mu} \le G_B\nu + \varepsilon p$. Letting $\varepsilon \downarrow 0$, we get $R^A_{G_B\mu} \le G_B\nu$ wherever $p < \infty$, that is, almost everywhere. Thus $\widehat{R}^A_{G_B\mu} \le G_B\nu$ almost everywhere, and hence everywhere by the ball-average formula preceding Proposition 5.2.4, since both functions are superharmonic. Conversely, $G_B\mu_n \le G_B\mu$, so Proposition 5.2.4 gives $\widehat{R}^A_{G_B\mu} \ge \widehat{R}^A_{G_B\mu_n} = G_B\mu_n^A$ for every n, and therefore $\widehat{R}^A_{G_B\mu} \ge G_B\nu$. Consequently $\widehat{R}^A_{G_B\mu} = G_B\nu$, and we put $\mu^A = \nu$; this measure is unique because a measure is determined by its Green potential.

It remains to collect the other conclusions. The inequality $G_B\mu^A \le G_B\mu$, the equality quasi-everywhere on A, the mass bound and the orthogonality relation have been proved above. If ν_1 and ν_2 are positive measures on A with finite Green energy satisfying (5.1) in place of μ^A, then $I_B(\nu_1,\tau) = I_B(\mu,\tau) = I_B(\nu_2,\tau)$ for $\tau = \nu_1$ and $\tau = \nu_2$, so $\nu_1 = \nu_2$ by the uniqueness criterion at the beginning of this section; thus (5.1) characterizes μ^A. If μ_1 and μ_2 have finite Green energy, so has $\mu_1 + \mu_2$, because the mutual energy $I_B(\mu_1,\mu_2)$ is finite; and $\mu_1^A + \mu_2^A$ is a positive measure on A with finite Green energy which satisfies (5.1) for the source $\mu_1 + \mu_2$. By the characterization, $(\mu_1+\mu_2)^A = \mu_1^A + \mu_2^A$, and in the same way $(c\mu_1)^A = c\mu_1^A$ for $c \ge 0$. Taking $\tau = \mu^A$ in (5.1) gives $I_B(\mu,\mu^A) = I_B(\mu^A)$, and then

$$I_B(\mu - \mu^A) = I_B(\mu) - 2I_B(\mu,\mu^A) + I_B(\mu^A) = I_B(\mu) - I_B(\mu^A),$$

which is nonnegative by Proposition 5.2.2. Finally, if μ is supported on A, then μ itself satisfies (5.1), so $\mu^A = \mu$ by the characterization. □

Example 5.3.3 Let $A \subset B$ be a compact set such that $W = B \setminus A$ is a regular domain, and let $x \in W$. The Dirac measure δ_x satisfies the hypothesis of Theorem 5.3.1, since $G_B\delta_x = G_B(x,\cdot)$ is continuous on $B \setminus \{x\}$ and therefore bounded on the compact set A. Its swept potential $r_x = G_B\delta_x^A$ is

$$r_x(y) = \begin{cases} G_B(x,y), & y \in A, \\ G_B(x,y) - G_W(x,y), & y \in W, \end{cases}$$

where at $y = x$ the difference $G_B(x,y) - G_W(x,y)$ is understood as the value of its harmonic extension across the common pole. Equivalently, let f_x be the function on $\partial W = \partial A \cup \partial B$ equal to $G_B(x,\cdot)$ on ∂A and to zero on ∂B. In terms of the harmonic

measure ω_y^W of W (Proposition 2.5.1),

$$r_x(y) = \int_{\partial W} f_x(\zeta)\, \mathrm{d}\omega_y^W(\zeta), \qquad y \in W.$$

The swept measure δ_x^A is supported on ∂A and has mass at most one. For example, if A is a closed ball concentric with B, then W is a regular domain.

Proof We first check that ∂W is the disjoint union of ∂A and ∂B. Since $W = B \cap (\mathbb{R}^d \setminus A)$ is the intersection of two open sets, $\partial W \subset \partial B \cup \partial A$, and $\partial A \subset A \subset B$ does not meet ∂B. Conversely, no point of $\partial B \cup \partial A$ lies in W. A point of ∂B is a limit of points of $B \setminus A = W$, because the compact set $A \subset B$ has positive distance from ∂B. Every small ball around a point of ∂A lies in B and contains points outside A, that is, points of W. Consequently ∂A and ∂B are disjoint compact sets whose union is ∂W, and f_x is continuous on ∂W, because $G_B(x,\cdot)$ is continuous on $B \setminus \{x\} \supset A$.

Let $H = H_W f_x$ be the solution of the Dirichlet problem on W with boundary values f_x; it is continuous on $\overline{W}$ since W is regular, and $H \geq 0$ because $f_x \geq 0$ and H_W preserves order. The function $G_B(x,\cdot) = G_B\delta_x$ is superharmonic on B by part (2) of Proposition 4.3.2, it is continuous on $B \setminus \{x\}$, and it tends to zero at ∂B (Theorem 2.5.2); hence its boundary limits on ∂W are the values of f_x. Therefore the maximum principle (Proposition 3.1.4), applied to the subharmonic function $H - G_B(x,\cdot)$ on the bounded open set W, gives $H \leq G_B(x,\cdot)$ on W. Define $r = H$ on W and $r = G_B(x,\cdot)$ on A. Then r is continuous on B, because H and $G_B(x,\cdot)$ have the same values on ∂A, and $0 \leq r \leq G_B(x,\cdot)$.

To see that r is superharmonic, we argue as in Step 2 of the proof of Theorem 5.2.5. Let E be a ball with $\overline{E} \subset B$, and let q be harmonic on E and continuous on $\overline{E}$ with $q \leq r$ on ∂E. Then $q \leq G_B(x,\cdot)$ on ∂E, hence on E by superharmonicity of $G_B(x,\cdot)$; in particular $r \geq q$ on $E \cap A$. On $E \cap W$ the harmonic function $H - q$ has nonnegative boundary limits: at points of ∂E because H extends continuously to $r \geq q$ there, and at points of $E \cap \partial A$ because H tends to $G_B(x,\cdot) \geq q$ there. The minimum principle gives $H \geq q$ on $E \cap W$. Hence $r \geq q$ on E, and r is superharmonic by Theorem 3.1.3.

The function r is admissible for $R^A_{G_B(x,\cdot)}$: it is nonnegative and superharmonic on B and equals $G_B(x,\cdot)$ on A. Let h be any admissible function. Then $h \geq r$ on A. On W, the superharmonic function h satisfies $\varliminf_{y\to\zeta} h(y) \geq h(\zeta) \geq f_x(\zeta)$ at points $\zeta \in \partial A$, by lower semicontinuity, and $\varliminf_{y\to\zeta} h(y) \geq 0 = f_x(\zeta)$ at points $\zeta \in \partial B$, since $h \geq 0$. Applying Proposition 3.1.4 to the subharmonic function $H - h$ on W, we get $h \geq H$ there. Hence $h \geq r$ on B, and $R^A_{G_B(x,\cdot)} = r$ pointwise. Since r is continuous, $\widehat{R}^A_{G_B(x,\cdot)} = r$, that is, $r_x = r$.

On W, both $G_B(x,\cdot)$ and $G_W(x,\cdot)$ are locally integrable and satisfy $-\Delta = \delta_x$ there (Theorem 2.5.2). Their difference is therefore distributionally harmonic on W; by Weyl's lemma (Theorem 2.1.4) it has a harmonic representative, which coincides with the difference off x. This harmonic function tends to f_x at ∂W, because $G_W(x,\cdot)$ tends to zero there. By uniqueness of the solution of the Dirichlet problem (Proposition 2.1.3), it equals H. The harmonic-measure formula is Proposition 2.5.1 applied to $H = H_W f_x$.

Finally, the measure $\delta_x^A = -\Delta r_x$ vanishes on W, where $r_x = H$ is harmonic, and on $\operatorname{Int} A$, where $r_x = G_B(x,\cdot)$ is harmonic because $x \notin A$. It is therefore supported on ∂A, and its mass is at most $\delta_x(B) = 1$ by Theorem 5.3.1.

For the last assertion, write $B = B(c,R)$ and let $A = \overline{B(c,\rho)}$ with $0 < \rho < R$, so that $W = \{y : \rho < |y-c| < R\}$ is a bounded open annulus. It is connected: every point of W is joined by a radial segment in W to the sphere $\partial B(c,(\rho+R)/2)$, which is path-connected because $d \geq 2$. By Theorem 2.4.6, it suffices to find, for every $\zeta \in \partial W$, a closed ball that meets $\overline{W}$ only at ζ. If $|\zeta - c| = R$, the closed ball $\overline{B(2\zeta - c, R)}$ meets $\overline{B(c,R)} \supset \overline{W}$ only at ζ, as noted at the beginning of Section 2.5. If $|\zeta - c| = \rho$, let $m = (c+\zeta)/2$. Every $y \in \overline{B(m,\rho/2)}$ satisfies $|y-c| \leq |y-m| + |m-c| \leq \rho$, with equality only if $y - m = (\zeta - c)/2$, that is, $y = \zeta$. Since $|y-c| \geq \rho$ on $\overline{W}$, the closed ball $\overline{B(m,\rho/2)}$ meets $\overline{W}$ only at ζ. Hence every boundary point of W is regular, and W is a regular domain. □

5.4 Countable covers and quasi-open sets

The failure of countability in Example 5.1.7 takes place on a polar set. We now show that, outside polar sets, Choquet's lemma recovers the countable operations needed for measures and capacity.

Theorem 5.4.1 *Let $(U_i)_{i\in I}$ be a family of finely open subsets of Ω. Then there is an at most countable subset $J \subset I$ such that*

$$\left(\bigcup_{i\in I} U_i\right) \setminus \left(\bigcup_{j\in J} U_j\right)$$

is polar. Moreover, every finely open set U contains a set V such that $U \setminus V$ is polar and V is the union of countably many sets $Q_k \cap \{u_k > a_k\}$, where Q_k is an open ball with $\overline{Q_k} \subset \Omega$, u_k is subharmonic on an open neighborhood of $\overline{Q_k}$ in Ω, and $a_k \in \mathbb{R}$; in particular, V is a Borel set. Consequently, U is measurable for the completion of Lebesgue measure, and for the completion of every finite positive measure of compact support with finite energy.

Proof We first reduce to sets of a special form attached to balls with rational centers and radii; for each such ball, Choquet's lemma then selects countably many of these sets. Let $U = \bigcup_{i\in I} U_i$. Call a set *basic* if it has the form $Q \cap \{u > a\}$, where Q is a ball with rational center and rational radius such that $\overline{Q} \subset \Omega$, u is subharmonic on an open neighborhood of $\overline{Q}$ in Ω, and $a \in \mathbb{R}$. Basic sets are finely open. Every point $x \in U$ lies in a basic set contained in some U_i. Indeed, if $x \in U_i$, then Proposition 5.1.3 provides a Euclidean open set V, a function $u \in \mathrm{SH}(V)$ and $a < u(x)$ with $\{y \in V : u(y) > a\} \subset U_i$, and we choose a rational ball Q with $x \in Q$ and $\overline{Q} \subset V$. Let $\mathcal{B}$ be the family of basic sets contained in some U_i; thus $U = \bigcup \mathcal{B}$.

Fix a rational ball Q, and let $\mathcal{B}_Q$ consist of the members of $\mathcal{B}$ of the form $Q \cap \{u > a\}$; we may assume that $\mathcal{B}_Q$ is nonempty. For every pair (u, a) as in the definition of basic sets with $Q \cap \{u > a\} \in \mathcal{B}_Q$, put

$$v = \frac{u - a}{1 + \max\{0, \sup_Q (u - a)\}} \quad \text{on } Q.$$

The supremum is less than $+\infty$, because u is upper semicontinuous on a neighborhood of the compact set $\overline{Q}$ and hence bounded above on $\overline{Q}$. Thus v is subharmonic on Q, $v \le 1$, and $\{v > 0\} = Q \cap \{u > a\}$.

Let F be the supremum of all the functions v so obtained. By Choquet's lemma (Lemma 3.4.1), there is a sequence (v_k) among them such that $F' = \sup_k v_k$ satisfies $(F')^* = F^*$. By part (2) of Theorem 4.4.4, the sets $\{F \ne F^*\}$ and $\{F' \ne (F')^*\}$ are polar; let P_Q be their union. If $y \in Q \setminus P_Q$ and $v(y) > 0$ for some v, then $F'(y) = (F')^*(y) = F^*(y) = F(y) > 0$, so $v_k(y) > 0$ for some k. Hence the union of $\mathcal{B}_Q$ is contained in the union of P_Q and the countably many basic sets $\{v_k > 0\}$.

Doing this for each of the countably many rational balls Q, we obtain countably many members of $\mathcal{B}$ whose union covers U outside the set $P = \bigcup_Q P_Q$, which is polar by Proposition 4.2.3. For each selected member choose one index i such that the member is contained in U_i. The set J of chosen indices is at most countable, and $U \setminus \bigcup_{j \in J} U_j \subset P$. This proves the first assertion.

For the second, let U be finely open, and apply the first assertion to the family of all basic sets contained in U, whose union is U by Proposition 5.1.3. We obtain countably many basic sets whose union $V \subset U$ is such that $U \setminus V$ is polar. Each basic set is Borel, since subharmonic functions are Borel measurable, so V is Borel.

For the measurability assertions, it suffices to show that the polar set $U \setminus V$ is contained in a Borel set N that is null for Lebesgue measure and for every finite positive measure of compact support with finite energy. Since $\mathbb{R}^d$ has a countable base, $U \setminus V$ is covered by countably many connected open sets V_k carrying nontrivial subharmonic functions ψ_k with $(U \setminus V) \cap V_k \subset \{\psi_k = -\infty\}$. The Borel set $N = \bigcup_k \{\psi_k = -\infty\}$ is polar by Proposition 4.2.3; it is Lebesgue null by Proposition 3.1.5, and null for measures of finite energy by Theorem 4.3.3. □

Proposition 5.4.2 *Let $U \subset \Omega$ be finely open, let $B \subset \mathbb{R}^d$ be an open ball, let $L \subset B$ be compact, and let $\eta > 0$. Then there is a Euclidean open set $O \Subset B$ with $C_B(O) < \eta$ such that $(U \cup O) \cap L$ is relatively open in L. Equivalently, $U \cap (L \setminus O)$ is relatively open in $L \setminus O$. In this sense, every finely open set is locally quasi-open.*

Proof We construct O as the union of an open set covering the polar part of U near L and of open sets outside which the functions defining the rest of U are continuous on L. By the second assertion of Theorem 5.4.1, there are at most countably many sets $V_k = Q_k \cap \{u_k > a_k\} \subset U$, with Q_k, u_k and a_k as there, whose union V is such that $U \setminus V$ is contained in a polar set P. Discarding the empty ones, we may assume that every V_k is nonempty and that k ranges over a set $\mathcal{K} \subset \{1, 2, \ldots\}$. Since $V_k \ne \varnothing$, we may assume that u_k is defined on an open ball $Q_k' \supset \overline{Q_k}$ and is nontrivial there; by Proposition 3.1.5, it is then locally integrable, hence not identically $-\infty$ on any ball in Q_k'. Fix an open set W_0 with $L \subset W_0 \Subset B$.

First we cover the polar part. The set $P \cap W_0$ is polar and its closure is a compact subset of B, so $C_B^*(P \cap W_0) = 0$ by Corollary 4.4.2. Hence there is an open set O_0 with $P \cap W_0 \subset O_0 \subset W_0$ and $C_B(O_0) < \eta/2$.

Next we deal with the discontinuities of the functions u_k. For each $k \in \mathcal{K}$, the compact set $\overline{Q_k} \cap L$ lies in the open set $Q_k' \cap B$. Each of its points therefore lies in a ball D' such that $D' \Subset D \Subset Q_k' \cap B$ for another ball D, and finitely many such balls D' cover $\overline{Q_k} \cap L$. Enumerate all the pairs of balls obtained in this way, for all $k \in \mathcal{K}$, as (D_j', D_j) with j in a set $\mathcal{J}$ of positive integers, and let $k(j)$ be the index k for which the pair (D_j', D_j) was chosen. For each $j \in \mathcal{J}$, the quasi-continuity theorem (Theorem 4.5.2), applied to $u_{k(j)}$ on $Q_{k(j)}'$ with the ball $D_j \Subset Q_{k(j)}'$ and the compact set $\overline{D_j'} \cap L \subset D_j$, provides an open set $O_j \Subset D_j$ with $C_{D_j}(O_j) < \eta 2^{-j-1}$ such that $u_{k(j)}$ is finite and continuous on $(\overline{D_j'} \cap L) \setminus O_j$. By Proposition 5.2.3, $C_B(O_j) = C_B^*(O_j) \le C_{D_j}^*(O_j) = C_{D_j}(O_j) < \eta 2^{-j-1}$.

Put $O = O_0 \cup \bigcup_{j \in \mathcal{J}} (O_j \cap W_0)$. This is an open subset of W_0, so $O \Subset B$, and countable subadditivity and monotonicity of the outer capacity (Theorem 4.3.5) give

$$C_B(O) \le C_B(O_0) + \sum_{j \in \mathcal{J}} C_B(O_j) < \frac{\eta}{2} + \sum_{j \ge 1} \eta 2^{-j-1} = \eta.$$

It remains to show that $U \cap F$ is relatively open in $F = L \setminus O$. Since $L \subset W_0$, the set $(U \setminus V) \cap F$ is contained in $(P \cap W_0) \setminus O_0 = \varnothing$, so $U \cap F = V \cap F$. Let $y \in V \cap F$. Then $y \in V_k$ for some $k \in \mathcal{K}$; since $y \in \overline{Q_k} \cap L$, we have $y \in D_j'$ for some j with $k(j) = k$. Since $L \subset W_0$, we have $O_j \cap L \subset O$, so $D_j' \cap F \subset (\overline{D_j'} \cap L) \setminus O_j$, and the function u_k is finite and continuous on $D_j' \cap F$. Hence

$$\{z \in D_j' \cap Q_k \cap F : u_k(z) > a_k\}$$

is relatively open in F, contains y, and is contained in $V_k \cap F$. Therefore $U \cap F = V \cap F$ is relatively open in F.

Write $U \cap F = T \cap F$ with T open in $\mathbb{R}^d$. Then $(U \cup O) \cap L = (U \cap F) \cup (O \cap L) = (T \cup O) \cap L$ is relatively open in L. Conversely, if $(U \cup O) \cap L$ is relatively open in L, then its intersection with F, which is $U \cap F$, is relatively open in F. This proves the equivalence of the two formulations. □

Fine continuity should not be confused with quasi-continuity: a subharmonic function is continuous on all of Ω for the fine topology, whereas Theorem 4.5.2 gives Euclidean continuity only after an open set of small capacity has been removed.

5.5 Complex lines and the choice of topology

On $\mathbb{C}^n$, identified with $\mathbb{R}^{2n}$, the classical fine topology is generated by all subharmonic functions of $2n$ real variables. In Part II we shall instead use the upper semicontinuous functions whose restriction to each connected component of the intersection of

their domain with an affine complex line is subharmonic or identically $-\infty$. On domains these are, apart from the constant $-\infty$, the plurisubharmonic functions of Definition 8.1.1, and we shall see in Example 10.3.4 that they generate the plurifine topology of Definition 10.3.1 in the same way. In complex dimension one the line condition is planar subharmonicity, so the two topologies coincide; in higher dimensions they differ.

Proposition 5.5.1 *Let $n \geq 2$ and $H = \{z \in \mathbb{C}^n : z_1 = 0\}$. Then the classical fine topology of $\mathbb{C}^n = \mathbb{R}^{2n}$ induces the discrete topology on H. On the other hand, let τ be the coarsest topology on $\mathbb{C}^n$ that contains the Euclidean topology and makes continuous every upper semicontinuous function $u\colon V \to [-\infty, \infty)$, where $V \subset \mathbb{C}^n$ is open, such that for every affine complex line L the restriction of u to each connected component of $L \cap V$ is subharmonic or identically $-\infty$. Then no point of H is isolated in the topology induced on H by τ.*

Proof By Example 3.2.4, the function $w \mapsto \log|w|$ is subharmonic on $\mathbb{C}$. Hence $z \mapsto \log|z_1|$, a function of the first two real coordinates, is subharmonic on $\mathbb{R}^{2n}$ by Lemma 5.1.6; it is finite off H and equals $-\infty$ on H. Thus H is polar, and Proposition 5.1.5 shows that the classical fine topology induces the discrete topology on H.

Now let $x \in H$, and let N' be a neighborhood of x for τ. As in Section 5.1, upper semicontinuity turns every condition of the form $u < b$ into a Euclidean open condition, and restrictions to smaller open sets preserve the complex-line condition; hence N' contains a set

$$N = V \cap \{u_1 > a_1\} \cap \cdots \cap \{u_m > a_m\},$$

where V is a Euclidean open neighborhood of x, the u_i are upper semicontinuous functions on V satisfying the complex-line condition in the statement, and $u_i(x) > a_i$. Choose an affine complex line $\ell \subset H$ through x, which is possible since $n \geq 2$, and identify ℓ with $\mathbb{C}$. On the connected component of $\ell \cap V$ containing x, each restriction $u_i|_\ell$ is subharmonic and is not identically $-\infty$, because $u_i(x)$ is finite. Hence $\{u_i|_\ell > a_i\}$ is open in the planar fine topology, and the intersection of these sets with that component is a planar fine neighborhood of x contained in $N \cap \ell$. By Proposition 5.1.4, applied in the plane, it contains points other than x. These points lie in $H \cap N'$, so x is not isolated in H. □

Chapter 6
Fourier analysis and incidence of complex lines

This chapter develops the Fourier analysis on Euclidean space that the complex theory of Part II requires, and applies it to functions of several complex variables. Let $D \subset \mathbb{C}^n$ be open, with $n \geq 2$, and let u be a function on D whose restriction to every complex line is subharmonic. On each line u is upper semicontinuous, and hence bounded above on compact subsets of that line, but the hypothesis gives no control of these bounds as the line varies. We prove that a Borel function with this property is nevertheless locally bounded above on D (Corollary 6.5.4). In Chapter 8 this is the step that makes upper semicontinuity automatic for functions that are subharmonic on complex lines (Theorem 8.1.3).

The proof is by contradiction. Suppose that $u(x_j) \to \infty$ along a sequence $x_j \to p$. By the submean inequality, $u \geq u(x_j)$ on a subset of positive length of every sufficiently small circle centered at x_j in a complex line through x_j. Choosing one such point on each circle, we obtain a probability measure on an annulus about x_j, carried by the set where $u \geq u(x_j)$, whose Riesz energy of order $2n - 2$ is bounded independently of j (Theorem 6.4.2).

By Plancherel's theorem, energy of this order is a constant multiple of the average, over all complex hyperplanes H through the origin, of the squared L^2 norms of the densities of the measures projected onto H (Theorem 6.5.2). By the Cauchy–Schwarz inequality, a projected measure with a density of controlled L^2 norm cannot be concentrated on a set of small measure. An incidence argument then produces a single complex line that meets infinitely many of the sets $\{u \geq u(x_j)\}$ near p (Theorem 6.5.3), which contradicts the upper bound of u on that line. The Fourier analysis is developed in Section 6.1, and the invariant measures used to average over directions and over hyperplanes are constructed in Section 6.3.

The same Fourier methods give L^p estimates for second derivatives. In Section 6.2 we show that, for $1 < p < \infty$, the L^p norm of the Laplacian of a test function bounds the L^p norm of its full real Hessian. Together with Morrey's inequality, these estimates supply the real regularity of functions with bounded Laplacian that is used in Chapter 16 to compute Monge–Ampère measures of envelopes below nonsmooth obstacles.

Historical remarks and references for this chapter are collected in Appendix B.6.

6.1 Fourier transform and energy

Let $d \geq 1$. We develop the Fourier transform on $\mathbb{R}^d$ as far as Plancherel's theorem, and use it to express the Riesz energy of a measure as a weighted L^2 norm of its Fourier transform (Theorem 6.1.4). For $f \in L^1(\mathbb{R}^d)$ and a finite positive Borel measure μ on $\mathbb{R}^d$ we use the Fourier transforms

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x) \mathrm{e}^{-2\pi \mathrm{i} x \cdot \xi} \, \mathrm{d}x, \qquad \widehat{\mu}(\xi) = \int \mathrm{e}^{-2\pi \mathrm{i} x \cdot \xi} \, \mathrm{d}\mu(x).$$

Both functions are bounded, by $\|f\|_1$ and by $\mu(\mathbb{R}^d)$, respectively, and they are continuous by dominated convergence.

The Schwartz space $\mathcal{S}(\mathbb{R}^d)$ consists of the smooth complex functions f on $\mathbb{R}^d$ such that $x^\alpha \partial^\beta f$ is bounded for all multi-indices α, β. By the Leibniz rule, $\mathcal{S}(\mathbb{R}^d)$ is stable under differentiation and under multiplication by polynomials, and $C_c^\infty(\mathbb{R}^d) \subset \mathcal{S}(\mathbb{R}^d)$. Every derivative $\partial^\beta f$ of a Schwartz function is integrable: $(1 + |x|^2)^d \partial^\beta f$ is a linear combination of the bounded functions $x^\alpha \partial^\beta f$, so $|\partial^\beta f| \leq C(1 + |x|^2)^{-d}$, and $(1 + |x|^2)^{-d}$ is integrable by polar coordinates (2.1), because $r^{d-1}(1 + r^2)^{-d} \leq \min\{r^{d-1}, r^{-d-1}\}$.

Lemma 6.1.1 *The Fourier transform maps $\mathcal{S}(\mathbb{R}^d)$ into itself, and for $f \in \mathcal{S}(\mathbb{R}^d)$ and $1 \leq j \leq d$,*

$$\widehat{\partial_j f}(\xi) = 2\pi \mathrm{i} \xi_j \widehat{f}(\xi), \qquad \partial_{\xi_j} \widehat{f} = \widehat{(-2\pi \mathrm{i} x_j f)}. \tag{6.1}$$

Moreover, for $t > 0$,

$$\widehat{\mathrm{e}^{-\pi t |x|^2}}(\xi) = t^{-d/2} \mathrm{e}^{-\pi |\xi|^2 / t}.$$

Proof Let $f \in \mathcal{S}$. We first prove the second rule in (6.1). By the mean value theorem, the difference quotients of $\xi \mapsto \mathrm{e}^{-2\pi \mathrm{i} x \cdot \xi}$ in the direction of ξ_j are bounded by $2\pi |x_j|$, and $x_j f$ is integrable. Hence dominated convergence allows us to differentiate under the integral sign: $\partial_{\xi_j} \widehat{f}$ exists and is the transform of $-2\pi \mathrm{i} x_j f$, which is continuous. Since $x_j f \in \mathcal{S}$, induction shows that $\widehat{f}$ is smooth and that $\partial^\beta \widehat{f}$ is the transform of $(-2\pi \mathrm{i} x)^\beta f$.

For the first rule, Fubini's theorem allows us to integrate in x_j before the other coordinates. For almost every value of the other coordinates, f and $\partial_j f$ are integrable on the corresponding line, and an integration by parts over $[-a, a]$ followed by $a \to \infty$ gives

$$\int_{\mathbb{R}} \partial_j f(x) \, \mathrm{e}^{-2\pi \mathrm{i} x \cdot \xi} \, \mathrm{d}x_j = 2\pi \mathrm{i} \xi_j \int_{\mathbb{R}} f(x) \, \mathrm{e}^{-2\pi \mathrm{i} x \cdot \xi} \, \mathrm{d}x_j,$$

because the boundary terms are bounded by $|f(x)|$ at $x_j = \pm a$, and $f(x) \to 0$ as $|x| \to \infty$. Integrating over the other coordinates proves the first rule. Combining the two rules, we find that $\xi^\alpha \partial^\beta \widehat{f}$ is a constant multiple of the transform of $\partial^\alpha (x^\beta f)$. This function belongs to $\mathcal{S}$ and is therefore integrable, so $\xi^\alpha \partial^\beta \widehat{f}$ is bounded by a multiple of its L^1 norm. Hence $\widehat{f} \in \mathcal{S}$.

For the Gaussian, consider first $d = 1$, put $g(x) = \mathrm{e}^{-\pi x^2}$, which belongs to $\mathcal{S}(\mathbb{R})$, and set $F = \widehat{g}$. Since $-2\pi \mathrm{i} x g = \mathrm{i} g'$, the two rules give $F' = \mathrm{i}\,\widehat{g'} = \mathrm{i}(2\pi\mathrm{i}\xi)F = -2\pi\xi F$, so $F(\xi)\mathrm{e}^{\pi\xi^2}$ is constant. Moreover $F(0) = 1$: squaring the positive integral $F(0)$ and using polar coordinates in the plane, where $s_1 = 2\pi$, gives $F(0)^2 = 2\pi \int_0^\infty r\mathrm{e}^{-\pi r^2}\,\mathrm{d}r = 1$. Hence $F(\xi) = \mathrm{e}^{-\pi\xi^2}$. In dimension d, Fubini's theorem writes the transform of $\mathrm{e}^{-\pi|x|^2}$ as a product of one-dimensional transforms, which proves the formula for $t = 1$. For general t, the linear substitution $x = y/\sqrt{t}$ gives

$$\int \mathrm{e}^{-\pi t|x|^2}\mathrm{e}^{-2\pi\mathrm{i}x\cdot\xi}\,\mathrm{d}x = t^{-d/2}\int \mathrm{e}^{-\pi|y|^2}\mathrm{e}^{-2\pi\mathrm{i}y\cdot\xi/\sqrt{t}}\,\mathrm{d}y = t^{-d/2}\mathrm{e}^{-\pi|\xi|^2/t}.$$

Theorem 6.1.2 (Plancherel) *Every $f \in \mathcal{S}(\mathbb{R}^d)$ satisfies the inversion formula*

$$f(x) = \int \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\cdot\xi}\,\mathrm{d}\xi, \qquad x \in \mathbb{R}^d.$$

The Fourier transform on $\mathcal{S}(\mathbb{R}^d)$ extends uniquely to a bounded operator on $L^2(\mathbb{R}^d)$. This extension is unitary, its inverse is given on Schwartz functions by the integral with the exponential $\mathrm{e}^{2\pi\mathrm{i}x\cdot\xi}$, and on $L^1 \cap L^2$ it agrees almost everywhere with the integral transform.

Proof Let $f \in \mathcal{S}$ and $\varepsilon > 0$. Insert the factor $\mathrm{e}^{-\pi\varepsilon^2|\xi|^2}$ into the inversion integral. By Fubini's theorem and Lemma 6.1.1 with $t = \varepsilon^2$,

$$\int \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\cdot\xi}\mathrm{e}^{-\pi\varepsilon^2|\xi|^2}\,\mathrm{d}\xi = f * g_\varepsilon(x), \qquad g_\varepsilon(y) = \varepsilon^{-d}\mathrm{e}^{-\pi|y|^2/\varepsilon^2}.$$

The functions g_ε are positive with integral one, and the substitution $y = \varepsilon y'$ shows that their mass outside a fixed ball $B(0,\delta)$ tends to zero as $\varepsilon \to 0$. Since f is bounded, and uniformly continuous because its gradient is bounded, splitting the convolution integral into the regions $|y| < \delta$ and $|y| \geq \delta$ shows that $f * g_\varepsilon \to f$ uniformly. On the left-hand side, $\widehat{f} \in \mathcal{S} \subset L^1$ by Lemma 6.1.1, so dominated convergence lets $\varepsilon \to 0$ under the frequency integral. This proves the inversion formula.

Write $\mathcal{F}$ for the Fourier transform on $\mathcal{S}$, $\mathcal{F}^*$ for the inverse integral transform $\mathcal{F}^*g(x) = \int g(\xi)\mathrm{e}^{2\pi\mathrm{i}x\cdot\xi}\,\mathrm{d}\xi$, and $Rf(x) = f(-x)$. The substitution $x \mapsto -x$ shows that $\mathcal{F}^* = R\mathcal{F} = \mathcal{F}R$ on $\mathcal{S}$. In particular $\mathcal{F}^*$ maps $\mathcal{S}$ into itself, and both compositions $\mathcal{F}^*\mathcal{F}$ and $\mathcal{F}\mathcal{F}^*$ equal $R\mathcal{F}\mathcal{F}$, which is the identity on $\mathcal{S}$ by the inversion formula. For $f, g \in \mathcal{S}$, applying inversion to g and Fubini's theorem, we obtain Parseval's identity

$$\int f\overline{g}\,\mathrm{d}x = \int \widehat{f}\,\overline{\widehat{g}}\,\mathrm{d}\xi;$$

the interchange of integrals is justified because the double integral of the absolute value is at most $\|f\|_1\|\widehat{g}\|_1$.

By Lemma 1.4.1(5), applied to Lebesgue measure, $C_c^\infty(\mathbb{R}^d)$, and hence $\mathcal{S}$, is dense in L^2. By Parseval's identity, $\mathcal{F}$ maps sequences in $\mathcal{S}$ that are Cauchy in L^2

to Cauchy sequences, and L^2 is complete; therefore $\mathcal{F}$ extends uniquely from $\mathcal{S}$ to a bounded operator on L^2, and this extension is an isometry. Since $\mathcal{F}^* = R\mathcal{F}$, the inverse integral transform extends in the same way. Both compositions of the two extensions are continuous and equal to the identity on the dense subspace $\mathcal{S}$, hence everywhere. Thus the extension is unitary, with the stated inverse.

Finally let $f \in L^1 \cap L^2$. We choose $f_j \in C_c^\infty$ converging to f both in L^1 and in L^2: first replace f by $f\mathbf{1}_{B(0,R)}$ with R large, and then mollify; by Lemma 1.4.1(2) the mollifications converge in both norms, since their supports lie in a fixed compact set. The integral transforms of the f_j converge uniformly to that of f, because the difference is bounded by $\|f_j - f\|_1$, while by isometry they converge in L^2 to the L^2 transform of f. A subsequence of an L^2 convergent sequence converges almost everywhere: choose it so that the squared L^2 errors are summable, and apply Tonelli's theorem to the sum of the squared errors. Hence the two transforms of f agree almost everywhere. □

We continue to write $\widehat{f}$ for the L^2 transform of $f \in L^2(\mathbb{R}^d)$, and we call its inverse the inverse L^2 transform.

Proposition 6.1.3 *Let μ be a finite positive Borel measure on $\mathbb{R}^d$ such that $\widehat{\mu} \in L^2(\mathbb{R}^d)$. Then $\mu = g\,\mathrm{d}x$ for a nonnegative function $g \in L^1 \cap L^2$, and $\|g\|_2 = \|\widehat{\mu}\|_2$.*

Proof Let $g \in L^2$ be the inverse L^2 transform of $\widehat{\mu}$, so that $\widehat{g} = \widehat{\mu}$. For $\phi \in \mathcal{S}$ we have

$$\int \phi \,\mathrm{d}\mu = \int \widehat{\phi}(\xi)\widehat{\mu}(-\xi)\,\mathrm{d}\xi = \int \phi g \,\mathrm{d}x.$$

The first equality follows from the inversion formula for ϕ and Fubini's theorem. For the second, we apply Parseval's identity to ϕ and $\overline{g}$; it extends from $\mathcal{S}$ to L^2 because the L^2 transform is unitary. We also use that the transform of $\overline{g}$ at ξ is $\overline{\widehat{g}(-\xi)}$; this holds for Schwartz functions by definition, and it extends to L^2 because the maps $g \mapsto \widehat{\overline{g}}$ and $g \mapsto \overline{\widehat{g}(-\,\cdot\,)}$ are continuous from L^2 to L^2.

Consequently, for every test function ϕ the number $\int \phi g \,\mathrm{d}x = \int \phi \,\mathrm{d}\mu$ is real when ϕ is real and nonnegative when $\phi \geq 0$. By the Cauchy–Schwarz inequality on compact sets, g is locally integrable, so it defines a positive distribution. By Proposition 1.4.3(3) its mollifications $g * \rho_\varepsilon$ are nonnegative, and they converge to g in L^1 on compact sets by Lemma 1.4.1(2). Hence g is real and nonnegative almost everywhere. The measures μ and $g\,\mathrm{d}x$ are Radon by Proposition 1.1.2, and they define the same distribution; by Proposition 1.4.3(4), $\mu = g\,\mathrm{d}x$. In particular $\int g\,\mathrm{d}x = \mu(\mathbb{R}^d) < \infty$, and $\|g\|_2 = \|\widehat{\mu}\|_2$ because the L^2 transform is unitary. □

For $0 < s < d$, the Riesz s-energy of a finite positive Borel measure μ on $\mathbb{R}^d$ is

$$I_s(\mu) = \iint |x - y|^{-s}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \in [0, \infty].$$

The kernel $|x - y|^{-s}$ carries no normalizing constant. For comparison, if $d \geq 3$ and μ has compact support, the energy $I(\mu)$ of Section 4.1, defined with the normalized

kernel Γ_d, satisfies $(d-2)s_{d-1}I(\mu) = I_{d-2}(\mu)$, where s_{d-1} is the area of the unit sphere S^{d-1} (Section 2.1).

Theorem 6.1.4 *Let μ be a finite positive Borel measure on $\mathbb{R}^d$, and let $0 < s < d$. Then*

$$I_s(\mu) = b_{d,s} \int_{\mathbb{R}^d} |\widehat{\mu}(\xi)|^2 |\xi|^{s-d}\, \mathrm{d}\xi, \qquad b_{d,s} = \pi^{s-d/2} \frac{\Gamma((d-s)/2)}{\Gamma(s/2)},$$

where $\Gamma(a) = \int_0^\infty v^{a-1}\mathrm{e}^{-v}\,\mathrm{d}v$, $a > 0$, is Euler's Gamma function. Both sides may be infinite.

Proof For $z \in \mathbb{R}^d$, the substitution $v = \pi t|z|^2$ gives

$$|z|^{-s} = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^\infty t^{s/2-1}\mathrm{e}^{-\pi t|z|^2}\,\mathrm{d}t,$$

and both sides are $+\infty$ when $z = 0$. By Lemma 6.1.1 with t replaced by $1/t$, we have $\mathrm{e}^{-\pi t|z|^2} = t^{-d/2}\int \mathrm{e}^{-\pi|\xi|^2/t}\mathrm{e}^{-2\pi \mathrm{i} z\cdot\xi}\,\mathrm{d}\xi$. Taking $z = x - y$ and integrating with respect to $\mu \times \mu$, we obtain

$$\iint \mathrm{e}^{-\pi t|x-y|^2}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) = t^{-d/2}\int \mathrm{e}^{-\pi|\xi|^2/t}|\widehat{\mu}(\xi)|^2\,\mathrm{d}\xi;$$

Fubini's theorem applies here because the Gaussian in ξ is integrable and $\mu \times \mu$ is finite. Now insert the integral formula for $|x-y|^{-s}$ into $I_s(\mu)$. All integrands are nonnegative, so Tonelli's theorem allows us to integrate first in (x, y), then use the last identity, and finally integrate in t before ξ, even when the integrals are infinite:

$$I_s(\mu) = \frac{\pi^{s/2}}{\Gamma(s/2)} \int |\widehat{\mu}(\xi)|^2 \int_0^\infty t^{(s-d)/2-1}\mathrm{e}^{-\pi|\xi|^2/t}\,\mathrm{d}t\,\mathrm{d}\xi.$$

For $\xi \neq 0$, the substitution $v = \pi|\xi|^2/t$ evaluates the inner integral as

$$\int_0^\infty t^{(s-d)/2-1}\mathrm{e}^{-\pi|\xi|^2/t}\,\mathrm{d}t = \pi^{(s-d)/2}|\xi|^{s-d}\Gamma((d-s)/2),$$

and the point $\xi = 0$ is a Lebesgue null set. Multiplying the constants gives the formula. □

Example 6.1.5 The area of the unit sphere $S^{d-1} \subset \mathbb{R}^d$ is

$$s_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \tag{6.2}$$

For $0 < s < d$, the probability measure $\mathrm{d}\mu = \mathrm{e}^{-\pi|x|^2}\,\mathrm{d}x$ on $\mathbb{R}^d$ satisfies

$$I_s(\mu) = \left(\frac{\pi}{2}\right)^{s/2} \frac{\Gamma((d-s)/2)}{\Gamma(d/2)}.$$

Proof By Lemma 6.1.1 with $\xi = 0$ and $t = 1$, the Gaussian $\mathrm{e}^{-\pi|x|^2}$ has integral one. Integrating in polar coordinates and substituting $v = \pi r^2$, we obtain

$$1 = \int \mathrm{e}^{-\pi|x|^2}\,\mathrm{d}x = s_{d-1}\int_0^\infty \mathrm{e}^{-\pi r^2} r^{d-1}\,\mathrm{d}r = \frac{s_{d-1}\Gamma(d/2)}{2\pi^{d/2}},$$

which proves (6.2). By Lemma 6.1.1, $\widehat{\mu}(\xi) = \mathrm{e}^{-\pi|\xi|^2}$. Hence Theorem 6.1.4 and polar coordinates give

$$I_s(\mu) = b_{d,s}\int \mathrm{e}^{-2\pi|\xi|^2}|\xi|^{s-d}\,\mathrm{d}\xi = b_{d,s}s_{d-1}\int_0^\infty \mathrm{e}^{-2\pi r^2} r^{s-1}\,\mathrm{d}r.$$

The substitution $v = 2\pi r^2$ evaluates the radial integral as $\Gamma(s/2)/(2(2\pi)^{s/2})$. Inserting the values of $b_{d,s}$ and s_{d-1} proves the formula. □

6.2 Second derivatives and singular integrals

Let $d \geq 1$. For a function v on $\mathbb{R}^d$ we write $D^2 v = (\partial_a \partial_b v)_{a,b=1}^d$ for its real Hessian. In L^p norms of vector-valued and matrix-valued functions, vectors carry the Euclidean norm and matrices the Hilbert–Schmidt norm $|M| = (\sum_{a,b} |M_{ab}|^2)^{1/2}$. Constants denoted by C_d and $C_{d,p}$ depend only on the indicated parameters and may change from line to line.

The Laplacian Δv is the trace of $D^2 v$. If $d \geq 2$, a bound for Δv does not bound $D^2 v$ pointwise: a test function equal to $x_1^2 - x_2^2$ near the origin has vanishing Laplacian and nonzero Hessian there. We show that Δv does bound $D^2 v$ in L^p for $1 < p < \infty$ (Theorem 6.2.3). On the Fourier side, $\partial_a \partial_b v$ is obtained from $-\Delta v$ by multiplication with the bounded function $-\xi_a \xi_b / |\xi|^2$, so the case $p = 2$ follows from Plancherel's theorem. For the other exponents we represent the corresponding operator, away from the support of a function, by a kernel that is smooth off the diagonal. We then use the Calderón–Zygmund decomposition: on each of its cubes the function minus its average has mean zero, and this cancellation makes the image of that part integrable far from the cube. The section ends with Sobolev spaces, on open subsets of $\mathbb{R}^d$ and on compact manifolds, and with Morrey's inequality, which turns L^p bounds for second derivatives into Hölder continuity of first derivatives.

Lemma 6.2.1 *Let $1 \leq a, b \leq d$, and define an operator T_{ab} on $L^2(\mathbb{R}^d)$ by*

$$\widehat{T_{ab} f}(\xi) = -\frac{\xi_a \xi_b}{|\xi|^2}\,\widehat{f}(\xi), \qquad \xi \neq 0,$$

the multiplier being zero at the origin. Then T_{ab} is self-adjoint and has norm at most one. Moreover:

(1) *There is a smooth real function K_{ab} on $\mathbb{R}^d \setminus \{0\}$ such that*

$$|K_{ab}(z)| \le C_d|z|^{-d}, \qquad |\nabla K_{ab}(z)| \le C_d|z|^{-d-1}.$$

(2) *If $f \in L^2(\mathbb{R}^d)$ vanishes almost everywhere outside a compact set F, then f is integrable, and for almost every $x \in \mathbb{R}^d \setminus F$,*

$$T_{ab}f(x) = \int K_{ab}(x-y)f(y)\,\mathrm{d}y.$$

(3) *For $v \in C_c^\infty(\mathbb{R}^d)$ we have $\partial_a\partial_b v = T_{ab}(-\Delta v)$.*

Proof The multiplier is real and bounded by one in absolute value, so Plancherel's theorem shows that T_{ab} is self-adjoint with norm at most one. For $v \in C_c^\infty$, the first rule in (6.1) (Lemma 6.1.1), applied twice, gives $\widehat{-\Delta v}(\xi) = 4\pi^2|\xi|^2\widehat{v}(\xi)$ and $\widehat{\partial_a\partial_b v}(\xi) = -4\pi^2\xi_a\xi_b\widehat{v}(\xi)$. Multiplying the first identity by the multiplier of T_{ab} gives the second, which proves (3).

To construct the kernel we use the heat kernel $H_t(z) = (4\pi t)^{-d/2}\mathrm{e}^{-|z|^2/(4t)}$, $t > 0$. By Lemma 6.1.1, $\widehat{H}_t(\xi) = \mathrm{e}^{-4\pi^2 t|\xi|^2}$, and hence $\widehat{\partial_a\partial_b H_t}(\xi) = -4\pi^2\xi_a\xi_b\mathrm{e}^{-4\pi^2 t|\xi|^2}$. For $0 < \varepsilon < R < \infty$ put

$$k_{\varepsilon,R} = \int_\varepsilon^R \partial_a\partial_b H_t\,\mathrm{d}t.$$

Since $\partial_a\partial_b H_t(z) = \big(z_a z_b/(4t^2) - \delta_{ab}/(2t)\big)H_t(z)$, where δ_{ab} is the Kronecker symbol, we have

$$|\partial_a\partial_b H_t(z)| \le C_d t^{-d/2-1}(1 + |z|^2/t)\mathrm{e}^{-|z|^2/(4t)}.$$

Hence $\sup_z|\partial_a\partial_b H_t(z)| \le C_d t^{-d/2-1}$, and the substitution $z = \sqrt{t}\,w$ gives $\int|\partial_a\partial_b H_t(z)|\,\mathrm{d}z \le C_d t^{-1}$. Therefore the kernel $k_{\varepsilon,R}$ is bounded and integrable, and Fubini's theorem, applied with the integrable majorant $|\partial_a\partial_b H_t(z)|$ on $\mathbb{R}^d \times [\varepsilon, R]$, gives its Fourier transform:

$$\widehat{k_{\varepsilon,R}}(\xi) = -\frac{\xi_a\xi_b}{|\xi|^2}\big(\mathrm{e}^{-4\pi^2\varepsilon|\xi|^2} - \mathrm{e}^{-4\pi^2 R|\xi|^2}\big), \qquad \xi \ne 0.$$

Let $f \in L^1 \cap L^2$. Then $k_{\varepsilon,R} * f$ is bounded and integrable, hence in L^2, and by Fubini's theorem and the last assertion of Theorem 6.1.2 its Fourier transform is $\widehat{k_{\varepsilon,R}}\widehat{f}$. As $\varepsilon \downarrow 0$ and $R \uparrow \infty$, these multipliers are bounded by one and converge to $-\xi_a\xi_b/|\xi|^2$ for $\xi \ne 0$. By dominated convergence on the Fourier side and Plancherel's theorem, $k_{\varepsilon,R} * f \to T_{ab}f$ in L^2.

For $z \ne 0$ define

$$K_{ab}(z) = \int_0^\infty \partial_a\partial_b H_t(z)\,\mathrm{d}t.$$

Since $H_t(z) = t^{-d/2}H_1(z/\sqrt{t})$, every derivative satisfies

$$\partial^\gamma H_t(z) = t^{-(d+|\gamma|)/2}(\partial^\gamma H_1)(z/\sqrt{t}),$$

and induction on $|\gamma|$ shows that $\partial^\gamma H_1(w) = P_\gamma(w)\mathrm{e}^{-|w|^2/4}$ for a polynomial P_γ, of degree m_γ say. Fix $0 < c \le c'$, and let $c \le |z| \le c'$ and $|\gamma| \ge 2$. Since $|P_\gamma(w)| \le C_\gamma(1+|w|)^{m_\gamma}$ and $\mathrm{e}^{-|z|^2/(4t)} \le \mathrm{e}^{-c^2/(4t)}$, we obtain $|\partial^\gamma H_t(z)| \le g_\gamma(t)$, where

$$g_\gamma(t) = \begin{cases} C_\gamma t^{-(d+|\gamma|)/2}(1+c't^{-1/2})^{m_\gamma}\mathrm{e}^{-c^2/(4t)}, & 0 < t \le 1, \\ C_\gamma(1+c')^{m_\gamma}t^{-(d+|\gamma|)/2}, & t \ge 1. \end{cases}$$

This function is integrable on $(0,\infty)$ and does not depend on z: near $t = 0$ the factor $\mathrm{e}^{-c^2/(4t)}$ dominates every power of t^{-1}, and for $t \ge 1$ the exponent satisfies $(d+|\gamma|)/2 \ge 3/2 > 1$. By the mean value theorem and dominated convergence, we may therefore differentiate under the integral sign any number of times. Hence K_{ab} is smooth on $\mathbb{R}^d \setminus \{0\}$, and its derivatives are the integrals of the corresponding derivatives of $\partial_a\partial_b H_t$. Moreover, the integrals $\int_0^\varepsilon$ and $\int_R^\infty$ of $|\partial_a\partial_b H_t(z)|$ tend to zero as $\varepsilon \downarrow 0$ and $R \uparrow \infty$, uniformly for $c \le |z| \le c'$.

Differentiating the identity $H_{r^2 s}(rz) = r^{-d}H_s(z)$ twice in z and substituting $t = r^2 s$, we find that $K_{ab}(rz) = r^{-d}K_{ab}(z)$ for $r > 0$. Thus K_{ab} is homogeneous of degree $-d$ and ∇K_{ab} is homogeneous of degree $-d-1$, and the bounds in (1) follow from the boundedness of the continuous functions K_{ab} and ∇K_{ab} on the unit sphere. The same scaling shows that the function $z \mapsto \int_0^\infty |\partial_a\partial_b H_t(z)|\,\mathrm{d}t$ is homogeneous of degree $-d$; it is bounded on the unit sphere by $\int_0^\infty g_\gamma(t)\,\mathrm{d}t$, where $\partial^\gamma = \partial_a\partial_b$ and $c = c' = 1$, so it is at most $C_d|z|^{-d}$.

To prove (2), let f and F be as there. By the Cauchy–Schwarz inequality on F, the function f is integrable, so $f \in L^1 \cap L^2$. Let L be a compact set disjoint from F. For $x \in L$,

$$k_{\varepsilon,R} * f(x) = \int f(y) \int_\varepsilon^R \partial_a\partial_b H_t(x-y)\,\mathrm{d}t\,\mathrm{d}y,$$

where the outer integral may be taken over F. The differences $x - y$, with $x \in L$ and $y \in F$, form a compact set not containing the origin, on which $c \le |x-y| \le c'$ for some $0 < c \le c'$. By the uniform estimates just proved, the inner integral converges to $K_{ab}(x-y)$ uniformly on this set, and it is bounded by $\int_0^\infty |\partial_a\partial_b H_t(x-y)|\,\mathrm{d}t \le C_d c^{-d}$. Hence, since f is integrable, $k_{\varepsilon,R} * f$ converges uniformly on L to $\int K_{ab}(x-y)f(y)\,\mathrm{d}y$. Since it also converges to $T_{ab}f$ in L^2, a subsequence converges almost everywhere to $T_{ab}f$, and the two limits agree almost everywhere on L. Covering the open set $\mathbb{R}^d \setminus F$ by countably many such compact sets proves (2). For $d = 1$ the kernel vanishes, in accordance with $T_{11} = -1$: a direct differentiation gives $\partial_z^2 H_t = \big(z^2/(4t^2) - 1/(2t)\big)H_t = \partial_t H_t$, so $K_{11}(z) = \lim_{t\uparrow\infty} H_t(z) - \lim_{t\downarrow 0} H_t(z) = 0$ for $z \ne 0$. □

Lemma 6.2.2 *Let T_{ab} be the operator of Lemma 6.2.1, let $f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ and let $\lambda > 0$. Then*

$$|\{x : |T_{ab}f(x)| > \lambda\}| \le \frac{C_d}{\lambda}\|f\|_1.$$

Proof We use the Calderón–Zygmund decomposition of f at height λ into a bounded part g, handled by the L^2 bound for T_{ab}, and parts b_Q of mean zero on disjoint cubes Q, handled by Lemma 6.2.1. The dyadic cubes are the cubes $2^k(m + [0,1)^d)$ with

$k \in \mathbb{Z}$ and $m \in \mathbb{Z}^d$. Two dyadic cubes are either disjoint or nested, and each dyadic cube Q lies in a unique dyadic cube of twice its side length, its parent, which has volume $2^d|Q|$. Call a dyadic cube Q selected if the average of $|f|$ over Q exceeds λ, while the average over every strictly larger dyadic cube containing Q does not. Every dyadic cube on which the average of $|f|$ exceeds λ is contained in a selected cube, because the averages over its successive ancestors are at most $\|f\|_1$ divided by their volumes and therefore tend to zero. The selected cubes are pairwise disjoint. For each of them, the average over its parent is at most λ, and hence

$$\lambda < \frac{1}{|Q|}\int_Q |f| \le 2^d\lambda, \qquad \sum_Q |Q| \le \frac{1}{\lambda}\sum_Q \int_Q |f| \le \frac{1}{\lambda}\|f\|_1.$$

Moreover, $|f| \le \lambda$ almost everywhere outside the union of the selected cubes. Indeed, let x be a point outside this union at which the conclusion of Theorem 1.7.3 holds. Every dyadic cube containing x has average of $|f|$ at most λ, since otherwise it would lie in a selected cube. The dyadic cube of side 2^k containing x lies in the cube $x + (-2^k, 2^k)^d$, whose volume is 2^d times larger. Hence the average of $|f - f(x)|$ over it tends to zero as $k \to -\infty$, so its average of $|f|$ tends to $|f(x)|$, and therefore $|f(x)| \le \lambda$.

For a selected cube Q, let $f_Q = |Q|^{-1}\int_Q f$ be the average of f. Define $g = f$ outside the selected cubes and $g = f_Q$ on each selected cube Q, and put $b_Q = (f - f_Q)\mathbf{1}_Q$. Each b_Q has integral zero, and

$$\|g\|_1 \le \|f\|_1, \quad \|g\|_\infty \le 2^d\lambda, \quad \|g\|_2^2 \le 2^d\lambda\|f\|_1, \qquad \sum_Q \|b_Q\|_1 \le 2\|f\|_1.$$

By the Cauchy–Schwarz inequality, $|Q||f_Q|^2 \le \int_Q |f|^2$, so $\|b_Q\|_2 \le 2\|f\mathbf{1}_Q\|_2$. Since the b_Q have disjoint supports, the series $\sum_Q b_Q$ converges to $f - g$ in L^2.

Let c_Q be the center of Q, and let Q^* be the cube with center c_Q and $4\sqrt{d}$ times the side length of Q. If $x \notin Q^*$ and $y \in Q$, then $|x - c_Q| \ge 2\operatorname{diam} Q$ and $|y - c_Q| \le \operatorname{diam} Q/2$, so every point of the segment from $x - c_Q$ to $x - y$ has norm at least $|x - c_Q|/2$. The function $b_Q \in L^2$ vanishes outside the compact set $\overline{Q}$, which does not meet $\mathbb{R}^d \setminus Q^*$, and it has integral zero. Therefore parts (1) and (2) of Lemma 6.2.1 and the mean value theorem give, for almost every $x \notin Q^*$,

$$\begin{aligned} |T_{ab}b_Q(x)| &= \left|\int_Q \big(K_{ab}(x - y) - K_{ab}(x - c_Q)\big)b_Q(y)\,\mathrm{d}y\right| \\ &\le C_d\frac{\operatorname{diam} Q}{|x - c_Q|^{d+1}}\|b_Q\|_1. \end{aligned}$$

Integrating in polar coordinates about c_Q over the region $|x - c_Q| \ge 2\operatorname{diam} Q$, which contains $\mathbb{R}^d \setminus Q^*$, we obtain

$$\int_{\mathbb{R}^d\setminus Q^*} |T_{ab}b_Q| \le C_d\|b_Q\|_1.$$

Put $B^* = \bigcup_Q Q^*$. Then $|B^*| \le (4\sqrt{d})^d \sum_Q |Q| \le C_d \lambda^{-1} \|f\|_1$. Enumerate the selected cubes, and let S_N be the partial sums of $\sum_Q b_Q$. Since $S_N \to f - g$ in L^2, also $T_{ab} S_N \to T_{ab}(f-g)$ in L^2, and a subsequence converges almost everywhere. Moreover $\int_{\mathbb{R}^d \setminus B^*} |T_{ab} S_N| \le \sum_Q \int_{\mathbb{R}^d \setminus Q^*} |T_{ab} b_Q| \le C_d \|f\|_1$, so Fatou's lemma gives

$$\int_{\mathbb{R}^d \setminus B^*} |T_{ab}(f-g)| \le C_d \|f\|_1 .$$

Finally, $T_{ab} f = T_{ab} g + T_{ab}(f-g)$, and the set where $|T_{ab} f| > \lambda$ is contained in the union of B^*, the set where $|T_{ab} g| > \lambda/2$, and the set of $x \notin B^*$ with $|T_{ab}(f-g)(x)| > \lambda/2$. By Chebyshev's inequality and the L^2 bound for T_{ab}, the second set has measure at most $4\lambda^{-2}\|g\|_2^2 \le 2^{d+2}\lambda^{-1}\|f\|_1$, and the third has measure at most $2\lambda^{-1} \int_{\mathbb{R}^d \setminus B^*} |T_{ab}(f-g)| \le C_d \lambda^{-1} \|f\|_1$. Together with the bound for $|B^*|$, this proves the lemma. □

Theorem 6.2.3 *Let $1 < p < \infty$, and let T_{ab} be the operators of Lemma 6.2.1. There is a constant $C_{d,p}$ such that $\|T_{ab} f\|_p \le C_{d,p} \|f\|_p$ for all a, b and all $f \in L^p \cap L^2$; consequently each T_{ab} extends uniquely to a bounded operator on $L^p(\mathbb{R}^d)$. In particular*

$$\|D^2 v\|_p \le C_{d,p} \|\Delta v\|_p , \qquad v \in C_c^\infty(\mathbb{R}^d). \tag{6.3}$$

Proof Assume first that $1 < p < 2$, and let $f \in L^p \cap L^2$. Fix $\lambda > 0$ and split $f = f_1 + f_2$ with $f_1 = f \mathbf{1}_{\{|f| > \lambda\}}$ and $f_2 = f \mathbf{1}_{\{|f| \le \lambda\}}$. Since $|f_1| \le \lambda^{1-p} |f|^p$, we have $f_1 \in L^1 \cap L^2$, and $f_2 \in L^2$. Apply Lemma 6.2.2 to f_1, and Chebyshev's inequality with the L^2 bound to f_2, both with threshold $\lambda/2$. This gives

$$|\{|T_{ab} f| > \lambda\}| \le \frac{C_d}{\lambda} \int_{\{|f| > \lambda\}} |f| + \frac{C_d}{\lambda^2} \int_{\{|f| \le \lambda\}} |f|^2 .$$

For a measurable function h, Tonelli's theorem applied to $|h(x)|^p = \int_0^{|h(x)|} p\lambda^{p-1}\, \mathrm{d}\lambda$ expresses $\|h\|_p^p$ as the integral of $p\lambda^{p-1}$ times the distribution function $\lambda \mapsto |\{|h| > \lambda\}|$, also when the norm is infinite. We apply this to $h = T_{ab} f$: multiply the last inequality by $p\lambda^{p-1}$ and integrate in λ. By Tonelli's theorem again, the first term on the right contributes $C_d \int |f| \int_0^{|f|} p\lambda^{p-2}\, \mathrm{d}\lambda\, \mathrm{d}x = C_d p \|f\|_p^p/(p-1)$, and the second contributes $C_d \int |f|^2 \int_{|f|}^\infty p\lambda^{p-3}\, \mathrm{d}\lambda\, \mathrm{d}x = C_d p \|f\|_p^p/(2-p)$; the inner integrals converge precisely because $p > 1$ and $p < 2$. Therefore

$$\|T_{ab} f\|_p^p \le C_d p \left(\frac{1}{p-1} + \frac{1}{2-p} \right) \|f\|_p^p .$$

For $p = 2$ the bound is Plancherel's theorem.

Now let $2 < p < \infty$, put $p' = p/(p-1) \in (1,2)$, let $f \in L^p \cap L^2$, and write $h = T_{ab} f \in L^2$. For integers $N \ge 1$, put $E_N = B(0,N) \cap \{|h| \le N\}$ and $g_N = \mathbf{1}_{E_N} h |h|^{p-2}$. Then $|g_N| \le N^{p-1} \mathbf{1}_{E_N}$, so $g_N \in L^2 \cap L^{p'}$, and $|g_N|^{p'} = \mathbf{1}_{E_N} |h|^p$.

Since T_{ab} is self-adjoint on L^2, the case p' already proved and Hölder's inequality give

$$\int_{E_N} |h|^p = \int h\overline{g_N} = \int f\overline{T_{ab}g_N} \le C_{d,p'}\|f\|_p \left(\int_{E_N} |h|^p\right)^{1/p'}.$$

The integral on the left is finite. If it is nonzero, dividing by its $(1/p')$th power gives $(\int_{E_N} |h|^p)^{1/p} \le C_{d,p'}\|f\|_p$; if it is zero, this bound is trivial. The sets E_N increase to $\{|h| < \infty\}$, which has full measure since $h \in L^2$, so monotone convergence gives $\|T_{ab}f\|_p \le C_{d,p'}\|f\|_p$.

By Lemma 1.4.1(5), $C_c^\infty \subset L^p \cap L^2$ is dense in L^p, and L^p is complete, so T_{ab} has a unique bounded extension from $L^p \cap L^2$ to L^p. Finally, for $v \in C_c^\infty$, Lemma 6.2.1(3) gives $\|\partial_a\partial_b v\|_p = \|T_{ab}\Delta v\|_p \le C_{d,p}\|\Delta v\|_p$, and summing over a and b proves (6.3). □

For an integer $k \ge 0$, $1 \le p < \infty$ and an open set $U \subset \mathbb{R}^d$, the Sobolev space $W^{k,p}(U)$ consists of the functions in $L^p(U)$ whose distributional derivatives of order at most k belong to $L^p(U)$, with the norm $\|v\|_{W^{k,p}(U)} = \sum_{|\alpha|\le k} \|\partial^\alpha v\|_{L^p(U)}$. We write $W^{k,p}_{\mathrm{loc}}(U)$ for the space of functions whose restrictions to every open set $V \Subset U$ belong to $W^{k,p}(V)$. If $v \in W^{k,p}_{\mathrm{loc}}(U)$, $V \Subset U$ is open and $|\alpha| \le k$, then $\partial^\alpha(v * \rho_\varepsilon) = (\partial^\alpha v) * \rho_\varepsilon \to \partial^\alpha v$ in $L^p(V)$ as $\varepsilon \to 0$, by Proposition 1.4.3(1) and Lemma 1.4.1(2).

Lemma 6.2.4 *Let $k \ge 0$ and $1 \le p < \infty$, let $U, V \subset \mathbb{R}^d$ be open, and let $v \in W^{k,p}_{\mathrm{loc}}(U)$.*

(1) *If $\chi \in C^\infty(U)$, then $\chi v \in W^{k,p}_{\mathrm{loc}}(U)$, and $\partial^\alpha(\chi v) = \sum_{\beta\le\alpha} \binom{\alpha}{\beta} \partial^{\alpha-\beta}\chi\, \partial^\beta v$ for $|\alpha| \le k$. If moreover $\operatorname{Supp}\chi$ is a compact subset of U, then χv, extended by zero, belongs to $W^{k,p}(\mathbb{R}^d)$, its derivatives are given by the same formula extended by zero, and $\|\chi v\|_{W^{k,p}(\mathbb{R}^d)} \le C\|v\|_{W^{k,p}(W)}$ for every open set W with $\operatorname{Supp}\chi \subset W \subset U$, where C depends only on d, k and χ.*

(2) *If $\Phi\colon V \to U$ is a C^∞ diffeomorphism, then $v \circ \Phi \in W^{k,p}_{\mathrm{loc}}(V)$. If moreover v vanishes almost everywhere outside a compact set $K \subset U$, then $v \circ \Phi$ vanishes almost everywhere outside the compact set $\Phi^{-1}(K)$, and $\|v \circ \Phi\|_{W^{k,p}(V)} \le C\|v\|_{W^{k,p}(U)}$, where C depends only on k, p, Φ and K.*

Proof (1) For $k = 0$ the first assertion holds because χ is bounded on compact sets. Let $k \ge 1$ and $1 \le j \le d$. For a test function ϕ on U, the function $\chi\phi$ is a test function with $\partial_j(\chi\phi) = \chi\partial_j\phi + \phi\partial_j\chi$, so

$$\int \chi v\, \partial_j\phi\, \mathrm{d}x = \int v\, \partial_j(\chi\phi)\, \mathrm{d}x - \int v\, \partial_j\chi\, \phi\, \mathrm{d}x = -\int \big(\chi\partial_j v + v\partial_j\chi\big)\phi\, \mathrm{d}x.$$

Thus $\partial_j(\chi v) = \chi\partial_j v + (\partial_j\chi)v$. The functions $\partial_j v$ and v belong to $W^{k-1,p}_{\mathrm{loc}}(U)$, so by induction on k both terms belong to $W^{k-1,p}_{\mathrm{loc}}(U)$; hence $\chi v \in W^{k,p}_{\mathrm{loc}}(U)$, and iterating

the formula for first derivatives gives the Leibniz formula, exactly as for smooth functions.

Now let $\operatorname{Supp}\chi$ be a compact subset of U, and let W be as in the statement. By the Leibniz formula, χv and its derivatives of order at most k vanish almost everywhere outside $\operatorname{Supp}\chi$, and there they are bounded by $C\sum_{|\beta|\le k}|\partial^\beta v|$, where C depends on d, k and the derivatives of χ of order at most k. Hence their L^p norms are at most $C\|v\|_{W^{k,p}(W)}$, which is finite when $W \Subset U$. To identify the derivatives of the extension by zero, let ϕ be a test function on $\mathbb{R}^d$, and choose a smooth cutoff function ζ for $\operatorname{Supp}\chi$ in U which equals 1 on a neighborhood N of $\operatorname{Supp}\chi$ (Lemma 1.4.1(4)). Then $\zeta\phi$ is a test function on U, and $\partial^\alpha(\zeta\phi) = \partial^\alpha\phi$ on N. Since χv and $\partial^\alpha(\chi v)$ vanish almost everywhere outside $\operatorname{Supp}\chi$,

$$\begin{aligned}\int \chi v\,\partial^\alpha\phi\,\mathrm{d}x &= \int \chi v\,\partial^\alpha(\zeta\phi)\,\mathrm{d}x\\ &= (-1)^{|\alpha|}\int \partial^\alpha(\chi v)\,\zeta\phi\,\mathrm{d}x = (-1)^{|\alpha|}\int \partial^\alpha(\chi v)\,\phi\,\mathrm{d}x.\end{aligned}$$

This proves (1).

(2) The composite $v\circ\Phi$ is defined almost everywhere: if $Z\subset U$ is a Borel null set, the change of variables formula for the diffeomorphism Φ^{-1} gives $|\Phi^{-1}(Z)| = \int_Z |\det D\Phi^{-1}(x)|\,\mathrm{d}x = 0$. For a smooth function w on U and $1\le|\alpha|\le k$, the classical chain rule gives

$$\partial^\alpha(w\circ\Phi) = \sum_{1\le|\beta|\le|\alpha|} c_{\alpha\beta}\,(\partial^\beta w)\circ\Phi \qquad \text{on } V,$$

with smooth functions $c_{\alpha\beta}$ on V that depend only on Φ: for $|\alpha|=1$ this is $\partial_j(w\circ\Phi) = \sum_i \partial_j\Phi_i\,(\partial_i w)\circ\Phi$, and differentiating the formula once more, with the product rule and the chain rule applied to each $(\partial^\beta w)\circ\Phi$, gives the induction step.

Assume first that v vanishes almost everywhere outside the compact set K. By (1), applied with a smooth cutoff function equal to 1 near K, the extension of v by zero belongs to $W^{k,p}(\mathbb{R}^d)$. Its derivatives of order at most k vanish almost everywhere outside K, because v vanishes almost everywhere on the open set $\mathbb{R}^d\setminus K$ and hence so do its distributional derivatives. Choose $\delta>0$ such that the compact set $K' = \{x : \operatorname{dist}(x,K)\le\delta\}$ lies in U. For $\varepsilon<\delta$, the mollification $v_\varepsilon = v * \rho_\varepsilon$ is smooth and vanishes outside K', and $\partial^\beta v_\varepsilon = (\partial^\beta v)*\rho_\varepsilon \to \partial^\beta v$ in $L^p(\mathbb{R}^d)$ for $|\beta|\le k$, by the remark preceding the lemma, applied on $\mathbb{R}^d$ to a bounded open set containing K', outside of which both sides vanish.

Let J be the maximum of $|\det D\Phi^{-1}|$ on K', and let $M\ge 1$ bound the functions $|c_{\alpha\beta}|$ on the compact set $\Phi^{-1}(K')$. By the change of variables formula,

$$\int_V |g\circ\Phi|^p\,\mathrm{d}y = \int_U |g(x)|^p\,|\det D\Phi^{-1}(x)|\,\mathrm{d}x \le J\|g\|_p^p$$

for every measurable function g on U that vanishes outside K'. The smooth functions $w_\varepsilon = v_\varepsilon\circ\Phi$ vanish outside $\Phi^{-1}(K')$. Applying this inequality to $v_\varepsilon - v$ and its

derivatives, we find that $w_\varepsilon \to v \circ \Phi$ and, for $1 \le |\alpha| \le k$,

$$\partial^\alpha w_\varepsilon \to G_\alpha = \sum_{1 \le |\beta| \le |\alpha|} c_{\alpha\beta}\, (\partial^\beta v) \circ \Phi \qquad \text{in } L^p(V),$$

with $\|G_\alpha\|_p \le M J^{1/p} \sum_{1 \le |\beta| \le |\alpha|} \|\partial^\beta v\|_p$. For a test function ϕ on V, we obtain

$$\begin{aligned}\int (v \circ \Phi)\, \partial^\alpha \phi \, \mathrm{d}y &= \lim_{\varepsilon \to 0} \int w_\varepsilon\, \partial^\alpha \phi \, \mathrm{d}y \\ &= (-1)^{|\alpha|} \lim_{\varepsilon \to 0} \int \partial^\alpha w_\varepsilon\, \phi \, \mathrm{d}y = (-1)^{|\alpha|} \int G_\alpha \phi \, \mathrm{d}y.\end{aligned}$$

Hence $\partial^\alpha (v \circ \Phi) = G_\alpha \in L^p(V)$. These functions, and $v \circ \Phi$, vanish almost everywhere outside $\Phi^{-1}(K)$, and summing the bounds over $|\alpha| \le k$ gives $\|v \circ \Phi\|_{W^{k,p}(V)} \le C \|v\|_{W^{k,p}(U)}$, with C depending only on k, p, J and M, that is, on k, p, Φ and K once δ has been fixed in terms of K and U.

In general, let $V' \Subset V$ be open. Choose a smooth cutoff function χ for the compact set $\Phi(\overline{V'})$ in U (Lemma 1.4.1(4)). By (1), $\chi v \in W^{k,p}(U)$, and it vanishes outside $\operatorname{Supp} \chi$; by the case just treated, $(\chi v) \circ \Phi \in W^{k,p}(V)$. Since $(\chi v) \circ \Phi = v \circ \Phi$ on V', we obtain $v \circ \Phi \in W^{k,p}(V')$. Hence $v \circ \Phi \in W^{k,p}_{\mathrm{loc}}(V)$. □

Let X be a compact smooth manifold of dimension d. Fix finitely many coordinate charts $\varphi_i : U_i \to \varphi_i(U_i) \subset \mathbb{R}^d$ covering X and a smooth partition of unity subordinate to them, that is, smooth functions $\chi_i \ge 0$ on X with $\sum_i \chi_i = 1$ such that $\operatorname{Supp} \chi_i$ is a compact subset of U_i. Such functions exist. By Lemma 1.4.1(4), applied in the charts, every point of X has a smooth nonnegative function on X that is positive at the point and has compact support in some U_i. By compactness, finitely many of these functions have a positive sum s. Assigning each of them to one U_i that contains its support, we let χ_i be the sum of those assigned to U_i, divided by s.

Let v be a function on X defined outside a set whose images in the charts are Lebesgue null; by the change of variables formula, this condition does not depend on the charts. We say that v belongs to $W^{k,p}(X)$ if $(\chi_i v) \circ \varphi_i^{-1} \in W^{k,p}(\varphi_i(U_i))$ for every i, and we let $\|v\|_{W^{k,p}(X)}$ be the sum of these chart norms.

Other choices of charts and partitions of unity give the same space with equivalent norms. Indeed, let $\psi_j : V_j \to \psi_j(V_j)$ and η_j be another choice. Since $\eta_j v = \sum_i \eta_j \chi_i v$, it suffices to show that each function $(\eta_j \chi_i v) \circ \psi_j^{-1}$, extended by zero, belongs to $W^{k,p}(\psi_j(V_j))$, with norm at most $C \|(\chi_i v) \circ \varphi_i^{-1}\|_{W^{k,p}(\varphi_i(U_i))}$ for a constant C independent of v. Summing over i and j then gives

$$\sum_j \|(\eta_j v) \circ \psi_j^{-1}\|_{W^{k,p}(\psi_j(V_j))} \le C \|v\|_{W^{k,p}(X)},$$

and by symmetry the two norms are equivalent.

To prove the claim about $(\eta_j \chi_i v) \circ \psi_j^{-1}$, fix i and j. The compact set $\operatorname{Supp} \chi_i \cap \operatorname{Supp} \eta_j$ lies in $U_i \cap V_j$, so by Lemma 1.4.1(4) in the chart φ_i there is a smooth function ϑ on X, with compact support in $U_i \cap V_j$, that equals 1 on a neighborhood of it. Then

$\eta_j\chi_i v = (\vartheta\eta_j)\chi_i v$, and $(\vartheta\eta_j)\circ\varphi_i^{-1}$ is smooth with compact support in $\varphi_i(U_i\cap V_j)$. By Lemma 6.2.4(1), the function $w = (\eta_j\chi_i v)\circ\varphi_i^{-1}$ belongs to $W^{k,p}(\varphi_i(U_i))$, vanishes outside the compact set $\varphi_i(\operatorname{Supp}\vartheta)$, and satisfies $\|w\|_{W^{k,p}(\varphi_i(U_i))} \le C\|(\chi_i v)\circ\varphi_i^{-1}\|_{W^{k,p}(\varphi_i(U_i))}$.

We now change charts. By Lemma 6.2.4(2), applied to the restriction of w to $\varphi_i(U_i\cap V_j)$ and to the diffeomorphism $\Phi = \varphi_i\circ\psi_j^{-1}$ from $\psi_j(U_i\cap V_j)$ onto $\varphi_i(U_i\cap V_j)$, the function $(\eta_j\chi_i v)\circ\psi_j^{-1} = w\circ\Phi$ belongs to $W^{k,p}(\psi_j(U_i\cap V_j))$, with norm at most $C\|w\|_{W^{k,p}(\varphi_i(U_i))}$, and vanishes outside the compact set $\psi_j(\operatorname{Supp}\vartheta)$. By Lemma 6.2.4(1), applied with a smooth cutoff function equal to 1 near this set, its extension by zero belongs to $W^{k,p}(\psi_j(V_j))$, with norm at most a constant multiple of the previous one.

Lemma 6.2.5 *Let $1 \le p < \infty$ and $\varepsilon > 0$. There is a constant $C_{d,\varepsilon}$, depending only on d and ε, such that*

$$\|\nabla v\|_p \le \varepsilon\|D^2 v\|_p + C_{d,\varepsilon}\|v\|_p, \qquad v \in C_c^\infty(\mathbb{R}^d).$$

Proof Let e_a be the ath coordinate vector and $h > 0$. Taylor's formula with integral remainder gives

$$h\partial_a v(x) = v(x+he_a) - v(x) - \int_0^h (h-t)\partial_a^2 v(x+te_a)\,\mathrm{d}t.$$

The L^p norm is invariant under translation. Moreover, $\omega(t)\,\mathrm{d}t = 2h^{-2}(h-t)\,\mathrm{d}t$ is a probability measure on $[0,h]$, so Hölder's inequality gives $|\int_0^h g\,\omega\,\mathrm{d}t|^p \le \int_0^h |g|^p\omega\,\mathrm{d}t$ for every measurable function g on $[0,h]$. Applying this to $g(t) = \partial_a^2 v(x+te_a)$, integrating in x and using Tonelli's theorem, we obtain

$$\Big\|\int_0^h (h-t)\partial_a^2 v(\cdot+te_a)\,\mathrm{d}t\Big\|_p \le \frac{h^2}{2}\Big(\int_0^h \|\partial_a^2 v(\cdot+te_a)\|_p^p\,\omega(t)\,\mathrm{d}t\Big)^{1/p} = \frac{h^2}{2}\|\partial_a^2 v\|_p.$$

Hence $h\|\partial_a v\|_p \le 2\|v\|_p + (h^2/2)\|\partial_a^2 v\|_p$. Since $|\nabla v| \le \sum_a |\partial_a v|$ and $|\partial_a^2 v| \le |D^2 v|$, summing over a yields $\|\nabla v\|_p \le 2dh^{-1}\|v\|_p + (dh/2)\|D^2 v\|_p$. The choice $h = 2\varepsilon/d$ proves the lemma with $C_{d,\varepsilon} = d^2/\varepsilon$. □

For a compact set $F\subset\mathbb{R}^d$ and $0 < \alpha \le 1$, the Hölder norm of a function v on F is

$$\|v\|_{C^{0,\alpha}(F)} = \sup_F |v| + \sup_{x,y\in F,\ x\ne y} \frac{|v(x)-v(y)|}{|x-y|^\alpha}.$$

A continuously differentiable function on an open set U is of class $C^{1,\alpha}$ on a compact set $F\subset U$ if its first derivatives have finite $C^{0,\alpha}(F)$ norm, and it belongs to $C^{1,\alpha}_{\mathrm{loc}}(U)$ if it is of class $C^{1,\alpha}$ on every compact subset of U.

Lemma 6.2.6 (Morrey's inequality) *Let $d < p < \infty$, put $\alpha = 1 - d/p$, and let $B' \Subset B$ be Euclidean balls in $\mathbb{R}^d$. Every $v \in W^{1,p}(B)$ has a representative that is continuous on B and satisfies*

$$\|v\|_{C^{0,\alpha}(\overline{B'})} \le C\|v\|_{W^{1,p}(B)},$$

where C depends only on d, p, B', B. Consequently, for an open set $U \subset \mathbb{R}^d$, every function in $W^{2,p}_{\mathrm{loc}}(U)$ has a representative in $C^{1,\alpha}_{\mathrm{loc}}(U)$.

Proof We prove the estimate for smooth v, pass to general v by mollification, and then apply the result to first derivatives. Assume first that v is smooth on B. Let $Q \subset B$ be a ball of radius r, let $v_Q = |Q|^{-1}\int_Q v$ be the average of v over Q, and let $x \in Q$. We claim that

$$|v(x) - v_Q| \le C_d \int_Q |\nabla v(z)|\,|x - z|^{1-d}\,\mathrm{d}z.$$

For $z \in Q$, write $z = x + s\theta$ with $s = |z - x|$ and $\theta \in S^{d-1}$; then $|v(z) - v(x)| \le \int_0^s |\nabla v(x + t\theta)|\,\mathrm{d}t$, and $|v(x) - v_Q|$ is at most the average over $z \in Q$ of these segment integrals. Since Q is convex and open and contains x, for each θ the radii $s > 0$ with $x + s\theta \in Q$ form an interval $0 < s < R(\theta)$, and $R(\theta) \le 2r$. In polar coordinates about x ((2.1) on $B(x, 2r) \supset Q$), the average of the segment integrals is

$$\frac{1}{|Q|}\int_{S^{d-1}}\int_0^{R(\theta)} s^{d-1}\int_0^s |\nabla v(x + t\theta)|\,\mathrm{d}t\,\mathrm{d}s\,\mathrm{d}S(\theta).$$

Interchanging the integrals in s and t produces the factor $|Q|^{-1}\int_t^{R(\theta)} s^{d-1}\,\mathrm{d}s$, which is at most $(2r)^d/(d|Q|)$ and hence bounded by a constant depending only on d. The remaining integral $\int_{S^{d-1}}\int_0^{R(\theta)} |\nabla v(x + t\theta)|\,\mathrm{d}t\,\mathrm{d}S(\theta)$ is the integral on the right side of the claim, written in polar coordinates. This proves the claim.

By Hölder's inequality with $p' = p/(p-1)$, the right side of the claim is at most $C_{d,p} r^{1-d/p}\|\nabla v\|_{L^p(Q)}$, because

$$\int_Q |x - z|^{(1-d)p'}\,\mathrm{d}z \le \int_{B(x,2r)} |x - z|^{(1-d)p'}\,\mathrm{d}z = C_{d,p} r^{(1-d)p' + d},$$

and this integral is finite exactly when $(1 - d)p' > -d$, that is, when $p > d$. (For $d = 1$ the exponent $(1 - d)p'$ is zero.)

Let $\delta > 0$ be the distance from B' to $\mathbb{R}^d \setminus B$. If $x, y \in B'$ and $0 < |x - y| < \delta/2$, the ball $Q = B(x, 2|x - y|)$ lies in B and contains x and y. Applying the preceding bound at both points gives

$$|v(x) - v(y)| \le C_{d,p}|x - y|^{1-d/p}\|\nabla v\|_{L^p(B)}.$$

For $x \in B'$, the same bound on the ball $Q = B(x, \delta/2) \subset B$, together with the estimate $|v_Q| \le |Q|^{-1/p}\|v\|_{L^p(B)}$ from Hölder's inequality, gives $\sup_{B'} |v| \le C\|v\|_{W^{1,p}(B)}$ with C depending on d, p, δ. If $|x - y| \ge \delta/2$, then $|v(x) - v(y)|/|x - y|^\alpha \le 2(\delta/2)^{-\alpha}\sup_{B'} |v|$. These bounds prove the estimate for smooth v on B', and hence on $\overline{B'}$, since v is continuous on $B \supset \overline{B'}$.

For general $v \in W^{1,p}(B)$, choose a ball B'', depending only on B' and B, with $B' \Subset B'' \Subset B$, and let $v_\varepsilon = v * \rho_\varepsilon$, which is smooth on B'' for small ε. By the remark preceding Lemma 6.2.4, $\partial_a v_\varepsilon = (\partial_a v)_\varepsilon$, and $v_\varepsilon \to v$ and $\partial_a v_\varepsilon \to \partial_a v$ in $L^p(B'')$. The estimate for smooth functions, applied on $B' \Subset B''$ to the differences of the v_ε, shows that they form a Cauchy family in the norm of $C^{0,\alpha}(\overline{B'})$. Their uniform limit is continuous on $\overline{B'}$, agrees almost everywhere with v, and satisfies the asserted estimate, because $\|v\|_{W^{1,p}(B'')} \le \|v\|_{W^{1,p}(B)}$. Two continuous functions that agree almost everywhere on an open set agree everywhere on it. Hence the continuous representatives obtained from different balls $B' \Subset B$ agree on their overlaps and define a single continuous representative on B.

Finally, let $v \in W^{2,p}_{\mathrm{loc}}(U)$. On every ball $B \Subset U$, both v and its first derivatives $\partial_a v$ belong to $W^{1,p}(B)$. By the preceding paragraph, on each smaller ball $B' \Subset B$ the mollifications v_ε and $\partial_a v_\varepsilon = (\partial_a v)_\varepsilon$ converge uniformly to the continuous representatives $\tilde v$ of v and $\tilde w_a$ of $\partial_a v$. Passing to the limit in

$$v_\varepsilon(x + h e_a) - v_\varepsilon(x) = \int_0^h \partial_a v_\varepsilon(x + t e_a)\, \mathrm{d}t$$

shows that $\tilde v$ has the continuous partial derivatives $\partial_a \tilde v = \tilde w_a$. Hence $\tilde v$ is continuously differentiable, and by the first part its first derivatives are Hölder continuous with exponent α on each $\overline{B'}$. Every compact subset of U is covered by finitely many such balls B'; pairs of points closer than a Lebesgue number of this cover lie in a common ball, and for the remaining pairs the supremum bound suffices. Hence $\tilde v \in C^{1,\alpha}_{\mathrm{loc}}(U)$.□

6.3 Unitarily invariant measures

The incidence argument outlined at the beginning of the chapter averages over complex directions and over complex hyperplanes, so we need invariant probability measures on complex projective space and on the Grassmannian of hyperplanes. We construct them from the Gaussian measure on matrices, compute the measure of small balls in projective space, and introduce orthonormal frames on hyperplanes that depend on the hyperplane in a Borel way.

We identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ and write $\langle z, w\rangle = \sum_j z_j \overline{w_j}$ for the Hermitian inner product, so that the Euclidean inner product is $\operatorname{Re}\langle z, w\rangle$. The unitary group $\mathrm{U}(n)$ consists of the complex $n \times n$ matrices U with $U^* U = 1$, where U^* is the conjugate transpose; it is compact, being closed and bounded in the space of matrices.

Proposition 6.3.1 *There is a Borel probability measure η on $\mathrm{U}(n)$ that is invariant under left and right multiplication by elements of $\mathrm{U}(n)$. Let Y be a nonempty metric space on which $\mathrm{U}(n)$ acts continuously and transitively. Then Y is compact, and it carries a unique $\mathrm{U}(n)$-invariant Borel probability measure; for every $y_0 \in Y$, this measure is the image of η under the map $g \mapsto g y_0$. In particular, the unit sphere of*

$\mathbb{C}^n$ and the complex Grassmannians described below carry unique unitarily invariant probability measures.

Proof Identify the space of complex $n \times n$ matrices with $\mathbb{R}^{2n^2}$ through the real and imaginary parts of the entries, and let γ be the measure on it with density $e^{-\pi \operatorname{tr}(X^*X)}$ with respect to Lebesgue measure. Since $\operatorname{tr}(X^*X) = \sum_{j,k} |X_{jk}|^2$, this density is the Gaussian $e^{-\pi|x|^2}$ on $\mathbb{R}^{2n^2}$, so γ is a probability measure by Lemma 6.1.1 (with $\xi = 0$ and $t = 1$). The singular matrices form a γ-null set: they are the zero set of the real polynomial $|\det X|^2$, which is not identically zero since $\det 1 = 1$, and the zero set of such a polynomial is Lebesgue null.

To prove the last fact, let P be a real polynomial in $m \geq 1$ variables that is not identically zero; we argue by induction on m. If $m = 1$, the zero set is finite. If $m \geq 2$, write $P = \sum_k c_k(x') x_m^k$ as a polynomial in the last variable, with coefficients c_k that are polynomials in $x' = (x_1, \ldots, x_{m-1})$. Some c_k is not identically zero, so by induction the common zero set Z' of the coefficients is null in $\mathbb{R}^{m-1}$. For $x' \notin Z'$, the polynomial $P(x', \cdot)$ is not identically zero and has finitely many zeros. By Tonelli's theorem applied to the indicator function of the closed set $\{P = 0\}$, this set is null.

For invertible X define

$$Q(X) = X(X^*X)^{-1/2},$$

where $(X^*X)^{1/2}$ is the positive square root of the positive definite matrix X^*X. By the spectral theorem for Hermitian matrices, this square root exists and is unique. It depends continuously on the matrix. Indeed, let $A_k \to A$ be positive definite matrices. The square roots $A_k^{1/2}$ are bounded, since $\operatorname{tr}(A_k^{1/2} A_k^{1/2}) = \operatorname{tr} A_k$. Every convergent subsequence of them has a positive semidefinite limit S with $S^2 = A$; then S is invertible, hence positive definite, and $S = A^{1/2}$ by uniqueness. A bounded sequence of matrices all of whose convergent subsequences have the same limit converges to it, so $A_k^{1/2} \to A^{1/2}$. Since inversion is continuous on invertible matrices, Q is continuous on the open set of invertible matrices, and $Q(X)^*Q(X) = (X^*X)^{-1/2} X^*X (X^*X)^{-1/2} = 1$. Let η be the image under Q of the restriction of γ to the invertible matrices; it is a Borel probability measure on $\mathrm{U}(n)$.

Fix $U, V \in \mathrm{U}(n)$. The map $X \mapsto UXV$ is real linear and preserves $\operatorname{tr}(X^*X)$, so it is orthogonal on $\mathbb{R}^{2n^2}$ and preserves γ. By uniqueness of positive square roots, $(V^*X^*XV)^{1/2} = V^*(X^*X)^{1/2}V$, and therefore $Q(UXV) = UQ(X)V$. Consequently the image of η under $g \mapsto UgV$ is the image of γ under $X \mapsto Q(UXV)$, which is η because $X \mapsto UXV$ preserves γ. This proves the invariance of η on both sides.

Now let Y be as in the statement and fix $y_0 \in Y$. By transitivity, Y is the image of the compact space $\mathrm{U}(n)$ under the continuous map $g \mapsto gy_0$, so Y is compact, and hence separable. Let σ be the image of η under $g \mapsto gy_0$. For $h \in \mathrm{U}(n)$, the image of σ under h is the image of η under $g \mapsto hgy_0$, which is σ by left invariance; thus σ is invariant. Let f be continuous on Y. Every $y \in Y$ has the form hy_0, by transitivity, and right invariance gives

$$\int f(gy)\, \mathrm{d}\eta(g) = \int f(ghy_0)\, \mathrm{d}\eta(g) = \int f\, \mathrm{d}\sigma.$$

Let τ be any invariant Borel probability measure on Y. We integrate the last identity against τ and apply Fubini's theorem to the bounded continuous function $(g, y) \mapsto f(gy)$ on $\mathrm{U}(n) \times Y$, which is product-measurable by Lemma 1.1.8 because $\mathrm{U}(n)$ and Y are separable metric spaces. We obtain

$$\int f \,\mathrm{d}\sigma = \int_{\mathrm{U}(n)} \int_Y f(gy) \,\mathrm{d}\tau(y) \,\mathrm{d}\eta(g) = \int f \,\mathrm{d}\tau.$$

Thus σ and τ agree on continuous functions. For a nonempty closed set $F \subset Y$, the continuous functions $\max\{0, 1 - j \operatorname{dist}(y, F)\}$ decrease to $\mathbf{1}_F$, so dominated convergence gives $\sigma(F) = \tau(F)$. The closed sets are stable under finite intersections and generate the Borel σ-algebra, so Lemma 1.1.6(2) gives $\sigma = \tau$. In particular σ does not depend on y_0.

Finally, $\mathrm{U}(n)$ acts continuously on the unit sphere of $\mathbb{C}^n$, and transitively: every unit vector is the first column of a unitary matrix, obtained by completing it to an orthonormal basis with the Gram–Schmidt process. For the Grassmannians, these properties are verified below, where they are introduced. □

For an integer $0 \le k \le n$, we realize the complex Grassmannian $\mathrm{Gr}_{\mathbb{C}}(k, n)$ of k-dimensional complex subspaces of $\mathbb{C}^n$ as the set of Hermitian matrices P with $P^2 = P$ and $\operatorname{tr} P = k$. By the spectral theorem, such a matrix has the eigenvalues 0 and 1 only, so it is the orthogonal projection onto its image, which has dimension $\operatorname{tr} P = k$; conversely, the orthogonal projection onto a k-dimensional subspace is such a matrix. The set $\mathrm{Gr}_{\mathbb{C}}(k, n)$ is closed and bounded, since $\operatorname{tr}(P^*P) = \operatorname{tr} P = k$, and hence compact. The unitary group acts continuously by $P \mapsto UPU^*$, which replaces the image of P by its image under U. The action is transitive: given two k-dimensional subspaces, choose orthonormal bases of $\mathbb{C}^n$ whose first k vectors span them; the unitary matrix carrying one basis to the other maps one subspace onto the other.

Complex projective space $\mathbb{P}^{n-1}(\mathbb{C})$ is the case $k = 1$. For $\xi \in \mathbb{P}^{n-1}(\mathbb{C})$ we write P_ξ for the corresponding rank-one projection, and we denote the invariant probability measure on $\mathbb{P}^{n-1}(\mathbb{C})$ by σ. The Hopf map $q\colon S^{2n-1} \to \mathbb{P}^{n-1}(\mathbb{C})$ sends a unit vector e to the complex line it spans, that is, to the projection ee^*.

For $\xi, \xi' \in \mathbb{P}^{n-1}(\mathbb{C})$ with unit representatives e, e', set

$$d(\xi, \xi') = \arccos |\langle e, e' \rangle| \in [0, \pi/2];$$

this does not depend on the choice of representatives. We claim that d is a metric inducing the topology of $\mathbb{P}^{n-1}(\mathbb{C})$. Let $\delta(e, e') = \arccos \operatorname{Re}\langle e, e' \rangle \in [0, \pi]$ be the angle between two unit vectors, that is, their great-circle distance on S^{2n-1}. Since the maximum of $\operatorname{Re}\langle \mathrm{e}^{\mathrm{i}\theta} e, e' \rangle$ over θ is $|\langle e, e' \rangle|$ and arccos is decreasing, the number $d(\xi, \xi')$ is the minimum of δ over the unit representatives of ξ and ξ'.

The angle δ satisfies the triangle inequality. Indeed, let e, e', e'' be unit vectors, put $a = \delta(e, e')$ and $b = \delta(e', e'')$, and write $e = \cos a \, e' + \sin a \, u$ and $e'' = \cos b \, e' + \sin b \, w$ with unit vectors u, w orthogonal to e' for the Euclidean inner product. Then $\operatorname{Re}\langle e, e'' \rangle = \cos a \cos b + \sin a \sin b \operatorname{Re}\langle u, w \rangle \ge \cos(a + b)$ by the

Cauchy–Schwarz inequality. Hence $\delta(e,e'') \le a+b$ when $a+b \le \pi$, and this inequality is trivial when $a+b > \pi$. To deduce the triangle inequality for d, fix a unit representative e' of ξ', and choose the representatives e of ξ and e'' of ξ'' such that $\delta(e,e') = d(\xi,\xi')$ and $\delta(e',e'') = d(\xi',\xi'')$. Then

$$d(\xi,\xi'') \le \delta(e,e'') \le d(\xi,\xi') + d(\xi',\xi'').$$

The function d is symmetric, and $d(\xi,\xi') = 0$ forces equality in the Cauchy–Schwarz inequality $|\langle e,e'\rangle| \le 1$, hence $\xi = \xi'$. Finally, $\cos^2 d(\xi,\xi') = |\langle e,e'\rangle|^2 = \operatorname{tr}(P_\xi P_{\xi'})$ depends continuously on the two projections. Therefore the identity map from $\mathbb{P}^{n-1}(\mathbb{C})$, with its compact matrix topology, to the metric space $(\mathbb{P}^{n-1}(\mathbb{C}), d)$ is a continuous bijection onto a Hausdorff space, hence a homeomorphism. Since q is a continuous surjection from a compact space onto a Hausdorff space, this topology is also the quotient topology of the Hopf map.

Lemma 6.3.2 *Let $n \ge 1$, $0 < a \le \pi/2$ and $\xi_0 \in \mathbb{P}^{n-1}(\mathbb{C})$. Then*

$$\sigma\{\xi : d(\xi,\xi_0) < a\} = (\sin a)^{2n-2}.$$

In particular, this ball has σ-measure at most a^{2n-2}.

Proof If $n = 1$, then $\mathbb{P}^0(\mathbb{C})$ is a single point, and both sides equal 1; so let $n \ge 2$. Let γ be the probability measure on $\mathbb{C}^n = \mathbb{R}^{2n}$ with density $\mathrm{e}^{-\pi|z|^2}$. Unitary maps are orthogonal on $\mathbb{R}^{2n}$, so they preserve Lebesgue measure and $|z|$; hence γ is unitarily invariant. Moreover $\gamma(\{0\}) = 0$. The map $z \mapsto \mathbb{C}z$ from $\mathbb{C}^n \setminus \{0\}$ to $\mathbb{P}^{n-1}(\mathbb{C})$, that is, $z \mapsto zz^*/|z|^2$, is continuous and commutes with the action of $\mathrm{U}(n)$. Hence the image of γ under this map is a unitarily invariant Borel probability measure, and by the uniqueness in Proposition 6.3.1 it is σ.

The metric d is unitarily invariant, because unitary matrices preserve $\langle\cdot,\cdot\rangle$. Since σ is invariant and the unitary group acts transitively, we may therefore take $\xi_0 = \mathbb{C}e_1$, the first coordinate line. Write $z = (z_1, z')$ with $z' \in \mathbb{C}^{n-1}$. For the line $\xi = \mathbb{C}z$, the condition $d(\xi,\xi_0) < a$ means $|z_1| > \cos a\,|z|$, because arccos is strictly decreasing; that is, $|z_1|^2 \sin^2 a > \cos^2 a\,|z'|^2$, or $|z_1|^2 > \cot^2 a\,|z'|^2$, where $\cot(\pi/2) = 0$. Thus the preimage of the ball under $z \mapsto \mathbb{C}z$ is the set of $z \ne 0$ with $|z_1|^2 > \cot^2 a\,|z'|^2$. We compute its γ-measure by Fubini's theorem, integrating first in z_1. For $c \ge 0$, polar coordinates in $\mathbb{C}$ give

$$\int_{\{w\in\mathbb{C}:|w|^2>c\}} \mathrm{e}^{-\pi|w|^2}\,\mathrm{d}w = 2\pi\int_{\sqrt{c}}^{\infty} r\mathrm{e}^{-\pi r^2}\,\mathrm{d}r = \mathrm{e}^{-\pi c},$$

and we apply this with $c = \cot^2 a\,|z'|^2$. Using Lemma 6.1.1 in $\mathbb{C}^{n-1} = \mathbb{R}^{2n-2}$ with $\xi = 0$ and $t = 1+\cot^2 a$, we obtain

$$\sigma\{\xi : d(\xi,\xi_0) < a\} = \int_{\mathbb{C}^{n-1}} \mathrm{e}^{-\pi(1+\cot^2 a)|z'|^2}\,\mathrm{d}z' = (1+\cot^2 a)^{-(n-1)} = (\sin a)^{2n-2}.$$

The last assertion follows from $\sin a \le a$. □

We write $\mathcal{H} = \mathrm{Gr}_{\mathbb{C}}(n-1, n)$ for the space of complex hyperplanes through the origin in $\mathbb{C}^n$, and ν for its invariant probability measure. For $H \in \mathcal{H}$, the orthogonal projection P_H onto H is the matrix representing H, and the orthogonal complement $H^\perp$ is a complex line. Let m_H be the Lebesgue measure of H, that is, the image of Lebesgue measure on $\mathbb{C}^{n-1} = \mathbb{R}^{2n-2}$ under any complex linear isometry $\mathbb{C}^{n-1} \to H$.

To integrate functions that depend on H, we use orthonormal frames of H that depend on H in a Borel way. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and order the $(n-1)$-element subsets I of $\{1, \dots, n\}$. For each I, let W_I be the set of hyperplanes H for which the vectors $P_H e_i$, $i \in I$, are linearly independent. Since $\langle P_H e_i, P_H e_j \rangle = \langle P_H e_i, e_j \rangle$, the Gram determinant of these vectors is the principal minor of P_H with indices in I, so W_I is open. These sets cover $\mathcal{H}$, because the n vectors $P_H e_1, \dots, P_H e_n$ span the $(n-1)$-dimensional space H, so that $n-1$ of them form a basis. Let W_I' be the Borel set of hyperplanes H for which I is the first index with $H \in W_I$; these finitely many sets partition $\mathcal{H}$. For $H \in W_I'$, the Gram–Schmidt process applied to $(P_H e_i)_{i \in I}$ gives an orthonormal basis of H, and hence a complex linear isometry $T_H \colon \mathbb{C}^{n-1} \to H$.

The Gram–Schmidt process depends continuously on linearly independent families, so the map $(H, w) \mapsto T_H w$ is continuous on each set $W_I' \times \mathbb{C}^{n-1}$. Hence the preimage of an open subset of $\mathbb{C}^n$ under this map is a finite union of relatively open subsets of these Borel sets, and $(H, w) \mapsto T_H w$ is a Borel map from $\mathcal{H} \times \mathbb{C}^{n-1}$ to $\mathbb{C}^n$. A map into a product of two separable metric spaces whose components are Borel is Borel, because the preimage of a product of Borel sets is Borel and these products generate the Borel sets of the product (Lemma 1.1.8). In particular, $(H, w) \mapsto (H, T_H w)$ is a Borel map into $\mathcal{H} \times \mathbb{C}^n$. Consequently, for every nonnegative Borel function f on the closed set $\{(H, c) \in \mathcal{H} \times \mathbb{C}^n : c \in H\}$, the function

$$H \longmapsto \int_H f(H, c)\, \mathrm{d}m_H(c) = \int_{\mathbb{C}^{n-1}} f(H, T_H w)\, \mathrm{d}w$$

is Borel, by Tonelli's theorem (Proposition 1.1.7), applied with respect to ν and Lebesgue measure to the Borel function $(H, w) \mapsto f(H, T_H w)$, which is product-measurable by Lemma 1.1.8, since $\mathcal{H}$ and $\mathbb{C}^{n-1}$ are separable metric spaces. The value of the integral does not depend on the frame, because a change of orthonormal frame preserves Lebesgue measure.

6.4 Circle selection and high-value sets

An affine complex line in $\mathbb{C}^n$ is a set $L = a + \mathbb{C}b$ with $a \in \mathbb{C}^n$ and $b \in \mathbb{C}^n \setminus \{0\}$. For an open set $D \subset \mathbb{C}^n$, we identify the section $D \cap L$ with the open set $\{\zeta \in \mathbb{C} : a + \zeta b \in D\}$, and we call a function on a connected component of $D \cap L$ subharmonic if its composite with $\zeta \mapsto a + \zeta b$ is subharmonic in the sense of Definition 3.1.1.

In the plane, the spherical means of (3.1) are averages with respect to arc length: the surface measure S on the unit circle (Section 2.1) is the image of Lebesgue measure on $[0, 2\pi)$ under $\theta \mapsto \mathrm{e}^{\mathrm{i}\theta}$. Indeed, by the change of variables formula for $(t, \theta) \mapsto t\mathrm{e}^{\mathrm{i}\theta}$,

whose Jacobian determinant is t, the cone over an arc $\{e^{i\theta} : \theta_1 \le \theta < \theta_2\}$ with $0 \le \theta_1 \le \theta_2 \le 2\pi$ has area $(\theta_2 - \theta_1)/2$, so the arc has S-measure $\theta_2 - \theta_1$. These arcs are closed under finite intersections, and every open subset of the circle is a countable union of them, so they generate the Borel sets of the circle. Since the whole circle is such an arc, Lemma 1.1.6(2) identifies the two finite measures. Hence

$$M_w(\zeta, r) = \frac{1}{2\pi} \int_0^{2\pi} w(\zeta + re^{i\theta})\, d\theta$$

for an upper semicontinuous function w on an open subset of $\mathbb{C}$ containing $\overline{B(\zeta, r)}$.

Consequently, subharmonicity on $D \cap L$ does not depend on the parametrization. Two parametrizations differ by a map $\psi(\zeta) = \alpha\zeta + \beta$ with $\alpha \ne 0$, which is a homeomorphism and maps the circle of radius r about ζ onto the circle of radius $|\alpha|r$ about $\psi(\zeta)$. Since $\psi(\zeta + re^{i\theta}) = \psi(\zeta) + |\alpha| re^{i(\theta + \arg \alpha)}$ and Lebesgue measure on the circle is invariant under rotations, the arc-length formula gives $M_{w\circ\psi}(\zeta, r) = M_w(\psi(\zeta), |\alpha|r)$.

In the proof of Theorem 6.4.2 we must choose, measurably in the circle, a point of each circle at which a given function is large. The sets of admissible points will be closed and have positive length, and for such sets the arc-length median provides an explicit Borel choice.

Lemma 6.4.1 *Let T be a separable metric space, and let $F \subset T \times [0, 2\pi]$ be a Borel set whose sections $F_t = \{\theta : (t, \theta) \in F\}$ are closed and have positive Lebesgue measure for every $t \in T$. Then there is a Borel map $a : T \to [0, 2\pi]$ such that $a(t) \in F_t$ for every $t \in T$.*

Proof Write $|\cdot|$ for Lebesgue measure on $[0, 2\pi]$, and put $\ell(t) = |F_t|$ and $A_t(b) = |F_t \cap [0, b]|$ for $b \in [0, 2\pi]$. We first show that, for fixed b, the function $t \mapsto A_t(b)$ is Borel; with $b = 2\pi$, so is ℓ. Since T and $[0, 2\pi]$ are separable metric spaces, the Borel sets of $T \times [0, 2\pi]$ form the σ-algebra generated by the products of Borel sets (Lemma 1.1.8). Hence the sections G_t of a Borel set G are Borel, because the sets all of whose sections are Borel form a σ-algebra containing these products. The class of Borel sets $G \subset T \times [0, 2\pi]$ for which $t \mapsto |G_t \cap [0, b]|$ is Borel contains the products $B_1 \times B_2$ of Borel sets, for which this function is $\mathbf{1}_{B_1}(t)|B_2 \cap [0, b]|$. It is closed under complements, since the complement of G gives $b - |G_t \cap [0, b]|$, and under countable disjoint unions, which give sums. Since the products of Borel sets are closed under finite intersections, Lemma 1.1.6(1) shows that the class contains every Borel set.

For fixed t, the function A_t is nondecreasing and 1-Lipschitz, with $A_t(0) = 0$ and $A_t(2\pi) = \ell(t) > 0$. Define

$$a(t) = \inf\{b \in \mathbb{Q} \cap [0, 2\pi] : A_t(b) \ge \ell(t)/2\}.$$

Because A_t is continuous and nondecreasing and $0 < \ell(t)/2 < \ell(t)$, the set of all $b \in [0, 2\pi]$ with $A_t(b) \ge \ell(t)/2$ is an interval $[b_t, 2\pi]$ with $0 < b_t < 2\pi$ and $A_t(b_t) = \ell(t)/2$. The rational numbers are dense in this interval, so $a(t) = b_t$ and $A_t(a(t)) = \ell(t)/2$. The map a is Borel, since for every $c \in (0, 2\pi]$

$$\{t : a(t) < c\} = \bigcup_{b \in \mathbb{Q} \cap [0,c)} \{t : A_t(b) \geq \ell(t)/2\}.$$

Finally, suppose that $a(t) \notin F_t$. Since F_t is closed and $a(t) > 0$, some interval $[a(t) - \varepsilon, a(t)]$ with $\varepsilon > 0$ misses F_t, and A_t is constant on it. A rational number b in this interval with $b < a(t)$ then satisfies $A_t(b) = \ell(t)/2$, contradicting the definition of $a(t)$. Hence $a(t) \in F_t$. □

Theorem 6.4.2 *Let $D \subset \mathbb{C}^n$ be open, with $n \geq 2$, and let $x \in D$ and $R > 0$ satisfy $\overline{B(x,R)} \subset D$. Let $u : D \to [-\infty, \infty)$ be a Borel function such that, for every complex line L through x, the restriction of u to the connected component of $D \cap L$ containing x is subharmonic, and put $A = u(x)$. Then there is a Borel probability measure μ carried by the Borel set*

$$E = \{y \in \mathbb{C}^n : R/2 \leq |y - x| \leq R,\ u(y) \geq A\}$$

such that $I_{2n-2}(\mu) \leq C_n R^{-(2n-2)}$, where the constant C_n depends only on n.

Proof We obtain μ by choosing, in a Borel way, a point of E on each circle of radius $r \in [R/2, R]$ centered at x in a complex line through x, and then we estimate its energy. For $\xi \in \mathbb{P}^{n-1}(\mathbb{C})$ we first choose a unit vector $e_0(\xi) \in \xi$ in a Borel way: if j is the first index with $(P_\xi)_{jj} > 0$, put $e_0(\xi) = P_\xi e_j / \sqrt{(P_\xi)_{jj}}$. Since $|P_\xi e_j|^2 = \langle P_\xi e_j, e_j \rangle = (P_\xi)_{jj}$, this is a unit vector in ξ. The set of ξ with a given first index j is Borel, and e_0 is continuous on it.

Let $r \in [R/2, R]$ and $\xi \in \mathbb{P}^{n-1}(\mathbb{C})$. The circle $\theta \mapsto x + r\mathrm{e}^{\mathrm{i}\theta} e_0(\xi)$ lies in the closed disc $\{x + \zeta e_0(\xi) : |\zeta| \leq r\}$, which is connected, contains x and is contained in D; hence it lies in the connected component containing x of the section of D by the line $x + \xi$. On this component u is subharmonic; in particular it is upper semicontinuous and bounded above on the circle, and by Theorem 3.1.3(2) and the arc-length formula for circle means at the beginning of this section, it satisfies the submean inequality

$$A = u(x) \leq \frac{1}{2\pi} \int_0^{2\pi} u(x + r\mathrm{e}^{\mathrm{i}\theta} e_0(\xi))\, \mathrm{d}\theta.$$

Let

$$F_{r,\xi} = \{\theta \in [0, 2\pi] : u(x + r\mathrm{e}^{\mathrm{i}\theta} e_0(\xi)) \geq A\}.$$

This set is closed by upper semicontinuity, and it has positive length. If $A = -\infty$, it is the whole interval $[0, 2\pi]$. If A is finite and $F_{r,\xi}$ were a null set, then $A - u$ would be positive almost everywhere on the circle, so the circle mean of u would be strictly less than A, contradicting the submean inequality.

The set of (r, ξ, θ) with $\theta \in F_{r,\xi}$ is Borel, because u is Borel and the map $(r, \xi, \theta) \mapsto x + r\mathrm{e}^{\mathrm{i}\theta} e_0(\xi)$ into $\overline{B(x,R)}$ is Borel, being continuous on each of the finitely many Borel sets on which e_0 is continuous. Since $[R/2, R] \times \mathbb{P}^{n-1}(\mathbb{C})$ is a compact, hence separable, metric space, Lemma 6.4.1 gives a Borel map $(r, \xi) \mapsto \alpha(r, \xi)$ with $\alpha(r, \xi) \in F_{r,\xi}$. Put $e(r, \xi) = \mathrm{e}^{\mathrm{i}\alpha(r,\xi)} e_0(\xi)$ and $\Psi(r, \xi) = x + re(r, \xi)$. Then Ψ is Borel, hence product-measurable (Lemma 1.1.8), and it takes its values in E. Let μ

be the image under Ψ of the product of normalized Lebesgue measure on $[R/2, R]$ and σ. It is a Borel probability measure carried by E.

It remains to estimate the energy; the estimate does not depend on the choice of the phases $\alpha(r,\xi)$. For $r, r' \in [R/2, R]$ and unit vectors e, e' spanning ξ, ξ',

$$|re - r'e'|^2 = (r-r')^2 + rr'|e-e'|^2 \geq (r-r')^2 + \frac{R^2}{4}\bigl(2 - 2|\langle e, e'\rangle|\bigr),$$

where the identity follows by expanding both sides with $|e| = |e'| = 1$ and $|e-e'|^2 = 2 - 2\,\mathrm{Re}\langle e, e'\rangle$, and the inequality holds since $rr' \geq R^2/4$ and $\mathrm{Re}\langle e, e'\rangle \leq |\langle e, e'\rangle|$. With $a = d(\xi, \xi') \in [0, \pi/2]$, we have $2 - 2|\langle e, e'\rangle| = 4\sin^2(a/2) \geq 4a^2/\pi^2$, because $\sin y \geq 2y/\pi$ for $0 \leq y \leq \pi/2$ by concavity of the sine on this interval. Hence $|re - r'e'|^2 \geq (r-r')^2 + R^2a^2/\pi^2$.

Write $r = Rt$ with $t \in [1/2, 1]$. Equip $[1/2, 1] \times \mathbb{P}^{n-1}(\mathbb{C})$ with the metric ρ defined by $\rho((t,\xi),(t',\xi'))^2 = (t-t')^2 + d(\xi,\xi')^2$, and with the product probability m of normalized Lebesgue measure and σ, which is defined on all Borel sets of this separable metric space by Lemma 1.1.8. Then μ is the image of m under $(t,\xi) \mapsto \Psi(Rt,\xi)$, and the lower bound for $|re - r'e'|^2$, together with $\pi > 1$, gives

$$|\Psi(Rt,\xi) - \Psi(Rt',\xi')| \geq \frac{R}{\pi}\,\rho((t,\xi),(t',\xi')).$$

A ρ-ball of radius $s \leq 1$ is contained in the product of an interval of length $2s$, whose normalized measure is at most $4s$, and a ball of radius s for the metric d. Since $s \leq 1 < \pi/2$, Lemma 6.3.2 shows that its m-measure is at most $4s^{2n-1}$. In particular m has no atoms, so the diagonal, a closed and hence product-measurable set (Lemma 1.1.8), is null for $m \times m$ by Tonelli's theorem. Fix $z \in [1/2, 1] \times \mathbb{P}^{n-1}(\mathbb{C})$. Splitting the integral into the set where $\rho(z,z') \geq 1$ and the shells where $2^{-k-1} \leq \rho(z,z') < 2^{-k}$, $k \geq 0$, we obtain

$$\int \rho(z,z')^{-(2n-2)}\,\mathrm{d}m(z') \leq 1 + \sum_{k=0}^{\infty} 2^{(k+1)(2n-2)} \cdot 4 \cdot 2^{-k(2n-1)} = 1 + 2^{2n+1}.$$

Since μ is the image of m, the product $\mu \times \mu$ is the image of $m \times m$ under the product map, because both measures agree on rectangles (Proposition 1.1.7). Therefore the lower bound for $|\Psi(Rt,\xi) - \Psi(Rt',\xi')|$, valid off the $m \times m$-null diagonal, and an integration in z give

$$I_{2n-2}(\mu) \leq \Bigl(\frac{\pi}{R}\Bigr)^{2n-2} \iint \rho(z,z')^{-(2n-2)}\,\mathrm{d}m(z')\,\mathrm{d}m(z) \leq \pi^{2n-2}(1+2^{2n+1})R^{-(2n-2)}.$$

This is the assertion, with $C_n = \pi^{2n-2}(1 + 2^{2n+1})$. □

6.5 Projection densities and incidence

We now show that a measure of finite Riesz $(2n-2)$-energy on $\mathbb{C}^n$, $n \geq 2$, projects onto almost every complex hyperplane with a square-integrable density, and we deduce the incidence theorem. The first step expresses averages over hyperplanes as a weighted integral over $\mathbb{C}^n$.

Lemma 6.5.1 *Let $n \geq 2$. For every nonnegative Borel function f on $\mathbb{C}^n$,*

$$\int_{\mathcal{H}} \int_H f(\xi)\, \mathrm{d}m_H(\xi)\, \mathrm{d}\nu(H) = a_n \int_{\mathbb{C}^n} f(\xi)|\xi|^{-2}\, \mathrm{d}\xi, \qquad a_n = \frac{s_{2n-3}}{s_{2n-1}}.$$

Proof We write the inner integral in polar coordinates on H and show that the normalized surface measures of the unit spheres of the hyperplanes H average to the normalized surface measure of S^{2n-1}. For $H \in \mathcal{H}$, let S_H be the unit sphere of H, and let τ_H be the image under T_H of the normalized surface measure $\mathrm{d}S/s_{2n-3}$ of the unit sphere S^{2n-3} of $\mathbb{C}^{n-1} = \mathbb{R}^{2n-2}$. We regard τ_H as a Borel probability measure on S^{2n-1} carried by S_H; it does not depend on the frame, since surface measure is invariant under orthogonal transformations (Section 2.1). Polar coordinates (2.1) on all of $\mathbb{C}^{n-1}$, transported to H by T_H, give

$$\int_H f\, \mathrm{d}m_H = s_{2n-3} \int_0^\infty r^{2n-3} \int f(r\theta)\, \mathrm{d}\tau_H(\theta)\, \mathrm{d}r.$$

Measurability. Let h be a nonnegative Borel function on $(0,\infty) \times S^{2n-1}$. By the definition of τ_H,

$$\int h(r,\theta)\, \mathrm{d}\tau_H(\theta) = \frac{1}{s_{2n-3}} \int_{S^{2n-3}} h(r, T_H\omega)\, \mathrm{d}S(\omega). \tag{6.4}$$

The map $(H, r, \omega) \mapsto (r, T_H\omega)$ is Borel by the frame construction of Section 6.3, since its two components are Borel. Hence $(H, r, \omega) \mapsto h(r, T_H\omega)$ is a nonnegative Borel function on the product of the separable metric spaces $\mathcal{H} \times (0,\infty)$ and S^{2n-3}, and by Lemma 1.1.8 and Tonelli's theorem the right side of (6.4) is a Borel function of (H, r). In particular, taking $h(r,\theta) = g(\theta)$, we see that $H \mapsto \int g\, \mathrm{d}\tau_H$ is Borel for every nonnegative Borel function g on S^{2n-1}; for instance, $H \mapsto \tau_H(B)$ is Borel for every Borel set $B \subset S^{2n-1}$.

Averaging. For a Borel set $B \subset S^{2n-1}$ put $\tau(B) = \int \tau_H(B)\, \mathrm{d}\nu(H)$. By monotone convergence, τ is countably additive, so it is a Borel probability measure on S^{2n-1}. For a unitary matrix U, the map UT_H is a complex linear isometry from $\mathbb{C}^{n-1}$ onto UH, so U maps τ_H to τ_{UH}, that is, $\tau_H(U^{-1}B) = \tau_{UH}(B)$. Since ν is invariant, it follows that $\tau(U^{-1}B) = \int \tau_{UH}(B)\, \mathrm{d}\nu(H) = \tau(B)$, so τ is unitarily invariant. The normalized surface measure $\mathrm{d}S/s_{2n-1}$ of S^{2n-1} is also unitarily invariant, because unitary matrices are orthogonal transformations of $\mathbb{R}^{2n}$. By the uniqueness in Proposition 6.3.1, $\tau = \mathrm{d}S/s_{2n-1}$. By the definition of τ for indicator functions, by linearity for simple

functions, and by monotone convergence in general, $\int_{\mathcal{H}} \int g \, d\tau_H \, d\nu(H) = \int g \, d\tau$ for every nonnegative Borel function g on S^{2n-1}.

Conclusion. Integrate the polar-coordinate identity over H with respect to ν. The function $(r,\theta) \mapsto f(r\theta)$ is Borel on $(0,\infty) \times S^{2n-1}$, so by the measurability step $(H,r) \mapsto r^{2n-3} \int f(r\theta) \, d\tau_H(\theta)$ is a nonnegative Borel function on $\mathcal{H} \times (0,\infty)$, which is product-measurable by Lemma 1.1.8. Tonelli's theorem therefore allows us to integrate first over H, and the averaging identity with $g(\theta) = f(r\theta)$ shows that the left side of the lemma equals

$$s_{2n-3} \int_0^\infty r^{2n-3} \int_{S^{2n-1}} f(r\theta) \, \frac{dS(\theta)}{s_{2n-1}} \, dr.$$

In polar coordinates $\xi = r\theta$ on all of $\mathbb{C}^n = \mathbb{R}^{2n}$, (2.1) gives $|\xi|^{-2} \, d\xi = r^{2n-3} \, dr \, dS(\theta)$, so this is the right side. All integrals may be infinite. □

Theorem 6.5.2 *Let $n \geq 2$, and let μ be a finite positive Borel measure on $\mathbb{C}^n$ with $I_{2n-2}(\mu) < \infty$. Then for ν-almost every $H \in \mathcal{H}$, the projected measure $(P_H)_*\mu$ has a density $g_H \in L^2(H, m_H)$ with respect to m_H, and*

$$\int_{\mathcal{H}} \|g_H\|_2^2 \, d\nu(H) = \beta_n I_{2n-2}(\mu), \qquad \beta_n = \frac{a_n}{b_{2n,2n-2}} = \frac{\Gamma(n)}{\pi^{n-1}} > 0.$$

Proof On H we use the Fourier transform defined by the Euclidean inner product $\mathrm{Re}\langle \cdot, \cdot \rangle$ and the measure m_H; in an orthonormal frame it is the transform of Section 6.1 on $\mathbb{R}^{2n-2}$. For $y \in \mathbb{C}^n$ and $\xi \in H$ we have $\mathrm{Re}\langle P_H y, \xi \rangle = \mathrm{Re}\langle y, \xi \rangle$, because $y - P_H y$ is orthogonal to H. Hence the Fourier transform of $(P_H)_*\mu$ at $\xi \in H$ is $\int e^{-2\pi i \, \mathrm{Re}\langle P_H y, \xi \rangle} \, d\mu(y) = \widehat{\mu}(\xi)$; that is, it is the restriction of $\widehat{\mu}$ to H. Apply Lemma 6.5.1 to the continuous function $|\widehat{\mu}|^2$, and then Theorem 6.1.4 with $d = 2n$ and $s = 2n-2$, for which $|\xi|^{s-d} = |\xi|^{-2}$. We obtain

$$\int_{\mathcal{H}} \int_H |\widehat{\mu}(\xi)|^2 \, dm_H(\xi) \, d\nu(H) = a_n \int_{\mathbb{C}^n} |\widehat{\mu}(\xi)|^2 |\xi|^{-2} \, d\xi = \beta_n I_{2n-2}(\mu) < \infty.$$

The inner integral is a Borel function of H, by the frame construction in Section 6.3; hence it is finite for ν-almost every H. For each such H, we apply Proposition 6.1.3 on $\mathbb{C}^{n-1} = \mathbb{R}^{2n-2}$ to the image of μ under $T_H^{-1} P_H$. Since T_H preserves $\mathrm{Re}\langle \cdot, \cdot \rangle$, the Fourier transform of this image is $w \mapsto \widehat{\mu}(T_H w)$, whose squared L^2 norm is $\int_H |\widehat{\mu}|^2 \, dm_H < \infty$. Hence the image has a density in $L^1 \cap L^2$, and composing it with T_H^{-1} we obtain a density g_H of $(P_H)_*\mu$ with respect to m_H, with $\|g_H\|_2^2 = \int_H |\widehat{\mu}|^2 \, dm_H$. In particular, $H \mapsto \|g_H\|_2^2$ is Borel, being equal to this Fourier integral, so no measurable choice of the densities themselves is needed. Integrating over H proves the identity.

To compute β_n, recall from (6.2) that $s_{k-1} = 2\pi^{k/2}/\Gamma(k/2)$ for every $k \geq 1$. Hence $a_n = \Gamma(n)/(\pi\Gamma(n-1))$, while $b_{2n,2n-2} = \pi^{n-2}/\Gamma(n-1)$, and therefore $\beta_n = \Gamma(n)/\pi^{n-1}$. □

Theorem 6.5.3 *Let $n \geq 2$, let $K \subset \mathbb{C}^n$ be bounded, and let $E_j \subset K$, $j \geq 1$, be Borel sets. Suppose that there are Borel probability measures μ_j on $\mathbb{C}^n$ such that*

$\mu_j(E_j) = 1$ and $I_{2n-2}(\mu_j) \le C$ for a constant $C < \infty$ independent of j. Then some affine complex line meets E_j for infinitely many j.

Proof We parametrize the affine complex lines that can meet K by a compact set carrying a finite measure λ, and bound from below, independently of j, the λ-measure of a set of lines meeting E_j. Choose $M > 0$ with $K \subset B(0, M)$. Every affine complex line can be written as $c + H^\perp$ with $H \in \mathcal{H}$ and $c \in H$, and $c = P_H y$ for every point y of the line. Indeed, if the line is $a + \mathbb{C}b$, let H be the hyperplane orthogonal to b, so that $H^\perp = \mathbb{C}b$, and put $c = P_H a$; then $P_H(a + \zeta b) = c$ for every $\zeta \in \mathbb{C}$. If the line meets K at a point y, then $|c| = |P_H y| \le |y| < M$. Hence the lines that can meet K are parametrized by the set

$$\mathcal{L}_M = \{(H, c) : H \in \mathcal{H},\ c \in H,\ |c| \le M\},$$

which is compact, being closed in $\mathcal{H} \times \overline{B(0, M)}$ because $c \in H$ means $P_H c = c$. We equip $\mathcal{L}_M$ with the finite measure λ defined by

$$\int h \, \mathrm{d}\lambda = \int_{\mathcal{H}} \int_{H \cap \overline{B(0,M)}} h(H, c) \, \mathrm{d}m_H(c) \, \mathrm{d}\nu(H)$$

for nonnegative Borel functions h. In the frames of Section 6.3, λ is the image of the product of ν and Lebesgue measure on the closed ball of radius M in $\mathbb{C}^{n-1}$. Thus it is a Borel measure, and its total mass is the volume of a ball of radius M in $\mathbb{R}^{2n-2}$.

By inner regularity (Proposition 1.1.2), choose compact sets $F_j \subset E_j$ with $\mu_j(F_j) \ge 1/2$. The set

$$J_j = \{(H, c) : H \in \mathcal{H},\ c \in P_H F_j\} \subset \mathcal{L}_M$$

parametrizes the lines meeting F_j. It is compact, as the image of $\mathcal{H} \times F_j$ under the continuous map $(H, y) \mapsto (H, P_H y)$. Its section over H is $P_H F_j$, so $\lambda(J_j) = \int m_H(P_H F_j) \, \mathrm{d}\nu(H)$. Let $g_{j,H}$ be the densities given by Theorem 6.5.2 for μ_j. Since $F_j \subset P_H^{-1}(P_H F_j)$, the Cauchy–Schwarz inequality gives, for ν-almost every H,

$$\frac{1}{2} \le \mu_j(F_j) \le (P_H)_* \mu_j(P_H F_j) = \int_{P_H F_j} g_{j,H} \, \mathrm{d}m_H \le \|g_{j,H}\|_2 \, m_H(P_H F_j)^{1/2}.$$

Integrate this inequality over H with respect to the probability measure ν, and apply the Cauchy–Schwarz inequality once more. Squaring the result and using Theorem 6.5.2, we obtain

$$\frac{1}{4} \le \lambda(J_j) \int_{\mathcal{H}} \|g_{j,H}\|_2^2 \, \mathrm{d}\nu(H) \le \lambda(J_j) \beta_n C.$$

In particular $C > 0$, and $\lambda(J_j) \ge \delta = 1/(4\beta_n C)$ for every j. The sets $\bigcup_{j \ge m} J_j$ decrease in m and have measure at least δ. Since λ is finite, continuity from above gives

$$\lambda\left(\bigcap_m \bigcup_{j\ge m} J_j\right) \ge \delta > 0,$$

so this set is nonempty. If (H, c) belongs to it, then $c \in P_H F_j$ for infinitely many j. For each such j there is $y \in F_j$ with $c = P_H y$, so $y \in c + H^\perp$. Hence the line $c + H^\perp$ meets F_j, and therefore E_j, for infinitely many j. □

Corollary 6.5.4 *Let $D \subset \mathbb{C}^n$ be open, with $n \ge 1$, and let $u\colon D \to [-\infty, \infty)$ be a Borel function whose restriction to every connected component of every affine complex-line section of D is subharmonic. Then u is locally bounded above on D.*

Proof If $n = 1$, the only affine complex line is $\mathbb{C}$ itself. Thus u is subharmonic on every component of D, hence upper semicontinuous and locally bounded above.

Let $n \ge 2$, and suppose that u is not bounded above on any neighborhood of some point $p \in D$. Then there are points $x_j \to p$ with $A_j = u(x_j) \to \infty$. Choose $R > 0$ with $\overline{B(p, 2R)} \subset D$; discarding finitely many terms, we may assume that $|x_j - p| < R$ for all j. Then $\overline{B(x_j, R)} \subset B(p, 2R) \subset D$, and Theorem 6.4.2 gives Borel probability measures μ_j carried by the sets

$$E_j = \{y \in \mathbb{C}^n : R/2 \le |y - x_j| \le R,\ u(y) \ge A_j\},$$

with $I_{2n-2}(\mu_j) \le C_n R^{-(2n-2)}$ for every j. All the sets E_j lie in the bounded set $B(p, 2R)$. By Theorem 6.5.3, some affine complex line L meets E_j for all j in an infinite set J. Choose $y_j \in L \cap E_j$ for $j \in J$. Passing to a subsequence, we may assume that $y_j \to y \in L \cap \overline{B(p, 2R)} \subset D$. A small closed disc in L centered at y lies in D, so for large j the points y_j lie in the same connected component of $D \cap L$ as y. On this component u is subharmonic, hence upper semicontinuous, and therefore $\overline{\lim}_j u(y_j) \le u(y) < \infty$. This contradicts $u(y_j) \ge A_j \to \infty$. □

Borel measurability enters this argument through the construction of the measures in Theorem 6.4.2. In the proof of Theorem 8.1.3, measurability is first derived from the line hypothesis, and only then is Corollary 6.5.4 applied.

Part II
Pluripotential theory

Introduction to Part II

In several complex variables a positive Laplacian carries too little information. It is only the trace of the complex Hessian, whereas plurisubharmonicity requires positivity in every complex direction. Complex differential forms retain this directional information in a form that is invariant under holomorphic changes of coordinates, and currents extend the calculus to singular potentials. The first task of this part (Chapters 7 and 8) is to connect these invariant derivatives with the pointwise submean inequalities of Part I.

Once plurisubharmonic functions are available, the main difficulty is nonlinear: measures cannot in general be multiplied. For locally bounded potentials, the recursive construction of Bedford and Taylor (Chapter 9) defines products of their curvature currents, and continuity along decreasing sequences yields the symmetry of these products and the comparison principle. Plurisubharmonic functions are quasi-continuous with respect to the associated Monge–Ampère capacity, and the Dirichlet problem on a ball can be solved for increasingly singular measures as data. The capacity detects pluripolar sets (Chapter 10); extremal functions and locality in the plurifine topology are developed there before the boundedness hypothesis is removed, because all later truncation arguments depend on them.

A second route into singularities passes through holomorphic functions (Chapters 11 to 13). Solving a weighted $\bar{\partial}$-equation corrects the error introduced by a cutoff function, and a singular weight forces the correction to satisfy additional conditions. Extension from slices then controls integrability at a point and along a submanifold. These estimates lead to coherent multiplier ideals, to the analyticity of the level sets of Lelong numbers and to strong openness. Weighted Bergman kernels turn the same information into approximations with analytic singularities. In this way the algebraic description of a singularity stays tied to the analytic estimates that justify it.

The passage to compact Kähler manifolds (Chapter 14) introduces a global constraint. Local potentials of a closed positive current differ by pluriharmonic functions on overlaps, and a cohomology class need not contain a smooth positive representative. Global potentials, positive cohomology classes and singular Hermitian metrics provide the framework in which the local theory can still be used. Approximation and extension are established with explicit curvature losses and uniform constants before

they enter the global constructions; the chapter ends with a global form of Josefson's theorem on compact Hermitian manifolds, where no Kähler assumption is needed.

Non-pluripolar products (Chapter 15) are obtained by truncating the local potentials and keeping the product only where all of them are finite. Plurifine locality makes the truncations compatible across truncation levels and on overlaps, and mass bounds on compact manifolds make their limit finite. Discarding the pole sets changes the nature of the mass problem: even a nonzero closed positive current can have zero non-pluripolar mass in degree one. Comparison, monotonicity and convergence in capacity must therefore be proved anew for these products; they do not follow from Stokes' formula for bounded potentials.

Envelopes (Chapter 16) select potentials subject to an obstacle and to a constraint on their singularities. Their contact properties lead to mass inequalities for rooftop envelopes and to model potentials. Envelopes of obstacles with bounded metric Laplacian inherit this regularity, and on the contact set the non-pluripolar measure of a potential is then computed from the second derivatives of the obstacle. Relative capacities make integration by parts possible for potentials sharing an unbounded singularity, and the relative energy (Chapter 17) links the theory of envelopes with Monge–Ampère equations and Perron segments. The identity between ceilings and singularity envelopes of [PSWZ26, Theorem 1.2], including its zero-mass case, is proved once this theory of equations is in place. The last chapter (Chapter 18) develops convex duality, real Monge–Ampère measures and the geometry of ample toric varieties, as needed for the toric part of [Xia].

Two related questions recur throughout this part: which information about a singularity is retained by holomorphic integrability, and which by nonlinear mass? The local estimates and geometric constructions prepare both questions, and the global products and envelopes show that their answers differ. The classification of singularities built on these foundations is the subject of [Xia].

Chapter 7
Complex differential forms and positive currents

This chapter sets up the calculus on which Part II is built. The central object of the following chapters is the complex Hessian $(\partial^2 u/\partial z_j \partial \bar{z}_k)$ of a function u on an open subset of $\mathbb{C}^n$, or rather the $(1,1)$-form $\mathrm{dd}^{\mathrm{c}} u$ that it defines. Two features distinguish it from the Laplacian of Part I: its positivity is a condition in every complex direction, and it transforms naturally under holomorphic changes of coordinates. Complex differential forms express both features, and currents, that is, differential forms with distribution coefficients, allow us to apply dd^{c} to locally integrable functions such as $\log|f|^2$ with f holomorphic.

We begin with the elementary calculus of holomorphic functions, complex manifolds and differential forms of type (p,q). Positivity of a current is defined by duality from positivity of forms, which is a matter of linear algebra (Section 7.2); along the way we prove a pointwise form of the Hodge index theorem for mixed discriminants (Lemma 7.2.3). The basic analytic fact is Theorem 7.2.5: the coefficients of a positive current are measures, all dominated by a single positive measure, its trace measure. Consequently, on a manifold with a countable basis, every sequence of positive currents with locally bounded trace measures has a weakly convergent subsequence.

A closed positive $(1,1)$-current T is locally of the form $\mathrm{dd}^{\mathrm{c}} u$, with u unique up to pluriharmonic functions (Theorem 7.3.2), and positivity of $\mathrm{dd}^{\mathrm{c}} u$ survives composition with holomorphic maps. The fundamental example is the Poincaré–Lelong formula $\mathrm{dd}^{\mathrm{c}} \log|f|^2 = [\mathrm{div}(f)]$ of Theorem 7.4.5, which identifies the complex Hessian of $\log|f|^2$ with the current of integration over the zero set of f, counted with multiplicities. The chapter closes with pushforward and pullback of currents and with Hermitian metrics, which give trace measures independent of coordinates.

Historical remarks and references for this chapter are collected in Appendix B.7.

7.1 Complex manifolds and differential forms

Let $n \geq 0$ be an integer. On $\mathbb{C}^n$ we write $z_j = x_j + \mathrm{i}y_j$ for $1 \leq j \leq n$, and we use the complex partial derivatives and differentials

$$\partial_{z_j} = \tfrac{1}{2}(\partial_{x_j} - \mathrm{i}\partial_{y_j}), \qquad \partial_{\bar{z}_j} = \tfrac{1}{2}(\partial_{x_j} + \mathrm{i}\partial_{y_j}), \qquad \mathrm{d}z_j = \mathrm{d}x_j + \mathrm{i}\mathrm{d}y_j.$$

A smooth function f on an open subset of $\mathbb{C}^n$ is holomorphic if $\partial_{\bar{z}_j} f = 0$ for $1 \leq j \leq n$. Substituting the definitions, we find $\mathrm{d}f = \sum_j (\partial_{z_j} f \, \mathrm{d}z_j + \partial_{\bar{z}_j} f \, \mathrm{d}\bar{z}_j)$ for every smooth f. On each tangent space the forms $\mathrm{d}z_j$ are complex linear and the forms $\mathrm{d}\bar{z}_j$ are conjugate linear, and a real-linear form which is both complex linear and conjugate linear vanishes; hence f is holomorphic if and only if its real derivative is complex linear at every point. By the chain rule, compositions of holomorphic maps, that is, of maps with holomorphic components, are therefore holomorphic. A holomorphic function is holomorphic in each variable separately in the sense of one-variable theory, since it satisfies the Cauchy–Riemann equations in each variable, and its derivatives $\partial_{z_j} f$ are again holomorphic, because partial derivatives of smooth functions commute.

Let $n \geq 1$, and let f be holomorphic on a neighborhood of the closed polydisc $\{|z_j - a_j| \leq r,\ 1 \leq j \leq n\}$, where $a \in \mathbb{C}^n$ and $r > 0$. Applying the one-variable Cauchy formula in each variable in turn, we obtain the Cauchy formula on a polydisc: for $|z_j - a_j| < r$,

$$f(z) = \frac{1}{(2\pi\mathrm{i})^n} \int_{|\zeta_1 - a_1| = r} \cdots \int_{|\zeta_n - a_n| = r} \frac{f(\zeta)\, \mathrm{d}\zeta_1 \cdots \mathrm{d}\zeta_n}{\prod_j (\zeta_j - z_j)}.$$

For $|z_j - a_j| \leq s < r$, the geometric series $(\zeta_j - z_j)^{-1} = \sum_{k \geq 0} (z_j - a_j)^k (\zeta_j - a_j)^{-k-1}$ converge uniformly for $|\zeta_j - a_j| = r$. The product of these series is a multiple series which converges absolutely and uniformly on the torus $\{|\zeta_j - a_j| = r\}$; integrating it term by term, we obtain the power series expansion

$$f(z) = \sum_{\alpha \in \mathbb{N}^n} c_\alpha (z - a)^\alpha, \qquad |c_\alpha| \leq \frac{M}{r^{|\alpha|}}, \tag{7.1}$$

with

$$c_\alpha = \frac{1}{(2\pi\mathrm{i})^n} \int_{|\zeta_1 - a_1| = r} \cdots \int_{|\zeta_n - a_n| = r} \frac{f(\zeta)\, \mathrm{d}\zeta_1 \cdots \mathrm{d}\zeta_n}{\prod_j (\zeta_j - a_j)^{\alpha_j + 1}},$$

where $(z - a)^\alpha = \prod_j (z_j - a_j)^{\alpha_j}$, $|\alpha| = \alpha_1 + \cdots + \alpha_n$, and M is the maximum of $|f|$ on the torus $\{|\zeta_j - a_j| = r\}$. The series converges normally on every polydisc $\{|z_j - a_j| \leq s\}$ with $s < r$, since $\sum_\alpha (s/r)^{|\alpha|} = (1 - s/r)^{-n}$. Differentiating the Cauchy formula under the integral sign, we obtain integral formulas for all derivatives of f, and in particular $\partial_{z_1}^{\alpha_1} \cdots \partial_{z_n}^{\alpha_n} f(a) = \alpha_1! \cdots \alpha_n!\, c_\alpha$.

Grouping the terms of (7.1) by degree, we write $f(a + w) = \sum_{k \geq 0} P_k(w)$ for $|w| < r$, where P_k is a homogeneous polynomial of degree k; this is the homogeneous expansion of f at a. If $P_k = 0$ for all $k < m$, then

$$|f(a+w) - P_m(w)| \le 2^{m+n+1} M \Big(\frac{|w|}{r}\Big)^{m+1} \qquad \text{for } |w| \le \frac{r}{2}. \tag{7.2}$$

Indeed, with $t = |w|/r \le \frac{1}{2}$ we have $|w^\alpha| \le |w|^{|\alpha|}$ and $t^{|\alpha|} \le t^{m+1} 2^{m+1-|\alpha|}$ for $|\alpha| \ge m+1$, so the terms of degree at least $m+1$ contribute at most $M t^{m+1} 2^{m+1} \sum_\alpha 2^{-|\alpha|} = 2^{m+n+1} M t^{m+1}$. The expansion (7.1) also gives the identity theorem: a holomorphic function on a domain that vanishes on a nonempty open subset vanishes identically. Indeed, the set of points at which all derivatives $\partial_{z_1}^{\alpha_1} \cdots \partial_{z_n}^{\alpha_n} f$ vanish is nonempty, and it is closed because these derivatives are continuous. It is also open: at each of its points a all coefficients c_α vanish, by the formula for the derivatives at a, so f vanishes on a polydisc around a. Hence this set is the whole domain.

Definition 7.1.1 A complex manifold of dimension n is a paracompact Hausdorff space with an atlas of homeomorphisms onto open subsets of $\mathbb{C}^n$, whose coordinate changes are holomorphic with holomorphic inverses. A map between complex manifolds is holomorphic if its coordinate expressions are holomorphic. A complex submanifold is a closed subset that is locally a coordinate plane in suitable holomorphic coordinates.

The underlying real manifold has dimension $2n$. If $F = (F_1, \dots, F_m)$ is a holomorphic map from an open subset of $\mathbb{C}^n$ to $\mathbb{C}^m$, then $\mathrm{d}F_j = \sum_k \partial_{z_k} F_j \, \mathrm{d}z_k$, so the real derivative of F at a point a is the complex-linear map given by the complex Jacobian matrix $(\partial_{z_k} F_j(a))_{j,k}$. Consequently the inverse function theorem has the following holomorphic version.

Proposition 7.1.2 (1) *Let $U \subset \mathbb{C}^n$ be open, let $F\colon U \to \mathbb{C}^n$ be holomorphic, and let $a \in U$ be a point at which the complex Jacobian matrix $(\partial_{z_k} F_j(a))_{j,k}$ is invertible. Then there are open neighborhoods $U' \subset U$ of a and $V' \subset \mathbb{C}^n$ of $F(a)$ such that F maps U' bijectively onto V' and the inverse map $V' \to U'$ is holomorphic.*

(2) *Let $U \subset \mathbb{C}^m$ be open, let $a \in U$, and let $f_1, \dots, f_r$, $0 \le r \le m$, be holomorphic functions on U whose complex Jacobian matrix $(\partial_{z_k} f_j(a))_{1\le j\le r,\, 1\le k\le m}$ at a has rank r. Then there are indices $k_1 < \cdots < k_{m-r}$ such that the map $(f_1, \dots, f_r, z_{k_1}, \dots, z_{k_{m-r}})$ is biholomorphic from an open neighborhood of a onto an open subset of $\mathbb{C}^m$. Thus $f_1, \dots, f_r$ are the first r coordinates of a holomorphic chart near a.*

(3) *Let X be a complex manifold of dimension m, let $a \in X$, and let $f_1, \dots, f_r$ be holomorphic functions on a neighborhood of a whose complex Jacobian matrix at a, computed in a holomorphic chart, has rank r. Then $f_1, \dots, f_r$ are the first r coordinates of a holomorphic chart of X near a.*

Proof (1) The real derivative of F at a is the complex-linear map given by the invertible matrix $(\partial_{z_k} F_j(a))$, so it is invertible. By the real inverse function theorem there are open neighborhoods U' of a and V' of $F(a)$ such that F maps U' bijectively onto V', the inverse map $G\colon V' \to U'$ is smooth, and $\mathrm{d}G_w = (\mathrm{d}F_{G(w)})^{-1}$ for $w \in V'$. Since F is holomorphic, $\mathrm{d}F_{G(w)}$ is complex linear, and hence so is its inverse. Thus the real derivative of G is complex linear at every point, that is, G is holomorphic.

(2) If $r = 0$, the identity map has the required property. Otherwise some $r \times r$ minor of the matrix $(\partial_{z_k} f_j(a))$ is nonzero; let $k_1 < \cdots < k_{m-r}$ be the indices of the columns not used by this minor. The map $\Phi = (f_1, \ldots, f_r, z_{k_1}, \ldots, z_{k_{m-r}})$ is holomorphic on U. The last $m - r$ rows of its complex Jacobian matrix at a are the unit row vectors with entry 1 in the columns $k_1, \ldots, k_{m-r}$, so, expanding the determinant along these rows, we see that it equals the chosen minor up to sign; hence it is nonzero. By (1), Φ maps an open neighborhood of a bijectively onto an open subset of $\mathbb{C}^m$, with holomorphic inverse.

(3) The rank of the complex Jacobian matrix does not depend on the chart, by the chain rule, since coordinate changes have invertible complex Jacobian matrices. Let $\kappa\colon U \to \kappa(U) \subset \mathbb{C}^m$ be a holomorphic chart around a on which the f_j are defined. By (2), applied to the functions $f_j \circ \kappa^{-1}$ at $\kappa(a)$, there is a map Φ from an open neighborhood $V \subset \kappa(U)$ of $\kappa(a)$ onto an open subset of $\mathbb{C}^m$, biholomorphic and with first components $f_j \circ \kappa^{-1}$. Then $\Phi \circ \kappa$ is a homeomorphism of $\kappa^{-1}(V)$ onto an open subset of $\mathbb{C}^m$ whose first r components are $f_1, \ldots, f_r$. For every chart λ of the atlas, the coordinate changes $(\Phi \circ \kappa) \circ \lambda^{-1} = \Phi \circ (\kappa \circ \lambda^{-1})$ and $\lambda \circ (\Phi \circ \kappa)^{-1} = (\lambda \circ \kappa^{-1}) \circ \Phi^{-1}$ are compositions of holomorphic maps, so $\Phi \circ \kappa$ is a holomorphic chart. □

A complex submanifold Z of a complex manifold X that is locally a coordinate plane of dimension r is itself a complex manifold of dimension r. Indeed, the restrictions to Z of the charts in which Z is a coordinate plane $\mathbb{C}^r \times \{0\}$, followed by the projection to $\mathbb{C}^r$, form an atlas of Z whose coordinate changes are restrictions of holomorphic maps to coordinate planes. Moreover Z is paracompact, that is, every open cover of Z has a locally finite open refinement: given an open cover of Z by sets $Z \cap O_i$ with O_i open in X, a locally finite open refinement of the cover of X by the O_i and by $X \setminus Z$ restricts to a locally finite open refinement on Z.

Every connected component of a complex manifold X is open, since manifolds are locally connected, and it has a countable basis of its topology. To see this, refine the cover of X by all relatively compact coordinate balls to a locally finite open cover (U_a). Each U_a lies in a coordinate ball, hence in a single connected component, and its closure is compact, so it meets only finitely many members of the cover. Fix a member U_{a_0}, and call a member reachable if it is joined to U_{a_0} by a finite chain of members, each meeting the next. By the finiteness just noted, there are only countably many reachable members. The union of the reachable members and the union of the remaining ones are disjoint open sets covering X. Hence the first union is open and closed; it contains the connected component C of U_{a_0}, and it is contained in C because each member lies in a single component. Thus C is covered by countably many coordinate balls, and therefore has a countable basis.

In particular, connected manifolds and compact manifolds have a countable basis, whereas a general complex manifold may have uncountably many components, for instance the product of $\mathbb{C}^n$ with an uncountable discrete space. We state the countability of a basis as a hypothesis whenever a global subsequence or another countable selection requires it.

A differential k-form is a smooth field of alternating k-linear forms on the real tangent spaces, with complex values. In a chart with real coordinates $t_1, \ldots, t_N$, where N is the real dimension, every k-form is uniquely $\alpha = \sum_J \alpha_J \, \mathrm{d}t_J$ with smooth

coefficients α_J; here $J = (j_1 < \cdots < j_k)$ runs over the increasing multi-indices of length k, and $\mathrm{d}t_J = \mathrm{d}t_{j_1} \wedge \cdots \wedge \mathrm{d}t_{j_k}$. The pullback of α by a smooth map F is defined pointwise by $(F^*\alpha)_x(v_1, \ldots, v_k) = \alpha_{F(x)}(\mathrm{d}F_x v_1, \ldots, \mathrm{d}F_x v_k)$; it is multiplicative for the exterior product, and $F^*\mathrm{d}t_j = \mathrm{d}F_j$ for the components $F_j = t_j \circ F$. The exterior derivative is defined in coordinates by

$$\mathrm{d}\Big(\sum_J \alpha_J \,\mathrm{d}t_J\Big) = \sum_J \sum_l \partial_{t_l} \alpha_J \,\mathrm{d}t_l \wedge \mathrm{d}t_J.$$

It satisfies

$$\mathrm{d}(\alpha \wedge \gamma) = \mathrm{d}\alpha \wedge \gamma + (-1)^k \alpha \wedge \mathrm{d}\gamma, \qquad \mathrm{d}^2 = 0, \qquad F^*\mathrm{d}\alpha = \mathrm{d}F^*\alpha \tag{7.3}$$

for a k-form α, every form γ and every smooth map F between open subsets of coordinate spaces. Indeed, the Leibniz rule follows from the product rule for the coefficients, the sign $(-1)^k$ arising when $\mathrm{d}t_l$ is moved past the k factors of α. Next, $\mathrm{d}^2(f\,\mathrm{d}t_J) = \sum_{l,l'} \partial_{t_{l'}} \partial_{t_l} f \,\mathrm{d}t_{l'} \wedge \mathrm{d}t_l \wedge \mathrm{d}t_J = 0$, because the second derivatives of f are symmetric in l and l', whereas $\mathrm{d}t_{l'} \wedge \mathrm{d}t_l$ is antisymmetric. Finally, for a function f the chain rule gives $\mathrm{d}(f \circ F) = \sum_j (\partial_{t_j} f \circ F)\,\mathrm{d}F_j = F^*\mathrm{d}f$. Since $\mathrm{d}(\mathrm{d}F_{j_1} \wedge \cdots \wedge \mathrm{d}F_{j_k}) = 0$ by the Leibniz rule and $\mathrm{d}^2 = 0$, it follows that

$$\mathrm{d}F^*(f\,\mathrm{d}t_J) = \mathrm{d}\big((f \circ F)\,\mathrm{d}F_{j_1} \wedge \cdots \wedge \mathrm{d}F_{j_k}\big) = F^*\mathrm{d}f \wedge F^*\mathrm{d}t_J = F^*\mathrm{d}(f\,\mathrm{d}t_J).$$

Applied to coordinate changes, the last identity shows that the exterior derivative does not depend on the chart. Hence d is defined for forms on manifolds, and the rules (7.3) hold there, for smooth maps F between manifolds.

On a complex manifold of dimension n, in holomorphic coordinates $z_j = x_j + \mathrm{i}y_j$, the 1-forms $\mathrm{d}z_j$ and $\mathrm{d}\bar{z}_j$ form a basis of the complex-valued real-linear forms on each tangent space, so every k-form α decomposes uniquely as

$$\alpha = \sum_{p+q=k} \alpha^{p,q}, \qquad \alpha^{p,q} = \sum_{|I|=p,\,|J|=q} \alpha_{I,J}\,\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J,$$

where, for an increasing multi-index $I = (i_1 < \cdots < i_p)$, we write $\mathrm{d}z_I = \mathrm{d}z_{i_1} \wedge \cdots \wedge \mathrm{d}z_{i_p}$, and similarly $\mathrm{d}\bar{z}_J$. The form $\alpha^{p,q}$ is the component of bidegree, or type, (p,q) of α. A holomorphic change of coordinates $w = w(z)$ gives $\mathrm{d}w_j = \sum_k \partial_{z_k} w_j \,\mathrm{d}z_k$ and $\mathrm{d}\bar{w}_j = \sum_k \overline{\partial_{z_k} w_j}\,\mathrm{d}\bar{z}_k$, so it transforms the $\mathrm{d}z_j$ among themselves and the $\mathrm{d}\bar{z}_j$ among themselves. Hence this decomposition does not depend on the chart, and, for the same reason, pullback by holomorphic maps preserves bidegrees. Since $\mathrm{d}(\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J) = 0$ and $\mathrm{d}f = \sum_l (\partial_{z_l} f\,\mathrm{d}z_l + \partial_{\bar{z}_l} f\,\mathrm{d}\bar{z}_l)$, we have $\mathrm{d}(f\,\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J) = \sum_l (\partial_{z_l} f\,\mathrm{d}z_l + \partial_{\bar{z}_l} f\,\mathrm{d}\bar{z}_l) \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$. Thus, for a form α of type (p,q), the form $\mathrm{d}\alpha$ is the sum of a form $\partial\alpha$ of type $(p+1,q)$ and a form $\bar{\partial}\alpha$ of type $(p,q+1)$. Extending ∂ and $\bar{\partial}$ linearly to all forms, we have $\mathrm{d} = \partial + \bar{\partial}$. Comparing types in $\mathrm{d}^2 = 0$ gives

$$\partial^2 = \bar{\partial}^2 = 0, \qquad \partial\bar{\partial} = -\bar{\partial}\partial.$$

Complex conjugation exchanges the types (p,q) and (q,p), and a form is real if it equals its conjugate. We orient a complex manifold by the form $\mathrm{d}x_1 \wedge \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}x_n \wedge \mathrm{d}y_n$ in each holomorphic chart. These orientations agree on overlapping charts. Indeed, since $\frac{\mathrm{i}}{2}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j = \mathrm{d}x_j \wedge \mathrm{d}y_j$ and 2-forms commute with all forms, this form equals $c_n\,\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar{z}_1 \wedge \cdots \wedge \mathrm{d}\bar{z}_n$ for a constant $c_n \neq 0$ depending only on n. If w is a second holomorphic chart, then $\mathrm{d}w_1 \wedge \cdots \wedge \mathrm{d}w_n = \det(\partial_{z_k} w_j)\,\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$ by multilinearity and antisymmetry of the exterior product, and the conjugate identity holds for the $\mathrm{d}\bar{w}_j$. Hence the form defined by w is $|\det(\partial_{z_k} w_j)|^2 > 0$ times the form defined by z.

Let X be a complex manifold of dimension n, and let α be a $2n$-form on X with compact support whose coefficients are locally integrable, that is, Lebesgue integrable on compact subsets of charts. If $\operatorname{Supp}\alpha$ lies in a holomorphic chart, in which $\alpha = g\,\mathrm{d}x_1 \wedge \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}x_n \wedge \mathrm{d}y_n$, we put $\int_X \alpha = \int g\,\mathrm{d}V_{2n}$, the Lebesgue integral in the coordinates. This does not depend on the chart. Indeed, by the preceding paragraph, the coefficient $\tilde{g}$ of α in a second chart w satisfies $g = (\tilde{g} \circ w)\,|\det(\partial_{z_k} w_j)|^2$, and $|\det(\partial_{z_k} w_j)|^2$ is the real Jacobian determinant of the coordinate change $z \mapsto w$, so the change of variables formula gives $\int g\,\mathrm{d}V_{2n}(z) = \int \tilde{g}\,\mathrm{d}V_{2n}(w)$. For general α, we choose a partition of unity (χ_a) subordinate to a cover of X by charts (Proposition 7.1.3 below). Only finitely many χ_a are not identically zero on the compact set $\operatorname{Supp}\alpha$, and we put $\int_X \alpha = \sum_a \int_X \chi_a \alpha$. If (ψ_b) is a second such partition of unity, then, by the case of a single chart,

$$\sum_a \int_X \chi_a \alpha = \sum_{a,b} \int_X \chi_a \psi_b \alpha = \sum_b \int_X \psi_b \alpha,$$

so the integral is well defined. It is linear, and $\int_X \alpha \geq 0$ if α is a nonnegative multiple of the volume form at every point. The same definition applies to complex submanifolds, which are complex manifolds with their own complex orientation.

Proposition 7.1.3 *Let X be a complex manifold of dimension n. Then X admits smooth partitions of unity subordinate to every open cover and consisting of functions with compact support. For every compact set $K \subset X$ and every open set $U \supset K$ there is a smooth function $\chi\colon X \to [0,1]$ with compact support in U that equals one on a neighborhood of K. If $n \geq 1$ and α is a smooth $(2n-1)$-form with compact support on X, then $\int_X \mathrm{d}\alpha = 0$.*

Proof Since the connected components of X are open, it suffices to construct partitions of unity on each component; the union of the families so obtained is locally finite on X. So let X be connected. By the preceding discussion, X is covered by a sequence of coordinate balls $B_1, B_2, \ldots$ with compact closures. Put $K_1 = \overline{B_1}$, and, once the compact set K_j is defined, cover it by finitely many of the balls B_i and let K_{j+1} be the union of their closures and of $\overline{B_1}, \ldots, \overline{B_{j+1}}$. We obtain compact sets $K_1 \subset \operatorname{Int} K_2 \subset K_2 \subset \operatorname{Int} K_3 \subset \cdots$ with union X; put $K_j = \varnothing$ for $j \leq 0$.

Now let an open cover of X be given. For $j \geq 1$, the compact set $K_j \setminus \operatorname{Int} K_{j-1}$ lies in the open set $\operatorname{Int} K_{j+1} \setminus K_{j-2}$. Cover it by finitely many coordinate balls B' whose concentric enlargements B have compact closures contained in a member of

the given cover and in $\operatorname{Int} K_{j+1} \setminus K_{j-2}$. For each such pair choose, in the chart, a smooth function with values in $[0,1]$ and compact support in B that equals one on B' (Lemma 1.4.1), and extend it by zero to X. The resulting family is locally finite, since a point of $\operatorname{Int} K_i$ has the neighborhood $\operatorname{Int} K_i$, which meets the supports chosen at stage j only if $j \le i+1$. Its sum is therefore smooth, and it is positive, because the balls B' cover X. Dividing each function by this sum gives the required partition of unity.

For the cutoff function, cover K by finitely many relatively compact coordinate balls with closures in U, and let V be their union; thus $K \subset V$ and $\overline{V}$ is a compact subset of U. Let (χ_a) be a partition of unity with compact supports subordinate to the cover $\{V, X \setminus K\}$, and let χ be the sum of those χ_a whose support is not contained in $X \setminus K$. Each of these has its support in V; by local finiteness, only finitely many supports meet the compact set $\overline{V}$, so χ is a finite sum, smooth, with values in $[0,1]$ and compact support in $\overline{V} \subset U$. The supports of the remaining χ_a lie in $X \setminus K$, and their union F is closed, as a locally finite union of closed sets. On the neighborhood $X \setminus F$ of K we have $\chi = \sum_a \chi_a = 1$.

For the last assertion, consider first a compactly supported $(2n-1)$-form γ in a single chart. Then $\mathrm{d}\gamma$ is a sum of terms $\pm\partial_{t_l} g \, \mathrm{d}t_1 \wedge \cdots \wedge \mathrm{d}t_{2n}$, where $t_1, \ldots, t_{2n}$ are the real coordinates $x_1, y_1, \ldots, x_n, y_n$ and g is smooth with compact support; the integral of each term vanishes by Fubini's theorem and the fundamental theorem of calculus on lines parallel to the t_l-axis. Now let (χ_a) be a partition of unity subordinate to a cover by charts. Only finitely many χ_a are nonzero on the compact set $\operatorname{Supp}\alpha$, so $\alpha = \sum_a \chi_a \alpha$ is a finite sum of compactly supported forms in single charts, and $\int_X \mathrm{d}\alpha = \sum_a \int_X \mathrm{d}(\chi_a \alpha) = 0$. □

Cutoff functions also embed a compact manifold in a Euclidean space; later chapters use the resulting metric to measure distances to closed sets.

Lemma 7.1.4 *Let X be a compact complex manifold. Then there are an integer $N \ge 0$ and a smooth injective map $F\colon X \to \mathbb{R}^N$ which is a homeomorphism of X onto $F(X)$. Consequently, $d(x,y) = |F(x) - F(y)|$ is a metric on X which induces the topology of X.*

Proof If X is empty, take $N = 0$. Otherwise let n be the dimension of X. For each $x \in X$ choose a holomorphic chart $\kappa_x\colon U_x \to \mathbb{C}^n = \mathbb{R}^{2n}$ with $x \in U_x$. By Proposition 7.1.3, applied to the compact set $\{x\}$ and the open set U_x, there is a smooth function $\chi_x\colon X \to [0,1]$ with compact support in U_x which equals one on an open neighborhood W_x of x. Since X is compact, finitely many of the sets W_x cover X; we denote the corresponding charts, cutoff functions and neighborhoods by $\kappa_i\colon U_i \to \mathbb{R}^{2n}$, χ_i and W_i, $1 \le i \le k$. The map $\chi_i\kappa_i$, defined as the product on U_i and as 0 on $X \setminus U_i$, is smooth: it is smooth on U_i, it vanishes on $X \setminus \operatorname{Supp}\chi_i$, which is open because the compact set $\operatorname{Supp}\chi_i$ is closed in the Hausdorff space X, and these two open sets cover X because $\operatorname{Supp}\chi_i \subset U_i$. Put $N = k + 2nk$ and

$$F = (\chi_1, \ldots, \chi_k, \chi_1\kappa_1, \ldots, \chi_k\kappa_k)\colon X \to \mathbb{R}^N,$$

a smooth map. It is injective. Indeed, let $F(x) = F(y)$, and choose i with $x \in W_i$. Then $\chi_i(y) = \chi_i(x) = 1$, so $y \in \operatorname{Supp}\chi_i \subset U_i$, and $\chi_i\kappa_i(x) = \chi_i\kappa_i(y)$ gives $\kappa_i(x) = \kappa_i(y)$; since κ_i is injective on U_i, we obtain $x = y$.

If $C \subset X$ is closed, then C is compact, so $F(C)$ is compact and hence closed in $\mathbb{R}^N$. Thus the continuous bijection $F\colon X \to F(X)$ maps closed sets to closed sets, so its inverse is continuous, and F is a homeomorphism onto $F(X)$. Hence d, the Euclidean metric of $F(X)$ transported to X by F, is a metric on X that induces its topology. □

For computations in $\mathbb{C}^n$ we use the Euclidean Kähler form

$$\beta = \frac{\mathrm{i}}{2}\sum_{j=1}^{n} \mathrm{d}z_j \wedge \mathrm{d}\bar z_j, \qquad \frac{\beta^n}{n!} = \mathrm{d}V_{2n}. \tag{7.4}$$

Here $\mathrm{d}V_{2n} = \mathrm{d}x_1 \wedge \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}x_n \wedge \mathrm{d}y_n$ is the Euclidean volume form, which we identify with Lebesgue measure on $\mathbb{C}^n = \mathbb{R}^{2n}$; for $n = 0$, $\mathrm{d}V_0$ is the unit mass on the singleton $\mathbb{C}^0$. The second identity holds because $(\mathrm{i}/2)\mathrm{d}z_j \wedge \mathrm{d}\bar z_j = \mathrm{d}x_j \wedge \mathrm{d}y_j$ and these 2-forms commute and have square zero, so that in the expansion of β^n only the $n!$ products containing each of them once survive, each equal to $\mathrm{d}V_{2n}$.

The operator $\mathrm{dd^c}$ is normalized as in the Conventions:

$$\mathrm{dd^c}u = \frac{\mathrm{i}}{2\pi}\partial\bar\partial u = \frac{\mathrm{i}}{2\pi}\sum_{j,k} u_{j\bar k}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_k, \qquad \Delta u = 4\sum_j u_{j\bar j}. \tag{7.5}$$

Here $u_{j\bar k} = \partial_{z_j}\partial_{\bar z_k}u$, and Δ is the Laplacian of $\mathbb{R}^{2n}$; the matrix $(u_{j\bar k})$ is the complex Hessian of u. The same formulas define $\mathrm{dd^c}u$ for a distribution u on an open subset of $\mathbb{C}^n$, with distributional derivatives $u_{j\bar k}$; then $\mathrm{dd^c}u$ is a current in the sense of the next section. We also use the real operator

$$\mathrm{d^c} = \frac{\mathrm{i}}{4\pi}(\bar\partial - \partial) = \frac{\partial - \bar\partial}{4\pi\mathrm{i}}, \qquad \text{so that} \qquad \mathrm{dd^c} = \frac{\mathrm{i}}{2\pi}\partial\bar\partial = \mathrm{dd^c}. \tag{7.6}$$

Indeed, since $\partial^2 = \bar\partial^2 = 0$ and $\bar\partial\partial = -\partial\bar\partial$, we have $(\partial + \bar\partial)(\bar\partial - \partial) = \partial\bar\partial - \bar\partial\partial = 2\partial\bar\partial$. Moreover $\overline{\mathrm{d^c}\alpha} = \mathrm{d^c}\bar\alpha$, because conjugation exchanges ∂ and $\bar\partial$ and changes the sign of i; hence $\mathrm{d^c}$ maps real forms to real forms.

7.2 Currents and positivity

Let X be a complex manifold of dimension n. A test form on X is a smooth complex differential form with compact support; the test forms of degree zero are the test functions. In a chart with real coordinates $t_1, \ldots, t_{2n}$ we write test forms as $\varphi = \sum_J \varphi_J\,\mathrm{d}t_J$ and measure them by the seminorms $p_{K,m}$ of Section 1.3 applied to their coefficients.

For an integer $0 \le k \le 2n$, a current of degree k on X is a complex-linear functional T on the test forms of degree $2n-k$ with the following continuity property: for every coordinate chart and every compact set K in it, there are $C > 0$ and $m \in \mathbb{N}$ such that $|T(\varphi)| \le C \sum_J p_{K,m}(\varphi_J)$ for all test forms φ of degree $2n-k$ supported in K. Equivalently, in every chart and for every multi-index J with $|J| = 2n-k$, the functional $\chi \mapsto T(\chi\, \mathrm{d}t_J)$ is a distribution. In a chart there are therefore unique distributions T_J, $|J| = k$, such that

$$T(\varphi) = \sum_J \langle T_J, g_J \rangle, \qquad \text{where } \mathrm{d}t_J \wedge \varphi = g_J\, \mathrm{d}t_1 \wedge \cdots \wedge \mathrm{d}t_{2n},$$

for all test forms φ supported in the chart. Namely, if J' is the increasing multi-index complementary to J and $\mathrm{d}t_J \wedge \mathrm{d}t_{J'} = \varepsilon_J\, \mathrm{d}t_1 \wedge \cdots \wedge \mathrm{d}t_{2n}$ with $\varepsilon_J = \pm 1$, then $\langle T_J, \chi \rangle = \varepsilon_J T(\chi\, \mathrm{d}t_{J'})$, because $g_J = \varepsilon_J \varphi_{J'}$ for $\varphi = \sum_L \varphi_L\, \mathrm{d}t_L$. We write $T = \sum_J T_J\, \mathrm{d}t_J$ and call the T_J the coefficients of T; more generally, for multi-indices M that are not increasing, $S\, \mathrm{d}t_M$ denotes the current $\varphi \mapsto \langle S, g \rangle$, where $\mathrm{d}t_M \wedge \varphi = g\, \mathrm{d}t_1 \wedge \cdots \wedge \mathrm{d}t_{2n}$. In holomorphic coordinates we use the basis $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ in the same way. Thus locally a current is a differential form with distribution coefficients. A locally integrable k-form α, that is, a form with locally integrable coefficients, defines the current $T_\alpha(\varphi) = \int_X \alpha \wedge \varphi$ of degree k, which we usually denote again by α.

A current of degree k has bidegree (p, q), where $p + q = k$, if it vanishes on all test forms of bidegree other than $(n-p, n-q)$; in holomorphic coordinates it is then $T = \sum_{|I|=p, |J|=q} T_{IJ}\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$. Its bidimension is $(n-p, n-q)$. Every current is the sum of its components of pure bidegree. Forms and currents of a degree or bidegree outside the admissible range are taken to be zero. The conjugate of a current T is $\overline{T}(\varphi) = \overline{T(\bar{\varphi})}$, and T is real if $\overline{T} = T$, that is, $T(\bar{\varphi}) = \overline{T(\varphi)}$ for every test form φ.

A sequence, or more generally a net, of currents T_i of degree k converges weakly to T if $T_i(\varphi) \to T(\varphi)$ for every test form φ of degree $2n-k$. For currents of degree $2n$ given by positive measures, weak convergence is the vague convergence of these measures; see the discussion following Definition 7.2.4.

As for distributions (Section 1.3), a current T vanishes on an open set $W \subset X$ if $T(\varphi) = 0$ for every test form φ with $\operatorname{Supp}\varphi \subset W$, and the support $\operatorname{Supp} T$ of T is the complement of the union of all open sets on which T vanishes. If T vanishes on each member of a family of open sets, it vanishes on their union, because a partition of unity subordinate to the cover formed by these sets and by $X \setminus \operatorname{Supp}\varphi$ (Proposition 7.1.3) writes every test form φ supported in the union as a finite sum of test forms supported in single members. Hence $X \setminus \operatorname{Supp} T$ is the largest open set on which T vanishes, and $T(\varphi) = 0$ whenever $\operatorname{Supp}\varphi \cap \operatorname{Supp} T = \varnothing$.

Let T be a current of degree k and α a smooth form. We define

$$(\mathrm{d}T)(\varphi) = (-1)^{k+1} T(\mathrm{d}\varphi), \qquad (T \wedge \alpha)(\varphi) = T(\alpha \wedge \varphi),$$

and, in the same way, $(\partial T)(\varphi) = (-1)^{k+1} T(\partial\varphi)$ and $(\bar{\partial} T)(\varphi) = (-1)^{k+1} T(\bar{\partial}\varphi)$. These are currents, because d, ∂, $\bar{\partial}$ and multiplication by α map test forms supported in a compact set K of a chart to test forms supported in K, and the seminorms

$p_{K,m}$ of the new coefficients are bounded by a constant times the seminorms $p_{K,m+1}$ of the old ones. For a smooth form T these definitions agree with the exterior derivative and the exterior product: by the Leibniz rule and Proposition 7.1.3, $\int_X \mathrm{d}T \wedge \varphi = (-1)^{k+1} \int_X T \wedge \mathrm{d}\varphi$, and applying this identity to the components of pure bidegree of T and φ, of which only the products of bidegree (n,n) have nonzero integrals, we obtain the analogous identities for ∂ and $\bar\partial$.

In a chart, these operations act on the distribution coefficients exactly as on the coefficients of smooth forms. Indeed, let $t = (x_1, y_1, \dots, x_n, y_n)$ be the real coordinates of a holomorphic chart, let φ be a test form of degree $2n-k-1$ supported in the chart, and write $\mathrm{d}t_I \wedge \mathrm{d}t_J \wedge \varphi = g_{I,J}\, \mathrm{d}t_1 \wedge \cdots \wedge \mathrm{d}t_{2n}$ and $\mathrm{d}t_J \wedge \mathrm{d}\varphi = h_J\, \mathrm{d}t_1 \wedge \cdots \wedge \mathrm{d}t_{2n}$. For every smooth function a on the chart, the identity just proved for the smooth form $a\,\mathrm{d}t_J$ reads $\sum_l \int \partial_{t_l} a\, g_{l,J}\, \mathrm{d}V_{2n} = (-1)^{k+1} \int a h_J\, \mathrm{d}V_{2n}$. Integrating by parts on the left, we find $(-1)^{k+1} h_J = -\sum_l \partial_{t_l} g_{l,J}$. Hence, for a current $T = \sum_J T_J\, \mathrm{d}t_J$,

$$(\mathrm{d}T)(\varphi) = (-1)^{k+1} \sum_J \langle T_J, h_J \rangle = -\sum_{J,l} \langle T_J, \partial_{t_l} g_{l,J} \rangle = \sum_{J,l} \langle \partial_{t_l} T_J, g_{l,J} \rangle,$$

that is, $\mathrm{d}T = \sum_{J,l} \partial_{t_l} T_J\, \mathrm{d}t_l \wedge \mathrm{d}t_J$ with distributional derivatives. The same computation applies to ∂ and $\bar\partial$ in the basis $\mathrm{d}z_I \wedge \mathrm{d}\bar z_J$, and directly from the definitions $T \wedge \alpha = \sum_{J,L} \alpha_L T_J\, \mathrm{d}t_J \wedge \mathrm{d}t_L$ for $\alpha = \sum_L \alpha_L\, \mathrm{d}t_L$. If α has degree r, then, by the Leibniz rule for $\alpha \wedge \varphi$,

$$\begin{aligned}(\mathrm{d}T \wedge \alpha)(\varphi) + (-1)^k (T \wedge \mathrm{d}\alpha)(\varphi) &= (-1)^{k+1} T\big(\mathrm{d}(\alpha \wedge \varphi)\big) + (-1)^k T(\mathrm{d}\alpha \wedge \varphi) \\ &= (-1)^{k+r+1} T(\alpha \wedge \mathrm{d}\varphi),\end{aligned}$$

so that $\mathrm{d}(T \wedge \alpha) = \mathrm{d}T \wedge \alpha + (-1)^k T \wedge \mathrm{d}\alpha$, and similarly for ∂ and $\bar\partial$. In general, the product of two currents is not defined. We put $\mathrm{dd^c}T = (\mathrm{i}/2\pi)\partial\bar\partial T$ and $\mathrm{d^c}T = (\mathrm{i}/4\pi)(\bar\partial T - \partial T)$. Since $(\partial\bar\partial T)(\varphi) = (-1)^{k+2}(\bar\partial T)(\partial\varphi) = -T(\bar\partial\partial\varphi) = T(\partial\bar\partial\varphi)$, and since the identities $\mathrm{d} = \partial + \bar\partial$, $\partial^2 = \bar\partial^2 = 0$ and $\bar\partial\partial = -\partial\bar\partial$ pass from test forms to currents by duality, we obtain

$$(\mathrm{dd^c}T)(\varphi) = T(\mathrm{dd^c}\varphi), \qquad \mathrm{dd^c}T = \frac{\mathrm{i}}{4\pi}(\partial + \bar\partial)(\bar\partial - \partial)T = \frac{\mathrm{i}}{4\pi}(\partial\bar\partial T - \bar\partial\partial T) = \mathrm{dd^c}T.$$

All these operations are continuous for weak convergence, directly from the definitions. A current T is closed if $\mathrm{d}T = 0$. For a current of pure bidegree this is equivalent to $\partial T = \bar\partial T = 0$, since ∂T and $\bar\partial T$ are the components of $\mathrm{d}T$ of two different bidegrees.

We now turn to positivity of forms, which is a question of linear algebra. Let V be a complex vector space of dimension n, identified with $\mathbb{C}^n$ by linear coordinates $z_1, \dots, z_n$, and let β be the form (7.4) in these coordinates; the forms below are constant forms on V. For an integer $0 \le p \le n$, a real (p,p)-form is *strongly positive* if it is a finite sum, with nonnegative coefficients, of the forms

$$\mathrm{i}^{p^2} \theta_1 \wedge \cdots \wedge \theta_p \wedge \bar\theta_1 \wedge \cdots \wedge \bar\theta_p = \bigwedge_{j=1}^{p} (\mathrm{i}\theta_j \wedge \bar\theta_j), \tag{7.7}$$

where $\theta_1, \dots, \theta_p \in V^*$ are complex-linear forms, that is, forms of type $(1,0)$. We call the forms (7.7) generators. The identity in (7.7) holds because rearranging the factors of the right-hand side produces the sign $(-1)^{p(p-1)/2} = \mathrm{i}^{p^2-p}$, and the generators are real because each factor $\mathrm{i}\theta_j \wedge \bar\theta_j$ is real. A real (p,p)-form is *weakly positive* if its exterior product with every strongly positive $(n-p, n-p)$-form is a nonnegative multiple of the volume form $\beta^n/n!$. In degree zero both notions mean a nonnegative real number. A form on a complex manifold is strongly or weakly positive if it is so at every point, as a form on the tangent space. Finally, the restriction of a (p,p)-form to a complex subspace W of dimension p is a form of top degree on W; we call it nonnegative if it is a nonnegative multiple of the volume form of W in its complex orientation.

Proposition 7.2.1 *Let $0 \le p \le n$. The strongly positive (p,p)-forms on V form a closed convex cone, which spans the real vector space of real (p,p)-forms and contains β^p in its interior. A real (p,p)-form is weakly positive if and only if its restriction to every complex p-dimensional subspace of V is nonnegative. Pullback by complex-linear maps, and hence by holomorphic maps between complex manifolds, preserves strong and weak positivity. The product of strongly positive forms is strongly positive, and the product of a weakly positive form with a strongly positive form is weakly positive when its bidegree is at most (n,n). Moreover, strongly positive forms are weakly positive.*

Proof We use the Hermitian inner product on V^* for which $\mathrm{d}z_1, \dots, \mathrm{d}z_n$ are orthonormal. The form β is invariant under unitary changes of coordinates: if $\mathrm{d}w_j = \sum_k U_{jk}\,\mathrm{d}z_k$ with a unitary matrix U, then $\sum_j \mathrm{d}w_j \wedge \mathrm{d}\bar w_j = \sum_{k,l} \bigl(\sum_j U_{jk}\overline{U_{jl}}\bigr)\mathrm{d}z_k \wedge \mathrm{d}\bar z_l = \sum_k \mathrm{d}z_k \wedge \mathrm{d}\bar z_k$. Let $\Lambda^{p,0}V^*$ be the space of $(p,0)$-forms. For $\theta \in \Lambda^{p,0}V^*$ put $g(\theta) = \mathrm{i}^{p^2}\theta \wedge \bar\theta$, so that $g(c\theta) = |c|^2 g(\theta)$ and the generators are the forms $g(\theta)$ with θ decomposable.

Closedness. Let G be the set of the generators $g(e_1 \wedge \cdots \wedge e_p)$ with $e_1, \dots, e_p$ orthonormal in V^*. It is the continuous image of the compact set of orthonormal p-tuples, so G is compact. Every nonzero generator $g(\theta_1 \wedge \cdots \wedge \theta_p)$ is a positive multiple of an element of G. Indeed, the θ_j are linearly independent, and the Gram–Schmidt process gives an orthonormal family $e_1, \dots, e_p$ such that each θ_j is a linear combination of $e_1, \dots, e_j$; hence $\theta_1 \wedge \cdots \wedge \theta_p = c\, e_1 \wedge \cdots \wedge e_p$, where $c \neq 0$ is the product of the diagonal entries of the triangular transition matrix, and $g(\theta_1 \wedge \cdots \wedge \theta_p) = |c|^2 g(e_1 \wedge \cdots \wedge e_p)$.

Define a linear functional τ on real (p,p)-forms by $u \wedge \beta^{n-p} = \tau(u)\beta^n/n!$. If $e_1, \dots, e_p$ is orthonormal, complete it to an orthonormal basis $e_1, \dots, e_n$ of V^* and use the unitary coordinates w_j with $\mathrm{d}w_j = e_j$. Then $\beta = (\mathrm{i}/2)\sum_j \mathrm{d}w_j \wedge \mathrm{d}\bar w_j$, only one term of β^{n-p} contributes, and $\tau(g(e_1 \wedge \cdots \wedge e_p)) = 2^p(n-p)!$. Thus τ is a positive constant c_0 on G. Let H be the convex hull of G in the real vector space of real (p,p)-forms, of dimension N say. Then H is compact, by Carathéodory's argument: if a point is a convex combination of more than $N+1$ points of G, these points are affinely dependent, and subtracting a suitable multiple of an affine dependence relation makes one coefficient zero while the others stay nonnegative. Repeating this, we see that H is the continuous image of the compact set $G^{N+1} \times \Delta_N$, where Δ_N is

the standard simplex of dimension N. Every strongly positive form can be written as $u = sh$ with $h \in H$ and $s = \tau(u)/c_0 \geq 0$. If strongly positive forms $u_k = s_k h_k$ converge to u, then $s_k = \tau(u_k)/c_0$ converges to some $s \geq 0$, and a subsequence of (h_k) converges to some $h \in H$; hence $u = sh$ is strongly positive. The cone is convex by definition.

Spanning. Let λ be a real-linear functional on the real (p,p)-forms that vanishes on all generators; we show that $\lambda = 0$. Every (p,p)-form is uniquely $u + \mathrm{i}v$ with u, v real, so λ extends complex-linearly to all (p,p)-forms. The form $B(a,b) = \lambda(\mathrm{i}^{p^2} a \wedge \bar{b})$ on $\Lambda^{p,0}V^*$ is sesquilinear, and $B(a,a) = 0$ for every decomposable a. We prove by induction on $r = 0, \ldots, p$ that $B(a_1 \wedge \cdots \wedge a_p, b_1 \wedge \cdots \wedge b_p) = 0$ whenever $a_j = b_j$ for all $j > r$. For $r = 0$ this is the hypothesis. Assume it for $r-1$, and fix all vectors except the rth ones. The resulting form $B_r(x,y) = B(\cdots \wedge x \wedge \cdots, \cdots \wedge y \wedge \cdots)$, with x and y in the rth slots, is sesquilinear and vanishes for $x = y$ by the induction hypothesis. By the polarization identity $B_r(x,y) = \frac{1}{4}\sum_{k=0}^{3} \mathrm{i}^k B_r(x + \mathrm{i}^k y, x + \mathrm{i}^k y)$, valid for every sesquilinear form, B_r vanishes identically. For $r = p$ we obtain $B = 0$ on decomposable vectors, which span $\Lambda^{p,0}V^*$, so $B = 0$. Since the forms $a \wedge \bar{b}$ with $a, b \in \Lambda^{p,0}V^*$ span all (p,p)-forms, $\lambda = 0$. Hence the generators span the real (p,p)-forms over $\mathbb{R}$.

Interior. Since the generators span the real (p,p)-forms and every nonzero generator is a positive multiple of an element of G, we can choose finitely many elements $q_1, \ldots, q_M$ of G spanning the real (p,p)-forms. For a permutation σ of $\{1, \ldots, n\}$ and signs $\varepsilon \in \{\pm 1\}^n$, consider the linear map $z \mapsto (\varepsilon_1 z_{\sigma(1)}, \ldots, \varepsilon_n z_{\sigma(n)})$; its pullback maps generators to generators. Let $\widehat{q}_r$ be the average of the pullbacks of q_r under all these maps.

To compute $\widehat{q}_r$, write $q_r = \mathrm{i}^{p^2} \sum_{I,J} c_{IJ}\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$. The sign changes multiply c_{IJ} by $\prod_{i \in I} \varepsilon_i \prod_{j \in J} \varepsilon_j$, whose average over ε vanishes unless $I = J$. A permutation maps $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_I$ to $\mathrm{d}z_{I'} \wedge \mathrm{d}\bar{z}_{I'}$, where I' is the permuted index set in increasing order, because reordering the factors produces the same sign for $\mathrm{d}z_{I'}$ and for $\mathrm{d}\bar{z}_{I'}$; since every p-element set is mapped to every other one by the same number of permutations, the average makes all diagonal coefficients equal to $\binom{n}{p}^{-1} \sum_I c_{II}$. Since $q_r = g(\theta)$ with $\theta = \sum_I \theta_I\, \mathrm{d}z_I \neq 0$, we have $c_{IJ} = \theta_I \overline{\theta_J}$ and $s_r = \sum_I c_{II} = \sum_I |\theta_I|^2 > 0$. Therefore, by (7.7) and the expansion $\beta^p = p! \sum_{|I|=p} \bigwedge_{i \in I} (\frac{\mathrm{i}}{2} \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_i)$,

$$\widehat{q}_r = s_r \binom{n}{p}^{-1} \mathrm{i}^{p^2} \sum_{|I|=p} \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_I = \frac{2^p s_r}{p! \binom{n}{p}} \beta^p .$$

Consequently β^p is a positive multiple of $\sum_r \widehat{q}_r$, which is a combination of generators with strictly positive coefficients in which every q_r occurs. A real (p,p)-form v of small norm can be written as $\sum_r t_r(v) q_r$ with small coefficients depending linearly on v. Adding these coefficients to the positive coefficients of the q_r keeps all of them positive, so $\beta^p + v$ is strongly positive. Thus β^p is an interior point.

Restriction criterion. Let W be a complex subspace of dimension p, and choose linear coordinates in which $W = \{z_{p+1} = \cdots = z_n = 0\}$. For a real (p,p)-form u, the product $u \wedge \bigwedge_{j>p} (\mathrm{i}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j)$ involves only the coefficient of $\mathrm{d}z_{(1,\ldots,p)} \wedge \mathrm{d}\bar{z}_{(1,\ldots,p)}$ in

u, which is also the coefficient of the restriction $u|_W$ in the coordinates $z_1, \dots, z_p$ of W. Hence this product is a nonnegative multiple of the volume form if and only if $u|_W$ is nonnegative. Conversely, every generator $\bigwedge_{j=1}^{n-p} (\mathrm{i}\theta_j \wedge \bar\theta_j)$ of bidegree $(n-p, n-p)$ is zero if the θ_j are linearly dependent, and otherwise it is of the above form, with $z_{p+j} = \theta_j$ and $W = \{\theta_1 = \cdots = \theta_{n-p} = 0\}$. Since weak positivity need only be tested on generators, the restriction criterion follows.

Pullbacks and products. Let $F\colon V' \to V$ be complex linear. The pullback of the generator $\bigwedge_j (\mathrm{i}\theta_j \wedge \bar\theta_j)$ is the generator $\bigwedge_j (\mathrm{i}F^*\theta_j \wedge \overline{F^*\theta_j})$, so F^* preserves strong positivity. Let u be weakly positive on V and let $W' \subset V'$ be a complex subspace of dimension p. If $F|_{W'}$ is injective, it is a complex-linear isomorphism onto the p-dimensional subspace $F(W')$ and preserves the complex orientations, so $(F^*u)|_{W'} = (F|_{W'})^*(u|_{F(W')})$ is nonnegative. Otherwise $F(W')$ has dimension less than p, and $(F^*u)|_{W'} = 0$ because it factors through the (p,p)-forms on $F(W')$. By the restriction criterion, F^*u is weakly positive. For a holomorphic map, apply this to its derivative at each point.

A product of generators is a generator, so products of strongly positive forms are strongly positive. If u is a weakly positive (p,p)-form and s a strongly positive (r,r)-form with $p + r \le n$, then $s \wedge t$ is strongly positive for every strongly positive $(n-p-r, n-p-r)$-form t, so $u \wedge s \wedge t$ is a nonnegative multiple of the volume form; hence $u \wedge s$ is weakly positive. Finally, an (n,n)-generator equals $|\det(\theta_1, \dots, \theta_n)|^2 \bigwedge_j (\mathrm{i}\mathrm{d}z_j \wedge \mathrm{d}\bar z_j)$, a nonnegative multiple of the volume form, so strongly positive forms are weakly positive. □

In bidegree $(1,1)$ the two notions coincide. Write a real $(1,1)$-form as $u = \mathrm{i}\sum a_{j\bar k}\mathrm{d}z_j \wedge \mathrm{d}\bar z_k$, where reality means that the matrix $A = (a_{j\bar k})$ is Hermitian. The restriction of u to the complex line spanned by a vector ξ is $(\sum_{j,k} a_{j\bar k}\xi_j\bar\xi_k)\,\mathrm{i}\mathrm{d}t \wedge \mathrm{d}\bar t$ in the coordinate t of the line $\{t\xi : t \in \mathbb{C}\}$. Hence, by the restriction criterion of Proposition 7.2.1, u is weakly positive if and only if A is positive semidefinite. In that case a unitary diagonalization of A writes u as $\sum_r \lambda_r \mathrm{i}\theta_r \wedge \bar\theta_r$ with $\lambda_r \ge 0$, a sum of generators.

By duality, the two notions also coincide in bidegree $(n-1, n-1)$. Let u be a real $(n-1,n-1)$-form. For $\theta, \eta \in V^*$ define $B_u(\theta,\eta) \in \mathbb{C}$ by $u \wedge \mathrm{i}\theta \wedge \bar\eta = B_u(\theta,\eta)\,\beta^n/n!$. Then B_u is sesquilinear, and $B_u(\eta,\theta) = \overline{B_u(\theta,\eta)}$ because u is real; thus $H_u(\theta) = B_u(\theta,\theta)$ is a Hermitian quadratic form on V^*, and it depends linearly on u. Since the strongly positive $(1,1)$-forms are the finite sums of the forms $\mathrm{i}\theta \wedge \bar\theta$, the form u is weakly positive if and only if $H_u \ge 0$.

Moreover, H_u determines u. Indeed, H_u determines B_u by the polarization identity $B_u(\theta,\eta) = \frac{1}{4}\sum_{k=0}^{3} \mathrm{i}^k H_u(\theta + \mathrm{i}^k\eta)$, which holds for every form that is linear in the first and conjugate-linear in the second variable (expand $H_u(\theta + \mathrm{i}^k\eta) = H_u(\theta) + \mathrm{i}^{-k}B_u(\theta,\eta) + \mathrm{i}^k B_u(\eta,\theta) + H_u(\eta)$ and use $\sum_{k=0}^3 \mathrm{i}^k = \sum_{k=0}^3 \mathrm{i}^{2k} = 0$). Hence H_u determines the products $u \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar z_k$; and if $\hat\jmath$ denotes the increasing multi-index of length $n-1$ omitting j, then $u \wedge \mathrm{d}z_j \wedge \mathrm{d}\bar z_k = c_{jk}u_{\hat\jmath\hat k}\,\beta^n/n!$, where $u_{\hat\jmath\hat k}$ is the coefficient of $\mathrm{d}z_{\hat\jmath} \wedge \mathrm{d}\bar z_{\hat k}$ in u and $c_{jk} \ne 0$ is a constant.

Now let u be weakly positive. Diagonalizing H_u with respect to the inner product on V^* used in the proof of Proposition 7.2.1, we find unitary coordinates $w_1, \dots, w_n$ and

numbers $\lambda_j \geq 0$ such that $H_u(\sum_j \xi_j \mathrm{d}w_j) = \sum_j \lambda_j |\xi_j|^2$. In these coordinates β keeps its form, and for $\theta = \sum_l \xi_l \, \mathrm{d}w_l$ we have $\bigwedge_{l\neq j}(\mathrm{i}\mathrm{d}w_l \wedge \mathrm{d}\bar{w}_l) \wedge \mathrm{i}\theta \wedge \bar{\theta} = 2^n |\xi_j|^2 \beta^n / n!$. Hence the strongly positive form $s = 2^{-n} \sum_j \lambda_j \bigwedge_{l\neq j}(\mathrm{i}\mathrm{d}w_l \wedge \mathrm{d}\bar{w}_l)$ satisfies $H_s = H_u$, and therefore $u = s$ is strongly positive.

In general the two notions differ. Let $n = 4$ and $\theta = \mathrm{d}z_1 \wedge \mathrm{d}z_2 + \mathrm{d}z_3 \wedge \mathrm{d}z_4$. The $(2,2)$-form $\theta \wedge \bar{\theta} = \mathrm{i}^4 \theta \wedge \bar{\theta}$ is real, since 2-forms commute, and it is weakly positive by the restriction criterion of Proposition 7.2.1: if w_1, w_2 are linear coordinates on a complex 2-plane W, the restriction of θ to W is $c \, \mathrm{d}w_1 \wedge \mathrm{d}w_2$ for some $c \in \mathbb{C}$, and then the restriction of $\theta \wedge \bar{\theta}$ is $|c|^2 \bigwedge_{j=1}^{2}(\mathrm{i}\mathrm{d}w_j \wedge \mathrm{d}\bar{w}_j)$, by (7.7).

However, $\theta \wedge \bar{\theta}$ is not strongly positive. To see this, identify a real (p,p)-form $\mathrm{i}^{p^2} \sum_{I,J} c_{IJ} \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ with the Hermitian matrix (c_{IJ}). A generator $\mathrm{i}^{p^2} \eta \wedge \bar{\eta}$, with $\eta = \sum_I \eta_I \mathrm{d}z_I$ decomposable, corresponds to the rank-one matrix $(\eta_I \bar{\eta}_J)$. If $\theta \wedge \bar{\theta}$ were a sum of generators $\mathrm{i}^4 \eta_r \wedge \bar{\eta}_r$ with nonzero decomposable η_r, the corresponding matrices would satisfy $(\theta_I \bar{\theta}_J) = \sum_r (\eta_{r,I} \bar{\eta}_{r,J})$. Let $\langle a, b \rangle = \sum_I a_I \overline{b_I}$ for $a = \sum_I a_I \, \mathrm{d}z_I$ and $b = \sum_I b_I \, \mathrm{d}z_I$ in $\Lambda^{2,0} V^*$. For every $v \in \Lambda^{2,0} V^*$ with $\langle \theta, v \rangle = 0$, evaluating the Hermitian forms of both matrices at the vector (v_I) gives $0 = |\langle \theta, v \rangle|^2 = \sum_r |\langle \eta_r, v \rangle|^2$, so $\langle \eta_r, v \rangle = 0$ for every r; hence every η_r lies in the orthogonal complement of the orthogonal complement of θ, that is, it is a multiple of θ. This is impossible: $\theta \wedge \theta = 2 \, \mathrm{d}z_1 \wedge \mathrm{d}z_2 \wedge \mathrm{d}z_3 \wedge \mathrm{d}z_4 \neq 0$, whereas $\eta \wedge \eta = 0$ for every decomposable 2-form η.

Since the two notions agree in bidegrees $(1,1)$ and $(n-1, n-1)$, we speak simply of positive forms in these bidegrees. Such a form is *strictly positive* if it lies in the interior of the cone of positive forms, and a form on a complex manifold is strictly positive if it is so at every point. For example, β and β^{n-1} are strictly positive by Proposition 7.2.1. A real $(1,1)$-form $\mathrm{i} \sum a_{j\bar{k}} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ is strictly positive if and only if the matrix $A = (a_{j\bar{k}})$ is positive definite. Indeed, if $\xi^* A \xi \geq c|\xi|^2$ for all $\xi \in \mathbb{C}^n$, with $c > 0$, then $A + E$ is positive semidefinite for every Hermitian matrix E with $|\xi^* E \xi| \leq c|\xi|^2$ for all ξ; conversely, if A is an interior point of the cone of positive semidefinite matrices, then $A - \varepsilon I$ is positive semidefinite for some $\varepsilon > 0$. Positive $(1,1)$-forms are also called semipositive, to stress that they need not be strictly positive.

If $\omega_3, \ldots, \omega_n$ are fixed positive $(1,1)$-forms, the exterior product $\omega \wedge \omega' \wedge \omega_3 \wedge \cdots \wedge \omega_n$ is a multiple of the volume form that depends symmetrically and bilinearly on the real $(1,1)$-forms ω and ω'. We now describe this bilinear form in terms of matrices and determine its signature. For Hermitian $n \times n$ matrices $A_1, \ldots, A_n$, the *mixed discriminant* $D(A_1, \ldots, A_n)$ is $1/n!$ times the coefficient of $t_1 \cdots t_n$ in the polynomial $\det(\sum_j t_j A_j)$. For a Hermitian matrix A put $\omega_A = \mathrm{i} \sum_{j,k} A_{j\bar{k}} \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$; thus $\omega_I = 2\beta$ for the identity matrix I.

Lemma 7.2.2 *Let $n \geq 1$, and let $A, A_1, \ldots, A_n$ be Hermitian $n \times n$ matrices.*

(1) *The mixed discriminant is real, symmetric and real-multilinear in its arguments, and $D(A, \ldots, A) = \det A$. More precisely,*

$$n! \, D(A_1, \ldots, A_n) = \sum_{\sigma} \det\left(A_{\sigma(1)} e_1, \ldots, A_{\sigma(n)} e_n\right),$$

the sum running over all permutations σ of $\{1,\dots,n\}$, where Ae_l is the lth column of A.

(2) *We have*

$$\omega_A^n = \det A\, \omega_I^n, \qquad \omega_{A_1} \wedge \cdots \wedge \omega_{A_n} = D(A_1,\dots,A_n)\, \omega_I^n,$$

where $\omega_I^n = n! \bigwedge_l (\mathrm{i}\mathrm{d}z_l \wedge \mathrm{d}\bar z_l) = 2^n \beta^n$ is a positive multiple of the volume form.

(3) *If $A_1,\dots,A_n$ are positive semidefinite, then $D(A_1,\dots,A_n) \geq 0$. If they are positive definite, then $D(A_1,\dots,A_n) > 0$.*

Proof (1) The lth column of $\sum_j t_j A_j$ is $\sum_j t_j A_j e_l$. Expanding the determinant by multilinearity in its columns, we obtain

$$\det\Big(\sum_j t_j A_j\Big) = \sum_\tau t_{\tau(1)} \cdots t_{\tau(n)} \det\big(A_{\tau(1)}e_1, \dots, A_{\tau(n)}e_n\big),$$

the sum running over all maps $\tau\colon \{1,\dots,n\} \to \{1,\dots,n\}$. The monomial $t_{\tau(1)}\cdots t_{\tau(n)}$ equals $t_1 \cdots t_n$ exactly when τ is a permutation, which gives the displayed formula. A permutation of $A_1,\dots,A_n$ permutes the terms of this sum, so D is symmetric. Each term is multilinear in $(A_1,\dots,A_n)$, since each A_j occurs in exactly one column, so D is multilinear. For $A_1 = \cdots = A_n = A$ all $n!$ terms equal $\det A$, so $D(A,\dots,A) = \det A$. Finally, D is real: for real t_j the matrix $\sum_j t_j A_j$ is Hermitian, so its determinant is real, being the product of its real eigenvalues; hence the polynomial $\det(\sum_j t_j A_j)$ is real on $\mathbb{R}^n$, and so are its coefficients, which are its partial derivatives at $t = 0$ divided by factorials.

(2) In the expansion of ω_A^n, the products $\bigwedge_l (\mathrm{d}z_{j_l} \wedge \mathrm{d}\bar z_{k_l})$ vanish unless $(j_1,\dots,j_n)$ and $(k_1,\dots,k_n)$ are permutations of $(1,\dots,n)$. Since the 2-forms $\mathrm{d}z_j \wedge \mathrm{d}\bar z_k$ commute, such a product equals $\bigwedge_l (\mathrm{d}z_l \wedge \mathrm{d}\bar z_{\pi(l)})$ for a permutation π, which arises from $n!$ ordered products, and $\bigwedge_l (\mathrm{d}z_l \wedge \mathrm{d}\bar z_{\pi(l)}) = \operatorname{sign}(\pi) \bigwedge_l (\mathrm{d}z_l \wedge \mathrm{d}\bar z_l)$, because the two products differ by a permutation of the factors $\mathrm{d}\bar z_k$. Therefore

$$\omega_A^n = n! \sum_\pi \operatorname{sign}(\pi) \prod_l A_{l\overline{\pi(l)}} \bigwedge_l (\mathrm{i}\mathrm{d}z_l \wedge \mathrm{d}\bar z_l) = n! \det A \bigwedge_l (\mathrm{i}\mathrm{d}z_l \wedge \mathrm{d}\bar z_l) = \det A\, \omega_I^n,$$

where the last equality is the case $A = I$ of the preceding one. Since $\mathrm{i}\mathrm{d}z_l \wedge \mathrm{d}\bar z_l = 2\,\mathrm{d}x_l \wedge \mathrm{d}y_l$, we get $\omega_I^n = 2^n n!\, \mathrm{d}V_{2n} = 2^n \beta^n$ by (7.4). Now let $t_1,\dots,t_n$ be real, and apply the first identity to the Hermitian matrix $\sum_j t_j A_j$, for which $\omega_{\sum_j t_j A_j} = \sum_j t_j \omega_{A_j}$. Since 2-forms commute, the coefficient of $t_1 \cdots t_n$ in $(\sum_j t_j \omega_{A_j})^n$ is $n!\, \omega_{A_1} \wedge \cdots \wedge \omega_{A_n}$, while in $\det(\sum_j t_j A_j)\, \omega_I^n$ it is $n!\, D(A_1,\dots,A_n)\, \omega_I^n$. Two polynomials that agree on $\mathbb{R}^n$ have the same coefficients, and we obtain the second identity.

(3) A real $(1,1)$-form ω_A with A positive semidefinite is strongly positive, by the discussion following Proposition 7.2.1: a unitary diagonalization of A writes it as a sum of generators with nonnegative coefficients. By Proposition 7.2.1, the product $\omega_{A_1} \wedge \cdots \wedge \omega_{A_n}$ is then strongly positive, hence weakly positive, that is, a nonnegative multiple of the volume form. By (2) and since ω_I^n is a positive multiple of the volume form, $D(A_1,\dots,A_n) \geq 0$. If all A_j are positive definite, choose $\varepsilon > 0$ smaller than the least eigenvalue of every A_j, so that every $A_j - \varepsilon I$ is positive

semidefinite, and expand $D(A_1,\dots,A_n) = D\big((A_1-\varepsilon I)+\varepsilon I,\dots,(A_n-\varepsilon I)+\varepsilon I\big)$ by multilinearity. Every term is the mixed discriminant of positive semidefinite matrices, hence nonnegative, and one of them is $D(\varepsilon I,\dots,\varepsilon I) = \varepsilon^n \det I = \varepsilon^n > 0$. □

Lemma 7.2.3 *Let $n \geq 2$, and let $A_3,\dots,A_n$ be positive definite Hermitian $n\times n$ matrices. The real symmetric bilinear form*

$$q(M,N) = D(M,N,A_3,\dots,A_n)$$

on the real vector space of Hermitian $n\times n$ matrices, which has dimension n^2, has signature $(1,n^2-1)$. For every positive definite H, the restriction of q to the hyperplane $\{M : q(M,H)=0\}$ is negative definite.

Proof We first show that, for a given tuple $A_3,\dots,A_n$, the second assertion follows from the first. Let H be positive definite. Then $q(H,H) > 0$ by Lemma 7.2.2(3), so the space of Hermitian matrices is the q-orthogonal direct sum of $\mathbb{R}H$ and $H^\perp = \{M : q(M,H)=0\}$. The restriction of q to $H^\perp$ is nondegenerate, since a vector of $H^\perp$ orthogonal to $H^\perp$ would be orthogonal to the whole space. By Sylvester's law of inertia it has signature $(0,n^2-1)$; that is, it is negative definite.

We prove the signature assertion by induction on n. For $n=2$ there are no fixed matrices, and expanding $\det(t_1M+t_2N)$ gives $q(M,N) = (\operatorname{tr}M\operatorname{tr}N - \operatorname{tr}(MN))/2$. The decomposition of a matrix into a scalar matrix and a trace-free matrix is q-orthogonal, $q(I,I)=1$, and $q(M,M) = -\operatorname{tr}(M^2)/2 < 0$ for a nonzero trace-free M, since $\operatorname{tr}(M^2)$ is the sum of the squares of the real eigenvalues of M. Hence the signature is $(1,3)$.

Let $n\geq 3$, and assume both assertions in dimension $n-1$. We first show that q is nondegenerate for every tuple of positive definite matrices. Replacing every matrix X by G^*XG multiplies D by $|\det G|^2$, by the definition of D through the determinant, and preserves positive definiteness; since $X\mapsto G^*XG$ is a linear automorphism of the Hermitian matrices, we may take $G = A_n^{-1/2}$ and assume that $A_n = I$. For a Hermitian matrix X, let $X^{(i)}$ be the $(n-1)\times(n-1)$ matrix obtained by deleting the ith row and column, and let P_i be the diagonal matrix with entry 1 in position (i,i) and zeros elsewhere. Expanding along the ith row, $\det(tP_i+Y) = \det Y + t\det Y^{(i)}$; comparing coefficients of $t_1\cdots t_n$ in $\det(t_1P_i + \sum_{j\geq 2} t_jB_j)$ gives, for Hermitian matrices $B_2,\dots,B_n$,

$$D(P_i,B_2,\dots,B_n) = \frac{1}{n} D_{n-1}(B_2^{(i)},\dots,B_n^{(i)}),$$

where the factor $1/n = (n-1)!/n!$ comes from the normalizations of D and of the mixed discriminant D_{n-1} of size $n-1$. Put

$$q_i(B,C) = D_{n-1}(B,C,A_3^{(i)},\dots,A_{n-1}^{(i)});$$

the matrices $A_j^{(i)}$ and $I^{(i)}$ are positive definite, as principal submatrices of positive definite matrices. Using the last formula, and writing $I = \sum_i P_i$ in the last slot of $q(M,M)$, we obtain

$$q(M,P_i)=\frac{1}{n}q_i(M^{(i)},I^{(i)}),\qquad q(M,M)=\frac{1}{n}\sum_{i=1}^{n}q_i(M^{(i)},M^{(i)}).$$

Now let M lie in the kernel of q. Then $q(M,P_i)=0$, so $M^{(i)}$ lies in the hyperplane $\{q_i(\,\cdot\,,I^{(i)})=0\}$, on which q_i is negative definite by the induction hypothesis. Since also $q(M,M)=0$, every term of the sum vanishes, and $M^{(i)}=0$ for every i. As $n\ge 3$, every entry of M lies in some $M^{(i)}$, so $M=0$.

Finally, the set of tuples $A=(A_3,\dots,A_n)$ of positive definite matrices is convex, hence connected, and we write q_A for the corresponding form. By Sylvester's law of inertia, the number $k(A)$ of positive squares of q_A is the largest dimension of a subspace on which q_A is positive definite. We show that k is locally constant. Fix a tuple A^0. Since q_{A^0} is nondegenerate, the space of Hermitian matrices is a direct sum $E_+\oplus E_-$, where q_{A^0} is positive definite on E_+, negative definite on E_-, and $\dim E_+=k(A^0)$. The function $(A,M)\mapsto q_A(M,M)$ is continuous, and it is positive on $\{A^0\}\times S_+$ and negative on $\{A^0\}\times S_-$, where $S_\pm$ is the unit sphere of $E_\pm$ for some norm. These spheres are compact, so for all tuples A near A^0 the form q_A is still positive definite on E_+ and negative definite on E_-. Then $k(A)\ge\dim E_+$, and $k(A)\le\dim E_+$ because a subspace of larger dimension meets E_- in a nonzero vector. Hence $k(A)=k(A^0)$ near A^0, and k is constant on the connected set of tuples.

It therefore suffices to compute the signature at $A_3=\cdots=A_n=I$. With $s=t_3+\cdots+t_n$, the coefficient of s^{n-2} in $\det(t_1M+t_2N+sI)$ is the sum of the principal 2×2 minors of t_1M+t_2N, that is, $((\operatorname{tr}X)^2-\operatorname{tr}(X^2))/2$ for $X=t_1M+t_2N$, in which t_1t_2 has the coefficient $\operatorname{tr}M\operatorname{tr}N-\operatorname{tr}(MN)$; moreover, the coefficient of $t_3\cdots t_n$ in s^{n-2} is $(n-2)!$. Hence

$$q(M,N)=\frac{\operatorname{tr}M\operatorname{tr}N-\operatorname{tr}(MN)}{n(n-1)},$$

and, as in the case $n=2$, the decomposition into scalar and trace-free matrices shows that the signature is $(1,n^2-1)$. □

Definition 7.2.4 Let $0\le p\le n$. A current T of bidegree (p,p) on X is positive if $T(\varphi)\ge 0$ for every test form φ of bidegree $(n-p,n-p)$ that is strongly positive at every point. We then write $T\ge 0$, and $T\ge S$ if $T-S\ge 0$. If T is positive and U is a coordinate chart, the trace measure of T in U is the positive Radon measure

$$\mu_T=T\wedge\beta^{n-p},\qquad\text{that is,}\qquad \mu_T(\chi)=T(\chi\beta^{n-p})\quad\text{for test functions }\chi\text{ on }U,$$

where β is the form (7.4) in the coordinates of U.

The functional $\chi\mapsto T(\chi\beta^{n-p})$ is indeed a Radon measure. It is nonnegative on nonnegative test functions, because $\chi\beta^{n-p}$ is then strongly positive, and such a distribution is real: a real test function χ is the difference of the nonnegative test functions $\chi+\|\chi\|_\infty\eta$ and $\|\chi\|_\infty\eta$, where η is a smooth cutoff function for $\operatorname{Supp}\chi$ in U (Lemma 1.4.1). Hence it is a positive distribution, and Theorem 1.4.5 applies.

The trace measure depends on the chart; a version defined by a Hermitian metric is introduced at the end of Section 7.5.

A current S of degree $2n$ acts on test functions, and by Definition 7.2.4 with $p = n$ it is positive if $S(\chi) \geq 0$ for every nonnegative test function χ. In a chart, its only coefficient is the distribution $\chi \mapsto S(\chi)$ itself. Positive currents of degree $2n$ are measures on X. We call a Borel measure μ on X a *positive Radon measure* if it is finite on compact sets and inner regular, that is, $\mu(E) = \sup\{\mu(K) : K \subset E \text{ compact}\}$ for every Borel set $E \subset X$; on open subsets of $\mathbb{C}^n$ these are the Radon measures of Definition 1.1.1, by Proposition 1.1.2. The restriction of μ to the Borel subsets of a chart is a Radon measure in the coordinates, by the same proposition, and $\chi \mapsto \int \chi \, \mathrm{d}\mu$ is a positive current of degree $2n$. Conversely, every positive current S of degree $2n$ is given in this way by a unique positive Radon measure, which we denote again by S.

For uniqueness, note that a compact set $K \subset X$ is the disjoint union of finitely many Borel sets, each contained in a compact subset of a chart. On each of them μ is determined by S, by the uniqueness in Theorem 1.4.5. Hence $\mu(K)$ is determined by S, and so is μ, by inner regularity.

For existence, let μ_U be the Radon measure representing S in a chart U (Theorem 1.4.5); by the same uniqueness, μ_U and $\mu_{U'}$ agree on the Borel subsets of $U \cap U'$. Choose a partition of unity (χ_a) with compact supports subordinate to a cover of X by charts U_a (Proposition 7.1.3), and put $\mu(E) = \sum_a \int_{E \cap U_a} \chi_a \, \mathrm{d}\mu_{U_a}$, a sum of nonnegative terms, defined as the supremum of its finite partial sums. Since such sums can be rearranged, μ is countably additive. If E is a Borel subset of a chart U, then only countably many χ_a are not identically zero on U, because U is a countable union of compact sets, each meeting only finitely many supports; since $\mu_{U_a} = \mu_U$ on $U_a \cap U$, monotone convergence gives $\mu(E) = \int_E \sum_a \chi_a \, \mathrm{d}\mu_U = \mu_U(E)$. Hence μ is finite on compact sets, which are finite unions of compact subsets of charts, and $\int \chi \, \mathrm{d}\mu = S(\chi)$ for test functions χ supported in a chart, and then for all test functions, by a partition of unity.

Finally, μ is inner regular: given a Borel set E, finitely many indices a and $\varepsilon > 0$, inner regularity of the measures μ_{U_a} gives compact sets $K_a \subset E \cap \operatorname{Supp} \chi_a$ with $\int_{K_a} \chi_a \, \mathrm{d}\mu_{U_a} > \int_{E \cap U_a} \chi_a \, \mathrm{d}\mu_{U_a} - \varepsilon$, and the compact set $K = \bigcup_a K_a \subset E$ satisfies $\mu(K) \geq \sum_a \int_{K_a} \chi_a \, \mathrm{d}\mu_{U_a}$; since $\mu(E)$ is the supremum of the finite partial sums of $\sum_a \int_{E \cap U_a} \chi_a \, \mathrm{d}\mu_{U_a}$, it is the supremum of the numbers $\mu(K)$.

A net of positive Radon measures μ_i on X converges vaguely to μ if $\int f \, \mathrm{d}\mu_i \to \int f \, \mathrm{d}\mu$ for every $f \in C_c(X)$. Applying Proposition 1.5.1 in charts and writing $f \in C_c(X)$ as a finite sum $\sum_a \chi_a f$ with a partition of unity, we see that this is the same as weak convergence of the corresponding currents; for this reason, vague convergence of measures is called weak convergence in this part of the book.

A smooth real (p, p)-form α, regarded as a current, is positive if and only if it is weakly positive at every point. Indeed, if α is weakly positive, then $\alpha \wedge \varphi$ is a nonnegative multiple of the volume form for every strongly positive test form φ, so that $\int_X \alpha \wedge \varphi \geq 0$. Conversely, let α be positive as a current, let s be a constant strongly positive $(n - p, n - p)$-form in a chart, and write $\alpha \wedge s = f \, \mathrm{d}V_{2n}$ with f continuous

and real. For every nonnegative test function χ in the chart, $\int f\chi \, dV_{2n} = \alpha(\chi s) \geq 0$, hence $f \geq 0$. Since every strongly positive form on the tangent space at a point of the chart is the value of such a constant form s, the form α is weakly positive at every point. (By Theorem 7.2.5 below, a smooth form that is positive as a current is automatically real.)

Positive currents have measure coefficients. To state this precisely we need measures with complex values. Let $\Omega \subset \mathbb{R}^d$ be open. A complex Radon measure on Ω is a linear functional on $C_c(\Omega)$ of the form $\sigma = \sigma_1 - \sigma_2 + i(\sigma_3 - \sigma_4)$ with positive Radon measures $\sigma_1, \ldots, \sigma_4$; it assigns the complex number $\sigma(A) = \sigma_1(A) - \sigma_2(A) + i(\sigma_3(A) - \sigma_4(A))$ to every relatively compact Borel set $A \subset \Omega$. This number does not depend on the decomposition: if $\sigma = \sigma_1' - \sigma_2' + i(\sigma_3' - \sigma_4')$ is another one, then, evaluating on real functions and taking real and imaginary parts, we find $\sigma_1 + \sigma_2' = \sigma_1' + \sigma_2$ and $\sigma_3 + \sigma_4' = \sigma_3' + \sigma_4$ as positive linear functionals on $C_c(\Omega, \mathbb{R})$, hence as measures, by the uniqueness in Theorem 1.2.2. Its total variation $|\sigma|$ is defined, as for real signed measures in Section 1.1, by

$$|\sigma|(A) = \sup \sum_{i=1}^{m} |\sigma(A_i)|,$$

the supremum being taken over all finite Borel partitions of the relatively compact Borel set A. The argument given there for real signed measures shows that $|\sigma|$ is countably additive on relatively compact Borel sets, and $|\sigma| \leq \sigma_1 + \sigma_2 + \sigma_3 + \sigma_4$. Exhausting Ω by relatively compact open sets, we obtain a Borel measure $|\sigma|$ on Ω that is finite on compact sets, hence a positive Radon measure by Proposition 1.1.2. Approximation by simple functions gives $|\int f \, d\sigma| \leq \int |f| \, d|\sigma|$ for bounded Borel functions f with compact support.

A current T on X has *order zero* if, in every chart, its coefficients are complex Radon measures; we call them the coefficient measures of T. To compare two charts, let t and t' be the real coordinates of holomorphic charts U and U', regard functions on $U \cap U'$ as functions of either set of coordinates, write $dt_J = \sum_{J'} a_{JJ'} \, dt'_{J'}$ on $U \cap U'$ with smooth functions $a_{JJ'}$, and let $\delta > 0$ be the real Jacobian determinant of the coordinate change, so that $dt'_1 \wedge \cdots \wedge dt'_{2n} = \delta \, dt_1 \wedge \cdots \wedge dt_{2n}$. If $dt'_{J'} \wedge \varphi = g'_{J'} \, dt'_1 \wedge \cdots \wedge dt'_{2n}$, then $dt_J \wedge \varphi = \delta \sum_{J'} a_{JJ'} g'_{J'} \, dt_1 \wedge \cdots \wedge dt_{2n}$. Comparing the two expressions for $T(\varphi)$, in which the functions $g'_{J'}$ are arbitrary test functions on $U \cap U'$, we obtain

$$\langle T'_{J'}, h \rangle_{t'} = \Big\langle \sum_J \delta \, a_{JJ'} T_J, h \Big\rangle_t \tag{7.8}$$

for test functions h on $U \cap U'$, where the subscripts indicate the coordinates in which the pairings are computed. Hence, if the T_J are complex Radon measures on $U \cap U'$, then so are the $T'_{J'}$, namely the measures $\sum_J \delta a_{JJ'} T_J$ transported to the coordinates t'; and for every compact set $K \subset U \cap U'$ we have $|T'_{J'}|(K) \leq C_K \sum_J |T_J|(K)$, where C_K depends only on the charts and on K. Moreover, a distribution that agrees near every point with a complex Radon measure is a complex Radon measure. Indeed, take a partition of unity (χ_a) subordinate to a cover by open sets on which the

distribution agrees with complex Radon measures σ_a. Then the distribution is the locally finite sum of the complex measures $\chi_a\sigma_a$. Each of these is a combination of four positive Radon measures with compact support, and the locally finite sums of such measures are Borel measures finite on compact sets, hence Radon by Proposition 1.1.2. Consequently, it suffices to check the condition of order zero in the charts of an atlas.

A family (T_i) of currents of order zero has locally uniformly bounded total variation if the total variations of the coefficient measures of the T_i are uniformly bounded on every compact subset of every chart, and $T_i \to T$ in total variation on compact sets if the total variations of the coefficient measures of $T_i - T$ tend to zero on every such compact set.

A current of order zero extends, by the formula $T(\varphi) = \sum_J \int g_J \, \mathrm{d}T_J$ in charts and a partition of unity, to forms φ with bounded Borel coefficients and compact support. The result does not depend on the choices: (7.8) persists for bounded Borel functions h with compact support in $U \cap U'$, because both sides are integrals against complex Radon measures which agree on test functions, hence on $C_c(U \cap U')$ by Lemma 1.4.1, hence as measures; and two partitions of unity give the same value by the computation used for integrals of forms. If f is a locally bounded Borel function on X, multiplying all coefficient measures by f defines a current fT of order zero; this is compatible with coordinate changes. In particular $\mathbf{1}_E T$ is defined for every Borel set $E \subset X$. We say that T gives E zero mass, or does not charge E, if $\mathbf{1}_E T = 0$, that is, if all coefficient measures of T vanish on E in every chart.

Theorem 7.2.5 *Let T be a positive current of bidegree (p,p) on a complex manifold X of dimension n. Then T is real and of order zero. More precisely, in every coordinate chart the coefficient measures of $T = \sum_{|I|=|J|=p} T_{IJ} \, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ satisfy*

$$|T_{IJ}| \le C_{n,p} \, \mu_T,$$

where μ_T is the trace measure and $C_{n,p}$ depends only on n and p. Moreover, suppose that X has a countable basis, and let (T_j) be a sequence of positive currents of bidegree (p,p) whose trace measures are locally uniformly bounded: $\sup_j \mu_{T_j}(K) < \infty$ for every compact subset K of every chart. Then a subsequence converges weakly to a positive current, which is closed if all T_j are closed.

Proof Fix a chart and put $q = n - p$. By Proposition 7.2.1, β^q is an interior point of the cone of strongly positive (q,q)-forms. Hence for every constant real (q,q)-form τ there is $C_\tau > 0$ such that the forms $C_\tau\beta^q \pm \tau$ are strongly positive. For a nonnegative test function χ, the forms $\chi(C_\tau\beta^q \pm \tau)$ are strongly positive test forms, so positivity of T gives $T(\chi(C_\tau\beta^q \pm \tau)) \ge 0$. Therefore $T(\chi\tau)$ is real and

$$|T(\chi\tau)| \le C_\tau \mu_T(\chi).$$

The distributions $C_\tau\mu_T \pm T(\,\cdot\,\tau)$ are nonnegative on nonnegative test functions, hence positive Radon measures by the remark after Definition 7.2.4 and Theorem 1.4.5. Consequently $S_\tau = T(\,\cdot\,\tau)$ is a real signed Radon measure, the difference of the positive measures $(C_\tau\mu_T \pm S_\tau)/2$, and $|S_\tau(A)| \le C_\tau\mu_T(A)$ for every relatively compact Borel set A, so that $|S_\tau| \le C_\tau\mu_T$.

Choose real constant (q,q)-forms $\tau_1,\dots,\tau_N$ forming a basis of the real (q,q)-forms; they also form a complex basis of all constant (q,q)-forms, and they can be chosen depending only on n and p. Up to a constant factor depending only on n, I and J, the coefficient T_{IJ} is the distribution $\chi \mapsto T(\chi\, \mathrm{d}z_{I^c} \wedge \mathrm{d}\bar z_{J^c})$, where I^c and J^c are the complementary multi-indices. Expanding $\mathrm{d}z_{I^c} \wedge \mathrm{d}\bar z_{J^c} = \sum_r c_r \tau_r$, we see that T_{IJ} is a fixed complex linear combination of the measures S_{τ_r}, and the coefficient bound follows from $|S_{\tau_r}| \le C_{\tau_r}\mu_T$. To prove that T is real, let φ be a test form supported in the chart; only its (q,q)-component contributes, and we write it as $\sum_r f_r\tau_r$ with complex test functions f_r. Since $T(f\tau_r) = \int f\,\mathrm{d}S_{\tau_r}$ is real for real f, we obtain $T(\bar\varphi) = \overline{T(\varphi)}$, and a partition of unity extends this identity to all test forms.

Now let X have a countable basis, and let (T_j) be as in the statement. Then X is covered by countably many charts $U_1, U_2, \dots$: for each basic open set contained in some chart, choose one such chart. On U_1 the positive Radon measures $C_{\tau_r}\mu_{T_j} \pm S^j_{\tau_r}$, where $S^j_\tau = T_j(\,\cdot\,\tau)$, have mass at most $2C_{\tau_r}\sup_j \mu_{T_j}(K)$ on every compact $K \subset U_1$. Applying Theorem 1.5.5 successively to these finitely many sequences, we find a subsequence along which all of them converge vaguely on U_1; then each $S^j_{\tau_r}$ converges vaguely, and $T_j(\varphi)$ converges for every test form φ supported in U_1. Repeating the extraction for $U_2, U_3, \dots$ and passing to the diagonal subsequence, we may assume that $T_j(\varphi)$ converges for every test form φ supported in some U_k. A partition of unity subordinate to (U_k) writes every test form as a finite sum of such forms, so $T(\varphi) = \lim_j T_j(\varphi)$ exists for every test form φ.

The functional T is linear, and it is a current. Indeed, if $\varphi = \sum_r f_r\tau_r$ is supported in a compact set $K \subset U_k$, then $|T_j(\varphi)| \le \sum_r C_{\tau_r}\|f_r\|_\infty \sup_i \mu_{T_i}(K)$ for all j, and this bound passes to the limit. On compact subsets of the intersection of two charts, the coefficients of a form in the two charts are related by a matrix of continuous functions. Hence, covering a compact subset of an arbitrary chart by finitely many U_k and using a partition of unity, we obtain the estimate in the definition of a current, with $m = 0$. Thus $T_j \to T$ weakly. If φ is a strongly positive test form, then $T(\varphi) = \lim_j T_j(\varphi) \ge 0$; and if all T_j are closed, then $T(\mathrm{d}\psi) = \lim_j T_j(\mathrm{d}\psi) = 0$ for every test form ψ. □

Currents can be regularized in coordinates, and the regularization preserves positivity and closedness. Let T be a current on an open set $\Omega \subset \mathbb{C}^n$ with coefficients T_{IJ}, and let ρ_ε and Ω_ε be as in Section 1.4. The smooth form $T_\varepsilon = \sum_{I,J}(T_{IJ})_\varepsilon\,\mathrm{d}z_I \wedge \mathrm{d}\bar z_J$ on Ω_ε satisfies $T_\varepsilon(\varphi) = T(\varphi * \rho_\varepsilon)$ for test forms φ supported in Ω_ε, where $\varphi * \rho_\varepsilon$ is obtained by mollifying the coefficients of φ; this is (1.3), applied to each coefficient. By Proposition 1.4.3, $T_\varepsilon \to T$ weakly as $\varepsilon \to 0$, in the sense that $T_\varepsilon(\varphi) \to T(\varphi)$ for every test form φ on Ω; here $T_\varepsilon(\varphi)$ is defined as soon as $\operatorname{Supp}\varphi \subset \Omega_\varepsilon$.

If T is positive, so is T_ε on Ω_ε. Indeed, let φ be a strongly positive test form supported in Ω_ε. Since the frame $\mathrm{d}z_I \wedge \mathrm{d}\bar z_J$ is invariant under translations, the translates $x \mapsto \varphi(x-h)$ are strongly positive, and $(\varphi * \rho_\varepsilon)(x) = \int \rho_\varepsilon(h)\varphi(x-h)\,\mathrm{d}V_{2n}(h)$ is a limit of Riemann sums, which are nonnegative combinations of strongly positive forms; it is strongly positive because the cone is closed (Proposition 7.2.1). Moreover, $\varphi * \rho_\varepsilon$ is a test form on Ω, supported in the set of points at distance at most ε from $\operatorname{Supp}\varphi$. Hence $T_\varepsilon(\varphi) = T(\varphi * \rho_\varepsilon) \ge 0$. Finally, if T is closed, so is T_ε, since d acts

on the coefficients by differentiation, and derivatives commute with mollification (Proposition 1.4.3).

If T is positive of bidegree (p,p) and α is a smooth strongly positive (s,s)-form with $p+s \le n$, then $T \wedge \alpha$ is positive, because $\alpha \wedge \varphi$ is a strongly positive test form for every strongly positive test form φ (Proposition 7.2.1). The trace measure $T \wedge \beta^{n-p}$ is an example.

Currents on a complex manifold X can be glued from local pieces; we use this in Section 7.5 to define pullbacks of currents. For an open set $W \subset X$, the restriction of T to W is the current on W given by the values of T on test forms supported in W; two currents agree on W if their restrictions to W coincide.

Lemma 7.2.6 *Let $(W_a)_{a\in A}$ be an open cover of X, and for each a let T_a be a current of degree k on W_a, such that T_a and T_b agree on $W_a \cap W_b$ for all $a, b \in A$. Then there is a unique current T of degree k on X which agrees with T_a on W_a for every a. If all T_a have bidegree (p,q), then so has T; if all T_a are closed, then so is T; and if $p = q$ and all T_a are positive, then so is T.*

Proof If two currents on X agree with T_a on W_a for every a, their difference vanishes on every W_a, hence on X, since a current that vanishes on each member of a family of open sets vanishes on their union, as shown at the beginning of this section. For existence, choose a partition of unity $(\chi_c)_{c\in C}$ with compact supports subordinate to the cover (Proposition 7.1.3), and for each c an index $a(c)$ with $\operatorname{Supp}\chi_c \subset W_{a(c)}$. For a test form φ of degree $2n-k$ on X, only finitely many χ_c are not identically zero on the compact set $\operatorname{Supp}\varphi$, and $\chi_c\varphi$ is a test form supported in $W_{a(c)}$; we put

$$T(\varphi) = \sum_c T_{a(c)}(\chi_c\varphi).$$

This is a linear functional, and it is a current. Indeed, let K be a compact subset of a chart domain U. Only finitely many indices $c_1, \dots, c_N$ have $\chi_{c_i} \not\equiv 0$ on K, and for test forms φ supported in K the form $\chi_{c_i}\varphi$ is supported in the compact subset $K \cap \operatorname{Supp}\chi_{c_i}$ of the chart domain $U \cap W_{a(c_i)}$ of $W_{a(c_i)}$. By the Leibniz rule, its seminorms $p_{K,m}$ are bounded by a constant times those of φ, so the estimates for the currents $T_{a(c_i)}$ give the estimate in the definition of a current for T. If φ is supported in some W_b, then each $\chi_c\varphi$ is supported in $W_{a(c)} \cap W_b$, where $T_{a(c)}$ and T_b agree, so that $T(\varphi) = \sum_c T_b(\chi_c\varphi) = T_b(\varphi)$. Thus T agrees with T_b on W_b.

Multiplication by χ_c preserves bidegrees, and $\chi_c\varphi$ is strongly positive at every point if φ is, because $\chi_c \ge 0$; so the assertions about bidegrees and positivity follow from the formula for $T(\varphi)$. Finally, if all T_a are closed and ψ is a test form supported in W_a, then so is $\mathrm{d}\psi$, and $(\mathrm{d}T)(\psi) = (-1)^{k+1}T(\mathrm{d}\psi) = (-1)^{k+1}T_a(\mathrm{d}\psi) = (\mathrm{d}T_a)(\psi) = 0$; hence $\mathrm{d}T$ vanishes on every W_a, and therefore $\mathrm{d}T = 0$. □

Example 7.2.7 Let Z be a complex submanifold of X of dimension r, with its complex orientation. Then

$$[Z](\varphi) = \int_Z \varphi|_Z$$

defines a closed positive current of bidegree $(n-r, n-r)$ on X, the current of integration over Z. Since Z is closed, $Z \cap \operatorname{Supp}\varphi$ is compact for every test form φ, so the integral is defined. In a chart in which $Z = \{z_{r+1} = \cdots = z_n = 0\}$, the value $[Z](\varphi)$ is a constant times the integral over this coordinate plane of the coefficient of $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_I$ in φ, where $I = (1, \ldots, r)$, because the pullbacks of $\mathrm{d}z_j$ and $\mathrm{d}\bar{z}_j$ to Z vanish for $j > r$. Hence $[Z]$ is a current, and its coefficients in such a chart are constant multiples of Lebesgue measure on the plane. Since $[Z]$ vanishes on $X \setminus Z$, the charts of this kind and the charts contained in $X \setminus Z$ form an atlas in which $[Z]$ has measure coefficients, and $[Z]$ is a current of order zero. If φ is strongly positive, so is its pullback $\varphi|_Z$ to Z, which is therefore a nonnegative multiple of the volume form of Z at each point (Proposition 7.2.1); thus $[Z](\varphi) \geq 0$. Finally, since pullback commutes with d by (7.3), $[Z](\mathrm{d}\psi) = \int_Z \mathrm{d}(\psi|_Z) = 0$ by Proposition 7.1.3 applied on Z when $r \geq 1$; when $r = 0$ the current $[Z]$ has degree $2n$, and $\mathrm{d}[Z] = 0$ because currents of degree $2n+1$ are zero.

7.3 Local potentials for closed (1, 1)-currents

Coordinate computations in this section take place on an open set $\Omega \subset \mathbb{C}^n$ with $n \geq 1$, for instance a coordinate domain. By Theorem 7.2.5, a positive $(1,1)$-current on Ω is real and has measure coefficients. We normalize them as

$$T = \frac{\mathrm{i}}{2\pi} \sum_{j,k} t_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k,$$

so that $T = \mathrm{dd}^{\mathrm{c}} u$ means $t_{j\bar{k}} = u_{j\bar{k}}$, by (7.5).

Positivity of T is expressed by the following matrix criterion. The coefficient matrix $(t_{j\bar{k}})$ is Hermitian, that is, $t_{k\bar{j}} = \overline{t_{j\bar{k}}}$, because T is real, and it is positive semidefinite in the distributional sense: $\sum_{j,k} t_{j\bar{k}} \xi_j \bar{\xi}_k$ is a positive measure for every $\xi \in \mathbb{C}^n$. To see this, we may assume that $|\xi| = 1$ and choose a unitary matrix U whose first row is $\bar{\xi}$. In the coordinates $w = Uz$ the coefficient of $\mathrm{d}w_1 \wedge \mathrm{d}\bar{w}_1$ in T is $(\mathrm{i}/2\pi) \sum_{j,k} t_{j\bar{k}} \xi_j \bar{\xi}_k$, and evaluating T on the strongly positive test forms $\chi \bigwedge_{j\geq 2} (\mathrm{i}\mathrm{d}w_j \wedge \mathrm{d}\bar{w}_j)$ with $\chi \geq 0$ gives a positive constant times $\langle \sum_{j,k} t_{j\bar{k}} \xi_j \bar{\xi}_k, \chi \rangle$, which is therefore nonnegative. Conversely, a real $(1,1)$-current whose coefficient matrix is positive semidefinite in this sense is positive. For smooth coefficients this is the $(1,1)$-case discussed after Proposition 7.2.1. In general the mollified currents T_ε have pointwise positive semidefinite coefficient matrices, because mollifications of positive measures are nonnegative (Proposition 1.4.3); hence they are positive, and so is their weak limit T.

Taking for ξ the coordinate vectors, we see that every diagonal coefficient $t_{j\bar{j}}$, and hence the trace $\mu = \sum_j t_{j\bar{j}}$, is a positive measure; expanding β^{n-1} shows that the trace measure of T is $T \wedge \beta^{n-1} = ((n-1)!/\pi)\, \mu$. Since ∂T and $\bar{\partial} T$ are computed by differentiating the coefficients, T is closed if and only if

$$\partial_{z_l} t_{j\bar{k}} = \partial_{z_j} t_{l\bar{k}}, \qquad \partial_{\bar{z}_l} t_{j\bar{k}} = \partial_{\bar{z}_k} t_{j\bar{l}} \qquad \text{for all } j, k, l. \tag{7.9}$$

The construction of local potentials uses the following smooth local exactness results, which we prove by explicit integral formulas.

Lemma 7.3.1 *Let $\Delta \subset \mathbb{C}^n$, $n \geq 1$, be a polydisc, and let g be a smooth $(0,1)$-form on Δ with $\bar\partial g = 0$. Then for every polydisc $\Delta' \Subset \Delta$ there is a smooth function f on Δ' with $\bar\partial f = g$. Moreover, every smooth real closed $(1,1)$-form R on an open subset of $\mathbb{C}^n$ is locally of the form $\mathrm{dd^c} v$: each point has a neighborhood on which $R = \mathrm{dd^c} v$ for a smooth real function v. Finally, every smooth closed k-form γ with $k \geq 1$, with real or complex coefficients, on a ball $B \subset \mathbb{C}^n = \mathbb{R}^{2n}$ is exact on B: $\gamma = \mathrm{d}\alpha$ for a smooth $(k-1)$-form α on B, which is real if γ is real.*

Proof We solve the $\bar\partial$-equation in one variable by a Cauchy integral, then on a polydisc one variable at a time, and combine it with the Poincaré lemma.

Step 1: one variable. The function $K(z) = 1/(\pi z)$ on $\mathbb{C}$ is locally integrable, and $\partial_{\bar z} K = \delta_0$ in the sense of distributions. Indeed, for a test function φ, dominated convergence gives $\langle \partial_{\bar z} K, \varphi\rangle = -\pi^{-1} \lim_{\varepsilon\to 0} \int_{|z|>\varepsilon} (\partial_{\bar z}\varphi/z)\, \mathrm{d}V_2$. In polar coordinates $z = r\mathrm{e}^{\mathrm{i}\vartheta}$ we have $\partial_{\bar z} = \frac{1}{2}\mathrm{e}^{\mathrm{i}\vartheta}(\partial_r + \frac{\mathrm{i}}{r}\partial_\vartheta)$ and $\mathrm{d}V_2 = r\,\mathrm{d}r\,\mathrm{d}\vartheta$. Hence, by Fubini's theorem,

$$\int_{|z|>\varepsilon} \frac{\partial_{\bar z}\varphi}{z}\,\mathrm{d}V_2 = \frac{1}{2}\int_\varepsilon^\infty \int_0^{2\pi} \Big(\partial_r\varphi + \frac{\mathrm{i}}{r}\partial_\vartheta\varphi\Big)\,\mathrm{d}\vartheta\,\mathrm{d}r = -\frac{1}{2}\int_0^{2\pi} \varphi(\varepsilon \mathrm{e}^{\mathrm{i}\vartheta})\,\mathrm{d}\vartheta,$$

because $\int_0^{2\pi} \partial_\vartheta\varphi\,\mathrm{d}\vartheta = 0$ by periodicity and $\int_\varepsilon^\infty \partial_r\varphi\,\mathrm{d}r = -\varphi(\varepsilon\mathrm{e}^{\mathrm{i}\vartheta})$, φ having compact support. As $\varepsilon \to 0$, the right-hand side tends to $-\pi\varphi(0)$, since φ is continuous at 0; so $\langle \partial_{\bar z} K, \varphi\rangle = \varphi(0)$. Consequently, for a smooth function a with compact support in $\mathbb{C}$, the function

$$f(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{a(\zeta)}{z-\zeta}\,\mathrm{d}V_2(\zeta) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{a(z-w)}{w}\,\mathrm{d}V_2(w)$$

is smooth, since in the second expression all derivatives may be taken under the integral sign by dominated convergence, and $\partial_{\bar z} f(z) = \int K(w)(\partial_{\bar z} a)(z-w)\,\mathrm{d}V_2(w) = \langle \partial_{\bar w} K, a(z-\cdot)\rangle = a(z)$, because $\partial_{\bar w}[a(z-w)] = -(\partial_{\bar z} a)(z-w)$. More generally, let $a(\zeta, s)$ be smooth on $\mathbb{C}\times S$, where S is an open subset of some $\mathbb{R}^N$, and suppose that a vanishes for ζ outside a fixed compact set. Then $f(z,s) = \pi^{-1}\int a(z-w,s)w^{-1}\,\mathrm{d}V_2(w)$ is smooth on $\mathbb{C}\times S$, and it is holomorphic in those complex parameters in which a is holomorphic, again by differentiation under the integral sign: for (z,s) in a compact set, the integrand vanishes for w outside a fixed bounded set, and its derivatives are bounded by constants times the integrable function $|w|^{-1}$.

Step 2: the $\bar\partial$-equation on a polydisc. Write $\Delta = \Delta_1\times\cdots\times\Delta_n$ and $\Delta' = \Delta_1'\times\cdots\times\Delta_n'$, and write $g = \sum_j a_j\,\mathrm{d}\bar z_j$. For each j choose a smooth function χ_j with compact support in Δ_j that equals one on an open neighborhood D_j of $\overline{\Delta_j'}$ (Lemma 1.4.1), and put $P_k = D_1\times\cdots\times D_k\times\Delta_{k+1}\times\cdots\times\Delta_n$. We construct smooth functions $f_1,\dots,f_n$ such that, for $k = 0,\dots,n$, the $\bar\partial$-closed form $g^{(k)} = g - \bar\partial(f_1+\cdots+f_k) = \sum_j a_j^{(k)}\mathrm{d}\bar z_j$ satisfies $a_j^{(k)} = 0$ on P_k for $j \leq k$. For $k = 0$ there is nothing to prove. Let $k \geq 1$.

Since $g^{(k-1)}$ is $\bar\partial$-closed, $\partial_{\bar z_i} a_j^{(k-1)} = \partial_{\bar z_j} a_i^{(k-1)} = 0$ on P_{k-1} for $i < k$; thus the coefficients of $g^{(k-1)}$ are holomorphic in $z_1, \dots, z_{k-1}$ there. Define

$$f_k(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{\chi_k(\zeta)\, a_k^{(k-1)}(z_1, \dots, z_{k-1}, \zeta, z_{k+1}, \dots, z_n)}{z_k - \zeta}\, \mathrm{d}V_2(\zeta).$$

By Step 1, applied with the parameters z_i, $i \neq k$, the function f_k is smooth on Δ and holomorphic in $z_1, \dots, z_{k-1}$ on P_{k-1}, and $\partial_{\bar z_k} f_k = \chi_k a_k^{(k-1)} = a_k^{(k-1)}$ on P_k. Hence $g^{(k)} = g^{(k-1)} - \bar\partial f_k$ has the required property. For $k = n$ we obtain $g = \bar\partial(f_1 + \cdots + f_n)$ on $P_n \supset \Delta'$.

Step 3: the Poincaré lemma and real closed $(1,1)$*-forms.* In this step we write points of $\mathbb{C}^n = \mathbb{R}^{2n}$ as $x = (x_1, \dots, x_{2n})$, and we may assume that R is defined on a ball B centered at 0. For a smooth k-form γ on B, $k \geq 1$, define the $(k-1)$-form $H\gamma$ by

$$(H\gamma)_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \gamma_{tx}(x, v_1, \dots, v_{k-1})\, \mathrm{d}t,$$

where x is regarded as a tangent vector at $tx \in B$; the form $H\gamma$ is smooth, by differentiation under the integral sign. We claim that $\mathrm{d}H\gamma + H\mathrm{d}\gamma = \gamma$. Since both sides are linear in γ, it suffices to verify this for $\gamma = c\, \mathrm{d}x_I$ with $|I| = k$. Let ι_x denote contraction with x, and write $\partial_l = \partial/\partial x_l$. Then $H\gamma = F\, \iota_x \mathrm{d}x_I$ with $F(x) = \int_0^1 t^{k-1} c(tx)\, \mathrm{d}t$, and $H\mathrm{d}\gamma = \sum_l \partial_l F\, \iota_x(\mathrm{d}x_l \wedge \mathrm{d}x_I)$, because $\partial_l F(x) = \int_0^1 t^k (\partial_l c)(tx)\, \mathrm{d}t$. Using $\mathrm{d}(\iota_x \mathrm{d}x_I) = k\, \mathrm{d}x_I$ and $\iota_x(\mathrm{d}x_l \wedge \mathrm{d}x_I) = x_l\, \mathrm{d}x_I - \mathrm{d}x_l \wedge \iota_x \mathrm{d}x_I$, we find

$$\mathrm{d}H\gamma + H\mathrm{d}\gamma = \Big(kF + \sum_l x_l \partial_l F\Big)\mathrm{d}x_I = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\big(t^k c(tx)\big)\, \mathrm{d}t\, \mathrm{d}x_I = \gamma,$$

where we used that $t^k c(tx)$ vanishes at $t = 0$, since $k \geq 1$. In particular, a closed k-form γ on B satisfies $\gamma = \mathrm{d}(H\gamma)$, and $H\gamma$ is real if γ is real; after a translation this proves the last assertion of the lemma for every ball. Applied to the closed real 2-form R, this gives $R = \mathrm{d}\alpha$ with the real 1-form $\alpha = HR$. Since α is real, $\alpha^{1,0} = \overline{\alpha^{0,1}}$. The $(0,2)$-component of $\mathrm{d}\alpha = R$ is $\bar\partial \alpha^{0,1} = 0$, so by Step 2, applied to a polydisc in B centered at 0, there is a smooth function f with $\alpha^{0,1} = \bar\partial f$ on a smaller polydisc; then $\alpha^{1,0} = \partial \bar f$. Hence

$$R = \mathrm{d}\alpha = \bar\partial\partial \bar f + \partial\bar\partial f = \partial\bar\partial(f - \bar f) = 2\mathrm{i}\partial\bar\partial \operatorname{Im} f = \mathrm{dd}^{\mathrm{c}}(4\pi \operatorname{Im} f),$$

and $v = 4\pi \operatorname{Im} f$ has the required properties. □

Theorem 7.3.2 *Let T be a closed positive $(1,1)$-current on a complex manifold of positive dimension. Every point has a coordinate neighborhood U on which $T = \mathrm{dd}^{\mathrm{c}} u$ for a real function $u \in L^1_{\mathrm{loc}}(U)$. Since $\Delta u = 4 \sum_j u_{j\bar j} \geq 0$ by the matrix criterion, the class of u has a canonical subharmonic representative in the coordinates of U (Theorem 3.2.2), which we call a canonical local potential of T. Two canonical local*

potentials u and $\widetilde{u}$ of T on the same coordinate neighborhood differ by a smooth pluriharmonic function h, that is, a real smooth function with $\mathrm{dd^c}h = 0$: we have $\widetilde{u} = u + h$ at every point.

Proof Fix a point and a coordinate chart around it, identified with an open set $\Omega \subset \mathbb{C}^n$, and write T as above with trace $\mu = \sum_j t_{j\bar{j}}$, a positive Radon measure on Ω. Choose a test function χ on Ω with $0 \le \chi \le 1$ that equals one on a neighborhood W of the point (Lemma 1.4.1), and let E_{2n} be the fundamental solution of Proposition 2.2.1. By Proposition 3.2.1, the function $u_0 = 4E_{2n} * (\chi\mu)$ is locally integrable and subharmonic, and $\Delta u_0 = 4\chi\mu$; in particular $\Delta u_0 = 4\mu$ on W. The current $R = T - \mathrm{dd^c}u_0$ on W is real and closed, and its coefficients $r_{j\bar{k}} = t_{j\bar{k}} - (u_0)_{j\bar{k}}$ have trace $\sum_l r_{l\bar{l}} = \mu - \Delta u_0/4 = 0$, by (7.5). Using both relations (7.9) for R, we find

$$\tfrac{1}{4}\Delta r_{j\bar{k}} = \sum_l \partial_{z_l}\partial_{\bar{z}_l} r_{j\bar{k}} = \sum_l \partial_{\bar{z}_k}\partial_{z_l} r_{j\bar{l}} = \partial_{z_j}\partial_{\bar{z}_k} \sum_l r_{l\bar{l}} = 0.$$

By Weyl's lemma (Theorem 2.1.4) every coefficient $r_{j\bar{k}}$ is smooth, so R is a smooth real closed $(1,1)$-form on W. By Lemma 7.3.1, $R = \mathrm{dd^c}v$ on a coordinate neighborhood $U \subset W$ of the point, with v smooth and real. Then $u = u_0 + v$ satisfies $\mathrm{dd^c}u = T$ on U. Moreover $\Delta v = 4\sum_l r_{l\bar{l}} = 0$, so v is harmonic. Since the sum of a subharmonic function and a real harmonic function is subharmonic (see the discussion following Theorem 3.1.3), u is a locally integrable subharmonic function. By the uniqueness in Theorem 3.2.2, it is the canonical subharmonic representative of its class.

Now let u and $\widetilde{u}$ be canonical local potentials of T on the same coordinate neighborhood U. Their classes in $L^1_{\mathrm{loc}}(U)$ differ by a real distribution with vanishing complex Hessian, hence with vanishing Laplacian. By Weyl's lemma, this difference is a smooth real harmonic function h, and $\mathrm{dd^c}h = 0$. Hence, as above, $u + h$ is a locally integrable subharmonic function, and it agrees almost everywhere with $\widetilde{u}$. By the uniqueness in Theorem 3.2.2, $\widetilde{u} = u + h$ at every point of U, including the points where $u = -\infty$. □

A smooth real function h is pluriharmonic if and only if it is locally the real part of a holomorphic function. If $h = \mathrm{Re}\, g$ with g holomorphic, then $\bar{\partial} g = 0$ and $\partial \bar{g} = \overline{\bar{\partial} g} = 0$, so $\partial\bar{\partial} h = \frac{1}{2}(\partial\bar{\partial} g - \bar{\partial}\partial\bar{g}) = 0$. Conversely, suppose that $\mathrm{dd^c}h = 0$ on a ball B, which after a translation is centered at 0. The $(1,0)$-form $\partial h = \sum_j h_{z_j} \mathrm{d}z_j$ has holomorphic coefficients, since $\partial_{\bar{z}_k} h_{z_j} = h_{j\bar{k}} = 0$, and it is closed, since $\partial_{z_k} h_{z_j}$ is symmetric in j and k. The function

$$g(z) = 2\int_0^1 \sum_j z_j h_{z_j}(tz)\, \mathrm{d}t$$

is holomorphic on B, by differentiation under the integral sign. Since the functions h_{z_k} are holomorphic, the chain rule and the symmetry give

$$\partial_{z_k} g(z) = 2\int_0^1 \Big(h_{z_k}(tz) + t\sum_j z_j(\partial_{z_j} h_{z_k})(tz)\Big)\mathrm{d}t$$
$$= 2\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\big(t h_{z_k}(tz)\big)\,\mathrm{d}t = 2h_{z_k}(z).$$

Hence $\partial(\operatorname{Re} g - h) = \frac{1}{2}\partial g - \partial h = 0$. Since $\operatorname{Re} g - h$ is real, also $\bar\partial(\operatorname{Re} g - h) = 0$. Thus its gradient vanishes, so $\operatorname{Re} g - h$ is constant on the ball B (Corollary 1.4.4); subtracting this constant from g gives $h = \operatorname{Re} g$ on B. In particular, if f is holomorphic and nowhere zero, then $\log|f|^2$ is pluriharmonic: near each point z_0, a branch of the logarithm on a disc around $f(z_0)$ not containing 0 gives a holomorphic function L with $\mathrm{e}^L = f$, and $\log|f|^2 = 2\operatorname{Re} L$.

Proposition 7.3.3 *Let $\Omega \subset \mathbb{C}^n$, $n \geq 1$, be open, and let u be the canonical subharmonic representative of a real function in $L^1_{\mathrm{loc}}(\Omega)$ with $\mathrm{dd}^{\mathrm{c}} u \geq 0$. Let $D \subset \mathbb{C}^m$, $m \geq 1$, be open and let $F\colon D \to \Omega$ be holomorphic. Then on each connected component of D the composition $u \circ F$ is either identically $-\infty$, or locally integrable and subharmonic with $\mathrm{dd}^{\mathrm{c}}(u \circ F) \geq 0$. In particular, the restriction of u to each connected component of the intersection of Ω with an affine complex line has this alternative.*

Proof We first work near a point. Fix $w_0 \in D$, a ball $B \Subset \Omega$ centered at $F(w_0)$, and a connected open neighborhood W of w_0 with $F(W) \subset B$. The mollifications $u_\varepsilon = u * \rho_\varepsilon$, with the radial mollifier ρ_ε of Section 1.4, are defined on B for small ε, and by (3.3) they decrease pointwise to u as $\varepsilon \downarrow 0$. Their complex Hessians are positive semidefinite at every point: for $\xi \in \mathbb{C}^n$, the function $\sum_{j,k}(u_\varepsilon)_{j\bar k}\xi_j\bar\xi_k$ is the mollification of the positive measure $\sum_{j,k} u_{j\bar k}\xi_j\bar\xi_k$ (matrix criterion), hence nonnegative by Proposition 1.4.3. Since F is holomorphic, the chain rule gives

$$(u_\varepsilon \circ F)_{a\bar b} = \sum_{j,k}(u_\varepsilon)_{j\bar k}(F)\,\frac{\partial F_j}{\partial w_a}\,\overline{\frac{\partial F_k}{\partial w_b}},$$

so the complex Hessian of $u_\varepsilon \circ F$ is positive semidefinite, and in particular its Laplacian is nonnegative. A smooth function with nonnegative Laplacian is subharmonic, because by Proposition 2.1.1 its spherical means increase with the radius and tend to its value at the center. Thus, along a sequence $\varepsilon_k \downarrow 0$, the smooth subharmonic functions $u_{\varepsilon_k} \circ F$ decrease to $u \circ F$ on W. By Theorem 3.1.6, $u \circ F$ is subharmonic on W, and it is either identically $-\infty$ there or locally integrable, in which case $u_{\varepsilon_k} \circ F \to u \circ F$ in $L^1_{\mathrm{loc}}(W)$. In the second case the currents $\mathrm{dd}^{\mathrm{c}}(u_{\varepsilon_k} \circ F)$, which are positive by the matrix criterion, converge weakly to $\mathrm{dd}^{\mathrm{c}}(u \circ F)$, and positivity passes to the limit; so $\mathrm{dd}^{\mathrm{c}}(u \circ F) \geq 0$ on W.

It remains to pass from these neighborhoods to the connected components of D. Let D_0 be a component and let A be the set of points of D_0 having a neighborhood on which $u \circ F \equiv -\infty$. The set A is open. It is also closed in D_0: if $w_0 \in D_0$ lies in the closure of A, the connected neighborhood W of w_0 constructed above meets A, so $u \circ F = -\infty$ on a nonempty open subset of W; since a locally integrable function is finite almost everywhere, the alternative on W forces $u \circ F \equiv -\infty$ on W, and $w_0 \in A$.

Hence either $A = D_0$, or $A = \varnothing$; in the latter case $u \circ F$ is locally integrable and subharmonic with positive complex Hessian near every point of D_0, hence on D_0.□

Applied to holomorphic changes of coordinates, Proposition 7.3.3 shows that the canonical subharmonic representative of a class with positive complex Hessian does not depend on the chart. Here we use that the current $\mathrm{dd^c}u$ of a locally integrable function u, given by $(\mathrm{dd^c}u)(\varphi) = \int u\,\mathrm{dd^c}\varphi$, is defined without reference to coordinates, and that in every chart its coefficients are $(\mathrm{i}/2\pi)u_{j\bar{k}}$, as computed in Section 7.2. Now let F be a biholomorphic map between open subsets of $\mathbb{C}^n$. Since u is finite almost everywhere and biholomorphic maps preserve Lebesgue null sets, $u \circ F$ is not identically $-\infty$ on any component; hence it is locally integrable and subharmonic, and it represents the composition of the class of u with F. By the uniqueness in Theorem 3.2.2, it is the canonical representative in the new coordinates. In particular, canonical local potentials of a closed positive $(1,1)$-current are well defined on open subsets of the manifold, and by Theorem 7.3.2 any two of them differ by a smooth pluriharmonic function on the intersection of their domains.

7.4 Holomorphic zeros and the Poincaré–Lelong formula

Let f be a holomorphic function on a domain in $\mathbb{C}^n$, not identically zero. Away from the zero set of f, the function $\log|f|^2$ is pluriharmonic, so the current $\mathrm{dd^c}\log|f|^2$ is supported on the zero set. We show that it is the current of integration over the zero set, each smooth sheet counted with the vanishing order of f along it. The main difficulty is the singular part of the zero set, where several sheets may meet. Using the Weierstrass preparation theorem in several directions, we show that in suitable linear coordinates this part lies over a Lebesgue null set in every coordinate projection, which is enough to see that it carries no mass.

Lemma 7.4.1 *Let f be a holomorphic function on a domain $\Omega \subset \mathbb{C}^n$, $n \geq 1$, that is not identically zero. Then the zero set of f has Lebesgue measure zero. Let $P \subset \Omega$ be a polydisc, let $1 \leq j \leq n$, and write $P = P' \times D$ and points of P as (z', z_j), where $D \subset \mathbb{C}$ is a disc in the variable z_j and z' consists of the other coordinates. Then the set of points $z' \in P'$ such that $z_j \mapsto f(z', z_j)$ vanishes identically on D is a Lebesgue null subset of $\mathbb{C}^{n-1}$; for $n = 1$ it is empty. Moreover, for every zero a of f there is a complex linear coordinate system centered at a such that f does not vanish identically on any of the coordinate axes through a.*

Proof We prove the first two assertions together by induction on n. For $n = 1$, every zero of f is isolated: at a zero a, the Taylor expansion has a first nonzero term $c(z-a)^m$, since otherwise f would vanish near a and hence on Ω by the identity theorem; so $f(z) = (z-a)^m g(z)$ with $g(a) \neq 0$. Hence the zero set meets every compact subset of Ω in a finite set, and since Ω is a countable union of compact sets, it is countable, hence null. Moreover, f does not vanish identically on a disc $D \subset \Omega$, by the identity theorem, which proves the second assertion for $n = 1$.

Let $n \geq 2$, and let $P = P' \times D$ be as in the statement; after permuting the coordinates, which preserves Lebesgue measure, we may assume that $j = n$, and we write $w = z_n$ and w_0 for the center of D. The functions $c_k(z') = \partial_w^k f(z', w_0)/k!$ are holomorphic on P', and, by the one-variable Taylor expansion of $w \mapsto f(z', w)$ on the disc D centered at w_0, this function vanishes identically on D if and only if all $c_k(z')$ vanish. Since f does not vanish identically on the nonempty open set P, by the identity theorem, some c_k is not identically zero on the polydisc P', and its zero set $N \subset P'$ is null by the first assertion in dimension $n-1$. This proves the second assertion in dimension n, since the set in question is contained in N. For $z' \notin N$, the zero set of $w \mapsto f(z', w)$ in D is null by the case $n = 1$. By Tonelli's theorem, applied to the indicator function of the relatively closed zero set of f in P, this zero set is null; and countably many such polydiscs, for instance those with rational centers and radii contained in Ω, cover Ω.

For the last assertion, translate a to the origin and let P_m be the first nonzero term of the homogeneous expansion of f at 0; it exists because f does not vanish identically near 0, and $m \geq 1$ because $f(0) = 0$. The set $\{P_m \neq 0\}$ is open and dense in $\mathbb{C}^n$, because a polynomial vanishing on a nonempty open set is zero: its derivatives at a point b of that set vanish, and the polynomial is the sum of its finite Taylor expansion at b. Choose $v_1 \in \{P_m \neq 0\}$, and inductively $v_j \in \{P_m \neq 0\}$ outside the linear span of $v_1, \dots, v_{j-1}$; this is possible because the complement of a proper linear subspace is open and dense, and the intersection of two dense open sets is dense. Then $v_1, \dots, v_n$ is a basis, and $f(tv_j) = t^m P_m(v_j) + O(t^{m+1})$ by (7.2), so $t \mapsto f(tv_j)$ does not vanish identically near $t = 0$. Taking $v_1, \dots, v_n$ as coordinate axes proves the assertion. □

Theorem 7.4.2 (Weierstrass preparation) *Let f be holomorphic on a neighborhood U of 0 in $\mathbb{C}^{n-1} \times \mathbb{C}$, $n \geq 1$, and suppose that $w \mapsto f(0, w)$ has a zero of order $m \geq 1$ at $w = 0$. Then there are polydiscs $P' \times \{|w| < \rho\} \subset U$ centered at 0 with arbitrarily small radii, and numbers $\rho_1 \in (0, \rho)$, such that on $P' \times \{|w| < \rho\}$*

$$f(z', w) = g(z', w)W(z', w), \qquad W = w^m + a_1(z')w^{m-1} + \cdots + a_m(z'),$$

where g is holomorphic and nowhere zero, the coefficients a_j are holomorphic on P' with $a_j(0) = 0$, and for every $z' \in P'$ the m roots of $W(z', \cdot)$ lie in $\{|w| < \rho_1\}$.

Proof Since the zeros of $f(0, \cdot)$ are isolated, there is $\rho_0 > 0$ such that $\{0\} \times \{|w| \leq \rho_0\} \subset U$ and $f(0, w) \neq 0$ for $0 < |w| \leq \rho_0$. Let $0 < \rho < \rho_0$, and choose $\rho_1 < \rho < \rho_2 \leq \rho_0$. Then $f(0, w) \neq 0$ on the compact annulus $\{\rho_1 \leq |w| \leq \rho_2\}$. Since $|f(0, \cdot)|$ has a positive minimum on this annulus and f is uniformly continuous on a compact neighborhood of $\{0\} \times \{\rho_1 \leq |w| \leq \rho_2\}$ in U, there is a polydisc P' centered at 0, as small as we wish, such that $P' \times \{|w| \leq \rho_2\} \subset U$ and $f(z', w) \neq 0$ for $z' \in P'$ and $\rho_1 \leq |w| \leq \rho_2$. By the argument principle, the number of zeros of $f(z', \cdot)$ in $\{|w| < \rho\}$, counted with multiplicity, is

$$\frac{1}{2\pi \mathrm{i}} \int_{|w|=\rho} \frac{f_w(z', w)}{f(z', w)} \, \mathrm{d}w,$$

a continuous integer-valued function of z' on the connected set P'; hence it equals its value m at $z' = 0$. Denote these zeros by $\lambda_1(z'), \dots, \lambda_m(z')$, in any order; they lie in $\{|w| < \rho_1\}$.

By the residue theorem, their power sums are

$$p_k(z') = \sum_{l=1}^{m} \lambda_l(z')^k = \frac{1}{2\pi \mathrm{i}} \int_{|w|=\rho} w^k \frac{f_w(z', w)}{f(z', w)} \, \mathrm{d}w, \qquad 1 \le k \le m,$$

which are holomorphic functions on P', by differentiation under the integral sign. The elementary symmetric functions e_k of the roots satisfy Newton's identities

$$k e_k = \sum_{j=1}^{k} (-1)^{j-1} e_{k-j} p_j, \qquad e_0 = 1.$$

To prove them, put $E(t) = \prod_l (1 + t\lambda_l) = \sum_k e_k t^k$ and compare the coefficients of t^{k-1} in $E'(t) = E(t) \sum_{j \ge 1} (-1)^{j-1} p_j t^{j-1}$, which is the expansion of the logarithmic derivative $E'/E = \sum_l \lambda_l/(1 + t\lambda_l)$. By induction on k, every e_k is holomorphic on P'. Hence $W(z', w) = \prod_l (w - \lambda_l(z'))$ is a monic polynomial in w with holomorphic coefficients $a_k = (-1)^k e_k$, and $a_k(0) = 0$ because all $\lambda_l(0) = 0$.

It remains to show that f/W extends to a nowhere vanishing holomorphic function on $P' \times \{|w| < \rho\}$. For fixed $z' \in P'$, the functions $f(z', \cdot)$ and $W(z', \cdot)$ have the same zeros, with the same multiplicities, in $\{|w| < \rho_2\}$; so f/W and W/f extend to holomorphic functions of w there. On the annulus $\{\rho_1 < |w| < \rho_2\}$ both quotients are holomorphic in (z', w). The Cauchy integral

$$G(z', w) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=\rho} \frac{(f/W)(z', \zeta)}{\zeta - w} \, \mathrm{d}\zeta, \qquad |w| < \rho,$$

is holomorphic in (z', w), by differentiation under the integral sign, and equals f/W by Cauchy's formula in w. Defining $\widetilde{G}$ in the same way from W/f, we obtain $G\widetilde{G} = 1$. Thus $g = G$ is holomorphic and nowhere zero, and $f = gW$. □

Proposition 7.4.3 *Let f be holomorphic on a domain $\Omega \subset \mathbb{C}^n$, $n \ge 1$, and not identically zero, and put $\log|f|^2 = -\infty$ on the zero set of f. Then $\log|f|^2$ is locally integrable, it is the canonical subharmonic representative of its class, and $\mathrm{dd^c} \log|f|^2 \ge 0$. Moreover, let $P \subset \Omega$ be a polydisc, let $1 \le j \le n$, and write points of P as (z', z_j), where z' consists of the coordinates other than z_j. Then for every test function φ on P,*

$$\left\langle (\log|f|^2)_{j\bar{j}}, \varphi \right\rangle = \pi \int_{\mathbb{C}^{n-1}} \sum_{a:\, f(z', a)=0} m(z', a) \varphi(z', a) \, \mathrm{d}V_{2n-2}(z'), \tag{7.10}$$

where the sum runs over the zeros a of $z_j \mapsto f(z', z_j)$ in the fiber of P over z', and $m(z', a)$ is the order of that zero; the integrand is defined outside the null set of those z' over which f vanishes identically on the fiber. In particular, if π_j is the projection

of P omitting z_j, the positive measure $(\log|f|^2)_{j\bar{j}}$ gives zero mass to $\pi_j^{-1}(N)$ for every Lebesgue null set N.

Proof *Local integrability.* Near a point where $f \neq 0$, the function $\log|f|^2$ is smooth. Let a be a zero of f. After a translation and a complex-linear change of coordinates, which preserve local integrability, we may assume that $a = 0$ and that $f(0, w)$ does not vanish identically in the last variable w (Lemma 7.4.1). Then $f(0, \cdot)$ has a zero of some order $m \geq 1$ at 0, and by Theorem 7.4.2, $f = gW$ on a polydisc $P' \times \{|w| < \rho\}$, where g is a holomorphic unit and $W(z', w) = \prod_{l=1}^{m}(w - \lambda_l(z'))$ with all roots in $\{|w| < \rho_1\}$ for some $\rho_1 < \rho$. Choose $r \in (\rho_1, \rho)$ and replace P' by a smaller concentric polydisc with compact closure in P'; since g is continuous and nowhere zero, $\log|g|^2$ is then bounded on $P' \times \{|w| < r\}$. For $|\lambda| \leq \rho_1$ we have

$$\int_{|w|<r} \big|\log|w - \lambda|\big|\, dV_2(w) \leq C,$$

with C independent of λ: the positive part of $\log|w-\lambda|$ is at most $\max\{\log(r+\rho_1), 0\}$, and after the translation $\zeta = w - \lambda$ the integral of its negative part is at most $\int_{|\zeta|<1}(-\log|\zeta|)\, dV_2(\zeta)$. Since $|\log|W(z', w)|| \leq \sum_l |\log|w - \lambda_l(z')||$, it follows that $\int_{|w|<r} |\log|W(z', w)||\, dV_2(w) \leq mC$ for every $z' \in P'$. (The estimate holds for each fiber separately, so we need not choose the roots as measurable functions of z'.) By Tonelli's theorem $\log|W|^2$, and hence $\log|f|^2$, is integrable on $P' \times \{|w| < r\}$, a neighborhood of $a = 0$.

The slice formula. For $n = 1$, Example 3.2.4 gives $\Delta\log|f|^2 = 4\pi\sum_a m(a)\delta_a$ locally, that is, $(\log|f|^2)_{z\bar{z}} = \pi\sum_a m(a)\delta_a$; this is (7.10) with $\mathbb{C}^0$ a point. In general, write $P = P' \times D$ with D a disc in the variable z_j. By Lemma 7.4.1, the set N_0 of points $z' \in P'$ over which f vanishes identically on D is null. By Fubini's theorem,

$$\begin{aligned}\big\langle (\log|f|^2)_{j\bar{j}}, \varphi\big\rangle &= \int_P \log|f|^2\, \partial_{z_j}\partial_{\bar{z}_j}\varphi\, dV_{2n} \\ &= \int_{P'} \Big(\int_D \log|f(z', w)|^2\, \partial_w\partial_{\bar{w}}\varphi(z', w)\, dV_2(w)\Big) dV_{2n-2}(z'),\end{aligned}$$

the inner integral being an integrable function of z'. For $z' \notin N_0$, the one-variable case applied to $f(z', \cdot)$ on D shows that the inner integral equals $\pi\sum_a m(z', a)\varphi(z', a)$. This proves (7.10); the integrand on the right is measurable and integrable, being equal almost everywhere to the inner integral.

Positivity. By (7.10), $(\log|f|^2)_{j\bar{j}}$ is nonnegative on nonnegative test functions supported in polydiscs contained in Ω, hence on all nonnegative test functions, which are finite sums of such functions by Lemma 1.4.2. This holds in every system of complex linear coordinates, since f remains holomorphic after a linear change of coordinates. Applying this after a unitary change of coordinates, as in the matrix criterion of Section 7.3, we see that $\sum_{j,k}(\log|f|^2)_{j\bar{k}}\xi_j\bar{\xi}_k$ is a positive measure for every $\xi \in \mathbb{C}^n$. By the matrix criterion, $dd^c\log|f|^2 \geq 0$, and by Theorem 7.2.5 its coefficients are measures.

Null sets. Let $v = (\log|f|^2)_{j\bar{j}}$ on $P = P' \times D$, and let $\eta \geq 0$ be a test function on D. By (7.10), for every nonnegative test function ψ on P',

$$\int_P \psi(z')\eta(z_j)\,\mathrm{d}v = \int_{P'} \psi\, h_\eta\,\mathrm{d}V_{2n-2}, \qquad h_\eta(z') = \pi \sum_a m(z',a)\eta(a) \geq 0.$$

Both sides define positive Radon measures on P' in ψ, the function h_η being locally integrable by the identity itself. By the uniqueness in Theorem 1.4.5, the image measure $A \mapsto \int_{\pi_j^{-1}(A)} \eta\,\mathrm{d}v$ of ηv under π_j is therefore $h_\eta\,\mathrm{d}V_{2n-2}$. Every null set is contained in a Borel null set, by outer regularity of Lebesgue measure, so we may assume that N is a Borel set. Then $\int_{\pi_j^{-1}(N)} \eta\,\mathrm{d}v = \int_N h_\eta\,\mathrm{d}V_{2n-2} = 0$. Letting η increase to 1 on D along a sequence, we obtain $v(\pi_j^{-1}(N)) = 0$ by monotone convergence.

The canonical representative. The trace of $\mathrm{dd^c}\log|f|^2$ is positive, so by Theorem 3.2.2 the class of $\log|f|^2$ has a canonical subharmonic representative v, given by the limits of ball averages in (3.3). At a point where $f \neq 0$, the function $\log|f|^2$ is continuous, so v agrees with it there. At a zero a, there is C with $|f(z)| \leq C|z-a|$ near a, since f is smooth and $f(a) = 0$; hence the average of $\log|f|^2$ over $B(a,r)$ is at most $\log C^2 + 2\log r$, which tends to $-\infty$ as $r \to 0$. Thus $v(a) = -\infty = \log|f(a)|^2$, and $v = \log|f|^2$ everywhere. □

To describe $\mathrm{dd^c}\log|f|^2$, let f be as in Proposition 7.4.3, and consider the zero set $V = f^{-1}(0)$. Its regular part V_{reg} is the set of points $p \in V$ having a neighborhood on which

$$f = g\,a^m, \qquad g \neq 0, \quad \mathrm{d}a \neq 0, \quad m \geq 1,$$

with a and g holomorphic. In a chart around p we have $\mathrm{d}a = \sum_k \partial_{z_k} a\,\mathrm{d}z_k$, so the complex Jacobian matrix of a at p is a nonzero row, and Proposition 7.1.2(2), applied with $r = 1$, shows that a is the first coordinate of a holomorphic chart near p, in which $V = \{a = 0\}$ is a coordinate hyperplane. The integer m depends only on f and p. Indeed, let $f = \widetilde{g}\,\widetilde{a}^{\widetilde{m}}$ be a second such representation near p. Since g and $\widetilde{g}$ do not vanish and $m, \widetilde{m} \geq 1$, both a and $\widetilde{a}$ vanish exactly on V near p. We use holomorphic coordinates (t, x') centered at p with $t = a$, defined on a polydisc. Since $\widetilde{a}(0,x') = 0$, we have $\widetilde{a}(t,x') = t\,b(t,x')$ with the holomorphic function $b(t,x') = \int_0^1 \partial_t \widetilde{a}(st,x')\,\mathrm{d}s$, and $\mathrm{d}\widetilde{a}_p = b(p)\,\mathrm{d}a_p$, so that $b(p) \neq 0$. On the line $x' = 0$, the function $t \mapsto f(t,0)$ is $g(t,0)t^m$ and also $\widetilde{g}(t,0)b(t,0)^{\widetilde{m}}t^{\widetilde{m}}$, with nonvanishing factors in front of the powers of t near $t = 0$; comparing the orders of vanishing at $t = 0$ gives $\widetilde{m} = m$.

We call m the multiplicity, or transverse vanishing order, of f along V at p; it is locally constant on V_{reg}. The set V_{reg} is open in V, so $S = V \setminus V_{\mathrm{reg}}$ is closed, and V_{reg} is a complex hypersurface, that is, a submanifold of dimension $n-1$, of the open set $\Omega \setminus S$. These definitions are local and invariant under holomorphic changes of coordinates, so they apply equally to holomorphic functions on complex manifolds.

Lemma 7.4.4 *Let f be as in Proposition 7.4.3, and let z_0 be a zero of f. There are complex linear coordinates centered at z_0 and a polydisc P centered at z_0 such that,*

for every $j = 1, \dots, n$, the projection π_j of P omitting the coordinate z_j satisfies $S \cap P \subset \pi_j^{-1}(N_j)$ for a Lebesgue null set N_j.

Proof Choose coordinates as in Lemma 7.4.1, so that f does not vanish identically on any coordinate axis through $z_0 = 0$. Fix j, and write points as (z', w) with $w = z_j$. By Theorem 7.4.2 in the variable w, there is a polydisc $P_j = P_j' \times \{|w| < \rho\}$ in the domain of f on which $f = gW$, where g is a holomorphic unit and W is a monic polynomial in w of some degree $m \geq 1$ whose coefficients are holomorphic on P_j' and whose roots lie in $\{|w| < \rho_1\}$ for some $\rho_1 < \rho$. We shall find a holomorphic function h on P_j', not identically zero, such that every zero of f in P_j lying over $\{h \neq 0\}$ belongs to V_{reg}. Then $S \cap P_j \subset \pi_j^{-1}(N_j)$ with $N_j = \{h = 0\}$, which is null by Lemma 7.4.1 if $n \geq 2$, and empty if $n = 1$, when P_j' is a point. Any polydisc P centered at 0 and contained in $P_1 \cap \cdots \cap P_n$ then has the required property.

For $z' \in P_j'$, let $\lambda_1(z'), \dots, \lambda_m(z')$ be the roots of $W(z', \cdot)$, repeated according to multiplicity, and put $p_k(z') = \sum_l \lambda_l(z')^k$ for $k \geq 0$. By the residue theorem,

$$p_k(z') = \frac{1}{2\pi \mathrm{i}} \int_{|w|=\rho'} w^k \frac{\partial_w W(z', w)}{W(z', w)} \, \mathrm{d}w \qquad (\rho_1 < \rho' < \rho),$$

so p_k is holomorphic on P_j', by differentiation under the integral sign. Let s be the largest number of distinct roots of $W(z', \cdot)$ for $z' \in P_j'$, and let h be the determinant of the $s \times s$ matrix $(p_{a+b})_{0 \leq a, b \leq s-1}$; it is holomorphic on P_j'. For fixed z', this matrix is $A^{\mathsf{T}} A$, where A is the $m \times s$ matrix with entries $A_{la} = \lambda_l^a$.

To compute $h(z')$, note that the bth column of $A^{\mathsf{T}} A$ is $\sum_l \lambda_l^b v_l$, where $v_l = (\lambda_l^0, \dots, \lambda_l^{s-1})$ is the lth row of A. Expanding the determinant by multilinearity in the columns, discarding the terms in which some v_l occurs twice, and grouping the remaining terms according to the set L of indices that occur, we find that each group contributes $(\det A_L)^2$. Indeed, if $L = \{l_1 < \cdots < l_s\}$, the terms of this group are indexed by the permutations π of $\{1, \dots, s\}$, the term of π taking $\lambda_{l_{\pi(b+1)}}^b v_{l_{\pi(b+1)}}$ from the bth column. Reordering the columns multiplies the determinant by $\operatorname{sign}(\pi)$, so the group contributes $\det A_L \sum_\pi \operatorname{sign}(\pi) \prod_{b=0}^{s-1} \lambda_{l_{\pi(b+1)}}^b$, and this sum is the Leibniz expansion of $\det A_L$. Thus we obtain the Cauchy–Binet formula

$$h(z') = \det(A^{\mathsf{T}} A) = \sum_L (\det A_L)^2 = \sum_L \prod_{l < l',\ l, l' \in L} (\lambda_l - \lambda_{l'})^2,$$

where L runs over the s-element subsets of $\{1, \dots, m\}$, A_L is the square matrix formed by the rows of A with indices in L, and the last equality is the Vandermonde formula. If $W(z', \cdot)$ has fewer than s distinct roots, every set L contains two indices with equal roots, and $h(z') = 0$. If $W(z', \cdot)$ has exactly s distinct roots $\mu_1, \dots, \mu_s$, of multiplicities $k_1, \dots, k_s$, then the nonzero terms are those of the $k_1 \cdots k_s$ sets L containing exactly one index of each root, and each of them equals $\prod_{c < c'} (\mu_c - \mu_{c'})^2 \neq 0$; hence $h(z') \neq 0$. By the choice of s, the function h is not identically zero.

Now let $h(z_0') \neq 0$, and let $(z_0', w_0) \in P_j$ be a zero of f. Then $W(z_0', w_0) = 0$, since g is a unit, and $W(z_0', \cdot)$ has exactly s distinct roots $\mu_1, \dots, \mu_s$, of multiplicities

$k_1, \dots, k_s$; say $w_0 = \mu_1$. Choose disjoint closed discs E_c centered at μ_c in $\{|w| < \rho\}$, and a connected neighborhood Y of z_0' in P_j' such that $W(z', \cdot)$ has no zeros on the boundary circles ∂E_c for $z' \in Y$; this is possible because $W(z_0', \cdot)$ has no zeros on these compact circles and W is continuous. By the argument principle, the number of roots of $W(z', \cdot)$ in E_c, counted with multiplicity, is $(2\pi\mathrm{i})^{-1} \int_{\partial E_c} \partial_w W/W \, \mathrm{d}w$, a continuous integer-valued function on Y, hence equal to k_c. Thus each E_c contains at least one root of $W(z', \cdot)$. Since $W(z', \cdot)$ has at most s distinct roots, each E_c contains exactly one, $\mu_c(z')$, of multiplicity k_c, and since $k_1 + \cdots + k_s = m$, there are no other roots. By the residue theorem,

$$\mu_c(z') = \frac{1}{2\pi\mathrm{i} k_c} \int_{\partial E_c} w \, \frac{\partial_w W(z', w)}{W(z', w)} \, \mathrm{d}w,$$

which is holomorphic on Y, and $W(z', w) = \prod_c (w - \mu_c(z'))^{k_c}$ for $z' \in Y$, both sides being monic polynomials in w with the same roots and multiplicities. Near (z_0', w_0) the function $U = \prod_{c \geq 2} (w - \mu_c(z'))^{k_c}$ does not vanish, so $f = (gU)(w - \mu_1(z'))^{k_1}$, where gU is a holomorphic unit and $\mathrm{d}(w - \mu_1(z')) \neq 0$. That is, $(z_0', w_0) \in V_{\mathrm{reg}}$. □

Theorem 7.4.5 (Poincaré–Lelong) *Let f be a holomorphic function on a complex manifold X that is not identically zero on any connected component of X, and let V, V_{reg} and S be as above. Then the current $m[V_{\mathrm{reg}}]$ on $X \setminus S$, the integration over V_{reg} weighted by the multiplicity m, has locally finite mass near S: its coefficient measures have finite total variation on $K \setminus S$ for every compact subset K of a chart. Extending its coefficient measures by zero on S, we obtain a closed positive $(1,1)$-current $[\mathrm{div}(f)]$ on X, and*

$$\mathrm{dd^c} \log |f|^2 = [\mathrm{div}(f)]. \tag{7.11}$$

Proof If X has dimension zero, then f has no zeros and both sides vanish. Let X have dimension $n \geq 1$. The set of points near which f vanishes identically is open, and it is closed by the identity theorem, applied in connected coordinate neighborhoods; since f does not vanish identically on any component of X, this set is empty. Hence Proposition 7.4.3 applies in every connected chart: the function $\log |f|^2$ is locally integrable, and $T = \mathrm{dd^c} \log |f|^2$, given by $T(\varphi) = \int_X \log |f|^2 \, \mathrm{dd^c}\varphi$ independently of coordinates, is a positive current with measure coefficients. It is closed because $\mathrm{ddd^c} = \mathrm{ddd^c} = 0$. We show that T gives zero mass to S and agrees with $m[V_{\mathrm{reg}}]$ outside S.

Step 1: T gives S zero mass. Let $x_0 \in S$, and choose coordinates and a polydisc P centered at x_0 as in Lemma 7.4.4. For each j, the last assertion of Proposition 7.4.3 shows that the diagonal coefficient $(\log |f|^2)_{j\bar{j}}$ vanishes on $S \cap P \subset \pi_j^{-1}(N_j)$. By the computation of the trace measure in Section 7.3, the sum of these diagonal coefficients is a positive multiple of the trace measure of T in these coordinates, which therefore vanishes on $S \cap P$; by Theorem 7.2.5 so do all coefficient measures of T in these coordinates. By (7.8), the vanishing of the coefficient measures on a Borel set does not depend on the chart. Every compact subset of $S \cap U$, for a chart U, is covered by finitely many such polydiscs P, so by inner regularity all coefficient measures of T vanish on $S \cap U$.

Step 2: $T = m[V_{\mathrm{reg}}]$ on $X \setminus S$. Since T is defined without reference to coordinates, we may compute it in any holomorphic coordinates. Away from V the function $\log|f|^2$ is pluriharmonic, so $T = 0$ there. Near a point $p \in V_{\mathrm{reg}}$ write $f = g\,a^m$ as above, and use holomorphic coordinates (t, x') with $t = a$. Then $\log|f|^2 = \log|g|^2 + m\log|t|^2$, where $\log|g|^2$ is pluriharmonic. In one variable, $\partial_t\partial_{\bar t}\log|t|^2 = \pi\delta_0$ by Example 3.2.4. Since $\log|t|^2$ does not depend on x', Fubini's theorem shows that its distributional derivatives involving x' or $\bar x'$ vanish and that $\langle(\log|t|^2)_{t\bar t}, \psi\rangle = \pi\int \psi(0,x')\,\mathrm{d}V_{2n-2}(x')$ for test functions ψ; thus $\mathrm{dd^c}\log|t|^2 = (\mathrm{i}/2\pi)(\log|t|^2)_{t\bar t}\,\mathrm{d}t\wedge\mathrm{d}\bar t$. Let φ be a test form of bidegree $(n-1, n-1)$, and let $\varphi_0 = c(t,x')\bigwedge_l(\frac{\mathrm{i}}{2}\mathrm{d}x'_l\wedge\mathrm{d}\bar x'_l)$ be its component without $\mathrm{d}t$ and $\mathrm{d}\bar t$, the only one that contributes. Since $(\mathrm{i}/2\pi)\,\mathrm{d}t\wedge\mathrm{d}\bar t\wedge\varphi_0 = \pi^{-1}c\,\mathrm{d}V_{2n}$, we obtain

$$(\mathrm{dd^c}\log|t|^2)(\varphi) = \int c(0,x')\,\mathrm{d}V_{2n-2}(x') = \int_{\{t=0\}}\varphi|_{\{t=0\}},$$

that is, $\mathrm{dd^c}\log|t|^2 = [\{t=0\}]$ in the sense of Example 7.2.7, the hypersurface carrying its complex orientation. Therefore $T = m[V_{\mathrm{reg}}]$ near p.

Step 3: extension across S. Let U be a chart and $K \subset U$ compact. By Step 2, the trace measure of the positive current $m[V_{\mathrm{reg}}]$ on $U\setminus S$ is the restriction of μ_T, so its mass on $K\setminus S$ is at most $\mu_T(K) < \infty$. By the coefficient bound of Theorem 7.2.5, applied on $U\setminus S$, the coefficient measures of $m[V_{\mathrm{reg}}]$, a positive current of bidegree $(1,1)$, have total variation at most $C_{n,1}\mu_T(K)$ on $K\setminus S$. Hence they extend by zero on S to complex Radon measures on U. Indeed, each coefficient measure σ is a complex combination of the positive and negative variations of $\mathrm{Re}\,\sigma$ and $\mathrm{Im}\,\sigma$, which are positive Radon measures on $U\setminus S$ bounded by $|\sigma|$; extended by zero on S, they become Borel measures on U that are finite on compact sets, hence Radon by Proposition 1.1.2. By Step 2 the coefficient measures of T agree with those of $m[V_{\mathrm{reg}}]$ on $U\setminus S$, and by Step 1 they vanish on S. Thus the extensions by zero are the coefficient measures of the current T of order zero; we denote this current by $[\mathrm{div}(f)]$. This proves (7.11), and $[\mathrm{div}(f)] = T$ is positive and closed. □

Example 7.4.6 Let $1 \le r \le n$ and $f(z) = z_1^{a_1}\cdots z_r^{a_r}$ on $\mathbb{C}^n$, with positive integers a_j. Then $\log|f|^2 = \sum_j a_j\log|z_j|^2$ everywhere, both sides being $-\infty$ on the zero set, and Theorem 7.4.5 applied to each z_j gives $\mathrm{dd^c}\log|f|^2 = \sum_{j=1}^r a_j[\{z_j=0\}]$ on all of $\mathbb{C}^n$, including the intersections of the coordinate hyperplanes. More generally, let g and h be holomorphic and not identically zero on a connected coordinate neighborhood. The function $\log|g|^2 - \log|h|^2$, defined outside the null set where g or h vanishes, is locally integrable, and its complex Hessian is $\mathrm{dd^c}(\log|g|^2 - \log|h|^2) = [\mathrm{div}(g)] - [\mathrm{div}(h)]$. This current depends only on the quotient g/h: if g', h' is another such pair with $gh' = g'h$, the two logarithmic differences agree almost everywhere, hence as distributions. We regard the difference only as an element of L^1_{loc}; in particular, it is assigned no values at the common zeros of g and h.

7.5 Maps of currents and local mass

Let $F\colon X \to Y$ be a holomorphic map between complex manifolds, with $\dim_{\mathbb{C}} X = m$ and $\dim_{\mathbb{C}} Y = n$. Smooth forms on Y can be pulled back to X by means of the derivative of F; the pullback commutes with d, by (7.3), and also with ∂ and $\bar\partial$, because pullback by a holomorphic map preserves types. By duality, currents on X can be pushed forward to Y when F is proper on their support, and currents on Y can be pulled back when F is a submersion. Closed positive $(1,1)$-currents can also be pulled back through their local potentials, as long as these do not become identically $-\infty$.

Proposition 7.5.1 *Let T be a current of bidegree (p,q) on X, and assume that the restriction of F to $\operatorname{Supp} T$ is proper, that is, $\operatorname{Supp} T \cap F^{-1}(L)$ is compact for every compact set $L \subset Y$. Then the pushforward F_*T defined below is a current of bidegree $(p+n-m, q+n-m)$ on Y; it is the zero current if this bidegree lies outside the range $0,\dots,n$. The pushforward commutes with* d*, ∂ and $\bar\partial$, and if $p=q$ and T is positive, then F_*T is positive.*

Proof Let φ be a test form on Y. By properness, the set $\operatorname{Supp} T \cap F^{-1}(\operatorname{Supp}\varphi)$ is compact. Choose a smooth function $\chi \geq 0$ with compact support on X that equals one on a neighborhood of this set (Proposition 7.1.3), and define

$$(F_*T)(\varphi) = T(\chi F^*\varphi).$$

The value does not depend on χ: if χ' is another choice, then $(\chi-\chi')F^*\varphi$ is a test form whose support is disjoint from $\operatorname{Supp} T$, since $\chi = \chi'$ near $\operatorname{Supp} T \cap F^{-1}(\operatorname{Supp}\varphi)$ and $F^*\varphi = 0$ outside $F^{-1}(\operatorname{Supp}\varphi)$; so T vanishes on this form. For all test forms supported in a fixed compact set $L \subset Y$ a single cutoff serves, namely one equal to one near the compact set $\operatorname{Supp} T \cap F^{-1}(L)$; in particular, F_*T is linear. Now let L lie in a chart of Y, fix such a cutoff χ, cover $\operatorname{Supp}\chi$ by finitely many charts of X and choose a partition of unity subordinate to this cover near $\operatorname{Supp}\chi$. By the chain rule, the coefficients of the pieces of $\chi F^*\varphi$ in these charts, together with their derivatives of order at most m', are bounded by a constant times $\sum_J p_{L,m'}(\varphi_J)$, where the φ_J are the coefficients of φ (at points mapped outside $\operatorname{Supp}\varphi$ they vanish). The estimates for T in these charts therefore give the estimate in the definition of a current for F_*T.

Pullback preserves bidegrees, and T acts on test forms of bidegree $(m-p, m-q)$, so F_*T has bidegree $(n-(m-p), n-(m-q))$. For differentiation, note that $\mathrm{d}(\chi F^*\varphi) = \chi F^*\mathrm{d}\varphi + \mathrm{d}\chi \wedge F^*\varphi$, and T vanishes on the last term because its support is disjoint from $\operatorname{Supp} T$. Since the degrees of T and F_*T differ by the even integer $2(m-n)$, the signs in the definitions of $\mathrm{d}T$ and $\mathrm{d}F_*T$ agree, and $\mathrm{d}(F_*T) = F_*(\mathrm{d}T)$; the same argument applies to ∂ and $\bar\partial$. If φ is strongly positive, so is $\chi F^*\varphi$ by Proposition 7.2.1, which proves the last assertion. □

For example, the current $[Z]$ of Example 7.2.7 is the pushforward of the constant function 1, regarded as a current of degree zero on Z, under the inclusion $Z \hookrightarrow X$, which is proper. A current with compact support can be pushed forward by every holomorphic map.

Proposition 7.5.2 *Assume that F is a submersion, that is, its complex derivative has rank n at every point of X. Then every current T on Y has a pullback F^*T, a current on X of the same bidegree, which agrees with the pullback of forms when T is smooth, and with the locally integrable function $a \circ F$ when T is given by a locally integrable function a on Y. The map $T \mapsto F^*T$ is continuous for weak convergence, commutes with* d*, ∂ and $\bar\partial$, and maps positive currents to positive currents.*

Proof *Local construction.* Let $x \in X$, and choose holomorphic coordinates $z = (z_1, \dots, z_n)$ near $F(x)$. Since F is a submersion, the complex Jacobian matrix at x of the n components of $z \circ F$ has rank n in every chart around x. By Proposition 7.1.2(3), these components are the first n coordinates of holomorphic coordinates $(z, w) \in \mathbb{C}^n \times \mathbb{C}^{m-n}$ near x, in which F is the projection $(z, w) \mapsto z$; shrinking, we may assume that the coordinate domain is a product $U \times W$. For a distribution a on U, define the distribution F^*a on $U \times W$ by

$$\langle F^*a, \psi \rangle = \left\langle a, \int_W \psi(\cdot, w)\, \mathrm{d}V_{2m-2n}(w) \right\rangle .$$

Integration in w maps test functions on $U \times W$ continuously to test functions on U: the z-derivatives of $\int_W \psi(\cdot, w)\, \mathrm{d}V_{2m-2n}(w)$ are the integrals of the z-derivatives of ψ, by differentiation under the integral sign, and its support lies in the projection of $\operatorname{Supp}\psi$. Hence F^*a is a distribution, and if a is locally integrable, then F^*a is the function $(z, w) \mapsto a(z)$, by Fubini's theorem. For $T = \sum T_{IJ}\, \mathrm{d}z_I \wedge \mathrm{d}\bar z_J$ on U put $F^*T = \sum F^*T_{IJ}\, \mathrm{d}z_I \wedge \mathrm{d}\bar z_J$. This is a linear map from currents on U to currents on $U \times W$ preserving bidegrees, continuous for weak convergence, and equal to the pullback of forms for smooth T. It commutes with ∂ and $\bar\partial$, since these act on the coefficients by differentiation, $\partial_{z_l} F^*a = F^*\partial_{z_l} a$, and $\partial_{w_l} F^*a = 0$ because $\int_W \partial_{w_l}\psi(\cdot, w)\, \mathrm{d}V_{2m-2n}(w) = 0$ by the fundamental theorem of calculus; similarly for the conjugate derivatives. Continuity for weak convergence holds because the coefficients of a current depend continuously on it, by their explicit description in Section 7.2.

Independence and gluing. Two such local constructions near x agree. Indeed, mollifying the coefficients of T in a chart near $F(x)$, we obtain smooth forms $T_\varepsilon \to T$ weakly on a neighborhood of $F(x)$, as in the discussion following Theorem 7.2.5. Both constructions are weakly continuous and agree with the pullback of forms on each T_ε, so they agree on T near x. Consequently the local pullbacks agree on the overlaps of their domains, and by Lemma 7.2.6 they glue to a unique current F^*T on X that agrees with each local pullback on its domain. Explicitly, if (χ_c) is a partition of unity with compact supports subordinate to the cover of X by the domains of the local pullbacks, and $(F^*T)_c$ denotes a local pullback whose domain contains $\operatorname{Supp}\chi_c$, then $(F^*T)(\varphi) = \sum_c (F^*T)_c(\chi_c\varphi)$.

The local pullbacks are linear, preserve bidegrees and commute with d, ∂ and $\bar\partial$; they give the pullback of forms if T is smooth, and the function $a \circ F$ if T is given by a locally integrable function a. By the uniqueness in Lemma 7.2.6, F^*T has the same properties. Continuity for weak convergence follows from the explicit formula, since each of the finitely many nonzero terms $(F^*T)_c(\chi_c\varphi)$ depends continuously on T.

Positivity. If T is positive, the forms T_ε are positive near $F(x)$, hence weakly positive at every point. By Proposition 7.2.1 their pullbacks are weakly positive at every point, hence positive currents, and positivity passes to the weak limit, the local pullback of T near x. Hence F^*T is positive, by Lemma 7.2.6. □

On currents of order zero, the submersion pullback is described by product measures. Let $U \subset \mathbb{C}^n$ and $W \subset \mathbb{C}^k$ be open, and let λ_W be Lebesgue measure $\mathrm{d}V_{2k}$ on W, which is the unit mass on the point $W = \mathbb{C}^0$ if $k = 0$ and $W \neq \varnothing$. Every open subset of $U \times W$ is a countable union of products of open boxes, so the Borel sets of $U \times W$ form the σ-algebra generated by the products of Borel subsets of U and of W. For a positive Radon measure σ on U, which is σ-finite by Proposition 1.1.2, Tonelli's theorem therefore defines the product measure $\sigma \otimes \lambda_W$ on the Borel sets of $U \times W$, and

$$\int_{U\times W} h \,\mathrm{d}(\sigma \otimes \lambda_W) = \int_U \Big(\int_W h(z,w)\,\mathrm{d}V_{2k}(w) \Big) \mathrm{d}\sigma(z) \tag{7.12}$$

for every nonnegative Borel function h on $U \times W$, the inner integral being a Borel function of z. A compact subset of $U \times W$ lies in the product of its two projections, which are compact, so $\sigma \otimes \lambda_W$ is finite on compact sets and hence a positive Radon measure, by Proposition 1.1.2. For a complex Radon measure $\sigma = \sigma_1 - \sigma_2 + \mathrm{i}(\sigma_3 - \sigma_4)$ on U we put $\sigma \otimes \lambda_W = \sigma_1 \otimes \lambda_W - \sigma_2 \otimes \lambda_W + \mathrm{i}(\sigma_3 \otimes \lambda_W - \sigma_4 \otimes \lambda_W)$. By linearity, (7.12) holds for every bounded Borel function h with compact support. Its right-hand side is the integral against σ of the bounded Borel function $\widehat{h}(z) = \int_W h(z,w)\,\mathrm{d}V_{2k}(w)$, which has compact support, so it does not depend on the decomposition of σ; neither does $\sigma \otimes \lambda_W$, which is determined by its integrals of continuous functions with compact support.

Corollary 7.5.3 *Let $F\colon X \to Y$ be a submersion, and let T be a current of order zero on Y.*

(1) *Suppose that $X = U \times W$ and $Y = U$, where $U \subset \mathbb{C}^n$ and $W \subset \mathbb{C}^{m-n}$ are open, and that F is the projection $(z,w) \mapsto z$. Write $T = \sum_{I,J} T_{IJ}\,\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$. Then, with the product measures $T_{IJ} \otimes \lambda_W$ defined above,*

$$F^*T = \sum_{I,J} (T_{IJ} \otimes \lambda_W)\,\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J;$$

that is, in the coordinates (z,w) the coefficient measure of F^*T at $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ is $T_{IJ} \otimes \lambda_W$, and the coefficients at the basis forms containing some $\mathrm{d}w_l$ or $\mathrm{d}\bar{w}_l$ vanish.

(2) *In general, F^*T has order zero, and*

$$F^*(fT) = (f \circ F)\,F^*T$$

for every locally bounded Borel function f on Y. In particular, $F^*(\mathbf{1}_E T) = \mathbf{1}_{F^{-1}(E)} F^*T$ for every Borel set $E \subset Y$.

Proof (1) The local construction in the proof of Proposition 7.5.2 can be carried out with the coordinates z on U and (z, w) on $U \times W$, and it is then defined on all of $U \times W$. By the independence of the local constructions shown in that proof, F^*T agrees with it near every point, hence on $U \times W$. Thus the coefficients of F^*T at the basis forms containing some $\mathrm{d}w_l$ or $\mathrm{d}\bar{w}_l$ vanish, and the coefficient at $\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ is the distribution $\psi \mapsto \langle T_{IJ}, \widehat{\psi}\rangle$, where $\widehat{\psi}(z) = \int_W \psi(z, w)\, \mathrm{d}V_{2m-2n}(w)$ is a test function on U. Since T_{IJ} is a complex Radon measure, (7.12) gives $\langle T_{IJ}, \widehat{\psi}\rangle = \int_U \widehat{\psi}\, \mathrm{d}T_{IJ} = \int \psi\, \mathrm{d}(T_{IJ} \otimes \lambda_W)$ for every test function ψ on $U \times W$. Hence this coefficient is the complex Radon measure $T_{IJ} \otimes \lambda_W$.

(2) Let $x \in X$. As in the proof of Proposition 7.5.2, there are holomorphic coordinates z on a chart of Y around $F(x)$ and holomorphic coordinates (z, w) on a neighborhood N of x, with coordinate domain a product $U \times W$, in which F is the projection $(z, w) \mapsto z$. On N, the current F^*T is the pullback of the restriction of T to the open set U by this projection. By (1), the coefficients of F^*T in the chart N are complex Radon measures. The charts N obtained in this way form an atlas of X, so F^*T has order zero, by the discussion following (7.8).

Now let f be a locally bounded Borel function on Y; then $f \circ F$ is a locally bounded Borel function on X, since F is continuous. In the coordinates above, the coefficient measures of fT are the measures fT_{IJ}, where f is written in the coordinates z, so by (1) the coefficient measures of $F^*(fT)$ on N are the measures $(fT_{IJ}) \otimes \lambda_W$, while those of $(f \circ F)F^*T$ are the measures $(f \circ F)(T_{IJ} \otimes \lambda_W)$; the coefficients at the basis forms containing some $\mathrm{d}w_l$ or $\mathrm{d}\bar{w}_l$ vanish for both currents. Let g be a continuous function with compact support in $U \times W$. The function $g \cdot (f \circ F)$ is bounded, Borel and compactly supported, and $\widehat{g}(z) = \int_W g(z, w)\, \mathrm{d}V_{2m-2n}(w)$ is a bounded Borel function on U with compact support. Applying (7.12) twice, we obtain

$$\int g\, (f \circ F)\, \mathrm{d}(T_{IJ} \otimes \lambda_W) = \int_U \widehat{g}\, f\, \mathrm{d}T_{IJ} = \int_U \widehat{g}\, \mathrm{d}(fT_{IJ}) = \int g\, \mathrm{d}\big((fT_{IJ}) \otimes \lambda_W\big).$$

Hence the two currents have the same coefficient measures on N, and therefore agree on N. Since the sets N cover X, the currents $F^*(fT)$ and $(f \circ F)F^*T$ coincide. The last assertion is the case $f = \mathbf{1}_E$, since $\mathbf{1}_E \circ F = \mathbf{1}_{F^{-1}(E)}$. □

Proposition 7.5.4 *Let T be a closed positive $(1,1)$-current on Y, and let $F\colon X \to Y$ be holomorphic. Suppose that every point of X has a neighborhood W such that $F(W)$ lies in the domain of a canonical local potential u of T and $u \circ F$ is not identically $-\infty$ on any connected component of W. Then the currents $\mathrm{dd}^{\mathrm{c}}(u \circ F)$ on these neighborhoods glue to a closed positive current*

$$F^*T = \mathrm{dd}^{\mathrm{c}}(u \circ F)$$

on X, which does not depend on the chosen potentials. It agrees with the pullback of forms if T is smooth, and with the pullback of Proposition 7.5.2 if F is a submersion.

Proof Canonical local potentials are defined on manifolds of positive dimension (Theorem 7.3.2), so Y has positive dimension. If X has dimension zero, every $(1,1)$-current on X vanishes, and $F^*T = 0$. Assume that X has positive dimension, let W

and u be as in the hypothesis, and let W_0 be a connected component of W. Each point of W_0 has a connected neighborhood $W' \subset W_0$ in a chart of X such that $F(W')$ lies in a chart of Y contained in the domain of u. In that chart, u is the canonical subharmonic representative of its class, by the remark following Proposition 7.3.3, so by that proposition $u \circ F$ is either identically $-\infty$ on W', or locally integrable on W' with $\mathrm{dd^c}(u \circ F) \geq 0$. No point has neighborhoods W' of both kinds, because a locally integrable function is finite almost everywhere. Hence the points of W_0 having a neighborhood of the first kind and those having one of the second kind form two disjoint open sets covering W_0. Since W_0 is connected and $u \circ F$ is not identically $-\infty$ on W_0, the first set is empty. Thus $u \circ F$ is locally integrable on W, and $\mathrm{dd^c}(u \circ F) \geq 0$ there.

Let $(\widetilde{W}, \widetilde{u})$ be another pair as in the hypothesis. Near each point of the intersection of the domains of u and $\widetilde{u}$ we have $\widetilde{u} = u + h$ with h smooth and pluriharmonic, by Theorem 7.3.2 and the remark following Proposition 7.3.3. Locally $h = \mathrm{Re}\, g$ with g holomorphic, so $h \circ F = \mathrm{Re}(g \circ F)$ is smooth and pluriharmonic, and $\widetilde{u} \circ F = u \circ F + h \circ F$ near each point of $W \cap \widetilde{W}$. Consequently $\mathrm{dd^c}(\widetilde{u} \circ F) = \mathrm{dd^c}(u \circ F)$ near each point of $W \cap \widetilde{W}$, hence on $W \cap \widetilde{W}$, since a current that vanishes near each point vanishes. By Lemma 7.2.6, the local currents $\mathrm{dd^c}(u \circ F)$, which are positive and closed because $\mathrm{ddd^c} = 0$, glue to a unique closed positive current F^*T on X. It is independent of the choices: the currents obtained from two systems of choices agree near each point, by what we have just shown, and hence coincide by the uniqueness in Lemma 7.2.6.

If T is smooth, Lemma 7.3.1 gives smooth real local potentials v with $T = \mathrm{dd^c} v$; they are subharmonic, by the matrix criterion and Proposition 2.1.1, hence their own canonical representatives, so we may use them as the potentials u. For a smooth potential u the chain rule gives $\mathrm{dd^c}(u \circ F) = F^* \mathrm{dd^c} u = F^*T$. Now let F be a submersion, and let V be the domain of u. The restriction of F to $F^{-1}(V)$ is a submersion to V, and its submersion pullback agrees on $F^{-1}(V)$ with that of F, since both are glued from the same local constructions. By Proposition 7.5.2, the locally integrable function $u \circ F$ represents the submersion pullback F^*u, and since this pullback commutes with ∂ and $\bar{\partial}$, we obtain $\mathrm{dd^c}(u \circ F) = \mathrm{dd^c} F^*u = F^* \mathrm{dd^c} u = F^*T$ on $F^{-1}(V)$, in particular on W. □

Example 7.5.5 On $\mathbb{C}^2$, the current $T = [\{z_1 = 0\}]$ has the canonical local potential $\log |z_1|^2$, by Theorem 7.4.5 and Proposition 7.4.3. For $F(w) = (0, w)$, the composition $\log |z_1|^2 \circ F$ is identically $-\infty$, and so is the composition of every canonical local potential, since these differ from $\log |z_1|^2$ by smooth functions; thus Proposition 7.5.4 does not define F^*T. For $G(w) = (w, 0)$, the composition is $\log |w|^2$, and $G^*T = [0]$ is the Dirac mass at the origin. Thus the current of integration over a line can be restricted to a transverse line, but not to the line itself.

We finally compare trace measures in different charts by means of a Hermitian metric. A Hermitian metric on X is a smooth field of positive definite Hermitian quadratic forms on its tangent spaces. In coordinates we write it as $h = \sum h_{j\bar{k}} \mathrm{d}z_j \otimes \mathrm{d}\bar{z}_k$, with $(h_{j\bar{k}})$ a positive definite Hermitian matrix at each point, and we associate with it

the real $(1,1)$-form $\omega = (\mathrm{i}/2)\sum h_{j\bar{k}}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$; for the Euclidean metric $\sum_j \mathrm{d}z_j \otimes \mathrm{d}\bar{z}_j$ we obtain $\omega = \beta$. Every complex manifold carries Hermitian metrics: a partition of unity combines the Euclidean metrics of charts into a locally finite convex combination, which is positive definite.

By the matrix description of strict positivity in Section 7.2, the forms ω obtained in this way are exactly the smooth real $(1,1)$-forms that are strictly positive at every point; we call them *Hermitian forms*. A Hermitian form is a *Kähler form* if it is closed, and a complex manifold carrying a Kähler form is a *Kähler manifold*. For instance, β is a Kähler form on $\mathbb{C}^n$, but in general $\mathrm{d}\omega \neq 0$: on $\mathbb{C}^2$, the Hermitian form $\omega = \beta + \frac{\mathrm{i}}{2}|z_1|^2\mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_2$ has $\mathrm{d}\omega = \frac{\mathrm{i}}{2}(\bar{z}_1\mathrm{d}z_1 + z_1\mathrm{d}\bar{z}_1) \wedge \mathrm{d}z_2 \wedge \mathrm{d}\bar{z}_2 \neq 0$. We usually denote Hermitian and Kähler forms by ω; in Part I the same letter denoted other objects, such as points of the unit sphere and harmonic measures.

For a positive current T of bidegree (p,p) on X, its trace measure with respect to ω is

$$\sigma_{T,\omega} = T \wedge \frac{\omega^{n-p}}{(n-p)!}.$$

Since ω is positive, ω^{n-p} is a smooth strongly positive form (Proposition 7.2.1), so $T \wedge \omega^{n-p}$ is a positive current of degree $2n$; hence $\sigma_{T,\omega}$ is a positive Radon measure on X, by the discussion following Definition 7.2.4, and unlike μ_T it does not depend on a chart. It is comparable with the trace measures in charts. Let K be a compact subset of a chart, choose a compact neighborhood K' of K in the chart, and let $0 < c \leq C$ be lower and upper bounds for the eigenvalues of $(h_{j\bar{k}})$ on K'. At a point of K', a unitary change of coordinates w puts ω in the form $(\mathrm{i}/2)\sum_j \lambda_j\,\mathrm{d}w_j \wedge \mathrm{d}\bar{w}_j$ with $c \leq \lambda_j \leq C$, while β keeps its form. Expanding,

$$\frac{\omega^{n-p}}{(n-p)!} = \sum_{|L|=n-p} \Big(\prod_{l\in L}\lambda_l\Big) \bigwedge_{l\in L}\Big(\frac{\mathrm{i}}{2}\mathrm{d}w_l \wedge \mathrm{d}\bar{w}_l\Big),$$

$$\frac{\beta^{n-p}}{(n-p)!} = \sum_{|L|=n-p} \bigwedge_{l\in L}\Big(\frac{\mathrm{i}}{2}\mathrm{d}w_l \wedge \mathrm{d}\bar{w}_l\Big).$$

Hence $\omega^{n-p} - c^{n-p}\beta^{n-p}$ and $C^{n-p}\beta^{n-p} - \omega^{n-p}$ are strongly positive at every point of K', and positivity of T gives

$$c^{n-p}\mu_T \leq (n-p)!\,\sigma_{T,\omega} \leq C^{n-p}\mu_T \qquad \text{on } \operatorname{Int} K' \supset K.$$

Consequently, for a family of positive currents, locally uniform bounds for the trace measures in charts, as in Theorem 7.2.5, are equivalent to uniform bounds for $\sigma_{T,\omega}(K)$ on every compact set $K \subset X$, for any Hermitian metric.

For a complex submanifold Z of X of dimension r, the restriction of $\omega^r/r!$ to Z is the volume form of the Riemannian metric $\operatorname{Re} h$ restricted to Z, that is, the $2r$-form on Z that takes the value 1 on positively oriented bases of the tangent spaces which are orthonormal for $\operatorname{Re} h$. Indeed, at a point of Z choose a basis $e_1, \ldots, e_r$ of the complex tangent space of Z that is orthonormal for h. Then $e_1, \mathrm{i}e_1, \ldots, e_r, \mathrm{i}e_r$ is a positively oriented orthonormal basis for $\operatorname{Re} h$, and in the dual real coordinates the restriction of

ω is $\sum_{j=1}^{r} \mathrm{d}x_j \wedge \mathrm{d}y_j$, whose rth power divided by $r!$ is $\mathrm{d}x_1 \wedge \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}x_r \wedge \mathrm{d}y_r$. Consequently the trace measure $\sigma_{[Z],\omega}$, given by $\sigma_{[Z],\omega}(\chi) = \int_Z \chi\, \omega^r/r!$, is the Riemannian volume measure of Z. Finally, in dimension zero all forms of positive degree and all $(1,1)$-currents vanish, and integration currents and currents of degree zero are sums over the discrete points of the manifold.

Chapter 8
Plurisubharmonic functions

Plurisubharmonic functions are the basic objects of pluripotential theory. On a domain $\Omega \subset \mathbb{C}^n$, a plurisubharmonic function is an upper semicontinuous function whose restriction to every complex line is subharmonic. By (7.5), the Laplacian of a function on Ω, viewed as an open subset of $\mathbb{R}^{2n}$, is four times the trace of its complex Hessian, so subharmonicity controls only this trace. Subharmonicity on all complex lines is the pointwise counterpart of the positivity of the whole complex Hessian, $\mathrm{dd}^{\mathrm{c}} u \geq 0$.

Since $\mathrm{dd}^{\mathrm{c}} u \geq 0$ determines u only almost everywhere, the pointwise definition and the distributional condition must be kept apart. We first show that the upper semicontinuity required in the definition follows from the condition on complex lines (Theorem 8.1.3); the proof combines the incidence estimate of Chapter 6 with the distributional theory of Chapter 3. We then identify the plurisubharmonic functions with the canonical representatives, in the sense of Theorem 3.2.2, of the distributions with positive complex Hessian (Theorem 8.2.1). This makes the potential theory of Part I available. A smooth function is plurisubharmonic exactly when its complex Hessian is positive semidefinite, and mollification approximates every plurisubharmonic function from above by smooth ones. Plurisubharmonicity is invariant under holomorphic maps, hence makes sense on complex manifolds, and it is preserved by decreasing limits, regularized upper envelopes and L^1_{loc} limits. Moreover, plurisubharmonic functions belong to L^p_{loc} for every finite p, and near each point $\mathrm{e}^{-\alpha u}$ is integrable for small $\alpha > 0$.

In the last two sections we extend plurisubharmonic functions across closed sets that are locally contained in zero sets of holomorphic functions or are small in the sense of Hausdorff measure, and we prove Kiselman's minimum principle: although an infimum of plurisubharmonic functions is in general not plurisubharmonic, a partial infimum over real variables is plurisubharmonic or identically $-\infty$ under convexity assumptions, provided that the function does not depend on the corresponding imaginary parts.

Historical remarks and references for this chapter are collected in Appendix B.8.

8.1 The line definition and upper semicontinuity

We identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ through $z_j = x_j + \mathrm{i}y_j$, and we call a function on an open subset of $\mathbb{C}^n$ *subharmonic* if it is subharmonic in the sense of Definition 3.1.1 on the corresponding open subset of $\mathbb{R}^{2n}$. Recall from Section 6.4 that an affine complex line is a set $L = \{a + \zeta b : \zeta \in \mathbb{C}\}$ with $a \in \mathbb{C}^n$ and $b \in \mathbb{C}^n \setminus \{0\}$. For an open set $\Omega \subset \mathbb{C}^n$, the section $L \cap \Omega$ is identified with the open set $\{\zeta \in \mathbb{C} : a + \zeta b \in \Omega\}$, and a function on a connected component of $L \cap \Omega$ is called subharmonic if its composite with $\zeta \mapsto a + \zeta b$ is subharmonic; this does not depend on the parametrization of L. Accordingly, the restriction of a function u on Ω to $L \cap \Omega$ is the function $\zeta \mapsto u(a + \zeta b)$.

Definition 8.1.1 Let $\Omega \subset \mathbb{C}^n$ be a domain. If $n \geq 1$, a function $u\colon \Omega \to [-\infty, \infty)$ is *plurisubharmonic* if $u \not\equiv -\infty$, u is upper semicontinuous, and for every affine complex line L the restriction of u to each connected component of $L \cap \Omega$ is subharmonic or identically $-\infty$. The set of plurisubharmonic functions on Ω is denoted by $\mathrm{PSH}(\Omega)$. In dimension zero a domain is a single point, and the plurisubharmonic functions on it are the real-valued functions.

We often abbreviate *plurisubharmonic* to *psh*. Since the constant $-\infty$ is subharmonic by Definition 3.1.1, the alternative "identically $-\infty$" in the line condition is stated only for emphasis; it occurs, for instance, for $\log|z_1|^2$ on the line $\{z_1 = 0\}$. In dimension one the only affine complex line is $\mathbb{C}$ itself, so $\mathrm{PSH}(\Omega)$ is the set of nontrivial subharmonic functions on Ω, that is, of those not identically $-\infty$. Whenever a limit of plurisubharmonic functions may degenerate to the function identically $-\infty$, we state this alternative explicitly.

We keep upper semicontinuity in Definition 8.1.1, but it is redundant: Theorem 8.1.3 below shows that it follows from the other two conditions. The proof of that theorem uses the following slicing lemma, which we apply again in Theorems 8.2.1 and 8.4.3: for a locally integrable function, subharmonicity on the complex lines parallel to a fixed direction makes the second derivative in that direction a positive distribution.

Lemma 8.1.2 *Let $\Omega \subset \mathbb{C}^n$ be open, $n \geq 1$, let $u\colon \Omega \to [-\infty, \infty)$ be a locally integrable function, let $a \in \mathbb{C}^n$ be a unit vector, and let $\chi \in C_c^\infty(\Omega)$ be nonnegative. Write the points of $\mathbb{C}^n$ as $v + ta$ with $v \in (\mathbb{C}a)^\perp$ and $t \in \mathbb{C}$, and put $K_v = \{t \in \mathbb{C} : v + ta \in \operatorname{Supp}\chi\}$. Assume that for almost every $v \in (\mathbb{C}a)^\perp$ there is an open set $G_v \subset \{t \in \mathbb{C} : v + ta \in \Omega\}$ containing K_v such that the restriction of $t \mapsto u(v + ta)$ to each connected component of G_v is subharmonic, possibly identically $-\infty$. Then*

$$\Big\langle \sum_{j,k} a_j \bar{a}_k u_{j\bar{k}}, \chi \Big\rangle \geq 0.$$

Consequently, if this hypothesis holds for every unit vector a and every nonnegative $\chi \in C_c^\infty(\Omega)$, then $\mathrm{dd^c}u \geq 0$. This is the case, in particular, if the restriction of u to each connected component of each affine complex-line section of Ω is subharmonic.

Proof The map $(v,t) \mapsto v + ta$ is a unitary isomorphism of $(\mathbb{C}a)^\perp \times \mathbb{C}$ onto $\mathbb{C}^n$, so it preserves Lebesgue measure, and by the chain rule $\partial_t \partial_{\bar t}[\chi(v+ta)] = \sum_{j,k} a_j \bar a_k \chi_{j\bar k}(v+ta)$. We extend u and χ by zero outside Ω. Since u is locally integrable and the derivatives of χ are bounded and supported in $\operatorname{Supp}\chi$, Fubini's theorem gives

$$\Big\langle \sum_{j,k} a_j \bar a_k u_{j\bar k}, \chi \Big\rangle = \int_{(\mathbb{C}a)^\perp} I(v)\, dV_{2n-2}(v),$$

where

$$I(v) = \int_{\mathbb{C}} u(v+ta)\, \partial_t \partial_{\bar t}[\chi(v+ta)]\, dV_2(t).$$

Applied to $|u|\mathbf{1}_{\operatorname{Supp}\chi}$, Fubini's theorem also shows that for almost every v the function $t \mapsto u(v+ta)$ is integrable on K_v. It therefore suffices to show that $I(v) \geq 0$ for every v for which this integrability holds and an open set G_v as in the hypothesis exists.

Fix such a v. The compact set K_v is covered by the disjoint open connected components of G_v, so it meets only finitely many of them, say $U_1, \dots, U_N$, and each set $K_v \cap U_i$ is compact, being the complement in K_v of the open set $K_v \cap \bigcup_{l \neq i} U_l$. Hence the function $\chi_i(t) = \chi(v+ta)$, $t \in U_i$, is a nonnegative test function on U_i, and

$$I(v) = \sum_{i=1}^{N} \int_{U_i} u(v+ta)\, \partial_t \partial_{\bar t} \chi_i(t)\, dV_2(t).$$

If $\chi_i \equiv 0$, the i-th term vanishes. Otherwise $\chi_i > 0$ on a nonempty open subset of U_i, contained in K_v, on which $t \mapsto u(v+ta)$ is integrable and hence finite almost everywhere. Thus the subharmonic function $w_i(t) = u(v+ta)$ on the domain U_i is not identically $-\infty$, and it is locally integrable by Proposition 3.1.5. Since $\partial_t \partial_{\bar t} = \frac{1}{4}\Delta$, the i-th term equals $\frac{1}{4}\langle \Delta w_i, \chi_i \rangle$, which is nonnegative by Theorem 3.2.2(1). Hence $I(v) \geq 0$.

If the hypothesis holds for every unit vector a and every χ, then $\sum_{j,k} a_j \bar a_k u_{j\bar k}$ is a positive distribution for every unit vector a, and by homogeneity for every $a \in \mathbb{C}^n$. Since $dd^c u$ is a real $(1,1)$-current, the matrix criterion for positivity in Section 7.3 shows that $dd^c u \geq 0$. For the last assertion, take $G_v = \{t \in \mathbb{C} : v + ta \in \Omega\}$ for every v; its connected components are the connected components of the affine complex-line sections of Ω in the direction a. □

The proof of the next theorem also uses polar coordinates adapted to the complex lines through a point. Let $n \geq 1$, let τ be the normalized surface measure on the unit sphere $S^{2n-1} \subset \mathbb{C}^n$, and let s_{2n-1} be the area of this sphere. Let $x \in \mathbb{C}^n$ and $r > 0$. For every Borel function $F\colon B(x,r) \to [0,\infty]$, integration in polar coordinates (2.1), in which $dS = s_{2n-1}\, d\tau$, gives the first equality in

$$\begin{aligned} \int_{B(x,r)} F\, dV_{2n} &= s_{2n-1} \int_{S^{2n-1}} \int_0^r F(x+\rho a) \rho^{2n-1}\, d\rho\, d\tau(a) \\ &= \frac{s_{2n-1}}{2\pi} \int_{S^{2n-1}} \int_{|t|<r} F(x+ta)|t|^{2n-2}\, dV_2(t)\, d\tau(a). \end{aligned} \tag{8.1}$$

For the second, we replace a by $e^{i\theta}a$, which does not change τ, average over $\theta \in [0, 2\pi]$, and use polar coordinates $t = \rho e^{i\theta}$ in $\mathbb{C}$; Tonelli's theorem justifies the exchanges of integrals.

Theorem 8.1.3 *Let $\Omega \subset \mathbb{C}^n$ be a domain, $n \geq 1$, and let $u\colon \Omega \to [-\infty, \infty)$ be a function that is not identically $-\infty$. Assume that for every affine complex line L the restriction of u to each connected component of $L \cap \Omega$ is subharmonic or identically $-\infty$. Then u is upper semicontinuous, and hence $u \in \mathrm{PSH}(\Omega)$.*

Proof We first treat finite nonnegative functions, in Steps 1–3, and then reduce the general case to this one by truncation in Step 4. For the first part we prove by induction on $n \geq 1$ the following assertion: *if $\Omega \subset \mathbb{C}^n$ is a domain and $u\colon \Omega \to [0, \infty)$ is a finite function whose restriction to each connected component of each affine complex-line section of Ω is subharmonic, then u is upper semicontinuous.* For $n = 1$ the only affine complex line is $\mathbb{C}$, so u is subharmonic on Ω and in particular upper semicontinuous.

Let $n \geq 2$, assume the assertion in dimension $n-1$, and let u be as in the assertion. Write $z = (z', w)$ with $z' \in \mathbb{C}^{n-1}$ and $w \in \mathbb{C}$. For fixed w, the restriction of $u(\,\cdot\,, w)$ to each connected component of the open set $\{z' : (z', w) \in \Omega\}$ satisfies the hypothesis in dimension $n-1$. Indeed, affine complex lines in $\mathbb{C}^{n-1} \times \{w\}$ are affine complex lines in $\mathbb{C}^n$, a connected component of a line section of the slice lies in a connected component of the corresponding line section of Ω, and subharmonicity is inherited by open subsets. By the induction hypothesis, $u(\,\cdot\,, w)$ is therefore upper semicontinuous on this open set. For fixed z', the function $u(z', \,\cdot\,)$ is subharmonic on each connected component of $\{w : (z', w) \in \Omega\}$.

Step 1: u is Borel measurable. We need joint measurability of u before we can integrate it. Fix a ball $B' \subset \mathbb{C}^{n-1}$, a point $w_0 \in \mathbb{C}$ and $\rho > 0$ with $B' \times B(w_0, 2\rho) \subset \Omega$. For integers $N \geq 1$ put $u_N = \min\{u, N\}$, and for $0 < r < \rho$ and $(z', w) \in B' \times B(w_0, \rho)$ define

$$F_r^N(z', w) = \frac{1}{\pi r^2} \int_{B(w,r)} u_N(z', \eta)\, dV_2(\eta).$$

The integral exists, since $u(z', \,\cdot\,)$ is subharmonic, hence upper semicontinuous, on $B(w_0, 2\rho) \supset B(w, r)$, so the integrand is a Borel function with values in $[0, N]$. For fixed w, the function $F_r^N(\,\cdot\,, w)$ is upper semicontinuous: if $z'_l \to z'$ in B', the reverse Fatou lemma, applicable because $0 \leq u_N \leq N$, gives

$$\varlimsup_{l\to\infty} F_r^N(z'_l, w) \leq \frac{1}{\pi r^2} \int_{B(w,r)} \varlimsup_{l\to\infty} u_N(z'_l, \eta)\, dV_2(\eta) \leq F_r^N(z', w),$$

where the last inequality holds because each $u_N(\,\cdot\,, \eta)$ is upper semicontinuous on B'. For fixed z', the function $F_r^N(z', \,\cdot\,)$ is continuous: for $|w - v| \leq r$ the symmetric difference $B(w, r) \triangle B(v, r)$ lies in the annulus $\{r - |w - v| \leq |\eta - w| < r + |w - v|\}$, whose area is $4\pi r|w - v|$, so that $|F_r^N(z', w) - F_r^N(z', v)| \leq 4N|w - v|/r$.

A function on $B' \times B(w_0, \rho)$ that is Borel in z' and continuous in w is Borel; we check this for F_r^N. For $k \geq 1$ partition $B(w_0, \rho)$ into countably many Borel sets $E_{k,i}$ of diameter less than $1/k$, choose a point $w_{k,i}$ in each nonempty $E_{k,i}$, and put

$F_k(z',w) = F_r^N(z',w_{k,i})$ for $w \in E_{k,i}$. Each F_k is Borel, because each $F_r^N(\,\cdot\,,w_{k,i})$ is upper semicontinuous, hence Borel, and $F_k(z',w) \to F_r^N(z',w)$ for every (z',w), because $F_r^N(z',\,\cdot\,)$ is continuous and the chosen points $w_{k,i}$ with $w \in E_{k,i}$ tend to w. Thus F_r^N is Borel, and so is $G_N = \overline{\lim}_{k\to\infty} F^N_{\rho/(k+1)}$.

Now fix $(z',w) \in B' \times B(w_0,\rho)$. The upper semicontinuous function $u(z',\,\cdot\,)$ is bounded above on the closed disc $\overline{B(w,\rho/2)}$ by some M. For $N \geq M$ the truncation does not change $u(z',\,\cdot\,)$ on this disc, so $F^N_{\rho/(k+1)}(z',w)$ is the mean of $u(z',\,\cdot\,)$ over $B(w,\rho/(k+1))$. The function $u(z',\,\cdot\,)$ is subharmonic and finite on $B(w_0,2\rho)$, hence locally integrable there by Proposition 3.1.5, and by (3.3) its means over the discs $B(w,\rho/(k+1))$ converge to $u(z',w)$ as $k \to \infty$. Hence $G_N(z',w) = u(z',w)$ for all $N \geq M$, and $u = \lim_N G_N$ is Borel on $B' \times B(w_0,\rho)$. Countably many such product sets cover Ω, so u is Borel on Ω.

Step 2: the canonical representative. By Corollary 6.5.4, the finite Borel function u is locally bounded above. Since $u \geq 0$, it follows that $u \in L^1_{\rm loc}(\Omega)$. Since the restriction of u to each connected component of each affine complex-line section of Ω is subharmonic, Lemma 8.1.2 gives $\mathrm{dd^c}u \geq 0$. In particular $\Delta u = 4\sum_j u_{j\bar{j}} \geq 0$, so by Theorem 3.2.2 u has a canonical subharmonic representative $\widetilde{u}$, a locally integrable subharmonic function on Ω equal to u almost everywhere. Its complex Hessian is the same positive current $\mathrm{dd^c}u$. Applying Proposition 7.3.3 to the parametrizations of affine complex lines, we see that the restriction of $\widetilde{u}$ to each connected component of each affine complex-line section of Ω is subharmonic or identically $-\infty$. Since $\widetilde{u}$ is upper semicontinuous and agrees almost everywhere with the finite function u, we conclude that $\widetilde{u} \in \mathrm{PSH}(\Omega)$.

Step 3: $u = \widetilde{u}$ everywhere. Fix $x \in \Omega$ and $r > 0$ with $\overline{B(x,r)} \subset \Omega$. The set $Z = \{u \neq \widetilde{u}\}$ is Borel, by Step 1, and Lebesgue null, so the left-hand side of (8.1) vanishes for $F = \mathbf{1}_Z$. Since the weight $|t|^{2n-2}$ is positive for $t \neq 0$, for τ-almost every a the set $\{t : |t| < r,\ x + ta \in Z\}$ has planar measure zero. Fix such an a. On the disc $\{|t| < r\}$, the function $t \mapsto u(x+ta)$ is finite and subharmonic, while $t \mapsto \widetilde{u}(x+ta)$ is subharmonic or identically $-\infty$; the latter is impossible, because the two functions agree almost everywhere on the disc. By Proposition 3.1.5 both are locally integrable subharmonic functions on the disc that agree almost everywhere, so they agree everywhere by Theorem 3.2.2. At $t = 0$ this gives $u(x) = \widetilde{u}(x)$. As x was arbitrary, $u = \widetilde{u}$ is upper semicontinuous. This completes the induction.

Step 4: truncation. Let u be as in the theorem, and for integers $m \geq 1$ put $u_m = \max\{u,-m\} + m$. This is a finite nonnegative function. Its restriction to each connected component of each affine complex-line section of Ω is subharmonic: if the restriction of u is subharmonic, so is its maximum with the constant $-m$, and if the restriction of u is identically $-\infty$, the restriction of u_m is the constant 0. By the case already proved, each u_m, and hence each $\max\{u,-m\}$, is upper semicontinuous. For every real c we have $\{u < c\} = \bigcup_m \{\max\{u,-m\} < c\}$, which is open. Thus u is upper semicontinuous, and together with the hypotheses this gives $u \in \mathrm{PSH}(\Omega)$. □

8.2 Positive Hessians and holomorphic maps

We now compare plurisubharmonicity with the distributional condition $\mathrm{dd^c}u \geq 0$, where $(u_{j\bar{k}})$ is the complex Hessian of (7.5). The next theorem shows that every almost-everywhere class satisfying this condition contains exactly one plurisubharmonic function, namely its canonical representative. We then treat smooth functions, regularization and holomorphic maps, and define plurisubharmonic functions on complex manifolds.

Theorem 8.2.1 *Let $\Omega \subset \mathbb{C}^n$ be a domain, $n \geq 1$. Every $u \in \mathrm{PSH}(\Omega)$ is subharmonic and locally integrable on Ω, and $\mathrm{dd^c}u \geq 0$. Conversely, a real distribution T on Ω with $\mathrm{dd^c}T \geq 0$ has a unique plurisubharmonic representative, that is, there is exactly one $u \in \mathrm{PSH}(\Omega)$ with $\langle T, \chi\rangle = \int_\Omega u\chi \, \mathrm{d}V_{2n}$ for every test function χ. This representative is the canonical subharmonic representative of T given by Theorem 3.2.2, and it satisfies (3.3).*

Proof Let $u \in \mathrm{PSH}(\Omega)$. Being upper semicontinuous, u is bounded above on compact sets. Fix $x \in \Omega$ and $r > 0$ with $\overline{B(x,r)} \subset \Omega$, and let M be an upper bound for u on $\overline{B(x,r)}$. For every unit vector $a \in \mathbb{C}^n$ the closed disc $\{x + ta : |t| \leq r\}$ lies in Ω, and the restriction $t \mapsto u(x+ta)$ is subharmonic, possibly identically $-\infty$, on a neighborhood of this disc. The monotonicity of circle means in Theorem 3.1.3 therefore gives

$$u(x) \leq \frac{1}{2\pi}\int_0^{2\pi} u(x + r\mathrm{e}^{\mathrm{i}\theta}a)\,\mathrm{d}\theta.$$

With τ as in (8.1), we integrate this inequality over $a \in S^{2n-1}$ with respect to τ; by Tonelli's theorem, applied to the nonnegative function $M-u$, and by the invariance of τ under $a \mapsto \mathrm{e}^{\mathrm{i}\theta}a$, the right-hand side becomes the sphere mean $\int_{S^{2n-1}} u(x+ra)\,\mathrm{d}\tau(a)$. Thus u satisfies the submean inequality of Definition 3.1.1 at every point, so u is subharmonic. Since Ω is connected and $u \not\equiv -\infty$, Proposition 3.1.5 shows that $u \in L^1_{\mathrm{loc}}(\Omega)$. Since, moreover, the restriction of u to each connected component of each affine complex-line section of Ω is subharmonic, possibly identically $-\infty$, Lemma 8.1.2 gives $\mathrm{dd^c}u \geq 0$.

Conversely, let T be a real distribution with $\mathrm{dd^c}T \geq 0$. Then $\Delta T = 4\sum_j T_{j\bar{j}} \geq 0$, so by Theorem 3.2.2 T has a canonical subharmonic representative v, which is locally integrable and satisfies (3.3). Since $\mathrm{dd^c}v = \mathrm{dd^c}T \geq 0$, Proposition 7.3.3 shows that the restriction of v to each connected component of each affine complex-line section of Ω is subharmonic or identically $-\infty$. As v is upper semicontinuous and locally integrable, hence not identically $-\infty$, it is plurisubharmonic. If w is another plurisubharmonic representative of T, then w is a locally integrable subharmonic function by the first part, and $w = v$ almost everywhere; by the uniqueness in Theorem 3.2.2, $w = v$ everywhere. □

The condition $\mathrm{dd^c}u \geq 0$ depends only on the almost-everywhere class of u, whereas plurisubharmonicity is a pointwise property, and Theorem 8.2.1 singles out one function in each class. For example, the function equal to 1 at one point x_0 and to

0 elsewhere is upper semicontinuous and has zero distributional complex Hessian, but it violates the submean inequality on complex lines at x_0; the plurisubharmonic representative of its class is the zero function.

For smooth functions, plurisubharmonicity is a pointwise condition on the complex Hessian, and every plurisubharmonic function is, locally, the decreasing limit of smooth plurisubharmonic functions. In the next proposition, ρ_ε is the radial mollifier of Section 1.4 and, as there, $\Omega_\varepsilon = \{x \in \Omega : \operatorname{dist}(x, \mathbb{C}^n \setminus \Omega) > \varepsilon\}$ for an open set $\Omega \subset \mathbb{C}^n$ and $\varepsilon > 0$.

Proposition 8.2.2 *Let $\Omega \subset \mathbb{C}^n$ be open, $n \geq 1$.*

(1) *A real function u of class C^2 on Ω is plurisubharmonic on each connected component of Ω if and only if the Hermitian matrix $(u_{j\bar{k}}(z))$ is positive semidefinite at every point $z \in \Omega$.*

(2) *Let u be a function on Ω which is plurisubharmonic on each connected component of Ω, and let $\varepsilon > 0$. Then $u * \rho_\varepsilon$ is smooth on Ω_ε and plurisubharmonic on each connected component of Ω_ε, and $u * \rho_\varepsilon(x)$ decreases to $u(x)$ as $\varepsilon \downarrow 0$, for every $x \in \Omega$.*

Proof (1) Let Ω_0 be a connected component of Ω. We show that both conditions are equivalent to $\mathrm{dd}^{\mathrm{c}} u \geq 0$ on Ω_0. Since u is of class C^2, integration by parts shows that its distributional derivatives $u_{j\bar{k}}$ are the continuous functions $\partial_{z_j}\partial_{\bar{z}_k} u$, so for $\xi \in \mathbb{C}^n$ the distribution $\sum_{j,k} u_{j\bar{k}}\xi_j\bar{\xi}_k$ is the continuous real function $z \mapsto \sum_{j,k} u_{j\bar{k}}(z)\xi_j\bar{\xi}_k$. A continuous real function is a positive distribution exactly when it is nonnegative: if it were negative at a point, it would be negative on a ball around that point, and its integral against a nonnegative test function supported in this ball and positive at its center would be negative. Hence, by the matrix criterion of Section 7.3, $\mathrm{dd}^{\mathrm{c}} u \geq 0$ on Ω_0 if and only if $(u_{j\bar{k}}(z))$ is positive semidefinite at every $z \in \Omega_0$.

If u is plurisubharmonic on Ω_0, then $\mathrm{dd}^{\mathrm{c}} u \geq 0$ there by Theorem 8.2.1. Conversely, if $\mathrm{dd}^{\mathrm{c}} u \geq 0$ on Ω_0, then by Theorem 8.2.1 the class of u has a plurisubharmonic representative v on Ω_0 satisfying the ball-average formula in (3.3). Since $v = u$ almost everywhere and u is continuous, $v(x)$ is the limit of the averages of u over the balls $B(x, r)$ as $r \to 0$, that is, $v(x) = u(x)$. Hence $u = v$ is plurisubharmonic on Ω_0.

(2) By Theorem 8.2.1, applied on each connected component of Ω, the function u is subharmonic and locally integrable on Ω, and $\mathrm{dd}^{\mathrm{c}} u \geq 0$. By Proposition 1.4.3(1), $u * \rho_\varepsilon$ is smooth on Ω_ε, and its derivatives are the mollifications of the distributional derivatives of u. Thus, for $\xi \in \mathbb{C}^n$, the function $\sum_{j,k} (u*\rho_\varepsilon)_{j\bar{k}}\xi_j\bar{\xi}_k$ is the mollification of the distribution $\sum_{j,k} u_{j\bar{k}}\xi_j\bar{\xi}_k$, which is positive by the matrix criterion; so it is nonnegative by Proposition 1.4.3(3). By (1), $u * \rho_\varepsilon$ is plurisubharmonic on each connected component of Ω_ε. Finally, since u is subharmonic on Ω, Theorem 3.2.2(2) shows that $u * \rho_\varepsilon(x) \downarrow u(x)$ as $\varepsilon \downarrow 0$, for every $x \in \Omega$. □

Let u be a smooth real function on an open set $\Omega \subset \mathbb{C}^n$. We call u *strictly plurisubharmonic* if its complex Hessian matrix $(u_{j\bar{k}})$ is positive definite everywhere. For a general u, strict plurisubharmonicity means that every point has a neighborhood on which $u - \varepsilon|z|^2$ is plurisubharmonic for some $\varepsilon > 0$; on a complex manifold we

require this in local holomorphic coordinates around each point, and ε may depend on the point and the coordinates.

For smooth u the two definitions agree. Since $(|z|^2)_{j\bar{k}} = \delta_{jk}$, the complex Hessian of $u - \varepsilon|z|^2$ is $(u_{j\bar{k}}) - \varepsilon I$. The smallest eigenvalue $\lambda(z) = \min_{|\xi|=1} \sum_{j,k} u_{j\bar{k}}(z)\xi_j\bar{\xi}_k$ of the matrix $(u_{j\bar{k}}(z))$ is a continuous function of z, as the minimum over the compact unit sphere of a jointly continuous function of (z, ξ). If the matrix is positive definite everywhere, then λ is positive, hence bounded below by some $\varepsilon > 0$ on a closed ball around each point, and on the open ball the smooth function $u - \varepsilon|z|^2$ has a positive semidefinite complex Hessian, so it is plurisubharmonic by Proposition 8.2.2(1). Conversely, the plurisubharmonicity of $u - \varepsilon|z|^2$ on a ball around a point gives $(u_{j\bar{k}}) \geq \varepsilon I$ there, by the same criterion.

Proposition 8.2.3 *Let $D \subset \mathbb{C}^m$ and $\Omega \subset \mathbb{C}^n$ be domains, let $F\colon D \to \Omega$ be a holomorphic map, and let $u \in \mathrm{PSH}(\Omega)$. Then $u \circ F$ is either identically $-\infty$ or plurisubharmonic on D. More generally, if $D \subset \mathbb{C}^m$ is only assumed to be open, this alternative holds on each connected component of D.*

Proof The statement for open D follows by applying the statement for domains to each connected component of D. If $n = 0$, then Ω is a point, u is a real constant, and so is $u \circ F$. If $m = 0$, then D is a point and $u \circ F$ is a single value in $[-\infty, \infty)$; it is either $-\infty$ or a real number, that is, a plurisubharmonic function on the point. Now let $m, n \geq 1$. By Theorem 8.2.1, u is the canonical subharmonic representative of a locally integrable function with $\mathrm{dd}^c u \geq 0$, so Proposition 7.3.3 applies: $u \circ F$ is either identically $-\infty$, or locally integrable and subharmonic, in particular upper semicontinuous. In the second case, the restriction of $u \circ F$ to a connected component of an affine complex-line section of D is the composition of u with a holomorphic map from an open subset of $\mathbb{C}$, so it is subharmonic or identically $-\infty$ by the same proposition. Hence $u \circ F$ is plurisubharmonic. □

Let X be a complex manifold. A function $u\colon X \to [-\infty, \infty)$ is *plurisubharmonic* if its expression in every holomorphic chart with connected domain is plurisubharmonic in the sense of Definition 8.1.1; we write $\mathrm{PSH}(X)$ for the set of such functions.

On a domain $\Omega \subset \mathbb{C}^n$, plurisubharmonicity is a local property: if every point of Ω has an open neighborhood on which u is plurisubharmonic, then $u \in \mathrm{PSH}(\Omega)$. Indeed, upper semicontinuity is a local condition. Let Λ be a connected component of an affine complex-line section of Ω. Near each point of Λ, the restriction of u to Λ is subharmonic or identically $-\infty$, hence subharmonic; since subharmonicity is a local property, the restriction of u to Λ is subharmonic. Finally, u is locally integrable by Theorem 8.2.1, hence not identically $-\infty$.

It follows that it suffices to check the condition on X for the charts of one atlas with connected chart domains. Indeed, the transition maps are biholomorphic, so by Proposition 8.2.3 the expression of u in another chart with connected domain is, on a connected neighborhood of each point, plurisubharmonic or identically $-\infty$. The second alternative would make the expression of u in a chart of the atlas identically $-\infty$ on a nonempty open set, which is impossible because plurisubharmonic functions are locally integrable. Hence the expression of u in the other chart is plurisubharmonic near each point, and therefore on its connected domain.

Consequently plurisubharmonicity is also a local property on X: if every point of X has a neighborhood on which u is plurisubharmonic, then $u \in \mathrm{PSH}(X)$, because the expression of u in a chart with connected domain is then plurisubharmonic near each point. Since a plurisubharmonic function on X is locally integrable in charts, it is not identically $-\infty$ on any connected component of X. On an open set $\Omega \subset \mathbb{C}^n$, regarded as a complex manifold, a function is plurisubharmonic exactly when it is plurisubharmonic on each connected component of Ω: each component is a connected chart domain, and conversely the locality just proved applies.

Limits of plurisubharmonic functions may degenerate to the constant $-\infty$. On a connected manifold, whether this happens can be decided locally.

Lemma 8.2.4 *Let X be a connected complex manifold, and let $v\colon X \to [-\infty, \infty)$ be a function such that every point of X has an open neighborhood on which v is plurisubharmonic or identically $-\infty$. Then v is plurisubharmonic on X or identically $-\infty$ on X.*

Proof Let A be the set of points of X having an open neighborhood on which $v \equiv -\infty$, and B the set of points having an open neighborhood on which v is plurisubharmonic. Both sets are open, and they cover X by hypothesis. They are disjoint: if $x \in A \cap B$, with open neighborhoods W_1 on which $v \equiv -\infty$ and W_2 on which v is plurisubharmonic, then the restriction of v to the nonempty open set $W_1 \cap W_2$ is plurisubharmonic and identically $-\infty$, which is impossible because plurisubharmonic functions are locally integrable. Since X is connected, either $X = A$, and then $v \equiv -\infty$, or $X = B$, and then v is plurisubharmonic near every point, hence on X, because plurisubharmonicity is a local property. □

Every closed positive $(1,1)$-current T on a complex manifold of positive dimension has local plurisubharmonic potentials: every point has a neighborhood on which $T = \mathrm{dd}^{\mathrm{c}} u$ for a plurisubharmonic function u. Indeed, on a coordinate ball inside the neighborhood given by Theorem 7.3.2, the canonical local potential is the canonical subharmonic representative of a class with positive complex Hessian, hence plurisubharmonic by Theorem 8.2.1.

Proposition 8.2.5 *Let X be a complex manifold. Nonnegative multiples, finite sums and maxima of nonempty finite families of plurisubharmonic functions on X are plurisubharmonic; here the zero multiple of a function is the zero function. Moreover, let $u \in \mathrm{PSH}(X)$, let $D \subset X$ be open, and let $v \in \mathrm{PSH}(D)$. If*

$$\overline{\lim_{D \ni y \to x}}\, v(y) \le u(x) \qquad \text{for every } x \in \partial D,$$

where ∂D is the boundary of D in X, then the function equal to $\max\{u, v\}$ on D and to u on $X \setminus D$ is plurisubharmonic on X. This boundary inequality holds in particular if

$$\underline{\lim_{\substack{D \ni y \to x \\ v(y) > -\infty}}}\, (u(y) - v(y)) \ge 0 \qquad \text{for every } x \in \partial D.$$

Proof In local coordinates, upper semicontinuity and the submean inequality on complex lines are preserved by nonnegative multiples, sums and maxima; for sums we use that circle means are additive on functions with values in $[-\infty, \infty)$. It remains to check that the resulting functions are not identically $-\infty$ on any component. For cu with $c > 0$ this holds because $cu = -\infty$ exactly where $u = -\infty$; a sum of plurisubharmonic functions is locally integrable by Theorem 8.2.1, and a maximum dominates each of its members. This proves the first assertion.

Let w denote the glued function. On D it equals $\max\{u, v\}$, which is plurisubharmonic by the first part, and on the open set $X \setminus \overline{D}$ it equals u. At $x \in \partial D$ we have $w(x) = u(x)$, and the upper semicontinuity of u together with the boundary inequality gives

$$\overline{\lim_{y\to x}}\, w(y) \le \max\Big\{\overline{\lim_{y\to x}}\, u(y),\ \overline{\lim_{D\ni y\to x}}\, v(y)\Big\} \le u(x) = w(x).$$

Hence w is upper semicontinuous. To verify the line condition, work in a chart and restrict w to a connected component of an affine complex-line section. At points of D and of $X \setminus \overline{D}$, the restriction satisfies the submean inequality on small circles, because w is plurisubharmonic near such points. At a point $x \in \partial D$ we have $w \ge u$ everywhere and $w(x) = u(x)$, so each small circle mean of w is at least the corresponding circle mean of u, which is at least $u(x) = w(x)$; if the restriction of u is identically $-\infty$, then $w(x) = u(x) = -\infty$ and there is nothing to prove. Thus the restriction of w is upper semicontinuous and satisfies the submean inequality on small circles at every point; that is, it is subharmonic. Finally $w \ge u$, so w is not identically $-\infty$ in any chart, and w is plurisubharmonic.

For the sufficient condition, fix $x \in \partial D$ and $\varepsilon > 0$. By hypothesis, x has a neighborhood W such that $v(y) < u(y) + \varepsilon$ for all $y \in W \cap D$ with $v(y) > -\infty$. Hence $v \le u + \varepsilon$ on $W \cap D$, the inequality being trivial where $v = -\infty$, and the upper semicontinuity of u gives

$$\overline{\lim_{D\ni y\to x}}\, v(y) \le \overline{\lim_{y\to x}}\, u(y) + \varepsilon \le u(x) + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives the boundary inequality. □

8.3 Limits, envelopes and local integrability

This section treats limits and envelopes of plurisubharmonic functions and their local integrability. Throughout, X is a complex manifold of positive dimension, except in Theorem 8.3.7, which also allows dimension zero. By Theorem 8.2.1, plurisubharmonic functions are locally integrable, and we regard each of them as the distinguished representative of its class. Local L^p norms on X are computed with Euclidean volume in local coordinates, and "almost everywhere" refers to Lebesgue measure in charts. Any smooth positive volume form gives the same local

L^p spaces and the same L^p_{loc} convergence, since in coordinates its density is bounded above and below by positive constants on compact sets. By the discussion following Definition 7.1.1, each connected component of X has a countable basis; X itself need not have one, and we assume this explicitly when a global countable selection is needed.

We call a subset of an open set in $\mathbb{C}^n$ *real polar* if it is polar in the sense of Definition 3.5.1 when $\mathbb{C}^n$ is identified with $\mathbb{R}^{2n}$. The adjective distinguishes these sets from the pluripolar sets of Chapter 10. Real polar sets are Lebesgue null, by the remark following Definition 3.5.1.

Theorem 8.3.1 *Let $(u_i)_{i\in I}$ be a decreasing net in $\mathrm{PSH}(X)$, that is, $u_j \le u_i$ whenever $i \le j$, and let $u = \lim_i u_i = \inf_i u_i$ be its pointwise limit. Then on each connected component X_0 of X, exactly one of the following holds: either u is plurisubharmonic on X_0 and $u_i \to u$ in $L^1_{\mathrm{loc}}(X_0)$, or $u \equiv -\infty$ on X_0 and $u_i \to -\infty$ locally uniformly on X_0.*

Proof The limit u is upper semicontinuous, since $\{u < c\} = \bigcup_i \{u_i < c\}$ is open for every real c. Fix a connected component X_0 of X; it has a countable basis.

Step 1: a decreasing sequence with the same limit. Let $i_0 \in I$. We show that there are indices $i_0 \le i_1 \le i_2 \le \cdots$ such that $u_{i_l} \downarrow u$ pointwise on X_0. Replacing I by the cofinal directed set $\{i \in I : i \ge i_0\}$, which does not change the limit, we only have to find indices $i_1 \le i_2 \le \cdots$. For each rational c, the open sets $\{u_i < c\} \cap X_0$ cover $\{u < c\} \cap X_0$, and countably many of them suffice: for each basis element of X_0 contained in some set $\{u_i < c\}$, choose one such index i; every point of $\{u < c\} \cap X_0$ lies in such a basis element. Let $k_1, k_2, \ldots$ be an enumeration, with repetitions allowed, of the indices chosen for all rational c. Since I is directed, we can choose inductively $i_1 \ge k_1$ and $i_{l+1} \ge i_l, k_{l+1}$. The sequence (u_{i_l}) is decreasing and bounded below by u. If $u(x) < c$ with c rational and $x \in X_0$, then $u_{k_l}(x) < c$ for some l, hence $u_{i_l}(x) \le u_{k_l}(x) < c$. Therefore $u_{i_l} \downarrow u$ on X_0.

Step 2: the limit. Let $V \subset X_0$ be a connected coordinate neighborhood, identified with a domain in $\mathbb{C}^n$. On each connected component of each affine complex-line section of V, the restrictions of the u_{i_l} form a decreasing sequence of subharmonic functions, whose limit is subharmonic by Theorem 3.1.6. Since u is upper semicontinuous, it is either plurisubharmonic on V or identically $-\infty$ on V, by Definition 8.1.1. As X_0 is connected, Lemma 8.2.4 shows that either u is plurisubharmonic on X_0 or $u \equiv -\infty$ on X_0.

Step 3: convergence. Suppose first that u is plurisubharmonic on X_0. Fix $i_0 \in I$, take the sequence of Step 1, and let K be a compact subset of a coordinate neighborhood in X_0. Then $0 \le u_{i_l} - u \le u_{i_0} - u$ almost everywhere, namely wherever u is finite, and $u_{i_0} - u$ is integrable on K. By dominated convergence, $\|u_{i_l} - u\|_{L^1(K)} \to 0$. Given $\varepsilon > 0$, choose l with $\|u_{i_l} - u\|_{L^1(K)} < \varepsilon$; for every $i \ge i_l$ we have $0 \le u_i - u \le u_{i_l} - u$ almost everywhere, hence $\|u_i - u\|_{L^1(K)} < \varepsilon$. Thus the net converges to u in $L^1(K)$, and covering an arbitrary compact subset of X_0 by finitely many such sets K gives convergence in $L^1_{\mathrm{loc}}(X_0)$.

Suppose now that $u \equiv -\infty$ on X_0, and let $K \subset X_0$ be compact and $A > 0$. The open sets $\{u_i < -A\}$ cover K, so finitely many of them, say those with indices $j_1, \ldots, j_p$,

already cover K. If $i \geq j_1, \ldots, j_p$, then $u_i \leq u_{j_q}$ for each q, and hence $u_i < -A$ on K. This is locally uniform convergence to $-\infty$. □

Recall that a function F with values in $[-\infty, \infty)$ that is locally bounded above has an upper semicontinuous regularization F^*, the smallest upper semicontinuous function that dominates F. On X it is given by $F^*(x) = \inf_W \sup_W F$, where W ranges over the neighborhoods of x, and in a chart it is the regularization of Section 3.4.

Theorem 8.3.2 *Let $(u_i)_{i\in I}$ be a nonempty family in* $\mathrm{PSH}(X)$ *that is locally uniformly bounded above, and let $F = \sup_i u_i$ be its upper envelope. Then the upper semicontinuous regularization F^* is plurisubharmonic, and $F = F^*$ almost everywhere; more precisely, in every coordinate chart the set $\{F < F^*\}$ is real polar. If X has a countable basis, there is a countable subset $J \subset I$ such that $(\sup_{i\in J} u_i)^* = F^*$.*

Proof Since F^* is defined by a local formula and plurisubharmonicity is a local property, it suffices to prove that F^* is plurisubharmonic on every connected coordinate neighborhood V, which we identify with a domain in $\mathbb{C}^n$. By Theorem 3.4.2, F^* is subharmonic on V, and F is Lebesgue measurable and equal to F^* almost everywhere on V. By Choquet's lemma, Lemma 3.4.1, there is a countable set $J_V \subset I$ with $(\sup_{i\in J_V} u_i)^* = F^*$ on V.

We next show that $\mathrm{dd^c} F^* \geq 0$ on V. Fix $i_0 \in I$, enumerate $J_V \cup \{i_0\}$ as $i_0, i_1, i_2, \ldots$, with repetitions if this set is finite, and put $v_j = \max\{u_{i_0}, \ldots, u_{i_j}\}$. The functions v_j are plurisubharmonic by Proposition 8.2.5, increase with j, are locally uniformly bounded above, and dominate the locally integrable function u_{i_0}. Their limit $G = \sup_j v_j$ lies between $\sup_{i\in J_V} u_i$ and F, so $G^* = F^*$, and hence $G = F^*$ almost everywhere by Theorem 3.4.2 applied to the countable family. On a compact set $K \subset V$ on which every u_i is bounded above by M, we have $0 \leq v_j - u_{i_0} \leq M - u_{i_0}$, so dominated convergence gives $v_j \to F^*$ in $L^1(K)$. Therefore $\mathrm{dd^c} F^* \geq 0$, as a limit of positive currents. Since F^* is subharmonic and dominates the locally integrable function u_{i_0}, it is locally integrable and hence the canonical representative of its class, and Theorem 8.2.1 shows that F^* is plurisubharmonic on V.

If X has a countable basis, cover X by countably many connected coordinate neighborhoods $V_1, V_2, \ldots$ and put $J = \bigcup_m J_{V_m}$. On each V_m we have $F^* = (\sup_{i\in J_{V_m}} u_i)^* \leq (\sup_{i\in J} u_i)^* \leq F^*$, and since the regularization is local, $(\sup_{i\in J} u_i)^* = F^*$ on X.

Finally, in each chart the functions u_i are subharmonic, so the second assertion of Theorem 4.4.4 shows that $\{F < F^*\}$ is polar in the corresponding open subset of $\mathbb{R}^{2n}$, that is, real polar. □

Proposition 8.3.3 *Let $\Omega \subset \mathbb{C}^n$ be a domain, $n \geq 1$, and let $g = (g_1, \ldots, g_r)$, $r \geq 1$, be a tuple of holomorphic functions on Ω, not all identically zero. Then the function $\log \sum_j |g_j|^2$, with the value $-\infty$ at the common zeros of the g_j, is plurisubharmonic on Ω. In particular, for every $a \in \mathbb{C}^n$ the function $\log |z-a|^2$ is plurisubharmonic on $\mathbb{C}^n$.*

Proof By the Cauchy–Schwarz inequality, with equality for λ proportional to $(\overline{g_1(z)}, \dots, \overline{g_r(z)})$, we have

$$\log \sum_j |g_j(z)|^2 = \sup_{\sum_j |\lambda_j|^2 = 1} \log \left| \sum_j \lambda_j g_j(z) \right|^2$$

for every $z \in \Omega$, both sides being $-\infty$ at a common zero. The supremum does not change if we omit the λ for which $\sum_j \lambda_j g_j$ vanishes identically: at a point z that is not a common zero, the maximizing λ gives $\sum_j \lambda_j g_j(z) \neq 0$, and at a common zero all terms equal $-\infty$. Some g_j is not identically zero, so the remaining family is nonempty. Its members are plurisubharmonic by Proposition 7.4.3 and Theorem 8.2.1, and they are locally uniformly bounded above by the left-hand side. Their supremum $\log \sum_j |g_j|^2$ is already upper semicontinuous: it is continuous as a map to $[-\infty, \infty)$, because $\sum_j |g_j|^2$ is continuous and its logarithm tends to $-\infty$ at a common zero. Hence it equals its regularization, which is plurisubharmonic by Theorem 8.3.2. The particular case is the tuple $(z_1 - a_1, \dots, z_n - a_n)$. □

Theorem 8.3.4 *Let $\Omega \subset \mathbb{C}^n$ be a domain, $n \geq 1$. Every sequence in* $\mathrm{PSH}(\Omega)$ *that is locally uniformly bounded above has a subsequence that either converges locally uniformly to $-\infty$ or converges in $L^1_{\mathrm{loc}}(\Omega)$ to a plurisubharmonic function. If a sequence (u_j) in* $\mathrm{PSH}(\Omega)$ *converges in $L^1_{\mathrm{loc}}(\Omega)$, then its limit has a plurisubharmonic representative u, and the following holds: for every compact set $K \subset \Omega$ and every lower semicontinuous function $f \colon K \to \mathbb{R}$ with $u < f$ on K, for instance every real continuous function on K, one has $u_j < f$ on K for all sufficiently large j. In particular $\overline{\lim}_j u_j(x) \leq u(x)$ for every $x \in \Omega$.*

Proof Plurisubharmonic functions are subharmonic by Theorem 8.2.1. Hence, by the compactness theorem (Theorem 3.3.2), a sequence as in the first assertion either tends to $-\infty$ locally uniformly, or has a subsequence converging in $L^1_{\mathrm{loc}}(\Omega)$ to a locally integrable subharmonic function v. In the second case $\mathrm{dd}^{\mathrm{c}} v \geq 0$, as a distributional limit of positive currents, and v is the canonical representative of its class; so v is plurisubharmonic by Theorem 8.2.1.

Now let (u_j) converge in $L^1_{\mathrm{loc}}(\Omega)$ to T. As above, $\mathrm{dd}^{\mathrm{c}} T \geq 0$, so T has a plurisubharmonic representative u, namely its canonical subharmonic representative. The u_j are locally integrable subharmonic functions converging in $L^1_{\mathrm{loc}}(\Omega)$, and u is the canonical representative of their limit, so the remaining assertions are Hartogs' lemma for subharmonic functions, Theorem 3.3.3, which requires no upper bound on the sequence; the last one is its first assertion with the constant sequence $x_j = x$. □

We next show that plurisubharmonic functions belong to L^p_{loc} for every finite p, and that L^1_{loc} convergence of plurisubharmonic functions implies L^p_{loc} convergence. We first treat a subharmonic function $v \leq 0$ on the unit disc. By the Riesz decomposition (Theorem 3.2.3), v is locally the sum of a harmonic function and the fundamental potential $E_2 * \mu$ of a measure, and the kernel $E_2(z) = (2\pi)^{-1} \log |z|$ belongs locally to every L^p with $p < \infty$; this gives an L^p bound in terms of the value of v at the

center. Applying this bound in one complex variable at a time then gives the estimate for plurisubharmonic functions on polydiscs.

We write $D(a,r) = \{z \in \mathbb{C} : |z-a| < r\}$ and $D = D(0,1)$, and for $t > 0$ we write $tD = D(0,t)$; thus D^n is the unit polydisc in $\mathbb{C}^n$ and $(tD)^n$ the polydisc of radius t centered at 0. Integrals over discs and polydiscs are taken with respect to Lebesgue measure $\mathrm{d}V$.

Lemma 8.3.5 *There is a constant $C_0 \geq 1$ with the following property. Let $v \leq 0$ be a subharmonic function on D with $v(0) > -\infty$. Then there are a finite positive Borel measure ν with support in $D(0,9/10)$ and a harmonic function h on $D(0,4/5)$ such that*

$$v(z) = P(z) + h(z) \quad (z \in D(0,4/5)), \qquad P(z) = \frac{1}{2\pi}\int \log|z-\zeta|\,\mathrm{d}\nu(\zeta),$$
$$\nu(D) \leq C_0|v(0)|, \qquad \sup_{D(0,1/2)} |h| \leq C_0|v(0)|. \tag{8.2}$$

Moreover, for every $1 \leq p < \infty$ there is a constant $C_p > 0$ such that every such v satisfies

$$\|v\|_{L^p(D(0,1/2))} \leq C_p|v(0)|.$$

Proof In this proof C denotes constants that depend only on two cutoff functions χ and η fixed below, and C' constants that depend in addition on p. Since $v(0) > -\infty$, the function v is not identically $-\infty$, and hence it is locally integrable by Proposition 3.1.5. For $0 < r < 1$, the ball submean inequality (Theorem 3.1.3) and $v \leq 0$ give

$$\|v\|_{L^1(D(0,r))} = -\int_{D(0,r)} v\,\mathrm{d}V \leq \pi r^2|v(0)|. \tag{8.3}$$

Let $\mu = \Delta v$ be the Riesz measure of v, and choose a smooth cutoff $\chi \geq 0$ supported in $D(0,19/20)$ and equal to one on $D(0,9/10)$. By the definition of the distributional Laplacian and (8.3),

$$\mu(D(0,9/10)) \leq \int \chi\,\mathrm{d}\mu = \int v\Delta\chi\,\mathrm{d}V \leq \sup|\Delta\chi|\,\|v\|_{L^1(D(0,19/20))} \leq C|v(0)|.$$

Next choose a smooth cutoff $0 \leq \eta \leq 1$ supported in $D(0,9/10)$ and equal to one on a neighborhood of $\overline{D(0,4/5)}$. The fundamental solution in the plane is $(2\pi)^{-1}\log|z|$, so Theorem 3.2.3, applied with the measure $\nu = \eta\mu$, which is finite and supported in $\operatorname{Supp}\eta \subset D(0,9/10)$, gives

$$v(z) = P(z) + h(z), \qquad P(z) = \frac{1}{2\pi}\int \log|z-\zeta|\,\mathrm{d}\nu(\zeta),$$

for every $z \in D(0,4/5)$, with h harmonic on $D(0,4/5)$.

We estimate P. For $|\zeta| \leq 9/10$, the translated disc $D(0,4/5) - \zeta$ lies in $D(0,2)$, so

$$\int_{D(0,4/5)} |\log|z-\zeta||^p\,\mathrm{d}V(z) \leq L_p = \int_{D(0,2)} |\log|z||^p\,\mathrm{d}V(z) < \infty;$$

the integral L_p is finite because its part over D equals $2\pi\int_0^1 r|\log r|^p\,\mathrm{d}r$, whose integrand is bounded on $(0,1]$, while $|\log|z|| \le \log 2$ on the rest of $D(0,2)$. Let $m = \nu(D) = \int \eta\,\mathrm{d}\mu$, so that $m \le \mu(D(0,9/10)) \le C|v(0)|$. If $m = 0$, then $P = 0$. Otherwise, $|P(z)| \le \frac{m}{2\pi}\int |\log|z-\zeta||\,\mathrm{d}\nu(\zeta)/m$, and Hölder's inequality for the probability measure ν/m gives

$$|P(z)|^p \le \left(\frac{m}{2\pi}\right)^p \int |\log|z-\zeta||^p\,\frac{\mathrm{d}\nu(\zeta)}{m},$$

and integrating over $z \in D(0,4/5)$ with Tonelli's theorem yields $\|P\|_{L^p(D(0,4/5))} \le L_p^{1/p} m/(2\pi) \le C'|v(0)|$.

We now estimate h. The case $p = 1$ of the bound for P, in which L_1 is an absolute constant, and (8.3) give $\|h\|_{L^1(D(0,4/5))} \le C|v(0)|$. For $|z| \le 1/2$, the closed disc $\overline{D(z,1/4)}$ lies in $D(0,4/5)$, and the ball mean-value property of the harmonic function h (Proposition 2.1.1) gives $|h(z)| \le (16/\pi)\|h\|_{L^1(D(0,4/5))}$. We have thus found a constant $C_0 \ge 1$, depending only on the cutoff functions χ and η, such that $\nu(D) = m \le C_0|v(0)|$ and $\sup_{\overline{D(0,1/2)}}|h| \le C_0|v(0)|$; this proves (8.2). Adding the estimates for P and h on $D(0,1/2)$, we obtain

$$\|v\|_{L^p(D(0,1/2))} \le \|P\|_{L^p(D(0,4/5))} + (\pi/4)^{1/p}C_0|v(0)| \le C'|v(0)|,$$

which proves the last assertion. □

Proposition 8.3.6 *Let $n \ge 1$ and $1 \le p < \infty$. There are constants $C_{n,p}, A_{n,p} > 0$ such that every plurisubharmonic function $v \le 0$ on the unit polydisc D^n satisfies*

$$\|v\|_{L^p((\frac{1}{2}D)^n)} \le C_{n,p}|v(0)| \quad \text{if } v(0) > -\infty,$$

and

$$\|v\|_{L^p((\frac{1}{4}D)^n)} \le A_{n,p}\|v\|_{L^1(D^n)}. \tag{8.4}$$

Here $\|v\|_{L^1(D^n)}$ may be infinite, in which case there is nothing to prove.

Proof We prove the first estimate by induction on n. For $n = 1$ it is Lemma 8.3.5. Let $n \ge 2$, write $z = (z_1, z')$ with $z' \in \mathbb{C}^{n-1}$, and let v be as in the statement with $v(0) > -\infty$. The function $w(z') = v(0,z')$ is the composition of v with the holomorphic map $z' \mapsto (0,z')$, so by Proposition 8.2.3 it is plurisubharmonic or identically $-\infty$ on D^{n-1}. Since $w(0) = v(0) > -\infty$, it is plurisubharmonic, hence locally integrable and finite outside a Lebesgue null set $E \subset D^{n-1}$. For $z' \in D^{n-1}\setminus E$, the function $z_1 \mapsto v(z_1,z')$ is subharmonic and nonpositive on D, with the finite value $w(z')$ at the center, so Lemma 8.3.5 gives

$$\int_{\frac{1}{2}D} |v(z_1,z')|^p\,\mathrm{d}V(z_1) \le C_p^p|w(z')|^p.$$

We integrate over $z' \in (\frac{1}{2}D)^{n-1}$ using Tonelli's theorem; the null set E, on which the inner integral may be infinite, does not contribute. The induction hypothesis applied

to w then gives

$$\|v\|^p_{L^p((\frac{1}{2}D)^n)} \le C_p^p \|w\|^p_{L^p((\frac{1}{2}D)^{n-1})} \le C_p^p C_{n-1,p}^p |v(0)|^p,$$

which is the first estimate with $C_{n,p} = C_p C_{n-1,p}$.

For the second estimate, let $Q = (\frac{1}{10}D)^n$ and $|Q|$ its volume. If $\|v\|_{L^1(D^n)} = 0$, then $v = 0$ almost everywhere, hence $v \equiv 0$ by Theorem 8.2.1, and there is nothing to prove; we may also assume that $\|v\|_{L^1(D^n)} < \infty$. Put $\lambda = 2|Q|^{-1}\|v\|_{L^1(D^n)} > 0$. By Chebyshev's inequality, the set of points of Q with $|v| > \lambda$ has measure at most $\lambda^{-1}\int_Q |v|\,dV \le |Q|/2$, so its complement in Q has positive measure. Since v is finite almost everywhere, there is a point $a \in Q$ with $v(a) > -\infty$ and

$$|v(a)| \le 2|Q|^{-1}\|v\|_{L^1(D^n)}.$$

The polydisc of radius $4/5$ centered at a lies in D^n, and the polydisc of radius $2/5$ centered at a contains $(\frac{1}{4}D)^n$, since $\frac{1}{4} + \frac{1}{10} < \frac{2}{5}$. We apply the first estimate to the plurisubharmonic function $\zeta \mapsto v(a + \frac{4}{5}\zeta) \le 0$ on D^n and change variables:

$$\|v\|_{L^p((\frac{1}{4}D)^n)} \le \left(\tfrac{4}{5}\right)^{2n/p} C_{n,p}|v(a)| \le 2\left(\tfrac{4}{5}\right)^{2n/p} C_{n,p}|Q|^{-1}\|v\|_{L^1(D^n)}.$$

This is (8.4). □

Theorem 8.3.7 *Let X be a complex manifold and $1 \le p < \infty$. Every plurisubharmonic function on X belongs to $L^p_{\mathrm{loc}}(X)$. If a sequence of plurisubharmonic functions on X converges in $L^1_{\mathrm{loc}}(X)$, then it converges in $L^p_{\mathrm{loc}}(X)$ to the same limit.*

Proof If X has dimension zero, every compact subset of X is finite, plurisubharmonic functions are real-valued, and convergence in L^p_{loc} is pointwise convergence; so both assertions hold. Let X have dimension $n \ge 1$. Every compact subset of X is covered by finitely many coordinate polydiscs, so it suffices to work near a point $x \in X$ in a chart in which $x = 0$ and the closed polydisc $\overline{(RD)^n}$ lies in the chart, for some $R > 0$.

Let v be plurisubharmonic. Being upper semicontinuous, v is bounded above on $\overline{(RD)^n}$ by some constant A. The function $\zeta \mapsto v(R\zeta) - A$ is plurisubharmonic and nonpositive on D^n, and its L^1 norm on D^n is finite because v is locally integrable. By (8.4), v belongs to L^p on the polydisc $(\frac{R}{4}D)^n$. This proves the first assertion.

Now let (u_j) be a sequence of plurisubharmonic functions converging in $L^1_{\mathrm{loc}}(X)$. For $p = 1$ there is nothing to prove, so let $p > 1$. By Theorem 8.3.4 applied in charts, together with the uniqueness in Theorem 8.2.1, the limit is represented by a plurisubharmonic function u. The norms $\|u_j\|_{L^1((RD)^n)}$ are bounded because the sequence converges. If $y \in (\frac{R}{2}D)^n$, the ball $B(y, R/2)$ lies in $(RD)^n$, so the ball submean inequality gives $u_j(y) \le |B(y,R/2)|^{-1}\|u_j\|_{L^1((RD)^n)}$. Hence there is a constant M, independent of j, with $u_j \le M$ and $u \le M$ on $(\frac{R}{2}D)^n$. Applying (8.4) to $\zeta \mapsto u_j(\frac{R}{2}\zeta) - M$ and to $\zeta \mapsto u(\frac{R}{2}\zeta) - M$, whose L^1 norms on D^n are bounded independently of j, we obtain a bound for the L^{2p} norms of u_j and u on $(\frac{R}{8}D)^n$

that is uniform in j. By a finite covering, for every compact set $K \subset X$ the norms $\|u_j\|_{L^{2p}(K)}$ and $\|u\|_{L^{2p}(K)}$ are bounded independently of j.

We interpolate between L^1 and L^{2p}. For a function f on K, write $|f|^p = |f|^{p/(2p-1)} \cdot |f|^{p(2p-2)/(2p-1)}$ and apply Hölder's inequality with the conjugate exponents $(2p-1)/p$ and $(2p-1)/(p-1)$; since $\frac{p(2p-2)}{2p-1} \cdot \frac{2p-1}{p-1} = 2p$, this gives

$$\int_K |f|^p \, dV \le \Big(\int_K |f| \, dV \Big)^{p/(2p-1)} \Big(\int_K |f|^{2p} \, dV \Big)^{(p-1)/(2p-1)},$$

that is, $\|f\|_{L^p(K)} \le \|f\|_{L^1(K)}^{1/(2p-1)} \|f\|_{L^{2p}(K)}^{(2p-2)/(2p-1)}$. With $f = u_j - u$, whose $L^{2p}(K)$ norms are bounded independently of j, we obtain

$$\|u_j - u\|_{L^p(K)} \le \|u_j - u\|_{L^1(K)}^{1/(2p-1)} \|u_j - u\|_{L^{2p}(K)}^{(2p-2)/(2p-1)} \longrightarrow 0.$$

The same Riesz decomposition gives exponential integrability: if $v \le 0$ is plurisubharmonic on a ball, then $\mathrm{e}^{-\alpha v}$ is integrable on a smaller concentric ball for every sufficiently small $\alpha > 0$, and the admissible α and the bound for the integral depend only on a bound for the L^1 norm of v. In one variable this follows from Jensen's inequality applied to the fundamental potential in the Riesz decomposition. In several variables we integrate the one-variable estimates over all complex lines through a suitable point. For $t > 0$ we write tB for the ball of radius t centered at 0 in $\mathbb{C}^n$; thus B is the unit ball.

Lemma 8.3.8 *There are constants $\beta, C > 0$ such that every subharmonic function $v \le 0$ on D with $v(0) \ge -1$ satisfies*

$$\int_{D(0,1/2)} \mathrm{e}^{-\beta v} \, dV \le C.$$

Proof Since $v(0) \ge -1 > -\infty$, Lemma 8.3.5 gives a finite positive measure ν with support in $D(0, 9/10)$ and a harmonic function h on $D(0, 4/5)$ such that

$$v = P + h, \qquad P(z) = \frac{1}{2\pi} \int \log|z - \zeta| \, d\nu(\zeta)$$

at every point of $D(0, 4/5)$. Since $|v(0)| \le 1$, the bounds (8.2) give $M = \nu(D) \le C_0$ and $\sup_{D(0,1/2)} |h| \le C_0$, where $C_0 \ge 1$ is the universal constant of that lemma. Fix $\beta \in (0, 1]$ with $\beta C_0 \le 2\pi$.

We show that $\int_{D(0,1/2)} \mathrm{e}^{-\beta P} \, dV \le 8\pi$. If $M = 0$, then $P = 0$, and this integral is $\pi/4$. Assume $M > 0$ and put $b = \beta M/(2\pi) \in (0, 1]$. Fix $z \in D(0, 1/2)$. For $N \ge 1$ the Borel function $f_N(\zeta) = -b \max\{\log|z - \zeta|, -N\}$ is bounded on $D(0, 9/10)$, so Jensen's inequality (Lemma 1.1.4), applied to the convex function exp on $\mathbb{R}$ and the probability measure ν/M, gives

$$\exp\Big(\int f_N \, \frac{d\nu}{M} \Big) \le \int \mathrm{e}^{f_N} \, \frac{d\nu}{M} \le \frac{1}{M} \int |z - \zeta|^{-b} \, d\nu(\zeta).$$

As $N \to \infty$, the functions f_N increase to $-b \log|z-\zeta|$, which may take the value $+\infty$, and they are bounded below by the bounded function f_1. By monotone convergence, $\int f_N \, d\nu/M$ increases to $\int -b \log|z-\zeta| \, d\nu(\zeta)/M = -\beta P(z)$, and therefore

$$e^{-\beta P(z)} \le \frac{1}{M} \int |z-\zeta|^{-b} \, d\nu(\zeta)$$

for every $z \in D(0,1/2)$, with the convention $e^{\infty} = \infty$. For every $\zeta \in D(0,9/10)$ the translated disc $D(0,1/2) - \zeta$ lies in $D(0,2)$, and $r^{-b} \le 1 + r^{-1}$ for $r > 0$ because $0 < b \le 1$; hence

$$\int_{D(0,1/2)} |z-\zeta|^{-b} \, dV(z) \le \int_{D(0,2)} (1 + |z|^{-1}) \, dV(z) = 8\pi.$$

Integrating the bound for $e^{-\beta P(z)}$ over $z \in D(0,1/2)$ with Tonelli's theorem, we obtain $\int_{D(0,1/2)} e^{-\beta P} \, dV \le 8\pi$. Finally, since $e^{-\beta v} = e^{-\beta P} e^{-\beta h} \le e^{\beta C_0} e^{-\beta P}$ on $D(0,1/2)$, the lemma follows with $C = 8\pi e^{C_0}$. □

Proposition 8.3.9 *Let $n \ge 1$, and let B be the unit ball of $\mathbb{C}^n$. There are constants $\alpha_n, C_n > 0$ such that every plurisubharmonic function $v \le 0$ on B with $\|v\|_{L^1(B)} \le 1$ satisfies*

$$\int_{\frac{1}{4}B} e^{-\alpha_n v} \, dV \le C_n. \tag{8.5}$$

More generally, if $\|v\|_{L^1(B)} \le M$ for some $M > 0$, the same bound holds with the exponent α_n/M in place of α_n.

Proof Put $A_n = \max\{1, 2/|\frac{1}{8}B|\}$. The mean of $|v|$ over $\frac{1}{8}B$ is at most $1/|\frac{1}{8}B| < A_n$. Hence the set of points of $\frac{1}{8}B$ with $|v| \le A_n$ has positive measure, since otherwise $|v| > A_n$ almost everywhere on $\frac{1}{8}B$ and the mean would be at least A_n; so there is a point $a \in \frac{1}{8}B$ with $|v(a)| \le A_n$. For each unit vector $e \in \mathbb{C}^n$, the disc $\{a + \frac{3}{4}\zeta e : |\zeta| < 1\}$ lies in B, so the function

$$\zeta \longmapsto A_n^{-1} v(a + \tfrac{3}{4}\zeta e)$$

is subharmonic and nonpositive on D, with value at least -1 at 0. Set $\alpha_n = \beta/A_n$, with β as in Lemma 8.3.8. That lemma and the change of variables $\zeta \mapsto \frac{3}{4}\zeta$ give a bound, independent of e and v, for

$$\int_{|\zeta|<3/8} e^{-\alpha_n v(a+\zeta e)} \, dV(\zeta).$$

It remains to pass from these integrals over discs in complex lines to an integral over a ball. Let τ and s_{2n-1} be as in (8.1). By that identity, applied with center a, and since $|\zeta|^{2n-2} \le r^{2n-2}$ for $|\zeta| < r$, every Borel function $F \ge 0$, possibly taking the value $+\infty$, satisfies

$$\int_{|z-a|<r} F(z)\,\mathrm{d}V(z) \le \frac{s_{2n-1}r^{2n-2}}{2\pi}\int_{S^{2n-1}}\int_{|\zeta|<r} F(a+\zeta e)\,\mathrm{d}V(\zeta)\,\mathrm{d}\tau(e)$$

for every $r > 0$. We apply this with $F = \mathrm{e}^{-\alpha_n v}$ and $r = 3/8$. Since $\frac{1}{4}B \subset B(a, 3/8)$, the uniform bound for the line integrals proves (8.5). Finally, if $\|v\|_{L^1(B)} \le M$ with $M > 0$, then v/M is a nonpositive plurisubharmonic function with $\|v/M\|_{L^1(B)} \le 1$, and (8.5) for v/M is the last assertion. □

Consequently, every plurisubharmonic function u on a complex manifold X is locally exponentially integrable: every point $x \in X$ has a neighborhood W for which there is $\alpha_0 > 0$ such that $\mathrm{e}^{-\alpha u}$ is integrable on W for $0 < \alpha \le \alpha_0$, the integrals being computed in local coordinates. In dimension zero we may take $W = \{x\}$. In dimension $n \ge 1$, choose a chart in which $x = 0$ and the closed ball $\overline{B(0,R)}$ lies in the chart domain, for some $R > 0$, and let c be an upper bound for u on this ball. The function $v(\zeta) = u(R\zeta) - c$ is plurisubharmonic and nonpositive on the unit ball B, and since u is locally integrable there is $A > 0$ with $\|v\|_{L^1(B)} \le A$. Put $\alpha_0 = \alpha_n/A$ and $W = B(0, R/4)$. For $0 < \alpha \le \alpha_0$ we have $\mathrm{e}^{-\alpha v} \le \mathrm{e}^{-\alpha_0 v}$, because $v \le 0$, so the change of variables $z = R\zeta$ and the last assertion of Proposition 8.3.9 give

$$\int_W \mathrm{e}^{-\alpha u}\,\mathrm{d}V = R^{2n}\mathrm{e}^{-\alpha c}\int_{\frac{1}{4}B} \mathrm{e}^{-\alpha v}\,\mathrm{d}V \le R^{2n}\mathrm{e}^{-\alpha c}C_n < \infty.$$

8.4 Extension across closed exceptional sets

Let S be a relatively closed subset of a domain $\Omega \subset \mathbb{C}^n$. We give two conditions under which plurisubharmonic functions on $\Omega \setminus S$ extend across S. In the first, S is locally contained in the zero set of a holomorphic function, and the function is assumed to be locally bounded above near S; a logarithmic barrier built from the holomorphic function then gives the extension. In the second, S is small in the sense of Hausdorff measure, and the local upper bound is not assumed but follows from the smallness of S by an argument with complex lines.

Theorem 8.4.1 *Let $\Omega \subset \mathbb{C}^n$ be a domain, $n \ge 1$, and let $S \subset \Omega$ be relatively closed. Assume that every point of S has a connected neighborhood $W \subset \Omega$ with a holomorphic function $f \not\equiv 0$ on W such that $S \cap W \subset \{f = 0\}$. Let u be a plurisubharmonic function on $\Omega \setminus S$ that is locally bounded above near S, that is, every point of S has a neighborhood W such that u is bounded above on $W \setminus S$. Then u has a unique plurisubharmonic extension $\widetilde{u}$ to Ω, and its value at $x \in S$ is*

$$\widetilde{u}(x) = \varlimsup_{\Omega\setminus S \ni y \to x} u(y).$$

Proof Define $\widetilde{u}$ by u on $\Omega \setminus S$ and by the displayed formula on S. Since plurisubharmonicity is a local property, it suffices to show that every point $x_0 \in S$ has a

neighborhood on which $\widetilde{u}$ is plurisubharmonic; uniqueness is proved at the end. By hypothesis there are a connected neighborhood W_0 of x_0 with a holomorphic function $f \not\equiv 0$ on W_0 such that $S \cap W_0 \subset \{f = 0\}$, and a neighborhood W_1 of x_0 such that u is bounded above on $W_1 \setminus S$. Choose a ball W about x_0 with $\overline{W} \subset W_0 \cap W_1$. By the identity theorem, f does not vanish identically on W, and u is bounded above on $W \setminus S$. Multiplying f by a nonzero constant, we may assume that $|f| \le 1$, that is, $\log|f|^2 \le 0$, on W. For $\varepsilon > 0$ define u_ε on W by

$$u_\varepsilon = u + \varepsilon \log|f|^2 \quad \text{on } W \setminus S, \qquad u_\varepsilon = -\infty \quad \text{on } S \cap W.$$

Step 1: $u_\varepsilon \in \mathrm{PSH}(W)$. At a point $x \in S \cap W$ we have $f(x) = 0$, so $\log|f(y)|^2 \to -\infty$ as $y \to x$, while u is bounded above near x; hence $u_\varepsilon(y) \to -\infty$ as $y \to x$, and u_ε is upper semicontinuous at x. On $W \setminus S$, u_ε is upper semicontinuous as a sum of upper semicontinuous functions.

Now let Λ be a connected component of $L \cap W$ for an affine complex line L. If f vanishes identically on Λ, then $u_\varepsilon \equiv -\infty$ on Λ. Otherwise the zeros of f on the connected set Λ are isolated, by the identity theorem in one variable. Since S is relatively closed, $A = S \cap \Lambda$ is therefore a closed discrete subset of Λ, hence countable, because each of the countably many compact sets exhausting Λ contains only finitely many of its points. It is polar, because points are polar in the plane and countable unions of polar sets are polar by Proposition 4.2.3.

Each connected component of $\Lambda \setminus A$ is a connected subset of $L \cap (\Omega \setminus S)$ and therefore lies in a connected component of this line section, on which the restriction of u is subharmonic or identically $-\infty$. Hence the restriction of u to $\Lambda \setminus A$ is subharmonic, the constant $-\infty$ being subharmonic. The function $\log|f|^2$ is subharmonic on Λ by Example 3.2.4. Thus u_ε is subharmonic on $\Lambda \setminus A$, and it is bounded above there by $\sup_{W\setminus S} u$, because $\log|f|^2 \le 0$. By Theorem 3.5.2, it has a subharmonic extension to Λ whose value at $x \in A$ is $\overline{\lim}_{\Lambda\setminus A \ni y \to x} u_\varepsilon(y) = -\infty$, which is the value of u_ε at x. Hence the restriction of u_ε to Λ is subharmonic. Finally, u_ε is not identically $-\infty$, because u is finite almost everywhere on $W \setminus S$ and $\{f = 0\}$ is Lebesgue null by Lemma 7.4.1. Thus $u_\varepsilon \in \mathrm{PSH}(W)$.

Step 2: the extension. As $\varepsilon \downarrow 0$ the functions u_ε increase, because $\log|f|^2 \le 0$, and they are bounded above on W by $\sup_{W\setminus S} u$. Let $F = \sup_{\varepsilon>0} u_\varepsilon$. By Theorem 8.3.2, $F^* \in \mathrm{PSH}(W)$ and $F^* = F$ almost everywhere. Since $F = u$ outside the null set $\{f = 0\}$, we get $F^* = u$ almost everywhere on $W \setminus S$; both functions are locally integrable and subharmonic there, so $F^* = u$ everywhere on $W \setminus S$ by Theorem 3.2.2.

It remains to show that $F^* = \widetilde{u}$ on $S \cap W$, which is Lebesgue null as a subset of $\{f = 0\}$. Let $x \in S \cap W$. Since $F^* = u$ on $W \setminus S$, the upper semicontinuity of F^* gives $\widetilde{u}(x) = \overline{\lim}_{W\setminus S \ni y \to x} u(y) \le F^*(x)$. Conversely, for small $r > 0$ the ball submean inequality for F^* gives $F^*(x) \le |B(x,r)|^{-1} \int_{B(x,r)\setminus S} F^* \, dV \le \sup_{B(x,r)\setminus S} u$, because removing the null set S does not change the integral, and letting $r \to 0$ gives $F^*(x) \le \widetilde{u}(x)$. Hence $\widetilde{u} = F^*$ on W, and $\widetilde{u}$ is plurisubharmonic on W.

Uniqueness. The set S is Lebesgue null, since it is covered by countably many neighborhoods W as in the hypothesis and $S \cap W \subset \{f = 0\}$ is null. Two plurisubhar-

monic extensions of u therefore agree almost everywhere, and hence everywhere by Theorem 8.2.1. □

For the second criterion we measure the size of S by Hausdorff measure. Let $q > 0$. A subset E of a metric space is *$\mathcal{H}^q$-null* if for all $\delta, \eta > 0$ it can be covered by countably many balls of radii $r_l < \delta$ with $\sum_l r_l^q < \eta$. This means that the q-dimensional Hausdorff measure of E, which is defined by coverings with sets of small diameter, vanishes, whatever its normalization: a set of diameter ℓ lies in the ball of radius ℓ' about any of its points for every $\ell' > \ell$, and a ball of radius r has diameter at most $2r$. Every subset of an $\mathcal{H}^q$-null set is $\mathcal{H}^q$-null, and so is every countable union of $\mathcal{H}^q$-null sets $E_1, E_2, \ldots$: cover E_l by balls of radii less than δ with sum of the q-th powers of the radii less than $2^{-l}\eta$.

We use two further elementary properties. First, if E is $\mathcal{H}^q$-null and Φ is a Lipschitz map with constant $L > 0$ from E to a metric space, then $\Phi(E)$ is $\mathcal{H}^q$-null. To see this, discard the covering balls that do not meet E, and replace each remaining ball $B(c, r)$ by $B(c', 2r)$ with $c' \in E \cap B(c, r)$; then $\Phi(E \cap B(c', 2r)) \subset B(\Phi(c'), 2Lr)$, so $\Phi(E)$ is covered by balls of radii $2Lr_l < 2L\delta$ with $\sum_l (2Lr_l)^q < (2L)^q \eta$. Second, if $q \le b$, every $\mathcal{H}^q$-null subset of $\mathbb{R}^b$ is Lebesgue null, because the total volume of the covering balls is at most a constant times $\sum_l r_l^b \le \delta^{b-q} \sum_l r_l^q$.

For $n \ge 2$, let σ be the unitarily invariant probability measure on $\mathbb{P}^{n-1}(\mathbb{C})$ and d the metric on $\mathbb{P}^{n-1}(\mathbb{C})$ introduced in Section 6.3. For $v \in \mathbb{C}^n \setminus \{0\}$ we write $[v] \in \mathbb{P}^{n-1}(\mathbb{C})$ for the complex line spanned by v; thus the restriction of $v \mapsto [v]$ to the unit sphere is the Hopf map of Section 6.3. For $y \ne x$ in $\mathbb{C}^n$, the point $[y - x]$ is the *direction* of y from x.

Lemma 8.4.2 *Let $n \ge 2$ and let $K \subset \mathbb{C}^n$ be a compact $\mathcal{H}^{2n-3}$-null set. Then:*

(1) *K is real polar;*
(2) *for every $x \in \mathbb{C}^n$, the set of directions from x of the points of $K \setminus \{x\}$ has σ-measure zero;*
(3) *the orthogonal projection of K onto any complex hyperplane is Lebesgue null.*

Proof (1) Suppose that K is not polar. By Theorem 4.3.3, K carries a nonzero positive measure of finite energy; dividing it by its mass, we obtain a probability measure μ on K of finite energy $I(\mu)$ for the Newtonian kernel $k(x, y) = |x-y|^{2-2n}/((2n-2)s_{2n-1})$ of Section 4.1 in $\mathbb{R}^{2n}$. Its potential U^μ is lower semicontinuous, and it is finite μ-almost everywhere because $\int U^\mu \, d\mu = I(\mu) < \infty$. The sets $E_M = K \cap \{U^\mu \le M\}$, $M \in \mathbb{N}$, are compact, because U^μ is lower semicontinuous, and they increase to a set of full μ-measure; hence $\mu(E_M) > 0$ for M large enough. Let ν be the restriction of μ to E_M. For $a \in E_M$ and $r > 0$,

$$\frac{r^{2-2n}\, \nu(B(a,r))}{(2n-2)s_{2n-1}} \le \int_{B(a,r)} k(a, y)\, d\nu(y) \le U^\mu(a) \le M,$$

so $\nu(B(a,r)) \le C_n M r^{2n-2}$ with $C_n = (2n-2)s_{2n-1}$. Now cover E_M by balls $B(c_l, r_l)$ with $r_l < 1$ and $\sum_l r_l^{2n-3} < \eta$, and discard those not meeting E_M. Each remaining ball lies in $B(a_l, 2r_l)$ for some $a_l \in E_M$, so

$$\nu(E_M) \le \sum_l \nu(B(a_l, 2r_l)) \le 2^{2n-2} C_n M \sum_l r_l^{2n-2} \le 2^{2n-2} C_n M \eta,$$

where we used $r_l < 1$. Since $\eta > 0$ is arbitrary, $\nu(E_M) = 0$, a contradiction. Hence K is polar.

(2) Fix x. For $m \ge 1$ the set $K_m = K \cap \{|y-x| \ge 1/m\}$ is compact and $\mathcal{H}^{2n-3}$-null, and $K \setminus \{x\} = \bigcup_m K_m$. The direction map $\Phi(y) = [y-x]$ is Lipschitz on K_m. Indeed, $\Phi(y) = [(y-x)/|y-x|]$. The normalization $y \mapsto (y-x)/|y-x|$ has Lipschitz constant at most $2m$ on $\{|y-x| \ge 1/m\}$: for nonzero vectors a, b,

$$\Big|\frac{a}{|a|} - \frac{b}{|b|}\Big| \le \frac{|a-b|}{|a|} + |b|\Big|\frac{1}{|a|} - \frac{1}{|b|}\Big| = \frac{|a-b| + \big||b| - |a|\big|}{|a|} \le \frac{2|a-b|}{|a|},$$

and we apply this with $a = y - x$, $b = y' - x$ and $|a| \ge 1/m$. The Hopf map is Lipschitz with constant $\pi/2$ for the metric d. Indeed, let e, e' be unit vectors and $\delta = \arccos \mathrm{Re}\langle e, e'\rangle \in [0, \pi]$ their spherical distance. Since d is the minimum of the spherical distance over the unit representatives (Section 6.3), we have $d([e], [e']) \le \delta$. Moreover $|e - e'|^2 = 2 - 2\cos\delta = 4\sin^2(\delta/2)$, and $\sin s \ge 2s/\pi$ for $0 \le s \le \pi/2$, so $|e - e'| \ge 2\delta/\pi$. Hence

$$d([e], [e']) \le \delta \le \tfrac{\pi}{2}|e - e'|.$$

Since Lipschitz images of $\mathcal{H}^{2n-3}$-null sets are $\mathcal{H}^{2n-3}$-null, the compact set $\Phi(K_m)$ is $\mathcal{H}^{2n-3}$-null in $(\mathbb{P}^{n-1}(\mathbb{C}), d)$. Cover it by balls of radii $r_l < 1$ with $\sum_l r_l^{2n-3} < \eta$. By Lemma 6.3.2, and since $r_l < 1$, we have $\sigma(\Phi(K_m)) \le \sum_l r_l^{2n-2} < \eta$, so $\sigma(\Phi(K_m)) = 0$. Hence the set of directions $\Phi(K \setminus \{x\}) = \bigcup_m \Phi(K_m)$ is a Borel set of σ-measure zero.

(3) An orthogonal projection onto a complex hyperplane $H \cong \mathbb{R}^{2n-2}$ is 1-Lipschitz. Hence the projection of K is $\mathcal{H}^{2n-3}$-null in H, and therefore Lebesgue null because $2n - 3 < 2n - 2$. □

Theorem 8.4.3 *Let $n \ge 2$, let $\Omega \subset \mathbb{C}^n$ be a domain, and let $S \subset \Omega$ be relatively closed and locally $\mathcal{H}^{2n-3}$-null, that is, every point of S has a neighborhood W such that $S \cap W$ is $\mathcal{H}^{2n-3}$-null. Then every plurisubharmonic function on $\Omega \setminus S$ extends uniquely to a plurisubharmonic function on Ω, given on S by the same upper-limit formula as in Theorem 8.4.1.*

Proof Let u be a plurisubharmonic function on $\Omega \setminus S$. For every closed ball $\overline{B} \subset \Omega$, the compact set $S \cap \overline{B}$ is covered by finitely many neighborhoods W as in the hypothesis, and is therefore $\mathcal{H}^{2n-3}$-null.

Step 1: u is locally bounded above near S. Fix $p \in S$ and $r > 0$ with $\overline{B} = \overline{B(p,r)} \subset \Omega$, and write $B = B(p,r)$. By Lemma 8.4.2(2) applied with $x = p$, almost every complex line through p meets $S \cap \overline{B}$ only at p; fix such a line $L = \{p + te : t \in \mathbb{C}\}$, with $|e| = 1$. The circle $\Gamma = L \cap \partial B$ is a compact subset of $\Omega \setminus S$, so the upper semicontinuous function u is bounded above on Γ by a finite constant C, and the open set $O = \{y \in \Omega \setminus S : u(y) < C + 1\}$ contains Γ.

Suppose that u is not bounded above near p. Then there are points $x_j \in B \setminus S$ with $x_j \to p$ and $u(x_j) \to +\infty$. For each j, Lemma 8.4.2(2) applied with $x = x_j$ shows that the directions of the lines through x_j meeting $S \cap \overline{B}$ form a σ-null set. Every nonempty open ball in $\mathbb{P}^{n-1}(\mathbb{C})$ has positive σ-measure by Lemma 6.3.2, so the complement of this null set is dense. We can therefore choose directions ξ_j outside this null set with $d(\xi_j, [e]) < 1/j$. Since d is the minimum of the spherical distance over the unit representatives, ξ_j has a unit representative e_j whose spherical distance to e is less than $1/j$, and then $|e_j - e| < 1/j$, because a chord is shorter than the arc. Thus $e_j \to e$, and the lines $L_j = \{x_j + te_j : t \in \mathbb{C}\}$ do not meet $S \cap \overline{B}$.

We claim that $L_j \cap \partial B \subset O$ for all large j. Let c_j be the point of L_j closest to p, that is, the orthogonal projection of p to L_j; then $|c_j - p| \le |x_j - p| \to 0$. Since L_j is a complex line, $c_j = x_j + \langle p - x_j, e_j\rangle e_j$ and $\langle c_j - p, e_j\rangle = 0$ for the Hermitian inner product, so $\operatorname{Re}\langle c_j - p, se_j\rangle = 0$ and $|c_j + se_j - p|^2 = |c_j - p|^2 + |s|^2$ for $s \in \mathbb{C}$. Hence $L_j \cap \partial B$ is the circle $\{c_j + \rho_j e^{i\theta} e_j\}$ with $\rho_j = (r^2 - |c_j - p|^2)^{1/2} \to r$. Since $|c_j + \rho_j e^{i\theta} e_j - p - re^{i\theta} e| \le |c_j - p| + |\rho_j - r| + r|e_j - e|$, these circles converge uniformly in θ to $\Gamma = \{p + re^{i\theta} e\}$. The distance from the compact set Γ to the closed complement of the open set O is positive, so the circles lie in O for large j.

For such j, the closed disc $\Delta_j = L_j \cap \overline{B}$ lies in $\Omega \setminus S$, and the restriction of u to the connected component of $L_j \cap (\Omega \setminus S)$ containing Δ_j is subharmonic or identically $-\infty$. Its values on the boundary circle of Δ_j are less than $C + 1$, so the maximum principle, that is, Theorem 3.1.3 with the constant harmonic function $C + 1$, gives $u \le C + 1$ on Δ_j. Since $x_j \in B \cap L_j \subset \Delta_j$, this gives $u(x_j) \le C + 1$, contradicting $u(x_j) \to +\infty$. Hence u is bounded above near every point of S.

Step 2: the subharmonic extension. By Lemma 8.4.2(1), every compact set $S \cap \overline{B}$ as above is real polar. Countably many such sets cover S, so S is real polar by Proposition 4.2.3. By Step 1 and the upper semicontinuity of u on $\Omega \setminus S$, the function u is bounded above on $K \setminus S$ for every compact $K \subset \Omega$. Moreover u is subharmonic on $\Omega \setminus S$ by Theorem 8.2.1. By Theorem 3.5.2, u has a subharmonic extension $\widetilde{u}$ to Ω, given on S by the upper-limit formula. Since S is Lebesgue null and u is finite almost everywhere on $\Omega \setminus S$, the function $\widetilde{u}$ is not identically $-\infty$ on the domain Ω, so it is locally integrable.

Step 3: $\widetilde{u}$ is plurisubharmonic. We verify the hypothesis of Lemma 8.1.2 for the locally integrable function $\widetilde{u}$. Fix a unit vector $a \in \mathbb{C}^n$ and a nonnegative $\chi \in C_c^\infty(\Omega)$, and write $z = v + ta$ with $v \in (\mathbb{C}a)^\perp$ and $t \in \mathbb{C}$, as in that lemma. The compact set $S \cap \operatorname{Supp}\chi$ is $\mathcal{H}^{2n-3}$-null, since it is covered by finitely many of the sets $S \cap \overline{B}$ of the first paragraph, so by Lemma 8.4.2(3) its orthogonal projection onto $(\mathbb{C}a)^\perp$ is Lebesgue null. For v outside this projection, the line $\{v + ta : t \in \mathbb{C}\}$ meets $\operatorname{Supp}\chi$ only in $\Omega \setminus S$. Thus the open set $G_v = \{t : v + ta \in \Omega \setminus S\}$ contains $K_v = \{t : v + ta \in \operatorname{Supp}\chi\}$, and its connected components are the connected components of an affine complex-line section of $\Omega \setminus S$, on each of which the restriction of $\widetilde{u} = u$ is subharmonic or identically $-\infty$. As a and χ are arbitrary, Lemma 8.1.2 gives $\mathrm{dd}^c\widetilde{u} \ge 0$. Since $\widetilde{u}$ is a locally integrable subharmonic function, it is the canonical representative of its class, and it is plurisubharmonic by Theorem 8.2.1. Uniqueness follows from the same theorem, since two plurisubharmonic extensions agree outside the Lebesgue null set S. □

8.5 Kiselman's minimum principle

An infimum of plurisubharmonic functions need not be plurisubharmonic. For instance, the function $v(z,w) = |w|^2 + 2\,\mathrm{Re}(zw) = |w+\bar z|^2 - |z|^2$ is plurisubharmonic on $\mathbb{C}^2$, since $2\,\mathrm{Re}(zw)$ is pluriharmonic, but $\inf_{w\in\mathbb{C}} v(z,w) = -|z|^2$ is not. Kiselman's minimum principle shows that, under convexity assumptions on the domain, the infimum is plurisubharmonic or identically $-\infty$ if it is taken only over the real parts of some variables and the function does not depend on their imaginary parts. The reason is visible for a smooth function $u(z,w)$ that depends only on z and $\mathrm{Re}\,w$. Its complex Hessian in w is a quarter of its real Hessian in $\mathrm{Re}\,w$, and at a nondegenerate minimum over $\mathrm{Re}\,w$, the complex Hessian of the infimum is the Schur complement of the w-block in the complex Hessian of u, which is positive semidefinite.

Throughout this section, points of $\mathbb{C}^m \times \mathbb{C}^k$ are written (z,w), and $w = t + is$ with $t = \mathrm{Re}\,w$ and $s = \mathrm{Im}\,w$ in $\mathbb{R}^k$. A set $\Omega \subset \mathbb{C}^m \times \mathbb{C}^k$ is *invariant under imaginary translations* if $(z, w + is') \in \Omega$ for all $(z,w) \in \Omega$ and $s' \in \mathbb{R}^k$. A function u on such a set is *invariant under imaginary translations* if $u(z, w+is') = u(z,w)$ for all such (z,w) and s', that is, if $u(z,w)$ depends only on z and $\mathrm{Re}\,w$. For such Ω and u, we identify $t \in \mathbb{R}^k$ with $t + i0 \in \mathbb{C}^k$, so that $u(z,t)$ is defined. We write

$$V_z = \{t \in \mathbb{R}^k : (z,t) \in \Omega\}, \qquad \Omega_{\mathbb{R}} = \{(z,t) \in \mathbb{C}^m \times \mathbb{R}^k : t \in V_z\}$$

for the real fiber of Ω over $z \in \mathbb{C}^m$ and the real slice of Ω, and $p(\Omega) = \{z : V_z \neq \varnothing\}$ for the projection of Ω to $\mathbb{C}^m$; it is the image of Ω under $(z,w) \mapsto z$, because $(z, \mathrm{Re}\,w) \in \Omega$ for every $(z,w) \in \Omega$ by invariance. Since this projection is continuous and open, $p(\Omega)$ is open if Ω is open, and connected if Ω is connected.

Theorem 8.5.1 (Kiselman's minimum principle) *Let $m, k \geq 1$, and let $\Omega \subset \mathbb{C}^m \times \mathbb{C}^k$ be a domain invariant under imaginary translations whose nonempty real fibers V_z are convex. Assume that there is a plurisubharmonic function ψ on Ω, invariant under imaginary translations, such that for every $c \in \mathbb{R}$ the set*

$$\{(z,t) \in \Omega_{\mathbb{R}} : \psi(z,t) \leq c\}$$

is compact. If $u \in \mathrm{PSH}(\Omega)$ is invariant under imaginary translations, then the function

$$v(z) = \inf_{t \in V_z} u(z,t)$$

is plurisubharmonic or identically $-\infty$ on $p(\Omega)$.

Proof We first treat smooth functions with positive definite real Hessian in t that tend to $+\infty$ at the boundary of the real fibers (Step 1). We then approximate u by such functions on subdomains exhausting Ω (Step 2) and pass to the limit (Step 3).

Step 1: a smooth case. In this step, Ω may be any open set invariant under imaginary translations whose real fibers are convex. Suppose that u is smooth and plurisubharmonic, invariant under imaginary translations, and that for every $z \in p(\Omega)$ the function $t \mapsto u(z,t)$ has positive definite real Hessian on V_z and tends to $+\infty$

at the boundary of V_z and, if V_z is unbounded, as $|t| \to \infty$. We show that v is plurisubharmonic on each connected component of $p(\Omega)$.

Fix $z \in p(\Omega)$ and put $f(t) = u(z,t)$ for $t \in V_z$. Since its Hessian is positive definite, f is strictly convex on every segment in the open convex set V_z, hence on V_z. Its sublevel sets $\{t \in V_z : f(t) \le c\}$ are compact: they are bounded, because $f \to +\infty$ as $|t| \to \infty$ when V_z is unbounded, and the limit in $\mathbb{R}^k$ of a convergent sequence in such a set cannot lie on the boundary of V_z, because $f \to +\infty$ there, so it lies in V_z and, since f is continuous on V_z, in the sublevel set. Hence f attains its minimum on the nonempty set V_z, and by strict convexity at a unique point $g(z)$.

This point is the only critical point of f: if $\nabla f(t_1) = 0$, then the convex function $s \mapsto f(t_1 + s(t_2 - t_1))$ has derivative zero at $s = 0$, hence $f(t_2) \ge f(t_1)$ for every $t_2 \in V_z$, and $t_1 = g(z)$. Fix $z_0 \in p(\Omega)$. Since the t-Hessian is invertible, the implicit function theorem gives a smooth solution $t = G(z)$ of $\partial_t u(z,t) = 0$ near z_0 with $G(z_0) = g(z_0)$, and $G = g$ near z_0 because $g(z)$ is the only critical point. Hence g and $v(z) = u(z, g(z))$ are smooth on $p(\Omega)$.

At the point $(z, g(z))$, let $H = (\partial^2 u/\partial t_a \partial t_b)$, $B = (\partial^2 u/\partial z_j \partial t_a)$ and $C = (\partial^2 u/\partial z_j \partial \bar{z}_l)$. Thus H is a real symmetric positive definite $k \times k$ matrix and B is a complex $m \times k$ matrix; since u is real, $(\partial^2 u/\partial t_a \partial \bar{z}_j) = B^*$, the conjugate transpose of B. Differentiating the identity $\partial_t u(z, g(z)) = 0$ with respect to $\bar{z}$ gives $B^* + H g_{\bar{z}} = 0$, where $g_{\bar{z}} = (\partial g_a/\partial \bar{z}_j)$; hence $g_{\bar{z}} = -H^{-1}B^*$. By the chain rule and the same identity, $v_{z_j}(z) = u_{z_j}(z, g(z))$, and differentiating once more with respect to $\bar{z}$ gives

$$v_{z\bar{z}} = C + B g_{\bar{z}} = C - B H^{-1} B^*.$$

On the other hand, for functions of z and $\operatorname{Re} w$ we have $\partial_{w_a} = \partial_{\bar{w}_a} = \frac{1}{2}\partial_{t_a}$. Hence the complex Hessian of u with respect to (z, w), which is positive semidefinite by Proposition 8.2.2(1), is

$$\begin{pmatrix} C & B/2 \\ B^*/2 & H/4 \end{pmatrix} \ge 0.$$

Evaluating the associated Hermitian quadratic form at $(\xi, -2H^{-1}B^*\xi)$, $\xi \in \mathbb{C}^m$, gives $\xi^*(C - BH^{-1}B^*)\xi \ge 0$. Thus $v_{z\bar{z}} \ge 0$, and the smooth function v is plurisubharmonic on each component of $p(\Omega)$ by Proposition 8.2.2(1).

Step 2: regularization. Now let u be as in the theorem. For $\varepsilon > 0$ let $u_\varepsilon = u * \rho_\varepsilon$ and $\psi_\varepsilon = \psi * \rho_\varepsilon$ on $\Omega_\varepsilon = \{x \in \Omega : \operatorname{dist}(x, \mathbb{C}^{m+k} \setminus \Omega) > \varepsilon\}$, where ρ_ε is the radial mollifier of Section 1.4. Imaginary translations are isometries preserving Ω, and convolution commutes with translations, so Ω_ε, u_ε and ψ_ε are invariant under imaginary translations. By Proposition 8.2.2, the functions u_ε and ψ_ε are smooth and plurisubharmonic on each connected component of Ω_ε, so their complex Hessians are positive semidefinite at every point, and they decrease to u and ψ as $\varepsilon \downarrow 0$; in particular $u_\varepsilon \ge u$ and $\psi_\varepsilon \ge \psi$.

The real fibers of Ω_ε are convex. Indeed, $(z,t) \in \Omega_\varepsilon$ if and only if the closed ball of radius ε about (z,t) lies in Ω (if it does, its distance to the closed complement of Ω is positive, because it is compact); by invariance this means that $(z + y, t + y') \in \Omega_{\mathbb{R}}$ for all $(y, y') \in \mathbb{C}^m \times \mathbb{R}^k$ with $|y|^2 + |y'|^2 \le \varepsilon^2$. Hence the real fiber of Ω_ε over z is the intersection of the convex sets $V_{z+y} - y'$ over these (y, y'). Moreover, on each real

fiber of Ω_ε the function $t \mapsto \psi_\varepsilon(z,t)$ is convex, since by invariance its real Hessian is four times the complex Hessian of ψ_ε in w, which is positive semidefinite, and a smooth function with positive semidefinite Hessian on an open convex set is convex along every segment.

For $c \in \mathbb{R}$ let $K_c = \{(z,t) \in \Omega_{\mathbb{R}} : \psi(z,t) \le c\}$; it is a compact subset of Ω. Choose $\varepsilon_j \downarrow 0$, $j \ge 1$, such that $\operatorname{dist}(K_{j+1}, \mathbb{C}^{m+k} \setminus \Omega) > 4\varepsilon_j$ whenever $K_{j+1} \neq \varnothing$. Since ψ and the distance to the complement of Ω are invariant under imaginary translations, the same bound holds for the set $\{x \in \Omega : \psi(x) \le j+1\} = K_{j+1} + \mathrm{i}\mathbb{R}^k$. Define

$$G_j = \{(z,w) \in \Omega_{\varepsilon_j} : \psi_{\varepsilon_j}(z,w) < j\}, \qquad U_j = u_{\varepsilon_j} + \frac{|\operatorname{Re} w|^2}{j} + \frac{1}{j - \psi_{\varepsilon_j}} \quad \text{on } G_j.$$

The open set G_j is invariant under imaginary translations, and its real fiber over z is a sublevel set of the convex function $\psi_{\varepsilon_j}(z,\cdot)$ on the convex real fiber of Ω_{ε_j}, hence convex. Since $\psi \le \psi_{\varepsilon_j}$, the real slice of G_j is contained in the compact set $K_j \subset K_{j+1}$, which lies in Ω_{ε_j} at positive distance from its complement. Consequently, for each z, the closure of the real fiber of G_j over z is a compact subset of the real fiber of Ω_{ε_j}, and $\psi_{\varepsilon_j}(z,\cdot) = j$ on its boundary.

The function U_j is smooth, plurisubharmonic and invariant under imaginary translations. Indeed, by the chain rule the complex Hessian of $1/(j - \psi_{\varepsilon_j})$ has the entries

$$\frac{(\psi_{\varepsilon_j})_{a\bar b}}{(j-\psi_{\varepsilon_j})^2} + \frac{2(\psi_{\varepsilon_j})_a (\psi_{\varepsilon_j})_{\bar b}}{(j-\psi_{\varepsilon_j})^3},$$

where the subscripts denote derivatives with respect to the coordinates of (z,w). Since $j - \psi_{\varepsilon_j} > 0$ on G_j and $(\psi_{\varepsilon_j})_{\bar b} = \overline{(\psi_{\varepsilon_j})_b}$, this matrix is the sum of positive multiples of the complex Hessian of ψ_{ε_j} and of a positive semidefinite matrix of rank at most one, hence positive semidefinite. Moreover, $|\operatorname{Re} w|^2$ has the positive semidefinite complex Hessian in (z,w) given by the diagonal matrix with entries 0 in the z-directions and $\frac12$ in the w-directions. Hence U_j is plurisubharmonic on each connected component of G_j, by Proposition 8.2.2(1).

The real t-Hessian of U_j is positive definite: the t-Hessians of u_{ε_j} and $1/(j-\psi_{\varepsilon_j})$ are positive semidefinite, as for ψ_ε above, and that of $|t|^2/j$ is $(2/j)I$. On each real fiber of G_j, which is bounded, $U_j(z,t) \to +\infty$ at the boundary, because $1/(j-\psi_{\varepsilon_j}) \to +\infty$ there while the other two terms are bounded below on the compact closure of the fiber. Hence Step 1 applies to U_j on G_j, and

$$v_j(z) = \inf\{U_j(z,t) : (z,t) \in G_j\}$$

is plurisubharmonic on each connected component of $p(G_j)$.

Step 3: passage to the limit. The sets G_j increase to Ω. Indeed, $\varepsilon_{j+1} \le \varepsilon_j$ gives $\Omega_{\varepsilon_j} \subset \Omega_{\varepsilon_{j+1}}$ and $\psi_{\varepsilon_{j+1}} \le \psi_{\varepsilon_j} < j < j+1$ on G_j. Every point $x \in \Omega$ lies in Ω_{ε_j} for large j, and $\psi_{\varepsilon_j}(x) \to \psi(x) < \infty$, so $x \in G_j$ for large j. On G_j we have $U_{j+1} \le U_j$, since $u_{\varepsilon_{j+1}} \le u_{\varepsilon_j}$, $|\operatorname{Re} w|^2/(j+1) \le |\operatorname{Re} w|^2/j$ and $0 < j - \psi_{\varepsilon_j} \le j+1-\psi_{\varepsilon_{j+1}}$. Hence the functions v_j decrease on the increasing open sets $p(G_j)$. They satisfy $v_j \ge v$, since $U_j \ge u_{\varepsilon_j} \ge u$. Conversely, let $z \in p(\Omega)$ and $t \in V_z$. Then $(z,t) \in G_j$ for

large j, and $U_j(z,t) \to u(z,t)$, also when $u(z,t) = -\infty$, because $u_{\varepsilon_j}(z,t) \to u(z,t)$ and the other two terms tend to zero. Hence $\lim_j v_j(z) \le u(z,t)$ for every $t \in V_z$, and therefore $v_j \downarrow v$ on $p(\Omega)$.

Let $B' \Subset p(\Omega)$ be a ball. Since $\overline{B'}$ is compact and covered by the increasing open sets $p(G_j)$, we have $B' \subset p(G_j)$ for large j, and the v_j with such j form a decreasing sequence of plurisubharmonic functions on B'. By Theorem 8.3.1, v is plurisubharmonic or identically $-\infty$ on B'. Since every point of the connected open set $p(\Omega)$ lies in such a ball, Lemma 8.2.4 shows that v is plurisubharmonic or identically $-\infty$ on $p(\Omega)$; this proves the theorem. □

Corollary 8.5.2 *Let $m, k \ge 1$, let $D \subset \mathbb{C}^m$ be a domain and let $V \subset \mathbb{R}^k$ be a nonempty open convex set. If $u \in \mathrm{PSH}(D \times (V + i\mathbb{R}^k))$ is invariant under imaginary translations, then the function $z \mapsto \inf_{t \in V} u(z,t)$ is plurisubharmonic or identically $-\infty$ on D. The same holds for $z \mapsto \inf_{t \in V}(u(z,t) - \ell(t))$, for every real affine function ℓ on $\mathbb{R}^k$.*

Proof It suffices to prove the assertion on every ball $B(z_0, r) \Subset D$: since D is connected, Lemma 8.2.4 then gives it on D. We construct a function ψ as in Theorem 8.5.1 for the domain $B(z_0, r) \times (V + i\mathbb{R}^k)$, which is invariant under imaginary translations and has real fibers V.

We first need a continuous convex function $b \ge 0$ on V with compact sublevel sets. If $V = \mathbb{R}^k$, take $b(t) = |t|^2$. Otherwise let $\delta(t) = \mathrm{dist}(t, \mathbb{R}^k \setminus V)$, which is continuous and positive on V. It is concave on V: if the balls $B(t_1, r_1)$ and $B(t_2, r_2)$ lie in V and $0 \le \lambda \le 1$, then by convexity V contains $\lambda B(t_1, r_1) + (1-\lambda) B(t_2, r_2) = B(\lambda t_1 + (1-\lambda)t_2, \lambda r_1 + (1-\lambda) r_2)$, so $\delta(\lambda t_1 + (1-\lambda)t_2) \ge \lambda r_1 + (1-\lambda) r_2$, and letting $r_i \uparrow \delta(t_i)$ gives concavity. Consequently $1/\delta$ is convex: since $x \mapsto 1/x$ is decreasing and convex on $(0, \infty)$,

$$\frac{1}{\delta(\lambda t_1 + (1-\lambda)t_2)} \le \frac{1}{\lambda \delta(t_1) + (1-\lambda)\delta(t_2)} \le \frac{\lambda}{\delta(t_1)} + \frac{1-\lambda}{\delta(t_2)}.$$

Put $b(t) = |t|^2 + 1/\delta(t)$. For $c > 0$ the set $\{b \le c\}$ is closed in V and contained in the compact set $\{t : |t|^2 \le c,\ \delta(t) \ge 1/c\} \subset V$, hence compact.

A continuous convex function f on V defines a plurisubharmonic function $f(\mathrm{Re}\, w)$ on every domain of the form $D' \times (V + i\mathbb{R}^k)$. Indeed, $f(\mathrm{Re}\, w)$ is continuous, and its restriction to an affine complex line is $f \circ A$ for a real affine map A from an open subset of $\mathbb{C}$ to V. If the closed disc of radius r about ζ_0 lies in the domain of A, then, since A is affine, $A(\zeta_0)$ is the midpoint of $A(\zeta_0 + re^{i\theta})$ and $A(\zeta_0 - re^{i\theta})$ for every θ, and the convexity of f gives

$$f(A(\zeta_0)) \le \tfrac{1}{2}\big(f(A(\zeta_0 + re^{i\theta})) + f(A(\zeta_0 - re^{i\theta}))\big).$$

Averaging over $\theta \in [0, 2\pi]$, we obtain the submean inequality for the continuous function $f \circ A$ on this circle.

Now put $\psi(z,w) = (r^2 - |z - z_0|^2)^{-1} + b(\mathrm{Re}\, w)$. This function is continuous, positive and invariant under imaginary translations. It is plurisubharmonic as a sum of two plurisubharmonic functions (Proposition 8.2.5). The second term is

plurisubharmonic by the preceding paragraph. For the first, $x \mapsto (r^2 - x)^{-1}$ is convex and increasing on $(-\infty, r^2)$ and $|z - z_0|^2$ has the complex Hessian I, so by the chain rule the complex Hessian of $(r^2 - |z - z_0|^2)^{-1}$ is positive semidefinite, and Proposition 8.2.2(1) applies. For $c > 0$, the set $\{(z,t) \in B(z_0,r) \times V : \psi(z,t) \le c\}$ is closed in $B(z_0,r) \times V$ and, since both terms of ψ are nonnegative, contained in the compact set $\{|z - z_0|^2 \le r^2 - 1/c\} \times \{b \le c\}$; it is therefore compact, and for $c \le 0$ it is empty. Hence Theorem 8.5.1 applies on $B(z_0,r) \times (V + \mathrm{i}\mathbb{R}^k)$, and shows that $\inf_{t \in V} u(z,t)$ is plurisubharmonic or identically $-\infty$ on $B(z_0,r)$.

Finally, if ℓ is real affine, then $\ell(\operatorname{Re} w)$ is the real part of a complex affine function of w. Hence $-\ell(\operatorname{Re} w)$ is smooth with vanishing complex Hessian, so it is plurisubharmonic by Proposition 8.2.2(1), and $u - \ell(\operatorname{Re} w)$ is plurisubharmonic by Proposition 8.2.5. It is invariant under imaginary translations, and the first part applies to it. □

Chapter 9
Bedford–Taylor products and local capacity

Let u be a plurisubharmonic function on an open set $\Omega \subset \mathbb{C}^n$. By Theorem 8.2.1, $\mathrm{dd^c} u$ is a closed positive $(1,1)$-current, but in general its coefficients are only measures. Since products of measures have no general meaning, the exterior powers of $\mathrm{dd^c} u$ cannot be formed by multiplying coefficients. For $u \in C^2(\Omega)$ the top power is the Monge–Ampère measure $(\mathrm{dd^c} u)^n = n!\,\pi^{-n} \det(u_{j\bar{k}})\, \mathrm{d}V_{2n}$. This fully nonlinear expression in the second derivatives of u plays in pluripotential theory the role that the Laplacian plays in classical potential theory. Bedford and Taylor showed that for locally bounded u these powers can be defined by induction, and so can the mixed products $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T$ with a closed positive current T. Indeed, since T has measure coefficients and u is a locally bounded Borel function, the product uT makes sense, and one puts $\mathrm{dd^c} u \wedge T = \mathrm{dd^c}(uT)$. This chapter develops the local theory of these products, whose factors are always locally bounded; products of unbounded potentials are treated in Chapter 15.

The first two sections establish the basic properties of the products: positivity, the Chern–Levine–Nirenberg mass bounds, continuity along decreasing sequences and symmetry (Theorem 9.2.5), and the comparison principle. The products then define the Monge–Ampère capacity, a complex analogue of the relative capacity of Chapter 4. Locally, pole sets of plurisubharmonic functions have outer capacity zero; plurisubharmonic functions are quasi-continuous with respect to this capacity, and the products are continuous under convergence in capacity and along increasing sequences. Solving the homogeneous Monge–Ampère equation on a ball, we show that a locally bounded plurisubharmonic function is maximal if and only if its Monge–Ampère measure vanishes. These are the tools with which Chapter 10 characterizes pluripolar sets.

The remaining sections, needed for non-pluripolar products and envelopes on compact manifolds in Chapters 15 and 16, solve the equation $(\mathrm{dd^c} u)^n = \mu$ on a ball for increasingly singular measures μ: continuous densities, bounded densities, and finally compactly supported measures dominated by the Monge–Ampère measure of a bounded plurisubharmonic function. The main application is a lower bound for mixed products: if $(\mathrm{dd^c} u_i)^n \geq f_i \mu$ for $1 \leq i \leq n$, then $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n \geq (f_1 \cdots f_n)^{1/n} \mu$

(Theorem 9.11.1). The chapter closes with the locality of complex Hessians on contact sets and a chain rule for gradients against positive currents.

Historical remarks and references for this chapter are collected in Appendix B.9.

9.1 Bounded potentials and recursive products

Throughout this section and the next, $\Omega \subset \mathbb{C}^n$ is open, $n \geq 1$, and T is a closed positive current on Ω of bidimension (r, r), where $0 \leq r \leq n$ is an integer; thus T has bidegree $(n-r, n-r)$. Let $\beta = (\mathrm{i}/2)\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$ be the form of (7.4). For a positive current S of bidimension (s, s) on Ω and a compact set $K \subset \Omega$ we write

$$\|S\|_K = \int_K S \wedge \frac{\beta^s}{s!}$$

for the mass of S on K; it is $1/s!$ times the value on K of the trace measure of S (Definition 7.2.4). By Theorem 7.2.5, S is a current of order zero, and the total variations of its coefficient measures are bounded by a constant multiple, depending only on n and s, of its trace measure. Since plurisubharmonic functions are upper semicontinuous, they are bounded above on compact sets; hence a plurisubharmonic function is locally bounded if and only if it is bounded below on compact sets.

Definition 9.1.1 Suppose that $r \geq 1$, and let u be a locally bounded plurisubharmonic function on Ω. Let uT be the current of order zero obtained by multiplying the coefficient measures of T by the locally bounded Borel function u (Section 7.2), and define

$$\mathrm{dd^c}\, u \wedge T = \mathrm{dd^c}\,(uT).$$

For an integer $1 \leq q \leq r$ and locally bounded plurisubharmonic functions $u_1, \ldots, u_q$ on Ω, define recursively

$$\mathrm{dd^c}\, u_1 \wedge \cdots \wedge \mathrm{dd^c}\, u_q \wedge T = \mathrm{dd^c}\big(u_1 \mathrm{dd^c}\, u_2 \wedge \cdots \wedge \mathrm{dd^c}\, u_q \wedge T\big).$$

Here $\mathrm{dd^c}\, u_2 \wedge \cdots \wedge \mathrm{dd^c}\, u_q \wedge T$ is a closed positive current, hence of order zero, by Proposition 9.1.2 below applied to $q-1$ factors; so its product with u_1 is defined as before. The product with no factors is T. When $T = 1$ (so that $r = n$), $q = n$ and $u_1 = \cdots = u_n = u$, the resulting positive current of bidegree (n, n) is a positive measure, the *Monge–Ampère measure* $(\mathrm{dd^c}\, u)^n$ of u.

Proposition 9.1.2 *Every product* $\mathrm{dd^c}\, u_1 \wedge \cdots \wedge \mathrm{dd^c}\, u_q \wedge T$ *of Definition 9.1.1 is a closed positive current, and it has the following properties.*

(1) *If* $u_1, \ldots, u_q$ *are smooth, it coincides with the exterior product of the smooth forms* $\mathrm{dd^c}\, u_j$ *with* T.
(2) *The construction is local: if* $u_j = v_j$ *for every* j *and* $T = T'$ *on an open set* $U \subset \Omega$, *then the two products agree on* U.

(3) *It is additive and positively homogeneous in each u_j and in T, whenever the summands are locally bounded plurisubharmonic functions, respectively closed positive currents of bidimension (r,r), and it does not change if a real constant is added to some u_j.*

(4) *If γ is a smooth closed form of bidegree (s,s), $q \le r-s$, such that $T\wedge\gamma$ is positive (for instance, γ strongly positive), then*

$$\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge(T\wedge\gamma) = (\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge T)\wedge\gamma.$$

(5) *It is invariant under biholomorphic maps: if $F\colon\Omega'\to\Omega$ is biholomorphic and F^* denotes the pullback of currents of Proposition 7.5.2, then*

$$F^*\big(\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge T\big) = \mathrm{dd^c}(u_1\circ F)\wedge\cdots\wedge\mathrm{dd^c}(u_q\circ F)\wedge F^*T.$$

Proof We first record two consequences of the Leibniz rule of Section 7.2. Let R be a current of even degree on an open subset of $\mathbb{C}^n$ such that $\partial R = \bar\partial R = 0$; this holds if R is closed and of pure bidegree. If f is a smooth function, then $\mathrm{d}(fR) = \mathrm{d}f\wedge R$ and $\mathrm{d^c}(fR) = \mathrm{d^c}f\wedge R$, and applying d to the latter gives

$$\mathrm{dd^c}(fR) = \mathrm{dd^c}f\wedge R. \tag{9.1}$$

If γ is a smooth closed form of pure bidegree, then $\partial\gamma$ and $\bar\partial\gamma$ are the components of $\mathrm{d}\gamma = 0$ of two different bidegrees, so they vanish, and the Leibniz rule gives $\mathrm{dd^c}(R'\wedge\gamma) = \mathrm{dd^c}R'\wedge\gamma$ for every current R'.

We now prove positivity and closedness, first for a single factor u. By Proposition 8.2.2, the regularizations $u_\varepsilon = u * \rho_\varepsilon$ by the radial mollifiers of Section 1.4 are smooth plurisubharmonic functions on Ω_ε with positive semidefinite complex Hessians, and they decrease to u as $\varepsilon\downarrow 0$. On each compact set the u_ε are bounded uniformly in ε, because $u\le u_\varepsilon\le u_{\varepsilon_0}$ for $\varepsilon\le\varepsilon_0$ and u is locally bounded. Dominated convergence with respect to the total variations of the coefficient measures of T therefore gives $u_\varepsilon T\to uT$ weakly. By (9.1), $\mathrm{dd^c}(u_\varepsilon T) = \mathrm{dd^c}u_\varepsilon\wedge T$. Since weak and strong positivity agree in bidegree $(1,1)$ (Section 7.2), the positive smooth form $\mathrm{dd^c}u_\varepsilon$ is strongly positive, so its product with the positive current T is positive; it is also closed. As $\mathrm{dd^c}$ is continuous for weak convergence, $\mathrm{dd^c}u_\varepsilon\wedge T\to\mathrm{dd^c}(uT)$ weakly, and positivity and closedness pass to the limit. Thus $\mathrm{dd^c}u\wedge T$ is positive and closed. Applying this with T replaced by the closed positive current $\mathrm{dd^c}u_2\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge T$, we obtain the first assertion by induction on q.

(1) If the u_j are smooth, (9.1) shows at each step of the recursion that the product is the exterior product of smooth forms with T.

(2), (3) The definition uses only multiplication of coefficient measures by functions and the operator $\mathrm{dd^c}$; both operations are local and linear, and no product of two measures occurs. This gives locality, and additivity and positive homogeneity in u_1 and in T; for the other factors they follow by induction on q, since multiplication by u_1 and $\mathrm{dd^c}$ are linear. If c is a constant and $R = \mathrm{dd^c}u_2\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge T$, then $\mathrm{dd^c}((u_1+c)R) = \mathrm{dd^c}(u_1R) + c\,\mathrm{dd^c}R = \mathrm{dd^c}(u_1R)$, since R is closed; for the other factors the invariance under constants follows again by induction.

(4) The product $T \wedge \gamma$ is closed, and multiplication of coefficient measures by u_1 commutes with the exterior product by γ. By the identity $\mathrm{dd^c}(R' \wedge \gamma) = \mathrm{dd^c} R' \wedge \gamma$ and induction on q, every step of the recursion commutes with $\wedge\gamma$.

(5) The pullback F^*T is closed and positive and commutes with ∂ and $\bar\partial$, hence with $\mathrm{dd^c}$ (Proposition 7.5.2), and the functions $u_j \circ F$ are locally bounded and plurisubharmonic (Proposition 8.2.3). We claim that $F^*(uT) = (u\circ F)F^*T$ for a locally bounded plurisubharmonic u. For a smooth function f we have $F^*(fT) = (f\circ F)F^*T$: both sides are weakly continuous in T, they agree when T is smooth, and on every relatively compact open subset of Ω the current T is the weak limit of its smooth regularizations (discussion following Theorem 7.2.5). Applying this to the regularizations u_ε on relatively compact open subsets, and letting $\varepsilon \downarrow 0$, we obtain the claim: $u_\varepsilon T \to uT$ as shown above, so $F^*(u_\varepsilon T) \to F^*(uT)$, while $(u_\varepsilon \circ F)F^*T \to (u\circ F)F^*T$ by dominated convergence with respect to the coefficient measures of the positive current F^*T. Consequently every step of the recursion commutes with F^*, which proves (5). In particular, by (2) and (5), products of locally bounded plurisubharmonic functions with closed positive currents on a complex manifold can be computed in holomorphic charts: the products defined in two charts agree on their overlap. □

Let $u \in C^2(\Omega)$ be real, and write $a_{jk} = u_{j\bar k}$, so that $\mathrm{dd^c} u = \frac{\mathrm{i}}{2\pi}\sum_{j,k} a_{jk}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_k$ by (7.5). The 2-forms $\alpha_{jk} = \mathrm{d}z_j \wedge \mathrm{d}\bar z_k$ commute with each other, and $\alpha_{j_1k_1} \wedge \cdots \wedge \alpha_{j_nk_n} = 0$ unless $(j_1,\dots,j_n)$ and $(k_1,\dots,k_n)$ are permutations of $(1,\dots,n)$. If $j_i = \sigma(i)$ and $k_i = \tau(i)$ for permutations σ,τ, we may reorder the commuting factors so that the first indices increase; with $\kappa = \tau\circ\sigma^{-1}$ we get $\mathrm{d}z_1\wedge\mathrm{d}\bar z_{\kappa(1)}\wedge\cdots\wedge\mathrm{d}z_n\wedge\mathrm{d}\bar z_{\kappa(n)}$, and permuting the 1-forms $\mathrm{d}\bar z_{\kappa(i)}$ among themselves gives the sign $\operatorname{sgn}\kappa$. Hence

$$\Big(\sum_{j,k} a_{jk}\alpha_{jk}\Big)^n = \sum_{\sigma,\tau}\prod_{i=1}^n a_{\sigma(i)\tau(i)}\ \operatorname{sgn}(\tau\sigma^{-1})\bigwedge_{j=1}^n \alpha_{jj} = n!\det(a_{jk})\bigwedge_{j=1}^n \alpha_{jj},$$

because for each of the $n!$ permutations σ the sum over τ is $\sum_\kappa \operatorname{sgn}\kappa \prod_j a_{j\kappa(j)} = \det(a_{jk})$. Since $\mathrm{i}\,\mathrm{d}z_j\wedge\mathrm{d}\bar z_j = 2\,\mathrm{d}x_j\wedge\mathrm{d}y_j$, we have $\bigwedge_j(\mathrm{i}\alpha_{jj}) = 2^n \mathrm{d}V_{2n}$, and therefore

$$(\mathrm{dd^c}u)^n = \Big(\frac{\mathrm{i}}{2\pi}\Big)^n n!\det(u_{j\bar k})\bigwedge_{j=1}^n \alpha_{jj} = \frac{n!}{\pi^n}\det(u_{j\bar k})\,\mathrm{d}V_{2n}. \tag{9.2}$$

Here $(\mathrm{dd^c}u)^n$ is the exterior power of the continuous form $\mathrm{dd^c}u$; when u is smooth and plurisubharmonic, it is the Monge–Ampère measure of u, by Proposition 9.1.2(1), and the same holds for plurisubharmonic $u \in C^2(\Omega)$ (see the remark after Proposition 9.1.4). In particular $\mathrm{dd^c}|z|^2 = \frac{\mathrm{i}}{2\pi}\sum_j \mathrm{d}z_j\wedge\mathrm{d}\bar z_j = \beta/\pi$ and $(\mathrm{dd^c}|z|^2)^n = n!\pi^{-n}\,\mathrm{d}V_{2n}$. (The formula also holds for $n = 0$: the empty product is the constant 1, and both sides are the unit mass $\mathrm{d}V_0$ on the point $\mathbb{C}^0$.)

Theorem 9.1.3 (Chern–Levine–Nirenberg) *Let $K, L \subset \Omega$ be compact sets with $K \subset \operatorname{Int} L$, and let $0 \le q \le r$ be an integer. There is a constant $C_{K,L,q}$, depending only on K, L, n and q, such that*

$$\|\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T\|_K \le C_{K,L,q}\|T\|_L \prod_{j=1}^{q} \|u_j\|_{L^\infty(L)}$$

for every closed positive current T of bidimension (r,r) on Ω and all locally bounded plurisubharmonic functions $u_1,\dots,u_q$ on Ω. Here $\|u_j\|_{L^\infty(L)} = \sup_L |u_j|$ is the pointwise supremum.

Proof For $q = 0$ the estimate holds with constant 1, since $K \subset L$. Let $q = 1$ and write $u = u_1$. By Lemma 1.4.1 there is a smooth function $0 \le \chi \le 1$ with compact support in $\operatorname{Int} L$ which equals 1 on a neighborhood of K; it depends only on K and L. Since $\mathrm{dd^c}u \wedge T \wedge \beta^{r-1}$ is a positive measure, β is closed, and $(\mathrm{dd^c}S)(\chi) = S(\mathrm{dd^c}\chi)$ for currents S of bidimension $(1,1)$, we have

$$\begin{aligned}\int_K \mathrm{dd^c}u \wedge T \wedge \beta^{r-1} &\le \int \chi\, \mathrm{dd^c}(uT) \wedge \beta^{r-1} = \big(\mathrm{dd^c}(uT \wedge \beta^{r-1})\big)(\chi)\\ &= \int uT \wedge \mathrm{dd^c}\chi \wedge \beta^{r-1} \le C\|u\|_{L^\infty(L)}\|T\|_L.\end{aligned}$$

The form $\mathrm{dd^c}\chi \wedge \beta^{r-1}$ is smooth with support in L, though in general not positive; its coefficients are bounded by a constant depending only on χ and n. The last inequality therefore holds because the total variations of the coefficient measures of T on L are bounded by a constant multiple of $\|T\|_L$, the multiple depending only on n and r (Theorem 7.2.5). Dividing by $(r-1)!$ and taking the largest of the constants obtained for $r = 1,\dots,n$, we obtain the case $q = 1$ with a constant depending only on K, L and n.

For $q > 1$, let $\delta > 0$ be the distance from K to $\mathbb{C}^n \setminus \operatorname{Int} L$ (if $K = \varnothing$ there is nothing to prove), and put $L_j = \{z : \operatorname{dist}(z,K) \le j\delta/(q+1)\}$ for $0 \le j < q$ and $L_q = L$. These are compact sets, depending only on K, L and q, with $K = L_0$ and $L_{j-1} \subset \operatorname{Int} L_j$ for $1 \le j \le q$. By Proposition 9.1.2, the currents $S_j = \mathrm{dd^c}u_{j+1} \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T$, $0 \le j \le q$, are positive and closed, and $S_{j-1} = \mathrm{dd^c}u_j \wedge S_j$. The case $q = 1$, applied to S_j and the pair $L_{j-1} \subset \operatorname{Int} L_j$, gives $\|S_{j-1}\|_{L_{j-1}} \le C_j\|u_j\|_{L^\infty(L)}\|S_j\|_{L_j}$ for $j = 1,\dots,q$, since $\|u_j\|_{L^\infty(L_j)} \le \|u_j\|_{L^\infty(L)}$. Combining these q inequalities, we obtain the estimate with $C_{K,L,q} = C_1 \cdots C_q$. □

For continuous potentials, uniform convergence is enough to pass to the limit in the products.

Proposition 9.1.4 *Let $T_k \to T$ weakly, where T_k and T are closed positive currents on Ω of bidimension (r,r), and let $0 \le q \le r$. For $1 \le j \le q$, let $(u_{j,k})_{k\ge1}$ be locally bounded plurisubharmonic functions on Ω converging locally uniformly, as $k \to \infty$, to a continuous function u_j, and let $(u_{0,k})_{k\ge1}$ be locally bounded Borel functions on Ω converging locally uniformly to a continuous function u_0. Then $u_1,\dots,u_q$ are plurisubharmonic, and*

$$\begin{aligned}\mathrm{dd^c}u_{1,k} \wedge \cdots \wedge \mathrm{dd^c}u_{q,k} \wedge T_k &\longrightarrow \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T,\\ u_{0,k}\mathrm{dd^c}u_{1,k} \wedge \cdots \wedge \mathrm{dd^c}u_{q,k} \wedge T_k &\longrightarrow u_0\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge T\end{aligned}$$

weakly as $k \to \infty$.

Proof We first show that u_j is plurisubharmonic for $1 \le j \le q$. Since plurisubharmonicity is a local property, it suffices to prove this on every ball $W \Subset \Omega$. Choose $k_1 < k_2 < \cdots$ such that $|u_{j,k_m} - u_j| \le 2^{-m}$ on W, and put $w_m = u_{j,k_m} + 3 \cdot 2^{-m}$. These functions are plurisubharmonic on W, and they decrease, because

$$w_{m+1} \le u_j + 2^{-m-1} + 3 \cdot 2^{-m-1} = u_j + 2^{1-m} \le u_{j,k_m} + 2^{-m} + 2^{1-m} = w_m.$$

Their limit u_j is finite, so it is plurisubharmonic on W by Theorem 8.3.1.

Since the $u_{j,k}$ converge locally uniformly to continuous functions, they are bounded on each compact set uniformly in k. Since $T_k \to T$ weakly, the masses $\|T_k\|_L$ are bounded for each compact set L: if $\chi \in C_c^\infty(\Omega)$ satisfies $0 \le \chi \le 1$ and $\chi = 1$ on L, then $\|T_k\|_L \le T_k(\chi\beta^r)/r!$, and the right-hand side converges. By Theorem 9.1.3, the masses of all the products considered below are therefore bounded on compact sets uniformly in k, and so are the total variations of their coefficient measures (Theorem 7.2.5).

We claim: if $S_k \to S$ weakly, where S_k and S are closed positive currents whose masses are bounded on compact sets uniformly in k, and if locally bounded Borel functions v_k converge locally uniformly to a continuous function v, then $v_k S_k \to vS$ weakly. Applying this to $S_k = \mathrm{dd^c} u_{2,k} \wedge \cdots \wedge \mathrm{dd^c} u_{q,k} \wedge T_k$ and $v_k = u_{1,k}$, and then $\mathrm{dd^c}$, which is weakly continuous, proves the first assertion by induction on q; the case $q = 0$ is the hypothesis. The second assertion is the claim with $v_k = u_{0,k}$.

To prove the claim, fix a test form φ with compact support K and $\varepsilon > 0$. By Lemma 1.4.1(1), applied to v on a relatively compact open neighborhood of K, there is a smooth function v_ε on a neighborhood of K with $|v - v_\varepsilon| \le \varepsilon$ on K. Write

$$v_k S_k - vS = (v_k - v)S_k + (v - v_\varepsilon)S_k + v_\varepsilon(S_k - S) + (v_\varepsilon - v)S.$$

Evaluated on φ, a term fS_k with a bounded Borel function f is at most $C_\varphi \sup_K |f|$ times the total variation on K of the coefficient measures of S_k, where C_φ bounds the coefficients of φ; the same holds for S. Hence the first term tends to zero, since $v_k \to v$ uniformly on K; the second and fourth terms are bounded by a constant times ε, uniformly in k; and the third tends to zero as $k \to \infty$, since $v_\varepsilon \varphi$ is a test form. Letting first $k \to \infty$ and then $\varepsilon \downarrow 0$ proves the claim. □

If $u \in C^2(\Omega)$ is plurisubharmonic, its Monge–Ampère measure is given by (9.2). Indeed, the regularizations u_ε are smooth and plurisubharmonic on Ω_ε (Proposition 8.2.2) and converge to u locally uniformly (Lemma 1.4.1(1)), and their second derivatives are the regularizations of the continuous second derivatives of u (Proposition 1.4.3), so they also converge locally uniformly. Hence $\det((u_\varepsilon)_{j\bar{k}}) \to \det(u_{j\bar{k}})$ locally uniformly. Applying (9.2) to u_ε and Proposition 9.1.4 on each Ω_{ε_0}, along a sequence $\varepsilon_k \downarrow 0$ with $\varepsilon_k < \varepsilon_0$, we obtain the formula for u on every Ω_{ε_0}, hence on Ω.

9.2 Decreasing continuity and symmetry

A bounded plurisubharmonic function need not be continuous (see the example after Proposition 9.2.7), so in general it is not a uniform limit of continuous functions; it is, however, the limit of the decreasing sequence of its smooth regularizations. The main result of this section, Theorem 9.2.5, states that the products are continuous along decreasing sequences and symmetric in their factors. Upper semicontinuity bounds every weak limit of weighted products from above (Lemma 9.2.2). To prove equality, we first modify all functions so that they coincide with one fixed smooth function near the boundary of a ball, and then compare total masses by an integration by parts (Lemma 9.2.1). As applications, we prove that the products are local on sets of the form $\{u > v\}$ (Proposition 9.2.7) and derive the comparison principle (Theorem 9.2.8).

We use the real operator d^{c} of (7.6), for which $\mathrm{dd}^{\mathrm{c}} = \mathrm{dd}^{\mathrm{c}}$. If a and b are smooth real functions, the $(1,1)$-part of $\mathrm{d}a \wedge \mathrm{d}^{\mathrm{c}} b$ is $\frac{\mathrm{i}}{4\pi}(\partial a \wedge \bar\partial b + \partial b \wedge \bar\partial a)$, which is symmetric in a and b. Only this part survives multiplication by a current R of bidegree $(n-1, n-1)$, so

$$\mathrm{d}a \wedge \mathrm{d}^{\mathrm{c}} b \wedge R = \mathrm{d}b \wedge \mathrm{d}^{\mathrm{c}} a \wedge R.$$

Moreover, the $(2,0)$- and $(0,2)$-parts of $\mathrm{d}a \wedge \mathrm{d}^{\mathrm{c}} a$ are multiples of $\partial a \wedge \partial a = 0$ and $\bar\partial a \wedge \bar\partial a = 0$, so

$$\mathrm{d}a \wedge \mathrm{d}^{\mathrm{c}} a = \frac{\mathrm{i}}{2\pi} \partial a \wedge \bar\partial a; \tag{9.3}$$

this is a positive $(1,1)$-form, because $\bar\partial a = \overline{\partial a}$ for real a.

If B is a ball, a *collar* of ∂B is an open neighborhood of the sphere ∂B.

Several proofs below make bounded plurisubharmonic functions coincide on a common collar as follows. Let $B = B(z_0, s)$, let U be an open neighborhood of $\overline{B}$, put $\psi = |z - z_0|^2 - s^2$, and let $N \geq 1$ and $\delta > 0$. If v is plurisubharmonic on U and $-N \leq v \leq -1$ there, then the function

$$\max\{v, A\psi\}, \qquad A = N/\delta, \tag{9.4}$$

is plurisubharmonic on U (Proposition 8.2.5). It equals v on $\{\psi < -\delta\}$, where $A\psi < -N$, and it equals $A\psi$ on the collar $\{z \in U : \psi(z) > -1/A\}$ of ∂B, where $A\psi > -1$; on B, where $\psi < 0$, it takes values in $[-N, 0)$. Since $t \mapsto \max\{t, A\psi(z)\}$ is increasing and continuous, the modification preserves inequalities between functions, monotone sequences, and convergence at each point.

Lemma 9.2.1 *Let B be a ball and R a closed positive current of bidimension $(1,1)$ on a neighborhood of $\overline{B}$. Let a be a smooth function and b a bounded plurisubharmonic function on a neighborhood of $\overline{B}$, and suppose that $a = b$ on a collar of ∂B. Then*

$$\int_B a \, \mathrm{dd}^{\mathrm{c}} b \wedge R = \int_B b \, \mathrm{dd}^{\mathrm{c}} a \wedge R.$$

In particular one may take $R = S \wedge \gamma$, where S is a closed positive current, γ is a smooth closed form of pure bidegree, and $S \wedge \gamma$ is positive of bidimension $(1,1)$.

Proof Both integrals are finite, since a and b are bounded on B and $\mathrm{dd^c}a \wedge R$ and $\mathrm{dd^c}b \wedge R$ are measures of finite total variation near $\overline{B}$. The function $h = b - a$ is bounded and Borel, and it vanishes on a collar V of ∂B; hence its restriction to B vanishes outside the compact set $K_0 = \overline{B} \setminus V \subset B$. On B we have $bR = aR + hR$, and since R is closed, $\mathrm{dd^c}(aR) = \mathrm{dd^c}a \wedge R$ by (9.1); thus

$$\mathrm{dd^c}b \wedge R = \mathrm{dd^c}a \wedge R + \mathrm{dd^c}(hR) \quad \text{on } B,$$

where $\mathrm{dd^c}(hR)$ is a signed measure with support in K_0. Choose $\chi \in C_c^\infty(B)$ equal to 1 on a neighborhood of K_0. By the definition of $\mathrm{dd^c}$ on currents,

$$\int_B a\,\mathrm{dd^c}(hR) = \int \chi a\,\mathrm{dd^c}(hR) = \int h\,\mathrm{dd^c}(\chi a) \wedge R = \int_B h\,\mathrm{dd^c}a \wedge R.$$

Consequently

$$\int_B a\,\mathrm{dd^c}b \wedge R = \int_B a\,\mathrm{dd^c}a \wedge R + \int_B h\,\mathrm{dd^c}a \wedge R = \int_B b\,\mathrm{dd^c}a \wedge R.$$

The last assertion holds because $S \wedge \gamma$ is then closed, by the Leibniz rule, and positive of bidimension $(1,1)$; moreover $\mathrm{dd^c}b \wedge (S \wedge \gamma) = (\mathrm{dd^c}b \wedge S) \wedge \gamma$ by Proposition 9.1.2(4). □

Lemma 9.2.2 *Let $\mu_j \to \mu$ weakly, where μ_j and μ are positive Radon measures on Ω, and let f_j be real-valued upper semicontinuous functions on Ω which decrease to a locally bounded function f. If $f_{j_k}\mu_{j_k} \to \nu$ weakly along a subsequence, then $\nu \le f\mu$.*

Proof Let $\chi \ge 0$ be a continuous function with compact support, let N be a compact neighborhood of $\operatorname{Supp}\chi$, and put $W = \operatorname{Int} N$. The function f_1 is upper semicontinuous, hence bounded above on N, and f is bounded below on N; since $f \le f_j \le f_1$, there is $c \ge 0$ with $|f_j| \le c$ on N for all j. The positive Radon measures $(f_{j_k} + c)\mu_{j_k}$ on W converge weakly to $\nu + c\mu$. By Proposition 1.5.1, $\nu + c\mu$ is a positive Radon measure on W, and the convergence is vague. Since also $\mu_j \to \mu$ vaguely, we obtain $\int \psi f_{j_k}\,\mathrm{d}\mu_{j_k} \to \int \psi\,\mathrm{d}\nu$ for every continuous function ψ with compact support in W.

Fix l. By Lemma 3.1.2, there are bounded continuous functions g_m on N decreasing to f_l. For $j_k \ge l$ we have $f_{j_k} \le f_l \le g_m$ on N, so

$$\int \chi f_{j_k}\,\mathrm{d}\mu_{j_k} \le \int \chi g_m\,\mathrm{d}\mu_{j_k}.$$

Letting $k \to \infty$ gives $\int \chi\,\mathrm{d}\nu \le \int \chi g_m\,\mathrm{d}\mu$. Since $-c \le f_l \le g_m \le g_1$ on N and $\mu(N) < \infty$, bounded convergence as $m \to \infty$ gives $\int \chi\,\mathrm{d}\nu \le \int \chi f_l\,\mathrm{d}\mu$, and bounded convergence as $l \to \infty$ gives $\int \chi\,\mathrm{d}\nu \le \int \chi f\,\mathrm{d}\mu$. Since $\chi \ge 0$ was arbitrary, $\nu \le f\mu$.□

To smooth the functions (9.4) without changing them on a collar, we replace the maximum by a smooth convex function of two variables.

Lemma 9.2.3 *There are smooth convex functions $M_\varepsilon\colon \mathbb{R}^2 \to \mathbb{R}$, $\varepsilon > 0$, increasing in each variable, such that $M_\varepsilon \downarrow \max$ as $\varepsilon \downarrow 0$, $\max \le M_\varepsilon \le \max + \varepsilon$, and $M_\varepsilon(s,t) = \max\{s,t\}$ when $|s-t| > 2\varepsilon$. If a and b are smooth plurisubharmonic functions, so is $M_\varepsilon(a,b)$.*

Proof Let ρ_ε be the mollifier of Section 1.4 on $\mathbb{R}^2$; it is a nonnegative even radial probability density supported in the disc of radius ε. Put $M_\varepsilon = \max * \rho_\varepsilon$, that is, $M_\varepsilon(x) = \int \max(x-y)\rho_\varepsilon(y)\,\mathrm{d}y$, where $\max(s,t) = \max\{s,t\}$; it is smooth, since ρ_ε is (Section 1.4). Since $\rho_\varepsilon \ge 0$ has integral 1, and $x \mapsto \max(x-y)$ is convex and increasing in each variable for each y, the function M_ε is convex and increasing in each variable. Since ρ_ε is even, $\int y_i \rho_\varepsilon(y)\,\mathrm{d}y = 0$ for $i = 1, 2$; integrating $\max(s-y_1, t-y_2) \ge s - y_1$ and $\max(s-y_1, t-y_2) \ge t - y_2$ against ρ_ε therefore gives $M_\varepsilon(s,t) \ge s$ and $M_\varepsilon(s,t) \ge t$, that is, $M_\varepsilon \ge \max$. Since $|\max(x-y) - \max(x)| \le |y| \le \varepsilon$ on the support of ρ_ε, we have $M_\varepsilon \le \max + \varepsilon$. If $s - t > 2\varepsilon$ and $|y| \le \varepsilon$, then $(s-y_1) - (t-y_2) \ge s - t - 2\varepsilon > 0$, so $M_\varepsilon(s,t) = \int (s-y_1)\rho_\varepsilon(y)\,\mathrm{d}y = s$; similarly $M_\varepsilon(s,t) = t$ if $t - s > 2\varepsilon$.

It remains to show that M_ε increases with ε. For fixed $x \in \mathbb{R}^2$, the function $g(\varepsilon) = \int \max(x - \varepsilon y)\rho_1(y)\,\mathrm{d}y$, defined for all real ε, equals $M_\varepsilon(x)$ for $\varepsilon > 0$ and $\max(x)$ for $\varepsilon = 0$; it is convex, since $\varepsilon \mapsto \max(x - \varepsilon y)$ is convex, and even, since ρ_1 is even. For $0 \le \varepsilon_1 < \varepsilon_2$, writing $\varepsilon_1 = \lambda(-\varepsilon_2) + (1-\lambda)\varepsilon_2$ with $\lambda \in (0,1]$, we get $g(\varepsilon_1) \le \lambda g(-\varepsilon_2) + (1-\lambda) g(\varepsilon_2) = g(\varepsilon_2)$. This proves the monotonicity, and together with $\max \le M_\varepsilon \le \max + \varepsilon$ this gives $M_\varepsilon \downarrow \max$ as $\varepsilon \downarrow 0$.

Finally, let a and b be smooth and plurisubharmonic, and write $(a_1, a_2) = (a, b)$. Denoting by $\partial_k M_\varepsilon$ and $\partial_k \partial_l M_\varepsilon$ the partial derivatives evaluated at (a,b), the chain rule gives

$$\mathrm{i}\partial\bar\partial M_\varepsilon(a,b) = \sum_{k=1}^{2} \partial_k M_\varepsilon\, \mathrm{i}\partial\bar\partial a_k + \mathrm{i} \sum_{k,l=1}^{2} \partial_k\partial_l M_\varepsilon\, \partial a_k \wedge \bar\partial a_l.$$

The first sum is positive because $\partial_k M_\varepsilon \ge 0$, M_ε being increasing in each variable, and the forms $\mathrm{i}\partial\bar\partial a_k$ are positive, since the smooth plurisubharmonic functions a_k have positive semidefinite complex Hessians (Proposition 8.2.2). For the second, note that the real Hessian $H = (\partial_k \partial_l M_\varepsilon)$ is positive semidefinite: for $v \in \mathbb{R}^2$, the function $t \mapsto M_\varepsilon(x + tv)$ is convex, so its second derivative $v^t H v$ is nonnegative. Writing $H = \sum_m \lambda_m v_m v_m^t$ with $\lambda_m \ge 0$ and real orthonormal vectors v_m, and putting $\theta_m = \sum_k (v_m)_k \partial a_k$, we get $\mathrm{i}\sum_{k,l} \partial_k\partial_l M_\varepsilon\, \partial a_k \wedge \bar\partial a_l = \sum_m \lambda_m \mathrm{i}\theta_m \wedge \bar\theta_m$, because $\bar\partial a_l = \overline{\partial a_l}$ for real a_l; this is a positive $(1,1)$-form. Hence the complex Hessian of $M_\varepsilon(a,b)$ is positive semidefinite, and $M_\varepsilon(a,b)$ is plurisubharmonic by Proposition 8.2.2. □

We also note that weak convergence of currents is a local property, so that it can be proved near each point.

Lemma 9.2.4 *Let S_j, $j \geq 1$, and S be currents of the same degree on an open set $\Omega \subset \mathbb{C}^n$. If every point of Ω has an open neighborhood $U \subset \Omega$ such that $S_j \to S$ weakly on U, then $S_j \to S$ weakly on Ω.*

Proof Let φ be a test form on Ω of complementary degree. Its support is compact, so it is covered by finitely many open sets $U_1, \dots, U_m$ as in the hypothesis. By Lemma 1.4.2, there are smooth functions $\eta_1, \dots, \eta_m$ with compact supports in $U_1, \dots, U_m$, respectively, such that $\sum_i \eta_i = 1$ on a neighborhood of $\operatorname{Supp} \varphi$. Then $\varphi = \sum_i \eta_i \varphi$, where $\eta_i \varphi$ is a test form supported in U_i, and therefore $S_j(\varphi) = \sum_i S_j(\eta_i \varphi) \to \sum_i S(\eta_i \varphi) = S(\varphi)$. □

Theorem 9.2.5 (Bedford–Taylor) *Let T be a closed positive current on Ω of bidimension (r, r), and let $0 \leq q \leq r$ be an integer. For $0 \leq l \leq q$, let $(u_{l,j})_{j \geq 1}$ be a decreasing sequence of locally bounded plurisubharmonic functions on Ω whose limit u_l is locally bounded (and therefore plurisubharmonic, by Theorem 8.3.1). Then*

$$\mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{q,j} \wedge T \longrightarrow \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T,$$
$$u_{0,j} \mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{q,j} \wedge T \longrightarrow u_0 \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T$$

weakly as $j \to \infty$. Moreover, the product $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T$ of locally bounded plurisubharmonic functions is symmetric in $u_1, \dots, u_q$.

Proof We argue by induction on q. Step 1 derives the unweighted assertion and the symmetry from the weighted assertion for fewer factors. The other steps prove the weighted assertion on a ball, after a modification near its boundary: every weak limit of the weighted products is at most the expected limit (Step 4), and repeated integration by parts gives the reverse inequality for the total masses (Step 5).

Step 1: the induction. We prove the second (weighted) assertion by induction on q, simultaneously for all open sets $\Omega \subset \mathbb{C}^n$, all $r \geq q$, all closed positive currents T of bidimension (r, r) on Ω and all decreasing sequences as in the statement. For $q = 0$ it states that $u_{0,j} T \to u_0 T$; this follows from dominated convergence with respect to the total variations of the coefficient measures of T, since $u_0 \leq u_{0,j} \leq u_{0,1}$. Let $q \geq 1$ and assume the weighted assertion for fewer than q factors. Applying it to the $q - 1$ factors $u_{2,j}, \dots, u_{q,j}$ with the weight $u_{1,j}$, and then applying the weakly continuous operator $\mathrm{dd^c}$, we obtain the unweighted assertion for q factors.

Next we prove symmetry for q factors. Let $u_1, \dots, u_q$ be locally bounded and plurisubharmonic, let $\varepsilon > 0$, and choose $\varepsilon_k \downarrow 0$ with $\varepsilon_k < \varepsilon$. The regularizations $u_l * \rho_{\varepsilon_k}$ are smooth plurisubharmonic functions on Ω_ε decreasing to u_l (Proposition 8.2.2). By the unweighted assertion, the product $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T$ on Ω_ε is therefore the weak limit of exterior products of smooth forms with T (Proposition 9.1.2(1)), and these are symmetric in the factors because 2-forms commute. Since the sets Ω_ε exhaust Ω, products with q factors are symmetric. It remains to prove the weighted assertion for q factors; in doing so we may use the unweighted assertion and the symmetry for q factors.

Step 2: reduction to a common collar. By Lemma 9.2.4, it suffices to prove the convergence near every point. Fix a ball $B = B(a, R)$, a larger ball B_1 with

$\overline{B} \subset B_1 \Subset \Omega$, and a compact set $K \subset B$; we prove the convergence on a neighborhood of K. Since $u_l \le u_{l,j} \le u_{l,1}$, all the functions $u_{l,j}$ and u_l are bounded on B_1 uniformly in j. Subtracting a constant from $u_{l,j}$ and u_l changes neither the Hessians nor, since the unweighted assertion is already known, the validity of the weighted assertion. We may therefore assume that $-N \le u_{l,j}, u_l \le -1$ on B_1 for all l and j, where $N \ge 1$. Put $\psi = |z-a|^2 - R^2$, choose $\delta > 0$ with $K \subset \{\psi < -\delta\}$, and replace every function v among the $u_{l,j}$ and u_l by the function $\max\{v, A\psi\}$ of (9.4), with $U = B_1$ and $A = N/\delta$. The new functions are plurisubharmonic on B_1, and the new sequences still decrease to the new limits. The functions are unchanged on the neighborhood $\{\psi < -\delta\}$ of K, and by locality so are their products. From now on, $u_{l,j}$ and u_l denote the new functions; they take values in $[-N, 0)$ on B and coincide with $A\psi$ on the collar $V = \{z \in B_1 : \psi(z) > -1/A\}$ of ∂B.

Step 3: smooth approximation with the same collar. Fix $\eta > 0$ such that $\{\psi \le \eta\} \subset B_1$, and let $V' = \{z : -1/(2A) < \psi(z) < \eta\}$; it is a collar of ∂B contained in V. Fix l and j, and let $\tilde{u}$ be the function $u_{l,j}$ before the modification of Step 2, so that $\tilde{u} \le -1$ on B_1 and $u_{l,j} = \max\{\tilde{u}, A\psi\}$. For $0 < \varepsilon < 1/4$ smaller than the distance from $\{\psi \le \eta\}$ to ∂B_1, put

$$u^{\varepsilon}_{l,j} = M_\varepsilon(\tilde{u} * \rho_\varepsilon, A\psi),$$

with M_ε as in Lemma 9.2.3. The functions $\tilde{u} * \rho_\varepsilon$ are smooth and plurisubharmonic on $\{\psi < \eta\}$ and decrease to $\tilde{u}$ as $\varepsilon \downarrow 0$ (Proposition 8.2.2), so the $u^{\varepsilon}_{l,j}$ are smooth and plurisubharmonic on $\{\psi < \eta\}$ by Lemma 9.2.3. Since $\tilde{u} * \rho_\varepsilon \le -1$, on V' we have $A\psi - \tilde{u} * \rho_\varepsilon > 1/2 > 2\varepsilon$, and therefore $u^{\varepsilon}_{l,j} = A\psi$. Since M_ε decreases to max, is increasing in each variable and satisfies $\max \le M_\varepsilon \le \max + \varepsilon$, the functions $u^{\varepsilon}_{l,j}$ decrease to $\max\{\tilde{u}, A\psi\} = u_{l,j}$. In particular $u^{\varepsilon}_{l,j} \ge u_{l,j} \ge u_l$, and all the functions $u_{l,j}$, u_l and $u^{\varepsilon}_{l,j}$ coincide with $A\psi$ on V'.

Step 4: an upper bound for weak limits. Put

$$S_j = \mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{q,j} \wedge T, \qquad S = \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T.$$

By Step 1, $S_j \to S$ weakly. On B the weights $u_{0,j}$ take values in $[-N, 0)$, so the currents $-u_{0,j} S_j$ are positive there, and by Theorem 9.1.3, since $|u_{l,j}| \le N$ on B, their masses on compact subsets of B are bounded uniformly in j. By the compactness part of Theorem 7.2.5, every subsequence of $(u_{0,j} S_j)$ has a further subsequence converging weakly on B. Let Θ be the limit of such a subsequence. Let φ be a strongly positive test form on B of complementary bidegree. The measures $\mu_j = S_j \wedge \varphi$ are positive and converge weakly to $S \wedge \varphi$, and along the chosen subsequence $u_{0,j}\mu_j \to \Theta \wedge \varphi$. Hence Lemma 9.2.2, applied on B with the real-valued upper semicontinuous functions $f_j = u_{0,j}$, gives $\Theta \wedge \varphi \le u_0 S \wedge \varphi$. Evaluating this inequality of measures on a nonnegative test function equal to 1 on $\operatorname{Supp} \varphi$, we get $(u_0 S - \Theta)(\varphi) \ge 0$; thus $u_0 S - \Theta$ is a positive current on B. By locality (Proposition 9.1.2), all the currents $u_{0,j} S_j$, and also $u_0 S$, coincide on V' with the fixed current $A\psi\, (A\mathrm{dd^c}\psi)^q \wedge T$; hence so does Θ, and $u_0 S - \Theta$ has compact support in B.

Step 5: comparison of masses. Set $\gamma = \beta^{r-q}$ and

$$I = \int_B u_0 S \wedge \gamma, \qquad I_j = \int_B u_{0,j} S_j \wedge \gamma.$$

We claim that $I \le I_j$ for every j. Fix j and parameters $\varepsilon_0, \dots, \varepsilon_q$, and abbreviate $w_l = u_{l,j}^{\varepsilon_l}$. For $0 \le k \le q$ put

$$J_k = \int_B u_k \, \mathrm{dd^c} w_0 \wedge \cdots \wedge \mathrm{dd^c} w_{k-1} \wedge \mathrm{dd^c} u_{k+1} \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T \wedge \gamma,$$

so that $J_0 = I$. The measure in J_k is positive and $u_k \le w_k$, so J_k does not decrease if u_k is replaced by w_k. If $k < q$, symmetry of products with q factors and Proposition 9.1.2(4) let us write that measure as $\mathrm{dd^c} u_{k+1} \wedge R_k$, where

$$R_k = \mathrm{dd^c} w_0 \wedge \cdots \wedge \mathrm{dd^c} w_{k-1} \wedge \mathrm{dd^c} u_{k+2} \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T \wedge \gamma$$

is a closed positive current of bidimension $(1,1)$ on $\{\psi < \eta\}$; then Lemma 9.2.1, with $a = w_k$ and $b = u_{k+1}$, which both equal $A\psi$ on the collar V', gives

$$J_k \le \int_B w_k \, \mathrm{dd^c} u_{k+1} \wedge R_k = \int_B u_{k+1} \, \mathrm{dd^c} w_k \wedge R_k = J_{k+1}.$$

Hence

$$I = J_0 \le J_1 \le \cdots \le J_q \le \int_B w_q \, \mathrm{dd^c} w_0 \wedge \cdots \wedge \mathrm{dd^c} w_{q-1} \wedge T \wedge \gamma.$$

Now let $\varepsilon_q \downarrow 0$, then $\varepsilon_{q-1} \downarrow 0$, and so on down to $\varepsilon_1 \downarrow 0$, keeping ε_0 fixed. We claim that the right-hand side of the last inequality then tends to

$$\int_B u_{q,j} \, \mathrm{dd^c} w_0 \wedge \mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{q-1,j} \wedge T \wedge \gamma.$$

When $\varepsilon_q \downarrow 0$, the measure is fixed and the bounded weights w_q decrease to $u_{q,j}$, so dominated convergence applies. When $\varepsilon_k \downarrow 0$ for some $1 \le k \le q-1$, the integrand is, after moving the smooth factors next to T by symmetry,

$$u_{q,j} \, \mathrm{dd^c} u_{k,j}^{\varepsilon_k} \wedge \mathrm{dd^c} u_{k+1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{q-1,j} \wedge T_k \wedge \gamma,$$

where $T_k = \mathrm{dd^c} w_0 \wedge \cdots \wedge \mathrm{dd^c} w_{k-1} \wedge T$ is a fixed closed positive current on $\{\psi < \eta\}$ of bidimension $(r-k, r-k)$. There are $q-k \le q-1$ Hessian factors and the fixed weight $u_{q,j}$. Along any sequence $\varepsilon_k \downarrow 0$, the functions $u_{k,j}^{\varepsilon_k}$ decrease to $u_{k,j}$ (Step 3), so the induction hypothesis, applied on $\{\psi < \eta\}$ with the base current T_k, constant sequences for the other factors and the constant weight $u_{q,j}$, shows that these currents converge weakly; so do their exterior products with γ.

In each case the convergence holds for the integrals over B, and not only against compactly supported test forms, because all the currents involved coincide on the

collar V' with the same fixed current: if $\chi \in C_c^\infty(B)$ equals 1 on $B \setminus V'$, the integrals against χ converge, and the integrals against $1-\chi$ do not depend on the parameters.

By Lemma 9.2.1, with $a = w_0$ and $b = u_{q,j}$, and by symmetry of products with q factors, the limit integral equals

$$\int_B w_0 \, \mathrm{dd^c} u_{q,j} \wedge \mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{q-1,j} \wedge T \wedge \gamma = \int_B u_{0,j}^{\varepsilon_0} S_j \wedge \gamma.$$

Hence $I \le \int_B u_{0,j}^{\varepsilon_0} S_j \wedge \gamma$ for every ε_0, and dominated convergence with respect to the finite measure $S_j \wedge \gamma$ on B gives $I \le I_j$ as $\varepsilon_0 \downarrow 0$.

Step 6: conclusion. Along the subsequence defining Θ, splitting the integrals with a cutoff as in Step 5 shows that $I_j \to \int_B \Theta \wedge \gamma$. Therefore

$$\int_B (u_0 S - \Theta) \wedge \gamma = I - \lim_j I_j \le 0.$$

Since $u_0 S - \Theta$ is positive with compact support in B, its trace measure $(u_0 S - \Theta) \wedge \gamma$ is a positive measure of total mass at most zero, hence zero; by the coefficient bounds of Theorem 7.2.5, $u_0 S - \Theta = 0$ on B. Thus every subsequence of $(u_{0,j} S_j)$ has a further subsequence converging weakly to $u_0 S$ on B. Hence the whole sequence converges to $u_0 S$ on B: otherwise there would be a test form φ on B and a subsequence along which $|(u_{0,j} S_j - u_0 S)(\varphi)|$ stays bounded away from zero, and no further subsequence of it could converge to $u_0 S$. The replacement by maxima in Step 2 changes neither the functions nor, by locality, the weighted products on the neighborhood $\{\psi < -\delta\}$ of K, and the subtraction of constants does not affect the convergence; hence the original weighted products converge on this neighborhood. As K and B were arbitrary, this proves the weighted assertion for q factors on Ω and completes the induction. □

Corollary 9.2.6 *Let P be a real polynomial in m variables and, for $1 \le i \le m$, let $(v_{i,j})_{j \ge 1}$ be a decreasing sequence of locally bounded plurisubharmonic functions on Ω with locally bounded limit v_i. Then in Theorem 9.2.5 the weight $u_{0,j}$ may be replaced by $P(v_{1,j}, \ldots, v_{m,j})$, and the limit is then*

$$P(v_1, \ldots, v_m) \, \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T.$$

Moreover, the identity of Lemma 9.2.1 also holds when a and b are both bounded plurisubharmonic functions on a neighborhood of $\overline{B}$ which coincide on a collar of ∂B.

Proof Since weak convergence is a local property (Lemma 9.2.4), it suffices to prove the convergence on every relatively compact open set $W \Subset \Omega$. On W all the functions $v_{i,j}$ and v_i are bounded, because $v_i \le v_{i,j} \le v_{i,1}$. Adding a constant c_i to each $v_{i,j}$ and to v_i replaces P by the polynomial $P(x_1 - c_1, \ldots, x_m - c_m)$; so we may assume that all these functions are nonnegative on W. Since the weighted products depend linearly on the weight, it suffices to treat a monomial. A constant weight is covered by the unweighted assertion of the theorem. A monomial of degree $N \ge 1$

is a product $a_1 \cdots a_N$, where each a_l is one of the v_i, repetitions being allowed; we write $a_{l,j}$ for the corresponding function $v_{i,j}$. For real numbers $x_1, \dots, x_N$ we have the polarization identity

$$N!\, x_1 \cdots x_N = \sum_{I \subset \{1,\dots,N\}} (-1)^{N-|I|} \Big(\sum_{l \in I} x_l \Big)^N . \tag{9.5}$$

To prove it, expand $(\sum_{l\in I} x_l)^N = \sum_\alpha \frac{N!}{\alpha!} x^\alpha$, the sum running over the multi-indices $\alpha \in \mathbb{N}^N$ with $|\alpha| = N$ and $\alpha_l = 0$ for $l \notin I$. Fix such an α, and let $J = \{l : \alpha_l > 0\}$ have k elements. The monomial x^α occurs exactly in the terms with $I \supset J$, so its total coefficient on the right-hand side is

$$\frac{N!}{\alpha!} \sum_{I \supset J} (-1)^{N-|I|} = \frac{N!}{\alpha!} \sum_{i=0}^{N-k} \binom{N-k}{i} (-1)^{N-k-i} = \frac{N!}{\alpha!} (1-1)^{N-k}.$$

This vanishes unless $k = N$, in which case $\alpha = (1, \dots, 1)$ and the coefficient is $N!$.

Each sum $s = \sum_{l\in I} a_l$ is a bounded nonnegative plurisubharmonic function on W, and so is s^N. For $N = 1$ there is nothing to prove; for $N \geq 2$ and smooth s this follows from

$$\mathrm{dd^c}(s^N) = N s^{N-1} \mathrm{dd^c} s + N(N-1) s^{N-2}\, \mathrm{d}s \wedge \mathrm{d^c} s,$$

in which both terms are positive $(1,1)$-forms, the first because $s \geq 0$ and s has a positive semidefinite complex Hessian, and the second by (9.3); hence s^N is plurisubharmonic, by both directions of Proposition 8.2.2. In general, the regularizations $s * \rho_\varepsilon$ are smooth, plurisubharmonic and nonnegative, and they decrease to s (Proposition 8.2.2); since $t \mapsto t^N$ is increasing on $[0, \infty)$, the plurisubharmonic functions $(s * \rho_\varepsilon)^N$ decrease to s^N, which is therefore plurisubharmonic by Theorem 8.3.1. Likewise the sums $s_j = \sum_{l\in I} a_{l,j}$ are nonnegative and decrease to s, so the s_j^N are bounded plurisubharmonic functions on W decreasing to s^N. The weighted assertion of the theorem therefore applies with the weights s_j^N, and by (9.5), with $x_l = a_{l,j}$, it applies to the monomial weights $a_{1,j} \cdots a_{N,j}$.

For the second assertion, let a and b be bounded plurisubharmonic functions on a neighborhood U of $\overline{B}$ which are equal on a collar V of ∂B. Let $d > 0$ be smaller than the distances from ∂B to $\mathbb{C}^n \setminus V$ and from $\overline{B}$ to $\mathbb{C}^n \setminus U$, and put $V' = \{z : \mathrm{dist}(z, \partial B) < d/2\}$. For $\varepsilon = \varepsilon_k \downarrow 0$ with $\varepsilon_k < d/2$, the regularizations $a_\varepsilon = a * \rho_\varepsilon$ and $b_\varepsilon = b * \rho_\varepsilon$ are smooth and plurisubharmonic on a neighborhood of $\overline{B}$ and decrease to a and b (Proposition 8.2.2); moreover $a_\varepsilon = b_\varepsilon$ on the collar V', because $a = b$ on the ball $B(z, \varepsilon) \subset V$ for $z \in V'$. By Lemma 9.2.1, applied to the smooth function a_ε and the bounded plurisubharmonic function b_ε,

$$\int_B a_\varepsilon\, \mathrm{dd^c} b_\varepsilon \wedge R = \int_B b_\varepsilon\, \mathrm{dd^c} a_\varepsilon \wedge R.$$

Choose $\chi \in C_c^\infty(B)$ equal to 1 on $B \setminus V'$. On V' the measures $a_\varepsilon \mathrm{dd^c} b_\varepsilon \wedge R$ and $b_\varepsilon \mathrm{dd^c} a_\varepsilon \wedge R$ coincide, by locality, so the identity is equivalent to $\int \chi a_\varepsilon \, \mathrm{dd^c} b_\varepsilon \wedge R = \int \chi b_\varepsilon \, \mathrm{dd^c} a_\varepsilon \wedge R$. By the weighted assertion of the theorem with one factor and the fixed base current R, this passes to the limit: $\int \chi a \, \mathrm{dd^c} b \wedge R = \int \chi b \, \mathrm{dd^c} a \wedge R$. Since $a = b$ on V, the measures $a \, \mathrm{dd^c} b \wedge R$ and $b \, \mathrm{dd^c} a \wedge R$ also coincide on V', and adding the integrals against $1 - \chi$ over B gives the identity for a and b. □

Proposition 9.2.7 *Let T be a closed positive current on Ω of bidimension (r, r), let u and v be locally bounded plurisubharmonic functions on Ω, and let $0 \le p \le r$ be an integer. Then*

$$\mathbf{1}_{\{u>v\}} (\mathrm{dd^c} \max\{u, v\})^p \wedge T = \mathbf{1}_{\{u>v\}} (\mathrm{dd^c} u)^p \wedge T.$$

Here the indicator function multiplies the coefficient measures of these currents of order zero, as in Section 7.2; the set $\{u > v\}$ is Borel, but in general not open (see the example following the proof).

Proof The assertion is local, so it suffices to prove it on each Ω_{ε_0}, $\varepsilon_0 > 0$. Let $u_j = u * \rho_{\varepsilon_j}$ with $\varepsilon_j \downarrow 0$, $\varepsilon_j < \varepsilon_0$; these are smooth plurisubharmonic functions on Ω_{ε_0} decreasing to u (Proposition 8.2.2). Since u_j is continuous and v is upper semicontinuous, the set $\{u_j > v\}$ is open, and on it $w_j = \max\{u_j, v\}$ equals u_j; by locality, $(\mathrm{dd^c} w_j)^p \wedge T = (\mathrm{dd^c} u_j)^p \wedge T$ there. The nonnegative Borel function $f_j = w_j - v$ vanishes outside $\{u_j > v\}$, so

$$f_j (\mathrm{dd^c} w_j)^p \wedge T = f_j (\mathrm{dd^c} u_j)^p \wedge T.$$

The functions w_j decrease to $w = \max\{u, v\}$. Both sides of the last identity are differences of weighted products, for instance $f_j (\mathrm{dd^c} w_j)^p \wedge T = w_j (\mathrm{dd^c} w_j)^p \wedge T - v (\mathrm{dd^c} w_j)^p \wedge T$. Applying Theorem 9.2.5 to them, with the weights w_j and the constant weight v, and with the factors w_j on the left and u_j on the right, we obtain $f (\mathrm{dd^c} w)^p \wedge T = f (\mathrm{dd^c} u)^p \wedge T$, where $f = w - v \ge 0$. Multiply the coefficient measures of both sides by $1/(f + \varepsilon)$ and let $\varepsilon \downarrow 0$. Since $f/(f + \varepsilon)$ increases to the indicator of $\{f > 0\} = \{u > v\}$, dominated convergence with respect to the locally finite total variations of the coefficient measures gives the assertion. □

To see that $\{u > v\}$ need not be open, let $D = \{z \in \mathbb{C}^n : |z_1| < 1/2\}$, put $a_j = \mathrm{e}^{-j}$ for $j \ge 1$, and let $w = \sum_{j \ge 1} 2^{-j} \log |z_1 - a_j|$ on D. Each term is a positive multiple of $\log |z_1 - a_j|^2$, which is plurisubharmonic by Proposition 8.3.3, and it is negative on D, because $|z_1 - a_j| < 1/2 + \mathrm{e}^{-1} < 1$. Hence the partial sums form a decreasing sequence of plurisubharmonic functions (Proposition 8.2.5), and since $w(0) = -\sum_j j 2^{-j} = -2$ is finite, w is plurisubharmonic on the domain D by Theorem 8.3.1. The function $u = \max\{w, -4\}$ is plurisubharmonic with values in $[-4, 0]$, and for the constant $v = -3$ the set $\{u > v\} = \{w > -3\}$ contains the origin but none of the points $(a_j, 0, \ldots, 0)$, at which $w = -\infty$; these points converge to the origin. Since $u(0) = -2$ while $u = -4$ at these points, u is also an example of a bounded plurisubharmonic function which is not continuous.

Theorem 9.2.8 *Let $D \subset \mathbb{C}^n$ be a bounded open set, and let u and v be locally bounded plurisubharmonic functions on D such that $\varliminf_{D \ni z \to x}(u(z) - v(z)) \geq 0$ for every $x \in \partial D$. Then*

$$\int_{\{u<v\}} (\mathrm{dd^c} v)^n \leq \int_{\{u<v\}} (\mathrm{dd^c} u)^n,$$

where both sides may be infinite. If $(\mathrm{dd^c} u)^n = 0$, then $u \geq v$ on D.

Proof Let $\varepsilon > 0$ and $w = \max\{u, v - \varepsilon\}$. We show that $(\mathrm{dd^c} w)^n$ and $(\mathrm{dd^c} u)^n$ have the same mass near a compact set outside which $w = u$, and then identify $(\mathrm{dd^c} w)^n$ by Proposition 9.2.7. Every $x \in \partial D$ has an open neighborhood U_x with $u > v - \varepsilon$ on $U_x \cap D$, and finitely many of them cover the compact set ∂D; let U be their union. Then $F = D \setminus U = \overline{D} \setminus U$ is a compact subset of D, it contains $\{u \leq v - \varepsilon\}$, and $w = u$ on $D \setminus F$. For $0 \leq l \leq n-1$ put $R_l = (\mathrm{dd^c} w)^l \wedge (\mathrm{dd^c} u)^{n-1-l}$. By additivity and symmetry of the products (Proposition 9.1.2 and Theorem 9.2.5),

$$\begin{aligned} \mathrm{dd^c}\big((w-u)R_l\big) &= \mathrm{dd^c} w \wedge R_l - \mathrm{dd^c} u \wedge R_l \\ &= (\mathrm{dd^c} w)^{l+1} \wedge (\mathrm{dd^c} u)^{n-1-l} - (\mathrm{dd^c} w)^l \wedge (\mathrm{dd^c} u)^{n-l}, \end{aligned}$$

and summing over l, the right-hand sides telescope:

$$(\mathrm{dd^c} w)^n - (\mathrm{dd^c} u)^n = \mathrm{dd^c}\Big((w-u)\sum_{l=0}^{n-1} R_l\Big).$$

The current in parentheses vanishes outside F, since $w - u$ does. Hence the signed measure $(\mathrm{dd^c} w)^n - (\mathrm{dd^c} u)^n$ is supported in F, and if $\chi \in C_c^\infty(D)$ equals 1 on a neighborhood of F, then its total mass is

$$\int \chi \, \mathrm{d}\big((\mathrm{dd^c} w)^n - (\mathrm{dd^c} u)^n\big) = \Big((w-u)\sum_l R_l\Big)(\mathrm{dd^c}\chi) = 0,$$

because $\mathrm{dd^c}\chi = 0$ on a neighborhood of F. Choose a relatively compact open set $D' \subset D$ containing F. Both measures are finite on D', and $\int_{D'}(\mathrm{dd^c} w)^n = \int_{D'}(\mathrm{dd^c} u)^n$. By Proposition 9.2.7, $(\mathrm{dd^c} w)^n$ coincides with $(\mathrm{dd^c} v)^n$ on $\{u < v - \varepsilon\}$ and with $(\mathrm{dd^c} u)^n$ on $\{u > v - \varepsilon\}$; here we use that adding the constant $-\varepsilon$ does not change $(\mathrm{dd^c} v)^n$. Since $\{u < v - \varepsilon\}$ and $D' \cap \{u > v - \varepsilon\}$ are disjoint Borel subsets of D', and $\{u \leq v - \varepsilon\} \subset F \subset D'$,

$$\begin{aligned} \int_{\{u<v-\varepsilon\}} (\mathrm{dd^c} v)^n &\leq \int_{D'} (\mathrm{dd^c} w)^n - \int_{D' \cap \{u>v-\varepsilon\}} (\mathrm{dd^c} u)^n \\ &= \int_{\{u \leq v-\varepsilon\}} (\mathrm{dd^c} u)^n \leq \int_{\{u<v\}} (\mathrm{dd^c} u)^n. \end{aligned}$$

As $\varepsilon \downarrow 0$, the sets $\{u < v - \varepsilon\}$ increase to $\{u < v\}$, and monotone convergence proves the comparison inequality.

The last assertion is the case of Corollary 9.2.9 below in which the roles of u and v are interchanged: if $(\mathrm{dd^c}u)^n = 0$, then $v \le u$ holds almost everywhere for $(\mathrm{dd^c}u)^n$. The proof of that corollary uses only the comparison inequality. □

Corollary 9.2.9 *Let $D \subset \mathbb{C}^n$ be a bounded open set, $n \ge 1$, and let u and v be locally bounded plurisubharmonic functions on D. Suppose that*

$$\varliminf_{D \ni z \to x} (v(z) - u(z)) \ge 0 \quad (x \in \partial D), \qquad u \le v \quad \text{almost everywhere for } (\mathrm{dd^c}v)^n.$$

Then $u \le v$ everywhere on D.

Proof Choose R with $D \subset B(0,R)$, and for $\delta > 0$ put $u_\delta = u + \delta(|z|^2 - R^2)$, a locally bounded plurisubharmonic function. Since $u_\delta \le u$ on D, the boundary condition still holds for v and u_δ, and the set $A = \{v < u_\delta\}$ is contained in $\{v < u\}$. By additivity, positive homogeneity and symmetry of the products (Proposition 9.1.2 and Theorem 9.2.5), $(\mathrm{dd^c}u_\delta)^n = \sum_{k=0}^n \binom{n}{k} \delta^k (\mathrm{dd^c}u)^{n-k} \wedge (\mathrm{dd^c}|z|^2)^k$ is a sum of positive measures, and its term with $k = n$ is $\delta^n n!\, \pi^{-n}\, \mathrm{d}V_{2n}$, by (9.2). The comparison inequality of Theorem 9.2.8, applied to v and u_δ, therefore gives

$$\delta^n \frac{n!}{\pi^n} |A| \le \int_A (\mathrm{dd^c}u_\delta)^n \le \int_A (\mathrm{dd^c}v)^n = 0,$$

where the last equality holds because $A \subset \{v < u\}$ and $u \le v$ almost everywhere for $(\mathrm{dd^c}v)^n$. Thus A is a Lebesgue null set, and $\max\{u_\delta, v\} = v$ almost everywhere. Both functions are plurisubharmonic, so the uniqueness of plurisubharmonic representatives in Theorem 8.2.1, applied on each connected component of D, gives $\max\{u_\delta, v\} = v$ everywhere, that is, $u_\delta \le v$. Letting $\delta \downarrow 0$ gives $u \le v$. □

9.3 Weighted estimates and Monge–Ampère capacity

We now use the products to define a capacity adapted to plurisubharmonic functions. The Monge–Ampère capacity of a set E is the supremum of the masses that the measures $(\mathrm{dd^c}v)^n$ give to E when v ranges over the plurisubharmonic functions with values in $[0,1]$. To control exceptional sets with it, we need integral estimates that are uniform over this whole class. The first one is a version of the Chern–Levine–Nirenberg inequality with an unbounded plurisubharmonic weight.

Theorem 9.3.1 *Let $\Omega \subset \mathbb{C}^n$ be open, $n \ge 1$, let $K, L \subset \Omega$ be compact sets with $K \subset \operatorname{Int} L$, and let $0 \le q \le n$ be an integer. There is a constant $C_{K,L,q}$, depending only on K, L, n and q, such that for every $V \in \mathrm{PSH}(\Omega)$ and all locally bounded plurisubharmonic functions $u_1, \dots, u_q$ on Ω,*

$$\int_K |V|\, \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \beta^{n-q} \le C_{K,L,q} \|V\|_{L^1(L)} \prod_{l=1}^q \|u_l\|_{L^\infty(L)}.$$

In particular, the function V, with its values $-\infty$ on the pole set $\{V=-\infty\}$, is locally integrable with respect to the trace measure $\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge\beta^{n-q}$.

Proof For $q\le n-1$ we integrate V by parts against a cutoff function on a ball, after making the factors coincide near its boundary (Steps 1–4); the case $q=n$ follows by adding one variable (Step 5).

Step 1: reductions. By Theorem 8.2.1, V is locally integrable, so $\|V\|_{L^1(L)}<\infty$. For $q=0$ the left-hand side equals $n!\int_K|V|\,dV_{2n}\le n!\|V\|_{L^1(L)}$, by (7.4). Let $q\ge 1$. If $\|u_l\|_{L^\infty(L)}=0$ for some l, then $u_l=0$ on $\operatorname{Int}L$, so the product vanishes on $\operatorname{Int}L\supset K$ by locality. Otherwise we replace each u_l by $u_l/(2\|u_l\|_{L^\infty(L)})-3/2$, which takes values in $[-2,-1]$ on L; by Proposition 9.1.2(3) this multiplies the left-hand side by $\prod_l(2\|u_l\|_{L^\infty(L)})^{-1}$. For each $a\in K$ choose $R_a>0$ with $\overline{B(a,R_a)}\subset\operatorname{Int}L$; finitely many of the balls $B(a,R_a/2)$ cover K, and this choice depends only on K and L. Hence it suffices to prove the following: for concentric balls $B'=B(a,R')$ and $B=B(a,R)$ with $R'<R$ and $\overline{B}\subset\operatorname{Int}L$, there is a constant C, depending only on B', B, L, n and q, such that

$$\int_{B'}|V|\,\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_q\wedge\beta^{n-q}\le C\|V\|_{L^1(L)}$$

whenever $u_1,\dots,u_q$ are plurisubharmonic on Ω with values in $[-2,-1]$ on L.

Step 2: a common collar. Put $\psi=|z-a|^2-R^2$ and $b=\mathrm{dd^c}\psi=\beta/\pi$, and choose $\delta>0$ with $\overline{B'}\subset\{\psi<-\delta\}$. Replace each u_l by $\max\{u_l,A\psi\}$, where $A=2/\delta$; this function is plurisubharmonic on Ω (Proposition 8.2.5). By (9.4), with $N=2$ and $U=\operatorname{Int}L$, it is unchanged on $\{\psi<-\delta\}$, it equals $A\psi$ on the collar $V_0=\{z\in\operatorname{Int}L:\psi(z)>-\delta/2\}$ of ∂B, and it takes values in $[-2,0)$ on B. Let $S=\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_q$ be the product of the modified functions. Put $\tau=\delta/4$ and $B_\tau=\{\psi<-\tau\}$. Then $\overline{B'}\subset B_\tau$, and on the neighborhood V_0 of $\overline{B}\setminus B_\tau$ we have $S=(A\,\mathrm{dd^c}\psi)^q=A^qb^q$ by Proposition 9.1.2(1),(2). By Lemma 1.4.1(4) there is $\theta\in C_c^\infty(B)$ with $0\le\theta\le1$ and $\theta=1$ on a neighborhood of $\overline{B_\tau}$; the function $\chi=-\psi\theta$ is smooth, nonnegative, has compact support in B, and equals $-\psi$ on a neighborhood of $\overline{B_\tau}$. All these choices depend only on B', B and L.

Step 3: smooth V and $q\le n-1$. Suppose that $q\le n-1$ and that V is smooth and plurisubharmonic on a neighborhood of $\overline{B}$ with $V\le0$ on B_τ. The signed measure $S\wedge b^{n-q-1}\wedge\mathrm{dd^c}\chi$ has compact support in B. Splitting its integral against V over B_τ, where $\mathrm{dd^c}\chi=-b$, and over $B\setminus B_\tau\subset V_0$, where $S=A^qb^q$, we obtain

$$\begin{aligned}\int_{B_\tau}(-V)S\wedge b^{n-q}&=\int_B VS\wedge b^{n-q-1}\wedge\mathrm{dd^c}\chi-A^q\int_{B\setminus B_\tau}Vb^{n-1}\wedge\mathrm{dd^c}\chi\\&=\int_B\chi\,\mathrm{dd^c}V\wedge S\wedge b^{n-q-1}-A^q\int_{B\setminus B_\tau}Vb^{n-1}\wedge\mathrm{dd^c}\chi.\end{aligned}$$

The second equality holds because the first integral is $(VS\wedge b^{n-q-1})(\mathrm{dd^c}\chi)=\big(\mathrm{dd^c}(VS\wedge b^{n-q-1})\big)(\chi)$, and $\mathrm{dd^c}(VS\wedge b^{n-q-1})=\mathrm{dd^c}V\wedge S\wedge b^{n-q-1}$ by (9.1), the current $S\wedge b^{n-q-1}$ being closed. The smooth form $\mathrm{dd^c}V$ is closed, and it is strongly positive, since its coefficient matrix is positive semidefinite (Proposition 8.2.2) and

weak and strong positivity agree in bidegree $(1,1)$ (Section 7.2); so by Proposition 9.1.2(4), with $T = 1$ and $\gamma = \mathrm{dd^c}V$, the current $\mathrm{dd^c}V \wedge S$ is the product $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \mathrm{dd^c}V$ with the closed positive base current $\mathrm{dd^c}V$ of bidimension $(n-1, n-1)$.

Choose a compact set $L' \subset B$ whose interior contains $\operatorname{Supp}\chi$. The first term is at most $\pi^{q+1-n}\sup\chi$ times the mass of the trace measure of $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \mathrm{dd^c}V$ on $\operatorname{Supp}\chi$. Since the modified u_l take values in $[-2,0)$ on B, Theorem 9.1.3, applied to $\operatorname{Supp}\chi \subset \operatorname{Int}L'$, shows that the first term is at most $C\|\mathrm{dd^c}V\|_{L'}$, with C depending only on χ, L', n and q. Let $\chi' \in C_c^\infty(B)$ satisfy $0 \le \chi' \le 1$ and $\chi' = 1$ on L'. Since $\mathrm{dd^c}V \wedge \beta^{n-1} \ge 0$,

$$\|\mathrm{dd^c}V\|_{L'} \le \int \chi'\,\mathrm{dd^c}V \wedge \frac{\beta^{n-1}}{(n-1)!} = \int V\,\mathrm{dd^c}\chi' \wedge \frac{\beta^{n-1}}{(n-1)!} \le C\int_B |V|\,\mathrm{d}V_{2n},$$

where C bounds the coefficient of the smooth (n,n)-form $\mathrm{dd^c}\chi' \wedge \beta^{n-1}/(n-1)!$. The collar term is bounded by the same quantity, because $b^{n-1} \wedge \mathrm{dd^c}\chi$ has bounded smooth coefficients. Hence

$$\int_{B_\tau} (-V)S \wedge b^{n-q} \le C\int_B |V|\,\mathrm{d}V_{2n}.$$

Step 4: general V and $q \le n-1$. Let $V \in \mathrm{PSH}(\Omega)$, and choose $r_0 > 0$ with $2r_0 < \operatorname{dist}(\overline{B}, \mathbb{C}^n \setminus \operatorname{Int}L)$. The function V is subharmonic (Theorem 8.2.1). If x lies within distance r_0 of $\overline{B}$, then $\overline{B(x,r_0)} \subset \operatorname{Int}L$, and the ball submean inequality of Theorem 3.1.3(2) gives

$$V(x) \le \frac{1}{|B(x,r_0)|}\int_{B(x,r_0)} V\,\mathrm{d}V_{2n} \le c_0\|V\|_{L^1(L)}, \qquad c_0 = |B(0,r_0)|^{-1};$$

put $c = c_0\|V\|_{L^1(L)}$. For $0 < \varepsilon < r_0$ the regularization $V_\varepsilon = V * \rho_\varepsilon$ is smooth and plurisubharmonic on the neighborhood $\{z : \operatorname{dist}(z,\overline{B}) < r_0\}$ of $\overline{B}$ (Proposition 8.2.2), since this neighborhood lies in Ω_ε. It satisfies $V_\varepsilon \le c$ on $\overline{B}$, since $V_\varepsilon(x)$ is an average of values of V at points within distance ε of x, and $\int_B |V_\varepsilon|\,\mathrm{d}V_{2n} \le \int \rho_\varepsilon(y)\int_B |V(x-y)|\,\mathrm{d}V_{2n}(x)\,\mathrm{d}V_{2n}(y) \le \|V\|_{L^1(L)}$ by Tonelli's theorem, since $B - y \subset L$ for $|y| \le \varepsilon$. Applying Step 3 to $V_\varepsilon - c$ gives

$$\int_{B_\tau} (c - V_\varepsilon)\,S \wedge b^{n-q} \le C\big(\|V\|_{L^1(L)} + c|B|\big) \le C'\|V\|_{L^1(L)}.$$

As $\varepsilon \downarrow 0$, the nonnegative functions $c - V_\varepsilon$ increase to $c - V$, by (3.3), so monotone convergence gives the same bound for $\int_{B_\tau}(c-V)\,S \wedge b^{n-q}$. Since $V \le c$ on B_τ, we have $|V| \le (c-V) + c$ there, and the mass of $S \wedge b^{n-q}$ on B_τ is bounded by a constant depending only on B', B, n and q, by Theorem 9.1.3 applied to $\overline{B_\tau} \subset B = \operatorname{Int}\overline{B}$, since the modified u_l take values in $[-2,0]$ on $\overline{B}$. Since S coincides with the original product on $B' \subset B_\tau$, by locality, this proves the estimate of Step 1 for $q \le n-1$. By

the finite cover and the normalizations of Step 1, the theorem holds for $q \le n-1$, in every dimension.

Step 5: the case $q = n$. Consider $\mathbb{C}^{n+1}$ with coordinates (z, w), the unit disc $\Delta \subset \mathbb{C}$, and the projection $p(z,w) = z$; let β_z, β_w be the forms (7.4) in the variables z and w, so that $\beta_z + \beta_w$ is the form β of $\mathbb{C}^{n+1}$, and $\beta_w = \mathrm{d}V_2(w)$. By Proposition 8.2.3, the functions $u_l \circ p$ are locally bounded and plurisubharmonic on $\Omega \times \Delta$, and $V \circ p$ is plurisubharmonic there, since it is not identically $-\infty$ on any connected component $\Omega_0 \times \Delta$ of $\Omega \times \Delta$. Let $\mu = \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n$, a positive measure on Ω. We claim that

$$\mathrm{dd^c}(u_1 \circ p) \wedge \cdots \wedge \mathrm{dd^c}(u_n \circ p) \wedge (\beta_z + \beta_w) = \mu \otimes \mathrm{d}V_2.$$

If the u_l are smooth, then $\mathrm{dd^c}(u_l \circ p) = p^* \mathrm{dd^c} u_l$, so the product of these n factors is the pullback $p^*\mu$ of the (n,n)-form $\mu = g\,\mathrm{d}V_{2n}$ (Proposition 9.1.2(1)). Its exterior product with β_z is $p^*(\mu \wedge \beta) = 0$, since $\mu \wedge \beta$ has bidegree $(n+1, n+1)$ on $\mathbb{C}^n$, and its exterior product with β_w is $g(z)\,\mathrm{d}V_{2n}(z) \wedge \mathrm{d}V_2(w)$, the density of $\mu \otimes \mathrm{d}V_2$.

In general, let $u_{l,\varepsilon} = u_l * \rho_\varepsilon$, which are smooth plurisubharmonic functions on Ω_{ε_0} for $\varepsilon < \varepsilon_0$, decreasing to u_l as $\varepsilon \downarrow 0$ (Proposition 8.2.2). By Theorem 9.2.5, applied on $\Omega_{\varepsilon_0} \times \Delta$ to the functions $u_{l,\varepsilon} \circ p$, which decrease to $u_l \circ p$, the left-hand sides for $u_{l,\varepsilon}$ converge weakly to the left-hand side for u_l. By the same theorem on Ω_{ε_0}, the measures $\mu_\varepsilon = \mathrm{dd^c} u_{1,\varepsilon} \wedge \cdots \wedge \mathrm{dd^c} u_{n,\varepsilon}$ converge weakly to μ; hence $\mu_\varepsilon \otimes \mathrm{d}V_2 \to \mu \otimes \mathrm{d}V_2$, because for a test function φ on $\Omega_{\varepsilon_0} \times \Delta$ we have $\int \varphi\, \mathrm{d}(\mu_\varepsilon \otimes \mathrm{d}V_2) = \int \Phi\, \mathrm{d}\mu_\varepsilon$ with $\Phi(z) = \int_\Delta \varphi(z,w)\,\mathrm{d}V_2(w)$, a test function on Ω_{ε_0} (differentiate under the integral sign). Since ε_0 is arbitrary, the claim follows from the smooth case.

Now apply the case already proved in dimension $n+1$, with $q = n \le (n+1) - 1$, to the compact sets $\widetilde{K} = K \times \{|w| \le 1/2\}$ and $\widetilde{L} = L \times \{|w| \le 3/4\}$ in $\Omega \times \Delta$; note that $\widetilde{K} \subset \operatorname{Int} \widetilde{L}$. By the claim and Fubini's theorem, the left-hand side of the estimate for $V \circ p$ and the $u_l \circ p$ is $\frac{\pi}{4} \int_K |V|\,\mathrm{d}\mu$, while $\|V \circ p\|_{L^1(\widetilde{L})} = \frac{9\pi}{16} \|V\|_{L^1(L)}$ and $\|u_l \circ p\|_{L^\infty(\widetilde{L})} = \|u_l\|_{L^\infty(L)}$. Hence the estimate holds in $\mathbb{C}^n$ with the constant $\frac{9}{4} C_{\widetilde{K},\widetilde{L},n}$, which depends only on K, L and n. This proves the case $q = n$. □

The base current in Theorem 9.3.1 is $T = 1$, and for a general closed positive current T no such estimate holds, already for $q = 0$. Let $n \ge 1$, let $H = \{z_1 = 0\} \subset \mathbb{C}^n$, and let $T = [H]$ be the current of integration of Example 7.2.7, a closed positive current of bidimension $(n-1, n-1)$. In the coordinates $z_2, \ldots, z_n$ of H, the restriction of β to H is the form (7.4) of $\mathbb{C}^{n-1}$, so $T \wedge \beta^{n-1}/(n-1)!$ is Lebesgue measure on H. The function $V = \log|z_1|$ is plurisubharmonic (Proposition 8.3.3), hence $\|V\|_{L^1(L)} < \infty$ for every compact L, but $V = -\infty$ on H. Therefore $\int_K |V|\, T \wedge \beta^{n-1} = \infty$ for the closed unit ball K, which meets H in a set of positive Lebesgue measure on H.

Definition 9.3.2 Let $D \subset \mathbb{C}^n$, $n \ge 1$, be a bounded open set. The *Monge–Ampère capacity* of a Borel set $E \subset D$ is

$$\mathrm{Cap}_D(E) = \sup \left\{ \int_E (\mathrm{dd^c} v)^n : v \in \mathrm{PSH}(D),\ 0 \le v \le 1 \right\}.$$

The functions v in this supremum are called *competitors* for Cap_D. The *outer capacity* of an arbitrary set $A \subset D$ is

$$\mathrm{Cap}_D^*(A) = \inf\{\mathrm{Cap}_D(G) : A \subset G \subset D,\ G \text{ open}\}.$$

On a complex manifold we use these capacities in coordinate charts, taking for D a bounded open set, usually a ball, which is relatively compact in the chart.

Proposition 9.3.3 *Let $D \subset \mathbb{C}^n$ be a bounded open set. The set functions Cap_D and Cap_D^* are increasing with respect to inclusion and countably subadditive. Moreover:*

(1) *$\mathrm{Cap}_D(K) < \infty$ for every compact $K \subset D$, and $\mathrm{Cap}_D(E) = \sup\{\mathrm{Cap}_D(K) : K \subset E$ compact$\}$ for every Borel set $E \subset D$;*
(2) *if $E_j \subset D$ are Borel sets increasing to E, then $\mathrm{Cap}_D(E_j) \to \mathrm{Cap}_D(E)$;*
(3) *there is a constant C_D such that $|E| \le C_D\,\mathrm{Cap}_D(E)$ for every Borel set $E \subset D$, where $|E|$ is Lebesgue measure;*
(4) *if $B \Subset D$ is a ball and $K \subset B$ is compact, there is a constant C such that, for every Borel set $E \subset K$,*

$$\mathrm{Cap}_D(E) \le \mathrm{Cap}_B(E) \le C\,\mathrm{Cap}_D(E),$$

and the same inequalities hold for the outer capacities of arbitrary subsets of K;
(5) *both capacities are invariant under biholomorphisms between bounded open sets: if $F\colon D \to D'$ is biholomorphic, then $\mathrm{Cap}_{D'}(F(E)) = \mathrm{Cap}_D(E)$ for Borel sets $E \subset D$ and $\mathrm{Cap}_{D'}^*(F(A)) = \mathrm{Cap}_D^*(A)$ for arbitrary sets $A \subset D$.*

Proof For each competitor v, the measure $(\mathrm{dd^c}v)^n$ is increasing and countably subadditive on Borel sets: $\int_{\bigcup_j E_j}(\mathrm{dd^c}v)^n \le \sum_j \int_{E_j}(\mathrm{dd^c}v)^n \le \sum_j \mathrm{Cap}_D(E_j)$. Taking the supremum over v gives the same properties for Cap_D. Outer capacity is increasing by definition. For countable subadditivity, given sets A_j with $\sum_j \mathrm{Cap}_D^*(A_j) < \infty$ and $\varepsilon > 0$, choose open sets $G_j \supset A_j$ with $\mathrm{Cap}_D(G_j) < \mathrm{Cap}_D^*(A_j) + \varepsilon 2^{-j}$; then $\mathrm{Cap}_D^*(\bigcup_j A_j) \le \mathrm{Cap}_D(\bigcup_j G_j) \le \sum_j \mathrm{Cap}_D^*(A_j) + \varepsilon$.

Finiteness on a compact set K follows from Theorem 9.1.3 with $T = 1$, applied to K and a compact neighborhood L of K in D, since $\|v\|_{L^\infty(L)} \le 1$ for every competitor. Inner regularity of each Radon measure $(\mathrm{dd^c}v)^n$ (Definition 1.1.1) gives $\int_E (\mathrm{dd^c}v)^n = \sup_K \int_K (\mathrm{dd^c}v)^n \le \sup_K \mathrm{Cap}_D(K)$ over compact $K \subset E$, and taking the supremum over v gives (1). For (2), $\mathrm{Cap}_D(E_j) \le \mathrm{Cap}_D(E)$ by monotonicity, and for every competitor v monotone convergence gives $\int_E (\mathrm{dd^c}v)^n = \lim_j \int_{E_j}(\mathrm{dd^c}v)^n \le \lim_j \mathrm{Cap}_D(E_j)$; taking the supremum over v gives (2). For (3), choose $a \in D$ and $d > 0$ with $D \subset B(a,d)$; the function $|z-a|^2/d^2$ is a competitor, and by (9.2) and Proposition 9.1.2(3) its Monge–Ampère measure is $n!\pi^{-n}d^{-2n}\,\mathrm{d}V_{2n}$. Hence (3) holds with $C_D = \pi^n d^{2n}/n!$.

For (4), the first inequality holds because the restriction to B of a competitor for D is a competitor for B, with the same Monge–Ampère measure on B. For the second, write $B = B(a,r)$ and choose $\lambda > 0$ such that $\rho = \lambda(|z-a|^2 - r^2) < -1$ on a neighborhood of K. Given a competitor v for B, let w be the function equal to $\max\{v-1, \rho\}$ on B and to ρ on $D \setminus B$. Since $v - 1 \le 0$ on B and $\rho = 0$ on ∂B, Proposition 8.2.5 shows that w is plurisubharmonic on D. It equals $v-1$ near K, satisfies $w \ge -1$ on D, and is bounded above by $\Lambda = \max\{0, \sup_D \rho\}$, which

does not depend on v. Thus $(w+1)/(\Lambda+1)$ is a competitor for D, and by locality its Monge–Ampère measure near K is $(\Lambda+1)^{-n}(\mathrm{dd^c}v)^n$. This gives the second inequality with $C=(\Lambda+1)^n$; the first inequality holds for all Borel subsets of B.

For the outer capacities in (4), let K' be a compact neighborhood of K in B; the constant obtained for K' also serves for K, and we use it. If $A\subset K$ and G is an open subset of D or of B containing A, then $G\cap \operatorname{Int} K'$ is open in both B and D, contains A, lies in K', and has no larger capacity than G. Applying the inequalities to these sets and taking infima over G gives them for the outer capacities of A.

For (5), let $F\colon D\to D'$ be biholomorphic. By Proposition 8.2.3, applied to F and to F^{-1}, the map $v\mapsto v\circ F$ is a bijection from the competitors for D' onto the competitors for D, and by Proposition 9.1.2(5), $(\mathrm{dd^c}(v\circ F))^n = F^*\big((\mathrm{dd^c}v)^n\big)$. We claim that for every positive Radon measure μ on D', regarded as a current of bidegree (n,n), the pullback $F^*\mu$ is the image of μ under F^{-1}, that is, $\int \psi\,\mathrm{d}(F^*\mu)=\int \psi\circ F^{-1}\,\mathrm{d}\mu$ for every test function ψ on D.

If $\mu = g\,\mathrm{d}V_{2n}$ with g smooth, then $F^*\mu = (g\circ F)\,|\det F'|^2\,\mathrm{d}V_{2n}$, because $F^*(\mathrm{d}z_1\wedge\cdots\wedge\mathrm{d}z_n)=\det F'\,\mathrm{d}z_1\wedge\cdots\wedge\mathrm{d}z_n$ and similarly for the conjugate forms; since $|\det F'|^2$ is the Jacobian determinant of F as a map of $\mathbb{R}^{2n}$, the change of variables formula gives the claim. Both sides of the claimed identity are weakly continuous in μ (Proposition 7.5.2), and on relatively compact open subsets of D' every μ is the weak limit of its smooth regularizations, so the claim holds in general. Consequently $(F^*\mu)(F^{-1}(E')) = \mu(E')$ for Borel sets $E'\subset D'$, since two Radon measures that agree on test functions agree on Borel sets (Proposition 1.4.3(4)). Applying this to $\mu=(\mathrm{dd^c}v)^n$ and $E'=F(E)$ and taking the supremum over v gives $\operatorname{Cap}_D(E)=\operatorname{Cap}_{D'}(F(E))$. Since F is a homeomorphism, it maps the open sets containing a set A bijectively onto the open sets containing $F(A)$, so the outer capacities agree as well. □

Proposition 9.3.4 *Let $D\subset\mathbb{C}^n$ be a bounded open set, let $K,L\subset D$ be compact sets with $K\subset\operatorname{Int}L$, and let $V\in\mathrm{PSH}(D)$. There is a constant $C_{K,L}$, depending only on K, L and n, such that*

$$\operatorname{Cap}^*_D(K\cap\{V<-s\})\le\frac{C_{K,L}}{s}\|V\|_{L^1(L)},\qquad s>0.$$

Moreover, let $E\subset D$ be a set locally contained in pole sets of plurisubharmonic functions, that is, every point of D has an open neighborhood $W\subset D$ and a plurisubharmonic function V_W on W with $E\cap W\subset\{V_W=-\infty\}$. (Such sets are called pluripolar in Chapter 10.) Then $\operatorname{Cap}^*_D(E\cap K)=0$ for every compact set $K\subset D$, and E is contained in a Borel set which has measure zero for every measure $\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_n$ with locally bounded plurisubharmonic functions $u_1,\dots,u_n$ on D.

Proof Choose an open neighborhood U of K with compact closure in $\operatorname{Int}L$. For every competitor v, Theorem 9.3.1, applied to $\overline{U}\subset\operatorname{Int}L$ with $q=n$ and $u_1=\cdots=u_n=v$, gives

$$\int_{U\cap\{V<-s\}}(\mathrm{dd^c}v)^n\le s^{-1}\int_{\overline{U}}|V|\,(\mathrm{dd^c}v)^n\le Cs^{-1}\|V\|_{L^1(L)}.$$

Since V is upper semicontinuous, $U\cap\{V<-s\}$ is an open set containing $K\cap\{V<-s\}$; taking the supremum over v proves the estimate. Since V is locally integrable (Theorem 8.2.1), $\|V\|_{L^1(L)}<\infty$, and letting $s\to\infty$ gives $\mathrm{Cap}^*_D(K\cap\{V=-\infty\})=0$.

Now let E be as in the statement. Let $x\in D$, let W and V_W be as in the hypothesis, and choose $r>0$ with $\overline{B(x,3r)}\subset W$; put $B_x=B(x,3r)$, $K_x=\overline{B(x,r)}$ and $L_x=\overline{B(x,2r)}$. Then $E\cap B_x\subset\{V_W=-\infty\}$, and the restriction of V_W to B_x is plurisubharmonic. The estimate just proved, applied in B_x to this restriction and to $K_x\subset\operatorname{Int}L_x$, gives $\mathrm{Cap}^*_{B_x}(E\cap K_x)=0$; since every open set in B_x is open in D, the first inequality of Proposition 9.3.3(4) gives $\mathrm{Cap}^*_D(E\cap K_x)=0$. A compact set $K\subset D$ is covered by finitely many interiors $\operatorname{Int}K_x$, and subadditivity gives $\mathrm{Cap}^*_D(E\cap K)=0$. Finally, by Theorem 9.3.1, applied on B_x to $K_x\subset\operatorname{Int}L_x$ with $q=n$, the function V_W is integrable on K_x for every measure $\mathrm{dd^c}u_1\wedge\cdots\wedge\mathrm{dd^c}u_n$ as in the statement, so the Borel set $K_x\cap\{V_W=-\infty\}$ has measure zero for all these measures. Since D has a countable basis, countably many sets $\operatorname{Int}K_x$ cover D, and the union of the corresponding sets $K_x\cap\{V_W=-\infty\}$ is a Borel set with the required property. □

The next estimate controls, uniformly over all competitors, the set where two bounded plurisubharmonic functions with the same boundary values differ.

Proposition 9.3.5 (Błocki) *Let B be a ball, and let u and h be bounded plurisubharmonic functions on a neighborhood of $\overline{B}$ with $u\le h$ on B and $u=h$ on a collar of ∂B. Let v be a bounded plurisubharmonic function on a neighborhood of $\overline{B}$, and put $M=\sup_B v-\inf_B v$. Then*

$$\int_B(h-u)^{n+1}(\mathrm{dd^c}v)^n\le(n+1)!M^n\int_B(h-u)(\mathrm{dd^c}u)^n.$$

Proof First suppose that u, h and v are smooth; then $\mathrm{dd^c}u$, $\mathrm{dd^c}h$ and $\mathrm{dd^c}v$ are positive $(1,1)$-forms, by Proposition 8.2.2. Subtracting $\sup_B v$ from v, we may assume $-M\le v\le 0$ on B. Put $w=h-u\ge 0$; it vanishes on a collar of ∂B. Let $p\ge 2$, and let R be a smooth closed strongly positive form of bidegree $(n-1,n-1)$; the exterior product of R with a positive $(1,1)$-form is then a nonnegative multiple of the volume form, by Proposition 7.2.1, since positive $(1,1)$-forms are strongly positive. Since $\mathrm{d}w\wedge\mathrm{d^c}w$ is a positive $(1,1)$-form by (9.3) and $\mathrm{dd^c}h\wedge R\ge 0$, the chain rule $\mathrm{dd^c}(w^p)=pw^{p-1}\mathrm{dd^c}w+p(p-1)w^{p-2}\mathrm{d}w\wedge\mathrm{d^c}w$ gives

$$\begin{aligned}\mathrm{dd^c}(w^p)\wedge R&=\bigl(pw^{p-1}(\mathrm{dd^c}h-\mathrm{dd^c}u)+p(p-1)w^{p-2}\mathrm{d}w\wedge\mathrm{d^c}w\bigr)\wedge R\\&\ge -pw^{p-1}\mathrm{dd^c}u\wedge R.\end{aligned}$$

Since w^p has compact support in B and R is closed, (9.1) gives $\int_B w^p\mathrm{dd^c}v\wedge R=(\mathrm{dd^c}(vR))(w^p)=(vR)(\mathrm{dd^c}(w^p))$. As $v\le 0$ and $-v\le M$, the preceding inequality therefore yields

$$\int_B w^p\mathrm{dd^c}v\wedge R=\int_B v\,\mathrm{dd^c}(w^p)\wedge R\le pM\int_B w^{p-1}\mathrm{dd^c}u\wedge R.$$

For $k = 1, \dots, n$, apply this with $p = n + 2 - k$ and $R = (\mathrm{dd^c} v)^{n-k} \wedge (\mathrm{dd^c} u)^{k-1}$, a product of positive $(1,1)$-forms, hence strongly positive (Proposition 7.2.1) and closed. This gives

$$\int_B w^{n+2-k} (\mathrm{dd^c} v)^{n+1-k} \wedge (\mathrm{dd^c} u)^{k-1} \le (n+2-k) M \int_B w^{n+1-k} (\mathrm{dd^c} v)^{n-k} \wedge (\mathrm{dd^c} u)^k,$$

and chaining these n inequalities from $k = 1$ to $k = n$ gives the asserted inequality, since $(n+1)n \cdots 2 = (n+1)!$.

In general, write $B = B(a, r)$ and let V be a collar of ∂B on which $u = h$. Since ∂B is compact, there is $r'' < r$ with $\{z : r'' \le |z - a| \le r\} \subset V$; choose $r' \in (r'', r)$ and put $B' = B(a, r')$. Then $h - u$ vanishes on $B \setminus \overline{B(a, r'')}$. Convolve u, h and v with the same radial kernel ρ_ε, with $\varepsilon < \min\{(r' - r'')/2, r - r'\}$ so small that the regularizations are defined, smooth and plurisubharmonic on a neighborhood of $\overline{B'}$ (Proposition 8.2.2). At points of $\overline{B'}$ they are averages over balls of radius ε contained in B, where $u \le h$; hence $u_\varepsilon \le h_\varepsilon$ on $\overline{B'}$. Moreover $h_\varepsilon - u_\varepsilon$ vanishes on the collar $\{z : r'' + \varepsilon < |z - a| < r - \varepsilon\}$ of $\partial B'$ and outside $\overline{B(a, r'' + \varepsilon)}$ in B', hence outside the compact set $\overline{B(a, (r' + r'')/2)} \subset B'$. The values of v_ε on B' are averages of values of v on B, so the oscillation of v_ε on B' is at most M.

Apply the smooth case on B' and let $\varepsilon = \varepsilon_k \downarrow 0$. The weights $(h_\varepsilon - u_\varepsilon)^{n+1}$ and $h_\varepsilon - u_\varepsilon$ are polynomials in decreasing sequences of bounded plurisubharmonic functions, and they, as well as their limits $(h - u)^{n+1}$ and $h - u$, vanish in B' outside the fixed compact set $\overline{B(a, (r' + r'')/2)}$. Testing against a function $\chi \in C_c^\infty(B')$ equal to 1 on this set, Corollary 9.2.6 lets both sides pass to the limit. Since $h - u$ vanishes on $B \setminus B'$, the limiting inequality on B' is the asserted inequality on B. □

In the next two sections we use the proposition through a capacity bound. Let u and h be as in Proposition 9.3.5, and let $D \subset \mathbb{C}^n$ be a bounded open set with $\overline{B} \subset D$. Every competitor v for Cap_D has oscillation at most 1 on B. Hence Chebyshev's inequality and the proposition give, for every $\delta > 0$,

$$\begin{aligned} \mathrm{Cap}_D\big(B \cap \{h - u > \delta\}\big) &\le \sup_v \frac{1}{\delta^{n+1}} \int_B (h - u)^{n+1} (\mathrm{dd^c} v)^n \\ &\le \frac{(n+1)!}{\delta^{n+1}} \int_B (h - u) (\mathrm{dd^c} u)^n, \end{aligned} \tag{9.6}$$

where the supremum is taken over all competitors v.

9.4 Quasi-continuity and convergence in capacity

Throughout this section and the next, $D \subset \mathbb{C}^n$, $n \ge 1$, is a bounded open set. By Theorem 4.5.2, locally integrable subharmonic functions are quasi-continuous in the sense of Part I: they are continuous outside open sets of arbitrarily small relative capacity. We now define quasi-continuity with respect to the Monge–Ampère

capacity instead, which is the meaning of the term in the rest of the book, prove that plurisubharmonic functions are quasi-continuous in this sense, and then show that the products are continuous under convergence in capacity.

Definition 9.4.1 A Borel function $f\colon D \to [-\infty,\infty]$ is *quasi-continuous* (for Cap_D) if for every compact set $K \subset D$ and every $\varepsilon > 0$ there is an open set $G \subset D$ with $\mathrm{Cap}_D(G) < \varepsilon$ such that the restriction of f to $K \setminus G$ is continuous. Here $[-\infty,\infty]$ carries its order topology.

The same definition applies to a function defined on an open subset $W \subset D$, with compact sets $K \subset W$ and still with Cap_D. The sum $f_1 + f_2$ and the difference $f_1 - f_2$ of two real-valued quasi-continuous functions f_1, f_2 are quasi-continuous, and so is $\Phi \circ f_1$ for every continuous function $\Phi\colon \mathbb{R} \to \mathbb{R}$. Indeed, these functions are Borel. Given K and ε, choose open sets $G_i \subset D$ with $\mathrm{Cap}_D(G_i) < \varepsilon/2$ such that f_i is continuous on $K \setminus G_i$, $i = 1, 2$. Then $G = G_1 \cup G_2$ satisfies $\mathrm{Cap}_D(G) < \varepsilon$ by subadditivity (Proposition 9.3.3), and $f_1 + f_2$ and $f_1 - f_2$ are continuous on $K \setminus G$, as sums and differences of continuous real functions there; similarly, $\Phi \circ f_1$ is continuous on $K \setminus G_1$.

Theorem 9.4.2 *Let $u \in \mathrm{PSH}(D)$. For every $\varepsilon > 0$ there is an open set $G \subset D$ with $\mathrm{Cap}_D(G) < \varepsilon$ which contains the pole set $\{u = -\infty\}$ and such that the restriction of u to $D \setminus G$ is finite and continuous. In particular u is quasi-continuous.*

Proof *Step 1: locally bounded u.* Let u be locally bounded, $K \subset D$ compact and $\varepsilon > 0$; we find an open set $G \subset D$ with $\mathrm{Cap}_D(G) < \varepsilon$ such that u is continuous on $K \setminus G$. For each $x \in K$ choose $r_x > 0$ with $\overline{B(x, 3r_x)} \subset D$; finitely many balls $B(x_i, r_{x_i})$, $1 \le i \le m$, cover K. Suppose that for each i we find an open set G_i with $\mathrm{Cap}_D(G_i) < \varepsilon/m$ such that u is continuous on $\overline{B(x_i, r_{x_i})} \setminus G_i$. Then $G = \bigcup_i G_i$ satisfies $\mathrm{Cap}_D(G) < \varepsilon$, and u is continuous on $K \setminus G$, because the finitely many sets $(K \setminus G) \cap \overline{B(x_i, r_{x_i})}$ are closed in $K \setminus G$ and cover it, and u is continuous on each of them. We may therefore assume that $K = \overline{B'}$, where $B' = B(x, r)$, $B = B(x, 2r)$ and $B_1 = B(x, 3r)$ satisfy $\overline{B_1} \subset D$.

Subtracting a constant, we may assume that $-N \le u \le -1$ on B_1. Let $j > 2/r$. Since $U = B(x, 5r/2)$, a neighborhood of $\overline{B}$, lies at distance $r/2 > 1/j$ from $\mathbb{C}^n \setminus B_1$, the regularization $u * \rho_{1/j}$ is smooth and plurisubharmonic on U and takes values in $[-N, -1]$ there; moreover, $u * \rho_{1/j}$ decreases to u as $j \to \infty$ (Proposition 8.2.2). Put $\psi = |z - x|^2 - 4r^2$, choose $\delta > 0$ with $\overline{B'} \subset \{\psi < -\delta\}$, and let $g = \max\{u, A\psi\}$ and $g_j = \max\{u * \rho_{1/j}, A\psi\}$, with $A = N/\delta$, be the functions (9.4) on U. The g_j are continuous and plurisubharmonic on U and decrease to g; moreover, $g = u$ on $\{\psi < -\delta\} \supset \overline{B'}$, and $g_j = g = A\psi$ on a collar of ∂B.

For $\delta' > 0$ put $G_{j,\delta'} = \{z \in B : g_j(z) - g(z) > \delta'\}$; it is open, because g_j is continuous and g is upper semicontinuous. By (9.6), applied with $h = g_j$, $u = g$ and δ' in place of δ,

$$\mathrm{Cap}_D(G_{j,\delta'}) \le \frac{(n+1)!}{\delta'^{n+1}} \int_B (g_j - g)(\mathrm{dd^c} g)^n.$$

The right-hand side tends to zero as $j \to \infty$ by dominated convergence: $(\mathrm{dd}^c g)^n$ has finite mass on B, since g is bounded and plurisubharmonic near $\overline{B}$, and the functions $g_j - g \geq 0$ are bounded on B uniformly in j and decrease to zero. Choose $j_1 < j_2 < \cdots$ such that $G_l = G_{j_l, 2^{-l}}$ satisfies $\mathrm{Cap}_D(G_l) < \varepsilon 2^{-l}$, and let $G = \bigcup_l G_l$. Then $\mathrm{Cap}_D(G) < \varepsilon$, and on $B \setminus G$ we have $0 \leq g_{j_l} - g \leq 2^{-l}$ for every l. Thus g is a uniform limit of continuous functions on $B \setminus G$, hence continuous there; in particular $u = g$ is continuous on $\overline{B'} \setminus G$.

Step 2: arbitrary u. Let $K \subset D$ be compact and $\varepsilon > 0$. By Proposition 9.3.4, applied to K and a compact neighborhood L of K in D, there is $s > 0$ with $\mathrm{Cap}^*_D(K \cap \{u < -s\}) < \varepsilon/2$, and hence an open set $G_1 \supset K \cap \{u < -s\}$ with $\mathrm{Cap}_D(G_1) < \varepsilon/2$. By Step 1, applied to the locally bounded plurisubharmonic function $\max\{u, -s\}$, there is an open set G_2 with $\mathrm{Cap}_D(G_2) < \varepsilon/2$ such that $\max\{u, -s\}$ is continuous on $K \setminus G_2$. On $K \setminus (G_1 \cup G_2)$ we have $u \geq -s$, hence $u = \max\{u, -s\}$. Thus $G = G_1 \cup G_2$ contains $K \cap \{u = -\infty\}$, $\mathrm{Cap}_D(G) < \varepsilon$, and u is finite and continuous on $K \setminus G$.

Step 3: a single exceptional set. Choose compact sets K_l with $K_l \subset \operatorname{Int} K_{l+1}$ and $\bigcup_l K_l = D$. By Step 2 there are open sets $G^{(l)} \supset K_l \cap \{u = -\infty\}$ with $\mathrm{Cap}_D(G^{(l)}) < \varepsilon 2^{-l}$ such that u is finite and continuous on $K_l \setminus G^{(l)}$. Put $G = \bigcup_l G^{(l)}$. Then $\mathrm{Cap}_D(G) < \varepsilon$, and G contains the pole set of u. Every $x \in D \setminus G$ lies in some $\operatorname{Int} K_l$, and on the neighborhood $\operatorname{Int} K_l \setminus G$ of x in $D \setminus G$ the function u is a restriction of the continuous function $u|_{K_l \setminus G^{(l)}}$. Hence $u|_{D \setminus G}$ is finite and continuous. □

In the proofs below we extend continuous functions from compact sets, using the following elementary case of Tietze's extension theorem.

Lemma 9.4.3 *Let (X, d) be a metric space, let $F \subset X$ be compact, and let $f \colon F \to \mathbb{R}$ be continuous with $|f| \leq a$. Then f has a continuous extension $g \colon X \to \mathbb{R}$ with $|g| \leq a$.*

Proof If $F = \varnothing$ or $a = 0$, take $g = 0$. Otherwise, for $L > 0$ put

$$f_L(x) = \max\Big\{-a, \min\big\{a, \inf_{y \in F}\big(f(y) + L\, d(x, y)\big)\big\}\Big\}, \qquad x \in X.$$

The infimum is finite, since $f \geq -a$, and as an infimum of functions that are Lipschitz with constant L it is Lipschitz with constant L; hence f_L is continuous and $|f_L| \leq a$. We claim that $f_L \to f$ uniformly on F as $L \to \infty$. Taking $y = x$ shows $f_L \leq f$ on F. Given $\eta > 0$, uniform continuity of f on the compact set F gives $r > 0$ with $|f(y) - f(x)| < \eta$ for $x, y \in F$ with $d(x, y) < r$. Let $Lr > 2a$ and $x \in F$. Points $y \in F$ with $d(x, y) \geq r$ give $f(y) + L\, d(x, y) \geq -a + Lr > a \geq f(x)$, and the other points give $f(y) + L\, d(x, y) > f(x) - \eta$. Hence the infimum is at least $f(x) - \eta$, and so is $f_L(x)$, because $a \geq f(x) > f(x) - \eta$ and $-a \leq f(x)$. This proves the claim. Consequently there is a continuous g_1 on X with $|g_1| \leq a$ and $|f - g_1| \leq a/2$ on F. Applying the same construction to $f - g_1$ on F, we find a continuous g_2 with $|g_2| \leq a/2$ and $|f - g_1 - g_2| \leq a/4$ on F, and inductively continuous g_k with $|g_k| \leq 2^{1-k} a$ and $|f - g_1 - \cdots - g_k| \leq 2^{-k} a$ on F. The series $\sum_k g_k$ converges

uniformly on X; its sum is continuous and equals f on F, and clamping it to $[-a, a]$, that is, replacing it by $\max\{-a, \min\{a, \sum_k g_k\}\}$, does not change its values on F. □

Lemma 9.4.4 *Let $K, L \subset D$ be compact sets with $K \subset \operatorname{Int} L$, and let f be a real Borel function which is defined and quasi-continuous on a neighborhood of L. Then, for every $\delta > 0$,*

$$\operatorname{Cap}^*_D(K \cap \{|f| > \delta\}) \le \operatorname{Cap}_D(L \cap \{|f| > \delta/2\}).$$

Proof Let $\eta > 0$. By quasi-continuity, applied to the compact set L, there is an open set $G \subset D$ with $\operatorname{Cap}_D(G) < \eta$ such that f is continuous on $L \setminus G$; let g be a continuous extension of $f|_{L\setminus G}$ to $\mathbb{C}^n$ (Lemma 9.4.3); here $f|_{L\setminus G}$ is bounded, being continuous on a compact set. The open set $G \cup (\operatorname{Int} L \cap \{|g| > \delta/2\})$ contains $K \cap \{|f| > \delta\}$, since $K \subset \operatorname{Int} L$ and $g = f$ on $K \setminus G$. Its part outside G is a Borel set contained in $L \cap \{|f| > \delta/2\}$, because $g = f$ on $L \setminus G$. By monotonicity and subadditivity,

$$\operatorname{Cap}^*_D(K \cap \{|f| > \delta\}) \le \eta + \operatorname{Cap}_D(L \cap \{|f| > \delta/2\}).$$

Let $\eta \downarrow 0$. □

Definition 9.4.5 A net (f_a) of Borel functions $f_a : D \to [-\infty, +\infty]$ *converges locally in capacity* to a Borel function $f : D \to [-\infty, +\infty]$ if, for every compact set $K \subset D$ and every $\delta > 0$,

$$\operatorname{Cap}^*_D(K \cap \{f_a > f + \delta\}) \longrightarrow 0, \qquad \operatorname{Cap}^*_D(K \cap \{f_a < f - \delta\}) \longrightarrow 0.$$

Here $\pm\infty + \delta = \pm\infty$ and $\pm\infty - \delta = \pm\infty$.

For finite functions, $K \cap \{|f_a - f| > \delta\}$ is the union of the two sets in the definition, so by monotonicity and subadditivity of outer capacity the two conditions together are equivalent to $\operatorname{Cap}^*_D(K \cap \{|f_a - f| > \delta\}) \to 0$. Convergence locally in capacity is a local property: it holds on D if and only if, for every ball $B \Subset D$, it holds on B with Cap_B in place of Cap_D. Indeed, if it holds on D and $K \subset B$ is compact, then Proposition 9.3.3(4) gives $\operatorname{Cap}^*_B(A) \le C \operatorname{Cap}^*_D(A)$ for all $A \subset K$. Conversely, if it holds on every ball $B \Subset D$, cover a compact set $K \subset D$ by finitely many balls $B(x_i, r_i)$ with $\overline{B(x_i, 2r_i)} \subset D$; for $A \subset K_i = K \cap \overline{B(x_i, r_i)}$, Proposition 9.3.3(4) gives $\operatorname{Cap}^*_D(A) \le \operatorname{Cap}^*_{B(x_i,2r_i)}(A)$, and subadditivity over i gives the condition on D. In particular, for functions on an open set $W \subset D$, convergence locally in capacity with Cap_W is equivalent to the same condition, for compact subsets of W, with Cap_D, since both are equivalent to the condition on balls $B \Subset W$.

Proposition 9.4.6 *Let u_j be locally bounded plurisubharmonic functions on D decreasing to a locally bounded function u. Then $u_j \to u$ locally in capacity. Moreover, a net of real Borel functions on D which is uniformly bounded on each compact subset of D and converges locally in capacity to a locally bounded real Borel function also converges in $L^1_{\mathrm{loc}}(D)$.*

Proof Let $K \subset D$ be compact and $\delta > 0$. Since $u_j \geq u$, the sets $\{u_j < u - \delta\}$ are empty, and we must show that $\mathrm{Cap}_D^*(K \cap \{u_j - u > \delta\}) \to 0$. By Theorem 8.3.1, u is plurisubharmonic. Covering K by finitely many balls $B(x,r)$ with $\overline{B(x,4r)} \subset D$ and using subadditivity of outer capacity, we may assume that $K \subset \overline{B(x,r)}$ for such a ball; put $L = \overline{B(x,3r/2)}$, $B = B(x,3r)$ and $B_1 = B(x,4r)$.

We first make the functions coincide near ∂B. Since $u \leq u_j \leq u_1$, all these functions are bounded on B_1 uniformly in j, so there are constants c and $N \geq 1$ with $-N \leq u_j - c$, $u - c \leq -1$ on B_1. Put $\psi = |z-x|^2 - 9r^2$, choose $\delta_0 > 0$ with $L \subset \{\psi < -\delta_0\}$, and let $\hat{u}_j = \max\{u_j - c, A\psi\}$ and $\hat{u} = \max\{u - c, A\psi\}$, with $A = N/\delta_0$, be the functions (9.4) on B_1, with δ_0 in place of δ. They are bounded and plurisubharmonic on B_1, $\hat{u}_j \downarrow \hat{u}$, $\hat{u}_j - \hat{u} = u_j - u$ on L, and $\hat{u}_j = \hat{u}$ on a collar of ∂B. By (9.6), applied with $u = \hat{u}$ and $h = \hat{u}_j$,

$$\mathrm{Cap}_D\big(B \cap \{\hat{u}_j - \hat{u} > \delta/2\}\big) \leq (n+1)!\,(2/\delta)^{n+1} \int_B (\hat{u}_j - \hat{u})(\mathrm{dd^c}\hat{u})^n,$$

and the right-hand side tends to zero by dominated convergence, since $(\mathrm{dd^c}\hat{u})^n$ has finite mass on B and $0 \leq \hat{u}_j - \hat{u} \leq \hat{u}_1 - \hat{u}$ decreases to zero. The real function $w_j = u_j - u$ is quasi-continuous on D, as a difference of two real quasi-continuous functions (Theorem 9.4.2 and the remark following Definition 9.4.1), and $w_j = \hat{u}_j - \hat{u}$ on L. Hence Lemma 9.4.4 gives

$$\mathrm{Cap}_D^*(K \cap \{w_j > \delta\}) \leq \mathrm{Cap}_D(L \cap \{w_j > \delta/2\}) \leq \mathrm{Cap}_D\big(B \cap \{\hat{u}_j - \hat{u} > \delta/2\}\big) \longrightarrow 0.$$

For the second assertion, let $f_a \to f$ be as in the statement, let $K \subset D$ be compact, and let M bound $|f_a|$ and $|f|$ on K. Splitting K into the Borel set where $|f_a - f| > \delta$ and its complement, and using Proposition 9.3.3(3) together with $\mathrm{Cap}_D \leq \mathrm{Cap}_D^*$ on Borel sets, we get

$$\int_K |f_a - f|\,\mathrm{d}V_{2n} \leq 2MC_D\,\mathrm{Cap}_D^*(K \cap \{|f_a - f| > \delta\}) + \delta|K|.$$

Taking the limit along the net and then letting $\delta \downarrow 0$ proves L^1 convergence on K. □

Lemma 9.4.7 *Let $B \subset \mathbb{C}^n$ be a ball, and let $\mathcal{U}$ be a family of plurisubharmonic functions on B which is uniformly bounded on each compact subset of B.*

(1) *For every compact set $K \subset B$ there is a constant C such that, for all integers $0 \leq q \leq n$, all $u_1, \ldots, u_q \in \mathcal{U}$ and all Borel sets $E \subset K$, the trace measure $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \beta^{n-q}$, and the total variation of every coefficient measure of $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q$, give E mass at most $C\,\mathrm{Cap}_B(E)$.*

(2) *Let $S_j = \mathrm{dd^c}u_{1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j}$ with $u_{l,j} \in \mathcal{U}$, and suppose that $S_j \to S$ weakly. Let f_j be real Borel functions on B, uniformly bounded on each compact subset of B, which converge locally in capacity (for Cap_B) to a real Borel function f that is locally bounded and quasi-continuous. Then $f_j S_j \to fS$ weakly, and the same holds for the trace measures of S_j and S. This applies in particular to a fixed weight $f_j = f$.*

Proof (1) Let $B = B(a,R)$, fix K, and choose $R' < R$ with $K \subset B' = B(a,R')$. Let $M \geq 1$ be a common bound for $|u|$ on B', $u \in \mathcal{U}$. For $u_1, \dots, u_q \in \mathcal{U}$ put $v_l = (u_l + M)/(2M)$ and $\phi = |z-a|^2/R'^2$; these functions take values in $[0,1]$ on B', and so does

$$F = \frac{1}{n}\big(v_1 + \cdots + v_q + (n-q)\phi\big),$$

which is therefore a competitor for $\mathrm{Cap}_{B'}$. Expand $(\mathrm{dd^c} F)^n$ by additivity, positive homogeneity and symmetry (Proposition 9.1.2 and Theorem 9.2.5). All terms are positive measures, and those in which each $\mathrm{dd^c} v_l$ occurs once and $\mathrm{dd^c}\phi$ occurs $n-q$ times add up to $c_{n,q}\,\mathrm{dd^c} v_1 \wedge \cdots \wedge \mathrm{dd^c} v_q \wedge (\mathrm{dd^c}\phi)^{n-q}$, where $c_{n,q} = n^{-n}\,n!\,(n-q)^{n-q}/(n-q)! > 0$ (with $0^0 = 1$) is the corresponding multinomial coefficient. Hence $\mathrm{dd^c} v_1 \wedge \cdots \wedge \mathrm{dd^c} v_q \wedge (\mathrm{dd^c}\phi)^{n-q} \leq c_{n,q}^{-1}(\mathrm{dd^c} F)^n$. Since $\mathrm{dd^c}\phi = \beta/(\pi R'^2)$ and $\mathrm{dd^c} u_l = 2M\,\mathrm{dd^c} v_l$, it follows that $\int_E \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge \beta^{n-q} \leq C\,\mathrm{Cap}_{B'}(E)$ for Borel $E \subset B'$. The bound for the coefficient measures follows from Theorem 7.2.5, and Proposition 9.3.3(4), applied to $B' \Subset B$ and K, replaces $\mathrm{Cap}_{B'}$ by Cap_B on subsets of K.

(2) The weak limit S is a positive current (Theorem 7.2.5). Let σ_j be either the trace measure or a coefficient measure of S_j, and σ the corresponding measure of S; then $\int \psi\,\mathrm{d}\sigma_j \to \int \psi\,\mathrm{d}\sigma$ for every test function ψ, since this integral is the value of S_j on a test form. Fix a continuous function χ with compact support in B and a compact set $K \subset B$ with $\mathrm{Supp}\,\chi \subset \mathrm{Int}\,K$, and let C be the constant of (1) for K; in particular $|\sigma_j|(K) \leq C\,\mathrm{Cap}_B(K)$, which is finite by Proposition 9.3.3(1). A similar bound holds for σ on open subsets $U \subset \mathrm{Int}\,K$: for the trace measure, which is the vague limit of the trace measures of S_j (Section 7.2), Proposition 1.5.2 gives $\sigma(U) \leq \varliminf_j \sigma_j(U) \leq C\,\mathrm{Cap}_B(U)$, and the total variations of the coefficient measures of the positive current S are bounded by a constant multiple of its trace measure.

Since the total variations $|\sigma_j|(K)$ are bounded, the convergence $\int \psi\,\mathrm{d}\sigma_j \to \int \psi\,\mathrm{d}\sigma$ extends to continuous functions ψ with support in $\mathrm{Int}\,K$: approximating ψ uniformly by test functions supported in a fixed compact subset of $\mathrm{Int}\,K$ (Lemma 1.4.1(5)) changes the integrals by at most $\sup|\psi - \psi_\varepsilon|$ times a bound for $|\sigma_j|(K)$ and $|\sigma|(K)$.

Consider first the fixed weight f, and let $\eta > 0$. Since f is quasi-continuous, there is an open set $G \subset B$ with $\mathrm{Cap}_B(G) < \eta$ such that f is continuous on $K \setminus G$; let g be a continuous extension of $f|_{K\setminus G}$ to $\mathbb{C}^n$ with $\sup|g| \leq \sup_K |f|$ (Lemma 9.4.3). Then

$$\Big|\int \chi f\,\mathrm{d}\sigma_j - \int \chi g\,\mathrm{d}\sigma_j\Big| \leq 2\sup_K |f|\sup|\chi|\,|\sigma_j|(G \cap \mathrm{Supp}\,\chi) \leq C'\eta,$$

and similarly for σ, using the open set $G \cap \mathrm{Int}\,K$. Since χg is continuous with support in $\mathrm{Int}\,K$, we have $\int \chi g\,\mathrm{d}\sigma_j \to \int \chi g\,\mathrm{d}\sigma$. Hence $\varlimsup_j \big|\int \chi f\,\mathrm{d}\sigma_j - \int \chi f\,\mathrm{d}\sigma\big| \leq 2C'\eta$ for every $\eta > 0$, which proves $f\sigma_j \to f\sigma$.

For varying weights, let M_K bound $|f_j|$ and $|f|$ on K, and let $\delta > 0$. By (1),

$$\Big|\int \chi(f_j - f)\,\mathrm{d}\sigma_j\Big| \leq \sup|\chi|\Big(\delta\,C\,\mathrm{Cap}_B(K) + 2M_K C\,\mathrm{Cap}_B^*\big(K \cap \{|f_j - f| > \delta\}\big)\Big).$$

As $j \to \infty$, the second term tends to zero by convergence in capacity; then we let $\delta \downarrow 0$. Together with the fixed-weight case, this proves $f_j\sigma_j \to f\sigma$, and hence $f_jS_j \to fS$. □

Theorem 9.4.8 *Let $0 \le q \le n$ be an integer. For $1 \le l \le q$, let $(u_{l,j})_{j\ge1}$ be a sequence of plurisubharmonic functions on D, uniformly bounded on each compact subset of D, which converges locally in capacity to a locally bounded plurisubharmonic function u_l. Then*

$$\mathrm{dd^c}u_{1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j} \longrightarrow \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q$$

weakly. More generally, if f_j are real Borel functions on D, uniformly bounded on each compact subset of D, which converge locally in capacity to a locally bounded, quasi-continuous real Borel function f, then $f_j\mathrm{dd^c}u_{1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j} \to f\,\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q$ weakly.

Proof Weak convergence is local (Lemma 9.2.4), so it suffices to prove the convergence on every ball $B \Subset D$. By the remark following Definition 9.4.5, the convergence in capacity holds with Cap_B in place of Cap_D on every ball $B \Subset D$. The limit f is quasi-continuous for Cap_B as well: if $K \subset B$ is compact, K' is a compact neighborhood of K in B, and $G \subset D$ is open with f continuous on $K' \setminus G$, then $G \cap \operatorname{Int} K'$ is open in B, f is continuous on $K \setminus (G \cap \operatorname{Int} K') = K \setminus G$, and $\mathrm{Cap}_B(G \cap \operatorname{Int} K') \le C\,\mathrm{Cap}_D(G)$ by Proposition 9.3.3(4) applied to K'. Finally, the functions $u_{l,j}$ and u_l, restricted to B, form a family which is uniformly bounded on compact subsets of B. We may therefore work on such a ball and use Lemma 9.4.7.

We argue by induction on q. For $q = 0$ the product is 1, and the weighted assertion is the second part of the lemma with $S_j = 1$. Let $q \ge 1$, and assume that $S_j = \mathrm{dd^c}u_{2,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j}$ converges weakly to $S = \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q$. The functions $u_{1,j}$ converge in capacity to u_1, which is quasi-continuous by Theorem 9.4.2; hence the second part of the lemma gives $u_{1,j}S_j \to u_1S$, and applying $\mathrm{dd^c}$ proves the unweighted assertion for q factors. The weighted assertion then follows from the second part of the lemma once more. □

Unlike Theorem 9.2.5, Theorem 9.4.8 concerns only products with the base current $T = 1$.

9.5 Increasing limits and exceptional envelopes

The supremum of an increasing sequence of plurisubharmonic functions need not be upper semicontinuous, and it may differ from its upper semicontinuous regularization on an exceptional set. For instance, on the unit ball of $\mathbb{C}^n$ the functions $\max\{j^{-1}\log|z_1|, -1\}$ are plurisubharmonic (Propositions 8.2.5 and 8.3.3) and increase with j, since $\log|z_1| < 0$ there; their supremum equals 0 where $z_1 \ne 0$ and

-1 on the hyperplane $\{z_1 = 0\}$, while its upper semicontinuous regularization is 0. We show that the products are nevertheless continuous along increasing sequences. The proof follows the pattern of the decreasing case: the inequality $u_{0,j} \le u_0$ bounds weak limits from above, while the quasi-continuity of the fixed weights, through Lemma 9.4.7, supplies the lower bound for the masses. As a consequence, the exceptional set of an upper envelope has outer capacity zero.

Theorem 9.5.1 *Let $0 \le q \le n$ be an integer. For $0 \le l \le q$, let $(u_{l,j})_{j\ge 1}$ be an increasing sequence of locally bounded plurisubharmonic functions on D which converges almost everywhere to a locally bounded plurisubharmonic function u_l. Then*

$$\mathrm{dd^c}u_{1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j} \longrightarrow \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q,$$
$$u_{0,j}\mathrm{dd^c}u_{1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j} \longrightarrow u_0\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q$$

weakly. Moreover $u_{l,j} \to u_l$ locally in capacity for every l.

Proof We first note that $u_{l,j} \le u_l$ everywhere. Indeed, $u_{l,j} \le u_l$ almost everywhere, so the plurisubharmonic functions $\max\{u_{l,j}, u_l\}$ and u_l agree almost everywhere, and hence everywhere, by the uniqueness of plurisubharmonic representatives in Theorem 8.2.1, applied on each connected component of D. Consequently $u_{l,1} \le u_{l,j} \le u_l$, and the sequences are uniformly bounded on each compact subset of D.

As in the proof of Theorem 9.2.5, we prove the weighted assertion by induction on q, for all bounded open sets D and all sequences as in the statement; convergence in capacity is proved at the end. For $q = 0$ it follows from dominated convergence with respect to Lebesgue measure. Let $q \ge 1$ and assume the weighted assertion for $q-1$ factors; applying $\mathrm{dd^c}$ to it, with the weight $u_{1,j}$, gives the unweighted assertion for q factors. Put $S_j = \mathrm{dd^c}u_{1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{q,j}$ and $S = \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q$, so that $S_j \to S$ weakly.

Weak convergence is a local property (Lemma 9.2.4). Fix $x \in D$ and $r > 0$ with $\overline{B(x,4r)} \subset D$, and put $B = B(x,3r)$ and $B_1 = B(x,4r)$; we prove the convergence on $B(x,2r)$. Subtracting constants does not affect the assertions, since the unweighted assertion for q factors is known, so we may assume that $-N \le u_{l,j}, u_l \le -1$ on B_1 for all l and j, where $N \ge 1$. Put $\psi = |z-x|^2 - 9r^2$, choose $\delta_0 > 0$ such that $\overline{B(x,2r)} \subset \{\psi < -\delta_0\}$, and replace each function v among the $u_{l,j}$ and u_l by the function $\max\{v, A\psi\}$ of (9.4), with $U = B_1$, δ_0 in place of δ and $A = N/\delta_0$. The new functions are plurisubharmonic on B_1 and take values in $[-N, 0)$ on B; the new sequences still increase, satisfy $u_{l,j} \le u_l$ and converge almost everywhere. The functions do not change on $B(x,2r)$, and by locality neither do the products. On the collar $V = \{z \in B_1 : \psi(z) > -1/A\}$ of ∂B the new functions coincide with $A\psi$.

By locality, on V all the currents considered below coincide with fixed currents, independent of j and k. Hence, if $\chi \in C_c^\infty(B)$ equals 1 on $B \setminus V$, the integral over B of each of these currents splits into its integral against χ, to which weak convergence applies, and its integral against $(1-\chi)\mathbf{1}_B$, which only involves values on V and therefore does not change; we use this without further comment. The currents $-u_{0,j}S_j$ are positive on B, and their masses on compact subsets of B are bounded uniformly in j by Theorem 9.1.3; by the compactness part of Theorem 7.2.5, every subsequence

of $(u_{0,j}S_j)$ has a further subsequence converging weakly on B. Let Θ be the limit of such a subsequence.

Since $u_{0,j} \le u_0$, the currents $(u_0 - u_{0,j})S_j$ are positive. Let φ be a strongly positive test form on B of complementary bidegree. The positive measures $-u_0 S_j \wedge \varphi$ have bounded masses on compact sets, so by Theorem 1.5.5 a further subsequence of $u_0 S_j \wedge \varphi$ converges weakly to some ν, and Lemma 9.2.2, applied to the constant sequence $f_j = u_0$ and the measures $S_j \wedge \varphi \to S \wedge \varphi$, gives $\nu \le u_0 S \wedge \varphi$. The measures $(u_0 - u_{0,j})S_j \wedge \varphi$ are positive and converge along this subsequence to $\nu - \Theta \wedge \varphi$, which is therefore positive. Hence $\Theta \wedge \varphi \le u_0 S \wedge \varphi$. Evaluating this inequality of measures on a nonnegative test function equal to 1 on $\operatorname{Supp}\varphi$, we get $(u_0 S - \Theta)(\varphi) \ge 0$. Thus $u_0 S - \Theta$ is a positive current on B, and it vanishes on V, since all the currents $u_{0,j}S_j$ and $u_0 S$ coincide there with $A\psi(A\mathrm{dd^c}\psi)^q$. Write $\gamma = \beta^{n-q}$.

For fixed k and all $j \ge k$, positivity of $S_j \wedge \gamma$ and $u_{0,j} \ge u_{0,k}$ give

$$\int_B u_{0,j} S_j \wedge \gamma \ge \int_B u_{0,k} S_j \wedge \gamma.$$

Let $j \to \infty$ along the subsequence defining Θ. The left-hand side tends to $\int_B \Theta \wedge \gamma$. The fixed weight $u_{0,k}$ is quasi-continuous for Cap_B by Theorem 9.4.2, and the functions $u_{l,j}$, u_l form a family which is uniformly bounded on B; so by Lemma 9.4.7(2) the right-hand side tends to $\int_B u_{0,k} S \wedge \gamma$. By the second assertion of Corollary 9.2.6, applied with $a = u_{0,k}$, $b = u_1$ and $R = \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \gamma$, and by symmetry,

$$\int_B u_{0,k} S \wedge \gamma = \int_B u_1 \, \mathrm{dd^c}u_{0,k} \wedge \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \gamma.$$

As $k \to \infty$, the unweighted assertion for q factors, already established, gives $\mathrm{dd^c}u_{0,k} \wedge \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q \to \mathrm{dd^c}u_0 \wedge \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_q$ weakly, and Lemma 9.4.7, with the fixed quasi-continuous weight u_1, passes the integral to the limit. A second application of Corollary 9.2.6 identifies the limit with $\int_B u_0 \, \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge \gamma = \int_B u_0 S \wedge \gamma$. We have shown that $\int_B \Theta \wedge \gamma \ge \int_B u_0 S \wedge \gamma$.

Since $u_0 S - \Theta$ is positive with compact support in B, its trace measure has total mass $\int_B (u_0 S - \Theta) \wedge \gamma \le 0$ and therefore vanishes, and $\Theta = u_0 S$ by Theorem 7.2.5. Thus every subsequence of $(u_{0,j}S_j)$ has a further subsequence converging weakly to $u_0 S$ on B. Hence the whole sequence converges to $u_0 S$ on B: otherwise there would be a test form φ on B and a subsequence along which $|(u_{0,j}S_j - u_0 S)(\varphi)|$ stays bounded away from zero, and no further subsequence of it could converge to $u_0 S$. Since the modifications change neither the functions nor the weighted products on $B(x, 2r)$, the original weighted products converge on $B(x, 2r)$; this completes the induction.

Now that both assertions are known for every $q \le n$, we prove convergence in capacity. Fix l, a compact set $K \subset D$ and $\delta > 0$. Since $u_{l,j} \le u_l$, the sets $\{u_{l,j} > u_l + \delta\}$ are empty. Covering K by finitely many balls and using subadditivity of outer capacity, we may assume that $K \subset \overline{B(x, r)}$ with $\overline{B(x, 4r)} \subset D$; put $L = \overline{B(x, 3r/2)}$. Let $\hat{u}_{l,j}$ and $\hat{u}_l$ be the functions obtained by the modification above, on $B = B(x, 3r)$; on $\{\psi < -\delta_0\} \supset L$ we have $\hat{u}_l - \hat{u}_{l,j} = u_l - u_{l,j}$. These functions are bounded and

plurisubharmonic on B_1, they satisfy $\hat u_{l,j} \le \hat u_l$, and both coincide with $A\psi$ on the collar V of ∂B. By (9.6), applied with $u = \hat u_{l,j}$ and $h = \hat u_l$,

$$\operatorname{Cap}_D\bigl(B \cap \{\hat u_l - \hat u_{l,j} > \delta/2\}\bigr) \le \frac{(n+1)!\,2^{n+1}}{\delta^{n+1}} \int_B (\hat u_l - \hat u_{l,j})(\mathrm{dd^c}\hat u_{l,j})^n.$$

The right-hand side tends to zero: $\int_B \hat u_l(\mathrm{dd^c}\hat u_{l,j})^n \to \int_B \hat u_l(\mathrm{dd^c}\hat u_l)^n$ by the unweighted assertion and Lemma 9.4.7(2) with the fixed quasi-continuous weight $\hat u_l$, and $\int_B \hat u_{l,j}(\mathrm{dd^c}\hat u_{l,j})^n \to \int_B \hat u_l(\mathrm{dd^c}\hat u_l)^n$ by the weighted assertion with all factors and the weight equal to $\hat u_{l,j}$; as before, the integrals over B are split by a cutoff. The real function $u_l - u_{l,j}$ is quasi-continuous on D (Theorem 9.4.2 and the remark following Definition 9.4.1), so Lemma 9.4.4 gives

$$\begin{aligned}\operatorname{Cap}_D^*\bigl(K \cap \{u_l - u_{l,j} > \delta\}\bigr) &\le \operatorname{Cap}_D\bigl(L \cap \{u_l - u_{l,j} > \delta/2\}\bigr)\\ &\le \operatorname{Cap}_D\bigl(B \cap \{\hat u_l - \hat u_{l,j} > \delta/2\}\bigr) \longrightarrow 0.\end{aligned}$$

Corollary 9.5.2 *Let $\mathcal{F}$ be a nonempty family of plurisubharmonic functions on D which is locally uniformly bounded above, and let $F = \sup_{v\in\mathcal{F}} v$ be its pointwise supremum and F^* its upper semicontinuous regularization. Then* $\operatorname{Cap}_D^*(K \cap \{F < F^*\}) = 0$ *for every compact set $K \subset D$.*

Proof By Theorem 8.3.2, $u = F^*$ is plurisubharmonic, and there is a countable subfamily $\{w_1, w_2, \ldots\} \subset \mathcal{F}$ with $(\sup_k w_k)^* = u$. The functions $v_j = \max\{w_1, \ldots, w_j\}$ are plurisubharmonic (Proposition 8.2.5) and increase to $G = \sup_k w_k$, and $G = G^* = u$ almost everywhere by Theorem 8.3.2, applied to the countable family. Fix an integer $m \ge 1$. The functions $\max\{v_j, -m\}$ are locally bounded and plurisubharmonic on D, and they increase almost everywhere to the locally bounded plurisubharmonic function $\max\{u, -m\}$, so by Theorem 9.5.1 they converge to it locally in capacity. For rational $\delta > 0$, the set where $\sup_j \max\{v_j, -m\} < \max\{u, -m\} - \delta$ is contained in $\{\max\{v_k, -m\} < \max\{u, -m\} - \delta\}$ for every k, and the outer capacities of the intersections of the latter sets with a compact set tend to zero as $k \to \infty$. Taking the union over rational δ, countable subadditivity shows that the set

$$E_m = \Bigl\{\sup_j \max\{v_j, -m\} < \max\{u, -m\}\Bigr\}$$

meets every compact set in a set of outer capacity zero.

If $\sup_j v_j(x) < u(x)$, then $u(x) > -\infty$, so $u(x) > -m$ for some integer m, and then $x \in E_m$. Hence $\{\sup_j v_j < u\} \subset \bigcup_m E_m$ also meets every compact set in a set of outer capacity zero. Finally, $\sup_j v_j \le F \le u$, so $\{F < F^*\} \subset \{\sup_j v_j < u\}$. (Since outer capacity is defined for all sets, we need not know whether $\{F < F^*\}$ is Borel.) □

9.6 The homogeneous Dirichlet problem and balayage

A plurisubharmonic function is maximal if it cannot be increased on a compact set without losing plurisubharmonicity (Definition 9.6.5). We show in Theorem 9.6.6 that, for locally bounded functions, maximality is equivalent to $(\mathrm{dd}^c u)^n = 0$. The proof replaces u on a ball by a solution H of the homogeneous equation $(\mathrm{dd}^c H)^n = 0$ with the same boundary values (Corollary 9.6.4); in Chapter 10 the same replacement shows that relative extremal functions have vanishing Monge–Ampère measure away from the sets on which they are constrained. We construct H as an upper envelope. To show that it solves the equation, we bound its second differences by composing with automorphisms of the ball (Lemma 9.6.1), which makes it twice differentiable almost everywhere (Lemma 9.6.2).

Lemma 9.6.1 *Let $B \subset \mathbb{C}^n$ be the unit ball. For $a \in B$ and $h \in \mathbb{C}^n$ with $|h|$ small, depending on a, there are biholomorphic automorphisms $S_{a,h}$ of B such that $S_{a,0}$ is the identity and $S_{a,h}(a) = a + h$. Each $S_{a,h}$ extends to a holomorphic map on a neighborhood of $\overline{B}$ which maps ∂B onto ∂B. For every compact set $K \subset B$ there is $h_0 > 0$ such that the map $(a, h, z) \mapsto S_{a,h}(z)$ is defined and smooth on an open subset of $\mathbb{C}^n \times \mathbb{C}^n \times \mathbb{C}^n$ containing $K \times \overline{B(0, h_0)} \times \overline{B}$, and holomorphic in z there. In particular, there is a constant C such that, for $a \in K$, $|h| \le h_0$ and $z \in \overline{B}$, the first and second derivatives of $S_{a,h}(z)$ with respect to the real variable $h \in \mathbb{C}^n = \mathbb{R}^{2n}$ are bounded by C.*

Proof Use the Hermitian product $\langle z, a\rangle = \sum z_j \bar{a}_j$, put $s_a = \sqrt{1 - |a|^2}$, and define

$$T_a(z) = \frac{s_a z + \langle z, a\rangle a/(1 + s_a) - a}{1 - \langle z, a\rangle}.$$

This expression depends smoothly on (a, z) in the open set $\{(a, z) : |a| < 1,\ |a||z| < 1\}$, where the denominator does not vanish, and it is holomorphic in z; moreover T_0 is the identity. If U is a unitary matrix, then $\langle Uz, Ua\rangle = \langle z, a\rangle$ and $s_{Ua} = s_a$, so $T_{Ua}(Uz) = UT_a(z)$. Given $a \neq 0$, choose U with $Ua = (t, 0, \ldots, 0)$, $t = |a|$; it therefore suffices to verify the following identities for $a = (t, 0, \ldots, 0)$ with $0 \le t < 1$, and also for $-a = (-t, 0, \ldots, 0)$. Put $s = s_a = \sqrt{1 - t^2}$. Then $\langle z, a\rangle = tz_1$, and the first component of the numerator of $T_a(z)$ is $sz_1 + t^2 z_1/(1 + s) - t = z_1 - t$, because $s(1 + s) + t^2 = s + s^2 + t^2 = 1 + s$. Hence

$$T_a(z) = \left(\frac{z_1 - t}{1 - tz_1}, \frac{s\, z_2}{1 - tz_1}, \ldots, \frac{s\, z_n}{1 - tz_1}\right),$$

$$T_{-a}(w) = \left(\frac{w_1 + t}{1 + tw_1}, \frac{s\, w_2}{1 + tw_1}, \ldots, \frac{s\, w_n}{1 + tw_1}\right),$$

the second formula by the same computation with t replaced by $-t$. In particular $T_a(a) = 0$ and $T_{-a}(0) = a$. If $w = T_a(z)$, then

$$w_1 + t = \frac{(1-t^2)z_1}{1-tz_1}, \qquad 1 + tw_1 = \frac{1-t^2}{1-tz_1},$$

so the first component of $T_{-a}(w)$ is z_1 and the others are $s \cdot sz_k/(1-tz_1) \cdot (1-tz_1)/(1-t^2) = z_k$; thus $T_{-a} \circ T_a = \mathrm{id}$, and by symmetry $T_a \circ T_{-a} = \mathrm{id}$. Finally,

$$|1-tz_1|^2 - |z_1 - t|^2 - (1-t^2)\sum_{k\geq 2}|z_k|^2 = (1-t^2)(1-|z_1|^2) - (1-t^2)\sum_{k\geq 2}|z_k|^2,$$

because the terms $\mp 2t \operatorname{Re} z_1$ cancel; dividing by $|1-tz_1|^2$ gives

$$1 - |T_a(z)|^2 = \frac{(1-|a|^2)(1-|z|^2)}{|1-\langle z, a\rangle|^2}.$$

Hence T_a maps B into B and the sphere ∂B into itself, and the same holds for T_{-a}; since T_{-a} is inverse to T_a, the map T_a is a biholomorphic automorphism of B which maps ∂B onto itself.

Set $S_{a,h} = T_{-(a+h)} \circ T_a$ for $|a+h| < 1$. It maps a to $T_{-(a+h)}(0) = a + h$ and is the identity for $h = 0$. The map T_a is holomorphic on the neighborhood $\{z : |a||z| < 1\}$ of $\overline{B}$, and, as we have just seen, it maps $\overline{B}$ into $\overline{B}$ and ∂B onto ∂B; the same holds for $T_{-(a+h)}$. Hence $S_{a,h}$ is holomorphic on the neighborhood $\{z : |a||z| < 1,\ |a+h||T_a(z)| < 1\}$ of $\overline{B}$, and it maps ∂B onto ∂B. Let $K \subset B$ be compact (we may assume $K \neq \varnothing$), put $t_K = \max_K |a| < 1$, and choose $h_0 > 0$ with $t_K + 3h_0 < 1$. On the open set O of all (a, h, z) with $|a| < t_K + h_0$, $|h| < 2h_0$, $|a||z| < 1$ and $(t_K + 3h_0)|T_a(z)| < 1$, we have $|a+h| < 1$ and $|a+h||T_a(z)| < 1$, so $(a, h, z) \mapsto S_{a,h}(z)$ is defined and smooth on O, and holomorphic in z. Since $|T_a(z)| \leq 1$ for $|z| \leq 1$, the set O contains the compact set $K \times \overline{B(0, h_0)} \times \overline{B}$, on which the continuous first and second derivatives of $S_{a,h}(z)$ with respect to h are therefore bounded by a constant C. □

A plurisubharmonic function whose second differences are bounded above is locally of class $C^{1,1}$, and its Monge–Ampère measure is then given by the determinant of its complex Hessian.

Lemma 9.6.2 *Let $U \subset \mathbb{C}^n$ be open, $n \geq 1$, and let H be a locally bounded plurisubharmonic function on U. Suppose that for every compact set $K \subset U$ there are constants $C_K \geq 0$ and $h_K > 0$ such that, for all $a \in K$ and all $h \in \mathbb{C}^n$ with $|h| \leq h_K$, the points $a \pm h$ lie in U and*

$$H(a+h) + H(a-h) - 2H(a) \leq 2C_K|h|^2 \qquad (a \in K,\ |h| \leq h_K). \tag{9.7}$$

Then H is continuous, $H \in C^{1,1}_{\mathrm{loc}}(U)$, and the distributional real Hessian of H is a matrix $A = (A_{ab})$ of measurable functions which are bounded on compact subsets of U. Almost every $x \in U$ satisfies

$$H(x+h) = H(x) + \nabla H(x) \cdot h + \tfrac{1}{2}h^t A(x)h + o(|h|^2) \qquad (h \to 0).$$

Moreover, let $H_{j\bar{k}}$ be the coefficients of the distributional complex Hessian of H, which are fixed complex linear combinations of the A_{ab}. Then the matrix $(H_{j\bar{k}})$ is positive semidefinite almost everywhere, and

$$(\mathrm{dd^c}H)^n = \frac{n!}{\pi^n}\det(H_{j\bar{k}})\,\mathrm{d}V_{2n}.$$

Proof Let $H_\varepsilon = H{*}\rho_\varepsilon$. Fix $a_0 \in U$ and $r > 0$ with $\overline{B(a_0, 4r)} \subset U$, put $K = \overline{B(a_0, 3r)}$, and let $M_0 = \sup_K |H|$, which is finite because H is locally bounded. For $0 < \varepsilon < r$, the regularization H_ε is smooth and plurisubharmonic on a neighborhood of $\overline{B(a_0, 2r)}$ (Proposition 8.2.2), and $|H_\varepsilon| \le M_0$ on $\overline{B(a_0, 2r)}$, since its values there are averages of values of H on K. For $x \in \overline{B(a_0, 2r)}$ and $|h| \le h_K$,

$$\begin{aligned} &H_\varepsilon(x+h) + H_\varepsilon(x-h) - 2H_\varepsilon(x) \\ &\quad = \int \big(H(x-y+h) + H(x-y-h) - 2H(x-y)\big)\rho_\varepsilon(y)\,\mathrm{d}y \le 2C_K|h|^2, \end{aligned}$$

by (9.7), because $x - y \in K$ for $|y| \le \varepsilon$. Replacing h by th with $0 < t \le 1$, dividing by t^2 and letting $t \to 0$, we get $D^2H_\varepsilon(x)[h,h] \le 2C_K|h|^2$ for $|h| \le h_K$, and hence, by homogeneity, for all $h \in \mathbb{C}^n = \mathbb{R}^{2n}$. On the other hand, the complex Hessian of the smooth plurisubharmonic function H_ε is positive semidefinite (Proposition 8.2.2), and $D^2H_\varepsilon[h,h] + D^2H_\varepsilon[\mathrm{i}h, \mathrm{i}h] = 4\sum_{j,k}(H_\varepsilon)_{j\bar{k}}h_j\bar{h}_k \ge 0$; hence $D^2H_\varepsilon(x)[h,h] \ge -2C_K|h|^2$ as well. Thus the second derivatives of the H_ε are bounded on $\overline{B(a_0, 2r)}$ by a constant Λ_0 independent of ε. For $x \in \overline{B(a_0, r)}$ and a unit vector e, Taylor's formula

$$H_\varepsilon(x+re) = H_\varepsilon(x) + r\,\partial_e H_\varepsilon(x) + \int_0^r (r-t)D^2H_\varepsilon(x+te)[e,e]\,\mathrm{d}t$$

gives $|\partial_e H_\varepsilon(x)| \le 2M_0/r + \Lambda_0 r = \Lambda$. By the mean value theorem on the convex set $B(a_0, r)$, $|H_\varepsilon(x) - H_\varepsilon(y)| \le \Lambda|x-y|$ for $x, y \in B(a_0, r)$. Since $H_\varepsilon \to H$ pointwise as $\varepsilon \downarrow 0$, by (3.3), the same inequality holds for H. Hence H is continuous on $B(a_0, r)$, and, as a_0 was arbitrary, on U.

Moreover, $H_\varepsilon \to H$ in L^1 on compact subsets of U, by dominated convergence, and hence in the sense of distributions; and every ball $B_0 \Subset U$ is covered by finitely many balls $B(a_0, 2r)$ as above, so the second derivatives of the H_ε are bounded on B_0 uniformly in small ε. By the converse part of Proposition 1.7.5, the distributional real Hessian of H has locally bounded measurable coefficients A_{ab}; hence $H \in C^{1,1}_{\mathrm{loc}}(U)$, and almost every $x \in U$ satisfies the second-order expansion of the statement.

The functions $H_{j\bar{k}}$ are bounded measurable functions on compact subsets of U. By Proposition 1.4.3, the complex Hessian of H_ε has the coefficients $(H_{j\bar{k}})_\varepsilon$; these are bounded on compact sets uniformly in small ε, and they converge to $H_{j\bar{k}}$ at every Lebesgue point of $H_{j\bar{k}}$, hence almost everywhere (Theorem 1.7.3). In particular, the matrix $(H_{j\bar{k}})$ is positive semidefinite almost everywhere, as an almost everywhere limit of the positive semidefinite complex Hessians of the smooth plurisubharmonic

functions H_ε. Since H is continuous, $H_\varepsilon \to H$ locally uniformly (Lemma 1.4.1(1)), so $(\mathrm{dd^c} H_\varepsilon)^n \to (\mathrm{dd^c} H)^n$ weakly on every ball $B_0 \Subset U$ as $\varepsilon \downarrow 0$ along a sequence, by Proposition 9.1.4 applied on B_0. By (9.2), $(\mathrm{dd^c} H_\varepsilon)^n = n!\pi^{-n} \det((H_{j\bar{k}})_\varepsilon)\, \mathrm{d}V_{2n}$, and the determinants converge almost everywhere and are bounded on compact sets uniformly in small ε. Hence, for every test function χ supported in such a ball B_0, dominated convergence gives

$$\int \chi\, (\mathrm{dd^c} H)^n = \frac{n!}{\pi^n} \int \chi \det(H_{j\bar{k}})\, \mathrm{d}V_{2n}.$$

By Lemma 1.4.2, every test function on U is a finite sum of test functions supported in balls $B_0 \Subset U$, so this identity holds for all test functions on U, and the two Radon measures coincide by Proposition 1.4.3(4). □

Theorem 9.6.3 *Let $B \subset \mathbb{C}^n$ be a ball and φ a real continuous function on ∂B. There is a unique $H \in \mathrm{PSH}(B) \cap C(\overline{B})$ such that*

$$(\mathrm{dd^c} H)^n = 0 \quad \text{on } B, \qquad H|_{\partial B} = \varphi.$$

It is the upper envelope of the class of all $v \in \mathrm{PSH}(B)$ with $\overline{\lim}_{B \ni z \to x} v(z) \le \varphi(x)$ for every $x \in \partial B$. If φ is the restriction of a C^2 function on a neighborhood of ∂B, then $H \in C^{1,1}_{\mathrm{loc}}(B)$.

Proof After an affine change of coordinates $z \mapsto z_0 + Rz$ with $R > 0$, we may assume that B is the unit ball; plurisubharmonicity, the boundary conditions, C^2 data and the vanishing of the Monge–Ampère measure are preserved by such a change, by Proposition 8.2.3 and Proposition 9.1.2(5). We call the functions v in the statement the *subsolutions for φ* of the homogeneous equation. If v is a subsolution for φ and $\varphi \le \varphi' + c$ on ∂B with a constant c, then $v - c$ is a subsolution for φ'. Every subsolution is subharmonic on B (Theorem 8.2.1), so the comparison principle (Theorem 3.1.3 and the remark following it), applied with the harmonic function $P\varphi$, the Poisson integral of φ given by Theorem 2.3.1, gives $v \le P\varphi$, and $P\varphi \le \max_{\partial B} \varphi$ because the Poisson kernel is positive with integral 1. Hence, as soon as the class of subsolutions is nonempty, its upper envelope H is defined, and $H \le P\varphi$.

Step 1: the envelope for C^2 data. Suppose that φ is the restriction of a C^2 function on a neighborhood of ∂B. Multiplying this function by a cutoff equal to 1 near ∂B, we obtain a C^2 function $\tilde{\varphi}$ with compact support in $\mathbb{C}^n$ and $\tilde{\varphi} = \varphi$ on ∂B. Its complex Hessian is continuous with compact support, so there is $\lambda > 0$ with $\sum_{j,k} \tilde{\varphi}_{j\bar{k}} \xi_j \bar{\xi}_k \ge -\lambda |\xi|^2$ everywhere. The function $s = \tilde{\varphi} + \lambda(|z|^2 - 1)$ is then of class C^2 with positive semidefinite complex Hessian, so it is plurisubharmonic by Proposition 8.2.2. As $s = \varphi$ on ∂B, it is a subsolution. Hence $s \le H \le P\varphi$. By Theorem 8.3.2, the upper semicontinuous regularization H^* is plurisubharmonic, and $s \le H^* \le P\varphi$ since $P\varphi$ is continuous. As s and $P\varphi$ are continuous on $\overline{B}$ with boundary values φ, we have $H^*(z) \to \varphi(x)$ as $z \to x \in \partial B$; so H^* is a subsolution, $H^* \le H$, and $H = H^*$ is plurisubharmonic. Moreover $H(z) \to \varphi(x)$ as $z \to x \in \partial B$, and H is bounded.

Step 2: second differences. Let $K \subset B$ be compact, and let $S_{a,h}$, h_0 and C be as in Lemma 9.6.1. For $x \in \partial B$ and $a \in K$, the function $g(h) = \tilde{\varphi}(S_{a,h}(x))$, $|h| \le h_0$, is of class C^2, and by the chain rule its second derivatives are bounded by a constant $2C_K$ independent of a and x, since the first and second derivatives of $\tilde{\varphi}$ are bounded and those of $S_{a,h}(x)$ with respect to h are bounded by C. Since $g(0) = \varphi(x)$, Taylor's formula with integral remainder gives $g(h) + g(-h) - 2g(0) = \int_0^1 (1-t)\big(D^2 g(th)[h,h] + D^2 g(-th)[h,h]\big)\,\mathrm{d}t \le 2C_K|h|^2$. As $S_{a,\pm h}$ maps ∂B to ∂B, where $\tilde{\varphi} = \varphi$, this reads

$$\tfrac{1}{2}\big(\varphi \circ S_{a,h} + \varphi \circ S_{a,-h}\big) \le \varphi + C_K|h|^2 \quad \text{on } \partial B \qquad (a \in K,\ |h| \le h_0).$$

The function $w = \frac{1}{2}(H \circ S_{a,h} + H \circ S_{a,-h}) - C_K|h|^2$ is plurisubharmonic on B, by Propositions 8.2.3 and 8.2.5, since $S_{a,\pm h}$ are automorphisms of B and H is bounded. If $y \in \partial B$ and $z \to y$ in B, then $S_{a,\pm h}(z) \to S_{a,\pm h}(y) \in \partial B$, so $H(S_{a,\pm h}(z)) \to \varphi(S_{a,\pm h}(y))$ by Step 1, and the preceding inequality gives $\overline{\lim}_{z\to y} w(z) \le \varphi(y)$. Thus w is a subsolution, so $w \le H$. Evaluating at $z = a$, where $S_{a,\pm h}(a) = a \pm h$, we obtain

$$H(a+h) + H(a-h) - 2H(a) \le 2C_K|h|^2 \qquad (a \in K,\ |h| \le h_0).$$

Since K was arbitrary and the points $a \pm h = S_{a,\pm h}(a)$ lie in B, Lemma 9.6.2 applies on $U = B$. Thus H is continuous on B and belongs to $C^{1,1}_{\mathrm{loc}}(B)$, and its distributional real Hessian A and complex Hessian $(H_{j\bar{k}})$ have the properties stated there: the second-order expansion holds almost everywhere, $(H_{j\bar{k}})$ is positive semidefinite almost everywhere, and $(\mathrm{dd^c}H)^n = n!\,\pi^{-n}\det(H_{j\bar{k}})\,\mathrm{d}V_{2n}$.

Step 3: the equation. Suppose that $\det(H_{j\bar{k}}) > 0$ on a set of positive measure, and choose a point a of this set at which $(H_{j\bar{k}}(a))$ is positive semidefinite and the second-order expansion holds. The Taylor polynomial $Q(z) = H(a) + \nabla H(a) \cdot (z-a) + \frac{1}{2}(z-a)^t A(a)(z-a)$ is a real quadratic polynomial with real Hessian $A(a)$, so its complex Hessian is the constant matrix $(H_{j\bar{k}}(a))$; this matrix is positive semidefinite with positive determinant, hence positive definite. Let $\eta > 0$ be smaller than its smallest eigenvalue; then $Q - \eta|z-a|^2$ has positive semidefinite complex Hessian and is plurisubharmonic (Proposition 8.2.2). For sufficiently small $r > 0$ we have $\overline{B(a,r)} \subset B$ and, by the expansion, $|H - Q| < \eta r^2/2$ on the sphere $|z-a| = r$. The polynomial $\tilde{Q} = Q - \eta|z-a|^2 + \eta r^2/4$ is then strictly smaller than H on that sphere, since there $\tilde{Q} < H + \eta r^2/2 - 3\eta r^2/4$, and $\tilde{Q}(a) = H(a) + \eta r^2/4 > H(a)$. By Proposition 8.2.5, applied with $D = B(a,r)$ and $v = \tilde{Q}$, whose boundary inequality holds because $\tilde{Q}$ and H are continuous, the function equal to $\max\{H, \tilde{Q}\}$ on $B(a,r)$ and to H elsewhere is plurisubharmonic on B. It coincides with H near ∂B, so it is a subsolution, and it exceeds H at a, a contradiction. Hence $\det(H_{j\bar{k}}) = 0$ almost everywhere, and $(\mathrm{dd^c}H)^n = 0$. This proves the theorem for C^2 data, including the $C^{1,1}$ regularity, except for uniqueness.

Step 4: continuous data and uniqueness. For continuous φ, the function $z \mapsto \varphi(z/|z|)$ is continuous on the annulus $\{1/2 < |z| < 2\}$, and its mollifications φ_j, $j \ge 4$, are smooth on the neighborhood $\{3/4 < |z| < 3/2\}$ of ∂B and converge to φ uniformly on ∂B (Lemma 1.4.1(1)). Let H_j be the envelope for $\varphi_j|_{\partial B}$; by Steps 1–3 it is

continuous on $\overline{B}$ with boundary values φ_j, plurisubharmonic, and $(\mathrm{dd^c} H_j)^n = 0$. Put $c_j = \max_{\partial B} |\varphi_j - \varphi|$. Since $H_j - c_j$ is a subsolution for φ, the class of subsolutions for φ is nonempty, and its envelope H satisfies $H_j - c_j \le H$; symmetrically, $H - c_j \le H_j$. Therefore $\sup_B |H_j - H| \le c_j \to 0$, so H is a uniform limit of continuous functions on B; it is continuous, extends continuously to $\overline{B}$ with the boundary values φ, and is plurisubharmonic, being equal to its regularization H^*, which is plurisubharmonic by Theorem 8.3.2. By Proposition 9.1.4, $(\mathrm{dd^c} H)^n = \lim_j (\mathrm{dd^c} H_j)^n = 0$. Finally, if H' and H'' are two solutions, then $H' - H'' \to 0$ at ∂B, and the last assertion of Theorem 9.2.8, applied in both directions, gives $H' = H''$. □

Corollary 9.6.4 *Let u be a locally bounded plurisubharmonic function on an open set $\Omega \subset \mathbb{C}^n$, and let $B \Subset \Omega$ be a ball. There is a locally bounded plurisubharmonic function $\widehat{u}$ on Ω which equals u outside B, satisfies $\widehat{u} \ge u$ on B, and $(\mathrm{dd^c}\widehat{u})^n = 0$ on B. The function $\widehat{u}$ is increasing in u, and its restriction to B is the largest plurisubharmonic function v on B with $\overline{\lim}_{B \ni z \to x} v(z) \le u(x)$ for every $x \in \partial B$.*

Proof Let $u_j = u * \rho_{\varepsilon_j}$, with $\varepsilon_j \downarrow 0$ smaller than the distance from $\overline{B}$ to $\mathbb{C}^n \setminus \Omega$, be radial regularizations of u; they are smooth plurisubharmonic functions on a neighborhood of $\overline{B}$ decreasing to u (Proposition 8.2.2). Let H_j be the solution of Theorem 9.6.3 with boundary values $u_j|_{\partial B}$, that is, the upper envelope of the subsolutions for $u_j|_{\partial B}$. Since u_j is such a subsolution, $u \le u_j \le H_j$ on B. Since $u_{j+1} \le u_j$ on ∂B, every subsolution for $u_{j+1}|_{\partial B}$ is one for $u_j|_{\partial B}$, so the envelopes decrease: $H_{j+1} \le H_j$. Let $H = \lim_j H_j$. The function H_1 is subharmonic on B and continuous on $\overline{B}$ with boundary values u_1, so $H_1 \le \max_{\partial B} u_1$ by the comparison principle (Theorem 3.1.3). Hence $\inf_{\overline{B}} u \le H \le \max_{\partial B} u_1$, so H is bounded; it is plurisubharmonic by Theorem 8.3.1, and Theorem 9.2.5 gives $(\mathrm{dd^c} H)^n = \lim_j (\mathrm{dd^c} H_j)^n = 0$. For $x \in \partial B$, since $H \le H_j$ and H_j is continuous on $\overline{B}$ with $H_j = u_j$ on ∂B,

$$\overline{\lim_{B \ni z \to x}} H(z) \le u_j(x) \quad \text{for every } j, \qquad \text{hence} \qquad \overline{\lim_{B \ni z \to x}} H(z) \le u(x).$$

By Proposition 8.2.5, the function $\widehat{u}$ equal to $\max\{u, H\} = H$ on B and to u outside B is plurisubharmonic on Ω. It is locally bounded, since H is bounded and u is locally bounded; it equals u outside B, satisfies $\widehat{u} \ge u$ on B, and $(\mathrm{dd^c}\widehat{u})^n = (\mathrm{dd^c} H)^n = 0$ on B by locality.

Let v be plurisubharmonic on B with $\overline{\lim}_{B \ni z \to x} v(z) \le u(x)$ for $x \in \partial B$. Since $u \le u_j$, the function v is a subsolution for the boundary values $u_j|_{\partial B}$, so $v \le H_j$ by Theorem 9.6.3. Letting $j \to \infty$ gives $v \le H$. Conversely, H belongs to this class by the displayed estimate, so H is its largest member. If $u \le u'$, the class for u is contained in the class for u', which proves that $\widehat{u} \le \widehat{u'}$. □

Definition 9.6.5 Let $\Omega \subset \mathbb{C}^n$ be open. A plurisubharmonic function u on Ω is *maximal* if every plurisubharmonic function v on Ω with $v \le u$ outside a compact subset of Ω satisfies $v \le u$ everywhere.

Theorem 9.6.6 *A locally bounded plurisubharmonic function u on an open set $\Omega \subset \mathbb{C}^n$ is maximal if and only if $(\mathrm{dd}^{\mathrm{c}} u)^n = 0$. In particular, maximality is a local property for locally bounded plurisubharmonic functions.*

Proof Suppose that $(\mathrm{dd}^{\mathrm{c}} u)^n = 0$, and let v be plurisubharmonic on Ω with $v \le u$ outside a compact set K. Choose a bounded open set D with $K \subset D \Subset \Omega$. By upper semicontinuity, v is bounded above on $\overline{D}$; since u is bounded on $\overline{D}$, we may choose M with $-M < \inf_D u$. The function $\max\{v, -M\}$ is bounded and plurisubharmonic on D, and it is at most u on $D \setminus K$; since every point of D near ∂D lies in $D \setminus K$, we get $\varliminf_{D \ni z \to x} (u(z) - \max\{v(z), -M\}) \ge 0$ for $x \in \partial D$. The last assertion of Theorem 9.2.8 therefore gives $\max\{v, -M\} \le u$ on D. Hence $v \le u$ everywhere, and u is maximal.

Conversely, suppose that u is maximal, and let $B \Subset \Omega$ be a ball. The function $\widehat{u}$ of Corollary 9.6.4 is plurisubharmonic, satisfies $\widehat{u} \ge u$, and equals u outside the compact set $\overline{B}$. Maximality gives $\widehat{u} \le u$, so $\widehat{u} = u$ and $(\mathrm{dd}^{\mathrm{c}} u)^n = (\mathrm{dd}^{\mathrm{c}} \widehat{u})^n = 0$ on B. Since the balls $B \Subset \Omega$ cover Ω, and a distribution vanishing on each member of a family of open sets vanishes on their union (Lemma 1.4.2), $(\mathrm{dd}^{\mathrm{c}} u)^n = 0$ on Ω. For the last assertion, let every point of Ω have an open neighborhood U such that $u|_U$ is maximal on U. By the first part applied on U, and by locality of the products (Proposition 9.1.2), $(\mathrm{dd}^{\mathrm{c}} u)^n$ vanishes on each such U, hence on Ω, and u is maximal on Ω. Conversely, if u is maximal on Ω, then $(\mathrm{dd}^{\mathrm{c}} u)^n = 0$ on every open set $U \subset \Omega$, so $u|_U$ is maximal on U. □

9.7 Density lower bounds and mixed products

We now turn to the equation $(\mathrm{dd}^{\mathrm{c}} u)^n = \mu$ with a prescribed measure μ on a ball, and to the lower bounds that mixed products inherit from lower bounds of the pure Monge–Ampère measures. From here to the end of Section 9.11, $\Omega \subset \mathbb{C}^n$ is open, $n \ge 1$, and

$$\eta = \mathrm{dd}^{\mathrm{c}} |z|^2 = \beta/\pi, \qquad \text{so that} \qquad \eta^n = \frac{n!}{\pi^n}\, \mathrm{d}V_{2n}$$

by (9.2). Densities of Monge–Ampère measures are taken with respect to η^n; thus, by (9.2) and the remark following Proposition 9.1.4, a plurisubharmonic function $u \in C^2(\Omega)$ satisfies $(\mathrm{dd}^{\mathrm{c}} u)^n = \det(u_{j\bar{k}})\, \eta^n$.

A lower bound $(\mathrm{dd}^{\mathrm{c}} u)^n \ge f \eta^n$ is a nonlinear condition on u, and it is not evident that it is preserved by convolution. We show that, for a continuous density f, it can be tested pointwise with C^2 functions touching u from above, and that it is equivalent to a family of linear elliptic inequalities, one for each positive definite Hermitian matrix of determinant one (Proposition 9.7.3). Two lemmas prepare it: an expression of $(\det A)^{1/n}$ as an infimum of traces, and a passage from test functions to distributions. Since linear inequalities are preserved by convolution, the determinant inequalities for mixed products of smooth functions then pass to locally bounded potentials

(Corollaries 9.7.4 and 9.7.6). The same characterization will make subsolutions of the Dirichlet problem stable under the operations used in Section 9.8.

Let u be a real function on an open subset of $\mathbb{R}^d$. A real C^2 function q defined near a point x *touches u from above at x* if $q(x) = u(x)$ and $q \geq u$ on a neighborhood of x; we then call q an *upper test function* for u at x.

Lemma 9.7.1 *Let A be a positive semidefinite Hermitian $n \times n$ matrix. Then*

$$(\det A)^{1/n} = \inf\left\{ \frac{\operatorname{tr}(BA)}{n} : \ B \text{ positive definite Hermitian}, \ \det B = 1 \right\}.$$

If C is another positive semidefinite Hermitian $n \times n$ matrix, define the numbers $M_k(A, C)$, $0 \leq k \leq n$, by

$$\det(tA + sC) = \sum_{k=0}^{n} \binom{n}{k} M_k(A, C)\, t^k s^{n-k} \qquad (t, s \in \mathbb{R}).$$

Then $M_n(A, C) = \det A$, $M_0(A, C) = \det C$, and for each integer $1 \leq k \leq n-1$,

$$M_k(A, C) \geq (\det A)^{k/n} (\det C)^{(n-k)/n}.$$

Proof Let B be positive definite with $\det B = 1$. The positive semidefinite matrix $B^{1/2} A B^{1/2}$ has trace $\operatorname{tr}(BA)$ and determinant $\det A$, so the arithmetic–geometric mean inequality for its eigenvalues gives $\operatorname{tr}(BA)/n \geq (\det A)^{1/n}$. If A is positive definite, equality holds for $B = (\det A)^{1/n} A^{-1}$. If A is singular, the matrices $B_\varepsilon = \det(A + \varepsilon I)^{1/n} (A + \varepsilon I)^{-1}$, $\varepsilon > 0$, are admissible and

$$0 \leq \operatorname{tr}(B_\varepsilon A) \leq \operatorname{tr}\big(B_\varepsilon (A + \varepsilon I)\big) = n \det(A + \varepsilon I)^{1/n} \longrightarrow 0$$

as $\varepsilon \downarrow 0$.

The formulas for M_n and M_0 follow from $\det(tA) = t^n \det A$ and $\det(sC) = s^n \det C$. Suppose first that A and C are positive definite, and let $\lambda_1, \ldots, \lambda_n > 0$ be the eigenvalues of $A^{-1/2} C A^{-1/2}$. Since $\det(tA + sC) = \det A \prod_{i=1}^{n} (t + s\lambda_i)$, comparison of coefficients gives

$$M_k(A, C) = \frac{\det A}{\binom{n}{k}} \sum_{|J| = n-k} \prod_{j \in J} \lambda_j,$$

the sum being over the subsets $J \subset \{1, \ldots, n\}$ with $n - k$ elements. Each eigenvalue occurs in exactly $\binom{n-1}{n-k-1}$ of these $\binom{n}{n-k}$ products, so the geometric mean of the products is $(\lambda_1 \cdots \lambda_n)^{(n-k)/n}$. Since $\lambda_1 \cdots \lambda_n = \det C / \det A$, the arithmetic–geometric mean inequality gives

$$M_k(A, C) \geq \det A \left(\frac{\det C}{\det A} \right)^{(n-k)/n} = (\det A)^{k/n} (\det C)^{(n-k)/n}.$$

For semidefinite A and C, apply this to $A + \varepsilon I$ and $C + \varepsilon I$ and let $\varepsilon \downarrow 0$; both sides depend continuously on the matrices. □

Lemma 9.7.2 *Let $d \geq 1$, let (c_{kl}) be a real symmetric positive definite $d \times d$ matrix, and put $L = \sum_{k,l=1}^{d} c_{kl}\partial_k\partial_l$. Let $U \subset \mathbb{R}^d$ be open, let $w\colon U \to \mathbb{R}$ be upper semicontinuous, and let $F\colon U \to \mathbb{R}$ be continuous. Suppose that, for every $x \in U$, every upper test function q for w at x satisfies $Lq(x) \geq F(x)$. Then w is locally integrable, and $Lw \geq F$ in the sense of distributions on U.*

Proof We reduce to the Laplacian, treat the case $F = 0$ by a maximum principle argument, and reduce the general case to it.

Step 1. Reduction to $L = \Delta$. Let $C = (c_{kl})$, let P be the positive definite square root of C^{-1}, so that $PCP = I$, and put $\tilde{U} = P(U)$, $\tilde{w} = w \circ P^{-1}$ and $\tilde{F} = F \circ P^{-1}$. If q is a C^2 function near $x \in U$ and $\tilde{q} = q \circ P^{-1}$, then $\partial_k\partial_l q(x) = \sum_{a,b} P_{ak}P_{bl}(\partial_a\partial_b\tilde{q})(Px)$ by the chain rule, and hence

$$(Lq)(x) = \sum_{a,b}(PCP)_{ab}(\partial_a\partial_b\tilde{q})(Px) = (\Delta\tilde{q})(Px).$$

The map $q \mapsto \tilde{q}$ is a bijection from the upper test functions for w at x onto those for $\tilde{w}$ at Px, so $\tilde{w}$, Δ and $\tilde{F}$ satisfy the hypothesis of the lemma on $\tilde{U}$. Suppose that the lemma holds for them. Then $w = \tilde{w} \circ P$ is locally integrable, and for every nonnegative test function χ on U the change of variables $y = Px$, with $\tilde{\chi} = \chi \circ P^{-1}$ and $L\chi = (\Delta\tilde{\chi}) \circ P$, gives

$$\langle Lw, \chi\rangle = \int_U w\, L\chi \,\mathrm{d}x = \frac{1}{|\det P|}\int_{\tilde{U}} \tilde{w}\,\Delta\tilde{\chi}\,\mathrm{d}y \geq \frac{1}{|\det P|}\int_{\tilde{U}} \tilde{F}\tilde{\chi}\,\mathrm{d}y = \int_U F\chi\,\mathrm{d}x;$$

the first equality is the definition of distributional derivatives. We may therefore assume that $L = \Delta$.

Step 2. The case $F = 0$. We show that w satisfies assertion (1) of Theorem 3.1.3 for every ball $D = B(y, r)$ with $\overline{B(y,r)} \subset U$; by the converse part of that theorem, w is then subharmonic on U. Let h be a real function which is harmonic on $B(y, r)$ and continuous on $\overline{B(y,r)}$, with $\overline{\lim}_{B(y,r)\ni x\to\zeta} w(x) \leq h(\zeta)$ for every $\zeta \in \partial B(y, r)$, and suppose that $w(x_1) > h(x_1)$ for some $x_1 \in B(y, r)$. Choose $\varepsilon > 0$ so small that the upper semicontinuous function $g(x) = w(x) - h(x) - \varepsilon(r^2 - |x - y|^2)$ on $B(y, r)$ satisfies $g(x_1) > 0$.

Since h is continuous on the closed ball, $\overline{\lim}_{B(y,r)\ni x\to\zeta} g(x) \leq 0$ for every $\zeta \in \partial B(y, r)$. Hence every $\zeta \in \partial B(y, r)$ has an open neighborhood N_ζ with $g < g(x_1)/2$ on $N_\zeta \cap B(y, r)$; finitely many of these neighborhoods cover the sphere, and the complement K of their union in $\overline{B(y,r)}$ is a compact subset of $B(y, r)$ containing x_1. The upper semicontinuous function g attains its maximum over K at a point x_0, and $c = g(x_0) \geq g(x_1) > 0$; since $g < g(x_1)/2$ outside K, we have $g \leq c$ on $B(y, r)$. The function $q = h + \varepsilon(r^2 - |x - y|^2) + c$ is smooth on $B(y, r)$, satisfies $q - w = c - g \geq 0$ there and $q(x_0) = w(x_0)$, so it is an upper test function for w at x_0. But $\Delta q = -2d\varepsilon < 0$, which contradicts the hypothesis. Hence $w \leq h$ on $B(y, r)$, and

w is subharmonic on U. Being real-valued, it is not identically $-\infty$ on any connected component of U, so Theorem 3.2.2(1) shows that w is locally integrable and $\Delta w \geq 0$.

Step 3. The general case. Let $V \Subset U$ be open. Every compact subset of U lies in such a set V, so it suffices to show that w is locally integrable on V and that $\Delta w \geq F$ on V. Choose an open set V' with $\overline{V} \subset V' \Subset U$. By Lemma 1.4.1(1), the mollifications of F converge to F uniformly on $\overline{V'}$; subtracting from each of them its maximal deviation from F on $\overline{V'}$, we obtain smooth functions F_j on a neighborhood of $\overline{V'}$ such that $F_j \leq F$ on $\overline{V'}$ and $F_j \to F$ uniformly there. Let $\vartheta \in C_c^\infty(V')$ be equal to one on an open set $W \supset \overline{V}$ (Lemma 1.4.1(4)). By Proposition 2.2.1, the convolution $h_j = E_d * (\vartheta F_j)$, where ϑF_j is extended by zero, is a smooth function on $\mathbb{R}^d$ with $\Delta h_j = \vartheta F_j$; in particular $\Delta h_j = F_j$ on W.

If q is an upper test function for $w - h_j$ at $x \in V$, then $q + h_j$ is an upper test function for w at x, so that $\Delta q(x) + F_j(x) \geq F(x)$, that is, $\Delta q(x) \geq F(x) - F_j(x) \geq 0$. Step 2, applied to the real upper semicontinuous function $w - h_j$ on V, shows that $w - h_j$ is locally integrable on V and $\Delta(w - h_j) \geq 0$ there. Hence w is locally integrable on V and $\Delta w \geq F_j$ on V. Testing against a nonnegative test function on V and letting $j \to \infty$, we obtain $\Delta w \geq F$ on V. □

For a real C^2 function q on an open subset of $\mathbb{C}^n$, we write $\mathcal{H}(q) = (q_{j\bar{k}})$ for its complex Hessian matrix, normalized as in (7.5). For a positive definite Hermitian matrix B with $\det B = 1$, put

$$L_B q = \operatorname{tr}\bigl(B\mathcal{H}(q)\bigr).$$

This is a real operator with constant coefficients, since B and $\mathcal{H}(q)$ are Hermitian, and it is elliptic. Indeed, write $B = V \operatorname{diag}(b_1, \dots, b_n) V^*$ with V unitary and $b_1, \dots, b_n > 0$, and introduce coordinates ζ by $z = \bar{V}\zeta$. By the chain rule, the complex Hessian of $\tilde{q}(\zeta) = q(\bar{V}\zeta)$ is $V^*\mathcal{H}(q)V$, so that

$$L_B q = \operatorname{tr}\bigl(\operatorname{diag}(b_1, \dots, b_n)\, V^*\mathcal{H}(q)V\bigr) = \sum_{j=1}^n b_j \tilde{q}_{j\bar{j}} = \frac{1}{4}\sum_{j=1}^n b_j \Delta_j \tilde{q}$$

by (7.5), where Δ_j is the Laplacian in the variable ζ_j. The real-linear map $z \mapsto \zeta$ is orthogonal on $\mathbb{C}^n = \mathbb{R}^{2n}$; let O be its matrix. By the chain rule, in the real coordinates of $\mathbb{C}^n$ the operator L_B takes the form $\sum_{k,l} c_{kl}\partial_k\partial_l$ with $(c_{kl}) = O^t \Lambda O$, where Λ is the diagonal matrix with entries $b_1/4, b_1/4, \dots, b_n/4, b_n/4$. This matrix is symmetric and positive definite, so Lemma 9.7.2 applies to L_B.

Proposition 9.7.3 *Let u be a locally bounded plurisubharmonic function on Ω, and let $f \geq 0$ be a continuous function on Ω. The following conditions are equivalent:*

(1) $(\mathrm{dd}^c u)^n \geq f\eta^n$ *as measures on* Ω*;*
(2) $\det \mathcal{H}(q)(x) \geq f(x)$ *for every* $x \in \Omega$ *and every upper test function q for u at x;*
(3) $L_B u \geq n f^{1/n}$ *in the sense of distributions on* Ω*, for every positive definite Hermitian matrix B with* $\det B = 1$.

If these conditions hold, then for every $\varepsilon > 0$ the mollification $u_\varepsilon = u * \rho_\varepsilon$ satisfies $\det \mathcal{H}(u_\varepsilon) \geq (f^{1/n} * \rho_\varepsilon)^n$ *on* Ω_ε.

Proof We first observe that the complex Hessian $\mathcal{H}(q)(x)$ of an upper test function q for u at x is positive semidefinite. Indeed, let $\xi \in \mathbb{C}^n \setminus \{0\}$, and let $r > 0$ be so small that the closed disc $\{x + \zeta\xi : |\zeta| \le r\}$ lies in the neighborhood of x on which $q \ge u$. The function $\zeta \mapsto u(x + \zeta\xi)$ is subharmonic near this disc and finite at $\zeta = 0$, so Theorem 3.1.3(2) gives

$$q(x) = u(x) \le \frac{1}{2\pi}\int_0^{2\pi} u(x + re^{i\theta}\xi)\,d\theta \le \frac{1}{2\pi}\int_0^{2\pi} q(x + re^{i\theta}\xi)\,d\theta.$$

Write $q_j = \partial_{z_j} q$ and $q_{jk} = \partial_{z_j}\partial_{z_k} q$. Since q is real and of class C^2, Taylor's formula gives

$$q(x+h) = q(x) + 2\,\mathrm{Re}\sum_j q_j(x)h_j + \mathrm{Re}\sum_{j,k} q_{jk}(x)h_j h_k + \sum_{j,k} q_{j\bar{k}}(x)h_j\bar{h}_k + o(|h|^2).$$

For $h = re^{i\theta}\xi$ the terms containing $e^{i\theta}$ and $e^{2i\theta}$ have mean zero over $\theta \in [0, 2\pi]$, so the right-hand side of the preceding display equals $q(x) + r^2\sum_{j,k} q_{j\bar{k}}(x)\xi_j\bar{\xi}_k + o(r^2)$. Dividing by r^2 and letting $r \downarrow 0$, we obtain $\sum_{j,k} q_{j\bar{k}}(x)\xi_j\bar{\xi}_k \ge 0$.

(1) implies (2). Assume (1), let q be an upper test function for u at x, and suppose that $\det \mathcal{H}(q)(x) < f(x)$. Since $\mathcal{H}(q)(x)$ is positive semidefinite, we can choose $\delta > 0$ so small that $q_\delta = q + \delta|z-x|^2$ satisfies $\mathcal{H}(q_\delta)(x) = \mathcal{H}(q)(x) + \delta I > 0$ and still $\det \mathcal{H}(q_\delta)(x) < f(x)$. Since q is of class C^2 and f is continuous, there is $r > 0$ such that $\overline{B(x,r)} \subset \Omega$, $q \ge u$ on $\overline{B(x,r)}$, the complex Hessian $\mathcal{H}(q_\delta)$ is positive definite on $\overline{B(x,r)}$, and $\det \mathcal{H}(q_\delta) < f$ on $\overline{B(x,r)}$; in particular, q_δ is plurisubharmonic on $B(x,r)$ by Proposition 8.2.2. Put $b = q_\delta - \delta r^2/2$. Then $b = q + \delta r^2/2 > u$ on the sphere $\partial B(x,r)$, while $b(x) = u(x) - \delta r^2/2 < u(x)$.

The Borel set $S = \{z \in B(x,r) : b(z) < u(z)\}$ has positive Lebesgue measure. Otherwise $\max\{u, b\} = b$ almost everywhere on $B(x,r)$; since both functions are plurisubharmonic, the uniqueness assertion of Theorem 8.2.1 gives $\max\{u, b\} = b$ everywhere, contradicting $b(x) < u(x)$. The function $b - u$ is lower semicontinuous and positive on $\partial B(x,r)$, so $\underline{\lim}_{z\to\zeta}(b(z) - u(z)) > 0$ for every $\zeta \in \partial B(x,r)$. The comparison principle Theorem 9.2.8, applied on $B(x,r)$ with the bounded plurisubharmonic functions b and u in place of u and v, therefore gives

$$\int_S f\eta^n \le \int_S (dd^c u)^n \le \int_S (dd^c b)^n = \int_S \det \mathcal{H}(q_\delta)\,\eta^n.$$

Here $(dd^c b)^n = \det \mathcal{H}(q_\delta)\,\eta^n$ on $B(x,r)$ because b is a plurisubharmonic C^2 function there (see the beginning of this section), and the integrals are finite because $(dd^c u)^n$ is a Radon measure on Ω and $\overline{B(x,r)}$ is a compact subset of Ω. Since $\det \mathcal{H}(q_\delta) < f$ on S and S has positive measure, the right side is strictly smaller than the left side, a contradiction. Thus (2) holds.

(2) implies (3). Let q be an upper test function for u at x, and let B be positive definite with $\det B = 1$. Since $\mathcal{H}(q)(x)$ is positive semidefinite, the infimum formula of Lemma 9.7.1 and (2) give

$$L_B q(x) = \operatorname{tr}\big(B\mathcal{H}(q)(x)\big) \geq n\big(\det \mathcal{H}(q)(x)\big)^{1/n} \geq n f(x)^{1/n}.$$

Since u is real-valued and upper semicontinuous, Lemma 9.7.2, applied to L_B and the continuous function $nf^{1/n}$, gives (3).

(3) implies (1). The mollifications $u_\varepsilon = u * \rho_\varepsilon$ of Section 1.4 are smooth plurisubharmonic functions on Ω_ε which decrease to u as $\varepsilon \downarrow 0$ (Proposition 8.2.2). Fix B as in (3). The distribution $L_B u - nf^{1/n}$ is positive, so its mollification is a nonnegative function on Ω_ε (Proposition 1.4.3(3)). Since L_B has constant coefficients, Proposition 1.4.3(1) identifies the mollification of $L_B u$ with $L_B u_\varepsilon$, and we obtain the pointwise inequality

$$\operatorname{tr}\big(B\mathcal{H}(u_\varepsilon)\big) \geq n\,(f^{1/n} * \rho_\varepsilon) \quad \text{on } \Omega_\varepsilon \qquad (B > 0,\ \det B = 1).$$

Since $\mathcal{H}(u_\varepsilon)$ is positive semidefinite (Proposition 8.2.2), taking the infimum over B and using Lemma 9.7.1, we obtain $\det \mathcal{H}(u_\varepsilon) \geq (f^{1/n} * \rho_\varepsilon)^n$, which is the last assertion of the proposition. By the formula recalled at the beginning of this section, it reads

$$(\mathrm{dd}^{\mathrm{c}} u_\varepsilon)^n \geq (f^{1/n} * \rho_\varepsilon)^n \eta^n \quad \text{on } \Omega_\varepsilon.$$

As $\varepsilon \downarrow 0$, the density on the right converges locally uniformly to f by Lemma 1.4.1(1), and $(\mathrm{dd}^{\mathrm{c}} u_\varepsilon)^n \to (\mathrm{dd}^{\mathrm{c}} u)^n$ weakly by Theorem 9.2.5, applied along any sequence $\varepsilon_j \downarrow 0$. Testing against nonnegative functions in $C_c(\Omega)$ gives (1). □

Corollary 9.7.4 *Let u, v be locally bounded plurisubharmonic functions on Ω, and let $f, g \geq 0$ be continuous functions on Ω such that $(\mathrm{dd}^{\mathrm{c}} u)^n \geq f\eta^n$ and $(\mathrm{dd}^{\mathrm{c}} v)^n \geq g\eta^n$. Then, for each integer $1 \leq k \leq n-1$,*

$$(\mathrm{dd}^{\mathrm{c}} u)^k \wedge (\mathrm{dd}^{\mathrm{c}} v)^{n-k} \geq f^{k/n} g^{(n-k)/n} \eta^n.$$

For $k = 0$ and $k = n$ the same inequality is the hypothesis.

Proof Let $u_\varepsilon = u * \rho_\varepsilon$ and $v_\varepsilon = v * \rho_\varepsilon$, with the same mollifiers for both functions. By the last assertion of Proposition 9.7.3, applied to u with the density f and to v with the density g,

$$\det \mathcal{H}(u_\varepsilon) \geq (f^{1/n} * \rho_\varepsilon)^n, \qquad \det \mathcal{H}(v_\varepsilon) \geq (g^{1/n} * \rho_\varepsilon)^n.$$

For real t and s, (9.2), applied to the smooth function $tu_\varepsilon + sv_\varepsilon$, gives $(t\mathrm{dd}^{\mathrm{c}} u_\varepsilon + s\mathrm{dd}^{\mathrm{c}} v_\varepsilon)^n = \det\big(t\mathcal{H}(u_\varepsilon) + s\mathcal{H}(v_\varepsilon)\big)\eta^n$ for the exterior powers of smooth forms. Since 2-forms commute, the left side can be expanded by the binomial formula. Comparing the coefficients of $\binom{n}{k} t^k s^{n-k}$, and recalling that Bedford–Taylor products of smooth functions are exterior products of smooth forms (Proposition 9.1.2(1)), we find

$$(\mathrm{dd}^{\mathrm{c}} u_\varepsilon)^k \wedge (\mathrm{dd}^{\mathrm{c}} v_\varepsilon)^{n-k} = M_k\big(\mathcal{H}(u_\varepsilon), \mathcal{H}(v_\varepsilon)\big)\eta^n \geq (f^{1/n} * \rho_\varepsilon)^k (g^{1/n} * \rho_\varepsilon)^{n-k} \eta^n,$$

where the inequality is the second assertion of Lemma 9.7.1, applied to the matrices $\mathcal{H}(u_\varepsilon)$ and $\mathcal{H}(v_\varepsilon)$, which are positive semidefinite by Proposition 8.2.2. As $\varepsilon \downarrow 0$,

the density on the right converges locally uniformly to $f^{k/n}g^{(n-k)/n}$, while the left side converges weakly to $(\mathrm{dd^c}u)^k \wedge (\mathrm{dd^c}v)^{n-k}$ by Theorem 9.2.5, since u_ε and v_ε are smooth plurisubharmonic functions decreasing to u and v (Proposition 8.2.2). □

9.7.1 Several independent factors

Corollary 9.7.4 concerns products in which only two Hessians occur, each repeated several times. For a product $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_n$ of n arbitrary potentials, the coefficients M_k of Lemma 9.7.1 are replaced by the mixed discriminant $D(A_1, \ldots, A_n)$ of Hermitian matrices. Recall from the paragraph before Lemma 7.2.2 that $D(A_1, \ldots, A_n)$ is $1/n!$ times the coefficient of $t_1 \cdots t_n$ in $\det(\sum_j t_j A_j)$; it is symmetric and real multilinear, and $D(A, \ldots, A) = \det A$ (Lemma 7.2.2(1)). We prove the geometric-mean inequality for D by means of a reverse Cauchy–Schwarz inequality, which follows from the signature of D computed in Lemma 7.2.3. Convolution then extends it to locally bounded potentials as before.

Lemma 9.7.5 *Let $n \geq 1$. For positive semidefinite Hermitian $n \times n$ matrices $A_1, \ldots, A_n$,*

$$D(A_1, \ldots, A_n) \geq \prod_{i=1}^{n} (\det A_i)^{1/n}.$$

Proof For $n = 1$ both sides equal A_1. Let $n \geq 2$, and assume first that all A_i are positive definite. By Lemma 7.2.2(3), D is positive on positive definite matrices.

We first derive the reverse Cauchy–Schwarz inequality. Fix positive definite matrices $A_3, \ldots, A_n$, and consider the symmetric bilinear form

$$q(A, B) = D(A, B, A_3, \ldots, A_n)$$

on Hermitian matrices. Let A, B be positive definite, put $c = q(A, B)/q(B, B)$ and $M = A - cB$. Then $q(M, B) = 0$, so $q(M, M) \leq 0$ by Lemma 7.2.3. Since $q(M, M) = q(A, A) - q(A, B)^2/q(B, B)$, we obtain

$$q(A, B)^2 \geq q(A, A)q(B, B). \tag{9.8}$$

Put $\widetilde{A}_i = (\det A_i)^{-1/n} A_i$, so that $\det \widetilde{A}_i = 1$. By multilinearity it suffices to prove that $D(\widetilde{A}_1, \ldots, \widetilde{A}_n) \geq 1$. Consider the finitely many ordered n-tuples whose entries belong to the set $\{\widetilde{A}_1, \ldots, \widetilde{A}_n\}$, and let $F > 0$ be the smallest mixed discriminant of such a tuple. A tuple whose entries all equal $\widetilde{A}_i$ has discriminant $\det \widetilde{A}_i = 1$, so $F \leq 1$. The multiplicity of a matrix in a tuple is the number of entries equal to it. Among the tuples with discriminant F, choose one for which the sum of the squares of the multiplicities is largest.

Suppose that this tuple contains two different matrices A and B. By symmetry of D we may place them in the first two slots, so that $F = q(A, B)$, where q is formed with the remaining entries of the tuple. Replacing the pair (A, B) by (A, A) or by

(B,B) gives tuples with discriminants $q(A,A) \ge F$ and $q(B,B) \ge F$, whereas (9.8) gives $q(A,A)q(B,B) \le F^2$. Hence $q(A,A) = q(B,B) = F$, and both new tuples are minimizing. If A and B have multiplicities r and s, the two replacements change the sum of the squared multiplicities by $2(r-s)+2$ and $2(s-r)+2$, respectively. These two numbers add up to 4, so one of them is positive, which contradicts the choice of the tuple. Therefore all entries of the chosen tuple are equal, and $F = 1$. In particular $D(\widetilde{A}_1, \dots, \widetilde{A}_n) \ge 1$.

Finally, for positive semidefinite A_i, apply the inequality to $A_i + \varepsilon I$ and let $\varepsilon \downarrow 0$. Both sides are continuous: D is a polynomial in the matrix entries, and $(\det A)^{1/n}$ is continuous on positive semidefinite matrices. □

Corollary 9.7.6 *Let $u_1, \dots, u_n$ be locally bounded plurisubharmonic functions on an open set $\Omega \subset \mathbb{C}^n$, $n \ge 1$, and let $f_1, \dots, f_n \ge 0$ be continuous functions on Ω such that*

$$(\mathrm{dd^c} u_i)^n \ge f_i \eta^n \quad (1 \le i \le n), \qquad \eta = \mathrm{dd^c}|z|^2.$$

Then

$$\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n \ge \left(\prod_{i=1}^n f_i^{1/n} \right) \eta^n.$$

Proof For $n = 1$ this is the hypothesis, so let $n \ge 2$. Put $u_{i,\varepsilon} = u_i * \rho_\varepsilon$, with the same radial mollifiers for all i. By the last assertion of Proposition 9.7.3, applied to each u_i with the density f_i,

$$\det \mathcal{H}(u_{i,\varepsilon}) \ge (f_i^{1/n} * \rho_\varepsilon)^n \qquad (1 \le i \le n).$$

By (9.2), applied to the smooth function $\sum_i t_i u_{i,\varepsilon}$, the exterior powers of smooth forms satisfy

$$\Big(\mathrm{dd^c} \sum_{i=1}^n t_i u_{i,\varepsilon} \Big)^n = \det\Big(\sum_{i=1}^n t_i \mathcal{H}(u_{i,\varepsilon}) \Big) \eta^n \qquad (t_1, \dots, t_n \in \mathbb{R}).$$

Both sides are polynomials in $t_1, \dots, t_n$. Since 2-forms commute, the coefficient of $t_1 \cdots t_n$ on the left is $n!\, \mathrm{dd^c} u_{1,\varepsilon} \wedge \cdots \wedge \mathrm{dd^c} u_{n,\varepsilon}$, and on the right it is $n!\, D(\mathcal{H}(u_{1,\varepsilon}), \dots, \mathcal{H}(u_{n,\varepsilon}))\eta^n$. Since the Bedford–Taylor product of smooth functions is the exterior product of the forms $\mathrm{dd^c} u_{i,\varepsilon}$ (Proposition 9.1.2(1)), we obtain

$$\mathrm{dd^c} u_{1,\varepsilon} \wedge \cdots \wedge \mathrm{dd^c} u_{n,\varepsilon} = D\big(\mathcal{H}(u_{1,\varepsilon}), \dots, \mathcal{H}(u_{n,\varepsilon})\big)\eta^n \ge \prod_{i=1}^n (f_i^{1/n} * \rho_\varepsilon)\, \eta^n,$$

where the inequality is Lemma 9.7.5, applied to the complex Hessians of the smooth plurisubharmonic functions $u_{i,\varepsilon}$, which are positive semidefinite (Proposition 8.2.2). As $\varepsilon \downarrow 0$, the density on the right converges locally uniformly to $\prod_i f_i^{1/n}$, and the left side converges weakly to $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n$ by Theorem 9.2.5, since $u_{i,\varepsilon} \downarrow u_i$ for each i (Proposition 8.2.2). □

9.8 The ball Dirichlet problem with continuous density

We now solve the Dirichlet problem

$$(\mathrm{dd^c}u)^n = f\eta^n \quad \text{on } B, \qquad u = \varphi \quad \text{on } \partial B,$$

on a ball B, for a continuous density $f \geq 0$ and continuous boundary values φ. As for the homogeneous equation in Theorem 9.6.3, the solution is the upper envelope of subsolutions. By the linear characterization of density lower bounds in Proposition 9.7.3, subsolutions are stable under maxima, convex combinations and the addition of multiples of $|z|^2$ (Lemma 9.8.1). Stability under maxima makes the envelope a subsolution, and stability under averages allows us to compose with the automorphisms of Lemma 9.6.1 once more to bound its second differences; since a biholomorphism multiplies the density by the squared modulus of its Jacobian determinant, this factor now enters the estimate.

Lemma 9.8.1 *Let u, v be locally bounded plurisubharmonic functions on Ω, and let $F, G \geq 0$ be continuous functions on Ω with $(\mathrm{dd^c}u)^n \geq F^n\eta^n$ and $(\mathrm{dd^c}v)^n \geq G^n\eta^n$. Then, for $0 \leq t \leq 1$ and $c \geq 0$,*

$$\big(\mathrm{dd^c}\big((1-t)u + tv + c|z|^2\big)\big)^n \geq \big((1-t)F + tG + c\big)^n\eta^n.$$

Moreover, $(\mathrm{dd^c}\max\{u, v\})^n \geq \min\{F, G\}^n\eta^n$.

Proof Let B be a positive definite Hermitian matrix with $\det B = 1$. By the implication from (1) to (3) in Proposition 9.7.3, $L_B u \geq nF$ and $L_B v \geq nG$ in the sense of distributions. Moreover $L_B|z|^2 = \operatorname{tr} B \geq n(\det B)^{1/n} = n$ by the arithmetic–geometric mean inequality for the eigenvalues of B. Hence

$$L_B\big((1-t)u + tv + c|z|^2\big) \geq n\big((1-t)F + tG\big) + c\operatorname{tr} B \geq n\big((1-t)F + tG + c\big),$$

and the implication from (3) to (1), applied with the continuous density $((1-t)F + tG + c)^n$, gives the first assertion.

For the second, let q be an upper test function for $\max\{u, v\}$ at a point x. If $u(x) \geq v(x)$, then $q(x) = u(x)$ and $q \geq \max\{u, v\} \geq u$ near x, so q is an upper test function for u at x, and the implication from (1) to (2) gives $\det \mathcal{H}(q)(x) \geq F(x)^n \geq \min\{F, G\}(x)^n$. The case $v(x) \geq u(x)$ is symmetric. The implication from (2) to (1) now gives the assertion. □

Theorem 9.8.2 *Let $B = B(a, R) \subset \mathbb{C}^n$ with $n \geq 1$ and $R > 0$. Let $f \in C(\overline{B})$ be nonnegative, and let $\varphi \in C(\partial B)$ be real. There is a unique $H_{f,\varphi} \in \mathrm{PSH}(B) \cap C(\overline{B})$ such that*

$$(\mathrm{dd^c}H_{f,\varphi})^n = f\eta^n \quad \text{on } B, \qquad H_{f,\varphi}|_{\partial B} = \varphi.$$

It is the pointwise supremum of the bounded plurisubharmonic functions v on B such that $(\mathrm{dd^c}v)^n \geq f\eta^n$ and $\overline{\lim}_{B\ni z\to\zeta} v(z) \leq \varphi(\zeta)$ for every $\zeta \in \partial B$. If $g \in C(\overline{B})$ is another nonnegative density and $\chi \in C(\partial B)$ another real boundary function, then

$$\|H_{f,\varphi} - H_{g,\chi}\|_{L^\infty(B)} \le \|\varphi - \chi\|_{L^\infty(\partial B)} + R^2 \|f^{1/n} - g^{1/n}\|_{L^\infty(B)}. \tag{9.9}$$

If $f^{1/n}$ is the restriction of a real C^2 function on a neighborhood of $\overline{B}$ and φ that of a real C^2 function on a neighborhood of ∂B, then $H_{f,\varphi} \in C^{1,1}_{\mathrm{loc}}(B)$.

Proof For data of class C^2 (defined below) we show that the envelope of subsolutions is a locally $C^{1,1}$ solution (Steps 1–3). A stability estimate then gives a solution for continuous data (Step 4), and a comparison argument identifies every continuous solution with the envelope (Step 5).

The affine map $\tau(\xi) = a + R\xi$ carries the unit ball onto B. Since $|\tau(\xi)|^2 - R^2|\xi|^2 = 2R\,\mathrm{Re}\sum_j \xi_j \bar{a}_j + |a|^2$ is the real part of a holomorphic polynomial plus a constant, its complex Hessian vanishes, so τ pulls $\eta = \mathrm{dd^c}|z|^2$ back to $R^2\eta$ and η^n back to $R^{2n}\eta^n$. Since Bedford–Taylor products are invariant under biholomorphisms (Proposition 9.1.2(5)), a function u on B satisfies $(\mathrm{dd^c}u)^n \ge f\eta^n$ if and only if $\xi \mapsto u(a + R\xi)$ satisfies the same inequality with the density $R^{2n} f(a + R\xi)$, whose nth root is $R^2 f^{1/n}(a + R\xi)$; the same holds with equality. It therefore suffices to prove the theorem for the unit ball $B = B(0,1)$, with $R = 1$ in (9.9). We write z again for the variable and put

$$\psi = |z|^2 - 1, \qquad F = f^{1/n}.$$

In this proof a *subsolution for* (f, φ) is a bounded plurisubharmonic function v on B with $(\mathrm{dd^c}v)^n \ge f\eta^n$ and $\overline{\lim}_{B \ni z \to \zeta} v(z) \le \varphi(\zeta)$ for all $\zeta \in \partial B$. We say that the data (f, φ) are *of class* C^2 if F and φ are restrictions of real C^2 functions on neighborhoods of $\overline{B}$ and ∂B, respectively.

Step 1. The envelope of subsolutions for data of class C^2. Assume that (f, φ) is of class C^2. We first construct a subsolution with boundary values φ. Multiplying a C^2 extension of φ by a smooth cutoff function for ∂B in its domain which equals one near ∂B (Lemma 1.4.1(4)), we obtain a real C^2 function b on $\mathbb{C}^n$ with compact support and $b = \varphi$ on ∂B. Let $\lambda \ge 0$ bound the operator norm of the complex Hessian of b, and choose $c_0 > \lambda$ with $(c_0 - \lambda)^n \ge \max_{\overline{B}} f$. The complex Hessian of $s = b + c_0\psi$ is at least $(c_0 - \lambda)I$, so s is plurisubharmonic (Proposition 8.2.2) and $\det(s_{j\bar{k}}) \ge (c_0 - \lambda)^n \ge f$ on B; by the formula for plurisubharmonic C^2 functions at the beginning of Section 9.7, $(\mathrm{dd^c}s)^n \ge f\eta^n$ on B. Moreover s is continuous on $\overline{B}$ and equals φ on ∂B; in particular, s is a subsolution. On the other hand, every subsolution is subharmonic (Theorem 8.2.1) with boundary upper limits at most φ, so the comparison principle (Theorem 3.1.3 and the remark following it) bounds it above by the harmonic extension $P\varphi$ of φ, its Poisson integral (Theorem 2.3.1).

Let H be the pointwise supremum of all subsolutions, so that $s \le H \le P\varphi$. By Lemma 9.8.1, the maximum of finitely many subsolutions is a subsolution. By Theorem 8.3.2, some countable family of subsolutions has the same upper semicontinuous regularization H^* as the whole family, and its supremum equals H^* almost everywhere. Taking finite maxima of s and the members of this family, we obtain subsolutions v_j which increase to a function equal to H^* almost everywhere, and $s \le v_j \le P\varphi$. By Theorem 9.5.1, $(\mathrm{dd^c}v_j)^n \to (\mathrm{dd^c}H^*)^n$ weakly, and hence $(\mathrm{dd^c}H^*)^n \ge f\eta^n$. Since $P\varphi$ is continuous, $s \le H \le H^* \le P\varphi$; as s and $P\varphi$ are continuous on $\overline{B}$ and equal φ on ∂B, the function H^* has boundary limits φ. Thus

H^* is a subsolution, so $H^* \le H$ and $H = H^*$. In particular, H is a subsolution with $\lim_{B \ni z \to \zeta} H(z) = \varphi(\zeta)$ for all $\zeta \in \partial B$.

Step 2. Second differences for data of class C^2. We show that H satisfies the hypothesis of Lemma 9.6.2 on B. Fix a compact set $K \subset B$, and let $h_0 > 0$ and the automorphisms $S_{a,h}$, $a \in K$, $|h| \le h_0$, be as in Lemma 9.6.1. Thus $(a, h, z) \mapsto S_{a,h}(z)$ is smooth, and holomorphic in z, on an open set O containing the compact set $L = K \times \overline{B(0, h_0)} \times \overline{B}$, and so is the complex Jacobian determinant $J_{a,h}(z) = \det_{\mathbb{C}} DS_{a,h}(z)$ of $z \mapsto S_{a,h}(z)$. Since $S_{a,0}$ is the identity, $J_{a,0} = 1$ on B, and hence on $\overline{B}$, as $z \mapsto J_{a,0}(z)$ is continuous. The open set $W = \{(a, h, z) \in O : |J_{a,h}(z) - 1| < 1/2\}$ therefore contains the compact set $K \times \{0\} \times \overline{B}$, and hence all points within some positive distance of it; decreasing h_0, we may assume that $L \subset W$. Then $|J_{a,h}|^{2/n} = (J_{a,h}\overline{J_{a,h}})^{1/n}$ is smooth and positive on the neighborhood W of L.

Since $S_{a,h}$ is holomorphic, $S_{a,h}^*(\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n) = J_{a,h}\, \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$, and the conjugate identity holds for $\mathrm{d}\bar{z}_1 \wedge \cdots \wedge \mathrm{d}\bar{z}_n$. As η^n is a constant multiple of $\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar{z}_1 \wedge \cdots \wedge \mathrm{d}\bar{z}_n$, it follows that $S_{a,h}^* \eta^n = |J_{a,h}|^2 \eta^n$. Pulling back the inequality $(\mathrm{dd^c} H)^n \ge F^n \eta^n$ by $S_{a,h}$, and using again the invariance of Bedford–Taylor products under biholomorphisms, we obtain

$$\big(\mathrm{dd^c}(H \circ S_{a,h})\big)^n \ge (F \circ S_{a,h})^n |J_{a,h}|^2 \eta^n = \big((F \circ S_{a,h})|J_{a,h}|^{2/n}\big)^n \eta^n.$$

Put

$$\Phi_{a,h} = \tfrac{1}{2}\big((F \circ S_{a,h})|J_{a,h}|^{2/n} + (F \circ S_{a,-h})|J_{a,-h}|^{2/n}\big),$$

a nonnegative continuous function on $\overline{B}$. We claim that there are constants $C_1, C_2 \ge 0$ such that, for all $a \in K$ and $|h| \le h_0$,

$$\begin{aligned} \Phi_{a,h} &\ge F - C_1|h|^2 \quad \text{on } \overline{B}, \\ \tfrac{1}{2}(\varphi \circ S_{a,h} + \varphi \circ S_{a,-h}) &\le \varphi + C_2|h|^2 \quad \text{on } \partial B. \end{aligned}$$

To prove the first inequality, fix $a \in K$ and $z \in \overline{B}$, and consider $\gamma(h) = F(S_{a,h}(z))|J_{a,h}(z)|^{2/n}$ as a function of the real variable $h \in \mathbb{C}^n = \mathbb{R}^{2n}$, $|h| \le h_0$. Since $S_{a,h}(\overline{B}) = \overline{B}$, F is of class C^2 on a neighborhood of $\overline{B}$, and the maps $(a, h, z) \mapsto S_{a,h}(z)$ and $(a, h, z) \mapsto |J_{a,h}(z)|^{2/n}$ are smooth near the compact set L, the second derivatives of γ are bounded by a constant $2C_1$ which depends neither on a nor on z. Taylor's formula with integral remainder, applied at h and at $-h$, gives $|\gamma(h) + \gamma(-h) - 2\gamma(0)| \le 2C_1|h|^2$, since the first-order terms cancel. As $\gamma(0) = F(z)$, it follows that $\Phi_{a,h}(z) = \frac{1}{2}(\gamma(h) + \gamma(-h)) \ge F(z) - C_1|h|^2$. The second inequality follows in the same way from a C^2 extension of φ to a neighborhood of ∂B, since $S_{a,\pm h}$ map ∂B onto itself and $S_{a,0}$ is the identity. The constants depend only on K, h_0, n and the C^2 extensions of F and φ.

Now consider

$$V_{a,h} = \tfrac{1}{2}(H \circ S_{a,h} + H \circ S_{a,-h}) + C_1|h|^2\psi - C_2|h|^2.$$

Since additive constants do not change Monge–Ampère measures, Lemma 9.8.1, applied with $t = 1/2$ and the quadratic term $C_1|h|^2|z|^2$, gives

$$(\mathrm{dd^c} V_{a,h})^n \geq \bigl(\Phi_{a,h} + C_1|h|^2\bigr)^n \eta^n \geq F^n \eta^n = f\eta^n.$$

Since $S_{a,\pm h}$ preserve ∂B and H has boundary limits φ, the boundary limits of $V_{a,h}$ are $\frac{1}{2}(\varphi \circ S_{a,h} + \varphi \circ S_{a,-h}) - C_2|h|^2 \leq \varphi$. Thus $V_{a,h}$ is a subsolution, and $V_{a,h} \leq H$. Evaluating at a, where $S_{a,\pm h}(a) = a \pm h$, and using $-1 \leq \psi(a) \leq 0$, we obtain

$$H(a+h) + H(a-h) - 2H(a) \leq 2(C_1 + C_2)|h|^2$$

for all $a \in K$ and $|h| \leq h_0$; here the points $a \pm h = S_{a,\pm h}(a)$ lie in B.

Step 3. The equation for data of class C^2. Since K was arbitrary and H is bounded and plurisubharmonic, Lemma 9.6.2 applies on B. Thus H is continuous and locally $C^{1,1}$ on B, and its distributional real Hessian is a locally bounded measurable matrix A. Almost every $x \in B$ satisfies

$$H(x+h) = H(x) + \nabla H(x) \cdot h + \tfrac{1}{2} h^t A(x) h + o(|h|^2) \qquad (h \to 0),$$

and the complex Hessian $(H_{j\bar{k}})$ computed from A is positive semidefinite almost everywhere and satisfies $(\mathrm{dd^c} H)^n = n!\pi^{-n} \det(H_{j\bar{k}})\, \mathrm{d}V_{2n} = \det(H_{j\bar{k}})\, \eta^n$. Since $(\mathrm{dd^c} H)^n \geq f\eta^n$, we have $\det(H_{j\bar{k}}) \geq f$ almost everywhere, and it remains to prove the reverse inequality.

Suppose, to the contrary, that $\det(H_{j\bar{k}}(x)) > f(x)$ at a point x at which the Taylor expansion holds and $(H_{j\bar{k}}(x))$ is positive semidefinite. The quadratic Taylor polynomial $Q(x+h) = H(x) + \nabla H(x) \cdot h + \frac{1}{2} h^t A(x) h$ has the constant complex Hessian $(H_{j\bar{k}}(x))$, which is positive semidefinite with positive determinant, hence positive definite. Choose $\varepsilon > 0$ so small that the matrix $(H_{j\bar{k}}(x)) - \varepsilon I$ is positive definite and its determinant exceeds $f(x)$. By the continuity of f and the Taylor expansion at x, there is $r > 0$ such that $\overline{B(x,r)} \subset B$, this determinant exceeds f on $B(x,r)$, and $|H - Q| < \varepsilon r^2/2$ on $\partial B(x,r)$. The function

$$q = Q - \varepsilon|z - x|^2 + \varepsilon r^2/4$$

is plurisubharmonic (Proposition 8.2.2), and $(\mathrm{dd^c} q)^n = \det\bigl((H_{j\bar{k}}(x)) - \varepsilon I\bigr)\eta^n \geq f\eta^n$ on $B(x,r)$. Moreover $q < H - \varepsilon r^2/4$ on $\partial B(x,r)$, while $q(x) = H(x) + \varepsilon r^2/4 > H(x)$.

Let $\widehat{H} = \max\{H, q\}$ on $B(x,r)$ and $\widehat{H} = H$ on $B \setminus B(x,r)$. Since q and H are continuous and $q < H$ on the sphere $\partial B(x,r)$, we have $q < H$ on a neighborhood of this sphere. Hence $\widehat{H}$ is plurisubharmonic by Proposition 8.2.5, and it coincides with H near $\partial B(x,r)$. On $B(x,r)$, Lemma 9.8.1, applied with $F = G = f^{1/n}$, gives $(\mathrm{dd^c}\widehat{H})^n \geq f\eta^n$. Every other point of B has a neighborhood on which $\widehat{H} = H$, so $(\mathrm{dd^c}\widehat{H})^n \geq f\eta^n$ near it as well. Since every nonnegative test function on B is a finite sum of nonnegative test functions supported in such neighborhoods (Lemma 1.4.2), the inequality holds on all of B. Thus $\widehat{H}$ is a subsolution with $\widehat{H}(x) > H(x)$, a contradiction. Consequently $\det(H_{j\bar{k}}) = f$ almost everywhere, that

is, $(\mathrm{dd^c}H)^n = f\eta^n$. As H is continuous on B with boundary limits φ (Step 1), it extends to a solution in $\mathrm{PSH}(B)\cap C(\overline{B})$, which is locally $C^{1,1}$ on B.

Step 4. Stability and continuous data. Let (f,φ) and (g,χ) be data of class C^2, with envelopes $H_{f,\varphi}$ and $H_{g,\chi}$ as in Step 1. Put $\delta = \|\varphi-\chi\|_{L^\infty(\partial B)}$ and $\sigma = \|f^{1/n}-g^{1/n}\|_{L^\infty(B)}$. By Lemma 9.8.1,

$$\bigl(\mathrm{dd^c}(H_{g,\chi}+\sigma\psi-\delta)\bigr)^n \geq (g^{1/n}+\sigma)^n\eta^n \geq f\eta^n,$$

and $H_{g,\chi}+\sigma\psi-\delta$ has boundary limits $\chi-\delta\leq\varphi$. It is therefore a subsolution for (f,φ), and since $\psi\geq -1$ we get $H_{g,\chi}-\sigma-\delta\leq H_{f,\varphi}$. Exchanging the roles of the two data gives (9.9) with $R=1$ for data of class C^2.

Now let f and φ be continuous. Extend F to $\mathbb{C}^n$ by $F(z)=F(z/|z|)$ for $|z|>1$ and convolve with mollifiers; this gives smooth functions $F_j\geq 0$ with $F_j\to F$ uniformly on $\overline{B}$ (Lemma 1.4.1(1)). Similarly, extending φ to a neighborhood of ∂B by $\varphi(z)=\varphi(z/|z|)$ and mollifying, we obtain smooth functions φ_j on a neighborhood of ∂B with $\varphi_j\to\varphi$ uniformly on ∂B. The data (F_j^n,φ_j) are of class C^2, and by Steps 1–3 the corresponding envelopes $H_j = H_{F_j^n,\varphi_j}$ are solutions which are continuous on $\overline{B}$ with boundary values φ_j. By the estimate just proved,

$$\|H_j-H_k\|_{L^\infty(B)} \leq \|\varphi_j-\varphi_k\|_{L^\infty(\partial B)} + \|F_j-F_k\|_{L^\infty(B)},$$

so (H_j) is a uniform Cauchy sequence on $\overline{B}$. Its uniform limit U is continuous on $\overline{B}$ and equal to φ on ∂B. By Proposition 9.1.4, U is plurisubharmonic on B and $(\mathrm{dd^c}H_j)^n\to(\mathrm{dd^c}U)^n$ weakly; since $F_j^n\to f$ uniformly on $\overline{B}$, we obtain $(\mathrm{dd^c}U)^n = f\eta^n$.

Step 5. The envelope, uniqueness and stability. Let U be a solution of the problem that is continuous on $\overline{B}$, for instance the one constructed in Step 4, and let v be a subsolution for (f,φ). For $\varepsilon>0$ put $v_\varepsilon = v+\varepsilon\psi$. By Lemma 9.8.1, $(\mathrm{dd^c}v_\varepsilon)^n\geq(F+\varepsilon)^n\eta^n$, while $(\mathrm{dd^c}U)^n = F^n\eta^n$. Since $\varliminf_{z\to\zeta}(U(z)-v_\varepsilon(z))\geq 0$ for every $\zeta\in\partial B$, the comparison principle Theorem 9.2.8 gives

$$\int_{\{U<v_\varepsilon\}}(F+\varepsilon)^n\eta^n \leq \int_{\{U<v_\varepsilon\}}(\mathrm{dd^c}v_\varepsilon)^n \leq \int_{\{U<v_\varepsilon\}}F^n\eta^n.$$

These integrals are finite, because F is bounded on B, and $(F+\varepsilon)^n-F^n\geq\varepsilon^n$, so the Borel set $\{U<v_\varepsilon\}$ is Lebesgue null. Hence $\max\{U,v_\varepsilon\}=U$ almost everywhere, and the uniqueness assertion of Theorem 8.2.1 gives $U\geq v_\varepsilon$ everywhere. Letting $\varepsilon\downarrow 0$, we get $U\geq v$. Since U is itself a subsolution, it is the pointwise supremum of all subsolutions. In particular, any two solutions in $\mathrm{PSH}(B)\cap C(\overline{B})$ coincide with this supremum, which proves uniqueness. For data of class C^2 the solution is therefore the envelope of Step 1, which is locally $C^{1,1}$ by Step 3.

Finally, if (f,φ) and (g,χ) are continuous data, approximate both as in Step 4. By uniqueness, the corresponding solutions converge uniformly to $H_{f,\varphi}$ and $H_{g,\chi}$, and (9.9) for the approximating data of class C^2 passes to the limit. Undoing the affine change of variables proves all assertions of the theorem. □

9.9 Bounded measurable densities

We now allow an arbitrary bounded nonnegative measurable density f, approximate it by continuous densities f_j obtained by convolution, and pass to the limit in the solutions given by Theorem 9.8.2. Since f_j need not converge to f uniformly, the stability estimate (9.9) is of no use here. However, the f_j have a common bound and converge in L^1, and we show that this suffices for convergence of the solutions in capacity. A comparison argument (Lemma 9.9.1) bounds the capacity of the set where an approximate solution lies far below the limit in terms of the common bound and the L^1 convergence, while Hartogs' lemma and quasi-continuity control the set where it lies far above. The convergence theorem (Theorem 9.4.8) then identifies the Monge–Ampère measure of the limit.

Lemma 9.9.1 *Let $D \subset \mathbb{C}^n$ be a bounded open set, let u, v be locally bounded plurisubharmonic functions on D, and let $s \in \mathbb{R}$ and $t > 0$. If $\{u < v - s\} \Subset D$, then*

$$t^n \operatorname{Cap}_D(\{u < v - s - t\}) \le \int_{\{u<v-s\}} (\mathrm{dd^c} u)^n.$$

Proof Let $w \in \mathrm{PSH}(D)$ with $-1 \le w \le 0$. Then

$$\{u < v - s - t\} \subset \{u < v - s + tw\} \subset \{u < v - s\},$$

so the middle set is relatively compact in D. Hence every $x \in \partial D$ has a neighborhood on whose intersection with D we have $u \ge v - s + tw$, and in particular $\underline{\lim}_{D \ni z \to x} \big(u(z) - (v(z) - s + tw(z))\big) \ge 0$. Expanding $(\mathrm{dd^c} v + t\,\mathrm{dd^c} w)^n$ by additivity and symmetry of the products (Proposition 9.1.2 and Theorem 9.2.5), and using that all the mixed products of the locally bounded plurisubharmonic functions v and w are positive, we get $\big(\mathrm{dd^c}(v - s + tw)\big)^n \ge t^n (\mathrm{dd^c} w)^n$. The comparison principle Theorem 9.2.8, applied to u and $v - s + tw$, therefore gives

$$\begin{aligned} t^n \int_{\{u<v-s-t\}} (\mathrm{dd^c} w)^n &\le \int_{\{u<v-s+tw\}} \big(\mathrm{dd^c}(v - s + tw)\big)^n \\ &\le \int_{\{u<v-s+tw\}} (\mathrm{dd^c} u)^n \le \int_{\{u<v-s\}} (\mathrm{dd^c} u)^n. \end{aligned}$$

As w ranges over these functions, $w + 1$ ranges over the competitors for Cap_D (Definition 9.3.2). Taking the supremum over w proves the lemma. □

Theorem 9.9.2 *Let $B = B(a, R) \subset \mathbb{C}^n$ with $n \ge 1$ and $R > 0$, let $f \in L^\infty(B)$ be nonnegative, and let $\varphi \in C(\partial B)$ be real. There is a unique bounded plurisubharmonic function $U_{f,\varphi}$ on B such that*

$$(\mathrm{dd^c} U_{f,\varphi})^n = f\eta^n, \qquad \lim_{B \ni z \to x} U_{f,\varphi}(z) = \varphi(x) \quad (x \in \partial B).$$

It is the pointwise supremum of the bounded plurisubharmonic functions v on B such that $(\mathrm{dd}^c v)^n \geq f\eta^n$ and $\overline{\lim}_{B\ni z\to x} v(z) \leq \varphi(x)$ for every $x \in \partial B$. The stability estimate (9.9) also holds with $U_{f,\varphi}$ and $U_{g,\chi}$ in place of $H_{f,\varphi}$ and $H_{g,\chi}$, for every nonnegative $g \in L^\infty(B)$ and real $\chi \in C(\partial B)$, the norm of $f^{1/n} - g^{1/n}$ being the essential supremum.

Moreover, let $f_j \in L^\infty(B)$ and $\varphi_j \in C(\partial B)$ be such that $0 \leq f_j \leq M$ almost everywhere for a constant M, $f_j \to f$ in $L^1(B)$ and $\varphi_j \to \varphi$ uniformly on ∂B. Then

$$U_{f_j,\varphi_j} \longrightarrow U_{f,\varphi} \quad \text{in } L^1(B) \text{ and locally in capacity.}$$

Proof Throughout the proof, $H_{g,\chi}$ denotes the solution of Theorem 9.8.2 for continuous data (g,χ).

Step 1. Approximation and barriers. Extend f by zero outside B, let $M = \|f\|_{L^\infty(B)}$, and convolve f with mollifiers. The resulting smooth functions f_j on $\mathbb{C}^n$ satisfy $0 \leq f_j \leq M$, and $f_j \to f$ in $L^1(B)$ by Lemma 1.4.1(2), applied on $\mathbb{C}^n$ with the compact set $\overline{B}$; thus, with $\varphi_j = \varphi$, they satisfy the hypotheses of the second part of the theorem. Let $u_j = H_{f_j,\varphi}$. The constant density M and the zero density give two barriers. Indeed, since $M\eta^n \geq f_j\eta^n \geq 0$, the function $H_{M,\varphi}$ belongs to the family whose supremum is u_j in Theorem 9.8.2, and u_j belongs to the family whose supremum is $H_{0,\varphi}$. Hence

$$H_{M,\varphi} \leq u_j \leq H_{0,\varphi}.$$

Both barriers are continuous on $\overline{B}$ and equal φ on ∂B.

By Theorem 8.3.4, after passing to a subsequence (not relabeled), u_j converges in $L^1_{\mathrm{loc}}(B)$ to a plurisubharmonic function U; convergence to $-\infty$ is excluded by the lower barrier. For every compact set $K \subset B$ we have $(H_{M,\varphi} - U)_+ \leq |u_j - U|$ and $(U - H_{0,\varphi})_+ \leq |u_j - U|$, so the integrals of these two functions over K are bounded by $\|u_j - U\|_{L^1(K)} \to 0$ and hence vanish. Thus $H_{M,\varphi} \leq U \leq H_{0,\varphi}$ almost everywhere, and hence everywhere by the uniqueness assertion of Theorem 8.2.1, applied to $\max\{U, H_{M,\varphi}\}$ and U, and to $\max\{U, H_{0,\varphi}\}$ and $H_{0,\varphi}$. Consequently U is bounded and $\lim_{B\ni z\to x} U(z) = \varphi(x)$ for every $x \in \partial B$. Since u_j and U lie between the two bounded barriers, $|u_j - U|$ is bounded independently of j, so the convergence also holds in $L^1(B)$: given $\varepsilon > 0$, choose a compact set $K \subset B$ with $|B \setminus K| < \varepsilon$ and use the convergence in $L^1(K)$.

Step 2. Convergence in capacity. Fix a compact set $K \subset B$ and $\delta > 0$. We first show that the outer capacity of $K \cap \{u_j > U + \delta\}$ tends to zero. Let $\varepsilon > 0$. By Theorem 9.4.2, there is an open set $G \subset B$ with $\mathrm{Cap}_B(G) < \varepsilon$ such that the restriction of U to $B \setminus G$ is continuous. We claim that $u_j < U + \delta$ on the compact set $K \setminus G$ for all sufficiently large j. Let $x \in K \setminus G$. By upper semicontinuity, $U < U(x) + \delta/4$ on a neighborhood of a closed ball $\overline{B'}$ centered at x, so Hartogs' lemma (Theorem 8.3.4), applied with the constant majorant $U(x) + \delta/4$, gives $u_j < U(x) + \delta/4$ on $\overline{B'}$ for all large j. Shrinking B', we may also assume that $U > U(x) - \delta/4$ on $(K \setminus G) \cap B'$, since the restriction of U to $B \setminus G$ is continuous. Hence $u_j - U < \delta/2$ on $(K \setminus G) \cap B'$ for all large j, and a finite cover of $K \setminus G$ by such balls proves the claim. Consequently $K \cap \{u_j > U + \delta\} \subset G$

for all large j, so that $\overline{\lim}_j \operatorname{Cap}_B^*(K \cap \{u_j > U + \delta\}) \le \operatorname{Cap}_B(G) < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, the limit is zero.

Next we consider the set where u_j lies below U. Since $H_{M,\varphi} \le u_j$ and $U \le H_{0,\varphi}$, the set $\{u_j < U - \delta\}$ is contained in $\{H_{M,\varphi} - H_{0,\varphi} \le -\delta\}$, which is a compact subset of B because the continuous function $H_{M,\varphi} - H_{0,\varphi}$ on $\overline{B}$ vanishes on ∂B. Lemma 9.9.1, applied with $s = t = \delta$ to u_j and U, gives

$$\begin{aligned}\delta^n \operatorname{Cap}_B(\{u_j < U - 2\delta\}) &\le \int_{\{u_j < U-\delta\}} f_j \eta^n \\ &\le \frac{M}{\delta} \int_B (U - u_j)_+ \eta^n \longrightarrow 0,\end{aligned}$$

because $(\mathrm{dd}^{\mathrm{c}} u_j)^n = f_j \eta^n \le M\eta^n$, $U - u_j > \delta$ on the domain of integration, $\eta^n = n!\pi^{-n}\,\mathrm{d}V_{2n}$, and $u_j \to U$ in $L^1(B)$. To pass from capacity to the outer capacity of Definition 9.4.5, we use quasi-continuity. The function $(U - u_j)_+$ is bounded and quasi-continuous: U and u_j are real and quasi-continuous by Theorem 9.4.2, so their difference, and its composition with the continuous map $t \mapsto t_+$, are quasi-continuous by the remark following Definition 9.4.1. Let $L \subset B$ be a compact neighborhood of K. By Lemma 9.4.4,

$$\begin{aligned}\operatorname{Cap}_B^*(K \cap \{u_j < U - 4\delta\}) &\le \operatorname{Cap}_B(L \cap \{(U - u_j)_+ > 2\delta\}) \\ &\le \operatorname{Cap}_B(\{u_j < U - 2\delta\}) \longrightarrow 0.\end{aligned}$$

As K and δ are arbitrary, $u_j \to U$ locally in capacity in the sense of Definition 9.4.5.

Step 3. The equation, the envelope and uniqueness. The functions u_j are uniformly bounded and converge to U locally in capacity, so Theorem 9.4.8 gives $(\mathrm{dd}^{\mathrm{c}} u_j)^n \to (\mathrm{dd}^{\mathrm{c}} U)^n$ weakly. Since $f_j \to f$ in $L^1(B)$, also $f_j\eta^n \to f\eta^n$ weakly, and therefore $(\mathrm{dd}^{\mathrm{c}} U)^n = f\eta^n$.

Let v be a bounded plurisubharmonic function on B with $(\mathrm{dd}^{\mathrm{c}} v)^n \ge f\eta^n$ and boundary upper limits at most φ, and put $v_\varepsilon = v + \varepsilon(|z-a|^2 - R^2)$ for $\varepsilon > 0$. Expanding $(\mathrm{dd}^{\mathrm{c}} v + \varepsilon\eta)^n$ and using positivity of the mixed products, we get $(\mathrm{dd}^{\mathrm{c}} v_\varepsilon)^n \ge (f + \varepsilon^n)\eta^n$. Since $\underline{\lim}_{z \to x}(U(z) - v_\varepsilon(z)) \ge 0$ for every $x \in \partial B$, the comparison principle Theorem 9.2.8 gives

$$\int_{\{U < v_\varepsilon\}} (f + \varepsilon^n)\eta^n \le \int_{\{U < v_\varepsilon\}} f\eta^n.$$

These integrals are finite, so $\{U < v_\varepsilon\}$ is Lebesgue null. By the uniqueness assertion of Theorem 8.2.1, applied to $\max\{U, v_\varepsilon\}$ and U, we obtain $U \ge v_\varepsilon$ everywhere, and letting $\varepsilon \downarrow 0$ gives $U \ge v$. Since U itself is such a function v, it is the pointwise supremum in the statement; in particular, it does not depend on the subsequence chosen in Step 1. If U' is another bounded solution with boundary limits φ, then $U' \le U$, and the same argument with U' in place of U gives $U \le U'$. This proves existence and uniqueness, and we write $U_{f,\varphi} = U$.

Step 4. Stability. Let f_j, φ_j and M be as in the second part of the theorem, and suppose first that $\varphi_j = \varphi$ for all j. By the envelope characterization just proved, $H_{M,\varphi} \le U_{f_j,\varphi} \le H_{0,\varphi}$. Steps 1–3 apply without change to the sequence $U_{f_j,\varphi}$, since they used only these barrier inequalities, the equations $(\mathrm{dd^c} U_{f_j,\varphi})^n = f_j \eta^n$ with $0 \le f_j \le M$, and the convergence $f_j \to f$ in $L^1(B)$. Hence every subsequence of $U_{f_j,\varphi}$ has a further subsequence converging in $L^1(B)$ and locally in capacity to a bounded solution with data (f, φ), that is, to $U_{f,\varphi}$. Therefore the whole sequence converges to $U_{f,\varphi}$ in $L^1(B)$ and locally in capacity.

For varying boundary data we use that, for a fixed density f, the function $U_{f,\chi} - \|\varphi - \chi\|_{L^\infty(\partial B)}$ belongs to the family whose supremum is $U_{f,\varphi}$; by symmetry, $\|U_{f,\varphi} - U_{f,\chi}\|_{L^\infty(B)} \le \|\varphi - \chi\|_{L^\infty(\partial B)}$. Applied with f_j in place of f, this gives $\|U_{f_j,\varphi_j} - U_{f_j,\varphi}\|_{L^\infty(B)} \le \|\varphi_j - \varphi\|_{L^\infty(\partial B)} \to 0$. Since B is bounded, it follows that $U_{f_j,\varphi_j} - U_{f_j,\varphi} \to 0$ in $L^1(B)$. Moreover, for every $\delta > 0$ and all large j, the set $\{U_{f_j,\varphi_j} > U_{f,\varphi} + \delta\}$ is contained in $\{U_{f_j,\varphi} > U_{f,\varphi} + \delta/2\}$, and similarly for the sets where the functions lie below $U_{f,\varphi} - \delta$. The general case therefore follows from the case of fixed boundary data.

Finally, we prove (9.9) for bounded measurable densities. Let $f, g \in L^\infty(B)$ be nonnegative and $\varphi, \chi \in C(\partial B)$. Extend $f^{1/n}$ and $g^{1/n}$ by zero outside B, and convolve them with the same mollifiers. The resulting smooth functions $F_j, G_j \ge 0$ satisfy $\|F_j - G_j\|_{L^\infty(B)} \le \|f^{1/n} - g^{1/n}\|_{L^\infty(B)}$. Moreover $F_j \to f^{1/n}$ and $G_j \to g^{1/n}$ in $L^1(B)$ with a common bound, so $F_j^n \to f$ and $G_j^n \to g$ in $L^1(B)$ with a common bound. By Theorem 9.8.2,

$$\|H_{F_j^n,\varphi} - H_{G_j^n,\chi}\|_{L^\infty(B)} \le c, \qquad c = \|\varphi - \chi\|_{L^\infty(\partial B)} + R^2 \|f^{1/n} - g^{1/n}\|_{L^\infty(B)}.$$

By uniqueness, $H_{F_j^n,\varphi} = U_{F_j^n,\varphi}$ and $H_{G_j^n,\chi} = U_{G_j^n,\chi}$, and by the stability just proved these functions converge in $L^1(B)$ to $U_{f,\varphi}$ and $U_{g,\chi}$. Since $|H_{F_j^n,\varphi} - H_{G_j^n,\chi}| \le c$ on B, the triangle inequality gives

$$\bigl(|U_{f,\varphi} - U_{g,\chi}| - c\bigr)_+ \le |U_{f,\varphi} - H_{F_j^n,\varphi}| + |H_{G_j^n,\chi} - U_{g,\chi}|,$$

and integrating over B and letting $j \to \infty$, we obtain $|U_{f,\varphi} - U_{g,\chi}| \le c$ almost everywhere. The uniqueness assertion of Theorem 8.2.1, applied to $\max\{U_{f,\varphi}, U_{g,\chi} + c\}$ and $U_{g,\chi} + c$, and to $\max\{U_{g,\chi}, U_{f,\varphi} + c\}$ and $U_{f,\varphi} + c$, shows that this inequality holds everywhere. □

9.10 Local energy moments

To solve $(\mathrm{dd^c} u)^n = \mu$ for a measure μ which need not be absolutely continuous, we approximate μ by the measures μ_j obtained by averaging μ over the cubes of finer and finer grids, and we solve the equations for μ_j by Theorem 9.9.2. Each μ_j has a bounded density, but the bounds are not uniform in j, so the arguments of the preceding section do not apply.

A bounded subsolution provides a substitute. If $\mu \le (\mathrm{dd^c} w)^n$ for a bounded plurisubharmonic function w, then μ integrates the powers $(-\varphi)^p$ of plurisubharmonic functions φ with zero boundary values, with a bound in terms of the local energy moments $\int (-\varphi)^p (\mathrm{dd^c}\varphi)^n$ (Theorem 9.10.2). The cube averages inherit this control (Proposition 9.10.3). In the subsections that follow, this control replaces the common density bound. It yields convergence of the energy integrals and convergence in capacity of the solutions for μ_j, and hence the solution of the Dirichlet problem for compactly supported measures μ of this kind (Theorem 9.10.12). The section closes with an inequality for mixed energies (Theorem 9.10.13), which is needed in Section 9.11. Throughout this section, $B \subset \mathbb{C}^n$ is a ball, $n \ge 1$.

Definition 9.10.1 Let $B \subset \mathbb{C}^n$, $n \ge 1$, be a ball. We denote by $\mathcal{E}_0(B)$ the class of bounded plurisubharmonic functions $\varphi \le 0$ on B such that

$$\lim_{B \ni z \to \zeta} \varphi(z) = 0 \quad (\zeta \in \partial B), \qquad \int_B (\mathrm{dd^c}\varphi)^n < \infty.$$

For $\varphi \in \mathcal{E}_0(B)$ and real $p \ge 1$, the *local energy moment* of order p is

$$e_p(\varphi) = \int_B (-\varphi)^p (\mathrm{dd^c}\varphi)^n.$$

This class is due to Cegrell; see Appendix B.9. The zero function belongs to $\mathcal{E}_0(B)$. Since φ is bounded and its Monge–Ampère measure has finite mass, every moment $e_p(\varphi)$ is finite. Since ∂B is compact, the boundary limit is uniform: for every $s > 0$ there is a compact set $K_s \subset B$ such that $\varphi > -s$ on $B \setminus K_s$. Indeed, every $\zeta \in \partial B$ has an open neighborhood N_ζ with $\varphi > -s$ on $N_\zeta \cap B$; finitely many of these neighborhoods cover ∂B, and we may take for K_s the complement of their union in $\overline{B}$. In particular, each set $\{\varphi \le -s\}$, $s > 0$, lies in a compact subset of B.

Theorem 9.10.2 *Let $B \subset \mathbb{C}^n$ be a ball, and let μ be a finite positive Borel measure on B such that $\mu \le (\mathrm{dd^c} w)^n$ for some bounded plurisubharmonic function w on B. Put $m = \mu(B)$ and $M = \sup_B w - \inf_B w$, and for real $p \ge 1$ put*

$$\kappa_{n,p} = 2^{n+p}(n+1)!M^n, \qquad A_p = \Bigl(1 + \frac{p}{n}\Bigr)\kappa_{n,p}^{p/(n+p)} m^{n/(n+p)}.$$

Then, for every $\varphi \in \mathcal{E}_0(B)$, every real $p \ge 1$ and every $t > 0$,

$$\mu(\{\varphi < -t\}) \le \min\{m, \kappa_{n,p} e_p(\varphi) t^{-(n+p)}\}. \tag{9.10}$$

Consequently,

$$\int_B (-\varphi)^p \, \mathrm{d}\mu \le A_p \, e_p(\varphi)^{p/(n+p)}. \tag{9.11}$$

The constant A_p depends only on n, p, m and M.

Proof We prove the tail bound by applying Błocki's estimate (Proposition 9.3.5) to truncations of φ, and then integrate it. For $s > 0$ let

$$h_s = \max\{\varphi, -s\}, \qquad d_s = h_s - \varphi = (-\varphi - s)_+.$$

By the remark following Definition 9.10.1, there is a compact set $K_s \subset B$ with $\varphi > -s$ on $B \setminus K_s$, and since K_s is compact there is a ball $B_s \Subset B$, concentric with B, which contains K_s. Thus $h_s = \varphi$ on $B_s \setminus K_s$, which is a collar of ∂B_s, and d_s vanishes outside K_s. The functions φ, h_s and w are bounded and plurisubharmonic near $\overline{B_s}$, $\varphi \le h_s$, and the oscillation of w on B_s is at most M. Błocki's estimate on B_s, together with $\mu \le (\mathrm{dd^c} w)^n$, therefore gives

$$\int_B d_s^{n+1}\,\mathrm{d}\mu \le \int_{B_s} d_s^{n+1}(\mathrm{dd^c} w)^n \le (n+1)!M^n \int_{B_s} d_s(\mathrm{dd^c}\varphi)^n.$$

Now take $s = t/2$. On $\{\varphi < -t\}$ we have $d_s > t/2$. On the other hand, $0 \le d_s \le -\varphi$, and $d_s = 0$ where $-\varphi \le t/2$; since $p \ge 1$, we have $-\varphi \le (t/2)^{1-p}(-\varphi)^p$ where $-\varphi > t/2$, and hence

$$\int_{B_s} d_s(\mathrm{dd^c}\varphi)^n \le \int_{\{-\varphi > t/2\}} (-\varphi)(\mathrm{dd^c}\varphi)^n \le (t/2)^{1-p} e_p(\varphi).$$

Combining the two estimates and dividing by $(t/2)^{n+1}$, we obtain

$$\mu(\{\varphi < -t\}) \le (n+1)!M^n (t/2)^{-n-p} e_p(\varphi) = \kappa_{n,p} e_p(\varphi) t^{-(n+p)}.$$

Since also $\mu(\{\varphi < -t\}) \le m$, this proves (9.10).

If $m = 0$, both sides of (9.11) vanish. If $\kappa_{n,p} e_p(\varphi) = 0$, then (9.10) shows that $\mu(\{\varphi < -t\}) = 0$ for every $t > 0$, so the left side of (9.11) vanishes. Otherwise put

$$T = \left(\frac{\kappa_{n,p} e_p(\varphi)}{m}\right)^{1/(n+p)}.$$

The first equality below is the layer-cake formula, which follows from Tonelli's theorem applied to the nonnegative function $(z,t) \mapsto pt^{p-1}\mathbf{1}_{\{\varphi(z) < -t\}}$ on $B \times (0,\infty)$. Using (9.10) with the bound m for $t < T$ and the second bound for $t \ge T$, we obtain

$$\begin{aligned}\int_B (-\varphi)^p\,\mathrm{d}\mu &= p\int_0^\infty t^{p-1}\mu(\{\varphi < -t\})\,\mathrm{d}t \le mT^p + p\kappa_{n,p} e_p(\varphi)\int_T^\infty t^{-n-1}\,\mathrm{d}t \\ &= mT^p + \frac{p}{n}\kappa_{n,p} e_p(\varphi) T^{-n}.\end{aligned}$$

By the choice of T,

$$mT^p = \kappa_{n,p} e_p(\varphi) T^{-n} = \kappa_{n,p}^{p/(n+p)} m^{n/(n+p)} e_p(\varphi)^{p/(n+p)},$$

so the right-hand side equals $(1 + p/n) mT^p = A_p e_p(\varphi)^{p/(n+p)}$. This proves (9.11).□

We now show that the cube averages of a compactly supported measure μ as in Theorem 9.10.2 satisfy the same moment estimate up to a constant depending only on n and p. We call the solutions U_j of $(\mathrm{dd^c} U_j)^n = \mu_j$ with zero boundary values

constructed in the following proposition the *canonical approximants* of μ (for the chosen grids).

Proposition 9.10.3 *Let μ, w, m and A_p be as in Theorem 9.10.2, and assume moreover that the support K of μ is a compact subset of B. Extend μ by zero to $\mathbb{C}^n$. Let $h_j \downarrow 0$, and for each j let $\mathcal{Q}_j$ be a partition of $\mathbb{C}^n = \mathbb{R}^{2n}$ into half-open cubes of side length h_j. Put*

$$\mu_j = \sum_{Q \in \mathcal{Q}_j} \frac{\mu(Q)}{|Q|} \mathbf{1}_Q \, \mathrm{d}V,$$

where $|Q| = h_j^{2n}$ is the volume of Q and $\mathrm{d}V = \mathrm{d}V_{2n}$ is Lebesgue measure. For $z \in \mathbb{C}^n$ and $r > 0$ let $P(z,r) = \{\zeta \in \mathbb{C}^n : |\zeta_k - z_k| < r,\ 1 \le k \le n\}$ be the polydisc of radius r centered at z. Then $\mu_j \to \mu$ weakly. Moreover, there are an index j_0 and, for every real $p \ge 1$, a constant $D_{n,p} > 0$ depending only on n and p, such that the following hold for all $j \ge j_0$ and $p \ge 1$.

(1) *The measure μ_j has a bounded density and compact support in B, and $\mu_j(B) = m$.*

(2) *If $Q \in \mathcal{Q}_j$ is a cube which meets K and $z \in Q$, then $P(z, 4\sqrt{2}\, h_j) \subset B$, and every plurisubharmonic function $v \le 0$ on $P(z, 4\sqrt{2}\, h_j)$ satisfies*

$$\frac{1}{|Q|} \int_Q (-v)^p \, \mathrm{d}V \le D_{n,p} (-v(z))^p, \tag{9.12}$$

where the right side is $+\infty$ if $v(z) = -\infty$.

(3) *Every $\varphi \in \mathcal{E}_0(B)$ satisfies*

$$\int_B (-\varphi)^p \, \mathrm{d}\mu_j \le D_{n,p} A_p e_p(\varphi)^{p/(n+p)}. \tag{9.13}$$

(4) *The bounded plurisubharmonic solution U_j of $(\mathrm{dd}^c U_j)^n = \mu_j$ with zero boundary values, given by Theorem 9.9.2, belongs to $\mathcal{E}_0(B)$ and satisfies $e_p(U_j) \le (D_{n,p} A_p)^{(n+p)/n}$.*

Finally, the sequence $(U_j)_{j \ge j_0}$ is relatively compact in $L^1(B)$, and the limit of every subsequence converging in $L^1(B)$ has a nonpositive plurisubharmonic representative.

Proof If $m = 0$, then $\mu_j = 0$ and $U_j = 0$ for all j, and all assertions hold. Assume that $m > 0$, put $d = \operatorname{dist}(K, \mathbb{C}^n \setminus B) > 0$, and choose j_0 such that $h_j < d/(6\sqrt{2n})$ for $j \ge j_0$; in the rest of the proof we consider only $j \ge j_0$. Since μ is a finite Borel measure, it is a Radon measure (Proposition 1.1.2), so μ vanishes outside its support K. Hence a cube Q with $\mu(Q) > 0$ meets K, and as its diameter is $\sqrt{2n}\, h_j < d/6$, it lies in the $(d/6)$-neighborhood of K, which is contained in B. These cubes are disjoint, each has volume h_j^{2n}, and they lie in a bounded set; so there are only finitely many of them. Therefore μ_j is a finite sum of measures with constant densities on cubes contained in B, so it has a bounded density and compact support in B, and $\mu_j(B) = \sum_Q \mu(Q) = m$. This proves (1).

For the weak convergence, let g be a continuous function on $\mathbb{C}^n$. Then

$$\int g\,\mathrm{d}\mu_j - \int g\,\mathrm{d}\mu = \sum_{Q\in\mathcal{Q}_j} \int_Q \left(\frac{1}{|Q|} \int_Q g\,\mathrm{d}V - g \right) \mathrm{d}\mu,$$

and on every cube with $\mu(Q) > 0$ the integrand is bounded by the modulus of continuity of g at $\sqrt{2n}\,h_j$ on the closed $(d/6)$-neighborhood of K. Hence the difference tends to zero.

To prove (2), fix a cube $Q \in \mathcal{Q}_j$ which meets K and a point $z \in Q$, and put $\varrho = 4\sqrt{2}\,h_j$. Every point of Q differs from z by less than $\sqrt{2}\,h_j$ in each complex coordinate, so $Q \subset P(z, \varrho/2)$. Moreover $P(z,\varrho)$ lies in the Euclidean ball of radius $4\sqrt{2n}\,h_j$ about z, and $\operatorname{dist}(z,K) \le \sqrt{2n}\,h_j$, so this polydisc lies in the $(5d/6)$-neighborhood of K and hence in B. Let $v \le 0$ be a plurisubharmonic function on $P(z,\varrho)$; if $v(z) = -\infty$, there is nothing to prove, so we assume that $v(z) > -\infty$. The function $\zeta \mapsto v(z + \varrho\zeta)$ is plurisubharmonic and nonpositive on the unit polydisc (Proposition 8.2.3), with the finite value $v(z)$ at the origin. The first estimate of Proposition 8.3.6, with its constant $C_{n,p}$, and the change of variables $\zeta \mapsto z + \varrho\zeta$, whose Jacobian is ϱ^{2n}, give

$$\int_Q (-v)^p\,\mathrm{d}V \le \int_{P(z,\varrho/2)} (-v)^p\,\mathrm{d}V \le \varrho^{2n} C_{n,p}^p (-v(z))^p.$$

Since $|Q| = h_j^{2n}$ and $(\varrho/h_j)^{2n} = 32^n$, we obtain (9.12) with $D_{n,p} = 32^n C_{n,p}^p$.

For (3), let $\varphi \in \mathcal{E}_0(B)$. Since $\varphi \le 0$ is plurisubharmonic on B, (9.12) applies to $v = \varphi$ at every point z of every cube with $\mu(Q) > 0$. Integrating over $z \in Q$ with respect to μ and summing over the cubes, we obtain

$$\int_B (-\varphi)^p\,\mathrm{d}\mu_j = \sum_{Q\in\mathcal{Q}_j} \frac{\mu(Q)}{|Q|} \int_Q (-\varphi)^p\,\mathrm{d}V \le D_{n,p} \int_B (-\varphi)^p\,\mathrm{d}\mu,$$

and (9.13) follows from (9.11).

For (4), note that μ_j equals $f_j\eta^n$ for a bounded density $f_j \ge 0$, so the solution U_j exists by Theorem 9.9.2, applied with zero boundary values. Since U_j is subharmonic (Theorem 8.2.1) with boundary limits 0, the maximum principle (Proposition 3.1.4) gives $U_j \le 0$, and its Monge–Ampère measure has finite mass $\mu_j(B) = m$; hence $U_j \in \mathcal{E}_0(B)$. Taking $\varphi = U_j$ in (9.13), we get

$$e_p(U_j) = \int_B (-U_j)^p\,\mathrm{d}\mu_j \le D_{n,p} A_p e_p(U_j)^{p/(n+p)}.$$

If $e_p(U_j) > 0$, dividing by $e_p(U_j)^{p/(n+p)}$ gives the asserted bound; if $e_p(U_j) = 0$, there is nothing to prove.

Finally write $B = B(a,R)$ and $\rho(z) = |z-a|^2 - R^2$. The function ρ is bounded and plurisubharmonic on B, and $(\mathrm{dd^c}\rho)^n = \eta^n = n!\pi^{-n}\,\mathrm{d}V$. Applying Theorem 9.10.2 to the finite measure $(\mathrm{dd^c}\rho)^n$ on B, with $p = 2$, and using the uniform bound for $e_2(U_j)$, we obtain $S = \sup_j \int_B U_j^2\,\mathrm{d}V < \infty$. Since $U_j \le 0$, Theorem 8.3.4 shows that every subsequence has a further subsequence converging in $L^1_{\mathrm{loc}}(B)$ to a plurisubharmonic

function u; convergence to $-\infty$ is excluded by the L^2 bound. Exhausting B by compact sets and passing to a diagonal subsequence, we may assume that $U_j \to u$ almost everywhere on B, so Fatou's lemma gives $\int_B u^2 \, \mathrm{d}V \le S$. For a compact set $L \subset B$, the Cauchy–Schwarz inequality gives

$$\int_{B\setminus L} |U_j - u| \, \mathrm{d}V \le 2S^{1/2} |B \setminus L|^{1/2},$$

which is small when $|B \setminus L|$ is small. Since $U_j \to u$ in $L^1(L)$, it follows that $U_j \to u$ in $L^1(B)$ along this subsequence. In particular, if a subsequence of (U_j) converges in $L^1(B)$, its limit is represented by such a function u. Finally $u \le 0$ almost everywhere, hence everywhere, by the uniqueness assertion of Theorem 8.2.1 applied to $\max\{u, 0\}$ and 0. □

9.10.1 Convergence of energy integrals

We shall identify the limit u of a convergent subsequence of canonical approximants as a solution of $(\mathrm{dd^c} u)^n = \mu$. For this we must pass to the limit in integrals such as $\int U_j \, \mathrm{d}\mu_j$, in which both the integrand and the measure depend on j, and the energy bounds of Proposition 9.10.3 control only the large negative values of the integrand. We first prove that L^1_{loc} convergence of uniformly bounded plurisubharmonic functions implies L^1 convergence against any fixed finite measure that does not charge sets of outer capacity zero (Lemma 9.10.4). Writing integrals against μ_j as integrals of cube averages against μ, we then transfer this continuity to the varying measures μ_j (Theorem 9.10.5).

Lemma 9.10.4 *Let $D \subset \mathbb{C}^n$ be a bounded open set, and let ν be a finite positive Borel measure on D such that $\nu(E) = 0$ for every Borel set $E \subset D$ satisfying*

$$\mathrm{Cap}^*_D(E \cap K) = 0 \qquad \textit{for every compact } K \subset D.$$

Let $L \ge 0$, let v_j be plurisubharmonic functions on D with $|v_j| \le L$, and suppose that $v_j \to v$ in $L^1_{\mathrm{loc}}(D)$, where v is the plurisubharmonic representative of the limit (Theorem 8.3.4, applied on each connected component of D). Then

$$\int_D |v_j - v| \, \mathrm{d}\nu \longrightarrow 0.$$

The hypothesis on ν holds if $\nu \le (\mathrm{dd^c} w)^n$ for a bounded plurisubharmonic function w on D.

Proof We show that $\overline{\lim}_j v_j = v$ ν-almost everywhere (Step 1) and that $\int v_j \, \mathrm{d}\nu \to \int v \, \mathrm{d}\nu$ (Step 2), and combine the two facts (Step 3). First, the bounds $-L \le v \le L$ hold almost everywhere: for every compact set $K \subset D$ we have $(|v| - L)_+ \le |v - v_j|$,

so $\int_K (|v| - L)_+ \, dV \le \|v_j - v\|_{L^1(K)} \to 0$. They hold everywhere by the uniqueness assertion of Theorem 8.2.1, applied on each connected component of D to the plurisubharmonic functions $\max\{v, -L\}$ and v, and to $\max\{v, L\}$ and the constant L.

Step 1. The upper limit of v_j. We show that

$$\overline{\lim_{j\to\infty}} \, v_j = v \qquad \nu\text{-almost everywhere.} \tag{9.14}$$

For $\ell \ge 1$ let $b_\ell = (\sup_{j\ge\ell} v_j)^*$ be the upper semicontinuous regularization of the supremum. By Theorem 8.3.2, these are plurisubharmonic functions with values in $[-L, L]$, and they decrease as ℓ increases. We claim that $b_\ell \downarrow v$ everywhere. Let $z \in D$ and $\varepsilon > 0$. By upper semicontinuity, $v < v(z) + \varepsilon$ on a neighborhood of a closed ball $\overline{B'} \subset D$ centered at z, and Hartogs' lemma (Theorem 8.3.4), applied on the connected component of D containing z with the constant majorant $v(z) + \varepsilon$, gives $v_j < v(z) + \varepsilon$ on $\overline{B'}$ for all large j. Hence $\sup_{j\ge\ell} v_j \le v(z) + \varepsilon$ on B' for all large ℓ, so $b_\ell(z) \le v(z) + \varepsilon$, and the decreasing limit of b_ℓ is at most v.

Conversely, $b_\ell \ge v_j$ for all $j \ge \ell$, so $(v - b_\ell)_+ \le |v - v_j|$ for $j \ge \ell$; integrating over a compact set $K \subset D$ and letting $j \to \infty$, we find that $b_\ell \ge v$ almost everywhere on K, hence almost everywhere on D. The decreasing limit of the b_ℓ is bounded below by $-L$, so it is plurisubharmonic by Theorem 8.3.1; it is at most v and equals v almost everywhere, hence everywhere by the uniqueness of plurisubharmonic representatives, applied on each connected component of D.

The sets $N_\ell = \{\sup_{j\ge\ell} v_j < b_\ell\}$ are Borel, and by Corollary 9.5.2 they satisfy $\mathrm{Cap}^*_D(N_\ell \cap K) = 0$ for every compact $K \subset D$. By countable subadditivity of outer capacity (Proposition 9.3.3), the union $N = \bigcup_\ell N_\ell$ has the same property, so $\nu(N) = 0$ by hypothesis. For $z \notin N$ we have $\sup_{j\ge\ell} v_j(z) = b_\ell(z)$ for every ℓ, and therefore $\overline{\lim}_j \, v_j(z) = \lim_\ell b_\ell(z) = v(z)$. This proves (9.14). The argument applies to every sequence of plurisubharmonic functions with values in $[-L, L]$ that converges to v in $L^1_{\mathrm{loc}}(D)$.

Step 2. Convergence of $\int v_j \, d\nu$. We first construct convex combinations of the v_j whose integrals converge to $\int v \, d\nu$. Let C_ℓ be the set of finite convex combinations of $v_\ell, v_{\ell+1}, \ldots$; its elements are plurisubharmonic functions with values in $[-L, L]$ (Proposition 8.2.5). Let

$$d_\ell = \inf_{r\in C_\ell} \|r\|_{L^2(\nu)}.$$

Since $C_{\ell+1} \subset C_\ell$, the numbers d_ℓ increase, and they are bounded by $L\,\nu(D)^{1/2}$; let d be their limit. Choose $r_\ell \in C_\ell$ with

$$\|r_\ell\|^2_{L^2(\nu)} \le d_\ell^2 + \ell^{-1}.$$

For $k \ge \ell$ the midpoint $(r_k + r_\ell)/2$ belongs to the convex set C_ℓ, so its norm is at least d_ℓ, and the parallelogram identity gives

$$\|r_k - r_\ell\|^2_{L^2(\nu)} = 2\|r_k\|^2_{L^2(\nu)} + 2\|r_\ell\|^2_{L^2(\nu)} - 4\left\|\frac{r_k + r_\ell}{2}\right\|^2_{L^2(\nu)} \le 2(d^2 - d_\ell^2) + 4\ell^{-1}.$$

The right-hand side tends to zero as $\ell \to \infty$, so we can choose $\ell_1 < \ell_2 < \cdots$ with $\|r_{\ell_{i+1}} - r_{\ell_i}\|_{L^2(\nu)} \le 2^{-i}$. By the triangle inequality in $L^2(\nu)$ and the monotone convergence theorem, the function $g = \sum_i |r_{\ell_{i+1}} - r_{\ell_i}|$ satisfies $\|g\|_{L^2(\nu)} \le 1$. Hence $g < \infty$ ν-almost everywhere, and at every point where g is finite the sequence (r_{ℓ_i}) converges.

On the other hand, the r_{ℓ_i} are plurisubharmonic with values in $[-L, L]$, and $r_\ell \to v$ in $L^1_{\mathrm{loc}}(D)$, since $\|r_\ell - v\|_{L^1(K)} \le \sup_{j \ge \ell} \|v_j - v\|_{L^1(K)}$ for every compact $K \subset D$. By Step 1, applied to the sequence (r_{ℓ_i}), we have $\overline{\lim}_i r_{\ell_i} = v$ ν-almost everywhere. Therefore $r_{\ell_i} \to v$ ν-almost everywhere, and dominated convergence, with the constant bound L and the finite measure ν, gives $\int r_{\ell_i}\, \mathrm{d}\nu \to \int v\, \mathrm{d}\nu$.

We now show that $\int v_j\, \mathrm{d}\nu \to \int v\, \mathrm{d}\nu$. Otherwise there are $c > 0$ and a subsequence (v_{j_k}) such that either $\int v_{j_k}\, \mathrm{d}\nu \ge \int v\, \mathrm{d}\nu + c$ for all k, or $\int v_{j_k}\, \mathrm{d}\nu \le \int v\, \mathrm{d}\nu - c$ for all k. The subsequence satisfies the hypotheses of the lemma, so the construction above, applied to it, provides convex combinations of its terms whose integrals converge to $\int v\, \mathrm{d}\nu$. This is impossible, since every convex combination of the v_{j_k} satisfies the same inequality as the v_{j_k} themselves.

Step 3. Conclusion. By Step 1, $\overline{\lim}_j v_j \le v$ everywhere, so $(v_j - v)_+ \to 0$ pointwise. These functions are bounded by $2L$, so $\int (v_j - v)_+\, \mathrm{d}\nu \to 0$ by dominated convergence. Since

$$|v_j - v| = 2(v_j - v)_+ - (v_j - v),$$

Step 2 proves the assertion.

Finally, suppose that $\nu \le (\mathrm{dd^c} w)^n$ with w bounded and plurisubharmonic on D, and put $M = \sup_D w - \inf_D w$. If $M = 0$, then w is constant and $\nu = 0$. Otherwise $(w - \inf_D w)/M$ is a competitor for Cap_D (Definition 9.3.2), whose Monge–Ampère measure is $M^{-n}(\mathrm{dd^c} w)^n$. Hence $\nu(A) \le M^n \mathrm{Cap}_D(A)$ for every Borel set $A \subset D$, and $\mathrm{Cap}_D(A) \le \mathrm{Cap}^*_D(A)$ because Cap_D is increasing. If E is as in the hypothesis, then $\nu(E \cap K) = 0$ for every compact $K \subset D$, and exhausting D by countably many compact sets gives $\nu(E) = 0$. □

Theorem 9.10.5 *Let μ, μ_j and U_j be as in Proposition 9.10.3. Suppose that a subsequence of (U_j), relabeled (U_j) together with the corresponding measures μ_j, converges in $L^1(B)$, and let u be the plurisubharmonic representative of its limit, which is nonpositive by Proposition 9.10.3. Then $u \in L^2(\mu)$ and*

$$\int_B |U_j - u|\, \mathrm{d}\mu \longrightarrow 0, \qquad \int_B |U_j - u|\, \mathrm{d}\mu_j \longrightarrow 0. \tag{9.15}$$

In particular,

$$\int_B U_j (\mathrm{dd^c} U_j)^n \longrightarrow \int_B u\, \mathrm{d}\mu.$$

Proof If $m = 0$, then $\mu = \mu_j = 0$ and there is nothing to prove. We assume that $m > 0$, and use the cubes $Q \in \mathcal{Q}_j$, the polydiscs $P(z, r)$ and the estimate (9.12) of Proposition 9.10.3; recall that K denotes the support of μ.

Step 1. Convergence against the fixed measure μ. By Theorem 9.10.2 with $p = 2$ and the uniform bound for $e_2(U_j)$ in Proposition 9.10.3, we have $S = \sup_j \int_B U_j^2 \, d\mu < \infty$. Fix $T > 0$. The truncations $\max\{U_j, -T\}$ are plurisubharmonic with values in $[-T, 0]$, and they converge to $\max\{u, -T\}$ in $L^1_{\mathrm{loc}}(B)$, since $s \mapsto \max\{s, -T\}$ is 1-Lipschitz. The measure μ satisfies the hypothesis of Lemma 9.10.4, because $\mu \le (dd^c w)^n$, so the truncations converge in $L^1(\mu)$. Since $|a^2 - b^2| \le 2T|a - b|$ for $a, b \in [-T, 0]$, it follows that $\int \max\{u, -T\}^2 \, d\mu \le S$. Letting $T \to \infty$, monotone convergence gives $\int u^2 \, d\mu \le S$; in particular $u \in L^2(\mu)$, and u is finite μ-almost everywhere. For every measurable $v \le 0$ we have $\int_B |v - \max\{v, -T\}| \, d\mu \le T^{-1} \int_B v^2 \, d\mu$, because $0 \le \max\{v, -T\} - v \le v^2/T$. Applying this to $v = U_j$ and $v = u$, and using the convergence of the truncations in $L^1(\mu)$, we get $\overline{\lim}_j \int_B |U_j - u| \, d\mu \le 2S/T$, and letting $T \to \infty$ proves the first convergence in (9.15).

Step 2. Cube averages. For $z \in K$ let $Q_j(z)$ be the cube of $\mathcal{Q}_j$ containing z, and write

$$\mathcal{A}_j v(z) = \frac{1}{|Q_j(z)|} \int_{Q_j(z)} v \, dV$$

for the average over this cube of a function v integrable on it. If v is μ_j-integrable, then v is integrable on every cube Q with $\mu(Q) > 0$; on a cube with $\mu(Q) = 0$ on which v is not integrable we set $\mathcal{A}_j v = 0$, which does not affect integrals against μ. Since μ is carried by K, the definition of μ_j gives

$$\int_B v \, d\mu_j = \int_K \mathcal{A}_j v \, d\mu. \tag{9.16}$$

For a plurisubharmonic function $v \le 0$ on B and $z \in K$, the Cauchy–Schwarz inequality for the probability measure $dV/|Q_j(z)|$ on $Q_j(z)$ and (9.12) with $p = 2$ give

$$|\mathcal{A}_j v(z)|^2 \le \mathcal{A}_j(v^2)(z) \le D_{n,2} |v(z)|^2; \tag{9.17}$$

if $v(z) = -\infty$, the right side is infinite. This applies in particular to the possibly unbounded function u.

Fix $z \in K$ with $u(z) > -\infty$ and $\varepsilon > 0$, and put $c = u(z) + 2\varepsilon$. By upper semicontinuity of u and Hartogs' lemma (Theorem 8.3.4), there is a neighborhood of z on which $u \le c$ and $U_j \le c$ for all large j. Since the polydiscs $P(z, 4\sqrt{2}\, h_j)$ shrink to z, for $v = u$ or $v = U_j$ and all large j the function $v - c$ is plurisubharmonic and nonpositive on $P(z, 4\sqrt{2}\, h_j)$. Then (9.12) with $p = 1$, applied to $v - c$, gives $\mathcal{A}_j(c - v)(z) \le D_{n,1}(c - v(z))$, while $\mathcal{A}_j v(z) \le c$ because $v \le c$ on $Q_j(z)$. Thus, for all large j,

$$c - D_{n,1}(c - v(z)) \le \mathcal{A}_j v(z) \le c, \qquad v = u \text{ or } v = U_j.$$

Consequently $\mathcal{A}_j u(z) \to u(z)$, and $\mathcal{A}_j U_j(z) \to u(z)$ whenever $U_j(z) \to u(z)$. Indeed, in both cases the lower bound tends to $u(z) + 2\varepsilon(1 - D_{n,1})$ as $j \to \infty$, the upper bound is $u(z) + 2\varepsilon$, and $\varepsilon > 0$ is arbitrary.

We use the following elementary fact: if Borel functions F_j on K converge to zero μ-almost everywhere and are bounded in $L^2(\mu)$, then $\int_K |F_j|\,\mathrm{d}\mu \to 0$. Indeed, for $\varepsilon > 0$ the Cauchy–Schwarz inequality gives

$$\int_K |F_j|\,\mathrm{d}\mu \le \varepsilon m + \|F_j\|_{L^2(\mu)}\mu(\{|F_j| > \varepsilon\})^{1/2},$$

and $\mu(\{|F_j| > \varepsilon\}) = \int_K \mathbf{1}_{\{|F_j|>\varepsilon\}}\,\mathrm{d}\mu \to 0$ by dominated convergence; we let first $j \to \infty$ and then $\varepsilon \downarrow 0$.

By (9.17) and Step 1, the sequences $(\mathcal{A}_j U_j - u)$ and $(\mathcal{A}_j u - u)$ are bounded in $L^2(\mu)$. Since u is finite μ-almost everywhere, we have shown that $\mathcal{A}_j u \to u$ μ-almost everywhere, so $\mathcal{A}_j u \to u$ in $L^1(\mu)$ by the elementary fact. By the first convergence in (9.15), every subsequence of (U_j) has a further subsequence (U_{j_k}) with $\sum_k \int_B |U_{j_k} - u|\,\mathrm{d}\mu < \infty$. By the monotone convergence theorem, $\sum_k |U_{j_k} - u|$ is μ-integrable, hence finite μ-almost everywhere, so $U_{j_k} \to u$ μ-almost everywhere. By the pointwise convergence proved above, applied along this subsequence, $\mathcal{A}_{j_k} U_{j_k} \to u$ μ-almost everywhere, and hence in $L^1(\mu)$. Consequently the whole sequence $\mathcal{A}_j U_j$ converges to u in $L^1(\mu)$. The function $U_j - u$ is μ_j-integrable, since u is locally integrable and μ_j has a bounded density with compact support. Therefore (9.16) gives

$$\int_B (U_j - u)\,\mathrm{d}\mu_j = \int_K (\mathcal{A}_j U_j - \mathcal{A}_j u)\,\mathrm{d}\mu \longrightarrow 0. \tag{9.18}$$

Step 3. The absolute difference. Put $g_j = \mathcal{A}_j((U_j - u)_+)$ on K. Let $z \in K$ with $u(z) > -\infty$, and let c be as in Step 2. For all large j we have $U_j \le c$ and $u \le c$ on $Q_j(z)$, so $(U_j - u)_+ \le c - u$ there, and hence $0 \le g_j(z) \le c - \mathcal{A}_j u(z)$. Since $\mathcal{A}_j u(z) \to u(z)$, letting $j \to \infty$ and then $\varepsilon \downarrow 0$ shows that $g_j \to 0$ μ-almost everywhere. The Cauchy–Schwarz inequality and (9.17) give

$$g_j(z)^2 \le 2\mathcal{A}_j(U_j^2)(z) + 2\mathcal{A}_j(u^2)(z) \le 2D_{n,2}\big(U_j(z)^2 + u(z)^2\big),$$

so the g_j are bounded in $L^2(\mu)$ by Step 1. The elementary fact of Step 2 gives $\int_K g_j\,\mathrm{d}\mu \to 0$, and by (9.16) this integral equals $\int_B (U_j - u)_+\,\mathrm{d}\mu_j$. Combining this with (9.18) and the identity $|U_j - u| = 2(U_j - u)_+ - (U_j - u)$, we obtain the second convergence in (9.15).

Finally, since $(\mathrm{dd^c} U_j)^n = \mu_j$, we have $\int_B U_j(\mathrm{dd^c} U_j)^n = \int_B (U_j - u)\,\mathrm{d}\mu_j + \int_B u\,\mathrm{d}\mu_j$. The first term tends to zero by the second convergence in (9.15), and by (9.16) and Step 2 the second term equals $\int_K \mathcal{A}_j u\,\mathrm{d}\mu$, which tends to $\int_B u\,\mathrm{d}\mu$. This proves the last assertion. □

9.10.2 Capacity convergence and truncated measures

In Theorem 9.9.2, a common bound for the densities forced the solutions to converge in capacity, and Theorem 9.4.8 then identified the limiting equation. The canonical

approximants U_j of Proposition 9.10.3 have no such bound unless μ itself has a bounded density, since a bound $\mu_j \le M\,dV$ for all j would pass to the weak limit μ; nor are the functions U_j uniformly bounded. We show that the convergence of the energy integrals in Theorem 9.10.5 takes the place of the density bound once we pass to the uniformly bounded truncations $\max\{U_j, -T\}$, whose Monge–Ampère measures differ from μ_j by an error controlled by the energy (Lemma 9.10.7). Thus a subsequence of canonical approximants that converges in $L^1(B)$ also converges locally in capacity, and the Monge–Ampère measures of the truncations of its limit approximate μ in total variation (Theorem 9.10.9).

As throughout this section, B is a ball in $\mathbb{C}^n$, $n \ge 1$. For finite positive Borel measures ν_1, ν_2 on B we use the total variation distance

$$\|\nu_1 - \nu_2\|_{\mathrm{TV}} = \sup\left\{\left|\int_B f\,d(\nu_1 - \nu_2)\right| : f \text{ real Borel},\ |f| \le 1\right\}.$$

The supremum does not change if f ranges only over the real functions in $C_c(B)$ with $|f| \le 1$. Indeed, given a real Borel function f with $|f| \le 1$, Proposition 1.2.3 provides functions in $C_c(B)$ converging to f in $L^1(\nu_1 + \nu_2)$, which we may take real by passing to real parts; clamping them to $[-1, 1]$ keeps them in $C_c(B)$ and does not increase their distance from f.

Let $T > 0$ and $\delta > 0$, and let f, g be functions on B with values in $[-\infty, \infty)$. Then

$$\begin{aligned} \big\{\max\{f, -T\} > \max\{g, -T\} + \delta\big\} &\subset \{f > g + \delta\}, \\ \big\{\max\{f, -T\} < \max\{g, -T\} - \delta\big\} &\subset \{f < g - \delta\}. \end{aligned} \tag{9.19}$$

Indeed, at a point of the first set we have $\max\{f, -T\} > -T$, so $f = \max\{f, -T\} > \max\{g, -T\} + \delta \ge g + \delta$; the second inclusion is the first with f and g interchanged. Consequently, truncation at a fixed level preserves convergence locally in capacity (Definition 9.4.5): if $f_j \to f$, then $\max\{f_j, -T\} \to \max\{f, -T\}$.

Modifying a locally bounded plurisubharmonic function on a compact set does not change its total Monge–Ampère mass.

Lemma 9.10.6 *Let $\Omega \subset \mathbb{C}^n$ be open, and let φ, ψ be locally bounded plurisubharmonic functions on Ω which coincide outside a compact set $K \subset \Omega$. Then $(dd^c\varphi)^n$ and $(dd^c\psi)^n$ agree on $\Omega \setminus K$ and assign the same finite mass to K. In particular $\int_\Omega (dd^c\varphi)^n = \int_\Omega (dd^c\psi)^n$, where both sides may be infinite.*

Proof The two measures agree on the open set $\Omega \setminus K$, because the products are local (Proposition 9.1.2). For $0 \le k \le n-1$ put $S_k = (dd^c\varphi)^k \wedge (dd^c\psi)^{n-1-k}$. By the symmetry of the products (Theorem 9.2.5) and Definition 9.1.1,

$$dd^c(\varphi S_k) = (dd^c\varphi)^{k+1} \wedge (dd^c\psi)^{n-1-k}, \qquad dd^c(\psi S_k) = (dd^c\varphi)^k \wedge (dd^c\psi)^{n-k},$$

so the sum over k telescopes and

$$(dd^c\varphi)^n - (dd^c\psi)^n = dd^c R, \qquad R = (\varphi - \psi)\sum_{k=0}^{n-1} S_k.$$

The coefficient measures of the current R of order zero vanish on $\Omega \setminus K$, so $\operatorname{Supp} R \subset K$. By Lemma 1.4.1 there is $\chi \in C_c^\infty(\Omega)$ with $0 \le \chi \le 1$ which equals one on a neighborhood of K. Then $\mathrm{dd^c}\chi$ vanishes on a neighborhood of $\operatorname{Supp} R$, and therefore

$$\int \chi\,(\mathrm{dd^c}\varphi)^n - \int \chi\,(\mathrm{dd^c}\psi)^n = R(\mathrm{dd^c}\chi) = 0,$$

where both integrals are finite. Since $\chi = 1$ on K and the two measures agree on $\Omega \setminus K$, it follows that $(\mathrm{dd^c}\varphi)^n(K) = (\mathrm{dd^c}\psi)^n(K) < \infty$, and adding the equal masses of $\Omega \setminus K$ gives the last assertion. □

Lemma 9.10.7 *Let $\varphi \in \mathcal{E}_0(B)$ and $T > 0$, and put $\varphi_T = \max\{\varphi, -T\}$. Then $\varphi_T \in \mathcal{E}_0(B)$, and for every $p \ge 1$,*

$$\int_B (\mathrm{dd^c}\varphi_T)^n = \int_B (\mathrm{dd^c}\varphi)^n, \qquad e_p(\varphi_T) \le e_p(\varphi),$$

$$\|(\mathrm{dd^c}\varphi_T)^n - (\mathrm{dd^c}\varphi)^n\|_{\mathrm{TV}} \le 2e_p(\varphi)T^{-p}. \tag{9.20}$$

Moreover, for every Borel set $E \subset B$,

$$\int_E (\mathrm{dd^c}\varphi)^n \le T^n \operatorname{Cap}_B(E) + e_p(\varphi)T^{-p}. \tag{9.21}$$

Proof Since φ tends to zero uniformly at ∂B, there is an open neighborhood V of ∂B such that $\varphi > -T$ on $V \cap B$; there $\varphi_T = \varphi$. Thus φ_T is bounded, nonpositive and plurisubharmonic, it has zero boundary values, and it coincides with φ outside the compact set $\overline{B} \setminus V \subset B$. By Lemma 9.10.6, the measures $(\mathrm{dd^c}\varphi_T)^n$ and $(\mathrm{dd^c}\varphi)^n$ have the same total mass, which is finite; in particular $\varphi_T \in \mathcal{E}_0(B)$.

Put $F_T = \{\varphi \le -T\}$. By maximum locality (Proposition 9.2.7), the two measures agree on the complement $\{\varphi > -T\}$. Since their total masses are equal and finite, they also assign the same mass m_T to F_T, and Chebyshev's inequality gives

$$m_T \le T^{-p} \int_{F_T} (-\varphi)^p (\mathrm{dd^c}\varphi)^n \le e_p(\varphi)T^{-p}.$$

As $\varphi_T = -T$ on F_T, we get

$$e_p(\varphi_T) = \int_{\{\varphi > -T\}} (-\varphi)^p (\mathrm{dd^c}\varphi)^n + T^p m_T \le e_p(\varphi).$$

The two measures differ only on F_T, where each has mass m_T; this proves (9.20). Finally, $(\varphi_T + T)/T$ is plurisubharmonic with values in $[0,1]$, so it is a competitor for Cap_B (Definition 9.3.2), and therefore $\int_E (\mathrm{dd^c}\varphi_T)^n \le T^n \operatorname{Cap}_B(E)$. Since the two measures agree on $E \setminus F_T$, we conclude

$$\int_E (\mathrm{dd^c}\varphi)^n \le \int_{E\setminus F_T} (\mathrm{dd^c}\varphi_T)^n + m_T \le T^n \operatorname{Cap}_B(E) + e_p(\varphi)T^{-p}.$$

Proposition 9.10.8 *Let $p \geq 1$.*

(1) *For every $\varphi \in \mathcal{E}_0(B)$ and every $T > 0$,*

$$\operatorname{Cap}_B(\{\varphi < -T\}) \leq 2^{n+p}(n+1)!\, e_p(\varphi) T^{-(n+p)}.$$

(2) *Let $\varphi_j \in \mathcal{E}_0(B)$ converge in $L^1_{\mathrm{loc}}(B)$, let u be the plurisubharmonic representative of the limit (Theorem 8.3.4), and suppose that $E_p = \sup_j e_p(\varphi_j) < \infty$. For every compact set $K \subset B$ there is a constant $C_{K,p}$, depending only on K, p and the ball B, such that*

$$\operatorname{Cap}^*_B(K \cap \{u < -T\}) \leq C_{K,p} E_p^{p/(n+p)} T^{-p} \qquad (T > 0). \tag{9.22}$$

In particular, $\operatorname{Cap}^*_B(K \cap \{u = -\infty\}) = 0$ for every compact set $K \subset B$.

Proof (1) Let h be a competitor for Cap_B, that is, $h \in \mathrm{PSH}(B)$ and $0 \leq h \leq 1$. The set $\{\varphi < -T\}$ is open, because φ is upper semicontinuous, and it is relatively compact in B, because φ has zero boundary values. Hence $\nu = \mathbf{1}_{\{\varphi<-T\}}(\mathrm{dd^c} h)^n$ is a finite measure, and it is dominated by $(\mathrm{dd^c} h)^n$. The tail estimate (9.10) of Theorem 9.10.2, applied to ν and $w = h$, involves the constant $\kappa_{n,p} = 2^{n+p}(n+1)!M^n$, where $M \leq 1$ is the oscillation of h on B, but not the mass of ν; it gives

$$\int_{\{\varphi<-T\}} (\mathrm{dd^c} h)^n = \nu(\{\varphi < -T\}) \leq 2^{n+p}(n+1)!\, e_p(\varphi) T^{-(n+p)}.$$

Taking the supremum over h proves (1).

(2) Choose an open set U and compact sets $L, L' \subset B$ with $K \subset U \Subset \operatorname{Int} L$ and $L \subset \operatorname{Int} L'$. By the Chern–Levine–Nirenberg inequality (Theorem 9.1.3, with the base current 1 and the pair $L \subset \operatorname{Int} L'$), there is a constant C_L with $\int_L (\mathrm{dd^c} h)^n \leq C_L$ for every competitor h, since $\|h\|_{L^\infty(L')} \leq 1$. The finite measure $\mathbf{1}_L(\mathrm{dd^c} h)^n$ is dominated by $(\mathrm{dd^c} h)^n$ and has mass at most C_L, so the constant A_p of Theorem 9.10.2 for this measure and $w = h$ is at most

$$C_{K,p} = \left(1 + \frac{p}{n}\right)\left(2^{n+p}(n+1)!\right)^{p/(n+p)} C_L^{n/(n+p)}.$$

Hence the moment estimate (9.11), applied to φ_j, gives

$$\int_L (-\varphi_j)^p (\mathrm{dd^c} h)^n \leq C_{K,p} E_p^{p/(n+p)}$$

for all j and all competitors h.

We pass this bound to the limit u. Since the φ_j are nonpositive, $u \leq 0$ almost everywhere, and hence everywhere by the submean inequality. Fix $S > 0$. The truncations $\max\{\varphi_j, -S\}$ take values in $[-S, 0]$ and converge to the plurisubharmonic function $\max\{u, -S\}$ in $L^1_{\mathrm{loc}}(B)$, since $t \mapsto \max\{t, -S\}$ is 1-Lipschitz. Since $\mathbf{1}_L(\mathrm{dd^c} h)^n \leq (\mathrm{dd^c} h)^n$, Lemma 9.10.4 shows that they also converge in $L^1(\mathbf{1}_L(\mathrm{dd^c} h)^n)$. As $t \mapsto t^p$ is Lipschitz on $[0, S]$ and $(-\max\{\varphi_j, -S\})^p \leq (-\varphi_j)^p$, it follows that

$$\int_L (-\max\{u,-S\})^p (\mathrm{dd^c} h)^n = \lim_{j\to\infty} \int_L (-\max\{\varphi_j,-S\})^p (\mathrm{dd^c} h)^n \le C_{K,p} E_p^{p/(n+p)}.$$

Letting $S \to \infty$, monotone convergence gives the same bound for $\int_L (-u)^p (\mathrm{dd^c} h)^n$. The set $U \cap \{u < -T\}$ is open, because u is upper semicontinuous, and it contains $K \cap \{u < -T\}$. By Chebyshev's inequality its $(\mathrm{dd^c} h)^n$-measure is at most $C_{K,p} E_p^{p/(n+p)} T^{-p}$. Taking the supremum over h bounds $\mathrm{Cap}_B(U \cap \{u < -T\})$, and hence the outer capacity in (9.22). Finally, $K \cap \{u = -\infty\}$ is contained in $K \cap \{u < -T\}$ for every T, so letting $T \to \infty$ shows that it has zero outer capacity.□

Theorem 9.10.9 *Under the hypotheses of Theorem 9.10.5, the relabeled subsequence U_j converges to u locally in capacity. Moreover, let $E_2 = \sup_j e_2(U_j)$, which is finite by Proposition 9.10.3. For every $T > 0$, the Monge–Ampère measure of the bounded plurisubharmonic function $u_T = \max\{u, -T\}$ has total mass at most m, and*

$$\|(\mathrm{dd^c} u_T)^n - \mu\|_{\mathrm{TV}} \le 2E_2 T^{-2}. \tag{9.23}$$

Proof We estimate the capacities of the sets where the truncation $\max\{U_j, -T\}$ lies far below or far above u_T (Steps 1 and 2) and deduce convergence in capacity (Step 3). The estimate for $(\mathrm{dd^c} u_T)^n$ then follows from the convergence of the truncated measures (Step 4). Fix $T > 0$ and write $a_j = \max\{U_j, -T\}$ and $b = u_T$. These plurisubharmonic functions take values in $[-T, 0]$, and $a_j \to b$ in $L^1_{\mathrm{loc}}(B)$ because truncation is 1-Lipschitz. By Theorem 9.10.5, $\varepsilon_j = \int_B |U_j - u| \, \mathrm{d}\mu_j \to 0$.

Step 1. The sets where $a_j < b - \delta$. Let $\delta > 0$. The set $\{a_j < b - \delta/2\}$ is relatively compact in B, because $b \le 0$ while a_j tends to zero at ∂B. Applying Lemma 9.9.1 with $s = t = \delta/2$ and then Chebyshev's inequality, we obtain

$$\mathrm{Cap}_B(\{a_j < b - \delta\}) \le \left(\frac{2}{\delta}\right)^n \int_{\{a_j < b - \delta/2\}} (\mathrm{dd^c} a_j)^n \le 2^{n+1} \delta^{-(n+1)} \int_B (b - a_j)_+ (\mathrm{dd^c} a_j)^n.$$

The weight $(b - a_j)_+$ takes values in $[0, T]$, and it is at most $|U_j - u|$, since truncation does not increase distances. By (9.20) with $p = 2$, $\|(\mathrm{dd^c} a_j)^n - \mu_j\|_{\mathrm{TV}} \le 2E_2 T^{-2}$, where $\mu_j = (\mathrm{dd^c} U_j)^n$. Hence

$$\int_B (b - a_j)_+ (\mathrm{dd^c} a_j)^n \le \int_B (b - a_j)_+ \, \mathrm{d}\mu_j + T \cdot 2E_2 T^{-2} \le \varepsilon_j + \frac{2E_2}{T},$$

and therefore $\mathrm{Cap}_B(\{a_j < b - \delta\}) \le 2^{n+1} \delta^{-(n+1)} (\varepsilon_j + 2E_2/T)$.

Step 2. The sets where $a_j > b + \delta$. Let $L \subset B$ be compact. By Theorem 9.4.2, for every $\sigma > 0$ there is an open set $G \subset B$ with $\mathrm{Cap}_B(G) < \sigma$ such that b is continuous on $B \setminus G$. We claim that $L \cap \{a_j > b + \delta\} \subset G$ for all sufficiently large j. Let $z \in L \setminus G$. Since b is upper semicontinuous, $b < b(z) + \delta/3$ on a closed ball $\overline{B(z, r)} \subset B$. Hartogs'

lemma (Theorem 8.3.4), applied to the sequence a_j with the constant majorant $b(z)+\delta/3$, gives $a_j < b(z)+\delta/3$ on $\overline{B(z,r)}$ for all large j. Since b is continuous on $B \setminus G$, after shrinking r we also have $b > b(z)-\delta/3$ on $(L \setminus G) \cap B(z,r)$. Hence $a_j - b < 2\delta/3$ on $(L\setminus G)\cap B(z,r)$ for all large j. As $L\setminus G$ is compact, finitely many such balls cover it, which proves the claim. Consequently $\overline{\lim}_j \operatorname{Cap}_B(L\cap\{a_j > b+\delta\}) \le \sigma$ for every $\sigma > 0$, that is, $\operatorname{Cap}_B(L \cap \{a_j > b+\delta\}) \to 0$.

Step 3. Convergence in capacity. Let $K \subset B$ be compact, and apply Step 2 to a compact set $L \subset B$ with $K \subset \operatorname{Int} L$. The bounded functions $a_j - b$ are quasi-continuous by Theorem 9.4.2 and the remark following Definition 9.4.1. Therefore Lemma 9.4.4, together with Steps 1 and 2 applied with $\delta/2$ in place of δ, gives

$$\begin{aligned}
&\overline{\lim_{j\to\infty}} \operatorname{Cap}_B^*(K \cap \{|a_j - b| > \delta\}) \\
&\qquad \le \overline{\lim_{j\to\infty}} \bigl(\operatorname{Cap}_B(L \cap \{a_j > b+\delta/2\}) + \operatorname{Cap}_B(\{a_j < b-\delta/2\})\bigr) \\
&\qquad \le 2^{2n+3}\delta^{-(n+1)}\frac{E_2}{T}.
\end{aligned}$$

It remains to remove the truncation. Outside $\{U_j < -T\} \cup \{u < -T\}$ we have $a_j = U_j$ and $b = u$. Hence both sets $K \cap \{U_j > u+\delta\}$ and $K \cap \{U_j < u-\delta\}$ in Definition 9.4.5 are contained in

$$\bigl(K \cap \{|a_j - b| > \delta\}\bigr) \cup \{U_j < -T\} \cup \bigl(K \cap \{u < -T\}\bigr).$$

By Proposition 9.10.8 with $p = 2$, applied in part (1) to U_j, whose energy is at most E_2, and in part (2) to the sequence (U_j),

$$\operatorname{Cap}_B(\{U_j < -T\}) \le \frac{2^{n+2}(n+1)!\,E_2}{T^{n+2}}, \qquad \operatorname{Cap}_B^*(K\cap\{u < -T\}) \le \frac{C_{K,2}E_2^{2/(n+2)}}{T^2},$$

and the capacity of the open set $\{U_j < -T\}$ equals its outer capacity. By subadditivity of the outer capacity, the upper limits as $j \to \infty$ of the outer capacities of the two sets in Definition 9.4.5 are bounded by the sum of three terms, each of which tends to zero as $T \to \infty$. Since these upper limits do not depend on T, they vanish. Thus $U_j \to u$ locally in capacity.

Step 4. The truncated limit measure. Fix $T > 0$ again. By Step 3 and (9.19), the uniformly bounded functions a_j converge to b locally in capacity, so $(\mathrm{dd^c}a_j)^n \to (\mathrm{dd^c}u_T)^n$ weakly on B by Theorem 9.4.8. By Lemma 9.10.7, each $(\mathrm{dd^c}a_j)^n$ has total mass $\mu_j(B) = m$. Testing with functions $\chi \in C_c(B)$, $0 \le \chi \le 1$, and letting χ increase to one, we see that $(\mathrm{dd^c}u_T)^n$ has total mass at most m. Moreover, (9.20) gives

$$\left|\int_B \chi\,(\mathrm{dd^c}a_j)^n - \int_B \chi\,\mathrm{d}\mu_j\right| \le 2E_2T^{-2}$$

for every real $\chi \in C_c(B)$ with $|\chi| \le 1$. Since $\mu_j \to \mu$ weakly by Proposition 9.10.3, letting $j \to \infty$ yields the same bound with $(\mathrm{dd^c}u_T)^n$ and μ in place of $(\mathrm{dd^c}a_j)^n$ and μ_j. Both limit measures are finite, so by the remark following the definition of the

total variation distance, the supremum over such χ equals $\|(\mathrm{dd^c}u_T)^n - \mu\|_{\mathrm{TV}}$. This proves (9.23). □

9.10.3 A bounded subsolution and the limiting equation

We can now solve the Dirichlet problem

$$(\mathrm{dd^c}U)^n = \mu \quad \text{on } B, \qquad U = 0 \quad \text{on } \partial B,$$

for a compactly supported measure μ dominated by the Monge–Ampère measure of a bounded plurisubharmonic function, and show that the whole sequence of canonical approximants of μ converges to the solution (Theorem 9.10.12). By Theorem 9.10.9, the limit u of a subsequence of canonical approximants converging in $L^1(B)$ satisfies $(\mathrm{dd^c}\max\{u, -T\})^n \to \mu$ as $T \to \infty$, so it remains to show that u is bounded and has zero boundary values. For this we build from the bounded subsolution a barrier with zero boundary values and compare the approximants with a strict perturbation of the barrier. The comparison involves the masses that the varying measures μ_j give to varying sublevel sets, and we first prove a convergence lemma for such masses.

Lemma 9.10.10 *Let $K_0 \subset B$ be compact, and let ν_j, ν be finite positive Borel measures on B, supported in K_0, with $\nu_j \to \nu$ weakly. Assume that there is a nondecreasing function $\omega\colon [0,+\infty] \to [0,+\infty]$ with $\omega(t) \to 0$ as $t \downarrow 0$ such that*

$$\nu_j(E) \le \omega(\mathrm{Cap}^*_B(E))$$

for all j and every Borel set $E \subset B$. Let f_j, f be real Borel functions on B bounded by a common constant, with f quasi-continuous on a neighborhood of K_0, and suppose that $\mathrm{Cap}^*_B(K_0 \cap \{|f_j - f| > \delta\}) \to 0$ for every $\delta > 0$. Then

$$\int_B f_j \,\mathrm{d}\nu_j \longrightarrow \int_B f \,\mathrm{d}\nu.$$

Moreover, for every $c \in \mathbb{R}$ with $\nu(\{f = c\}) = 0$,

$$\nu_j(\{f_j < c\}) \longrightarrow \nu(\{f < c\}). \tag{9.24}$$

Proof Let C be a common bound for $|f_j|$ and $|f|$. Since ω is nonnegative and nondecreasing with $\omega(t) \to 0$ as $t \downarrow 0$, we have $\omega(0) = 0$ and $\omega(t_j) \to 0$ whenever $t_j \ge 0$ and $t_j \to 0$. Choose $\chi \in C_c(B)$ with $0 \le \chi \le 1$ and $\chi = 1$ on K_0. Since the measures are carried by K_0, we have $\nu_j(B) = \int \chi \,\mathrm{d}\nu_j \to \int \chi \,\mathrm{d}\nu = \nu(B)$, so the masses of the ν_j and of ν have a common bound M. The capacity bound passes to ν on open sets: if $G \subset B$ is open, then $\mathrm{Cap}^*_B(G) = \mathrm{Cap}_B(G)$, and Proposition 1.5.2 gives $\nu(G) \le \varliminf_j \nu_j(G) \le \omega(\mathrm{Cap}_B(G))$.

We first consider the fixed weight f. Let $L \subset B$ be a compact neighborhood of K_0 on which f is quasi-continuous, and let $\kappa > 0$. Choose an open set $G \subset B$ with

$\mathrm{Cap}_B(G) < \kappa$ such that f is continuous on the compact set $L \setminus G$. By Lemma 9.4.3, this restriction extends to a continuous function on $\mathbb{C}^n$ bounded by C. Multiplying the extension by a cutoff in $C_c(\operatorname{Int} L)$ with values in $[0,1]$ and equal to one on K_0, we obtain $g \in C_c(B)$ with $|g| \le C$ and $g = f$ on $K_0 \setminus G$. Since ν_j and ν are carried by K_0,

$$\left| \int_B f \, \mathrm{d}\nu_j - \int_B g \, \mathrm{d}\nu_j \right| \le 2C\nu_j(G) \le 2C\omega(\kappa),$$

and similarly for ν, since $\nu(G) \le \omega(\mathrm{Cap}_B(G)) \le \omega(\kappa)$. Since $\int g \, \mathrm{d}\nu_j \to \int g \, \mathrm{d}\nu$, we get $\overline{\lim}_j \left| \int f \, \mathrm{d}\nu_j - \int f \, \mathrm{d}\nu \right| \le 4C\omega(\kappa)$. Letting $\kappa \downarrow 0$ proves the assertion for the fixed weight.

For the varying weights, let $\delta > 0$. Since $|f_j - f| \le 2C$ and ν_j is carried by K_0,

$$\int_B |f_j - f| \, \mathrm{d}\nu_j \le \delta M + 2C\omega\left(\mathrm{Cap}_B^*(K_0 \cap \{|f_j - f| > \delta\})\right).$$

As $j \to \infty$, the capacity on the right tends to zero, and hence so does the second term. Letting $\delta \downarrow 0$ gives $\int |f_j - f| \, \mathrm{d}\nu_j \to 0$, which together with the fixed-weight case proves the first assertion.

For the level sets, fix $s > 0$ and let $\sigma_s^-, \sigma_s^+ \colon \mathbb{R} \to [0,1]$ be the continuous piecewise linear functions such that σ_s^- equals one on $(-\infty, c-s]$, zero on $[c, \infty)$ and is linear in between, and σ_s^+ equals one on $(-\infty, c]$, zero on $[c+s, \infty)$ and is linear in between. Then

$$\mathbf{1}_{(-\infty, c-s)} \le \sigma_s^- \le \mathbf{1}_{(-\infty, c)} \le \sigma_s^+ \le \mathbf{1}_{(-\infty, c+s]}.$$

Since $\sigma_s^\pm$ are Lipschitz, $\sigma_s^\pm \circ f_j$ converge to $\sigma_s^\pm \circ f$ in the same sense as $f_j \to f$, and $\sigma_s^\pm \circ f$ is quasi-continuous on a neighborhood of K_0 (remark following Definition 9.4.1). The first assertion therefore gives

$$\nu(\{f < c - s\}) \le \underline{\lim_{j\to\infty}} \, \nu_j(\{f_j < c\}) \le \overline{\lim_{j\to\infty}} \, \nu_j(\{f_j < c\}) \le \nu(\{f \le c + s\}).$$

As $s \downarrow 0$, the two outer terms tend to $\nu(\{f < c\})$ and $\nu(\{f \le c\})$, which are equal since $\nu(\{f = c\}) = 0$. This proves (9.24). □

Corollary 9.10.11 *Let $\varphi_j \in \mathcal{E}_0(B)$ with $\sup_j e_2(\varphi_j) < \infty$ converge in $L^1_{\mathrm{loc}}(B)$, and let u be the plurisubharmonic representative of the limit (Theorem 8.3.4). If ν is a finite positive Borel measure on B with $\nu \le (\mathrm{dd}^{\mathrm{c}} w)^n$ for some bounded $w \in \mathrm{PSH}(B)$, then $u \in L^2(\nu)$ and $\varphi_j \to u$ in $L^1(\nu)$.*

Proof By the moment estimate (9.11) with $p = 2$, the integrals $\int_B \varphi_j^2 \, \mathrm{d}\nu$ are bounded uniformly in j. Since the φ_j are nonpositive, so is u, by the submean inequality. Fix $T > 0$. By Lemma 9.10.4, the truncations $\max\{\varphi_j, -T\}$ converge to $\max\{u, -T\}$ in $L^1(\nu)$, and so do their squares, since squaring is Lipschitz on $[-T, 0]$. Hence $\int_B \max\{u, -T\}^2 \, \mathrm{d}\nu \le \sup_j \int_B \varphi_j^2 \, \mathrm{d}\nu$, and letting $T \to \infty$ shows that $u \in L^2(\nu)$ with the same bound. For a function $v \le 0$ we have $0 \le \max\{v, -T\} - v \le v^2/T$, so the

$L^1(\nu)$ distances between φ_j, u and their truncations are at most T^{-1} times a bound independent of j. Letting first $j \to \infty$ and then $T \to \infty$ proves that $\varphi_j \to u$ in $L^1(\nu)$. □

Theorem 9.10.12 *Let $B \subset \mathbb{C}^n$ be a ball, $n \geq 1$, and let μ be a finite positive Borel measure with compact support in B. Suppose that $\mu \leq (\mathrm{dd^c} w)^n$ for some bounded $w \in \mathrm{PSH}(B)$. Then there is a unique bounded plurisubharmonic function U on B with*

$$(\mathrm{dd^c} U)^n = \mu, \qquad \lim_{B \ni z \to \zeta} U(z) = 0 \quad (\zeta \in \partial B).$$

It belongs to $\mathcal{E}_0(B)$, and it is the pointwise supremum of the bounded plurisubharmonic functions v on B such that $(\mathrm{dd^c} v)^n \geq \mu$ and $\overline{\lim}_{B \ni z \to \zeta} v(z) \leq 0$ for every $\zeta \in \partial B$. Moreover, for every choice of grids, the whole sequence of canonical approximants U_j of μ (Proposition 9.10.3) converges to U in $L^1(B)$ and locally in capacity, and

$$\int_B |U_j - U| \, \mathrm{d}\mu + \int_B |U_j - U| \, \mathrm{d}\mu_j \longrightarrow 0.$$

Proof If $\mu = 0$, then all μ_j and all U_j vanish, and $U = 0$ is a solution. By the maximum principle (Proposition 3.1.4), every plurisubharmonic function on B whose upper boundary limits are at most zero is nonpositive, so $U = 0$ is the stated supremum, and the last assertion of Theorem 9.2.8, applied in both directions, shows that any two bounded solutions coincide. We may therefore assume that $m = \mu(B) > 0$. We construct a barrier W (Step 1) and, along a convergent subsequence of canonical approximants, prove that the masses of certain sublevel sets converge (Step 2). The comparison theorem then shows that the limit lies above W and is a solution (Step 3), and uniqueness gives the convergence of the whole sequence (Step 4).

Step 1. A barrier with zero boundary values. Write $B = B(a, R)$ and $\rho(z) = |z - a|^2 - R^2$, let $M = \sup_B w - \inf_B w$, and put $w_0 = w - \sup_B w - 1$, so that $-M - 1 \leq w_0 \leq -1$. Since $\operatorname{Supp} \mu$ is a compact subset of B, the function ρ is bounded away from zero on an open neighborhood $N \Subset B$ of $\operatorname{Supp} \mu$, and we can choose $A > 0$ with $A\rho < -M - 2$ on N. Put

$$W = \max\{w_0, A\rho\},$$

a plurisubharmonic function on B (Proposition 8.2.5). Then $W = w_0$ on N, while $W = A\rho$ near ∂B, because $A\rho \to 0$ at ∂B and $w_0 \leq -1$. Thus W is bounded and negative and has zero boundary values. By locality on the open set N (Proposition 9.1.2), $(\mathrm{dd^c} W)^n = (\mathrm{dd^c} w)^n \geq \mu$ there; as μ is carried by N, we get $(\mathrm{dd^c} W)^n \geq \mu$ on B.

For $\varepsilon > 0$ put $W_\varepsilon = W + \varepsilon\rho$. It equals $(A + \varepsilon)\rho$ near ∂B, so its Monge–Ampère measure is a smooth measure there and is finite on the remaining compact part of B; hence it has finite total mass. Expanding $(\mathrm{dd^c} W + \varepsilon \mathrm{dd^c} \rho)^n$ and dropping the mixed terms, which are positive, we obtain

$$\nu_\varepsilon := (\mathrm{dd^c} W_\varepsilon)^n \geq \mu + \varepsilon^n (\mathrm{dd^c} \rho)^n. \tag{9.25}$$

Step 2. Convergence of the masses of sublevel sets. By Proposition 9.10.3, a subsequence of the canonical approximants, still denoted U_j (together with the corresponding measures μ_j), converges in $L^1(B)$; let u be the nonpositive plurisubharmonic representative of the limit. By Theorem 9.10.9, $U_j \to u$ locally in capacity, and $E_2 = \sup_j e_2(U_j) < \infty$. By Corollary 9.10.11 with $\nu = \nu_\varepsilon$, we have $U_j \to u$ in $L^1(\nu_\varepsilon)$ for every $\varepsilon > 0$.

We check that Lemma 9.10.10 applies to the measures μ_j and μ. Let K be the support of μ and $d = \operatorname{dist}(K, \mathbb{C}^n \setminus B) > 0$, and let h_j be the side lengths of the cubes defining μ_j. A cube Q with $\mu_j(Q) > 0$ satisfies $\mu(Q) > 0$, so it meets K. Since $h_j \to 0$, after discarding finitely many indices every such cube has diameter $\sqrt{2n}\, h_j < d/2$, and then μ_j is supported in the compact set $K_0 = \{z : \operatorname{dist}(z, K) \le d/2\} \subset B$, which also contains the support of μ. By Proposition 9.10.3, $\mu_j \to \mu$ weakly. By (9.21) with $p = 2$, and since $\operatorname{Cap}_B(E) \le \operatorname{Cap}_B^*(E)$, we have $\mu_j(E) \le T^n \operatorname{Cap}_B^*(E) + E_2 T^{-2}$ for every Borel set $E \subset B$ and every $T > 0$. Let $\omega(t)$ be the supremum of $\mu_j(E)$ over all j and all Borel sets $E \subset B$ with $\operatorname{Cap}_B^*(E) \le t$. Then ω is nondecreasing with values in $[0, m]$, and $\omega(t) \le T^n t + E_2 T^{-2}$ for every $T > 0$. Choosing first T large and then t small shows that $\omega(t) \to 0$ as $t \downarrow 0$.

Fix $\varepsilon > 0$. The level sets $\{u - W_\varepsilon = -\delta\}$, $0 < \delta < 1$, are pairwise disjoint Borel sets, and $\mu + \nu_\varepsilon$ is a finite measure, so by Lemma 1.1.5 all but countably many of these sets are null for $\mu + \nu_\varepsilon$, hence for both measures. Choose $\delta \in (0, 1)$ such that $\{u - W_\varepsilon = -\delta\}$ is null for both μ and ν_ε, and fix $T > \|W_\varepsilon\|_\infty + 2$. Since $W_\varepsilon - \delta > -T$, the sets $\{U_j < W_\varepsilon - \delta\}$ and $\{u < W_\varepsilon - \delta\}$ do not change when U_j and u are replaced by $a_j = \max\{U_j, -T\}$ and $b = \max\{u, -T\}$.

We apply Lemma 9.10.10 to the functions $f_j = a_j - W_\varepsilon$ and $f = b - W_\varepsilon$, which are bounded by $T + \|W_\varepsilon\|_\infty$ and quasi-continuous by Theorem 9.4.2 and the remark following Definition 9.4.1. Since $U_j \to u$ locally in capacity, (9.19) shows that $a_j \to b$ locally in capacity, and hence $f_j \to f$, because W_ε is finite. Moreover, $\mu(\{f = -\delta\}) = 0$ by the choice of δ, since $\{f = -\delta\} = \{u - W_\varepsilon = -\delta\}$: indeed, $b = u$ wherever $u \ge -T$, while $u - W_\varepsilon < -2$ and $b - W_\varepsilon \le -T + \|W_\varepsilon\|_\infty < -2$ wherever $u < -T$. Thus (9.24), with $c = -\delta$, gives

$$\mu_j(\{U_j < W_\varepsilon - \delta\}) \longrightarrow \mu(\{u < W_\varepsilon - \delta\}).$$

The analogous convergence holds for the fixed measure ν_ε. Indeed, for $s > 0$, outside $\{|U_j - u| > s\}$, the indicators of $\{U_j < W_\varepsilon - \delta\}$ and $\{u < W_\varepsilon - \delta\}$ can differ only on $\{|u - W_\varepsilon + \delta| \le s\}$. The ν_ε-mass of the first set tends to zero as $j \to \infty$, by the convergence in $L^1(\nu_\varepsilon)$, and that of the second set tends to $\nu_\varepsilon(\{u - W_\varepsilon = -\delta\}) = 0$ as $s \downarrow 0$. Hence

$$\nu_\varepsilon(\{U_j < W_\varepsilon - \delta\}) \longrightarrow \nu_\varepsilon(\{u < W_\varepsilon - \delta\}).$$

Step 3. The limit is a bounded solution. Since U_j and W_ε have zero boundary values, the difference $U_j - (W_\varepsilon - \delta)$ has boundary limits $\delta > 0$. The comparison theorem (Theorem 9.2.8) therefore gives

$$\nu_\varepsilon(\{U_j < W_\varepsilon - \delta\}) \le \mu_j(\{U_j < W_\varepsilon - \delta\}).$$

Let $j \to \infty$ and put $S = \{u < W_\varepsilon - \delta\}$. By (9.25),

$$\mu(S) + \varepsilon^n \int_S (\mathrm{dd^c}\rho)^n \le \nu_\varepsilon(S) \le \mu(S).$$

Since $\mu(S)$ is finite, S has Lebesgue measure zero. Thus $u \ge W_\varepsilon - \delta$ almost everywhere, and hence everywhere, by the uniqueness assertion of Theorem 8.2.1 applied to $\max\{u, W_\varepsilon - \delta\}$ and u. Letting $\delta \downarrow 0$ through admissible values and then $\varepsilon \downarrow 0$, we obtain $W \le u \le 0$.

As W has zero boundary values, u is bounded and has zero boundary values. For $T > \|u\|_\infty$ the truncation $\max\{u, -T\}$ is u itself, so (9.23) gives $\|(\mathrm{dd^c}u)^n - \mu\|_{\mathrm{TV}} \le 2E_2T^{-2}$ for all large T. Hence $(\mathrm{dd^c}u)^n = \mu$. In particular $(\mathrm{dd^c}u)^n$ has finite mass m, and $u \in \mathcal{E}_0(B)$.

Step 4. Uniqueness and convergence of the whole sequence. Let v be a bounded plurisubharmonic function on B with $(\mathrm{dd^c}v)^n \ge \mu$ and upper boundary limits at most zero, and put $v_\varepsilon = v + \varepsilon\rho$ for $\varepsilon > 0$. As in Step 1, $(\mathrm{dd^c}v_\varepsilon)^n \ge \mu + \varepsilon^n(\mathrm{dd^c}\rho)^n$. Since $u - v_\varepsilon$ has nonnegative lower boundary limits, the comparison theorem, applied to u and v_ε, gives

$$\mu(\{u < v_\varepsilon\}) + \varepsilon^n \int_{\{u<v_\varepsilon\}} (\mathrm{dd^c}\rho)^n \le \mu(\{u < v_\varepsilon\}).$$

The two μ-masses are finite, so the set $\{u < v_\varepsilon\}$ is Lebesgue null. Therefore $u \ge v_\varepsilon$ everywhere, by the uniqueness assertion of Theorem 8.2.1 applied to $\max\{u, v_\varepsilon\}$ and u. Letting $\varepsilon \downarrow 0$ gives $u \ge v$. Since u is itself such a function, it is the pointwise supremum in the statement. Any bounded solution with zero boundary values is such a function, hence is at most u; exchanging the roles of the two solutions proves uniqueness. We denote the solution by U.

Finally, by Proposition 9.10.3, every subsequence of the canonical approximants has a further subsequence converging in $L^1(B)$. By Step 3 and uniqueness, its limit is U, and Theorems 9.10.5 and 9.10.9 give the asserted convergence of the integrals and convergence in capacity along it. If one of the asserted convergences failed for the whole sequence, some subsequence would stay away from the stated limit, and no further subsequence could converge as just described. This contradiction proves all the assertions. □

9.10.4 Mixed energy and truncation

In the next section we approximate several potentials at once by canonical approximants and pass to the limit in their mixed Monge–Ampère products. The approximants converge in capacity and have bounded energies, but we have no uniform bound for them, so Theorem 9.4.8 does not apply to them directly. We therefore compare the mixed products with those of the truncated potentials, and for this we need to know that the sets where one of the potentials is very negative carry little mass of a mixed

product. This follows from the mixed energy inequality of Theorem 9.10.13, which bounds a mixed energy by a constant multiple of the geometric mean of the energies e_1 of Definition 9.10.1.

Theorem 9.10.13 *Let $u_0, \dots, u_n \in \mathcal{E}_0(B)$. Then*

$$\int_B (-u_0)\, \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n \le C_n \prod_{l=0}^{n} e_1(u_l)^{1/(n+1)}, \qquad C_n = 2^{n+2}. \tag{9.26}$$

In particular the integral is finite, and it vanishes if one of the energies $e_1(u_l)$ vanishes.

Proof Write $B = B(a, R)$ and $\rho(z) = |z - a|^2 - R^2$. We say that $v \in \mathcal{E}_0(B)$ has a *quadratic collar* if $v = c\rho$ on a neighborhood of ∂B in B for some constant $c > 0$. We first prove the inequality with constant one for potentials with quadratic collars (Steps 1 and 2), and then approximate arbitrary elements of $\mathcal{E}_0(B)$ by such potentials while keeping their energies under control (Steps 3 and 4). For functions $b_0, \dots, b_n$ which either all have quadratic collars, or are smooth real functions on a neighborhood of $\overline{B}$ with $b_2, \dots, b_n$ plurisubharmonic, put

$$I(b_0, \dots, b_n) = \int_B (-b_0)\, \mathrm{dd^c} b_1 \wedge \cdots \wedge \mathrm{dd^c} b_n.$$

The integral is finite: in the first case every mixed product of the b_l is smooth near ∂B and locally finite on the rest of B, and in the second the integrand is a smooth form on a neighborhood of $\overline{B}$.

Step 1. Smooth functions. Let $b_0, \dots, b_n$ be smooth real functions on a neighborhood of $\overline{B}$, with $b_2, \dots, b_n$ plurisubharmonic and $b_0 = 0$ on ∂B, and put $S = \mathrm{dd^c} b_2 \wedge \cdots \wedge \mathrm{dd^c} b_n$, with $S = 1$ if $n = 1$. With $\mathrm{d^c}$ as in (7.6), we claim that

$$I(b_0, b_1, b_2, \dots, b_n) = \int_B \mathrm{d}b_0 \wedge \mathrm{d^c} b_1 \wedge S. \tag{9.27}$$

Fix a smooth function $\theta\colon \mathbb{R} \to [0, 1]$ with $\theta = 1$ on $(-\infty, -2]$ and $\theta = 0$ on $[-1, \infty)$, and put $\chi_k = \theta(k\rho)$ for $k \ge 1$; then $\chi_k \in C_c^\infty(B)$. The $(2n-1)$-form $\chi_k b_0\, \mathrm{d^c} b_1 \wedge S$ is smooth with compact support in B, and S is closed, so Proposition 7.1.3 and the Leibniz rule give

$$0 = \int_B \chi_k\, \mathrm{d}b_0 \wedge \mathrm{d^c} b_1 \wedge S + \int_B b_0\, \mathrm{d}\chi_k \wedge \mathrm{d^c} b_1 \wedge S + \int_B \chi_k b_0\, \mathrm{dd^c} b_1 \wedge S.$$

As $k \to \infty$, $\chi_k \to 1$ pointwise on B, and the forms $\mathrm{d}b_0 \wedge \mathrm{d^c} b_1 \wedge S$ and $b_0\, \mathrm{dd^c} b_1 \wedge S$ have bounded coefficients on B; by dominated convergence, the first and the last integral tend to $\int_B \mathrm{d}b_0 \wedge \mathrm{d^c} b_1 \wedge S$ and $-I(b_0, \dots, b_n)$. In the middle integral, $\mathrm{d}\chi_k = k\theta'(k\rho)\, \mathrm{d}\rho$ vanishes outside the set where $-2 \le k\rho \le -1$. Let Λ be the maximum of $|\nabla b_0|$ on $\overline{B}$. If $z \in B$ and $\zeta \in \partial B$ is a nearest boundary point, then $b_0(\zeta) = 0$ and $|z - \zeta| = R - |z - a|$, so the mean value theorem on the segment from z to ζ gives

$$|b_0(z)| \le \Lambda(R - |z-a|) \le \frac{\Lambda}{R}(R^2 - |z-a|^2) = \frac{\Lambda}{R}|\rho(z)|.$$

Hence $|b_0\, k\theta'(k\rho)| \le (\Lambda/R)|k\rho|\,|\theta'(k\rho)| \le 2\Lambda \sup|\theta'|/R$ on B, uniformly in k, and $b_0\, k\theta'(k\rho) \to 0$ pointwise on B. Since $\mathrm{d}\rho \wedge \mathrm{d}^{\mathrm{c}} b_1 \wedge S$ has bounded coefficients on B, dominated convergence shows that the middle integral tends to zero. This proves (9.27).

As observed at the beginning of Section 9.2, $\mathrm{d}b \wedge \mathrm{d}^{\mathrm{c}} c \wedge S = \mathrm{d}c \wedge \mathrm{d}^{\mathrm{c}} b \wedge S$ for smooth real b, c. Moreover, $\mathrm{d}b \wedge \mathrm{d}^{\mathrm{c}} b = (\mathrm{i}/2\pi)\partial b \wedge \bar\partial b$ is a positive $(1,1)$-form, and so are the forms $\mathrm{dd}^{\mathrm{c}} b_l$, $l \ge 2$, since the complex Hessian of a smooth plurisubharmonic function is positive semidefinite (Proposition 8.2.2). Positive $(1,1)$-forms are strongly positive (Section 7.2), so by Proposition 7.2.1 the (n,n)-form $\mathrm{d}b \wedge \mathrm{d}^{\mathrm{c}} b \wedge S$ is a nonnegative multiple of the volume form. Consequently the real bilinear form

$$Q(b,c) = \int_B \mathrm{d}b \wedge \mathrm{d}^{\mathrm{c}} c \wedge S,$$

defined for smooth real functions b, c on a neighborhood of $\overline{B}$ vanishing on ∂B, is symmetric and positive semidefinite, and by (9.27) we have $Q(b,c) = I(b, c, b_2, \dots, b_n)$. The Cauchy–Schwarz inequality for Q, which follows from the nonnegativity of the quadratic polynomial $t \mapsto Q(b + tc, b + tc)$, therefore gives

$$I(b, c, b_2, \dots, b_n)^2 \le I(b, b, b_2, \dots, b_n) I(c, c, b_2, \dots, b_n). \tag{9.28}$$

The symmetry of Q allows us to interchange the first two arguments of I, and the symmetry of the products allows us to permute the last n arguments. Hence I is symmetric in all $n+1$ arguments when they are smooth plurisubharmonic functions on a neighborhood of $\overline{B}$ vanishing on ∂B.

Step 2. Quadratic collars. Let $u_0, \dots, u_n$ have quadratic collars, say $u_l = c_l\rho$ near ∂B. We first show that the assertions of Step 1 persist for them. Extend each u_l by $c_l\rho$ to a plurisubharmonic function on a neighborhood of $\overline{B}$. Let ϑ_ε be the radial mollifiers of Section 1.4, denoted ρ_ε there; we change the letter because ρ is now the quadratic function above. Because ρ is a quadratic polynomial and ϑ_ε is radial, $\rho * \vartheta_\varepsilon = \rho + \gamma\varepsilon^2$ with the constant $\gamma = \int |y|^2 \vartheta_1(y)\, \mathrm{d}V(y)$. Hence

$$u_{l,\varepsilon} = u_l * \vartheta_\varepsilon - c_l\gamma\varepsilon^2$$

equals $c_l\rho$ on a fixed neighborhood of ∂B for all small ε. By Proposition 8.2.2, the convolutions $u_l * \vartheta_\varepsilon$ are smooth and plurisubharmonic on a neighborhood of $\overline{B}$ for small ε, and they decrease to u_l as $\varepsilon \downarrow 0$. Hence each $u_{l,\varepsilon}$, which differs from $u_l * \vartheta_\varepsilon$ by a constant, is smooth and plurisubharmonic on a neighborhood of $\overline{B}$ and vanishes on ∂B, although the $u_{l,\varepsilon}$ need not decrease. By Theorem 9.2.5,

$$(-u_0 * \vartheta_\varepsilon)\, \mathrm{dd}^{\mathrm{c}} u_{1,\varepsilon} \wedge \cdots \wedge \mathrm{dd}^{\mathrm{c}} u_{n,\varepsilon} \longrightarrow (-u_0)\, \mathrm{dd}^{\mathrm{c}} u_1 \wedge \cdots \wedge \mathrm{dd}^{\mathrm{c}} u_n$$

weakly, and the constant $c_0\gamma\varepsilon^2$ changes the integral against a fixed test function only by an amount tending to zero, by the local mass bounds of Theorem 9.1.3.

Near ∂B, all the $u_{l,\varepsilon}$ and u_l coincide with $c_l\rho$. Splitting the integrals over B by a cutoff in $C_c(B)$ which equals one outside this neighborhood of ∂B, we conclude that $I(u_{0,\varepsilon},\dots,u_{n,\varepsilon}) \to I(u_0,\dots,u_n)$, and likewise for every rearrangement of the potentials. Passing to the limit in the smooth case proves that I is symmetric and satisfies (9.28) for potentials with quadratic collars.

We now prove the inequality with constant one for these potentials. Since $u_l = c_l\rho$ near ∂B with $c_l > 0$, the energy $e_1(u_l)$ is positive: already its integral over a neighborhood of ∂B is positive. Since I is positively homogeneous of degree one in each argument and $e_1(tv) = t^{n+1}e_1(v)$ for $t > 0$, both sides of the inequality are multiplied by the same factor when each u_l is replaced by $e_1(u_l)^{-1/(n+1)}u_l$. We may therefore assume that $e_1(u_l) = 1$ for every l. Consider the finitely many ordered $(n+1)$-tuples whose entries are taken from the functions $u_0,\dots,u_n$, and let F be the largest value of I on these tuples. Since $I(u_l,\dots,u_l) = e_1(u_l) = 1$, we have $F \geq 1$. Among the tuples on which I equals F, choose one for which the sum of the squares of the multiplicities of its distinct entries is largest.

Suppose that this tuple contains two distinct functions b and c, with multiplicities r and s. By symmetry we may place them in the first two slots. In (9.28), the left side is F^2, while the two factors on the right are nonnegative values of I on tuples of our family, hence at most F. Therefore both factors equal F. Replacing one entry c by b changes the sum of the squares of the multiplicities by $(r+1)^2+(s-1)^2-r^2-s^2 = 2(r-s)+2$, and replacing one entry b by c changes it by $2(s-r)+2$. One of these numbers is positive, which contradicts the choice of the tuple. Hence the chosen tuple consists of a single function repeated $n+1$ times, and $F = 1$. In particular $I(u_0,\dots,u_n) \leq 1$, which is the inequality with constant one.

Step 3. Approximation with controlled energy. Let $u \in \mathcal{E}_0(B)$, put $m_\rho = \int_B (\mathrm{dd^c}\rho)^n$, and for $j \geq 1$ let

$$\delta_j = j^{-n-1}, \qquad v_j = \max\{u-\delta_j, j\rho\}.$$

Since $u-\delta_j \leq -\delta_j$ while $j\rho > -\delta_j$ near ∂B, the function v_j equals $j\rho$ on a neighborhood of ∂B in B. Since v_j and $j\rho$ coincide outside a compact subset of B, Lemma 9.10.6 gives $\int_B (\mathrm{dd^c}v_j)^n = j^n m_\rho$, and hence $v_j \in \mathcal{E}_0(B)$. Moreover, $v_j \to u$ uniformly on B. Indeed, let $s > 0$, and choose a neighborhood V of ∂B with $u > -s$ on $V \cap B$. Since $u-\delta_j \leq v_j \leq 0$, we have $|v_j - u| \leq s+\delta_j$ on $V \cap B$. On the set $B \setminus V$, whose closure is a compact subset of B, the function ρ is bounded away from zero and u is bounded, so $j\rho < u-\delta_j$, and hence $v_j = u-\delta_j$, there for all large j.

Next we estimate $e_1(v_j)$. Let $t > 4\delta_j$. Since $u \leq v_j+\delta_j$ and $v_j \leq 0$,

$$\{v_j < -t\} \subset \{u < v_j/2 - t/4\} \subset \{u < -t/4\};$$

for the first inclusion note that on $\{v_j < -t\}$ we have $v_j/2 < -t/2 < -t/4-\delta_j$, hence $u \leq v_j+\delta_j < v_j/2 - t/4$. As u tends to 0 and $v_j/2-t/4$ tends to $-t/4$ at ∂B, the comparison theorem (Theorem 9.2.8) applies to u and $v_j/2-t/4$ and gives

$$\int_{\{v_j<-t\}} (\mathrm{dd^c}v_j)^n \leq 2^n \int_{\{u<v_j/2-t/4\}} \big(\mathrm{dd^c}(v_j/2)\big)^n \leq 2^n \int_{\{u<-t/4\}} (\mathrm{dd^c}u)^n.$$

For $0 < t \le 4\delta_j$ we use the total mass $j^n m_\rho$ instead. Integrating over t by the layer-cake formula $e_1(v) = \int_0^\infty (\mathrm{dd^c} v)^n(\{v < -t\})\,\mathrm{d}t$, we obtain

$$e_1(v_j) \le 4\delta_j j^n m_\rho + 2^{n+2} e_1(u) = \frac{4m_\rho}{j} + C_n e_1(u). \tag{9.29}$$

Step 4. The general case. Apply the construction of Step 3 to each u_l, obtaining functions $v_{l,j} \in \mathcal{E}_0(B)$ which equal $j\rho$ near ∂B. They are uniformly bounded, since $u_l - 1 \le v_{l,j} \le 0$, and they converge uniformly, hence locally in capacity, to u_l. Let $\chi \in C_c(B)$ with $0 \le \chi \le 1$. The weight $-u_0$ is bounded and quasi-continuous (Theorem 9.4.2), so the weighted form of Theorem 9.4.8 and Step 2 give

$$\begin{aligned}\int_B \chi(-u_0)\,\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n &= \lim_{j\to\infty} \int_B \chi(-v_{0,j})\,\mathrm{dd^c} v_{1,j} \wedge \cdots \wedge \mathrm{dd^c} v_{n,j} \\ &\le \varlimsup_{j\to\infty} \prod_{l=0}^{n} e_1(v_{l,j})^{1/(n+1)}.\end{aligned}$$

By (9.29), the right side is at most

$$\lim_{j\to\infty} \prod_{l=0}^{n} \left(\frac{4m_\rho}{j} + C_n e_1(u_l)\right)^{1/(n+1)} = C_n \prod_{l=0}^{n} e_1(u_l)^{1/(n+1)}.$$

Letting χ increase to one proves (9.26). Vanishing energies need no separate treatment, since the energies of the $v_{l,j}$ are positive. □

Corollary 9.10.14 *Let $E \ge 0$, and for $1 \le l \le n$ and $j \ge 1$ let $u_{l,j} \in \mathcal{E}_0(B)$ with $e_1(u_{l,j}) \le E$. Suppose that, for each l, $u_{l,j} \to u_l$ locally in capacity as $j \to \infty$, where u_l is a locally bounded plurisubharmonic function on B. Then*

$$\mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{n,j} \longrightarrow \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n$$

weakly on B.

Proof We compare the products with those of the truncations at a level $-T$: the latter converge by Theorem 9.4.8, and by Theorem 9.10.13 the two differ by a term of order $1/T$. For $T > 0$ put $a_{l,j} = \max\{u_{l,j}, -T\}$, and write

$$R_j = \mathrm{dd^c} u_{1,j} \wedge \cdots \wedge \mathrm{dd^c} u_{n,j}, \qquad R_{j,T} = \mathrm{dd^c} a_{1,j} \wedge \cdots \wedge \mathrm{dd^c} a_{n,j}.$$

By Lemma 9.10.7, $a_{l,j} \in \mathcal{E}_0(B)$ and $e_1(a_{l,j}) \le E$. For each l, Theorem 9.10.13 with $u_0 = u_{l,j}$, respectively $u_0 = a_{l,j}$, gives

$$\int_B (-u_{l,j})\,R_j \le C_n E, \qquad \int_B (-a_{l,j})\,R_{j,T} \le C_n E.$$

Let $D_{j,T} = \bigcup_{l=1}^n \{u_{l,j} \le -T\}$. On $\{u_{l,j} \le -T\}$ both $-u_{l,j}$ and $-a_{l,j}$ are at least T, so Chebyshev's inequality gives

$$R_j(D_{j,T}) \le nC_nE/T, \qquad R_{j,T}(D_{j,T}) \le nC_nE/T.$$

On the complement of $D_{j,T}$, every $u_{l,j}$ is greater than $-T$. For $0 \le k \le n$ put

$$P_k = \mathrm{dd^c}a_{1,j} \wedge \cdots \wedge \mathrm{dd^c}a_{k,j} \wedge \mathrm{dd^c}u_{k+1,j} \wedge \cdots \wedge \mathrm{dd^c}u_{n,j},$$

so that $P_0 = R_j$ and $P_n = R_{j,T}$. By the symmetry of the products, $P_k = \mathrm{dd^c}a_{k,j} \wedge \Theta_k$ and $P_{k-1} = \mathrm{dd^c}u_{k,j} \wedge \Theta_k$ with the same closed positive current Θ_k formed by the other $n-1$ factors, and maximum locality (Proposition 9.2.7 with $p = 1$, $u = u_{k,j}$, $v = -T$ and base current Θ_k) gives $P_k = P_{k-1}$ on $\{u_{k,j} > -T\}$. Hence $R_j = R_{j,T}$ on the complement of $D_{j,T}$, and for every $\chi \in C_c(B)$,

$$\left| \int_B \chi \, (R_j - R_{j,T}) \right| \le 2nC_nE\|\chi\|_\infty/T.$$

Now fix $\chi \in C_c(B)$, choose an open set $B' \Subset B$ containing $\operatorname{Supp}\chi$, and let $T > \max_l \sup_{B'} |u_l|$. By (9.19), the uniformly bounded functions $a_{l,j}$ converge to the locally bounded plurisubharmonic function $\max\{u_l, -T\}$ locally in capacity, and Theorem 9.4.8 shows that $R_{j,T} \to \bigwedge_l \mathrm{dd^c}\max\{u_l, -T\}$ weakly on B. On B' we have $\max\{u_l, -T\} = u_l$, so by locality (Proposition 9.1.2) this limit equals $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_n$ there. Consequently

$$\varlimsup_{j\to\infty} \left| \int_B \chi \, R_j - \int_B \chi \, \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_n \right| \le 2nC_nE\|\chi\|_\infty/T$$

for all large T, and letting $T \to \infty$ completes the proof. □

9.11 Mixed products for a common measure

Corollary 9.7.6 bounds a mixed Monge–Ampère product from below when the pure Monge–Ampère measures of the factors have continuous lower densities with respect to Lebesgue measure. For non-pluripolar products in Chapter 15 we need the same inequality relative to an arbitrary positive Radon measure μ, which may be singular with respect to Lebesgue measure. Instead of regularizing the potentials by convolution, as in the proof of that corollary, we restrict each pure Monge–Ampère measure to a compact set and replace it by its cube averages, as in Proposition 9.10.3. The averaged measures have piecewise constant densities, so the case of Lebesgue measure applies on each cube, and Hölder's inequality compares the resulting bound with the cube averages of the expected lower bound. The convergence results of Section 9.10 then allow us to pass to the limit.

Theorem 9.11.1 *Let $\Omega \subset \mathbb{C}^n$ be open, $n \ge 1$, and let $u_1, \ldots, u_n$ be locally bounded plurisubharmonic functions on Ω. Let μ be a positive Radon measure on Ω, and let $f_1, \ldots, f_n$ be nonnegative Borel functions on Ω. If*

$$(\mathrm{dd}^{\mathrm{c}} u_i)^n \geq f_i \mu \qquad (1 \leq i \leq n),$$

then

$$\mathrm{dd}^{\mathrm{c}} u_1 \wedge \cdots \wedge \mathrm{dd}^{\mathrm{c}} u_n \geq \left(\prod_{i=1}^{n} f_i^{1/n} \right) \mu. \tag{9.30}$$

For $n = 1$ the assertion is the hypothesis.

Proof Let $n \geq 2$, and put $\lambda = (\prod_i f_i^{1/n})\mu$. We first prove the theorem when Ω is a ball $B = B(a_0, R)$ and all u_i belong to $\mathcal{E}_0(B)$ (Steps 1 and 2), and then reduce the general case to this one by modifying the potentials near the boundary of a ball (Step 3). For a ball $B = B(a_0, R)$ we write $\rho(z) = |z - a_0|^2 - R^2$.

We use the generalized Hölder inequality

$$\int_A \prod_{i=1}^{n} g_i^{1/n} \, \mathrm{d}\nu \leq \prod_{i=1}^{n} \Big(\int_A g_i \, \mathrm{d}\nu \Big)^{1/n} \tag{9.31}$$

for a positive measure ν, a measurable set A and nonnegative measurable functions g_i. If one of the integrals $c_i = \int_A g_i \, \mathrm{d}\nu$ vanishes, then $g_i = 0$ ν-almost everywhere on A and both sides vanish, with the convention $0 \cdot \infty = 0$; if none vanishes and one is infinite, the right side is infinite. Otherwise, integrating the arithmetic–geometric mean inequality $\prod_i (g_i/c_i)^{1/n} \leq \frac{1}{n} \sum_i g_i/c_i$ over A gives $\int_A \prod_i (g_i/c_i)^{1/n} \, \mathrm{d}\nu \leq 1$, which is (9.31). For every compact $K \subset \Omega$ we have $\int_K f_i \, \mathrm{d}\mu \leq (\mathrm{dd}^{\mathrm{c}} u_i)^n(K) < \infty$ by hypothesis, so (9.31) shows that λ is a positive Radon measure.

Step 1. Exhaustion by compactly supported measures. Choose closed balls $K_l \subset B$ concentric with B, increasing with l and exhausting B, and set

$$\alpha_{i,l} = \mathbf{1}_{K_l} (\mathrm{dd}^{\mathrm{c}} u_i)^n.$$

These measures are finite since $u_i \in \mathcal{E}_0(B)$, they have compact support in B, and they are dominated by the Monge–Ampère measure of the bounded plurisubharmonic function u_i. By Theorem 9.10.12, the equation $(\mathrm{dd}^{\mathrm{c}} A)^n = \alpha_{i,l}$ has a bounded solution $A_{i,l} \in \mathcal{E}_0(B)$ with zero boundary values, and $A_{i,l}$ is the largest bounded plurisubharmonic function v on B with $(\mathrm{dd}^{\mathrm{c}} v)^n \geq \alpha_{i,l}$ and upper boundary limits at most zero. The function u_i has these properties for every l, and so has $A_{i,l+1}$, because $\alpha_{i,l+1} \geq \alpha_{i,l}$. Therefore

$$u_i \leq A_{i,l+1} \leq A_{i,l} \leq 0.$$

We claim that $A_{i,l} \downarrow u_i$ as $l \to \infty$. The decreasing limit $A_{i,\infty}$ satisfies $u_i \leq A_{i,\infty} \leq 0$, so it is a bounded plurisubharmonic function (Theorem 8.3.1) with zero boundary values. By Theorem 9.2.5, $(\mathrm{dd}^{\mathrm{c}} A_{i,\infty})^n$ is the weak limit of $\alpha_{i,l}$, that is, $(\mathrm{dd}^{\mathrm{c}} A_{i,\infty})^n = (\mathrm{dd}^{\mathrm{c}} u_i)^n$, and this measure has finite total mass. Let $\varepsilon > 0$ and $S = \{u_i < A_{i,\infty} + \varepsilon\rho\}$. Since $u_i - (A_{i,\infty} + \varepsilon\rho)$ tends to zero at ∂B, the comparison theorem (Theorem 9.2.8) applies. Expanding $(\mathrm{dd}^{\mathrm{c}}(A_{i,\infty} + \varepsilon\rho))^n$ and dropping the mixed terms, which are positive, we obtain

$$\int_S (\mathrm{dd^c} u_i)^n + \varepsilon^n \int_S (\mathrm{dd^c}\rho)^n \le \int_S \bigl(\mathrm{dd^c}(A_{i,\infty} + \varepsilon\rho)\bigr)^n \le \int_S (\mathrm{dd^c} u_i)^n.$$

The integrals of $(\mathrm{dd^c} u_i)^n$ are finite, so S is Lebesgue null. Hence $u_i \ge A_{i,\infty} + \varepsilon\rho$ almost everywhere, and therefore everywhere on B, by the uniqueness assertion of Theorem 8.2.1 applied to $\max\{u_i, A_{i,\infty} + \varepsilon\rho\}$ and u_i. Letting $\varepsilon \downarrow 0$ gives $A_{i,\infty} = u_i$, as claimed.

Step 2. A common grid. Fix l. Choose a sequence of partitions of $\mathbb{C}^n = \mathbb{R}^{2n}$ into half-open cubes whose side lengths decrease to zero, the same for all i. Let $\alpha_{i,l,j}$ be the corresponding cube averages of $\alpha_{i,l}$, as in Proposition 9.10.3, and let $A_{i,l,j} \in \mathcal{E}_0(B)$ be the corresponding canonical approximants of $\alpha_{i,l}$, that is, the zero-boundary solutions of $(\mathrm{dd^c} A_{i,l,j})^n = \alpha_{i,l,j}$; they are defined for all large j. Discarding finitely many indices j, we may assume that they are defined for all j and all i, and that every cube meeting K_l has its closure in B. By Theorem 9.10.12, $A_{i,l,j} \to A_{i,l}$ locally in capacity as $j \to \infty$. By Proposition 9.10.3, applied with the bounded subsolution u_i, the energies $e_1(A_{i,l,j})$ are bounded uniformly in j; since there are only finitely many indices i, they have a common bound. (If $\alpha_{i,l} = 0$, all these solutions vanish.) Therefore Corollary 9.10.14 gives

$$\bigwedge_{i=1}^n \mathrm{dd^c} A_{i,l,j} \longrightarrow \bigwedge_{i=1}^n \mathrm{dd^c} A_{i,l} \tag{9.32}$$

weakly on B.

Let $\lambda_l = \mathbf{1}_{K_l}\lambda$, a finite measure, and let $\lambda_{l,j}$ be its cube averages. For every cube Q, the hypothesis gives $\alpha_{i,l}(Q) \ge \int_{Q\cap K_l} f_i \, \mathrm{d}\mu$, and (9.31) then gives

$$\prod_{i=1}^n \left(\frac{\alpha_{i,l}(Q)}{|Q|}\right)^{1/n} \ge \frac{1}{|Q|}\int_{Q\cap K_l} \prod_{i=1}^n f_i^{1/n}\, \mathrm{d}\mu = \frac{\lambda_l(Q)}{|Q|}. \tag{9.33}$$

Let Q be a cube with $\lambda_l(Q) > 0$; it meets K_l, so its interior lies in B. There the measures $(\mathrm{dd^c} A_{i,l,j})^n = \alpha_{i,l,j}$ have the constant densities $\alpha_{i,l}(Q)/|Q|$ with respect to Lebesgue measure $\mathrm{d}V$. Recall that $\eta^n = c_n \, \mathrm{d}V$ with $\eta = \mathrm{dd^c}|z|^2$ and $c_n = n!\pi^{-n}$, by (9.2). Applying Corollary 9.7.6 on the open set $\operatorname{Int} Q$ to the constant densities $c_n^{-1}\alpha_{i,l}(Q)/|Q|$ with respect to η^n, and using (9.33), we obtain on $\operatorname{Int} Q$

$$\bigwedge_{i=1}^n \mathrm{dd^c} A_{i,l,j} \ge \prod_{i=1}^n \left(\frac{\alpha_{i,l}(Q)}{c_n|Q|}\right)^{1/n} \eta^n = \prod_{i=1}^n \left(\frac{\alpha_{i,l}(Q)}{|Q|}\right)^{1/n} \mathrm{d}V \ge \lambda_{l,j}.$$

The factor c_n cancels because the n exponents $1/n$ sum to one. The measure $\lambda_{l,j}$ vanishes on the cubes with $\lambda_l(Q) = 0$ and does not charge the cube boundaries, which are Lebesgue null. Since the mixed product is positive, we conclude that $\bigwedge_i \mathrm{dd^c} A_{i,l,j} \ge \lambda_{l,j}$ on B.

As $j \to \infty$, $\lambda_{l,j} \to \lambda_l$ weakly. Indeed, for $\chi \in C_c(B)$, replacing the values of χ by its cube averages changes $\int \chi \, \mathrm{d}\lambda_l$ by at most $\lambda_l(B)$ times the modulus of continuity of χ at the cube diameter. Letting $j \to \infty$ and using (9.32), we obtain $\bigwedge_i \mathrm{dd^c} A_{i,l} \ge \lambda_l$.

Finally let $l \to \infty$. By Step 1, $A_{i,l} \downarrow u_i$ with $u_i \le A_{i,l} \le 0$, so Theorem 9.2.5 gives $\bigwedge_i \mathrm{dd^c} A_{i,l} \to \bigwedge_i \mathrm{dd^c} u_i$ weakly, while λ_l increases to λ. Testing with nonnegative functions in $C_c(B)$ and using monotone convergence on the right, we obtain (9.30) for potentials in $\mathcal{E}_0(B)$.

Step 3. Localization. Now let $u_1, \dots, u_n$ be locally bounded on Ω. Choose a ball $B = B(a_0, R)$ with $\overline{B} \subset \Omega$ and a compact set $K \subset B$. Since adding constants does not change the products, we may assume that $u_i \le -1$ on B. With constants $C_i > 0$ so large that $C_i\rho < \inf_B u_i$ on a neighborhood of K, put

$$\widetilde{u}_i = \max\{u_i, C_i\rho\}.$$

Then $\widetilde{u}_i = u_i$ on a neighborhood of K, and $\widetilde{u}_i = C_i\rho$ near ∂B, because $C_i\rho \to 0$ at ∂B while $u_i \le -1$. Thus $\widetilde{u}_i \in \mathcal{E}_0(B)$: its Monge–Ampère measure is smooth near ∂B and locally finite on the rest of B. By locality of the products on the open neighborhood of K where $\widetilde{u}_i = u_i$, we have $(\mathrm{dd^c}\widetilde{u}_i)^n \ge f_i \mathbf{1}_K \mu$ on B. The case already proved, applied to the measure $\mathbf{1}_K\mu$, gives $\bigwedge_i \mathrm{dd^c}\widetilde{u}_i \ge \mathbf{1}_K\lambda$, and on a neighborhood of K the left side equals $\bigwedge_i \mathrm{dd^c} u_i$ by locality. Thus $P = \bigwedge_i \mathrm{dd^c} u_i$ satisfies $P(E) \ge \lambda(E)$ for every Borel set $E \subset K$. The interiors of such compact sets K cover Ω, and since Ω has a countable basis, countably many of them, say $\operatorname{Int} L_1, \operatorname{Int} L_2, \dots$, already cover Ω. A Borel set $E \subset \Omega$ is the disjoint union of the Borel sets $E_m = (E \cap L_m) \setminus \bigcup_{m'<m} L_{m'}$, and therefore $P(E) = \sum_m P(E_m) \ge \sum_m \lambda(E_m) = \lambda(E)$. □

Corollary 9.11.2 *Let u, v be locally bounded plurisubharmonic functions on an open set $\Omega \subset \mathbb{C}^n$, $n \ge 1$, and let μ be a positive Radon measure on Ω. If f, g are nonnegative Borel functions on Ω and*

$$(\mathrm{dd^c} u)^n \ge f\mu, \qquad (\mathrm{dd^c} v)^n \ge g\mu,$$

then, for each integer $1 \le k \le n-1$,

$$(\mathrm{dd^c} u)^k \wedge (\mathrm{dd^c} v)^{n-k} \ge f^{k/n} g^{(n-k)/n} \mu. \tag{9.34}$$

For $k = 0$ and $k = n$ the corresponding inequalities are the hypotheses; for $n = 1$ only these occur.

Proof Apply Theorem 9.11.1 to k copies of u with density f and $n-k$ copies of v with density g. □

Corollary 9.11.3 *Under the hypotheses of Corollary 9.11.2,*

$$(\mathrm{dd^c}(u+v))^n \ge (f^{1/n} + g^{1/n})^n \mu.$$

Proof By additivity and symmetry of the products, $(\mathrm{dd^c}(u+v))^n = \sum_{k=0}^n \binom{n}{k} (\mathrm{dd^c} u)^k \wedge (\mathrm{dd^c} v)^{n-k}$. Bound the terms with $1 \le k \le n-1$ by Corollary 9.11.2 and the terms with $k = 0, n$ by the hypotheses; the binomial formula then gives the assertion. The density on the right is locally μ-integrable, since $f^{k/n} g^{(n-k)/n} \le f + g$ by the

weighted arithmetic–geometric mean inequality, and f and g are locally μ-integrable because $f\mu \le (\mathrm{dd}^c u)^n$ and $g\mu \le (\mathrm{dd}^c v)^n$. □

9.12 Bounded complex Hessians and contact sets

Monge–Ampère measures of envelopes are computed on contact sets: if a potential u lies below an obstacle f and $u = f$ on a set E, one expects the Monge–Ampère measure of u on E to be determined by the Hessian of f. A contact set may have empty interior, and then the locality of the products on open sets (Proposition 9.1.2) says nothing about it. We show that weak derivatives are local on arbitrary measurable sets: two functions whose weak derivatives up to order two are locally integrable have the same such derivatives almost everywhere on the set where they agree. For potentials with bounded complex Hessians this yields a formula for the Bedford–Taylor product restricted to a contact set (Proposition 9.12.4), which is used for envelopes in Chapter 16.

We use the Sobolev spaces $W^{k,p}_{\mathrm{loc}}(D)$ of Section 6.2, for open sets $D \subset \mathbb{R}^d$, integers $k \ge 0$ and $p \in \{1, 2\}$; thus a function belongs to $W^{k,p}_{\mathrm{loc}}(D)$ if it and its distributional derivatives of order at most k belong to $L^p_{\mathrm{loc}}(D)$. Since $L^2_{\mathrm{loc}}(D) \subset L^1_{\mathrm{loc}}(D)$, we have $W^{k,2}_{\mathrm{loc}}(D) \subset W^{k,1}_{\mathrm{loc}}(D)$.

Lemma 9.12.1 *Let $D \subset \mathbb{R}^d$ be open and let $g \in W^{1,1}_{\mathrm{loc}}(D)$ be real. Then $\nabla g = 0$ almost everywhere on $\{g = 0\}$. If moreover $g \in W^{2,1}_{\mathrm{loc}}(D)$, then all first and second weak derivatives of g vanish almost everywhere on $\{g = 0\}$. Here $\{g = 0\}$ may be formed with any measurable representative of g, since changing g on a null set changes this set only by a null set.*

Proof We first prove the chain rule

$$\partial_a \Phi(g) = \Phi'(g)\, \partial_a g \qquad (1 \le a \le d)$$

for a C^1 function $\Phi\colon \mathbb{R} \to \mathbb{R}$ with bounded derivative; note that $\Phi(g) \in L^1_{\mathrm{loc}}(D)$, since $|\Phi(g)| \le |\Phi(0)| + \|\Phi'\|_\infty |g|$. Let $V \Subset D$ be open. The mollifications $g_\varepsilon = g * \rho_\varepsilon$ converge to g in $L^1(V)$, together with their first derivatives $\partial_a g_\varepsilon = (\partial_a g) * \rho_\varepsilon$, by the remark preceding Lemma 6.2.4; passing to a sequence $\varepsilon_k \to 0$, we may also assume that $g_{\varepsilon_k} \to g$ almost everywhere on V. Since Φ is Lipschitz, $\Phi(g_{\varepsilon_k}) \to \Phi(g)$ in $L^1(V)$. Moreover,

$$\Phi'(g_{\varepsilon_k})\partial_a g_{\varepsilon_k} - \Phi'(g)\partial_a g = \Phi'(g_{\varepsilon_k})(\partial_a g_{\varepsilon_k} - \partial_a g) + \big(\Phi'(g_{\varepsilon_k}) - \Phi'(g)\big)\partial_a g$$

tends to zero in $L^1(V)$: the first term is bounded by $\|\Phi'\|_\infty |\partial_a g_{\varepsilon_k} - \partial_a g|$, and the second tends to zero by dominated convergence, since it tends to zero almost everywhere, Φ' being continuous, and is bounded by $2\|\Phi'\|_\infty |\partial_a g|$. For $\psi \in C^\infty_c(V)$ the classical chain rule gives $\int \Phi(g_{\varepsilon_k})\partial_a \psi = -\int \Phi'(g_{\varepsilon_k})\partial_a g_{\varepsilon_k}\, \psi$, and letting

$k \to \infty$ we obtain $\int \Phi(g)\partial_a\psi = -\int \Phi'(g)\partial_a g\,\psi$. As V was arbitrary, this proves the chain rule.

Choose a smooth function χ on $\mathbb{R}$ supported in $(-2,2)$, with $0 \le \chi \le 1$ and $\chi = 1$ on $[-1,1]$, and put $\Phi_\varepsilon(t) = \int_0^t \chi(s/\varepsilon)\,\mathrm{d}s$ for $\varepsilon > 0$. Then Φ_ε is smooth with $0 \le \Phi_\varepsilon' \le 1$, $|\Phi_\varepsilon| \le 2\varepsilon$, and by the chain rule

$$\nabla\Phi_\varepsilon(g) = \chi(g/\varepsilon)\nabla g \longrightarrow \mathbf{1}_{\{g=0\}}\nabla g \quad \text{in } L^1_{\mathrm{loc}}(D)$$

as $\varepsilon \to 0$, by dominated convergence, since $\chi(g/\varepsilon) \to \mathbf{1}_{\{g=0\}}$ pointwise and $|\chi(g/\varepsilon)\nabla g| \le |\nabla g|$. On the other hand, $\Phi_\varepsilon(g) \to 0$ uniformly, so $\nabla\Phi_\varepsilon(g) \to 0$ in the sense of distributions. Since convergence in $L^1_{\mathrm{loc}}(D)$ implies convergence in the sense of distributions, the locally integrable function $\mathbf{1}_{\{g=0\}}\nabla g$ defines the zero distribution, and hence vanishes almost everywhere (Proposition 1.4.3). This is the first assertion.

If $g \in W^{2,1}_{\mathrm{loc}}(D)$, then each $\partial_a g$ belongs to $W^{1,1}_{\mathrm{loc}}(D)$ and, by the first assertion, vanishes almost everywhere on $\{g = 0\}$. Applying the first assertion to $\partial_a g$, formed with any measurable representative, shows that $\nabla\partial_a g = 0$ almost everywhere on $\{\partial_a g = 0\}$, a set which contains $\{g = 0\}$ up to a null set. Since there are finitely many derivatives, the union of the exceptional null sets is null. □

Lemma 9.12.2 *Let $D \subset \mathbb{R}^d$ be open, and let $f \in L^2_{\mathrm{loc}}(D)$. If the distributional Laplacian Δf belongs to $L^2_{\mathrm{loc}}(D)$, then $f \in W^{2,2}_{\mathrm{loc}}(D)$. In particular, this holds if f is continuous and Δf is a locally bounded measurable function.*

Proof Considering real and imaginary parts separately, we may assume that f is real. Let $f_\varepsilon = f * \rho_\varepsilon$ be the mollifications of f. For every compact set $L \subset D$ they are smooth on a neighborhood of L for small ε, and by Proposition 1.4.3 and Lemma 1.4.1, $f_\varepsilon \to f$ and $\Delta f_\varepsilon = (\Delta f) * \rho_\varepsilon \to \Delta f$ in $L^2(L)$ as $\varepsilon \to 0$.

We first show that the gradients converge. Let χ be a real test function, and let h be a real smooth function on a neighborhood of $\operatorname{Supp}\chi$. Since $\int \partial_a(\chi^2 h\,\partial_a h) = 0$ for each a, by Fubini's theorem and the fundamental theorem of calculus, we have

$$\begin{aligned}\int \chi^2|\nabla h|^2 &= -\int \chi^2 h\Delta h - 2\int \chi h\nabla\chi\cdot\nabla h \\ &\le \tfrac{1}{2}\int \chi^2|\nabla h|^2 + \int \chi^2|h\Delta h| + 2\int h^2|\nabla\chi|^2,\end{aligned}$$

where we used $2xy \le x^2/2 + 2y^2$. Hence

$$\int \chi^2|\nabla h|^2 \le 2\|\chi\|_\infty^2\|h\|_{L^2(S)}\|\Delta h\|_{L^2(S)} + 4\|\nabla\chi\|_\infty^2\|h\|^2_{L^2(S)}, \qquad S = \operatorname{Supp}\chi.$$

Let $L \subset D$ be compact, and choose $\chi \in C^\infty_c(D)$ equal to one on L (Lemma 1.4.1). Apply the last inequality to $h = f_\varepsilon - f_\delta$. As $\varepsilon, \delta \to 0$, the norms $\|f_\varepsilon - f_\delta\|_{L^2(\operatorname{Supp}\chi)}$ tend to zero, while the norms $\|\Delta f_\varepsilon - \Delta f_\delta\|_{L^2(\operatorname{Supp}\chi)}$ stay bounded. Hence the gradients ∇f_ε form a Cauchy family in $L^2(L)$. For $\psi \in C^\infty_c(\operatorname{Int} L)$ we have $\int \partial_a f_\varepsilon\,\psi =$

$-\int f_\varepsilon\,\partial_a\psi \to -\int f\,\partial_a\psi$, so the L^2 limits are the distributional first derivatives of f on $\operatorname{Int} L$. Since L was arbitrary, the first derivatives of f belong to $L^2_{\mathrm{loc}}(D)$, and $\nabla f_\varepsilon \to \nabla f$ in $L^2(L)$ for every compact $L \subset D$.

For the second derivatives, let $L \subset D$ be compact, choose $\chi \in C_c^\infty(D)$ equal to one on a neighborhood N of L, and for small ε put $v_\varepsilon = \chi f_\varepsilon$, extended by zero to $\mathbb{R}^d$. These are smooth functions with compact support, and

$$\Delta v_\varepsilon = \chi\Delta f_\varepsilon + 2\nabla\chi\cdot\nabla f_\varepsilon + f_\varepsilon\Delta\chi$$

is a Cauchy family in $L^2(\mathbb{R}^d)$, by the first part applied on $\operatorname{Supp}\chi$. For every real $v \in C_c^\infty(\mathbb{R}^d)$, integrating by parts twice gives

$$\sum_{a,b=1}^{d} \|\partial_a\partial_b v\|_2^2 = \sum_{a,b=1}^{d}\int (\partial_a^2 v)(\partial_b^2 v) = \|\Delta v\|_2^2.$$

Applying this identity to $v_\varepsilon - v_\delta$, we see that all second derivatives of v_ε form Cauchy families in $L^2(\mathbb{R}^d)$. Testing against compactly supported smooth functions identifies their limits with the distributional second derivatives of χf, which agree with those of f on N, where $\chi f = f$. Thus the second derivatives of f are square-integrable on L. Since L was arbitrary, $f \in W^{2,2}_{\mathrm{loc}}(D)$. The last assertion holds because continuous functions and locally bounded measurable functions are locally square-integrable.□

Let $\Omega \subset \mathbb{C}^n$ be open. We say that a continuous function f on Ω has a *bounded complex Hessian* if its distributional derivatives $f_{j\bar k} = \partial^2 f/\partial z_j\partial\bar z_k$ are locally bounded measurable functions. Since the Laplacian of $\mathbb{C}^n = \mathbb{R}^{2n}$ is $4\sum_j \partial^2/\partial z_j\partial\bar z_j$ by (7.5), such a function belongs to $W^{2,2}_{\mathrm{loc}}(\Omega)$ by Lemma 9.12.2.

Proposition 9.12.3 *Let $f, g \in W^{2,1}_{\mathrm{loc}}(\Omega)$ be real, and put $E = \{f = g\}$, formed with any measurable representatives of f and g. Then the real weak first and second derivatives of f and g agree almost everywhere on E. In particular,*

$$\mathbf{1}_E f_{j\bar k} = \mathbf{1}_E g_{j\bar k} \quad \textit{almost everywhere, for all } j, k.$$

Proof Apply Lemma 9.12.1 to $f - g \in W^{2,1}_{\mathrm{loc}}(\Omega)$, whose zero set, formed with the chosen representatives, is E. The complex derivatives $f_{j\bar k}$ and $g_{j\bar k}$ are fixed linear combinations of the real second derivatives, which gives the second assertion. □

As in Section 9.7.1, $D(A_1,\dots,A_n)$ denotes the mixed discriminant of Hermitian $n\times n$ matrices (Section 7.2); in dimension zero we put $D() = 1$.

Proposition 9.12.4 *Let $u_1,\dots,u_n$ be continuous plurisubharmonic functions with bounded complex Hessians on Ω, and let $A_j = ((u_j)_{a\bar b})$ be their complex Hessian matrices. Then*

$$\mathrm{d}^{\mathrm{c}}u_1 \wedge \cdots \wedge \mathrm{d}^{\mathrm{c}}u_n = \frac{n!}{\pi^n} D(A_1,\dots,A_n)\,\mathrm{d}V_{2n}.$$

If $v_1, \dots, v_n \in W^{2,1}_{\mathrm{loc}}(\Omega)$ are real, with weak complex Hessian matrices $B_j = ((v_j)_{a\bar{b}})$, and $E = \bigcap_j \{u_j = v_j\}$, formed with any Borel representatives of the v_j, then

$$\mathbf{1}_E \, \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n = \frac{n!}{\pi^n} \mathbf{1}_E D(B_1, \dots, B_n) \, \mathrm{d}V_{2n}.$$

In particular, the density $\mathbf{1}_E D(B_1, \dots, B_n)$ is locally bounded. (In dimension zero, where $\mathbb{C}^0$ is a point and $\mathrm{d}V_0$ is the unit point mass, both sides of each identity equal $\mathrm{d}V_0$.)

Proof For smooth u_j we polarize (9.2), as in the proof of Corollary 9.7.6. Applied to the smooth function $\sum_j t_j u_j$ with $t_1, \dots, t_n \in \mathbb{R}$, it gives the identity

$$\Big(\mathrm{dd^c} \sum_j t_j u_j\Big)^n = \frac{n!}{\pi^n} \det\Big(\sum_j t_j A_j\Big) \, \mathrm{d}V_{2n}$$

between exterior powers of smooth forms, and both sides are polynomials in $t_1, \dots, t_n$. We compare the coefficients of $t_1 \cdots t_n$. Since 2-forms commute, on the left this coefficient is $n!$ times the exterior product $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_n$, which is the Bedford–Taylor product by Proposition 9.1.2(1); on the right it is $(n!)^2 \pi^{-n} D(A_1, \dots, A_n) \, \mathrm{d}V_{2n}$, by the definition of the mixed discriminant. This proves the first formula for smooth u_j.

For continuous u_j, let $u_{j,\varepsilon} = u_j * \rho_\varepsilon$ be their radial mollifications, which are smooth and plurisubharmonic on Ω_ε (Proposition 8.2.2). They converge to u_j uniformly on compact sets (Lemma 1.4.1), so their products converge weakly by Proposition 9.1.4. By Proposition 1.4.3, the complex Hessian matrix of $u_{j,\varepsilon}$ is the mollification $A_{j,\varepsilon}$ of the matrix A_j of locally bounded functions. Let $K \subset \Omega$ be compact. For small ε, the entries of $A_{j,\varepsilon}$ are bounded on K by the bounds of the entries of A_j on a fixed compact neighborhood of K, and by Lemma 1.4.1 they converge to those of A_j in $L^1(K)$. Products of functions which are bounded on K uniformly in ε and converge in $L^1(K)$ converge in $L^1(K)$, because $|ab - a'b'| \le |a||b - b'| + |b'||a - a'|$. Since D is a polynomial in the matrix entries, it follows that $D(A_{1,\varepsilon}, \dots, A_{n,\varepsilon}) \to D(A_1, \dots, A_n)$ in $L^1(K)$. By the first part, $(n!/\pi^n) D(A_{1,\varepsilon}, \dots, A_{n,\varepsilon}) \, \mathrm{d}V_{2n}$ is the product of the $\mathrm{dd^c} u_{j,\varepsilon}$, so the weak limit of these products is $(n!/\pi^n) D(A_1, \dots, A_n) \, \mathrm{d}V_{2n}$. This proves the first formula in general.

By Lemma 9.12.2, each u_j belongs to $W^{2,2}_{\mathrm{loc}}(\Omega) \subset W^{2,1}_{\mathrm{loc}}(\Omega)$, so Proposition 9.12.3, applied to u_j and the chosen representative of v_j, gives $A_j = B_j$ almost everywhere on $\{u_j = v_j\}$, hence on E. Substituting into the first formula proves the second, and also shows that $\mathbf{1}_E D(B_1, \dots, B_n) = \mathbf{1}_E D(A_1, \dots, A_n)$ is locally bounded. □

On a complex manifold, the second identity is applied in holomorphic charts to functions u_j which become plurisubharmonic after the addition of a smooth local potential g_j of a closed real $(1,1)$-form. If $g_j + u_j$ is plurisubharmonic with bounded complex Hessian, we apply Proposition 9.12.4 to $g_j + u_j$ and $g_j + v_j$: adding the same smooth function to u_j and v_j does not change the contact set, and $g_j + v_j \in W^{2,1}_{\mathrm{loc}}$ whenever $v_j \in W^{2,1}_{\mathrm{loc}}$.

9.13 Gradients against positive currents

Several later arguments differentiate the product $v\Theta$ of a bounded plurisubharmonic function v and a closed positive current Θ; examples are the proof that non-pluripolar products are closed (Theorem 15.2.1) and cutoff arguments on compact manifolds. For smooth v the derivative is $\mathrm{d}v \wedge \Theta$, and its size is controlled by the gradient measure $\mathrm{d}v \wedge \mathrm{d}^{\mathrm{c}}v \wedge \Theta$, with d^{c} as in (7.6). We extend both objects, together with the chain rule, to locally bounded plurisubharmonic functions by decreasing approximation. In this section η denotes a real one-form, not the form $\mathrm{dd}^{\mathrm{c}}|z|^2$ of the preceding sections.

Throughout, $U \subset \mathbb{C}^n$ is open and Θ is a closed positive current of bidimension $(1,1)$ on U, with trace measure $\mu = \Theta \wedge \beta$. We first describe Θ by a matrix of densities with respect to μ. For $1 \le j,k \le n$ let ν_{jk} be the complex Radon measure $\frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_k \wedge \mathrm{d}\bar z_j \wedge \Theta$. Up to a constant factor, ν_{jk} is a coefficient measure of Θ, so by Theorem 7.2.5 there is a constant C, depending only on n, with $|\nu_{jk}| \le C\mu$. Hence $C\mu + \operatorname{Re}\nu_{jk}$ and $C\mu + \operatorname{Im}\nu_{jk}$ are positive measures bounded by $2C\mu$, and Lemma 1.7.1, applied to them, gives bounded Borel functions h_{jk} with

$$\frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_k \wedge \mathrm{d}\bar z_j \wedge \Theta = h_{jk}\,\mu \qquad (1 \le j,k \le n).$$

Since Θ is real (Theorem 7.2.5), the complex conjugate of ν_{jk} is $\frac{-\mathrm{i}}{2\pi}\,\mathrm{d}\bar z_k \wedge \mathrm{d}z_j \wedge \Theta = \nu_{kj}$; by the uniqueness of densities in Lemma 1.7.1, $h_{kj} = \overline{h_{jk}}$ μ-almost everywhere, so the matrix $H = (h_{jk})$ is Hermitian μ-almost everywhere. Since $\sum_j \frac{\mathrm{i}}{2\pi}\,\mathrm{d}z_j \wedge \mathrm{d}\bar z_j = \beta/\pi$, summing the identities with $j = k$ gives $\operatorname{tr} H = 1/\pi$ μ-almost everywhere.

We identify a $(1,0)$-form $\xi = \sum_k \xi_k\,\mathrm{d}z_k$ with its coefficient vector $(\xi_1,\dots,\xi_n) \in \mathbb{C}^n$, and write $\langle a,b\rangle = \sum_j a_j\bar b_j$ on $\mathbb{C}^n$. For a constant $(1,0)$-form ξ, the measure $(\mathrm{i}/2\pi)\,\xi\wedge\bar\xi\wedge\Theta = \langle H\xi,\xi\rangle\mu$ is positive, because $\mathrm{i}\xi\wedge\bar\xi$ is a strongly positive form and Θ is positive; so $\langle H\xi,\xi\rangle \ge 0$ μ-almost everywhere. Applying this to the countably many ξ with rational real and imaginary parts, and using that $\xi \mapsto \langle H\xi,\xi\rangle$ is continuous, we see that H is positive semidefinite μ-almost everywhere. Changing H on a μ-null set, we may assume that it is Hermitian and positive semidefinite with trace $1/\pi$ everywhere. Its eigenvalues then lie in $[0,1/\pi]$, so that $\langle H\xi,\xi\rangle \le |\xi|^2/\pi$.

Let $H^{1/2}$ be the positive semidefinite square root of H. It is a bounded Borel function on U. Indeed, by the Weierstrass approximation theorem there are real polynomials p_m with $p_m(t) \to \sqrt{t}$ uniformly on $[0,1/\pi]$. Diagonalizing $H(x)$ by a unitary matrix, we see that every entry of $p_m(H(x)) - H(x)^{1/2}$ is bounded by $\sup_{[0,1/\pi]} |p_m(t) - \sqrt{t}|$. The entries of $p_m(H)$ are polynomials in the h_{jk}, hence Borel functions, and so are their pointwise limits, the entries of $H^{1/2}$. Moreover $|H^{1/2}\xi|^2 = \langle H\xi,\xi\rangle \le |\xi|^2/\pi$.

For a real C^1 function w, the coefficient vector of the form ∂w is $(\partial w/\partial z_k)_{k=1}^n$, and we denote it also by ∂w. Since $\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}}w = (\mathrm{i}/2\pi)\,\partial w \wedge \bar\partial w$ by (9.3), it follows that

$$\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}}w \wedge \Theta = \bigl|H^{1/2}\partial w\bigr|^2\mu. \tag{9.35}$$

Thus, for a compact set $K \subset U$, the square of the *energy seminorm* $\|H^{1/2}\partial w\|_{L^2(K,\mu)}$ is the mass that $\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}}w \wedge \Theta$ gives to K.

Similarly, let η be a real one-form with continuous coefficients and $(1,0)$-part $\sum_k e_k\,\mathrm{d}z_k$, and put $e = (e_1, \dots, e_n)$ and $w_k = \partial w/\partial z_k$. Only the $(1,1)$-part

$$\sum_{j,k} (e_k \bar{w}_j - \bar{e}_j w_k)\,\mathrm{d}z_k \wedge \mathrm{d}\bar{z}_j$$

of $\eta \wedge \mathrm{d}w$ contributes to $\eta \wedge \mathrm{d}w \wedge \Theta$, so that $\eta \wedge \mathrm{d}w \wedge \Theta = -2\pi\mathrm{i}\big(\langle He, \partial w\rangle - \langle H\partial w, e\rangle\big)\mu$. Since H is Hermitian, the second inner product is the complex conjugate of the first, and therefore

$$\eta \wedge \mathrm{d}w \wedge \Theta = 4\pi\,\mathrm{Im}\big\langle H^{1/2}e, H^{1/2}\partial w\big\rangle\,\mu. \tag{9.36}$$

Lemma 9.13.1 *Let Θ be a closed positive current of bidimension $(1,1)$ on an open set $U \subset \mathbb{C}^n$, and let v be a locally bounded plurisubharmonic function on U. Then the current $\mathrm{d}(v\Theta)$ has measure coefficients; we denote it by $\mathrm{d}v \wedge \Theta$. The current*

$$E_\Theta(v) = \mathrm{d}v \wedge \mathrm{d}^{\mathrm{c}}v \wedge \Theta = \tfrac{1}{2}\mathrm{d}\mathrm{d}^{\mathrm{c}}(v^2\Theta) - v\,\mathrm{d}\mathrm{d}^{\mathrm{c}}v \wedge \Theta \tag{9.37}$$

is a positive measure, the gradient measure *of v with respect to Θ. If smooth plurisubharmonic functions v_l decrease to v on a neighborhood of a compact set $K \subset U$, then their gradients ∂v_l converge in the energy seminorm on K, that is, $H^{1/2}\partial v_l$ converges in $L^2(K,\mu)$. For every $f \in C^1(\mathbb{R})$,*

$$\mathrm{d}(f(v)\Theta) = f'(v)\,\mathrm{d}v \wedge \Theta. \tag{9.38}$$

Finally, let η be a compactly supported smooth real one-form on U, and let $\tau \in C_c(U)$, $\tau \geq 0$, be equal to one on Supp η. *There is a constant C_η, depending only on η, such that for every bounded Borel function a,*

$$\left|\int a\eta \wedge \mathrm{d}v \wedge \Theta\right|^2 \leq C_\eta \left(\int \tau E_\Theta(v)\right)\left(\int_{\mathrm{Supp}\,\eta} |a|^2\,\Theta \wedge \beta\right). \tag{9.39}$$

Proof We approximate v by smooth plurisubharmonic functions v_l and pass to the limit, first in the gradients (Step 1) and then in each assertion of the lemma (Steps 2–5). All assertions are local, so we fix a compact set $K \subset U$ and an open neighborhood $W \subset U$ of K on which smooth plurisubharmonic functions v_l decrease to v. For the convergence assertion, W and the v_l are given. Otherwise we take for W a relatively compact open neighborhood of K in U and for v_l the radial mollifications $v * \rho_{\varepsilon_l}$, where $\varepsilon_l \downarrow 0$ and ε_1 is less than the distance from $\overline{W}$ to $\mathbb{C}^n \setminus U$; they are smooth and plurisubharmonic on W and decrease to v (Proposition 8.2.2). Choose a compact set $K_1 \subset W$ with $K \subset \mathrm{Int}\,K_1$, and a function $\sigma \in C_c^\infty(W)$ with $\sigma \geq 0$ which equals one on a neighborhood of K_1. Since $v \leq v_l \leq v_1$, the v_l are uniformly bounded on $\mathrm{Supp}\,\sigma$.

Step 1. Convergence of the gradients. For a smooth real function w, we have $\mathrm{d}\mathrm{d}^{\mathrm{c}}(w^2) = 2\,\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}}w + 2w\,\mathrm{d}\mathrm{d}^{\mathrm{c}}w$, and $\mathrm{d}\mathrm{d}^{\mathrm{c}}(w^2\Theta) = \mathrm{d}\mathrm{d}^{\mathrm{c}}(w^2) \wedge \Theta$ because Θ is closed. Hence

$$\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}} w \wedge \Theta = \tfrac{1}{2}\mathrm{d}\mathrm{d}^{\mathrm{c}}(w^2\Theta) - w\,\mathrm{d}\mathrm{d}^{\mathrm{c}} w \wedge \Theta.$$

Apply this identity to $w = v_l - v_m$ and integrate against σ:

$$\int \sigma\,\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}} w \wedge \Theta = \tfrac{1}{2}\int (v_l - v_m)^2 \mathrm{d}\mathrm{d}^{\mathrm{c}}\sigma \wedge \Theta - \int \sigma(v_l - v_m)\,\mathrm{d}\mathrm{d}^{\mathrm{c}}(v_l - v_m) \wedge \Theta.$$

Both terms on the right tend to zero as $l, m \to \infty$. For the first, $|v_l - v_m| \le v_N - v$ for $l, m \ge N$, and $v_N - v \to 0$ pointwise and boundedly on $\operatorname{Supp}\sigma$; since $\mathrm{d}\mathrm{d}^{\mathrm{c}}\sigma \wedge \Theta$ is a signed measure with total variation bounded by a multiple of μ on $\operatorname{Supp}\sigma$, dominated convergence applies. For the second, expand it into the four integrals $\int \sigma v_p\,\mathrm{d}\mathrm{d}^{\mathrm{c}} v_q \wedge \Theta$ with $p, q \in \{l, m\}$, taken with the signs $+, -, -, +$. Given indices $l_k, m_k \to \infty$, pass to a subsequence along which both are strictly increasing; then v_{l_k} and v_{m_k} decrease to v, and the weighted form of Theorem 9.2.5 on W, with the base current Θ, shows that each of the four integrals tends to $\int \sigma v\,\mathrm{d}\mathrm{d}^{\mathrm{c}} v \wedge \Theta$. Their signed sum therefore tends to zero, and since this holds for a subsequence of every such pair of index sequences, the second term tends to zero as $l, m \to \infty$. Since $\sigma \ge 0$, $\sigma = 1$ on K_1 and $\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}} w \wedge \Theta$ is the positive measure $|H^{1/2}\partial w|^2\mu$, by (9.35), the left side bounds $\|H^{1/2}\partial v_l - H^{1/2}\partial v_m\|^2_{L^2(K_1,\mu)}$. Consequently $H^{1/2}\partial v_l$ is a Cauchy sequence in $L^2(K_1, \mu)$; let $\Gamma \in L^2(K_1, \mu)$ be its limit. In particular, $H^{1/2}\partial v_l$ converges in $L^2(K, \mu)$, which is the convergence assertion.

Step 2. The first derivative. Let η be a smooth real one-form with compact support in $\operatorname{Int} K_1$, with $(1,0)$-part $\sum_k e_k\,\mathrm{d}z_k$. By (9.36),

$$\int \eta \wedge \mathrm{d}v_l \wedge \Theta = 4\pi \int \operatorname{Im}\big\langle H^{1/2}e, H^{1/2}\partial v_l\big\rangle\,\mathrm{d}\mu \longrightarrow 4\pi \int \operatorname{Im}\big\langle H^{1/2}e, \Gamma\big\rangle\,\mathrm{d}\mu,$$

since $H^{1/2}e$ is bounded and vanishes outside K_1, and $\mu(K_1) < \infty$. On the other hand, $\mathrm{d}v_l \wedge \Theta = \mathrm{d}(v_l\Theta)$ since Θ is closed, and $v_l\Theta \to v\Theta$ weakly on W by dominated convergence on the coefficient measures; hence $\mathrm{d}(v_l\Theta) \to \mathrm{d}(v\Theta)$ weakly. Therefore the pairing of $\mathrm{d}(v\Theta)$ with η, written $\int \eta \wedge \mathrm{d}(v\Theta)$, equals the last integral. Write $\langle H^{1/2}e, \Gamma\rangle = \sum_k e_k G_k$, where $G_k = \sum_j (H^{1/2})_{jk}\bar{\Gamma}_j$ belongs to $L^2(K_1, \mu) \subset L^1(K_1, \mu)$, so that

$$4\pi \operatorname{Im}\big\langle H^{1/2}e, \Gamma\big\rangle = \sum_k \big(-2\pi\mathrm{i}\,G_k\,e_k + 2\pi\mathrm{i}\,\bar{G}_k\,\bar{e}_k\big).$$

Thus, on real test one-forms η supported in $\operatorname{Int} K_1$, the current $\mathrm{d}(v\Theta)$ acts by integrating against μ the coefficients e_k and $\bar{e}_k$ of η, multiplied by the functions $-2\pi\mathrm{i}\,G_k$ and $2\pi\mathrm{i}\,\bar{G}_k$, which are integrable on K_1. Every test one-form is a complex linear combination of two real ones, so on $\operatorname{Int} K_1$ the current $\mathrm{d}(v\Theta)$ has measure coefficients, which are absolutely continuous with respect to μ and finite on K_1. In particular, for a bounded Borel function a,

$$\int a\eta \wedge \mathrm{d}v \wedge \Theta = 4\pi \int a \operatorname{Im}\big\langle H^{1/2}e, \Gamma\big\rangle\,\mathrm{d}\mu.$$

Step 3. The gradient measure. Since $H^{1/2}\partial v_l \to \Gamma$ in $L^2(K_1,\mu)$, the densities $|H^{1/2}\partial v_l|^2$ converge to $|\Gamma|^2$ in $L^1(K_1,\mu)$, so the measures $\mathrm{d}v_l \wedge \mathrm{d}^{\mathrm{c}} v_l \wedge \Theta = |H^{1/2}\partial v_l|^2\mu$ converge weakly on $\operatorname{Int} K_1$ to the positive measure $|\Gamma|^2\mu$. On the other hand, by the identity of Step 1,

$$\mathrm{d}v_l \wedge \mathrm{d}^{\mathrm{c}} v_l \wedge \Theta = \tfrac{1}{2}\mathrm{d}\mathrm{d}^{\mathrm{c}}(v_l^2\Theta) - v_l\,\mathrm{d}\mathrm{d}^{\mathrm{c}} v_l \wedge \Theta \longrightarrow \tfrac{1}{2}\mathrm{d}\mathrm{d}^{\mathrm{c}}(v^2\Theta) - v\,\mathrm{d}\mathrm{d}^{\mathrm{c}} v \wedge \Theta$$

weakly on W: the first term converges by dominated convergence on the coefficient measures of $v_l^2\Theta$ and the weak continuity of $\mathrm{d}\mathrm{d}^{\mathrm{c}}$, and the second by Theorem 9.2.5. Hence the right side of (9.37) is the positive measure $|\Gamma|^2\mu$ on $\operatorname{Int} K_1$. Since K was arbitrary and $K \subset \operatorname{Int} K_1$, the current $E_\Theta(v)$, which does not involve the approximations, is a positive measure on U.

Step 4. The Cauchy–Schwarz inequality. Let η, τ and a be as in the statement, and apply Steps 1–3 with $K = \operatorname{Supp}\tau$; then $\operatorname{Supp}\eta \subset K \subset \operatorname{Int} K_1$. For a smooth real function w on W, (9.36) and the Cauchy–Schwarz inequality give

$$\left|\int a\eta \wedge \mathrm{d}w \wedge \Theta\right|^2 \le 16\pi^2 \int_{\operatorname{Supp}\eta} |a|^2 \left|H^{1/2}e\right|^2 \mathrm{d}\mu \int_{\operatorname{Supp}\eta} \left|H^{1/2}\partial w\right|^2 \mathrm{d}\mu.$$

Since $|H^{1/2}e|^2 \le |e|^2/\pi$, we have $16\pi^2|H^{1/2}e|^2 \le C_\eta$ with $C_\eta = 16\pi \sup|e|^2$, which depends only on η. The second integral is at most $\int \tau\,\mathrm{d}w \wedge \mathrm{d}^{\mathrm{c}} w \wedge \Theta$, because $\tau \ge 0$ and $\tau = 1$ on $\operatorname{Supp}\eta$. Apply this to $w = v_l$ and let $l \to \infty$. The left side converges to $|\int a\eta \wedge \mathrm{d}v \wedge \Theta|^2$ by Step 2, since $a\,\operatorname{Im}\langle H^{1/2}e, H^{1/2}\partial v_l\rangle \to a\,\operatorname{Im}\langle H^{1/2}e, \Gamma\rangle$ in $L^1(K_1,\mu)$, and $\int \tau\,\mathrm{d}v_l \wedge \mathrm{d}^{\mathrm{c}} v_l \wedge \Theta \to \int \tau E_\Theta(v)$ by Step 3, since τ is continuous with compact support in $\operatorname{Int} K_1$. This proves (9.39).

Step 5. The chain rule. Both sides of (9.38) are complex-linear in f, so we may assume that f is real. Let first f be smooth. The functions v_l and v take values in a compact interval J on $\operatorname{Supp}\sigma$. Since $f(v_l) \to f(v)$ boundedly, $f(v_l)\Theta \to f(v)\Theta$ weakly on W by dominated convergence, and hence $\mathrm{d}(f(v_l)\Theta) \to \mathrm{d}(f(v)\Theta)$. By the chain rule for smooth functions and the closedness of Θ, $\mathrm{d}(f(v_l)\Theta) = f'(v_l)\,\mathrm{d}v_l \wedge \Theta$. Moreover $f'(v_l)H^{1/2}\partial v_l \to f'(v)\Gamma$ in $L^2(K_1,\mu)$: the difference is

$$f'(v_l)\big(H^{1/2}\partial v_l - \Gamma\big) + \big(f'(v_l) - f'(v)\big)\Gamma,$$

where the first term tends to zero because f' is bounded on J, and the second by dominated convergence. Now let η be as in Step 2. Identity (9.36), multiplied by the continuous function $f'(v_l)$, and Step 2 with $a = f'(v)$ give

$$\int \eta\wedge f'(v_l)\,\mathrm{d}v_l\wedge\Theta = 4\pi \int \operatorname{Im}\big\langle H^{1/2}e, f'(v_l)H^{1/2}\partial v_l\big\rangle\,\mathrm{d}\mu \longrightarrow \int \eta\wedge f'(v)\,\mathrm{d}v\wedge\Theta.$$

Since the left side also tends to $\int \eta\wedge\mathrm{d}(f(v)\Theta)$, the currents $\mathrm{d}(f(v)\Theta)$ and $f'(v)\,\mathrm{d}v\wedge\Theta$ agree on $\operatorname{Int} K_1$. As K was arbitrary, this proves (9.38) for smooth f.

Now let $f \in C^1(\mathbb{R})$, and let $f_m = f * \rho_{1/m}$ be its mollifications on $\mathbb{R}$. By Proposition 1.4.3 and Lemma 1.4.1, $f_m \to f$ and $f_m' = f' * \rho_{1/m} \to f'$ uniformly on compact subsets of $\mathbb{R}$. Since v is locally bounded, $f_m(v) \to f(v)$ and $f_m'(v) \to f'(v)$

uniformly on compact subsets of U. As Θ and $\mathrm{d}v \wedge \Theta$ are currents of order zero, it follows that $f_m(v)\Theta \to f(v)\Theta$ and $f_m'(v)\,\mathrm{d}v \wedge \Theta \to f'(v)\,\mathrm{d}v \wedge \Theta$ weakly. Since d is weakly continuous, the identity $\mathrm{d}(f_m(v)\Theta) = f_m'(v)\,\mathrm{d}v \wedge \Theta$ passes to the limit. □

Chapter 10
Pluripolar sets and the plurifine topology

The upper semicontinuous regularization of the supremum of a family of plurisubharmonic functions is plurisubharmonic, but it may differ from the supremum on an exceptional set (in dimension one, see Example 3.4.3 with $d = 2$). In Chapter 9 we showed that, locally, this exceptional set has outer Monge–Ampère capacity zero (Corollary 9.5.2), and that the same holds for the pole set of every plurisubharmonic function (Proposition 9.3.4). The first aim of this chapter is the converse: every subset of a ball with outer capacity zero lies in the pole set of a single plurisubharmonic function on the ball. Sets that are locally contained in pole sets are called pluripolar; they play in pluripotential theory the role that polar sets play in the classical theory of Chapters 3 to 5.

The link between capacity and pole sets is the relative extremal function H_E of a set E in a ball B, the regularized upper envelope of the plurisubharmonic functions on B that are nonpositive and at most -1 on E. For compact sets and for relatively compact open sets, the capacity of E is the total Monge–Ampère mass of H_E (Propositions 10.1.2 and 10.1.3). If E has outer capacity zero, the extremal functions of small open neighborhoods of E therefore have small mass, and the comparison principle shows that $H_E = 0$; a convergent series of functions from the envelope then has poles on all of E (Theorem 10.1.4). It follows that countable unions of pluripolar sets are pluripolar and that the exceptional sets of upper envelopes are pluripolar (Corollary 10.2.2). On $\mathbb{C}^n$ the local functions can even be glued, with the help of small multiples of $\log^+ |z|$, into a single plurisubharmonic function on the whole space; this is Josefson's theorem (Theorem 10.2.1).

The second half of the chapter studies the plurifine topology, the analogue of the fine topology of Chapter 5 defined by plurisubharmonic functions. Unbounded potentials force us to use it: the set $\{u > -k\}$, on which a plurisubharmonic function u coincides with its truncation $\max\{u, -k\}$, is plurifine open but in general not open. We prove that Bedford–Taylor products of locally bounded potentials are local for the plurifine topology (Theorem 10.4.3), by combining their locality on sets of the form $\{u > v\}$ (Proposition 9.2.7) with a quasi-Lindelöf theorem: every union of plurifine open sets coincides with a countable subunion outside a pluripolar set (Theorem 10.4.1).

Consequently, for plurisubharmonic functions $u_1, \dots, u_q$, the products of the truncations $\max\{u_l, -k\}$ do not depend on $k \geq m$ on the set $\bigcap_l \{u_l > -m\}$ (Proposition 10.4.4); this compatibility is the starting point of the non-pluripolar products of Chapter 15. Finally, the difference of two plurisubharmonic functions is plurifinely continuous off their common pole set. Its quasi-supremum and quasi-infimum on a plurifine open set, which disregard pluripolar sets, are therefore its ordinary supremum and infimum off any pluripolar set containing the common poles (Corollary 10.5.3).

Historical remarks and references for this chapter are collected in Appendix B.10.

10.1 Pluripolar sets and relative extremal functions

We define pluripolar sets on complex manifolds and then introduce, on a ball, the relative extremal functions that connect them with the Monge–Ampère capacity of Section 9.3.

Definition 10.1.1 Let X be a complex manifold. A subset $E \subset X$ is *pluripolar*, or *locally pluripolar*, if every point of X has a connected coordinate neighborhood U carrying a function $V \in \mathrm{PSH}(U)$ with $E \cap U \subset \{V = -\infty\}$. If X is connected, a subset $E \subset X$ is *globally pluripolar* in X if there is a single $V \in \mathrm{PSH}(X)$ with $E \subset \{V = -\infty\}$. A property holds *quasi-everywhere* if it holds outside a pluripolar set.

Pluripolar sets are not required to be Borel, and subsets of pluripolar sets are pluripolar. If $E \subset X$ is pluripolar and $W \subset X$ is open, then $E \cap W$ is pluripolar in W: restrict the functions of the definition to small coordinate balls in W. Every pluripolar set E has Lebesgue measure zero in every chart. Indeed, a chart domain C has a countable basis, so it is covered by countably many of the sets $U \cap C$, with U as in the definition, and each set $E \cap U \cap C$ lies in the pole set of the corresponding function V. In the coordinates of the chart, V is plurisubharmonic on each connected component of $U \cap C$, hence locally integrable there (Theorem 8.2.1), so its pole set in $U \cap C$ is Lebesgue null. In dimension zero every plurisubharmonic function is finite, so the only pluripolar set is the empty set. From now on the complex dimension n is positive.

Let $\Omega \subset \mathbb{C}^n$ be open. By Theorem 8.2.1, the plurisubharmonic functions in Definition 10.1.1 are nontrivial subharmonic functions when $\mathbb{C}^n$ is identified with $\mathbb{R}^{2n}$, so every pluripolar subset of Ω is real polar in the sense of Section 8.3. For $n \geq 2$ the converse fails: Example 10.2.3 exhibits a real polar subset of $\mathbb{C}^n$ that is not pluripolar. For $n = 1$ the converse holds, so that the pluripolar subsets of Ω are exactly its polar subsets in the sense of Definition 3.5.1. Indeed, let $A \subset \Omega$ be polar, and let D be a disc with $\overline{D} \subset \Omega$. By Proposition 4.2.3, applied to $A \cap D$, there is a fundamental potential on $\mathbb{C}$ equal to $-\infty$ on $A \cap D$; it is a nontrivial subharmonic function by Proposition 3.2.1, hence plurisubharmonic on D. Since every point of Ω, whether in A or not, lies in such a disc, A is pluripolar.

This changes the meaning of *quasi-everywhere*. In Definition 5.2.1 it meant outside a polar set of $\mathbb{R}^d$; from now on, throughout Part II, it means outside a pluripolar set, as in Definition 10.1.1. Since pluripolar sets are real polar, a property that holds quasi-everywhere on Ω in the new sense also holds outside a real polar set; for $n = 1$ the two notions agree, whereas for $n \geq 2$ the new notion is strictly stronger.

For the rest of this section, B denotes an open ball in $\mathbb{C}^n$. Recall the capacity Cap_B and the outer capacity Cap_B^* of Definition 9.3.2; the competitors for Cap_B are the functions $v \in \mathrm{PSH}(B)$ with $0 \leq v \leq 1$. For a set $E \subset B$ we define

$$h_E(z) = \sup\{v(z) : v \in \mathrm{PSH}(B),\ v \leq 0 \text{ on } B,\ v \leq -1 \text{ on } E\}, \qquad H_E = h_E^*.$$

The functions v in this supremum are called *competitors for* E; we always name the set, to distinguish them from the competitors for Cap_B. The constant -1 is a competitor for E, and so is the maximum of a competitor for E with -1; hence we may restrict the supremum to competitors with values in $[-1, 0]$, and $-1 \leq h_E \leq H_E \leq 0$. Since every competitor for E is at most -1 on E, we also have $h_E = -1$ on E. If v is a competitor for E with values in $[-1, 0]$, then $v + 1$ is a competitor for Cap_B. By Theorem 8.3.2, H_E is plurisubharmonic; we call it the *relative extremal function* of E with respect to B. It is the plurisubharmonic counterpart of the relative extremal function u_A of Section 4.3, which is defined by the same constraints on subharmonic functions.

We shall use the following properties of these functions repeatedly. If $E \subset F \subset B$, then every competitor for F is a competitor for E, so $h_F \leq h_E$ and $H_F \leq H_E$. If $B = B(a, R)$ and $\overline{E}$ is a compact subset of B, then for a sufficiently large constant $A > 0$ the function $A(|z - a|^2 - R^2)$ is at most -1 on E, hence it is a competitor for E, and

$$A(|z - a|^2 - R^2) \leq h_E \leq H_E \leq 0 \quad \text{on } B; \tag{10.1}$$

in particular $H_E(z) \to 0$ as $z \to \partial B$.

Next, Choquet's lemma (Lemma 3.4.1) gives competitors $w_1, w_2, \ldots$ for E with $(\sup_k w_k)^* = H_E$. The finite maxima $v_j = \max\{-1, w_1, \ldots, w_j\}$ are competitors for E, and their limit $g = \lim_j v_j$ satisfies $\sup_k w_k \leq g \leq h_E$; taking regularizations, we get $g^* = H_E$. By Theorem 8.3.2, applied to the family (v_j), we have $g = g^*$ almost everywhere. Thus there are competitors v_j for E with

$$-1 \leq v_1 \leq v_2 \leq \cdots \leq 0, \qquad \lim_j v_j = H_E \quad \text{almost everywhere on } B. \tag{10.2}$$

Finally, by Corollary 9.5.2, for every compact set $L \subset B$ the set $L \cap \{h_E < H_E\}$ has outer capacity zero in B. Exhausting B by countably many compact sets and using the countable subadditivity of outer capacity (Proposition 9.3.3), we obtain

$$\mathrm{Cap}_B^*(\{h_E < H_E\}) = 0. \tag{10.3}$$

A set $N \subset B$ with $\mathrm{Cap}_B^*(N) = 0$ has outer measure zero for $(\mathrm{dd^c} v)^n$ whenever v is a competitor for Cap_B, since $\int_G (\mathrm{dd^c} v)^n \leq \mathrm{Cap}_B(G)$ for every open set $G \supset N$.

Proposition 10.1.2 *Let $B \subset \mathbb{C}^n$ be an open ball, and let $K \subset B$ be compact. Then $H_K(z) \to 0$ as $z \to \partial B$, the Monge–Ampère measure $(\mathrm{dd^c} H_K)^n$ is carried by $K \cap \{H_K = -1\}$, and*

$$\mathrm{Cap}_B(K) = \int_B (\mathrm{dd^c} H_K)^n.$$

If (K_j) is a decreasing sequence of compact subsets of B with intersection K, then the functions H_{K_j} increase and converge to H_K almost everywhere, and $\mathrm{Cap}_B(K_j)$ decreases to $\mathrm{Cap}_B(K)$. In particular $\mathrm{Cap}^*_B(K) = \mathrm{Cap}_B(K)$.

Proof The boundary behavior of H_K follows from (10.1). Since H_K is bounded, its Monge–Ampère measure is defined (Definition 9.1.1).

We first show that $(\mathrm{dd^c} H_K)^n$ vanishes on $B \setminus K$. Choose competitors v_j for K as in (10.2), and let D be a ball with $\overline{D} \subset B \setminus K$. Let $\widehat{v}_j$ be the function obtained from v_j by Corollary 9.6.4: it is plurisubharmonic on B, equals v_j outside D, and satisfies $\widehat{v}_j \geq v_j$ and $(\mathrm{dd^c}\widehat{v}_j)^n = 0$ on D. Each $\widehat{v}_j$ is again a competitor for K. Indeed, $\widehat{v}_j = v_j \leq -1$ on K. Moreover $\widehat{v}_j \leq 0$: outside D because $\widehat{v}_j = v_j$, and on D by the maximum principle (Proposition 3.1.4), since by upper semicontinuity the upper limit of $\widehat{v}_j$ at each point $x \in \partial D$ is at most $\widehat{v}_j(x) = v_j(x) \leq 0$.

Since $\widehat{v}_j$ is increasing in v_j by Corollary 9.6.4, the functions $\widehat{v}_j$ increase with j, and $v_j \leq \widehat{v}_j \leq h_K \leq H_K$ shows that $\widehat{v}_j \to H_K$ almost everywhere. They take values in $[-1, 0]$, so Theorem 9.5.1 gives $(\mathrm{dd^c}\widehat{v}_j)^n \to (\mathrm{dd^c} H_K)^n$ weakly, and therefore $(\mathrm{dd^c} H_K)^n = 0$ on D. Covering the open set $B \setminus K$ by countably many such balls D, we conclude that $(\mathrm{dd^c} H_K)^n$ is carried by K.

As $h_K = -1$ on K, the Borel set $K \cap \{H_K > -1\}$ is contained in $\{h_K < H_K\}$, which has outer capacity zero by (10.3). Because $H_K + 1$ is a competitor for Cap_B, this set is therefore null for $(\mathrm{dd^c}(H_K + 1))^n = (\mathrm{dd^c} H_K)^n$. Thus $(\mathrm{dd^c} H_K)^n$ is carried by $K \cap \{H_K = -1\}$.

We now prove the capacity formula. As $H_K + 1$ is a competitor for Cap_B and $(\mathrm{dd^c} H_K)^n$ is carried by K, we have $\int_B (\mathrm{dd^c} H_K)^n = \int_K (\mathrm{dd^c}(H_K + 1))^n \leq \mathrm{Cap}_B(K)$. Conversely, let v be a competitor for Cap_B, let $0 < \varepsilon < 1/2$, and put $w = (1 - 2\varepsilon)(v - 1) - \varepsilon$. Then $-1 + \varepsilon \leq w \leq -\varepsilon$. In particular $w > -1$, so $K \cap \{H_K \geq w\} \subset K \cap \{H_K > -1\}$; this set has outer capacity zero and is therefore null for $(\mathrm{dd^c} w)^n = (1 - 2\varepsilon)^n (\mathrm{dd^c} v)^n$. Moreover $H_K(z) \to 0$ as $z \to \partial B$ while $w \leq -\varepsilon$, so $\underline{\lim}_{z \to x}(H_K(z) - w(z)) \geq \varepsilon > 0$ for every $x \in \partial B$, and the comparison principle (Theorem 9.2.8) applies to H_K and w on B. We obtain

$$\begin{aligned}(1 - 2\varepsilon)^n \int_K (\mathrm{dd^c} v)^n &= \int_{K \cap \{H_K < w\}} (\mathrm{dd^c} w)^n \leq \int_{\{H_K < w\}} (\mathrm{dd^c} w)^n \\ &\leq \int_{\{H_K < w\}} (\mathrm{dd^c} H_K)^n \leq \int_B (\mathrm{dd^c} H_K)^n.\end{aligned}$$

Letting $\varepsilon \downarrow 0$ and taking the supremum over v, we get $\mathrm{Cap}_B(K) \leq \int_B (\mathrm{dd^c} H_K)^n$.

Now let (K_j) be compact subsets of B decreasing to K. Since $K \subset K_{j+1} \subset K_j$, the functions H_{K_j} increase and are bounded above by H_K. Put $H = (\sup_j H_{K_j})^*$. By Theorem 8.3.2, H is plurisubharmonic and equals $\sup_j H_{K_j}$ almost everywhere, and

$H \le H_K$ because H_K is upper semicontinuous. For the reverse inequality, let v be a competitor for K and let $0 < \varepsilon < 1$. The open set $G = \{v < -1 + \varepsilon\}$ contains K. The compact sets $K_j \setminus G$ decrease and have empty intersection, so one of them is empty; hence $K_j \subset G$ for all large j. For such j the function $v/(1-\varepsilon)$ is a competitor for K_j, and hence $v \le (1-\varepsilon)H_{K_j} \le (1-\varepsilon)H$. Letting $\varepsilon \downarrow 0$ gives $v \le H$, and taking the supremum over all competitors v for K gives $h_K \le H$. As H is upper semicontinuous, $H_K \le H$. Therefore $H = H_K$, and $H_{K_j} \to H_K$ almost everywhere.

By Theorem 9.5.1, $(\mathrm{dd^c} H_{K_j})^n \to (\mathrm{dd^c} H_K)^n$ weakly. All these measures are carried by the fixed compact set K_1, so testing with a function in $C_c(B)$ equal to one on K_1 shows that their total masses converge. By the capacity formula, $\mathrm{Cap}_B(K_j) \to \mathrm{Cap}_B(K)$, and this convergence is decreasing because capacity is increasing.

Finally, $\mathrm{Cap}_B(K) \le \mathrm{Cap}_B^*(K)$ because capacity is increasing. For the converse we may assume $K \ne \varnothing$, since both capacities of the empty set vanish. Choose $\delta > 0$ such that the compact sets $K_j = \{z : \mathrm{dist}(z, K) \le \delta/j\}$ lie in B. They decrease to K, and each contains K in its interior. Hence $\mathrm{Cap}_B^*(K) \le \mathrm{Cap}_B(\mathrm{Int}\, K_j) \le \mathrm{Cap}_B(K_j)$, and the right-hand side tends to $\mathrm{Cap}_B(K)$ by what we have just proved. □

Proposition 10.1.3 *Let $B \subset \mathbb{C}^n$ be an open ball, and let G be an open set with $G \Subset B$. Then $h_G = H_G$, $H_G = -1$ on G, $H_G(z) \to 0$ as $z \to \partial B$, the measure $(\mathrm{dd^c} H_G)^n$ is carried by $\overline{G}$, and*

$$\mathrm{Cap}_B(G) = \int_B (\mathrm{dd^c} H_G)^n.$$

Proof Choose compact sets K_j with $K_j \subset \mathrm{Int}\, K_{j+1}$ and $\bigcup_j K_j = G$. Since the K_j increase, the functions H_{K_j} decrease. They take values in $[-1, 0]$, so their limit H is plurisubharmonic by Theorem 8.3.1. Since $h_{K_j} = -1$ on K_j and the upper regularization at a point only involves the values of h_{K_j} near that point, we have $H_{K_j} = -1$ on $\mathrm{Int}\, K_j$. Every point of G lies in $\mathrm{Int}\, K_j$ for all large j, so $H = -1$ on G. Thus H is a competitor for G, and $H \le h_G$. Conversely, every competitor for G is a competitor for each K_j, hence bounded above by every H_{K_j} and thus by H; so $h_G \le H$. We have shown $h_G = H$. As H is upper semicontinuous, $H_G = h_G^* = H$; in particular $H_G = -1$ on G. The boundary behavior follows from (10.1).

By Theorem 9.2.5, $(\mathrm{dd^c} H_{K_j})^n \to (\mathrm{dd^c} H_G)^n$ weakly. By Proposition 10.1.2, these measures are carried by $K_j \subset \overline{G}$. Hence the limit measure vanishes on the open set $B \setminus \overline{G}$, that is, it is carried by the compact set $\overline{G}$, and testing with a function in $C_c(B)$ equal to one on $\overline{G}$ shows that the total masses converge. Since the Borel sets K_j increase to G, Proposition 9.3.3(2) gives $\mathrm{Cap}_B(K_j) \to \mathrm{Cap}_B(G)$. Together with the capacity formula of Proposition 10.1.2 for the K_j, this yields $\mathrm{Cap}_B(G) = \lim_j \int_B (\mathrm{dd^c} H_{K_j})^n = \int_B (\mathrm{dd^c} H_G)^n$. □

Theorem 10.1.4 *Let E be a subset of an open ball $B \subset \mathbb{C}^n$. The following conditions are equivalent:*

(1) $\mathrm{Cap}_B^*(E) = 0$;

(2) $E \subset \{V = -\infty\}$ *for some* $V \in \mathrm{PSH}(B)$ *with* $V \le 0$*;*
(3) $H_E = 0$ *on* B.

Every pluripolar subset of B satisfies these conditions; in particular it is globally pluripolar in B. Consequently, on every complex manifold X, a countable union of pluripolar sets is pluripolar, and if $W \subset X$ is open, every subset of W that is pluripolar in W is pluripolar in X.

Proof We prove (2)⇒(1), then (1)⇒(3) for relatively compact sets E and (3)⇒(2), deduce (1)⇒(2) in general by exhaustion, and finally prove (2)⇒(3). The remaining assertions then follow by localization.

(2)⇒(1). We show more generally that every pluripolar set $E \subset B$ satisfies (1); a set as in (2) is pluripolar. By Definition 10.1.1, every point of B has an open neighborhood $W \subset B$ carrying a plurisubharmonic function whose pole set contains $E \cap W$. The second assertion of Proposition 9.3.4, applied with $D = B$, therefore gives $\mathrm{Cap}^*_B(E \cap L) = 0$ for every compact set $L \subset B$. Exhausting B by countably many compact sets and using countable subadditivity, we obtain $\mathrm{Cap}^*_B(E) = 0$.

(1)⇒(3) when $E \Subset B$. Fix an open set O with $\overline{E} \subset O \Subset B$. Since $\mathrm{Cap}^*_B(E) = 0$, there are open sets $G'_j \supset E$ with $\mathrm{Cap}_B(G'_j) < 1/j$. The open sets $G_j = O \cap G'_1 \cap \cdots \cap G'_j$ decrease, contain E, lie in O, and satisfy $\mathrm{Cap}_B(G_j) \to 0$. Since the G_j decrease, the functions H_{G_j} increase; put $H = (\sup_j H_{G_j})^*$. By Theorem 8.3.2, H is plurisubharmonic, takes values in $[-1, 0]$, and is the almost-everywhere limit of the H_{G_j}. By Proposition 10.1.3, $\int_B (\mathrm{dd^c} H_{G_j})^n = \mathrm{Cap}_B(G_j) \to 0$, and by Theorem 9.5.1 these measures converge weakly to $(\mathrm{dd^c} H)^n$. Hence $\int \chi\, (\mathrm{dd^c} H)^n = \lim_j \int \chi\, (\mathrm{dd^c} H_{G_j})^n \le \lim_j \mathrm{Cap}_B(G_j) = 0$ for every $\chi \in C_c(B)$ with $0 \le \chi \le 1$, and therefore $(\mathrm{dd^c} H)^n = 0$.

Since every G_j lies in O, the barrier (10.1), with a constant A chosen for O, bounds every H_{G_j}, and hence H, from below by one function tending to zero at ∂B; thus $H(z) \to 0$ as $z \to \partial B$. The last assertion of the comparison principle (Theorem 9.2.8), applied to $u = H$ and $v = 0$, gives $H \ge 0$, and therefore $H = 0$. Finally, $E \subset G_j$ gives $H_{G_j} \le H_E$ for all j; since H_E is upper semicontinuous, $0 = H \le H_E \le 0$.

(3)⇒(2). Suppose $H_E = 0$, and choose competitors v_j for E as in (10.2). Fix a point $a \in B$ with $v_j(a) \to 0$, and choose indices $j_1 < j_2 < \cdots$ with $v_{j_l}(a) > -2^{-l}$. Since every term is nonpositive, the partial sums of $V = \sum_l v_{j_l}$ form a decreasing sequence of nonpositive plurisubharmonic functions, and $V(a) \ge -1$. By Theorem 8.3.1, V is plurisubharmonic on B; moreover $V \le 0$, and $V = -\infty$ on E because every term is at most -1 there.

(1)⇒(2) in general. Let $\mathrm{Cap}^*_B(E) = 0$, and let (L_l) be an exhaustion of B by compact sets. Each set $E \cap L_l$ has outer capacity zero and is relatively compact in B, so the two preceding steps give $V_l \in \mathrm{PSH}(B)$ with $V_l \le 0$ and $E \cap L_l \subset \{V_l = -\infty\}$. The pole sets of the V_l are Lebesgue null, so there is a point $a \in B$ at which all V_l are finite. Choose constants $c_l > 0$ with $c_l |V_l(a)| \le 2^{-l}$. The partial sums of $\sum_l c_l V_l$ decrease and are at least -1 at a, so, as in the previous step, $V = \sum_l c_l V_l$ is plurisubharmonic by Theorem 8.3.1, and it is nonpositive. Every point of E lies in

some L_l, where $c_l V_l = -\infty$ while all other terms are nonpositive; hence $V = -\infty$ on E.

(2)⇒(3). If V is as in (2), then εV is a competitor for E for every $\varepsilon > 0$, so $\varepsilon V \le h_E \le 0$. Letting $\varepsilon \downarrow 0$, we get $h_E = 0$ on the set $\{V > -\infty\}$. This set has full measure and is therefore dense in B, so every neighborhood of a point contains points where $h_E = 0$. Hence $H_E = h_E^* \ge 0$, and $H_E = 0$. This completes the proof of the equivalences.

By the proof of (2)⇒(1), every pluripolar subset of B satisfies (1), and hence all three conditions; in particular, it is globally pluripolar in B. Now let X be a complex manifold and $E = \bigcup_m E_m$, where each $E_m \subset X$ is pluripolar. Let $x \in X$, and let B be a coordinate ball around x, identified with an open ball in $\mathbb{C}^n$. Each $E_m \cap B$ is pluripolar in B, so $\mathrm{Cap}_B^*(E_m \cap B) = 0$ for every m, and $\mathrm{Cap}_B^*(E \cap B) = 0$ by countable subadditivity. By (1)⇒(2), the set $E \cap B$ lies in the pole set of a plurisubharmonic function on the connected coordinate neighborhood B of x. Hence E is pluripolar.

Finally, let $W \subset X$ be open and let $E \subset W$ be pluripolar in W. That E is pluripolar in X is not immediate from Definition 10.1.1 at the boundary points of W, where a single plurisubharmonic function on a neighborhood must have poles at all nearby points of E. Let $x \in X$, and let U be a coordinate ball around x, identified with an open ball in $\mathbb{C}^n$. The open set $W \cap U$ is a countable union of balls D_k with $\overline{D_k} \subset W \cap U$. Each set $E \cap D_k$ is pluripolar in U: at points of $\overline{D_k} \subset W$ we may use, after restriction to small balls in U, the functions provided by the pluripolarity of E in W, and every point of $U \setminus \overline{D_k}$ has a neighborhood disjoint from D_k. By the previous paragraph, applied in U, the countable union $E \cap U = \bigcup_k (E \cap D_k)$ is pluripolar in U, and hence, by the first part of the theorem, it lies in the pole set of a plurisubharmonic function on U. Thus E is pluripolar in X. □

10.2 Global poles on Euclidean space

Let $E \subset \mathbb{C}^n$ be pluripolar. By Theorem 10.1.4, for every ball D the set $E \cap D$ lies in the pole set of a plurisubharmonic function on D. To obtain a single function on $\mathbb{C}^n$, we cannot simply extend such a function by zero outside the ball, since the extended function need not satisfy the submean inequality at the boundary of the ball. For instance, extending $|z|^2 - 1$ from the unit ball by zero gives $\min\{|z|^2 - 1, 0\}$. If $|a| = 1$ and $0 < r < 2$, its values on the circle $\{(1 + re^{i\theta})a\}$ are $\min\{2r\cos\theta + r^2, 0\}$; they are nonpositive, and negative for $\cos\theta < -r/2$, so its mean over this circle is negative, whereas its value at a is 0.

Instead, we use the vanishing of relative extremal functions. It provides functions u_j with values in $[-1, 0]$ that equal -1 on a bounded piece of E and have small integral. We glue each of them to a small multiple of $\log^+|z| - \log R_j$, where $\log^+ t = \max\{\log t, 0\}$ and $R_j > 1$; this function is negative on the ball $B(0, R_j)$ and vanishes on its boundary. The small integrals keep the sum of the glued functions locally integrable, while every point of E lies in infinitely many of the pieces.

Theorem 10.2.1 (Josefson) *Let $E \subset \mathbb{C}^n$ be pluripolar. Then there is $V \in \mathrm{PSH}(\mathbb{C}^n)$ with $E \subset \{V = -\infty\}$ and $V(z) \le \log^+ |z|$ for all $z \in \mathbb{C}^n$.*

Proof *Step 1: the glued functions.* Choose compact sets $L_1, L_2, \ldots$ whose interiors cover $\mathbb{C}^n$, put $E_m = E \cap L_m$, and choose a sequence $(m_j)_{j\ge1}$ in which every positive integer occurs infinitely often. Since each L_m is bounded, we can choose radii $R_j > 1$ increasing to ∞ such that

$$\rho_j(z) = 2^{-j}(\log^+ |z| - \log R_j) \le -1 \quad \text{on } L_{m_j};$$

in particular L_{m_j} lies in the ball $B_j = B(0, R_j)$. The set $E \cap B_j$ is pluripolar in B_j, so by Theorem 10.1.4 the relative extremal function of its subset E_{m_j} with respect to B_j vanishes identically. The competitors of (10.2) for E_{m_j} in B_j therefore converge to zero almost everywhere on B_j, and by bounded convergence their integrals over B_j tend to zero. Since a competitor for E_{m_j} with values in $[-1, 0]$ equals -1 on E_{m_j}, one of these competitors, which we call u_j, satisfies

$$-1 \le u_j \le 0, \qquad u_j = -1 \text{ on } E_{m_j}, \qquad \int_{B_j} (-u_j)\, dV_{2n} < 2^{-j}.$$

The function $\log^+ |z| = \max\{\frac{1}{2}\log |z|^2, 0\}$ is plurisubharmonic on $\mathbb{C}^n$ by Propositions 8.2.5 and 8.3.3, and so is ρ_j. On ∂B_j we have $|z| = R_j > 1$, hence $\rho_j = 0$, while the upper limit of u_j at every point of ∂B_j is at most 0. By Proposition 8.2.5, the function V_j equal to $\max\{u_j, \rho_j\}$ on B_j and to ρ_j on $\mathbb{C}^n \setminus B_j$ is plurisubharmonic on $\mathbb{C}^n$. Since $\rho_j \le -1 = u_j$ on E_{m_j}, we have $V_j = -1$ on E_{m_j}. On B_j both u_j and ρ_j are nonpositive, so $V_j \le 0$ there, while $V_j = \rho_j$ outside B_j; hence

$$\rho_j \le V_j \le 2^{-j} \log^+ |z| \quad \text{on } \mathbb{C}^n.$$

In particular V_j is locally bounded.

Step 2: the sum. Fix $r > 0$ and choose j_0 with $R_{j_0} > r$. For $j \ge j_0$ the ball $B(0, r)$ lies in B_j, so $V_j \le 0$ on $B(0, r)$, and since $V_j \ge u_j$ there,

$$\int_{B(0,r)} (-V_j)\, dV_{2n} \le \int_{B_j} (-u_j)\, dV_{2n} < 2^{-j}.$$

Hence the partial sums of $\sum_{j\ge j_0} V_j$ decrease on $B(0, r)$, and by monotone convergence their limit W_r satisfies $\int_{B(0,r)} (-W_r)\, dV_{2n} \le \sum_j 2^{-j} < \infty$. Thus W_r is not identically $-\infty$, and $W_r \in \mathrm{PSH}(B(0, r))$ by Theorem 8.3.1. At every point of $\mathbb{C}^n$ all but finitely many terms of the series $V = \sum_j V_j$ are nonpositive, so V is well defined with values in $[-\infty, \infty)$. On $B(0, r)$ it equals $\sum_{j<j_0} V_j + W_r$, a sum of finitely many finite plurisubharmonic functions and a plurisubharmonic function; hence V is plurisubharmonic on $B(0, r)$. As r is arbitrary and plurisubharmonicity is a local property, $V \in \mathrm{PSH}(\mathbb{C}^n)$.

Let $z \in E$, so that $z \in E_m$ for some m. Since $m = m_j$ for infinitely many j, infinitely many terms $V_j(z)$ equal -1, while all terms with $R_j > |z|$ are nonpositive. Hence

$V(z) = -\infty$. Finally, summing the upper bounds of Step 1 gives $V \le \sum_j 2^{-j} \log^+ |z| = \log^+ |z|$. □

Corollary 10.2.2 *Let X be a complex manifold of dimension n.*

(1) *If $(u_i)_{i\in I}$ is a nonempty family in $\mathrm{PSH}(X)$ that is locally uniformly bounded above and $F = \sup_i u_i$, then the set $\{F < F^*\}$ is pluripolar.*
(2) *If (u_j) is an increasing sequence in $\mathrm{PSH}(X)$ that is locally uniformly bounded above, then $\lim_j u_j = (\sup_j u_j)^*$ quasi-everywhere; here $(\sup_j u_j)^*$ is plurisubharmonic by Theorem 8.3.2.*
(3) *If $u_1, \dots, u_q$, $1 \le q \le n$, are locally bounded plurisubharmonic functions on X, then the product $\mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q$ does not charge any Borel pluripolar set. More precisely, in every chart, every pluripolar set has outer measure zero for the trace measure of this product and for the total variations of its coefficient measures.*

Proof (1) Let $x \in X$, and let B be a coordinate ball around x, identified with an open ball in $\mathbb{C}^n$. On B the regularization F^* is the regularization of $F|_B$, so by Corollary 9.5.2 the set $L \cap \{F < F^*\}$ has outer capacity zero in B for every compact set $L \subset B$. Exhausting B by countably many compact sets, we obtain $\mathrm{Cap}^*_B(B \cap \{F < F^*\}) = 0$. By Theorem 10.1.4, $B \cap \{F < F^*\}$ lies in the pole set of a plurisubharmonic function on B. Hence $\{F < F^*\}$ is pluripolar.

(2) For an increasing sequence, $\lim_j u_j = \sup_j u_j$, so this is the special case of (1) for the family (u_j).

(3) Let E be pluripolar, and work in a chart, identified with an open set $\Omega \subset \mathbb{C}^n$. Let $B' \Subset B \Subset \Omega$ be concentric open balls. The set $E \cap B$ is pluripolar in B, so $\mathrm{Cap}^*_B(E \cap B) = 0$ by Theorem 10.1.4.

Let μ be the trace measure of the product, or the total variation of one of its coefficient measures. The restrictions of $u_1, \dots, u_q$ to B form a family that is uniformly bounded on each compact subset of B, since $\overline{B} \subset \Omega$ is compact; so Lemma 9.4.7(1), applied to this family and the compact set $\overline{B'}$, gives a constant C with $\mu(A) \le C\,\mathrm{Cap}_B(A)$ for every Borel set $A \subset \overline{B'}$. Given $\varepsilon > 0$, choose an open set $G \supset E \cap B$ with $\mathrm{Cap}_B(G) < \varepsilon$; then the outer measure of $E \cap B'$ satisfies $\mu^*(E \cap B') \le \mu(G \cap B') \le C\,\mathrm{Cap}_B(G \cap B') \le C\,\mathrm{Cap}_B(G) < C\varepsilon$. Hence $\mu^*(E \cap B') = 0$. Since Ω is covered by countably many such balls B', the set $E \cap \Omega$ has outer measure zero for μ. For a Borel set E this means that the product does not charge E. □

Example 10.2.3 The torus $\mathbb{T}^2 = \{z \in \mathbb{C}^2 : |z_1| = |z_2| = 1\}$ is real polar, in the sense of Section 8.3, but not pluripolar.

To see that $\mathbb{T}^2$ is real polar, let μ be the probability measure on $\mathbb{T}^2$ obtained as the image of $(2\pi)^{-2}\,\mathrm{d}\theta$ on $[-\pi, \pi)^2$ under $\theta \mapsto (\mathrm{e}^{\mathrm{i}\theta_1}, \mathrm{e}^{\mathrm{i}\theta_2})$, and put

$$u(z) = -\int_{\mathbb{T}^2} |z - \zeta|^{-2}\,\mathrm{d}\mu(\zeta).$$

If we identify $\mathbb{C}^2$ with $\mathbb{R}^4$, then u is a positive multiple of the fundamental potential $E_4 * \mu$, so it is subharmonic on $\mathbb{R}^4$ by Proposition 3.2.1. It is finite off $\mathbb{T}^2$, since for $z \notin \mathbb{T}^2$ the distances $|z - \zeta|$, $\zeta \in \mathbb{T}^2$, are bounded below by the distance from z to the compact set $\mathbb{T}^2$. Now let $z = (e^{i\alpha_1}, e^{i\alpha_2}) \in \mathbb{T}^2$. By rotation invariance, μ is also the image of $(2\pi)^{-2}\, d\theta$ under $\theta \mapsto \zeta(\theta) = (e^{i(\alpha_1+\theta_1)}, e^{i(\alpha_2+\theta_2)})$, and

$$|z - \zeta(\theta)|^2 = |1 - e^{i\theta_1}|^2 + |1 - e^{i\theta_2}|^2 \le |\theta|^2.$$

Hence

$$-u(z) \ge \frac{1}{(2\pi)^2} \int_{|\theta|<\pi} \frac{d\theta}{|\theta|^2} = \frac{1}{2\pi} \int_0^\pi \frac{dr}{r} = \infty,$$

so $\mathbb{T}^2 \subset \{u = -\infty\}$, and $\mathbb{T}^2$ is real polar.

Suppose that $\mathbb{T}^2$ were pluripolar. By Theorem 10.2.1 there would be $V \in \mathrm{PSH}(\mathbb{C}^2)$ with $\mathbb{T}^2 \subset \{V = -\infty\}$. Fix $c \in \mathbb{C}$ with $|c| = 1$. By Proposition 8.2.3, the function $f(\zeta) = V(\zeta, c)$ is either identically $-\infty$ or subharmonic on $\mathbb{C}$, and $f = -\infty$ on the unit circle. In the second case, upper semicontinuity gives $\overline{\lim}_{\zeta\to\zeta_0} f(\zeta) \le f(\zeta_0) = -\infty$ at every point ζ_0 of the unit circle, and the maximum principle (Proposition 3.1.4), applied to $f + M$ on the unit disc for every $M > 0$, shows that $f = -\infty$ on the unit disc. Hence $V = -\infty$ on $\{|z_1| \le 1,\ |z_2| = 1\}$. Now fix z_1 with $|z_1| < 1$; the function $w \mapsto V(z_1, w)$ equals $-\infty$ on the unit circle, and the same argument shows that it equals $-\infty$ on the unit disc. Thus $V = -\infty$ on the open bidisc $\{|z_1| < 1,\ |z_2| < 1\}$, contradicting the local integrability of V (Theorem 8.2.1).

For $n \ge 3$, the same arguments show that $\mathbb{T}^2 \times \mathbb{C}^{n-2} \subset \mathbb{C}^n$ is real polar but not pluripolar. The function $U(z) = u(z_1, z_2)$ is subharmonic on $\mathbb{R}^{2n}$, by Lemma 5.1.6 with $k = 4$ and $d = 2n$. The function U is finite off $\mathbb{T}^2 \times \mathbb{C}^{n-2}$ and equal to $-\infty$ on it. If the set were pluripolar, Theorem 10.2.1 would give $V \in \mathrm{PSH}(\mathbb{C}^n)$ with poles on it, and the argument above, with the last $n - 2$ coordinates fixed, would show that $V = -\infty$ on the product of the open bidisc with $\mathbb{C}^{n-2}$, which is again impossible.

10.3 Thinness and plurifine neighborhoods

The classical fine topology of Definition 5.1.1 is the coarsest topology that contains the Euclidean topology and makes all subharmonic functions continuous. Replacing subharmonic by plurisubharmonic functions, we obtain the plurifine topology, whose terminology below parallels that of Section 5.1. On an open subset of $\mathbb{C}^n = \mathbb{R}^{2n}$ every plurisubharmonic function is subharmonic (Theorem 8.2.1), so every plurifine open set is finely open. The main results of this section, a base of plurifine neighborhoods and the positive volume of plurifine open sets, are the plurifine counterparts of Propositions 5.1.3 and 5.1.4.

For $n = 1$ the two topologies coincide. Indeed, let $V \subset \mathbb{C}$ be open, let $u \in \mathrm{SH}(V)$, and let V' be the union of the connected components of V on which u is not identically $-\infty$. Then $u|_{V'}$ is a plurisubharmonic function on V', and $u = -\infty$ on $V \setminus V'$; hence for every $t \in \mathbb{R}$ the superlevel set $\{u > t\}$ is a superlevel set of

$u|_{V'}$, which is plurifine open, while the sublevel sets $\{u < s\}$ are Euclidean open. Thus every subharmonic function is continuous for the plurifine topology, and every finely open set is plurifine open. For $n \geq 2$ the plurifine topology is strictly coarser (Proposition 5.5.1 and Example 10.3.4).

Definition 10.3.1 Let X be a complex manifold. The *plurifine topology* on X is the coarsest topology on X that contains the topology of X, which we call the Euclidean topology, and makes every $u \in \mathrm{PSH}(V)$, for every open set $V \subset X$, continuous as a map from V with the induced topology to $[-\infty, \infty)$. Sets that are open in this topology are called *plurifine open*, and neighborhoods in this topology are called *plurifine neighborhoods*. A set $E \subset X$ is *plurithin* at a point $a \in X$ if either a does not belong to the Euclidean closure of $E \setminus \{a\}$, or there is a plurisubharmonic function ψ on a Euclidean open neighborhood of a such that $\psi(a)$ is finite and

$$\overline{\lim_{E\setminus\{a\}\ni z\to a}} \psi(z) < \psi(a),$$

where the upper limit is taken with respect to the Euclidean topology.

Since the sublevel sets $\{u < s\}$ of upper semicontinuous functions are Euclidean open, the plurifine topology has a subbasis consisting of the Euclidean open sets and the sets $\{z \in V : u(z) > t\}$, where $V \subset X$ is open, $u \in \mathrm{PSH}(V)$ and $t \in \mathbb{R}$. Because $\{u > t\} = \{\max\{u, t-1\} > t\}$, it suffices to use locally bounded functions u here.

On $\mathbb{C}^n$ one may even use finite plurisubharmonic functions defined on all of $\mathbb{C}^n$. Indeed, let u be plurisubharmonic on a neighborhood of a closed ball $\overline{B(a,r)}$, with $m \leq u \leq M$ on $B(a,r)$. Choose $A > 0$ so large that the function $q(z) = m - 1 + A(|z-a|^2 - r^2/16)$ exceeds M where $|z-a| \geq r/2$, and note that $q < m$ on $B(a, r/4)$. By Proposition 8.2.5, the function equal to $\max\{u, q\}$ on $B(a,r)$ and to q outside $B(a,r)$ is plurisubharmonic on $\mathbb{C}^n$; it is finite and coincides with u on $B(a, r/4)$. Hence, for every $t \in \mathbb{R}$, the set $B(a, r/4) \cap \{u > t\}$ is the intersection of a ball with a superlevel set of a finite plurisubharmonic function on $\mathbb{C}^n$, and every subbasic set $\{z \in V : u(z) > t\}$ with locally bounded u is a union of such sets. Since, moreover, the functions $\pm \operatorname{Re} z_j$ and $\pm \operatorname{Im} z_j$ are finite and plurisubharmonic on $\mathbb{C}^n$ and generate the Euclidean topology, the plurifine topology of $\mathbb{C}^n$ is the coarsest topology that makes all finite plurisubharmonic functions on $\mathbb{C}^n$ continuous.

The superlevel sets $\{u > t\}$ need not be Euclidean open. For example, let $D = \{z \in \mathbb{C}^n : |z_1| < 1/2\}$, put $a_j = \mathrm{e}^{-j}$ for $j \geq 1$, and let $u = \sum_{j\geq 1} 2^{-j} \log|z_1 - a_j|$ on D. Each term is a positive multiple of the function $\log|z_1 - a_j|^2$, which is plurisubharmonic on the domain D by Proposition 8.3.3, and each term is negative on D, because $|z_1 - a_j| < 1/2 + \mathrm{e}^{-1} < 1$. Hence the partial sums form a decreasing sequence of plurisubharmonic functions (Proposition 8.2.5), and since $u(0) = -\sum_j j2^{-j} = -2$ is finite, $u \in \mathrm{PSH}(D)$ by Theorem 8.3.1. The point 0 lies in the plurifine open set $\{u > -3\}$, whereas the points $(a_j, 0, \ldots, 0)$, at which $u = -\infty$, converge to 0. Thus $\{u > -3\}$ is not a Euclidean neighborhood of 0.

Proposition 10.3.2 *Let X be a complex manifold and $a \in X$.*

(1) *A set $U \subset X$ containing a is a plurifine neighborhood of a if and only if $X \setminus U$ is plurithin at a.*
(2) *In a chart around a, the sets $B \cap \{\psi > 0\}$ containing a, where B is a sufficiently small coordinate ball centered at a and ψ is a bounded plurisubharmonic function on a neighborhood of $\overline{B}$, form a base of plurifine neighborhoods of a.*
(3) *The plurifine topology of an open subset of X is the topology induced by the plurifine topology of X. Biholomorphic maps are homeomorphisms for the plurifine topologies, and holomorphic maps between complex manifolds are continuous for the plurifine topologies.*

Proof We begin with the first two assertions of (3). Let $W \subset X$ be open. Every subbasic set of the plurifine topology of W is a subbasic set of the plurifine topology of X. Conversely, the intersection of a subbasic set of X with W is subbasic for W, because the restriction of a function $u \in \mathrm{PSH}(V)$ to $V \cap W$ is plurisubharmonic. Since the sets $S \cap W$, with S subbasic for X, form a subbasis of the induced topology, the plurifine topology of W is the induced topology.

Next, if F is a biholomorphism, then $F^{-1}(\{u > t\}) = \{u \circ F > t\}$, and $u \circ F$ is plurisubharmonic whenever u is. Indeed, by Proposition 8.2.3, in connected charts $u \circ F$ is plurisubharmonic or identically $-\infty$, and the second alternative would make u identically $-\infty$ on a nonempty open set, which is impossible because u is locally integrable (Theorem 8.2.1). Applying this to F and to F^{-1}, we see that both map subbasic sets to subbasic sets. In particular, we may compute plurifine neighborhoods of a in a chart, as if X were an open subset of $\mathbb{C}^n$.

(2) Each set $B \cap \{\psi > 0\}$ as in (2) is plurifine open, as the intersection of a Euclidean open set with a subbasic set. Now let N be a plurifine neighborhood of a. By the definition of the topology, N contains a set

$$N_0 = W_0 \cap \{u_1 > c_1\} \cap \cdots \cap \{u_k > c_k\}$$

containing a, where W_0 is a Euclidean open neighborhood of a, $k \geq 0$, each u_i is plurisubharmonic on a Euclidean open neighborhood V_i of a, and $c_i \in \mathbb{R}$. If $k = 0$, every coordinate ball $B \subset W_0$ centered at a works, with $\psi = 1$. Let $k \geq 1$. Replacing u_i by $\max\{u_i, c_i - 1\}$, which does not change $\{u_i > c_i\}$, we may assume that the u_i are locally bounded. Put $\delta_i = u_i(a) - c_i > 0$. By upper semicontinuity there is a coordinate ball B' centered at a, contained in W_0 and in every V_i, on which every u_i is bounded and satisfies $u_i < u_i(a) + \delta_i/(2k)$. Let B be any coordinate ball centered at a with $\overline{B} \subset B'$, and put

$$\psi = \frac{1}{4} + \sum_{i=1}^{k} \frac{u_i - u_i(a)}{\delta_i}.$$

This is a bounded plurisubharmonic function on B' with $\psi(a) = 1/4 > 0$. If $z \in B$ and $u_i(z) \leq c_i$ for some i, then the i-th term of the sum is at most -1 and each of the other $k - 1$ terms is less than $1/(2k)$, so $\psi(z) \leq 1/4 - 1 + (k-1)/(2k) < 0$. Hence $B \cap \{\psi > 0\} \subset N_0 \subset N$, which proves (2).

(1) Let U be a plurifine neighborhood of a, and put $E = X \setminus U$; note that $a \notin E$. By (2), U contains a set $B \cap \{\psi > 0\}$ with $\psi(a) \geq 0$, so $\psi \leq 0$ on $E \cap B$. Hence either a is not in the Euclidean closure of E, or $\overline{\lim}_{E \ni z \to a} \psi(z) \leq 0 < \psi(a)$; in both cases E is plurithin at a. Conversely, suppose that $E = X \setminus U$ is plurithin at $a \in U$. If a is not in the Euclidean closure of E, then U contains a Euclidean neighborhood of a. Otherwise let ψ be as in Definition 10.3.1, and choose c with $\overline{\lim}_{E \ni z \to a} \psi(z) < c < \psi(a)$. Then $\psi < c$ on $E \cap B$ for a small ball B centered at a, so the plurifine open set $B \cap \{\psi > c\}$ contains a and is contained in U.

(3) It remains to prove that a holomorphic map $F\colon Y \to X$ is continuous for the plurifine topologies. Preimages of Euclidean open sets are Euclidean open. Let $u \in \mathrm{PSH}(V)$ with $V \subset X$ open, let $t \in \mathbb{R}$, and let $y \in F^{-1}(\{u > t\})$. Choose a connected coordinate neighborhood Y' of y such that $F(Y')$ lies in a connected coordinate neighborhood contained in V. By Proposition 8.2.3, $u \circ F$ is either identically $-\infty$ or plurisubharmonic on Y'; the first alternative is excluded because $u(F(y)) > t$. Thus $Y' \cap \{u \circ F > t\}$ is a plurifine open neighborhood of y contained in $F^{-1}(\{u > t\})$. Hence preimages of subbasic sets are plurifine open. □

Proposition 10.3.3 *Let X be a complex manifold.*

(1) *Every nonempty plurifine open subset of X contains a Borel set of positive Lebesgue measure in some chart.*
(2) *The complement of every pluripolar set is dense for the plurifine topology.*
(3) *For every $u \in \mathrm{PSH}(X)$, the set $\{u > -\infty\}$ is plurifine open.*

Proof (1) Let O be a nonempty plurifine open set and $a \in O$. By Proposition 10.3.2(2), O contains a set $A = B \cap \{\psi > 0\}$ with $a \in A$, where B is a coordinate ball and ψ is a bounded plurisubharmonic function on a neighborhood of $\overline{B}$. This set is Borel because ψ is upper semicontinuous. If A had measure zero, then $\max\{\psi, 0\} = 0$ almost everywhere on B; since both functions are plurisubharmonic on B, they would coincide everywhere by the uniqueness in Theorem 8.2.1, contradicting $\psi(a) > 0$.

(2) A pluripolar set is Lebesgue null in every chart, so by (1) it contains no nonempty plurifine open set. Equivalently, its complement meets every nonempty plurifine open set.

(3) The set $\{u > -\infty\} = \bigcup_m \{u > -m\}$ is a union of subbasic plurifine open sets.□

Example 10.3.4 Let $n \geq 2$ and $H = \{z \in \mathbb{C}^n : z_1 = 0\}$. Then H is pluripolar, being the pole set of $\log |z_1|^2$ (Proposition 8.3.3), but no point of H is isolated in the topology induced on H by the plurifine topology. This is the second assertion of Proposition 5.5.1, because the topology τ of that proposition is the plurifine topology. Indeed, every $u \in \mathrm{PSH}(V)$, with $V \subset \mathbb{C}^n$ open, satisfies the conditions in the definition of τ, so every plurifine open set is open for τ. Conversely, let $u\colon V \to [-\infty, \infty)$ be as in the definition of τ. By Definition 8.1.1, u is plurisubharmonic on the union V' of the connected components of V on which it is not identically $-\infty$. Hence its superlevel sets $\{u > t\} = \{z \in V' : u(z) > t\}$, $t \in \mathbb{R}$, are plurifine open, while its sublevel sets are Euclidean open. Thus u is continuous for the plurifine topology, and every τ-open set is plurifine open.

Consequently, neither conclusion of Proposition 5.1.5 carries over to pluripolar sets: H is not plurifinely discrete, and for $a \in H$ the pluripolar set $H \setminus \{a\}$ is not plurifine closed, since every plurifine neighborhood of a meets it.

Example 10.3.5 For every $n \geq 1$ there is a plurifine open subset of $\mathbb{C}^n$ that is not a Borel set.

Let $K \subset [0, 1/10] \subset \mathbb{C}$ be the compact polar set and μ the atom-free probability measure carried by K of Proposition 4.2.4, and put $F(x) = \mu(K \cap (-\infty, x])$ for $x \in \mathbb{R}$. The function F is nondecreasing, $F = 0$ on $(-\infty, 0)$ and $F = 1$ on $[1/10, \infty)$. It is continuous: it is right continuous because μ is finite, and its jump at x is $\mu(K \cap \{x\}) = 0$, because μ has no atoms. For $y \in (0, 1)$ the set $\{x \in \mathbb{R} : F(x) \geq y\}$ is closed, nonempty and bounded below, and we let $g(y)$ be its smallest element. Then $F(g(y)) \geq y$, and $F(x) < y$ for $x < g(y)$, so $F(g(y)) = y$ because F is continuous at $g(y)$. Hence g is injective; it is also nondecreasing, so g is strictly increasing and in particular Borel measurable. Moreover $g(y) \in K$. Otherwise, since K is closed, there is $\delta > 0$ with $[g(y) - \delta, g(y)] \cap K = \varnothing$; then $F(g(y) - \delta) = F(g(y)) = y$, which contradicts the minimality of $g(y)$.

Let $N \subset (0, 1)$ be a set that is not Lebesgue measurable, and put $A = g(N) \subset K$. Since g is injective, $N = g^{-1}(A)$; so if A were a Borel subset of $\mathbb{C}$, then $A = A \cap \mathbb{R}$ would be a Borel subset of $\mathbb{R}$, and N would be a Borel set. Hence A is not a Borel subset of $\mathbb{C}$. As a subset of the polar set K, the set A is polar, hence finely closed in $\mathbb{C} = \mathbb{R}^2$ by Proposition 5.1.5. Since the fine and the plurifine topologies of $\mathbb{C}$ coincide, as shown at the beginning of this section, $W_1 = \mathbb{C} \setminus A$ is a plurifine open subset of $\mathbb{C}$ which is not Borel. For $n \geq 2$, the set $W = W_1 \times \mathbb{C}^{n-1}$ is the preimage of W_1 under the holomorphic projection $z \mapsto z_1$, so it is plurifine open by Proposition 10.3.2(3). It is not Borel, because W_1 is its preimage under the continuous map $\zeta \mapsto (\zeta, 0, \ldots, 0)$.

10.4 Countable selections and plurifine locality

The plurifine topology is in general not Lindelöf: in dimension one it is the classical fine topology of $\mathbb{C} = \mathbb{R}^2$, which is not Lindelöf by Example 5.1.7. As in Theorem 5.4.1, countability is recovered outside an exceptional set, which is now pluripolar. Since Bedford–Taylor products of locally bounded potentials do not charge pluripolar sets (Corollary 10.2.2(3)), this allows us to pass from the local identity of Lemma 10.4.2 to locality on arbitrary plurifine open sets. Applied to the truncations $\max\{u, -k\}$ of unbounded potentials, plurifine locality shows that their products for different levels $k \geq m$ agree on the set where all the potentials exceed $-m$.

Theorem 10.4.1 *Let X be a complex manifold with a countable basis.*

(1) *For every family $(U_i)_{i \in I}$ of plurifine open subsets of X, there is a countable set $J \subset I$ such that $\bigcup_{i \in I} U_i \setminus \bigcup_{j \in J} U_j$ is pluripolar.*

(2) *Every plurifine open set $U \subset X$ contains countably many sets $A_k = Q_k \cap \{\psi_k > 0\}$, where each Q_k is a coordinate ball and ψ_k is a bounded plurisubharmonic function on Q_k, such that $U \setminus \bigcup_k A_k$ is pluripolar. In particular, U contains the Borel set $V = \bigcup_k A_k$, and $U \setminus V$ is pluripolar. Consequently, if $C \subset X$ is a chart domain and μ is a positive Borel measure on C for which every pluripolar subset of C has outer measure zero, then $U \cap C$ is measurable for the completion of μ.*

Proof Since X has a countable basis, it is covered by countably many charts; in each chart, the coordinate balls with rational centers and radii and with closure in the chart domain form a countable basis. Let $\mathcal{Q}$ be the union of these countable families; it is a countable basis of X.

(1) Let $U = \bigcup_i U_i$; if I is empty, take J empty. We call a set *basic* if it has the form $Q \cap \{\psi > 0\}$, where $Q \in \mathcal{Q}$ and ψ is a bounded plurisubharmonic function on a neighborhood of $\overline{Q}$. Every point $a \in U$ lies in a basic set contained in some U_i: if $a \in U_i$, then Proposition 10.3.2(2) gives a ball B and a bounded plurisubharmonic function ψ on a neighborhood of $\overline{B}$ with $a \in B \cap \{\psi > 0\} \subset U_i$, and it suffices to choose $Q \in \mathcal{Q}$ with $a \in Q \subset B$.

Fix $Q \in \mathcal{Q}$, and let Ψ_Q be the set of bounded plurisubharmonic functions ψ on neighborhoods of $\overline{Q}$ for which the basic set $Q \cap \{\psi > 0\}$ is contained in some U_i. If Ψ_Q is empty, we put $P_Q = \varnothing$ and select nothing for this Q. Otherwise, we may assume that all members of Ψ_Q take values in $[-1, 1]$ on Q, since replacing ψ by ψ/c, where $c > 0$ bounds $|\psi|$ on Q, does not change $Q \cap \{\psi > 0\}$.

Let F_Q be the pointwise supremum of the members of Ψ_Q on Q. By Choquet's lemma (Lemma 3.4.1) there are $\psi_{Q,1}, \psi_{Q,2}, \ldots$ in Ψ_Q whose supremum F'_Q satisfies $(F'_Q)^* = F^*_Q$. By Corollary 10.2.2(1) and the last assertion of Theorem 10.1.4, the set $P_Q = \{F_Q < F^*_Q\} \cup \{F'_Q < (F'_Q)^*\}$ is pluripolar in Q, and hence in X. If $z \in Q \setminus P_Q$ and $\psi(z) > 0$ for some $\psi \in \Psi_Q$, then $F'_Q(z) = F^*_Q(z) = F_Q(z) > 0$, so $\psi_{Q,k}(z) > 0$ for some k. Therefore

$$\bigcup_{\psi \in \Psi_Q} \big(Q \cap \{\psi > 0\}\big) \subset P_Q \cup \bigcup_k \big(Q \cap \{\psi_{Q,k} > 0\}\big).$$

For each Q and k choose an index $i(Q,k) \in I$ with $Q \cap \{\psi_{Q,k} > 0\} \subset U_{i(Q,k)}$, and let J be the countable set of these indices. Every point of U lies in some basic set $Q \cap \{\psi > 0\}$ with $\psi \in \Psi_Q$, so $U \setminus \bigcup_{j \in J} U_j$ is contained in $\bigcup_Q P_Q$. This countable union of pluripolar sets is pluripolar by Theorem 10.1.4.

(2) Apply the construction of (1) to the family consisting of the single set U. It selects countably many sets $Q \cap \{\psi_{Q,k} > 0\}$, where $Q \in \mathcal{Q}$ is a coordinate ball and $\psi_{Q,k}$ restricts to a bounded plurisubharmonic function on Q; these sets are contained in U, and their union V satisfies $U \setminus V \subset \bigcup_Q P_Q$, which is pluripolar. Each selected set is Borel, because $\psi_{Q,k}$ is upper semicontinuous, so the countable union V is Borel. Finally, let C and μ be as in the statement. The set $(U \setminus V) \cap C$ is pluripolar in C, so it has outer μ-measure zero and is a null set for the completion of μ. Since $U \cap C = (V \cap C) \cup \big((U \setminus V) \cap C\big)$ and $V \cap C$ is Borel, $U \cap C$ is measurable for the completion of μ. □

We now turn to Bedford–Taylor products. The first step is their locality on a single superlevel set of a plurisubharmonic function, which we derive from their locality on the sets $\{u > v\}$ (Proposition 9.2.7).

Lemma 10.4.2 *Let $\Omega \subset \mathbb{C}^n$ be open, let T be a closed positive current of bidimension (r,r) on Ω, let u and v be locally bounded plurisubharmonic functions on Ω, and let $t \in \mathrm{PSH}(\Omega)$. If $u = v$ on $\{t > 0\}$, then, for every integer $0 \le p \le r$,*

$$\mathbf{1}_{\{t>0\}} (\mathrm{dd^c} u)^p \wedge T = \mathbf{1}_{\{t>0\}} (\mathrm{dd^c} v)^p \wedge T.$$

Proof Replacing t by $\max\{t, 0\}$, which is plurisubharmonic (Proposition 8.2.5) and has the same set $\{t > 0\}$, we may assume that $t \ge 0$. Then t is locally bounded, because plurisubharmonic functions are bounded above on compact sets, and $u + \varepsilon t$ is a locally bounded plurisubharmonic function for every $\varepsilon > 0$. For $\varepsilon > 0$ put $w_\varepsilon = \max\{u + \varepsilon t, v\}$. On $\{t > 0\}$ we have $u + \varepsilon t > u = v$, so $\{t > 0\} \subset \{u + \varepsilon t > v\}$, and Proposition 9.2.7, applied to $u + \varepsilon t$ and v, gives

$$\mathbf{1}_{\{t>0\}} (\mathrm{dd^c} w_\varepsilon)^p \wedge T = \mathbf{1}_{\{t>0\}} (\mathrm{dd^c} (u + \varepsilon t))^p \wedge T.$$

Multiplying by t, which vanishes outside $\{t > 0\}$, we obtain $t(\mathrm{dd^c} w_\varepsilon)^p \wedge T = t(\mathrm{dd^c}(u + \varepsilon t))^p \wedge T$. As ε decreases to zero along a sequence, w_ε decreases to $\max\{u, v\}$ and $u + \varepsilon t$ decreases to u, because $t \ge 0$. The weighted form of Theorem 9.2.5, with the constant sequence of weights equal to t, therefore yields

$$t(\mathrm{dd^c} \max\{u, v\})^p \wedge T = t(\mathrm{dd^c} u)^p \wedge T.$$

Interchanging the roles of u and v, we also get $t(\mathrm{dd^c} \max\{u, v\})^p \wedge T = t(\mathrm{dd^c} v)^p \wedge T$, and hence $t(\mathrm{dd^c} u)^p \wedge T = t(\mathrm{dd^c} v)^p \wedge T$. Finally, multiply the coefficient measures of both sides by $1/(t+\delta)$ and let $\delta \downarrow 0$. Since $t/(t+\delta) \to \mathbf{1}_{\{t>0\}}$ pointwise and boundedly, dominated convergence with respect to the total variations of the coefficient measures gives the assertion. □

Bedford–Taylor products are local in the plurifine topology, in the following sense.

Theorem 10.4.3 *Let X be a complex manifold of dimension n, let $1 \le q \le n$ be an integer, and let $u_1, \dots, u_q$ and $v_1, \dots, v_q$ be locally bounded plurisubharmonic functions on X. If $u_l = v_l$ on a plurifine open set $W \subset X$ for every l, then*

$$\mathbf{1}_W \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q = \mathbf{1}_W \mathrm{dd^c} v_1 \wedge \cdots \wedge \mathrm{dd^c} v_q.$$

Here W need not be Borel (Example 10.3.5). It is measurable for the completions of the total variations of the coefficient measures of both products, and $\mathbf{1}_W$ acts on these completed measures; up to null sets of these measures, W coincides with a Borel set.

Proof The identity and the measurability of W are local properties, so we may assume that X is an open ball in $\mathbb{C}^n$; by Proposition 10.3.2(3), the intersection of W with this ball is plurifine open in the ball. Write $S_u = \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q$ and $S_v = \mathrm{dd^c} v_1 \wedge \cdots \wedge \mathrm{dd^c} v_q$. We first prove the identity on sets of the form $Q \cap \{\psi > 0\}$,

and then cover W by countably many such sets up to a pluripolar set, which the products do not charge.

Step 1: basic sets. We show that $\mathbf{1}_A S_u = \mathbf{1}_A S_v$ whenever $Q \subset X$ is open, $\psi \in \mathrm{PSH}(Q)$ and $A = Q \cap \{\psi > 0\} \subset W$. For every nonempty set $I \subset \{1, \dots, q\}$, the functions $u_I = \sum_{l \in I} u_l$ and $v_I = \sum_{l \in I} v_l$ are locally bounded and plurisubharmonic on Q, and they coincide on A, because $u_l = v_l$ on W. By Lemma 10.4.2, applied on Q with $t = \psi$, $T = 1$ and $p = q$, we have $\mathbf{1}_A (\mathrm{dd^c} u_I)^q = \mathbf{1}_A (\mathrm{dd^c} v_I)^q$.

We now recover S_u from the products $(\mathrm{dd^c} u_I)^q$ by polarization. For a multi-index $\alpha \in \mathbb{N}^q$ with $|\alpha| = q$, let S_u^α be the product of q factors in which $\mathrm{dd^c} u_l$ occurs α_l times; it does not depend on the order of the factors, because products of locally bounded plurisubharmonic functions are symmetric (Theorem 9.2.5). Since these products are also additive in each factor (Proposition 9.1.2(3)), expanding $(\mathrm{dd^c} u_I)^q$ with $\mathrm{dd^c} u_I = \sum_{l \in I} \mathrm{dd^c} u_l$ gives, exactly as in the multinomial expansion of $(\sum_{l \in I} x_l)^q$,

$$(\mathrm{dd^c} u_I)^q = \sum_\alpha \frac{q!}{\alpha!} S_u^\alpha,$$

where α ranges over the multi-indices with $|\alpha| = q$ and $\alpha_l = 0$ for $l \notin I$. Multiplying by $(-1)^{q-|I|}$ and summing over the nonempty sets $I \subset \{1, \dots, q\}$, we obtain $\sum_\alpha c_\alpha S_u^\alpha$, summed over all α with $|\alpha| = q$, where c_α is the coefficient of the monomial x^α on the right-hand side of the polarization identity (9.5) with $N = q$ (in which the term with $I = \varnothing$ vanishes). Both sides of that identity are polynomials in $x_1, \dots, x_q$ that agree on $\mathbb{R}^q$, so they have the same coefficients: $c_\alpha = q!$ if $\alpha = (1, \dots, 1)$, and $c_\alpha = 0$ otherwise. Since $S_u^{(1,\dots,1)} = S_u$, we obtain

$$q!\, S_u = \sum_{\varnothing \neq I \subset \{1,\dots,q\}} (-1)^{q-|I|} (\mathrm{dd^c} u_I)^q \quad \text{on } Q.$$

The same identity holds for the v_l. Multiplying both identities by $\mathbf{1}_A$ and using $\mathbf{1}_A (\mathrm{dd^c} u_I)^q = \mathbf{1}_A (\mathrm{dd^c} v_I)^q$, we obtain $\mathbf{1}_A S_u = \mathbf{1}_A S_v$.

Step 2: countable selection. By Theorem 10.4.1(2), applied to the ball X and its plurifine open subset W, there are countably many sets $A_k = Q_k \cap \{\psi_k > 0\} \subset W$, with $Q_k \subset X$ open and $\psi_k \in \mathrm{PSH}(Q_k)$, whose union V is Borel and such that $W \setminus V$ is pluripolar. By Corollary 10.2.2(3), every pluripolar subset of X has outer measure zero for the total variations of the coefficient measures of S_u and S_v. Hence, by the last assertion of Theorem 10.4.1(2), W is measurable for their completions; moreover $W \setminus V$ is a null set for them, so $\mathbf{1}_W S_u = \mathbf{1}_V S_u$ and $\mathbf{1}_W S_v = \mathbf{1}_V S_v$. The Borel sets $A'_k = A_k \setminus (A_1 \cup \cdots \cup A_{k-1})$ partition V, and Step 1 gives $\mathbf{1}_{A'_k} S_u = \mathbf{1}_{A'_k} \mathbf{1}_{A_k} S_u = \mathbf{1}_{A'_k} \mathbf{1}_{A_k} S_v = \mathbf{1}_{A'_k} S_v$. Summing over k, by countable additivity of the coefficient measures, we obtain $\mathbf{1}_V S_u = \mathbf{1}_V S_v$, which proves the theorem. □

Step 1 of this proof works, with the same argument, for products $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge T$ with an arbitrary closed positive current T, because Lemma 10.4.2 does. Step 2 does not: such a current may charge pluripolar sets, and then the exceptional set $W \setminus V$ cannot be neglected. For instance, if $H = \{z \in \mathbb{C}^n : z_1 = 0\}$, the current of integration $[H]$ of Example 7.2.7 has the trace measure $[H] \wedge \beta^{n-1}$, which is $(n-1)!$

times the $(2n-2)$-dimensional Lebesgue measure of H, because the restriction of $\beta^{n-1}/(n-1)!$ to H is the Euclidean volume form of H; so it gives positive mass to H, which is pluripolar as the pole set of $\log|z_1|^2$.

Proposition 10.4.4 *Let $\Omega \subset \mathbb{C}^n$ be open, let $1 \le q \le n$ be an integer, and let $u_1, \dots, u_q \in \mathrm{PSH}(\Omega)$. For every integer $m \ge 1$ put*

$$u_l^{(m)} = \max\{u_l, -m\}, \qquad E_m = \bigcap_{l=1}^{q} \{u_l > -m\}, \qquad S_m = \mathrm{dd^c} u_1^{(m)} \wedge \cdots \wedge \mathrm{dd^c} u_q^{(m)}.$$

Then, for all integers $k \ge m \ge 1$,

$$\mathbf{1}_{E_m} S_k = \mathbf{1}_{E_m} S_m.$$

The restrictions $\mu_m = \mathbf{1}_{E_m}\mu_{S_m}$ of the trace measures μ_{S_m} (Definition 7.2.4) increase with m, and every pluripolar subset of Ω has outer measure zero for each μ_m. Their increasing limit $\mu = \lim_m \mu_m$ is a Borel measure on Ω, possibly of infinite mass on compact sets, for which every pluripolar subset of Ω also has outer measure zero.

Proof Each set $\{u_l > -m\}$ is a subbasic plurifine open set, so E_m is plurifine open; it is also Borel. On E_m we have $u_l > -m$, and hence $u_l^{(k)} = u_l = u_l^{(m)}$ for all l and all $k \ge m$. The truncations are locally bounded plurisubharmonic functions, so Theorem 10.4.3 gives $\mathbf{1}_{E_m} S_k = \mathbf{1}_{E_m} S_m$. Taking trace measures, and using $E_m \subset E_k$ and the positivity of μ_{S_k}, we obtain $\mu_m = \mathbf{1}_{E_m}\mu_{S_k} \le \mathbf{1}_{E_k}\mu_{S_k} = \mu_k$. By Corollary 10.2.2(3), every pluripolar subset of Ω has outer measure zero for μ_{S_m}, and hence for μ_m.

For a Borel set $A \subset \Omega$ put $\mu(A) = \lim_m \mu_m(A) \in [0,\infty]$. If A is the disjoint union of Borel sets A_i, then

$$\mu(A) = \lim_m \sum_i \mu_m(A_i) = \sum_i \lim_m \mu_m(A_i) = \sum_i \mu(A_i),$$

by monotone convergence for series of nonnegative terms that increase with m. Thus μ is a Borel measure. Finally, let $N \subset \Omega$ be pluripolar. Since N has outer measure zero for each μ_m, there are Borel sets $G_{m,i} \supset N$ with $\mu_m(G_{m,i}) < 1/i$ for all $m, i \ge 1$. The Borel set $G = \bigcap_{m,i} G_{m,i}$ contains N and satisfies $\mu_m(G) = 0$ for every m, hence $\mu(G) = 0$. Thus N has outer measure zero for μ. □

In general μ is not locally finite (Example 15.1.6), so a separate local mass estimate is needed before the restrictions $\mathbf{1}_{E_m} S_m$ can define a current of order zero in the limit; this is the subject of Chapter 15.

10.5 Common poles and quasi-extrema

Let φ and ψ be plurisubharmonic functions on an open set $\Omega \subset \mathbb{C}^n$. Their difference $\varphi - \psi$ is defined, with values in $[-\infty,\infty]$, outside the common pole set $\{\varphi = \psi = -\infty\}$,

on which it would be the meaningless expression $-\infty-(-\infty)$. We show that $\varphi-\psi$ is plurifinely continuous off the common pole set, and that its supremum and infimum over a plurifine open set do not change when an arbitrary pluripolar set containing the common poles is removed.

Proposition 10.5.1 *Let $\Omega\subset\mathbb{C}^n$ be open, let $\varphi,\psi\in\mathrm{PSH}(\Omega)$, and put $P=\{\varphi=\psi=-\infty\}$. Then P is plurifine closed and pluripolar, and the function $\varphi-\psi$ is continuous from the plurifine open set $\Omega\setminus P$, with the plurifine topology, to $[-\infty,\infty]$. The same holds for $\varphi-\psi+b$, for every continuous real function b on Ω.*

Proof The function φ is plurifinely continuous with values in $[-\infty,\infty)$, and $\{-\infty\}$ is closed in $[-\infty,\infty)$; hence the pole set $\{\varphi=-\infty\}$ is plurifine closed. The same holds for ψ, so P is plurifine closed. It is pluripolar because it lies in the pole set of φ, which is pluripolar by Definition 10.1.1, applied with small balls in Ω as neighborhoods.

The subtraction $(s,t)\mapsto s-t$ is continuous from $[-\infty,\infty)^2\setminus\{(-\infty,-\infty)\}$ to $[-\infty,\infty]$. At points where s and t are finite, this is the continuity of subtraction on $\mathbb{R}^2$. If $s=-\infty$ and t is finite, then for every $M>0$ we have $s'-t'<-M$ whenever $t'>t-1$ and $s'<t-1-M$; this is continuity with the value $-\infty$. Symmetrically, if s is finite and $t=-\infty$, then $s'-t'>M$ whenever $s'>s-1$ and $t'<s-1-M$. Since $z\mapsto(\varphi(z),\psi(z))$ is plurifinely continuous on $\Omega\setminus P$ with values in this domain, $\varphi-\psi$ is plurifinely continuous on $\Omega\setminus P$. Finally, b is continuous for the Euclidean topology, hence for the plurifine topology, and addition is continuous from $[-\infty,\infty]\times\mathbb{R}$ to $[-\infty,\infty]$. □

Definition 10.5.2 Let X be a complex manifold, let $W\subset X$ be a nonempty plurifine open set, let $Z\subset X$ be pluripolar, and let $f\colon W\setminus Z\to[-\infty,\infty]$. The *quasi-supremum* and the *quasi-infimum* of f on W are

$$\operatorname*{q\text{-}sup}_W f=\inf_N\sup_{W\setminus N} f,\qquad \operatorname*{q\text{-}inf}_W f=\sup_N\inf_{W\setminus N} f,$$

where N ranges over the pluripolar subsets of X containing Z, and all suprema and infima are taken in $[-\infty,\infty]$.

The sets $W\setminus N$ in this definition are nonempty: by Proposition 10.3.3(1), W contains a Borel set of positive Lebesgue measure in some chart, whereas N is Lebesgue null in every chart.

Corollary 10.5.3 *Let $\Omega\subset\mathbb{C}^n$ be open, let $\varphi,\psi\in\mathrm{PSH}(\Omega)$, let b be a continuous real function on Ω, and let $f=\varphi-\psi+b$ on $\Omega\setminus P$, where $P=\{\varphi=\psi=-\infty\}$. Let $Z\subset\Omega$ be a pluripolar set containing P, and let $W\subset\Omega$ be a nonempty plurifine open set. Then*

$$\operatorname*{q\text{-}sup}_W f=\sup_{W\setminus Z} f,\qquad \operatorname*{q\text{-}inf}_W f=\inf_{W\setminus Z} f.$$

In particular, the right-hand sides do not depend on the choice of Z.

Proof The set $W' = W \setminus P$ is plurifine open, and f is plurifinely continuous on W', by Proposition 10.5.1. Let N be a pluripolar set containing P. We claim that $\sup_{W'\setminus N} f = \sup_{W'} f$. The inequality $\le$ holds because $W' \setminus N \subset W'$. Conversely, if $s \in \mathbb{R}$ and $s < \sup_{W'} f$, then $\{z \in W' : f(z) > s\}$ is a nonempty plurifine open set. It meets $\Omega \setminus N$, which is plurifine dense by Proposition 10.3.3(2), and hence $\sup_{W'\setminus N} f > s$. This proves the claim, and the same argument with the sets $\{f < s\}$ gives $\inf_{W'\setminus N} f = \inf_{W'} f$.

Since $W \setminus N = W' \setminus N$ whenever $N \supset P$, every pluripolar set $N \supset Z$ gives $\sup_{W\setminus N} f = \sup_{W\setminus P} f$; taking $N = Z$, we also get $\sup_{W\setminus Z} f = \sup_{W\setminus P} f$. Hence $\text{q-sup}_W f = \sup_{W\setminus Z} f = \sup_{W\setminus P} f$, and similarly $\text{q-inf}_W f = \inf_{W\setminus Z} f = \inf_{W\setminus P} f$. The common values do not involve Z. □

Chapter 11
Weighted L^2 estimates and holomorphic extension

Many constructions in the rest of the book need a holomorphic function F on a domain $\Omega \subset \mathbb{C}^n$ with a prescribed value at a point, or a prescribed restriction to a hyperplane section, together with a bound for the weighted L^2 norm $\int_\Omega |F|^2 \mathrm{e}^{-\varphi}\, \mathrm{d}V$, where φ is a plurisubharmonic weight. Gluing by a partition of unity does not preserve holomorphicity: if f is holomorphic near the support of a smooth cutoff function χ, then χf is a global smooth function, but $\bar\partial(\chi f) = f\,\bar\partial\chi$ need not vanish. If we can solve $\bar\partial u = f\,\bar\partial\chi$ with an estimate in a weighted L^2 norm, then $F = \chi f - u$ is holomorphic. The weight has two tasks: it must make the equation solvable with a bound on u, and, where values are prescribed, it must be so singular that the finiteness of the weighted norm forces u to vanish there.

The basic existence theorem is Hörmander's estimate (Theorems 11.3.3 and 11.3.4). By the Hilbert space duality of Section 11.1, a solution of $\bar\partial u = \alpha$ with a norm bound exists as soon as an a priori inequality holds for the adjoint operator. For compactly supported forms this inequality follows from an integration-by-parts identity, the Bochner–Kodaira–Morrey–Kohn identity, in which the complex Hessian of the weight appears as a curvature term. To pass from compactly supported forms to the whole domain of the adjoint without any assumption on the boundary, we work on pseudoconvex open sets and modify the weight near the boundary by means of a smooth strictly plurisubharmonic exhaustion function, whose existence is proved in Section 11.2.

In Section 11.4 we apply these estimates. Adding the logarithmic pole $n \log|z-a|^2$ to the weight prescribes the value of a holomorphic function at the point a (Proposition 11.4.1). For extension from a hyperplane, a bounded twist of the weight and an explicit one-variable weight of log log type yield the Ohsawa–Takegoshi extension theorem, with a constant that depends only on the width of the domain in the normal direction (Theorem 11.4.4). Its consequences for point values and for integrable germs (Corollaries 11.4.5 and 11.4.6) are the analytic input for the multiplier ideals of Chapter 12 and the Bergman kernel approximation of Chapter 13.

Historical remarks and references for this chapter are collected in Appendix B.11.

11.1 Hilbert spaces and closed operator complexes

A complex *Hilbert space* is a complex vector space with an inner product $\langle \cdot, \cdot \rangle$ that is complete for the norm $\|x\| = \langle x, x \rangle^{1/2}$; our inner products are linear in the first variable and conjugate-linear in the second. A *real* Hilbert space is defined in the same way, with a real vector space and a real inner product, which is then symmetric and bilinear. All results of this section hold for real Hilbert spaces, with the same proofs: the completeness, proved below, of the spaces $L^2(\mu)$ of real-valued functions, and Propositions 11.1.1 and 11.1.2, Lemma 11.1.3, and Theorem 11.1.4. Indeed, these proofs use only the axioms of an inner product and completeness. For real spaces one omits all complex conjugations, reads "conjugate-linear" as "linear", uses only real values of t in the proof of Proposition 11.1.1, and replaces the coefficients in $\mathbb{Q} + \mathrm{i}\mathbb{Q}$ in the proof of Proposition 11.1.2 by rational ones.

The Hilbert spaces of this chapter are weighted L^2 spaces. For a positive measure μ on a measurable space, $L^2(\mu)$ carries the inner product $\langle f, g \rangle = \int f \bar{g} \, \mathrm{d}\mu$, and it is complete. Indeed, let (f_j) be a Cauchy sequence in $L^2(\mu)$, and choose $j_1 < j_2 < \cdots$ with $\|f_{j_{k+1}} - f_{j_k}\| \le 2^{-k}$. By the triangle inequality and the monotone convergence theorem, $g = \sum_k |f_{j_{k+1}} - f_{j_k}|$ satisfies $\|g\| \le 1$, so $g < \infty$ almost everywhere. Hence (f_{j_k}) converges almost everywhere to a function f, and the bound $|f - f_{j_k}| \le \sum_{l \ge k} |f_{j_{l+1}} - f_{j_l}|$ gives $\|f - f_{j_k}\| \le 2^{1-k}$. A Cauchy sequence with a convergent subsequence converges, so $f_j \to f$ in $L^2(\mu)$. Applying this to each coefficient, we see that spaces of L^2 differential forms, whose elements are finite tuples of functions in $L^2(\mu)$, are complete as well.

Proposition 11.1.1 *Let H be a complex Hilbert space. Every nonempty closed convex subset of H contains a unique point of smallest distance to a given point of H. If $M \subset H$ is a closed linear subspace, then $H = M \oplus M^\perp$ and $(M^\perp)^\perp = M$. Every bounded linear functional L on H has the form $L(x) = \langle x, y \rangle$ for a unique $y \in H$, and every bounded conjugate-linear functional ℓ on H has the form $\ell(x) = \langle y, x \rangle$ for a unique $y \in H$; in both cases the norm of the functional is $\|y\|$.*

Proof Let C be a nonempty closed convex set and $x \in H$. Replacing C by $C - x$, we may assume that $x = 0$. Let $d = \inf_{z \in C} \|z\|$, and let (x_j) be a sequence in C with $\|x_j\| \to d$. Since $(x_j + x_k)/2 \in C$ by convexity, the parallelogram identity gives

$$\|x_j - x_k\|^2 = 2\|x_j\|^2 + 2\|x_k\|^2 - 4\|(x_j + x_k)/2\|^2 \le 2\|x_j\|^2 + 2\|x_k\|^2 - 4d^2.$$

The right side tends to zero as $j, k \to \infty$, so (x_j) is a Cauchy sequence. Its limit lies in C, because C is closed, and has norm d. If $x, x' \in C$ both have norm d, the same inequality with $x_j = x$ and $x_k = x'$ gives $\|x - x'\|^2 \le 0$, which proves uniqueness.

Now let M be a closed linear subspace and $x \in H$. Let p be the point of M closest to x, and put $r = x - p$. For $v \in M$ and $t \in \mathbb{C}$ we have $p + tv \in M$, and hence

$$0 \le \|r - tv\|^2 - \|r\|^2 = -2\,\mathrm{Re}\big(\bar{t}\langle r, v \rangle\big) + |t|^2 \|v\|^2.$$

Taking small real t of both signs gives $\operatorname{Re}\langle r, v\rangle = 0$, and taking $t \in \mathrm{i}\mathbb{R}$ gives $\operatorname{Im}\langle r, v\rangle = 0$. Thus $r \in M^\perp$ and $x = p + r$. Since $M \cap M^\perp = \{0\}$, this decomposition is unique, so $H = M \oplus M^\perp$. Every element of M is orthogonal to $M^\perp$, so $M \subset (M^\perp)^\perp$. Conversely, if $x \in (M^\perp)^\perp$ and $x = p + r$ as above, then $r = x - p$ lies in the linear subspace $(M^\perp)^\perp$ and in $M^\perp$, so $\|r\|^2 = \langle r, r\rangle = 0$ and $x = p \in M$. We call p the *orthogonal projection* of x onto M; the map $P\colon x \mapsto p$ is linear by the uniqueness of the decomposition, and $\|Px\| \le \|x\|$, because $\|x\|^2 = \|p\|^2 + \|r\|^2$.

Finally, let L be a bounded linear functional. If $L = 0$ we take $y = 0$. Otherwise $\ker L$ is a proper closed subspace. Choose $x_0 \notin \ker L$, and let e be its component in $(\ker L)^\perp$; then $e \neq 0$ and $L(e) = L(x_0) \neq 0$. For every $x \in H$ the vector $x - (L(x)/L(e))e$ lies in $\ker L$ and is therefore orthogonal to e, so that $\langle x, e\rangle = (L(x)/L(e))\|e\|^2$. With $y = \overline{L(e)}\, e/\|e\|^2$ this reads $L(x) = \langle x, y\rangle$. The Cauchy–Schwarz inequality shows that L has norm at most $\|y\|$, and $L(y) = \|y\|^2$ shows equality. If y' also represents L, then $\langle x, y - y'\rangle = 0$ for all x, and $x = y - y'$ gives $y = y'$. If ℓ is a bounded conjugate-linear functional, then $\overline{\ell}\colon x \mapsto \overline{\ell(x)}$ is a bounded linear functional with the same norm, and its representation $\overline{\ell(x)} = \langle x, y\rangle$ reads $\ell(x) = \langle y, x\rangle$. □

Proposition 11.1.2 *Every bounded sequence in a Hilbert space H has a weakly convergent subsequence, that is, a subsequence (x_j) for which there is $x \in H$ with $\langle x_j, v\rangle \to \langle x, v\rangle$ for every $v \in H$. If (x_j) converges weakly to x, then $\|x\| \le \varliminf_j \|x_j\|$.*

Proof Let (x_j) be a sequence with $\|x_j\| \le C$, and let V be the closed linear span of its terms. The finite linear combinations of the x_j with coefficients in $\mathbb{Q} + \mathrm{i}\mathbb{Q}$ form a countable dense subset of V. Applying the Gram–Schmidt process to it, and discarding the vectors that depend linearly on earlier ones, we obtain a finite or countable orthonormal family (e_k) whose linear span is dense in V.

For each k the sequence $(\langle x_j, e_k\rangle)_j$ is bounded by C, so by a diagonal argument we may pass to a subsequence, still denoted (x_j), such that $\langle x_j, e_k\rangle$ converges to a number a_k for every k. For every N, Bessel's inequality gives $\sum_{k\le N} |\langle x_j, e_k\rangle|^2 \le C^2$; letting $j \to \infty$ and then $N \to \infty$, we get $\sum_k |a_k|^2 \le C^2$. Since $\|\sum_{N<k\le N'} a_k e_k\|^2 = \sum_{N<k\le N'} |a_k|^2$ by Pythagoras' theorem, the partial sums of the series $x = \sum_k a_k e_k$ form a Cauchy sequence; the series therefore converges in the closed, hence complete, subspace V, and $\|x\|^2 = \sum_k |a_k|^2 \le C^2$.

By the continuity of the inner product and the orthonormality of the e_k, we have $\langle x, e_k\rangle = a_k = \lim_j \langle x_j, e_k\rangle$ for every k; hence $\langle x_j, v\rangle \to \langle x, v\rangle$ for every v in the linear span of the e_k. If $v \in V$ and v' lies in that span with $\|v - v'\| < \delta$, then $|\langle x_j - x, v\rangle| \le |\langle x_j - x, v'\rangle| + 2C\delta$, so the convergence extends to all $v \in V$. For an arbitrary $v \in H$, write $v = v_1 + v_2$ with $v_1 \in V$ and $v_2 \in V^\perp$ by Proposition 11.1.1; since x_j and x lie in V, we get $\langle x_j - x, v\rangle = \langle x_j - x, v_1\rangle \to 0$.

For the last assertion we may assume that $x \neq 0$. By the Cauchy–Schwarz inequality, $\|x\|^2 = \lim_j \langle x_j, x\rangle \le \varliminf_j \|x_j\|\,\|x\|$, and we divide by $\|x\|$. □

Differential operators are defined only on subspaces of L^2 spaces, so we need the following notions. Let H_0 and H_1 be Hilbert spaces. An *operator* T from H_0 to H_1 is a linear map $T\colon \operatorname{Dom} T \to H_1$ defined on a linear subspace $\operatorname{Dom} T \subset H_0$,

its *domain*. We write $\ker T$ for its kernel, $\operatorname{Ran} T = T(\operatorname{Dom} T)$ for its range, and $\operatorname{Gr} T = \{(x, Tx) : x \in \operatorname{Dom} T\}$ for its graph, a subspace of the Hilbert space $H_0 \oplus H_1$ with the inner product $\langle (x, y), (x', y') \rangle = \langle x, x' \rangle + \langle y, y' \rangle$. The operator T is *densely defined* if $\operatorname{Dom} T$ is dense in H_0, and *closed* if $\operatorname{Gr} T$ is closed; equivalently, $x_j \in \operatorname{Dom} T$, $x_j \to x$ and $Tx_j \to y$ imply $x \in \operatorname{Dom} T$ and $Tx = y$. In particular, the kernel of a closed operator is a closed subspace.

If T is densely defined, its *Hilbert adjoint* T^*, an operator from H_1 to H_0, is defined as follows: $y \in \operatorname{Dom} T^*$ if the linear functional $x \mapsto \langle Tx, y \rangle$ on $\operatorname{Dom} T$ is bounded with respect to the norm of H_0, and then $T^* y$ is the vector determined by

$$\langle Tx, y \rangle = \langle x, T^* y \rangle \qquad (x \in \operatorname{Dom} T).$$

This vector is well defined: since $\operatorname{Dom} T$ is dense, the functional extends uniquely to a bounded linear functional on H_0, which Proposition 11.1.1 represents by a unique vector.

Lemma 11.1.3 *Let T be a densely defined operator from H_0 to H_1. Then T^* is closed and $\ker T^* = (\operatorname{Ran} T)^\perp$. If T is also closed, then T^* is densely defined and $T^{**} = T$.*

Proof If $y_j \in \operatorname{Dom} T^*$, $y_j \to y$ and $T^* y_j \to w$, then for every $x \in \operatorname{Dom} T$

$$\langle Tx, y \rangle = \lim_j \langle Tx, y_j \rangle = \lim_j \langle x, T^* y_j \rangle = \langle x, w \rangle;$$

hence $y \in \operatorname{Dom} T^*$ and $T^* y = w$, so T^* is closed. A vector $y \in H_1$ is orthogonal to $\operatorname{Ran} T$ exactly when the functional $x \mapsto \langle Tx, y \rangle$ vanishes, that is, when $y \in \operatorname{Dom} T^*$ and $T^* y = 0$. This proves $\ker T^* = (\operatorname{Ran} T)^\perp$. In the same way, $(a, b) \in H_0 \oplus H_1$ is orthogonal to $\operatorname{Gr} T$ if and only if $\langle Tx, b \rangle = \langle x, -a \rangle$ for all $x \in \operatorname{Dom} T$, so

$$(\operatorname{Gr} T)^\perp = \{(-T^* y, y) : y \in \operatorname{Dom} T^*\}.$$

Now let T be closed, and let $v \in H_1$ be orthogonal to $\operatorname{Dom} T^*$. Since $\operatorname{Gr} T$ is closed, Proposition 11.1.1 and the description of $(\operatorname{Gr} T)^\perp$ decompose $(0, v)$ as $(x, Tx) + (-T^* y, y)$ with $x \in \operatorname{Dom} T$ and $y \in \operatorname{Dom} T^*$. Thus $x = T^* y$ and $v = Tx + y$. Since v is orthogonal to $y \in \operatorname{Dom} T^*$, we obtain

$$0 = \langle v, y \rangle = \langle Tx, y \rangle + \|y\|^2 = \langle x, T^* y \rangle + \|y\|^2 = \|x\|^2 + \|y\|^2,$$

so $x = y = 0$ and $v = 0$. Hence $\operatorname{Dom} T^*$ is dense, and T^{**} is defined. Let $J : H_1 \oplus H_0 \to H_0 \oplus H_1$ and $J' : H_0 \oplus H_1 \to H_1 \oplus H_0$ be the unitary maps $J(b, a) = (-a, b)$ and $J'(a, b) = (-b, a)$. The description of the orthogonal complement of a graph, applied to T and to the densely defined operator T^*, reads $(\operatorname{Gr} T)^\perp = J(\operatorname{Gr} T^*)$ and $(\operatorname{Gr} T^*)^\perp = J'(\operatorname{Gr} T^{**})$. For a closed subspace M we have $(M^\perp)^\perp = M$ by Proposition 11.1.1, and unitary maps carry orthogonal complements to orthogonal complements. Therefore

$$\operatorname{Gr} T = \big((\operatorname{Gr} T)^\perp\big)^\perp = J\big((\operatorname{Gr} T^*)^\perp\big) = JJ'(\operatorname{Gr} T^{**}) = \operatorname{Gr} T^{**},$$

because $JJ' = -\mathrm{id}$ and graphs are linear subspaces. Thus $T^{**} = T$. □

The next theorem, the abstract form of Hörmander's method, produces a solution of $Tu = g$ from an inequality for the adjoint operator. The second operator S allows us to require the inequality only on $\operatorname{Dom} T^* \cap \operatorname{Dom} S$; in the application in Section 11.3, smooth compactly supported forms are dense in this space, and for them the inequality follows from an integration by parts.

Theorem 11.1.4 *Let H_0, H_1, H_2 be Hilbert spaces, let T from H_0 to H_1 be a closed densely defined operator, and let S from H_1 to H_2 be a closed operator with $\operatorname{Ran} T \subset \ker S$. Let $g \in \ker S$ and $0 \le M < \infty$, and assume that*

$$|\langle g, f\rangle|^2 \le M\big(\|T^* f\|^2 + \|Sf\|^2\big) \qquad (f \in \operatorname{Dom} T^* \cap \operatorname{Dom} S).$$

Then the equation $Tu = g$ has a solution $u \in \operatorname{Dom} T$ with $\|u\|^2 \le M$. Among all solutions there is a unique one of smallest norm, and it is orthogonal to $\ker T$.

Proof Since S is closed, $\ker S$ is a closed subspace, and $H_1 = \ker S \oplus (\ker S)^\perp$ by Proposition 11.1.1. Since $\operatorname{Ran} T \subset \ker S$, Lemma 11.1.3 gives $(\ker S)^\perp \subset (\operatorname{Ran} T)^\perp = \ker T^*$. Let $f \in \operatorname{Dom} T^*$, and write $f = f_0 + f_1$ with $f_0 \in \ker S$ and $f_1 \in (\ker S)^\perp$. Then $f_1 \in \operatorname{Dom} T^*$ and $T^* f_1 = 0$, so $f_0 \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$, $T^* f_0 = T^* f$ and $S f_0 = 0$. Moreover $\langle g, f\rangle = \langle g, f_0\rangle$, because $g \in \ker S$. The hypothesis applied to f_0 therefore gives

$$|\langle g, f\rangle| \le \sqrt{M}\, \|T^* f\| \qquad (f \in \operatorname{Dom} T^*).$$

Consequently the map $\lambda\colon T^* f \mapsto \langle g, f\rangle$ is well defined on $\operatorname{Ran} T^*$: if $T^* f = T^* f'$, then $|\langle g, f - f'\rangle| \le \sqrt{M}\, \|T^*(f - f')\| = 0$. It is conjugate-linear and bounded by $\sqrt{M}$, hence uniformly continuous, so it extends uniquely to a conjugate-linear functional $\tilde\lambda$ on the closed subspace $W = \overline{\operatorname{Ran} T^*}$, again bounded by $\sqrt{M}$. If P_W is the orthogonal projection onto W, then $x \mapsto \tilde\lambda(P_W x)$ is a conjugate-linear functional on H_0 of norm at most $\sqrt{M}$, because $\|P_W x\| \le \|x\|$. By the representation of conjugate-linear functionals in Proposition 11.1.1, there is $u \in H_0$ with $\|u\| \le \sqrt{M}$ and

$$\langle u, T^* f\rangle = \langle g, f\rangle \qquad (f \in \operatorname{Dom} T^*).$$

Taking complex conjugates, we get $\langle T^* f, u\rangle = \langle f, g\rangle$ for all $f \in \operatorname{Dom} T^*$. Thus the functional $f \mapsto \langle T^* f, u\rangle$ is bounded on $\operatorname{Dom} T^*$, so $u \in \operatorname{Dom} T^{**}$ and $T^{**} u = g$. Since $T^{**} = T$ by Lemma 11.1.3, we conclude that $u \in \operatorname{Dom} T$ and $Tu = g$.

The solutions of $Tu = g$ form the affine subspace $u + \ker T$, which is closed because $\ker T$ is closed. If P denotes the orthogonal projection onto $\ker T$, then $\|u + k\|^2 = \|u - Pu\|^2 + \|Pu + k\|^2$ for $k \in \ker T$. Hence $u - Pu$ is the unique solution of smallest norm; it is orthogonal to $\ker T$, and its norm is at most $\|u\| \le \sqrt{M}$. □

11.2 Pseudoconvexity and boundary distance

We shall prove Hörmander's estimate on open sets that are exhausted by the compact sublevel sets of a continuous plurisubharmonic function. These pseudoconvex open

sets are characterized by the plurisubharmonicity of minus the logarithm of the boundary distance, and they carry smooth strictly plurisubharmonic exhaustion functions, which Section 11.3 uses to localize the estimate. For $r > 0$ we write $\Delta_r = \{t \in \mathbb{C} : |t| < r\}$ for the open disc of radius r in $\mathbb{C}$, and $\Delta = \Delta_1$ for the unit disc; in this chapter the letter Δ always denotes a disc, never the Laplacian.

Definition 11.2.1 Let $n \geq 1$ and let $\Omega \subset \mathbb{C}^n$ be open. An *exhaustion function* of Ω is a function $\rho\colon \Omega \to \mathbb{R}$ such that the sublevel set $\{z \in \Omega : \rho(z) \leq c\}$ is compact for every $c \in \mathbb{R}$. The open set Ω is *pseudoconvex* if it has a continuous plurisubharmonic exhaustion function.

A continuous exhaustion function attains its minimum on each nonempty sublevel set, so it is bounded below; adding a constant, we may take it nonnegative. An open set Ω is pseudoconvex if and only if each of its connected components is. Indeed, the restriction of a continuous plurisubharmonic exhaustion function of Ω to a component is a continuous plurisubharmonic exhaustion function of that component, because the component is open and closed in Ω. Conversely, the components of Ω are open, and each of them contains a point with rational real coordinates, so there are at most countably many of them, say $\Omega_1, \Omega_2, \ldots$. If $\rho_k \geq 0$ is a continuous plurisubharmonic exhaustion function of Ω_k, then the function equal to $\rho_k + k$ on Ω_k is a continuous plurisubharmonic exhaustion function of Ω: it is continuous and plurisubharmonic because the components are open, and its sublevel set $\{\leq c\}$ is the finite union of the compact sets $\{\rho_k \leq c - k\}$, $k \leq c$.

Recall from Proposition 8.2.2 that a real function u of class C^2 on an open set is plurisubharmonic if and only if its complex Hessian matrix $(u_{j\bar{k}})$ is positive semidefinite at every point. Here, as in Section 8.2, a function on an open set is plurisubharmonic if it is plurisubharmonic on each connected component. Recall also, from the discussion following Proposition 8.2.2, that a function u is strictly plurisubharmonic if every point has a neighborhood on which $u - \varepsilon|z|^2$ is plurisubharmonic for some $\varepsilon > 0$. For smooth u this means that $(u_{j\bar{k}})$ is positive definite at every point, and then its smallest eigenvalue is a positive continuous function.

To recognize plurisubharmonic functions we use the following criterion, in which the harmonic majorants in the submean inequality are restricted to real parts of polynomials.

Lemma 11.2.2 *Let $U \subset \mathbb{C}^N$ be open and let $v\colon U \to [-\infty, \infty)$ be upper semicontinuous. Assume that for all $z_0 \in U$ and $a \in \mathbb{C}^N$ with $\{z_0 + ta : |t| \leq 1\} \subset U$, and for every polynomial P in one complex variable with $v(z_0 + ta) \leq \operatorname{Re} P(t)$ whenever $|t| = 1$, we have $v(z_0) \leq \operatorname{Re} P(0)$. Then*

$$v(z_0) \leq \frac{1}{2\pi} \int_0^{2\pi} v(z_0 + \mathrm{e}^{\mathrm{i}\theta} a)\, \mathrm{d}\theta$$

for all such z_0 and a.

Proof Fix z_0 and a, and put $g(\theta) = v(z_0 + \mathrm{e}^{\mathrm{i}\theta} a)$, an upper semicontinuous function on the circle. By Lemma 3.1.2 there are continuous real functions g_m decreasing to g

whose integrals decrease to the integral of g, which may be $-\infty$. Fix m and $\delta > 0$. We approximate g_m uniformly by real trigonometric polynomials, using the Fejér kernel

$$F_N(\theta) = \frac{1}{N}\Big|\sum_{k=0}^{N-1} \mathrm{e}^{\mathrm{i}k\theta}\Big|^2 = \sum_{|k|<N} \Big(1 - \frac{|k|}{N}\Big)\mathrm{e}^{\mathrm{i}k\theta}, \qquad N \geq 1.$$

The two expressions agree because the product $\sum_{k,l=0}^{N-1} \mathrm{e}^{\mathrm{i}(k-l)\theta}$ contains exactly $N-|h|$ pairs (k,l) with $k-l=h$, for each integer h with $|h| < N$. By the first expression $F_N \geq 0$, and summing the geometric series gives $F_N(\theta) = \sin^2(N\theta/2)/(N\sin^2(\theta/2)) \leq 1/(N\sin^2(\theta/2))$ for $0 < |\theta| \leq \pi$; by the second, F_N has integral one with respect to the normalized measure $\mathrm{d}\theta/(2\pi)$ on $[-\pi,\pi]$.

Let $q_N = g_m * F_N$ be the convolution with respect to $\mathrm{d}\theta/(2\pi)$, that is, $q_N(\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} g_m(\theta-\tau)F_N(\tau)\,\mathrm{d}\tau$. For $0 < \delta' \leq \pi$ let $\omega(\delta')$ be the largest oscillation of g_m over arcs of length at most δ', which tends to zero with δ' by uniform continuity. Splitting the integral at $|\tau| = \delta'$, we obtain

$$|q_N(\theta) - g_m(\theta)| \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |g_m(\theta-\tau) - g_m(\theta)|F_N(\tau)\,\mathrm{d}\tau \leq \omega(\delta') + \frac{2\max|g_m|}{N\sin^2(\delta'/2)},$$

so $q_N \to g_m$ uniformly as $N \to \infty$. Inserting the second expression for F_N and substituting $\sigma = \theta - \tau$, we see that $q_N(\theta) = \sum_{|k|<N} c_k \mathrm{e}^{\mathrm{i}k\theta}$ with $c_k = (1 - |k|/N)\frac{1}{2\pi}\int_{-\pi}^{\pi} g_m(\sigma)\mathrm{e}^{-\mathrm{i}k\sigma}\,\mathrm{d}\sigma$. Since g_m is real, $c_{-k} = \bar{c}_k$, so $q_N(\theta) = \operatorname{Re} P(\mathrm{e}^{\mathrm{i}\theta})$ for the polynomial $P(t) = c_0 + 2\sum_{0<k<N} c_k t^k$, and $\operatorname{Re} P(0) = c_0$ is the mean of g_m. Choose N with $|q_N - g_m| < \delta$. Then $\operatorname{Re}(P(t) + \delta) \geq g_m \geq g$ on the circle, and the hypothesis gives

$$v(z_0) \leq c_0 + \delta = \frac{1}{2\pi}\int_0^{2\pi} g_m\,\mathrm{d}\theta + \delta.$$

Letting $\delta \to 0$ and then $m \to \infty$, we obtain the asserted inequality. □

Richberg's lemma approximates a continuous strictly plurisubharmonic function from above by smooth strictly plurisubharmonic functions, at the cost of half of the lower bound for the complex Hessian. Convolution does this locally; the local pieces are glued by a regularized maximum.

Lemma 11.2.3 (Richberg) *Let $\Omega \subset \mathbb{C}^n$ be open, let u be a continuous function on Ω, and let $a > 0$ be such that $u - a|z|^2$ is plurisubharmonic. Then for every continuous function $\varepsilon\colon \Omega \to (0,\infty)$ there is a smooth function v on Ω with*

$$u \leq v \leq u + \varepsilon, \qquad v - \tfrac{a}{2}|z|^2 \in \mathrm{PSH}(\Omega).$$

Proof *Step 1: local approximations.* Let $(K_m)_{m\geq 1}$ be a compact exhaustion of Ω (Section 1.1), so that $K_m \subset \operatorname{Int} K_{m+1}$ and $\bigcup_m \operatorname{Int} K_m = \Omega$, and put $K_0 = K_{-1} = \varnothing$. For each $m \geq 1$, the compact set $K_m \setminus \operatorname{Int} K_{m-1}$ is covered by finitely many balls $B(c, r/3)$ with $c \in K_m \setminus \operatorname{Int} K_{m-1}$ and $B(c,r) \Subset \operatorname{Int} K_{m+1} \setminus K_{m-2}$. Collecting these balls for all m, we obtain a sequence of balls $B_i = B(c_i, r_i) \Subset \Omega$ whose inner balls

$B(c_i, r_i/3)$ cover Ω. The family (B_i) is locally finite: a compact subset of Ω lies in some K_m, and a ball chosen at a stage $m' \geq m+2$ lies in the complement of $K_{m'-2} \supset K_m$, so the compact set meets only the finitely many balls chosen at the stages $m' \leq m+1$.

The function $u = (u - a|z|^2) + a|z|^2$ is continuous and plurisubharmonic by Proposition 8.2.5. Let ρ_h, $h > 0$, be the radial mollifiers of Section 1.4. By Proposition 8.2.2(2), we have $u * \rho_h \geq u$, and $(u - a|z|^2) * \rho_h = u * \rho_h - a|z|^2 * \rho_h$ is smooth and plurisubharmonic, hence has positive semidefinite complex Hessian by Proposition 8.2.2(1); by Lemma 1.4.1, $u * \rho_h \to u$ locally uniformly as $h \to 0$. Since ρ_h is radial, $\int y \rho_h(y)\, \mathrm{d}V(y) = 0$, and expanding $|z-y|^2 = |z|^2 - 2\,\mathrm{Re}\langle z, y\rangle + |y|^2$ gives $|z|^2 * \rho_h = |z|^2 + \int |y|^2 \rho_h(y)\, \mathrm{d}V(y)$, whose complex Hessian matrix is I. Hence the complex Hessian matrix of $u * \rho_h$ is at least aI.

For each i choose δ_i with $0 < \delta_i < a/2$ and $\delta_i r_i^2 < \min_{\overline{B}_i} \varepsilon$, put $\eta_i = \delta_i r_i^2/6$, and choose $h_i > 0$ so small that $u * \rho_{h_i}$ is defined near $\overline{B}_i$ and satisfies $0 \leq u * \rho_{h_i} - u < \eta_i$ there. Define

$$u_i = u * \rho_{h_i} + \delta_i\big((2r_i/3)^2 - |z - c_i|^2\big) \quad \text{on } B_i.$$

The quadratic term is at least $\delta_i r_i^2/3 = 2\eta_i$ on the inner ball $B(c_i, r_i/3)$, is less than $-2\eta_i$ where $|z - c_i| > \sqrt{7}\, r_i/3$, and is at most $4\delta_i r_i^2/9$ everywhere. Therefore

$$\begin{aligned} &u_i > u + \eta_i && \text{on } B(c_i, r_i/3), \\ &u_i < u - \eta_i && \text{where } \sqrt{7}\, r_i/3 < |z - c_i| < r_i, \\ &u_i + \eta_i < u + \varepsilon && \text{on } B_i, \end{aligned}$$

the last because $2\eta_i + 4\delta_i r_i^2/9 = 7\delta_i r_i^2/9 < \varepsilon$ on B_i. The complex Hessian matrix of u_i is at least $(a - \delta_i)I \geq (a/2)I$.

Step 2: a regularized maximum. Fix an even smooth function $\theta \geq 0$ on $\mathbb{R}$ with support in $(-1, 1)$ and integral one, for instance the mollifier $\rho_{1/2}$ of Section 1.4 in dimension $d = 1$. For an integer $p \geq 1$, positive numbers $\eta_1, \ldots, \eta_p$ and $t = (t_1, \ldots, t_p) \in \mathbb{R}^p$, define

$$M_\eta(t) = \int_{\mathbb{R}^p} \max_i (t_i + \eta_i x_i) \prod_i \theta(x_i)\, \mathrm{d}x.$$

The substitution $y_i = \eta_i x_i$ and the evenness of θ show that M_η is the convolution of the continuous function $t \mapsto \max_i t_i$ with the smooth compactly supported function $y \mapsto \prod_i \eta_i^{-1} \theta(y_i/\eta_i)$, so M_η is smooth by differentiation under the integral sign (Section 1.4). It has the following properties.

(1) M_η is convex and nondecreasing in each variable, since for fixed x the function $t \mapsto \max_i (t_i + \eta_i x_i)$ is.

(2) $\max_i t_i \leq M_\eta(t) \leq \max_i (t_i + \eta_i)$. The upper bound holds because $|x_i| < 1$ on the support of the integrand. For the lower bound, integrate $\max_j (t_j + \eta_j x_j) \geq t_i + \eta_i x_i$ and use $\int x_i \theta(x_i)\, \mathrm{d}x_i = 0$.

(3) $M_\eta(t_1+c,\dots,t_p+c) = M_\eta(t)+c$ for $c\in\mathbb{R}$. Differentiating in c, together with the first property, gives $\partial M_\eta/\partial t_i \ge 0$ and $\sum_i \partial M_\eta/\partial t_i = 1$.
(4) If $t_i+\eta_i \le \max_{j\ne i}(t_j-\eta_j)$ for some i, then deleting the variable t_i and the parameter η_i does not change the value of M_η. Indeed, for $|x_j|<1$ we have $t_i+\eta_i x_i < t_i+\eta_i \le \max_{j\ne i}(t_j-\eta_j) < \max_{j\ne i}(t_j+\eta_j x_j)$, so the i-th term never realizes the maximum, and integrating out x_i removes it.

For smooth real functions $w_1,\dots,w_p$, the chain rule gives

$$\partial_{z_j}\partial_{\bar z_k} M_\eta(w) = \sum_i \frac{\partial M_\eta}{\partial t_i}(w)\,\partial_{z_j}\partial_{\bar z_k} w_i + \sum_{i,l}\frac{\partial^2 M_\eta}{\partial t_i\partial t_l}(w)\,\partial_{z_j} w_i\,\overline{\partial_{z_k} w_l}.$$

The real Hessian matrix $G = (\partial^2 M_\eta/\partial t_i\partial t_l)$ is positive semidefinite, because the second derivative of the convex function $s\mapsto M_\eta(t+s\tau)$ is nonnegative for every $\tau\in\mathbb{R}^p$. Hence the matrix given by the second sum is positive semidefinite: for $b\in\mathbb{C}^n$ its quadratic form equals $\sum_{i,l} G_{il}\zeta_i\bar\zeta_l$ with $\zeta_i = \sum_j \partial_{z_j} w_i\, b_j$, and writing $\zeta = \xi + \mathrm{i}\xi'$ with $\xi,\xi'\in\mathbb{R}^p$, this is $\xi^{\mathsf T}G\xi + \xi'^{\mathsf T}G\xi' \ge 0$ by the symmetry of G. Therefore, by the third property, if each w_i has complex Hessian matrix at least $(a/2)I$, so does $M_\eta(w_1,\dots,w_p)$.

Step 3: gluing. For $z\in\Omega$ let $I(z) = \{i : z\in B_i\}$, a finite nonempty set, and define $v(z) = M_\eta\big((u_i(z))_{i\in I(z)}\big)$ with the parameters $(\eta_i)_{i\in I(z)}$. We show that v is smooth near a fixed point z_0. Let $I_0 = \{i : z_0\in B_i\}$ and $I_1 = \{i : z_0\in\partial B_i\}$, finite sets by local finiteness, and choose j with $z_0\in B(c_j, r_j/3)$. There is a neighborhood W of z_0 that is contained in $B(c_j,r_j/3)$ and in B_i for $i\in I_0$, that meets $\overline{B}_i$ only for $i\in I_0\cup I_1$, and whose intersection with B_i, for $i\in I_1$, lies in the shell $\sqrt{7}\,r_i/3 < |z-c_i| < r_i$. For $z\in W$ we have $I_0\subset I(z)\subset I_0\cup I_1$, $j\in I_0$, and every $i\in I(z)\cap I_1$ satisfies

$$u_i(z)+\eta_i < u(z) < u_j(z)-\eta_j.$$

By the fourth property of M_η, applied to these indices one at a time, $v = M_\eta\big((u_i)_{i\in I_0}\big)$ on W, which is smooth. Thus v is smooth on Ω, and by Step 2 its complex Hessian matrix is at least $(a/2)I$; hence $v - \frac{a}{2}|z|^2$ is smooth with positive semidefinite complex Hessian, and so it is plurisubharmonic by Proposition 8.2.2(1). Finally, every z lies in some inner ball $B(c_j,r_j/3)$, and the second property of M_η gives $v(z)\ge u_j(z) > u(z)$ and $v(z)\le \max_{i\in I(z)}(u_i(z)+\eta_i) < u(z)+\varepsilon(z)$. □

Theorem 11.2.4 *Let $n\ge 1$ and let $\Omega\subsetneq\mathbb{C}^n$ be an open set. If Ω is pseudoconvex, then $-\log\operatorname{dist}(z,\mathbb{C}^n\setminus\Omega)$ is a continuous plurisubharmonic function on Ω. Conversely, if this function is plurisubharmonic, then Ω is pseudoconvex. Every pseudoconvex open set in $\mathbb{C}^n$ has a nonnegative smooth strictly plurisubharmonic exhaustion function.*

Proof Steps 1 and 2 deduce the plurisubharmonicity of $-\log\operatorname{dist}(\cdot,\mathbb{C}^n\setminus\Omega)$ from a submean inequality for the distance to the boundary in a fixed direction; Step 3 uses this function to construct a continuous exhaustion function, which Lemma 11.2.3 smooths.

Step 1: distances in a fixed direction. Let Ω be pseudoconvex, with a continuous plurisubharmonic exhaustion function ρ. For $z \in \Omega$ and $\xi \in \mathbb{C}^n$ put

$$\delta(z,\xi) = \sup\{r > 0 : z + \xi\Delta_r \subset \Omega\} \in (0,\infty],$$

so that $\delta(z,0) = \infty$. The function δ is lower semicontinuous on $\Omega\times\mathbb{C}^n$: if $r < \delta(z,\xi)$, then the compact disc $z + \xi\overline{\Delta}_r$ lies in Ω, hence so does $z' + \xi'\overline{\Delta}_r$ for all (z',ξ') close to (z,ξ), and $\delta(z',\xi') \geq r$ for these points. Thus $V = -\log\delta$ is an upper semicontinuous function on $\Omega\times\mathbb{C}^n$ with values in $[-\infty,\infty)$. We verify the hypothesis of Lemma 11.2.2 for V on the open set $\Omega\times\mathbb{C}^n \subset \mathbb{C}^{2n}$.

Let (z_0+ta, ξ_0+tb), $|t| \leq 1$, be a closed disc in $\Omega\times\mathbb{C}^n$, and let P be a polynomial with $V(z_0+ta,\xi_0+tb) \leq \operatorname{Re} P(t)$ for $|t| = 1$, that is, $\delta(z_0+ta,\xi_0+tb) \geq |e^{-P(t)}|$. Consider the holomorphic map

$$h(t,w) = z_0 + ta + we^{-P(t)}(\xi_0 + tb), \qquad (t,w) \in \mathbb{C}^2.$$

We have $h(t,0) = z_0 + ta \in \Omega$ for $|t| \leq 1$; and $h(t,w) \in \Omega$ for $|t| = 1$ and $|w| < 1$, because then $|we^{-P(t)}| < \delta(z_0+ta,\xi_0+tb)$. Let R be the supremum of the numbers $r \geq 0$ with $h(\overline{\Delta}\times\overline{\Delta}_r) \subset \Omega$. Since $h(\overline{\Delta}\times\{0\})$ is a compact subset of Ω and h is continuous, $R > 0$. We show that $R \geq 1$.

Suppose that $R < 1$. Then $h(\partial\Delta\times\overline{\Delta}_R)$ is a compact subset of Ω; let c be the maximum of ρ on it. For $|w| < R$ we have $h(\overline{\Delta}\times\{w\}) \subset \Omega$, because every radius smaller than an admissible one is admissible; since this set is compact and h is continuous, there is $r' > 1$ such that $t \mapsto h(t,w)$ maps the disc $\Delta_{r'}$ into Ω, and hence into one connected component of Ω. The function $t \mapsto \rho(h(t,w))$ is finite, hence subharmonic on $\Delta_{r'}$ by Proposition 8.2.3, and it is at most c on $\partial\Delta$. By upper semicontinuity, $\overline{\lim}_{t\to\zeta}(\rho(h(t,w)) - c) \leq 0$ for every $\zeta \in \partial\Delta$, and the maximum principle of Proposition 3.1.4 on the disc Δ gives $\rho(h(t,w)) \leq c$ for $|t| < 1$, hence for $|t| \leq 1$. Thus $h(\overline{\Delta}\times\Delta_R)$ lies in the compact set $\{\rho \leq c\}$, and so does its closure $h(\overline{\Delta}\times\overline{\Delta}_R)$. This compact subset of Ω has positive distance from $\mathbb{C}^n \setminus \Omega$, and the uniform continuity of h on compact sets gives $h(\overline{\Delta}\times\overline{\Delta}_{R'}) \subset \Omega$ for some $R' > R$, contradicting the definition of R.

Hence $R \geq 1$. In particular $h(0,w) = z_0 + we^{-P(0)}\xi_0 \in \Omega$ for $|w| < 1$, which means that $\delta(z_0,\xi_0) \geq e^{-\operatorname{Re} P(0)}$, that is, $V(z_0,\xi_0) \leq \operatorname{Re} P(0)$. By Lemma 11.2.2, V satisfies the circle submean inequality on every closed complex disc in $\Omega\times\mathbb{C}^n$.

Step 2: the boundary distance. Now let $\Omega \neq \mathbb{C}^n$ and put $d(z) = \operatorname{dist}(z, \mathbb{C}^n \setminus \Omega)$. For a unit vector ξ we have $\delta(z,\xi) \geq d(z)$, because $z + \xi\Delta_{d(z)}$ lies in the ball $B(z,d(z)) \subset \Omega$. Equality holds if ξ points from z to a nearest point of the closed set $\mathbb{C}^n \setminus \Omega$, since that point, $z + d(z)\xi$, lies in $z + \xi\Delta_r$ for every $r > d(z)$. Hence

$$-\log d(z) = \max_{|\xi|=1} V(z,\xi).$$

For fixed ξ, the function $z \mapsto V(z,\xi)$ satisfies the circle submean inequality on every closed disc $\{z_0 + ta : |t| \leq 1\} \subset \Omega$, by Step 1 with $b = 0$, and it is bounded above by $-\log d$. Therefore

$$V(z_0,\xi) \le \frac{1}{2\pi}\int_0^{2\pi} V(z_0+\mathrm{e}^{\mathrm{i}\theta}a,\xi)\,\mathrm{d}\theta \le \frac{1}{2\pi}\int_0^{2\pi} -\log d(z_0+\mathrm{e}^{\mathrm{i}\theta}a)\,\mathrm{d}\theta,$$

and taking the maximum over ξ gives the submean inequality for $-\log d$. Since d is positive and 1-Lipschitz, $-\log d$ is continuous. Its restriction to each connected component of each affine complex-line section of Ω is therefore subharmonic (Definition 3.1.1): at a point $z_0 + t_0 a$ of such a section, the circle means over $\{z_0 + (t_0 + r\mathrm{e}^{\mathrm{i}\theta})a\}$ dominate the value for all small $r > 0$. Hence $-\log d$, which is finite, is plurisubharmonic on each connected component of Ω by Definition 8.1.1, that is, on Ω.

Step 3: smooth exhaustion functions. Let $\Omega \ne \mathbb{C}^n$ be an open set on which $-\log d$ is plurisubharmonic, and put

$$u(z) = |z|^2 + \max\{-\log d(z), 0\}.$$

Then u is continuous, $u - |z|^2$ is plurisubharmonic by Proposition 8.2.5, and u is an exhaustion function of Ω: the set $\{u \le c\}$ is contained in the ball $\{|z|^2 \le c\}$ and in $\{d \ge \mathrm{e}^{-c}\}$, so it is a closed and bounded subset of $\mathbb{C}^n$ contained in Ω, hence compact. In particular, Ω is pseudoconvex, which proves the converse statement.

By Lemma 11.2.3 with $a = 1$ and $\varepsilon = 1$, there is a smooth function s with $u \le s \le u + 1$ such that $s - |z|^2/2$ is plurisubharmonic. Then $s \ge u \ge 0$, and the complex Hessian matrix of s is at least $I/2$ by Proposition 8.2.2(1) applied to $s - |z|^2/2$, so that s is strictly plurisubharmonic. Moreover, since $s \ge u$, the sublevel set $\{s \le c\}$ is a closed subset of the compact set $\{u \le c\}$. Thus s is a nonnegative smooth strictly plurisubharmonic exhaustion function. If $\Omega \ne \mathbb{C}^n$ is pseudoconvex, Step 2 shows that $-\log d$ is plurisubharmonic, so this construction applies; for $\Omega = \mathbb{C}^n$ we take $s = |z|^2$. □

Corollary 11.2.5 *Products of pseudoconvex open sets are pseudoconvex. If s is a smooth plurisubharmonic exhaustion function of an open set $\Omega \subset \mathbb{C}^n$, then each sublevel set $\{s < c\}$, $c \in \mathbb{R}$, is pseudoconvex and relatively compact in Ω. In particular, every pseudoconvex open set $\Omega \subset \mathbb{C}^n$ is the union of an increasing sequence of pseudoconvex open sets $\Omega_j \Subset \Omega$.*

Proof Let $\Omega \subset \mathbb{C}^n$ and $\Omega' \subset \mathbb{C}^m$ be pseudoconvex, with nonnegative continuous plurisubharmonic exhaustion functions ρ and ρ'. The function $(z,w) \mapsto \rho(z) + \rho'(w)$ is continuous and plurisubharmonic on $\Omega \times \Omega'$, by Proposition 8.2.3 applied to the two projections and by Proposition 8.2.5. Its sublevel set $\{\le c\}$ is a closed subset of the compact set $\{\rho \le c\} \times \{\rho' \le c\}$, so $\Omega \times \Omega'$ is pseudoconvex.

Next let s be as in the statement and $c \in \mathbb{R}$. The function $x \mapsto 1/(c-x)$ is convex and increasing on $(-\infty, c)$, so the complex Hessian matrix of $1/(c-s)$ on the open set $\{s < c\}$, namely $(c-s)^{-2}(s_{j\bar{k}}) + 2(c-s)^{-3}\big(\partial_{z_j} s\,\overline{\partial_{z_k} s}\big)$ by the chain rule, is positive semidefinite. Hence $|z|^2 + 1/(c-s)$ is a smooth plurisubharmonic function on $\{s < c\}$, by Proposition 8.2.2(1). For $C \le 0$ its sublevel set $\{\le C\}$ is empty. For $C > 0$ this sublevel set is closed in $\{s < c\}$ and contained in the set $\{s \le c - 1/C\}$, which is a compact subset of $\{s < c\}$; hence it is compact, and

$\{s < c\}$ is pseudoconvex. Moreover $\{s < c\}$ is contained in the compact set $\{s \le c\}$, so it is relatively compact in Ω.

Finally, if Ω is pseudoconvex, Theorem 11.2.4 provides a smooth plurisubharmonic exhaustion function s, and the sets $\Omega_j = \{s < j\}$, $j \ge 1$, are pseudoconvex, relatively compact in Ω, increasing, and have union Ω. (Some of them may be empty or disconnected; Definition 11.2.1 allows arbitrary open sets.) □

With the boundary distance at hand, the invariant exhaustion function required in Kiselman's minimum principle exists automatically on pseudoconvex domains.

Corollary 11.2.6 *Let $m, k \ge 1$, and let $\Omega \subset \mathbb{C}^m \times \mathbb{C}^k$ be a pseudoconvex domain that is invariant under imaginary translations and whose nonempty real fibers V_z are convex, in the notation of Section 8.5. If $u \in \mathrm{PSH}(\Omega)$ is invariant under imaginary translations, then the function $v(z) = \inf_{t \in V_z} u(z,t)$ is plurisubharmonic or identically $-\infty$ on $p(\Omega)$.*

Proof We construct a function ψ with the properties required in Theorem 8.5.1; the corollary then follows from that theorem. If $\Omega = \mathbb{C}^{m+k}$, let $\psi(z,w) = |z|^2 + |\operatorname{Re} w|^2$. Otherwise let $d = \operatorname{dist}(\cdot, \mathbb{C}^{m+k} \setminus \Omega)$ and put

$$\psi(z,w) = |z|^2 + |\operatorname{Re} w|^2 + \max\{-\log d(z,w), 0\}.$$

Imaginary translations are isometries of $\mathbb{C}^{m+k}$ that preserve Ω and its complement, so d and ψ are invariant under them. The function ψ is plurisubharmonic by Proposition 8.2.5: $-\log d$ is plurisubharmonic by Theorem 11.2.4, and $|z|^2 + |\operatorname{Re} w|^2$ is plurisubharmonic by Proposition 8.2.2(1), since its complex Hessian matrix is the diagonal matrix with entries 1 in the z-directions and $\frac{1}{2}$ in the w-directions, by the identity $(\operatorname{Re} w_l)^2 = \frac{1}{2}|w_l|^2 + \frac{1}{2}\operatorname{Re}(w_l^2)$ for $1 \le l \le k$. Finally, let $c \in \mathbb{R}$. Since ψ is continuous, the set $\{(z,t) \in \Omega_{\mathbb{R}} : \psi(z,t) \le c\}$ is closed in $\Omega_{\mathbb{R}}$. It is contained in $\{|z|^2 + |t|^2 \le c\}$ and, if $\Omega \ne \mathbb{C}^{m+k}$, in the set $\{d \ge \mathrm{e}^{-c}\}$, which is closed in $\mathbb{C}^{m+k}$ and contained in Ω; it is therefore a closed and bounded subset of $\mathbb{C}^m \times \mathbb{R}^k$, hence compact. □

11.3 Interior density and the weighted estimate

We now prove Hörmander's estimate, first for smooth weights (Theorem 11.3.3) and then, by regularization, for arbitrary plurisubharmonic weights (Theorem 11.3.4). Throughout, $\Omega \subset \mathbb{C}^n$ is open, $n \ge 1$, and $\mathrm{d}V = \mathrm{d}V_{2n}$ is Lebesgue measure. For a real function q of class C^2 we write $H_q = (q_{j\bar{k}})$ for its complex Hessian matrix, with $q_{j\bar{k}} = \partial_{z_j}\partial_{\bar{z}_k} q$ as in (7.5), and we express our estimates through H_q. This is the matrix denoted $\mathcal{H}(q)$ in Section 9.7; we use the lighter notation here because $\mathcal{H}_0$, $\mathcal{H}_1$ and $\mathcal{H}_2$ will denote Hilbert spaces. By (7.5), the inequality $\mathrm{dd}^{\mathrm{c}} q \ge \varepsilon\beta$, with β as in (7.4), is equivalent to $H_q \ge \pi\varepsilon I$.

We also fix a smooth nonincreasing function κ_0 on $\mathbb{R}$ with $\kappa_0 = 1$ on $(-\infty, 0]$ and $\kappa_0 = 0$ on $[1, \infty)$. For instance, if $\rho_{1/2}$ is the mollifier of Section 1.4 in dimension $d = 1$, then $\theta_0(x) = \rho_{1/2}(x - 1/2)$ is smooth and nonnegative, has integral one and vanishes outside $[0, 1]$, and $\kappa_0(x) = 1 - \int_{-\infty}^{x} \theta_0(y)\,dy$ has these properties. The cutoff functions of one real variable used below are obtained from κ_0 by affine changes of the variable.

We give $(0,1)$- and $(0,2)$-forms the pointwise norms

$$\Big|\sum_j f_j\,d\bar z_j\Big|^2 = \sum_j |f_j|^2, \qquad \Big|\sum_{j<k} g_{jk}\,d\bar z_j \wedge d\bar z_k\Big|^2 = \sum_{j<k} |g_{jk}|^2;$$

for $n = 1$ there are no nonzero $(0,2)$-forms. For a measurable function $\varphi\colon \Omega \to [-\infty, \infty)$, which we call a *weight*, we write $L^2(\Omega, e^{-\varphi})$ for the Hilbert space of measurable functions u with $\int_\Omega |u|^2 e^{-\varphi}\,dV < \infty$, modulo null functions, with the inner product $\langle u, v\rangle_\varphi = \int_\Omega u\bar v\, e^{-\varphi}\,dV$ and the norm $\|u\|_\varphi$. Similarly, $L^2_{(0,1)}(\Omega, e^{-\varphi})$ and $L^2_{(0,2)}(\Omega, e^{-\varphi})$ consist of the $(0,1)$- and $(0,2)$-forms f with measurable coefficients and $\int_\Omega |f|^2 e^{-\varphi}\,dV < \infty$, with the analogous inner products. We write $L^2_{\mathrm{loc},(0,1)}(\Omega)$ for the $(0,1)$-forms with coefficients in $L^2_{\mathrm{loc}}(\Omega)$. The operator $\bar\partial$ is always taken in the sense of distributions.

Fix smooth real functions $\varphi_1, \varphi_2, \varphi_3$ on Ω and put

$$\mathcal{H}_0 = L^2(\Omega, e^{-\varphi_1}), \quad \mathcal{H}_1 = L^2_{(0,1)}(\Omega, e^{-\varphi_2}), \quad \mathcal{H}_2 = L^2_{(0,2)}(\Omega, e^{-\varphi_3}).$$

Let $T = \bar\partial$ from $\mathcal{H}_0$ to $\mathcal{H}_1$ and $S = \bar\partial$ from $\mathcal{H}_1$ to $\mathcal{H}_2$ be defined on their maximal domains

$$\operatorname{Dom} T = \{u \in \mathcal{H}_0 : \bar\partial u \in \mathcal{H}_1\}, \qquad \operatorname{Dom} S = \{f \in \mathcal{H}_1 : \bar\partial f \in \mathcal{H}_2\}.$$

Since the weights are continuous, convergence in $\mathcal{H}_0$, $\mathcal{H}_1$ or $\mathcal{H}_2$ implies convergence in L^2 on compact sets, and hence in the sense of distributions; therefore T and S are closed. Since the measures $e^{-\varphi_k}\,dV$ are Radon measures, Lemma 1.4.1 shows that smooth compactly supported functions and forms are dense in $\mathcal{H}_0$ and in $\mathcal{H}_1$; they belong to $\operatorname{Dom} T$ and $\operatorname{Dom} S$, so T and S are densely defined. Finally $\bar\partial\bar\partial = 0$ on distributions, so $\operatorname{Ran} T \subset \ker S$. Thus T and S satisfy the hypotheses on the operators in Theorem 11.1.4, with $\mathcal{H}_k$ in place of H_k.

Let $f = \sum_j f_j\,d\bar z_j \in \operatorname{Dom} T^*$. For every $u \in C^\infty_c(\Omega)$ we have

$$\sum_j \int_\Omega \partial_{\bar z_j} u\ \overline{f_j}\, e^{-\varphi_2}\,dV = \langle Tu, f\rangle_{\mathcal{H}_1} = \langle u, T^* f\rangle_{\mathcal{H}_0} = \int_\Omega u\,\overline{T^* f}\, e^{-\varphi_1}\,dV.$$

Integrating by parts on the left in the sense of distributions, we obtain $e^{-\varphi_1} T^* f = -\sum_j \partial_{z_j}(f_j e^{-\varphi_2})$, that is,

$$T^* f = -e^{\varphi_1 - \varphi_2} \sum_j \big(\partial_{z_j} f_j - (\partial_{z_j}\varphi_2) f_j\big) \tag{11.1}$$

in the sense of distributions. Conversely, if f is a smooth compactly supported $(0,1)$-form, the same computation is valid for every $u \in \operatorname{Dom} T$, since it only pairs the distribution $\bar\partial u$ with the test form $f\mathrm{e}^{-\varphi_2}$; hence $f \in \operatorname{Dom} T^*$, and T^*f is given by (11.1). For a general f, however, (11.1) describes T^*f but does not characterize $\operatorname{Dom} T^*$: membership in $\operatorname{Dom} T^*$ also encodes a boundary condition, which compactly supported test functions cannot detect. The following lemma shows that this boundary condition does not matter for the estimate, provided the weights are suitably related.

Lemma 11.3.1 *Let φ and ψ be smooth real functions on Ω, and put $\varphi_k = \varphi + (k-3)\psi$ for $k = 1,2,3$, so that $\varphi_1 = \varphi - 2\psi$, $\varphi_2 = \varphi - \psi$ and $\varphi_3 = \varphi$. Assume that there are functions $\chi_\nu \in C_c^\infty(\Omega)$, $\nu \in \mathbb{N}$, with $0 \le \chi_\nu \le 1$ and $|\bar\partial\chi_\nu|^2 \le \mathrm{e}^{\psi}$, such that for every compact set $K \subset \Omega$ we have $\chi_\nu = 1$ on a neighborhood of K for all large ν. Then, for the operators T and S defined by these weights, the smooth compactly supported $(0,1)$-forms are dense in $\operatorname{Dom} T^* \cap \operatorname{Dom} S$ for the graph norm $\|f\|_{\mathcal{H}_1} + \|T^*f\|_{\mathcal{H}_0} + \|Sf\|_{\mathcal{H}_2}$.*

Proof *Step 1: cutting off.* Let $f \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$, and write $\chi = \chi_\nu$. Since χ is real, $|\partial_{z_j}\chi| = |\partial_{\bar z_j}\chi|$, so the Cauchy–Schwarz inequality gives $|\sum_j (\partial_{z_j}\chi) f_j|^2 \le |\bar\partial\chi|^2 |f|^2$. Lagrange's identity $\sum_{j<k} |a_j b_k - a_k b_j|^2 = |a|^2|b|^2 - |\sum_j a_j \bar b_j|^2$ gives likewise $|\bar\partial\chi \wedge f|^2 \le |\bar\partial\chi|^2|f|^2$. Since $\varphi_1 - 2\varphi_2 = -\varphi = -\varphi_3$ and $|\bar\partial\chi|^2\mathrm{e}^{-\varphi} \le \mathrm{e}^{-\varphi_2}$, the form $\bar\partial\chi \wedge f$ lies in $\mathcal{H}_2$, the function $\mathrm{e}^{\varphi_1-\varphi_2}\sum_j(\partial_{z_j}\chi)f_j$ lies in $\mathcal{H}_0$, and both of their squared norms are at most

$$\int_\Omega |f|^2 |\bar\partial\chi_\nu|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V \le \|f\|_{\mathcal{H}_1}^2. \tag{11.2}$$

Let $u \in \operatorname{Dom} T$. Then $\chi u \in \operatorname{Dom} T$ and $T(\chi u) = \chi Tu + u\,\bar\partial\chi$, because $u\,\bar\partial\chi \in \mathcal{H}_1$, the weights being bounded on the support of χ. As χ is real and $\overline{\partial_{\bar z_j}\chi} = \partial_{z_j}\chi$, we get

$$\langle Tu, \chi f\rangle_{\mathcal{H}_1} = \langle T(\chi u), f\rangle_{\mathcal{H}_1} - \langle u\,\bar\partial\chi, f\rangle_{\mathcal{H}_1} = \Big\langle u, \chi T^*f - \mathrm{e}^{\varphi_1-\varphi_2}\sum_j (\partial_{z_j}\chi) f_j\Big\rangle_{\mathcal{H}_0}.$$

Hence $\chi f \in \operatorname{Dom} T^*$, and the distributional product rule shows that $\chi f \in \operatorname{Dom} S$, with

$$T^*(\chi f) - \chi T^* f = -\mathrm{e}^{\varphi_1-\varphi_2}\sum_j (\partial_{z_j}\chi) f_j,$$
$$S(\chi f) - \chi S f = \bar\partial\chi \wedge f.$$

By (11.2), the squared norms of both differences are bounded by $\int |f|^2|\bar\partial\chi_\nu|^2\mathrm{e}^{-\varphi}\,\mathrm{d}V$. As $\nu \to \infty$, $\bar\partial\chi_\nu$ tends to zero pointwise, since it vanishes near each point for large ν, and the integrand is dominated by the integrable function $|f|^2\mathrm{e}^{-\varphi_2}$. By dominated convergence, both differences tend to zero. Again by dominated convergence, $\chi_\nu f \to f$, $\chi_\nu T^*f \to T^*f$ and $\chi_\nu Sf \to Sf$. Therefore $\chi_\nu f \to f$ in the graph norm.

Step 2: smoothing. It remains to approximate a form $f \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$ with compact support $K \subset \Omega$. By (11.1),

$$\sum_j \partial_{z_j} f_j = -\mathrm{e}^{\varphi_2-\varphi_1} T^* f + \sum_j (\partial_{z_j}\varphi_2) f_j \in L^2_{\mathrm{loc}}(\Omega),$$

and $\bar\partial f = Sf \in L^2_{\mathrm{loc}}(\Omega)$. Let $f^\varepsilon = \sum_j (f_j * \rho_\varepsilon)\,\mathrm{d}\bar z_j$, with ρ_ε the mollifier of Section 1.4; for small ε these forms are smooth and supported in a fixed compact neighborhood K' of K in Ω. Convolution commutes with derivatives with constant coefficients (Proposition 1.4.3), so $\bar\partial f^\varepsilon = (\bar\partial f) * \rho_\varepsilon$ and $\sum_j \partial_{z_j} f_j^\varepsilon = (\sum_j \partial_{z_j} f_j) * \rho_\varepsilon$. By Lemma 1.4.1, f^ε, $\sum_j \partial_{z_j} f_j^\varepsilon$ and $\bar\partial f^\varepsilon$ converge in $L^2(K')$ to f, $\sum_j \partial_{z_j} f_j$ and $\bar\partial f$. Since (11.1) applies to both f^ε and f, and its remaining coefficients are smooth, $T^* f^\varepsilon \to T^* f$ in $L^2(K')$. The weights are bounded above and below on K', so $f^\varepsilon \to f$ in the graph norm. □

For compactly supported smooth forms, the inequality we need follows from an integration-by-parts identity, the Bochner–Kodaira–Morrey–Kohn identity in its flat weighted form.

Proposition 11.3.2 *Let φ be a smooth real function on Ω, and let δ_j be the differential operator $\delta_j g = -\partial_{z_j} g + (\partial_{z_j}\varphi) g$, $1 \le j \le n$. Then every smooth compactly supported $(0,1)$-form $f = \sum_j f_j\,\mathrm{d}\bar z_j$ on Ω satisfies*

$$\int \Big|\sum_j \delta_j f_j\Big|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V + \int |\bar\partial f|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V = \int \sum_{j,k} \varphi_{j\bar k} f_j \bar f_k\,\mathrm{e}^{-\varphi}\,\mathrm{d}V + \int \sum_{j,k} |\partial_{\bar z_k} f_j|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V.$$

Consequently, if ψ is a smooth real function and T and S are the operators defined by the weights $\varphi_k = \varphi + (k-3)\psi$ of Lemma 11.3.1, then for every $t > 0$

$$\int \sum_{j,k} \varphi_{j\bar k} f_j \bar f_k \mathrm{e}^{-\varphi}\,\mathrm{d}V \le (1+t^{-1})\|T^* f\|^2_{\mathcal{H}_0} + \|Sf\|^2_{\mathcal{H}_2} + (1+t)\int |f|^2 |\bar\partial\psi|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V. \tag{11.3}$$

Proof For smooth compactly supported functions g and w, integration by parts (Fubini's theorem and the fundamental theorem of calculus on lines) gives

$$\langle \partial_{\bar z_j} g, w\rangle_\varphi = -\int g\,\partial_{\bar z_j}\big(\bar w \mathrm{e}^{-\varphi}\big)\,\mathrm{d}V = \int g\,\overline{\big(-\partial_{z_j} w + (\partial_{z_j}\varphi) w\big)}\,\mathrm{e}^{-\varphi}\,\mathrm{d}V = \langle g, \delta_j w\rangle_\varphi,$$

because φ is real; thus δ_j is the formal adjoint of $\partial_{\bar z_j}$ for the inner product $\langle\cdot,\cdot\rangle_\varphi$, and taking complex conjugates we also have $\langle \delta_j w, g\rangle_\varphi = \langle w, \partial_{\bar z_j} g\rangle_\varphi$. Since $\partial_{\bar z_k}\big((\partial_{z_j}\varphi) g\big) = \varphi_{j\bar k} g + (\partial_{z_j}\varphi)\partial_{\bar z_k} g$ and the derivatives ∂_{z_j} and $\partial_{\bar z_k}$ commute, we obtain the commutation relation

$$\partial_{\bar z_k}(\delta_j g) - \delta_j(\partial_{\bar z_k} g) = \varphi_{j\bar k}\, g.$$

By the adjoint relation, then the commutation relation, and then the conjugate adjoint relation,

$$\begin{aligned}\Big\|\sum_j \delta_j f_j\Big\|_\varphi^2 &= \sum_{j,k}\langle \delta_j f_j, \delta_k f_k\rangle_\varphi = \sum_{j,k}\langle \partial_{\bar z_k}\delta_j f_j, f_k\rangle_\varphi \\ &= \sum_{j,k}\langle \varphi_{j\bar k} f_j, f_k\rangle_\varphi + \sum_{j,k}\langle \delta_j \partial_{\bar z_k} f_j, f_k\rangle_\varphi \\ &= \int \sum_{j,k} \varphi_{j\bar k} f_j \bar f_k \mathrm{e}^{-\varphi}\,\mathrm{d}V + \sum_{j,k}\langle \partial_{\bar z_k} f_j, \partial_{\bar z_j} f_k\rangle_\varphi .\end{aligned}$$

On the other hand, $\bar\partial f = \sum_{j<k}(\partial_{\bar z_j} f_k - \partial_{\bar z_k} f_j)\,\mathrm{d}\bar z_j \wedge \mathrm{d}\bar z_k$, and expanding the squares gives

$$\|\bar\partial f\|_\varphi^2 = \sum_{j\neq k}\|\partial_{\bar z_k} f_j\|_\varphi^2 - \sum_{j\neq k}\langle \partial_{\bar z_k} f_j, \partial_{\bar z_j} f_k\rangle_\varphi .$$

When we add the two identities, the terms with $j \neq k$ in the last sums cancel, and the remaining terms $\|\partial_{\bar z_j} f_j\|_\varphi^2$ combine with $\sum_{j\neq k}\|\partial_{\bar z_k} f_j\|_\varphi^2$ to give $\sum_{j,k}\|\partial_{\bar z_k} f_j\|_\varphi^2$. This proves the identity.

For the estimate, we have $\varphi_1 - \varphi_2 = -\psi$ and $\partial_{z_j}\varphi_2 = \partial_{z_j}\varphi - \partial_{z_j}\psi$, so (11.1) reads

$$\sum_j \delta_j f_j = \mathrm{e}^{\psi} T^* f + \sum_j (\partial_{z_j}\psi) f_j .$$

By the elementary inequality $|x+y|^2 \le (1+t^{-1})|x|^2 + (1+t)|y|^2$ for $x, y \in \mathbb{C}$,

$$\Big\|\sum_j \delta_j f_j\Big\|_\varphi^2 \le (1+t^{-1})\int |T^* f|^2 \mathrm{e}^{2\psi-\varphi}\,\mathrm{d}V + (1+t)\int \Big|\sum_j (\partial_{z_j}\psi) f_j\Big|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V .$$

The first integral is $\|T^* f\|_{\mathcal{H}_0}^2$, because $\varphi_1 = \varphi - 2\psi$, and $|\sum_j (\partial_{z_j}\psi) f_j|^2 \le |\bar\partial\psi|^2 |f|^2$ by the Cauchy–Schwarz inequality, since $|\partial_{z_j}\psi| = |\partial_{\bar z_j}\psi|$ for real ψ. Moreover $\|\bar\partial f\|_\varphi^2 = \|Sf\|_{\mathcal{H}_2}^2$, because $\varphi_3 = \varphi$. Inserting these bounds into the identity and dropping its last, nonnegative, term, we obtain (11.3). □

The right-hand side of $\bar\partial u = \alpha$ is measured by a norm dual to the complex Hessian of the weight. Let $A = (A_{j\bar k})$ be a positive semidefinite Hermitian $n \times n$ matrix, and write $A[b] = \sum_{j,k} A_{j\bar k} b_j \bar b_k$ for $b \in \mathbb{C}^n$. For $\alpha \in \mathbb{C}^n$, identified with the $(0,1)$-covector $\sum_j \alpha_j\,\mathrm{d}\bar z_j$, the *dual energy* of α with respect to A is

$$|\alpha|_{A^{-1}}^2 = \sup_{b\neq 0} \frac{|\sum_j \alpha_j \bar b_j|^2}{A[b]} \in [0,\infty],$$

where a quotient with zero denominator is 0 if its numerator vanishes and $+\infty$ otherwise. Directly from the definition, $|\sum_j \alpha_j \bar b_j|^2 \le |\alpha|_{A^{-1}}^2 A[b]$ whenever $|\alpha|_{A^{-1}} < \infty$. If (e_l) is an orthonormal basis of $\mathbb{C}^n$ in which the Hermitian quadratic form $A[\cdot]$ is diagonal, with $A[\sum_l c_l e_l] = \sum_l \lambda_l |c_l|^2$, then $|\alpha|_{A^{-1}}^2 = \sum_{\lambda_l>0} |\langle \alpha, e_l\rangle|^2/\lambda_l$ if $\langle \alpha, e_l\rangle = 0$ for every l with $\lambda_l = 0$, and $|\alpha|_{A^{-1}}^2 = \infty$ otherwise. In particular, the dual energy is a Hermitian quadratic form on the subspace of vectors where it is finite; it decreases when A increases; and $|\alpha|_{A^{-1}}^2 \le \lambda^{-1}|\alpha|^2$ if $A \ge \lambda I$ with $\lambda > 0$.

We shall integrate dual energies over Ω. Let A be a continuous function on Ω with values in the positive semidefinite Hermitian matrices, and let α be a $(0,1)$-form on Ω with measurable coefficients. Then $x \mapsto |\alpha(x)|^2_{A(x)^{-1}}$ is measurable. To see this, let Q be a countable dense subset of $\mathbb{C}^n \setminus \{0\}$, and for $b \neq 0$ let $\sigma_x(b)$ be the quotient in the definition, formed with $\alpha(x)$ and $A(x)$. For fixed b, its numerator is a measurable function of x and its denominator a continuous one, so $x \mapsto \sigma_x(b)$ is measurable. It therefore suffices to show that $\sup_{b\in Q} \sigma_x(b) = \sup_{b\neq 0} \sigma_x(b)$ for each x. Let $b \neq 0$ and choose $b_k \in Q$ with $b_k \to b$. If $A(x)[b] > 0$, then $\sigma_x(b_k) \to \sigma_x(b)$, because the numerator and the denominator are continuous functions of b. If $A(x)[b] = 0$ and $\sum_j \alpha_j(x)\bar b_j \neq 0$, then the numerators of $\sigma_x(b_k)$ tend to a positive number while the denominators tend to 0, so $\sigma_x(b_k) \to \infty = \sigma_x(b)$. If both vanish, then $\sigma_x(b) = 0 \leq \sigma_x(b_k)$.

The largest eigenvalue $\|A(x)\| = \max_{|b|=1} A(x)[b]$ depends continuously on x, as the maximum over the unit sphere of a jointly continuous function, and $A \leq (1+\|A\|)I$. Since the dual energy decreases when the matrix increases, we obtain

$$|\alpha|^2 \leq (1 + \|A\|)\, |\alpha|^2_{A^{-1}}. \tag{11.4}$$

Consequently, if φ is a weight that is bounded above on every compact subset of Ω and $\int_\Omega |\alpha|^2_{A^{-1}} e^{-\varphi}\, dV < \infty$, then $\alpha \in L^2_{\mathrm{loc},(0,1)}(\Omega)$: on a compact set $K \subset \Omega$ we have $1 + \|A\| \leq C_K$ and $\varphi \leq m_K$ for real constants C_K and m_K, so that $|\alpha|^2 \leq C_K e^{m_K} |\alpha|^2_{A^{-1}} e^{-\varphi}$ on K by (11.4), and $\int_K |\alpha|^2\, dV < \infty$. In particular, $\bar\partial\alpha$ is defined in the sense of distributions.

Theorem 11.3.3 (Hörmander) *Let $\Omega \subset \mathbb{C}^n$ be a pseudoconvex open set, $n \geq 1$, and let φ be a smooth plurisubharmonic function on Ω. Let α be a $(0,1)$-form on Ω with measurable coefficients such that*

$$M = \int_\Omega |\alpha|^2_{H_\varphi^{-1}} e^{-\varphi}\, dV < \infty$$

and $\bar\partial\alpha = 0$. Then there is $u \in L^2(\Omega, e^{-\varphi})$ with $\bar\partial u = \alpha$ and $\int_\Omega |u|^2 e^{-\varphi}\, dV \leq M$.

Proof The idea is to modify the weight far out in Ω so that Lemma 11.3.1 applies and the error term in (11.3) is absorbed by additional convexity, and then to let the modification recede to the boundary.

Step 1: auxiliary functions. The matrix H_φ is positive semidefinite because φ is plurisubharmonic, and by the discussion preceding the theorem, applied with $A = H_\varphi$, the finiteness of M implies that $\alpha \in L^2_{\mathrm{loc},(0,1)}(\Omega)$. By Theorem 11.2.4, Ω has a nonnegative smooth strictly plurisubharmonic exhaustion function s. Fix an integer $t \geq 1$. Let κ_0 be the function fixed at the beginning of this section, put $c_\kappa = \sup |\kappa_0'|$, and for integers $\nu \geq 1$ put $\chi_\nu = \kappa_0(s - t - 1 - \nu)$. Then $0 \leq \chi_\nu \leq 1$, and $\chi_\nu \in C^\infty_c(\Omega)$, because $\{s \leq t+2+\nu\}$ is compact; $\chi_\nu = 1$ on $\{s \leq t+1+\nu\}$, hence on a neighborhood of any given compact set for large ν; and $\bar\partial\chi_\nu = \kappa_0'(s - t - 1 - \nu)\, \bar\partial s$ vanishes on $\{s < t + 2\}$ and satisfies $|\bar\partial\chi_\nu|^2 \leq c_\kappa^2 |\bar\partial s|^2$. We put

$$\psi = \big(1 - \kappa_0(s - t - 1)\big) \log\big(1 + c_\kappa^2 |\bar\partial s|^2\big).$$

Then ψ is smooth and nonnegative; ψ and $\bar\partial\psi$ vanish on $\{s \le t+1\}$, since $1-\kappa_0(x-t-1)$ and its derivative vanish for $x \le t+1$; and $|\bar\partial\chi_\nu|^2 \le \mathrm{e}^{\psi}$ for every ν, because $\mathrm{e}^{\psi} = 1 + c_\kappa^2|\bar\partial s|^2$ on $\{s \ge t+2\}$ and $\bar\partial\chi_\nu = 0$ on $\{s < t+2\}$.

Next we choose a smooth convex nondecreasing function $\gamma\colon \mathbb{R} \to [0,\infty)$ with $\gamma = 0$ on $(-\infty, t]$ such that

$$\gamma(s) \ge 2\psi, \qquad H_{\gamma(s)} \ge (1+t)|\bar\partial\psi|^2 I, \qquad \int_\Omega |\alpha|^2 \mathrm{e}^{-\varphi-\gamma(s)+\psi}\,\mathrm{d}V < \infty. \tag{11.5}$$

For integers $l \ge 1$ the sets $K_l = \{t+l \le s \le t+l+1\}$ are compact. The smallest eigenvalue of H_s is a positive continuous function on Ω (see the discussion following Proposition 8.2.2), so its minimum λ_l over K_l is positive (if $K_l = \varnothing$, put $\lambda_l = 1$ and $m_l = P_l = I_l = 0$), and the numbers $m_l = (1+t)\max_{K_l}|\bar\partial\psi|^2$, $P_l = 2\max_{K_l}\psi$ and $I_l = \int_{K_l}|\alpha|^2\mathrm{e}^{-\varphi+\psi}\,\mathrm{d}V$ are finite, since $\alpha \in L^2_{\mathrm{loc},(0,1)}(\Omega)$. Choose a smooth nondecreasing function $g\colon \mathbb{R} \to [0,\infty)$ that vanishes on $(-\infty, t]$ and satisfies, for every $l \ge 1$,

$$g \ge m_l/\lambda_l \text{ on } [t+l, t+l+1], \qquad \int_{t+l-1}^{t+l} g(x)\,\mathrm{d}x \ge \max\bigl\{P_l, \log^+(2^l I_l)\bigr\}.$$

For instance, let $\kappa_1(x) = 1-\kappa_0(2x)$, a smooth nondecreasing function with $\kappa_1 = 0$ on $(-\infty, 0]$ and $\kappa_1 = 1$ on $[1/2,\infty)$, and put $g(x) = \sum_{l\ge1} c_l\kappa_1(x-t-l+1)$ with $c_l = \max\{m_l/\lambda_l, 2P_l, 2\log^+(2^l I_l)\}$. The sum is locally finite, since only the terms with $l < x-t+1$ are nonzero at x, and all its terms vanish for $x \le t$. The l-th term equals c_l on $[t+l-1/2,\infty)$, so $g \ge c_l$ on $[t+l, t+l+1]$ and $\int_{t+l-1}^{t+l} g(x)\,\mathrm{d}x \ge c_l/2$.

Put $\gamma(x) = \int_{-\infty}^{x} g(y)\,\mathrm{d}y$. Then γ is smooth, nondecreasing and convex, because $\gamma' = g$ is nondecreasing, and $\gamma = 0$ on $(-\infty, t]$. Since Ω is the union of $\{s \le t+1\}$ and the sets K_l, we may check the first two conditions in (11.5) on each of these sets. On $\{s \le t+1\}$ they hold because ψ and $\bar\partial\psi$ vanish there while $\gamma \ge 0$ and $H_{\gamma(s)} \ge 0$. On K_l we have $\gamma(s) \ge \gamma(t+l) \ge P_l \ge 2\psi$, and the chain rule

$$H_{\gamma(s)} = \gamma'(s)H_s + \gamma''(s)\bigl(\partial_{z_j} s\,\overline{\partial_{z_k} s}\bigr)_{j,k} \ge \gamma'(s)H_s$$

gives $H_{\gamma(s)} \ge (m_l/\lambda_l)\lambda_l I \ge (1+t)|\bar\partial\psi|^2 I$. Finally, the integral in (11.5) over the compact set $\{s \le t+1\}$ is finite, and its integral over K_l is at most $\mathrm{e}^{-\gamma(t+l)} I_l \le 2^{-l}$, because $\gamma(s) \ge \gamma(t+l) \ge \log^+(2^l I_l)$ there; summing over l, we obtain the third condition.

Step 2: the estimate. Put $\varphi_0 = \varphi + \gamma(s)$ and $\varphi_k = \varphi_0 + (k-3)\psi$ for $k = 1, 2, 3$, and let T and S be the corresponding operators. By the last condition in (11.5), $\alpha \in \mathcal{H}_1 = L^2_{(0,1)}(\Omega, \mathrm{e}^{-\varphi_2})$, and $\alpha \in \ker S$. Let f be a smooth compactly supported $(0,1)$-form. We apply (11.3) with φ_0 in place of φ and with the integer t fixed in Step 1. Since $H_{\varphi_0} = H_\varphi + H_{\gamma(s)}$, the second condition in (11.5) absorbs the last term of (11.3), and we obtain

$$\int H_\varphi[f]\,\mathrm{e}^{-\varphi_0}\,\mathrm{d}V \le (1+t^{-1})\|T^* f\|^2_{\mathcal{H}_0} + \|Sf\|^2_{\mathcal{H}_2}.$$

Moreover $\varphi - 2\varphi_2 = -\varphi_0 - \gamma(s) + 2\psi \le -\varphi_0$ by the first condition in (11.5). Since $M < \infty$, the dual energy of α is finite almost everywhere, and the inequality $|\sum_j \alpha_j \bar f_j|^2 \le |\alpha|^2_{H_\varphi^{-1}} H_\varphi[f]$ together with the Cauchy–Schwarz inequality gives

$$|\langle \alpha, f\rangle_{\mathcal{H}_1}|^2 \le \int |\alpha|^2_{H_\varphi^{-1}} \mathrm{e}^{-\varphi}\,\mathrm{d}V \cdot \int H_\varphi[f]\,\mathrm{e}^{\varphi-2\varphi_2}\,\mathrm{d}V \le M\int H_\varphi[f]\,\mathrm{e}^{-\varphi_0}\,\mathrm{d}V$$
$$\le M(1+t^{-1})\big(\|T^*f\|^2_{\mathcal{H}_0} + \|Sf\|^2_{\mathcal{H}_2}\big).$$

Both sides are continuous for the graph norm of Lemma 11.3.1, whose hypotheses hold by Step 1; hence the inequality extends to all $f \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$. By Theorem 11.1.4 there is $u_t \in \mathcal{H}_0$ with $\bar\partial u_t = \alpha$ and $\|u_t\|^2_{\mathcal{H}_0} \le M(1+t^{-1})$. On $\{s < t\}$ both $\gamma(s)$ and ψ vanish, so $\varphi_1 = \varphi$ there and

$$\int_{\{s<t\}} |u_t|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V \le M(1+t^{-1}).$$

Step 3: passage to the limit. For each integer $l \ge 1$ the functions u_t with $t \ge l$ have squared norms at most $2M$ in $L^2(\{s < l\}, \mathrm{e}^{-\varphi})$. By Proposition 11.1.2 and a diagonal argument, there is a sequence $t_i \to \infty$ such that u_{t_i} converges weakly in $L^2(\{s < l\}, \mathrm{e}^{-\varphi})$ for every l. The limits on different sublevel sets agree on their intersections, because restriction from $\{s < l+1\}$ to $\{s < l\}$ is a bounded linear map, and they define a function u on Ω. Weak convergence in these spaces implies convergence in the sense of distributions: a test function χ has support in some set $\{s < l\}$, and $\int u_{t_i}\chi\,\mathrm{d}V = \langle u_{t_i}, \bar\chi \mathrm{e}^{\varphi}\rangle_\varphi$ with $\bar\chi \mathrm{e}^{\varphi} \in L^2(\{s < l\}, \mathrm{e}^{-\varphi})$. Hence $\bar\partial u = \alpha$. By the lower semicontinuity of the norm in Proposition 11.1.2, $\int_{\{s<l\}} |u|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V \le \varliminf_i M(1+t_i^{-1}) = M$ for every l, and the monotone convergence theorem gives $\int_\Omega |u|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V \le M$. □

We now allow arbitrary plurisubharmonic weights. We write the weight as $p + q$, where p is plurisubharmonic, possibly singular, and q is smooth. The complex Hessian of p is discarded, and the right-hand side is measured by the dual energy with respect to H_q alone, which need not be positive definite.

Theorem 11.3.4 *Let $\Omega \subset \mathbb{C}^n$ be a pseudoconvex open set, $n \ge 1$, let $p \in \mathrm{PSH}(\Omega)$, and let q be a smooth real function on Ω whose complex Hessian matrix $A = H_q$ is positive semidefinite. Let α be a $(0,1)$-form on Ω with measurable coefficients such that*

$$M = \int_\Omega |\alpha|^2_{A^{-1}} \mathrm{e}^{-p-q}\,\mathrm{d}V < \infty$$

and $\bar\partial\alpha = 0$. Then the equation $\bar\partial u = \alpha$ has a solution with $\int_\Omega |u|^2 \mathrm{e}^{-p-q}\,\mathrm{d}V \le M$. In particular, if $A \ge \lambda I$ for a positive continuous function λ on Ω, then M may be replaced by $\int_\Omega \lambda^{-1}|\alpha|^2 \mathrm{e}^{-p-q}\,\mathrm{d}V$.

Proof We regularize p on relatively compact pseudoconvex subsets of Ω, apply Theorem 11.3.3 there, and pass to a weak limit. Since the weight $p + q$ is bounded above on compact sets, the discussion preceding Theorem 11.3.3 shows that $\alpha \in$

$L^2_{\mathrm{loc},(0,1)}(\Omega)$, so that $\bar\partial\alpha$ is defined. By Corollary 11.2.5 there are pseudoconvex open sets $\Omega_1 \subset \Omega_2 \subset \cdots$ with $\Omega_j \Subset \Omega$ and $\bigcup_j \Omega_j = \Omega$. Let ρ_ε be the radial mollifier of Section 1.4, and choose $\varepsilon_j \downarrow 0$ with $\varepsilon_j < \operatorname{dist}(\overline{\Omega}_j, \mathbb{C}^n \setminus \Omega)$. By Proposition 8.2.2(2), $p_j = p * \rho_{\varepsilon_j}$ is defined, smooth and plurisubharmonic on the open neighborhood $\{z \in \Omega : \operatorname{dist}(z, \mathbb{C}^n \setminus \Omega) > \varepsilon_j\}$ of $\overline{\Omega}_j$, so its complex Hessian is positive semidefinite there, and $p_j \downarrow p$ pointwise as $j \to \infty$. Put $\varphi_j = p_j + q$. Then φ_j is smooth with complex Hessian matrix $H_{\varphi_j} \geq A \geq 0$ near $\overline{\Omega}_j$, so it is plurisubharmonic there by Proposition 8.2.2(1), and $\varphi_j \geq p + q$. Since the dual energy decreases when the matrix increases,

$$\int_{\Omega_j} |\alpha|^2_{H_{\varphi_j}^{-1}} \mathrm{e}^{-\varphi_j}\, \mathrm{d}V \leq \int_{\Omega_j} |\alpha|^2_{A^{-1}} \mathrm{e}^{-p-q}\, \mathrm{d}V \leq M.$$

By Theorem 11.3.3 there is u_j on Ω_j with $\bar\partial u_j = \alpha$ and $\int_{\Omega_j} |u_j|^2 \mathrm{e}^{-\varphi_j}\, \mathrm{d}V \leq M$.

Let $K \subset \Omega$ be compact. Then $K \subset \Omega_k$ for some k, and for $j \geq k$ we have $\varphi_j \leq \varphi_k$ on K, because p_j decreases. Hence $\int_K |u_j|^2\, \mathrm{d}V \leq \mathrm{e}^{\max_K \varphi_k} M$ for $j \geq k$, so the sequence (u_j) is eventually bounded in $L^2(K)$ for every compact set K. Let (K_m) be a compact exhaustion of Ω (Section 1.1). By Proposition 11.1.2 and a diagonal argument, we find a subsequence that converges weakly in $L^2(K_m)$ for every m. The weak limits are compatible, because restriction from K_{m+1} to K_m is a bounded linear map, and they define a function $u \in L^2_{\mathrm{loc}}(\Omega)$; since every compact subset of Ω lies in some K_m, the subsequence converges weakly to u in $L^2(K)$ for every compact $K \subset \Omega$. Then $u_j \to u$ in the sense of distributions, and $\bar\partial u = \alpha$.

Now let $K \subset \Omega$ be compact, and let k be so large that $K \subset \Omega_k$. On $L^2(K)$ the norm with weight $\mathrm{e}^{-\varphi_k}$ is equivalent to the unweighted norm, so the subsequence also converges weakly for it, and the lower semicontinuity in Proposition 11.1.2 gives

$$\int_K |u|^2 \mathrm{e}^{-\varphi_k}\, \mathrm{d}V \leq \varliminf_j \int_K |u_j|^2 \mathrm{e}^{-\varphi_k}\, \mathrm{d}V \leq \varliminf_j \int_K |u_j|^2 \mathrm{e}^{-\varphi_j}\, \mathrm{d}V \leq M.$$

This holds for all large k, and $\mathrm{e}^{-\varphi_k}$ increases to e^{-p-q} as $k \to \infty$; so the monotone convergence theorem gives $\int_K |u|^2 \mathrm{e}^{-p-q}\, \mathrm{d}V \leq M$, and exhausting Ω by compact sets proves the estimate. The last assertion follows from the pointwise inequality $|\alpha|^2_{A^{-1}} \leq \lambda^{-1}|\alpha|^2$. □

The density e^{-p-q} in Theorem 11.3.4 takes values in $(0, \infty]$. It is infinite exactly on the pole set $\{p = -\infty\}$, which is a Lebesgue null set because p is locally integrable (Theorem 8.2.1), and it need not be locally integrable. Nevertheless, $L^2(\Omega, \mathrm{e}^{-p-q})$ is a Hilbert space, and since $p + q$ is bounded above on compact sets, convergence in it implies convergence in $L^2_{\mathrm{loc}}(\Omega)$; we show both facts in the proof of Lemma 11.3.5(2) below. That lemma also provides a distinguished solution of $\bar\partial u = \alpha$ in such weighted spaces. We write $\mathcal{O}(\Omega)$ for the space of holomorphic functions on an open set $\Omega \subset \mathbb{C}^n$.

Lemma 11.3.5 *Let $\Omega \subset \mathbb{C}^n$ be open, $n \geq 1$.*

(1) *Every distribution h on Ω with $\bar\partial h = 0$ is a holomorphic function: there is $g \in \mathcal{O}(\Omega)$ with $\langle h, \chi\rangle = \int_\Omega g\chi \,\mathrm{d}V$ for every test function χ.*
(2) *Let $\varphi\colon \Omega \to [-\infty, \infty)$ be a measurable function that is bounded above on every compact subset of Ω. Then the elements of $L^2(\Omega, \mathrm{e}^{-\varphi})$ represented by holomorphic functions form a closed subspace $\mathcal{K}$. Let α be a $(0,1)$-form with distribution coefficients such that the equation $\bar\partial u = \alpha$ has a solution in $L^2(\Omega, \mathrm{e}^{-\varphi})$. Then among these solutions there is exactly one of smallest norm, and a solution is of smallest norm if and only if it is orthogonal to $\mathcal{K}$.*

Proof (1) The real and imaginary parts of h, given by $\mathrm{Re}\langle h, \chi\rangle$ and $\mathrm{Im}\langle h, \chi\rangle$ for real test functions χ, are real distributions. Since derivatives of distributions commute, (7.5) shows that the Laplacian of h is $4\sum_j \partial_{z_j}\partial_{\bar z_j} h = 0$. As the Laplacian has real coefficients, the real and imaginary parts of h are harmonic distributions. By Weyl's lemma (Theorem 2.1.4) they are given by smooth functions, and hence h is given by a smooth function g. By integration by parts, the classical derivatives $\partial_{\bar z_j} g$ represent the distributions $\partial_{\bar z_j} h = 0$, so they vanish almost everywhere by Proposition 1.4.3(4), and hence everywhere, being continuous functions. Thus g is holomorphic.

(2) Let $L \subset \Omega$ be compact, and let m_L be a real upper bound for φ on L. For $u \in L^2(\Omega, \mathrm{e}^{-\varphi})$ we have $\int_L |u|^2 \,\mathrm{d}V \le \mathrm{e}^{m_L} \int_L |u|^2 \mathrm{e}^{-\varphi}\,\mathrm{d}V$. Hence $L^2(\Omega, \mathrm{e}^{-\varphi}) \subset L^2_{\mathrm{loc}}(\Omega)$, so that $\bar\partial u$ is defined in the sense of distributions, and convergence in $L^2(\Omega, \mathrm{e}^{-\varphi})$ implies convergence in $L^2_{\mathrm{loc}}(\Omega)$, hence in the sense of distributions. The space $L^2(\Omega, \mathrm{e}^{-\varphi})$ is a Hilbert space, by the completeness argument at the beginning of Section 11.1, applied to the positive measure $\mathrm{e}^{-\varphi}\,\mathrm{d}V$. The set $\mathcal{K}$ is a linear subspace. It is closed: if $g_j \in \mathcal{K}$ converge to g in $L^2(\Omega, \mathrm{e}^{-\varphi})$, then $g_j \to g$ in the sense of distributions, so $\bar\partial g = \lim_j \bar\partial g_j = 0$, and by (1) g agrees almost everywhere with a holomorphic function, that is, $g \in \mathcal{K}$.

Let u_0 be a solution of $\bar\partial u = \alpha$ in $L^2(\Omega, \mathrm{e}^{-\varphi})$. By (1), an element u of this space is a solution if and only if $\bar\partial(u - u_0) = 0$, that is, $u - u_0 \in \mathcal{K}$. Thus the solutions form the nonempty closed convex set $u_0 + \mathcal{K}$, which by Proposition 11.1.1 contains a unique element $u_{\min}$ of smallest norm. If a solution u is orthogonal to $\mathcal{K}$, then $\|u + k\|^2 = \|u\|^2 + \|k\|^2 \ge \|u\|^2$ for every $k \in \mathcal{K}$, so u has smallest norm, that is, $u = u_{\min}$. Conversely, if P is the orthogonal projection onto $\mathcal{K}$ (Proposition 11.1.1), then $u_{\min} - Pu_{\min}$ is a solution orthogonal to $\mathcal{K}$, hence equal to $u_{\min}$; so $u_{\min}$ is orthogonal to $\mathcal{K}$. □

The solution of smallest norm in (2) is called the *minimal solution* of $\bar\partial u = \alpha$ in $L^2(\Omega, \mathrm{e}^{-\varphi})$.

11.4 Interpolation and extension across a hyperplane

We now construct holomorphic functions with a prescribed value at a point, or a prescribed restriction to a hyperplane, and with controlled weighted norms. In both cases, a smooth function with the right values is corrected by a solution of a

$\bar\partial$-equation whose weight has a singular term forcing the correction to vanish where the values are prescribed.

Proposition 11.4.1 *Let $r, L > 0$, let $\Omega \subset \mathbb{C}^n$, $n \geq 1$, be a pseudoconvex open set of diameter at most L, let $a \in \mathbb{C}^n$ with $B(a, 2r) \subset \Omega$, and let $p \in \mathrm{PSH}(\Omega)$. If h is holomorphic on $B(a, 2r)$ and $\int_{B(a,2r)} |h|^2 \mathrm{e}^{-p}\, \mathrm{d}V < \infty$, then there is $F \in \mathcal{O}(\Omega)$ such that*

$$F(a) = h(a), \qquad \int_\Omega |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V \leq C(n, r, L) \int_{B(a,2r)} |h|^2 \mathrm{e}^{-p}\, \mathrm{d}V,$$

where the constant $C(n, r, L)$ depends only on n, r and L, and not on Ω, p, h or a.

Proof Fix a smooth function κ on $\mathbb{R}$ with $0 \leq \kappa \leq 1$, $\kappa = 1$ on $(-\infty, 1]$ and $\kappa = 0$ on $[3/2, \infty)$, for instance $\kappa(x) = \kappa_0(2x - 2)$ with the function κ_0 fixed at the beginning of Section 11.3, and put $\chi(z) = \kappa(|z - a|/r)$. Then χ is smooth, because κ is constant near 0; it has compact support in $B(a, 2r)$, equals one on $B(a, r)$, and satisfies $|\bar\partial\chi| \leq c_0/r$ with $c_0 = \sup |\kappa'|/2$, because $|\bar\partial|z - a|| = 1/2$ for $z \neq a$. Extend χh by zero to Ω. Then $\alpha = \bar\partial(\chi h) = h\,\bar\partial\chi$ is a smooth $\bar\partial$-closed form supported in the shell $r \leq |z - a| \leq 2r$.

The function $p + n \log |z - a|^2$ is plurisubharmonic on Ω by Propositions 8.2.5 and 8.3.3. We apply Theorem 11.3.4 with this function in place of p and with $q = |z - a|^2$, for which $A = I$. Since $\mathrm{e}^{-|z-a|^2} \leq 1$ and $|z - a|^{-2n} \leq r^{-2n}$ on the support of α, we obtain a solution of $\bar\partial u = \alpha$ with

$$\int_\Omega |u|^2 |z - a|^{-2n} \mathrm{e}^{-p-|z-a|^2}\, \mathrm{d}V \leq c_0^2 r^{-2n-2} \int_{B(a,2r)} |h|^2 \mathrm{e}^{-p}\, \mathrm{d}V.$$

On $B(a, r)$ we have $\bar\partial u = \alpha = 0$, so u is holomorphic there, by Lemma 11.3.5(1). Since p is bounded above on $B(a, r)$, the factor $\mathrm{e}^{-p-|z-a|^2}$ is bounded below there by a positive constant. If $u(a)$ were nonzero, then $|u|$ would be bounded below by a positive constant near a, and the integral on the left would diverge, because $\int_{B(a,\rho)} |z - a|^{-2n}\, \mathrm{d}V$ is a positive multiple of $\int_0^\rho \tau^{-2n} \tau^{2n-1}\, \mathrm{d}\tau = \infty$ by integration in polar coordinates (2.1). Thus $u(a) = 0$. The function $F = \chi h - u$ satisfies $\bar\partial F = 0$, so it is holomorphic by the same lemma, and $F(a) = h(a)$. Since $|z - a| < L$ on Ω, we have $|z - a|^{2n} \mathrm{e}^{|z-a|^2} \leq L^{2n} \mathrm{e}^{L^2}$, and therefore

$$\int_\Omega |u|^2 \mathrm{e}^{-p}\, \mathrm{d}V \leq c_0^2 L^{2n} \mathrm{e}^{L^2} r^{-2n-2} \int_{B(a,2r)} |h|^2 \mathrm{e}^{-p}\, \mathrm{d}V.$$

Finally, $|F|^2 \leq 2|\chi h|^2 + 2|u|^2$ and $|\chi h| \leq |h| \mathbf{1}_{B(a,2r)}$ give the assertion with $C(n, r, L) = 2 + 2c_0^2 L^{2n} \mathrm{e}^{L^2} r^{-2n-2}$. □

For extension from a hyperplane we use the same method, with the pole $\log |z_n|^2$ along the hyperplane $\{z_n = 0\}$. Now, however, the cutoff must be concentrated in a thin neighborhood $\{|z_n| < \varepsilon\}$ of the hyperplane, and then its $\bar\partial$-derivative has size of

order ε^{-1}. On the support of this derivative the factor $\mathrm{e}^{-\log|z_n|^2}$ of the weight has size of order ε^{-2}, while the support has area of order ε^2 in the normal direction. With a fixed smooth function q, the bound of Theorem 11.3.4 is therefore of order ε^{-2}. The remedy is a bounded twist of the weight. Let ψ be a smooth bounded plurisubharmonic function such that the dual energy h of $\bar\partial\psi$ with respect to H_ψ is less than one. The next lemma gives a solution of $\bar\partial u = \alpha$ whose norm with weight $(1-h)\mathrm{e}^{\psi-p}$ is bounded by a multiple of $\int |\alpha|^2_{H_\psi^{-1}}\mathrm{e}^{\psi-p}\,\mathrm{d}V$, and this integral is small if the complex Hessian of ψ is large near the hyperplane.

Lemma 11.4.2 *Let $\Omega \subset \mathbb{C}^n$ be a pseudoconvex open set, $n \geq 1$, let $p \in \mathrm{PSH}(\Omega)$, and let ψ be a smooth bounded plurisubharmonic function on Ω. Put $A = H_\psi$ and $h = |\bar\partial\psi|^2_{A^{-1}}$, where $\bar\partial\psi$ is identified with the vector $(\partial_{\bar z_j}\psi)_j$, and assume that $h < 1$ almost everywhere. Let α be a $(0,1)$-form on Ω with measurable coefficients such that $E = \int_\Omega |\alpha|^2_{A^{-1}}\mathrm{e}^{\psi-p}\,\mathrm{d}V < \infty$ and $\bar\partial\alpha = 0$, let $0 \leq a < 1$, and assume that $h \leq a$ almost everywhere on $\{\alpha \neq 0\}$. Then $\bar\partial u = \alpha$ has a solution with*

$$\int_\Omega (1-h)|u|^2\mathrm{e}^{\psi-p}\,\mathrm{d}V \leq \frac{1+\sqrt{a}}{1-\sqrt{a}}\,E.$$

Proof The idea is that, if u is the minimal solution in $L^2(\Omega,\mathrm{e}^{-p})$, then $u\mathrm{e}^{\psi}$ is the minimal solution of a twisted equation in $L^2(\Omega,\mathrm{e}^{-p-\psi})$, to which we apply Hörmander's estimate. Put $m = \sup_\Omega|\psi| < \infty$. Since $\mathrm{e}^{-m} \leq \mathrm{e}^{\pm\psi} \leq \mathrm{e}^{m}$, the spaces $L^2(\Omega,\mathrm{e}^{-p})$ and $L^2(\Omega,\mathrm{e}^{-p-\psi})$ consist of the same functions, with equivalent norms, and so contain the same holomorphic functions. By the discussion preceding Theorem 11.3.3, applied with the weight $p-\psi$, which is bounded above on compact sets, $\alpha \in L^2_{\mathrm{loc},(0,1)}(\Omega)$; so $\bar\partial\alpha$ is defined. Since $\mathrm{e}^{-p-\psi} \leq \mathrm{e}^{2m}\mathrm{e}^{\psi-p}$, we have $\int_\Omega |\alpha|^2_{A^{-1}}\mathrm{e}^{-p-\psi}\,\mathrm{d}V \leq \mathrm{e}^{2m}E < \infty$. Hence Theorem 11.3.4, applied with the plurisubharmonic function p and the smooth function $q = \psi$, whose complex Hessian matrix A is positive semidefinite, shows that the equation $\bar\partial u = \alpha$ has a solution in $L^2(\Omega,\mathrm{e}^{-p-\psi}) = L^2(\Omega,\mathrm{e}^{-p})$. Let u be the minimal solution in $L^2(\Omega,\mathrm{e}^{-p})$, given by Lemma 11.3.5(2).

The function $v = u\mathrm{e}^{\psi}$ belongs to $L^2(\Omega,\mathrm{e}^{-p-\psi})$ and solves

$$\bar\partial v = \beta = (\alpha + u\,\bar\partial\psi)\mathrm{e}^{\psi}.$$

For every holomorphic g in $L^2(\Omega,\mathrm{e}^{-p-\psi})$ we have $\int v\bar g\,\mathrm{e}^{-p-\psi}\,\mathrm{d}V = \int u\bar g\,\mathrm{e}^{-p}\,\mathrm{d}V = 0$, so v is the minimal solution of $\bar\partial v = \beta$ in $L^2(\Omega,\mathrm{e}^{-p-\psi})$, by Lemma 11.3.5(2).

We now estimate v by Theorem 11.3.4, applied again with p and $q = \psi$. The form $\beta = \bar\partial v$ is $\bar\partial$-closed, and it lies in $L^2_{\mathrm{loc},(0,1)}(\Omega)$ because α and u are locally square integrable and ψ is smooth. Since $E < \infty$ and $h < 1$, the dual energies of α and $\bar\partial\psi$ are finite almost everywhere, and at such points both vectors lie in the subspace on which the dual energy is a Hermitian quadratic form. The triangle inequality $|\alpha + u\,\bar\partial\psi|_{A^{-1}} \leq |\alpha|_{A^{-1}} + |u|h^{1/2}$ therefore gives

$$\int |\beta|^2_{A^{-1}}\mathrm{e}^{-p-\psi}\,\mathrm{d}V = \int |\alpha+u\,\bar\partial\psi|^2_{A^{-1}}\mathrm{e}^{\psi-p}\,\mathrm{d}V \leq 2E+2\sup_\Omega \mathrm{e}^{\psi}\int |u|^2\mathrm{e}^{-p}\,\mathrm{d}V < \infty.$$

So Theorem 11.3.4 yields a solution of $\bar\partial w = \beta$ whose squared norm in $L^2(\Omega, \mathrm{e}^{-p-\psi})$ is at most this integral, and the minimal solution v has no larger norm. As $\int |v|^2 \mathrm{e}^{-p-\psi}\,\mathrm{d}V = \int |u|^2 \mathrm{e}^{\psi-p}\,\mathrm{d}V$, we obtain

$$\int |u|^2 \mathrm{e}^{\psi-p}\,\mathrm{d}V \le \int |\alpha + u\,\bar\partial\psi|^2_{A^{-1}} \mathrm{e}^{\psi-p}\,\mathrm{d}V.$$

At almost every point, let $\langle\cdot,\cdot\rangle_{A^{-1}}$ be the Hermitian inner product associated with the dual energy on the subspace where it is finite; by the polarization identity $\langle x, y\rangle_{A^{-1}} = \frac{1}{4}\sum_{k=0}^{3} \mathrm{i}^k |x + \mathrm{i}^k y|^2_{A^{-1}}$, the function $\langle \alpha, \bar\partial\psi\rangle_{A^{-1}}$ is measurable. We expand

$$|\alpha + u\,\bar\partial\psi|^2_{A^{-1}} = |\alpha|^2_{A^{-1}} + 2\,\mathrm{Re}\big(\bar u \langle\alpha, \bar\partial\psi\rangle_{A^{-1}}\big) + h|u|^2, \qquad |\langle\alpha, \bar\partial\psi\rangle_{A^{-1}}| \le |\alpha|_{A^{-1}} h^{1/2},$$

the second inequality being the Cauchy–Schwarz inequality. The function $h|u|^2\mathrm{e}^{\psi-p}$ is integrable, since $h < 1$ almost everywhere and $u \in L^2(\Omega, \mathrm{e}^{-p})$; moving its integral to the left, we find that $J = \int (1-h)|u|^2\mathrm{e}^{\psi-p}\,\mathrm{d}V$, which is finite, satisfies

$$J \le E + 2\int |u|\,|\alpha|_{A^{-1}} h^{1/2} \mathrm{e}^{\psi-p}\,\mathrm{d}V.$$

If $a = 0$, then $h = 0$ almost everywhere on $\{\alpha \ne 0\}$, the last integral vanishes, and $J \le E$, as claimed. Let $a > 0$. The last integrand vanishes outside $\{\alpha \ne 0\}$, where $h \le a$ and hence $h/(1-h) \le a/(1-a)$. For $t > 1$, the inequality $2xy \le t^{-1}x^2 + ty^2$ with $x = (1-h)^{1/2}|u|$ and $y = (h/(1-h))^{1/2}|\alpha|_{A^{-1}}$ gives there

$$2|u|\,|\alpha|_{A^{-1}} h^{1/2} \le t^{-1}(1-h)|u|^2 + \frac{ta}{1-a}|\alpha|^2_{A^{-1}}.$$

Integrating, we obtain

$$J \le \left(1 + \frac{ta}{1-a}\right) E + t^{-1} J.$$

Since J is finite, this gives $J \le \frac{t}{t-1}\big(1 + \frac{ta}{1-a}\big)E$. Choose $t = 1 + a^{-1/2}$. Then $1 + ta/(1-a) = (1+\sqrt{a})/(1-a) = 1/(1-\sqrt{a})$ and $t/(t-1) = 1 + \sqrt{a}$, so $J \le (1+\sqrt{a})E/(1-\sqrt{a})$. □

The twisting function used for extension depends only on the normal variable. The next lemma records the one-variable calculus behind it.

Lemma 11.4.3 *Let $0 < \varepsilon < 1/4$. For $\zeta \in \mathbb{C}$ with $|\zeta| < 1/4$ put*

$$s = -\log(|\zeta|^2 + \varepsilon^2), \qquad \eta = s + \log s, \qquad \psi_\varepsilon = -\log\eta.$$

Then ψ_ε is a smooth, bounded, strictly subharmonic function on the disc $\Delta_{1/4}$. Let $h = |\partial_{\bar\zeta}\psi_\varepsilon|^2/(\psi_\varepsilon)_{\zeta\bar\zeta}$. Then

$$(1-h)\mathrm{e}^{\psi_\varepsilon} \geq (6s^2)^{-1},$$
$$h \leq \frac{2}{-\log(2\varepsilon^2)} \quad \textit{for } |\zeta| \leq \varepsilon,$$
$$\frac{\mathrm{e}^{\psi_\varepsilon}}{|\zeta|^2(\psi_\varepsilon)_{\zeta\bar\zeta}} \leq 25/4 \quad \textit{for } \varepsilon/2 \leq |\zeta| \leq \varepsilon.$$

In particular $h < 1$ everywhere, and the bound for h on the disc $|\zeta| \leq \varepsilon$ tends to zero as $\varepsilon \to 0$.

Proof Since $|\zeta|^2 + \varepsilon^2 < 1/8$, we have $s > \log 8 > 2$, so $\eta > s > 2$ and $\psi_\varepsilon < 0$ is well defined. For fixed ε, s ranges in a bounded interval, so ψ_ε is bounded; and $\psi_\varepsilon = G(|\zeta|^2)$ for a function G that is smooth near $[0, 1/16)$, so ψ_ε is smooth.

For $\zeta \neq 0$ put $x = |\zeta|^2$, $t = \log x$ and $b = x/(x+\varepsilon^2) \in (0,1)$, and denote derivatives with respect to t by primes. Then $s' = -b$ and $b' = b(1-b)$, so $s'' = -b(1-b)$. With $D = 1 + 1/s$ we have $\eta' = s' + s'/s = -bD$ and $D' = -s'/s^2 = b/s^2$. Hence $\psi_\varepsilon' = -\eta'/\eta = bD/\eta$ and $\psi_\varepsilon'' = (b'D + bD')/\eta - bD\eta'/\eta^2$, that is,

$$\psi_\varepsilon' = \frac{bD}{\eta}, \qquad \psi_\varepsilon'' = A + B, \quad A = \frac{b^2D^2}{\eta^2}, \quad B = \frac{b(1-b)D}{\eta} + \frac{b^2}{\eta s^2} > 0.$$

For a function g of $t = \log|\zeta|^2$ we have $\partial_{\bar\zeta} g = g'/\bar\zeta$ and $\partial_\zeta \partial_{\bar\zeta} g = g''/|\zeta|^2$. Hence $(\psi_\varepsilon)_{\zeta\bar\zeta} = \psi_\varepsilon''/x > 0$ and $h = (\psi_\varepsilon')^2/\psi_\varepsilon'' = A/(A+B) < 1$ for $\zeta \neq 0$. In general $\partial_{\bar\zeta}\psi_\varepsilon = G'(|\zeta|^2)\zeta$ and $(\psi_\varepsilon)_{\zeta\bar\zeta} = G''(|\zeta|^2)|\zeta|^2 + G'(|\zeta|^2)$, so at $\zeta = 0$ we have $\partial_{\bar\zeta}\psi_\varepsilon = 0$ and $(\psi_\varepsilon)_{\zeta\bar\zeta} = G'(0)$, where $G'(x) = \psi_\varepsilon'/x = D/(\eta(x+\varepsilon^2)) > 0$; so $(\psi_\varepsilon)_{\zeta\bar\zeta} > 0$ and $h = 0$ there. Thus ψ_ε is strictly subharmonic, and $h < 1$ everywhere.

For the first estimate let $\zeta \neq 0$. Since $B \geq b^2/(\eta s^2)$, we have $A/B \leq D^2 s^2/\eta = (s+1)^2/\eta$, and therefore

$$(1-h)\mathrm{e}^{\psi_\varepsilon} = \frac{B}{\eta(A+B)} \geq \frac{1}{\eta + (s+1)^2} \geq \frac{1}{6s^2},$$

because $\eta \leq 2s \leq 2s^2$ and $(s+1)^2 \leq 4s^2$ for $s \geq 1$. At $\zeta = 0$ we have $(1-h)\mathrm{e}^{\psi_\varepsilon} = 1/\eta \geq 1/(6s^2)$ as well. For the second estimate let $0 < x \leq \varepsilon^2$. Then $b \leq 1/2$, and $B \geq b(1-b)D/\eta$ gives

$$h \leq \frac{A}{B} \leq \frac{bD}{\eta(1-b)} \leq \frac{D}{\eta} = \frac{s+1}{s\eta} \leq \frac{2}{s},$$

since $s + 1 \leq 2s \leq 2\eta$; moreover $s \geq -\log(2\varepsilon^2)$, because $x + \varepsilon^2 \leq 2\varepsilon^2$. Together with $h(0) = 0$, this proves the second estimate. For the third, let $\varepsilon/2 \leq |\zeta| \leq \varepsilon$. Then $1/5 \leq b \leq 1/2$, so $\eta\psi_\varepsilon'' \geq \eta B \geq b(1-b)D \geq b(1-b) \geq 4/25$, and $\mathrm{e}^{\psi_\varepsilon}/(|\zeta|^2(\psi_\varepsilon)_{\zeta\bar\zeta}) = 1/(\eta\psi_\varepsilon'') \leq 25/4$. □

Theorem 11.4.4 (Ohsawa–Takegoshi) *There is an absolute constant C with the following property. Let $n \geq 1$ and $R > 0$, let $\Omega \subset \mathbb{C}^{n-1} \times \Delta_R$ be a pseudoconvex*

open set, where $\Delta_R \subset \mathbb{C}$ is the disc of radius R centered at 0, and let $p \in \mathrm{PSH}(\Omega)$. Put $\Omega_0 = \Omega \cap \{z_n = 0\}$, which we identify with an open subset of $\mathbb{C}^{n-1}$. Then every holomorphic function f on Ω_0 with

$$I = \int_{\Omega_0} |f|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2} < \infty$$

has an extension $F \in \mathcal{O}(\Omega)$ such that

$$F|_{\Omega_0} = f, \qquad \int_{\Omega} |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n} \le CR^2 I.$$

Here, for $n = 1$, Ω_0 is a point or empty, and I is the weighted value of $|f|^2$ at that point, the integral with respect to the counting measure $\mathrm{d}V_0$. If $\Omega_0 = \varnothing$ we take $F = 0$. On a connected component of Ω_0 on which the restriction of p is identically $-\infty$, finiteness of I forces $f = 0$, and this component contributes zero to I.

Proof After a normalization, Step 1 treats smooth weights on relatively compact pseudoconvex subsets, by correcting a cutoff of f with Lemmas 11.4.2 and 11.4.3, and Step 2 regularizes p and passes to the limit.

Normalization. The substitution $z_n = 4R\zeta$ maps $\mathbb{C}^{n-1} \times \Delta_{1/4}$ onto $\mathbb{C}^{n-1} \times \Delta_R$, preserves the hyperplane and the integral I, and multiplies Lebesgue measure on $\mathbb{C}^n$ by $16R^2$. Hence it suffices to show the following: if $\Omega \subset \mathbb{C}^{n-1} \times \Delta_{1/4}$, then f has an extension $F \in \mathcal{O}(\Omega)$ with $\int_\Omega |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V \le C_0 I$ for an absolute constant C_0; then $C = 16C_0$ works. From now on we write the points of Ω as (z', ζ) with $z' \in \mathbb{C}^{n-1}$ and $|\zeta| < 1/4$; for $n = 1$ the variable z' is absent. We may assume $\Omega_0 \neq \varnothing$.

Step 1: smooth weights on a relatively compact set. Let $D \Subset \Omega$ be a pseudoconvex open set, let $U \subset \Omega$ be an open neighborhood of $\overline{D}$, and let φ be a smooth plurisubharmonic function on U. We show that there are an open set $P \Subset \Omega_0$ with $\overline{P} \times \{0\} \subset U$ and a function $F \in \mathcal{O}(D)$ with $F(z', 0) = f(z')$ whenever $(z', 0) \in D$, such that

$$\int_D |F|^2 \mathrm{e}^{-\varphi}\, \mathrm{d}V \le C_0 \int_P |f(z')|^2 \mathrm{e}^{-\varphi(z',0)}\, \mathrm{d}V_{2n-2}(z'), \tag{11.6}$$

where C_0 is an absolute constant; we write I_P for the integral over P on the right.

The set $K_0 = \{z' : (z', 0) \in \overline{D}\}$ is a compact subset of the open set $U_0 = \{z' : (z', 0) \in U\} \subset \Omega_0$; choose an open set P with $K_0 \subset P \Subset U_0$. There is $\varepsilon_0 \in (0, 1/4)$ such that every $(z', \zeta) \in D$ with $|\zeta| \le \varepsilon_0$ satisfies $z' \in P$: otherwise there would be points $(z'_k, \zeta_k) \in D$ with $\zeta_k \to 0$ and $z'_k \notin P$, and a subsequence would converge, D being bounded, to a point $(z', 0) \in \overline{D}$ with $z' \notin P$, which is impossible. Decreasing ε_0, we may also assume that $\overline{P} \times \overline{\Delta}_{\varepsilon_0} \subset U$ and, by the uniform continuity of φ on this compact set, that

$$\mathrm{e}^{-\varphi(z',\zeta)} \le 2\mathrm{e}^{-\varphi(z',0)} \qquad (z' \in \overline{P},\ |\zeta| \le \varepsilon_0). \tag{11.7}$$

Fix a smooth function χ on $\mathbb{R}$ with $0 \le \chi \le 1$, $\chi = 1$ on $(-\infty, 1/4]$ and $\chi = 0$ on $[1, \infty)$, for instance $\chi(x) = \kappa_0((4x - 1)/3)$ with the function κ_0 fixed at the

beginning of Section 11.3, and put $c_\chi = \sup|\chi'|^2$. For $0 < \varepsilon \le \varepsilon_0$ define g_ε on D by $g_\varepsilon(z',\zeta) = f(z')\chi(|\zeta|^2/\varepsilon^2)$ if $z' \in P$, and $g_\varepsilon = 0$ otherwise. The open sets $D \cap (P \times \mathbb{C})$ and $D \cap \{|\zeta| > \varepsilon\}$ cover D by the choice of ε_0, and g_ε is smooth on each of them (on the second it vanishes); hence g_ε is smooth on D, and $g_\varepsilon(z',0) = f(z')$ for $(z',0) \in D$. Since f is holomorphic,

$$\alpha_\varepsilon = \bar\partial g_\varepsilon = f(z')\chi'(|\zeta|^2/\varepsilon^2)\frac{\zeta}{\varepsilon^2}\,\mathrm{d}\bar\zeta$$

is a $\bar\partial$-closed smooth form supported in $\{\varepsilon/2 \le |\zeta| \le \varepsilon\}$.

We solve $\bar\partial u = \alpha_\varepsilon$ on D by Lemma 11.4.2, with the function $\widetilde\varphi = \varphi + \log|\zeta|^2$ in place of p, which is plurisubharmonic on D by Propositions 8.2.5 and 8.3.3, and with $\psi = \psi_\varepsilon(\zeta)$ from Lemma 11.4.3, a smooth bounded function on D which is plurisubharmonic by Proposition 8.2.3, applied to the projection $(z',\zeta) \mapsto \zeta$. The complex Hessian matrix $A = H_\psi$ has the single nonzero entry $(\psi_\varepsilon)_{\zeta\bar\zeta}$, so $|\bar\partial\psi|^2_{A^{-1}}$ is the function h of Lemma 11.4.3; thus $h < 1$, and $h \le a_\varepsilon = 2/(-\log(2\varepsilon^2))$ on $\{\alpha_\varepsilon \ne 0\} \subset \{|\zeta| \le \varepsilon\}$. The dual energy of α_ε is $|f|^2|\chi'|^2|\zeta|^2\varepsilon^{-4}/(\psi_\varepsilon)_{\zeta\bar\zeta}$, so the third estimate of Lemma 11.4.3 and $|\zeta| \le \varepsilon$ give, on its support,

$$|\alpha_\varepsilon|^2_{A^{-1}}\mathrm{e}^{\psi_\varepsilon - \widetilde\varphi} = |f(z')|^2|\chi'|^2\varepsilon^{-4}\frac{\mathrm{e}^{\psi_\varepsilon}}{(\psi_\varepsilon)_{\zeta\bar\zeta}}\mathrm{e}^{-\varphi} \le \frac{25}{4}c_\chi\varepsilon^{-2}|f(z')|^2\mathrm{e}^{-\varphi(z',\zeta)}.$$

By (11.7) and Fubini's theorem, since every point of the support has $z' \in P$,

$$E = \int_D |\alpha_\varepsilon|^2_{A^{-1}}\mathrm{e}^{\psi-\widetilde\varphi}\,\mathrm{d}V \le \frac{25}{4}c_\chi\varepsilon^{-2}\cdot 2\pi\varepsilon^2 I_P = \frac{25\pi}{2}c_\chi I_P.$$

If ε is so small that $a_\varepsilon \le 1/4$, then $(1+\sqrt{a_\varepsilon})/(1-\sqrt{a_\varepsilon}) \le 3$, and Lemma 11.4.2 gives u_ε with $\bar\partial u_\varepsilon = \alpha_\varepsilon$ and $\int_D (1-h)|u_\varepsilon|^2\mathrm{e}^{\psi_\varepsilon-\widetilde\varphi}\,\mathrm{d}V \le 3E$. By the first estimate of Lemma 11.4.3, $(1-h)\mathrm{e}^{\psi_\varepsilon} \ge 1/(6\log^2(|\zeta|^2+\varepsilon^2))$, and $\mathrm{e}^{-\widetilde\varphi} = \mathrm{e}^{-\varphi}|\zeta|^{-2}$. Hence, with $C_1 = 225\pi c_\chi$,

$$\int_D \frac{|u_\varepsilon|^2\mathrm{e}^{-\varphi}}{|\zeta|^2\log^2(|\zeta|^2+\varepsilon^2)}\,\mathrm{d}V \le 18E \le C_1 I_P. \tag{11.8}$$

On $D \cap \{|\zeta| < \varepsilon/2\}$ we have $\bar\partial u_\varepsilon = 0$, so u_ε is holomorphic there, by Lemma 11.3.5(1). Suppose that $u_\varepsilon(z_0',0) \ne 0$ for some $(z_0',0) \in D$. On a small polydisc around this point, $|u_\varepsilon|$ is bounded below by a positive constant, φ is bounded above, and $\log^2(|\zeta|^2+\varepsilon^2) \le \log^2(\varepsilon^2)$; so the integrand in (11.8) is at least a positive constant times $|\zeta|^{-2}$, whose integral over a disc around $\zeta = 0$ diverges. This contradicts (11.8). Hence $u_\varepsilon = 0$ on $D \cap \{\zeta = 0\}$, and $F_\varepsilon = g_\varepsilon - u_\varepsilon$ is holomorphic on D, by the same lemma, because $\bar\partial F_\varepsilon = 0$; moreover $F_\varepsilon(z',0) = f(z')$.

It remains to estimate F_ε. For $0 < x = |\zeta|^2 < 1/16$ we have $x < x+\varepsilon^2 < 1/8$, so $x\log^2(x+\varepsilon^2) \le x\log^2 x \le 4\mathrm{e}^{-2} < 1$. Thus (11.8) gives $\int_D |u_\varepsilon|^2\mathrm{e}^{-\varphi}\,\mathrm{d}V \le C_1 I_P$. By (11.7) and Fubini's theorem, $\int_D |g_\varepsilon|^2\mathrm{e}^{-\varphi}\,\mathrm{d}V \le 2\pi\varepsilon^2 I_P \le I_P$. Therefore $\int_D |F_\varepsilon|^2\mathrm{e}^{-\varphi}\,\mathrm{d}V \le 2(1+C_1)I_P$, which is (11.6) with $C_0 = 2(1+C_1)$.

Step 2: the general case. By Corollary 11.2.5 there are pseudoconvex open sets $D_1 \subset D_2 \subset \cdots$ with $D_j \Subset \Omega$ and $\bigcup_j D_j = \Omega$. Choose $\varepsilon_j \downarrow 0$ with $\varepsilon_j < \operatorname{dist}(\overline{D}_j, \mathbb{C}^n \setminus \Omega)$, and let ρ_ε be the radial mollifier of Section 1.4; then, by Proposition 8.2.2(2), $p_j = p * \rho_{\varepsilon_j}$ is a smooth plurisubharmonic function on the open neighborhood $U_j = \{z \in \Omega : \operatorname{dist}(z, \mathbb{C}^n \setminus \Omega) > \varepsilon_j\}$ of $\overline{D}_j$, and $p_j \downarrow p$ pointwise. Step 1, applied to D_j, U_j and $\varphi = p_j$, gives $F_j \in \mathcal{O}(D_j)$ with $F_j(z', 0) = f(z')$ for $(z', 0) \in D_j$ and

$$\int_{D_j} |F_j|^2 \mathrm{e}^{-p_j}\, \mathrm{d}V \le C_0 \int_{P_j} |f(z')|^2 \mathrm{e}^{-p_j(z',0)}\, \mathrm{d}V_{2n-2}(z') \le C_0 I,$$

because $p_j \ge p$. (On a component of Ω_0 on which $p(\cdot, 0) \equiv -\infty$ we have $f = 0$, and the last inequality holds with the convention $0 \cdot \infty = 0$.)

We show that a subsequence of (F_j) converges locally uniformly on Ω. Let $K \subset \Omega$ be compact. Then $K \subset D_k$ for some k, and for $j \ge k$ we have $p_j \le p_k$ on K, so $\int_K |F_j|^2\, \mathrm{d}V \le \mathrm{e}^{\max_K p_k} C_0 I$. It follows that (F_j) is uniformly bounded on compact subsets of Ω. Indeed, for a holomorphic function G the complex Hessian matrix of $|G|^2$ is $\bigl(\partial_{z_\mu} G\, \overline{\partial_{z_\nu} G}\bigr)_{\mu,\nu}$, which is positive semidefinite; so $|G|^2$ is plurisubharmonic by Proposition 8.2.2(1), and hence subharmonic by Theorem 8.2.1. If $K \subset \Omega$ is compact, $\delta > 0$ and $K_\delta = \{z : \operatorname{dist}(z, K) \le \delta\} \subset \Omega$, then the mean value inequality over balls in Theorem 3.1.3, applied to $|F_j|^2$, gives $|F_j(z)|^2 \le |B(z, \delta)|^{-1} \int_{K_\delta} |F_j|^2\, \mathrm{d}V$ for $z \in K$ and large j, and the right side is bounded independently of j by the preceding estimate applied to K_δ.

Next, let $K \subset \Omega$ be compact, and choose $r > 0$ such that the compact set K' of points at distance at most $2\sqrt{n}\, r$ from K lies in Ω. For $z \in K_0' = \{z : \operatorname{dist}(z, K) \le \sqrt{n}\, r\}$, the closed polydisc of radius r centered at z lies in K', and the coefficient bound in (7.1) with $a = z$ gives $|\partial_{z_l} F_j(z)| \le r^{-1} \max_{K'} |F_j|$ for $1 \le l \le n$ and all large j (those with $K' \subset D_j$). Since $\mathrm{d}F_j = \sum_l \partial_{z_l} F_j\, \mathrm{d}z_l$, it follows that these F_j are Lipschitz with a common constant on every ball of radius $\sqrt{n}\, r$ centered at a point of K, so the sequence (F_j) is equicontinuous on K from some index on. By Proposition 1.6.4 on a compact exhaustion of Ω and a diagonal argument, a subsequence converges locally uniformly on Ω to a continuous function F. Since $\bar\partial F_j = 0$ and $F_j \to F$ in the sense of distributions, $\bar\partial F = 0$, and F is holomorphic by Lemma 11.3.5(1). Moreover $F(z', 0) = \lim_j F_j(z', 0) = f(z')$ for $z' \in \Omega_0$.

Finally, let $K \subset \Omega$ be compact, and let k be so large that $K \subset D_k$. For $j \ge k$ we have $p_j \le p_k$ on K, so locally uniform convergence gives

$$\int_K |F|^2 \mathrm{e}^{-p_k}\, \mathrm{d}V = \lim_j \int_K |F_j|^2 \mathrm{e}^{-p_k}\, \mathrm{d}V \le \varliminf_j \int_{D_j} |F_j|^2 \mathrm{e}^{-p_j}\, \mathrm{d}V \le C_0 I.$$

This holds for all large k, and e^{-p_k} increases to e^{-p} as $k \to \infty$; so the monotone convergence theorem gives $\int_K |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V \le C_0 I$, and exhausting Ω by compact sets gives $\int_\Omega |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V \le C_0 I$. □

Applying the theorem once for each normal direction, we obtain extension from affine subspaces of any codimension, in particular from points.

Corollary 11.4.5 *Let C be the constant of Theorem 11.4.4. Let $\Omega \subset \mathbb{C}^n$, $n \geq 1$, be a pseudoconvex open set of diameter at most $L > 0$, and let $p \in \mathrm{PSH}(\Omega)$. Then for every $a \in \Omega$ with $p(a) > -\infty$ there is $F \in \mathcal{O}(\Omega)$ such that*

$$F(a) = 1, \qquad \int_\Omega |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V \leq C^n L^{2n} \mathrm{e}^{-p(a)}.$$

More generally, let $V \subset \mathbb{C}^n$ be an affine complex subspace of codimension r, $0 \leq r \leq n$, and let f be holomorphic on $\Omega \cap V$ with finite slice norm $\int_{\Omega\cap V} |f|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2r}$, computed with Lebesgue measure on V. Then there is $F \in \mathcal{O}(\Omega)$ with $F|_{\Omega\cap V} = f$ and $\int_\Omega |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V \leq C^r L^{2r} \int_{\Omega\cap V} |f|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2r}$.

Proof We prove the second assertion; the first is the case $V = \{a\}$, $r = n$ and $f = 1$, whose slice norm with respect to the counting measure $\mathrm{d}V_0$ is $\mathrm{e}^{-p(a)}$. If $r = 0$ we take $F = f$, and if $\Omega \cap V = \varnothing$ we take $F = 0$.

Assume now that $r \geq 1$ and $\Omega \cap V \neq \varnothing$. Choose $b \in \Omega \cap V$ and unitary affine coordinates $w = (w_1, \ldots, w_n)$ on $\mathbb{C}^n$ centered at b with $V = \{w_1 = \cdots = w_r = 0\}$, and put $\Omega_k = \Omega \cap \{w_1 = \cdots = w_k = 0\}$ for $0 \leq k \leq r$, an open subset of a space isomorphic to $\mathbb{C}^{n-k}$. Unitary coordinates preserve Lebesgue measure. Each Ω_k with $k < n$ is pseudoconvex: the restriction of a continuous plurisubharmonic exhaustion function of Ω to the closed subset Ω_k of Ω is continuous and plurisubharmonic by Proposition 8.2.3, applied to the linear embedding of $\{w_1 = \cdots = w_k = 0\}$, and its sublevel sets are compact. By the discussion following Definition 11.2.1, every connected component of Ω_k is pseudoconvex as well. Since Ω is open and has diameter at most L, we have $|z - b| < L$ for $z \in \Omega$ (a point $z \in \Omega$ with $|z - b| = L$ could be moved slightly away from b inside Ω), so the coordinate w_k takes its values in Δ_L on Ω_{k-1}.

Put $f_r = f$, let $1 \leq k \leq r$, and assume that $f_k \in \mathcal{O}(\Omega_k)$ has finite norm $\int_{\Omega_k} |f_k|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2k}$. Let W be a connected component of Ω_{k-1}; ordering the coordinates as $(w_{k+1}, \ldots, w_n, w_k)$, we regard W as an open subset of $\mathbb{C}^{n-k} \times \Delta_L$ whose slice by $\{w_k = 0\}$ is the open subset $W \cap \Omega_k$ of Ω_k. By Proposition 8.2.3, the restriction of p to W is plurisubharmonic or identically $-\infty$. In the second case $\mathrm{e}^{-p} = \infty$ on $W \cap \Omega_k$, so the finiteness of the norm of f_k forces $f_k = 0$ almost everywhere on $W \cap \Omega_k$, hence everywhere there, since f_k is continuous, and we put $f_{k-1} = 0$ on W. In the first case, Theorem 11.4.4, applied to W with normal coordinate w_k, gives a holomorphic function f_{k-1} on W extending the restriction of f_k to $W \cap \Omega_k$, with

$$\int_W |f_{k-1}|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2k+2} \leq CL^2 \int_{W\cap\Omega_k} |f_k|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2k}.$$

The components W are open, disjoint and cover Ω_{k-1}, and the sets $W \cap \Omega_k$ are disjoint and cover Ω_k. Hence these functions define $f_{k-1} \in \mathcal{O}(\Omega_{k-1})$ extending f_k, and summing over W gives

$$\int_{\Omega_{k-1}} |f_{k-1}|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2k+2} \leq CL^2 \int_{\Omega_k} |f_k|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n-2k}.$$

Proceeding in this way for $k = r, r-1, \dots, 1$ and multiplying the r estimates, we see that $F = f_0$ has the required properties. □

Corollary 11.4.6 *Let X be a complex manifold, let $Y \subset X$ be a complex submanifold, let $a \in Y$, and let p be a plurisubharmonic function on a neighborhood of a in X. Let f be a holomorphic germ on Y at a such that $|f|^2 e^{-p|_Y}$ is integrable on some neighborhood of a in Y. Then there is a holomorphic germ F on X at a with $F|_Y = f$ such that $|F|^2 e^{-p}$ is integrable on some neighborhood of a in X. Here integrability refers to smooth positive volume forms on Y and X; since any two such forms are locally comparable, the notion does not depend on their choice.*

Proof Let $k = \dim Y$ and $n = \dim X$ near a. By the definition of a complex submanifold (Definition 7.1.1), there are holomorphic coordinates $w = (w', w'') \in \mathbb{C}^k \times \mathbb{C}^{n-k}$ centered at a in which $Y = \{w'' = 0\}$; by Proposition 8.2.3, p is plurisubharmonic near 0 in these coordinates. On a compact neighborhood of a, the densities of smooth positive volume forms with respect to coordinate Lebesgue measure are bounded above and below by positive constants. Hence we may test integrability with Lebesgue measure dV_{2k} on $Y = \mathbb{C}^k \times \{0\}$ and dV_{2n} in the coordinates w.

Choose a ball B around 0 so small that p is plurisubharmonic on a neighborhood of $\overline{B}$, that f is defined on $B \cap Y$, and that $\int_{B\cap Y} |f|^2 e^{-p}\, dV_{2k} < \infty$. The ball B is a bounded pseudoconvex domain, by Corollary 11.2.5 applied to the exhaustion function $|w|^2$ of $\mathbb{C}^n$, and $B \cap Y$ is its intersection with the linear subspace $\{w'' = 0\}$ of codimension $n - k$. By Corollary 11.4.5 there is $F \in \mathcal{O}(B)$ with $F|_{B\cap Y} = f$ and $\int_B |F|^2 e^{-p}\, dV_{2n} < \infty$. (If $k = 0$, the integral over Y is the weighted value $|f(a)|^2 e^{-p(a)}$; if $p|_Y$ is identically $-\infty$ near a, then $f = 0$ and $F = 0$, in accordance with the conventions of Theorem 11.4.4.) □

Chapter 12
Lelong numbers and multiplier ideals

A plurisubharmonic function can have a complicated pole set, but near each point its growth is governed by a single logarithmic rate. The *Lelong number* $\nu(u,a)$ of a plurisubharmonic function u at a point a is the largest $c \geq 0$ such that $u(z) \leq c \log|z-a|^2 + O(1)$ near a; with this normalization $\nu(\log|z|^2, 0) = 1$. By the Jensen–Lelong formula, $\nu(u,a)$ is also the limiting density at a of the positive current $\mathrm{dd}^c u$, which links the pointwise growth of u with the measures studied in the preceding chapters. Lelong numbers do not, however, determine the integrability properties of a singular weight φ (Example 12.3.3). These are recorded by the *multiplier ideal* $\mathcal{I}(\varphi)$, which consists of the holomorphic germs f for which $|f|^2 \mathrm{e}^{-\varphi}$ is locally integrable.

The L^2 estimates and extension theorems of Chapter 11 make multiplier ideals accessible. Solving a $\bar\partial$-equation with a weight that has a logarithmic pole at a point, we approximate every germ of $\mathcal{I}(\varphi)$, to arbitrarily high order, by holomorphic functions on a polydisc that are square integrable against $\mathrm{e}^{-\varphi}$; with the Krull intersection theorem and the strong Noetherian property, this shows that $\mathcal{I}(\varphi)$ is a coherent sheaf (Theorem 12.2.7). Comparing the vanishing orders of the generators of $\mathcal{I}(m\varphi)$ with the growth of φ (Lemma 12.3.1), we prove Siu's theorem that the sets where the Lelong number is at least $c > 0$ are analytic. The same comparison shows that $\mathrm{e}^{-t\varphi}$ is integrable near a point x for every $t > 0$ if $\nu(\varphi, x) = 0$, a fact used in Chapter 15 for an example on the projective line.

The Ohsawa–Takegoshi theorem gives the restriction and subadditivity properties of multiplier ideals. Since its constant does not depend on the weight, it can be applied along a moving family of hyperplanes; comparing the resulting extensions on one fixed holomorphic curve, we prove that multiplier ideals are stable under increasing limits of weights and deduce the strong openness theorem $\mathcal{I}(\varphi) = \bigcup_{\varepsilon>0} \mathcal{I}((1+\varepsilon)\varphi)$ (Corollary 12.5.4).

The last section studies integrability on modifications. For a weight with analytic singularities, the change of variables formula on a log resolution reduces membership in the multiplier ideal to explicit one-variable integrals (Theorem 12.6.5). For a general weight, the ideals $\mathcal{I}(m\varphi)$ recover the generic Lelong numbers of φ along all prime divisors over the ambient manifold, and strong openness gives a valuative

criterion for membership in $\mathcal{I}(\varphi)$ in which one positive parameter serves for all divisors (Theorem 12.6.9). These results are the local foundation for comparing singularities through their multiplier ideals in later chapters.

Historical remarks and references for this chapter are collected in Appendix B.12.

12.1 Logarithmic growth and the Jensen formula

Proposition 12.1.1 *Let $n \geq 1$ and let u be a plurisubharmonic function on a neighborhood of $a \in \mathbb{C}^n$. For sufficiently negative t put*

$$N(t) = \sup_{|z-a|<\exp(t/2)} u(z).$$

Then N is finite, nondecreasing and convex, $N(t)$ is also the supremum of u over the sphere $|z-a| = \exp(t/2)$, and the limit

$$\nu(u,a) = \lim_{t\to-\infty} \frac{N(t)}{t} \tag{12.1}$$

exists in $[0,\infty)$; it is zero if $u(a) > -\infty$. For every $r_0 > 0$ such that $\overline{B(a,r_0)}$ lies in the domain of u,

$$u(z) \leq \nu(u,a) \log \frac{|z-a|^2}{r_0^2} + \sup_{B(a,r_0)} u, \qquad 0 < |z-a| < r_0. \tag{12.2}$$

Consequently $\nu(u,a)$ is the largest $c \geq 0$ for which $u(z) \leq c \log|z-a|^2 + O(1)$ near a; moreover, if $c \geq 0$ and $u(z) \leq c \log|z-a|^2 + C$ for all z outside a Lebesgue null set in some neighborhood of a, then $\nu(u,a) \geq c$. The Lelong number does not change under a holomorphic change of coordinates, nor under the addition of a bounded function, provided the resulting function is plurisubharmonic.

Proof We may assume $a = 0$. Let $R_* \in (0,\infty]$ be the supremum of the radii r for which $\overline{B(0,r)}$ lies in the domain of u; since closed balls are compact, $\overline{B(0,r)}$ lies in the domain exactly when $0 < r < R_*$. For such r put $S(r) = \sup_{|z|=r} u(z)$. This number is less than $+\infty$ because u is upper semicontinuous and hence bounded above on the compact sphere. It is also greater than $-\infty$. Otherwise, for every unit vector e, the function $\zeta \mapsto u(\zeta e)$, which is subharmonic or identically $-\infty$ near the closed disc $|\zeta| \leq r$, would be $-\infty$ on the circle $|\zeta| = r$ and hence, by upper semicontinuity, have upper limits $-\infty$ there; the maximum principle (Proposition 3.1.4), applied to $\zeta \mapsto u(\zeta e) + k$ for every $k > 0$, would give $u = -\infty$ on the disc. Thus u would be identically $-\infty$ on $B(0,r)$, which is impossible because u is locally integrable (Theorem 8.2.1).

Let $0 < r_1 < r_2 < r_3 < R_*$, and let $h(\zeta) = A + B\log|\zeta|^2$ be the harmonic function on $\mathbb{C}\setminus\{0\}$ equal to $S(r_1)$ on $|\zeta| = r_1$ and to $S(r_3)$ on $|\zeta| = r_3$. For every unit vector e, the function $\zeta \mapsto u(\zeta e) - h(\zeta)$ is subharmonic or identically $-\infty$ near

the closed annulus $r_1 \le |\zeta| \le r_3$, and by upper semicontinuity its upper limits at the two boundary circles are at most 0; by the maximum principle (Proposition 3.1.4) it is at most 0 on the annulus. Taking the supremum over e on the circle $|\zeta| = r_2$ gives $S(r_2) \le h(r_2)$. Since $h(r_2)$ is the value at $\log r_2^2$ of the affine function of $\log r^2$ that equals $S(r_1)$ and $S(r_3)$ at $\log r_1^2$ and $\log r_3^2$, this says that S is a convex function of $\log r^2$ on $(0, R_*)$.

Applying the maximum principle in the same way to $\zeta \mapsto u(\zeta e) - S(r)$ on the disc $|\zeta| \le r$, we obtain $u(\zeta e) \le S(r)$ for $|\zeta| \le r$; hence S is nondecreasing and $u(0) \le S(r)$. A finite convex function on an open interval is continuous, since on a compact subinterval its difference quotients are bounded between two fixed difference quotients taken outside the subinterval. Hence $\sup_{B(0,r)} u = \sup_{\rho<r} S(\rho) = S(r)$, so that $N(t) = S(\mathrm{e}^{t/2})$ for $t < \log R_*^2$; this proves the assertions about N.

Fix r_0 as in the statement and put $t_0 = \log r_0^2$. Since N is convex and nondecreasing on $(-\infty, t_0]$, the difference quotients $(N(t_0) - N(t))/(t_0 - t)$ are nonnegative and decrease as $t \downarrow -\infty$; let $c \in [0, \infty)$ be their limit. Writing $N(t) = N(t_0) - (t_0 - t) \cdot \frac{N(t_0)-N(t)}{t_0-t}$ and dividing by t, we obtain $N(t)/t \to c$ as $t \to -\infty$. In particular c does not depend on t_0, so for every fixed $s \le t_0$ the quotients $(N(s) - N(s'))/(s - s')$ also tend to c as $s' \to -\infty$. If $s' < s < t \le t_0$, convexity gives $(N(t) - N(s))/(t - s) \ge (N(s) - N(s'))/(s - s')$; letting $s' \to -\infty$, we see that every difference quotient of N on $(-\infty, t_0]$ is at least c. Hence $N(t) - ct$ is nondecreasing, and $N(t) \le c(t - t_0) + N(t_0)$ for $t < t_0$. Since $u(z) \le S(|z|) = N(\log |z|^2)$, this is (12.2).

Since $u(0) \le S(r)$ for all r, we have $N(t) \ge u(0)$, hence $N(t)/t \le u(0)/t$ for $t < 0$, and $c = 0$ if $u(0) > -\infty$. Conversely, if $c' \ge 0$ and $u(z) \le c' \log |z|^2 + C$ for $0 < |z| < r'$, then $N(t) = S(\mathrm{e}^{t/2}) \le c't + C$ for all sufficiently negative t, and division by $t < 0$ gives $c \ge c'$. This proves the characterization of $\nu(u, a)$ as the largest admissible coefficient.

Now let $c \ge 0$, and suppose that $u(z) \le c \log |z|^2 + C$ for all $z \in B(0, r) \setminus A$, where $\overline{B(0, r)}$ lies in the domain of u and A is a Lebesgue null set. Let $0 < |z| < r/2$ and $0 < \delta < |z|/2$. The function u is subharmonic (Theorem 8.2.1), so Theorem 3.1.3 gives $u(z) \le |B(z, \delta)|^{-1} \int_{B(z,\delta)} u \, \mathrm{d}V$. Every $y \in B(z, \delta) \setminus A$ satisfies $u(y) \le c \log(|z| + \delta)^2 + C$, because $c \ge 0$ and $|y| \le |z| + \delta$; since A is a null set, we obtain $u(z) \le c \log(|z| + \delta)^2 + C$. Letting $\delta \downarrow 0$, we get $u(z) \le c \log |z|^2 + C$ for $0 < |z| < r/2$, and the converse part of the characterization gives $\nu(u, 0) \ge c$.

Let F be a biholomorphic map from a neighborhood of a onto a neighborhood of $F(a)$; by Proposition 8.2.3, $u \circ F^{-1}$ is plurisubharmonic near $F(a)$. The maps F and F^{-1} are continuously differentiable, hence Lipschitz near a and $F(a)$, so $\log |F(z) - F(a)|^2 - \log |z - a|^2$ is bounded near a, $z \ne a$. Since adding a bounded function also changes u by a bounded amount, in both cases the characterization just proved shows that the Lelong number is unchanged. □

Definition 12.1.2 The number $\nu(u, a)$ in (12.1) is the *Lelong number* of u at a. If u is plurisubharmonic on a complex manifold X that has positive dimension at a, we compute $\nu(u, a)$ in a holomorphic chart at a; by Proposition 12.1.1 the result does not depend on the chart. If X has dimension zero at a, where plurisubharmonic functions are finite, we set $\nu(u, a) = 0$. For a closed positive $(1, 1)$-current T on

X we put $\nu(T,a) = \nu(u,a)$, where u is a plurisubharmonic potential of T near a, that is, $\mathrm{dd^c}u = T$. Such potentials exist, and two of them differ near a by a smooth pluriharmonic function: by Theorem 8.2.1, the plurisubharmonic potentials of T on a coordinate ball are exactly its canonical local potentials, to which Theorem 7.3.2 applies. Since a smooth function is bounded near a, $\nu(T,a)$ does not depend on the choice of the potential.

The Lelong number can also be read off from the mass of $\mathrm{dd^c}u$ on small balls. We use the form

$$\mathrm{dd^c}|z|^2 = \frac{\beta}{\pi} = \frac{1}{\pi}\sum_{j=1}^{n} \mathrm{d}x_j \wedge \mathrm{d}y_j,$$

where β is the Euclidean Kähler form (7.4); note that $\mathrm{dd^c}|z-a|^2 = \mathrm{dd^c}|z|^2$. By (6.2) with $d = 2n$, and since $\Gamma(n) = (n-1)!$, the area of the unit sphere $S^{2n-1} \subset \mathbb{R}^{2n}$ is $s_{2n-1} = 2\pi^n/(n-1)!$.

Let u be plurisubharmonic on a neighborhood of a closed ball $\overline{B(a,R)} \subset \mathbb{C}^n$, $n \geq 1$. By Theorems 3.2.2 and 8.2.1, $\mu = \Delta u$ is a positive Radon measure. For $0 < r < R$ define

$$D_u(a,r) = \frac{\mu(B(a,r))}{2s_{2n-1}r^{2n-2}} = \frac{1}{r^{2n-2}}\int_{B(a,r)} \mathrm{dd^c}u \wedge (\mathrm{dd^c}|z|^2)^{n-1}. \tag{12.3}$$

Here $\mathrm{dd^c}u \wedge (\mathrm{dd^c}|z|^2)^{n-1}$ is the product of the current $\mathrm{dd^c}u$ with a smooth form, as defined in Chapter 7. To verify the second equality, assume first that u is smooth. When $(\mathrm{dd^c}|z|^2)^{n-1}$ is expanded, only the diagonal coefficients $u_{j\bar{j}}$ of $\mathrm{dd^c}u$ contribute to the product, and since $\frac{\mathrm{i}}{2}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j = \mathrm{d}x_j \wedge \mathrm{d}y_j$ and $\Delta u = 4\sum_j u_{j\bar{j}}$, we obtain

$$\mathrm{dd^c}u \wedge (\mathrm{dd^c}|z|^2)^{n-1} = \frac{(n-1)!}{\pi^n}\sum_{j=1}^{n} u_{j\bar{j}}\,\mathrm{d}V = \frac{(n-1)!}{4\pi^n}\Delta u\,\mathrm{d}V.$$

Both sides are obtained from u by the same constant-coefficient differential operator, so the identity holds for every distribution u. Since $2s_{2n-1} = 4\pi^n/(n-1)!$, this gives (12.3).

Theorem 12.1.3 (Jensen–Lelong) *Let $n \geq 1$, $R > 0$, and let u be plurisubharmonic on a neighborhood of $\overline{B(a,R)} \subset \mathbb{C}^n$. Then the spherical mean $M_u(a,r)$ of u over $\partial B(a,r)$ is finite for every $0 < r < R$, the function $r \mapsto D_u(a,r)$ is nondecreasing on $(0,R)$, and*

$$M_u(a,r_2) - M_u(a,r_1) = \int_{\log r_1^2}^{\log r_2^2} D_u(a,\exp(t/2))\,\mathrm{d}t, \qquad 0 < r_1 < r_2 < R. \tag{12.4}$$

In particular,

$$\nu(u,a) = \lim_{r\downarrow 0} D_u(a,r) = \lim_{r\downarrow 0}\frac{M_u(a,r)}{\log r^2}. \tag{12.5}$$

Proof *Step 1: smooth functions.* Suppose first that u is smooth. By Proposition 2.1.1 in $\mathbb{R}^{2n}$ and the substitution $r = \mathrm{e}^{t/2}$, for which $\mathrm{d}r/\mathrm{d}t = r/2$,

$$\frac{\mathrm{d}}{\mathrm{d}t} M_u(a, \mathrm{e}^{t/2}) = \frac{r}{2} \cdot \frac{\mu(B(a,r))}{s_{2n-1} r^{2n-1}} = D_u(a, \mathrm{e}^{t/2}),$$

and (12.4) follows by integration, since $\mu = \Delta u \, \mathrm{d}V$ has a continuous density, so that the right side is continuous in t. To see that $D_u(a,\cdot)$ is nondecreasing, we show that $t \mapsto M_u(a, \mathrm{e}^{t/2})$ is a convex function on $(-\infty, \log R^2)$; the derivative of a differentiable convex function is nondecreasing. For a unit vector $\omega \in S^{2n-1}$, the function $v_\omega(\zeta) = u(a + \zeta\omega)$ is smooth on a neighborhood of the closed disc $|\zeta| \le R$ in $\mathbb{C}$, and by (7.5) its Laplacian is

$$\Delta v_\omega(\zeta) = 4 \sum_{j,k} u_{j\bar{k}}(a + \zeta\omega)\, \omega_j \bar{\omega}_k \ge 0,$$

because the complex Hessian of the smooth plurisubharmonic function u is positive semidefinite (Proposition 8.2.2(1)). Let $m_\omega(r) = (2\pi)^{-1} \int_0^{2\pi} v_\omega(r\mathrm{e}^{\mathrm{i}\theta})\, \mathrm{d}\theta$ be the circle means of v_ω about 0; they are the spherical means of v_ω in $\mathbb{R}^2$, because surface measure on the unit circle is arc length (Section 6.4). By Proposition 2.1.1 in $\mathbb{R}^2$, where $s_1 = 2\pi$, and the same substitution,

$$\frac{\mathrm{d}}{\mathrm{d}t} m_\omega(\mathrm{e}^{t/2}) = \frac{1}{4\pi} \int_{|\zeta| < \mathrm{e}^{t/2}} \Delta v_\omega \, \mathrm{d}V_2,$$

which is nondecreasing in t because $\Delta v_\omega \ge 0$. By the mean value theorem, a differentiable function with nondecreasing derivative is convex; so $t \mapsto m_\omega(\mathrm{e}^{t/2})$ is convex. Surface measure on S^{2n-1} is invariant under the orthogonal transformations $\omega \mapsto \mathrm{e}^{\mathrm{i}\theta}\omega$ of $\mathbb{R}^{2n}$ (Section 2.1), so $\int_{S^{2n-1}} u(a + r\omega)\, \mathrm{d}S(\omega) = \int_{S^{2n-1}} u(a + r\mathrm{e}^{\mathrm{i}\theta}\omega)\, \mathrm{d}S(\omega)$ for every θ. Averaging over $\theta \in [0, 2\pi]$ and using Fubini's theorem for the continuous integrand, we obtain

$$M_u(a,r) = \frac{1}{s_{2n-1}} \int_{S^{2n-1}} m_\omega(r)\, \mathrm{d}S(\omega), \qquad 0 < r < R.$$

An average of convex functions is convex, so $t \mapsto M_u(a, \mathrm{e}^{t/2})$ is convex, and $D_u(a,\cdot)$ is nondecreasing on $(0, R)$.

Step 2: approximation. For general u, let $u_\varepsilon = u * \rho_\varepsilon$ be the regularizations of (3.3). For small ε they are defined on a neighborhood of the compact set $\overline{B(a,R)}$, and by Proposition 8.2.2(2) they are smooth and plurisubharmonic there and decrease to u as $\varepsilon \downarrow 0$. Subtracting a constant, which changes neither side of (12.4), nor D_u, nor the Lelong number, we may assume $u_\varepsilon \le 0$ on $\overline{B(a,R)}$ for all small ε. For $0 \le \eta < 1$ put $\kappa_\eta = (1-\eta)/(1+\eta)^{2n-1}$. If $0 < r < R$ and $|x - a| \le \eta r$, the explicit formula (2.9) shows that the Poisson kernel P of $B(a,r)$ satisfies $P(x,\zeta) \ge \kappa_\eta P(a,\zeta) = \kappa_\eta/(s_{2n-1} r^{2n-1})$ for $\zeta \in \partial B(a,r)$.

Fix $0 < r < R$ and choose $b \in B(a,r)$ with $u(b) > -\infty$, which is possible since the pole set of u has measure zero; let $\eta = |b-a|/r$. By the maximum principle (Proposition 3.1.4), applied to u_ε minus the Poisson integral of its boundary values (Theorem 2.3.1), we have $u_\varepsilon(x) \le \int_{\partial B(a,r)} P(x,\zeta) u_\varepsilon(\zeta)\, \mathrm{d}S(\zeta)$ for $x \in B(a,r)$. At $x = b$, since $u_\varepsilon \le 0$, this gives

$$u(b) \le u_\varepsilon(b) \le \kappa_\eta M_{u_\varepsilon}(a,r).$$

Thus the integrals of the nonnegative functions $-u_\varepsilon$ over the sphere are bounded independently of ε. As $-u_\varepsilon$ increases to $-u$, the monotone convergence theorem shows that $M_u(a,r)$ is finite and $M_{u_\varepsilon}(a,r) \to M_u(a,r)$. For fixed $x \in B(a,r)$ the kernel $P(x,\cdot)$ is positive and bounded on $\partial B(a,r)$, so the same argument, applied to the functions $-P(x,\cdot)u_\varepsilon$, lets us pass to the limit in the Poisson inequality, and we obtain $u(x) \le \int_{\partial B(a,r)} P(x,\zeta) u(\zeta)\, \mathrm{d}S(\zeta)$ for $x \in B(a,r)$.

Since the functions u_ε decrease to the locally integrable function u, the monotone convergence theorem, applied to $u_{\varepsilon_0} - u_\varepsilon$ for a fixed ε_0, shows that $u_\varepsilon \to u$ in L^1_{loc}; hence the positive measures Δu_ε converge to μ in the sense of distributions, hence vaguely (Proposition 1.5.1). By Proposition 1.5.2, $\Delta u_\varepsilon(B(a,r)) \to \mu(B(a,r))$ for every radius r with $\mu(\partial B(a,r)) = 0$; since the spheres $\partial B(a,r)$, $0 < r < R$, are disjoint and μ is finite on $B(a,R)$, only countably many radii are excluded (Lemma 1.1.5). Moreover, if $r_2 < R$ and $\chi \in C_c^\infty$ is a cutoff function with values in $[0,1]$, equal to one on $B(a,r_2)$, with support in a neighborhood of $\overline{B(a,R)}$ on which all u_ε with small ε are defined, then $\Delta u_\varepsilon(B(a,r_2)) \le \int \chi\, \Delta u_\varepsilon \to \int \chi\, \mathrm{d}\mu$, so these masses are bounded uniformly in ε.

By Step 1 each $D_{u_\varepsilon}(a,\cdot)$ is nondecreasing; letting $\varepsilon \downarrow 0$, we see that $D_u(a,\cdot)$ is nondecreasing on the dense set of radii with $\mu(\partial B(a,r)) = 0$, and hence on $(0,R)$, because $r \mapsto \mu(B(a,r))$ is left continuous. Finally, (12.4) holds for each u_ε by Step 1. As $\varepsilon \downarrow 0$, its left side converges to $M_u(a,r_2) - M_u(a,r_1)$, while the integrands on the right converge for all but countably many t and are bounded uniformly in ε; dominated convergence gives (12.4) for u.

Step 3: the limits. The nondecreasing nonnegative function $D_u(a,\cdot)$ has a limit $\ell = \lim_{r\downarrow 0} D_u(a,r) \in [0,\infty)$. Put $g(t) = M_u(a, \mathrm{e}^{t/2})$ and fix $t_0 < \log R^2$. Given $\delta > 0$, choose $s_\delta < \min\{t_0, 0\}$ with $D_u(a, \mathrm{e}^{s/2}) \le \ell + \delta$ for $s \le s_\delta$. For $t < s_\delta$, (12.4) and the monotonicity of $D_u(a,\cdot)$ give

$$\ell(t_0 - t) \le g(t_0) - g(t) \le (\ell + \delta)(s_\delta - t) + (t_0 - s_\delta) D_u(a, \mathrm{e}^{t_0/2}).$$

Dividing by $-t > 0$ and letting $t \to -\infty$, we see that all limit points of $g(t)/t$ lie in $[\ell, \ell+\delta]$. Hence $M_u(a,r)/\log r^2 \to \ell$ as $r \downarrow 0$. It remains to show that $\ell = \nu(u,a)$. Let $0 < \eta < 1$. For $x \in B(a,\eta r)$, the Poisson inequality of Step 2 and the kernel bound give, since $u \le 0$,

$$u(x) \le \int_{\partial B(a,r)} P(x,\zeta) u(\zeta)\, \mathrm{d}S(\zeta) \le \kappa_\eta M_u(a,r).$$

Together with $M_u(a,r) \le \sup_{\partial B(a,r)} u = N(\log r^2)$, where N is as in Proposition 12.1.1, we obtain

$$N(\log(\eta^2 r^2)) \le \kappa_\eta M_u(a,r), \qquad M_u(a,r) \le N(\log r^2).$$

We divide by $\log r^2 < 0$ and let $r \downarrow 0$. Since $\log(\eta^2 r^2)/\log r^2 \to 1$, this gives $\nu(u,a) \ge \kappa_\eta \ell$ and $\ell \ge \nu(u,a)$. Letting $\eta \downarrow 0$, so that $\kappa_\eta \to 1$, we conclude that $\ell = \nu(u,a)$, which proves (12.5). □

Proposition 12.1.4 *Let $n \ge 1$, let u and v be plurisubharmonic on a neighborhood U of $a \in \mathbb{C}^n$, and let $b > 0$. Then*

$$\nu(u+v,a) = \nu(u,a) + \nu(v,a), \qquad \nu(bu,a) = b\nu(u,a),$$
$$\nu(\max\{u,v\},a) = \min\{\nu(u,a), \nu(v,a)\}.$$

If $(u_i)_{i\in I}$ is a nonempty family of plurisubharmonic functions on U that is locally uniformly bounded above, and $u = (\sup_i u_i)^*$, then $\nu(u,a) = \inf_i \nu(u_i,a)$. If u_j, $j \ge 1$, and u are plurisubharmonic on U, $u_j \to u$ in $L^1_{\mathrm{loc}}(U)$, and $a_j \to a$, then

$$\overline{\lim_j}\, \nu(u_j,a_j) \le \nu(u,a). \tag{12.6}$$

Finally, for a holomorphic germ $f \not\equiv 0$ at a, $\nu(\log|f|^2,a) = \operatorname{ord}_a f$, where the order $\operatorname{ord}_a f$ *of f at a is the degree of the first nonzero homogeneous term of its Taylor expansion at a (Appendix A.1); moreover $\nu(\log|z-a|^2,a) = 1$.*

Proof By (12.5), $\nu(w,a) = \lim_{r\downarrow 0} M_w(a,r)/\log r^2$ for every plurisubharmonic w near a, and the spherical means involved are finite and linear in w; this proves the formulas for $u+v$ and bu. If $w_1 \le w_2 + O(1)$ near a, then $\nu(w_1,a) \ge \nu(w_2,a)$ by the characterization in Proposition 12.1.1; hence $\nu(\max\{u,v\},a) \le \min\{\nu(u,a),\nu(v,a)\}$. Conversely, choose $r_0 > 0$ with $\overline{B(a,r_0)} \subset U$. By (12.2), and since the logarithm $\log(|z-a|^2/r_0^2)$ is negative on $B(a,r_0)\setminus\{a\}$, both u and v, and hence $\max\{u,v\}$, are bounded there by $c\log(|z-a|^2/r_0^2) + C$ with $c = \min\{\nu(u,a),\nu(v,a)\}$ and a constant C. This gives the reverse inequality.

For the family, u is plurisubharmonic by Theorem 8.3.2. Choose r_0 as above, a common upper bound C for the u_i on $B(a,r_0)$, and put $c = \inf_i \nu(u_i,a)$. By (12.2), again using that the logarithm is negative, $u_i(z) \le c\log(|z-a|^2/r_0^2) + C$ for $0 < |z-a| < r_0$ and all i. The supremum of the u_i satisfies the same bound, and so does its upper semicontinuous regularization u on $B(a,r_0)\setminus\{a\}$, because the bound is continuous there. Hence $\nu(u,a) \ge c$ by the characterization in Proposition 12.1.1. Conversely $u \ge u_i$ gives $\nu(u,a) \le \nu(u_i,a)$ for every i.

For (12.6), note that $\Delta u_j \to \Delta u$ in the sense of distributions, because $u_j \to u$ in $L^1_{\mathrm{loc}}(U)$; hence the positive measures Δu_j converge vaguely to Δu (Proposition 1.5.1). Fix $0 < r < R$ with $\overline{B(a,R)} \subset U$ and $\Delta u(\partial B(a,r)) = 0$, and let $0 < \rho < r$. Then $B(a_j,\rho) \subset B(a,r)$ for all large j. Since $D_{u_j}(a_j,\cdot)$ is nondecreasing with limit $\nu(u_j,a_j)$ at 0 (Theorem 12.1.3), and $\overline{\lim}_j \Delta u_j(K) \le \Delta u(K)$ for the compact set

$K = \overline{B(a,r)}$ (Proposition 1.5.2), we obtain

$$\overline{\lim_j}\, \nu(u_j, a_j) \le \overline{\lim_j}\, \frac{\Delta u_j(\overline{B(a,r)})}{2s_{2n-1}\rho^{2n-2}} \le \frac{\Delta u(\overline{B(a,r)})}{2s_{2n-1}\rho^{2n-2}}.$$

Let $\rho \uparrow r$; as $\Delta u(\partial B(a,r)) = 0$, the right side becomes $D_u(a,r)$. By Lemma 1.1.5, applied to the finite measure Δu on $B(a,R)$ and the disjoint spheres $\partial B(a,r)$, all but countably many radii $r \in (0,R)$ satisfy $\Delta u(\partial B(a,r)) = 0$. Letting $r \downarrow 0$ through such radii and using (12.5) proves (12.6).

Finally, let $k = \operatorname{ord}_a f$, and let $P_k \not\equiv 0$ be the homogeneous term of degree k in the expansion of f at a; the function $\log|f|^2$ is plurisubharmonic near a by Proposition 8.3.3. By (7.2) there is C with $|f(a+w) - P_k(w)| \le C|w|^{k+1}$ for small w. Put $c_1 = \max_{|\omega|=1} |P_k(\omega)|$, which is positive because a homogeneous polynomial vanishing on the unit sphere vanishes identically. Since $|P_k(w)| \le c_1|w|^k$, with equality for some w on every sphere $|w| = r$, we obtain $r^k(c_1 - Cr) \le \sup_{|w|=r} |f(a+w)| \le r^k(c_1 + Cr)$. By Proposition 12.1.1, $N(\log r^2)$ is the supremum of $\log|f|^2$ over the sphere $|z-a| = r$, so $N(\log r^2) = k\log r^2 + O(1)$ as $r \downarrow 0$, and $\nu(\log|f|^2, a) = k$. For the last assertion, recall that $\log|z-a|^2$ is plurisubharmonic (Proposition 8.3.3). Since $\sup_{B(a,r)} \log|z-a|^2 = \log r^2$, we have $N(t) = t$ for this function, and (12.1) gives $\nu(\log|z-a|^2, a) = 1$. □

12.2 Coherent ideals from weighted integrability

From now on X denotes a complex manifold, and $n \ge 0$ its dimension at the point under consideration. We also narrow a term of Chapter 11: a *weight* on X is now a plurisubharmonic function on X, a special case of the measurable weights of Section 11.3; in particular, a weight is not identically $-\infty$ on any connected component of X. Restrictions of weights to submanifolds may be identically $-\infty$ on some components, and Definition 12.2.5 allows this case. Local integrability on X refers to Lebesgue measure in holomorphic charts, or equivalently to any smooth positive volume form, since the densities of two such measures with respect to each other are bounded above and below by positive constants on compact subsets of a chart.

To pass from a Hilbert space of holomorphic functions to a coherent ideal sheaf we need two facts from local algebra, the Krull intersection theorem and the strong Noetherian property of coherent sheaves. We prove them first, the second by induction on the dimension with the help of Weierstrass polynomials.

Lemma 12.2.1 *Let $x \in X$, let $R = \mathcal{O}_{X,x}$ with maximal ideal $\mathfrak{m}$, and let E be a submodule of a finitely generated R-module F. Then*

$$\bigcap_{k\ge 1} (E + \mathfrak{m}^k F) = E.$$

In particular, if a sequence of holomorphic functions on a neighborhood of x converges locally uniformly and the germs at x of all its members belong to an ideal $J \subset R$, then the germ of the limit belongs to J.

Proof We derive the intersection formula from the Artin–Rees lemma, after collecting the facts about Noetherian rings that we need. A commutative ring A is *Noetherian* if all its ideals are finitely generated. Let A be Noetherian.

(i) Every submodule N of A^r, $r \geq 1$, is finitely generated. For $r = 1$ this is the definition. Let $r \geq 2$. The last coordinates of the elements of N form an ideal of A; choose $v_1, \dots, v_q \in N$ whose last coordinates generate it, and, by induction on r, finitely many generators of the submodule $N \cap (A^{r-1} \times \{0\})$, which is a submodule of A^{r-1}. If $v \in N$, subtracting from v a combination of the v_i with the same last coordinate leaves an element of $N \cap (A^{r-1} \times \{0\})$; hence these finitely many elements generate N.

(ii) Every submodule N of a finitely generated A-module M is finitely generated. Indeed, let $\pi\colon A^r \to M$ map the standard basis to generators of M. By (i), the submodule $\pi^{-1}(N)$ of A^r is finitely generated, and the images of its generators generate $N = \pi(\pi^{-1}(N))$.

(iii) The polynomial ring $A[t]$ is Noetherian (Hilbert's basis theorem). Let $I \subset A[t]$ be a nonzero ideal. The leading coefficients of the nonzero elements of I, together with 0, form an ideal of A: if $f, g \in I$ have degrees $d \leq e$ and leading coefficients a, b with $a + b \neq 0$, then $t^{e-d} f + g \in I$ has leading coefficient $a + b$, and if $c \in A$ with $ca \neq 0$, then $cf \in I$ has leading coefficient ca. Choose $p_1, \dots, p_k \in I$ whose leading coefficients generate this ideal, and let d_0 be the largest of their degrees. If $f \in I$ has degree $d \geq d_0$, its leading coefficient is $\sum_i a_i c_i$ for some $a_i \in A$, where c_i is the leading coefficient of p_i, and $f - \sum_i a_i t^{d - \deg p_i} p_i$ is an element of I of smaller degree. Repeating this, we see that f differs from an element of $I \cap (A + At + \cdots + At^{d_0-1})$ by an element of the ideal generated by $p_1, \dots, p_k$. The set $I \cap (A + At + \cdots + At^{d_0-1})$ is a submodule of the A-module $A + At + \cdots + At^{d_0-1} \cong A^{d_0}$, finitely generated by (i). Hence I is generated by $p_1, \dots, p_k$ and finitely many further elements.

(iv) If $I \subset A$ is an ideal, then A/I is Noetherian, since the ideals of A/I are the images of the ideals of A containing I. Hence, by (iii) and induction on the number of variables, every quotient of a polynomial ring in finitely many variables over A is Noetherian.

All this applies to $A = R$, which is Noetherian by Proposition A.1.2, since a holomorphic chart at x identifies R with $\mathcal{O}_n$.

Next we prove the Artin–Rees lemma. If $\mathfrak{m} = 0$, which happens only in dimension zero, the lemma is trivial; so let $\mathfrak{m} \neq 0$, and choose generators $\xi_1, \dots, \xi_p$ of $\mathfrak{m}$. Consider the subring $\mathcal{R} = \bigoplus_{k \geq 0} \mathfrak{m}^k t^k$ of $R[t]$, consisting of the polynomials whose coefficient of t^k lies in $\mathfrak{m}^k$, where $\mathfrak{m}^0 = R$. Since $\mathfrak{m}^k$ is generated by the monomials of degree k in $\xi_1, \dots, \xi_p$, the ring $\mathcal{R}$ is the image of the homomorphism $R[T_1, \dots, T_p] \to R[t]$, $T_i \mapsto \xi_i t$, and it is Noetherian by (iii) and (iv).

In the $\mathcal{R}$-module of polynomials in t with coefficients in F, consider the submodule $\bigoplus_{k \geq 0} \mathfrak{m}^k F t^k$. It is generated over $\mathcal{R}$ by generators of F placed in degree zero, and it contains the submodule $\mathcal{E} = \bigoplus_{k \geq 0} (E \cap \mathfrak{m}^k F) t^k$, because $\mathfrak{m}^j (E \cap \mathfrak{m}^k F) \subset$

$E \cap \mathfrak{m}^{j+k}F$. By (ii), $\mathcal{E}$ is finitely generated. The homogeneous components of the elements of $\mathcal{E}$ again belong to $\mathcal{E}$, so the homogeneous components of finitely many generators generate $\mathcal{E}$; let these be $e_1 t^{d_1}, \dots, e_l t^{d_l}$ with $e_i \in E \cap \mathfrak{m}^{d_i}F$, and let $s = \max_i d_i$. For $k \geq s$ this gives the Artin–Rees equality

$$E \cap \mathfrak{m}^k F = \mathfrak{m}^{k-s}(E \cap \mathfrak{m}^s F).$$

Indeed, let $y \in E \cap \mathfrak{m}^k F$, and write $y t^k = \sum_i \rho_i e_i t^{d_i}$ with $\rho_i \in \mathcal{R}$. Comparing the coefficients of t^k, we may replace each ρ_i by its term of degree $k - d_i$, which is $a_i t^{k-d_i}$ with $a_i \in \mathfrak{m}^{k-d_i}$. Thus $y = \sum_i a_i e_i$, where $a_i \in \mathfrak{m}^{k-d_i} = \mathfrak{m}^{k-s}\mathfrak{m}^{s-d_i}$ and $\mathfrak{m}^{s-d_i} e_i \subset E \cap \mathfrak{m}^s F$; hence $y \in \mathfrak{m}^{k-s}(E \cap \mathfrak{m}^s F)$. The reverse inclusion is immediate.

Now let Q be a finitely generated R-module and $G = \bigcap_k \mathfrak{m}^k Q$. By (ii), G is finitely generated. Let s be the index of the Artin–Rees equality for $G \subset Q$. Since $G \subset \mathfrak{m}^k Q$ for all k, this equality with $k = s + 1$ gives $G = G \cap \mathfrak{m}^{s+1}Q = \mathfrak{m}(G \cap \mathfrak{m}^s Q) = \mathfrak{m}G$. If $g_1, \dots, g_r$ generate G, write $g_i = \sum_j a_{ij} g_j$ with $a_{ij} \in \mathfrak{m}$. Multiplying this system of equations by the adjugate of the matrix $(\delta_{ij} - a_{ij})$ gives $\det(\delta_{ij} - a_{ij})\, g_i = 0$ for every i. Expanding the determinant, we see that it is congruent to 1 modulo $\mathfrak{m}$, hence a unit of the local ring R, and therefore $G = 0$. Applied to $Q = F/E$, for which $\mathfrak{m}^k Q = (E + \mathfrak{m}^k F)/E$, this proves the intersection formula.

For the last assertion, let $f_l \to f$ locally uniformly near x, with germs $f_l \in J$, and work in a holomorphic chart centered at x. By Appendix A.1, a germ belongs to $\mathfrak{m}^k$ if and only if its Taylor coefficients of order less than k vanish; hence these coefficients identify $R/\mathfrak{m}^k$ with a finite-dimensional vector space. By the integral formula for the coefficients in (7.1), the Taylor coefficients of f_l at x converge to those of f, so the images of the f_l in $R/\mathfrak{m}^k$ converge to the image of f. The image of J there is a linear subspace of a finite-dimensional vector space, hence closed, so $f \in J + \mathfrak{m}^k$ for every k. The intersection formula with $F = R$ and $E = J$ gives $f \in J$. □

Recall from Appendix A.1 that a sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ is locally finitely generated if every point has a neighborhood U with sections $g_1, \dots, g_r \in \mathcal{I}(U)$ whose germs generate $\mathcal{I}_y$ for every $y \in U$. For such sheaves the identity theorem takes the following form.

Lemma 12.2.2 *Let $\mathcal{I}$ be a locally finitely generated sheaf of ideals on a connected complex manifold W. If $\mathcal{I}_y = 0$ for some $y \in W$, then $\mathcal{I}_z = 0$ for every $z \in W$.*

Proof The set of points z with $\mathcal{I}_z = 0$ is open: near such a point, choose sections $g_1, \dots, g_r$ generating all stalks of $\mathcal{I}$ on a neighborhood; their germs at z vanish, so they vanish on a neighborhood U' of z, and $\mathcal{I}$ has zero stalks on U'. The set of points with $\mathcal{I}_z \neq 0$ is open as well: a section with nonzero germ at z, defined on a connected neighborhood of z, has nonzero germs at all points of this neighborhood, by the identity theorem. Since W is connected, one of the two sets is empty. □

In the proof of the strong Noetherian property we reduce the number of variables by the following finiteness lemma for Weierstrass polynomials. For an open set $W' \subset \mathbb{C}^{n-1}$, a disc $\Delta \subset \mathbb{C}$, the projection $\pi\colon W' \times \Delta \to W'$, and a sheaf $\mathcal{G}$ of $\mathcal{O}_{W' \times \Delta}$-modules, the *direct image* $\pi_* \mathcal{G}$ is the sheaf $V' \mapsto \mathcal{G}(V' \times \Delta)$ on W'; it is a

sheaf of $\mathcal{O}_{W'}$-modules, a holomorphic function c on V' acting by multiplication with $c \circ \pi$.

Lemma 12.2.3 *Let $n \geq 1$, let $W' \subset \mathbb{C}^{n-1}$ be open, let $\Delta = \{w \in \mathbb{C} : |w| < r\}$, and let*

$$P(z', w) = w^\mu + a_1(z')w^{\mu-1} + \cdots + a_\mu(z'), \qquad \mu \geq 1,$$

with $a_1, \ldots, a_\mu$ holomorphic on W', such that for some $\rho < r$ all roots of $P(z', \cdot)$, $z' \in W'$, lie in the disc $|w| \leq \rho$. Put $D = W' \times \Delta$ and $Z = \{P = 0\} \subset D$, let $\pi \colon D \to W'$ be the projection, and let $P\mathcal{O}_D$ be the ideal generated by P.

(1) *Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_D$-modules with $\mathcal{G}_z = 0$ for $z \in D \setminus Z$. Then for every $b \in W'$ the map*

$$(\pi_*\mathcal{G})_b \to \bigoplus_{z \in Z \cap \pi^{-1}(b)} \mathcal{G}_z,$$

which sends the germ at b of a section over $V' \times \Delta$ to its germs at the points of the fiber, is bijective.

(2) *The morphism $\Phi \colon \mathcal{O}_{W'}^\mu \to \pi_*(\mathcal{O}_D/P\mathcal{O}_D)$ that sends $(c_0, \ldots, c_{\mu-1})$ to the class of $\sum_k c_k w^k$ is an isomorphism.*

(3) *Let $\mathcal{I} \subset \mathcal{O}_D$ be a coherent ideal with $P \in \mathcal{I}(D)$, and let $\mathcal{I}/P\mathcal{O}_D$ denote its image in $\mathcal{O}_D/P\mathcal{O}_D$. Then $\mathcal{M}(\mathcal{I}) = \Phi^{-1}(\pi_*(\mathcal{I}/P\mathcal{O}_D))$ is a coherent subsheaf of $\mathcal{O}_{W'}^\mu$. If $\mathcal{I}'$ is a second such ideal and $\mathcal{M}(\mathcal{I}) = \mathcal{M}(\mathcal{I}')$ on an open set $V' \subset W'$, then $\mathcal{I} = \mathcal{I}'$ on $V' \times \Delta$.*

Proof For $z' \in W'$, the polynomial $P(z', \cdot)$ has at most μ roots, all in Δ; so the fibers of Z over W' are finite. We first describe Z near a fiber. Let $b \in W'$, let $w_1, \ldots, w_l$ be the distinct roots of $P(b, \cdot)$, and put $z_i = (b, w_i)$. Let $\delta > 0$ be so small that the closed discs $|w - w_i| \leq 2\delta$ are pairwise disjoint and contained in Δ. The function $|P|$ is continuous and positive on the compact set $\{b\} \times K$, where $K = \{|w| \leq \rho\} \setminus \bigcup_i \{|w - w_i| < \delta\}$; hence it is positive on $V_b \times K$ for a neighborhood V_b of b, and for $z' \in V_b$ all roots of $P(z', \cdot)$ lie in $\bigcup_i \{|w - w_i| < \delta\}$. Put $B_i = V_b \times \{|w - w_i| < 2\delta\}$. These open sets are pairwise disjoint, and $Z \cap (V_b \times \Delta) \subset \bigcup_i B_i$. Moreover, given neighborhoods N_i of the points z_i, we can achieve $V_b \times \{|w - w_i| \leq 2\delta\} \subset N_i$ for all i: first decrease δ so that $\{b\} \times \{|w - w_i| \leq 2\delta\} \subset N_i$, and then decrease V_b, using the compactness of these discs. We call such δ and V_b adapted to b; a smaller neighborhood of b remains adapted.

(1) Let s be a section of $\mathcal{G}$ over $V' \times \Delta$ whose germs at all the z_i vanish. Then s vanishes on neighborhoods N_i of the z_i; choose δ and $V_b \subset V'$ adapted to b with $B_i \subset N_i$. The open sets $B_1, \ldots, B_l$ and $(V_b \times \Delta) \setminus Z$ cover $V_b \times \Delta$, and s vanishes on each of them, on the last one because all stalks of $\mathcal{G}$ vanish there. Hence $s = 0$ on $V_b \times \Delta$, and the map is injective. Conversely, let $g_i \in \mathcal{G}_{z_i}$ be given, represented by sections t_i on neighborhoods N_i of the z_i, and choose δ and V_b adapted to b with $B_i \subset N_i$. On the open cover of $V_b \times \Delta$ formed by the B_i and $(V_b \times \Delta) \setminus Z$, consider the sections $t_i|_{B_i}$ and 0. They agree on the overlaps: the B_i are pairwise disjoint, and t_i vanishes on $B_i \setminus Z$, since $\mathcal{G}$ has zero stalks there. Gluing them, we obtain a section over $V_b \times \Delta$ with germs g_i at the z_i. Hence the map is surjective.

(2) The map Φ is a morphism of sheaves of $\mathcal{O}_{W'}$-modules. It is injective on sections: suppose that the class of $R = \sum_{k<\mu} c_k w^k$ vanishes on $V' \times \Delta$, and let $z' \in V'$. If w_0 is a root of $P(z', \cdot)$ of multiplicity m, then $R = Pq$ near (z', w_0) with q holomorphic, so the polynomial $R(z', \cdot) = P(z', \cdot)q(z', \cdot)$ vanishes to order at least m at w_0. Thus $R(z', \cdot)$, a polynomial of degree less than μ, has at least μ zeros counted with multiplicity, and therefore vanishes; that is, $c_k(z') = 0$ for all k. To prove surjectivity on stalks, let s be a section of $\mathcal{O}_D/P\mathcal{O}_D$ over $V' \times \Delta$ and $b \in V'$. Near each z_i, s is the class of a holomorphic function f_i on a neighborhood N_i of z_i. Choose δ and $V_b \subset V'$ adapted to b with $V_b \times \{|w - w_i| \le 2\delta\} \subset N_i$. On the circles $|\zeta - w_i| = 2\delta$ the function $P(z', \zeta)$ does not vanish for $z' \in V_b$, since these circles have distance at least δ from all roots. For $z' \in V_b$ and $w \in \mathbb{C}$ put

$$R(z', w) = \frac{1}{2\pi \mathrm{i}} \sum_{i=1}^{l} \int_{|\zeta - w_i| = 2\delta} \frac{f_i(z', \zeta)}{P(z', \zeta)} \frac{P(z', \zeta) - P(z', w)}{\zeta - w} \, \mathrm{d}\zeta.$$

The difference quotient is a polynomial in w of degree $\mu - 1$ whose coefficients are polynomials in ζ and in the $a_k(z')$; hence R is a polynomial in w of degree less than μ whose coefficients are holomorphic on V_b, by differentiation under the integral sign. For $z' \in V_b$ and $|w - w_i| < 2\delta$ put

$$q_i(z', w) = \frac{1}{2\pi \mathrm{i}} \sum_{j=1}^{l} \int_{|\zeta - w_j| = 2\delta} \frac{f_j(z', \zeta)}{P(z', \zeta)(\zeta - w)} \, \mathrm{d}\zeta,$$

a holomorphic function on B_i. Adding the two integrals, we obtain

$$P(z', w)q_i(z', w) + R(z', w) = \frac{1}{2\pi \mathrm{i}} \sum_{j=1}^{l} \int_{|\zeta - w_j| = 2\delta} \frac{f_j(z', \zeta)}{\zeta - w} \, \mathrm{d}\zeta = f_i(z', w),$$

since the term $j = i$ equals $f_i(z', w)$ by Cauchy's formula, and the terms $j \neq i$ vanish by Cauchy's theorem, the point w lying outside the closed disc $|\zeta - w_j| \le 2\delta$. Hence $f_i \equiv R$ modulo P on B_i, so s and the class of R agree on each B_i; on $(V_b \times \Delta) \setminus Z$ both vanish, since P is a unit there. Thus $s = \Phi(c)$ on $V_b \times \Delta$, where c is the tuple of coefficients of R. Being injective, and surjective on all stalks, Φ is an isomorphism.

(3) The sheaf $\mathcal{M}(\mathcal{I})$ is a subsheaf of $\mathcal{O}_{W'}$-modules of $\mathcal{O}_{W'}^{\mu}$. Since $\mathcal{O}_{W'}$ is coherent (Theorem A.1.5), the last assertion of Lemma A.1.4 shows that it suffices to prove that $\mathcal{M}(\mathcal{I})$ is locally finitely generated. Note that $\mathcal{I}/P\mathcal{O}_D$ has zero stalks off Z, because P is a unit there; so (1) applies to it as well as to $\mathcal{O}_D/P\mathcal{O}_D$.

Fix $b \in W'$, with z_i as above. Since $\mathcal{I}$ is coherent, there are neighborhoods N_i of z_i and sections $f_{i1}, \ldots, f_{i\kappa}$ of $\mathcal{I}$ over N_i generating all stalks of $\mathcal{I}$ on N_i. Choose δ and V_b adapted to b with $B_i \subset N_i$. For $1 \le i \le l$, $1 \le \lambda \le \kappa$ and $0 \le m < \mu$, let $\sigma_{i\lambda m}$ be the section of $\mathcal{I}/P\mathcal{O}_D$ over $V_b \times \Delta$ that equals the class of $w^m f_{i\lambda}$ on B_i and 0 on the other B_j and on $(V_b \times \Delta) \setminus Z$. As in (1), these local sections glue, since the B_j are pairwise disjoint and the class of $w^m f_{i\lambda}$ vanishes on $B_i \setminus Z$, where P is a unit.

We claim that the sections $\Phi^{-1}(\sigma_{i\lambda m})$ of $\mathcal{M}(\mathcal{I})$ over V_b generate all its stalks. Let $b' \in V_b$, and let $y_1, \dots, y_p$ be the points of Z over b'; each of them lies in exactly one B_i. By (1), every element of $\pi_*(\mathcal{I}/P\mathcal{O}_D)_{b'}$ is a sum of elements e whose germs vanish at all points of the fiber except one point y_q, where the germ is some $g \in (\mathcal{I}/P\mathcal{O}_D)_{y_q}$. Let $y_q \in B_i$, and write g as the class of $\sum_\lambda h_\lambda f_{i\lambda}$ with germs h_λ at y_q. By (1) for $\mathcal{O}_D/P\mathcal{O}_D$ and by (2), for each λ there is $c^{(\lambda)} \in \mathcal{O}^{\mu}_{W',b'}$ such that $R_\lambda = \sum_m c_m^{(\lambda)} w^m$ is congruent to h_λ modulo P at y_q and to 0 modulo P at the other points of the fiber. Then the element $\sum_{\lambda,m} c_m^{(\lambda)} \sigma_{i\lambda m}$ of $\pi_*(\mathcal{I}/P\mathcal{O}_D)_{b'}$ has the germ g at y_q and the germ 0 at every other point of the fiber: at the other points in B_i its germ is the class of $\sum_\lambda R_\lambda f_{i\lambda} \in P\mathcal{O}_D$, and at the points outside B_i it vanishes by construction. By the injectivity in (1), it equals e. This proves the claim, and $\mathcal{M}(\mathcal{I})$ is coherent.

Finally, let $\mathcal{M}(\mathcal{I}) = \mathcal{M}(\mathcal{I}')$ on V', and let $z \in V' \times \Delta$. If $z \notin Z$, then $\mathcal{I}_z = \mathcal{O}_{D,z} = \mathcal{I}'_z$, since P is a unit at z. Let $z \in Z$, $b = \pi(z)$, and $f \in \mathcal{I}_z$. By (1), there is an element of $\pi_*(\mathcal{O}_D/P\mathcal{O}_D)_b$ whose germ at z is the class of f and whose germs at the other points of the fiber vanish. It belongs to $\pi_*(\mathcal{I}/P\mathcal{O}_D)_b = \Phi(\mathcal{M}(\mathcal{I})_b) = \Phi(\mathcal{M}(\mathcal{I}')_b) = \pi_*(\mathcal{I}'/P\mathcal{O}_D)_b$, so its germ at z belongs to $(\mathcal{I}'/P\mathcal{O}_D)_z$. Hence $f \in \mathcal{I}'_z + P\mathcal{O}_{D,z} = \mathcal{I}'_z$. By symmetry, $\mathcal{I}_z = \mathcal{I}'_z$. □

Lemma 12.2.4 (Strong Noetherian property) *Let $\mathcal{F}$ be a coherent analytic sheaf on X. Every increasing sequence $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots$ of coherent subsheaves of $\mathcal{F}$ is stationary near every point: each $x \in X$ has a neighborhood U such that $\mathcal{F}_j|_U = \mathcal{F}_{j_0}|_U$ for all $j \geq j_0$, for some index j_0. Consequently, on each compact subset of X the sequence is stationary after one sufficiently large index. In particular, a sheaf of ideals generated by countably many holomorphic functions on X is coherent, and near each point it is generated by finitely many of these functions, forming an initial segment of the sequence.*

Proof The assertion on compact sets follows from the local one by covering a compact set with finitely many of the neighborhoods U and taking the largest of the corresponding indices. To prove the local assertion at a point x, note that $\mathcal{F}$, being locally finitely generated, is near x the image of a surjective morphism $\phi\colon \mathcal{O}^r \to \mathcal{F}$ defined by finitely many generators. The inverse images $\phi^{-1}(\mathcal{F}_j)$ form an increasing sequence of subsheaves of $\mathcal{O}^r$, and they are coherent by Lemma A.1.4, being the kernels of the morphisms $\mathcal{O}^r \to \mathcal{F}/\mathcal{F}_j$ between coherent sheaves; here $\mathcal{O}^r$ is coherent by Theorem A.1.5. Since $\mathcal{F}_j = \phi(\phi^{-1}(\mathcal{F}_j))$, the sequence $(\mathcal{F}_j)$ is stationary wherever $(\phi^{-1}(\mathcal{F}_j))$ is.

We may therefore assume that $\mathcal{F} = \mathcal{O}^r$, and we reduce to the case of ideals, $r = 1$, by induction on r. Let $p\colon \mathcal{O}^r \to \mathcal{O}$ be the projection to the last coordinate. The images $p(\mathcal{F}_j)$ form an increasing sequence of coherent ideals, and the kernels $\mathcal{F}_j \cap (\mathcal{O}^{r-1} \oplus 0)$ of $p|_{\mathcal{F}_j}$ form an increasing sequence of coherent subsheaves of $\mathcal{O}^{r-1}$, by Lemma A.1.4. Suppose that both sequences are constant for $j \geq j_0$ on a neighborhood U of x. If σ is a germ of $\mathcal{F}_j$ at a point of U, with $j \geq j_0$, choose a germ σ' of $\mathcal{F}_{j_0}$ with $p(\sigma') = p(\sigma)$; then $\sigma - \sigma'$ lies in the kernel for the index j, which equals the kernel for j_0, so $\sigma \in \mathcal{F}_{j_0}$. Thus $\mathcal{F}_j = \mathcal{F}_{j_0}$ on U.

For ideals $\mathcal{I}_1 \subset \mathcal{I}_2 \subset \cdots$ we argue by induction on the dimension n of X at x; by the reduction just made, the induction hypothesis applies to coherent subsheaves of $\mathcal{O}^\mu$ in dimension $n-1$. In dimension zero, x is an open point of X and the only ideals of $\mathcal{O}_{X,x} = \mathbb{C}$ are 0 and $\mathbb{C}$, so the increasing sequence $(\mathcal{I}_{j,x})$ is eventually constant, and $U = \{x\}$ has the required property.

Let $n \geq 1$, and let W be a connected coordinate neighborhood of x. If all stalks $\mathcal{I}_{j,x}$ vanish, then all $\mathcal{I}_j$ vanish on W by Lemma 12.2.2, and there is nothing to prove. Otherwise some $\mathcal{I}_{j_1}$ has a nonzero germ g at x. After a linear change of coordinates $z = (z', w)$ centered at x, in which g is distinguished in w (Appendix A.1), the Weierstrass preparation theorem (Theorem 7.4.2) shows that g is a unit times a Weierstrass polynomial $P(z', w)$ of some degree μ, so that the germ of P belongs to $\mathcal{I}_{j_1,x}$. If $\mu = 0$, then g is a unit, the section $1 = g \cdot g^{-1}$ belongs to $\mathcal{I}_{j_1}$ near x, and $\mathcal{I}_j$ is the unit ideal near x for all $j \geq j_1$.

Let $\mu \geq 1$. By (A.1) the roots of $P(z', \cdot)$ tend to 0 as $z' \to 0$, so there is a product $D = W' \times \Delta \subset W$ of a polydisc $W' \subset \mathbb{C}^{n-1}$ centered at 0 and a disc $\Delta = \{|w| < r\}$ such that the coefficients of P are holomorphic on W', P is a section of $\mathcal{I}_{j_1}$ over D, and all roots of $P(z', \cdot)$, $z' \in W'$, lie in $\{|w| \leq \rho\}$ for some $\rho < r$. For $j \geq j_1$ the ideals $\mathcal{I}_j|_D$ contain P, and by Lemma 12.2.3(3) the sheaves $\mathcal{M}(\mathcal{I}_j|_D)$ form an increasing sequence of coherent subsheaves of $\mathcal{O}^\mu_{W'}$. By the induction hypothesis, they are constant for $j \geq j_0$ on a neighborhood V' of 0, for some $j_0 \geq j_1$, and Lemma 12.2.3(3) gives $\mathcal{I}_j = \mathcal{I}_{j_0}$ on the neighborhood $V' \times \Delta$ of x for all $j \geq j_0$. This proves the local assertion.

Finally, let $h_1, h_2, \ldots$ be holomorphic functions on X. The ideals $\mathcal{J}_j$ generated by $h_1, \ldots, h_j$ are locally finitely generated subsheaves of $\mathcal{O}$, hence coherent by Theorem A.1.5 and Lemma A.1.4, and the stalk at y of the ideal $\mathcal{J}$ generated by all the h_l is the union of the stalks $\mathcal{J}_{j,y}$. By the local assertion, every point has a neighborhood U and an index j_0 with $\mathcal{J}_j = \mathcal{J}_{j_0}$ on U for $j \geq j_0$; then $\mathcal{J} = \mathcal{J}_{j_0}$ on U. Hence $\mathcal{J}$ is coherent, and near each point it is generated by $h_1, \ldots, h_{j_0}$. □

Definition 12.2.5 Let φ be a weight on X. Its *multiplier ideal* is the subsheaf $\mathcal{I}(\varphi) \subset \mathcal{O}_X$ with stalks

$$\mathcal{I}(\varphi)_x = \{f \in \mathcal{O}_{X,x} : |f|^2 \mathrm{e}^{-\varphi} \text{ is integrable near } x\}.$$

Here $|f|^2\mathrm{e}^{-\varphi}$ is taken to be 0 where $f = 0$, so the zero germ always belongs to $\mathcal{I}(\varphi)_x$. The same definition applies to a function φ that is plurisubharmonic or identically $-\infty$ on each component, as happens for restrictions of weights; where φ is identically $-\infty$, the multiplier ideal is zero.

These stalks define a subsheaf of $\mathcal{O}_X$: if $|f|^2\mathrm{e}^{-\varphi}$ is integrable on a neighborhood V of x, then the germs of f at all points of V belong to $\mathcal{I}(\varphi)$. Each $\mathcal{I}(\varphi)_x$ is an ideal, because holomorphic germs are locally bounded and $|f+g|^2 \leq 2|f|^2 + 2|g|^2$. If $\varphi \leq \psi + C$ near x for a constant C, then $\mathrm{e}^{-\psi} \leq \mathrm{e}^C \mathrm{e}^{-\varphi}$, and therefore $\mathcal{I}(\varphi)_x \subset \mathcal{I}(\psi)_x$. In particular, adding a bounded function to φ does not change its multiplier ideal.

We use two elementary estimates for holomorphic functions.

Lemma 12.2.6 *Let F be a holomorphic function on an open set $D \subset \mathbb{C}^n$, $n \geq 1$.*

(1) *If $\overline{B(y,\delta)} \subset D$, then*

$$|F(y)|^2 \leq \frac{1}{|B(y,\delta)|} \int_{B(y,\delta)} |F|^2 \, dV. \tag{12.7}$$

(2) *Let $x \in D$, suppose that the germ of F at x is not zero, and let $P_k \neq 0$ be the first nonzero homogeneous term of the expansion of F at x. Then for all sufficiently small $t > 0$*

$$\int_{S^{2n-1}} |F(x+t\omega)|^2 \, dS(\omega) \geq t^{2k} \int_{S^{2n-1}} |P_k|^2 \, dS, \qquad \int_{S^{2n-1}} |P_k|^2 \, dS > 0. \tag{12.8}$$

Proof (1) The function F is smooth with $\Delta F = 4 \sum_j \partial_{z_j} \partial_{\bar{z}_j} F = 0$, by (7.5); so its real and imaginary parts are harmonic and have the ball mean value property (Proposition 2.1.1). Thus $F(y)$ is the mean of F over $B(y,\delta)$, and the Cauchy–Schwarz inequality gives (12.7).

(2) Homogeneous polynomials P_l, $P_{l'}$ of different degrees are orthogonal in L^2 of the unit sphere: $P_l(e^{i\theta}\omega) = e^{il\theta} P_l(\omega)$, and surface measure is invariant under the orthogonal maps $\omega \mapsto e^{i\theta}\omega$ (Section 2.1), so $\int_{S^{2n-1}} P_l \overline{P_{l'}} \, dS = e^{i(l-l')\theta} \int_{S^{2n-1}} P_l \overline{P_{l'}} \, dS$ for every θ. For small $t > 0$ the homogeneous expansion $F(x+t\omega) = \sum_l t^l P_l(\omega)$ converges uniformly in $\omega \in S^{2n-1}$, by (7.1); integrating term by term, we obtain $\int_{S^{2n-1}} |F(x+t\omega)|^2 \, dS(\omega) = \sum_{l \geq k} t^{2l} \int_{S^{2n-1}} |P_l|^2 \, dS$, which implies the first inequality. The last integral is positive, because P_k does not vanish identically on the sphere, by homogeneity, and nonempty relatively open subsets of the sphere have positive surface measure. □

Theorem 12.2.7 *For every weight φ on X, the sheaf $\mathcal{I}(\varphi)$ is coherent. More precisely, let $\Omega \subset X$ be an open set that a holomorphic chart identifies with a bounded open subset of $\mathbb{C}^n$, pseudoconvex if $n \geq 1$; for instance, Ω may be a bounded polydisc in a chart (Corollary 11.2.5). Let ϱ be a measurable function on Ω with $c_\varrho \leq \varrho \leq C_\varrho$ for positive constants c_ϱ and C_ϱ, and let*

$$\mathcal{H}_\varphi(\Omega, \varrho) = \Big\{ F \in \mathcal{O}(\Omega) : \|F\|^2_{\varphi,\varrho} = \int_\Omega |F|^2 e^{-\varphi} \varrho \, dV < \infty \Big\},$$

with the inner product $\langle F, G \rangle_{\varphi,\varrho} = \int_\Omega F \bar{G} \, e^{-\varphi} \varrho \, dV$, where dV is Lebesgue measure in the chart (the counting measure if $n = 0$). For $\varrho = 1$ we write $\mathcal{H}_\varphi(\Omega)$, $\|\cdot\|_\varphi$ and $\langle \cdot, \cdot \rangle_\varphi$. Then the following hold.

(1) *For every compact set $K \subset \Omega$ there is a constant C_K such that $|F(z)| \leq C_K \|F\|_{\varphi,\varrho}$ for all $F \in \mathcal{H}_\varphi(\Omega, \varrho)$ and $z \in K$.*

(2) *The space $\mathcal{H}_\varphi(\Omega, \varrho)$ is a separable Hilbert space. It has orthonormal bases, that is, orthonormal families whose linear span is dense, and each of them is finite or countable. If $(e_j)_{j \geq 1}$ is an enumeration of an orthonormal basis and $F \in \mathcal{H}_\varphi(\Omega, \varrho)$, then the partial sums $\sum_{j \leq N} \langle F, e_j \rangle_{\varphi,\varrho} e_j$ converge to F in norm and locally uniformly on Ω, and $\sum_j |\langle F, e_j \rangle_{\varphi,\varrho}|^2 = \|F\|^2_{\varphi,\varrho}$.*

(3) *The sheaf $\mathcal{I}(\varphi)|_\Omega$ is generated by the germs of the elements of $\mathcal{H}_\varphi(\Omega,\varrho)$. In fact, if $(e_j)_{j\ge1}$ is as in (2), then the functions e_j generate $\mathcal{I}(\varphi)|_\Omega$, and near each point of Ω already an initial segment $e_1,\dots,e_k$ generates it.*

Proof If $n=0$, then Ω is empty, and there is nothing to prove, or a point x; in the latter case $\varphi(x)$ is finite, $\mathcal{H}_\varphi(\Omega,\varrho)=\mathbb{C}$, and $\mathcal{I}(\varphi)_x=\mathbb{C}$ is generated by every nonzero element, so all assertions are immediate. Assume therefore $n\ge1$, and write $\mathcal{H}$, $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ for $\mathcal{H}_\varphi(\Omega,\varrho)$, its norm and its inner product. In the chart, $\varphi\in\mathrm{PSH}(\Omega)$ (Section 8.2). We prove (1) and (2) by Hilbert space arguments, and (3) by solving a $\bar\partial$-equation with a weight that has a logarithmic pole.

We first prove (1). Let $K\subset\Omega$ be compact, let $r>0$ be half the distance from K to $\mathbb{C}^n\setminus\Omega$, and let M be the maximum of the upper semicontinuous function φ on the compact set $K_r=\{z:\operatorname{dist}(z,K)\le r\}\subset\Omega$. Since $\mathrm{e}^{-\varphi}\varrho\ge c_\varrho\mathrm{e}^{-M}$ on K_r, the inequality (12.7) on $\overline{B(z,r)}\subset K_r$ gives, for $z\in K$ and $F\in\mathcal{H}$,

$$|F(z)|^2\le\frac{1}{|B(z,r)|}\int_{B(z,r)}|F|^2\,\mathrm{d}V\le\frac{\mathrm{e}^M}{c_\varrho|B(z,r)|}\|F\|^2.$$

This proves (1).

Next we show that $\mathcal{H}$ is a separable Hilbert space. The measurable function $\varphi-\log\varrho$ is bounded above on compact subsets of Ω, and $L^2(\Omega,\mathrm{e}^{-\varphi+\log\varrho})$ is the L^2 space of the measure $\mathrm{e}^{-\varphi}\varrho\,\mathrm{d}V$. By Lemma 11.3.5(2), applied to this function, the elements of this space that are represented by holomorphic functions form a closed subspace. A holomorphic function is determined by its class, since continuous functions that agree almost everywhere agree everywhere; hence $\mathcal{H}$ is identified with this closed subspace and is a Hilbert space.

Multiplication by $\mathrm{e}^{-\varphi/2}\varrho^{1/2}$ maps $L^2(\Omega,\mathrm{e}^{-\varphi+\log\varrho})$ isometrically into $L^2(\Omega)$, since φ is locally integrable (Theorem 8.2.1) and hence finite almost everywhere. The space $L^2(\Omega)$ is separable: by Lemma 1.4.1(5), $C_c(\Omega)$ is dense in it. Let $\mathcal{S}$ be the countable set of functions that take constant values in $\mathbb{Q}+\mathrm{i}\mathbb{Q}$ on finitely many half-open cubes $\prod_{i=1}^{2n}[m_i2^{-k},(m_i+1)2^{-k})$, $m_i\in\mathbb{Z}$, $k\ge1$, whose closures lie in Ω, and vanish elsewhere. Let $g\in C_c(\Omega)$. For large k, the cubes of side 2^{-k} of this form that meet $\operatorname{Supp}g$ lie in a fixed compact subset K' of Ω. Assigning to each of them a value in $\mathbb{Q}+\mathrm{i}\mathbb{Q}$ within 2^{-k} of the value of g at one of its points, we obtain $s_k\in\mathcal{S}$ vanishing outside K' with $\sup|g-s_k|\le2^{-k}+\sup\{|g(y)-g(y')|:|y-y'|\le\sqrt{2n}\,2^{-k}\}$, which tends to zero by uniform continuity of g; hence $s_k\to g$ in $L^2(\Omega)$, and $\mathcal{S}$ is dense in $L^2(\Omega)$. A subset of a separable metric space is separable, so $\mathcal{H}$ is separable.

Orthonormal bases of $\mathcal{H}$ exist: applying the Gram–Schmidt process to a countable dense sequence and discarding the vectors that depend linearly on the preceding ones, we obtain an orthonormal family whose linear span contains the dense sequence. Every orthonormal family is finite or countable, because two of its members have distance $\sqrt2$, so that the balls of radius $1/2$ around them are disjoint, and each of these balls contains a member of a fixed countable dense set. Let $(e_j)_{j\ge1}$ be an enumeration of an orthonormal basis, let $F\in\mathcal{H}$, and put $c_j=\langle F,e_j\rangle$ and $S_N=\sum_{j\le N}c_je_j$. Since $F-S_N$ is orthogonal to the span V_N of $e_1,\dots,e_N$, every $v\in V_N$ satisfies $\|F-v\|^2=\|F-S_N\|^2+\|S_N-v\|^2\ge\|F-S_N\|^2$. Given $\varepsilon>0$, there is v in some

V_{N_0} with $\|F - v\| < \varepsilon$, and since $V_{N_0} \subset V_N$ for $N \geq N_0$, we get $\|F - S_N\| < \varepsilon$ for all $N \geq N_0$. Thus $S_N \to F$ in $\mathcal{H}$, and hence locally uniformly on Ω, by (1). Taking $v = 0$ above gives $\|F\|^2 = \|F - S_N\|^2 + \sum_{j \leq N} |c_j|^2$, and letting $N \to \infty$ we obtain $\sum_j |c_j|^2 = \|F\|^2$. This proves (2).

We turn to (3). Let $\mathcal{J}$ be the ideal sheaf on Ω generated by the e_j. By Lemma 12.2.4 it is coherent and near each point generated by an initial segment $e_1, \ldots, e_k$. Every $F \in \mathcal{H}$ is the locally uniform limit of the partial sums S_N, whose germs lie in $\mathcal{J}$, so the germs of F lie in $\mathcal{J}$ by Lemma 12.2.1. Thus $\mathcal{J}$ is the ideal generated by $\mathcal{H}$, independently of the basis, and $\mathcal{J} \subset \mathcal{I}(\varphi)|_\Omega$ because $\int_\Omega |F|^2 \mathrm{e}^{-\varphi}\, \mathrm{d}V \leq c_\varrho^{-1} \|F\|^2$ for $F \in \mathcal{H}$.

It remains to prove that $\mathcal{I}(\varphi)_x \subset \mathcal{J}_x$ for $x \in \Omega$. Fix $f \in \mathcal{I}(\varphi)_x$ and an integer $s \geq 0$; we construct $F \in \mathcal{H}$ with $f - F \in \mathfrak{m}_x^{s+1}$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. Choose a cutoff function $\chi \in C_c^\infty(\Omega)$, equal to one near x, such that f is holomorphic and $|f|^2 \mathrm{e}^{-\varphi}$ is integrable on a neighborhood of $\operatorname{Supp} \chi$. The $(0,1)$-form $\alpha = f \bar\partial \chi$, extended by zero, is smooth, $\bar\partial$-closed and vanishes near x.

We apply Theorem 11.3.4 on the pseudoconvex set Ω with the plurisubharmonic function $p = \varphi + (n+s) \log |z - x|^2$ (Propositions 8.2.5 and 8.3.3) and the smooth function $q = |z|^2$, whose complex Hessian is the identity matrix. Since $|z - x|^{-2(n+s)}$ and $|\bar\partial \chi|$ are bounded on the support of α, we have $\int_\Omega |\alpha|^2 \mathrm{e}^{-p-q}\, \mathrm{d}V \leq C \int_{\operatorname{Supp}\chi} |f|^2 \mathrm{e}^{-\varphi}\, \mathrm{d}V < \infty$. Hence there is u with $\bar\partial u = \alpha$ and

$$\int_\Omega |u|^2 \mathrm{e}^{-\varphi - |z|^2} |z - x|^{-2(n+s)}\, \mathrm{d}V < \infty.$$

Since e^{-p-q} is bounded below by a positive constant on compact sets, u is locally square integrable. Near x we have $\bar\partial u = 0$, so, by Lemma 11.3.5(1), u agrees almost everywhere near x with a holomorphic function; we replace u by this function near x. We claim that u vanishes to order at least $s+1$ at x. Since φ is bounded above near x, the last display gives $\int_{B(x,r)} |u|^2 |z - x|^{-2(n+s)}\, \mathrm{d}V < \infty$ for small $r > 0$. We may assume that the germ of u at x is not zero, and we let $P_{k_0} \neq 0$ be the first nonzero homogeneous term of its expansion at x. Applying (12.8) of Lemma 12.2.6 to u and integrating in polar coordinates (2.1), we obtain

$$\int_{B(x,r)} |u|^2 |z - x|^{-2(n+s)}\, \mathrm{d}V \geq c \int_0^r t^{2k_0 - 2s - 1}\, \mathrm{d}t$$

with $c > 0$. This is finite only if $k_0 \geq s + 1$, which proves the claim.

Since Ω is bounded, $\mathrm{e}^{-|z|^2} |z - x|^{-2(n+s)}$ is bounded below on Ω by a positive constant, so $|u|^2 \mathrm{e}^{-\varphi}$ is integrable on Ω. The function $\chi f - u$ satisfies $\bar\partial(\chi f - u) = \alpha - \alpha = 0$, so it agrees almost everywhere with a holomorphic function F on Ω, by Lemma 11.3.5(1). Since $|F|^2 \leq 2|\chi f|^2 + 2|u|^2$ almost everywhere, $|\chi f|^2 \mathrm{e}^{-\varphi}$ is integrable by the choice of χ, and $\varrho \leq C_\varrho$, we have $F \in \mathcal{H}$. Near x the continuous functions $f - F$ and u agree almost everywhere, hence everywhere, so $f - F = u \in \mathfrak{m}_x^{s+1}$ by Appendix A.1. Thus $f \in \mathcal{J}_x + \mathfrak{m}_x^{s+1}$ for every s, and Lemma 12.2.1 gives $f \in \mathcal{J}_x$. Hence $\mathcal{I}(\varphi)|_\Omega = \mathcal{J}$, which proves (3).

Finally, every point of X has a neighborhood Ω of the kind considered, namely a bounded polydisc in a chart. If $n \geq 1$, such a polydisc is pseudoconvex by Corollary 11.2.5, being a product of discs, each of which is a sublevel set of the smooth plurisubharmonic exhaustion function $|\zeta - \zeta_0|^2$ of $\mathbb{C}$, where ζ_0 is its center. By (3), $\mathcal{I}(\varphi)$ is generated near each point by finitely many holomorphic functions, so it is coherent by Theorem A.1.5 and the last assertion of Lemma A.1.4. □

Lemma 12.2.8 *Let φ, Ω, ϱ and $\mathcal{H}_\varphi(\Omega, \varrho)$ be as in Theorem 12.2.7, let $x \in \Omega$, and let $g_1, \ldots, g_r \in \mathcal{H}_\varphi(\Omega, \varrho)$ be elements whose germs generate $\mathcal{I}(\varphi)$ on a neighborhood of x. Then there are a neighborhood V of x and a constant C such that every $F \in \mathcal{H}_\varphi(\Omega, \varrho)$ can be written as $F = \sum_i g_i h_i$ on V with $h_i \in \mathcal{O}(V)$ and*

$$\sup_V |h_i| \leq C \|F\|_{\varphi,\varrho}.$$

Consequently, if $m > 0$ is real and $g_1, \ldots, g_r \in \mathcal{H}_{m\varphi}(\Omega)$ generate $\mathcal{I}(m\varphi)$ on a neighborhood of x, then there is a constant C_m such that

$$\varphi \leq \frac{1}{m} \log \sum_i |g_i|^2 + C_m \tag{12.9}$$

on a neighborhood of x, with the convention $\log 0 = -\infty$.

Proof We write $\mathcal{H} = \mathcal{H}_\varphi(\Omega, \varrho)$ and $\|\cdot\| = \|\cdot\|_{\varphi,\varrho}$. Choose balls $V_1 \supset V_2 \supset \cdots$ centered at x, with radii tending to zero, inside the neighborhood on which the g_i generate $\mathcal{I}(\varphi)$. For positive integers j and k let $S_{j,k}$ be the set of $F \in \mathcal{H}$ that can be written as $F = \sum_i g_i h_i$ on V_j with $h_i \in \mathcal{O}(V_j)$ and $\sup_{V_j} |h_i| \leq k$. These sets cover $\mathcal{H}$: by Theorem 12.2.7(3), the germ at x of each $F \in \mathcal{H}$ lies in $\mathcal{I}(\varphi)_x$, so $F = \sum_i g_i h_i$ on some neighborhood of x with holomorphic h_i, and these coefficients are bounded on a smaller ball V_j. We show that each $S_{j,k}$ is closed and then apply Baire's theorem.

Let $F_l \in S_{j,k}$ converge to F in $\mathcal{H}$, with coefficients $h_{i,l}$. Then $F_l \to F$ locally uniformly on Ω, by Theorem 12.2.7(1). The coefficients are bounded by k on V_j. Let $K \subset V_j$ be compact, and let $\varepsilon > 0$ be such that the closed polydiscs of radius 2ε centered at points of K lie in V_j. For $y \in K$ and $y' \in B(y, \varepsilon)$, the closed polydisc of radius ε centered at y' then lies in V_j, and the coefficient bound in (7.1) on this polydisc bounds all first partial derivatives of all $h_{i,l}$ at y' by k/ε. So the $h_{i,l}$ are Lipschitz with a common constant on the balls of radius ε centered at points of K, and they are equicontinuous on K. By Proposition 1.6.4, applied on a compact exhaustion of V_j, and a diagonal argument, a subsequence converges locally uniformly on V_j to continuous functions h_i with $|h_i| \leq k$. Locally uniform convergence implies convergence in the sense of distributions, so $\bar\partial h_i = 0$, and the continuous functions h_i are holomorphic by Lemma 11.3.5(1) (if $n = 0$, every function on V_j is holomorphic). Passing to the limit, we obtain $F = \sum_i g_i h_i$ on V_j, that is, $F \in S_{j,k}$.

By Baire's theorem, a complete metric space is not a countable union of closed subsets with empty interior. (If all the sets had empty interior, we could choose closed balls of radii at most 2^{-l}, each contained in the interior of the previous one and

disjoint from the lth set, since the interior of a ball minus a closed set with empty interior is a nonempty open set; their centers would form a Cauchy sequence whose limit lies in every ball but in none of the sets.) Hence some $S_{j,k}$ contains a ball $\{F : \|F - F_0\| < \delta\}$. If $\|F\| < \delta$, then F_0 and $F_0 + F$ both lie in $S_{j,k}$, and subtracting their representations writes F on V_j with coefficients bounded by $2k$. For $F \neq 0$ we apply this to $\delta F/(2\|F\|)$ and obtain the first assertion with $V = V_j$ and $C = 4k/\delta$; for $F = 0$ we take zero coefficients.

For the second assertion, apply the first part with $\varrho = 1$ to the weight $m\varphi$, and let V and C be the resulting neighborhood and constant. Let $a \in V$ with $\varphi(a) > -\infty$. By Corollary 11.4.5, applied on the pseudoconvex set Ω to the plurisubharmonic function $m\varphi$, there is $F \in \mathcal{H}_{m\varphi}(\Omega)$ with $F(a) = 1$ and $\|F\|^2_{m\varphi} \leq C_\Omega \mathrm{e}^{-m\varphi(a)}$, where C_Ω depends only on n and the diameter of Ω; if $n = 0$, then $F = 1$ has these properties with $C_\Omega = 1$. Writing $F = \sum_i g_i h_i$ on V and using the Cauchy–Schwarz inequality, we obtain

$$1 = |F(a)|^2 \leq rC^2 \|F\|^2_{m\varphi} \sum_i |g_i(a)|^2 \leq rC^2 C_\Omega \mathrm{e}^{-m\varphi(a)} \sum_i |g_i(a)|^2.$$

Taking logarithms gives (12.9) at a with $C_m = m^{-1} \log(rC^2 C_\Omega)$. At points where $\varphi = -\infty$ the left side is $-\infty$; by the last display, these points include all common zeros of the g_i. □

12.3 Analytic Lelong level sets and generic values

We now compare multiplier ideals with Lelong numbers. Recall from Proposition 12.1.4 the order $\operatorname{ord}_x f$ of a nonzero germ f at x, computed in local coordinates; since $\operatorname{ord}_x f \geq k$ if and only if $f \in \mathfrak{m}_x^k$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$ (Appendix A.1), it does not depend on the coordinates. For a coherent ideal $\mathfrak{a}$ and a point x with $\mathfrak{a}_x \neq 0$, the *order* of $\mathfrak{a}$ at x is $\operatorname{ord}_x \mathfrak{a} = \min\{\operatorname{ord}_x f : f \in \mathfrak{a}_x,\ f \neq 0\}$. If $g_1, \ldots, g_r$ generate $\mathfrak{a}$ near x, then $\operatorname{ord}_x \mathfrak{a} = \min_i \operatorname{ord}_x g_i$, the minimum being taken over the g_i with nonzero germ at x, since every element $\sum_i h_i g_i$ has at least this order. By Lemma 12.2.2, a coherent ideal on a connected complex manifold that has a nonzero stalk at one point has nonzero stalks everywhere, so its order is finite at every point.

Lemma 12.3.1 *Let φ be a weight on X, let $x \in X$ be a point at which X has dimension $n \geq 1$, and let $m > 0$ be a real number. Then*

$$m\nu(\varphi, x) - n < \operatorname{ord}_x \mathcal{I}(m\varphi) \leq m\nu(\varphi, x). \tag{12.10}$$

Proof We work on a bounded polydisc Ω around x in a chart on which φ is bounded above. Subtracting a constant changes neither side of (12.10), so we may assume $\varphi \leq 0$ on Ω, and we write $\nu = \nu(\varphi, x)$. The space $\mathcal{H}_{m\varphi}(\Omega)$ is not zero: choose $a \in \Omega$ with $\varphi(a) > -\infty$ and apply Corollary 11.4.5. By Theorem 12.2.7(3), $\mathcal{I}(m\varphi)_x$ is

generated by germs of elements of $\mathcal{H}_{m\varphi}(\Omega)$; since Ω is connected, a nonzero element of $\mathcal{H}_{m\varphi}(\Omega)$ has a nonzero germ at x, by the identity theorem. Hence the order k of $\mathcal{I}(m\varphi)$ at x is the smallest vanishing order at x of the nonzero elements of $\mathcal{H}_{m\varphi}(\Omega)$.

For the lower bound, let $F \in \mathcal{H}_{m\varphi}(\Omega)$ vanish to order exactly k at x, and let $P_k \neq 0$ be its homogeneous term of degree k. By (12.2), computed in the coordinates of the chart, and since $\varphi \leq 0$, we have $\varphi(z) \leq \nu \log(|z-x|^2/r_0^2)$ for $0 < |z-x| < r_0$ and a suitable $r_0 > 0$; hence $\mathrm{e}^{-m\varphi(z)} \geq c|z-x|^{-2m\nu}$ there, with $c = r_0^{2m\nu}$. By (12.8) and polar coordinates (2.1), for small $r > 0$

$$\int_{B(x,r)} |F|^2 \mathrm{e}^{-m\varphi}\, \mathrm{d}V \geq c' \int_0^r t^{2k-2m\nu+2n-1}\, \mathrm{d}t$$

with $c' > 0$. The left side is finite, so $2k - 2m\nu + 2n > 0$, that is, $k > m\nu - n$.

For the upper bound, we use that every element of $\mathcal{H}_{m\varphi}(\Omega)$ vanishes to order at least k at x. Choose $R > 0$ such that the polydisc Δ of polyradius $(2R, \ldots, 2R)$ centered at x is relatively compact in Ω. Let $\delta > 0$ be half the distance from $\overline{\Delta}$ to the complement of Ω. Put $A = |B(0,\delta)|^{-1/2}$. For $G \in \mathcal{H}_{m\varphi}(\Omega)$, (12.7) gives $\sup_{\overline{\Delta}} |G| \leq A\|G\|_{L^2(\Omega)} \leq A\|G\|_{m\varphi}$, and the coefficient bound in (7.1) gives $|c_\alpha| \leq (2R)^{-|\alpha|} \sup_{\overline{\Delta}} |G|$ for the Taylor coefficients c_α of G at x. Since $c_\alpha = 0$ for $|\alpha| < k$ and $|(a-x)^\alpha| \leq |a-x|^{|\alpha|}$, for $|a-x| < R$ we obtain

$$|G(a)| \leq A\|G\|_{m\varphi} \sum_{|\alpha|\geq k} \Big(\frac{|a-x|}{2R}\Big)^{|\alpha|} \leq C_k |a-x|^k \|G\|_{m\varphi}.$$

For the second inequality, note that there are $\binom{d+n-1}{n-1}$ multi-indices of length d; with $\rho = |a-x|/(2R) \leq 1/2$ we get $\sum_{d\geq k} \binom{d+n-1}{n-1}\rho^d \leq \rho^k \sum_{e\geq 0} \binom{e+k+n-1}{n-1} 2^{-e}$, a convergent series. Now let $|a-x| < R$ with $\varphi(a) > -\infty$, and let G be given by Corollary 11.4.5, so that $G(a) = 1$ and $\|G\|_{m\varphi}^2 \leq C_\Omega \mathrm{e}^{-m\varphi(a)}$, where C_Ω depends only on n and the diameter of Ω. Then $1 \leq C_k^2 C_\Omega |a-x|^{2k} \mathrm{e}^{-m\varphi(a)}$, that is,

$$\varphi(a) \leq \frac{k}{m} \log|a-x|^2 + C';$$

here $C' = m^{-1}\log(C_k^2 C_\Omega)$, and the inequality holds trivially where $\varphi(a) = -\infty$. Since it holds for all a with $|a-x| < R$, the characterization in Proposition 12.1.1 gives $k/m \leq \nu$. □

Corollary 12.3.2 *Let $x \in X$, and let u be a function on a neighborhood of x that near x is the sum $u = v + h$ of a plurisubharmonic function v and a bounded measurable function h. Put $\nu(u,x) = \nu(v,x)$; by Proposition 12.1.1 this does not depend on the decomposition. If $\nu(u,x) = 0$, then e^{-tu} is integrable on a neighborhood of x for every $t > 0$. This applies in particular to a function u that is* quasi-plurisubharmonic *near x, that is, the sum of a plurisubharmonic function and a smooth real function near x. For a weight φ on X with $\nu(\varphi,x) = 0$ it says that $\mathcal{I}(t\varphi)_x = \mathcal{O}_{X,x}$ for every $t > 0$.*

Proof If $u = v + h = v' + h'$ are two such decompositions near x, then v and v' take the value $-\infty$ at the same points, and $v \le v' + C$ and $v' \le v + C$ near x, where C bounds $|h| + |h'|$; so $\nu(v,x) = \nu(v',x)$ by Proposition 12.1.1. If X has dimension zero at x, then $v(x)$ is finite and there is nothing to prove. Otherwise let U be a connected neighborhood of x on which v is plurisubharmonic and $|h| \le M$ for a constant M. Applying Lemma 12.3.1 to the weight v on U and $m = t$, we obtain $\operatorname{ord}_x \mathcal{I}(tv) \le t\nu(v,x) = 0$. Hence $\mathcal{I}(tv)_x$ contains a germ that does not vanish at x; this germ is a unit, so $\mathcal{I}(tv)_x = \mathcal{O}_{X,x}$, that is, e^{-tv} is integrable on a neighborhood of x. Since $\mathrm{e}^{-tu} \le \mathrm{e}^{tM}\mathrm{e}^{-tv}$ on U, the same holds for e^{-tu}. For a weight φ, the integrability of $\mathrm{e}^{-t\varphi}$ near x means that $1 \in \mathcal{I}(t\varphi)_x$. □

By Lemma 12.3.1, the Lelong number of φ at x determines the order of $\mathcal{I}(m\varphi)$ at x up to an error less than n. The Lelong numbers do not determine the multiplier ideals themselves, even if they are known at all points, as the following example shows.

Example 12.3.3 On $\mathbb{C}^2$ consider the plurisubharmonic functions (Proposition 8.3.3)

$$\psi = \log\bigl(|z_1|^2 + |z_2|^2\bigr), \qquad \varphi = \log\bigl(|z_1|^2 + |z_2|^4\bigr).$$

They have the same Lelong numbers at every point. Off the origin both functions are finite, so their Lelong numbers vanish there by Proposition 12.1.1. At the origin, $\nu(\psi,0) = 1$ by Proposition 12.1.4. For $|z_2| \le 1$ we have $\varphi \le \psi$, so $\nu(\varphi,0) \ge 1$; conversely, the supremum of φ over the ball $|z| < \mathrm{e}^{t/2}$ is at least $\sup_{|z_1|<\mathrm{e}^{t/2}} \log|z_1|^2 = t$, so $\nu(\varphi,0) \le 1$ by (12.1). Nevertheless $\mathcal{I}(\frac{3}{2}\psi)_0 \ne \mathcal{I}(\frac{3}{2}\varphi)_0$. Indeed, $\mathrm{e}^{-3\psi/2} = |z|^{-3}$ is integrable near $0 \in \mathbb{R}^4$ by polar coordinates, since $\int_0^r t^{-3}t^3\,\mathrm{d}t < \infty$; so $\mathcal{I}(\frac{3}{2}\psi)_0 = \mathcal{O}_{\mathbb{C}^2,0}$. On the other hand, on the set $A_\delta = \{0 < |z_2| < \delta,\ |z_1| < |z_2|^2\}$ we have $|z_1|^2 + |z_2|^4 < 2|z_2|^4$, hence $\mathrm{e}^{-3\varphi/2} > 2^{-3/2}|z_2|^{-6}$, and by Fubini's theorem

$$\int_{A_\delta} \mathrm{e}^{-3\varphi/2}\,\mathrm{d}V \ge 2^{-3/2}\int_{0<|z_2|<\delta} \pi|z_2|^4\,|z_2|^{-6}\,\mathrm{d}V_2(z_2) = 2^{-3/2}\pi\int_0^\delta 2\pi t^{-1}\,\mathrm{d}t = \infty.$$

Since every neighborhood of 0 contains A_δ for small $\delta > 0$, the function $\mathrm{e}^{-3\varphi/2}$ is not integrable near 0, and $1 \notin \mathcal{I}(\frac{3}{2}\varphi)_0$.

Theorem 12.3.4 (Siu) *Let φ be a weight on X and let $c > 0$. Then*

$$E_c(\varphi) = \{x \in X : \nu(\varphi,x) \ge c\}$$

is an analytic subset of X, and its intersection with every connected component of X of positive dimension is a proper subset of that component. The same holds for the Lelong numbers of a closed positive $(1,1)$-current on X.

Proof Lelong numbers vanish on components of dimension zero, so we may assume that X is connected of dimension $n \ge 1$. We describe $E_c(\varphi)$ as an intersection of analytic sets defined by the orders of the ideals $\mathcal{I}(m\varphi)$ and then show that such an

intersection is analytic. Let $\mathfrak{a}$ be a coherent ideal and k an integer. The set of points y where $\operatorname{ord}_y \mathfrak{a} \geq k$ (with $\operatorname{ord}_y \mathfrak{a} = \infty$ if $\mathfrak{a}_y = 0$) is analytic. Indeed, let $g_1, \ldots, g_r$ generate $\mathfrak{a}$ on a coordinate neighborhood W. A germ has order at least k at y if and only if its derivatives of order less than k vanish at y, since these derivatives are multiples of its Taylor coefficients, by (7.1). Hence for $y \in W$ we have $\operatorname{ord}_y \mathfrak{a} \geq k$ if and only if $\partial^\alpha g_i(y) = 0$ for all i and all $|\alpha| < k$: the condition is necessary because the g_i belong to $\mathfrak{a}_y$, and sufficient because, by the Leibniz rule, every element $\sum_i h_i g_i$ then vanishes to order at least k at y. This is a finite system of holomorphic equations, and for $k \leq 0$ the set is all of W. By (12.10),

$$E_c(\varphi) = \bigcap_{m \geq 1} \{x : \operatorname{ord}_x \mathcal{I}(m\varphi) \geq \lfloor mc - n \rfloor + 1\}\,. \tag{12.11}$$

Indeed, if $\nu(\varphi, x) \geq c$, the lower bound in (12.10) gives the integer inequality $\operatorname{ord}_x \mathcal{I}(m\varphi) > mc - n$, that is, $\operatorname{ord}_x \mathcal{I}(m\varphi) \geq \lfloor mc - n \rfloor + 1$, for every m. If $\nu(\varphi, x) < c$, the upper bound gives $\operatorname{ord}_x \mathcal{I}(m\varphi) \leq m\nu(\varphi, x) \leq mc - n < \lfloor mc - n \rfloor + 1$ as soon as $m(c - \nu(\varphi, x)) \geq n$.

Thus $E_c(\varphi)$ is an intersection of analytic subsets of X. An arbitrary intersection $A = \bigcap_\lambda A_\lambda$ of analytic subsets of X is analytic. Since A is closed, it suffices to show that each $p \in A$ has a neighborhood W such that $A \cap W$ is the intersection of W with finitely many of the A_λ. Let $I_\lambda \subset \mathcal{O}_{X,p}$ be the ideal generated by the germs at p of local defining functions of A_λ. By Proposition A.1.2 the sum $\sum_\lambda I_\lambda$ is finitely generated, so it equals $I_{\lambda_1} + \cdots + I_{\lambda_q}$ for finitely many indices; put $B = A_{\lambda_1} \cap \cdots \cap A_{\lambda_q}$. Every local defining function of each A_λ is then a combination of defining functions of $A_{\lambda_1}, \ldots, A_{\lambda_q}$ near p, so A_λ contains a neighborhood of p in B. This neighborhood may depend on λ, and we now remove this dependence.

By Corollary A.2.13, the irreducible components of the analytic set B form a locally finite family; so some neighborhood N of p meets only finitely many of them. Let $B_1, \ldots, B_l$ be those that contain p. The union of the other components meeting N is closed and does not contain p, so there is an open neighborhood $W' \subset N$ of p with $B \cap W' \subset B_1 \cup \cdots \cup B_l$. Fix λ and i. The regular locus $(B_i)_{\mathrm{reg}}$ is a connected manifold, dense in B_i (Corollary A.2.13), and its analytic subset $A_\lambda \cap (B_i)_{\mathrm{reg}}$ has nonempty interior in it, because A_λ contains a neighborhood of p in B_i and p lies in the closure of $(B_i)_{\mathrm{reg}}$. By Lemma A.1.6, $A_\lambda \cap (B_i)_{\mathrm{reg}}$ is therefore all of $(B_i)_{\mathrm{reg}}$, and $B_i \subset A_\lambda$ because A_λ is closed. Therefore $B \cap W' \subset A_\lambda$ for every λ, and $A \cap W' = B \cap W'$. Applied to (12.11), this shows that $E_c(\varphi)$ is analytic.

By Proposition 12.1.1, $\nu(\varphi, x) = 0$ at every point where $\varphi(x) > -\infty$. Thus $E_c(\varphi)$ is contained in the pole set of φ, which has measure zero because φ is locally integrable (Theorem 8.2.1), and therefore has no interior points; so $E_c(\varphi) \neq X$.

Finally, let T be a closed positive $(1,1)$-current on X. Every point of X at which X has positive dimension has a coordinate ball U on which $T = \mathrm{dd}^{\mathrm{c}} u$ with u plurisubharmonic, and $\nu(T, y) = \nu(u, y)$ for $y \in U$ (Definition 12.1.2). By what we have proved, applied to the weight u on U, the set $\{y \in U : \nu(T, y) \geq c\}$ is an analytic subset of U with empty interior. Since the components of X of dimension zero do not meet $\{y : \nu(T, y) \geq c\}$, and analyticity is a local property, this set is an

analytic subset of X whose intersection with every component of positive dimension is proper. □

Proposition 12.3.5 *Let φ be a weight on X and let $Z \subset X$ be an irreducible analytic subset. The* generic Lelong number *of φ along Z is*

$$\nu(\varphi, Z) = \inf_{x \in Z} \nu(\varphi, x).$$

It is attained at every point of Z outside a countable union of proper analytic subsets of Z. Moreover, the complement in Z of every countable union of proper analytic subsets of Z is dense in Z; in particular, $\nu(\varphi, Z)$ is attained on a dense subset of Z. The same definition and conclusions apply to the Lelong numbers of a closed positive $(1,1)$-current T on X, and give the generic Lelong number $\nu(T, Z)$. If $\pi\colon Y \to X$ is a smooth model of X (Appendix A.7), that is, a proper bimeromorphic holomorphic map from a complex manifold Y, and $E \subset Y$ is a prime divisor, that is, an irreducible hypersurface, we define

$$\nu(\varphi, E) = \nu(\varphi \circ \pi, E).$$

This number depends only on the prime divisor over X represented by E (Definition A.7.4), not on the model.

Proof Put $\lambda = \nu(\varphi, Z)$. For every integer $j \geq 1$, Theorem 12.3.4 shows that $Z \cap E_{\lambda+1/j}(\varphi)$ is an analytic subset of Z, and it is a proper subset by the definition of λ. At every point of Z outside the union of these sets, $\nu(\varphi, x) < \lambda + 1/j$ for all j, hence $\nu(\varphi, x) = \lambda$.

We now show that the complement in Z of an arbitrary countable union of proper analytic subsets of Z is dense in Z; applied to the sets $Z \cap E_{\lambda+1/j}(\varphi)$, this shows that $\nu(\varphi, Z)$ is attained on a dense subset of Z. The density is clear if Z is a point, for then the proper analytic subsets of Z are empty. Let $\dim Z > 0$. By Corollary A.2.13, applied to the family consisting of Z alone, Z is an irreducible component of itself; so its regular locus Z_{reg} is a connected manifold, dense in Z.

Let B be a proper analytic subset of Z. Then $B \cap Z_{\mathrm{reg}}$ is an analytic subset of Z_{reg}, and it is proper, since otherwise the closed set B would contain the closure Z of Z_{reg}. By Lemma A.1.6, every point of $B \cap Z_{\mathrm{reg}}$ has a connected open neighborhood U in Z_{reg} with a holomorphic function $g \not\equiv 0$ on U such that $B \cap U \subset \{g = 0\}$; shrinking U to a coordinate ball, on which $g \not\equiv 0$ by the identity theorem, we see from Lemma 7.4.1 that $B \cap U$ is a null set in the chart. Covering a chart N of Z_{reg} by countably many such balls and by open sets disjoint from B, we see that $B \cap N$ is a null set. Consequently, a countable union of proper analytic subsets of Z meets every chart of Z_{reg} in a null set, which has no interior points. Its complement is therefore dense in Z_{reg}, and hence in Z. For a closed positive $(1,1)$-current the same argument applies, since Theorem 12.3.4 holds for its Lelong numbers.

For the second assertion, note first that $\varphi \circ \pi$ is a weight on Y: by Proposition 8.2.3 it is plurisubharmonic or identically $-\infty$ on each component of Y, and the second alternative is excluded. Indeed, since π is bimeromorphic (Appendix A.3), there is a nowhere dense analytic subset $A \subset X$ such that $\pi^{-1}(A)$ is nowhere dense in Y and π

maps $Y \setminus \pi^{-1}(A)$ biholomorphically onto $X \setminus A$. If $\varphi \circ \pi$ were identically $-\infty$ on a component Y_0 of Y, then φ would be $-\infty$ on the nonempty open set $\pi(Y_0 \setminus \pi^{-1}(A))$, which is impossible because φ is locally integrable.

Now let $E \subset Y$ and $E' \subset Y'$ be prime divisors on smooth models $\pi\colon Y \to X$ and $\pi'\colon Y' \to X$ that represent the same prime divisor over X, and let $\rho\colon Y'' \to Y$ and $\rho'\colon Y'' \to Y'$ be maps from a common smooth model on which the strict transforms of E and E' coincide, say with E''. By Proposition A.7.3, ρ is biholomorphic over $Y \setminus S$ for an analytic subset $S \subset Y$ of codimension at least two, so it maps the nonempty open subset $E'' \cap \rho^{-1}(Y \setminus S)$ of E'' onto the nonempty open subset $E \setminus S$ of E, and $\nu(\varphi \circ \pi \circ \rho, q) = \nu(\varphi \circ \pi, \rho(q))$ for q in this set, because Lelong numbers are invariant under biholomorphisms (Proposition 12.1.1). By the first part, the generic value along E is attained on a dense subset of E, so it equals the infimum of the Lelong numbers over any nonempty open subset of E; the same holds for E''. Hence $\nu(\varphi \circ \pi, E) = \nu(\varphi \circ \pi \circ \rho, E'')$. The same argument for ρ' gives $\nu(\varphi \circ \pi', E') = \nu(\varphi \circ \pi \circ \rho, E'')$, since $\pi \circ \rho = \pi' \circ \rho'$. □

Proposition 12.3.6 *Let φ be a weight on X, let $Z \subset X$ be a connected closed complex submanifold of positive codimension, which is an irreducible analytic subset by Corollary A.2.13, and let $z = (z', z'')$ be local coordinates near a point of Z in which $Z = \{z' = 0\}$. Then near that point*

$$\varphi(z) \le \nu(\varphi, Z) \log |z'|^2 + C, \qquad z' \neq 0,$$

for a constant C, and no bound of this form with a coefficient larger than $\nu(\varphi, Z)$ holds near any point of Z. If $\pi\colon Y \to X$ is the blowup of a point x at which X has positive dimension (Proposition A.5.3), and $E = \pi^{-1}(x)$, then $\nu(\varphi, E) = \nu(\varphi, x)$.

Proof Let $p \in Z$ be the given point and $\lambda = \nu(\varphi, Z)$. Choose $r_0 > 0$, a constant C and a neighborhood W of p in Z such that for every $y \in W$ the closed ball $\overline{B(y, r_0)}$ lies in the coordinate domain and $\varphi \le C$ on it. By (12.2) at y, we have $\varphi(z) \le \nu(\varphi, y) \log(|z - y|^2/r_0^2) + C$ for $0 < |z - y| < r_0$. Take $z = (z', z'')$ with $y = (0, z'') \in W$ and $0 < |z'| < r_0$. Then $|z - y| = |z'|$, the logarithm is negative and $\nu(\varphi, y) \ge \lambda$, so $\varphi(z) \le \lambda \log |z'|^2 - \lambda \log r_0^2 + C$. This is the asserted bound, with a constant independent of z''. Conversely, suppose that $\varphi(z) \le c \log |z'|^2 + C'$ for $z' \neq 0$ near a point q of Z, where we may assume $c \ge 0$. For $y \in Z$ near q we have $|z'| \le |z - y|$, so $\varphi(z) \le c \log |z - y|^2 + C'$ for $z \notin Z$ near y. Since $Z = \{z' = 0\}$ is a Lebesgue null set in the chart, Proposition 12.1.1 gives $\nu(\varphi, y) \ge c$. By Proposition 12.3.5 there are points y arbitrarily close to q with $\nu(\varphi, y) = \lambda$, so $c \le \lambda$.

Now let π be the blowup of x, and let $n = \dim_x X$. If $n = 1$, then π is the identity and $E = \{x\}$, so there is nothing to prove. Let $n \ge 2$ and choose coordinates centered at x. The blowup is covered by n charts; in the kth chart, $z_k = w_k$ and $z_j = w_k w_j$ for $j \neq k$, so that

$$|z|^2 = |w_k|^2 \Big(1 + \sum_{j \neq k} |w_j|^2\Big), \qquad E = \{w_k = 0\}.$$

By (12.2), $\varphi \le \nu(\varphi,x)\log|z|^2 + C$ near x. Pulling back, we get $\varphi\circ\pi \le \nu(\varphi,x)\log|w_k|^2 + C'$ on compact subsets of the kth chart near E; since $|w_k|$ is at most the distance to any point of E in that chart, $\nu(\varphi\circ\pi, q) \ge \nu(\varphi,x)$ for every $q \in E$, and $\nu(\varphi,E) \ge \nu(\varphi,x)$. Conversely, $E \cong \mathbb{P}^{n-1}$ is a compact connected submanifold of Y, so the first part of the proposition, applied to the weight $\varphi\circ\pi$ and E, gives $\varphi\circ\pi \le \nu(\varphi,E)\log|w_k|^2 + C$ near each point of E, in the coordinates of a chart containing it. Cover E by finitely many such neighborhoods, on each of which $\log|w_k|^2 - \log|z|^2$ is bounded. Since π is proper, the inverse image of a small neighborhood of x lies in their union, so $\varphi(z) \le \nu(\varphi,E)\log|z|^2 + C''$ near x, for $z \ne x$. Hence $\nu(\varphi,E) \le \nu(\varphi,x)$. □

12.4 Restriction, products and integral closure

The Ohsawa–Takegoshi theorem and the Hilbert spaces of Theorem 12.2.7 determine how multiplier ideals behave under restriction, products and holomorphic pullback; we also show that multiplier ideals are integrally closed.

For a holomorphic map $f\colon X \to Y$ and a coherent ideal $\mathfrak{a} \subset \mathcal{O}_Y$, we denote by $f^{-1}\mathfrak{a}\cdot\mathcal{O}_X$ the ideal of $\mathcal{O}_X$ generated by the compositions $g\circ f$ of local sections g of $\mathfrak{a}$; it is locally generated by the compositions of local generators of $\mathfrak{a}$ with f. It is the ideal written $\mathfrak{a}\mathcal{O}_X$ in Appendix A.5; we keep f in the notation when several maps occur. For a complex submanifold $Z \subset X$ with ideal sheaf $\mathcal{I}_Z$ and an ideal $\mathfrak{a} \subset \mathcal{O}_X$, the *restriction* of $\mathfrak{a}$ to Z is the image

$$\mathfrak{a}|_Z^{\mathrm{im}} = \mathfrak{a}/(\mathfrak{a}\cap\mathcal{I}_Z) \subset \mathcal{O}_X/\mathcal{I}_Z = \mathcal{O}_Z,$$

that is, the ideal of restrictions $g|_Z$ of local sections g of $\mathfrak{a}$. Here $\mathcal{O}_X/\mathcal{I}_Z$ is identified with $\mathcal{O}_Z$ through restriction: in local coordinates (v,w) with $Z = \{w = 0\}$, every holomorphic function on Z is the restriction of a function independent of w, and a function vanishing on Z belongs to $\mathcal{I}_Z$ by definition. We use the image rather than the tensor product $\mathfrak{a}\otimes_{\mathcal{O}_X}\mathcal{O}_Z$, whose natural map to $\mathcal{O}_Z$ need not be injective. For instance, for $X = \mathbb{C}$, $Z = \{0\}$ and $\mathfrak{a} = (z)$, the element $z\otimes 1$ of the stalk at 0 is mapped to $z|_Z = 0$, but it is not zero, since $(zf, g) \mapsto f(0)g(0)$ is a well-defined $\mathcal{O}_{X,0}$-balanced bilinear map $\mathfrak{a}_0\times\mathcal{O}_{Z,0} \to \mathbb{C}$ with value 1 at $(z,1)$.

Theorem 12.4.1 *Let $Z \subset X$ be a complex submanifold and φ a weight on X. Then*

$$\mathcal{I}(\varphi|_Z) \subset \mathcal{I}(\varphi)|_Z^{\mathrm{im}}; \tag{12.12}$$

on a connected component of Z on which $\varphi|_Z \equiv -\infty$, the left side is zero. If φ_1 and φ_2 are weights on complex manifolds X_1 and X_2, and $p_i\colon X_1\times X_2 \to X_i$ are the projections, then

$$\mathcal{I}(\varphi_1\circ p_1 + \varphi_2\circ p_2) = (p_1^{-1}\mathcal{I}(\varphi_1)\cdot\mathcal{O})(p_2^{-1}\mathcal{I}(\varphi_2)\cdot\mathcal{O}). \tag{12.13}$$

In particular, for weights φ and ψ on X,

$$\mathcal{I}(\varphi+\psi) \subset \mathcal{I}(\varphi)\mathcal{I}(\psi). \tag{12.14}$$

Proof The inclusion (12.12) is a restatement of Corollary 11.4.6: if $f \in \mathcal{I}(\varphi|_Z)_x$, that corollary provides a holomorphic germ F at x with $F|_Z = f$ and $F \in \mathcal{I}(\varphi)_x$.

The product formula is local. Near a point (x_1, x_2) we choose bounded polydiscs Ω_i around x_i in charts of X_i on which φ_i is bounded above; subtracting constants, which changes none of the multiplier ideals, we may assume that $\varphi_1, \varphi_2 \le 0$. We write $\Phi = \varphi_1 \circ p_1 + \varphi_2 \circ p_2$, a nonpositive weight on the polydisc $\Omega_1 \times \Omega_2$. Let (e_j) and (g_k) be orthonormal bases of $\mathcal{H}_{\varphi_1}(\Omega_1)$ and $\mathcal{H}_{\varphi_2}(\Omega_2)$, which exist by Theorem 12.2.7(2). By Fubini's theorem the functions $e_j(z)g_k(w)$ form an orthonormal family in $\mathcal{H}_\Phi(\Omega_1 \times \Omega_2)$. We show that its linear span is dense. Let $F \in \mathcal{H}_\Phi(\Omega_1 \times \Omega_2)$. By Fubini's theorem, for almost every $w \in \Omega_2$ we have $\varphi_2(w) > -\infty$ and $F(\cdot, w) \in \mathcal{H}_{\varphi_1}(\Omega_1)$, and for such w we put

$$a_j(w) = \int_{\Omega_1} F(z,w)\overline{e_j(z)}\mathrm{e}^{-\varphi_1(z)}\,\mathrm{d}V_z.$$

We claim that each a_j agrees almost everywhere with an element of $\mathcal{H}_{\varphi_2}(\Omega_2)$ of norm at most $\|F\|_\Phi$. Choose compact sets $K_\ell \uparrow \Omega_1$ and put $E_\ell = K_\ell \cap \{\mathrm{e}^{-\varphi_1} \le \ell\}$; these sets increase and exhaust Ω_1 up to the pole set of φ_1, which has measure zero. The function $a_{j,\ell}$ obtained by integrating over E_ℓ instead of Ω_1 is holomorphic on Ω_2. Indeed, let $L \subset \Omega_2$ be compact. For $z \in E_\ell \subset K_\ell$ and $w \in L$, the integrand $F(z,w)\overline{e_j(z)}\mathrm{e}^{-\varphi_1(z)}$ and all its partial derivatives in w are bounded, because F and its derivatives are continuous on $K_\ell \times L$, e_j is bounded on K_ℓ, and $\mathrm{e}^{-\varphi_1} \le \ell$ on E_ℓ. Hence we may differentiate under the integral sign any number of times: $a_{j,\ell}$ is smooth, and $\partial a_{j,\ell}/\partial \bar{w}_m = 0$ for all m, because F is holomorphic in w. By the Cauchy–Schwarz inequality in z, followed by integration in w against $\mathrm{e}^{-\varphi_2}$,

$$\|a_j - a_{j,\ell}\|^2_{\varphi_2} \le \|F\|^2_\Phi \int_{\Omega_1 \setminus E_\ell} |e_j|^2 \mathrm{e}^{-\varphi_1}\,\mathrm{d}V_z \longrightarrow 0$$

by dominated convergence, since the sets $\Omega_1 \setminus E_\ell$ decrease to a null set and $|e_j|^2\mathrm{e}^{-\varphi_1}$ is integrable. Since $\varphi_2 \le 0$, the holomorphic functions $a_{j,\ell}$ therefore converge to a_j in $L^2(\Omega_2)$, and hence in the sense of distributions; so $\bar\partial a_j = 0$, and a_j agrees almost everywhere with a holomorphic function, by Lemma 11.3.5(1). The same Cauchy–Schwarz estimate over all of Ω_1 gives $\|a_j\|_{\varphi_2} \le \|F\|_\Phi$, which proves the claim.

Now suppose that F is orthogonal to all the products $e_j g_k$. By Fubini's theorem, whose use is justified by the Cauchy–Schwarz inequality, $\langle a_j, g_k\rangle_{\varphi_2} = \langle F, e_j g_k\rangle_\Phi = 0$ for all k. Since the linear span of the g_k is dense in $\mathcal{H}_{\varphi_2}(\Omega_2)$, the element a_j of this space is orthogonal to itself, so $a_j = 0$ almost everywhere, for every j. Hence for almost every w all the coefficients $\langle F(\cdot, w), e_j\rangle_{\varphi_1}$ vanish, and $F(\cdot, w) = 0$, because $F(\cdot, w)$ is the limit of the partial sums of its expansion (Theorem 12.2.7(2)). Thus $F = 0$ almost everywhere, and therefore everywhere, since F is continuous.

Consequently, the closure M of the linear span of the products $e_j g_k$ satisfies $M^\perp = 0$, and $M = \mathcal{H}_\Phi(\Omega_1 \times \Omega_2)$ by Proposition 11.1.1.

We can now prove (12.13). Denote its right side by $\mathfrak{b}$; it is the ideal generated by the products $(f_1 \circ p_1)(f_2 \circ p_2)$ of local sections f_i of $\mathcal{I}(\varphi_i)$. Each product $e_j g_k$ is a section of $\mathfrak{b}$. Every $F \in \mathcal{H}_\Phi(\Omega_1 \times \Omega_2)$ is the norm limit, hence by (12.7) the locally uniform limit, of a sequence of finite linear combinations of these products, so its germs lie in $\mathfrak{b}$ by Lemma 12.2.1. Since $\mathcal{I}(\Phi)$ is generated by the germs of the elements of $\mathcal{H}_\Phi(\Omega_1 \times \Omega_2)$ (Theorem 12.2.7(3)), we obtain $\mathcal{I}(\Phi) \subset \mathfrak{b}$. Conversely, if $|f_i|^2 \mathrm{e}^{-\varphi_i}$ is integrable near x_i for $i = 1, 2$, then $|f_1(z) f_2(w)|^2 \mathrm{e}^{-\Phi(z,w)}$ is integrable near (x_1, x_2) by Fubini's theorem; since $\mathcal{I}(\Phi)$ is an ideal, $\mathfrak{b} \subset \mathcal{I}(\Phi)$.

Finally, let φ and ψ be weights on X, and apply (12.12) to the weight $\varphi \circ p_1 + \psi \circ p_2$ on $X \times X$ and the diagonal, which we identify with X. The restriction of the weight is $\varphi + \psi$, and by (12.13) the restriction of its multiplier ideal is generated by the products $f_1 f_2$ of local sections f_1 of $\mathcal{I}(\varphi)$ and f_2 of $\mathcal{I}(\psi)$, that is, it equals $\mathcal{I}(\varphi)\mathcal{I}(\psi)$. This proves (12.14). □

Corollary 12.4.2 *Let $f\colon X \to Y$ be a holomorphic map of complex manifolds and φ a weight on Y. Then*

$$\mathcal{I}(\varphi \circ f) \subset f^{-1}\mathcal{I}(\varphi) \cdot \mathcal{O}_X,$$

with equality if f is a submersion. On a connected component of X on which $\varphi \circ f \equiv -\infty$, the left side is zero.

Proof Suppose first that f is a submersion, and let $a \in X$. Choose holomorphic coordinates y on Y centered at $f(a)$. Since the differential of f at a is surjective, the complex Jacobian matrix at a of the components of $y \circ f$, computed in a chart of X, has rank $\dim_{f(a)} Y$. By the holomorphic inverse function theorem (Proposition 7.1.2(3)), these components are the first coordinates of a holomorphic chart $(z, w) = (y \circ f, w)$ of X near a. In these coordinates f is the projection $(z, w) \mapsto z$. Thus $\varphi \circ f = \varphi \circ p_1 + 0 \circ p_2$, and (12.13), together with $\mathcal{I}(0) = \mathcal{O}$, gives $\mathcal{I}(\varphi \circ f) = p_1^{-1}\mathcal{I}(\varphi) \cdot \mathcal{O} = f^{-1}\mathcal{I}(\varphi) \cdot \mathcal{O}_X$.

In general, the graph $\Gamma = \{(x, f(x))\}$ is a complex submanifold of $X \times Y$, since in local coordinates (x, y) the functions $(x, y - f(x))$ are holomorphic coordinates in which $\Gamma = \{y - f(x) = 0\}$; and $\iota(x) = (x, f(x))$ is a biholomorphism of X onto Γ. The projection $q\colon X \times Y \to Y$ is a submersion, so $\mathcal{I}(\varphi \circ q) = q^{-1}\mathcal{I}(\varphi) \cdot \mathcal{O}_{X\times Y}$ by the first case. Since $\varphi \circ q \circ \iota = \varphi \circ f$, (12.12) applied to Γ gives $\mathcal{I}(\varphi \circ f) \subset (q^{-1}\mathcal{I}(\varphi) \cdot \mathcal{O}_{X\times Y})|_\Gamma^{\mathrm{im}}$ after transport by ι. The right side is generated by the functions $g \circ q \circ \iota = g \circ f$ with g a local section of $\mathcal{I}(\varphi)$, so it equals $f^{-1}\mathcal{I}(\varphi) \cdot \mathcal{O}_X$. □

Proposition 12.4.3 *Every multiplier ideal is integrally closed: if a germ $F \in \mathcal{O}_{X,x}$ satisfies an equation $F^d + a_1 F^{d-1} + \cdots + a_d = 0$ with $a_j \in \mathcal{I}(\varphi)_x^j$, then $F \in \mathcal{I}(\varphi)_x$. More explicitly, if $g_1, \ldots, g_r \in \mathcal{I}(\varphi)_x$ and a holomorphic germ F satisfies $|F| \le C(\sum_j |g_j|^2)^{1/2}$ near x for a constant C, then $F \in \mathcal{I}(\varphi)_x$.*

Proof The second assertion is immediate, since $|F|^2 \mathrm{e}^{-\varphi} \le C^2 \sum_j |g_j|^2 \mathrm{e}^{-\varphi}$ near x. For the first, choose generators $g_1, \ldots, g_r$ of $\mathcal{I}(\varphi)_x$ (Theorem 12.2.7), representatives

of the g_j, of F and of the a_j on a common neighborhood, and put $|g| = (\sum_j |g_j|^2)^{1/2}$. Each a_j is a finite sum of products of j generators with holomorphic coefficients, which are bounded on a smaller neighborhood; hence $|a_j| \le C_j|g|^j$ there. At a point where $|g| > 0$, put $\lambda = |F|/|g|$. Dividing the equation by $|g|^d$ gives $\lambda^d \le \sum_j C_j \lambda^{d-j}$, which forces $\lambda \le C$ with $C = \max\{1, \sum_j C_j\}$. At a point where $|g| = 0$, all the a_j vanish, so $F^d = 0$ and $F = 0$. Thus $|F| \le C|g|$ near x, and the second assertion applies. □

12.5 Strong openness by extension from moving slices

The constant in the Ohsawa–Takegoshi theorem (Theorem 11.4.4) depends only on the width of the normal direction, not on the weight. This uniformity allows us to extend a germ from a sequence of parallel hyperplanes approaching a point, with weights that vary along the sequence. If a germ were not in the expected ideal, we would obtain extensions whose norms are too small to be compatible with their behavior along a fixed holomorphic curve. The two lemmas below provide this curve and the corresponding norm estimate.

Lemma 12.5.1 *Let $n \ge 1$, let $F, g_1, \dots, g_r$, $r \ge 0$, be holomorphic functions near $0 \in \mathbb{C}^n$ such that the germ of F at 0 is not zero, and put $|g| = (\sum_{j=1}^r |g_j|^2)^{1/2}$, with $|g| = 0$ if $r = 0$. Suppose that there are no neighborhood of 0 and constant C such that $|F| \le C|g|$ on that neighborhood. Then, after a linear change of coordinates, there are an integer $q \ge 1$, a constant C and a holomorphic map*

$$\gamma(t) = (t^q, \gamma_2(t), \dots, \gamma_n(t)), \qquad |\gamma_j(t)| \le C|t|^q,$$

defined on a disc around $0 \in \mathbb{C}$, such that the function $F \circ \gamma$ has no zeros other than $t = 0$, and each quotient $(g_j/F) \circ \gamma$ extends holomorphically across $t = 0$ with value 0 there.

Proof We first construct a nonconstant holomorphic map δ from a disc around $0 \in \mathbb{C}$ to $\mathbb{C}^n$ with $\delta(0) = 0$ such that $F \circ \delta$ has no zeros other than $\tau = 0$ and each $(g_j/F) \circ \delta$ extends holomorphically across $\tau = 0$ with value 0.

If $r = 0$, choose a vector v such that $\zeta \mapsto F(\zeta v)$ does not vanish identically near 0; for instance, v may be any vector at which the first nonzero homogeneous term of F at 0 does not vanish. The zeros of this one-variable function are isolated, so on a small disc it vanishes at most at 0, and $\delta(\tau) = \tau v$ has the required properties.

Let $r \ge 1$. The bound $|F| \le i|g|$ fails on the ball $B(0, 1/i)$ for every large integer i, so there are points z_i with $|z_i| < 1/i$ and $|F(z_i)| > i|g(z_i)|$; in particular $F(z_i) \ne 0$ and $|g(z_i)|/|F(z_i)| < 1/i$. We may take $z_i \ne 0$: if $z_i = 0$, the strict inequality $|F| > i|g|$ at 0 persists at points $z \ne 0$ near 0, because F and $|g|$ are continuous.

Let U be a connected neighborhood of 0 on which all functions are defined, and let $\sigma\colon \widetilde{U} \to U$ be the blowup of the ideal $(F, g_1, \dots, g_r)$, which is not identically

zero on U (Definition A.5.1). Thus $\widetilde{U} \subset U \times \mathbb{P}^r$ is the closure of the graph of $z \mapsto [F(z) : g_1(z) : \cdots : g_r(z)]$ over the complement of the common zeros, and it is a complex space by Proposition A.5.2. The graph points $\tilde{z}_i = (z_i, [F(z_i) : g(z_i)])$ converge to $b = (0, [1 : 0 : \cdots : 0])$, because $g(z_i)/F(z_i) \to 0$. On the part of $\widetilde{U}$ lying over the chart $\{\xi_0 \neq 0\}$ of $\mathbb{P}^r$, the functions ξ_j/ξ_0 are holomorphic and vanish at b, and the identities $g_j \circ \sigma = (\xi_j/\xi_0)(F \circ \sigma)$ hold on the graph, hence, both sides being continuous, on its closure, which is this part of $\widetilde{U}$.

The germ of $\widetilde{U}$ at b is the union of finitely many irreducible branches (Proposition A.2.1); passing to a subsequence, we may assume that all $\tilde{z}_i$ lie on a representative W of one branch. The analytic subgerm of (W, b) formed by $\sigma^{-1}(0) \cup \{F \circ \sigma = 0\}$ is different from (W, b), because W contains the points $\tilde{z}_i$, arbitrarily close to b, at which $z_i \neq 0$ and $F(z_i) \neq 0$; for the same reason (W, b) is not the point b and has positive dimension. By Proposition A.4.2 there is a nonconstant holomorphic map c from a disc into W with $c(0) = b$ such that $c(\tau) \notin \sigma^{-1}(0)$ and $F(\sigma(c(\tau))) \neq 0$ for $\tau \neq 0$; shrinking the disc, we may assume that c takes values over the chart $\{\xi_0 \neq 0\}$. Put $\delta = \sigma \circ c$. Then $\delta(0) = 0$, and for $\tau \neq 0$ we have $\delta(\tau) \neq 0$ and $F(\delta(\tau)) \neq 0$; so δ is not constant, and $F \circ \delta$ has no zeros other than $\tau = 0$. For $\tau \neq 0$ the identities above give $(g_j/F)(\delta(\tau)) = (\xi_j/\xi_0)(c(\tau))$, and the right side is holomorphic on the whole disc with value $(\xi_j/\xi_0)(b) = 0$ at $\tau = 0$.

Finally we choose the coordinates and the parameter. Let q be the smallest vanishing order at 0 of the coordinate functions of δ; it is finite because δ is not constant, and $q \geq 1$ because $\delta(0) = 0$. Then $\delta(\tau) = \tau^q e + O(\tau^{q+1})$ with a vector $e \neq 0$. Choose a linear form ℓ with $\ell(e) \neq 0$ and linear coordinates in which ℓ is the first coordinate; then $\delta_1(\tau) = \tau^q u(\tau)$ with u holomorphic and $u(0) = \ell(e) \neq 0$, while the other coordinates satisfy $|\delta_j(\tau)| \leq C|\tau|^q$ near 0. On a disc around 0 on which u has no zeros, u has a holomorphic logarithm h, and $\kappa = \mathrm{e}^{h/q}$ is a holomorphic qth root of u. The function $s(\tau) = \tau\kappa(\tau)$ satisfies $s'(0) = \kappa(0) \neq 0$, so by the holomorphic inverse function theorem in one variable (Proposition 7.1.2(1) with $n = 1$) it maps a neighborhood of 0 biholomorphically onto a disc around 0. Its inverse s^{-1} is holomorphic and vanishes at 0, so that $|s^{-1}(t)| \leq C'|t|$ after shrinking the disc. Put $\gamma = \delta \circ s^{-1}$ on this disc. Then $\gamma_1(t) = s^{-1}(t)^q u(s^{-1}(t)) = s(s^{-1}(t))^q = t^q$, and $|\gamma_j(t)| \leq CC'^q|t|^q$ for $j \geq 2$. Since s^{-1} maps the punctured disc onto a punctured neighborhood of 0, $F \circ \gamma$ has no zeros other than $t = 0$, and each $(g_j/F) \circ \gamma = ((g_j/F) \circ \delta) \circ s^{-1}$ extends holomorphically across $t = 0$ with value 0. □

Lemma 12.5.2 *Let $q \geq 1$, and let $\gamma(t) = (t^q, \gamma_2(t), \ldots, \gamma_n(t))$ be a holomorphic map from a disc around $0 \in \mathbb{C}$ to $\mathbb{C}^n$ with $\gamma(0) = 0$, for instance a map as in Lemma 12.5.1. Let $\Omega = \Delta_R \times P \subset \mathbb{C} \times \mathbb{C}^{n-1}$, where $\Delta_R = \{|z_1| < R\}$, $R > 0$, and P is a polydisc centered at 0, and let F be holomorphic on Ω with $F(\gamma(t)) \neq 0$ for small $t \neq 0$. Then there are $r, c > 0$, depending only on γ, F and Ω, with the following property. If $0 < |a| < r$ and $H \in \mathcal{O}(\Omega)$ satisfies*

$$H(a, \cdot) = F(a, \cdot) \text{ on } P, \qquad (H/F) \circ \gamma \text{ is holomorphic at } 0 \text{ with value } 0,$$

then

$$\int_\Omega |H|^2 \, \mathrm{d}V \geq c|a|^{-2}. \tag{12.15}$$

For $n = 1$, the polydisc P is a point and carries counting measure.

Proof We bound the norm of H from below by a weighted norm of $H \circ \gamma$ on a disc, and then use that $H \circ \gamma = F \circ \gamma$ at the points t with $t^q = a$.

Choose $\rho > 0$ with $\rho^q < R$ such that γ is defined on a neighborhood of the closed disc $|t| \leq \rho$, $F \circ \gamma$ has no zeros for $0 < |t| \leq \rho$, and, since γ is continuous with $\gamma(0) = 0$, $(\gamma_2(t), \dots, \gamma_n(t))$ lies in a fixed polydisc $P' \Subset P$ centered at 0 for $|t| \leq \rho$. Let $\delta > 0$ be half the distance from P' to the complement of P. For $0 < |z_1| < \rho^q$, each of the q points $\gamma(t)$ with $t^q = z_1$ has the form (z_1, w) with $w \in P'$, and (12.7) for the holomorphic function $H(z_1, \cdot)$ on the ball $B(w, \delta) \subset P$ gives $|H(z_1, w)|^2 \leq C_\delta \int_P |H(z_1, w')|^2 \, \mathrm{d}V(w')$, where C_δ^{-1} is the volume of a ball of radius δ in $\mathbb{C}^{n-1}$. Summing over the q points, we obtain $\sum_{t^q = z_1} |H(\gamma(t))|^2 \leq qC_\delta \int_P |H(z_1, w')|^2 \, \mathrm{d}V(w')$.

Now we integrate over $|z_1| < \rho^q$. The map $t \mapsto t^q$ maps each of the q half-open sectors $\{0 < |t| < \rho,\ 2\pi k/q \leq \arg t < 2\pi(k+1)/q\}$ bijectively onto the punctured disc $0 < |z_1| < \rho^q$, and on the interior of the sector it is a diffeomorphism onto the complement of a ray, with real Jacobian $|qt^{q-1}|^2$. By the change of variables formula, applied on each sector, $\int_{|t|<\rho} \Phi(t) q^2 |t|^{2q-2} \, \mathrm{d}V_2(t) = \int_{|z_1|<\rho^q} \sum_{t^q = z_1} \Phi(t) \, \mathrm{d}V_2(z_1)$ for every nonnegative measurable Φ. Since $\Delta_{\rho^q} \times P \subset \Omega$, it follows that

$$\int_\Omega |H|^2 \, \mathrm{d}V \geq c_0 \int_{|t|<\rho} |H(\gamma(t))|^2 q^2 |t|^{2q-2} \, \mathrm{d}V_2(t)$$

with $c_0 = 1/(qC_\delta)$. If $n = 1$, the same inequality holds with $c_0 = 1/q$, without the submean step.

Write $F \circ \gamma(t) = t^m v(t)$ with an integer $m \geq 0$ and v without zeros on $|t| \leq \rho$, and put $h = (H/F) \circ \gamma$. This function is holomorphic for $0 < |t| < \rho$, because $F \circ \gamma$ has no zeros there, and by hypothesis it extends holomorphically across 0 with $h(0) = 0$; thus $h(t) = \sum_{\ell \geq 1} h_\ell t^\ell$ for $|t| < \rho$. Then $H \circ \gamma = t^m v h$, and $|v|$ is bounded below by a positive constant on the compact disc $|t| \leq \rho$, being continuous and without zeros there. On each circle $|t| = s < \rho$ the power series of h converges uniformly, and the monomials t^ℓ are orthogonal there, so $\int_0^{2\pi} |h(s\mathrm{e}^{\mathrm{i}\theta})|^2 \, \mathrm{d}\theta = 2\pi \sum_\ell |h_\ell|^2 s^{2\ell}$. Integrating in polar coordinates and using the monotone convergence theorem, we obtain

$$\int_\Omega |H|^2 \, \mathrm{d}V \geq c_1 \sum_{\ell \geq 1} \frac{|h_\ell|^2 \rho^{2(\ell+m+q)}}{\ell + m + q}$$

with $c_1 = \pi q^2 c_0 \min_{|t| \leq \rho} |v(t)|^2$. Now let $0 < |a| < \rho^q$. For each of the q roots b of $b^q = a$, the point $\gamma(b)$ lies on the slice $\{z_1 = a\}$, where $H = F$, and $F(\gamma(b)) \neq 0$; hence $h(b) = 1$. The average $q^{-1} \sum_{b^q = a} b^\ell$ equals $a^{\ell/q}$ if q divides ℓ and 0 otherwise, so averaging h over these roots gives $\sum_{j \geq 1} h_{qj} a^j = 1$. By the Cauchy–Schwarz inequality,

$$1 \le c_1^{-1}\|H\|^2_{L^2(\Omega)} \sum_{j\ge 1} \frac{(qj+m+q)|a|^{2j}}{\rho^{2(qj+m+q)}}.$$

If $|a| \le \rho^q/2$, the last sum equals $\rho^{-2(m+q)} \sum_{j\ge1}(qj+m+q)x^j$ with $x = |a|^2\rho^{-2q} \le 1/4$, and this series is at most a constant times x. Hence (12.15) holds for $0 < |a| < r = \rho^q/2$, with c independent of a and H. □

Theorem 12.5.3 *Let $(\psi_j)_{j\ge1}$ be an increasing sequence of weights on X that converges almost everywhere to a weight φ. Then at every $x \in X$,*

$$\bigcup_j \mathcal{I}(\psi_j)_x = \mathcal{I}(\varphi)_x, \tag{12.16}$$

and $\mathcal{I}(\psi_j)_x = \mathcal{I}(\varphi)_x$ for all sufficiently large j; the required index may depend on x and on the sequence.

Proof Since the ψ_j increase to φ almost everywhere, $\psi_j \le \varphi$ almost everywhere. The plurisubharmonic functions $\max\{\psi_j, \varphi\}$ and φ therefore agree almost everywhere, hence everywhere by Theorem 8.2.1; that is, $\psi_j \le \varphi$ everywhere. Consequently $\mathcal{I}(\psi_j) \subset \mathcal{I}(\varphi)$ for every j, and the left side of (12.16) is contained in the right side. We prove the reverse inclusion by induction on the dimension n of X at x. If $n = 0$, all the weights are finite at x and all the ideals equal $\mathcal{O}_{X,x}$. Let $n \ge 1$, assume the theorem in dimension $n-1$, and take $x = 0$ in a coordinate chart. We argue by contradiction, extending a germ that is missing from $\bigcup_j \mathcal{I}(\psi_j)_0$ from slices $\{z_1 = a_i\}$ with $a_i \to 0$, as outlined at the beginning of this section.

Step 1: reduction to a curve. The union $J = \bigcup_j \mathcal{I}(\psi_j)_0$ is an ideal, because the ideals increase with j. By Proposition A.1.2 it is generated by finitely many germs $g_1, \dots, g_k$, $k \ge 0$, and these all belong to $\mathcal{I}(\psi_{j_0})_0$ for some j_0. Suppose, to reach a contradiction, that some $F \in \mathcal{I}(\varphi)_0$ does not belong to J; in particular $F \ne 0$. Choose representatives of F and of the g_l on a neighborhood U of 0 on which $|g_l|^2 e^{-\psi_{j_0}}$ and $|F|^2 e^{-\varphi}$ are integrable. If $|F| \le C|g|$, with $|g| = (\sum_l |g_l|^2)^{1/2}$, held on a neighborhood of 0, then $F \in \mathcal{I}(\psi_{j_0})_0 \subset J$ by Proposition 12.4.3; so no such bound holds. Lemma 12.5.1 now provides linear coordinates and a curve $\gamma(t) = (t^q, \gamma_2(t), \dots, \gamma_n(t))$ such that $F \circ \gamma$ has no zeros for small $t \ne 0$ and each $(g_l/F) \circ \gamma$ extends holomorphically with value 0 at $t = 0$. In these coordinates choose polydiscs $\Delta_{R_0} \times P_0 \Subset U$ and $\Omega = \Delta_R \times P \Subset \Delta_{R_0} \times P_0$ centered at 0, and let $r, c > 0$ be the constants of Lemma 12.5.2 for γ, F and Ω. Subtracting the same constant from φ and all ψ_j, we may assume that $\varphi \le 0$ on $\Delta_{R_0} \times P_0$; then also $\psi_j \le 0$ there.

Step 2: choice of slices. By Fubini's theorem, the function

$$G(a) = \int_{P_0} |F(a,w)|^2 e^{-\varphi(a,w)}\, dV_w$$

is integrable on Δ_{R_0}. Hence, given any null set $N \subset \Delta_{R_0}$, there is a sequence of points $a_i \notin N$, $a_i \ne 0$, with $a_i \to 0$ and $|a_i|^2 G(a_i) \to 0$; otherwise $G(a) \ge c'|a|^{-2}$ for almost every a in some punctured disc around 0, and G would not be integrable. We take for N the union of three sets: the set where $G = \infty$; the set of a for which one of

the countably many functions $\varphi(a,\cdot)$, $\psi_j(a,\cdot)$ is identically $-\infty$ on P_0; and the set of a for which $\psi_j(a,\cdot) \to \varphi(a,\cdot)$ fails on a subset of P_0 of positive measure. The first set is null because G is integrable, and the other two are null by Fubini's theorem, because pole sets of weights have measure zero and $\psi_j \to \varphi$ almost everywhere. Then the functions $\psi_j(a_i,\cdot)$ and $\varphi(a_i,\cdot)$ are weights on P_0 (Proposition 8.2.3), and the former increase to the latter almost everywhere on P_0. Moreover $G(a_i) > 0$ for all large i, because $F(a_i,\cdot)$ does not vanish identically on P_0: for small $|a_i|$, the slice $\{a_i\} \times P_0$ contains the points $\gamma(b)$ with $b^q = a_i$, at which $F \neq 0$. Discarding finitely many terms, we assume that this holds for all i.

Step 3: integrability on a fixed inner slice. Fix i. Since $G(a_i) < \infty$, the germ of $F(a_i,\cdot)$ at every point of P_0 belongs to $\mathcal{I}(\varphi(a_i,\cdot))$. By the induction hypothesis, applied on P_0, each point of the compact set $\overline{P} \subset P_0$ has a neighborhood on which $|F(a_i,\cdot)|^2 \mathrm{e}^{-\psi_j(a_i,\cdot)}$ is integrable for some j. Covering $\overline{P}$ by finitely many such neighborhoods and using that the ψ_j increase, we find one index j for which $\int_P |F(a_i,w)|^2 \mathrm{e}^{-\psi_j(a_i,w)}\,\mathrm{d}V_w < \infty$. For larger j these integrands decrease and converge almost everywhere to $|F(a_i,\cdot)|^2 \mathrm{e}^{-\varphi(a_i,\cdot)}$, so by dominated convergence their integrals over P converge to a number at most $G(a_i)$. Since $G(a_i) > 0$, we can choose an index j_i with

$$\int_P |F(a_i,w)|^2 \mathrm{e}^{-\psi_{j_i}(a_i,w)}\,\mathrm{d}V_w \le 2G(a_i).$$

The polydisc P does not depend on i; this is what makes the estimates of the next step uniform.

Step 4: extension and contradiction. For large i we have $|a_i| < \min\{r, R\}$, so that the normal coordinate $z_1 - a_i$ takes values in a disc of radius $2R$ on Ω. The polydisc Ω is pseudoconvex by Corollary 11.2.5, being a product of discs, each of which is a sublevel set of a function $|\zeta - \zeta_0|^2$ on $\mathbb{C}$. Applying Theorem 11.4.4, after reordering the coordinates, with the normal coordinate $z_1 - a_i$, the radius $2R$ and the weight ψ_{j_i}, and using the last display of Step 3, we obtain $H_i \in \mathcal{O}(\Omega)$ with $H_i(a_i,\cdot) = F(a_i,\cdot)$ on P and

$$\int_\Omega |H_i|^2 \mathrm{e}^{-\psi_{j_i}}\,\mathrm{d}V \le 8C_0R^2G(a_i),$$

where C_0 is the absolute constant of that theorem. In particular the germ of H_i at 0 lies in $\mathcal{I}(\psi_{j_i})_0 \subset J$, so $H_i = \sum_l b_l g_l$ near 0 with holomorphic b_l, and $(H_i/F) \circ \gamma = \sum_l (b_l \circ \gamma) \cdot (g_l/F) \circ \gamma$ extends holomorphically across $t = 0$ with value 0. Since $\psi_{j_i} \le 0$, the last display also bounds $\int_\Omega |H_i|^2\,\mathrm{d}V$. Now Lemma 12.5.2 gives $c|a_i|^{-2} \le 8C_0R^2G(a_i)$ for all large i, contradicting $|a_i|^2 G(a_i) \to 0$. Hence $F \in J$, which proves (12.16).

Finally, $\mathcal{I}(\varphi)_x$ is generated by finitely many germs, which by (12.16) all lie in $\mathcal{I}(\psi_j)_x$ for one index j. Then $\mathcal{I}(\varphi)_x \subset \mathcal{I}(\psi_j)_x$, and equality holds for this and every later index, since the ideals increase and are contained in $\mathcal{I}(\varphi)_x$. □

Corollary 12.5.4 (Strong openness) *Let φ and χ be weights on X, and let $x \in X$. Then*

$$\mathcal{I}(\varphi)_x = \bigcup_{\varepsilon>0} \mathcal{I}((1+\varepsilon)\varphi)_x \tag{12.17}$$

and, more generally,

$$\mathcal{I}(\varphi)_x = \bigcup_{\varepsilon>0} \mathcal{I}(\varphi+\varepsilon\chi)_x.$$

Moreover, there is a single $\varepsilon > 0$ with $\mathcal{I}((1+\varepsilon)\varphi)_x = \mathcal{I}(\varphi)_x$, and similarly a single $\varepsilon > 0$ with $\mathcal{I}(\varphi+\varepsilon\chi)_x = \mathcal{I}(\varphi)_x$.

Proof The assertions are local. On a relatively compact neighborhood of x the weights are bounded above, and subtracting constants we may assume $\varphi \le 0$ and $\chi \le 0$ there; this changes $(1+\varepsilon)\varphi$ and $\varphi+\varepsilon\chi$ only by bounded amounts and hence changes none of the ideals. Then $(1+\varepsilon)\varphi$ and $\varphi+\varepsilon\chi$ decrease as ε increases, so the corresponding ideals decrease with ε and are contained in $\mathcal{I}(\varphi)_x$. The sequences $(1+1/j)\varphi$ and $\varphi+\chi/j$ increase with j and converge to φ wherever φ and χ are finite, hence almost everywhere. By Theorem 12.5.3, $\mathcal{I}((1+1/j)\varphi)_x = \mathcal{I}(\varphi)_x$ and $\mathcal{I}(\varphi+\chi/j)_x = \mathcal{I}(\varphi)_x$ for all large j, which proves both assertions. □

Example 12.5.5 Let $c \ge 0$ and $\varphi = c\log|z|^2$ on a disc around $0 \in \mathbb{C}$, where $\varphi = 0$ if $c = 0$. A germ of order k at 0 is z^k times a unit, so by polar coordinates it belongs to $\mathcal{I}(\varphi)_0$ if and only if $\int_0^r t^{2k-2c+1}\,\mathrm{d}t < \infty$, that is, $k+1 > c$. Hence $\mathcal{I}(\varphi) = (z^{\lfloor c\rfloor})$. When c is a positive integer, the germ z^{c-1} just fails, because the radial integral diverges logarithmically. This strict threshold shows why Theorem 12.5.3 concerns increasing sequences: on the unit disc, the weights $\varphi_j = (1-1/j)\log|z|^2$, $j \ge 2$, decrease to $\log|z|^2$, yet $\mathcal{I}(\varphi_j) = \mathcal{O}$ for every j, whereas $\mathcal{I}(\log|z|^2) = (z)$.

12.6 Birational change of variables and logarithmic thresholds

On a modification of X, the change of variables formula turns membership in a multiplier ideal into an integrability condition twisted by the Jacobian (Proposition 12.6.4), which becomes explicit on a log resolution.

Let $\pi\colon Y \to X$ be a proper bimeromorphic holomorphic map between complex manifolds of dimension n. Thus there is a nowhere dense analytic subset $A \subset X$ such that $\pi^{-1}(A)$ is nowhere dense in Y and π maps $Y \setminus \pi^{-1}(A)$ biholomorphically onto $X \setminus A$ (Appendix A.3). In local coordinates w on Y and z on X, the complex Jacobian determinant $J_\pi = \det(\partial z_i/\partial w_j)$ is holomorphic, and it does not vanish identically on any nonempty open set, since it has no zeros on the dense open set $Y \setminus \pi^{-1}(A)$. By the chain rule, a change of coordinates on either side multiplies J_π by a nowhere vanishing holomorphic function; equivalently, $\pi^*(\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n) = J_\pi\,\mathrm{d}w_1 \wedge \cdots \wedge \mathrm{d}w_n$. Hence the functions J_π are the local equations of an effective Cartier divisor $K_{Y/X}$ on Y (Appendix A.5), the *relative canonical divisor*, and $\mathcal{O}_Y(K_{Y/X})$ is generated locally by J_π^{-1}.

We measure the vanishing of holomorphic functions along hypersurfaces as follows. Let $E \subset Y$ be a prime divisor, that is, an irreducible hypersurface. By Corollary A.2.13, applied to the family consisting of E alone, its regular locus E_{reg} is a connected manifold, dense in E. Let $p \in E_{\mathrm{reg}}$, choose coordinates $w = (w_1, w')$ at p with $E = \{w_1 = 0\}$ near p, and let h be holomorphic near p. The *order of h along E at p* is the supremum, in $\{0, 1, 2, \ldots\} \cup \{+\infty\}$, of the integers $k \geq 0$ such that h is divisible by w_1^k near p, that is, $h = w_1^k h_1$ near p with h_1 holomorphic. By the next lemma, it does not depend on the coordinates, and for a function defined near E it does not depend on the point of E_{reg}.

Lemma 12.6.1 *Let E be a prime divisor on a complex manifold Y, let $p \in E_{\mathrm{reg}}$, let $w = (w_1, w')$ be holomorphic coordinates at p with $E = \{w_1 = 0\}$ near p, and let h be holomorphic near p. Let $\Delta = \Delta_1 \times \Delta'$ be a polydisc centered at p, where Δ_1 is a disc in the w_1-plane, so small that $E \cap \Delta = \{w_1 = 0\} \cap \Delta$ and that (7.1) gives the expansion*

$$h = \sum_{l \geq 0} a_l(w') w_1^l \quad \text{on } \Delta, \qquad a_l(w') = \frac{1}{l!} \partial_{w_1}^l h(0, w'),$$

where the functions a_l are holomorphic on Δ'.

(1) *Let $k \geq 0$ be an integer. If h is divisible by w_1^k near some point of $E \cap \Delta$, then $a_l = 0$ for all $l < k$; conversely, if $a_l = 0$ for all $l < k$, then h is divisible by w_1^k near p. In particular, a holomorphic function that vanishes on E near p is divisible by w_1 near p, and if $\tilde{w} = (\tilde{w}_1, \tilde{w}')$ are other coordinates at p with $E = \{\tilde{w}_1 = 0\}$ near p, then $\tilde{w}_1$ is a unit times w_1 near p.*

(2) *The order of h along E at p does not depend on the coordinates. It is the smallest l with $a_l \neq 0$, and it is $+\infty$ if and only if h vanishes near p. In particular, h is not divisible by w_1 near p if and only if $h(0, \cdot)$ does not vanish identically near p. Moreover, the order of h along E is the same at all points of $E \cap \Delta$.*

(3) *If h is holomorphic on a neighborhood of E, then its order along E is the same at all points of E_{reg}.*

(4) *Orders add under products: if g is also holomorphic near p, then the order of hg along E at p is the sum of the orders of h and g.*

Proof (1) If $h = w_1^k h_1$ near a point $(0, w_0')$ of $E \cap \Delta$, with h_1 holomorphic, then the Leibniz rule gives $\partial_{w_1}^l h(0, w') = 0$ for $l < k$ and w' near w_0'; so $a_l = 0$ near w_0', and hence on the connected set Δ' by the identity theorem. Conversely, if $a_l = 0$ for $l < k$, then $h = w_1^k \sum_{l \geq k} a_l w_1^{l-k}$ near p; the last series is obtained from the Taylor series of h at p by lowering the exponent of w_1 in each term by k, so it converges on the same polydiscs and defines a holomorphic function (Appendix A.1). If h vanishes on E near p, then $a_0 = h(0, \cdot)$ vanishes near the center of Δ', hence on Δ', and the case $k = 1$ shows that h is divisible by w_1 near p. Applying this to $\tilde{w}_1$, and to w_1 with the roles of the two coordinate systems exchanged, we obtain $\tilde{w}_1 = c\, w_1$ and $w_1 = \tilde{c}\, \tilde{w}_1$ near p with holomorphic c and $\tilde{c}$. Then $c\tilde{c} = 1$ off E; since the continuous function $c\tilde{c}$ equals 1 on the dense set $\{w_1 \neq 0\}$, it equals 1 near p, and c is a unit.

(2) By (1), h is divisible by w_1^k near p if and only if it is divisible by $\tilde{w}_1^k$ near p, so the order does not depend on the coordinates. Let k_0 be the smallest l with $a_l \neq 0$, and $k_0 = +\infty$ if all a_l vanish. By (1), h is divisible by w_1^k near p if and only if $k \leq k_0$, so the order of h along E at p is k_0. If all a_l vanish, then $h = 0$ on Δ; conversely, if h vanishes near p, then all a_l vanish near the center of Δ', hence on Δ'. The case $k_0 = 0$ gives the assertion about $h(0, \cdot) = a_0$, because a_0 vanishes identically on Δ' if and only if it vanishes near the center of Δ'. Finally, let $q = (0, w'_q) \in E \cap \Delta$. The coordinates w are also coordinates at q with $E = \{w_1 = 0\}$ near q, and by their formula the coefficients of the expansion of h on a small polydisc centered at q are the restrictions of the same functions a_l. Since Δ' is connected, a_l vanishes near w'_q if and only if it vanishes on Δ', by the identity theorem. Hence the orders of h along E at q and at p are equal.

(3) By the last assertion of (2), the order of h along E is a locally constant function on E_{reg}, which is connected (Corollary A.2.13). Hence it is constant.

(4) Shrinking Δ, we may assume that g is defined on Δ; let b_j and e_m be the coefficients of the expansions of g and hg on Δ. By the Leibniz rule, $e_m = \sum_{i+j=m} a_i b_j$. If h or g has infinite order, then hg vanishes near p and has infinite order by (2). Otherwise let k and l be the orders of h and g. Then $e_m = 0$ for $m < k + l$, because each term $a_i b_j$ with $i + j < k + l$ has $i < k$ or $j < l$, and $e_{k+l} = a_k b_l$. Since a_k and b_l are not identically zero and the ring of holomorphic functions on the connected set Δ' is an integral domain, by the identity theorem, $e_{k+l} \neq 0$. By (2), the order of hg is $k + l$. □

If h is holomorphic on a neighborhood of E, we call the common value in Lemma 12.6.1(3) the *order* $\operatorname{ord}_E h$ of h along E. The order of J_π along a prime divisor $E \subset Y$ is defined in the same way, because local Jacobians differ by units, which have order zero, and orders add under products (Lemma 12.6.1(4)); it is the coefficient $\operatorname{ord}_E K_{Y/X}$ of E in $K_{Y/X}$.

For divisors with simple normal crossings support we use local monomial equations.

Lemma 12.6.2 *Let w be holomorphic coordinates centered at a point p of a complex manifold, let $0 \leq \ell \leq n$, and let h be holomorphic near p, not identically zero near p, with zero set near p contained in $\{w_1 \cdots w_\ell = 0\}$. For $i \leq \ell$ let k_i be the largest integer such that h is divisible by $w_i^{k_i}$ near p. Then $h = u \prod_{i \leq \ell} w_i^{k_i}$ near p, with u holomorphic and $u(p) \neq 0$. Moreover, a holomorphic function near p that is divisible by $w_i^{a_i}$ near p for each $i \leq \ell$ is divisible by $\prod_{i \leq \ell} w_i^{a_i}$.*

Proof Let $g = \sum_\alpha c_\alpha w^\alpha$ be the power series at p of a holomorphic function, as in (7.1). If $g = w_i^a g'$ near p, then multiplying the series of g' by w_i^a shows that $c_\alpha = 0$ whenever $\alpha_i < a$. Conversely, if $c_\alpha = 0$ whenever $\alpha_i < a_i$ for some $i \leq \ell$, then $g = \prod_i w_i^{a_i} \cdot \sum_\alpha c_\alpha w^{\alpha - \sum_i a_i e_i}$, where e_i is the ith unit vector, and the last series converges on the same polydiscs as that of g. This proves the last assertion, and it shows that $k_i < \infty$, since otherwise all c_α of h would vanish. Hence $h = u \prod_{i \leq \ell} w_i^{k_i}$ with u holomorphic near p, and u is not divisible by any w_i, $i \leq \ell$, by the maximality of k_i.

It remains to show that $u(p) \neq 0$. Suppose, to the contrary, that $u(p) = 0$. The zero set of u near p lies in that of h, hence in $\{w_1 \cdots w_\ell = 0\}$; so $\ell \geq 1$. We work in the coordinates w, with $p = 0$. Let P be the first nonzero homogeneous term of u; it has positive degree m. Choose $v \in \mathbb{C}^n$ with $P(v) \neq 0$ and $v_i \neq 0$ for all i, which is possible since the polynomial $P(v)v_1 \cdots v_n$ is not identically zero. Then $t \mapsto u(tv)$ has a zero of order m at $t = 0$; choose $\varepsilon > 0$ such that it has no other zeros in $|t| \leq \varepsilon$, and such that all points $y + tv$ considered below lie in the neighborhood of 0 where the hypothesis on the zero set holds. Since $(y, t) \mapsto u(y + tv)$ is continuous and does not vanish on the compact set $\{0\} \times \{|t| = \varepsilon\}$, there is a connected neighborhood N of 0 such that $u(y + tv) \neq 0$ for $y \in N$ and $|t| = \varepsilon$. By the argument principle, the number of zeros of $t \mapsto u(y + tv)$ in $|t| < \varepsilon$, counted with multiplicity, is

$$\frac{1}{2\pi \mathrm{i}} \int_{|t|=\varepsilon} \frac{\partial_t [u(y + tv)]}{u(y + tv)} \, \mathrm{d}t,$$

a continuous integer-valued function of $y \in N$; it equals $m \geq 1$ at $y = 0$, hence everywhere on N. So for every $y \in N$ there is t with $u(y+tv) = 0$, and then $y_i + tv_i = 0$ for some $i \leq \ell$, that is, $u(\sigma_i(y)) = 0$ for the linear projection $\sigma_i(y) = y - (y_i/v_i)v$ onto the hyperplane $H_i = \{y_i = 0\}$. Hence the product of the holomorphic functions $u \circ \sigma_i$, $i \leq \ell$, vanishes on N. If none of them vanished identically on N, each would be nonzero on a dense open subset of N, by the identity theorem, and so would be their product. Thus some $u \circ \sigma_i$ vanishes on N; since $\sigma_i(y) = y$ for $y \in H_i$, the function u vanishes on $H_i \cap N$, which is a neighborhood of 0 in H_i. By the first paragraph, applied with $a = 1$, u is then divisible by w_i near 0: all coefficients c_α of u with $\alpha_i = 0$ vanish, since they are the coefficients of the restriction of u to H_i. This contradiction shows that $u(p) \neq 0$. □

Let D be an effective divisor on Y with simple normal crossings support, and let $E_1, E_2, \ldots$ be the irreducible components of its support; they are smooth, and connected because their regular loci are (Corollary A.2.13). Near a point p choose coordinates w in which the support of D is $\{w_1 \cdots w_\ell = 0\}$, with $E_i = \{w_i = 0\}$ for the components through p. By Lemma 12.6.2, a local equation of D at p is a unit times $\prod_{i \leq \ell} w_i^{d_i}$, where $d_i \geq 1$ is its order along E_i. Two local equations differ by a unit on the overlap of their domains, so d_i does not depend on the point of E_i; it is the *multiplicity* of E_i in D, and we write $D = \sum_i d_i E_i$. Conversely, for integers $a_i \geq 0$ we write $\sum_i a_i E_i$ for the effective divisor whose local equations are the monomials $\prod_{i \leq \ell} w_i^{a_i}$. If A and B are effective divisors on Y with local equations s_A and s_B, then $\mathcal{O}_Y(A - B) = \mathcal{O}_Y(A)\mathcal{O}_Y(-B)$ is the sheaf of functions on the complement of the support of A that are locally of the form $g\, s_B/s_A$ with g holomorphic.

Definition 12.6.3 Let $E \subset Y$ be a prime divisor. Its *center* on X is $\pi(E)$, which is an analytic subset of X by Theorem A.6.2; it is connected, because E is. Let f be a holomorphic function on an open set $V \subset X$ containing the center, and let V_0 be the connected component of V containing it. If f vanishes identically on V_0, for instance if $f = 0$, we set $\mathrm{ord}_E f = +\infty$. Otherwise $f \circ \pi$ does not vanish identically on any nonempty open subset of $\pi^{-1}(V_0)$, because such a set contains a nonempty open set

mapped biholomorphically onto an open subset of V_0, on which f does not vanish identically by the identity theorem; since $E \subset \pi^{-1}(V_0)$, the *order* of f along E is $\operatorname{ord}_E f = \operatorname{ord}_E(f \circ \pi) < \infty$. This order along a divisor differs from the order at a point of Proposition 12.1.4, although the two agree for the exceptional divisor of the blowup of a point (Example 12.6.10). For a holomorphic germ f at a point x, this defines $\operatorname{ord}_E f$ for the prime divisors E on smooth models of open neighborhoods of x on which f is defined; this is how orders of germs enter the results below. The *log discrepancy* of E is

$$A_X(E) = 1 + \operatorname{ord}_E K_{Y/X},$$

where $\operatorname{ord}_E K_{Y/X} = \operatorname{ord}_E J_\pi$ is the coefficient of E in $K_{Y/X}$. Since J_π is holomorphic, $A_X(E) \geq 1$, and a prime divisor on X itself, with π the identity, has log discrepancy one. The center, the order and the log discrepancy depend only on the prime divisor over X represented by E.

To see that these notions depend only on the prime divisor over X, let $\rho\colon Y' \to Y$ be a map from a common smooth model and E' the strict transform of E. By Proposition A.7.3, ρ is biholomorphic over $Y \setminus S$ for an analytic subset S of codimension at least two, and it maps the nonempty open subset $E' \cap \rho^{-1}(Y \setminus S)$ of E' isomorphically onto the dense open subset $E \setminus S$ of E; hence $\rho(E') = E$, since $\rho(E')$ is closed by properness. At a point q' of $E' \cap \rho^{-1}(Y \setminus S)$ that is regular on E' and maps to a regular point of E, the biholomorphism ρ near q' transforms local equations of E into local equations of E', so $\operatorname{ord}_{E'}(f \circ \pi \circ \rho) = \operatorname{ord}_E(f \circ \pi)$ and $\operatorname{ord}_{E'}(J_\pi \circ \rho) = \operatorname{ord}_E J_\pi$, while J_ρ does not vanish near q'. By the chain rule $J_{\pi\circ\rho} = (J_\pi \circ \rho)J_\rho$, and since orders add under products (Lemma 12.6.1(4)), $\operatorname{ord}_{E'} K_{Y'/X} = \operatorname{ord}_E K_{Y/X}$. Thus E' has the same center $\pi(\rho(E')) = \pi(E)$ as E, and the same orders and log discrepancy. Since two representatives of the same prime divisor over X have the same strict transform on a common smooth model, this proves the claim.

Proposition 12.6.4 *Let φ be a weight on X, $x \in X$, and $f \in \mathcal{O}_{X,x}$. Then $f \in \mathcal{I}(\varphi)_x$ if and only if, near every point of the compact fiber $\pi^{-1}(x)$, the function*

$$|f \circ \pi|^2 |J_\pi|^2 \mathrm{e}^{-\varphi\circ\pi} \tag{12.18}$$

is integrable with respect to Lebesgue measure in local coordinates on Y. Moreover, $\pi_*\mathcal{O}_Y(K_{Y/X}) = \mathcal{O}_X$: if $U \subset X$ is open, then every section of $\mathcal{O}_Y(K_{Y/X})$ over $\pi^{-1}(U)$, that is, every function that is locally of the form g/J_π with g holomorphic, equals $f \circ \pi$ for a unique $f \in \mathcal{O}(U)$. Under this identification

$$\mathcal{I}(\varphi) = \pi_*\big(\mathcal{O}_Y(K_{Y/X})\mathcal{I}(\varphi \circ \pi)\big), \tag{12.19}$$

where $\mathcal{O}_Y(K_{Y/X})\mathcal{I}(\varphi \circ \pi)$ is the sheaf of functions that are locally of the form g/J_π with g a section of $\mathcal{I}(\varphi \circ \pi)$.

Proof By Proposition A.7.3, π is biholomorphic over $X \setminus S$ for an analytic subset $S \subset X$ of codimension at least two; in particular S is nowhere dense. The analytic subset $\pi^{-1}(S)$ of Y is nowhere dense as well, because every nonempty open subset

of Y contains a nonempty open set that π maps biholomorphically onto an open subset of X, and such a set is not contained in S. Near each of their points, nowhere dense analytic subsets lie in the zero set of a holomorphic function that does not vanish identically on a connected neighborhood (Appendix A.1); so S and $\pi^{-1}(S)$ are Lebesgue null sets in charts, by Lemma 7.4.1.

In holomorphic coordinates the real Jacobian determinant of π, that is, the factor by which π^* multiplies the volume form $\mathrm{d}x_1 \wedge \mathrm{d}y_1 \wedge \cdots \wedge \mathrm{d}x_n \wedge \mathrm{d}y_n$, is $|J_\pi|^2$. Indeed, since $\frac{\mathrm{i}}{2}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j = \mathrm{d}x_j \wedge \mathrm{d}y_j$, this form equals $c_n \, \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n \wedge \mathrm{d}\bar{z}_1 \wedge \cdots \wedge \mathrm{d}\bar{z}_n$ with $c_n = (\mathrm{i}/2)^n(-1)^{n(n-1)/2}$, the sign coming from moving each $\mathrm{d}z_j$ to the left past $\mathrm{d}\bar{z}_1, \ldots, \mathrm{d}\bar{z}_{j-1}$, and similarly in the coordinates w. Since $\pi^*(\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n) = J_\pi \, \mathrm{d}w_1 \wedge \cdots \wedge \mathrm{d}w_n$ and the conjugate identity holds for the $\mathrm{d}\bar{z}_j$, the claim follows. The change of variables formula for nonnegative measurable functions, applied to the biholomorphism π from $W \setminus \pi^{-1}(S)$ onto its image, therefore gives

$$\int_{\pi(W\setminus\pi^{-1}(S))} g \, \mathrm{d}V_z = \int_W (g \circ \pi)|J_\pi|^2 \, \mathrm{d}V_w$$

for every nonnegative measurable g and every open set W in a coordinate chart of Y that is mapped into a coordinate chart of X, with J_π computed in these coordinates.

Now suppose that $|f|^2\mathrm{e}^{-\varphi}$ is integrable on a coordinate neighborhood V of x. Applying the formula with $g = |f|^2\mathrm{e}^{-\varphi}$ to small coordinate neighborhoods $W \subset \pi^{-1}(V)$, we see that (12.18) is integrable near every point of $\pi^{-1}(V)$. Conversely, suppose that (12.18) is integrable near every point of $\pi^{-1}(x)$, where f is defined on a coordinate neighborhood V_0 of x. Cover the compact fiber by finitely many coordinate neighborhoods $W_1, \ldots, W_k \subset \pi^{-1}(V_0)$ on which it is integrable. Since π is proper, $\pi(Y \setminus (W_1 \cup \cdots \cup W_k))$ is closed and does not contain x, so there is a neighborhood $V \subset V_0$ of x with $\pi^{-1}(V) \subset W_1 \cup \cdots \cup W_k$. The sets $\pi(W_l \setminus \pi^{-1}(S))$ cover $V \setminus S$, and S is a null set, so the formula gives $\int_V |f|^2\mathrm{e}^{-\varphi} \, \mathrm{d}V \le \sum_l \int_{W_l} |f \circ \pi|^2 |J_\pi|^2 \mathrm{e}^{-\varphi\circ\pi} \, \mathrm{d}V_w < \infty$.

Next let $U \subset X$ be open and h a section of $\mathcal{O}_Y(K_{Y/X})$ over $\pi^{-1}(U)$; we may assume that U is connected. Over $U \setminus S$ the map π is biholomorphic and J_π has no zeros, so h is holomorphic on $\pi^{-1}(U \setminus S)$, and $h = f \circ \pi$ there for the holomorphic function $f = h \circ (\pi|_{\pi^{-1}(U\setminus S)})^{-1}$ on $U \setminus S$, which is connected by Lemma A.1.7. If $f \not\equiv 0$, the function $\log|f|^2$ is plurisubharmonic on $U \setminus S$ (Proposition 8.3.3), so by Corollary A.2.15 it is locally bounded above near S; hence f is locally bounded near S, and Lemma A.1.7 extends f holomorphically to U. Locally $h = g/J_\pi$ with g holomorphic, and $g = (f \circ \pi)J_\pi$ on the dense open set $\pi^{-1}(U \setminus S)$, hence everywhere; so $h = f \circ \pi$. The function f is unique, since it is determined on $U \setminus S$, which is dense in U.

Finally, let $U \subset X$ be open and $f \in \mathcal{O}(U)$. Since $\mathcal{O}_Y(K_{Y/X})$ is locally generated by J_π^{-1}, the function $f \circ \pi$ is a section of $\mathcal{O}_Y(K_{Y/X})\mathcal{I}(\varphi \circ \pi)$ over $\pi^{-1}(U)$ if and only if $(f \circ \pi)J_\pi$ is a section of $\mathcal{I}(\varphi \circ \pi)$ there, that is, if and only if (12.18) is integrable near every point of $\pi^{-1}(U)$. By the first part, applied at every point of U, this holds if and only if f is a section of $\mathcal{I}(\varphi)$ over U. Together with the second part, which shows that every section of the right side of (12.19) over $\pi^{-1}(U)$ has the form $f \circ \pi$, this proves (12.19). □

Log resolutions put weights of the form $c \log \sum_j |g_j|^2$ into monomial form. Let $g_1, \dots, g_r$ be holomorphic functions on X that generate an ideal $\mathfrak{a}$ with $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-D)$, where $D = \sum_i d_i E_i$ is an effective divisor with simple normal crossings support. Let $p \in Y$, and choose coordinates w at p in which the support of D is a union of coordinate hyperplanes; after renumbering, $E_1, \dots, E_\ell$ are the components of D through p, with $E_i = \{w_i = 0\}$ near p. By Lemma 12.6.2, the ideal $\mathfrak{a}\mathcal{O}_Y$ is generated near p by $s = \prod_{i=1}^{\ell} w_i^{d_i}$, so $g_j \circ \pi = s h_j$ with holomorphic h_j. Writing $s = \sum_j \beta_j (g_j \circ \pi) = s \sum_j \beta_j h_j$ with holomorphic β_j and cancelling s, which does not vanish identically on any open set, we get $\sum_j \beta_j h_j = 1$. Hence the h_j have no common zero near p, the function $\sum_j |h_j|^2$ is bounded above and below by positive constants on a smaller neighborhood of p, and

$$\log \sum_{j=1}^{r} |g_j \circ \pi|^2 = \sum_{i=1}^{\ell} d_i \log |w_i|^2 + O(1) \qquad \text{near } p. \tag{12.20}$$

Theorem 12.6.5 *Let φ be a weight on X such that near x*

$$\varphi = c \log \left(\sum_{j=1}^{r} |g_j|^2 \right) + b, \qquad c > 0, \quad b \text{ bounded},$$

where the holomorphic functions g_j generate a nonzero ideal $\mathfrak{a}$. Let V be an open neighborhood of x on which this holds, and let $\pi \colon Y \to V$ be a log resolution of $\mathfrak{a}$ (Appendix A.5), as supplied by Theorems A.5.4 and A.5.5, with $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-D)$, where $D = \sum_i d_i E_i$ is an effective divisor with simple normal crossings support, the E_i being prime divisors and the d_i positive integers. Then a holomorphic function f on V belongs to $\mathcal{I}(\varphi)_x$ if and only if

$$\operatorname{ord}_{E_i} f + A_X(E_i) > c d_i \tag{12.21}$$

for every component E_i of D that meets $\pi^{-1}(x)$, exceptional or not. Since π restricts to a log resolution of $\mathfrak{a}$ over every smaller open neighborhood V' of x, the criterion applies to every germ at x, with the components of the restriction of D to $\pi^{-1}(V')$ in place of the E_i. Equivalently, with $\lfloor cD \rfloor = \sum_i \lfloor c d_i \rfloor E_i$, near x

$$\mathcal{I}(\varphi) = \pi_* \mathcal{O}_Y(K_{Y/X} - \lfloor cD \rfloor). \tag{12.22}$$

For rational c, this computes the multiplier ideals of weights with analytic singularities (Chapter 13).

Proof Let $p \in \pi^{-1}(V)$, and choose coordinates w at p as before (12.20), so that $E_1, \dots, E_\ell$ are the components of D through p and $E_i = \{w_i = 0\}$ near p. Since b is bounded, (12.20), applied to $\pi \colon Y \to V$, gives

$$\varphi \circ \pi = c \sum_{i=1}^{\ell} d_i \log |w_i|^2 + O(1)$$

near p.

Let h be holomorphic near p; we decide when $|h|^2 \prod_{i\le\ell} |w_i|^{-2cd_i}$, or equivalently $|h|^2 e^{-\varphi\circ\pi}$, is integrable near p. Let k_i be the order of h along E_i at p, which is finite if h does not vanish identically near p; if it does, the integrability is trivial and $k_i = +\infty$. We claim that the integrability holds if and only if $k_i + 1 > cd_i$ for $i = 1, \dots, \ell$. If these inequalities hold, write $h = \prod_i w_i^{k_i} \tilde{h}$ with $\tilde{h}$ holomorphic, by Lemma 12.6.2; the integrand is at most a constant times $\prod_i |w_i|^{2k_i - 2cd_i}$, which is integrable near p by Fubini's theorem and polar coordinates in each w_i, since $2k_i - 2cd_i > -2$. Conversely, suppose that $k_1 + 1 \le cd_1$, say, and write $h = w_1^{k_1}\tilde{h}$ with $\tilde{h}$ not divisible by w_1. Then $\tilde{h}$ does not vanish identically on $\{w_1 = 0\}$ near p, by Lemma 12.6.1(2), so arbitrarily close to p there are an open set W' of points $w' = (w_2, \dots, w_n)$ and $\delta, \rho > 0$ with $|\tilde{h}(w_1, w')| \ge \delta$ for $|w_1| < \rho$ and $w' \in W'$. There the integrand is at least a positive constant times $|w_1|^{2k_1 - 2cd_1}$, since the other factors $|w_i|^{-2cd_i}$ are bounded below; as $2k_1 - 2cd_1 \le -2$, its integral in w_1 diverges, and by Fubini's theorem the integrand is not integrable near p.

Now let f be holomorphic on V, and apply this to $h = (f \circ \pi) J_\pi$, with J_π computed in local coordinates near p. The function $f \circ \pi$ is holomorphic on $\pi^{-1}(V)$, which contains E_i, so its order along E_i at p is $\operatorname{ord}_{E_i} f$, by Lemma 12.6.1(3) and Definition 12.6.3; the order of J_π along E_i at p is $\operatorname{ord}_{E_i} J_\pi$. Since orders add under products (Lemma 12.6.1(4)), $k_i = \operatorname{ord}_{E_i} f + \operatorname{ord}_{E_i} J_\pi = \operatorname{ord}_{E_i} f + A_X(E_i) - 1$. Thus, for $p \in \pi^{-1}(x)$, the condition at p is (12.21) for the components through p. As p ranges over $\pi^{-1}(x)$, these are exactly the components meeting the fiber, and Proposition 12.6.4 proves the first assertion.

For (12.22), let $U \subset V$ be open and $f \in \mathcal{O}(U)$, and put $h = (f \circ \pi) J_\pi$ on $\pi^{-1}(U)$. For an integer k, the strict inequality $k + 1 > cd_i$ is equivalent to $k \ge \lfloor cd_i \rfloor$. By Proposition 12.6.4 and the claim, applied at every point $p \in \pi^{-1}(U)$, we have $f \in \mathcal{I}(\varphi)(U)$ if and only if at every such p the order of h at p along each component E_i through p is at least $\lfloor cd_i \rfloor$. By Lemma 12.6.2, this means that h is divisible near every point of $\pi^{-1}(U)$ by the local equation $\prod_i w_i^{\lfloor cd_i \rfloor}$ of $\lfloor cD \rfloor$, that is, that $f \circ \pi = h/J_\pi$ is a section of $\mathcal{O}_Y(K_{Y/X} - \lfloor cD \rfloor)$ over $\pi^{-1}(U)$. Since $\lfloor cD \rfloor$ is effective, every section of $\mathcal{O}_Y(K_{Y/X} - \lfloor cD \rfloor)$ over $\pi^{-1}(U)$ is a section of $\mathcal{O}_Y(K_{Y/X})$, and hence of the form $f \circ \pi$ with $f \in \mathcal{O}(U)$, by Proposition 12.6.4. This proves (12.22) on V.□

For general weights we combine the change of variables with an estimate on one-dimensional slices transversal to a divisor.

Lemma 12.6.6 *Let ψ be a weight on a complex manifold Y, let $E \subset Y$ be a prime divisor, let $p \in E_{\mathrm{reg}}$, and let h be holomorphic near p with $|h|^2 e^{-\psi}$ integrable near p. If k is the order of h along E at p, then $k + 1 > \nu(\psi, E)$, where $\nu(\psi, E)$ is the generic Lelong number of Proposition 12.3.5.*

Proof We may assume $k < \infty$. Choose coordinates $w = (w_1, w')$ at p with $E = \{w_1 = 0\}$, and write $h = w_1^k v$ with v holomorphic and not divisible by w_1; then $v(0, \cdot)$ does not vanish identically near p, by Lemma 12.6.1(2). Moving p slightly along E, we may assume that $v(p) \ne 0$, so that $|v| \ge \delta > 0$ on a neighborhood $\{|w_1| < \rho\} \times W'$ of p on which $|h|^2 e^{-\psi}$ is integrable. By Fubini's theorem, for almost

every $w' \in W'$ the function $w_1 \mapsto |w_1|^{2k}|v|^2 \mathrm{e}^{-\psi(w_1,w')}$ is integrable on $|w_1| < \rho$; moreover, for almost every w' the function $u = \psi(\cdot, w')$ is not identically $-\infty$ on this disc, since the pole set of ψ is a null set.

Fix such a w'. By Proposition 8.2.3, u is subharmonic on the disc, and its Lelong number at 0 is at least $\nu(\psi, (0,w'))$: restricting the bound $\psi(w) \le c\log\bigl(|w - (0,w')|^2/r_0^2\bigr) + C$ of (12.2), with $c = \nu(\psi,(0,w'))$, to the line through $(0,w')$ in the w_1-direction gives $u(w_1) \le c\log(|w_1|^2/r_0^2) + C$. Moreover $\nu(\psi,(0,w')) \ge \nu(\psi,E)$, the latter being an infimum over E. Hence, by (12.2) for u, there are $r_1 \in (0,\rho]$ and $c' > 0$ with $\mathrm{e}^{-u(w_1)} \ge c'|w_1|^{-2\nu(\psi,E)}$ for $0 < |w_1| < r_1$. Therefore

$$\begin{aligned}\infty > \int_{|w_1|<r_1} |w_1|^{2k}|v|^2\mathrm{e}^{-u}\,\mathrm{d}V_2 &\ge \delta^2 c' \int_{|w_1|<r_1} |w_1|^{2k-2\nu(\psi,E)}\,\mathrm{d}V_2 \\ &= 2\pi\delta^2 c' \int_0^{r_1} t^{2k-2\nu(\psi,E)+1}\,\mathrm{d}t,\end{aligned}$$

which forces $2k - 2\nu(\psi,E) + 2 > 0$, that is, $k + 1 > \nu(\psi,E)$. □

Proposition 12.6.7 *Let φ be a weight on X and $f \in \mathcal{I}(\varphi)_x$. Then there are $\varepsilon > 0$ and a neighborhood V of x on which f is defined such that*

$$\operatorname{ord}_E f + A_X(E) > (1+\varepsilon)\nu(\varphi,E) \tag{12.23}$$

for every prime divisor E over an open subset of V; in particular, this holds for every prime divisor over an open neighborhood of x contained in V whose center contains x.

Proof By Corollary 12.5.4 there are $\varepsilon > 0$ and a neighborhood V of x on which f is defined such that $|f|^2\mathrm{e}^{-(1+\varepsilon)\varphi}$ is integrable on V. Let E be a prime divisor over an open set $U \subset V$, represented on a smooth model $\pi\colon Y \to U$; we may assume that $\operatorname{ord}_E f < \infty$. Since $f \in \mathcal{I}((1+\varepsilon)\varphi)_y$ for every $y \in U$, Proposition 12.6.4, applied to the weight $(1+\varepsilon)\varphi$ on U, shows that $|h|^2\mathrm{e}^{-(1+\varepsilon)\varphi\circ\pi}$, with $h = (f\circ\pi)J_\pi$, is integrable near every point of Y. Let $p \in E_{\mathrm{reg}}$. Since orders add under products, the order of h along E at p is $\operatorname{ord}_E f + \operatorname{ord}_E J_\pi = \operatorname{ord}_E f + A_X(E) - 1$. By Lemma 12.6.6, applied to the weight $(1+\varepsilon)\varphi\circ\pi$, whose generic Lelong number along E is $(1+\varepsilon)\nu(\varphi,E)$ by Proposition 12.1.4, we obtain (12.23). □

Corollary 12.6.8 *Let φ be a weight on X, let E be a prime divisor over X, and let $m > 0$ be a real number. Then*

$$m\nu(\varphi,E) - A_X(E) < \operatorname{ord}_E \mathcal{I}(m\varphi) \le m\nu(\varphi,E). \tag{12.24}$$

Here, for a smooth model $\pi\colon Y \to X$ on which E lies, $\operatorname{ord}_E \mathcal{I}(m\varphi)$ is the order along E of the pulled-back ideal $\mathcal{I}(m\varphi)\mathcal{O}_Y$, defined as follows. Near a point $p \in E_{\mathrm{reg}}$, let $g_1,\dots,g_r$ generate $\mathcal{I}(m\varphi)$ near $\pi(p)$. The minimum of the orders of the functions $g_i\circ\pi$ along E at p does not depend on the choice of generators, since the members of a second system of generators are combinations of the g_i with holomorphic coefficients

near $\pi(p)$, and conversely. This minimum is locally constant along E_{reg}, because the orders of the $g_i \circ \pi$ are, by Lemma 12.6.1(2); hence it is constant on the connected manifold E_{reg}, and it is $\operatorname{ord}_E \mathcal{I}(m\varphi)$. *As for the order of a function in Definition 12.6.3, it depends only on the prime divisor over X represented by E. In particular,*

$$\nu(\varphi, E) = \lim_{m\to\infty} \frac{1}{m} \operatorname{ord}_E \mathcal{I}(m\varphi).$$

Proof Let $k = \operatorname{ord}_E \mathcal{I}(m\varphi)$, and choose a point $p \in E_{\mathrm{reg}}$. Put $y = \pi(p)$, choose a bounded polydisc Ω around y in a chart on which φ is bounded above, and choose $g_1, \dots, g_r \in \mathcal{H}_{m\varphi}(\Omega)$ generating $\mathcal{I}(m\varphi)$ near y (Theorem 12.2.7(3)). Near p, in coordinates $w = (w_1, w')$ with $E = \{w_1 = 0\}$, let k_i be the order of $g_i \circ \pi$ along E at p, so that $k = \min_i k_i$; since $\mathcal{I}(m\varphi)_y \neq 0$, this minimum is finite.

For the lower bound, choose i with $k_i = k$. Since $|g_i|^2 \mathrm{e}^{-m\varphi}$ is integrable on Ω, Proposition 12.6.4, applied to the weight $m\varphi$, shows that $|h|^2 \mathrm{e}^{-m\varphi\circ\pi}$, with $h = (g_i \circ \pi) J_\pi$, is integrable near p. The order of h along E at p is $k + A_X(E) - 1$, because orders add under products. By Lemma 12.6.6, applied to the weight $m\varphi \circ \pi$, whose generic Lelong number along E is $m\nu(\varphi, E)$ by Proposition 12.1.4, we obtain $k + A_X(E) > m\nu(\varphi, E)$. This is the left inequality.

For the upper bound, Lemma 12.2.8 gives $m\varphi \leq \log \sum_i |g_i|^2 + mC_m$ on a neighborhood of y. Near p we have $g_i \circ \pi = w_1^{k_i} u_i$ with u_i holomorphic, so $|g_i \circ \pi| \leq C|w_1|^{k_i} \leq C|w_1|^k$ where $|w_1| \leq 1$; it follows that $m\varphi\circ\pi \leq k \log |w_1|^2 + C''$ on a neighborhood N of p. For $q \in E \cap N$ we have $|w_1| \leq |w - q|$, so $m\varphi \circ \pi(w) \leq k \log |w-q|^2 + C''$ near q, and Propositions 12.1.1 and 12.1.4 give $\nu(\varphi \circ \pi, q) \geq k/m$. Since the generic value $\nu(\varphi \circ \pi, E)$ is attained at points of E arbitrarily close to p (Proposition 12.3.5), we conclude that $m\nu(\varphi, E) \geq k$. Dividing (12.24) by m and letting $m \to \infty$ gives the limit formula. □

Theorem 12.6.9 *Let φ be a weight on X, $x \in X$, and $f \in \mathcal{O}_{X,x}$. Then $f \in \mathcal{I}(\varphi)_x$ if and only if there are $\varepsilon > 0$ and a neighborhood V of x on which f is defined such that*

$$\operatorname{ord}_E f + A_X(E) \geq (1 + \varepsilon)\nu(\varphi, E) \tag{12.25}$$

for every prime divisor E over an open neighborhood of x contained in V whose center contains x. The criterion remains valid if $\geq$ is replaced by $>$. For a weight with analytic singularities, Theorem 12.6.5 needs instead only the finitely many strict inequalities (12.21) on one log resolution, without a parameter ε.

Proof Necessity, even with strict inequality, is Proposition 12.6.7. Since the strict condition implies the non-strict one, it suffices to show that the non-strict condition implies $f \in \mathcal{I}(\varphi)_x$; we may assume $f \neq 0$. We compare φ with a weight φ_m with analytic singularities defined by generators of $\mathcal{I}(m\varphi)$, apply Theorem 12.6.5 to a multiple of φ_m, and then return to φ. Put $\lambda = 1 + \varepsilon/2$ and choose an integer $m \geq 2$ with $\lambda \geq m/(m-1)$, that is, $(m-1)\lambda \geq m$. On a bounded polydisc $\Omega \subset V$ around x on which φ is bounded above, choose $g_1, \dots, g_r \in \mathcal{H}_{m\varphi}(\Omega)$ generating $\mathcal{I}(m\varphi)$ near x, and put

$$\varphi_m = \frac{1}{m} \log \sum_i |g_i|^2.$$

This is plurisubharmonic by Proposition 8.3.3, since $\mathcal{I}(m\varphi)_x \neq 0$ by Lemma 12.3.1, so that the g_i do not all vanish identically. By (12.9), $\varphi \leq \varphi_m + C_m$ on a neighborhood $W \subset V$ of x. Pulling this back to a smooth model and using the characterization of Lelong numbers in Proposition 12.1.1 at every point, we see that $\nu(\varphi_m, E) \leq \nu(\varphi, E)$ for every prime divisor E over an open subset of W. Hence, for every prime divisor E over an open neighborhood of x contained in W whose center contains x,

$$\operatorname{ord}_E f + A_X(E) \geq (1+\varepsilon)\nu(\varphi, E) \geq (1+\varepsilon)\nu(\varphi_m, E),$$

and this is strictly larger than $\lambda\nu(\varphi_m, E)$: when $\nu(\varphi_m, E) > 0$ because $1+\varepsilon > \lambda$, and when $\nu(\varphi_m, E) = 0$ because the left side is at least $A_X(E) \geq 1$.

Now let $\pi\colon Y \to W'$ be a log resolution of the ideal $(g_1, \dots, g_r)$ over an open neighborhood $W' \subset W$ of x, with $(g_1, \dots, g_r)\mathcal{O}_Y = \mathcal{O}_Y(-D)$ and $D = \sum_i d_i E_i$. Let $q \in E_i$, and choose coordinates w at q as before (12.20), so that the components of D through q are coordinate hyperplanes and $E_i = \{w_i = 0\}$. By (12.20), $\log \sum_l |g_l \circ \pi|^2 = \sum_j d_j \log |w_j|^2 + O(1) \leq d_i \log |w_i|^2 + C$ near q, where j runs over the components through q; since $|w_i| \leq |w - q|$, the characterization in Proposition 12.1.1 gives $\nu(\varphi_m \circ \pi, q) \geq d_i/m$. If q lies on no other component of D, then $\varphi_m \circ \pi = (d_i/m) \log |w_i|^2 + O(1)$ near q, and $\nu(\varphi_m \circ \pi, q) = d_i/m$ by Proposition 12.1.4. Such points exist, because E_i is not contained in the union of the other components of the support of D (Corollary A.2.13). Hence $\nu(\varphi_m, E_i) = d_i/m$. Every component E_i meeting $\pi^{-1}(x)$ has a center containing x, so the strict inequality above reads $\operatorname{ord}_{E_i} f + A_X(E_i) > (\lambda/m)d_i$ for these components, and Theorem 12.6.5, applied on W' to the weight $\lambda\varphi_m$ with $c = \lambda/m$ and $b = 0$, gives $f \in \mathcal{I}(\lambda\varphi_m)_x$.

It remains to pass from $\lambda\varphi_m$ to φ. By the definition of the generators,

$$\mathrm{e}^{m(\varphi_m - \varphi)} = \sum_i |g_i|^2 \mathrm{e}^{-m\varphi} \in L^1_{\mathrm{loc}} \tag{12.26}$$

near x. Shrinking the neighborhood and subtracting a constant from φ_m, we may assume $\varphi_m \leq 0$; this multiplies the functions in question by constants and changes none of the integrability statements. We split a small neighborhood of x into the set where $\varphi \geq \lambda\varphi_m$ and its complement, ignoring the null set formed by the poles of φ and the common zeros of the g_i, so that all exponents below are finite. On the first set, $|f|^2\mathrm{e}^{-\varphi} \leq |f|^2\mathrm{e}^{-\lambda\varphi_m}$, which is integrable because $f \in \mathcal{I}(\lambda\varphi_m)_x$. On the complement we have $(m-1)\varphi < (m-1)\lambda\varphi_m$, since $m \geq 2$; as $(m-1)\lambda \geq m$ and $\varphi_m \leq 0$, it follows that

$$(m-1)\varphi - m\varphi_m \leq ((m-1)\lambda - m)\varphi_m \leq 0,$$

and therefore

$$|f|^2\mathrm{e}^{-\varphi} = |f|^2\mathrm{e}^{m(\varphi_m - \varphi)}\mathrm{e}^{(m-1)\varphi - m\varphi_m} \leq C\mathrm{e}^{m(\varphi_m - \varphi)},$$

where C is a local bound for $|f|^2$. By (12.26) the right side is integrable near x. Hence $|f|^2 e^{-\varphi}$ is integrable near x, that is, $f \in \mathcal{I}(\varphi)_x$. □

Example 12.6.10 Let $c > 0$ and $\varphi = c \log |z_1|^2$ on a polydisc in $\mathbb{C}^n$. The ideal (z_1) is already principal with smooth zero set, so the identity map is a log resolution, with $D = E = \{z_1 = 0\}$, $d = 1$ and $A_X(E) = 1$. By Theorem 12.6.5, a germ f lies in $\mathcal{I}(\varphi)$ if and only if $\operatorname{ord}_E f + 1 > c$. Here the log resolution has no exceptional divisors, and the only condition comes from the non-exceptional component E; this is why Theorem 12.6.5 involves all components of D meeting the fiber, exceptional or not.

For the blowup π of the origin in dimension $n \geq 2$, consider the chart $z_1 = w_1$, $z_j = w_1 w_j$ for $j > 1$ of Proposition A.5.3. The Jacobian matrix $(\partial z_i / \partial w_j)$ is lower triangular with diagonal entries $1, w_1, \ldots, w_1$, so $J_\pi = w_1^{n-1}$, and the exceptional divisor $E = \{w_1 = 0\}$ has $A_X(E) = n$. Let f be holomorphic near 0 with homogeneous expansion $f = \sum_{k \geq k_0} P_k$ and $P_{k_0} \neq 0$. Then $f \circ \pi(w) = \sum_{k \geq k_0} w_1^k P_k(1, w_2, \ldots, w_n) = w_1^{k_0} g(w)$ for small w_1 and bounded $(w_2, \ldots, w_n)$, where $g(0, w') = P_{k_0}(1, w')$; this polynomial is not identically zero, since $P_{k_0}(z) = z_1^{k_0} P_{k_0}(1, z_2/z_1, \ldots, z_n/z_1)$ for $z_1 \neq 0$. Hence $\operatorname{ord}_E f = \operatorname{ord}_0 f$.

For $\varphi = c \log |z|^2$, this blowup is a log resolution of the maximal ideal, with $D = E$ and $d = 1$, by Proposition A.5.3, so Theorem 12.6.5 gives $f \in \mathcal{I}(\varphi)_0$ if and only if $\operatorname{ord}_0 f + n > c$. This agrees with a direct computation in polar coordinates: since $|f(z)| \leq C|z|^{k_0}$ near 0 by (7.2), and $\int_{S^{2n-1}} |f(t\omega)|^2 \, dS(\omega) \geq c' t^{2k_0}$ for small $t > 0$, with $c' > 0$, by (12.8), the integral of $|f|^2 |z|^{-2c}$ over a small ball $B(0, r)$ is finite if and only if $\int_0^r t^{2k_0 - 2c + 2n - 1} \, dt < \infty$, by (2.1), that is, if and only if $k_0 + n > c$.

Chapter 13
Analytic singularities and Demailly approximation

Let u be a plurisubharmonic function on a domain in $\mathbb{C}^n$. Convolution with a smoothing kernel approximates u by smooth plurisubharmonic functions, but these approximants have lost all singularities: their Lelong numbers vanish and their multiplier ideals are trivial. Many problems require approximants whose singularities are still visible but as simple as possible, namely functions locally of the form $c \log \sum_i |g_i|^2 + O(1)$ with holomorphic g_i and rational $c > 0$; such functions are said to have analytic singularities. On a compact Kähler manifold a second difficulty appears: convolution is defined only in charts, and gluing local approximations costs positivity. This chapter presents Demailly's answer to both problems, in which convolution is replaced by weighted Bergman kernels.

For u plurisubharmonic on a bounded pseudoconvex domain Ω and an integer $m \geq 1$, the holomorphic functions F on Ω with $\int_\Omega |F|^2 \mathrm{e}^{-mu} < \infty$ form a Hilbert space, which generates the multiplier ideal $\mathcal{I}(mu)$ of Chapter 12. If $K_m(z)$ denotes the supremum of $|F(z)|^2$ over its unit ball, then $u_m = m^{-1} \log K_m$ is plurisubharmonic and has analytic singularities described by $\mathcal{I}(mu)$. The Ohsawa–Takegoshi estimate of Chapter 11 bounds u_m below by $u - O(1/m)$, and the mean-value inequality bounds it above by the supremum of u over a small ball, up to an error that tends to zero as $m \to \infty$. Hence $u_m \to u$, and the Lelong numbers of u_m differ from those of u by at most n/m (Theorem 13.2.3).

On a compact Kähler manifold (X, ω) we apply this construction in finitely many coordinate balls to a function φ with $\theta + \mathrm{dd}^{\mathrm{c}}\varphi \geq 0$, where θ is a smooth closed real $(1,1)$-form. A $\bar\partial$-estimate with a logarithmic pole shows that the kernels of two overlapping balls are comparable up to a factor that depends only on the charts (Lemma 13.3.2). Gluing the local potentials by a maximum gives the quantitative regularization of Theorem 13.3.3, with a curvature loss of order $1/m$ and a lower bound by the generators of $\mathcal{I}(m\varphi)$. Gluing them instead by a sum with smooth cutoffs gives approximants G_m with gentle analytic singularities (Lemma 13.4.1): $\exp(mG_m)$ is smooth, and G_m has a smooth remainder on every log resolution of $\mathcal{I}(m\varphi)$. Comparing the kernels at the scales m and $2m$ by extension from the diagonal, we show that the dyadic sequence G_{2^j}, corrected by small constants, decreases to φ and that its multiplier ideals approximate those of φ, so that it is a quasi-equisingular

approximation of φ in the sense of Definition 13.4.4 (Theorem 13.4.5). Any two quasi-equisingular approximations of φ dominate each other up to an arbitrarily small loss in the coefficient (Proposition 13.5.2).

The first section collects the elementary properties of quasi-plurisubharmonic functions, introduces θ-plurisubharmonic functions and analytic singularities, and shows, using metrics on line bundles, that blowups with smooth centers preserve the Kähler property. The last section studies the ideal of holomorphic germs f with $|f|^2 \le Ce^v$: for v with analytic singularities it is coherent, and on a compact manifold the ideal attached to kv contains $\mathcal{I}(k(1+\varepsilon)v)$ for all large k.

Historical remarks and references for this chapter are collected in Appendix B.13.

13.1 Smooth backgrounds and analytic singularities

Let X be a complex manifold of dimension $n \ge 0$. Recall from Section 7.5 that a Hermitian form on X is a smooth real $(1,1)$-form that is strictly positive at every point, that is, the form associated with a Hermitian metric. It is a Kähler form if it is closed, and X is a Kähler manifold if it carries a Kähler form. A Hermitian form ω defines a Riemannian metric on X and the positive volume form $\omega^n/n!$.

Continuous real $(1,1)$-forms are controlled by a Hermitian form on compact sets.

Lemma 13.1.1 *Let ω be a Hermitian form on a complex manifold X, let $K \subset X$ be compact, and let α be a continuous real $(1,1)$-form on a neighborhood of K. Then there is a constant $C \ge 0$ with $\alpha \ge -C\omega$ at every point of K.*

Proof If $K = \varnothing$ we take $C = 0$. Otherwise cover K by finitely many compact sets K_ι, each contained in a chart and in the domain of α, and write $\alpha = \mathrm{i}\sum_{j,k} a_{jk}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ and $\omega = \mathrm{i}\sum_{j,k} w_{jk}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$ there, with Hermitian matrices (a_{jk}) and (w_{jk}). On K_ι the smallest eigenvalue of (w_{jk}), a continuous positive function, is at least some $c_\iota > 0$, and the largest eigenvalue of $-(a_{jk})$ is at most some $M_\iota \ge 0$. Hence $(a_{jk}) \ge -M_\iota I \ge -(M_\iota/c_\iota)(w_{jk})$ on K_ι. By the matrix criterion for positivity (Section 7.3), it follows that $\alpha \ge -C\omega$ on K with $C = \max_\iota M_\iota/c_\iota$. □

The following lemma provides smooth local potentials for smooth closed forms on whole coordinate balls, not only on small polydiscs as in Lemma 7.3.1.

Lemma 13.1.2 *Let B be a coordinate ball and let θ be a smooth closed real $(1,1)$-form defined on a neighborhood of $\overline{B}$. Then there is a smooth real function h on B with $\mathrm{dd^c}h = \theta$. Moreover, every real pluriharmonic function on a coordinate ball is, up to an additive constant, the real part of a holomorphic function on that ball.*

Proof In dimension zero there is nothing to prove, so let $n \ge 1$. Choose concentric balls $B \Subset B' \Subset B''$ such that θ is defined on a neighborhood of $\overline{B''}$. By the last assertion of Lemma 7.3.1, applied to the closed real 2-form θ on the ball B'', there is a smooth real one-form α on B'' with $\mathrm{d}\alpha = \theta$. Since θ has type $(1,1)$, the $(0,2)$-part $\bar{\partial}\alpha^{0,1}$ of $\mathrm{d}\alpha$ vanishes. The ball B' is pseudoconvex by Corollary 11.2.5, and $\alpha^{0,1}$

is bounded on B'. Hence Theorem 11.3.4, applied on B' with $p = 0$ and $q = |z|^2$, whose complex Hessian is the identity, gives a function $f \in L^2(B')$ with $\bar\partial f = \alpha^{0,1}$.

The function f is smooth. Indeed, let $y \in B'$ and choose polydiscs $P' \Subset P \Subset B'$ centered at y. By Lemma 7.3.1, applied to the smooth $\bar\partial$-closed form $\alpha^{0,1}$ on P, there is a smooth function f_0 on P' with $\bar\partial f_0 = \alpha^{0,1}$. Then $\bar\partial(f - f_0) = 0$ on P' in the sense of distributions, so $\Delta(f - f_0) = 4\sum_j \partial_{z_j}\partial_{\bar z_j}(f - f_0) = 0$ there, and by Theorem 2.1.4 the locally integrable function $f - f_0$ agrees almost everywhere on P' with a smooth function. Thus f agrees almost everywhere near each point with a smooth function; two such smooth functions coincide on the overlap of their domains, being continuous and equal almost everywhere, so we may take f smooth on B'. Since α is real, $\alpha^{1,0} = \overline{\alpha^{0,1}} = \partial\bar f$, and the $(1,1)$-part of $\mathrm{d}\alpha = \theta$ gives

$$\theta = \partial\alpha^{0,1} + \bar\partial\alpha^{1,0} = \partial\bar\partial f + \bar\partial\partial\bar f = \partial\bar\partial(f - \bar f) = \mathrm{dd^c}(4\pi\,\mathrm{Im}\, f).$$

Thus $h = 4\pi\,\mathrm{Im}\, f$ has the required property on $B \subset B'$.

For the last assertion, let h be real and pluriharmonic on a coordinate ball. Then ∂h is a $(1,0)$-form with holomorphic coefficients, since $\bar\partial\partial h = 0$, and it is closed. By the last assertion of Lemma 7.3.1, applied to the closed one-form $2\partial h$ on this ball, there is a smooth function F on the whole ball with $\mathrm{d}F = 2\partial h$. As $\mathrm{d}F$ has type $(1,0)$, F is holomorphic, and $\mathrm{d}(\mathrm{Re}\, F) = \partial h + \bar\partial h = \mathrm{d}h$. Hence $h - \mathrm{Re}\, F$ is constant. □

Definition 13.1.3 Let X be a complex manifold. A function $\varphi\colon X \to [-\infty, \infty)$ is *quasi-plurisubharmonic* if it is quasi-plurisubharmonic near every point of X in the sense of Corollary 12.3.2, that is, if every point of X has a neighborhood V on which $\varphi = u + f$ pointwise, with $u \in \mathrm{PSH}(V)$ and f a smooth real function on V. For a smooth closed real $(1,1)$-form θ on X we put

$$\mathrm{PSH}(X, \theta) = \{\varphi \text{ quasi-plurisubharmonic on } X : \theta + \mathrm{dd^c}\varphi \geq 0\},$$

and call its elements *θ-plurisubharmonic* (*θ-psh*) functions.

We collect the elementary properties of these classes.

Lemma 13.1.4 *Let X be a complex manifold and let θ be a smooth closed real $(1,1)$-form on X.*

(1) *Every quasi-plurisubharmonic function on X is locally integrable, hence finite almost everywhere. If X has positive dimension, a quasi-plurisubharmonic function is, at every point, the limit of its averages over small coordinate balls centered at that point. Consequently, if φ and ψ are quasi-plurisubharmonic functions on an open set $V \subset X$ with $\varphi \leq \psi$ almost everywhere on V, then $\varphi \leq \psi$ everywhere on V; in particular, two quasi-plurisubharmonic functions on V that agree almost everywhere on V agree everywhere on V.*

(2) *Let $V \subset X$ be open and let h be a smooth real function on V with $\mathrm{dd^c}h = \theta$. Then a quasi-plurisubharmonic function φ on V satisfies $\theta + \mathrm{dd^c}\varphi \geq 0$ on V if and only if $\varphi + h$ is plurisubharmonic on V.*

(3) *If $\varphi_1, \ldots, \varphi_k \in \mathrm{PSH}(X, \theta)$, $k \geq 1$, then $\max_i \varphi_i \in \mathrm{PSH}(X, \theta)$.*

(4) *If X is compact and ω is a Hermitian form on X, then for every quasi-plurisubharmonic function φ on X there is a constant $C \geq 0$ with* $\mathrm{dd}^c\varphi \geq -C\omega$.
(5) *Let ω be a Hermitian form on X, and let $\varphi \in L^1_{\mathrm{loc}}(X)$ satisfy* $\mathrm{dd}^c\varphi \geq -C\omega$ *in the sense of currents for a constant C. Then φ agrees almost everywhere with a quasi-plurisubharmonic function on X.*

Proof At points where X has dimension zero, which are isolated, all functions are smooth and dd^c vanishes, so every assertion is trivial there; we may therefore assume that X has positive dimension.

(1) Near each point, $\varphi = u + f$ with u plurisubharmonic and f smooth, and u is locally integrable by Theorem 8.2.1. Moreover u is subharmonic, by the same theorem, so at every point it is the limit of its averages over small balls, by (3.3); the continuous function f has the same property, and hence so has φ. Now let φ and ψ be quasi-plurisubharmonic on V with $\varphi \leq \psi$ almost everywhere, and let $x \in V$. Since both functions are locally integrable, their averages over small coordinate balls centered at x, in a chart at x, are finite, and those of φ are at most those of ψ; letting the radius tend to zero gives $\varphi(x) \leq \psi(x)$. Applying this in both directions, we see that two quasi-plurisubharmonic functions that agree almost everywhere agree everywhere.

(2) Let φ be quasi-plurisubharmonic on V. If $\varphi + h$ is plurisubharmonic, then $\theta + \mathrm{dd}^c\varphi = \mathrm{dd}^c(\varphi + h) \geq 0$ by Theorem 8.2.1. Conversely, suppose that $\mathrm{dd}^c(\varphi + h) \geq 0$ on V. On a coordinate ball $B \subset V$, Theorem 8.2.1 gives a plurisubharmonic function on B that agrees almost everywhere with $\varphi + h$; both are quasi-plurisubharmonic on B, so they agree everywhere on B by (1). Hence $\varphi + h$ is plurisubharmonic on V.

(3) Let B be a coordinate ball and h a smooth real function on B with $\mathrm{dd}^c h = \theta$ (Lemma 13.1.2). By (2) the functions $\varphi_i + h$ are plurisubharmonic on B, hence so is $\max_i \varphi_i + h = \max_i(\varphi_i + h)$, by Proposition 8.2.5. Thus $\max_i \varphi_i$ is quasi-plurisubharmonic on B, and $\theta + \mathrm{dd}^c \max_i \varphi_i \geq 0$ there by (2). Such balls cover X.

(4) By compactness there are finitely many open sets $V_\kappa \Subset V_\kappa^+$ such that the V_κ cover X and $\varphi = u_\kappa + f_\kappa$ on V_κ^+, with u_κ plurisubharmonic and f_κ smooth. By Lemma 13.1.1, applied to the compact set $\overline{V_\kappa}$ and the form $\mathrm{dd}^c f_\kappa$, there are constants $C_\kappa \geq 0$ with $\mathrm{dd}^c f_\kappa \geq -C_\kappa\omega$ on V_κ, and $\mathrm{dd}^c u_\kappa \geq 0$ by Theorem 8.2.1. Hence $\mathrm{dd}^c\varphi \geq -C\omega$ with $C = \max_\kappa C_\kappa$.

(5) Replacing C by $\max\{C, 0\}$, we may assume that $C \geq 0$. Let B be a coordinate ball whose closure lies in a chart. By Lemma 13.1.1, applied in the chart to the compact set $\overline{B}$, the Hermitian form $\mathrm{dd}^c|z|^2$ and the form $-\omega$, there is C' with $\omega \leq C'\mathrm{dd}^c|z|^2$ on B, so $\mathrm{dd}^c(\varphi + A|z|^2) \geq 0$ on B for $A = CC'$. By Theorem 8.2.1, $\varphi + A|z|^2$ agrees almost everywhere on B with a plurisubharmonic function, and subtracting $A|z|^2$ gives a quasi-plurisubharmonic function φ_B on B that agrees almost everywhere with φ. If B' is a second such ball, then φ_B and $\varphi_{B'}$ agree almost everywhere on $B \cap B'$, and both are quasi-plurisubharmonic, so they agree everywhere on $B \cap B'$ by (1). Since such balls cover X, the functions φ_B define a quasi-plurisubharmonic function on X that agrees almost everywhere with φ. □

Multiplier ideals of a quasi-plurisubharmonic weight are defined as in Definition 12.2.5. Adding a smooth function to the weight multiplies the integrand by a locally bounded positive factor with locally bounded inverse, so the multiplier ideal of $\varphi = u + f$ near a point is that of u; in particular it is coherent, and the local results of Chapter 12 apply to it. Lelong numbers of quasi-plurisubharmonic functions are defined as in Corollary 12.3.2, by $\nu(\varphi, x) = \nu(u, x)$.

Definition 13.1.5 A quasi-plurisubharmonic function φ on X has *analytic singularities* if every point of X has a connected coordinate neighborhood V on which

$$\varphi = c \log \sum_{i=1}^{r} |g_i|^2 + b, \qquad c \in \mathbb{Q}_{>0}, \quad b \text{ bounded}, \tag{13.1}$$

where $r \geq 1$ is an integer and $g_1, \ldots, g_r \in \mathcal{O}(V)$ are not all identically zero. It has *neat analytic singularities* if such presentations exist with b smooth. Let $c \in \mathbb{Q}_{>0}$ and let $\mathfrak{a} \subset \mathcal{O}_X$ be a coherent ideal sheaf. We say that φ has analytic singularities of *type* $(c, \mathfrak{a})$ if presentations (13.1) exist with this coefficient c, where on each V the functions $g_1, \ldots, g_r$ generate $\mathfrak{a}$.

The type condition does not depend on the choice of local generators: if (g_i) and (g'_k) both generate $\mathfrak{a}$ on V, then near each point of V each tuple is a holomorphic matrix times the other, and both matrices are bounded on smaller neighborhoods, so $\log \sum |g_i|^2 - \log \sum |g'_k|^2$ is locally bounded. A type is additional data and is not unique; for example, a function of type $(c, \mathfrak{a})$ also has type $(c/2, \mathfrak{a}^2)$.

The Lelong numbers of a function with analytic singularities are determined by its type.

Lemma 13.1.6 *Let x be a point of a complex manifold X at which X has positive dimension, let $\mathfrak{a} \subset \mathcal{O}_X$ be a coherent ideal sheaf with $\mathfrak{a}_x \neq 0$, generated near x by $g_1, \ldots, g_s$, and let $c > 0$. If w is quasi-plurisubharmonic near x and $w = c \log \sum_i |g_i|^2 + O(1)$ near x, then $\nu(w, x) = c \operatorname{ord}_x \mathfrak{a}$.*

Proof Choose a connected coordinate neighborhood V of x on which the g_i are defined and generate $\mathfrak{a}$, on which $w = v + f$ with v plurisubharmonic and f smooth and bounded, and on which $|w - c \log \sum_i |g_i|^2| \leq M$ for a constant M. Let I be the set of indices i for which the germ of g_i at x is not zero. It is not empty, since these germs generate $\mathfrak{a}_x \neq 0$, and for $i \notin I$ the function g_i vanishes identically on V, by the identity theorem. Since $0 \leq \log \sum_{i \in I} |g_i|^2 - \max_{i \in I} \log |g_i|^2 \leq \log s$, the function v differs from $c \max_{i \in I} \log |g_i|^2$ by a bounded function on V. The latter function is plurisubharmonic by Propositions 8.2.5 and 8.3.3, so Proposition 12.1.1 and Proposition 12.1.4 give

$$\nu(w, x) = \nu(v, x) = c\, \nu\Big(\max_{i \in I} \log |g_i|^2, x\Big) = c \min_{i \in I} \operatorname{ord}_x g_i = c \operatorname{ord}_x \mathfrak{a},$$

where the first equality is the definition of Lelong numbers of quasi-plurisubharmonic functions (Corollary 12.3.2) and the last one is the description of $\operatorname{ord}_x \mathfrak{a}$ by generators in Section 12.3. □

For a holomorphic map $\pi\colon Y \to X$ and an ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$, we write $\pi^*\mathfrak{a} = \mathfrak{a} \cdot \mathcal{O}_Y$ for the ideal generated by the pullbacks of local generators of $\mathfrak{a}$. Let $\mathfrak{a}$ be a coherent ideal on X, nonzero on each component. Since X is smooth, the resolution step in the construction of log resolutions at the end of Appendix A.5 is not needed, and Theorem A.5.5, applied with the empty divisor, produces log resolutions of $\mathfrak{a}$ by blowups with smooth centers. From now on a *log resolution* of $\mathfrak{a}$ always means a map of this form: a composite $\pi\colon Y \to X$ of blowups along closed submanifolds of positive codimension, finite over each relatively compact open subset of X, such that $\pi^*\mathfrak{a} = \mathcal{O}_Y(-D)$ for an effective divisor D whose support has simple normal crossings. We also call π a *principalization* of $\mathfrak{a}$, and we write s_D for a local holomorphic equation of D.

Proposition 13.1.7 *Analytic singularities are preserved by multiplication by positive rational numbers, by sums and finite maxima, and by pullback under a holomorphic map $f\colon Y \to X$ of complex manifolds whenever the composition is not identically $-\infty$ on any component of Y. If φ has type $(c, \mathfrak{a})$ and $\pi\colon Y \to X$ is a log resolution of $\mathfrak{a}$ with $\pi^*\mathfrak{a} = \mathcal{O}_Y(-D)$, then locally on Y*

$$\varphi \circ \pi = c \log |s_D|^2 + O(1),$$

and the remainder is smooth if the presentations of type $(c, \mathfrak{a})$ are neat.

Proof Multiplication by $q \in \mathbb{Q}_{>0}$ replaces c by qc. Sums and finite maxima of quasi-plurisubharmonic functions are quasi-plurisubharmonic. Indeed, let $\varphi_i = u_i + \sigma_i$ near a point, with u_i plurisubharmonic and σ_i smooth. On a small coordinate ball about the point, the smooth functions $\sigma_i + N|z|^2$ have positive semidefinite complex Hessians for a large constant N, hence are plurisubharmonic by Proposition 8.2.2(1). By Proposition 8.2.5, $\varphi_1 + \varphi_2$ is a plurisubharmonic function minus $2N|z|^2$, and $\max_i \varphi_i$ is a plurisubharmonic function minus $N|z|^2$. For a sum or a maximum of two functions with analytic singularities, choose presentations about a given point, with coefficients c and d, and a connected coordinate neighborhood V of the point contained in both domains; by the identity theorem, neither tuple vanishes identically on V. Since c and d are rational, we can choose $c_0 \in \mathbb{Q}_{>0}$ such that c/c_0 and d/c_0 are positive integers. For an integer $k \geq 1$ the multinomial formula gives

$$\Bigl(\sum_i |g_i|^2\Bigr)^k = \sum_{|\alpha|=k} \Bigl|\sqrt{k!/\alpha!}\, g^\alpha\Bigr|^2,$$

again a sum of squared moduli of holomorphic functions, not all identically zero on V. We may therefore assume that both presentations have the coefficient c_0, say $c_0 \log A + b$ and $c_0 \log B + b'$ with $A = \sum_i |g_i|^2$ and $B = \sum_l |g'_l|^2$. The sum is $c_0 \log(AB) + b + b'$, and $AB = \sum_{i,l} |g_i g'_l|^2$, where some product $g_i g'_l$ is not identically zero on the connected set V. For the maximum we use

$$\max(\log A, \log B) \leq \log(A + B) \leq \max(\log A, \log B) + \log 2,$$

which shows that $\max(c_0 \log A + b, c_0 \log B + b')$ differs from $c_0 \log(A + B)$ by a bounded function.

For the pullback, let $f\colon Y \to X$ be holomorphic, and suppose that $\varphi \circ f$ is not identically $-\infty$ on any connected component of Y. Near each point of Y, write $\varphi \circ f = u \circ f + \sigma \circ f$ on a connected coordinate neighborhood W, where $\varphi = u + \sigma$ near the image point with u plurisubharmonic and σ smooth; by Proposition 8.2.3, $u \circ f$ is plurisubharmonic or identically $-\infty$ on W. Let S be the open set of points of Y near which $\varphi \circ f$ is identically $-\infty$. If y lies in the closure of S and W is as above for y, then $u \circ f = -\infty$ on the nonempty open set $W \cap S$; since plurisubharmonic functions are locally integrable, $u \circ f$ is identically $-\infty$ on W, so $y \in S$. Thus S is open and closed, and by hypothesis it contains no component of Y; hence $S = \varnothing$, and $\varphi \circ f$ is quasi-plurisubharmonic. If $\varphi = c \log \sum_i |g_i|^2 + b$ on V, then on every connected coordinate neighborhood $W \subset f^{-1}(V)$ we have $\varphi \circ f = c \log \sum_i |g_i \circ f|^2 + b \circ f$, with $b \circ f$ bounded, and the functions $g_i \circ f$ do not all vanish identically on W, because $S = \varnothing$.

Now let π be a log resolution of $\mathfrak{a}$ as in the statement. If $g_1, \dots, g_r$ generate $\mathfrak{a}$, their pullbacks generate $\mathcal{O}_Y(-D)$, so locally $g_i \circ \pi = s_D h_i$ with holomorphic h_i, and $s_D = \sum_i k_i (g_i \circ \pi)$ for holomorphic k_i. Since s_D is not a zero divisor, $\sum_i k_i h_i = 1$, so the h_i have no common zero and $\log \sum_i |h_i|^2$ is smooth. Hence

$$\varphi \circ \pi = c \log |s_D|^2 + c \log \sum_i |h_i|^2 + b \circ \pi,$$

which proves both assertions about the remainder. □

If φ has analytic singularities of type $(c, \mathfrak{a})$, we also call a log resolution of $\mathfrak{a}$ a *log resolution of* φ; since types are not unique, this refers to the chosen type. On such a resolution $\varphi \circ \pi = c \log |s_D|^2 + O(1)$ locally, by Proposition 13.1.7. A merely bounded remainder need not become smooth on a log resolution. For instance, on the unit disc in $\mathbb{C}$ the function $\log |z|^2 + |\operatorname{Re} z|$ is subharmonic, since $|\operatorname{Re} z| = \max(\operatorname{Re} z, -\operatorname{Re} z)$ is a maximum of harmonic functions (Proposition 8.2.5), and it has analytic singularities of type $(1, (z))$. In dimension one a closed submanifold of positive codimension is a hypersurface, so every blowup in a log resolution is the identity by Proposition A.5.3. The remainder $|\operatorname{Re} z|$ therefore stays as it is, and it is not smooth on the imaginary axis. This is why neat singularities, and the gentle singularities defined below on compact Kähler manifolds, are separate notions.

When X is compact, a log resolution consists of finitely many blowups with smooth centers, and if X is moreover Kähler, so is the resolved manifold (Proposition 13.1.10). The proof uses metrics on line bundles and their curvature, which we now introduce.

Definition 13.1.8 A holomorphic line bundle L on X is given by an open cover (V_i) of X and local frames e_i over V_i with transition functions $e_i = g_{ij} e_j$ on $V_i \cap V_j$, where g_{ij} is holomorphic and nowhere zero and $g_{ij} g_{jk} = g_{ik}$ on triple overlaps. A smooth *quadratic Hermitian metric* on L assigns to each frame a smooth positive squared norm $|e_i|^2$ on V_i, such that $|e_i|^2 = |g_{ij}|^2 |e_j|^2$ on $V_i \cap V_j$. Its *curvature* is the closed real $(1,1)$-form equal to $-\mathrm{dd}^{\mathrm{c}} \log |e_i|^2$ on V_i.

When several metrics occur, we write $|e_i|_h^2$ for the squared norms of a metric h. Smooth metrics always exist, and any two of them differ by a smooth positive factor.

Lemma 13.1.9 *Let L be a holomorphic line bundle on a complex manifold X, given by frames e_i over an open cover (V_i) with transition functions g_{ij} as in Definition 13.1.8.*

(1) *L carries a smooth quadratic Hermitian metric.*
(2) *The curvature of a smooth quadratic Hermitian metric on L is well defined, and it is a smooth closed real $(1,1)$-form on X.*
(3) *Let h_0 be a smooth quadratic Hermitian metric on L with curvature η_0. For every smooth real function f on X, the squared norms $|e_i|_{h_0}^2 \mathrm{e}^{-f}$ define a smooth quadratic Hermitian metric $h_0\mathrm{e}^{-f}$ on L, whose curvature is $\eta_0 + \mathrm{dd^c} f$. Conversely, every smooth quadratic Hermitian metric on L equals $h_0\mathrm{e}^{-f}$ for exactly one smooth real function f on X.*

Proof (1) By Proposition 7.1.3 there is a smooth partition of unity (χ_a) subordinate to the cover (V_i); for each a choose an index $k(a)$ with $\operatorname{Supp}\chi_a \subset V_{k(a)}$, and put $\rho_k = \sum_{k(a)=k} \chi_a$. Since the family (χ_a) is locally finite, each ρ_k is smooth, the family (ρ_k) is locally finite, $\sum_k \rho_k = 1$, and $\operatorname{Supp}\rho_k \subset V_k$, because a locally finite union of closed sets is closed. For each i put

$$|e_i|^2 = \sum_k \rho_k |g_{ik}|^2 \qquad \text{on } V_i,$$

where the kth term, defined on $V_i \cap V_k$, is extended by zero to the rest of V_i. Each term is smooth on V_i, since it is smooth on $V_i \cap V_k$ and vanishes on the open set $V_i \setminus \operatorname{Supp}\rho_k$, and these two open sets cover V_i. By local finiteness, $|e_i|^2$ is smooth on V_i, and it is positive, because the ρ_k are nonnegative with sum one and every g_{ik} is nowhere zero. On $V_i \cap V_j$, the cocycle relation $g_{ik} = g_{ij}g_{jk}$ holds on $V_i \cap V_j \cap V_k$, hence wherever $\rho_k \neq 0$, so

$$|e_i|^2 = \sum_k \rho_k |g_{ij}|^2 |g_{jk}|^2 = |g_{ij}|^2 |e_j|^2.$$

Thus these squared norms form a smooth quadratic Hermitian metric on L.

(2) On $V_i \cap V_j$ we have $\log|e_i|^2 - \log|e_j|^2 = \log|g_{ij}|^2$, which is pluriharmonic, since g_{ij} is holomorphic and nowhere zero (Section 7.3). Hence the forms $-\mathrm{dd^c}\log|e_i|^2$ on V_i agree on the overlaps and define a $(1,1)$-form η on X. It is smooth, because each $\log|e_i|^2$ is a smooth real function. Moreover $\eta = -\mathrm{d}(\mathrm{d^c}\log|e_i|^2)$ on V_i, by (7.6), and $\mathrm{d^c}$ maps real forms to real forms; hence η is real, and it is closed because $\mathrm{d}^2 = 0$.

(3) The functions $|e_i|_{h_0}^2 \mathrm{e}^{-f}$ are smooth and positive on V_i, and they satisfy the compatibility condition of Definition 13.1.8 because the functions $|e_i|_{h_0}^2$ do. The curvature of $h_0\mathrm{e}^{-f}$ is $-\mathrm{dd^c}\log\bigl(|e_i|_{h_0}^2 \mathrm{e}^{-f}\bigr) = -\mathrm{dd^c}\log|e_i|_{h_0}^2 + \mathrm{dd^c} f = \eta_0 + \mathrm{dd^c} f$ on V_i. Conversely, let h_1 be a smooth quadratic Hermitian metric on L. On V_i the function $|e_i|_{h_1}^2 / |e_i|_{h_0}^2$ is smooth and positive, and on $V_i \cap V_j$ it equals $|e_j|_{h_1}^2 / |e_j|_{h_0}^2$, since both numerator and denominator are multiplied by $|g_{ij}|^2$. Hence these quotients define a

smooth positive function on X, which we write as e^{-f} with f smooth and real; then $h_1 = h_0 e^{-f}$. The function f is unique, since e^{-f} must equal these quotients. □

Proposition 13.1.10 *Let X be a compact Kähler manifold with Kähler form ω, let $W \subset X$ be a closed submanifold of positive codimension, and let $\pi\colon Y \to X$ be the blowup of X along W, with exceptional divisor $E = \pi^{-1}(W)$. Then Y is a compact complex manifold, and there are a smooth function $\rho\colon X \to [0,\infty)$ with $\rho^{-1}(0) = W$ and a smooth closed real $(1,1)$-form η on Y with $\eta = \mathrm{dd}^c \log(\rho\circ\pi)$ on $Y \setminus E$, such that $\pi^*\omega + \varepsilon\eta$ is a Kähler form on Y for every sufficiently small $\varepsilon > 0$. The form η is the curvature of a smooth quadratic Hermitian metric on the line bundle $\mathcal{O}_Y(-E)$, whose local frames are the function 1 on $Y \setminus E$ and the local equations of E near E. In particular Y is a compact Kähler manifold. Consequently every log resolution of a coherent ideal on X, as above, is a compact Kähler manifold.*

Proof By Proposition A.5.3, Y is a complex manifold and E is a smooth hypersurface with $\mathcal{I}_W\mathcal{O}_Y = \mathcal{O}_Y(-E)$; moreover Y is compact, since π is proper (Proposition A.5.2), and π restricts to a biholomorphism $Y \setminus E \to X \setminus W$. The last assertion follows from the preceding ones by induction: since X is compact, a log resolution is a composite of finitely many blowups along closed submanifolds of positive codimension, and each intermediate manifold is compact Kähler by what precedes.

We construct η as the curvature of a metric on $\mathcal{O}_Y(-E)$ and show that it is positive definite along the fibers of $E \to W$ (Step 1). Since $\pi^*\omega$ is positive in the remaining directions, $\pi^*\omega + \varepsilon\eta$ is then positive definite near E for small $\varepsilon > 0$ (Step 2), and also away from E (Step 3). We use the charts of Proposition A.5.3. Near a point of W at which W has codimension r, choose coordinates (z,w) on X, $w = (w_1,\dots,w_r)$, with $W = \{w = 0\}$. For each pivot i the blowup has a chart with coordinates (z,t,v), $v = (v_j)_{j\neq i}$, in which

$$\pi(z,t,v) = (z, t\tilde{v}), \qquad \tilde{v}_i = 1, \quad \tilde{v}_j = v_j \ (j \neq i);$$

in it $E = \{t = 0\}$, and the fibers of $E \to W$ are the sets $\{z = \text{const},\ t = 0\}$. If $r = 1$ there is no variable v, and π is the identity near the point.

Step 1: a closed $(1,1)$-form that is positive along the fibers of E. Choose finitely many charts $V_1,\dots,V_N$ of X covering W, with coordinates (z^α, w^α) such that $W \cap V_\alpha = \{w^\alpha = 0\}$, and put $V_0 = X \setminus W$. Grouping the members of a partition of unity subordinate to this cover (Proposition 7.1.3) according to a set V_α containing their supports, we obtain a smooth partition of unity $(\lambda_\alpha)_{0\le\alpha\le N}$ on X with $\operatorname{Supp}\lambda_\alpha \subset V_\alpha$. Put

$$\rho = \lambda_0 + \sum_{\alpha=1}^{N} \lambda_\alpha |w^\alpha|^2,$$

each term extended by zero outside V_α. This is a smooth nonnegative function on X that vanishes on W, because $\lambda_0 = 0$ there, and is positive on $X \setminus W$: at a point $x \notin W$ some $\lambda_\alpha(x)$ is positive, and $w^\alpha(x) \neq 0$ if $\alpha \ge 1$.

We describe ρ near a point $x_0 \in W$, in coordinates (z,w) as above on a polydisc Δ centered at x_0. Shrinking Δ, we may assume that Δ does not meet $\operatorname{Supp}\lambda_0$, nor

Supp λ_α for the indices α with $x_0 \notin V_\alpha$, and that $\Delta \subset V_\alpha$ for the others. For such an α, each component w_k^α vanishes on $W = \{w = 0\}$, so

$$w_k^\alpha(z,w) = \sum_{l=1}^{r} w_l \int_0^1 \frac{\partial w_k^\alpha}{\partial w_l}(z,sw)\,\mathrm{d}s,$$

that is, $w^\alpha = A_\alpha w$ with a holomorphic $r \times r$ matrix A_α on Δ. At points of W the matrix $A_\alpha(z,0) = (\partial w_k^\alpha/\partial w_l)(z,0)$ is invertible: since w^α vanishes on W, its derivatives in the z-directions vanish there, and the Jacobian matrix of the coordinate change $(z,w) \mapsto (z^\alpha, w^\alpha)$ is invertible. Hence, on Δ,

$$\rho = \sum_\alpha \lambda_\alpha |A_\alpha w|^2 = w^* Q\, w, \qquad Q = \sum_\alpha \lambda_\alpha A_\alpha^* A_\alpha,$$

where the sums run over the indices $\alpha \geq 1$ with $x_0 \in V_\alpha$, and w^* is the conjugate transpose of the column vector w. The Hermitian matrix Q is smooth and positive semidefinite, and it is positive definite at x_0, because the coefficients $\lambda_\alpha(x_0)$ of the invertible matrices $A_\alpha(x_0)$ add up to one; shrinking Δ, we may assume that Q is positive definite on Δ. In a chart (z,t,v) over Δ we obtain

$$\rho \circ \pi = |t|^2 h_E, \qquad h_E(z,t,v) = \tilde{v}^* Q(z,t\tilde{v})\,\tilde{v},$$

and h_E is smooth and positive, since $\tilde{v} \neq 0$.

Define a real $(1,1)$-form η on Y by $\eta = \mathrm{dd^c}\log(\rho\circ\pi)$ on $Y \setminus E$ and $\eta = \mathrm{dd^c}\log h_E$ on each chart (z,t,v) as above. These definitions agree. Off E we have $\log(\rho\circ\pi) = \log|t|^2 + \log h_E$, and $\log|t|^2$ is pluriharmonic there. On the overlap of two charts, the local equations t and t' of E generate the same ideal $\mathcal{O}_Y(-E)$, so $t' = ut$ with u holomorphic and nowhere zero. Off E the corresponding functions h_E and h'_E satisfy $\log h_E - \log h'_E = \log|u|^2$. Both sides are continuous on the overlap and E is nowhere dense, so this identity holds on the whole overlap, and $\log|u|^2$ is pluriharmonic. Thus η is a smooth closed real $(1,1)$-form on Y. It is the curvature, in the sense of Definition 13.1.8, of the metric on the line bundle $\mathcal{O}_Y(-E)$ with the frame 1 on $Y \setminus E$, the frames t on the charts and the transition functions t and t'/t, for which $|1|^2 = 1/(\rho\circ\pi)$ and $|t|^2 = 1/h_E$. These squared norms are smooth and positive, and they satisfy the compatibility condition of that definition because $\rho\circ\pi = |t|^2 h_E$ and $h'_E = |t/t'|^2 h_E$.

We show that η is positive definite along the fibers of $E \to W$. Fix a point $(z_0,0,v_0)$ of E in a chart, and put $Q_0 = Q(z_0,0)$. On the fiber $\{z = z_0,\ t = 0\}$ we have $h_E = \tilde{v}^* Q_0 \tilde{v}$. Write $Q_0 = \Gamma^*\Gamma$ with an invertible matrix Γ, for instance the positive square root of Q_0, and let $e_1,\dots,e_r$ be the standard basis of $\mathbb{C}^r$. Then $h_E = |F(v)|^2$ on the fiber, where $F(v) = \Gamma\tilde{v} = \Gamma e_i + \sum_{j\neq i} v_j \Gamma e_j$ is an affine map whose linear part $\Lambda\xi = \sum_{j\neq i}\xi_j\Gamma e_j$ is injective, and $F(v)$ never lies in the image of Λ, because the columns $\Gamma e_1,\dots,\Gamma e_r$ are linearly independent. Writing $\langle a,b\rangle = \sum_k a_k\bar{b}_k$ on $\mathbb{C}^r$, we have $\partial_{\bar{v}_l}|F|^2 = \langle F, \Gamma e_l\rangle$ and $\partial_{v_k}|F|^2 = \langle \Gamma e_k, F\rangle$, so for $\xi \in \mathbb{C}^{r-1}$

$$\sum_{k,l\neq i} \frac{\partial^2 \log|F|^2}{\partial v_k \partial \bar v_l}\,\xi_k\bar\xi_l = \frac{|\Lambda\xi|^2}{|F|^2} - \frac{|\langle \Lambda\xi, F\rangle|^2}{|F|^4} = \frac{|F|^2|\Lambda\xi|^2 - |\langle\Lambda\xi,F\rangle|^2}{|F|^4}.$$

By the Cauchy–Schwarz inequality this is nonnegative, and it vanishes only if $\Lambda\xi = \mu F$ for some $\mu \in \mathbb{C}$. Since F does not lie in the image of Λ, this forces $\mu = 0$, so $\Lambda\xi = 0$ and $\xi = 0$. By the matrix criterion, the restriction of η to each fiber of $E \to W$ is positive definite.

Step 2: positivity near E. The form $P = \pi^*\omega$ is a smooth closed semipositive $(1,1)$-form on Y. We regard the real $(1,1)$-forms P and η in a chart (z,t,v) as Hermitian forms on holomorphic tangent vectors, given by their coefficient matrices, so that $P(\tau,\tau) = \omega(\mathrm{d}\pi\,\tau, \mathrm{d}\pi\,\tau)$ by the chain rule; norms of tangent vectors are taken in the coordinates of the chart. Let V be spanned by the vectors $\partial/\partial v_k$ and H by the vectors $\partial/\partial z_l$ and $\partial/\partial t$, and write P and η in block form with respect to the decomposition of the tangent space into V and H. At points of E we have $\mathrm{d}\pi(\partial/\partial v_k) = t\,\partial/\partial w_k = 0$, while $\mathrm{d}\pi(\partial/\partial z_l) = \partial/\partial z_l$ and $\mathrm{d}\pi(\partial/\partial t) = \sum_j \bar v_j\,\partial/\partial w_j$ are linearly independent. Since ω is positive definite, $P_{HV} = 0$ and P_{HH} is positive definite at points of E, while η_{VV} is positive definite there by Step 1.

Choose finitely many compact sets K_ι whose interiors cover E, each contained in a chart (z,t,v). The smallest eigenvalues of P_{HH} and of η_{VV} are continuous and positive on the compact sets $E \cap K_\iota$, so there are $a, c > 0$ such that they exceed $2c$ and $2a$, respectively, on $E \cap K_\iota$ for every ι. Let $M > 0$ bound the operator norm of η on every K_ι, so that $|\eta(\tau,\tau')| \le M|\tau||\tau'|$, and choose $\ell \in (0,1]$ with $2M\ell + M\ell^2 \le a/2$. Where P_{HH} is invertible, the linear map $L = P_{HH}^{-1}P_{HV}\colon V \to H$ depends continuously on the point, and it vanishes on E. Hence every point of $E \cap K_\iota$ has a neighborhood on whose intersection with K_ι we have $P_{HH} \ge c$, $\eta_{VV} \ge a$ and $\|L\| \le \ell$. Let O_ι be the union of these neighborhoods. Then

$$\mathcal{N} = \mathrm{Int}\Big(\bigcup_\iota K_\iota\Big) \cap \bigcap_\iota \big(O_\iota \cup (Y \setminus K_\iota)\big)$$

is an open neighborhood of E with $\mathcal{N} \cap K_\iota \subset O_\iota$ for every ι. On $\mathcal{N} \cap K_\iota$ we have, for $\xi \in V$,

$$\eta(\xi - L\xi, \xi - L\xi) \ge \big(a - 2M\|L\| - M\|L\|^2\big)|\xi|^2 \ge \frac{a}{2}|\xi|^2, \qquad |\xi - L\xi| \le 2|\xi|.$$

Write a tangent vector at a point of $\mathcal{N} \cap K_\iota$ as $\tau = \xi + \zeta$ with $\xi \in V$ and $\zeta \in H$, and put $\sigma = \zeta + L\xi \in H$. Completing the square,

$$P(\tau,\tau) = P_{HH}(\sigma,\sigma) + S(\xi,\xi), \qquad S = P_{VV} - P_{VH}P_{HH}^{-1}P_{HV},$$

and $S \ge 0$ because $P \ge 0$ (take $\zeta = -L\xi$, so that $\sigma = 0$). Hence $P(\tau,\tau) \ge c|\sigma|^2$. Since $\tau = (\xi - L\xi) + \sigma$,

$$\eta(\tau,\tau) \ge \frac{a}{2}|\xi|^2 - 4M|\xi||\sigma| - M|\sigma|^2.$$

Therefore, for $\varepsilon > 0$,

$$(P+\varepsilon\eta)(\tau,\tau) \geq (c-\varepsilon M)|\sigma|^2 + \frac{\varepsilon a}{2}|\xi|^2 - 4\varepsilon M|\xi||\sigma|.$$

The right-hand side is a quadratic form in $(|\sigma|, |\xi|)$, which is positive definite as soon as $\varepsilon M < c$ and $(c-\varepsilon M)a > 8\varepsilon M^2$; and $(\sigma,\xi) \neq 0$ for $\tau \neq 0$. Thus $\pi^*\omega + \varepsilon\eta$ is positive definite on the fixed neighborhood $\mathcal{N}$ of E for all sufficiently small $\varepsilon > 0$.

Step 3: positivity away from E. On the compact set $Y \setminus \mathcal{N} \subset Y \setminus E$ the map π is a local biholomorphism, so $\pi^*\omega$ is positive definite at every point. Cover $Y \setminus \mathcal{N}$ by finitely many compact sets contained in charts. On each of them the smallest eigenvalue of the coefficient matrix of $\pi^*\omega$ is bounded below by a positive constant c', and the operator norm of the coefficient matrix of η is bounded by a constant M', with c' and M' independent of the set. Hence $\pi^*\omega + \varepsilon\eta$ is positive definite on $Y \setminus \mathcal{N}$ if $\varepsilon M' < c'$. Together with Step 2, this shows that $\pi^*\omega + \varepsilon\eta$ is positive definite on all of Y for every sufficiently small $\varepsilon > 0$. It is closed, since both terms are, so it is a Kähler form on Y. □

Definition 13.1.11 Let X be a compact Kähler manifold. A quasi-plurisubharmonic function φ on X has *gentle analytic singularities of type* $(c, \mathfrak{a})$ if it has analytic singularities of type $(c, \mathfrak{a})$, the function $\exp(\varphi/c)$ is smooth on X, including at its zeros, and on every log resolution π of $\mathfrak{a}$, with $\pi^*\mathfrak{a} = \mathcal{O}(-D)$, one has locally

$$\varphi \circ \pi = c \log |s_D|^2 + h, \qquad h \text{ smooth}.$$

The log resolutions in this definition are compact Kähler manifolds by Proposition 13.1.10. The smoothness of $\exp(\varphi/c)$ on X and that of the remainder on the log resolutions are separate requirements. We require the local form on every log resolution, not only on one, because later proofs use it on a resolution chosen in the course of the argument. Both requirements hold if φ has presentations (13.1) of type $(c, \mathfrak{a})$ with smooth remainders b, since then $\exp(\varphi/c) = \mathrm{e}^{b/c}\sum_i |g_i|^2$ locally and Proposition 13.1.7 applies; they also hold for the cutoff sums of Lemma 13.4.1.

13.2 Local Bergman kernels and ideal generators

In this section $\Omega \subset \mathbb{C}^n$, $n \geq 1$, is a bounded pseudoconvex open set, $u \in \mathrm{PSH}(\Omega)$, and integrals are taken with respect to Lebesgue measure unless another measure is specified. For an integer $m \geq 1$ we put

$$H_m = \Big\{F \in \mathcal{O}(\Omega) : \|F\|_m^2 = \int_\Omega |F|^2 \mathrm{e}^{-mu} < \infty\Big\}, \qquad K_m(z) = \sup_{\|F\|_m \leq 1} |F(z)|^2.$$

Thus H_m is the space $\mathcal{H}_{mu}(\Omega)$ of Theorem 12.2.7, for the complex manifold Ω and the weight mu, and the function K_m is the *Bergman kernel* of H_m on the diagonal.

We need that H_m is a Hilbert space of holomorphic functions that generates the multiplier ideal $\mathcal{I}(mu)$, with bounded coefficients near each point. These facts were

proved in Chapter 12 on bounded pseudoconvex open sets, also for measures $\varrho\,dV$ comparable with Lebesgue measure (Theorem 12.2.7 and Lemma 12.2.8), and the next lemma restates them in the notation of this section. Throughout this section, ϱ denotes a measurable function on Ω with $c_\varrho \le \varrho \le C_\varrho$ for positive constants c_ϱ and C_ϱ. Such measures arise in Section 13.3, where the kernels are computed with the volume form of a Kähler form.

Lemma 13.2.1 *Let $\Omega \subset \mathbb{C}^n$, $n \ge 1$, be a bounded pseudoconvex open set, let $\psi \in \mathrm{PSH}(\Omega)$, and let ϱ be a measurable function on Ω with $c_\varrho \le \varrho \le C_\varrho$ for positive constants c_ϱ and C_ϱ. Put*

$$\mathcal{H} = \Big\{F \in \mathcal{O}(\Omega) : \|F\|^2 = \int_\Omega |F|^2 \mathrm{e}^{-\psi}\varrho\,dV < \infty\Big\}, \qquad \langle F, G\rangle = \int_\Omega F\bar{G}\,\mathrm{e}^{-\psi}\varrho\,dV.$$

(1) *For every compact set $K \subset \Omega$ there is a constant C_K such that $|F(z)| \le C_K\|F\|$ for all $F \in \mathcal{H}$ and $z \in K$. The space $\mathcal{H}$ is a separable Hilbert space, and it has an orthonormal basis, that is, an orthonormal family (e_ℓ), finite or countable, whose linear span is dense. For every such family and every $F \in \mathcal{H}$, the partial sums of $\sum_\ell \langle F, e_\ell\rangle e_\ell$ converge to F in norm and locally uniformly on Ω, and $\sum_\ell |\langle F, e_\ell\rangle|^2 = \|F\|^2$.*

(2) *The germs of the elements of $\mathcal{H}$ generate $\mathcal{I}(\psi)$ at every point of Ω, and near each point finitely many elements of $\mathcal{H}$ already generate $\mathcal{I}(\psi)$.*

(3) *Let $x \in \Omega$, and let $g_1, \dots, g_s \in \mathcal{H}$ generate $\mathcal{I}(\psi)$ on a neighborhood of x. Then there are a neighborhood V of x and a constant C such that every $F \in \mathcal{H}$ can be written as $F = \sum_i g_i h_i$ on V with $h_i \in \mathcal{O}(V)$ and $\sup_V |h_i| \le C\|F\|$.*

Proof We regard Ω as a complex manifold, with the single chart given by the identity map. Then ψ is a weight on Ω in the sense of Section 12.2, the set Ω satisfies the hypotheses of Theorem 12.2.7, and $\mathcal{H}$ is the space $\mathcal{H}_\psi(\Omega, \varrho)$ of that theorem, with the same inner product. Hence (1) is contained in Theorem 12.2.7(1) and (2), and (2) follows from Theorem 12.2.7(3), because an initial segment of an enumerated orthonormal basis is a finite family of elements of $\mathcal{H}$. Finally, (3) is the first assertion of Lemma 12.2.8, applied to the weight ψ on Ω. □

By Lemma 13.2.1, applied with $\psi = mu$ and $\varrho = 1$, the space H_m is a Hilbert space on which evaluation at each point is a bounded linear functional; in particular K_m is finite. Let (e_ℓ) be an orthonormal basis of H_m. Then

$$K_m = \sum_\ell |e_\ell|^2. \tag{13.2}$$

Indeed, let F lie in the unit ball of H_m and put $c_\ell = \langle F, e_\ell\rangle$. By Lemma 13.2.1(1), $F(z) = \sum_\ell c_\ell e_\ell(z)$ and $\sum_\ell |c_\ell|^2 = \|F\|_m^2 \le 1$, so the Cauchy–Schwarz inequality gives $|F(z)|^2 \le \sum_\ell |e_\ell(z)|^2$. Conversely, for each N the function $F_N = \sum_{\ell\le N} \overline{e_\ell(z)}\,e_\ell$ satisfies $F_N(z) = \|F_N\|_m^2 = \sum_{\ell\le N} |e_\ell(z)|^2$; if this number is positive, then $F_N/\|F_N\|_m$ lies in the unit ball, and $\sum_{\ell\le N} |e_\ell(z)|^2 = |F_N(z)|^2/\|F_N\|_m^2 \le K_m(z)$. The same argument applies to the kernel of the space defined with $\varrho\,dV$ in place of Lebesgue measure, for an orthonormal basis of that space.

Lemma 13.2.2 *The function K_m is real analytic on Ω. Near each point of Ω it is bounded above and below by positive constant multiples of $\sum_i |g_i|^2$, where $g_1, \dots, g_s$ are finitely many generators of $\mathcal{I}(mu)$ near that point. If $\pi : Y \to \Omega$ is a holomorphic map from a complex manifold Y such that $\pi^*\mathcal{I}(mu) = \mathcal{O}_Y(-D)$ for an effective divisor D on Y, for instance a log resolution of $\mathcal{I}(mu)$, then locally on Y*

$$K_m \circ \pi = |s_D|^2 b_m, \qquad b_m \text{ smooth and strictly positive},$$

where s_D is a local equation of D. The same conclusions hold if Lebesgue measure is replaced by $\varrho\, \mathrm{d}V$ in the definition of H_m.

Proof We give the proof for the measure $\varrho\, \mathrm{d}V$; Lebesgue measure is the case $\varrho = 1$. By Lemma 13.2.1, applied with $\psi = mu$, the space H_m is a Hilbert space with an orthonormal basis (e_ℓ), evaluation is uniformly bounded on compact subsets of Ω, and (13.2) holds.

We show that the map $z \mapsto (e_\ell(z))_\ell$ into ℓ^2 is given locally by a convergent power series with coefficients in ℓ^2. Fix a point z_0 and a polydisc P of multiradius ρ about z_0 with $\overline{P} \subset \Omega$, and let C be the evaluation bound on $\overline{P}$. For a finitely supported sequence (c_ℓ) with $\sum_\ell |c_\ell|^2 \le 1$, the function $F = \sum_\ell c_\ell e_\ell$ has norm at most one, so $|F| \le C$ on P, and the Cauchy estimates give $|\partial^\alpha F(z_0)/\alpha!| \le C\rho^{-\alpha}$. Since $\partial^\alpha F(z_0)/\alpha! = \sum_\ell c_\ell A_{\alpha,\ell}$ with $A_{\alpha,\ell} = \partial^\alpha e_\ell(z_0)/\alpha!$, taking the supremum over such sequences shows that the sequence $A_\alpha = (A_{\alpha,\ell})_\ell$ has ℓ^2 norm $\|A_\alpha\| \le C\rho^{-\alpha}$. Let $0 < \lambda < 1$, and let P_λ be the concentric polydisc of multiradius $\lambda\rho$. For $z \in P_\lambda$ we have $\sum_\alpha \|A_\alpha\||(z-z_0)^\alpha| \le C \sum_\alpha \lambda^{|\alpha|} < \infty$, so the ℓ^2-valued series $\sum_\alpha A_\alpha (z-z_0)^\alpha$ converges absolutely, and its ℓth component is the Taylor series of e_ℓ, which converges to $e_\ell(z)$. By (13.2),

$$K_m(z) = \Big\| \sum_\alpha A_\alpha (z-z_0)^\alpha \Big\|^2 = \sum_{\alpha,\beta} \langle A_\alpha, A_\beta \rangle (z-z_0)^\alpha \overline{(z-z_0)^\beta}, \qquad z \in P_\lambda,$$

where $|\langle A_\alpha, A_\beta\rangle| \le C^2 \rho^{-\alpha-\beta}$. Hence this double power series, and each series obtained from it by differentiating term by term, converge uniformly on $P_{\lambda'}$ for every $\lambda' < \lambda$, since a derivative of order k only introduces factors bounded by a polynomial of degree k in $|\alpha| + |\beta|$. Therefore K_m is real analytic, in particular smooth.

Next choose members $g_1, \dots, g_s$ of H_m generating $\mathcal{I}(mu)$ near a given point, by Lemma 13.2.1(2); by the remark following Definition 13.1.5, any other finite system of generators gives a comparable sum of squares. By the definition of K_m, $|g_i|^2 \le \|g_i\|_m^2 K_m$, so $\sum_i |g_i|^2 \le \big(\sum_i \|g_i\|_m^2\big) K_m$. Conversely, Lemma 13.2.1(3) gives, on a fixed smaller neighborhood, a representation $F = \sum_i g_i h_i$ with $|h_i| \le C'$ for every F in the unit ball of H_m. The Cauchy–Schwarz inequality gives $|F|^2 \le sC'^2 \sum_i |g_i|^2$, and hence $K_m \le sC'^2 \sum_i |g_i|^2$ there. These constants may depend on m, u and ϱ.

Let π and D be as in the statement, let $y \in Y$, let V be a neighborhood of $\pi(y)$ on which the bounded representation of the previous paragraph holds, and let s_D be a local equation of D on a neighborhood $W \subset \pi^{-1}(V)$ of y. Since $\pi^*\mathcal{I}(mu)$ is generated by s_D on W, we may write $g_i \circ \pi = s_D k_i$ and $s_D = \sum_i l_i\, (g_i \circ \pi)$ with holomorphic k_i and l_i, after shrinking W; since s_D does not vanish identically on

any open set, $\sum_i l_i k_i = 1$. For $F \in H_m$, the representation $F = \sum_i g_i h_i$ on V gives $F \circ \pi = s_D \sum_i (h_i \circ \pi) k_i$ on W, so

$$\frac{F \circ \pi}{s_D} = \sum_i (h_i \circ \pi)\, k_i, \qquad \left| \frac{F \circ \pi}{s_D} \right| \le C' \|F\|_m \sum_i |k_i|.$$

Hence evaluation of $(F \circ \pi)/s_D$ is bounded on each compact subset of W, uniformly for F in the unit ball of H_m. The power-series argument above therefore applies, in local coordinates on Y, to the holomorphic functions $(e_\ell \circ \pi)/s_D$, and shows that $b_m = \sum_\ell |(e_\ell \circ \pi)/s_D|^2$ is smooth; by (13.2), $K_m \circ \pi = |s_D|^2 b_m$.

To see that b_m is strictly positive, let F lie in the unit ball of H_m, with coefficients $c_\ell = \langle F, e_\ell \rangle$ and partial sums S_N. The estimate in the last display, applied to $F - S_N$, shows that $(S_N \circ \pi)/s_D \to (F \circ \pi)/s_D$ pointwise on W; hence $(F \circ \pi)/s_D = \sum_\ell c_\ell (e_\ell \circ \pi)/s_D$, and the Cauchy–Schwarz inequality gives $|(F \circ \pi)/s_D|^2 \le b_m$. Taking $F = g_i / \|g_i\|_m$ for the indices with $g_i \not\equiv 0$, we obtain $b_m \ge |k_i|^2 / \|g_i\|_m^2$, and the functions k_i have no common zero. □

Theorem 13.2.3 *Let Ω have diameter at most L, and put $u_m = m^{-1} \log K_m$. Then u_m is plurisubharmonic on Ω and has analytic singularities of type $(m^{-1}, \mathcal{I}(mu))$. There are constants $C_1, C_2 > 0$, depending only on n and L, such that*

$$u(z) - \frac{\log C_1}{m} \le u_m(z) \le \sup_{B(z,r)} u + \frac{1}{m} \log \frac{C_2}{r^{2n}} \tag{13.3}$$

for all $z \in \Omega$ and $r > 0$ with $B(z,r) \Subset \Omega$. As $m \to \infty$, $u_m \to u$ pointwise and in $L^1_{\mathrm{loc}}(\Omega)$. For every $z \in \Omega$,

$$\nu(u,z) - \frac{n}{m} \le \nu(u_m, z) \le \nu(u,z). \tag{13.4}$$

Finally, for each fixed m, the function $\exp(m(u_m - u))$ is locally integrable on Ω. All these assertions remain true if Lebesgue measure is replaced by $\varrho\, dV$ in the definition of H_m, provided C_1 and C_2 are replaced by $C_\varrho C_1$ and C_2 / c_ϱ.

Proof Let Ω_0 be a connected component of Ω, and choose $a \in \Omega_0$ with $u(a) > -\infty$. By Corollary 11.4.5, applied on Ω to the weight mu, some element of H_m takes the value 1 at a, so H_m contains a function that does not vanish identically on Ω_0. For every F in the unit ball of H_m that is not identically zero on Ω_0, $\log |F|^2$ is plurisubharmonic on Ω_0 by Proposition 8.3.3, and $\log K_m$ is the supremum of these functions on Ω_0. Since K_m is continuous by Lemma 13.2.2, this family is locally uniformly bounded above, and $\log K_m$ is upper semicontinuous with values in $[-\infty, \infty)$, so it equals its upper semicontinuous regularization. Hence Theorem 8.3.2 shows that u_m is plurisubharmonic. The comparison in Lemma 13.2.2 gives $u_m = m^{-1} \log \sum_i |g_i|^2 + O(1)$ locally, with generators g_i of $\mathcal{I}(mu)$; this is the asserted type.

For the lower bound in (13.3), let $u(z) > -\infty$; at points of the pole set of u there is nothing to prove. By Corollary 11.4.5, applied on Ω to the weight mu,

there is $F \in H_m$ with $F(z) = 1$ and $\|F\|_m^2 \le C_1 \mathrm{e}^{-mu(z)}$, where C_1 depends only on n and L. Hence $K_m(z) \ge |F(z)|^2/\|F\|_m^2 \ge C_1^{-1}\mathrm{e}^{mu(z)}$. For the upper bound, let $\|F\|_m \le 1$. The mean-value inequality (12.7) of Lemma 12.2.6 for $|F|^2$ on $B(z,r)$, where $|B(z,r)| = \pi^n r^{2n}/n!$, gives

$$|F(z)|^2 \le C_2 r^{-2n} \int_{B(z,r)} |F|^2 \le C_2 r^{-2n} \exp\Bigl(m \sup_{B(z,r)} u\Bigr) \int_{B(z,r)} |F|^2 \mathrm{e}^{-mu} \le C_2 r^{-2n} \exp\Bigl(m \sup_{B(z,r)} u\Bigr),$$

with $C_2 = n!/\pi^n$, and taking the supremum over F proves the upper bound.

Now fix z and r, and let $m \to \infty$ in (13.3): we obtain $u(z) \le \underline{\lim}_m u_m(z) \le \overline{\lim}_m u_m(z) \le \sup_{B(z,r)} u$. Letting $r \downarrow 0$ and using the upper semicontinuity of u gives $u_m(z) \to u(z)$, also at points of the pole set. On a compact set $K \subset \Omega$, a fixed $r < \operatorname{dist}(K, \partial\Omega)$ in the upper bound gives a uniform upper bound for all u_m, while $u_m \ge u - |\log C_1|$. Since $u \in L^1(K)$, dominated convergence gives $u_m \to u$ in $L^1(K)$.

For the Lelong numbers, let $g_1, \dots, g_s$ generate $\mathcal{I}(mu)$ near z. The stalk $\mathcal{I}(mu)_z$ is not zero: by the first paragraph, applied to the component of Ω containing z, it contains the germ of an element of H_m that does not vanish identically on that component, and this germ is not zero by the identity theorem. Since $u_m = m^{-1} \log \sum_i |g_i|^2 + O(1)$ near z, Lemma 13.1.6 gives $\nu(u_m, z) = m^{-1} \operatorname{ord}_z \mathcal{I}(mu)$, and (13.4) follows from Lemma 12.3.1.

Finally, near each point $K_m \le C \sum_i |g_i|^2$ with $g_i \in H_m$, by Lemma 13.2.2, and each $|g_i|^2 \mathrm{e}^{-mu}$ is integrable on Ω. Hence $\exp(m(u_m - u)) = K_m \mathrm{e}^{-mu}$ is locally integrable.

Now replace Lebesgue measure by $\varrho\,\mathrm{d}V$, and let K_m^ϱ be the corresponding kernel. The space of holomorphic functions is the same, and the two norms satisfy $c_\varrho \|F\|_m^2 \le \|F\|_{m,\varrho}^2 \le C_\varrho \|F\|_m^2$. If $\|F\|_m \le 1$, then $F/C_\varrho^{1/2}$ lies in the unit ball for ϱ, and if $\|F\|_{m,\varrho} \le 1$, then $c_\varrho^{1/2} F$ lies in the unit ball for Lebesgue measure; hence

$$C_\varrho^{-1} K_m \le K_m^\varrho \le c_\varrho^{-1} K_m.$$

The proof of plurisubharmonicity applies verbatim to $m^{-1} \log K_m^\varrho$, with the unit ball taken for ϱ and the continuity of K_m^ϱ given by Lemma 13.2.2. The lower and upper bounds follow from the last display, with the constants $C_\varrho C_1$ and C_2/c_ϱ. The convergence and the Lelong numbers then follow as before, since Lemma 13.2.2 applies to K_m^ϱ. Finally, $K_m^\varrho \mathrm{e}^{-mu} \le c_\varrho^{-1} K_m \mathrm{e}^{-mu}$ is locally integrable. □

The kernels at the scales m and $2m$ are compared by extending a function from the diagonal of $\Omega \times \Omega$.

Lemma 13.2.4 *Suppose that the spaces H_m are defined with respect to $\varrho\,\mathrm{d}V$; for instance $\varrho = 1$. There is a constant $C_\Omega > 0$, depending only on n, on a bound L for the diameter of Ω and on c_ϱ and C_ϱ, and in particular independent of m and u, such that*

$$K_{2m} \le C_\Omega K_m^2 \qquad \text{on } \Omega.$$

Proof We write $\|\cdot\|_m$ for the norm of H_m. By the definition of K_m, every $h \in H_m$ satisfies $|h(\zeta)|^2 \le \|h\|_m^2 K_m(\zeta)$ for all $\zeta \in \Omega$; if $\|h\|_m = 0$, then h vanishes almost everywhere, hence identically.

Let $F \in H_{2m}$. We first extend F from the diagonal to $\Omega \times \Omega$. By Corollary 11.2.5, $\Omega \times \Omega$ is a bounded pseudoconvex open set in $\mathbb{C}^{2n}$, of diameter at most $\sqrt{2}L$, and the function $p(z,w) = mu(z) + mu(w)$ is plurisubharmonic on it (Propositions 8.2.3 and 8.2.5). The diagonal $\Lambda = \{(z,w) \in \mathbb{C}^{2n} : z = w\}$ is a complex linear subspace of codimension n, and the map $z \mapsto (z,z)$ multiplies Euclidean lengths by $\sqrt{2}$, so Lebesgue measure on Λ corresponds to 2^n times Lebesgue measure on $\mathbb{C}^n$. On $\Lambda \cap (\Omega \times \Omega)$, identified with Ω, the weight p restricts to $2mu$, so

$$\int_{\Lambda\cap(\Omega\times\Omega)} |F|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{2n} = 2^n \int_\Omega |F|^2 \mathrm{e}^{-2mu}\, \mathrm{d}V \le 2^n c_\varrho^{-1} \|F\|_{2m}^2.$$

The extension statement of Corollary 11.4.5, applied on $\Omega \times \Omega$ to the weight p and the subspace Λ, gives $G \in \mathcal{O}(\Omega\times\Omega)$ with $G(z,z) = F(z)$ and $\int_{\Omega\times\Omega} |G|^2 \mathrm{e}^{-p}\, \mathrm{d}V_{4n} \le C^n (\sqrt{2}L)^{2n} 2^n c_\varrho^{-1} \|F\|_{2m}^2$, where C is the constant of Theorem 11.4.4. Hence

$$\|G\|^2 = \int_{\Omega\times\Omega} |G(z,w)|^2 \mathrm{e}^{-mu(z)-mu(w)} \varrho(z)\varrho(w)\, \mathrm{d}V(z)\, \mathrm{d}V(w) \le C_\Omega \|F\|_{2m}^2$$

with $C_\Omega = 4^n C^n L^{2n} C_\varrho^2 / c_\varrho$.

We now bound G on the diagonal by K_m. For $w \in \Omega$ put

$$N(w) = \int_\Omega |G(z,w)|^2 \mathrm{e}^{-mu(z)} \varrho(z)\, \mathrm{d}V(z) \in [0,\infty].$$

By Tonelli's theorem, N is measurable and $\int_\Omega N(w)\mathrm{e}^{-mu(w)}\varrho(w)\, \mathrm{d}V(w) = \|G\|^2 < \infty$; since $\mathrm{e}^{-mu}\varrho > 0$ almost everywhere, $N(w) < \infty$ for almost every w. For such w, the function $G(\cdot, w)$ belongs to H_m with $\|G(\cdot,w)\|_m^2 = N(w)$, so $|G(z,w)|^2 \le N(w) K_m(z)$ for every z. Fix $z \in \Omega$. Integrating this inequality in w, we obtain

$$\int_\Omega |G(z,w)|^2 \mathrm{e}^{-mu(w)} \varrho(w)\, \mathrm{d}V(w) \le K_m(z) \|G\|^2,$$

so $G(z,\cdot) \in H_m$ with $\|G(z,\cdot)\|_m^2 \le K_m(z)\|G\|^2$, and therefore $|G(z,w)|^2 \le K_m(z)K_m(w)\|G\|^2$ for every w. Taking $w = z$,

$$|F(z)|^2 = |G(z,z)|^2 \le K_m(z)^2 \|G\|^2 \le C_\Omega \|F\|_{2m}^2 K_m(z)^2.$$

Taking the supremum over the unit ball of H_{2m} proves the lemma. □

13.3 Overlap interpolation and quantitative patching

From now on (X, ω) is a compact Kähler manifold of dimension n. If $n = 0$, then X is a finite set, every quasi-plurisubharmonic function on X is real-valued and hence smooth, and every multiplier ideal is the unit ideal; the constant sequence $\Phi_j = \varphi$ then has all the approximation properties proved below. We therefore assume $n \geq 1$.

We fix, once and for all, finitely many concentric coordinate balls

$$U_j \Subset U_j' \Subset B_j \Subset B_j^+, \qquad j = 1, \dots, J, \qquad X = \bigcup_j U_j,$$

and smooth functions $\chi_j \colon X \to [0,1]$ with compact support in U_j' and $\chi_j = 1$ on U_j. We require in addition that whenever $\overline{U_j'} \cap \overline{U_l'} \neq \varnothing$, the union $B_j^+ \cup B_l^+$ is contained in a coordinate ball $\mathcal{B}_{jl}$ whose closure lies in a coordinate chart. By Lemma 13.1.1, applied to the forms $\mathrm{dd^c}\chi_j$ on X, there is a constant $C_\chi \geq 0$ with $\mathrm{dd^c}\chi_j \geq -C_\chi\omega$ for all j.

Such balls and cutoff functions exist. To see this, we first construct a metric d defining the topology of X. Choose finitely many charts $\kappa_k \colon V_k \to \mathbb{C}^n$, $1 \leq k \leq N$, and compact sets $L_k \subset V_k$ whose interiors cover X, and, by Proposition 7.1.3, smooth functions $\psi_k \colon X \to [0,1]$ with compact support in V_k and $\psi_k = 1$ on L_k. The map $\Psi = (\psi_k, \psi_k\kappa_k)_{1\leq k\leq N}$ from X to $\mathbb{R}^N \times \mathbb{C}^{nN}$, where $\psi_k\kappa_k$ is extended by zero outside V_k, is continuous and injective: if $\Psi(x) = \Psi(y)$ and $x \in L_k$, then $\psi_k(y) = \psi_k(x) = 1$, so $y \in V_k$ and $\kappa_k(y) = \kappa_k(x)$, whence $y = x$. A continuous injective map from a compact space to a Hausdorff space is a homeomorphism onto its image, so $d(x,y) = |\Psi(x) - \Psi(y)|$ is a metric defining the topology of X.

Now cover X by finitely many coordinate balls whose closures lie in coordinate charts, and let $\delta > 0$ be a Lebesgue number of this cover for d, so that every subset of X of d-diameter less than δ lies in one of these balls. About every point of X choose concentric coordinate balls $U \Subset U' \Subset B \Subset B^+$, with $\overline{B^+}$ contained in a coordinate chart and of d-diameter less than $\delta/2$. By compactness finitely many of the balls U cover X; these give the U_j, and Proposition 7.1.3 gives the cutoff functions χ_j. If $\overline{U_j'} \cap \overline{U_l'} \neq \varnothing$, then $B_j^+ \cup B_l^+$ is a union of two intersecting sets of diameter less than $\delta/2$, so it has diameter less than δ and lies in one ball $\mathcal{B}_{jl}$ of the cover.

All the quantities attached to these choices are now fixed: the bounds for the densities of $\omega^n/n!$ with respect to Lebesgue measure in the charts, the coordinate changes, the diameters of the balls, the distances from $\overline{U_j'}$ to ∂B_j, and the derivatives of the cutoffs. We say that a constant *depends only on the fixed geometry* if it depends only on these data, hence only on (X, ω) once the choices have been made.

Lemma 13.3.1 *Let θ be a smooth closed real $(1,1)$-form on X, let $a \geq 0$, and let $v_j \in \mathrm{PSH}(B_j, \theta)$ for $j = 1, \dots, J$ satisfy $v_j \leq v_l + a$ on $\overline{U_j'} \cap \overline{U_l'}$ for all j, l. Then the function*

$$v(x) = \max_{j\colon x \in \overline{U_j'}} \bigl(v_j(x) + a\chi_j(x)\bigr)$$

is quasi-plurisubharmonic on X and satisfies $\theta+\mathrm{dd^c}v \geq -aC_\chi\omega$ and $v_j \leq v \leq v_j+2a$ on U_j.

Proof Since the U_i cover X, the maximum is taken over a nonempty set at every point. Fix $x_0 \in X$ and choose i with $x_0 \in U_i$. Indices j with $x_0 \notin \overline{U'_j}$ do not occur near x_0. If $x_0 \in \partial U'_j$, then $\chi_j = 0$ near x_0, because χ_j has compact support in U'_j; so near x_0 the jth term, where it is defined, equals $v_j \leq v_i + a = v_i + a\chi_i$, since $\chi_i = 1$ on the neighborhood U_i of x_0. Such a term never exceeds the ith term and can be omitted. Hence on a small neighborhood of x_0, v is the maximum of $v_j + a\chi_j$ over the fixed set of indices j with $x_0 \in U'_j$. Each of these functions satisfies $\theta + aC_\chi\omega + \mathrm{dd^c}(v_j + a\chi_j) \geq 0$. Adding a local potential of the closed form $\theta + aC_\chi\omega$ on a coordinate ball about x_0 (Lemma 13.1.2), which makes these functions plurisubharmonic by Lemma 13.1.4(2), and applying Proposition 8.2.5, we obtain that v is quasi-plurisubharmonic near x_0 with $\theta + \mathrm{dd^c}v \geq -aC_\chi\omega$.

On U_j the jth term is $v_j + a \geq v_j$, and every term with index l is at most $v_l + a \leq v_j + 2a$. This proves the two bounds, also at points of the pole sets. □

Now fix a smooth closed real $(1,1)$-form θ on X. By Lemma 13.1.2 we can choose smooth real functions h_j on B_j^+ with $\mathrm{dd^c}h_j = \theta$. Let $\varphi \in \mathrm{PSH}(X,\theta)$ and let $m \geq 1$ be an integer. Then $\varphi + h_j$ is plurisubharmonic on B_j by Lemma 13.1.4(2), and we let $K_{j,m}$ be the Bergman kernel of Section 13.2 on the ball B_j for the weight $m(\varphi + h_j)$ and the volume form $\omega^n/n!$:

$$K_{j,m}(x) = \sup\Big\{|F(x)|^2 : F \in \mathcal{O}(B_j),\ \int_{B_j} |F|^2 \mathrm{e}^{-m(\varphi+h_j)}\frac{\omega^n}{n!} \leq 1\Big\}.$$

We put

$$a_{j,m} = \mathrm{e}^{-mh_j}K_{j,m}, \qquad v_{j,m} = m^{-1}\log a_{j,m} = m^{-1}\log K_{j,m} - h_j.$$

The closure of B_j lies in a chart, so in the coordinates of B_j the density of $\omega^n/n!$ with respect to Lebesgue measure is bounded above and below by positive constants on B_j. Hence Theorem 13.2.3, in the version with the measure $\varrho\,\mathrm{d}V = \omega^n/n!$, applies on B_j: $m^{-1}\log K_{j,m}$ is plurisubharmonic, and $v_{j,m} \in \mathrm{PSH}(B_j,\theta)$.

Lemma 13.3.2 *There is a constant $C_B \geq 0$, depending only on the fixed geometry and in particular independent of θ, φ and m, such that*

$$a_{j,m} \leq \mathrm{e}^{C_B}a_{l,m} \qquad \text{on } \overline{U'_j} \cap \overline{U'_l}$$

for all j,l. In particular $a_{j,m}$ and $a_{l,m}$ have the same zeros on $\overline{U'_j} \cap \overline{U'_l}$, and $|v_{j,m} - v_{l,m}| \leq C_B/m$ there outside these common zeros.

Proof Fix j,l with $\overline{U'_j} \cap \overline{U'_l} \neq \varnothing$, and work in the coordinates of the ball $\mathcal{B}_{jl}$. By Lemma 13.1.2 choose a smooth function g on $\mathcal{B}_{jl}$ with $\mathrm{dd^c}g = \theta$. The functions $h_j - g$ and $h_l - g$ are pluriharmonic on the coordinate balls B_j^+ and B_l^+, so by the same lemma $h_j - g = \mathrm{Re}\,b_j$ and $h_l - g = \mathrm{Re}\,b_l$ with b_j, b_l holomorphic on these balls

(the constants are absorbed into b_j, b_l). Put $u = \varphi + g$, a plurisubharmonic function on $\mathcal{B}_{jl}$. Since $|\mathrm{e}^{-mb_j/2}|^2 = \mathrm{e}^{-m\operatorname{Re} b_j}$, multiplication by $\mathrm{e}^{-mb_j/2}$ is an isometry from the weighted space defining $K_{j,m}$ onto the space of holomorphic functions on B_j with weight mu and volume form $\omega^n/n!$. If $K^{mu}_{B_j}$ denotes the Bergman kernel of the latter space, then $K^{mu}_{B_j} = \mathrm{e}^{-m\operatorname{Re} b_j} K_{j,m}$, and therefore

$$a_{j,m} = \mathrm{e}^{-mg} K^{mu}_{B_j}, \qquad a_{l,m} = \mathrm{e}^{-mg} K^{mu}_{B_l}.$$

Hence it suffices to prove $K^{mu}_{B_j} \le C_0^2 K^{mu}_{B_l}$ on $\overline{U'_j} \cap \overline{U'_l}$ with $C_0 \ge 1$ depending only on the fixed geometry.

Choose a smooth function $\rho\colon X \to [0,1]$ with compact support in $B_j \cap B_l$, equal to one on a neighborhood of the compact set $\overline{U'_j} \cap \overline{U'_l}$ (Proposition 7.1.3); this choice belongs to the fixed geometry. Let $r > 0$ be the distance, in the coordinates of $\mathcal{B}_{jl}$, from $\overline{U'_j} \cap \overline{U'_l}$ to the support of $\bar\partial\rho$, and let $c > 0$ and C'' be lower and upper bounds for the density of $\omega^n/n!$ with respect to Lebesgue measure on $B_j \cup B_l$ in these coordinates. The ball B_l is pseudoconvex also in the coordinates of $\mathcal{B}_{jl}$, since a biholomorphism transports a continuous plurisubharmonic exhaustion function (Proposition 8.2.3).

To compare the kernels at a point $x \in \overline{U'_j} \cap \overline{U'_l}$, we correct ρf, for f holomorphic on B_j, to a holomorphic function on B_l with the same value at x. Let f be holomorphic on B_j with norm at most one in the weighted space, and extend ρf by zero to a smooth function on B_l. We apply Theorem 11.3.4 on B_l to the smooth $\bar\partial$-closed form $f\bar\partial\rho$, with the plurisubharmonic function $p = mu + n\log|z-x|^2$ and the smooth function $q = |z|^2$, whose complex Hessian matrix is the identity. On the support of $\bar\partial\rho$ we have $|z-x|^{-2n} \le r^{-2n}$, so

$$\int_{B_l} |f\bar\partial\rho|^2 \mathrm{e}^{-p-q}\,\mathrm{d}V \le r^{-2n} \sup|\bar\partial\rho|^2 \int_{B_j} |f|^2 \mathrm{e}^{-mu}\,\mathrm{d}V \le \frac{\sup|\bar\partial\rho|^2}{cr^{2n}}.$$

Hence there is a solution of $\bar\partial w = f\bar\partial\rho$ on B_l with

$$\int_{B_l} |w|^2 |z-x|^{-2n} \mathrm{e}^{-mu-|z|^2}\,\mathrm{d}V \le C = \frac{\sup|\bar\partial\rho|^2}{cr^{2n}},$$

and C depends only on the fixed geometry. The function $|z-x|^{2n}\mathrm{e}^{|z|^2}$ is bounded on the bounded set B_l by a constant of the same kind, and the density of $\omega^n/n!$ is at most C''. Hence the squared weighted norm of w, for the weight mu and the volume form $\omega^n/n!$, is at most a constant C' depending only on the fixed geometry.

Near x we have $\bar\partial w = f\bar\partial\rho = 0$, because $\rho = 1$ there; hence $\Delta w = 0$ near x, and by Theorem 2.1.4 w is smooth, hence holomorphic, near x. Suppose that $w(x) \ne 0$. Then $|w|^2 \ge |w(x)|^2/4$ on a ball $B(x,\delta)$ on which $u \le M$ for some constant M, since u is upper semicontinuous, and the last integral is at least

$$\frac{|w(x)|^2}{4}\,\mathrm{e}^{-mM-\sup_{B_l}|z|^2} \int_{B(x,\delta)} |z-x|^{-2n}\,\mathrm{d}V = \infty,$$

because $\int_{B(x,\delta)} |z-x|^{-2n}\,\mathrm{d}V = s_{2n-1}\int_0^\delta t^{-1}\,\mathrm{d}t$ in polar coordinates (2.1). Hence $w(x) = 0$. The function $F = \rho f - w$ satisfies $\bar\partial F = 0$ on B_l, so it is holomorphic by Theorem 2.1.4, and $F(x) = f(x)$. Since $0 \le \rho \le 1$ and ρ vanishes outside B_j, the weighted norm of F on B_l is at most $1 + C'^{1/2} = C_0$. By the definition of the kernel, $|f(x)|^2 = |F(x)|^2 \le C_0^2 K^{mu}_{B_l}(x)$, and taking the supremum over f gives $K^{mu}_{B_j}(x) \le C_0^2 K^{mu}_{B_l}(x)$.

No bound on u, θ or m enters these constants. Taking C_B to be the maximum of $2\log C_0$ over the finitely many pairs (j,l), with the roles of j and l also exchanged, proves the lemma. □

Theorem 13.3.3 *There is a constant $c > 0$, depending only on (X,ω) through the fixed geometry, with the following property. Let θ be a smooth closed real $(1,1)$-form on X, let $\varphi \in \mathrm{PSH}(X,\theta)$, and let $m \ge 1$ be an integer. Then there is a quasi-plurisubharmonic function φ_m on X such that*

$$\theta + \mathrm{dd^c}\varphi_m \ge -\frac{c}{m}\omega, \qquad \varphi_m \ge \frac{1}{m}\log\sum_i |g_i|^2 + O(1) \quad \textit{locally},$$

where the g_i are local generators of $\mathcal{I}(m\varphi)$. The bounded term may depend on m, θ, φ and the point.

Proof By Lemma 13.3.2, the functions $v_{j,m} \in \mathrm{PSH}(B_j,\theta)$ satisfy $v_{j,m} \le v_{l,m} + C_B/m$ on $\overline{U'_j} \cap \overline{U'_l}$, also at their common zeros. Hence Lemma 13.3.1 with $a = C_B/m$ gives a quasi-plurisubharmonic function φ_m with $\theta + \mathrm{dd^c}\varphi_m \ge -(C_B C_\chi/m)\omega$, so the first conclusion holds with $c = C_B C_\chi$, increased if necessary to make it positive. On U_j the function φ_m dominates $v_{j,m} = m^{-1}\log K_{j,m} - h_j$. Since h_j is smooth, $\mathcal{I}(m(\varphi + h_j)) = \mathcal{I}(m\varphi)$, and the comparison of Lemma 13.2.2 gives $v_{j,m} \ge m^{-1}\log\sum_i |g_i|^2 + O(1)$ near each point of U_j. This proves the second conclusion. □

13.4 Smooth cutoff sums and decreasing approximation

The maximum in Theorem 13.3.3 destroys smoothness, and that theorem does not compare the functions φ_m for different values of m. To obtain the regularity required of gentle singularities and, later, a decreasing sequence, we glue the local kernels instead by a sum with smooth cutoffs and control the curvature of its logarithm by a direct computation. We keep the notation of the previous section.

Lemma 13.4.1 (1) *Let θ be a smooth closed real $(1,1)$-form on X, let $\varphi \in \mathrm{PSH}(X,\theta)$, let $m \ge 1$ be an integer, and put*

$$R_m = \sum_j \chi_j^2 a_{j,m}, \qquad G_m = m^{-1}\log R_m,$$

each summand being extended by zero outside the support of χ_j. There is a constant $c' > 0$, depending only on the fixed geometry, such that $\theta + \mathrm{dd^c} G_m \geq -(c'/m)\omega$. Moreover R_m is smooth on X, and G_m has gentle analytic singularities of type $(m^{-1}, \mathcal{I}(m\varphi))$.

(2) *Let W be an open subset of a complex manifold, let $N \geq 1$, and let $w_1, \ldots, w_N$ and $\chi_1, \ldots, \chi_N$ be smooth real functions on W such that $S = \sum_{j=1}^N \chi_j^2 \mathrm{e}^{w_j}$ is positive on W. Put $\alpha_j = \chi_j \partial w_j + 2\partial\chi_j$. Then, on W,*

$$\begin{aligned} \mathrm{dd^c} \log S = \frac{\mathrm{i}}{2\pi S^2} \sum_{j<l} \mathrm{e}^{w_j+w_l} (\chi_j\alpha_l - \chi_l\alpha_j) \wedge \overline{(\chi_j\alpha_l - \chi_l\alpha_j)} \\ + \frac{\sum_j \chi_j^2 \mathrm{e}^{w_j} \mathrm{dd^c} w_j}{S} + 2\frac{\sum_j \mathrm{e}^{w_j} \big(\chi_j \mathrm{dd^c}\chi_j - \frac{\mathrm{i}}{2\pi}\partial\chi_j \wedge \bar\partial\chi_j\big)}{S}, \end{aligned} \tag{13.5}$$

and the first term on the right of (13.5) is a semipositive $(1,1)$-form.

Proof *Step 1: proof of* (2). Let w_j, χ_j, S and α_j be as in (2), and put $f_j = \chi_j^2 \mathrm{e}^{w_j}$, so that $S = \sum_j f_j$. Then $\partial f_j = \chi_j \mathrm{e}^{w_j}(\chi_j \partial w_j + 2\partial\chi_j) = \chi_j \mathrm{e}^{w_j} \alpha_j$, so that $\partial S = \sum_j \chi_j \mathrm{e}^{w_j} \alpha_j$. Since χ_j and w_j are real, $\bar\partial f_j = \overline{\partial f_j} = \chi_j \mathrm{e}^{w_j} \bar\alpha_j$ with $\bar\alpha_j = \chi_j \bar\partial w_j + 2\bar\partial\chi_j$, and

$$\begin{aligned} \partial\bar\partial f_j &= \partial(\chi_j \mathrm{e}^{w_j}) \wedge \bar\alpha_j + \chi_j \mathrm{e}^{w_j}\, \partial\bar\alpha_j \\ &= \mathrm{e}^{w_j} \big((\partial\chi_j + \chi_j \partial w_j) \wedge (\chi_j \bar\partial w_j + 2\bar\partial\chi_j) \\ &\qquad + \chi_j \partial\chi_j \wedge \bar\partial w_j + \chi_j^2 \partial\bar\partial w_j + 2\chi_j \partial\bar\partial\chi_j \big) \\ &= \mathrm{e}^{w_j} \big(\alpha_j \wedge \bar\alpha_j + \chi_j^2 \partial\bar\partial w_j + 2\chi_j \partial\bar\partial\chi_j - 2\partial\chi_j \wedge \bar\partial\chi_j \big), \end{aligned}$$

where the last line follows by expanding $\alpha_j \wedge \bar\alpha_j = \chi_j^2 \partial w_j \wedge \bar\partial w_j + 2\chi_j \partial w_j \wedge \bar\partial\chi_j + 2\chi_j \partial\chi_j \wedge \bar\partial w_j + 4\partial\chi_j \wedge \bar\partial\chi_j$. Next, $\partial\bar\partial \log S = \partial\bar\partial S/S - \partial S \wedge \bar\partial S/S^2$, and since $\sum_l \chi_l^2 \mathrm{e}^{w_l} = S$,

$$\sum_j \frac{\mathrm{e}^{w_j} \alpha_j \wedge \bar\alpha_j}{S} - \frac{\partial S \wedge \bar\partial S}{S^2} = \frac{1}{S^2} \sum_{j,l} \mathrm{e}^{w_j+w_l} \big(\chi_l^2\, \alpha_j \wedge \bar\alpha_j - \chi_j\chi_l\, \alpha_j \wedge \bar\alpha_l \big).$$

In the last sum the terms with $j = l$ vanish, and for $j < l$ the terms with indices (j,l) and (l,j) add up to $\mathrm{e}^{w_j+w_l}(\chi_j\alpha_l - \chi_l\alpha_j) \wedge \overline{(\chi_j\alpha_l - \chi_l\alpha_j)}$. Multiplying by $\mathrm{i}/2\pi$ and using $\mathrm{dd^c} = \frac{\mathrm{i}}{2\pi}\partial\bar\partial$, we obtain (13.5). The first term on the right of (13.5) is semipositive, because the coefficients $\mathrm{e}^{w_j+w_l}/(2\pi S^2)$ are positive and, for a $(1,0)$-form $\gamma = \sum_k \gamma_k \, \mathrm{d}z_k$, the coefficient matrix $(\gamma_k \bar\gamma_{k'})$ of $\mathrm{i}\gamma \wedge \bar\gamma$ is positive semidefinite.

We now prove (1) in three steps.

Step 2: a curvature estimate. By Lemma 13.1.1, applied on the compact manifold X to the finitely many smooth forms below, there is a constant $A > 0$, depending only on the fixed geometry, such that $\chi_j \mathrm{dd^c}\chi_j - \frac{\mathrm{i}}{2\pi}\partial\chi_j \wedge \bar\partial\chi_j \geq -A\omega$ for every j. Let $W \subset X$ be open, and for the indices j with $W \cap \operatorname{Supp}\chi_j \neq \varnothing$ let w_j be smooth real functions on W such that

$$\mathrm{dd^c} w_j \geq -m\theta \quad \text{on } W, \qquad \mathrm{e}^{w_j} \leq \mathrm{e}^{C_B} \mathrm{e}^{w_i} \quad \text{on } W \cap U_i \cap \operatorname{Supp}\chi_j$$

for all such i, j. We claim that $S = \sum_j \chi_j^2 \mathrm{e}^{w_j}$ is positive on W and that

$$\theta + \frac{1}{m}\mathrm{dd^c} \log S \geq -\frac{2JA\mathrm{e}^{C_B}}{m}\omega \qquad \text{on } W.$$

Let $x \in W$ and choose i with $x \in U_i$. Then $\chi_i(x) = 1$, so $S(x) \geq \mathrm{e}^{w_i(x)} > 0$, and $\mathrm{e}^{w_j}/S \leq \mathrm{e}^{C_B}$ at x for every j with $\chi_j(x) \neq 0$. Since $S > 0$ on W, the identity (13.5) of (2) holds on W. If $\chi_j(x) = 0$, the jth summand of the last term in (13.5) vanishes at x, because the nonnegative function χ_j attains its minimum at x, so that $\partial\chi_j(x) = 0$. Hence the last term is at least $-2JA\mathrm{e}^{C_B}\omega$ at x. The coefficients $\chi_j^2\mathrm{e}^{w_j}/S$ are nonnegative with sum one, so the middle term is at least $-m\theta$, and the first term is semipositive. This proves the claim.

Step 3: G_m is quasi-plurisubharmonic and satisfies the curvature bound. Since the kernels $a_{j,m}$ may vanish, we apply Step 2 to smooth approximations of $\log a_{j,m}$ and pass to the limit. Fix $x_0 \in X$ and a small coordinate ball W about x_0 such that $W \subset U_j'$ whenever $x_0 \in \operatorname{Supp}\chi_j$, and $W \cap \operatorname{Supp}\chi_j = \varnothing$ otherwise; then the indices j with $W \cap \operatorname{Supp}\chi_j \neq \varnothing$ are those with $x_0 \in \operatorname{Supp}\chi_j$. Choose a smooth function g on W with $\mathrm{dd^c} g = \theta$, and for each such j write $h_j - g = \operatorname{Re} b_j$ on W with b_j holomorphic, by Lemma 13.1.2. Since $h_j = g + \operatorname{Re} b_j$ on W, we have $a_{j,m} = \mathrm{e}^{-mh_j}K_{j,m} = \mathrm{e}^{-mg}k_j$ on W, where, by (13.2) and since $|\mathrm{e}^{-mb_j/2}|^2 = \mathrm{e}^{-m\operatorname{Re} b_j}$, $k_j = \mathrm{e}^{-m\operatorname{Re} b_j}K_{j,m} = \sum_\ell |\mathrm{e}^{-mb_j/2}e_\ell|^2$ for an orthonormal basis (e_ℓ) of the space defining $K_{j,m}$.

For $\delta > 0$ the function $\log(k_j + \delta)$ is smooth, by Lemma 13.2.2, and plurisubharmonic: it is the increasing limit of the logarithms of the finite partial sums of these squared moduli together with $|\sqrt{\delta}|^2$, which are plurisubharmonic by Proposition 8.3.3, and since the limit is continuous, Theorem 8.3.2 applies. Put $w_j^\delta = \log(k_j + \delta) - mg$, so that $\mathrm{dd^c} w_j^\delta \geq -m\theta$. On $W \cap U_i \cap \operatorname{Supp}\chi_j$, which lies in $\overline{U_j'} \cap \overline{U_i'}$, Lemma 13.3.2 and the identities $a_{j,m} = \mathrm{e}^{-mg}k_j$, $a_{i,m} = \mathrm{e}^{-mg}k_i$ give $k_j \leq \mathrm{e}^{C_B}k_i$, and since $\mathrm{e}^{C_B} \geq 1$ also $k_j + \delta \leq \mathrm{e}^{C_B}(k_i + \delta)$. Thus the functions w_j^δ satisfy the hypotheses of Step 2 on W, and the function

$$\frac{1}{m}\log \sum_j \chi_j^2 \mathrm{e}^{w_j^\delta} + g'$$

is plurisubharmonic on W, where g' is a smooth function on W with $\mathrm{dd^c} g' = \theta + (2JA\mathrm{e}^{C_B}/m)\omega$, given by Lemma 13.1.2. Indeed, by Step 2 this function is smooth with positive semidefinite complex Hessian, so Proposition 8.2.2(1) applies.

As $\delta \downarrow 0$, the sums $\sum_j \chi_j^2 \mathrm{e}^{w_j^\delta} = \mathrm{e}^{-mg}\sum_j \chi_j^2(k_j + \delta)$ decrease to R_m, so these functions decrease to $G_m + g'$, which is not identically $-\infty$, since $G_m \geq v_{i,m}$ on U_i. By Theorem 8.3.1, $G_m + g'$ is plurisubharmonic on W. Hence G_m is quasi-plurisubharmonic, and the curvature bound holds with $c' = 2JA\mathrm{e}^{C_B}$, which depends only on the fixed geometry.

Step 4: the singularities of G_m are gentle. Each summand of R_m is smooth by Lemma 13.2.2. On U_i we have $a_{i,m} \leq R_m \leq J\mathrm{e}^{C_B}a_{i,m}$ by Lemma 13.3.2, and $a_{i,m}$ is comparable with $\sum_k |g_k|^2$ for local generators g_k of $\mathcal{I}(m(\varphi + h_i)) = \mathcal{I}(m\varphi)$. Thus

G_m has analytic singularities of type $(m^{-1}, \mathcal{I}(m\varphi))$, and $\exp(mG_m) = R_m$ is smooth on X.

Now let $\pi\colon Y \to X$ be any log resolution of $\mathcal{I}(m\varphi)$, with $\pi^*\mathcal{I}(m\varphi) = \mathcal{O}(-D)$. Its restriction $\pi^{-1}(B_j) \to B_j$ satisfies $\pi^*\mathcal{I}(m(\varphi + h_j)) = \mathcal{O}(-D)$ on $\pi^{-1}(B_j)$, so by Lemma 13.2.2, applied with the volume form $\omega^n/n!$, the pulled-back kernel $K_{j,m} \circ \pi$ is locally $|s_D|^2$ times a smooth strictly positive function on $\pi^{-1}(B_j)$. The factors $\mathrm{e}^{-mh_j} \circ \pi$ and $\chi_j^2 \circ \pi$ are smooth, and the latter vanishes outside the closed set $\pi^{-1}(\operatorname{Supp}\chi_j) \subset \pi^{-1}(B_j)$; hence $(R_m \circ \pi)/|s_D|^2$ is smooth on Y, and it is strictly positive because at every point some $\chi_i \circ \pi$ equals one. Therefore $G_m \circ \pi$ is locally the sum of $m^{-1}\log|s_D|^2$ and a smooth function. These are exactly the requirements of Definition 13.1.11. □

Lemma 13.4.2 *Let θ be a smooth closed real $(1,1)$-form on X and $\varphi \in \mathrm{PSH}(X,\theta)$. As $m \to \infty$, $G_m \to \varphi$ pointwise on X and in $L^1(X)$. For every integer $m \geq 1$ and every $x \in X$,*

$$\nu(\varphi, x) - \frac{n}{m} \leq \nu(G_m, x) \leq \nu(\varphi, x), \qquad \mathrm{e}^{m(G_m - \varphi)} \in L^1(X).$$

Moreover, there is a constant $D \geq 0$, depending only on the fixed geometry, such that $G_{2m} \leq G_m + D/m$ for every integer $m \geq 1$.

Proof We first bound G_m from below. Let $x \in U_i$. Then $R_m(x) \geq a_{i,m}(x)$, and the lower bound in (13.3), in the version of Theorem 13.2.3 with the measure $\omega^n/n!$, gives $m^{-1}\log K_{i,m}(x) \geq \varphi(x) + h_i(x) - C/m$. Since h_i cancels, $G_m \geq v_{i,m} \geq \varphi - C/m$ on U_i, where $C \geq 0$ depends only on the fixed geometry.

For an upper bound, let $r > 0$ be smaller than the distance, in the coordinates of B_j, from $\operatorname{Supp}\chi_j$ to ∂B_j for every j, and for $x \in \operatorname{Supp}\chi_j$ let $B_j(x,r) \subset B_j$ denote the ball of radius r about x in these coordinates. If $\chi_j(x) \neq 0$, the upper bound in (13.3) on B_j, again in the version with the measure $\omega^n/n!$, gives

$$v_{j,m}(x) = \frac{1}{m}\log K_{j,m}(x) - h_j(x) \leq \sup_{B_j(x,r)} \varphi + \Bigl(\sup_{B_j(x,r)} h_j - h_j(x)\Bigr) + \frac{1}{m}\log\frac{C_2}{r^{2n}},$$

where C_2 depends only on the fixed geometry. By the mean value theorem, $\sup_{B_j(x,r)} h_j - h_j(x) \leq C_h r$, where C_h bounds the gradients of the finitely many functions h_j on the compact sets $\overline{B_j}$ and therefore depends on θ. Since $R_m(x)$ is a sum of at most J terms, each at most $\max_j a_{j,m}(x)$ over the indices with $\chi_j(x) \neq 0$,

$$G_m(x) \leq \max_{j\colon \chi_j(x)\neq 0} \sup_{B_j(x,r)} \varphi + C_h r + \frac{1}{m}\log\frac{JC_2}{r^{2n}}.$$

Fix x. Letting first $m \to \infty$ and then $r \downarrow 0$, and using that the balls $B_j(x,r)$ shrink to x, the upper semicontinuity of φ and the lower bound, we obtain $G_m(x) \to \varphi(x)$, also at points of the pole set. For a fixed r, the upper bound shows that $G_m \leq \sup_X \varphi + C_h r + |\log(JC_2 r^{-2n})|$ on X for every m, where $\sup_X \varphi < \infty$ because φ is

upper semicontinuous on the compact space X. Since moreover $\varphi - C \le G_m$ and $\varphi \in L^1(X)$, dominated convergence gives $G_m \to \varphi$ in $L^1(X)$.

By Lemma 13.4.1(1), $G_m = m^{-1}\log\sum_k |g_k|^2 + O(1)$ near x with generators g_k of $\mathcal{I}(m\varphi)$. Since $G_m \ge \varphi - C/m$ and φ is locally integrable (Lemma 13.1.4(1)), G_m is not identically $-\infty$ near x, so the germ of some g_k at x is not zero and $\mathcal{I}(m\varphi)_x \ne 0$. Hence Lemma 13.1.6 gives $\nu(G_m, x) = m^{-1}\operatorname{ord}_x \mathcal{I}(m\varphi)$. Write $\varphi = u + f$ on a connected neighborhood of x, with u plurisubharmonic and f smooth; then $\mathcal{I}(m\varphi) = \mathcal{I}(mu)$ and $\nu(\varphi, x) = \nu(u, x)$ near x, and the Lelong bounds follow from Lemma 12.3.1, applied to the weight u.

For the integrability, note that $\mathrm{e}^{m(G_m-\varphi)} = R_m\mathrm{e}^{-m\varphi} = \sum_j \chi_j^2 K_{j,m}\mathrm{e}^{-m(\varphi+h_j)}$. Cover the compact set $\operatorname{Supp}\chi_j$ by finitely many open sets on which $K_{j,m} \le C_m\sum_i |g_i|^2$ with members g_i of the weighted space on B_j, by Lemma 13.2.2. Since these g_i have finite weighted norm, every summand is integrable.

For the doubling inequality, Lemma 13.2.4, applied on each B_j with $u = \varphi + h_j$ and the measure $\omega^n/n!$, gives $K_{j,2m} \le C_* K_{j,m}^2$, hence $a_{j,2m} \le C_* a_{j,m}^2$, with C_* depending only on the fixed geometry. Let $x \in U_i$. For every j with $\chi_j(x) \ne 0$, Lemma 13.3.2 gives $a_{j,m}(x) \le \mathrm{e}^{C_B} a_{i,m}(x)$, and $R_m(x) \ge a_{i,m}(x)$. Consequently

$$R_{2m} \le C_* \sum_j \chi_j^2 a_{j,m}^2 \le C_* J \mathrm{e}^{2C_B} R_m^2$$

at x, also when both sides vanish. Taking logarithms and dividing by $2m$ gives $G_{2m} \le G_m + D/m$ with $D = \frac{1}{2}\log\max(1, C_* J\mathrm{e}^{2C_B})$. □

To obtain inclusions of multiplier ideals from an integrability estimate, we use the following elementary lemma.

Lemma 13.4.3 *Let u, v be quasi-plurisubharmonic near a point x of a complex manifold, and let $m > 0$ be such that $\mathrm{e}^{m(v-u)}$ is integrable near x. If $0 < \lambda < \lambda'$ and $m \ge \lambda\lambda'/(\lambda' - \lambda)$, then*

$$\mathcal{I}(\lambda' v)_x \subseteq \mathcal{I}(\lambda u)_x.$$

Proof After shrinking the neighborhood we may subtract a constant from v and assume $v \le 0$; this changes neither the ideals nor the integrability of $\mathrm{e}^{m(v-u)}$. Put $p = \lambda'/\lambda > 1$, and note that $m \ge \lambda p/(p-1) > \lambda$. Let $f \in \mathcal{I}(\lambda' v)_x$, defined on a small neighborhood on which $|f|^2 \le C$ for a constant C. All the following inequalities are taken outside the pole sets of u and v, which have measure zero. Where $u \ge pv$, we have $\lambda u \ge \lambda' v$ and hence $|f|^2\mathrm{e}^{-\lambda u} \le |f|^2\mathrm{e}^{-\lambda' v}$. Where $u < pv$, since $m - \lambda > 0$, $v \le 0$ and $(m-\lambda)p - m = m(p-1) - \lambda p \ge 0$,

$$(m-\lambda)u - mv \le ((m-\lambda)p - m)v \le 0,$$

that is, $-\lambda u \le m(v-u)$; hence $|f|^2\mathrm{e}^{-\lambda u} \le C\mathrm{e}^{m(v-u)}$. Both bounds are integrable near x, so $f \in \mathcal{I}(\lambda u)_x$. □

Definition 13.4.4 Let θ be a smooth closed real $(1,1)$-form on X and $\varphi \in \mathrm{PSH}(X, \theta)$. A *quasi-equisingular approximation* of φ is a sequence $(\Phi_j)_{j\ge 1}$ of

quasi-plurisubharmonic functions with analytic singularities, decreasing pointwise to φ, such that $\theta + \mathrm{dd^c}\Phi_j \geq -\varepsilon_j\omega$ for some numbers $\varepsilon_j \downarrow 0$, and such that for all $0 < \lambda < \lambda'$

$$\mathcal{I}(\lambda'\Phi_j) \subseteq \mathcal{I}(\lambda\varphi) \quad \text{for every sufficiently large } j.$$

Theorem 13.4.5 *Let θ be a smooth closed real $(1,1)$-form on X. Every $\varphi \in \mathrm{PSH}(X,\theta)$ admits a quasi-equisingular approximation (Φ_j) in which each Φ_j has gentle analytic singularities of type $(2^{-j}, \mathcal{I}(2^j\varphi))$, and which can be chosen with $\varepsilon_j = c'2^{-j}$, where c' is the constant of Lemma 13.4.1(1). Moreover $\Phi_j \to \varphi$ in $L^1(X)$, and the Lelong numbers of Φ_j satisfy the bounds of Lemma 13.4.2 with $m = 2^j$.*

Proof Put $\Phi_j = G_{2^j} + 2D/2^j$, with D as in Lemma 13.4.2. The doubling inequality gives

$$\Phi_{j+1} = G_{2\cdot 2^j} + \frac{D}{2^j} \leq G_{2^j} + \frac{D}{2^j} + \frac{D}{2^j} = \Phi_j,$$

so the sequence is decreasing, and by Lemma 13.4.2 its pointwise and L^1 limit is φ. Adding a constant changes neither the curvature, nor the analytic type, nor the Lelong numbers. Moreover $\exp(2^j\Phi_j) = \mathrm{e}^{2D}R_{2^j}$ is smooth, and the remainder on a log resolution changes only by a constant. Hence Φ_j has gentle analytic singularities of type $(2^{-j}, \mathcal{I}(2^j\varphi))$ by Lemma 13.4.1(1), and $\theta + \mathrm{dd^c}\Phi_j \geq -c'2^{-j}\omega$.

It remains to prove the inclusions of multiplier ideals. Let $0 < \lambda < \lambda'$. The function $\exp(2^j(\Phi_j - \varphi)) = \mathrm{e}^{2D}\mathrm{e}^{2^j(G_{2^j}-\varphi)}$ is integrable on X by Lemma 13.4.2. If j is so large that $2^j \geq \lambda\lambda'/(\lambda' - \lambda)$, then Lemma 13.4.3, with $u = \varphi$, $v = \Phi_j$ and $m = 2^j$, gives $\mathcal{I}(\lambda'\Phi_j)_x \subseteq \mathcal{I}(\lambda\varphi)_x$ at every point $x \in X$. Since this threshold for j depends only on λ and λ', the inclusion holds as an inclusion of sheaves on X for all such j.□

13.5 Comparison of analytic singularities

For functions $u, v\colon X \to [-\infty, \infty)$ we write $u \preceq v$, and say that u is *more singular* than v, if $u \leq v + C$ on X for some constant C. Since X is compact, it suffices that every point have a neighborhood on which such an inequality holds. We now show that two quasi-equisingular approximations of the same function are comparable for $\preceq$, up to an arbitrarily small loss in the coefficient. The key is a growth estimate for the members of the multiplier ideals of a function with analytic singularities.

Lemma 13.5.1 *Let v be a quasi-plurisubharmonic function with analytic singularities on a compact complex manifold X, and let $0 < \eta < 1$. There is $t_0 > 0$ such that for every real $t \geq t_0$ and every germ $f \in \mathcal{I}(tv)_x$, $x \in X$,*

$$\log|f|^2 \leq t(1-\eta)v + O(1)$$

near x, where the bounded term may depend on f. Moreover, for every quasi-plurisubharmonic function u and every $t > 0$, local generators g_i of $\mathcal{I}(tu)$ satisfy

$$u \le t^{-1} \log \sum_i |g_i|^2 + O(1).$$

Proof For the first assertion we pull v back to log resolutions of its local presentations, where Theorem 12.6.5 bounds the vanishing orders of f along the components of the divisor from below. Cover X by finitely many open sets V_κ^+ on which v has a presentation (13.1), and choose open sets $V_\kappa \Subset V_\kappa^+$ that still cover X. By Theorem A.5.5, the ideal generated by the presentation on V_κ^+ has a log resolution $\pi_\kappa \colon Y_\kappa \to V_\kappa^+$, a sequence of blowups that is finite over V_κ; it is proper and surjective, so $\pi_\kappa^{-1}(\overline{V_\kappa})$ and the fibers of π_κ are compact. On Y_κ we have locally $v \circ \pi_\kappa = \sum_E a_E \log |s_E|^2 + O(1)$, where $\pi_\kappa^*(g_1, \dots, g_r) = O(-\sum_E d_E E)$, E runs over the components of this divisor, s_E is a local equation of E, and $a_E = cd_E > 0$. Let A_E be the log discrepancy of E (Definition 12.6.3). Only finitely many components E meet the compact set $\pi_\kappa^{-1}(\overline{V_\kappa})$, so altogether there are finitely many pairs (a_E, A_E) with E meeting one of these sets. Choose $t_0 > 0$ with $t_0 \eta a_E \ge A_E$ for all of them.

Now let $x \in V_\kappa$, let $t \ge t_0$, and let $f \in \mathcal{I}(tv)_x$ be defined on an open neighborhood $V' \subset V_\kappa$ of x. Near x the function tv differs by a smooth function from a plurisubharmonic function with the same multiplier ideal and of the form required in Theorem 12.6.5, with coefficient tc. We apply that theorem over V', shrunk if necessary, to the log resolution obtained by restricting π_κ. Each component E' of the restricted divisor is a connected component of $E \cap \pi_\kappa^{-1}(V')$ for a component E as above, and it has the same coefficient d_E and the same log discrepancy A_E, as we show below. Since $tc\, d_E = ta_E$, the theorem gives

$$\operatorname{ord}_{E'} f > ta_E - A_E \ge t(1-\eta)a_E$$

for every component E' of the restricted divisor that meets $\pi_\kappa^{-1}(x)$; the second inequality holds because $t \ge t_0$ and E meets $\pi_\kappa^{-1}(\overline{V_\kappa})$.

The coefficient and the log discrepancy of E' are those of E because the vanishing orders along E of a local generator of $\pi_\kappa^*(g_1, \dots, g_r)$ and of a local Jacobian determinant of π_κ do not depend on the point of E at which they are computed. Indeed, two local generators of this invertible ideal, like two local Jacobian determinants, differ by a unit, which has order zero along E; since orders add under products, they have the same order along E at every point of E where both are defined (Lemma 12.6.1(2),(4)). Moreover, the order along E of a holomorphic function is locally constant on E_{reg}, by the last assertion of Lemma 12.6.1(2), and E_{reg} is connected (Corollary A.2.13); here $E_{\mathrm{reg}} = E$, since the components of a simple normal crossings divisor are smooth.

If $f = 0$ there is nothing to prove, so let $f \ne 0$. Shrinking V', we may assume that V' is connected; then f does not vanish identically on any nonempty open subset of V'. Hence $f \circ \pi_\kappa$ does not vanish identically on any nonempty open subset of $\pi_\kappa^{-1}(V')$, because π_κ is a biholomorphism off the nowhere dense set $\pi_\kappa^{-1}(V(g_1, \dots, g_r))$; in particular, the orders of f along the components of the restricted divisor are finite. Let $p \in \pi_\kappa^{-1}(x)$, and choose coordinates w on a neighborhood of p in $\pi_\kappa^{-1}(V')$ in which the components of the divisor through p are $\{w_k = 0\}$, $k = 1, \dots, \ell$, with coefficients $a_k = a_E$ for the corresponding components E. Then $v \circ \pi_\kappa = \sum_k a_k \log |w_k|^2 + O(1)$ near p. For each k, let E'_k be the component of the restricted divisor that contains

$\{w_k = 0\}$ near p, and put $o_k = \operatorname{ord}_{E'_k} f$, an integer with $o_k > t(1-\eta)a_k$ by the inequality above. Since $f \circ \pi_\kappa$ is holomorphic on a neighborhood of E'_k, its order along E'_k is the same at all points of the connected submanifold E'_k, by Lemma 12.6.1(3); hence $f \circ \pi_\kappa$ is divisible by $w_k^{o_k}$ near p, by Lemma 12.6.1(1),(2).

By the last assertion of Lemma 12.6.2, divisibility by each $w_k^{o_k}$ separately implies divisibility by their product: $f \circ \pi_\kappa = \prod_k w_k^{o_k} \cdot h$ with h holomorphic near p, hence bounded on a smaller neighborhood of p. Since $|w_k| \le 1$ there, we obtain $|f \circ \pi_\kappa|^2 \le C \prod_k |w_k|^{2t(1-\eta)a_k} \le C' \exp(t(1-\eta)v \circ \pi_\kappa)$. Finitely many such neighborhoods cover the compact fiber $\pi_\kappa^{-1}(x)$, and since π_κ is proper, their union contains the inverse image of a neighborhood of x. As π_κ is surjective, the inequality $|f|^2 \le C' \exp(t(1-\eta)v)$ holds near x.

For the second assertion, fix a point and write $u = u_0 - \sigma$ near it, with u_0 plurisubharmonic and σ smooth. Then $\mathcal{I}(tu) = \mathcal{I}(tu_0)$ near the point. On a bounded polydisc Ω about the point on which u_0 is bounded above, Theorem 12.2.7(3) provides finitely many elements of $\mathcal{H}_{tu_0}(\Omega)$ that generate $\mathcal{I}(tu_0)$ near the point, and (12.9) of Lemma 12.2.8, applied with $m = t$ to the weight u_0, gives $u_0 \le t^{-1} \log \sum_i |g_i|^2 + O(1)$ for these generators. Since σ is bounded near the point, and any two finite systems of local generators give comparable sums of squares (the remark following Definition 13.1.5), the assertion follows. □

Proposition 13.5.2 *Let X be a compact complex manifold, let φ and Φ be quasi-plurisubharmonic functions on X with $\varphi \le \Phi$, where Φ has analytic singularities, and let $(\Psi_k)_{k\ge 1}$ be quasi-plurisubharmonic functions on X such that for all $0 < \lambda < \lambda'$*

$$\mathcal{I}(\lambda' \Psi_k) \subseteq \mathcal{I}(\lambda \varphi) \quad \textit{for every sufficiently large } k.$$

Then for each $0 < \varepsilon < 1$,

$$\Psi_k \preceq (1-\varepsilon)\Phi \quad \textit{for every sufficiently large } k.$$

In particular, if (X, ω) is a compact Kähler manifold, θ is a smooth closed real $(1,1)$-form on X, $\varphi \in \mathrm{PSH}(X, \theta)$, and (Φ_j) and (Ψ_k) are quasi-equisingular approximations of φ, then for each j and each $0 < \varepsilon < 1$ we have $\Psi_k \preceq (1-\varepsilon)\Phi_j$ for every sufficiently large k.

Proof Choose $0 < \eta < \varepsilon$ and $\delta > 0$ so small that $(1-\eta)/(1+\delta) \ge 1 - \varepsilon$. Fix $t \ge t_0$, where t_0 is given by Lemma 13.5.1 for $v = \Phi$ and η. By the hypothesis on (Ψ_k), applied with $\lambda = t$ and $\lambda' = t(1+\delta)$, and since $\varphi \le \Phi$,

$$\mathcal{I}(t(1+\delta)\Psi_k) \subseteq \mathcal{I}(t\varphi) \subseteq \mathcal{I}(t\Phi)$$

for every sufficiently large k. Fix such a k and choose local generators g_i of $\mathcal{I}(t(1+\delta)\Psi_k)$ near a point of X. They belong to $\mathcal{I}(t\Phi)$, so the first part of Lemma 13.5.1 gives $\log |g_i|^2 \le t(1-\eta)\Phi + O(1)$ near the point, and hence $\log \sum_i |g_i|^2 \le t(1-\eta)\Phi + O(1)$. Its second part, applied to $u = \Psi_k$ and $t(1+\delta)$, then gives

$$\Psi_k \le \frac{1}{t(1+\delta)} \log \sum_i |g_i|^2 + O(1) \le \frac{1-\eta}{1+\delta}\Phi + O(1).$$

The upper semicontinuous function Φ is bounded above on the compact space X, say by M. Since $\Phi - M \le 0$ and $(1-\eta)/(1+\delta) \ge 1-\varepsilon$, we have $\frac{1-\eta}{1+\delta}(\Phi - M) \le (1-\varepsilon)(\Phi - M)$, and therefore $\Psi_k \le (1-\varepsilon)\Phi + O(1)$ near every point. By compactness $\Psi_k \preceq (1-\varepsilon)\Phi$; the constant may depend on k.

For the last assertion, recall from Definition 13.4.4 that each Φ_j is quasi-plurisubharmonic with analytic singularities and that $\varphi \le \Phi_j$, since the sequence decreases to φ; and (Ψ_k) satisfies the hypothesis on the inclusions of multiplier ideals. Hence the first part applies with $\Phi = \Phi_j$. □

Example 13.5.3 Let $u = c \log |z|^2$ on a disc about $0 \in \mathbb{C}$, with $c > 0$. A germ $z^k h$ with $h(0) \neq 0$ lies in $\mathcal{I}(mu)_0$ if and only if $|z|^{2k-2mc}$ is integrable near 0, that is, $k > mc - 1$; hence $\mathcal{I}(mu)$ is generated by $z^{\lfloor mc \rfloor}$ (Example 12.5.5). A potential of type $(m^{-1}, \mathcal{I}(mu))$, such as the Bergman potential u_m of Theorem 13.2.3, is therefore $(\lfloor mc \rfloor / m) \log |z|^2 + O(1)$ near 0. Its coefficient $\lfloor mc \rfloor / m$ need not equal c for any m, and then such an approximation does not reproduce the multiplier ideals of u at a fixed scale. For instance, let c be irrational, so that $\lfloor mc \rfloor / m < c$ for every m, and take the scale $1/c$. Since u_m/c differs by a bounded function from $c_m \log |z|^2$ with $c_m = \lfloor mc \rfloor / (mc) < 1$, the same computation gives $\mathcal{I}(u_m/c)_0 = \mathcal{O}_0$ for every m, whereas $\mathcal{I}(u/c)_0 = \mathcal{I}(\log |z|^2)_0$ is generated by z. This is why Definition 13.4.4 compares two different scales $\lambda' > \lambda$ instead of requiring equality of multiplier ideals.

13.6 Bounded growth ideals

A germ f belongs to the multiplier ideal $\mathcal{I}(v)$ when $|f|^2 \mathrm{e}^{-v}$ is integrable, a condition that allows f to vanish to a slightly lower order than the growth of e^v suggests. The stricter condition $|f|^2 \le C\mathrm{e}^v$ defines a smaller ideal. For a weight v with analytic singularities we prove that this ideal sheaf is coherent (Theorem 13.6.2) and that, on a compact manifold, the ideal attached to kv contains $\mathcal{I}(k(1+\varepsilon)v)$ for all large k (Corollary 13.6.3). In the proof of coherence, the condition $|f| \le C|g|^{p/q}$ for a tuple g of holomorphic functions becomes a holomorphic one: on the normalization of a root cover of a cone over the blowup of the ideal (g), it is the holomorphy of a function homogeneous of weight p.

Definition 13.6.1 Let v be a quasi-plurisubharmonic function on a complex manifold X. The *bounded growth ideal* $\mathcal{I}_\infty(v)$ is the sheaf of holomorphic germs f satisfying

$$\log |f|^2 \le v + O(1), \quad \text{equivalently} \quad |f|^2 \le C \exp(v),$$

on a neighborhood of their base point, for some finite constant C.

The zero germ belongs to $\mathcal{I}_\infty(v)$, and $\mathcal{I}_\infty(v)$ does not change if v is changed by a bounded function. It is a sheaf of ideals, because $|f+g|^2 \le 2|f|^2 + 2|g|^2$ and holomorphic coefficients are locally bounded. It is contained in $\mathcal{I}(v)$, since $|f|^2 e^{-v}$ is locally bounded for $f \in \mathcal{I}_\infty(v)$.

Theorem 13.6.2 *Let v be a quasi-plurisubharmonic function with analytic singularities on a complex manifold X. Then $\mathcal{I}_\infty(v)$ is a coherent ideal sheaf. More precisely, let $g = (g_1, \dots, g_r)$, with $r \ge 1$, be a tuple of holomorphic functions on a coordinate neighborhood U, put $|g| = (\sum_j |g_j|^2)^{1/2}$, and let p, q be positive integers. Then the ideal sheaf*

$$\mathcal{J}_{p/q}(g) = \{f \in \mathcal{O}_U : |f| \le C|g|^{p/q} \text{ near the base point of } f, \text{ for some } C\}$$

is coherent.

Proof Coherence is a local property. Fix $x \in U$ and shrink U to a connected neighborhood of x. If all g_j vanish identically there, then $\mathcal{J}_{p/q}(g)$ is the zero ideal near x. Otherwise choose an index i with g_i not identically zero, and put $U_i = \{g_i \ne 0\}$. By the identity theorem U_i is dense in U, and by Lemma A.1.7 it is connected. Steps 1 and 2 construct the normal space described at the beginning of this section, Steps 3 to 5 show that its functions of weight p are finitely generated over the holomorphic functions of z, and Step 6 identifies their coefficients with the germs of $\mathcal{J}_{p/q}(g)$.

Step 1: a cone over the blowup. Let $B \to U$ be the blowup of the ideal $(g_1, \dots, g_r)$, constructed in Proposition A.5.2. Thus $B \subset U \times \mathbb{P}^{r-1}$, and a point $(z, [\xi])$ of B determines the line $\mathbb{C}\xi \subset \mathbb{C}^r$. Let $\Lambda = \{(z, [\xi], a) \in B \times \mathbb{C}^r : a \in \mathbb{C}\xi\}$ be the total space of this line bundle, and consider the map $\Lambda \to U \times \mathbb{C}^r$, $(z, [\xi], a) \mapsto (z, a)$. It is proper: the inverse image of a compact set is closed and contained in the product of a compact subset of B, namely the inverse image of its projection to U, with a bounded set of vectors. By Theorem A.6.2, its image $C \subset U \times \mathbb{C}^r$ is analytic, and we regard it as a reduced complex space. It is invariant under the scalar multiplications $a \mapsto \lambda a$. Over U_i the blowup is the graph of $z \mapsto [g(z)]$, so the part of C over U_i is the smooth family of lines

$$a_j = \tau g_j(z), \qquad z \in U_i, \quad \tau \in \mathbb{C}.$$

This family is dense in C. Indeed, the graph over U_i is dense in B, since U_i is dense in $U \setminus V(g)$ and the graph over $U \setminus V(g)$ is dense in B. Hence its part of Λ is dense in Λ: if $(z, [\xi], \lambda\xi) \in \Lambda$, choose graph points $(z_k, [\xi_k]) \to (z, [\xi])$ over U_i with representatives $\xi_k \to \xi$, which is possible in the affine charts of $\mathbb{P}^{r-1}$; then $(z_k, [\xi_k], \lambda\xi_k) \to (z, [\xi], \lambda\xi)$. The image of a dense set under the continuous surjection $\Lambda \to C$ is dense.

This family is the part of C over the open set U_i, hence open in C, and it is the graph of the holomorphic map $(z, a_i) \mapsto (a_i g_l(z)/g_i(z))_{l \ne i}$ on the connected manifold $U_i \times \mathbb{C}$, hence a connected complex manifold. Therefore C is irreducible. Indeed, suppose that C is the union of two analytic subsets different from C. Their traces on this connected manifold are analytic subsets covering it, so one of them has nonempty interior in it and hence, by the identity theorem, contains all of it.

Since this manifold is dense in C and analytic subsets are closed, that analytic subset contains C, a contradiction. Taking $\tau = 0$, we see that C contains $U_i \times \{0\}$, hence $U \times \{0\}$, since C is closed and U_i is dense in U.

Step 2: a root cover and its normalization. In $C \times \mathbb{C}_b$ consider the analytic set defined by

$$b^q = a_i g_i^{q-1},$$

and let C' be the closure of its part over U_i. The set C' is analytic by Lemma A.2.5, applied in $U \times \mathbb{C}^r \times \mathbb{C}_b$ to this analytic set and to the analytic set $\{g_i(z) = 0\}$; near each of its points it is the union of the branches of that set not contained in $\{g_i(z) = 0\}$. The projection $C' \to C$ is finite: the monic equation bounds b over compact subsets of C and allows at most q values of b. Over U_i we have $a = \tau g(z)$ and $b^q = \tau g_i(z)^q$, so $b = t g_i(z)$ with $t^q = \tau$. Hence the part of C' over U_i is the image of the injective immersion

$$U_i \times \mathbb{C} \to C', \qquad (z, t) \longmapsto (z,\ a = t^q g(z),\ b = t g_i(z)),$$

whose inverse is $t = b/g_i(z)$; it is a dense open connected submanifold of C', so C' is irreducible by the argument just given for C.

Let $Y \to C'$ be the normalization (Theorem A.3.6), and let $\nu\colon Y \to C$ be its composition with the projection; ν is finite. Over U_i the normalization is an isomorphism, and we identify the corresponding open subset of Y with $U_i \times \mathbb{C}_t$; we call it the *dense open family*. Its complement in Y is the zero set of g_i composed with ν, an analytic subset, and it is nowhere dense. Indeed, by Theorem A.3.6 the map $Y \to C'$ is bimeromorphic: it restricts to a homeomorphism from the complement of a nowhere dense analytic subset of Y onto the complement of a nowhere dense analytic subset of C'. The part of C' over U_i is open and dense in C', so its part in the second complement is dense there, and the inverse image of this part is dense in the first complement, hence in Y.

Step 3: finite generation over C. For an open set $W \subset C$ let $\mathcal{F}(W)$ be the ring of holomorphic functions on $\nu^{-1}(W)$; holomorphic functions on W act on it by multiplication after composition with ν. We show that every point $w_0 \in C$ has a neighborhood W in C with the following property: there are finitely many $s_1, \dots, s_N \in \mathcal{F}(W)$ such that for every $w \in W$ and every holomorphic function s on a neighborhood of $\nu^{-1}(w)$ there are holomorphic functions $k_1, \dots, k_N$ of (z, a) on a neighborhood of w in $U \times \mathbb{C}^r$ with $s = \sum_j (k_j \circ \nu) s_j$ near $\nu^{-1}(w)$.

The tool is division by monic polynomials whose roots need not coincide. Let $\Phi(\xi, u) = u^\mu + c_1(\xi) u^{\mu-1} + \cdots + c_\mu(\xi)$, with coefficients holomorphic on an open subset of a complex vector space, let ξ_1 be a point of this set, let Γ be the union of small disjoint circles about the distinct roots of $\Phi(\xi_1, \cdot)$, and let Δ be the union of the open discs they bound. Choose a relatively compact neighborhood A_1 of ξ_1 such that $\Phi \neq 0$ on a neighborhood of $\overline{A_1} \times \Gamma$. Let H be holomorphic on a neighborhood of $\overline{A_1} \times \overline{\Delta}$. The two integrals (A.2) of Proposition A.1.1, with f, P and w replaced by H, Φ and u, and taken over Γ instead of a single circle,

$$Q(\xi,u)=\frac{1}{2\pi\mathrm{i}}\int_\Gamma\frac{H(\xi,\zeta)\,\mathrm{d}\zeta}{\Phi(\xi,\zeta)(\zeta-u)},$$
$$R(\xi,u)=\frac{1}{2\pi\mathrm{i}}\int_\Gamma\frac{H(\xi,\zeta)}{\Phi(\xi,\zeta)}\,\frac{\Phi(\xi,\zeta)-\Phi(\xi,u)}{\zeta-u}\,\mathrm{d}\zeta,$$

define, by differentiation under the integral sign, a holomorphic function Q on $A_1\times\Delta$ and, since $(\Phi(\xi,\zeta)-\Phi(\xi,u))/(\zeta-u)$ is a polynomial in u of degree less than μ whose coefficients are polynomials in ζ and the $c_k(\xi)$, a polynomial $R=\sum_{k<\mu}r_k(\xi)u^k$ with coefficients holomorphic on a neighborhood of $\overline{A_1}$. They satisfy $H=\Phi Q+R$ on $A_1\times\Delta$: for $u\in\Delta$, Cauchy's formula on the disc containing u gives $H(\xi,u)=\frac{1}{2\pi\mathrm{i}}\int_\Gamma H(\xi,\zeta)(\zeta-u)^{-1}\,\mathrm{d}\zeta$, since the circles that do not surround u contribute nothing, and writing $1=\big(\Phi(\xi,u)+(\Phi(\xi,\zeta)-\Phi(\xi,u))\big)/\Phi(\xi,\zeta)$ in this integral gives $\Phi(\xi,u)Q(\xi,u)+R(\xi,u)$. If H depends holomorphically on further variables, so do Q and the r_k.

Applying this successively to variables $u_1,\dots,u_m$ and monic polynomials $\Phi_1(\xi,u_1),\dots,\Phi_m(\xi,u_m)$, with the corresponding sets $\Delta_1,\dots,\Delta_m$ and a common A_1, we find: a function holomorphic on a neighborhood of $\overline{A_1}\times\overline{\Delta_1}\times\cdots\times\overline{\Delta_m}$ agrees, at the points of $A_1\times\Delta_1\times\cdots\times\Delta_m$ where all $\Phi_l(\xi,u_l)$ vanish, with a polynomial $\sum_\beta r_\beta(\xi)u^\beta$, $\beta_l<\deg\Phi_l$, whose coefficients are holomorphic on A_1.

Next we embed Y locally by functions that are integral over the functions of (z,a). Let $y\in Y$, let y' be its image in C', and let $w=\nu(y)$. By Corollary A.3.7, applied to the analytic subset C' of the domain $U\times\mathbb{C}^r\times\mathbb{C}_b$ and to the point y', a neighborhood of y in Y is identified, compatibly with the maps to C', with the closure of the graph of finitely many weakly holomorphic functions $f_1,\dots,f_m$ over the regular part of the branch of (C',y') that corresponds to y. This closure is an analytic subset of $\Omega'\times\mathbb{C}^m$ for a neighborhood Ω' of y' in $U\times\mathbb{C}^r\times\mathbb{C}_b$, embedded by the coordinates (z,a,b) and by the coordinates $f_1,\dots,f_m$ of $\mathbb{C}^m$. Each f_j satisfies on it a monic equation whose coefficients are holomorphic functions of (z,a,b) on Ω', and every holomorphic function on Y near a point of this neighborhood of y is the restriction of a holomorphic function of $(z,a,b,f_1,\dots,f_m)$.

We first reduce the coefficients of the monic equations of the f_j to polynomials in b. Divide each of these coefficients, extended by zero near the roots of $b^q-a_ig_i(z)^{q-1}$ different from the b-coordinate of y', by this monic polynomial in b. Since the polynomial vanishes on C', each coefficient agrees on Y near y with a polynomial in b of degree less than q whose coefficients are holomorphic functions of (z,a) near w.

Let $\mathcal{O}_w$ be the ring of germs at w of holomorphic functions of (z,a), acting through ν. Reducing powers of b by $b^q=a_ig_i^{q-1}$ and the top power of f_j by its equation, we see that the $\mathcal{O}_w$-module spanned by the germs at y of the products $b^kf_j^l$, with $k<q$ and l less than the degree of the equation of f_j, contains 1 and is stable under multiplication by f_j. Thus, if $v_1,\dots,v_M$ are these products, then $f_jv_\kappa=\sum_\lambda e_{\kappa\lambda}v_\lambda$ with $e_{\kappa\lambda}\in\mathcal{O}_w$; in other words, the column vector $v=(v_\lambda)$ satisfies $(f_jI-(e_{\kappa\lambda}))v=0$. Multiplying by the adjugate matrix of $f_jI-(e_{\kappa\lambda})$, whose product with this matrix is $\det(f_jI-(e_{\kappa\lambda}))\,I$ over every commutative ring, we find that $\det(f_jI-(e_{\kappa\lambda}))$ annihilates every v_κ, in particular 1. Expanding this determinant gives a monic equation for f_j with coefficients in $\mathcal{O}_w$. Consequently, after shrinking

the neighborhood of y, each coordinate of $u = (b, f_1, \dots, f_m)$ satisfies a monic equation $\Phi_l(z, a; u_l) = 0$ whose coefficients are holomorphic near w in $U \times \mathbb{C}^r$.

Now fix $w_0 \in C$, and let $y_1, \dots, y_\ell$ be the points of the finite fiber $\nu^{-1}(w_0)$. Choose embeddings as above on pairwise disjoint neighborhoods $N_1, \dots, N_\ell$ of these points, with coordinates $u^{(\iota)}$ on N_ι and monic equations $\Phi_{\iota,l}$. Since ν is proper, there is a neighborhood W of w_0 in C with $\nu^{-1}(W) \subset N_1 \cup \cdots \cup N_\ell$, and we may assume that $W = C \cap A$ for an open set $A \subset U \times \mathbb{C}^r$ on which the coefficients of all the $\Phi_{\iota,l}$ are holomorphic. For each ι and each multi-index β with $\beta_l < \deg \Phi_{\iota,l}$, let $s_{\iota,\beta} \in \mathcal{F}(W)$ be the function equal to the monomial $(u^{(\iota)})^\beta$ on $N_\iota \cap \nu^{-1}(W)$ and to zero on the other sets $N_{\iota'} \cap \nu^{-1}(W)$.

We show that these functions have the required property. Let $w \in W$, let s be holomorphic near $\nu^{-1}(w)$, and fix ι. The coordinates $u^{(\iota)}$ of the points of $\nu^{-1}(w) \cap N_\iota$ are distinct points of the finite product of the sets of roots of the polynomials $\Phi_{\iota,l}(w, \cdot)$, and near each of them s is the restriction of a holomorphic function of $(z, a, u^{(\iota)})$. Let H be the function, defined on a neighborhood of $\{w\}$ times this finite product, that equals such an extension near the coordinates of the points of $\nu^{-1}(w) \cap N_\iota$ and zero near the other points of the product. By the iterated division, H agrees with a polynomial $\sum_\beta r_{\iota,\beta}(z, a)(u^{(\iota)})^\beta$, with coefficients holomorphic near w, on the common zero set of the $\Phi_{\iota,l}$ near these points. This zero set contains $Y \cap N_\iota$ near $\nu^{-1}(w) \cap N_\iota$, so $s = \sum_\beta (r_{\iota,\beta} \circ \nu) s_{\iota,\beta}$ there. Summing over ι, we obtain the claim with the finitely many functions $s_{\iota,\beta}$.

Step 4: homogeneous components. For $x' \in U$ and $\varepsilon > 0$ with $\{|z - x'| \le \varepsilon\} \subset U$ put

$$W_\varepsilon(x') = \{(z, a) \in C : |z - x'| < \varepsilon,\ |a_l| < \varepsilon \text{ for } l = 1, \dots, r\}.$$

These sets form a basis of neighborhoods of $(x', 0)$ in C, and since C is closed in $U \times \mathbb{C}^r$, the closure of $W_{\varepsilon'}(x')$ is a compact subset of $W_\varepsilon(x')$ for $\varepsilon' < \varepsilon$. The part G of $\nu^{-1}(W_\varepsilon(x'))$ in the dense open family consists of the points (z, t) with $z \in U_i$, $|z - x'| < \varepsilon$ and $|t^q g_l(z)| < \varepsilon$ for all l. For fixed z these t form a disc about 0, and G is invariant under the rotations $t \mapsto \mathrm{e}^{\mathrm{i}\vartheta} t$. Let $s \in \mathcal{F}(W_\varepsilon(x'))$. On G we expand $s(z, t) = \sum_{e \ge 0} c_e(z) t^e$, where $c_e(z) = \partial_t^e s(z, 0)/e!$ is holomorphic on $U_i \cap \{|z - x'| < \varepsilon\}$, and for an integer d we define

$$(P_d s)(z, t) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}d\vartheta} s(z, \mathrm{e}^{\mathrm{i}\vartheta} t)\, \mathrm{d}\vartheta = \begin{cases} c_d(z) t^d, & d \ge 0, \\ 0, & d < 0, \end{cases}$$

integrating the power series term by term. For $\varepsilon' < \varepsilon$ the circle through a point of $G \cap \nu^{-1}(W_{\varepsilon'}(x'))$ stays in this set, so $|P_d s|$ is bounded there by the maximum of $|s|$ on the compact set $\nu^{-1}(\overline{W_{\varepsilon'}(x')})$. Since every point of $\nu^{-1}(W_\varepsilon(x'))$ lies in such a set $\nu^{-1}(W_{\varepsilon'}(x'))$, the holomorphic function $P_d s$ on G is locally bounded near every point of $\nu^{-1}(W_\varepsilon(x'))$. Since Y is normal and the complement of G in $\nu^{-1}(W_\varepsilon(x'))$ is a nowhere dense analytic subset, Lemma A.3.4, applied on each of the open sets $\nu^{-1}(W_{\varepsilon'}(x'))$, extends $P_d s$ holomorphically to $\nu^{-1}(W_\varepsilon(x'))$; the extensions on these sets agree on their overlaps by density of G. We denote the extension again by $P_d s$.

We say that s is *homogeneous of weight d* if $s(z, \mathrm{e}^{\mathrm{i}\vartheta}t) = \mathrm{e}^{\mathrm{i}d\vartheta}s(z,t)$ on G, that is, comparing the power series in t, if $c_e = 0$ for every $e \neq d$; then $P_d s = s$ on G, hence on $\nu^{-1}(W_\varepsilon(x'))$ by density of G. In particular, $P_d s$ is homogeneous of weight d, a function homogeneous of negative weight vanishes, and since $a = t^q g(z)$ on G, the function a^α is homogeneous of weight $q|\alpha|$. The operators P_d commute with restriction to smaller sets $W_{\varepsilon_1}(x'') \subset W_\varepsilon(x')$.

Let k be the restriction to $W_\varepsilon(x')$ of a holomorphic function K on $\{|z - x'| < \varepsilon\} \times \{|a_l| < \varepsilon \text{ for all } l\}$, with Taylor expansion $K(z,a) = \sum_\alpha K_\alpha(z)a^\alpha$ in a. At a point of G we have $k(\nu(z, \mathrm{e}^{\mathrm{i}\vartheta}t)) = \sum_\alpha K_\alpha(z)\mathrm{e}^{\mathrm{i}q|\alpha|\vartheta}a^\alpha$ with $a = t^q g(z)$, and $\sum_\alpha |K_\alpha(z)a^\alpha| < \infty$, while s is bounded on the circle. Integrating term by term, we obtain, first on G and then on $\nu^{-1}(W_\varepsilon(x'))$ by density,

$$P_p(ks) = \sum_{q|\alpha| \le p} K_\alpha(z)\, a^\alpha\, P_{p-q|\alpha|}s$$

for every integer p, where $K_\alpha(z)$ and a^α are composed with ν; the terms with $q|\alpha| > p$ vanish because $P_e s = 0$ for $e < 0$.

Step 5: functions of weight p. Let W and $s_1, \dots, s_N$ be as in Step 3 for $w_0 = (x, 0)$, and choose $\varepsilon > 0$ with $\{|z - x| \le \varepsilon\} \subset U$ and $W_\varepsilon(x) \subset W$. Let $|x' - x| < \varepsilon/2$, let $0 < \varepsilon_1 \le \varepsilon/2$, and let $h \in \mathcal{F}(W_{\varepsilon_1}(x'))$ be homogeneous of weight p. Then $W_{\varepsilon_1}(x') \subset W_\varepsilon(x)$. By Step 3, $h = \sum_j (k_j \circ \nu)s_j$ near $\nu^{-1}(x', 0)$, with k_j holomorphic near $(x', 0)$ in $U \times \mathbb{C}^r$. Since ν is proper, after shrinking ε_1 this identity holds on $\nu^{-1}(W_{\varepsilon_1}(x'))$, and each k_j is holomorphic on $\{|z - x'| < \varepsilon_1\} \times \{|a_l| < \varepsilon_1 \text{ for all } l\}$. We apply P_p to the identity. Since $P_p h = h$, Step 4 gives

$$h = \sum_{j=1}^{N} \sum_{q|\alpha| \le p} k_{j,\alpha}(z)\, a^\alpha\, P_{p-q|\alpha|}s_j \qquad \text{on } \nu^{-1}(W_{\varepsilon_1}(x')),$$

where the $k_{j,\alpha}$ are the Taylor coefficients of k_j in a. Thus every function of weight p near $\nu^{-1}(x', 0)$ is a combination, with coefficients holomorphic in z near x', of the finitely many functions

$$a^\alpha P_e s_j, \qquad 1 \le j \le N, \quad e \ge 0, \quad q|\alpha| + e = p,$$

which are holomorphic on $\nu^{-1}(W_\varepsilon(x))$ and homogeneous of weight p.

Step 6: identification with $\mathcal{J}_{p/q}(g)$. By Step 4, each function $a^\alpha P_e s_j$ of Step 5 equals $f_{j,\alpha}(z)t^p$ on the dense open family, with $f_{j,\alpha}$ holomorphic on $U_i \cap \{|z - x| < \varepsilon\}$. Let M be the maximum of the moduli of these finitely many functions on the compact set $\nu^{-1}(L)$, where $L = \{(z,a) \in C : |z - x| \le \varepsilon/2,\ |a_l| \le \varepsilon/2 \text{ for all } l\} \subset W_\varepsilon(x)$. For z with $|z - x| < \varepsilon/2$ and $g_i(z) \neq 0$, choose t with

$$|t|^q |g(z)| = \varepsilon/2.$$

Then $|t^q g_l(z)| \le \varepsilon/2$ for all l, so $(z,t) \in \nu^{-1}(L)$, and $|f_{j,\alpha}(z)||t|^p \le M$, that is,

$$|f_{j,\alpha}(z)| \le M(\varepsilon/2)^{-p/q}|g(z)|^{p/q}.$$

In particular $f_{j,\alpha}$ is bounded near $\{g_i = 0\}$, so it extends holomorphically to the ball $\{|z - x| < \varepsilon/2\}$ by Lemma A.1.7, and the inequality persists on $\{g_i = 0\}$, since both sides are continuous and U_i is dense. Hence the finitely many functions $f_{j,\alpha}$ belong to $\mathcal{J}_{p/q}(g)$ on this ball.

Conversely, let f be a germ of $\mathcal{J}_{p/q}(g)$ at a point x' with $|x' - x| < \varepsilon/2$, say $|f| \le C_f|g|^{p/q}$ on $\{|z - x'| < \rho\}$, and let $0 < \varepsilon_1 \le \min(\rho, \varepsilon/2)$. On the part of $\nu^{-1}(W_{\varepsilon_1}(x'))$ in the dense open family, the function $f(z)t^p$ satisfies

$$|f(z)t^p| \le C_f|t^q g(z)|^{p/q} = C_f|a|^{p/q}.$$

It is therefore bounded, and since Y is normal it extends holomorphically to $\nu^{-1}(W_{\varepsilon_1}(x'))$ by Lemma A.3.4; the extension h is homogeneous of weight p. By Step 5, after shrinking ε_1, h is a combination of the functions $a^\alpha P_e s_j$ with coefficients $k_{j,\alpha}(z)$ holomorphic near x'. Comparing the coefficients of t^p on the dense open family gives $f = \sum k_{j,\alpha} f_{j,\alpha}$ on U_i near x', hence near x', since both sides are continuous and U_i is dense. Thus the functions $f_{j,\alpha}$ generate $\mathcal{J}_{p/q}(g)$ on $\{|z - x| < \varepsilon/2\}$, and $\mathcal{J}_{p/q}(g)$ is coherent by Theorem A.1.5 and Lemma A.1.4.

Finally, if $v = (p/q)\log\sum_j |g_j|^2 + O(1)$ is a local presentation of v, then $\mathcal{I}_\infty(v) = \mathcal{J}_{p/q}(g)$ on its domain. Since coherence is a local property, $\mathcal{I}_\infty(v)$ is coherent. (If $g(x) \neq 0$, both ideals are the unit ideal near x; in dimension zero every finite weight gives the unit ideal.) □

Corollary 13.6.3 *Let v be a quasi-plurisubharmonic function with analytic singularities on a compact complex manifold X. For every $\varepsilon > 0$ there is $k_0 > 0$ such that for every real $k \ge k_0$*

$$\mathcal{I}(k(1+\varepsilon)v) \subseteq \mathcal{I}_\infty(kv).$$

Proof Apply Lemma 13.5.1 with $\eta = \varepsilon/(1+\varepsilon)$ and $t = k(1+\varepsilon)$, so that $t(1-\eta) = k$. The lemma provides one threshold t_0 for all germs on X; put $k_0 = t_0/(1+\varepsilon)$. For $k \ge k_0$, every germ $f \in \mathcal{I}(k(1+\varepsilon)v)$ satisfies $\log|f|^2 \le kv + O(1)$ near its base point, that is, $f \in \mathcal{I}_\infty(kv)$. In dimension zero all these ideals are the unit ideal. □

Chapter 14
Global potentials and positive cohomology classes

Let X be a connected compact Kähler manifold, and let T be a closed positive $(1,1)$-current on X. By Theorem 7.3.2, every point of X has a coordinate neighborhood on which $T = \mathrm{dd}^{\mathrm{c}} u$ with u plurisubharmonic. On the overlap of two such neighborhoods the two local potentials differ by a pluriharmonic function, and in general no choice of the local potentials makes them agree. For instance, let X have positive dimension and let $T = \omega$ be a Kähler form. A global potential u would be plurisubharmonic on X, so it would attain its maximum, being upper semicontinuous on the compact space X. The set where it does so is closed, and it is also open by the maximum principle (Proposition 3.1.4), applied on coordinate balls, where u is subharmonic (Theorem 8.2.1). Thus u would be constant and $\omega = \mathrm{dd}^{\mathrm{c}} u = 0$. The obstruction is the cohomology class of T: by the dd^{c}-lemma (Theorem 14.1.7), if θ is a smooth closed form in the class of T, then $T = \theta + \mathrm{dd}^{\mathrm{c}}\varphi$ for a quasi-plurisubharmonic function φ on X, unique up to an additive constant. We prove this by solving a Laplace equation for the trace of T, after developing the elliptic theory that this requires from the beginning; the same argument shows that the real $(1,1)$-cohomology of X is finite dimensional. Closed positive currents in the class of θ therefore correspond, up to additive constants, to the functions in $\mathrm{PSH}(X,\theta)$ (Definition 13.1.3), and the rest of the book works with such potentials.

Once the additive constant is fixed by normalizing the supremum, the local compactness and integrability theorems for plurisubharmonic functions become uniform statements on X, and the upper envelope V_θ of the nonpositive potentials has minimal singularities. With these tools and the Demailly approximation of Chapter 13 we study the positivity of a class (pseudo-effective, nef, big or Kähler) and the places where it fails. Siu's decomposition (Theorem 14.3.7) splits off the divisorial part of a current. The minimal multiplicities of a class and its non-nef locus locate the points at which the positive currents in the class are forced to be singular, and the non-Kähler locus does the same for Kähler currents; we also compare these notions on X and on a modification of X. A mixed version of the scalar equation gives the mixed Hodge index theorem (Theorem 14.3.3) and inequalities between intersection numbers.

The second half of the chapter concerns a holomorphic line bundle L on X. If h_0 is a smooth metric on L with curvature form $c_1(L,h_0)$, a singular metric $h_0 \mathrm{e}^{-\varphi}$

with $\varphi \in \mathrm{PSH}(X, c_1(L, h_0))$ is the global form of a plurisubharmonic weight; the metrics defined by finitely many holomorphic sections of powers of L are examples with analytic singularities. We prove the L^2 estimate for $\bar\partial$ on line bundles over Kähler manifolds and combine it with the local extension theorems of Chapter 11 to construct global sections with a prescribed value at a point (Theorem 14.6.7) or a prescribed restriction to a compact submanifold (Theorem 14.7.8), with constants that depend only on the geometry. We also describe the weighted spaces of global sections through their Bergman kernels. A related gluing construction extends strictly positive potentials from a submanifold (Theorem 14.7.12). In the last section we leave the Kähler setting and prove Josefson's theorem on a compact complex manifold: every pluripolar set lies in the pole set of a single quasi-plurisubharmonic function (Theorem 14.8.7).

Historical remarks and references for this chapter are collected in Appendix B.14.

14.1 The trace equation and global potentials

Throughout this chapter, except in Section 14.8, (X, ω) is a connected compact Kähler manifold of dimension $n \geq 1$. (In dimension zero all the notions below are trivial: $H^{1,1}(X, \mathbb{R})$ and all positivity cones are $\{0\}$, both positivity loci are empty and normalized potentials vanish. On a disconnected compact Kähler manifold the results apply to each of its finitely many components, with normalizations imposed on each component.) The goal of this section is the $\mathrm{dd^c}$-lemma (Theorem 14.1.7). We first develop the Sobolev spaces of X and the elliptic theory of its Laplacians (Lemma 14.1.3), and then express the trace of $\mathrm{dd^c} f$ through the Laplacian of f (Lemma 14.1.6).

Let J be the complex structure of X. The Kähler form defines the Riemannian metric $g(v, w) = \omega(v, Jw)$ and the volume form $\mathrm{d}V = \omega^n/n!$. Forms carry the pointwise inner product induced by g, and the L^2 inner product $\langle u, v\rangle = \int_X \langle u, v\rangle \, \mathrm{d}V$. The Hodge star operator is the complex-linear bundle map characterized by $u \wedge *\overline{v} = \langle u, v\rangle \, \mathrm{d}V$ for forms u, v of the same degree. Explicitly, if $e_1, \ldots, e_{2n}$ is a positively oriented orthonormal basis of the real cotangent space at a point and I' denotes the increasing multi-index complementary to I, then $*e_I = \varepsilon_I e_{I'}$, where $e_I \wedge e_{I'} = \varepsilon_I e_1 \wedge \cdots \wedge e_{2n}$ and $\varepsilon_I = \pm 1$; indeed $e_I \wedge \varepsilon_J e_{J'}$ is δ_{IJ} times the volume form. Hence $*$ is a real isometry, and since $e_{I'} \wedge e_I = (-1)^{k(2n-k)} e_I \wedge e_{I'}$ for $|I| = k$, we have $** = (-1)^{k(2n-k)} = (-1)^k$ on k-forms, the real dimension $2n$ being even.

Now let u be a $(k-1)$-form and v a k-form, one of them with compact support. Since $\mathrm{d}*\overline{v}$ has degree $2n-k+1$, on which $** = (-1)^{k-1}$, we have $u \wedge \mathrm{d}*\overline{v} = (-1)^{k-1} u \wedge *\overline{(*\mathrm{d} * v)} = (-1)^{k-1}\langle u, *\mathrm{d} * v\rangle \, \mathrm{d}V$. Integrating $\mathrm{d}(u \wedge *\overline{v}) = \mathrm{d}u \wedge *\overline{v} + (-1)^{k-1} u \wedge \mathrm{d}*\overline{v}$ by Stokes' formula (Proposition 7.1.3), we obtain $\int_X \langle \mathrm{d}u, v\rangle \, \mathrm{d}V = -\int_X \langle u, *\mathrm{d} * v\rangle \, \mathrm{d}V$. Thus the formal adjoint of d is $\mathrm{d}^* = - * \mathrm{d} *$. More generally, a linear differential operator P on forms with smooth coefficients has a formal adjoint P^*, a differential operator of the same kind characterized by $\langle P\alpha, \phi\rangle = \langle \alpha, P^*\phi\rangle$ for smooth forms

α, ϕ; in a chart it is obtained by integrating by parts, and the local expressions agree on overlaps by this characterization.

A *distributional k-form* is a current of degree k (Section 7.2). It is paired with smooth k-forms ϕ by $\langle u, \phi\rangle = u(*\overline{\phi})$, which equals $\int_X \langle u, \phi\rangle \, dV$ when u is a locally integrable form. For a scalar distribution f and a smooth function χ the pairing reads $\langle f, \chi\rangle = f(\overline{\chi}\, dV)$, and we say that f has *mean zero* if $\langle f, 1\rangle = 0$. We let P act on distributional forms by $\langle Pu, \phi\rangle = \langle u, P^*\phi\rangle$. For $P = d$ this is the exterior derivative of currents, by the formula for d^*; for $*$, for the exterior product with a smooth form and for its pointwise adjoint, it is the action on the distribution coefficients in local frames. Since $(PQ)^* = Q^*P^*$, and since an operator is determined by its formal adjoint, every identity between such operators that holds on smooth forms also holds on distributional forms.

On functions we use the Laplace–Beltrami operator $L = d^*d$. For the Euclidean metric on $\mathbb{R}^{2n}$ we have $L = -\sum_j \partial^2/\partial x_j^2$, so L *is the negative of the Laplacian* Δ *of Part I*; we use this sign convention throughout the chapter because it makes L a nonnegative operator: $\int_X fLf \, dV = \int_X |df|^2 \, dV$ for smooth real f. On forms we use the Hodge Laplacian $\mathcal{L} = dd^* + d^*d$, and a form or current u is *harmonic* if $\mathcal{L}u = 0$. For smooth u, $\langle \mathcal{L}u, u\rangle = \|du\|^2 + \|d^*u\|^2$, so u is harmonic if and only if $du = 0$ and $d^*u = 0$. On functions $\mathcal{L} = L$, so for the Euclidean metric this notion of harmonic function agrees with that of Section 2.1; on the compact connected manifold X, however, the only solutions of $Lf = 0$ are the constants (Lemma 14.1.3), and the harmonic objects of interest here are forms.

We use the following Sobolev spaces. Choose a finite cover of X by coordinate charts, frames of the bundle of differential forms over them, and smooth functions $\chi_k \geq 0$ with compact supports in the charts and $\sum_k \chi_k = 1$ (Proposition 7.1.3). Using the frames, $\chi_k u$ is a vector-valued function on $\mathbb{R}^{2n}$ for every form u. For an integer $r \geq 0$, let H^r be the completion of the space of smooth forms for the norm whose square is

$$\sum_k \int_{\mathbb{R}^{2n}} (1 + |\xi|^2)^r \, |\widehat{\chi_k u}(\xi)|^2 \, d\xi,$$

and let H^{-r} be the space of distributional forms u for which $\phi \mapsto \langle u, \phi\rangle$ is bounded on smooth forms for the norm of H^r, with the norm of this functional. By Lemma 14.1.1 below, these spaces do not depend on the choices made or on ω. The same notation is used for functions.

Lemma 14.1.1 *Let $r, l \geq 0$ be integers. For a $\mathbb{C}^N$-valued function v on an open set $W \subset \mathbb{R}^{2n}$ whose distributional derivatives of order at most r are square integrable, put $\|v\|^2_{(r),W} = \sum_{|\alpha|\leq r} \|\partial^\alpha v\|^2_{L^2(W)}$, and $\|v\|_{(r)} = \|v\|_{(r),\mathbb{R}^{2n}}$. For a form u of class C^l, let $\|u\|_{C^l}$ be the maximum over k of the suprema on* $\operatorname{Supp} \chi_k$ *of the derivatives of order at most l of the coefficients of u in the kth chart and frame.*

(1) *On smooth forms, the norm of H^r is equivalent to the norm $\left(\sum_k \|\chi_k u\|^2_{(r)}\right)^{1/2}$, and other choices of the charts, frames and cutoff functions give equivalent norms. Multiplication by a smooth function, or by a smooth endomorphism of the bundle*

of forms, is bounded for the norm of H^r, and $\|\phi\|_{H^r} \le C\|\phi\|_{C^r}$ for smooth forms ϕ.

(2) *The norm of H^r dominates a multiple of the L^2 norm, and the continuous extension to H^r of the identity map of smooth forms is an injective map into the space of L^2 forms; we regard H^r as a subspace of the L^2 forms. An L^2 form u belongs to H^r if and only if, for every k, the distributional derivatives of order at most r of $\chi_k u$ are square integrable, and then $\big(\sum_k \|\chi_k u\|^2_{(r)}\big)^{1/2}$ is equivalent to $\|u\|_{H^r}$. In particular, H^0 is the space of all L^2 forms, with an equivalent norm, and the subspace H^r, with its norm up to equivalence, does not depend on the choices of the charts, frames and cutoffs. Moreover $H^{r'} \subset H^r$ for $r \le r'$, and if W is an open set whose closure is a compact subset of a chart with a frame, then the coefficient vector of every $u \in H^r$ on W satisfies $\|u\|_{(r),W} \le C_W\|u\|_{H^r}$.*

(3) *Let $r > l + n$. Every $u \in H^r$ agrees almost everywhere with a form of class C^l, and for this form $\|u\|_{C^l} \le C\|u\|_{H^r}$. Likewise, a compactly supported $\mathbb{C}^N$-valued function v on $\mathbb{R}^{2n}$ with $\|v\|_{(r)} < \infty$ agrees almost everywhere with a function of class C^l whose derivatives of order at most l are bounded by $C\|v\|_{(r)}$. Consequently, a form that lies in every H^r is smooth.*

(4) *For $u \in H^{-r}$ the functional $\phi \mapsto \langle u, \phi\rangle$ extends uniquely to a bounded conjugate-linear functional on H^r, with norm $\|u\|_{H^{-r}}$, and every bounded conjugate-linear functional on H^r arises in this way from a unique $u \in H^{-r}$; thus H^{-r} is the dual space of H^r. Multiplication by smooth functions and endomorphisms is bounded on H^{-r}, and $H^{-r} \subset H^{-r'}$ for $r \le r'$. The space H^{-r}, with its norm up to equivalence, depends neither on the choices of the charts, frames and cutoffs nor on the Kähler form ω that defines the pairing; neither does H^r.*

(5) *Every distributional form belongs to H^{-r} for some r.*

Proof In this proof, the constants C depend on r, l and the data indicated, but not on the forms or functions involved. We first compare the coefficient vectors of a form in two charts, and then prove (1)–(5) in turn.

(a) A change of charts. Let $\kappa\colon U \to \mathbb{R}^{2n}$ and $\kappa'\colon U' \to \mathbb{R}^{2n}$ be charts of X with frames of the bundle of forms over U and U', let $K \subset U$ be compact, and let $W \subset U'$ be an open set whose closure is a compact subset of U'. For a smooth form w that vanishes outside K, let w_κ and $w_{\kappa'}$ be its coefficient vectors in the two charts and frames, the first one extended by zero to $\mathbb{R}^{2n}$. We claim that

$$\|w_{\kappa'}\|_{(r),\kappa'(W)} \le C\|w_\kappa\|_{(r)}, \tag{14.1}$$

with C depending on the charts, frames, K and W. Indeed, on $\kappa'(U \cap U')$ we have $w_{\kappa'}(y) = A(y)\, w_\kappa(\tau(y))$, where $\tau = \kappa \circ \kappa'^{-1}$ and A is the smooth matrix of the change of frames. By the chain rule, the Leibniz rule and induction on $|\alpha|$,

$$\partial^\alpha w_{\kappa'}(y) = \sum_{|\beta|\le|\alpha|} c_{\alpha\beta}(y)\, (\partial^\beta w_\kappa)(\tau(y)), \qquad y \in \kappa'(U \cap U'),$$

where the matrices $c_{\alpha\beta}$ are polynomials in the derivatives of order at most $|\alpha|$ of A and of τ; they are smooth on $\kappa'(U \cap U')$ and therefore bounded on its compact subset

$\kappa'(\overline{W} \cap K)$. Since w vanishes on the open set $X \setminus K$, all derivatives of $w_{\kappa'}$ vanish on $\kappa'(W \setminus K)$, and every point of $\kappa'(W)$ lies in this set or in $\kappa'(W \cap K)$. Hence, for $|\alpha| \le r$,

$$\begin{aligned}\int_{\kappa'(W)} |\partial^\alpha w_{\kappa'}|^2 \,\mathrm{d}y &\le C \sum_{|\beta|\le r} \int_{\kappa'(W\cap K)} |(\partial^\beta w_\kappa)(\tau(y))|^2 \,\mathrm{d}y \\ &\le C \sum_{|\beta|\le r} \int_{\mathbb{R}^{2n}} |\partial^\beta w_\kappa(x)|^2 \,\mathrm{d}x,\end{aligned}$$

by the change of variables $x = \tau(y)$, whose Jacobian determinant $|\det D\tau^{-1}|$ is bounded on the compact set $\kappa(\overline{W} \cap K)$. This proves (14.1). We also use the Leibniz rule in the following form. If M is a smooth matrix-valued function on an open set $V \subset \mathbb{R}^{2n}$ whose derivatives of order at most r are bounded on V, then $\|Mv\|_{(r),V} \le C\|v\|_{(r),V}$ for smooth v on V, with C depending on these bounds. If v vanishes outside a compact set $L \subset V$, it suffices that M be smooth on V, the bounds being taken on L, and then $\|Mv\|_{(r)} \le C\|v\|_{(r)}$ for the extensions by zero.

(b) Proof of (1). For a test function v on $\mathbb{R}^{2n}$ we have $\widehat{\partial^\beta v}(\xi) = (2\pi \mathrm{i}\xi)^\beta \widehat{v}(\xi)$, and $(1 + |\xi|^2)^r = \sum_{|\beta|\le r} c_\beta |\xi^\beta|^2$ with positive multinomial coefficients c_β. Hence Plancherel's theorem (Theorem 6.1.2) gives

$$\int_{\mathbb{R}^{2n}} (1 + |\xi|^2)^r |\widehat{v}(\xi)|^2 \,\mathrm{d}\xi = \sum_{|\beta|\le r} c_\beta (2\pi)^{-2|\beta|} \|\partial^\beta v\|_{L^2}^2, \tag{14.2}$$

and the norm of H^r is equivalent on smooth forms to $\big(\sum_k \|\chi_k u\|_{(r)}^2\big)^{1/2}$. Let charts with domains U'_m, frames over them, and cutoffs χ'_m with $\sum_m \chi'_m = 1$ be other choices, and choose open sets W_m with $\operatorname{Supp} \chi'_m \subset W_m$ and $\overline{W_m}$ compact in U'_m. For a smooth form u we have $\chi'_m u = \sum_k \chi'_m \chi_k u$, and the form $\chi'_m \chi_k u$ vanishes outside the compact set $\operatorname{Supp} \chi_k$ and outside $\operatorname{Supp} \chi'_m \subset W_m$. By (14.1), with $K = \operatorname{Supp} \chi_k$ and $W = W_m$, its norm $\|\cdot\|_{(r)}$ in the mth new chart is at most C times its norm in the kth old chart, and by the Leibniz rule, applied to the factor χ'_m on the compact set $\kappa_k(\operatorname{Supp} \chi_k)$, where κ_k is the kth chart, the latter is at most $C\|\chi_k u\|_{(r)}$. Summing over k and m, we see that the new norm is bounded by a constant times the old one, and by symmetry the two norms are equivalent.

If M is a smooth function or endomorphism, then $\chi_k M u = M \chi_k u$ vanishes outside $\operatorname{Supp} \chi_k$, so the Leibniz rule gives $\|\chi_k M u\|_{(r)} \le C\|\chi_k u\|_{(r)}$. Finally, $\|\chi_k \phi\|_{(r)}$ is bounded, by the Leibniz rule and since $\kappa_k(\operatorname{Supp} \chi_k)$ has finite measure, by a constant times the suprema on $\operatorname{Supp} \chi_k$ of the derivatives of order at most r of the coefficients of ϕ; hence $\|\phi\|_{H^r} \le C\|\phi\|_{C^r}$.

(c) Proof of (2). For a smooth form u we have $|u| \le \sum_k |\chi_k u|$ pointwise, since $\chi_k \ge 0$ and $\sum_k \chi_k = 1$. On the compact sets $\operatorname{Supp} \chi_k$ the pointwise norm of forms is comparable with the Euclidean norm of the coefficient vectors, and $\mathrm{d}V$ with Lebesgue measure in the charts; hence $\|u\|_{L^2} \le C \sum_k \|\chi_k u\|_{(0)} \le C\|u\|_{H^r}$, and conversely $\|\chi_k u\|_{(0)} \le C\|u\|_{L^2}$ because $0 \le \chi_k \le 1$. Therefore the identity map of smooth forms extends to a bounded map j from H^r to the space of L^2 forms. Let

$u \in H^r$ be the limit in H^r of smooth forms u_i. By (1), for every k and $|\alpha| \le r$ the functions $\partial^\alpha(\chi_k u_i)$ form a Cauchy sequence in $L^2(\mathbb{R}^{2n})$, with a limit $g_{k,\alpha}$, while $\chi_k u_i \to \chi_k j(u)$ in L^2. For every test function ϕ on $\mathbb{R}^{2n}$,

$$\int g_{k,\alpha}\phi \,\mathrm{d}x = \lim_i \int \partial^\alpha(\chi_k u_i)\phi \,\mathrm{d}x = \lim_i (-1)^{|\alpha|} \int \chi_k u_i \,\partial^\alpha\phi \,\mathrm{d}x$$
$$= (-1)^{|\alpha|} \int \chi_k j(u)\,\partial^\alpha\phi \,\mathrm{d}x,$$

so $g_{k,\alpha}$ is the distributional derivative $\partial^\alpha(\chi_k j(u))$. Hence these derivatives are square integrable, and $\sum_k \|\chi_k j(u)\|^2_{(r)} = \lim_i \sum_k \|\chi_k u_i\|^2_{(r)}$, which by (1) lies between two positive multiples of $\lim_i \|u_i\|^2_{H^r} = \|u\|^2_{H^r}$. In particular, $j(u) = 0$ implies $u = 0$, so j is injective, and we identify u with $j(u)$. This proves the first assertion, the necessity of the condition and the equivalence of the norms.

Conversely, let u be an L^2 form such that every $\chi_k u$ has square integrable distributional derivatives of order at most r, and let ρ_ε be the mollifier of Section 1.4. Since $\chi_k u$ vanishes outside the compact set $\kappa_k(\operatorname{Supp}\chi_k)$, the functions $v_{k,\varepsilon} = \rho_\varepsilon * (\chi_k u)$ are smooth and, for small ε, vanish outside a fixed compact subset L_k of $\kappa_k(U_k)$, where U_k is the domain of the kth chart; moreover $\partial^\alpha v_{k,\varepsilon} = \rho_\varepsilon * \partial^\alpha(\chi_k u)$ by Proposition 1.4.3, and this converges to $\partial^\alpha(\chi_k u)$ in L^2 as $\varepsilon \to 0$, by Lemma 1.4.1, for $|\alpha| \le r$. Let $w_{k,\varepsilon}$ be the smooth form on X whose coefficient vector in the kth chart and frame is $v_{k,\varepsilon}$, extended by zero outside U_k, and put $u_\varepsilon = \sum_k w_{k,\varepsilon}$. Choose open sets W_m with $\operatorname{Supp}\chi_m \subset W_m$ and $\overline{W_m}$ compact in U_m. By (14.1), with $K = \kappa_k^{-1}(L_k)$ and $W = W_m$, and by the Leibniz rule for the factor χ_m, we have $\|\chi_m(w_{k,\varepsilon} - w_{k,\varepsilon'})\|_{(r)} \le C\|v_{k,\varepsilon} - v_{k,\varepsilon'}\|_{(r)}$. By (1), (u_ε) is therefore a Cauchy sequence in H^r as $\varepsilon \to 0$ along a sequence, and it converges in L^2 to $\sum_k \chi_k u = u$, by the comparison of pointwise norms on the compact sets $\kappa_k^{-1}(L_k) \cup \operatorname{Supp}\chi_k \subset U_k$. Hence $u \in H^r$.

For $r = 0$ the condition holds for every L^2 form, so H^0 is the space of all L^2 forms, and its norm is equivalent to the L^2 norm by the first paragraph. Other choices of the charts, frames and cutoffs give, by (1), equivalent norms on smooth forms, hence the same Cauchy sequences and the same limits in L^2; so they give the same subspace H^r with an equivalent norm. If $r \le r'$, then $\|\phi\|_{H^r} \le \|\phi\|_{H^{r'}}$ for smooth ϕ, so every Cauchy sequence in $H^{r'}$ is a Cauchy sequence in H^r with the same limit in L^2; hence $H^{r'} \subset H^r$.

Finally, let W be open with closure compact in a chart with a frame. For smooth u, we have $u = \sum_m \chi_m u$, and (14.1), applied to $w = \chi_m u$ with $K = \operatorname{Supp}\chi_m$, gives $\|u\|_{(r),W} \le C\sum_m \|\chi_m u\|_{(r)} \le C\|u\|_{H^r}$. If $u \in H^r$ is the limit in H^r of smooth forms u_i, the coefficient vectors of the u_i on W form a Cauchy sequence for $\|\cdot\|_{(r),W}$ and converge in $L^2(W)$ to that of u. Testing against test functions on W as above, we find that their derivatives of order at most r converge in $L^2(W)$ to those of the coefficient vector of u, and the bound passes to the limit.

(d) Proof of (3). Let first v be a smooth $\mathbb{C}^N$-valued function with compact support, and let $|\alpha| \le l$. The inversion formula of Theorem 6.1.2, applied to $\partial^\alpha v$, whose Fourier transform is $(2\pi i\xi)^\alpha \widehat{v}$, and the Cauchy–Schwarz inequality give

$$\sup|\partial^\alpha v| \le \int_{\mathbb{R}^{2n}} |2\pi\xi|^{|\alpha|}|\widehat{v}(\xi)|\,d\xi$$
$$\le (2\pi)^l\Big(\int_{\mathbb{R}^{2n}}(1+|\xi|^2)^{l-r}\,d\xi\Big)^{1/2}\Big(\int_{\mathbb{R}^{2n}}(1+|\xi|^2)^r|\widehat{v}(\xi)|^2\,d\xi\Big)^{1/2},$$

where the first integral on the right is finite, by polar coordinates, because $2(r-l) > 2n$. By (14.2), $\sup|\partial^\alpha v| \le C\|v\|_{(r)}$. If v is only compactly supported with $\|v\|_{(r)} < \infty$, then, as in (c), the mollifications $v_\varepsilon = \rho_\varepsilon * v$ are smooth with compact support and $\|v_\varepsilon - v\|_{(r)} \to 0$ as $\varepsilon \to 0$. Along a sequence $\varepsilon_i \to 0$, the estimate shows that the derivatives of order at most l of the v_{ε_i} converge uniformly; hence the v_{ε_i} converge uniformly to a function of class C^l whose derivatives of order at most l are bounded by $C\|v\|_{(r)}$. This function agrees with v almost everywhere, since $v_{\varepsilon_i} \to v$ in L^2 and a subsequence converges almost everywhere.

Now let u be a smooth form. On $\operatorname{Supp}\chi_k$, the coefficient vector of u in the kth chart and frame is the sum over m of the coefficient vectors of the forms $\chi_m u$. We apply the formula for $\partial^\alpha w_{\kappa'}$ in (a) with κ the mth chart, κ' the kth chart and $K = \operatorname{Supp}\chi_m$. The derivatives of the coefficient vector of $\chi_m u$ in the kth chart vanish outside $\kappa_k(\operatorname{Supp}\chi_m)$, and the coefficients $c_{\alpha\beta}$ are bounded on the compact set $\kappa_k(\operatorname{Supp}\chi_k \cap \operatorname{Supp}\chi_m)$. Hence the derivatives of order at most l of this coefficient vector are bounded on $\kappa_k(\operatorname{Supp}\chi_k)$ by C times the suprema of the derivatives of order at most l of the coefficient vector of $\chi_m u$ in the mth chart, and the Euclidean estimate and (1) give $\|u\|_{C^l} \le C\sum_m \|\chi_m u\|_{(r)} \le C\|u\|_{H^r}$.

If $u \in H^r$ is the limit in H^r of smooth forms u_i, then $\|u_i - u_{i'}\|_{C^l} \le C\|u_i - u_{i'}\|_{H^r}$, so in every chart the coefficients of the u_i and their derivatives of order at most l converge uniformly on $\operatorname{Supp}\chi_k$, and hence on the open sets $\{\chi_k > 0\}$, which cover X. The limit is a form of class C^l with $\|\cdot\|_{C^l} \le C\|u\|_{H^r}$, and it agrees with u almost everywhere, because $u_i \to u$ in L^2. Finally, if u lies in every H^r, then for each l it agrees almost everywhere with a form of class C^l; these continuous forms agree almost everywhere, hence everywhere, so u agrees almost everywhere with a smooth form.

(e) Proof of (4). Let $u \in H^{-r}$. Since smooth forms are dense in H^r, the bounded functional $\phi \mapsto \langle u, \phi\rangle$ on smooth forms extends uniquely to a bounded conjugate-linear functional on H^r, with the same norm. Conversely, let ℓ be a bounded conjugate-linear functional on H^r. Since $*$ is a real bundle isomorphism with smooth coefficients, the formula $T(\alpha) = \ell(\overline{*^{-1}\alpha})$ defines a complex-linear functional T on smooth forms. If α is supported in a compact subset K of a chart, then, by (1) and the chain rule on the compact sets $K \cap \operatorname{Supp}\chi_k$, we have $|T(\alpha)| \le \|\ell\|\,\|*^{-1}\overline{\alpha}\|_{H^r} \le C\|*^{-1}\overline{\alpha}\|_{C^r} \le C_K\sum_J p_{K,r}(\alpha_J)$ in the notation of Section 7.2. Hence the components of T of the various degrees are currents, and T is a distributional form u with $\langle u, \phi\rangle = u(*\overline{\phi}) = \ell(\phi)$ for smooth ϕ. Thus $u \in H^{-r}$ and $\|u\|_{H^{-r}} = \|\ell\|$. It is unique, because $*$ maps the smooth forms onto the smooth forms, so that a distributional form is determined by its pairings with smooth forms.

For a smooth function or endomorphism M, with pointwise adjoint $M^\dagger$, we have $\langle Mu, \phi\rangle = \langle u, M^\dagger\phi\rangle$ for distributional forms u and smooth ϕ, so $|\langle Mu, \phi\rangle| \le \|u\|_{H^{-r}}\|M^\dagger\phi\|_{H^r} \le C\|u\|_{H^{-r}}\|\phi\|_{H^r}$ by (1). The inclusion $H^{-r} \subset H^{-r'}$ for $r \le r'$

follows from $\|\phi\|_{H^r} \le \|\phi\|_{H^{r'}}$, and by (1) other choices of the charts, frames and cutoffs change the norm of H^r on smooth forms, and hence H^{-r}, only up to equivalence.

The definition of H^r, $r \ge 0$, does not involve ω. Let ω' be another Kähler form, with pointwise inner product $\langle\cdot,\cdot\rangle'$ and volume form $\mathrm{d}V'$. At each point, the Hermitian form $\langle a, b\rangle' \,\mathrm{d}V'/\mathrm{d}V$ on forms is $\langle a, Sb\rangle$ for a unique endomorphism S, which is positive definite, hence invertible, and depends smoothly on the point. For smooth u and ϕ we then have $u \wedge *'\overline{\phi} = \langle u, \phi\rangle' \,\mathrm{d}V' = \langle u, S\phi\rangle \,\mathrm{d}V = u \wedge *\overline{S\phi}$, where $*'$ is the Hodge star of ω'; since u is arbitrary, $*'\overline{\phi} = *\overline{S\phi}$. Hence the pairing defined by ω' is $u(*'\overline{\phi}) = \langle u, S\phi\rangle$ for every distributional form u. Since multiplication by S and by S^{-1} is bounded on H^r by (1), the condition defining H^{-r} is the same for both pairings, and the two norms are equivalent.

(f) Proof of (5). Let u be a distributional form. Since $\sum_k \chi_k = 1$, we have $\langle u, \phi\rangle = \sum_k u(\chi_k * \overline{\phi})$ for smooth ϕ. The forms $\chi_k * \overline{\phi}$ are supported in the compact sets $\operatorname{Supp}\chi_k$ of the charts, so the definition of currents (Section 7.2) gives integers m_k and constants C_k with $|u(\chi_k * \overline{\phi})| \le C_k \sum_J p_{\operatorname{Supp}\chi_k, m_k}\big((\chi_k * \overline{\phi})_J\big)$. By the Leibniz rule, and since $*$ and the change from the coordinate frames to the given frames have smooth coefficients, the right side is at most $C\|\phi\|_{C^l}$ with $l = \max_k m_k$. By (3), $\|\phi\|_{C^l} \le C\|\phi\|_{H^r}$ for $r > l + n$, and therefore $u \in H^{-r}$. □

The elliptic regularity used below is a local statement for second-order systems in divergence form; it applies to the Hodge Laplacian and to the scalar operators of Lemma 14.1.4.

Lemma 14.1.2 *Let $U \subset \mathbb{R}^{2n}$ be a bounded open set and let $N \ge 1$. For an open set $W \subset \mathbb{R}^{2n}$ and an integer $k \ge 0$ let $H^k(W)$ be the space of $\mathbb{C}^N$-valued functions on W whose distributional derivatives of order at most k are square integrable on W, with the norm $\|v\|_{(k),W}$ of Lemma 14.1.1. For $v \in H^1(U)$ and $\mathbb{C}^N$-valued $w \in C^\infty_c(U)$ put*

$$\mathcal{A}(v,w) = \int_U \Big(\sum a^{ij}_{IJ}\, \partial_j v_I\, \overline{\partial_i w_J} + \sum b^j_{IJ}\, \partial_j v_I\, \overline{w_J} + \sum c_{IJ}\, v_I\, \overline{w_J} \Big) \mathrm{d}x,$$

where the sums run over $1 \le i, j \le 2n$ and $1 \le I, J \le N$, the coefficients are smooth on a neighborhood of $\overline{U}$, and the principal coefficients satisfy, for some $\lambda > 0$,

$$\sum a^{ij}_{IJ}(x)\, \zeta_{jI}\, \overline{\zeta_{iJ}} \ge \lambda |\zeta|^2 \qquad \text{for all } x \in U \text{ and } \zeta \in \mathbb{C}^{2n\times N}.$$

(A term of the form $\int e^i_{IJ} v_I \overline{\partial_i w_J}$ can be integrated by parts into terms of the last two kinds, so it is also allowed.) Let $k \ge 0$, $F \in H^k(U)$ and $v \in H^1(U)$, and suppose that $\mathcal{A}(v,w) = \int_U \sum_J F_J \overline{w_J}\,\mathrm{d}x$ for all $w \in C^\infty_c(U)$. Then, for every $\chi \in C^\infty_c(U)$, the function χv, extended by zero, belongs to $H^{k+2}(\mathbb{R}^{2n})$, and $\|\chi v\|_{H^{k+2}(\mathbb{R}^{2n})} \le C(\|F\|_{H^k(U)} + \|v\|_{H^1(U)})$, where C depends only on χ, k, λ and bounds for the coefficients and their derivatives.

Proof In this proof H^j stands for $H^j(\mathbb{R}^{2n})$. We prove the case $k = 0$ with difference quotients and then argue by induction on k. Let $0 \le \chi \le 1$, let h_0 be half the

distance from $\operatorname{Supp}\chi$ to ∂U, and for a coordinate direction e_l and $0<|h|<h_0$ put $\delta_h v(x)=(v(x+he_l)-v(x))/h$. For smooth f we have $\delta_h f(x)=\int_0^1 \partial_l f(x+the_l)\,\mathrm{d}t$, so the Cauchy–Schwarz inequality in t and Fubini's theorem bound the L^2 norm of $\delta_h f$ on a set by that of $\partial_l f$ on a slightly larger set; by mollification the same holds for $f\in H^1$. The function $w=-\delta_{-h}(\chi^2\delta_h v)$ lies in H^1 and has compact support in U. Its mollifications are test functions converging to w in H^1 (Lemma 1.4.1), and both sides of the equation are continuous in w for the norm of H^1, so the equation also holds for w.

Discrete integration by parts, $\int f\,\overline{\delta_{-h}g}=-\int(\delta_h f)\,\overline{g}$, turns the principal term into $\int\sum\delta_h(a^{ij}_{IJ}\partial_j v_I)\,\overline{\partial_i(\chi^2\delta_h v_J)}$. Here $\delta_h(a\,\partial_j v)=a(\cdot+he_l)\,\partial_j\delta_h v+(\delta_h a)\,\partial_j v$ and $\partial_i(\chi^2\delta_h v)=\chi^2\partial_i\delta_h v+2\chi\,\partial_i\chi\,\delta_h v$. The ellipticity condition at $x+he_l$ bounds the main term $\int\sum a^{ij}_{IJ}(x+he_l)\chi^2\partial_j\delta_h v_I\overline{\partial_i\delta_h v_J}$ from below by $\lambda\|\chi\nabla\delta_h v\|^2_{L^2}$. The remaining terms, the lower-order terms and the right-hand side are estimated with $|\delta_h a|\le C$, with $\|\delta_h v\|_{L^2(\operatorname{Supp}\chi)}\le\|\partial_l v\|_{L^2(U)}$, and with $\|w\|_{L^2}\le\|\partial_l(\chi^2\delta_h v)\|_{L^2}\le\|\chi\nabla\delta_h v\|_{L^2}+C\|v\|_{H^1(U)}$, where the difference quotients are bounded as above. This gives

$$\lambda\|\chi\nabla\delta_h v\|^2_{L^2}\le C\big(\|F\|_{L^2(U)}+\|v\|_{H^1(U)}\big)\big(\|\chi\nabla\delta_h v\|_{L^2}+\|v\|_{H^1(U)}\big),$$

and hence $\|\chi\nabla\delta_h v\|_{L^2}\le C(\|F\|_{L^2(U)}+\|v\|_{H^1(U)})$, independently of h.

If $\phi\in C^\infty_c$ is supported in the interior of $\{\chi=1\}$, then $-\int\partial_j v\,\partial_l\phi=\lim_{h\to0}\int(\delta_h\partial_j v)\,\phi$ is bounded by this constant times $\|\phi\|_{L^2}$. By the Riesz representation theorem in L^2 (Proposition 11.1.1), applied to the extension of $\phi\mapsto-\int\partial_j v\,\partial_l\phi$ to L^2, the distributional derivatives $\partial_l\partial_j v$ are square integrable there, with the same bound. Since χ may be chosen equal to one near the support of any given cutoff, this proves the lemma for $k=0$.

Suppose the lemma holds for some $k\ge0$, let $F\in H^{k+1}(U)$, and fix χ and an open set $U'\Subset U$ containing $\operatorname{Supp}\chi$. By the lemma for k, $v\in H^{k+2}(U')$, with norm at most $C(\|F\|_{H^k(U)}+\|v\|_{H^1(U)})$. Fix m and test the equation with $-\partial_m w$, $w\in C^\infty_c(U')$. Integrating by parts in x_m, we find that $u=\partial_m v\in H^1(U')$ satisfies $\mathcal{A}(u,w)=\int\sum_J F'_J\overline{w_J}\,\mathrm{d}x$ for all $w\in C^\infty_c(U')$, where, with the indices I,J suppressed,

$$F'=\partial_m F+\sum_{i,j}\partial_i\big((\partial_m a^{ij})\partial_j v\big)-\sum_j(\partial_m b^j)\partial_j v-(\partial_m c)v.$$

Since $v\in H^{k+2}(U')$, we have $F'\in H^k(U')$ with $\|F'\|_{H^k(U')}\le C(\|F\|_{H^{k+1}(U)}+\|v\|_{H^{k+2}(U')})$. The lemma for k on U' gives $\chi\,\partial_m v\in H^{k+2}$ for every m, hence $\chi v\in H^{k+3}$ with the asserted bound. □

Lemma 14.1.3 *On X the following hold.*

(1) *Every distributional form u with $\mathcal{L}u=0$ is smooth, and the space of harmonic forms is finite dimensional.*

(2) *For every scalar distribution a of mean zero there is a unique distribution f of mean zero with $Lf = a$. If a is smooth, then f is smooth, and if a is real, then f is real.*
(3) *For every constant $c > 0$, the operator $\mathcal{L} + c$ is a bijection of the space of distributional forms onto itself, and it maps H^{r+2} isomorphically onto H^r for every integer r.*
(4) *There is a constant C such that $\|v\|_{L^2} \le C\|\mathrm{d}v\|_{L^2}$ for every scalar function $v \in H^1$ of mean zero.*

Proof Throughout, we use the properties of the spaces H^r proved in Lemma 14.1.1. We show that the inclusion $H^1 \subset L^2$ is compact, prove Gårding's inequality, solve the equation $(\mathcal{L} + c)v = b$ weakly with the Riesz representation theorem, and obtain the regularity of the solution from Lemma 14.1.2. This gives (3); we then prove (1), (4) and (2) in turn.

Step 1: the inclusion $H^1 \subset L^2$ is compact. Let (u_j) be bounded in H^1, fix k, and put $v_j = \chi_k u_j$, a coefficient vector in the kth chart that vanishes outside a fixed compact set S. For $R > 0$, the part of $\widehat{v_j}$ on $|\xi| > R$ has L^2 norm at most $R^{-1}\|v_j\|_{H^1} \le C/R$. The inverse Fourier transforms $w_{j,R}$ of the parts on $|\xi| \le R$ are uniformly bounded with uniformly bounded derivatives, by the Cauchy–Schwarz inequality on the ball $|\xi| \le R$, so by the Arzelà–Ascoli theorem (Proposition 1.6.4) a subsequence converges uniformly on S. Since v_j vanishes outside S,

$$\|v_j - v_i\|_{L^2} \le \|w_{j,R} - w_{i,R}\|_{L^2(S)} + 2CR^{-1}.$$

A diagonal subsequence over $R = 1, 2, \ldots$ is therefore Cauchy in L^2. Repeating this for the finitely many charts and using $u_j = \sum_k \chi_k u_j$ proves compactness.

Step 2: Gårding's inequality. We first compute $\mathcal{L}$ in a chart with real coordinates $x_1, \ldots, x_{2n}$ and the coordinate frame $\mathrm{d}x_I$ of the forms. Let (g^{ij}) be the inverse of the matrix $(g(\partial_i, \partial_j))$, so that g^{ij} is the inner product of $\mathrm{d}x_i$ and $\mathrm{d}x_j$, write the pointwise inner product of forms as $\sum_{I,J} G_{IJ} u_I \overline{v_J}$ and the volume form as $\rho\,\mathrm{d}x$, and let ϵ_i be the exterior multiplication by $\mathrm{d}x_i$ and $\epsilon_i^\dagger$ its pointwise adjoint.

If $e_1, \ldots, e_{2n}$ is an orthonormal basis of the cotangent space at a point, then exterior multiplication $\epsilon(e_a)$ by e_a and its adjoint $\epsilon(e_a)^\dagger$, which maps $e_a \wedge e_K$ to e_K for $a \notin K$ and kills the basis forms e_J with $a \notin J$, satisfy $\epsilon(e_a)\epsilon(e_b)^\dagger + \epsilon(e_b)^\dagger\epsilon(e_a) = \delta_{ab}$. Indeed, for $a = b$, exactly one of the two terms maps a given basis form e_J to itself, namely the first if $a \in J$ and the second if $a \notin J$, and the other term kills it. For $a \ne b$, both terms kill e_J unless $b \in J$ and $a \notin J$; in that case, writing $e_J = \pm e_b \wedge e_K$, they map $e_b \wedge e_K$ to $e_a \wedge e_K$ and to $\epsilon(e_b)^\dagger(e_a \wedge e_b \wedge e_K) = -e_a \wedge e_K$, respectively. Writing $\mathrm{d}x_i = \sum_a M_{ia} e_a$, so that $\sum_a M_{ia} M_{ja} = g^{ij}$, we obtain by bilinearity

$$\epsilon_i \epsilon_j^\dagger + \epsilon_j^\dagger \epsilon_i = g^{ij}. \tag{14.3}$$

In the coordinate frame $\mathrm{d}u = \sum_i \epsilon_i \partial_i u$, where ∂_i acts on the coefficients. Integrating by parts in $\int \sum_{I,J} G_{IJ}(\epsilon_i \partial_i u)_I \overline{v_J}\,\rho\,\mathrm{d}x$, we find $\mathrm{d}^* v = -\sum_i \epsilon_i^\dagger \partial_i v + B_0 v$, where B_0 is a matrix whose entries are linear combinations, with smooth coefficients, of first derivatives of the G_{IJ} and of ρ. Since the ϵ_i have constant coefficients in the

coordinate frame, (14.3) gives

$$\mathcal{L} = \mathrm{d}\mathrm{d}^* + \mathrm{d}^*\mathrm{d} = -\sum_{i,j} g^{ij}\partial_i\partial_j + B_1$$

on each coefficient, where B_1 is a differential operator of order at most one with smooth coefficients. On functions d^* vanishes, so that

$$Lf = \mathrm{d}^*\mathrm{d}f = -\sum_{i,j} g^{ij}\partial_i\partial_j f + B_0\Big(\sum_i \partial_i f\,\mathrm{d}x_i\Big); \tag{14.4}$$

at a point where the first derivatives of the metric coefficients vanish, $Lf = -\sum_{i,j} g^{ij}\partial_i\partial_j f$ there.

Now let u be smooth with compact support in the chart. Since $\langle \mathcal{L}u, u\rangle = \|\mathrm{d}u\|^2 + \|\mathrm{d}^*u\|^2$, integrating the second-order term by parts once gives

$$\|\mathrm{d}u\|^2 + \|\mathrm{d}^*u\|^2 = \int \sum_{i,j,I,J} g^{ij} G_{IJ}\,\partial_j u_I\,\overline{\partial_i u_J}\,\rho\,\mathrm{d}x + E(u),$$

where $|E(u)| \le C\|u\|_{L^2}\|u\|_{H^1}$, with C depending on a compact set containing the support of u. For $\zeta = (\zeta_{jI}) \in \mathbb{C}^{2n\times N}$, where N is the number of multi-indices I, write $\zeta_j = (\zeta_{jI})_I$ and $G(\zeta_j,\zeta_i) = \sum_{I,J} G_{IJ}\zeta_{jI}\overline{\zeta_{iJ}}$. Diagonalizing the positive definite real symmetric matrix (g^{ij}) by an orthogonal matrix O, we get $\sum_{i,j} g^{ij}G(\zeta_j,\zeta_i) = \sum_a d_a G(\eta_a,\eta_a)$ with eigenvalues $d_a > 0$ and $\eta_a = \sum_j O_{aj}\zeta_j$; hence this Hermitian form is positive definite, uniformly on compact subsets of the chart, and the integral is at least $\lambda\|\nabla u\|^2_{L^2}$ for some $\lambda > 0$. Applying this to the $\chi_k u$, whose images under d and d^* differ from $\chi_k\,\mathrm{d}u$ and $\chi_k\,\mathrm{d}^*u$ by terms bounded by $C|u|$, summing over k, using $\sum_k \chi_k^2 \le 1$ and absorbing the error terms by $ab \le \tau a^2 + b^2/(4\tau)$ with small $\tau > 0$, we obtain

$$\|u\|^2_{H^1} \le C\big(\|\mathrm{d}u\|^2_{L^2} + \|\mathrm{d}^*u\|^2_{L^2} + \|u\|^2_{L^2}\big) \tag{14.5}$$

for smooth forms u on X, and hence for all $u \in H^1$, since both sides are continuous for the norm of H^1.

Step 3: weak solutions. Fix $c > 0$ and put $P = \mathcal{L} + c$, with the quadratic form $Q(v,w) = \langle \mathrm{d}v, \mathrm{d}w\rangle + \langle \mathrm{d}^*v, \mathrm{d}^*w\rangle + c\langle v, w\rangle$. Since d and d^* are first-order operators with smooth coefficients, $|Q(v,w)| \le C\|v\|_{H^1}\|w\|_{H^1}$ for smooth forms, so Q extends to a bounded sesquilinear form on H^1, and by (14.5) the norm $Q(u,u)^{1/2}$ is equivalent to the norm of H^1. For $b \in H^{-1}$, the map $w \mapsto \langle b, w\rangle$ is a bounded conjugate-linear functional on H^1, so the Riesz representation theorem (Proposition 11.1.1) gives a unique $v \in H^1$ with $Q(v,w) = \langle b, w\rangle$ for all $w \in H^1$, and $\|v\|_{H^1} \le C\|b\|_{H^{-1}}$. For smooth ϕ we have $Q(v,\phi) = \langle v, P\phi\rangle$, first for smooth v and then for all $v \in H^1$, since both sides are continuous in v for the norm of H^1; hence $Pv = b$ as distributional forms. We write $v = P^{-1}b$.

Step 4: interior regularity. We apply Lemma 14.1.2 to P. In a chart, with the notation of Step 2, one integration by parts in the expression of $\mathcal{L}$ shows

that $\langle Pv, w\rangle = \mathcal{A}(v, w)$ for smooth v and w, the latter with compact support in the chart, where $\mathcal{A}$ is a form as in Lemma 14.1.2 with principal coefficients $a_{IJ}^{ij} = g^{ij} G_{IJ} \rho$ and smooth lower-order coefficients. These principal coefficients satisfy the ellipticity condition on compact subsets of the chart, as shown in Step 2. Both sides are continuous in $v \in H^1$, so the equation $Q(v, w) = \langle b, w\rangle$ of Step 3 reads $\mathcal{A}(v, w) = \int \sum_J F_J \overline{w_J}\, \mathrm{d}x$ with $F_J = \rho \sum_I G_{IJ} b_I$. Let $b \in H^r$, $r \geq 0$, and $v = P^{-1} b$. Applying Lemma 14.1.2 with $\chi = \chi_k$ on relatively compact open subsets of the finitely many charts, and using Lemma 14.1.1(2), we obtain $\|v\|_{H^{r+2}} \leq C(\|b\|_{H^r} + \|v\|_{H^1})$, and since $\|v\|_{H^1} \leq C\|b\|_{H^{-1}} \leq C\|b\|_{L^2}$,

$$\|P^{-1} b\|_{H^{r+2}} \leq C_r \|b\|_{H^r}, \qquad r \geq 0.$$

In particular, if b is smooth, then $P^{-1} b$ lies in every H^r and is therefore smooth by Lemma 14.1.1(3), and $PP^{-1} b = b$; and if u is smooth, then $P^{-1} Pu = u$, because $Q(u, w) = \langle Pu, w\rangle$ for $w \in H^1$ and the solution in Step 3 is unique. For smooth b and ϕ, with $\psi = P^{-1}\phi$, we get $\langle P^{-1} b, \phi\rangle = \langle P^{-1} b, P\psi\rangle = Q(P^{-1} b, \psi) = \langle b, P^{-1}\phi\rangle$.

We therefore extend P^{-1} to all distributional forms by $\langle P^{-1} u, \phi\rangle = \langle u, P^{-1}\phi\rangle$ for smooth ϕ. If $u \in H^{-s}$ with $s \geq 2$, then $|\langle u, P^{-1}\phi\rangle| \leq \|u\|_{H^{-s}} \|P^{-1}\phi\|_{H^s} \leq C\|u\|_{H^{-s}} \|\phi\|_{H^{s-2}}$, so $P^{-1} u \in H^{-s+2}$ and P^{-1} is bounded from H^r to H^{r+2} for $r \leq -2$; for $u \in H^{-1}$ the new definition agrees with that of Step 3, by the same computation with $b = u$. Since P is formally self-adjoint, $\langle PP^{-1} u, \phi\rangle = \langle u, P^{-1} P\phi\rangle = \langle u, \phi\rangle$ and $\langle P^{-1} Pu, \phi\rangle = \langle u, PP^{-1}\phi\rangle = \langle u, \phi\rangle$ for every distributional form u, so P is a bijection of the space of distributional forms onto itself with inverse P^{-1}. Conversely, P is bounded from H^{r+2} to H^r for every integer r: for $r \geq 0$ because it is a differential operator of order two with smooth coefficients, for $r = -1$ because $|\langle Pv, \phi\rangle| = |Q(v, \phi)| \leq C\|v\|_{H^1} \|\phi\|_{H^1}$, and for $r \leq -2$ by transposition, since $|\langle Pu, \phi\rangle| = |\langle u, P\phi\rangle| \leq C\|u\|_{H^{r+2}} \|\phi\|_{H^{-r}}$. Hence P maps H^{r+2} isomorphically onto H^r. As $c > 0$ was arbitrary, this proves (3).

Step 5: harmonic forms. If $\mathcal{L} u = 0$ distributionally, then $Pu = cu$, so $u = cP^{-1} u$. By Lemma 14.1.1(5), $u \in H^{-r}$ for some r; iterating this identity raises the Sobolev order by two at each step, so u lies in every H^r and is smooth by Lemma 14.1.1(3). For smooth harmonic u, (14.5) gives $\|u\|_{H^1} \leq C\|u\|_{L^2}$. An L^2 limit of harmonic forms is distributionally harmonic, hence smooth and harmonic, so the closed unit ball of the space of harmonic forms in L^2 is bounded in H^1 and therefore compact by Step 1. A normed space with compact closed unit ball is finite dimensional: an infinite orthonormal sequence would have no convergent subsequence. This proves (1).

Step 6: the scalar equation. We first prove (4). If it fails, there are scalar functions $v_j \in H^1$ of mean zero with $\|v_j\|_{L^2} = 1$ and $\mathrm{d}v_j \to 0$ in L^2. Since d^* vanishes on functions, (14.5) shows that (v_j) is bounded in H^1, so by Step 1 a subsequence converges in L^2 to some v of norm one and mean zero, and $\mathrm{d}v = \lim_j \mathrm{d}v_j = 0$ in the sense of distributions. In every chart v is locally constant by Corollary 1.4.4, so v is constant on the connected manifold X, which contradicts $\|v\|_{L^2} = 1$ and mean zero. This proves (4).

Consequently $\|\mathrm{d}v\|_{L^2}$ is an equivalent Hilbert norm on the closed subspace of H^1 of functions of mean zero. Let a be smooth of mean zero. The Riesz representation theorem (Proposition 11.1.1) gives f in this subspace with $\langle \mathrm{d}f, \mathrm{d}w\rangle = \langle a, w\rangle$ for all w of mean zero, hence for all $w \in H^1$, since both sides vanish for constant w. Testing with $w = f$ gives $\|\mathrm{d}f\|^2 = \langle a, f\rangle \le \|a\|\,\|f\| \le C\|a\|\,\|\mathrm{d}f\|$, hence $\|f\|_{H^1} \le C\|a\|_{L^2}$. With $c = 1$ the function f satisfies $Q(f, w) = \langle a + f, w\rangle$ for all $w \in H^1$, so $f = P^{-1}(a + f)$ by the uniqueness in Step 3; if $f \in H^r$, $r \ge 0$, then Step 4 gives $f \in H^{r+2}$ with $\|f\|_{H^{r+2}} \le C(\|a\|_{H^r} + \|f\|_{H^r})$. By induction f lies in every H^r, so it is smooth, and $\|f\|_{H^{r+2}} \le C_r\|a\|_{H^r}$ for every $r \ge 0$.

Let $\Pi a = \langle a, 1\rangle / \int_X \mathrm{d}V$ be the average of a, and let $\mathcal{G}$ be the operator $a \mapsto f$ applied to $a - \Pi a$, for smooth a. Then $\|\mathcal{G}a\|_{H^{r+2}} \le C_r\|a\|_{H^r}$ for $r \ge 0$, and $L\mathcal{G} = 1 - \Pi$ by construction. If a is smooth, then La has mean zero, because $\langle La, 1\rangle = \langle a, L1\rangle = 0$, and both $\mathcal{G}La$ and $a - \Pi a$ are smooth solutions f of mean zero of $Lf = La$. The difference f_0 of two such solutions satisfies $\int_X |\mathrm{d}f_0|^2\,\mathrm{d}V = \langle Lf_0, f_0\rangle = 0$, so it is constant, and it vanishes since it has mean zero; hence $\mathcal{G}L = 1 - \Pi$. For smooth a and b, since $\mathcal{G}a$ has mean zero and L is formally self-adjoint,

$$\langle \mathcal{G}a, b\rangle = \langle \mathcal{G}a, L\mathcal{G}b + \Pi b\rangle = \langle L\mathcal{G}a, \mathcal{G}b\rangle = \langle a - \Pi a, \mathcal{G}b\rangle = \langle a, \mathcal{G}b\rangle,$$

so $\mathcal{G}$ is formally self-adjoint, and so is Π. Extend $\mathcal{G}$ to distributions by transposition, $\langle \mathcal{G}a, \chi\rangle = \langle a, \mathcal{G}\chi\rangle$. This defines a distribution: if $a \in H^{-s}$ with $s \ge 2$, then $|\langle a, \mathcal{G}\chi\rangle| \le C\|a\|_{H^{-s}}\|\chi\|_{H^{s-2}}$. The two identities $L\mathcal{G} = \mathcal{G}L = 1 - \Pi$ remain valid, by transposition of the same identities for smooth functions. Hence $f = \mathcal{G}a$ solves $Lf = a$ for every distribution a of mean zero, and $\langle \mathcal{G}a, 1\rangle = \langle a, \mathcal{G}1\rangle = 0$. If $Lf = 0$ for a distribution f, then $\mathcal{L}f = 0$, so f is smooth by Step 5, and $\int_X |\mathrm{d}f|^2\,\mathrm{d}V = \langle Lf, f\rangle = 0$ shows that f is constant; this proves uniqueness. If a is real, then $\overline{f}$ is another solution of mean zero, since L has real coefficients, so f is real. □

The next lemma solves a scalar equation in which the Laplacian is replaced by a mixed trace with respect to several Kähler forms; it is used for the mixed Hodge index theorem (Theorem 14.3.3). We use the real operator d^{c} of (7.6), for which $\mathrm{dd}^{\mathrm{c}} = \mathrm{dd}^{\mathrm{c}}$.

Lemma 14.1.4 *Let $\omega_1, \dots, \omega_{n-1}$ be Kähler forms on X and put $A = \omega_1 \wedge \cdots \wedge \omega_{n-1}$, with $A = 1$ when $n = 1$. For every smooth real (n, n)-form η with $\int_X \eta = 0$, there is a unique smooth real function f of mean zero such that*

$$\mathrm{dd}^{\mathrm{c}} f \wedge A = \eta.$$

Proof For smooth real functions f, g put $B(f, g) = \int_X \mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}}g \wedge A$. Only the $(1, 1)$-part of $\mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}}g$ contributes, and this part equals $\frac{\mathrm{i}}{4\pi}(\partial f \wedge \bar\partial g + \partial g \wedge \bar\partial f)$, which is symmetric in f and g; in particular $\mathrm{d}f \wedge \mathrm{d}^{\mathrm{c}}f$ has $(1, 1)$-part $\frac{\mathrm{i}}{2\pi}\partial f \wedge \bar\partial f$. Thus B is symmetric. Since A is closed, Stokes' formula (Proposition 7.1.3) applied to $\mathrm{d}(g\,\mathrm{d}^{\mathrm{c}}f \wedge A)$ gives

$$B(f, g) = B(g, f) = -\int_X g\,\mathrm{dd}^{\mathrm{c}} f \wedge A.$$

On the compact manifold there are constants $0 < c \le C$ with $c\,\omega \le \omega_j \le C\omega$ for all j, by Lemma 13.1.1. For $(1,1)$-forms γ_j, γ_j' we have the telescoping identity

$$\gamma_1\wedge\cdots\wedge\gamma_{n-1} - \gamma_1'\wedge\cdots\wedge\gamma_{n-1}' = \sum_{m=1}^{n-1} \gamma_1\wedge\cdots\wedge\gamma_{m-1}\wedge(\gamma_m - \gamma_m')\wedge\gamma_{m+1}'\wedge\cdots\wedge\gamma_{n-1}'. \tag{14.6}$$

Applying it with $\gamma_j = \omega_j$, $\gamma_j' = c\,\omega$ and with $\gamma_j = C\omega$, $\gamma_j' = \omega_j$, we see that the forms $A - c^{n-1}\omega^{n-1}$ and $C^{n-1}\omega^{n-1} - A$ are sums of products of semipositive $(1,1)$-forms. Such products, and their products with the semipositive form $\mathrm{i}\partial f\wedge\bar\partial f$, are strongly positive (Proposition 7.2.1). Hence $B(f,f)$ lies between c^{n-1} and C^{n-1} times $\int_X \frac{\mathrm{i}}{2\pi}\partial f\wedge\bar\partial f\wedge\omega^{n-1}$. At a point, if $\theta_1,\dots,\theta_n$ is a unitary coframe, that is, $\omega = \mathrm{i}\sum_j \theta_j\wedge\bar\theta_j$, and $\partial f = \sum_j a_j\theta_j$, then $\mathrm{i}\partial f\wedge\bar\partial f\wedge\omega^{n-1}/(n-1)! = \sum_j |a_j|^2\omega^n/n!$ and $|\mathrm{d}f|^2 = 2\sum_j |a_j|^2$; so the last integral is $\frac{(n-1)!}{4\pi}\|\mathrm{d}f\|^2_{L^2}$. By Lemma 14.1.3(4), B is therefore an equivalent inner product on the closed subspace of real functions of mean zero in H^1.

The sesquilinear extension $B_{\mathbb{C}}(f_1 + \mathrm{i}f_2, g_1 + \mathrm{i}g_2) = B(f_1,g_1) + B(f_2,g_2) + \mathrm{i}\big(B(f_2,g_1) - B(f_1,g_2)\big)$ to complex-valued functions is, by the symmetry of B, a Hermitian form with $B_{\mathbb{C}}(f,f) = B(f_1,f_1) + B(f_2,f_2)$, hence an equivalent inner product on the complex Hilbert space of functions of mean zero in H^1. The Riesz representation theorem (Proposition 11.1.1), applied to the bounded conjugate-linear functional $g \mapsto -\int_X \overline{g}\,\eta$, gives F in this space with $B_{\mathbb{C}}(F,g) = -\int_X \overline{g}\,\eta$ for all g. For real g the right side is real and $B_{\mathbb{C}}(F,g) = B(\operatorname{Re}F, g) + \mathrm{i}B(\operatorname{Im}F, g)$, so $f = \operatorname{Re}F$ is a real function of mean zero in H^1 with $B(f,g) = -\int_X g\eta$ for every real mean-zero $g \in H^1$. Since $\int_X \eta = 0$ and $B(f,1) = 0$, the identity holds for all $g \in H^1$; this is the equation $\mathrm{dd^c}f\wedge A = \eta$ in the weak sense.

It remains to prove that f is smooth. In a chart, with $\eta = \tilde\eta\,\mathrm{d}x$, the weak equation reads $\mathcal{A}(f,w) = -\int\tilde\eta\,\overline{w}\,\mathrm{d}x$ for $w \in C_c^\infty$, where $\mathcal{A}(f,w) = B(f,\overline{w}) = \int\sum_{i,j} a^{ij}\partial_j f\,\overline{\partial_i w}\,\mathrm{d}x$ and the real symmetric matrix (a^{ij}) is the density of the quadratic form $\mathrm{d}f\wedge\mathrm{d^c}f\wedge A$. By the bounds just proved, (a^{ij}) is uniformly positive definite on compact subsets of the chart. This is a form of the kind considered in Lemma 14.1.2, with $N = 1$, and since $\tilde\eta$ is smooth, that lemma, applied with the cutoffs χ_k on relatively compact open subsets of the charts, shows that every $\chi_k f$ has square integrable distributional derivatives of all orders. Hence f lies in every H^r by Lemma 14.1.1(2), and it is smooth by Lemma 14.1.1(3). If f and f' are two solutions, then $B(f-f', f-f') = 0$, so $f - f'$ is constant on the connected manifold X, and it vanishes since it has mean zero. □

Let Λ be the pointwise adjoint of the Lefschetz operator $u \mapsto \omega\wedge u$; thus $\Lambda\omega = |\omega|^2 = n$. Let $\theta_1,\dots,\theta_n$ be a unitary coframe of $(1,0)$-forms at a point, so that $\omega = \mathrm{i}\sum_j \theta_j\wedge\bar\theta_j$ there. A real $(1,1)$-form $T = \mathrm{i}\sum_{j,k} T_{j\bar k}\theta_j\wedge\bar\theta_k$ corresponds to the Hermitian matrix $(T_{j\bar k})$, and $\Lambda T = \sum_j T_{j\bar j}$ is its trace. Indeed, the covectors $\theta_j, \bar\theta_k$ form an orthonormal basis of the complexified cotangent space: in linear coordinates with $\mathrm{d}z_j = \sqrt{2}\,\theta_j$ at the point we have $\omega = \frac{\mathrm{i}}{2}\sum_j \mathrm{d}z_j\wedge\mathrm{d}\bar z_j$ there, so g is Euclidean in the real coordinates at the point and $\langle \mathrm{d}x_j + \mathrm{i}\,\mathrm{d}y_j, \mathrm{d}x_k \pm \mathrm{i}\,\mathrm{d}y_k\rangle = (1\pm1)\delta_{jk}$.

Hence the forms $\mathrm{i}\theta_j \wedge \bar\theta_k$ are orthonormal, and $\Lambda T = \langle T, \omega\rangle$ for 2-forms T. Since $\omega^{n-1}/(n-1)!$ is the sum of the products of $n-1$ distinct forms $\mathrm{i}\theta_l \wedge \bar\theta_l$, we have $\mathrm{i}\theta_j \wedge \bar\theta_k \wedge \omega^{n-1}/(n-1)! = \delta_{jk}\,\omega^n/n!$; equivalently,

$$T \wedge \frac{\omega^{n-1}}{(n-1)!} = (\Lambda T)\frac{\omega^n}{n!}.$$

The last formula defines ΛT for a current T, as a distribution; it is the action of Λ on currents described at the beginning of this section.

To compute the trace of $\mathrm{dd^c} f$, we use holomorphic coordinates adapted to a Kähler form at a point.

Lemma 14.1.5 *Let σ be a Kähler form on a complex manifold M of dimension $n \geq 1$, and let $x \in M$. There are holomorphic coordinates w centered at x in which $\sigma = \frac{\mathrm{i}}{2}\sum_{k,l} \tilde g_{k\bar l}\,\mathrm{d}w_k \wedge \mathrm{d}\bar w_l$ with $\tilde g_{k\bar l}(0) = \delta_{kl}$ and with all first derivatives of the functions $\tilde g_{k\bar l}$ vanishing at 0.*

Proof Choose holomorphic coordinates z centered at x. Since the matrix of σ at x is positive definite Hermitian, after a linear change of coordinates $\sigma = \frac{\mathrm{i}}{2}\sum_{k,l} g_{k\bar l}\,\mathrm{d}z_k \wedge \mathrm{d}\bar z_l$ with $g_{k\bar l}(0) = \delta_{kl}$; moreover $\overline{g_{l\bar k}} = g_{k\bar l}$, since σ is real. The $(2,1)$-part of $\mathrm{d}\sigma = 0$ is $\partial\sigma = \frac{\mathrm{i}}{2}\sum_{k,l,m} \partial_m g_{k\bar l}\,\mathrm{d}z_m \wedge \mathrm{d}z_k \wedge \mathrm{d}\bar z_l = 0$, which gives $\partial_m g_{k\bar l} = \partial_k g_{m\bar l}$.

By the holomorphic inverse function theorem (Proposition 7.1.2(1)), the equations $z_l = w_l - \frac{1}{2}\sum_{k,m} \partial_m g_{k\bar l}(0) w_k w_m$ define holomorphic coordinates w near 0, since the Jacobian matrix of this map at 0 is the identity. Substituting $\mathrm{d}z_l = \mathrm{d}w_l - \sum_{k,m} \partial_m g_{k\bar l}(0) w_m\,\mathrm{d}w_k$, which uses the symmetry in k and m, and $g_{k\bar l}(z) = \delta_{kl} + \sum_m \partial_m g_{k\bar l}(0) w_m + \sum_m \partial_{\bar m} g_{k\bar l}(0)\bar w_m + O(|w|^2)$, we find that the coefficient of $\frac{\mathrm{i}}{2}\mathrm{d}w_k \wedge \mathrm{d}\bar w_l$ in σ is δ_{kl} plus terms of order two in w: the terms $\sum_m \partial_m g_{k\bar l}(0) w_m$ of $g_{k\bar l}$ cancel against the correction in $\mathrm{d}z_k$, and their conjugates $\sum_m \partial_{\bar m} g_{k\bar l}(0)\bar w_m = \sum_m \overline{\partial_m g_{l\bar k}(0)}\,\bar w_m$ cancel against the correction in $\mathrm{d}\bar z_l$. Hence the first derivatives of the new coefficients vanish at 0. □

Lemma 14.1.6 *For a scalar distribution f and a real $(1,1)$-current T on X,*

$$\Lambda \mathrm{dd^c} f = -\frac{Lf}{4\pi}, \qquad *T = (\Lambda T)\frac{\omega^{n-1}}{(n-1)!} - T \wedge \frac{\omega^{n-2}}{(n-2)!} \quad (n \geq 2). \tag{14.7}$$

If $\mathrm{d}T = 0$ and ΛT is constant, then $\mathcal{L}T = 0$.

Proof Both sides of each identity are differential operators, or bundle maps, with smooth coefficients, acting on distributions and currents as explained at the beginning of this section. Hence it suffices to verify the identities for smooth f and T, at an arbitrary point. By Lemma 14.1.5, there are holomorphic coordinates centered at the point in which $\omega = \frac{\mathrm{i}}{2}\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar z_j$ at the point and the first derivatives of the coefficients of ω, hence of the metric coefficients, vanish there. Thus (14.4) gives $Lf = -\sum_j (\partial^2_{x_j} + \partial^2_{y_j}) f = -4\sum_j \partial_j \partial_{\bar j} f$ there, while $\theta_j = \mathrm{d}z_j/\sqrt{2}$ is a unitary coframe and

$$\mathrm{dd^c} f = \frac{\mathrm{i}}{2\pi} \sum_{j,k} f_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k = \mathrm{i} \sum_{j,k} \frac{f_{j\bar{k}}}{\pi}\, \theta_j \wedge \bar{\theta}_k .$$

Hence $\Lambda \mathrm{dd^c} f = \pi^{-1} \sum_j f_{j\bar{j}} = -Lf/(4\pi)$. For the second identity, diagonalize the Hermitian matrix of T by a unitary change of coframe; by linearity it suffices to treat $T = \mathrm{i}\theta_1 \wedge \bar{\theta}_1$. Write $\sqrt{2}\theta_j = e_{2j-1} + \mathrm{i} e_{2j}$ with real covectors e_a; they form an orthonormal basis, positively oriented since $\omega^n/n! = e_1 \wedge \cdots \wedge e_{2n}$, and $\omega_j = \mathrm{i}\theta_j \wedge \bar{\theta}_j = e_{2j-1} \wedge e_{2j}$. By the explicit formula for $*$, the star of ω_1 is $\omega_2 \wedge \cdots \wedge \omega_n$. On the other hand $\omega^{n-1}/(n-1)!$ is the sum of all products of $n-1$ distinct ω_j, while $\omega_1 \wedge \omega^{n-2}/(n-2)!$ is the sum of those products that contain ω_1; their difference is exactly $\omega_2 \wedge \cdots \wedge \omega_n$, and $\Lambda\omega_1 = 1$.

If $\mathrm{d}T = 0$ and ΛT is constant, the second identity and $\mathrm{d}\omega = 0$ give $\mathrm{d} * T = 0$, hence $\mathrm{d}^* T = - * \mathrm{d} * T = 0$, and $\mathcal{L}T = \mathrm{dd}^* T + \mathrm{d}^* \mathrm{d} T = 0$. In dimension one a $(1,1)$-current is $T = (\Lambda T)\omega$, so $*T = \Lambda T$ is constant and the same conclusion holds. □

Since ω is an arbitrary Kähler form on X, Lemmas 14.1.3 and 14.1.6 hold for every Kähler form ω' on X in place of ω, with the same Sobolev spaces H^r; we write $L_{\omega'}$ and $\Lambda_{\omega'}$ for the corresponding operators. We record a consequence for the complex Laplacian, which is used in Chapter 16. Write ω' in local holomorphic coordinates as $\omega' = \frac{\mathrm{i}}{2\pi} \sum_{j,k} g_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, let $(g^{j\bar{k}})$ be the inverse matrix, normalized by $\sum_k g^{j\bar{k}} g_{l\bar{k}} = \delta_{jl}$, and put $D_g f = \sum_{j,k} g^{j\bar{k}} \partial_j \partial_{\bar{k}} f$. Since $\mathrm{dd^c} f = \frac{\mathrm{i}}{2\pi} \sum_{j,k} f_{j\bar{k}}\, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, the function $D_g f$ is the trace of $\mathrm{dd^c} f$ relative to ω', and (14.7) for ω' gives

$$D_g f = \Lambda_{\omega'} \mathrm{dd^c} f = -\frac{L_{\omega'} f}{4\pi}. \tag{14.8}$$

Consequently, by Lemma 14.1.3 for ω', for every constant $\beta > 0$ the operator $D_g - \beta = -(4\pi)^{-1}(L_{\omega'} + 4\pi\beta)$ is a bijection of the space of distributions onto itself and maps H^{r+2} isomorphically onto H^r for every integer r. For $\beta = 0$, the equation $D_g f = a$ for a distribution a has a solution if and only if a has mean zero with respect to the volume form $\omega'^n/n!$ (that is, $\int_X a\, \omega'^n = 0$ when a is integrable); the solution is then unique up to an additive constant, and it is smooth if a is smooth. The condition is necessary because $\langle L_{\omega'} f, 1\rangle = \langle f, L_{\omega'} 1\rangle = 0$, with the pairing through $\omega'^n/n!$.

Theorem 14.1.7 (The $\mathrm{dd^c}$-lemma) *Every closed real $(1,1)$-current T on X has a unique decomposition*

$$T = H + \mathrm{dd^c} f, \tag{14.9}$$

where H is a smooth harmonic real $(1,1)$-form and f is a real distribution of mean zero. If T is smooth, then f is smooth. If T is positive, then f has a canonical quasi-plurisubharmonic representative. Moreover, T is exact as a current if and only if $H = 0$.

Proof Put $a = \int_X T \wedge \omega^{n-1} / \int_X \omega^n$. Since $T \wedge \omega^{n-1} = (\Lambda T)\omega^n/n$, the distribution $\Lambda T - na$ has mean zero, and Lemma 14.1.3 gives a real distribution f of mean zero with $Lf = -4\pi(\Lambda T - na)$. By Lemma 14.1.6, $\Lambda \mathrm{dd^c} f = \Lambda T - na$, so the closed

current $U = T - a\omega - \mathrm{dd^c} f$ has $\Lambda U = 0$. By Lemma 14.1.6 again, U is harmonic, and by Lemma 14.1.3 it is smooth. The Kähler form is harmonic as well, since $\mathrm{d}\omega = 0$ and $*\omega = \omega^{n-1}/(n-1)!$ is closed. Hence $H = a\omega + U$ and f give the decomposition (14.9). If T is smooth, then ΛT is smooth, and so is f by Lemma 14.1.3.

An exact harmonic form B vanishes: if $B = \mathrm{d}S$ for a current S, then

$$\|B\|_{L^2}^2 = \int_X \mathrm{d}S \wedge *B = \pm S(\mathrm{d} * B) = 0,$$

since $\mathrm{d}*B = 0$. If $T = H' + \mathrm{dd^c} f'$ is a second decomposition, then $H - H' = \mathrm{d}\,\mathrm{d^c}(f' - f)$ is exact and harmonic, hence zero. Then $\mathrm{dd^c}(f - f') = 0$, so $L(f - f') = 0$ by Lemma 14.1.6; thus $f - f'$ is constant by Lemma 14.1.3, and it vanishes because both have mean zero. The same argument proves the exactness criterion: if T is exact, so is $H = T - \mathrm{d}\,\mathrm{d^c} f$, whence $H = 0$; conversely, if $H = 0$, then $T = \mathrm{d}\,\mathrm{d^c} f$.

Suppose now that $T \geq 0$. By Theorems 7.3.2 and 8.2.1 and Lemma 13.1.2, every point of X has a coordinate ball on which $T = \mathrm{dd^c} u$ with u plurisubharmonic and $H = \mathrm{dd^c} h$ with h smooth. Then $\mathrm{dd^c}(f + h - u) = 0$, so the scalar Laplacian of the distribution $f + h - u$ vanishes, and Weyl's lemma (Theorem 2.1.4) shows that it is a smooth pluriharmonic function p. Thus $f = u - h + p$ is, on the ball, the class of a plurisubharmonic function minus a smooth function. On the overlap of two balls, the two local representatives $u_1 - h_1 + p_1$ and $u_2 - h_2 + p_2$ agree almost everywhere; after adding $h_1 - p_1$ to both, both become plurisubharmonic, because $h_1 - p_1 - h_2 + p_2$ is smooth with $\mathrm{dd^c}(h_1 - p_1 - h_2 + p_2) = H - H = 0$, and plurisubharmonic functions that agree almost everywhere agree everywhere (Theorem 8.2.1). Hence the local representatives define a global quasi-plurisubharmonic representative of f. It is the only one, since two quasi-plurisubharmonic functions that agree almost everywhere agree everywhere (Lemma 13.1.4(1)). □

Definition 14.1.8 The real $(1,1)$-cohomology space of X is

$$H^{1,1}(X, \mathbb{R}) = \frac{\{\text{closed smooth real } (1,1)\text{-forms}\}}{\{\mathrm{dd^c} f : f \in C^\infty(X, \mathbb{R})\}}.$$

The class of a closed smooth real $(1,1)$-form θ is denoted $\{\theta\}$. For a closed real $(1,1)$-current T we put $\{T\} = \{H\}$, where H is the harmonic form in (14.9).

By Theorem 14.1.7, every class contains exactly one harmonic form, so $H^{1,1}(X, \mathbb{R})$ is isomorphic to the space of harmonic real $(1,1)$-forms; in particular it is finite dimensional, and we give it its vector-space topology. The real de Rham cohomology in degree two is the quotient of the closed smooth real 2-forms by the forms $\mathrm{d}\beta$, with β a smooth real 1-form. A closed smooth real $(1,1)$-form that is exact in this sense has $H = 0$ and is therefore of the form $\mathrm{dd^c} f$ with f smooth; hence $H^{1,1}(X, \mathbb{R})$ is identified with the subspace of de Rham classes represented by $(1,1)$-forms.

Choosing a basis of harmonic forms gives a linear choice $\alpha \mapsto \theta_\alpha$ of smooth representatives. Each basis form H_k satisfies $-c_k\omega \leq H_k \leq c_k\omega$ for some c_k, by Lemma 13.1.1, applied with $K = X$ to H_k and to $-H_k$, and all norms on the finite-dimensional space $H^{1,1}(X, \mathbb{R})$ are equivalent; hence for every norm there is a

constant C such that

$$-C\|\alpha\|\omega \le \theta_\alpha \le C\|\alpha\|\omega. \tag{14.10}$$

The class of a closed current does not change when $\mathrm{dd^c}$ of a distribution is added to it, by the uniqueness in Theorem 14.1.7. If $H_1, \dots, H_N$ is an L^2-orthonormal basis of harmonic real $(1,1)$-forms, the coefficients of the harmonic part of T are the pairings $\int_X T \wedge *H_k$, because $*H_k$ is closed and therefore $\int_X \mathrm{dd^c} f \wedge *H_k = 0$. Hence the map $T \mapsto \{T\}$ on closed currents is continuous for weak convergence.

Positive currents in a class are thus described by quasi-plurisubharmonic potentials.

Corollary 14.1.9 *Let θ be a smooth closed real $(1,1)$-form on X. A closed positive $(1,1)$-current T on X satisfies $\{T\} = \{\theta\}$ if and only if $T = \theta + \mathrm{dd^c}\varphi$ for some $\varphi \in \mathrm{PSH}(X,\theta)$, and then φ is unique up to an additive constant. In particular, the class $\{\theta\}$ contains a closed positive current if and only if $\mathrm{PSH}(X,\theta) \neq \varnothing$.*

Proof Let $T \ge 0$ be closed with $\{T\} = \{\theta\}$. Write $T = H + \mathrm{dd^c} f$ as in (14.9); since $\{T\} = \{\theta\}$, also $\theta = H + \mathrm{dd^c} g$ with g smooth. By Theorem 14.1.7, f has a canonical quasi-plurisubharmonic representative, and the difference φ of this representative and g satisfies $T = \theta + \mathrm{dd^c}\varphi$; thus $\varphi \in \mathrm{PSH}(X,\theta)$. Conversely, if $\varphi \in \mathrm{PSH}(X,\theta)$, then $\theta + \mathrm{dd^c}\varphi$ is a closed positive current, and it lies in $\{\theta\}$ because adding $\mathrm{dd^c}$ of a distribution does not change the class. If $\theta + \mathrm{dd^c}\varphi = \theta + \mathrm{dd^c}\varphi'$ with $\varphi, \varphi' \in \mathrm{PSH}(X,\theta)$, then $\varphi - \varphi'$ is constant almost everywhere by the uniqueness part of Theorem 14.1.7, say equal to c, and the quasi-plurisubharmonic functions φ and $\varphi' + c$, which agree almost everywhere, agree everywhere by Lemma 13.1.4(1). □

If θ is replaced by $\theta + \mathrm{dd^c} h$ with h smooth, then φ is replaced by $\varphi - h$, up to a constant. Consequently the pole set, the Lelong numbers and the comparison of singularities of potentials of currents in $\{\theta\}$ do not depend on the choice of the smooth representative θ.

14.2 Normalized potentials and minimal singularities

Fix a smooth closed real $(1,1)$-form θ on X. A function $\varphi \in \mathrm{PSH}(X,\theta)$ is determined by its current $\theta + \mathrm{dd^c}\varphi$ up to an additive constant, which we fix by normalizing the supremum. The local compactness and integrability theorems then become uniform statements for normalized potentials, and the upper envelope of the nonpositive potentials has minimal singularities. We also compare potentials on X and on a modification of X.

Integrals over X are taken with respect to $\mathrm{d}V$, or equivalently ω^n, unless stated otherwise. A θ-plurisubharmonic function is integrable on the compact manifold X, being locally integrable. If φ is quasi-plurisubharmonic and ψ is quasi-plurisubharmonic or continuous, an inequality between φ and ψ that holds almost everywhere holds everywhere. Indeed, by Lemma 13.1.4(1) the value of a quasi-plurisubharmonic function at a point is the limit of its averages over small coordinate balls centered

at that point, a continuous function has the same property, and the inequality holds between these averages. In particular, a θ-plurisubharmonic function is determined by its class in $L^1(X)$, and we regard $\mathrm{PSH}(X,\theta)$ as a subset of $L^1(X)$.

Proposition 14.2.1 *Let $a \le b$ be real numbers. The set of $\varphi \in \mathrm{PSH}(X,\theta)$ with $a \le \sup_X \varphi \le b$ is compact in $L^1(X)$, and the map $\varphi \mapsto \sup_X \varphi$ is continuous on $\mathrm{PSH}(X,\theta)$ for the L^1 topology. In particular, the functions $\varphi \in \mathrm{PSH}(X,\theta)$ with $\sup_X \varphi = 0$ are uniformly bounded in $L^1(X)$. The same statements hold for a sequence of backgrounds: if smooth closed real $(1,1)$-forms θ_j converge smoothly to θ and $\varphi_j \in \mathrm{PSH}(X,\theta_j)$ satisfy $a \le \sup_X \varphi_j \le b$, then a subsequence converges in $L^1(X)$ to some $\varphi \in \mathrm{PSH}(X,\theta)$ with $a \le \sup_X \varphi \le b$, and $\sup_X \varphi_j \to \sup_X \varphi$ whenever $\varphi_j \to \varphi \in \mathrm{PSH}(X,\theta)$ in $L^1(X)$. The assertions for a fixed background hold more generally on every connected compact complex manifold X' of positive dimension, for every smooth closed real $(1,1)$-form θ' on X', with $L^1(X')$ taken with respect to any smooth positive volume form.*

Proof Choose finitely many coordinate balls B_k covering X, smaller concentric balls B'_k still covering X, and smooth functions h_k near $\overline{B_k}$ with $\mathrm{dd^c} h_k = \theta$ (Lemma 13.1.2, applied on slightly larger concentric balls). For $\varphi \in \mathrm{PSH}(X,\theta)$ the functions $\varphi + h_k$ are plurisubharmonic on B_k.

Let (φ_j) be a sequence with $a \le \sup_X \varphi_j \le b$. The plurisubharmonic functions $\varphi_j + h_k$ are uniformly bounded above on each B_k. Since φ_j is upper semicontinuous on the compact manifold X, it attains its supremum at a point x_j, and after passing to a subsequence $x_j \to x$, since X is compact and metrizable. Choose k with $x \in B'_k$. On B_k the functions $\varphi_j + h_k$ cannot converge locally uniformly to $-\infty$, because their values at x_j are bounded below by $a + \inf_{B'_k} h_k$ for large j. By Theorem 8.3.4, a further subsequence converges in $L^1_{\mathrm{loc}}(B_k)$. If B_l meets B_k, the functions $\varphi_j + h_l$ cannot tend to $-\infty$ locally uniformly on B_l, since on the nonempty open set $B_k \cap B_l$ they differ by the smooth function $h_l - h_k$ from a sequence that converges in L^1_{loc}; so again a subsequence converges in $L^1_{\mathrm{loc}}(B_l)$. Since X is connected, every ball is joined to B_k by a finite chain of overlapping balls, and after finitely many extractions the sequence converges in $L^1(B'_l)$ for every l, hence in $L^1(X)$, to some φ. The limit satisfies $\theta + \mathrm{dd^c}\varphi \ge 0$ as a distributional limit of positive currents, and its canonical representative belongs to $\mathrm{PSH}(X,\theta)$.

Now let $\varphi_j \to \varphi$ in $L^1(X)$, all in $\mathrm{PSH}(X,\theta)$; we show that $\sup_X \varphi_j \to \sup_X \varphi$. The submean inequality for $\varphi_j + h_k$ on small balls shows that the φ_j are uniformly bounded above. Let $x_j \to x$; we claim that $\overline{\lim}_j \varphi_j(x_j) \le \varphi(x)$. Indeed, choose k with $x \in B_k$, let $c > \varphi(x)$ be real, and use upper semicontinuity to choose a closed ball $K \subset B_k$ around x on which $\varphi < c$. The function $c + h_k$ is continuous near K and $\varphi + h_k < c + h_k$ on K, so the second part of Theorem 8.3.4 gives $\varphi_j + h_k < c + h_k$ on K for large j; hence $\varphi_j(x_j) < c$ for large j.

Choose a subsequence along which $\sup_X \varphi_j$ tends to $\overline{\lim}_j \sup_X \varphi_j$ and points x_j with $\varphi_j(x_j) = \sup_X \varphi_j$ converge to some x; the claim gives $\overline{\lim}_j \sup_X \varphi_j \le \varphi(x) \le \sup_X \varphi$. Conversely, choose a subsequence along which $\sup_X \varphi_j$ tends to $\underline{\lim}_j \sup_X \varphi_j$ and $\sum_j \|\varphi_j - \varphi\|_{L^1} < \infty$; by monotone convergence $\sum_j |\varphi_j - \varphi|$ is

integrable, hence finite almost everywhere, so $\varphi_j \to \varphi$ almost everywhere along it. We get $\varphi \le \varlimsup_j \sup_X \varphi_j$ almost everywhere, hence everywhere, as noted at the beginning of this section. The two inequalities prove that the supremum is continuous; in particular, the limit in the previous paragraph satisfies $a \le \sup_X \varphi \le b$. Thus the set in question is sequentially compact, hence compact, in the metric space $L^1(X)$. The continuous function $\|\cdot\|_{L^1}$ is bounded on the compact set of normalized functions.

The arguments so far use only that X is a connected compact complex manifold of positive dimension and that the volume form defining $L^1(X)$ is smooth and positive; hence they also prove the last assertion of the proposition.

For smoothly convergent backgrounds, choose $C \ge 0$ with $\theta_j \le C\omega$ for all j. Then $C\omega + \mathrm{dd^c}\varphi_j \ge 0$, so all φ_j lie in $\mathrm{PSH}(X, C\omega)$, and the first part, applied to the closed form $C\omega$, gives an L^1-convergent subsequence and the convergence of the suprema. Since $\theta_j + \mathrm{dd^c}\varphi_j \to \theta + \mathrm{dd^c}\varphi$ as distributions, the limit belongs to $\mathrm{PSH}(X, \theta)$. □

Corollary 14.2.2 *For every $1 \le p < \infty$ there is a constant M_p, depending only on X, θ, ω and p, such that*

$$\|u\|_{L^p(X,\omega^n)} \le M_p \quad \textit{for all } u \in \mathrm{PSH}(X,\theta) \textit{ with } \sup_X u = 0.$$

Proof (If $\mathrm{PSH}(X, \theta)$ is empty, there is nothing to prove.) Choose finitely many coordinate polydiscs D_k such that the polydiscs $\frac{1}{4}D_k$ with the same centers and a quarter of the radii cover X, and smooth functions h_k near $\overline{D_k}$ with $\mathrm{dd^c}h_k = \theta$. For u as in the statement, $v_k = u + h_k - \sup_{D_k} h_k$ is a nonpositive plurisubharmonic function on D_k. By Proposition 14.2.1 and the boundedness of the h_k and of the coordinate volume densities, the norms $\|v_k\|_{L^1(D_k)}$ are bounded independently of u. After rescaling D_k to the unit polydisc, (8.4) bounds $\|v_k\|_{L^p(\frac{1}{4}D_k)}$ independently of u. Since $u = v_k - h_k + \sup_{D_k} h_k$ on D_k, the bounded functions h_k and the bounded volume densities give the assertion. □

Corollary 14.2.3 *There are $\alpha, C > 0$, depending only on X, θ and ω, such that*

$$\int_X \mathrm{e}^{-\alpha u}\omega^n \le C \quad \textit{for all } u \in \mathrm{PSH}(X,\theta) \textit{ with } \sup_X u = 0. \tag{14.11}$$

Proof (If $\mathrm{PSH}(X, \theta)$ is empty, any $\alpha, C > 0$ will do.) Choose finitely many coordinate unit balls B_j such that the balls $\frac{1}{4}B_j$ with the same centers and a quarter of the radii cover X, and smooth functions ρ_j near $\overline{B_j}$ with $\mathrm{dd^c}\rho_j = \theta$. For u as in the statement put $v_j = u + \rho_j - \sup_{B_j} \rho_j$ on B_j; then v_j is plurisubharmonic and $v_j \le u \le 0$. By Proposition 14.2.1 and comparison of the coordinate volume densities, $\|v_j\|_{L^1(B_j)} \le M_j$ for a constant $M_j \ge 1$ independent of u. Applying Proposition 8.3.9 to v_j/M_j, we see that $\int_{\frac{1}{4}B_j} \mathrm{e}^{-\alpha_j v_j}$ is bounded independently of u, where $\alpha_j = \alpha_n/M_j$. Put $\alpha = \min_j \alpha_j > 0$. Since $v_j \le u \le 0$ and $\alpha \le \alpha_j$, we have $\mathrm{e}^{-\alpha u} \le \mathrm{e}^{-\alpha_j u} \le \mathrm{e}^{-\alpha_j v_j}$ on B_j. Summing over j and comparing the coordinate volume densities with ω^n gives (14.11). □

Recall from Definition 10.1.1 that a subset of X is pluripolar if it is locally contained in the pole set of a plurisubharmonic function.

Corollary 14.2.4 *Let $K \subset X$ be a compact set that is not pluripolar. Then there is a constant C such that*

$$\sup_X \varphi \le \sup_K \varphi + C \qquad \text{for all } \varphi \in \mathrm{PSH}(X,\theta).$$

Consequently, every subset of $\mathrm{PSH}(X,\theta)$ *on which* $\sup_K \varphi$ *is bounded above and below is relatively compact in* $L^1(X)$.

Proof Suppose that no such constant exists. Since $\sup_X \varphi - \sup_K \varphi$ does not change when a constant is added to φ, there are $\varphi_j \in \mathrm{PSH}(X,\theta)$ with $\sup_X \varphi_j = 0$ and $\sup_K \varphi_j \le -2^j$. Choose $C \ge 0$ with $\theta \le C\omega$ (Lemma 13.1.1) and put $\psi_N = \sum_{j=1}^N 2^{-j}\varphi_j$. Since $\sum_{j\le N} 2^{-j} \le 1$, we have

$$C\omega + \mathrm{dd^c}\psi_N = \Big(1 - \sum_{j\le N} 2^{-j}\Big) C\omega + \sum_{j\le N} 2^{-j}(C\omega + \mathrm{dd^c}\varphi_j) \ge \sum_{j\le N} 2^{-j}(\theta + \mathrm{dd^c}\varphi_j) \ge 0.$$

The sequence (ψ_N) is decreasing because $\varphi_j \le 0$, and $\|\psi_N\|_{L^1} \le \sup_j \|\varphi_j\|_{L^1} < \infty$ by Proposition 14.2.1. By monotone convergence the limit ψ is integrable, so it is not identically $-\infty$, and Theorem 8.3.1, applied on coordinate balls after adding local potentials of $C\omega$, shows that $\psi \in \mathrm{PSH}(X, C\omega)$. On K every term $2^{-j}\varphi_j$ is at most -1, so $\psi = -\infty$ on K. On every coordinate ball U with a smooth function h such that $\mathrm{dd^c}h = C\omega$, the function $\psi + h$ is plurisubharmonic and equals $-\infty$ on $K \cap U$. Hence K is pluripolar, a contradiction. The last assertion follows: bounds for $\sup_K \varphi$ give bounds for $\sup_X \varphi$, and Proposition 14.2.1 applies. □

Corollary 14.2.5 *Let (φ_i) be a decreasing net in* $\mathrm{PSH}(X,\theta)$ *whose suprema* $\sup_X \varphi_i$ *are bounded below. Then its pointwise infimum φ belongs to* $\mathrm{PSH}(X,\theta)$, $\varphi_i \to \varphi$ *in* $L^1(X)$, *and* $\sup_X \varphi_i \to \sup_X \varphi$.

Proof Fix an index i_0. For $i \ge i_0$ the suprema are bounded above by $\sup_X \varphi_{i_0}$, so the tail of the net lies in a compact subset of $L^1(X)$ by Proposition 14.2.1, and a subnet converges in L^1 to some $\psi \in \mathrm{PSH}(X,\theta)$. For each fixed i, the members of the subnet are eventually below φ_i, so $\psi \le \varphi_i$ almost everywhere, and therefore everywhere, as noted at the beginning of this section. Hence the pointwise infimum φ is at least ψ and is not identically $-\infty$. Applying Theorem 8.3.1 on the coordinate balls of a finite cover, after adding local potentials of θ, we find that $\varphi \in \mathrm{PSH}(X,\theta)$ and $\varphi_i \to \varphi$ in $L^1(X)$. The continuity of the supremum in Proposition 14.2.1 gives the last assertion. □

Definition 14.2.6 Let $\varphi, \psi \in \mathrm{PSH}(X,\theta)$. Recall from Section 13.5 that φ is *more singular* than ψ, written $\varphi \preceq \psi$, if $\varphi \le \psi + C$ on X for some constant C; we then also say that ψ is *less singular* than φ. A potential $\psi \in \mathrm{PSH}(X,\theta)$ has *minimal singularities* if every $\varphi \in \mathrm{PSH}(X,\theta)$ is more singular than ψ. If $\mathrm{PSH}(X,\theta) \neq \varnothing$, put

$$V_\theta = \big(\sup\{\varphi \in \mathrm{PSH}(X,\theta) : \varphi \le 0\}\big)^*,$$

where * denotes the upper semicontinuous regularization.

Lemma 14.2.7 *Suppose that* $\mathrm{PSH}(X,\theta) \neq \varnothing$. *Then* $V_\theta \in \mathrm{PSH}(X,\theta)$, $\sup_X V_\theta = 0$, *and* $\varphi \le V_\theta + \sup_X \varphi$ *for every* $\varphi \in \mathrm{PSH}(X,\theta)$; *in particular,* V_θ *has minimal singularities. Any two potentials with minimal singularities differ by a bounded function on the set where both are finite, and the notion of minimal singularities depends only on the class* $\{\theta\}$.

Proof The family whose upper envelope defines V_θ in Definition 14.2.6 is nonempty, because $\varphi - \sup_X \varphi$ belongs to it for every $\varphi \in \mathrm{PSH}(X,\theta)$, and it is bounded above by 0. Theorem 8.3.2, applied on coordinate balls after adding smooth local potentials of θ, therefore gives $V_\theta \in \mathrm{PSH}(X,\theta)$ and $V_\theta \le 0$; and $\sup_X V_\theta = 0$, because $V_\theta \ge \varphi - \sup_X \varphi$ for every φ. The same inequality reads $\varphi \le V_\theta + \sup_X \varphi$, so V_θ has minimal singularities. Any two potentials with minimal singularities are each more singular than the other, so their difference is bounded on the set where both are finite. Replacing θ by $\theta + \mathrm{dd}^{\mathrm{c}} h$ with h smooth replaces $\mathrm{PSH}(X,\theta)$ by $\mathrm{PSH}(X,\theta) - h$; since h is bounded, the notion of minimal singularities depends only on the class $\{\theta\}$. □

Lelong numbers of quasi-plurisubharmonic functions are defined pointwise as in Corollary 12.3.2: if $\varphi = v + g$ near y, with v plurisubharmonic and g smooth, then $\nu(\varphi, y) = \nu(v, y)$, which does not depend on the decomposition. Hence the rules of Proposition 12.1.4, the characterization of $\nu(\varphi, y)$ as the largest $c \ge 0$ with $\varphi \le c \log|z - z(y)|^2 + O(1)$ near y (Proposition 12.1.1), and the upper semicontinuity (12.6) along sequences $\varphi_j = v_j - g$ with a common smooth g, all carry over.

Recall from Proposition 12.3.5 the generic Lelong number of a plurisubharmonic function along an irreducible analytic subset, for instance along a prime divisor (an irreducible hypersurface). For a quasi-plurisubharmonic function φ on a complex manifold and an irreducible analytic subset Z, we define in the same way the generic Lelong number $\nu(\varphi, Z) = \inf_{y \in Z} \nu(\varphi, y)$ of φ along Z. The infimum is attained at every point of Z outside a countable union of proper analytic subsets of Z. Indeed, $\lambda = \nu(\varphi, Z)$ is finite, and for every integer $j \ge 1$ the set $A_j = \{y \in Z : \nu(\varphi, y) \ge \lambda + 1/j\}$ is an analytic subset of Z: near each point of Z it is the intersection of Z with the set $\{\nu(v, \cdot) \ge \lambda + 1/j\}$ for a local plurisubharmonic function $v = \varphi - g$ as above, which is analytic by Theorem 12.3.4. It is a proper subset of Z by the definition of λ, and $\nu(\varphi, y) = \lambda$ at every point $y \in Z \setminus \bigcup_j A_j$. By Proposition 12.3.5, the complement in Z of a countable union of proper analytic subsets of Z is dense in Z; in particular, $\nu(\varphi, Z)$ is attained on a dense subset of Z. If $\pi : Y \to X$ is a proper bimeromorphic holomorphic map from a complex manifold Y and $E \subset Y$ is a prime divisor, we put $\nu(\varphi, E) = \nu(\varphi \circ \pi, E)$ whenever $\varphi \circ \pi$ is quasi-plurisubharmonic, as in Proposition 12.3.5.

Proposition 14.2.8 *Let* $\pi : Y \to X$ *be a proper bimeromorphic holomorphic map from a complex manifold* Y. *Then* Y *is compact and connected, and* $\varphi \mapsto \varphi \circ \pi$ *is a bijection from* $\mathrm{PSH}(X,\theta)$ *onto* $\mathrm{PSH}(Y, \pi^*\theta)$, *which is continuous for convergence in*

$L^1(X)$ and $L^1(Y)$, the latter taken with respect to any smooth positive volume form. For every prime divisor E on Y, the generic Lelong numbers $\nu(\varphi, E) = \nu(\varphi \circ \pi, E)$ are bounded above uniformly for $\varphi \in \mathrm{PSH}(X, \theta)$.

Proof The manifold $Y = \pi^{-1}(X)$ is compact, because π is proper. By the definition of bimeromorphic maps (Appendix A.3), there is a nowhere dense analytic subset $E_0 \subset X$ such that $\pi^{-1}(E_0)$ is nowhere dense and π restricts to an isomorphism of $Y \setminus \pi^{-1}(E_0)$ onto $X \setminus E_0$. The set $X \setminus E_0$ is connected by Lemma A.1.7, hence so is its dense preimage, and therefore Y is connected. The image $\pi(Y)$ is closed, since proper maps are closed, and it contains the dense set $X \setminus E_0$; so π is surjective. By Proposition A.7.3 there is an analytic subset $A \subset X$ of codimension at least two such that π is an isomorphism over $X \setminus A$. Then $\pi^{-1}(A)$ is an analytic subset of Y different from Y, because π is surjective. Since Y is connected, $\pi^{-1}(A)$ is locally contained in the zero set of a holomorphic function that does not vanish identically on a connected neighborhood (Appendix A.1), so it is nowhere dense and a null set (Lemma 7.4.1). The same holds for A in X.

If $\varphi \in \mathrm{PSH}(X, \theta)$, let B be a coordinate ball and h a smooth function on B with $\mathrm{dd}^{\mathrm{c}} h = \theta$ (Lemma 13.1.2). On every connected coordinate neighborhood $W \subset \pi^{-1}(B)$ the function $(\varphi + h) \circ \pi$ is plurisubharmonic or identically $-\infty$, by Proposition 8.2.3, and the second case is excluded because $\varphi + h$ is finite almost everywhere on the image under the biholomorphism π of the nonempty open set $W \setminus \pi^{-1}(A)$. Hence $\varphi \circ \pi = (\varphi + h) \circ \pi - h \circ \pi$ is quasi-plurisubharmonic, with $\pi^*\theta + \mathrm{dd}^{\mathrm{c}}(\varphi \circ \pi) = \mathrm{dd}^{\mathrm{c}}((\varphi + h) \circ \pi) \geq 0$ on $\pi^{-1}(B)$. Thus $\varphi \circ \pi \in \mathrm{PSH}(Y, \pi^*\theta)$. If $\varphi \circ \pi = \varphi' \circ \pi$, then $\varphi = \varphi'$ on $X \setminus A$, hence almost everywhere, and therefore everywhere, as noted at the beginning of this section; this proves injectivity.

Conversely, let $\psi \in \mathrm{PSH}(Y, \pi^*\theta)$. Over $X \setminus A$ it descends to the function $\psi \circ \pi^{-1}$, which is θ-plurisubharmonic there. It is bounded above, because ψ is upper semicontinuous on the compact manifold Y. On a coordinate ball B with $\mathrm{dd}^{\mathrm{c}} h = \theta$ on B, the plurisubharmonic function $\psi \circ \pi^{-1} + h$ on $B \setminus A$ is therefore locally bounded above near A, and it extends to a plurisubharmonic function on B by Theorem 8.4.1. The extensions on two balls agree outside the null set A, hence everywhere, so they define $\varphi \in \mathrm{PSH}(X, \theta)$. The pullback $\varphi \circ \pi$ equals ψ outside the null set $\pi^{-1}(A)$, hence everywhere. This proves surjectivity.

For continuity, we use Proposition 14.2.1 on X and on Y; by its last assertion, its assertions for a fixed background hold on Y. Let $\varphi_j \to \varphi$ in $L^1(X)$. By the continuity of the supremum, it suffices to treat normalized functions, $\sup_X \varphi_j = \sup_X \varphi = 0$, since subtracting the suprema preserves L^1 convergence. The pullbacks have the same suprema, because π is surjective, so they are relatively compact in $L^1(Y)$, and every cluster value lies in $\mathrm{PSH}(Y, \pi^*\theta)$. On compact subsets of $Y \setminus \pi^{-1}(A)$, a change of variables shows that every L^1 cluster value of $(\varphi_j \circ \pi)$ coincides almost everywhere with $\varphi \circ \pi$; since $\pi^{-1}(A)$ is a null set, every cluster value is $\varphi \circ \pi$. A relatively compact sequence in a metric space with a single cluster value converges to it; hence $\varphi_j \circ \pi \to \varphi \circ \pi$ in $L^1(Y)$.

Finally, $\nu(\varphi \circ \pi, E)$ is the infimum over $y \in E$ of the point Lelong numbers $\nu(\varphi \circ \pi, y)$. Near y choose a smooth g with $\mathrm{dd}^{\mathrm{c}} g = \pi^*\theta$, so that the functions $\varphi \circ \pi + g$ are plurisubharmonic. By (12.6), applied to them, each $\nu(\varphi \circ \pi, y)$ is

upper semicontinuous for L^1 convergence of $\varphi \circ \pi$, hence, by the continuity just proved, for L^1 convergence of φ; and an infimum of upper semicontinuous functions is upper semicontinuous. Since Lelong numbers do not change when a constant is added, we may restrict to the normalized functions, which form a compact set by Proposition 14.2.1. An upper semicontinuous function with finite values attains its maximum on a compact set, which gives the uniform bound. □

Corollary 14.2.9 *Let B' be a coordinate ball in X and $B \Subset B'$ a ball in the same coordinates, and let $\varphi \in \mathrm{PSH}(X,\theta)$ be bounded near $\overline{B'}$. There is $\psi \in \mathrm{PSH}(X,\theta)$ with $\psi \geq \varphi$, $\psi = \varphi$ on $X \setminus B$, ψ bounded near $\overline{B}$, and $(\theta + \mathrm{dd}^{\mathrm{c}}\psi)^n = 0$ on B. It is the largest function in $\mathrm{PSH}(X,\theta)$ with these properties, and any two functions with these properties that are continuous on $\overline{B}$ coincide.*

Proof Choose a ball B'' in the same coordinates with $\overline{B} \subset B''$ and $\overline{B''} \subset B'$. Write $\theta = \mathrm{dd}^{\mathrm{c}} h$ on B'', with h smooth (Lemma 13.1.2), and apply Corollary 9.6.4 to the locally bounded plurisubharmonic function $\varphi + h$ on B''. Subtracting h from the result on B'' and keeping φ outside B'' gives ψ, since the result equals $\varphi + h$ on $B'' \setminus B$; here $(\theta + \mathrm{dd}^{\mathrm{c}}\psi)^n = (\mathrm{dd}^{\mathrm{c}}(\psi + h))^n$ is the Bedford–Taylor product of the bounded plurisubharmonic function $\psi + h$. If ψ' is another function with the stated properties, then by upper semicontinuity $\overline{\lim}_{z \to \zeta,\, z \in B} \psi'(z) \leq \psi'(\zeta) = \varphi(\zeta)$ for every $\zeta \in \partial B$. Hence $\psi' + h$ belongs to the family whose upper envelope is $\psi + h$ in Corollary 9.6.4, and $\psi' \leq \psi$ on B. If ψ' is continuous on $\overline{B}$, then its boundary values $\varphi|_{\partial B}$ are continuous, and $\psi' + h$ is the solution of the homogeneous Dirichlet problem with the continuous boundary values $(\varphi + h)|_{\partial B}$; by the uniqueness in Theorem 9.6.3, any two such continuous functions coincide. □

14.3 Positivity cones and divisorial decomposition

Let $\alpha \in H^{1,1}(X,\mathbb{R})$ and fix a smooth representative $\theta \in \alpha$. By Theorem 14.1.7, the closed currents in α are exactly the currents $\theta + \mathrm{dd}^{\mathrm{c}}\varphi$ with φ a real distribution, so positivity properties of α become inequalities for $\theta + \mathrm{dd}^{\mathrm{c}}\varphi$. Besides the positivity cones and the mixed Hodge index theorem, we prove Siu's decomposition of a closed positive current (Theorem 14.3.7), which leads to modified nef classes.

Definition 14.3.1 A class $\alpha \in H^{1,1}(X,\mathbb{R})$ is

- *pseudo-effective* if it contains a closed positive current;
- *Kähler* if it contains a Kähler form;
- *nef* if for every $\varepsilon > 0$ it contains a smooth form θ_ε with $\theta_\varepsilon \geq -\varepsilon\omega$;
- *big* if it contains a closed current T with $T \geq \delta\omega$ for some $\delta > 0$.

A closed real $(1,1)$-current T with $T \geq \delta\omega$ for some $\delta > 0$ is called a *Kähler current*. The cones of pseudo-effective, Kähler, nef and big classes are denoted $\mathcal{E}$, $\mathcal{K}$, $\mathcal{N}$ and $\mathcal{B}$. The zero class belongs to the pseudo-effective cone and to the nef cone.

These notions do not depend on ω, because any two Hermitian forms on the compact manifold X are bounded by constant multiples of each other (Lemma 13.1.1). By Corollary 14.1.9, α is pseudo-effective if and only if $\mathrm{PSH}(X,\theta)\neq\varnothing$.

Proposition 14.3.2 *The pseudo-effective cone $\mathcal{E}$ is closed and convex, and the Kähler cone $\mathcal{K}$ is open and convex. Moreover $\mathcal{N}=\overline{\mathcal{K}}\subset\mathcal{E}$, the Kähler cone $\mathcal{K}$ is the interior of $\mathcal{N}$, and the big cone $\mathcal{B}$ is the interior of $\mathcal{E}$.*

Proof Sums and positive multiples of positive currents, Kähler forms or nef representatives are of the same kind, which proves convexity of all four cones.

Let $\alpha_j\to\alpha$ be pseudo-effective classes, with positive currents $T_j\in\alpha_j$. Since the $\mathrm{dd^c}$ of a distribution pairs to zero with the closed form ω^{n-1}, the trace masses are

$$\int_X T_j\wedge\omega^{n-1}=\int_X\theta_{\alpha_j}\wedge\omega^{n-1},$$

where θ_{α_j} are the representatives of (14.10); they are bounded because the classes converge. Let K be a compact subset of a chart, χ a test function on the chart with $0\le\chi\le1$ and $\chi=1$ on K, and $c'>0$ such that $\beta\le c'\omega$ on $\operatorname{Supp}\chi$ (Lemma 13.1.1), where β is the Euclidean form (7.4) of the chart. By (14.6), applied with $\gamma_j=c'\omega$ and $\gamma'_j=\beta$, the form $(c'\omega)^{n-1}-\beta^{n-1}$ is a sum of products of semipositive $(1,1)$-forms on $\operatorname{Supp}\chi$, hence strongly positive (Proposition 7.2.1). Therefore $c'^{n-1}\omega^{n-1}-\chi\beta^{n-1}=(1-\chi)c'^{n-1}\omega^{n-1}+\chi\big(c'^{n-1}\omega^{n-1}-\beta^{n-1}\big)$, with $\chi\beta^{n-1}$ extended by zero outside the chart, is a smooth strongly positive form on X, so the trace measures satisfy $\mu_{T_j}(K)\le T_j(\chi\beta^{n-1})\le c'^{n-1}\int_X T_j\wedge\omega^{n-1}$, which is bounded independently of j. By the compactness assertion of Theorem 7.2.5, a subsequence of (T_j) converges weakly to a closed positive current T, and $\{T\}=\alpha$ because the class map is weakly continuous. Hence $\mathcal{E}$ is closed.

If $\omega_0\in\alpha$ is a Kähler form, then $\omega_0\ge c\omega$ for some $c>0$, and by (14.10) the form $\omega_0+\theta_{\alpha'-\alpha}\in\alpha'$ is Kähler whenever $C\|\alpha'-\alpha\|<c$; hence $\mathcal{K}$ is open. If α is nef and $\varepsilon>0$, a representative $\theta_\varepsilon\ge-\varepsilon\omega$ gives the Kähler form $\theta_\varepsilon+2\varepsilon\omega\in\alpha+2\varepsilon\{\omega\}$, so $\alpha\in\overline{\mathcal{K}}$. Conversely, let Kähler classes α_j with Kähler forms $\omega_j\in\alpha_j$ converge to α. Then $\omega_j+\theta_{\alpha-\alpha_j}\in\alpha$ is at least $-C\|\alpha-\alpha_j\|\omega$, so α is nef; thus $\mathcal{N}=\overline{\mathcal{K}}$, and $\mathcal{N}\subset\mathcal{E}$ because $\mathcal{K}\subset\mathcal{E}$ and $\mathcal{E}$ is closed.

Since $\mathcal{K}$ is open and contained in $\mathcal{N}$, it lies in the interior of $\mathcal{N}$. If α is nef and $t>0$, then $\alpha+t\{\omega\}$ is Kähler: take a representative $\theta_\varepsilon\ge-\varepsilon\omega$ with $\varepsilon<t$. If α lies in the interior of $\mathcal{N}$, then $\alpha-t\{\omega\}$ is nef for small $t>0$, and therefore $\alpha=(\alpha-t\{\omega\})+t\{\omega\}$ is Kähler. Thus $\mathcal{K}$ is the interior of $\mathcal{N}$.

If $T\in\alpha$ and $T\ge\delta\omega$, then $T+\theta_{\alpha'-\alpha}\in\alpha'$ is positive whenever $C\|\alpha'-\alpha\|\le\delta$, so a neighborhood of α lies in $\mathcal{E}$. Conversely, if α lies in the interior of $\mathcal{E}$, then $\alpha-\delta\{\omega\}\in\mathcal{E}$ for some $\delta>0$, and adding $\delta\omega$ to a positive current in $\alpha-\delta\{\omega\}$ gives a Kähler current in α. Hence $\mathcal{B}$ is the interior of $\mathcal{E}$. □

If $\theta_1,\dots,\theta_n$ are smooth closed real $(1,1)$-forms, Stokes' formula shows that $\int_X\theta_1\wedge\cdots\wedge\theta_n$ is unchanged when one θ_i is replaced by $\theta_i+\mathrm{dd^c}f$ with f smooth. We therefore write $\int_X\alpha_1\cdots\alpha_n=\int_X\theta_1\wedge\cdots\wedge\theta_n$ for the *intersection number* of

classes $\alpha_i = \{\theta_i\}$, and use the usual notation for powers, such as $\int_X \alpha^n$. In the mixed Hodge index theorem below, the mixed scalar equation of Lemma 14.1.4 turns a cohomological orthogonality condition into a pointwise one, to which the signature lemma for Hermitian matrices applies.

Theorem 14.3.3 (The mixed Hodge index theorem) *Let $n \geq 2$, let $\omega_3, \dots, \omega_n$ be Kähler forms on X, and put $\Omega = \omega_3 \wedge \cdots \wedge \omega_n$, with $\Omega = 1$ when $n = 2$. The symmetric bilinear form*

$$Q(\alpha, \beta) = \int_X a \wedge b \wedge \Omega, \qquad a \in \alpha, \quad b \in \beta,$$

on $H^{1,1}(X, \mathbb{R})$ has signature $(1, \dim H^{1,1}(X, \mathbb{R}) - 1)$. For every Kähler class γ, it is negative definite on the hyperplane $\{\alpha : Q(\alpha, \gamma) = 0\}$. For Kähler classes α, β,

$$Q(\alpha, \beta)^2 \geq Q(\alpha, \alpha) Q(\beta, \beta). \tag{14.12}$$

Proof Let γ be a Kähler class with a Kähler form $h \in \gamma$, and let α satisfy $Q(\alpha, \gamma) = 0$. Choose a smooth closed real form $a \in \alpha$; then $\int_X a \wedge h \wedge \Omega = 0$. Apply Lemma 14.1.4 with $A = h \wedge \Omega$, a product of $n - 1$ Kähler forms, and $\eta = -a \wedge h \wedge \Omega$. The resulting f makes $a' = a + \mathrm{dd}^{\mathrm{c}} f \in \alpha$ satisfy $a' \wedge h \wedge \Omega = 0$ at every point. At a point, in holomorphic coordinates, the wedge products of real $(1,1)$-forms are expressed by mixed discriminants of their Hermitian matrices, so Lemma 7.2.3 gives $a' \wedge a' \wedge \Omega \leq 0$, with equality only if a' vanishes at that point. By Stokes' formula,

$$Q(\alpha, \alpha) = \int_X a' \wedge a' \wedge \Omega \leq 0.$$

If this integral vanishes, the smooth nonpositive density $a' \wedge a' \wedge \Omega$ vanishes everywhere, so $a' = 0$ and $\alpha = 0$. Thus Q is negative definite on the hyperplane $\{Q(\cdot, \gamma) = 0\}$.

Since $Q(\gamma, \gamma) = \int_X h^2 \wedge \Omega > 0$, every class splits uniquely as a multiple of γ plus a class in that hyperplane. The space $H^{1,1}(X, \mathbb{R})$ is finite dimensional by Theorem 14.1.7, and Q is positive on the line spanned by γ and negative definite on its Q-orthogonal complement; this gives the signature. Finally, let α, β be Kähler classes and write $\beta = c\alpha + \delta$ with $Q(\delta, \alpha) = 0$. Negative definiteness gives $Q(\beta, \beta) = c^2 Q(\alpha, \alpha) + Q(\delta, \delta) \leq c^2 Q(\alpha, \alpha)$, while $Q(\alpha, \beta) = c Q(\alpha, \alpha)$. Since $Q(\alpha, \alpha) > 0$, we obtain $Q(\alpha, \beta)^2 = c^2 Q(\alpha, \alpha)^2 \geq Q(\alpha, \alpha) Q(\beta, \beta)$. □

Corollary 14.3.4 *Let $r \geq 1$, let $\alpha_1, \dots, \alpha_r$ be nef classes, and let $k_1, \dots, k_r \geq 0$ be integers with $\sum_i k_i = n$. Then*

$$\int_X \alpha_1^{k_1} \cdots \alpha_r^{k_r} \geq \prod_{i : k_i > 0} \left(\int_X \alpha_i^n \right)^{k_i / n}.$$

In particular, for nef classes $\alpha_1, \dots, \alpha_n$,

$$\int_X \alpha_1 \cdots \alpha_n \geq \prod_{i=1}^{n} \left(\int_X \alpha_i^n \right)^{1/n}.$$

Proof Assume first that all α_i are Kähler. Then all the intersection numbers $\int_X \prod_i \alpha_i^{k_i}$ are positive, and we put $b(k) = \log \int_X \prod_i \alpha_i^{k_i}$ for $k = (k_1, \ldots, k_r)$ with $\sum_i k_i = n$. If $n = 1$, only one k_i is positive and there is nothing to prove; let $n \geq 2$. Let $i \neq j$ with $k_i, k_j \geq 1$, let Ω be the wedge product of Kähler representatives of the remaining $n-2$ factors, counted with multiplicity, and let e_i be the ith coordinate vector. Then (14.12) for α_i, α_j reads

$$2b(k) \geq b(k + e_i - e_j) + b(k - e_i + e_j).$$

Fix the coordinates other than i, j and put $s = k_i + k_j$. The inequality says that the finite sequence $l \mapsto b(\ldots, l, \ldots, s-l, \ldots)$, with l in the ith and $s-l$ in the jth place, $0 \leq l \leq s$, is concave: its consecutive differences decrease. A concave sequence lies above the linear interpolation between its endpoints, since the average of its first l differences is at least the average of all s differences. Consequently

$$b(k) \geq \frac{k_i}{s} b(k^{(i)}) + \frac{k_j}{s} b(k^{(j)}),$$

where $k^{(i)}$ is obtained from k by putting all s factors in the ith place and $k^{(j)}$ by putting them in the jth place.

We now prove $b(k) \geq \sum_i (k_i/n) \log \int_X \alpha_i^n$ by induction on the number of positive coordinates of k. If only one coordinate is positive, this is an equality. Otherwise choose $i \neq j$ with $k_i, k_j \geq 1$. The vectors $k^{(i)}$ and $k^{(j)}$ have one positive coordinate fewer, and inserting the induction hypothesis for them into the last display gives the claim: for $m \neq i, j$ the coefficient of $\log \int_X \alpha_m^n$ is $(k_i/s + k_j/s)\, k_m/n = k_m/n$, and the coefficients of $\log \int_X \alpha_i^n$ and $\log \int_X \alpha_j^n$ are $(k_i/s)(s/n) = k_i/n$ and k_j/n. Exponentiating proves the corollary for Kähler classes.

For nef classes, $\alpha_i + \varepsilon\{\omega\}$ is Kähler for every $\varepsilon > 0$ by Proposition 14.3.2, so the inequality holds for these classes. Each intersection number involved is a polynomial in ε and tends to the corresponding number for the α_i as $\varepsilon \downarrow 0$; in particular the limits $\int_X \alpha_i^n$ are nonnegative. Since $t \mapsto t^{k_i/n}$ is continuous on $[0, \infty)$, the inequality passes to the limit. □

The next proposition uses the cutoff-sum approximations G_m of Lemma 13.4.1: for $\varphi \in \mathrm{PSH}(X, \theta)$ and integers $m \geq 1$, G_m has gentle analytic singularities (Definition 13.1.11) of type $(m^{-1}, \mathcal{I}(m\varphi))$ and satisfies $\theta + \mathrm{dd}^{\mathrm{c}} G_m \geq -(c/m)\omega$, where $c > 0$ (the constant c' of that lemma) depends only on (X, ω) and not on θ, φ or m. In particular G_m is smooth near every point where $\mathcal{I}(m\varphi)$ is the unit ideal.

Proposition 14.3.5 *A class that contains a closed positive current with locally bounded potentials is nef. A class that contains a Kähler current with locally bounded potentials is Kähler. Every big class contains a Kähler current with gentle analytic singularities.*

Proof Write the given current as $\theta + \mathrm{dd}^{\mathrm{c}}\varphi$ with $\varphi \in \mathrm{PSH}(X,\theta)$. If φ is bounded, then every $\mathcal{I}(m\varphi)$ is the unit ideal, so the approximations G_m are smooth and satisfy $\theta + \mathrm{dd}^{\mathrm{c}}G_m \geq -(c/m)\omega$; hence the class is nef. If moreover $\theta + \mathrm{dd}^{\mathrm{c}}\varphi \geq \delta\omega$, then $\varphi \in \mathrm{PSH}(X, \theta - \delta\omega)$, and the approximations for the background $\theta - \delta\omega$ satisfy $\theta + \mathrm{dd}^{\mathrm{c}}G_m \geq (\delta - c/m)\omega$; for large m the smooth form $\theta + \mathrm{dd}^{\mathrm{c}}G_m$ is a Kähler form in the class. For the last assertion, let $\theta + \mathrm{dd}^{\mathrm{c}}\varphi \geq \delta\omega$ be a Kähler current in a big class, with φ not necessarily bounded. The same inequality shows that $\theta + \mathrm{dd}^{\mathrm{c}}G_m$ is a Kähler current for large m, and G_m has gentle analytic singularities of type $(m^{-1}, \mathcal{I}(m\varphi))$ by Lemma 13.4.1. □

Corollary 14.3.6 *If a closed positive current with minimal singularities is a Kähler current, then its class is Kähler. Moreover, in a Kähler class every potential with minimal singularities is bounded.*

Proof Write the current as $T = \theta + \mathrm{dd}^{\mathrm{c}}\varphi \geq \delta\omega$, where φ has minimal singularities, and normalize $\varphi \leq 0$. Choose $C' \geq 0$ with $-\theta \leq C'\omega$ (Lemma 13.1.1). For $0 < t \leq \delta/(\delta + C')$ the current

$$(1-t)T + t\theta = \theta + \mathrm{dd}^{\mathrm{c}}\big((1-t)\varphi\big) \geq \big((1-t)\delta - tC'\big)\omega \geq 0$$

is positive, so $(1-t)\varphi \in \mathrm{PSH}(X,\theta)$. By minimality, $(1-t)\varphi \leq \varphi + C$ for a constant C, so $\varphi \geq -C/t$ at every point where φ is finite, hence almost everywhere. Since φ is quasi-plurisubharmonic, the lower bound holds everywhere, as noted at the beginning of Section 14.2. Thus φ is bounded, and Proposition 14.3.5 shows that the class is Kähler. For the second assertion, suppose that the class contains a Kähler form $\theta + \mathrm{dd}^{\mathrm{c}}u$. Then u is smooth and $u \in \mathrm{PSH}(X,\theta)$, so a potential φ with minimal singularities satisfies $u \leq \varphi + C$; hence φ is bounded below, and it is bounded above as every quasi-plurisubharmonic function on X. □

Recall from Definition 12.1.2 that the Lelong number $\nu(T,x)$ of a closed positive $(1,1)$-current T at $x \in X$ is the Lelong number at x of any local plurisubharmonic potential of T. If $T = \theta + \mathrm{dd}^{\mathrm{c}}\varphi$ with $\varphi \in \mathrm{PSH}(X,\theta)$, then $\varphi + h$ is such a potential wherever h is smooth with $\mathrm{dd}^{\mathrm{c}}h = \theta$, so $\nu(T,x) = \nu(\varphi,x)$ in the sense recalled before Proposition 14.2.8; in particular, these numbers do not depend on the background θ. For a prime divisor $D \subset X$, the generic Lelong number of T along D (Proposition 12.3.5) is therefore

$$\nu(T,D) = \inf_{x\in D} \nu(T,x) = \inf_{x\in D} \nu(\varphi,x) = \nu(\varphi,D),$$

and by that proposition the infimum is attained at every point of D outside a countable union of proper analytic subsets of D. We write $[D]$ for the current of integration along D (Lemma A.2.16); its coefficient measures are given by integration over D_{reg} and give no mass to D_{sing}.

Theorem 14.3.7 (Siu's divisorial decomposition) *Let T be a closed positive $(1,1)$-current on X. Then $\nu(T,D) > 0$ for at most countably many prime divisors D, the series below converges weakly, and*

$$T = R + \sum_D \nu(T, D)[D], \tag{14.13}$$

where R is a closed positive current with $\nu(R, D) = 0$ for every prime divisor D. In particular $T - \nu(T, D)[D] \geq 0$ for every D. The decomposition is unique: if $T = R' + \sum_D b_D[D]$ for a weakly convergent series with coefficients $b_D \geq 0$, at most countably many of them nonzero, and a closed positive current R' with $\nu(R', D) = 0$ for every D, then $b_D = \nu(T, D)$ for every D and $R' = R$.

Proof We first subtract one divisor, then show that only countably many divisors occur and that the series converges, and finally identify the residual current and prove uniqueness.

Step 1: one divisor. Let D be a prime divisor and $c = \nu(T, D)$; we show that $T - c[D] \geq 0$. Near a regular point of D choose coordinates with $D = \{z_1 = 0\}$ on a coordinate polydisc W with $|z_1| < 1$, and a plurisubharmonic potential u of T on W. Since the Lelong numbers of u at the points of $D \cap W$ are at least c, Proposition 12.3.6 gives $u \leq c \log |z_1|^2 + C$ near the point. Off D the function $u - c \log |z_1|^2$ is plurisubharmonic, and it is bounded above near D; by Theorem 8.4.1 it extends to a plurisubharmonic function v. Since $v = u - c \log |z_1|^2$ almost everywhere and $[D] = \mathrm{dd^c} \log |z_1|^2$ near the point (Theorem 7.4.5), we get $\mathrm{dd^c} v = T - c[D]$ there. Hence $T - c[D]$ is positive on a neighborhood of D_{reg}, and it equals T on $X \setminus D$; so it is positive on the open set $X \setminus D_{\mathrm{sing}}$.

To extend the positivity across D_{sing}, let ϕ be a strongly positive test form of bidegree $(n-1, n-1)$. Since T and $[D]$ have measure coefficients (Theorem 7.2.5), there is a real measure m_ϕ on X, a combination of the coefficient measures with the coefficients of ϕ as densities, such that $(T - c[D])(\chi\phi) = \int \chi \, \mathrm{d}m_\phi$ for test functions χ. The restriction of m_ϕ to the open set $X \setminus D_{\mathrm{sing}}$ is a positive measure, because $(T - c[D])(\chi\phi) \geq 0$ for test functions $\chi \geq 0$ supported there; and its restriction to D_{sing} is that of the positive measure $\chi \mapsto T(\chi\phi)$, because $[D]$ gives no mass to D_{sing}. Hence $(T - c[D])(\phi) = m_\phi(X) \geq 0$, that is, $T - c[D] \geq 0$.

Step 2: the series. By Theorem 12.3.4, each set $E_j = \{x : \nu(T, x) \geq 1/j\}$ is an analytic subset of X different from X. Being compact, E_j has finitely many irreducible components Z by Corollary A.2.13. We first show:

$$\begin{gathered}\text{a prime divisor } D \text{ contained in an analytic subset } E \neq X \\ \text{is an irreducible component of } E.\end{gathered} \tag{14.14}$$

An analytic subset of the connected manifold X with an interior point is all of X, since the set of its interior points is open and, by the identity theorem applied to local defining functions, closed. Hence E has no interior points, so its germs, and their branches, have dimension at most $n-1$, and E_{sing} has dimension at most $n-2$ at every point (Proposition A.2.7 and Corollary A.2.10). At a point $y \in D_{\mathrm{reg}}$ the germ of D has dimension $n-1$, so by monotonicity of dimension (see Proposition A.2.7 and the remarks following it) it is not contained in the germ of E_{sing}; thus there are points $y' \in D_{\mathrm{reg}} \cap E_{\mathrm{reg}}$. Near such a point E is a complex submanifold of dimension at most

$n-1$ containing the $(n-1)$-dimensional submanifold D, so D and E coincide near y'.

Let F be the connected component of E_{reg} containing y'. The analytic subset $D \cap F$ of the connected manifold F has an interior point, so it is all of F, and D contains the closure Z of F, an irreducible component of E (Corollary A.2.13). Since D is compact and the components of E form a locally finite family, D meets only finitely many of them and is the union of its intersections with these. By induction on the number of sets, an irreducible set is not a finite union of analytic subsets different from itself; hence $D = D \cap Z'$ for a component Z'. Then $Z \subset D \subset Z'$, and since no component is contained in another, $D = Z$. This proves (14.14). In particular, two distinct prime divisors are not contained in one another, and every prime divisor D with $\nu(T, D) \geq 1/j$, being contained in E_j, is one of its finitely many components. Therefore at most countably many prime divisors have $\nu(T, D) > 0$. Let $D_1, D_2, \ldots$ be an enumeration of them and put $S_N = \sum_{k \leq N} \nu(T, D_k)[D_k]$.

We claim that $T - S_N \geq 0$ for every N, and argue by induction; suppose that $T - S_{N-1} \geq 0$. Choose a point $x \in D_N$ outside the proper analytic subsets $D_N \cap D_k$, $k < N$, and outside the exceptional sets of Proposition 12.3.5 for T and $T - S_{N-1}$ along D_N. Such points exist, because by that proposition the complement in D_N of a countable union of proper analytic subsets of D_N is dense in D_N. The currents $[D_k]$, $k < N$, vanish near x, so $T - S_{N-1} = T$ near x, and therefore $\nu(T - S_{N-1}, D_N) = \nu(T - S_{N-1}, x) = \nu(T, x) = \nu(T, D_N)$. Step 1, applied to $T - S_{N-1}$, gives $T - S_N \geq 0$. Consequently the trace measures of the S_N in every chart are bounded by that of T, and the masses $\int_X S_N \wedge \omega^{n-1}$ are bounded by $\int_X T \wedge \omega^{n-1}$.

We now show that the sequence (S_N) converges weakly. Every test form is a finite linear combination of test forms $g\gamma$, where $g \geq 0$ is a test function supported in a chart and γ is a strongly positive form with constant coefficients in that chart. To see this, use a partition of unity (Proposition 7.1.3), the fact that strongly positive forms span the real forms (Proposition 7.2.1), and write a real test function g as $(g + M\psi) - M\psi$ with $\psi \geq 0$ a test function equal to one on $\operatorname{Supp} g$ and $M \geq \sup |g|$. For such $g\gamma$ the numbers $S_N(g\gamma)$ increase with N, since $S_{N+1} - S_N \geq 0$, and they are bounded by $T(g\gamma)$; hence they converge, and so does $S_N(\phi)$ for every test form ϕ. By the compactness assertion of Theorem 7.2.5, a subsequence converges weakly to a closed positive current S, so $S_N(\phi) \to S(\phi)$ for every ϕ. Finally $R = T - S$ is closed and positive, as a weak limit of the closed positive currents $T - S_N$.

Step 3: properties of R and uniqueness. Fix a prime divisor D and write $T = R + \nu(T, D)[D] + S_D$, where S_D is the sum of the terms of the series for the other divisors. Lelong numbers are additive and nonnegative (Proposition 12.1.4), and $\nu([D], x) = \nu(\log |f|^2, x) = 1$ at regular points x of D, where f is a local coordinate defining D. Hence, at a point of D_{reg} outside the countable union of proper analytic subsets where one of the currents T, R, S_D has a nongeneric Lelong number (such points exist by Proposition 12.3.5, since D_{sing} is also a proper analytic subset of D),

$$\nu(T, D) = \nu(R, D) + \nu(T, D) + \nu(S_D, D) \geq \nu(R, D) + \nu(T, D).$$

Hence $\nu(R, D) = 0$.

Now consider any decomposition $T = R' + \sum_D b_D[D]$ as in the statement, fix D, and let S'_D be the sum of the terms for the other divisors. We first show that the coefficient measures of S'_D give no mass to D. If $D' \neq D$ is another prime divisor, then $D \cap D'_{\mathrm{reg}}$ is an analytic subset of the connected manifold D'_{reg} (Corollary A.2.13) which is not all of it, since otherwise $D' \subset D$, and then $D' = D$ by (14.14). Hence $D \cap D'_{\mathrm{reg}}$ is a null set for the surface measure of D'_{reg} (Lemma 7.4.1, in local coordinates of D'_{reg}), and $[D']$ gives no mass to D. Now let $K \subset D$ be a compact subset of a chart, and for large m let χ_m be test functions with $0 \leq \chi_m \leq 1$, equal to one on K and supported in the $1/m$-neighborhood of K (Lemma 1.4.1), all supported in a fixed compact subset of the chart, on which a test function ψ equals one. For the trace measures in the chart, the weak convergence of the series gives $\mu_{S'_D}(\chi_m) = \sum_{D'} b_{D'} \mu_{[D']}(\chi_m)$. As $m \to \infty$, $\chi_m \to \mathbf{1}_K$ pointwise, and dominated convergence, applied to the measure $\mu_{S'_D}$ and to the series, whose terms are bounded by $b_{D'}\mu_{[D']}(\psi)$ with $\sum_{D'} b_{D'}\mu_{[D']}(\psi) = \mu_{S'_D}(\psi) < \infty$, gives $\mu_{S'_D}(K) = \sum_{D'} b_{D'}\mu_{[D']}(K) = 0$. Since D is covered by countably many such compact sets, the trace measures of S'_D, and hence its coefficient measures (Theorem 7.2.5), vanish on D.

By Step 1, $S'_D - \nu(S'_D, D)[D] \geq 0$; comparing the trace measures of both sides on D, where $[D]$ has positive trace mass, forces $\nu(S'_D, D) = 0$. Additivity of Lelong numbers at generic points of D, applied to $T = R' + b_D[D] + S'_D$, gives $\nu(T, D) = 0 + b_D + 0$. Thus $b_D = \nu(T, D)$ for every D, and $R' = R$. □

Definition 14.3.8 A class α is *modified nef* if for every $\varepsilon > 0$ there is a closed current $T_\varepsilon \in \alpha$ with $T_\varepsilon \geq -\varepsilon\omega$ and $\nu(T_\varepsilon + \varepsilon\omega, D) = 0$ for every prime divisor D. We write $\mathcal{MN}$ for the cone of modified nef classes. A closed positive $(1,1)$-current T with $\nu(T, D) = 0$ for every prime divisor D is called *non-divisorial*.

The set $\mathcal{MN}$ is a convex cone. Indeed, multiplying T_ε by $t > 0$ shows that $t\alpha$ is modified nef, with $tT_{\varepsilon/t}$ for the allowance ε. If α, α' are modified nef, with currents $T_\varepsilon, T'_\varepsilon$, then $T_{\varepsilon/2} + T'_{\varepsilon/2} \geq -\varepsilon\omega$ lies in $\alpha + \alpha'$. At a point of a prime divisor D at which the Lelong numbers of the positive currents $T_{\varepsilon/2} + \varepsilon\omega/2$ and $T'_{\varepsilon/2} + \varepsilon\omega/2$ attain their generic values 0 (Proposition 12.3.5), the additivity of Lelong numbers (Proposition 12.1.4) shows that their sum has Lelong number 0; hence $\nu(T_{\varepsilon/2} + T'_{\varepsilon/2} + \varepsilon\omega, D) = 0$.

The cone $\mathcal{MN}$ contains $\mathcal{N}$, since smooth forms have zero Lelong numbers, and it is contained in $\mathcal{E}$: if α is modified nef, then $\alpha + \varepsilon\{\omega\}$ contains the positive current $T_\varepsilon + \varepsilon\omega$, and $\mathcal{E}$ is closed. It is also closed. Indeed, let $\alpha_j \to \alpha$ with $\alpha_j \in \mathcal{MN}$, and let $\varepsilon > 0$. By (14.10) we may choose j such that $\alpha - \alpha_j$ has a smooth representative η with $-\varepsilon\omega/2 \leq \eta \leq \varepsilon\omega/2$. If $T' \in \alpha_j$ satisfies $T' \geq -\varepsilon\omega/2$ and $\nu(T' + \varepsilon\omega/2, D) = 0$ for all D, then $T' + \eta \in \alpha$ satisfies $T' + \eta \geq -\varepsilon\omega$, and $\nu(T' + \eta + \varepsilon\omega, D) = 0$ for all D, since adding smooth forms does not change Lelong numbers.

As for the classes of Definition 14.3.1, the definition of modified nef classes does not depend on ω. Indeed, let ω' be another Kähler form and $C \geq 1$ with $\omega \leq C\omega'$ (Lemma 13.1.1). For every $\varepsilon > 0$ the current $T = T_{\varepsilon/C}$ satisfies $T \geq -(\varepsilon/C)\omega \geq -\varepsilon\omega'$, and the positive currents $T + \varepsilon\omega'$ and $T + (\varepsilon/C)\omega$ differ by a smooth form, so that $\nu(T + \varepsilon\omega', D) = \nu(T + (\varepsilon/C)\omega, D) = 0$ for every prime divisor D.

In the definition of modified nef classes we may take T_ε with gentle analytic singularities whose pole set has codimension at least two. Indeed, write $T_{\varepsilon/2} = \theta + \mathrm{dd^c}\varphi$ with $\varphi \in \mathrm{PSH}(X, \theta + \varepsilon\omega/2)$ and apply Lemma 13.4.1 to φ with the background $\theta + \varepsilon\omega/2$, choosing m with $c/m < \varepsilon/2$. Then $\theta + \mathrm{dd^c}G_m \geq -\varepsilon\omega$. Since $\nu(\varphi, D) = \nu(T_{\varepsilon/2} + \varepsilon\omega/2, D) = 0$ for every prime divisor D, at a point x of D where $\nu(\varphi, x) = \nu(\varphi, D)$ the ideal $\mathcal{I}(m\varphi)_x$ is the unit ideal by Corollary 12.3.2. Hence the pole set of G_m, which is the zero set of $\mathcal{I}(m\varphi)$, contains no prime divisor. It is not all of X, since $\mathcal{I}(m\varphi)_x$ is the unit ideal at every point x with $\varphi(x) > -\infty$, where $\nu(\varphi, x) = 0$ (Proposition 12.1.1); so its irreducible components have codimension at least one, those of codimension one would be prime divisors, and it has codimension at least two. Moreover G_m is smooth near such a point x, so the Lelong numbers of $\theta + \mathrm{dd^c}G_m + \varepsilon\omega$ vanish there, and $\nu(\theta + \mathrm{dd^c}G_m + \varepsilon\omega, D) = 0$ for every prime divisor D. Thus $T_\varepsilon = \theta + \mathrm{dd^c}G_m$ has the required properties.

Corollary 14.3.9 *Let $\varphi \in \mathrm{PSH}(X, \theta)$, let $D_1, \ldots, D_N$ be distinct prime divisors and $a_1, \ldots, a_N > 0$ real numbers, and put $D = \sum_i a_i D_i$. Suppose that φ has logarithmic singularities along D with bounded remainder: near each point, $\varphi = \sum_i a_i \log|s_i|^2 + v$ with v bounded, where s_i is a local equation of D_i (Lemma A.2.16), and $s_i = 1$ if the point does not lie on D_i. Then $\{\theta\} - \{D\}$ is nef, where $\{D\} = \sum_i a_i\{[D_i]\}$. If $\theta + \mathrm{dd^c}\varphi$ is a Kähler current, then $\{\theta\} - \{D\}$ is Kähler.*

Proof Near a point, let h be smooth with $\mathrm{dd^c}h = \theta$. Off the support of D the function $v + h = (\varphi + h) - \sum_i a_i \log|s_i|^2$ is plurisubharmonic, since $\log|s_i|^2$ is pluriharmonic there, and it is bounded above. On a connected coordinate neighborhood of the point, the support of D lies in the zero set of the holomorphic function $\prod_i s_i \not\equiv 0$, so Theorem 8.4.1 extends $v + h$ to a plurisubharmonic function $\widetilde{v}$. The values of $\widetilde{v}$ on the support of D are upper limits of values of $v + h$, so $\widetilde{v}$ is bounded, and $\widetilde{v} = v + h$ outside the null set $\mathrm{Supp}\, D$. Since $\mathrm{dd^c}\log|s_i|^2 = [D_i]$ (Lemma A.2.16), the current $T = \theta + \mathrm{dd^c}\varphi$ satisfies $T - [D] = \mathrm{dd^c}\widetilde{v} \geq 0$ near the point, where $[D] = \sum_i a_i[D_i]$. Thus $T - [D]$ is a positive current in $\{\theta\} - \{D\}$ with locally bounded potentials. If $T \geq \delta\omega$, apply the same argument to $T - \delta\omega$, whose potential has the same logarithmic singularities; then $T - [D] \geq \delta\omega$. In both cases Proposition 14.3.5 gives the assertion. □

14.4 Minimal multiplicities and positivity loci

Let α be a class in the pseudo-effective cone $\mathcal{E}$ and $\theta \in \alpha$ a smooth representative. For every $\varepsilon > 0$ the class $\alpha + \varepsilon\{\omega\}$ is big, since it contains $T + \varepsilon\omega$ for every positive current $T \in \alpha$, and its potential $V_{\theta+\varepsilon\omega}$ with minimal singularities measures the positivity of α that remains after the curvature is allowed to drop by $\varepsilon\omega$. The loci where positivity fails are described in Theorem 14.4.4, and at the end of the section we compare positivity on X and on a modification of X.

Definition 14.4.1 For $\alpha \in \mathcal{E}$, $x \in X$ and a prime divisor $D \subset X$ put

$$\nu(\alpha,x)=\sup_{\varepsilon>0}\nu(V_{\theta+\varepsilon\omega},x),\qquad \nu(\alpha,D)=\sup_{\varepsilon>0}\nu(V_{\theta+\varepsilon\omega},D).$$

These are the *minimal multiplicities* of α. The *non-nef locus* of α is

$$E_{\mathrm{nn}}(\alpha)=\{x\in X:\nu(\alpha,x)>0\}.$$

If α is big, its *non-Kähler locus* is

$$E_{\mathrm{nK}}(\alpha)=\bigcap_T\{\varphi_T=-\infty\},$$

where $T=\theta+\mathrm{dd^c}\varphi_T$ ranges over the Kähler currents in α for which φ_T has analytic singularities. For a class $\alpha\in H^{1,1}(X,\mathbb{R})$ that is not big we put $E_{\mathrm{nK}}(\alpha)=X$, and for a class that is not pseudo-effective we put $E_{\mathrm{nn}}(\alpha)=X$.

If $0<\varepsilon<\varepsilon'$, then $\mathrm{PSH}(X,\theta+\varepsilon\omega)\subset\mathrm{PSH}(X,\theta+\varepsilon'\omega)$, so $V_{\theta+\varepsilon\omega}\le V_{\theta+\varepsilon'\omega}$. Hence the minimal potentials decrease as $\varepsilon\downarrow 0$, their Lelong numbers increase, and the suprema in the definition may be taken along $\varepsilon=1/j$. Replacing θ by another smooth representative $\theta+\mathrm{dd^c}g$ changes $V_{\theta+\varepsilon\omega}$ only by a bounded function, because $V_{\theta+\varepsilon\omega}-g$ and $V_{\theta+\mathrm{dd^c}g+\varepsilon\omega}$ both have minimal singularities in $\mathrm{PSH}(X,\theta+\mathrm{dd^c}g+\varepsilon\omega)$; and replacing ω by another Kähler form ω' with $C^{-1}\omega\le\omega'\le C\omega$ places $V_{\theta+\varepsilon\omega'}$ between $V_{\theta+C^{-1}\varepsilon\omega}$ and $V_{\theta+C\varepsilon\omega}$. Thus the minimal multiplicities and the two loci depend only on α.

If $T=\theta+\mathrm{dd^c}\varphi$ is a positive current in α, then $\varphi\in\mathrm{PSH}(X,\theta+\varepsilon\omega)$ for every $\varepsilon>0$, so $\varphi\le V_{\theta+\varepsilon\omega}+\sup_X\varphi$; hence

$$\nu(\alpha,x)\le\nu(T,x),\qquad \nu(\alpha,D)\le\nu(T,D),$$

and in particular both multiplicities are finite. For each j, the Lelong number $\nu(V_{\theta+\omega/j},x)$ equals its generic value $\nu(V_{\theta+\omega/j},D)$ at every point $x\in D$ outside a countable union of proper analytic subsets of D (Proposition 12.3.5 and the remarks before Proposition 14.2.8). At points of D outside the union of these countably many exceptional sets, which form a dense subset of D by Proposition 12.3.5, we therefore have $\nu(\alpha,x)=\nu(\alpha,D)$, while $\nu(\alpha,x)\ge\nu(\alpha,D)$ at every point of D. Consequently

$$\nu(\alpha,D)=\inf_{x\in D}\nu(\alpha,x),$$

and the infimum is attained at every point of D outside a countable union of proper analytic subsets.

Proposition 14.4.2 *Let $\alpha\in\mathcal{E}$. Then $\nu(\alpha,x)\le\nu(V_\theta,x)$ and $\nu(\alpha,D)\le\nu(V_\theta,D)$ for every $x\in X$ and every prime divisor D, with equality if α is big. Moreover, α is modified nef if and only if $\nu(\alpha,D)=0$ for every prime divisor D.*

Proof The positive current $T_0=\theta+\mathrm{dd^c}V_\theta\in\alpha$ gives the inequalities, by the bound preceding the proposition. Suppose that α is big, and choose $S\in\alpha$ with $S\ge\delta\omega$, $\delta>0$. For $\varepsilon>0$ put $T_\varepsilon=\theta+\mathrm{dd^c}V_{\theta+\varepsilon\omega}\ge-\varepsilon\omega$ and $t=\varepsilon/(\delta+\varepsilon)$. Then

$(1-t)T_\varepsilon + tS \geq \big(-(1-t)\varepsilon + t\delta\big)\omega = 0$ is a positive current in α, whose potential $(1-t)V_{\theta+\varepsilon\omega} + t\varphi_S$, with $S = \theta + \mathrm{dd^c}\varphi_S$, is more singular than V_θ. Since Lelong numbers are additive and positively homogeneous (Proposition 12.1.4) and do not decrease when a function is replaced by a more singular one, we obtain

$$\nu(V_\theta, x) \leq (1-t)\nu(V_{\theta+\varepsilon\omega}, x) + t\nu(S, x) \leq (1-t)\nu(\alpha, x) + t\nu(S, x).$$

As $\varepsilon \downarrow 0$ we have $t \to 0$, and since $\nu(S, x)$ is finite we obtain $\nu(V_\theta, x) \leq \nu(\alpha, x)$. For a prime divisor D, evaluate the same inequality at a point $x \in D$ at which the Lelong numbers of $V_{\theta+\varepsilon\omega}$ and of S attain their generic values along D (Proposition 12.3.5); it gives $\nu(V_\theta, D) \leq \nu(V_\theta, x) \leq (1-t)\nu(\alpha, D) + t\nu(S, D)$, and letting $\varepsilon \downarrow 0$ proves the equality for D.

If α is modified nef and T_ε is as in Definition 14.3.8, then $T_\varepsilon = \theta + \mathrm{dd^c}\varphi_\varepsilon$ with $\varphi_\varepsilon \in \mathrm{PSH}(X, \theta + \varepsilon\omega)$, so $\nu(V_{\theta+\varepsilon\omega}, D) \leq \nu(\varphi_\varepsilon, D) = 0$ for every $\varepsilon > 0$; hence $\nu(\alpha, D) = 0$. Conversely, if $\nu(\alpha, D) = 0$ for every D, then the currents $\theta + \mathrm{dd^c}V_{\theta+\varepsilon\omega}$ satisfy the conditions of Definition 14.3.8. □

Proposition 14.4.3 *For a fixed point $x \in X$ or a fixed prime divisor D, the minimal multiplicity $\alpha \mapsto \nu(\alpha, x)$, respectively $\alpha \mapsto \nu(\alpha, D)$, is positively homogeneous, convex and lower semicontinuous on the pseudo-effective cone $\mathcal{E}$, and continuous on the big cone $\mathcal{B}$.*

Proof We treat $\nu(\cdot, x)$. The proof for D is the same, except that the additivity of Lelong numbers along D is applied at a point of D at which all the functions involved attain their generic Lelong numbers along D (Proposition 12.3.5). For $t > 0$ we have $\mathrm{PSH}(X, t\theta + \varepsilon\omega) = t\mathrm{PSH}(X, \theta + (\varepsilon/t)\omega)$, hence $V_{t\theta+\varepsilon\omega} = tV_{\theta+(\varepsilon/t)\omega}$, and taking suprema over ε gives $\nu(t\alpha, x) = t\nu(\alpha, x)$. If $\alpha_1, \alpha_2 \in \mathcal{E}$ have representatives θ_1, θ_2, the sum of the minimal currents with allowance ε in the two classes is a current $\theta_1 + \theta_2 + \mathrm{dd^c}(V_{\theta_1+\varepsilon\omega} + V_{\theta_2+\varepsilon\omega}) \geq -2\varepsilon\omega$ in $\alpha_1 + \alpha_2$. By minimality and additivity of Lelong numbers,

$$\nu(V_{\theta_1+\theta_2+2\varepsilon\omega}, x) \leq \nu(V_{\theta_1+\varepsilon\omega}, x) + \nu(V_{\theta_2+\varepsilon\omega}, x) \leq \nu(\alpha_1, x) + \nu(\alpha_2, x),$$

and taking the supremum over ε gives subadditivity. Together with homogeneity this proves convexity.

Let $\alpha_j \to \alpha$ in $\mathcal{E}$, with representatives θ_j, and fix $\varepsilon > 0$. For large j, by (14.10), the class $\alpha - \alpha_j$ has a smooth representative η_j with $-\varepsilon\omega \leq \eta_j \leq \varepsilon\omega$. The current $\theta_j + \mathrm{dd^c}V_{\theta_j+\varepsilon\omega} + \eta_j \in \alpha$ is at least $-2\varepsilon\omega$, and its potential relative to θ differs from $V_{\theta_j+\varepsilon\omega}$ by a smooth function. By minimality,

$$\nu(V_{\theta+2\varepsilon\omega}, x) \leq \nu(V_{\theta_j+\varepsilon\omega}, x) \leq \nu(\alpha_j, x)$$

for large j, so $\nu(V_{\theta+2\varepsilon\omega}, x) \leq \varliminf_j \nu(\alpha_j, x)$. Taking the supremum over ε proves lower semicontinuity.

Finally, a convex function f with finite values on an open convex subset U of a finite-dimensional real vector space is continuous. Indeed, let $Q \subset U$ be a small closed cube; by convexity the function is bounded above on Q by the

maximum M of its values at the vertices. If c is the center of Q and $c \pm v \in Q$, then $f(c+v) \geq 2f(c) - f(c-v) \geq 2f(c) - M$, so f is also bounded below on Q. If y, z lie in the concentric cube of half the size, extend the segment from z to y beyond y to the point w at distance s from y, where s is half the side length of the smaller cube, so that $w \in Q$. The slope inequality for convex functions gives $f(y) - f(z) \leq |y-z|(f(w) - f(y))/|w-y| \leq |y-z|(M - \inf_Q f)/s$, and by symmetry f is Lipschitz on the smaller cube. Applied to the open convex cone $\mathcal{B}$, this proves continuity there. □

Theorem 14.4.4 *Let $\alpha \in \mathcal{E}$. Then the non-nef locus $E_{\mathrm{nn}}(\alpha)$ is a countable union of proper analytic subsets of X, and it is empty if and only if α is nef. If α is big, then $E_{\mathrm{nK}}(\alpha)$ is the pole set of the potential of a single Kähler current in α with analytic singularities; it is a proper analytic subset of X, it contains $E_{\mathrm{nn}}(\alpha)$, and it is empty if and only if α is Kähler. Finally, for every class $\alpha \in H^{1,1}(X, \mathbb{R})$,*

$$E_{\mathrm{nK}}(\alpha) = \bigcap_{\varepsilon > 0} E_{\mathrm{nn}}(\alpha - \varepsilon\{\omega\}). \tag{14.15}$$

Proof *The non-nef locus.* By the monotonicity in ε, $E_{\mathrm{nn}}(\alpha)$ is the union over positive integers j, k of the sets $\{x : \nu(V_{\theta+\omega/j}, x) \geq 1/k\}$. Each of them is analytic by Theorem 12.3.4, and proper, because it is contained in the pole set of $V_{\theta+\omega/j}$. If α is nef, then $\alpha + \varepsilon\{\omega\}$ is Kähler for every $\varepsilon > 0$, so $V_{\theta+\varepsilon\omega}$ is bounded by Corollary 14.3.6 and all these Lelong numbers vanish. Conversely, suppose that $\nu(V_{\theta+\varepsilon\omega}, x) = 0$ for all $x \in X$ and $\varepsilon > 0$. By Corollary 12.3.2, $\mathcal{I}(mV_{\theta+\varepsilon\omega})$ is the unit ideal at every point, so the approximations G_m of Lemma 13.4.1 for the background $\theta + \varepsilon\omega$ are smooth and satisfy $\theta + \mathrm{dd^c} G_m \geq -(\varepsilon + c/m)\omega$. Hence α is nef.

Realization of the non-Kähler locus. Let α be big. Kähler currents with analytic singularities exist in α by Proposition 14.3.5, and their pole sets are closed analytic subsets of X. Since X has a countable basis, countably many of the open complements of these pole sets already have the same union as all of them: for every element of a countable basis that is contained in one of these complements, choose one such complement. The intersection of the corresponding countably many pole sets $Z_1, Z_2, \ldots$ is therefore $E_{\mathrm{nK}}(\alpha)$. Let $\mathcal{I}_{Z_k}$ be the ideal sheaf of Z_k, which is coherent by Theorem A.2.8. The increasing sequence of coherent ideals $\mathcal{I}_{Z_1} + \cdots + \mathcal{I}_{Z_k}$, whose zero sets are $Z_1 \cap \cdots \cap Z_k$, is stationary on the compact manifold X by Lemma 12.2.4. Hence $E_{\mathrm{nK}}(\alpha) = Z_1 \cap \cdots \cap Z_N$ for some N.

Let $T_k = \theta + \mathrm{dd^c}\varphi_k$, $k \leq N$, be the corresponding Kähler currents, and choose $\delta > 0$ with $T_k \geq \delta\omega$ for all $k \leq N$. Then $\varphi = \max_k \varphi_k$ belongs to $\mathrm{PSH}(X, \theta - \delta\omega)$, it has analytic singularities by Proposition 13.1.7, and its pole set is exactly $Z_1 \cap \cdots \cap Z_N = E_{\mathrm{nK}}(\alpha)$. Thus $\theta + \mathrm{dd^c}\varphi$ is a Kähler current realizing $E_{\mathrm{nK}}(\alpha)$, which is a proper analytic subset because φ is finite almost everywhere. If $\nu(\alpha, x) > 0$, then $\nu(\varphi, x) \geq \nu(V_\theta, x) = \nu(\alpha, x) > 0$ by minimality of V_θ and Proposition 14.4.2, so $\varphi(x) = -\infty$; hence $E_{\mathrm{nn}}(\alpha) \subset E_{\mathrm{nK}}(\alpha)$.

If $E_{\mathrm{nK}}(\alpha) = \varnothing$, the potential φ just constructed has analytic singularities and no poles, so it is bounded, and Proposition 14.3.5 shows that α is Kähler. Conversely, if

α contains a Kähler form $\theta + \mathrm{dd^c}u$, then u is smooth and has analytic singularities without poles, so $E_{\mathrm{nK}}(\alpha) = \varnothing$.

The identity (14.15). Suppose first that $x \notin E_{\mathrm{nK}}(\alpha)$. Then α is big, and there is a Kähler current $\theta + \mathrm{dd^c}\varphi \geq \delta\omega$ with analytic singularities and $\varphi(x) > -\infty$; in particular φ is bounded near x and $\nu(\varphi, x) = 0$. For $0 < \varepsilon < \delta$, the current $\theta - \varepsilon\omega + \mathrm{dd^c}\varphi$ is positive and lies in $\alpha - \varepsilon\{\omega\}$, which is therefore pseudo-effective, and the bound $\nu(\alpha - \varepsilon\{\omega\}, x) \leq \nu(\varphi, x) = 0$ shows that $x \notin E_{\mathrm{nn}}(\alpha - \varepsilon\{\omega\})$.

Conversely, suppose that $x \notin E_{\mathrm{nn}}(\alpha - \varepsilon\{\omega\})$ for some $\varepsilon > 0$. By the convention in Definition 14.4.1, the class $\alpha - \varepsilon\{\omega\}$ is pseudo-effective. For $0 < \eta < \varepsilon/2$ put $u = V_{\theta - (\varepsilon - \eta)\omega}$; then $\nu(u, x) \leq \nu(\alpha - \varepsilon\{\omega\}, x) = 0$. By Corollary 12.3.2, $\mathcal{I}(mu)$ is the unit ideal near x for every m, and the approximation G_m of Lemma 13.4.1 for u and the background $\theta - (\varepsilon - \eta)\omega$ is smooth near x and satisfies

$$\theta + \mathrm{dd^c}G_m \geq (\varepsilon - \eta - c/m)\omega.$$

For large m this is a Kähler current in α with analytic singularities whose potential is finite at x. Hence α is big and $x \notin E_{\mathrm{nK}}(\alpha)$. This proves (14.15); the argument also shows that α is big as soon as the right-hand side of (14.15) is not all of X, so the formula is consistent with the convention $E_{\mathrm{nK}}(\alpha) = X$ for classes that are not big. □

We now turn to modifications. Let $\pi\colon Y \to X$ be a proper bimeromorphic holomorphic map from a Kähler manifold Y, with Kähler form ω_Y. By Proposition 14.2.8, Y is compact and connected, so the results above apply on Y. Pullback of smooth forms induces a linear map $\pi^*\colon H^{1,1}(X, \mathbb{R}) \to H^{1,1}(Y, \mathbb{R})$. Pushforward of currents, $(\pi_* S)(\chi) = S(\pi^*\chi)$, maps closed real $(1,1)$-currents on Y to closed real $(1,1)$-currents on X, preserves positivity, and satisfies $\pi_* \mathrm{dd^c} f = \mathrm{dd^c}\pi_* f$ for distributions f (Proposition 7.5.1); hence it induces a linear map $\pi_*\colon H^{1,1}(Y, \mathbb{R}) \to H^{1,1}(X, \mathbb{R})$ with $\pi_*\{S\} = \{\pi_* S\}$. If $T = \theta + \mathrm{dd^c}\varphi$ is a closed positive current on X with $\varphi \in \mathrm{PSH}(X, \theta)$, the pullbacks $(\varphi + h) \circ \pi$ of its local potentials, where $\mathrm{dd^c}h = \theta$, are not identically $-\infty$, because $\varphi \circ \pi \in \mathrm{PSH}(Y, \pi^*\theta)$ by Proposition 14.2.8. Hence the pullback π^*T of Proposition 7.5.4 is defined, and

$$\pi^*T = \pi^*\theta + \mathrm{dd^c}(\varphi \circ \pi), \qquad \{\pi^*T\} = \pi^*\{T\}.$$

For a prime divisor $E \subset Y$, the image $\pi(E)$ is an analytic subset of X by Remmert's theorem (Theorem A.6.2), and it is irreducible: if $\pi(E) = A_1 \cup A_2$ with analytic subsets A_1, A_2, then $E = (E \cap \pi^{-1}(A_1)) \cup (E \cap \pi^{-1}(A_2))$, so $E \subset \pi^{-1}(A_i)$ for some i and $\pi(E) = A_i$. The divisor E is *exceptional* for π if $\pi(E)$ has codimension at least two in X.

Fix an analytic subset $A \subset X$ of codimension at least two outside which π is an isomorphism (Proposition A.7.3). We show that a prime divisor E of Y is exceptional if and only if $E \subset \pi^{-1}(A)$. If $E \subset \pi^{-1}(A)$, then $\pi(E) \subset A$ and E is exceptional. Otherwise $E \cap \pi^{-1}(A)$ is an analytic subset of E different from E; it contains no nonempty open subset of the connected manifold E_{reg}, by the identity theorem, so it is nowhere dense in E. Then π maps the dense open subset $E \setminus \pi^{-1}(A)$ of E, a complex hypersurface of $Y \setminus \pi^{-1}(A)$, isomorphically onto $\pi(E) \setminus A$, which is

therefore a complex hypersurface of $X \setminus A$ and is dense in $\pi(E)$, because $\pi(E)$ is the image of the closure of $E \setminus \pi^{-1}(A)$ under the closed map π. Hence $\pi(E)$ is an irreducible analytic subset of codimension one, that is, a prime divisor D, and E is not exceptional; this proves the claim. Moreover, E is the strict transform of D, since the closure of $\pi^{-1}(D \setminus A) = E \setminus \pi^{-1}(A)$ is E. By the claim and (14.14), applied on Y, every exceptional prime divisor is an irreducible component of $\pi^{-1}(A)$, while every other prime divisor E of Y is the strict transform of the prime divisor $\pi(E)$ of X, and π is an isomorphism near the points of E outside $\pi^{-1}(A)$.

Proposition 14.4.5 *Let $\pi\colon Y \to X$ be as above. Then π^* maps nef classes to nef classes, and π_* maps modified nef classes to modified nef classes. The pushforward of a non-divisorial closed positive current on Y is non-divisorial.*

Proof By Lemma 13.1.1 on the compact manifold Y, $\pi^*\omega \le C\omega_Y$ for a constant C. If $\theta_\varepsilon \ge -\varepsilon\omega$ are nef representatives of a class on X, then $\pi^*\theta_\varepsilon \ge -C\varepsilon\omega_Y$, so the pullback class is nef. Let S be a non-divisorial closed positive current on Y and $D \subset X$ a prime divisor, with strict transform D'. Over $X \setminus A$ the map π is a biholomorphism, which identifies π_*S with S and $D \setminus A$ with $D' \setminus \pi^{-1}(A)$, and Lelong numbers are invariant under biholomorphisms (Proposition 12.1.1). The generic value of the Lelong numbers of S along D' is attained on a dense subset of D' (Proposition 12.3.5), hence at some point y of the nonempty open subset $D' \setminus \pi^{-1}(A)$. Therefore $\nu(\pi_*S, D) \le \nu(\pi_*S, \pi(y)) = \nu(S, y) = \nu(S, D') = 0$. Hence π_*S is non-divisorial.

Let β be a modified nef class on Y and $t > 0$. The class $\beta + t\{\omega_Y\}$ is big, and it is modified nef because $\mathcal{MN}$ is a convex cone containing the Kähler classes. By Proposition 14.4.2, a positive current with minimal singularities in this class is non-divisorial. Its pushforward is a non-divisorial positive current in $\pi_*\beta + t\pi_*\{\omega_Y\}$, so this class is modified nef: the definition is satisfied with T_ε equal to this current for every ε. These classes tend to $\pi_*\beta$ as $t \downarrow 0$, and $\mathcal{MN}$ is closed, so $\pi_*\beta$ is modified nef. □

Lemma 14.4.6 *Let $\pi\colon Y \to X$ be as above, and let S be a non-divisorial closed positive $(1,1)$-current on Y. Then*

$$\pi^*\pi_*S = S + \sum_E c_E[E], \qquad c_E \ge 0,$$

where E ranges over the finitely many exceptional prime divisors of π. For every closed positive $(1,1)$-current T on X, $\pi_*\pi^*T = T$. For a prime divisor E on Y, $\pi_*[E] = 0$ if E is exceptional, and $\pi_*[E] = [\pi(E)]$ otherwise.

Proof Let A be as above. The positive current $\pi^*\pi_*S$, defined through local potentials as above, agrees with S on $Y \setminus \pi^{-1}(A)$. Their difference is a closed real current with measure coefficients supported on the proper analytic subset $\pi^{-1}(A)$ of Y. By Theorem A.2.17, it is a finite real linear combination of the currents of integration along the irreducible components of codimension one of the compact analytic set $\pi^{-1}(A)$, which are exactly the exceptional prime divisors, as shown above; there are finitely many of them by Corollary A.2.13.

It remains to show that the coefficients c_E are nonnegative. Near a regular point of an exceptional prime divisor E that lies on no other component, the difference is $c_E[E] = \mathrm{dd}^{\mathrm{c}}(c_E \log|s_E|^2)$, where s_E is a local equation of E (Lemma A.2.16), so local potentials u' of $\pi^*\pi_*S$ and u of S satisfy $u' = u + c_E \log|s_E|^2 + p$ with p pluriharmonic. The additivity of Lelong numbers (Proposition 12.1.4), applied to this identity if $c_E \geq 0$ and to $u = u' + |c_E| \log|s_E|^2 - p$ otherwise, together with $\nu(\log|s_E|^2, y) = 1$, gives $\nu(u', y) = \nu(u, y) + c_E$ at such points y. At points that are moreover generic for both currents, which exist because the points of E excluded by these conditions form a countable union of proper analytic subsets of E (Proposition 12.3.5), this reads $\nu(\pi^*\pi_*S, E) = \nu(S, E) + c_E$; hence $c_E = \nu(\pi^*\pi_*S, E) \geq 0$, since S is non-divisorial.

Since π is a biholomorphism over $X \setminus A$, the positive current $\pi_*\pi^*T$ agrees with T on $X \setminus A$. Hence $\pi_*\pi^*T - T$ is a closed real current with measure coefficients (Theorem 7.2.5) supported on A, which has codimension at least two, so it vanishes by Theorem A.2.17. For a prime divisor E on Y, the current $\pi_*[E]$ is positive, closed and supported on the analytic set $\pi(E)$. If E is exceptional, $\pi(E)$ has codimension at least two and $\pi_*[E] = 0$ by the same theorem. Otherwise π maps $E \setminus \pi^{-1}(A)$ isomorphically onto $\pi(E) \setminus A$, so $\pi_*[E] = [\pi(E)]$ on $X \setminus A$, and the closed current $\pi_*[E] - [\pi(E)]$ of order zero is supported on $\pi(E) \cap A$, of codimension at least two in X; it vanishes by the same theorem. □

Proposition 14.4.7 *Let $\pi : Y \to X$ be as above. Then π^* and π_* map big classes to big classes.*

Proof Choose C with $\pi^*\omega \leq C\omega_Y$. Since π_* preserves positivity, Lemma 14.4.6 gives $\pi_*\omega_Y \geq C^{-1}\pi_*\pi^*\omega = C^{-1}\omega$. Hence a current $S \geq \delta\omega_Y$ on Y has $\pi_*S \geq \delta C^{-1}\omega$, which proves the assertion for π_*.

Let α be big on X and $T \in \alpha$ with $T \geq \delta\omega$. Choose a smooth closed representative η of the class $\{\pi_*\omega_Y\}$, a constant $B > 0$ with $\eta \leq B\omega$ (Lemma 13.1.1), and $t > 0$ with $tB \leq \delta$, so that $T - t\eta \geq 0$. The positive current $T - t\eta + t\pi_*\omega_Y$ lies in α. By Lemma 14.4.6, applied to the smooth, hence non-divisorial, current ω_Y, its pullback is

$$\pi^*(T - t\eta) + t\omega_Y + t\sum_E c_E[E] \geq t\omega_Y.$$

This is a Kähler current in $\pi^*\alpha$, so $\pi^*\alpha$ is big. □

Example 14.4.8 The zero class $\alpha = 0$ is nef, so $E_{\mathrm{nn}}(0) = \varnothing$ by Theorem 14.4.4. It is not big: a Kähler current $T \geq \delta\omega$ would have $\int_X T \wedge \omega^{n-1} \geq \delta \int_X \omega^n > 0$, whereas this integral vanishes for every closed current in the zero class. Hence $E_{\mathrm{nK}}(0) = X$. More generally, $E_{\mathrm{nK}}(\alpha) = X$ for every class that is not big, while $E_{\mathrm{nn}}(\alpha) = \varnothing$ for every nef class; on the boundary of the pseudo-effective cone the two loci therefore detect different failures of positivity. This is also why the minimal multiplicities are defined with the allowances $\varepsilon\omega$ rather than through V_θ: for a class that is not big, Proposition 14.4.2 gives only the inequality $\nu(\alpha, x) \leq \nu(V_\theta, x)$, since the proof of equality uses a Kähler current in α.

14.5 Curvature and metrics defined by sections

This section collects the notions for line bundles on X used in the rest of the chapter, from the curvature class to the Fubini–Study metrics defined by holomorphic sections, and shows that spaces of global holomorphic sections are finite dimensional.

Let $L \to X$ be a holomorphic line bundle, given as in Definition 13.1.8 by frames e_i over an open cover (V_i) of X, and let h_0 be a smooth quadratic Hermitian metric on L. Its curvature form, which we denote by $c_1(L, h_0)$, equals $-\mathrm{dd^c} \log |e_i|^2_{h_0}$ on V_i. It also equals $-\mathrm{dd^c} \log |e|^2_{h_0}$ for every local holomorphic frame e of L: on V_i we have $e = ge_i$ with g holomorphic and nowhere zero, so $\log |e|^2_{h_0} = \log |g|^2 + \log |e_i|^2_{h_0}$, and $\log |g|^2$ is pluriharmonic (Section 7.3). Any other smooth metric is $h_1 = h_0 \mathrm{e}^{-f}$ with f smooth and real, and then $c_1(L, h_1) = c_1(L, h_0) + \mathrm{dd^c} f$ (Lemma 13.1.9(3)). Hence the class $c_1(L) = \{c_1(L, h_0)\} \in H^{1,1}(X, \mathbb{R})$ does not depend on the metric. A holomorphic section of L is given by holomorphic functions f_i on V_i with $f_i e_i = f_j e_j$ on $V_i \cap V_j$.

If L and L' are given by frames e_i and e'_i over a common cover, with transition functions g_{ij} and g'_{ij}, their tensor product $L \otimes L'$ is given by the frames $e_i \otimes e'_i$ with the transition functions $g_{ij} g'_{ij}$, and the dual L^{-1} of L by the dual frames e^*_i with the transition functions g^{-1}_{ij}. Metrics on L and L' induce metrics with $|e_i \otimes e'_i|^2 = |e_i|^2 |e'_i|^2$ and $|e^*_i|^2 = |e_i|^{-2}$. The curvature of a tensor product of metrics is the sum of the curvatures, and the curvature of a pullback metric is the pullback of the curvature, because the logarithms of the squared norms of local frames add, respectively compose.

Line bundles are trivial over coordinate balls; we use this to work in a single frame on balls of a fixed size.

Lemma 14.5.1 *Let L be a holomorphic line bundle on a complex manifold M, and let B be a coordinate ball whose closure is contained in the domain of its chart. Then L has a holomorphic frame over B, that is, a nowhere vanishing holomorphic section of L on B.*

Proof We make the metric flat on B and obtain the frame by parallel transport along the segments issuing from the center of B. Choose a smooth metric h_0 on L (Lemma 13.1.9(1)). By Lemma 13.1.2 there is a smooth real function p on B with $\mathrm{dd^c} p = c_1(L, h_0)$, and the metric $h_1 = h_0 \mathrm{e}^p$ on $L|_B$ has curvature $c_1(L, h_0) - \mathrm{dd^c} p = 0$, by Lemma 13.1.9(3). We call a holomorphic frame of h_1-length one over an open subset of B a *unit frame*. Unit frames exist near every point of B. Indeed, every point of B has a ball neighborhood $V \subset B$ over which L has a holomorphic frame e, and $\log |e|^2_{h_1}$ is smooth and pluriharmonic, since $\mathrm{dd^c} \log |e|^2_{h_1} = -c_1(L, h_1) = 0$, so by Lemma 13.1.2 it equals $\mathrm{Re}\, F$ for a holomorphic function F on V, and $\mathrm{e}^{-F/2} e$ is a unit frame. Two unit frames on a connected open set differ by a holomorphic factor g with $|g| = 1$; then $\bar{g} = 1/g$ is both holomorphic and antiholomorphic, so $\mathrm{d}\bar{g} = 0$ and g is constant.

Let c be the center of B and fix a unit vector $e_c \in L_c$. For $z \in B$, a *flat lift* along the segment from c to z is a continuous map $t \mapsto \varepsilon(t) \in L_{c+t(z-c)}$, $t \in [0,1]$, with $\varepsilon(0) = e_c$, which near every t is a constant multiple of a unit frame. There is at most one flat lift: if ε' is another one, then $\varepsilon' = g\varepsilon$ for a continuous function g on $[0,1]$ which is locally constant, by the preceding paragraph, and equals 1 at $t = 0$, so $g = 1$.

It remains to show that flat lifts exist and that their values at $t = 1$ depend holomorphically on z. Let $z_0 \in B$, and choose $\delta_0 > 0$ with $\overline{B(z_0, \delta_0)} \subset B$. The set $\Gamma_0 = \{c + t(z-c) : t \in [0,1],\ |z - z_0| \le \delta_0\}$ is a compact subset of the convex set B. Cover it by finitely many balls $V_i \subset B$ carrying unit frames e_i, let $\lambda > 0$ be a Lebesgue number of this cover, choose $\delta \le \delta_0$ with $2\delta < \lambda/2$, and choose $0 = t_0 < t_1 < \cdots < t_N = 1$ with $(t_k - t_{k-1})(|z_0 - c| + \delta_0) < \lambda/2$. For $W = B(z_0, \delta)$, each set $\Gamma_k = \{c + t(z-c) : t \in [t_{k-1}, t_k],\ z \in W\}$ has diameter less than λ, because $|(c + t(z-c)) - (c + t'(z'-c))| \le |t - t'|\,|z - c| + |z - z'|$; so Γ_k lies in some V_{i_k}. The convex set $V_{i_k} \cap V_{i_{k+1}}$ is connected and contains $c + t_k(z_0 - c)$, so $e_{i_k} = \gamma_k e_{i_{k+1}}$ there for a constant γ_k. Let a be the number with $e_c = a\, e_{i_1}(c)$. For $z \in W$, the map equal to $a\gamma_1 \cdots \gamma_{k-1} e_{i_k}(c + t(z-c))$ for $t \in [t_{k-1}, t_k]$ is well defined at the points t_k and is a flat lift along the segment from c to z. Hence, on W, the value $E(z)$ of the unique flat lift at $t = 1$ is $a\gamma_1 \cdots \gamma_{N-1} e_{i_N}(z)$, a constant multiple of the unit frame e_{i_N}. Since z_0 was arbitrary, E is a nowhere vanishing holomorphic section of L on B. □

Families of holomorphic sections that are bounded in L^2 are relatively compact for uniform convergence.

Lemma 14.5.2 (1) *Let $U \subset \mathbb{C}^n$ be open, $n \ge 1$, and let (f_j) be a sequence of holomorphic functions on U with $\sup_j \int_U |f_j|^2\, \mathrm{d}V < \infty$. Then a subsequence converges locally uniformly on U to a holomorphic function.*

(2) *Let E be a holomorphic line bundle on X with a smooth quadratic metric h_E, and let (s_j) be a sequence of holomorphic sections of E over X with $\sup_j \int_X |s_j|^2_{h_E}\, \mathrm{d}V < \infty$. Then a subsequence converges uniformly on X to a holomorphic section s of E, that is, $\sup_X |s_j - s|_{h_E} \to 0$ along the subsequence.*

(3) *For every holomorphic line bundle E on X, the space $H^0(X, E)$ of holomorphic sections of E over X is finite dimensional.*

Assertions (2) and (3) hold more generally on every compact complex manifold X' of positive dimension in place of X, with $\mathrm{d}V$ in (2) replaced by any smooth positive volume form on X'.

Proof (1) Let M be the supremum in the statement. If $\overline{B(z,\delta)} \subset U$, then $|f_j(z)|^2 \le |B(z,\delta)|^{-1} M$ by (12.7). Hence the f_j are uniformly bounded on every compact subset of U: if $K \subset U$ is compact and $\delta > 0$ is smaller than the distance from K to $\mathbb{C}^n \setminus U$ (any $\delta > 0$ if $U = \mathbb{C}^n$), then $|f_j|^2 \le |B(0,\delta)|^{-1} M$ on K.

We next show that the sequence is equicontinuous on compact sets. Let $K \subset U$ be compact, and choose $r > 0$ such that the compact set K' of points at distance at most $2\sqrt{n}\, r$ from K lies in U; let M' be a bound for all $|f_j|$ on K'. For z at distance at most $\sqrt{n}\, r$ from K, the closed polydisc of radius r centered at z lies in K', and the coefficient bound in (7.1) gives $|\partial_{z_l} f_j(z)| \le M'/r$ for $1 \le l \le n$. Since $\mathrm{d}f_j = \sum_l \partial_{z_l} f_j\, \mathrm{d}z_l$, the

f_j are Lipschitz with a common constant on every ball of radius $\sqrt{n}\,r$ centered at a point of K, so the sequence is equicontinuous on K. By Proposition 1.6.4, applied on the compact sets $K_m = \{z \in U : |z| \le m,\ \operatorname{dist}(z, \mathbb{C}^n \setminus U) \ge 1/m\}$, which exhaust U, and a diagonal argument, a subsequence converges locally uniformly on U to a continuous function f. Then $\bar\partial f = 0$ in the sense of distributions, as a limit of $\bar\partial f_j = 0$, so f agrees almost everywhere with a holomorphic function by Lemma 11.3.5(1); both being continuous, f is holomorphic.

(2) Choose finitely many coordinate balls B_k, with closures in the domains of their charts, such that the concentric balls B_k' of half the radius cover X, and let B_k'' be the concentric balls of three quarters of the radius. By Lemma 14.5.1, E has a holomorphic frame e_k over B_k; write $s_j = f_{j,k} e_k$ on B_k. On the compact set $\overline{B_k''}$ the function $|e_k|^2_{h_E}$ is bounded below by a positive constant and $\mathrm{d}V$ is bounded below by a positive multiple of Lebesgue measure in the chart, so the integrals of $|f_{j,k}|^2$ over B_k'' with respect to Lebesgue measure in the chart are bounded independently of j. Applying (1) successively for the finitely many k, we obtain a subsequence along which $f_{j,k}$ converges locally uniformly on B_k'', hence uniformly on $\overline{B_k'}$, to a holomorphic function f_k, for every k. On $B_k'' \cap B_l''$ the sections $f_k e_k$ and $f_l e_l$ are both the pointwise limit of s_j, so they define a holomorphic section s of E over X, and $\sup_{B_k'} |s_j - s|_{h_E} \le \sup_{\overline{B_k'}} |e_k|_{h_E} \cdot \sup_{\overline{B_k'}} |f_{j,k} - f_k| \to 0$ for every k.

(3) Fix a smooth quadratic metric h_E on E (Lemma 13.1.9(1)). The integral $\|s\|^2 = \int_X |s|^2_{h_E}\,\mathrm{d}V$ defines an inner product on $H^0(X, E)$: if $\|s\| = 0$, then the continuous function $|s|_{h_E}$ vanishes almost everywhere, hence everywhere. If $H^0(X, E)$ were infinite dimensional, the Gram–Schmidt process would give an orthonormal sequence (s_j). By (2), a subsequence converges uniformly on X, hence in L^2, since X has finite volume; but $\|s_i - s_j\|^2 = 2$ for $i \ne j$. This contradiction proves (3).

The proofs of (2) and (3) use only that X is a compact complex manifold of positive dimension and that $\mathrm{d}V$ is a smooth positive volume form; hence they also prove the last assertion of the lemma. □

A line bundle is *positive* if it admits a smooth metric with positive definite curvature form. It is *very ample* if its full space of global holomorphic sections defines a holomorphic embedding into a projective space: this space is finite dimensional by Lemma 14.5.2(3), and for a basis $s_0, \dots, s_N$ we require that the sections have no common zero and that $x \mapsto [s_0(x) : \cdots : s_N(x)]$ is a holomorphic embedding of X into $\mathbb{P}^N$. A compact complex manifold is *projective* if it admits a holomorphic embedding into a projective space.

Definition 14.5.3 A *plurisubharmonic metric* on L is a quadratic metric of the form $h = h_0 \mathrm{e}^{-\varphi}$ with $\varphi \in \mathrm{PSH}(X, c_1(L, h_0))$. It assigns the value zero to the zero vector, also over the pole set of φ, and the value $+\infty$ to a nonzero vector over a pole of φ. Its *Chern current* is

$$c_1(L, h) = c_1(L, h_0) + \mathrm{dd}^{\mathrm{c}}\varphi.$$

This definition uses the pointwise function φ, that is, the canonical quasi-plurisubharmonic representative; for an arbitrary measurable weight, the distributional

curvature inequality would determine the metric only almost everywhere. If h_0 is replaced by $h_0 e^{-f}$, then φ is replaced by $\varphi - f$, so the class of plurisubharmonic metrics and the pointwise metric h do not depend on h_0. In particular, L admits a plurisubharmonic metric if and only if $c_1(L)$ is pseudo-effective. The tensor product of two plurisubharmonic metrics is defined by adding their local weights; the norm of a nonzero vector is infinite over a pole of either factor. The pullback of a plurisubharmonic metric by a holomorphic map is a plurisubharmonic metric provided the pulled-back weights are not identically $-\infty$ on any connected component; the restriction to a submanifold is defined under the same condition.

Proposition 14.5.4 *For a nonzero holomorphic section s of L and a smooth metric h_0,*

$$c_1(L, h_0) + \mathrm{dd^c} \log |s|^2_{h_0} = [Z(s)], \tag{14.16}$$

where $[Z(s)]$ is the current of integration along the zero divisor of s, counted with multiplicities. In particular $[Z(s)]$ represents $c_1(L)$.

Proof In a local frame write $s = fe$. Since $\log |s|^2_{h_0} = \log |f|^2 + \log |e|^2_{h_0}$, the left side equals $\mathrm{dd^c} \log |f|^2$, which is the current of integration along the zero divisor of f by Theorem 7.4.5. These local divisors agree on overlaps, because the transition functions have no zeros. □

Example 14.5.5 Let $N \geq 0$. The projective space $\mathbb{P}^N$ is the quotient of $\mathbb{C}^{N+1} \setminus \{0\}$ by the action of $\mathbb{C}^*$ by multiplication, with the quotient topology; we write q for the quotient map and $[Z] = [Z_0 : \cdots : Z_N]$ for the class of Z. For $0 \leq i \leq N$ the sets $U_i = \{[Z] : Z_i \neq 0\}$ are open and cover $\mathbb{P}^N$, and the map $\phi_i \colon U_i \to \mathbb{C}^N$, $[Z] \mapsto (Z_k/Z_i)_{k \neq i}$, is a homeomorphism: it is continuous because $\phi_i \circ q$ is, and its inverse is q composed with the map inserting the coordinate 1 in the ith place. On $\phi_i(U_i \cap U_j) = \{w_j \neq 0\}$, where we put $w_i = 1$, the transition map $\phi_j \circ \phi_i^{-1}$ is $w \mapsto (w_k/w_j)_{k \neq j}$, which is holomorphic, and so is its inverse $\phi_i \circ \phi_j^{-1}$. The space $\mathbb{P}^N$ is compact and connected, as the image under q of the unit sphere of $\mathbb{C}^{N+1}$. It is Hausdorff, because the continuous map $[Z] \mapsto Z\overline{Z}^{\mathsf{T}}/|Z|^2$ to the space of complex matrices is injective, the matrix being the orthogonal projection onto the line $\mathbb{C}Z$; and being compact, it is paracompact. Thus $\mathbb{P}^N$ is a connected compact complex manifold of dimension N (Definition 7.1.1).

The *tautological line bundle* $\mathcal{O}_{\mathbb{P}^N}(-1)$ has the fiber $\mathbb{C}Z$ over $[Z]$. In the sense of Definition 13.1.8 it is given by the frames $e_i([Z]) = Z/Z_i$ over U_i, the points of the line with ith coordinate one. Since $e_i = (Z_j/Z_i)e_j$ on $U_i \cap U_j$, its transition functions are $g_{ij} = Z_j/Z_i$, which are holomorphic and nowhere zero on $U_i \cap U_j$ and satisfy $g_{ij}g_{jk} = g_{ik}$. The Euclidean norm of $\mathbb{C}^{N+1}$ gives the smooth metric $|e_i|^2 = |Z|^2/|Z_i|^2$, which satisfies $|e_i|^2 = |g_{ij}|^2|e_j|^2$. The dual line bundle $\mathcal{O}_{\mathbb{P}^N}(1)$ has the dual frames e_i^*, with $e_i^*(e_i) = 1$ and transition functions g_{ij}^{-1}, and the dual metric $|e_i^*|^2 = |e_i|^{-2} = |Z_i|^2/|Z|^2$, the *Fubini–Study metric* h_{FS}. A linear form ℓ on $\mathbb{C}^{N+1}$ defines a holomorphic section s_ℓ of $\mathcal{O}_{\mathbb{P}^N}(1)$, the restriction of ℓ to the fibers of $\mathcal{O}_{\mathbb{P}^N}(-1)$: on U_i it is $s_\ell = \ell(e_i)e_i^* = (\ell(Z)/Z_i)\, e_i^*$, and these expressions agree

on overlaps, because $\ell(e_i)e_i^* = \ell(g_{ij}e_j)\,g_{ij}^{-1}e_j^*$. For the coordinate forms we write $X_k = s_{Z_k}$. Thus $X_i = e_i^*$ on U_i, and

$$|X_k|^2_{\mathrm{FS}}([Z]) = \frac{|Z_k|^2}{|Z|^2}.$$

The tensor power $\mathcal{O}_{\mathbb{P}^N}(k) = \mathcal{O}_{\mathbb{P}^N}(1)^{\otimes k}$, $k \geq 1$, has the frames $(e_i^*)^{\otimes k}$, the transition functions g_{ij}^{-k} and the metric h_{FS}^k; products of k sections of $\mathcal{O}_{\mathbb{P}^N}(1)$, such as the monomials $X_0^{a_0}\cdots X_N^{a_N}$ with $a_0+\cdots+a_N = k$, are holomorphic sections of $\mathcal{O}_{\mathbb{P}^N}(k)$.

Example 14.5.6 Let $\mathcal{O}_{\mathbb{P}^N}(1)$ carry the Fubini–Study metric h_{FS} of Example 14.5.5, dual to the Euclidean norm on the tautological line. On the chart $U_0 = \{[1 : z_1 : \cdots : z_N]\}$ the tautological line is spanned by $(1, z)$, so the frame $X_0 = e_0^*$ has squared norm $|X_0(1,z)|^2/|(1,z)|^2 = (1+|z|^2)^{-1}$, and the curvature is

$$\omega_{\mathrm{FS}} = \mathrm{dd^c}\log(1+|z|^2).$$

The matrix of $\partial_j\partial_{\bar{k}}\log(1+|z|^2)$ is $\big((1+|z|^2)\delta_{jk} - \bar{z}_j z_k\big)/(1+|z|^2)^2$, which is positive definite by the Cauchy–Schwarz inequality $|\sum_j \bar{z}_j\xi_j|^2 \leq |z|^2|\xi|^2$. The same computation in the chart U_i, with the frame X_i, shows that ω_{FS} is a Kähler form on $\mathbb{P}^N$; thus $\mathbb{P}^N$ is a compact Kähler manifold. On $\mathbb{P}^1$, integration in polar coordinates gives $\int_{\mathbb{P}^1}\omega_{\mathrm{FS}} = \frac{1}{\pi}\int_{\mathbb{C}}(1+|z|^2)^{-2}\,\mathrm{d}V = 1$. The function $\log|X_1|^2_{\mathrm{FS}} = \log\big(|z|^2/(1+|z|^2)\big)$, which vanishes at the point at infinity, is the weight of a singular plurisubharmonic metric whose Chern current is the Dirac current $[0]$ at the point $[1:0]$, by (14.16).

Definition 14.5.7 A *Fubini–Study metric* on L is a metric $h_0\mathrm{e}^{-\varphi}$ with

$$\varphi = \frac{1}{m}\max_{1\leq i\leq N}\big(\log|s_i|^2_{h_0^m} + \lambda_i\big), \qquad m, N \in \mathbb{Z}_{\geq 1}, \quad \lambda_i \in \mathbb{Q},$$

where $s_1, \ldots, s_N$ are holomorphic sections of L^m, not all zero. We denote the set of these metrics by $\mathrm{FS}(L)$. The set of *generalized Fubini–Study metrics* is the smallest set of plurisubharmonic metrics that contains $\mathrm{FS}(L)$ and is closed under minima of metrics and under convex combinations of weights with rational coefficients. A convex combination of weights corresponds to the weighted geometric mean of the quadratic metrics, with value zero on the zero vector and the value $+\infty$ on a nonzero vector over a pole of a factor with positive coefficient.

The classical Fubini–Study metric of Example 14.5.6, whose weight is the logarithm of a sum of squares rather than a maximum, does not belong to $\mathrm{FS}(\mathcal{O}_{\mathbb{P}^N}(1))$ when $N \geq 1$, since its curvature form ω_{FS} is positive definite, whereas the curvature current of a metric in $\mathrm{FS}(L)$ vanishes on a nonempty open set. Indeed, one of the finitely many closed sets on which a single term attains the maximum has interior points. There, away from the zeros of the corresponding section s_i, the induced metric on L^m gives s_i the constant norm $\mathrm{e}^{-\lambda_i/2}$, so the curvature vanishes by Proposition 14.5.4. The classical metric differs, however, only by a bounded factor from a metric in $\mathrm{FS}(\mathcal{O}_{\mathbb{P}^N}(1))$: with $h_0 = h_{\mathrm{FS}}$, the coordinate sections $X_0, \ldots, X_N$, $m = 1$ and $\lambda_i = 0$

give the metric $h_{\mathrm{FS}}\mathrm{e}^{-\varphi}$ with $\varphi = \max_i \log|X_i|^2_{\mathrm{FS}}$, and $1 \le \mathrm{e}^{-\varphi} \le N+1$ because $|X_i|^2_{\mathrm{FS}}([Z]) = |Z_i|^2/|Z|^2$.

Proposition 14.5.8 *Fubini–Study metrics do not depend on the choice of h_0, they have analytic singularities, and* $\mathrm{FS}(L)$ *is closed under minima and rational convex combinations of weights. Consequently the generalized Fubini–Study metrics are exactly the metrics in* $\mathrm{FS}(L)$*. The set* $\mathrm{FS}(L)$ *is empty if no positive tensor power of L has a nonzero holomorphic section, even when L admits plurisubharmonic metrics.*

Proof In a local frame e write $s_i = f_i e^m$. The local weight $-\log|e|^2_h = \varphi - \log|e|^2_{h_0}$ of the metric equals $m^{-1}\max_i(\log|f_i|^2 + \lambda_i)$, which does not involve h_0; it is plurisubharmonic and not identically $-\infty$, since X is connected and some $s_i \neq 0$. It differs by a bounded function from $m^{-1}\log\sum_i \mathrm{e}^{\lambda_i}|f_i|^2$, so it has analytic singularities with the rational coefficient $1/m$; the pole set is the common zero set of the s_i.

Consider two such metrics, given by sections s_i of L^m with constants λ_i and sections r_j of L^l with constants μ_j. Replacing s_i by $s_i^{M/m}$ and λ_i by $(M/m)\lambda_i$, and similarly for the r_j, where M is a common multiple of m and l, does not change the potentials, so we may assume that both are given in degree M. The maximum of the two potentials, which corresponds to the minimum of the metrics, is then given by the union of the two lists. For a rational $t = a/b \in (0,1)$, the combination $t\varphi + (1-t)\psi$ of the two maxima is the maximum over all pairs (i,j) of the corresponding combinations, that is,

$$t\varphi + (1-t)\psi = \frac{1}{Mb}\max_{i,j}\Big(\log\big|s_i^a r_j^{b-a}\big|^2_{h_0^{Mb}} + a\lambda_i + (b-a)\mu_j\Big),$$

where the sections $s_i^a r_j^{b-a}$ of L^{Mb} are not all zero and the constants are rational. Thus $\mathrm{FS}(L)$ is closed under both operations, and by induction under every finite sequence of them, so the generated set is $\mathrm{FS}(L)$ itself. If no positive tensor power of L has a nonzero section, no presentation as in Definition 14.5.7 exists. □

14.6 Compact interpolation and global Bergman kernels

The local extension theorems of Chapter 11 produce holomorphic functions with prescribed values on pseudoconvex domains. On the compact manifold X we obtain global holomorphic sections in the same way, by cutting off a local solution and correcting it with an L^2 estimate for $\bar\partial$ on line bundles; the curvature must dominate the cost of a logarithmic pole at the interpolation point. Since the weights are singular, we prove the L^2 estimate (Lemma 14.6.5) on complements of analytic subsets of X, starting from the Bochner–Kodaira identity. After the interpolation theorem at a point (Theorem 14.6.7) we describe the weighted spaces of global sections and their Bergman kernels.

All norms use the volume $\mathrm{d}V = \omega^n/n!$. The *canonical bundle* $K_X = \Lambda^{n,0}T^*X$ is the holomorphic line bundle of $(n,0)$-forms; we give it the metric induced by ω, and

its dual K_X^{-1} the dual metric $h_{K^{-1},\omega}$. Recall that $H^0(X,L)$ denotes the vector space of global holomorphic sections of L (Lemma 14.5.2).

We begin with the pointwise linear algebra behind the L^2 estimates of this chapter. Let σ be a Hermitian form on a complex manifold M of dimension $n \geq 1$. Forms on M carry the pointwise inner product defined by the Riemannian metric $\sigma(\cdot, J\cdot)$ as in Section 14.1; for a holomorphic line bundle F with a smooth quadratic metric h, F-valued forms carry the product of this inner product with h, written $\langle\cdot,\cdot\rangle_{\sigma,h}$ or $\langle\cdot,\cdot\rangle_\sigma$, and $\mathrm{d}V_\sigma = \sigma^n/n!$. Let Λ_σ be the pointwise adjoint of the Lefschetz operator $u \mapsto \sigma \wedge u$, and for a real $(1,1)$-form γ put $[\gamma, \Lambda_\sigma]u = \gamma \wedge \Lambda_\sigma u - \Lambda_\sigma(\gamma \wedge u)$. A basis $\theta_1, \dots, \theta_n$ of the $(1,0)$-covectors at a point x is a *σ-unitary coframe* if $\sigma = \mathrm{i}\sum_j \theta_j \wedge \bar\theta_j$ at x. By the spectral theorem for Hermitian matrices, for every real $(1,1)$-form γ there is a σ-unitary coframe at x in which $\gamma = \mathrm{i}\sum_j \lambda_j \theta_j \wedge \bar\theta_j$ at x, with real numbers λ_j, the eigenvalues of γ with respect to σ at x.

Lemma 14.6.1 *Let σ, F and h be as above, let γ be a real $(1,1)$-form on M, let $x \in M$, and let $\theta_1, \dots, \theta_n$ be a σ-unitary coframe at x in which $\gamma = \mathrm{i}\sum_j \lambda_j \theta_j \wedge \bar\theta_j$. Put $e = \theta_1 \wedge \cdots \wedge \theta_n$.*

(1) *For an F-valued $(n,1)$-covector $v = \sum_j v_j\, e \wedge \bar\theta_j$ at x we have $[\gamma, \Lambda_\sigma]v = \sum_j \lambda_j v_j\, e \wedge \bar\theta_j$ and $\langle [\gamma, \Lambda_\sigma]v, v\rangle_{\sigma,h} = \sum_j \lambda_j |v_j|_h^2$. In particular, $[\gamma, \Lambda_\sigma]$ is positive semidefinite on $(n,1)$-covectors at x if $\gamma \geq 0$ at x, and positive definite if γ is positive definite at x.*

(2) *If α is a $(1,0)$-covector and u an F-valued $(n,0)$-covector at x, then $|\langle \bar\alpha \wedge u, v\rangle_{\sigma,h}|^2 \leq |u|^2_{\sigma,h}\, \langle [\mathrm{i}\alpha \wedge \bar\alpha, \Lambda_\sigma]v, v\rangle_{\sigma,h}$ for every F-valued $(n,1)$-covector v at x.*

(3) *Let γ be positive definite at x, and let g be an F-valued $(n,1)$-covector at x. Then the (n,n)-covector $\langle [\gamma, \Lambda_\sigma]^{-1} g, g\rangle_{\sigma,h}\, \mathrm{d}V_\sigma$ does not depend on the Hermitian form σ, and neither does $|u|^2_{\sigma,h}\, \mathrm{d}V_\sigma$ for an F-valued $(n,0)$-covector u at x. If $\gamma \geq a\sigma$ at x for a constant $a > 0$, then $\langle [\gamma, \Lambda_\sigma]^{-1} g, g\rangle_{\sigma,h} \leq a^{-1}|g|^2_{\sigma,h}$. If σ' is a second Hermitian form with $\sigma' \geq c\sigma$ at x for a constant $c > 0$, then $|g|^2_{\sigma',h}\, \mathrm{d}V_{\sigma'} \leq c^{-1}|g|^2_{\sigma,h}\, \mathrm{d}V_\sigma$.*

Proof In linear coordinates at x with $\mathrm{d}z_j = \sqrt{2}\,\theta_j$ we have $\sigma = \frac{\mathrm{i}}{2}\sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar z_j = \sum_j \mathrm{d}x_j \wedge \mathrm{d}y_j$ at x, so the Riemannian metric of σ at x is Euclidean in the real coordinates, and $\langle \mathrm{d}x_j + \mathrm{i}\,\mathrm{d}y_j, \mathrm{d}x_k \pm \mathrm{i}\,\mathrm{d}y_k\rangle = (1 \pm 1)\delta_{jk}$ shows that the covectors $\theta_j, \bar\theta_k$ form an orthonormal basis of the complexified cotangent space at x. Hence the products $\theta_I \wedge \bar\theta_J$ form an orthonormal basis of the complex covectors at x. Let ι_j and $\bar\iota_j$ be the adjoints of the exterior products with θ_j and $\bar\theta_j$; thus ι_j maps $\theta_j \wedge \theta_K \wedge \bar\theta_L$ to $\theta_K \wedge \bar\theta_L$ for $j \notin K$ and annihilates the basis covectors that do not contain θ_j, and similarly for $\bar\iota_j$. Since $\sigma\wedge$ is $\mathrm{i}\sum_j (\theta_j \wedge)(\bar\theta_j \wedge)$, its adjoint is $\Lambda_\sigma = -\mathrm{i}\sum_j \bar\iota_j \iota_j$.

(1) The covector $\gamma \wedge v$ has bidegree $(n+1, 2)$ and vanishes, so $[\gamma, \Lambda_\sigma]v = \gamma \wedge \Lambda_\sigma v$. We have $\iota_j(e \wedge \bar\theta_k) = (-1)^{j-1}\theta_1 \wedge \cdots \wedge \widehat{\theta_j} \wedge \cdots \wedge \theta_n \wedge \bar\theta_k$, and $\bar\iota_j$ maps this to $(-1)^{j+n}\delta_{jk}\,\theta_1 \wedge \cdots \wedge \widehat{\theta_j} \wedge \cdots \wedge \theta_n$, the factor $\bar\theta_k$ standing in the nth place. Hence $\Lambda_\sigma v = -\mathrm{i}\sum_j (-1)^{j+n} v_j\, \theta_1 \wedge \cdots \wedge \widehat{\theta_j} \wedge \cdots \wedge \theta_n$. In the product of the jth term with γ only $\mathrm{i}\lambda_j \theta_j \wedge \bar\theta_j$ contributes, and $\theta_j \wedge \bar\theta_j \wedge \theta_1 \wedge \cdots \wedge \widehat{\theta_j} \wedge \cdots \wedge \theta_n = (-1)^{n-j} e \wedge \bar\theta_j$;

so $\gamma \wedge \Lambda_\sigma v = \sum_j \lambda_j v_j\, e \wedge \bar\theta_j$. Since the covectors $e \wedge \bar\theta_j$ are orthonormal, this gives the formula for $\langle [\gamma, \Lambda_\sigma] v, v\rangle_{\sigma,h}$, and the last assertion follows because the λ_j are nonnegative if $\gamma \ge 0$ and positive if γ is positive definite.

(2) We may assume that $\alpha \ne 0$. Since the assertion does not involve γ, we may choose the σ-unitary coframe with $\theta_1 = \alpha/|\alpha|_\sigma$, by completing this unit covector to an orthonormal basis of the $(1,0)$-covectors. Then $\mathrm{i}\alpha \wedge \bar\alpha = \mathrm{i}|\alpha|^2_\sigma \theta_1 \wedge \bar\theta_1$, so (1) gives $\langle [\mathrm{i}\alpha \wedge \bar\alpha, \Lambda_\sigma] v, v\rangle_{\sigma,h} = |\alpha|^2_\sigma |v_1|^2_h$. Writing $u = fe$, we have $\bar\alpha \wedge u = (-1)^n |\alpha|_\sigma f\, e \wedge \bar\theta_1$, hence $|\langle \bar\alpha \wedge u, v\rangle_{\sigma,h}| \le |\alpha|_\sigma |f|_h |v_1|_h$, and $|f|_h = |u|_{\sigma,h}$.

(3) Write $g = e \wedge \tau = \sum_j \tau_j\, e \wedge \bar\theta_j$ with an F-valued $(0,1)$-covector $\tau = \sum_j \tau_j \bar\theta_j$. By (1), $\langle [\gamma, \Lambda_\sigma]^{-1} g, g\rangle_{\sigma,h} = \sum_j |\tau_j|^2_h/\lambda_j = |\tau|^2_\gamma$, the squared norm of τ in the metric induced by γ on $(0,1)$-covectors, which does not involve σ. Moreover $\mathrm{d}V_\sigma = \sigma^n/n! = c_n\, e \wedge \bar e$ with $c_n = \mathrm{i}^n (-1)^{n(n-1)/2}$. Hence $\langle [\gamma, \Lambda_\sigma]^{-1} g, g\rangle_{\sigma,h}\, \mathrm{d}V_\sigma = c_n |\tau|^2_\gamma\, e \wedge \bar e$. If $\kappa = \mu^{-1} e$, $\mu \ne 0$, is another nonzero $(n,0)$-covector, then $g = \kappa \wedge \mu\tau$ and $c_n |\mu\tau|^2_\gamma\, \kappa \wedge \bar\kappa = c_n |\tau|^2_\gamma\, e \wedge \bar e$. Thus the density equals $c_n |\tau'|^2_\gamma\, \kappa \wedge \bar\kappa$ whenever $g = \kappa \wedge \tau'$, an expression that does not involve σ. In the same way, $|u|^2_{\sigma,h}\, \mathrm{d}V_\sigma = c_n |f|^2_h\, \kappa \wedge \bar\kappa$ for $u = f\kappa$. If $\gamma \ge a\sigma$, all λ_j are at least a, and $\sum_j |\tau_j|^2_h/\lambda_j \le a^{-1} \sum_j |\tau_j|^2_h = a^{-1} |g|^2_{\sigma,h}$. Finally, the eigenvalues of σ' with respect to σ' itself are all equal to one, so $[\sigma', \Lambda_{\sigma'}]$ is the identity on $(n,1)$-covectors by (1), and the independence just proved gives

$$\begin{aligned} |g|^2_{\sigma',h}\, \mathrm{d}V_{\sigma'} &= \langle [\sigma', \Lambda_{\sigma'}]^{-1} g, g\rangle_{\sigma',h}\, \mathrm{d}V_{\sigma'} \\ &= \langle [\sigma', \Lambda_\sigma]^{-1} g, g\rangle_{\sigma,h}\, \mathrm{d}V_\sigma \le c^{-1} |g|^2_{\sigma,h}\, \mathrm{d}V_\sigma \end{aligned}$$

if $\sigma' \ge c\sigma$. □

The L^2 estimates of this chapter rest on the following integral identity.

Lemma 14.6.2 *Let M be a complex manifold of dimension $n \ge 1$ with a Kähler form σ, and let $F \to M$ be a holomorphic line bundle with a smooth quadratic metric h. In a holomorphic frame of F with squared norm e^{-p}, let $D' = \partial - \partial p \wedge$ and $D'' = \bar\partial$, acting on the coefficients of F-valued forms; these are the components of the Chern connection, and they do not depend on the frame. Let D'^*, D''^* be their formal adjoints for σ and h, and put $B_\sigma = [2\pi c_1(F,h), \Lambda_\sigma]$. Then, for every compactly supported smooth F-valued $(n,1)$-form v,*

$$\|D''^* v\|^2_\sigma + \|D'' v\|^2_\sigma = \|D'^* v\|^2_\sigma + \int_M \langle B_\sigma v, v\rangle_\sigma\, \mathrm{d}V_\sigma. \tag{14.17}$$

Proof If $e' = ge$ is a second holomorphic frame, with g holomorphic and nowhere zero, then its weight is $p' = p - \log|g|^2$, and for a form u' we have $\partial(gu') - \partial p \wedge gu' = g(\partial u' - \partial p' \wedge u')$, since $\partial \log|g|^2 = g^{-1}\partial g$; so D' does not depend on the frame. In a frame we have $D'D'' + D''D' = \partial\bar\partial p \wedge$, and $\mathrm{i}\partial\bar\partial p = 2\pi c_1(F,h)$. We prove the Kähler identities at a point, in coordinates and a frame adapted to σ and h, and deduce the Bochner–Kodaira formula from them.

Fix a point, choose holomorphic coordinates centered at it in which $\sigma = \mathrm{i} \sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar z_j$ at the point and the first derivatives of the coefficients of σ vanish there

(Lemma 14.1.5, after replacing the coordinates z by $z/\sqrt{2}$), and a frame with $\partial p = 0$ at the point: replacing a frame e by $e\exp(\sum_j \partial_j p(0) z_j)$ replaces p by $p - 2\operatorname{Re}\sum_j \partial_j p(0) z_j$. Let $\epsilon_k, \bar\epsilon_k$ denote exterior multiplication by $\mathrm{d}z_k, \mathrm{d}\bar z_k$, and $\iota_k, \bar\iota_k$ their pointwise adjoints, the contractions. At the point they satisfy $\iota_j\epsilon_k + \epsilon_k\iota_j = \delta_{jk}$ and $\bar\iota_j\bar\epsilon_k + \bar\epsilon_k\bar\iota_j = \delta_{jk}$, while operators of different types anticommute; moreover $\Lambda_\sigma = -\mathrm{i}\sum_j \bar\iota_j\iota_j$, as the adjoint of $\mathrm{i}\sum_j \epsilon_j\bar\epsilon_j$.

The formal adjoints of $\sum_k \epsilon_k\partial_k$ and $\sum_k \bar\epsilon_k\partial_{\bar k}$ are $-\sum_k \iota_k\partial_{\bar k}$ and $-\sum_k \bar\iota_k\partial_k$, with $\iota_k, \bar\iota_k$ computed for the metric at each point, plus zero-order terms whose coefficients are linear in the first derivatives of the coefficients of σ and of e^{-p}. Indeed, in the frame $\mathrm{d}z_I \wedge \mathrm{d}\bar z_J$ the operators ϵ_k have constant matrices, and the pointwise inner product of F-valued forms times the density of $\mathrm{d}V_\sigma$ is $\overline{w}^{\mathsf{T}} H u$ for a smooth positive definite Hermitian matrix H whose entries are smooth functions of the coefficients of σ times e^{-p}. For a constant matrix A and compactly supported smooth u, w, integration by parts and $\partial_k\overline{w} = \overline{\partial_{\bar k} w}$ give $\int \overline{w}^{\mathsf{T}} H A\,\partial_k u = -\int \overline{(\partial_{\bar k} w)}^{\mathsf{T}} H A u - \int \overline{w}^{\mathsf{T}}(\partial_k H) A u$, so the formal adjoint of $A\partial_k$ is $-A^\dagger\partial_{\bar k} - H^{-1}A^*(\partial_{\bar k} H)$, where $A^\dagger = H^{-1}A^*H$ is the pointwise adjoint of A and A^* the conjugate transpose; the same computation applies with ∂_k and $\partial_{\bar k}$ exchanged. Since these first derivatives, and ∂p, vanish at the point, there

$$D' = \sum_k \epsilon_k\partial_k, \quad D'' = \sum_k \bar\epsilon_k\partial_{\bar k}, \quad D'^* = -\sum_k \iota_k\partial_{\bar k}, \quad D''^* = -\sum_k \bar\iota_k\partial_k,$$

where $\partial_k, \partial_{\bar k}$ act on the coefficients. The anticommutation rules give $\bar\iota_j\iota_j\epsilon_k - \epsilon_k\bar\iota_j\iota_j = \delta_{jk}\bar\iota_j$ and $\bar\iota_j\iota_j\bar\epsilon_k - \bar\epsilon_k\bar\iota_j\iota_j = -\delta_{jk}\iota_j$, hence $[\Lambda_\sigma, D'] = -\mathrm{i}\sum_k \bar\iota_k\partial_k$ and $[\Lambda_\sigma, D''] = \mathrm{i}\sum_k \iota_k\partial_{\bar k}$ at the point, since the coefficients of Λ_σ in the frame $\mathrm{d}z_I \wedge \mathrm{d}\bar z_J$ have vanishing first derivatives there. These are the Kähler identities

$$D''^* = -\mathrm{i}[\Lambda_\sigma, D'], \qquad D'^* = \mathrm{i}[\Lambda_\sigma, D''],$$

which therefore hold everywhere, the point being arbitrary. Substituting them into $\Delta'' = D''D''^* + D''^*D''$ and $\Delta' = D'D'^* + D'^*D'$, the mixed compositions $D''\Lambda_\sigma D'$ and $D'\Lambda_\sigma D''$ cancel in the difference, and the curvature identity gives the Bochner–Kodaira formula

$$\Delta'' - \Delta' = \mathrm{i}\big[D'D'' + D''D', \Lambda_\sigma\big] = [2\pi c_1(F,h), \Lambda_\sigma],$$

that is, $\Delta'' - \Delta' = B_\sigma$. For a compactly supported smooth $(n,1)$-form v, the form $D'v$ vanishes for degree reasons, and integrating $\langle(\Delta'' - \Delta')v, v\rangle_\sigma$ over M, using the definition of the formal adjoints, gives (14.17). □

To pass from this identity to an estimate, we need the density of compactly supported forms in the domain of the Hilbert space adjoint of $\bar\partial$; we allow three metrics for later use in Section 14.7.

Lemma 14.6.3 *Let M be a complex manifold of dimension $n \geq 1$ with a Hermitian form σ, and suppose that there are functions $\chi_R \in C_c^\infty(M)$, $R \geq 1$, with $0 \leq \chi_R \leq 1$, $\chi_R \to 1$ pointwise as $R \to \infty$, and $|\mathrm{d}\chi_R|_\sigma \leq C/R$ for a constant C. Let $F \to M$ be*

a holomorphic line bundle with smooth quadratic metrics h_0, h_1, h_2 such that the positive functions h_1/h_0 and h_2/h_1 are bounded on M, and let

$$T\colon L^2_{(n,0)}(\sigma, h_0) \to L^2_{(n,1)}(\sigma, h_1), \qquad S\colon L^2_{(n,1)}(\sigma, h_1) \to L^2_{(n,2)}(\sigma, h_2)$$

be the operators $D'' = \bar\partial$ with maximal distributional domains, where the spaces $L^2_{(n,q)}(\sigma, h_k)$ of F-valued (n,q)-forms are defined by the pointwise norms of σ and h_k and the volume form $\mathrm{d}V_\sigma = \sigma^n/n!$. Then T and S are closed and densely defined, and $ST = 0$. Every compactly supported smooth F-valued $(n,1)$-form v lies in $\operatorname{Dom} T^* \cap \operatorname{Dom} S$, and $T^*v = \delta v$, where δ is the formal adjoint of $\bar\partial$ for the pointwise inner products of (σ, h_0) on $(n,0)$-forms and of (σ, h_1) on $(n,1)$-forms. Compactly supported smooth F-valued $(n,1)$-forms are dense in $\operatorname{Dom} T^* \cap \operatorname{Dom} S$ for the graph norm $\|v\| + \|T^*v\| + \|Sv\|$.

Proof On a compact subset of a coordinate chart over which F has a holomorphic frame, the pointwise norms of σ and h_k are comparable with the Euclidean norms of the coefficients, and $\mathrm{d}V_\sigma$ with Lebesgue measure. Hence convergence in $L^2_{(n,q)}(\sigma, h_k)$ implies L^2 convergence of the coefficients on such sets, and therefore convergence in the sense of distributions; since $\bar\partial$ is continuous for distributional convergence, T and S are closed. Compactly supported smooth forms belong to $\operatorname{Dom} T$ and $\operatorname{Dom} S$, and they are dense in $L^2_{(n,0)}(\sigma, h_0)$ and in $L^2_{(n,1)}(\sigma, h_1)$: if (ψ_a) is a partition of unity with compact supports in such charts (Proposition 7.1.3), then an L^2 form u is the L^2 limit of the finite partial sums of $\sum_a \psi_a u$, by dominated convergence, and each $\psi_a u$ is the L^2 limit of the forms whose coefficients are the mollifications of those of $\psi_a u$, by Lemma 1.4.1. Thus T and S are densely defined, and $ST = 0$ because $\bar\partial\bar\partial = 0$ on distributions.

If v is a compactly supported smooth $(n,1)$-form and $u \in \operatorname{Dom} T$, then $\langle Tu, v\rangle$ is the value of the distribution $\bar\partial u$ on a smooth compactly supported form built from v, and the definition of the distributional derivative gives $\langle Tu, v\rangle = \langle u, \delta v\rangle$; hence $v \in \operatorname{Dom} T^*$ and $T^*v = \delta v$. For an arbitrary $v \in \operatorname{Dom} T^*$, testing with compactly supported smooth u shows in the same way that $T^*v = \delta v$ in the sense of distributions, and $Sv = \bar\partial v$ in the sense of distributions for $v \in \operatorname{Dom} S$ by definition.

For the density, multiplication by the cutoffs χ_R first reduces the problem to forms with compact support. Let $v \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$ and $\chi = \chi_R$. For $u \in \operatorname{Dom} T$, the Leibniz rule for distributions gives $\chi u \in \operatorname{Dom} T$ and $T(\chi u) = \chi Tu + \bar\partial\chi \wedge u$, so that $\langle Tu, \chi v\rangle = \langle T(\chi u), v\rangle - \langle \bar\partial\chi \wedge u, v\rangle = \langle u, \chi T^*v - (\bar\partial\chi\wedge)^* v\rangle$, where $(\bar\partial\chi\wedge)^*$ is the adjoint of the bounded operator $\bar\partial\chi\wedge$ from $L^2_{(n,0)}(\sigma, h_0)$ to $L^2_{(n,1)}(\sigma, h_1)$. Hence $\chi v \in \operatorname{Dom} T^*$ and $T^*(\chi v) = \chi T^*v - (\bar\partial\chi\wedge)^* v$; and the Leibniz rule gives $\chi v \in \operatorname{Dom} S$ with $S(\chi v) = \chi Sv + \bar\partial\chi \wedge v$. Since $|\bar\partial\chi_R|_\sigma \le |\mathrm{d}\chi_R|_\sigma \le C/R$ and the ratios h_1/h_0 and h_2/h_1 are bounded, the error terms have norm at most $C'\|v\|/R$, and $\chi_R v \to v$, $\chi_R T^*v \to T^*v$, $\chi_R Sv \to Sv$ by dominated convergence.

Second, a compactly supported form is approximated by local convolutions, after a finite partition of unity (Proposition 7.1.3). For a first-order operator $P = \sum_l a_l(x)\partial_l + b(x)$ with smooth matrix coefficients and a mollifier ρ_t, the commutator $[P, \rho_t * \cdot\,]$ applied to v equals

$$\sum_l \int (a_l(x) - a_l(y))\, \partial_l \rho_t(x-y)\, v(y)\, \mathrm{d}y$$
$$+ \sum_l \int (\partial_l a_l)(y)\, \rho_t(x-y)\, v(y)\, \mathrm{d}y + \big(b\,(\rho_t * v) - \rho_t * (bv)\big).$$

On a fixed compact set, the mean-value bound $|a_l(x) - a_l(y)| \le C|x-y|$ and the scaling of ρ_t bound the kernels of these integral operators by $k_t(x-y)$, where $\int k_t$ is bounded independently of t. An integral operator with a kernel bounded by $k(x-y)$, $k \ge 0$ integrable, has norm at most $\int k$ on L^2, since by the Cauchy–Schwarz inequality $|\int k(x-y)v(y)\,\mathrm{d}y|^2 \le \int k \cdot \int k(x-y)|v(y)|^2\,\mathrm{d}y$; so these operators are bounded on L^2 uniformly in t. For smooth v the commutator tends to zero in L^2, and by the uniform bound and the density of smooth functions in L^2 it tends to zero for every $v \in L^2$.

Now let $v \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$ have compact support, and let (ψ_a) be a finite partition of unity near $\operatorname{Supp} v$ by functions with compact supports in charts over which F has frames. By the product rules above, with ψ_a in place of χ, each $w = \psi_a v$ lies in $\operatorname{Dom} T^* \cap \operatorname{Dom} S$. In the chart and frame, δ and $\bar\partial$ are first-order operators with smooth matrix coefficients, and $T^*w = \delta w$ and $Sw = \bar\partial w$ lie in L^2. The forms w_t whose coefficients are the mollifications of those of w are smooth with compact support, so $T^*w_t = \delta w_t = \rho_t * (\delta w) + [\delta, \rho_t * \cdot\,]w$ and $Sw_t = \rho_t * (\bar\partial w) + [\bar\partial, \rho_t * \cdot\,]w$. By Lemma 1.4.1 and the commutator bound, $w_t \to w$ in the graph norm as $t \to 0$, which proves the density. □

The cutoff functions required in Lemma 14.6.3 exist, for a suitable Kähler form, on the complement of the pole set of a quasi-plurisubharmonic function that is smooth outside its poles.

Lemma 14.6.4 *Let ψ be a quasi-plurisubharmonic function on X that is smooth outside its pole set $A = \{\psi = -\infty\}$. Then there are a Kähler form $\sigma \ge \omega/2$ on $X \setminus A$ and functions $\chi_R \in C_c^\infty(X \setminus A)$, $R \ge 1$, with $0 \le \chi_R \le 1$, $\chi_R \to 1$ pointwise as $R \to \infty$, and $|\mathrm{d}\chi_R|_\sigma \le C/R$ for a constant C. This applies in particular when ψ is a finite sum of quasi-plurisubharmonic functions with gentle analytic singularities and of quasi-plurisubharmonic functions that are smooth outside a point x_k and equal to $c_k \log|z - z(x_k)|^2$ plus a smooth function near x_k, in local coordinates, with $c_k > 0$; then A is the union of their pole sets.*

Proof The set A is a null set, since ψ is locally integrable, and in particular $A \ne X$. Choose $b \ge 0$ with $\mathrm{dd^c}\psi \ge -b\omega$, which is possible by Lemma 13.1.4(4), and subtract a constant so that $\psi < -2b - 2$; this is possible because ψ is upper semicontinuous on the compact manifold X. Put $\rho = -\log(-\psi)$ on $X \setminus A$. The chain rule gives

$$\mathrm{dd^c}\rho = -\frac{1}{\psi}\mathrm{dd^c}\psi + \frac{\mathrm{i}}{2\pi\psi^2}\partial\psi \wedge \bar\partial\psi.$$

Since $0 < -1/\psi < 1/(2b+2)$, the first term is at least $-\omega/2$, and since $\partial\rho = -\partial\psi/\psi$, the form $\sigma = \omega + \mathrm{dd^c}\rho$ satisfies

$$\sigma \geq \frac{\omega}{2} + \frac{\mathrm{i}}{2\pi}\partial\rho \wedge \bar\partial\rho.$$

It is a Kähler form on $X \setminus A$. The rank-one form $\frac{\mathrm{i}}{2\pi}\partial\rho \wedge \bar\partial\rho$ has, in a σ-unitary coframe, the single eigenvalue $|\partial\rho|^2_\sigma/(2\pi)$, so the last inequality gives $|\partial\rho|^2_\sigma \leq 2\pi$; hence $|\mathrm{d}\rho|_\sigma$ is bounded by a numerical constant.

The function $-\rho = \log(-\psi)$ tends to $+\infty$ at A, because ψ is upper semicontinuous and equal to $-\infty$ on A, and its sublevel sets $\{-\rho \leq t\} = \{\psi \geq -\mathrm{e}^t\}$ are closed in X, because ψ is upper semicontinuous, and disjoint from A; so they are compact subsets of $X \setminus A$. Choose a smooth function χ on $\mathbb{R}$ equal to one on $(-\infty, 1]$ and to zero on $[2, \infty)$, with $0 \leq \chi \leq 1$, and put $\chi_R = \chi(-\rho/R)$. Then χ_R has compact support in $X \setminus A$, equals one on $\{-\rho \leq R\}$, and $|\mathrm{d}\chi_R|_\sigma \leq \sup|\chi'|\,|\mathrm{d}\rho|_\sigma/R$. The same gradient bound shows that every path in $X \setminus A$ that eventually leaves every compact subset of $X \setminus A$ has infinite σ-length, since $-\rho$ tends to $+\infty$ along it, by the compactness of its sublevel sets.

For the last assertion, recall that a function φ with gentle analytic singularities of type $(c, \mathfrak{a})$ is smooth outside its pole set, where it equals c times the logarithm of the smooth positive function $\exp(\varphi/c)$ (Definition 13.1.11). Hence a finite sum as in the statement is quasi-plurisubharmonic and smooth outside the union of the pole sets of its terms, and this union is its pole set, since all terms are bounded above. □

Lemma 14.6.5 *Let M be a complex manifold of dimension $n \geq 1$ with a Hermitian form ω and the volume form $\mathrm{d}V = \omega^n/n!$, and let $F \to M$ be a holomorphic line bundle with a smooth quadratic metric h such that $c_1(F, h) \geq a\omega$ for a constant $a > 0$. Suppose that M carries a Kähler form $\sigma \geq \omega/2$ and functions χ_R with the properties stated in Lemma 14.6.4. Let g be an F-valued $(n, 1)$-form with $\bar\partial g = 0$ in the sense of distributions and $\int_M |g|^2_{\omega,h}\,\mathrm{d}V < \infty$. Then there is an F-valued $(n, 0)$-form u with $\bar\partial u = g$ and*

$$\int_M |u|^2_{\omega,h}\,\mathrm{d}V \leq \frac{1}{2\pi a}\int_M |g|^2_{\omega,h}\,\mathrm{d}V. \tag{14.18}$$

Proof Let $B_\sigma = [2\pi c_1(F, h), \Lambda_\sigma]$. Since $c_1(F, h)$ is positive definite, B_σ is positive definite on $(n, 1)$-forms by Lemma 14.6.1(1). Let $T = D'' \colon L^2_{(n,0)} \to L^2_{(n,1)}$ and $S = D'' \colon L^2_{(n,1)} \to L^2_{(n,2)}$ be the operators with maximal distributional domains, for the norms defined by σ and h. By Lemma 14.6.3, applied with $h_0 = h_1 = h_2 = h$, they are closed and densely defined, $ST = 0$, and compactly supported smooth forms are dense in $\operatorname{Dom} T^* \cap \operatorname{Dom} S$ for the graph norm $\|v\| + \|T^*v\| + \|Sv\|$.

Let v be a compactly supported smooth $(n, 1)$-form. We integrate the pointwise Cauchy–Schwarz inequality $|\langle g, v\rangle_\sigma|^2 \leq \langle B_\sigma^{-1}g, g\rangle_\sigma \langle B_\sigma v, v\rangle_\sigma$ for the positive operator B_σ and use (14.17), in which $D''^*v = T^*v$ (Lemma 14.6.3) and $D''v = Sv$. This gives

$$\Big|\int_M \langle g, v\rangle_\sigma\,\mathrm{d}V_\sigma\Big|^2 \leq \int_M \langle B_\sigma^{-1}g, g\rangle_\sigma\,\mathrm{d}V_\sigma \cdot \big(\|T^*v\|^2_\sigma + \|Sv\|^2_\sigma\big).$$

By Lemma 14.6.1(3), the density $\langle B_\sigma^{-1} g, g\rangle_\sigma \, dV_\sigma$ does not depend on σ. Computing it with ω in place of σ, and using that $2\pi c_1(F,h) \geq 2\pi a\omega$, so that $\langle [2\pi c_1(F,h), \Lambda_\omega]^{-1} g, g\rangle_{\omega,h} \leq (2\pi a)^{-1} |g|^2_{\omega,h}$ by the same lemma, we find that the first integral is at most the right side of (14.18).

The form g lies in $\ker S$, and in $L^2_{(n,1)}$ for σ and h, because $\sigma \geq \omega/2$ gives $|g|^2_{\sigma,h} \, dV_\sigma \leq 2|g|^2_{\omega,h} \, dV$, again by Lemma 14.6.1(3). Hence both sides of the displayed inequality converge along a graph-convergent sequence, and by the density of compactly supported smooth forms the inequality extends to all $v \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$. Theorem 11.1.4 therefore gives $u \in \operatorname{Dom} T$ with $Tu = g$ and $\|u\|^2_\sigma$ at most the first integral. Finally, $|u|^2_{\sigma,h} \, dV_\sigma = |u|^2_{\omega,h} \, dV$ by Lemma 14.6.1(3), which gives exactly (14.18). □

The sections obtained from this estimate are holomorphic only outside an analytic set; the following removable singularity theorem extends them.

Lemma 14.6.6 *Let A be a nowhere dense analytic subset of a complex manifold U. A holomorphic function on $U \setminus A$ which, extended arbitrarily to A, belongs to $L^2_{\mathrm{loc}}(U)$ extends holomorphically to U. The same holds for sections of a holomorphic line bundle.*

Proof The question is local. We show that the given function f satisfies $\bar\partial f = 0$ in the sense of distributions near each point of A, by a one-variable argument on complex lines. Near a point of A, the set A is the common zero set of finitely many holomorphic functions on a connected coordinate neighborhood, and one of them, b, does not vanish identically, since otherwise A would contain a nonempty open set. The lowest-order nonzero homogeneous term P of b at the point is a nonzero polynomial, so its zero set is closed and nowhere dense; hence we may choose linearly independent vectors $v_1, \ldots, v_n$ with $P(v_k) \neq 0$, choosing each v_k outside the union of this zero set and the span of $v_1, \ldots, v_{k-1}$. We use linear coordinates centered at the point in which the v_k are the coordinate unit vectors. Then $t \mapsto b(tv_k)$ has a zero of finite order at $t = 0$, so by the Weierstrass preparation theorem (Theorem 7.4.2), on a sufficiently small polydisc, b is a unit times a monic polynomial in the kth coordinate with holomorphic coefficients in the others, for each k. Hence every complex line parallel to a coordinate axis meets the zero set of b in this polydisc in finitely many points.

By Fubini's theorem, f is square integrable on almost every such line. On such a line, near each of the finitely many punctures, the Laurent expansion of f has no terms of negative order: by orthogonality of the powers on circles, a term of order $-j$ with $j \geq 1$ would make the integral of $|f|^2$ diverge like $\int_0^r t^{1-2j} \, dt$. Hence the restriction of f to almost every line extends holomorphically, and its $\bar\partial$-derivative along the line vanishes as a distribution. Integrating against compactly supported test functions and using Fubini's theorem, we find that every derivative $\partial f/\partial \bar z_k$ vanishes as a distribution on the polydisc. A distribution with $\bar\partial f = 0$ satisfies $\Delta f = 4 \sum_k \partial_{z_k} \partial_{\bar z_k} f = 0$, so it is smooth by Weyl's lemma (Theorem 2.1.4) and therefore holomorphic. This holomorphic function agrees with f on the complement

of A, and the local extensions agree on overlaps. For sections of a line bundle, apply the result in local holomorphic frames. □

Theorem 14.6.7 *There are constants $N, C > 0$, depending only on (X, ω), with the following property. Let L be a holomorphic line bundle with a smooth quadratic metric h_0 and a plurisubharmonic metric $h = h_0 \mathrm{e}^{-\varphi}$ such that $c_1(L,h) \geq N\omega$. Let $x \in X$ with $\varphi(x) > -\infty$ and $v \in L_x$. Then there is $s \in H^0(X, L)$ such that*

$$s(x) = v, \qquad \int_X |s|_h^2 \, \mathrm{d}V \leq C|v|_h^2. \tag{14.19}$$

Proof If $v = 0$, take $s = 0$; assume $v \neq 0$. We cut off a local extension of v and correct the error with Lemma 14.6.5 outside x and the poles of gentle approximations of φ; a logarithmic pole at x in the weight makes the correction vanish at x.

Step 1: data independent of x. Choose finitely many coordinate charts and, inside them, balls B_k of radii ϱ_k, with closures in the charts, whose concentric balls of half the radius cover X. Put $r_0 = \min_k \varrho_k/8$. Then for every $x \in X$ there is k with $x \in \frac{1}{2}B_k$, and the coordinate ball B of radius $4r_0$ centered at x in the kth chart lies in B_k. Let λ be a smooth function on $\mathbb{R}^{2n}$ with values in $[0,1]$, equal to one on the ball of radius $2r_0$ and zero outside the ball of radius $3r_0$, and put

$$\psi_x = \lambda(z - z(x))\, n \log \frac{|z - z(x)|^2}{16 r_0^2} \ \text{ on } B, \qquad \psi_x = 0 \ \text{ outside } B,$$

where z are the chart coordinates. Then $\psi_x \leq 0$, ψ_x is plurisubharmonic where $\lambda(z - z(x)) = 1$, and on the annulus where $\lambda(z - z(x))$ is not constant the logarithm and its first two derivatives are bounded independently of x. Hence there is $b_0 \geq 0$, independent of x, with

$$\mathrm{dd^c}\psi_x \geq -b_0\omega, \qquad \psi_x = n \log |z - z(x)|^2 + O(1) \text{ near } x.$$

Choose $b_1 \geq 0$ with $c_1(K_X^{-1}, h_{K^{-1},\omega}) \geq -b_1\omega$ (Lemma 13.1.1), and put $N = b_0 + b_1 + 2$.

Step 2: a local extension and its cutoff. By Lemma 14.5.1, L has a holomorphic frame e over B. With $|e|_{h_0}^2 = \mathrm{e}^{-p_0}$, the local weight $p_0 + \varphi$ of h is plurisubharmonic on B, because $\mathrm{dd^c}(p_0 + \varphi) = c_1(L,h) \geq 0$. By Corollary 11.4.5, applied in this frame on the ball of radius r_0 around x, whose diameter $2r_0$ is fixed, there is a holomorphic section f of L over this ball with $f(x) = v$ and

$$\int_{|z - z(x)| < r_0} |f|_h^2 \, \mathrm{d}V \leq C_0 |v|_h^2;$$

here C_0 does not depend on x, because the volume densities are uniformly comparable on the finite cover. Let χ be a cutoff equal to one for $|z - z(x)| < r_0/2$ and zero for $|z - z(x)| > 3r_0/4$, obtained by translating a fixed one. Put $G = \chi f$, extended by zero, and $g = \bar\partial G = f\, \bar\partial\chi$. The form g is supported in the annulus $r_0/2 \leq |z - z(x)| \leq 3r_0/4$, on which ψ_x and $\bar\partial\chi$ are bounded independently of x.

Step 3: global sections for the approximations of φ. Apply Theorem 13.4.5 to φ with the background $c_1(L,h_0)-N\omega$. It gives functions $\varphi_j\downarrow\varphi$ with gentle analytic singularities, analytic pole sets A_j, and $c_1(L,h_0)+\mathrm{dd}^{\mathrm{c}}\varphi_j\geq(N-\varepsilon_j)\omega$ with $\varepsilon_j\to 0$. Since $\varphi_j(x)\geq\varphi(x)>-\infty$, no A_j contains x. Put

$$F=L\otimes K_X^{-1},\qquad h_{F,j}=h_0\mathrm{e}^{-\varphi_j-\psi_x}\otimes h_{K^{-1},\omega}.$$

For j so large that $\varepsilon_j\leq 1$, this metric is smooth on $M_j=X\setminus(A_j\cup\{x\})$ and its curvature is at least $(N-\varepsilon_j-b_0-b_1)\omega\geq\omega$. By Lemma 14.6.4, applied to $\varphi_j+\psi_x$, the manifold M_j carries a Kähler form $\sigma\geq\omega/2$ and cutoffs as required in Lemma 14.6.5. The canonical identification of L-valued $(0,q)$-forms with F-valued (n,q)-forms preserves the pointwise norms for ω, because the $(n,0)$-covector and its dual cancel. Under this identification g is $\bar\partial$-closed, and since $\varphi_j\geq\varphi$,

$$\int_{M_j}|g|^2_{h_0}\mathrm{e}^{-\varphi_j-\psi_x}\,\mathrm{d}V\leq C_1\int_{|z-z(x)|<r_0}|f|^2_h\,\mathrm{d}V\leq C_1C_0|v|^2_h,$$

with C_1 depending only on the fixed cutoffs and the bounds for ψ_x on the annulus. Lemma 14.6.5, with $a=1$, gives u_j on M_j with $\bar\partial u_j=g$ and

$$\int_{M_j}|u_j|^2_{h_0}\mathrm{e}^{-\varphi_j-\psi_x}\,\mathrm{d}V\leq\frac{C_1C_0}{2\pi}|v|^2_h.$$

The section $s_j=G-u_j$ is holomorphic on M_j. The weights $\mathrm{e}^{-\varphi_j-\psi_x}$ are bounded below by a positive constant, since $\varphi_j\leq\varphi_1$ is bounded above and $\psi_x\leq 0$; hence s_j is square integrable for h_0 near $A_j\cup\{x\}$, and Lemma 14.6.6 extends it to a global holomorphic section. Near x we have $g=0$ and no point of A_j, so $u_j=G-s_j$ is holomorphic near x. Moreover $|u_j|^2_{h_0}\mathrm{e}^{-\varphi_j-\psi_x}\geq c\,|u_j|^2_{h_0}|z-z(x)|^{-2n}$ near x for a constant $c>0$, by the upper bound for φ_j and since $\psi_x=n\log|z-z(x)|^2+O(1)$. Since $\int_{|z-z(x)|<r}|z-z(x)|^{-2n}\,\mathrm{d}V=\infty$ for every $r>0$, by polar coordinates in $\mathbb{R}^{2n}$, the finiteness of the weighted integral of the continuous function u_j forces $u_j(x)=0$. Hence $s_j(x)=G(x)=v$.

Since $\psi_x\leq 0$, dropping the factor $\mathrm{e}^{-\psi_x}$ only decreases the integrals, and with $|G|^2_{h_0}\mathrm{e}^{-\varphi_j}\leq|f|^2_h$ on the ball we obtain

$$\int_X|s_j|^2_{h_0}\mathrm{e}^{-\varphi_j}\,\mathrm{d}V\leq C_2|v|^2_h,$$

where C_2 depends only on the fixed geometric cutoffs and on the constant of the local point estimate.

Step 4: the limit. Since $\varphi_j\leq\varphi_1\leq\sup_X\varphi_1$, the sections s_j are bounded in L^2 for h_0, and by Lemma 14.5.2(2) a subsequence converges uniformly to a holomorphic section s of L, which satisfies $s(x)=v$. Since $\varphi_j\downarrow\varphi$, Fatou's lemma gives (14.19) with $C=C_2$. □

Lemma 14.6.8 *Let L and T be holomorphic line bundles on X with smooth quadratic metrics h_0 and h_T, and let $\mathrm{d}V$ be a smooth positive volume form on X. For an integer $k \geq 1$ and a quasi-plurisubharmonic function φ on X, the space*

$$\mathcal{H}_k = \left\{ s \in H^0(X, L^k \otimes T) : \int_X |s|^2_{h_0^k \otimes h_T} \mathrm{e}^{-k\varphi} \, \mathrm{d}V < \infty \right\}$$

is a finite-dimensional Hilbert space for the displayed squared norm $\|s\|_k^2$. If $\mathcal{H}_k \neq \{0\}$ and $s_1, \ldots, s_{d_k}$ is an orthonormal basis, then

$$\varphi_k = \frac{1}{k} \log \sup_{\|s\|_k \leq 1} |s|^2_{h_0^k \otimes h_T} = \frac{1}{k} \log \sum_{i=1}^{d_k} |s_i|^2_{h_0^k \otimes h_T}. \tag{14.20}$$

Writing $\theta = c_1(L, h_0)$ and $\theta_T = c_1(T, h_T)$, the function φ_k belongs to $\mathrm{PSH}(X, \theta + k^{-1}\theta_T)$ and has gentle analytic singularities with coefficient $1/k$. Except for the gentle analytic singularities, which are defined only on compact Kähler manifolds (Definition 13.1.11), these assertions hold more generally on every compact complex manifold X' of positive dimension in place of X.

Proof By Lemma 14.5.2(3), $H^0(X, L^k \otimes T)$ and its linear subspace $\mathcal{H}_k$ are finite dimensional. The weighted integral defines an inner product on $\mathcal{H}_k$: the function φ is finite almost everywhere, so the weight is positive almost everywhere, and a section of norm zero vanishes almost everywhere and hence identically. A finite-dimensional inner product space is complete.

At a point y, with the coefficients c_i of $s = \sum_i c_i s_i$, the Cauchy–Schwarz inequality gives $|s(y)|^2 \leq \sum_i |c_i|^2 \sum_i |s_i(y)|^2$, with equality when the c_i are proportional to the complex conjugates of the values of the s_i at y in a local frame. This proves the equality in (14.20). In local frames with $|e|^2_{h_0} = \mathrm{e}^{-p_0}$ and $|e_T|^2_{h_T} = \mathrm{e}^{-p_T}$, write $s_i = f_i e^k \otimes e_T$. Then

$$\varphi_k = \frac{1}{k} \log \sum_i |f_i|^2 - p_0 - \frac{p_T}{k},$$

and $\theta + k^{-1}\theta_T + \mathrm{dd^c}\varphi_k = k^{-1}\mathrm{dd^c} \log \sum_i |f_i|^2 \geq 0$ by Proposition 8.3.3. Hence $\varphi_k \in \mathrm{PSH}(X, \theta + k^{-1}\theta_T)$, with analytic singularities of type $(1/k, \mathfrak{b})$, where $\mathfrak{b}$ is the ideal sheaf generated locally by the f_i. This ideal sheaf does not depend on the frames, because a change of frames multiplies all f_i by the same nowhere vanishing holomorphic function, and it is coherent by Lemma 12.2.4. Moreover $\mathrm{e}^{k\varphi_k} = \sum_i |s_i|^2$ is smooth on X. The local expressions for φ_k above are presentations with the smooth remainders $-p_0 - p_T/k$, so Proposition 13.1.7 shows that on every log resolution π of $\mathfrak{b}$, with $\pi^*\mathfrak{b} = \mathcal{O}(-D)$, locally $\varphi_k \circ \pi$ is $k^{-1} \log |s_D|^2$ plus a smooth function. These are the conditions of Definition 13.1.11.

Apart from this appeal to Definition 13.1.11, the proof uses only that X is a compact complex manifold of positive dimension, since the last assertion of Lemma 14.5.2 gives its part (3) in this generality; hence it also proves the last assertion of the lemma. □

Example 14.6.9 On $\mathbb{P}^1$ take $L = O(1)$ with its Fubini–Study metric, T trivial, $\varphi = 0$ and $\mathrm{d}V = \omega_{\mathrm{FS}}$. A section of L^k is $f(z)X_0^k$ on the chart $[1 : z]$ with f entire, and in the chart at infinity, with $\zeta = 1/z$, it equals $\zeta^k f(1/\zeta)X_1^k$; holomorphicity at $\zeta = 0$ and Cauchy's estimates show that f is a polynomial of degree at most k. Integration over circles makes the monomials orthogonal, and with $t = |z|^2$,

$$\|z^j\|_k^2 = \int_0^\infty \frac{t^j}{(1+t)^{k+2}}\,\mathrm{d}t = \frac{j!(k-j)!}{(k+1)!}, \qquad 0 \le j \le k;$$

the last equality follows from the substitution $u = t/(1+t)$ and repeated integration by parts. Thus the sections $\sqrt{(k+1)\binom{k}{j}}X_0^{k-j}X_1^j$, $0 \le j \le k$, form an orthonormal basis, and the sum of their squared norms is

$$\frac{k+1}{(1+|z|^2)^k}\sum_{j=0}^{k}\binom{k}{j}|z|^{2j} = k+1.$$

Consequently $\varphi_k = k^{-1}\log(k+1)$, which tends to zero uniformly.

14.7 Extension from a compact submanifold

In Theorem 14.6.7 the error $\bar\partial(\chi f)$ of the cutoff is supported away from the interpolation point. To extend a section from a submanifold Y of positive dimension, we must glue local extensions, and the gluing error then meets Y itself. We control it in two steps. First, local extensions with a logarithmic gain (Lemma 14.7.3) make the gluing error integrable against the critical weight R_Y^{-q}, where q is the codimension of Y and R_Y is a smooth function vanishing to second order exactly along Y. Second, for weights with gentle analytic singularities, a computation on a resolution identifies the limit of the cutoff error with the norm of the prescribed section on Y (Lemma 14.7.7). This yields an estimate whose constant depends only on the geometry, and decreasing approximation then gives the result for arbitrary singular metrics (Theorem 14.7.8). In Section 14.7.1 a related gluing construction, by maxima instead of sums, extends potentials of Kähler currents from Y to X.

We solve the $\bar\partial$-equations of this section with the following twisted form of Lemma 14.6.5. The auxiliary function a contributes the term $-\mathrm{i}\partial\bar\partial a$ to the curvature, at the price of the weight $1/(a+b)$ in the estimate for the solution.

Lemma 14.7.1 *Let M be a complex manifold of dimension $n \ge 1$ with a Hermitian form ω and the volume form $\mathrm{d}V = \omega^n/n!$, carrying a Kähler form $\sigma \ge \omega/2$ and functions χ_R with the properties stated in Lemma 14.6.4. Let $F \to M$ be a holomorphic line bundle with a smooth quadratic metric h, and let a, b be smooth functions on M that are bounded above and below by positive constants. Suppose that the real $(1,1)$-form*

$$H = a\,2\pi c_1(F,h) - \mathrm{i}\partial\bar\partial a - \frac{\mathrm{i}}{b}\partial a\wedge\bar\partial a$$

is positive definite, and put $B_\omega = [H,\Lambda_\omega]$ *on* $(n,1)$*-forms. Let* g *be an* F*-valued* $(n,1)$*-form with* $\bar\partial g = 0$ *in the sense of distributions,* $g \in L^2_{(n,1)}(\omega,h)$*, and*

$$M_g = \int_M \langle B_\omega^{-1} g, g\rangle_{\omega,h}\,\mathrm{d}V < \infty.$$

Then there is an F*-valued* $(n,0)$*-form* u *with* $\bar\partial u = g$ *and*

$$\int_M \frac{|u|^2_{\omega,h}}{a+b}\,\mathrm{d}V \le M_g. \tag{14.21}$$

Proof The Bochner–Kodaira identity (14.17) of Lemma 14.6.2 holds for every smooth metric on F; we apply it to the metric ah. Let $\delta_h = D''^*$ be the formal adjoint for the metric h and the Kähler form σ, and $C_a = (\bar\partial a\wedge)^*$ the contraction with the dual of $\bar\partial a$. In a frame, the weight of ah is $p - \log a$, so

$$\delta_{ah} = \delta_h - a^{-1}C_a, \qquad a\,2\pi c_1(F,ah) = a\,2\pi c_1(F,h) - \mathrm{i}\partial\bar\partial a + \frac{\mathrm{i}}{a}\partial a\wedge\bar\partial a.$$

On $(n,1)$-forms, the commutator with the last term is $[\frac{\mathrm{i}}{a}\partial a\wedge\bar\partial a,\Lambda_\sigma] = a^{-1}C_a^*C_a$. Indeed, where $\partial a \ne 0$, choose a σ-unitary coframe with $\partial a = |\partial a|\theta_1$, so that $\bar\partial a = |\partial a|\bar\theta_1$, and put $e = \theta_1\wedge\cdots\wedge\theta_n$. By Lemma 14.6.1(1), the left side multiplies the coefficient v_1 of $e\wedge\bar\theta_1$ by $|\partial a|^2/a$ and annihilates the other coefficients; so does the right side, because $\bar\partial a\wedge fe = (-1)^n|\partial a|f\,e\wedge\bar\theta_1$, whence $C_a v = (-1)^n|\partial a|v_1 e$. Where $\partial a = 0$, both sides vanish. Writing (14.17) for ah in terms of h, expanding $a|\delta_h v - a^{-1}C_a v|^2$, cancelling the term $a^{-1}|C_a v|^2$ that appears on both sides, and dropping the nonnegative term $\|D'^* v\|^2$, we obtain for compactly supported smooth v, with all integrals taken with respect to h, σ and $\mathrm{d}V_\sigma$,

$$\int_M a\big(|\delta_h v|^2 + |D''v|^2\big) \ge \int_M \langle [a\,2\pi c_1(F,h) - \mathrm{i}\partial\bar\partial a,\Lambda_\sigma]v,v\rangle + 2\,\mathrm{Re}\int_M \langle \delta_h v, C_a v\rangle.$$

The inequality $2\,\mathrm{Re}\langle A,B\rangle \ge -b|A|^2 - b^{-1}|B|^2$ and the identity $b^{-1}|C_a v|^2 = \langle [\frac{\mathrm{i}}{b}\partial a\wedge\bar\partial a,\Lambda_\sigma]v,v\rangle$ consequently give

$$\int_M \langle [H,\Lambda_\sigma]v,v\rangle \le \int_M (a+b)|\delta_h v|^2 + a|D''v|^2. \tag{14.22}$$

Consider the Hilbert spaces of $(n,0)$-, $(n,1)$- and $(n,2)$-forms with the respective weights $h/(a+b)$, h and ah, and the operators $T = D''$ and $S = D''$ with maximal domains between them. Testing against compactly supported forms shows that $T^* = (a+b)\delta_h$, so $\|T^* v\|^2$ is the first term on the right of (14.22), and $\|Sv\|^2$ is the second. Compactly supported smooth forms are dense in $\mathrm{Dom}\,T^* \cap \mathrm{Dom}\,S$ for the

graph norm, by Lemma 14.6.3, applied with the metrics $h/(a+b)$, h and ah, whose ratios $a+b$ and a are bounded.

The form g lies in $\ker S$, and in the second space by Lemma 14.6.1(3), because $\sigma \geq \omega/2$. The operator $[H, \Lambda_\sigma]$ is positive definite on $(n,1)$-forms by Lemma 14.6.1(1). As in the proof of Lemma 14.6.5, the pointwise Cauchy–Schwarz inequality for this operator, (14.22) and the density of compactly supported smooth forms give the hypothesis of Theorem 11.1.4, which yields a solution of $\bar\partial u = g$ whose squared norm in the first space is at most $\int \langle [H,\Lambda_\sigma]^{-1} g, g\rangle \, \mathrm{d}V_\sigma$. This integral, and the squared norm of $(n,0)$-forms times the volume form, do not depend on the Hermitian form used, by Lemma 14.6.1(3); hence the bound is M_g, and we obtain (14.21). □

In the applications of Lemma 14.7.1, the functions a and b are the following functions of $\log(\varepsilon + R)$, where R is a smooth function vanishing to second order along a submanifold, considered on the complement of the submanifold.

Lemma 14.7.2 *Let R be a smooth function on a complex manifold M with $0 < R < \mathrm{e}^{-4}$, let $0 < \varepsilon < \mathrm{e}^{-4}$, and put*

$$\psi = \log R, \qquad \psi_\varepsilon = \log(\varepsilon + R), \qquad \mu(t) = \varepsilon - t + \log(1-t) \quad (t<1),$$
$$a_\varepsilon = \mu(\psi_\varepsilon), \qquad b_\varepsilon = (2-\psi_\varepsilon)^2.$$

(1) *We have $\psi < \psi_\varepsilon < -3$, $a_\varepsilon \geq -\psi_\varepsilon > 3$, $1 \leq -\mu'(\psi_\varepsilon) \leq 2$ and $a_\varepsilon + b_\varepsilon \leq 5\psi_\varepsilon^2 \leq 5\psi^2$, and a_ε and b_ε are bounded above and below by positive constants depending only on ε.*

(2) *For every real $(1,1)$-form Θ on M,*

$$a_\varepsilon \Theta - \mathrm{i}\partial\bar\partial a_\varepsilon - \frac{\mathrm{i}}{b_\varepsilon}\partial a_\varepsilon \wedge \bar\partial a_\varepsilon = a_\varepsilon \Theta - \mu'(\psi_\varepsilon)\,\mathrm{i}\partial\bar\partial\psi_\varepsilon.$$

(3) *The logarithmic Hessians satisfy*

$$\mathrm{i}\partial\bar\partial\psi_\varepsilon = \frac{R}{\varepsilon+R}\mathrm{i}\partial\bar\partial\psi + \frac{\varepsilon}{R}\,\mathrm{i}\partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon. \tag{14.23}$$

Proof (1) Since $\varepsilon + R < 2\mathrm{e}^{-4} < \mathrm{e}^{-3}$, we have $\log\varepsilon < \psi_\varepsilon < -3$, and $\psi < \psi_\varepsilon$ because $\varepsilon > 0$. Next, $a_\varepsilon = \varepsilon - \psi_\varepsilon + \log(1-\psi_\varepsilon) \geq -\psi_\varepsilon$, and $\mu'(t) = -1 - (1-t)^{-1}$ lies in $[-2,-1]$ for $t<0$. With $s = -\psi_\varepsilon > 3$ we have $a_\varepsilon \leq 1 + s + \log(1+s) \leq 1 + 2s$ and $b_\varepsilon = (2+s)^2 \leq 4s^2$, so $a_\varepsilon + b_\varepsilon \leq 1 + 2s + 4s^2 \leq 5s^2$, since $s^2 - 2s - 1 > 0$ for $s \geq 3$; and $s^2 \leq \psi^2$ because $\psi < \psi_\varepsilon < 0$. Finally, $\log\varepsilon < \psi_\varepsilon < -3$ gives $3 < a_\varepsilon \leq \varepsilon - \log\varepsilon + \log(1-\log\varepsilon)$ and $25 < b_\varepsilon \leq (2-\log\varepsilon)^2$.

(2) We have $\mu''(t) = -(1-t)^{-2}$, hence $-\mu'(t)^2/\mu''(t) = \big((1-t)+1\big)^2 = (2-t)^2$, so $b_\varepsilon = -\mu'(\psi_\varepsilon)^2/\mu''(\psi_\varepsilon)$. By the chain rule, $\mathrm{i}\partial\bar\partial a_\varepsilon = \mu'(\psi_\varepsilon)\mathrm{i}\partial\bar\partial\psi_\varepsilon + \mu''(\psi_\varepsilon)\mathrm{i}\partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon$ and $b_\varepsilon^{-1}\mathrm{i}\partial a_\varepsilon \wedge \bar\partial a_\varepsilon = b_\varepsilon^{-1}\mu'(\psi_\varepsilon)^2\mathrm{i}\partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon = -\mu''(\psi_\varepsilon)\mathrm{i}\partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon$. The gradient terms cancel, which proves (2).

(3) For $c = \varepsilon$ and for $c = 0$ we have $\partial\bar\partial\log(c+R) = (c+R)^{-1}\partial\bar\partial R - (c+R)^{-2}\partial R \wedge \bar\partial R$. Hence $\partial\bar\partial\psi_\varepsilon - \frac{R}{\varepsilon+R}\partial\bar\partial\psi = \big(R^{-1}(\varepsilon+R)^{-1} - (\varepsilon+R)^{-2}\big)\partial R \wedge \bar\partial R = \frac{\varepsilon}{R}(\varepsilon+R)^{-2}\partial R \wedge \bar\partial R$, and $(\varepsilon+R)^{-2}\partial R \wedge \bar\partial R = \partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon$. □

Lemma 14.7.3 *Let $m \geq 0$ and $q \geq 1$ be integers, and let $D = B' \times B'' \Subset \mathbb{C}^m \times \mathbb{C}^q$ be a product of balls centered at the origin, of radii r' and r'', with $r'' < \mathrm{e}^{-2}$; when $m = 0$, $B' = \{0\}$. We write points of D as (z', w). Let p be plurisubharmonic on a neighborhood of $\overline{D}$, and let f be holomorphic near $\overline{B'}$ with $\int_{B'} |f|^2 \mathrm{e}^{-p(z',0)}\, \mathrm{d}V_{B'} < \infty$, where $|f|^2 \mathrm{e}^{-p(z',0)}$ is taken to be 0 where $f = 0$. Then there is a holomorphic function F on D with $F(z', 0) = f(z')$ and*

$$\int_D \frac{|F|^2 \mathrm{e}^{-p}}{|w|^{2q}(-\log|w|^2)^2}\, \mathrm{d}V \leq C \int_{B'} |f|^2 \mathrm{e}^{-p(z',0)}\, \mathrm{d}V_{B'}, \tag{14.24}$$

where both volumes are Euclidean and C depends only on m, q, r' and r''. When $m = 0$, the right side is $C|f|^2 \mathrm{e}^{-p(0)}$.

Proof If $f = 0$, take $F = 0$. Put $n = m + q$, and let β be the Euclidean Kähler form on $\mathbb{C}^n$. We first treat smooth weights: we cut off the trivial extension of f near $\{w = 0\}$, correct it with Lemma 14.7.1 on $D \setminus \{w = 0\}$, and let the width ε of the cutoff tend to zero. So assume that p is smooth near $\overline{D}$, and set $R = |w|^2$, $\psi = \log R$ and $\psi_\varepsilon = \log(\varepsilon + R)$ for small $\varepsilon > 0$. The function $P = p + \pi|z|^2$, where $z = (z', w)$, satisfies $\mathrm{dd^c} P \geq \beta$, and replacing p by P changes both sides of (14.24) by factors bounded in terms of r' and r'' only; we prove the estimate for P.

The twisted form. On $D \setminus \{w = 0\}$ consider the trivial line bundle with the metric $\mathrm{e}^{-P - q\psi}$, and let μ, a_ε and b_ε be the functions of Lemma 14.7.2, for $0 < \varepsilon < \mathrm{e}^{-4}$; that lemma applies because $0 < R < \mathrm{e}^{-4}$ on $D \setminus \{w = 0\}$, since $r'' < \mathrm{e}^{-2}$. By its part (1), $a_\varepsilon \geq 1$ and $-\mu'(\psi_\varepsilon) \geq 1$, and for fixed ε the functions a_ε and b_ε are bounded above and below by positive constants. By its part (2), applied with $\Theta = \mathrm{i}\partial\bar\partial(P + q\psi)$, the curvature form of this metric times 2π, the gradient terms cancel in the form H of Lemma 14.7.1, and

$$H_\varepsilon = a_\varepsilon\bigl(\mathrm{i}\partial\bar\partial P + q\,\mathrm{i}\partial\bar\partial\psi\bigr) - \mu'(\psi_\varepsilon)\,\mathrm{i}\partial\bar\partial\psi_\varepsilon.$$

Both logarithmic Hessians $\mathrm{i}\partial\bar\partial\psi$ and $\mathrm{i}\partial\bar\partial\psi_\varepsilon$ are semipositive, so (14.23) gives $\mathrm{i}\partial\bar\partial\psi_\varepsilon \geq (\varepsilon/R)\,\mathrm{i}\partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon$. Since also $\mathrm{i}\partial\bar\partial P = 2\pi\mathrm{dd^c} P \geq 2\pi\beta$, the bounds $a_\varepsilon \geq 1$ and $-\mu'(\psi_\varepsilon) \geq 1$ give $H_\varepsilon \geq 2\pi\beta + (\varepsilon/R)\,\mathrm{i}\partial\psi_\varepsilon \wedge \bar\partial\psi_\varepsilon$.

Complete cutoffs. The manifold $D \setminus \{w = 0\}$ carries a Kähler form and cutoffs as required in Lemma 14.7.1. Indeed, put $\tau_0 = -\log(r'^2 - |z'|^2) - \log(r''^2 - |w|^2)$, omitting the first term if $m = 0$, and $\tau_1 = -\log(-\psi)$. Each of the three functions τ formed by the two terms of τ_0 and by τ_1 is of the form $\tau = -\log g$ with $g > 0$ smooth and $-\mathrm{dd^c} g \geq 0$, namely $g = r'^2 - |z'|^2$, $g = r''^2 - |w|^2$ and $g = -\psi$; hence

$$\mathrm{dd^c}\tau = -\frac{\mathrm{dd^c} g}{g} + \frac{\mathrm{i}}{2\pi g^2}\partial g \wedge \bar\partial g \geq \frac{\mathrm{i}}{2\pi}\partial\tau \wedge \bar\partial\tau \geq 0.$$

Thus τ_0 and τ_1 are plurisubharmonic, and $\sigma = \beta + \mathrm{dd^c}\tau_0 + \mathrm{dd^c}\tau_1 \geq \beta$ is a Kähler form. Moreover $\sigma \geq \mathrm{dd^c}\tau \geq \frac{\mathrm{i}}{2\pi}\partial\tau \wedge \bar\partial\tau$ for each of the three functions; the eigenvalue of the rank-one form $\frac{\mathrm{i}}{2\pi}\partial\tau \wedge \bar\partial\tau$ with respect to σ is $|\partial\tau|^2_\sigma/(2\pi)$, so $|\partial\tau|^2_\sigma \leq 2\pi$, and these functions have bounded gradients for σ. The function $\tau_0 - \tau_1$ tends to

$+\infty$ at the boundary of D and along $\{w=0\}$, and its sublevel sets are compact in $D\setminus\{w=0\}$. If $\lambda\in C^\infty(\mathbb{R})$ equals one on $(-\infty,1]$ and zero on $[2,\infty)$, the functions $\chi_j=\lambda((\tau_0-\tau_1)/j)$ are compactly supported, tend to one, and have gradients $O(1/j)$. The canonical bundle of $\mathbb{C}^n$ is trivialized by $\mathrm{d}z_1\wedge\cdots\wedge\mathrm{d}z_n$, whose norm for β is constant, so we identify functions with $(n,0)$-forms, as in the proof of Theorem 14.6.7; this multiplies all the pointwise norms below by the same constant factor.

The correction. Let $G(z',w)=f(z')$. Choose $\rho\in C^\infty(\mathbb{R})$ with $0\le\rho\le1$, equal to one on $(-\infty,1/2]$ and to zero on $[1,\infty)$, and put $G_\varepsilon=\rho(R/\varepsilon)G$. Since $\bar\partial R/\varepsilon=(1+R/\varepsilon)\bar\partial\psi_\varepsilon$,

$$g_\varepsilon=\bar\partial G_\varepsilon=\rho'(R/\varepsilon)(1+R/\varepsilon)\bar\partial\psi_\varepsilon\,G,$$

which is supported in $\{\varepsilon/2\le R\le\varepsilon\}$. For an $(n,1)$-form v, the rank-one part of the lower bound for H_ε and Lemma 14.6.1(1),(2), applied with $\alpha=\partial\psi_\varepsilon$, give

$$|\langle\bar\partial\psi_\varepsilon\wedge G,v\rangle|^2\le\frac{R}{\varepsilon}\,|G|^2\mathrm{e}^{-P}R^{-q}\langle[H_\varepsilon,\Lambda_\beta]v,v\rangle,$$

where $|G|$ is the modulus of the function G and the inner products are those of β and the metric $\mathrm{e}^{-P-q\psi}=\mathrm{e}^{-P}R^{-q}$. Hence, on the support of g_ε, where $R/\varepsilon\le1$, we have $\langle[H_\varepsilon,\Lambda_\beta]^{-1}g_\varepsilon,g_\varepsilon\rangle\le C|G|^2\mathrm{e}^{-P}R^{-q}$, and the integral M_{g_ε} of Lemma 14.7.1 is at most

$$C\int_{\{\varepsilon/2\le R\le\varepsilon\}}|G|^2\mathrm{e}^{-P}R^{-q}\,\mathrm{d}V.$$

In polar coordinates in w, the integral of R^{-q} over the shell $\varepsilon/2\le|w|^2\le\varepsilon$ is a fixed multiple of $\int_{\sqrt{\varepsilon/2}}^{\sqrt{\varepsilon}}\mathrm{d}t/t=\frac12\log2$. Since P is uniformly continuous on $\overline{D}$ and G does not depend on w, the last integral tends to a fixed multiple of $\int_{B'}|f|^2\mathrm{e}^{-P(z',0)}\,\mathrm{d}V_{B'}$ as $\varepsilon\to0$. Lemma 14.7.1 gives u_ε on $D\setminus\{w=0\}$ with $\bar\partial u_\varepsilon=g_\varepsilon$ and

$$\int_D\frac{|u_\varepsilon|^2\mathrm{e}^{-P}R^{-q}}{a_\varepsilon+b_\varepsilon}\,\mathrm{d}V\le C\int_{\{\varepsilon/2\le R\le\varepsilon\}}|G|^2\mathrm{e}^{-P}R^{-q}\,\mathrm{d}V.$$

For fixed ε the denominator is bounded and $\mathrm{e}^{-P}R^{-q}$ is bounded below, so u_ε is square integrable, and $F_\varepsilon=G_\varepsilon-u_\varepsilon$, which is holomorphic off $\{w=0\}$, extends holomorphically to D by Lemma 14.6.6. The difference $u_\varepsilon=G_\varepsilon-F_\varepsilon$ is then smooth near $\{w=0\}$, and it vanishes there: a nonzero value at a point $(z_0',0)$ would make $\int|u_\varepsilon|^2|w|^{-2q}$ diverge near that point, since $\int|w|^{-2q}\,\mathrm{d}V(w)$ diverges logarithmically at $w=0$ in $\mathbb{C}^q$. Hence $F_\varepsilon(z',0)=G_\varepsilon(z',0)=f(z')$.

The limit $\varepsilon\to0$. By Lemma 14.7.2(1), $a_\varepsilon+b_\varepsilon\le5(-\log(\varepsilon+R))^2$, so the correction estimate bounds u_ε in the weaker weight $(\varepsilon+R)^{-q}(-\log(\varepsilon+R))^{-2}\mathrm{e}^{-P}$. In this weight the squared norm of G_ε is at most $C_P(-\log\varepsilon)^{-2}$, because its support has normal volume $O(\varepsilon^q)$ and $(\varepsilon+R)^{-q}\le\varepsilon^{-q}$ there; the constant C_P is finite for the fixed smooth weight P. Consequently the squared norms of F_ε in this weight have upper limit, as $\varepsilon\to0$, at most a constant, depending only on m,q,r',r'', times $\int_{B'}|f|^2\mathrm{e}^{-P(z',0)}$. Since $t^{-q}(-\log t)^{-2}$ is bounded below on $(0,\mathrm{e}^{-3}]$, the F_ε

are bounded in $L^2(D)$, and by Lemma 14.5.2(1) a sequence $\varepsilon_k \to 0$ gives functions converging locally uniformly to a holomorphic F with $F(z',0) = f(z')$. By Fatou's lemma, F satisfies (14.24) with P in place of p, and removing the bounded factor $\mathrm{e}^{-\pi|z|^2}$ gives (14.24) for smooth p.

General p. Take smooth plurisubharmonic functions $p_l \downarrow p$ on a neighborhood of $\overline{D}$, obtained by convolution: by (3.3) the convolutions $p * \rho_{1/l}$ decrease to p, they are smooth, and they are plurisubharmonic because $\mathrm{dd^c}(p * \rho_{1/l}) = (\mathrm{dd^c} p) * \rho_{1/l} \geq 0$ (Theorem 8.2.1). Their restrictions $p_l(z',0)$ decrease to $p(z',0)$, so the right sides of (14.24) for p_l are at most the right side for p, and the smooth case gives extensions F_l satisfying (14.24) for p_l with the same constant. Since $p_l \leq p_1$, the weights have a common positive lower bound, so the F_l are bounded in $L^2(D)$ and, by Lemma 14.5.2(1), a subsequence converges locally uniformly on D, including on $\{w = 0\}$. The limit F satisfies $F(z',0) = f(z')$, and Fatou's lemma gives (14.24). □

Lemma 14.7.4 *Let $Y \subset X$ be a nonempty connected closed complex submanifold of codimension $q \geq 1$. There is a smooth function $R_Y \geq 0$ on X, vanishing exactly on Y, with the following properties.*

(1) *Let $y \in Y$. Since R_Y attains its minimum at y, its differential vanishes there, so $H_y(v,v) = (R_Y \circ \gamma)''(0)$, for smooth curves γ with $\gamma(0) = y$ and $\gamma'(0) = v$, is a well-defined quadratic form on the real tangent space T_yX, the real Hessian of R_Y at y. It satisfies $H_y(v,v) > 0$ for every $v \in T_yX \setminus T_yY$.*

(2) *There is a constant $b \geq 0$, depending only on X, ω and R_Y, such that*

$$\mathrm{dd^c} \log R_Y \geq -b\omega, \qquad \mathrm{dd^c} \log(R_Y + \varepsilon) \geq -b\omega \quad (\varepsilon > 0).$$

(3) *Let (z',w) be holomorphic coordinates on an open set $U \subset X$ with $Y \cap U = \{w = 0\}$, and write $(0,w)$ for the tangent vector with components 0 in the z'-directions and w in the w-directions. Then $Q_{z'}(w) = \frac{1}{2} H_{(z',0)}\big((0,w),(0,w)\big)$ is a positive definite Hermitian form on $\mathbb{C}^q$ depending continuously on z', and every point of $Y \cap U$ has a neighborhood on which*

$$\big|R_Y(z',w) - Q_{z'}(w)\big| \leq C|w|^3, \qquad c|w|^2 \leq R_Y(z',w) \leq C|w|^2, \tag{14.25}$$

for constants $c, C > 0$.

(4) *The function $\log R_Y$ has gentle analytic singularities of type $(1, \mathcal{I}_Y)$, where $\mathcal{I}_Y$ is the ideal sheaf of Y.*

Multiplying R_Y by a small positive constant, we may assume that $\max_X R_Y < \mathrm{e}^{-4}$.

Proof Cover a neighborhood of Y by finitely many coordinate charts with coordinates (z',w), $w \in \mathbb{C}^q$, in which $Y = \{w = 0\}$, and add one open set whose closure does not meet Y. Choose smooth functions $\chi_i \geq 0$ with compact supports in these sets and $\sum_i \chi_i^2 = 1$ on X; for instance, divide a finite family of bump functions by the square root of the sum of their squares. On a normal chart put $K_i = \sum_{l=1}^q |w_l|^2$, on the additional set put $K_0 = 1$, and define $R_Y = \sum_i \chi_i^2 K_i$, each term being extended by zero outside its chart.

The functions K_i are uniformly comparable on the overlaps where they occur. Indeed, let (z', w) and $(\tilde{z}', \tilde{w})$ be the coordinates of two normal charts, and let $y \in Y$ lie in both. In the coordinates (z', w) choose a polydisc $P' \times P''$ centered at y and contained in both charts. Each $\tilde{w}_l$ vanishes on $Y \cap (P' \times P'') = \{w = 0\}$, and since P'' is a polydisc centered at 0,

$$\tilde{w}_l(z', w) = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} \tilde{w}_l(z', tw)\, \mathrm{d}t = \sum_{k=1}^{q} w_k\, a_{lk}(z', w),$$

where the functions $a_{lk}(z', w) = \int_0^1 (\partial \tilde{w}_l / \partial w_k)(z', tw)\, \mathrm{d}t$ are holomorphic, by differentiation under the integral sign, and bounded on smaller polydiscs. Hence $\sum_l |\tilde{w}_l|^2 \le C \sum_k |w_k|^2$ near y, and by symmetry the reverse inequality holds. The compact set $\operatorname{Supp} \chi_i \cap \operatorname{Supp} \chi_j \cap Y$ is covered by finitely many such smaller polydiscs, and on the remaining part of $\operatorname{Supp} \chi_i \cap \operatorname{Supp} \chi_j$, a compact set disjoint from Y, the functions K_i and K_j are continuous and positive, hence bounded above and below; the same holds on $\operatorname{Supp} \chi_0 \cap \operatorname{Supp} \chi_i$, which does not meet Y. Hence there is one constant $C \ge 1$ with $K_j \le C K_i$ on $\operatorname{Supp} \chi_i \cap \operatorname{Supp} \chi_j$ for all i, j. At every point some $\chi_i^2 \ge 1/J$, where J is the number of sets in the cover, so $K_j / R_Y \le JC$ wherever K_j contributes and $R_Y > 0$, and likewise $(K_j + \varepsilon)/(R_Y + \varepsilon) \le JC$.

To prove (2), let $x \in X$. The terms with $x \notin \operatorname{Supp} \chi_j$ vanish near x, and for the remaining indices j the point x lies in the chart of K_j. On a small neighborhood of x we may therefore apply the identity (13.5) of Lemma 13.4.1(2) to these indices, with $K_j + \varepsilon$ in place of the functions denoted there by e^{w_j}, so that $S = R_Y + \varepsilon$. The term with pairwise differences is semipositive, and so is the term with the Hessians $\mathrm{dd^c} \log(K_j + \varepsilon)$, since $\log(\sum_l |w_l|^2 + \varepsilon)$ is plurisubharmonic. The remaining term is

$$2 \sum_j \frac{K_j + \varepsilon}{R_Y + \varepsilon} \left(\chi_j \mathrm{dd^c} \chi_j - \frac{\mathrm{i}}{2\pi} \partial \chi_j \wedge \bar{\partial} \chi_j \right),$$

which is bounded below by $-b\omega$ because the cutoffs have bounded first and second derivatives and the ratios are bounded by JC. This proves the second inequality in (2), and the first follows as $\varepsilon \downarrow 0$, since $\log(R_Y + \varepsilon)$ decreases to $\log R_Y$.

To prove (1), let $y \in Y$, and let γ be a smooth curve with $\gamma(0) = y$ and $\gamma'(0) = v$. In coordinates the chain rule gives $(R_Y \circ \gamma)''(0) = D^2 R_Y(y)(v, v)$, because $D R_Y(y) = 0$, so H_y is well defined. The term $\chi_0^2 K_0$ vanishes near Y, and a term $\chi_i^2 K_i$ with $y \notin \operatorname{Supp} \chi_i$ vanishes near y. For a normal chart with $y \in \operatorname{Supp} \chi_i$ and normal coordinates w, we have $w(\gamma(t)) = t\, \mathrm{d}w_y(v) + O(t^2)$, hence $(\chi_i^2 K_i)(\gamma(t)) = t^2 \chi_i(y)^2 |\mathrm{d}w_y(v)|^2 + O(t^3)$. Therefore

$$H_y(v, v) = 2 \sum_i \chi_i(y)^2 |\mathrm{d}w_y^{(i)}(v)|^2,$$

the sum running over the normal charts, with normal coordinates $w^{(i)}$. Since the kernel of $\mathrm{d}w_y^{(i)}$ on $T_y X$ is $T_y Y$ and $\sum_i \chi_i(y)^2 = 1$, we get $H_y(v, v) > 0$ for $v \notin T_y Y$.

To prove (3), let (z', w) be coordinates on U as in the statement, and let $y = (z', 0) \in Y \cap U$. Since $\mathrm{d}w_y^{(i)}$ is complex linear, the formula for H_y shows that $Q_{z'}(w) = \sum_i \chi_i(y)^2 |\mathrm{d}w_y^{(i)}(0, w)|^2$ is a Hermitian form in w, and it is positive definite because $(0, w) \notin T_y Y$ for $w \neq 0$. In the coordinates, $Q_{z'}(w)$ is half the second derivative of R_Y at $(z', 0)$ in the direction $(0, w)$, so it depends continuously on z'. Let $V' \times V''$ be a product of closed balls in the coordinates, centered at a point of $Y \cap U$ and contained in U. Taylor's formula in the variable w at $(z', 0)$, where R_Y and its first derivatives vanish, gives $|R_Y(z', w) - Q_{z'}(w)| \leq C|w|^3$ on $V' \times V''$, with C depending on a bound for the third derivatives of R_Y there. If $c_0 > 0$ and C_0 are the minimum and the maximum of $Q_{z'}(w)$ for $z' \in V'$ and $|w| = 1$, then $c_0|w|^2 - C|w|^3 \leq R_Y \leq C_0|w|^2 + C|w|^3$ on $V' \times V''$, and (14.25) follows on the set where $|w| \leq c_0/(2C)$.

Finally, for (4), note that on the blowup of Y, or on any log resolution of $\mathcal{I}_Y$, every normal tuple is a local equation of the exceptional divisor times a tuple without common zeros, because the tuples generate $\mathcal{I}_Y$; so $\log R_Y$ is the logarithm of the squared modulus of that equation plus a smooth function. Since moreover $\exp(\log R_Y) = R_Y$ is smooth on X, $\log R_Y$ has gentle analytic singularities of type $(1, \mathcal{I}_Y)$. Multiplying R_Y by a constant $\lambda > 0$ adds $\log \lambda$ to $\log R_Y$ and replaces $\log(R_Y + \varepsilon)$ by $\log \lambda + \log(R_Y + \varepsilon/\lambda)$, so the inequalities persist. □

The proof of Lemma 14.7.7 below uses the following computation with log discrepancies (Definition 12.6.3). Let Z be a complex manifold and $E \subset Z$ a smooth divisor. We consider finite sequences of blowups $Z_k \to Z_{k-1} \to \cdots \to Z_0 = Z$ with smooth centers in which the *boundary* of Z_0 is E, the boundary of Z_i is the union of the inverse image of the boundary of Z_{i-1} and the exceptional divisor of $Z_i \to Z_{i-1}$, and each center has normal crossings with the boundary of the model on which it lies. For a prime divisor D on the final model, put $b_D = \mathrm{ord}_D(E)$, the vanishing order along D of the pullback of a local equation of E, and $A_D = A_Z(D)$. We call $A_D - b_D$ the *margin* of D.

Lemma 14.7.5 *In this situation, $A_D - b_D = 0$ if D is the strict transform of E, and $A_D - b_D > 0$ for every other prime divisor D in the boundary.*

Proof We argue by induction on the number of blowups. On Z itself, $A_E = b_E = 1$, and every other prime divisor D of Z has $A_D = 1$ and $b_D = 0$. Strict transforms keep their log discrepancies and orders, hence their margins. Since every exceptional divisor lies in the boundary, a prime divisor of a model outside the boundary is the strict transform of a prime divisor of Z other than E, so it has $A_D = 1$ and $b_D = 0$. Suppose that the statement holds on a model. Blowing up a center of codimension one is an isomorphism and creates no new divisor; it only adds the center to the boundary, and the center then has margin one if it was not a boundary component. It therefore suffices to show that the exceptional divisor E_{new} of the blowup of a center C of codimension $c \geq 2$ has a positive margin.

Suppose that C is contained in exactly r boundary components, near its generic point. Since C has normal crossings with the boundary, $r \leq c$. Near such a point choose coordinates (z, w), $w = (w_1, \ldots, w_c)$, in which $C = \{w = 0\}$ and the

boundary components through the point are coordinate hyperplanes; those containing C are then hyperplanes $\{w_j = 0\}$. By Proposition A.5.3, in the coordinates (z, t, v) of (A.6) we have $E_{\text{new}} = \{t = 0\}$, and the Jacobian determinant of the blowup is $\pm t^{c-1}$, so it vanishes to order $c-1$ along E_{new}. A local equation of a component $\{w_j = 0\}$ containing C is a unit times w_j (Lemma 12.6.1(1)), and w_j pulls back to t or to tv_j, which vanishes to order one along E_{new}, because v_j does not vanish identically on $\{t = 0\}$ (Lemma 12.6.1(2),(4)). Thus the pullback of a local equation of each component containing C vanishes to order one along E_{new}. Hence $A_{\text{new}} = c + \sum_{D \supset C}(A_D - 1)$ and $b_{\text{new}} = \sum_{D \supset C} b_D$, the sums running over the boundary components containing C (other divisors through C have $A_D = 1$ and $b_D = 0$ and contribute nothing). Therefore

$$A_{\text{new}} - b_{\text{new}} = c - r + \sum_{D \supset C} (A_D - b_D).$$

All terms on the right are nonnegative by the induction hypothesis. If one of the components containing C has a positive margin, so does E_{new}. Otherwise only the strict transform of E can contain C, so $r \le 1$ and $c - r \ge 1$, and again the margin of E_{new} is positive. □

Lemma 14.7.6 *Let $L \to X$ be a holomorphic line bundle with a smooth quadratic metric h_0 and a plurisubharmonic metric $h = h_0 \mathrm{e}^{-\Phi}$. Let Y be a nonempty connected closed complex submanifold of codimension $q \ge 1$ such that the restriction of h to Y is defined, and let R_Y be as in Lemma 14.7.4, with $\max_X R_Y < \mathrm{e}^{-4}$. If f is a holomorphic section of $L|_Y$ with $\int_Y |f|_h^2 \, \mathrm{d}V_Y < \infty$, there are $\kappa > 0$, finitely many open sets $U_i \subset X$, holomorphic sections F_i of L over U_i with $F_i = f$ on $Y \cap U_i$, and smooth functions η_i with compact support in U_i and $\sum_i \eta_i = 1$ on a neighborhood of Y, such that*

$$\int_{U_i} \frac{|F_i|_{h_0}^2 \mathrm{e}^{-(1+\kappa)\Phi}}{R_Y^q (-\log R_Y)^2} \, \mathrm{d}V < \infty \quad \textit{for every } i, \tag{14.26}$$

and such that $G = \sum_i \eta_i F_i$, each term extended by zero, is a smooth global section of L with $G|_Y = f$, $\bar\partial G|_Y = 0$, and

$$\int_X \frac{(|G|_{h_0}^2 + |\bar\partial G|_{\omega, h_0}^2) \mathrm{e}^{-(1+\kappa)\Phi}}{R_Y^q (-\log R_Y)^2} \, \mathrm{d}V < \infty. \tag{14.27}$$

In particular

$$\int_{\{R_Y < \varepsilon\}} |\bar\partial G|_{\omega, h}^2 R_Y^{-q} \, \mathrm{d}V \longrightarrow 0 \qquad (\varepsilon \downarrow 0). \tag{14.28}$$

Here $\mathrm{d}V_Y = \omega|_Y^{n-q}/(n-q)!$; if Y is a point, $\mathrm{d}V_Y$ is the unit mass at that point.

Proof In a holomorphic frame of L the local weight p of h is plurisubharmonic, and its restriction to Y is plurisubharmonic, because the restriction is defined and Y is connected. If Y is a point, the value of p there is finite and we take any $\kappa > 0$. Otherwise, by the strong openness theorem (Corollary 12.5.4), every point of Y has a

neighborhood in Y on which f is square integrable with the weight $\mathrm{e}^{-(1+\kappa)p|_Y}$ for some $\kappa > 0$; since Y is compact, finitely many coordinate products $B_i' \times B_i''$ of the form of Lemma 14.7.3, with tangential balls slightly larger than needed and each contained in a neighborhood on which (14.25) holds, cover Y, and one $\kappa > 0$ works for all of them.

Lemma 14.7.3, applied in a frame with the plurisubharmonic weight $(1+\kappa)p$, gives holomorphic extensions F_i of f on these products whose critical integrals with the logarithmic gain are finite. Let U_i be smaller concentric products that still cover Y. On U_i, the smooth factors coming from the frames and the comparison $c|w|^2 \le R_Y \le C|w|^2$ of (14.25) preserve this finiteness, which gives (14.26); for the logarithmic factor we use that $-\log R_Y \ge c'(-\log|w|^2)$ for a constant $c' > 0$, since $R_Y \le C|w|^2$ and both R_Y and $|w|^2$ are less than e^{-4}.

We now glue the local extensions. Choose smooth functions η_i with compact support in U_i whose sum is one on a neighborhood of Y; they are obtained by grouping the members of a partition of unity with compact supports subordinate to the cover of X by the sets U_i and $X \setminus Y$ (Proposition 7.1.3). Put $G = \sum_i \eta_i F_i$, each term extended by zero. Then $G|_Y = f$. Since all F_i are holomorphic with the same restriction f, we have $\bar\partial G = \sum_i (\bar\partial \eta_i) F_i$, and on Y this equals $f\,\bar\partial \sum_i \eta_i = 0$. The bounds on the cutoffs and their derivatives and (14.26) give (14.27).

Put $\gamma = |\bar\partial G|^2_{\omega,h_0}$. Since $\bar\partial G$ is smooth and vanishes on Y, we have $\gamma \le C|w|^2$ in normal coordinates near each point of Y, hence $\gamma \le C R_Y$ near Y, by (14.25) and the compactness of Y. Let $A = \{R_Y < \varepsilon\}$; all integrals below are taken with $\mathrm{d}V$. Hölder's inequality with the exponents $1+\kappa$ and $(1+\kappa)/\kappa$ gives

$$\int_A \gamma\, \mathrm{e}^{-\Phi} R_Y^{-q} \le \left(\int_A \frac{\gamma \mathrm{e}^{-(1+\kappa)\Phi}}{R_Y^q (-\log R_Y)^2} \right)^{1/(1+\kappa)} \left(\int_A \gamma R_Y^{-q} (-\log R_Y)^{2/\kappa} \right)^{\kappa/(1+\kappa)} .$$

The first factor is bounded by (14.27). In normal polar coordinates, using (14.25) and $\gamma = O(R_Y)$, the integrability of $\gamma R_Y^{-q}(-\log R_Y)^{2/\kappa}$ near Y reduces to $\int_0^r (-\log t)^{2/\kappa}\,\mathrm{d}t < \infty$. Hence the second factor tends to zero as $\varepsilon \downarrow 0$, by the absolute continuity of the integral, which proves (14.28). □

Lemma 14.7.7 *Let $L \to X$ be a holomorphic line bundle with a smooth quadratic metric h_0 and a plurisubharmonic metric $h = h_0 \mathrm{e}^{-\Phi}$, where Φ has gentle analytic singularities. Let Y be a nonempty connected closed complex submanifold of codimension $q \ge 1$ to which h restricts, let f be a holomorphic section of $L|_Y$ with $\int_Y |f|^2_h\,\mathrm{d}V_Y < \infty$, and let G be constructed from f as in Lemma 14.7.6. Then*

$$\varlimsup_{\varepsilon \downarrow 0} \int_{\{\varepsilon/2 \le R_Y \le \varepsilon\}} |G|^2_h R_Y^{-q}\,\mathrm{d}V \le C_Y \int_Y |f|^2_h\,\mathrm{d}V_Y, \qquad \mathrm{d}V_Y = \omega|_Y^{n-q}/(n-q)!, \tag{14.29}$$

where C_Y depends only on Y, R_Y and ω, and not on Φ or f.

Proof We compute on a log resolution adapted to Φ and Y; by Lemma 14.7.5, only the strict transform of the exceptional divisor of the blowup of Y contributes to the limit.

A resolution adapted to Φ *and* Y. Let $\mathrm{Bl}_Y X \to X$ be the blowup of Y, with exceptional divisor E; for $q = 1$ this is the identity and $E = Y$. By Proposition A.5.3, its Jacobian determinant is $\pm t^{q-1}$ in the charts of (A.6), where $E = \{t = 0\}$, so its relative canonical divisor is $(q-1)E$. Let $\mathfrak{a}$ be the ideal in the type of Φ. We apply Theorem A.5.5 on the compact manifold $\mathrm{Bl}_Y X$ to the pullback of $\mathfrak{a}$, which is nonzero, and to the divisor E. This gives a finite sequence of blowups with smooth centers, each having normal crossings with the accumulated divisor of Theorem A.5.5, which starts with E; this accumulated divisor is the boundary in the sense of Lemma 14.7.5. On the final model the boundary has simple normal crossings, and the pullback of $\mathfrak{a}$ is the ideal of an effective divisor supported in the boundary. Denote the composite map to X by π. It is a composite of blowups along closed submanifolds that principalizes $\mathfrak{a}$, that is, a log resolution of $\mathfrak{a}$ in the sense of Section 13.1. Since Definition 13.1.11 requires a smooth remainder on every log resolution, $\Phi \circ \pi$ has a smooth remainder.

Divisor orders. In a normal-crossing chart on the final model write

$$\Phi \circ \pi = \sum_D a_D \log |s_D|^2 + v, \qquad R_Y \circ \pi = \prod_D |s_D|^{2b_D} r,$$

with v smooth, r smooth and strictly positive, and $a_D, b_D \geq 0$. The Jacobian factor of the pulled-back volume is $\prod_D |s_D|^{2(A_D-1)}$ times a smooth positive function, where $A_D = A_X(D)$ is the log discrepancy. Let F_i be one of the holomorphic extensions used in G, and suppose that the chart lies over the open set U_i of Lemma 14.7.6. Then $F_i \circ \pi$ vanishes to some integral order $e_{i,D} \geq 0$ along each coordinate divisor D of the chart. By (14.26), the integral of $|F_i|_{h_0}^2 \mathrm{e}^{-(1+\kappa)\Phi} R_Y^{-q} (-\log R_Y)^{-2}$ over U_i is finite, and this forces

$$e_{i,D} + A_D \geq (1+\kappa)a_D + qb_D \quad (b_D > 0), \qquad e_{i,D} + A_D > (1+\kappa)a_D \quad (b_D = 0). \tag{14.30}$$

Indeed, fix the other coordinates away from their divisors and integrate radially in s_D: the integrand behaves like $|s_D|^{2(e_{i,D}+A_D-(1+\kappa)a_D-qb_D)-2}(-\log|s_D|^2)^{-2}$ when $b_D > 0$, and the squared logarithm in the denominator makes the exponent -1 of the radial variable integrable but cannot compensate a smaller exponent. If $b_D = 0$, the logarithmic factor is bounded above and below near generic points of D, and the inequality must be strict. This is the computation of Theorem 12.6.5 with one additional logarithmic factor.

Shrinking the chart, we may assume that for each i either the chart lies over U_i or $\eta_i \circ \pi$ vanishes on it, because $\operatorname{Supp} \eta_i$ is a compact subset of the open set U_i. Then, on a smaller chart, the pullback of $G = \sum_i \eta_i F_i$ is the monomial $\prod_D s_D^{e_D}$, with $e_D = \min_i e_{i,D}$ over the indices i of the first kind, times a smooth bounded function. The density of $|G|_h^2 R_Y^{-q}\, \mathrm{d}V$ is therefore bounded by a constant times $\prod_D |s_D|^{2(\gamma_D-1)}$, where

$$\gamma_D = e_D - a_D - qb_D + A_D.$$

If $a_D > 0$, (14.30) gives $\gamma_D \geq \kappa a_D > 0$, and if $a_D = b_D = 0$, it gives $\gamma_D > 0$. If $a_D = 0$ and $b_D > 0$, the formula for the relative canonical divisor of the first blowup gives $A_D = A_{\mathrm{Bl}_Y X}(D) + (q-1)\operatorname{ord}_D(E)$, and $b_D = \operatorname{ord}_D(E)$ because $\mathcal{I}_Y$ pulls back

to the ideal of E; hence

$$A_D - qb_D = A_{\mathrm{Bl}_Y X}(D) - \mathrm{ord}_D(E).$$

By Lemma 14.7.5, applied with $Z = \mathrm{Bl}_Y X$, this is positive unless D is the strict transform of E. For that divisor $b_E = 1$ and $A_E = q$, and $a_E = 0$: since the restriction of h to Y is defined, generic points of Y lie outside the pole set of Φ. Hence $\gamma_E = e_E \geq 0$, and the strict transform of E is the only divisor along which γ_D can vanish.

The annular limit. By a finite partition of unity on the compact inverse image of Y, it suffices to consider the integral over a small chart. If no coordinate divisor of the chart is the strict transform of E with $\gamma_E = 0$, the density is integrable, the shells $\{\varepsilon/2 \leq R_Y \leq \varepsilon\}$ shrink to the null set $\pi^{-1}(Y)$, and dominated convergence shows that the integral over the shell tends to zero. Otherwise let t be the coordinate defining the strict transform of E. The density is $|t|^{-2}$ times a smooth bounded coefficient and the factors $|s_D|^{2(\gamma_D - 1)}$, $D \neq E$, which are integrable since $\gamma_D > 0$. For fixed values of the remaining coordinates s, the shell is $\varepsilon/2 \leq |t|^2 m(s) r(t,s) \leq \varepsilon$, with $m(s) = \prod_{D \neq E} |s_D|^{2b_D}$. Since r is bounded above and below, the shell lies in a range of $\log|t|$ of bounded length, so the integral of $|t|^{-2}$ over it is bounded independently of ε and s; this gives an integrable dominating function. As $\varepsilon \to 0$, the shell shrinks to $t = 0$, the smooth coefficient tends to its value at $t = 0$, and the integral of $|t|^{-2}$ over the shell tends to $\pi \log 2$. By dominated convergence, only the restriction to $t = 0$ of the coefficient of $|t|^{-2}$ in the density contributes to the limit.

Identification of the limit. The argument just given, applied after multiplying the density by a continuous function with compact support, shows that the integrals over the shells converge to $\pi \log 2$ times the total mass of a measure μ on the strict transform of E; in a chart as above, μ has the density obtained by restricting to $t = 0$ the smooth coefficient of $|t|^{-2}$, multiplied by the factors $|s_D|^{2(\gamma_D - 1)}$, $D \neq E$. Since μ is determined by the limits of integrals over the shells $\{\varepsilon/2 \leq R_Y \leq \varepsilon\}$, it does not depend on the coordinates in which it is computed, and since its density in every chart is integrable, it gives no mass to proper analytic subsets. We may therefore compute μ on the dense open subset of the strict transform of E lying over the complement of the pole set of Φ in Y and over which the further blowups are isomorphisms, using coordinates of the first blowup $\mathrm{Bl}_Y X$.

Let (z', w) be coordinates on X in which $Y = \{w = 0\}$, and write $\mathrm{d}V = J(z', w)\, \mathrm{d}V(z')\, \mathrm{d}V(w)$ with J smooth and positive. By Lemma 14.7.4, $R_Y = Q_{z'}(w) + O(|w|^3)$ locally uniformly, where $Q_{z'}$ is a positive definite Hermitian form on $\mathbb{C}^q$ depending continuously on z'. Over this chart, $\mathrm{Bl}_Y X$ is covered by charts with coordinates (z', t, ξ), $\xi \in \mathbb{C}^{q-1}$, in which, after renumbering the coordinates w, we have $w = t(1, \xi)$ and $E = \{t = 0\}$ (for $q = 1$ there is no ξ, and $t = w$). There $\mathrm{d}V(w) = |t|^{2(q-1)}\, \mathrm{d}V(t)\, \mathrm{d}V(\xi)$ and $R_Y = |t|^2 \big(Q_{z'}(1, \xi) + O(|t|)\big)$, so away from the poles of Φ the density of $|G|_h^2 R_Y^{-q}\, \mathrm{d}V$ is $|t|^{-2}$ times the smooth function

$$|G|_{h_0}^2 \mathrm{e}^{-\Phi} \big(Q_{z'}(1, \xi) + O(|t|)\big)^{-q} J,$$

whose restriction to $t = 0$ equals $|f(z')|_h^2 Q_{z'}(1,\xi)^{-q} J(z',0)$, because $G|_Y = f$. The points of the projective normal fibers not covered by such a chart form a hyperplane in each fiber and are negligible. Integrating over ξ, we find that the image of μ under the projection to Y has the density $c(z')|f(z')|_h^2$ with respect to $\mathrm{d}V(z')$, where

$$c(z') = J(z',0) \int_{\mathbb{C}^{q-1}} Q_{z'}(1,\xi)^{-q}\, \mathrm{d}V(\xi).$$

Since $Q_{z'}(1,\xi) \geq c_0(1+|\xi|^2)$ with $c_0 > 0$ locally uniformly in z', and $(1+|\xi|^2)^{-q}$ is integrable on $\mathbb{C}^{q-1}$, the function c is locally bounded. It depends only on ω, R_Y and the chart. Covering the compact manifold Y by finitely many such charts and comparing $\mathrm{d}V(z')$ with $\mathrm{d}V_Y$, we obtain (14.29). □

Theorem 14.7.8 *Let $Y \subset X$ be a nonempty closed connected complex submanifold, and put $\mathrm{d}V_Y = \omega|_Y^{\dim Y}/(\dim Y)!$, the unit mass if Y is a point. There are constants $N, C > 0$, depending only on X, ω and Y, with the following property. Let L be a holomorphic line bundle with a smooth quadratic metric h_0 and a plurisubharmonic metric $h = h_0 \mathrm{e}^{-\varphi}$ such that $c_1(L,h) \geq N\omega$, and suppose that the restriction of h to Y is defined, that is, the local weights of h are not identically $-\infty$ on Y. Then every $f \in H^0(Y, L|_Y)$ with $\int_Y |f|_h^2\, \mathrm{d}V_Y < \infty$ has an extension $s \in H^0(X,L)$ such that*

$$s|_Y = f, \qquad \int_X |s|_h^2\, \mathrm{d}V \leq C \int_Y |f|_h^2\, \mathrm{d}V_Y. \tag{14.31}$$

Proof If $Y = X$, take $s = f$, any N and $C = 1$. Suppose that Y has codimension $q \geq 1$; if $f = 0$, take $s = 0$. As in the proof of Theorem 14.6.7, we cut off a smooth extension of f and correct the error, now with Lemma 14.7.1; a pole of order q along Y in the weight makes the correction vanish on Y. Fix $R = R_Y$ and the constant b of Lemma 14.7.4, with $\max_X R < \mathrm{e}^{-4}$, and choose $b_K \geq 0$ with $c_1(K_X^{-1}, h_{K^{-1},\omega}) \geq -b_K\omega$ (Lemma 13.1.1). In Steps 1–4 we prove the theorem for a weight Φ with gentle analytic singularities, as for the approximations of Theorem 13.4.5, under the curvature bound

$$c_1(L, h_0\mathrm{e}^{-\Phi}) \geq N_0\omega, \qquad N_0 = b_K + (q+2)b + 2.$$

In these steps h denotes the metric $h_0\mathrm{e}^{-\Phi}$, and A the pole set of Φ. Step 5 treats arbitrary weights.

Step 1: the twisted curvature form. For the given f choose G and κ by Lemma 14.7.6. On $M = X \setminus (A \cup Y)$ consider $F = L \otimes K_X^{-1}$ with the metric $h \otimes h_{K^{-1},\omega}\mathrm{e}^{-q\psi}$, where $\psi = \log R$; the canonical identification of L-valued functions with F-valued $(n,0)$-forms preserves the squared norms times the volume. The manifold M carries a Kähler form and cutoffs as required in Lemma 14.7.1, by Lemma 14.6.4 applied to $\Phi + \psi$, which is quasi-plurisubharmonic, has pole set $A \cup Y$ and is smooth on M. The resulting Kähler form depends on Φ, but the estimate of Lemma 14.7.1 does not involve it.

Let $\psi_\varepsilon = \log(\varepsilon + R)$, μ, a_ε and b_ε be the functions of Lemma 14.7.2, applied on M to the smooth function R, which satisfies $0 < R < \mathrm{e}^{-4}$ there, for $0 < \varepsilon < \mathrm{e}^{-4}$. For

fixed ε, a_ε and b_ε are bounded above and below by positive constants, $a_\varepsilon \geq 1$ and $1 \leq -\mu'(\psi_\varepsilon) \leq 2$. By part (2) of that lemma, applied with 2π times the curvature form of the metric of F,

$$H_\varepsilon = a_\varepsilon\bigl(2\pi c_1(L,h) + 2\pi c_1(K_X^{-1}, h_{K^{-1},\omega}) + q\,\mathrm{i}\partial\bar\partial\psi\bigr) - \mu'(\psi_\varepsilon)\,\mathrm{i}\partial\bar\partial\psi_\varepsilon.$$

Since $\mathrm{i}\partial\bar\partial\psi = 2\pi\mathrm{dd}^{\mathrm{c}}\psi \geq -2\pi b\omega$, $0 \leq R/(\varepsilon+R) \leq 1$ and $-\mu' \leq 2$, the identity (14.23) gives

$$H_\varepsilon \geq 2\pi\bigl(a_\varepsilon(N_0 - b_K - qb) - 2b\bigr)\omega + \frac{\varepsilon}{R}\,\mathrm{i}\partial\psi_\varepsilon\wedge\bar\partial\psi_\varepsilon \geq 2\pi\omega + \frac{\varepsilon}{R}\,\mathrm{i}\partial\psi_\varepsilon\wedge\bar\partial\psi_\varepsilon, \tag{14.32}$$

where in the last step we used $a_\varepsilon \geq 1$ and $N_0 - b_K - qb = 2b + 2$. All constants in this display depend only on the geometry.

Step 2: the correction. Let $\rho \in C^\infty(\mathbb{R})$ satisfy $0 \leq \rho \leq 1$, $\rho = 1$ on $(-\infty, 1/2]$ and $\rho = 0$ on $[1,\infty)$, and put $G_\varepsilon = \rho(R/\varepsilon)G$. Since $\bar\partial R/\varepsilon = (1 + R/\varepsilon)\bar\partial\psi_\varepsilon$, the $\bar\partial$-closed form $g_\varepsilon = \bar\partial G_\varepsilon$ splits as

$$g_\varepsilon = g_\varepsilon^{(1)} + g_\varepsilon^{(2)}, \qquad g_\varepsilon^{(1)} = \rho'(R/\varepsilon)(1 + R/\varepsilon)\bar\partial\psi_\varepsilon\, G, \qquad g_\varepsilon^{(2)} = \rho(R/\varepsilon)\bar\partial G,$$

regarded as F-valued $(n,1)$-forms. As in the proof of Lemma 14.7.3, the rank-one part of (14.32) and Lemma 14.6.1(1),(2) give

$$|\langle g_\varepsilon^{(1)}, v\rangle|^2 \leq 4\sup|\rho'|^2\,|G|^2\langle[H_\varepsilon,\Lambda_\omega]v, v\rangle$$

pointwise on the shell $\{\varepsilon/2 \leq R \leq \varepsilon\}$, where $R/\varepsilon \leq 1$, the norms being those of the metric of F. Hence the integral $M_{g_\varepsilon^{(1)}}$ of Lemma 14.7.1, formed with H_ε, is at most $C_1\int_{\{\varepsilon/2\leq R\leq\varepsilon\}} |G|_h^2 R^{-q}\,\mathrm{d}V$, whose upper limit as $\varepsilon \to 0$ is at most $C_2\int_Y |f|_h^2\,\mathrm{d}V_Y$ by Lemma 14.7.7. Since $H_\varepsilon \geq 2\pi\omega$, Lemma 14.6.1(3) bounds $M_{g_\varepsilon^{(2)}}$ by

$$C_3\int_{\{R<\varepsilon\}} |\bar\partial G|_{\omega,h}^2 R^{-q}\,\mathrm{d}V,$$

which tends to zero by (14.28).

Both forms are square integrable for the metric of F. For $g_\varepsilon^{(1)}$, the derivative of the cutoff is bounded for fixed ε, and (14.27) bounds the integral of $|G|_h^2 R^{-q}$ over the shell, where the logarithm is bounded and $\mathrm{e}^{-\Phi} \leq \mathrm{e}^{\kappa\sup_X\Phi}\mathrm{e}^{-(1+\kappa)\Phi}$; for $g_\varepsilon^{(2)}$ this follows from (14.28). Since $M_{g'+g''} \leq 2(M_{g'} + M_{g''})$, by the Cauchy–Schwarz inequality for the positive form $\langle[H_\varepsilon,\Lambda_\omega]^{-1}\cdot,\cdot\rangle$, Lemma 14.7.1 gives u_ε on M with

$$\bar\partial u_\varepsilon = g_\varepsilon, \qquad \varlimsup_{\varepsilon\downarrow 0}\int_M \frac{|u_\varepsilon|_h^2 R^{-q}}{a_\varepsilon + b_\varepsilon}\,\mathrm{d}V \leq C_4\int_Y |f|_h^2\,\mathrm{d}V_Y. \tag{14.33}$$

The constants $C_1,\dots,C_4$ depend only on the geometry and the fixed cutoff ρ; neither the exponent κ nor the derivatives of the local extensions F_i enter them.

Step 3: holomorphic sections with the trace f. For fixed ε the denominator $a_\varepsilon + b_\varepsilon$ is bounded, and $h \geq \mathrm{e}^{-\sup_X \Phi} h_0$ because Φ is bounded above on the compact manifold X. Hence $s_\varepsilon = G_\varepsilon - u_\varepsilon$, which is holomorphic on M, is square integrable for h_0, and it extends to a global holomorphic section by Lemma 14.6.6. Then $u_\varepsilon = G_\varepsilon - s_\varepsilon$ is smooth on X.

It remains to show that $s_\varepsilon|_Y = f$. Let $y \in Y \setminus A$, and choose holomorphic coordinates (z', w) near y with $Y = \{w = 0\}$. Near y the weight Φ is smooth and $a_\varepsilon + b_\varepsilon$ is bounded, so $\int |u_\varepsilon|^2_{h_0} R^{-q}\, \mathrm{d}V$ is finite over a neighborhood of y. Since $R \leq C|w|^2$ near y by Lemma 14.7.4, and $\int_{|w|<r} |w|^{-2q}\, \mathrm{d}V(w) = \infty$ in $\mathbb{C}^q$ for every $r > 0$, by polar coordinates, the continuous function u_ε vanishes at y. Hence $u_\varepsilon = 0$ on $Y \setminus A$, and $s_\varepsilon = G = f$ there. Since the restriction of h to Y is defined, $A \cap Y$ is a proper analytic subset of the connected manifold Y, and the identity theorem gives $s_\varepsilon|_Y = f$.

Step 4: the norm bound. For all sufficiently small ε,

$$(a_\varepsilon + b_\varepsilon) R^q \leq C_5 R^q (-\log R)^2 \leq C_6,$$

because $a_\varepsilon + b_\varepsilon \leq 5\psi^2$ by Lemma 14.7.2(1), and because $R^q(-\log R)^2$ is bounded on X. Consequently (14.33) bounds $\int_X |u_\varepsilon|^2_h\, \mathrm{d}V$ by C_6 times the integral there. The section G has finite h-norm, since (14.27) controls it, Φ being bounded above and $R^q(-\log R)^2$ bounded; therefore $\|G_\varepsilon\|_h \to 0$ as $\varepsilon \to 0$, by absolute continuity of the integral on the shrinking supports. The triangle inequality gives

$$\varlimsup_{\varepsilon \downarrow 0} \|s_\varepsilon\|^2_h \leq C_7 \int_Y |f|^2_h\, \mathrm{d}V_Y,$$

with $C_7 = C_6 C_4$. The sections s_ε are bounded in L^2 for h_0, so, by Lemma 14.5.2(2), a sequence $\varepsilon_k \to 0$ gives sections converging uniformly to a holomorphic section s. Its restriction to Y is f, and Fatou's lemma gives $\|s\|^2_h \leq C_7 \int_Y |f|^2_h\, \mathrm{d}V_Y$. This proves the theorem for gentle weights with the constant C_7, which depends only on the geometry.

Step 5: arbitrary weights. Put $N = N_0 + 1$. By Theorem 13.4.5, applied to φ with the background $c_1(L, h_0) - N\omega$, there are $\Phi_j \downarrow \varphi$ with gentle analytic singularities of the kind required in the gentle case and $c_1(L, h_0 \mathrm{e}^{-\Phi_j}) \geq (N - \delta_j)\omega$, where $\delta_j \to 0$; for large j the curvature is at least $N_0\omega$. Since $\Phi_j \geq \varphi$, the restrictions of the metrics $h_0 \mathrm{e}^{-\Phi_j}$ to Y are defined, and the norms of f for them are at most its norm for h. The gentle case gives $s_j \in H^0(X, L)$ with $s_j|_Y = f$ and $\int_X |s_j|^2_{h_0} \mathrm{e}^{-\Phi_j}\, \mathrm{d}V \leq C_7 \int_Y |f|^2_h\, \mathrm{d}V_Y$. Since $\Phi_j \leq \Phi_1$, the s_j are bounded in L^2 for h_0, and by Lemma 14.5.2(2) a subsequence converges uniformly to a holomorphic section s with $s|_Y = f$. The weights $\mathrm{e}^{-\Phi_j}$ converge to $\mathrm{e}^{-\varphi}$ wherever φ is finite, hence almost everywhere, and Fatou's lemma gives (14.31) with $C = C_7$. □

Corollary 14.7.9 *Let L and T be holomorphic line bundles on X with smooth quadratic metrics h_0 and h_T, put $\theta = c_1(L, h_0)$, and choose $b_T \geq 0$ with $c_1(T, h_T) \geq -b_T\omega$. Let φ be quasi-plurisubharmonic with $\theta + \mathrm{dd}^{\mathrm{c}}\varphi \geq \delta\omega$ for some $\delta > 0$, and*

let $Y \subset X$ be a nonempty closed connected complex submanifold such that $\varphi|_Y$ is not identically $-\infty$. Let N and C be the constants of Theorem 14.7.8. Then for every integer $k \geq 1$ with $k\delta - b_T \geq N$, every holomorphic section f of $(T \otimes L^k)|_Y$ with finite norm for $h_T \otimes h_0^k \mathrm{e}^{-k\varphi|_Y}$ extends to a holomorphic section s of $T \otimes L^k$ on X with

$$\int_X |s|^2_{h_T \otimes h_0^k} \mathrm{e}^{-k\varphi}\, \mathrm{d}V \leq C \int_Y |f|^2_{h_T \otimes h_0^k} \mathrm{e}^{-k\varphi|_Y}\, \mathrm{d}V_Y.$$

Proof The metric $h_T \otimes h_0^k \mathrm{e}^{-k\varphi}$ on $T \otimes L^k$ is plurisubharmonic, its curvature current is $c_1(T, h_T) + k(\theta + \mathrm{dd^c}\varphi) \geq (k\delta - b_T)\omega \geq N\omega$, and its restriction to Y is defined. Hence Theorem 14.7.8 applies. Note that the finiteness of the norm on Y means that f lies locally in the multiplier ideal of $k\varphi|_Y$, and that the constant C is multiplied by $n!/(\dim Y)!$ if the volumes ω^n and $\omega|_Y^{\dim Y}$ are used instead of $\mathrm{d}V$ and $\mathrm{d}V_Y$. □

14.7.1 Extending strictly positive potentials

We keep the compact Kähler manifold (X, ω), but now we extend potentials instead of sections. Let $Y \subset X$ be a closed complex submanifold, and let $\varphi \in \mathrm{PSH}(Y, \omega|_Y)$ be such that $\omega|_Y + \mathrm{dd^c}\varphi$ is a Kähler current on Y, that is, dominates a positive multiple of $\omega|_Y$. We look for $\widetilde{\varphi} \in \mathrm{PSH}(X, \omega)$ that restricts exactly to φ and for which $\omega + \mathrm{dd^c}\widetilde{\varphi}$ is again a Kähler current (Theorem 14.7.12).

Near Y, local extensions are obtained by pulling φ back by a holomorphic projection onto Y and adding a large multiple of a function vanishing to second order along Y. The difficulty is to glue them, since two projections give different pullbacks. We take a maximum after lowering each local extension by a cutoff penalty near the edge of its chart. This maximum is ω-plurisubharmonic near Y and keeps the prescribed values on Y, provided that near every point of Y on the edge of a chart the penalized extension from that chart is dominated by the extension from another chart. To obtain this comparison, we add to e^{φ} the exponential of a small logarithmic pole along Y: when e^{φ} is Hölder continuous, the difference between two pullbacks of e^{φ} is small compared with this exponential near Y. The logarithm of such a sum is again quasi-plurisubharmonic (Lemma 14.7.11), and the Hölder continuity survives the operations of the construction (Lemma 14.7.10).

Hölder continuity on a manifold is defined in charts. Let M be a smooth manifold. A function $f\colon M \to \mathbb{R}$ is *Hölder continuous* if every point of M has a coordinate neighborhood V, with coordinates $x\colon V \to \mathbb{R}^d$, on which there are constants $C \geq 0$ and $\alpha \in (0, 1]$ such that

$$|f(a) - f(b)| \leq C|x(a) - x(b)|^{\alpha} \qquad \text{for all } a, b \in V.$$

We then say that f is Hölder continuous with *exponent* α on V, in the coordinates x. Such a function is continuous on V, and hence bounded near each point of V.

Lemma 14.7.10 *Let M and M' be smooth manifolds.*

(1) *Let f be Hölder continuous with exponent α on a coordinate neighborhood V in the coordinates x, let $p \in V$, and let y be coordinates on a neighborhood of p. Then there is a neighborhood of p on which f is Hölder continuous in the coordinates y with exponent α' for every $\alpha' \in (0, \alpha]$.*
(2) *If M is compact and f is Hölder continuous on M, then there are $\alpha \in (0, 1]$ and finitely many coordinate neighborhoods covering M on each of which f is Hölder continuous with exponent α.*
(3) *Smooth functions are Hölder continuous. If f and g are Hölder continuous on M, then so are $f + g$, fg and $\max\{f, g\}$. If $f \geq 0$ is Hölder continuous and $0 < \gamma \leq 1$, then f^γ is Hölder continuous.*
(4) *If $\Phi\colon M' \to M$ is a smooth map and f is Hölder continuous on M, then $f \circ \Phi$ is Hölder continuous on M'.*

Proof (1) Choose a closed ball N in the y-coordinates, centered at $y(p)$, whose preimage lies in V. The map $x \circ y^{-1}$ is smooth on the convex compact set N, so by the mean value inequality it is Lipschitz on N with some constant L. Moreover N has some diameter D, and $t^\alpha \leq D^{\alpha-\alpha'} t^{\alpha'}$ for $0 \leq t \leq D$. Hence, for a, b in the preimage of N, which is a neighborhood of p,

$$\begin{aligned} |f(a) - f(b)| &\leq C|x(a) - x(b)|^\alpha \leq CL^\alpha |y(a) - y(b)|^\alpha \\ &\leq CL^\alpha D^{\alpha-\alpha'} |y(a) - y(b)|^{\alpha'}. \end{aligned}$$

(2) By (1), applied with $y = x$, every point of M has a coordinate neighborhood on which f is Hölder continuous with some exponent, and on which it remains so with every smaller exponent. Finitely many of these neighborhoods cover M, and the smallest of their exponents serves on all of them.

(3) A smooth function is Lipschitz on every closed coordinate ball, by the mean value inequality, so it is Hölder continuous with exponent one. Let f and g be Hölder continuous and $p \in M$. By (1) there are coordinates x on a neighborhood V of p and $\alpha \in (0, 1]$ such that f and g are Hölder continuous with exponent α on V in these coordinates, say with the constant C; shrinking V, we may assume that $|f|, |g| \leq B$ on V. For $a, b \in V$ we have $|(f + g)(a) - (f + g)(b)| \leq 2C|x(a) - x(b)|^\alpha$,

$$\begin{aligned} |f(a)g(a) - f(b)g(b)| &\leq |f(a)|\,|g(a) - g(b)| + |g(b)|\,|f(a) - f(b)| \\ &\leq 2BC|x(a) - x(b)|^\alpha, \end{aligned}$$

and $|\max\{f, g\}(a) - \max\{f, g\}(b)| \leq \max\{|f(a) - f(b)|, |g(a) - g(b)|\}$.

For the power, note first that $(s + t)^\gamma \leq s^\gamma + t^\gamma$ for $s, t \geq 0$: if $s + t > 0$, divide by $(s + t)^\gamma$ and use $\lambda^\gamma \geq \lambda$ for $\lambda \in [0, 1]$. Hence, if $s \geq t \geq 0$, then $s^\gamma \leq (s - t)^\gamma + t^\gamma$, that is, $|s^\gamma - t^\gamma| \leq |s - t|^\gamma$ for all $s, t \geq 0$. If $f \geq 0$ has exponent α on V with the constant C, it follows that $|f(a)^\gamma - f(b)^\gamma| \leq C^\gamma |x(a) - x(b)|^{\alpha\gamma}$.

(4) Let $p' \in M'$, let V be a coordinate neighborhood of $\Phi(p')$, with coordinates x, on which f is Hölder continuous with exponent α and constant C, and let N be a closed coordinate ball around p', in coordinates x', with $\Phi(N) \subset V$. The map $x \circ \Phi$, written in the coordinates x', is smooth on the convex compact set N, hence Lipschitz

on N with some constant L. Therefore $|f(\Phi(a)) - f(\Phi(b))| \le CL^\alpha |x'(a) - x'(b)|^\alpha$ for $a, b \in N$. □

Lemma 14.7.11 *Let U be an open subset of a complex manifold, let θ be a smooth closed real $(1,1)$-form on U, and let $a, b \in \mathrm{PSH}(U, \theta)$. Then*

$$\log(\mathrm{e}^a + \mathrm{e}^b) \in \mathrm{PSH}(U, \theta),$$

where $\mathrm{e}^{-\infty} = 0$ and $\log 0 = -\infty$, so that this function equals $-\infty$ precisely at the common poles of a and b. Moreover

$$\max\{a, b\} \le \log(\mathrm{e}^a + \mathrm{e}^b) \le \max\{a, b\} + \log 2.$$

Proof We begin with the inequalities, which are pointwise. Fix a point and suppose that $a \ge b$ there; the other case is symmetric. If $a = -\infty$, then $b = -\infty$ as well, and all three expressions equal $-\infty$. Otherwise $\log(\mathrm{e}^a + \mathrm{e}^b) = a + \log(1 + \mathrm{e}^{b-a})$, and $0 \le \log(1 + \mathrm{e}^{b-a}) \le \log 2$.

Put $H(x, y) = \log(\mathrm{e}^x + \mathrm{e}^y)$ for $x, y \in [-\infty, \infty)$, with the same conventions. On $\mathbb{R}^2$ the function H is smooth and increasing in each variable: its partial derivatives are p and $1 - p$, where $p = \mathrm{e}^x/(\mathrm{e}^x + \mathrm{e}^y)$, and its Hessian matrix is $p(1-p)\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. On $[-\infty, \infty)^2$ it is still increasing in each variable, and it is continuous, with values in $[-\infty, \infty)$. Indeed, by the inequalities just proved, $\max\{x, y\} \le H(x, y) \le \max\{x, y\} + \log 2$, so $H(x, y) \to -\infty$ when $\max\{x, y\} \to -\infty$; and $H(x, y) = y + \log(1 + \mathrm{e}^{x-y}) \to y_0$ when $x \to -\infty$ and $y \to y_0 \in \mathbb{R}$.

The first assertion is local. Let $B \Subset U$ be a coordinate ball, let s be a smooth real function on B with $\mathrm{dd^c} s = \theta$ (Lemma 13.1.2), and let $B' \Subset B$ be a smaller concentric ball. Then $a + s$ and $b + s$ are plurisubharmonic on B, by Lemma 13.1.4(2), and $\log(\mathrm{e}^a + \mathrm{e}^b) + s = H(a + s, b + s)$. If f, g are smooth plurisubharmonic functions, the chain rule gives

$$\mathrm{dd^c} H(f, g) = p\,\mathrm{dd^c} f + (1 - p)\,\mathrm{dd^c} g + p(1 - p)\,\frac{\mathrm{i}}{2\pi}\,\partial(f - g) \wedge \bar\partial(f - g) \ge 0,$$

with p evaluated at (f, g); hence $H(f, g)$ is plurisubharmonic, by Proposition 8.2.2(1). In general, let f_k and g_k be the regularizations of $a + s$ and $b + s$ by radial mollifiers whose radii decrease to zero and are smaller than the distance from B' to ∂B. By Proposition 8.2.2(2), they are smooth plurisubharmonic functions on B', and they decrease to $a + s$ and $b + s$. Since H is continuous and increasing in each variable on $[-\infty, \infty)^2$, the functions $H(f_k, g_k)$ decrease to $H(a + s, b + s)$. This limit is at least $a + s$, so it is not identically $-\infty$, and it is plurisubharmonic on B' by Theorem 8.3.1. Consequently $\log(\mathrm{e}^a + \mathrm{e}^b) = H(a + s, b + s) - s$ is quasi-plurisubharmonic on B' and $\theta + \mathrm{dd^c} \log(\mathrm{e}^a + \mathrm{e}^b) = \mathrm{dd^c} H(a + s, b + s) \ge 0$ there. Every point of U is the center of such a ball B', so the first assertion follows. □

Theorem 14.7.12 *Let $Y \subset X$ be a closed complex submanifold, and let $\varphi \in \mathrm{PSH}(Y, \omega|_Y)$ satisfy*

$$\omega|_Y + \mathrm{dd^c}\varphi \ge \delta\,\omega|_Y$$

for some $\delta > 0$. Suppose that e^{φ}, defined to be zero at the poles of φ, is Hölder continuous on Y. Then there are $\widetilde{\varphi} \in \mathrm{PSH}(X, \omega)$ and $\varepsilon > 0$ such that

$$\widetilde{\varphi}|_Y = \varphi, \qquad \omega + dd^c \widetilde{\varphi} \geq \varepsilon\omega.$$

Moreover $\widetilde{\varphi}$ is continuous on $X \setminus Y$, and $e^{\widetilde{\varphi}}$ is Hölder continuous on X. If φ has analytic singularities, then so does $\widetilde{\varphi}$.

Proof We construct local extensions near Y (Steps 1 and 2), compare them on overlaps (Step 3), glue them (Step 4), extend the result from a neighborhood of Y to X (Step 5), and finally treat analytic singularities (Step 6).

Step 0. Reductions. If $Y = \varnothing$, the function $\widetilde{\varphi} = 0$ and $\varepsilon = 1$ have all the stated properties; $0 = \log|1|^2$ has analytic singularities. Let $Y \neq \varnothing$. The connected components of Y are open in the compact set Y, so there are finitely many of them, say $Y_1, \ldots, Y_r$; each Y_k is closed in X, hence a compact connected complex submanifold, and different components may have different dimensions. If some Y_k has dimension n, then it is open in X, because near each of its points it is a coordinate plane of full dimension; being also closed, it equals X, as X is connected. Then $Y = X$, and $\widetilde{\varphi} = \varphi$ with $\varepsilon = \delta$ has the stated properties. We may therefore assume that every component of Y has positive codimension; components of dimension zero are allowed.

Fix ε with $0 < \varepsilon < \min\{\delta/4, 1/4\}$. By Lemma 14.7.10(2), e^{φ} is Hölder continuous with one exponent $\alpha \in (0, 1]$ on finitely many coordinate neighborhoods covering Y. By Lemma 14.7.10(1), every point $p \in Y$ then has, in every system of coordinates of Y near p, a neighborhood on which e^{φ} is Hölder continuous with exponent α.

Step 1. A small normal pole. For each k let $R_k = R_{Y_k}$ be the function of Lemma 14.7.4 for the submanifold Y_k, and let $\kappa_k \geq 0$ be the constant called b there, so that $dd^c \log R_k \geq -\kappa_k\omega$. Put

$$R = R_1 \cdots R_r, \qquad \kappa = \kappa_1 + \cdots + \kappa_r.$$

Then $R \geq 0$ is smooth and vanishes exactly on Y. The function $\log R = \sum_k \log R_k$ is quasi-plurisubharmonic, satisfies $dd^c \log R \geq -\kappa\omega$, and has analytic singularities by Proposition 13.1.7, as a sum of functions with analytic singularities. Choose a rational number c with $0 < c < \alpha$ and $c\kappa/2 \leq 1 - 4\varepsilon$; this is possible because $1 - 4\varepsilon > 0$. Put $F = (c/2)\log R$. Then

$$\omega + dd^c F \geq \Big(1 - \frac{c\kappa}{2}\Big)\omega \geq 4\varepsilon\omega.$$

The function F is smooth on $X \setminus Y$, equals $-\infty$ on Y, and has analytic singularities, since $c/2$ is rational. Its exponential $e^F = R^{c/2}$ is Hölder continuous on X by Lemma 14.7.10(3).

We need a lower bound for R near Y. Let $p \in Y_k$, and let (z', w) be holomorphic coordinates near p in which $Y = \{w = 0\}$. The function $P = \prod_{l \neq k} R_l$ is smooth and positive near p, since the closed sets Y_l, $l \neq k$, do not contain p. On a neighborhood of p that does not meet these sets we have $Y_k = Y = \{w = 0\}$, so the second inequality in (14.25) of Lemma 14.7.4, applied to Y_k, bounds R_k below by a positive multiple

of $|w|^2$ near p. Since $R = R_k P$ and P is bounded away from zero near p, there are a closed ball K centered at p and a constant $c_0 > 0$ such that

$$R \geq c_0|w|^2, \qquad \mathrm{e}^F = R^{c/2} \geq c_0^{c/2}|w|^c \qquad \text{on } K. \tag{14.34}$$

Moreover, at the points $(z', 0)$ of the interior of K, the real Hessian $D_w^2 R(z', 0)$ of $w \mapsto R(z', w)$ at $w = 0$ is positive definite. Indeed, $R \geq 0$ attains its minimum 0 at such a point, so its first derivatives vanish there, and Taylor's formula gives $R(z', t\xi) = \frac{t^2}{2} D_w^2 R(z', 0)[\xi, \xi] + O(t^3)$ as $t \to 0$, for every ξ; comparing with $R(z', t\xi) \geq c_0 t^2 |\xi|^2$ for small t, we obtain $D_w^2 R(z', 0)[\xi, \xi] \geq 2c_0|\xi|^2$.

Step 2. Local extensions. Every point of Y has holomorphic coordinates $(z', w) \in \mathbb{C}^m \times \mathbb{C}^{n-m}$, centered at that point, in which $Y = \{w = 0\}$ on the coordinate domain, where $m < n$ is the dimension of the component of Y through the point. A product of two small balls centered at the origin is then a chart $\widehat{W} = \widehat{B}' \times \widehat{B}''$ with $Y \cap \widehat{W} = \widehat{B}' \times \{0\}$. The products $B' \times B''$ of the concentric balls of half the radii cover Y, and since Y is compact, finitely many of them suffice. We thus obtain finitely many charts $W_j = B'_j \times B''_j$, relatively compact in charts $\widehat{W}_j = \widehat{B}'_j \times \widehat{B}''_j$ of the same form, with $Y \cap W_j = B'_j \times \{0\}$ and with dimensions $m_j < n$. Let $\pi_j \colon \widehat{W}_j \to Y$, $\pi_j(z', w) = (z', 0)$, be the holomorphic projection.

Below we shrink the normal factors B''_j several times, keeping them balls centered at the origin; this does not change $Y \cap W_j$, so the charts keep covering Y. If a property holds on an open set $O \subset \widehat{W}_j$ containing the compact set $\overline{B'_j} \times \{0\}$, then this compact set has positive distance from the complement of O in the coordinates, so O contains W_j after the normal factor has been shrunk. We claim that there is $A > 0$ such that, after such a shrinking,

$$a_j = \varphi \circ \pi_j + AR, \qquad b_j = F + AR$$

belong to $\mathrm{PSH}(W_j, (1 - 2\varepsilon)\omega)$ for every j.

Since $4\varepsilon \leq \delta$, the hypothesis gives $\mathrm{dd^c}\varphi \geq -(1-\delta)\omega|_Y \geq -(1-4\varepsilon)\omega|_Y$. Choose a ball $\widetilde{B}'_j$, concentric with B'_j, with $\overline{B'_j} \subset \widetilde{B}'_j \Subset \widehat{B}'_j$, and a smooth function s on $\widetilde{B}'_j \times \{0\}$ with $\mathrm{dd^c} s = (1 - 4\varepsilon)\omega|_Y$ there (Lemma 13.1.2; if $m_j = 0$, take $s = 0$). Then $\varphi + s$ is plurisubharmonic on $\widetilde{B}'_j \times \{0\}$, by Lemma 13.1.4(2). By Proposition 8.2.3, $(\varphi + s) \circ \pi_j$ is plurisubharmonic or identically $-\infty$ on the connected set $\widetilde{B}'_j \times \widehat{B}''_j$, and the second alternative is excluded because π_j maps this set onto $\widetilde{B}'_j \times \{0\}$, where $\varphi + s$ is not identically $-\infty$. Since $\mathrm{dd^c}(s \circ \pi_j) = \pi_j^* \mathrm{dd^c} s$ for the smooth function s, we obtain that $\varphi \circ \pi_j$ is quasi-plurisubharmonic on $\widetilde{B}'_j \times \widehat{B}''_j \supset \overline{W_j}$ and

$$\mathrm{dd^c}(\varphi \circ \pi_j) \geq -(1 - 4\varepsilon)\pi_j^*(\omega|_Y).$$

For a_j it therefore suffices that the smooth form

$$Q_j = (1 - 2\varepsilon)\omega - (1 - 4\varepsilon)\pi_j^*(\omega|_Y) + A\mathrm{dd^c}R$$

be positive on W_j.

We first examine Q_j at the points of Y. Write a real $(1,1)$-form on W_j as $\mathrm{i}\sum_{\mu,\nu} g_{\mu\nu}\,\mathrm{d}\zeta_\mu \wedge \mathrm{d}\bar\zeta_\nu$, where $\zeta = (z', w)$, and split its Hermitian coefficient matrix $(g_{\mu\nu})$ into blocks with respect to the tangential coordinates z' and the normal coordinates w; the form is positive definite at a point if and only if this matrix is. The form $\pi_j^*(\omega|_Y)$ involves only $\mathrm{d}z'$ and $\mathrm{d}\bar z'$, and at a point $(z', 0)$ its coefficients are those of $\omega|_Y$ at that point. Hence its mixed and normal blocks vanish, and at points of Y its tangential block is the tangential block Ω_T of ω.

The function R vanishes on Y together with its first derivatives. Differentiating the identities $R(z', 0) = 0$ and $\partial R/\partial\bar w_\nu(z', 0) = 0$ with respect to z' and $\bar z'$ shows that $\mathrm{dd^c} R$ has zero tangential and mixed blocks along Y. Its normal block is $(2\pi)^{-1}$ times the matrix $N_R = (\partial^2 R/\partial w_\mu \partial\bar w_\nu)$, which is positive definite at points of Y: for $\xi \in \mathbb{C}^{n-m_j} \setminus \{0\}$, applying the identity $\Delta = 4\partial_t\partial_{\bar t}$ to $t \mapsto R(z', t\xi)$ at $t = 0$ gives

$$\sum_{\mu,\nu} \frac{\partial^2 R}{\partial w_\mu \partial \bar w_\nu}(z', 0)\xi_\mu\bar\xi_\nu = \frac{1}{4}\big(D_w^2 R(z', 0)[\xi, \xi] + D_w^2 R(z', 0)[\mathrm{i}\xi, \mathrm{i}\xi]\big) > 0$$

by Step 1.

Consequently, at a point of Y, the tangential block of Q_j is $2\varepsilon\Omega_T$, which is positive definite, its mixed block is the mixed block Ω_M of $(1 - 2\varepsilon)\omega$, and its normal block is $\Omega_N + AN_R/(2\pi)$, where Ω_N is the normal block of $(1 - 2\varepsilon)\omega$. By the Schur complement criterion for Hermitian block matrices whose upper left block is positive definite, Q_j is positive definite at a point of Y if and only if

$$\Omega_N + \frac{A}{2\pi}N_R - \Omega_M^*\big(2\varepsilon\,\Omega_T\big)^{-1}\Omega_M$$

is positive definite there. (If $m_j = 0$, there is no tangential block, and the condition reads $\Omega_N + AN_R/(2\pi) > 0$.) All blocks depend continuously on the point, N_R is positive definite, and the sets $\overline{B'_j} \times \{0\}$ are compact and finite in number. Hence the smallest eigenvalue of N_R on them has a positive lower bound and the other two terms are bounded, so one A makes this matrix positive definite at every point of every $\overline{B'_j} \times \{0\}$. Fix such an A. Since the coefficients of Q_j are continuous on $\widehat{W}_j$ and positive definiteness is an open condition, Q_j is positive definite on an open set containing $\overline{B'_j} \times \{0\}$, and we shrink the normal factor B''_j so that W_j lies in this set. This proves the claim for a_j.

Along Y the form $\mathrm{dd^c} R$ is positive semidefinite, so $A\mathrm{dd^c} R + \varepsilon\omega$ is positive definite on an open set containing $\overline{B'_j} \times \{0\}$; after shrinking the normal factors once more we have $A\mathrm{dd^c} R \geq -\varepsilon\omega$ on every W_j. Together with $\mathrm{dd^c} F \geq -(1 - 4\varepsilon)\omega$ this gives

$$(1 - 2\varepsilon)\omega + \mathrm{dd^c} b_j \geq (1 - 2\varepsilon)\omega - (1 - 4\varepsilon)\omega - \varepsilon\omega = \varepsilon\omega \geq 0,$$

which proves the claim for b_j.

By Lemma 14.7.11, applied with $\theta = (1 - 2\varepsilon)\omega$, the functions

$$\varphi_j = \log(\mathrm{e}^{a_j} + \mathrm{e}^{b_j}) = \log(\mathrm{e}^{\varphi\circ\pi_j} + \mathrm{e}^F) + AR$$

belong to $\mathrm{PSH}(W_j, (1-2\varepsilon)\omega)$. On Y we have $F = -\infty$, $R = 0$ and $\varphi \circ \pi_j = \varphi$, so $\varphi_j = \varphi$ on $Y \cap W_j$, also at the poles of φ. Off Y the function e^F is positive and continuous and $\mathrm{e}^{\varphi\circ\pi_j}$ is continuous, so φ_j is continuous on $W_j \setminus Y$. Finally $\mathrm{e}^{\varphi_j} = (\mathrm{e}^{\varphi} \circ \pi_j + \mathrm{e}^F)\,\mathrm{e}^{AR}$ is Hölder continuous on W_j by Lemma 14.7.10(3),(4).

Step 3. Comparison on overlaps. We claim that for every $p \in Y \cap W_i \cap W_j$ and every $\eta > 0$ there is a neighborhood of p on which

$$\varphi_i \le \varphi_j + \eta. \tag{14.35}$$

Work in the coordinates (z', w) of W_i, and write $p = (z'_p, 0)$. The coordinate z' restricts to coordinates of Y on $Y \cap W_i$, so by Step 0 there are a neighborhood V_p of p in $Y \cap W_i$ and a constant C_H with $|\mathrm{e}^{\varphi(y_1)} - \mathrm{e}^{\varphi(y_2)}| \le C_H |z'(y_1) - z'(y_2)|^\alpha$ for $y_1, y_2 \in V_p$. Since π_i and π_j are continuous and fix p, there is a closed ball $K \subset W_i \cap W_j$ centered at p such that $\pi_i(K) \cup \pi_j(K) \subset V_p$ and (14.34) holds on K in these coordinates. The map $z' \circ \pi_j$ is smooth on the convex compact set K, so it is Lipschitz there with some constant L. Let $x = (z', w) \in K$, and put $y = (z', 0) = \pi_i(x)$. Then $|y - p| \le |x - p|$, so $y \in K \cap Y$, and $\pi_j(y) = y$ because π_j is the identity on $Y \cap W_j$. Hence

$$|z'(\pi_i(x)) - z'(\pi_j(x))| = |z'(\pi_j(y)) - z'(\pi_j(x))| \le L|y - x| = L|w|,$$

and therefore

$$\bigl|\mathrm{e}^{\varphi\circ\pi_i}(x) - \mathrm{e}^{\varphi\circ\pi_j}(x)\bigr| \le C_H L^\alpha |w|^\alpha.$$

Since $c < \alpha$, we have $C_H L^\alpha |w|^\alpha \le (\mathrm{e}^\eta - 1) c_0^{c/2} |w|^c$ whenever $|w|$ is sufficiently small, and then (14.34) gives $C_H L^\alpha |w|^\alpha \le (\mathrm{e}^\eta - 1)\mathrm{e}^{F(x)}$. At such points $x \in K$, including the points of Y,

$$\mathrm{e}^{\varphi\circ\pi_i} + \mathrm{e}^F \le \mathrm{e}^{\varphi\circ\pi_j} + \mathrm{e}^\eta \mathrm{e}^F \le \mathrm{e}^\eta \bigl(\mathrm{e}^{\varphi\circ\pi_j} + \mathrm{e}^F\bigr).$$

The logarithm is increasing on $[0, \infty)$, with $\log 0 = -\infty$, so taking logarithms and adding AR to both sides gives (14.35) on the neighborhood of p formed by the interior points x of K with $|w(x)|$ sufficiently small.

Step 4. Gluing near Y. Every point of Y has an open neighborhood whose closure is contained in some W_j, and finitely many of these neighborhoods cover Y; let W'_j be the union of those chosen inside W_j. Then $W'_j \Subset W_j$ and the W'_j cover Y. Choose open sets U_j and T_j with $W'_j \Subset U_j \Subset T_j \Subset W_j$, for instance sets of points whose coordinate distance from $\overline{W'_j}$ is less than suitable small constants. The compact set $\overline{T_j} \setminus U_j$ is contained in the open set $W_j \setminus \overline{W'_j}$, so by Proposition 7.1.3 there is a smooth function $f_j \colon X \to [0,1]$ with compact support in $W_j \setminus \overline{W'_j}$ which equals one on a neighborhood of $\overline{T_j} \setminus U_j$; in particular $f_j = 0$ on the neighborhood $X \setminus \operatorname{Supp} f_j$ of $\overline{W'_j}$. Fix $\eta > 0$ so small that $\eta \mathrm{dd}^{\mathrm{c}} f_j \le \varepsilon\omega$ for every j. Since $(1-\varepsilon)\omega + \mathrm{dd}^{\mathrm{c}}\varphi_j \ge \varepsilon\omega$, the functions

$$\widehat{\varphi}_j = \varphi_j - \eta f_j$$

belong to $\mathrm{PSH}(W_j, (1-\varepsilon)\omega)$. On $W = \bigcup_j T_j$ define

$$\psi(x) = \max\{\widehat{\varphi}_j(x) : x \in T_j\}.$$

The set of indices in this maximum changes with x. We show that near Y the maximum is nevertheless taken over a fixed set of indices. Let $p \in Y$, and choose j with $p \in W_j'$; then $f_j = 0$ near p, and j occurs in the maximum near p. Let l be an index with $p \in \partial T_l$. Then $p \in \overline{T_l} \setminus U_l$, so $f_l = 1$ near p. Since $p \in Y \cap W_l \cap W_j$, the inequality (14.35) with $\eta/2$ in place of η gives, near p,

$$\widehat{\varphi}_l = \varphi_l - \eta \le \varphi_j - \eta/2 \le \widehat{\varphi}_j.$$

These are inequalities between functions with values in $[-\infty, \infty)$, valid also at common poles. As there are finitely many indices, p has a neighborhood O_p on which the indices l with $p \in \partial T_l$ never affect the maximum, the indices with $p \notin \overline{T_l}$ do not occur, and the indices with $p \in T_l$ occur at every point. Thus, on O_p, the function ψ is the maximum of the $\widehat{\varphi}_l$ over the fixed finite set of indices l with $p \in T_l$.

Let $W_0 = \bigcup_{p\in Y} O_p$. A finite maximum of functions in $\mathrm{PSH}(O_p, (1-\varepsilon)\omega)$ belongs to this class, by Lemma 13.1.4(3). Hence $\psi \in \mathrm{PSH}(W_0, (1-\varepsilon)\omega)$; moreover ψ is continuous on $W_0 \setminus Y$, and e^{ψ}, which is locally a finite maximum of the functions $\mathrm{e}^{\widehat{\varphi}_l} = \mathrm{e}^{\varphi_l}\mathrm{e}^{-\eta f_l}$, is Hölder continuous on W_0 by Lemma 14.7.10(3). On Y every $\widehat{\varphi}_l$ restricts to $\varphi - \eta f_l \le \varphi$, while $\widehat{\varphi}_j = \varphi$ near p for the index j chosen above. Consequently $\psi|_Y = \varphi$.

Step 5. The global extension. Choose an open neighborhood G of Y with $\overline{G} \subset W_0$, for instance a finite union of coordinate balls with closures in W_0 covering the compact set Y. Its boundary ∂G is compact and disjoint from Y, and both ψ and F are finite and continuous near ∂G. Choose a constant M with $M > \max_{\partial G}(\psi - F)$ (if $\partial G = \varnothing$, every M will do); since $F + M - \psi$ is continuous on $W_0 \setminus Y$ and positive on ∂G, it is positive on an open neighborhood $G_1 \subset W_0 \setminus Y$ of ∂G. Set

$$\widetilde{\varphi} = \begin{cases} \max\{\psi, F + M\} & \text{on } G, \\ F + M & \text{on } X \setminus G. \end{cases}$$

The two expressions agree on G_1, so $\widetilde{\varphi} = F + M$ on the open set $G_1 \cup (X \setminus \overline{G})$, which contains $X \setminus G$.

The function $F + M$ belongs to $\mathrm{PSH}(X, (1-4\varepsilon)\omega) \subset \mathrm{PSH}(X, (1-\varepsilon)\omega)$, and $\psi \in \mathrm{PSH}(W_0, (1-\varepsilon)\omega)$. Let B be a coordinate ball whose closure lies in a chart, and h a smooth function on B with $\mathrm{dd^c}h = (1-\varepsilon)\omega$ (Lemma 13.1.2). Then $F + M + h \in \mathrm{PSH}(B)$ and $\psi + h \in \mathrm{PSH}(G \cap B)$. Every point x of the boundary of $G \cap B$ in B lies in ∂G, and $\overline{\lim}_{G \ni y \to x}(\psi + h)(y) \le F(x) + M + h(x)$, because $\psi < F + M$ on G_1 and $F + M + h$ is continuous at x. By Proposition 8.2.5, $\widetilde{\varphi} + h$ is plurisubharmonic on B. Since such balls cover X, $\widetilde{\varphi}$ is quasi-plurisubharmonic and $(1-\varepsilon)\omega + \mathrm{dd^c}\widetilde{\varphi} \ge 0$, that is, $\widetilde{\varphi} \in \mathrm{PSH}(X, \omega)$ and $\omega + \mathrm{dd^c}\widetilde{\varphi} \ge \varepsilon\omega$. Since $F = -\infty$ on Y, we have $\widetilde{\varphi}|_Y = \psi|_Y = \varphi$. The function $\widetilde{\varphi}$ is continuous on $X \setminus Y$, because ψ and F are continuous on $W_0 \setminus Y$ and $X \setminus Y$, respectively. Its exponential equals

$\max\{e^{\psi}, e^{M}e^{F}\}$ on G and $e^{M}e^{F}$ on the open set $G_1 \cup (X \setminus \overline{G})$; these two open sets cover X, so $e^{\widetilde{\varphi}}$ is Hölder continuous on X by Lemma 14.7.10(3).

Step 6. Analytic singularities. Finally suppose that φ has analytic singularities (Definition 13.1.5). By Proposition 13.1.7, so does the holomorphic pullback $\varphi \circ \pi_j$, which is not identically $-\infty$, and the function F has analytic singularities by Step 1. By the same proposition, $\max\{\varphi \circ \pi_j, F\}$ has analytic singularities; adding the smooth function AR changes the bounded remainder in its local presentations (13.1) by a locally bounded function, so $\max\{\varphi \circ \pi_j, F\} + AR$ has analytic singularities. By Lemma 14.7.11,

$$\max\{\varphi \circ \pi_j, F\} + AR \le \varphi_j \le \max\{\varphi \circ \pi_j, F\} + AR + \log 2.$$

Hence a local presentation (13.1) of $\max\{\varphi \circ \pi_j, F\} + AR$ yields one of φ_j, with the same holomorphic functions and a bounded remainder changed by at most $\log 2$; so φ_j has analytic singularities. Subtracting the smooth function ηf_j changes the remainders by bounded functions, and the locally fixed finite maxima of Step 4 preserve analytic singularities by Proposition 13.1.7; so ψ has analytic singularities on W_0. On G, the function $\widetilde{\varphi}$ is the maximum of ψ and $F + M$, and on the open set $G_1 \cup (X \setminus \overline{G})$ it equals $F + M$; both functions have analytic singularities. Therefore $\widetilde{\varphi}$ has analytic singularities on X. □

14.8 Global poles on compact complex manifolds

We end the chapter by dropping the Kähler assumption. Recall that a subset of a complex manifold is pluripolar if it is locally contained in pole sets of plurisubharmonic functions (Definition 10.1.1), and that by Josefson's theorem (Theorem 10.2.1) a pluripolar subset of $\mathbb{C}^n$ lies in the pole set of a single plurisubharmonic function on $\mathbb{C}^n$. On a compact connected complex manifold, however, every plurisubharmonic function is constant: the set where it attains its maximum is nonempty and closed, by upper semicontinuity, and open, by the maximum principle (Proposition 3.1.4) applied in coordinate balls, where plurisubharmonic functions are subharmonic (Theorem 8.2.1). A global pole function must therefore be quasi-plurisubharmonic, and we prove that every pluripolar subset of a compact complex manifold, Kähler or not, lies in the pole set of one quasi-plurisubharmonic function (Theorem 14.8.7).

Throughout this section X is a connected compact complex manifold of dimension $n \ge 1$, and ω is a Hermitian form on X (Section 7.5), which need not be closed. Then Definition 13.1.3 does not apply to ω, but its formula still makes sense: we write $\mathrm{PSH}(X, \omega)$ for the set of quasi-plurisubharmonic functions u on X with $\omega + dd^c u \ge 0$ in the sense of currents, and call its elements *ω-plurisubharmonic*; for an open set $D \subset X$ the class $\mathrm{PSH}(D, \omega)$ is defined in the same way. When ω is Kähler, this is the class of Definition 13.1.3. We put $\omega_u = \omega + dd^c u$. By Lemma 13.1.4(4), every quasi-plurisubharmonic function Φ on X satisfies $dd^c\Phi \ge -C\omega$ for some constant C, which we may take positive. Consequently $t\Phi \in \mathrm{PSH}(X, \omega)$ for $0 < t \le 1/C$.

In local coordinates β denotes the Euclidean form (7.4), and for a point a of a chart, $\omega(a)$ denotes the form with constant coefficients equal to those of ω at a. A real $(1,1)$-form with constant coefficients $\gamma = \frac{\mathrm{i}}{2}\sum_{j,k} c_{jk}\,\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$, where the matrix (c_{jk}) is Hermitian, is dd^{c} of the real quadratic polynomial $\pi\sum_{j,k} c_{jk} z_j \bar{z}_k$, because $\partial\bar{\partial}(z_j\bar{z}_k) = \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k$.

Since every plurisubharmonic function on X is constant, only the empty set is globally pluripolar in X in the sense of Definition 10.1.1. In this section we therefore call a set $A \subset X$ *globally pluripolar* if $A \subset \{\Phi = -\infty\}$ for some quasi-plurisubharmonic function Φ on X. Subsets of globally pluripolar sets are again globally pluripolar.

The proof adapts the capacity argument of Section 10.1. For a set $A \subset X$, the regularization of the global extremal function T_A, the upper envelope of the ω-plurisubharmonic functions that are nonpositive on A, is bounded unless A is globally pluripolar (Lemma 14.8.4). If a piece of a pluripolar set were not globally pluripolar, we would approximate it by compact sublevel sets of small capacity. The Monge–Ampère measure of the extremal function of a compact set is carried by that set, and its mass is controlled by the capacity (Proposition 14.8.6), so in the limit we would obtain a bounded potential with zero Monge–Ampère mass. Since ω need not be closed, Stokes' formula does not compute the total mass $\int_X (\omega + \mathrm{dd}^{\mathrm{c}} u)^n$; its positivity follows from a comparison principle for small gaps (Proposition 14.8.3), in which the error terms produced by $\mathrm{d}\omega$ and $\mathrm{dd}^{\mathrm{c}}\omega$ are absorbed by the size of the gap. To show that the Monge–Ampère measure of an extremal function vanishes off its set, we freeze the coefficients of ω on small balls and use the homogeneous balayage of Corollary 9.6.4 and a local estimate for nearby bounded potentials (Lemma 14.8.5).

We first define products of the currents ω_u for bounded u. They are built from Bedford–Taylor products (Definition 9.1.1), which do not depend on the chart in which they are computed (Proposition 9.1.2(2),(5)) and are symmetric in their factors (Theorem 9.2.5). We use the following rule for adding smooth functions to the potentials. Let $v_1, \dots, v_k$, $0 \le k \le n$, be locally bounded plurisubharmonic functions on an open set $\Omega \subset \mathbb{C}^n$, and let $g_1, \dots, g_k$ be smooth real functions on Ω such that the functions $v_l + g_l$ are plurisubharmonic. Then

$$\bigwedge_{l=1}^{k} \mathrm{dd}^{\mathrm{c}}(v_l + g_l) = \sum_{I \subset \{1,\dots,k\}} \Big(\bigwedge_{l\in I} \mathrm{dd}^{\mathrm{c}} v_l\Big) \wedge \bigwedge_{l\notin I} \mathrm{dd}^{\mathrm{c}} g_l, \tag{14.36}$$

where the products of the currents $\mathrm{dd}^{\mathrm{c}} v_l$ and $\mathrm{dd}^{\mathrm{c}}(v_l + g_l)$ are Bedford–Taylor products, and the forms $\mathrm{dd}^{\mathrm{c}} g_l$ are smooth.

We prove (14.36) by induction on k; for $k = 0$ both sides equal 1. Let $k \ge 1$ and assume (14.36) for $k-1$ factors, so that $R = \bigwedge_{l\ge 2} \mathrm{dd}^{\mathrm{c}}(v_l + g_l) = \sum_I S_I \wedge \Gamma_I$, where I runs over the subsets of $\{2, \dots, k\}$, $S_I = \bigwedge_{l\in I} \mathrm{dd}^{\mathrm{c}} v_l$, and $\Gamma_I = \bigwedge_{l\in\{2,\dots,k\}\setminus I} \mathrm{dd}^{\mathrm{c}} g_l$ is a smooth closed form of pure bidegree. By definition, $\mathrm{dd}^{\mathrm{c}}(v_1 + g_1) \wedge R = \mathrm{dd}^{\mathrm{c}}(v_1 R) + \mathrm{dd}^{\mathrm{c}}(g_1 R)$, and $\mathrm{dd}^{\mathrm{c}}(g_1 R) = \mathrm{dd}^{\mathrm{c}} g_1 \wedge R$ by (9.1). Multiplication of the coefficient measures by v_1 commutes with the exterior product by Γ_I. Moreover, $\partial\Gamma_I$ and $\bar{\partial}\Gamma_I$ vanish, being the components of two different bidegrees of $\mathrm{d}\Gamma_I = 0$, so the Leibniz rule for currents (Section 7.2) gives $\mathrm{dd}^{\mathrm{c}}(R' \wedge \Gamma_I) = \mathrm{dd}^{\mathrm{c}} R' \wedge \Gamma_I$ for every

current R'. Hence $\mathrm{dd^c}(v_1 R) = \sum_I \mathrm{dd^c}(v_1 S_I) \wedge \Gamma_I = \sum_I (\mathrm{dd^c} v_1 \wedge S_I) \wedge \Gamma_I$. Adding the two expansions gives (14.36) for k factors.

Now let D be an open subset of a coordinate chart, identified with an open subset of $\mathbb{C}^n$, and let q be a smooth function on D with $\mathrm{dd^c} q \geq \omega$, for instance a large multiple of $|z|^2$ if D is a ball whose closure lies in the chart. If $u \in \mathrm{PSH}(X, \omega)$ is bounded, then $u + q$ is quasi-plurisubharmonic with $\mathrm{dd^c}(u + q) = \omega_u + (\mathrm{dd^c} q - \omega) \geq 0$, so it is a locally bounded plurisubharmonic function on D (Lemma 13.1.4(2), with $\theta = 0$), and $\omega_u = \mathrm{dd^c}(u + q) + \sigma$ with the smooth form $\sigma = \omega - \mathrm{dd^c} q$. For bounded $u_1, \dots, u_p \in \mathrm{PSH}(X, \omega)$, where $1 \leq p \leq n$, we define on D

$$\omega_{u_1} \wedge \cdots \wedge \omega_{u_p} = \sum_{J \subset \{1,\dots,p\}} \Big(\bigwedge_{l \in J} \mathrm{dd^c}(u_l + q) \Big) \wedge \sigma^{p-|J|}. \tag{14.37}$$

This does not depend on q. Let q' be another such function on an open set D'. On $D \cap D'$ put $g = q' - q$ and $\gamma = \mathrm{dd^c} g$, so that $\sigma' = \omega - \mathrm{dd^c} q' = \sigma - \gamma$. By (14.36), applied to $v_l = u_l + q$ and $g_l = g$, the right side of (14.37) for q' equals

$$\begin{aligned} &\sum_J \sum_{I \subset J} S_I \wedge \gamma^{|J|-|I|} \wedge (\sigma - \gamma)^{p-|J|} \\ &\qquad = \sum_I S_I \wedge \sum_{K \subset \{1,\dots,p\} \setminus I} \gamma^{|K|} \wedge (\sigma - \gamma)^{p-|I|-|K|} = \sum_I S_I \wedge \sigma^{p-|I|}, \end{aligned}$$

where $S_I = \bigwedge_{l \in I} \mathrm{dd^c}(u_l + q)$ and the last step is the binomial formula for the commuting forms γ and $\sigma - \gamma$. Hence the local definitions agree on overlaps, and with a partition of unity subordinate to a cover of X by such sets D (Proposition 7.1.3) they define a current of order zero on X.

When the u_l are smooth, Proposition 9.1.2(1) shows that this current is the exterior product of the forms ω_{u_l}. For $u_p = 0$, since $\mathrm{dd^c}(0 + q) \wedge S = \mathrm{dd^c} q \wedge S$ for every Bedford–Taylor product S by (9.1), the expansion (14.37) becomes $(\omega_{u_1} \wedge \cdots \wedge \omega_{u_{p-1}}) \wedge (\mathrm{dd^c} q + \sigma)$; thus a factor ω in the product is the exterior product with the smooth form ω. The product is symmetric in its factors, because the Bedford–Taylor products are. It is affine in each potential. Indeed, let $u_1, u_1' \in \mathrm{PSH}(X, \omega)$ be bounded and $0 \leq t \leq 1$. Then $tu_1 + (1-t)u_1' + q = t(u_1 + q) + (1-t)(u_1' + q)$, and Bedford–Taylor products are additive and positively homogeneous in each factor (Proposition 9.1.2(3)). Hence each term of (14.37) with $1 \in J$ is the corresponding combination of the terms for u_1 and u_1', while the terms with $1 \notin J$ do not involve the first potential, and

$$\omega_{tu_1+(1-t)u_1'} \wedge \omega_{u_2} \wedge \cdots \wedge \omega_{u_p} = t\, \omega_{u_1} \wedge \omega_{u_2} \wedge \cdots \wedge \omega_{u_p} + (1-t)\, \omega_{u_1'} \wedge \omega_{u_2} \wedge \cdots \wedge \omega_{u_p}.$$

For $u_1 = \cdots = u_p = u$ we write ω_u^p, and similarly $\omega_u^j \wedge \omega_v^k \wedge \omega^{n-j-k}$ for mixed products. By Lemma 14.8.1(2) below, ω_u^n is a positive measure; we call it the *Monge–Ampère measure* of u with respect to ω. It is the global counterpart of the measure $(\mathrm{dd^c} u)^n$ of Definition 9.1.1.

Lemma 14.8.1 *The class* $\mathrm{PSH}(X,\omega)$ *and the products of the currents* ω_u *have the following properties.*

(1) *A finite maximum of functions in* $\mathrm{PSH}(X,\omega)$ *belongs to* $\mathrm{PSH}(X,\omega)$*, and so does the upper semicontinuous regularization of the supremum of a nonempty family in* $\mathrm{PSH}(X,\omega)$ *that is uniformly bounded above. If* $u \in \mathrm{PSH}(X,\omega)$*,* $D \subset X$ *is open,* $v \in \mathrm{PSH}(D,\omega)$*, and* $\overline{\lim}_{D\ni y\to x} v(y) \le u(x)$ *for every* $x \in \partial D$*, then the function equal to* $\max\{u,v\}$ *on* D *and to* u *on* $X \setminus D$ *belongs to* $\mathrm{PSH}(X,\omega)$*.*

(2) *Let* $u_1,\dots,u_p \in \mathrm{PSH}(X,\omega)$ *be bounded,* $1 \le p \le n$*. The current* $\omega_{u_1}\wedge\cdots\wedge\omega_{u_p}$ *is positive. The total mass of its trace measure with respect to* ω *(Section 7.5) is bounded by a constant depending only on* (X,ω) *and* $\max_l \sup_X |u_l|$*, and every pluripolar set has outer measure zero for this trace measure. Moreover, if* $u \in \mathrm{PSH}(X,\omega)$ *is bounded and* g *is a smooth function on an open subset* D *of a coordinate chart with* $\mathrm{dd^c} g \ge \omega$*, then* $(\mathrm{dd^c}(u+g))^n \ge \omega_u^n$ *on* D*.*

(3) *For* $1 \le l \le p$ *let* $(u_{l,j})_{j\ge1}$ *be sequences in* $\mathrm{PSH}(X,\omega)$ *with* $\sup_{l,j}\sup_X |u_{l,j}| < \infty$*. Suppose that either each sequence decreases pointwise to a function* u_l*, or each sequence increases and* u_l *is the upper semicontinuous regularization of its limit. Then the* u_l *are bounded functions in* $\mathrm{PSH}(X,\omega)$*, and* $\omega_{u_{1,j}}\wedge\cdots\wedge\omega_{u_{p,j}} \to \omega_{u_1}\wedge\cdots\wedge\omega_{u_p}$ *weakly.*

(4) *Let* $L^1(X)$ *be taken with respect to the volume form* ω^n*. The set of* $u \in \mathrm{PSH}(X,\omega)$ *with* $\sup_X u = 0$ *is compact in* $L^1(X)$*, hence bounded in* $L^1(X)$*, and* $u \mapsto \sup_X u$ *is continuous on* $\mathrm{PSH}(X,\omega)$ *for the* L^1 *topology.*

Proof (1) We freeze the coefficients of ω on small balls. Fix $\varepsilon > 0$, a coordinate chart and a compact subset L of it. By uniform continuity of the coefficients of ω on a compact neighborhood of L in the chart, there is $r > 0$ such that, for every $a \in L$, the ball $B(a,r)$ lies in the chart and the form $\gamma_a = \omega(a) + \frac{\varepsilon}{2}\beta$ satisfies $\omega \le \gamma_a \le \omega + \varepsilon\beta$ on $B(a,r)$. Let g_a be the real quadratic polynomial with $\mathrm{dd^c} g_a = \gamma_a$ described at the beginning of this section. If $u \in \mathrm{PSH}(X,\omega)$, then $u + g_a$ is quasi-plurisubharmonic with $\mathrm{dd^c}(u+g_a) = \omega_u + (\gamma_a - \omega) \ge 0$, so it is plurisubharmonic on $B(a,r)$; in the same way $v + g_a$ is plurisubharmonic on $D \cap B(a,r)$.

Let w be the function in question: a finite maximum $\max_i u_i$, a regularized supremum $(\sup_i u_i)^*$, or the pasted function. We show that $w + g_a$ is plurisubharmonic on $B(a,r)$. In the first case $w + g_a$ is the maximum of the plurisubharmonic functions $u_i + g_a$, which is plurisubharmonic by Proposition 8.2.5. In the second case $w + g_a = (\sup_i (u_i + g_a))^*$, because g_a is continuous, and it is plurisubharmonic by Theorem 8.3.2, since the family $(u_i + g_a)$ is uniformly bounded above on $B(a,r)$. In the third case $w + g_a$ equals $\max\{u+g_a, v+g_a\}$ on $D \cap B(a,r)$ and $u + g_a$ elsewhere; every boundary point x of $D \cap B(a,r)$ in $B(a,r)$ lies in ∂D, and $\overline{\lim}_{y\to x}(v+g_a)(y) \le u(x) + g_a(x)$ because g_a is continuous, so Proposition 8.2.5 applies on $B(a,r)$. Hence w is quasi-plurisubharmonic on $B(a,r)$ and $\omega + \mathrm{dd^c} w \ge \omega - \gamma_a \ge -\varepsilon\beta$ there. The balls $B(a,r)$, $a \in L$, cover a neighborhood of L, and positivity of currents is a local property, so $\omega + \mathrm{dd^c} w \ge -\varepsilon\beta$ on a neighborhood of L. Letting $\varepsilon \downarrow 0$ gives $\omega + \mathrm{dd^c} w \ge 0$ on the interior of L, and since the chart and L were arbitrary, $w \in \mathrm{PSH}(X,\omega)$.

(2) We first prove positivity on a coordinate ball B whose closure lies in a chart, choosing $q = C|z|^2$ in (14.37). Let $B' \Subset B$ be a smaller concentric ball, let ρ_k be radial mollifiers whose radii decrease to zero and are smaller than the distance from B' to ∂B, and put $u_{l,k} = (u_l+q)*\rho_k - q$ on B'. Since $\mathrm{dd^c} q$ has constant coefficients and $\int \rho_k = 1$, we have $\mathrm{dd^c} u_{l,k} = (\mathrm{dd^c} u_l) * \rho_k$, where the convolution acts on the coefficients, and $(\omega + \mathrm{dd^c} u_l) * \rho_k \geq 0$ because mollification preserves positivity (Section 7.2). By uniform continuity of the coefficients of ω, there are $\tau_k \to 0$ with $\omega * \rho_k \leq \omega + \tau_k \beta$ on B'. Hence the smooth $(1,1)$-forms $\omega_{u_{l,k}} + \tau_k\beta = (\omega + \mathrm{dd^c} u_l) * \rho_k + (\omega + \tau_k\beta - \omega * \rho_k)$ are positive, and so is their exterior product, by Proposition 7.2.1. Expanding this product of the forms $\mathrm{dd^c}((u_l + q) * \rho_k) + (\sigma + \tau_k\beta)$, we obtain

$$\sum_J \Big(\bigwedge_{l \in J} \mathrm{dd^c}\big((u_l + q) * \rho_k\big) \Big) \wedge (\sigma + \tau_k\beta)^{p-|J|}.$$

As $k \to \infty$, the smooth plurisubharmonic functions $(u_l + q) * \rho_k$ decrease to $u_l + q$ by Proposition 8.2.2(2), so the Bedford–Taylor products converge weakly by Theorem 9.2.5; their masses are bounded on compact subsets of B' uniformly in k by Theorem 9.1.3, and the coefficients of $(\sigma + \tau_k\beta)^{p-|J|}$ converge uniformly to those of $\sigma^{p-|J|}$. Hence each term converges weakly to the corresponding term of (14.37), the product is a weak limit of positive forms on B', and it is positive. Since such balls B' cover X, the product is positive on X.

Cover X by finitely many such balls B', with q defined on the larger balls B. On compact subsets of B, Theorem 9.1.3 bounds the trace masses of the Bedford–Taylor products in (14.37) by a constant times a product of the numbers $\sup_B |u_l + q|$, and Theorem 7.2.5 bounds the total variations of their coefficient measures in the same way. Since σ and ω are smooth, the measure $(\omega_{u_1} \wedge \cdots \wedge \omega_{u_p}) \wedge \omega^{n-p}$ is a sum of the coefficient measures of these Bedford–Taylor products, and of a smooth density for $J = \varnothing$, multiplied by bounded smooth functions. Summing over the finite cover gives the asserted bound for the total mass of the trace measure. By Corollary 10.2.2(3), every pluripolar set has outer measure zero for the total variations of the coefficient measures of Bedford–Taylor products of locally bounded plurisubharmonic functions; it is also Lebesgue null (remarks following Definition 10.1.1). The trace measure is a positive measure bounded on each B' by a finite sum of such total variations and of a multiple of Lebesgue measure, so every pluripolar set has outer measure zero for it.

Finally, let $\mathrm{dd^c} g \geq \omega$ on D. We may take $q = g$ in (14.37) on D. Put $A = \mathrm{dd^c}(u+g)$ and $\gamma = \mathrm{dd^c} g - \omega \geq 0$, so that $\sigma = -\gamma$ and (14.37) reads $\omega_u^k = \sum_{i=0}^k \binom{k}{i} A^i \wedge (-\gamma)^{k-i}$, where A^i is a Bedford–Taylor power. Using $\binom{n}{k}\binom{k}{i} = \binom{n}{i}\binom{n-i}{k-i}$ and $\sum_{m=0}^{n-i} \binom{n-i}{m}(-1)^m = 0$ for $i < n$, we obtain

$$\sum_{k=0}^n \binom{n}{k} \omega_u^k \wedge \gamma^{n-k} = \sum_{i=0}^n \binom{n}{i} A^i \wedge \gamma^{n-i} \sum_{k=i}^n \binom{n-i}{k-i} (-1)^{k-i} = A^n.$$

The currents ω_u^k are positive and the forms γ^{n-k} are strongly positive, so every term on the left is a positive measure, and $(\mathrm{dd^c}(u+g))^n = A^n \geq \omega_u^n$.

(3) On a coordinate ball, add a function q as in (14.37). In the decreasing case, u_l is bounded, the functions $u_{l,j}+q$ are plurisubharmonic and decrease to u_l+q, which is bounded below, so u_l+q is plurisubharmonic and $u_{l,j}+q \to u_l+q$ in L^1_{loc} by Theorem 8.3.1. Hence u_l is quasi-plurisubharmonic, and $\omega+\mathrm{dd}^{\mathrm{c}}u_l$ is the weak limit of the positive currents $\omega+\mathrm{dd}^{\mathrm{c}}u_{l,j}$, so $u_l \in \mathrm{PSH}(X,\omega)$. The Bedford–Taylor products of the functions $u_{l,j}+q$ converge weakly by Theorem 9.2.5. In the increasing case, $u_l \in \mathrm{PSH}(X,\omega)$ by (1), and u_l is bounded. The plurisubharmonic functions $u_{l,j}+q$ increase, and u_l+q is the upper semicontinuous regularization of their limit, because q is continuous; so $u_{l,j}+q \to u_l+q$ almost everywhere by Theorem 8.3.2, and Theorem 9.5.1, applied on the ball, gives the weak convergence of the Bedford–Taylor products. In both cases we multiply by the fixed smooth forms $\sigma^{p-|J|}$ and sum; a partition of unity subordinate to finitely many balls gives weak convergence on X.

(4) Choose finitely many coordinate balls B_k, with closures in charts, that cover X, concentric balls $B'_k \Subset B_k$ that still cover X, and smooth functions q_k on neighborhoods of $\overline{B_k}$ with $\mathrm{dd}^{\mathrm{c}}q_k \geq \omega$. For every $u \in \mathrm{PSH}(X,\omega)$ the functions $u+q_k$ are plurisubharmonic on B_k. Two quasi-plurisubharmonic functions that agree almost everywhere on an open set agree everywhere on it, by Lemma 13.1.4(1). We may therefore regard $\mathrm{PSH}(X,\omega)$ as a subset of $L^1(X)$; on compact subsets of a chart, ω^n is bounded above and below by positive multiples of Lebesgue measure.

Continuity of the supremum. Let $u_j, u \in \mathrm{PSH}(X,\omega)$ with $u_j \to u$ in $L^1(X)$; then $u_j+q_k \to u+q_k$ in $L^1_{\mathrm{loc}}(B_k)$ for every k. Choose points x_j with $u_j(x_j)=\sup_X u_j$, which exist because u_j is upper semicontinuous on the compact manifold X, and a subsequence along which $\sup_X u_j \to \varlimsup_j \sup_X u_j$ and $x_j \to x$. Choose k with $x \in B_k$ and a real number $c > u(x)$, and a closed ball $K \subset B_k$ centered at x on which $u < c$. Since $c+q_k$ is continuous and $u+q_k < c+q_k$ on K, the second part of Theorem 8.3.4 gives $u_j+q_k < c+q_k$ on K for large j; hence $u_j(x_j) < c$ for large j. Letting $c \downarrow u(x)$ we obtain $\varlimsup_j \sup_X u_j \leq u(x) \leq \sup_X u$. Conversely, choose a subsequence along which $\sup_X u_j$ tends to $s = \varliminf_j \sup_X u_j$ and $u_j \to u$ almost everywhere. Then $u \leq s$ almost everywhere, so $s > -\infty$ because u is integrable, and $u \leq s$ everywhere by Lemma 13.1.4(1), applied to u and the constant s. Thus $\sup_X u \leq \varliminf_j \sup_X u_j$, and the supremum is continuous.

Compactness. Let (u_j) be a sequence in $\mathrm{PSH}(X,\omega)$ with $\sup_X u_j = 0$. On each B_k the plurisubharmonic functions u_j+q_k are bounded above by $\sup_{B_k} q_k$. Choose x_j with $u_j(x_j)=0$, pass to a subsequence with $x_j \to x$, and choose k_0 with $x \in B'_{k_0}$. On B_{k_0} the functions $u_j+q_{k_0}$ do not converge locally uniformly to $-\infty$, since for large j their values at the points $x_j \in \overline{B'_{k_0}}$ are at least $\min_{\overline{B'_{k_0}}} q_{k_0}$. By Theorem 8.3.4, a subsequence converges in $L^1_{\mathrm{loc}}(B_{k_0})$.

If B_l meets a ball B_k on which (u_j+q_k) converges in $L^1_{\mathrm{loc}}(B_k)$, then on the nonempty open set $B_k \cap B_l$ the functions $u_j+q_l = (u_j+q_k)+(q_l-q_k)$ converge in L^1_{loc} to a locally integrable function, so they do not tend to $-\infty$ locally uniformly on B_l, and a further subsequence converges in $L^1_{\mathrm{loc}}(B_l)$. Since X is connected, every ball is joined to B_{k_0} by a finite chain of balls, each meeting the next, so after finitely many extractions (u_j+q_k) converges in $L^1_{\mathrm{loc}}(B_k)$ to a plurisubharmonic function v_k

for every k (Theorem 8.3.4). Then (u_j) converges in $L^1(B'_k)$ for every k, hence in $L^1(X)$.

It remains to identify the limit. The quasi-plurisubharmonic functions $v_k - q_k$ and $v_l - q_l$ agree almost everywhere on $B_k \cap B_l$, hence everywhere, so they define a quasi-plurisubharmonic function u on X, and $\omega + \mathrm{dd^c} u \geq 0$ as a distributional limit of the positive currents $\omega + \mathrm{dd^c} u_j$. Thus $u \in \mathrm{PSH}(X, \omega)$, $u_j \to u$ in $L^1(X)$, and $\sup_X u = \lim_j \sup_X u_j = 0$ by the continuity just proved. The set of normalized functions is therefore sequentially compact, hence compact, in the metric space $L^1(X)$, and the continuous function $u \mapsto \|u\|_{L^1(X)}$ is bounded on it. □

The comparison principle for small gaps rests on the next lemma, which controls the error term $\mathrm{dd^c} T$ produced by integration by parts against a product T of currents ω_u; such a product need not be closed, since ω need not be.

Lemma 14.8.2 *Let $u, v \in \mathrm{PSH}(X, \omega)$ be bounded, put $E = \{u < v\}$, and let T be a product of $n-1$ factors, each equal to ω, ω_u or ω_v. Then $\mathrm{dd^c} T$ is a real measure on X, that is, a real current of order zero and degree $2n$, and*

$$\int_E (\omega_v - \omega_u) \wedge T \leq \int_E (v - u)\, \mathrm{dd^c} T, \tag{14.38}$$

where $(\omega_v - \omega_u) \wedge T = \omega_v \wedge T - \omega_u \wedge T$. Moreover, there is a constant C depending only on (X, ω) such that the total variation $|\mathrm{dd^c} T|$ of this measure satisfies

$$|\mathrm{dd^c} T| \leq C\Big(\sum_i T_{\hat{\imath}} \wedge \omega^2 + \sum_{i<i'} T_{\hat{\imath}\hat{\imath}'} \wedge \omega^3\Big). \tag{14.39}$$

Here $T_{\hat{\imath}}$ denotes the product of the factors of T other than the ith one, $T_{\hat{\imath}\hat{\imath}'}$ the product of those other than the ith and the i'th, and empty sums are zero.

Proof After computing $\mathrm{dd^c} T$ (Step 1), we prove a version of (14.38) in which $\mathbf{1}_E$ is smoothed by a convex cutoff, integrating by parts against regularizations of u and v (Step 2). The products T, $T_{\hat{\imath}}$ and $T_{\hat{\imath}\hat{\imath}'}$ are positive currents: for products with at least one factor this is Lemma 14.8.1(2), and the empty product is the constant 1. (If $n = 1$, then $T = 1$, $\mathrm{dd^c} T = 0$, and both sums in (14.39) are empty; the sum over pairs is empty also if $n = 2$.)

Step 1. The measure $\mathrm{dd^c} T$. Work on an open subset of a chart with q and σ as in (14.37). Each factor of T has the form $\Theta_i + \sigma$, where Θ_i is $\mathrm{dd^c} q$, $\mathrm{dd^c}(u+q)$ or $\mathrm{dd^c}(v+q)$; thus $T = \sum_J \Theta_J \wedge \sigma^{n-1-|J|}$, where Θ_J is the Bedford–Taylor product of the Θ_i with $i \in J$. The currents Θ_J are closed of pure bidegree, so $\partial\Theta_J = \bar\partial\Theta_J = 0$, and hence $\mathrm{d}\Theta_J = \mathrm{d^c}\Theta_J = 0$, where $\mathrm{d^c}$ is the operator (7.6). By the Leibniz rule for currents (Section 7.2), $\mathrm{dd^c}(\Theta_J \wedge \sigma^m) = \Theta_J \wedge \mathrm{dd^c}(\sigma^m)$. Since the closed $(1,1)$-form $\mathrm{dd^c} q$ satisfies $\partial\mathrm{dd^c} q = \bar\partial\mathrm{dd^c} q = 0$, we have $\mathrm{d}\sigma = \mathrm{d}\omega$, $\mathrm{d^c}\sigma = \mathrm{d^c}\omega$ and $\mathrm{dd^c}\sigma = \mathrm{dd^c}\omega$, and the Leibniz rule gives

$$\mathrm{dd^c}(\sigma^m) = m\,\sigma^{m-1} \wedge \mathrm{dd^c}\omega + m(m-1)\,\sigma^{m-2} \wedge \mathrm{d}\omega \wedge \mathrm{d^c}\omega.$$

Expand $T_{\hat\imath}$ and $T_{\hat\imath\hat\imath'}$ in the same way. For a fixed J, with $m = n-1-|J|$, there are m indices $i \notin J$ and $m(m-1)/2$ pairs $i < i'$ outside J; comparing coefficients, we obtain

$$\mathrm{dd^c}T = \sum_i \mathrm{dd^c}\omega \wedge T_{\hat\imath} + 2\sum_{i<i'} \mathrm{d}\omega \wedge \mathrm{d^c}\omega \wedge T_{\hat\imath\hat\imath'}. \tag{14.40}$$

In particular $\mathrm{d}T = \sum_J \Theta_J \wedge \mathrm{d}(\sigma^{n-1-|J|})$, $\mathrm{d^c}T$ and $\mathrm{dd^c}T$ are currents of order zero, being sums of Bedford–Taylor products multiplied by smooth forms. The form $\mathrm{dd^c}\omega$ is real of bidegree $(2,2)$. Since $\partial\omega$ and $\bar\partial\omega$ have odd degree, $\partial\omega\wedge\partial\omega = \bar\partial\omega\wedge\bar\partial\omega = 0$ and $\bar\partial\omega\wedge\partial\omega = -\partial\omega\wedge\bar\partial\omega$, so $\mathrm{d}\omega\wedge\mathrm{d^c}\omega = \frac{\mathrm{i}}{2\pi}\partial\omega\wedge\bar\partial\omega$ is real of bidegree $(3,3)$; these forms vanish if $n<2$ and $n<3$, respectively.

We claim that there is C such that $C\omega^2 \pm \mathrm{dd^c}\omega$ and $C\omega^3 \pm \mathrm{d}\omega\wedge\mathrm{d^c}\omega$ are strongly positive at every point of X. Consider ω^2 and $\mathrm{dd^c}\omega$, with $n \geq 2$; the other case, with $n \geq 3$, is the same. Let D be a coordinate ball whose closure lies in a chart, and let C be the cone of strongly positive $(2,2)$-forms on $\mathbb{C}^n$. For $x \in \overline{D}$, a complex linear change of coordinates takes $\omega(x)$ to β; since such changes preserve strong positivity in both directions, they map the interior of C onto itself, and β^2 is an interior point of C (Proposition 7.2.1). Hence $\omega(x)^2$ lies in the interior of C. As $x \mapsto \omega(x)^2$ is continuous, these forms make up a compact subset of the interior of C, which has a positive distance d_0 from the closed complement of the interior, for a fixed norm on the real $(2,2)$-forms. The coefficients of $\mathrm{dd^c}\omega$ are bounded on $\overline{D}$, so for C large the forms $\omega(x)^2 \pm C^{-1}\mathrm{dd^c}\omega(x)$ lie at distance less than d_0 from $\omega(x)^2$, hence in C, for all $x \in \overline{D}$. Covering the compact manifold X by finitely many such balls proves the claim.

The currents $T_{\hat\imath}$ and $T_{\hat\imath\hat\imath'}$ are positive, so $-CT_{\hat\imath}\wedge\omega^2 \leq \mathrm{dd^c}\omega\wedge T_{\hat\imath} \leq CT_{\hat\imath}\wedge\omega^2$, and similarly for the terms with $\mathrm{d}\omega\wedge\mathrm{d^c}\omega$. A real measure ν with $-\mu \leq \nu \leq \mu$ for a positive measure μ satisfies $|\nu| \leq \mu$, so (14.39), with the constant $2C$, follows from (14.40).

Step 2. A convex cutoff. Put $w = v - u$, a bounded Borel function on X. Fix a smooth increasing function $\lambda\colon \mathbb{R} \to [0,1]$ with $\lambda = 0$ on $(-\infty, 0]$ and $\lambda = 1$ on $[1,\infty)$, and for $\delta > 0$ put $F_\delta(t) = \int_0^t \lambda(s/\delta)\,\mathrm{d}s$. Then F_δ is smooth and convex, $F_\delta = 0$ on $(-\infty, 0]$, $0 \leq F'_\delta \leq 1$, and as $\delta \downarrow 0$ we have $F_\delta(t) \to t^+$ and $F'_\delta(t) \to \mathbf{1}_{(0,\infty)}(t)$ for every real t; indeed $|F_\delta(t) - t| \leq \delta$ for $t > 0$. We shall prove that, for every $\delta > 0$,

$$\int_X F'_\delta(w)\,(\omega_v - \omega_u) \wedge T \leq \int_X F_\delta(w)\,\mathrm{dd^c}T. \tag{14.41}$$

Granting this, let $\delta \downarrow 0$. The integrands converge pointwise and boundedly to $\mathbf{1}_E$ and to $(v-u)\mathbf{1}_E$; note that $F'_\delta(0) = 0$, so the first limit is the indicator of $\{w > 0\} = E$. Both measures have finite total variation, so dominated convergence turns (14.41) into (14.38).

To prove (14.41), choose finitely many coordinate balls B_k covering X, smooth functions q_k with $\mathrm{dd^c}q_k \geq \omega$ on a neighborhood of $\overline{B_k}$, and a smooth partition of unity (χ_k) on X with $\operatorname{Supp}\chi_k \subset B_k$ (Proposition 7.1.3). Fix k, write $\sigma_k = \omega - \mathrm{dd^c}q_k$, and choose an open set Ω_k with $\operatorname{Supp}\chi_k \subset \Omega_k \Subset B_k$. Convolution of $u + q_k$ and $v + q_k$ with radial mollifiers whose radii decrease to zero and are smaller than the

distance from Ω_k to ∂B_k gives smooth plurisubharmonic functions on Ω_k, which we write as $u_l + q_k$ and $v_l + q_k$. They decrease to $u + q_k$ and $v + q_k$ as $l \to \infty$, by Proposition 8.2.2(2), and they take values in a fixed compact interval $[-M', M']$, because $u + q_k \le u_l + q_k \le u_1 + q_k$ and similarly for v. Then $w_l = v_l - u_l \to w$ pointwise on Ω_k, with $|w_l| \le 2M'$.

The function $f = F_\delta(w_l)$ is smooth on Ω_k, which contains the support of $\chi_k T$, so the definition of derivatives of currents and the Leibniz rule give

$$\int f \,\mathrm{dd^c}(\chi_k T) = \int \chi_k \,\mathrm{dd^c} f \wedge T, \qquad \mathrm{dd^c}(\chi_k T) = \chi_k \mathrm{dd^c} T + \Lambda_k,$$

where $\Lambda_k = \mathrm{dd^c}\chi_k \wedge T + \mathrm{d}\chi_k \wedge \mathrm{d^c} T - \mathrm{d^c}\chi_k \wedge \mathrm{d}T$, and the chain rule gives

$$\mathrm{dd^c} f = F'_\delta(w_l)\,\mathrm{dd^c} w_l + F''_\delta(w_l)\,\frac{\mathrm{i}}{2\pi}\,\partial w_l \wedge \bar\partial w_l.$$

The last term is a smooth strongly positive $(1,1)$-form, since $F''_\delta \ge 0$, and T is positive, so the product of this term with $\chi_k T$ is a positive measure. Consequently

$$\int \chi_k F'_\delta(w_l)\,\mathrm{dd^c} w_l \wedge T \le \int F_\delta(w_l)\big(\chi_k \mathrm{dd^c} T + \Lambda_k\big). \tag{14.42}$$

Now let $l \to \infty$. By Step 1, $\mathrm{dd^c}T$ and Λ_k are measures of finite total variation, so the right side tends to $\int F_\delta(w)(\chi_k \mathrm{dd^c} T + \Lambda_k)$ by dominated convergence.

On the left, write $T = \sum_J \Theta_J \wedge \sigma_k^{m_J}$ as in Step 1, with $q = q_k$ and $m_J = n - 1 - |J|$. The exterior product of the smooth form $\mathrm{dd^c}(v_l + q_k)$ with Θ_J is the Bedford–Taylor product $\mathrm{dd^c}(v_l + q_k) \wedge \Theta_J$, by (9.1), so the left side of (14.42) is the sum over J of

$$\int \chi_k F'_\delta(w_l)\,\mathrm{dd^c}(v_l + q_k) \wedge \Theta_J \wedge \sigma_k^{m_J} - \int \chi_k F'_\delta(w_l)\,\mathrm{dd^c}(u_l + q_k) \wedge \Theta_J \wedge \sigma_k^{m_J}.$$

Fix J and consider the first integral; the second is treated in the same way. Let $\vartheta > 0$. By the Weierstrass approximation theorem there is a real polynomial P of one variable with $|F'_\delta - P| \le \vartheta$ on $[-2M', 2M']$. The weights $P(w_l) = P\big((v_l + q_k) - (u_l + q_k)\big)$ are polynomials in two decreasing sequences of bounded plurisubharmonic functions on Ω_k, so Corollary 9.2.6, applied on Ω_k with the closed positive base current Θ_J and the decreasing factors $v_l + q_k$, gives the weak convergence of $P(w_l)\,\mathrm{dd^c}(v_l + q_k) \wedge \Theta_J$ to $P(w)\,\mathrm{dd^c}(v + q_k) \wedge \Theta_J$, which we test with the smooth form $\chi_k \sigma_k^{m_J}$ with compact support in Ω_k. On the other hand, by Theorem 9.1.3 and Theorem 7.2.5, the total variations on $\operatorname{Supp}\chi_k$ of the coefficient measures of the positive currents $\mathrm{dd^c}(v_l + q_k) \wedge \Theta_J$ and $\mathrm{dd^c}(v + q_k) \wedge \Theta_J$ are bounded uniformly in l, so replacing $F'_\delta(w_l)$ by $P(w_l)$, or $F'_\delta(w)$ by $P(w)$, changes the integrals by at most a constant times ϑ, with a constant independent of l. As ϑ was arbitrary,

$$\int \chi_k F'_\delta(w_l)\,\mathrm{dd^c}(v_l + q_k) \wedge \Theta_J \wedge \sigma_k^{m_J} \longrightarrow \int \chi_k F'_\delta(w)\,\mathrm{dd^c}(v + q_k) \wedge \Theta_J \wedge \sigma_k^{m_J}.$$

By (14.37), $\omega_v \wedge T = \sum_J \mathrm{dd^c}(v+q_k)\wedge\Theta_J\wedge\sigma_k^{m_J} + \sigma_k \wedge T$ on B_k, and similarly for u. Summing over J, we conclude that the left side of (14.42) tends to $\int \chi_k F'_\delta(w)\,(\omega_v - \omega_u)\wedge T$. Finally we sum the limiting inequalities over k. Since $\sum_k \chi_k = 1$, we have $\sum_k \Lambda_k = \mathrm{dd^c}T - \sum_k \chi_k \mathrm{dd^c}T = 0$, and (14.41) follows. □

Proposition 14.8.3 *There are constants $s_0, C > 0$, depending only on (X,ω), with the following property. Let $u, v \in \mathrm{PSH}(X,\omega)$ be bounded with $\omega_v \geq \frac{1}{2}\omega$, put $E = \{u < v\}$, and suppose that $v - u \leq s$ on E for some $s \in (0, s_0)$. Then*

$$\int_E \omega_v^n \leq (1 + Cs)\int_E \omega_u^n.$$

In particular $\int_X \omega_u^n > 0$ for every bounded $u \in \mathrm{PSH}(X,\omega)$.

Proof We replace the factors ω_v of ω_v^n by ω_u one at a time, using Lemma 14.8.2; the errors are traces of products with fewer factors different from ω, and Steps 1 and 2 bound them by multiples of $\int_E \omega_u^n$. For a product T of $n-1$ factors chosen among $\omega, \omega_u, \omega_v$, let $\mathcal{P}(T)$ be the set of products obtained from T by deleting one factor and inserting ω^2, or by deleting two factors and inserting ω^3. The right side of (14.39) has at most n^2 terms, each of which is an element of $\mathcal{P}(T)$, and $0 < v - u \leq s$ on E. Hence Lemma 14.8.2 gives

$$\int_E \omega_v \wedge T - \int_E \omega_u \wedge T \leq s\,|\mathrm{dd^c}T|(E) \leq sB \sum_{P\in\mathcal{P}(T)} \int_E P, \tag{14.43}$$

with a constant $B \geq 1$ depending only on (X,ω).

Step 1. Pure traces. For $0 \leq k \leq n$ put $a_k = \int_E \omega_u^k \wedge \omega^{n-k} \geq 0$, and apply (14.43) to $T = \omega_u^k \wedge \omega^{n-k-1}$, $0 \leq k < n$. The function $2v$ belongs to $\mathrm{PSH}(X,\omega)$, because $\omega + \mathrm{dd^c}(2v) = 2\omega_v - \omega \geq 0$, and it is bounded. Since the products are affine in each potential and $v = \frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 2v$, we have $\omega_v \wedge T = \frac{1}{2}\omega\wedge T + \frac{1}{2}\omega_{2v}\wedge T$, and $\omega_{2v}\wedge T$ is positive by Lemma 14.8.1(2); hence $\int_E \omega_v \wedge T \geq \frac{1}{2}a_k$. Moreover $\int_E \omega_u \wedge T = a_{k+1}$, and the elements of $\mathcal{P}(T)$ are among the products $\omega_u^h \wedge \omega^{n-h}$ with $h \in \{k-2, k-1, k\}$. Thus (14.43) gives

$$\tfrac{1}{2}a_k \leq a_{k+1} + sB\sum_{h=0}^{k} a_h, \qquad 0 \leq k < n. \tag{14.44}$$

Choose $s_0 > 0$ with $s_0 B(n+1)4^n \leq 1/4$; in particular $sB \leq 1$. We prove $a_k \leq 4a_{k+1}$ for $0 \leq k < n$ by induction on k. If it holds for all smaller indices, then $a_h \leq 4^{k-h}a_k$ for $h \leq k$, so $\sum_{h=0}^k a_h \leq (k+1)4^k a_k$, and the error term in (14.44) is at most $s_0B(n+1)4^n a_k \leq \frac{1}{4}a_k$. Hence $\frac{1}{2}a_k \leq a_{k+1} + \frac{1}{4}a_k$, that is, $a_k \leq 4a_{k+1}$. Consequently

$$a_k \leq 4^{n-k}a_n, \qquad 0 \leq k \leq n.$$

Step 2. Mixed traces. For $j, k \geq 0$ with $j+k \leq n$ put $a_{j,k} = \int_E \omega_u^j \wedge \omega_v^k \wedge \omega^{n-j-k} \geq 0$, so that $a_{j,0} = a_j$. If $k \geq 1$, apply (14.43) to $T = \omega_u^j \wedge \omega_v^{k-1} \wedge \omega^{n-j-k}$. Then

$\int_E \omega_v \wedge T = a_{j,k}$ and $\int_E \omega_u \wedge T = a_{j+1,k-1}$, while every element of $\mathcal{P}(T)$ has fewer than $j+k$ factors different from ω. Thus

$$a_{j,k} \le a_{j+1,k-1} + sB \sum_{p+q<j+k} a_{p,q} \qquad (k \ge 1).$$

We show by induction on $m = j+k$ that $a_{j,k} \le C_m \sum_{h=0}^{m} a_h$ with C_m depending only on n. For $k = 0$ this holds with $C_m = 1$. For $k \ge 1$, since $sB \le 1$, applying the last inequality k times, which replaces the factors ω_v one at a time by ω_u, gives

$$a_{j,k} \le a_{j+k,0} + k \sum_{p+q<m} a_{p,q},$$

and by the induction hypothesis the last sum, which has at most $(m+1)^2$ terms, is at most a constant depending only on n times $\sum_{h<m} a_h$. Together with Step 1 this shows $a_{j,k} \le C'_n a_n$ for all j,k, where C'_n depends only on n.

Step 3. Conclusion. By symmetry of the products,

$$\omega_v^n - \omega_u^n = \sum_{i=0}^{n-1} (\omega_v - \omega_u) \wedge \omega_v^i \wedge \omega_u^{n-1-i},$$

since the sum telescopes. Apply (14.43) to each of the n products $T = \omega_v^i \wedge \omega_u^{n-1-i}$. The resulting error terms are sB times sums of at most n^2 traces $a_{p,q}$, hence at most a constant times sa_n by Step 2. Since $a_n = \int_E \omega_u^n$, this proves $\int_E \omega_v^n \le (1 + Cs) \int_E \omega_u^n$ with C depending only on (X, ω).

Step 4. Positivity. Let $u \in \mathrm{PSH}(X,\omega)$ be bounded, put $c = \inf_X u$, fix $s \in (0, s_0)$, and apply the inequality to the constant function $v = c + s$. Then $\omega_v = \omega \ge \frac{1}{2}\omega$, and $E = \{u < c + s\}$ is nonempty by the definition of the infimum and open because u is upper semicontinuous; moreover $0 < v - u \le s$ on E since $u \ge c$. Hence $0 < \int_E \omega^n \le (1 + Cs) \int_E \omega_u^n$, and therefore $\int_X \omega_u^n \ge \int_E \omega_u^n > 0$. □

We now introduce the extremal functions and the capacity. For a set $A \subset X$, the *global extremal function* of A is

$$T_A = \sup\{u \in \mathrm{PSH}(X,\omega) : u \le 0 \text{ on } A\},$$

and the functions u in this supremum are called *competitors* for T_A. For a Borel set $A \subset X$, the *Monge–Ampère capacity* of A with respect to ω is

$$c_\omega(A) = \sup\Big\{ \int_A \omega_v^n : v \in \mathrm{PSH}(X,\omega),\ 0 \le v \le 1 \Big\}.$$

This is the global analogue of the capacity Cap_D of a bounded open set $D \subset \mathbb{C}^n$ (Definition 9.3.2), with ω-plurisubharmonic functions on X in place of plurisubharmonic functions on D and ω_v^n in place of $(\mathrm{dd}^{\mathrm{c}} v)^n$. Unlike for Cap_D, constants contribute: $v = 0$ gives $c_\omega(A) \ge \int_A \omega^n$. By Lemma 14.8.1(2), $c_\omega(A) \le c_\omega(X) < \infty$.

Lemma 14.8.4 *For every set $A \subset X$, exactly one of the following holds: A is globally pluripolar, or T_A^* is a bounded nonnegative function in $\mathrm{PSH}(X,\omega)$. A subset of X with nonempty interior is not globally pluripolar. A countable union of globally pluripolar sets is globally pluripolar.*

Proof Let $\mathcal{F}_A$ be the family of competitors for T_A. It contains the constant 0, so $T_A \geq 0$.

We first show that if $\psi_j \in \mathrm{PSH}(X,\omega)$ and $\sup_X \psi_j = 0$ for $j \geq 1$, then $\Psi = \sum_{j\geq 1} 2^{-j}\psi_j$ belongs to $\mathrm{PSH}(X,\omega)$. By Lemma 14.8.1(4), there is a constant M with $\|\psi_j\|_{L^1(X)} \leq M$ for all j. The partial sums $\Psi_N = \sum_{j=1}^N 2^{-j}\psi_j$ decrease with N, since $\psi_j \leq 0$, and $\mathrm{dd^c}\Psi_N \geq -\sum_j 2^{-j}\omega \geq -\omega$. On a coordinate ball with a function q as in (14.37), the functions $\Psi_N + q$ are plurisubharmonic and decrease, so by Theorem 8.3.1 their limit $\Psi + q$ is plurisubharmonic or identically $-\infty$ on the ball. By monotone convergence $\Psi_N \to \Psi$ in $L^1(X)$ and $\int_X |\Psi|\,\omega^n \leq \sum_j 2^{-j}M \leq M$, so Ψ is finite almost everywhere and $\Psi + q$ is not identically $-\infty$ on any ball. Hence Ψ is quasi-plurisubharmonic, and $\Psi \in \mathrm{PSH}(X,\omega)$ because the inequality $\omega + \mathrm{dd^c}\Psi_N \geq 0$ passes to the limit.

Suppose now that $\sup\{\sup_X u : u \in \mathcal{F}_A\} = \infty$. Choose $u_j \in \mathcal{F}_A$ with $b_j = \sup_X u_j \geq 2^j$ for $j \geq 1$, and put $\psi_j = u_j - b_j$. Then $\psi_j \in \mathrm{PSH}(X,\omega)$, $\sup_X \psi_j = 0$ and $\psi_j \leq -2^j$ on A, so $\Psi = \sum_j 2^{-j}\psi_j$ belongs to $\mathrm{PSH}(X,\omega)$. Moreover $\Psi = -\infty$ on A, because every term $2^{-j}\psi_j$ is at most -1 there and all terms are nonpositive. Thus A is globally pluripolar.

Otherwise $T_A \leq C$ for a constant C, and by Lemma 14.8.1(1) the regularization T_A^* belongs to $\mathrm{PSH}(X,\omega)$ and satisfies $0 \leq T_A^* \leq C$. Thus at least one of the alternatives holds. They exclude each other: if $A \subset \{\Phi = -\infty\}$ for a quasi-plurisubharmonic Φ, choose $t > 0$ with $t\Phi \in \mathrm{PSH}(X,\omega)$; then $t\Phi + c \in \mathcal{F}_A$, hence $T_A \geq t\Phi + c$, for every real c, so T_A^* is not bounded.

A quasi-plurisubharmonic function is locally integrable (Lemma 13.1.4(1)), so its pole set has Lebesgue measure zero in every chart and contains no nonempty open set. Therefore a set with nonempty interior is not globally pluripolar.

Finally let $A_j \subset \{\varphi_j = -\infty\}$, $j \geq 1$, with φ_j quasi-plurisubharmonic. After multiplying by small positive constants and subtracting constants, we may assume that $\varphi_j \in \mathrm{PSH}(X,\omega)$ and $\sup_X \varphi_j = 0$. By the claim proved at the start, $\sum_j 2^{-j}\varphi_j$ belongs to $\mathrm{PSH}(X,\omega)$; it equals $-\infty$ on every A_j, because one term is $-\infty$ there and all terms are nonpositive. (A finite union is the case in which $A_j = \varnothing$ for all large j.)□

The next estimate is purely local: it compares the mixed Monge–Ampère masses of two bounded plurisubharmonic functions on a ball whose difference is small compared with the square of the radius. In Proposition 14.8.6 it is applied to a potential and its homogeneous balayage. We use cutoff functions with controlled complex Hessians. If $0 < t_0 < t_1$ and $\eta\colon \mathbb{R} \to [0,1]$ is a smooth function with $\eta = 1$ on $(-\infty, t_0^2]$ and $\eta = 0$ on $[t_1^2, \infty)$, then $\chi(z) = \eta(|z-a|^2/r^2)$ is a smooth function with support in the closed ball $\overline{B(a,t_1 r)}$, equal to 1 on $B(a,t_0 r)$, and

$$\mathrm{dd^c}\chi = \frac{\eta'}{r^2}\,\mathrm{dd^c}|z-a|^2 + \frac{\eta''}{r^4}\,\frac{\mathrm{i}}{2\pi}\,\partial|z-a|^2 \wedge \bar\partial|z-a|^2,$$

with η' and η'' evaluated at $|z-a|^2/r^2$. Here $\mathrm{dd^c}|z-a|^2 = \beta/\pi$, and if $\partial|z-a|^2 = \sum_k \bar{c}_k\,\mathrm{d}z_k$ with $c = z-a$, then $0 \le \mathrm{i}\,\partial|z-a|^2 \wedge \bar{\partial}|z-a|^2 \le 2|c|^2\beta$, because the Hermitian matrix $(\bar{c}_j c_k)$ is at most $|c|^2$ times the identity. Since $|z-a| \le t_1 r$ on the support of χ, we obtain

$$-C_\eta r^{-2}\beta \le \mathrm{dd^c}\chi \le C_\eta r^{-2}\beta, \qquad C_\eta = \pi^{-1}\bigl(\sup|\eta'| + t_1^2 \sup|\eta''|\bigr), \tag{14.45}$$

with a constant that depends only on η.

Lemma 14.8.5 *Let U, V be bounded plurisubharmonic functions on a ball $B(a,r) \subset \mathbb{C}^n$, $n \ge 1$, with $|V-U| \le \kappa r^2$ for some $\kappa \ge 0$. Put $A = \mathrm{dd^c}U$ and $D = \mathrm{dd^c}V$. For every integer $0 \le k \le n$,*

$$\int_{B(a,r/2)} \sum_{j=0}^{k} A^j \wedge D^{k-j} \wedge \beta^{n-k} \le C_n \sum_{l=0}^{k} \kappa^{k-l} \int_{B(a,r)} A^l \wedge \beta^{n-l},$$

where C_n depends only on n and $\kappa^0 = 1$, also when $\kappa = 0$.

Proof Put $\rho_i = r/2 + ir/(2n)$ and $B_i = B(a,\rho_i)$ for $0 \le i \le n$, so that $B_0 = B(a,r/2)$ and $B_n = B(a,r)$. For $0 \le i < n$ fix a smooth function $\eta_i : \mathbb{R} \to [0,1]$ with $\eta_i = 1$ on $(-\infty, (\rho_i/r)^2]$ and $\eta_i = 0$ on $[((\rho_i+\rho_{i+1})/(2r))^2, \infty)$; the numbers ρ_i/r depend only on n and i, so η_i can be chosen depending only on n and i. By (14.45), the functions $\chi_i(z) = \eta_i(|z-a|^2/r^2)$ satisfy $\chi_i \in C_c^\infty(B_{i+1})$, $0 \le \chi_i \le 1$, $\chi_i = 1$ on B_i and $-C'r^{-2}\beta \le \mathrm{dd^c}\chi_i \le C'r^{-2}\beta$, where $C' = \max_i C_{\eta_i}$ depends only on n. For integers $j, m \ge 0$ with $j+m \le n$ and $0 \le i \le n$, put

$$I_i(j,m) = \int_{B_i} A^j \wedge D^m \wedge \beta^{n-j-m}.$$

Let $m \ge 1$, $i < n$, and let $R = A^j \wedge D^{m-1} \wedge \beta^{n-j-m}$, a closed positive current by Proposition 9.1.2. By the definition and the symmetry of Bedford–Taylor products, $D \wedge R = A^j \wedge D^m \wedge \beta^{n-j-m}$ and $D \wedge R - A \wedge R = \mathrm{dd^c}((V-U)R)$, so

$$\int \chi_i\, D \wedge R = \int \chi_i\, A \wedge R + \int (V-U)\, R \wedge \mathrm{dd^c}\chi_i.$$

Since R is positive, the forms $C'r^{-2}\beta \pm \mathrm{dd^c}\chi_i$ give positive measures $R \wedge (C'r^{-2}\beta \pm \mathrm{dd^c}\chi_i)$; as $|V-U| \le \kappa r^2$ and $\operatorname{Supp}\chi_i \subset B_{i+1}$, the last integral is at most $C'\kappa \int_{B_{i+1}} R\wedge\beta$ in absolute value. Using $\mathbf{1}_{B_i} \le \chi_i \le \mathbf{1}_{B_{i+1}}$ and the positivity of the currents involved, we obtain

$$I_i(j,m) \le I_{i+1}(j+1, m-1) + C'\kappa\, I_{i+1}(j, m-1).$$

Both terms on the right have one factor D fewer. Applying this inequality m times, starting from $i = 0$, we find

$$I_0(j,m) \le \sum_{t=0}^{m} \binom{m}{t} (C'\kappa)^{m-t} I_m(j+t, 0) \le \sum_{t=0}^{m} \binom{m}{t} (C'\kappa)^{m-t} \int_{B(a,r)} A^{j+t} \wedge \beta^{n-j-t}.$$

Take $m = k - j$ and sum over $0 \le j \le k$. With $l = j + t$ we have $m - t = k - l$, so every term with A^l carries the factor κ^{k-l}, and the coefficients depend only on n. This proves the lemma; for $l = k$ the factor is $\kappa^0 = 1$. □

Proposition 14.8.6 *Let $K \subset X$ be compact, and suppose that $u = T_K^*$ is bounded. Then the measure ω_u^n is carried by K. Moreover, with $S = \max\{1, \sup_X u\}$,*

$$0 < \int_X \omega_u^n \le S^n c_\omega(K). \tag{14.46}$$

Proof Steps 1–4 show that ω_u^n vanishes on $X \setminus K$: on small balls outside K we compare u with a homogeneous balayage computed with the coefficients of ω frozen. Step 5 proves the mass bound.

Step 1. Increasing competitors. The competitors for T_K are bounded above by $\sup_X u$, and $u \in \mathrm{PSH}(X, \omega)$ by Lemma 14.8.1(1). Cover X by finitely many coordinate charts. Applying Choquet's lemma (Lemma 3.4.1) in each chart, we obtain for each chart a countable family of competitors whose supremum has, on that chart, the same upper semicontinuous regularization as T_K. Let $v_1, v_2, \ldots$ be an enumeration of the union of these families. On each chart, $\sup_k v_k$ lies between the supremum of the family chosen for that chart and T_K; since the regularization is computed locally, $(\sup_k v_k)^* = u$. The functions $u_j = \max\{0, v_1, \ldots, v_j\}$ belong to $\mathrm{PSH}(X, \omega)$ by Lemma 14.8.1(1), increase with j, vanish on K, and are competitors; hence $0 \le u_j \le T_K \le u$, and $(\sup_j u_j)^* = u$, because $\sup_k v_k \le \sup_j u_j \le T_K$.

Step 2. Frozen balayage. Let $G \subset X \setminus K$ be the domain of a coordinate chart, identified with an open subset of $\mathbb{C}^n$, and let $L \subset G$ be compact. Since $X \setminus K$ is open in the compact manifold X, it is a countable union of such sets L, so it suffices to show that $\omega_u^n(L) = 0$. Choose open sets $G_2 \Subset G_1 \Subset G_0 \Subset G$ with $L \subset G_2$. For $a \in G_2$ let h_a be the real quadratic polynomial with $\mathrm{dd^c} h_a = \omega(a)$ described at the beginning of this section. Choose $r_0 > 0$ with $2r_0 < \mathrm{dist}(G_2, \mathbb{C}^n \setminus G_1)$. The coefficients of ω have bounded first derivatives on G_1, so by the mean value inequality on balls there is C_1 such that, for $a \in G_2$ and $0 < r \le r_0$, the ball $B(a, 2r)$ lies in G_1 and

$$-\delta\beta \le \omega - \omega(a) \le \delta\beta \quad \text{on } B(a, 2r), \qquad \delta = C_1 r;$$

indeed a Hermitian matrix whose entries are bounded by τ lies between $-n\tau$ and $n\tau$ times the identity.

Fix such a and r, put $q = \pi|z - a|^2$, so that $\mathrm{dd^c} q = \beta$ by (7.5), and let $B = B(a, r)$. The functions

$$U_j = u_j + h_a + \delta q$$

are bounded and plurisubharmonic on $B(a, 2r)$, since they are quasi-plurisubharmonic and $\mathrm{dd^c} U_j = \omega_{u_j} + (\omega(a) - \omega + \delta\beta) \ge 0$. Let V_j be the homogeneous balayage of U_j in B given by Corollary 9.6.4, applied on $B(a, 2r)$: it equals U_j outside B, satisfies $V_j \ge U_j$ and $(\mathrm{dd^c} V_j)^n = 0$ in B, and its restriction to B, being a member of the family described there, has upper boundary limits at most U_j on ∂B.

We compare V_j with u. Put $\rho = q - \pi r^2$, which is nonpositive on B and vanishes on ∂B, and let $Z_j = V_j - h_a - \delta q + 2\delta\rho$ on B. Then Z_j is quasi-plurisubharmonic and

$$\omega + \mathrm{dd^c} Z_j = \mathrm{dd^c} V_j + (\omega - \omega(a) + \delta\beta) \geq 0,$$

so $Z_j \in \mathrm{PSH}(B,\omega)$, and for $x \in \partial B$ we have $\overline{\lim}_{B \ni y \to x} Z_j(y) \leq U_j(x) - h_a(x) - \delta q(x) = u_j(x)$. By Lemma 14.8.1(1), the function equal to $\max\{u_j, Z_j\}$ on B and to u_j on $X \setminus B$ belongs to $\mathrm{PSH}(X,\omega)$. It vanishes on K, because $B \cap K = \varnothing$, so it is a competitor for T_K; hence $\max\{u_j, Z_j\} \leq T_K \leq u$ on B. Therefore

$$U_j \leq V_j \leq u + h_a + \delta q - 2\delta\rho \quad \text{on } B.$$

The balayage is increasing in the data, so V_j increases with j. Let V be the upper semicontinuous regularization of $\lim_j V_j$ on B. Since $U_1 \leq V_j \leq u + h_a + \delta q + 2\pi\delta r^2$, the V_j are uniformly bounded, V is bounded and plurisubharmonic, and $V_j \to V$ almost everywhere by Theorem 8.3.2; hence $(\mathrm{dd^c} V)^n = 0$ on B by Theorem 9.5.1. Put $U = u + h_a + \delta q$. Since $\sup_j U_j = \sup_j u_j + h_a + \delta q$, $h_a + \delta q$ is continuous and $(\sup_j u_j)^* = u$, upper regularization gives $V \geq (\sup_j U_j)^* = U$. The upper bound for V_j is upper semicontinuous, so it also holds for V. As $-2\delta\rho \leq 2\pi\delta r^2$ on B, we conclude

$$0 \leq V - U \leq 2\pi\delta r^2 \quad \text{on } B. \tag{14.47}$$

Step 3. A local mass estimate. Put $A = \mathrm{dd^c} U$ and $D = \mathrm{dd^c} V$ on B. The function $g = h_a + \delta q$ satisfies $\mathrm{dd^c} g = \omega(a) + \delta\beta \geq \omega$ on $B(a,2r)$, so Lemma 14.8.1(2) gives $A^n = (\mathrm{dd^c}(u+g))^n \geq \omega_u^n$ on B. By (14.45), with a fixed function η equal to 1 on $(-\infty, 1/16]$ and to 0 on $[1/5, \infty)$, there is $\chi \in C_c^\infty(B(a,r/2))$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $B(a,r/4)$ and $-C_2 r^{-2}\beta \leq \mathrm{dd^c}\chi \leq C_2 r^{-2}\beta$, where C_2 does not depend on a and r. By the definition and the symmetry of Bedford–Taylor products, $A^n - D^n = \sum_{j=0}^{n-1} (A-D) \wedge A^j \wedge D^{n-1-j} = \mathrm{dd^c}\big((U-V)\sum_{j=0}^{n-1} A^j \wedge D^{n-1-j}\big)$. Using $D^n = 0$, integration by parts against χ, the positivity of the currents $A^j \wedge D^{n-1-j}$ and (14.47), we obtain

$$\int_{B(a,r/4)} \omega_u^n \leq \int \chi A^n = \int (U-V)\,\mathrm{dd^c}\chi \wedge \sum_{j=0}^{n-1} A^j \wedge D^{n-1-j}$$

$$\leq 2\pi C_2 \delta \int_{B(a,r/2)} \sum_{j=0}^{n-1} A^j \wedge D^{n-1-j} \wedge \beta.$$

Apply Lemma 14.8.5 with $k = n-1$ and $\kappa = 2\pi\delta = 2\pi C_1 r$; for $r \leq 1/(2\pi C_1)$ we have $\kappa \leq 1$, and the last integral is at most $C_n \sum_{l=0}^{n-1} \int_{B(a,r)} A^l \wedge \beta^{n-l}$.

To sum these estimates we replace A by a current that does not depend on a and r. Choose C_0 such that $C_0\beta \geq \omega$ on G_0 and $C_0\beta \geq \omega(a) + \delta\beta$ for all $a \in G_2$ and $r \leq r_0$. Then $\Gamma = \mathrm{dd^c} u + C_0\beta = \mathrm{dd^c}(u + C_0\pi|z|^2)$, where $u + C_0\pi|z|^2$ is bounded and plurisubharmonic on G_0 because $\mathrm{dd^c}(u + C_0\pi|z|^2) = \omega_u + (C_0\beta - \omega) \geq 0$. On B the difference $\Gamma - A$ is $\mathrm{dd^c}$ of the smooth function $C_0\pi|z|^2 - h_a - \delta q$, and it equals the form $\gamma = C_0\beta - \omega(a) - \delta\beta \geq 0$ with constant coefficients. By (14.36), $\Gamma^l = \sum_{i=0}^{l} \binom{l}{i} A^i \wedge \gamma^{l-i}$, and all terms are positive currents, so $A^l \wedge \beta^{n-l} \leq \Gamma^l \wedge \beta^{n-l}$. Altogether, for $a \in G_2$ and $r \leq \min\{r_0, 1/(2\pi C_1)\}$,

$$\int_{B(a,r/4)} \omega_u^n \le C_3\, r \sum_{l=0}^{n-1} \int_{B(a,r)} \Gamma^l \wedge \beta^{n-l},$$

with $C_3 = 2\pi C_1 C_2 C_n$ independent of a and r.

Step 4. Zero mass off K. Let r be as above and also smaller than $4\,\mathrm{dist}(L, \mathbb{C}^n \setminus G_2)$, and consider the cubical grid $h\mathbb{Z}^{2n} \subset \mathbb{C}^n = \mathbb{R}^{2n}$ of mesh $h = r/(4\sqrt{2n})$. The grid points a within distance $r/4$ of L are finitely many and belong to G_2. Every point of $\mathbb{C}^n$ lies within distance $\sqrt{2n}\,h/2 = r/8$ of a grid point, so the balls $B(a, r/4)$ centered at these grid points cover L. The centers of the balls $B(a,r)$ containing a given point x lie in the cube of side $2r$ centered at x, which contains at most $(2r/h+1)^{2n} = (8\sqrt{2n}+1)^{2n} = N_n$ grid points; so each point lies in at most N_n of these balls. Summing the local estimate over these balls, which lie in G_1, gives

$$\omega_u^n(L) \le C_3 N_n\, r \sum_{l=0}^{n-1} \int_{\overline{G_1}} \Gamma^l \wedge \beta^{n-l}.$$

The integrals on the right are finite by Theorem 9.1.3, applied in G_0, and do not depend on r. Letting $r \downarrow 0$ gives $\omega_u^n(L) = 0$. Hence ω_u^n is carried by K.

Step 5. The mass bound. Since $T_K \ge 0$, we have $0 \le u \le S$. The function $v = u/S$ satisfies $0 \le v \le 1$ and belongs to $\mathrm{PSH}(X,\omega)$ because $S \ge 1$: indeed $\omega + \mathrm{dd^c} v = S^{-1}\omega_u + (1 - S^{-1})\omega \ge 0$. Since $v = S^{-1}u + (1 - S^{-1}) \cdot 0$ and the products are affine in each potential, expanding ω_v^n factor by factor gives $\omega_v^n = \sum_{k=0}^n \binom{n}{k} S^{-k}(1 - S^{-1})^{n-k} \omega_u^k \wedge \omega^{n-k}$, and all terms are positive measures; hence $\omega_v^n \ge S^{-n}\omega_u^n$. As ω_u^n is carried by K,

$$\int_X \omega_u^n = \int_K \omega_u^n \le S^n \int_K \omega_v^n \le S^n c_\omega(K).$$

Finally $\int_X \omega_u^n > 0$ by Proposition 14.8.3. □

Theorem 14.8.7 *Let X be a compact complex manifold, and let $A \subset X$ be pluripolar (Definition 10.1.1). Then there is a quasi-plurisubharmonic function Φ on X with $A \subset \{\Phi = -\infty\}$; that is, A is globally pluripolar.*

Proof The compact manifold X has finitely many connected components, which are open, and it suffices to construct Φ on each of them, since the intersection of A with a component is pluripolar in that component. On a component of dimension zero, which is a point, pluripolar sets are empty and $\Phi = 0$ works. We may therefore assume that X is connected of dimension $n \ge 1$, and we fix a Hermitian form ω on X (one exists by Section 7.5), so that the notation of this section applies.

Every point of X has a coordinate ball U and a plurisubharmonic function on U whose pole set contains $A \cap U$, by Definition 10.1.1; choose a smaller ball $U_1 \Subset U$ around the point. By compactness finitely many of the balls U_1 cover X. By the last assertion of Lemma 14.8.4 it therefore suffices to prove the following: if U is a coordinate ball, φ is plurisubharmonic on U and $U_1 \Subset U$ is a smaller ball, then $E = \{\varphi = -\infty\} \cap U_1$ is globally pluripolar.

If E is empty, there is nothing to prove, so let $E \neq \varnothing$. Choose balls $U_1 \Subset U_2 \Subset U_3 \Subset U$ and subtract a constant from φ so that $\varphi \le 0$ on U_3. Suppose, to reach a contradiction, that E is not globally pluripolar. By Lemma 14.8.4, there is a constant C with $0 \le T_E^* \le C$. From the extremal functions of compact sublevel sets of regularizations of φ, which have small capacity, we shall obtain a bounded function in $\mathrm{PSH}(X,\omega)$ with zero Monge–Ampère mass.

Step 1. Sublevel sets and their extremal functions. Convolution with radial mollifiers of radii decreasing to zero and smaller than the distance from U_2 to ∂U_3 gives smooth plurisubharmonic functions φ_j on U_2 that decrease to φ as $j \to \infty$, by Proposition 8.2.2(2). Since $\varphi \le 0$ on U_3, we have $\varphi \le \varphi_j \le 0$ on U_2, so $\|\varphi_j\|_{L^1(U_2)} \le \|\varphi\|_{L^1(U_2)} < \infty$. For integers $N, j \ge 1$ put

$$K_{j,N} = \{\varphi_j \le -N\} \cap \overline{U_1}, \qquad O_{j,N} = \{\varphi_j < -N\} \cap U_1.$$

Then $K_{j,N}$ is compact and contains the open set $O_{j,N}$. Fix $x_0 \in E$. Since $\varphi_j(x_0) \downarrow -\infty$, there is $j_0(N)$ such that $x_0 \in O_{j,N}$ for $j \ge j_0(N)$; we choose $j_0(N)$ nondecreasing in N. For these j the set $K_{j,N}$ has nonempty interior, so $u_{j,N} = T^*_{K_{j,N}}$ is a bounded nonnegative function in $\mathrm{PSH}(X,\omega)$ by Lemma 14.8.4. Every competitor for $T_{K_{j,N}}$ is nonpositive on the open set $O_{j,N}$; hence so is $T_{K_{j,N}}$, and its upper semicontinuous regularization satisfies $u_{j,N} = 0$ on $O_{j,N}$.

For fixed N, the sets $K_{j,N}$ increase with j, because φ_j decreases. A competitor for $T_{K_{j+1,N}}$ is a competitor for $T_{K_{j,N}}$, so $u_{j,N}$ decreases with $j \ge j_0(N)$; let u_N be its limit. By Lemma 14.8.1(3), $u_N \in \mathrm{PSH}(X,\omega)$, and $0 \le u_N \le u_{j_0(N),N}$ is bounded. If $x \in U_1$ and $\varphi(x) < -N$, then $x \in O_{j,N}$ for all large j, so $u_N(x) = 0$. In particular $u_N = 0$ on E, so u_N is a competitor for T_E, and

$$0 \le u_N \le T_E^* \le C.$$

For fixed j, the sets $K_{j,N}$ decrease as N increases, so $u_{j,N} \le u_{j,N+1}$ for $j \ge j_0(N+1)$; letting $j \to \infty$ gives $u_N \le u_{N+1}$. Let u_∞ be the upper semicontinuous regularization of $\lim_N u_N$. By Lemma 14.8.1(1), $u_\infty \in \mathrm{PSH}(X,\omega)$ and $0 \le u_\infty \le C$.

Step 2. Convergence of the masses. By Lemma 14.8.1(3), applied to the uniformly bounded decreasing sequence $(u_{j,N})_{j \ge j_0(N)}$ and to the uniformly bounded increasing sequence $(u_N)_{N \ge 1}$, and testing the weak convergence with the constant function 1 on the compact manifold X, we obtain

$$\int_X \omega_{u_{j,N}}^n \longrightarrow \int_X \omega_{u_N}^n \quad (j \to \infty), \qquad \int_X \omega_{u_N}^n \longrightarrow \int_X \omega_{u_\infty}^n \quad (N \to \infty).$$

Moreover $\int_X \omega_{u_\infty}^n > 0$ by Proposition 14.8.3, since u_∞ is bounded.

Step 3. The capacity of $K_{j,N}$. Choose a smooth function h on U_2 with $\mathrm{dd^c} h \ge \omega$ and $\sup_{U_2} |h| < \infty$, for instance a large multiple of $|z|^2$. If $v \in \mathrm{PSH}(X,\omega)$ and $0 \le v \le 1$, then $v + h$ is plurisubharmonic on U_2 with $|v + h| \le 1 + \sup_{U_2} |h|$, and $\omega_v^n \le (\mathrm{dd^c}(v+h))^n$ on U_2 by Lemma 14.8.1(2). Since $K_{j,N} \subset \overline{U_1}$ and $-\varphi_j \ge N$ on $K_{j,N}$, while $-\varphi_j \ge 0$ on U_2, the weighted Chern–Levine–Nirenberg inequality

(Theorem 9.3.1), applied in U_2 to φ_j, the compact set $\overline{U_1}$ and a compact neighborhood $L \subset U_2$ of $\overline{U_1}$, gives

$$\int_{K_{j,N}} \omega_v^n \le \frac{1}{N} \int_{\overline{U_1}} (-\varphi_j)\, (\mathrm{dd^c}(v+h))^n \le \frac{C'}{N}.$$

The constant C' depends on $\|\varphi_j\|_{L^1(L)} \le \|\varphi\|_{L^1(U_2)}$ and on the bound for $|v+h|$, but not on v, j or N. Taking the supremum over v, we obtain $c_\omega(K_{j,N}) \le C'/N$.

Step 4. Conclusion. By Proposition 14.8.6,

$$\int_X \omega_{u_{j,N}}^n \le \max\{1, \sup_X u_{j,N}\}^n \frac{C'}{N} \qquad (j \ge j_0(N)).$$

For fixed N, the uniformly bounded functions $u_{j,N}$ decrease to u_N, so $u_{j,N} \to u_N$ in $L^1(X)$ by dominated convergence, and $\sup_X u_{j,N} \to \sup_X u_N$ by Lemma 14.8.1(4). Letting $j \to \infty$ and using $u_N \le C$, we obtain $\int_X \omega_{u_N}^n \le \max\{1, C\}^n C'/N$. Letting $N \to \infty$ gives $\int_X \omega_{u_\infty}^n = 0$, which contradicts Step 2. Hence E is globally pluripolar, and the theorem is proved. □

Chapter 15
Non-pluripolar products

In Chapter 9 we multiplied the currents $\mathrm{dd^c} u_1, \dots, \mathrm{dd^c} u_p$ of locally bounded plurisubharmonic functions by the Bedford–Taylor recursion $\mathrm{dd^c} u_1 \wedge T = \mathrm{dd^c}(u_1 T)$. The potentials that arise in global geometry are often unbounded: by Corollary 14.1.9, a closed positive $(1,1)$-current on a compact Kähler manifold is $\theta + \mathrm{dd^c}\varphi$ with θ a smooth form and φ quasi-plurisubharmonic, and $\varphi = -\infty$ wherever the current has a positive Lelong number, for instance along a hypersurface carrying a current of integration. Indeed, at a point where φ is finite, the spherical means of a local potential are bounded below by its value at that point, so the Lelong number vanishes there by (12.5). For such potentials the recursion breaks down, since it multiplies the measure coefficients of a current by an unbounded function. In this chapter we construct a product for arbitrary plurisubharmonic potentials which keeps positivity, closedness, multilinearity and locality, at the price of ignoring everything that happens on pluripolar sets.

The construction truncates the potentials and discards their poles. We replace each u_j by $\max\{u_j, -k\}$, take the Bedford–Taylor product, and keep only its restriction to the set where all $u_j > -k$, on which the truncated potentials coincide with the original ones. Since bounded products are local in the plurifine topology (Theorem 10.4.3), the restrictions obtained for different k are compatible and increase with k. Their limit is the *non-pluripolar product* $\langle \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle$, which by construction gives no mass to pluripolar sets. Two phenomena without counterpart in the bounded theory appear. The limit may have infinite mass on compact sets (Example 15.1.6). And the mass carried by the poles is lost: in degree one, $\langle \mathrm{dd^c} u \rangle$ is the restriction of $\mathrm{dd^c} u$ to $\{u > -\infty\}$ (Proposition 15.3.6), so the current of integration at a point of $\mathbb{P}^1$ has zero non-pluripolar product, and even a potential with zero Lelong numbers and trivial multiplier ideals can have zero non-pluripolar mass (Example 15.3.8). When the local masses are finite, the product is a closed positive current (Theorem 15.2.1).

On a compact Kähler manifold X the local masses are always finite: every tuple of closed positive $(1,1)$-currents has a non-pluripolar product, and its total mass is bounded by a cohomological quantity (Theorem 15.4.3). The total mass is, however, no longer determined by the cohomology classes of the factors; it depends on the singularities of the potentials. The central result of the second half of the chapter

is that it is monotone in the singularities: if θ is a smooth closed real $(1,1)$-form, $\varphi, \psi \in \mathrm{PSH}(X,\theta)$, and ψ is more singular than φ in the sense of Definition 14.2.6, then the mass of φ is at least that of ψ (Theorem 15.7.4). When all potentials are locally bounded outside the closed pole set of a fixed quasi-plurisubharmonic function, this follows from Stokes' formula; arbitrary singularities are reduced to this case by an auxiliary potential on $X \times \mathbb{P}^N$, whose total mass is an average of the masses along a segment of potentials. Along the way we prove that non-pluripolar products are lower semicontinuous under convergence in capacity, and continuous when no mass is lost in the limit (Theorem 15.6.5).

Monotonicity has several consequences that are used in the rest of the book. Mixed masses do not change under bounded perturbations of the potentials, and in a big class the potentials with minimal singularities have the largest masses; these maximal masses define the positive intersection numbers and the volume of pseudo-effective classes. The full-mass classes and the comparison and domination principles at the end of the chapter are basic tools for the envelopes of Chapter 16, and the fiber formula of Section 15.5 is used for relative energy in Chapter 17.

Historical remarks and references for this chapter are collected in Appendix B.15.

15.1 Truncation and plurifine locality

Let X be a complex manifold of dimension $n \geq 1$, let $U \subset X$ be a coordinate domain, and fix an integer $1 \leq p \leq n$. Let $u_1, \dots, u_p$ be plurisubharmonic functions on U, and recall that $\{u_j = -\infty\}$ is the pole set of u_j. For each integer $k \geq 1$ put

$$u_{j,k} = \max\{u_j, -k\}, \qquad O_k = \bigcap_{j=1}^{p} \{u_j > -k\},$$

$$S_k = \mathbf{1}_{O_k} \mathrm{dd^c} u_{1,k} \wedge \cdots \wedge \mathrm{dd^c} u_{p,k}.$$

The functions $u_{j,k}$ are locally bounded and plurisubharmonic, so the product in S_k is the Bedford–Taylor product of Definition 9.1.1, and multiplication by $\mathbf{1}_{O_k}$ restricts its coefficient measures to O_k, where every $u_{j,k}$ coincides with u_j. By Proposition 10.4.4, the trace measures of the currents S_k increase with k to a Borel measure μ for which every pluripolar subset of U has outer measure zero, but μ may be infinite on compact sets (Example 15.1.6 below).

The set O_k is Borel, because plurisubharmonic functions are upper semicontinuous, and it is plurifine open, because they are plurifine continuous; it need not be open in the ordinary topology. For instance, the countable set A of points with rational coordinates in the unit disc $\Delta \subset \mathbb{C}$ is polar, each point a being the pole of $\log|z-a|$, so by Proposition 4.2.3 there is a finite positive measure ν with compact support such that the fundamental potential $E_2 * \nu$ equals $-\infty$ on A. The function $u_1(z) = (E_2 * \nu)(z_1)$ is plurisubharmonic on $\mathbb{C}^n$ by Propositions 3.2.1 and 8.2.3, and its pole set contains the dense subset $A \times \mathbb{C}^{n-1}$ of $\Delta \times \mathbb{C}^{n-1}$. For $p = 1$ and large k, the set $O_k \cap (\Delta \times \mathbb{C}^{n-1})$

is nonempty, because $E_2 * \nu$ is locally integrable, but it has empty interior, so O_k is not open. We shall often restrict positive currents to Borel sets such as O_k, and we first note that this preserves positivity.

Lemma 15.1.1 *Let R be a positive current of bidegree (p,p) on an open set $W \subset \mathbb{C}^n$, and let $E \subset W$ be a Borel set. Then the current $\mathbf{1}_E R$, obtained by multiplying the coefficient measures of R by $\mathbf{1}_E$, is positive. If R is strongly positive, that is, $R(\varphi) \geq 0$ for every test form φ that is weakly positive at every point, then so is $\mathbf{1}_E R$.*

Proof By Theorem 7.2.5, R has measure coefficients. Let φ be a test form of bidegree $(n-p, n-p)$ that is strongly positive, respectively weakly positive, at every point. The complex measure $R \wedge \varphi$, whose integral against a test function χ is $R(\chi\varphi)$, defines a positive distribution, because $\chi\varphi$ has the same positivity as φ whenever $\chi \geq 0$; by the uniqueness in Theorem 1.4.5, it is a positive measure. Hence $(\mathbf{1}_E R)(\varphi) = (R \wedge \varphi)(E) \geq 0$. □

The next lemma shows that local finiteness of μ is exactly what is needed for the currents S_k to converge.

Lemma 15.1.2 *For $l \geq k$ the current $S_l - S_k$ is positive, and $\mathbf{1}_{O_k} S_l = S_k$. The currents S_k converge weakly to a positive current S if and only if*

$$\sup_k \int_K S_k \wedge \beta^{n-p} < \infty \quad \text{for every compact set } K \subset U, \tag{15.1}$$

where β is the Euclidean Kähler form (7.4) of the coordinates. In that case $\mathbf{1}_{O_k} S = S_k$ for every k, the coefficient measures of S_k converge to those of S in total variation on compact subsets of U, and the trace measure of S is the measure μ above. Every pluripolar subset of U has outer measure zero for μ and for the total variations of the coefficient measures of S; consequently every plurifine open subset of U is measurable for the completions of these measures. Moreover, if $E_k \subset O_k$ are Borel sets such that $\mathbf{1}_{E_k} \to 1$ almost everywhere for the trace measure of S, then the coefficient measures of $\mathbf{1}_{E_k} S_k$ converge to those of S when integrated against any bounded Borel function with compact support in U. This is the case, in particular, if the sets E_k increase and $U \setminus \bigcup_k E_k$ is pluripolar.

Proof Let $l \geq k$. By Proposition 10.4.4, the level-l and level-k Bedford–Taylor products have the same restriction to O_k; since $O_k \subset O_l$, this is the identity $\mathbf{1}_{O_k} S_l = S_k$. Consequently $S_l - S_k = \mathbf{1}_{O_l \setminus O_k} \mathrm{dd^c} u_{1,l} \wedge \cdots \wedge \mathrm{dd^c} u_{p,l}$, and this current is positive by Lemma 15.1.1.

Suppose that (15.1) holds. The trace measures $S_k \wedge \beta^{n-p}$ increase with k and, by (15.1), converge to a locally finite measure. Hence the trace mass of the positive current $S_l - S_k$ on a compact set K tends to zero as $k, l \to \infty$. By Theorem 7.2.5, the total variation on K of each coefficient measure of $S_l - S_k$ is at most a fixed multiple of this trace mass. Therefore each coefficient measure of S_k is a Cauchy sequence in total variation on compact sets, and the limits are the coefficients of a current S, which is positive as a weak limit of positive currents. Conversely, suppose

that $S_k \to S$ weakly for a positive current S. For every smooth compactly supported $\chi \geq 0$, the numbers $\int \chi S_k \wedge \beta^{n-p}$ increase with k and converge to $\int \chi S \wedge \beta^{n-p}$, which therefore bounds them; choosing $\chi = 1$ on K gives (15.1).

Assume now that (15.1) holds. Letting $l \to \infty$ in $\mathbf{1}_{O_k} S_l = S_k$ gives $\mathbf{1}_{O_k} S = S_k$. By the convergence in total variation, the trace measure of S is the increasing limit of the trace measures of the S_k, which is μ. By Proposition 10.4.4, every pluripolar subset of U has outer μ-measure zero. The total variation of each coefficient measure of S is at most $C_{n,p}\mu$ (Theorem 7.2.5), so pluripolar sets have outer measure zero for these total variations as well, and Theorem 10.4.1(2), applied to μ and to these total variations, shows that plurifine open subsets of U are measurable for their completions. In particular S vanishes on $U \setminus \bigcup_k O_k$, the union of the pole sets of the u_j, which is a pluripolar Borel set.

Finally, let E_k be as in the statement. Since $E_k \subset O_k$, we have $\mathbf{1}_{E_k} S_k = \mathbf{1}_{E_k} S$. By Theorem 7.2.5, the total variation of each coefficient measure of S is locally finite and absolutely continuous with respect to the trace measure. Hence $\mathbf{1}_{E_k} \to 1$ almost everywhere for each of these variations, and dominated convergence proves the assertion. If the E_k increase and $U \setminus \bigcup_k E_k$ is pluripolar, then $\mathbf{1}_{E_k} \to 1$ outside this pluripolar Borel set, which S does not charge. □

Definition 15.1.3 If (15.1) holds, we say that the non-pluripolar product of $u_1, \dots, u_p$ is *locally finite*, and we call the limit S of Lemma 15.1.2 the *non-pluripolar product*

$$\langle \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle = S.$$

The angle brackets distinguish this construction from a product that retains mass on pluripolar sets. The empty product ($p = 0$) is 1.

If all u_j are locally bounded, then every relatively compact open subset of U is contained in O_k for large k. In this case the non-pluripolar product is locally finite and coincides with the Bedford–Taylor product, and we omit the brackets.

Proposition 15.1.4 *Every locally finite non-pluripolar product S is strongly positive: $S(\varphi) \geq 0$ for every compactly supported smooth weakly positive form φ of bidegree $(n-p, n-p)$.*

Proof First let all potentials be locally bounded. Near each point they are limits of decreasing sequences of smooth plurisubharmonic functions, for instance of their mollifications (Proposition 8.2.2(2)). The complex Hessians of these functions are semipositive $(1,1)$-forms (Proposition 8.2.2(1)), hence strongly positive, since the two notions of positivity coincide in bidegree $(1,1)$; by Proposition 7.2.1, their exterior products are strongly positive, and the product of such a product with a weakly positive form of complementary bidegree is a nonnegative multiple of the volume form. Thus the smooth products pair nonnegatively with every smooth compactly supported weakly positive form. By Theorem 9.2.5, the Bedford–Taylor product is their weak limit, and therefore it pairs nonnegatively with these forms as well.

By Lemma 15.1.1, applied to the bounded products in the definition of S_k, every S_k is strongly positive. Since $S_k \to S$ weakly by Lemma 15.1.2, so is S. □

Proposition 15.1.5 *Let $u_1, \dots, u_p$ and $v_1, \dots, v_p$ be plurisubharmonic on U. If $\mathrm{dd^c} u_j = \mathrm{dd^c} v_j$ for every j, then the non-pluripolar product of the u_j is locally finite if and only if that of the v_j is, and then the two products are equal; thus local finiteness and the product depend only on the currents $\mathrm{dd^c} u_j$. The product is symmetric in its factors and local in the plurifine topology: if $u_j = v_j$ on a plurifine open set O for every j and both products are locally finite, then they have the same restriction to O. Finally, the product is multilinear: if u_1' is plurisubharmonic on U, then the product of $u_1 + u_1', u_2, \dots, u_p$ is locally finite if and only if the products of $u_1, u_2, \dots, u_p$ and of $u_1', u_2, \dots, u_p$ are, and in that case*

$$\langle \mathrm{dd^c}(u_1 + u_1') \wedge \mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle = \langle \mathrm{dd^c} u_1 \wedge \mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle + \langle \mathrm{dd^c} u_1' \wedge \mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle.$$

Similarly, for real $c > 0$ the product of $cu_1, u_2, \dots, u_p$ is locally finite if and only if that of $u_1, u_2, \dots, u_p$ is, and then it is c times the latter.

Proof In this proof we write $P^{(m)}(w_1, \dots, w_p) = \mathrm{dd^c} \max\{w_1, -m\} \wedge \cdots \wedge \mathrm{dd^c} \max\{w_p, -m\}$ for plurisubharmonic $w_1, \dots, w_p$ and an integer $m \geq 1$.

Symmetry and locality. Each truncated product S_k is symmetric in its factors, hence so is the limit. Suppose that $u_j = v_j$ on the plurifine open set O for every j, and that both products are locally finite. By Lemma 15.1.2, plurifine open subsets of U are measurable for the completions of the total variations of the coefficient measures of both products, so the restrictions of the products to O and to the plurifine open sets $W_k = O \cap \bigcap_j \{u_j > -k\} = O \cap \bigcap_j \{v_j > -k\}$ are well defined; the sets W_k increase with k. On W_k the truncations $\max\{u_j, -k\}$ and $\max\{v_j, -k\}$ coincide, so the two level-k products agree on W_k by Theorem 10.4.3. Since W_k is contained in the level-k sets of both tuples, Lemma 15.1.2 shows that the two non-pluripolar products also agree on W_k. The union of the W_k is O minus the union of the pole sets of the u_j, a pluripolar set, which has outer measure zero for these total variations by the same lemma. Letting $k \to \infty$, we conclude that the two products have the same restriction to O.

Dependence on the currents $\mathrm{dd^c} u_j$. Suppose that $\mathrm{dd^c} u_j = \mathrm{dd^c} v_j$ for every j. The distribution $v_j - u_j$ then has zero Laplacian, so by Weyl's lemma (Theorem 2.1.4) it is a smooth function w_j, and its complex Hessian vanishes: w_j is pluriharmonic. The plurisubharmonic functions v_j and $u_j + w_j$ agree almost everywhere, hence everywhere, since both are subharmonic (Theorems 3.2.2 and 8.2.1). Fix a relatively compact open set $V \subset U$ and an integer C such that $|w_j| \leq C$ on V for all j. Write $O_k(u), S_k(u)$ and $O_k(v), S_k(v)$ for the level sets and truncated products of the two tuples. On $V \cap O_k(v)$ we have $u_j = v_j - w_j > -k - C$, hence

$$\max\{v_j, -k\} = v_j = \max\{u_j, -k - C\} + w_j.$$

The functions $\max\{v_j, -k\}$ and $\max\{u_j, -k - C\} + w_j$ are bounded and plurisubharmonic on V and coincide on $V \cap O_k(v)$. Moreover, adding the pluriharmonic function w_j to a factor does not change Bedford–Taylor products. Indeed, by additivity (Proposition 9.1.2) and symmetry (Theorem 9.2.5), the difference is a sum of products

having $\mathrm{dd^c} w_j$ as first factor; by Proposition 9.1.2(1), such a product is the exterior product of the remaining closed positive current with the smooth form $\mathrm{dd^c} w_j = 0$. As $V \cap O_k(v) \subset O_{k+C}(u)$, plurifine locality gives on V

$$S_k(v) = \mathbf{1}_{O_k(v)} P^{(k+C)}(u_1, \dots, u_p) = \mathbf{1}_{O_k(v)} S_{k+C}(u) \le S_{k+C}(u).$$

Thus (15.1) for the u_j implies (15.1) for the v_j on compact subsets of V, and exchanging the two tuples (that is, replacing w_j by $-w_j$) gives the converse. If both products are locally finite, the display and Lemma 15.1.2 give $\mathbf{1}_{O_k(v)} \langle \mathrm{dd^c} v_1 \wedge \cdots \wedge \mathrm{dd^c} v_p \rangle = \mathbf{1}_{O_k(v)} \langle \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle$ on V for every k. The sets $O_k(v)$ exhaust the complement of the union of the pole sets, which are the same for the two tuples and carry no mass for either product. Hence the two products coincide.

Additivity. Let u_1' be plurisubharmonic and put

$$E_k = \{u_1 > -k\} \cap \{u_1' > -k\} \cap \bigcap_{j=2}^{p} \{u_j > -k\}.$$

These plurifine open sets increase with k and exhaust the complement of the union of the pole sets of $u_1, u_1', u_2, \dots, u_p$. On E_k we have $\max\{u_1 + u_1', -2k\} = u_1 + u_1' = \max\{u_1, -k\} + \max\{u_1', -k\}$ and $\max\{u_j, -2k\} = \max\{u_j, -k\}$ for $j \ge 2$. Plurifine locality and the additivity of bounded products (Proposition 9.1.2) therefore give

$$\begin{aligned} \mathbf{1}_{E_k} P^{(2k)}(u_1 + u_1', u_2, \dots, u_p) = {} & \mathbf{1}_{E_k} P^{(k)}(u_1, u_2, \dots, u_p) \\ & + \mathbf{1}_{E_k} P^{(k)}(u_1', u_2, \dots, u_p). \end{aligned} \tag{15.2}$$

Before local finiteness is known we argue with trace measures. By Proposition 10.4.4, for each of the three tuples the trace measures of the truncated products increase to a Borel measure, possibly infinite on compact sets, for which every pluripolar set has outer measure zero; for a locally finite product it is the trace measure of the product. The set E_k is contained in the level-$2k$ set of the first tuple and in the level-k sets of the other two, so by the identity $\mathbf{1}_{O_k} S_l = S_k$ of Lemma 15.1.2, the restriction of each limit measure to E_k is the restriction of the corresponding trace measure in (15.2). Hence, on E_k, the limit measure of the first tuple is the sum of those of the other two. Letting $k \to \infty$, monotone convergence gives this equality on every Borel subset of $\bigcup_k E_k$, and hence everywhere, because the complement of $\bigcup_k E_k$ is pluripolar. Since all three measures are positive, the first is locally finite if and only if the other two are; this is the asserted equivalence of local finiteness.

When the products are locally finite, (15.2) and Lemma 15.1.2 give the additivity identity on each E_k, and letting $k \to \infty$ proves it. For $c > 0$ the same argument applies to the sets $\{u_1 > -k\} \cap \{cu_1 > -k\} \cap \bigcap_{j \ge 2} \{u_j > -k\}$, on which $\max\{cu_1, -k\} = c \max\{u_1, -k\}$, and proves the homogeneity assertion. □

Let now $T_1, \dots, T_p$ be closed positive $(1,1)$-currents on the complex manifold X. By Theorem 7.3.2, every point of X has a coordinate neighborhood on which $T_j = \mathrm{dd^c} u_j$ with u_j plurisubharmonic. By Proposition 15.1.5, local finiteness and the

local non-pluripolar product do not depend on the choice of these potentials, and the construction is compatible with restriction to smaller open sets.

They do not depend on the coordinates either. Indeed, let F be a biholomorphism between coordinate domains. Then $\max\{u_j, -k\} \circ F = \max\{u_j \circ F, -k\}$ and $O_k(u \circ F) = F^{-1}(O_k(u))$. Since F is a submersion, the pullback F^* of Proposition 7.5.2 satisfies $F^*(\mathbf{1}_E S) = \mathbf{1}_{F^{-1}(E)} F^* S$ for every current S of order zero and every Borel set E (Corollary 7.5.3(2)), so Proposition 9.1.2(5) gives $F^* S_k(u) = S_k(u \circ F)$ for the truncated products. To compare trace measures, note that if α and α' are smooth semipositive $(1,1)$-forms on an open set with $\alpha' \le c\alpha$, then

$$(c\alpha)^{n-p} - \alpha'^{n-p} = (c\alpha - \alpha') \wedge \sum_{i=0}^{n-p-1} (c\alpha)^i \wedge \alpha'^{n-p-1-i}$$

is a sum of products of semipositive $(1,1)$-forms, hence strongly positive (Proposition 7.2.1). Since the product of a positive current with a smooth strongly positive form of complementary bidegree is a positive measure, it follows that

$$R \wedge \alpha'^{n-p} \le c^{n-p} R \wedge \alpha^{n-p} \tag{15.3}$$

on that open set, for every positive current R of bidegree (p,p). On a neighborhood of a compact set, the Euclidean forms β and β' of the two coordinate systems satisfy $\beta' \le c\beta$ for some c (Lemma 13.1.1), so by (15.3) the trace measures of a positive current in the two coordinate systems are comparable on compact sets, and local finiteness does not depend on the coordinates. Since F^* is weakly continuous, $F^*\langle \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle = \langle \mathrm{dd^c}(u_1 \circ F) \wedge \cdots \wedge \mathrm{dd^c}(u_p \circ F) \rangle$.

We say that the non-pluripolar product of $T_1, \dots, T_p$ is *locally finite* if this holds near every point; the local products then glue to a positive current $\langle T_1 \wedge \cdots \wedge T_p \rangle$ on X, which is closed by Theorem 15.2.1 below. When X is compact we also say simply that the product is *finite*. Repeated factors are written as powers, as in $\langle T^j \wedge S^{r-j} \rangle$, and for $T_j = \theta_j + \mathrm{dd^c}\varphi_j$ we also write $\langle (\theta_1 + \mathrm{dd^c}\varphi_1) \wedge \cdots \wedge (\theta_p + \mathrm{dd^c}\varphi_p) \rangle$. The assertions of Proposition 15.1.5 carry over to these products, with multilinearity in the factors T_j.

Example 15.1.6 Let $U = \{(x,y) \in \mathbb{C}^2 : |x|^2 < 1/2,\ |y| < \mathrm{e}^{-1}\}$, put $L = -\log|y|$, and define

$$u(x,y) = (|x|^2 - 1)\sqrt{L} \quad (y \neq 0), \qquad u(x,0) = -\infty.$$

On $U \setminus \{y = 0\}$ we have $L > 1$, the function u is smooth, and its complex second derivatives are

$$u_{x\bar{x}} = \sqrt{L}, \qquad u_{y\bar{y}} = \frac{1 - |x|^2}{16|y|^2 L^{3/2}}, \qquad u_{x\bar{y}} = -\frac{\bar{x}}{4\bar{y}\sqrt{L}}, \qquad \det(u_{j\bar{k}}) = \frac{1 - 2|x|^2}{16|y|^2 L}.$$

Since $u_{x\bar{x}} > 0$ and $\det(u_{j\bar{k}}) > 0$, the complex Hessian is positive definite, so u is plurisubharmonic off the axis $\{y = 0\}$ (Proposition 8.2.2(1)). Moreover $u < -\sqrt{L}/2$ is

bounded above and tends to $-\infty$ at the axis. By Theorem 8.4.1, the plurisubharmonic extension across the axis takes the value $\overline{\lim}\, u = -\infty$ there, which is the displayed value; hence u is plurisubharmonic on U.

Off the axis, $(\mathrm{dd^c} u)^2$ is the smooth measure with density a fixed positive multiple of $\det(u_{j\bar{k}})$, by (9.2). The sets $O_k = \{u > -k\}$ are open, since u is continuous as an extended-real function, and they increase to $U \setminus \{y = 0\}$; on O_k the truncation $\max\{u, -k\}$ equals u, so $S_k = \mathbf{1}_{O_k} (\mathrm{dd^c} u)^2$. Take $K = \{|x|^2 \le 1/4,\ |y| \le r\}$ with $0 < r < \mathrm{e}^{-1}$. On K we have $1 - 2|x|^2 \ge 1/2$, and polar coordinates in y give

$$\int_{|y|\le r} \frac{\mathrm{d}A(y)}{|y|^2 L} = 2\pi \int_0^r \frac{\mathrm{d}t}{t(-\log t)} = \infty.$$

Hence $\sup_k \int_K S_k = \int_{K\setminus\{y=0\}} (\mathrm{dd^c} u)^2 = \infty$, and (15.1) fails. Thus the non-pluripolar product of an arbitrary plurisubharmonic function need not be locally finite.

15.2 Closedness and cutoff gradients

The indicators $\mathbf{1}_{O_k}$ cannot be differentiated, so closedness of the non-pluripolar product does not follow directly from closedness of the truncated products. We replace the indicators by bounded plurisubharmonic cutoff functions and control the resulting error with the gradient estimates of Section 9.13. Here $\mathrm{d^c}$ is the real operator of (7.6), so that $\mathrm{dd^c} = \mathrm{dd^c}$.

Theorem 15.2.1 *Every locally finite non-pluripolar product is closed.*

Proof Closedness is a local property, so we may shrink U and subtract constants to assume that $u_j \le 0$ for every j. If $p = n$ the product has top degree and is closed for degree reasons, so let $p < n$. Fix a constant strongly positive $(n-p-1, n-p-1)$-form γ and put

$$\Theta_k = \gamma \wedge \bigwedge_j \mathrm{dd^c} u_{j,k}, \qquad S_\gamma = \gamma \wedge \Big\langle \bigwedge_j \mathrm{dd^c} u_j \Big\rangle.$$

The Θ_k are closed positive currents of bidimension $(1,1)$. By Lemma 15.1.2, $\mathbf{1}_{O_k}\Theta_k = \mathbf{1}_{O_k} S_\gamma$ increases to S_γ, and in particular these restrictions have uniformly bounded mass on compact sets. We shall prove that $\mathrm{d}S_\gamma = 0$.

Cutoff functions. Put $u = \sum_j u_j \le 0$ and $w_k = \max\{1 + u/k, 0\}$. By Proposition 8.2.5, w_k is plurisubharmonic, and $0 \le w_k \le 1$. As $k \to \infty$, w_k increases to 1 at every point where $u > -\infty$, because u/k increases to 0 there, while $w_k = 0$ on the pole set of u. Moreover $w_k = 0$ outside O_k: if $u_j \le -k$ for some j, then $u \le u_j \le -k$, since all $u_i \le 0$. Choose a smooth function ζ on $\mathbb{R}$ with $0 \le \zeta \le 1$, $\zeta = 0$ on $(-\infty, 1/2]$ and $\zeta = 1$ near 1. Then $\zeta(w_k)$ and $\zeta'(w_k)$ vanish outside O_k, so that $\zeta(w_k)\Theta_k = \zeta(w_k) S_\gamma$. At every point where $u > -\infty$ we have $\zeta(w_k) \to 1$ and $\zeta'(w_k) \to 0$, and S_γ gives the pole set of u zero mass. Dominated convergence therefore gives

$$\zeta(w_k)\Theta_k \longrightarrow S_\gamma, \qquad \zeta'(w_k) \longrightarrow 0 \quad S_\gamma\text{-almost everywhere.}$$

Gradient bound. Let η be a compactly supported smooth real one-form, and let $\tau \geq 0$ be a smooth compactly supported function equal to one near $\operatorname{Supp}\eta$. By (9.37), the gradient measure of w_k with respect to Θ_k is $E_{\Theta_k}(w_k) = \frac{1}{2}\mathrm{dd^c}(w_k^2\Theta_k) - w_k\mathrm{dd^c}w_k \wedge \Theta_k$, and the last term is a positive measure because $w_k \geq 0$. Hence

$$\int \tau E_{\Theta_k}(w_k) \leq \tfrac{1}{2}\int w_k^2\mathrm{dd^c}\tau \wedge \Theta_k \leq C\int_{\operatorname{Supp}\tau} \mathbf{1}_{O_k}\Theta_k \wedge \beta \leq C'.$$

The second inequality uses $w_k = 0$ outside O_k, $w_k \leq 1$, and the bound of the coefficients of $\mathrm{dd^c}\tau \wedge \Theta_k$ by a constant times the trace measure (Theorem 7.2.5). The constant C' is independent of k because $\mathbf{1}_{O_k}\Theta_k \leq S_\gamma$ and S_γ is locally finite.

Conclusion. By the chain rule (9.38), $\mathrm{d}(\zeta(w_k)\Theta_k) = \zeta'(w_k)\,\mathrm{d}w_k \wedge \Theta_k$. We apply (9.39) with $a = \zeta'(w_k)$. The constant C_η there depends only on η, so it is the same for every k. The first factor on the right is bounded by the gradient bound, and the second tends to zero, since

$$|\zeta'(w_k)|^2\Theta_k \wedge \beta = |\zeta'(w_k)|^2\mathbf{1}_{O_k}\Theta_k \wedge \beta \leq |\zeta'(w_k)|^2 S_\gamma \wedge \beta$$

and the right side tends to zero on $\operatorname{Supp}\eta$ by dominated convergence. Hence the pairing of $\mathrm{d}(\zeta(w_k)\Theta_k)$ with η tends to zero. Since $\zeta(w_k)\Theta_k \to S_\gamma$, the pairing of $\mathrm{d}S_\gamma$ with η vanishes, so $\mathrm{d}S_\gamma = 0$.

Finally, $\gamma \wedge \mathrm{d}\langle\bigwedge_j \mathrm{dd^c}u_j\rangle = \mathrm{d}S_\gamma = 0$ for every constant strongly positive γ. These forms span the constant real $(n-p-1, n-p-1)$-forms (Proposition 7.2.1), and their products with smooth functions and the coordinate one-forms $\mathrm{d}z_i, \mathrm{d}\bar{z}_i$ span the test forms of the two bidegrees against which $\mathrm{d}\langle\bigwedge_j \mathrm{dd^c}u_j\rangle$ is evaluated. Therefore the non-pluripolar product is closed. □

15.3 Bounded factors and the first-degree product

Let v be locally bounded and plurisubharmonic, and let $S = \langle \mathrm{dd^c}u_2 \wedge \cdots \wedge \mathrm{dd^c}u_p\rangle$ be locally finite. Since S is closed and positive, the recursion of Definition 9.1.1 defines $\mathrm{dd^c}v \wedge S$, and we expect this to be the non-pluripolar product of $v, u_2, \ldots, u_p$. To prove it we must know that $\mathrm{dd^c}v \wedge S$, like S, gives no mass to the pole sets of $u_2, \ldots, u_p$. This follows from a weighted form of the Chern–Levine–Nirenberg inequality. We then identify the non-pluripolar product in degree one, which requires a measurable form of Fubini's theorem for Bedford–Taylor products.

Lemma 15.3.1 *Let T be a closed positive current of bidimension (r,r), $r \geq 1$, on a coordinate domain $U \subset \mathbb{C}^n$, and let K, L be compact sets with $K \Subset \operatorname{Int} L \subset L \Subset U$. Let $V \leq 0$ be plurisubharmonic on U, and let v be locally bounded and plurisubharmonic on U. Then*

$$\int_K (-V)\,\mathrm{dd^c} v \wedge T \wedge \beta^{r-1} \le C_{K,L}\|v\|_{L^\infty(L)} \int_L (-V)T \wedge \beta^r,$$

where the right side may be infinite and the constant $C_{K,L}$ depends only on K, L and the dimension, and not on T, V or v.

Proof We treat $r \ge 2$ and smooth V by an integration by parts on small balls, then general V by monotone approximation, and reduce $r = 1$ to $r = 2$ by adding a variable.

Step 1: $r \ge 2$ and V smooth. Here $\|v\|_{L^\infty(L)} = \sup_L |v|$, as in Theorem 9.1.3. If it vanishes, then $v = 0$ on L, so $\mathrm{dd^c} v = 0$ on $\operatorname{Int} L \supset K$, and the left side vanishes. Otherwise both sides are homogeneous of degree one in v, and the left side does not change when a constant is added to v; replacing v by $v/(2\|v\|_{L^\infty(L)}) - 3/2$, it therefore suffices to prove that $\int_K (-V)\,\mathrm{dd^c} v \wedge T \wedge \beta^{r-1} \le C \int_L (-V)T \wedge \beta^r$ for plurisubharmonic functions v satisfying $-2 \le v \le -1$ on $\operatorname{Int} L$. Cover K by finitely many balls $B(a, R_1)$ such that $\overline{B(a,R)} \subset \operatorname{Int} L$ for some $R > R_1$; it suffices to bound the integral over one ball $B(a, R_1)$.

Put $B = B(a, R)$, $\psi = |z - a|^2 - R^2$ and $b = \mathrm{dd^c}\psi = \beta/\pi$. Fix $R_1 < R_2 < R$, and let $\widetilde{v} = \max\{v, A\psi\}$, with $A = 2/(R^2 - R_2^2)$, be the collar modification (9.4) of v on $\operatorname{Int} L$, with $N = 2$ and with $R^2 - R_2^2$ in place of δ. Thus $\widetilde{v}$ is plurisubharmonic on $\operatorname{Int} L$ and takes values in $[-2, 0)$ on B; it equals v on $B(a, R_2)$, and $A\psi$ on the collar $\{R^2 - 1/A < |z - a|^2 < R^2\}$. Choose $0 < \delta < 1/A$ and put $B_\delta = \{\psi < -\delta\}$, so that $\widetilde{v} = A\psi$ on $B \setminus B_\delta$ and $B(a, R_1) \subset B_\delta$. Choose a smooth function κ with $0 \le \kappa \le R^2$, with compact support in B and equal to $-\psi$ on a neighborhood of $\overline{B_\delta}$; this is possible because $0 < -\psi \le R^2$ on B. Put $S = \mathrm{dd^c}\widetilde{v} \wedge T$ and $Q = \mathrm{dd^c} V \wedge T$; since V is smooth and plurisubharmonic, Q is a closed positive current of bidimension $(r-1, r-1)$. Since $\mathrm{dd^c}\kappa = -b$ near $\overline{B_\delta}$ and $S = Ab \wedge T$ on $B \setminus B_\delta$,

$$\begin{aligned}\int_{B_\delta} (-V)S \wedge b^{r-1} &= \int_B VS \wedge b^{r-2} \wedge \mathrm{dd^c}\kappa - A\int_{B\setminus B_\delta} VT \wedge b^{r-1} \wedge \mathrm{dd^c}\kappa \\ &= \int_B \kappa\,\mathrm{dd^c}\widetilde{v} \wedge Q \wedge b^{r-2} - A\int_{B\setminus B_\delta} VT \wedge b^{r-1} \wedge \mathrm{dd^c}\kappa.\end{aligned}$$

For the second line we integrate by parts against the smooth compactly supported function κ: since V is smooth and $\mathrm{dd^c}\widetilde{v} \wedge T \wedge b^{r-2}$ is closed, $\mathrm{dd^c}(V\mathrm{dd^c}\widetilde{v} \wedge T \wedge b^{r-2}) = \mathrm{dd^c} V \wedge \mathrm{dd^c}\widetilde{v} \wedge T \wedge b^{r-2}$, and the Bedford–Taylor product is symmetric with respect to the smooth factor.

We bound the two terms on the right. Choose a compact set L' with $\overline{B} \subset \operatorname{Int} L'$ and $L' \subset \operatorname{Int} L$. Since $0 \le \kappa \le R^2$, the Chern–Levine–Nirenberg inequality (Theorem 9.1.3) applied to Q bounds the first term by a constant times the trace mass of Q on $\overline{B}$. To bound this mass, test Q against a smooth cutoff $0 \le \chi \le 1$ equal to one on $\overline{B}$ and supported in L':

$$\int \chi\,\mathrm{dd^c} V \wedge T \wedge \beta^{r-1} = \int V\,T \wedge \mathrm{dd^c}\chi \wedge \beta^{r-1} \le C\int_L (-V)T \wedge \beta^r,$$

since $V \le 0$ and the coefficients of $T \wedge \mathrm{dd^c}\chi \wedge \beta^{r-1}$ are bounded by a fixed multiple of the trace measure. The collar term has the same bound, because $\mathrm{dd^c}\kappa$ is a fixed smooth form. Finally, $S = \mathrm{dd^c}v \wedge T$ on $B(a,R_1)$ and $-V \ge 0$, so the left side of the display bounds $\pi^{1-r}\int_{B(a,R_1)}(-V)\mathrm{dd^c}v \wedge T \wedge \beta^{r-1}$. Summing over the finitely many balls proves the estimate, with a constant depending only on the chosen radii.

Step 2: $r \ge 2$ and V arbitrary. Choose a compact set L_1 with $K \subset \operatorname{Int} L_1$ and $L_1 \subset \operatorname{Int} L$, depending only on K and L. The mollifications $V_l = V * \rho_{\varepsilon_l}$, with $\varepsilon_l \downarrow 0$, are defined, smooth and plurisubharmonic on a neighborhood of L_1 for large l, and they decrease to V (Proposition 8.2.2(2)); in particular $V \le V_l \le 0$, so that $0 \le -V_l \le -V$. Step 1 only involves the weight on a neighborhood of the larger compact set, so it applies to V_l and the pair (K, L_1) and gives

$$\int_K (-V_l)\,\mathrm{dd^c}v \wedge T \wedge \beta^{r-1} \le C_{K,L_1}\|v\|_{L^\infty(L)} \int_L (-V)T \wedge \beta^r .$$

As $l \to \infty$, $-V_l$ increases to $-V$, and monotone convergence against the positive measure $\mathrm{dd^c}v \wedge T \wedge \beta^{r-1}$ proves the estimate.

Step 3: $r = 1$. Let t be the coordinate of $\mathbb{C}$, put $\beta_t = \frac{\mathrm{i}}{2}\mathrm{d}t \wedge \mathrm{d}\bar{t}$, let $D' \Subset D'' \Subset D$ be the discs about 0 of radii $1, 2, 3$, and let $P\colon U \times D \to U$ be the projection. By Proposition 7.5.2, P^*T is a closed positive current of bidimension $(2,2)$ on $U \times D$. Moreover, by Corollary 7.5.3(1), the pullback of a current $R = \sum R_{IJ}\,\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$ of order zero on U is $P^*R = \sum (R_{IJ} \otimes \lambda_D)\,\mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$, where λ_D is Lebesgue measure on D. Since v is a locally bounded Borel function, $P^*(vT) = (v \circ P)P^*T$ by Corollary 7.5.3(2), and since P^* commutes with ∂ and $\bar{\partial}$,

$$\mathrm{dd^c}(v \circ P) \wedge P^*T = \mathrm{dd^c}\big(P^*(vT)\big) = P^*(\mathrm{dd^c}v \wedge T).$$

The functions $V \circ P \le 0$ and $v \circ P$ are plurisubharmonic (Proposition 8.2.3), and $\|v \circ P\|_{L^\infty(L \times \overline{D''})} = \|v\|_{L^\infty(L)}$. We apply Step 2 with $r = 2$ to these data, to the Euclidean form $\beta + \beta_t$ and to the compact sets $K \times \overline{D'} \subset \operatorname{Int} L \times D'' = \operatorname{Int}(L \times \overline{D''})$. Since $T \wedge \beta^2 = 0$ and $\mathrm{dd^c}v \wedge T \wedge \beta = 0$ for bidegree reasons, the trace measures in Step 2 are

$$P^*T \wedge (\beta+\beta_t)^2 = 2P^*(T \wedge \beta) \wedge \beta_t, \qquad \mathrm{dd^c}(v \circ P) \wedge P^*T \wedge (\beta+\beta_t) = P^*(\mathrm{dd^c}v \wedge T) \wedge \beta_t,$$

that is, the products of the measures $2T \wedge \beta$ and $\mathrm{dd^c}v \wedge T$ with Lebesgue measure on D. By Fubini's theorem (Proposition 1.1.7), the inequality of Step 2 therefore reads

$$\pi \int_K (-V)\,\mathrm{dd^c}v \wedge T \le C\|v\|_{L^\infty(L)} \cdot 8\pi \int_L (-V)T \wedge \beta,$$

where C is the constant of Step 2 for $K \times \overline{D'}$ and $L \times \overline{D''}$ in dimension $n+1$, which depends only on K, L and n. This is the case $r = 1$, with $C_{K,L} = 8C$. □

Lemma 15.3.2 *Let T be a closed positive current of bidimension (r,r), $r \ge 1$, on a coordinate domain, whose trace measure gives zero mass to every pluripolar set.*

Then for every locally bounded plurisubharmonic v, the trace measure of $\mathrm{dd^c}v \wedge T$ also gives zero mass to every pluripolar set. Consequently the same holds for every Bedford–Taylor product $\mathrm{dd^c}v_1 \wedge \cdots \wedge \mathrm{dd^c}v_q \wedge T$, $1 \le q \le r$, of locally bounded plurisubharmonic functions with base T.

Proof Let E be a pluripolar subset of the coordinate domain U on which T lives. By Definition 10.1.1 and since U has a countable basis, E is covered by countably many compact sets K, each having a compact neighborhood L with $K \subset \operatorname{Int} L$, together with an open neighborhood W of L and a function $V \in \mathrm{PSH}(W)$ such that $E \cap K \subset \{V = -\infty\}$; shrinking W and subtracting a constant, we may assume $V \le 0$. The set $\{V = -\infty\}$ is a pluripolar Borel set, so it suffices to show that $K \cap \{V = -\infty\}$ has measure zero for the trace measure of $\mathrm{dd^c}v \wedge T$.

Put $\mu = \mathbf{1}_L T \wedge \beta^r$. This is a finite measure that does not charge $\{V = -\infty\}$, so V is finite μ-almost everywhere. Hence $\min\{-V/R, 1\} \to 0$ μ-almost everywhere as $R \to \infty$, and by dominated convergence we may choose numbers $R_j \ge 2^j$ such that

$$\int \min\{-V/R_j, 1\}\, \mathrm{d}\mu \le 2^{-j} \qquad (j \ge 1).$$

Put $\widetilde{V} = \sum_{j \ge 1} \max\{V/R_j, -1\}$. The partial sums are plurisubharmonic by Proposition 8.2.5, and they decrease, because every term is nonpositive. Their limit satisfies $\widetilde{V} \ge \sum_j V/R_j \ge V$, since $V \le 0$ and $\sum_j 1/R_j \le 1$; hence it is not identically $-\infty$ on any component of W, and it is plurisubharmonic by Theorem 8.3.1. Moreover $\widetilde{V} \le 0$, $\widetilde{V} = -\infty$ on $\{V = -\infty\}$, where every term equals -1, and by monotone convergence

$$\int_L (-\widetilde{V})\, T \wedge \beta^r = \sum_{j \ge 1} \int \min\{-V/R_j, 1\}\, \mathrm{d}\mu \le 1.$$

By Lemma 15.3.1, applied on W, the function $-\widetilde{V}$ is integrable on K for the trace measure $\mathrm{dd^c}v \wedge T \wedge \beta^{r-1}$ of $\mathrm{dd^c}v \wedge T$. Hence this measure gives zero mass to $K \cap \{\widetilde{V} = -\infty\} \supset K \cap \{V = -\infty\}$, which proves the first assertion. The second follows by induction on q, since $\mathrm{dd^c}v_1 \wedge \cdots \wedge \mathrm{dd^c}v_q \wedge T = \mathrm{dd^c}v_1 \wedge (\mathrm{dd^c}v_2 \wedge \cdots \wedge \mathrm{dd^c}v_q \wedge T)$, where the base current in parentheses is closed and positive of bidimension $(r - q + 1, r - q + 1)$. □

Proposition 15.3.3 *Let $u_1, \ldots, u_p$ be plurisubharmonic on a coordinate domain, $1 \le p \le n$, and suppose that $u_1, \ldots, u_q$ are locally bounded for some $0 \le q \le p$. If $S = \langle \mathrm{dd^c}u_{q+1} \wedge \cdots \wedge \mathrm{dd^c}u_p \rangle$ is locally finite, then the non-pluripolar product of $u_1, \ldots, u_p$ is locally finite and*

$$\langle \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_p \rangle = \mathrm{dd^c}u_1 \wedge \cdots \wedge \mathrm{dd^c}u_q \wedge S,$$

where the right side is the Bedford–Taylor product of Definition 9.1.1 with base S.

Proof For $q = 0$ there is nothing to prove. Inserting the bounded factors $u_q, u_{q-1}, \ldots, u_1$ one at a time, we reduce to the case $q = 1$. So let $v = u_1$ be

locally bounded and $S = \langle \mathrm{dd^c} u_2 \wedge \cdots \wedge \mathrm{dd^c} u_p \rangle$ locally finite. By Theorem 15.2.1, S is a closed positive current, so $R = \mathrm{dd^c} v \wedge S$ is defined; it is closed and positive, and by Lemma 15.3.2 it gives zero mass to pluripolar sets, because S does. If $p = 1$, then $S = 1$ and the assertion is that of bounded Bedford–Taylor theory. Let $p \geq 2$. The assertion is local, so we may assume $u_j \leq 0$ for $j \geq 2$ and put

$$O_k = \bigcap_{j \geq 2} \{u_j > -k\}, \qquad \Theta_k = \bigwedge_{j \geq 2} \mathrm{dd^c} \max\{u_j, -k\},$$
$$w_k = \max\Big\{1 + \frac{1}{k} \sum_{j \geq 2} u_j,\, 0\Big\}.$$

The functions w_k are plurisubharmonic by Proposition 8.2.5, and they take values in $[0, 1]$ because all $u_j \leq 0$. They vanish outside O_k: if $u_i \leq -k$ for some $i \geq 2$, then $\sum_{j \geq 2} u_j \leq u_i \leq -k$. The sets $\{w_k > 0\} = \{\sum_{j \geq 2} u_j > -k\}$ increase to the set where all u_j, $j \geq 2$, are finite.

Take smooth plurisubharmonic functions v_l decreasing to v near a given compact set. By Lemma 15.1.2, S and Θ_k have the same restriction to O_k; since w_k vanishes outside O_k and $\mathrm{dd^c} v_l$ is smooth,

$$w_k \mathrm{dd^c} v_l \wedge S = w_k \mathrm{dd^c} v_l \wedge \Theta_k.$$

The weighted form of Theorem 9.2.5, applied with the constant sequence of weights w_k and the fixed base currents S on the left and Θ_k on the right, gives $w_k R = w_k \mathrm{dd^c} v \wedge \Theta_k$ in the limit $l \to \infty$. Multiplying by the bounded Borel functions $\mathbf{1}_{\{w_k > \varepsilon\}}/w_k$ and letting $\varepsilon \downarrow 0$, we see that R and $\mathrm{dd^c} v \wedge \Theta_k$ have the same restriction to $\{w_k > 0\}$. Now fix an integer $m \geq 1$. For $k \geq m$, the bounded functions $\max\{u_i, -k\}$ and $\max\{u_i, -m\}$ coincide on O_m, so $\mathrm{dd^c} v \wedge \Theta_k = \mathrm{dd^c} v \wedge \Theta_m$ on O_m by Theorem 10.4.3. Since the sets $\{w_k > 0\} \cap O_m$, $k \geq m$, exhaust O_m, we conclude that $\mathbf{1}_{O_m} R = \mathbf{1}_{O_m} \mathrm{dd^c} v \wedge \Theta_m$.

Consider finally the full tuple $v, u_2, \ldots, u_p$ on a relatively compact open set on which $|v| < m_0$. For $m \geq m_0$, its level-m set is O_m and v needs no truncation, so its truncated product at level m is $\mathbf{1}_{O_m} \mathrm{dd^c} v \wedge \Theta_m = \mathbf{1}_{O_m} R$. These restrictions are bounded by R, hence have locally bounded mass, and they increase to R because R gives the pole sets of $u_2, \ldots, u_p$ zero mass. Therefore the non-pluripolar product of $v, u_2, \ldots, u_p$ is locally finite and equals $R = \mathrm{dd^c} v \wedge S$. □

To identify the first-degree product we shall integrate planar measures over almost every fiber of a coordinate projection. We first record the measure theory of such fiber integrals, and then prove the fiber formula for bounded potentials, with the Borel weights needed for truncation.

Lemma 15.3.4 *Let $Q \subset \mathbb{C}^r$ and $D \subset \mathbb{C}^m$ be open sets, $r, m \geq 1$, let $G \subset D$ be a Borel set, and for each $z \in G$ let μ_z be a positive Radon measure on Q. Assume that $z \mapsto \int a \, \mathrm{d}\mu_z$ is Borel on G for every continuous function a with compact support in Q. Then $z \mapsto \mu_z(E_z)$ is Borel on G for every Borel set $E \subset Q \times D$, where $E_z = \{x : (x, z) \in E\}$. Moreover, for every nonnegative Borel function f on D,*

$$\lambda(E) = \int_G \mu_z(E_z) f(z)\, \mathrm{d}V(z)$$

is a measure on the Borel sets of $Q \times D$, and

$$\int h\, \mathrm{d}\lambda = \int_G \Big(\int_Q h(x,z)\, \mathrm{d}\mu_z(x) \Big) f(z)\, \mathrm{d}V(z)$$

for every nonnegative Borel function h on $Q \times D$, the inner integral being a Borel function of $z \in G$.

Proof Fix open sets $Q'' \Subset Q$ and $D'' \Subset D$, so that $\mu_z(Q'') < \infty$ for every z. If $A \subset Q''$ is open, the continuous functions $a_l(x) = \min\{1, \max\{l\, \mathrm{dist}(x, \mathbb{C}^r \setminus A) - 1, 0\}\}$ have compact support in A and increase to $\mathbf{1}_A$, so $z \mapsto \mu_z(A)$ is Borel by monotone convergence. Let $\mathcal{D}$ be the family of Borel sets $A \subset Q''$ for which $z \mapsto \mu_z(A)$ is Borel on G. It contains the open subsets of Q'', which are closed under finite intersections and generate the Borel σ-algebra of Q''. It is closed under complements in Q'', because $\mu_z(Q'' \setminus A) = \mu_z(Q'') - \mu_z(A)$ with finite values, and under countable disjoint unions. By Dynkin's theorem (Lemma 1.1.6), $\mathcal{D}$ contains every Borel subset of Q''.

Next, let $\mathcal{D}'$ be the family of Borel sets $E \subset Q'' \times D''$ for which $z \mapsto \mu_z(E_z)$ is Borel on G. It contains the products $A \times A'$ of Borel sets $A \subset Q''$ and $A' \subset D''$, since $\mu_z((A \times A')_z) = \mu_z(A) \mathbf{1}_{A'}(z)$; these products are closed under finite intersections and generate the Borel σ-algebra of $Q'' \times D''$, because every open subset of $Q'' \times D''$ is a countable union of products of open balls. As before, $\mathcal{D}'$ is closed under complements and countable disjoint unions, so it contains every Borel subset of $Q'' \times D''$. Exhausting Q and D by increasing sequences of such sets Q'' and D'', and using monotone convergence, we obtain the first assertion.

By the first assertion, $\lambda(E)$ is defined, and λ is countably additive by monotone convergence, applied first to the measures μ_z and then to the integral over G. The formula for $\int h\, \mathrm{d}\lambda$ holds for indicator functions of Borel sets by definition, hence for nonnegative simple functions by linearity, and for nonnegative Borel functions by monotone convergence, applied in the same order; the inner integrals are Borel in z as increasing limits of Borel functions. □

Lemma 15.3.5 *Let $r, m \geq 0$ be integers, let $Q \subset \mathbb{C}^r$ and $D \subset \mathbb{C}^m$ be open sets with coordinates x and z, and let $p_D \colon Q \times D \to D$ be the projection. Let $u_1, \ldots, u_r$ be locally bounded plurisubharmonic functions on $Q \times D$, and put*

$$M = \mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_r, \qquad M_z = \bigwedge_{j=1}^{r} \mathrm{dd^c_x} u_j(\,\cdot\,, z),$$

the Bedford–Taylor products on $Q \times D$ and on the fibers $Q \times \{z\}$; here $u_j(\,\cdot\,, z)$ is locally bounded and plurisubharmonic on Q by Proposition 8.2.3. For every smooth real (m,m)-form η on D and every bounded Borel function h on $Q \times D$ with compact support, the fiber integral $z \mapsto \int_Q h(x,z)\, M_z(x)$ is Borel measurable and

$$\int_{Q\times D} h\, p_D^*\eta \wedge M = \int_D \left(\int_Q h(x,z)\, M_z(x) \right) \eta(z). \tag{15.4}$$

If $\eta \geq 0$, the identity holds for every nonnegative Borel function h, with possibly infinite values.

Proof If $r = 0$ or $m = 0$, one factor of $Q \times D$ is a point, carrying counting measure, and the corresponding product is the empty product 1; then the two sides of (15.4) are the same integral. Let $r, m \geq 1$. For smooth potentials the formula is Fubini's theorem; we pass to the limit for continuous weights h, and then extend the formula to Borel weights by the uniqueness of measures.

Approximation. Let $Q' \Subset Q$ and $D' \Subset D$ be open. For $\varepsilon_l \downarrow 0$, the mollifications $u_{j,l} = u_j * \rho_{\varepsilon_l}$ are smooth and plurisubharmonic on a neighborhood of $\overline{Q' \times D'}$ for large l, they decrease to u_j (Proposition 8.2.2(2)), and they are bounded there by a constant B, since $u_j \leq u_{j,l} \leq u_{j,l_0}$. Let $M^{(l)}$ and $M_z^{(l)}$ be the corresponding products. By Theorem 9.2.5, $M^{(l)} \to M$ weakly on $Q' \times D'$, and for every $z \in D'$, $M_z^{(l)} \to M_z$ weakly on Q', since $u_{j,l}(\,\cdot\,, z)$ decreases to $u_j(\,\cdot\,, z)$. By Theorem 9.1.3, for every compact set $A \subset Q'$ the masses $M_z^{(l)}(A)$ and $M_z(A)$ are bounded by a constant depending only on A, Q' and B, uniformly in l and $z \in D'$; similarly for $M^{(l)}$ and M on compact subsets of $Q' \times D'$. For positive measures with locally uniformly bounded masses, weak convergence implies convergence of the integrals of continuous functions with compact support, because such functions are uniform limits of test functions supported in a fixed compact set (Lemma 1.4.1).

Continuous weights. Let $\eta = f\, dV_z \geq 0$, where dV_z is the Euclidean volume form of $\mathbb{C}^m$ and $f \geq 0$ is smooth, and let h be continuous with compact support in $Q' \times D'$. In the smooth exterior product $M^{(l)}$, wedging with the form $p_D^*\eta$ of top degree in z kills every term containing a differential dz_i or $d\bar{z}_i$, so that $p_D^*\eta \wedge M^{(l)} = f(z)\, M_z^{(l)}(x) \wedge dV_z$, and Fubini's theorem (Proposition 1.1.7) gives (15.4) for the smooth approximations. As $l \to \infty$, the left side converges to $\int h\, p_D^*\eta \wedge M$, by the approximation step applied to the positive measures $p_D^*\eta \wedge M^{(l)}$. On the right, the inner integrals are continuous functions of z, they converge to $\int_Q h(x,z)\, M_z(x)$ for every z, and they are bounded by $\sup |h|$ times a constant, uniformly in l and z; hence the limit is a Borel function of z, and dominated convergence proves (15.4). A finite smooth partition of unity (Lemma 1.4.2), subordinate to a cover of $\operatorname{Supp} h$ by such products $Q' \times D'$, treats arbitrary continuous h with compact support.

Measurability. For a continuous function a with compact support in Q, the function $z \mapsto \int a\, M_z$ is Borel on D: if the support of a lies in Q', then on D' it is the pointwise limit of the continuous functions $z \mapsto \int a\, M_z^{(l)}$, and D is covered by countably many such sets D'. Hence Lemma 15.3.4, applied with $G = D$ and $\mu_z = M_z$, shows that $z \mapsto \int_Q h(x,z)\, M_z(x)$ is Borel for every nonnegative Borel function h, and hence for every bounded Borel function h with compact support.

Borel weights. Let $\eta = f\, dV_z \geq 0$ as before, and let λ be the measure of Lemma 15.3.4 for $\mu_z = M_z$ and this function f, so that

$$\lambda(E) = \int_D M_z(E_z)\,\eta(z)$$

for Borel sets $E \subset Q \times D$. The uniform local bound for the fiber masses makes λ finite on compact sets, so it is a Radon measure by Proposition 1.1.2. By Lemma 15.3.4, $\int h\,\mathrm{d}\lambda = \int_D \big(\int_Q h(x,z)\,M_z(x)\big)\,\eta(z)$ for every nonnegative Borel function h, hence for every continuous h with compact support. By the step on continuous weights, λ and $p_D^*\eta \wedge M$ therefore have the same integral against every test function, so they are equal by the uniqueness in Theorem 1.4.5. This proves the formula for nonnegative Borel h, and then for bounded real Borel h with compact support, by splitting h into its positive and negative parts.

Finally, let $\eta = f\,\mathrm{d}V_z$ be a smooth real form and h a bounded Borel function supported in $K \times K_D$, with $K \subset Q$ and $K_D \subset D$ compact. Choose $\kappa \in C_c^\infty(D)$ with $0 \le \kappa \le 1$ and $\kappa = 1$ on K_D, and put $c = \sup_{\operatorname{Supp}\kappa} |f|$. Replacing η by $\kappa\eta$ changes neither side of (15.4), and $\kappa\eta = \kappa(f+c)\,\mathrm{d}V_z - c\kappa\,\mathrm{d}V_z$ is the difference of two smooth nonnegative forms. This proves the formula for signed η. □

Proposition 15.3.6 *Let u be plurisubharmonic on a coordinate domain. Then its non-pluripolar product of degree one is locally finite and*

$$\langle \mathrm{dd^c} u \rangle = \mathbf{1}_{\{u>-\infty\}}\,\mathrm{dd^c} u, \tag{15.5}$$

where $\mathrm{dd^c} u$ on the right is the distributional complex Hessian. In particular the current $\mathbf{1}_{\{u>-\infty\}}\mathrm{dd^c} u$ is closed.

Proof *Upper bound.* The assertion is local, so we may assume $u \le 0$. The truncations $u_l = \max\{u, -l\}$ decrease to u and converge to it in L^1_{loc} (Theorem 8.3.1), so $\mathrm{dd^c} u_l \to \mathrm{dd^c} u$ weakly. For fixed k and every $l \ge k$, plurifine locality (Theorem 10.4.3) and Lemma 15.1.1 give $\mathbf{1}_{\{u>-k\}}\mathrm{dd^c} u_k = \mathbf{1}_{\{u>-k\}}\mathrm{dd^c} u_l \le \mathrm{dd^c} u_l$, and letting $l \to \infty$ yields $\mathbf{1}_{\{u>-k\}}\mathrm{dd^c} u_k \le \mathrm{dd^c} u$. Hence the truncated products have locally bounded mass, the non-pluripolar product $S = \langle \mathrm{dd^c} u \rangle$ is locally finite, and $0 \le S \le \mathrm{dd^c} u$. Since S gives no mass to the pole set of u, it follows that $S \le \mathbf{1}_{\{u>-\infty\}}\mathrm{dd^c} u$. It remains to prove the reverse inequality.

One variable. Let $n = 1$ and put $\mu = \mathrm{dd^c} u$, a positive measure. Let V be a relatively compact open subset of the domain, and choose a smooth compactly supported function $\kappa \ge 0$ equal to one on V. With $\nu = \kappa\mu$ and the fundamental solution E_2 of (2.7), Theorem 3.2.3 gives

$$u = 4\pi E_2 * \nu + h \quad \text{on } V,$$

where h is harmonic; the factor 4π appears because $\mathrm{dd^c} u = (\Delta u)/(4\pi)$ as planar measures, while the Riesz measure is Δu. Put $p = E_2 * \nu$ and $\nu_f = \mathbf{1}_{\{p>-\infty\}}\nu$; on V we have $\{p > -\infty\} = \{u > -\infty\}$. The kernel E_2 is bounded above on the compact set of differences of points of $\operatorname{Supp}\nu$, so removing positive mass from ν lowers its potential on $\operatorname{Supp}\nu$ by at most a constant: $E_2 * \nu_f \ge p - C$ there. Hence $E_2 * \nu_f$ is finite ν_f-almost everywhere, and Theorem 4.2.2 writes $\nu_f = \sum_i \nu_i$ with compactly supported measures ν_i whose potentials are finite and continuous.

For each N, the measure $\nu^{(N)} = \sum_{i\le N} \nu_i$ has the continuous subharmonic potential $v_N = 4\pi E_2 * \nu^{(N)}$, with $\mathrm{dd^c} v_N = \nu^{(N)}$. On V we have $\mu = \nu \ge \nu_f \ge \nu^{(N)}$, so the distribution $u - v_N$ has positive Laplacian on V. By Theorem 3.2.2(3), this distribution has a subharmonic representative s on V, and by part (2) of that theorem the value of s at each point is the limit of its averages over small discs. These averages are also those of the function $u - v_N$, and they converge to $u - v_N$ at every point, by part (2) applied to u and the continuity of v_N. Hence $s = u - v_N$ everywhere on V, and $u - v_N$ is subharmonic on V. Since v_N is locally bounded, multilinearity (Proposition 15.1.5) gives

$$S = \langle \mathrm{dd^c} v_N \rangle + \langle \mathrm{dd^c}(u - v_N) \rangle \ge \mathrm{dd^c} v_N = \nu^{(N)} \quad \text{on } V.$$

Letting $N \to \infty$ yields $S \ge \nu_f = \mathbf{1}_{\{u>-\infty\}}\mu$ on V, which proves the proposition for $n = 1$.

Several variables. Let $n \ge 2$. The assertion is local, so we may work on a polydisc P such that $u \le 0$ is plurisubharmonic on a neighborhood of $\overline{P}$. Fix one of the coordinates, denote it by x, and write $z \in \mathbb{C}^{n-1}$ for the others, so that $P = \Delta \times Z$ with a disc $\Delta \subset \mathbb{C}$ and a polydisc $Z \subset \mathbb{C}^{n-1}$. Let $\mathrm{d}V_z$ be the Euclidean volume form of the z-coordinates, viewed as an $(n-1, n-1)$-form, and $\mathrm{d}A$ the area measure on Δ. We disintegrate $\mathrm{dd^c} u \wedge \mathrm{d}V_z$ along the fibers $\Delta \times \{z\}$ and apply the one-variable case on each fiber. Since u is integrable on $\Delta \times Z$, Fubini's theorem (Proposition 1.1.7) shows that $G = \{z \in Z : \int_\Delta |u(x,z)|\,\mathrm{d}A(x) < \infty\}$ is a Borel set of full measure in Z. For $z \in G$ the function $u_z = u(\,\cdot\,, z)$ is integrable, hence not identically $-\infty$, so it is subharmonic on Δ by Proposition 8.2.3; let $\mu_z = \mathrm{dd^c_x} u_z$ be its Hessian measure.

For a test function a on Δ, we have $\int a\,\mathrm{d}\mu_z = \int_\Delta u_z \mathrm{dd^c_x} a$, and Fubini's theorem, applied to the positive and negative parts of the integrand, shows that this is a Borel function of $z \in G$. Uniform approximation by test functions supported in a fixed compact set extends this to continuous functions a with compact support, since each μ_z is finite on compact sets. Hence Lemma 15.3.4, with $f = 1$, gives a measure λ on $\Delta \times Z$ with $\int h\,\mathrm{d}\lambda = \int_G (\int h(\,\cdot\,, z)\,\mathrm{d}\mu_z)\,\mathrm{d}V(z)$ for nonnegative Borel h. If $\chi \in C_c^\infty(\Delta)$ satisfies $0 \le \chi \le 1$ and equals one on a disc $\Delta' \Subset \Delta$, then $\mu_z(\Delta') \le \int u_z \mathrm{dd^c_x}\chi \le C_\chi \int_\Delta |u_z|\,\mathrm{d}A$, which is integrable over G; so λ is finite on compact sets. For a test function φ on $\Delta \times Z$, only the derivative $\partial_x \partial_{\bar{x}} \varphi$ survives in $\mathrm{dd^c}\varphi \wedge \mathrm{d}V_z$, and Fubini's theorem gives

$$\int \varphi\,\mathrm{dd^c} u \wedge \mathrm{d}V_z = \int u\,\mathrm{dd^c}\varphi \wedge \mathrm{d}V_z = \int_G \Big(\int_\Delta u_z\,\mathrm{dd^c_x}\varphi(\,\cdot\,, z) \Big) \mathrm{d}V(z) = \int \varphi\,\mathrm{d}\lambda.$$

By the uniqueness in Theorem 1.4.5, the positive measure $\mathrm{dd^c} u \wedge \mathrm{d}V_z$ equals λ, that is,

$$\int g\,\mathrm{dd^c} u \wedge \mathrm{d}V_z = \int_G \Big(\int g(\,\cdot\,, z)\,\mathrm{d}\mu_z \Big) \mathrm{d}V(z) \tag{15.6}$$

for every nonnegative Borel function g on $\Delta \times Z$.

Now let $g \ge 0$ be a bounded Borel function with compact support in $\Delta \times Z$, and apply Lemma 15.3.5, with $r = 1$, $Q = \Delta$, $D = Z$ and $\eta = \mathrm{d}V_z$, to the bounded function $u_k = \max\{u, -k\}$ and the weight $g\mathbf{1}_{\{u>-k\}}$:

$$\int g\mathbf{1}_{\{u>-k\}}\mathrm{dd^c}u_k\wedge\mathrm{d}V_z=\int\Big(\int g\mathbf{1}_{\{u_z>-k\}}\mathrm{dd^c_x}\max\{u_z,-k\}\Big)\mathrm{d}V_z.$$

As $k\to\infty$, the left side increases to $\int gS\wedge\mathrm{d}V_z$ by Lemma 15.1.2, since $\mathbf{1}_{\{u>-k\}}\mathrm{dd^c}u_k$ is the truncated product S_k of u. For $z\in G$, the inner integral on the right is the integral of $g(\,\cdot\,,z)$ against the truncated product of u_z at level k; by Lemma 15.1.2 and the one-variable case, it increases to $\int g\,\mathrm{d}\langle\mathrm{dd^c_x}u_z\rangle=\int g\mathbf{1}_{\{u_z>-\infty\}}\,\mathrm{d}\mu_z$. Since $Z\setminus G$ is a null set, monotone convergence in z and (15.6), applied to $g\mathbf{1}_{\{u>-\infty\}}$, give $\int gS\wedge\mathrm{d}V_z=\int g\mathbf{1}_{\{u>-\infty\}}\mathrm{dd^c}u\wedge\mathrm{d}V_z$, so that $S\wedge\mathrm{d}V_z=\mathbf{1}_{\{u>-\infty\}}\mathrm{dd^c}u\wedge\mathrm{d}V_z$.

Since $\beta^{n-1}=(n-1)!\sum_{j=1}^n\mathrm{d}V_{(j)}$, where $\mathrm{d}V_{(j)}$ is the volume form of the coordinates other than the jth, summing over the n coordinate directions shows that the trace measures of S and of $\mathbf{1}_{\{u>-\infty\}}\mathrm{dd^c}u$ coincide. Thus $\mathbf{1}_{\{u>-\infty\}}\mathrm{dd^c}u-S$ is a positive current with zero trace measure, and it vanishes by Theorem 7.2.5. This proves (15.5); closedness follows from Theorem 15.2.1. □

Proposition 15.3.7 *Let μ be a probability measure supported on a compact set $K\subset\mathbb{C}\subset\mathbb{P}^1$, and let ω_{FS} be the Fubini–Study form of Example 14.5.6, which has total mass one on $\mathbb{P}^1$. Then*

$$q(z)=\int_K\log\frac{|z-a|^2}{(1+|z|^2)(1+|a|^2)}\,\mathrm{d}\mu(a)$$

defines a nonpositive ω_{FS}-plurisubharmonic function on $\mathbb{P}^1$ with $\omega_{\mathrm{FS}}+\mathrm{dd^c}q=\mu$. Its Lelong number at every point $x\in\mathbb{P}^1$ is $\mu(\{x\})$, where μ is regarded as a measure on $\mathbb{P}^1$.

Proof In the affine chart put $\rho(z)=\log(1+|z|^2)$, so that $\omega_{\mathrm{FS}}=\mathrm{dd^c}\rho$. The integrand is nonpositive, since $|z-a|^2\le(|z|+|a|)^2\le(1+|z|^2)(1+|a|^2)$ by the Cauchy–Schwarz inequality. On the affine chart,

$$q+\rho=p_\mu-c,\qquad p_\mu(z)=\int_K\log|z-a|^2\,\mathrm{d}\mu(a),\qquad c=\int_K\log(1+|a|^2)\,\mathrm{d}\mu(a).$$

Since $\log|z-a|^2=4\pi E_2(z-a)$ with the fundamental solution E_2 of (2.7), the function $p_\mu=4\pi E_2*\mu$ is subharmonic and locally integrable with $\Delta p_\mu=4\pi\mu$, by Proposition 3.2.1. Since $\mathrm{dd^c}=\Delta/(4\pi)$ on functions of one complex variable, as planar measures, this gives $\omega_{\mathrm{FS}}+\mathrm{dd^c}q=\mathrm{dd^c}p_\mu=\mu$ on the affine chart, and $q=p_\mu-c-\rho$ is quasi-plurisubharmonic there.

Near infinity use the coordinate $w=1/z$, in which the integrand becomes

$$\log|1-aw|^2-\log(1+|w|^2)-\log(1+|a|^2).$$

Let $R_K=\max_{a\in K}|a|$. For $|w|<1/(2R_K+2)$ we have $|1-aw|\ge1/2$, so this expression is a smooth function of (a,w) on $K\times\{|w|<1/(2R_K+2)\}$ whose derivatives in w of every order are bounded uniformly in $a\in K$; at $w=0$ it takes the value $-\log(1+|a|^2)$. Differentiating under the integral sign, we see that q extends to a smooth function near infinity, and that $q+\log(1+|w|^2)=\int_K\log|1-aw|^2\,\mathrm{d}\mu(a)-c$

is harmonic there, because each $\log|1 - aw|^2$ is harmonic in w. Since $\omega_{\mathrm{FS}} = \mathrm{dd^c}\log(1+|w|^2)$ in this chart and μ vanishes near infinity, $\omega_{\mathrm{FS}} + \mathrm{dd^c}q = 0 = \mu$ there. Hence q is quasi-plurisubharmonic on $\mathbb{P}^1$, $\omega_{\mathrm{FS}} + \mathrm{dd^c}q = \mu \geq 0$ globally, and q is a nonpositive ω_{FS}-plurisubharmonic function.

At infinity both the Lelong number and the point mass vanish, because q is smooth there and $K \subset \mathbb{C}$. At an affine point x, the Lelong number of q is that of the local subharmonic function $q + \rho$ (Corollary 12.3.2), and by (12.5) and (12.3) it equals $\lim_{r\downarrow 0} \mu(B(x,r)) = \mu(\{x\})$. □

Example 15.3.8 Let ω_{FS} be the Fubini–Study form on $\mathbb{P}^1$, as in Proposition 15.3.7. There is a nonpositive ω_{FS}-plurisubharmonic function q on $\mathbb{P}^1$ such that

$$\nu(q,x) = 0 \quad \text{for every } x, \qquad \mathcal{I}(tq) = \mathcal{O}_{\mathbb{P}^1} \quad (t > 0), \qquad \langle \omega_{\mathrm{FS}} + \mathrm{dd^c}q \rangle = 0.$$

Let μ be the atom-free probability measure on the compact polar set K of Proposition 4.2.4, and let q be its potential from Proposition 15.3.7. Since μ has no atoms, all Lelong numbers of q vanish, and Corollary 12.3.2 gives the assertion on multiplier ideals. On the affine chart, q differs from the function $p(z) = \int_K \log|z-a|^2\,\mathrm{d}\mu(a)$ of Proposition 4.2.4 by the smooth function $-\log(1+|z|^2)$ and a constant, and $p = -\infty$ on K; hence $q = -\infty$ on K. Applying Proposition 15.3.6 to local potentials of $\omega_{\mathrm{FS}} + \mathrm{dd^c}q$, we obtain

$$\langle \omega_{\mathrm{FS}} + \mathrm{dd^c}q \rangle = \mathbf{1}_{\{q > -\infty\}}\mu = 0,$$

because μ is carried by K. By contrast, the potential 0 has the same multiplier ideals and non-pluripolar mass one. Thus exponential integrability at every scale does not determine the non-pluripolar mass.

15.4 Global finiteness and bounded cohomological mass

From now on, X is a connected compact Kähler manifold of dimension $n \geq 1$ with a fixed Kähler form ω, and θ and θ_j always denote smooth closed real $(1,1)$-forms on X. (We exclude dimension zero, where the only product is the empty one.) If $\varphi \in \mathrm{PSH}(X,\theta)$ (Definition 13.1.3) is locally bounded, then near each point $\theta + \mathrm{dd^c}\varphi = \mathrm{dd^c}(h+\varphi)$, where h is a smooth local potential of θ and $h+\varphi$ is bounded and plurisubharmonic. By Proposition 15.1.5, the Bedford–Taylor products of such local potentials do not depend on the choice of h; hence they glue to closed positive currents $\bigwedge_j(\theta_j + \mathrm{dd^c}\varphi_j)$ on X, which can be multiplied by smooth forms. We first prove Stokes' formula for these products and then show that non-pluripolar products on X are always finite.

Proposition 15.4.1 *Let $1 \leq p \leq n$, let $\varphi_j \in \mathrm{PSH}(X,\theta_j)$ be locally bounded for $1 \leq j \leq p$, and let η be a smooth closed $(n-p,n-p)$-form on X. Then*

$$\int_X \bigwedge_{j=1}^{p} (\theta_j + \mathrm{dd^c}\varphi_j) \wedge \eta = \int_X \theta_1 \wedge \cdots \wedge \theta_p \wedge \eta.$$

Proof Fix j, let R be the closed positive current $\bigwedge_{i>j}(\theta_i + \mathrm{dd^c}\varphi_i)$, and let η' be a smooth closed form of pure complementary bidegree. Locally $\theta_j = \mathrm{dd^c} h_j$ with h_j smooth, and since R is closed, the recursion gives

$$(\theta_j + \mathrm{dd^c}\varphi_j) \wedge R = \mathrm{dd^c}((h_j + \varphi_j)R) = \theta_j \wedge R + \mathrm{dd^c}(\varphi_j R).$$

Here $\mathrm{dd^c}(h_j R) = \mathrm{dd^c} h_j \wedge R$ because h_j is smooth and R is closed of pure bidegree, so that $\partial R = \bar\partial R = 0$. Both sides are globally defined, because φ_j is a global bounded function. Moreover $\int_X \mathrm{dd^c}(\varphi_j R) \wedge \eta' = \int_X \varphi_j R \wedge \mathrm{dd^c}\eta' = 0$, since η' is closed of pure bidegree, so that $\partial\eta' = \bar\partial\eta' = 0$ and $\mathrm{dd^c}\eta' = 0$. Hence $\int_X (\theta_j + \mathrm{dd^c}\varphi_j) \wedge R \wedge \eta' = \int_X \theta_j \wedge R \wedge \eta'$. Apply this successively for $j = 1, 2, \ldots, p$, with $\eta' = \theta_1 \wedge \cdots \wedge \theta_{j-1} \wedge \eta$ at the jth step, which is smooth and closed. After p steps the integral becomes $\int_X \theta_1 \wedge \cdots \wedge \theta_p \wedge \eta$. □

Lemma 15.4.2 *Let R be a closed positive current of bidimension $(1,1)$ on X, and let a, b be bounded quasi-plurisubharmonic functions on X, or differences of such functions. Then $\mathrm{dd^c} b \wedge R = \mathrm{dd^c}(bR)$ and $\mathrm{dd^c} a \wedge R = \mathrm{dd^c}(aR)$ are finite signed measures, and*

$$\int_X a\,\mathrm{dd^c} b \wedge R = \int_X b\,\mathrm{dd^c} a \wedge R = -\int_X \mathrm{d}a \wedge \mathrm{d^c} b \wedge R.$$

Here the mixed gradient

$$\mathrm{d}a \wedge \mathrm{d^c} b \wedge R = \tfrac{1}{2}\big(\mathrm{dd^c}(abR) - a\,\mathrm{dd^c} b \wedge R - b\,\mathrm{dd^c} a \wedge R\big)$$

is a finite signed measure on X, symmetric in a and b. It is the exterior product of $\mathrm{d}a \wedge \mathrm{d^c} b$ with R when a and b are smooth, and on an open set where $a = b$ is plurisubharmonic it is the gradient measure $E_R(a)$ of (9.37).

Proof *The measures.* Near each point of X, a bounded quasi-plurisubharmonic function b is the difference $b_1 - b_2$ of two bounded plurisubharmonic functions: if $b + g$ is plurisubharmonic on a coordinate ball with g smooth, and $q = C|z|^2$ with C so large that $g + q$ is plurisubharmonic on a smaller ball, take $b_1 = b + g + q$ and $b_2 = g + q$. Hence $\mathrm{dd^c}(bR) = \mathrm{dd^c} b_1 \wedge R - \mathrm{dd^c} b_2 \wedge R$ is locally the difference of two Bedford–Taylor products, which are positive measures. Covering the compact manifold X by finitely many such neighborhoods, we see that $\mathrm{dd^c}(bR)$ is a finite signed measure; the same holds for a.

For bounded plurisubharmonic functions A and B on an open set, additivity of Bedford–Taylor products (Proposition 9.1.2) and the formula (9.37) give, after expanding $(A + B)^2$,

$$E_R(A + B) = E_R(A) + E_R(B) + \mathrm{dd^c}(ABR) - A\,\mathrm{dd^c} B \wedge R - B\,\mathrm{dd^c} A \wedge R,$$

so that $\frac{1}{2}(\mathrm{dd^c}(ABR) - A\,\mathrm{dd^c}B\wedge R - B\,\mathrm{dd^c}A\wedge R) = \frac{1}{2}(E_R(A+B) - E_R(A) - E_R(B))$ is a signed measure, by Lemma 9.13.1. Since the expression defining the mixed gradient is bilinear in (a,b), writing $a = a_1 - a_2$ and $b = b_1 - b_2$ locally shows that the mixed gradient is a signed measure near each point, hence a finite signed measure on X. It is symmetric by definition. For $a = b$ plurisubharmonic it equals $\frac{1}{2}\mathrm{dd^c}(a^2R) - a\,\mathrm{dd^c}a\wedge R = E_R(a)$. For smooth a and b we have $\mathrm{dd^c}(ab) = a\,\mathrm{dd^c}b + b\,\mathrm{dd^c}a + \mathrm{d}a\wedge\mathrm{d^c}b + \mathrm{d}b\wedge\mathrm{d^c}a$, and the $(1,1)$-parts of $\mathrm{d}a\wedge\mathrm{d^c}b$ and $\mathrm{d}b\wedge\mathrm{d^c}a$ both equal $\frac{\mathrm{i}}{4\pi}(\partial a\wedge\bar\partial b + \partial b\wedge\bar\partial a)$; since only $(1,1)$-parts contribute in products with R and $\mathrm{dd^c}(abR) = \mathrm{dd^c}(ab)\wedge R$, the mixed gradient is $\mathrm{d}a\wedge\mathrm{d^c}b\wedge R$.

A local identity. Let χ be a test function with support in a coordinate ball W, and let A and B be bounded plurisubharmonic functions on a neighborhood of $\overline{W}$. We claim that

$$\int \chi A\,\mathrm{dd^c}(BR) - \int \chi B\,\mathrm{dd^c}(AR) = -\int AB\,\mathrm{dd^c}\chi\wedge R - 2\,\mathrm{d}(BR)(A\,\mathrm{d^c}\chi), \quad (15.7)$$

where $\mathrm{d}(BR)$ is the current of order zero of Lemma 9.13.1 and its value on the bounded Borel one-form $A\,\mathrm{d^c}\chi$ with compact support is computed with its coefficient measures. Suppose first that A is smooth. Then $\int \chi A\,\mathrm{dd^c}(BR) = (BR)(\mathrm{dd^c}(\chi A))$ and

$$\mathrm{dd^c}(\chi A) = \chi\,\mathrm{dd^c}A + A\,\mathrm{dd^c}\chi + \mathrm{d}\chi\wedge\mathrm{d^c}A + \mathrm{d}A\wedge\mathrm{d^c}\chi.$$

The first term contributes $\int \chi B\,\mathrm{dd^c}A\wedge R$. Since BR has bidegree $(n-1,n-1)$, it sees only the $(1,1)$-parts of the last two terms, which agree as above; so these terms contribute $2(BR)(\mathrm{d}A\wedge\mathrm{d^c}\chi)$. Finally $\mathrm{d}A\wedge\mathrm{d^c}\chi = \mathrm{d}(A\,\mathrm{d^c}\chi) - A\,\mathrm{dd^c}\chi$, and $(BR)(\mathrm{d}\psi) = -\mathrm{d}(BR)(\psi)$ for one-forms ψ, because BR has even degree. Collecting terms gives (15.7).

In general, let A_l be smooth plurisubharmonic functions decreasing to A on a neighborhood of $\overline{W}$, for instance mollifications (Proposition 8.2.2(2)). Then (15.7) holds for A_l, and every term converges to the corresponding term for A. For the first, third and last terms this follows by dominated convergence against the fixed signed measures $\mathrm{dd^c}(BR)$, $\mathrm{dd^c}\chi\wedge R$ and the coefficient measures of $\mathrm{d}(BR)$, since the A_l are uniformly bounded on $\operatorname{Supp}\chi$ and converge pointwise to A. For the second term it follows from the weighted form of Theorem 9.2.5, with the constant sequence of weights B. This proves the claim. Since both sides of (15.7) are bilinear in (A,B), the identity also holds for differences of bounded plurisubharmonic functions, and in particular for $A = a$ and $B = b$ near $\overline{W}$.

Conclusion. Let (χ_i) be a finite smooth partition of unity on X subordinate to coordinate balls as above (Proposition 7.1.3). Summing (15.7) for $\chi = \chi_i$, $A = a$ and $B = b$ over i, and using that abR, $\mathrm{d}(bR)$ and $\mathrm{dd^c}(aR)$, $\mathrm{dd^c}(bR)$ are global currents, we obtain

$$\int_X a\,\mathrm{dd^c}b\wedge R - \int_X b\,\mathrm{dd^c}a\wedge R = -\int_X ab\,\mathrm{dd^c}\Big(\sum_i \chi_i\Big)\wedge R - 2\,\mathrm{d}(bR)\Big(a\,\mathrm{d^c}\sum_i \chi_i\Big) = 0,$$

because $\sum_i \chi_i = 1$. This is the first equality. For the second, note that $\mathrm{dd^c}(abR)$ is a finite signed measure, being a combination of the mixed gradient and the measures

$a\,\mathrm{dd^c}\,b\wedge R$ and $b\,\mathrm{dd^c}\,a\wedge R$, and that its total integral is $\mathrm{dd^c}\,(abR)(1) = (abR)(\mathrm{dd^c}\,1) = 0$, because the constant function 1 is a test function on the compact manifold X. Integrating the definition of the mixed gradient over X and using the first equality, we get $\int_X \mathrm{d}a\wedge\mathrm{d^c}\,b\wedge R = -\int_X a\,\mathrm{dd^c}\,b\wedge R$. □

Theorem 15.4.3 *Let $T_1,\dots,T_p$, $1\le p\le n$, be closed positive $(1,1)$-currents on the compact Kähler manifold X. Then their non-pluripolar product is finite. More precisely, write $T_j=\theta_j+\mathrm{dd^c}\varphi_j$ with $\varphi_j\in\mathrm{PSH}(X,\theta_j)$, and choose $C\ge 0$ such that the forms $\tau_j=\theta_j+C\omega$ are semipositive. Then*

$$\int_X \langle T_1\wedge\cdots\wedge T_p\rangle\wedge\omega^{n-p} \le \int_X \tau_1\wedge\cdots\wedge\tau_p\wedge\omega^{n-p}. \tag{15.8}$$

In particular, if the forms θ_j are semipositive, we may take $C=0$.

Proof The decomposition $T_j=\theta_j+\mathrm{dd^c}\varphi_j$ exists by Theorem 14.1.7. Since $\tau_j\ge\theta_j$ and $\tau_j\ge 0$, φ_j is τ_j-plurisubharmonic, and so is every constant; hence $\max\{\varphi_j,-k\}$ is a bounded τ_j-plurisubharmonic function, and

$$R_{j,k}=\tau_j+\mathrm{dd^c}\max\{\varphi_j,-k\}$$

is a closed positive current. By Proposition 15.4.1, the total mass $\int_X R_{1,k}\wedge\cdots\wedge R_{p,k}\wedge\omega^{n-p}$ equals the right side of (15.8) for every k.

Local finiteness. Let W be a relatively compact coordinate chart, and choose smooth functions h_j and h near $\overline{W}$ with $\mathrm{dd^c}\,h_j=\theta_j$ and $\mathrm{dd^c}\,h=\omega$. Then $u_j=\varphi_j+h_j$ is a plurisubharmonic potential of T_j on W. Let B be an integer with $h_j\le B$ on W for all j, fix an integer $l\ge 1$, and put $O_l(u)=\bigcap_j\{u_j>-l\}$. On $W\cap O_l(u)$ we have $\varphi_j>-l-B$, so for every integer $k\ge l+B$ the bounded plurisubharmonic functions $\max\{u_j,-l\}+Ch$ and $\max\{\varphi_j,-k\}+h_j+Ch$ both equal u_j+Ch there; the second is a local potential of $R_{j,k}$. By plurifine locality,

$$\mathbf{1}_{O_l(u)}\bigwedge_j(\mathrm{dd^c}\max\{u_j,-l\}+C\omega)=\mathbf{1}_{O_l(u)}\bigwedge_j R_{j,k}\quad\text{on } W.$$

Expanding the left side into positive terms gives

$$\mathbf{1}_{O_l(u)}\bigwedge_j\mathrm{dd^c}\max\{u_j,-l\}\le\mathbf{1}_{O_l(u)}\bigwedge_j R_{j,k}\le\bigwedge_j R_{j,k}.$$

Let $K\subset W$ be compact. On a neighborhood of K in W we have $\beta\le c\,\omega$ for some $c>0$ (Lemma 13.1.1), so by the last display and (15.3), the trace masses on K of the truncated local products are at most c^{n-p} times the right side of (15.8), uniformly in l. Hence the non-pluripolar product is locally finite everywhere, and $\langle T_1\wedge\cdots\wedge T_p\rangle$ is a closed positive current by Theorem 15.2.1.

Mass bound. By the same argument, the currents $T_j+C\omega=\tau_j+\mathrm{dd^c}\varphi_j$, with local potentials u_j+Ch, have a locally finite non-pluripolar product. The sets $G_k=\bigcap_j\{\varphi_j>-k\}$ increase and exhaust the complement of the union of the pole

sets of the φ_j. On G_k the global truncations $\max\{\varphi_j, -k\}$ coincide with φ_j. If B' is an integer with $h_j + Ch \geq -B'$ on W for all j, then on $G_k \cap W$ all local potentials satisfy $u_j + Ch > -k - B'$, so they coincide there with their truncations at level $k + B'$, and also with the local potentials $\max\{\varphi_j, -k\} + h_j + Ch$ of $R_{j,k}$. By plurifine locality and Lemma 15.1.2, $\mathbf{1}_{G_k} \bigwedge_j R_{j,k} = \mathbf{1}_{G_k} \langle \bigwedge_j (T_j + C\omega) \rangle$, and these restrictions increase to $\langle \bigwedge_j (T_j + C\omega) \rangle$. Their total masses against ω^{n-p} are at most the right side of (15.8), hence so is that of the limit. Finally, expanding $\langle \bigwedge_j (T_j + C\omega) \rangle$ by multilinearity (Proposition 15.1.5) into a sum of positive currents, one of which is $\langle \bigwedge_j T_j \rangle$, we obtain $\langle \bigwedge_j T_j \rangle \leq \langle \bigwedge_j (T_j + C\omega) \rangle$, which proves (15.8). □

Example 15.4.4 On $\mathbb{P}^1$, the current of integration $[0]$ at the point 0 satisfies $\langle [0] \rangle = 0$. Indeed, near 0 it has the potential $\log |z|^2$ (Theorem 7.4.5), whose pole set is $\{0\}$, so $\langle [0] \rangle = \mathbf{1}_{\{z \neq 0\}} [0] = 0$ there by Proposition 15.3.6; away from 0 the current vanishes. Thus $[0]$ has mass one while $\langle [0] \rangle$ has mass zero: discarding the poles can change the cohomology class, even in degree one.

15.4.1 Mixed lower bounds

Lower bounds $\langle T_i^n \rangle \geq f_i \nu$ by densities with respect to a common measure ν pass to mixed products, as they do for bounded potentials (Theorem 9.11.1): we apply the bounded inequality to the truncated potentials and restrict to the set where no truncation is active. A pluripolar set charged by ν causes no difficulty, because there the densities f_i vanish ν-almost everywhere.

Proposition 15.4.5 *Let $T_1, \ldots, T_n$ be closed positive $(1,1)$-currents on a compact Kähler manifold X of dimension $n \geq 1$, not necessarily distinct. Let ν be a positive Radon measure on X, and let $f_1, \ldots, f_n$ be nonnegative Borel functions on X such that*

$$\langle T_i^n \rangle \geq f_i \nu \qquad (1 \leq i \leq n).$$

Then

$$\langle T_1 \wedge \cdots \wedge T_n \rangle \geq \left(\prod_{i=1}^{n} f_i^{1/n} \right) \nu.$$

All the products are finite by Theorem 15.4.3; in particular each f_i is ν-integrable, and so is $\prod_i f_i^{1/n}$.

Proof For $n = 1$ the conclusion is the hypothesis, so let $n \geq 2$. Work on a relatively compact coordinate domain on which $T_i = \mathrm{dd}^{\mathrm{c}} a_i$ with plurisubharmonic potentials a_i bounded above, and put

$$a_{i,j} = \max\{a_i, -j\}, \qquad G_j = \bigcap_{i=1}^{n} \{a_i > -j\}.$$

Since G_j is contained in the level-j set $\{a_i > -j\}$ of the tuple $(a_i, \dots, a_i)$, Lemma 15.1.2 shows that the bounded measure $(\mathrm{dd^c} a_{i,j})^n$ agrees with $\langle T_i^n \rangle$ on G_j. By positivity, therefore, on the whole coordinate domain

$$(\mathrm{dd^c} a_{i,j})^n \geq \mathbf{1}_{G_j} (\mathrm{dd^c} a_{i,j})^n \geq \mathbf{1}_{G_j} f_i \nu \qquad (1 \leq i \leq n).$$

Now Theorem 9.11.1, applied to the densities $\mathbf{1}_{G_j} f_i$, gives $\mathrm{dd^c} a_{1,j} \wedge \cdots \wedge \mathrm{dd^c} a_{n,j} \geq \mathbf{1}_{G_j} \prod_i f_i^{1/n} \nu$. Since G_j is the level-j set of the tuple $(a_1, \dots, a_n)$, the left side agrees on G_j with the non-pluripolar product of the T_i, by Lemma 15.1.2. Restricting to G_j and using positivity (Lemma 15.1.1), we obtain

$$\langle T_1 \wedge \cdots \wedge T_n \rangle \geq \mathbf{1}_{G_j} \prod_i f_i^{1/n} \nu.$$

The complement of $\bigcup_j G_j$ is the union P of the pole sets of the a_i. Every $\langle T_i^n \rangle$ gives zero mass to the whole pluripolar set P, not only to the pole set of a_i, so the inequality $\langle T_i^n \rangle \geq f_i \nu$ forces $f_i = 0$ ν-almost everywhere on P, for every i. Hence $\mathbf{1}_{G_j} \prod_i f_i^{1/n}$ increases to $\prod_i f_i^{1/n}$ ν-almost everywhere as $j \to \infty$, and monotone convergence on every Borel subset of the coordinate domain removes the indicator. Since finitely many such coordinate domains cover X, and an inequality of measures that holds on each set of a finite open cover holds on X (decompose a Borel set into Borel pieces, each contained in one set of the cover), the asserted inequality follows.

Finally, $\int_X f_i \, \mathrm{d}\nu \leq \int_X \langle T_i^n \rangle < \infty$. To integrate the product, we use the inequality between weighted geometric and arithmetic means: for $y_1, \dots, y_m \in [0, \infty]$ and weights $\lambda_i > 0$ with $\sum_i \lambda_i = 1$,

$$\prod_i y_i^{\lambda_i} \leq \sum_i \lambda_i y_i. \tag{15.9}$$

If all y_i are positive and finite, this is Lemma 1.1.4 for the exponential function on the probability space $\{1, \dots, m\}$ with masses λ_i, on which the integrand takes the values $\log y_i$; otherwise the left side vanishes or the right side is infinite. With $\lambda_i = 1/n$, it gives $\int_X \prod_i f_i^{1/n} \, \mathrm{d}\nu \leq \frac{1}{n} \sum_i \int_X f_i \, \mathrm{d}\nu < \infty$. □

Corollary 15.4.6 *Let T, S be closed positive $(1,1)$-currents on a compact Kähler manifold X of dimension $n \geq 1$. Let ν be a positive Radon measure on X, and let f, g be nonnegative Borel functions on X such that*

$$\langle T^n \rangle \geq f \nu, \qquad \langle S^n \rangle \geq g \nu.$$

Then for every integer $1 \leq k \leq n-1$,

$$\langle T^k \wedge S^{n-k} \rangle \geq f^{k/n} g^{(n-k)/n} \nu.$$

The endpoint cases $k = 0$ and $k = n$ are the hypotheses; for $n = 1$ there are no other cases.

Proof Apply Proposition 15.4.5 with k factors equal to T, with density f, and $n-k$ factors equal to S, with density g. □

Corollary 15.4.7 *Under the hypotheses of Corollary 15.4.6, for every* $0 \le t \le 1$,

$$\big\langle \big((1-t)T + tS\big)^n \big\rangle \ge \big((1-t) f^{1/n} + t g^{1/n}\big)^n \nu.$$

Proof For $t = 0$ and $t = 1$ this is the hypothesis. Otherwise, multilinearity (Proposition 15.1.5) expands the left side as $\sum_{k=0}^{n} \binom{n}{k} (1-t)^k t^{n-k} \langle T^k \wedge S^{n-k} \rangle$. Bounding the terms with $1 \le k \le n-1$ by Corollary 15.4.6 and the terms with $k = 0, n$ by the hypotheses, we obtain

$$\big\langle \big((1-t)T + tS\big)^n \big\rangle \ge \sum_{k=0}^{n} \binom{n}{k} (1-t)^k t^{n-k} f^{k/n} g^{(n-k)/n} \nu = \big((1-t) f^{1/n} + t g^{1/n}\big)^n \nu$$

by the binomial formula. The measure on the right is finite, because f and g are ν-integrable by the hypotheses and each density $f^{k/n} g^{(n-k)/n}$ is at most $\frac{k}{n} f + \frac{n-k}{n} g \le f + g$: for $k = 0$ and $k = n$ this is an equality, and for $1 \le k \le n-1$ it is (15.9). □

15.5 Fiber integration and modifications

We now pass the fiber formula of Lemma 15.3.5 through the truncations that define non-pluripolar products on a product of compact Kähler manifolds. The resulting identity, weighted by a bounded difference of potentials, is the form needed for relative energy in Chapter 17. We also show that non-pluripolar products commute with modifications.

Corollary 15.5.1 *Let X and Z be connected compact Kähler manifolds of dimensions r and m, with projections p_X, p_Z from $X \times Z$. Let θ be a smooth closed real $(1,1)$-form on X, let $\varphi \in \mathrm{PSH}(X, \theta)$, and let $\gamma \ge 0$ be a smooth closed $(1,1)$-form on Z. Put*

$$\Theta = p_X^* \theta + p_Z^* \gamma, \qquad \Phi = p_X^* \varphi, \qquad T = \Theta + \mathrm{dd^c} W, \quad S = \Theta + \mathrm{dd^c} \Phi,$$

where $W \in \mathrm{PSH}(X \times Z, \Theta)$ satisfies $\Phi - C \le W \le \Phi + C$ for a constant $C \ge 0$. Then for every $z \in Z$ the function $W_z = W(\,\cdot\,, z)$ belongs to $\mathrm{PSH}(X, \theta)$, and for every integer $0 \le j \le r$ and every smooth real (m,m)-form η on Z,

$$\begin{aligned} &\int_{X \times Z} (W - \Phi)\, p_Z^* \eta \wedge \langle T^j \wedge S^{r-j} \rangle \\ &\qquad = \int_Z \left(\int_X (W_z - \varphi) \langle (\theta + \mathrm{dd^c} W_z)^j \wedge (\theta + \mathrm{dd^c} \varphi)^{r-j} \rangle \right) \eta(z). \end{aligned} \tag{15.10}$$

The same identity holds with both difference weights replaced by 1. *The weights are undefined only on the common pole set of the two potentials involved, where they may be given any bounded values.*

Proof In product charts choose smooth local potentials $a(x)$ of θ and $b(z)$ of γ. Then $W + a + b$ is plurisubharmonic, so by Proposition 8.2.3 its restriction $W_z + a + b(z)$ to a fiber is plurisubharmonic in x or identically $-\infty$; the second alternative is excluded, because $W_z \geq \varphi - C$ and φ is not identically $-\infty$ on any open set. Hence W_z is quasi-plurisubharmonic with $\theta + \mathrm{dd^c} W_z \geq 0$, that is, $W_z \in \mathrm{PSH}(X, \theta)$. All the products in (15.10) are finite by Theorem 15.4.3, and (15.8) bounds the fiber masses uniformly in z, because the shifted background can be chosen from the fixed form θ, independently of W_z.

Put $w = W + a + b$, $v = \Phi + a + b$ and $O_k = \{w > -k\} \cap \{v > -k\}$ in a product chart. Let $\rho \geq 0$ be a smooth function with compact support in the chart, let $G \geq 0$ be bounded and Borel, and let $\eta \geq 0$. Apply (15.4) to the bounded functions $\max\{w, -k\}$ and $\max\{v, -k\}$, with multiplicities j and $r - j$, and to the weight $\rho G \mathbf{1}_{O_k}$. The set O_k is contained in the level-k set of the tuple formed by j copies of w and $r - j$ copies of v (if $j = 0$ or $j = r$, one of the two conditions defining O_k concerns a potential that does not occur in the tuple). Hence, by Lemma 15.1.2, the restriction to O_k of the bounded mixed product equals the restriction to O_k of the non-pluripolar product, on the total space and, with $(O_k)_z$ in place of O_k, on every fiber. These restrictions increase with k, and they increase to the non-pluripolar products, because the complements of $\bigcup_k O_k$ and of $\bigcup_k (O_k)_z$ are contained in the pole sets of w, v and of w_z, v_z, which are pluripolar and carry no mass. Monotone convergence, on the total space and in z, therefore gives the fiber formula for the non-pluripolar products with weight ρG.

Taking $G = 1$ gives the unweighted identity. Since $\Phi - C \leq W \leq \Phi + C$, the potentials W and Φ have the same pole set, and so do W_z and φ; take $G = W - \Phi + C \in [0, 2C]$ off this set and $G = C$ on it, and subtract C times the unweighted identity. Summing over a finite partition of unity by such functions ρ (Proposition 7.1.3), and writing a signed η as a difference of nonnegative forms, gives (15.10). All integrals are finite by the uniform mass bound and $|W - \Phi| \leq C$, and the values assigned on the common pole set do not matter, because this set is pluripolar and carries no mass for any of the products. □

Proposition 15.5.2 *Let $\pi\colon \widetilde{X} \to X$ be a modification between compact Kähler manifolds, that is, a proper bimeromorphic holomorphic map, and let $1 \leq p \leq \dim X$. For closed positive $(1,1)$-currents $T_1, \ldots, T_p$ on X,*

$$\pi_* \langle \pi^* T_1 \wedge \cdots \wedge \pi^* T_p \rangle = \langle T_1 \wedge \cdots \wedge T_p \rangle,$$

and for closed positive $(1,1)$-currents $R_1, \ldots, R_p$ on $\widetilde{X}$,

$$\pi_* \langle R_1 \wedge \cdots \wedge R_p \rangle = \langle \pi_* R_1 \wedge \cdots \wedge \pi_* R_p \rangle.$$

In particular, in top degree the corresponding total masses agree.

Proof The pullback $\pi^* T_j$ is the closed positive current of Proposition 7.5.4, defined by pulling back local plurisubharmonic potentials; a pulled-back potential is not identically $-\infty$ on any open set, because π is biholomorphic over a dense open set

whose preimage is dense (Proposition A.7.3 and the definition of bimeromorphic maps in Appendix A.3). The pushforward $\pi_* R$ of a closed positive current is closed and positive, since pullback of smooth forms commutes with d and preserves strong positivity (Proposition 7.2.1). All products in the statement are therefore defined, and they are finite by Theorem 15.4.3.

Choose a nowhere dense analytic subset $A \subset X$ such that $\pi^{-1}(A)$ is nowhere dense and π restricts to a biholomorphism $\pi^{-1}(X \setminus A) \to X \setminus A$. Both A and $\pi^{-1}(A)$ are pluripolar: near each point, a nowhere dense analytic set is contained in the zero set of a holomorphic function f that does not vanish identically on a connected neighborhood, hence in the pole set of $\log |f|^2$. For a point $\widetilde{x} \in \pi^{-1}(X \setminus A)$, use on a neighborhood of $\widetilde{x}$ the coordinates $z \circ \pi$, where z are coordinates near $\pi(\widetilde{x})$. In these coordinates π is the identity, and the local potentials $u_j \circ \pi$ of $\pi^* T_j$ are the local potentials u_j of T_j. Since non-pluripolar products do not depend on the choice of coordinates or of local potentials, the two sides of the first identity agree on $X \setminus A$.

It remains to show that neither side charges A. The right side gives the pluripolar set A zero mass. The product $S = \langle \pi^* T_1 \wedge \cdots \wedge \pi^* T_p \rangle$ gives $\pi^{-1}(A)$ zero mass, and hence $\pi_* S$ gives A zero mass. Indeed, in a chart of X, the trace measure of $\pi_* S$ is the image under π of the positive measure $S \wedge \pi^* \beta^{n-p}$, which does not charge $\pi^{-1}(A)$, and the coefficient measures of $\pi_* S$ are bounded by a multiple of this trace measure (Theorem 7.2.5). Hence the two sides of the first identity have the same coefficient measures on $X \setminus A$ and on A, and they are equal. The same argument on $X \setminus A$, where now the local potentials of $\pi_* R_j$ are those of R_j in the coordinates $z \circ \pi$, proves the second identity. In top degree, the total mass of $\pi_* S$ is $(\pi_* S)(1) = S(\pi^* 1) = S(1)$, which proves the assertion on masses. □

15.6 Capacity convergence and loss of mass

Bedford–Taylor products of locally uniformly bounded potentials are continuous under convergence in capacity (Theorem 9.4.8). For unbounded potentials this fails, because mass can escape into the poles in the limit (Example 15.6.6). What survives is a lower semicontinuity statement, which becomes continuity when the total masses do not drop in the limit. Throughout, capacities are the local Monge–Ampère capacities Cap_D of Definition 9.3.2, used in coordinate charts as in Definition 9.4.5, and not the global capacity c_ω of Section 14.8. Accordingly, a Borel function on X is quasi-continuous (Definition 9.4.1), and a net of Borel functions on X converges locally in capacity (Definition 9.4.5), if this holds in the charts of a finite atlas whose charts have bounded images; by the following lemma, this does not depend on the atlas. We then extend convergence in capacity to functions with poles.

Lemma 15.6.1 *Let (W, z) and (W', z') be holomorphic charts of a complex manifold whose images $D = z(W)$ and $D' = z'(W')$ are bounded, and let $K \subset W \cap W'$ be compact. Then there is a constant C such that*

$$\mathrm{Cap}^*_{D'}(z'(E)) \le C\,\mathrm{Cap}^*_D(z(E)) \qquad \textit{for every set } E \subset K,$$

and $\mathrm{Cap}_{D'}(z'(E)) \le C\,\mathrm{Cap}_D(z(E))$ *for every Borel set* $E \subset K$. *Consequently, for a net of Borel functions on a complex manifold, convergence locally in capacity in the charts of an atlas whose charts have bounded images implies the same in every chart with bounded image; similarly for quasi-continuity of a Borel function.*

Proof For every $x \in K$ choose $r_x > 0$ such that the closed ball $\overline{B(z(x), 2r_x)}$ lies in $z(W \cap W')$. Finitely many of the sets $z^{-1}(B(z(x_i), r_i))$, $r_i = r_{x_i}$, cover K; put $B_i = B(z(x_i), 2r_i)$ and $K_i = K \cap z^{-1}(\overline{B(z(x_i), r_i)})$. The map $F_i = z' \circ z^{-1}$ is a biholomorphism from B_i onto the bounded open set $B'_i = F_i(B_i) \subset D'$. Let $E \subset K$ and $E_i = E \cap K_i$. Every competitor for $\mathrm{Cap}_{D'}$ restricts to a competitor for $\mathrm{Cap}_{B'_i}$, so $\mathrm{Cap}_{D'}(G) \le \mathrm{Cap}_{B'_i}(G)$ for Borel sets $G \subset B'_i$, and taking infima over open sets, $\mathrm{Cap}^*_{D'} \le \mathrm{Cap}^*_{B'_i}$ on subsets of B'_i. By Proposition 9.3.3(5) and (4), applied to the ball $B_i \Subset D$ and the compact set $\overline{B(z(x_i), r_i)} \subset B_i$, there is a constant C_i with

$$\begin{aligned}\mathrm{Cap}^*_{D'}(z'(E_i)) &\le \mathrm{Cap}^*_{B'_i}(z'(E_i)) = \mathrm{Cap}^*_{B_i}(z(E_i)) \\ &\le C_i\,\mathrm{Cap}^*_D(z(E_i)) \le C_i\,\mathrm{Cap}^*_D(z(E)).\end{aligned}$$

By subadditivity of $\mathrm{Cap}^*_{D'}$ (Proposition 9.3.3), the assertion holds with $C = \sum_i C_i$. If E is a Borel set, then so are E_i, $z(E_i)$ and $z'(E_i)$, because z and z' are homeomorphisms onto their images. The restriction argument above and Proposition 9.3.3(4),(5) also apply to the capacities of Borel sets, so the same chain of inequalities holds with Cap in place of Cap^*, and the subadditivity of $\mathrm{Cap}_{D'}$ gives the second assertion with the same constant.

For the consequences, let (W', z') be a chart with bounded image and $K' \subset W'$ compact. Cover K' by finitely many compact sets $K'_i \subset W' \cap W_i$, where the (W_i, z_i) are charts of the given atlas. Suppose that (f_a) converges to f locally in capacity in the charts W_i. By subadditivity, the outer capacity in $D' = z'(W')$ of the image of $K' \cap \{f_a > f + \delta\}$ is at most the sum over i of the outer capacities of the images of $K'_i \cap \{f_a > f + \delta\}$. By the first part, each of these is at most a constant times the corresponding outer capacity in $z_i(W_i)$, which tends to zero; similarly for $\{f_a < f - \delta\}$.

If f is quasi-continuous in the charts W_i and $\varepsilon > 0$, choose open sets $G_i \subset z_i(W_i)$ of small capacity such that f is continuous on $K'_i \setminus z_i^{-1}(G_i)$. By the first part, $z'(K'_i \cap z_i^{-1}(G_i))$ has outer capacity at most $C_i\,\mathrm{Cap}_{z_i(W_i)}(G_i)$ in D', so it is contained in an open set $G'_i \subset D'$ with $\mathrm{Cap}_{D'}(G'_i) < \varepsilon/N$, where N is the number of sets K'_i, provided the G_i are chosen small enough. Then $G' = \bigcup_i G'_i$ satisfies $\mathrm{Cap}_{D'}(G') < \varepsilon$, and the set $F = K' \setminus z'^{-1}(G')$ is the union of the finitely many relatively closed sets $F \cap K'_i$, on each of which f is continuous; hence f is continuous on F. □

Definition 15.6.2 For $s, t \in [-\infty, \infty)$ put $d_*(s,t) = |\arctan s - \arctan t|$, where $\arctan(-\infty) = -\pi/2$. A net (u_a) of plurisubharmonic functions on a bounded open set $D \subset \mathbb{C}^n$ converges in *extended capacity* to a plurisubharmonic function u if, for every compact set $K \subset D$ and every $\delta > 0$, the outer capacity of $K \cap \{d_*(u_a, u) >$

$\delta\}$ tends to zero, the capacity being taken as in Definition 9.4.5. A net of θ-plurisubharmonic functions on a compact manifold converges in extended capacity if the local plurisubharmonic functions obtained by adding smooth local potentials of θ do so on the charts of a finite coordinate cover by charts with bounded images. This does not depend on the choice of the local potentials or of the cover, by Lemma 15.6.3 below.

Lemma 15.6.3 *Let (u_k) be a sequence of plurisubharmonic functions on a bounded open set $D \subset \mathbb{C}^n$ that is locally uniformly bounded above, and let u be plurisubharmonic on D. Then $u_k \to u$ in extended capacity if and only if $\max\{u_k, -C\} \to \max\{u, -C\}$ locally in capacity for every $C > 0$. A decreasing sequence of plurisubharmonic functions with plurisubharmonic limit converges to it in extended capacity. Extended capacity convergence of θ-plurisubharmonic functions on a compact manifold does not depend on the choices made in Definition 15.6.2. If a sequence (u_k) in $\mathrm{PSH}(X, \theta)$ converges in extended capacity to an element of $\mathrm{PSH}(X, \theta)$, then the suprema $\sup_X u_k$ are bounded above.*

Proof Fix a compact set K and let M be an upper bound for all u_k and u on K. On $[-\infty, M]$, the function $\max\{s, -C\}$ is a Lipschitz function of $\arctan s$: it is constant for $\arctan s \leq \arctan(-C)$, and the derivative of the tangent is bounded on the compact interval $[\arctan(-C), \arctan M]$. Hence extended capacity convergence implies capacity convergence of the truncations. Conversely, $|\arctan s - \arctan \max\{s, -C\}| \leq \pi/2 - \arctan C$ for all s, so that

$$d_*(u_k, u) \leq |\max\{u_k, -C\} - \max\{u, -C\}| + 2(\pi/2 - \arctan C),$$

where we also used $|\arctan s - \arctan t| \leq |s - t|$ for finite s, t. Choosing C with $2(\pi/2 - \arctan C) < \delta/2$, the set $\{d_*(u_k, u) > \delta\}$ is contained in $\{|\max\{u_k, -C\} - \max\{u, -C\}| > \delta/2\}$, which proves the equivalence.

Adding the same bounded real function h to s and t changes d_* by at most a bounded factor: since $1+\sigma^2 \leq 2(1+h^2)(1+(\sigma+h)^2)$, the formula $\arctan t - \arctan s = \int_s^t \mathrm{d}\sigma/(1+\sigma^2)$ gives $d_*(s+h, t+h) \leq 2(1+h^2)d_*(s,t)$. Now let (W_i, h_i) and (W'_l, h'_l) be two choices of a finite cover of X by charts with bounded images and of smooth local potentials h_i of θ on W_i and h'_l on W'_l, and suppose that $u_k + h_i \to u + h_i$ in extended capacity on every W_i. Fix l, a compact set $K \subset W'_l$ and $\delta > 0$, and cover K by finitely many compact sets $K_i \subset W_i \cap W'_l$. The smooth function $h'_l - h_i$ is bounded on K_i, say by H_i, so the inequality just proved, applied with $h = h'_l - h_i$, gives

$$K_i \cap \{d_*(u_k + h'_l, u + h'_l) > \delta\} \subset K_i \cap \{d_*(u_k + h_i, u + h_i) > \delta/(2 + 2H_i^2)\}.$$

By Lemma 15.6.1, the outer capacity of the left side, computed in the chart W'_l, is at most a constant times the outer capacity of the right side, computed in W_i, which tends to zero. By subadditivity of outer capacity, the outer capacity of $K \cap \{d_*(u_k + h'_l, u + h'_l) > \delta\}$ in W'_l tends to zero. Thus extended capacity convergence does not depend on the choice of the cover or of the local potentials.

If u_k decreases to u, then for every C the truncations decrease to the bounded plurisubharmonic function $\max\{u, -C\}$, so they converge in capacity by Proposition 9.4.6; the equivalence just proved gives extended capacity convergence.

For the global assertion, put $m_k = \sup_X u_k$ and $v_k = u_k - m_k$. By Proposition 14.2.1, $\int_X |v_k|\,\omega^n \le M_0$ for a constant M_0. Suppose that $m_k \to +\infty$ along a subsequence, to which we pass. By Markov's inequality, the ω^n-volume of $\{u_k < m_k/2\} = \{v_k < -m_k/2\}$ is at most $2M_0/m_k$, which tends to zero. On the other hand, let (W, z) be one of the charts of Definition 15.6.2, with smooth local potential h of θ, and let $K \subset W$ be a compact set of positive volume, with $|h| \le H$ and $u + h \le M$ on K. On $K \cap \{u_k \ge m_k/2\}$ we have $u_k + h \ge m_k/2 - H$, so for large k this set is contained in $K \cap \{d_*(u_k + h, u + h) > \delta_0\}$, where $\delta_0 = (\pi/2 - \arctan M)/2$. The outer capacity in $z(W)$ of the image of the latter set tends to zero, and hence so does its Lebesgue measure, by Proposition 9.3.3(3). Since Lebesgue measure in the chart and the ω^n-volume are comparable on K, the volume of $K \cap \{u_k \ge m_k/2\}$ tends to zero, whereas by the first observation it tends to the volume of K, which is positive. This contradiction shows that the suprema are bounded above. □

The lower semicontinuity of non-pluripolar products under convergence in capacity is proved with the following cutoff weights. They vanish where one of the potentials is at most $-C$, and multiplied by them, non-pluripolar products become Bedford–Taylor products of bounded potentials.

Lemma 15.6.4 *Let $D \subset \mathbb{C}^n$ be a bounded open set and $1 \le p \le n$. For $1 \le j \le p$ let $(v_{j,k})_{k\ge1}$ be a sequence of plurisubharmonic functions on D that is locally uniformly bounded above and converges in extended capacity to a plurisubharmonic function v_j, and suppose that the non-pluripolar products $R_k = \langle \mathrm{dd^c} v_{1,k} \wedge \cdots \wedge \mathrm{dd^c} v_{p,k}\rangle$ and $R = \langle \mathrm{dd^c} v_1 \wedge \cdots \wedge \mathrm{dd^c} v_p\rangle$ are locally finite. For $C, \varepsilon > 0$ put*

$$v_{j,k}^C = \max\{v_{j,k}, -C\}, \qquad a_{j,k} = \frac{v_{j,k}^C + C}{v_{j,k}^C + C + \varepsilon}, \qquad a_k = \prod_j a_{j,k},$$

and define v_j^C, a_j and a from the v_j in the same way. Then:

(1) *the functions a_k and a are Borel and take values in $[0, 1]$, the function a is quasi-continuous, and $a_k \to a$ locally in capacity as $k \to \infty$;*
(2) *$a_k R_k = a_k\,\mathrm{dd^c} v_{1,k}^C \wedge \cdots \wedge \mathrm{dd^c} v_{p,k}^C$ and $aR = a\,\mathrm{dd^c} v_1^C \wedge \cdots \wedge \mathrm{dd^c} v_p^C$, where the products on the right are Bedford–Taylor products;*
(3) *for fixed C, the weight a increases to the indicator function of $\bigcap_j \{v_j > -C\}$ as $\varepsilon \downarrow 0$, and these indicator functions increase to the indicator function of the complement of the union of the pole sets of the v_j as $C \to \infty$.*

Proof (1) The weights are Borel functions of the Borel functions $v_{j,k}$ and v_j, and they take values in $[0, 1]$ because $v_{j,k}^C + C \ge 0$ and $v_j^C + C \ge 0$. Let $K \subset D$ be compact and $\delta > 0$. By Theorem 9.4.2 there are open sets $G_j \subset D$ with $\mathrm{Cap}_D(G_j) < \delta/p$ such that v_j is continuous on $K \setminus G_j$. Then a, a continuous function of $v_1^C, \ldots, v_p^C$, is continuous on $K \setminus \bigcup_j G_j$, and $\mathrm{Cap}_D(\bigcup_j G_j) < \delta$ by subadditivity; thus a is quasi-continuous.

By Lemma 15.6.3, $v_{j,k}^C \to v_j^C$ locally in capacity. Since $s \mapsto (s+C)/(s+C+\varepsilon)$ is Lipschitz with constant $1/\varepsilon$ on $[-C,\infty)$, and since $|\prod_j b_j - \prod_j b'_j| \le \sum_j |b_j - b'_j|$ for numbers in $[0,1]$, the set where $|a_k - a| > \delta$ is contained in the union over j of the sets where $|v_{j,k}^C - v_j^C| > \varepsilon\delta/p$. By subadditivity of outer capacity, $a_k \to a$ locally in capacity.

(2) Fix k, and let $O = \bigcap_j \{v_{j,k} > -C\}$, a plurifine open set. For every integer $m \ge C$, the set O is contained in the level-m set $O_m = \bigcap_j \{v_{j,k} > -m\}$ of the tuple $(v_{j,k})_j$, and on O the truncations $\max\{v_{j,k}, -m\}$ coincide with $v_{j,k}$, hence with $v_{j,k}^C$. By Lemma 15.1.2, $\mathbf{1}_{O_m} R_k$ is the Bedford–Taylor product of the truncations at level m restricted to O_m, and by Theorem 10.4.3 this product coincides on O with $\bigwedge_j \mathrm{dd^c} v_{j,k}^C$. Hence $\mathbf{1}_O R_k = \mathbf{1}_O \bigwedge_j \mathrm{dd^c} v_{j,k}^C$. The weight a_k vanishes outside O, because $a_{j,k} = 0$ wherever $v_{j,k} \le -C$, and multiplying by a_k gives the first identity. The same argument, applied to the tuple $(v_j)_j$, gives the second.

(3) Where $v_j > -C$ we have $v_j^C + C > 0$, so a_j increases to 1 as $\varepsilon \downarrow 0$; where $v_j \le -C$ we have $a_j = 0$. Hence a increases to the indicator function of $\bigcap_j \{v_j > -C\}$, and these sets increase to $\bigcap_j \{v_j > -\infty\}$ as $C \to \infty$. □

Theorem 15.6.5 *Let $T_{j,k} = \theta_j + \mathrm{dd^c} u_{j,k}$ and $T_j = \theta_j + \mathrm{dd^c} u_j$ be closed positive currents on X, for $1 \le j \le p \le n$ and $k \ge 1$, and suppose that $u_{j,k} \to u_j$ in extended capacity as $k \to \infty$, for every j. Put $R_k = \langle T_{1,k} \wedge \cdots \wedge T_{p,k} \rangle$ and $R = \langle T_1 \wedge \cdots \wedge T_p \rangle$. Then for every smooth strongly positive $(n-p, n-p)$-form η on X,*

$$\varliminf_k \int_X R_k \wedge \eta \ge \int_X R \wedge \eta.$$

If moreover $\varlimsup_k \int_X R_k \wedge \omega^{n-p} \le \int_X R \wedge \omega^{n-p}$, then $R_k \to R$ weakly. The lower inequality also holds locally, on any complex manifold and for compactly supported η, whenever the local products are finite and the potentials converge in extended capacity and are locally uniformly bounded above.

Proof *Lower inequality.* Work on a coordinate ball B such that $\overline{B}$ lies in a larger coordinate ball, add fixed smooth local potentials of the θ_j, defined near $\overline{B}$, and denote the resulting plurisubharmonic functions by $v_{j,k}$ and v_j. By Lemma 15.6.3 (including its independence assertion, which allows us to use the chart B), $v_{j,k} \to v_j$ in extended capacity on B, and the $v_{j,k}$ are uniformly bounded above on B, because the suprema of the $u_{j,k}$ are bounded above. The local non-pluripolar products of the $v_{j,k}$ and of the v_j are the restrictions of R_k and R to B, and they are finite by Theorem 15.4.3. For $C, \varepsilon > 0$ let $v_{j,k}^C$, v_j^C, a_k and a be as in Lemma 15.6.4. By that lemma, the weights take values in $[0,1]$, a is quasi-continuous, $a_k \to a$ locally in capacity, and

$$a_k R_k = a_k \bigwedge_j \mathrm{dd^c} v_{j,k}^C, \qquad a R = a \bigwedge_j \mathrm{dd^c} v_j^C$$

on B. Moreover, the plurisubharmonic functions $v_{j,k}^C \ge -C$ are uniformly bounded on B, and they converge to v_j^C locally in capacity, by Lemma 15.6.3.

Let $\chi \geq 0$ be a smooth function with compact support in B. By the weighted capacity continuity of Theorem 9.4.8,

$$\int \chi\, aR \wedge \eta = \lim_k \int \chi\, a_k R_k \wedge \eta \leq \varliminf_k \int \chi R_k \wedge \eta,$$

where the inequality holds because $0 \leq a_k \leq 1$ and $R_k \wedge \chi\eta$ is a positive measure. Now let $\varepsilon \downarrow 0$ and then $C \to \infty$. By Lemma 15.6.4(3), the weights a increase to the indicator function of the complement of the union of the pole sets of the v_j, which R does not charge; so monotone convergence gives $\int \chi R \wedge \eta \leq \varliminf_k \int \chi R_k \wedge \eta$. This is the local lower inequality; it used only the finiteness of the local products, the extended capacity convergence and the local upper bounds. A finite partition of unity by such functions χ (Proposition 7.1.3) gives the global inequality.

Convergence. By (15.8), the masses $\int_X R_k \wedge \omega^{n-p}$ are bounded, since the shifted backgrounds depend only on the fixed forms θ_j. Near a compact subset L of a chart with Euclidean form β, we have $\beta \leq c\,\omega$ for some $c > 0$ (Lemma 13.1.1), and hence $R_k \wedge \beta^{n-p} \leq c^{n-p} R_k \wedge \omega^{n-p}$ by (15.3). Therefore the trace measures of the R_k are locally uniformly bounded, and by Theorem 7.2.5 every subsequence of (R_k) has a weakly convergent subsequence. Its limit R' is positive, and $R' - R$ is positive, by the lower inequality applied to all smooth strongly positive forms η on X. The additional hypothesis gives $\int_X (R' - R) \wedge \omega^{n-p} \leq 0$, because ω^{n-p} is a test form on the compact manifold X. Hence the positive measure $(R' - R) \wedge \omega^{n-p}$ vanishes, so the trace measures of $R' - R$, which are bounded on compact subsets of charts by multiples of it, vanish as well, and $R' - R = 0$ by Theorem 7.2.5. Thus every subsequence of (R_k) has a further subsequence converging weakly to R, which proves $R_k \to R$. □

Example 15.6.6 On $\mathbb{P}^1$ let ω_{FS} be the Fubini–Study form of Example 14.5.6, which has total mass one, and put $u = \log|z|^2 - \log(1 + |z|^2)$ and $u_k = \max\{u, -k\}$. As computed in that example, u extends smoothly across infinity and $\omega_{\mathrm{FS}} + \mathrm{dd^c} u = [0]$. Each u_k is bounded and ω_{FS}-plurisubharmonic, so $\omega_{\mathrm{FS}} + \mathrm{dd^c} u_k$ has mass one by Proposition 15.4.1. The sequence u_k decreases to u, hence converges to it in extended capacity (Lemma 15.6.3), and the measures $\omega_{\mathrm{FS}} + \mathrm{dd^c} u_k$ converge weakly to $[0]$, because $u_k \to u$ in L^1. However, $\langle [0] \rangle = 0$ by Example 15.4.4. The weak limit retains the mass that the non-pluripolar product discards at the pole, so the mass hypothesis in Theorem 15.6.5 cannot be omitted.

15.6.1 Ordered error sets, monotone nets and weights

Convergence locally in capacity in the sense of Definition 9.4.5 controls the two one-sided error sets $\{u_a > u + \delta\}$ and $\{u_a < u - \delta\}$, while extended capacity convergence controls the arctangent distance. For plurisubharmonic functions the two notions coincide, even when the limit has poles; the comparison uses the quasi-continuity of the limit. We then show that monotone nets of plurisubharmonic functions converge in this sense, and we extend Theorem 15.6.5 to varying bounded weights.

Proposition 15.6.7 *Let (u_a) be a net of plurisubharmonic functions and u a plurisubharmonic function on a bounded open set $D \subset \mathbb{C}^n$, $n \geq 1$. Then $u_a \to u$ locally in capacity in the sense of Definition 9.4.5 if and only if $u_a \to u$ in extended capacity. A decreasing net whose pointwise limit u is plurisubharmonic converges to u in this sense. So does an increasing net whose regularized upper envelope $u = (\sup_a u_a)^*$ is plurisubharmonic; in particular, an increasing net that converges almost everywhere to a plurisubharmonic function u converges to u in this sense.*

Proof *Equivalence.* Fix a compact set $K \subset D$ and $\varepsilon > 0$. By Theorem 9.4.2 there is an open set G of outer capacity less than ε, containing the poles of u near K, such that u is finite and continuous on the compact set $F = K \setminus G$. Then u is bounded on F, say $-C \leq u \leq M$. For $\delta > 0$, the two continuous positive functions

$$s \longmapsto \arctan(s+\delta) - \arctan s, \qquad s \longmapsto \arctan s - \arctan(s-\delta)$$

have a common positive lower bound κ on $[-C, M]$. Hence, at a point of F, an ordered error of size δ in either direction implies $d_*(u_a, u) > \kappa/2$. By subadditivity, the outer capacities of the two ordered error sets in K are at most ε plus the outer capacity of $K \cap \{d_*(u_a, u) > \kappa/2\}$. Thus extended capacity convergence, followed by $\varepsilon \downarrow 0$, gives convergence in the ordered sense. Conversely, $|\arctan s - \arctan t| \leq |s - t|$ for finite s, t; if one value is $-\infty$ and the other finite, there is an ordered error for every finite threshold; and equal infinite values have extended distance zero. Hence the set where $d_*(u_a, u) > \delta$ is contained in the union of the two ordered error sets of size δ, and subadditivity proves the converse.

Decreasing nets. Let u_a decrease to u, so that $u_a \geq u$. With F and G as above, the functions $u_a - u$ are finite and upper semicontinuous on the compact set F, and decrease pointwise to zero. Hence the closed sets $\{u_a - u \geq \delta\} \cap F$ have empty intersection, so finitely many of them already have empty intersection, and since the net is decreasing, all sets with a sufficiently large index are empty. Consequently, for large a the only ordered error in K lies in G, whose capacity is less than ε. This proves convergence. Only compactness is used, so the argument applies to nets and not only to sequences.

Increasing nets. Let the net increase, with $u = (\sup_a u_a)^*$ plurisubharmonic. By Theorem 8.3.2, there are countably many indices $b_1, b_2, \ldots$ such that $(\sup_i u_{b_i})^* = u$. Since the index set is directed, there is an increasing sequence of indices a_k with $a_k \geq b_1, \ldots, b_k$, so that $u_{a_k} \geq \max\{u_{b_1}, \ldots, u_{b_k}\}$. Since every $u_a \leq \sup_a u_a \leq u$, the regularized supremum of the sequence (u_{a_k}) is u, and by Theorem 8.3.2 the increasing limit $\lim_k u_{a_k} = \sup_k u_{a_k}$ equals u almost everywhere. Hence, for every $C > 0$, the truncations $\max\{u_{a_k}, -C\}$, which lie between $-C$ and $\max\{u, -C\}$, are locally uniformly bounded and increase almost everywhere to $\max\{u, -C\}$, so they converge in capacity by Theorem 9.5.1. By Lemma 15.6.3 and the equivalence just proved, the sequence (u_{a_k}) converges to u locally in capacity in the ordered sense. For $a \geq a_k$, monotonicity gives

$$\{u_a < u - \delta\} \subseteq \{u_{a_k} < u - \delta\}, \qquad \{u_a > u + \delta\} = \varnothing,$$

so the estimates for the sequence give convergence of the whole net; the indices a_k need not be cofinal. Finally, suppose that the increasing net converges to a plurisubharmonic function u outside a Lebesgue null set N. For $x \notin N$ we have $\sup_a u_a(x) = \lim_a u_a(x) = u(x)$, so $u_a \le u$ almost everywhere, and hence everywhere: $\max\{u_a, u\}$ and u are plurisubharmonic and agree almost everywhere, so they agree everywhere (Theorems 3.2.2 and 8.2.1). Thus $v = \sup_a u_a \le u$, so that $v^* \le u$, as u is upper semicontinuous. Moreover v^* is plurisubharmonic and equals v, hence u, almost everywhere, by Theorem 8.3.2; therefore $v^* = u$, and the previous case applies. □

Corollary 15.6.8 *Let $T_{j,k}$, T_j, $1 \le j \le p \le n$, and R_k, R be as in Theorem 15.6.5. Let χ_k and χ be Borel functions on X with $0 \le \chi_k, \chi \le A < \infty$, where χ is quasi-continuous, and suppose that $\chi_k \to \chi$ locally in capacity. Then for every smooth strongly positive $(n-p, n-p)$-form η on X,*

$$\varliminf_k \int_X \chi_k R_k \wedge \eta \ge \int_X \chi R \wedge \eta.$$

If moreover

$$\varlimsup_k \int_X R_k \wedge \omega^{n-p} \le \int_X R \wedge \omega^{n-p},$$

then $\chi_k R_k \to \chi R$ weakly as currents, and weakly as Radon measures when $p = n$; this holds in particular when $\int_X R_k \wedge \omega^{n-p} \to \int_X R \wedge \omega^{n-p}$. The lower inequality also holds locally for finite local products whose potentials are locally uniformly bounded above. By Proposition 15.6.7, the hypothesis of extended capacity convergence is satisfied when, for each j, the potentials $u_{j,k}$ decrease to u_j, or increase to u_j almost everywhere.

Proof *Lower inequality.* Let B be a coordinate ball such that $\overline{B}$ lies in a larger coordinate ball, add fixed smooth local potentials of the θ_j, defined near $\overline{B}$, and denote the resulting plurisubharmonic functions on B by $v_{j,k}$ and v_j. By Lemma 15.6.3, $v_{j,k} \to v_j$ in extended capacity on B, and the $v_{j,k}$ are uniformly bounded above on B, because the suprema of the $u_{j,k}$ are bounded above; the local products are finite by Theorem 15.4.3. For $C, \varepsilon > 0$ let $v_{j,k}^C$, v_j^C and a_k, a be the truncations and cutoff weights of Lemma 15.6.4; thus $0 \le a_k, a \le 1$, a is quasi-continuous and $a_k \to a$ locally in capacity. We may assume $A > 0$. The bounded Borel weights $\chi_k a_k$ converge locally in capacity to the bounded quasi-continuous function χa, because $|\chi_k a_k - \chi a| \le A|a_k - a| + |\chi_k - \chi|$, so that the set where $|\chi_k a_k - \chi a| > \delta$ is contained in the union of the sets where $|a_k - a| > \delta/(2A)$ and where $|\chi_k - \chi| > \delta/2$. The product χa is quasi-continuous, since χ and a are continuous outside open sets of arbitrarily small capacity and capacity is subadditive.

By Lemma 15.6.4(2) and $0 \le a_k \le 1$,

$$\chi_k R_k \ge \chi_k a_k R_k = \chi_k a_k \bigwedge_{j=1}^{p} \mathrm{dd^c} v_{j,k}^C.$$

The functions $v^C_{j,k} \geq -C$ are uniformly bounded on B and converge to v^C_j locally in capacity (Lemma 15.6.3). By the weighted capacity continuity of bounded products (Theorem 9.4.8), the right side converges weakly to $\chi a \bigwedge_j \mathrm{dd^c} v^C_j = \chi a R$, the last identity by Lemma 15.6.4(2). Testing against $g\eta$ for a nonnegative test function g on B, we obtain $\int g\chi a R \wedge \eta \leq \varliminf_k \int g\chi_k R_k \wedge \eta$, and we let $\varepsilon \downarrow 0$ and then $C \to \infty$. By Lemma 15.6.4(3), the weight a increases to one outside the union of the pole sets, which R does not charge, so monotone convergence gives the local lower inequality, and a finite partition of unity (Proposition 7.1.3) gives the global one. The local argument uses only the local upper bounds, the extended capacity convergence and the finiteness of the local products.

Convergence. Under the additional mass hypothesis, Theorem 15.6.5 gives $R_k \to R$ weakly, and in particular $\int_X R_k \wedge \eta \to \int_X R \wedge \eta$. Apply the lower inequality also to the weights $A - \chi_k$, which converge in capacity to $A - \chi$. Since

$$\int_X \chi_k R_k \wedge \eta = A \int_X R_k \wedge \eta - \int_X (A - \chi_k) R_k \wedge \eta,$$

this gives the matching upper limit, so $\int_X \chi_k R_k \wedge \eta \to \int_X \chi R \wedge \eta$.

Every smooth real $(n-p, n-p)$-form η on X is the difference of two smooth strongly positive forms. Indeed, if (ρ_i) is a finite partition of unity subordinate to coordinate charts (Proposition 7.1.3), then $\rho_i \eta = \sum_l f_l e_l$ with smooth real functions f_l supported in $\operatorname{Supp} \rho_i$ and constant strongly positive forms e_l, which span the real forms by Proposition 7.2.1. If $\kappa_i \geq 0$ is a test function of the chart equal to one on $\operatorname{Supp} \rho_i$ and $c_l \geq \sup |f_l|$, then $f_l e_l = (f_l + c_l \kappa_i) e_l - c_l \kappa_i e_l$ is a difference of smooth strongly positive forms. Hence the convergence holds for all smooth real forms, and by linearity for all smooth forms, so $\chi_k R_k \to \chi R$ weakly as currents. Since $\chi_k \leq A$ and the trace masses of R_k are bounded, the coefficient measures of $\chi_k R_k$ have bounded total variation; when $p = n$, approximating continuous functions on the compact manifold X uniformly by smooth ones, using a partition of unity and Lemma 1.4.1, upgrades the convergence to weak convergence of Radon measures. Finally, for monotone sequences the extended capacity hypothesis follows from Proposition 15.6.7, applied to the local potentials. □

15.7 Mass comparison and projective fibers

Recall from Definition 14.2.6 that $\psi \in \mathrm{PSH}(X, \theta)$ is more singular than $\varphi \in \mathrm{PSH}(X, \theta)$, written $\psi \preceq \varphi$, if $\psi \leq \varphi + C$ for a constant C; we also write this as $\varphi \geq \psi - O(1)$. We now prove that the total mass of a non-pluripolar product does not increase when the potentials become more singular. If all potentials are locally bounded outside a closed set A that is the pole set of a quasi-plurisubharmonic function, this follows from Stokes' formula, because the relevant difference of potentials can be arranged to have compact support in $X \setminus A$. The general case is reduced to this one by a construction on $X \times \mathbb{P}^N$: the new potential is bounded

outside a fixed pole set, and its total mass is an average of masses along a segment of potentials, with weights that concentrate at the endpoint as $N \to \infty$.

Lemma 15.7.1 *Let $\varphi_j, \psi_j \in \mathrm{PSH}(X, \theta_j)$ for $1 \le j \le p \le n$, and put $T_j = \theta_j + \mathrm{dd^c}\varphi_j$ and $T'_j = \theta_j + \mathrm{dd^c}\psi_j$. Suppose that $\varphi_j \ge \psi_j - O(1)$ for every j. Assume moreover that there is a quasi-plurisubharmonic function τ on X such that $A = \{\tau = -\infty\}$ is closed and τ and all the φ_j, ψ_j are locally bounded on $X \setminus A$. Then*

$$\int_X \langle T_1 \wedge \cdots \wedge T_p \rangle \wedge \omega^{n-p} \ge \int_X \langle T'_1 \wedge \cdots \wedge T'_p \rangle \wedge \omega^{n-p}.$$

If every difference $\varphi_j - \psi_j$ is bounded, equality holds.

Proof We replace the factors T_j by T'_j one at a time; the intermediate tuples satisfy the same hypotheses, so it suffices to treat one step, in which a single factor changes. Write it as $\theta + \mathrm{dd^c}\varphi$ before and $\theta + \mathrm{dd^c}\psi$ after the change, and let Θ be the non-pluripolar product of the remaining factors (some primed, some not) wedged with ω^{n-p}. By Theorems 15.2.1 and 15.4.3, Θ is a closed positive current of bidimension $(1,1)$, and it gives zero mass to the set A, which is pluripolar and Borel, being closed and locally contained in the pole sets of plurisubharmonic functions.

On the open set $X \setminus A$ all potentials are locally bounded, so there the non-pluripolar products of the remaining factors are Bedford–Taylor products, and by Proposition 15.3.3 the products of the two tuples, wedged with ω^{n-p}, are the Bedford–Taylor products $(\theta + \mathrm{dd^c}\varphi) \wedge \Theta$ and $(\theta + \mathrm{dd^c}\psi) \wedge \Theta$ with base Θ; here $(\theta + \mathrm{dd^c}\varphi) \wedge \Theta = \mathrm{dd^c}(h + \varphi) \wedge \Theta$ for smooth local potentials h of θ. More generally, if Ω' is a smooth closed real $(1,1)$-form and $\varrho \in \mathrm{PSH}(X, \Omega')$ is locally bounded on $X \setminus A$, then the measure $(\Omega' + \mathrm{dd^c}\varrho) \wedge \Theta$ on $X \setminus A$ is the restriction of a global non-pluripolar product wedged with ω^{n-p}, so it has finite mass by Theorem 15.4.3. Since none of these products charges A, it suffices to prove

$$\int_{X\setminus A} (\theta + \mathrm{dd^c}\psi) \wedge \Theta \le \int_{X\setminus A} (\theta + \mathrm{dd^c}\varphi) \wedge \Theta.$$

We prove it by applying Stokes' formula on $X \setminus A$ to functions that equal $\varphi - k$ near A. After adding constants we may assume $\varphi \ge \psi$ and $\tau \le 0$. Since τ is quasi-plurisubharmonic and X is compact, there is $C_\tau \ge 0$ with $C_\tau\omega + \mathrm{dd^c}\tau \ge 0$ (Lemma 13.1.4(4)). Put $\Omega = (C_\tau+1)\omega$. Fix $0 < \varepsilon \le 1$ and put $f_k = \max\{\psi+\varepsilon\tau, \varphi-k\}$. Since $\theta + \mathrm{dd^c}\psi \ge 0$ and $\varepsilon(C_\tau\omega + \mathrm{dd^c}\tau) \ge 0$, both $\psi + \varepsilon\tau$ and $\varphi - k$ are $(\theta + \Omega)$-plurisubharmonic, hence so is f_k, and f_k is locally bounded on $X \setminus A$. Since $\psi + \varepsilon\tau - \varphi \le \varepsilon\tau$, we have $f_k = \varphi - k$ on the open neighborhood $\{\tau < -k/\varepsilon\}$ of A. Hence on $X \setminus A$ the function $g_k = f_k - \varphi + k = \max\{\psi + \varepsilon\tau - \varphi + k, 0\}$ takes values in $[0, k]$, and it vanishes outside the compact set $L_k = \{\tau \ge -k/\varepsilon\} \subset X \setminus A$.

Stokes' formula. With a smooth local potential h of $\theta + \Omega$, the Bedford–Taylor recursion gives, on $X \setminus A$,

$$(\theta+\Omega+\mathrm{dd^c}f_k)\wedge\Theta-(\theta+\Omega+\mathrm{dd^c}\varphi)\wedge\Theta = \mathrm{dd^c}\big((f_k+h)\Theta\big)-\mathrm{dd^c}\big((\varphi+h)\Theta\big) = \mathrm{dd^c}(g_k\Theta),$$

because $\mathrm{dd^c}(k\Theta) = 0$. The signed measure $\lambda_k = \mathrm{dd^c}(g_k\Theta)$ vanishes outside L_k. Let κ be a smooth function with compact support in $X \setminus A$ that equals one on a neighborhood of L_k. Then $\lambda_k(X \setminus A) = \int \kappa \, \mathrm{d}\lambda_k = (g_k\Theta)(\mathrm{dd^c}\kappa) = 0$, because $\mathrm{dd^c}\kappa$ vanishes near the support of $g_k\Theta$. Since the measures on the left have finite mass, we obtain

$$\int_{X\setminus A} (\theta + \Omega + \mathrm{dd^c} f_k) \wedge \Theta = \int_{X\setminus A} (\theta + \Omega + \mathrm{dd^c}\varphi) \wedge \Theta.$$

Passage to the limit. On $X \setminus A$ the functions f_k decrease to the locally bounded function $\psi + \varepsilon\tau$ as $k \to \infty$. By decreasing continuity with the fixed base Θ (Theorem 9.2.5), applied locally, the positive measures $(\theta + \Omega + \mathrm{dd^c} f_k) \wedge \Theta$ converge weakly on $X \setminus A$ to $(\theta + \Omega + \mathrm{dd^c}(\psi + \varepsilon\tau)) \wedge \Theta$. The mass of the limit on the open set $X \setminus A$ is at most the lower limit of the masses, because it is the supremum of the integrals of test functions $0 \le \kappa \le 1$ supported in $X \setminus A$, and each such integral is the limit of the corresponding integrals for the approximating measures. Therefore

$$\int_{X\setminus A} (\theta + \Omega + \mathrm{dd^c}(\psi + \varepsilon\tau)) \wedge \Theta \le \int_{X\setminus A} (\theta + \Omega + \mathrm{dd^c}\varphi) \wedge \Theta.$$

By additivity of Bedford–Taylor products, the measure on the left is the sum of the positive measures $(\theta + \mathrm{dd^c}\psi) \wedge \Theta$, $\varepsilon(C_\tau\omega + \mathrm{dd^c}\tau) \wedge \Theta$ and $(\Omega - \varepsilon C_\tau\omega) \wedge \Theta$, and the measure on the right is the sum of $(\theta + \mathrm{dd^c}\varphi) \wedge \Theta$ and $\Omega \wedge \Theta$; all these measures have finite mass on $X \setminus A$. Dropping the second term on the left, we get

$$\int_{X\setminus A} (\theta + \mathrm{dd^c}\psi) \wedge \Theta \le \int_{X\setminus A} (\theta + \mathrm{dd^c}\varphi) \wedge \Theta + \varepsilon C_\tau \int_X \omega \wedge \Theta,$$

and letting $\varepsilon \downarrow 0$ proves the desired inequality. If the differences are bounded, the same argument with the roles of φ and ψ exchanged gives equality. □

Lemma 15.7.2 *Let $U \subset \mathbb{C}^n$ be open, let u be smooth and plurisubharmonic and v plurisubharmonic on U, and let $N \ge 1$ be an integer. Put $\Sigma_N = \{x \in \mathbb{R}^N : x_i \ge 0,\ t(x) = \sum_i x_i \le 1\}$, and define on $U \times \mathbb{C}^N$, with coordinates (w, z),*

$$F = \sup_{x\in\Sigma_N} \left((1 - t(x))u + t(x)v + \sum_i x_i \log|z_i|^2 - \sum_i x_i^2 \right),$$

with the convention that a zero coefficient times $-\infty$ is zero. Then F is plurisubharmonic and locally bounded on $U \times \mathbb{C}^N$, and $F \ge u$. If v is locally bounded, then, with $p\colon U \times \mathbb{C}^N \to U$ the projection,

$$p_*(\mathrm{dd^c}F)^{n+N} = \binom{n+N}{n} N \int_0^1 (\mathrm{dd^c}((1-t)u + tv))^n t^{N-1} \, \mathrm{d}t, \tag{15.11}$$

in the sense that both sides have the same integral against every bounded Borel function with compact support in U; in particular the pushforward is locally finite.

Moreover, for such v, if $v - u \geq -C$ on an open set $E \subset U$, then over E the measure $(\mathrm{dd^c} F)^{n+N}$ is carried by $\{|z_i| \leq \mathrm{e}^{1+C/2} \text{ for all } i\}$.

Proof After showing that F is plurisubharmonic, we compute $(\mathrm{dd^c} F)^{n+N}$ explicitly for smooth v, and we treat locally bounded v by decreasing approximation. Points of $U \times \mathbb{C}^N$ are also denoted by q.

Step 1: F is plurisubharmonic. For fixed $x \in \Sigma_N$, the function

$$\phi_x = (1 - t(x))u + t(x)v + \sum_i x_i \log |z_i|^2 - |x|^2$$

is plurisubharmonic on $U \times \mathbb{C}^N$, as a sum of nonnegative multiples of plurisubharmonic functions and a constant (Proposition 8.2.5). Since $\phi_0 = u$ and $\phi_x \leq \max\{u, v\} + \sum_i \max\{\log |z_i|^2, 0\}$ for all x, the family (ϕ_x) is locally uniformly bounded above, and F lies between u and the right side of this inequality; hence F is locally bounded. The function $(x, q) \mapsto \phi_x(q)$ on $\Sigma_N \times (U \times \mathbb{C}^N)$ is upper semicontinuous. Indeed, if $c \geq 0$ is a continuous function of x and f is an upper semicontinuous function of q with values in $[-\infty, M]$ for some $M \in \mathbb{R}$, then cf is upper semicontinuous, with the convention $0 \cdot (-\infty) = 0$: at a point where $c = 0$, we have $cf \leq cM^+ \to 0$ nearby, and at a point where $c > 0$ this follows from the upper semicontinuity of f. This applies to the terms $(1 - t(x))u$, $t(x)v$ and $x_i \log |z_i|^2$, and a finite sum of upper semicontinuous functions with values in $[-\infty, \infty)$ is upper semicontinuous. Since Σ_N is compact, F is upper semicontinuous: if $q_k \to q$ and $x_k \in \Sigma_N$ satisfy $\phi_{x_k}(q_k) \geq F(q_k) - 1/k$, then along a subsequence with $\varlimsup_k F(q_k)$ as limit, $x_k \to x$, and $\varlimsup_k F(q_k) \leq \phi_x(q) \leq F(q)$. Hence F coincides with its upper semicontinuous regularization, which is plurisubharmonic by Theorem 8.3.2.

Step 2: regularity of F for smooth v. Let v be smooth, and work on $G = U \times (\mathbb{C}^*)^N$. Put $a = v - u$, $y_i = \log |z_i|^2$ and $b_i = (y_i + a)/2$. Since

$$(1 - t(x))u + t(x)v + \sum_i x_i y_i - \sum_i x_i^2 = u + |b|^2 - |x - b|^2,$$

we have $F = u + \gamma(b)$ with $\gamma(b) = \max_{x \in \Sigma_N}(2x \cdot b - |x|^2) = |b|^2 - |b - P(b)|^2$, where $P(b)$ is the closest point of Σ_N to b; it exists by compactness and is unique because $x \mapsto |x - b|^2$ is strictly convex on the convex set Σ_N. Differentiating $\sigma \mapsto |b - P(b) - \sigma(x' - P(b))|^2$ at $\sigma = 0^+$ for $x' \in \Sigma_N$ gives the variational inequality $(b - P(b)) \cdot (x' - P(b)) \leq 0$. Applying it to b with $x' = P(b')$ and to b' with $x' = P(b)$ and adding, we get $|P(b) - P(b')|^2 \leq (b - b') \cdot (P(b) - P(b'))$, so P is 1-Lipschitz. Moreover $\gamma(b') \geq 2P(b) \cdot b' - |P(b)|^2 = \gamma(b) + 2P(b) \cdot (b' - b)$, and symmetrically, so that

$$0 \leq \gamma(b') - \gamma(b) - 2P(b) \cdot (b' - b) \leq 2(P(b') - P(b)) \cdot (b' - b) \leq 2|b' - b|^2.$$

Hence γ is differentiable, with the Lipschitz gradient $2P$. Consequently F is of class C^1 on G with locally Lipschitz first derivatives, and

$$\mathrm{d}F = \mathrm{d}u + \sum_i P_i(b)\,(\mathrm{d}y_i + \mathrm{d}a) = (1-t)\,\mathrm{d}u + t\,\mathrm{d}v + \sum_i x_i\,\mathrm{d}y_i, \qquad x = P(b),\ t = t(x). \tag{15.12}$$

The projection is explicit. Let $\lambda(b) = 0$ if $\sum_i \max\{b_i, 0\} \le 1$; otherwise let $\lambda(b) > 0$ be the unique solution of $\sum_i \max\{b_i - \lambda, 0\} = 1$, which exists because the left side is continuous, nonincreasing, and strictly decreasing where it is positive. The point x with coordinates $\max\{b_i - \lambda(b), 0\}$ lies in Σ_N and satisfies the variational inequality: if I is the set of indices with $b_i > \lambda(b)$, then for $x' \in \Sigma_N$ the left side of the inequality is $\sum_{i\in I} \lambda(b)(x_i' - x_i) + \sum_{i\notin I} b_i x_i' \le \lambda(b)(t(x') - t(x)) \le 0$, since $t(x) = 1$ when $\lambda(b) > 0$. A point $x \in \Sigma_N$ satisfying the variational inequality for all $x' \in \Sigma_N$ is the closest point, because $|b - x'|^2 = |b-x|^2 + |x'-x|^2 - 2(b-x)\cdot(x'-x) \ge |b-x|^2$; hence $P_i(b) = \max\{b_i - \lambda(b), 0\}$.

For a nonempty set $I \subset \{1, \dots, N\}$ put $\lambda_I(b) = (\sum_{i\in I} b_i - 1)/|I|$, and consider the open sets

$$\Omega_{I,0} = \{b :\ b_i > 0\ (i \in I),\ b_i < 0\ (i \notin I),\ \textstyle\sum_{i\in I} b_i < 1\},$$
$$\Omega_{I,+} = \{b :\ \lambda_I(b) > 0,\ b_i > \lambda_I(b)\ (i \in I),\ b_i < \lambda_I(b)\ (i \notin I)\},$$

and $\Omega_{\varnothing,0} = \{b : b_i < 0 \text{ for all } i\}$. On $\Omega_{I,0}$ we have $\lambda(b) = 0$ and $P(b) = (b_i \mathbf{1}_I(i))_i$, and on $\Omega_{I,+}$ we have $\lambda(b) = \lambda_I(b)$, by the uniqueness of λ, and $P_i(b) = (b_i - \lambda_I(b))\mathbf{1}_I(i)$; in either case P is affine there. Let $\mathcal{H}$ be the finite family of affine hyperplanes $\{b_i = 0\}$, $\{\sum_{i\in I} b_i = 1\}$ and $\{b_i = \lambda_I(b)\}$, for $1 \le i \le N$ and nonempty I, omitting the empty sets $\{b_i = \lambda_{\{i\}}(b)\}$. Every b outside $\bigcup \mathcal{H}$ lies in one of the sets $\Omega_{I,0}$, $\Omega_{I,+}$: if $\lambda(b) = 0$, it lies in $\Omega_{I,0}$ with $I = \{i : b_i > 0\}$, and if $\lambda(b) > 0$, then $\lambda(b) = \lambda_I(b)$ for $I = \{i : b_i > \lambda(b)\}$, and $b \in \Omega_{I,+}$ because no b_i equals $\lambda_I(b)$.

Let $Z \subset G$ be the closed set of points with $b \in \bigcup \mathcal{H}$. For a fixed base point, each hyperplane of $\mathcal{H}$ is given by a nonconstant affine equation in $y = (\log|z_i|^2)_i$; in polar coordinates $z_i = r_i e^{i\alpha_i}$, the corresponding set is the product of the image of a hyperplane of $\mathbb{R}^N$ under the diffeomorphism $y \mapsto (e^{y_i/2})_i$ with the torus of angles, so it is a Lebesgue null set. By Fubini's theorem, Z is a Lebesgue null set. On $G \setminus Z$ the function $F = u + |b|^2 - |b - P(b)|^2$ is smooth, because P is locally affine there.

Step 3: for smooth v, $(\mathrm{dd^c}F)^{n+N}$ is carried by the interior region. On the *interior region* $\Omega_{\{1,\dots,N\},0} = \{b : b_i > 0,\ \sum_i b_i < 1\}$, the interior of Σ_N, we have $x = P(b) = b$ and $F = u + \sum_i x_i^2$. On every other region Ω, the complex Hessian of F, which is positive semidefinite by Proposition 8.2.2(1), has a vertical null vector, and hence zero determinant. Indeed, if $j \notin I$, then $x_j = P_j(b) = 0$ on Ω, so $\partial F/\partial z_j = x_j/z_j = 0$ there by (15.12), and all second derivatives $\partial^2 F/\partial z_j \partial \bar{z}_k$ vanish. If instead $I = \{1, \dots, N\}$ and $\lambda > 0$, then $\lambda = \bar{b} - 1/N$ with $\bar{b} = N^{-1}\sum_i b_i$, and since $|b|^2 = \sum_i (b_i - \bar{b})^2 + N\bar{b}^2$,

$$F = u + |b|^2 - N\lambda^2 = v + \frac{1}{N}\sum_i y_i - \frac{1}{N} + \sum_i (b_i - \bar{b})^2.$$

For $\zeta \in \mathbb{C}$ near 0, the point $(w, e^{\zeta} z_1, \dots, e^{\zeta} z_N)$ stays in the region, every y_i increases by $2\,\mathrm{Re}\,\zeta$, and the differences $b_i - \bar{b} = (y_i - \bar{y})/2$ do not change, so $F(w, e^{\zeta} z) = F(w, z) + 2\,\mathrm{Re}\,\zeta$ is harmonic in ζ. Its Laplacian at $\zeta = 0$ is a positive

multiple of the value of the complex Hessian of F on the vector $\xi = \sum_i z_i\,\partial/\partial z_i$, which therefore vanishes; since the Hessian is positive semidefinite, ξ is a null vector by the Cauchy–Schwarz inequality.

We now identify $(\mathrm{dd^c}F)^{n+N}$ on G. Let D be a first derivative of F with respect to a real coordinate; by Step 2 it has a Lipschitz constant L on a given ball $B \Subset G$. For a test function $\chi \geq 0$ supported in a smaller ball and a real coordinate q_l,

$$|\langle \partial_{q_l} D, \chi\rangle| = \left|\lim_{h\to 0} \int \frac{D(q + he_l) - D(q)}{h}\chi(q)\,\mathrm{d}q\right| \leq L \int \chi\,\mathrm{d}q,$$

so by Corollary 1.7.2 the distributional derivative $\partial_{q_l} D$ is a locally bounded function. Hence the distributional second derivatives of F are locally bounded functions on G, and on the open set $G \setminus Z$ they coincide almost everywhere with the classical ones, since F is smooth there. As Z is a null set, the Hessian determinant $\det(F_{j\bar k})$, formed with these functions, is a locally bounded function that equals the classical determinant on $G \setminus Z$.

Let $F_\varepsilon = F * \rho_\varepsilon$ be the mollifications, defined, smooth and plurisubharmonic on a relatively compact open set $G' \Subset G$ for small ε (Proposition 8.2.2(2)). Their second derivatives are the mollifications of those of F (Proposition 1.4.3); by Lemma 1.4.1 they converge in $L^1_{\mathrm{loc}}(G')$ and are locally uniformly bounded. Since the determinant is a polynomial in the entries, $\det(F_{\varepsilon,j\bar k}) \to \det(F_{j\bar k})$ in $L^1_{\mathrm{loc}}(G')$. On the other hand $F_\varepsilon \to F$ locally uniformly, because F is continuous, so $(\mathrm{dd^c}F_\varepsilon)^{n+N} \to (\mathrm{dd^c}F)^{n+N}$ weakly by Proposition 9.1.4. With (9.2) for the smooth functions F_ε, we conclude that $(\mathrm{dd^c}F)^{n+N} = \frac{(n+N)!}{\pi^{n+N}}\det(F_{j\bar k})\,\mathrm{d}V$ on G. In particular this measure does not charge the null set Z, and it vanishes on all regions except the interior one. It does not charge the pluripolar complement of G either (Corollary 10.2.2), so $(\mathrm{dd^c}F)^{n+N}$ is carried by the set $G_0 = \{(w,z) \in G : b(w,z) \in \operatorname{Int}\Sigma_N\}$ over the interior region.

Step 4: the formula for smooth v. On G_0 we have $F = u + \sum_i x_i^2$ with $x_i = (y_i + a)/2$. Since $\mathrm{dd^c}(x_i^2) = 2x_i\mathrm{dd^c}x_i + 2\,\mathrm{d}x_i \wedge \mathrm{d^c}x_i$ and $\mathrm{dd^c}x_i = \mathrm{dd^c}a/2$, the functions y_i being pluriharmonic on G,

$$\mathrm{dd^c}F = (1-t)\mathrm{dd^c}u + t\mathrm{dd^c}v + \sum_i \Gamma_i, \qquad \Gamma_i = 2\,\mathrm{d}x_i \wedge \mathrm{d^c}x_i,$$

with $t = t(x)$. The form $\Lambda = (1-t)\mathrm{dd^c}u + t\mathrm{dd^c}v$ contains only differentials of the base variables, so $\Lambda^{n+1} = 0$, and $\Gamma_i \wedge \Gamma_i = 0$. Since forms of even degree commute, the binomial expansion leaves only the term $\binom{n+N}{n}\Lambda^n \wedge (\sum_i \Gamma_i)^N$, and $(\sum_i \Gamma_i)^N = N!\,\Gamma_1 \wedge \cdots \wedge \Gamma_N$. The map $(w,z) \mapsto (w,x,\alpha)$, with $\alpha_i = \arg z_i$, is an orientation-preserving diffeomorphism of G_0 onto $U \times \operatorname{Int}\Sigma_N \times (\mathbb{R}/2\pi\mathbb{Z})^N$, with inverse $z_i = \mathrm{e}^{x_i - a(w)/2}\mathrm{e}^{\mathrm{i}\alpha_i}$. With our normalization, $\mathrm{d^c}y_i = \mathrm{d}\alpha_i/(2\pi)$, so $\Gamma_i = \mathrm{d}x_i \wedge \mathrm{d}\alpha_i/(2\pi) + \mathrm{d}x_i \wedge \mathrm{d^c}a$, and the second term contains a differential of the base variables and disappears in the product with Λ^n. Hence, in these coordinates,

$$(\mathrm{dd^c}F)^{n+N} = \binom{n+N}{n} N!\,\big(\mathrm{dd^c}((1-t(x))u + t(x)v)\big)^n(w) \wedge \bigwedge_i \Big(\mathrm{d}x_i \wedge \frac{\mathrm{d}\alpha_i}{2\pi}\Big).$$

Let $g \geq 0$ be a bounded Borel function with compact support in U. Since $(\mathrm{dd^c}F)^{n+N}$ is carried by G_0, integrating $g(w)$ against this measure in the coordinates (w, x, α), first over the angles by Fubini's theorem, we obtain

$$\int (g \circ p)\,(\mathrm{dd^c}F)^{n+N} = \binom{n+N}{n} N! \int_{\Sigma_N} \Phi(t(x))\,\mathrm{d}x,$$

where $\Phi(t) = \int_U g\,(\mathrm{dd^c}((1-t)u + tv))^n$. By multilinearity of Bedford–Taylor products, Φ is a polynomial in t. The image of Lebesgue measure on Σ_N under t is $t^{N-1}\,\mathrm{d}t/(N-1)!$ on $[0,1]$, because $\{x \in \Sigma_N : t(x) \leq t_0\} = t_0\Sigma_N$ has volume $t_0^N|\Sigma_N|$ and $|\Sigma_N| = 1/N!$ by iterated integration, and finite measures on $[0,1]$ that agree on the intervals $[0,t_0]$ agree (Lemma 1.1.6). This proves (15.11) for smooth v; for signed bounded Borel g split g into its positive and negative parts.

Step 5: support for smooth v. If $a \geq -C$ at a base point, then at the points of G_0 over it we have $y_i = 2x_i - a \leq 2 + C$, because $x_i < 1$; hence $|z_i| \leq \mathrm{e}^{1+C/2}$. Since $(\mathrm{dd^c}F)^{n+N}$ is carried by G_0, this proves the support assertion for smooth v.

Step 6: general v. Let v be locally bounded, let $U' \Subset U$ be open, and let v_l be smooth plurisubharmonic functions on a neighborhood of $\overline{U'}$ decreasing to v, for instance mollifications (Proposition 8.2.2(2)). Let F_l be the corresponding functions. For each x, the branch of F_l exceeds that of F by $t(x)(v_l - v) \in [0, v_l - v]$, so $F \leq F_l \leq F + (v_l - v)$, and the F_l decrease pointwise to F; by Step 1 they are locally uniformly bounded. By Theorem 9.2.5, $(\mathrm{dd^c}F_l)^{n+N} \to (\mathrm{dd^c}F)^{n+N}$ weakly on $U' \times \mathbb{C}^N$. If $v - u \geq -C$ on an open set E, then also $v_l - u \geq -C$ there, so by Step 5 the measures $(\mathrm{dd^c}F_l)^{n+N}$ give no mass to the open set of points over $E \cap U'$ at which some $|z_i| > \mathrm{e}^{1+C/2}$. The weak limit gives no mass to this open set either, since the mass of an open set is the supremum of the integrals of test functions $0 \leq \chi \leq 1$ supported in it. This proves the support assertion.

Now let g be continuous with compact support in U'. Since v is bounded below and u is bounded above on a relatively compact neighborhood E of $\operatorname{Supp} g$, there is C with $v - u \geq -C$ on E, so all the measures $(\mathrm{dd^c}F_l)^{n+N}$ and $(\mathrm{dd^c}F)^{n+N}$ over E are carried by $E \times \{|z_i| \leq \mathrm{e}^{1+C/2}\}$. Multiplying $g \circ p$ by a test function of z equal to one on this polydisc, we see that $\int (g \circ p)(\mathrm{dd^c}F_l)^{n+N} \to \int (g \circ p)(\mathrm{dd^c}F)^{n+N}$; here we use that weakly convergent positive measures with locally bounded masses converge against continuous functions with compact support.

On the right side of (15.11), the functions $(1-t)u + tv_l$ decrease to $(1-t)u + tv$ for each t, so $\int g\,(\mathrm{dd^c}((1-t)u + tv_l))^n \to \int g\,(\mathrm{dd^c}((1-t)u + tv))^n$ by Theorem 9.2.5; these integrals are bounded uniformly in $t \in [0,1]$ and l by Theorem 9.1.3, and dominated convergence in t applies. Hence (15.11) holds on U' when tested against continuous functions with compact support. Both sides are positive measures that are finite on compact subsets of U', so they agree as measures by the uniqueness in Theorem 1.4.5, and therefore also against bounded Borel functions with compact support. Exhausting U by such sets U' proves the lemma. □

Proposition 15.7.3 *Let $\varphi_0 \in \mathrm{PSH}(X, \theta)$, and assume that its pole set $P = \{\varphi_0 = -\infty\}$ is closed and that φ_0 is smooth on $X \setminus P$. Fix an integer $N \geq 1$, let $p_X, p_{\mathbb{P}^N}$ be the*

projections of $X \times \mathbb{P}^N$, and let ω_{FS} be the Fubini–Study form of Example 14.5.6. For $\varphi \in \mathrm{PSH}(X, \theta)$ and homogeneous coordinates $[Z_0 : \cdots : Z_N]$ put

$$s_i = \log \frac{|Z_i|^2}{\sum_{l=0}^N |Z_l|^2}, \qquad \widetilde{\theta} = p_X^* \theta + p_{\mathbb{P}^N}^* \omega_{\mathrm{FS}},$$

and let Φ be the upper semicontinuous regularization of

$$\Phi_0 = \sup_{x \in \Sigma_N} \left((1 - t(x))(\varphi_0 + s_0) + t(x)\varphi + \sum_{i=1}^N x_i s_i - \sum_{i=1}^N x_i^2 \right),$$

with the convention that a zero coefficient times $-\infty$ is zero. Then Φ is $\widetilde{\theta}$-plurisubharmonic and locally bounded outside the closed set $A = (P \times \mathbb{P}^N) \cup (X \times \{Z_0 = 0\})$, which is the pole set of the quasi-plurisubharmonic function $\varphi_0 + s_0$, and

$$\begin{aligned} &\int_{X \times \mathbb{P}^N} \langle (\widetilde{\theta} + \mathrm{dd^c}\Phi)^{n+N} \rangle \\ &\quad = \binom{n+N}{n} N \int_0^1 \left(\int_X \langle (\theta + \mathrm{dd^c}((1-t)\varphi_0 + t\varphi))^n \rangle \right) t^{N-1} \, \mathrm{d}t. \end{aligned} \tag{15.13}$$

If $\{\theta\}$ is big, a potential φ_0 with these properties exists.

Proof *The potential Φ.* Each function in the supremum is quasi-plurisubharmonic and $\widetilde{\theta}$-plurisubharmonic: its coefficients of the θ-plurisubharmonic functions φ_0 and φ are nonnegative and add up to one, and so do its coefficients of the functions s_i, since $(1 - t(x)) + \sum_i x_i = 1$. The s_i are ω_{FS}-plurisubharmonic: in the standard chart $\{Z_l \neq 0\}$, where $\omega_{\mathrm{FS}} = \mathrm{dd^c} \log \sum_m |Z_m/Z_l|^2$ as in Example 14.5.6, the function $s_i + \log \sum_m |Z_m/Z_l|^2 = \log |Z_i/Z_l|^2$ is plurisubharmonic. The family is uniformly bounded above, because φ_0 and φ are bounded above on the compact manifold X and $s_i \leq 0$. Hence, by Theorem 8.3.2 applied after adding smooth local potentials of $\widetilde{\theta}$, its regularized supremum Φ is $\widetilde{\theta}$-plurisubharmonic. The branch $x = 0$ is $\tau = \varphi_0 + s_0$, which is quasi-plurisubharmonic, equal to $-\infty$ exactly on A, and locally bounded outside A, because φ_0 is smooth on $X \setminus P$ and s_0 is smooth on $\{Z_0 \neq 0\}$. Since $\Phi \geq \Phi_0 \geq \tau$ and Φ is bounded above, Φ is locally bounded outside A; the set A is closed because P is.

Local form. Let $W \subset X \setminus P$ be a coordinate chart, choose a smooth function h on W with $\mathrm{dd^c} h = \theta$, and use the affine coordinates $z_i = Z_i/Z_0$ on $\{Z_0 \neq 0\} = \mathbb{C}^N$. There $s_0 = -\log(1 + |z|^2)$ and $s_i = \log |z_i|^2 - \log(1 + |z|^2)$, and $\log(1 + |z|^2)$ is a potential of ω_{FS}. Since the coefficients of the s_i add up to one, adding the smooth function $h + \log(1 + |z|^2)$ to each function in the supremum turns Φ_0 into the function F of Lemma 15.7.2, formed with $u = \varphi_0 + h$, which is smooth and plurisubharmonic on W, and $v = \varphi + h$, which is plurisubharmonic on W. By that lemma, F is plurisubharmonic, hence upper semicontinuous, so Φ_0 is upper semicontinuous on the open set $(X \setminus P) \times \{Z_0 \neq 0\} = (X \times \mathbb{P}^N) \setminus A$, and $\Phi = \Phi_0$ there. Thus F is a locally

bounded plurisubharmonic potential of $\widetilde{\theta} + \mathrm{dd^c}\Phi$ on $W \times \mathbb{C}^N$, and the non-pluripolar product $\langle(\widetilde{\theta} + \mathrm{dd^c}\Phi)^{n+N}\rangle$ is the Bedford–Taylor measure $(\mathrm{dd^c}F)^{n+N}$ there.

The mass formula. For $C > 0$ put $v_C = \max\{v, u - C\}$, a locally bounded plurisubharmonic function on W, and let F_C be the function of Lemma 15.7.2 formed with u and v_C. At base points w with $v(w) > u(w) - C$ we have $v_C(w) = v(w)$, so $F_C = F$ on the set $Y_C = \{(w, z) : v(w) > u(w) - C\}$. This set is plurifine open in $W \times \mathbb{C}^N$, because $(w, z) \mapsto v(w)$ is plurisubharmonic and u is continuous. By plurifine locality (Theorem 10.4.3), $(\mathrm{dd^c}F)^{n+N}$ and $(\mathrm{dd^c}F_C)^{n+N}$ agree on Y_C.

Similarly, for $0 \le t \le 1$, the function $(1-t)u + tv$ is a local potential of the closed positive current $\theta + \mathrm{dd^c}((1-t)\varphi_0 + t\varphi)$, whose non-pluripolar products are finite by Theorem 15.4.3, and on the plurifine open set $\{v > u - C\}$ it coincides with the bounded function $(1-t)u + tv_C$; so by Proposition 15.1.5,

$$\mathbf{1}_{\{v>u-C\}}(\mathrm{dd^c}((1-t)u + tv_C))^n = \mathbf{1}_{\{v>u-C\}}\langle(\mathrm{dd^c}((1-t)u + tv))^n\rangle.$$

Let $g \ge 0$ be a bounded Borel function with compact support in W. Applying Lemma 15.7.2 to F_C with the Borel weight $g\mathbf{1}_{\{v>u-C\}}$, we obtain

$$\int_{Y_C} (g \circ p_X)(\mathrm{dd^c}F)^{n+N} = \binom{n+N}{n} N \int_0^1 \Big(\int_W g\, \mathbf{1}_{\{v>u-C\}} \langle(\mathrm{dd^c}((1-t)u + tv))^n\rangle \Big) t^{N-1}\, \mathrm{d}t.$$

Let $C \to \infty$. The sets $\{v > u - C\}$ increase to $\{v > -\infty\}$, whose complement is the pole set of v; this set and its product with $\mathbb{C}^N$ are pluripolar Borel sets, which carry no mass for the non-pluripolar products on W and for the Bedford–Taylor measure $(\mathrm{dd^c}F)^{n+N}$ (Corollary 10.2.2). By monotone convergence, first in the inner integrals and then in t,

$$\int_{W\times\mathbb{C}^N} (g \circ p_X)\, \langle(\widetilde{\theta} + \mathrm{dd^c}\Phi)^{n+N}\rangle = \binom{n+N}{n} N \int_0^1 \Big(\int_X g\, \langle(\theta + \mathrm{dd^c}((1-t)\varphi_0 + t\varphi))^n\rangle \Big) t^{N-1}\, \mathrm{d}t.$$

The non-pluripolar product on $X \times \mathbb{P}^N$ gives zero mass to the pluripolar Borel set A, and the products on X give zero mass to P. Hence the left side is the integral of $g \circ p_X$ over $X \times \mathbb{P}^N$. Now cover $X \setminus P$ by countably many compact sets K_k, each contained in a chart $W_k \subset X \setminus P$, and put $B_k = K_k \setminus \bigcup_{j<k} K_j$. For a nonnegative Borel function g on X, applying the identity to the functions $\min\{g, M\}\mathbf{1}_{B_k}$ and using monotone convergence in M and in the sum over k, we obtain the identity for $g\mathbf{1}_{X\setminus P}$, and hence for g, since both sides give P zero mass. Taking $g = 1$ gives (15.13).

Existence of φ_0. If $\{\theta\}$ is big, it contains a Kähler current $\theta + \mathrm{dd^c}\psi \ge \delta\omega$ with $\delta > 0$, that is, $\psi \in \mathrm{PSH}(X, \theta - \delta\omega)$. By Lemma 13.4.1(1), applied to ψ with the background $\theta - \delta\omega$, for every integer $m \ge 1$ there is a quasi-plurisubharmonic function $G_m = m^{-1}\log R_m$ with R_m smooth and $\theta - \delta\omega + \mathrm{dd^c}G_m \ge -(c'/m)\omega$,

where the constant $c' > 0$ does not depend on m. For $m \geq c'/\delta$, the function $\varphi_0 = G_m$ is θ-plurisubharmonic; its pole set $\{R_m = 0\}$ is closed, and φ_0 is smooth on its complement, where $R_m > 0$. □

Theorem 15.7.4 *Let $\varphi, \psi \in \mathrm{PSH}(X, \theta)$ with $\varphi \geq \psi - O(1)$. Then*

$$\int_X \langle (\theta + \mathrm{dd^c}\varphi)^n \rangle \geq \int_X \langle (\theta + \mathrm{dd^c}\psi)^n \rangle.$$

Proof Suppose first that $\{\theta\}$ is big. Choose a reference potential φ_0 as in Proposition 15.7.3, which exists because the class is big, and construct the corresponding potentials Φ_N from φ and Ψ_N from ψ. If $\varphi \geq \psi - C$, then for every $x \in \Sigma_N$ the branch defining Φ_N is at least the branch defining Ψ_N minus $t(x)C \leq C$; hence the suprema and their regularizations satisfy $\Phi_N \geq \Psi_N - C$, with a constant independent of N. By Proposition 15.7.3, both are locally bounded outside the same closed set A, which is the pole set of the quasi-plurisubharmonic function $\varphi_0 + s_0$, and this function is locally bounded outside A. The product $X \times \mathbb{P}^N$ is a connected compact Kähler manifold of dimension $n + N$, with the Kähler form $p_X^*\omega + p_{\mathbb{P}^N}^*\omega_{\mathrm{FS}}$, so the results of this section apply to it.

For $0 \leq t \leq 1$ put

$$f_\varphi(t) = \int_X \langle (\theta + \mathrm{dd^c}((1-t)\varphi_0 + t\varphi))^n \rangle,$$

and define f_ψ in the same way. Applying Lemma 15.7.1 on $X \times \mathbb{P}^N$, with all $n + N$ factors equal to $\widetilde{\theta} + \mathrm{dd^c}\Phi_N$, respectively $\widetilde{\theta} + \mathrm{dd^c}\Psi_N$, and cancelling the common binomial coefficient in (15.13), we obtain

$$\int_0^1 f_\varphi(t)\, N t^{N-1}\, \mathrm{d}t \geq \int_0^1 f_\psi(t)\, N t^{N-1}\, \mathrm{d}t.$$

Writing $\theta + \mathrm{dd^c}((1-t)\varphi_0 + t\varphi) = (1-t)T_0 + tT_\varphi$ with $T_0 = \theta + \mathrm{dd^c}\varphi_0$ and $T_\varphi = \theta + \mathrm{dd^c}\varphi$, multilinearity shows that f_φ is a polynomial in t, in particular continuous on $[0, 1]$, and $f_\varphi(1)$ is the mass of φ. The probability measures $N t^{N-1}\, \mathrm{d}t$ concentrate at 1 as $N \to \infty$, since their mass on $[0, 1 - \delta]$ is $(1 - \delta)^N$. By uniform continuity the two integrals tend to $f_\varphi(1)$ and $f_\psi(1)$, which proves the theorem for big classes.

In general, $\{\theta\}$ is pseudo-effective, because it contains $\theta + \mathrm{dd^c}\varphi$. For $\varepsilon > 0$ the class of $\theta + \varepsilon\omega$ is big, since it contains the Kähler current $\theta + \varepsilon\omega + \mathrm{dd^c}\varphi \geq \varepsilon\omega$, and φ, ψ are $(\theta + \varepsilon\omega)$-plurisubharmonic. By the big case, the mass of φ with respect to $\theta + \varepsilon\omega$ is at least that of ψ. By multilinearity and Theorem 15.4.3, both masses are polynomials in ε with finite coefficients, and letting $\varepsilon \downarrow 0$ proves the theorem. □

Corollary 15.7.5 *Let $u_j, v_j \in \mathrm{PSH}(X, \theta_j)$ for $1 \leq j \leq p \leq n$, and put $T_j = \theta_j + \mathrm{dd^c}u_j$ and $T'_j = \theta_j + \mathrm{dd^c}v_j$. If every difference $u_j - v_j$ is bounded, then*

$$\int_X \langle T_1 \wedge \cdots \wedge T_p \rangle \wedge \omega^{n-p} = \int_X \langle T'_1 \wedge \cdots \wedge T'_p \rangle \wedge \omega^{n-p}.$$

If only $u_j \geq v_j - O(1)$ for every j, then the left side is at least the right side.

Proof *Bounded differences.* For real numbers $a_0, a_1, \ldots, a_p > 0$, apply Theorem 15.7.4 in both directions to the potentials $\sum_j a_j u_j$ and $\sum_j a_j v_j$, whose difference is bounded, with respect to the background $a_0\omega + \sum_j a_j \theta_j$. The two resulting masses are equal. By multilinearity, both are homogeneous polynomials of degree n in $(a_0, \ldots, a_p)$, and they agree on an open set; fixing all but one variable and using the uniqueness of polynomials in one variable successively, we see that their coefficients agree. By Proposition 15.3.3, the coefficient of $a_0^{n-p} a_1 \cdots a_p$ is $n!/(n-p)!$ times the mixed mass in the statement, which proves equality.

The ordered case. Put $v_{j,k} = \max\{u_j - k, v_j\}$. Since $v_j \leq u_j + C_j$ for constants C_j, we have $u_j - k \leq v_{j,k} \leq u_j + C_j$, so each tuple $(v_{j,k})_j$ has the same mixed mass as $(u_j)_j$, by the first part. As $k \to \infty$, $v_{j,k}$ decreases to v_j and therefore converges to it in extended capacity (Lemma 15.6.3). The lower inequality of Theorem 15.6.5, with test form ω^{n-p}, bounds the mixed mass of the v_j by these equal masses. □

Corollary 15.7.6 *Suppose that $u_{j,k} \in \mathrm{PSH}(X, \theta_j)$ increase almost everywhere to $u_j \in \mathrm{PSH}(X, \theta_j)$ as $k \to \infty$, for $1 \leq j \leq p \leq n$. Then the non-pluripolar products $\langle \bigwedge_j (\theta_j + \mathrm{dd^c} u_{j,k}) \rangle$ converge weakly to $\langle \bigwedge_j (\theta_j + \mathrm{dd^c} u_j) \rangle$, and their mixed masses against ω^{n-p} converge.*

Proof The inequality $u_{j,k} \leq u_j$ holds almost everywhere, hence everywhere, since $\max\{u_{j,k}, u_j\}$ and u_j are quasi-plurisubharmonic functions that agree almost everywhere. On coordinate charts add smooth local potentials and truncate below at a fixed level $-C$. The resulting locally uniformly bounded plurisubharmonic sequences increase almost everywhere to the truncations of the limits, so they converge in capacity by Theorem 9.5.1; by Lemma 15.6.3, the original sequences converge in extended capacity. Since $u_{j,k} \leq u_j$, Corollary 15.7.5 bounds every mixed mass of the sequence by that of the limit, and Theorem 15.6.5 gives weak convergence and convergence of the masses. This applies in particular to the regularized supremum of an increasing sequence in $\mathrm{PSH}(X, \theta_j)$ that is bounded above: it belongs to $\mathrm{PSH}(X, \theta_j)$ by Theorem 8.3.2, and it differs from the pointwise supremum only on a pluripolar set, by Corollary 10.2.2, hence only on a Lebesgue null set, because plurisubharmonic functions are locally integrable. □

15.8 Maximal mass and comparison

By the monotonicity just proved, potentials with minimal singularities have the largest non-pluripolar masses in a big class, and these maximal masses are invariants of the classes. We extend them to pseudo-effective classes by perturbing with a Kähler class, and then derive the comparison and domination principles for non-pluripolar Monge–Ampère measures. Recall that V_θ denotes the extremal potential of Definition 14.2.6.

Definition 15.8.1 For big classes $\alpha_j = \{\theta_j\}$, $1 \le j \le p \le n$, the *positive mixed intersection number* relative to the Kähler class $\{\omega\}$ is

$$I_\omega(\alpha_1, \dots, \alpha_p) = \int_X \left\langle \bigwedge_{j=1}^{p} (\theta_j + \mathrm{dd^c} V_{\theta_j}) \right\rangle \wedge \omega^{n-p}.$$

For pseudo-effective classes α_j that are not all big, we define $I_\omega(\alpha_1, \dots, \alpha_p)$ as the limit of the numbers $I_\omega(\alpha_1 + \varepsilon\{\omega\}, \dots, \alpha_p + \varepsilon\{\omega\})$, which decrease as $\varepsilon \downarrow 0$ (Proposition 15.8.2); when all α_j are big, the same limit equals the number defined by the display. In top degree we put $\mathrm{vol}(\alpha) = I_\omega(\alpha, \dots, \alpha)$, the *volume* of α; the factor ω^{n-p} is then absent.

Proposition 15.8.2 *The numbers of Definition 15.8.1 do not depend on the chosen smooth representatives or minimal potentials, nor on the Kähler form within its class. The limit in Definition 15.8.1 exists for all pseudo-effective classes, does not depend on the Kähler class used for the perturbation, and equals the number defined by the display when all α_j are big. For closed positive currents $T_j \in \alpha_j$,*

$$\int_X \langle T_1 \wedge \cdots \wedge T_p \rangle \wedge \omega^{n-p} \le I_\omega(\alpha_1, \dots, \alpha_p).$$

The numbers are positively homogeneous in each class and increase when a pseudo-effective class is added to any argument. For nef classes they equal the cohomological intersection numbers $\int_X \theta_1 \wedge \cdots \wedge \theta_p \wedge \omega^{n-p}$. Big classes have positive volume.

Proof *Big classes.* Any two potentials with minimal singularities in a fixed big class differ by a bounded function, and a change of smooth representative changes each potential by a smooth function. Hence Corollary 15.7.5 shows that I_ω does not depend on these choices. The same corollary and minimality give the upper bound for closed positive currents in big classes. If $\omega' = \omega + \mathrm{dd^c} f$ is another Kähler form in the class of ω, with f smooth, then $\omega'^{n-p} - \omega^{n-p} = \mathrm{d}\big(\mathrm{d^c} f \wedge \sum_{i=0}^{n-p-1} \omega'^{i} \wedge \omega^{n-p-1-i}\big)$ is a smooth exact form. Its integral against the closed current $\langle T_1 \wedge \cdots \wedge T_p \rangle$ vanishes, because $S(\mathrm{d}\psi) = \pm(\mathrm{d}S)(\psi) = 0$ for a closed current S and a smooth form ψ on the compact manifold X. Hence the numbers for big classes do not depend on ω within its class, and neither do their limits, since the perturbing classes $\varepsilon\{\omega\}$ are the same.

Let α be big and γ pseudo-effective, represented by a closed positive current S. Adding S to a current with minimal singularities in α gives a current in the big class $\alpha + \gamma$, which by the upper bound has mixed mass at most the number with α replaced by $\alpha + \gamma$. By multilinearity and positivity, its mixed mass is at least the number for α. Hence, when all arguments are big, replacing an argument α by $\alpha + \gamma$ increases the number. Homogeneity for big classes follows by scaling a minimal potential together with its background: if $c > 0$, then cV_θ has minimal singularities in $\mathrm{PSH}(X, c\theta)$, and the product is homogeneous in each factor by Proposition 15.1.5.

Pseudo-effective classes. By this monotonicity for big classes, the perturbed numbers decrease as $\varepsilon \downarrow 0$; they are nonnegative, so the limit exists. Every Kähler

form κ satisfies $c^{-1}\omega \le \kappa \le c\omega$ for some $c \ge 1$ (Lemma 13.1.1), so monotonicity places the perturbation by $\varepsilon\kappa$ between the perturbations by $c^{-1}\varepsilon\omega$ and $c\varepsilon\omega$, with the same fixed test form ω^{n-p}; this proves independence of the perturbing class. The same comparison shows that perturbing the arguments by different fixed multiples of $\varepsilon\omega$ gives the same limit, which extends homogeneity to pseudo-effective classes. Monotonicity also extends: if γ is pseudo-effective, then for every $\varepsilon > 0$ the number with an argument $\alpha + \varepsilon\{\omega\}$ replaced by $\alpha + \gamma + \varepsilon\{\omega\}$ is at least the original one, by the big case, and we let $\varepsilon \downarrow 0$.

If all α_j are big, choose $\delta > 0$ such that each α_j contains a Kähler current $R_j \ge \delta\omega$. Then $(1+\varepsilon/\delta)\alpha_j - (\alpha_j + \varepsilon\{\omega\}) = (\varepsilon/\delta)\{R_j - \delta\omega\}$ is pseudo-effective, so monotonicity and homogeneity give

$$I_\omega(\alpha_1, \dots, \alpha_p) \le I_\omega(\alpha_1 + \varepsilon\{\omega\}, \dots, \alpha_p + \varepsilon\{\omega\}) \le (1+\varepsilon/\delta)^p I_\omega(\alpha_1, \dots, \alpha_p),$$

and the limit is $I_\omega(\alpha_1, \dots, \alpha_p)$. For arbitrary closed positive currents $T_j \in \alpha_j$, the upper bound in the big classes $\alpha_j + \varepsilon\{\omega\}$, applied to $T_j + \varepsilon\omega$, and multilinearity give $\int_X \langle \bigwedge_j T_j \rangle \wedge \omega^{n-p} \le I_\omega(\alpha_1 + \varepsilon\{\omega\}, \dots)$, and letting $\varepsilon \downarrow 0$ proves the upper bound.

Nef and big classes. If the α_j are nef, their perturbations are Kähler classes: $\alpha_j + \varepsilon\{\omega\}$ contains $\theta_{j,\varepsilon/2} + \varepsilon\omega \ge \varepsilon\omega/2$, where $\theta_{j,\varepsilon/2} \ge -(\varepsilon/2)\omega$ is a smooth representative of α_j given by Definition 14.3.1. If $\theta_j + \varepsilon\omega + \mathrm{dd^c} f$ is a Kähler form with f smooth, then $f \in \mathrm{PSH}(X, \theta_j + \varepsilon\omega)$, so $V_{\theta_j+\varepsilon\omega} \ge f - \sup_X f$ by Lemma 14.2.7; since also $V_{\theta_j+\varepsilon\omega} \le 0$ by the same lemma, the minimal potentials are bounded. By Proposition 15.4.1, the perturbed numbers are the integrals $\int_X \bigwedge_j (\theta_j + \varepsilon\omega) \wedge \omega^{n-p}$, and their limits are the cohomological intersection numbers. Finally, let α be big and $T_0 \in \alpha$ a Kähler current with $T_0 \ge \delta\omega$. Expanding $T_0 = \delta\omega + (T_0 - \delta\omega)$ by multilinearity gives $\langle T_0^n \rangle \ge \delta^n \omega^n$, so $\mathrm{vol}(\alpha) \ge \delta^n \int_X \omega^n > 0$ by the upper bound. □

Conversely, a pseudo-effective class of positive volume is big. The proof requires the construction of a Kähler current in the class and is given in Section 17.4.4, after Monge–Ampère equations with prescribed measures have been solved.

Definition 15.8.3 A closed positive $(1,1)$-current $T \in \alpha$ has *full Monge–Ampère mass* if $\int_X \langle T^n \rangle = \mathrm{vol}(\alpha)$. We write $\mathcal{E}(X, \theta)$ for the set of $u \in \mathrm{PSH}(X, \theta)$ such that $\theta + \mathrm{dd^c} u$ has full Monge–Ampère mass. For $\phi \in \mathrm{PSH}(X, \theta)$, the class $\mathcal{E}(X, \theta, \phi)$ consists of the potentials $u \in \mathrm{PSH}(X, \theta)$ with $u \le \phi + O(1)$ and

$$\int_X \langle (\theta + \mathrm{dd^c} u)^n \rangle = \int_X \langle (\theta + \mathrm{dd^c} \phi)^n \rangle;$$

these potentials have *full mass relative to* ϕ. In a big class, $\mathcal{E}(X, \theta) = \mathcal{E}(X, \theta, V_\theta)$, because every $u \in \mathrm{PSH}(X, \theta)$ satisfies $u \le V_\theta + \sup_X u$ and $\mathrm{vol}(\{\theta\})$ is the mass of V_θ.

The class $\mathcal{E}(X, \theta)$ should not be confused with the local Cegrell class $\mathcal{E}_0(B)$ of Definition 9.10.1, nor with the pseudo-effective cone of Definition 14.3.1, a cone of cohomology classes.

Corollary 15.8.4 *Let $u_k \in \mathrm{PSH}(X,\theta)$ satisfy $u_k \le \phi + C_k$ for constants C_k, let $u \in \mathcal{E}(X,\theta,\phi)$, and suppose that $u_k \to u$ in extended capacity. Then*

$$\langle(\theta + \mathrm{dd^c} u_k)^n\rangle \longrightarrow \langle(\theta + \mathrm{dd^c} u)^n\rangle$$

weakly, and the masses converge. The same holds, without the constants C_k, if $u \in \mathcal{E}(X,\theta)$; in particular it holds for every sequence in $\mathrm{PSH}(X,\theta)$ that decreases to an element of $\mathcal{E}(X,\theta)$.

Proof By Theorem 15.7.4, the mass of each u_k is at most that of ϕ, which equals the mass of u. Hence Theorem 15.6.5, with $p = n$, gives weak convergence, and its lower inequality together with this bound gives convergence of the masses. If $u \in \mathcal{E}(X,\theta)$, then by Proposition 15.8.2 the mass of each u_k is at most $\mathrm{vol}(\{\theta\})$, which is the mass of u, and the same argument applies. Finally, decreasing sequences converge in extended capacity by Lemma 15.6.3. □

Proposition 15.8.5 *For $u \in \mathrm{PSH}(X,\theta)$ write $\mu_u = \langle(\theta + \mathrm{dd^c} u)^n\rangle$. Then for all $u, v \in \mathrm{PSH}(X,\theta)$,*

$$\int_{\{u<v\}} \mu_v \le \int_{\{u<v\}} \mu_u + \mathrm{vol}(\{\theta\}) - \mu_u(X).$$

The last two terms may be omitted if u has full mass or if $u \ge v - O(1)$.

Proof Let $\varepsilon > 0$. The function $w = \max\{u, v - \varepsilon\}$ is θ-plurisubharmonic. The sets $\{u < v - \varepsilon\}$ and $\{u > v - \varepsilon\}$ are disjoint and plurifine open. Indeed, near each point we have $u - v = (u + h) - (v + h)$ for a smooth local potential h of θ, and $u + h$, $v + h$ are plurisubharmonic, so by Proposition 10.5.1 the function $u - v$ is plurifinely continuous off the common pole set of u and v, which is disjoint from both sets. Moreover w equals $v - \varepsilon$ on the first set and u on the second. By plurifine locality (Proposition 15.1.5),

$$\mu_w(X) \ge \mu_v(\{u < v - \varepsilon\}) + \mu_u(\{u > v - \varepsilon\}).$$

The left side is at most $\mathrm{vol}(\{\theta\})$ by Proposition 15.8.2. The complement of $\{u > v-\varepsilon\}$ consists of $\{u \le v - \varepsilon\}$, which is contained in $\{u < v\}$ up to the common pole set of u and v, and that pluripolar set carries no mass. Hence

$$\mu_v(\{u < v - \varepsilon\}) \le \mathrm{vol}(\{\theta\}) - \mu_u(X) + \mu_u(\{u < v\}).$$

As $\varepsilon \downarrow 0$, the sets $\{u < v - \varepsilon\}$ increase to $\{u < v\}$, which proves the inequality. If u has full mass, the last two terms cancel. If $u \ge v - O(1)$, then $w - u$ is bounded, so $\mu_w(X) = \mu_u(X)$ by Corollary 15.7.5, and the same computation with $\mu_u(X)$ in place of the volume gives the inequality without these terms. □

Corollary 15.8.6 *Let $u, v, \rho \in \mathrm{PSH}(X,\theta)$ with $u \ge v - O(1)$ and $u \ge \rho - O(1)$. If $u \ge v$ almost everywhere for μ_u, then $u \ge v$ almost everywhere for μ_ρ. If, in addition, the class $\{\theta\}$ is big and u has minimal singularities, then $u \ge v$ everywhere on X.*

Proof After adding a constant to ρ we may assume $\rho \le u$. For $0 < t < 1$ put $q_t = (1-t)v + t\rho$. Then $u \ge q_t - O(1)$, and $\{u < q_t\} \subset \{u < v\}$, because $u < (1-t)v + t\rho \le (1-t)v + tu$ forces $u < v$. Multilinearity and Proposition 15.8.5 therefore give

$$t^n \mu_\rho(\{u < q_t\}) \le \mu_{q_t}(\{u < q_t\}) \le \mu_u(\{u < q_t\}) \le \mu_u(\{u < v\}) = 0.$$

As $t \downarrow 0$, every point where $u < v$ and $\rho > -\infty$ eventually lies in $\{u < q_t\}$, so Fatou's lemma, applied along a sequence $t \downarrow 0$, gives $\mu_\rho(\{u < v\} \cap \{\rho > -\infty\}) = 0$. Since μ_ρ gives no mass to the pole set of ρ, this proves the first assertion.

Now let the class be big and u have minimal singularities. The class contains a Kähler current $\theta + \mathrm{dd^c}\rho \ge \delta\omega$ with $\delta > 0$. Then $u \ge \rho - O(1)$ by minimality, and $\mu_\rho \ge \delta^n \omega^n$, because multilinearity expands $\langle(\theta+\mathrm{dd^c}\rho)^n\rangle = \langle(\delta\omega+(\theta+\mathrm{dd^c}\rho-\delta\omega))^n\rangle$ into positive terms, one of which is $\delta^n\omega^n$. The first assertion gives $u \ge v$ almost everywhere with respect to ω^n, that is, almost everywhere for Lebesgue measure in charts. Near each point, u and v become plurisubharmonic after adding a common smooth local potential h of θ, and then $\max\{u, v\} + h$ and $u + h$ are plurisubharmonic functions that agree almost everywhere; by Theorems 3.2.2 and 8.2.1 they agree everywhere, that is, $u \ge v$ everywhere. □

Chapter 16
Envelopes and model potentials

Throughout this chapter X is a connected compact Kähler manifold of dimension $n \geq 1$ with a fixed Kähler form ω, and θ is a smooth closed real $(1,1)$-form on X, as in the two preceding chapters. Many potentials are defined as envelopes: given an obstacle f, one takes the largest θ-plurisubharmonic function that lies below f outside a pluripolar set. For instance, the extremal potential V_θ of Definition 14.2.6 is the envelope of the obstacle 0. Two features of the global setting require care. The obstacle is often only a Borel function, so we must keep track of which inequalities hold everywhere and which only quasi-everywhere. And there may be no θ-plurisubharmonic function below the obstacle at all: two θ-plurisubharmonic functions need not have a common θ-plurisubharmonic minorant (Example 16.1.7).

The basic property of an envelope is that its non-pluripolar Monge–Ampère measure is carried by the contact set, where the envelope touches the obstacle; for quasi-continuous obstacles in a big class this is Theorem 16.2.5. Away from the contact set, the envelope of a lower semicontinuous obstacle can be replaced locally by a maximal plurisubharmonic function, so its measure vanishes there, and quasi-continuity lets us approximate a general obstacle by lower semicontinuous ones. Regularity of the obstacles also passes to the envelope. If finitely many obstacles have bounded Laplacian with respect to the Riemannian metric of ω, then so does the envelope of their minimum. For a Kähler background we prove this with an exponentially penalized Monge–Ampère equation (Theorem 16.2.24), which also shows that envelopes of finite minima of $C^{1,1}$ obstacles are $C^{1,1}$ (Theorem 16.2.26); in a big class the bound holds locally on the ample locus, the complement of the non-Kähler locus (Theorem 16.2.29). On the contact set of such an obstacle with an arbitrary potential lying below it, the non-pluripolar measure of the potential is then computed from the second derivatives of the obstacle (Theorem 16.2.31).

The second half of the chapter studies envelopes that prescribe singularities rather than values. The singularity envelope $P_\theta[u]$ of a θ-plurisubharmonic function u is the upper envelope of all nonpositive θ-plurisubharmonic functions v with $v \leq u + C$ for some constant C, which may depend on v. It has the same non-pluripolar masses as u but may be less singular, and its fixed points of positive mass are called model potentials. The main tool is extrapolation (Theorem 16.4.4): if u is more singular than

v and has almost the same mass, then for some $a > 1$ the function $v + a(u - v)$, which continues the segment from v to u beyond u, still has a θ-plurisubharmonic minorant of positive mass. From this we derive the domination principle for potentials of full mass relative to a reference potential (Theorem 16.4.6) and the characterization of $P_\theta[u]$, for u of positive mass, as the largest nonpositive potential that is less singular than u and has the same mass (Theorem 16.4.7). Rooftop envelopes satisfy a mass inequality (Theorem 16.5.5), and upper envelopes preserve the comparison of singularities through multiplier ideals (Section 16.6).

Finally we introduce the Monge–Ampère capacity $\mathrm{Cap}_{\theta,\phi}$ relative to a reference potential ϕ, prove that it is comparable with the capacity of a Kähler form when ϕ has positive mass (Theorem 16.9.8), and use it to measure the deep sublevel sets $\{u < \phi - s\}$ of potentials of full mass relative to ϕ. These estimates are what the next chapter needs in order to integrate by parts with singular potentials. Singularity envelopes of potentials of zero mass, and contact concentration in arbitrary pseudo-effective classes, depend on the Monge–Ampère equations solved there (Sections 17.4.4 and 17.5). Historical remarks and references for this chapter are collected in Appendix B.16.

16.1 Obstacles and rooftops

The fixed Kähler form ω serves as a reference: volumes, norms and the Laplacians of obstacles are taken with respect to ω, also when an envelope is formed with respect to a different background. We call the elements of $\mathrm{PSH}(X, \theta)$ (Definition 13.1.3) *potentials*; they are pointwise defined functions, not identically $-\infty$. Recall from Definition 14.2.6 that u is more singular than v, written $u \preceq v$, if $u \le v + C$ for some constant C. We write $u \sim v$, and say that u and v have the same *singularity type*, if $u \preceq v$ and $v \preceq u$.

Two facts are used constantly. First, pluripolar sets have Lebesgue measure zero (see the remarks after Definition 10.1.1). Second, if two quasi-plurisubharmonic functions satisfy $u \le v$ almost everywhere, then $u \le v$ everywhere, by Lemma 13.1.4(1). Consequently, an inequality between quasi-plurisubharmonic functions that holds quasi-everywhere, that is, outside a pluripolar set (Definition 10.1.1), holds everywhere. In particular, if $u \sim v$, then u and v have the same pole set and $u - v$ is bounded outside it.

Quasi-continuity on X is the notion of Definition 9.4.1, read in local charts as in Section 15.6: a Borel function $f\colon X \to [-\infty, +\infty]$ is quasi-continuous if its restriction to every coordinate chart, identified with a bounded open set $D \subset \mathbb{C}^n$, is quasi-continuous for the Monge–Ampère capacity Cap_D of Definition 9.3.2. The following lemma shows that it suffices to check this on the charts of one atlas; in Proposition 16.9.9 we show that quasi-continuous functions are also continuous outside open sets of small global capacity.

Lemma 16.1.1 *Let D and D' be coordinate charts of X, identified with bounded open subsets of $\mathbb{C}^n$.*

(1) *For every compact set $L \subset D \cap D'$ there is a constant c such that* $\mathrm{Cap}_D(E) \le c\,\mathrm{Cap}_{D'}(E)$ *for every Borel set $E \subset L$.*
(2) *If $E \subset X$ is pluripolar, then* $\mathrm{Cap}^*_D(E \cap D) = 0$.
(3) *If a Borel function $f\colon X \to [-\infty, +\infty]$ is quasi-continuous on each chart of an atlas of X, then it is quasi-continuous on every coordinate chart.*

Proof Assertions (1) and (3) are contained in Lemma 15.6.1: (1) is its inequality for Borel sets, applied with the roles of D and D' exchanged, and (3) is its final assertion.

For (2) we first observe that if $W \subset D$ is open, then every competitor v for Cap_D restricts to a plurisubharmonic function on W with values in $[0, 1]$, and the restriction of the Bedford–Taylor measure $(\mathrm{dd^c}v)^n$ to W is the Bedford–Taylor measure of this restriction. Hence $\mathrm{Cap}_D(E) \le \mathrm{Cap}_W(E)$ for every Borel set $E \subset W$, and, taking infima over the open sets G with $A \subset G \subset W$, also $\mathrm{Cap}^*_D(A) \le \mathrm{Cap}^*_W(A)$ for every set $A \subset W$.

Now cover D by countably many open balls B_k, in the coordinates of D, with $\overline{B_k} \subset D$. Since the condition in Definition 10.1.1 is local, $E \cap B_k$ is pluripolar in B_k, so $\mathrm{Cap}^*_{B_k}(E \cap B_k) = 0$ by Theorem 10.1.4. By the observation above, $\mathrm{Cap}^*_D(E \cap B_k) = 0$, and the countable subadditivity of Cap^*_D (Proposition 9.3.3) gives $\mathrm{Cap}^*_D(E \cap D) = 0$. □

Quasi-plurisubharmonic functions are quasi-continuous, and quasi-continuity is preserved by the following operations.

Lemma 16.1.2 *Let θ' be a smooth closed real $(1,1)$-form on X.*

(1) *Every $u \in \mathrm{PSH}(X, \theta')$ is quasi-continuous. In particular, every potential $u \in \mathrm{PSH}(X, \theta)$ is quasi-continuous.*
(2) *Finite maxima and minima of quasi-continuous functions on X are quasi-continuous.*
(3) *If f is quasi-continuous on X and $g\colon X \to [-\infty, +\infty]$ is a Borel function that agrees with f outside a pluripolar set, then g is quasi-continuous.*

Proof (1) By Lemma 7.3.1 every point of X has a neighborhood on which $\theta' = \mathrm{dd^c}h$ for a smooth function h, so X is covered by finitely many coordinate balls D_i on which $\theta' = \mathrm{dd^c}h_i$ with h_i smooth near $\overline{D_i}$. On D_i the function $u + h_i$ is plurisubharmonic by Lemma 13.1.4(2), hence quasi-continuous by Theorem 9.4.2, and so is $u = (u+h_i) - h_i$, because h_i is continuous and real. By Lemma 16.1.1(3), u is quasi-continuous on every chart.

(2) Let $f_1, \dots, f_k$ be quasi-continuous, let K be a compact subset of a chart D, and let $\varepsilon > 0$. Choose open sets $G_i \subset D$ with $\mathrm{Cap}_D(G_i) < \varepsilon/k$ such that f_i is continuous on $K \setminus G_i$. By subadditivity, $G = G_1 \cup \cdots \cup G_k$ has capacity less than ε, and all the f_i, hence also their maximum and minimum, are continuous on $K \setminus G$.

(3) Let D be a chart, $K \subset D$ compact and $\varepsilon > 0$. The set $\{f \ne g\}$ is pluripolar, so by Lemma 16.1.1(2) its intersection with D is contained in an open set $G' \subset D$ with $\mathrm{Cap}_D(G') < \varepsilon/2$. If $G \subset D$ is an open set with $\mathrm{Cap}_D(G) < \varepsilon/2$ such that f is continuous on $K \setminus G$, then $\mathrm{Cap}_D(G \cup G') < \varepsilon$, and $g = f$ is continuous on $K \setminus (G \cup G')$. □

Definition 16.1.3 Let $f\colon X \to [-\infty, +\infty]$ be a Borel function, called an *obstacle*, and put

$$\mathcal{A}_\theta(f) = \{v \in \mathrm{PSH}(X,\theta) : v \le f \text{ quasi-everywhere}\}.$$

The members of $\mathcal{A}_\theta(f)$ are called the *competitors* for f. If $\mathcal{A}_\theta(f)$ is nonempty and uniformly bounded above, the *obstacle envelope* of f is

$$P_\theta(f) = \big(\sup \mathcal{A}_\theta(f)\big)^*,$$

where * denotes the upper semicontinuous regularization. If $\mathcal{A}_\theta(f)$ is empty, we put $P_\theta(f) = -\infty$. If $\mathcal{A}_\theta(f)$ is nonempty but not uniformly bounded above, $P_\theta(f)$ is not defined, and whenever we use $P_\theta(f)$ for an obstacle that is not bounded above, we verify the upper bound first. If $f \le b$ quasi-everywhere for a constant b, the upper bound is automatic: every competitor satisfies $v \le b$ quasi-everywhere, hence almost everywhere, and therefore everywhere by Lemma 13.1.4(1).

As for the capacity Cap_D (Definition 9.3.2) and the relative extremal functions of Chapter 10, a competitor is a member of the family whose supremum defines the object under consideration; we always say which obstacle, envelope or capacity is meant.

Proposition 16.1.4 *Let f be a Borel obstacle.*

(1) *If $\mathcal{A}_\theta(f)$ is nonempty and uniformly bounded above, then $P_\theta(f) \in \mathcal{A}_\theta(f)$; thus $P_\theta(f)$ is the largest competitor for f.*
(2) *If f is upper semicontinuous, then every competitor v for f satisfies $v \le f$ everywhere.*
(3) *The envelope does not change when f is modified on a pluripolar set, and $P_\theta(f + c) = P_\theta(f) + c$ for $c \in \mathbb{R}$. If g is a Borel obstacle with $f \le g$ quasi-everywhere and $\mathcal{A}_\theta(g)$ is uniformly bounded above, then $P_\theta(f) \le P_\theta(g)$.*
(4) *Suppose that $\mathcal{A}_\theta(f)$ is nonempty and uniformly bounded above, and let g be a Borel obstacle with $f - c \le g \le f + c$ quasi-everywhere for a constant $c \ge 0$. Then $\mathcal{A}_\theta(g)$ is nonempty and uniformly bounded above, and*

$$P_\theta(f) - c \le P_\theta(g) \le P_\theta(f) + c.$$

In particular the two envelopes have the same pole set and differ by at most c outside it.
(5) *Let Borel obstacles f_j decrease pointwise to f, and suppose that $\mathcal{A}_\theta(f_1)$ is uniformly bounded above. Then $P_\theta(f_j)$ decreases pointwise to $P_\theta(f)$. Here the families may be empty; in particular, $P_\theta(f_j)$ decreases to the constant $-\infty$ if and only if $\mathcal{A}_\theta(f) = \varnothing$.*

Proof (1) Cover X by finitely many coordinate balls on which θ has smooth local potentials h (Lemma 7.3.1). Applying Theorem 8.3.2 on each ball to the plurisubharmonic functions $v + h$, $v \in \mathcal{A}_\theta(f)$, and taking the union of the resulting countable subfamilies, we find competitors $v_1, v_2, \dots$ with $(\sup_k v_k)^* = P_\theta(f)$. The finite maxima $w_k = \max\{v_1, \dots, v_k\}$ are again competitors, and they increase with k.

Each w_k satisfies $w_k \le f$ outside a pluripolar set N_k, and $N = \bigcup_k N_k$ is pluripolar by Theorem 10.1.4. By Corollary 10.2.2, $\sup_k w_k = P_\theta(f)$ outside a pluripolar set N'. Hence $P_\theta(f) \le f$ outside $N \cup N'$. Since the family is nonempty and uniformly bounded above, $P_\theta(f)$ is θ-plurisubharmonic by Theorem 8.3.2. Thus $P_\theta(f)$ is a competitor, and by construction it dominates every competitor.

(2) Let v be a competitor, let $x \in X$, and choose a coordinate ball $B(x,r)$ and a smooth function h with $\mathrm{dd^c} h = \theta$ near its closure. Since $v + h$ is plurisubharmonic and $v \le f$ outside a Lebesgue null set, the sub-mean value inequality gives

$$v(x) + h(x) \le \frac{1}{|B(x,r)|} \int_{B(x,r)} (v + h)\, \mathrm{d}V_{2n} \le \sup_{B(x,r)} (f + h).$$

As $r \to 0$, the right side tends to at most $f(x) + h(x)$, because $f + h$ is upper semicontinuous. Hence $v(x) \le f(x)$.

(3) A modification of f on a pluripolar set does not change $\mathcal{A}_\theta(f)$, since the union of two pluripolar sets is pluripolar. Moreover $\mathcal{A}_\theta(f + c) = \mathcal{A}_\theta(f) + c$, and $\mathcal{A}_\theta(f) \subset \mathcal{A}_\theta(g)$ if $f \le g$ quasi-everywhere. These relations between the families of competitors give the three assertions.

(4) By (3), $\mathcal{A}_\theta(f) - c \subset \mathcal{A}_\theta(g) \subset \mathcal{A}_\theta(f) + c$. Hence $\mathcal{A}_\theta(g)$ is nonempty and uniformly bounded above, and the two inequalities follow from (3).

(5) By (3) the envelopes $P_\theta(f_j)$ decrease; if some family $\mathcal{A}_\theta(f_j)$ is empty, then so are all later ones and $\mathcal{A}_\theta(f)$. Let v be the pointwise limit of $P_\theta(f_j)$. Suppose first that $v(x_0) > -\infty$ for some x_0. Then $\sup_X P_\theta(f_j) \ge v(x_0)$ for every j, so $v \in \mathrm{PSH}(X,\theta)$ by Corollary 14.2.5. By (1), $v \le P_\theta(f_j) \le f_j$ outside a pluripolar set N_j, so $v \le \inf_j f_j = f$ outside the pluripolar set $\bigcup_j N_j$. Thus v is a competitor for f and $v \le P_\theta(f)$; the reverse inequality follows from $P_\theta(f) \le P_\theta(f_j)$. Suppose next that $v \equiv -\infty$. A competitor w for f would be a competitor for every f_j, so $w \le P_\theta(f_j)$ for all j and $w \le v$, which is impossible. Hence $\mathcal{A}_\theta(f) = \varnothing$ and $P_\theta(f) = -\infty = v$.□

Definition 16.1.5 For $u, v \in \mathrm{PSH}(X,\theta)$, the *rooftop envelope* of u and v is

$$P_\theta(u,v) = P_\theta(\min\{u,v\}).$$

We also put $P_\theta(u,-\infty) = P_\theta(-\infty,u) = -\infty$ for $u \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}$.

The obstacle $\min\{u,v\}$ is upper semicontinuous and bounded above. By Proposition 16.1.4, a potential is therefore a competitor for it exactly when it lies below both u and v everywhere. Thus $P_\theta(u,v)$ is the largest common minorant of u and v in $\mathrm{PSH}(X,\theta)$ if they have one, and $P_\theta(u,v) = -\infty$ if they have none.

Proposition 16.1.6 *The rooftop envelope is commutative, associative, increasing in both arguments and idempotent, that is, $P_\theta(u,u) = u$. For a constant c, $P_\theta(u + c, v + c) = P_\theta(u,v) + c$. If $u_j, v_j \in \mathrm{PSH}(X,\theta)$ decrease pointwise to u and v, where the limits may be identically $-\infty$, then $P_\theta(u_j, v_j)$ decreases pointwise to $P_\theta(u,v)$. Finally, for $u \in \mathrm{PSH}(X,\theta)$ and every constant $C \in \mathbb{R}$, the rooftop envelope $P_\theta(u + C, V_\theta)$ is a potential with $P_\theta(u + C, V_\theta) \sim u$.*

Proof Commutativity, monotonicity and idempotence follow directly from the description of $P_\theta(u,v)$ as the largest common minorant. For associativity, let also $w \in \mathrm{PSH}(X,\theta) \cup \{-\infty\}$. A potential lies below $P_\theta(u,v)$ and w if and only if it lies below u, v and w: when $P_\theta(u,v) \neq -\infty$ this is the maximality of $P_\theta(u,v)$, and otherwise neither condition can be satisfied. Hence $P_\theta(P_\theta(u,v),w)$ and $P_\theta(u,P_\theta(v,w))$ are both the largest common minorant of u, v and w, or both equal $-\infty$ if there is none. The behavior under constant shifts and decreasing limits follows from Proposition 16.1.4, applied to the obstacles $\min\{u_j,v_j\}$, which are bounded above by $\sup_X u_1$ and decrease to $\min\{u,v\}$. If u or v is identically $-\infty$, this limit obstacle has no competitor, in accordance with our convention.

For the last assertion put $B = \max\{\sup_X u, -C\}$. Then $u - B \leq 0$, so $u - B \leq V_\theta$, and also $u - B \leq u + C$. Hence $u - B$ is a common minorant of $u + C$ and V_θ, and

$$u - B \leq P_\theta(u + C, V_\theta) \leq u + C.$$

Example 16.1.7 On $\mathbb{P}^1$ let $\theta = \omega_{\mathrm{FS}} = \mathrm{dd^c}\log(1+|z|^2)$ be the Fubini–Study form of Example 14.5.6, in the affine coordinate z, so that $\int_{\mathbb{P}^1}\theta = 1$. Put

$$u_0 = \log\frac{|z|^2}{1+|z|^2}, \qquad u_\infty = -\log(1+|z|^2).$$

In the coordinate $\zeta = 1/z$ near ∞ we have $\log(1+|z|^2) = \log(1+|\zeta|^2) - \log|\zeta|^2$, so $\theta = \mathrm{dd^c}\log(1+|\zeta|^2)$ there, while $u_0 = -\log(1+|\zeta|^2)$ and $u_\infty = \log(|\zeta|^2/(1+|\zeta|^2))$. Thus the substitution $z \leftrightarrow \zeta$ exchanges the two functions. On $\mathbb{C}$ we have $\theta + \mathrm{dd^c}u_0 = \mathrm{dd^c}\log|z|^2 = [0]$ by the Poincaré–Lelong formula (Theorem 7.4.5), and u_0 is smooth near ∞ with $\theta + \mathrm{dd^c}u_0 = 0$ there. Hence $u_0, u_\infty \in \mathrm{PSH}(\mathbb{P}^1,\theta)$, with $\theta + \mathrm{dd^c}u_0 = [0]$ and $\theta + \mathrm{dd^c}u_\infty = [\infty]$, the currents of integration at the two points.

Suppose that $w \in \mathrm{PSH}(\mathbb{P}^1,\theta)$ is a common minorant of u_0 and u_∞. Near 0 the function u_0 is $\log|z|^2$ plus a smooth function, so $\nu(u_0,0) = 1$ by Proposition 12.1.4, and $w \leq u_0$ gives $\nu(w,0) \geq 1$ by the description of Lelong numbers through logarithmic growth in Proposition 12.1.1; likewise $\nu(w,\infty) \geq 1$. In dimension one, the Lelong number of a subharmonic function s at a point is the mass that the measure $\mathrm{dd^c}s$ gives to that point (Theorem 12.1.3). Hence the positive measure $\theta + \mathrm{dd^c}w$ has atoms of mass at least one at 0 and at ∞, whereas its total mass is $\int_{\mathbb{P}^1}\theta = 1$, because $\langle \mathrm{dd^c}w, 1\rangle = \langle w, \mathrm{dd^c}1\rangle = 0$. This contradiction shows that $P_\theta(u_0,u_\infty) = -\infty$, although u_0 and u_∞ are both θ-plurisubharmonic.

16.2 Measures on contact sets

For $u \in \mathrm{PSH}(X,\theta)$ we write $\theta_u = \theta + \mathrm{dd^c}u$ and denote by $\mu_u = \langle\theta_u^n\rangle$ the non-pluripolar Monge–Ampère measure of u, as in Proposition 15.8.5; its total mass $\mu_u(X) = \int_X\langle\theta_u^n\rangle$ is called the *mass* of u. When a different background is used, we say

so or indicate it by a subscript. A *contact set* is a set $\{u = f\}$ on which two functions agree; for Borel functions it is a Borel set, but it need not be closed (for instance, if $U \subset X$ is open, the functions 0 and $\mathbf{1}_{X\setminus U}$ agree exactly on U).

Three properties of these measures are used throughout. They give no mass to pluripolar sets (Lemma 15.1.2), in particular to pole sets, so points where a potential equals $-\infty$ can be ignored in measure-theoretic statements. They are local in the plurifine topology (Proposition 15.1.5): if two potentials agree on a plurifine open set O, their measures agree on O; and for potentials u, v and $c \in \mathbb{R}$ the set $\{u > v + c\}$ is plurifine open by Proposition 10.5.1. Finally, multilinearity and positivity give

$$\mu_{(1-t)u+tv} \geq t^n \mu_v \qquad (u, v \in \mathrm{PSH}(X, \theta),\ 0 \leq t \leq 1),$$

and the mass is monotone: $\int_X \langle \theta_u^n \rangle \leq \int_X \langle \theta_v^n \rangle$ if $u \preceq v$, with equality if $u \sim v$, by Corollary 15.7.5.

Inequalities between measures will usually be obtained from weak limits, that is, by testing against continuous functions. The following lemma justifies this step, and it records the inner regularity of finite measures on X.

Lemma 16.2.1 *Let λ and λ' be finite positive Borel measures on X.*

(1) *For every Borel set $E \subset X$, $\lambda(E) = \sup\{\lambda(K) : K \subset E \text{ compact}\}$.*
(2) *Let $W \subset X$ be open, and suppose that $\int_X \chi \, \mathrm{d}\lambda \leq \int_X \chi \, \mathrm{d}\lambda'$ for every continuous function $\chi \geq 0$ with compact support in W. Then $\lambda(E) \leq \lambda'(E)$ for every Borel set $E \subset W$.*

Proof (1) Cover X by finitely many coordinate charts $U_1, \dots, U_N$, with homeomorphisms κ_i of U_i onto open sets $\Omega_i \subset \mathbb{R}^{2n}$, and let $E \subset X$ be Borel and $\varepsilon > 0$. The Borel sets $E_i = (E \cap U_i) \setminus (U_1 \cup \cdots \cup U_{i-1})$ partition E. The image of the restriction of λ to U_i under κ_i is a finite Borel measure on Ω_i, hence a Radon measure by Proposition 1.1.2, so there is a compact set $L_i \subset \kappa_i(E_i)$ whose complement in $\kappa_i(E_i)$ has measure less than ε/N. Then $K = \bigcup_i \kappa_i^{-1}(L_i)$ is a compact subset of E with $\lambda(E \setminus K) < \varepsilon$.

(2) Fix a metric d on X that induces its topology (Lemma 7.1.4). Let $K \subset W$ be compact and nonempty; by compactness there is $r > 0$ such that every point at distance at most r from K lies in W. For $k > 1/r$ the continuous functions $\chi_k = \max\{0, 1 - k\, d(\cdot, K)\}$ take values in $[0, 1]$, equal 1 on K, have compact support in W, and decrease pointwise to $\mathbf{1}_K$ as $k \to \infty$. By dominated convergence for the finite measures λ and λ',

$$\lambda(K) = \lim_{k\to\infty} \int_X \chi_k \, \mathrm{d}\lambda \leq \lim_{k\to\infty} \int_X \chi_k \, \mathrm{d}\lambda' = \lambda'(K).$$

If $E \subset W$ is Borel, then by (1), $\lambda(E) = \sup_K \lambda(K) \leq \sup_K \lambda'(K) \leq \lambda'(E)$, the suprema being taken over the compact sets $K \subset E$. □

16.2.1 Concentration on the contact set

We first show that the measure of an envelope is carried by its contact set with the obstacle, beginning with a comparison of the measures of two ordered potentials on the set where they agree.

Lemma 16.2.2 *Let $u, v \in \mathrm{PSH}(X,\theta)$ with $u \le v$. Then*

$$\mathbf{1}_{\{u=v\}}\mu_u \le \mathbf{1}_{\{u=v\}}\mu_v.$$

Proof For $\varepsilon > 0$ put $w_\varepsilon = \max\{u, v-\varepsilon\}$. As $\varepsilon \downarrow 0$, w_ε increases to $\max\{u,v\} = v$, so $\mu_{w_\varepsilon} \to \mu_v$ weakly along every sequence $\varepsilon \downarrow 0$, by Corollary 15.7.6. On the plurifine open set $\{u > v-\varepsilon\}$ we have $w_\varepsilon = u$, hence $\mathbf{1}_{\{u>v-\varepsilon\}}\mu_{w_\varepsilon} = \mathbf{1}_{\{u>v-\varepsilon\}}\mu_u$. This set contains every point of $\{u=v\}$ at which u is finite, and μ_u does not charge the pole set of u. Therefore

$$\mu_{w_\varepsilon} \ge \mathbf{1}_{\{u>v-\varepsilon\}}\mu_u \ge \mathbf{1}_{\{u=v\}}\mu_u.$$

The right side is a fixed finite measure. Integrating a continuous function $\chi \ge 0$ against both sides and letting $\varepsilon \downarrow 0$, we obtain $\int_X \chi\,\mu_v \ge \int_X \chi \mathbf{1}_{\{u=v\}}\,\mu_u$, so $\mu_v \ge \mathbf{1}_{\{u=v\}}\mu_u$ by Lemma 16.2.1. Restricting both sides to $\{u=v\}$ proves the lemma. □

Lemma 16.2.3 *Suppose that $\{\theta\}$ is big. There is a proper analytic subset $A \subset X$, the pole set of a θ-plurisubharmonic function with analytic singularities, such that every potential with minimal singularities is locally bounded on $X \setminus A$. The set A is closed and pluripolar; in particular $\mu_u(A) = 0$ for every $u \in \mathrm{PSH}(X,\theta)$.*

Proof By Proposition 14.3.5, the big class contains a Kähler current $\theta + \mathrm{dd}^c\psi$ whose potential $\psi \in \mathrm{PSH}(X,\theta)$ has analytic singularities, and we put $A = \{\psi = -\infty\}$. Every point of X has a connected coordinate neighborhood V on which $\psi = c\log\sum_i |g_i|^2 + b$, with $c > 0$, b bounded and holomorphic functions g_i that do not all vanish identically (Definition 13.1.5). Thus $A \cap V$ is the common zero set of the g_i, so A is an analytic subset of X, different from X, and it is closed and pluripolar, being locally the pole set of the plurisubharmonic function $\log\sum_i |g_i|^2$. On $V \setminus A$ the continuous function $\sum_i |g_i|^2$ is positive, so ψ is locally bounded on $X \setminus A$. If φ has minimal singularities, then $\psi \le \varphi + C'$ for a constant C', so φ is locally bounded below on $X \setminus A$; it is bounded above, being upper semicontinuous on the compact space X. The last assertion holds because non-pluripolar products give no mass to pluripolar sets. □

Lemma 16.2.4 *Suppose that $\{\theta\}$ is big and that f is bounded and lower semicontinuous. Then $u = P_\theta(f)$ has minimal singularities, and μ_u is carried by $\{u = f\}$.*

Proof Adding a constant to f, we may assume that $-C \le f \le 0$. Then $V_\theta - C \le f$, so $V_\theta - C$ is a competitor, while every competitor is nonpositive and hence at most V_θ. Thus $V_\theta - C \le u \le V_\theta$, and u has minimal singularities.

Let A be the proper analytic subset of Lemma 16.2.3. Since u has minimal singularities, it is locally bounded on $X \setminus A$, and $\mu_u(A) = 0$.

Let $x \in X \setminus A$ with $u(x) < f(x)$, and choose $\delta > 0$ with $u(x) < f(x) - 3\delta$. Since u is upper semicontinuous and f is lower semicontinuous, there is a coordinate ball $B \ni x$ with $\overline{B} \subset X \setminus A$ such that $u < u(x) + \delta$ and $f > f(x) - \delta$ on $\overline{B}$; thus $\sup_{\overline{B}} u \le \inf_{\overline{B}} f - \delta$. Shrinking B, we may assume that $B \Subset B'$ for a coordinate ball B' with $\overline{B'} \subset X \setminus A$ and that $\theta = \mathrm{dd}^{\mathrm{c}} h$ near $\overline{B'}$ for a smooth function h (Lemma 7.3.1), and then, shrinking B further, that the oscillation $\operatorname{osc}_{\overline{B}} h = \sup_{\overline{B}} h - \inf_{\overline{B}} h$ is less than δ.

We show that μ_u vanishes on B. Since u is bounded near $\overline{B'}$, Corollary 14.2.9 gives $\widetilde{u} \in \mathrm{PSH}(X, \theta)$ with $\widetilde{u} \ge u$, $\widetilde{u} = u$ on $X \setminus B$ and $(\theta + \mathrm{dd}^{\mathrm{c}} \widetilde{u})^n = 0$ on B. The function $\widetilde{u} + h$ is plurisubharmonic, hence subharmonic, on B, and at every point $\zeta \in \partial B$ its upper limit from B is at most $(\widetilde{u} + h)(\zeta) = (u + h)(\zeta) \le \sup_{\overline{B}} (u + h)$, by upper semicontinuity. The maximum principle (Proposition 3.1.4) therefore gives

$$\widetilde{u} \le \sup_{\overline{B}} (u + h) - h \le \sup_{\overline{B}} u + \operatorname{osc}_{\overline{B}} h < \inf_{\overline{B}} f \qquad \text{on } B.$$

Hence $\widetilde{u} \le f$ quasi-everywhere on X, so $\widetilde{u}$ is a competitor and $\widetilde{u} \le u$. Therefore $\widetilde{u} = u$, and $\mu_u = (\theta + \mathrm{dd}^{\mathrm{c}} \widetilde{u})^n$ vanishes on B.

The set $\{u < f\}$ is open, because $f - u$ is lower semicontinuous. Covering $\{u < f\} \setminus A$ by countably many balls as above, we obtain $\mu_u(\{u < f\} \setminus A) = 0$. Moreover $u \le f$ quasi-everywhere by Proposition 16.1.4, so the set $\{u > f\}$ is pluripolar. Since μ_u gives no mass to A or to $\{u > f\}$, it is carried by $\{u = f\}$. □

Theorem 16.2.5 *Suppose that $\{\theta\}$ is big, and let f be a quasi-continuous Borel obstacle such that $\mathcal{A}_\theta(f)$ is nonempty and uniformly bounded above. Then $\mu_{P_\theta(f)}$ is carried by the contact set $\{P_\theta(f) = f\}$, that is,*

$$\mu_{P_\theta(f)}(\{P_\theta(f) \ne f\}) = 0.$$

Proof Write $u = P_\theta(f)$, which is a potential by Proposition 16.1.4, and let A be the proper analytic subset of Lemma 16.2.3: every potential with minimal singularities is locally bounded on $X \setminus A$, and A is closed and pluripolar. After reducing to obstacles that are bounded above, we treat bounded obstacles by approximating them with lower semicontinuous ones (Step 1), and then obstacles that are only bounded above by truncating them from below (Step 2).

Choose a constant $C_0 > \sup_X u$ and put $\widetilde{f} = \min\{f, C_0\}$, a quasi-continuous Borel function that is bounded above (Lemma 16.1.2(2)). By Proposition 16.1.4, $u \le f$ quasi-everywhere, and $u < C_0$ everywhere, so $u \in \mathcal{A}_\theta(\widetilde{f}) \subset \mathcal{A}_\theta(f)$. Since u is the largest member of $\mathcal{A}_\theta(f)$, it follows that $P_\theta(\widetilde{f}) = u$; moreover $\{u = \widetilde{f}\} = \{u = f\}$, because $u < C_0$. Replacing f by $\widetilde{f}$, we may therefore assume that f is bounded above.

Step 1: bounded obstacles. Suppose first that $|f| \le C'$ quasi-everywhere for a constant C'. Then $\{|f| > C'\}$ is a pluripolar Borel set, and replacing f by 0 on it changes neither $P_\theta(f)$ (Proposition 16.1.4) nor the quasi-continuity of f (Lemma 16.1.2(3)); it changes the contact set only on a pluripolar set, which μ_u does

not charge. After this modification and the addition of a constant, we may therefore assume that $-C \le f \le 0$ everywhere.

We first construct open sets of small capacity outside which f is continuous. Fix finitely many coordinate balls $D_1, \dots, D_N$, compact sets $K_i \subset D_i$ with $\bigcup_i K_i = X$, and open sets D_i' with $K_i \subset D_i' \Subset D_i$. For all i and $k \ge 1$, the quasi-continuity of f on D_i gives an open set $G_{ik} \subset D_i'$ with $\operatorname{Cap}_{D_i}(G_{ik}) < 2^{-k}$ such that f is continuous on $K_i \setminus G_{ik}$; intersecting with D_i' the open set provided by Definition 9.4.1 does not change $K_i \setminus G_{ik}$. Put

$$G_j = \bigcap_{k=1}^{j} \bigcup_{i=1}^{N} G_{ik}.$$

These open sets decrease, and $X \setminus G_j$ is the union of the finitely many closed sets $K_i \setminus \bigcup_{i'} G_{i'k}$, $1 \le i \le N$, $1 \le k \le j$. Since f is continuous on each of them, it is continuous on the closed set $X \setminus G_j$. Moreover, if D is a coordinate ball and $L \subset D$ is compact, then $\operatorname{Cap}_D(G_j \cap L) \to 0$. Indeed, $L \cap \overline{D_i'}$ is a compact subset of $D \cap D_i$, so Lemma 16.1.1(1) gives a constant c_i with $\operatorname{Cap}_D(E) \le c_i \operatorname{Cap}_{D_i}(E)$ for all Borel sets $E \subset L \cap \overline{D_i'}$. Since $G_j \cap L \subset \bigcup_i (G_{ij} \cap L)$ and $G_{ij} \subset D_i'$, it follows that

$$\operatorname{Cap}_D(G_j \cap L) \le \sum_{i=1}^{N} c_i \operatorname{Cap}_{D_i}(G_{ij}) \le 2^{-j} \sum_{i=1}^{N} c_i.$$

Next we approximate f by lower semicontinuous obstacles that agree with f outside G_j. Fix a metric on X that induces its topology (Lemma 7.1.4). By the Tietze extension theorem in the form of Lemma 9.4.3, applied to the restriction of $f + C/2$ to the compact set $X \setminus G_j$, which takes values in $[-C/2, C/2]$, the restriction of f to $X \setminus G_j$ extends to a continuous function $g_j \colon X \to [-C, 0]$. Put

$$h_j = \sup_{k \ge j} g_k.$$

These functions are lower semicontinuous, take values in $[-C, 0]$, and decrease with j. Since $G_k \subset G_j$ for $k \ge j$, we have $h_j = f$ on $X \setminus G_j$. The intersection $\bigcap_j G_j$ is pluripolar: if D is a coordinate ball and $U \Subset D$ is open, then $U \cap \bigcap_j G_j$ is contained in the open sets $U \cap G_j$, whose capacities Cap_D tend to zero, so it has outer capacity zero and lies in the pole set of a plurisubharmonic function on D by Theorem 10.1.4. Outside $\bigcap_j G_j$ we have $h_j = f$ for all large j. Hence the limit h of the h_j equals f quasi-everywhere.

Finally we pass to the limit in the contact property of the envelopes $w_j = P_\theta(h_j)$. By Proposition 16.1.4, w_j decreases to $P_\theta(h) = P_\theta(f) = u$. Since the obstacles h_j and f take values in $[-C, 0]$, the potential $V_\theta - C$ is a competitor for each of them, and every competitor is nonpositive, hence at most V_θ; thus $V_\theta - C \le w_j \le V_\theta$ and $V_\theta - C \le u \le V_\theta$. Fix a coordinate ball $D \Subset X \setminus A$ with a smooth potential ρ of θ near $\overline{D}$ (such balls exist around every point of $X \setminus A$, by Lemma 7.3.1), and a continuous function $\chi \ge 0$ with compact support in D. On D the functions w_j and u are bounded uniformly in j. By Lemma 16.2.4, μ_{w_j} is carried by $\{w_j = h_j\}$, so

$$\int_X \chi\bigl(1-\mathrm{e}^{w_j-h_j}\bigr)\,\mu_{w_j}=0.$$

The integrands with h_j and with f are bounded uniformly in j on the support of χ, and they coincide outside G_j. By Lemma 9.4.7, applied to the bounded plurisubharmonic functions $\rho+w_j$, the measures μ_{w_j} of $G_j\cap\operatorname{Supp}\chi$ are at most a fixed constant times $\operatorname{Cap}_D(G_j\cap\operatorname{Supp}\chi)$, which tends to zero, as shown above. Consequently

$$\lim_{j\to\infty}\int_X \chi\bigl(1-\mathrm{e}^{w_j-f}\bigr)\,\mu_{w_j}=0.$$

The weights $1-\mathrm{e}^{w_j-f}$ are bounded and quasi-continuous on D, and they converge in capacity to $1-\mathrm{e}^{u-f}$. Indeed, the uniformly bounded potentials w_j decrease to u and therefore converge in capacity (Proposition 9.4.6), and $|\mathrm{e}^{w_j-f}-\mathrm{e}^{u-f}|\le \mathrm{e}^C|w_j-u|$, because w_j-f and $u-f$ are at most C. The weighted form of Theorem 9.4.8, applied to the bounded plurisubharmonic functions $\rho+w_j$ on D, therefore gives

$$\int_X \chi\bigl(1-\mathrm{e}^{u-f}\bigr)\,\mu_u=0.$$

The integrand is nonnegative outside the pluripolar set $\{u>f\}$, which μ_u does not charge, and it is positive exactly at the points of D where $u<f$. Since χ is arbitrary, Lemma 16.2.1(2), applied on $W=D$ to the positive measure $\max\{1-\mathrm{e}^{u-f},0\}\mu_u$ and the zero measure, shows that this measure vanishes on D; hence $\mu_u(\{u<f\}\cap D)=0$. Covering $X\setminus A$ by such balls, and recalling that μ_u does not charge the pluripolar sets A and $\{u>f\}$, we conclude that μ_u is carried by $\{u=f\}$.

Step 2: obstacles bounded above. Adding a constant, we may assume that $f\le 0$. Put $f_j=\max\{f,-j\}$ and $u_j=P_\theta(f_j)$. The obstacles f_j are quasi-continuous by Lemma 16.1.2(2), satisfy $-j\le f_j\le 0$, and decrease to f; their families of competitors contain $V_\theta-j$ and are bounded above by 0. By Proposition 16.1.4, u_j decreases to the potential u, and by Step 1, μ_{u_j} is carried by $\{u_j=f_j\}$. Moreover $V_\theta-j\le u_j\le V_\theta$. The potentials u_j are not uniformly bounded below, so we pass to the limit on a fixed ball with truncated potentials, and with weights that vanish where the truncation is active.

Fix a coordinate ball $D\Subset X\setminus A$ as in Step 1 and constants $M,\varepsilon>0$, and put

$$v_j^M=\max\{u_j,V_\theta-M\},\qquad a_j=\frac{v_j^M-V_\theta+M}{v_j^M-V_\theta+M+\varepsilon},$$

and define v^M and a in the same way with u in place of u_j. On D these truncated potentials are bounded uniformly in j and decrease to v^M, so they converge in capacity (Proposition 9.4.6). The weights satisfy $0\le a_j\le 1$ and $a_j=0$ on $\{u_j\le V_\theta-M\}$. They are quasi-continuous, and they converge in capacity to a, since $s\mapsto(s+M)/(s+M+\varepsilon)$ is Lipschitz on $[-M,0]$.

Next put $b_j=\mathrm{e}^{u_j-f_j}$ at the points where u_j and f_j are finite and $u_j\le f_j$, and $b_j=1$ elsewhere; define b from u and f in the same way. Since $f\le 0$ everywhere,

$u \le f$ and $u_j \le f_j$ quasi-everywhere, and u_j and f_j are finite on D, the set of points of D at which some b_j or b is given the value 1 is contained in the union of the pole set of u and countably many pluripolar sets, so it is pluripolar.

We claim that $b_j \to b$ in capacity on compact subsets of D and that b is quasi-continuous. Let $K \subset D$ be compact and $\varepsilon' > 0$. Cover the exceptional pluripolar set by an open set of capacity less than ε', which is possible by Theorem 10.1.4, and enlarge it, using the quasi-continuity of u (Lemma 16.1.2(1)) and of f, to an open set G of capacity less than $2\varepsilon'$ such that u and f are continuous and finite on the compact set $F = K \setminus G$. Then $u, f \ge -l$ on F for some integer l, so $f_j = f$ on F for $j \ge l$. Since the exponential is 1-Lipschitz on $(-\infty, 0]$, we get $|b_j - b| \le |u_j - u|$ on F for $j \ge l$. As $u \le u_j$ and $u \ge -l$ on F, we have $u_j - u = \max\{u_j, -l\} - \max\{u, -l\}$ there, and these truncations converge in capacity because they decrease. Hence, for every $\delta > 0$, the outer capacity of $K \cap \{|b_j - b| > \delta\}$ is at most $2\varepsilon'$ plus a quantity that tends to zero, and b is continuous on F. This proves the claim.

Since a_j vanishes outside the plurifine open set $\{u_j > V_\theta - M\}$, on which $v_j^M = u_j$, and since μ_{u_j} is carried by $\{u_j = f_j\}$, where $b_j = 1$ up to a pluripolar set, we have

$$a_j(1 - b_j)\mu_{v_j^M} = a_j(1 - b_j)\mu_{u_j} = 0.$$

The weights $a_j(1 - b_j)$ are bounded and converge in capacity to the quasi-continuous function $a(1 - b)$. The weighted form of Theorem 9.4.8, applied to the bounded plurisubharmonic functions $\rho + v_j^M$, gives $a(1 - b)\mu_{v^M} = 0$ on D, and plurifine locality turns this into $a(1 - b)\mu_u = 0$ on D. Let $\varepsilon \downarrow 0$ and then $M \to \infty$. Since V_θ is finite on D, the weights a tend to 1 at every point of D where u is finite. They are bounded by 1, and μ_u does not charge the pole set of u, so dominated convergence gives $(1 - b)\mu_u = 0$ on D. Thus $u = f$ holds μ_u-almost everywhere on D. Covering $X \setminus A$ by such balls and using $\mu_u(A) = 0$ completes the proof. □

In Corollary 17.4.21 of the next chapter, the assumption that $\{\theta\}$ is big is removed.

Corollary 16.2.6 *Suppose that $\{\theta\}$ is big, let $u, v \in \mathrm{PSH}(X, \theta)$, and suppose that $w = P_\theta(u, v) \ne -\infty$. Then*

$$\mu_w \le \mathbf{1}_{\{w=u\}}\mu_u + \mathbf{1}_{\{w=v\}}\mu_v.$$

More generally, for finitely many potentials $u_1, \dots, u_r$ such that $w = P_\theta(\min_j u_j) \ne -\infty$, the measure μ_w is carried by $\bigcup_j \{w = u_j\}$.

Proof By Lemma 16.1.2, potentials are quasi-continuous, and hence so is the obstacle $\min\{u, v\}$. Its family of competitors is nonempty, because $w \ne -\infty$, and bounded above by $\sup_X u$. By Theorem 16.2.5, μ_w is carried by $\{w = \min\{u, v\}\} \subset \{w = u\} \cup \{w = v\}$. Since $w \le u$ and $w \le v$ everywhere, Lemma 16.2.2 gives $\mathbf{1}_{\{w=u\}}\mu_w \le \mathbf{1}_{\{w=u\}}\mu_u$ and $\mathbf{1}_{\{w=v\}}\mu_w \le \mathbf{1}_{\{w=v\}}\mu_v$. Adding these inequalities proves the first assertion; the intersection of the two contact sets is counted twice, which is harmless for an upper bound. The same argument, applied to the obstacle $\min_j u_j$, proves the last assertion. □

16.2.2 Obstacles with bounded metric Laplacian

The results above show that the Monge–Ampère measure of an envelope is carried by the contact set, but they do not identify the measure there. For obstacles with bounded Laplacian we now compute the measure on the contact set from the second derivatives of the obstacle. Recall from Section 14.1 that $L = \mathrm{d}^*\mathrm{d}$ is the Laplace–Beltrami operator of the Riemannian metric defined by the fixed Kähler form ω; with our sign convention it is nonnegative, and for the Euclidean metric it equals $-\Delta$. We call L the *metric Laplacian*. A function f on an open set $U \subset X$ has *(locally) bounded metric Laplacian* if the distribution Lf on U is a (locally) bounded measurable function.

By the trace identity $\Lambda \mathrm{dd}^{\mathrm{c}} f = -Lf/(4\pi)$ of Lemma 14.1.6, where Λ is the trace with respect to ω, a bound for Lf is a bound for the trace of the complex Hessian of f. In local holomorphic coordinates, write $\omega = \frac{\mathrm{i}}{2\pi}\sum_{i,j} h_{i\bar{j}}\,\mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$ and let $(h^{i\bar{j}})$ be the inverse matrix, normalized by $\sum_j h^{i\bar{j}} h_{l\bar{j}} = \delta_{il}$; then (14.8), applied with $\omega' = \omega$, gives

$$Lf = -4\pi \sum_{i,j} h^{i\bar{j}} f_{i\bar{j}} \tag{16.1}$$

for every distribution f in the coordinate domain. For a θ-psh function u the current $\theta + \mathrm{dd}^{\mathrm{c}} u$ is positive, and positivity turns this trace bound into bounds for all complex Hessian coefficients. A general obstacle has no such positivity, and if $n \geq 2$ its complex Hessian need not be bounded (Example 16.2.7). Nevertheless it has square-integrable weak second derivatives (Lemma 16.2.8), and on the set where it agrees with a potential that has bounded metric Laplacian, these coincide almost everywhere with the second derivatives of the potential. This gives the contact formula of Proposition 16.2.9. The L^p estimate of Proposition 16.2.11 shows moreover that such an obstacle has weak second derivatives in every L^p with $p < \infty$, so that wedge products of the corresponding weak forms are integrable on all of X (Corollary 16.2.12).

Example 16.2.7 Let $n \geq 2$ and $x_0 \in X$. Choose holomorphic coordinates z centered at x_0 in which $h_{i\bar{j}}(0) = \delta_{ij}$, which is possible after a complex-linear change of coordinates, and choose $0 < r_0 < \mathrm{e}^{-1}$ such that the coordinates are defined on a neighborhood of $\{|z| \leq r_0\}$. On $0 < |z| < 1$ put

$$\ell(z) = \log(-\log|z|), \qquad P(z) = |z_1|^2 - |z_2|^2, \qquad \phi = P\ell,$$

and $\phi(0) = 0$. Write $r = |z|$. Since $\partial_{z_i} \log r = \bar{z}_i/(2r^2)$ and $\partial_{\bar{z}_j}(r^2 \log r) = z_j(\log r + \frac{1}{2})$, we have

$$\partial_{z_i}\ell = \frac{\bar{z}_i}{2r^2 \log r}, \qquad \partial_{z_i}\partial_{z_j}\ell = -\frac{(\log r + \frac{1}{2})\bar{z}_i\bar{z}_j}{2r^4(\log r)^2},$$

$$\ell_{i\bar{j}} = \frac{\delta_{ij}}{2r^2 \log r} - \frac{(\log r + \frac{1}{2})\bar{z}_i z_j}{2r^4(\log r)^2}.$$

For $0 < r < r_0$ we have $\log r < -1$, hence $|\log r + \frac{1}{2}| \le |\log r|$, and these formulas give $|\nabla \ell| \le C/(r|\log r|)$ and $|D^2\ell| \le C/(r^2|\log r|)$ for the real gradient and Hessian, with a constant C depending only on n. Moreover $|P| \le r^2$, $|\nabla P| \le 2r$, the real Hessian of P is constant, $P_{1\bar 1} = 1$, $P_{2\bar 2} = -1$, and all other $P_{i\bar j}$ vanish. By the Leibniz rule, on $0 < r < r_0$,

$$|\nabla \phi| \le 2r\ell + \frac{Cr}{|\log r|}, \qquad |D^2\phi - \ell D^2 P| \le \frac{C}{|\log r|}, \qquad \Big|\phi_{i\bar j} - \ell P_{i\bar j}\Big| \le \frac{C}{|\log r|}.$$

In particular ϕ is continuously differentiable on $\{|z| < r_0\}$, with $\nabla\phi(0) = 0$, since $|\phi| \le r^2\ell$ and $\nabla\phi \to 0$ as $z \to 0$; and its pointwise second derivatives off the origin are bounded by $C(1+\ell) \le C(1 - \log|z|)$, hence locally integrable.

These pointwise derivatives are the distributional second derivatives of ϕ. Indeed, let η be a smooth function on $\mathbb{R}$ with $\eta = 0$ on $(-\infty, 1]$ and $\eta = 1$ on $[2, \infty)$, put $\eta_\varepsilon(z) = \eta(|z|/\varepsilon)$, and let ψ be a test function on $\{|z| < r_0\}$. Since ϕ is C^1, we have $\int \phi\, \partial_a\partial_b\psi = -\int \partial_a\phi\, \partial_b\psi$ for real coordinate derivatives ∂_a, ∂_b. Splitting $\psi = \eta_\varepsilon\psi + (1-\eta_\varepsilon)\psi$ and integrating by parts once more in the first term, where ϕ is smooth, we get

$$\int \phi\, \partial_a\partial_b\psi = \int \partial_b\partial_a\phi\, \eta_\varepsilon\psi - \int \partial_a\phi\, \partial_b\big((1-\eta_\varepsilon)\psi\big).$$

The last integral is bounded by $\sup_{|z|\le 2\varepsilon} |\nabla\phi| \cdot C_\psi(\varepsilon^{-1}+1)\varepsilon^{2n}$, which tends to zero, and the first tends to $\int \partial_b\partial_a\phi\,\psi$ by dominated convergence.

Choose a smooth function χ on X with support in $\{|z| < r_0\}$ that equals 1 on $\{|z| \le r_0/2\}$, and put $f = \chi\phi$, extended by zero. Then f is continuous. On $\{|z| < r_0/2\}$, (16.1) and the estimates above give

$$\Big|\frac{Lf}{4\pi}\Big| \le \ell\,|h^{1\bar 1} - h^{2\bar 2}| + \frac{C'}{|\log r|} \le C''\big(r\ell + 1\big),$$

because $h^{1\bar 1} - h^{2\bar 2}$ is smooth and vanishes at the origin; the right side is bounded, since $r\ell \to 0$ as $r \to 0$, and elsewhere f is smooth. Hence f has bounded metric Laplacian. On the other hand $f_{1\bar 1} = \phi_{1\bar 1} \ge \ell - C/|\log r|$ near x_0, which is unbounded. Thus the complex Hessian of f is not locally bounded.

We use the Sobolev spaces $W^{k,p}$ of Section 6.2. For an open set $U \subset X$, a function on U belongs to $W^{k,p}_{\mathrm{loc}}(U)$ if it belongs to $W^{k,p}_{\mathrm{loc}}$ in every holomorphic chart of U. Since membership in $W^{k,p}_{\mathrm{loc}}$ is a local property, Lemma 6.2.4(2), applied to the coordinate changes, shows that it suffices to check this in the charts of one atlas of U. To see that the property is local, let $D \subset \mathbb{R}^d$ be open, covered by open sets $D_1, D_2, \ldots$ (an open cover of D has a countable subcover), and let $v \in W^{k,p}_{\mathrm{loc}}(D_i)$ for every i. For $|\alpha| \le k$, the L^p_{loc} functions representing $\partial^\alpha v$ on D_i and on D_j agree almost everywhere on $D_i \cap D_j$ (Proposition 1.4.3(4)), so they define a measurable function g_α on D. It belongs to $L^p_{\mathrm{loc}}(D)$, because every compact subset of D is covered by finitely many

compact balls, each contained in some D_i. Finally, the distribution $\partial^\alpha v - g_\alpha$ vanishes on every D_i, hence on D (Lemma 1.4.2).

Let (U_i, φ_i) and (χ_i) be the charts and the partition of unity that define the norm of $W^{k,p}(X)$. For an open set $U' \subset X$ we put $\|f\|_{W^{k,p}(U')} = \sum_i \|(\chi_i f) \circ \varphi_i^{-1}\|_{W^{k,p}(\varphi_i(U' \cap U_i))}$. Every $v \in W^{k,p}(X)$ belongs to $W^{k,p}_{\mathrm{loc}}(X)$, with a bound: for every holomorphic chart $\psi\colon U_0 \to \mathbb{C}^n$ and every open set $U' \Subset U_0$,

$$\|v \circ \psi^{-1}\|_{W^{k,p}(\psi(U'))} \le C\|v\|_{W^{k,p}(X)}, \tag{16.2}$$

with C independent of v. Indeed, let η be a smooth function on X with compact support in U_0 that equals 1 on U' (Proposition 7.1.3). The charts $\psi, \varphi_1, \varphi_2, \dots$ with the partition of unity $\eta, (1-\eta)\chi_1, (1-\eta)\chi_2, \dots$ are another admissible choice in Section 6.2, so the norm they define is at most a constant times $\|v\|_{W^{k,p}(X)}$; one of its terms is $\|(\eta v) \circ \psi^{-1}\|_{W^{k,p}(\psi(U_0))}$, which dominates the left side of (16.2).

Finally, the Hilbert spaces H^r, $r \ge 0$, of Section 14.1 are contained in $W^{r,2}(X)$, with $\|u\|_{W^{r,2}(X)} \le C\|u\|_{H^r}$. Indeed, the charts and the cutoff functions that define H^r are an admissible choice in Section 6.2, so, up to a constant factor, the norm of $W^{r,2}(X)$ is the sum of the $W^{r,2}$ norms of the products of u with these cutoff functions, written in the corresponding charts. By Lemma 14.1.1(2), for $u \in H^r$ these products have square-integrable distributional derivatives of order at most r, and this sum is at most $C\|u\|_{H^r}$.

The following lemma is the variable-coefficient analogue of Lemma 9.12.2.

Lemma 16.2.8 *Let $U \subset X$ be open, and let $f \in L^2_{\mathrm{loc}}(U)$ be such that its distributional metric Laplacian Lf belongs to $L^2_{\mathrm{loc}}(U)$. Then $f \in W^{2,2}_{\mathrm{loc}}(U)$. More precisely, for open sets $U' \Subset V \Subset U$ there is a constant C, depending only on U', V and ω (and on the fixed charts defining the norms), such that*

$$\|f\|_{W^{2,2}(U')} \le C\big(\|f\|_{L^2(V)} + \|Lf\|_{L^2(V)}\big).$$

In particular, every continuous function on U with locally bounded metric Laplacian belongs to $W^{2,2}_{\mathrm{loc}}(U)$.

Proof Fix $c > 0$. By Lemma 14.1.3(3), applied to functions, on which $\mathcal{L} = L$, the operator $P = L + c$ is a bijection of the space of distributions on X onto itself, and its inverse maps H^{-1} boundedly into H^1 and $H^0 = L^2$ (Lemma 14.1.1(2)) boundedly into H^2. Since no derivative of f is known in advance, we gain regularity in two steps, using two cutoff functions.

For a smooth function χ on X, the Leibniz rules $\mathrm{d}(\chi u) = \chi\,\mathrm{d}u + u\,\mathrm{d}\chi$ and $\mathrm{d}^*(\chi\alpha) = \chi\,\mathrm{d}^*\alpha - \langle\alpha, \mathrm{d}\chi\rangle$, for smooth functions u and real 1-forms α, give

$$L(\chi u) = \chi L u + [L, \chi]u, \qquad [L, \chi]u = -2\langle \mathrm{d}\chi, \mathrm{d}u\rangle + (L\chi)u, \tag{16.3}$$

where $\langle\cdot,\cdot\rangle$ is the pointwise inner product of real 1-forms, extended bilinearly. Since L and multiplication by χ are formally self-adjoint, the transpose of $[L, \chi]$ is $-[L, \chi]$, and (16.3) extends to distributions u, with $\langle [L, \chi]u, \phi\rangle = -\langle u, [L, \chi]\phi\rangle$ for smooth ϕ.

By Proposition 7.1.3, choose smooth functions χ_0, χ_1 on X with compact supports in V such that $\chi_1 = 1$ on a neighborhood of $\overline{U'}$ and $\chi_0 = 1$ on a neighborhood of $\operatorname{Supp}\chi_1$. Extend $\chi_0 f$ by zero to a function on X. By (16.3),

$$P(\chi_0 f) = \chi_0(Lf + cf) + [L, \chi_0]f.$$

The first term lies in L^2, with norm at most $\|Lf\|_{L^2(V)} + c\|f\|_{L^2(V)}$, hence in H^{-1} with a comparable norm. For smooth ϕ, the function $[L, \chi_0]\phi$ is a first-order expression in ϕ with smooth coefficients supported in V, so

$$|\langle [L, \chi_0]f, \phi\rangle| = |\langle f, [L, \chi_0]\phi\rangle| \le C\|f\|_{L^2(V)}\|\phi\|_{H^1},$$

because the norm of H^1 dominates the L^2 norms of the first derivatives on compact subsets of the charts (Lemma 14.1.1(2)). Hence $[L, \chi_0]f \in H^{-1}$, and the right side of the last identity has H^{-1} norm at most $C(\|Lf\|_{L^2(V)} + \|f\|_{L^2(V)})$. Applying P^{-1}, we obtain $\chi_0 f \in H^1$ with the same bound.

In particular f has square-integrable first derivatives on the interior of $\{\chi_0 = 1\}$, which contains $\operatorname{Supp}\chi_1$, with L^2 norm bounded in the same way. Therefore the distribution $[L, \chi_1]f = -2\langle \mathrm{d}\chi_1, \mathrm{d}f\rangle + (L\chi_1)f$ is a square-integrable function, and in the identity

$$P(\chi_1 f) = \chi_1(Lf + cf) + [L, \chi_1]f$$

the right side lies in L^2, with norm at most $C(\|Lf\|_{L^2(V)} + \|f\|_{L^2(V)})$. Since P^{-1} is bounded from L^2 to H^2, it follows that $\chi_1 f \in H^2 \subset W^{2,2}(X)$, with this bound for its norm in $W^{2,2}(X)$. Since $\chi_1 = 1$ on U', this proves the estimate.

Finally, let $\psi : U_0 \to \mathbb{C}^n$ be a holomorphic chart with $U_0 \subset U$, and let $U'' \Subset U_0$ be open. Applying the preceding argument with $U' = U''$ and some open V with $U'' \Subset V \Subset U$, we find $\chi_1 f \in W^{2,2}(X)$ with $\chi_1 = 1$ on U'', so $f \circ \psi^{-1} \in W^{2,2}(\psi(U''))$ by (16.2). Hence $f \in W^{2,2}_{\mathrm{loc}}(U)$. The last assertion holds because continuous functions and locally bounded measurable functions are locally square-integrable. □

For the contact formula we work in a coordinate ball. Recall that $D(A_1, \dots, A_n)$ denotes the mixed discriminant of Hermitian $n \times n$ matrices, defined in Section 7.2 (see Lemma 7.2.2 for its basic properties), that $\mathrm{d}V_{2n}$ is Lebesgue measure in the coordinates, and that $(u_{a\bar{b}})$, with $u_{a\bar{b}} = \partial_{z_a}\partial_{\bar{z}_b}u$, is the complex Hessian matrix of u (Section 7.1).

Proposition 16.2.9 *Let $U \subset X$ be open. For $1 \le j \le n$, let θ_j be a closed smooth real $(1,1)$-form on U, let u_j be a continuous θ_j-psh function on U with locally bounded metric Laplacian, and let f_j be a real Borel function on U that belongs to $W^{2,1}_{\mathrm{loc}}(U)$. By Lemma 16.2.8, this holds, for instance, if f_j is continuous and has locally bounded metric Laplacian. Put*

$$E = \bigcap_{j=1}^{n} \{u_j = f_j\}.$$

Let $W \Subset U$ be a coordinate ball, choose smooth real functions ρ_j on W with $\mathrm{dd}^{\mathrm{c}}\rho_j = \theta_j$, and let

$$A_j = \big((\rho_j + u_j)_{a\bar b}\big)_{a,b}, \qquad B_j = \big((\rho_j + f_j)_{a\bar b}\big)_{a,b}$$

be the weak complex Hessian matrices, defined almost everywhere on W. Then each A_j is positive semidefinite almost everywhere and locally bounded, and on W

$$\bigwedge_{j=1}^{n} (\theta_j + \mathrm{dd^c} u_j) = \frac{n!}{\pi^n} D(A_1, \dots, A_n)\, \mathrm{d}V_{2n},$$
$$\mathbf{1}_E \bigwedge_{j=1}^{n} (\theta_j + \mathrm{dd^c} u_j) = \frac{n!}{\pi^n} \mathbf{1}_E D(B_1, \dots, B_n)\, \mathrm{d}V_{2n}.$$

In particular the density $\mathbf{1}_E D(B_1, \dots, B_n)$ is nonnegative and locally bounded, although the f_j need not be θ_j-psh.

Proof The potentials ρ_j exist by Lemma 13.1.2. By Lemma 16.2.8, all u_j belong to $W^{2,2}_{\mathrm{loc}}(U) \subset W^{2,1}_{\mathrm{loc}}(U)$, so the matrices A_j and B_j are defined almost everywhere, and the positive current $T_j = \theta_j + \mathrm{dd^c} u_j = \mathrm{dd^c}(\rho_j + u_j)$ has coefficients in L^2_{loc}.

We first show that A_j is positive semidefinite almost everywhere. For each $\xi \in \mathbb{C}^n$ the distribution $\sum_{a,b} (\rho_j + u_j)_{a\bar b} \xi_a \bar\xi_b$ is positive, because T_j is a positive current; since it is a locally integrable function, it is nonnegative almost everywhere, by Proposition 1.4.3(3),(4) and Lemma 1.4.1(2) (its mollifications are nonnegative and converge to it in L^1_{loc}). We apply this to the countably many ξ with rational real and imaginary parts. Outside the union of the corresponding null sets, $\sum_{a,b} (A_j)_{ab} \xi_a \bar\xi_b \geq 0$ for all such ξ, hence for all $\xi \in \mathbb{C}^n$, because these ξ are dense and a quadratic form on $\mathbb{C}^n$ is continuous.

Next we bound the coefficients. By the trace identity (14.7),

$$\Lambda T_j = \Lambda \theta_j - \frac{L u_j}{4\pi},$$

which is bounded on W, since $L u_j$ is locally bounded and $\overline{W}$ is compact. Since $T_j = \frac{\mathrm{i}}{2\pi} \sum_{a,b} (A_j)_{ab}\, \mathrm{d}z_a \wedge \mathrm{d}\bar z_b$, the function ΛT_j is the trace $\sum_{a,b} h^{a\bar b} (A_j)_{ab} = \operatorname{tr}(H^{-1} A_j)$ of A_j relative to h, as in (14.8), where $H = (h_{a\bar b})$. As $\overline{W}$ is compact, $H \leq CI$ on W for a constant C, so $H^{-1} \geq C^{-1} I$; since A_j is positive semidefinite, $\operatorname{tr}(H^{-1} A_j) = \operatorname{tr}(A_j^{1/2} H^{-1} A_j^{1/2}) \geq C^{-1} \operatorname{tr} A_j$ almost everywhere on W. For a positive semidefinite Hermitian matrix every diagonal entry is at most the trace, and every entry satisfies $|a_{ab}|^2 \leq a_{aa} a_{bb}$, by the positivity of the principal 2×2 minors. Therefore A_j is bounded on W.

The functions $\rho_j + u_j$ are continuous and plurisubharmonic on W, and they have bounded complex Hessian in the sense of Section 9.12. The first identity is therefore the first part of Proposition 9.12.4. The functions $\rho_j + f_j$ are real Borel functions in $W^{2,1}_{\mathrm{loc}}(W)$, and adding ρ_j to both u_j and f_j does not change the contact set $\{u_j = f_j\}$. Hence the second part of Proposition 9.12.4 gives the second identity. Moreover Proposition 9.12.3, applied to $\rho_j + u_j$ and $\rho_j + f_j$, shows that $A_j = B_j$ almost everywhere on $\{u_j = f_j\}$, hence on E. Consequently $\mathbf{1}_E D(B_1, \dots, B_n) =$

$\mathbf{1}_E D(A_1, \dots, A_n)$ almost everywhere. This density is nonnegative, since the mixed discriminant of positive semidefinite matrices is nonnegative (Lemma 7.2.2(3)), and it is bounded, since D is a polynomial in the entries of the bounded matrices A_j. □

The contact formula involves the weak complex Hessians of an obstacle only on the contact set. Off this set the second derivatives of an obstacle with bounded metric Laplacian may be unbounded, and to give a meaning to the wedge products of the forms $\theta_j + \mathrm{dd}^c f_j$ on all of X we need more integrability. We now prove that all weak second derivatives of such an obstacle lie in L^p for every $p < \infty$. This belongs to the L^p theory of the elliptic operator L on the compact manifold X, and we derive it from the Euclidean estimate (6.3) by freezing coefficients. We write $C^{1,\alpha}(X)$ for the space of functions on X that belong to $C^{1,\alpha}_{\mathrm{loc}}$ in every chart (Section 6.2). As there, matrices carry the Hilbert–Schmidt norm, and for a matrix $A = (A^{\mu\sigma})$ of functions and a function w we write $A : D^2 w = \sum_{\mu,\sigma} A^{\mu\sigma} \partial_\mu \partial_\sigma w$.

Lemma 16.2.10 *Let $d \geq 1$, $1 < p < \infty$ and $\lambda > 0$. There are constants $\delta_0 > 0$ and C, depending only on d, p and λ, with the following property. Let $B \subset \mathbb{R}^d$ be a ball with center x_B, and let A be a measurable function on B with values in the real symmetric $d \times d$ matrices such that*

$$A(x_B) \geq \lambda I, \qquad |A(x) - A(x_B)| \leq \delta_0 \quad \text{for } x \in B.$$

Then every $w \in W^{2,p}(\mathbb{R}^d)$ that vanishes almost everywhere outside a compact subset of B satisfies

$$\|D^2 w\|_p \leq C \|A : D^2 w\|_{L^p(B)}.$$

Proof First let A_0 be a constant real symmetric matrix with $A_0 \geq \lambda I$, and put $M = A_0^{-1/2}$, a symmetric matrix of operator norm at most $\lambda^{-1/2}$. For $v \in C_c^\infty(\mathbb{R}^d)$ let $\tilde{v}(y) = v(M^{-1}y)$, so that $v(x) = \tilde{v}(Mx)$. Then $D^2 v(x) = M\, D^2\tilde{v}(Mx)\, M$ and $A_0 : D^2 v(x) = \mathrm{tr}(M A_0 M\, D^2\tilde{v}(Mx)) = \Delta\tilde{v}(Mx)$. Since $|MNM| \leq \lambda^{-1}|N|$ for every matrix N, the change of variables $y = Mx$ and (6.3) give

$$\begin{aligned} \|D^2 v\|_p &\leq \lambda^{-1} |\det M|^{-1/p} \|D^2\tilde{v}\|_p \\ &\leq \lambda^{-1} C_{d,p} |\det M|^{-1/p} \|\Delta\tilde{v}\|_p = \lambda^{-1} C_{d,p} \|A_0 : D^2 v\|_p. \end{aligned}$$

This estimate extends to every $w \in W^{2,p}(\mathbb{R}^d)$ that vanishes almost everywhere outside a compact set K. Indeed, for $0 < \varepsilon \leq 1$ the mollifications $w * \rho_\varepsilon$ are test functions supported in the compact set $K' = \{x : \mathrm{dist}(x, K) \leq 1\}$, their second derivatives are $(\partial_\mu \partial_\sigma w) * \rho_\varepsilon$, by Proposition 1.4.3(1), and these converge to $\partial_\mu \partial_\sigma w$ in $L^p(K')$, by Lemma 1.4.1(2), hence in $L^p(\mathbb{R}^d)$, since all these functions vanish almost everywhere outside K'. Applying the estimate to $v = w * \rho_\varepsilon$ and letting $\varepsilon \to 0$, we obtain it for w.

Now let A and w be as in the statement, put $C_1 = \lambda^{-1} C_{d,p}$, $\delta_0 = 1/(2C_1)$, and $A_0 = A(x_B)$. Since $|\sum_{\mu,\sigma} m_{\mu\sigma} m'_{\mu\sigma}| \leq |m|\,|m'|$ for matrices m and m', we have $|A_0 : D^2 w - A : D^2 w| \leq \delta_0 |D^2 w|$ almost everywhere on B, and both sides vanish outside B. Hence

$$\|D^2 w\|_p \le C_1 \|A_0 : D^2 w\|_p \le C_1 \|A : D^2 w\|_{L^p(B)} + \tfrac{1}{2}\|D^2 w\|_p,$$

and since $\|D^2 w\|_p < \infty$ we obtain the lemma with $C = 2C_1$. □

We apply the lemma to complex Hessian operators. Let $U \subset \mathbb{C}^n = \mathbb{R}^{2n}$ be open, with real coordinates $x_{2i-1} = \operatorname{Re} z_i$, $x_{2i} = \operatorname{Im} z_i$, and let $(a^{i\bar{j}})$ be a function on U with values in $n \times n$ matrices such that $\overline{a^{i\bar{j}}} = a^{j\bar{i}}$. Since $\partial_{z_i} = \frac{1}{2}(\partial_{2i-1} - \mathrm{i}\partial_{2i})$ and $\partial_{\bar{z}_j} = \frac{1}{2}(\partial_{2j-1} + \mathrm{i}\partial_{2j})$, the operator $\sum_{i,j} a^{i\bar{j}} \partial_{z_i} \partial_{\bar{z}_j}$ equals $\sum_{\mu,\sigma} c^{\mu\sigma} \partial_\mu \partial_\sigma$ with complex coefficients $c^{\mu\sigma}$. For $\xi \in \mathbb{R}^{2n}$ they satisfy $\sum_{\mu,\sigma} c^{\mu\sigma} \xi_\mu \xi_\sigma = \frac{1}{4} \sum_{i,j} a^{i\bar{j}} \zeta_i \bar{\zeta}_j$, where $\zeta_i = \xi_{2i-1} - \mathrm{i}\xi_{2i}$. The operator maps real functions to real functions, because $\overline{a^{i\bar{j}} \partial_{z_i} \partial_{\bar{z}_j} w} = a^{j\bar{i}} \partial_{\bar{z}_i} \partial_{z_j} w$ for real w. Hence, on real functions,

$$\sum_{i,j} a^{i\bar{j}} \partial_{z_i} \partial_{\bar{z}_j} w = A : D^2 w, \qquad A\xi \cdot \xi = \frac{1}{4} \sum_{i,j} a^{i\bar{j}} \zeta_i \bar{\zeta}_j, \tag{16.4}$$

where A is the symmetric part of the real part of $(c^{\mu\sigma})$. Thus A is real symmetric, $|A(x) - A(y)| \le |a(x) - a(y)|$, and since $|\zeta| = |\xi|$, a lower bound $\sum a^{i\bar{j}} \zeta_i \bar{\zeta}_j \ge \lambda |\zeta|^2$ gives $A \ge (\lambda/4) I$. By (16.1), in a holomorphic chart the metric Laplacian acts on real functions as $L = -4\pi\, A : D^2$, where A is obtained in this way from $a^{i\bar{j}} = h^{i\bar{j}}$; in particular L has no terms of order less than two.

Proposition 16.2.11 *Let $1 < p < \infty$, and let $f \in L^p(X)$ be such that its distributional metric Laplacian Lf belongs to $L^p(X)$. Then $f \in W^{2,p}(X)$, and*

$$\|f\|_{W^{2,p}(X)} \le C_{X,\omega,p} \big(\|Lf\|_{L^p(X)} + \|f\|_{L^p(X)} \big), \tag{16.5}$$

where the constant $C_{X,\omega,p}$ depends only on X, ω and p (and on the fixed charts defining the norm). In particular, a continuous function on X with bounded metric Laplacian belongs to $W^{2,p}(X)$ for every finite $p \ge 1$ and to $C^{1,\alpha}(X)$ for every $0 < \alpha < 1$.

Proof We first treat smooth functions and then pass to the general case through the inverse of $L + c$.

Step 1: the estimate for smooth real functions. Let $d = 2n$. Fix finitely many holomorphic charts and compact subsets K_ι of their domains whose interiors cover X. By (16.4), in these charts $L = -4\pi\, A : D^2$ on real functions, with smooth real symmetric matrices A; by compactness there is $\lambda > 0$ with $A \ge \lambda I$ on every K_ι. Let δ_0 and C be the constants of Lemma 16.2.10 for d, p and λ. Since A is uniformly continuous on a compact neighborhood of each K_ι, there is $r > 0$ such that, for every $x \in K_\iota$, the ball $B(x, r)$ lies in that neighborhood and $|A(y) - A(x)| \le \delta_0$ for $y \in B(x, r)$. Cover X by finitely many such balls B_ν, and take a smooth partition of unity (χ_ν) with $\operatorname{Supp} \chi_\nu$ a compact subset of B_ν (Proposition 7.1.3).

Let f be smooth and real, and let $v_\nu = \chi_\nu f$, written in the chart of B_ν and extended by zero. By Lemma 16.2.10, $\|D^2 v_\nu\|_p \le C \|A : D^2 v_\nu\|_p = (C/4\pi) \|L v_\nu\|_p$. By (16.3), $L v_\nu = \chi_\nu L f + [L, \chi_\nu] f$, where the commutator involves only f and its first derivatives. Summation therefore gives

$$S_2 := \sum_\nu \|D^2 v_\nu\|_p \le C\big(\|Lf\|_p + \|\nabla f\|_p + \|f\|_p\big).$$

Here and below the L^p norms of f and ∇f on X may be computed in the charts or with the metric of ω; on the fixed finite cover these choices give equivalent norms. Since $f = \sum_\nu v_\nu$, the norm $\|\nabla f\|_p$ is at most a fixed multiple of $\sum_\nu \|\nabla v_\nu\|_p$. By Lemma 6.2.5, for every $\varepsilon > 0$,

$$\sum_\nu \|\nabla v_\nu\|_p \le \varepsilon S_2 + C_\varepsilon \|f\|_p.$$

Combining the last three estimates and choosing ε small, we can absorb S_2, which is finite because f is smooth. This gives $S_2 \le C(\|Lf\|_p + \|f\|_p)$ and then $\|\nabla f\|_p \le C(\|Lf\|_p + \|f\|_p)$. Since the norm of $W^{2,p}(X)$ may be computed, up to equivalence, with the charts of the balls B_ν and the partition of unity (χ_ν) (Section 6.2), this proves (16.5) for smooth real f.

Step 2: an L^p bound for the inverse. Fix $c > 0$ and put $P = L + c$. For a smooth real function ψ, Lemma 14.1.3(3) gives a unique distributional solution $u = P^{-1}\psi$, which lies in every H^r and is therefore smooth, by Lemma 14.1.1(3); it is real since P is a real operator. For $\eta > 0$ put

$$\phi_\eta(s) = s(s^2{+}\eta)^{(p-2)/2}, \qquad \phi_\eta'(s) = \big((p{-}1)s^2{+}\eta\big)(s^2{+}\eta)^{(p-4)/2} \ge 0, \qquad s \in \mathbb{R}.$$

The function ϕ_η is smooth, so we may multiply the equation $Lu + cu = \psi$ by $\phi_\eta(u)$ and integrate. Since d^* is the formal adjoint of d, we have $\int_X (Lu)\phi_\eta(u)\,\mathrm{d}V = \int_X \phi_\eta'(u)|\mathrm{d}u|^2\,\mathrm{d}V \ge 0$, and therefore

$$c\int_X u\phi_\eta(u)\,\mathrm{d}V \le \int_X \psi\phi_\eta(u)\,\mathrm{d}V.$$

As $\eta \downarrow 0$, $u\phi_\eta(u) \to |u|^p$ and $\phi_\eta(u) \to u|u|^{p-2}$ pointwise, the latter being 0 where $u = 0$. For $p \ge 2$ and $\eta \le 1$ both functions are bounded by $(1+u^2)^{p/2}$, which is bounded since u is continuous on the compact manifold X; for $1 < p < 2$ we have $(u^2+\eta)^{(p-2)/2} \le |u|^{p-2}$ where $u \ne 0$, so they are bounded by $|u|^p$ and $|u|^{p-1}$. Dominated convergence and Hölder's inequality therefore give $c\|u\|_p^p \le \|\psi\|_p \|u\|_p^{p-1}$, that is (if $u \ne 0$; otherwise there is nothing to prove),

$$\|P^{-1}\psi\|_p \le c^{-1}\|\psi\|_p.$$

Applying Step 1 to u, for which $Lu = \psi - cu$, we conclude that $\|P^{-1}\psi\|_{W^{2,p}} \le C\|\psi\|_p$.

Step 3: the general case. Let f be real, put $\psi = Lf + cf \in L^p(X)$, and choose smooth real functions $\psi_j \to \psi$ in $L^p(X)$, using chart cutoffs and mollification (Lemma 1.4.1). By Step 2 the solutions $u_j = P^{-1}\psi_j$ form a Cauchy sequence in $W^{2,p}(X)$. This space is complete: in each chart the derivatives of order at most two converge in L^p, and testing against smooth functions identifies the limits with the weak derivatives of the limit function. Let u be the limit. Then $Pu_j \to Pu$ in the sense of distributions, so $Pu = \psi$. Thus $P(u - f) = 0$, and since P is injective on

distributions (Lemma 14.1.3(3)), $u = f$. Passing to the limit in the estimate of Step 2 gives $\|f\|_{W^{2,p}} \le C\|\psi\|_p \le C(\|Lf\|_p + \|f\|_p)$. For complex f we apply this to the real and imaginary parts, since L is a real operator.

Step 4: continuous functions with bounded Laplacian. If f is continuous and Lf is bounded, then f and Lf lie in every $L^p(X)$ with $p < \infty$, which proves the assertion for $1 < p < \infty$; the case $p = 1$ follows from the case $p = 2$, since X has finite volume. Given $0 < \alpha < 1$, choose $p > d/(1-\alpha)$. In every chart, f belongs to $W^{2,p}_{\mathrm{loc}}$ by (16.2), so Morrey's inequality (Lemma 6.2.6) gives a representative of f in $C^{1,1-d/p}_{\mathrm{loc}}$, hence in $C^{1,\alpha}_{\mathrm{loc}}$. This representative agrees with the continuous function f almost everywhere, hence everywhere. □

Consequently the Hessian forms of obstacles with bounded metric Laplacian can be multiplied on all of X, the products being taken pointwise almost everywhere.

Corollary 16.2.12 *Let $\theta_1, \dots, \theta_n$ be closed smooth real $(1,1)$-forms on X, and let $f_1, \dots, f_n$ be continuous real functions on X with bounded metric Laplacian. Then each $\theta_j + \mathrm{dd^c} f_j$ is a real $(1,1)$-form whose coefficients, computed from the weak second derivatives of f_j, lie in L^p for every $p < \infty$. Their pointwise wedge product is defined almost everywhere and satisfies*

$$\bigwedge_{j=1}^{n} (\theta_j + \mathrm{dd^c} f_j) = F \, \mathrm{d}V, \qquad F \in L^q(X) \quad \textit{for every } 1 \le q < \infty;$$

in particular it is a finite signed Radon measure. On a coordinate ball, with smooth real ρ_j such that $\mathrm{dd^c}\rho_j = \theta_j$, it equals $n!\pi^{-n} D(B_1, \dots, B_n)\, \mathrm{d}V_{2n}$, where $B_j = ((\rho_j + f_j)_{a\bar{b}})_{a,b}$. If moreover u_j are continuous θ_j-psh functions on X with bounded metric Laplacian, then on $E = \bigcap_j \{u_j = f_j\}$ this signed measure agrees with the positive measure $\mathbf{1}_E \bigwedge_j (\theta_j + \mathrm{dd^c} u_j)$ of Proposition 16.2.9.

Proof By Proposition 16.2.11 and (16.2), every f_j belongs to $W^{2,p}_{\mathrm{loc}}$ in every chart, for every finite p. In a chart, the current $\theta_j + \mathrm{dd^c} f_j$ is therefore given by integration against the form $\theta_j + \frac{\mathrm{i}}{2\pi} \sum_{a,b} (f_j)_{a\bar{b}} \, \mathrm{d}z_a \wedge \mathrm{d}\bar{z}_b$, where the $(f_j)_{a\bar{b}}$ are the weak derivatives, as in (7.5); its coefficients lie in L^p_{loc}. On the overlap of two charts, the forms obtained in the two charts represent the same current. For a current with locally integrable coefficients, (7.8), in which δ is the Jacobian determinant relating the two Lebesgue measures, says that the coefficients in the two charts are related, as distributions and hence almost everywhere (Proposition 1.4.3(4)), by the pointwise change of variables of forms. Thus the two forms agree almost everywhere as pointwise forms. Hence the pointwise wedge product of the forms $\theta_j + \mathrm{dd^c} f_j$ is a well-defined (n,n)-form, almost everywhere on X, and we write it as $F \, \mathrm{d}V$.

A coefficient of the wedge product in a chart is a sum of products of n coefficients of the factors and of smooth functions; by Hölder's inequality, with each factor in L^{nq} on compact subsets of the chart, these products lie in L^q there, for every $1 \le q < \infty$. Covering X by finitely many compact subsets of charts, we obtain $F \in L^q(X)$. A real function in $L^1(X)$ defines a finite signed Radon measure through its positive and negative parts.

On a coordinate ball, $\theta_j + \mathrm{dd^c} f_j = \mathrm{dd^c}(\rho_j + f_j) = \frac{\mathrm{i}}{2\pi} \sum_{a,b} (B_j)_{ab}\, \mathrm{d}z_a \wedge \mathrm{d}\bar{z}_b$ almost everywhere, and the identity $\omega_{A_1} \wedge \cdots \wedge \omega_{A_n} = D(A_1, \dots, A_n)\, \omega_I^n$ of Lemma 7.2.2(2), applied pointwise with $A_j = B_j/(2\pi)$, together with $\omega_I^n = n!\, 2^n\, \mathrm{d}V_{2n}$, gives $\bigwedge_j (\theta_j + \mathrm{dd^c} f_j) = n! \pi^{-n} D(B_1, \dots, B_n)\, \mathrm{d}V_{2n}$. Finally, on every coordinate ball W the second identity of Proposition 16.2.9, applied with $U = X$, shows that $\mathbf{1}_E \bigwedge_j (\theta_j + \mathrm{dd^c} u_j) = \mathbf{1}_E F\, \mathrm{d}V$ on W. Since coordinate balls cover X, the two measures agree on X. □

16.2.3 Penalized equations and metric-Laplacian envelopes

We now show that, for a Kähler background, the envelope of the minimum of finitely many obstacles with bounded metric Laplacian again has bounded metric Laplacian (Theorem 16.2.24). Throughout this subsection κ is a fixed Kähler form on X. It is the background of the envelopes, while ω remains the metric with respect to which the Laplacians of obstacles are measured. Constants are κ-psh, so for a continuous obstacle f the envelope $P_\kappa(f)$ is a bounded κ-psh function with $\min_X f \le P_\kappa(f) \le f$ everywhere, by Proposition 16.1.4(1),(2). For smooth f we approximate $P_\kappa(f)$ by the smooth solutions a_β of the Monge–Ampère equation with exponential penalty

$$(\kappa + \mathrm{dd^c} a)^n = \mathrm{e}^{\beta(a-f)} \kappa^n,$$

where $\beta > 0$ is a large parameter. On the set where $a > f$ the right side is very large, which forces the solution to lie essentially below f; on the set where $a < f$ it is very small, just as the Monge–Ampère measure of the envelope vanishes off the contact set. As $\beta \to \infty$ the solutions converge uniformly to $P_\kappa(f)$, with an explicit rate (Proposition 16.2.22).

The main estimate is a bound for the Laplacian of a_β that is uniform in β and depends on the obstacle f only through its supremum norm and an *upper* bound for the trace $\Lambda \mathrm{dd^c} f = -Lf/(4\pi)$ of its complex Hessian, that is, a *lower* bound for its metric Laplacian (Lemma 16.2.15). This one-sided dependence is what makes finite minima accessible. The complex Hessian of $\min_j f_j$ can have a negative singular part on the set where two of the f_j cross. For instance, in a chart, with $x_1 = \operatorname{Re} z_1$, the function $\min\{x_1, -x_1\} = -|x_1|$ satisfies $\partial_{z_1} \partial_{\bar{z}_1} (-|x_1|) = -\frac{1}{4} \partial_{x_1}^2 |x_1| = -\frac{1}{2} \sigma$, where σ is Lebesgue measure on the hyperplane $\{x_1 = 0\}$. However, the smooth approximations $-\delta \log \sum_j \mathrm{e}^{-f_j/\delta}$ of the minimum have complex Hessians whose traces are bounded above independently of δ (Lemma 16.2.23).

We solve the penalized equation by the continuity method (Theorem 16.2.21). Besides the trace bound, the a priori estimates require control of the third derivatives of the solutions. We obtain it from Calabi's maximum principle for the squared norm of the difference of two Chern connections (Lemma 16.2.17). The higher derivatives, and the smoothness of the solutions produced by perturbation, are then controlled by the elliptic L^p estimates of Section 16.2.2 (Lemma 16.2.19).

In local holomorphic coordinates we put $\partial_i = \partial_{z_i}$ and $\partial_{\bar{\jmath}} = \partial_{\bar{z}_j}$, sum over repeated indices, and write

$$\omega = \frac{\mathrm{i}}{2\pi} h_{i\bar{j}}\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j, \qquad \kappa = \frac{\mathrm{i}}{2\pi} k_{i\bar{j}}\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j,$$

so that $h = (h_{i\bar{j}})$ and $k = (k_{i\bar{j}})$ are the coefficient matrices of $2\pi\omega$ and $2\pi\kappa$. As in (14.8), let $(h^{i\bar{j}})$ be the inverse matrix of h, normalized by $h^{i\bar{j}} h_{l\bar{j}} = \delta_{il}$ (then also $h^{j\bar{i}} h_{j\bar{l}} = \delta_{il}$), and for a Hermitian matrix $m = (m_{i\bar{j}})$ put $\operatorname{tr}_h m = h^{i\bar{j}} m_{i\bar{j}}$. For a real function a write $a_{i\bar{j}} = \partial_i \partial_{\bar{j}} a$. The operator $D_h = h^{i\bar{j}} \partial_i \partial_{\bar{j}}$ is the operator D_g of (14.8) for the Kähler form $\omega' = \omega$, so that

$$D_h a = h^{i\bar{j}} a_{i\bar{j}} = \Lambda \mathrm{d}\mathrm{d}^{\mathrm{c}} a = -\frac{La}{4\pi}.$$

Hence D_h is the metric Laplacian up to the factor $-1/(4\pi)$; in particular, an upper bound for $D_h f$ is a lower bound for Lf. Since ω is an arbitrary Kähler form on X, everything in this subsection holds with ω replaced by any Kähler form, for instance by κ. As X is compact, there is $\gamma > 0$ with $k \geq \gamma h$.

For a smooth real function a put $g_{i\bar{j}} = k_{i\bar{j}} + a_{i\bar{j}}$, so that $\kappa + \mathrm{d}\mathrm{d}^{\mathrm{c}} a = \frac{\mathrm{i}}{2\pi} g_{i\bar{j}}\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$. When $g > 0$, that is, when $\kappa + \mathrm{d}\mathrm{d}^{\mathrm{c}} a$ is a Kähler form, we define $D_g = g^{i\bar{j}} \partial_i \partial_{\bar{j}}$ and $\operatorname{tr}_g m = g^{i\bar{j}} m_{i\bar{j}}$ in the same way; by (14.8), $D_g = -L_{\kappa + \mathrm{d}\mathrm{d}^{\mathrm{c}} a}/(4\pi)$. For a $(1,0)$-form $\xi = \xi_i\, \mathrm{d}z_i$ we write $|\xi|_g^2 = g^{i\bar{j}} \xi_i \bar{\xi}_j$ and $|\xi|_h^2 = h^{i\bar{j}} \xi_i \bar{\xi}_j$, and we set

$$H = \operatorname{tr}_h g = \operatorname{tr}_h k + D_h a.$$

The *eigenvalues of g relative to h* are the eigenvalues of the matrix $h^{-1}g$; they are positive, their sum is H, the sum of their inverses is $\operatorname{tr}_g h$, and their product is $\det g / \det h$.

Under a holomorphic change of coordinates $z = z(w)$ with Jacobian matrix $J = (\partial z_a / \partial w_b)$, the coefficient matrices change by $k \mapsto J^{\mathrm{T}} k \bar{J}$, so $\det k$ is multiplied by $|\det J|^2$, and $\log \det k$ changes by the function $\log |\det J|^2$. This function is pluriharmonic: for a nonvanishing holomorphic function F, $\partial_{\bar{j}} \log |F|^2 = \overline{\partial_j F} / \bar{F}$ is antiholomorphic, so $\partial_i \partial_{\bar{j}} \log |F|^2 = 0$. Consequently the ratios $\det k / \det h$ and $\det g / \det k$ are smooth positive functions on X, the functions $D_h \log \det k$ and $D_h \log \det g$ do not depend on the coordinates, and the functions $\operatorname{Ric}(k)_{i\bar{j}} = -\partial_i \partial_{\bar{j}} \log \det k$ are the coefficients of a well-defined smooth real $(1,1)$-form $\frac{\mathrm{i}}{2\pi} \operatorname{Ric}(k)_{i\bar{j}}\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$ on X.

By Lemma 7.2.2(2), $\omega_A^n = \det A\, \omega_I^n$ for every Hermitian $n \times n$ matrix A, where $\omega_A = \mathrm{i} \sum_{i,j} A_{i\bar{j}}\, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$; since $\kappa + \mathrm{d}\mathrm{d}^{\mathrm{c}} a = \omega_{g/2\pi}$ and $\kappa = \omega_{k/2\pi}$, it follows that $(\kappa + \mathrm{d}\mathrm{d}^{\mathrm{c}} a)^n / \kappa^n = \det g / \det k$. We also use Jacobi's formula $\frac{\mathrm{d}}{\mathrm{d}\tau} \log \det M(\tau) = \operatorname{tr}(M^{-1} M')$ for a differentiable family of matrices with positive determinant. It follows from the identities $\det(M + \varepsilon N) = \det M \det(I + \varepsilon M^{-1} N)$ and $\det(I + \varepsilon N') = 1 + \varepsilon \operatorname{tr} N' + O(\varepsilon^2)$. With our normalization of the inverse matrix, it reads $\partial \log \det g = g^{i\bar{j}} \partial g_{i\bar{j}}$ for every first-order derivative ∂.

The maximum principle enters through the following observation. If a smooth real function b has a local maximum at a point x, then its complex Hessian is negative semidefinite at x: for $\xi \in \mathbb{C}^n$ the function $s \mapsto b(x + s\xi)$ of one complex variable has a local maximum at $s = 0$, so its Laplacian $4 b_{i\bar{j}} \xi_i \bar{\xi}_j$ is nonpositive there. Consequently

$D_g b \le 0$ at x, since the trace of the product of a positive definite and a negative semidefinite Hermitian matrix is nonpositive.

Finally $\operatorname{osc}_X f = \max_X f - \min_X f$ denotes the oscillation of a continuous function f. For smooth f, Stokes' formula gives $\int_X D_h f \, \mathrm{d}V = -(4\pi)^{-1} \int_X L f \, \mathrm{d}V = -(4\pi)^{-1} \int_X \langle \mathrm{d}f, \mathrm{d}1 \rangle \, \mathrm{d}V = 0$, so that

$$\sup_X D_h f \ge 0. \tag{16.6}$$

Moreover, two-sided bounds for $D_h v$ in the sense of distributions are pointwise bounds. Let v be a real function in $L^1(X)$; since $D_h = -L/(4\pi)$ is formally self-adjoint, $\langle D_h v, \chi \rangle = \int_X v \, D_h \chi \, \mathrm{d}V$ for smooth χ. Suppose that there are constants $c_1, c_2 \ge 0$ such that

$$-c_1 \int_X \chi \, \mathrm{d}V \le \langle D_h v, \chi \rangle \le c_2 \int_X \chi \, \mathrm{d}V \qquad \text{for every smooth function } \chi \ge 0. \tag{16.7}$$

Then $D_h v$, and hence $Lv = -4\pi D_h v$, is a bounded function, with $|D_h v| \le c = \max\{c_1, c_2\}$. Indeed, let $\mathrm{d}V = \varrho \, \mathrm{d}V_{2n}$ in a chart, with ϱ smooth and positive. Pairing the real distribution $D_h v$ with a test function χ through Lebesgue measure amounts to pairing it with χ/ϱ through $\mathrm{d}V$, so $|\langle D_h v, \chi \rangle| \le c \int \chi \, \mathrm{d}V_{2n}$ for nonnegative test functions χ, and Corollary 1.7.2(1) applies.

The maximum principle arguments below are carried out at a point, in holomorphic coordinates centered there in which the coefficient matrix of a Kähler form equals the identity matrix at the point and has vanishing first derivatives there. Such coordinates exist by Lemma 14.1.5, after a linear rescaling that adapts its normalization to ours. Since our estimates must be uniform in the point, we use the following explicit choice, for which all derivatives of the coefficients at the center are bounded uniformly.

Lemma 16.2.13 *Let $U \subset \mathbb{C}^n$ be open, and let $e = (e_{i\bar{j}})$ be a smooth function on U with values in the positive definite Hermitian matrices such that the form $\frac{\mathrm{i}}{2\pi} e_{i\bar{j}} \, \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$ is closed. For every $z_0 \in U$ there is a polynomial map $\Phi_{z_0} \colon \mathbb{C}^n \to \mathbb{C}^n$ of degree at most two, with $\Phi_{z_0}(0) = z_0$, which is a biholomorphism from a neighborhood of 0 onto a neighborhood of z_0, such that the coefficient matrix $\tilde{e}$ of the form in the coordinates w given by $z = \Phi_{z_0}(w)$ satisfies $\tilde{e}(0) = I$ and $\mathrm{d}\tilde{e}(0) = 0$. For every compact set $K \subset U$ and every integer $q \ge 0$, the derivatives of order at most q of $\tilde{e}$ at 0 are bounded independently of $z_0 \in K$. These properties persist if Φ_{z_0} is composed with a unitary linear map.*

Proof The form is closed if and only if its $(2,1)$-part $\frac{\mathrm{i}}{2\pi} \partial_l e_{i\bar{j}} \, \mathrm{d}z_l \wedge \mathrm{d}z_i \wedge \mathrm{d}\bar{z}_j$ and the conjugate $(1,2)$-part vanish, that is, if and only if $\partial_l e_{i\bar{j}} = \partial_i e_{l\bar{j}}$; this holds in all holomorphic coordinates. If $z = \Phi(w)$ with Φ holomorphic and $J = D\Phi$ its complex Jacobian matrix, then $\mathrm{d}z_i = J_{ia} \, \mathrm{d}w_a$, so the new coefficients are $\tilde{e}(w) = J(w)^{\mathrm{T}} e(\Phi(w)) \overline{J(w)}$.

Let $A = \overline{e(z_0)^{-1/2}}$, where $e(z_0)^{-1/2}$ is the positive square root of the inverse matrix; its operator norm is $\lambda(z_0)^{-1/2}$, where $\lambda(z) = \min_{|\xi|=1} \sum_{i,j} e_{i\bar{j}}(z) \xi_i \bar{\xi}_j$ is the smallest eigenvalue of $e(z)$, and $A^{\mathrm{T}} e(z_0) \bar{A} = e(z_0)^{-1/2} e(z_0) e(z_0)^{-1/2} = I$. In the coordinates

v with $z = z_0 + Av$ the coefficient matrix e' satisfies $e'(0) = I$. Put $c^m_{ij} = \partial_{v_i} e'_{j\bar{m}}(0)$, which is symmetric in i and j by the closedness, and define $v = w - \frac{1}{2}c(w,w)$, that is, $v_m = w_m - \frac{1}{2}c^m_{ij} w_i w_j$. Then $J_{ia} = \partial v_i/\partial w_a = \delta_{ia} - c^i_{aj} w_j$, so $J(0) = I$ and $\partial_{w_p} J_{ia}(0) = -c^i_{ap}$. Differentiating $\tilde{e}_{a\bar{b}} = J_{ia} e'_{i\bar{j}}(v(w))\overline{J_{jb}}$ at $w = 0$, and using that $\overline{J_{jb}}$ and $\bar{v}$ are antiholomorphic in w, we find

$$\partial_{w_p} \tilde{e}_{a\bar{b}}(0) = -c^b_{ap} + \partial_{v_p} e'_{a\bar{b}}(0) = -\partial_{v_a} e'_{p\bar{b}}(0) + \partial_{v_p} e'_{a\bar{b}}(0) = 0,$$

by the closedness, and $\partial_{\bar{w}_p} \tilde{e}_{a\bar{b}}(0) = \overline{\partial_{w_p} \tilde{e}_{b\bar{a}}(0)} = 0$. Put $\Phi_{z_0}(w) = z_0 + A(w - \frac{1}{2}c(w,w))$. Its derivative at 0 is the invertible matrix A, so it is a biholomorphism near 0 by the holomorphic inverse function theorem (Proposition 7.1.2(1)).

By the chain rule, the derivatives of $\tilde{e}$ at 0 are polynomials in the derivatives of e at z_0 and in the entries of A, $\bar{A}$ and c, and the entries of c are polynomials in the first derivatives of e at z_0 and in the entries of A and $\bar{A}$. The derivatives of e are bounded on compact sets, and so are the entries of A, because the function λ is continuous and positive, as a minimum over the unit sphere of a function that is continuous in (z,ξ); hence these polynomials are bounded for $z_0 \in K$. Finally, if Q is unitary, replacing $\Phi_{z_0}(w)$ by $\Phi_{z_0}(Qw)$ replaces $\tilde{e}(w)$ by $Q^{\mathrm{T}} \tilde{e}(Qw)\bar{Q}$, which equals $Q^{\mathrm{T}}\bar{Q} = \overline{Q^* Q} = I$ at 0, still has vanishing differential at 0, and has derivatives at 0 bounded by a constant depending only on n times those of $\tilde{e}$. □

We first compute the Laplacian of the trace $H = \operatorname{tr}_h g$ with respect to g, for an arbitrary Kähler form $\kappa + \mathrm{dd}^{\mathrm{c}} a$.

Lemma 16.2.14 *There is a constant C_E, depending only on X and ω, with the following property. Let a be a smooth real function on X with $\kappa + \mathrm{dd}^{\mathrm{c}} a > 0$, let $g = (k_{i\bar{j}} + a_{i\bar{j}})$ be the coefficient matrix of $2\pi(\kappa + \mathrm{dd}^{\mathrm{c}} a)$, and put $H = \operatorname{tr}_h g$. Then $H \operatorname{tr}_g h \geq n^2$ on X. Moreover, every $x \in X$ is the center of holomorphic coordinates in which $h_{i\bar{j}} = \delta_{ij}$ and all first derivatives of h vanish at x, and g is diagonal at x, with eigenvalues $\lambda_1, \dots, \lambda_n > 0$, such that at x*

$$D_g H = D_h \log\det g + \sum_{i,p,q} \frac{|\partial_p g_{i\bar{q}}|^2}{\lambda_i \lambda_q} + E, \qquad |E| \leq C_E H \operatorname{tr}_g h, \tag{16.8}$$

and

$$D_g \log H \geq \frac{1}{H}\big(D_h \log\det g + E\big). \tag{16.9}$$

Proof At every point the eigenvalues of g relative to h are positive, their sum is H and the sum of their inverses is $\operatorname{tr}_g h$, so the Cauchy–Schwarz inequality gives $H \operatorname{tr}_g h \geq n^2$.

Fix finitely many charts and compact subsets of their domains whose union is X, and let $x \in X$. By Lemma 16.2.13, applied to h in one of these charts, followed by a unitary change of coordinates that diagonalizes g at x, there are coordinates as in the statement. In them, the derivatives of order at most two of h at x are bounded by

a constant depending only on X and ω. All the following computations are at x, in these coordinates. Locally κ and $\kappa + \mathrm{dd^c} a$ have Kähler potentials (Lemma 13.1.2): $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} \varphi$ with φ smooth. Hence the derivatives $\partial_p g_{i\bar{q}}$ are symmetric in p and i, and the second derivatives $\partial_p \partial_{\bar{l}} g_{i\bar{q}}$ are symmetric in p, i and in $\bar{l}, \bar{q}$. Differentiating $H = h^{i\bar{q}} g_{i\bar{q}}$ twice, and using that the first derivatives of h, hence of h^{-1}, vanish at x, we get

$$D_g H = \sum_p \lambda_p^{-1} \sum_i \partial_p \partial_{\bar{p}} g_{i\bar{i}} + E, \qquad E = \sum_p \lambda_p^{-1} \big(\partial_p \partial_{\bar{p}} h^{i\bar{q}}\big) g_{i\bar{q}}.$$

Since g is diagonal with positive eigenvalues, $|E| \le C_E H \operatorname{tr}_g h$, where C_E bounds the second derivatives of h^{-1} at x. As $h = I$ and $\mathrm{d}h = 0$ at x, the formula $\partial(h^{-1}) = -h^{-1}(\partial h)h^{-1}$ shows that these are the second derivatives of $-h$, up to transposition; so C_E depends only on X and ω.

On the other hand, differentiating the identity $\partial_p \log\det g = g^{i\bar{q}} \partial_p g_{i\bar{q}}$ and using $\partial(g^{-1}) = -g^{-1}(\partial g) g^{-1}$ gives

$$\partial_p \partial_{\bar{p}} \log\det g = \sum_i \lambda_i^{-1} \partial_p \partial_{\bar{p}} g_{i\bar{i}} - \sum_{i,q} \frac{|\partial_p g_{i\bar{q}}|^2}{\lambda_i \lambda_q}.$$

By the symmetry of the second derivatives, $\partial_p \partial_{\bar{p}} g_{i\bar{i}} = \partial_i \partial_{\bar{i}} g_{p\bar{p}}$. Summing over p, comparing with the formula for $D_g H$, and noting that $\sum_p \partial_p \partial_{\bar{p}} = D_h$ at x, we obtain the identity (16.8), with the bound for E just proved.

For the inequality for $\log H$, note that $\partial_p H = \sum_i \partial_p g_{i\bar{i}} = \sum_i \partial_i g_{p\bar{i}}$ by the symmetry of the first derivatives. The Cauchy–Schwarz inequality with weights λ_i therefore gives

$$|\partial H|_g^2 = \sum_p \frac{1}{\lambda_p} \Big| \sum_i \partial_i g_{p\bar{i}} \Big|^2 \le H \sum_{p,i} \frac{|\partial_i g_{p\bar{i}}|^2}{\lambda_p \lambda_i} \le H \sum_{i,p,q} \frac{|\partial_p g_{i\bar{q}}|^2}{\lambda_i \lambda_q},$$

the last sum containing the previous one (take $p = q$ in the term of index (i, p, q) and rename). Since $D_g \log H = H^{-1} D_g H - H^{-2} |\partial H|_g^2$, the quadratic term in $D_g H$ absorbs the negative term, and (16.9) follows. □

For a solution of the penalized equation, the identity $\log\det g = \log\det k + \beta(a - f)$ controls the term $D_h \log\det g$, and the maximum principle then bounds H.

Lemma 16.2.15 *Let f be a smooth real function on X and $\beta > 0$, and let a be a smooth real function satisfying*

$$(\kappa + \mathrm{dd^c} a)^n = \mathrm{e}^{\beta(a-f)} \kappa^n, \qquad \kappa + \mathrm{dd^c} a > 0. \tag{16.10}$$

Then $\min_X f \le a \le \max_X f$, and the metrics g and h are uniformly equivalent: $C^{-1} h \le g \le C h$ on X, with C depending only on X, ω, κ, β and upper bounds for $\|f\|_\infty$ and $\sup_X D_h f$. Moreover, there is $\beta_0 > 0$, depending only on X, ω and κ, such that for $\beta > \beta_0$

$$H \le 2B\, \mathrm{e}^{A \operatorname{osc}_X f} \quad \text{on } X, \qquad B = \sup_X \operatorname{tr}_h k + \sup_X D_h f,$$

where $A > 0$ depends only on X, ω and κ. In particular, for $\beta > \beta_0$ the trace H is bounded by a constant depending only on X, ω, κ and upper bounds for $\|f\|_\infty$ and $\sup_X D_h f$.

Proof *Step 1: the value bounds.* Since $(\kappa + \mathrm{dd}^{\mathrm{c}} a)^n / \kappa^n = \det g / \det k$, the equation says that $\det g = \mathrm{e}^{\beta(a-f)} \det k$. At a maximum point of a the complex Hessian $(a_{i\bar{j}})$ is negative semidefinite, so $0 < g \le k$ and therefore $\det g \le \det k$ there, the eigenvalues of $k^{-1/2} g k^{-1/2}$ lying in $(0, 1]$. Hence $a \le f \le \max_X f$ at that point, and therefore everywhere. At a minimum point of a the inequalities are reversed, and $a \ge \min_X f$.

Step 2: a differential inequality for $\log H$. Let $x \in X$, and take the coordinates of Lemma 16.2.14 at x. By that lemma, (16.9) holds at x, with $|E| \le C_E H \operatorname{tr}_g h$, and $H \operatorname{tr}_g h \ge n^2$. Now we use the equation in the form $\log \det g = \log \det k + \beta(a - f)$, valid in every chart. Since $D_h a = H - \operatorname{tr}_h k$,

$$D_h \log \det g = D_h \log \det k + \beta\big(H - \operatorname{tr}_h k - D_h f\big).$$

The function $D_h \log \det k$ is a smooth function on X, so it is bounded by a constant depending only on X, ω and κ. Dividing by H and using $H^{-1} \le \operatorname{tr}_g h / n^2$ and $|E|/H \le C_E \operatorname{tr}_g h$, we arrive at

$$D_g \log H \ge \beta \left(1 - \frac{\operatorname{tr}_h k + D_h f}{H}\right) - C_0 \operatorname{tr}_g h, \tag{16.11}$$

where C_0 depends only on X, ω and κ. Both sides are independent of the coordinates, so (16.11) holds everywhere on X.

Step 3: the maximum principle. By (16.6) we have $B > 0$. Let $B' \ge B$ be a constant, so that $\operatorname{tr}_h k + D_h f \le B'$ on X. For a constant $A > 0$, to be chosen, let x_0 be a maximum point of $Q = \log H - Aa$. Since

$$D_g a = g^{i\bar{j}} (g_{i\bar{j}} - k_{i\bar{j}}) = n - \operatorname{tr}_g k \quad \text{and} \quad \operatorname{tr}_g k \ge \gamma \operatorname{tr}_g h,$$

inequality (16.11) gives at x_0, where the complex Hessian of Q is negative semidefinite,

$$0 \ge D_g Q \ge \beta \left(1 - \frac{B'}{H}\right) + (A\gamma - C_0) \operatorname{tr}_g h - An. \tag{16.12}$$

We first prove the uniform equivalence of g and h. Let β be fixed, let $M \ge \|f\|_\infty$ and $N \ge \sup_X D_h f$ be given, take $B' = \sup_X \operatorname{tr}_h k + N$, and $A = (C_0 + \beta B'/n^2 + 1)/\gamma$. Since $1/H \le \operatorname{tr}_g h / n^2$, the right side of (16.12) is at least $\beta + \operatorname{tr}_g h - An$, so $\operatorname{tr}_g h \le An$ at x_0. Thus every eigenvalue of g relative to h is at least $(An)^{-1}$ at x_0. By Step 1, $|a - f| \le \operatorname{osc}_X f \le 2M$, and therefore, by the equation, the product of the eigenvalues satisfies

$$\left(\inf_X \frac{\det k}{\det h}\right) \mathrm{e}^{-2\beta M} \le \frac{\det g}{\det h} = \frac{\det k}{\det h}\, \mathrm{e}^{\beta(a-f)} \le \left(\sup_X \frac{\det k}{\det h}\right) \mathrm{e}^{2\beta M}$$

everywhere on X. Together with the lower bound for the eigenvalues at x_0, the upper bound for their product bounds each eigenvalue above at x_0, and hence bounds $H(x_0)$. For every $x \in X$, since $Q(x) \le Q(x_0)$ and $\min_X f \le a \le \max_X f$,

$$\log H(x) \le \log H(x_0) + A\big(a(x) - a(x_0)\big) \le \log H(x_0) + A \operatorname{osc}_X f.$$

Thus H is bounded on X, so every eigenvalue of g relative to h is bounded above on X, and the lower bound for their product then bounds every eigenvalue below. All the constants depend only on X, ω, κ, β, M and N, which proves the uniform equivalence of g and h.

Next we prove the bound for H that is uniform in β. Take $B' = B$, $A = (C_0+1)/\gamma$ and $\beta_0 = 2An$; both A and β_0 depend only on X, ω and κ. Suppose that $\beta > \beta_0$, and let x_0 be a maximum point of $Q = \log H - Aa$ for this A. If $H(x_0) > 2B$, then $1 - B/H > 1/2$ at x_0, and the right side of (16.12) is at least $\beta/2 + \operatorname{tr}_g h - An > 0$, a contradiction. Hence $H(x_0) \le 2B$, and the same argument as above gives $H \le 2B\, \mathrm{e}^{A \operatorname{osc}_X f}$ on X. Since $\operatorname{osc}_X f \le 2\|f\|_\infty$ and $B \le \sup_X \operatorname{tr}_h k + N$, this proves the last assertion. □

Along the continuity path of Theorem 16.2.21 we also need a bound for the first derivatives of g, that is, for the third derivatives of a. Its proof uses a little tensor calculus, which we now set up.

Let $z = z(w)$ be a holomorphic change of coordinates on an open subset of X, with Jacobian matrix $J = (J_{ab})$, $J_{ab} = \partial z_a/\partial w_b$. A *tensor* is given, in every holomorphic chart, by functions $T^{m_1\cdots m_r \bar n_1 \cdots \bar n_{r'}}_{i_1 \cdots i_s \bar j_1 \cdots \bar j_{s'}}$ of upper and lower, holomorphic and antiholomorphic indices in $\{1, \dots, n\}$. Its components in the coordinates w are obtained from those in z by contracting each upper holomorphic index m with $(J^{-1})_{m'm}$, each lower holomorphic index i with $J_{ii'}$, and each upper or lower antiholomorphic index with the complex conjugate of the corresponding factor; for instance $T'^{m'}_{i'\bar j'} = (J^{-1})_{m'm} J_{ii'} \overline{J_{jj'}}\, T^m_{i\bar j}$. Examples are the first derivatives $\partial_i u$ of a function, the matrices $h_{i\bar j}$, $k_{i\bar j}$, $g_{i\bar j}$, the complex Hessian $u_{i\bar j}$ of a function and $\operatorname{Ric}(k)_{i\bar j}$ (all with lower indices, since $\mathrm{d}z_i = J_{ii'}\, \mathrm{d}w_{i'}$), the inverse matrices $h^{i\bar j}$, $g^{i\bar j}$ (with upper indices), and the identity δ^i_j.

The complex conjugate of a tensor is a tensor, with the roles of holomorphic and antiholomorphic indices exchanged; products of tensors are tensors, and so are contractions of an upper with a lower index of the same kind. For a Hermitian metric g and a tensor T, we define the pointwise norm $|T|_g$ by letting $|T|^2_g$ be the complete contraction of T with $\overline{T}$ in which the lower indices are paired by $g^{\cdot\bar\cdot}$ and the upper ones by $g_{\cdot\bar\cdot}$; for instance $|T|^2_g = g^{i\bar a} g_{m\bar c} T^m_i \overline{T^c_a}$. At a point, in coordinates in which $g_{i\bar j} = \delta_{ij}$ there, $|T|^2_g$ is the sum of the squared moduli of the components. Hence the contraction of one index of T with one index of U satisfies $|T \cdot U|_g \le |T|_g |U|_g$, and contraction of two indices of T with $g^{p\bar q}$ or $g_{p\bar q}$ changes the norm at most by the factor $\sqrt{n}$. Diagonalizing g and h simultaneously, we see that $C^{-1}h \le g \le Ch$ implies $|T|_g \le C^{N/2}|T|_h$ for a tensor with N indices.

Let g be the coefficient matrix of $2\pi\eta$, where η is a Kähler form on an open subset of X. Its *Chern connection* has the coefficients

$$\Gamma^m_{ij} = g^{m\bar l}\partial_i g_{j\bar l},$$

which are symmetric in i and j because $\partial_i g_{j\bar l} = \partial_j g_{i\bar l}$. For a tensor T, the covariant derivative $\nabla_p T$ is obtained from $\partial_p T$ by adding, for each upper holomorphic index m, the term Γ^m_{pa} contracted with T in that index, and subtracting, for each lower holomorphic index i, the term Γ^a_{pi} contracted with T in that index; antiholomorphic indices receive no term. For instance $\nabla_p T^m_{i\bar j} = \partial_p T^m_{i\bar j} + \Gamma^m_{pa} T^a_{i\bar j} - \Gamma^a_{pi} T^m_{a\bar j}$. We put $\nabla_{\bar q} T = \overline{\nabla_q \overline{T}}$; thus $\nabla_{\bar q}$ acts by $\partial_{\bar q}$ and conjugate terms for the antiholomorphic indices only. We write $\widehat\Gamma$, $\widehat\nabla$ for the Chern connection of h.

Lemma 16.2.16 *Let g be as above, on an open set $O \subset X$.*

(1) *If T is a tensor, then $\nabla_p T$ and $\nabla_{\bar q} T$ are tensors, with one more lower index p, respectively $\bar q$. The covariant derivatives satisfy the Leibniz rule for products, they commute with contractions, and $\nabla g = 0$, $\nabla g^{-1} = 0$. Consequently, if S is a complete contraction of a product of tensors, then $\partial_p S$ is the sum of the contractions in which one factor is replaced by its covariant derivative ∇_p, and similarly for $\partial_{\bar q}$.*

(2) *The functions $\Psi^m_{ij} = \Gamma^m_{ij} - \widehat\Gamma^m_{ij}$ form a tensor, and $\Psi^m_{ij} = g^{m\bar l}\widehat\nabla_i g_{j\bar l}$.*

(3) *The functions $R^m{}_{i\bar q j} = -\partial_{\bar q}\Gamma^m_{ij}$ form a tensor, symmetric in i and j. The* Ricci tensor *$R_{i\bar j} = R^m{}_{i\bar j m}$ equals $-\partial_i\partial_{\bar j}\log\det g$, and with $R_i{}^m = g^{m\bar l}R_{i\bar l}$ we have*

$$g^{p\bar q}R^m{}_{i\bar q p} = R_i{}^m, \qquad g^{p\bar q}\nabla_p R^m{}_{i\bar q j} = \nabla_i R_j{}^m.$$

(4) *If T has only holomorphic indices, then $\nabla_{\bar q}T = \partial_{\bar q}T$, and, for instance for $T = (T^m_{ij})$,*

$$(\nabla_{\bar q}\nabla_p - \nabla_p\nabla_{\bar q})T^m_{ij} = -R^m{}_{a\bar q p}T^a_{ij} + R^a{}_{i\bar q p}T^m_{aj} + R^a{}_{j\bar q p}T^m_{ia}.$$

Proof (1) In the coordinates w the metric and its inverse have the components

$$g'_{a\bar b} = J_{ia}g_{i\bar j}\overline{J_{jb}}, \qquad g'^{d\bar b} = (J^{-1})_{dm}\overline{(J^{-1})_{bl}}g^{m\bar l}.$$

By these rules and the chain rule $\partial_{w_c}(F\circ z) = J_{kc}\,(\partial_{z_k}F)\circ z$, which holds because $\partial_{w_c}\bar z_k = 0$, we have $\partial_{w_c}g'_{a\bar b} = (\partial_{w_c}J_{ia})g_{i\bar j}\overline{J_{jb}} + J_{ia}J_{kc}\,\partial_k g_{i\bar j}\,\overline{J_{jb}}$, and hence

$$\Gamma'^d_{ca} = (J^{-1})_{dm}J_{kc}J_{ia}\Gamma^m_{ki} + (J^{-1})_{dm}\partial_{w_c}J_{ma}. \tag{16.13}$$

Let T be a tensor, and differentiate its transformation rule by ∂_{w_c}. By the product rule, this gives the transformed array $J_{kc}\partial_k T$, plus one term for each holomorphic index, in which ∂_{w_c} falls on the Jacobian factor of that index; the factors of the antiholomorphic indices are antiholomorphic and are not differentiated. For an upper holomorphic index, the term contains $\partial_{w_c}(J^{-1})_{dm} = -(J^{-1})_{dm'}(\partial_{w_c}J_{m'b})(J^{-1})_{bm}$, and it cancels against the contribution of the second term of (16.13) to $\Gamma'^d_{cb}T'^{\cdots b\cdots}$; for a lower holomorphic index a, the term contains $\partial_{w_c}J_{ia}$, and it cancels against the contribution of the second term of (16.13) to $-\Gamma'^b_{ca}T'_{\cdots b\cdots}$. The contributions of the

first term of (16.13) are the transforms of the connection terms of T. Hence $\nabla_c T'$ is the transform of $\nabla_k T$, contracted with J_{kc}, so $\nabla_p T$ is a tensor; by conjugation, so is $\nabla_{\bar q} T$.

The Leibniz rule follows from the product rule for ∂_p and the additivity of the connection terms over the indices. If an upper and a lower holomorphic index of T are contracted, their connection terms $\Gamma^m_{pa} T^{a\cdots}{}_{\cdots m}$ and $-\Gamma^a_{pm} T^{m\cdots}{}_{\cdots a}$ cancel, so ∇ commutes with contractions. Next,

$$\nabla_p g_{i\bar j} = \partial_p g_{i\bar j} - \Gamma^a_{pi} g_{a\bar j} = \partial_p g_{i\bar j} - g^{a\bar l}\partial_p g_{i\bar l}\, g_{a\bar j} = 0,$$

because $g^{a\bar l} g_{a\bar j} = \delta_{lj}$, and $\nabla_{\bar q} g_{i\bar j} = \overline{\nabla_q g_{j\bar i}} = 0$, since $\overline{g_{i\bar j}} = g_{j\bar i}$. The connection terms of the identity tensor cancel, so $\nabla\delta = 0$, and the Leibniz rule applied to $g^{i\bar j} g_{l\bar j} = \delta^i_l$ gives $(\nabla g^{i\bar j}) g_{l\bar j} = 0$, hence $\nabla g^{-1} = 0$. For a complete contraction S, a function, $\nabla_p S = \partial_p S$, and the last assertion follows from the Leibniz rule and the commutation with contractions.

(2) By (16.13) for g and for h, the inhomogeneous terms of Γ' and $\widehat\Gamma'$ coincide, so the difference transforms as a tensor. Moreover $\widehat\nabla_i g_{j\bar l} = \partial_i g_{j\bar l} - \widehat\Gamma^p_{ij} g_{p\bar l}$, and contracting with $g^{m\bar l}$ gives $\Gamma^m_{ij} - \widehat\Gamma^m_{ij}$.

(3) The second term of (16.13) is holomorphic in w, so it is annihilated by $\partial_{\bar w_e} = \overline{J_{le}}\,\partial_{\bar z_l}$, and R transforms as a tensor; it is symmetric in i and j because Γ is. Since $\Gamma^m_{im} = g^{m\bar l}\partial_i g_{m\bar l} = \partial_i \log\det g$, we get $R_{i\bar j} = -\partial_{\bar j}\partial_i \log\det g$. Both sides of each of the two identities are tensors, so it suffices to prove them at a point x in coordinates in which $g_{i\bar j}(x) = \delta_{ij}$ and $\mathrm{d}g(x) = 0$, which exist by Lemma 16.2.13. There $\Gamma = 0$, so covariant derivatives at x are ordinary derivatives, and $\partial g^{-1} = -g^{-1}(\partial g)g^{-1}$ and $\bar\partial g^{-1}$ vanish at x. Using a local Kähler potential φ, $g_{i\bar j} = \partial_i\partial_{\bar j}\varphi$, whose derivatives commute, we find at x

$$g^{p\bar q} R^m{}_{i\bar q p} = -\sum_p \partial_i\partial_{\bar p} g_{p\bar m} = -\sum_p \partial_i\partial_{\bar m} g_{p\bar p} = R_i{}^m,$$

and

$$g^{p\bar q}\nabla_p R^m{}_{i\bar q j} = -\sum_p \partial_p\partial_{\bar p}\partial_i g_{j\bar m} = -\sum_p \partial_i\partial_j\partial_{\bar m} g_{p\bar p} = \nabla_i R_j{}^m;$$

in each case all terms containing a first derivative of g or of g^{-1} vanish at x.

(4) A tensor with only holomorphic indices receives no connection term from $\nabla_{\bar q}$, so $\nabla_{\bar q} T = \partial_{\bar q} T$. The tensor $\nabla_p T$ again has only holomorphic indices, so $\nabla_{\bar q}\nabla_p T^m_{ij}$ is the $\partial_{\bar q}$-derivative of $\partial_p T^m_{ij} + \Gamma^m_{pa} T^a_{ij} - \Gamma^a_{pi} T^m_{aj} - \Gamma^a_{pj} T^m_{ia}$. On the other hand, $\nabla_{\bar q} T = \partial_{\bar q} T$ has the additional antiholomorphic index $\bar q$, which receives no term from ∇_p, so $\nabla_p\nabla_{\bar q} T^m_{ij} = \partial_p\partial_{\bar q} T^m_{ij} + \Gamma^m_{pa}\partial_{\bar q} T^a_{ij} - \Gamma^a_{pi}\partial_{\bar q} T^m_{aj} - \Gamma^a_{pj}\partial_{\bar q} T^m_{ia}$. The difference consists of the terms in which $\partial_{\bar q}$ falls on Γ; since $\partial_{\bar q}\Gamma^m_{pa} = -R^m{}_{a\bar q p}$ by the symmetry of R, this is the asserted formula. □

Lemma 16.2.17 *Let f be a smooth real function on X, let $\beta \in \mathbb{R}$ and $0 \le t \le 1$, and let a be a smooth real function with $\kappa + \mathrm{dd}^c a > 0$ and*

$$(\kappa + \mathrm{dd^c} a)^n = \mathrm{e}^{\beta(a - tf)} \kappa^n.$$

Suppose that $C_0^{-1} h \le g \le C_0 h$ on X for a constant $C_0 \ge 1$, and let Ψ be the tensor of Lemma 16.2.16(2) for g. Then

$$S = |\Psi|_g^2 = g^{i\bar{p}} g^{j\bar{q}} g_{m\bar{l}} \Psi^m_{ij} \overline{\Psi^l_{pq}} \le C,$$

where C depends only on X, ω, κ, β, C_0 and f, but not on t or a. Consequently, in each chart of a fixed finite atlas, the first derivatives of the functions $g_{i\bar{j}}$ are bounded on every compact subset by a constant of the same kind.

Proof In this proof $C, C_1, C_2, \ldots$ denote constants depending only on X, ω, κ, β, C_0 and f. We show that $D_g S \ge -C_1(S+1)$ and $D_g H \ge C_0^{-1} S - C_2$, where $H = \mathrm{tr}_h\, g$, and apply the maximum principle to $S + KH$ for a large constant K.

By Lemma 16.2.16(1),(2), S is a smooth function on X. The fixed smooth tensors $\widehat{R}$ (the tensor of Lemma 16.2.16(3) for h), $\widehat{\nabla}\widehat{R}$, $\tau = \mathrm{Ric}(k) + \beta k + \beta t(f_{i\bar{j}})$ and $\widehat{\nabla}\tau$ have h-norms bounded on X independently of $t \in [0,1]$, because they are smooth on the compact manifold X and depend affinely on t; by the assumption on g, their g-norms are bounded as well.

The Laplacian of Ψ. Let R be the tensor of Lemma 16.2.16(3) for g. Since Ψ has only holomorphic indices,

$$\nabla_{\bar{q}} \Psi^m_{ij} = \partial_{\bar{q}} \Psi^m_{ij} = \widehat{R}^m{}_{i\bar{q}j} - R^m{}_{i\bar{q}j}.$$

Put $D_g T = g^{p\bar{q}} \nabla_p \nabla_{\bar{q}} T$ and $\overline{D}_g T = g^{p\bar{q}} \nabla_{\bar{q}} \nabla_p T$ for tensors T. On a function u both operators reduce to $D_g u = g^{p\bar{q}} \partial_p \partial_{\bar{q}} u$, since ∇_p adds no term to the antiholomorphic index of $\partial_{\bar{q}} u$, and $\nabla_{\bar{q}}$ none to the holomorphic index of $\partial_p u$. Applying $g^{p\bar{q}} \nabla_p$ to the last identity and using $\nabla g^{-1} = 0$ and Lemma 16.2.16(3), we obtain

$$D_g \Psi^m_{ij} = g^{p\bar{q}} \nabla_p \widehat{R}^m{}_{i\bar{q}j} - \nabla_i R_j{}^m. \tag{16.14}$$

By Lemma 16.2.16(4) and the first identity of (3), $\overline{D}_g \Psi = D_g \Psi + \mathcal{R}\Psi$, where

$$(\mathcal{R}\Psi)^m_{ij} = -R_a{}^m \Psi^a_{ij} + R_i{}^a \Psi^m_{aj} + R_j{}^a \Psi^m_{ia}.$$

The Laplacian of S. By Lemma 16.2.16(1) and $\nabla_r \overline{\Psi^l_{pq}} = \overline{\nabla_{\bar{r}} \Psi^l_{pq}}$, which holds because ∇_r adds no term to antiholomorphic indices,

$$\partial_r S = g^{i\bar{p}} g^{j\bar{q}} g_{m\bar{l}} \big(\nabla_r \Psi^m_{ij} \, \overline{\Psi^l_{pq}} + \Psi^m_{ij} \, \overline{\nabla_{\bar{r}} \Psi^l_{pq}} \big).$$

The right side is a tensor with one lower holomorphic index r, on which $\nabla_{\bar{s}}$ acts as $\partial_{\bar{s}}$. Applying Lemma 16.2.16(1) once more and contracting with $g^{r\bar{s}}$, we obtain, with the inner products and norms defined by g,

$$D_g S = |\nabla \Psi|_g^2 + |\overline{\nabla} \Psi|_g^2 + \langle \overline{D}_g \Psi, \Psi \rangle_g + \langle \Psi, D_g \Psi \rangle_g,$$

where $\nabla\Psi = (\nabla_r \Psi^m_{ij})$ and $\overline{\nabla}\Psi = (\nabla_{\bar s}\Psi^m_{ij})$; for the last term we used $\overline{g^{r\bar s}} = g^{s\bar r}$. Inserting $\overline{D}_g\Psi = D_g\Psi + \mathcal{R}\Psi$, we get

$$D_g S = |\nabla\Psi|^2_g + |\overline{\nabla}\Psi|^2_g + 2\,\mathrm{Re}\langle D_g\Psi, \Psi\rangle_g + \langle \mathcal{R}\Psi, \Psi\rangle_g; \tag{16.15}$$

in particular $\langle \mathcal{R}\Psi, \Psi\rangle_g$ is real.

Bounds from the equation. In every chart, $\log\det g = \log\det k + \beta(a - tf)$, so

$$R_{i\bar j} = -\partial_i\partial_{\bar j}\log\det g = \mathrm{Ric}(k)_{i\bar j} - \beta a_{i\bar j} + \beta t f_{i\bar j} = \tau_{i\bar j} - \beta g_{i\bar j}.$$

Hence $R_i{}^m = g^{m\bar l}\tau_{i\bar l} - \beta\delta^m_i$ is bounded in g-norm, and so $|\langle \mathcal{R}\Psi, \Psi\rangle_g| \le CS$. Moreover $\nabla_i R_j{}^m = g^{m\bar l}\nabla_i \tau_{j\bar l}$, because $\nabla g^{-1} = 0$ and $\nabla\delta = 0$. For a tensor the difference $\nabla_p - \widehat{\nabla}_p$ consists of the connection terms with $\Gamma - \widehat{\Gamma} = \Psi$ in place of Γ; thus $\nabla_i\tau_{j\bar l} = \widehat{\nabla}_i\tau_{j\bar l} - \Psi^a_{ij}\tau_{a\bar l}$ and

$$\nabla_p \widehat{R}^m{}_{i\bar q j} = \widehat{\nabla}_p \widehat{R}^m{}_{i\bar q j} + \Psi^m_{pa}\widehat{R}^a{}_{i\bar q j} - \Psi^a_{pi}\widehat{R}^m{}_{a\bar q j} - \Psi^a_{pj}\widehat{R}^m{}_{i\bar q a}.$$

By (16.14) it follows that $|D_g\Psi|_g \le C(1 + S^{1/2})$. Discarding the two nonnegative gradient terms in (16.15), we conclude that

$$D_g S \ge -2C(1 + S^{1/2})S^{1/2} - CS \ge -C_1(S + 1).$$

A compensating term. We use the trace $H = \mathrm{tr}_h\, g$. Let $x \in X$, and take the coordinates of Lemma 16.2.14 at x. By that lemma, identity (16.8) holds at x, with $|E| \le C_E H\,\mathrm{tr}_g\, h \le C$ by the assumption on g, and $D_h \log\det g = D_h \log\det k + \beta(H - \mathrm{tr}_h\, k - tD_h f)$ is bounded by C. At x the coefficients $\widehat{\Gamma}$ vanish, so $\Psi^m_{ij} = \lambda_m^{-1}\partial_i g_{j\bar m}$ and

$$S = \sum_{i,p,q} \frac{|\partial_p g_{i\bar q}|^2}{\lambda_p\lambda_i\lambda_q}.$$

Hence the quadratic sum in (16.8) is at least $(\min_p \lambda_p)S \ge C_0^{-1}S$, and $D_g H \ge C_0^{-1}S - C_2$ at x. Both sides are functions on X, so this inequality holds everywhere.

The maximum principle. Choose K with $KC_0^{-1} \ge C_1 + 1$, and let x_0 be a maximum point of $S + KH$. There the complex Hessian of $S + KH$ is negative semidefinite, so

$$0 \ge D_g(S + KH) \ge (KC_0^{-1} - C_1)S - C_1 - KC_2 \ge S - C_1 - KC_2.$$

Thus $S \le C_1 + KC_2$ at x_0. Since $H \le nC_0$ on X, the function $S + KH$, and hence S, is bounded on X.

First derivatives of g. In a chart of the fixed atlas, $\partial_p g_{i\bar q} = \widehat{\nabla}_p g_{i\bar q} + \widehat{\Gamma}^l_{pi} g_{l\bar q} = g_{m\bar q}\Psi^m_{pi} + \widehat{\Gamma}^l_{pi} g_{l\bar q}$ by Lemma 16.2.16(2). On a compact subset of the chart, h lies between two positive multiples of the identity matrix and $\widehat{\Gamma}$ is bounded, so both terms are bounded, by the bounds for S and g. Finally $\partial_{\bar p} g_{i\bar q} = \overline{\partial_p g_{q\bar i}}$, so all first derivatives of the $g_{i\bar j}$ in the underlying real coordinates are bounded. □

The interior regularity of the solutions of the penalized equation is proved with difference quotients. For a coordinate unit vector e of $\mathbb{R}^d$ and a real number $s \neq 0$ we write $\tau_s Z(x) = Z(x + se)$ and $\delta_s Z = (\tau_s Z - Z)/s$.

Lemma 16.2.18 *Let $D' \Subset D$ be open subsets of $\mathbb{R}^d$, let e be a coordinate unit vector, and put $s_0 = \operatorname{dist}(D', \mathbb{R}^d \setminus D)$, with $s_0 = \infty$ if $D = \mathbb{R}^d$.*

(1) *Let $1 \le p < \infty$, and let $Z \in L^1_{\mathrm{loc}}(D)$ be such that the distribution $\partial_e Z$ belongs to $L^p(D)$. Then, for $0 < |s| < s_0$, $\|\delta_s Z\|_{L^p(D')} \le \|\partial_e Z\|_{L^p(D)}$, and $\delta_s Z \to \partial_e Z$ in $L^p(D')$ as $s \to 0$.*

(2) *Let $2 \le p < \infty$, let $Z \in L^p(D)$, and suppose that $\|\delta_s Z\|_{L^p(D')} \le M$ for $0 < |s| < s_1$, where $0 < s_1 \le s_0$. Then $\partial_e Z \in L^p(D')$ and $\|\partial_e Z\|_{L^p(D')} \le M$.*

Proof (1) Fix s with $0 < |s| < s_0$. We claim that $\delta_s Z(x) = \int_0^1 \partial_e Z(x + tse)\,\mathrm{d}t$ for almost every $x \in D'$. For smooth Z this is the fundamental theorem of calculus. In general, let $K \subset D'$ be compact; the set $K' = \{x + tse : x \in K,\ 0 \le t \le 1\}$ is a compact subset of D. The mollifications $Z_\varepsilon = Z * \rho_\varepsilon$ are smooth near K' for small ε, they converge to Z in $L^1(K')$, and $\partial_e Z_\varepsilon = (\partial_e Z) * \rho_\varepsilon \to \partial_e Z$ in $L^1(K')$, by Proposition 1.4.3(1) and Lemma 1.4.1(2). Hence the identity for Z_ε passes to the limit in $L^1(K)$; for the right side we use $\int_K \int_0^1 |F(x + tse)|\,\mathrm{d}t\,\mathrm{d}x \le \int_{K'} |F|$, by Tonelli's theorem. This proves the claim. By Hölder's inequality on $[0, 1]$ and Tonelli's theorem,

$$\int_{D'} |\delta_s Z|^p \le \int_0^1 \int_{D'} |\partial_e Z(x + tse)|^p \,\mathrm{d}x\,\mathrm{d}t \le \|\partial_e Z\|^p_{L^p(D)}.$$

Similarly, since $\delta_s Z(x) - \partial_e Z(x) = \int_0^1 \big(\partial_e Z(x + tse) - \partial_e Z(x)\big)\,\mathrm{d}t$ for almost every $x \in D'$, Hölder's inequality and Tonelli's theorem give, with F the extension of $\partial_e Z$ by zero to $\mathbb{R}^d$,

$$\|\delta_s Z - \partial_e Z\|_{L^p(D')} \le \sup_{|r| \le |s|} \|F(\cdot + re) - F\|_{L^p(\mathbb{R}^d)},$$

which tends to zero as $s \to 0$ by the continuity of translation in $L^p(\mathbb{R}^d)$ (Lemma 1.4.1(2)).

(2) Let $\chi \in C^\infty_c(D')$ with $0 \le \chi \le 1$, extend χZ by zero, and let $0 < |s| < \min\{s_1, \operatorname{dist}(\operatorname{Supp}\chi, \mathbb{R}^d \setminus D')\}$. Then $\delta_s(\chi Z)$ vanishes outside D', and on D' it equals $(\tau_s\chi)\delta_s Z + (\delta_s\chi)Z$. Since D' is bounded and $p \ge 2$, Hölder's inequality gives

$$\|\delta_s(\chi Z)\|_{L^2(\mathbb{R}^d)} \le |D'|^{1/2 - 1/p}\big(M + \|\nabla\chi\|_\infty \|Z\|_{L^p(D)}\big) =: M'.$$

The function χZ lies in $L^1 \cap L^2$, so its L^2 Fourier transform is given by the integral (Theorem 6.1.2), and $\widehat{\delta_s(\chi Z)} = m_s \widehat{\chi Z}$ with $m_s(\xi) = (\mathrm{e}^{2\pi \mathrm{i} s\xi_e} - 1)/s$, where ξ_e is the component of ξ in the direction e. By Plancherel's theorem, $\int |m_s|^2 |\widehat{\chi Z}|^2 \le M'^2$, and as $m_s(\xi) \to 2\pi\mathrm{i}\xi_e$, Fatou's lemma gives $\int 4\pi^2 \xi_e^2 |\widehat{\chi Z}|^2 \le M'^2$. Let $F \in L^2$ be the inverse L^2 transform of $2\pi\mathrm{i}\xi_e \widehat{\chi Z}$. For a test function ϕ we have $\widehat{\partial_e \phi} = 2\pi\mathrm{i}\xi_e \widehat{\phi}$, by

integration by parts, so Parseval's identity, which extends to L^2 because the transform is unitary, gives

$$\int F\,\overline{\phi} = \int 2\pi i\xi_e \widehat{\chi Z}\,\overline{\widehat{\phi}} = -\int \widehat{\chi Z}\,\overline{\widehat{\partial_e \phi}} = -\int \chi Z\,\overline{\partial_e \phi}.$$

Thus $\partial_e(\chi Z) = F \in L^2$. Since χ may equal 1 near any given compact subset of D', it follows that $\partial_e Z \in L^2_{\mathrm{loc}}(D')$.

Now let $D'' \Subset D''' \Subset D'$ be open. By (1), applied with D''' in place of D and $p = 2$, we have $\delta_s Z \to \partial_e Z$ in $L^2(D'')$, so along a sequence $s_j \to 0$ the functions $\delta_{s_j} Z$ converge to $\partial_e Z$ almost everywhere on D''. Fatou's lemma gives $\|\partial_e Z\|_{L^p(D'')} \le \varliminf_j \|\delta_{s_j} Z\|_{L^p(D'')} \le M$. Exhausting D' by such sets D'' and using monotone convergence, we obtain $\|\partial_e Z\|_{L^p(D')} \le M$. □

We now prove this regularity in a chart, with estimates for the higher derivatives. For a function $G = (G_{i\bar{j}})$ with values in the positive definite Hermitian matrices we write $(g^{i\bar{j}})$ for its inverse in the normalization above and $D_G w = g^{i\bar{j}}\partial_i \partial_{\bar{j}} w$; by Jacobi's formula, $\partial \log\det G = g^{i\bar{j}}\partial G_{i\bar{j}} = \mathrm{tr}(G^{-1}\partial G)$ for every first-order derivative ∂.

Lemma 16.2.19 *Let $U \subset \mathbb{C}^n$ be open, let $k = (k_{i\bar{j}})$ be a smooth function on U with values in the Hermitian matrices, let ϕ_0 be a smooth real function on U, and let $\beta \in \mathbb{R}$. Let $u \in C^2(U)$ be real, suppose that $G = (k_{i\bar{j}} + u_{i\bar{j}})$ is positive definite on U, and that*

$$\log\det G = \phi_0 + \beta u \qquad \text{on } U. \tag{16.16}$$

Then u is smooth. Moreover, let $V \Subset U$ be open, let $\ell \ge 3$ be an integer and $2 \le p < \infty$, and suppose that $\lambda I \le G \le \Lambda I$ on U for constants $0 < \lambda \le \Lambda$, that $|G(x) - G(y)| \le \varpi(|x-y|)$ for $x, y \in U$, where $\varpi(r) \to 0$ as $r \to 0$, and that $\|u\|_{C^1(U)} + \|D^2 u\|_{L^q(U)} \le M$, where $q = 2^\ell(p+2n)$. Then $\|u\|_{W^{\ell,p}(V)} \le C$, where C depends only on n, ℓ, p, V, U, λ, Λ, ϖ, M, $|\beta|$ and bounds on U for the derivatives of order at most ℓ of k and ϕ_0.

Proof For the quantitative assertion, C denotes constants depending only on the quantities listed there; $O' \Subset O \Subset O_1 \Subset U$ denote open sets, and the constants may also depend on them. The qualitative assertion follows from the quantitative one. Indeed, if $V \Subset U' \Subset U$ with U' open, then on U' the matrix G is bounded above and below and uniformly continuous, and u has finite C^1 norm and bounded second derivatives. The quantitative assertion, applied on U', therefore gives $u \in W^{\ell,p}(V)$ for all ℓ and p, and Morrey's inequality (Lemma 6.2.6), applied to the derivatives of u, shows that all derivatives of u are continuous on V.

We prove the estimate by bootstrapping: an interior L^p estimate (Step 1), applied to difference quotients, raises the order of differentiability by one at a time (Steps 2 and 3), on smaller domains and with smaller exponents, and Step 4 chains these steps.

Step 1: an interior estimate. Let $(a^{i\bar{j}})$ be a function on O with $\overline{a^{i\bar{j}}} = a^{j\bar{i}}$, $\Lambda^{-1}|\zeta|^2 \le a^{i\bar{j}}\zeta_i\bar{\zeta}_j \le \lambda^{-1}|\zeta|^2$ and $|a(x) - a(y)| \le \varpi'(|x-y|)$, where $\varpi'(r) \to 0$ as $r \to 0$. We claim that every real $w \in W^{2,p}_{\mathrm{loc}}(O)$ with $w \in W^{1,p}(O)$ satisfies

$$\|D^2 w\|_{L^p(O')} \le C\big(\|a^{i\bar j}\partial_i\partial_{\bar j} w\|_{L^p(O)} + \|w\|_{W^{1,p}(O)}\big),$$

with C depending also on ϖ'. Indeed, by (16.4), $a^{i\bar j}\partial_i\partial_{\bar j} w = A : D^2 w$ with a real symmetric $A \ge (4\Lambda)^{-1} I$, $|A| \le C\lambda^{-1}$ and $|A(x) - A(y)| \le \varpi'(|x-y|)$. Let δ_0 and C_1 be the constants of Lemma 16.2.10 for $d = 2n$, p and $(4\Lambda)^{-1}$, and choose $r > 0$ with $\varpi'(\varrho) \le \delta_0$ for $0 < \varrho < r$ and $r < \operatorname{dist}(O', \mathbb{R}^{2n} \setminus O)$. Cover $\overline{O'}$ by finitely many balls $B(x_\nu, r/2)$ with $x_\nu \in \overline{O'}$, and let $\chi_\nu = \chi(\cdot - x_\nu)$, where $\chi \in C_c^\infty(B(0,r))$ equals 1 on $B(0, r/2)$. By Lemma 6.2.4(1) and Lemma 16.2.10,

$$\|D^2(\chi_\nu w)\|_p \le C_1 \|A : D^2(\chi_\nu w)\|_{L^p(B(x_\nu, r))},$$

and $A : D^2(\chi_\nu w) = \chi_\nu\, A : D^2 w + 2\sum_{\mu,\sigma} A^{\mu\sigma}\partial_\mu\chi_\nu\,\partial_\sigma w + w\, A : D^2\chi_\nu$. Summing over ν proves the claim.

Step 2: $u \in W^{3,p}_{\mathrm{loc}}$. Let e be a real coordinate unit vector, and let $0 < |s| < \operatorname{dist}(O, \mathbb{R}^{2n} \setminus O_1)$, with the notation τ_s, δ_s of Lemma 16.2.18. For $x \in O$ and $0 \le t \le 1$ the matrix $G_t(x) = (1-t)G(x) + tG(x+se)$ satisfies $\lambda I \le G_t \le \Lambda I$; let $(g_t^{i\bar j})$ be its inverse and $a_s^{i\bar j} = \int_0^1 g_t^{i\bar j}\,\mathrm{d}t$. By Jacobi's formula, $\frac{\mathrm{d}}{\mathrm{d}t}\log\det G_t = g_t^{i\bar j}(G_{i\bar j}(x+se) - G_{i\bar j}(x))$, so integration over $0 \le t \le 1$ gives $\delta_s \log\det G = a_s^{i\bar j}\delta_s G_{i\bar j}$. Since $\delta_s G_{i\bar j} = \delta_s k_{i\bar j} + \partial_i\partial_{\bar j}\delta_s u$, equation (16.16) gives

$$a_s^{i\bar j}\partial_i\partial_{\bar j}(\delta_s u) = \delta_s\phi_0 + \beta\,\delta_s u - a_s^{i\bar j}\delta_s k_{i\bar j} \qquad \text{on } O.$$

The right side is bounded by $\sup_U|\nabla\phi_0| + |\beta|\sup_U|\nabla u| + C\lambda^{-1}\sup_U|\nabla k|$. The matrices a_s satisfy the hypotheses of Step 1 with $\varpi' = 2\lambda^{-2}\varpi$, because they are averages of inverses of matrices between λI and ΛI, and $|M^{-1} - N^{-1}| = |M^{-1}(N - M)N^{-1}| \le \lambda^{-2}|M - N|$ for such matrices. By Lemma 16.2.18(1), $\|\delta_s u\|_{W^{1,p}(O)} \le \|u\|_{W^{2,p}(O_1)}$. Step 1 therefore gives $\|D^2\delta_s u\|_{L^p(O')} \le C$, uniformly in s. Since $\delta_s\partial_\mu\partial_\sigma u = \partial_\mu\partial_\sigma\delta_s u$, Lemma 16.2.18(2), applied to $Z = \partial_\mu\partial_\sigma u$, shows that $\partial_e\partial_\mu\partial_\sigma u \in L^p(O')$, with norm at most C. Thus $\|u\|_{W^{3,p}(O')} \le C$, where C depends on M through $\|u\|_{C^1(U)}$ and $\|D^2 u\|_{L^p(O_1)}$.

Step 3: from W^ℓ to $W^{\ell+1}$. Let $\ell \ge 3$ and suppose that $u \in W^{\ell,q'}(O_1)$ for some $q' \ge 2p + 2n$. By Morrey's inequality (Lemma 6.2.6), applied to the derivatives of u of order at most $\ell - 2$, which lie in $W^{2,q'}(O_1)$, the function u is of class $C^{\ell-1}$ on O_1, and on a slightly smaller set, which we again call O_1, its derivatives of order at most $\ell - 1$ are bounded by $C\|u\|_{W^{\ell,q'}}$. Thus $G \in C^{\ell-3}(O_1)$, its derivatives of order at most $\ell - 3$ lie in $W^{1,2p}(O_1)$, and those of order $\ell - 2$ lie in $L^{2p}(O_1)$.

We first record a formula for the derivatives of $\log\det G$. For a smooth function G with values in the positive definite Hermitian matrices and a multi-index γ with $|\gamma| \ge 1$,

$$\partial^\gamma \log\det G = \operatorname{tr}(G^{-1}\partial^\gamma G) + P_\gamma(G),$$

where $P_\gamma(G)$ is a linear combination, with coefficients depending only on γ, of functions $\operatorname{tr}(G^{-1}\partial^{\beta_1}G\,G^{-1}\partial^{\beta_2}G\cdots G^{-1}\partial^{\beta_r}G)$ with $r \ge 2$, $|\beta_i| \ge 1$ and $\beta_1 + \cdots + \beta_r = \gamma$; in particular $|\beta_i| \le |\gamma| - 1$. For $|\gamma| = 1$ this is Jacobi's formula, with $P_\gamma = 0$. If it holds for γ, differentiate once more: by the Leibniz rule and $\partial_\mu(G^{-1}) =$

$-G^{-1}(\partial_\mu G)G^{-1}$, the derivative of each term $\operatorname{tr}(G^{-1}\partial^{\beta_1}G\cdots)$ is a sum of terms of the same form, with r or $r+1$ factors and total multi-index $\gamma+e_\mu$, and the derivative of $\operatorname{tr}(G^{-1}\partial^\gamma G)$ is $\operatorname{tr}(G^{-1}\partial^{\gamma+e_\mu}G) - \operatorname{tr}(G^{-1}\partial_\mu G\, G^{-1}\partial^\gamma G)$. Note that $\operatorname{tr}(G^{-1}\partial^\gamma G) = g^{i\bar j}\partial^\gamma G_{i\bar j}$.

Now fix γ with $|\gamma| = \ell - 2 \ge 1$. We claim that, almost everywhere on O_1,

$$\operatorname{tr}(G^{-1}\partial^\gamma G) + P_\gamma(G) = \partial^\gamma\phi_0 + \beta\,\partial^\gamma u,$$

where $\partial^\gamma G$ is formed with weak derivatives. Let $K \subset O_1$ be compact, and let $u_\varepsilon = u * \rho_\varepsilon$ and $G_\varepsilon = k + (\partial_i\partial_{\bar j}u_\varepsilon)$. By Proposition 1.4.3(1) and Lemma 1.4.1(1),(2), as $\varepsilon \to 0$ the derivatives of u_ε of order at most $\ell-1$ converge uniformly on K to those of u, and those of order ℓ converge in $L^{2p}(K)$. Hence G_ε is positive definite on K for small ε, its derivatives of order at most $\ell-3$ converge uniformly on K to those of G, and $\partial^\gamma G_\varepsilon \to \partial^\gamma G$ in $L^{2p}(K)$. The formula above holds for the smooth functions G_ε on the interior of K. Its left side converges to $\partial^\gamma \log\det G = \partial^\gamma\phi_0 + \beta\partial^\gamma u$ in the sense of distributions, since $\log\det G_\varepsilon \to \log\det G$ uniformly on K and $\log\det G = \phi_0 + \beta u$, and its right side converges in $L^1(K)$, since every factor of $P_\gamma(G_\varepsilon)$ involves derivatives of G_ε of order at most $\ell-3$. Hence the identity holds in the sense of distributions on the interior of K, and therefore almost everywhere (Proposition 1.4.3(4)). This proves the claim.

Put $v = \partial^\gamma u \in W^{2,2p}(O_1)$. Since $\operatorname{tr}(G^{-1}\partial^\gamma G) = D_G v + g^{i\bar j}\partial^\gamma k_{i\bar j}$, the claim says that

$$D_G v = R := \partial^\gamma\phi_0 + \beta v - g^{i\bar j}\partial^\gamma k_{i\bar j} - P_\gamma(G) \qquad \text{almost everywhere on } O_1.$$

For $x \in O$ and small s, as in Step 2, the discrete product rule $\delta_s(FF') = (\tau_s F)\delta_s F' + (\delta_s F)F'$ gives

$$(\tau_s g^{i\bar j})\,\partial_i\partial_{\bar j}(\delta_s v) = \delta_s R - (\delta_s g^{i\bar j})\,\partial_i\partial_{\bar j}v \qquad \text{almost everywhere on } O.$$

We bound the right side in $L^p(O)$ uniformly in s, using Lemma 16.2.18(1) on $O \Subset O_1$. First, $\delta_s(G^{-1}) = -(\tau_s G)^{-1}(\delta_s G)G^{-1}$, so $\|\delta_s G^{-1}\|_{L^{2p}(O)} \le \lambda^{-2}\|\partial_e G\|_{L^{2p}(O_1)} \le C$, as $\partial_e G$ involves derivatives of u of order three. Hence the last term is bounded by $\|\delta_s G^{-1}\|_{L^{2p}}\|\partial\bar\partial v\|_{L^{2p}} \le C$. Next, $\|\delta_s\partial^\gamma\phi_0\|_\infty \le C$ and $\|\delta_s v\|_{L^p(O)} \le \|\partial_e v\|_{L^p(O_1)} \le C$. Finally, $g^{i\bar j}\partial^\gamma k_{i\bar j}$ and $P_\gamma(G)$ are sums of products of bounded continuous factors, namely entries of G^{-1}, derivatives of k, and derivatives of G of order at most $\ell-3$, each of which has difference quotients bounded in $L^p(O)$; by the discrete product rule their difference quotients are bounded in $L^p(O)$. Thus the right side is bounded in $L^p(O)$, and $\|\delta_s v\|_{W^{1,p}(O)} \le \|\partial_e v\|_{W^{1,p}(O_1)} \le C$. The matrices $\tau_s G^{-1}$ satisfy the hypotheses of Step 1 with $\varpi' = \lambda^{-2}\varpi$. By Step 1, $\|D^2\delta_s v\|_{L^p(O')} \le C$ uniformly in s, and Lemma 16.2.18(2), applied to the second derivatives of v, shows that the third derivatives of v lie in $L^p(O')$, with norm at most C. Every multi-index of length $\ell+1$ is the sum of a multi-index of length $\ell-2$ and of three coordinate unit vectors, so $\|u\|_{W^{\ell+1,p}(O')} \le C$, where C depends on u only through $\|u\|_{W^{\ell,q'}(O_1)}$.

Step 4: conclusion. Choose open sets $V = O_{3\ell} \Subset O_{3\ell-1} \Subset \cdots \Subset O_0 = U$, depending only on V, U and ℓ, and put $p_m = 2^{\ell-m}(p+2n) - 2n$ for $3 \le m \le \ell$. Then $p_\ell = p \ge 2$, $p_m = 2p_{m+1} + 2n$, and $p_3 \le q$. Step 2, applied with the exponent p_3 in place of p and with the sets O_1, O_2, O_3 in the roles of O_1, O, O', bounds $\|u\|_{W^{3,p_3}(O_3)}$, since $\|D^2u\|_{L^{p_3}(O_1)} \le |O_1|^{1/p_3-1/q}\|D^2u\|_{L^q(U)}$. For $m = 3, \dots, \ell-1$ we then apply Step 3 with ℓ replaced by m, with the exponent p_{m+1} in place of p and $q' = p_m$, and with the sets O_{3m} (shrunk to O_{3m+1}), O_{3m+2}, O_{3m+3} in the roles of O_1, O, O'; this bounds $\|u\|_{W^{m+1,p_{m+1}}(O_{3m+3})}$ in terms of $\|u\|_{W^{m,p_m}(O_{3m})}$. For $m = \ell - 1$ we obtain the bound for $\|u\|_{W^{\ell,p}(V)}$. □

The openness part of the continuity method is a perturbation argument in Sobolev spaces that are algebras.

Lemma 16.2.20 *Let $d \ge 1$, let $s > d/2$ be an integer, and for $v \in \mathcal{S}(\mathbb{R}^d)$ put*

$$\|v\|_{\langle s\rangle} = \Big(\int_{\mathbb{R}^d} (1+|\xi|^2)^s |\widehat{v}(\xi)|^2 \,\mathrm{d}\xi\Big)^{1/2}.$$

Then $\|vw\|_{\langle s\rangle} \le C_s\|v\|_{\langle s\rangle}\|w\|_{\langle s\rangle}$ for $v, w \in \mathcal{S}(\mathbb{R}^d)$, with C_s depending only on d and s.

Proof By the inversion formula of Theorem 6.1.2 and Fubini's theorem, $\widehat{vw} = \widehat{v} * \widehat{w}$. With $\langle\xi\rangle = (1+|\xi|^2)^{1/2}$ we have $\langle\xi\rangle \le \langle\eta\rangle + \langle\xi-\eta\rangle$, since $(1,\xi) = (1,\eta) + (0,\xi-\eta)$ in $\mathbb{R}^{1+d}$; hence $\langle\xi\rangle^s \le 2^s(\langle\eta\rangle^s + \langle\xi-\eta\rangle^s)$, and therefore

$$\langle\xi\rangle^s |\widehat{v} * \widehat{w}|(\xi) \le 2^s\Big(\big(\langle\cdot\rangle^s|\widehat{v}|\big) * |\widehat{w}| + |\widehat{v}| * \big(\langle\cdot\rangle^s|\widehat{w}|\big)\Big)(\xi).$$

Young's inequality $\|F * G\|_2 \le \|F\|_2\|G\|_1$ follows from the Cauchy–Schwarz inequality $|F*G(x)|^2 \le \|G\|_1 \int |F(x-y)|^2|G(y)|\,\mathrm{d}y$ by integration in x and Fubini's theorem. It therefore remains to bound $\|\widehat{w}\|_1$ and $\|\widehat{v}\|_1$ by the norms $\|\cdot\|_{\langle s\rangle}$. This follows from the Cauchy–Schwarz inequality, $\|\widehat{w}\|_1 \le \|\langle\cdot\rangle^{-s}\|_2\|\langle\cdot\rangle^s\widehat{w}\|_2$, since $\langle\xi\rangle^{-2s}$ is integrable on $\mathbb{R}^d$ for $2s > d$. □

Theorem 16.2.21 *For every smooth real function f on X and every $\beta > 0$, equation (16.10) has a unique smooth real solution a_β.*

Proof For $0 \le t \le 1$ consider the equation

$$(\kappa + \mathrm{dd^c}a)^n = \mathrm{e}^{\beta(a-tf)}\kappa^n, \qquad \kappa + \mathrm{dd^c}a > 0,$$

and let $T \subset [0,1]$ be the set of t for which it has a smooth real solution. For $t = 0$ the solution is $a = 0$, so $T \ne \varnothing$. We show that T is open and closed in $[0,1]$; then $T = [0,1]$, since $[0,1]$ is connected, and the case $t = 1$ gives existence. Uniqueness is proved at the end.

Step 1: a priori estimates. Let a be a smooth solution for some $t \in [0,1]$, with metric g. In this step, constants depend only on X, ω, κ, β and f, but not on t or on the solution. Since $\|tf\|_\infty \le \|f\|_\infty$ and, by (16.6), $\sup_X D_h(tf) = t\sup_X D_h f \le \sup_X D_h f$,

Lemma 16.2.15, applied to tf, gives $|a| \le \|f\|_\infty$ and $C_0^{-1}h \le g \le C_0 h$ with C_0 independent of t. Hence $La = -4\pi D_h a = -4\pi(H - \operatorname{tr}_h k)$ is bounded, and Proposition 16.2.11 bounds a in $W^{2,p}(X)$ for every finite p; by (16.2) and Morrey's inequality, this bounds $\|a\|_{C^1}$ and $\|D^2 a\|_{L^p}$ on compact subsets of charts. By Lemma 16.2.17, the first derivatives of the $g_{i\bar j}$ are bounded on compact subsets of the charts of a fixed finite atlas.

Fix finitely many coordinate balls U_ν whose closures lie in chart domains, and concentric balls $V'_\nu \Subset V_\nu \Subset U_\nu$ such that the V'_ν cover X. In the coordinates of U_ν, the function a solves (16.16) with $\phi_0 = \log\det k - \beta t f$, whose derivatives are bounded independently of t. By the preceding paragraph, on U_ν the matrix $G = g$ lies between two fixed positive multiples of the identity, it is Lipschitz with a fixed constant, so that we may take $\varpi(r) = Cr$, and $\|a\|_{C^1(U_\nu)} + \|D^2 a\|_{L^p(U_\nu)}$ is bounded for every finite p. By Lemma 16.2.19, $\|a\|_{W^{\ell,p}(V_\nu)}$ is bounded for all $\ell \ge 3$ and $p < \infty$, and by Morrey's inequality every derivative of a is bounded on V'_ν. Thus every C^ℓ norm of a, computed in these coordinates on the sets V'_ν, is bounded independently of t.

Step 2: openness. Let $t_0 \in T$, with smooth solution a and metric g; thus $\kappa + \mathrm{dd^c} a$ is a Kähler form. For t near t_0 we look for a solution of the form $a + b$, and we find b by the contraction mapping principle in a Sobolev space. Fix an integer $s > n$. We use the spaces H^r of Section 14.1, defined with the coordinate balls U_ν of Step 1 and a partition of unity (χ_ν) subordinate to them; by Lemma 14.1.1(1), other choices give equivalent norms. For a smooth function v with compact support in U_ν, written in the coordinates of U_ν, let $\|v\|_{\langle r\rangle}$ be defined by the formula of Lemma 16.2.20 with $d = 2n$, so that $\|u\|^2_{H^r} = \sum_\nu \|\chi_\nu u\|^2_{\langle r\rangle}$ for smooth u. Choose smooth functions $\tilde\chi_\nu$, $\check\chi_\nu$ on X with values in $[0,1]$ and compact supports in U_ν such that $\tilde\chi_\nu = 1$ near $\operatorname{Supp}\chi_\nu$ and $\check\chi_\nu = 1$ near $\operatorname{Supp}\tilde\chi_\nu$. We use three facts.

(i) Since $s > n$, Lemma 14.1.1(3) gives continuous inclusions $H^{s+2} \subset C^2(X)$ and $H^s \subset C^0(X)$.

(ii) For a smooth function ψ with compact support in U_ν, multiplication by ψ is bounded for $\|\cdot\|_{\langle s\rangle}$, by Lemma 16.2.20; and $\|\partial_\mu v\|_{\langle r\rangle} \le 2\pi\|v\|_{\langle r+1\rangle}$ for real coordinate derivatives, since $\widehat{\partial_\mu v} = 2\pi i\xi_\mu \widehat v$.

(iii) $\|\check\chi_\nu b\|_{\langle s+2\rangle} \le C\|b\|_{H^{s+2}}$ for smooth b. Indeed, $\Sigma = \sum_\iota \check\chi_\iota \ge 1$, because $\check\chi_\iota = 1$ wherever $\chi_\iota > 0$; so the functions $\check\chi_\iota/\Sigma$ form a partition of unity subordinate to the same charts, and by Lemma 14.1.1(1) the norm of H^{s+2} computed with them is equivalent to the given one. Moreover $\check\chi_\nu b = \eta_\nu \cdot (\check\chi_\nu/\Sigma) b$, where η_ν is a smooth function with compact support in U_ν equal to Σ near $\operatorname{Supp}\check\chi_\nu$, and multiplication by η_ν is bounded for $\|\cdot\|_{\langle s+2\rangle}$ by Lemma 16.2.20.

For a real function $b \in C^2(X)$, let Y be the tensor $Y^m{}_j = g^{m\bar l} b_{j\bar l}$; its powers $(Y^q)^m{}_j = Y^m{}_{a_1} Y^{a_1}{}_{a_2} \cdots Y^{a_{q-1}}{}_j$ are tensors, so the traces $\operatorname{tr}(Y^q)$ are functions on X. In a chart, with the matrices $G = (g_{i\bar j})$ and $B = (b_{i\bar j})$, the traces of the powers of Y are those of BG^{-1}, which is similar to the Hermitian matrix $G^{-1/2} B G^{-1/2}$; let $\lambda_1, \dots, \lambda_n$ be their real eigenvalues. Then $\det(G + B) = \det G \prod_i (1 + \lambda_i)$ and $\operatorname{tr} Y = \sum_i \lambda_i = g^{i\bar j} b_{i\bar j} = D_g b$. Since $|\lambda_i| \le \|G^{-1}\|\,|B|$, and G is bounded below on compact subsets of the charts of a finite atlas, (i) gives $r_0 > 0$ such that $\|b\|_{H^{s+2}} \le r_0$ implies $|\lambda_i| \le 1/2$; then $G + B > 0$, that is, $\kappa + \mathrm{dd^c}(a + b) > 0$, and

$$\begin{aligned} Q(b) &:= \log\det(G+B) - \log\det G - D_g b \\ &= \sum_i \bigl(\log(1+\lambda_i) - \lambda_i\bigr) = \sum_{q\ge 2} \frac{(-1)^{q+1}}{q}\operatorname{tr}(Y^q). \end{aligned}$$

Put $\mathcal{N}_t(a+b) = \log(\det(G+B)/\det k) - \beta(a+b-tf)$, so that $a+b$ solves the equation for t exactly when $\mathcal{N}_t(a+b)=0$. Since $\log(\det G/\det k) = \beta(a-t_0 f)$,

$$\mathcal{N}_t(a+b) = \beta(t-t_0)f + \mathcal{D}b + Q(b), \qquad \mathcal{D} = D_g - \beta.$$

We estimate Q in H^s. Let b be smooth and real with $\|b\|_{H^{s+2}} \le r_0$. In the chart U_ν let $Y_\nu = \tilde\chi_\nu Y$, a matrix of smooth functions with compact support, and $\|Y_\nu\|_{\langle s\rangle} = \max_{m,j}\|\tilde\chi_\nu Y^m{}_j\|_{\langle s\rangle}$. Since $\tilde\chi_\nu Y^m{}_j = (\tilde\chi_\nu g^{m\bar l})\,\partial_j\partial_{\bar l}(\check\chi_\nu b)$, facts (ii) and (iii) give $\|Y_\nu\|_{\langle s\rangle} \le C_Y\|b\|_{H^{s+2}}$. For matrices M, N of smooth compactly supported functions, Lemma 16.2.20 gives $\|(MN)_{mj}\|_{\langle s\rangle} \le nC_s \max\|M_{\cdot\cdot}\|_{\langle s\rangle}\max\|N_{\cdot\cdot}\|_{\langle s\rangle}$, and therefore

$$\|\chi_\nu \operatorname{tr}(Y_\nu^q)\|_{\langle s\rangle} \le C_\chi n(nC_s)^{q-1}\|Y_\nu\|_{\langle s\rangle}^q,$$

where C_χ bounds multiplication by χ_ν. Since $Y_\nu = Y$ on $\operatorname{Supp}\chi_\nu$, the function $\chi_\nu Q(b)$ is the sum of the series $\sum_{q\ge2}\frac{(-1)^{q+1}}{q}\chi_\nu\operatorname{tr}(Y_\nu^q)$, which converges uniformly, with supports in $\operatorname{Supp}\chi_\nu$. Decreasing r_0, we may assume that $nC_sC_Yr_0 \le 1/2$. Then the series converges absolutely for $\|\cdot\|_{\langle s\rangle}$; since its partial sums converge in L^2, their Fourier transforms converge in L^2 (Theorem 6.1.2), a subsequence converges almost everywhere, and Fatou's lemma gives

$$\|\chi_\nu Q(b)\|_{\langle s\rangle} \le \sum_{q\ge2}\frac{1}{q}C_\chi n(nC_s)^{q-1}\|Y_\nu\|_{\langle s\rangle}^q \le 2C_\chi n^2C_s\|Y_\nu\|_{\langle s\rangle}^2.$$

Summing over ν, we obtain $\|Q(b)\|_{H^s} \le C_Q\|b\|_{H^{s+2}}^2$. In the same way, using $Y^q - Y'^q = \sum_{i=0}^{q-1} Y^i(Y-Y')Y'^{q-1-i}$, we obtain

$$\|Q(b)-Q(b')\|_{H^s} \le C_Q\bigl(\|b\|_{H^{s+2}} + \|b'\|_{H^{s+2}}\bigr)\|b-b'\|_{H^{s+2}}$$

for two such functions b, b'.

Both estimates extend to real elements of H^{s+2}. Let $H^{s+2}_{\mathbb{R}}$ be the closed subspace of real elements of H^{s+2}. Smooth real functions are dense in it, since taking real parts does not increase the norms: $|\widehat{\chi\operatorname{Re}v}(\xi)|^2 \le \frac12(|\widehat{\chi v}(\xi)|^2 + |\widehat{\chi v}(-\xi)|^2)$ for real χ. If $b\in H^{s+2}_{\mathbb{R}}$ with $\|b\|_{H^{s+2}} < r_0$, and smooth real b_j converge to b in H^{s+2}, with norms less than r_0, then the $Q(b_j)$ form a Cauchy sequence in H^s; by (i), $b_j\to b$ in C^2, so $Q(b_j)\to Q(b)$ uniformly, and, again by (i), the limit in H^s is $Q(b)$. Hence the two estimates for Q hold on the ball $\|b\|_{H^{s+2}} < r_0$ of $H^{s+2}_{\mathbb{R}}$.

The operator $\mathcal{D}\colon H^{s+2}\to H^s$ is invertible, with bounded inverse. Indeed, $\kappa + \mathrm{dd^c}a = \frac{\mathrm{i}}{2\pi}g_{i\bar j}\,\mathrm{d}z_i\wedge\mathrm{d}\bar z_j$ is a Kähler form, and by (14.8), applied with $\omega' = \kappa + \mathrm{dd^c}a$, we have $D_g - \beta = -(4\pi)^{-1}(L_{\omega'} + 4\pi\beta)$. Since ω is an arbitrary Kähler form on X, Lemma 14.1.3(3) holds for ω'; applied to functions, on which the Hodge Laplacian

of ω' is $L_{\omega'}$, with $c = 4\pi\beta > 0$, it shows that $D_g - \beta$ maps H^{s+2} isomorphically onto H^s, because these spaces do not depend on the Kähler form (Lemma 14.1.1(4)). Let $M_{\mathcal{D}}$ be the norm of its inverse. Since $\mathcal{D}$ is a real operator, its inverse maps real elements to real elements. For $b \in H^{s+2}$, the element $\mathcal{D}b$ of H^s is the continuous function $D_g b - \beta b$, by approximation with smooth functions and (i).

The solutions $b \in H^{s+2}_{\mathbb{R}}$ of $\mathcal{N}_t(a+b) = 0$ with $\|b\|_{H^{s+2}} < r_0$ are the fixed points of

$$\Phi_t(b) = -\mathcal{D}^{-1}\big(\beta(t-t_0)f + Q(b)\big).$$

Choose $0 < r < r_0$ with $4M_{\mathcal{D}}C_Q r \le 1$, and suppose that $|t - t_0| \le \delta$, where $2M_{\mathcal{D}}\beta\delta\|f\|_{H^s} \le r$. On the closed ball $\overline{B}_r$ of radius r in $H^{s+2}_{\mathbb{R}}$ we then have $\|\Phi_t(b)\|_{H^{s+2}} \le M_{\mathcal{D}}(\beta\delta\|f\|_{H^s} + C_Q r^2) \le r/2 + r/4$ and $\|\Phi_t(b) - \Phi_t(b')\|_{H^{s+2}} \le 2M_{\mathcal{D}}C_Q r\|b - b'\|_{H^{s+2}} \le \frac{1}{2}\|b - b'\|_{H^{s+2}}$. The iterates $b^{(0)} = 0$, $b^{(j+1)} = \Phi_t(b^{(j)})$ therefore stay in $\overline{B}_r$, their successive differences decrease geometrically, so they form a Cauchy sequence, and their limit b is a fixed point of Φ_t, by the Lipschitz bound. Thus $\mathcal{D}b = -\beta(t-t_0)f - Q(b)$ in H^s, hence pointwise by (i), that is, $\mathcal{N}_t(a+b) = 0$. The function $u = a + b$ is of class C^2, satisfies $\kappa + \mathrm{dd}^{\mathrm{c}}u > 0$, and in every chart it solves (16.16) with $\phi_0 = \log\det k - \beta t f$. By Lemma 16.2.19, u is smooth. Hence T contains $[t_0 - \delta, t_0 + \delta] \cap [0,1]$, and T is open in $[0,1]$.

Step 3: closedness. Let $t_j \in T$ converge to $t_* \in [0,1]$, with solutions a_j. By Step 1, every C^ℓ norm of a_j on the compact sets $\overline{V'_\nu}$ is bounded independently of j. Hence, for every ℓ, the derivatives of order at most ℓ of the a_j are uniformly bounded on each $\overline{V'_\nu}$, and they are equicontinuous there by the mean value theorem, since the derivatives of order $\ell + 1$ are bounded and the balls V'_ν are convex. By the Arzelà–Ascoli theorem (Proposition 1.6.4) and a diagonal argument over ℓ and ν, a subsequence converges, together with all its derivatives, uniformly on each $\overline{V'_\nu}$. The limit a_* is smooth, and its derivatives are the limits of the derivatives of the a_j, because uniform convergence of functions and of their first derivatives on a ball implies that the limit is continuously differentiable with the limit derivatives. The uniform lower bound $g_j \ge C_0^{-1}h$ of Step 1 gives $\kappa + \mathrm{dd}^{\mathrm{c}}a_* > 0$, and a_* solves the equation for t_*. Thus $t_* \in T$, and T is closed.

Step 4: uniqueness. Let a and a' be smooth solutions of (16.10). At a maximum point of $a - a'$ the complex Hessian of $a - a'$ is negative semidefinite, so $0 < k + (a_{i\bar{j}}) \le k + (a'_{i\bar{j}})$, and the equations give $e^{\beta(a-f)} \le e^{\beta(a'-f)}$ there. Hence $a - a' \le 0$ at that point, and therefore everywhere. Interchanging a and a' proves that $a = a'$. □

As $\beta \to \infty$, the solutions a_β converge to the envelope of f. The upper bound follows from the maximum principle, the lower bound from the comparison principle applied to a slightly lowered convex combination of the envelope and a constant.

Proposition 16.2.22 *Let f be a smooth real function on X, and for $\beta > 0$ let a_β be the solution of* (16.10) *given by Theorem 16.2.21. Put*

$$B = \sup_X \mathrm{tr}_h\, k + \sup_X D_h f, \qquad d_0 = \inf_X \frac{\det k}{\det h}, \qquad K_0 = \frac{B^n}{n^n d_0}.$$

Then $K_0 \geq 1$, $a_\beta \leq P_\kappa(f) + \beta^{-1} \log K_0$ for every $\beta > 0$, and for every $\beta \geq 1$

$$P_\kappa(f) - \frac{\operatorname{osc}_X f + n \log \beta}{\beta} \leq a_\beta \leq P_\kappa(f) + \frac{\log K_0}{\beta}.$$

In particular $a_\beta \to P_\kappa(f)$ uniformly on X as $\beta \to \infty$. Moreover $P_\kappa(f)$ has bounded metric Laplacian: with the constant A of Lemma 16.2.15,

$$-\sup_X \operatorname{tr}_h k \leq D_h P_\kappa(f) \leq 2B\, \mathrm{e}^{2A\|f\|_\infty} \tag{16.17}$$

in the sense of distributions, that is, (16.7) holds for $v = P_\kappa(f)$ with $c_1 = \sup_X \operatorname{tr}_h k$ and $c_2 = 2B\mathrm{e}^{2A\|f\|_\infty}$. Finally, $P_\kappa(f) \in C^{1,\alpha}(X)$ for every $0 < \alpha < 1$.

Proof At each point, the eigenvalues of k relative to h are positive, and the arithmetic–geometric mean inequality gives $\operatorname{tr}_h k \geq n(\det k / \det h)^{1/n} \geq n d_0^{1/n}$. Since $\sup_X D_h f \geq 0$ by (16.6), it follows that $B \geq n d_0^{1/n}$, that is, $K_0 \geq 1$.

The upper bound. Let $\beta > 0$, write $a = a_\beta$ and $g = k + (a_{i\bar{j}})$, and let x_0 be a maximum point of $a - f$. There the complex Hessian of $a - f$ is negative semidefinite, so $0 < g \leq k + (f_{i\bar{j}})$. In particular $k + (f_{i\bar{j}})$ is positive definite at x_0, and the arithmetic–geometric mean inequality for its eigenvalues relative to h gives

$$\mathrm{e}^{\beta(a-f)} = \frac{\det g}{\det k} \leq \frac{\det(k + (f_{i\bar{j}}))}{\det h} \cdot \frac{\det h}{\det k} \leq \Big(\frac{\operatorname{tr}_h k + D_h f}{n}\Big)^n \frac{1}{d_0} \leq K_0$$

at x_0. Hence $a - f \leq \log K_0/\beta$ on X. The function $a - \log K_0/\beta$ is therefore a smooth κ-psh function below f, that is, a competitor for $P_\kappa(f)$, and so $a \leq P_\kappa(f) + \log K_0/\beta$.

The lower bound. Let $\beta \geq 1$ and put $v = P_\kappa(f)$; as noted at the beginning of this subsection, $\min_X f \leq v \leq f$. For $0 < t \leq 1$ put

$$w = (1-t)v + t \min_X f - \frac{n \log(1/t)}{\beta}.$$

This is a bounded κ-psh function, and we show that $w \leq a$.

For a bounded κ-psh function b write $\mu_b = \langle (\kappa + \mathrm{dd^c} b)^n \rangle$, the non-pluripolar Monge–Ampère measure, which is the Bedford–Taylor measure since b is bounded (Proposition 15.3.3, applied with $q = p = n$ to the sums of b with smooth local potentials of κ), and equals the smooth form $(\kappa + \mathrm{dd^c} b)^n$ if b is smooth (Proposition 9.1.2(1)). Locally $\kappa = \mathrm{dd^c}\rho$ with ρ smooth, and $\kappa + \mathrm{dd^c} w = \mathrm{dd^c}\big((1-t)(\rho + v) + t\rho\big)$. Expanding the nth power by multilinearity and positive homogeneity (Proposition 15.1.5), and discarding the mixed terms, which are positive measures, we get $\mu_w \geq t^n \kappa^n$. On the other hand $w \leq f - n \log(1/t)/\beta$, so $\mathrm{e}^{\beta(w-f)} \leq t^n$.

Let $E = \{a < w\}$, a Borel set. Since a and w are bounded, $a \geq w - O(1)$, so Proposition 15.8.5 applies without the correction terms; together with the two inequalities just proved and the equation for a it gives

$$\int_E \mathrm{e}^{\beta(w-f)} \kappa^n \leq \int_E \mu_w \leq \int_E \mu_a = \int_E \mathrm{e}^{\beta(a-f)} \kappa^n.$$

On E the first integrand is strictly larger than the last one, so E has volume zero. Thus $w \le a$ almost everywhere, and hence everywhere by Lemma 13.1.4(1), since both functions are quasi-plurisubharmonic.

Now take $t = 1/\beta$, which is allowed since $\beta \ge 1$. Since $\min_X f - v \ge \min_X f - \max_X f = -\operatorname{osc}_X f$, we obtain

$$a \ge w = v + \frac{\min_X f - v}{\beta} - \frac{n \log \beta}{\beta} \ge v - \frac{\operatorname{osc}_X f + n \log \beta}{\beta}.$$

The Laplacian bound. Let β_0 be the constant of Lemma 16.2.15. For $\beta > \beta_0$ that lemma, together with $D_h a_\beta = H - \operatorname{tr}_h k$, $H > 0$ and $\operatorname{osc}_X f \le 2\|f\|_\infty$, gives

$$-\sup_X \operatorname{tr}_h k \le D_h a_\beta \le 2B\, \mathrm{e}^{2A\|f\|_\infty}.$$

Since $a_\beta \to v$ uniformly, $\langle D_h v, \chi\rangle = \int_X v\, D_h\chi \, \mathrm{d}V = \lim_{\beta\to\infty} \int_X (D_h a_\beta)\chi \, \mathrm{d}V$ for smooth χ, which proves (16.17). By the remark following (16.7), $Lv = -4\pi D_h v$ is therefore a bounded function, with a bound depending only on X, ω, κ and upper bounds for $\|f\|_\infty$ and $\sup_X D_h f$. Finally v is continuous, as a uniform limit of continuous functions, so by Proposition 16.2.11 it belongs to $C^{1,\alpha}(X)$ for every $0 < \alpha < 1$. □

We can now remove the smoothness of the obstacle and pass to finite minima. A finite minimum of smooth functions is approximated by smooth functions whose complex Hessians have traces bounded above by those of the given functions.

Lemma 16.2.23 *Let $r \ge 1$ be an integer, let $f_1, \dots, f_r$ be smooth real functions on X, let $\delta > 0$, and put*

$$F = -\delta \log \sum_{j=1}^{r} \mathrm{e}^{-f_j/\delta}.$$

Then F is a smooth real function with $\min_j f_j - \delta \log r \le F \le \min_j f_j$, and $D_h F \le \max_j \sup_X D_h f_j$ on X.

Proof Put

$$p_j = \frac{\mathrm{e}^{-f_j/\delta}}{\sum_l \mathrm{e}^{-f_l/\delta}}.$$

The weights p_j are positive with $\sum_j p_j = 1$, and $\min_j f_j - \delta \log r \le F \le \min_j f_j$, because the sum lies between $\mathrm{e}^{-\min_j f_j/\delta}$ and r times this number. In local holomorphic coordinates, the chain rule gives $\partial_i F = \sum_j p_j \partial_i f_j$ and $\partial_{\bar m} p_j = -\delta^{-1} p_j (\partial_{\bar m} f_j - \sum_l p_l \partial_{\bar m} f_l)$; differentiating once more and contracting with $h^{i\bar m}$, we get

$$D_h F = \sum_j p_j D_h f_j - \frac{1}{\delta}\left(\sum_j p_j |\partial f_j|_h^2 - \left| \sum_j p_j \partial f_j \right|_h^2 \right).$$

The term in parentheses equals $\sum_j p_j |\partial f_j - \sum_l p_l \partial f_l|_h^2 \geq 0$, the variance of the vectors ∂f_j with respect to the probability weights p_j. Hence $D_h F \leq \sum_j p_j D_h f_j \leq \max_j \sup_X D_h f_j$. □

Theorem 16.2.24 *Let $r \geq 1$ be an integer, and let $f_1, \dots, f_r$ be continuous real functions on X with bounded metric Laplacian. Then the envelope*

$$u = P_\kappa\big(\min_{1 \leq j \leq r} f_j\big)$$

has bounded metric Laplacian, with a bound depending only on X, ω, κ, and on bounds for $\|f_j\|_\infty$ and $\|Lf_j\|_{L^\infty}$, $1 \leq j \leq r$. In particular u belongs to $C^{1,\alpha}(X)$ for every $0 < \alpha < 1$, and the positive current $\kappa + \mathrm{dd}^{\mathrm{c}} u$ has bounded coefficients.

Proof We approximate $\min_j f_j$ uniformly by smooth obstacles to which (16.17) applies with uniform constants, and pass to the limit.

Step 1: smoothing the obstacles. By Proposition 16.2.11, each f_j belongs to $W^{2,p}(X)$ for every finite p, and it is of class C^1; fix $p > 2n$. Choose a finite smooth partition of unity (χ_ν) with each χ_ν supported in the interior of a coordinate chart. For small $\varepsilon > 0$, let $f_{j,\nu,\varepsilon}$ be the convolution of f_j, written in the chart of χ_ν, with the mollifier ρ_ε of Section 1.4; it is defined on a neighborhood of the support of χ_ν. Put $f_{j,\varepsilon} = \sum_\nu \chi_\nu f_{j,\nu,\varepsilon}$. These are smooth functions with $\|f_{j,\varepsilon}\|_\infty \leq \|f_j\|_\infty$, and $f_{j,\varepsilon} \to f_j$ in C^1, because the convolutions of the C^1 function f_j and of its first derivatives converge to them uniformly on compact sets (Lemma 1.4.1(1)), and $\partial_\mu(f_j * \rho_\varepsilon) = (\partial_\mu f_j) * \rho_\varepsilon$.

We bound the Laplacians of the approximations. By (16.4), in the real coordinates $x_1, \dots, x_{2n}$ underlying a holomorphic chart, with $\partial_\mu = \partial/\partial x_\mu$, we have $D_h = \sum_{\mu,\sigma} c^{\mu\sigma}(x) \partial_\mu \partial_\sigma$ on real functions, with smooth real coefficients; there are no first-order terms. Since $f_j \in W^{2,p}_{\mathrm{loc}}$ in the chart, by (16.2), and $\partial_\mu \partial_\sigma (f_j * \rho_\varepsilon) = (\partial_\mu \partial_\sigma f_j) * \rho_\varepsilon$, the commutator of D_h with convolution is

$$\begin{aligned} &D_h(f_j * \rho_\varepsilon)(x) - \big((D_h f_j) * \rho_\varepsilon\big)(x) \\ &\qquad = \int \sum_{\mu,\sigma} \big(c^{\mu\sigma}(x) - c^{\mu\sigma}(y)\big) \partial_\mu \partial_\sigma f_j(y)\, \rho_\varepsilon(x-y)\, \mathrm{d}y, \end{aligned}$$

where $D_h f_j = -Lf_j/(4\pi)$ is a bounded function. Since $|c^{\mu\sigma}(x) - c^{\mu\sigma}(y)| \leq C\varepsilon$ on the support of $\rho_\varepsilon(x - \cdot)$ and $\|\rho_\varepsilon\|_{L^{p'}} = \varepsilon^{-2n/p} \|\rho\|_{L^{p'}}$ with $p' = p/(p-1)$, Hölder's inequality and (16.2) bound this commutator by $C\varepsilon^{1-2n/p} \|f_j\|_{W^{2,p}(X)}$, which tends to zero because $p > 2n$. As $(D_h f_j) * \rho_\varepsilon$ lies between the essential infimum and the essential supremum of $D_h f_j$, since $\rho_\varepsilon \geq 0$ has integral one, we obtain on a neighborhood of the support of χ_ν

$$\operatorname*{ess\,inf}_X D_h f_j - o(1) \leq D_h f_{j,\nu,\varepsilon} \leq \operatorname*{ess\,sup}_X D_h f_j + o(1) \qquad (\varepsilon \to 0).$$

Next, $D_h f_{j,\varepsilon} = \sum_\nu \chi_\nu D_h f_{j,\nu,\varepsilon} + \sum_\nu [D_h, \chi_\nu] f_{j,\nu,\varepsilon}$, where each commutator $[D_h, \chi_\nu]$ is a first-order operator with coefficients supported in $\operatorname{Supp} \chi_\nu$. Since

$\sum_\nu \chi_\nu = 1$, we have $\sum_\nu [D_h, \chi_\nu] f_j = D_h f_j - D_h f_j = 0$, so the second sum equals $\sum_\nu [D_h, \chi_\nu](f_{j,\nu,\varepsilon} - f_j)$, which involves only $f_{j,\nu,\varepsilon} - f_j$ and its first derivatives and tends to zero uniformly. The first sum is a convex combination. Hence $|D_h f_{j,\varepsilon}| \le \|D_h f_j\|_{L^\infty} + 1$ for all sufficiently small ε.

Step 2: smoothing the minimum. For $\delta > 0$ set

$$F_{\varepsilon,\delta} = -\delta \log \sum_{j=1}^{r} \mathrm{e}^{-f_{j,\varepsilon}/\delta}.$$

By Lemma 16.2.23,

$$\min_j f_{j,\varepsilon} - \delta \log r \le F_{\varepsilon,\delta} \le \min_j f_{j,\varepsilon}, \qquad \sup_X D_h F_{\varepsilon,\delta} \le \max_j \sup_X D_h f_{j,\varepsilon}.$$

Choose sequences $\varepsilon_m, \delta_m \to 0$ with ε_m small as in Step 1, and put $F_m = F_{\varepsilon_m,\delta_m}$. Then $F_m \to \min_j f_j$ uniformly on X, since $|\min_j f_{j,\varepsilon_m} - \min_j f_j| \le \max_j \|f_{j,\varepsilon_m} - f_j\|_\infty$, while $\|F_m\|_\infty \le \max_j \|f_j\|_\infty + \delta_m \log r$ and $\sup_X D_h F_m \le \max_j \|D_h f_j\|_{L^\infty} + 1$ remain bounded.

Step 3: passage to the limit. By (16.17), applied to the smooth obstacles F_m, we have $-\sup_X \mathrm{tr}_h\, k \le D_h P_\kappa(F_m) \le 2B_m \mathrm{e}^{2A\|F_m\|_\infty}$ in the sense of distributions, where A depends only on X, ω, κ, and $B_m = \sup_X \mathrm{tr}_h\, k + \sup_X D_h F_m$. By Step 2 these bounds are uniform in m and depend only on the asserted data. By the stability of envelopes in Proposition 16.1.4(4), $\|P_\kappa(F_m) - u\|_\infty \le \|F_m - \min_j f_j\|_\infty \to 0$. Therefore $D_h P_\kappa(F_m) \to D_h u$ in the sense of distributions, and u satisfies (16.7) with $c_1 = \sup_X \mathrm{tr}_h\, k$ and $c_2 = \sup_m 2B_m \mathrm{e}^{2A\|F_m\|_\infty}$. By the remark following (16.7), $D_h u$, and hence $Lu = -4\pi D_h u$, is a bounded function with the asserted bound.

The function u is continuous, as a uniform limit of the functions $P_\kappa(F_m)$, which are continuous by Proposition 16.2.22; so Proposition 16.2.11 shows that $u \in C^{1,\alpha}(X)$ for every $0 < \alpha < 1$. Finally, u is a continuous κ-psh function with bounded metric Laplacian, so by Proposition 16.2.9, applied with $U = X$, $\theta_j = \kappa$, and u in place of both u_j and f_j for every j, the coefficients of $\kappa + \mathrm{dd}^{\mathrm{c}} u$ are bounded on every coordinate ball $W \Subset X$, hence on X. □

16.2.4 Real Hessian bounds for Kähler envelopes

A bounded metric Laplacian bounds the complex Hessian of a potential, by positivity, but the full real Hessian also contains the $(2,0)$ and $(0,2)$ parts, and bounding these requires a different argument. We now show that, for a Kähler background, the envelope of a finite minimum of $C^{1,1}$ obstacles is again $C^{1,1}$ (Theorem 16.2.26). The main step is an a priori bound for the full real Hessian of the solutions of the penalized equation (Lemma 16.2.25), obtained from the maximum principle applied to the largest eigenvalue of the real Hessian. Differentiating this eigenvalue twice

produces positive third-derivative terms, and we keep them: they absorb the negative term that comes from differentiating its logarithm.

We first recall the covariant Hessian of a Riemannian metric. Let g be a smooth Riemannian metric on X. For smooth vector fields Y, Z, W on an open set $U \subset X$ let $\mathcal{K}_{Y,Z}(W)$ be the right side of the Koszul formula

$$\begin{aligned} 2g(\nabla_Y Z, W) = {} & Y\, g(Z,W) + Z\, g(Y,W) - W\, g(Y,Z) \\ & + g([Y,Z],W) - g([Y,W],Z) - g([Z,W],Y). \end{aligned}$$

If W is replaced by ϕW for a smooth function ϕ, the terms of $\mathcal{K}_{Y,Z}(\phi W)$ containing derivatives of ϕ cancel in pairs, so $\mathcal{K}_{Y,Z}(\phi W) = \phi\, \mathcal{K}_{Y,Z}(W)$. In a chart, therefore, $\mathcal{K}_{Y,Z}(W) = \sum_s W^s \mathcal{K}_{Y,Z}(\partial_s)$, and since g is nondegenerate there is a unique smooth vector field $\nabla_Y Z$ satisfying the Koszul formula for all W; by uniqueness, the vector fields obtained in different charts agree on overlaps. This is the *Levi-Civita connection* of g. Replacing Y or Z by ϕY or ϕZ in the formula, the terms containing derivatives of ϕ cancel, respectively add up to $2(Y\phi)g(Z,W)$, so $\nabla_{\phi Y} Z = \phi \nabla_Y Z$ and $\nabla_Y(\phi Z) = (Y\phi)Z + \phi \nabla_Y Z$; and interchanging Y and Z changes the right side by $-2g([Y,Z],W)$, so $\nabla_Y Z - \nabla_Z Y = [Y,Z]$.

In coordinates, with $g_{uv} = g(\partial_u, \partial_v)$, the Koszul formula for coordinate vector fields gives $\nabla_{\partial_u}\partial_v = \sum_t \Gamma^t_{uv}\partial_t$, where the Christoffel symbols are determined by $\sum_t g_{st}\Gamma^t_{uv} = \frac{1}{2}(\partial_u g_{vs} + \partial_v g_{us} - \partial_s g_{uv})$; they are symmetric in u and v and vanish at a point where the first derivatives of the g_{uv} vanish. For a smooth function b, the *covariant Hessian* $\nabla^2 b(Y,Z) = Y(Zb) - (\nabla_Y Z)b$ is, by these rules, given in coordinates by

$$\nabla^2 b(Y,Z) = \sum_{u,v} Y^u Z^v \Big(\partial_u \partial_v b - \sum_t \Gamma^t_{uv} \partial_t b \Big).$$

Hence $\nabla^2 b$ is a symmetric bilinear form on each tangent space, and at a point where the first derivatives of the metric vanish it is the matrix of second derivatives of b. If g' is another smooth Riemannian metric, the difference $\nabla'_Y Z - \nabla_Y Z$ is linear over smooth functions in Y and in Z, by the rules above, so the two covariant Hessians of b differ by a smooth tensor applied to $\mathrm{d}b$. For a C^1 function b the same coordinate formula, with distributional second derivatives, defines the *weak covariant Hessian*.

Recall that a function is of class $C^{1,1}$ if it is C^1 and its first derivatives are locally Lipschitz in smooth real coordinates (Proposition 1.7.5). On the compact manifold X we measure such a function by the supremum norms of its value, its gradient and its weak covariant Hessian with respect to a fixed smooth Riemannian metric; the sum of these three norms is its $C^{1,1}$ *norm*.

The two descriptions agree: the first derivatives of a C^1 function are locally Lipschitz exactly when its weak second derivatives are locally bounded. Indeed, if a first derivative $\partial_a f$ is Lipschitz with constant M on a coordinate ball, its difference quotients in a direction e_b are bounded by M; pairing them with a test function φ and passing to the limit gives $|\langle \partial_b \partial_a f, \varphi\rangle| \le M \int |\varphi|\, \mathrm{d}x$. By Corollary 1.7.2(1), $\partial_b \partial_a f$ is therefore a bounded function, with $|\partial_b \partial_a f| \le M$. Conversely, by Proposition 1.7.5,

a continuous function whose weak second derivatives are bounded by M on a ball has a gradient that is Lipschitz on this ball with constant $2nM$, the real dimension being $2n$. The Christoffel symbols of a metric and the derivatives of coordinate changes are bounded on compact subsets of the charts, and they multiply only first derivatives, so different smooth metrics give equivalent $C^{1,1}$ norms. In the applications below, the C^1 property of the envelope will be known in advance from Proposition 16.2.11.

Lemma 16.2.25 *Let κ be a Kähler form on X, and let g_0 be a smooth Riemannian metric on X with Levi-Civita connection ∇. Let $B > 0$, and let f be a smooth real function such that*

$$\|f\|_\infty + \|\nabla f\|_\infty \le B, \qquad \nabla^2 f \le B g_0$$

as real quadratic forms, and for $\beta > 0$ let a_β be the solution of (16.10) given by Theorem 16.2.21. Then there are constants β_0 and C, depending only on B and on the fixed geometry (X, κ and g_0), such that

$$\|a_\beta\|_\infty + \|\nabla a_\beta\|_\infty + \|\nabla^2 a_\beta\|_\infty \le C \qquad (\beta \ge \beta_0).$$

Proof Step 1 reduces to a metric defined by κ and collects the bounds for a_β and its gradient. Steps 2 and 3 apply the maximum principle to the logarithm of the largest eigenvalue of $\nabla^2 a_\beta$, corrected by functions of the gradient and of a_β, and Step 4 absorbs the negative term with third derivatives.

Step 1: reductions and first bounds. Any two smooth Riemannian metrics on X are uniformly equivalent, and, as observed before the lemma, the Hessians of a function for the two metrics differ by a bounded tensor applied to its gradient. Changing g_0 therefore changes B and the constants only by factors depending on the two metrics, and we may assume that g_0 is the following Riemannian metric defined by κ. In local holomorphic coordinates write $\kappa = \frac{\mathrm{i}}{2\pi} k_{i\bar\jmath}\, \mathrm{d}z_i \wedge \mathrm{d}\bar z_j$, so that k is the coefficient matrix of $2\pi\kappa$ as in Section 16.2.3, and let $g_0 = 2\,\mathrm{Re}(k_{i\bar\jmath}\, \mathrm{d}z_i \otimes \mathrm{d}\bar z_j)$; this is 2π times the Riemannian metric $\kappa(\cdot, J\cdot)$, where J is the complex structure. At the center of normal coordinates for κ, given by Lemma 16.2.13, we have $k_{i\bar\jmath} = \delta_{ij}$ and $\mathrm{d}k = 0$; hence the Christoffel symbols of g_0 vanish and $g_0 = 2\sum_i (\mathrm{d}x_i^2 + \mathrm{d}y_i^2)$. Comparing coordinate derivatives at such a point, we obtain for every smooth function b

$$\mathrm{tr}_{g_0} \nabla^2 b = 2 D_k b, \qquad D_k b = k^{i\bar\jmath} b_{i\bar\jmath}. \tag{16.18}$$

In particular the upper bound for $\nabla^2 f$ gives $D_k f \le nB$.

Write $a = a_\beta$, $G = (k_{i\bar\jmath} + a_{i\bar\jmath})$ and $D_G = G^{i\bar\jmath}\partial_i\partial_{\bar\jmath}$; thus G is the matrix denoted g in Section 16.2.3. We apply Lemma 16.2.15 with κ also as the measuring metric. It bounds $|a|$ by $\|f\|_\infty$ and, for all sufficiently large β, bounds $\mathrm{tr}_k G$ in terms of B and the fixed geometry. Since $G > 0$ and $\mathrm{tr}_k G$ is bounded, all coefficients of G, and hence of the complex Hessian of a, are bounded; in particular the metric Laplacian La for the fixed Kähler form ω, which is a constant multiple of the trace of $\mathrm{dd}^{\mathrm{c}} a$ with respect to ω (Lemma 14.1.6), is uniformly bounded. Fix $p > 2n$. By Proposition 16.2.11, $\|a\|_{W^{2,p}(X)} \le C(\|La\|_{L^p} + \|a\|_{L^p})$ is uniformly bounded, and Morrey's inequality

(Lemma 6.2.6), applied in the charts defining the norm to the first derivatives of the products of a with the cutoff functions, bounds ∇a uniformly. From now on C denotes a constant depending only on these bounds, on B and on the fixed geometry; it may increase from line to line, but it never depends on β.

Step 2: the auxiliary function. Let ℓ be the largest eigenvalue of the real covariant Hessian $\nabla^2 a$ with respect to g_0. By (16.18), the sum of all eigenvalues is $2D_k a = 2\operatorname{tr}_k G - 2n \geq -2n$. Since each eigenvalue is at most ℓ, it is at least $-2n - (2n-1)\ell$; hence every eigenvalue of $\nabla^2 a$ has absolute value at most $2n(\max\{\ell, 0\} + 1)$. It therefore suffices to bound ℓ from above, and we do so by the maximum principle. Put

$$s = |\partial a|_k^2, \quad S = \sup_X s, \quad \tau = (1 + 2\sup_X |\partial f|_k^2)^{-1}, \quad \eta(s) = -\frac{\tau}{2}\log(1 + S - s).$$

Then $S \leq C$, the number $\tau \in (0, 1]$ has a uniform positive lower bound, and since $0 \leq s \leq S$,

$$\frac{\tau}{2(1+S)} \leq \eta' \leq \frac{\tau}{2}, \qquad \eta'' = \frac{2(\eta')^2}{\tau} \geq 2(\eta')^2. \tag{16.19}$$

For a constant $A > 0$ to be chosen later, consider the function $Q = \log \ell + \eta(s) - Aa$ on the open set $\{\ell > 0\}$. If this set is empty, then $\ell \leq 0$ and there is nothing to prove. Otherwise, since a and $\eta(s)$ are bounded, Q tends to $-\infty$ where ℓ tends to zero, so by compactness of X it attains its maximum at some point x_0 with $\ell(x_0) > 0$. For the same reason a uniform bound for $\ell(x_0)$ bounds ℓ on all of X. We may therefore assume that $\ell(x_0) \geq 1$, and that $\ell(x_0)$ exceeds any uniform threshold used below.

Choose normal coordinates for κ at x_0 in which $G(x_0)$ is diagonal, with diagonal entries $r_i > 0$: by Lemma 16.2.13, applied to k in one of finitely many fixed charts that contains x_0 in a fixed compact subset, and followed by a unitary change of coordinates that diagonalizes $G(x_0)$, there are such coordinates in which the derivatives of k of every fixed order at x_0 are bounded independently of x_0. Since $\sum_i r_i = \operatorname{tr}_k G \leq C$, we have $r_i \leq C$, hence $r_i^{-1} \geq c$ for a uniform $c > 0$. Write $T = \sum_i r_i^{-1}$; thus $T \geq c$. Take a g_0-orthonormal basis $V_1, \ldots, V_{2n}$ of the real tangent space at x_0 consisting of eigenvectors of $\nabla^2 a(x_0)$, with V_1 belonging to the eigenvalue $\ell(x_0)$, and extend these vectors to the chart as vector fields with constant real coordinate coefficients.

The largest eigenvalue ℓ may be multiple and need not be smooth, so we perturb the Hessian. Let B_0 be the symmetric bilinear form with constant coefficients in this chart which at x_0 is the g_0-orthogonal projection onto $V_1^\perp$, that is, $B_0(Y, Z) = g_0(PY, PZ)$ at x_0 for this projection P, and set

$$\Phi = g_0^{-1}(\nabla^2 a - B_0).$$

The form B_0 is positive semidefinite everywhere in the chart. Hence the largest eigenvalue $\lambda_1(\Phi)$ of Φ is at most ℓ, and it equals ℓ at x_0. Denote the eigenvalues of Φ at x_0 by $\lambda = \ell(x_0), \lambda_2, \ldots, \lambda_{2n}$, with eigenvectors $V_1, \ldots, V_{2n}$. For $\alpha > 1$, λ_α is the Hessian eigenvalue belonging to V_α minus one, so $\lambda - \lambda_\alpha \geq 1$. In particular $\lambda_1(\Phi)$ is a smooth function near x_0. Indeed, λ is a simple root of the characteristic polynomial of $\Phi(x_0)$, so by the implicit function theorem the characteristic polynomial of Φ has, near x_0, exactly one root close to λ, and this root depends smoothly on the point; it equals

$\lambda_1(\Phi)$, which is continuous as the maximum of $(\nabla^2 a - B_0)(Y,Y)$ over g_0-unit vectors Y, and it is a simple eigenvalue. The smooth function $\widehat{Q} = \log\lambda_1(\Phi) + \eta(s) - Aa$ satisfies $\widehat{Q} \le Q \le Q(x_0) = \widehat{Q}(x_0)$, so it has a local maximum at x_0.

Step 3: the inequality at the maximum point. From now on all quantities are evaluated at x_0. Write $a_{V_\alpha V_\gamma} = \nabla^2 a(V_\alpha, V_\gamma)$, a smooth function near x_0, and define the nonnegative quantities

$$U = 2\sum_{i,\alpha>1} \frac{|\partial_i a_{V_\alpha V_1}|^2}{r_i\lambda(\lambda-\lambda_\alpha)}, \qquad R = \sum_{p,q} \frac{|V_1 G_{p\bar q}|^2}{r_p r_q \lambda},$$
$$N = \sum_{i} \frac{|\partial_i a_{V_1 V_1}|^2}{r_i\lambda^2}, \qquad K = \sum_{i,p} \frac{|a_{ip}|^2 + |a_{i\bar p}|^2}{r_i}.$$

We first compute $D_G \log\lambda_1(\Phi)$, keeping the third-derivative terms. Let $M(t)$ be a smooth family of real symmetric matrices, and let $\lambda(t)$ be a simple eigenvalue with unit eigenvector V_1 at $t=0$, the other eigenvalues being λ_α with orthonormal eigenvectors V_α. Then

$$\lambda' = \langle M'V_1, V_1\rangle, \qquad \lambda'' = \langle M''V_1, V_1\rangle + 2\sum_{\alpha>1} \frac{|\langle M'V_1, V_\alpha\rangle|^2}{\lambda-\lambda_\alpha}.$$

Indeed, near $t=0$ the simple eigenvalue $\lambda(t)$ is a smooth function of t, by the implicit function theorem applied to the characteristic polynomial, and the columns of the adjugate matrix of $\lambda(t)I - M(t)$, which has rank one, span the eigenspace; normalizing a column that does not vanish at $t=0$ gives a smooth unit eigenvector $y(t)$ with $y(0) = \pm V_1$, and we may take the sign $+$. Differentiate the eigenvector equation $M(t)y(t) = \lambda(t)y(t)$. Pairing with V_1 gives the first formula, and pairing with V_α shows that the derivative of the eigenvector has component $\langle M'V_1, V_\alpha\rangle/(\lambda-\lambda_\alpha)$ along V_α, and it has no component along V_1, because $|y(t)| = 1$. Differentiating the identity $\lambda'(t) = \langle M'(t)y(t), y(t)\rangle$, where $y(t)$ is the normalized eigenvector with $y(0) = V_1$, and substituting these components gives the second formula.

We apply these formulas to the symmetric matrix M with entries $(\nabla^2 a - B_0)(e_\alpha, e_\gamma)$, where $e_1,\dots,e_{2n}$ is the g_0-orthonormal frame obtained from the constant vector fields $V_1,\dots,V_{2n}$ by the Gram–Schmidt process; M has the same eigenvalues as Φ. The frame depends smoothly on the point, equals $V_1,\dots,V_{2n}$ at x_0, and its first derivatives vanish at x_0, because those of g_0 do. We differentiate along the real coordinate directions and use $\partial_i\partial_{\bar\imath} = \frac14(\partial^2_{x_i} + \partial^2_{y_i})$. Since B_0 is constant, the first derivatives of M at x_0 are those of the functions $\nabla^2 a(V_\alpha, V_\gamma)$. In the second derivatives, the second derivatives of the frame multiply $\nabla^2 a - B_0$, which is bounded by $C(\lambda+1)$; after weighting with r_i^{-1} and summing they contribute at most $C\lambda T$. Since $|\partial_i b|^2 = \frac14(|\partial_{x_i} b|^2 + |\partial_{y_i} b|^2)$ for a real function b, we obtain

$$D_G\lambda_1(\Phi) \ge 2\sum_{i,\alpha>1} \frac{|\partial_i a_{V_\alpha V_1}|^2}{r_i(\lambda-\lambda_\alpha)} + \sum_i r_i^{-1} V_1 V_1 G_{i\bar\imath} - C\lambda T.$$

Here the middle term comes from $\sum_i r_i^{-1}\partial_i\partial_{\bar\imath}a_{V_1V_1}$ after commuting derivatives, and we check that this commutation costs at most $C\lambda T$. In real coordinates $a_{V_1V_1} = V_1V_1a - \Gamma^t_{uv}V_1^uV_1^v\partial_t a$, where the Christoffel symbols Γ of g_0 vanish at x_0. When we apply $\partial_i\partial_{\bar\imath}$, the term containing third derivatives of a is multiplied by $\Gamma(x_0) = 0$; the remaining terms are a first derivative of Γ times a second derivative of a, or a second derivative of Γ times a first derivative of a. By the Hessian and gradient bounds above they are at most $C(\lambda+1)$. Moreover $\partial_i\partial_{\bar\imath}V_1V_1a = V_1V_1a_{i\bar\imath} = V_1V_1G_{i\bar\imath} - V_1V_1k_{i\bar\imath}$, and the derivatives of the fixed matrix k are bounded. Summing with the weights r_i^{-1} gives an error at most $C\lambda T$, and no third derivative of a enters it.

Next we use the equation. Taking logarithms in (16.10) gives $\log\det G = \log\det k + \beta(a-f)$. Differentiating twice along V_1 and using $\partial(G^{-1}) = -G^{-1}(\partial G)G^{-1}$, we obtain at x_0

$$\sum_i r_i^{-1}V_1V_1G_{i\bar\imath} = \sum_{p,q}\frac{|V_1G_{p\bar q}|^2}{r_pr_q} + V_1V_1\log\det k + \beta(\lambda - f_{V_1V_1}).$$

Here we used that at the center of normal coordinates directional and covariant second derivatives agree, so that $V_1V_1a = \lambda$ and $V_1V_1f = f_{V_1V_1}$. If $\lambda \ge 2\max\{1, \sup_{X,|Y|_{g_0}=1} f_{YY}\}$, which is a uniform threshold because $\nabla^2 f \le Bg_0$, the last term is at least $\beta\lambda/2$. The term $V_1V_1\log\det k$ is bounded, hence at least $-CT$ because $T \ge c$. By the first variation formula, $\partial_i\lambda_1(\Phi) = \partial_i a_{V_1V_1}$ at x_0. Since $D_G\log\lambda_1 = \lambda^{-1}D_G\lambda_1 - \sum_i r_i^{-1}|\partial_i\lambda_1|^2/\lambda^2$, we conclude that

$$D_G\log\lambda_1(\Phi) \ge U + R - N - CT + \beta/2.$$

Next we differentiate the logarithmic equation once and apply the product rule to $s = k^{p\bar q}a_pa_{\bar q}$. At x_0,

$$\partial_i\partial_{\bar\imath}s = \sum_p\big(|a_{pi}|^2 + |a_{p\bar\imath}|^2\big) + 2\,\mathrm{Re}\sum_p a_{\bar p}\,\partial_p a_{i\bar\imath} + (\partial_i\partial_{\bar\imath}k^{p\bar q})a_pa_{\bar q}.$$

The last term involves only the bounded gradient. In the middle term the third derivatives occur only in the contraction $\sum_i r_i^{-1}\partial_p a_{i\bar\imath} = \partial_p\log\det G = \beta(a_p - f_p)$, because $\mathrm{d}k(x_0) = 0$; so we need no estimate for third derivatives. Weighting with r_i^{-1} and summing, we get

$$D_Gs \ge K - CT + 2\beta\,\mathrm{Re}\sum_p a_{\bar p}(a_p - f_p) \ge K - CT + \beta s - \beta|\partial f|_k^2.$$

Also $D_Ga = \sum_i r_i^{-1}(r_i - 1) = n - T$. At the local maximum x_0 we have $D_G\widehat{Q} \le 0$, where

$$D_G\widehat{Q} = D_G\log\lambda_1(\Phi) + \eta' D_Gs + \eta''\sum_i\frac{|s_i|^2}{r_i} - AD_Ga.$$

Since $0 < \eta' \le \frac12$ and $\eta'|\partial f|_k^2 \le \frac{\tau}{2}\sup_X|\partial f|_k^2 \le \frac14$, the three preceding estimates yield

$$0 \geq U + R - N + \eta' K + \eta'' \sum_i \frac{|s_i|^2}{r_i} + (A - C_0)T - An + \beta\eta' s + \beta/4, \qquad (16.20)$$

where C_0 is a uniform constant that does not depend on A.

Step 4: absorption of the third-order term. It remains to absorb the negative term $-N$ completely. Fix $0 < \varepsilon < 1/2$, to be chosen below, and assume that $\lambda \geq C/\varepsilon^2$. Let J be the complex structure of X, and define a $(1,0)$-vector W and real numbers μ_α by

$$W = (V_1 - \mathrm{i}JV_1)/\sqrt{2} = \sum_q \nu_q \partial_q, \qquad JV_1 = \sum_{\alpha>1} \mu_\alpha V_\alpha.$$

Since $k(x_0)$ is the identity, $\sum_q |\nu_q|^2 = g_0(V_1, V_1) = 1$; since JV_1 is a g_0-unit vector orthogonal to V_1, also $\sum_{\alpha>1} \mu_\alpha^2 = 1$. The coefficients ν_q, μ_α, and J itself, are constant in our coordinate system.

Writing $V_1 = \sqrt{2}\,\overline{W} - \mathrm{i}JV_1$ in the outer derivative of $\partial_i a_{V_1 V_1}$, we obtain

$$\partial_i a_{V_1 V_1} = \sqrt{2} \sum_q \overline{\nu}_q V_1 G_{i\bar{q}} - \mathrm{i} \sum_{\alpha>1} \mu_\alpha \partial_i a_{V_1 V_\alpha} + E_i, \qquad |E_i| \leq C. \qquad (16.21)$$

Indeed, the third coordinate derivatives of the scalar function a commute, and $\partial_i \overline{W} = \sum_q \overline{\nu}_q \partial_i \partial_{\bar{q}}$. Replacing coordinate second derivatives by covariant ones before the last differentiation introduces only first derivatives of Γ times the bounded gradient of a, since $\Gamma(x_0) = 0$; and the derivatives of the fixed matrix k are bounded. Thus the error E_i involves only the gradient bound.

We use twice the inequality $|b + c|^2 \leq (1 + t)|b|^2 + (1 + t^{-1})|c|^2$, $t > 0$. First take $t = \varepsilon/(1-2\varepsilon)$ to separate E_i; then $(1-2\varepsilon)(1+t) = 1-\varepsilon$ and $(1-2\varepsilon)(1+t^{-1}) \leq \varepsilon^{-1}$, so the contribution of E_i to $(1 - 2\varepsilon)N$ is at most $CT/(\varepsilon\lambda^2) \leq T$. Then take $t = \varepsilon$ for the two principal terms (with the factor $1 + \varepsilon$ on the second one). This gives

$$\begin{aligned}(1 - 2\varepsilon)N \leq (1 - \varepsilon)(1 + \varepsilon^{-1}) \sum_i \frac{2|\sum_q \overline{\nu}_q V_1 G_{i\bar{q}}|^2}{r_i \lambda^2} \\ + (1 - \varepsilon^2) \sum_i \frac{|\sum_{\alpha>1} \mu_\alpha \partial_i a_{V_1 V_\alpha}|^2}{r_i \lambda^2} + T.\end{aligned}$$

By the weighted Cauchy–Schwarz inequality and $\sum_q r_q |\nu_q|^2 \leq C$, the first line is at most $CR/(\varepsilon\lambda) \leq R$. For the second line the Cauchy–Schwarz inequality and $\sum_{\alpha>1} \mu_\alpha^2 = 1$ give

$$\left| \sum_{\alpha>1} \mu_\alpha \partial_i a_{V_1 V_\alpha} \right|^2 \leq \left(\lambda - \sum_{\alpha>1} \lambda_\alpha \mu_\alpha^2 \right) \sum_{\alpha>1} \frac{|\partial_i a_{V_1 V_\alpha}|^2}{\lambda - \lambda_\alpha}.$$

To bound the first factor we use the positivity of G. Since the Hessian eigenvalue of V_α is $\lambda_\alpha + 1$ (this is where the perturbation by B_0 enters) and $\partial\bar{\partial} a(W, \overline{W}) = \frac{1}{2}\big(a_{V_1 V_1} + a_{JV_1 JV_1}\big)$ at x_0, we have

$$0 < G(W, \overline{W}) = 1 + \tfrac{1}{2}\left(\lambda + \sum_{\alpha>1}(\lambda_\alpha + 1)\mu_\alpha^2\right).$$

Thus $\lambda - \sum_{\alpha>1}\lambda_\alpha\mu_\alpha^2 \le 2\lambda + 3$. If $\lambda \ge 3/(2\varepsilon^2)$, then $(1-\varepsilon^2)(2\lambda+3) \le 2\lambda$, so the second line is at most U. Finally, the critical point equation for $\widehat{Q}$ at x_0 reads

$$\frac{\partial_i a_{V_1 V_1}}{\lambda} = Aa_i - \eta' s_i.$$

Together with $|b+c|^2 \le 2|b|^2 + 2|c|^2$ and $4\varepsilon < 2$, it bounds the remaining part of N:

$$2\varepsilon N \le 4\varepsilon A^2 \sum_i \frac{|a_i|^2}{r_i} + 2(\eta')^2 \sum_i \frac{|s_i|^2}{r_i}.$$

Combining the three bounds, we obtain

$$N \le U + R + T + 4\varepsilon A^2 \sum_i \frac{|a_i|^2}{r_i} + 2(\eta')^2 \sum_i \frac{|s_i|^2}{r_i}. \tag{16.22}$$

Now choose $A = C_0 + 3$, after enlarging C_0 so that it is positive, and $\varepsilon = [4A^2(S+1)]^{-1} < 1/2$. Both choices are uniform, and so are the thresholds for λ used above. By (16.19), the η'' term of (16.20) absorbs the last term of (16.22), and $4\varepsilon A^2 \sum_i |a_i|^2/r_i \le sT/(S+1) \le T$. Substituting (16.22) into (16.20) and dropping its nonnegative β terms, we get

$$0 \ge \eta' K + T - An.$$

By the lower bound for η' in (16.19) and $r_i^{-1} \ge c$, this bounds every a_{ip} and $a_{i\bar{p}}$ at x_0, and hence every real second derivative of a there; at x_0 these coincide with the covariant ones. Thus $\lambda = \ell(x_0)$ is uniformly bounded, either by the thresholds assumed above or by this estimate. As observed at the beginning, the maximum property of Q then bounds ℓ everywhere, and the lower bound for the trace bounds the remaining eigenvalues of $\nabla^2 a$. This proves the lemma. □

Theorem 16.2.26 *Let κ be a Kähler form on X, let $r \ge 1$ be an integer, and let $f_1, \dots, f_r$ be real $C^{1,1}$ functions on X. Then the envelope*

$$P_\kappa\big(\min_{1\le j\le r} f_j\big)$$

is $C^{1,1}$. Its $C^{1,1}$ norm is bounded by a constant depending only on the fixed geometry and on a common bound for the $C^{1,1}$ norms of the obstacles.

Proof First let $r = 1$, and suppose that $f = f_1$ is smooth. By Lemma 16.2.25, the solutions a_β have uniformly bounded values, gradients and full real Hessians for $\beta \ge \beta_0$, and by Proposition 16.2.22 they converge uniformly to $P_\kappa(f)$. The limit $P_\kappa(f)$ is therefore continuous, and by the same proposition it has bounded metric

Laplacian; hence it belongs to $C^{1,\alpha}(X)$ by Proposition 16.2.11, and in particular it is C^1.

We now pass to the limit in the Hessian bounds. For a smooth vector field Y, the function $\nabla^2 a_\beta(Y,Y) = YYa_\beta - (\nabla_Y Y)a_\beta$ is obtained from a_β by a second-order differential operator. Since $a_\beta \to P_\kappa(f)$ uniformly, $\nabla^2 a_\beta(Y,Y) \to \nabla^2 P_\kappa(f)(Y,Y)$ in the sense of distributions, where the right side is the weak covariant Hessian, and for every nonnegative test function φ we obtain $|\langle \nabla^2 P_\kappa(f)(Y,Y), \varphi\rangle| \le C\int |Y|^2_{g_0}\varphi$. By Corollary 1.7.2(1), the weak quadratic components $\nabla^2 P_\kappa(f)(Y,Y)$ are bounded functions, with $|\nabla^2 P_\kappa(f)(Y,Y)| \le C|Y|^2_{g_0}$ almost everywhere; by polarization, so are all components of the weak covariant Hessian.

The gradient is treated in the same way: $Ya_\beta \to YP_\kappa(f)$ in the sense of distributions, and $|Ya_\beta| \le C|Y|_{g_0}$; since $YP_\kappa(f)$ is continuous, pairing with nonnegative test functions of integral one supported in small balls about a point shows that $|YP_\kappa(f)| \le C|Y|_{g_0}$ everywhere, so $|\nabla P_\kappa(f)|_{g_0} \le C$. In a chart, the weak second derivatives of $P_\kappa(f)$ differ from the components of its weak covariant Hessian by the Christoffel symbols times its first derivatives, so they are bounded on compact subsets of the chart. Hence $P_\kappa(f)$ is $C^{1,1}$, by Proposition 1.7.5, and the bounds just proved show that its $C^{1,1}$ norm is bounded by a constant depending only on the fixed geometry and on the constant B of Lemma 16.2.25 for f.

Now let the f_j be arbitrary $C^{1,1}$ functions. Using convolution in finitely many charts and a smooth partition of unity, we obtain smooth functions $f_{j,\rho}$ that converge to f_j in C^1 as $\rho \to 0$ and have uniformly bounded covariant Hessians. Indeed, the second derivatives of such a partitioned convolution are the cutoffs times the convolutions of the bounded weak second derivatives of f_j, plus terms that involve only the values and first derivatives of the local convolutions; all of these are uniformly bounded. For $\delta > 0$ put

$$F_{\rho,\delta} = -\delta \log \sum_j \mathrm{e}^{-f_{j,\rho}/\delta}, \qquad p_j = \frac{\mathrm{e}^{-f_{j,\rho}/\delta}}{\sum_k \mathrm{e}^{-f_{k,\rho}/\delta}}.$$

The weights p_j are positive with sum one, $\min_j f_{j,\rho} - \delta \log r \le F_{\rho,\delta} \le \min_j f_{j,\rho}$, and the gradient of $F_{\rho,\delta}$ is the convex combination $\sum_j p_j \nabla f_{j,\rho}$. For every real tangent vector Y the chain rule gives

$$\begin{aligned}\nabla^2 F_{\rho,\delta}(Y,Y) &= \sum_j p_j \nabla^2 f_{j,\rho}(Y,Y) \\ &\quad - \frac{1}{\delta}\left(\sum_j p_j (\mathrm{d}f_{j,\rho}(Y))^2 - \left(\sum_j p_j \mathrm{d}f_{j,\rho}(Y)\right)^2\right) \\ &\le C|Y|^2_{g_0},\end{aligned}$$

because the term in parentheses is a variance and hence nonnegative. Let $\rho, \delta \to 0$ with $\delta \log r \le 1$. Then the smooth obstacles $F_{\rho,\delta}$ have uniformly bounded values and gradients and a uniform upper bound for their Hessians, although their Hessians have no uniform lower bound as $\delta \to 0$; this is why Lemma 16.2.25 uses only an

upper bound. Since the smooth case bounds the $C^{1,1}$ norm of the envelope in terms of the fixed geometry and the constant B of that lemma, the envelopes $P_\kappa(F_{\rho,\delta})$ have uniformly bounded $C^{1,1}$ norms. The obstacles converge uniformly to $\min_j f_j$, so the envelopes converge uniformly to $P_\kappa(\min_j f_j)$ by the stability estimate of Proposition 16.1.4(4). The limit is C^1 by Theorem 16.2.24, since $C^{1,1}$ functions have bounded metric Laplacian. The same passage to the limit as in the smooth case, with the smooth functions a_β replaced by the $C^{1,1}$ functions $P_\kappa(F_{\rho,\delta})$, whose gradients and weak covariant Hessians are bounded by a uniform constant, now proves the theorem, together with the bound for the norm. □

16.2.5 Smoothing and its joint complex Hessian

We now turn to envelopes in a big class $\{\theta\}$. The background θ is no longer positive, so the penalized equations of Section 16.2.3 are not available. Instead we compare the envelope u with a smoothing $\Psi_u(x,r)$, an average of u over a neighborhood of x of size r, and the metric-Laplacian bound will follow from an estimate $\Psi_u(x,r) - u(x) = O(r^2)$. Convolution in coordinates does not define a global family on X, so we build the smoothing kernel from finite holomorphic jets of the exponential map of ω. Smoothing loses a small amount of positivity. This loss will be corrected by minimizing over the radius, as in Kiselman's minimum principle, and for that we need the complex Hessian of the smoothing jointly in the base point and the radius. This joint estimate is Lemma 16.2.27.

For the calculations in this and the next two subsections we put

$$\mathcal{D} = \mathrm{i}\partial\bar\partial = 2\pi\mathrm{dd^c}, \qquad \alpha = 2\pi\theta, \qquad \widehat{\omega} = 2\pi\omega.$$

Thus $\alpha+\mathcal{D}u \geq 0$ is precisely the condition that u is θ-psh, and multiplying forms by 2π only changes the constants in the estimates. Let h be the coefficient matrix of $\widehat{\omega}$ in local holomorphic coordinates, as in Section 16.2.3, and write $|v|_h$ for the corresponding norm of a tangent vector v. For a real $(1,1)$-form $T = \mathrm{i}\sum_{l,m} t_{l\bar m}\,\mathrm{d}z_l \wedge \mathrm{d}\bar z_m$ we write $T[y]^2 = \sum_{l,m} t_{l\bar m} y_l \bar y_m$ for the value of the associated Hermitian quadratic form on a vector y; thus $\widehat{\omega}[v]^2 = |v|_h^2$. For a real $(1,1)$-form on a product of an open subset of X with an open subset of $\mathbb{C}$, we write $T[v,\eta]^2$ for its value on the product tangent vector (v,η). If $P = (p_{l\bar m})$ and $M = (M_{lm})$ are Hermitian matrices, we write $P[M] = \sum_{l,m} p_{l\bar m} M_{lm}$ for their contraction, so that $P[y]^2 = P[yy^*]$.

Lemma 16.2.27 *There are $r_0 > 0$ and, for every integrable function u on X, a function $\Psi_u(x,w)$ of $x \in X$ and $w \in \mathbb{C}$ with $0 < |w| < r_0$, obtained by integrating u against a positive kernel of total mass one, such that Ψ_u is smooth, invariant under rotations of w and linear in u, and has the following properties. The kernel is built from the radial weight $\chi(t) = C_n(1-t)^{-2}\mathrm{e}^{-1/(1-t)}$ for $t < 1$, $\chi(t) = 0$ for $t \geq 1$, where the constant $C_n > 0$ is chosen so that $\int_{\mathbb{C}^n} \chi(|\zeta|^2)\,\mathrm{d}\zeta = 1$. We write $\Psi_u(x,r)$ for the value of Ψ_u at $|w| = r$. In the assertions about H_u and its joint Levi form we*

assume that $u \in \mathrm{PSH}(X,\theta)$; constants denoted by C may then depend on the fixed geometry, $\sup_X u$ and $\|u\|_{L^1}$, but not on the radius, and the constant A depends only on the fixed geometry.

For a sufficiently large constant K of the same kind as C, put

$$H_u(x,r) = \Psi_u(x,r) + Kr^2, \qquad \lambda_u(x,r) = r\partial_r H_u(x,r).$$

Then $H_u(x,\mathrm{e}^s)$ is a strictly convex nondecreasing function of s which tends to $u(x)$ as $s \to -\infty$; in particular $\lambda_u \geq 0$. If u is bounded near x, then $\lambda_u(x,r) \to 0$ as $r \to 0$.

At each point x we use holomorphic normal coordinates for ω centered at x (Lemma 16.2.13), defined on a ball of a fixed size (Step 1 of the proof). Let q be a smooth function there with $\mathcal{D}q = \alpha$, so that $p = u + q$ is plurisubharmonic, and let μ_p be the trace of the complex Hessian of p in these coordinates, a positive measure. Put $d = 2n$ and

$$m_u(x,r) = 1 + r^{-d}\mu_p(B(x,4r)).$$

All estimates below hold uniformly over these fixed normal charts. For a tangent vector v at x and a tangent vector $\eta \in \mathbb{C}$ in the radius variable, the joint Levi form of H_u in (x,w) satisfies

$$\begin{aligned}(\alpha + \mathcal{D}_{x,w}H_u)[v,\eta]^2 \geq\ & -\big(A\lambda_u + C(r^2 + r^3 m_u)\big)|v|_h^2 \\ & - C(r + r^2 m_u)|v|_h|\eta| - Cr^2|\eta|^2, \qquad r = |w|. \end{aligned} \tag{16.23}$$

Since $m_u \leq Cr^{-2}$, as we show in the proof, this gives

$$(\alpha + \mathcal{D}_{x,w}H_u)[v,\eta]^2 \geq -(A\lambda_u + Cr)|v|_h^2 - C|v|_h|\eta| - C|\eta|^2. \tag{16.24}$$

If $m_u(x,r) \leq M(x)/r$ with $M(x) \geq 1$, then the stronger estimate

$$(\alpha + \mathcal{D}_{x,w}H_u)[v,\eta]^2 \geq -(A\lambda_u + CM(x)r^2)|v|_h^2 - CM(x)r|v|_h|\eta| - CM(x)r|\eta|^2 \tag{16.25}$$

holds. Moreover, if u is bounded on the ball $B(x,2r)$ of the normal coordinates at x, then $\Psi_u(x,r)$ differs from the ordinary radial convolution $\int u(x + r\zeta)\chi(|\zeta|^2)\,\mathrm{d}\zeta$ in these coordinates by at most $Cr^2 \sup_{B(x,2r)} |u|$, where C depends only on the fixed geometry. For a C^1 function f with bounded weak complex Hessian,

$$|\Psi_f(x,r) - f(x)| \leq C_f r^2 \tag{16.26}$$

uniformly on X, where C_f is controlled by the bounds for the value, the gradient and the complex Hessian of f. Finally, for every integrable function u on X,

$$\frac{\Psi_u(\cdot,r) - u}{r^2} \longrightarrow a_\chi D_h u \quad \text{in the sense of distributions as } r \to 0, \tag{16.27}$$

where $D_h u = \sum_{i,j} h^{i\bar{j}} u_{i\bar{j}}$ is the trace of the complex Hessian, as in Section 16.2.3, and $a_\chi = \frac{1}{n}\int |\zeta|^2 \chi(|\zeta|^2)\,\mathrm{d}\zeta > 0$.

Proof We construct the smoothing in Step 1. In Steps 2–4 we compute, for a smooth plurisubharmonic local potential p, the joint complex Hessian of Ψ_p up to a remainder of high order in the radius; the result is the identity (16.37). From this identity we derive the joint estimates, first for smooth potentials (Step 5) and then in general (Step 6), and Step 7 proves the remaining assertions. All bounds are uniform over the centers, because the constructions are made in the charts of a fixed finite atlas, with coefficients bounded uniformly in the center.

Step 1: construction of the smoothing. We construct a smooth map $F(x,\zeta) \in X$, defined for small tangent vectors $\zeta \in T_x X$, from the holomorphic parts of the jets of geodesics, first in coordinates and then globally; Ψ_u is then an average of $u \circ F$.

We begin with uniform normal coordinates. Fix the jet order $N = 8$. For $x \in X$ we use the normal coordinates for ω at x of Lemma 16.2.13, applied in a chart of a fixed finite atlas that contains x in a fixed compact subset; in them $h(0) = I$ and $\mathrm{d}h(0) = 0$. The coordinate change there is a polynomial map of degree at most two. Its linear part L_x satisfies $L_x^{\mathrm{T}} h(x) \overline{L_x} = I$, where $h(x)$ is the coefficient matrix in the chart of the atlas, so L_x and L_x^{-1} are bounded uniformly in x. The condition $\mathrm{d}h(0) = 0$ is a linear equation for the quadratic part, whose coefficient matrix $h(x)\overline{L_x}$ is invertible and whose right side is built from L_x and the first derivatives of h at x; so the quadratic part is bounded uniformly in x as well. Hence these coordinates are defined on balls of a fixed size, and all finite jets of h in them are bounded uniformly in x.

Next we define the jet map, in arbitrary holomorphic coordinates. Let Γ_{jl}^m be given by $\sum_m \Gamma_{jl}^m h_{m\bar{k}} = \partial_j h_{l\bar{k}}$. Thus $\Gamma(z)$ is a bilinear map $\mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}^n$, symmetric because $\widehat{\omega}$ is closed, depending smoothly on z, and $\Gamma(0) = 0$ in normal coordinates. Consider the geodesic equation

$$\ddot{\gamma} + \Gamma(\gamma)(\dot{\gamma}, \dot{\gamma}) = 0$$

for formal power series $\gamma(t) = z + \sum_{k\geq 1} c_k t^k/k!$ in a real variable t, with $c_k \in \mathbb{C}^n$; here $\Gamma(\gamma(t))$ is the composition of the Taylor series of Γ at z with the series $\gamma(t) - z$, which has no constant term. Prescribe $c_1 = \zeta$. Comparing the coefficients of t^k in the equation determines c_{k+2} as a polynomial in $c_1, \dots, c_{k+1}$ whose coefficients are derivatives of Γ at z of order at most k. By induction, c_k is a polynomial in ζ and $\bar{\zeta}$ with coefficients depending smoothly on z, and it is homogeneous of degree k: for real λ the series $\gamma(\lambda t)$ is the formal solution with initial velocity $\lambda\zeta$, so that $c_k(\lambda\zeta) = \lambda^k c_k(\zeta)$. Let $c_k^{\mathrm{h}}(z,\zeta)$ be the sum of the monomials of c_k that contain no $\bar{\zeta}$, and define

$$J_z(\zeta) = z + \zeta + \sum_{k=2}^{N} \frac{c_k^{\mathrm{h}}(z,\zeta)}{k!},$$

a holomorphic polynomial in ζ with coefficients depending smoothly on z. Since $c_2 = -\Gamma(z)(\zeta,\zeta)$, in normal coordinates at x we have $J_0(\zeta) = \zeta + O(|\zeta|^3)$.

These jets do not depend on the coordinates. Let $y = \phi(z)$ be a holomorphic change of coordinates, with derivative $\Phi = D\phi$, and let $\widetilde{h}$, $\widetilde{\Gamma}$ and $\widetilde{J}$ be the corresponding objects in the coordinates y. Then $h_{j\bar{k}} = \sum_{a,b} \Phi_j^a \, \widetilde{h}_{a\bar{b}}(\phi) \, \overline{\Phi_k^b}$, and since Φ is holomorphic,

$$\partial_j h_{l\bar{k}} = \sum_{a,b} \Big(\partial_j \Phi^a_l + \sum_{c,e} \Phi^c_j \Phi^e_l \, \widetilde{\Gamma}^a_{ce}(\phi) \Big) \widetilde{h}_{a\bar{b}}(\phi) \, \overline{\Phi^b_k}.$$

Comparing with the definition of Γ, and using that the matrix $(\sum_b \widetilde{h}_{a\bar{b}} \overline{\Phi^b_k})_{a,k}$ is invertible, we obtain $\Phi\,\Gamma(\zeta,\zeta') = D^2\phi(\zeta,\zeta') + \widetilde{\Gamma}(\phi)(\Phi\zeta, \Phi\zeta')$ for all $\zeta, \zeta' \in \mathbb{C}^n$. Let γ be a formal solution in the coordinates z. The second derivative of $\phi \circ \gamma$ is $\Phi(\gamma)\ddot{\gamma} + D^2\phi(\gamma)(\dot{\gamma}, \dot{\gamma})$, because Φ is holomorphic, and therefore

$$\frac{\mathrm{d}^2(\phi\circ\gamma)}{\mathrm{d}t^2} + \widetilde{\Gamma}(\phi\circ\gamma)\Big(\frac{\mathrm{d}(\phi\circ\gamma)}{\mathrm{d}t}, \frac{\mathrm{d}(\phi\circ\gamma)}{\mathrm{d}t}\Big) = \Phi(\gamma)\big(\ddot{\gamma} + \Gamma(\gamma)(\dot{\gamma},\dot{\gamma})\big) = 0.$$

By uniqueness, $\phi \circ \gamma$ is the formal solution with initial point $\phi(z)$ and initial velocity $\Phi(z)\zeta$. By homogeneity we may put $t = 1$ degree by degree, and we obtain an identity between formal power series in ζ and $\bar{\zeta}$. Call a monomial in ζ and $\bar{\zeta}$ *discardable* if it contains a factor $\bar{\zeta}_m$ or has degree at least $N+1$; the product of a discardable monomial with any monomial is discardable. The two sides of this identity, the formal solution in the coordinates y and the series $\phi \circ \gamma$, differ from $\widetilde{J}_{\phi(z)}(\Phi(z)\zeta)$ and from $\phi(J_z(\zeta))$, respectively, by series of discardable monomials; for $\phi \circ \gamma$ we use the Taylor expansion of the holomorphic map ϕ at $J_z(\zeta)$, each of whose terms beyond the first contains the discardable difference as a factor. The two holomorphic functions of ζ just named therefore have the same Taylor coefficients of degree at most N at $\zeta = 0$, and Taylor's formula gives

$$\widetilde{J}_{\phi(z)}\big(\Phi(z)\zeta\big) = \phi\big(J_z(\zeta)\big) + O(|\zeta|^{N+1}), \tag{16.28}$$

where the remainder is a smooth function vanishing to order $N+1$ at $\zeta = 0$, with bounds for its derivatives that are locally uniform in z.

We now glue the local jet maps. Choose finitely many holomorphic charts $\kappa_\nu \colon \Omega_\nu \to \mathbb{C}^n$, $1 \le \nu \le \nu_0$, and open sets $V_\nu \Subset \Omega_\nu$ covering X. For $x \in \Omega_\nu$ and small $\zeta \in T_x X$ put $F_\nu(x,\zeta) = \kappa_\nu^{-1}\big(J^\nu_{\kappa_\nu(x)}(D\kappa_\nu(x)\zeta)\big)$, where J^ν is the jet map in the coordinates κ_ν. By (16.28), in every holomorphic chart, F_ν and F_μ differ by $O(|\zeta|^{N+1})$ where both are defined. We say that a smooth map $\widehat{F}(x,\zeta)$, defined for x in an open set O and $|\zeta|$ below a positive constant, has the *geodesic jets of order N* if in every chart κ_ν we have $\widehat{F} - F_\nu = O(|\zeta|^{N+1})$, with bounds that are uniform on compact subsets of $O \cap \Omega_\nu$.

We construct inductively such maps $\widehat{F}_k$ on open sets $O_k \supset \overline{V_1 \cup \cdots \cup V_k}$. Let $\widehat{F}_1 = F_1$ and $O_1 = \Omega_1$. Given $\widehat{F}_{k-1}$, choose an open set O' with $\overline{V_1 \cup \cdots \cup V_{k-1}} \subset O' \Subset O_{k-1}$, a function $\vartheta \in C^\infty_c(\Omega_k)$ with $0 \le \vartheta \le 1$ that equals one on a relatively compact open neighborhood $O'' \subset \Omega_k$ of the compact set $\overline{V_k} \setminus O'$ (Lemma 1.4.1(4)), and an open set Ω' with $\operatorname{Supp}\vartheta \subset \Omega' \Subset \Omega_k$. Put $O_k = O' \cup O''$, and define $\widehat{F}_k = \widehat{F}_{k-1}$ on $O' \setminus \operatorname{Supp}\vartheta$, $\widehat{F}_k = F_k$ on O'', and

$$\widehat{F}_k = \kappa_k^{-1}\big((1-\vartheta)\,\kappa_k \circ \widehat{F}_{k-1} + \vartheta\,\kappa_k \circ F_k\big) \qquad \text{on } O' \cap \Omega'.$$

For x in the compact set $\overline{O'} \cap \overline{\Omega'}$ and small ζ, the points $\widehat{F}_{k-1}(x,\zeta)$ and $F_k(x,\zeta)$ lie in a small ball about $\kappa_k(x)$ in the chart κ_k, so the last formula is defined. The three sets cover O_k, and the three definitions agree on the overlaps, because $\vartheta = 0$ outside $\operatorname{Supp}\vartheta$ and $\vartheta = 1$ on O''. In the chart κ_k the last formula gives $\widehat{F}_k - F_k = (1-\vartheta)(\widehat{F}_{k-1} - F_k) = O(|\zeta|^{N+1})$, so $\widehat{F}_k$ has the geodesic jets of order N.

The resulting map $F = \widehat{F}_{v_0}$ is defined on X for $|\zeta| < r_1$, with a fixed $r_1 > 0$. Since $F(x,0) = x$ and $D_\zeta F(x,0)$ is the identity, we may decrease r_1 so that, in each chart, $\|D_\zeta F - I\| \le \frac{1}{2}$ for $|\zeta| < r_1$; then each fiber map $\zeta \mapsto F(x,\zeta)$ is injective, because $|F(x,\zeta) - F(x,\zeta') - (\zeta - \zeta')| \le \frac{1}{2}|\zeta - \zeta'|$, and by the inverse function theorem it is a diffeomorphism onto its image. In the normal coordinates at a center x, (16.28) shows that $F(x',\zeta)$ differs from the jet map $J_{z'}(\zeta)$ of these coordinates by $O(|\zeta|^{N+1})$, where z' is the coordinate of x'. Since $J_0(\zeta) = \zeta + O(|\zeta|^3)$, we obtain at the center

$$F(x,\zeta) = x + \zeta + O(|\zeta|^3), \qquad D_\zeta F = I + O(|\zeta|^2). \tag{16.29}$$

Now we define the smoothing. Let χ be the weight of the statement, and let $\chi_1(t) = -C_n \mathrm{e}^{-1/(1-t)}$ for $t < 1$ and $\chi_1(t) = 0$ otherwise. Both functions are smooth on $\mathbb{R}$, $\chi_1' = \chi$, and $-\chi_1/\chi = (1-t)^2$ for $t < 1$. Define

$$\Psi_u(x,w) = \int_{T_xX} u(F(x,w\zeta))\chi(|\zeta|_h^2)\,\mathrm{d}\zeta_h,$$

where $\mathrm{d}\zeta_h$ is the Lebesgue measure of the Hermitian space (T_xX, h); in coordinates it is $\det h$ times the Lebesgue measure of the components of ζ, and in unitary coordinates of T_xX the kernel has total mass $\int \chi(|\zeta|^2)\,\mathrm{d}\zeta = 1$. The substitution $\zeta \mapsto \mathrm{e}^{\mathrm{i}t}\zeta$ shows that Ψ_u is invariant under rotations of w. For $w \ne 0$, the change of variables $y = F(x,w\zeta)$ in the fiber, whose inverse depends smoothly on (x,w,y) by the inverse function theorem, writes $\Psi_u(x,w)$, in a chart around x, as the integral of u against a smooth nonnegative kernel supported in a ball of radius $C|w|$ about x. Differentiation under the integral sign shows that Ψ_u is smooth for $w \ne 0$ even when u has poles.

Step 2: a holomorphic parametrization of the fiber. Throughout Steps 2–4 we fix a center, the origin of its normal coordinates z, and a smooth plurisubharmonic function p on a neighborhood of $\overline{B(0,3r)}$, where $r = |w|$ and $0 < |w| < r_0$; Step 6 removes the smoothness. In this step we rewrite Ψ_p so that p is composed with a map that is holomorphic in all variables. We write $F(z,\xi)$ for $F(x',\zeta)$ when x' has coordinate z and $\zeta = \sum_m \xi_m \partial/\partial z_m$. We use multi-indices I, J with $2 \le |I|, |J| \le N$ and write 1_l for the lth coordinate multi-index; a product such as $I_l s^{I-1_l}$ is zero when $I_l = 0$, and the same convention applies to conjugates and to J.

We first separate the holomorphic part of F. Let $a(z,\rho)$ be the polynomial formed by the holomorphic monomials of the Taylor expansion of F at $(0,0)$ that have degree at most two in z and at most N in ξ, with ξ renamed ρ. Since $F(z,0) = z$ and $D_\xi F(z,0)$ is the identity, $a(z,0) = z$ and $\partial_\rho a(z,0) = I$. Hence, for small z, the holomorphic map $a(z,\cdot)$ has a holomorphic inverse near 0, and we define

$$\rho(z,\xi) = a(z,\cdot)^{-1}\big(F(z,\xi)\big), \qquad \text{so that} \qquad F(z,\xi) = a\big(z,\rho(z,\xi)\big).$$

We extend the notion of a discardable monomial to the variables (z,ξ): a monomial in $z,\bar z,\xi,\bar\xi$ is discardable if it contains a factor $\bar z_j$ or $\bar\xi_m$, has degree at least three in z, or has degree at least $N+1$ in $(\xi,\bar\xi)$. Again the product of a discardable monomial with any monomial is discardable. By construction, the Taylor expansion of $F(z,\xi)-a(z,\xi)$ at $(0,0)$ consists of discardable monomials. Since $\rho(z,\xi)-\xi$ is the integral over $t\in[0,1]$ of the derivative of $a(z,\cdot)^{-1}$ at $a(z,\xi)+t(F(z,\xi)-a(z,\xi))$, applied to $F(z,\xi)-a(z,\xi)$, the same holds for $\rho(z,\xi)-\xi$. Moreover, for each fixed z, the Taylor polynomial of degree N of $F(z,\cdot)$ at 0 is holomorphic, because it is that of J_z; since $a(z,\cdot)^{-1}$ is holomorphic, the same holds for $\rho(z,\cdot)$. Together with $\rho(z,0)=0$ and $\partial_\xi\rho(z,0)=I$, Taylor's formula in ξ therefore gives

$$\rho(z,\xi) = \xi + \sum_I \rho_I(z)\xi^I + R(z,\xi), \tag{16.30}$$

with smooth vector-valued coefficients ρ_I and a smooth remainder R whose derivatives of order at most seven, with k of them in ξ or $\bar\xi$, are bounded by $C|\xi|^{N+1-k}$. The monomials ξ^I and $z_j\xi^I$ are not discardable, so $\rho_I(0)=0$ and $\partial_{z_j}\rho_I(0)=0$. We put

$$d_{Ik} = \partial_{\bar z_k}\rho_I(0), \qquad e_{Ijk} = \partial_{z_j}\partial_{\bar z_k}\rho_I(0),$$

vectors with components d_{Ikm} and e_{Ijkm}.

Put $\tilde\rho(z,w,\xi) = w^{-1}\rho(z,w\xi)$, so that

$$\tilde\rho(z,w,\xi) = \xi + \sum_I \rho_I(z)w^{|I|-1}\xi^I + \tilde R(z,w,\xi), \qquad \tilde R(z,w,\xi) = w^{-1}R(z,w\xi).$$

By the chain rule and the bounds for R, for $|\xi|\le 3$ every derivative of $\tilde R$ of order at most two in $(z,\bar z)$, at most three in $(\xi,\bar\xi)$ and $k\le 2$ in $(w,\bar w)$ is bounded by Cr^{N-k}. Since also $\rho_I(z)=O(|z|)$, we have $\|\partial_\xi\tilde\rho - I\| + \|\partial_{\bar\xi}\tilde\rho\| \le \frac14$ for $|\xi|\le 3$ when $|z|$ and r are small. As above, $\xi\mapsto\tilde\rho(z,w,\xi)$ is then a diffeomorphism of $\{|\xi|<3\}$ onto an open set, which contains $\{|s|\le 2\}$ by the contraction mapping principle, applied to $\xi\mapsto s+\xi-\tilde\rho(z,w,\xi)$ on $\{|\xi|\le 8/3\}$ (note that $\tilde\rho(z,w,0)=0$); we denote its inverse by $s\mapsto\xi(z,w,s)$. Since $a(z,w\tilde\rho(z,w,\xi)) = F(z,w\xi)$, the substitution $s=\tilde\rho(z,w,\xi)$ gives

$$\Psi_u(z,w) = \int u\big(a(z,ws)\big)W(z,w,s)\,\mathrm{d}s, \qquad W = \chi(Q)\mathcal{J},$$

where $Q = |\xi(z,w,s)|^2_{h(z)}$, $\mathcal{J}$ is $\det h(z)$ times the real Jacobian determinant of $s\mapsto\xi(z,w,s)$, and W is extended by zero outside the image. On the support of $\chi(|\xi|_h^2)$ we have $|\xi|<2$ and hence $|s|<2$ for small $|z|$ and r, so W is smooth and vanishes for $|s|\ge 2$. Thus the argument $a(z,ws)$ of u is holomorphic in (z,w,s), and the weight W carries the nonholomorphic dependence on the base point.

Step 3: the weight at the center. We compute the first and the mixed second derivatives of W in z at $z=0$, up to errors of order r^N, and write them as divergences

in s; the result is (16.32). For functions of (w,s) with $0<|w|<r_0$ and $|s|\le 2$, we write O_N for any function whose derivatives of order at most two in $(s,\bar s)$ and $k\le 2$ in $(w,\bar w)$ are bounded by Cr^{N-k}, where $r=|w|$. Put

$$\beta^k(w,s)=\sum_I d_{Ik}w^{|I|-1}s^I,\qquad \gamma^{jk}(w,s)=\sum_I e_{Ijk}w^{|I|-1}s^I,$$

vectors that are holomorphic in s; for a holomorphic vector field $b(s)$ write $\operatorname{div} b=\sum_m \partial_{s_m}b_m$, $b\cdot\bar s=\sum_m b_m\bar s_m$, and $b\cdot\bar b'=\sum_m b_m\overline{b'_m}$.

We begin with the derivatives of ξ. We differentiate the identity $\tilde\rho(z,w,\xi(z,w,s))=s$ in $\bar z_k$, in z_j, and then in z_j and $\bar z_k$, and evaluate at $z=0$, where $\rho_I=0$ and $\partial_{z_j}\rho_I=0$. In each case the derivative of ξ to be determined appears multiplied by $\partial_\xi\tilde\rho=I+\partial_\xi\tilde R$, and the remaining terms are the explicit terms $\partial_{\bar z_k}\rho_I w^{|I|-1}\xi^I$ and $\partial_{z_j}\partial_{\bar z_k}\rho_I w^{|I|-1}\xi^I$, terms containing a derivative of $\tilde R$, and terms containing the factor $\partial_{z_j}\xi$. The conjugates of these derivatives of ξ also occur, multiplied by $\partial_{\bar\xi}\tilde R$; since $\partial_\xi\tilde R$ and $\partial_{\bar\xi}\tilde R$ are bounded by Cr^N, each of these linear systems is a small perturbation of the identity, and its solutions are bounded uniformly. Using this, the bounds for $\tilde R$, and the fact that β^k and γ^{jk} have bounded derivatives, we find successively that at $z=0$

$$\xi=s+O_N,\qquad \partial_{z_j}\xi=O_N,\qquad \partial_{\bar z_k}\xi=-\beta^k+O_N,\qquad \partial_{z_j}\partial_{\bar z_k}\xi=-\gamma^{jk}+O_N; \tag{16.31}$$

the bounds for one further derivative in s follow in the same way, because $\tilde R$ has bounded derivatives of order three in ξ. Differentiating the identity in $\bar s$ instead, we find $\partial_{\bar s}\xi=-(\partial_\xi\tilde\rho)^{-1}\partial_{\bar\xi}\tilde R\,\overline{\partial_s\xi}$, because the coefficients of (16.30) do not involve $\bar\xi$; hence $\partial_{\bar s}\xi$ and its derivatives ∂_{z_j}, $\partial_{\bar z_k}$ and $\partial_{z_j}\partial_{\bar z_k}$ are O_N at $z=0$.

Next, $Q=\sum_{l,m}h_{l\bar m}(z)\xi_l\bar\xi_m$, where $h(0)=I$ and $\mathrm dh(0)=0$; we put $c_{jklm}=-\partial_{z_j}\partial_{\bar z_k}h_{l\bar m}(0)$. By the product rule, (16.31) and $\partial_{z_j}\bar\xi=\overline{\partial_{\bar z_j}\xi}$, at $z=0$

$$\begin{aligned}Q&=|s|^2+O_N,\qquad \partial_{\bar z_k}Q=-\beta^k\cdot\bar s+O_N,\\ \partial_{z_j}\partial_{\bar z_k}Q&=-\sum_{l,m}c_{jklm}s_l\bar s_m-\gamma^{jk}\cdot\bar s-\overline{\gamma^{kj}\cdot\bar s}+\beta^k\cdot\overline{\beta^j}+O_N,\end{aligned}$$

and $\partial_{z_j}Q=\overline{\partial_{\bar z_j}Q}$, since Q is real.

For the Jacobian factor $\mathcal J$, let $M=\partial_s\xi$. The real Jacobian determinant of $s\mapsto\xi$ is the determinant of the matrix $\left(\begin{smallmatrix}M & \partial_{\bar s}\xi\\ \overline{\partial_{\bar s}\xi} & \overline M\end{smallmatrix}\right)$, that is, $|\det M|^2$ plus a polynomial in the entries of these blocks each of whose terms contains an entry of $\partial_{\bar s}\xi$ or of its conjugate; as shown above, these entries and their relevant derivatives are O_N at $z=0$. Differentiating (16.31) in s, we have at $z=0$: $M=I+O_N$, $\partial_{z_j}M=O_N$, $\partial_{\bar z_k}M=-\partial_s\beta^k+O_N$ and $\partial_{z_j}\partial_{\bar z_k}M=-\partial_s\gamma^{jk}+O_N$. Jacobi's formula $\partial\det M=\det M\operatorname{tr}(M^{-1}\partial M)$, differentiated once more, gives $\partial_{\bar z_k}\det M=-\operatorname{div}\beta^k+O_N$, $\partial_{z_j}\det M=O_N$ and $\partial_{z_j}\partial_{\bar z_k}\det M=-\operatorname{div}\gamma^{jk}+O_N$. Writing $|\det M|^2=\det M\cdot\overline{\det M}$ and using $\det h(0)=1$, $\mathrm d\det h(0)=0$ and $\partial_{z_j}\partial_{\bar z_k}\det h(0)=-\sum_l c_{jkll}$, we obtain at $z=0$

$$\mathcal{J} = 1 + O_N, \qquad \partial_{\bar z_k}\mathcal{J} = -\operatorname{div}\beta^k + O_N,$$
$$\partial_{z_j}\partial_{\bar z_k}\mathcal{J} = -\sum_l c_{jkll} - \operatorname{div}\gamma^{jk} - \overline{\operatorname{div}\gamma^{kj}} + \operatorname{div}\beta^k\ \overline{\operatorname{div}\beta^j} + O_N.$$

Finally we differentiate $W = \chi(Q)\mathcal{J}$. By the chain rule, $\partial_{\bar z_k}W = \chi'(Q)\mathcal{J}\,\partial_{\bar z_k}Q + \chi(Q)\partial_{\bar z_k}\mathcal{J}$, and $\partial_{z_j}\partial_{\bar z_k}W$ is the sum of $\chi''\mathcal{J}\partial_{z_j}Q\,\partial_{\bar z_k}Q$, $\chi'\big(\mathcal{J}\partial_{z_j}\partial_{\bar z_k}Q + \partial_{z_j}Q\,\partial_{\bar z_k}\mathcal{J} + \partial_{\bar z_k}Q\,\partial_{z_j}\mathcal{J}\big)$ and $\chi\,\partial_{z_j}\partial_{\bar z_k}\mathcal{J}$, with χ and its derivatives evaluated at Q. Since $Q = |s|^2 + O_N$ at $z = 0$, we may evaluate them at $|s|^2$ instead, at the cost of an error O_N. From now on χ, χ', χ'' denote these functions evaluated at $|s|^2$, and we recall that $\partial_{s_m}|s|^2 = \bar s_m$. Inserting the formulas above, we find at $z = 0$ that $\partial_{\bar z_k}W = -\chi'\beta^k\cdot\bar s - \chi\operatorname{div}\beta^k + O_N$. In the mixed derivatives we group the terms. The terms containing c give $-\sum_{l,m} c_{jklm}(\chi' s_l\bar s_m + \chi\delta_{lm})$. The terms containing e give $-\chi'\gamma^{jk}\cdot\bar s - \chi\operatorname{div}\gamma^{jk}$ and its counterpart with $\overline{\gamma^{kj}}$. The terms quadratic in d are

$$\chi''\big(\overline{\beta^j}\cdot s\big)\big(\beta^k\cdot\bar s\big) + \chi'\beta^k\cdot\overline{\beta^j} + \chi'\big(\beta^k\cdot\bar s\big)\overline{\operatorname{div}\beta^j} + \chi'\big(\overline{\beta^j}\cdot s\big)\operatorname{div}\beta^k + \chi\operatorname{div}\beta^k\ \overline{\operatorname{div}\beta^j}.$$

Since β^k is holomorphic and $\overline{\beta^j}$ antiholomorphic in s, we have $\sum_l \partial_{\bar s_l}(\chi\beta^k_m\overline{\beta^j_l}) = \beta^k_m\big(\chi'\,\overline{\beta^j}\cdot s + \chi\overline{\operatorname{div}\beta^j}\big)$, and applying $\sum_m\partial_{s_m}$ to this produces exactly the five terms above. The first two groups are divergences as well: $\partial_{s_m}(\chi s_l) = \chi' s_l\bar s_m + \chi\delta_{lm}$, and, since γ^{jk} is holomorphic and $\overline{\gamma^{kj}}$ antiholomorphic in s, $\sum_m\partial_{s_m}(\chi\gamma^{jk}_m) = \chi'\gamma^{jk}\cdot\bar s + \chi\operatorname{div}\gamma^{jk}$ and $\sum_m\partial_{\bar s_m}(\chi\overline{\gamma^{kj}_m}) = \chi'\,\overline{\gamma^{kj}}\cdot s + \chi\overline{\operatorname{div}\gamma^{kj}}$. We have proved that at $z = 0$

$$W = \chi + O_N, \qquad \partial_{\bar z_k}W = -\sum_m \partial_{s_m}\big(\chi\beta^k_m\big) + O_N,$$
$$\begin{aligned}\partial_{z_j}\partial_{\bar z_k}W = &-\sum_{l,m} c_{jklm}\partial_{s_m}(\chi s_l) - \sum_m \partial_{s_m}\big(\chi\gamma^{jk}_m\big) - \sum_m \partial_{\bar s_m}\big(\chi\overline{\gamma^{kj}_m}\big)\\ &+\sum_{l,m}\partial_{s_m}\partial_{\bar s_l}\big(\chi\beta^k_m\overline{\beta^j_l}\big) + O_N,\end{aligned} \tag{16.32}$$

and $\partial_{z_j}W = \overline{\partial_{\bar z_j}W}$, since W is real. Here W and the explicit terms vanish for $|s| \ge 2$, so the remainders do as well.

Step 4: an identity for the joint Levi form. We compute $\mathcal{D}\Psi_p[v,\eta]^2$ at $z = 0$ by differentiating under the integral sign, inserting (16.32) and integrating by parts in s; the result is the identity (16.37). Put $G(z,w,s) = p(a(z,ws))$ and $\mathcal{L} = \sum_j v_j\partial_{z_j} + \eta\,\partial_w$, so that $\mathcal{D}f[v,\eta]^2 = \mathcal{L}\bar{\mathcal{L}}f$ for every smooth function f of (z,w). At $z = 0$ write $D_\rho a = \partial_\rho a(0,ws)$, let $\partial_{\rho_m}a$ denote its mth column, and put

$$Y = Da(v,\eta s) = \partial_z a(0,ws)v + \eta\,D_\rho a\,s;$$

these are holomorphic in s. Let $P = (p_{l\bar m}) \ge 0$ be the complex Hessian of p at $a(0,ws)$, and write $P(y,\bar y') = \sum_{l,m} p_{l\bar m}y_l\bar y'_m$, so that $P[y]^2 = P(y,\bar y)$. Since $a(z,ws)$ is holomorphic in (z,w,s), we have $\mathcal{L}G = \sum_l p_l(a)\,(\mathcal{L}a)_l$, where $p_l = \partial p/\partial z_l$, the

vector $\mathcal{L}a$ is holomorphic and equals Y at $z=0$, and $\partial_{s_m}[a(0,ws)] = w\,\partial_{\rho_m}a$. Hence, at $z=0$,

$$\begin{aligned} \mathcal{L}\bar{\mathcal{L}}G = P[Y]^2, \qquad \partial_{\bar{s}_l}(\mathcal{L}G) = \bar{w}\,P\big(Y, \overline{\partial_{\rho_l}a}\big), \\ \partial_{s_m}\partial_{\bar{s}_l}G = r^2 P\big(\partial_{\rho_m}a, \overline{\partial_{\rho_l}a}\big). \end{aligned} \tag{16.33}$$

Next we insert the expansion of the weight. The integrand GW is smooth with support in $\{|s|\le 2\}$, so we may differentiate under the integral sign. Since G and W are real, $\bar{\mathcal{L}}G\,\mathcal{L}W$ is the conjugate of $\mathcal{L}G\,\bar{\mathcal{L}}W$, and at $z=0$

$$\mathcal{D}\Psi_p[v,\eta]^2 = \int \Big(W\,\mathcal{L}\bar{\mathcal{L}}G + 2\,\mathrm{Re}\big(\mathcal{L}G\,\bar{\mathcal{L}}W\big) + G\,\mathcal{L}\bar{\mathcal{L}}W \Big)\mathrm{d}s.$$

Here, at $z=0$, $\bar{\mathcal{L}}W = \sum_k \bar{v}_k\partial_{\bar{z}_k}W + \bar{\eta}\partial_{\bar{w}}W$ and

$$\mathcal{L}\bar{\mathcal{L}}W = \sum_{j,k} v_j\bar{v}_k\partial_{z_j}\partial_{\bar{z}_k}W + \bar{\eta}\sum_j v_j\partial_{\bar{w}}\partial_{z_j}W + \eta\sum_k \bar{v}_k\partial_w\partial_{\bar{z}_k}W + |\eta|^2\partial_w\partial_{\bar{w}}W,$$

where the derivatives in w may be taken after putting $z=0$. We insert (16.32). The explicit terms there do not depend on w except through β^k and γ^{jk}, and $\partial_w\beta^k = \sum_I (|I|-1)d_{Ik}w^{|I|-2}s^I$; each derivative of a remainder in w costs one power of r. Thus at $z=0$

$$W = \chi + \delta_0, \qquad \bar{\mathcal{L}}W = -\sum_{k,m}\bar{v}_k\partial_{s_m}\big(\chi\beta^k_m\big) + \delta_1,$$

$$\mathcal{L}\bar{\mathcal{L}}W = \sum_{j,k} v_j\bar{v}_k\,[\partial_{z_j}\partial_{\bar{z}_k}W]^{\flat} - \eta\sum_{k,m}\bar{v}_k\partial_{s_m}\big(\chi\partial_w\beta^k_m\big) - \bar{\eta}\sum_{j,m}v_j\partial_{\bar{s}_m}\big(\chi\overline{\partial_w\beta^j_m}\big) + \delta_2,$$

where $[\partial_{z_j}\partial_{\bar{z}_k}W]^{\flat}$ denotes the explicit part of (16.32), and $\delta_0,\delta_1,\delta_2$ vanish for $|s|\ge 2$ and satisfy

$$|\delta_0| \le Cr^N, \qquad |\delta_1| + |\nabla_s\delta_1| \le Cr^{N-1}(|v|+|\eta|), \qquad |\delta_2| \le Cr^{N-2}(|v|+|\eta|)^2. \tag{16.34}$$

Accordingly $\mathcal{D}\Psi_p[v,\eta]^2 = T_1+T_2+T_3+E$, where $T_1 = \int P[Y]^2\chi\,\mathrm{d}s$, T_2 and T_3 are the contributions of the explicit parts of $\bar{\mathcal{L}}W$ and $\mathcal{L}\bar{\mathcal{L}}W$, and

$$E = \int \Big(P[Y]^2\delta_0 + 2\,\mathrm{Re}\big(\mathcal{L}G\,\delta_1\big) + G\,\delta_2 \Big)\mathrm{d}s. \tag{16.35}$$

We evaluate T_2 and T_3 by integrating by parts in s, using the following divergence formulas: for a multi-index I with $|I|\ge 1$,

$$\begin{aligned} \chi(|s|^2)\bar{s}^I &= \sum_m \partial_{s_m}\Big(\chi_1(|s|^2)\frac{I_m}{|I|}\bar{s}^{I-1_m}\Big), \\ \chi(|s|^2)s^I &= \sum_m \partial_{\bar{s}_m}\Big(\chi_1(|s|^2)\frac{I_m}{|I|}s^{I-1_m}\Big). \end{aligned} \tag{16.36}$$

Indeed, $\partial_{s_m}\chi_1(|s|^2) = \chi(|s|^2)\bar s_m$ because $\chi_1' = \chi$, the monomial $\bar s^{I-1_m}$ is antiholomorphic, and Euler's identity $\sum_m I_m \bar s_m \bar s^{I-1_m} = |I|\bar s^I$ gives the first formula; the second is its complex conjugate. Similarly $\chi(|s|^2)s_l = \partial_{\bar s_l}\chi_1(|s|^2)$. The integrations by parts are justified as follows. If f and g are continuously differentiable on a neighborhood of the support of g, and this support is compact, then $\int f\,\partial_{s_m} g\,\mathrm{d}s = -\int g\,\partial_{s_m} f\,\mathrm{d}s$, and similarly for $\partial_{\bar s_m}$. Since $\partial_{s_m} = \frac12(\partial_{\operatorname{Re} s_m} - \mathrm{i}\partial_{\operatorname{Im} s_m})$, this follows from Fubini's theorem and the fundamental theorem of calculus, applied to the compactly supported function fg. In all integrations by parts below, one of the two factors is a product of $\chi(|s|^2)$ or $\chi_1(|s|^2)$ with a polynomial, or one of the remainders in (16.34), hence smooth with support in $\{|s| \le 2\}$, so no boundary terms occur.

We now compute T_2. By (16.36),

$$\chi\beta_m^k = \sum_l \partial_{\bar s_l}(\chi_1\sigma_{ml}^k), \qquad \sigma_{ml}^k = \sum_I d_{Ikm} w^{|I|-1}\frac{I_l}{|I|}s^{I-1_l},$$

where σ_{ml}^k is holomorphic in s. Integrating by parts in $\bar s_l$ and using (16.33) and $\partial_{s_m}\chi_1(|s|^2) = \chi\bar s_m$, we get

$$\begin{aligned}-\int \mathcal{L}G\,\partial_{s_m}(\chi\beta_m^k)\,\mathrm{d}s &= \sum_l \int \partial_{\bar s_l}(\mathcal{L}G)\,\partial_{s_m}(\chi_1\sigma_{ml}^k)\,\mathrm{d}s\\ &= \bar w\sum_l \int P\big(Y, \overline{\partial_{\rho_l} a}\big)\big(\chi\bar s_m\sigma_{ml}^k + \chi_1\partial_{s_m}\sigma_{ml}^k\big)\,\mathrm{d}s.\end{aligned}$$

We multiply by $\bar v_k$ and sum over k and m. To write the result, we define vectors Ξ and Θ by

$$\begin{aligned}\Xi^m &= \sum_{I,j,l}\frac{I_m}{|I|}\overline{d}_{Ijl}\bar w^{|I|-2}s_l\bar s^{I-1_m}v_j,\\ \Theta^m &= \sum_{I,j,l}\frac{I_m}{|I|}(I-1_m)_l\,\overline{d}_{Ijl}\bar w^{|I|-2}\bar s^{I-1_m-1_l}v_j,\end{aligned}$$

where the factor $(I-1_m)_l\bar s^{I-1_m-1_l}$ is zero when $(I-1_m)_l = 0$. Then $\bar w\sum_{k,m}\bar v_k\bar s_m\sigma_{ml}^k = r^2\,\overline{\Xi^l}$ and $\bar w\sum_{k,m}\bar v_k\partial_{s_m}\sigma_{ml}^k = r^2\,\overline{\Theta^l}$, and P is antilinear in its second argument. Therefore

$$T_2 = 2r^2\operatorname{Re}\int P\big(Y, \overline{D_\rho a\,\Xi}\big)\chi\,\mathrm{d}s + 2r^2\operatorname{Re}\int P\big(Y, \overline{D_\rho a\,\Theta}\big)\chi_1\,\mathrm{d}s.$$

Next we compute T_3. By (16.36), each explicit term of $\mathcal{L}\bar{\mathcal{L}}W$ is a sum of second derivatives $\partial_{s_m}\partial_{\bar s_l}$ of weights, and two integrations by parts move these derivatives onto G. For the curvature term we use $\partial_{s_m}(\chi s_l) = \partial_{s_m}\partial_{\bar s_l}\chi_1$ and (16.33), and we let C^v be the matrix with entries $C_{ml}^v = \sum_{j,k} c_{jklm}v_j\bar v_k$:

$$-\sum_{j,k,l,m} v_j\bar v_k c_{jklm}\int G\,\partial_{s_m}(\chi s_l)\,\mathrm{d}s = -r^2\sum_{l,m}\int P\big(\partial_{\rho_m}a, \overline{\partial_{\rho_l}a}\big)\chi_1 C_{ml}^v\,\mathrm{d}s.$$

Since h is Hermitian, $\overline{c_{jklm}} = c_{kjml}$; hence C^v is Hermitian, and $\|C^v\| \le C|v|^2$ for its operator norm. For the terms containing γ^{jk} and $\partial_w \beta^k$, (16.36) writes $\chi\gamma^{jk}_m = \sum_l \partial_{\bar s_l}(\chi_1 \tau^{jk}_{ml})$ and $\chi \partial_w \beta^k_m = \sum_l \partial_{\bar s_l}(\chi_1 \partial_w \sigma^k_{ml})$, with σ^k_{ml} as above and

$$\tau^{jk}_{ml} = \sum_I e_{Ijkm} w^{|I|-1} \frac{I_l}{|I|} s^{I-1_l}, \qquad \partial_w \sigma^k_{ml} = \sum_I d_{Ikm}(|I|-1) w^{|I|-2} \frac{I_l}{|I|} s^{I-1_l}.$$

Two integrations by parts give the contribution

$$-r^2 \sum_{l,m} \int P\big(\partial_{\rho_m} a, \overline{\partial_{\rho_l} a}\big) \chi_1 B_{ml} \, \mathrm{d}s, \qquad B_{ml} = \sum_{j,k} v_j \bar v_k \tau^{jk}_{ml} + \eta \sum_k \bar v_k \partial_w \sigma^k_{ml}.$$

The terms containing $\overline{\gamma^{kj}}$ and $\overline{\partial_w \beta^j}$ are treated with the conjugate formula in (16.36), and after renaming $j \leftrightarrow k$ they give the same expression with the matrix B replaced by its conjugate transpose B^*. Finally, with $\beta = \sum_k \bar v_k \beta^k$, two integrations by parts turn the last term of (16.32) into

$$\sum_{j,k} v_j \bar v_k \sum_{l,m} \int G \, \partial_{s_m} \partial_{\bar s_l} \big(\chi \beta^k_m \overline{\beta^j_l}\big) \mathrm{d}s = r^2 \int P\big[D_\rho a \, \beta\big]^2 \chi \, \mathrm{d}s.$$

For a matrix M in the fiber variables write

$$P[Da(M)] = \sum_{l,m} M_{ml} P\big(\partial_{\rho_m} a, \overline{\partial_{\rho_l} a}\big),$$

so that $P[Da(yy'^*)] = P(D_\rho a \, y, \overline{D_\rho a \, y'})$ for vectors y and y'. Since χ and χ_1 vanish outside the unit ball, on which $\chi > 0$, we have shown that $T_3 = r^2 \int P[Da(U)] \chi \, \mathrm{d}s$, where, on the unit ball, $U = -\frac{\chi_1}{\chi}(C^v + B + B^*) + \beta\beta^*$, with the quotient χ_1/χ evaluated at $|s|^2$. Explicitly, U is the Hermitian part of the matrix $U' = -\frac{\chi_1}{\chi}(C^v + 2B) + \beta\beta^*$, whose entries are

$$\begin{aligned} U'_{ml} = \; & -\frac{\chi_1}{\chi}\Bigg(\sum_{j,k} c_{jklm} v_j \bar v_k + 2 \sum_{I,j,k} e_{Ijkm} w^{|I|-1} \frac{I_l}{|I|} s^{I-1_l} v_j \bar v_k \\ & \qquad + 2 \sum_{I,k} d_{Ikm}(|I|-1) w^{|I|-2} \frac{I_l}{|I|} s^{I-1_l} \eta \bar v_k \Bigg) \\ & + \sum_{I,J,j,k} d_{Ikm} \overline{d}_{Jjl} w^{|I|-1} \bar w^{|J|-1} s^I \bar s^J v_j \bar v_k. \end{aligned}$$

Finally we add T_1, T_2 and T_3. Completing the square, $P[Y]^2 + 2r^2 \operatorname{Re} P(Y, \overline{D_\rho a \, \Xi}) = P[Y + r^2 D_\rho a \, \Xi]^2 - r^4 P[D_\rho a \, \Xi]^2$, where $Y + r^2 D_\rho a \, \Xi = Da(v, \eta s + r^2 \Xi)$. Moreover, the real part of $P(Y, \overline{D_\rho a \, \Theta})$ equals that of $P(D_\rho a \, \Theta, \overline{Y})$, because P is Hermitian. We have therefore proved the identity

$$\mathcal{D}\Psi_p[v,\eta]^2 = \int P\big[Da(v,\eta s + r^2\Xi)\big]^2 \chi \,\mathrm{d}s + r^2 \int P\big[Da(U - r^2\Xi\Xi^*)\big]\chi \,\mathrm{d}s + 2r^2 \operatorname{Re} \int P\big(Da(0,\Theta), \overline{Da(v,\eta s)}\big)\chi_1 \,\mathrm{d}s + E. \tag{16.37}$$

Here Da is the differential of a at $(0, ws)$, $P[Da(\cdot)]$ is the contraction defined above, and the remainder E is given by (16.35).

Step 5: the joint estimate for smooth potentials. In this step we assume that $p = u + q$ is smooth near $\overline{B(0,3r)}$. We derive (16.23) from (16.37) and prove that H_u is a strictly convex function of $\log r$. We choose q as a fixed local potential of α on one of finitely many coordinate balls covering X (Lemma 7.3.1); by a Lebesgue number argument, the normal ball of a fixed radius about each center lies in one of them. This choice is harmless, because μ_p, and hence m_u, is the trace measure of $\alpha + \mathcal{D}u$ and does not depend on q.

We also use two elementary facts. For a nonnegative Hermitian matrix P and any matrix M we have $|\sum_{l,m} p_{l\bar{m}} M_{lm}| \le \|M\| \operatorname{tr} P$, by diagonalizing P; in particular $|P(y,\bar{y}')| \le |y||y'| \operatorname{tr} P$. Moreover, since $a(0,\rho)$ is the Taylor polynomial of degree N of $F(0,\cdot)$, which is that of J_0, we have $D_\rho a = I + O(r^2)$ and $\partial_z a(0,ws) = I + O(r)$, and $a(0,ws) = ws + O(r^3)$, for $|s| \le 2$. Hence the map $s \mapsto a(0,ws)$ is injective on $\{|s| \le 2\}$, has real Jacobian determinant at least $r^d/2$, and takes values in $B(0,3r)$.

We first bound the masses of p on small balls about the center; this part does not use the smoothness of p. By the Jensen–Lelong formula (Theorem 12.1.3), the normalized trace mass $D_p(x,t)$ of (12.3) is nondecreasing in t, and $\mu_p(B(x,t))$ is a fixed multiple of $t^{d-2} D_p(x,t)$. On a fixed larger ball the trace mass of p is bounded by $C\|p\|_{L^1}$ of a still larger ball: pair $\Delta p \ge 0$ with a cutoff function. Hence

$$\mu_p(B(x,t)) \le C t^{d-2}, \qquad t^{-d} \int_{B(x,t)} |u| \le C(1 + |\log t|).$$

For the second estimate, (12.4) and the bound for D_p give $M_p(x,\varrho) \ge M_p(x,R) - 2C\log(R/\varrho)$ for $\varrho < R$, where M_p denotes spherical means. Averaging over R in a fixed interval bounds the right side below by $-C(1 + |\log \varrho|)$, since the average of $|p|$ over a fixed annulus is bounded in terms of $\|u\|_{L^1}$. As $|p| = 2p^+ - p$ and p^+ is bounded in terms of $\sup_X u$, integration over spheres gives $\int_{B(x,t)} |p| \le C t^d (1 + |\log t|)$, and $|u| \le |p| + |q|$. In particular $m_u \le C r^{-2}$.

To bound E, note that $|Y| \le C(|v| + |\eta|)$ and $P[Y]^2 \le |Y|^2 \operatorname{tr} P$ for $|s| \le 2$. Moreover $\mathcal{L}G = \sum_m b_m \partial_{s_m} G$ with $b = w^{-1}(D_\rho a)^{-1} Y$, by (16.33) and its proof, and $|b| + |\nabla_s b| \le C r^{-1}(|v| + |\eta|)$; an integration by parts gives $\int \mathcal{L}G\, \delta_1 \,\mathrm{d}s = -\int G \sum_m \partial_{s_m}(b_m \delta_1)\,\mathrm{d}s$. By (16.34),

$$|E| \le C r^{N-2}(|v| + |\eta|)^2 \int_{|s|\le 2} \big(r^2 \operatorname{tr} P + |G|\big)\mathrm{d}s.$$

Changing variables by the map $s \mapsto a(0,ws)$ and using the two mass bounds, we obtain $|E| \le C r^{N-2}(1 + |\log r|)(|v|^2 + |\eta|^2) \le C r^2 (|v|^2 + |\eta|^2)$, since $N = 8$.

Now we bound the right side of (16.37) from below. Its first integral is nonnegative. The formulas for the coefficients give $|\Xi| \le C|v|$ and $|\Theta| \le (C_d + Cr)|v|$, where C_d is the norm of the linear map $v \mapsto \Theta$ formed by the terms with $|I| = 2$, which do not depend on s and w; the terms with $|I| \ge 3$ carry a factor $\bar w$. Every term of U except the last carries the factor $-\chi_1/\chi = (1 - |s|^2)^2 \in [0, 1]$; after this factor is removed, the curvature term has norm $\|C^v\| \le C_c|v|^2$, where C_c depends only on the c_{jklm}, the e term at most $Cr|v|^2$ and the mixed d term at most $C|v||\eta|$, while the last term $\beta\beta^*$ of U gives the nonnegative contribution $r^2 \int P[D_\rho a\, \beta]^2 \chi \, \mathrm{d}s$. Put

$$\lambda_0(x, r) = 2r^2 \int (-\chi_1(|s|^2)) \operatorname{tr} P(x + rs) \, \mathrm{d}s, \qquad m_0 = 1 + r^{-d} \mu_p(B(x, 3r)).$$

The substitution $s \mapsto w^{-1} a(0, ws) = s + O(r^2)$, whose derivative is $I + O(r^2)$, changes the kernel $-\chi_1(|s|^2)$ by $O(r^2)$ on $\{|s| \le 2\}$, and the unitary substitution $s \mapsto (w/r)s$ does not change it. Hence $r^2 \int (-\chi_1)|D_\rho a|^2 \operatorname{tr} P \, \mathrm{d}s \le \frac12 \lambda_0 + Cr^4 m_0$, and similarly $r^2 \int_{|s|\le 1} \operatorname{tr} P \, \mathrm{d}s \le Cr^2 m_0$. Consequently the second and third integrals of (16.37), the second including the term $-r^2 \Xi\Xi^*$ from the square, are bounded below by

$$-A\lambda_0(x, r)|v|^2 - C(r^3 m_0 + r^2)|v|^2 - Cr^2 m_0 |v||\eta|, \qquad A = \tfrac12 C_c + C_d.$$

Thus A depends only on the fixed geometry, and since also $|E| \le Cr^2(|v|^2 + |\eta|^2)$, we obtain

$$\mathcal{D}\Psi_p[v, \eta]^2 \ge -\big(A\lambda_0 + C(r^2 + r^3 m_0)\big)|v|^2 - Cr^2 m_0 |v||\eta| - Cr^2|\eta|^2. \tag{16.38}$$

For $v = 0$ the coefficients Ξ, Θ and U vanish, and (16.37) reads $\partial_w \partial_{\bar w} \Psi_p = \int P[D_\rho a\, s]^2 \chi \, \mathrm{d}s + E$.

It remains to subtract the smoothing of the smooth function q. Since q is smooth, $\Psi_q(z, w)$ is smooth also at $w = 0$, where it equals $q(z)$. By rotation invariance, the terms of degree one and the terms in w^2 and $\bar w^2$ of its Taylor expansion in w vanish, so $\Psi_q(z, w) - q(z)$ is $|w|^2$ times a smooth function of z, up to an error $O(|w|^3)$ whose derivatives lose one power of $|w|$ for each derivative in w. Hence, with $\mathcal{D}q = \alpha$,

$$\alpha[v]^2 - \mathcal{D}\Psi_q[v, \eta]^2 \ge -Cr^2|v|^2 - Cr|v||\eta| - C_q|\eta|^2.$$

Since $\Psi_u = \Psi_p - \Psi_q$ and $\partial_w \partial_{\bar w}(K|w|^2) = K$, we choose $K \ge C_q + 1$. Adding these estimates proves (16.23) with λ_0 in place of λ_u and m_0 in place of m_u. Moreover, for $v = 0$ it gives $\partial_w \partial_{\bar w} H_u \ge K - C_q - Cr^2 > 0$. Since $\partial_w \partial_{\bar w} g = \frac{1}{4r} \partial_r (r \partial_r g)$ for a function g of $r = |w|$, the second derivative of H_u with respect to $\log r$ is $(r\partial_r)^2 H_u = 4r^2 \partial_w \partial_{\bar w} H_u > 0$, so H_u is a strictly convex function of $\log r$.

Finally we compare λ_0 with λ_u. Let $\Psi_g^{\mathrm{e}}(r) = \int g(x + rs)\chi(|s|^2) \, \mathrm{d}s$ denote the Euclidean radial convolution in the normal coordinates. For a smooth function g, the radial derivative is

$$r\partial_r \Psi_g^{\mathrm{e}}(r) = -2r^2 \int \chi_1(|s|^2) \sum_m g_{m\bar{m}}(x+rs)\,\mathrm{d}s. \tag{16.39}$$

Indeed, $r\partial_r[g(x+rs)] = \sum_m \big(s_m\partial_{s_m} + \bar{s}_m\partial_{\bar{s}_m}\big)[g(x+rs)]$, and by $s_m\chi = \partial_{\bar{s}_m}\chi_1$ and an integration by parts, $\int s_m\partial_{s_m}[g(x+rs)]\chi\,\mathrm{d}s = -\int \partial_{s_m}\partial_{\bar{s}_m}[g(x+rs)]\chi_1\,\mathrm{d}s = -r^2\int g_{m\bar{m}}(x+rs)\chi_1\,\mathrm{d}s$; the terms with $\bar{s}_m$ are the complex conjugates. In particular $r\partial_r\Psi_p^{\mathrm{e}} = \lambda_0$, and $|r\partial_r\Psi_q^{\mathrm{e}}| \le Cr^2$.

By (16.29), the substitution $s \mapsto r^{-1}F(x,rs) = s + O(r^2)$, with derivative $I + O(r^2)$, shows that the kernel of $\Psi_u(x,r)$ differs from that of $\Psi_u^{\mathrm{e}}(r)$ by at most Cr^{2-d} on $B(x,2r)$, where $\Psi_u^{\mathrm{e}} = \Psi_p^{\mathrm{e}} - \Psi_q^{\mathrm{e}}$. Hence the radial function $\psi(r) = \Psi_u(x,r) - \Psi_u^{\mathrm{e}}(r)$ satisfies $|\psi(r)| \le Cr^{2-d}\int_{B(x,2r)}|u| \le Cr^2(1+|\log r|)$, so $\psi(r) \to 0$ as $r \to 0$; like the mass bounds, this holds for every $u \in \mathrm{PSH}(X,\theta)$. In the same way, by the case $v = 0$ of (16.37), the matrix-valued kernels against which P is integrated in $\partial_w\partial_{\bar{w}}\Psi_p$ and in $\partial_w\partial_{\bar{w}}\Psi_p^{\mathrm{e}} = \int P(x+ws)[s]^2\chi\,\mathrm{d}s$ differ in norm by at most Cr^{2-d} on $B(x,3r)$; since $|P[M]| \le \|M\|\,\mathrm{tr}\,P$, these vertical derivatives differ by at most $Cr^2m_0 + |E| \le C$. The smooth function q contributes bounded vertical derivatives as well. Thus $|\partial_r(r\partial_r\psi)| = 4r|\partial_w\partial_{\bar{w}}\psi| \le Cr$ for all small r.

Integrating from r' to r shows that $\ell = \lim_{r'\to 0} r'\psi'(r')$ exists and $|r\psi'(r) - \ell| \le Cr^2$. If $\ell \ne 0$, then $\psi'(r) = \ell/r + O(r)$, and $\psi(r) - \psi(r') = \ell\log(r/r') + O(r^2)$ would be unbounded as $r' \to 0$, contradicting $\psi \to 0$. Hence $\ell = 0$ and $|r\partial_r\psi| \le Cr^2$. Since $\lambda_u = r\partial_r\Psi_p^{\mathrm{e}} - r\partial_r\Psi_q^{\mathrm{e}} + r\partial_r\psi + 2Kr^2$, we obtain $|\lambda_u - \lambda_0| \le Cr^2$. Since $m_0 \le m_u$, this proves (16.23) for smooth p; we record that it holds with m_0 in place of m_u.

Step 6: nonsmooth potentials. Let p be plurisubharmonic near $\overline{B(0,4r)}$. We approximate p by its mollifications p_ε, $0 < \varepsilon < r$ (Section 1.4), and pass to the limit in the estimates of Step 5. By (3.3), applied to the subharmonic function p, we have $p \le p_\varepsilon \le \sup_{B(0,4r)} p$, so $|p_\varepsilon| \le |p| + C$ on $B(0,3r)$. The functions p_ε are smooth and plurisubharmonic on $B(0,3r)$, by Proposition 8.2.2(2). Since derivatives commute with mollification (Proposition 1.4.3(1)), the trace measure of p_ε is the mollification of μ_p, so $\mu_{p_\varepsilon}(B(0,3r)) \le \mu_p(B(0,4r))$, and the monotonicity of the normalized trace mass of p_ε gives $\mu_{p_\varepsilon}(B(0,t)) \le Ct^{d-2}$ for $t \le 3r$. Thus the mass bounds used above hold for p_ε on these balls, with constants independent of ε, and (16.23) holds for $u_\varepsilon = p_\varepsilon - q$ at the center, with $1 + r^{-d}\mu_{p_\varepsilon}(B(0,3r)) \le m_u$ in place of m_u; so do the inequalities $|\lambda_{u_\varepsilon} - \lambda_0(p_\varepsilon)| \le Cr^2$ and $\partial_w\partial_{\bar{w}}H_{u_\varepsilon} \ge K - C_q - Cr^2$.

Since $p_\varepsilon \to p$ in $L^1(B(0,3r))$ (Lemma 1.4.1(2)), and Ψ, its derivatives in (z,w) for $w \ne 0$ and Ψ^{e} are given by integration against smooth kernels, the joint Levi form of H_{u_ε} at $(0,w)$, λ_{u_ε} and $\lambda_0(p_\varepsilon) = r\partial_r\Psi_{p_\varepsilon}^{\mathrm{e}}$ converge to the corresponding quantities for u. Letting $\varepsilon \to 0$ proves (16.23) and the strict convexity of H_u in $\log r$ for every $u \in \mathrm{PSH}(X,\theta)$. Finally, inserting $m_u \le Cr^{-2}$, or the hypothesis $m_u \le M(x)/r$, and using $r < 1$, we obtain (16.24) and (16.25).

Step 7: the remaining assertions. We first study $H_u(x,r)$ as $r \to 0$; by Steps 5 and 6 it is a convex function of $\log r$. The kernel of Ψ_u is positive with mass one, so the upper bound for u bounds $H_u(x,r)$ from above as $r \to 0$. A convex function of $\log r$ that is bounded above as $r \to 0$ is nondecreasing: if its slope were negative at

some point, the tangent line there, which lies below the function, would tend to $+\infty$ as $r \to 0$. Hence its slope $\lambda_u = r\partial_r H_u$ is nonnegative.

To identify the limit, recall from Theorem 3.1.3(2), applied to the subharmonic function p (Theorem 8.2.1), that the spherical means of p about x decrease to $p(x)$ as the radius decreases to zero; since $\Psi^{\mathrm{e}}_p(r)$ is an average of these spherical means at radii at most r, monotone convergence gives $\Psi^{\mathrm{e}}_p(r) \to p(x)$, and $\Psi^{\mathrm{e}}_q(r) \to q(x)$. Since $\psi(r) \to 0$, we get $H_u(x,r) \to u(x)$ as $r \to 0$. If u is bounded near x, this limit is finite. The slope of a convex function is nondecreasing, so it has a limit as $r \to 0$; a positive limit would force $H_u(x,r) \to -\infty$. Therefore $\lambda_u(x,r) \to 0$.

Next we prove the two comparison statements. If u is bounded on $B(x,2r)$, the comparison of the kernels in Step 5 gives $|\Psi_u(x,r) - \Psi^{\mathrm{e}}_u(r)| \le Cr^2 \sup_{B(x,2r)} |u|$, which is the asserted comparison with radial convolution. Now let f be a C^1 function whose weak complex Hessian is bounded by M_f. In normal coordinates its Euclidean Laplacian $4\sum_m f_{m\bar m}$ is bounded by CM_f, and so is the Laplacian of its mollifications f_ε. By (16.39), $|\partial_r \Psi^{\mathrm{e}}_{f_\varepsilon}(r)| \le CM_f r$, and since $\Psi^{\mathrm{e}}_{f_\varepsilon}(r) \to f_\varepsilon(x)$ as $r \to 0$, we get $|\Psi^{\mathrm{e}}_{f_\varepsilon}(r) - f_\varepsilon(x)| \le CM_f r^2$. Letting $\varepsilon \to 0$, the same bound holds for f, by uniform convergence. Finally, by (16.29), $|f(F(x,r\zeta)) - f(x+r\zeta)| \le Cr^3 \|\nabla f\|_\infty$ for $|\zeta| \le 1$, and $h(0) = I$, so $|\Psi_f(x,r) - \Psi^{\mathrm{e}}_f(r)| \le Cr^3 \|\nabla f\|_\infty$. This proves (16.26).

Finally we prove (16.27). We work in a fixed coordinate chart with a smooth frame of TX that is unitary for h, obtained by the Gram–Schmidt process, and we describe a tangent vector by its vector $\zeta \in \mathbb{C}^n$ of coefficients in this frame. Then $\mathrm{d}\zeta_h = \mathrm{d}\zeta$, and the frame matrix $S(x)$, which satisfies $SS^* = (h^{i\bar\jmath})$, gives the vector $S(x)\zeta$ of coordinate components. In the chart, (16.28) shows that $F(x,\zeta)$ differs by $O(|\zeta|^{N+1})$ from $J_z(S(x)\zeta)$, where z is the coordinate of x and $J_z(\xi) = z + \xi - \frac12 \Gamma(z)(\xi,\xi) + O(|\xi|^3)$ is the jet map of the chart. Hence $F(x,\zeta) = x + S(x)\zeta + Q_x(S(x)\zeta, S(x)\zeta) + O(|\zeta|^3)$ with the holomorphic quadratic map $Q_x = -\frac12 \Gamma(z)$.

The first moments of the radial weight vanish, and so do the moments $\int \zeta_i \zeta_j \chi(|\zeta|^2)\,\mathrm{d}\zeta$, by the substitution $\zeta \mapsto \mathrm{e}^{\mathrm{i}t}\zeta$; by symmetry, $\int \zeta_i \bar\zeta_j \chi(|\zeta|^2)\,\mathrm{d}\zeta = a_\chi \delta_{ij}$. For smooth g, the second-order Taylor expansion of $g(F(x,r\zeta))$ in $r\zeta$, integrated against $\chi(|\zeta|^2)\,\mathrm{d}\zeta$, therefore gives

$$\Psi_g(x,r) = g(x) + a_\chi r^2 \sum_{i,j} h^{i\bar\jmath} g_{i\bar\jmath}(x) + O(r^3),$$

uniformly on compact subsets of the chart.

For the distributional statement let φ be a test function in the chart. For fixed ζ near 0 the map $\Phi_\zeta(x) = F(x,\zeta)$ is a diffeomorphism near $\operatorname{Supp}\varphi$, with Φ_0 the identity, and its inverse depends smoothly on (ζ, y), by the inverse function theorem. By Fubini's theorem and the change of variables $y = \Phi_{r\zeta}(x)$, the adjoint kernel is

$$\int \Psi_g(x,r)\varphi(x)\,\mathrm{d}x = \int g(y)\,\Psi_r^*\varphi(y)\,\mathrm{d}y,$$

$$\Psi_r^*\varphi(y) = \int \varphi\bigl(\Phi^{-1}_{r\zeta}(y)\bigr)\bigl|\det D\Phi^{-1}_{r\zeta}(y)\bigr|\chi(|\zeta|^2)\,\mathrm{d}\zeta.$$

The integrand is a smooth function of $(r\zeta, y)$, equal to $\varphi(y)$ at $r = 0$, so Taylor's formula in $r\zeta$ gives $\Psi_r^*\varphi = \varphi + r\varphi_1 + r^2\varphi_2 + O(r^3)$, with continuous functions φ_1, φ_2 supported in a fixed compact set and a remainder bounded uniformly in y. Pairing this with an arbitrary smooth g and comparing with the expansion of Ψ_g, we find $\int g\varphi_1 = 0$ and $\int g\varphi_2 = a_\chi \int \varphi \sum h^{i\bar{j}} g_{i\bar{j}}$ for every smooth g; hence $\varphi_1 = 0$ and $\varphi_2 = a_\chi \sum_{i,j} \partial_i \partial_{\bar{j}} (h^{i\bar{j}} \varphi)$. Pairing the expansion of $\Psi_r^*\varphi$ with a locally integrable u gives

$$\int \frac{\Psi_u(\cdot, r) - u}{r^2} \varphi \, \mathrm{d}x = \int u \, \frac{\Psi_r^*\varphi - \varphi}{r^2} \, \mathrm{d}x \longrightarrow a_\chi \int u \sum_{i,j} \partial_i \partial_{\bar{j}} (h^{i\bar{j}} \varphi) \, \mathrm{d}x,$$

which is (16.27), because the right side is $a_\chi \langle D_h u, \varphi \rangle$ by the definition of distributional derivatives. □

16.2.6 A minimum with a varying upper radius

Near the pole set of the potential of a Kähler current with analytic singularities the smoothing radius must be taken smaller, so the minimum over radii will be taken over an interval whose upper end depends on the base point. Kiselman's minimum principle (Theorem 8.5.1) concerns functions that are plurisubharmonic on a domain with open fibers. Here the minimum is taken over a closed interval and may be attained at its end point, and the joint Levi form is controlled only where the slope in the radius is small; the curvature of the end point function then enters. The following lemma records this curvature error explicitly.

Lemma 16.2.28 *Let $D \subset \mathbb{C}^n$ be a domain, let $\widehat{\omega}$ be a Hermitian form on D, and let b be a smooth real function on D with $\mathrm{i}\partial\bar{\partial} b \le C_b \widehat{\omega}$ for a constant $C_b \ge 0$. Let $H(z, s)$ be a smooth function, defined for $z \in D$ and s in a neighborhood of $(-\infty, b(z)]$, such that $H_s \ge 0$ and $H_{ss} > 0$, and such that H is bounded below, locally uniformly in z, as $s \to -\infty$. Regard H as a function of $(z, \zeta) \in D \times \mathbb{C}$ with $s = \operatorname{Re}\zeta$.*

Fix $c > 0$ and a smooth real $(1,1)$-form Λ on D, and let π_D denote the projection to D. Suppose that at every point with $s \le b(z)$ and $H_s \le c$ the joint Levi form satisfies

$$\mathrm{i}\partial\bar{\partial}_{z,\zeta} H \ge -\pi_D^*\Lambda.$$

Then the constrained minimum

$$v(z) = \inf_{s \le b(z)} \big(H(z, s) - cs \big)$$

is finite and continuous, and it satisfies, in the sense of distributions,

$$\mathrm{i}\partial\bar{\partial} v \ge -\Lambda - cC_b\widehat{\omega}.$$

In particular, if b is constant, or more generally plurisuperharmonic, we may take $C_b = 0$, and the upper end point produces no additional curvature error.

Proof For $\varepsilon > 0$ we minimize instead the penalized function

$$H_\varepsilon(z,s) = H(z,s) - cs - \varepsilon \log(b(z) - s), \qquad s < b(z).$$

For fixed z it tends to $+\infty$ at both ends of the interval: as $s \to -\infty$ the linear term $-cs$ dominates the logarithm, because H is bounded below. Its second derivative in s is $H_{ss} + \varepsilon/(b-s)^2 > 0$, so it has a unique minimizer $s_\varepsilon(z)$, a nondegenerate critical point. By the implicit function theorem, s_ε and the minimum value $v_\varepsilon(z) = H_\varepsilon(z, s_\varepsilon(z))$ are smooth. The critical equation is

$$H_s - c + \frac{\varepsilon}{b-s} = 0,$$

so at the minimizer $0 \le H_s \le c$ and $0 \le \sigma := \varepsilon/(b-s) \le c$. Since $\operatorname{Re}\zeta$ is pluriharmonic, the joint Levi form of the barrier $-\varepsilon \log(b(z) - \operatorname{Re}\zeta)$ is $\varepsilon\, \mathrm{i}\partial\phi \wedge \bar\partial\phi/\phi^2 - \sigma\, \mathrm{i}\partial\bar\partial b$, where $\phi = b(z) - \operatorname{Re}\zeta$; the first term is nonnegative, and the second is at least $-\sigma C_b \widehat\omega \ge -cC_b\widehat\omega$. Hence, at the minimizer, the joint Levi form of H_ε is bounded below by $-\pi_D^*(\Lambda + cC_b\widehat\omega)$.

We now use the Schur complement argument of Kiselman's minimum principle (Theorem 8.5.1); it needs the Levi bound only at the minimizer, not elsewhere in the fiber. At the minimizer write the joint complex Hessian of H_ε in block form $\left(\begin{smallmatrix} P & y/2 \\ y^*/2 & a/4 \end{smallmatrix}\right)$, where $P = (\partial^2 H_\varepsilon/\partial z_j \partial \bar z_k)$, $y = (\partial^2 H_\varepsilon/\partial z_j \partial s)$ and $a = \partial^2 H_\varepsilon/\partial s^2 > 0$; here we used that $\partial_\zeta = \partial_{\bar\zeta} = \frac12 \partial_s$ on functions of z and $s = \operatorname{Re}\zeta$. Since $\partial_s H_\varepsilon(z, s_\varepsilon(z)) = 0$, the chain rule gives $\partial_{z_j} v_\varepsilon(z) = (\partial_{z_j} H_\varepsilon)(z, s_\varepsilon(z))$, and differentiating the critical equation in $\bar z_k$ gives $\bar y_k + a\, \partial_{\bar z_k} s_\varepsilon = 0$, because H_ε is real. Hence $\partial_{z_j}\partial_{\bar z_k} v_\varepsilon = P_{jk} + y_j \partial_{\bar z_k} s_\varepsilon$, that is, the complex Hessian of v_ε is $P - a^{-1} y y^*$.

To bound it from below, let M be the coefficient matrix of $\Lambda + cC_b\widehat\omega = \mathrm{i}\sum_{j,k} M_{jk}\, \mathrm{d}z_j \wedge \mathrm{d}\bar z_k$. By the Levi bound at the minimizer, the block matrix $\left(\begin{smallmatrix} P+M & y/2 \\ y^*/2 & a/4 \end{smallmatrix}\right)$ is positive semidefinite, and evaluating its Hermitian form at the vector $(\xi, -2a^{-1}y^*\xi)$, $\xi \in \mathbb{C}^n$, gives $\xi^*(P + M - a^{-1} y y^*)\xi \ge 0$. Thus

$$\mathrm{i}\partial\bar\partial v_\varepsilon \ge -\Lambda - cC_b\widehat\omega.$$

It remains to pass to the limit $\varepsilon \to 0$. On a compact neighborhood in the base, b is bounded, and evaluating at $s = b - 1$ gives a uniform upper bound for v_ε when $0 < \varepsilon \le 1$. Since H is bounded below and $-cs - \log(b-s)$ grows linearly as $s \to -\infty$, the minimizers s_ε are bounded below uniformly; the minimizers of the unpenalized problem have the same lower bound. Both problems may therefore be restricted to a common compact interval below their varying upper end point. The unpenalized minimum v is finite, and it is continuous: after the substitution $s = b(z) - t$ the upper end point becomes fixed, the range of t is a fixed compact interval, and a minimum of a continuous function over a fixed compact interval depends continuously on the parameter.

On this range the penalty is bounded below by a quantity that tends uniformly to zero, so $v_\varepsilon \geq v - o(1)$. For an upper bound, evaluate H_ε at an unpenalized minimizer if it lies at least $\sqrt{\varepsilon}$ below b, and otherwise at $s = b - \sqrt{\varepsilon}$. In the first case the penalty is at most $\varepsilon|\log\sqrt{\varepsilon}|$; in the second, the unpenalized function changes by $o(1)$, uniformly, by its uniform continuity on the compact range. Consequently $v_\varepsilon \to v$ locally uniformly, and the inequalities for $\mathrm{i}\partial\bar\partial v_\varepsilon$ pass to the distributional limit. □

16.2.7 Big-class envelopes on the ample locus

We now prove that envelopes in a big class have locally bounded metric Laplacian away from the non-Kähler locus. The positivity that θ lacks is supplied by a Kähler current in $\{\theta\}$ with analytic singularities, whose pole set becomes a divisor after a modification. Near that divisor we shrink the smoothing radius of Lemma 16.2.27. A first bound for the admissible radius comes from the maximality of the envelope. High powers of the local equations of the divisor then turn the variable coefficients in the joint Hessian estimate into uniform ones, and a second application of maximality gives a quadratic bound for the smoothing error.

Theorem 16.2.29 *Suppose that $\{\theta\}$ is big, let $r \geq 1$ be an integer, and let $f_1, \dots, f_r$ be continuous real functions on X with bounded metric Laplacian. Let*

$$\mathrm{Amp}(\{\theta\}) = X \setminus E_{\mathrm{nK}}(\{\theta\})$$

be the ample locus *of $\{\theta\}$, the complement of its non-Kähler locus (Definition 14.4.1). Then*

$$u = P_\theta\big(\min_j f_j\big)$$

has bounded metric Laplacian on every compact subset of $\mathrm{Amp}(\{\theta\})$. On $\mathrm{Amp}(\{\theta\})$ its complex Hessian has locally bounded coefficients, $u \in W^{2,p}_{\mathrm{loc}}$ for every $1 \leq p < \infty$, and hence $u \in C^{1,\gamma}_{\mathrm{loc}}$ for every $0 < \gamma < 1$.

Proof After reducing to one obstacle and passing to a modification, we bound the smoothing error $H_u(x,r) - u(x)$ off the divisor, first linearly and then quadratically in r, as outlined above; the quadratic bound and (16.27) then bound $D_h u$, and the Sobolev regularity follows.

Reduction to one obstacle with bounded complex Hessian. Choose a Kähler form $\kappa \geq \theta$, for instance a large multiple of ω, and put $f = P_\kappa(\min_j f_j)$. By Theorem 16.2.24, f is C^1 and has bounded complex Hessian. Every θ-psh function below $\min_j f_j$ is κ-psh, and hence lies below f. Conversely $f \leq \min_j f_j$. These two inclusions of competitor families give $P_\theta(f) = u$.

Passage to a modification. Put $Z = E_{\mathrm{nK}}(\{\theta\})$. By Theorem 14.4.4, Z is the pole set of a function $\phi_1 \in \mathrm{PSH}(X,\theta)$ with analytic singularities such that $\theta + \mathrm{dd^c}\phi_1 \geq 2\delta_1\omega$ for some $\delta_1 > 0$. We apply Lemma 13.4.1 to ϕ_1, regarded as an element of

$\mathrm{PSH}(X, \theta - 2\delta_1\omega)$, with an integer m_0 so large that the constant c' of that lemma satisfies $c'/m_0 \le \delta_1$, and we let ϕ_0 be the resulting function. Then $\theta + \mathrm{dd^c}\phi_0 \ge \delta_1\omega$, and ϕ_0 has gentle analytic singularities (Definition 13.1.11) of type $(m_0^{-1}, \mathfrak{a})$, where $\mathfrak{a} = \mathcal{I}(m_0\phi_1)$. The pole set of ϕ_0 is the zero set of $\mathfrak{a}$, and it equals Z. Indeed, near a point where ϕ_1 is finite, ϕ_1 is bounded, because it has analytic singularities, so $\mathfrak{a}$ is the unit ideal there; thus the pole set of ϕ_0 is contained in Z. Conversely, $\theta + \mathrm{dd^c}\phi_0$ is a Kähler current whose potential has analytic singularities, so its pole set contains Z by the definition of the non-Kähler locus (Definition 14.4.1).

Let $\pi\colon Y \to X$ be a log resolution of $\mathfrak{a}$, that is, a principalization of $\mathfrak{a}$ by finitely many blowups with smooth centers (Theorem A.5.5), and write $\pi^*\mathfrak{a} = \mathcal{O}_Y(-D_0)$; the support of the effective divisor D_0 is $\pi^{-1}(Z)$. By Definition 13.1.11, locally on Y the function $\phi_0 \circ \pi$ is $m_0^{-1}\log|s_{D_0}|^2$ plus a smooth function. By Theorem A.5.5, π is an isomorphism over $X \setminus Z$, and all its centers lie over Z. If σ blows up a compact Kähler manifold with Kähler form ω' along a closed submanifold of positive codimension, with exceptional divisor E, then $\sigma^*\omega' + \varepsilon\eta$ is Kähler for every sufficiently small $\varepsilon > 0$, where η is the curvature of a suitable smooth quadratic Hermitian metric on $\mathcal{O}(-E)$ (Proposition 13.1.10). Applying this to the successive blowups of π, with rational ε, and pulling the forms back to Y, we obtain a Kähler form

$$\omega_Y = \pi^*\omega + \sum_\ell \varepsilon_\ell\eta_\ell > 0,$$

where η_ℓ is the curvature of a smooth metric on $\mathcal{O}(-E_\ell)$ for effective exceptional divisors E_ℓ, and the coefficients $\varepsilon_\ell > 0$ can be taken rational. The pulled-back current $\pi^*(\theta + \mathrm{dd^c}\phi_0)$ is at least $\delta_1\pi^*\omega$. Let s_{E_ℓ} be the canonical section of $\mathcal{O}(E_\ell)$, with norm $|s_{E_\ell}|$ computed in the dual metric. Choose a rational number $0 < \delta_2 \le \delta_1$ and add $\delta_2\sum_\ell \varepsilon_\ell \log|s_{E_\ell}|^2$ to the pulled-back potential $\phi_0 \circ \pi$. Since, by the Poincaré–Lelong formula (Theorem 7.4.5),

$$\mathrm{dd^c}\log|s_{E_\ell}|^2 = [E_\ell] + \eta_\ell,$$

the resulting current is at least $\delta_2\omega_Y$. Locally its potential is still of the form $\sum_j \beta_j \log|g_j|^2$ plus a smooth function, where the g_j are local equations of the components of an effective divisor with support $\pi^{-1}(Z)$ (the exceptional divisors lie over Z) and the β_j are positive rational numbers. Replacing ω_Y by $\delta_2\omega_Y$, we may assume that the lower bound is ω_Y itself.

By Proposition 14.2.8, pullback is a bijection from $\mathrm{PSH}(X, \theta)$ onto $\mathrm{PSH}(Y, \pi^*\theta)$. Since f is continuous, a potential lies below f if and only if its pullback lies below $f \circ \pi$, so $u \circ \pi = P_{\pi^*\theta}(f \circ \pi)$. The pullback $f \circ \pi$ is C^1 and has bounded complex Hessian. Indeed, off $\pi^{-1}(Z)$ the map π is a local biholomorphism, and since its differential is bounded, the chain rule bounds the coefficients of $\pi^*(\kappa + \mathrm{dd^c}f)$ there. Locally, $f \circ \pi$ plus a smooth local potential of $\pi^*\kappa$ is a bounded plurisubharmonic function. Applying Theorem 9.3.1 to it and to $\log|g|$, where g is a local equation of the hypersurface $\pi^{-1}(Z)$, shows that this logarithm is integrable for the trace measure of its complex Hessian, so the trace measure gives $\pi^{-1}(Z)$ zero mass. Hence the positive current $\pi^*\kappa + \mathrm{dd^c}(f \circ \pi)$, whose coefficients are bounded off $\pi^{-1}(Z)$, has

bounded coefficients everywhere. We may therefore work on Y with the pulled-back form and obstacle, and return through the isomorphism over $X \setminus Z$ at the end.

From now on we write X, θ, ω, f and u for $Y, \pi^*\theta$, the Kähler form just constructed, $f \circ \pi$ and $u \circ \pi$, and again put $\alpha = 2\pi\theta$ and $\widehat{\omega} = 2\pi\omega$. Adding a constant to the obstacle, we may assume that $0 \le f \le B$ with $B \ge 1$. Let ϕ be the potential $\phi_0 \circ \pi + \delta_2 \sum_\ell \varepsilon_\ell \log |s_{E_\ell}|^2$ constructed above, minus a constant chosen so that

$$\alpha + \mathcal{D}\phi \ge \widehat{\omega}, \qquad \phi \le -B - 1, \qquad \phi \le u \le f \le B; \tag{16.40}$$

here $\phi \le u$ holds because ϕ is a competitor for the envelope. Cover X by finitely many coordinate charts, with smaller charts still covering X, on each of which

$$\phi = \sum_j \beta_j \log |g_j|^2 + \sigma, \qquad \beta_j > 0, \quad |g_j| < \tfrac{1}{2},$$

where the g_j are local equations of the components of the divisor meeting the chart (there are none if the chart does not meet the divisor) and σ is smooth. Write D for the divisor and $U = X \setminus D$. In particular $\mathcal{D}\phi$ is uniformly bounded on U, although ϕ tends to $-\infty$ near D. All bounds below are uniform on this finite atlas; only the explicitly displayed functions of ϕ are allowed to grow.

A first-order smoothing bound. Let $H = H_u$ and $\lambda = \lambda_u$ be as in Lemma 16.2.27. The initial joint estimate (16.24) improves after adding $K_1 r$. Indeed, the vertical Levi form of $K_1|w|$ is $K_1/(4r)$, and Young's inequality absorbs the bounded mixed term at a horizontal cost Cr. Hence, for a fixed sufficiently large K_1 and small r,

$$\alpha + \mathcal{D}(H + K_1 r) \ge -(A\lambda + Cr)\widehat{\omega} \tag{16.41}$$

as joint forms; the right side is pulled back from X.

Fix a small $\delta_0 > 0$; it will be decreased finitely many times, each time depending only on the fixed data. For $0 < \delta < \delta_0$ and $c > 0$ define on U

$$\psi_{c,\delta}(x) = \inf_{0<t\le\delta} \{H(x,t) + K_1 t - K_1\delta - c\log(t/\delta)\}.$$

The infimum over t is finite and attained, because $H \ge u$, u is locally bounded on U, and $-c\log(t/\delta) \to +\infty$ as $t \to 0$. At a minimizing radius, including the upper end point, the logarithmic slope $\lambda + K_1 t$ of the nondecreasing part $H + K_1 t$ is at most c, so $\lambda \le c$. We apply Lemma 16.2.28 in the variable $s = \log t$, with the constant upper end point $\log\delta$, to $H(x, \mathrm{e}^s) + K_1\mathrm{e}^s$ plus a smooth local potential of α; by (16.41), the Levi bound required in the lemma holds with $\Lambda = (Ac + C\delta)\widehat{\omega}$. Moreover, since $u \le f$ and the kernel of Ψ is positive, evaluation at $t = \delta$ gives $\psi_{c,\delta} \le \Psi_u(x,\delta) + K\delta^2 \le \Psi_f(x,\delta) + K\delta^2$, and (16.26) bounds the right side by $f + C\delta^2$. Thus

$$\alpha + \mathcal{D}\psi_{c,\delta} \ge -(Ac + C\delta)\widehat{\omega}, \qquad \psi_{c,\delta} \le f + C\delta^2.$$

Put $e = Ac + C\delta$ and $\tau = e/(1+e)$. Then $(1-\tau)(\psi_{c,\delta} - C\delta^2) + \tau\phi$ is θ-psh on U, by (16.40), since $-(1-\tau)e + \tau = 0$; and it lies below $(1-\tau)f \le f$, because $\phi \le 0 \le f$. It is bounded above near D, so by Theorem 8.4.1, applied after adding smooth local potentials of α, it extends across D; the extension is given by upper limits and therefore still lies below the continuous obstacle. By the maximality of the envelope, this extension lies below u. Rearranging, and using $\psi_{c,\delta} \le B + C\delta^2$ so that no unbounded factor is divided by $1-\tau$, we obtain

$$\psi_{c,\delta} \le u + C\delta^2 + \tau(B - \phi + C\delta^2). \tag{16.42}$$

Now take $c = \delta$, so that e and τ are $O(\delta)$, and let t_0 be a minimizing radius for $\psi_{\delta,\delta}(x)$. The two nonnegative quantities $H(x,t_0) + K_1 t_0 - u(x)$ and $\delta \log(\delta/t_0)$ add up to $\psi_{\delta,\delta}(x) - u(x) + K_1\delta$, so (16.42) gives, with $R = C(1 + B + |\phi|)$ for a suitable uniform C,

$$t_0 \ge \delta \mathrm{e}^{-R(x)}, \qquad H(x,t_0) + K_1 t_0 - u(x) \le R(x)\delta.$$

Since H is nondecreasing in the radius, the choice $\delta = r\mathrm{e}^{R(x)}$ yields

$$0 \le H(x,r) - u(x) \le R(x)\mathrm{e}^{R(x)} r \quad \text{if } 0 < r < \delta_0 \mathrm{e}^{-R(x)}. \tag{16.43}$$

Since $\phi \le -B - 1$, we have $1 + B \le |\phi|$, so we may take $R(x) = B_0|\phi(x)|$ with a fixed large B_0.

This estimate also improves the bound for the trace mass. Enlarging B_0 if necessary, we may assume $B_0\beta_j \ge 1$ for all j; since $|g_j| < 1/2$, the formula for ϕ then gives $\mathrm{e}^{-R(x)} \le C|g_j(x)|^2$ for each local equation g_j. As the first derivatives of the g_j are bounded, there is a fixed small $S > 0$ such that, for $|y - x| \le S\mathrm{e}^{-R(x)}$, the ratios $|g_j(y)|/|g_j(x)|$ lie between $1/2$ and $3/2$. Hence $|\phi(y) - \phi(x)| \le C$ there. On a chart that does not meet the divisor, ϕ is smooth, and the same oscillation bound holds after decreasing S. By (16.40), $|u(y)| \le C(1 + |\phi(x)|)$ on that ball. The comparison with radial convolution in Lemma 16.2.27 therefore transfers (16.43) to the Euclidean radial mean of the local plurisubharmonic potential $p = u + q$, with an additional error $Cr^2(1 + |\phi(x)|)$.

We also need the following consequence of the Jensen–Lelong formula, with positive constants c_0 and $0 < c_1 < 1$ independent of x and r:

$$\int p(x + r\zeta)\chi(|\zeta|^2)\,\mathrm{d}\zeta - p(x) \ge c_0 r^{2-d}\mu_p(B(x, c_1 r)), \qquad d = 2n.$$

Indeed, the spherical means of p about x are nondecreasing in the radius and tend to $p(x)$. Letting $r_1 \to 0$ in (12.4) and using that $D_p(x,\cdot)$ is nondecreasing, we see that the spherical mean of radius ϱ exceeds $p(x)$ by at least $2\log(1/c_1)D_p(x, c_1\varrho)$, which by (12.3) is a fixed positive multiple of $\varrho^{2-d}\mu_p(B(x, c_1\varrho))$. It remains to integrate over the radii $\varrho = r\varrho'$ with ϱ' in a fixed subinterval of $(0,1)$ on which $\chi(\varrho'^2) > 0$, and to decrease c_1. We apply this inequality at a fixed multiple of r, decrease S, and use (16.43). This gives

$$m_u(x,r) \le \frac{M(x)}{r} \quad (0 < r < S\mathrm{e}^{-R(x)}), \qquad 1 \le M(x) \le CR(x)\mathrm{e}^{CR(x)}. \tag{16.44}$$

Thus the improved joint estimate (16.25) is available for these radii.

Uniform joint estimates in divisor coordinates. On each chart put $G = (\prod_j g_j)^m$, where an empty product equals one. By the formula for R and (16.44), M is bounded by a fixed product of negative powers of the $|g_j|$, and differentiating G loses only one power of one factor. Hence we can choose the integer m so large that $M|G|$ and $M|\mathrm{d}G|$ are uniformly bounded. Next, by the same formula for R, we can choose an integer $k \ge 1$ so large that

$$\frac{\mathrm{e}^{-kR(x)}}{|G(x)|} \le T$$

with T uniform on the finite cover. Put $\rho_\delta(x) = \delta\mathrm{e}^{-kR(x)}$, and take $\delta_0 < S$, so that these radii lie in the range of (16.44). In the holomorphic coordinate ξ defined by $w = G(x)\xi$, the range $0 < |w| \le \rho_\delta(x)$ corresponds to $|\xi| \le T\delta$.

The differential of the change of variables $(x,\xi) \mapsto (x, G(x)\xi)$ sends (v,η) to $(v, G\eta + \xi\,\mathrm{d}G(v))$. Substituting this in (16.25) and using the bounds just obtained, $|G| \le 1$ and a uniform bound for $|\mathrm{d}G|$, we get

$$\begin{aligned}(\alpha + \mathcal{D}H(x,|G\xi|))[v,\eta]^2 \ge\ & -(A\lambda + C_1|\xi|^2)|v|^2 \\ & - C_1|G||\xi||v||\eta| - C_1|G|^2|\xi||\eta|^2.\end{aligned} \tag{16.45}$$

For example, the purely horizontal error is bounded by a constant times $M(|G|^2 + |G||\mathrm{d}G| + |G||\mathrm{d}G|^2)|\xi|^2$, the mixed coefficient by $C|G||\xi|$, and the vertical coefficient by $C|G|^2|\xi|$. These estimates use only the boundedness of $M|G|$ and $M|\mathrm{d}G|$.

Now add $K_2|G\xi|^2$. Its Levi form is the square $K_2|G\eta + \xi\,\mathrm{d}G(v)|^2$, and the inequality $|a+b|^2 \ge \frac{1}{2}|a|^2 - |b|^2$ shows that it is at least $\frac{K_2}{2}|G|^2|\eta|^2$ minus a horizontal term $K_2|\mathrm{d}G|^2|\xi|^2|v|^2$. For δ_0 small and K_2 fixed and large, this absorbs the last term of (16.45), and Young's inequality absorbs its mixed term at a cost $C|\xi|^2|v|^2$. Since $|\xi| \le T\delta$, we have proved the uniform joint bound

$$\alpha + \mathcal{D}(H(x,|G\xi|) + K_2|G\xi|^2) \ge -(A\lambda + C_2\delta^2)\widehat{\omega}. \tag{16.46}$$

All constants here are independent of the chart and of the point of U.

A second-order smoothing bound. For $c > 0$ set

$$\psi(x) = \inf_{0<t\le\rho_\delta(x)} \{H(x,t) + K_2t^2 - c\log t\}.$$

In the variable $s = \log|\xi|$, so that $t = |G|\mathrm{e}^s$, the upper end point of the interval is $b(x) = \log\delta - kR(x) - \log|G(x)|$. The last summand is pluriharmonic on U, and $\mathcal{D}R = -B_0\mathcal{D}\phi$ is uniformly bounded there. Thus $|\mathcal{D}b| \le C_b\widehat{\omega}$ with a uniform constant C_b. The logarithmic slope of the nondecreasing part $H + K_2t^2$ is $\lambda + 2K_2t^2$, so at a minimizing radius it is at most c. Since $-c\log t = -cs - c\log|G|$ and $\log|G|$ is pluriharmonic on U, Lemma 16.2.28, applied to (16.46) (again after adding a smooth local potential of α), proves

$$\alpha + \mathcal{D}\psi \geq -C_0\widehat{\omega}, \qquad C_0 = A'c + C_2\delta^2, \quad A' = A + C_b. \tag{16.47}$$

Let $a = ckB_0$ and $C_3 = C_f + K + K_2$, with C_f from (16.26). Since $-c\log\rho_\delta = -c\log\delta - a\phi$, evaluation at $t = \rho_\delta(x) \leq \delta$, together with $\Psi_u \leq \Psi_f$ and (16.26), gives

$$Q := \psi + c\log\delta + a\phi - C_3\delta^2 \leq f, \qquad (1+a)\alpha + \mathcal{D}Q \geq -C_0\widehat{\omega};$$

the second inequality follows from (16.47) and $a(\alpha + \mathcal{D}\phi) \geq 0$, a consequence of (16.40). Put

$$\nu = \frac{C_0}{1 + a + C_0}, \qquad W = (1-\nu)Q + \nu(1+a)\phi.$$

Since $\nu(1+a) = (1-\nu)C_0$, we have $(1+a)\alpha + \mathcal{D}W \geq 0$, that is, W is $(1+a)\theta$-psh. It lies below f, because $Q \leq f$, $\phi \leq 0$ and $f \geq 0$. As before, W is bounded above near D, so it extends across D by Theorem 8.4.1, applied after adding smooth local potentials of $(1+a)\alpha$, and the extension, given by upper limits, still lies below the continuous function f. Consequently $W/(1+a)$, which lies below $f/(1+a) \leq f$, is a competitor for u, and $(1+a)u \geq W$. Since $u \leq B$ and $Q \leq B$, we can rearrange this without losing an unbounded positive term:

$$u \geq Q + \nu(1+a)\phi - (\nu + a)B. \tag{16.48}$$

Now take $c = \delta^2$, so that a and ν are $O(\delta^2)$, and let t_0 be a minimizing radius for $\psi(x)$. Writing $\psi(x)$ as the value at t_0 of the minimized expression and substituting it into (16.48), we obtain

$$t_0 \geq \delta e^{-R_2(x)}, \qquad 0 \leq H(x,t_0) + K_2 t_0^2 - u(x) \leq R_2(x)\delta^2, \qquad R_2 = C(1 + B + |\phi|). \tag{16.49}$$

Indeed, after rearrangement the left side consists of the two nonnegative terms $H(x,t_0) + K_2t_0^2 - u(x)$ and $-\delta^2\log(t_0/\delta)$ (recall $H \geq u$ and $t_0 \leq \delta$), while every term on the right is bounded by $C\delta^2(1 + B + |\phi|)$.

Fix a compact set $F \Subset U$ and a number $M_2 \geq \sup_F R_2$. For every sufficiently small r, choose $\delta = re^{M_2}$. Then $r \leq t_0$ at every $x \in F$, and since H is nondecreasing in the radius we obtain the uniform quadratic estimate

$$0 \leq H(x,r) - u(x) \leq M_2 e^{2M_2} r^2, \qquad x \in F. \tag{16.50}$$

Regularity. Divide (16.50) by r^2: on F we have $0 \leq (\Psi_u(x,r) - u(x))/r^2 + K \leq M_2e^{2M_2}$. Pairing with nonnegative test functions supported in the interior of F and letting $r \to 0$, (16.27) shows that, in the sense of distributions on the interior of F, $a_\chi D_h u + K$ lies between zero and the constant $M_2e^{2M_2}$. In particular $|\langle a_\chi D_h u, \varphi\rangle| \leq (K + M_2e^{2M_2})\int \varphi\,dx$ for these test functions $\varphi \geq 0$. Since every point of U lies in the interior of such a set F, Corollary 1.7.2(2), applied in the charts with Lebesgue measure, shows that $D_h u$ is an L^∞_{loc} function on U. The positive current $\alpha + \mathcal{D}u$ therefore has locally bounded h-trace, so in a fixed chart each of its diagonal coefficient measures is bounded by a constant times Lebesgue measure. The

Cauchy–Schwarz inequality for positive Hermitian matrices bounds the off-diagonal coefficient measures as well. Thus all complex coefficients of $\alpha + \mathcal{D}u$, and hence those of $\mathcal{D}u$, are locally bounded on U.

It remains to obtain the real Sobolev regularity; note that no weak first derivatives of u are known in advance. Let $V \Subset U$ be a coordinate ball. The complex second derivatives $u_{i\bar{j}}$ are bounded on V, so the Euclidean Laplacian $\Delta u = 4\sum_i u_{i\bar{i}}$ is a bounded function there. Let V' be a smaller concentric ball with $\overline{V'} \subset V$, choose $\vartheta \in C_c^\infty(V)$ equal to one near $\overline{V'}$, and let $N = E_{2n} * g$ be the fundamental potential of the bounded compactly supported function $g = \vartheta\Delta u$, where E_{2n} is the fundamental solution of Proposition 2.2.1. By Proposition 3.2.1, applied to the positive and negative parts of $g\,\mathrm{d}x$, we have $\Delta N = g$ in the sense of distributions. Hence $u - N$ is distributionally harmonic near $\overline{V'}$, and by Weyl's lemma (Theorem 2.1.4) it is smooth there. It therefore suffices to show that $N \in W^{2,p}(V')$ for $1 < p < \infty$.

The Hessian estimate of Theorem 6.2.3 concerns compactly supported smooth functions, so we mollify. Let g_ε be the mollifications of g, and put $N_\varepsilon = E_{2n} * g_\varepsilon$. For small ε the g_ε are smooth and supported in a fixed compact set, and $g_\varepsilon \to g$ in L^s for every $1 \le s < \infty$, by Lemma 1.4.1. By Proposition 2.2.1, N_ε is smooth and $\Delta N_\varepsilon = g_\varepsilon$. Moreover $\partial_a N_\varepsilon = E_{2n} * \partial_a g_\varepsilon = (\partial_a E_{2n}) * g_\varepsilon$, where the second identity follows by integration by parts outside a small ball about the singularity of E_{2n}, whose boundary term tends to zero. Fix q_0 with $1 < q_0 < 2n/(2n-1)$, and let $q_0' = q_0/(q_0-1)$. The functions E_{2n} and $\partial_a E_{2n}$, the latter bounded by $C|y|^{1-2n}$, belong to L^{q_0} on bounded sets, so Hölder's inequality gives

$$\sup_V |N_\varepsilon - N| + \sup_V \bigl|\partial_a N_\varepsilon - (\partial_a E_{2n}) * g\bigr| \le C\|g_\varepsilon - g\|_{L^{q_0'}} \longrightarrow 0.$$

Thus $N_\varepsilon \to N$ and $\partial_a N_\varepsilon \to (\partial_a E_{2n}) * g$ uniformly on V. Passing to the limit in the sense of distributions, we see that N is bounded on V and that its first distributional derivatives there are the bounded functions $(\partial_a E_{2n}) * g$. Choose $\vartheta_1 \in C_c^\infty(V)$ equal to one on V'. For small $\varepsilon, \varepsilon'$, the function $\vartheta_1(N_\varepsilon - N_{\varepsilon'})$ is smooth with compact support, and (6.3) gives

$$\bigl\|D^2\bigl(\vartheta_1(N_\varepsilon - N_{\varepsilon'})\bigr)\bigr\|_p \le C_{2n,p}\bigl\|\vartheta_1(g_\varepsilon - g_{\varepsilon'}) + 2\nabla\vartheta_1\cdot\nabla(N_\varepsilon - N_{\varepsilon'}) + (N_\varepsilon - N_{\varepsilon'})\Delta\vartheta_1\bigr\|_p.$$

By the convergences just proved, the right side tends to zero as $\varepsilon, \varepsilon' \to 0$. Hence, along a sequence $\varepsilon_k \to 0$, the second derivatives $\partial_a\partial_b(\vartheta_1 N_{\varepsilon_k})$ form a Cauchy sequence in $L^p(\mathbb{R}^{2n})$ and converge in L^p. The limits are supported in the fixed compact set $\operatorname{Supp}\vartheta_1$, so by Hölder's inequality the convergence also holds in the sense of distributions. On the other hand $\vartheta_1 N_{\varepsilon_k} \to \vartheta_1 N$ uniformly, so $\partial_a\partial_b(\vartheta_1 N_{\varepsilon_k}) \to \partial_a\partial_b(\vartheta_1 N)$ in the sense of distributions. Therefore $\partial_a\partial_b(\vartheta_1 N)$ is the L^p limit and belongs to L^p, and since $\vartheta_1 = 1$ on V', the second distributional derivatives of N belong to $L^p(V')$.

Hence $N \in W^{2,p}(V')$, and therefore $u \in W^{2,p}_{\mathrm{loc}}(U)$ for every $1 < p < \infty$; the case $p = 1$ follows on relatively compact sets because their volume is finite. Morrey's inequality (Lemma 6.2.6), with $p > 2n/(1-\gamma)$, gives a representative in $C^{1,\gamma}_{\mathrm{loc}}$ for every $\gamma < 1$. This continuous representative coincides with u: by Lemma 13.1.4(1), the quasi-plurisubharmonic function u is at every point the limit of its averages

over small coordinate balls centered there, and these averages converge to the continuous representative. Transferring these statements through the biholomorphism $Y \setminus \pi^{-1}(Z) \to X \setminus Z$, under which all smooth metrics are locally equivalent and local Sobolev spaces are preserved (Lemma 6.2.4(2)), proves the theorem. □

16.2.8 Contact with an arbitrary singular potential

Proposition 16.2.9 computes the contact density of a potential with bounded metric Laplacian. We now remove this regularity assumption on the potential and keep it only for the obstacle. Let u be an arbitrary quasi-plurisubharmonic potential below an obstacle f with bounded metric Laplacian. Between u and f we insert the envelopes of smooth functions decreasing to u; by Theorem 16.2.24 they have bounded metric Laplacian, so the regular theory applies to their contact with f. We then pass to the limit. The non-pluripolar measures of a decreasing sequence of unbounded potentials need not converge, so before passing to the limit we truncate at a fixed level and multiply by a bounded weight that vanishes below this level. The smooth functions decreasing to u are provided by the following lemma.

Lemma 16.2.30 *Let $u\colon X \to [-\infty, \infty)$ be upper semicontinuous. Then u is the pointwise limit of a decreasing sequence of smooth real functions ϕ_j on X.*

Proof If u is identically $-\infty$, the constants $\phi_j = -j$ will do, so we assume that u takes a finite value somewhere. Fix a distance d defining the topology of the compact manifold X, and put $M = \max_X u < \infty$. For each positive integer j define

$$v_j(x) = \max_{y \in X} \{u(y) - j d(x, y)\};$$

the maximum is attained because the function in braces is upper semicontinuous in y. The v_j are finite and j-Lipschitz: a point where u is finite bounds the maximum from below, and the triangle inequality bounds their differences. They decrease in j and dominate u (take $y = x$). If $u(x)$ is finite, a maximizing point y_j satisfies $j d(x, y_j) \le M - u(x)$, so $y_j \to x$, and since $v_j(x) \le u(y_j)$, upper semicontinuity gives $v_j(x) \downarrow u(x)$. If $u(x) = -\infty$, then for every N there is a small ball about x on which $u < -N$, while outside this ball $u(y) - j d(x, y) \le M - j d(x, y) < -N$ for large j. Thus $v_j(x) \downarrow u(x)$ in this case as well.

The v_j are only Lipschitz, so we smooth them while keeping monotonicity. Set $\phi_0 = M + 1$ and $\eta_j = 4^{-j}$, and suppose inductively that $\phi_{j-1} \ge u + \eta_{j-1}$. The continuous function

$$q_j = \min\{\phi_{j-1} - 2\eta_j,\ v_j + 2\eta_j\}$$

is at least $u + 2\eta_j$, since $\eta_{j-1} = 4\eta_j$. A continuous function on X can be approximated uniformly by smooth functions: convolution in a chart converges uniformly on compact subsets of the chart (Lemma 1.4.1(1)), and a finite smooth partition of unity

subordinate to charts (Proposition 7.1.3) sums these errors with total weight one. Choose a smooth real ϕ_j with $\|\phi_j - q_j\|_\infty < \eta_j$. Then

$$u + \eta_j \le \phi_j \le \phi_{j-1} - \eta_j, \qquad \phi_j \le v_j + 3\eta_j.$$

These bounds continue the induction, show that the sequence ϕ_j decreases, and, since $v_j + 3\eta_j \downarrow u$, show that $\phi_j \downarrow u$, also at the points where $u = -\infty$. □

Theorem 16.2.31 *Let $\theta_1, \dots, \theta_n$ be closed smooth real $(1,1)$-forms. For each j let $u_j \in \mathrm{PSH}(X, \theta_j)$, and let f_j be a continuous real function on X with bounded metric Laplacian. Suppose that $u_j \le f_j$ everywhere, and put $E = \bigcap_j \{u_j = f_j\}$. Then*

$$\mathbf{1}_E \left\langle \bigwedge_{j=1}^{n} (\theta_j + \mathrm{dd^c} u_j) \right\rangle = \mathbf{1}_E \bigwedge_{j=1}^{n} (\theta_j + \mathrm{dd^c} f_j). \tag{16.51}$$

Here the right side is the restriction to E of the finite signed measure of Corollary 16.2.12, defined by the weak second derivatives of the f_j; on E its density is nonnegative and bounded.

Proof We first prove the scalar case, in which all θ_j, u_j and f_j coincide, in three steps of increasing generality (Steps 1–3). The mixed case then follows by polarization (Step 4), and Step 5 proves the assertion about the density.

Step 1: a Kähler background and a plurisubharmonic obstacle. Let κ be a Kähler form, let u be κ-psh, and let $f \ge u$ be a continuous κ-psh function with bounded metric Laplacian. Write $F = \{u = f\}$ and $\mu_a = \langle (\kappa + \mathrm{dd^c} a)^n \rangle$ for κ-psh a. Since f is bounded, μ_f is the Bedford–Taylor product $(\kappa + \mathrm{dd^c} f)^n$ (Proposition 15.3.3, applied with smooth local potentials of κ). By Proposition 16.2.9, applied with $u_j = f_j = f$, this measure is given by the weak second derivatives of f, so it is the measure $(\kappa + \mathrm{dd^c} f)^n$ of Corollary 16.2.12; we must show that $\mathbf{1}_F \mu_u = \mathbf{1}_F \mu_f$. Take smooth functions $\phi_j \downarrow u$ as in Lemma 16.2.30, and set

$$\psi_j = P_\kappa(\min\{\phi_j, f\}).$$

Each ψ_j has bounded metric Laplacian by Theorem 16.2.24. Since $u \le \phi_j$ and $u \le f$, the function u is a competitor for $\min\{\phi_j, f\}$, and since this obstacle is continuous, Proposition 16.1.4(1),(2) give $u \le \psi_j \le f$. The obstacles $\min\{\phi_j, f\}$ decrease to $\min\{u, f\} = u$, so ψ_j decreases to $P_\kappa(u)$ by Proposition 16.1.4(5), and $P_\kappa(u) = u$, because u is itself a competitor and every competitor lies below the upper semicontinuous obstacle u by Proposition 16.1.4(2). Moreover $\psi_j = f$ on F, because $u \le \psi_j \le f$. The regular contact identity, Proposition 16.2.9, applied on the contact set $\{\psi_j = f\} \supset F$, therefore gives

$$\mu_{\psi_j} \ge \nu, \qquad \nu = \mathbf{1}_F \mu_f.$$

To pass to the limit, choose $C > 0$ with $\min_X f > -C$, fix $\varepsilon > 0$, and put

$$\psi_j^C = \max\{\psi_j, -C\}, \quad u^C = \max\{u, -C\},$$
$$b_j = \frac{\psi_j^C + C}{\psi_j^C + C + \varepsilon}, \qquad b = \frac{u^C + C}{u^C + C + \varepsilon}.$$

The bounded κ-psh functions ψ_j^C decrease to u^C, so they converge locally in capacity by Proposition 9.4.6. The weights b_j and b are bounded by one, and they are quasi-continuous, since ψ_j^C and u^C are, by Theorem 9.4.2 applied to their sums with smooth local potentials of κ; moreover $b_j \to b$ in capacity, because the map $t \mapsto t/(t+\varepsilon)$ on $[0,\infty)$ has Lipschitz constant at most $1/\varepsilon$. The weighted form of Theorem 9.4.8, applied with smooth local potentials of κ and a finite partition of unity, gives

$$b_j \mu_{\psi_j^C} \longrightarrow b\mu_{u^C} \quad \text{weakly on } X.$$

The measures $\mu_{\psi_j^C}$ and μ_{ψ_j} agree on the plurifine open set $\{\psi_j > -C\}$, by plurifine locality (Proposition 15.1.5), and b_j vanishes on $\{\psi_j \le -C\}$; hence $b_j\mu_{\psi_j^C} = b_j\mu_{\psi_j}$. In the same way $b\mu_{u^C} = b\mu_u$. On F all the weights take the same continuous value $b_f = (f+C)/(f+C+\varepsilon)$, which is strictly positive on X. Since ν is carried by F, we get $b_j\mu_{\psi_j} \ge b_j\nu = b_f\nu$ for every j, and in the weak limit $b\mu_u \ge b_f\nu$. Restricting to F, where $b = b_f$, and dividing by b_f, whose reciprocal is bounded on X, we obtain $\mathbf{1}_F\mu_u \ge \nu$. The reverse inequality is the ordered contact inequality of Lemma 16.2.2. This proves the scalar identity for a Kähler background and a plurisubharmonic obstacle.

Step 2: obstacles that are psh for a larger background. Let θ be any closed smooth real $(1,1)$-form, let $u \in \mathrm{PSH}(X,\theta)$, and suppose that $u \le f$ and that f is $(\theta + A\omega)$-psh for some $A > 0$. Increasing A, we may assume that $\theta + A\omega$ is Kähler. For every $t > A$, Step 1 applies to the Kähler background $\theta + t\omega$, with the same measuring metric ω. For a continuous real function g on X consider

$$Q_g(t) = \int_{\{u=f\}} g\,\langle(\theta + t\omega + \mathrm{dd^c}u)^n\rangle - \int_{\{u=f\}} g\,(\theta + t\omega + \mathrm{dd^c}f)^n.$$

This is a polynomial in t. Indeed, the first integrand expands by the multilinearity of the non-pluripolar product and its compatibility with smooth bounded factors (Propositions 15.1.5 and 15.3.3), and the second by expanding the wedge products of the weak forms. Every coefficient is a finite measure, by Theorem 15.4.3 and Corollary 16.2.12. Step 1 gives $Q_g(t) = 0$ for all $t > A$, so every coefficient vanishes, and in particular $Q_g(0) = 0$. Now a finite signed Borel measure σ on X with $\int g\,\mathrm{d}\sigma = 0$ for every continuous g vanishes. Indeed, write $\sigma = \sigma^+ - \sigma^-$ with finite positive Borel measures $\sigma^\pm$; then $\int g\,\mathrm{d}\sigma^+ = \int g\,\mathrm{d}\sigma^-$ for every continuous $g \ge 0$, so $\sigma^+ \le \sigma^-$ and $\sigma^- \le \sigma^+$ on Borel sets by Lemma 16.2.1(2), applied with $W = X$. Applied to the difference of the two measures in $Q_g(0)$, this proves the scalar identity whenever f is $(\theta + A\omega)$-psh for some A.

Step 3: general obstacles. Let f be any continuous real function with bounded metric Laplacian, and $u \le f$. Choose $A > 0$ such that $\kappa = \theta + A\omega$ is Kähler, and

insert the envelope $v = P_\kappa(f)$:

$$u \le v \le f.$$

Here $u \le v$ because u is κ-psh and lies below f. By Theorem 16.2.24, v is a continuous κ-psh function with bounded metric Laplacian. Step 2, with v as the obstacle, gives

$$\mathbf{1}_{\{u=v\}}\langle(\theta + \mathrm{dd^c}u)^n\rangle = \mathbf{1}_{\{u=v\}}(\theta + \mathrm{dd^c}v)^n.$$

Both v and f belong to $W^{2,p}$ for every finite p, by Proposition 16.2.11, and the locality of weak Hessians (Proposition 9.12.3) shows that their weak second derivatives agree almost everywhere on $\{v = f\}$. Since $\{u = f\} = \{u = v\} \cap \{v = f\}$, restricting the last identity to $\{u = f\}$ proves the scalar identity for f.

Step 4: the mixed identity. For $\lambda_1, \dots, \lambda_n > 0$ put $\theta_\lambda = \sum_j \lambda_j \theta_j$, $u_\lambda = \sum_j \lambda_j u_j$ and $f_\lambda = \sum_j \lambda_j f_j$. Then u_λ is θ_λ-psh, f_λ is continuous with bounded metric Laplacian, $u_\lambda \le f_\lambda$, and $E \subset \{u_\lambda = f_\lambda\}$. (Since the f_j are finite, E contains no pole of any u_j.) We restrict the scalar identity for u_λ and f_λ to E and expand both sides as measure-valued polynomials in λ, using, as in Step 2, the multilinearity of the non-pluripolar product (Proposition 15.1.5) and of the wedge product of the weak forms. Testing against a continuous function and comparing the coefficients of $\lambda_1 \cdots \lambda_n$, which are $n!$ times the integrals of this function against the two sides of (16.51), we see that these integrals agree; by the uniqueness proved in Step 2, we obtain (16.51).

Step 5: the density on E. For each j choose $A_j > 0$ such that $\theta_j + A_j\omega$ is Kähler, and insert the envelope $v_j = P_{\theta_j + A_j\omega}(f_j)$. Since u_j is $(\theta_j + A_j\omega)$-psh and lies below the continuous function f_j, we have $u_j \le v_j \le f_j$ by Proposition 16.1.4(1),(2), so $v_j = f_j$ on $\{u_j = f_j\}$; as in Step 3, Propositions 9.12.3 and 16.2.11 show that the weak Hessians of v_j and f_j agree almost everywhere there. The positive form $\theta_j + A_j\omega + \mathrm{dd^c}v_j$ has bounded coefficients by Theorem 16.2.24, so $\theta_j + \mathrm{dd^c}f_j$ has bounded coefficients almost everywhere on $\{u_j = f_j\}$. Hence the density of the mixed product of the weak forms is bounded on E. It is nonnegative there, because by (16.51) its restriction to E equals the restriction of a positive non-pluripolar product.□

Corollary 16.2.32 *Suppose that $\{\theta\}$ is big, and let f be a continuous real function with bounded metric Laplacian. Then*

$$\langle(\theta + \mathrm{dd^c}P_\theta(f))^n\rangle = \mathbf{1}_{\{P_\theta(f)=f\}}(\theta + \mathrm{dd^c}f)^n.$$

In particular $\langle(\theta + \mathrm{dd^c}V_\theta)^n\rangle = \mathbf{1}_{\{V_\theta=0\}}\theta^n$. The same identities hold in every pseudo-effective class; this is Corollary 17.4.22 in the next chapter, which adds a zero-mass argument for classes that are not big.

Proof A constant shift of V_θ lies below the bounded function f, so the competitor family of $P_\theta(f)$ is nonempty, and it is bounded above by $\max_X f$. Since f is continuous, $u = P_\theta(f)$ satisfies $u \le f$ everywhere, by Proposition 16.1.4. Theorem 16.2.31, with all $\theta_j = \theta$, $u_j = u$ and $f_j = f$, identifies the restriction of $\langle(\theta + \mathrm{dd^c}u)^n\rangle$ to $\{u = f\}$ with that of $(\theta + \mathrm{dd^c}f)^n$. By Theorem 16.2.5, applied to the continuous and hence quasi-continuous obstacle f, the measure $\langle(\theta + \mathrm{dd^c}u)^n\rangle$ is carried by $\{u = f\}$.

This proves the first identity. For $f = 0$ the envelope is V_θ by definition, which gives the second. □

16.3 Singularity envelopes

We now replace a pointwise obstacle by a constraint on singularities: we take the upper envelope of all nonpositive potentials that are more singular than a given potential u. Since the constant C in $v \le u + C$ may depend on the competitor v, the envelope can be less singular than u; this happens in Example 16.4.8 below.

Definition 16.3.1 For $u \in \mathrm{PSH}(X,\theta)$, the *singularity envelope* of u is

$$P_\theta[u] = \bigl(\sup\{v \in \mathrm{PSH}(X,\theta) : v \le 0,\ v \preceq u\}\bigr)^*.$$

The functions v in this supremum are the competitors for $P_\theta[u]$. A *model potential* is a potential ϕ with $P_\theta[\phi] = \phi$ and positive mass $\int_X \langle \theta_\phi^n \rangle > 0$. Since $u - \sup_X u$ is a competitor and all competitors are nonpositive, the singularity envelope is always defined, belongs to $\mathrm{PSH}(X,\theta)$, and satisfies $u - \sup_X u \le P_\theta[u] \le V_\theta$ and $\sup_X P_\theta[u] = 0$.

Proposition 16.3.2 *Let $u \in \mathrm{PSH}(X,\theta)$.*

(1) *$P_\theta[u]$ depends only on the singularity type of u, and $u \preceq u'$ implies $P_\theta[u] \le P_\theta[u']$.*
(2) *The rooftop envelopes $P_\theta(u + k, V_\theta)$ increase with k, and*

$$P_\theta[u] = \left(\lim_{k\to\infty} P_\theta(u + k, V_\theta)\right)^*.$$

(3) *The singularity envelope preserves all mixed non-pluripolar masses: if $\theta_1, \dots, \theta_p$, $1 \le p \le n$, are smooth closed real $(1,1)$-forms and $u_j \in \mathrm{PSH}(X,\theta_j)$, then*

$$\int_X \left\langle \bigwedge_j (\theta_j + \mathrm{dd^c} P_{\theta_j}[u_j]) \right\rangle \wedge \omega^{n-p} = \int_X \left\langle \bigwedge_j (\theta_j + \mathrm{dd^c} u_j) \right\rangle \wedge \omega^{n-p}.$$

In particular, $P_\theta[u]$ has the same mass as u.
(4) *If u has minimal singularities, then $P_\theta[u] = V_\theta$.*

Proof (1) The family of competitors depends only on the singularity type of u, and it increases when u is replaced by a less singular potential.

(2) The rooftop envelopes increase with k because the rooftop envelope is monotone. Every competitor v for $P_\theta[u]$ satisfies $v \le u + k$ for some integer k and $v \le V_\theta$, hence $v \le P_\theta(u + k, V_\theta)$. Conversely, each $P_\theta(u + k, V_\theta)$ is nonpositive and, by

Proposition 16.1.6, has the singularity type of u; so it is a competitor. Hence the two families have the same supremum, which proves the formula.

(3) By Proposition 16.1.6, the tuple of rooftop envelopes $P_{\theta_j}(u_j + k, V_{\theta_j})$ has the singularity types of the u_j, so it has the same mixed mass as the tuple (u_j) by Corollary 15.7.5. By (2) and Theorem 8.3.2, which shows that an increasing sequence differs from its regularized limit only on a Lebesgue null set, these rooftop envelopes increase almost everywhere to $P_{\theta_j}[u_j]$ as $k \to \infty$, and Corollary 15.7.6 shows that their mixed masses converge to the mixed mass of the envelopes.

(4) If u has minimal singularities, then $V_\theta \preceq u$, so V_θ is itself a competitor; it dominates all other competitors, since they are nonpositive. □

Proposition 16.3.3 *Suppose that $\{\theta\}$ is big, let $u \in \mathrm{PSH}(X, \theta)$, and put $w = P_\theta[u]$. Then*

$$\mu_w \le \mathbf{1}_{\{w=V_\theta\}}\mu_{V_\theta},$$

and μ_w is carried by $\{w = 0\}$.

Proof Put $w_k = P_\theta(u+k, V_\theta)$ for integers k; these are potentials by Proposition 16.1.6. By Corollary 16.2.6,

$$\mu_{w_k} \le \mathbf{1}_{\{w_k=u+k\}}\mu_u + \mathbf{1}_{\{w_k=V_\theta\}}\mu_{V_\theta} \le \mathbf{1}_{\{u+k\le V_\theta\}}\mu_u + \mathbf{1}_{\{w=V_\theta\}}\mu_{V_\theta};$$

for the second inequality we used $w_k \le V_\theta$ and $w_k \le w \le V_\theta$. Since $V_\theta \le 0$, the sets $\{u + k \le V_\theta\}$ decrease to the pole set of u, which is μ_u-null, and the first term tends to zero in total variation by dominated convergence. By Proposition 16.3.2, w_k increases almost everywhere to w, so $\mu_{w_k} \to \mu_w$ weakly by Corollary 15.7.6. Integrating a continuous function $\chi \ge 0$ and letting $k \to \infty$, we obtain $\int_X \chi\, \mu_w \le \int_X \chi \mathbf{1}_{\{w=V_\theta\}}\, \mu_{V_\theta}$, and Lemma 16.2.1 gives the asserted inequality. Finally, a potential lies below 0 quasi-everywhere if and only if it is nonpositive, so $V_\theta = P_\theta(0)$, and Theorem 16.2.5 shows that μ_{V_θ} is carried by $\{V_\theta = 0\}$. Hence μ_w is carried by $\{w = V_\theta = 0\}$. □

16.4 Extrapolation and model singularities

Throughout this section $\{\theta\}$ is big, and all potentials belong to $\mathrm{PSH}(X, \theta)$. Mass is monotone with respect to $\preceq$, but two potentials with the same mass need not have bounded difference (Example 16.4.8). The basic step reverses a convex combination: if $u \preceq v$ and the mass lost from v to u is small, then $u \ge a^{-1}\eta + (1 - a^{-1})v$ for some $a > 1$ and some potential η of positive mass. The contact concentration of Section 16.2.1 prevents the envelopes used to construct η from collapsing. This extrapolation yields the domination principle and identifies the singularity envelopes of potentials of positive mass as model potentials.

Lemma 16.4.1 *Let $\psi \in \mathrm{PSH}(X,\theta)$ with $\psi \le 0$ and $\int_X \langle \theta_\psi^n \rangle > 0$. Then there is a model potential $\phi \ge \psi$ with $\int_X \langle \theta_\phi^n \rangle = \int_X \langle \theta_\psi^n \rangle$.*

Proof Fix a smooth probability volume form $\mathrm{d}V$ on X, put $m = \int_X \langle \theta_\psi^n \rangle$, and consider

$$\mathcal{S} = \{v \in \mathrm{PSH}(X,\theta) : \psi \le v \le 0,\ \int_X \langle \theta_v^n \rangle = m\}, \qquad H(v) = \int_X v\,\mathrm{d}V.$$

The set $\mathcal{S}$ contains ψ, and H takes values in $[H(\psi), 0]$ on it. We construct an increasing sequence in $\mathcal{S}$ that almost maximizes H at every stage. Put $v_0 = \psi$. Given $v_k \in \mathcal{S}$, let

$$s_k = \sup\{H(w) : w \in \mathcal{S},\ w \ge v_k\},$$

and choose $v_{k+1} \in \mathcal{S}$ with $v_{k+1} \ge v_k$ and $H(v_{k+1}) \ge s_k - 2^{-k}$. The regularized limit $\phi = (\sup_k v_k)^*$ is a potential with $\psi \le \phi \le 0$, and $\int_X \langle \theta_\phi^n \rangle = m$ by Corollary 15.7.6; thus $\phi \in \mathcal{S}$. Since v_k increases to ϕ almost everywhere, between the integrable bounds ψ and 0, dominated convergence gives $H(v_k) \to H(\phi)$.

Let $w \in \mathcal{S}$ with $w \ge \phi$. Then $w \ge v_k$ for every k, so

$$H(w) \le s_k \le H(v_{k+1}) + 2^{-k},$$

and letting $k \to \infty$ gives $H(w) \le H(\phi)$. Since $w \ge \phi$, it follows that $w = \phi$ almost everywhere, hence everywhere. Now $P_\theta[\phi]$ lies between ϕ and 0, because $\phi \le 0$ is a competitor for it, and it has mass m by Proposition 16.3.2. Thus $P_\theta[\phi] \in \mathcal{S}$ and $P_\theta[\phi] \ge \phi$, so $P_\theta[\phi] = \phi$. Since $m > 0$, ϕ is a model potential. □

Lemma 16.4.2 *Let $\phi, \tau \in \mathrm{PSH}(X,\theta)$ with $P_\theta[\phi] = \phi$ and $\tau \preceq \phi$. Then*

$$\sup_{\{\phi > -\infty\}} (\tau - \phi) = \sup_X \tau.$$

In particular, if $\tau \le 0$, then $\tau \le \phi$.

Proof The potential $\tau - \sup_X \tau$ is nonpositive and more singular than ϕ, so it is a competitor for $P_\theta[\phi] = \phi$. Hence $\tau - \phi \le \sup_X \tau$ wherever ϕ is finite. Conversely, $\phi \le 0$, so $\tau - \phi \ge \tau$ wherever ϕ is finite. The pole set of ϕ is pluripolar, and removing a pluripolar set does not change the supremum of the quasi-plurisubharmonic function τ: an upper bound for τ off a Lebesgue null set holds everywhere. This proves the reverse inequality. □

The envelopes used for extrapolation have obstacles of the form $au + (1-a)v$ with $a > 1$. The following lemma collects their basic properties.

Lemma 16.4.3 *Let $u, v \in \mathrm{PSH}(X,\theta)$ with $u \le v$, let $a > 1$, and put $f = au + (1-a)v$, where $f = -\infty$ on the pole set of u, which contains that of v.*

(1) *The obstacle f is quasi-continuous, and $f \le u$. A potential η satisfies*

$$a^{-1}\eta + (1 - a^{-1})v \le u$$

if and only if $\eta \le f$ quasi-everywhere.

(2) *Suppose that f has a competitor. Then $\eta = P_\theta(f)$ is the largest potential satisfying this inequality, and $\eta \le u$. Moreover, if $\gamma = a^{-1}\eta + (1-a^{-1})v$ and $D = \{\gamma = u > -\infty\}$, then*

$$\mu_\eta \le a^n \mathbf{1}_D \mu_u.$$

Proof (1) Off the pole set of u, the functions u and v are finite and $f = u + (a-1)(u-v) \le u$, because $u \le v$; on the pole set, $f = -\infty$. Thus f is a Borel function with $f \le u$. Let K be a compact subset of a chart. Since u and v are quasi-continuous (Lemma 16.1.2(1)), there are open sets of arbitrarily small capacity outside which both are continuous on K, as functions with values in $[-\infty, \infty)$. On the complement in K of such a set, f is continuous at every point where u is finite, because $v \ge u$ is finite near such a point, and at a pole of u it is continuous with value $-\infty$, because $f \le u$. Hence f is quasi-continuous. Off the pole set of u, the inequalities $\eta \le f$ and $a^{-1}\eta + (1-a^{-1})v \le u$ are equivalent. Since the pole set is pluripolar, and an inequality between quasi-plurisubharmonic functions that holds quasi-everywhere holds everywhere, this proves the equivalence.

(2) By (1), the potentials satisfying the inequality are exactly the competitors for f; since $f \le u$, they lie below u quasi-everywhere, hence everywhere. Thus $\mathcal{A}_\theta(f)$ is nonempty, by assumption, and uniformly bounded above, and Proposition 16.1.4 shows that η is its largest member; in particular $\eta \le u$ and $\gamma \le u$. By Theorem 16.2.5, μ_η is carried by $\{\eta = f\}$. At a point of this set at which u is finite, $\eta = au + (1-a)v$ means $\gamma = u$; since μ_η does not charge the pole set of u, it is carried by D. Moreover $\mu_\gamma \ge a^{-n}\mu_\eta$, and Lemma 16.2.2, applied to $\gamma \le u$, gives $\mathbf{1}_D\mu_\gamma \le \mathbf{1}_D\mu_u$. Hence

$$\mu_\eta = \mathbf{1}_D\mu_\eta \le a^n \mathbf{1}_D\mu_\gamma \le a^n \mathbf{1}_D\mu_u.$$

Theorem 16.4.4 *Let $u, v \in \mathrm{PSH}(X,\theta)$ with $u \preceq v$ and $\int_X \langle\theta_u^n\rangle > 0$, and let $a > 1$ satisfy*

$$a^n \left(\int_X \langle\theta_v^n\rangle - \int_X \langle\theta_u^n\rangle \right) < \int_X \langle\theta_v^n\rangle. \tag{16.52}$$

Then there exists $\eta \in \mathrm{PSH}(X,\theta)$ with $\int_X \langle\theta_\eta^n\rangle > 0$ such that

$$a^{-1}\eta + (1-a^{-1})v \le u. \tag{16.53}$$

The largest such η is the obstacle envelope $E_a(u,v) = P_\theta(au + (1-a)v)$; here the obstacle $au + (1-a)v$ is defined to be $-\infty$ on the pole set of u, which contains that of v. If u and v have the same mass, then (16.52) holds for every $a > 1$, and $E_a(u,v)$ also has this mass.

Proof We first construct one potential satisfying (16.53) as the limit of a decreasing sequence of envelopes, and then show that $E_a(u,v)$ is the largest solution and has positive mass.

Reduction. Since $u \preceq v$, there are constants A, B with $u - A \le v - B \le 0$. If η' satisfies (16.53) for the pair $u - A$, $v - B$, then $\eta' + aA + (1-a)B$ satisfies it for

u, v; the envelopes of the two pairs differ by the same constant, and masses do not change. We may therefore assume that $u \le v \le 0$. Then $m = \int_X \langle \theta_v^n \rangle \ge \int_X \langle \theta_u^n \rangle > 0$, and Lemma 16.4.1 provides a model potential $\phi \ge v$ with $\int_X \langle \theta_\phi^n \rangle = m$. Since $1 - a < 0$, we have $au + (1-a)\phi \le au + (1-a)v$; we first construct a potential below $au + (1-a)\phi$.

A decreasing sequence of envelopes. For $j \ge 1$ put $u_j = \max\{u, \phi - j\}$, so that $\phi - j \le u_j \le \phi$; in particular, u_j and ϕ have the same pole set. Put

$$f_j = au_j + (1-a)\phi, \qquad \eta_j = P_\theta(f_j),$$

where $f_j = -\infty$ on the pole set of ϕ, as in Lemma 16.4.3 for the pair $u_j \le \phi$. Outside this pole set, $f_j = \phi + a(u_j - \phi)$ lies between $\phi - aj$ and ϕ. Hence the potential $\phi - aj$ is a competitor, and every competitor is at most ϕ, so η_j is a potential with

$$\phi - aj \le \eta_j \le \phi, \qquad \int_X \langle \theta_{\eta_j}^n \rangle = \int_X \langle \theta_{u_j}^n \rangle = m,$$

where the masses agree because $\eta_j \sim \phi \sim u_j$. Since u_j decreases with j, so do f_j and η_j.

A contact bound. Put $\gamma_j = a^{-1}\eta_j + (1 - a^{-1})\phi$ and $D_j = \{\gamma_j = u_j > -\infty\}$. By Lemma 16.4.3(2), applied to the pair $u_j \le \phi$, we have $\gamma_j \le u_j$ and

$$\mu_{\eta_j} \le a^n \mathbf{1}_{D_j} \mu_{u_j}. \tag{16.54}$$

A tail estimate. Fix $k > 0$ and let $j > k$. On D_j we have $\eta_j = \phi + a(u_j - \phi)$, so there $\eta_j \le \phi - ak$ if and only if $u_j \le \phi - k$, that is, if and only if $u \le \phi - k$. The plurifine open set $\{u > \phi - k\}$ is contained in $\{u > \phi - j\}$, on which $u_j = u$; hence $\mu_{u_j} = \mu_u$ on $\{u > \phi - k\}$, and $\mu_{u_j}(\{u \le \phi - k\}) = m - \mu_u(\{u > \phi - k\})$, pole sets having measure zero. Together with (16.54) this gives

$$\mu_{\eta_j}(\{\eta_j \le \phi - ak\}) \le a^n \left(m - \mu_u(\{u > \phi - k\})\right). \tag{16.55}$$

The envelopes do not collapse. Suppose that η_j decreases to $-\infty$ identically. Then $\sup_X \eta_j \to -\infty$ by Corollary 14.2.5. Since $\eta_j \preceq \phi$ and ϕ is a model potential, Lemma 16.4.2 gives $\eta_j - \phi \le \sup_X \eta_j$ off the pole set of ϕ. Hence, for each k, $\eta_j \le \phi - ak$ everywhere once j is large, and the left side of (16.55) equals $\int_X \langle \theta_{\eta_j}^n \rangle = m$. Since ϕ is finite wherever u is, $\mu_u(\{u > \phi - k\}) \to \int_X \langle \theta_u^n \rangle$ as $k \to \infty$, and (16.55) yields $m \le a^n \big(m - \int_X \langle \theta_u^n \rangle\big)$, which contradicts (16.52). Consequently η_j decreases to a potential η. Letting $j \to \infty$ in $\gamma_j \le u_j$ gives $a^{-1}\eta + (1 - a^{-1})\phi \le u$, and since $v \le \phi$, also (16.53).

The largest solution. By Lemma 16.4.3(1), applied to the pair $u \le v$, the potentials satisfying (16.53) are exactly the competitors for the obstacle $f = au + (1-a)v$. The potential η constructed above is one of them, so part (2) of the lemma shows that $E_a = E_a(u, v) = P_\theta(f)$ is the largest potential satisfying (16.53), and that $E_a \le u$.

Positive mass. Choose $b > a$ that still satisfies (16.52). The preceding steps, applied to b, give $E_b = E_b(u, v)$. The potential

$$q = \frac{a}{b}E_b + \Big(1 - \frac{a}{b}\Big)\,u$$

is a competitor for the obstacle f of E_a: quasi-everywhere, $E_b \le bu + (1-b)v$, and therefore

$$q \le au + (1-a)v + \Big(1 - \frac{a}{b}\Big)\,(u - v) \le au + (1-a)v.$$

Hence $q \le E_a$, and the monotonicity of mass together with $\mu_q \ge (1 - a/b)^n \mu_u$ yields

$$\int_X \langle \theta_{E_a}^n \rangle \ge \int_X \langle \theta_q^n \rangle \ge \Big(1 - \frac{a}{b}\Big)^n \int_X \langle \theta_u^n \rangle > 0.$$

If $\int_X \langle \theta_u^n \rangle = \int_X \langle \theta_v^n \rangle$, then (16.52) holds for every $a > 1$, so b can be taken arbitrarily large, and we obtain $\int_X \langle \theta_{E_a}^n \rangle \ge \int_X \langle \theta_u^n \rangle$. The reverse inequality follows from $E_a \le u$.□

Lemma 16.4.5 *Let $u \in \mathrm{PSH}(X,\theta)$ with $\int_X \langle \theta_u^n \rangle > 0$. Then there are $v \in \mathrm{PSH}(X,\theta)$ with $v \le u$ and $\delta > 0$ such that $\theta_v \ge \delta\omega$.*

Proof Put $m = \int_X \langle \theta_{V_\theta}^n \rangle$. Since $u \preceq V_\theta$ and $\int_X \langle \theta_u^n \rangle > 0$, we can choose $a > 1$ so close to one that $a^n\big(m - \int_X \langle \theta_u^n \rangle\big) < m$. Theorem 16.4.4 gives $\eta \in \mathrm{PSH}(X,\theta)$ with $a^{-1}\eta + (1-a^{-1})V_\theta \le u$. Since the class is big, it contains a closed current $T \ge \delta_0\omega$ for some $\delta_0 > 0$, and by Theorem 14.1.7 we can write $T = \theta + \mathrm{dd^c}\psi$ with ψ quasi-plurisubharmonic; after subtracting a constant, $\psi \in \mathrm{PSH}(X,\theta)$ satisfies $\psi \le 0$ and $\theta_\psi \ge \delta_0\omega$, and $\psi \le V_\theta$ because ψ is a competitor for V_θ. Therefore

$$v = a^{-1}\eta + (1-a^{-1})\psi \le u, \qquad \theta_v = a^{-1}\theta_\eta + (1-a^{-1})\theta_\psi \ge (1-a^{-1})\delta_0\omega.$$

Recall from Definition 15.8.3 that, for $\chi \in \mathrm{PSH}(X,\theta)$, the class $\mathcal{E}(X,\theta,\chi)$ of potentials of full mass relative to χ consists of the potentials $u \preceq \chi$ with $\int_X \langle \theta_u^n \rangle = \int_X \langle \theta_\chi^n \rangle$.

Theorem 16.4.6 (Domination principle) *Let $\chi \in \mathrm{PSH}(X,\theta)$ with $\int_X \langle \theta_\chi^n \rangle > 0$, and let $u, v \in \mathcal{E}(X,\theta,\chi)$. If $\mu_u(\{u < v\}) = 0$, then $u \ge v$ everywhere on X.*

Proof After reducing to the case $u \le v$, we show that the extrapolation envelopes $E_a(u,v)$ agree with u almost everywhere for μ_u. Hence they do not collapse as $a \to \infty$, while their limit equals $-\infty$ quasi-everywhere on $\{u < v\}$, so that this set is pluripolar.

The potential $\max\{u,v\}$ is more singular than χ and less singular than u, so

$$\int_X \langle \theta_\chi^n \rangle \ge \int_X \langle \theta_{\max\{u,v\}}^n \rangle \ge \int_X \langle \theta_u^n \rangle = \int_X \langle \theta_\chi^n \rangle;$$

thus $\max\{u,v\} \in \mathcal{E}(X,\theta,\chi)$. Since $\{u < \max\{u,v\}\} = \{u < v\}$, we may replace v by $\max\{u,v\}$ and assume that $u \le v$. Subtracting the same constant from u and v, we may also assume $v \le 0$. Let $m > 0$ be the common mass of u, v and χ.

For $a > 1$ let $\eta_a = E_a(u,v)$, $\gamma_a = a^{-1}\eta_a + (1-a^{-1})v$ and $D_a = \{\gamma_a = u > -\infty\}$. By Theorem 16.4.4, η_a is a potential of mass m; in particular its obstacle $au + (1-a)v$

has a competitor, and Lemma 16.4.3(2), applied to the pair $u \le v$, gives $\eta_a \le u$, $\gamma_a \le u$ and

$$\mu_{\eta_a} \le a^n \mathbf{1}_{D_a} \mu_u.$$

Since $u \le v$, the hypothesis means that μ_u is carried by $\{u = v\}$, so μ_{η_a} is carried by $D_a \cap \{u = v\}$. At a point of this set, $a^{-1}\eta_a + (1 - a^{-1})u = u$, that is, $\eta_a = u$. Hence μ_{η_a} is carried by $\{\eta_a = u\}$, and Lemma 16.2.2, applied to $\eta_a \le u$, gives $\mu_{\eta_a} \le \mathbf{1}_{\{\eta_a = u\}} \mu_u \le \mu_u$. Both measures have mass m, so they are equal, and $\eta_a = u$ holds μ_u-almost everywhere. In particular

$$\int_X \mathrm{e}^{\eta_a}\, \mu_u = \int_X \mathrm{e}^{u}\, \mu_u > 0,$$

where the last integral is positive because $m > 0$ and μ_u does not charge the pole set of u.

As a increases, the obstacle $v + a(u - v)$ decreases, so η_a decreases. If η_a decreased to $-\infty$ identically as $a \to \infty$ through the integers, dominated convergence, with $0 \le \mathrm{e}^{\eta_a} \le 1$, would make the left side of the last display tend to zero, a contradiction. Hence the limit η along the integers is a potential. Each η_a satisfies $\eta_a \le v + a(u - v)$ outside a pluripolar set; let N be the union of these sets over the integers $a \ge 2$. At a point outside N where u is finite and $u < v$, we get $\eta \le v + a(u - v)$ for every integer a, hence $\eta = -\infty$. Thus $\{u < v\}$ is contained in the union of N and the pole sets of u and η, which is pluripolar. Therefore $u \ge v$ quasi-everywhere, and hence everywhere. □

Theorem 16.4.7 *Let $u \in \mathrm{PSH}(X, \theta)$ with $\int_X \langle \theta_u^n \rangle > 0$. Then*

$$P_\theta[u] = \max\{v \in \mathrm{PSH}(X, \theta) : v \le 0,\ u \preceq v,\ \int_X \langle \theta_v^n \rangle = \int_X \langle \theta_u^n \rangle\}.$$

In particular $P_\theta[P_\theta[u]] = P_\theta[u]$, so $P_\theta[u]$ is a model potential of mass $\int_X \langle \theta_u^n \rangle$.

Proof Write $w = P_\theta[u]$, and let v belong to the displayed family. By Proposition 16.3.2, $w \le P_\theta[v]$, and the potentials w, u, v and $P_\theta[v]$ all have the same positive mass. Moreover $v \le P_\theta[v]$, because $v \le 0$ is a competitor for $P_\theta[v]$. Hence w and v both belong to $\mathcal{E}(X, \theta, P_\theta[v])$, a class whose reference potential has positive mass. By Proposition 16.3.3, μ_w is carried by $\{w = 0\}$, and since $v \le 0$, it gives no mass to $\{w < v\}$. The domination principle, Theorem 16.4.6, yields $w \ge v$. Finally, w itself belongs to the family: it is nonpositive, it satisfies $u - \sup_X u \le w$, and it has the same mass as u by Proposition 16.3.2. This proves the formula.

The potential $P_\theta[w]$ is nonpositive, satisfies $u \preceq w \preceq P_\theta[w]$, and has the same mass as u by Proposition 16.3.2. Hence it belongs to the family, and the formula gives $P_\theta[w] \le w$. Conversely, $w \le 0$ is a competitor for $P_\theta[w]$, so $P_\theta[w] = w$. Since the mass of w is positive, w is a model potential. □

Example 16.4.8 On $\mathbb{P}^1$ let $\omega = \mathrm{dd^c} \log(1 + |z|^2)$ in the affine coordinate z, so that $\int_{\mathbb{P}^1} \omega = 1$; since ω is Kähler, $V_\omega = 0$. Define

$$F(r) = \begin{cases} \max\{-\log(-r), r+2\}, & r \le -1, \\ r+2, & r \ge -1, \end{cases} \qquad u(z) = F(\log|z|^2) - \log(1+|z|^2).$$

The function F is continuous and nondecreasing. Since $-\log(-r) < r+2$ at $r=-1$, there is $\delta > 0$ with $F(r) = r+2$ for $r \ge -1-\delta$, and since $r+2+\log(-r) \to -\infty$ as $r \to -\infty$, we have $F(r) = -\log(-r)$ for all sufficiently negative r.

We first show that $F(\log|z|^2)$ is subharmonic on $\mathbb{C}\setminus\{0\}$. On the open set $\{|z|^2 > \mathrm{e}^{-1-\delta}\}$ it equals the harmonic function $\log|z|^2+2$. On the open set $\{0<|z|^2<\mathrm{e}^{-1}\}$ it equals $\max\{g(\log|z|^2), \log|z|^2+2\}$, where $g(t) = -\log(-t)$ for $t<0$. Since $g''(t) = t^{-2} > 0$, the function g is convex, so it is the supremum of its tangent lines $t \mapsto g(s)+g'(s)(t-s)$, $s<0$, whose slopes $g'(s) = -1/s$ are positive. Hence $g(\log|z|^2)$ is a continuous supremum of the harmonic functions $g(s)+g'(s)(\log|z|^2 - s)$; it therefore satisfies the submean inequality and is subharmonic, and so is its maximum with $\log|z|^2+2$. Since subharmonicity is a local property, $F(\log|z|^2)$ is subharmonic on $\mathbb{C}\setminus\{0\}$; it tends to $-\infty$ at 0 and extends subharmonically across the polar set $\{0\}$ by Theorem 3.5.2.

Next we check that u is ω-plurisubharmonic. For $|z|^2 \ge \mathrm{e}^{-1}$ we have $u = 2-\log(1+|z|^{-2})$. In the coordinate $\zeta = 1/z$ near ∞ this reads $u = 2-\log(1+|\zeta|^2)$, while $\omega = \mathrm{dd^c}\log(1+|\zeta|^2)$, because $\log(1+|z|^2) = \log(1+|\zeta|^2) - \log|\zeta|^2$ and $\log|\zeta|^2$ is pluriharmonic for $\zeta \ne 0$; thus u is smooth near ∞, with $u(\infty) = 2$ and $\omega + \mathrm{dd^c}u = 0$ there. Hence $u \in \mathrm{PSH}(\mathbb{P}^1, \omega)$, with $\omega + \mathrm{dd^c}u = \mathrm{dd^c}F(\log|z|^2)$ on $\mathbb{C}$, and its only pole is 0.

To compute the mass of u, note that its Lelong number at 0 is that of the subharmonic function $F(\log|z|^2)$, which differs from u by a smooth function, and since F is continuous and nondecreasing, $\sup_{|z|<\mathrm{e}^{t/2}} F(\log|z|^2) = F(t)$. By Proposition 12.1.1, the Lelong number is therefore

$$\lim_{t\to-\infty} \frac{F(t)}{t} = \lim_{t\to-\infty} \frac{-\log(-t)}{t} = 0.$$

By Theorem 12.1.3, in one variable the mass that $\mathrm{dd^c}F(\log|z|^2)$ gives to 0 is this Lelong number, namely the limit of the masses of the discs $B(0,r)$ as $r \downarrow 0$, so the current $\omega + \mathrm{dd^c}u$ has no atom at its pole. By Proposition 15.3.6, μ_u is the restriction of $\omega + \mathrm{dd^c}u$ to $\{u > -\infty\}$, hence the mass of u is $\int_{\mathbb{P}^1}\langle\omega+\mathrm{dd^c}u\rangle = \int_{\mathbb{P}^1}\omega = 1$.

The potential 0 is nonpositive, less singular than u, and has the same mass $\int_{\mathbb{P}^1}\omega = 1$, so Theorem 16.4.7 gives $P_\omega[u] \ge 0$; since also $P_\omega[u] \le V_\omega = 0$, we obtain $P_\omega[u] = 0$. Near 0, however, $u = -\log(-\log|z|^2) + O(1)$ is unbounded below. Thus the singularity envelope can change the singularity type of a potential while preserving its full mass.

Definition 16.4.9 Let $u, \phi \in \mathrm{PSH}(X,\theta)$ with $u \preceq \phi$. The *singularity envelope of u relative to ϕ* is

$$P_\theta[u](\phi) = \left(\lim_{C\to\infty} P_\theta(u+C, \phi)\right)^*.$$

Choose C_0 with $u - C_0 \le \phi$. For $C \ge -C_0$, the potential $u - C_0$ is a common minorant of $u + C$ and ϕ, so the rooftop envelope $P_\theta(u + C, \phi)$ lies between $u - C_0$ and $u + C$ and has the singularity type of u. These rooftop envelopes increase with C and are bounded above by ϕ, so the regularized limit is a potential between $u - C_0$ and ϕ.

Proposition 16.4.10 *Let $u, \phi \in \mathrm{PSH}(X, \theta)$ with $u \preceq \phi$ and $\int_X \langle \theta_\phi^n \rangle > 0$. Then the following are equivalent:*

(1) $u \in \mathcal{E}(X, \theta, \phi)$;
(2) $P_\theta[u](\phi) = \phi$;
(3) $P_\theta[u] = P_\theta[\phi]$.

Furthermore, the relative envelope $w = P_\theta[u](\phi)$ has the same mass as u, and

$$\mu_w \le \mathbf{1}_{\{w=\phi\}} \mu_\phi .$$

Proof Write $w_C = P_\theta(u + C, \phi)$. Each w_C with C large has the singularity type of u, hence mass $\int_X \langle \theta_u^n \rangle$, and w_C increases almost everywhere to w; by Corollary 15.7.6, $\int_X \langle \theta_w^n \rangle = \int_X \langle \theta_u^n \rangle$. By Corollary 16.2.6, together with $w_C \le \phi$ and $w_C \le w \le \phi$,

$$\mu_{w_C} \le \mathbf{1}_{\{u+C \le \phi\}} \mu_u + \mathbf{1}_{\{w=\phi\}} \mu_\phi .$$

As $C \to \infty$, the sets $\{u + C \le \phi\}$ decrease to the pole set of u, which is μ_u-null, because ϕ is bounded above; so the first term tends to zero in total variation by dominated convergence. Moreover $\mu_{w_C} \to \mu_w$ weakly by Corollary 15.7.6. Integrating continuous functions $\chi \ge 0$ and applying Lemma 16.2.1, we obtain the asserted measure bound.

If (1) holds, then w and ϕ have the same positive mass and both belong to $\mathcal{E}(X, \theta, \phi)$. By the measure bound, μ_w gives no mass to $\{w < \phi\}$, and the domination principle, Theorem 16.4.6, gives $w \ge \phi$; hence (2). Conversely, if (2) holds, then u has the mass of $w = \phi$, which is (1).

To prove that (2) implies (3), normalize $\phi \le 0$. This is harmless: replacing ϕ by $\phi + c$ replaces w_C by $P_\theta(u + C - c, \phi) + c$, hence w by $w + c$, so (2) is preserved, and $P_\theta[u]$ and $P_\theta[\phi]$ do not change. Every w_C is then a nonpositive potential with the singularity type of u, hence a competitor for $P_\theta[u]$, so $\phi = w \le P_\theta[u]$. Since (2) implies (1), u has the positive mass of ϕ, so $P_\theta[u]$ is a model potential by Theorem 16.4.7. Monotonicity and $u \preceq \phi$ now give

$$P_\theta[\phi] \le P_\theta[P_\theta[u]] = P_\theta[u] \le P_\theta[\phi].$$

Finally, if (3) holds, then by Proposition 16.3.2 the potential u has the mass of $P_\theta[u] = P_\theta[\phi]$, which is the mass of ϕ; this is (1). □

16.5 Rooftops and mass inequalities

Two potentials need not have a common minorant, as Example 16.1.7 shows. We prove that a strict mass inequality guarantees one, and that the masses of u, v, $\max\{u,v\}$ and $P_\theta(u,v)$ satisfy the inequality of Theorem 16.5.5. Contact concentration for model potentials gives these results for model potentials, and extrapolation transfers them to arbitrary potentials of full mass relative to model potentials. As in the preceding section, $\{\theta\}$ is big.

Proposition 16.5.1 *Let $u, v \in \mathrm{PSH}(X,\theta)$ satisfy*

$$\int_X \langle \theta_u^n \rangle + \int_X \langle \theta_v^n \rangle > \int_X \langle \theta^n_{\max\{u,v\}} \rangle .$$

Then $P_\theta(u,v) \neq -\infty$.

Proof Subtracting the same constant from u and v changes neither the hypothesis nor the conclusion, so we may assume that $u, v \leq 0$. Put $m = \int_X \langle \theta^n_{\max\{u,v\}} \rangle$ and $\phi = P_\theta[\max\{u,v\}]$. By monotonicity $\int_X \langle \theta_u^n \rangle \leq m$ and $\int_X \langle \theta_v^n \rangle \leq m$, so the strict inequality forces both masses to be positive. Hence $m > 0$, and by Theorem 16.4.7, ϕ is a model potential of mass m. Moreover $u, v \leq \max\{u,v\} \leq \phi$, because the nonpositive potential $\max\{u,v\}$ is a competitor for ϕ.

Since μ_u and μ_v do not charge pole sets, $\mu_u(\{u > \phi - k\}) \to \int_X \langle \theta_u^n \rangle$ and $\mu_v(\{v > \phi - k\}) \to \int_X \langle \theta_v^n \rangle$ as $k \to \infty$. Choose $k > 0$ so large that

$$\mu_u(\{u > \phi - k\}) + \mu_v(\{v > \phi - k\}) > m.$$

For $C > k$ put

$$u_C = \max\{u, \phi - C\}, \quad v_C = \max\{v, \phi - C\}, \quad w_C = P_\theta(u_C, v_C).$$

Since $\phi - C$ is a common minorant, all three potentials have the singularity type of ϕ and mass m. As $C \to \infty$, w_C decreases to $P_\theta(u,v)$ by Proposition 16.1.6. Suppose that $P_\theta(u,v) = -\infty$, so that w_C decreases to $-\infty$ identically. Then $\sup_X w_C \to -\infty$ by Corollary 14.2.5, and Lemma 16.4.2 gives $w_C \leq \phi - k$ for all large C. On $\{w_C = u_C\}$ we then have $u_C \leq \phi - k$, and similarly for v_C, so Corollary 16.2.6 gives

$$\begin{aligned} m = \int_X \langle \theta^n_{w_C} \rangle &\leq \mu_{u_C}(\{u_C \leq \phi - k\}) + \mu_{v_C}(\{v_C \leq \phi - k\}) \\ &= 2m - \mu_u(\{u > \phi - k\}) - \mu_v(\{v > \phi - k\}) < m. \end{aligned}$$

For the equality we used that, since $C > k$, $u_C = u$ on the plurifine open set $\{u > \phi - k\}$, so $\mu_{u_C} = \mu_u$ there, and similarly for v. This contradiction shows that $P_\theta(u,v) \neq -\infty$. □

Proposition 16.5.2 *Let $\phi, \psi \in \mathrm{PSH}(X,\theta)$ with $P_\theta[\phi] = \phi$ and $P_\theta[\psi] = \psi$, and suppose that $w = P_\theta(\phi,\psi) \neq -\infty$. Then $P_\theta[w] = w$. Thus the rooftop envelope of two model potentials is a model potential whenever its mass is positive.*

Proof Since $w \leq \phi$, monotonicity gives $P_\theta[w] \leq P_\theta[\phi] = \phi$, and likewise $P_\theta[w] \leq \psi$. Thus $P_\theta[w]$ is a common minorant of ϕ and ψ, and $P_\theta[w] \leq w$. Conversely, $w \leq \phi \leq 0$, so w is a competitor for $P_\theta[w]$. □

Proposition 16.5.3 *Let $\phi, \psi \in \mathrm{PSH}(X,\theta)$ be such that their rooftop envelope $w = P_\theta(\phi,\psi)$ is a potential of positive mass. If $u \in \mathcal{E}(X,\theta,\phi)$ and $v \in \mathcal{E}(X,\theta,\psi)$, then $P_\theta(u,v)$ is a potential with the same mass as w.*

Proof Shifting u and v by constants affects neither the existence of a common minorant nor the mass of $P_\theta(u,v)$. Indeed, a common minorant can be shifted down by the largest absolute change, and by Proposition 16.1.4(4) the rooftop envelopes of the original and of the shifted pair differ by a bounded amount, because their obstacles do. We may therefore assume that $u \leq \phi$ and $v \leq \psi$.

We first replace ϕ by u, and then ψ by v; both steps combine extrapolation with Proposition 16.5.1. Since $w \leq \phi$ and $w \leq \psi$, the masses of ϕ and ψ are at least that of w, hence positive. For $a > 1$ let $u_a = E_a(u,\phi)$. Since u and ϕ have the same positive mass, Theorem 16.4.4 shows that u_a also has this mass and that $a^{-1}u_a + (1-a^{-1})\phi \leq u$; moreover $u_a \leq u \leq \phi$, because the obstacle $u + (a-1)(u-\phi)$ of u_a is at most u. Since $\max\{u_a, w\} \leq \phi$ and $\int_X \langle \theta_w^n \rangle > 0$,

$$\int_X \langle \theta_{u_a}^n \rangle + \int_X \langle \theta_w^n \rangle > \int_X \langle \theta_\phi^n \rangle \geq \int_X \langle \theta^n_{\max\{u_a,w\}} \rangle.$$

By Proposition 16.5.1, the rooftop envelope $P_\theta(u_a, w)$ is a potential. By associativity and $u_a \leq \phi$, it equals $P_\theta(u_a,\psi)$. The potential

$$a^{-1}P_\theta(u_a,\psi) + (1-a^{-1})w$$

lies below $a^{-1}u_a + (1-a^{-1})\phi \leq u$ and below ψ, so it is a common minorant of u and ψ. Hence $z = P_\theta(u,\psi)$ is a potential, and positivity and monotonicity of mass give $\int_X \langle \theta_z^n \rangle \geq (1-a^{-1})^n \int_X \langle \theta_w^n \rangle$. On the other hand $z \leq P_\theta(\phi,\psi) = w$, so the mass of z is at most that of w. Letting $a \to \infty$, we see that z and w have the same positive mass.

Now let $v_a = E_a(v,\psi)$. In the same way, $v_a \leq v \leq \psi$, the mass of v_a is that of ψ, and $a^{-1}v_a + (1-a^{-1})\psi \leq v$. Since $\max\{v_a, z\} \leq \psi$ and z has positive mass, we have

$$\int_X \langle \theta_{v_a}^n \rangle + \int_X \langle \theta_z^n \rangle > \int_X \langle \theta_\psi^n \rangle \geq \int_X \langle \theta^n_{\max\{v_a,z\}} \rangle,$$

so Proposition 16.5.1 shows that $P_\theta(z, v_a)$ is a potential, and by associativity and $v_a \leq \psi$ it equals $P_\theta(u, v_a)$. The potential

$$a^{-1}P_\theta(u,v_a) + (1-a^{-1})z$$

lies below u and below $a^{-1}v_a + (1-a^{-1})\psi \le v$, and its mass is at least $(1-a^{-1})^n \int_X \langle \theta_z^n \rangle$. Hence $P_\theta(u,v)$ is a potential whose mass is at least $(1-a^{-1})^n \int_X \langle \theta_w^n \rangle$, for every $a > 1$. Since $P_\theta(u,v) \le w$, its mass is at most that of w, and letting $a \to \infty$ completes the proof. □

The proof of the mass inequality for rooftop envelopes passes to the limit along a decreasing sequence of potentials. The following lemma shows that the mass that nonpositive potentials put on their zero sets can only increase in such a limit.

Lemma 16.5.4 *Let $q_j \in \mathrm{PSH}(X,\theta)$ be nonpositive potentials decreasing to a potential q. Then*

$$\overline{\lim_{j\to\infty}}\, \mu_{q_j}(\{q_j = 0\}) \le \mu_q(\{q = 0\}).$$

Proof Put $\lambda_j = \mathbf{1}_{\{q_j=0\}}\mu_{q_j}$. Since $q_j \le V_\theta \le 0$, the set $\{q_j = 0\}$ is contained in $\{q_j = V_\theta\}$, and Lemma 16.2.2 gives $\lambda_j \le \mu_{V_\theta}$. Let A be the proper analytic subset of Lemma 16.2.3: potentials with minimal singularities are locally bounded on $X \setminus A$, and $\mu_{V_\theta}(A) = 0$.

For $t > 0$ put $b_j = \max\{1 + tq_j, 0\}$ and $b = \max\{1 + tq, 0\}$, with value zero at the poles. Fix a coordinate ball $D \Subset X \setminus A$ with a smooth potential ρ of θ near $\overline{D}$ (Lemma 7.3.1) and a number $L > 1/t$, and put $\widetilde{q}_j = \max\{q_j, V_\theta - L\}$ and $\widetilde{q} = \max\{q, V_\theta - L\}$. On D the functions $\widetilde{q}_j$ are uniformly bounded and decrease to $\widetilde{q}$, so they converge in capacity (Proposition 9.4.6). Since $V_\theta - L < -1/t$, the weight b_j vanishes on $\{q_j \le V_\theta - L\}$, and on the complementary plurifine open set $\widetilde{q}_j = q_j$; hence $b_j\mu_{q_j} = b_j\mu_{\widetilde{q}_j}$ by plurifine locality, and likewise $b\mu_q = b\mu_{\widetilde{q}}$. Moreover $b_j = \max\{1 + t\widetilde{q}_j, 0\}$ and $b = \max\{1 + t\widetilde{q}, 0\}$ on D, so the weights are bounded and quasi-continuous and converge in capacity to b. The weighted form of Theorem 9.4.8, applied to the bounded plurisubharmonic functions $\rho + \widetilde{q}_j$, therefore gives

$$b_j\mu_{q_j} = b_j\mu_{\widetilde{q}_j} \longrightarrow b\mu_{\widetilde{q}} = b\mu_q \quad \text{weakly on } D.$$

Let $\delta > 0$. Since $\mu_{V_\theta}(A) = 0$, Lemma 16.2.1(1) gives a compact set $K \subset X \setminus A$ with $\mu_{V_\theta}(X \setminus K) < \delta$; choose a continuous function $0 \le \zeta \le 1$ with compact support in $X \setminus A$ and $\zeta = 1$ on K (Proposition 7.1.3). Since $b_j = 1$ on $\{q_j = 0\}$ and $\lambda_j \le \mu_{V_\theta}$,

$$\lambda_j(X) \le \int_X \zeta b_j\, \mu_{q_j} + \delta.$$

Cover the support of ζ by finitely many balls D_l as above, and write $\zeta = \sum_l \zeta_l$ with continuous functions $\zeta_l \ge 0$ with compact support in D_l (for instance $\zeta_l = \zeta\psi_l / \sum_k \psi_k$, where $\psi_l \ge 0$ is continuous with compact support in D_l and $\sum_k \psi_k > 0$ on the support of ζ). The weak convergence on each D_l gives $\int_X \zeta b_j\, \mu_{q_j} \to \int_X \zeta b\, \mu_q \le \int_X b\, \mu_q$, and letting then $\delta \downarrow 0$ gives $\overline{\lim}_j\, \lambda_j(X) \le \int_X b\, \mu_q$. Finally, as $t \to \infty$ the weights b, which depend on t, decrease to $\mathbf{1}_{\{q=0\}}$, because $q \le 0$; dominated convergence for the finite measure μ_q proves the lemma. □

Theorem 16.5.5 *Let $u, v \in \mathrm{PSH}(X,\theta)$ with $w = P_\theta(u,v) \ne -\infty$. Then*

$$\int_X \langle \theta_u^n \rangle + \int_X \langle \theta_v^n \rangle \leq \int_X \langle \theta^n_{\max\{u,v\}} \rangle + \int_X \langle \theta_w^n \rangle.$$

Proof We first reduce to model potentials, and then apply contact concentration to the rooftop envelopes of their truncations.

If u has mass zero, the inequality follows from the monotonicity of mass, $\int_X \langle \theta_v^n \rangle \leq \int_X \langle \theta^n_{\max\{u,v\}} \rangle$, and similarly if v has mass zero. Otherwise, subtracting the same constant from u and v, which subtracts it from $\max\{u, v\}$ and w and changes no mass, we may assume that $u, v \leq 0$. Put $\phi = P_\theta[u]$ and $\psi = P_\theta[v]$; by Theorem 16.4.7, these are model potentials with the same masses as u and v, respectively, and $u \leq \phi$, $v \leq \psi$, since u and v are competitors for their own singularity envelopes. Put also $\gamma = \max\{\phi, \psi\}$ and $r = P_\theta(\phi, \psi)$.

The four masses in the inequality do not change when u, v are replaced by ϕ, ψ. Indeed, for every integer C the potential $\max\{P_\theta(u + C, V_\theta), P_\theta(v + C, V_\theta)\}$ has the singularity type of $\max\{u, v\}$, by Proposition 16.1.6, and by Proposition 16.3.2(2) these potentials increase almost everywhere to γ as $C \to \infty$; so γ has the same mass as $\max\{u, v\}$, by Corollary 15.7.6. Moreover w is a common minorant of ϕ and ψ, so $r \geq w$ is a potential. If r has mass zero, then so has w, by monotonicity. If the mass of r is positive, then w has the same mass as r by Proposition 16.5.3, applied to $u \in \mathcal{E}(X, \theta, \phi)$ and $v \in \mathcal{E}(X, \theta, \psi)$. It therefore suffices to prove the theorem for the model potentials ϕ and ψ, whose maximum is γ and whose rooftop envelope is r.

Put $m = \int_X \langle \theta_\gamma^n \rangle$, and for $C > 0$ let

$$\phi_C = \max\{\phi, \gamma - C\}, \quad \psi_C = \max\{\psi, \gamma - C\}, \quad r_C = P_\theta(\phi_C, \psi_C).$$

These potentials have the singularity type of γ, hence mass m, and r_C decreases to r as $C \to \infty$. By Proposition 16.3.3, the measure μ_ϕ is carried by $\{\phi = 0\}$, which is contained in the plurifine open set $\{\phi > \gamma - C\}$, where $\phi_C = \phi$. Plurifine locality therefore gives

$$\mu_{\phi_C} = \mu_\phi + \mathbf{1}_{\{\phi \leq \gamma - C\}} \mu_{\phi_C},$$

and the last summand has mass $m - \int_X \langle \theta_\phi^n \rangle$; the analogous decomposition holds for ψ_C. By Corollary 16.2.6, $\mu_{r_C} \leq \mathbf{1}_{\{r_C = \phi_C\}} \mu_{\phi_C} + \mathbf{1}_{\{r_C = \psi_C\}} \mu_{\psi_C}$. On the set $\{r_C < 0\}$, the summand μ_ϕ of μ_{ϕ_C} contributes nothing to this bound, because at a point of $\{\phi = 0\} \cap \{r_C = \phi_C\}$ we have $r_C = \phi_C = 0$; the same holds for μ_ψ. Consequently

$$\mu_{r_C}(\{r_C < 0\}) \leq 2m - \int_X \langle \theta_\phi^n \rangle - \int_X \langle \theta_\psi^n \rangle.$$

Since $r_C \leq 0$ and $\int_X \langle \theta_{r_C}^n \rangle = m$, it follows that

$$\mu_{r_C}(\{r_C = 0\}) \geq \int_X \langle \theta_\phi^n \rangle + \int_X \langle \theta_\psi^n \rangle - m.$$

Applying Lemma 16.5.4 along the integers $C \to \infty$ to the potentials r_C, which decrease to r, we find that

$$\int_X \langle \theta_r^n \rangle \geq \mu_r(\{r = 0\}) \geq \int_X \langle \theta_\phi^n \rangle + \int_X \langle \theta_\psi^n \rangle - m,$$

which is the assertion for ϕ and ψ. □

16.6 Divisorial constraints under envelopes

Multiplier ideals give another way of comparing singularities. By the valuative criterion of Theorem 12.6.9, inclusions between multiplier ideals are governed by generic Lelong numbers along divisors over X. We identify the resulting order and show that it is preserved by upper envelopes, which requires pulling envelopes back to smooth models.

In this section $\{\theta\}$ may be any pseudo-effective class. A prime divisor over X is a prime divisor E on a smooth model of X, as in Definition 12.6.3, and $\nu(u, E)$ denotes its generic Lelong number (Proposition 12.3.5). Since multiplier ideals are local, the valuative criterion uses prime divisors over small neighborhoods of a point; accordingly, in this section a prime divisor over X may also be a prime divisor over an open subset $U \subset X$, that is, a prime divisor on a smooth model of U, and $\nu(u, E)$ then refers to the restriction of u to U. Multiplier ideals and Lelong numbers of quasi-plurisubharmonic functions are computed from local plurisubharmonic functions that differ from them by smooth functions (see the definitions following Lemma 13.1.4), so the results of Chapter 12 apply to them.

Proposition 16.6.1 *For quasi-plurisubharmonic functions u, v on X, the following are equivalent:*

(1) $\mathcal{I}(tv) \subseteq \mathcal{I}(tu)$ *for every real $t > 0$;*
(2) $\mathcal{I}(tv) \subseteq \mathcal{I}(tu)$ *for every positive integer t;*
(3) $\nu(v, E) \geq \nu(u, E)$ *for every prime divisor E over X.*

We write $v \preceq_{\mathcal{I}} u$ when these conditions hold, and $v \sim_{\mathcal{I}} u$ when $v \preceq_{\mathcal{I}} u$ and $u \preceq_{\mathcal{I}} v$. In particular, equality of the multiplier ideals of all positive integer multiples is equivalent to equality for all positive real multiples, and to equality of all divisorial Lelong numbers.

Proof Condition (2) is a special case of (1). Assume (2) and let E be a prime divisor over an open subset $U \subset X$ (possibly $U = X$). The inclusions (2) restrict to U, and a smaller ideal has a larger order along E, so $\operatorname{ord}_E \mathcal{I}(tv) \geq \operatorname{ord}_E \mathcal{I}(tu)$ for every positive integer t. Dividing by t and letting $t \to \infty$, Corollary 12.6.8, applied on U, gives $\nu(v, E) \geq \nu(u, E)$; this is (3). Conversely, assume (3), let $t > 0$, and let f be a germ of $\mathcal{I}(tv)$ at a point x. The valuative criterion, Theorem 12.6.9, gives $\varepsilon > 0$ and a neighborhood V of x such that

$$\operatorname{ord}_E f + A_X(E) \geq (1 + \varepsilon) t \nu(v, E) \geq (1 + \varepsilon) t \nu(u, E)$$

for every prime divisor E over an open neighborhood of x contained in V whose center contains x; the second inequality is (3) for these divisors. Since $t\nu(u,E) = \nu(tu,E)$ by Proposition 12.1.4, the same criterion, applied to tu, shows that $f \in \mathcal{I}(tu)$; this is (1). Applying the equivalence in both directions proves the assertions about equalities. □

Lemma 16.6.2 *Let $(v_j)_{j\in J}$ be a nonempty family in $\mathrm{PSH}(X,\theta)$ that is uniformly bounded above, and let $v = (\sup_j v_j)^*$. Then*

$$\nu(v,E) = \inf_j \nu(v_j,E)$$

for every prime divisor E over X. Consequently, maxima, sums and positive multiples preserve the relation $\sim_{\mathcal{I}}$; and if two such families are termwise equivalent for $\sim_{\mathcal{I}}$, their upper envelopes are equivalent as well.

Proof Let $\pi\colon Y \to U$ be a smooth model of an open subset $U \subset X$ (possibly $U = X$) on which E is a divisor, and let $Z \subset Y$ be a nowhere dense analytic subset outside which π is a biholomorphism onto an open subset of U; such a set exists by the definition of a smooth model, and it has Lebesgue measure zero. We first show that $v \circ \pi$ is the regularized supremum of the family $(v_j \circ \pi)$. By Theorem 8.3.2, a countable subfamily has the same regularized supremum v, and its increasing finite maxima w_k satisfy $\sup_k w_k = v$ almost everywhere. The functions $w_k \circ \pi$ are $\pi^*\theta$-plurisubharmonic, increase with k, and are locally uniformly bounded above; let W be their regularized supremum, so that $W = \sup_k w_k \circ \pi$ almost everywhere by Theorem 8.3.2. If $N \subset U$ is a Lebesgue null set outside which $\sup_k w_k = v$, then $\pi^{-1}(N) \setminus Z$ is a null set, because π restricts to a biholomorphism of $Y \setminus Z$ onto an open set; hence $\sup_k w_k \circ \pi = v \circ \pi$ outside the null set $(\pi^{-1}(N) \setminus Z) \cup Z$. Thus the quasi-plurisubharmonic functions W and $v \circ \pi$ agree almost everywhere, and therefore everywhere. Since every v_j is at most v, we get $W \le (\sup_j v_j \circ \pi)^* \le v \circ \pi = W$, so $v \circ \pi$ is the regularized supremum of the whole family $(v_j \circ \pi)$.

It remains to compare Lelong numbers. Near a point $y \in E$, adding to all these functions the pullback of a smooth local potential of θ near $\pi(y)$ makes them plurisubharmonic, so Proposition 12.1.4 gives $\nu(v \circ \pi, y) = \inf_j \nu(v_j \circ \pi, y)$ at every point $y \in E$. Taking the infimum over $y \in E$, which defines the generic Lelong number (Proposition 12.3.5) and commutes with the infimum over j, we obtain the formula.

For the remaining assertions, let φ and ψ be quasi-plurisubharmonic functions on X, and recall from Proposition 12.1.4 that Lelong numbers at a point are additive and positively homogeneous, and that the Lelong number of a maximum is the minimum of the Lelong numbers. Taking infima over $y \in E$, we obtain $\nu(c\varphi, E) = c\nu(\varphi, E)$ for $c > 0$ and $\nu(\max\{\varphi,\psi\}, E) = \min\{\nu(\varphi,E), \nu(\psi,E)\}$. For sums we use that, by Proposition 12.3.5, the infima defining $\nu(\varphi,E)$, $\nu(\psi,E)$ and $\nu(\varphi+\psi,E)$ are attained at every point of E outside a countable union of proper analytic subsets of E. Such a point exists, since by the same proposition the complement in E of such a union is dense in E, and in particular nonempty. At a point of E where the three infima are

attained, the additivity at a point gives $\nu(\varphi+\psi,E)=\nu(\varphi,E)+\nu(\psi,E)$. By these three rules and Proposition 16.6.1, maxima, sums and positive multiples preserve $\sim_{\mathcal{I}}$, and by the formula just proved, so do upper envelopes of termwise equivalent families. □

16.7 Singularity constraints and relative capacity

We now combine a singularity constraint with an obstacle. For a bounded obstacle the resulting envelope stays within a bounded distance of the singularity envelope $P_\theta[u]$, and it satisfies a contact bound. We then introduce the capacity relative to a reference potential ϕ, defined through the potentials in the strip $\phi-1\le v\le\phi$, and use it to measure the deep sublevel sets of potentials of full mass relative to ϕ. Throughout this section $\{\theta\}$ is big.

Definition 16.7.1 For $u\in\mathrm{PSH}(X,\theta)$ and a Borel obstacle f, define

$$P_\theta[u](f)=\left(\sup\{v\in\mathrm{PSH}(X,\theta)\,:\,v\preceq u,\ v\le f\ \text{quasi-everywhere}\}\right)^*$$

when this family is nonempty and uniformly bounded above, and put $P_\theta[u](f)=-\infty$ when it is empty. For Borel obstacles g and f we write $P_\theta(g,f)=P_\theta(\min\{g,f\})$, under the same conditions on the family of competitors.

By the rooftop representation in Proposition 16.7.2 below, this is consistent with Definition 16.4.9 when f is a potential ϕ with $u\preceq\phi$.

Proposition 16.7.2 *Let $u\in\mathrm{PSH}(X,\theta)$, and let f be a Borel obstacle for which the family in Definition 16.7.1 is nonempty and uniformly bounded above.*

(1) *$P_\theta[u](f)\le f$ quasi-everywhere. The envelope increases with f and with the singularity type of u, does not change when f is modified on a pluripolar set, and satisfies $P_\theta[u](f+c)=P_\theta[u](f)+c$ for $c\in\mathbb{R}$.*

(2) *For all sufficiently large k, the rooftop envelopes $P_\theta(u+k,f)$ are potentials, and*

$$P_\theta[u](f)=\left(\lim_{k\to\infty}P_\theta(u+k,f)\right)^*.$$

(3) *If $|f|\le C$ quasi-everywhere, then*

$$P_\theta[u]-C\le P_\theta[u](f)\le P_\theta[u]+C,$$

and $P_\theta[u](f)$ has the same mass as u.

In particular, $P_\theta[u](0)=P_\theta[u]$ and $P_\theta[V_\theta](f)=P_\theta(f)$ whenever these envelopes are defined.

Proof We begin with the final identities. A competitor for $P_\theta[u](0)$ is a nonpositive potential more singular than u, exactly as in Definition 16.3.1; and every potential is more singular than V_θ, by minimality, so the families defining $P_\theta[V_\theta](f)$ and $P_\theta(f)$ coincide.

(1) The maximum of two members of the family in Definition 16.7.1 is again a member. By Theorem 8.3.2, applied on finitely many coordinate balls on which θ has smooth local potentials, there are members $v_1, v_2, \dots$ with $(\sup_k v_k)^* = P_\theta[u](f)$, and the maxima $w_k = \max\{v_1, \dots, v_k\}$ form an increasing sequence of members with the same regularized supremum. By Corollary 10.2.2, $\sup_k w_k = P_\theta[u](f)$ outside a pluripolar set, and each w_k lies below f outside a pluripolar set; since countable unions of pluripolar sets are pluripolar (Theorem 10.1.4), $P_\theta[u](f) \le f$ quasi-everywhere. (The envelope itself need not be more singular than u; see Example 16.4.8.) The other assertions follow from the corresponding properties of the families of competitors.

(2) A competitor h for the obstacle $\min\{u + k, f\}$ satisfies $h \le u + k$ quasi-everywhere, hence everywhere, and $h \le f$ quasi-everywhere; so it belongs to the family of Definition 16.7.1. In particular these competitors are uniformly bounded above, $P_\theta(u + k, f)$ is defined, and when it is a potential, it belongs to the family. Conversely, every member v of the family satisfies $v \le u + k$ for all sufficiently large integers k, hence $v \le \min\{u + k, f\}$ quasi-everywhere and $v \le P_\theta(u + k, f)$. Thus a fixed member shows that the rooftop envelopes, which increase with k, are potentials for all large k, and the family and the rooftop envelopes have the same supremum. This proves the representation.

(3) Normalize $u \le 0$. If $|f| \le C$ quasi-everywhere, then $u - C$ is a competitor, and for large k each rooftop envelope $P_\theta(u + k, f)$ lies between $u - C$ and $u + k$, so it has the singularity type of u and mass $\int_X \langle \theta_u^n \rangle$. By (2) and Corollary 15.7.6, $P_\theta[u](f)$ also has this mass. Comparing f with the constants $-C$ and C, and using (1) together with $P_\theta[u](0) = P_\theta[u]$, gives the two bounds. □

Proposition 16.7.3 *Let $u \in \mathrm{PSH}(X, \theta)$, and let f be a bounded quasi-continuous obstacle. Then $w = P_\theta[u](f)$ is a potential, and*

$$\mu_w \le \mathbf{1}_{\{w = P_\theta(f)\}} \mu_{P_\theta(f)}, \qquad \mu_w(X \setminus \{w = f\}) = 0.$$

In particular $\int_X (f - w)\, \mu_w = 0$, where the integrand may be given any value at the poles of w.

Proof Write $p = P_\theta(f)$ and choose C with $|f| \le C$. Then $V_\theta - C$ is a competitor for f, and every competitor v satisfies $v - C \le 0$, hence $v \le V_\theta + C$; thus $V_\theta - C \le p \le V_\theta + C$. Consequently p has minimal singularities, and its mass is that of V_θ, namely $\mathrm{vol}(\{\theta\}) > 0$, by Definition 15.8.1 and Proposition 15.8.2. A potential lies below f quasi-everywhere if and only if it lies below p: one direction is the maximality of p, the other holds because $p \le f$ quasi-everywhere. Hence the two constrained families coincide, and $w = P_\theta[u](p)$. Since $u \preceq p$, this is the relative envelope of Definition 16.4.9, by Proposition 16.7.2. The measure bound in Proposition 16.4.10, which applies because p has positive mass, gives the first inequality. By Theorem 16.2.5, μ_p is carried by $\{p = f\}$, so μ_w is carried by

$\{w = p\} \cap \{p = f\} \subset \{w = f\}$. Finally, $f - w$ is nonnegative quasi-everywhere and vanishes μ_w-almost everywhere, which gives the integral identity. □

Proposition 16.7.4 *Let $u, v, \phi \in \mathrm{PSH}(X, \theta)$, let p be an integer with $1 \le p \le n$, and let $T_1, \dots, T_{n-p}$ be closed positive $(1,1)$-currents on X. For $q \in \mathrm{PSH}(X, \theta)$ put*

$$\lambda_q = \left\langle (\theta + \mathrm{dd^c} q)^p \wedge T_1 \wedge \cdots \wedge T_{n-p} \right\rangle .$$

If $u, v \preceq \phi$, then

$$\lambda_v(\{u < v\}) \le \lambda_u(\{u < v\}) + \lambda_\phi(X) - \lambda_u(X).$$

If moreover $\int_X \langle \theta_\phi^n \rangle > 0$ and $u, v \in \mathcal{E}(X, \theta, \phi)$, then $\lambda_u(X) = \lambda_\phi(X)$, and the inequality also holds with $\{u \le v\}$ in place of $\{u < v\}$.

Proof For $\varepsilon > 0$ put $q = \max\{u, v - \varepsilon\}$. Write $T_i = \theta_i' + \mathrm{dd^c}\varphi_i$ with smooth closed real $(1,1)$-forms θ_i' and $\varphi_i \in \mathrm{PSH}(X, \theta_i')$ (Theorem 14.1.7). Since $q \preceq \phi$, the monotonicity of mixed masses in Corollary 15.7.5, applied with n factors to the potentials $q, \dots, q, \varphi_1, \dots, \varphi_{n-p}$ and $\phi, \dots, \phi, \varphi_1, \dots, \varphi_{n-p}$, gives $\lambda_q(X) \le \lambda_\phi(X)$. On the disjoint plurifine open sets $\{u < v - \varepsilon\}$ and $\{u > v - \varepsilon\}$, q equals $v - \varepsilon$ and u respectively, so plurifine locality gives

$$\lambda_v(\{u < v - \varepsilon\}) \le \lambda_\phi(X) - \lambda_u(\{u > v - \varepsilon\}).$$

Writing $\lambda_u(\{u > v - \varepsilon\}) = \lambda_u(X) - \lambda_u(\{u \le v - \varepsilon\})$, enlarging $\{u \le v - \varepsilon\}$ to $\{u < v\}$ up to the common pole set of u and v, which carries no mass, and letting $\varepsilon \downarrow 0$, we obtain the first inequality.

Now let $\int_X \langle \theta_\phi^n \rangle > 0$ and $u, v \in \mathcal{E}(X, \theta, \phi)$. By Proposition 16.4.10, $P_\theta[u] = P_\theta[v] = P_\theta[\phi]$. With the potentials φ_i of the currents T_i chosen above, the preservation of mixed masses in Proposition 16.3.2, applied to the n potentials $u, \dots, u, \varphi_1, \dots, \varphi_{n-p}$ and to the corresponding tuples for v and ϕ, gives $\lambda_u(X) = \lambda_v(X) = \lambda_\phi(X)$. Thus the last two terms in the inequality cancel. Applying the inequality with u and v interchanged, we get $\lambda_u(\{v < u\}) \le \lambda_v(\{v < u\})$, and subtracting this from $\lambda_v(X) = \lambda_u(X)$ gives $\lambda_v(\{u \le v\}) \le \lambda_u(\{u \le v\})$. □

Definition 16.7.5 For a reference potential $\phi \in \mathrm{PSH}(X, \theta)$ and a Borel set $A \subset X$, the *Monge–Ampère capacity relative to ϕ* is

$$\mathrm{Cap}_{\theta,\phi}(A) = \sup\{\mu_v(A) : v \in \mathrm{PSH}(X, \theta),\ \phi - 1 \le v \le \phi\}.$$

The potentials v in this supremum are called competitors for $\mathrm{Cap}_{\theta,\phi}$. For $\phi = V_\theta$ we obtain the *θ-capacity*.

Proposition 16.7.6 *Let $\phi \in \mathrm{PSH}(X, \theta)$. The relative capacity $\mathrm{Cap}_{\theta,\phi}$ is monotone, countably subadditive, zero on pluripolar Borel sets and continuous along increasing*

sequences of Borel sets. Its total value is $\mathrm{Cap}_{\theta,\phi}(X) = \int_X \langle \theta_\phi^n \rangle$*, and it is inner regular on Borel sets:*

$$\mathrm{Cap}_{\theta,\phi}(A) = \sup_{K \subset A \text{ compact}} \mathrm{Cap}_{\theta,\phi}(K).$$

It does not change when a constant is added to ϕ*. If* $\psi \in \mathrm{PSH}(X,\theta)$ *satisfies* $|\phi - \psi| \le C$ *outside the pole sets of* ϕ *and* ψ*, for a constant* $C \ge 0$*, then*

$$(2C+1)^{-n}\,\mathrm{Cap}_{\theta,\psi}(A) \le \mathrm{Cap}_{\theta,\phi}(A) \le (2C+1)^n\,\mathrm{Cap}_{\theta,\psi}(A).$$

Replacing (θ,ϕ) *by* $(\theta + \mathrm{dd^c} g, \phi - g)$ *for a smooth real function* g *leaves the capacity unchanged.*

Proof Every competitor has the singularity type of ϕ and hence mass $\int_X \langle \theta_\phi^n \rangle$; since ϕ itself is a competitor, the capacity is finite and $\mathrm{Cap}_{\theta,\phi}(X) = \int_X \langle \theta_\phi^n \rangle$. Monotonicity, countable subadditivity and continuity along increasing sequences hold for each measure μ_v and pass to the supremum; for the last property, interchange the supremum over the sequence with the supremum over competitors. Inner regularity follows in the same way from the inner regularity of each finite measure μ_v (Lemma 16.2.1): $\mathrm{Cap}_{\theta,\phi}(A) = \sup_v \sup_K \mu_v(K) = \sup_K \mathrm{Cap}_{\theta,\phi}(K)$, the inner suprema being taken over the compact sets $K \subset A$. Each μ_v gives no mass to pluripolar sets. Adding a constant c to ϕ corresponds to the bijection $v \mapsto v + c$ of competitors, which does not change μ_v.

For the comparison, note first that $\phi \le \psi + C$ and $\psi \le \phi + C$ hold almost everywhere, hence everywhere; in particular ϕ and ψ have the same pole set. Put $t = (2C+1)^{-1}$. If $\phi - 1 \le v \le \phi$, the potential $q = (1-t)\psi + tv - tC$ satisfies

$$q \le (1-t)\psi + t(\psi + C) - tC = \psi, \qquad q \ge (1-t)\psi + t(\psi - C - 1) - tC = \psi - 1,$$

so it is a competitor for $\mathrm{Cap}_{\theta,\psi}$, and $\mu_q \ge t^n \mu_v$. This proves the upper comparison, and exchanging ϕ and ψ proves the lower one. The last assertion follows from the bijection $v \mapsto v - g$ between the two families of competitors, which preserves the current $\theta + \mathrm{dd^c} v$. □

Proposition 16.7.7 *Let* $u, \phi \in \mathrm{PSH}(X,\theta)$ *with* $u \preceq \phi$*, and let* $s \in \mathbb{R}$ *and* $0 < t \le 1$*. Then*

$$t^n\,\mathrm{Cap}_{\theta,\phi}(\{u < \phi - s - t\}) \le \mu_u(\{u < \phi - s\}) + \int_X \langle \theta_\phi^n \rangle - \int_X \langle \theta_u^n \rangle.$$

In particular, if $u \in \mathcal{E}(X,\theta,\phi)$*, then* $\mathrm{Cap}_{\theta,\phi}(\{u < \phi - s\}) \to 0$ *as* $s \to \infty$*.*

Proof Let v be a competitor for $\mathrm{Cap}_{\theta,\phi}$, and put $q = (1-t)\phi + tv - s$. Then $\phi - s - t \le q \le \phi - s$, so

$$\{u < \phi - s - t\} \subset \{u < q\} \subset \{u < \phi - s\}.$$

Moreover $q \sim \phi$ and $\mu_q \ge t^n \mu_v$. Proposition 16.7.4, with $p = n$, applied to u and q, gives

$$t^n \mu_v(\{u < \phi - s - t\}) \le \mu_q(\{u < q\}) \le \mu_u(\{u < \phi - s\}) + \int_X \langle \theta_\phi^n \rangle - \int_X \langle \theta_u^n \rangle,$$

and taking the supremum over v proves the inequality. If $u \in \mathcal{E}(X, \theta, \phi)$, then $\int_X \langle \theta_u^n \rangle = \int_X \langle \theta_\phi^n \rangle$. With $t = 1$, the right side tends to zero as $s \to \infty$ by dominated convergence, because the sets $\{u < \phi - s\}$ decrease to the pole set of u, which is μ_u-null. □

16.8 Quantitative rooftop extrapolation

In Theorem 16.4.4 the potentials are ordered by $u \preceq v$. We now show that it suffices to order their singularity envelopes, $P_\theta[u] \le P_\theta[v]$. The obstacle $au + (1-a)v$ may then be unbounded above near the poles of v, so we must show separately that its competitors are uniformly bounded above. This follows from the uniform L^1 bound for normalized potentials, applied on a set of positive volume on which the obstacle is bounded. As before, $\{\theta\}$ is big.

Theorem 16.8.1 *Let $u, v \in \mathrm{PSH}(X, \theta)$ with $\int_X \langle \theta_u^n \rangle > 0$ and $P_\theta[u] \le P_\theta[v]$, and let $a > 1$ satisfy*

$$a^n \left(\int_X \langle \theta_v^n \rangle - \int_X \langle \theta_u^n \rangle \right) < \int_X \langle \theta_v^n \rangle.$$

Consider the obstacle $au + (1-a)v$, evaluated where u and v are both finite and given arbitrary Borel values on the remaining pluripolar set. Then $\eta = P_\theta(au + (1-a)v)$ is a potential of positive mass, and

$$a^{-1}\eta + (1 - a^{-1})v \le u, \qquad P_\theta[\eta] \le P_\theta[u].$$

If $P_\theta[u] = P_\theta[v]$, then every $a > 1$ is allowed, $P_\theta[\eta] = P_\theta[u]$, and η has the same mass as u.

Proof Write $\phi = P_\theta[u]$ and $\psi = P_\theta[v]$. Then $u \preceq \phi \le \psi$, and by Proposition 16.3.2 and the monotonicity of mass,

$$\int_X \langle \theta_v^n \rangle = \int_X \langle \theta_\psi^n \rangle \ge \int_X \langle \theta_\phi^n \rangle = \int_X \langle \theta_u^n \rangle > 0.$$

Hence Theorem 16.4.4, applied to u and ψ, gives a potential $\eta' = P_\theta(au + (1-a)\psi)$ of positive mass. Choose C with $v \le \psi + C$. Since $1 - a < 0$, we have $\eta' - (a-1)C \le au + (1-a)v$ quasi-everywhere, so $\eta' - (a-1)C$ is a competitor for the obstacle $au + (1-a)v$.

We show that the competitors are uniformly bounded above. Since u and v are finite almost everywhere, there is a Borel set K of positive ω^n-volume on which both are bounded; the obstacle is then bounded above on K by a constant D. Let h be a competitor. By Proposition 14.2.1, there is a constant B, depending only on X, θ and ω, such that

$$\int_X |h - \sup_X h|\,\omega^n \le B.$$

Since $h \le D$ almost everywhere on K, integration over K gives $\sup_X h \le D + B/\int_K \omega^n$. Hence the envelope η is a potential. It dominates $\eta' - (a-1)C$, so it has positive mass. The inequality $\eta \le au + (1-a)v$, which holds quasi-everywhere, is equivalent there to $a^{-1}\eta + (1-a^{-1})v \le u$; this inequality between quasi-plurisubharmonic functions therefore holds everywhere.

Next we compare singularity envelopes. Put $t = a^{-1}$. For all sufficiently large integers k, the potential $tP_\theta(\eta + k, 0) + (1-t)P_\theta(v + k, 0)$ is nonpositive and more singular than $t\eta + (1-t)v \le u$; here $P_\theta(\eta + k, 0) = P_\theta(\min\{\eta + k, 0\})$, as in Definition 16.7.1, and similarly for v. It is therefore a competitor for ϕ. By Proposition 16.7.2, $P_\theta(\eta + k, 0)$ and $P_\theta(v + k, 0)$ increase almost everywhere to $P_\theta[\eta]$ and ψ as $k \to \infty$. Hence

$$tP_\theta[\eta] + (1-t)\psi \le \phi$$

almost everywhere, and therefore everywhere. Since $\phi \le \psi$, subtracting $(1-t)\psi$ where ϕ is finite gives $P_\theta[\eta] \le \phi$ almost everywhere, hence everywhere.

Suppose finally that $\phi = \psi$. Then u and v have the same mass, so every $a > 1$ is allowed, and Theorem 16.4.4 shows that η' also has this mass. Since $\eta \ge \eta' - (a-1)C$, the mass of η is at least that of u, by monotonicity. Conversely, $P_\theta[\eta] \le \phi$ and the preservation of mass by singularity envelopes show that the mass of η is at most that of ϕ, which is the mass of u. Thus ϕ is nonpositive, has the same mass as η, and is less singular than η, because $\eta \preceq P_\theta[\eta] \le \phi$. The characterization of $P_\theta[\eta]$ in Theorem 16.4.7 therefore gives $\phi \le P_\theta[\eta]$, so $P_\theta[\eta] = \phi$. □

16.9 Comparison of relative capacities

We now compare the relative capacities $\mathrm{Cap}_{\theta,\phi}$ with each other and with the capacity of a Kähler form. The main tool is the envelope S of Lemma 16.9.3, an increasing limit of envelopes of smooth penalties that vanish exactly on a compact set K. Its Monge–Ampère measure is carried by K, and its height $\sup_X S$ controls the relative capacity of K from both sides.

All backgrounds are smooth closed real $(1,1)$-forms. In each statement, μ_q denotes the non-pluripolar Monge–Ampère measure of q for the background indicated there, and the mass of q is $\mu_q(X)$; when two backgrounds occur, we write the measures explicitly, as in $\langle(\kappa + \mathrm{dd^c} q)^n\rangle$. For a Kähler form κ we have $V_\kappa = 0$, since the constant 0 is κ-plurisubharmonic and $V_\kappa \le 0$; we write $\Omega = \int_X \kappa^n$ and $\mathrm{Cap}_\kappa = \mathrm{Cap}_{\kappa,0}$.

The capacity Cap_κ is the Monge–Ampère capacity c_κ of Section 14.8. Indeed, $v \mapsto v + 1$ maps the competitors for Cap_κ onto the functions in $\mathrm{PSH}(X, \kappa)$ with values in $[0, 1]$, and non-pluripolar products of bounded potentials are Bedford–Taylor products (Definition 15.1.3). Since κ is closed, the form σ in (14.37) is closed,

hence locally dd^{c} of a smooth function (Lemma 7.3.1), and the multilinearity of Bedford–Taylor products identifies the products defined there with the Bedford–Taylor products of local potentials.

Lemma 16.9.1 *Let κ be a Kähler form, and let $p \in \mathrm{PSH}(X,\kappa)$ be bounded. Then p is the pointwise limit of smooth functions $p_j \in \mathrm{PSH}(X,\kappa)$ with $\sup_j \sup_X |p_j| < \infty$.*

Proof Apply Theorem 13.4.5 with κ as the Kähler form of reference. It gives potentials Φ_j decreasing to p with $\kappa + \mathrm{dd}^{\mathrm{c}}\Phi_j \geq -\varepsilon_j\kappa$, where $\varepsilon_j \downarrow 0$, and with gentle analytic singularities of type $(2^{-j}, \mathcal{I}(2^j p))$. Since p is bounded, $\mathrm{e}^{-2^j p}$ is bounded, so $\mathcal{I}(2^j p)$ is the unit ideal. In the local presentations $\Phi_j = 2^{-j}\log\sum_i |g_i|^2 + b$ of Definition 13.1.5, the functions g_i therefore generate the unit ideal and have no common zero, so Φ_j is bounded. Hence the smooth function $\exp(2^j\Phi_j)$ has no zeros, and $\Phi_j = 2^{-j}\log\exp(2^j\Phi_j)$ is smooth. Put $p_j = \Phi_j/(1+\varepsilon_j)$. Then

$$\kappa + \mathrm{dd}^{\mathrm{c}} p_j = \frac{(1+\varepsilon_j)\kappa + \mathrm{dd}^{\mathrm{c}}\Phi_j}{1+\varepsilon_j} \geq 0.$$

The bounds $p \leq \Phi_j \leq \Phi_1$ give $|p_j| \leq |\Phi_j| \leq \max\{\sup_X|p|, \sup_X|\Phi_1|\}$, and $p_j \to p$ pointwise because $\Phi_j \to p$ and $\varepsilon_j \to 0$ (unlike the Φ_j, the p_j need not decrease). □

Lemma 16.9.2 *Let κ be a Kähler form, let $u, v \in \mathrm{PSH}(X,\kappa)$ satisfy $v \leq u \leq v + B$ for a constant $B \geq 0$, and let p be a bounded function in $\mathrm{PSH}(X,\kappa)$. Then*

$$\int_X p\,\mu_u \geq \int_X p\,\mu_v - nB\Omega. \tag{16.56}$$

Moreover, there is a constant C_κ, depending only on X and κ, such that for every model potential ϕ for the background κ, every $v \in \mathrm{PSH}(X,\kappa)$ with $\phi - 1 \leq v \leq \phi$, and every $p \in \mathrm{PSH}(X,\kappa)$ with $\sup_X p = 0$, bounded or not,

$$\int_X (-p)\,\mu_v \leq C_\kappa. \tag{16.57}$$

Proof Write $\kappa_q = \kappa + \mathrm{dd}^{\mathrm{c}} q$. We first prove (16.56) when $u = v$ near the poles of v and then reduce to this case; the second assertion follows from the first.

A special case. Assume first that there are $l_0 > 0$ and an open set $U \supset \{v < -l_0\}$ on which $u = v$. For $l > l_0$ put $u_l = \max\{u, -l\}$ and $v_l = \max\{v, -l\}$; these are bounded functions in $\mathrm{PSH}(X,\kappa)$ with $0 \leq u_l - v_l \leq B$. On U we have $u_l = v_l$ and $u = v$, so $\mu_{u_l} = \mu_{v_l}$ and $\mu_u = \mu_v$ there. The complement of U lies in the plurifine open set $\{v > -l\}$, on which $u_l = u$ and $v_l = v$, because $u \geq v$; by plurifine locality, $\mu_{u_l} = \mu_u$ and $\mu_{v_l} = \mu_v$ on this set. Hence $\mu_{u_l} - \mu_{v_l} = \mu_u - \mu_v$. Expanding $\kappa_{u_l}^n - \kappa_{v_l}^n$ and integrating by parts with Lemma 15.4.2, we obtain

$$\int_X p\,(\mu_u - \mu_v) = \int_X p(\mu_{u_l} - \mu_{v_l}) = \sum_{j=0}^{n-1} \int_X (u_l - v_l)\,\mathrm{dd^c} p \wedge \kappa_{u_l}^j \wedge \kappa_{v_l}^{n-1-j}$$
$$\geq -nB\Omega.$$

For the last inequality write $\mathrm{dd^c} p = (\kappa + \mathrm{dd^c} p) - \kappa$: the first part contributes a nonnegative amount, since $u_l - v_l \geq 0$, and the second contributes at least $-B \int_X \kappa \wedge \kappa_{u_l}^j \wedge \kappa_{v_l}^{n-1-j} = -B\Omega$, by Proposition 15.4.1.

The general case. Subtracting a common constant, we may assume $v \leq 0$. For $0 < a < 1$ put $v_a = av$ and $u_a = \max\{u, av\}$; these belong to $\mathrm{PSH}(X, \kappa)$ because $\kappa + \mathrm{dd^c}(av) = (1-a)\kappa + a\kappa_v$. Since $v \leq av$, we have $v_a \leq u_a \leq v_a + B$. On the open set $U_a = \{v < -B/(1-a)\}$ we have $av - v > B \geq u - v$, so $u_a = v_a$ there. With $l_0 = aB/(1-a) + 1$, the set $\{v_a < -l_0\} = \{v < -l_0/a\}$ is contained in U_a, so the special case applies to u_a and v_a.

As $a \uparrow 1$, v_a and u_a decrease to v and u. The masses of u_a and v_a agree, and by multilinearity $\int_X \langle \kappa_{v_a}^n \rangle$ is a polynomial in a that tends to $\int_X \langle \kappa_v^n \rangle = \int_X \langle \kappa_u^n \rangle$. Decreasing sequences converge in extended capacity (Lemma 15.6.3), so Theorem 15.6.5 shows that $\mu_{u_a} \to \mu_u$ and $\mu_{v_a} \to \mu_v$ weakly along every sequence $a \uparrow 1$. Passing to the limit in (16.56) for u_a and v_a, we obtain (16.56) for every continuous, in particular every smooth, $p \in \mathrm{PSH}(X, \kappa)$. For a bounded p, take the approximations p_j of Lemma 16.9.1; they converge pointwise with a common bound, and dominated convergence with respect to the finite measures μ_u and μ_v proves (16.56).

The second assertion. Let ϕ be a model potential for κ. Since $V_\kappa = 0$, Proposition 16.3.3 gives $\mu_\phi \leq \kappa^n$. If $\phi - 1 \leq v \leq \phi$, the first assertion, applied to $\phi \leq v + 1 \leq \phi + 1$, gives for every bounded p with $\sup_X p = 0$

$$\int_X (-p)\,\mu_v \leq \int_X (-p)\,\mu_\phi + n\Omega \leq \int_X (-p)\,\kappa^n + n\Omega.$$

By Proposition 14.2.1, applied on the compact Kähler manifold (X, κ) with background κ, the last integral is bounded by a constant depending only on X and κ. For an unbounded p with $\sup_X p = 0$, apply this bound to the truncations $\max\{p, -l\}$, which are bounded, belong to $\mathrm{PSH}(X, \kappa)$ and have supremum zero, and let $l \to \infty$ by monotone convergence. □

Lemma 16.9.3 *Suppose that $\{\theta\}$ is big, let ϕ be a model potential of mass $m > 0$, and let $K \subset X$ be a compact non-pluripolar set. There is a smooth function $g \geq 0$ on X whose zero set is K. For every such g, the envelopes $q_j = P_\theta(\phi + jg)$, $j \geq 1$, are potentials that increase with j, and their regularized limit $S = (\sup_j q_j)^*$ is a potential such that $L = \sup_X S \geq 0$ and*

$$\phi \leq S \leq \phi + L, \qquad S = \phi \text{ quasi-everywhere on } K,$$
$$\int_X \langle \theta_S^n \rangle = m, \qquad \mu_S(X \setminus K) = 0.$$

In particular,

$$\mathrm{Cap}_{\theta,\phi}(K) \geq \frac{m}{\max\{1, L\}^n}. \tag{16.58}$$

If $\theta = \kappa$ is a Kähler form, then also

$$\mathrm{Cap}_{\kappa,\phi}(K) \leq \frac{C}{\max\{1, L\}}, \tag{16.59}$$

where C depends only on X and κ.

Proof We first construct g. If $K = X$, take $g = 0$. Otherwise fix finitely many coordinate charts D_i and compact sets $K_i \subset D_i$ whose interiors cover X, and measure derivatives on K_i in the coordinates of D_i. Each point of $X \setminus K$ is the center of a coordinate ball whose closure lies in $X \setminus K$; since $X \setminus K$ has a countable basis, countably many of the concentric balls of half the radius, say $B'_1, B'_2, \ldots$, cover $X \setminus K$. Let B_l be the corresponding ball of full radius, and choose a smooth function $g_l \geq 0$ with compact support in B_l that is positive on $\overline{B'_l}$ (a smooth cutoff function, Lemma 1.4.1). Multiplying g_l by a small positive constant, we may assume that all its derivatives of order at most l are bounded by 2^{-l} on every K_i. Then the series $g = \sum_l g_l$ converges uniformly on every K_i together with all its derivatives, so g is smooth on the interior of every K_i, hence on X; it vanishes on K, and it is positive on $X \setminus K$, because every point there lies in some B'_l.

Now let g be any smooth nonnegative function with zero set K, and put $q_j = P_\theta(\phi + jg)$. Since ϕ is a competitor and every competitor is at most $\phi + j \sup_X g$, we have $\phi \leq q_j \leq \phi + j \sup_X g$, so $q_j \sim \phi$. The obstacle $\phi + jg$ is upper semicontinuous, so $q_j \leq \phi + jg$ everywhere by Proposition 16.1.4, and therefore $q_j = \phi$ on K. In particular $\sup_K q_j \leq 0$, and Corollary 14.2.4 bounds $\sup_X q_j$ uniformly in j. The envelopes q_j increase with j, since $g \geq 0$, so $S = (\sup_j q_j)^*$ is a potential; put $L = \sup_X S$. Then $L \geq 0$, because $S \geq \phi$ and $\sup_X \phi = 0$. Since $q_j \preceq \phi$ and $\sup_X q_j \leq \sup_X S = L$, Lemma 16.4.2 gives $q_j \leq \phi + L$, hence $\phi \leq S \leq \phi + L$. By Corollary 10.2.2, applied after adding smooth local potentials of θ, the regularization changes $\sup_j q_j$ only on a pluripolar set, so $S = \phi$ quasi-everywhere on K. Since $S \sim \phi$, $\int_X \langle \theta_S^n \rangle = m$.

The obstacle $\phi + jg$ is quasi-continuous, and its family of competitors is nonempty and bounded above by $j \sup_X g$, so Theorem 16.2.5 shows that μ_{q_j} is carried by the contact set $\{q_j = \phi + jg\}$. At a point of this set where ϕ is finite, the bound $q_j \leq \phi + L$ gives $jg \leq L$. Thus μ_{q_j} is carried by $\{g \leq L/j\}$. By Corollary 15.7.6, $\mu_{q_j} \to \mu_S$ weakly. If $\chi \geq 0$ is continuous with compact support in $X \setminus K$, then g has a positive lower bound c on the support of χ, so $\int_X \chi \, \mu_{q_j} = 0$ for $j > L/c$, and in the limit $\int_X \chi \, \mu_S = 0$. By Lemma 16.2.1(2), applied on $W = X \setminus K$ with the zero measure, $\mu_S(X \setminus K) = 0$.

Set $t = 1/\max\{1, L\}$ and $h = (1-t)\phi + tS - tL$. Then $\phi - 1 \leq \phi - tL \leq h \leq \phi$, so h is a competitor for $\mathrm{Cap}_{\theta,\phi}$, and $\mu_h \geq t^n \mu_S$. Hence $\mathrm{Cap}_{\theta,\phi}(K) \geq t^n \mu_S(K) = t^n m$, which is (16.58).

In the Kähler case apply (16.57) to the potential $p = S - L$, which has supremum zero. Quasi-everywhere on K we have $-p = L - \phi \geq L$, so $L\mu_v(K) \leq C_\kappa$ for every competitor v for $\mathrm{Cap}_{\kappa,\phi}$. If $L \geq 1$, this gives (16.59) with $C = C_\kappa$. If $L < 1$, use

$\mathrm{Cap}_{\kappa,\phi}(K) \le \mathrm{Cap}_{\kappa,\phi}(X) = m \le \Omega$; the last inequality is the monotonicity of mass, because $\phi \preceq V_\kappa = 0$ and the mass of 0 is $\int_X \kappa^n = \Omega$. Thus $C = \max\{C_\kappa, \Omega\}$ works.□

The capacity of a compact set controls the height L of its penalized envelope, and the uniform exponential integrability of normalized potentials converts this height into a volume estimate. The resulting constants do not depend on the model potential.

Proposition 16.9.4 *Suppose that $\{\theta\}$ is big, and let ϕ be a model potential of mass $m > 0$. There are $\alpha, C > 0$, depending only on X, θ and ω, such that every Borel set $E \subset X$ satisfies*

$$\int_E \omega^n \le C \exp\left[-\alpha\left(\frac{m}{\mathrm{Cap}_{\theta,\phi}(E)}\right)^{1/n}\right], \tag{16.60}$$

where the right side is interpreted as zero when the capacity is zero.

Proof Let α and C_0 be the constants of Corollary 14.2.3. First let K be compact and non-pluripolar, and let S, L be as in Lemma 16.9.3. The potential $p = S - L$ has supremum zero and satisfies $p \le -L$ quasi-everywhere on K, because $S = \phi \le 0$ there. Since pluripolar sets have zero volume, (14.11) gives

$$\int_K \omega^n \le \int_K \mathrm{e}^{-\alpha p - \alpha L}\,\omega^n \le C_0 \mathrm{e}^{-\alpha L}.$$

Write $c = \mathrm{Cap}_{\theta,\phi}(K)$, which is positive by (16.58). By the same inequality,

$$\left(\frac{m}{c}\right)^{1/n} \le \max\{1, L\} \le 1 + L,$$

so (16.60) holds for K with $C = C_0\mathrm{e}^{\alpha}$. If K is pluripolar, its volume and its capacity both vanish.

Now let E be a Borel set. If $\mathrm{Cap}_{\theta,\phi}(E) > 0$, the estimate for compact subsets $K \subset E$, together with the monotonicity of capacity and the inner regularity of the smooth volume, gives (16.60). If $\mathrm{Cap}_{\theta,\phi}(E) = 0$, every compact subset of E has zero capacity, hence is pluripolar by (16.58) and has zero volume; by inner regularity, E has zero volume. □

Corollary 16.9.5 *Under the hypotheses of Proposition 16.9.4, let $p > 1$, $0 \le f \in L^p(X, \omega^n)$ and $r > 0$. There is $B_r > 0$, depending only on X, θ, ω, p and r, such that every Borel set E satisfies*

$$\int_E f\omega^n \le B_r \|f\|_{L^p} \left(\frac{\mathrm{Cap}_{\theta,\phi}(E)}{m}\right)^r. \tag{16.61}$$

In particular, for every $r > 1$ the measure $f\omega^n$ is dominated by a constant multiple of $\mathrm{Cap}^r_{\theta,\phi}$.

Proof If $\mathrm{Cap}_{\theta,\phi}(E) = 0$, then E has zero volume by Proposition 16.9.4, and both sides vanish. Otherwise let $q = p/(p-1)$. By Hölder's inequality and (16.60), the integral is at most

$$\|f\|_{L^p} C^{1/q} \exp\left[-\frac{\alpha}{q}\left(\frac{m}{\mathrm{Cap}_{\theta,\phi}(E)}\right)^{1/n}\right].$$

Put $t = (m/\mathrm{Cap}_{\theta,\phi}(E))^{1/n}$; then $t \geq 1$, since the capacity is at most $\int_X \langle\theta_\phi^n\rangle = m$. The function $t^{nr}\mathrm{e}^{-\alpha t/q}$ is bounded on $[1,\infty)$: its logarithmic derivative is $nr/t - \alpha/q$, so its maximum is attained at 1 or at nrq/α. Hence $\mathrm{e}^{-\alpha t/q} \leq A\, t^{-nr}$ for a constant A depending only on n, r, q, α, and this proves (16.61). □

Proposition 16.9.6 *Let κ be a Kähler form and ϕ a model potential for κ of mass $m > 0$. There is $A \geq 1$ such that every Borel set $E \subset X$ satisfies*

$$A^{-1}\,\mathrm{Cap}_\kappa(E)^n \leq \mathrm{Cap}_{\kappa,\phi}(E) \leq A\,\mathrm{Cap}_\kappa(E)^{1/n}. \tag{16.62}$$

More generally, if $\theta \leq \kappa$, $\{\theta\}$ is big and ϕ is a model potential for θ of mass $m > 0$, then

$$\mathrm{Cap}_\kappa(E)^n \leq \frac{C^n}{m}\,\mathrm{Cap}_{\theta,\phi}(E), \tag{16.63}$$

where C depends only on X and κ.

Proof All capacities involved are inner regular (Proposition 16.7.6), so it suffices to treat compact sets K: the inequalities for a Borel set E follow by applying them to the compact subsets of E and taking suprema. If K is pluripolar, all capacities of K vanish. Otherwise fix a smooth function $g \geq 0$ with zero set K, and apply Lemma 16.9.3 with this g to (θ,ϕ) and to $(\kappa, 0)$; here $0 = V_\kappa$ is a model potential for κ of mass Ω, by Proposition 16.3.2(4). Denote the resulting envelopes by S_ϕ and S_0 and their suprema by L_ϕ and L_0. Every competitor for $P_\theta(\phi + jg)$ is κ-plurisubharmonic, because $\theta \leq \kappa$, and it is at most $\phi + jg \leq jg$ quasi-everywhere, so $P_\theta(\phi + jg) \leq P_\kappa(jg)$ for every j; hence $S_\phi \leq S_0$ and $L_\phi \leq L_0$. The two estimates of Lemma 16.9.3 give

$$\mathrm{Cap}_\kappa(K)^n \leq \frac{C^n}{\max\{1, L_0\}^n} \leq \frac{C^n}{\max\{1, L_\phi\}^n} \leq \frac{C^n}{m}\,\mathrm{Cap}_{\theta,\phi}(K).$$

This proves (16.63) and, for $\theta = \kappa$, the first inequality in (16.62).

For the upper bound let $\theta = \kappa$, and choose $b > 1$ with $b^n(\Omega - m) < \Omega$. By Theorem 16.4.4, applied to $\phi \preceq 0$, where the mass of 0 is $\int_X \kappa^n = \Omega$, the envelope $\eta = E_b(\phi, 0) = P_\kappa(b\phi)$ is a potential, and it satisfies $b^{-1}\eta \leq \phi$; in particular $\eta = -\infty$ at the poles of ϕ. Write $c = 1 - b^{-1} > 0$. For fixed j and $\delta > 0$ consider

$$v_j = c\,P_\kappa(jg) + b^{-1}\eta - \delta,$$

a convex combination of κ-plurisubharmonic functions minus a constant. The function $P_\kappa(jg)$ is upper semicontinuous and vanishes on K, because $0 \leq P_\kappa(jg) \leq jg$.

Hence $P_\kappa(jg) < \delta/c$ on a neighborhood U of K, and there $v_j < b^{-1}\eta \le \phi$. Outside the poles of ϕ, the difference $v_j - \phi$ is bounded above by $cj \sup_X g$, and g has a positive minimum on $X \setminus U$. Therefore $v_j \le \phi + lg$ for a sufficiently large integer l, so $v_j \le P_\kappa(\phi + lg) \le S_\phi$. Letting $\delta \downarrow 0$ and then $j \to \infty$, and using that the increasing limit of $P_\kappa(jg)$ equals S_0 almost everywhere, we obtain

$$cS_0 + b^{-1}\eta \le S_\phi$$

almost everywhere, hence everywhere.

Choose a Borel set G of positive volume on which $b^{-1}\eta \ge -D$ for a constant D; such a set exists because η is finite almost everywhere. By Proposition 14.2.1, there is a constant B, independent of K, with $\int_X |S_0 - L_0|\,\kappa^n \le B$. Hence $S_0 \ge L_0 - B/\int_G \kappa^n$ at some point $x \in G$, for otherwise $\int_G (L_0 - S_0)\,\kappa^n > B$. At this point the last display gives $L_\phi \ge S_\phi(x) \ge cL_0 - D'$ with $D' = cB/\int_G \kappa^n + D$, which does not depend on K. If $L_0 \ge \max\{1, 2D'/c\}$, then $L_\phi \ge cL_0/2$, and Lemma 16.9.3 gives

$$\mathrm{Cap}_{\kappa,\phi}(K) \le \frac{2C}{cL_0} \le \frac{2C}{c\,\Omega^{1/n}}\,\mathrm{Cap}_\kappa(K)^{1/n};$$

the last inequality is the lower bound (16.58) for the construction with reference potential 0, which gives $\Omega/L_0^n \le \mathrm{Cap}_\kappa(K)$. In the remaining range $L_0 < \max\{1, 2D'/c\}$, the same lower bound shows that $\mathrm{Cap}_\kappa(K)$ is bounded below by a positive constant independent of K, and we use $\mathrm{Cap}_{\kappa,\phi}(K) \le m$, enlarging A if necessary. The constant A in the upper bound depends on ϕ. □

Lemma 16.9.7 *Suppose that $\{\theta\}$ is big, let $\phi \in \mathrm{PSH}(X,\theta)$ have positive mass, and let $\psi \in \mathcal{E}(X,\theta,\phi)$. There is a continuous nondecreasing function $F\colon [0,\infty) \to [0,\infty)$ with $F(0) = 0$, which may depend on ψ and ϕ, such that*

$$\mathrm{Cap}_{\theta,\psi}(E) \le F(\mathrm{Cap}_{\theta,\phi}(E))$$

for every Borel set $E \subset X$.

Proof Since $\mathrm{Cap}_{\theta,\phi}$ does not change when a constant is added to ϕ (Proposition 16.7.6), we may assume that $\psi \le \phi$. Let u be a competitor for $\mathrm{Cap}_{\theta,\psi}$, so that $\psi - 1 \le u \le \psi$, and let $t \ge 1$. We estimate $\mu_u(E)$ separately on $\{u > \phi - 2t\}$ and on $\{u \le \phi - 2t\}$.

Put $u_t = \max\{u, \phi - 2t\}$, so that $\phi - 2t \le u_t \le \phi$. The potential $(1 - (2t)^{-1})\phi + (2t)^{-1}u_t$ is a competitor for $\mathrm{Cap}_{\theta,\phi}$, and its measure is at least $(2t)^{-n}\mu_{u_t}$. Since $u_t = u$ on the plurifine open set $\{u > \phi - 2t\}$, it follows that

$$\mu_u(E \cap \{u > \phi - 2t\}) \le (2t)^n\,\mathrm{Cap}_{\theta,\phi}(E).$$

The potentials ψ, u and $\psi - 1$ belong to $\mathcal{E}(X,\theta,\phi)$. So does the midpoint $q = (u+\phi)/2$: by Proposition 16.4.10, $P_\theta[u] = P_\theta[\phi]$, so by Proposition 16.3.2 each mixed mass $\int_X \langle \theta_u^k \wedge \theta_\phi^{n-k} \rangle$ equals the mass of ϕ. Expanding μ_q by multilinearity, we

see that q also has the mass of ϕ. On the set $\{u \le \phi - 2t\}$ we have $q - t \ge u \ge \psi - 1$, and on $\{\psi - 1 \le q - t\}$ we have $\psi - 1 \le (\psi + \phi)/2 - t$, that is, $\psi \le \phi - 2t + 2 \le \phi - t + 1$. Thus

$$\{u \le \phi - 2t\} \subset \{\psi - 1 \le q - t\} \subset \{\psi \le \phi - t + 1\}.$$

The non-strict comparison in Proposition 16.7.4, applied with $p = n$ to $\psi - 1$ and $q - t$, together with $\mu_q \ge 2^{-n}\mu_u$, gives

$$\begin{aligned}\mu_u(\{u \le \phi - 2t\}) &\le 2^n \mu_q(\{\psi - 1 \le q - t\}) \le 2^n \mu_\psi(\{\psi - 1 \le q - t\}) \\ &\le 2^n \mu_\psi(\{\psi \le \phi - t + 1\}).\end{aligned}$$

Adding the two estimates and taking the supremum over u, we obtain

$$\operatorname{Cap}_{\theta,\psi}(E) \le (2t)^n \operatorname{Cap}_{\theta,\phi}(E) + 2^n \mu_\psi(\{\psi \le \phi - t + 1\}) \qquad (t \ge 1). \quad (16.64)$$

The last term tends to zero as $t \to \infty$, since ψ and ϕ are finite outside a μ_ψ-null set.

It remains to replace the right side by a continuous function of the capacity. Put $m = \int_X \langle \theta_\psi^n \rangle$, and let $H(s)$ be the minimum of m and of the infimum over $t \ge 1$ of the right side of (16.64) with $\operatorname{Cap}_{\theta,\phi}(E)$ replaced by s. Since $\operatorname{Cap}_{\theta,\psi} \le \int_X \langle \theta_\psi^n \rangle = m$, we have $\operatorname{Cap}_{\theta,\psi}(E) \le H(\operatorname{Cap}_{\theta,\phi}(E))$. The function H is nondecreasing, and $H(s) \to 0$ as $s \downarrow 0$. Choose a strictly decreasing sequence $s_k \downarrow 0$ with $H(s_k) \le m2^{-k}$. Define $F(0) = 0$, $F(s_k) = m2^{1-k}$, F linear on each interval $[s_{k+1}, s_k]$, and $F = m$ on $[s_1, \infty)$. On $[s_{k+1}, s_k]$ we have $H(s) \le H(s_k) \le m2^{-k} = F(s_{k+1}) \le F(s)$, and $H \le m = F$ on $[s_1, \infty)$. Thus F is a continuous nondecreasing majorant of H with $F(0) = 0$. □

Theorem 16.9.8 *Suppose that $\{\theta\}$ is big and that $\psi \in \mathrm{PSH}(X, \theta)$ has positive mass. There are continuous nondecreasing functions $F, G\colon [0, \infty) \to [0, \infty)$, vanishing at zero, such that every Borel set $E \subset X$ satisfies*

$$\operatorname{Cap}_{\theta,\psi}(E) \le F(\operatorname{Cap}_\omega(E)), \qquad \operatorname{Cap}_\omega(E) \le G(\operatorname{Cap}_{\theta,\psi}(E)).$$

Consequently, any two capacities of this kind, for possibly different big backgrounds and potentials of positive mass, are bounded by such functions of each other. In particular, they define the same notion of convergence in capacity*, in which Borel functions f_j converge to f if the capacities of the sets $\{|f_j - f| > \varepsilon\}$ tend to zero for every $\varepsilon > 0$. They also define the same notion of* uniform absolute continuity *of a family of positive Borel measures: $\lambda(A)$ is small, uniformly over the members λ of the family, whenever the Borel set A has small capacity.*

Proof Choose $A \ge 1$ with $\theta \le \kappa = A\omega$. Every competitor v for $\operatorname{Cap}_{\theta,\psi}$ is a competitor for $\operatorname{Cap}_{\kappa,\psi}$, and its measure for κ dominates its measure for θ: since $\kappa + \mathrm{dd^c} v = (\theta + \mathrm{dd^c} v) + (\kappa - \theta)$ with a smooth closed semipositive form $\kappa - \theta$, multilinearity and positivity of non-pluripolar products give $\langle (\kappa + \mathrm{dd^c} v)^n \rangle \ge \langle (\theta + \mathrm{dd^c} v)^n \rangle$; hence $\operatorname{Cap}_{\theta,\psi} \le \operatorname{Cap}_{\kappa,\psi}$, and the mass of ψ for κ is at least its mass for θ, hence positive. Adding a constant to ψ does not change the capacities, so we may assume $\psi \le 0$. Let $\Phi = P_\kappa[\psi]$. By Theorem 16.4.7, Φ is a model potential for κ with $\psi \le \Phi$ and with the same mass for κ as ψ. Thus $\psi \in \mathcal{E}(X, \kappa, \Phi)$, and Lemma 16.9.7 and

Proposition 16.9.6 bound $\mathrm{Cap}_{\kappa,\psi}$ by a continuous nondecreasing function, vanishing at zero, of Cap_κ.

Finally, rescaling the competitors gives

$$\mathrm{Cap}_\omega \le \mathrm{Cap}_\kappa \le A^n \,\mathrm{Cap}_\omega :$$

every competitor for Cap_ω is one for Cap_κ, with a larger measure by the same argument, and if v is a competitor for Cap_κ, then v/A is a competitor for Cap_ω with measure $A^{-n}(\kappa + \mathrm{dd}^{\mathrm{c}} v)^n$. This proves the first inequality.

For the second inequality use the model potential $\phi = P_\theta[\psi]$ for θ, which has the same positive mass $m = \int_X \langle \theta_\psi^n \rangle$ as ψ. By (16.63), $\mathrm{Cap}_\kappa^n \le (C^n/m)\,\mathrm{Cap}_{\theta,\phi}$. By Theorem 16.8.1 with $a = 2$, applied to ψ and ϕ, whose singularity envelopes agree, the potential $\eta = P_\theta(2\psi - \phi)$ has mass m and satisfies $P_\theta[\eta] = \phi$ and $(\eta + \phi)/2 \le \psi$.

It remains to bound $\mathrm{Cap}_{\theta,\phi}$ by a function of $\mathrm{Cap}_{\theta,\psi}$. Put $u = (\eta + \phi)/2$. Since $P_\theta[\eta] = P_\theta[\phi]$, the preservation of mixed masses in Proposition 16.3.2 shows that u also has mass m. Since $u \le \psi$, this means that $u \in \mathcal{E}(X, \theta, \psi)$, and Lemma 16.9.7 bounds $\mathrm{Cap}_{\theta,u}$ by a continuous nondecreasing function, vanishing at zero, of $\mathrm{Cap}_{\theta,\psi}$. If v is a competitor for $\mathrm{Cap}_{\theta,\phi}$, then $h = (v + \eta)/2$ satisfies $u - 1/2 \le h \le u$, so it is a competitor for $\mathrm{Cap}_{\theta,u}$, and $\mu_h \ge 2^{-n}\mu_v$. Thus

$$\mathrm{Cap}_{\theta,\phi} \le 2^n \,\mathrm{Cap}_{\theta,u}\,.$$

Combining these inequalities with $\mathrm{Cap}_\omega \le \mathrm{Cap}_\kappa$ gives G; compositions, positive multiples and nth roots preserve continuity, monotonicity and the value zero at the origin. The remaining assertions follow by comparing both capacities with Cap_ω.□

Proposition 16.9.9 *Let $B_\alpha \Subset B'_\alpha$ be finitely many pairs of concentric coordinate balls such that each $\overline{B'_\alpha}$ lies in a coordinate chart and the balls B_α cover X, and let $\mathrm{Cap}_{B'_\alpha}$ be the capacities of Definition 9.3.2 in these coordinates. Then there is a constant $c \ge 1$ such that every Borel set $E \subset X$ satisfies*

$$c^{-1} \sum_\alpha \mathrm{Cap}_{B'_\alpha}(E \cap B_\alpha) \le \mathrm{Cap}_\omega(E) \le c \sum_\alpha \mathrm{Cap}_{B'_\alpha}(E \cap B_\alpha),$$

and moreover $\mathrm{Cap}_\omega(E) \le c\,\mathrm{Cap}_{B'_\alpha}(E)$ for every α and every Borel set $E \subset B'_\alpha$. Consequently, every relative capacity $\mathrm{Cap}_{\theta,\psi}$ as in Theorem 16.9.8 has the following properties.

(1) *A compact set has capacity zero if and only if it is pluripolar. Hence a Borel set has capacity zero if and only if all its compact subsets are pluripolar.*
(2) *If real Borel functions f_j converge to a real Borel function f locally in capacity on every ball B'_α (Definition 9.4.5), then $\mathrm{Cap}_{\theta,\psi}(\{|f_j - f| > \delta\}) \to 0$ for every $\delta > 0$.*
(3) *Every quasi-continuous function on X is continuous outside open sets of arbitrarily small capacity. Moreover, let θ' be any smooth closed real $(1,1)$-form on X. For every $u \in \mathrm{PSH}(X, \theta')$ and every $\varepsilon > 0$ there is an open set G with*

$\mathrm{Cap}_{\theta,\psi}(G) < \varepsilon$ which contains the pole set of u and such that u is finite and continuous on $X \setminus G$.

Proof Let v be a competitor for Cap_ω, so that $v \in \mathrm{PSH}(X,\omega)$ and $-1 \le v \le 0$. Fix α and write $B'_\alpha = B(0,R_\alpha)$ in its coordinates. Since $\overline{B'_\alpha}$ is a compact subset of a chart, there is $A_\alpha > 0$ with $\omega \le A_\alpha \mathrm{dd^c}|z|^2$ on B'_α (Lemma 13.1.1). Then $\mathrm{dd^c}(v+A_\alpha|z|^2) = (\omega+\mathrm{dd^c}v)+(A_\alpha\mathrm{dd^c}|z|^2-\omega) \ge 0$, so $v+A_\alpha|z|^2$ is plurisubharmonic on B'_α, by Lemma 13.1.4(2), applied on B'_α to the form $A_\alpha\mathrm{dd^c}|z|^2 = \mathrm{dd^c}(A_\alpha|z|^2)$, and the multilinearity and positivity of Bedford–Taylor products give $(\mathrm{dd^c}(v+A_\alpha|z|^2))^n \ge (\omega+\mathrm{dd^c}v)^n$ on B'_α. Put $c_\alpha = 1+A_\alpha R_\alpha^2$. The function $w = (v+A_\alpha|z|^2+1)/c_\alpha$ is plurisubharmonic on B'_α with values in $[0,1]$, and $(\omega+\mathrm{dd^c}v)^n \le c_\alpha^n(\mathrm{dd^c}w)^n$ there. Hence $\mu_v(E) \le c_\alpha^n \mathrm{Cap}_{B'_\alpha}(E)$ for every Borel set $E \subset B'_\alpha$, and taking the supremum over v gives $\mathrm{Cap}_\omega(E) \le c_\alpha^n \mathrm{Cap}_{B'_\alpha}(E)$. Applying this to the sets $E \cap B_\alpha$ and using the subadditivity of Cap_ω (Proposition 16.7.6), we obtain the upper bound.

For the lower bound fix α and write $B_\alpha = B(0,r)$ and $B'_\alpha = B(0,R)$ in the coordinates, with $r < R$. Choose $L > 0$ with $L(R^2-r^2) > 1$, and then $b > 0$ so small that $\omega \ge bL\mathrm{dd^c}|z|^2$ on $B(0,R)$ (Lemma 13.1.1) and that we can choose a number a with

$$b(1+Lr^2) < a < bLR^2, \qquad a \le 1.$$

Let w be plurisubharmonic on $B(0,R)$ with $-1 \le w \le 0$, and put $q = bw - bL|z|^2$. Then $\omega + \mathrm{dd^c}q \ge b\mathrm{dd^c}w \ge 0$ on $B(0,R)$. At the points of $B(0,R)$ where $|z|^2 > a/(bL)$, which include a neighborhood of the boundary sphere because $a < bLR^2$, we have $q \le -bL|z|^2 < -a$.

Define v to be $\max\{q,-a\}$ on $B(0,R)$ and $-a$ elsewhere. Then $v = -a$ on the open set U formed by $X \setminus \overline{B(0,R)}$ and the points of the chart where $|z|^2 > a/(bL)$. Since U and $B(0,R)$ cover X, and v is ω-plurisubharmonic on each of them, $v \in \mathrm{PSH}(X,\omega)$. It takes values in $[-1,0]$, and on $B(0,r)$ we have $q \ge -b(1+Lr^2) > -a$, so $v = q$ there. Since $\omega + \mathrm{dd^c}q$ is the sum of the semipositive form $\omega - bL\mathrm{dd^c}|z|^2$ and $b\mathrm{dd^c}w$, positivity of the mixed terms gives $(\omega+\mathrm{dd^c}v)^n \ge b^n(\mathrm{dd^c}w)^n$ on $B(0,r)$. The capacity $\mathrm{Cap}_{B(0,R)}$ may equally be computed with competitors taking values in $[-1,0]$, so it follows that

$$\mathrm{Cap}_\omega(E) \ge b^n \mathrm{Cap}_{B(0,R)}(E) \qquad (E \subset B(0,r) \text{ Borel}).$$

Summing over the finitely many balls gives the lower bound.

It remains to prove (1)–(3). Let F and G be the functions of Theorem 16.9.8 for $\mathrm{Cap}_{\theta,\psi}$; since they are continuous and vanish at zero, a sequence of Borel sets has capacities $\mathrm{Cap}_{\theta,\psi}$ tending to zero if and only if their capacities Cap_ω tend to zero.

(1) Pluripolar Borel sets have capacity zero by Proposition 16.7.6. Conversely, let K be a compact set that is not pluripolar. Since ω is Kähler, $0 = V_\omega$ is a model potential for ω of mass $\int_X \omega^n > 0$, by Proposition 16.3.2, and (16.58) for this model potential gives $\mathrm{Cap}_\omega(K) > 0$. Since $\mathrm{Cap}_\omega(K) \le G(\mathrm{Cap}_{\theta,\psi}(K))$ and $G(0) = 0$, it follows that $\mathrm{Cap}_{\theta,\psi}(K) > 0$. The assertion about Borel sets follows from the inner regularity in Proposition 16.7.6.

(2) Let $\delta > 0$ and $A_j = \{|f_j - f| > \delta\}$. By the upper bound, $\mathrm{Cap}_\omega(A_j) \le c \sum_\alpha \mathrm{Cap}_{B'_\alpha}(A_j \cap B_\alpha)$. Each term is at most the outer capacity $\mathrm{Cap}^*_{B'_\alpha}(\overline{B_\alpha} \cap A_j)$, which tends to zero because $\overline{B_\alpha}$ is a compact subset of B'_α. Hence $\mathrm{Cap}_\omega(A_j) \to 0$, and therefore $\mathrm{Cap}_{\theta,\psi}(A_j) \to 0$.

(3) Let f be quasi-continuous on X and $\eta > 0$. For each α there is an open set $W_\alpha \subset B'_\alpha$ with $\mathrm{Cap}_{B'_\alpha}(W_\alpha) < \eta$ such that f is continuous on the compact set $\overline{B_\alpha} \setminus W_\alpha$. Put $W = \bigcup_\alpha W_\alpha$. Then $X \setminus W$ is the union of the finitely many closed sets $\overline{B_\alpha} \setminus W$, on each of which f is continuous, so f is continuous on $X \setminus W$; and $\mathrm{Cap}_\omega(W) \le \sum_\alpha c\, \mathrm{Cap}_{B'_\alpha}(W_\alpha) < Nc\eta$, where N is the number of balls. Now let $u \in \mathrm{PSH}(X, \theta')$. By Lemma 16.1.2(1), u is quasi-continuous. Its pole set P is pluripolar, because near each point u is a plurisubharmonic function plus a smooth one (Definition 13.1.3). Hence, by Lemma 16.1.1(2) there are open sets $W'_\alpha \subset B'_\alpha$ with $P \cap B'_\alpha \subset W'_\alpha$ and $\mathrm{Cap}_{B'_\alpha}(W'_\alpha) < \eta$. Replacing W_α, chosen as above for $f = u$, by $W_\alpha \cup W'_\alpha$, we obtain an open set W with $\mathrm{Cap}_\omega(W) < 2Nc\eta$ which contains P, such that u is continuous on $X \setminus W$ and finite there. In both cases $\mathrm{Cap}_{\theta,\psi}(W) \le F(\mathrm{Cap}_\omega(W)) \le F(2Nc\eta)$, which is arbitrarily small when η is small. □

Chapter 17
Relative energy and Perron segments

Let (X, ω) be a connected compact Kähler manifold of dimension $n \geq 1$, and let θ be a smooth closed real $(1,1)$-form on X. In this chapter we study the Monge–Ampère energy of θ-plurisubharmonic potentials relative to a model potential ϕ (Definition 16.3.1); recall that model potentials have positive mass. The energy is a concave functional whose derivative is the non-pluripolar Monge–Ampère measure. We use it for two purposes: to solve Monge–Ampère equations with prescribed non-pluripolar measures by maximizing suitable functionals, and to show that the energy is affine along weak geodesics, which we construct as Perron envelopes of subgeodesics.

Everything rests on integration by parts for non-pluripolar products of potentials with bounded difference. Such potentials may be unbounded, so Stokes' formula for bounded potentials (Lemma 15.4.2) does not apply to them directly. In Section 17.1 we truncate them and control the truncation errors with the relative capacities of Chapter 16. This control requires a strictly positive background, a restriction that a polynomial expansion in an auxiliary Kähler form removes at the end. The resulting formula (Theorem 17.1.3) makes no assumption on the masses of the potentials.

The energy E_θ^ϕ is first defined for potentials with bounded difference from ϕ, where integration by parts computes its first and second variations, and is then extended by decreasing truncation to all potentials u with $u \leq \phi + O(1)$. It is finite exactly on the relative finite-energy class $\mathcal{E}^1(X, \theta, \phi)$ (Theorem 17.2.8), continuous along monotone sequences, and upper semicontinuous for L^1 convergence, so that sets of normalized potentials with energy bounded below are compact.

In Section 17.4 these properties turn maximizers of suitable functionals into solutions of Monge–Ampère equations, first for bounded densities and for the exponential equations $\mu_u = \mathrm{e}^{\lambda u}\nu$, and then, letting $\lambda \to 0$, for $\mu_u = \nu$ with an arbitrary non-pluripolar measure ν of the same mass as ϕ (Theorem 17.4.13). The solution is unique up to an additive constant (Theorem 17.4.14), and it has bounded difference from ϕ when ν has an L^p density with $p > 1$ (Corollary 17.4.17). Two further consequences are that a pseudo-effective class of positive volume is big (Theorem 17.4.20) and that mixed non-pluripolar masses dominate the geometric mean of the individual masses (Theorem 17.4.23).

With these equations we show in Section 17.5 that, in a big class, the singularity ceiling of a potential u (Definition 17.5.2) coincides with its singularity envelope $P_\theta[u]$, and that $P_\theta[P_\theta[u]] = P_\theta[u]$ (Theorem 17.5.3). This is [PSWZ26, Theorem 1.2]; it includes the case of zero mass, $\int_X \langle \theta_u^n \rangle = 0$, to which the characterization of $P_\theta[u]$ in Theorem 16.4.7 does not apply.

In the last three sections we return to a fixed model potential ϕ and join two potentials of full mass relative to ϕ by a Perron segment, the upper envelope of all subgeodesics lying below them at the endpoints. Rooftop envelopes show that the segment attains the prescribed endpoints. Its complexification is locally maximal on the product of X with a strip, so its top-degree non-pluripolar measure vanishes, and integration by parts on $X \times \mathbb{P}^1$ then shows that the energy is affine along the segment (Theorem 17.7.3). We conclude with estimates for the extremal slopes of a segment and for long segments; the latter produce profiles that depend only on the singularity type of the endpoint (Proposition 17.8.2).

Historical remarks and references for this chapter are collected in Appendices B.17 to B.19.

17.1 Integration by parts for bounded differences

As in Chapter 16, for a smooth closed real $(1,1)$-form θ on X and $q \in \mathrm{PSH}(X,\theta)$ we write

$$\theta_q = \theta + \mathrm{dd^c} q, \qquad \mu_q = \langle \theta_q^n \rangle,$$

so that $\mu_q(X) = \int_X \langle \theta_q^n \rangle$ is the total mass of q, and we indicate the background when several occur. Recall from Definition 14.2.6 that v is more singular than u, written $v \preceq u$, if $v \le u + C$ on X for some constant C, and that u and v have the same singularity type, written $u \sim v$, if $u \preceq v$ and $v \preceq u$. A difference $u - v$ of two quasi-psh functions is evaluated on their common finite locus $\{u > -\infty\} \cap \{v > -\infty\}$, whose complement is pluripolar. Non-pluripolar products do not charge pluripolar sets (Lemma 15.1.2), so the values assigned to such a difference on the pole sets never affect the integrals below.

A *bounded quasi-psh difference* is a bounded Borel function a on X such that $a = \varphi_1 - \varphi_2$ quasi-everywhere, that is, outside a pluripolar set, for two quasi-psh functions φ_1, φ_2 on X; we call (φ_1, φ_2) a *presentation* of a. The two functions then have the same singularity type. Indeed, if $|a| \le C$, then $\varphi_1 \le \varphi_2 + C$ quasi-everywhere, hence everywhere, because an inequality between quasi-psh functions that holds quasi-everywhere holds everywhere (Section 16.1); by symmetry, $\varphi_1 \sim \varphi_2$.

Let $T_1, \dots, T_{n-1}$ be closed positive $(1,1)$-currents on X and put $R = \langle T_1 \wedge \cdots \wedge T_{n-1} \rangle$, with $R = 1$ if $n = 1$. For a bounded quasi-psh difference a with presentation (φ_1, φ_2) we define the signed measure

$$\begin{aligned} \mathrm{dd^c} a \wedge R = {} & \langle (\kappa + \mathrm{dd^c}\varphi_1) \wedge T_1 \wedge \cdots \wedge T_{n-1} \rangle \\ & - \langle (\kappa + \mathrm{dd^c}\varphi_2) \wedge T_1 \wedge \cdots \wedge T_{n-1} \rangle , \end{aligned} \tag{17.1}$$

where κ is a multiple of ω so large that $\varphi_1, \varphi_2 \in \mathrm{PSH}(X, \kappa)$. Both terms are non-pluripolar products of n closed positive $(1,1)$-currents, hence finite measures by Theorem 15.4.3; for $n = 1$ they are first-degree non-pluripolar products.

The measure $\mathrm{dd^c} a \wedge R$ does not depend on the choices made. First, let θ be any smooth closed real $(1,1)$-form with $\varphi_1, \varphi_2 \in \mathrm{PSH}(X, \theta)$, and let $\kappa \geq \theta$ be a multiple of ω, so that $\varphi_1, \varphi_2 \in \mathrm{PSH}(X, \kappa)$. Multilinearity of non-pluripolar products (Proposition 15.1.5) gives

$$\begin{aligned}\left\langle (\kappa + \mathrm{dd^c}\varphi_j) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle = {} & \left\langle (\theta + \mathrm{dd^c}\varphi_j) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle \\ & + \left\langle (\kappa - \theta) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle\end{aligned}$$

for $j = 1, 2$. The last term cancels in the difference, so the right side of (17.1) does not change when κ is replaced by θ. Taking for θ a smaller multiple of ω with respect to which φ_1, φ_2 are psh, we see that the measure does not depend on κ; hence it may be computed with any form θ as above in place of κ. Finally, the measure does not depend on the presentation of a. Let (ψ_1, ψ_2) be another presentation and choose κ so that all four functions are κ-psh. Then $\varphi_1 + \psi_2 = \varphi_2 + \psi_1$ quasi-everywhere, hence everywhere, since both sides are 2κ-psh. Therefore the currents $2\kappa + \mathrm{dd^c}(\varphi_1 + \psi_2)$ and $2\kappa + \mathrm{dd^c}(\varphi_2 + \psi_1)$ coincide. By multilinearity,

$$\begin{aligned}&\left\langle (\kappa + \mathrm{dd^c}\varphi_1) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle + \left\langle (\kappa + \mathrm{dd^c}\psi_2) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle \\ &\quad = \left\langle (\kappa + \mathrm{dd^c}\varphi_2) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle + \left\langle (\kappa + \mathrm{dd^c}\psi_1) \wedge T_1 \wedge \cdots \wedge T_{n-1} \right\rangle .\end{aligned}$$

All four measures are finite, so the two presentations give the same measure (17.1). We extend the notation linearly to finite sums of such products.

The truncation argument below passes to the limit in integrals whose weights converge only in capacity. We use the capacity $\mathrm{Cap}_\omega = \mathrm{Cap}_{\omega,0}$ of Definition 16.7.5, relative to $V_\omega = 0$. Quasi-continuity of functions on X is understood as in Section 16.1; by Proposition 16.9.9, the exceptional open sets may be taken of arbitrarily small Cap_ω. Recall from Theorem 16.9.8 that Borel functions f_j converge in capacity to f if $\mathrm{Cap}_\omega(\{|f_j - f| > \varepsilon\}) \to 0$ for every $\varepsilon > 0$, and that a family of positive Borel measures on X is uniformly absolutely continuous with respect to Cap_ω if the supremum of $\lambda(A)$, over all members λ of the family and all Borel sets A with $\mathrm{Cap}_\omega(A) \leq \delta$, tends to zero as $\delta \downarrow 0$.

Lemma 17.1.1 *Let ν_j and ν be positive Borel measures on X such that $\nu_j \to \nu$ weakly, $\sup_j \nu_j(X) < \infty$, and the family $\{\nu, \nu_1, \nu_2, \ldots\}$ is uniformly absolutely continuous with respect to Cap_ω. Let f_j and f be Borel functions on X with $|f_j|, |f| \leq B$ for a constant B, and suppose that $f_j \to f$ in capacity and that f is continuous outside open sets of arbitrarily small capacity: for every $\eta > 0$ there is an open set $G \subset X$ with $\mathrm{Cap}_\omega(G) < \eta$ such that the restriction of f to $X \setminus G$ is continuous. (By Proposition 16.9.9(3), the last condition holds if f is quasi-continuous.) Then*

$$\int_X f_j \, \nu_j \longrightarrow \int_X f \, \nu .$$

Proof We write the difference of the two integrals as

$$\int_X (f_j - f)\,\nu_j + \left(\int_X f\,\nu_j - \int_X f\,\nu\right)$$

and show that both terms tend to zero. For the first term fix $\varepsilon > 0$ and put $A_j = \{|f_j - f| > \varepsilon\}$. Splitting the integral over A_j and its complement gives

$$\left|\int_X (f_j - f)\,\nu_j\right| \le 2B\nu_j(A_j) + \varepsilon \sup_k \nu_k(X).$$

Since $\mathrm{Cap}_\omega(A_j) \to 0$, uniform absolute continuity gives $\nu_j(A_j) \to 0$. Thus the upper limit of the first term is at most $\varepsilon \sup_k \nu_k(X)$, and ε is arbitrary.

For the second term let $\eta > 0$, and choose an open set G with $\mathrm{Cap}_\omega(G) < \eta$ such that the restriction of f to the compact set $X \setminus G$ is continuous. By Lemma 7.1.4, there is a metric d on X which induces its topology, and by Lemma 9.4.3, applied in the metric space (X, d), the restriction of f to $X \setminus G$ extends to a continuous function g on X with $|g| \le B$. Since X is compact, g has compact support, so weak convergence of measures (see the discussion following Definition 7.2.4) gives $\int_X g\,\nu_j \to \int_X g\,\nu$. Since $f = g$ outside G, we have $|\int_X f\,\lambda - \int_X g\,\lambda| \le 2B\lambda(G)$ for each $\lambda \in \{\nu, \nu_1, \nu_2, \ldots\}$. The upper limit of the second term is therefore at most $4B$ times the supremum of $\lambda(G)$ over the family, which tends to zero as $\eta \downarrow 0$ by uniform absolute continuity. □

Lemma 17.1.2 *Let θ be a smooth closed real $(1,1)$-form on X, and let $\varphi_1, \varphi_2, \psi_1, \psi_2 \in \mathrm{PSH}(X, \theta)$ satisfy $\varphi_1 \sim \varphi_2$ and $\psi_1 \sim \psi_2$. Put $a = \varphi_1 - \varphi_2$, $b = \psi_1 - \psi_2$, and*

$$S_j = \sum_{k=0}^{n-1} \left\langle \theta_{\varphi_j} \wedge \theta_{\psi_1}^k \wedge \theta_{\psi_2}^{n-1-k} \right\rangle, \qquad j = 1, 2.$$

Then

$$\int_X a(\mu_{\psi_1} - \mu_{\psi_2}) = \int_X b(S_1 - S_2). \tag{17.2}$$

Proof If the potentials were bounded, (17.2) would follow from Stokes' formula. We reduce to that case by truncation in three steps, assuming first that θ is a Kähler form, and remove this assumption in a fourth step.

Step 1: agreement on deep sublevel sets. Let $\theta = \kappa$ be a Kähler form, and assume that $\varphi_1 \le \varphi_2$, $\psi_1 \le \psi_2$, and that there is $L > 0$ such that $\varphi_1 = \varphi_2$ on $U = \{\varphi_1 < -L\}$ and $\psi_1 = \psi_2$ on $V = \{\psi_1 < -L\}$. These sets are open, because φ_1 and ψ_1 are upper semicontinuous. For an integer $l > L$ put $\varphi_j^l = \max\{\varphi_j, -l\}$ and $\psi_j^l = \max\{\psi_j, -l\}$. These are bounded κ-psh functions; let $a^l = \varphi_1^l - \varphi_2^l$, $b^l = \psi_1^l - \psi_2^l$, and let S_j^l be defined as S_j with the truncated functions. The algebraic identity $x^n - y^n = (x - y)\sum_k x^k y^{n-1-k}$ for the commuting Bedford–Taylor products gives

$$\mu_{\psi_1^l} - \mu_{\psi_2^l} = \mathrm{dd^c} b^l \wedge \Sigma^l, \qquad S_1^l - S_2^l = \mathrm{dd^c} a^l \wedge \Sigma^l, \qquad \Sigma^l = \sum_{k=0}^{n-1} \kappa_{\psi_1^l}^k \wedge \kappa_{\psi_2^l}^{n-1-k}.$$

Since Σ^l is a closed positive current, Stokes' formula for bounded potentials (Lemma 15.4.2) yields $\int_X a^l \,\mathrm{dd^c} b^l \wedge \Sigma^l = \int_X b^l \,\mathrm{dd^c} a^l \wedge \Sigma^l$, which is (17.2) for the truncated functions.

We show that both sides of this identity coincide with the corresponding sides of (17.2). On U we have $a^l = a = 0$. On the open set V we have $\psi_1^l = \psi_2^l$ and $\psi_1 = \psi_2$, so $\mu_{\psi_1^l} = \mu_{\psi_2^l}$ and $\mu_{\psi_1} = \mu_{\psi_2}$ there, by locality of Bedford–Taylor and non-pluripolar products. Hence both left sides are integrals over the closed set $F = X \setminus (U \cup V)$. On F we have $\varphi_2 \geq \varphi_1 \geq -L$ and $\psi_2 \geq \psi_1 \geq -L$. Thus F is contained in the plurifine open set where all four potentials exceed $-l$; there every truncation coincides with its potential, so the bounded products of the truncations coincide with the corresponding non-pluripolar products, by plurifine locality (Proposition 15.1.5). Consequently the two left sides are equal. The right sides are treated in the same way with the roles of U and V exchanged: $b^l = b = 0$ on V, while $S_1^l = S_2^l$ and $S_1 = S_2$ on U. This proves (17.2) under the assumptions of this step.

Step 2: a Kähler background with a margin. Let $\theta = \kappa$ be Kähler, and suppose that all four potentials are $\lambda\kappa$-psh for some $0 < \lambda < 1$. Adding a constant to one of the four potentials changes neither side of (17.2). Indeed, the measures are unchanged, and a constant added to a or b contributes nothing, since $\int_X (\mu_{\psi_1} - \mu_{\psi_2}) = 0$ and $\int_X (S_1 - S_2) = 0$ by Corollary 15.7.5. Subtracting suitable constants from φ_1 and ψ_1, and then a common constant from each pair, we may therefore assume that all four potentials are nonpositive and that

$$\varphi_1 \leq \varphi_2 \leq \varphi_1 + B, \qquad \psi_1 \leq \psi_2 \leq \psi_1 + B$$

for a constant $B \geq 1$. Take a sequence $\varepsilon \downarrow 0$ with $\lambda(1+\varepsilon) < 1$ and put

$$\varphi_{2,\varepsilon} = \max\{\varphi_1, (1+\varepsilon)\varphi_2\}, \qquad \psi_{2,\varepsilon} = \max\{\psi_1, (1+\varepsilon)\psi_2\}.$$

Since $(1+\varepsilon)\varphi_2$ is $(1+\varepsilon)\lambda\kappa$-psh, the function $\varphi_{2,\varepsilon}$ is κ-psh, and $\varphi_1 \leq \varphi_{2,\varepsilon} \leq \varphi_2 \leq \varphi_1 + B$ because $\varphi_2 \leq 0$. Where $\varphi_1 < -B - B/\varepsilon$,

$$(1+\varepsilon)\varphi_2 \leq (1+\varepsilon)(\varphi_1 + B) = \varphi_1 + B + \varepsilon(\varphi_1 + B) < \varphi_1,$$

so $\varphi_{2,\varepsilon} = \varphi_1$ there. The same holds for the ψ's. Step 1 therefore applies to the pairs $(\varphi_1, \varphi_{2,\varepsilon})$ and $(\psi_1, \psi_{2,\varepsilon})$ and gives (17.2) with $a_\varepsilon = \varphi_1 - \varphi_{2,\varepsilon}$ and $b_\varepsilon = \psi_1 - \psi_{2,\varepsilon}$ in place of a, b, and with the corresponding measures.

Step 3: the limit $\varepsilon \downarrow 0$. As $\varepsilon \downarrow 0$, the functions $(1+\varepsilon)\varphi_2$ increase to φ_2, so $\varphi_{2,\varepsilon}$ increases to $\varphi_{2,0} = \max\{\varphi_1, \varphi_2\} = \varphi_2$; likewise $\psi_{2,\varepsilon} \uparrow \psi_{2,0} = \psi_2$. By Corollary 15.7.6, every measure in the identity of Step 2 converges weakly to the corresponding measure in (17.2). To pass to the limit in the integrals we apply Lemma 17.1.1 to each of the finitely many terms, each of which is the integral of a bounded weight against a positive measure.

We first check uniform absolute continuity. For $\varepsilon \geq 0$ put

$$\zeta_\varepsilon = \tfrac{1}{5}(\varphi_1 + \varphi_{2,\varepsilon} + \psi_1 + \psi_{2,\varepsilon}), \qquad \chi = \zeta_0 - B.$$

Since $-B \leq \varphi_{2,\varepsilon} - \varphi_2 \leq 0$ and $-B \leq \psi_{2,\varepsilon} - \psi_2 \leq 0$, we have $\chi \leq \zeta_\varepsilon \leq \chi + B$ for all $\varepsilon \geq 0$. Each of the four potentials is $\lambda\kappa$-psh, so $\mathrm{dd^c}\chi \geq -\frac{4}{5}\lambda\kappa$, and $\kappa + \mathrm{dd^c}\chi$ is the sum of $(1 - 4\lambda/5)\kappa$ and the closed positive current $\frac{4}{5}\lambda\kappa + \mathrm{dd^c}\chi$. In particular χ is κ-psh, and expanding $\langle(\kappa + \mathrm{dd^c}\chi)^n\rangle$ by multilinearity and compatibility with smooth factors (Propositions 15.1.5 and 15.3.3) gives $\int_X \langle(\kappa + \mathrm{dd^c}\chi)^n\rangle \geq (1 - 4\lambda/5)^n \int_X \kappa^n > 0$.

Now let μ refer to the background κ. By the identity

$$5\kappa_{\zeta_\varepsilon} = \kappa_{\varphi_1} + \kappa_{\varphi_{2,\varepsilon}} + \kappa_{\psi_1} + \kappa_{\psi_{2,\varepsilon}} + \kappa$$

and multilinearity, each measure occurring in the identity of Step 2 (for $\varepsilon > 0$) or in (17.2) (for $\varepsilon = 0$) is a non-pluripolar product of n of the currents on the right, and hence is at most $5^n \mu_{\zeta_\varepsilon}$. The potential $q = B^{-1}\zeta_\varepsilon + (1 - B^{-1})\chi$ satisfies $\chi \leq q \leq \chi + 1$, so $q - 1$ is a competitor for $\mathrm{Cap}_{\kappa,\chi}$, and $\mu_q \geq B^{-n}\mu_{\zeta_\varepsilon}$ by multilinearity, since $B \geq 1$. Hence $\mu_{\zeta_\varepsilon}(A) \leq B^n \mathrm{Cap}_{\kappa,\chi}(A)$ for every Borel set A, and every measure in question, for every $\varepsilon \geq 0$, is bounded by $5^n B^n \mathrm{Cap}_{\kappa,\chi}$; in particular its total mass is at most $5^n B^n \mathrm{Cap}_{\kappa,\chi}(X) = 5^n B^n \int_X \langle(\kappa + \mathrm{dd^c}\chi)^n\rangle \leq 5^n B^n \int_X \kappa^n$, by Proposition 16.7.6 and Theorem 15.4.3. Since χ has positive mass, Theorem 16.9.8 bounds $\mathrm{Cap}_{\kappa,\chi}(A)$ by a continuous nondecreasing function of $\mathrm{Cap}_\omega(A)$ vanishing at zero. Uniform absolute continuity follows.

Next we check the weights. Assign the value zero to $a_\varepsilon, b_\varepsilon, a, b$ on the pole sets; these are Borel functions with values in $[-B, 0]$. Given $\eta > 0$, Proposition 16.9.9(3), applied to each of the four potentials with the bound $\eta/4$, and the countable subadditivity of Cap_ω (Proposition 16.7.6) give an open set G with $\mathrm{Cap}_\omega(G) < \eta$ which contains the pole sets of the four potentials and such that the four potentials are finite and continuous on the compact set $X \setminus G$. Then a and b are continuous on $X \setminus G$, so they satisfy the continuity hypothesis of Lemma 17.1.1. Moreover the four potentials are bounded in absolute value on $X \setminus G$ by a constant $C_\eta > 0$. Since $(1 + \varepsilon)\varphi_2 \leq \varphi_{2,\varepsilon} \leq \varphi_2$, we have on $X \setminus G$

$$|a_\varepsilon - a| = |\varphi_{2,\varepsilon} - \varphi_2| \leq |(1 + \varepsilon)\varphi_2 - \varphi_2| \leq \varepsilon C_\eta,$$

and likewise $|b_\varepsilon - b| \leq \varepsilon C_\eta$. Hence, for every $\delta > 0$ and $\varepsilon < \delta/C_\eta$, the sets $\{|a_\varepsilon - a| > \delta\}$ and $\{|b_\varepsilon - b| > \delta\}$ lie in G and have capacity less than η. Since η is arbitrary, $a_\varepsilon \to a$ and $b_\varepsilon \to b$ in capacity. Lemma 17.1.1 now passes the identity of Step 2 to the limit, which is (17.2).

Step 4: an arbitrary background. For all sufficiently large $s > 0$ we have $s\omega \geq \theta$ and the form $\kappa_s = \theta + s\omega$ is Kähler. Each $q \in \mathrm{PSH}(X, \theta)$ is then $\frac{1}{2}\kappa_s$-psh, because $\frac{1}{2}\kappa_s + \mathrm{dd^c}q = \theta_q + \frac{1}{2}(s\omega - \theta) \geq 0$. By Steps 2 and 3 with $\lambda = \frac{1}{2}$, (17.2) holds in the background κ_s for all large s. Since $\kappa_s + \mathrm{dd^c}q = \theta_q + s\omega$, multilinearity and compatibility with smooth factors (Propositions 15.1.5 and 15.3.3) show that every measure in the identity is a polynomial in s whose coefficients are finite measures of the form $\langle\theta_{q_1} \wedge \cdots \wedge \theta_{q_k}\rangle \wedge \omega^{n-k}$; the weights a, b do not depend on s. Both sides

of the identity are therefore polynomials in s, and since they agree for all large s, their coefficients agree. The constant coefficients are the two sides of (17.2) in the background θ. □

Theorem 17.1.3 *Let a, b be bounded quasi-psh differences on X, and let $T_1, \dots, T_{n-1}$ be closed positive $(1,1)$-currents on X. Then*

$$\int_X a\,\mathrm{dd^c} b \wedge \langle T_1 \wedge \cdots \wedge T_{n-1}\rangle = \int_X b\,\mathrm{dd^c} a \wedge \langle T_1 \wedge \cdots \wedge T_{n-1}\rangle, \tag{17.3}$$

where the Hessian factors are the signed measures defined in (17.1).

Proof *Step 1: a repeated factor and a common background.* Let (φ_1, φ_2) and (ψ_1, ψ_2) be presentations of a and b, and suppose that these four functions and a further potential η all belong to $\mathrm{PSH}(X, \theta)$ for one smooth closed real $(1,1)$-form θ. For $0 \le t \le 1$ the functions $\psi_j^t = (1-t)\eta + t\psi_j$ are θ-psh, and $\psi_1^t - \psi_2^t = tb$ is bounded. Lemma 17.1.2, applied to $\varphi_1, \varphi_2, \psi_1^t, \psi_2^t$, gives

$$\int_X a(\mu_{\psi_1^t} - \mu_{\psi_2^t}) = t\int_X b(S_1^t - S_2^t), \qquad S_j^t = \sum_{k=0}^{n-1} \left\langle \theta_{\varphi_j} \wedge \theta_{\psi_1^t}^k \wedge \theta_{\psi_2^t}^{n-1-k} \right\rangle.$$

Since $\theta_{\psi_j^t} = (1-t)\theta_\eta + t\theta_{\psi_j}$, multilinearity expands every measure here as a polynomial in t with finite measures as coefficients, so both sides are polynomials in t. We compare their coefficients of t. On the left this coefficient is

$$n\int_X a\left(\langle \theta_{\psi_1} \wedge \theta_\eta^{n-1}\rangle - \langle \theta_{\psi_2} \wedge \theta_\eta^{n-1}\rangle\right) = n\int_X a\,\mathrm{dd^c} b \wedge \langle \theta_\eta^{n-1}\rangle,$$

where we used that (17.1) may be computed in any smooth background with respect to which the presentation is psh. On the right it is $\int_X b(S_1^0 - S_2^0)$, and $S_j^0 = n\langle \theta_{\varphi_j} \wedge \theta_\eta^{n-1}\rangle$, so it equals $n\int_X b\,\mathrm{dd^c} a \wedge \langle \theta_\eta^{n-1}\rangle$. This proves (17.3) when $T_1 = \cdots = T_{n-1} = \theta_\eta$ and all potentials share the background θ. For $n = 1$ there are no currents T_j, and (17.3) is (17.2) for a Kähler background with respect to which the four functions are psh.

Step 2: arbitrary currents. Let $n > 1$. By Theorem 14.1.7, write $T_j = \theta_j + \mathrm{dd^c}\eta_j$ with θ_j smooth closed and $\eta_j \in \mathrm{PSH}(X, \theta_j)$. For positive numbers $r_1, \dots, r_{n-1}$ with $\sum_j r_j = 1$ put $\theta_r = \sum_j r_j\theta_j$ and $\eta_r = \sum_j r_j\eta_j$, so that $\sum_j r_j T_j = \theta_r + \mathrm{dd^c}\eta_r$. The potential η_r is $(\theta_r + s\omega)$-psh for every $s \ge 0$, since $\theta_r + s\omega + \mathrm{dd^c}\eta_r = \sum_j r_j T_j + s\omega \ge 0$. Let κ be a multiple of ω such that the presentations of a and b are κ-psh. Since the forms θ_r are convex combinations of the finitely many smooth forms θ_j, there is s_0 with $\theta_r + s\omega \ge \kappa$ for all r and all $s \ge s_0$, and then the presentations are $(\theta_r + s\omega)$-psh as well. Step 1 applied to the current $\sum_j r_j T_j + s\omega = (\theta_r + s\omega) + \mathrm{dd^c}\eta_r$ gives

$$\int_X a\,\mathrm{dd^c} b \wedge \left\langle \Big(\sum_j r_j T_j + s\omega\Big)^{n-1}\right\rangle = \int_X b\,\mathrm{dd^c} a \wedge \left\langle \Big(\sum_j r_j T_j + s\omega\Big)^{n-1}\right\rangle,$$

where, by the independence of (17.1) from the background, the Hessian factors may be computed in one fixed Kähler background. Expanding the repeated current

by multilinearity, both sides become polynomials in s, and comparing constant coefficients gives

$$\int_X a\,\mathrm{dd^c} b \wedge \Big\langle \Big(\sum_j r_j T_j\Big)^{n-1} \Big\rangle = \int_X b\,\mathrm{dd^c} a \wedge \Big\langle \Big(\sum_j r_j T_j\Big)^{n-1} \Big\rangle .$$

Both sides are homogeneous polynomials of degree $n-1$ in $(r_1,\dots,r_{n-1})$. Since they agree on the open simplex, homogeneity makes them agree on the open positive cone of $\mathbb{R}^{n-1}$. Hence their coefficients agree: a polynomial that vanishes on a nonempty open set is zero, because all its partial derivatives vanish at a point r_0 of that set and it equals its finite Taylor expansion at r_0. The coefficient of $r_1\cdots r_{n-1}$ is $(n-1)!$ times the corresponding side of (17.3). □

Lemma 17.1.4 *Let $u, v \in \mathrm{PSH}(X,\theta)$ satisfy $u \sim v$, let $T_1,\dots,T_{n-1}$ be closed positive $(1,1)$-currents on X, and put $R = \langle T_1 \wedge \cdots \wedge T_{n-1}\rangle$. Then*

$$\int_X (u-v)\,\mathrm{dd^c}(u-v) \wedge R \le 0.$$

Proof Put $\lambda_u = \langle \theta_u \wedge T_1 \wedge \cdots \wedge T_{n-1}\rangle$ and define λ_v in the same way, so that $\mathrm{dd^c}(u-v)\wedge R = \lambda_u - \lambda_v$ by (17.1). Since $u-v$ is bounded, Corollary 15.7.5, applied after writing $T_i = \theta_i + \mathrm{dd^c}\eta_i$ with smooth θ_i and $\eta_i \in \mathrm{PSH}(X,\theta_i)$ (Theorem 14.1.7), gives $\lambda_u(X) = \lambda_v(X)$. Hence the integral does not change if a constant is added to u, and we may assume $u \le v$. We must show that $\int_X (v-u)\lambda_v \le \int_X (v-u)\lambda_u$.

Let $t \ge 0$ and $\varepsilon > 0$, and put $q = \max\{u, v-t-\varepsilon\}$. Then $v-t-\varepsilon \le q \le v$, so $\lambda_q(X) = \lambda_v(X) = \lambda_u(X)$ by the same corollary. The sets $O_1 = \{u < v-t-\varepsilon\}$ and $O_2 = \{u > v-t-\varepsilon\}$ are disjoint and plurifine open, by Proposition 10.5.1 applied in charts after adding the same smooth local potential of θ to u and v; neither of them meets the common pole set of u and v. Moreover $q = v-t-\varepsilon$ on O_1 and $q = u$ on O_2, so plurifine locality (Proposition 15.1.5) gives $\lambda_q = \lambda_v$ on O_1 and $\lambda_q = \lambda_u$ on O_2, and hence

$$\lambda_v(O_1) + \lambda_u(O_2) \le \lambda_q(X) = \lambda_u(X).$$

Since pole sets carry no mass, it follows that

$$\lambda_v(\{u < v-t-\varepsilon\}) \le \lambda_u(\{u \le v-t-\varepsilon\}) \le \lambda_u(\{u < v-t\}).$$

As $\varepsilon \downarrow 0$ the sets $\{u < v-t-\varepsilon\}$ increase to $\{u < v-t\}$, and therefore

$$\lambda_v(\{u < v-t\}) \le \lambda_u(\{u < v-t\}) \qquad \text{for every } t \ge 0.$$

On the common finite locus the bounded function $v-u \ge 0$ equals $\int_0^\infty \mathbf{1}_{\{u<v-t\}}\,\mathrm{d}t$. Integrating the last inequality in t and using Tonelli's theorem, we obtain $\int_X (v-u)\lambda_v \le \int_X (v-u)\lambda_u$, as required. □

17.2 Relative Monge–Ampère energy

Fix a smooth closed real $(1,1)$-form θ whose class is big, and a model potential $\phi \in \mathrm{PSH}(X,\theta)$, that is, $P_\theta[\phi] = \phi$ and $m = \int_X \langle \theta_\phi^n \rangle > 0$ (Definition 16.3.1). In particular $\phi \le 0$ and $\sup_X \phi = 0$. Recall from Definition 15.8.3 that $\mathcal{E}(X,\theta,\phi)$ consists of the potentials $u \preceq \phi$ with $\int_X \langle \theta_u^n \rangle = m$. If $u_1, \dots, u_n \in \mathcal{E}(X,\theta,\phi)$, then

$$\int_X \langle \theta_{u_1} \wedge \cdots \wedge \theta_{u_n} \rangle = m.$$

Indeed $P_\theta[u_j] = P_\theta[\phi] = \phi$ by Proposition 16.4.10, and the singularity envelope preserves mixed masses by Proposition 16.3.2.

Definition 17.2.1 Put

$$\mathcal{E}^\infty(X,\theta,\phi) = \{u \in \mathrm{PSH}(X,\theta) : u \sim \phi\},$$
$$\mathcal{E}^1(X,\theta,\phi) = \left\{ u \in \mathcal{E}(X,\theta,\phi) : \int_X |u-\phi|\, \mu_u < \infty \right\}.$$

The potentials in $\mathcal{E}^\infty(X,\theta,\phi)$ have bounded difference from ϕ, but they need not be bounded functions. The class $\mathcal{E}^1(X,\theta,\phi)$ is the *relative finite-energy class*. For the potential $\phi = V_\theta$ with minimal singularities (Definition 14.2.6) we omit the reference from these notations.

If $u \sim \phi$, then $\int_X \langle \theta_u^n \rangle = m$ by Corollary 15.7.5, and $|u-\phi|$ is bounded. Hence

$$\mathcal{E}^\infty(X,\theta,\phi) \subset \mathcal{E}^1(X,\theta,\phi) \subset \mathcal{E}(X,\theta,\phi).$$

The potential V_θ is indeed a model potential: $P_\theta[V_\theta] = V_\theta$ by Proposition 16.3.2(4), and $\int_X \langle \theta_{V_\theta}^n \rangle = \mathrm{vol}(\{\theta\}) > 0$ by Definition 15.8.1 and Proposition 15.8.2. For $\phi = V_\theta$, the class $\mathcal{E}^\infty(X,\theta)$ consists of the potentials with minimal singularities, and $\mathcal{E}(X,\theta) = \mathcal{E}(X,\theta,V_\theta)$ is the class of potentials of full Monge–Ampère mass of Definition 15.8.3. These classes of global potentials are not related to the local class $\mathcal{E}_0(B)$ of bounded functions on a ball in Definition 9.10.1.

Definition 17.2.2 For $u \in \mathcal{E}^\infty(X,\theta,\phi)$ define

$$E_\theta^\phi(u) = \frac{1}{n+1} \sum_{j=0}^{n} \int_X (u-\phi)\, \langle \theta_u^j \wedge \theta_\phi^{n-j} \rangle. \tag{17.4}$$

All these integrals are finite, since the weight is bounded and every mixed measure has total mass m. For $\phi = V_\theta$ we write E_θ.

Proposition 17.2.3 *Let $u, v \in \mathcal{E}^\infty(X,\theta,\phi)$ and put $u_t = (1-t)u + tv$ for $0 \le t \le 1$. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t} E_\theta^\phi(u_t) = \int_X (v-u)\, \mu_{u_t}, \tag{17.5}$$

with one-sided derivatives at the endpoints, and

$$E_\theta^\phi(v) - E_\theta^\phi(u) = \frac{1}{n+1}\sum_{j=0}^{n}\int_X (v-u)\langle\theta_v^j \wedge \theta_u^{n-j}\rangle. \tag{17.6}$$

In particular, the energy is increasing, and $E_\theta^\phi(u+C) = E_\theta^\phi(u) + Cm$ *for every constant* C.

Proof Let $w \in \mathcal{E}^\infty(X,\theta,\phi)$. For $0 \le r \le 1$ the potential $\phi + r(w-\phi) = (1-r)\phi + rw$ again belongs to $\mathcal{E}^\infty(X,\theta,\phi)$, and multilinearity gives

$$\mu_{\phi+r(w-\phi)} = \sum_{j=0}^{n}\binom{n}{j} r^j(1-r)^{n-j}\langle\theta_w^j \wedge \theta_\phi^{n-j}\rangle.$$

Since $\binom{n}{j}\int_0^1 r^j(1-r)^{n-j}\,\mathrm{d}r = 1/(n+1)$, as one sees by integrating by parts j times, we obtain

$$E_\theta^\phi(w) = \int_0^1\int_X (w-\phi)\,\mu_{\phi+r(w-\phi)}\,\mathrm{d}r. \tag{17.7}$$

Now put $h = v-u$ and $q_{r,t} = \phi + r(u_t - \phi)$. Since $\theta_{q_{r,t}} = (1-r)\theta_\phi + r(1-t)\theta_u + rt\theta_v$, the measures $\mu_{q_{r,t}}$ are polynomials in (r,t) whose coefficients are mixed measures of total mass m, and the weights $u_t - \phi = (u-\phi) + th$ are bounded. All expressions below are therefore polynomials in (r,t) with finite coefficients, and the derivatives are derivatives of polynomials. Identities between such polynomials of measures follow from the corresponding identities between polynomials in commuting variables x, y, z, by substituting the mixed measure $\langle\theta_\phi^a \wedge \theta_u^b \wedge \theta_v^c\rangle$ for the monomial $x^a y^b z^c$; multilinearity shows that this substitution is compatible with the products occurring here. For instance, the identities $\partial_t \ell^n = nr(z-y)\ell^{n-1}$ and $\partial_r \ell^n = n\big((1-t)y + tz - x\big)\ell^{n-1}$ for $\ell = (1-r)x + r(1-t)y + rtz$ give

$$\partial_t \mu_{q_{r,t}} = nr\,\mathrm{dd^c}h \wedge \langle\theta_{q_{r,t}}^{n-1}\rangle, \qquad \partial_r \mu_{q_{r,t}} = n\,\mathrm{dd^c}(u_t - \phi) \wedge \langle\theta_{q_{r,t}}^{n-1}\rangle, \tag{17.8}$$

in the sense of (17.1). Differentiating (17.7) with $w = u_t$ and using the first of these identities, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}E_\theta^\phi(u_t) = \int_0^1\left[\int_X h\,\mu_{q_{r,t}} + nr\int_X (u_t-\phi)\,\mathrm{dd^c}h \wedge \langle\theta_{q_{r,t}}^{n-1}\rangle\right]\mathrm{d}r.$$

By Theorem 17.1.3, applied to the bounded quasi-psh differences $u_t - \phi$ and h, the second integral equals $\int_X h\,\mathrm{dd^c}(u_t-\phi) \wedge \langle\theta_{q_{r,t}}^{n-1}\rangle$, which is $\frac{1}{n}\partial_r \int_X h\,\mu_{q_{r,t}}$ by the second identity. The integrand in r is therefore $\partial_r\big[r\int_X h\,\mu_{q_{r,t}}\big]$, and since $q_{1,t} = u_t$,

$$\frac{\mathrm{d}}{\mathrm{d}t}E_\theta^\phi(u_t) = \int_X h\,\mu_{u_t}.$$

This is (17.5). Integrating it from $t=0$ to $t=1$, and expanding $\mu_{u_t}=\sum_j \binom{n}{j}t^j(1-t)^{n-j}\langle\theta_v^j\wedge\theta_u^{n-j}\rangle$ with the same beta integrals, gives (17.6). If $u\le v$, every term of the cocycle formula is nonnegative, so the energy is increasing. Every mixed measure has total mass m, so taking $v=u+C$ in the cocycle formula gives the last assertion.□

Proposition 17.2.4 *The energy is concave on $\mathcal{E}^\infty(X,\theta,\phi)$, and for $u,v\in\mathcal{E}^\infty(X,\theta,\phi)$,*

$$\int_X (v-u)\,\mu_v \le E_\theta^\phi(v)-E_\theta^\phi(u)\le \int_X (v-u)\,\mu_u. \tag{17.9}$$

Moreover $|E_\theta^\phi(v)-E_\theta^\phi(u)|\le m\sup|v-u|$, the supremum being taken over the common finite locus. If $\psi\in\mathrm{PSH}(X,\theta)$ and $\psi\sim\phi$, then the functional E_θ^ψ defined by (17.4) with ψ in place of ϕ satisfies

$$E_\theta^\psi(u)=E_\theta^\phi(u)-E_\theta^\phi(\psi).$$

Proof Let $u_t=(1-t)u+tv$ and $h=v-u$. Then $\partial_t\mu_{u_t}=n\,\mathrm{dd^c}h\wedge\langle\theta_{u_t}^{n-1}\rangle$. Indeed, by multilinearity and (17.1), both sides expand into combinations of the mixed measures $\langle\theta_u^a\wedge\theta_v^{n-a}\rangle$, obtained by substituting $\langle\theta_u^a\wedge\theta_v^{n-a}\rangle$ for y^az^{n-a} in the two sides of the identity $\partial_t((1-t)y+tz)^n=n(z-y)((1-t)y+tz)^{n-1}$ in commuting variables y,z. Differentiating (17.5) once more, again as a polynomial in t, and using this identity, we obtain

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}E_\theta^\phi(u_t)=n\int_X h\,\mathrm{dd^c}h\wedge\langle\theta_{u_t}^{n-1}\rangle\le 0$$

by Lemma 17.1.4. Thus $t\mapsto E_\theta^\phi(u_t)$ is concave, and its derivative decreases from $\int_X h\,\mu_u$ at $t=0$ to $\int_X h\,\mu_v$ at $t=1$. Integrating the derivative over $[0,1]$ gives the tangent bounds (17.9). Since $\int_X\langle\theta_{u_t}^n\rangle=m$, the first variation formula also gives $|E_\theta^\phi(v)-E_\theta^\phi(u)|\le m\sup|v-u|$. Finally, the cocycle formula (17.6) applied to the pair ψ,u reads $E_\theta^\phi(u)-E_\theta^\phi(\psi)=E_\theta^\psi(u)$. □

We now extend the energy to all potentials that are at least as singular as ϕ.

Definition 17.2.5 For $u\in\mathrm{PSH}(X,\theta)$ with $u\preceq\phi$ put

$$E_\theta^\phi(u)=\inf\{E_\theta^\phi(w):w\in\mathcal{E}^\infty(X,\theta,\phi),\ u\le w\}.$$

The family on the right is nonempty, since $\phi+C$ belongs to it for large C; hence $E_\theta^\phi(u)\in[-\infty,\infty)$. Because the energy is increasing on $\mathcal{E}^\infty(X,\theta,\phi)$, the new definition agrees with (17.4) there.

Proposition 17.2.6 *The extended energy is increasing, satisfies $E_\theta^\phi(u+C)=E_\theta^\phi(u)+Cm$, and*

$$E_\theta^\phi(u)=\lim_{k\to\infty}E_\theta^\phi(\max\{u,\phi-k\}),$$

where the limit exists in $[-\infty,\infty)$.

Proof If $u \le u'$, every majorant of u' in the definition is a majorant of u, so $E^\phi_\theta(u) \le E^\phi_\theta(u')$. Translating majorants by a constant and using the shift formula on $\mathcal{E}^\infty(X,\theta,\phi)$ gives the shift formula. The potentials $u_k = \max\{u, \phi - k\}$ belong to $\mathcal{E}^\infty(X,\theta,\phi)$, decrease with k and dominate u; hence $E^\phi_\theta(u_k)$ decreases and is bounded below by $E^\phi_\theta(u)$. Conversely, let w be any majorant in the definition, and choose C with $w \ge \phi - C$. For $k \ge C$ both u and $\phi - k$ lie below w, so $u_k \le w$ and $E^\phi_\theta(u_k) \le E^\phi_\theta(w)$. Letting $k \to \infty$ and taking the infimum over w gives $\lim_k E^\phi_\theta(u_k) \le E^\phi_\theta(u)$. □

Proposition 17.2.7 *Let $(u_i)_{i\in I}$ be a decreasing net in $\mathrm{PSH}(X,\theta)$ with $u_i \preceq \phi$, and suppose that $u = \inf_i u_i$ belongs to $\mathrm{PSH}(X,\theta)$. Then*

$$E^\phi_\theta(u_i) \downarrow E^\phi_\theta(u).$$

Consequently the extended energy is concave on $\{u \in \mathrm{PSH}(X,\theta) : u \preceq \phi\}$, and the set where it is finite is convex.

Proof First let the net be a sequence (u_j). By monotonicity $E^\phi_\theta(u_j)$ decreases to a limit $L \ge E^\phi_\theta(u)$. Let $w \in \mathcal{E}^\infty(X,\theta,\phi)$ with $w \ge u$, and put $v_j = \max\{u_j, w\}$. Then v_j decreases pointwise to $\max\{u,w\} = w$. Since $u_j \le u_1 \le \phi + C$ and $w \ge \phi - C$ for some constant C, the differences $v_j - w$ are uniformly bounded. The tangent bound (17.9) gives

$$0 \le E^\phi_\theta(v_j) - E^\phi_\theta(w) \le \int_X (v_j - w)\,\mu_w \longrightarrow 0$$

by bounded convergence. Since $E^\phi_\theta(u_j) \le E^\phi_\theta(v_j)$, we get $L \le E^\phi_\theta(w)$, and taking the infimum over w gives $L \le E^\phi_\theta(u)$.

For a general net, Corollary 14.2.5 shows that $u_i \to u$ in $L^1(X)$; its hypothesis holds because $\sup_X u_i \ge \sup_X u > -\infty$. Choose indices $i_1 \le i_2 \le \cdots$ with $\|u_{i_k} - u\|_{L^1} < 1/k$. The decreasing pointwise limit $\tilde u$ of (u_{i_k}) is a θ-psh function by Theorem 8.3.1, applied in charts after adding smooth local potentials of θ, since it dominates u and so is not identically $-\infty$. By monotone convergence, $\int_X (\tilde u - u)\,\omega^n = \lim_k \int_X (u_{i_k} - u)\,\omega^n = 0$, so $\tilde u = u$ almost everywhere, hence everywhere. The sequence case gives $E^\phi_\theta(u_{i_k}) \to E^\phi_\theta(u)$. The net $E^\phi_\theta(u_i)$ is decreasing, so its limit is its infimum, which is at most $\lim_k E^\phi_\theta(u_{i_k}) = E^\phi_\theta(u)$; the reverse inequality is monotonicity. (The selected indices need not be cofinal in I; this is why we compare infima.)

For concavity let $u, v \preceq \phi$ and $0 < t < 1$; we may assume that $E^\phi_\theta(u)$ and $E^\phi_\theta(v)$ are finite, since otherwise there is nothing to prove. The potentials $(1-t)\max\{u, \phi - k\} + t\max\{v, \phi - k\}$ belong to $\mathcal{E}^\infty(X,\theta,\phi)$ and decrease to $(1-t)u + tv$ as $k \to \infty$. Concavity on $\mathcal{E}^\infty(X,\theta,\phi)$ and decreasing continuity give

$$E^\phi_\theta((1-t)u + tv) \ge (1-t)E^\phi_\theta(u) + tE^\phi_\theta(v).$$

In particular the finite locus of the energy is convex. □

Theorem 17.2.8 *For $u \in \mathrm{PSH}(X,\theta)$ with $u \preceq \phi$,*

$$E^{\phi}_{\theta}(u) > -\infty \quad \Longleftrightarrow \quad u \in \mathcal{E}^1(X,\theta,\phi).$$

In particular, finite extended energy forces full mass relative to ϕ. If $u \in \mathcal{E}^1(X,\theta,\phi)$ and $u \le \phi$, then

$$\frac{1}{n+1}\int_X (\phi - u)\,\mu_u \le -E^{\phi}_{\theta}(u) \le \int_X (\phi - u)\,\mu_u. \tag{17.10}$$

Proof Both conditions are invariant under adding a constant to u, so we may assume $u \le \phi$. Put $u_k = \max\{u, \phi - k\}$ and $A_k = \{u > \phi - k\}$. The potentials u_k belong to $\mathcal{E}^{\infty}(X,\theta,\phi)$ and have mass m. Since $u_k \le \phi$, all terms in (17.4) are nonpositive, and keeping only the term $j = n$ gives

$$\int_X (\phi - u_k)\mu_{u_k} \le (n+1)\big(-E^{\phi}_{\theta}(u_k)\big).$$

Suppose that $E^{\phi}_{\theta}(u) > -\infty$. Since $E^{\phi}_{\theta}(u) \le E^{\phi}_{\theta}(u_k) \le E^{\phi}_{\theta}(\phi) = 0$, the right side is bounded by $C = -(n+1)E^{\phi}_{\theta}(u)$. On $X \setminus A_k$ the weight $\phi - u_k$ equals k outside a pluripolar set; on the plurifine open set A_k we have $u_k = u$, so $\mu_{u_k} = \mu_u$ there by locality. Hence

$$m - \mu_u(A_k) = \mu_{u_k}(X \setminus A_k) \le C/k.$$

The sets A_k increase to the set where u is finite, which carries μ_u, so $\int_X \langle \theta_u^n \rangle \ge m$; since $u \le \phi$, mass monotonicity (Theorem 15.7.4) gives the reverse inequality, and therefore $u \in \mathcal{E}(X,\theta,\phi)$. Locality also gives $\int_{A_k}(\phi - u)\mu_u \le C$, and monotone convergence yields $\int_X (\phi - u)\mu_u \le C$. This proves that $u \in \mathcal{E}^1(X,\theta,\phi)$ and the first inequality in (17.10).

Conversely, suppose $u \in \mathcal{E}^1(X,\theta,\phi)$, and let $I = \int_X (\phi - u)\mu_u < \infty$. The measures μ_{u_k} and μ_u have the same total mass m and agree on A_k, so $\mu_{u_k}(X \setminus A_k) = \mu_u(X \setminus A_k)$. Thus

$$\int_X (\phi - u_k)\mu_{u_k} = \int_X \min\{\phi - u, k\}\,\mu_u \le I.$$

The lower tangent bound in (17.9), applied to u_k and ϕ, gives $E^{\phi}_{\theta}(u_k) \ge -I$. Letting $k \to \infty$ and using Proposition 17.2.6, we obtain $E^{\phi}_{\theta}(u) \ge -I > -\infty$, which is the second inequality in (17.10). The first inequality was proved above. □

Example 17.2.9 Let X be a compact Kähler curve, so $n = 1$, and take $\theta = \omega$ and the reference $V_\omega = 0$. For a smooth ω-psh function u,

$$E_\omega(u) = \int_X u\,\omega - \tfrac{1}{2}\int_X du \wedge d^c u,$$

where d^c is the real operator of (7.6), so that $dd^c = dd^c$. Indeed, by definition $E_\omega(u) = \frac{1}{2}\big(\int_X u\,\omega + \int_X u(\omega + dd^c u)\big)$, and Stokes' formula (Lemma 15.4.2) gives

$\int_X u\,\mathrm{dd^c}u = -\int_X \mathrm{d}u\wedge\mathrm{d^c}u$. Since $\mathrm{d}u\wedge\mathrm{d^c}u = \frac{\mathrm{i}}{2\pi}\partial u\wedge\bar\partial u$, the second term is a multiple of the classical Dirichlet energy of u.

The unbounded potential u on $\mathbb{P}^1$ of Example 16.4.8, which has full mass, also lies in $\mathcal{E}^1(\mathbb{P}^1,\omega)$. Away from its only pole $z=0$ the function u is bounded, and μ_u is a finite measure. Near the pole write $r=\log|z|^2$. If G is a smooth function of one real variable, then $\bar\partial G(\log|z|^2) = G'(r)\,\mathrm{d}\bar z/\bar z$ and $\partial\bar\partial G(\log|z|^2) = G''(r)\,\mathrm{d}z\wedge\mathrm{d}\bar z/|z|^2$ on $\mathbb{C}\setminus\{0\}$; since $\mathrm{i}\,\mathrm{d}z\wedge\mathrm{d}\bar z = 2\,\mathrm{d}A$, where $\mathrm{d}A$ is Lebesgue measure, we get $\mathrm{dd^c}G(\log|z|^2) = \pi^{-1}G''(r)|z|^{-2}\,\mathrm{d}A$, which in polar coordinates $z=\rho\mathrm{e}^{\mathrm{i}\vartheta}$, with $r=2\log\rho$, is $(2\pi)^{-1}G''(r)\,\mathrm{d}r\,\mathrm{d}\vartheta$.

The function F of Example 16.4.8 satisfies $F(r) = -\log(-r)$ for $r\le r_0$, with r_0 sufficiently negative, and is therefore smooth there, with $F''(r)=r^{-2}$. On the punctured disc $\{0<|z|^2<\mathrm{e}^{r_0}\}$ the measure μ_u is $\mathrm{dd^c}F(\log|z|^2)$, and it does not charge the pole; hence the image of its restriction to $\{|z|^2<\mathrm{e}^{r_0}\}$ under $z\mapsto\log|z|^2$ is $F''(r)\,\mathrm{d}r = r^{-2}\,\mathrm{d}r$ on $(-\infty,r_0)$. Since $u-F(\log|z|^2) = -\log(1+|z|^2)$ is bounded near the pole, the finiteness of $\int|u|\,\mu_u$ near the pole reduces, after the substitution $t=-r$, to

$$\int_a^\infty \frac{\log t}{t^2}\,\mathrm{d}t = \frac{\log a+1}{a} < \infty \qquad (a>1).$$

Hence u is a potential of finite energy whose difference from the reference $V_\omega=0$ is unbounded.

Corollary 17.2.10 *The classes $\mathcal{E}^\infty(X,\theta,\phi)$, $\mathcal{E}^1(X,\theta,\phi)$ and $\mathcal{E}(X,\theta,\phi)$ are convex. If $u\in\mathcal{E}^1(X,\theta,\phi)$, $v\in\mathrm{PSH}(X,\theta)$ and $u\le v\preceq\phi$, then $v\in\mathcal{E}^1(X,\theta,\phi)$.*

Proof Convex combinations of potentials with the same singularity type as ϕ have that singularity type. For $\mathcal{E}^1(X,\theta,\phi)$, convexity follows from the concavity of the extended energy (Proposition 17.2.7) and Theorem 17.2.8. If $u,v\in\mathcal{E}(X,\theta,\phi)$ and $0<t<1$, then $(1-t)u+tv\preceq\phi$, and by multilinearity $\mu_{(1-t)u+tv}$ is a combination of the mixed measures $\langle\theta_u^j\wedge\theta_v^{n-j}\rangle$ with coefficients $\binom{n}{j}(1-t)^j t^{n-j}$. Each of these measures has total mass m, as noted at the beginning of this section, so $\int_X\langle\theta^n_{(1-t)u+tv}\rangle = m$. The last assertion follows from the monotonicity of the extended energy and Theorem 17.2.8.□

Lemma 17.2.11 *Let $u\in\mathcal{E}(X,\theta,\phi)$, let $v\in\mathrm{PSH}(X,\theta)$ satisfy $v\preceq\phi$, and let $\gamma\colon X\to[-\infty,\infty]$ be a Borel function, for instance a θ-psh function, with $\gamma\ge\max\{u,v\}$. Then, allowing infinite values,*

$$\int_X(\gamma-u)\mu_v \le 2\int_X(\gamma-u)\mu_u + 2\int_X(\gamma-v)\mu_v.$$

Consequently, if $u,v,w_1,\dots,w_n\in\mathcal{E}^1(X,\theta,\phi)$, then

$$\int_X|u-v|\,\langle\theta_{w_1}\wedge\cdots\wedge\theta_{w_n}\rangle<\infty.$$

Proof For $t\ge0$ we have

$$\{\gamma > u + 2t\} \subset \{\gamma > v + t\} \cup \{v > u + t\},$$

because $\gamma \le v + t$ and $v \le u + t$ imply $\gamma \le u + 2t$. The weights $\gamma - u$ and $\gamma - v$ are Borel functions with values in $[0, \infty]$ on the complements of the pole sets of u and v, respectively, and these pole sets carry no mass; the level sets above are taken on these complements. Since $u \in \mathcal{E}(X, \theta, \phi)$ and $v - t \preceq \phi$, the comparison principle of Proposition 16.7.4, applied with $p = n$ to u and $v - t$, gives

$$\mu_v(\{v > u + t\}) \le \mu_u(\{v > u + t\}) + \int_X \langle \theta_\phi^n \rangle - \int_X \langle \theta_u^n \rangle = \mu_u(\{v > u + t\}).$$

Integrating over level sets, we obtain

$$\begin{aligned} \int_X (\gamma - u)\mu_v &\le 2 \int_0^\infty \mu_v(\{\gamma > v + t\})\, \mathrm{d}t + 2 \int_0^\infty \mu_u(\{v > u + t\})\, \mathrm{d}t \\ &\le 2 \int_X (\gamma - v)\mu_v + 2 \int_X (\gamma - u)\mu_u, \end{aligned}$$

where in the last step we used $(v - u)_+ \le \gamma - u$, which holds because $\gamma \ge \max\{u, v\}$. All functions involved are nonnegative, so Tonelli's theorem applies without any integrability assumption.

For the second assertion, add constants to u and v so that $u, v \le \phi$; this changes $|u - v|$ by at most a constant, and the mixed measure has finite mass m. Put $w = \frac{1}{n} \sum_j w_j$, which lies in $\mathcal{E}^1(X, \theta, \phi)$ by Corollary 17.2.10. By multilinearity, $\langle \theta_{w_1} \wedge \cdots \wedge \theta_{w_n} \rangle \le n^n \mu_w$. Choose C with $w \le \phi + C$ and apply the first assertion to u, w and $\gamma = \phi + C$: its right side is finite because $u, w \in \mathcal{E}^1(X, \theta, \phi)$. Hence $\phi - u$ is integrable for μ_w, and so for the mixed measure. The same holds for $\phi - v$, and $|u - v| \le (\phi - u) + (\phi - v)$. □

Proposition 17.2.12 *For $u, v \in \mathcal{E}^1(X, \theta, \phi)$, the cocycle formula (17.6) and the tangent bounds (17.9) remain valid. In particular, (17.4) computes the extended energy on $\mathcal{E}^1(X, \theta, \phi)$.*

Proof All mixed measures involved have total mass m, so both assertions are unchanged if constants are added to u and v; we may therefore assume $u, v \le \phi$. Put $u_k = \max\{u, \phi - k\}$ and $v_k = \max\{v, \phi - k\}$. Fix $0 \le j \le n$ and write $\lambda_k = \langle \theta_{u_k}^j \wedge \theta_{v_k}^{n-j} \rangle$ and $\lambda = \langle \theta_u^j \wedge \theta_v^{n-j} \rangle$. We show that

$$\int_X (u_k - v_k)\lambda_k \longrightarrow \int_X (u - v)\lambda. \tag{17.11}$$

On the plurifine open set $A_k = \{u > \phi - k\} \cap \{v > \phi - k\}$ the measures λ_k and λ agree by locality, and $u_k - v_k = u - v$. By Lemma 17.2.11, $|u - v|$ is λ-integrable. The sets $X \setminus A_k$ decrease to the union of the pole sets of u, v and ϕ, which is pluripolar and therefore λ-null, so dominated convergence gives $\int_{X \setminus A_k} |u - v|\lambda \to 0$. It remains to show that the integral of $|u_k - v_k|$ against λ_k over $X \setminus A_k$ tends to zero.

For $z \in \{u, v\}$ put $D_k(z) = \{z \le \phi - k\}$, so that $X \setminus A_k = D_k(u) \cup D_k(v)$. We claim that

$$k\lambda_k(D_k(z)) \le 2^n k\mu_z(\{z \le \phi - k/2\}) \longrightarrow 0. \tag{17.12}$$

Consider $z = u$ and $j < n$, and put $q = (v_k + \phi - k)/2$. Since $\phi - k \le v_k \le \phi$, we have $q \le \phi - k/2$ on X, and $u_k = \phi - k \le q$ on $D_k(u)$. Moreover $\theta_{v_k} \le 2\theta_q$, since $2\theta_q - \theta_{v_k} = \theta_\phi \ge 0$, so multilinearity gives $\lambda_k \le 2^{n-j}\langle \theta_{u_k}^j \wedge \theta_q^{n-j}\rangle$. The potentials q and u_k belong to $\mathcal{E}(X,\theta,\phi)$, and the non-strict comparison principle of Proposition 16.7.4, with $n-j$ factors θ_q and the j fixed factors θ_{u_k}, gives

$$\lambda_k(D_k(u)) \le 2^{n-j}\mu_{u_k}(\{u_k \le q\}) \le 2^{n-j}\mu_{u_k}(\{u \le \phi - k/2\}).$$

For $j = n$ the same bound holds directly, since then $\lambda_k = \mu_{u_k}$ and $D_k(u) \subset \{u \le \phi - k/2\}$. The measures μ_{u_k} and μ_u have the same total mass and agree on $\{u > \phi - k/2\}$ by locality, so the last mass equals $\mu_u(\{u \le \phi - k/2\})$. Since exchanging u and v and replacing j by $n-j$ does not change λ_k, the case $z = v$ follows from the case $z = u$. Finally,

$$k\mu_z(\{z \le \phi - k/2\}) \le 2\int_{\{z \le \phi - k/2\}} (\phi - z)\mu_z \longrightarrow 0$$

because $\phi - z$ is μ_z-integrable. This proves (17.12).

Since $u_k - \phi$ and $v_k - \phi$ take values in $[-k, 0]$, we have $|u_k - v_k| \le k$, and (17.12) shows that the integral of $|u_k - v_k|$ against λ_k over $X \setminus A_k$ tends to zero. This proves (17.11). Now pass to the limit in the cocycle formula for u_k, v_k: its left side converges to $E_\theta^\phi(v) - E_\theta^\phi(u)$ by Proposition 17.2.6, and each term on the right converges by (17.11). The tangent bounds pass to the limit in the same way, using the terms $j = n$ and $j = 0$. Taking $v = \phi$ in the cocycle formula gives the last assertion. □

17.3 Continuity and compactness

We keep θ and the model potential ϕ of mass m. Convergence in $L^1(X)$ refers to the volume form ω^n, and a θ-psh potential is determined by its class in $L^1(X)$. The energy is continuous along increasing nets and upper semicontinuous for L^1 convergence, so that normalized potentials with energy bounded below form compact sets (Corollary 17.3.4), as the variational method of Section 17.4 requires. The finite-energy class is also stable under rooftop envelopes, which gives finite energy along the Perron segments of Section 17.6.

Proposition 17.3.1 *Let $(u_i)_{i\in I}$ be an increasing net in $\mathcal{E}^1(X,\theta,\phi)$ which is uniformly bounded above on X. Then its regularized upper envelope $u = (\sup_i u_i)^*$ belongs to $\mathcal{E}^1(X,\theta,\phi)$ and*

$$E_\theta^\phi(u_i) \uparrow E_\theta^\phi(u).$$

Proof By Lemma 16.4.2, $u_i \le \phi + \sup_X u_i$, and the suprema $\sup_X u_i$ are bounded above by hypothesis; hence $u \preceq \phi$. Since u dominates every u_i, monotonicity of the energy and Theorem 17.2.8 show that $u \in \mathcal{E}^1(X,\theta,\phi)$.

Bounded differences. Consider first an increasing sequence (u_j) and a constant $C \geq 1$ such that $\phi - C \leq u_j \leq u \leq \phi$ for all j. The pole sets of u_j and u coincide with that of ϕ. Let f_j be the function equal to $u - u_j$ outside this pole set and to 0 on it; it is a Borel function with values in $[0, C]$. By the tangent bound (17.9),

$$0 \leq E^{\phi}_{\theta}(u) - E^{\phi}_{\theta}(u_j) \leq \int_X (u - u_j)\, \mu_{u_j} = \int_X f_j\, \mu_{u_j},$$

and we show that the right side tends to zero by applying Lemma 17.1.1 with $\nu_j = \mu_{u_j}$, $\nu = \mu_u$, the weights f_j and the limit weight 0. For each j the potential $C^{-1}u_j + (1 - C^{-1})\phi$ lies between $\phi - 1$ and ϕ and its measure is at least $C^{-n}\mu_{u_j}$; hence $\mu_{u_j} \leq C^n \operatorname{Cap}_{\theta,\phi}$ on Borel sets, and likewise $\mu_u \leq C^n \operatorname{Cap}_{\theta,\phi}$. By Theorem 16.9.8, the measures μ_{u_j} and μ_u are therefore uniformly absolutely continuous with respect to $\operatorname{Cap}_{\omega}$. They converge weakly by Corollary 15.7.6, since u_j increases to u quasi-everywhere (Corollary 10.2.2).

It remains to show that $f_j \to 0$ in capacity. Fix finitely many pairs of concentric coordinate balls $B_\alpha \Subset B'_\alpha$ as in Proposition 16.9.9, chosen so that θ has a smooth potential h_α on a neighborhood of $\overline{B'_\alpha}$ (this is possible by Lemma 13.1.2, applied on slightly larger concentric balls), and let c be the constant of that proposition. Let $\delta, \eta > 0$. By Proposition 16.9.9(3) there is an open set G with $\operatorname{Cap}_{\omega}(G) < \eta$ which contains the pole set of ϕ and such that ϕ is finite and continuous on the compact set $X \setminus G$. Let L be a common bound for $|\phi|$ on $X \setminus G$ and for the functions $|h_\alpha|$ on B'_α. The functions

$$w_{\alpha,j} = \max\{u_j + h_\alpha, -2L - C\}, \qquad w_\alpha = \max\{u + h_\alpha, -2L - C\}$$

are bounded plurisubharmonic functions on B'_α, and $w_{\alpha,j}$ increases to w_α almost everywhere, because u_j increases to u quasi-everywhere and pluripolar sets have Lebesgue measure zero. By Theorem 9.5.1, $w_{\alpha,j} \to w_\alpha$ locally in capacity on B'_α. On $B_\alpha \setminus G$ we have $u_j + h_\alpha \geq \phi - C - L \geq -2L - C$, so the truncations are inactive there and $w_\alpha - w_{\alpha,j} = u - u_j$. Since the balls B_α cover X,

$$\{f_j > \delta\} \subset G \cup \bigcup_{\alpha} \big(B_\alpha \cap \{w_{\alpha,j} < w_\alpha - \delta\}\big).$$

By the countable subadditivity of $\operatorname{Cap}_{\omega}$ (Proposition 16.7.6) and the bound $\operatorname{Cap}_{\omega}(E) \leq c \operatorname{Cap}_{B'_\alpha}(E)$ for Borel sets $E \subset B'_\alpha$ in Proposition 16.9.9,

$$\operatorname{Cap}_{\omega}(\{f_j > \delta\}) \leq \eta + c \sum_{\alpha} \operatorname{Cap}^*_{B'_\alpha}\big(\overline{B_\alpha} \cap \{w_{\alpha,j} < w_\alpha - \delta\}\big),$$

where we also used that the capacity of a Borel set is at most its outer capacity. By Definition 9.4.5, the sum tends to zero as $j \to \infty$. Hence $\overline{\lim}_j \operatorname{Cap}_{\omega}(\{f_j > \delta\}) \leq \eta$, and since η is arbitrary, $f_j \to 0$ in capacity. Lemma 17.1.1 now gives $\int_X f_j\, \mu_{u_j} \to 0$, and hence $E^{\phi}_{\theta}(u_j) \to E^{\phi}_{\theta}(u)$.

Sequences. Now let (u_j) be an arbitrary increasing sequence; subtracting a common constant, we may assume $u_j \le u \le \phi$. Put $u_{j,k} = \max\{u_j, \phi - k\}$ and $u_k = \max\{u, \phi - k\}$. For fixed k, the sequence $u_{j,k}$ increases to $\max\{\sup_j u_j, \phi - k\}$, whose upper semicontinuous regularization is $\max\{(\sup_j u_j)^*, \phi - k\} = u_k$, because $\phi - k$ is upper semicontinuous; moreover $\phi - k \le u_{j,k} \le u_k \le \phi$. The preceding case, with $\max\{k, 1\}$ in place of C, gives $E_\theta^\phi(u_{j,k}) \to E_\theta^\phi(u_k)$ as $j \to \infty$. We bound the truncation error uniformly in j. The tangent bound of Proposition 17.2.12 gives

$$0 \le E_\theta^\phi(u_{j,k}) - E_\theta^\phi(u_j) \le \int_X (u_{j,k} - u_j)\, \mu_{u_j} = \int_k^\infty \mu_{u_j}(\{u_j < \phi - t\})\, \mathrm{d}t.$$

For $t \ge k$ put $q_t = (u_j + \phi - t)/2$. On $\{u_j < \phi - t\}$ we have $u_1 \le u_j < q_t$, and $q_t \le \phi - t/2$ on X. Since $\mu_{q_t} \ge 2^{-n}\mu_{u_j}$ by multilinearity, and $q_t, u_1 \in \mathcal{E}(X, \theta, \phi)$, the comparison principle of Proposition 16.7.4 gives

$$\begin{aligned}\mu_{u_j}(\{u_j < \phi - t\}) &\le 2^n \mu_{q_t}(\{u_1 < q_t\}) \\ &\le 2^n \mu_{u_1}(\{u_1 < q_t\}) \le 2^n \mu_{u_1}(\{u_1 < \phi - t/2\}).\end{aligned}$$

Consequently

$$\begin{aligned}0 &\le E_\theta^\phi(u_{j,k}) - E_\theta^\phi(u_j) \\ &\le 2^{n+1} \int_X (\phi - u_1 - k/2)_+\, \mu_{u_1} \longrightarrow 0\end{aligned} \tag{17.13}$$

as $k \to \infty$, uniformly in j; the limit is zero by dominated convergence, since $\phi - u_1$ is μ_{u_1}-integrable. Since $E_\theta^\phi(u_k) \ge E_\theta^\phi(u)$, letting first $j \to \infty$ and then $k \to \infty$ gives $\lim_j E_\theta^\phi(u_j) \ge E_\theta^\phi(u)$, and monotonicity gives the reverse inequality.

Nets. Let $(u_i)_{i \in I}$ be a net as in the statement. Cover X by finitely many coordinate charts D_β on which θ has smooth potentials h_β. By Theorem 8.3.2, applied to the functions $u_i + h_\beta$ on D_β, there are countable sets $J_\beta \subset I$ such that $(\sup_{i \in J_\beta} u_i)^* = u$ on D_β. The countable set $J = \bigcup_\beta J_\beta$ then satisfies $(\sup_{i \in J} u_i)^* = u$ on X, since on each D_β this envelope lies between the envelopes over J_β and over I. Since the index set is directed, we can choose indices $i_1 \le i_2 \le \cdots$ such that i_k dominates the first k elements of an enumeration of J. Since the net is increasing, $\sup_k u_{i_k} \ge \sup_{i \in J} u_i$, so the increasing sequence (u_{i_k}) has regularized upper envelope u, and the sequence case gives $E_\theta^\phi(u_{i_k}) \to E_\theta^\phi(u)$. Each of these energies is at most $\sup_i E_\theta^\phi(u_i)$, which in turn is at most $E_\theta^\phi(u)$ by monotonicity. Hence $\sup_i E_\theta^\phi(u_i) = E_\theta^\phi(u)$, and since the net of energies is increasing, its limit is this supremum. (The selected indices need not be cofinal in I, which is why we compare suprema.) □

Upper semicontinuity of the energy for L^1 convergence rests on the following description of L^1 limits of potentials by decreasing sequences.

Lemma 17.3.2 *Let θ be a smooth closed real $(1,1)$-form on X, and let $q_k, q \in \mathrm{PSH}(X, \theta)$ satisfy $q_k \to q$ in $L^1(X)$. Then the functions q_k are uniformly bounded above on X, the tail envelopes $a_j = (\sup_{k \ge j} q_k)^*$ belong to $\mathrm{PSH}(X, \theta)$, and a_j*

decreases pointwise to q as $j \to \infty$. In particular, if $\psi : X \to [-\infty, \infty)$ is upper semicontinuous and $q_k \le \psi$ for every k, then $q \le \psi$.

Proof By Proposition 14.2.1 the suprema $\sup_X q_k$ converge to $\sup_X q$, so they are bounded. Hence the tail envelopes a_j are θ-psh, by Theorem 8.3.2 applied in charts after adding smooth local potentials of θ, and they decrease as j increases. A subsequence of (q_k) converges to q almost everywhere, so $a_j \ge q$ almost everywhere, hence everywhere, since both functions are θ-psh (Section 16.1).

Conversely, let $x \in X$ and $\varepsilon > 0$, let h be a smooth local potential of θ on a coordinate ball Ω around x, and let $M > q(x) + h(x)$ be a real number. By upper semicontinuity there is a closed coordinate ball $K \subset \Omega$ around x on which $q + h < M$ and $|h - h(x)| < \varepsilon$. The plurisubharmonic functions $q_k + h$ converge to $q + h$ in $L^1_{\mathrm{loc}}(\Omega)$, so Hartogs' lemma in the form of Theorem 8.3.4, applied with the constant function M on K, gives $q_k + h < M$ on K for all large k. For such k we have $q_k < M - h(x) + \varepsilon$ on K. Since K is a neighborhood of x, it follows that $a_j(x) \le M - h(x) + \varepsilon$ for all large j. Letting $\varepsilon \downarrow 0$ and $M \downarrow q(x) + h(x)$, or $M \to -\infty$ if $q(x) = -\infty$, we obtain $\lim_j a_j(x) \le q(x)$. Thus $a_j \downarrow q$.

Finally, if $q_k \le \psi$ for all k with ψ upper semicontinuous, then $\sup_{k \ge j} q_k \le \psi$, so $a_j \le \psi$ for every j, and therefore $q = \lim_j a_j \le \psi$. □

Proposition 17.3.3 *The set $\{u \in \mathrm{PSH}(X, \theta) : u \preceq \phi\}$ is closed in $L^1(X)$, and E^ϕ_θ is upper semicontinuous on it, with values in $[-\infty, \infty)$: if $u_j \to u$ in $L^1(X)$, then*

$$E^\phi_\theta(u) \ge \overline{\lim_j} E^\phi_\theta(u_j).$$

Proof Let $u_j \preceq \phi$ converge to u in $L^1(X)$. By Lemma 17.3.2, the u_j are uniformly bounded above, so Lemma 16.4.2 gives a constant C with $u_j \le \phi + C$ for all j. Since $\phi + C$ is upper semicontinuous, the same lemma gives $u \le \phi + C$, which proves closedness. The lemma also shows that the tail envelopes $v_j = (\sup_{k \ge j} u_k)^*$ are θ-psh potentials which decrease pointwise to u; moreover $v_j \le \phi + C$, because $\phi + C$ is upper semicontinuous and dominates every u_k. Since $u_j \le v_j$, monotonicity and Proposition 17.2.7 give

$$\overline{\lim_j} E^\phi_\theta(u_j) \le \lim_j E^\phi_\theta(v_j) = E^\phi_\theta(u).$$

Since $L^1(X)$ is a metric space, sequential upper semicontinuity is upper semicontinuity. □

Corollary 17.3.4 *For $B \ge 0$, the set*

$$\{u \in \mathrm{PSH}(X, \theta) : u \le \phi, \quad E^\phi_\theta(u) \ge -B\}$$

is compact in $L^1(X)$ and is contained in $\mathcal{E}^1(X, \theta, \phi)$. The set

$$K_\phi = \{u \in \mathrm{PSH}(X, \theta) : u \le \phi, \quad \sup_X u = 0\}$$

is also compact in $L^1(X)$.

Proof For u in the first set, Lemma 16.4.2 gives $u \le \phi + \sup_X u$, so monotonicity and the shift formula give $-B \le E_\theta^\phi(u) \le m \sup_X u$. Also $\sup_X u \le \sup_X \phi = 0$. Hence the set lies in $\{u \in \mathrm{PSH}(X,\theta) : -B/m \le \sup_X u \le 0\}$, which is compact in $L^1(X)$ by Proposition 14.2.1. The set is closed in $L^1(X)$: the inequality $u \le \phi$ passes to L^1 limits by Lemma 17.3.2, since ϕ is upper semicontinuous, and the energy bound passes to limits by upper semicontinuity (Proposition 17.3.3). It is therefore compact. Its members have finite energy, so they lie in $\mathcal{E}^1(X,\theta,\phi)$ by Theorem 17.2.8.

The set K_ϕ is contained in $\{u \in \mathrm{PSH}(X,\theta) : \sup_X u = 0\}$, which is compact in $L^1(X)$ by Proposition 14.2.1, and it is closed in $L^1(X)$, because the supremum is continuous for L^1 convergence of potentials (by the same proposition) and the inequality $u \le \phi$ passes to L^1 limits by Lemma 17.3.2. Hence K_ϕ is compact. □

Proposition 17.3.5 *Let $\mathcal{F}$ be one of the three classes $\mathcal{E}(X,\theta,\phi)$, $\mathcal{E}^1(X,\theta,\phi)$ and $\mathcal{E}^\infty(X,\theta,\phi)$. The regularized upper envelope of a nonempty family in $\mathcal{F}$ which is uniformly bounded above on X belongs to $\mathcal{F}$.*

Proof By Lemma 16.4.2, every member of the family is bounded above by $\phi + C$ for a common constant C, and so is the regularized upper envelope U. Fix a member u_0 of the family; then $u_0 \le U \preceq \phi$. For $\mathcal{F} = \mathcal{E}(X,\theta,\phi)$, mass monotonicity gives $m = \int_X \langle \theta_{u_0}^n \rangle \le \int_X \langle \theta_U^n \rangle \le m$. For $\mathcal{F} = \mathcal{E}^1(X,\theta,\phi)$, the last assertion of Corollary 17.2.10, applied to $u_0 \le U$, gives $U \in \mathcal{F}$. For $\mathcal{F} = \mathcal{E}^\infty(X,\theta,\phi)$, the member u_0 supplies a lower bound $U \ge u_0 \ge \phi - C'$. □

Proposition 17.3.6 *Let $u, v \in \mathcal{E}(X,\theta,\phi)$. Then the rooftop envelope $w = P_\theta(u,v)$ of Definition 16.1.5 is not identically $-\infty$, and it belongs to $\mathcal{E}(X,\theta,\phi)$. If moreover $u, v \in \mathcal{E}^1(X,\theta,\phi)$, then $w \in \mathcal{E}^1(X,\theta,\phi)$, and if in addition $u, v \le \phi$, then*

$$\int_X (\phi - w)\,\mu_w \le \int_X (\phi - u)\,\mu_u + \int_X (\phi - v)\,\mu_v .$$

Proof Since $u \le \max\{u,v\} \preceq \phi$, mass monotonicity gives $\int_X \langle \theta_{\max\{u,v\}}^n \rangle = m$. Thus $\int_X \langle \theta_u^n \rangle + \int_X \langle \theta_v^n \rangle = 2m > \int_X \langle \theta_{\max\{u,v\}}^n \rangle$, and Proposition 16.5.1 shows that $w \not\equiv -\infty$. Theorem 16.5.5 gives $2m \le m + \int_X \langle \theta_w^n \rangle$, while $\int_X \langle \theta_w^n \rangle \le m$ by mass monotonicity, since $w \le u$. Hence $w \in \mathcal{E}(X,\theta,\phi)$.

Now let $u, v \in \mathcal{E}^1(X,\theta,\phi)$. The rooftop commutes with adding a common constant to u and v (Proposition 16.1.4(3)), so we may assume $u, v \le \phi$; then $w \le \phi$. The contact inequality of Corollary 16.2.6 gives

$$\mu_w \le \mathbf{1}_{\{w=u\}}\mu_u + \mathbf{1}_{\{w=v\}}\mu_v .$$

Integrate the nonnegative weight $\min\{\phi - w, N\}$ against both sides. On the first contact set the weight equals $\min\{\phi - u, N\}$, and on the second it equals $\min\{\phi - v, N\}$; pole sets carry no mass. Letting $N \to \infty$ gives the displayed bound. Both terms on the right are finite, so $w \in \mathcal{E}^1(X,\theta,\phi)$. □

Corollary 17.3.7 *For $u, v \in \mathcal{E}(X, \theta, \phi)$, the singularity-constrained envelope of Definition 16.7.1 satisfies $P_\theta[u](v) = v$.*

Proof By Proposition 17.3.6, the rooftop $w = P_\theta(u, v)$ belongs to $\mathcal{E}(X, \theta, \phi)$. Since $w \preceq v$ and $\int_X \langle \theta_w^n \rangle = \int_X \langle \theta_v^n \rangle > 0$, Proposition 16.4.10 gives $P_\theta[w](v) = v$. Every competitor for $P_\theta[w](v)$ is more singular than w, hence than u, so it is a competitor for $P_\theta[u](v)$; and $P_\theta[u](v) \le v$ quasi-everywhere by Proposition 16.7.2(1), hence everywhere. Therefore $v = P_\theta[w](v) \le P_\theta[u](v) \le v$. □

Proposition 17.3.8 *For $u, v \in \mathcal{E}^1(X, \theta, \phi)$ and $u_t = (1-t)u + tv$, the function $t \mapsto E_\theta^\phi(u_t)$ is continuously differentiable on $[0, 1]$, with one-sided derivatives at the endpoints, and*

$$\frac{\mathrm{d}}{\mathrm{d}t} E_\theta^\phi(u_t) = \int_X (v - u)\, \mu_{u_t}.$$

The derivative is a polynomial in t.

Proof By Lemma 17.2.11 every integral $\int_X (v-u)\langle \theta_v^a \wedge \theta_u^{n-a} \rangle$ is finite. Expanding μ_{u_t} by multilinearity shows that the right side is a polynomial in t with these integrals as coefficients. The potentials u_t lie in $\mathcal{E}^1(X, \theta, \phi)$ by Corollary 17.2.10. Apply the cocycle formula of Proposition 17.2.12 to u_{t+s} and u_t, with $s \neq 0$ and $t, t+s \in [0, 1]$, and use $u_{t+s} - u_t = s(v - u)$:

$$\frac{E_\theta^\phi(u_{t+s}) - E_\theta^\phi(u_t)}{s} = \frac{1}{n+1} \sum_{j=0}^{n} \int_X (v-u) \langle \theta_{u_{t+s}}^j \wedge \theta_{u_t}^{n-j} \rangle.$$

Each mixed integral on the right is a polynomial in s with the finite coefficients above, and at $s = 0$ each of the $n+1$ terms equals $\int_X (v-u)\mu_{u_t}$. Letting $s \to 0$ proves the formula for the derivative, including the one-sided derivatives at the endpoints, and its continuity. □

Lemma 17.3.9 *For $u, v \in \mathcal{E}^1(X, \theta, \phi)$,*

$$\int_X (v-u)\, \mu_v \le E_\theta^\phi(v) - E_\theta^\phi(u) \le \int_X (v-u)\, \mu_u.$$

If moreover $u \le v$, then

$$\frac{1}{n+1} \int_X (v-u)\, \mu_u \le E_\theta^\phi(v) - E_\theta^\phi(u).$$

Proof The first display is the tangent bound (17.9), which holds on $\mathcal{E}^1(X, \theta, \phi)$ by Proposition 17.2.12. By the same proposition, the cocycle formula (17.6) holds for u and v. If $u \le v$, all its terms are nonnegative, and retaining the term $j = 0$, whose measure is μ_u, gives the second display. □

Example 17.3.10 The energy is not continuous for L^1 convergence, even for smooth potentials on $\mathbb{P}^1$. Let $\omega = \mathrm{dd}^{\mathrm{c}} \log(1 + |z|^2)$, which has mass one. For $0 < a < 1$ put $\varepsilon_a = \exp(-a^{-2})$ and

$$u_a(z) = a \log \frac{|z|^2 + \varepsilon_a}{1 + |z|^2}.$$

These functions extend smoothly across infinity, and $\omega + \mathrm{dd^c} u_a = (1-a)\omega + a\mathrm{dd^c} \log(|z|^2+\varepsilon_a) > 0$. They tend to zero in L^1 as $a \downarrow 0$, since $a \log(|z|^2/(1+|z|^2)) \le u_a \le 0$ and the left side is a times a fixed integrable function.

We compute their Dirichlet energies. Write $r = \log|z|^2$ and $F(r) = \log(\mathrm{e}^r + \varepsilon) - \log(1+\mathrm{e}^r)$ with $\varepsilon = \varepsilon_a$, so that $u_a = aF(\log|z|^2)$. For a smooth function $g = G(\log|z|^2)$ we have $\partial g = G'(r)\,\mathrm{d}z/z$, hence $\mathrm{d}g \wedge \mathrm{d^c} g = \frac{\mathrm{i}}{2\pi}\partial g \wedge \bar\partial g = \pi^{-1} G'(r)^2 |z|^{-2}\,\mathrm{d}A$, where $\mathrm{d}A$ is Lebesgue measure; integration in polar coordinates gives $\int \mathrm{d}g \wedge \mathrm{d^c} g = \int_{-\infty}^{\infty} G'(r)^2\,\mathrm{d}r$. Therefore

$$\int_X \mathrm{d}u_a \wedge \mathrm{d^c} u_a = a^2 \int_{-\infty}^{\infty} F'(r)^2\,\mathrm{d}r = a^2(-\log\varepsilon_a + O(1)) \longrightarrow 1.$$

The bound $O(1)$ is uniform in a. Indeed, we have

$$F'(r) = \frac{\mathrm{e}^r}{\mathrm{e}^r+\varepsilon} - \frac{\mathrm{e}^r}{1+\mathrm{e}^r} \in [0,1], \qquad 1 - F'(r) = \frac{\varepsilon}{\mathrm{e}^r+\varepsilon} + \frac{\mathrm{e}^r}{1+\mathrm{e}^r} \le \varepsilon \mathrm{e}^{-r} + \mathrm{e}^r.$$

On $[\log\varepsilon, 0]$ the integral of $\varepsilon\mathrm{e}^{-r} + \mathrm{e}^r$ is at most 2, and $1 \ge F'^2 \ge 1 - 2(1-F')$, so $\int_{\log\varepsilon}^0 F'^2\,\mathrm{d}r = -\log\varepsilon + O(1)$. On the left tail $F'(r) \le \mathrm{e}^r/\varepsilon$, and on the right tail $F'(r) \le \mathrm{e}^{-r}$; the corresponding integrals of F'^2 are at most $\frac12$. Since $\int_X u_a\,\omega \to 0$, the formula of Example 17.2.9 gives $E_\omega(u_a) \to -\frac12$, whereas $E_\omega(0) = 0$.

17.4 Variational Monge–Ampère equations

We keep the big class of θ, the model potential ϕ and its mass $m = \int_X \langle\theta_\phi^n\rangle > 0$, and write $E = E_\theta^\phi$. We solve the equation $\mu_u = \nu$ for non-pluripolar measures ν of mass m (Definition 17.4.1) by a variational method. By the tangent bounds, a potential u with $\mu_u = \nu$ maximizes $v \mapsto E(v) - \int_X (v-\phi)\,\mathrm{d}\nu$. For the converse we perturb a maximizer u in the direction of a continuous function f. Since $u + tf$ need not be θ-psh, we replace it by its envelope $P_\theta(u+tf)$ and compute the derivative of the energy along these envelopes (Proposition 17.4.3).

Maximizers exist by the compactness of Corollary 17.3.4 when the integral term of the functional is continuous for L^1 convergence, as it is for measures with bounded densities. For an arbitrary non-pluripolar measure ν we replace the linear term by the exponential term $\lambda^{-1}\int_X \mathrm{e}^{\lambda v}\,\mathrm{d}\nu$, $\lambda > 0$, which is continuous for L^1 convergence of potentials, and then let $\lambda \to 0$. The remaining subsections prove uniqueness (Section 17.4.2) and the boundedness of $u - \phi$ for L^p densities (Section 17.4.3), and derive two consequences of the existence theorem (Sections 17.4.4 and 17.4.5).

17.4.1 Existence by maximization

Definition 17.4.1 A finite positive Radon measure ν on X is *non-pluripolar* if $\nu(A) = 0$ for every pluripolar Borel set $A \subset X$. Integrals against ν use its completion, so subsets of pluripolar sets are also ν-null.

For example, every measure of the form $g\,\omega^n$ with $0 \le g \in L^1(X, \omega^n)$ is non-pluripolar, since pluripolar sets have volume zero (see the remarks after Definition 10.1.1). Every θ-psh function is finite ν-almost everywhere, because its pole set is a pluripolar Borel set.

Finite positive Borel measures on X are compared through their integrals of continuous functions.

Lemma 17.4.2 *Let λ and σ be finite positive Borel measures on X.*

(1) *If $\int_X g\,\mathrm{d}\lambda \ge \int_X g\,\mathrm{d}\sigma$ for every continuous function $g \ge 0$ on X, then $\lambda(E) \ge \sigma(E)$ for every Borel set $E \subset X$. In particular, two finite positive Borel measures on X with the same integrals of all continuous functions coincide.*

(2) *If finite positive Borel measures λ_j converge weakly to λ and satisfy $\lambda_j \ge \sigma$ for every j, then $\lambda \ge \sigma$.*

Proof (1) This is Lemma 16.2.1(2) with $W = X$; the second assertion follows by applying the first in both directions.

(2) Since X is compact, weak convergence of measures on X is convergence of the integrals of all continuous functions (see the discussion following Definition 7.2.4). Hence $\int_X g\,\mathrm{d}\lambda = \lim_j \int_X g\,\mathrm{d}\lambda_j \ge \int_X g\,\mathrm{d}\sigma$ for every continuous function $g \ge 0$ on X, and (1) gives $\lambda \ge \sigma$. □

Proposition 17.4.3 *Let $u \in \mathcal{E}^1(X, \theta, \phi)$ and $f \in C(X, \mathbb{R})$, and put $u_t = P_\theta(u + tf)$ for $t \in \mathbb{R}$, the envelope of Definition 16.1.3. Then $u_0 = u$, and for every t we have $u_t \sim u$ and $u_t \le u + tf$ on X. Each u_t has finite energy, and*

$$\frac{\mathrm{d}}{\mathrm{d}t} E(u_t) = \int_X f\,\mu_{u_t}.$$

The derivative is continuous on $\mathbb{R}$.

Proof Let $B = \sup_X |f|$. The potential $u - |t|B$ is a competitor for the obstacle $u + tf$, and every competitor lies below $u + |t|B$, so $u - |t|B \le u_t \le u + |t|B$. Hence $u_t \sim u$, and $E(u_t) \ge E(u) - |t|Bm > -\infty$, so $u_t \in \mathcal{E}^1(X, \theta, \phi)$ by Theorem 17.2.8. The obstacle $u + tf$ is upper semicontinuous, so $u_t \le u + tf$ everywhere, by Proposition 16.1.4(1),(2). In particular $u_0 \le u$; since u is itself a competitor for the obstacle u, also $u \le u_0$, so $u_0 = u$.

Next we show that $\mu_{u_{t+s}} \to \mu_{u_t}$ weakly as $s \to 0$. The obstacles $u + tf$ and $u + (t+s)f$ differ by at most $|s|B$, so Proposition 16.1.4(4) gives $u_t - |s|B \le u_{t+s} \le u_t + |s|B$. In particular all u_t have the pole set of u, and $|u_{t+s} - u_t| \le |s|B$ outside it. Since $|\arctan a - \arctan b| \le |a - b|$ for real a, b, and adding the same smooth local potential

of θ to u_{t+s} and u_t does not change their difference, the distance d_* of Definition 15.6.2 between the local potentials of u_{t+s} and u_t is at most $|s|B$ everywhere. Hence, for every $\delta > 0$, the sets where it exceeds δ are empty once $|s|B < \delta$, and $u_{t+s} \to u_t$ in extended capacity as $s \to 0$. These potentials have full mass relative to ϕ, so Corollary 15.8.4, applied along every sequence $s_k \to 0$, gives the asserted weak convergence; in particular $\int_X f\,\mu_{u_{t+s}} \to \int_X f\,\mu_{u_t}$.

The obstacle $u + tf$ is bounded above, and it is quasi-continuous, because u is (Section 16.1) and f is continuous. Hence Theorem 16.2.5 shows that μ_{u_t} is carried by the contact set $\{u_t = u + tf\}$. On the contact set of u_{t+s} we have $u_{t+s} = u + (t+s)f$ and $u_t \le u + tf$, so $u_{t+s} - u_t \ge sf$ there. On the contact set of u_t we have $u_t = u + tf$ and $u_{t+s} \le u + (t+s)f$, so $u_{t+s} - u_t \le sf$ there. The differences are bounded and the pole sets carry no mass, so the tangent bounds of Lemma 17.3.9 give, for every real s,

$$s\int_X f\,\mu_{u_{t+s}} \le E(u_{t+s}) - E(u_t) \le s\int_X f\,\mu_{u_t}.$$

Dividing by $s > 0$, respectively $s < 0$, and letting $s \to 0$, we obtain the right and left derivatives from the weak convergence just proved. Both equal $\int_X f\,\mu_{u_t}$, which depends continuously on t by the same weak convergence. □

Theorem 17.4.4 *Let ν be a non-pluripolar measure on X, and suppose that $\mathcal{D}_\nu = \mathcal{E}^1(X,\theta,\phi) \cap L^1(\nu)$ is nonempty. Then $\phi \in L^1(\nu)$, and for the functional*

$$F_\nu(v) = E(v) - \int_X (v - \phi)\,\mathrm{d}\nu, \qquad v \in \mathcal{D}_\nu,$$

a potential $u \in \mathcal{D}_\nu$ maximizes F_ν if and only if $\mu_u = \nu$. In particular, F_ν has no maximizer unless $\nu(X) = m$.

Proof Let $v \in \mathcal{D}_\nu$. Then $v \le \phi + C$ for a constant C, and $\phi \le 0$. Thus the negative part of ϕ is at most that of v plus C, and the positive part of ϕ vanishes; hence $\phi \in L^1(\nu)$. If $\mu_u = \nu$, the tangent bound of Lemma 17.3.9 gives

$$E(v) - E(u) \le \int_X (v - u)\,\mathrm{d}\nu,$$

that is, $F_\nu(v) \le F_\nu(u)$.

Conversely, let u be a maximizer and $f \in C(X,\mathbb{R})$. By Proposition 17.4.3, the potentials $u_t = P_\theta(u + tf)$ have bounded difference from u and finite energy, so they belong to $\mathcal{D}_\nu$; moreover $u_0 = u$ and $u_t \le u + tf$ everywhere, hence ν-almost everywhere. Therefore

$$E(u_t) - E(u) - t\int_X f\,\mathrm{d}\nu \le F_\nu(u_t) - F_\nu(u) \le 0.$$

The left side vanishes at $t = 0$ and is differentiable there by Proposition 17.4.3; since it attains its maximum at $t = 0$, its derivative vanishes there. Thus $\int_X f\,\mu_u = \int_X f\,\mathrm{d}\nu$

for every $f \in C(X, \mathbb{R})$, and Lemma 17.4.2(1) gives $\mu_u = \nu$. Since $\mu_u(X) = m$, a maximizer can exist only if $\nu(X) = m$. □

Theorem 17.4.5 *Let* $\mathrm{d}V$ *be a smooth positive volume form on* X*, and let* $g \in L^\infty(\mathrm{d}V)$ *satisfy* $g \geq 0$ *and* $\int_X g\,\mathrm{d}V = m$*. Then there is* $u \in \mathcal{E}^1(X, \theta, \phi)$ *with*

$$\mu_u = g\,\mathrm{d}V, \qquad \sup_X u = 0.$$

It maximizes $F_{g\,\mathrm{d}V}$*.*

Proof Put $\nu = g\,\mathrm{d}V$, a non-pluripolar measure of mass m. Every θ-psh potential is integrable for $\mathrm{d}V$, hence for ν, so $\mathcal{D}_\nu = \mathcal{E}^1(X, \theta, \phi)$; we write $F = F_\nu$. By the shift formula for the energy and $\nu(X) = m$, the functional F is invariant under adding constants, so we may restrict it to potentials v with $\sup_X v = 0$. For such v, Lemma 16.4.2 gives $v \leq \phi$, hence $E(v) \leq 0$. Since $\mathrm{d}V \leq c_V\,\omega^n$ for a constant c_V, Proposition 14.2.1 shows that $\int_X |v|\,\mathrm{d}V$ is bounded independently of v, and $\phi \in L^1(\mathrm{d}V)$, so

$$0 \leq -\int_X (v - \phi) g\,\mathrm{d}V \leq C$$

with C independent of v. Consequently $\sup F \in [0, C]$, since $F(\phi) = 0$ and $F(v) \leq 0 + C$.

Let (v_j) be a normalized maximizing sequence with $F(v_j) \geq -1$. Then $E(v_j) = F(v_j) + \int_X (v_j - \phi) g\,\mathrm{d}V \geq -C - 1$. By Corollary 17.3.4 a subsequence converges in $L^1(X)$ to some $u \in \mathcal{E}^1(X, \theta, \phi)$, and $\sup_X u = 0$, because the supremum is continuous for L^1 convergence of potentials (Proposition 14.2.1). The linear term $v \mapsto \int_X (v - \phi) g\,\mathrm{d}V$ is continuous for L^1 convergence because g is bounded, and the energy is upper semicontinuous by Proposition 17.3.3. Hence $F(u) \geq \overline{\lim}_j F(v_j) = \sup F$, so u is a maximizer, and Theorem 17.4.4 gives $\mu_u = \nu$. □

Proposition 17.4.6 *Let* $\lambda > 0$ *and let* ν *be a finite positive non-pluripolar measure on* X*. If* $u, v \in \mathcal{E}(X, \theta, \phi)$ *satisfy*

$$\mu_u = \mathrm{e}^{\lambda u}\nu, \qquad \mu_v \geq \mathrm{e}^{\lambda v}\nu,$$

then $u \geq v$*. In particular, the equation* $\mu_u = \mathrm{e}^{\lambda u}\nu$ *has at most one solution in* $\mathcal{E}(X, \theta, \phi)$*.*

Proof Let $A = \{u < v\}$. The comparison principle of Proposition 16.7.4 gives

$$\int_A \mathrm{e}^{\lambda v}\,\mathrm{d}\nu \leq \mu_v(A) \leq \mu_u(A) = \int_A \mathrm{e}^{\lambda u}\,\mathrm{d}\nu \leq m.$$

Both potentials are finite ν-almost everywhere, and on A we have $\mathrm{e}^{\lambda v} > \mathrm{e}^{\lambda u}$. The difference $\mathrm{e}^{\lambda v} - \mathrm{e}^{\lambda u}$ is therefore positive on A and has nonpositive integral over A, so $\nu(A) = 0$. Hence $\mu_u(A) = 0$, and the domination principle (Theorem 16.4.6) gives $u \geq v$ everywhere. If u and v are two solutions, applying this in both directions gives $u = v$. □

Solutions of the exponential equations are obtained as maximizers of the functionals

$$F_\lambda(v) = E(v) - \lambda^{-1} \int_X \mathrm{e}^{\lambda v}\, \mathrm{d}\nu, \qquad v \in \mathcal{E}^1(X,\theta,\phi), \tag{17.14}$$

where $\lambda > 0$ and ν is a finite positive Borel measure on X. These functionals are finite: the energy is finite on $\mathcal{E}^1(X,\theta,\phi)$, and the Borel function $\mathrm{e}^{\lambda v}$ satisfies $0 \le \mathrm{e}^{\lambda v} \le \mathrm{e}^{\lambda \sup_X v}$, where $\mathrm{e}^{\lambda v} = 0$ on the pole set of v.

Lemma 17.4.7 *Let ν be a finite positive Borel measure on X and $\lambda > 0$. If $u \in \mathcal{E}^1(X,\theta,\phi)$ maximizes the functional F_λ of (17.14) on $\mathcal{E}^1(X,\theta,\phi)$, then $\mu_u = \mathrm{e}^{\lambda u}\nu$.*

Proof Let $f \in C(X,\mathbb{R})$ and $u_t = P_\theta(u + tf)$ for $t \in \mathbb{R}$. By Proposition 17.4.3, $u_t \in \mathcal{E}^1(X,\theta,\phi)$, $u_0 = u$ and $u_t \le u + tf$ everywhere. Hence $\mathrm{e}^{\lambda u_t} \le \mathrm{e}^{\lambda(u+tf)}$, and $F_\lambda(u_t) \le F_\lambda(u)$ gives

$$E(u_t) - E(u) - \lambda^{-1} \int_X \left(\mathrm{e}^{\lambda(u+tf)} - \mathrm{e}^{\lambda u}\right) \mathrm{d}\nu \le 0.$$

The left side vanishes at $t = 0$. Its energy term is differentiable, with derivative $\int_X f\, \mu_{u_t}$, by Proposition 17.4.3. For $|t| \le 1$ the t-derivative $\lambda f \mathrm{e}^{\lambda(u+tf)}$ of the integrand is bounded by $\lambda \sup_X |f|\, \mathrm{e}^{\lambda \sup_X u + \lambda \sup_X |f|}$, which is integrable for the finite measure ν; by the mean value theorem and dominated convergence, the integral is differentiable as well, with derivative $\lambda \int_X f \mathrm{e}^{\lambda(u+tf)}\, \mathrm{d}\nu$. Since the left side is maximal at $t = 0$, its derivative vanishes there: $\int_X f\, \mu_u = \int_X f \mathrm{e}^{\lambda u}\, \mathrm{d}\nu$. Since f is arbitrary and $\mathrm{e}^{\lambda u}\nu$ is a finite measure, Lemma 17.4.2(1) gives $\mu_u = \mathrm{e}^{\lambda u}\nu$. □

To find maximizers for an arbitrary non-pluripolar measure ν, we need the continuity of $v \mapsto \int_X \mathrm{e}^{\lambda v}\, \mathrm{d}\nu$ for L^1 convergence of potentials. This rests on the following two lemmas, in which θ is an arbitrary smooth closed real $(1,1)$-form.

Lemma 17.4.8 *Let θ be a smooth closed real $(1,1)$-form on X. If $q_j, q \in \mathrm{PSH}(X,\theta)$ and $q_j \to q$ in $L^1(X)$, then $\overline{\lim}_j\, q_j = q$ quasi-everywhere.*

Proof By Lemma 17.3.2, the tail envelopes $a_j = (\sup_{k \ge j} q_k)^*$ are θ-psh and decrease pointwise to q. For each j, Corollary 10.2.2, applied after adding smooth local potentials of θ, gives $a_j = \sup_{k \ge j} q_k$ outside a pluripolar set. The countable union of these sets is pluripolar (Theorem 10.1.4), and outside it $q = \lim_j a_j = \overline{\lim}_j\, q_j$. □

Lemma 17.4.9 *Let ν be a finite non-pluripolar measure on X, let θ be a smooth closed real $(1,1)$-form on X, and let $B \ge 0$. Suppose that $q_j, q \in \mathrm{PSH}(X,\theta)$ satisfy $|q_j|, |q| \le B$ and $q_j \to q$ in $L^1(X)$. Then $q_j \to q$ in $L^1(\nu)$.*

Proof By Lemma 17.4.8, $\overline{\lim}_j\, q_j = q$ outside a pluripolar set, hence ν-almost everywhere. Thus $(q_j - q)_+ \to 0$ ν-almost everywhere, and $\int_X (q_j - q)_+\, \mathrm{d}\nu \to 0$ by bounded convergence, since $0 \le (q_j - q)_+ \le 2B$ and ν is finite. Since $|q_j - q| = 2(q_j - q)_+ - (q_j - q)$, it remains to show that $\int_X q_j\, \mathrm{d}\nu \to \int_X q\, \mathrm{d}\nu$. We have just seen

that $\overline{\lim}_j \int_X (q_j - q)\,\mathrm{d}\nu \le 0$, so it suffices to prove that $L = \underline{\lim}_j \int_X q_j\,\mathrm{d}\nu$ is at least $\int_X q\,\mathrm{d}\nu$.

Choose a subsequence (q_{j_k}) with $\int_X q_{j_k}\,\mathrm{d}\nu \to L$. For bounded Borel functions y on X put $\|y\| = (\int_X y^2\,\mathrm{d}\nu)^{1/2}$. For $l \ge 1$ let C_l be the set of finite convex combinations of the functions q_{j_k} with $k \ge l$, and let $d_l = \inf\{\|y\| : y \in C_l\}$. Since $C_{l+1} \subset C_l$ and $\|y\| \le B\nu(X)^{1/2}$ for $y \in C_l$, the numbers d_l increase to a finite limit. Choose $y_l \in C_l$ with $\|y_l\|^2 \le d_l^2 + 1/l$. For $l \le l'$ the function $\frac{1}{2}(y_l + y_{l'})$ belongs to the convex set C_l, so its norm is at least d_l. The parallelogram identity, obtained by integrating the pointwise identity $(a-b)^2 + (a+b)^2 = 2a^2 + 2b^2$, therefore gives

$$\|y_l - y_{l'}\|^2 = 2\|y_l\|^2 + 2\|y_{l'}\|^2 - 4\left\|\tfrac{1}{2}(y_l + y_{l'})\right\|^2 \le 2(d_{l'}^2 - d_l^2) + \frac{2}{l} + \frac{2}{l'},$$

and the right side tends to zero as $l \to \infty$, uniformly in $l' \ge l$.

By the Cauchy–Schwarz inequality, $\int_X |y_l - y_{l'}|\,\mathrm{d}\nu \le \nu(X)^{1/2}\|y_l - y_{l'}\|$, so we can choose $l_1 < l_2 < \cdots$ with $\int_X |y_{l_{i+1}} - y_{l_i}|\,\mathrm{d}\nu \le 2^{-i}$. By monotone convergence the series $\sum_i |y_{l_{i+1}} - y_{l_i}|$ has finite integral, hence converges ν-almost everywhere, and therefore y_{l_i} converges ν-almost everywhere; let $h = \overline{\lim}_i y_{l_i}$, a Borel function with $|h| \le B$. By bounded convergence, $\int_X y_{l_i}\,\mathrm{d}\nu \to \int_X h\,\mathrm{d}\nu$. On the other hand, $\int_X y_l\,\mathrm{d}\nu$ is a convex combination of the numbers $\int_X q_{j_k}\,\mathrm{d}\nu$ with $k \ge l$, which tend to L, so $\int_X y_l\,\mathrm{d}\nu \to L$. Hence $L = \int_X h\,\mathrm{d}\nu$.

It remains to show that $h = q$ ν-almost everywhere. Each y_{l_i} is a convex combination of θ-psh functions, hence θ-psh, and $\|y_{l_i} - q\|_{L^1(X)} \le \sup_{k \ge l_i} \|q_{j_k} - q\|_{L^1(X)}$, which tends to zero. By Lemma 17.4.8, applied to the sequence $(y_{l_i})_i$, the function $h = \overline{\lim}_i y_{l_i}$ equals q outside a pluripolar set, hence ν-almost everywhere. Thus $L = \int_X q\,\mathrm{d}\nu$, as required. □

Corollary 17.4.10 *Let θ be a smooth closed real $(1,1)$-form on X, let $q_j, q \in \mathrm{PSH}(X,\theta)$ satisfy $q_j \to q$ in $L^1(X)$, and let ν be a finite non-pluripolar measure on X. Then for every $\lambda > 0$,*

$$\mathrm{e}^{\lambda q_j} \longrightarrow \mathrm{e}^{\lambda q} \quad \text{in } L^1(\nu).$$

Proof By Proposition 14.2.1 there is a constant C with $q_j, q \le C$. We show that the functions $\mathrm{e}^{\lambda q_j}$ and $\mathrm{e}^{\lambda q}$, with value zero on the pole sets, are $C_1\omega$-psh for a constant C_1 independent of j; the preceding lemma then applies to them.

Choose $C_0 \ge 0$ with $\theta \le C_0\omega$, and a finite cover of X by coordinate balls $W' \Subset W$ such that the smaller balls W' still cover X, together with smooth functions h_W near $\overline{W}$ with $\mathrm{dd}^{\mathrm{c}} h_W = \theta$. The function $q_j + h_W$ is plurisubharmonic on W, so for every $\delta > 0$ smaller than the distance from W' to the boundary of W, its regularization $g_\delta = (q_j + h_W) * \rho_\delta$ by the radial mollifier is a smooth plurisubharmonic function on the ball W', and g_δ decreases to $q_j + h_W$ as $\delta \downarrow 0$, by Proposition 8.2.2(2). Put $q_{j,\delta} = g_\delta - h_W$. Then $\mathrm{dd}^{\mathrm{c}} q_{j,\delta} \ge -\theta \ge -C_0\omega$. Moreover $g_\delta(x)$ is an average of values of $q_j + h_W \le C + h_W$ at points within distance δ of x, so by uniform continuity

of h_W on a compact neighborhood of $\overline{W'}$ in W we have $q_{j,\delta} \le C+1$ on W' for all small δ, independently of j.

For a smooth real function s, the chain rule $\partial\bar\partial\, \mathrm{e}^{\lambda s} = \lambda \mathrm{e}^{\lambda s}\partial\bar\partial s + \lambda^2 \mathrm{e}^{\lambda s}\partial s \wedge \bar\partial s$ and the identity $\frac{\mathrm{i}}{2\pi}\partial s \wedge \bar\partial s = \mathrm{d}s \wedge \mathrm{d}^{\mathrm{c}} s$, which follows from (7.6), give

$$\mathrm{dd}^{\mathrm{c}} \mathrm{e}^{\lambda q_{j,\delta}} = \lambda \mathrm{e}^{\lambda q_{j,\delta}} \mathrm{dd}^{\mathrm{c}} q_{j,\delta} + \lambda^2 \mathrm{e}^{\lambda q_{j,\delta}} \mathrm{d}q_{j,\delta} \wedge \mathrm{d}^{\mathrm{c}} q_{j,\delta} \ge -\lambda C_0 \mathrm{e}^{\lambda(C+1)}\omega,$$

because $\mathrm{d}q_{j,\delta} \wedge \mathrm{d}^{\mathrm{c}} q_{j,\delta}$ is a semipositive $(1,1)$-form. Thus $\mathrm{e}^{\lambda q_{j,\delta}}$ is $C_1\omega$-psh on W' with $C_1 = \lambda C_0 \mathrm{e}^{\lambda(C+1)}$, and so is its decreasing limit $\mathrm{e}^{\lambda q_j}$ as $\delta \downarrow 0$, by Theorem 8.3.1 applied after adding a smooth local potential of $C_1\omega$; the limit is not identically $-\infty$, since it is nonnegative. Since the balls W' cover X, the function $\mathrm{e}^{\lambda q_j}$ belongs to $\mathrm{PSH}(X, C_1\omega)$, and the same argument applies to $\mathrm{e}^{\lambda q}$.

The exponential is Lipschitz with constant $\lambda \mathrm{e}^{\lambda C}$ on $(-\infty, C]$, so $\mathrm{e}^{\lambda q_j} \to \mathrm{e}^{\lambda q}$ in $L^1(X)$. These functions take values in $[0, \mathrm{e}^{\lambda C}]$, and Lemma 17.4.9, with the background $C_1\omega$, gives the assertion. □

Theorem 17.4.11 *Let ν be a finite positive non-pluripolar measure on X with $\nu(X) > 0$, and let $\lambda > 0$. There is a unique $u \in \mathcal{E}(X,\theta,\phi)$ satisfying*

$$\mu_u = \mathrm{e}^{\lambda u}\nu.$$

This solution belongs to $\mathcal{E}^1(X,\theta,\phi)$ and maximizes $F_\lambda(v) = E(v) - \lambda^{-1}\int_X \mathrm{e}^{\lambda v}\,\mathrm{d}\nu$ on $\mathcal{E}^1(X,\theta,\phi)$.

Proof By Corollary 17.3.4, the set K_ϕ of potentials $w \in \mathrm{PSH}(X,\theta)$ with $w \le \phi$ and $\sup_X w = 0$ is compact in $L^1(X)$; it contains ϕ. By Corollary 17.4.10, the function $w \mapsto \int_X \mathrm{e}^{\lambda w}\,\mathrm{d}\nu$ is continuous on K_ϕ. Its values are positive, since the pole set of w is ν-null and $\nu(X) > 0$. Hence it has a positive minimum a on K_ϕ.

Any $v \in \mathcal{E}^1(X,\theta,\phi)$ can be written $v = w + c$ with $c = \sup_X v$ and $w \in K_\phi$, by Lemma 16.4.2. Since $E(w) \le 0$, the shift formula for the energy gives

$$F_\lambda(v) = E(w) + mc - \lambda^{-1}\mathrm{e}^{\lambda c}\int_X \mathrm{e}^{\lambda w}\,\mathrm{d}\nu \le mc - \lambda^{-1} a \mathrm{e}^{\lambda c}.$$

The right side tends to $-\infty$ as $c \to \pm\infty$. Since $F_\lambda(\phi)$ is finite, a maximizing sequence $v_j = w_j + c_j$ with $F_\lambda(v_j) \ge F_\lambda(\phi) - 1$ therefore has bounded constants c_j. Moreover $E(v_j) \ge F_\lambda(v_j)$, because the exponential term is nonnegative, so $E(w_j) = E(v_j) - mc_j$ is bounded below. By Corollary 17.3.4, after passing to a subsequence, c_j converges and w_j converges in $L^1(X)$; hence $v_j \to u$ in $L^1(X)$ for some $u \in \mathcal{E}^1(X,\theta,\phi)$. The energy is upper semicontinuous (Proposition 17.3.3), and $\int_X \mathrm{e}^{\lambda v_j}\,\mathrm{d}\nu \to \int_X \mathrm{e}^{\lambda u}\,\mathrm{d}\nu$ by Corollary 17.4.10, so $F_\lambda(u) \ge \overline{\lim}_j F_\lambda(v_j) = \sup F_\lambda$, and u is a maximizer. By Lemma 17.4.7, $\mu_u = \mathrm{e}^{\lambda u}\nu$. Uniqueness in $\mathcal{E}(X,\theta,\phi)$ is Proposition 17.4.6. □

For instance, the theorem applies to $\nu = g\,\mathrm{d}V$, where $\mathrm{d}V$ is a smooth positive volume form and $g \in L^\infty(\mathrm{d}V)$ satisfies $g \ge 0$ and $\int_X g\,\mathrm{d}V > 0$: this measure is finite, and it is non-pluripolar because pluripolar sets have volume zero.

To let $\lambda \to 0$, we need a lower bound for the measures of L^1 limits.

Lemma 17.4.12 *Let θ be a smooth closed real $(1,1)$-form on X, and suppose that $q_j, q \in \mathrm{PSH}(X,\theta)$ and $q_j \to q$ in $L^1(X)$. Let ν be a finite non-pluripolar measure on X, and suppose that $\mu_{q_j} \geq f_j\nu$, where $f_j \geq 0$ are measurable and $f_j \to f$ ν-almost everywhere. Then $\mu_q \geq f\nu$.*

Proof Put $g_j = \inf_{k\geq j} f_k$, so that $g_j \uparrow f$ ν-almost everywhere. We first prove that $\mu_{a_j} \geq g_j\nu$ for the tail envelopes $a_j = (\sup_{k\geq j} q_k)^*$, and then let $j \to \infty$. For $l \geq j$, the finite maximum $\max\{q_j,\dots,q_l\}$ has measure at least $g_j\nu$. Indeed, let $r_1, r_2 \in \mathrm{PSH}(X,\theta)$ satisfy $\mu_{r_1}, \mu_{r_2} \geq g_j\nu$, and put $r = \max\{r_1, r_2\}$. The set $\{r_1 \geq r_2\}$ is the contact set $\{r_1 = r\}$, so Lemma 16.2.2, applied to $r_1 \leq r$, gives $\mu_r \geq \mu_{r_1} \geq g_j\nu$ on $\{r_1 \geq r_2\}$; in the same way $\mu_r \geq g_j\nu$ on $\{r_2 > r_1\}$. Hence $\mu_r \geq g_j\nu$. Since $\mu_{q_k} \geq f_k\nu \geq g_j\nu$ for $k \geq j$, induction on l proves the claim.

The suprema $\sup_X q_k$ are bounded, by Proposition 14.2.1, so the regularized upper envelope a_j is θ-psh, and it equals $\sup_{k\geq j} q_k$ almost everywhere (Theorem 8.3.2, applied in charts after adding smooth local potentials of θ). The finite maxima $\max\{q_j,\dots,q_l\}$ therefore increase almost everywhere to a_j as $l \to \infty$, and their measures converge weakly to μ_{a_j} by Corollary 15.7.6. The fixed measure $g_j\nu \leq \mu_{q_j}$ is finite, so the inequality passes to the weak limit, by Lemma 17.4.2(2):

$$\mu_{a_j} \geq g_j\nu.$$

By Lemma 17.3.2, a_j decreases pointwise to q.

To let $j \to \infty$ we truncate. Fix $C > 0$ and an index j_0. The truncations $a_j^C = \max\{a_j, V_\theta - C\}$ have uniformly bounded difference from V_θ: they are at least $V_\theta - C$, and since the suprema of the a_j are bounded, minimality of V_θ bounds them above by $V_\theta + C'$. They decrease to $q^C = \max\{q, V_\theta - C\}$, so they converge to q^C in extended capacity, by Lemma 15.6.3 applied in charts after adding smooth local potentials of θ. Since q^C has bounded difference from V_θ, it has the mass of V_θ (Corollary 15.7.5), that is, $q^C \in \mathcal{E}(X,\theta,V_\theta)$. Corollary 15.8.4, with V_θ in place of ϕ, therefore gives $\mu_{a_j^C} \to \mu_{q^C}$ weakly.

On the set $A_C = \{q > V_\theta - C\}$ we have $a_j \geq q > V_\theta - C$, so $a_j^C = a_j$ on the plurifine open set $\{a_j > V_\theta - C\} \supset A_C$, and locality gives $\mu_{a_j^C} \geq \mathbf{1}_{A_C} g_{j_0}\nu$ for $j \geq j_0$. Since $\mathbf{1}_{A_C} g_{j_0}\nu$ is a fixed finite measure, the inequality passes to the weak limit, by Lemma 17.4.2(2), and gives $\mu_{q^C} \geq \mathbf{1}_{A_C} g_{j_0}\nu$. On the plurifine open set A_C we have $q^C = q$, so locality gives $\mu_q \geq \mathbf{1}_{A_C} g_{j_0}\nu$. Thus $\mu_q(E) \geq \int_{E\cap A_C} g_{j_0}\,\mathrm{d}\nu$ for every Borel set E. Let $j_0 \to \infty$ and then $C \to \infty$. The sets A_C increase to the set where q and V_θ are finite, whose complement is pluripolar and hence ν-null, so monotone convergence gives $\mu_q(E) \geq \int_E f\,\mathrm{d}\nu$, that is, $\mu_q \geq f\nu$. □

Theorem 17.4.13 *Let ν be a finite positive non-pluripolar measure on X with $\nu(X) = m$. Then there is*

$$u \in \mathcal{E}(X,\theta,\phi) \quad \textit{with} \quad \mu_u = \nu \quad \textit{and} \quad \sup_X u = 0.$$

Proof Let $\lambda_j \downarrow 0$, let u_j be the solution of $\mu_{u_j} = \mathrm{e}^{\lambda_j u_j}\nu$ given by Theorem 17.4.11, and write $u_j = w_j + c_j$ with $c_j = \sup_X u_j$. Since $u_j \preceq \phi$ and $\sup_X w_j = 0$, Lemma 16.4.2 gives $w_j \le \phi$, so w_j belongs to the set K_ϕ of Corollary 17.3.4, which is compact in $L^1(X)$. On K_ϕ the functions

$$H_j(w) = \int_X \mathrm{e}^{\lambda_j w}\,\mathrm{d}\nu$$

are continuous, by Corollary 17.4.10, and they increase pointwise to the constant m, by monotone convergence, since $w \le 0$ and the pole set of w is ν-null. The convergence is uniform on K_ϕ (Dini's theorem): for $\varepsilon > 0$ the compact sets $\{w \in K_\phi : m - H_j(w) \ge \varepsilon\}$ decrease and have empty intersection, so they are empty for large j.

Comparing total masses in the equation gives $m = \mathrm{e}^{\lambda_j c_j} H_j(w_j)$, so

$$\mathrm{e}^{\lambda_j c_j} = \frac{m}{H_j(w_j)} \longrightarrow 1.$$

Also $\int_X |1 - \mathrm{e}^{\lambda_j w_j}|\,\mathrm{d}\nu = m - H_j(w_j) \to 0$. Hence $r_j = \mathrm{e}^{\lambda_j u_j} = \mathrm{e}^{\lambda_j c_j}\mathrm{e}^{\lambda_j w_j} \to 1$ in $L^1(\nu)$, and $\mu_{w_j} = \mu_{u_j} = r_j\nu$. Pass to a subsequence along which $r_j \to 1$ ν-almost everywhere and, by compactness of K_ϕ, $w_j \to u$ in $L^1(X)$ for some $u \in K_\phi$. Lemma 17.4.12 gives $\mu_u \ge \nu$. Since $u \le \phi$, mass monotonicity gives $\mu_u(X) \le m = \nu(X)$. Hence $\mu_u = \nu$, u has full mass relative to ϕ, and $\sup_X u = 0$ because $u \in K_\phi$. □

17.4.2 Uniqueness by concentration of mass

We now show that the solution given by Theorem 17.4.13 is unique up to an additive constant. The proof uses existence once more. If two solutions u, v with $\mu_u = \mu_v = \mu$ do not differ by a constant μ-almost everywhere, then, after a shift of v, the set $U = \{u < v\}$ carries a proper fraction of the mass of μ. We solve the equation whose right-hand side is μ restricted to U and rescaled to total mass m. Mixing this auxiliary solution into v raises the density of the Monge–Ampère measure on U (Corollary 15.4.7), in contradiction with the comparison principle. The domination principle then turns equality μ-almost everywhere into equality everywhere.

Theorem 17.4.14 *Let $u, v \in \mathcal{E}(X, \theta, \phi)$ satisfy $\mu_u = \mu_v$. Then there is a constant $C \in \mathbb{R}$ such that $u = v + C$ everywhere on X. Consequently, every non-pluripolar measure of mass m is the Monge–Ampère measure μ_u of exactly one $u \in \mathcal{E}(X, \theta, \phi)$ with $\sup_X u = 0$.*

Proof Put $\mu = \mu_u = \mu_v$. Since $u \in \mathcal{E}(X, \theta, \phi)$, this is a finite measure of mass $\int_X \langle\theta_u^n\rangle = \int_X \langle\theta_\phi^n\rangle = m$, and it is non-pluripolar, because non-pluripolar products give no mass to pluripolar sets (Lemma 15.1.2). Hence the pole sets of u and v are μ-null,

and $u - v$ is defined and finite μ-almost everywhere. We first show that $u - v$ is constant μ-almost everywhere.

Suppose not. Let λ be the image of μ under $u - v$, a finite Borel measure on $\mathbb{R}$ of total mass $m > 0$, and put

$$c_- = \sup\{c \in \mathbb{R} : \lambda((-\infty, c)) = 0\}, \qquad c_+ = \inf\{c \in \mathbb{R} : \lambda((c, \infty)) = 0\},$$

with $\sup \varnothing = -\infty$ and $\inf \varnothing = +\infty$. The sets $(-\infty, c)$ increase to $\mathbb{R}$ as $c \to +\infty$, so $\lambda((-\infty, c)) \to m > 0$; hence $c_- < +\infty$, and similarly $c_+ > -\infty$. If $c_- > -\infty$, the λ-null sets $(-\infty, c)$ with $c < c_-$ increase to $(-\infty, c_-)$, so $\lambda((-\infty, c_-)) = 0$; similarly $\lambda((c_+, \infty)) = 0$ if $c_+ < +\infty$. Hence $c_+ < c_-$ is impossible, since $\mathbb{R} = (-\infty, c_-) \cup (c_+, \infty)$ would then be λ-null. If $c_- = c_+$, then λ is carried by the point c_-, that is, $u - v$ is constant μ-almost everywhere, contrary to our assumption. Thus $c_- < c_+$, and every $c \in (c_-, c_+)$ satisfies $\lambda((-\infty, c)) > 0$ and $\lambda((c, \infty)) > 0$, that is,

$$\mu(\{u - v < c\}) > 0, \qquad \mu(\{u - v > c\}) > 0.$$

The singletons $\{c\}$ with $\lambda(\{c\}) > 0$ are pairwise disjoint sets of positive measure, so there are only countably many of them by Lemma 1.1.5. Since the interval (c_-, c_+) is uncountable, we may choose c in it with $\mu(\{u - v = c\}) = 0$. Replacing v by $v + c$ changes neither the class $\mathcal{E}(X, \theta, \phi)$ nor the measure μ_v, so we may assume $c = 0$. Then $\mu(\{u < v\})$ and $\mu(\{u > v\})$ are positive, and the set $U = \{u < v\}$ satisfies

$$0 < \mu(U) < m, \qquad \mu(\{u = v\}) = 0.$$

Set $q = (m/\mu(U))^{1/n} > 1$. The measure $q^n \mathbf{1}_U \mu$ is non-pluripolar and has mass m, so Theorem 17.4.13 gives $h \in \mathcal{E}(X, \theta, \phi)$ with

$$\mu_h = q^n \mathbf{1}_U \mu.$$

For $0 < t < 1$ put $w_t = (1 - t)v + th$. By Corollary 17.2.10, $w_t \in \mathcal{E}(X, \theta, \phi)$. Since $\theta_{w_t} = (1 - t)\theta_v + t\theta_h$, Corollary 15.4.7, applied to the currents θ_v, θ_h, the common measure μ and the densities 1 and $q^n \mathbf{1}_U$, gives

$$\mu_{w_t} \geq \big((1 - t) + tq\mathbf{1}_U\big)^n \mu.$$

Let $U_t = \{u < w_t\}$, and put

$$d_t = (1 - t)^n, \qquad b_t = (1 - t + tq)^n.$$

The comparison principle for relative full mass, Proposition 16.7.4, applied to u and w_t, and the preceding lower bound give

$$\mu(U_t) = \mu_u(U_t) \geq \mu_{w_t}(U_t) \geq d_t \mu(U_t) + (b_t - d_t)\mu(U_t \cap U).$$

Since $1 - d_t > 0$, it follows that

$$\frac{b_t - d_t}{1 - d_t}\,\mu(U_t \cap U) \le \mu(U_t). \tag{17.15}$$

We let $t \downarrow 0$. At every point where u, v, h are finite and $u \neq v$, we have $w_t = v + t(h - v)$, so $\mathbf{1}_{U_t} \to \mathbf{1}_U$ there. These points have full μ-measure, because the three pole sets and the level set $\{u = v\}$ are μ-null. By dominated convergence,

$$\mu(U_t \cap U) \longrightarrow \mu(U), \qquad \mu(U_t) \longrightarrow \mu(U).$$

Moreover $b_t - d_t = ntq + O(t^2)$ and $1 - d_t = nt + O(t^2)$, so $(b_t - d_t)/(1 - d_t) \to q$. Passing to the limit in (17.15) gives $q\mu(U) \le \mu(U)$, which contradicts $q > 1$ and $\mu(U) > 0$.

Hence, for the original pair, there is a finite constant C such that $u = v + C$ μ-almost everywhere. Both u and $v + C$ belong to $\mathcal{E}(X, \theta, \phi)$, and their common measure μ gives zero mass to $\{u < v + C\}$ and to $\{v + C < u\}$. The domination principle (Theorem 16.4.6), applied in both directions, gives $u = v + C$ everywhere, including at the common poles. This proves the first assertion.

For the second, Theorem 17.4.13 provides a solution with supremum zero. If u and v are two such solutions, then $u = v + C$, and comparing suprema gives $C = 0$. $\square$

17.4.3 Bounded differences from the reference

The solutions constructed so far belong to $\mathcal{E}(X, \theta, \phi)$, but in general their difference from ϕ is unbounded. We now show that the difference is bounded when the measure is dominated by a power $1 + \delta > 1$ of the Monge–Ampère capacity $\mathrm{Cap}_{\theta,\phi}$ relative to ϕ (Definition 16.7.5). If $\mu_u \le A\,\mathrm{Cap}_{\theta,\phi}^{1+\delta}$, the sublevel estimate of Proposition 16.7.7 bounds the capacity of the deeper sublevel set $\{u < \phi - s - t\}$ by At^{-n} times the $(1+\delta)$th power of the capacity of $\{u < \phi - s\}$. Iterating with geometrically decreasing steps t, we reach a finite level beyond which the capacity vanishes. Measures with L^p densities, $p > 1$, satisfy such a bound by Corollary 16.9.5.

Theorem 17.4.15 *Let $u \in \mathcal{E}(X, \theta, \phi)$ with $u \le \phi$. Let $0 \le a < 1$, $A > 0$ and $\delta > 0$, and let ν be a positive Borel measure on X such that*

$$\mu_u \le \nu + a\mu_\phi, \qquad \nu(E) \le A\,\mathrm{Cap}_{\theta,\phi}(E)^{1+\delta} \quad \text{for every Borel set } E \subset X. \tag{17.16}$$

Then there is a finite constant $C \ge 0$ such that $\phi - C \le u \le \phi$ on X. If, in addition, $I = \int_X (\phi - u)\,\mathrm{d}\nu < \infty$, then one can take

$$C = 2 + \frac{I}{1 - a}\left(2\left(\frac{A}{1 - a}\right)^{1/n}\right)^{n/\delta} + \frac{1}{1 - 2^{-\delta}}. \tag{17.17}$$

Here $\phi - u$ and the sublevel sets $\{u < \phi - s\}$ are taken on the set where ϕ is finite; its complement is pluripolar, hence of zero capacity (Proposition 16.7.6) and therefore ν-null.

Proof We derive a recursive inequality for the capacities of the sublevel sets of $u - \phi$ and iterate it along levels that converge to a finite depth. For $s > 0$ put

$$E_s = \{u < \phi - s\}, \qquad g(s) = \operatorname{Cap}_{\theta,\phi}(E_s)^{1/n}, \qquad B = \left(\frac{A}{1-a}\right)^{1/n}.$$

The sets E_s decrease as s increases, so g is nonincreasing.

We first compare the measures on E_s. Both u and $\phi - s$ belong to $\mathcal{E}(X, \theta, \phi)$, so the comparison principle of Proposition 16.7.4, applied to u and $\phi - s$, gives $\mu_\phi(E_s) \le \mu_u(E_s)$. By (17.16),

$$\mu_u(E_s) \le \nu(E_s) + a\mu_\phi(E_s) \le \nu(E_s) + a\mu_u(E_s),$$

and hence $(1-a)\mu_u(E_s) \le \nu(E_s)$. Since u has full mass relative to ϕ, Proposition 16.7.7 gives $t^n \operatorname{Cap}_{\theta,\phi}(E_{s+t}) \le \mu_u(E_s)$ for $0 < t \le 1$. Combining this with the last inequality and the capacity bound in (17.16), we obtain $t^n g(s+t)^n \le B^n g(s)^{n(1+\delta)}$, that is,

$$t g(s+t) \le B g(s)^{1+\delta}, \qquad s > 0, \quad 0 < t \le 1. \tag{17.18}$$

By the second assertion of Proposition 16.7.7, $g(s) \to 0$ as $s \to \infty$. Choose $s_0 > 0$ with $2Bg(s_0)^\delta \le 1$, and put

$$C = s_0 + \frac{2B}{1 - 2^{-\delta}} g(s_0)^\delta. \tag{17.19}$$

We claim that $g(C) = 0$. If $g(s_0) = 0$, then $C = s_0$ and there is nothing to prove. Otherwise, as long as $g(s_j) > 0$, define inductively

$$t_j = 2Bg(s_j)^\delta, \qquad s_{j+1} = s_j + t_j.$$

Since g is nonincreasing, $t_j \le 2Bg(s_0)^\delta \le 1$, and (17.18) with $s = s_j$ and $t = t_j$ gives

$$g(s_{j+1}) \le \frac{Bg(s_j)^{1+\delta}}{t_j} = \frac{g(s_j)}{2}.$$

Hence $g(s_j) \le 2^{-j} g(s_0)$ and $t_j \le 2^{-j\delta} \cdot 2Bg(s_0)^\delta$, so that

$$s_j \le s_0 + 2Bg(s_0)^\delta \sum_{i=0}^{\infty} 2^{-i\delta} = C \qquad \text{for every } j.$$

If the construction stops because $g(s_j) = 0$ for some j, then $g(C) \le g(s_j) = 0$. Otherwise $g(C) \le g(s_j) \le 2^{-j} g(s_0)$ for every j, and again $g(C) = 0$. This proves the claim.

Thus $\mathrm{Cap}_{\theta,\phi}(E_C) = 0$. The capacity bound in (17.16) gives $\nu(E_C) = 0$. Since ϕ itself is a competitor for $\mathrm{Cap}_{\theta,\phi}$ (Definition 16.7.5), we also have $\mu_\phi(E_C) \le \mathrm{Cap}_{\theta,\phi}(E_C) = 0$. The measure inequality in (17.16) therefore gives $\mu_u(\{u < \phi - C\}) = 0$. As $\phi - C \in \mathcal{E}(X,\theta,\phi)$, the domination principle, Theorem 16.4.6, yields $u \ge \phi - C$ everywhere.

Now suppose that $I < \infty$. On E_s the weight $\phi - u$ exceeds s, so Markov's inequality gives $\nu(E_s) \le I/s$. Taking $t = 1$ in the sublevel estimate and using the measure inequality on E_s, we find

$$g(s+1)^n \le \mu_u(E_s) \le \frac{\nu(E_s)}{1-a} \le \frac{I}{(1-a)s}.$$

Choose $s = 1 + I(2B)^{n/\delta}/(1-a)$ and $s_0 = s+1$. Since $(1-a)s > I(2B)^{n/\delta}$, we get $g(s_0)^n \le (2B)^{-n/\delta}$, that is, $2Bg(s_0)^\delta \le 1$. Thus s_0 satisfies the condition imposed on it above, and the constant in (17.19) is at most $s_0 + (1-2^{-\delta})^{-1}$. Since $(2B)^{n/\delta} = \bigl(2(A/(1-a))^{1/n}\bigr)^{n/\delta}$, this is the right side of (17.17). The lower bound $\phi - C \le u$ remains true when C is increased, which proves (17.17). □

Corollary 17.4.16 *Let $u \in \mathcal{E}^1(X,\theta,\phi)$ with $u \le \phi$, and let $A > 0$ and $\delta > 0$ be such that*

$$\mu_u(E) \le A\,\mathrm{Cap}_{\theta,\phi}(E)^{1+\delta} \quad \text{for every Borel set } E \subset X.$$

If $L \ge 0$ and $E^\phi_\theta(u) \ge -L$, then

$$\phi - \left(2 + (n+1)L(2A^{1/n})^{n/\delta} + \frac{1}{1-2^{-\delta}}\right) \le u \le \phi.$$

Proof Apply Theorem 17.4.15 with $a = 0$ and $\nu = \mu_u$. By the first inequality in (17.10), $I = \int_X (\phi - u)\,\mu_u \le (n+1)\bigl(-E^\phi_\theta(u)\bigr) \le (n+1)L$, and (17.17) with $a = 0$ gives the stated bound. □

Corollary 17.4.17 *Let $p > 1$, let $0 \le f \in L^p(X,\omega^n)$ and $0 \le a < 1$, and let $u \in \mathcal{E}(X,\theta,\phi)$ satisfy $\sup_X u = 0$ and*

$$\mu_u \le f\omega^n + a\mu_\phi.$$

Then $\phi - C \le u \le \phi$, where C depends only on $X,\theta,\omega,p,m,\|f\|_{L^p}$ and $(1-a)^{-1}$. In particular, if $\int_X f\omega^n = m$, then the equation $\mu_u = f\omega^n$ has a solution $u \in \mathcal{E}(X,\theta,\phi)$ with $\sup_X u = 0$, and this solution has bounded difference from ϕ.

Proof Since $\sup_X u = 0$, Lemma 16.4.2 gives $u \le \phi$; recall also that $\sup_X \phi = 0$, by Definition 16.3.1. Let $q = p/(p-1)$ be the conjugate exponent. By Corollary 14.2.2, the $L^q(X,\omega^n)$ norms of u and ϕ are at most a constant M_q depending only on X,θ,ω,q. Hölder's inequality gives

$$I = \int_X (\phi - u) f\omega^n \le \|f\|_{L^p}\|\phi - u\|_{L^q} \le 2M_q\|f\|_{L^p}.$$

By Corollary 16.9.5 with $r = 2$, the measure $\nu = f\omega^n$ satisfies the capacity bound in (17.16) with $\delta = 1$ and $A = B_2\|f\|_{L^p}/m^2$, where B_2 depends only on X, θ, ω, p. (If $f = 0$ almost everywhere, take $A = 1$ instead.) With $\delta = 1$, the constant (17.17) equals $4 + 2^n AI/(1-a)^2$. Inserting the bounds for A and I, and recalling that the lower bound $\phi - C \le u$ survives an increase of C, we see that Theorem 17.4.15 gives the assertion with

$$C = 4 + \frac{2^{n+1} B_2 M_q \|f\|_{L^p}^2}{m^2(1-a)^2},$$

which has the stated dependence.

Finally, the measure $f\omega^n$ is non-pluripolar, as noted after Definition 17.4.1. If its mass is m, Theorem 17.4.13 gives a solution $u \in \mathcal{E}(X, \theta, \phi)$ of $\mu_u = f\omega^n$ with $\sup_X u = 0$, and the bound just proved, with $a = 0$, shows that u has bounded difference from ϕ. □

17.4.4 Positive mass and bigness

So far the background class has been big, and big classes have positive volume by Proposition 15.8.2. The equations solved above give the converse for pseudo-effective classes. In this subsection θ denotes an arbitrary smooth closed real $(1,1)$-form on X; its class need not be big, and no model potential is fixed.

Let $\alpha = \{\theta\}$ be a pseudo-effective class whose volume, in the sense of Definition 15.8.1, is positive. The classes $\alpha + \varepsilon\{\omega\}$, $\varepsilon > 0$, are big, and in each of them we solve the Monge–Ampère equation whose right-hand side is a fixed smooth measure rescaled to the total mass $\mathrm{vol}(\alpha + \varepsilon\{\omega\})$, which is at least $\mathrm{vol}(\alpha)$. A pointwise inequality between traces, integrated against these solutions, yields lower bounds for the pairings $\int_X \theta \wedge \tau$ with strictly positive dd^{c}-closed $(n-1,n-1)$-forms τ, uniformly in ε. Lamari's duality criterion, which we prove first, turns such bounds into a Kähler current in α.

Recall from Section 7.2 that weak and strong positivity coincide for real $(n-1,n-1)$-forms, so that we speak simply of positive forms, and that such a form is strictly positive if at every point it lies in the interior of the cone of positive forms. For instance, ω^{n-1} is strictly positive: at a point x, choose linear coordinates on the tangent space in which $\omega(x)$ is the standard form β; then $\omega(x)^{n-1} = \beta^{n-1}$ is an interior point of the cone by Proposition 7.2.1, and complex-linear isomorphisms preserve positivity. For real $(n-1,n-1)$-forms we write $\eta \le \tau$ if $\tau - \eta$ is positive.

Lemma 17.4.18 (Lamari's criterion) *Let θ be a smooth closed real $(1,1)$-form on X. Then $\mathrm{PSH}(X,\theta) \neq \varnothing$ if and only if*

$$\int_X \theta \wedge \tau \ge 0 \tag{17.20}$$

for every smooth strictly positive real $(n-1,n-1)$-form τ on X with $\mathrm{dd^c}\tau = 0$. Consequently, the class $\{\theta\}$ is big if and only if there is $\delta > 0$ such that every such form τ satisfies

$$\int_X \theta \wedge \tau \geq \delta \int_X \omega \wedge \tau. \tag{17.21}$$

Proof *Necessity.* If $u \in \mathrm{PSH}(X,\theta)$, then $T = \theta + \mathrm{dd^c}u \geq 0$. For τ as in the statement, the definition of $\mathrm{dd^c}u$ as a current gives $\int_X \mathrm{dd^c}u \wedge \tau = \int_X u\,\mathrm{dd^c}\tau = 0$, hence $\int_X \theta \wedge \tau = \int_X T \wedge \tau$, which is nonnegative by the definition of a positive current, since τ is positive, hence strongly positive.

Sufficiency. Conversely, assume (17.20) for all forms τ as in the statement. We extend the functional $\tau \mapsto \int_X \theta \wedge \tau$ from $\mathrm{dd^c}$-closed forms to a positive linear functional on all smooth real $(n-1,n-1)$-forms, regard the extension as a positive $(1,1)$-current T, and show that $T - \theta$ is $\mathrm{dd^c}$ of a distribution.

A positive extension. Let F be the real vector space of smooth real $(n-1,n-1)$-forms on X, let $S = \{\tau \in F : \mathrm{dd^c}\tau = 0\}$, and put $e = \omega^{n-1}$, with $e = 1$ if $n = 1$. Since ω is closed, $e \in S$, and e is strictly positive. Define $\ell_0(\tau) = \int_X \theta \wedge \tau$ for $\tau \in S$. By hypothesis $\ell_0(\tau) \geq 0$ when $\tau \in S$ is strictly positive. If $\tau \in S$ is merely positive, then $\tau + te \in S$ is strictly positive for every $t > 0$. Indeed, at each point x the cone of positive forms contains a ball about $e(x)$, for some norm; since the cone is stable under sums and positive multiples, it also contains the ball of t times that radius about $\tau(x) + te(x)$. Hence $\ell_0(\tau) + t\ell_0(e) \geq 0$, and letting $t \downarrow 0$ gives $\ell_0(\tau) \geq 0$.

We next compare the forms in F with e. Fix finitely many holomorphic charts $U_1, \ldots, U_N$ and compact sets $K_a \subset U_a$ with $\bigcup_a K_a = X$. In U_a write $\eta \in F$ as $\eta = \sum_{I,J} \eta^a_{IJ}\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J$, the sum running over increasing multi-indices of length $n-1$, and put

$$\|\eta\| = \max_{1 \leq a \leq N} \max_{I,J} \sup_{K_a} |\eta^a_{IJ}|.$$

We claim that there is a constant $M > 0$ such that

$$-M\|\eta\| e \leq \eta \leq M\|\eta\| e \qquad \text{for every } \eta \in F. \tag{17.22}$$

Let Z be the finite-dimensional real vector space of real $(n-1,n-1)$-forms with constant coefficients on $\mathbb{C}^n$, normed by the maximum modulus of the coefficients, and let $P \subset Z$ be the closed convex cone of positive forms (Proposition 7.2.1). For $x \in U_a$, the differential of the chart is a complex-linear isomorphism from the tangent space at x onto $\mathbb{C}^n$, under which $\mathrm{d}z_j$ corresponds to the jth coordinate form. Since complex-linear isomorphisms preserve positivity, a real $(n-1,n-1)$-form η is positive at x if and only if its coefficient form $\eta^a(x) = \sum_{I,J} \eta^a_{IJ}(x)\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J \in Z$ lies in P.

Because e is strictly positive, the continuous map $x \mapsto e^a(x)$ sends the compact set K_a into the interior of P; the complement of this interior is closed and disjoint from the compact image, so it has positive distance r_a from it. Put $r = \frac{1}{2}\min_a r_a$. If $\eta \in F$ satisfies $\|\eta\| \leq r$ and $x \in K_a$, then $e^a(x) \pm \eta^a(x)$ lies within distance $r < r_a$ of $e^a(x)$, hence in P; thus $e \pm \eta$ is positive at x. Since the sets K_a cover X, the forms $e \pm \eta$ are positive. Applying this to $r\eta/\|\eta\|$ for $\eta \neq 0$ proves (17.22) with $M = 1/r$.

For $\eta \in F$, a form $\tau \in S$ with $\tau \geq \eta$ is called an *S-majorant* of η, and we put

$$p(\eta) = \inf\{\ell_0(\tau) : \tau \text{ is an } S\text{-majorant of } \eta\}.$$

Let $\eta \in F$ and $C = M\|\eta\|$, so that $-Ce \leq \eta \leq Ce$ by (17.22). The form Ce is an S-majorant of η, so $p(\eta) \leq C\ell_0(e)$. If τ is an S-majorant of η, then $\tau + Ce \geq \eta + Ce \geq 0$, so $\tau + Ce \in S$ is positive and $\ell_0(\tau) \geq -C\ell_0(e)$. Thus p is finite. It is sublinear: the sum of S-majorants of η_1 and η_2 is an S-majorant of $\eta_1 + \eta_2$, and for $\lambda > 0$ the S-majorants of $\lambda\eta$ are λ times those of η. If $\eta \in S$, then η is its own S-majorant, and every S-majorant τ of η satisfies $\ell_0(\tau) - \ell_0(\eta) = \ell_0(\tau - \eta) \geq 0$ because $\tau - \eta \in S$ is positive. Hence $p = \ell_0$ on S.

Next we extend ℓ_0 to a linear functional on F dominated by p; this is the Hahn–Banach theorem, whose proof we recall. Let ℓ be a linear functional on a subspace $W \supset S$ which extends ℓ_0 and satisfies $\ell \leq p$ on W, and let $\eta_0 \in F \setminus W$. For $\eta_1, \eta_2 \in W$,

$$\ell(\eta_1) + \ell(\eta_2) = \ell(\eta_1 + \eta_2) \leq p(\eta_1 + \eta_2) \leq p(\eta_1 - \eta_0) + p(\eta_2 + \eta_0),$$

so

$$\sup_{\eta_1 \in W} \big(\ell(\eta_1) - p(\eta_1 - \eta_0)\big) \leq \inf_{\eta_2 \in W} \big(p(\eta_2 + \eta_0) - \ell(\eta_2)\big).$$

Taking $\eta_1 = \eta_2 = 0$ shows that the supremum is greater than $-\infty$ and the infimum is less than $+\infty$; hence both are finite, and we may choose a real number c between them. Define $\ell(\eta + t\eta_0) = \ell(\eta) + tc$ for $\eta \in W$ and $t \in \mathbb{R}$. For $t > 0$, the upper bound for c with $\eta_2 = \eta/t$, multiplied by t, gives $\ell(\eta) + tc \leq p(\eta + t\eta_0)$; for $t < 0$, the lower bound for c with $\eta_1 = \eta/|t|$, multiplied by $|t|$, gives the same inequality. Thus the extension to $W + \mathbb{R}\eta_0$ is still dominated by p. The union of a chain of such extensions is again such an extension, so Zorn's lemma gives a maximal one, and by the step just proved its domain is F. We denote it again by ℓ.

The functional ℓ is positive: if $\eta \in F$ is positive, then 0 is an S-majorant of $-\eta$, so $\ell(-\eta) \leq p(-\eta) \leq 0$. It is also bounded: by (17.22), the form $M\|\eta\|e$ is an S-majorant of both η and $-\eta$, so

$$|\ell(\eta)| \leq M\ell_0(e)\|\eta\| \qquad (\eta \in F). \tag{17.23}$$

We now turn ℓ into a current. Every smooth form on the compact manifold X is a test form. For a complex form φ of degree $2n-2$, let φ' be its component of bidegree $(n-1, n-1)$, and put $T(\varphi) = \ell(\operatorname{Re}\varphi') + \mathrm{i}\,\ell(\operatorname{Im}\varphi')$, where $\operatorname{Re}\varphi' = (\varphi' + \overline{\varphi'})/2$ and $\operatorname{Im}\varphi' = (\varphi' - \overline{\varphi'})/(2\mathrm{i})$ are real forms in F. This is a complex-linear functional: replacing φ by $\mathrm{i}\varphi$ replaces $(\operatorname{Re}\varphi', \operatorname{Im}\varphi')$ by $(-\operatorname{Im}\varphi', \operatorname{Re}\varphi')$. Moreover, T has the continuity property of a current of degree 2. Indeed, let Q be a compact subset of a chart U, and let φ be supported in Q, with coefficients φ_J in real coordinates on U. On the compact set $K_a \cap Q$, the coefficients of $\operatorname{Re}\varphi'$ and $\operatorname{Im}\varphi'$ in the chart U_a are linear combinations of the functions φ_J and $\overline{\varphi_J}$ with coefficients that are smooth on $U \cap U_a$, and they vanish on $K_a \setminus Q$. Hence $\|\operatorname{Re}\varphi'\| + \|\operatorname{Im}\varphi'\| \leq C_Q \sum_J \sup_Q |\varphi_J|$ with a constant C_Q independent of φ, and (17.23) gives $|T(\varphi)| \leq MC_Q\ell_0(e) \sum_J \sup_Q |\varphi_J|$.

By construction T vanishes on forms of bidegree other than $(n-1,n-1)$, so it is a current of bidegree $(1,1)$. It is real, because the $(n-1,n-1)$-component of $\overline{\varphi}$ is $\overline{\varphi'} = \operatorname{Re}\varphi' - \mathrm{i}\operatorname{Im}\varphi'$, so that $T(\overline{\varphi}) = \overline{T(\varphi)}$. It is positive, because a strongly positive test form φ of bidegree $(n-1,n-1)$ is real and positive, so that $T(\varphi) = \ell(\varphi) \geq 0$. Since ℓ extends ℓ_0, the real $(1,1)$-current $R = T - \theta$ annihilates S.

The current R is $\mathrm{dd^c}$*-exact.* We first show that R is closed. If ξ is a smooth form of bidegree $(n-2,n-1)$, then $\mathrm{dd^c}\partial\xi = \frac{\mathrm{i}}{2\pi}\partial\bar\partial\partial\xi = -\frac{\mathrm{i}}{2\pi}\partial\partial\bar\partial\xi = 0$, and $\mathrm{dd^c}\bar\partial\xi' = \frac{\mathrm{i}}{2\pi}\partial\bar\partial\bar\partial\xi' = 0$ for ξ' of bidegree $(n-1,n-2)$. Since $\mathrm{dd^c}$ is a real operator, the conjugate of a $\mathrm{dd^c}$-closed complex $(n-1,n-1)$-form σ is $\mathrm{dd^c}$-closed, so the real forms $\operatorname{Re}\sigma$ and $\operatorname{Im}\sigma$ lie in S; as R is complex-linear and annihilates S, it annihilates σ. In particular $R(\partial\xi) = R(\bar\partial\xi') = 0$. For a smooth $(2n-3)$-form γ, the $(n-1,n-1)$-component of $\mathrm{d}\gamma$ is $\partial\gamma^{n-2,n-1} + \bar\partial\gamma^{n-1,n-2}$; since R has bidegree $(1,1)$, it follows that $R(\mathrm{d}\gamma) = 0$. Therefore $\mathrm{d}R = 0$.

Since $e \in S$, we also have $\int_X R\wedge\omega^{n-1} = 0$. Let Λ denote contraction with ω, as in Lemma 14.1.6; the identity $R\wedge\omega^{n-1} = (\Lambda R)\omega^n/n$ shows that the trace ΛR has mean zero. By Lemma 14.1.3(2), applied to the Laplacian $L = \mathrm{d}^*\mathrm{d}$ of Section 14.1, there is a real distribution f with $Lf = -4\pi\Lambda R$. By Lemma 14.1.6, $\Lambda\mathrm{dd^c}f = \Lambda R$, so $R' = R - \mathrm{dd^c}f$ is a closed real $(1,1)$-current with zero trace. By the last assertion of Lemma 14.1.6, R' is harmonic, and by Lemma 14.1.3(1) it is smooth. If $n = 1$, a $(1,1)$-form with zero trace vanishes, so $R' = 0$. If $n \geq 2$, (14.7) gives $*R' = -R'\wedge\omega^{n-2}/(n-2)!$, a smooth closed $(n-1,n-1)$-form; hence $*R' \in S$. Since R annihilates S and $\mathrm{dd^c}*R' = 0$,

$$\|R'\|^2_{L^2} = \int_X R'\wedge *R' = R(*R') - \int_X f\,\mathrm{dd^c}*R' = 0.$$

Thus $R' = 0$ and $R = \mathrm{dd^c}f$ in every dimension.

Conclusion. The current $T = \theta + \mathrm{dd^c}f$ is closed and positive, and it lies in the class $\{\theta\}$, since it differs from θ by $\mathrm{dd^c}$ of a distribution. By Corollary 14.1.9, there is $u \in \mathrm{PSH}(X,\theta)$ with $\theta + \mathrm{dd^c}u = T$. This proves sufficiency.

For the last assertion, $\{\theta\}$ is big exactly when $\mathrm{PSH}(X,\theta-\delta\omega) \neq \varnothing$ for some $\delta > 0$. Indeed, if $T \geq \delta\omega$ is a Kähler current in $\{\theta\}$ (Definition 14.3.1), then $T - \delta\omega$ is a closed positive current in the class $\{\theta-\delta\omega\}$, so $T - \delta\omega = \theta - \delta\omega + \mathrm{dd^c}u$ with $u \in \mathrm{PSH}(X,\theta-\delta\omega)$, by Corollary 14.1.9. Conversely, if $u \in \mathrm{PSH}(X,\theta-\delta\omega)$, then $\theta + \mathrm{dd^c}u \geq \delta\omega$ is a Kähler current in $\{\theta\}$. Applying the first part to $\theta - \delta\omega$ shows that this holds if and only if (17.21) holds for that δ. □

The proof of the bigness criterion rests on the following lower bound for the product of two trace masses of a current with prescribed Monge–Ampère measure.

Lemma 17.4.19 *Let A be a closed subset of X of Lebesgue measure zero in every chart, and let S be a closed positive $(1,1)$-current on X whose local potentials are locally bounded on $X\setminus A$. Let τ be a smooth strictly positive real $(n-1,n-1)$-form, and put $\nu = \omega\wedge\tau$. If*

$$\langle S^n\rangle = m\nu \qquad \textit{for some } m \geq 0,$$

then

$$\left(\int_X S\wedge\tau\right)\left(\int_X \langle S^{n-1}\rangle\wedge\omega\right)\geq\frac{m}{n}\left(\int_X v\right)^2. \tag{17.24}$$

For $n=1$ the second factor is $\int_X\omega$.

Proof We prove a pointwise inequality for smooth forms, apply it to regularizations of the potentials of S on small balls in $X\setminus A$, and globalize with a partition of unity.

Step 1: a pointwise inequality. Let σ be a smooth positive definite real $(1,1)$-form. At a given point, choose linear coordinates on the tangent space that diagonalize the Hermitian matrices of ω and σ simultaneously, so that $\omega=\sum_j\gamma_j$ and $\sigma=\sum_j\lambda_j\gamma_j$ with $\lambda_j>0$, where $\gamma_j=\mathrm{i}\,\mathrm{d}z_j\wedge\mathrm{d}\bar z_j$, and put $\mathrm{d}V=\gamma_1\wedge\cdots\wedge\gamma_n$. Since τ is strictly positive (Section 7.2), the form $\tau-c\,\omega^{n-1}$ is positive at this point for some $c>0$. The product of the strongly positive form γ_j with this positive form is a nonnegative multiple of $\mathrm{d}V$ (Proposition 7.2.1), and $\gamma_j\wedge\omega^{n-1}=(n-1)!\,\mathrm{d}V$, so $\gamma_j\wedge\tau\geq c(n-1)!\,\mathrm{d}V$; thus $\gamma_j\wedge\tau=t_j\,\mathrm{d}V$ with $t_j>0$. Expanding the products, we obtain

$$\frac{\sigma\wedge\tau}{\omega\wedge\tau}=\frac{\sum_j\lambda_j t_j}{\sum_j t_j},\qquad\frac{\sigma^{n-1}\wedge\omega}{\sigma^n}=\frac{1}{n}\sum_j\lambda_j^{-1}.$$

Since $\lambda_j\sum_k\lambda_k^{-1}\geq 1$ for every j, the product of these two ratios is at least $1/n$. If σ is only semipositive, we apply this to $\sigma+s\omega$ and let $s\downarrow 0$. Thus, for every smooth semipositive real $(1,1)$-form σ, as densities with respect to any fixed smooth volume form,

$$(\sigma\wedge\tau)(\sigma^{n-1}\wedge\omega)\geq\frac{1}{n}\sigma^n v. \tag{17.25}$$

Step 2: regularization on a ball. Since the local potentials of S are locally bounded on the open set $X\setminus A$, every point of $X\setminus A$ is the center of a coordinate ball $B\Subset X\setminus A$ on which $S=\mathrm{dd^c}v$ with v plurisubharmonic and bounded. Fix such a ball, put $\eta=\mathrm{dd^c}|z|^2$ as in Section 9.7, and write $v=g\eta^n$ with g smooth and positive on B. Since v is bounded on B, the non-pluripolar powers of S on B are the Bedford–Taylor powers of v (see the remark following Definition 15.1.3); in particular $(\mathrm{dd^c}v)^n=mg\eta^n$.

Let $v_\varepsilon=v*\rho_\varepsilon$ be the regularizations of v by the radial mollifiers of Section 1.4. On each ball B' concentric with B and of smaller radius, the functions v_ε are defined for ε smaller than the difference of the radii, and by Proposition 8.2.2(2) they are smooth and plurisubharmonic on B' and decrease to v as $\varepsilon\downarrow 0$. As recalled at the beginning of Section 9.7, $(\mathrm{dd^c}v_\varepsilon)^n=\det\mathcal{H}(v_\varepsilon)\,\eta^n$. The last assertion of Proposition 9.7.3, applied on B to v and the continuous density mg, therefore gives

$$(\mathrm{dd^c}v_\varepsilon)^n=F_\varepsilon\eta^n,\qquad F_\varepsilon=\det\mathcal{H}(v_\varepsilon)\geq\big((mg)^{1/n}*\rho_\varepsilon\big)^n.$$

The right side converges to mg uniformly on compact subsets of B, by Lemma 1.4.1(1) applied to the continuous function $(mg)^{1/n}$. Hence $\sqrt{F_\varepsilon g}$ is bounded below by functions converging uniformly to $\sqrt{mg}$ on compact sets, and for every nonnegative $\chi\in C_c^\infty(B)$,

$$\varliminf_{\varepsilon\downarrow 0} \int \chi \sqrt{F_\varepsilon g}\, \eta^n \geq \sqrt{m} \int \chi \nu. \tag{17.26}$$

Moreover, for every $\chi \in C_c^\infty(B)$, as $\varepsilon \downarrow 0$ along any sequence,

$$\int \chi\, \mathrm{dd^c} v_\varepsilon \wedge \tau \longrightarrow \int \chi\, S \wedge \tau, \qquad \int \chi\, (\mathrm{dd^c} v_\varepsilon)^{n-1} \wedge \omega \longrightarrow \int \chi\, \langle S^{n-1} \rangle \wedge \omega.$$

For the first convergence, note that $\int \chi\, \mathrm{dd^c} v_\varepsilon \wedge \tau = \int v_\varepsilon\, \mathrm{dd^c}(\chi\tau)$, by the definition of $\mathrm{dd^c}$ on currents (Section 7.2); since $v_\varepsilon \to v$ in L^1 on $\operatorname{Supp}\chi$ (Lemma 1.4.1(2)), this tends to $\int v\, \mathrm{dd^c}(\chi\tau) = \int \chi\, S \wedge \tau$.

For the second convergence, choose a ball B' as above containing $\operatorname{Supp}\chi$, and a sequence $\varepsilon_j \downarrow 0$. By Theorem 9.2.5, applied on B' to the closed positive current ω of bidimension $(n-1, n-1)$ and to $n-1$ copies of the decreasing sequence (v_{ε_j}), the currents $(\mathrm{dd^c} v_{\varepsilon_j})^{n-1} \wedge \omega$ converge weakly to the Bedford–Taylor product $(\mathrm{dd^c} v)^{n-1} \wedge \omega$. By Proposition 9.1.2(1) and (4), these products are the exterior products with ω of the smooth forms $(\mathrm{dd^c} v_{\varepsilon_j})^{n-1}$ and of the current $(\mathrm{dd^c} v)^{n-1}$, which equals $\langle S^{n-1} \rangle$ on B. (For $n = 1$ both sides of the second convergence equal $\int \chi\, \omega$.)

Step 3: globalization. Let χ be a smooth function with compact support in $X \setminus A$ and $0 \leq \chi \leq 1$. Cover $\operatorname{Supp}\chi$ by finitely many balls B_l as in Step 2, and write $\chi = \sum_l \chi_l$ with nonnegative $\chi_l \in C_c^\infty(B_l)$; for instance, multiply χ by a partition of unity with compact supports subordinate to the cover of X by the balls B_l and $X \setminus \operatorname{Supp}\chi$ (Proposition 7.1.3). In each B_l we use the notation of Step 2 with an index l and put

$$a_{l,\varepsilon} = \int \chi_l\, \mathrm{dd^c} v_{l,\varepsilon} \wedge \tau, \qquad b_{l,\varepsilon} = \int \chi_l\, (\mathrm{dd^c} v_{l,\varepsilon})^{n-1} \wedge \omega.$$

Let $\alpha_{l,\varepsilon}$ and $\beta_{l,\varepsilon}$ be the densities of $\mathrm{dd^c} v_{l,\varepsilon} \wedge \tau$ and $(\mathrm{dd^c} v_{l,\varepsilon})^{n-1} \wedge \omega$ with respect to η_l^n. By (17.25) with $\sigma = \mathrm{dd^c} v_{l,\varepsilon}$, $\alpha_{l,\varepsilon}\beta_{l,\varepsilon} \geq F_{l,\varepsilon} g_l / n$, so the Cauchy–Schwarz inequality gives

$$\int \chi_l \sqrt{F_{l,\varepsilon} g_l}\, \eta_l^n \leq \sqrt{n} \int \chi_l \sqrt{\alpha_{l,\varepsilon}\beta_{l,\varepsilon}}\, \eta_l^n \leq \sqrt{n\, a_{l,\varepsilon} b_{l,\varepsilon}}.$$

Summing over l and applying the Cauchy–Schwarz inequality to the finite sum $\sum_l \sqrt{a_{l,\varepsilon} b_{l,\varepsilon}}$, we obtain

$$\left(\sum_l a_{l,\varepsilon} \right) \left(\sum_l b_{l,\varepsilon} \right) \geq \frac{1}{n} \left(\sum_l \int \chi_l \sqrt{F_{l,\varepsilon} g_l}\, \eta_l^n \right)^2.$$

Now let $\varepsilon \downarrow 0$ along a sequence. By Step 2, the two sums on the left converge to $\int_X \chi\, S \wedge \tau$ and $\int_X \chi \langle S^{n-1} \rangle \wedge \omega$. These are at most the corresponding integrals over X without χ, because $S \wedge \tau$ and $\langle S^{n-1} \rangle \wedge \omega$ are positive measures and $0 \leq \chi \leq 1$. By (17.26) and $g_l \eta_l^n = \nu$, the lower limit of the right side is at least $(m/n)(\int_X \chi \nu)^2$.

Therefore

$$\left(\int_X S\wedge\tau\right)\left(\int_X\langle S^{n-1}\rangle\wedge\omega\right)\geq\frac{m}{n}\left(\int_X\chi\nu\right)^2.$$

Finally, we remove the cutoff. Fix a metric d inducing the topology of X (Lemma 7.1.4), and for $k\geq 1$ let $K_k=\{x\in X : d(x,A)\geq 1/k\}$, with $K_k=X$ if $A=\varnothing$. These compact subsets of $X\setminus A$ increase to $X\setminus A$, because A is closed. By Proposition 7.1.3 there are smooth functions $\chi_k\colon X\to[0,1]$ with compact support in $X\setminus A$ and $\chi_k=1$ on K_k. Since ν has a smooth density and A has Lebesgue measure zero, $\nu(K_k)\to\nu(X\setminus A)=\int_X\nu$, so $\int_X\chi_k\nu\to\int_X\nu$. Applying the last inequality to $\chi=\chi_k$ and letting $k\to\infty$ proves (17.24). Both factors on its left are finite: the first is the pairing of a current with a smooth form on the compact manifold X, and the second is finite by Theorem 15.4.3. □

Theorem 17.4.20 *A pseudo-effective class $\alpha\in H^{1,1}(X,\mathbb{R})$ is big if and only if $\mathrm{vol}(\alpha)>0$. In particular, if a closed positive $(1,1)$-current T satisfies $\int_X\langle T^n\rangle>0$, then its class is big.*

Proof By Proposition 15.8.2, big classes have positive volume, and every closed positive $(1,1)$-current T satisfies $\int_X\langle T^n\rangle\leq\mathrm{vol}(\{T\})$. It therefore suffices to show that a pseudo-effective class α with $m=\mathrm{vol}(\alpha)>0$ is big, and we do so by verifying Lamari's criterion (Lemma 17.4.18). Write $\alpha=\{\theta\}$ with θ smooth and closed, choose $C>0$ such that $\theta+C\omega$ is positive definite, and put

$$C_0=\int_X(\theta+(C+1)\omega)^{n-1}\wedge\omega>0;$$

the integrand is a positive multiple of the volume form at every point, by the positivity of mixed discriminants of positive definite matrices (Lemma 7.2.2(2),(3)).

For $0<\varepsilon\leq 1$ let $\theta_\varepsilon=\theta+\varepsilon\omega$. Since α is pseudo-effective, there is $q\in\mathrm{PSH}(X,\theta)$ (Corollary 14.1.9), and $\theta_\varepsilon+\mathrm{dd^c}q\geq\varepsilon\omega$; thus $\{\theta_\varepsilon\}=\alpha+\varepsilon\{\omega\}$ is big. By Proposition 16.3.2(4), $P_{\theta_\varepsilon}[V_{\theta_\varepsilon}]=V_{\theta_\varepsilon}$, and the mass of V_{θ_ε} is

$$m_\varepsilon=\mathrm{vol}(\alpha+\varepsilon\{\omega\})\geq m>0;$$

the inequality holds because $\mathrm{vol}(\alpha+\varepsilon\{\omega\})$ decreases to $\mathrm{vol}(\alpha)$ as $\varepsilon\downarrow 0$, by Definition 15.8.1 and Proposition 15.8.2. Hence V_{θ_ε} is a model potential of mass m_ε.

Fix a smooth strictly positive real $(n-1,n-1)$-form τ with $\mathrm{dd^c}\tau=0$, normalized by $\int_X\omega\wedge\tau=1$. The measure $m_\varepsilon\omega\wedge\tau$ has a smooth density with respect to ω^n and total mass m_ε. By Corollary 17.4.17, applied in the big class $\{\theta_\varepsilon\}$ with the reference V_{θ_ε}, there are $u_\varepsilon\in\mathrm{PSH}(X,\theta_\varepsilon)$ and finite constants B_ε such that

$$S_\varepsilon=\theta_\varepsilon+\mathrm{dd^c}u_\varepsilon\geq 0,\qquad\langle S_\varepsilon^n\rangle=m_\varepsilon\omega\wedge\tau,$$
$$V_{\theta_\varepsilon}-B_\varepsilon\leq u_\varepsilon\leq V_{\theta_\varepsilon}.$$

The constants B_ε need not be bounded uniformly in ε; we only use that u_ε is locally bounded outside a closed set of measure zero. By Lemma 16.2.3, applied to the big

class $\{\theta_\varepsilon\}$, there is a closed pluripolar set $A_\varepsilon \subset X$ such that V_{θ_ε} is locally bounded on $X \setminus A_\varepsilon$; being pluripolar, A_ε has Lebesgue measure zero in every chart (see the remarks after Definition 10.1.1). Hence u_ε, which lies between $V_{\theta_\varepsilon} - B_\varepsilon$ and V_{θ_ε}, is locally bounded on $X \setminus A_\varepsilon$, and so are the local potentials of S_ε, which differ from u_ε by smooth local potentials of θ_ε (Lemma 7.3.1).

We may thus apply Lemma 17.4.19 to S_ε, with A_ε, $\nu = \omega \wedge \tau$ and m_ε in place of A, ν and m:

$$\left(\int_X S_\varepsilon \wedge \tau\right)\left(\int_X \langle S_\varepsilon^{n-1}\rangle \wedge \omega\right) \geq \frac{m_\varepsilon}{n} \geq \frac{m}{n}.$$

We bound the two factors on the left. Since $\mathrm{dd^c}\tau = 0$, the first factor equals the cohomological pairing $\int_X \theta_\varepsilon \wedge \tau = \int_X \theta \wedge \tau + \varepsilon$. If $n \geq 2$, the form $\theta_\varepsilon + C\omega$ is positive definite, so Theorem 15.4.3 gives

$$\int_X \langle S_\varepsilon^{n-1}\rangle \wedge \omega \leq \int_X (\theta + (C+\varepsilon)\omega)^{n-1} \wedge \omega \leq C_0;$$

the last inequality holds because $\theta + (C+1)\omega = (\theta + (C+\varepsilon)\omega) + (1-\varepsilon)\omega$ and products of positive $(1,1)$-forms are positive. If $n = 1$, the second factor equals $\int_X \omega = C_0$. As the first factor is nonnegative, in both cases

$$\left(\int_X \theta \wedge \tau + \varepsilon\right) C_0 \geq \frac{m}{n}.$$

Letting $\varepsilon \downarrow 0$ gives $\int_X \theta \wedge \tau \geq m/(nC_0)$. Removing the normalization of τ, we conclude that every smooth strictly positive $\mathrm{dd^c}$-closed $(n-1,n-1)$-form τ satisfies (17.21) with $\delta = m/(nC_0) > 0$. By Lemma 17.4.18, the class α is big. □

If $\{\theta\}$ is pseudo-effective but not big, the theorem shows that every θ-psh potential has zero non-pluripolar mass. This allows us to extend the contact theorems of Chapter 16 from big to pseudo-effective classes.

Corollary 17.4.21 *Let θ be a smooth closed real $(1,1)$-form, and let $f\colon X \to [-\infty,+\infty]$ be a quasi-continuous Borel function. Suppose that the family $\mathcal{A}_\theta(f)$ of Definition 16.1.3 is nonempty and uniformly bounded above. Then $u = P_\theta(f)$ satisfies*

$$\mu_u\big(X \setminus \{u = f\}\big) = 0.$$

If $\{\theta\}$ is not big, then $\mu_u = 0$. Moreover, if $r \geq 1$ and $u_1, \dots, u_r \in \mathrm{PSH}(X,\theta)$ have a rooftop $w = P_\theta(\min_{1\leq j\leq r} u_j)$ that is not identically $-\infty$, then

$$\mu_w \leq \sum_{j=1}^{r} \mathbf{1}_{\{w=u_j\}}\mu_{u_j}.$$

Proof By Proposition 16.1.4(1), $u \in \mathcal{A}_\theta(f)$; that is, $u \in \mathrm{PSH}(X,\theta)$ and $u \leq f$ quasi-everywhere. If $\{\theta\}$ is not big, then $\mu_u(X) = 0$ by Theorem 17.4.20, since otherwise $\theta + \mathrm{dd^c}u$ would be a closed positive current of positive non-pluripolar mass in $\{\theta\}$. Hence $\mu_u = 0$, and in particular the first assertion holds.

If $\{\theta\}$ is big, the first assertion is Theorem 16.2.5.

For the rooftop, the function $\min_j u_j$ is quasi-continuous, as a finite minimum of potentials (see the beginning of Section 16.1), and it is bounded above; its envelope is w. Since w is not identically $-\infty$, the family $\mathcal{A}_\theta(\min_j u_j)$ is nonempty, and it is uniformly bounded above because the obstacle is. By the first assertion, μ_w is carried by $\{w = \min_j u_j\} \subset \bigcup_j \{w = u_j\}$. The obstacle is upper semicontinuous, so $w \le u_j$ everywhere by Proposition 16.1.4(1),(2), and Lemma 16.2.2 gives $\mathbf{1}_{\{w=u_j\}}\mu_w \le \mathbf{1}_{\{w=u_j\}}\mu_{u_j}$ for every j. Summing over j proves the rooftop inequality.□

Corollary 17.4.22 *Let θ be a smooth closed real $(1,1)$-form with $\mathrm{PSH}(X,\theta) \neq \varnothing$, and let f be a continuous real function on X with bounded metric Laplacian. Then*

$$\langle(\theta + \mathrm{dd^c} P_\theta(f))^n\rangle = \mathbf{1}_{\{P_\theta(f)=f\}}(\theta + \mathrm{dd^c} f)^n,$$

where the right side is understood as in Theorem 16.2.31. In particular $\langle(\theta + \mathrm{dd^c} V_\theta)^n\rangle = \mathbf{1}_{\{V_\theta=0\}}\theta^n$. If $\{\theta\}$ is not big, both sides of these two identities vanish.

Proof The potential $V_\theta + \min_X f$ lies below f, so $\mathcal{A}_\theta(f)$ is nonempty, and it is bounded above by $\max_X f$. Put $u = P_\theta(f)$. Since f is continuous, $u \le f$ everywhere by Proposition 16.1.4(1),(2). The scalar case of Theorem 16.2.31, with all $\theta_j = \theta$, $u_j = u$ and $f_j = f$, gives

$$\mathbf{1}_{\{u=f\}}\langle\theta_u^n\rangle = \mathbf{1}_{\{u=f\}}(\theta + \mathrm{dd^c} f)^n,$$

and Corollary 17.4.21 shows that $\langle\theta_u^n\rangle$ is carried by $\{u = f\}$. This proves the first identity. For $f = 0$ the envelope $P_\theta(0)$ is V_θ, by Definition 14.2.6, which gives the second. If $\{\theta\}$ is not big, the left sides vanish by Corollary 17.4.21, and hence so do the right sides. □

17.4.5 Mixed masses and concavity

We now compare the mixed non-pluripolar mass of n closed positive currents, which may lie in different classes, with their individual masses. The idea is to replace each current by a solution of a Monge–Ampère equation with one and the same smooth right-hand side. Since singularity envelopes preserve mixed masses (Proposition 16.3.2(3)), such a replacement does not change the mixed mass, and for the solutions the pointwise mixed lower bound of Proposition 15.4.5 integrates to the desired inequality. The equations can be solved because, by Theorem 17.4.20, every current of positive mass lies in a big class.

Theorem 17.4.23 *Let X be a connected compact Kähler manifold of dimension $n \ge 1$, and let $T_1, \dots, T_n$ be closed positive $(1,1)$-currents on X. Then*

$$\int_X \langle T_1 \wedge \cdots \wedge T_n \rangle \geq \prod_{i=1}^{n} \left(\int_X \langle T_i^n \rangle \right)^{1/n}. \tag{17.27}$$

Consequently the functions

$$T \longmapsto \left(\int_X \langle T^n \rangle \right)^{1/n}, \qquad T \longmapsto \log \left(\int_X \langle T^n \rangle \right)$$

are concave on the convex cone of closed positive $(1,1)$*-currents on* X*, with the convention* $\log 0 = -\infty$ *for the second function.*

Proof All the masses involved are finite by Theorem 15.4.3. If one of the masses $\int_X \langle T_i^n \rangle$ vanishes, the right side of (17.27) is zero and there is nothing to prove. Otherwise put $m_i = \int_X \langle T_i^n \rangle > 0$. Both sides of (17.27) are positively homogeneous of degree one in each T_i, by the multilinearity of non-pluripolar products (Proposition 15.1.5), so after replacing T_i by $m_i^{-1/n} T_i$ we may assume $m_i = 1$ for all i.

By Theorem 17.4.20, the class of each T_i is big. Let θ_i be a smooth closed real $(1,1)$-form in the class of T_i (Definition 14.1.8). By Corollary 14.1.9 we can write $T_i = \theta_i + \mathrm{dd^c} u_i$ with $u_i \in \mathrm{PSH}(X, \theta_i)$. Set $\phi_i = P_{\theta_i}[u_i]$. By the mass preservation in Proposition 16.3.2(3) and by Theorem 16.4.7, ϕ_i is a model potential of mass one.

Let $\nu = \omega^n / \int_X \omega^n$. This smooth probability measure is non-pluripolar, as noted after Definition 17.4.1. In each big class $\{\theta_i\}$, Theorem 17.4.13 gives

$$w_i \in \mathcal{E}(X, \theta_i, \phi_i), \qquad S_i = \theta_i + \mathrm{dd^c} w_i \geq 0, \qquad \langle S_i^n \rangle = \nu.$$

By Proposition 16.4.10, $P_{\theta_i}[w_i] = P_{\theta_i}[\phi_i] = \phi_i$. Applying the mixed mass preservation of Proposition 16.3.2(3) first to the tuple $(w_1, \ldots, w_n)$ and then to the tuple $(u_1, \ldots, u_n)$, we obtain

$$\int_X \langle S_1 \wedge \cdots \wedge S_n \rangle = \int_X \left\langle \bigwedge_{i=1}^{n} (\theta_i + \mathrm{dd^c} \phi_i) \right\rangle = \int_X \langle T_1 \wedge \cdots \wedge T_n \rangle.$$

On the other hand, Proposition 15.4.5, with all densities equal to one, gives $\langle S_1 \wedge \cdots \wedge S_n \rangle \geq \nu$. Hence the common value of these mixed masses is at least $\nu(X) = 1$, which is the normalized inequality. Undoing the scalings proves (17.27).

For the concavity, let T, S be closed positive $(1,1)$-currents and $0 < t < 1$; for $t = 0, 1$ there is nothing to prove. Put $a = \int_X \langle T^n \rangle$ and $b = \int_X \langle S^n \rangle$. By multilinearity,

$$\int_X \langle ((1-t)T + tS)^n \rangle = \sum_{k=0}^{n} \binom{n}{k} (1-t)^k t^{n-k} \int_X \langle T^k \wedge S^{n-k} \rangle.$$

For $1 \leq k \leq n-1$, (17.27) with k factors T and $n-k$ factors S gives $\int_X \langle T^k \wedge S^{n-k} \rangle \geq a^{k/n} b^{(n-k)/n}$; the terms with $k = n$ and $k = 0$ are a and b, also when one of them vanishes. The binomial formula therefore yields

$$\int_X \langle ((1-t)T + tS)^n \rangle \geq \big((1-t)a^{1/n} + tb^{1/n}\big)^n.$$

Taking nth roots proves that the function $T \mapsto (\int_X \langle T^n \rangle)^{1/n}$ is concave. If a and b are both positive, then, since the logarithm is increasing and concave,

$$\begin{aligned} \log\Big(\int_X \langle ((1-t)T + tS)^n \rangle\Big) &\geq n \log\big((1-t)a^{1/n} + tb^{1/n}\big) \\ &\geq (1-t)\log a + t \log b. \end{aligned}$$

If one of them is zero, the right side is $-\infty$ for $0 < t < 1$, so there is nothing to prove. □

The connectedness of X cannot be omitted when $n \geq 2$. Let X be the disjoint union of two compact Kähler manifolds X', X'' of dimension $n \geq 2$, with Kähler forms ω' and ω''. Let T_1 equal ω' on X' and zero on X'', let T_2 equal zero on X' and ω'' on X'', and let $T_3, \ldots, T_n$ equal ω' on X' and ω'' on X''. Every mass $\int_X \langle T_i^n \rangle$ is positive, but $\langle T_1 \wedge \cdots \wedge T_n \rangle = 0$. The theorem applies separately on each connected component.

17.5 Singularity envelopes at arbitrary mass

Throughout this section θ is a smooth closed real $(1,1)$-form on X with big class, and we put $V = V_\theta$ and $\Omega = \int_X \omega^n$; volumes of Borel sets are taken with respect to ω^n. No model potential is fixed in this section.

For a potential u of positive mass, Theorem 16.4.7 describes $P_\theta[u]$ as the largest potential $v \leq 0$ with $u \preceq v$ and $\int_X \langle \theta_v^n \rangle = \int_X \langle \theta_u^n \rangle$, and it shows that $P_\theta[P_\theta[u]] = P_\theta[u]$. Its proof relies on the domination principle, which requires positive mass. For arbitrary u we replace the mass by the collection of all mixed masses of u against V; this leads to the singularity ceiling of Definition 17.5.2. We prove that the ceiling equals $P_\theta[u]$ and that $P_\theta[\cdot]$ is idempotent, without any assumption on the mass of u.

The main step, Proposition 17.5.1, perturbs u to $w_\varepsilon = (1-\varepsilon)u + \varepsilon V$, which has positive mass m_ε, and shows that $P_\theta[w_\varepsilon]$ decreases to $P_\theta[u]$. Since m_ε tends to zero when $\int_X \langle \theta_u^n \rangle = 0$, the estimates must not deteriorate as the mass degenerates. We therefore use a tail bound normalized by m_ε, which is uniform in ε, and convert it into a volume bound by means of constant-density solutions of Monge–Ampère equations.

Proposition 17.5.1 *Let $u \in \mathrm{PSH}(X,\theta)$ with $u \leq 0$. For $0 < \varepsilon < 1$ put*

$$w_\varepsilon = (1-\varepsilon)u + \varepsilon V, \qquad \phi_\varepsilon = P_\theta[w_\varepsilon].$$

Then ϕ_ε decreases to $P_\theta[u]$, pointwise and in $L^1(X)$, as $\varepsilon \downarrow 0$.

Proof The envelopes ϕ_ε decrease to a θ-psh limit $q \geq P_\theta[u]$; the point is the reverse inequality, which we obtain by showing that $\{q > P_\theta[u] + \delta\}$ has arbitrarily small volume for every $\delta > 0$.

Step 1: the limit exists. Put $\phi = P_\theta[u]$. Since $u \leq 0$, we have $u \leq V$, so $w_\varepsilon = u + \varepsilon(V - u)$ decreases to u as $\varepsilon \downarrow 0$ (at the poles of u all these functions equal $-\infty$). In particular $u \leq w_\varepsilon \leq 0$. By the monotonicity of $P_\theta[\cdot]$ in Proposition 16.3.2(1) and the bound $P_\theta[\cdot] \leq V$ of Definition 16.3.1, we get $\phi \leq \phi_\varepsilon \leq V$, and ϕ_ε decreases as ε decreases. Hence $q = \lim_{\varepsilon \downarrow 0} \phi_\varepsilon$ exists pointwise; as a decreasing limit of θ-psh potentials bounded below by ϕ, it is θ-psh (Corollary 14.2.5), and $\phi \leq q \leq V$. Since ϕ and V are integrable, dominated convergence gives $\phi_\varepsilon \to q$ in $L^1(X)$. It remains to prove that $q \leq \phi$.

Step 2: masses and a normalized tail bound. For $0 \leq k \leq n$ consider the finite non-pluripolar measures

$$\nu_k = \langle \theta_u^k \wedge \theta_V^{n-k} \rangle, \qquad a_k = \nu_k(X).$$

Since $\theta_{w_\varepsilon} = (1-\varepsilon)\theta_u + \varepsilon\theta_V$, multilinearity (Proposition 15.1.5) gives

$$\mu_{w_\varepsilon} = \sum_{k=0}^{n} \binom{n}{k} (1-\varepsilon)^k \varepsilon^{n-k} \nu_k,$$

$$m_\varepsilon := \int_X \langle \theta_{w_\varepsilon}^n \rangle = \sum_{k=0}^{n} \binom{n}{k} (1-\varepsilon)^k \varepsilon^{n-k} a_k.$$

Here $a_0 = \int_X \langle \theta_V^n \rangle > 0$ by Proposition 15.8.2, so $m_\varepsilon \geq \varepsilon^n a_0 > 0$. By the mass preservation in Proposition 16.3.2(3) and by Theorem 16.4.7, ϕ_ε is a model potential of mass m_ε.

In the rest of the proof, sets defined by inequalities between potentials, such as $\{u < V - A\}$, are understood pointwise, with values in $[-\infty, +\infty)$. The pole sets of all potentials involved are pluripolar, hence null for all the non-pluripolar measures below and for ω^n. For $A > 0$ put

$$r(A) = \max_{k:\, a_k > 0} \frac{\nu_k(\{u < V - A\})}{a_k}.$$

The maximum is over a finite set which contains $k = 0$. As $A \to \infty$, the sets $\{u < V - A\}$ decrease to a subset of the pole set of u, which is ν_k-null; since the measures ν_k are finite, continuity from above gives $r(A) \downarrow 0$. Moreover $w_\varepsilon - V = (1-\varepsilon)(u - V)$, so $\{w_\varepsilon < V - A\} \subset \{u < V - A\}$. Using the expansion of μ_{w_ε} and the bound $\nu_k(\{u < V - A\}) \leq r(A) a_k$, valid for every k (when $a_k = 0$ both sides vanish), we obtain the estimate

$$\frac{\mu_{w_\varepsilon}(\{w_\varepsilon < V - A\})}{m_\varepsilon} \leq r(A), \qquad 0 < \varepsilon < 1, \tag{17.28}$$

which is uniform in ε.

Step 3: rooftops. For fixed $A > 0$ put

$$p_{\varepsilon,A} = P_\theta(w_\varepsilon + A, V), \qquad p_A = P_\theta(u + A, V).$$

The potential w_ε is a common minorant of $w_\varepsilon + A$ and V, so $w_\varepsilon \le p_{\varepsilon,A}$. On the other hand $p_{\varepsilon,A} \le V \le 0$ and $p_{\varepsilon,A} \le w_\varepsilon + A$, so $p_{\varepsilon,A}$ is a competitor for $P_\theta[w_\varepsilon]$. Hence $w_\varepsilon \le p_{\varepsilon,A} \le \phi_\varepsilon$, and since w_ε and ϕ_ε both have mass m_ε, the monotonicity of mass (Theorem 15.7.4) shows that $p_{\varepsilon,A}$ has mass m_ε as well; thus $p_{\varepsilon,A} \in \mathcal{E}(X,\theta,\phi_\varepsilon)$. As $w_\varepsilon + A$ decreases to $u + A$, Proposition 16.1.6 gives $p_{\varepsilon,A} \downarrow p_A$ as $\varepsilon \downarrow 0$. Likewise $u \le p_A \le V \le 0$ and $p_A \le u + A$, so p_A is a competitor for $P_\theta[u]$, and $p_A \le \phi$.

Step 4: auxiliary solutions. Apply Theorem 17.4.5 with the model potential ϕ_ε of mass m_ε in place of ϕ, the volume form ω^n and the constant density m_ε/Ω. It gives $\chi_\varepsilon \in \mathcal{E}^1(X,\theta,\phi_\varepsilon)$ with

$$\mu_{\chi_\varepsilon} = \frac{m_\varepsilon}{\Omega}\,\omega^n, \qquad \sup_X \chi_\varepsilon = 0.$$

By Lemma 16.4.2, $\chi_\varepsilon \le \phi_\varepsilon$. By Proposition 14.2.1, normalized θ-psh potentials are uniformly bounded in L^1, so there is a constant C_1, independent of ε, with $\int_X (-\chi_\varepsilon)\,\omega^n \le C_1$.

Step 5: a volume bound uniform in ε. Fix $\delta > 0$ and $B > 1$, put $\tau = \min\{1/2, \delta/(4B)\}$, and define

$$\begin{aligned} G_{\varepsilon,B} &= \{\chi_\varepsilon > \phi_\varepsilon - B\},\\ \psi_\varepsilon &= (1-\tau)\phi_\varepsilon + \tau\chi_\varepsilon - \delta/2,\\ E_{\varepsilon,A,\delta} &= \{\phi_\varepsilon > p_{\varepsilon,A} + \delta\}. \end{aligned}$$

Since $\phi_\varepsilon \le 0$, the complement of $G_{\varepsilon,B}$ lies in $\{\chi_\varepsilon \le -B\}$, whose volume is at most C_1/B by Markov's inequality. Moreover,

$$\chi_\varepsilon - \delta/2 \le \psi_\varepsilon \le \phi_\varepsilon - \delta/2,$$

and since χ_ε and ϕ_ε have the same mass, the monotonicity of mass places ψ_ε in $\mathcal{E}(X,\theta,\phi_\varepsilon)$. On $G_{\varepsilon,B}$ we have $\psi_\varepsilon - \phi_\varepsilon = \tau(\chi_\varepsilon - \phi_\varepsilon) - \delta/2 > -\tau B - \delta/2 \ge -3\delta/4$, whereas $p_{\varepsilon,A} < \phi_\varepsilon - \delta$ on $E_{\varepsilon,A,\delta}$. Hence

$$E_{\varepsilon,A,\delta} \cap G_{\varepsilon,B} \subset \{p_{\varepsilon,A} < \psi_\varepsilon\}.$$

By multilinearity, $\mu_{\psi_\varepsilon} \ge \tau^n \mu_{\chi_\varepsilon} = \tau^n (m_\varepsilon/\Omega)\omega^n$. As $p_{\varepsilon,A}$ and ψ_ε both belong to $\mathcal{E}(X,\theta,\phi_\varepsilon)$, the comparison principle of Proposition 16.7.4 applies to them, and we obtain

$$\tau^n \frac{m_\varepsilon}{\Omega} \int_{E_{\varepsilon,A,\delta}\cap G_{\varepsilon,B}} \omega^n \le \int_{\{p_{\varepsilon,A}<\psi_\varepsilon\}} \mu_{\psi_\varepsilon} \le \int_{\{p_{\varepsilon,A}<\psi_\varepsilon\}} \mu_{p_{\varepsilon,A}}.$$

To estimate the right side, we use the contact inequality for rooftops, Corollary 16.2.6:

$$\mu_{p_{\varepsilon,A}} \le \mathbf{1}_{\{p_{\varepsilon,A}=w_\varepsilon+A\}}\mu_{w_\varepsilon} + \mathbf{1}_{\{p_{\varepsilon,A}=V\}}\mu_V.$$

The second contact set does not meet $\{p_{\varepsilon,A} < \psi_\varepsilon\}$: at a common point we would have $V < \psi_\varepsilon \le \phi_\varepsilon - \delta/2 \le V - \delta/2$. At a point of $\{p_{\varepsilon,A} < \psi_\varepsilon\}$ in the first contact set, $w_\varepsilon + A < \psi_\varepsilon \le V - \delta/2$, hence $w_\varepsilon < V - A$. Therefore, by (17.28),

$$\int_{\{p_{\varepsilon,A}<\psi_\varepsilon\}} \mu_{p_{\varepsilon,A}} \le \mu_{w_\varepsilon}(\{w_\varepsilon < V - A\}) \le m_\varepsilon r(A).$$

Dividing by the positive number $\tau^n m_\varepsilon/\Omega$ and adding the volume of the complement of $G_{\varepsilon,B}$, we arrive at

$$\int_{E_{\varepsilon,A,\delta}} \omega^n \le \frac{C_1}{B} + \Omega\tau^{-n} r(A). \tag{17.29}$$

Step 6: passage to the limit. Take a sequence $\varepsilon_j \downarrow 0$. If $q(x) > p_A(x) + \delta$, then $\phi_{\varepsilon_j}(x) \ge q(x)$ and $p_{\varepsilon_j,A}(x) \downarrow p_A(x)$, so $x \in E_{\varepsilon_j,A,\delta}$ for all large j. Thus

$$\{q > p_A + \delta\} \subset \varliminf_j E_{\varepsilon_j,A,\delta},$$

and Fatou's lemma together with (17.29) shows that the volume of $\{q > p_A + \delta\}$ is at most $C_1/B + \Omega\tau^{-n} r(A)$. Since $p_A \le \phi$, the same bound holds for the set $\{q > \phi + \delta\}$, which depends neither on A nor on B, whereas τ depends on B and δ but not on A. Letting first $A \to \infty$, so that $r(A) \to 0$, and then $B \to \infty$, we see that $\{q > \phi + \delta\}$ has zero volume. As $\delta > 0$ was arbitrary, $q \le \phi$ almost everywhere, and hence everywhere, because both are θ-psh (see the beginning of Section 16.1). Together with $\phi \le q$, this gives $q = \phi$. □

Definition 17.5.2 For $u \in \mathrm{PSH}(X,\theta)$ let $\mathcal{F}_u$ be the set of potentials $v \in \mathrm{PSH}(X,\theta)$ such that $v \le 0$, $u \preceq v$ and

$$\int_X \langle \theta_v^k \wedge \theta_V^{n-k} \rangle = \int_X \langle \theta_u^k \wedge \theta_V^{n-k} \rangle \qquad \text{for } 0 \le k \le n.$$

The *singularity ceiling* of u is the function $(\sup_{v\in\mathcal{F}_u} v)^*$.

The family $\mathcal{F}_u$ contains $u - \sup_X u$ and is bounded above by zero, so the singularity ceiling $(\sup_{v\in\mathcal{F}_u} v)^*$ is a θ-psh potential. Both the singularity ceiling of u and $P_\theta[u]$ are unchanged when a constant c is added to u. Indeed, $\theta_{u+c} = \theta_u$, and a potential v satisfies $u + c \preceq v$ if and only if $u \preceq v$; hence $\mathcal{F}_{u+c} = \mathcal{F}_u$, and $P_\theta[u + c]$ and $P_\theta[u]$ have the same competitors.

Theorem 17.5.3 *Let θ be a smooth closed real $(1,1)$-form on X with big class, and let $u \in \mathrm{PSH}(X,\theta)$. Then the singularity ceiling of u is its singularity envelope, that is,*

$$\Big(\sup_{v\in\mathcal{F}_u} v\Big)^* = P_\theta[u],$$

and $P_\theta[P_\theta[u]] = P_\theta[u]$. In particular, both identities hold when $\int_X \langle \theta_u^n \rangle = 0$.

Proof By the remark following Definition 17.5.2, both identities are unchanged when a constant is added to u, so we may assume $u \le 0$. Put $\phi = P_\theta[u]$.

We first show that ϕ lies below the singularity ceiling of u. By Definition 16.3.1, $\phi \le 0$, and $u \le \phi$ because u is itself a competitor. Applying the mixed mass preservation of Proposition 16.3.2(3) to the tuple consisting of k copies of u and $n-k$ copies of V, and using $P_\theta[V] = V$ (Proposition 16.3.2(4)), we see that ϕ has the same mixed masses against V as u. Hence $\phi \in \mathcal{F}_u$, and therefore $\phi \le (\sup_{v\in\mathcal{F}_u} v)^*$.

Conversely, let $v \in \mathcal{F}_u$, and for $0 < \varepsilon < 1$ put

$$w_\varepsilon = (1-\varepsilon)u + \varepsilon V, \qquad z_\varepsilon = (1-\varepsilon)v + \varepsilon V.$$

Expanding by multilinearity and using the mixed mass identities that define $\mathcal{F}_u$, we get

$$\int_X \langle \theta^n_{w_\varepsilon} \rangle = \int_X \langle \theta^n_{z_\varepsilon} \rangle \ge \varepsilon^n \int_X \langle \theta^n_V \rangle > 0.$$

Moreover $w_\varepsilon \preceq z_\varepsilon$, since $u \preceq v$. Let $\zeta_\varepsilon = P_\theta[z_\varepsilon]$. By Theorem 16.4.7, ζ_ε is a model potential with the same mass as z_ε, hence as w_ε. As $w_\varepsilon \preceq z_\varepsilon \le \zeta_\varepsilon$, the potential w_ε belongs to $\mathcal{E}(X,\theta,\zeta_\varepsilon)$, and Proposition 16.4.10 gives

$$P_\theta[w_\varepsilon] = P_\theta[\zeta_\varepsilon] = \zeta_\varepsilon = P_\theta[z_\varepsilon].$$

Now let $\varepsilon \downarrow 0$. Proposition 17.5.1, applied to u and to v, gives $P_\theta[u] = P_\theta[v]$. Since $v \le 0$, the potential v is itself a competitor for $P_\theta[v]$, so $v \le P_\theta[v] = \phi$. Taking the supremum over $v \in \mathcal{F}_u$ and its upper semicontinuous regularization, we obtain $(\sup_{v\in\mathcal{F}_u} v)^* \le \phi$.

For idempotence, let $\phi_\varepsilon = P_\theta[w_\varepsilon]$ be as in Proposition 17.5.1. Since $\phi \le \phi_\varepsilon$, the monotonicity of $P_\theta[\cdot]$ and Theorem 16.4.7, which applies to w_ε because $\int_X \langle \theta^n_{w_\varepsilon} \rangle > 0$, give

$$P_\theta[\phi] \le P_\theta[\phi_\varepsilon] = \phi_\varepsilon.$$

Letting $\varepsilon \downarrow 0$ and using the proposition, we obtain $P_\theta[\phi] \le \phi$. Conversely, $\phi \le 0$ is a competitor for $P_\theta[\phi]$, so $\phi \le P_\theta[\phi]$. This proves idempotence. □

17.6 Subgeodesics and Perron segments

We return to a smooth closed real $(1,1)$-form θ with big class and a model potential ϕ of mass m. Let $S = \{\zeta \in \mathbb{C} : 0 < \operatorname{Re}\zeta < 1\}$, and let $p\colon X \times S \to X$ be the projection. A family $(u_t)_{0<t<1}$ of functions on X defines the function $(x,\zeta) \mapsto u_{\operatorname{Re}\zeta}(x)$ on $X \times S$, which is invariant under imaginary translations in the sense of Section 8.5, and we ask that this function be $p^*\theta$-psh. This is a joint condition in x and ζ. It forces $t \mapsto u_t(x)$ to be convex (Lemma 17.6.3 below), but it is stronger than the convexity of these functions together with the θ-plurisubharmonicity of the slices u_t.

For instance, let θ be a Kähler form, and let g be a nonconstant smooth function with $\theta + t\mathrm{dd}^c g \geq 0$ for $0 \leq t \leq 1$ (any nonconstant smooth function multiplied by a small positive constant). The family $u_t = tg$ has θ-psh slices and affine functions $t \mapsto u_t(x)$, but $(x, \zeta) \mapsto \mathrm{Re}(\zeta) g(x)$ is not $p^*\theta$-psh near a point where $\mathrm{d}g \neq 0$. Indeed, in local coordinates $(z_1, \ldots, z_n, \zeta)$, the complex Hessian of $\mathrm{Re}(\zeta)g(z) + h(z)$, where h is a smooth local potential of θ, has vanishing $\zeta\bar{\zeta}$-entry and mixed entries $\partial^2/\partial z_j \partial \bar{\zeta} = \frac{1}{2}\partial g/\partial z_j$, which do not all vanish because g is real and $\mathrm{d}g \neq 0$. A positive semidefinite Hermitian matrix with a vanishing diagonal entry has vanishing entries in the corresponding row and column, as its 2×2 principal minors show; so this Hessian is not positive semidefinite, and the smooth function $\mathrm{Re}(\zeta)g(z) + h(z)$ is not plurisubharmonic on any neighborhood of the point, by Proposition 8.2.2(1).

Definition 17.6.1 Let $u_0, u_1 \in \mathrm{PSH}(X, \theta)$. A *subgeodesic* with endpoints u_0, u_1 is a family $(u_t)_{0<t<1}$ of functions on X whose *complexification*

$$U(x, \zeta) = u_{\mathrm{Re}\,\zeta}(x)$$

is $p^*\theta$-psh on $X \times S$ and such that $u_t \to u_0$ and $u_t \to u_1$ in $L^1(X)$ as $t \to 0$ and $t \to 1$, respectively.

Subgeodesics on an interval $[a, b]$ with $a < b$ are defined in the same way, after the affine change of parameter $t \mapsto a + (b - a)t$; a subgeodesic on a one-point interval consists of its given potential. The endpoints of a subgeodesic are uniquely determined, because a class in $L^1(X)$ has at most one θ-psh representative. The holomorphic covering $z = \mathrm{e}^{-\zeta}$ maps S onto the annulus $\{\mathrm{e}^{-1} < |z| < 1\}$, and U descends to the function $(x, z) \mapsto u_{-\log|z|}(x)$ on $X \times \{\mathrm{e}^{-1} < |z| < 1\}$, because U is invariant under imaginary translations. The map $(x, \zeta) \mapsto (x, \mathrm{e}^{-\zeta})$ is a local biholomorphism, and plurisubharmonicity is a local property invariant under biholomorphisms (Proposition 8.2.3). Hence U is $p^*\theta$-psh if and only if this annulus complexification is psh relative to the pullback of θ; we use both descriptions.

We need some elementary facts about convex functions of one real variable, allowing the value $-\infty$.

Lemma 17.6.2 *Let $I \subset \mathbb{R}$ be an open interval, and let $g\colon I \to [-\infty, \infty)$ be convex, that is, $g((1-\lambda)a + \lambda b) \leq (1-\lambda)g(a) + \lambda g(b)$ for $a, b \in I$ and $0 < \lambda < 1$.*

(1) *Either $g \equiv -\infty$, or g is finite and continuous on I.*
(2) *Suppose that g is finite, and for $a < b$ in I put $\sigma(a, b) = (g(b) - g(a))/(b - a)$. Then $\sigma(a, b)$ is nondecreasing in a and in b. For every $t \in I$ there is $c \in \mathbb{R}$ with $g(s) \geq g(t) + c(s - t)$ for all $s \in I$. Consequently, g is the pointwise maximum of the affine functions that it dominates, and it is bounded below on every bounded subinterval of I.*
(3) *Let $I = (0, 1)$, and suppose that g is finite and bounded above on $(0, \frac{1}{2})$. Then $\lim_{t \downarrow 0} g(t)$ exists and is finite. Moreover, for fixed $s \in (0, 1)$, the value $g(s) - s\sigma(t, s)$ at 0 of the affine function that agrees with g at t and s increases as $t \in (0, s)$ decreases, and it tends to $\lim_{t \downarrow 0} g(t)$ as $t \downarrow 0$.*

(4) *Let* $I = (0,1)$*, suppose that* $|g| \le C$ *on* I*, and let* $0 < \delta < \frac{1}{2}$*. Then* $|g(b) - g(a)| \le (2C/\delta)|b-a|$ *for all* $a, b \in [\delta, 1-\delta]$.

Proof (1) Suppose that $g(t_0) = -\infty$ for some $t_0 \in I$, and let $t \in I \setminus \{t_0\}$. Choose $s \in I$ such that t lies strictly between t_0 and s, and write $t = (1-\lambda)t_0 + \lambda s$ with $0 < \lambda < 1$. Since $g(s) < \infty$, convexity gives $g(t) \le (1-\lambda)g(t_0) + \lambda g(s) = -\infty$. Hence either $g \equiv -\infty$ or g is finite, and a finite convex function on I is continuous by Example 3.2.4.

(2) For $a < b < c$ in I we have $\sigma(a,b) \le \sigma(a,c) \le \sigma(b,c)$, as shown in Example 3.2.4; this gives the monotonicity of σ. Now let $t \in I$. For $r < t < r'$ in I we have $\sigma(r,t) \le \sigma(t,r')$, so $c = \sup_{r<t} \sigma(r,t)$ is finite and $c \le \sigma(t,r')$ for every $r' > t$. These two inequalities say that $g(s) \ge g(t) + c(s-t)$ for $s < t$ and for $s > t$. The affine function $s \mapsto g(t) + c(s-t)$ is therefore dominated by g and agrees with g at t, which proves the remaining assertions.

(3) Fix $s \in (0,1)$. For $0 < t < s$ we have $g(t) = g(s) - (s-t)\sigma(t,s)$. By (2), $\sigma(t,s)$ decreases as t decreases, so it has a limit $\sigma_0 \in [-\infty, \infty)$ as $t \downarrow 0$. If $\sigma_0 = -\infty$, then $g(t) \to +\infty$ as $t \downarrow 0$, which contradicts the upper bound. Hence σ_0 is finite and $g(t) \to g(s) - s\sigma_0$. The value at 0 of the affine function that agrees with g at t and s is $g(s) - s\sigma(t,s)$; it increases as t decreases and tends to $g(s) - s\sigma_0$.

(4) Let $\delta \le a < b \le 1-\delta$. For $b < r < 1$, (2) gives $\sigma(a,b) \le \sigma(b,r) \le 2C/(r-b)$, and letting $r \uparrow 1$ we get $\sigma(a,b) \le 2C/(1-b) \le 2C/\delta$. Similarly, for $0 < r < a$ we have $\sigma(a,b) \ge \sigma(r,a) \ge -2C/(a-r)$, and letting $r \downarrow 0$ we get $\sigma(a,b) \ge -2C/a \ge -2C/\delta$. □

Lemma 17.6.3 *Let* $(v_t)_{0<t<1}$ *be a family of functions on* X *whose complexification* $V(x,\zeta) = v_{\mathrm{Re}\,\zeta}(x)$ *is* $p^*\theta$*-psh on* $X \times S$*. Then:*

(1) *for every* $x \in X$*, the function* $t \mapsto v_t(x)$ *is convex on* $(0,1)$*, and it is either identically* $-\infty$ *or finite and continuous;*
(2) *for almost every* $x \in X$*, this function is finite;*
(3) *every slice* v_t *belongs to* $\mathrm{PSH}(X,\theta)$.

Proof Let $D \subset X$ be a connected coordinate chart, identified with a domain in $\mathbb{C}^n$, on which $\theta = \mathrm{dd}^c h$ for a smooth function h; by Lemma 13.1.2, such charts, for instance small coordinate balls, cover X. Then $V + h \circ p$ is plurisubharmonic on the domain $D \times S \subset \mathbb{C}^{n+1}$.

(1) Fix $x \in D$. The complex line $\{x\} \times \mathbb{C}$ meets $D \times S$ in $\{x\} \times S$, so by Definition 8.1.1 the function $\zeta \mapsto V(x,\zeta) + h(x)$ is subharmonic on S, possibly identically $-\infty$; hence so is $w(\zeta) = g(\mathrm{Re}\,\zeta)$, where $g(t) = v_t(x)$. If $w \equiv -\infty$, then $g \equiv -\infty$ is convex. Otherwise w is locally integrable on S by Proposition 3.1.5, and $\Delta w \ge 0$ by Theorem 3.2.2. For $0 < \delta < \frac{1}{2}$, the mollification $w_\delta = w * \rho_\delta$ is smooth on $S_\delta = \{\delta < \mathrm{Re}\,\zeta < 1-\delta\}$, and $\Delta w_\delta = (\Delta w)_\delta \ge 0$ there, by Proposition 1.4.3(1),(3). Since w is invariant under imaginary translations, so is w_δ; thus $w_\delta(\zeta) = g_\delta(\mathrm{Re}\,\zeta)$ for the smooth function $g_\delta(t) = w_\delta(t)$ on $(\delta, 1-\delta)$, and $g_\delta'' = \Delta w_\delta \ge 0$, so g_δ is convex by the mean value theorem. By (3.3), w_δ decreases to w as $\delta \downarrow 0$, so g_δ decreases to g on $(0,1)$. For $a, b \in (0,1)$ and $0 < \lambda < 1$, letting $\delta \downarrow 0$ in

$$g((1-\lambda)a+\lambda b) \le g_\delta((1-\lambda)a+\lambda b) \le (1-\lambda)g_\delta(a)+\lambda g_\delta(b)$$

shows that g is convex. The rest of (1) is Lemma 17.6.2(1).

(2) The plurisubharmonic function $V + h \circ p$ is locally integrable on $D \times S$ (Theorem 8.2.1). Let $K \subset D$ be compact and $[a,b] \subset (0,1)$. Since V does not depend on $\operatorname{Im}\zeta$, Tonelli's theorem on $K \times \{\zeta : a \le \operatorname{Re}\zeta \le b,\ 0 \le \operatorname{Im}\zeta \le 1\}$ gives

$$\int_K \int_a^b |v_t(x) + h(x)|\, \mathrm{d}t\, \mathrm{d}V_{2n}(x) < \infty.$$

Hence, for almost every $x \in K$, the value $v_t(x)$ is finite for almost every $t \in [a,b]$, and by (1) the function $t \mapsto v_t(x)$ is then finite on $(0,1)$. Since X is covered by countably many such compact sets K, this proves (2).

(3) Fix $t \in (0,1)$. The restriction of $V + h \circ p$ to $D \times \{t\}$ is the function $v_t + h$ on D. It is upper semicontinuous, and, by Definition 8.1.1 applied to the complex lines contained in $\mathbb{C}^n \times \{t\}$, its restriction to every connected component of the intersection of D with a complex line is subharmonic or identically $-\infty$. By (2), v_t is finite almost everywhere on D, so $v_t + h$ is not identically $-\infty$, and $v_t + h \in \mathrm{PSH}(D)$. Since such charts cover X, $v_t \in \mathrm{PSH}(X,\theta)$. □

Proposition 17.6.4 *Let (u_t) be a subgeodesic with endpoints u_0, u_1. Then for each $x \in X$ the function $t \mapsto u_t(x)$ is convex on $(0,1)$, possibly identically $-\infty$, and*

$$u_t \le (1-t)u_0 + tu_1, \qquad 0 < t < 1.$$

The infimum $w = \inf_{0<t<1} u_t$ is a θ-psh potential and satisfies $w \le P_\theta(u_0,u_1)$. The map $t \mapsto u_t$ is continuous from $[0,1]$ to $L^1(X)$, and its image is compact. For $j = 0, 1$, the limits $\lim_{t\to j} u_t(x)$ exist at every x, are at most $u_j(x)$, and equal $u_j(x)$ quasi-everywhere.

Proof By Lemma 17.6.3, for every $x \in X$ the function $t \mapsto u_t(x)$ is convex on $(0,1)$, and either identically $-\infty$ or finite and continuous; it is finite for almost every x; and every slice u_t is θ-psh. Since a sequence converging in $L^1(X)$ has a subsequence converging almost everywhere, we can choose sequences $t_j \downarrow 0$ and $s_j \uparrow 1$ with $u_{t_j} \to u_0$ and $u_{s_j} \to u_1$ almost everywhere. For fixed t and large j, convexity on $[t_j, s_j]$ gives

$$u_t \le \frac{s_j - t}{s_j - t_j} u_{t_j} + \frac{t - t_j}{s_j - t_j} u_{s_j},$$

and letting $j \to \infty$ gives $u_t \le (1-t)u_0 + tu_1$ almost everywhere. Both sides are θ-psh, so the inequality holds everywhere.

Let D be a connected coordinate chart of X on which $\theta = \mathrm{dd}^{\mathrm{c}} h$ with h smooth. The function $U + h \circ p$ is plurisubharmonic on $D \times S = D \times ((0,1) + \mathrm{i}\mathbb{R})$ and invariant under imaginary translations, so Corollary 8.5.2, with $V = (0,1)$, shows that $w + h = \inf_{0<t<1}(u_t + h)$ is plurisubharmonic or identically $-\infty$ on D. For almost every x, the function $t \mapsto u_t(x)$ is finite and convex, hence bounded below on $(0,1)$ by Lemma 17.6.2(2). Thus w is finite almost everywhere, the alternative $-\infty$ is

excluded on every chart, and w is θ-psh. Since $w \le u_t$ for all t and $u_t \to u_0$, $u_t \to u_1$ in $L^1(X)$, we get $w \le u_0$ and $w \le u_1$ almost everywhere, hence everywhere. Thus w is a competitor for the obstacle $\min\{u_0, u_1\}$, and $w \le P_\theta(u_0, u_1)$.

The bounds $w \le u_t \le \max\{u_0, u_1\}$ hold for all t, with integrable outer functions. For almost every x the function $t \mapsto u_t(x)$ is finite and continuous on $(0,1)$, so dominated convergence gives continuity of $t \mapsto u_t$ from $(0,1)$ to $L^1(X)$; continuity at the endpoints is part of the definition. The image of $[0,1]$ under a continuous map is compact. Now fix $x \in X$. If $t \mapsto u_t(x)$ is identically $-\infty$, its limits at 0 and 1 are $-\infty$. Otherwise it is finite and convex and bounded above by $\max\{u_0(x), u_1(x)\} < \infty$, so by Lemma 17.6.2(3), applied to $t \mapsto u_t(x)$ and to $t \mapsto u_{1-t}(x)$, it has finite limits at 0 and 1. By the displayed bound these limits are at most $u_0(x)$ and $u_1(x)$. Since the pointwise limits exist everywhere and $u_{t_j} \to u_0$, $u_{s_j} \to u_1$ almost everywhere, they equal u_0 and u_1 almost everywhere.

It remains to prove equality quasi-everywhere; we treat the endpoint 0, and the endpoint 1 follows by reversing the parameter. Fix $s \in (0,1)$ and a sequence $t_j \downarrow 0$ with $t_j < s/2$. Consider the value at $t = 0$ of the chord of $t \mapsto u_t$ through the parameters t_j and s:

$$a_j = \frac{u_{t_j} - (t_j/s)u_s}{1 - t_j/s}.$$

Where u_s is finite, the function $t \mapsto u_t(x)$ is finite and convex, so by Lemma 17.6.2(3) a_j increases with j and tends to $\lim_{t\to 0} u_t$; this limit equals u_0 almost everywhere and is at most u_0 everywhere. The functions

$$a_j + u_s = \frac{u_{t_j} + (1 - 2t_j/s)u_s}{1 - t_j/s}$$

are combinations of u_{t_j} and u_s with positive coefficients of sum two, hence 2θ-psh, since $2\theta + \mathrm{dd^c}(\alpha u_{t_j} + \beta u_s) = \alpha\theta_{u_{t_j}} + \beta\theta_{u_s}$ for $\alpha, \beta \ge 0$ with $\alpha + \beta = 2$. They increase with j and are bounded above by $u_0 + u_s$. Their pointwise limit is $\lim_{t\to 0} u_t + u_s$, which equals $u_0 + u_s$ almost everywhere; hence their regularized upper envelope, a 2θ-psh function by Theorem 8.3.2 which agrees almost everywhere with the pointwise limit, is $u_0 + u_s$ itself. By Corollary 10.2.2(2), applied in charts after adding smooth local potentials of 2θ, the pointwise limit $\lim_{t\to 0} u_t + u_s$ equals $u_0 + u_s$ quasi-everywhere. Removing the pole set of u_s proves the assertion at 0. □

Lemma 17.6.5 *Subgeodesics are preserved by restriction to a subinterval, by addition of an affine function of t, and by a change of background $\theta' = \theta + \mathrm{dd^c} g$, $u'_t = u_t - g$, with g smooth. If a subgeodesic on $[a,b] \subset [0,1]$ dominates a given subgeodesic on $[0,1]$ there and has the same values at a and b, then replacing the given subgeodesic on $[a,b]$ by it gives a subgeodesic on $[0,1]$.*

Proof Let $0 \le a < b \le 1$. The restriction of (u_t) to $[a,b]$, reparametrized as $t \mapsto u_{a+(b-a)t}$, has the complexification $U \circ F$, where $F(x,\zeta) = (x, a + (b-a)\zeta)$ is a holomorphic map of $X \times S$ into itself with $p \circ F = p$. In a chart D of X on which $\theta = \mathrm{dd^c} h$, we have $(U + h \circ p) \circ F = U \circ F + h \circ p$, which is plurisubharmonic

on $D \times S$ by Proposition 8.2.3; it is not identically $-\infty$, because its slices are the functions $u_{a+(b-a)t} + h$, which are not identically $-\infty$ by Lemma 17.6.3. Hence $U \circ F$ is $p^*\theta$-psh, and the restriction is a subgeodesic with endpoint potentials u_a and u_b, by the L^1 continuity of Proposition 17.6.4. An affine function $c_0 + c_1 \operatorname{Re}\zeta = \operatorname{Re}(c_0 + c_1\zeta)$ is pluriharmonic, so adding it preserves $p^*\theta$-plurisubharmonicity, and the change of background replaces U by $U - g \circ p$ and $p^*\theta$ by $p^*\theta + \mathrm{dd}^{\mathrm{c}}(g \circ p)$; this proves the first assertion.

For the replacement, let U be the complexification of the given subgeodesic, and V that of the new one on $W = X \times \{a < \operatorname{Re}\zeta < b\}$. We verify the boundary condition of Proposition 8.2.5 along each seam $\operatorname{Re}\zeta = a$ or $\operatorname{Re}\zeta = b$ lying in S; consider $a > 0$. Let $(x_k, \zeta_k) \in W$ converge to (x, ζ) with $\operatorname{Re}\zeta = a$, and put $t_k = \operatorname{Re}\zeta_k$. Denote the slices of the new subgeodesic by v_t, $a < t < b$; they converge to u_a in $L^1(X)$ as $t \downarrow a$.

We show that $\overline{\lim}_k V(x_k, \zeta_k) \le U(x, \zeta) = u_a(x)$. Since $v_{t_k} \to u_a$ in $L^1(X)$, Lemma 17.3.2 shows that the tail envelopes $w_l = (\sup_{k \ge l} v_{t_k})^*$ are θ-psh and decrease pointwise to u_a. For $k \ge l$ we have $V(x_k, \zeta_k) = v_{t_k}(x_k) \le w_l(x_k)$, and w_l is upper semicontinuous, so $\overline{\lim}_k V(x_k, \zeta_k) \le w_l(x)$ for every l. Letting $l \to \infty$ proves the claim. A seam $\operatorname{Re}\zeta = b$ with $b < 1$ is treated in the same way, with $t_k \uparrow b$. Hence, by Proposition 8.2.5 applied in charts after adding smooth local potentials of $p^*\theta$, the function equal to $\max\{U, V\} = V$ on W and to U elsewhere is $p^*\theta$-psh. Its endpoint limits at $t = 0$ and $t = 1$ are those of the given subgeodesic, or those of the new one if $a = 0$ or $b = 1$; in either case they are u_0 and u_1. □

Lemma 17.6.6 *Let $\{(u_t^\alpha)\}_{\alpha \in A}$ be a nonempty family of subgeodesics whose endpoint potentials u_0^α, u_1^α are uniformly bounded above. Then the regularized upper envelope U of their complexifications has slices*

$$U_t = \Big(\sup_\alpha u_t^\alpha\Big)^*, \qquad 0 < t < 1,$$

where the regularization on the right is taken on X. Moreover (U_t) is a subgeodesic whose endpoint potentials are the regularized upper envelopes $v_0 = (\sup_\alpha u_0^\alpha)^*$ and $v_1 = (\sup_\alpha u_1^\alpha)^*$ of the endpoints.

Proof Denote by U^α the complexification of (u_t^α). By Proposition 17.6.4, $u_t^\alpha \le (1-t)u_0^\alpha + t u_1^\alpha$, so the functions U^α are uniformly bounded above. Their regularized upper envelope U is therefore $p^*\theta$-psh (Theorem 8.3.2, applied in charts after adding smooth local potentials of $p^*\theta$), and it is invariant under imaginary translations, because the family is and these translations are homeomorphisms. Thus U is the complexification of the family of its slices U_t, which by Lemma 17.6.3 are θ-psh and have convex fibers $t \mapsto U_t(x)$.

We next reduce to a countable subfamily. Cover X by finitely many coordinate balls D_β on which θ has smooth potentials h_β (Lemma 13.1.2). By Theorem 8.3.2, applied to the plurisubharmonic functions $U^\alpha + h_\beta \circ p$ on $D_\beta \times S$ and to the functions $u_0^\alpha + h_\beta$ and $u_1^\alpha + h_\beta$ on D_β, there are countable sets $J_\beta \subset A$ such that, on $D_\beta \times S$ and on D_β respectively, the regularized suprema over $\alpha \in J_\beta$ of U^α, of u_0^α and of u_1^α equal U, v_0 and v_1. The countable set $J = \bigcup_\beta J_\beta$ then satisfies $(\sup_{\alpha \in J} U^\alpha)^* = U$,

$(\sup_{\alpha\in J} u_0^\alpha)^* = v_0$ and $(\sup_{\alpha\in J} u_1^\alpha)^* = v_1$, since on each $D_\beta \times S$, respectively D_β, these regularized suprema lie between the regularized suprema over J_β and over A.

Let $v_t = (\sup_\alpha u_t^\alpha)^*$; we show that $U_t = v_t$. Since U_t is upper semicontinuous on X and dominates every u_t^α, we have $v_t \le U_t$. For $a < t < b$, the inequality $u_t^\alpha \le \frac{b-t}{b-a}u_a^\alpha + \frac{t-a}{b-a}u_b^\alpha \le \frac{b-t}{b-a}v_a + \frac{t-a}{b-a}v_b$ and upper semicontinuity of the right side give $v_t \le \frac{b-t}{b-a}v_a + \frac{t-a}{b-a}v_b$. Thus $t \mapsto v_t(x)$ is convex for every x. In the same way, the bound $u_t^\alpha \le (1-t)u_0^\alpha + tu_1^\alpha$ gives $v_t \le (1-t)v_0 + tv_1$. Fix one member α_0 of the family and let $w_0 = \inf_t u_t^{\alpha_0}$, a θ-psh potential by Proposition 17.6.4. Since $w_0 \le v_t \le U_t$ and U is bounded above, all $v_t(x)$ and $U_t(x)$ are finite on the set where w_0 is finite, which has full measure.

The function $F_J = \sup_{\alpha\in J} U^\alpha$ is Borel, being a countable supremum of upper semicontinuous functions, and by Theorem 8.3.2 it agrees with its regularization U outside a Lebesgue null set. Both F_J and U are invariant under imaginary translations, so this exceptional set is $\{(x,\zeta) : (x, \operatorname{Re}\zeta) \in N\}$ for a Borel set $N \subset X\times(0,1)$, which is null by Fubini's theorem. Applying Fubini's theorem to N, we find that for almost every x we have $U_t(x) = \sup_{\alpha\in J} u_t^\alpha(x) \le v_t(x)$, hence $U_t(x) = v_t(x)$, for almost every $t \in (0,1)$. If moreover $w_0(x)$ is finite, both sides are finite convex functions of t, hence continuous (Lemma 17.6.2(1)), so they agree for every $t \in (0,1)$. Thus, for each t, $U_t = v_t$ almost everywhere, and both are θ-psh, so $U_t = v_t$ everywhere.

For the endpoints, note first that, since $U_t = v_t \le (1-t)v_0 + tv_1 \le \max\{v_0, v_1\}$, Lemma 17.6.2(1),(3) show that $\lim_{t\to 0} U_t(x)$ exists at every x, as an element of $[-\infty,\infty)$. For $\alpha \in J$ we have $U_t \ge u_t^\alpha$, so by Proposition 17.6.4 $\lim_{t\to 0} U_t \ge \lim_{t\to 0} u_t^\alpha = u_0^\alpha$ outside a pluripolar set. Outside a countable union of pluripolar sets this holds for all $\alpha \in J$, so $\lim_{t\to 0} U_t \ge \sup_{\alpha\in J} u_0^\alpha = v_0$ quasi-everywhere, by Corollary 10.2.2(1); countable unions of pluripolar sets are pluripolar by Theorem 10.1.4. The bound $U_t = v_t \le (1-t)v_0 + tv_1$ gives the reverse inequality. Hence $U_t \to v_0$ almost everywhere as $t \to 0$. Since $w_0 \le U_t \le \max\{v_0, v_1\}$, dominated convergence gives $U_t \to v_0$ in $L^1(X)$. The endpoint $t = 1$ is treated in the same way, after reversing the parameter. □

Definition 17.6.7 Let $u_0, u_1 \in \mathrm{PSH}(X,\theta)$. The subgeodesics whose endpoint potentials are at most u_0 and u_1, respectively, are called *competitors*. The *Perron segment* between u_0 and u_1 is the regularized upper envelope of the complexifications of all competitors; it is $-\infty$ if there are no competitors. When its endpoint potentials are u_0 and u_1, it is called the *weak geodesic* between u_0 and u_1. We write its slices as $G_t(u_0,u_1)$, $0 < t < 1$, and, for a weak geodesic, put $G_0(u_0,u_1) = u_0$ and $G_1(u_0,u_1) = u_1$.

By Lemma 17.6.6, a Perron segment which is not $-\infty$ is a subgeodesic, and its slices are the regularized upper envelopes of the slices of the competitors.

Theorem 17.6.8 *If $u_0, u_1 \in \mathcal{E}(X,\theta,\phi)$, then the Perron segment between them is the weak geodesic between u_0 and u_1, and*

$$P_\theta(u_0,u_1) \le G_t(u_0,u_1) \le (1-t)u_0 + tu_1, \qquad 0 < t < 1.$$

Every slice belongs to $\mathcal{E}(X,\theta,\phi)$. If $u_0,u_1\in\mathcal{E}^1(X,\theta,\phi)$, every slice belongs to $\mathcal{E}^1(X,\theta,\phi)$, and if $u_0,u_1\in\mathcal{E}^\infty(X,\theta,\phi)$, every slice belongs to $\mathcal{E}^\infty(X,\theta,\phi)$. The restriction of the weak geodesic to a subinterval $[a,b]\subset[0,1]$ is the weak geodesic between $G_a(u_0,u_1)$ and $G_b(u_0,u_1)$.

Proof The rooftop $w=P_\theta(u_0,u_1)$ belongs to $\mathcal{E}(X,\theta,\phi)$, by Proposition 17.3.6. The constant curve equal to w is a competitor. By Lemma 17.6.6, the Perron segment is therefore a subgeodesic, with endpoint potentials $a_0\le u_0$ and $a_1\le u_1$, and $w\le G_t\le(1-t)a_0+ta_1\le(1-t)u_0+tu_1$. Each slice is at most $\phi+C$ for some constant C and at least w, so mass monotonicity puts it in $\mathcal{E}(X,\theta,\phi)$.

We show that $a_0=u_0$. For $k\ge1$ the rooftop $q_k=P_\theta(u_0,u_1+k)$ is not identically $-\infty$, since it dominates w, and the affine curve q_k-kt is a competitor: its value at $t=0$ is at most u_0, and its value at $t=1$ is $q_k-k\le u_1$. Hence $G_t\ge q_k-kt$ for $0<t<1$, and letting $t\to0$ in $L^1(X)$ gives $a_0\ge q_k$ almost everywhere, hence everywhere, for every k. By the rooftop representation of Proposition 16.7.2 and Corollary 17.3.7,

$$\Big(\sup_k q_k\Big)^* = P_\theta[u_1](u_0)=u_0.$$

Since a_0 is upper semicontinuous and dominates every q_k, it dominates $(\sup_k q_k)^*=u_0$; hence $a_0=u_0$. Exchanging the roles of the endpoints, with the competitors $P_\theta(u_0+k,u_1)-k(1-t)$, gives $a_1=u_1$.

Let $[a,b]\subset[0,1]$. The restriction of the weak geodesic to $[a,b]$ is a subgeodesic between the slices $G_a,G_b\in\mathcal{E}(X,\theta,\phi)$. By the part already proved, the weak geodesic between G_a and G_b exists and dominates this restriction. By Lemma 17.6.5, replacing the restriction by this weak geodesic gives a subgeodesic on $[0,1]$ with endpoints u_0,u_1. It is a competitor for the original Perron segment, so its slices are at most G_t; hence the two coincide on $[a,b]$.

Finally, if $u_0,u_1\in\mathcal{E}^1(X,\theta,\phi)$, then $w\in\mathcal{E}^1(X,\theta,\phi)$ by Proposition 17.3.6, and $w\le G_t\preceq\phi$ gives $G_t\in\mathcal{E}^1(X,\theta,\phi)$ by Corollary 17.2.10. If $u_0,u_1\in\mathcal{E}^\infty(X,\theta,\phi)$, then $\phi-C$ is a common minorant for some C, so $G_t\ge w\ge\phi-C$, while the convex upper bound gives $G_t\le\phi+C$. □

Proposition 17.6.9 *Let $u_0,u_1\in\mathcal{E}(X,\theta,\phi)$. The weak geodesics of Theorem 17.6.8 have the following properties.*

(1) *They increase with either endpoint: if $u_0',u_1'\in\mathcal{E}(X,\theta,\phi)$ satisfy $u_0\le u_0'$ and $u_1\le u_1'$, then $G_t(u_0,u_1)\le G_t(u_0',u_1')$.*
(2) *For constants c_0,c_1, $G_t(u_0+c_0,u_1+c_1)=G_t(u_0,u_1)+(1-t)c_0+tc_1$, and $G_t(u_1,u_0)=G_{1-t}(u_0,u_1)$.*
(3) *If $u_0^k,u_1^k\in\mathcal{E}(X,\theta,\phi)$ decrease to u_0,u_1 as $k\to\infty$, then $G_t(u_0^k,u_1^k)$ decreases to $G_t(u_0,u_1)$ for every t.*
(4) *If $u_1\le u_0$, the slices $G_t(u_0,u_1)$ decrease with $t\in[0,1]$.*
(5) *If $|u_1-u_0|\le C$ quasi-everywhere, then on the common finite locus*

$$|G_t(u_0,u_1)-u_0|\le Ct,\qquad |G_t(u_0,u_1)-u_1|\le C(1-t).$$

(6) *For a constant c, $G_t(u_0, u_0 + c) = u_0 + tc$.*

Proof Assertion (1) holds because enlarging the endpoints enlarges the family of competitors, and (2) because adding the affine function $(1-t)c_0 + tc_1$, respectively reversing the parameter, is a bijection between the families of competitors.

For (3), the geodesics $G^k = G(u_0^k, u_1^k)$ decrease with k by (1) and dominate $G = G(u_0, u_1)$. Their complexifications decrease to a function which dominates the complexification of G and so is not identically $-\infty$ on any chart; by Theorem 8.3.1, applied in charts after adding smooth local potentials of $p^*\theta$, it is $p^*\theta$-psh. It is invariant under imaginary translations, and its slices $\widetilde{G}_t$ are the pointwise limits of G_t^k. For every k we have $G_t \le \widetilde{G}_t \le G_t^k \le (1-t)u_0^k + tu_1^k$, so $G_t \le \widetilde{G}_t \le (1-t)u_0 + tu_1$. As $t \to 0$, both outer functions converge to u_0 in $L^1(X)$, hence so does $\widetilde{G}_t$; similarly at $t = 1$. Thus $(\widetilde{G}_t)$ is a competitor for the Perron segment between u_0 and u_1, so $\widetilde{G} \le G$, and therefore $\widetilde{G} = G$.

For (4), let $u_1 \le u_0$ and $0 < r < s < 1$, and put $\delta = (s-r)/(1-r)$. The reparametrized curve $t \mapsto G_{\delta+(1-\delta)t}$ is a subgeodesic by Lemma 17.6.5. Its value at $t = 0$ is $G_\delta \le (1-\delta)u_0 + \delta u_1 \le u_0$, and its value at $t = 1$ is u_1. It is therefore a competitor for the segment between u_0 and u_1, and at $t = r$, where $\delta + (1-\delta)r = s$, it gives $G_s \le G_r$. For the endpoints, $G_t \le (1-t)u_0 + tu_1 \le u_0$, and the constant curve u_1 is a competitor, so $G_t \ge u_1$.

For (5), note that $u_1 \le u_0 + C$ and $u_0 \le u_1 + C$ everywhere, because inequalities between quasi-psh functions that hold quasi-everywhere hold everywhere (Section 16.1). Hence the affine curves $u_0 - Ct$ and $u_1 - C(1-t)$ are competitors, so that $G_t \ge \max\{u_0 - Ct, u_1 - C(1-t)\}$. On the other hand $G_t - u_0 \le t(u_1 - u_0) \le Ct$ and $G_t - u_1 \le (1-t)(u_0 - u_1) \le C(1-t)$ by the upper bound of Theorem 17.6.8. For (6), the upper bound $u_0 + tc$ is itself an affine competitor. □

Proposition 17.6.10 *For $u_0, u_1 \in \mathcal{E}(X, \theta, \phi)$ and $0 < t < 1$,*

$$G_t(u_0, u_1) = \left(\sup_{s\in\mathbb{R}}\{P_\theta(u_0, u_1 - s) + ts\}\right)^* .$$

The regularization may equally be taken on $X \times S$: the slices of the regularized supremum of the complexifications are the functions displayed here.

Proof For each s the potential $u_1 - s$ belongs to $\mathcal{E}(X, \theta, \phi)$, so the rooftop $P_\theta(u_0, u_1 - s)$ is not identically $-\infty$, by Proposition 17.3.6, and the affine curve $P_\theta(u_0, u_1 - s) + ts$ is a competitor: its value at $t = 0$ is at most u_0 and its value at $t = 1$ is at most u_1. Hence the right side is at most G_t.

Conversely, let (v_t) be a competitor and $s \in \mathbb{R}$. By Corollary 8.5.2, applied with the affine function $\ell(t) = ts$ in charts after adding smooth local potentials of $p^*\theta$, the function $q_s = \inf_{0<t<1}(v_t - ts)$ is θ-psh or identically $-\infty$ on each chart. By Lemma 17.6.3, for almost every x the function $t \mapsto v_t(x)$ is finite and convex, hence bounded below on $(0, 1)$ by an affine function (Lemma 17.6.2(2)), so q_s is finite almost everywhere and therefore θ-psh. Letting $t \to 0$ and $t \to 1$ in $q_s \le v_t - ts$, and using the L^1 convergence of v_t to its endpoint potentials, which are at most u_0

and u_1, we get $q_s \le u_0$ and $q_s \le u_1 - s$ almost everywhere, hence everywhere, and therefore $q_s \le P_\theta(u_0, u_1 - s)$.

Now fix x and $t_0 \in (0,1)$. By Lemma 17.6.3(1), the fiber $t \mapsto v_t(x)$ is either identically $-\infty$, in which case $v_{t_0}(x) = -\infty$, or finite. In the second case Lemma 17.6.2(2) gives $c \in \mathbb{R}$ with $v_t(x) \ge v_{t_0}(x) + c(t - t_0)$ for all t, so that $q_c(x) = \inf_t(v_t(x) - ct) = v_{t_0}(x) - ct_0$, and hence $v_{t_0}(x) = q_c(x) + ct_0 \le \sup_s(q_s(x) + t_0 s)$. In either case $v_{t_0}(x) \le \sup_s\{P_\theta(u_0, u_1 - s)(x) + t_0 s\}$, since $q_s \le P_\theta(u_0, u_1 - s)$. Taking the regularized supremum over all competitors and using Lemma 17.6.6 proves the formula. The same lemma, applied to the affine competitors, gives the last assertion. □

17.7 Energy along weak geodesics

We now show that the energy is affine along weak geodesics. The complexification of a weak geodesic lives on $X \times S$, which has one more complex dimension than X. We prove first that its local potentials are maximal plurisubharmonic functions, so that its top-degree non-pluripolar measure vanishes. To integrate by parts we then glue the complexification, over a smaller annulus, into a potential on the compact manifold $X \times \mathbb{P}^1$.

Lemma 17.7.1 *Let h be a maximal plurisubharmonic function on a domain $\Omega \subset \mathbb{C}^d$, $d \ge 1$, in the sense of Definition 9.6.5. Then the top-degree non-pluripolar product $\langle (\mathrm{dd^c} h)^d \rangle$ is locally finite and vanishes. Here h need not be locally bounded.*

Proof Fix a ball $B \Subset \Omega$ and a concentric ball B_1 with $B \Subset B_1 \Subset \Omega$. Put $h_j = \max\{h, -j\}$ and let $\widehat{h}_j$ be the function supplied by Corollary 9.6.4 for h_j and B. These are locally bounded plurisubharmonic functions with $\widehat{h}_j \ge h_j \ge h$, equal to h_j outside B, with $(\mathrm{dd^c}\widehat{h}_j)^d = 0$ on B; they decrease with j, because $\widehat{h}_j$ is increasing in h_j. Their decreasing limit v is plurisubharmonic by Theorem 8.3.1, since it dominates h, and it equals h outside B. Maximality gives $v \le h$, hence $v = h$.

Fix $C > 0$ and $\varepsilon > 0$, and put

$$z_j = \max\{\widehat{h}_j, -C\}, \quad a_j = \frac{z_j + C}{z_j + C + \varepsilon}, \quad z = \max\{h, -C\}, \quad a = \frac{z + C}{z + C + \varepsilon}.$$

The functions z_j are plurisubharmonic, locally uniformly bounded (between $-C$ and $\widehat{h}_1$), and decrease to z, so they converge to z locally in capacity on the bounded open set B_1 (Proposition 9.4.6). The weights take values in $[0,1]$, and since $s \mapsto (s + C)/(s + C + \varepsilon)$ is Lipschitz with constant ε^{-1} on $[-C, \infty)$, we have $|a_j - a| \le \varepsilon^{-1}|z_j - z|$, so $a_j \to a$ locally in capacity on B_1. The limit weight a is quasi-continuous on B_1, because z is (Theorem 9.4.2) and a is a continuous function of z. On B, the weight a_j vanishes where $\widehat{h}_j \le -C$, and on the plurifine open set $\{\widehat{h}_j > -C\}$ we have $z_j = \widehat{h}_j$; hence plurifine locality of Bedford–Taylor products (Theorem 10.4.3) gives

$$a_j(\mathrm{dd^c} z_j)^d = a_j(\mathrm{dd^c}\widehat{h}_j)^d = 0 \quad \text{on } B.$$

The last assertion of Theorem 9.4.8, applied on B_1 with d factors z_j and the weights a_j, gives

$$a_j(\mathrm{dd^c} z_j)^d \longrightarrow a(\mathrm{dd^c} z)^d \quad \text{weakly on } B_1,$$

so $a(\mathrm{dd^c} z)^d = 0$ on B. Since $a > 0$ exactly on $\{h > -C\}$ and $z = \max\{h, -C\}$, the measure $\mathbf{1}_{\{h>-C\}}(\mathrm{dd^c} z)^d$ vanishes on B. Taking $C = k$ for all integers k, the truncated products in the definition of $\langle(\mathrm{dd^c} h)^d\rangle$ all vanish on B; hence the non-pluripolar product is locally finite and zero on B. A cover of Ω by such balls proves the lemma.□

Lemma 17.7.2 *Let $u_0, u_1 \in \mathcal{E}(X, \theta, \phi)$, let D be a coordinate domain of $X \times S$ or of $X \times \{\mathrm{e}^{-1} < |z| < 1\}$, the latter with $t = -\log|z|$, and let a be a smooth function on D whose $\mathrm{dd^c}$ is the pullback of θ (such functions exist on every coordinate ball whose closure lies in a chart, by Lemma 13.1.2). Then the complexification of the weak geodesic between u_0 and u_1, plus a, is a maximal plurisubharmonic function on D.*

Proof Let Z be either $X \times S$, on which the group $\Gamma = \mathbb{R}$ acts by the imaginary translations $(x, \zeta) \mapsto (x, \zeta + \mathrm{i}y)$, or $X \times A$ with $A = \{\mathrm{e}^{-1} < |z| < 1\}$, on which $\Gamma = \mathbb{R}$ acts by the rotations $(x, z) \mapsto (x, \mathrm{e}^{\mathrm{i}y} z)$, according to where D lies. Put $\tau(x, \zeta) = \operatorname{Re}\zeta$ in the first case and $\tau(x, z) = -\log|z|$ in the second, and let θ_Z be the pullback of θ to Z. The action of Γ preserves θ_Z and τ, and the Γ-invariant functions on Z are exactly the functions whose value at $(x, w) \in Z$ is $v_{\tau(x,w)}(x)$, for a family $(v_t)_{0<t<1}$ of functions on X; by the discussion following Definition 17.6.1, such a function is θ_Z-psh if and only if the complexification of (v_t) on $X \times S$ is $p^*\theta$-psh. Let U be the Γ-invariant function on Z defined by the weak geodesic (G_t); it is θ_Z-psh and bounded above by $\max\{\sup_X u_0, \sup_X u_1\}$, since $G_t \le (1-t)u_0 + tu_1$.

Now $D \subset Z$ and $\mathrm{dd^c} a = \theta_Z$ on D, so that $h = U + a$ is plurisubharmonic on D; we show that it is maximal. Let v be plurisubharmonic on D with $v \le h$ outside a compact set $K \subset D$. Every boundary point of D in Z has a neighborhood disjoint from K, on whose intersection with D we have $v - a \le U$; since U is upper semicontinuous, the boundary condition of Proposition 8.2.5 holds, and the function $\widetilde{U}$ equal to $\max\{U, v - a\}$ on D and to U elsewhere is θ_Z-psh (apply the proposition in charts after adding smooth local potentials of θ_Z). Moreover $\widetilde{U} \ge U$, and $\widetilde{U} = U$ outside K. Since $v - a$ is bounded above on the compact set K, the function $\widetilde{U}$ is bounded above.

The functions $\widetilde{U} \circ \gamma$, $\gamma \in \Gamma$, are θ_Z-psh and uniformly bounded above, so their regularized upper envelope $\breve{U}$ is θ_Z-psh (Theorem 8.3.2, applied in charts after adding smooth local potentials); it is Γ-invariant and satisfies $\breve{U} \ge \widetilde{U}$. The compact set $\tau(K)$ lies in an interval $[r_1, r_2]$ with $0 < r_1 \le r_2 < 1$, so the Γ-orbit of K lies in $\{r_1 \le \tau \le r_2\}$. Outside this set every $\widetilde{U} \circ \gamma$ equals $U \circ \gamma = U$, so $\breve{U} = U$ on the open set $\{\tau < r_1\} \cup \{\tau > r_2\}$, because U is upper semicontinuous. Therefore $\breve{U}$ is defined by a subgeodesic whose slices coincide with G_t for $t < r_1$ and for $t > r_2$; in particular its endpoint potentials are u_0 and u_1. It is a competitor for the Perron segment between u_0 and u_1, so $\breve{U} \le U$. Consequently $\widetilde{U} \le U$, so $v - a \le U$ on D, that is, $v \le h$. This is maximality in the sense of Definition 9.6.5. □

Theorem 17.7.3 *For $u_0, u_1 \in \mathcal{E}^1(X,\theta,\phi)$,*

$$E^{\phi}_{\theta}(G_t(u_0,u_1)) = (1-t)E^{\phi}_{\theta}(u_0) + tE^{\phi}_{\theta}(u_1), \qquad 0 \le t \le 1.$$

Proof *Step 1: setting up a compact extension.* Assume first that $u_0, u_1 \in \mathcal{E}^\infty(X,\theta,\phi)$, and write $G_t = G_t(u_0,u_1)$. Let $A = \{\mathrm{e}^{-1} < |z| < 1\} \subset \mathbb{C} \subset \mathbb{P}^1$, and let $U(x,z) = G_{-\log|z|}(x)$ be the complexification on $X \times A$. Choose a constant C with $\phi - C \le u_0, u_1 \le \phi + C$. Then $\phi - C$ is a competitor for the rooftop envelope $P_\theta(u_0,u_1)$, so the bounds of Theorem 17.6.8 give $\phi - C \le U \le \phi + C$. Fix concentric annuli $A_0 \Subset A_1 \Subset A$. On $Y = X \times \mathbb{P}^1$, with projections $p_1 \colon Y \to X$ and $p_2 \colon Y \to \mathbb{P}^1$, let $\beta = p_2^*\omega_{\mathrm{FS}}$, where ω_{FS} is the Fubini–Study form of mass one, and put

$$\Theta = p_1^*\theta + B\beta, \qquad \Phi = p_1^*\phi,$$

with a constant $B > 0$ chosen below.

To glue U to Φ we use a smooth function q on $\mathbb{P}^1$ that is small on A_0 and large near the boundary of A_1. Write $A_0 = \{r_0 < |z| < r_0'\}$ and $A_1 = \{r_1 < |z| < r_1'\}$, so that $\mathrm{e}^{-1} < r_1 < r_0 < r_0' < r_1' < 1$, and choose $\varepsilon > 0$ with $r_1 + \varepsilon < r_0$ and $r_0' < r_1' - \varepsilon$. Let χ_0 be a smooth function on $\mathbb{R}$ such that $1 - \chi_0$ is a smooth cutoff function for $[r_0, r_0']$ in $(r_1+\varepsilon, r_1'-\varepsilon)$ (Lemma 1.4.1(4)); thus χ_0 takes values in $[0,1]$, vanishes on $[r_0, r_0']$, and equals 1 outside $(r_1+\varepsilon, r_1'-\varepsilon)$. The function $q(z) = (2C+4)\chi_0(|z|) - (C+2)$, with $q(\infty) = C+2$, is smooth on $\mathbb{P}^1$, because $|z|$ is smooth on $\mathbb{C} \setminus \{0\}$ and q is constant near 0 and near ∞. It equals $-C-2$ on A_0 and $C+2$ on a neighborhood of ∂A_1. Choose B so large that $B\omega_{\mathrm{FS}} + \mathrm{dd^c} q \ge 0$.

Then $\Phi + q$ is Θ-psh, and so is U on $X \times A_1$, since $\beta \ge 0$. Define $W = \max\{U, \Phi + q\}$ on $X \times A_1$ and $W = \Phi + q$ on $Y \setminus (X \times A_1)$. Near $X \times \partial A_1$ we have $\Phi + q = \Phi + C + 2 > U$ outside the pole set of Φ, on which both functions are $-\infty$; hence $W = \Phi + q$ on a collar inside $X \times A_1$. Thus, near every point of Y, W is either the maximum of two Θ-psh functions or equal to $\Phi + q$, and W is Θ-psh on Y. Moreover $W - \Phi$ is bounded, and $W = U$ on $X \times A_0$, since there $\Phi + q = \Phi - C - 2 < U$ outside the pole set of Φ.

Step 2: a current on $\mathbb{P}^1$. Put $T = \Theta + \mathrm{dd^c} W$ and $R = \Theta + \mathrm{dd^c}\Phi$, and define the current of order zero on $\mathbb{P}^1$

$$H = \frac{1}{n+1}(p_2)_*\left((W-\Phi)\sum_{j=0}^{n}\langle T^j \wedge R^{n-j}\rangle\right).$$

Let χ be a smooth function on $\mathbb{P}^1$. By Theorem 17.1.3 on the compact Kähler manifold Y, applied to the bounded quasi-psh differences $W - \Phi$ and $\chi \circ p_2$, we may move $\mathrm{dd^c}$ from $\chi \circ p_2$ to $W - \Phi$. By (17.1), $\mathrm{dd^c}(W-\Phi) \wedge \langle T^j \wedge R^{n-j}\rangle = \langle T^{j+1} \wedge R^{n-j}\rangle - \langle T^j \wedge R^{n-j+1}\rangle$, and the sum over j telescopes. Therefore

$$\mathrm{dd^c} H = \frac{1}{n+1}(p_2)_*(\langle T^{n+1}\rangle - \langle R^{n+1}\rangle). \tag{17.30}$$

On the other hand, Corollary 15.5.1, with $Z = \mathbb{P}^1$ and $\gamma = B\omega_{\mathrm{FS}}$, shows that H is given by the function $z \mapsto E_\theta^\phi(W(\cdot, z))$. In particular, on A_0, H is the function $z \mapsto E_\theta^\phi(G_{-\log|z|})$.

Step 3: $\mathrm{dd^c}H$ vanishes on A_0. On $X \times A_0$ put $L = p_1^*\theta + \mathrm{dd^c}U \geq 0$, so that $T = L + B\beta$ there. Locally T has a plurisubharmonic potential which is the sum of a potential of L and a smooth potential of $B\beta$. The product $\langle T^{n+1}\rangle$ is finite by Theorem 15.4.3, so multilinearity (Proposition 15.1.5), compatibility with smooth factors (Proposition 15.3.3) and $\beta \wedge \beta = 0$ give, on $X \times A_0$,

$$\langle T^{n+1}\rangle = \langle L^{n+1}\rangle + (n+1)B\beta \wedge \langle L^n\rangle, \qquad \beta \wedge \langle T^n\rangle = \beta \wedge \langle L^n\rangle,$$

with all terms locally finite. Every point of $X \times A_0$ has a coordinate ball $D \subset X \times A_0$ on which $p_1^*\theta = \mathrm{dd^c}a$ for a smooth function a (Lemma 13.1.2); there $L = \mathrm{dd^c}(U + a)$, and Lemmas 17.7.1 and 17.7.2 give $\langle L^{n+1}\rangle = 0$ on D. Hence $\langle L^{n+1}\rangle = 0$ on $X \times A_0$, so that $\langle T^{n+1}\rangle = (n+1)B\beta \wedge \langle T^n\rangle$ on $X \times A_0$. Every slice $W(\cdot, z)$ has bounded difference from ϕ and hence mass m, so Corollary 15.5.1 with weight one gives $(p_2)_*(\beta \wedge \langle T^n\rangle) = m\omega_{\mathrm{FS}}$.

The current R is treated in the same way. We have $R = p_1^*\theta_\phi + B\beta$, and $\langle (p_1^*\theta_\phi)^{n+1}\rangle = 0$. Indeed, the local potentials of $p_1^*\theta_\phi$ do not depend on the variable in $\mathbb{P}^1$, and so neither do their bounded truncations; each truncation is the decreasing limit of smooth plurisubharmonic functions of the n variables of X (Proposition 8.2.2(2)), whose $(n+1)$st powers of $\mathrm{dd^c}$ vanish, so the bounded products of the truncations vanish by Theorem 9.2.5, and with them the truncated products defining $\langle (p_1^*\theta_\phi)^{n+1}\rangle$. Hence $\langle R^{n+1}\rangle = (n+1)B\beta \wedge \langle R^n\rangle$, and the same fiber formula, applied to Φ in place of W, gives $(p_2)_*\langle R^{n+1}\rangle = (n+1)Bm\omega_{\mathrm{FS}}$. Hence the right side of (17.30) vanishes on A_0, that is, $\mathrm{dd^c}H = 0$ on A_0.

Step 4: the energy is affine. The function $\tau \mapsto E_\theta^\phi(G_\tau)$ is locally Lipschitz on $(0,1)$. Indeed, let $0 < \delta < \frac{1}{2}$. On the common finite locus the convex functions $\tau \mapsto G_\tau(x) - \phi(x)$ take values in $[-C, C]$, so by Lemma 17.6.2(4) their chord slopes on $[\delta, 1-\delta]$ are bounded by $2C/\delta$. Thus $G_s - (2C/\delta)|s-\tau| \leq G_\tau \leq G_s + (2C/\delta)|s-\tau|$ for $s, \tau \in [\delta, 1-\delta]$ on the common finite locus, hence everywhere, and monotonicity and the shift formula give $|E_\theta^\phi(G_\tau) - E_\theta^\phi(G_s)| \leq (2Cm/\delta)|s-\tau|$.

Writing $z = \xi + \mathrm{i}\eta$, we have $\mathrm{dd^c}f = (4\pi)^{-1}\Delta f \, \mathrm{d}\xi \wedge \mathrm{d}\eta$ for smooth functions f, and hence for distributions, so $\Delta H = 0$ on A_0 by Step 3, and Weyl's lemma (Theorem 2.1.4) shows that H is represented on A_0 by a harmonic function. By Step 2, H is also represented on A_0 by the function $z \mapsto E_\theta^\phi(G_{-\log|z|})$, which is continuous by the Lipschitz bound just proved. Two locally integrable functions defining the same distribution agree almost everywhere (Proposition 1.4.3(4)), and two continuous functions that agree almost everywhere agree everywhere, so this function is harmonic on A_0. A radial harmonic function on an annulus is an affine function of $\log|z|$: if $f(z) = g(\log|z|)$ is smooth, then g is smooth and the Laplacian in polar coordinates gives $\Delta f = |z|^{-2}g''(\log|z|)$, so $g'' = 0$. Since A_0 was arbitrary, $\tau \mapsto E_\theta^\phi(G_\tau)$ is affine on $(0,1)$.

Finally, $|u_1 - u_0| \leq 2C$ on the common finite locus, so Proposition 17.6.9(5) gives $|G_\tau - u_0| \leq 2C\tau$ and $|G_\tau - u_1| \leq 2C(1-\tau)$ there. By the Lipschitz bound

of Proposition 17.2.4, the energies converge to $E^{\phi}_{\theta}(u_0)$ and $E^{\phi}_{\theta}(u_1)$ as $\tau \to 0$ and $\tau \to 1$. This proves the theorem for $u_0, u_1 \in \mathcal{E}^{\infty}(X, \theta, \phi)$.

Step 5: finite energy. Let $u_0, u_1 \in \mathcal{E}^1(X, \theta, \phi)$. Adding the same constant c to both endpoints adds c to G_t, by Proposition 17.6.9(2), and adds cm to both sides of the asserted identity. We may therefore assume $u_0, u_1 \preceq \phi$. Put $u_{j,k} = \max\{u_j, \phi - k\}$ for $j = 0, 1$. By Proposition 17.6.9(3), $G_t(u_{0,k}, u_{1,k}) \downarrow G_t(u_0, u_1)$ as $k \to \infty$, and the limit has finite energy by Theorem 17.6.8. The identity holds for $u_{0,k}, u_{1,k}$ by the preceding steps, and Proposition 17.2.7, applied at the endpoints and at each interior slice, passes it to the limit. □

17.8 Extremal slopes and long Perron segments

We close the chapter with two estimates on how a weak geodesic depends on its endpoints. The first, Proposition 17.8.1, shows that the essential supremum of $G_t(u, v) - u$ is a linear function of t, and similarly for the essential infimum; essential suprema and infima are taken with respect to the volume form ω^n. The second, Proposition 17.8.2, concerns long segments from ϕ to the truncations of a singular potential.

Proposition 17.8.1 *Let $u, v \in \mathcal{E}(X, \theta, \phi)$ with $v \preceq u$, and put $C = \operatorname{ess\,sup}_X(v - u)$. Then C is finite and*

$$\operatorname*{ess\,sup}_X(G_t(u, v) - u) = tC, \qquad 0 \le t \le 1.$$

If moreover $u \sim v$ and $c = \operatorname{ess\,inf}_X(v - u)$, then $\operatorname{ess\,inf}_X(G_t(u, v) - u) = tc$. These essential extrema equal the corresponding extrema over the common finite locus, and they do not change if a pluripolar set is removed.

Proof The function $v - u$ is finite almost everywhere, and $v \preceq u$ gives $v - u \le C'$ for some constant C'; hence C is finite. The inequality $v \le u + C$ holds almost everywhere, hence everywhere, as recalled in Section 16.1. Thus the supremum of $v - u$ over the common finite locus is at most C; it is at least C because the common finite locus has full measure. The same argument applies to an infimum, and to the complement of any pluripolar set, which still has full measure. The same remarks apply to $G_t - u$.

By Theorem 17.6.8, $G_t \le (1-t)u + tv \le u + tC$. The affine curve $v - C(1-t)$ is a competitor, since $v - C \le u$, so $G_t \ge v - (1-t)C$. On the common finite locus $v - u = (v - G_t) + (G_t - u)$, and taking essential suprema gives $C \le (1-t)C + \operatorname{ess\,sup}_X(G_t - u)$. Together with the first bound, this proves the equality for the supremum.

Now let $u \sim v$. The affine curve $u + ct$ is a competitor, since $u + c \le v$, so $G_t - u \ge ct$. Applying the equality for the supremum to the reversed segment, $G_t(u, v) = G_{1-t}(v, u)$, gives $\operatorname{ess\,sup}_X(G_t - v) = (1-t)\operatorname{ess\,sup}_X(u - v) = -(1-t)c$, that is, $\operatorname{ess\,inf}_X(v - G_t) = (1-t)c$. Taking essential infima in $v - u = (v - G_t) + (G_t - u)$ gives $c \ge (1-t)c + \operatorname{ess\,inf}_X(G_t - u)$, which proves the opposite bound. □

We now turn to long segments. Given a potential $u \le \phi$ and $T > 0$, we join ϕ to the truncation $\max\{u, \phi - T\}$ by a weak geodesic parametrized by $[0, T]$. Changing u by at most a constant C changes this segment at time s by at most Cs/T, an error that tends to zero as $T \to \infty$. Hence the limits of these segments as $T \to \infty$ depend only on the singularity type of u.

Proposition 17.8.2 *Let $u \in \mathrm{PSH}(X, \theta)$ with $u \le \phi$. For $T > 0$ and $0 \le s \le T$ put*

$$u_T = \max\{u, \phi - T\}, \qquad L_s^{u,T} = G_{s/T}(\phi, u_T).$$

For fixed $s \ge 0$, the potentials $L_s^{u,T}$ increase with $T \ge s$ and satisfy $\phi - s \le L_s^{u,T} \le \phi$. Their regularized limit

$$R_s^u = \left(\sup_{T>s} L_s^{u,T}\right)^*$$

is θ-psh. If $v \in \mathrm{PSH}(X, \theta)$ satisfies $v \le \phi$, and $u \le v + C$ for a constant $C \ge 0$, then

$$L_s^{u,T} \le L_s^{v,T} + \frac{Cs}{T}, \qquad R_s^u \le R_s^v.$$

In particular, if $u \sim v$, then $R_s^u = R_s^v$ for every $s \ge 0$. Consequently, for an arbitrary $u \in \mathrm{PSH}(X, \theta)$ with $u \preceq \phi$, the potentials R_s^{u-c} do not depend on the constant c with $u - c \le \phi$. Denoting them by R_s^u, we have $R_s^u \le R_s^v$ whenever $u \preceq v \preceq \phi$.

Proof Since $\phi - T \le u_T \le \phi$, the potential u_T has bounded difference from ϕ; in particular $u_T \in \mathcal{E}(X, \theta, \phi)$. By Theorem 17.6.8, the Perron segment from ϕ to u_T is a weak geodesic. We regard it as a curve on $[0, T]$, so that its slice at time s is $L_s^{u,T}$.

Bounds. For $0 \le s \le T$ put $q_s = \max\{u, \phi - s\}$. After the affine change of parameter, the complexification of the curve $(q_s)_{0 \le s \le T}$ is $(x, \zeta) \mapsto \max\{u(x), \phi(x) - T \operatorname{Re} \zeta\}$. This is the maximum of two $p^*\theta$-psh functions, because $\operatorname{Re} \zeta$ is pluriharmonic, and hence it is $p^*\theta$-psh (Proposition 8.2.5, applied after adding smooth local potentials of $p^*\theta$, which exist by Lemma 7.3.1). Moreover $q_0 = \phi$, since $u \le \phi$, and $q_T = u_T$. As $|q_s - q_t| \le |s - t|$ on the finite locus of ϕ, the curve tends to ϕ and u_T in L^1 at the two ends. Thus (q_s) is a subgeodesic with endpoints ϕ and u_T, and by the definition of the Perron segment, $L_s^{u,T} \ge q_s \ge \phi - s$. On the other hand, Theorem 17.6.8 gives $L_s^{u,T} \le (1 - s/T)\phi + (s/T)u_T \le \phi$.

Monotonicity in T. Let $T' > T$. The same subgeodesic, now on $[0, T']$, shows that $L_T^{u,T'} \ge q_T = u_T$. By the last assertion of Theorem 17.6.8, the restriction of the segment of length T' to $[0, T]$ is the weak geodesic from ϕ to $L_T^{u,T'}$. Since weak geodesics increase with their endpoints (Proposition 17.6.9(1)), this restriction dominates the weak geodesic from ϕ to u_T. Hence $L_s^{u,T'} \ge L_s^{u,T}$ for $0 \le s \le T$.

The limit. Fix s. The family $(L_s^{u,T})_{T>s}$ is increasing, bounded above by ϕ and bounded below by $\phi - s$. By Theorem 8.3.2, applied after adding smooth local potentials of θ, its regularized supremum R_s^u is θ-psh, and $\phi - s \le R_s^u \le \phi$. Since the family is increasing, the supremum may be taken over $T \ge A$ for any $A > s$ without changing R_s^u. For $s = 0$, all the potentials $L_0^{u,T}$ and R_0^u equal ϕ.

Comparison. Suppose that $v \le \phi$ and $u \le v + C$ with $C \ge 0$. Then

$$u_T \le \max\{v + C, \phi - T\} \le v_T + C.$$

By the monotonicity of weak geodesics in their endpoints and the formula for constant shifts of the endpoints, Proposition 17.6.9(1) and (2), we have $G_t(\phi, u_T) \le G_t(\phi, v_T + C) = G_t(\phi, v_T) + tC$ for $0 \le t \le 1$. Taking $t = s/T$ gives the bound $L_s^{u,T} \le L_s^{v,T} + Cs/T$. Now fix s and $A > s$. For $T \ge A$ we get $L_s^{u,T} \le R_s^v + Cs/A$, and taking the regularized supremum over $T \ge A$ gives

$$R_s^u \le R_s^v + \frac{Cs}{A}.$$

Letting $A \to \infty$ yields $R_s^u \le R_s^v$.

If $u \sim v$, then $u \le v + C$ and $v \le u + C$ for some $C \ge 0$, and the comparison applied in both directions gives $R_s^u = R_s^v$. If $u \preceq \phi$ is arbitrary, any two constants c, c' with $u - c \le \phi$ and $u - c' \le \phi$ give potentials $u - c \sim u - c'$, so $R_s^{u-c} = R_s^{u-c'}$. Finally, if $u \preceq v \preceq \phi$, choose constants c, c' with $u - c \le \phi$ and $v - c' \le \phi$. Then $u - c \le v - c' + C$ for some $C \ge 0$, and the comparison gives $R_s^u = R_s^{u-c} \le R_s^{v-c'} = R_s^v$. □

Chapter 18
Convex analysis and toric potentials

Let the compact torus $(S^1)^n$ act on $(\mathbb{C}^*)^n$ by separate rotations of the coordinates, $(z_1, \dots, z_n) \mapsto (e^{i\theta_1} z_1, \dots, e^{i\theta_n} z_n)$. A plurisubharmonic function on $(\mathbb{C}^*)^n$ that is invariant under this action depends only on the logarithmic radii $x_j = \log |z_j|^2$, and we show in Corollary 18.1.6 that, as a function of $x \in \mathbb{R}^n$, it is precisely a finite convex function. The complex Hessian accounts for the convexity; conversely, the supporting affine functions of a convex function give the submean inequality on complex lines. Questions about the growth and the Monge–Ampère mass of invariant potentials thus become questions about convex functions on a real vector space, and we answer them in terms of the slopes of supporting affine functions.

The first part of the chapter develops the convex analysis that this requires. Separation of convex sets produces supporting affine functions, and Legendre duality identifies the closure of the set of slopes of a convex function F with the closure of the set where its conjugate F^* is finite. An upper bound of F, up to a constant, by the support function of a convex body P holds exactly when all slopes of F lie in P (Proposition 18.3.2); this leads to convex analogues of the classes of potentials with bounded growth, full mass and finite energy. The real Monge–Ampère measure of F assigns to a set E the volume of the set of slopes of F at points of E, multiplied by $n!$. Since the subgradient map is multivalued, we construct this measure by pushing Lebesgue measure forward under the gradient of F^*, which is defined almost everywhere (Theorem 18.4.1). Its total mass is $n!$ times the volume of the closed set of slopes, and it depends continuously on F for local uniform convergence (Theorem 18.4.2); when no mass escapes to infinity, the convergence also holds against bounded continuous functions. Finally, counting lattice points in the dilates kP relates these volumes to dimensions of spaces of sections.

The second part constructs, from a smooth lattice polytope P, a projective toric manifold X_P with a very ample line bundle L. Monomial charts at the vertices of P make its geometry explicit, and we read off from them the correspondence between torus orbits and faces of P, the orders of characters along the boundary divisors, and a basis of $H^0(X_P, L^k)$ indexed by the lattice points of kP (Theorem 18.8.5). The potential of Guillemin's reference metric on L is the Legendre transform of an explicit entropy function on P, whose gradient has logarithmic singularities along the facets.

In the same charts we show that these singularities cancel after the transformation, so that the metric is smooth, with positive definite curvature, also along the boundary orbits (Theorem 18.9.1). These are the foundations of convex analysis and of ample toric geometry used in [Xia, Part I].

Throughout the chapter, $N_{\mathbb{R}}$ is a real vector space of finite dimension $n \geq 0$ and $M_{\mathbb{R}}$ is its dual space, with pairing $\langle p, x\rangle$ for $p \in M_{\mathbb{R}}$ and $x \in N_{\mathbb{R}}$; we identify the dual of $M_{\mathbb{R}}$ with $N_{\mathbb{R}}$. Whenever an estimate requires a norm, we fix a Euclidean norm on $N_{\mathbb{R}}$ and give $M_{\mathbb{R}}$ the dual norm $|p| = \max_{|x|\leq 1}\langle p, x\rangle$, so that $|\langle p, x\rangle| \leq |p||x|$; the open and closed balls $B(x, r)$ and $\overline{B(x, r)}$ of Section 1.4 are taken with respect to these norms. When volumes are involved, we also fix a lattice $N \subset N_{\mathbb{R}}$, that is, the set of integral linear combinations of a basis of $N_{\mathbb{R}}$, and the dual lattice $M = \{p \in M_{\mathbb{R}} : \langle p, x\rangle \in \mathbb{Z} \text{ for all } x \in N\}$. Lebesgue measure on $M_{\mathbb{R}}$, written $\mathrm{d}p$ or vol, and Lebesgue measure $\mathrm{d}x$ on $N_{\mathbb{R}}$ are normalized so that a fundamental parallelepiped of M, respectively of N, has volume one; equivalently, they are Lebesgue measure in the coordinates of a lattice basis. (This volume of subsets of $M_{\mathbb{R}}$ is not to be confused with the volume of a cohomology class in Definition 15.8.1.) In dimension zero both spaces are a point and both measures are the unit point mass. A *convex body* is a nonempty compact convex set, possibly with empty interior.

Historical remarks and references for this chapter are collected in Appendix B.20.

18.1 Closed convex functions and supporting slopes

The conjugate of a finite convex function is in general finite only on a convex subset of the dual space. We therefore allow convex functions to take the value $+\infty$, and we impose lower semicontinuity as the regularity condition that controls their behavior at the boundary of the set where they are finite. The definitions and results of this section and the next are stated on $N_{\mathbb{R}}$; they apply equally to functions on $M_{\mathbb{R}}$, with the roles of the two spaces interchanged.

Definition 18.1.1 A function $F\colon N_{\mathbb{R}} \to [-\infty, +\infty]$ is *convex* if its epigraph

$$\operatorname{epi} F = \{(x, r) \in N_{\mathbb{R}} \times \mathbb{R} : r \geq F(x)\}$$

is convex. A convex function is *proper* if it does not take the value $-\infty$ and its *effective domain* $\operatorname{dom} F = \{x : F(x) < +\infty\}$ is nonempty; a proper convex function is *closed* if it is lower semicontinuous. The *relative interior* $\operatorname{ri} C$ of a convex set C is its interior in its affine span, and $\overline{C} \setminus \operatorname{ri} C$ is its *relative boundary*. For a proper convex function F and $x \in \operatorname{dom} F$, the *subgradient set* of F at x is

$$\partial F(x) = \{p \in M_{\mathbb{R}} : F(y) \geq F(x) + \langle p, y - x\rangle \text{ for every } y \in N_{\mathbb{R}}\}.$$

Its elements are the *subgradients*, or *supporting slopes*, of F at x: they are the slopes of the affine functions that lie below F and agree with F at x. We set $\partial F(x) = \varnothing$ for $x \notin \operatorname{dom} F$ and $\partial F(E) = \bigcup_{x\in E} \partial F(x)$ for $E \subset N_{\mathbb{R}}$.

For a function F with values in $(-\infty,+\infty]$, the epigraph is convex if and only if $F((1-\lambda)x+\lambda y) \le (1-\lambda)F(x)+\lambda F(y)$ for all $x, y \in N_{\mathbb{R}}$ and $0<\lambda<1$. Indeed, suppose that the epigraph is convex. If $F(x)$ and $F(y)$ are finite, the inequality expresses that the corresponding combination of $(x, F(x))$ and $(y, F(y))$ lies in epi F; otherwise its right side is $+\infty$. Conversely, for $(x,r), (y,s) \in \operatorname{epi} F$ the inequality gives $F((1-\lambda)x+\lambda y) \le (1-\lambda)r+\lambda s$. Thus Definition 18.1.1 extends the usual notion of a convex function, with the sign convention fixed in the Conventions, to functions with values in $[-\infty,+\infty]$. The effective domain of a convex function is convex, being the image of its epigraph under the projection to $N_{\mathbb{R}}$.

A convex set contains every convex combination $\sum_{i=1}^{q} \lambda_i x_i$ of its points, where $\lambda_i \ge 0$ and $\sum_i \lambda_i = 1$: for $q \ge 2$ and $\lambda_q < 1$ this follows by induction on q from $\sum_i \lambda_i x_i = (1-\lambda_q)\sum_{i<q} \frac{\lambda_i}{1-\lambda_q} x_i + \lambda_q x_q$. Consequently, the *convex hull* of a set S, that is, the smallest convex set containing S, is the set of all convex combinations of points of S, since this set is convex and contains S. Applied to the epigraph, the same fact gives the finite form of Jensen's inequality

$$F\Big(\sum_{i=1}^{q} \lambda_i x_i\Big) \le \sum_{i=1}^{q} \lambda_i F(x_i)$$

for every convex function F with values in $(-\infty,+\infty]$ and every convex combination.

A function with values in $(-\infty,+\infty]$ is lower semicontinuous if and only if its epigraph is closed in $N_{\mathbb{R}}\times\mathbb{R}$. Indeed, if F is lower semicontinuous and $(x_k, r_k) \in \operatorname{epi} F$ converge to (x,r), then $F(x) \le \underline{\lim}_k F(x_k) \le r$. Conversely, if the epigraph is closed and $x_k \to x$ with $\underline{\lim}_k F(x_k) < r < F(x)$ for some real r, then $(x_k, r) \in \operatorname{epi} F$ for infinitely many k, and in the limit $(x,r) \in \operatorname{epi} F$, which is impossible.

A *finite* convex function is a real-valued one. A finite convex function F on an open convex set $\Omega \subset N_{\mathbb{R}}$ is identified with the proper convex function on $N_{\mathbb{R}}$ that equals F on Ω and $+\infty$ elsewhere; this extension is convex, since its convexity inequality is trivial when x or y lies outside Ω. In particular, $\partial F(x)$ is defined for $x \in \Omega$. The subgradient notation ∂F is unrelated to the boundary ∂E of a set.

We shall need a Lipschitz bound on cubes. Let $Q' \subset Q$ be closed cubes with a common center and half-sides δ and 2δ in affine coordinates on an affine subspace of $N_{\mathbb{R}}$, and let $|\cdot|$ be the Euclidean norm of these coordinates. If a convex function F is finite on Q, with oscillation $\omega = \sup_Q F - \inf_Q F < \infty$, then

$$|F(z)-F(y)| \le \frac{\omega}{\delta}|z-y| \qquad (y, z \in Q'). \tag{18.1}$$

Indeed, for $y \ne z$ in Q' the point $w = z+\delta(z-y)/|z-y|$ lies in Q, and $z = (1-\lambda)y+\lambda w$ with $\lambda = |z-y|/(|z-y|+\delta) \le |z-y|/\delta$. Convexity gives $F(z)-F(y) \le \lambda(F(w)-F(y)) \le (\omega/\delta)|z-y|$, and we may exchange y and z.

Lemma 18.1.2 *Let W be a real vector space of finite dimension n, and let $C \subset W$ be convex.*

(1) *If C is closed and nonempty and $z \in W \setminus C$, there is a linear functional ℓ on W with* $\sup_C \ell < \ell(z)$.

(2) *If C is closed and z is a boundary point of C in W, there is a nonzero linear functional ℓ on W with $\sup_C \ell = \ell(z)$; that is, C has a supporting hyperplane at z.*

(3) *Every point of the convex hull of a set $S \subset W$ is a convex combination of affinely independent points of S, hence of at most $n+1$ points of S.*

(4) *If $C \neq \emptyset$, then $\operatorname{ri} C \neq \emptyset$, and $(1-t)x_0 + ty \in \operatorname{ri} C$ whenever $x_0 \in \operatorname{ri} C$, $y \in \overline{C}$ and $0 \leq t < 1$. Moreover,*

$$\overline{\operatorname{ri} C} = \overline{C}, \qquad \operatorname{ri} C = \operatorname{ri} \overline{C},$$

and the relative boundary of C has Lebesgue measure zero in the affine span of C.

Proof Fix a Euclidean inner product $y \cdot w$ on W, with norm $|y|$.

(1) Choose $y_k \in C$ with $|z - y_k| \to \inf_{w \in C} |z - w|$. The sequence (y_k) is bounded, so a subsequence converges to a point y, which lies in C because C is closed. Thus y minimizes the distance to z on C, and $y \neq z$. For $w \in C$ and $0 \leq t \leq 1$ the point $y + t(w-y)$ lies in C, so the function $t \mapsto |z - y - t(w-y)|^2$ on $[0,1]$ is minimal at $t = 0$, and its derivative there, $-2(z-y)\cdot(w-y)$, is nonnegative. Hence $\ell(w) = (z-y)\cdot w$ satisfies $\ell(w) \leq \ell(y) = \ell(z) - |z-y|^2 < \ell(z)$ for every $w \in C$.

(2) Since z is a boundary point of C, there are points $z_k \notin C$ with $z_k \to z$. By (1) there are unit vectors a_k with $a_k \cdot w < a_k \cdot z_k$ for all $w \in C$. A subsequence of (a_k) converges to a unit vector a, and in the limit $a \cdot w \leq a \cdot z$ for all $w \in C$. As $z \in C$, the functional $\ell(w) = a \cdot w$ has the required property.

(3) Let $x = \sum_{i=1}^{q} \lambda_i x_i$ with $x_i \in S$, $\lambda_i > 0$ and $\sum_i \lambda_i = 1$. If the points x_i are affinely dependent, the vectors $(x_i, 1) \in W \times \mathbb{R}$ are linearly dependent, say $\sum_i \mu_i (x_i, 1) = 0$ with not all μ_i zero. Since $\sum_i \mu_i = 0$, some μ_i is positive. Replacing each λ_i by $\lambda_i - s\mu_i$, where $s = \min\{\lambda_i/\mu_i : \mu_i > 0\}$, preserves $\sum_i \lambda_i$ and $\sum_i \lambda_i x_i$, keeps all coefficients nonnegative and makes at least one of them zero. We discard the points with zero coefficient and repeat. Eventually the remaining points are affinely independent, and an affinely independent subset of W has at most $n+1$ elements.

(4) Let A be the affine span of C, of dimension k, and write $B_A(x,r)$ for the open ball of A with center x and radius r. If $k = 0$, then C is a point, equal to its relative interior and to its closure, and its relative boundary is empty; so assume $k \geq 1$. A maximal affinely independent subset of C spans an affine subspace containing C, hence equal to A, so it consists of $k+1$ points $x'_0, \ldots, x'_k$. Every point $y \in A$ has unique barycentric coordinates $\lambda_i(y)$, which are affine functions of y with $\sum_i \lambda_i(y) = 1$ and $y = \sum_i \lambda_i(y) x'_i$. At the barycenter c of the simplex with vertices x'_i all of them equal $1/(k+1)$, so they are positive on a ball $B_A(c,r)$; every point of this ball is then a convex combination of the x'_i and lies in C. Hence $c \in \operatorname{ri} C$.

Now let $x_0 \in \operatorname{ri} C$, choose $r > 0$ with $B_A(x_0, r) \subset C$, and let $y \in \overline{C}$ and $0 \leq t < 1$. If $y \in C$, convexity gives $B_A((1-t)x_0 + ty, (1-t)r) = (1-t)B_A(x_0, r) + ty \subset C$. In general choose $y_k \in C$ with $y_k \to y$; then $B_A((1-t)x_0 + ty_k, (1-t)r) \subset C$, and for large k this ball contains $B_A((1-t)x_0 + ty, (1-t)r/2)$. Thus $(1-t)x_0 + ty \in \operatorname{ri} C$. Letting $t \uparrow 1$, we see that every point of $\overline{C}$ is a limit of points of $\operatorname{ri} C$, so $\overline{\operatorname{ri} C} = \overline{C}$. Next, $\operatorname{ri} C \subset \operatorname{ri} \overline{C}$, since both are interiors in the same affine space A. Conversely, if

$x \in \operatorname{ri}\overline{C}$, then $y = x + \varepsilon(x - c)$ lies in $\overline{C}$ for small $\varepsilon > 0$, and $x = (1 - t)c + ty$ with $t = 1/(1 + \varepsilon) < 1$; hence $x \in \operatorname{ri} C$.

For the measure assertion, translate so that $c = 0$ and identify A with $\mathbb{R}^k$ by a linear isomorphism, which changes Lebesgue measure on A only by a constant factor. Let $Z = \overline{C} \setminus \operatorname{ri} C$ be the relative boundary; it is closed in A, because $\operatorname{ri} C$ is open in A, hence Borel, and $0 \notin Z$. We claim that the dilates sZ, $s > 0$, are pairwise disjoint. Indeed, suppose that $sz = s'z'$ with $z, z' \in Z$ and $0 < s < s'$. Then $z' = (s/s')z$ is of the form $(1 - t) \cdot 0 + tz$ with $z \in \overline{C}$ and $0 < t < 1$, so $z' \in \operatorname{ri} C$ by what we have just shown, which is impossible. Now fix $R > 0$ and put $Z_R = Z \cap B(0, R)$. The sets sZ_R with s rational in $[\frac{1}{2}, 1]$ are pairwise disjoint subsets of $B(0, R)$, and each has measure $s^k|Z_R| \geq 2^{-k}|Z_R|$. Since infinitely many of them fit into a set of finite measure, $|Z_R| = 0$. Letting $R \to \infty$, we obtain $|Z| = 0$. □

Proposition 18.1.3 *Let F be a proper convex function on $N_{\mathbb{R}}$. Then F is locally Lipschitz on $\operatorname{ri}(\operatorname{dom} F)$, and $\partial F(x) \neq \varnothing$ for every $x \in \operatorname{ri}(\operatorname{dom} F)$. Now let F be a finite convex function on an open convex set $\Omega \subset N_{\mathbb{R}}$. Then the following hold.*

(1) *The sets $\partial F(x)$, $x \in \Omega$, are compact and bounded uniformly for x in each compact subset of Ω, and the graph $\{(x, p) : x \in \Omega,\ p \in \partial F(x)\}$ is closed in $\Omega \times M_{\mathbb{R}}$.*
(2) *For $x \in \Omega$ and $v \in N_{\mathbb{R}}$,*

$$F'(x; v) = \lim_{t \downarrow 0} \frac{F(x + tv) - F(x)}{t} = \max_{p \in \partial F(x)} \langle p, v \rangle.$$

(3) *F is differentiable at $x \in \Omega$ if and only if $\partial F(x)$ is a singleton; in that case $\partial F(x) = \{\nabla F(x)\}$, where $\nabla F(x) \in M_{\mathbb{R}}$ denotes the differential of F at x.*
(4) *F is differentiable almost everywhere on Ω. More precisely, the set Ω_F of points of Ω at which F is differentiable is a Borel set whose complement in Ω is a null set, and $\nabla F \colon \Omega_F \to M_{\mathbb{R}}$ is a Borel map.*

Proof We first show that F is bounded, hence Lipschitz, near each point of $\operatorname{ri}(\operatorname{dom} F)$, and find a subgradient there by separation; then we prove (1)–(4). Let A be the affine span of $\operatorname{dom} F$, of dimension k, and let $x \in \operatorname{ri}(\operatorname{dom} F)$. Choose affine coordinates on A centered at x, with the associated Euclidean norm $|\cdot|$; since all norms on A are equivalent, local Lipschitz continuity does not depend on this choice. For small $\delta > 0$ the cube Q of points of A whose coordinates are at most 2δ in absolute value lies in $\operatorname{dom} F$. Every point of Q is a convex combination of the 2^k vertices of Q; this follows by induction on k, since writing the last coordinate of a point as a convex combination of $\pm 2\delta$ exhibits the point as a convex combination of two points of the opposite faces of Q, which are cubes of dimension $k - 1$. By the finite form of Jensen's inequality, $F \leq C$ on Q, where C is the largest value of F at a vertex. If $y \in Q$, then also $2x - y \in Q$, and $F(x) \leq \frac{1}{2}(F(y) + F(2x - y))$ gives $F(y) \geq 2F(x) - C$. Thus F is bounded on Q, and by (18.1) it is Lipschitz on the cube of half-side δ centered at x.

Next we construct a subgradient at x. Since $\operatorname{dom} F \subset A$, the epigraph of F lies in $A \times \mathbb{R}$; let E be its closure, a closed convex subset of $A \times \mathbb{R}$. Because F is continuous at x as a function on A, the points $(x, F(x) - \varepsilon)$, $\varepsilon > 0$, do not belong to

E, so $(x, F(x))$ is a boundary point of E in $A \times \mathbb{R}$. By Lemma 18.1.2(2), applied in $A \times \mathbb{R}$ with origin $(x, F(x))$, there are a linear functional a on the linear subspace $A_0 = \{y - y' : y, y' \in A\}$ parallel to A and a real number c, not both zero, such that

$$\langle a, y - x\rangle + c\,(s - F(x)) \geq 0 \qquad \text{for all } (y, s) \in E.$$

Letting $s \to +\infty$ shows that $c \geq 0$. If $c = 0$, then $a \neq 0$ and $\langle a, y - x\rangle \geq 0$ for all $y \in \operatorname{dom} F$; but $\operatorname{dom} F$ contains a ball of A about x, and this ball contains points y with $\langle a, y - x\rangle < 0$. Hence $c > 0$, and taking $s = F(y)$ we obtain $F(y) \geq F(x) + \langle p_A, y - x\rangle$ for $y \in \operatorname{dom} F$, where $p_A = -a/c$. Extend p_A to a linear functional $p \in M_{\mathbb{R}}$. The inequality $F(y) \geq F(x) + \langle p, y - x\rangle$ holds for $y \in \operatorname{dom} F$, because then $y - x \in A_0$, and also for $y \notin \operatorname{dom} F$, where $F(y) = +\infty$. Thus $p \in \partial F(x)$.

From now on let F be a finite convex function on the open convex set Ω, so that $A = N_{\mathbb{R}}$ and $\operatorname{ri}(\operatorname{dom} F) = \Omega$. Fix a basis $e_1, \ldots, e_n$ of $N_{\mathbb{R}}$, let $K \subset \Omega$ be compact, and choose $r > 0$ such that the set $K_r = \{y + \sum_i s_i e_i : y \in K,\ |s_i| \leq r\}$ lies in Ω. If $x \in K$ and $p \in \partial F(x)$, the subgradient inequality at x, evaluated at the points $x \pm r e_i$, gives $\pm r\langle p, e_i\rangle \leq F(x \pm r e_i) - F(x)$, hence $|\langle p, e_i\rangle| \leq 2r^{-1} \max_{K_r} |F|$, where F is bounded on the compact set K_r because it is continuous. This is the uniform bound in (1). If $x_k \to x$ in Ω, $p_k \in \partial F(x_k)$ and $p_k \to p$, then letting $k \to \infty$ in $F(y) \geq F(x_k) + \langle p_k, y - x_k\rangle$ and using that F is continuous at x, we get $p \in \partial F(x)$. Thus the graph is closed, and each $\partial F(x)$ is closed and bounded, hence compact.

For (2) and (4) we use the following slope inequalities for a convex function ϕ on an interval: if $s_1 < s_2 < s_3$, then

$$\frac{\phi(s_2) - \phi(s_1)}{s_2 - s_1} \leq \frac{\phi(s_3) - \phi(s_1)}{s_3 - s_1} \leq \frac{\phi(s_3) - \phi(s_2)}{s_3 - s_2}. \tag{18.2}$$

Indeed, $s_2 = \lambda s_1 + (1 - \lambda) s_3$ with $\lambda = (s_3 - s_2)/(s_3 - s_1) \in (0, 1)$, and the convexity inequality $\phi(s_2) \leq \lambda\phi(s_1) + (1 - \lambda)\phi(s_3)$ is equivalent to each of the two inequalities. Now fix x and v, and apply (18.2) to $\phi(t) = F(x + tv)$, which is convex on the open interval of those t with $x + tv \in \Omega$: with $s_1 = 0$ it shows that the difference quotient in (2) is nondecreasing in $t > 0$, so its limit as $t \downarrow 0$ exists in $[-\infty, +\infty)$. Each $p \in \partial F(x)$ gives $F(x + tv) - F(x) \geq t\langle p, v\rangle$; hence $F'(x; v) \geq \max_{p \in \partial F(x)} \langle p, v\rangle$, the maximum being attained by compactness. Conversely, choose $p_t \in \partial F(x + tv)$. The subgradient inequality at $x + tv$, evaluated at x, gives $F(x + tv) - F(x) \leq t\langle p_t, v\rangle$. By (1) the vectors p_t stay bounded as $t \downarrow 0$, and each of their cluster points lies in $\partial F(x)$. Therefore $F'(x; v) \leq \overline{\lim}_{t \downarrow 0} \langle p_t, v\rangle \leq \max_{p \in \partial F(x)} \langle p, v\rangle$.

For (3), suppose that $\partial F(x) = \{p\}$, and choose $p_h \in \partial F(x + h)$. By (1), every cluster point of p_h as $h \to 0$ lies in $\partial F(x)$, so $p_h \to p$. The subgradient inequalities at x and at $x + h$ give

$$0 \leq F(x + h) - F(x) - \langle p, h\rangle \leq \langle p_h - p, h\rangle = o(|h|),$$

so F is differentiable at x, with $\nabla F(x) = p$. Conversely, if F is differentiable at x and $p \in \partial F(x)$, dividing $F(x + tv) - F(x) \geq t\langle p, v\rangle$ by t and letting $t \downarrow 0$

gives $\langle \nabla F(x), v\rangle \geq \langle p, v\rangle$ for every v, hence $p = \nabla F(x)$. As $\partial F(x) \neq \varnothing$, it equals $\{\nabla F(x)\}$.

For (4), fix i, and write $D_i^+F(x) = F'(x; e_i)$ and $D_i^-F(x) = -F'(x; -e_i)$ for the one-sided partial derivatives; they are finite by (2). Let ϕ be the restriction of F to the intersection of Ω with a line parallel to e_i, parametrized as $\phi(s) = F(y + se_i)$ on an open interval. By (18.2), ϕ has one-sided derivatives $\phi'_\pm$ at every point, $\phi'_-(s) \leq \phi'_+(s)$, and $\phi'_+(s) \leq \phi'_-(s')$ whenever $s < s'$, because both are bounded by the slope of the chord from s to s'. Hence the open intervals $(\phi'_-(s), \phi'_+(s))$ are pairwise disjoint; each nonempty one contains a rational number, so only countably many of them are nonempty, and $\phi'_- = \phi'_+$ outside a countable set. Next, D_i^+F is the pointwise limit on Ω of the Borel functions equal to $k(F(x + e_i/k) - F(x))$ where $x + e_i/k \in \Omega$ and to 0 elsewhere, so it is a Borel function, and similarly for D_i^-F. Thus the set $Z_i \subset \Omega$ where $D_i^+F \neq D_i^-F$ is a Borel set that meets each line parallel to e_i in a countable set, and Fubini's theorem, in the coordinates of the basis $e_1, \ldots, e_n$, shows that Z_i is a null set. At a point $x \notin Z_1 \cup \cdots \cup Z_n$, (2) gives

$$\max_{p \in \partial F(x)} \langle p, e_i\rangle = D_i^+F(x) = D_i^-F(x) = \min_{p \in \partial F(x)} \langle p, e_i\rangle$$

for every i, so all subgradients at x coincide, and F is differentiable at x by (3). Conversely, if F is differentiable at x, then $D_i^+F(x) = D_i^-F(x) = \langle \nabla F(x), e_i\rangle$ for every i. Therefore $\Omega_F = \Omega \setminus (Z_1 \cup \cdots \cup Z_n)$ is a Borel set with null complement, and the coordinates $\langle \nabla F, e_i\rangle = D_i^+F$ of ∇F are Borel functions on Ω_F. □

Proposition 18.1.4 *Let F be a proper convex function on $N_\mathbb{R}$, and let $\overline{F}$ be its lower semicontinuous regularization, that is, the largest lower semicontinuous function on $N_\mathbb{R}$ that is at most F. Then $\overline{F}$ is a closed proper convex function, and $\overline{F} = F$ on* $\operatorname{ri}(\operatorname{dom} F)$. *A closed proper convex function is determined by its values on the relative interior of its effective domain. Finally, if a sequence of finite convex functions on an open convex set $\Omega \subset N_\mathbb{R}$ converges to a finite limit at each point of a dense subset of Ω, then it converges locally uniformly on Ω to a finite convex function.*

Proof By Proposition 18.1.3, F has a subgradient p at some point $x_1 \in \operatorname{ri}(\operatorname{dom} F)$, so F lies above the affine function $\ell(y) = F(x_1) + \langle p, y - x_1\rangle$. The closure E of epi F is convex, and it lies above the graph of ℓ because ℓ is continuous. Hence, for each x, the set $\{r : (x, r) \in E\}$ is closed and bounded below by $\ell(x)$, and with each of its points r it contains $[r, +\infty)$, because epi F has this property. So this set is either empty or a closed half-line $[\widetilde{F}(x), +\infty)$ with $\widetilde{F}(x) \geq \ell(x) > -\infty$. Setting $\widetilde{F}(x) = +\infty$ in the first case, we see that E is the epigraph of a convex function $\widetilde{F}$. It is lower semicontinuous because its epigraph is closed, and $\widetilde{F} \leq F$, so $\operatorname{dom} \widetilde{F} \supset \operatorname{dom} F \neq \varnothing$. If $G \leq F$ is lower semicontinuous, then epi G is closed and contains epi F, hence also E, and so $G \leq \widetilde{F}$. Thus $\widetilde{F} = \overline{F}$, which is therefore closed, proper and convex.

Let $x \in \operatorname{ri}(\operatorname{dom} F)$. If $(y_k, r_k) \in \operatorname{epi} F$ converge to (x, r), then y_k lies in the affine span of $\operatorname{dom} F$, on which F is continuous at x; hence $r \geq \lim_k F(y_k) = F(x)$. This shows $\overline{F}(x) \geq F(x)$, and so $\overline{F}(x) = F(x)$.

Now let F be closed, fix $x_0 \in \operatorname{ri}(\operatorname{dom} F)$, and let $x \in \overline{\operatorname{dom} F}$. By Lemma 18.1.2(4), the points $(1 - t)x_0 + tx$ with $0 \leq t < 1$ lie in $\operatorname{ri}(\operatorname{dom} F)$, so the convex function

$\phi(t) = F((1-t)x_0+tx)$ is finite on $[0,1)$. Convexity gives $\phi(t) \le (1-t)\phi(0)+t\phi(1)$, hence $\overline{\lim}_{t\uparrow 1}\phi(t) \le \phi(1)$, while lower semicontinuity gives $\underline{\lim}_{t\uparrow 1}\phi(t) \ge \phi(1)$. Therefore

$$F(x) = \lim_{t\uparrow 1} F((1-t)x_0 + tx) \qquad (x \in \overline{\operatorname{dom} F}),$$

also when $F(x) = +\infty$. Since $F = +\infty$ outside $\overline{\operatorname{dom} F}$, which is the closure of $\operatorname{ri}(\operatorname{dom} F)$, the function F is determined by its restriction to $\operatorname{ri}(\operatorname{dom} F)$.

Finally, let (F_j) be a sequence of finite convex functions on Ω converging to a finite limit at every point of a dense set $S \subset \Omega$. We show that near each point of Ω the functions F_j are uniformly bounded, hence uniformly Lipschitz, so that convergence on S propagates to uniform convergence. Fix a basis of $N_{\mathbb{R}}$, write $Q(y,\rho)$ for the closed cube of half-side ρ centered at y in the corresponding coordinates, and let $x \in \Omega$ and $\rho > 0$ with $Q(x,2\rho) \subset \Omega$. For each vertex y of $Q(x,\rho)$ choose a point of S whose coordinates differ from those of y by less than $\rho/2$, and let $S_0 \subset S \cap Q(x,2\rho)$ be the finite set of these points. Its convex hull H is compact as the image of a simplex under a linear map, and it contains $Q(x,\rho/2)$. Indeed, let ℓ be a nonzero linear functional, $\ell(z) = \sum_i a_i z_i$ in the chosen coordinates, and put $\|a\| = \sum_i |a_i|$. At a vertex y^* of $Q(x,\rho)$ where ℓ is maximal, $\ell(y^*) = \ell(x) + \rho\|a\|$, and the point $s \in S_0$ chosen for y^* satisfies $\ell(s) > \ell(y^*) - \frac{\rho}{2}\|a\| = \max_{Q(x,\rho/2)}\ell$. Hence no point of $Q(x,\rho/2)$ can be separated from H as in Lemma 18.1.2(1).

The sequences $(F_j(s))_j$, $s \in S_0$, converge, so they are bounded above by some C, and the finite form of Jensen's inequality gives $F_j \le C$ on H for all j. Choose $s_0 \in S \cap Q(x,\rho/8)$. Then $Q(s_0,3\rho/8) \subset Q(x,\rho/2)$, and for $y \in Q(s_0,3\rho/8)$ the reflected point $2s_0 - y$ lies in the same cube, so that $F_j(y) \ge 2F_j(s_0) - C \ge 2\inf_k F_k(s_0) - C$. Hence $|F_j| \le B$ on $Q(x,\rho/4) \subset Q(s_0,3\rho/8)$ for all j, with a constant B. By (18.1), applied with $\delta = \rho/8$ and $\omega \le 2B$, the F_j are Lipschitz on $Q(x,\rho/8)$ with the common constant $L = 16B/\rho$, for the Euclidean norm $|\cdot|$ of the chosen coordinates.

Given $\varepsilon > 0$, choose a finite set $S_\varepsilon \subset S \cap Q(x,\rho/8)$ such that every point of $Q(x,\rho/8)$ lies within distance ε/L of S_ε. To this end, cover the compact cube $Q(x,\rho/8)$ by finitely many balls of radius $\varepsilon/(2L)$ centered in the cube, and pick a point of the dense set S in the intersection of each ball with the interior of the cube, a nonempty open set. For all sufficiently large j,k we have $|F_j - F_k| < \varepsilon$ on S_ε, hence $|F_j - F_k| < 3\varepsilon$ on $Q(x,\rho/8)$. Thus (F_j) is uniformly Cauchy on $Q(x,\rho/8)$. Finitely many such cubes cover any given compact subset of Ω, so (F_j) converges locally uniformly to a finite continuous function, which is convex because the convexity inequalities pass to pointwise limits. □

18.1.1 Invariant functions on tubes and complex tori

We now prove the correspondence between invariant plurisubharmonic functions and convex functions described at the beginning of the chapter. For a nonempty open convex set $V \subset \mathbb{R}^n$, $n \ge 1$, consider the tube

$$V + \mathrm{i}\mathbb{R}^n = \{t + \mathrm{i}s : t \in V,\ s \in \mathbb{R}^n\} \subset \mathbb{C}^n,$$

with coordinates $w = t + \mathrm{i}s$. Recall from Section 8.5, here without the variables z, that a function U on the tube is invariant under imaginary translations if $U(w + \mathrm{i}s') = U(w)$ for all $s' \in \mathbb{R}^n$, that is, if $U(t + \mathrm{i}s)$ depends only on t. As observed there, the complex Hessian of a smooth function of this kind is a quarter of its real Hessian in t, so plurisubharmonicity becomes convexity; the general case follows by smoothing. In addition, local integrability of U excludes the value $-\infty$.

Proposition 18.1.5 *Let $n \geq 1$, let $V \subset \mathbb{R}^n$ be a nonempty open convex set, and let $U \in \mathrm{PSH}(V + \mathrm{i}\mathbb{R}^n)$ be invariant under imaginary translations. Then there is a finite convex function $f : V \to \mathbb{R}$ such that $U(t + \mathrm{i}s) = f(t)$ for all $t \in V$ and $s \in \mathbb{R}^n$. Conversely, for every finite convex function f on V, the function $f(\operatorname{Re} w)$ is plurisubharmonic on $V + \mathrm{i}\mathbb{R}^n$ and invariant under imaginary translations.*

Proof Define $f(t) = U(t)$, identifying $t \in V$ with $t+\mathrm{i}0$; by invariance, $U(t+\mathrm{i}s) = f(t)$. By Theorem 8.2.1, U is locally integrable, so the set $\{U = -\infty\} = \{f = -\infty\} + \mathrm{i}\mathbb{R}^n$ is a Lebesgue null set in $\mathbb{C}^n$. Applying Fubini's theorem on products of compact boxes in t and in s, we see that $\{f = -\infty\}$ is a null set in V.

For $\varepsilon > 0$ let $V_\varepsilon = \{t \in V : \operatorname{dist}(t, \mathbb{R}^n \setminus V) > \varepsilon\}$. A point t lies in V_ε if and only if the closed ball $\overline{B(t,\varepsilon)}$ lies in V, so V_ε is the intersection of the convex sets $V - y$, $|y| \leq \varepsilon$, and is convex. The points of the tube at distance more than ε from its complement form the tube $V_\varepsilon + \mathrm{i}\mathbb{R}^n$. On this set let $U_\varepsilon = U * \rho_\varepsilon$, where ρ_ε is the radial mollifier of Section 1.4 on $\mathbb{C}^n = \mathbb{R}^{2n}$. The function U_ε is smooth, and its complex Hessian matrix $(\partial^2 U_\varepsilon / \partial w_j \partial \bar{w}_k)$ is positive semidefinite. Indeed, for $\xi \in \mathbb{C}^n$, the function $\sum_{j,k} (\partial^2 U_\varepsilon / \partial w_j \partial \bar{w}_k) \xi_j \bar{\xi}_k$ is, by Proposition 1.4.3(1), the mollification of the distribution $\sum_{j,k} (\partial^2 U / \partial w_j \partial \bar{w}_k) \xi_j \bar{\xi}_k$, which is positive by Theorem 8.2.1 and the matrix criterion of Section 7.3, so it is nonnegative by Proposition 1.4.3(3).

Moreover, $U_\varepsilon \downarrow U$ as $\varepsilon \downarrow 0$ by (3.3), because U is subharmonic and locally integrable (Theorem 8.2.1). Since imaginary translations commute with convolution, U_ε is invariant under them; write $U_\varepsilon(t + \mathrm{i}s) = f_\varepsilon(t)$. For a function of $t = \operatorname{Re} w$ alone we have $\partial/\partial w_j = \frac{1}{2}\partial/\partial t_j = \partial/\partial \bar{w}_j$, hence

$$\frac{\partial^2 U_\varepsilon}{\partial w_j \partial \bar{w}_k} = \frac{1}{4} \frac{\partial^2 f_\varepsilon}{\partial t_j \partial t_k}.$$

The left side is a positive semidefinite matrix, so f_ε has positive semidefinite real Hessian. Hence the restriction of f_ε to every segment in the convex set V_ε has nonnegative second derivative, and f_ε is convex on V_ε.

Suppose that $f(t_0) = -\infty$ for some $t_0 \in V$, and choose $r \in (0,1)$ with $\overline{B(t_0, 3r)} \subset V$. The upper semicontinuous function U is bounded above, say by C, on the compact set $\overline{B(t_0, 3r)} + \mathrm{i}[-1,1]^n$. Since $U_\varepsilon(w)$ is an average of U over the ball $\overline{B(w,\varepsilon)}$, we get $U_\varepsilon \leq C$ on $B(t_0, 2r)$ for $\varepsilon < r$; moreover $B(t_0, 2r) \subset V_\varepsilon$ for such ε. For $t \in B(t_0, r)$, the point $2t - t_0$ lies in $B(t_0, 2r)$, and t is the midpoint of t_0 and $2t - t_0$. Convexity of f_ε therefore gives

$$f_\varepsilon(t) \le \tfrac{1}{2} f_\varepsilon(t_0) + \tfrac{1}{2} f_\varepsilon(2t - t_0) \le \tfrac{1}{2} f_\varepsilon(t_0) + \tfrac{1}{2} C.$$

As $\varepsilon \downarrow 0$, $f_\varepsilon(t_0) \downarrow f(t_0) = -\infty$ and $f_\varepsilon(t) \downarrow f(t)$, so $f = -\infty$ on $B(t_0, r)$. This contradicts the fact that $\{f = -\infty\}$ is a null set. Hence f is finite. Finally, every segment $[a, b] \subset V$ is compact and therefore lies in V_ε for small ε; the convexity inequalities for f_ε on $[a, b]$ pass to the pointwise limit, so f is convex.

Conversely, let f be a finite convex function on V. By Proposition 18.1.3, f is continuous and has a subgradient $p \in \mathbb{R}^n$ at every point. Let $w_0 = t_0 + \mathrm{i}s_0$ be a point of the tube, $a \in \mathbb{C}^n \setminus \{0\}$, and $r > 0$ such that the circle $\{w_0 + r\mathrm{e}^{\mathrm{i}\theta} a : \theta \in [0, 2\pi]\}$ lies in the tube. If p is a subgradient of f at t_0, then

$$f\big(\operatorname{Re}(w_0 + r\mathrm{e}^{\mathrm{i}\theta} a)\big) \ge f(t_0) + r\,\big\langle p, \operatorname{Re}(\mathrm{e}^{\mathrm{i}\theta} a)\big\rangle = f(t_0) + r \operatorname{Re}\Big(\mathrm{e}^{\mathrm{i}\theta} \sum_j p_j a_j\Big).$$

The last term has mean zero over $\theta \in [0, 2\pi]$, so the mean of the left side over the circle is at least $f(t_0) = f(\operatorname{Re} w_0)$. Since w_0, a and r are arbitrary, the restriction of the continuous function $f(\operatorname{Re} w)$ to every affine complex line satisfies the submean inequality of Definition 3.1.1 at every point, and it is therefore subharmonic. As $f(\operatorname{Re} w)$ is finite and continuous and the tube is a domain, Definition 8.1.1 shows that $f(\operatorname{Re} w)$ is plurisubharmonic. It is invariant under imaginary translations by construction. □

Corollary 18.1.6 *Let $n \ge 1$, let $V \subset \mathbb{R}^n$ be a nonempty open convex set, and put*

$$\Omega_V = \{z \in (\mathbb{C}^*)^n : (\log|z_1|^2, \dots, \log|z_n|^2) \in V\}.$$

Let $\varphi \in \mathrm{PSH}(\Omega_V)$ be invariant under the action of the compact torus $(S^1)^n$ by separate rotations of the coordinates, $(z_1, \dots, z_n) \mapsto (\mathrm{e}^{\mathrm{i}\theta_1} z_1, \dots, \mathrm{e}^{\mathrm{i}\theta_n} z_n)$. Then there is a finite convex function f on V such that

$$\varphi(z) = f(\log|z_1|^2, \dots, \log|z_n|^2) \qquad (z \in \Omega_V).$$

Conversely, for every finite convex function f on V this formula defines an $(S^1)^n$-invariant plurisubharmonic function on Ω_V. In particular, an $(S^1)^n$-invariant plurisubharmonic function on $(\Delta^*)^n$ or on $(\mathbb{C}^*)^n$ is finite everywhere, where $\Delta^* = \{\zeta \in \mathbb{C} : 0 < |\zeta| < 1\}$.

Proof Consider the holomorphic map

$$E\colon \mathbb{C}^n \to (\mathbb{C}^*)^n, \qquad E(w) = (\exp(w_1/2), \dots, \exp(w_n/2)).$$

Since $\log|E_j(w)|^2 = \operatorname{Re} w_j$, it maps the tube $V + \mathrm{i}\mathbb{R}^n$ onto Ω_V; in particular, Ω_V is connected. By Proposition 8.2.3, $\varphi \circ E$ is either plurisubharmonic or identically $-\infty$ on the tube, and the second alternative is impossible because E is surjective and φ is nontrivial. Since $E(w + \mathrm{i}s') = (\mathrm{e}^{\mathrm{i}s_1'/2} E_1(w), \dots, \mathrm{e}^{\mathrm{i}s_n'/2} E_n(w))$, the rotation invariance of φ makes $\varphi \circ E$ invariant under imaginary translations. Proposition 18.1.5 now gives a finite convex function f with $\varphi(E(w)) = f(\operatorname{Re} w)$, which is the asserted formula.

Conversely, let f be finite and convex. Then $f(\operatorname{Re} w)$ is plurisubharmonic on the tube by Proposition 18.1.5. Near each point of Ω_V, the map E has the holomorphic local inverse $z \mapsto (2\log z_1, \dots, 2\log z_n)$, defined with local branches of the logarithm, and $f(\log|z_1|^2, \dots, \log|z_n|^2)$ is the composition of $f(\operatorname{Re} w)$ with this inverse. By Proposition 8.2.3 it is plurisubharmonic near that point, since it is finite. Plurisubharmonicity is a local property, so the function is plurisubharmonic on Ω_V, and it is $(S^1)^n$-invariant because it depends only on $|z_1|, \dots, |z_n|$. The domains $(\Delta^*)^n$ and $(\mathbb{C}^*)^n$ are Ω_V for $V = (-\infty, 0)^n$ and $V = \mathbb{R}^n$. □

18.2 Legendre duality and convex envelopes

The Legendre transform exchanges convex functions on $N_{\mathbb{R}}$ and on $M_{\mathbb{R}}$, and points with slopes: up to closure, the slopes of F are the points where F^* is finite. We prove the duality theorem and compute the conjugates of envelopes and sums.

Definition 18.2.1 The *Legendre transform*, or *convex conjugate*, of a function $F\colon N_{\mathbb{R}} \to [-\infty, +\infty]$ is the function on $M_{\mathbb{R}}$ given by

$$F^*(p) = \sup_{x \in N_{\mathbb{R}}} (\langle p, x\rangle - F(x)).$$

The conjugate of a function on $M_{\mathbb{R}}$ is defined in the same way and is a function on $N_{\mathbb{R}}$; we write $F^{**} = (F^*)^*$. The *convex indicator function* χ_C of a nonempty set C, in either space, is 0 on C and $+\infty$ elsewhere. The *support function* of a nonempty set $P \subset M_{\mathbb{R}}$ is

$$h_P(x) = \sup_{p \in P} \langle p, x\rangle \qquad (x \in N_{\mathbb{R}});$$

it is convex with values in $(-\infty, +\infty]$, and it is finite when P is bounded.

Theorem 18.2.2 *Let F be a proper convex function on $N_{\mathbb{R}}$. Then F^* is a closed proper convex function on $M_{\mathbb{R}}$, and*

$$F^{**} = \overline{F}, \qquad (\overline{F})^* = F^*.$$

If F is closed, then for $x \in N_{\mathbb{R}}$ and $p \in M_{\mathbb{R}}$ the following conditions are equivalent:

$$p \in \partial F(x), \qquad x \in \partial F^*(p), \qquad F(x) + F^*(p) = \langle p, x\rangle;$$

moreover,

$$\operatorname{ri}(\operatorname{dom} F^*) \subset \partial F(N_{\mathbb{R}}) \subset \operatorname{dom} F^*, \qquad \overline{\partial F(N_{\mathbb{R}})} = \overline{\operatorname{dom} F^*}.$$

Finally, for every nonempty closed convex set $P \subset M_{\mathbb{R}}$, the support function and the convex indicator function are conjugate: $h_P = \chi_P^*$ and $\chi_P = h_P^*$.

Proof The conjugate is the supremum of the continuous affine functions $p \mapsto \langle p, x\rangle - F(x), x \in \operatorname{dom} F$, so it is convex and lower semicontinuous. Since $\operatorname{dom} F \neq \varnothing$, we have $F^*(p) \geq \langle p, x_0\rangle - F(x_0) > -\infty$ for any $x_0 \in \operatorname{dom} F$. By Proposition 18.1.3, F has a subgradient p_0 at some point $x_1 \in \operatorname{ri}(\operatorname{dom} F)$, and then $F^*(p_0) = \langle p_0, x_1\rangle - F(x_1) < +\infty$. Thus F^* is closed and proper. The definition of F^* gives $F(x) \geq \langle p, x\rangle - F^*(p)$ for all p, that is, $F^{**} \leq F$.

We next show that $F^{**} = F$ when F is closed. An affine function $\langle p, \cdot\rangle - c$ lies below F if and only if $c \geq F^*(p)$. Hence F^{**} is the supremum of all affine minorants of F, and it suffices to find, for every x and every real $r < F(x)$, an affine minorant of F whose value at x exceeds r. The point (x, r) does not belong to the closed convex set $\operatorname{epi} F$, so by Lemma 18.1.2(1), applied in $N_{\mathbb{R}} \times \mathbb{R}$, there are $a \in M_{\mathbb{R}}$, $c \in \mathbb{R}$ and $\delta > 0$ with

$$\langle a, y\rangle + cs \geq \langle a, x\rangle + cr + \delta \qquad \text{for all } (y, s) \in \operatorname{epi} F.$$

Letting $s \to +\infty$ shows that $c \geq 0$. If $c > 0$, taking $s = F(y)$ gives the affine minorant $y \mapsto r + (\delta - \langle a, y - x\rangle)/c$, whose value at x is $r + \delta/c > r$. If $c = 0$, then $\langle a, y\rangle \geq \langle a, x\rangle + \delta$ for all $y \in \operatorname{dom} F$. Let L_0 be an affine minorant of F, which exists by the first paragraph. For $k > 0$ the affine functions

$$L_0(y) + k\big(\langle a, x\rangle + \delta/2 - \langle a, y\rangle\big)$$

are again minorants of F, since the added term is at most $-k\delta/2$ on $\operatorname{dom} F$, and their values at x tend to $+\infty$ as $k \to \infty$. In both cases the required minorant exists, so $F^{**} = F$. For a general proper convex F, a continuous affine function lies below F if and only if it lies below the largest lower semicontinuous minorant $\overline{F}$. Therefore $F^* = (\overline{F})^*$, and since $\overline{F}$ is closed and proper by Proposition 18.1.4, $F^{**} = (\overline{F})^{**} = \overline{F}$.

Let F be closed. By definition, $p \in \partial F(x)$ means that $x \in \operatorname{dom} F$ and $\langle p, y\rangle - F(y) \leq \langle p, x\rangle - F(x)$ for all y, that is, the supremum defining $F^*(p)$ is attained at x; equivalently, $F(x) + F^*(p) = \langle p, x\rangle$. (If $x \notin \operatorname{dom} F$, both conditions fail, since then $F(x) + F^*(p) = +\infty$.) Applying this to the closed proper function F^* and using $F^{**} = F$, we find that $x \in \partial F^*(p)$ if and only if $F^*(p) + F(x) = \langle p, x\rangle$. This proves the equivalences. If $p \in \operatorname{ri}(\operatorname{dom} F^*)$, then F^* has a subgradient x at p by Proposition 18.1.3, and $p \in \partial F(x)$ by the equivalences; this is the first inclusion. If $p \in \partial F(x)$, then $F^*(p) = \langle p, x\rangle - F(x) < +\infty$, which gives the second. The closures coincide because $\operatorname{ri}(\operatorname{dom} F^*)$ is dense in $\operatorname{dom} F^*$ by Lemma 18.1.2(4). Finally, $\chi_P^* = h_P$ by the definitions, and χ_P is a closed proper convex function, so $h_P^* = \chi_P^{**} = \chi_P$. □

Proposition 18.2.3 (1) *The pointwise supremum of a family of convex functions is convex. The pointwise limit of a decreasing net of convex functions is convex; it may take the value* $-\infty$.

(2) *Let* $(F_i)_{i\in I}$ *be a nonempty family of proper convex functions on* $N_{\mathbb{R}}$, *and put*

$$H(x) = \inf \sum_{j=1}^{q} \lambda_j F_{i_j}(x_j),$$

where the infimum is taken over the values $\sum_j \lambda_j F_{i_j}(x_j)$ *of all* admissible combinations *for* x*, that is, over all* $q \geq 1$*,* $i_j \in I$*,* $x_j \in \operatorname{dom} F_{i_j}$ *and* $\lambda_j > 0$ *with* $\sum_j \lambda_j = 1$ *and* $\sum_j \lambda_j x_j = x$*; we set* $H(x) = +\infty$ *if no such combination exists. Then* H*, the* convex envelope *of the family, is the largest convex function below* $\inf_i F_i$*, and* $H^* = \sup_i F_i^*$*. If* H *has an affine minorant, then its lower semicontinuous regularization* $\overline{H}$ *is the largest lower semicontinuous convex function below* $\inf_i F_i$*, and* $\overline{H} = (\sup_i F_i^*)^*$*.*

(3) *For proper convex functions* F, G *on* $N_{\mathbb{R}}$*, the* infimal convolution

$$(F \,\square\, G)(x) = \inf_{y+z=x} (F(y) + G(z))$$

is convex, and $(F \,\square\, G)^* = F^* + G^*$*.*

(4) *Let* F, G *be finite convex functions on* $N_{\mathbb{R}}$ *and* $p \in M_{\mathbb{R}}$*. Then*

$$(F + G)^*(p) = \min_{q+r=p} (F^*(q) + G^*(r))$$

whenever the right-hand side is finite, the minimum being attained; and $(F + G)^*(p) = +\infty$ *if* $F^*(q) + G^*(r) = +\infty$ *for every decomposition* $p = q + r$*.*

Proof (1) The epigraph of a supremum is the intersection of the epigraphs. Let (F_i) be a decreasing net of convex functions with limit $F = \inf_i F_i$, let $x, y \in N_{\mathbb{R}}$ and $0 < \lambda < 1$, and let $a > F(x)$ and $b > F(y)$ be real numbers. Choose indices i_1, i_2 with $F_{i_1}(x) < a$ and $F_{i_2}(y) < b$, and an index i following both. Then $F_i(x) < a$ and $F_i(y) < b$, and convexity of $\operatorname{epi} F_i$ gives

$$F((1-\lambda)x + \lambda y) \leq F_i((1-\lambda)x + \lambda y) \leq (1-\lambda)a + \lambda b.$$

Since a and b are arbitrary, this shows that $\operatorname{epi} F$ is convex; this also covers the case where $F(x)$ or $F(y)$ equals $-\infty$.

(2) If $H(x) < a$ and $H(x') < a'$, we can choose admissible combinations for x and x' with values below a and a'; mixing them with weights $1-\lambda$ and λ gives an admissible combination for $(1-\lambda)x + \lambda x'$ with value below $(1-\lambda)a + \lambda a'$. Hence H is convex. Combinations with $q = 1$ show $H \leq F_i$ for each i. If G is convex and $G \leq F_i$ for all i, then for every admissible combination the points $(x_j, F_{i_j}(x_j))$ lie in $\operatorname{epi} G$, and so does their convex combination with the weights λ_j, since a convex set contains the convex combinations of its points; that is,

$$G\Big(\sum_j \lambda_j x_j\Big) \leq \sum_j \lambda_j F_{i_j}(x_j).$$

Hence $G \leq H$. For the conjugate, the supremum defining $H^*(p)$ is the supremum over all admissible combinations of

$$\Big\langle p, \sum_j \lambda_j x_j \Big\rangle - \sum_j \lambda_j F_{i_j}(x_j) = \sum_j \lambda_j \big(\langle p, x_j\rangle - F_{i_j}(x_j)\big) \leq \sup_i F_i^*(p),$$

and combinations with $q = 1$ give equality. Since affine functions are convex, an affine function lies below H if and only if it lies below $\inf_i F_i$. Suppose that H has an affine minorant. Then H is proper, so $\overline{H}$ is a closed proper convex function by Proposition 18.1.4, and $\overline{H} = H^{**} = (\sup_i F_i^*)^*$ by Theorem 18.2.2. A lower semicontinuous convex function below $\inf_i F_i$ lies below H, and therefore below $\overline{H}$.

(3) Convexity of $F \square G$ follows as in (2), by mixing decompositions $x = y + z$ and $x' = y' + z'$ whose values $F(y) + G(z)$ are close to the infima. The two suprema over y and z in

$$(F \square G)^*(p) = \sup_{y,z}\big(\langle p, y\rangle - F(y) + \langle p, z\rangle - G(z)\big) = F^*(p) + G^*(p)$$

are independent, which gives the conjugate formula.

(4) If $n = 0$, then F and G are constants and both sides equal $-F(0) - G(0)$; so let $n \geq 1$. Let $J(p) = \inf_{q+r=p}(F^*(q) + G^*(r))$. Each conjugate is bounded below by minus its value at the origin: $F^* \geq -F(0)$ and $G^* \geq -G(0)$. Moreover, for every $R > 0$, choosing x with $|x| = R$ and $\langle q, x\rangle = R|q|$ gives

$$F^*(q) \geq R|q| - \max_{|x|\leq R} F(x),$$

and the analogous bound holds for G^*. Consequently, if p stays bounded and $F^*(q) + G^*(r)$ stays bounded above, then q and $r = p - q$ stay bounded: for instance $F^*(q) \leq F^*(q) + G^*(r) + G(0)$, and the last bound with $R = 1$ controls $|q|$. Lower semicontinuity of F^* and G^* therefore implies that the infimum defining $J(p)$ is attained whenever it is finite: a minimizing sequence of decompositions has a convergent subsequence. The same argument, applied to $p_k \to p$ and decompositions attaining $J(p_k)$, shows that J is lower semicontinuous. By (3), $J = F^* \square G^*$ is convex; it is bounded below by $-F(0) - G(0)$, and it is proper, since $J(q_0 + r_0) \leq F^*(q_0) + G^*(r_0) < +\infty$ for subgradients $q_0 \in \partial F(0)$ and $r_0 \in \partial G(0)$. Finite convex functions are continuous, so F and G are closed and $F^{**} = F$, $G^{**} = G$. By (3), $J^* = F^{**} + G^{**} = F + G$, and Theorem 18.2.2 gives $J = J^{**} = (F + G)^*$. □

18.3 Slope bodies and bounded growth

We now study the finite convex functions on $N_{\mathbb{R}}$ whose growth at infinity is controlled by the support function h_P of a convex body $P \subset M_{\mathbb{R}}$. By duality, such control is a condition on the set where the conjugate is finite, that is, on the slopes of the function.

Definition 18.3.1 Let F, G be finite convex functions on $N_{\mathbb{R}}$. As for functions on a compact manifold (Section 13.5), we write $F \preceq G$ if $F \leq G + C$ on $N_{\mathbb{R}}$ for some constant C, and $F \sim G$ if $F \preceq G$ and $G \preceq F$. The closed convex set

$$K_F = \overline{\operatorname{dom} F^*} \subset M_{\mathbb{R}}$$

is nonempty, and by Theorem 18.2.2 it is the closure of the set $\partial F(N_{\mathbb{R}})$ of all supporting slopes of F; when it is compact we call it the *slope body* of F. For a convex body $P \subset M_{\mathbb{R}}$, let $\mathcal{P}(N_{\mathbb{R}}, P)$ be the set of finite convex functions F on $N_{\mathbb{R}}$ with $F \preceq h_P$, and let the *bounded class* $\mathcal{E}^\infty(N_{\mathbb{R}}, P)$ be the set of those with $F \sim h_P$. If P has nonempty interior, the *full-mass class* $\mathcal{E}(N_{\mathbb{R}}, P)$ is the set of $F \in \mathcal{P}(N_{\mathbb{R}}, P)$ with $K_F = P$; if P has empty interior, we set $\mathcal{E}(N_{\mathbb{R}}, P) = \mathcal{P}(N_{\mathbb{R}}, P)$. Finally, the *finite-energy class* $\mathcal{E}^1(N_{\mathbb{R}}, P)$ is the set of $F \in \mathcal{E}(N_{\mathbb{R}}, P)$ with $\int_P F^* \, \mathrm{d}p < +\infty$. Since $F^* \geq -F(0)$ by the definition of F^*, this integral is well defined in $(-\infty, +\infty]$.

The notation is chosen by analogy with the classes $\mathcal{E}^\infty(X, \theta, \phi) \subset \mathcal{E}^1(X, \theta, \phi) \subset \mathcal{E}(X, \theta, \phi)$ of Definitions 15.8.3 and 17.2.1, with the support function h_P in the role of the model potential ϕ.

Proposition 18.3.2 *Let $P \subset M_{\mathbb{R}}$ be a convex body and F a finite convex function on $N_{\mathbb{R}}$. The following conditions are equivalent:*

$$F \preceq h_P, \qquad \operatorname{dom} F^* \subset P, \qquad \partial F(N_{\mathbb{R}}) \subset P.$$

If they hold, then $\max_x (F(x) - h_P(x)) = F(0)$, and F is Lipschitz on $N_{\mathbb{R}}$ with constant $\max_{p \in P} |p|$. Moreover, $F \sim h_P$ if and only if F^* is bounded on P and equal to $+\infty$ outside P, and

$$\mathcal{E}^\infty(N_{\mathbb{R}}, P) \subset \mathcal{E}^1(N_{\mathbb{R}}, P) \subset \mathcal{E}(N_{\mathbb{R}}, P) \subset \mathcal{P}(N_{\mathbb{R}}, P).$$

If F, G are finite convex functions on $N_{\mathbb{R}}$ and K_F is compact, then

$$K_{F+G} = K_F + K_G.$$

In particular, if $P' \subset M_{\mathbb{R}}$ is another convex body, $K_F = P$ and $K_G = P'$, then $F + G \in \mathcal{E}(N_{\mathbb{R}}, P + P')$.

Proof If $F \leq h_P + C$, conjugation reverses the inequality and gives $F^* \geq h_P^* - C = \chi_P - C$ by Theorem 18.2.2, so $\operatorname{dom} F^* \subset P$. Every subgradient of F lies in $\operatorname{dom} F^*$ by the same theorem, so the second condition implies the third. If all slopes of F lie in P and $p \in \partial F(x)$, the subgradient inequality at x, evaluated at 0, gives

$$F(x) - F(0) \leq \langle p, x \rangle \leq h_P(x).$$

Thus $F \leq h_P + F(0)$, which proves the first condition and, since $h_P(0) = 0$, the identity $\max_x (F(x) - h_P(x)) = F(0)$. If $p_x \in \partial F(x)$ and $p_y \in \partial F(y)$, the subgradient inequalities give $\langle p_y, y - x \rangle \geq F(y) - F(x) \geq \langle p_x, y - x \rangle$, so $|F(y) - F(x)| \leq \max_{p \in P} |p| \, |y - x|$.

Conjugation reverses inequalities, and $F^{**} = F$ because the finite convex function F is continuous. Hence $h_P - C \leq F \leq h_P + C$ holds if and only if $\chi_P - C \leq F^* \leq \chi_P + C$, that is, if and only if $F^* = +\infty$ outside P and $|F^*| \leq C$ on P. In that case $K_F = P$, so $F \in \mathcal{E}(N_{\mathbb{R}}, P)$ (by the convention if P has empty interior), and $\int_P F^* \, \mathrm{d}p \leq C \operatorname{vol}(P) < \infty$; thus $\mathcal{E}^\infty \subset \mathcal{E}^1$. The inclusions $\mathcal{E}^1 \subset \mathcal{E} \subset \mathcal{P}$ hold by definition. If P has empty interior and $n > 0$, then P lies in a proper affine subspace,

because its relative interior is nonempty and open in its affine span (Lemma 18.1.2(4)); hence $\mathrm{vol}(P) = 0$, every integral over P vanishes, and $\mathcal{E}^1 = \mathcal{E} = \mathcal{P}$.

Now let F, G be finite convex functions with K_F compact. By Proposition 18.2.3(4), $(F+G)^*(p)$ is finite exactly when $p = q + r$ with $q \in \operatorname{dom} F^*$ and $r \in \operatorname{dom} G^*$, so

$$\operatorname{dom}(F+G)^* = \operatorname{dom} F^* + \operatorname{dom} G^*.$$

It remains to pass to closures. Since addition $M_{\mathbb{R}} \times M_{\mathbb{R}} \to M_{\mathbb{R}}$ is continuous, $K_F + K_G$ lies in the closure of $\operatorname{dom} F^* + \operatorname{dom} G^*$. Conversely, if $q_k + r_k \to s$ with $q_k \in \operatorname{dom} F^*$ and $r_k \in \operatorname{dom} G^*$, then a subsequence of (q_k) converges to some $q \in K_F$, because K_F is compact, and along it $r_k \to s - q \in K_G$; hence $s \in K_F + K_G$. This proves $K_{F+G} = K_F + K_G$. Finally, if $K_F = P$ and $K_G = P'$, then $F \preceq h_P$ and $G \preceq h_{P'}$ by the first part, and $h_P + h_{P'} = h_{P+P'}$; hence $F + G \in \mathcal{P}(N_{\mathbb{R}}, P + P')$ and $K_{F+G} = P + P'$, so that $F + G \in \mathcal{E}(N_{\mathbb{R}}, P + P')$. □

For smooth approximation we use mollification, which preserves convexity and, on convex functions, is monotone in the radius.

Lemma 18.3.3 *Let A be a finite convex function on $N_{\mathbb{R}}$, and let $\rho_\varepsilon(y) = \varepsilon^{-n}\rho(y/\varepsilon)$ be the mollifiers of Section 1.4, in the coordinates of a basis of N. Then the functions $A * \rho_\varepsilon$, $\varepsilon > 0$, are smooth and convex, $A * \rho_\varepsilon(x)$ is a nondecreasing function of $\varepsilon > 0$ with $A * \rho_\varepsilon \geq A$ for every x, and $A * \rho_\varepsilon \to A$ locally uniformly as $\varepsilon \to 0$.*

Proof If $n = 0$, then $A * \rho_\varepsilon = A$ and there is nothing to prove; let $n \geq 1$. Recall that ρ is smooth, nonnegative, even, supported in the unit ball, and has integral one. Since A is continuous, being locally Lipschitz by Proposition 18.1.3, the functions $A * \rho_\varepsilon$ are smooth by Section 1.4, and they converge to A locally uniformly by Lemma 1.4.1(1). They are convex, because $A * \rho_\varepsilon(x) = \int A(x-y)\rho_\varepsilon(y)\,\mathrm{d}y$ and the convexity inequalities of the functions $x \mapsto A(x-y)$ can be integrated against $\rho_\varepsilon(y)\,\mathrm{d}y \geq 0$. For fixed x, consider the function

$$g(\varepsilon) = \int A(x - \varepsilon y)\rho(y)\,\mathrm{d}y \qquad (\varepsilon \in \mathbb{R}),$$

which is finite because A is continuous and ρ has compact support. The substitution $y \mapsto \varepsilon y$ shows that $g(\varepsilon) = A * \rho_\varepsilon(x)$ for $\varepsilon > 0$, and $g(0) = A(x)$. For each y, the function $\varepsilon \mapsto A(x - \varepsilon y)$ is convex, so g is convex; and g is even, because ρ is even. A convex even function on $\mathbb{R}$ is nondecreasing on $[0, +\infty)$: if $0 \leq a < b$, then $a = \lambda b + (1-\lambda)(-b)$ with $\lambda = (a+b)/(2b) \in (0, 1]$, so that $g(a) \leq \lambda g(b) + (1-\lambda)g(-b) = g(b)$. Hence $A * \rho_\varepsilon(x)$ is nondecreasing in $\varepsilon > 0$ and bounded below by $g(0) = A(x)$. □

Proposition 18.3.4 *Let $P \subset M_{\mathbb{R}}$ be a convex body. Every $F \in \mathcal{P}(N_{\mathbb{R}}, P)$ is the pointwise limit of a decreasing sequence of smooth functions $F_j \in \mathcal{E}^\infty(N_{\mathbb{R}}, P)$, and the convergence is locally uniform. If functions $F_j \colon N_{\mathbb{R}} \to (-\infty, +\infty]$ decrease pointwise to a function F, then the conjugates F_j^* increase pointwise to F^*. A sequence of finite convex functions on $N_{\mathbb{R}}$ converges pointwise to a finite function if and only if it converges locally uniformly.*

Proof Let $A_j = \max\{F, h_P - j\}$. These are convex functions, decreasing in j, with $A_j \sim h_P$, and $A_j = F$ on every compact set on which $h_P - F \le j$; hence $A_j \to F$ locally uniformly. With the mollifiers ρ_ε of Lemma 18.3.3, choose $\varepsilon_j \downarrow 0$ and set $F_j = A_j * \rho_{\varepsilon_j}$. By that lemma, these functions are smooth and convex, and the monotonicity in ε together with $A_{j+1} \le A_j$ gives $F_{j+1} = A_{j+1} * \rho_{\varepsilon_{j+1}} \le A_j * \rho_{\varepsilon_{j+1}} \le A_j * \rho_{\varepsilon_j} = F_j$, and $F_j \ge A_j \ge F$. By Proposition 18.3.2, F and h_P are Lipschitz with constant $\max_{p\in P} |p|$, hence so is A_j, and therefore $|F_j - A_j| \le C\varepsilon_j$ with $C = \max_{p\in P} |p| \int |y|\rho(y)\,\mathrm{d}y$. Consequently $F_j \sim A_j \sim h_P$, so $F_j \in \mathcal{E}^\infty(N_\mathbb{R}, P)$, and $F_j \to F$ locally uniformly.

If $F_j \downarrow F$ pointwise, then

$$\sup_j F_j^*(p) = \sup_j \sup_x \big(\langle p, x\rangle - F_j(x)\big) = \sup_x \sup_j \big(\langle p, x\rangle - F_j(x)\big) = F^*(p),$$

and the conjugates F_j^* increase, because conjugation reverses inequalities. Finally, if a sequence of finite convex functions on $N_\mathbb{R}$ converges pointwise to a finite function, it converges locally uniformly by Proposition 18.1.4, applied with Ω and the dense subset both equal to $N_\mathbb{R}$, and conversely local uniform convergence implies pointwise convergence. □

Example 18.3.5 The hypotheses $K_F = P$ and $K_G = P'$ in the last assertion of Proposition 18.3.2 cannot be replaced by $F \in \mathcal{E}(N_\mathbb{R}, P)$ and $G \in \mathcal{E}(N_\mathbb{R}, P')$: when P has empty interior, membership in $\mathcal{E}(N_\mathbb{R}, P)$ does not force a function to use every slope in P. Let $N_\mathbb{R} = M_\mathbb{R} = \mathbb{R}^2$ with the standard pairing, $P = [0,1] \times \{0\}$ and $P' = \{0\} \times [0,1]$. Both bodies have empty interior, and the zero function belongs to $\mathcal{P}(N_\mathbb{R}, P) = \mathcal{E}(N_\mathbb{R}, P)$ and to $\mathcal{E}(N_\mathbb{R}, P')$, since $0 \le h_P$ and $0 \le h_{P'}$. The sum of the zero function with itself is zero, whose slope body is $\{0\}$, whereas $P + P' = [0,1]^2$ has nonempty interior. Hence the zero function does not belong to $\mathcal{E}(N_\mathbb{R}, P + P')$.

Finally, we compare logarithms of sums of exponentials with linear functions.

Lemma 18.3.6 *Let $m \ge 1$ and $\beta_1, \dots, \beta_m, \alpha \in M_\mathbb{R}$. The function*

$$x \longmapsto \log \sum_{i=1}^m \mathrm{e}^{\langle \beta_i, x\rangle} - \langle \alpha, x\rangle$$

is bounded below on $N_\mathbb{R}$ if and only if α belongs to the convex hull of $\beta_1, \dots, \beta_m$. Suppose that all β_i and α lie in M, identify M with $\mathbb{Z}^n$ by a basis and $N_\mathbb{R}$ with $\mathbb{R}^n$ by the dual basis, and write $z^\alpha = z_1^{\alpha_1} \cdots z_n^{\alpha_n}$. Then this condition is also equivalent to the boundedness on $(\mathbb{C}^*)^n$ of

$$\frac{|z^\alpha|^2}{\sum_i |z^{\beta_i}|^2}.$$

Proof If $\alpha = \sum_i \lambda_i \beta_i$ is a convex combination, then $\langle \alpha, x\rangle \le \max_i \langle \beta_i, x\rangle \le \log \sum_i \mathrm{e}^{\langle \beta_i, x\rangle}$, so the function is nonnegative. Conversely, suppose that α is not in the convex hull. The hull is compact, being the image of the standard simplex under the linear map $(\lambda_i) \mapsto \sum_i \lambda_i \beta_i$, and convex, so Lemma 18.1.2(1), applied

in $M_{\mathbb{R}}$, gives $v \in N_{\mathbb{R}}$ with $\langle \alpha, v\rangle > \max_i \langle \beta_i, v\rangle$. Since the logarithmic sum differs from $\max_i \langle \beta_i, x\rangle$ by a number in $[0, \log m]$, the value of the function at $x = tv$ is at most $t(\max_i \langle \beta_i, v\rangle - \langle \alpha, v\rangle) + \log m$, which tends to $-\infty$ as $t \to +\infty$. For the last assertion, put $x_j = \log |z_j|^2$. Then $|z^\gamma|^2 = \mathrm{e}^{\langle \gamma, x\rangle}$ for every $\gamma \in \mathbb{Z}^n$, including exponents with negative entries, so the displayed quotient equals the exponential of minus the function above; and $z \mapsto x$ maps $(\mathbb{C}^*)^n$ onto $\mathbb{R}^n$. □

18.4 The real Monge–Ampère measure

Let F be a finite convex function on $N_{\mathbb{R}}$. Following Alexandrov, we measure a set $E \subset N_{\mathbb{R}}$ by the volume of the set $\partial F(E)$ of supporting slopes of F at points of E. For a smooth F with positive definite Hessian, $\partial F(E) = \nabla F(E)$, and the change of variables formula gives $\mathrm{vol}(\nabla F(E)) = \int_E \det(D^2 F)\, \mathrm{d}x$, so this is a weak form of the real Monge–Ampère operator (see Proposition 18.4.4). Two difficulties arise in general: the subgradient map is multivalued, and it is not clear a priori that $\partial F(E)$ is measurable. Both are resolved by looking at the inverse map. By duality, a slope p belongs to $\partial F(E)$ if and only if E meets $\partial F^*(p)$, and $\partial F^*(p)$ is a single point for almost every p in the domain of F^*.

Theorem 18.4.1 *Let F be a finite convex function on $N_{\mathbb{R}}$, let $G = F^*$, and let D be the set of points of the interior of* $\mathrm{dom}\, G$ *at which G is differentiable. For every Borel set $E \subset N_{\mathbb{R}}$, the set $\partial F(E)$ is measurable for the completed Lebesgue measure, and*

$$\mathrm{MA}_{\mathbb{R}}(F)(E) = n!\, \mathrm{vol}(\partial F(E))$$

defines a positive Radon measure on $N_{\mathbb{R}}$, the real Monge–Ampère measure *of F. It is given by*

$$\mathrm{MA}_{\mathbb{R}}(F) = (\nabla G)_*(n! \mathbf{1}_D\, \mathrm{d}p).$$

If $\mathrm{dom}\, G$ *has empty interior and $n > 0$, then $\mathrm{MA}_{\mathbb{R}}(F) = 0$. In every case*

$$\mathrm{MA}_{\mathbb{R}}(F)(N_{\mathbb{R}}) = n!\, \mathrm{vol}(K_F).$$

Proof The set $\mathrm{dom}\, G$ is convex, and its difference from its interior is a null set: if the interior is nonempty, this difference lies in the relative boundary, which is null by Lemma 18.1.2(4); otherwise $\mathrm{dom}\, G$ lies in a proper affine subspace. By Proposition 18.1.3(4), applied to the finite convex function G on the interior of its domain, D is a Borel set, the gradient $\nabla G \colon D \to N_{\mathbb{R}}$ is a Borel map, and the complement of D in the interior of $\mathrm{dom}\, G$ is a null set. Therefore the complement of D in $\mathrm{dom}\, G$ is a null set; it is a Borel set, because $\mathrm{dom}\, G = \bigcup_k \{G \le k\}$ is a countable union of closed sets, G being lower semicontinuous.

Let $p \in D$. Then $\partial G(p) = \{\nabla G(p)\}$. Indeed, $\partial G(p)$ is nonempty by Proposition 18.1.3, and every subgradient of G at p is also a subgradient at p of the restriction of G to the interior of $\mathrm{dom}\, G$, whose only subgradient at p is $\nabla G(p)$ by

Proposition 18.1.3(3). Hence Theorem 18.2.2 gives, for every $E \subset N_{\mathbb{R}}$,

$$p \in \partial F(E) \quad \Longleftrightarrow \quad \nabla G(p) \in E.$$

Every subgradient of F lies in $\operatorname{dom} G$, so $\partial F(E)$ differs from the Borel set $(\nabla G)^{-1}(E) \cap D$ by a subset of the fixed null set $\operatorname{dom} G \setminus D$. Hence $\partial F(E)$ is measurable for the completed measure, and $\operatorname{vol}(\partial F(E)) = \operatorname{vol}((\nabla G)^{-1}(E) \cap D)$. This proves the pushforward formula, which in turn shows that $E \mapsto n!\operatorname{vol}(\partial F(E))$ is countably additive on Borel sets.

If K is compact, $\partial F(K)$ is bounded by Proposition 18.1.3(1), and it is closed, since the graph of ∂F is closed and K is compact; so $\partial F(K)$ is compact and has finite volume. Thus the measure is finite on compact sets, and it is a Radon measure by Proposition 1.1.2. If $\operatorname{dom} G$ has empty interior and $n > 0$, then $D = \varnothing$ and the measure vanishes. In general, the total mass is $n!\operatorname{vol}(D) = n!\operatorname{vol}(\operatorname{dom} G)$, and $\operatorname{vol}(\operatorname{dom} G) = \operatorname{vol}(K_F)$ because $K_F \setminus \operatorname{dom} G$ lies in the relative boundary of $\operatorname{dom} G$, or in a proper affine subspace. In dimension zero, F is a constant, D is the single point of $M_{\mathbb{R}}$, and the formula gives unit mass, with $0! = 1$. □

Theorem 18.4.2 *Let F_j and F be finite convex functions on $N_{\mathbb{R}}$ such that $F_j \to F$ locally uniformly. Then $\mathrm{MA}_{\mathbb{R}}(F_j) \to \mathrm{MA}_{\mathbb{R}}(F)$ vaguely. Moreover, let $P \subset M_{\mathbb{R}}$ be a convex body. A function $F \in \mathcal{P}(N_{\mathbb{R}}, P)$ belongs to $\mathcal{E}(N_{\mathbb{R}}, P)$ if and only if*

$$\mathrm{MA}_{\mathbb{R}}(F)(N_{\mathbb{R}}) = n!\operatorname{vol}(P).$$

If F and H are finite convex functions with $F \preceq H$, then $K_F \subset K_H$, and the total mass of $\mathrm{MA}_{\mathbb{R}}(F)$ is at most that of $\mathrm{MA}_{\mathbb{R}}(H)$.

Proof Write $\mu_j = \mathrm{MA}_{\mathbb{R}}(F_j)$ and $\mu = \mathrm{MA}_{\mathbb{R}}(F)$. Steps 1 and 2 compare the masses of compact and of open sets, and Step 3 deduces vague convergence.

Step 1: compact sets. Let K be compact. If $x \in K$ and $0 \neq p \in \partial F_j(x)$ (in particular $n \geq 1$), choose $e \in N_{\mathbb{R}}$ with $|e| = 1$ and $\langle p, e\rangle = |p|$; the subgradient inequality at x, evaluated at $x + e$, gives $|p| \leq F_j(x+e) - F_j(x) \leq 2\sup\{|F_j(y)| : \operatorname{dist}(y, K) \leq 1\}$. By local uniform convergence, the sets $\partial F_j(K)$ are therefore contained in a fixed ball for all large j. If $x_j \in K$, $p_j \in \partial F_j(x_j)$, $x_j \to x$ and $p_j \to p$, then letting $j \to \infty$ in the subgradient inequalities gives $p \in \partial F(x)$. The set $\partial F(K)$ is compact, being bounded and closed by Proposition 18.1.3(1), since K is compact. Consequently, for every $\varepsilon > 0$, the set $\partial F_j(K)$ lies in the closed ε-neighborhood $(\partial F(K))^{\varepsilon}$ of $\partial F(K)$ for all large j. Otherwise, for infinitely many j there would be points $x_j \in K$ and $p_j \in \partial F_j(x_j)$ at distance at least ε from $\partial F(K)$; as K is compact and the p_j are bounded, a subsequence of (x_j, p_j) would converge to some (x, p) with $p \in \partial F(x)$, which is impossible. As $\varepsilon \downarrow 0$, these neighborhoods decrease to $\partial F(K)$, and continuity from above of Lebesgue measure on sets of finite measure gives

$$\overline{\lim_j}\, \mu_j(K) \leq \mu(K).$$

Step 2: open sets. Let U be open and let $G = F^*$, with D as in Theorem 18.4.1. For $p \in \partial F(U) \cap D$, the point $x = \nabla G(p)$ lies in U and is the only point with $p \in \partial F(x)$, so x is the unique minimum point of $F - \langle p, \cdot \rangle$. Choose $r > 0$ with $\overline{B(x,r)} \subset U$. Since the sphere $\{y : |y - x| = r\}$ is compact, $F - \langle p, \cdot \rangle$ exceeds its minimum by some $\delta > 0$ on it. For large j we have $|F_j - F| < \delta/3$ on $\overline{B(x,r)}$, so the minimum of $F_j - \langle p, \cdot \rangle$ on $\overline{B(x,r)}$ is attained at a point $x_j \in B(x,r)$. A local minimum of a convex function is a global minimum, because the function is convex along each segment starting at that point; hence $p \in \partial F_j(x_j) \subset \partial F_j(U)$ for all large j. Since $\partial F(U) \setminus D$ is a null set, Fatou's lemma, applied to the indicator functions of the sets $\partial F_j(U)$, gives

$$\mu(U) \le \varliminf_j \mu_j(U).$$

Step 3: vague convergence. Let $f \ge 0$ be continuous with compact support S. By Step 1 there is C with $\mu_j(S) \le C$ for all large j. For $t > 0$, the set $\{f \ge t\}$ is compact and $\{f > t\}$ is open. By Tonelli's theorem, $\int f \, d\nu = \int_0^\infty \nu(\{f > t\}) \, dt = \int_0^\infty \nu(\{f \ge t\}) \, dt$ for every positive measure ν. Hence Step 2 and Fatou's lemma give

$$\begin{aligned} \int f \, d\mu &= \int_0^\infty \mu(\{f > t\}) \, dt \\ &\le \varliminf_j \int_0^\infty \mu_j(\{f > t\}) \, dt = \varliminf_j \int f \, d\mu_j, \end{aligned}$$

and Step 1 and Fatou's lemma, applied to the nonnegative functions $t \mapsto C - \mu_j(\{f \ge t\})$ on $(0, \max f]$, give

$$\begin{aligned} \varlimsup_j \int f \, d\mu_j &= C \max f - \varliminf_j \int_0^{\max f} \mu_j(\{f \ge t\}) \, dt \\ &\le C \max f - \int_0^{\max f} \mu(\{f \ge t\}) \, dt = \int f \, d\mu. \end{aligned}$$

Hence $\int f \, d\mu_j \to \int f \, d\mu$. A general continuous function with compact support is the difference of its positive and negative parts, which proves vague convergence.

Full mass. Let $F \in \mathcal{P}(N_{\mathbb{R}}, P)$. Then $K_F \subset P$ by Proposition 18.3.2, and the total mass of $\mathrm{MA}_{\mathbb{R}}(F)$ is $n! \, \mathrm{vol}(K_F)$ by Theorem 18.4.1. Suppose that P has nonempty interior. If $K_F = P$, the two masses agree. If $K_F \ne P$, choose $p \in P \setminus K_F$ and a ball $B(p,r)$ disjoint from the closed set K_F. Since P is the closure of its interior by Lemma 18.1.2(4), there is an interior point q of P in $B(p, r/2)$, and a small ball about q lies in $P \setminus K_F$. Hence $\mathrm{vol}(K_F) < \mathrm{vol}(P)$. This proves the equivalence when P has nonempty interior. If P has empty interior, then $n > 0$, both $\mathrm{vol}(K_F)$ and $\mathrm{vol}(P)$ vanish, and $\mathcal{E}(N_{\mathbb{R}}, P) = \mathcal{P}(N_{\mathbb{R}}, P)$ by definition, so the equivalence holds as well.

Finally, if $F \le H + C$, then $F^* \ge H^* - C$, so $\mathrm{dom}\, F^* \subset \mathrm{dom}\, H^*$ and $K_F \subset K_H$. The inequality between the total masses follows from Theorem 18.4.1. □

Example 18.4.3 In Theorem 18.4.2, vague convergence cannot be replaced by narrow convergence without a condition that prevents mass from escaping to infinity. Let $n = 1$, identify $N_{\mathbb{R}}$ and $M_{\mathbb{R}}$ with $\mathbb{R}$ so that $N = M = \mathbb{Z}$, and put

$$F_j(x) = \max\{0, x - j\} \longrightarrow 0, \qquad \mathrm{MA}_{\mathbb{R}}(F_j) = \delta_j.$$

Indeed, $\partial F_j(x)$ is $\{0\}$ for $x < j$, $\{1\}$ for $x > j$, and $[0, 1]$ for $x = j$. All slopes lie in $[0, 1]$, and $F_j \to 0$ locally uniformly. As in Example 1.5.4, the measures δ_j converge vaguely to $\mathrm{MA}_{\mathbb{R}}(0) = 0$, while their total masses remain equal to one. Here each F_j has full mass for $P = [0, 1]$, but the limit does not. If the functions and their limit all have full mass for one fixed body, the measures converge narrowly by Lemma 18.5.2 below.

Proposition 18.4.4 *Let F be a convex function on $N_{\mathbb{R}}$ of class $C^{1,1}_{\mathrm{loc}}$, that is, differentiable with locally Lipschitz gradient. Then the distributional Hessian D^2F of F, computed in the coordinates of a basis of N, is a locally bounded measurable matrix-valued function, and*

$$\mathrm{MA}_{\mathbb{R}}(F) = n! \det(D^2F)\, \mathrm{d}x.$$

Moreover, there is a unique way to assign to every n-tuple $(F_1, \dots, F_n)$ of finite convex functions on $N_{\mathbb{R}}$ a positive Radon measure $\mathrm{MA}_{\mathbb{R}}(F_1, \dots, F_n)$, the mixed real Monge–Ampère measure*, which is symmetric in its arguments, additive and positively homogeneous in each argument, and satisfies $\mathrm{MA}_{\mathbb{R}}(F, \dots, F) = \mathrm{MA}_{\mathbb{R}}(F)$. It is continuous for the vague topology under local uniform convergence of each argument.*

Proof *Smooth functions.* Let F be smooth with positive definite Hessian. For $x \neq y$,

$$\langle \nabla F(y) - \nabla F(x), y - x \rangle = \int_0^1 D^2F(x + s(y - x))(y - x, y - x)\, \mathrm{d}s > 0,$$

so ∇F is injective, and it is a local diffeomorphism by the inverse function theorem. Since $\partial F(E) = \nabla F(E)$ by Proposition 18.1.3(3), the change of variables formula gives the asserted identity. Now let F be smooth and convex; its Hessian is positive semidefinite, because the second derivatives of F along lines are nonnegative. Apply the identity just proved to $F + \varepsilon|x|^2/2$, where $|x|$ is the Euclidean norm in the lattice coordinates. As $\varepsilon \downarrow 0$, these functions converge to F locally uniformly and their densities $n! \det(D^2F + \varepsilon I)$ converge to $n! \det(D^2F)$ locally uniformly; comparing the two vague limits, by Theorem 18.4.2, proves the identity for F.

Functions of class $C^{1,1}_{\mathrm{loc}}$. We work in the coordinates of a basis of N. Fix k and l, and let $g = \partial_k F$, which is locally Lipschitz. Let $B = B(x_0, r)$ be a ball, and let L_B be a Lipschitz constant of g on $\overline{B(x_0, 2r)}$. For a nonnegative test function χ supported in B and $0 < |h| < r$, the substitution $x \mapsto x - he_l$ in one of the two terms gives

$$\int g(x) \frac{\chi(x + he_l) - \chi(x)}{h}\, \mathrm{d}x = \int \frac{g(x - he_l) - g(x)}{h}\, \chi(x)\, \mathrm{d}x,$$

and the right side is at most $L_B \int \chi \, dx$ in absolute value. As $h \to 0$, the difference quotients of χ converge uniformly to $\partial_l \chi$, with supports in a fixed compact set, so the left side tends to $\int g \, \partial_l \chi \, dx = -\langle \partial_l g, \chi \rangle$. Hence $|\langle \partial_l \partial_k F, \chi \rangle| \le L_B \int \chi \, dx$, and Corollary 1.7.2(2), applied to Lebesgue measure, gives a Borel function H_{kl} that represents the distribution $\partial_l \partial_k F$ and satisfies $|H_{kl}| \le L_B$ almost everywhere on each such ball B. Covering $N_{\mathbb{R}}$ by countably many such balls and setting $H_{kl} = 0$ on the null set of points of some ball of the cover at which the bound fails, we may assume that H_{kl} is bounded on every compact set. This proves that $D^2 F = (H_{kl})$ is a locally bounded Borel matrix-valued function. Since distributional derivatives commute, $H_{kl} = H_{lk}$ almost everywhere, by the uniqueness in Corollary 1.7.2.

Next, $D^2 F(x)$ is positive semidefinite for almost every x. Indeed, let $v \in \mathbb{R}^n$ and let χ be a nonnegative test function. Taylor's formula shows that the second difference quotients $h^{-2}(\chi(x+hv) + \chi(x-hv) - 2\chi(x))$ converge uniformly to $\sum_{k,l} v_k v_l \partial_k \partial_l \chi$ as $h \to 0$, so the substitutions $x \mapsto x \mp hv$ give

$$\Big\langle \sum_{k,l} v_k v_l \partial_k \partial_l F, \chi \Big\rangle = \lim_{h \to 0} \int \frac{F(x-hv) + F(x+hv) - 2F(x)}{h^2} \, \chi(x) \, dx \ge 0,$$

because F is convex. Thus the function $\sum_{k,l} v_k v_l H_{kl}$ defines a positive distribution, so its mollifications are nonnegative by Proposition 1.4.3(3); since they converge to it at its Lebesgue points (Theorem 1.7.3), it is nonnegative almost everywhere. Applying this to a countable dense set of vectors v, and using the continuity of the quadratic form in v, we see that $D^2 F(x)$ is positive semidefinite for almost every x.

We now obtain the formula by mollification. With the mollifiers ρ_ε of Lemma 18.3.3, in the same coordinates, the functions $F_\varepsilon = F * \rho_\varepsilon$ are smooth and convex, and they converge to F locally uniformly, by that lemma. By Proposition 1.4.3(1), their second derivatives are $\partial_k \partial_l F_\varepsilon = H_{kl} * \rho_\varepsilon$. Hence, for $\varepsilon \le 1$, the Hessians $D^2 F_\varepsilon$ are bounded on each compact set K by the bound of $D^2 F$ on the set of points at distance at most 1 from K, and by Theorem 1.7.3 they converge to $D^2 F(x)$ at every point x that is a Lebesgue point of all the entries H_{kl}, that is, almost everywhere. By the smooth case, $\mathrm{MA}_{\mathbb{R}}(F_\varepsilon) = n! \det(D^2 F_\varepsilon) \, dx$. Let ε run through a sequence tending to zero. For $f \in C_c(N_{\mathbb{R}})$, dominated convergence gives $n! \int f \det(D^2 F_\varepsilon) \, dx \to n! \int f \det(D^2 F) \, dx$, while $\mathrm{MA}_{\mathbb{R}}(F_\varepsilon) \to \mathrm{MA}_{\mathbb{R}}(F)$ vaguely by Theorem 18.4.2. This proves the formula.

Mixed measures. For smooth convex $F_1, \dots, F_n$ define

$$\mathrm{MA}_{\mathbb{R}}(F_1, \dots, F_n) = n! \, D(D^2 F_1, \dots, D^2 F_n) \, dx,$$

where D is the mixed discriminant of Section 7.2, applied to the real symmetric, hence Hermitian, Hessian matrices. This measure is positive, because the mixed discriminant of positive semidefinite matrices is nonnegative (Lemma 7.2.2(3)), and it is symmetric and multilinear, with $D(A, \dots, A) = \det A$ (Lemma 7.2.2(1)). Expanding $\mathrm{MA}_{\mathbb{R}}(\sum_{j \in J} F_j, \dots, \sum_{j \in J} F_j)$ by multilinearity and using symmetry, we obtain the polarization identity

$$\mathrm{MA}_{\mathbb{R}}(F_1,\dots,F_n)=\frac{1}{n!}\sum_{J\subset\{1,\dots,n\}}(-1)^{n-|J|}\,\mathrm{MA}_{\mathbb{R}}\Bigl(\sum_{j\in J}F_j\Bigr),\tag{18.3}$$

where the empty sum is the zero function, whose measure is zero for $n>0$ (its only slope is 0). Indeed, after the expansion, a term $\mathrm{MA}_{\mathbb{R}}(F_{\sigma(1)},\dots,F_{\sigma(n)})$ with $\sigma\colon\{1,\dots,n\}\to\{1,\dots,n\}$ appears with total coefficient $\sum_{J\supset\sigma(\{1,\dots,n\})}(-1)^{n-|J|}$, which vanishes unless σ is a bijection, and the $n!$ bijections all give $\mathrm{MA}_{\mathbb{R}}(F_1,\dots,F_n)$ by symmetry. Any symmetric, multilinear assignment with the given diagonal values satisfies (18.3), by the same expansion; this proves uniqueness.

Now let $F_1,\dots,F_n$ be arbitrary finite convex functions, and let $F_{j,\varepsilon}=F_j*\rho_\varepsilon$ be their mollifications, which are smooth and convex by Lemma 18.3.3. For every J, $\sum_{j\in J}F_{j,\varepsilon}\to\sum_{j\in J}F_j$ locally uniformly as $\varepsilon\to0$ (Lemma 1.4.1(1)). Let ε run through a sequence tending to zero. By (18.3) for the smooth functions $F_{j,\varepsilon}$ and by Theorem 18.4.2, for every $f\in C_c(N_{\mathbb{R}},\mathbb{R})$ the integrals $\int f\,\mathrm{d}\,\mathrm{MA}_{\mathbb{R}}(F_{1,\varepsilon},\dots,F_{n,\varepsilon})$ converge to

$$\Lambda(f)=\frac{1}{n!}\sum_{J\subset\{1,\dots,n\}}(-1)^{n-|J|}\int f\,\mathrm{d}\,\mathrm{MA}_{\mathbb{R}}\Bigl(\sum_{j\in J}F_j\Bigr).$$

The functional Λ is linear, and it is positive as a limit of integrals against positive measures. By Theorem 1.2.2, there is a unique Radon measure $\mathrm{MA}_{\mathbb{R}}(F_1,\dots,F_n)$ with $\int f\,\mathrm{d}\,\mathrm{MA}_{\mathbb{R}}(F_1,\dots,F_n)=\Lambda(f)$ for every f; for smooth F_j it agrees with the previous definition, by (18.3). It is symmetric, because Λ is a symmetric function of $F_1,\dots,F_n$. Additivity and positive homogeneity in each argument hold for the smooth mixed measures; since mollification is linear, they pass to the vague limit. For $F_1=\cdots=F_n=F$, the smooth mixed measures are $n!\det(D^2F_\varepsilon)\,\mathrm{d}x=\mathrm{MA}_{\mathbb{R}}(F_\varepsilon)$, which converge vaguely to $\mathrm{MA}_{\mathbb{R}}(F)$; hence $\mathrm{MA}_{\mathbb{R}}(F,\dots,F)=\mathrm{MA}_{\mathbb{R}}(F)$. Finally, if $F_j^{(k)}\to F_j$ locally uniformly as $k\to\infty$ for each j, then $\sum_{j\in J}F_j^{(k)}\to\sum_{j\in J}F_j$ locally uniformly for every J, and the formula for Λ together with Theorem 18.4.2 shows that $\mathrm{MA}_{\mathbb{R}}(F_1^{(k)},\dots,F_n^{(k)})\to\mathrm{MA}_{\mathbb{R}}(F_1,\dots,F_n)$ vaguely. In dimension zero the mixed measure has no arguments, and it is the unit mass at the point. □

18.5 Compact convex envelopes and full-mass limits

We add two complements. First, the convex envelope of finitely many closed convex functions whose effective domains lie in a common compact set, such as the conjugates of finitely many functions in $\mathcal{P}(N_{\mathbb{R}},P)$, is closed, and the infimum defining it is attained. Second, vague convergence of real Monge–Ampère measures improves to narrow convergence when the total masses converge, as they do for functions of full mass.

Lemma 18.5.1 *Let $G_1,\dots,G_r$, with $r\ge1$, be closed proper convex functions on a finite-dimensional real vector space, whose effective domains lie in a common compact*

set K, and let H be the largest convex function below $\min_i G_i$, given by the formula of Proposition 18.2.3(2). Then H is closed and proper, and at every point x with $H(x) < +\infty$ the infimum defining $H(x)$ is attained. Consequently $H = (\max_i G_i^*)^*$. In particular, if $P \subset M_{\mathbb{R}}$ is a convex body and $F_1, \dots, F_r \in \mathcal{P}(N_{\mathbb{R}}, P)$, then the convex envelope of $F_1^*, \dots, F_r^*$ is closed and equals $(\max_i F_i)^*$.

Proof Each G_i is bounded below: the increasing open sets $\{G_i > -k\}$, $k \in \mathbb{N}$, cover the compact set K, because G_i is lower semicontinuous and does not take the value $-\infty$, so one of them contains K; and $G_i = +\infty$ outside K. Let B be a common lower bound. In a combination $\sum_j \lambda_j G_{i_j}(x_j)$ as in Proposition 18.2.3(2), the terms with the same function G_i can be merged: if Λ_i is the sum of their weights, the convexity of G_i gives $\sum_{j:i_j=i} \lambda_j G_i(x_j) \geq \Lambda_i G_i(y_i)$ with $y_i = \Lambda_i^{-1} \sum_{j:i_j=i} \lambda_j x_j$, and the barycenter is unchanged. Hence it suffices to consider combinations $\sum_{\lambda_i>0} \lambda_i G_i(x_i)$ with one point $x_i \in \operatorname{dom} G_i$ for each i and weights $\lambda_i \geq 0$ summing to one; for an index i with $\lambda_i = 0$ we let x_i be a fixed point of $\operatorname{dom} G_i$, and the corresponding term is omitted.

Let $x_\nu \to x$ with $\varliminf_\nu H(x_\nu) = L < +\infty$. Passing to a subsequence, we may assume that $H(x_\nu) \to L$, and we choose combinations for x_ν, with points $x_{i,\nu}$ and weights $\lambda_{i,\nu}$, whose values tend to L. Since K and the simplex of weights are compact, a further subsequence satisfies $x_{i,\nu} \to x_i \in K$ and $\lambda_{i,\nu} \to \lambda_i$ for every i. The limiting weights are nonnegative and sum to one, and since the $x_{i,\nu}$ are bounded, the terms with $\lambda_i = 0$ disappear in the limit of the barycenters, so $\sum_{\lambda_i>0} \lambda_i x_i = x$. The nonnegative numbers $\lambda_{i,\nu}(G_i(x_{i,\nu}) - B)$ have sum tending to $L - B$. For $\lambda_i > 0$, lower semicontinuity gives $\varliminf_\nu \lambda_{i,\nu}(G_i(x_{i,\nu}) - B) \geq \lambda_i(G_i(x_i) - B)$, and therefore

$$\sum_{\lambda_i>0} \lambda_i (G_i(x_i) - B) \leq \varliminf_\nu \sum_i \lambda_{i,\nu}(G_i(x_{i,\nu}) - B) = L - B.$$

In particular, all the values $G_i(x_i)$ with $\lambda_i > 0$ are finite, and the combination with points x_i and weights $\lambda_i > 0$ is admissible for x, with value at most L. Thus $H(x) \leq L$, which proves that H is lower semicontinuous. Moreover $B \leq H \leq G_1$, so H is proper. Applying the same argument to the constant sequence $x_\nu = x$ and admissible combinations whose values tend to $H(x) < +\infty$, we see that the infimum defining $H(x)$ is attained. By Proposition 18.2.3(2), $H^* = \max_i G_i^*$, and since H is closed and proper, Theorem 18.2.2 gives $H = H^{**} = (\max_i G_i^*)^*$.

For the last assertion, the conjugates F_i^* are closed proper convex functions whose effective domains lie in the compact set P, by Theorem 18.2.2 and Proposition 18.3.2. Since $F_i^{**} = F_i$, the first part gives the formula $(\max_i F_i)^*$ for their convex envelope.□

Lemma 18.5.2 *Let μ_j and μ be finite positive Radon measures on a finite-dimensional real vector space W such that $\mu_j \to \mu$ vaguely and $\mu_j(W) \to \mu(W)$. Then $\int h \, \mathrm{d}\mu_j \to \int h \, \mathrm{d}\mu$ for every bounded continuous function h on W, that is, $\mu_j \to \mu$ narrowly (Section 1.5). In particular, if $P \subset M_{\mathbb{R}}$ is a convex body, $F_j, F \in \mathcal{E}(N_{\mathbb{R}}, P)$ and $F_j \to F$ locally uniformly, then $\mathrm{MA}_{\mathbb{R}}(F_j) \to \mathrm{MA}_{\mathbb{R}}(F)$ narrowly.*

Proof Let $\varepsilon > 0$. By inner regularity there is a compact set S with $\mu(W \setminus S) < \varepsilon$; let η be a continuous function with compact support, $0 \leq \eta \leq 1$ and $\eta = 1$ on S. Since

$\int \eta \, d\mu_j \to \int \eta \, d\mu$ and the total masses converge, we have

$$\lim_j \int (1-\eta)\, d\mu_j = \int (1-\eta)\, d\mu < \varepsilon,$$

so that $\mu_j(W \setminus \operatorname{Supp}\eta) \le \int (1-\eta)\, d\mu_j < \varepsilon$ for all large j. Thus the sequence (μ_j) is eventually uniformly tight, and Proposition 1.5.3 gives narrow convergence. (If W is a point, every function on W has compact support, and vague and narrow convergence coincide.) For the real Monge–Ampère measures, Theorem 18.4.2 gives vague convergence and the common total mass $n!\operatorname{vol}(P)$. □

18.6 Lattice points and volume

By Theorem 18.8.5 below, the dimension of a space of sections on a projective toric manifold is the number of lattice points in a dilated polytope. We show that k^{-n} times the number of lattice points in kK tends to the volume of the convex body K, and that this remains true if the lattice points are prescribed only up to a sublinear distance from kK. The reason is that the boundary of a convex body has volume zero.

In this section we fix a basis of M. Its coordinates identify $M_{\mathbb{R}}$ with $\mathbb{R}^n$ and vol with Lebesgue measure; we use the Euclidean norm of these coordinates on $M_{\mathbb{R}}$ and the dual norm on $N_{\mathbb{R}}$. For a convex body $K \subset M_{\mathbb{R}}$, $k > 0$ and $\varepsilon > 0$ put $kK = \{kp : p \in K\}$ and

$$K^{\varepsilon} = \{p : \operatorname{dist}(p, K) \le \varepsilon\}, \qquad K_{\varepsilon} = \{p : \overline{B(p,\varepsilon)} \subset K\},$$

and $K^0 = K$. Since the distance to the compact set K is attained, the outer parallel body K^{ε} equals $K + \overline{B(0,\varepsilon)}$, the image of the compact convex set $K \times \overline{B(0,\varepsilon)}$ under addition; so it is a convex body. The inner set $K_{\varepsilon} = \bigcap_{|y|\le\varepsilon}(K - y)$ is compact and convex but may be empty; it is empty whenever K has empty interior and $n > 0$, because a closed ball of positive radius has interior points.

Definition 18.6.1 The *Hausdorff distance* between convex bodies $K, L \subset M_{\mathbb{R}}$ is

$$d_{\mathrm{H}}(K,L) = \max\left\{ \sup_{p\in K} \operatorname{dist}(p,L), \sup_{q\in L} \operatorname{dist}(q,K) \right\}.$$

Proposition 18.6.2 *If $n > 0$, then for convex bodies $K, L \subset M_{\mathbb{R}}$*

$$d_{\mathrm{H}}(K,L) = \max_{|x|=1} |h_K(x) - h_L(x)|.$$

If convex bodies K_j converge to a convex body K in this distance, then $\operatorname{vol}(K_j) \to \operatorname{vol}(K)$. *In particular,*

$$\lim_{\varepsilon\downarrow 0}\operatorname{vol}(K^{\varepsilon}) = \operatorname{vol}(K) = \lim_{\varepsilon\downarrow 0}\operatorname{vol}(K_{\varepsilon}).$$

In dimension zero the only convex body is the point, which has volume one.

Proof Let $n > 0$. If $d_{\mathrm{H}}(K,L) \le \varepsilon$, then every $p \in K$ has a point $q \in L$ with $|p-q| \le \varepsilon$, and for $|x| = 1$ we get $\langle p,x\rangle \le \langle q,x\rangle + \varepsilon \le h_L(x) + \varepsilon$; hence $h_K \le h_L + \varepsilon$ on the unit sphere, and by symmetry $|h_K - h_L| \le \varepsilon$ there. Conversely, suppose that $|h_K - h_L| \le \varepsilon$ on the unit sphere. The support function of the convex body $L + \overline{B(0,\varepsilon)}$ is $h_L(x) + \varepsilon|x|$. If some $p \in K$ were outside $L + \overline{B(0,\varepsilon)}$, Lemma 18.1.2(1) would give a unit vector x with $\langle p,x\rangle > h_L(x) + \varepsilon \ge h_K(x)$, which is impossible. Hence $K \subset L + \overline{B(0,\varepsilon)}$, and by symmetry $d_{\mathrm{H}}(K,L) \le \varepsilon$. This proves the formula. It shows that d_{H} satisfies the triangle inequality, and by Lemma 18.1.2(1) a convex body is determined by its support function; thus d_{H} is a metric.

Let $\varepsilon_j = d_{\mathrm{H}}(K_j,K) \to 0$. For large j all K_j lie in the bounded set $K + \overline{B(0,1)}$. A point p outside K has positive distance from K, so $p \notin K_j$ once $\varepsilon_j < \operatorname{dist}(p,K)$. If p is an interior point of K, choose $r > 0$ with $\overline{B(p,r)} \subset K$. Then, for $|x| = 1$,

$$h_{K_j}(x) \ge h_K(x) - \varepsilon_j \ge \langle p,x\rangle + r - \varepsilon_j.$$

If $\varepsilon_j < r$ and $p \notin K_j$, strict separation would give a unit vector x with $\langle p,x\rangle > h_{K_j}(x)$, contradicting this inequality; hence $p \in K_j$ for large j. Thus $\mathbf{1}_{K_j} \to \mathbf{1}_K$ outside the boundary of K. If K has nonempty interior, its boundary is a null set by Lemma 18.1.2(4); if K has empty interior, it lies in a proper affine subspace and is itself a null set, and $\mathbf{1}_{K_j} \to 0 = \mathbf{1}_K$ almost everywhere by the outside-point argument. In either case dominated convergence, with the dominating function $\mathbf{1}_{K+\overline{B(0,1)}}$, gives $\operatorname{vol}(K_j) \to \operatorname{vol}(K)$.

The outer parallel bodies K^{ε} have finite volume and decrease to the compact set K as $\varepsilon \downarrow 0$, so continuity of measure from above gives the first limit. The inner sets K_{ε} increase, as ε decreases, to the interior of K, which differs from K by a null set, as we saw in the previous paragraph; monotone convergence gives the second limit. In dimension zero both K^{ε} and K_{ε} equal the single point for every ε. □

Theorem 18.6.3 *Let $K \subset M_{\mathbb{R}}$ be a convex body, let $a_k \in M_{\mathbb{R}}$ be arbitrary, and let $r_k \ge 0$ satisfy $r_k/k \to 0$. Then for every continuous function $f\colon M_{\mathbb{R}} \to \mathbb{R}$,*

$$\lim_{k\to\infty} \frac{1}{k^n} \sum_{m \in (kK)^{r_k} \cap (M + a_k)} f(m/k) = \int_K f(p)\,\mathrm{d}p.$$

In particular $k^{-n}\#(kK \cap M) \to \operatorname{vol}(K)$. Here the exponent n and the measure are those of the ambient space, also when K has empty interior. The convergence is uniform with respect to the translations a_k.

Proof For $n = 0$ every set in question is the single point, and both sides equal its value under f. Suppose that $n > 0$, and let $Q = [0,1)^n$ be the half-open unit cube in the chosen coordinates. The cells

$$\frac{m}{k} + \frac{1}{k}Q, \qquad m \in M + a_k,$$

partition $M_{\mathbb{R}}$; each has volume k^{-n}, and its points lie within distance c/k of its anchor m/k, where $c = \sqrt{n}$. Let U_k be the union of the cells whose anchors m/k occur in the sum, that is, with $m/k \in K^{r_k/k}$. Then

$$K_{c/k} \subset U_k \subset K^{(r_k+c)/k}.$$

For the first inclusion, if $p \in K_{c/k}$, the anchor of the cell containing p lies within c/k of p, hence in K, so the cell is selected. For the second, every point of a selected cell lies within c/k of its anchor, which lies within r_k/k of K. Since $K_{c/k} \subset K \subset K^{(r_k+c)/k}$ and $(r_k + c)/k \to 0$, Proposition 18.6.2 gives $\operatorname{vol}(U_k \triangle K) \to 0$, and the volumes $\operatorname{vol}(U_k)$ stay bounded.

For large k the selected cells lie in the fixed compact set K^1. Let ω_f be the modulus of continuity of f on K^1. Since $k^{-n} f(m/k)$ is the integral of the constant $f(m/k)$ over the cell of m, we obtain

$$\left| k^{-n} \sum_m f(m/k) - \int_{U_k} f(p)\,\mathrm{d}p \right| \le \omega_f(c/k)\operatorname{vol}(U_k) \longrightarrow 0.$$

Since f is bounded on K^1, the symmetric-difference estimate gives $\int_{U_k} f\,\mathrm{d}p \to \int_K f\,\mathrm{d}p$. Taking $f = 1$, $r_k = 0$ and $a_k = 0$ proves the counting formula. All the estimates depend only on the size of the cells, and not on their position, so they hold uniformly in the translations a_k. □

Corollary 18.6.4 *Let $K \subset M_{\mathbb{R}}$ be a convex body, and suppose that subsets $S_k \subset M$ satisfy*

$$kK \cap M \subset S_k \subset (kK)^{r_k} \cap M, \qquad r_k \ge 0, \quad r_k/k \to 0.$$

Then $k^{-n}\#S_k \to \operatorname{vol}(K)$. In particular, a boundary error of bounded size does not affect the leading term of the count, even when K has zero volume.

Proof By Theorem 18.6.3 with $f = 1$ and $a_k = 0$, both $k^{-n}\#(kK \cap M)$ and $k^{-n}\#((kK)^{r_k} \cap M)$ tend to $\operatorname{vol}(K)$, and $\#S_k$ lies between these two counts. □

18.7 Smooth lattice polytopes and projective charts

So far a convex body $P \subset M_{\mathbb{R}}$ has served as a bound for the slopes of convex functions. When P is a lattice polytope, it also determines a compactification of the complex torus with character group M. If P is smooth, the primitive edge directions at each vertex form a basis of M, and the corresponding characters become holomorphic coordinates on a chart isomorphic to $\mathbb{C}^n$, which reaches across the boundary of the torus.

Definition 18.7.1 A *lattice polytope* in $M_{\mathbb{R}}$ is the convex hull of finitely many points of M. A *face* of a lattice polytope P is a set of the form $\{m \in P : \langle m, x\rangle = \min_{m' \in P}\langle m', x\rangle\}$ with $x \in N_{\mathbb{R}}$; faces are nonempty, and P itself is the face for $x = 0$. *Vertices*, *edges* and *facets* are the faces of dimension zero, of dimension one and of codimension one, respectively. A nonzero vector of a lattice is *primitive* if it is not of the form ky with y in the lattice and an integer $k \geq 2$. Let P have nonempty interior. Its vertices are lattice points, each edge is a segment $[v, v']$ between two vertices, and a vertex contained in an edge is one of its endpoints (Lemma 18.7.2). Thus an edge $[v, v']$ containing a vertex v determines the *primitive edge direction* at v, the unique primitive vector of M that is a positive multiple of $v' - v$ (see below). The polytope P is *smooth*, or *Delzant*, if at each vertex its primitive edge directions form a basis of M. In dimension zero the single point is smooth, with the empty basis.

Every nonzero vector y of a lattice is a positive multiple of exactly one primitive vector. Indeed, if d is the greatest common divisor of the coordinates of y in a basis of the lattice, then y/d is primitive, since its coordinates are coprime. If y' and y'' are primitive and $y' = cy''$ with $c > 0$, then c is a quotient of integer coordinates, say $c = a/b$ with coprime positive integers a and b, and $by' = ay''$ shows that b divides all coordinates of y'' and a all those of y'; hence $a = b = 1$.

The cone $\operatorname{cone}(S)$ generated by a set S consists of its finite nonnegative linear combinations. A *ray* is a half-line $\mathbb{R}_{\geq 0}y$ with $y \neq 0$, and a ray contained in a cone C is *extreme* if, whenever a vector of the ray is the sum of two vectors of C, both summands lie on the ray.

An affine function ℓ that is nonnegative on a convex set C and vanishes at a point $b \in \operatorname{ri} C$ vanishes on C. Indeed, for $y \in C$ the point $b - \varepsilon(y - b)$ lies in C for small $\varepsilon > 0$, and $0 = \ell(b)$ is a convex combination, with positive weights, of the nonnegative numbers $\ell(y)$ and $\ell(b - \varepsilon(y - b))$; hence $\ell(y) = 0$.

Lemma 18.7.2 *Let $P \subset M_{\mathbb{R}}$ be a lattice polytope; in* (2)–(4) *assume moreover that P has nonempty interior.*

(1) *The vertices of P are lattice points, P is their convex hull, every face of P is the convex hull of the vertices it contains, and P has finitely many faces. Every edge of P is a segment $[v, v']$ between two distinct vertices, and a vertex contained in an edge is one of its endpoints.*

(2) *P has an irredundant facet presentation*

$$P = \{m : \ell_\rho(m) = \langle m, u_\rho\rangle + a_\rho \geq 0 \text{ for every facet } \rho\},$$

where, for each facet ρ, the normal $u_\rho \in N$ is primitive and inward pointing, $a_\rho \in \mathbb{Z}$, and $\rho = \{m \in P : \ell_\rho(m) = 0\}$.

(3) *At a vertex v, the tangent cone*

$$C_v = \{y : \langle y, u_\rho\rangle \geq 0 \text{ for the facets } \rho \ni v\} = \operatorname{cone}(P - v)$$

is generated by its extreme rays, and these are the rays spanned by the edge directions at v.

(4) *If P is smooth and v is a vertex with primitive edge basis $e_1, \dots, e_n$, then exactly n facets contain v, and their normals $u_1, \dots, u_n$ form the dual basis of N, that is, $\langle e_i, u_j \rangle = \delta_{ij}$. Thus*

$$m - v = \sum_i t_i e_i, \qquad t_i = \ell_i(m) \geq 0 \quad (m \in P),$$

and $v + e_i \in P \cap M$ for each i.

Proof (1) Let P be the convex hull of a finite set $S \subset M$, and discard points of S one at a time as long as some point lies in the convex hull of the others; this does not change P. If only one point remains, P is that point, it is the only face, and all assertions hold. Otherwise, for a remaining point s, Lemma 18.1.2(1) gives $x \in N_{\mathbb{R}}$ with $\langle s, x \rangle < \langle s', x \rangle$ for all s' in the convex hull of $S \setminus \{s\}$. Every point of P is a convex combination $\lambda s + (1-\lambda)s''$ with s'' in that hull, so $\langle \cdot, x \rangle$ attains its minimum on P only at s, and $\{s\}$ is a vertex. Conversely, let Q be any face, the set where $\langle \cdot, x' \rangle$ is minimal for some $x' \in N_{\mathbb{R}}$, and write a point $m \in Q$ as a convex combination of points of S: all points with positive coefficient minimize $\langle \cdot, x' \rangle$, hence lie in Q. Thus Q is the convex hull of the points of S it contains. Applied to a vertex $\{v\}$, this shows $v \in S$. Hence the vertices are exactly the remaining points of S, and in particular lattice points. Every face is the convex hull of the vertices it contains, so there are only finitely many faces.

Let $\{v\}$ be a vertex, the set where some $\langle \cdot, x \rangle$ is minimal on P. Then v is not of the form $(1-t)a + tb$ with $a \neq b$ in P and $0 < t < 1$: the values $\langle a, x \rangle$ and $\langle b, x \rangle$ are at least $\langle v, x \rangle$, and their weighted mean equals it, so a and b would also be minimum points, that is, $a = b = v$. An edge is the convex hull of the finitely many vertices it contains, which lie on a line, so it is the segment between the two outermost ones; a vertex in the edge is not an interior point of the segment, hence it is an endpoint.

(2) We first show that every boundary point b of P lies in a facet. By Lemma 18.1.2(2), b lies in a face Q where some $x \neq 0$ is minimal; this face is proper, since $\langle \cdot, x \rangle$ is not constant on the full-dimensional set P. Suppose that Q has dimension $d < n-1$, and let c be the minimum of $\langle \cdot, x \rangle$ on P. The linear functionals vanishing on the directions of Q form a space of dimension $n - d \geq 2$, which contains x; choose y in it, independent of x, and put $x_s = x + sy$. On Q the functional $\langle \cdot, y \rangle$ is constant, say equal to c', and for each vertex $w \notin Q$ consider the gap $g_w(s) = \langle w, x_s \rangle - (c + sc') = (\langle w, x \rangle - c) + s(\langle w, y \rangle - c')$, which is positive at $s = 0$. Some vertex $w \notin Q$ has $\langle w, y \rangle \neq c'$, since otherwise y would be constant on P and hence zero; replacing y by $-y$ if necessary, we may assume that $\langle w, y \rangle < c'$ for some such w. Let $s_* > 0$ be the first value of $s > 0$ at which some gap vanishes. Then $x_{s_*} \neq 0$ by independence, all vertices of Q and at least one vertex $w \notin Q$ minimize $\langle \cdot, x_{s_*} \rangle$ on the vertices, and hence, by (1), on P. The new face contains Q and w, and w does not lie in the affine span of Q, because $\langle \cdot, x \rangle$ equals c on that span and is larger at w. So the dimension of the face has increased, and after finitely many steps we reach a facet containing b.

A facet ρ contains n affinely independent vertices, which are lattice points by (1). The linear functionals vanishing on the differences of these points form a line, which

is defined by linear equations with integral coefficients in the coordinates of bases of M and N; Gaussian elimination over $\mathbb{Q}$ produces a nonzero rational solution, and clearing denominators gives a nonzero vector of N on the line. Since ρ is a face, some $x \neq 0$ attains its minimum on P exactly on ρ; this x lies on the line, and we let u_ρ be the primitive vector of N that is a positive multiple of x, which exists and is unique by the observation following Definition 18.7.1. Then $a_\rho = -\langle v, u_\rho \rangle$, for any vertex $v \in \rho$, is an integer, $\ell_\rho \geq 0$ on P, and $\rho = \{m \in P : \ell_\rho(m) = 0\}$.

Next we show that P is the intersection of the half-spaces $\{\ell_\rho \geq 0\}$. Let $q \notin P$ and let p be an interior point of P. The set of $s \in [0,1]$ with $p + s(q-p) \in P$ is an interval $[0, s_0]$ with $s_0 < 1$, and $b = p + s_0(q-p)$ is a boundary point of P, hence lies in some facet ρ. The affine function $s \mapsto \ell_\rho(p + s(q-p))$ is positive at $s = 0$, because the nonconstant affine function ℓ_ρ, which is nonnegative on P, cannot attain its minimum at the interior point p. It vanishes at $s_0 < 1$, so $\ell_\rho(q) < 0$.

For irredundancy, let b be a point of the relative interior of a facet ρ, which exists by Lemma 18.1.2(4). If $\ell_{\rho'}(b) = 0$ for another facet ρ', then $\ell_{\rho'}$, which is nonnegative on ρ, vanishes on ρ by the observation preceding the lemma. Then $u_{\rho'}$ is constant on ρ, whose differences span a hyperplane, so $u_{\rho'}$ lies on the line of u_ρ, and $u_{\rho'} = \pm u_\rho$, because both vectors are primitive. If $u_{\rho'} = -u_\rho$, then $\ell_{\rho'} = -\ell_\rho$, since both vanish on ρ; as both are nonnegative on P, ℓ_ρ would vanish on P, which is impossible at an interior point. Hence $\ell_{\rho'} = \ell_\rho$ and $\rho' = \rho$. Thus all other facet functions are positive at b, and a point near b on the side where $\ell_\rho < 0$ violates only the inequality of ρ.

(3) If $m \in P$ and $v \in \rho$, then $\langle m - v, u_\rho \rangle = \ell_\rho(m) \geq 0$; hence $\operatorname{cone}(P - v) \subset C_v$. Conversely, if $y \in C_v$, then $v + sy \in P$ for all small $s > 0$, because the inequalities of the facets containing v hold along this half-line and the finitely many other facet functions are positive at v; hence $y \in \operatorname{cone}(P - v)$. The cone C_v contains no line: if $y \neq 0$ and $\pm y \in C_v$, then $v \pm sy \in P$ for small $s > 0$, and v would be the midpoint of two points of P, which is impossible for a vertex. Let u be the sum of the normals u_ρ of the facets containing v. If $y \in C_v$ and $\langle y, u \rangle = 0$, then $\langle y, u_\rho \rangle = 0$ for all these facets, so $-y \in C_v$ and $y = 0$. Consequently, the slice $\Sigma = \{y \in C_v : \langle y, u \rangle = 1\}$ is bounded (otherwise a limit of normalized unbounded points would give a nonzero $y \in C_v$ with $\langle y, u \rangle = 0$), and $C_v = \operatorname{cone}(\Sigma)$.

Call a point $y \in \Sigma$ *basic* if the normals u_ρ of the facets $\rho \ni v$ with $\langle y, u_\rho \rangle = 0$ span a space of dimension $n - 1$. Every point $y \in \Sigma$ is a convex combination of basic points. Indeed, if the active normals of y, together with u, span a space of dimension less than n, choose $d \neq 0$ annihilated by all of them. For small $|s|$ the points $y + sd$ remain in Σ, and since Σ is bounded there are maximal $s_\pm > 0$ with $y \pm s_\pm d \in \Sigma$. At $y \pm s_\pm d$ some new facet function $\langle \cdot, u_{\rho'} \rangle$ vanishes, and $\langle d, u_{\rho'} \rangle \neq 0$, so $u_{\rho'}$ is not in the span of the previous active normals and u. Thus y is a convex combination of two points with a larger span of active normals and u. Induction on the codimension of this span therefore proves the claim, because a point $y \in \Sigma$ whose active normals together with u span $N_\mathbb{R}$ is basic: its active normals annihilate $y \neq 0$, so they span a space of dimension at most $n - 1$, and at least $n - 1$, since adding u raises the dimension by at most one.

Next, $y \in \Sigma$ is basic if and only if $\mathbb{R}_{\geq 0} y$ is an extreme ray. If y is basic, its active normals annihilate exactly the line $\mathbb{R}y$; if $y = y_1 + y_2$ with $y_1, y_2 \in C_v$, then for

each active normal both pairings $\langle y_k, u_\rho\rangle$ are nonnegative with sum zero, hence zero, so $y_1, y_2 \in \mathbb{R}y \cap C_v = \mathbb{R}_{\geq 0}y$. Conversely, if $\mathbb{R}_{\geq 0}y$ is extreme, write y as a convex combination of basic points; extremality forces each of them to lie on the ray, hence to equal y. Therefore C_v is generated by its extreme rays.

It remains to identify the extreme rays with the edges at v. If $y \in \Sigma$ is basic, the set $\{m \in P : \ell_\rho(m) = 0 \text{ for the active facets of } y\}$ is a face, namely the set where the sum of the corresponding functionals $\langle \cdot, u_\rho\rangle$ is minimal; it contains $v + sy$ for small $s > 0$ and is contained in $v + (C_v \cap (\mathbb{R}y)) = v + \mathbb{R}_{\geq 0}y$. Hence it is an edge at v with direction y. Conversely, let E be an edge at v with direction y, so $y \in C_v$. If $y = y_1 + y_2$ with $y_1, y_2 \in C_v$, then for small $s > 0$ the point $v + sy \in E$ is the midpoint of $v + 2sy_1$ and $v + 2sy_2$, which lie in P. Since E is the set where some linear functional is minimal on P, both points lie in $E \subset v + \mathbb{R}_{\geq 0}y$, and so $y_1, y_2 \in \mathbb{R}_{\geq 0}y$. Thus the ray of y is extreme.

(4) Let P be smooth and v a vertex. By (3), $C_v = \operatorname{cone}(e_1, \dots, e_n)$. Let $e_1^*, \dots, e_n^* \in N$ be the dual basis, so that $C_v = \{y : \langle y, e_i^*\rangle \geq 0 \text{ for all } i\}$. If ρ is a facet containing v, then $u_\rho \geq 0$ on C_v, so $u_\rho = \sum_i c_i e_i^*$ with $c_i = \langle e_i, u_\rho\rangle \geq 0$. The set $\rho - v$ lies in $C_v \cap u_\rho^\perp = \operatorname{cone}\{e_i : c_i = 0\}$ and spans a space of dimension $n-1$, so exactly one coefficient c_i is positive. Since u_ρ and e_i^* are primitive, $u_\rho = e_i^*$. Distinct facets have distinct normals, because a facet is the set where its normal is minimal; so at most n facets contain v. If some e_i^* were not the normal of a facet through v, the vector $-e_i$ would satisfy all the inequalities defining C_v, although $-e_i \notin \operatorname{cone}(e_1, \dots, e_n)$. Hence exactly n facets contain v, and numbering them suitably, $u_i = e_i^*$. For $m \in P$ this gives $m - v = \sum_i \langle m - v, u_i\rangle e_i = \sum_i \ell_i(m) e_i$, because $\ell_i(v) = 0$. Finally, the edge in direction e_i is a segment $[v, v']$ with v' a lattice point, so $v' - v = k e_i$ for a positive integer k, and $v + e_i \in [v, v'] \subset P$. In dimension zero there are no facets and no edges, and the assertions are empty. □

For the rest of the chapter we fix a smooth lattice polytope $P \subset M_\mathbb{R}$, which by definition has nonempty interior, with the notation ℓ_ρ, u_ρ and a_ρ of Lemma 18.7.2, and we put $A = P \cap M$, a finite set containing the vertices. Let $T = \operatorname{Hom}(M, \mathbb{C}^*)$ be the group of homomorphisms from M to the multiplicative group $\mathbb{C}^*$, and $T_c = \operatorname{Hom}(M, S^1)$ its compact subgroup, where $S^1 = \{\zeta \in \mathbb{C} : |\zeta| = 1\}$. Each $m \in M$ defines a *character* $\chi^m : T \to \mathbb{C}^*$, $\chi^m(t) = t(m)$, so that $\chi^{m+m'} = \chi^m \chi^{m'}$. A basis $m_1, \dots, m_n$ of M identifies T with $(\mathbb{C}^*)^n$ through $t \mapsto (\chi^{m_1}(t), \dots, \chi^{m_n}(t))$, since a homomorphism on M is determined by its values on a basis, which may be arbitrary. The resulting structure of complex manifold on T does not depend on the basis, because a change of basis acts on the coordinates by a monomial map whose inverse is again monomial. In these coordinates χ^m becomes the Laurent monomial $z_1^{c_1} \cdots z_n^{c_n}$ for $m = \sum_j c_j m_j$, and T_c becomes $(S^1)^n$, acting on $T = (\mathbb{C}^*)^n$ by multiplication, that is, by separate rotations of the coordinates. The *tropicalization map* $\operatorname{Trop} : T \to N_\mathbb{R}$ is defined by

$$\langle m, \operatorname{Trop}(t)\rangle = \log |\chi^m(t)|^2 \qquad (m \in M);$$

in the coordinates above, with $N_\mathbb{R}$ identified with $\mathbb{R}^n$ by the dual basis, it is $z \mapsto (\log |z_1|^2, \dots, \log |z_n|^2)$. By Corollary 18.1.6, the T_c-invariant plurisubharmonic functions on T are exactly the functions $F \circ \operatorname{Trop}$ with F finite and convex on $N_\mathbb{R}$. (In

dimension zero, T and $N_{\mathbb{R}}$ are points, and by Definition 8.1.1 both classes consist of the real constants.)

Let $\mathbb{P}^{|A|-1}$ be the complex projective space of lines in $\mathbb{C}^A$, with homogeneous coordinates $Z = [Z_m]_{m\in A}$. By Example 14.5.5, with homogeneous coordinates indexed by the finite set A, it is a compact complex manifold of dimension $|A| - 1$, with the affine charts $\{Z_{m_0} \neq 0\}$, $m_0 \in A$, and the coordinates Z_m/Z_{m_0}, $m \neq m_0$. It carries the hyperplane bundle $\mathcal{O}_{\mathbb{P}^{|A|-1}}(1)$, dual to the tautological line bundle, with the Fubini–Study metric, and each linear form $Z \mapsto Z_m$ defines a holomorphic coordinate section of $\mathcal{O}_{\mathbb{P}^{|A|-1}}(1)$, which we denote again by Z_m. Define

$$\iota\colon T \longrightarrow \mathbb{P}^{|A|-1}, \qquad t \longmapsto [\chi^m(t)]_{m\in A}, \qquad X_P = \overline{\iota(T)}.$$

The torus T acts on $\mathbb{P}^{|A|-1}$ by $\tau \cdot [Z_m] = [\chi^m(\tau)Z_m]$. Since $\iota(\tau t) = \tau \cdot \iota(t)$, this action preserves $\iota(T)$ and hence its closure X_P.

Theorem 18.7.3 *The set X_P is an n-dimensional complex submanifold of $\mathbb{P}^{|A|-1}$; it is the common zero set of finitely many homogeneous polynomials and is irreducible, so it is a smooth projective variety of dimension n. The map ι is a biholomorphism of T onto a dense open subset of X_P. For each vertex v of P, with primitive edge basis $e_1, \dots, e_n$, the set $U_v = X_P \cap \{Z_v \neq 0\}$ is open in X_P, and*

$$w = (w_1, \dots, w_n), \qquad w_i = Z_{v+e_i}/Z_v,$$

is a biholomorphism of U_v onto $\mathbb{C}^n$ satisfying $w_i \circ \iota = \chi^{e_i}$. These vertex charts cover X_P. For $m \in A$ write $m - v = \sum_i b_i e_i$; then the b_i are nonnegative integers and $Z_m/Z_v = \prod_i w_i^{b_i}$ on U_v. Finally, the restriction $L = \mathcal{O}_{\mathbb{P}^{|A|-1}}(1)|_{X_P}$ of the hyperplane bundle is very ample, and the restriction of the Fubini–Study metric is a smooth quadratic metric on L with positive definite curvature form.

Proof *Every point of X_P has a nonzero vertex coordinate.* Let $m \in A$. By Lemma 18.7.2(1), m lies in the convex hull of the vertices of P, so by Lemma 18.1.2(3) it is a convex combination of affinely independent vertices. The coefficients of this combination are the unique solution of a linear system with integral coefficients, hence rational. Clearing denominators, we obtain

$$dm = \sum_v c_v v, \qquad c_v \in \mathbb{Z}_{\geq 0}, \quad \sum_v c_v = d \geq 1.$$

Since $\chi^{dm} = \prod_v (\chi^v)^{c_v}$, the homogeneous equation $Z_m^d = \prod_v Z_v^{c_v}$ of degree d holds on $\iota(T)$, hence on its closure X_P. If all vertex coordinates of a point of X_P vanished, these equations would force all its coordinates to vanish, which is impossible.

The vertex charts. Fix a vertex v. For $m \in A$, Lemma 18.7.2(4) gives $m - v = \sum_i b_i(m)e_i$ with $b_i(m) = \ell_i(m) = \langle m, u_i\rangle + a_i$, a nonnegative integer; we write $b(m) = (b_1(m), \dots, b_n(m))$. In particular $b(v + e_i)$ is the ith unit vector, and on $\iota(T)$

$$\frac{Z_m}{Z_v} = \chi^{m-v} = \prod_i (\chi^{e_i})^{b_i(m)} = \prod_i w_i^{b_i(m)}.$$

Let $\Phi_v\colon \mathbb{C}^n \to \{Z_v \neq 0\}$ be the map whose coordinates Z_m/Z_v are the monomials $\prod_i w_i^{b_i(m)}$. Its image Γ_v is the graph of a polynomial map γ from the coordinates $w_i = Z_{v+e_i}/Z_v$ of the affine chart $\{Z_v \neq 0\}$ to the remaining coordinates $y = (Z_m/Z_v)_{m\neq v, v+e_1,\dots,v+e_n}$. Hence Γ_v is closed in the chart, and in the holomorphic coordinates $(w, y-\gamma(w))$ of the chart it is the coordinate plane $\{y-\gamma(w)=0\}$; so Γ_v is a closed complex submanifold of the chart, and w is a biholomorphism of Γ_v onto $\mathbb{C}^n$, with inverse Φ_v. Since $e_1,\dots,e_n$ is a basis of M, the map $t \mapsto (\chi^{e_1}(t),\dots,\chi^{e_n}(t))$ is a bijection of T onto $(\mathbb{C}^*)^n$, and the last display shows that $\iota(T) = \Phi_v((\mathbb{C}^*)^n)$. As $(\mathbb{C}^*)^n$ is dense in $\mathbb{C}^n$ and Γ_v is closed in the chart, the closure of $\iota(T)$ in the chart is Γ_v; that is, $U_v = \Gamma_v$. By the first step the sets U_v cover X_P, so X_P is a complex submanifold of dimension n with the asserted charts. The coordinates w_i recover t from $\iota(t)$, so ι is injective with holomorphic inverse w on $\iota(T)$, and $\iota(T) = \Phi_v((\mathbb{C}^*)^n)$ is open in U_v. Thus ι is a biholomorphism of T onto a dense open subset of X_P.

Algebraicity and irreducibility. In addition to the equations of the first step, consider for each vertex v and each $m \in A \setminus \{v\}$ the homogeneous equation

$$Z_m Z_v^{\sum_i b_i(m)-1} = \prod_i Z_{v+e_i}^{b_i(m)},$$

where $\sum_i b_i(m) \geq 1$ because $m \neq v$. It holds on $\iota(T)$, hence on X_P. Let Y be the common zero set of all these equations in $\mathbb{P}^{|A|-1}$, so that $X_P \subset Y$. By the equations of the first step, every point of Y has a nonzero vertex coordinate Z_v, and on $\{Z_v \neq 0\}$ the equations for v say exactly that $Z_m/Z_v = \prod_i w_i^{b_i(m)}$; thus $Y \cap \{Z_v \neq 0\} \subset \Gamma_v = U_v$. Hence $Y = X_P$. To prove irreducibility, suppose that $X_P = Y_1 \cup Y_2$, where $Y_1, Y_2 \subsetneq X_P$ are common zero sets of homogeneous polynomials, and choose homogeneous polynomials f_1, f_2 with $f_k = 0$ on Y_k and $f_k \not\equiv 0$ on X_P. Then $g_k(t) = f_k((\chi^m(t))_{m\in A})$ is a Laurent polynomial in the coordinates of $T \cong (\mathbb{C}^*)^n$. It is not identically zero: otherwise f_k would vanish on $\iota(T)$, hence on its closure X_P, since the zero set of a homogeneous polynomial is closed. On the other hand $g_1 g_2 = 0$ on T, because $f_1 f_2 = 0$ on X_P. This is impossible. Indeed, since g_1 is continuous and not identically zero, it does not vanish on some nonempty open set $W \subset (\mathbb{C}^*)^n$, and g_2 vanishes on W. A suitable monomial multiple of g_2 is a polynomial that vanishes on W, so all its derivatives vanish at a point of W; being the sum of its Taylor expansion at that point, it is zero. Hence $g_2 = 0$, a contradiction. Thus X_P is irreducible.

The line bundle. The coordinate sections $s_m = Z_m|_{X_P}$, $m \in A$, of L have no common zero and define the inclusion $X_P \subset \mathbb{P}^{|A|-1}$. By Theorem 18.8.5 with $k = 1$, whose proof uses only the vertex charts constructed above, they form a basis of $H^0(X_P, L)$; in the notation of that theorem, $s_m = \chi^m s_0$. Hence the full space of sections of L defines an embedding, and L is very ample in the sense of Section 14.5. On U_v the section s_v has no zeros, and the Fubini–Study metric of Examples 14.5.5 and 14.5.6, restricted to L, gives it the squared norm

$$|s_v|^2 = \frac{|Z_v|^2}{\sum_{m\in A}|Z_m|^2} = \mathrm{e}^{-H_v}, \qquad H_v = \log \sum_{m\in A} |w^{b(m)}|^2,$$

where $w^b = \prod_i w_i^{b_i}$. The curvature on U_v is $\mathrm{dd^c} H_v$, by Definition 13.1.8. The complex Hessian of H_v is positive definite. Indeed, put $f_m = w^{b(m)}$ and $S = \sum_m |f_m|^2$, and for $\xi \in \mathbb{C}^n$ write $\xi f_m = \sum_j \xi_j \partial f_m / \partial w_j$. Then

$$\sum_{j,k} \frac{\partial^2 H_v}{\partial w_j \partial \bar{w}_k} \xi_j \bar{\xi}_k = \frac{S \sum_m |\xi f_m|^2 - \left|\sum_m \bar{f}_m \, \xi f_m\right|^2}{S^2} \geq 0$$

by the Cauchy–Schwarz inequality, with equality only if $(\xi f_m)_m = \lambda (f_m)_m$ for some $\lambda \in \mathbb{C}$. The constant monomial $f_v = 1$ forces $\lambda = 0$, and then the monomials $f_{v+e_i} = w_i$ give $\xi_i = 0$ for every i. Thus the restricted Fubini–Study metric is a smooth quadratic metric on L with positive definite curvature form. In dimension zero X_P is a point and there is nothing to prove. □

18.8 Fans, divisors and character sections

We now use the vertex charts to describe the complement of the torus in X_P and the global sections of the powers of L. The combinatorics of this description is organized by the inner normals of P. For a face Q of P, let $\sigma_Q \subset N_{\mathbb{R}}$ be the cone generated by the inward normals u_ρ of the facets ρ containing Q; thus $\sigma_P = \{0\}$. A *face* of a cone $\sigma \subset N_{\mathbb{R}}$ is a set $\{x \in \sigma : \langle m, x\rangle = 0\}$, where $m \in M_{\mathbb{R}}$ is nonnegative on σ.

Proposition 18.8.1 *For every face Q of P, a vector $x \in N_{\mathbb{R}}$ lies in σ_Q if and only if the minimum of $\langle m, x\rangle$ over $m \in P$ is attained at every point of Q, and Q is the intersection of P with the facets containing it. For faces Q_1, Q_2, the intersection Q of P with the facets containing $Q_1 \cup Q_2$ is the smallest face containing $Q_1 \cup Q_2$, and $\sigma_{Q_1} \cap \sigma_{Q_2} = \sigma_Q$. The cones σ_Q form a complete smooth fan, the* inner normal fan *of P: they cover $N_{\mathbb{R}}$, each is generated by part of a basis of N, every face of each σ_Q is one of the cones $\sigma_{Q'}$, and any two of them intersect in a face of each. If $n > 0$, there is a constant Λ, depending only on P and the fixed Euclidean norm, such that every $x \in N_{\mathbb{R}}$ with $|x| = 1$ can be written as*

$$x = \lambda \sum_i c_i u_{\rho_i}, \qquad 0 < \lambda \leq \Lambda, \quad c_i > 0, \quad \sum_i c_i = 1,$$

where all u_{ρ_i} lie in a single cone of this fan.

Proof Let Q be a face, choose a vertex $v \in Q$, and use the notation of Lemma 18.7.2(4). Since $u_1, \ldots, u_n$ is the basis dual to $e_1, \ldots, e_n$, every $x \in N_{\mathbb{R}}$ can be written as $x = \sum_i \alpha_i u_i$ with $\alpha_i = \langle e_i, x\rangle$, and $\langle m - v, x\rangle = \sum_i t_i \alpha_i$ for $m = v + \sum_i t_i e_i \in P$. If v minimizes $\langle \cdot, x\rangle$ on P, then $\alpha_i = \langle (v + e_i) - v, x\rangle \geq 0$, because $v + e_i \in P$. If moreover every point of Q minimizes it, and the ith facet at v does not contain Q, choose $m \in Q$ with $t_i = \ell_i(m) > 0$; then $\sum_j t_j \alpha_j = 0$ with nonnegative terms, so $\alpha_i = 0$. Hence x is a nonnegative combination of the normals of the facets containing Q, all of which

pass through v, that is, $x \in \sigma_Q$. Conversely, if $x = \sum_\rho c_\rho u_\rho$ with $c_\rho \geq 0$ and all ρ containing Q, then $\langle m, x\rangle = \sum_\rho c_\rho(\ell_\rho(m) - a_\rho)$, and each ℓ_ρ attains its minimum 0 on P at every point of Q; so does $\langle \cdot, x\rangle$. This proves the characterization, and it shows that σ_Q is generated by part of the basis $u_1, \ldots, u_n$ of N.

Every face Q is the intersection of P with the facets containing it. Indeed, if Q is the set where $\langle \cdot, x\rangle$ is minimal, then $x \in \sigma_Q$, and writing $x = \sum_\rho c_\rho u_\rho$ as above we see that the minimum is attained exactly where $\ell_\rho = 0$ for all ρ with $c_\rho > 0$; this set contains the intersection of the facets containing Q, which in turn contains Q.

The faces of σ_Q are the cones generated by subsets of its generators. Indeed, if $m \in M_\mathbb{R}$ is nonnegative on σ_Q and x is a nonnegative combination of the generators, then $\langle m, x\rangle$ is a sum of nonnegative terms, so the face $\{x \in \sigma_Q : \langle m, x\rangle = 0\}$ is generated by the generators annihilated by m. Conversely, since the generators are part of a basis of $N_\mathbb{R}$, for every subset of them there is an m that vanishes on this subset and equals 1 on the remaining generators, and m is nonnegative on σ_Q. If J is a set of facets containing v, the set $Q' = \{m \in P : \ell_\rho(m) = 0 \text{ for } \rho \in J\}$ is a face, namely the set where $\sum_{\rho\in J}\langle \cdot, u_\rho\rangle$ is minimal. It contains $v + se_i$ for small $s > 0$ whenever the ith facet at v is not in J, so the facets containing Q' are exactly those in J, and $\sigma_{Q'} = \operatorname{cone}\{u_\rho : \rho \in J\}$. Thus every face of σ_Q is in the family.

Every x attains its minimum on P on some face Q, and then $x \in \sigma_Q$ by the characterization; hence the cones cover $N_\mathbb{R}$. Let Q_1, Q_2 be faces, and let Q be the intersection of P with all facets containing both. It is a face (if no facet contains both, $Q = P$; otherwise it is the set where the sum of the corresponding functionals is minimal), and every face containing $Q_1 \cup Q_2$ contains Q, since it is the intersection of P with the facets containing it. Since $Q \supset Q_1 \cup Q_2$, it is the smallest face containing $Q_1 \cup Q_2$. By the characterization, $x \in \sigma_{Q_1} \cap \sigma_{Q_2}$ if and only if the face where $\langle \cdot, x\rangle$ is minimal contains $Q_1 \cup Q_2$, that is, if and only if it contains Q; so $\sigma_{Q_1} \cap \sigma_{Q_2} = \sigma_Q$. The generators of σ_Q, the normals of the facets containing $Q_1 \cup Q_2$, form a subset of the generators of σ_{Q_k}, so σ_Q is a face of σ_{Q_1} and of σ_{Q_2}.

For the last assertion, let $|x| = 1$ and choose a vertex v minimizing $\langle \cdot, x\rangle$ on P. The coefficients $\alpha_i = \langle e_i, x\rangle$ above are nonnegative and satisfy $\alpha_i \leq |e_i|$. Their sum λ is positive, because $x \neq 0$, and at most $\sum_i |e_i|$. Let Λ be the maximum of these finite sums over all vertices. Dividing by λ and omitting the zero coefficients gives the required expression, with all u_{ρ_i} in the cone $\sigma_{\{v\}}$. In dimension zero the fan consists of the zero cone, and there is no unit vector. □

Proposition 18.8.2 *The T-orbits in X_P are indexed by the faces Q of P, and the orbit indexed by Q has dimension* $\dim Q$. *The closure of the orbit indexed by Q is the union of the orbits indexed by the faces of P contained in Q. The complement of $\iota(T)$ is the union of the T-invariant prime divisors D_ρ, the closures of the orbits indexed by the facets ρ. For every $m \in M$ and every facet ρ, the character χ^m extends to a meromorphic function on X_P, with order*

$$\operatorname{ord}_{D_\rho} \chi^m = \langle m, u_\rho\rangle$$

along D_ρ. Finally, for every vertex v of P and every $m \in M$, the character χ^m is holomorphic on the chart U_v if and only if $\langle m, x\rangle \geq 0$ for all x in the cone $\sigma_{\{v\}}$ of Proposition 18.8.1.

Proof Fix a vertex v and use the chart $U_v \cong \mathbb{C}^n$ of Theorem 18.7.3. For $\tau \in T$ we have $w_i(\tau \cdot z) = \chi^{e_i}(\tau) w_i(z)$, and $\tau \mapsto (\chi^{e_i}(\tau))_i$ maps T onto $(\mathbb{C}^*)^n$; so T acts on U_v by independent scalings of the coordinates. The orbits in U_v are therefore the sets

$$O_{v,I} = \{w : w_i = 0 \text{ for } i \in I,\ w_j \neq 0 \text{ for } j \notin I\}, \qquad I \subset \{1, \dots, n\},$$

and $O_{v,I}$ has dimension $n - |I|$. Let $Q_{v,I} = \{m \in P : \ell_i(m) = 0 \text{ for } i \in I\}$. It is a face, namely the set where $\sum_{i\in I}\langle \cdot, u_i\rangle$ is minimal on P. It contains v and the points $v + se_j$ for $j \notin I$ and small $s > 0$, and it lies in $v + \operatorname{span}\{e_j : j \notin I\}$, because $m - v = \sum_i \ell_i(m) e_i$; so $\dim Q_{v,I} = n - |I|$. Since $Z_m/Z_v = \prod_i w_i^{\ell_i(m)}$, a homogeneous coordinate Z_m, $m \in A$, is nonzero on $O_{v,I}$ if and only if $\ell_i(m) = 0$ for all $i \in I$, that is, if and only if $m \in Q_{v,I}$.

The charts U_v are T-invariant, because the action of T multiplies Z_v by $\chi^v(\tau) \neq 0$. Hence every orbit O in X_P meets, and therefore lies in, some chart U_v, so it is one of the sets $O_{v,I}$. The set of $m \in A$ with $Z_m \neq 0$ on O does not depend on the chart, and it equals $Q_{v,I} \cap A$; since a face is the convex hull of its vertices, which are lattice points, $Q_{v,I}$ is the convex hull of this set. Thus each orbit O determines a face $Q(O)$, with $\dim Q(O) = \dim O$. Distinct orbits determine distinct faces: if $Q(O) = Q(O') = Q$ and v is a vertex of Q, then $Z_v \neq 0$ on both orbits, so both lie in U_v, where an orbit is determined by the set of indices i with $v + e_i \notin Q$. Every face Q arises: choose a vertex $v \in Q$ and let I be the set of indices of the facets at v that contain Q; then $Q_{v,I} = Q$, because Q is the intersection of P with the facets containing it (Proposition 18.8.1), and these facets all pass through v. We write O_Q for the orbit of the face Q; in particular $O_P = \iota(T)$.

The closure $\overline{O_Q}$ is T-invariant, hence a union of orbits, and it lies in the closed set $\{Z_m = 0 \text{ for } m \in A \setminus Q\}$. If $O_{Q'} \subset \overline{O_Q}$, then $Q' \cap A \subset Q$, and so $Q' \subset Q$. Conversely, let $Q' \subset Q$ be faces, let v be a vertex of Q', and let $I \subset I'$ be the index sets of Q and Q' at v. A point $w \in O_{Q'} = O_{v,I'}$ is the limit as $\varepsilon \downarrow 0$ of the points obtained from w by replacing the coordinates $w_i = 0$, $i \in I' \setminus I$, by ε, and these points lie in $O_{v,I} = O_Q$. Hence $O_{Q'} \subset \overline{O_Q}$.

Let ρ be a facet and $D_\rho = \overline{O_\rho}$. If $v \in \rho$ and ρ is the ith facet at v, the orbits in U_v indexed by faces contained in ρ are the sets $O_{v,I}$ with $i \in I$, so $D_\rho \cap U_v = \{w_i = 0\}$; if $v \notin \rho$, then $D_\rho \cap U_v = \varnothing$, because $Z_v = 0$ on D_ρ. Hence D_ρ is a T-invariant closed submanifold of codimension one. Every proper face is contained in a facet, being the intersection of the facets containing it; hence $X_P \setminus \iota(T)$, the union of the orbits of the proper faces, is the union of the divisors D_ρ.

Each D_ρ is irreducible, that is, not the union of two analytic subsets different from D_ρ. First, D_ρ is connected, being the closure of the orbit O_ρ: in a chart U_v in which ρ is the ith facet at v, the coordinates w_j, $j \neq i$, identify $O_\rho = O_{v,\{i\}}$ with the connected set $(\mathbb{C}^*)^{n-1}$. Next, let $Z \subsetneq D_\rho$ be an analytic subset. The set of interior points of Z in D_ρ is open, and it is closed in D_ρ. Indeed, if p lies in its closure, then $p \in Z$, and on a connected coordinate neighborhood of p in D_ρ the local holomorphic

functions defining Z vanish on a nonempty open set, hence identically by the identity theorem, so that p is an interior point. As D_ρ is connected and $Z \neq D_\rho$, the set Z has empty interior in D_ρ. If $D_\rho = Z_1 \cup Z_2$ with analytic subsets $Z_1, Z_2 \neq D_\rho$, then the nonempty open set $D_\rho \setminus Z_1$ would be contained in Z_2, which has empty interior.

Let $m \in M$. Since $m = \sum_j \langle m, u_j\rangle e_j$, the character χ^m is the Laurent monomial $\prod_j w_j^{\langle m,u_j\rangle}$ in the coordinates of U_v. Hence it extends to a meromorphic function on each U_v, and these extensions agree on the dense set $\iota(T)$, hence on overlaps. Near each point of the dense orbit $O_\rho \cap U_v$ of $D_\rho \cap U_v = \{w_i = 0\}$, it is $w_i^{\langle m,u_i\rangle}$ times a nowhere vanishing holomorphic function; so its order along D_ρ, the difference of the orders of numerator and denominator in the sense of Definition 12.6.3, is $\langle m, u_i\rangle = \langle m, u_\rho\rangle$. In particular, χ^m is holomorphic on $U_v \cong \mathbb{C}^n$ if and only if all the exponents $\langle m, u_j\rangle$ are nonnegative, that is, if and only if $\langle m, x\rangle \geq 0$ for every x in $\sigma_{\{v\}} = \operatorname{cone}(u_1, \ldots, u_n)$. □

The vertex charts are glued along open sets determined by the fan, as follows.

Proposition 18.8.3 *Let v and v' be vertices of P, with primitive edge bases $e_1, \ldots, e_n$ and $e'_1, \ldots, e'_n$ and vertex charts w and w', and let $u_1, \ldots, u_n$ be the inward normals of the facets through v, as in Lemma 18.7.2(4).*

(1) *A lattice point $m \in M$ satisfies $\langle m, x\rangle \geq 0$ for all $x \in \sigma_{\{v\}}$ if and only if $m = \sum_j c_j e_j$ with integers $c_j \geq 0$. Consequently, the characters that are holomorphic on U_v are exactly the monomials $w^c = \prod_j w_j^{c_j}$, $c \in \mathbb{Z}^n_{\geq 0}$.*

(2) *The character $\chi^{v'-v}$ is holomorphic on U_v, and $U_v \cap U_{v'}$ is the set of points of U_v at which it does not vanish. On this set*

$$w'_i = \prod_j w_j^{\langle e'_i, u_j\rangle} \qquad (1 \leq i \leq n),$$

where the factors with negative exponents do not vanish.

(3) *Let Q be the intersection of P with the facets containing both v and v', the smallest face of P containing v and v'. Then*

$$\sigma_{\{v\}} \cap \sigma_{\{v'\}} = \sigma_Q = \{x \in \sigma_{\{v\}} : \langle v' - v, x\rangle = 0\}.$$

Proof (1) Since $u_1, \ldots, u_n$ is the basis of N dual to $e_1, \ldots, e_n$, every $m \in M$ is $m = \sum_j \langle m, u_j\rangle e_j$ with integer coefficients, and $\sigma_{\{v\}} = \operatorname{cone}(u_1, \ldots, u_n)$. Hence $\langle m, \cdot\rangle \geq 0$ on $\sigma_{\{v\}}$ if and only if all coefficients $\langle m, u_j\rangle$ are nonnegative. The second assertion follows from the last assertion of Proposition 18.8.2, since $\chi^m = \prod_j w_j^{\langle m,u_j\rangle}$ on U_v.

(2) By Theorem 18.7.3, $Z_{v'}/Z_v = \prod_j w_j^{b_j(v')}$ on U_v, with $b_j(v') = \ell_j(v') = \langle v' - v, u_j\rangle \geq 0$; this holomorphic function is the extension of $\chi^{v'-v}$ to U_v. Since $U_v \cap U_{v'} = U_v \cap \{Z_{v'} \neq 0\}$, the first assertion follows. On $U_v \cap U_{v'}$ we have $w'_i = Z_{v'+e'_i}/Z_{v'} = w^{b(v'+e'_i)}/w^{b(v')}$, a quotient of monomials whose denominator does not vanish there, and $b_j(v' + e'_i) - b_j(v') = \langle e'_i, u_j\rangle$. A negative exponent

$\langle e_i', u_j\rangle$ forces $b_j(v') > 0$, so the corresponding factor w_j divides the denominator and does not vanish on $U_v \cap U_{v'}$.

(3) By Proposition 18.8.1, applied to the faces $\{v\}$ and $\{v'\}$, the set Q is the smallest face containing v and v', and $\sigma_{\{v\}} \cap \sigma_{\{v'\}} = \sigma_Q$. The facets containing Q are the facets ρ_j through v with $\ell_j(v') = 0$. If $x = \sum_j \alpha_j u_j \in \sigma_{\{v\}}$ with $\alpha_j \geq 0$, then $\langle v' - v, x\rangle = \sum_j \alpha_j \ell_j(v')$ is a sum of nonnegative terms, so it vanishes if and only if $\alpha_j = 0$ whenever $\ell_j(v') > 0$, that is, if and only if $x \in \operatorname{cone}\{u_j : \ell_j(v') = 0\} = \sigma_Q$.□

Thus X_P is obtained by gluing the affine spaces $U_v \cong \mathbb{C}^n$ along the open subsets $\{\chi^{v'-v} \neq 0\}$ by the monomial coordinate changes of Proposition 18.8.3(2). The holomorphic characters on U_v are the χ^m with m in the semigroup $\{m \in M : \langle m, \cdot\rangle \geq 0 \text{ on } \sigma_{\{v\}}\}$, and the open subset $\{\chi^{v'-v} \neq 0\}$ corresponds to the common face $\sigma_{\{v\}} \cap \sigma_{\{v'\}} = \sigma_{\{v\}} \cap (v' - v)^\perp$. These are the gluing data of the toric variety of the inner normal fan of P in [CLS11, Section 3.1], in which U_v is the affine toric variety of the cone $\sigma_{\{v\}}$; in this sense X_P is the toric variety of the inner normal fan of P.

Global sections will be computed from Laurent expansions on the torus.

Lemma 18.8.4 *Let $n \geq 1$, and let f be a holomorphic function on $(\mathbb{C}^*)^n$. For $m \in \mathbb{Z}^n$ and $r \in (0,\infty)^n$ put*

$$c_m(r) = \frac{1}{(2\pi)^n} \int_{[0,2\pi]^n} f(r_1 e^{i\theta_1}, \dots, r_n e^{i\theta_n})\, r^{-m} e^{-i\langle m,\theta\rangle}\, d\theta,$$

where $r^{-m} = \prod_j r_j^{-m_j}$ and $\langle m, \theta\rangle = \sum_j m_j\theta_j$. Then $c_m = c_m(r)$ does not depend on r, and the Laurent series $\sum_{m\in\mathbb{Z}^n} c_m z^m$, where $z^m = \prod_j z_j^{m_j}$, converges normally on every compact subset of $(\mathbb{C}^*)^n$, with sum f. If a series $\sum_m c_m' z^m$ converges normally to f on one torus $\{z : |z_j| = r_j \text{ for all } j\}$, then $c_m' = c_m$ for every m. If f extends holomorphically to $\mathbb{C}^n$, then $c_m = 0$ unless $m_j \geq 0$ for every j.

Proof Fix j and the variables other than z_j. The resulting function $g(\zeta)$ is holomorphic on $\mathbb{C}^*$, and for $k \in \mathbb{Z}$ the integral $\frac{1}{2\pi}\int_0^{2\pi} g(\varrho e^{i\theta})\varrho^{-k} e^{-ik\theta}\, d\theta = \frac{1}{2\pi i}\int_{|\zeta|=\varrho} g(\zeta)\zeta^{-k-1}\, d\zeta$ does not depend on $\varrho > 0$, by Cauchy's theorem on the annulus between two such circles. Integrating over the other angles, by Fubini's theorem, we see that $c_m(r)$ does not depend on r_j; hence it does not depend on r. The definition gives the Cauchy estimates $|c_m| \leq C(r)r^{-m}$, where $C(r)$ is the maximum of $|f|$ on the torus $\{|z_j| = r_j\}$. Let $K \subset (\mathbb{C}^*)^n$ be compact, and choose $0 < a \leq b$ with $a \leq |z_j| \leq b$ on K for all j. Given m, take $r_j = 2b$ if $m_j \geq 0$ and $r_j = a/2$ if $m_j < 0$. Then, for $z \in K$,

$$|c_m z^m| \leq C_K \prod_j \Big(\frac{|z_j|}{r_j}\Big)^{m_j} \leq C_K 2^{-(|m_1|+\cdots+|m_n|)},$$

where C_K is the largest of the numbers $C(r)$ for the 2^n tori with $r_j \in \{a/2, 2b\}$. Since $\sum_{m\in\mathbb{Z}^n} 2^{-(|m_1|+\cdots+|m_n|)} = 3^n$, the series converges normally on K.

We show by induction on n that its sum is f. For $n = 1$ this is the Laurent theorem of one-variable theory, with the coefficients given by the integral above. Let $n \geq 2$, and write $z = (z', z_n)$ with $z' \in (\mathbb{C}^*)^{n-1}$. For fixed z', the one-variable theorem gives $f(z', z_n) = \sum_k a_k(z') z_n^k$ with $a_k(z') = \frac{1}{2\pi} \int_0^{2\pi} f(z', \mathrm{e}^{\mathrm{i}\theta}) \mathrm{e}^{-\mathrm{i}k\theta} \, \mathrm{d}\theta$. Differentiation under the integral sign shows that a_k is smooth and satisfies the Cauchy–Riemann equations, so it is holomorphic on $(\mathbb{C}^*)^{n-1}$, and by Fubini's theorem its Laurent coefficients are the numbers $c_{(m',k)}$, $m' \in \mathbb{Z}^{n-1}$. By the induction hypothesis, $a_k(z') = \sum_{m'} c_{(m',k)} z'^{m'}$, so $f(z) = \sum_k \sum_{m'} c_{(m',k)} z'^{m'} z_n^k$. This double series converges absolutely, by the previous paragraph, so it may be summed as $\sum_m c_m z^m$.

For the uniqueness, integrate the normally convergent series $\sum_{m''} c'_{m''} r^{m''} \mathrm{e}^{\mathrm{i}\langle m'', \theta\rangle} = f(r_1 \mathrm{e}^{\mathrm{i}\theta_1}, \ldots)$ term by term against $(2\pi)^{-n} r^{-m} \mathrm{e}^{-\mathrm{i}\langle m, \theta\rangle} \, \mathrm{d}\theta$; since $\int_0^{2\pi} \mathrm{e}^{\mathrm{i}l\theta} \, \mathrm{d}\theta = 0$ for every integer $l \neq 0$, only the term $m'' = m$ survives, and we obtain $c'_m = c_m(r) = c_m$. Finally, if f is holomorphic on $\mathbb{C}^n$, its Taylor series at the origin converges normally on every polydisc by (7.1), in particular on every torus, so by uniqueness the coefficients c_m are the Taylor coefficients for $m \in \mathbb{Z}^n_{\geq 0}$ and vanish for all other m. □

Theorem 18.8.5 *There is a meromorphic section s_0 of L with divisor*

$$D_P = \sum_\rho a_\rho D_\rho,$$

so that L is isomorphic to the line bundle $\mathcal{O}_{X_P}(D_P)$ of this divisor. For every integer $k \geq 0$,

$$H^0(X_P, L^k) = \bigoplus_{m \in kP \cap M} \mathbb{C}\, \chi^m s_0^k,$$

where $0P = \{0\}$. Each summand is a line on which the natural action of T is given by a character, and the dimensions $h^0(X_P, L^k) = \dim H^0(X_P, L^k)$ satisfy

$$\lim_{k\to\infty} \frac{n!}{k^n} h^0(X_P, L^k) = n! \operatorname{vol}(P).$$

Proof For vertices v and v', the coordinate sections satisfy $s_v / s_{v'} = Z_v / Z_{v'} = \chi^{v - v'}$ on $\iota(T)$. Hence the section $s_0 = \chi^{-v} s_v$ of L over $\iota(T)$ does not depend on the vertex v. In the chart U_v, its coefficient χ^{-v} with respect to the frame s_v is a Laurent monomial in w, so s_0 extends to a meromorphic section on each U_v; the extensions agree on overlaps, because they agree on the dense set $\iota(T)$. By Proposition 18.8.2, if v is a vertex of the facet ρ, then $\operatorname{ord}_{D_\rho} s_0 = \operatorname{ord}_{D_\rho} \chi^{-v} = -\langle v, u_\rho\rangle = a_\rho$, since $\ell_\rho(v) = 0$. As s_0 has no zeros or poles on $\iota(T)$, its divisor is D_P. Similarly, the meromorphic function $f_v = \chi^{-v}$ on U_v has divisor $\sum_{\rho \ni v} a_\rho (D_\rho \cap U_v)$, which is the restriction of D_P to U_v, because $D_\rho \cap U_v = \varnothing$ for $v \notin \rho$. The line bundle $\mathcal{O}_{X_P}(D_P)$ of the divisor therefore has over U_v the local frame $1/f_v$, and its transition functions, in the sense of Definition 13.1.8, are $f_{v'}/f_v = \chi^{v - v'}$. These are the transition functions of L in the frames s_v, since $s_v = \chi^{v - v'} s_{v'}$. Hence the map sending s_v to $1/f_v$ for every vertex v is an isomorphism $L \cong \mathcal{O}_{X_P}(D_P)$.

Let s be a holomorphic section of L^k and write $s = f s_0^k$ on $\iota(T)$, with f holomorphic on $T \cong (\mathbb{C}^*)^n$. By Lemma 18.8.4, applied in the coordinates of a basis of M, we have $f = \sum_{m \in M} c_m \chi^m$, with normal convergence on compact subsets of T. The coefficients c_m do not depend on the basis: a change of basis replaces each character by the corresponding monomial in the new coordinates and preserves normal convergence on compact sets, so the uniqueness assertion of the lemma applies. Now fix a vertex v. The coefficient g_v of s with respect to the frame s_v^k is a holomorphic function on $U_v \cong \mathbb{C}^n$, and on $\iota(T)$

$$g_v = f \chi^{-kv} = \sum_m c_m \prod_i w_i^{\langle m - kv, u_i \rangle}.$$

Since $u_1, \dots, u_n$ is the basis of N dual to the basis $e_1, \dots, e_n$ of M, the map $m \mapsto (\langle m - kv, u_i \rangle)_i$ is a bijection of M onto $\mathbb{Z}^n$, so this is the Laurent expansion of g_v in the coordinates w. By the last assertion of Lemma 18.8.4, $c_m = 0$ unless $\langle m - kv, u_i \rangle \geq 0$ for every facet i at v. Since $\langle v, u_\rho \rangle = -a_\rho$ for $v \in \rho$, this condition reads $\langle m, u_\rho \rangle + k a_\rho \geq 0$. Every facet contains a vertex, so for $k > 0$ the conditions at all vertices say exactly that $m/k \in P$, that is, $m \in kP$. For $k = 0$ they say that $\langle m, u_\rho \rangle \geq 0$ for every ρ; then $p + sm \in P$ for $p \in P$ and all $s \geq 0$, so $m = 0$ by boundedness. Thus only the finitely many $m \in kP \cap M$ occur. Hence $s = \sum_{m \in kP \cap M} c_m \chi^m s_0^k$ on the dense set $\iota(T)$, and therefore on X_P, once we know that the sections on the right are holomorphic. Indeed, for $m \in kP \cap M$ all exponents $\langle m - kv, u_i \rangle$ are nonnegative, so $\chi^m s_0^k$ extends holomorphically to every chart. Finally, the uniqueness of Laurent coefficients shows that the sections $\chi^m s_0^k$ are linearly independent.

The action of T on $\mathbb{C}^A$ by $\tau \cdot (Z_m) = (\chi^m(\tau) Z_m)$ induces actions on the tautological line bundle (Example 14.5.5), whose fiber over $[Z]$ is the line $\mathbb{C}Z$, on its dual $\mathcal{O}(1)$, on L^k, and on $H^0(X_P, L^k)$ by $(\tau \cdot s)(x) = \tau \cdot s(\tau^{-1} \cdot x)$. It multiplies the linear form Z_v, and hence s_v, by $\chi^{-v}(\tau)$, and it maps the function χ^v to $\chi^v(\tau^{-1} \cdot) = \chi^{-v}(\tau) \chi^v$; therefore it fixes s_0 and multiplies $\chi^m s_0^k$ by $\chi^{-m}(\tau)$. So each summand is a line on which T acts through a character. Finally, $h^0(X_P, L^k) = \#(kP \cap M)$, and the limit follows from Theorem 18.6.3. In dimension zero there is one section, and the volume is one. □

18.9 The Guillemin reference metric

The Fubini–Study metric of Theorem 18.7.3 already gives a smooth metric on L with positive definite curvature form. In integrability calculations, however, it is useful to have a reference metric whose potential on $N_\mathbb{R}$ is the Legendre transform of an explicit function on P whose gradient has logarithmic singularities along the facets. Guillemin's metric, defined through the entropy function G_0 below, has this property; the content of the next theorem is that it is nevertheless smooth on all of X_P, with positive definite curvature form.

Theorem 18.9.1 *With the convention* $0 \log 0 = 0$*, define the closed proper convex function*

$$G_0(m) = \sum_\rho \ell_\rho(m) \log \ell_\rho(m) \quad (m \in P), \qquad G_0 = +\infty \quad \text{outside } P,$$

and put $F_0 = G_0^*$*. Then* F_0 *is smooth on* $N_\mathbb{R}$*, its Hessian is positive definite, and* $F_0 \sim h_P$*. The gradient* ∇G_0 *is a diffeomorphism of* $\operatorname{Int} P$ *onto* $N_\mathbb{R}$*. For each vertex* v*, the function*

$$H_v = F_0 \circ \operatorname{Trop} - \log |\chi^v|^2$$

on $\iota(T) \cong T$ *extends smoothly to* U_v*, and the squared norms* $|s_v|^2 = \mathrm{e}^{-H_v}$ *of the coordinate sections define a smooth quadratic metric on* L *with positive definite curvature form, invariant under* T_c*. Its curvature form is the unique smooth extension of* $\mathrm{dd}^\mathrm{c}(F_0 \circ \operatorname{Trop})$ *from* T *to* X_P*, and it represents* $c_1(L)$*.*

Proof The main step is to invert ∇G_0 near the boundary of P in the coordinates of a vertex chart (Step 2); the logarithmic singularities in H_v then cancel.

Step 1: the potentials on $N_\mathbb{R}$. The function $\lambda \mapsto \lambda \log \lambda$ is convex and continuous on $[0, +\infty)$, and each ℓ_ρ is affine and nonnegative on P. Hence G_0 is convex, continuous and bounded on P, and closed and proper on $M_\mathbb{R}$. On $\operatorname{Int} P$,

$$\nabla G_0 = \sum_\rho (\log \ell_\rho + 1) u_\rho, \qquad D^2 G_0 = \sum_\rho \frac{u_\rho \otimes u_\rho}{\ell_\rho}.$$

The normals u_ρ span $N_\mathbb{R}$, since a nonzero $y \in M_\mathbb{R}$ annihilated by all of them would give $P + \mathbb{R}y \subset P$ by the facet presentation, which is impossible because P is bounded; so $D^2 G_0(y, y) = \sum_\rho \langle y, u_\rho \rangle^2 / \ell_\rho > 0$ for $y \neq 0$, and the Hessian is positive definite.

Let $x \in N_\mathbb{R}$. The continuous function $G_0 - \langle \cdot, x \rangle$ attains its minimum on the compact set P, and not at a boundary point m_0. Indeed, let $p \in \operatorname{Int} P$ and move from m_0 towards p: for each facet with $\ell_\rho(m_0) = 0$ we have $\ell_\rho(m_0 + s(p - m_0)) = s\ell_\rho(p)$ with $\ell_\rho(p) > 0$, and the term $s\ell_\rho(p) \log(s\ell_\rho(p))$ has right derivative $-\infty$ at $s = 0$, while all other terms have finite derivatives; so the function decreases initially. The minimum is therefore attained in $\operatorname{Int} P$, at a solution of $\nabla G_0(m) = x$. Conversely, every solution of $\nabla G_0(m) = x$ in $\operatorname{Int} P$ is a minimum point, by convexity, and the minimum point is unique because G_0 is strictly convex on $\operatorname{Int} P$. Hence ∇G_0 is a smooth bijection of $\operatorname{Int} P$ onto $N_\mathbb{R}$ with invertible derivative, and so a diffeomorphism by the inverse function theorem. Writing $m(x) = (\nabla G_0)^{-1}(x)$, we have $F_0(x) = \langle m(x), x \rangle - G_0(m(x))$, so F_0 is smooth, and differentiating gives $\nabla F_0(x) = m(x)$ and $D^2 F_0(x) = (D^2 G_0(m(x)))^{-1}$, which is positive definite. Since G_0 is bounded on P, say $|G_0| \leq C$ there, we have $\chi_P - C \leq G_0 \leq \chi_P + C$, and conjugation gives $h_P - C \leq F_0 \leq h_P + C$.

Step 2: local inversion at a boundary point. Fix a vertex v, number the facets at v as in Lemma 18.7.2(4), and use on $M_\mathbb{R}$ the coordinates $t_k = \ell_k(m)$, so that $m = v + \sum_k t_k e_k$, and on $N_\mathbb{R}$ the dual coordinates $x_k = \langle e_k, x \rangle$, so that $x = \sum_k x_k u_k$ and $\langle m - v, x \rangle = \sum_k t_k x_k$. If $z \in \iota(T)$ and $x = \operatorname{Trop}(z)$, then $x_k = \log |\chi^{e_k}(z)|^2 = \log |w_k(z)|^2$. Let $w^0 \in U_v$, let $I = \{i : w_i^0 = 0\}$ and J its complement, and let

$Q = \{m \in P : t_i = 0 \text{ for } i \in I\}$ be the corresponding face (Proposition 18.8.2). Put $x^0_j = \log|w^0_j|^2$ for $j \in J$.

The facet functions other than t_i, $i \in I$, are positive on the relative interior of Q. Indeed, if such a function vanished at a point of the relative interior of Q, it would vanish on Q by the observation preceding Lemma 18.7.2, so its facet would contain Q and v. It would then be one of the facets at v containing Q, and these are the facets $i \in I$, because $v + se_j \in Q$ for $j \in J$ and small $s > 0$. Now consider the function $G_0 - \sum_{j\in J} t_j x^0_j$ on the compact set Q, on which the terms of the facets $i \in I$ vanish identically. As in Step 1, applied within the affine span of Q, its minimum is not attained on the relative boundary of Q. Indeed, if all facet functions other than t_i, $i \in I$, are positive at a point of Q, then they are positive on a neighborhood of this point in the affine span $\{t_I = 0\}$ of Q, which therefore lies in Q. Hence at a relative boundary point m_0 some facet function that is positive on the relative interior vanishes, and moving from m_0 towards a relative interior point decreases the function initially, as in Step 1. The minimum point q is unique, because on the relative interior of Q the Hessian of G_0 in the directions e_j, $j \in J$, of Q dominates the positive definite matrix $\operatorname{diag}(1/t_j)_{j\in J}$ coming from the terms $t_j \log t_j$.

Near q all facet functions except t_i, $i \in I$, are positive, so on a neighborhood W of q in $M_{\mathbb{R}}$

$$G_0(t) = \sum_{i\in I} t_i \log t_i + g(t) \qquad (t \in P \cap W),$$

where g, the sum of the remaining terms, is smooth on W. Write $g_k = \partial g/\partial t_k$ and $g_{jk} = \partial^2 g/\partial t_j \partial t_k$. The point q is characterized by $g_j(q) = x^0_j$ for $j \in J$, since on Q we have $G_0 = g$. At points of $\operatorname{Int} P \cap W$ the equation $\nabla G_0(t) = x$ reads

$$x_i = \log t_i + 1 + g_i(t) \quad (i \in I), \qquad x_j = g_j(t) \quad (j \in J).$$

Exponentiating the first equations and putting $y_i = \mathrm{e}^{x_i}$, which equals $|w_i|^2$ on $\iota(T)$, we obtain the system $\Psi(t) = (y_I, x_J)$ with

$$\Psi(t) = \Big(\big(t_i \exp(1 + g_i(t))\big)_{i\in I},\ \big(g_j(t)\big)_{j\in J}\Big),$$

a smooth map on all of W. At $t = q$, where $t_I = 0$, the Jacobian matrix of Ψ is block triangular: its $I\times I$ block is the diagonal matrix with entries $\exp(1+g_i(q)) > 0$, its $I\times J$ block vanishes, and its $J \times J$ block $(g_{jk}(q))$ is positive definite, because g contains the terms $t_j \log t_j$, $j \in J$, and its other terms are convex. Since $\Psi(q) = (0, x^0_J)$, the inverse function theorem gives a smooth local inverse $t = \Psi^{-1}(y_I, x_J)$ near $(0, x^0_J)$, with values near q. For $y_I \geq 0$ the solution satisfies

$$t_i = y_i \exp(-1 - g_i(t)) \geq 0 \quad (i \in I),$$

and its other facet functions are positive, so it lies in P. If $y_I > 0$, it lies in $\operatorname{Int} P$ and solves $\nabla G_0(t) = x$ with $x_i = \log y_i$; since ∇G_0 is injective on $\operatorname{Int} P$, $t = \nabla F_0(x)$.

Step 3: smooth extension. Let $z \in \iota(T)$ be close to w^0, put $x = \operatorname{Trop}(z)$, and let t be the coordinates of $\nabla F_0(x)$. Since $F_0(x) = \langle \nabla F_0(x), x\rangle - G_0(\nabla F_0(x))$ and

$\log|\chi^v(z)|^2 = \langle v, x\rangle$, we obtain, using the equations for $i \in I$,

$$H_v = \sum_k t_k x_k - \sum_{i\in I} t_i \log t_i - g(t) = \sum_{i\in I} t_i(1+g_i(t)) + \sum_{j\in J} t_j x_j - g(t).$$

The singular logarithms have canceled. By Step 2, $t = \Psi^{-1}(|w_I|^2, \log|w_J|^2)$, so the right-hand side is a smooth function $h(y_I, x_J)$ of $(y_I, x_J) = (|w_I|^2, \log|w_J|^2)$, which in turn is a smooth function of w near w^0, because $w_j^0 \neq 0$ for $j \in J$. Hence H_v extends smoothly to a neighborhood of w^0, and, $w^0 \in U_v$ being arbitrary, to U_v.

Step 4: positivity. On $\iota(T)$ we have $\partial F_0/\partial x_k = \langle \nabla F_0(x), u_k\rangle = \langle v, u_k\rangle + t_k$, hence $\partial H_v/\partial x_k = t_k$. In the variables (y_I, x_J) this says

$$\frac{\partial h}{\partial x_j} = t_j \quad (j \in J), \qquad \frac{\partial h}{\partial y_i} = \frac{t_i}{y_i} = \exp(-1 - g_i(t)) \quad (i \in I),$$

and both identities persist at $y_I = 0$, because both sides are continuous functions of (y_I, x_J). We now compute the complex Hessian of H_v at w^0, where $w_I = 0$. Since $\partial y_i/\partial \bar w_k = \delta_{ik} w_i$ for $i \in I$ and $\partial x_j/\partial \bar w_k = \delta_{jk}/\bar w_k$ for $j \in J$, the chain rule gives at w^0

$$\frac{\partial^2 H_v}{\partial w_i \partial \bar w_k} = \delta_{ik}\frac{\partial h}{\partial y_i} \quad (i \in I), \qquad \frac{\partial^2 H_v}{\partial w_j \partial \bar w_k} = \frac{1}{w_j \bar w_k}\frac{\partial^2 h}{\partial x_j \partial x_k} \quad (j,k \in J);$$

all other terms contain a factor w_i or $\bar w_i$ with $i \in I$. On $\{y_I = 0\}$ we have $t_I = 0$ and $x_J = g_J(t)$, so $\partial^2 h/\partial x_j \partial x_k = \partial t_j/\partial x_k$ is the inverse of the positive definite matrix $(g_{jk})_{j,k\in J}$. Hence the complex Hessian of H_v at w^0, where $t = q$, is block diagonal, with the positive numbers $\exp(-1 - g_i(q))$ on the I block and, on the J block, a positive definite matrix congruent to $(\partial t_j/\partial x_k)$ by the invertible diagonal matrix $\operatorname{diag}(1/w_j)$. It is therefore positive definite at w^0. Every point of U_v arises as such a w^0 (with $I = \varnothing$ on $\iota(T)$), so $\mathrm{dd^c} H_v$ is positive definite on U_v.

Step 5: the metric. On $\iota(T)$, for vertices v and v', we have $H_v - H_{v'} = \log|\chi^{v'-v}|^2$ and $s_v = \chi^{v-v'} s_{v'}$. The function $\chi^{v-v'} = s_v/s_{v'}$ is holomorphic and nowhere zero on $U_v \cap U_{v'}$, so both sides of the identity $\mathrm{e}^{-H_v} = |\chi^{v-v'}|^2 \mathrm{e}^{-H_{v'}}$ are continuous on $U_v \cap U_{v'}$; as they agree on the dense subset $\iota(T)$, the identity holds on $U_v \cap U_{v'}$. Thus the squared norms $|s_v|^2 = \mathrm{e}^{-H_v}$ satisfy the transition law of Definition 13.1.8 and define a smooth quadratic metric on L, with curvature $\mathrm{dd^c} H_v$ on U_v, which is positive definite by Step 4. For $\tau \in T_c$ we have $|\chi^m(\tau)| = 1$, so $\mathrm{Trop}(\tau z) = \mathrm{Trop}(z)$, the weights H_v are T_c-invariant, and T_c multiplies the frames s_v by factors of modulus one; hence the metric is T_c-invariant. On $\iota(T)$ the curvature is $\mathrm{dd^c} H_v = \mathrm{dd^c}(F_0 \circ \mathrm{Trop})$, because $\log|\chi^v|^2$ is pluriharmonic; since $\iota(T)$ is dense, the curvature form is the unique smooth extension of this form and represents $c_1(L)$ (Section 14.5). In dimension zero the facet sum and both potentials vanish, L is trivial over a point, and its curvature is zero. □

Appendix A
Complex spaces and modifications

Analytic sets enter pluripotential theory as the pole sets of potentials with analytic singularities, as the zero sets of the ideals that describe these poles, and as the exceptional divisors of modifications. This appendix collects the facts about analytic sets, reduced complex spaces and modifications that are used from Chapter 12 on: for multiplier ideals and Lelong numbers along divisors in Chapter 12, for analytic singularities and their log resolutions in Chapter 13, for currents supported on analytic sets in Chapter 14, and for the non-Kähler locus in Chapter 16. It can be consulted when these tools first appear. Besides results of the earlier chapters, such as the Weierstrass preparation theorem (Theorem 7.4.2), only two existence theorems are used without proof: the resolution of singularities and the principalization of coherent ideals, stated as Theorems A.5.4 and A.5.5.

The local theory rests on the Weierstrass division theorem. It shows that the ring of germs of holomorphic functions is Noetherian and, through Oka's theorem, that the sheaf of holomorphic functions is coherent, so that statements about one stalk extend to a neighborhood. The local parametrization theorem (Theorem A.2.3) presents an irreducible germ of an analytic set as a finite branched covering of a polydisc. From it we deduce the analytic Nullstellensatz, the basic properties of dimension, Cartan's theorem on the coherence of the ideal sheaf of an analytic set, and the decomposition of an analytic set into irreducible components. We also show that a closed real $(1,1)$-current with measure coefficients supported on a nowhere dense analytic set is a combination of currents of integration along hypersurfaces (Theorem A.2.17). One phenomenon requires care: an irreducible analytic set can have several local branches at one point. The normalization (Theorem A.3.6) separates these branches; its holomorphic functions are the weakly holomorphic functions, that is, the functions that are holomorphic on the regular locus and locally bounded near the singular locus. The cusp shows that a space with only one branch at each point need not be normal (Example A.3.10).

For modifications, explicit charts describe blowups along submanifolds and the transformation of ideals under them (Propositions A.5.2 and A.5.3); together with the two existence theorems they give log resolutions of coherent ideals. Remmert's theorem, proved together with the Remmert–Stein extension theorem, shows that

images of analytic sets under proper holomorphic maps, in particular the images of exceptional divisors, are analytic (Theorem A.6.2). We then prove a relative principalization theorem, which controls an ideal and its restriction to a submanifold simultaneously (Theorem A.6.5), show that proper bimeromorphic maps onto spaces with one local branch at each point have connected fibers (Theorem A.7.1), and prove that every prime divisor over a compact complex space appears on a model obtained by finitely many blowups with smooth centers (Theorem A.7.6).

The last section defines plurisubharmonic functions on reduced complex spaces by local extension to an ambient space, proves that they are constant on compact connected spaces, and shows that they are locally bounded above near analytic subsets of codimension at least two. Holomorphic curves through singular points, constructed in Appendix A.4, reduce these statements to the maximum principle in one variable. These results on reduced spaces are used in [Xia]. Historical remarks and references for this appendix are collected in Appendix B.21.

A.1 Division and finite analytic modules

Let $n \geq 0$. We write $\mathcal{O}_n$ for the ring of germs at $0 \in \mathbb{C}^n$ of holomorphic functions; two functions defined near 0 have the same germ if they agree on some neighborhood of 0, and identities between germs are understood in this sense. By the iterated Cauchy formula of Section 7.1, every germ is given near 0 by its Taylor series (7.1), which converges normally on a small polydisc. Conversely, a power series $\sum_\alpha c_\alpha z^\alpha$ with $\sum_\alpha |c_\alpha| r^{|\alpha|} < \infty$ for some $r > 0$ defines a holomorphic function on the polydisc $\{|z_j| < r\}$: since $|c_\alpha| \leq C r^{-|\alpha|}$, the series of partial derivatives of each order converge normally on every polydisc $\{|z_j| \leq s\}$ with $s < r$, so the sum is smooth, may be differentiated termwise, and satisfies the Cauchy–Riemann equations. Thus germs are the same as convergent power series, $\mathcal{O}_n = \mathbb{C}\{z_1, \ldots, z_n\}$, and $\mathcal{O}_0 = \mathbb{C}$.

The ring $\mathcal{O}_n$ is an integral domain: if $fg = 0$ near 0 and $f \neq 0$, then g vanishes on the nonempty open subset $\{f \neq 0\}$ of a connected neighborhood of 0, so $g = 0$ by the identity theorem. A germ that does not vanish at the origin has a holomorphic reciprocal near the origin; hence $\mathcal{O}_n$ is a local ring, whose maximal ideal $\mathfrak{m}$ consists of the germs vanishing at the origin. The *order* $\operatorname{ord}_0 f$ of a nonzero germ f is the degree of the first nonzero homogeneous term of its Taylor expansion. The order of a product is the sum of the orders. Indeed, if P_s and Q_t are the first nonzero homogeneous terms of f and g, then $P_s Q_t$ is the homogeneous term of degree $s + t$ of fg. It is not zero: a polynomial vanishing on a nonempty open set is zero, so the sets $\{P_s \neq 0\}$ and $\{Q_t \neq 0\}$ are open and dense, and therefore meet. Hence every element of the kth power $\mathfrak{m}^k$ has order at least k.

Conversely, let $f = \sum_\beta c_\beta z^\beta$ have order at least k. Assign to each β with $|\beta| \geq k$ the first multi-index $\alpha \leq \beta$ with $|\alpha| = k$ in the lexicographic order, and let $f_\alpha = \sum c_\beta z^{\beta-\alpha}$, the sum over the β assigned to α. These series converge normally wherever the series of f does, so $f_\alpha \in \mathcal{O}_n$, and $f = \sum_{|\alpha|=k} z^\alpha f_\alpha \in \mathfrak{m}^k$. Thus $\operatorname{ord}_0 f \geq k$ if and only if $f \in \mathfrak{m}^k$. A holomorphic change of coordinates fixing 0 induces a ring automorphism

of $\mathcal{O}_n$ that preserves $\mathfrak{m}$ and its powers; so the order does not depend on the coordinates. For a holomorphic function f near a point a of a complex manifold whose germ at a is nonzero, $\operatorname{ord}_a f$ is the order of its germ in local coordinates centered at a. This is the order used for Lelong numbers in Proposition 12.1.4; Chapter 12 also defines the order of a coherent ideal at a point and the order along a prime divisor (Definition 12.6.3).

Write points of $\mathbb{C}^n$, $n \geq 1$, as (z', w) with $z' \in \mathbb{C}^{n-1}$. A *Weierstrass polynomial* of degree $m \geq 0$ in w is a polynomial

$$P(z', w) = w^m + a_1(z')w^{m-1} + \cdots + a_m(z'), \qquad a_j \in \mathcal{O}_{n-1}, \quad a_j(0) = 0.$$

We shall often use the following bound for the roots t of a monic polynomial with complex coefficients:

$$t^m + d_1 t^{m-1} + \cdots + d_m = 0 \quad \Longrightarrow \quad |t| \leq 2 \max_{1 \leq j \leq m} |d_j|^{1/j}. \tag{A.1}$$

Indeed, otherwise $|d_j| < (|t|/2)^j$ for all j, and dividing the equation by t^m gives $1 \leq \sum_{j=1}^m |d_j/t^j| < \sum_{j=1}^m 2^{-j} < 1$. In particular, the roots of $P(z', \cdot)$ tend to 0 as $z' \to 0$.

A germ $f \in \mathcal{O}_n$ is *distinguished* in w if $f(0, w)$ is not identically zero. If $f(0, w)$ vanishes to order $m \geq 1$ at $w = 0$, the Weierstrass preparation theorem (Theorem 7.4.2) writes f as a unit times a Weierstrass polynomial of degree m; if $f(0) \neq 0$, then f is itself a unit, which is the case $m = 0$. Every nonzero germ becomes distinguished in the last coordinate after a linear change of coordinates. Indeed, let P_s be the first nonzero homogeneous term of f and choose v with $P_s(v) \neq 0$; in linear coordinates whose last basis vector is v, we have $f(0, w) = P_s(v)w^s + O(w^{s+1})$, which vanishes to order exactly s, the order of f. Since $P_s \neq 0$, the admissible vectors v form a dense open set. Hence the new coordinates can be chosen arbitrarily close to given ones, and finitely many nonzero germs become distinguished simultaneously if v avoids the zero set of the product of their first homogeneous terms.

Proposition A.1.1 (Weierstrass division) *Let $n \geq 1$ and $m \geq 0$, and let P be a Weierstrass polynomial of degree m in w. Then every germ $f \in \mathcal{O}_n$ can be written uniquely as*

$$f = Pq + R, \qquad q \in \mathcal{O}_n, \quad R \in \mathcal{O}_{n-1}[w], \quad \deg_w R < m.$$

Moreover, let $\Delta' \subset \mathbb{C}^{n-1}$ be a polydisc centered at 0 and $\Delta = \{|w| < r\}$ a disc such that the coefficients of P are holomorphic near $\overline{\Delta'}$ and all roots of $P(z', \cdot)$, $z' \in \Delta'$, lie in $\{|w| \leq \rho\}$ for some $\rho < r$; such polydiscs $\Delta' \times \Delta$ exist with arbitrarily small radii. Then every holomorphic function f on $\Delta' \times \Delta$ can be written uniquely as $f = Pq + R$, where q is holomorphic on $\Delta' \times \Delta$ and R is a polynomial in w of degree less than m with holomorphic coefficients on Δ'; for every $s \in (\rho, r)$, $z' \in \Delta'$ and $|w| < s$,

$$\begin{aligned} q(z',w) &= \frac{1}{2\pi\mathrm{i}} \int_{|\zeta|=s} \frac{f(z',\zeta)}{P(z',\zeta)(\zeta-w)}\,\mathrm{d}\zeta, \\ R(z',w) &= \frac{1}{2\pi\mathrm{i}} \int_{|\zeta|=s} \frac{f(z',\zeta)}{P(z',\zeta)} \frac{P(z',\zeta)-P(z',w)}{\zeta-w}\,\mathrm{d}\zeta. \end{aligned} \tag{A.2}$$

Division is a bounded linear operation on bounded holomorphic functions on $\Delta'\times\Delta$: there is a constant C, independent of f, with

$$\sup_{\Delta'\times\Delta} |q| + \sup_{\Delta'\times\Delta} |R| \le C \sup_{\Delta'\times\Delta} |f|.$$

Proof Polydiscs as in the statement exist with arbitrarily small radii, because the roots of $P(z',\cdot)$ tend to 0 as $z'\to 0$, by (A.1).

Let $\Delta'\times\Delta$ be such a polydisc and f a holomorphic function on it. For $\rho<s<r$ and $|w|<s$ define q and R by (A.2). For fixed z' and w, both integrands are holomorphic in ζ on the annulus $\{\rho<|\zeta|<r\}$, except for the pole of the first one at $\zeta=w$; by Cauchy's theorem the integrals do not depend on s as long as $|w|<s$, and the second one does not depend on $s\in(\rho,r)$ at all. Hence q and R are defined on $\Delta'\times\Delta$, and they are holomorphic by differentiation under the integral sign. The difference quotient $(P(z',\zeta)-P(z',w))/(\zeta-w)$ is a polynomial in w of degree $m-1$ whose coefficients are polynomials in ζ and in the $a_j(z')$; hence R is a polynomial in w of degree less than m with holomorphic coefficients on Δ'. Adding the two integrals, we obtain $Pq+R=(2\pi\mathrm{i})^{-1}\int_{|\zeta|=s} f(z',\zeta)(\zeta-w)^{-1}\,\mathrm{d}\zeta = f$ by Cauchy's formula.

For the estimate, let f be bounded, and fix $s_0=(\rho+r)/2$. For $|\zeta|\ge s_0$ we have $|P(z',\zeta)|=\prod_l|\zeta-\lambda_l(z')|\ge(s_0-\rho)^m$, where $\lambda_l(z')$ are the roots, and the coefficients of the difference quotient are bounded for $z'\in\Delta'$, $|\zeta|=s_0$ and $|w|<r$, since the a_j are bounded on Δ'. The integral over $|\zeta|=s_0$ therefore gives $|R|\le C_1\sup|f|$ on $\Delta'\times\Delta$. For $s_0\le|w|<r$ we have $|q|=|f-R|/|P|\le(1+C_1)(s_0-\rho)^{-m}\sup|f|$, and the maximum principle for the holomorphic functions $q(z',\cdot)$ on the discs $\{|w|\le s\}$, $s_0\le s<r$, extends this bound to all of $\Delta'\times\Delta$.

For uniqueness, suppose that $Pq+R=0$ on a polydisc as above, with R a polynomial in w of degree less than m. For each z', the polynomial $R(z',\cdot)=-P(z',\cdot)q(z',\cdot)$ vanishes at the m roots of $P(z',\cdot)$, counted with multiplicity, all of which lie in Δ. Its degree is less than m, so $R(z',\cdot)=0$. Then $P(z',\cdot)q(z',\cdot)=0$, and since $P(z',\cdot)$ has only finitely many zeros, the continuous function $q(z',\cdot)$ vanishes on a dense set, hence everywhere. Finally, every germ is represented by a bounded holomorphic function on some polydisc as above, and an identity between germs holds on some such polydisc; so the assertions for germs follow. □

The first consequence of division is the Noetherian property of $\mathcal{O}_n$.

Proposition A.1.2 *Every ideal of $\mathcal{O}_n$ is finitely generated, and every submodule of $\mathcal{O}_n^r$, $r\ge1$, is finitely generated. Consequently, every increasing sequence of ideals of $\mathcal{O}_n$, or of submodules of a finitely generated $\mathcal{O}_n$-module, is eventually constant.*

Proof We first recall three algebraic facts about a commutative ring Λ all of whose ideals are finitely generated. Every submodule N of Λ^r is finitely generated, by

induction on r: the projection of N to the last coordinate is an ideal; choose elements of N whose last coordinates generate it, and add generators of $N \cap (\Lambda^{r-1} \times \{0\})$, which exist by the induction hypothesis. A finitely generated module is a quotient of some Λ^r, and the inverse image of a submodule is a submodule of Λ^r; hence submodules of finitely generated modules are finitely generated. Finally, the union of an increasing sequence of submodules is a submodule; its finitely many generators already lie in one member of the sequence, which is therefore stationary from then on.

It remains to prove that the ideals of $\mathcal{O}_n$ are finitely generated, which we do by induction on n. The ring $\mathcal{O}_0 = \mathbb{C}$ is a field. Let $n \geq 1$ and let $I \subset \mathcal{O}_n$ be an ideal. If $I = 0$ or I contains a unit, it is generated by one element. Otherwise choose a nonzero germ in I, and coordinates in which it is distinguished in w. By the preparation theorem it is a unit times a Weierstrass polynomial P of some degree $m \geq 1$, and $P \in I$. Divide every $f \in I$ by P using Proposition A.1.1; the remainder $R_f = f - Pq$ again belongs to I. The remainders of degree less than m that lie in I form an $\mathcal{O}_{n-1}$-submodule of $\mathcal{O}_{n-1}[w]$, which we identify, through their coefficients, with a submodule of $\mathcal{O}_{n-1}^m$. By the induction hypothesis and the first paragraph, it is generated by finitely many remainders $R_1, \ldots, R_k$. Then $P, R_1, \ldots, R_k$ generate I.□

Division also factors a monic polynomial near each of its roots.

Lemma A.1.3 *Let $n \geq 1$, let $P(z', w) = w^\mu + a_1(z')w^{\mu-1} + \cdots + a_\mu(z')$ be a monic polynomial in w whose coefficients are holomorphic near a point $x' \in \mathbb{C}^{n-1}$, and let w_0 be a root of multiplicity $\mu' \geq 1$ of $P(x', \cdot)$. Then, near $x = (x', w_0)$,*

$$P = P_x Q_x,$$

where P_x is a Weierstrass polynomial of degree μ' in $w - w_0$ with respect to the coordinates $(z' - x', w - w_0)$, and Q_x is a monic polynomial of degree $\mu - \mu'$ in w with coefficients holomorphic near x' and $Q_x(x) \neq 0$.

Proof In the coordinates $(z' - x', w - w_0)$ centered at x, the function $w \mapsto P(x', w)$ has a zero of order μ' at w_0. By the Weierstrass preparation theorem (Theorem 7.4.2), P is, near x, a nowhere vanishing holomorphic function U times a Weierstrass polynomial P_x of degree μ' in $w - w_0$. Since P_x is monic in w, Euclidean division of the polynomial P by P_x, with coefficients holomorphic near x', gives $P = P_x Q + R$, where Q is a monic polynomial of degree $\mu - \mu'$ in w and R is a polynomial in w, equivalently in $w - w_0$, of degree less than μ'. The identities $P = P_x Q + R$ and $P = P_x U + 0$ are two divisions of the germ of P at x by P_x; by the uniqueness in Proposition A.1.1, applied in the coordinates centered at x, we get $Q = U$ near x and $R = 0$. Hence $Q_x = Q$ has the required properties, with $Q_x(x) = U(x) \neq 0$. □

We use sheaves in the following elementary form. A sheaf $\mathcal{F}$ of modules on a topological space assigns to each open set U a module $\mathcal{F}(U)$ of sections, with restriction maps, such that sections that agree locally are equal and compatible local sections glue. The stalk $\mathcal{F}_x$ consists of the germs at x of sections defined near x. Kernels and images of morphisms of sheaves are defined stalkwise, and a sequence

of morphisms is exact if it is exact on every stalk; in particular, a section of the image of a morphism is, near each point, the image of a section of the source. Similarly, the quotient $\mathcal{S}/\mathcal{F}$ of a sheaf of modules by a subsheaf has the stalks $\mathcal{S}_x/\mathcal{F}_x$, and its sections are the families of classes that are, near each point, the classes of a section of $\mathcal{S}$; so sections of $\mathcal{S}/\mathcal{F}$ can be lifted locally to sections of $\mathcal{S}$. The cokernel of a morphism is the quotient of its target by its image.

Let $\mathcal{A}$ be a sheaf of rings and $\mathcal{F}$ a sheaf of $\mathcal{A}$-modules. We say that $\mathcal{F}$ is *locally finitely generated* if every point has a neighborhood U and finitely many sections $s_1, \dots, s_p \in \mathcal{F}(U)$ whose germs generate $\mathcal{F}_y$ over $\mathcal{A}_y$ for every $y \in U$. This requires one family of generators on a whole neighborhood, not merely finite generation of each stalk. The *sheaf of relations* of sections $s_1, \dots, s_p \in \mathcal{F}(U)$ is the kernel of the morphism $\mathcal{A}^p|_U \to \mathcal{F}|_U$, $(a_i) \mapsto \sum_i a_i s_i$. The sheaf $\mathcal{F}$ is *coherent* if it is locally finitely generated and the sheaf of relations of every finite family of sections over every open set is locally finitely generated.

Lemma A.1.4 *Let $\mathcal{A}$ be a sheaf of rings. Kernels, images and cokernels of morphisms between coherent $\mathcal{A}$-modules are coherent. If two of the three sheaves in an exact sequence $0 \to \mathcal{F} \to \mathcal{S} \to \mathcal{G} \to 0$ of $\mathcal{A}$-modules are coherent, so is the third. Finite direct sums of coherent sheaves are coherent. If $\mathcal{A}$ is coherent as a module over itself, every locally finitely generated $\mathcal{A}$-submodule of $\mathcal{A}^r$ is coherent.*

Proof A locally finitely generated submodule of a coherent sheaf is coherent, because its sections have the same relations when they are regarded as sections of the larger sheaf. The image of a morphism from a coherent sheaf is locally generated by the images of local generators of the source, so the image of a morphism between coherent sheaves is coherent. For the kernel of a morphism $\phi\colon \mathcal{F} \to \mathcal{G}$ between coherent sheaves, choose local generators $F_1, \dots, F_p$ of $\mathcal{F}$ and local generators $R_j = (R_{j1}, \dots, R_{jp})$ of the relations of $\phi(F_1), \dots, \phi(F_p)$. If $\sum_i a_i F_i$ lies in the kernel, then (a_i) is a relation of the $\phi(F_i)$, hence a combination of the R_j; so the sections $\sum_i R_{ji} F_i$ generate the kernel, which is therefore coherent.

Let $\mathcal{F} \subset \mathcal{S}$ be coherent sheaves and consider $\mathcal{G} = \mathcal{S}/\mathcal{F}$; this covers cokernels, because images are coherent. Local generators of $\mathcal{S}$ map to local generators of $\mathcal{G}$. Given sections $G_1, \dots, G_p$ of $\mathcal{G}$, lift them locally to sections $S_1, \dots, S_p$ of $\mathcal{S}$, and choose local generators $F_1, \dots, F_k$ of $\mathcal{F}$. A tuple (a_i) is a relation of the G_i exactly when $\sum_i a_i S_i \in \mathcal{F}$, that is, when (a, b) is a relation of $(S_1, \dots, S_p, F_1, \dots, F_k)$ in $\mathcal{S}$ for some b. Hence the projections to the first p coordinates of local generators of the latter relations generate the relations of the G_i, and $\mathcal{G}$ is coherent.

Now let $\mathcal{F}$ and $\mathcal{G}$ be coherent. Local lifts of generators of $\mathcal{G}$, together with local generators of $\mathcal{F}$, generate $\mathcal{S}$. Given sections $S_1, \dots, S_p$ of $\mathcal{S}$, let $R_1, \dots, R_l$ generate the relations of their images in $\mathcal{G}$. The combinations $H_j = \sum_i R_{ji} S_i$ lie in $\mathcal{F}$; let $Q_1, \dots, Q_k$ generate their relations. Every relation (a_i) of the S_i is first a relation of their images, so $a = \sum_j c_j R_j$, and then $\sum_j c_j H_j = \sum_i a_i S_i = 0$, so c is a combination of $Q_1, \dots, Q_k$. Hence the tuples $\sum_j Q_{\nu j} R_j$, $1 \le \nu \le k$, which are relations of the S_i, generate all of them, and $\mathcal{S}$ is coherent. The remaining case of the exact sequence, with $\mathcal{S}$ and $\mathcal{G}$ coherent, is the kernel case. A finite direct sum is an

extension $0 \to \mathcal{F} \to \mathcal{F} \oplus \mathcal{G} \to \mathcal{G} \to 0$. Finally, if $\mathcal{A}$ is coherent, so is $\mathcal{A}^r$, and the last assertion follows from the first sentence of the proof. □

Theorem A.1.5 (Oka's coherence theorem) *The sheaf $\mathcal{O}_M$ of holomorphic functions on a complex manifold M is coherent. In particular, the kernel of the morphism $\mathcal{O}_U^q \to \mathcal{O}_U^p$ defined by a $p \times q$ matrix of holomorphic functions on an open set $U \subset M$ is locally finitely generated.*

Proof The sheaf $\mathcal{O}_M$ is generated by the constant 1, and the statement is local, so we must show that the relations of finitely many holomorphic functions $F_1, \dots, F_q$ near a point of $\mathbb{C}^n$, which we take to be the origin, are locally finitely generated. We argue by induction on n: we reduce to relations that are polynomials in one variable of bounded degree, and apply the induction hypothesis to their coefficients. For $n = 0$ the assertion is finite-dimensional linear algebra. Let $n \geq 1$. If the germ of some F_j at 0 is zero, then F_j vanishes on a connected neighborhood, and the relations of the F_i are those of the remaining functions, with an arbitrary jth coordinate; so we may assume that no germ F_j is zero (if all are zero, every tuple is a relation). Choose coordinates (z', w) in which all F_j are distinguished in w. By the preparation theorem, on a small polydisc each F_j is a holomorphic unit u_j times a Weierstrass polynomial; since (g_j) is a relation of the F_j exactly when $(u_j g_j)$ is a relation of the Weierstrass polynomials, we may assume that each F_j is a Weierstrass polynomial in w whose coefficients are holomorphic on a polydisc $\Delta' \subset \mathbb{C}^{n-1}$. Renumbering, let F_q have the maximal degree μ.

We claim that at every point $x = (x', w_0)$ of the polydisc, the stalk of relations is generated by relations whose coordinates are polynomials in w of degree at most μ with coefficients holomorphic near x'. If $F_q(x) \neq 0$, then for a relation $g = (g_1, \dots, g_q)$ we have $g_q = -\sum_{j<q} g_j F_j / F_q$, so $g = \sum_{j<q} (g_j/F_q)(F_q e_j - F_j e_q)$ is a combination of the elementary relations $F_q e_j - F_j e_q$, which are polynomial of degree at most μ. Suppose that $F_q(x) = 0$, and let $\mu' \geq 1$ be the order of the root w_0 of $F_q(x', \cdot)$. By Lemma A.1.3, near x the monic polynomial F_q is the product of a Weierstrass polynomial P_x of degree μ' in $w - w_0$ and a monic polynomial Q_x in w of degree $\mu - \mu'$:

$$F_q = P_x Q_x, \qquad Q_x(x) \neq 0.$$

Let g be a relation at x. Dividing each g_j, $j < q$, by P_x (Proposition A.1.1) and writing $P_x = F_q Q_x^{-1}$, we obtain $g_j = F_q t_j + r_j$ with $\deg_w r_j < \mu'$. Subtracting $\sum_{j<q} t_j (F_q e_j - F_j e_q)$ from g, we obtain a relation whose first $q-1$ coordinates are $r_1, \dots, r_{q-1}$; we call its last coordinate r_q. Then

$$P_x Q_x r_q = -\sum_{j<q} F_j r_j.$$

The right-hand side is a polynomial in w; dividing it by P_x as a polynomial and comparing with the uniqueness of division at x, we find that $Q_x r_q$ is a polynomial of degree less than μ. Each $Q_x r_j$, $j < q$, is also a polynomial of degree at most $(\mu - \mu') + (\mu' - 1) < \mu$. Thus $(Q_x r_1, \dots, Q_x r_q)$ is a polynomial relation of degree less than μ, and (r_j) is its product with the unit Q_x^{-1}. This proves the claim.

A tuple of polynomials $g_j = \sum_{k=0}^{\mu} c_{jk}(z')w^k$ is a relation exactly when the $2\mu+1$ coefficients of the polynomial $\sum_j F_j g_j$ vanish. These coefficients are given by a holomorphic matrix acting on the coefficient vectors (c_{jk}), with entries formed from the coefficients of the F_j. By the induction hypothesis and Lemma A.1.4, the kernel of this matrix is coherent, so it has finitely many generators on a smaller polydisc Δ_1', generating its stalk at every point of Δ_1'. The corresponding polynomial tuples are relations of the F_j, and they generate, over the rings of germs on the base, all polynomial relations of degree at most μ at every point over Δ_1'. By the claim, they generate all relations at every point of the polydisc over Δ_1'. This proves that $\mathcal{O}_M$ is coherent. The last assertion follows from Lemma A.1.4, since the kernel of a morphism between the coherent sheaves $\mathcal{O}_U^q$ and $\mathcal{O}_U^p$ is coherent. □

We now turn to the zero sets of holomorphic functions. An *analytic subset* of a complex manifold M (Definition 7.1.1) is a closed subset $A \subset M$ such that every point of M has a neighborhood U and finitely many holomorphic functions $g_1, \dots, g_k$ on U with $A \cap U = \{g_1 = \cdots = g_k = 0\}$. It is *proper* if $A \neq M$.

Lemma A.1.6 *Let A be a proper analytic subset of a connected complex manifold M. Then A is nowhere dense in M. Moreover, if $U \subset M$ is a nonempty connected open set and $g_1, \dots, g_k$ are holomorphic functions on U with $A \cap U = \{g_1 = \cdots = g_k = 0\}$, then some g_i is not identically zero on U. In particular, every point of A has a connected open neighborhood U with a holomorphic function $g \not\equiv 0$ on U such that $A \cap U \subset \{g = 0\}$.*

Proof The interior of A is open, and we show that it is closed in M. Let $a \in M$ be a limit of interior points of A, and choose a connected coordinate neighborhood U of a and functions $g_1, \dots, g_k$ with $A \cap U = \{g_1 = \cdots = g_k = 0\}$. Each g_i vanishes on a nonempty open subset of U, hence on U by the identity theorem; so $U \subset A$, and a is an interior point. Since M is connected and $A \neq M$, the interior of A is empty. In the second assertion, if all g_i vanished identically on U, the nonempty open set U would lie in A, contradicting the first. For the third, take for U a connected coordinate ball on which A is the common zero set of finitely many holomorphic functions, and for g one of them that is not identically zero, which exists by the second assertion. □

We recall some algebraic terminology. The *field of fractions* $\mathrm{Frac}(R)$ of an integral domain R consists of the formal quotients b/c with $b, c \in R$ and $c \neq 0$, where $b/c = b'/c'$ if $bc' = b'c$, with the usual addition and multiplication; it contains R through $b \mapsto b/1$. More generally, the *total ring of fractions* of a commutative ring R is defined in the same way, with denominators restricted to the nonzerodivisors of R. An element of a commutative ring containing R is *integral* over R if it is a root of a monic polynomial with coefficients in R, and R is *integrally closed* if every element of $\mathrm{Frac}(R)$ that is integral over R belongs to R.

Lemma A.1.7 *Let A be a proper analytic subset of a connected complex manifold M. Every holomorphic function on $M \setminus A$ which is locally bounded near A extends uniquely to a holomorphic function on M, and $M \setminus A$ is connected. Moreover, let $b, c \in \mathcal{O}_n$ with $c \neq 0$, and suppose that representatives satisfy $|b| \leq C|c|$ on a*

neighborhood of 0 *for a constant* C. *Then* $b \in c\,\mathcal{O}_n$. *In particular, the ring* $\mathcal{O}_n$ *is integrally closed.*

Proof Let f be holomorphic on $M \setminus A$ and locally bounded near A. If $\dim M = 0$, then $A = \varnothing$ and there is nothing to prove. Fix $a \in A$. By Lemma A.1.6 there is a holomorphic function $g \not\equiv 0$ on a connected neighborhood of a vanishing on A; choose linear coordinates (z', w) centered at a in which g is distinguished in w. By the Weierstrass preparation theorem and (A.1), there is a polydisc $\Delta' \times \Delta$ centered at a, with $\Delta = \{|w| < r\}$, on which g is a unit times a Weierstrass polynomial W and f is bounded off A, such that all roots of $W(z', \cdot)$, $z' \in \Delta'$, have modulus at most some $\rho < r$. For fixed $z' \in \Delta'$, the function $f(z', \cdot)$ is holomorphic on Δ outside the finitely many points of A in the fiber, which are roots of $W(z', \cdot)$, and it is bounded near them; by the removable singularity theorem in one variable it extends holomorphically to Δ. For $\rho < s < r$ the function

$$F(z', w) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=s} \frac{f(z', \zeta)}{\zeta - w}\, \mathrm{d}\zeta, \qquad z' \in \Delta',\ |w| < s,$$

is holomorphic, by differentiation under the integral sign, since f is holomorphic near $\Delta' \times \{|\zeta| = s\}$, a set disjoint from A. By Cauchy's formula for the extended functions $f(z', \cdot)$, it agrees with f outside A. Letting $s \uparrow r$, we obtain a holomorphic extension of f to $\Delta' \times \Delta$. Since A is nowhere dense, two holomorphic extensions on overlapping open sets agree, being continuous functions that coincide on a dense subset; so the local extensions glue to a holomorphic function on M, and this extension is unique.

If $M \setminus A$ were disconnected, the function equal to 0 on one nonempty union of its connected components and to 1 on the others would be bounded and holomorphic on $M \setminus A$. Its holomorphic extension is continuous and takes values in $\{0, 1\}$ on the dense set $M \setminus A$, hence everywhere; since M is connected, it is constant, a contradiction.

Now let $b, c \in \mathcal{O}_n$, $c \neq 0$, with $|b| \leq C|c|$ on a connected neighborhood U of the origin on which both are defined. The set $\{c = 0\}$ is a proper analytic subset of U, and the holomorphic function b/c on $U \setminus \{c = 0\}$ is bounded by C. By the first part it extends to a holomorphic function h on U, and $ch = b$ on the dense set $U \setminus \{c = 0\}$, hence on U; so $b \in c\,\mathcal{O}_n$. Finally, let $b/c \in \mathrm{Frac}(\mathcal{O}_n)$, with $c \neq 0$, satisfy a monic equation $(b/c)^m + a_1(b/c)^{m-1} + \cdots + a_m = 0$ with $a_j \in \mathcal{O}_n$. Multiplying by c^m, we obtain the identity $b^m + a_1 b^{m-1} c + \cdots + a_m c^m = 0$ between germs, which holds on a connected neighborhood of 0 on which the a_j are bounded. At the points of this neighborhood where $c \neq 0$, the number b/c is a root of a monic polynomial with bounded coefficients, so $|b/c|$ is bounded there by (A.1); at the points where $c = 0$ the identity gives $b^m = 0$. Hence $|b| \leq C|c|$ on this neighborhood for some constant C, and by what we have just proved, $b/c \in \mathcal{O}_n$, which proves that $\mathcal{O}_n$ is integrally closed. □

A.2 Analytic sets and their local branches

The local study of an analytic set rests on the parametrization of an irreducible germ as a branched covering of a polydisc (Theorem A.2.3). From it we derive the Nullstellensatz, the dimension theory, Cartan's coherence theorem, the structure of the singular locus and the irreducible components. The section ends with hypersurfaces and the currents supported on analytic sets.

Let M be a complex manifold and $x \in M$. Two subsets of M define the same *germ* at x if they agree on a neighborhood of x; an *analytic germ* at x is the germ (A, x) of an analytic subset A of a neighborhood of x. Holomorphic coordinates identify the ring $\mathcal{O}_{M,x}$ of germs of holomorphic functions at x with $\mathcal{O}_n$. For an analytic germ (A, x) let $I_{A,x} \subset \mathcal{O}_{M,x}$ be the ideal of germs vanishing on (A, x). Conversely, for an ideal $I \subset \mathcal{O}_{M,x}$ let $V(I)$ be the germ at x of the common zero set of finitely many generators of I, which exist by Proposition A.1.2; it does not depend on the choice of generators. Both operations reverse inclusions, and $V(I_{A,x}) = (A, x)$, because local defining functions of A belong to $I_{A,x}$. Finite unions and finite intersections of analytic subsets are analytic: they are defined locally by the products of the defining functions and by the union of the families of defining functions, respectively. A nonempty analytic germ is *irreducible* if it is not the union of two analytic subgerms different from it.

A point a of an analytic subset $A \subset M$ is *regular* if A is a complex submanifold of M near a; the dimension of this submanifold is the dimension of A at the regular point a. The regular points form an open subset A_{reg} of A, the *regular locus*, and $A_{\mathrm{sing}} = A \setminus A_{\mathrm{reg}}$ is the *singular locus*.

Proposition A.2.1 *Every analytic germ is the union of finitely many irreducible analytic germs, and this decomposition is unique up to order if none of its members is contained in another. A nonempty analytic germ is irreducible if and only if its ideal is prime. The members of the decomposition are called the* irreducible branches*, or* local branches*, of the germ.*

Proof A decreasing sequence of analytic germs is eventually constant, because the corresponding ideals increase and are eventually constant by Proposition A.1.2, and $V(I_{A,x}) = (A, x)$. Split a reducible germ into two analytic subgerms different from it, split each reducible piece again, and so on. If this process did not terminate, the resulting tree, in which every vertex has at most two children, would be infinite, and choosing at each level a child with infinitely many descendants would produce a strictly decreasing infinite sequence of analytic germs, which is impossible. Hence a finite decomposition into irreducible germs exists; removing members contained in others, we may assume that no member is contained in another. An irreducible germ contained in a finite union of analytic germs is contained in one of them, since it is the union of its intersections with them. Now let $\bigcup_i A_i = \bigcup_j B_j$ be two such decompositions. Each A_i is contained in some B_j, which in turn is contained in some A_k; then $A_i \subset A_k$ forces $k = i$ and $A_i = B_j$. By symmetry the two decompositions agree.

Let (A,x) be irreducible. Its ideal is proper, since $x \in A$. If fg vanishes on (A,x), then (A,x) is the union of its intersections with $\{f=0\}$ and $\{g=0\}$, so one of them is all of (A,x), and f or g belongs to $I_{A,x}$; thus $I_{A,x}$ is prime. Conversely, suppose that (A,x) is the union of two analytic subgerms (B,x) and (C,x) different from it. Since $V(I_{B,x}) = (B,x) \neq (A,x)$, there is a germ $f \in I_{B,x} \setminus I_{A,x}$, and similarly $g \in I_{C,x} \setminus I_{A,x}$. Then fg vanishes on (A,x) while neither factor does, so $I_{A,x}$ is not prime. □

Multiple roots of polynomials are detected by the discriminant, which we define by a polynomial formula in the coefficients. For a monic polynomial $P(T) = T^q + a_1 T^{q-1} + \cdots + a_q$, $q \geq 1$, with coefficients in a commutative ring, let C_P be its *companion matrix*, the $q \times q$ matrix with entries 1 just below the diagonal, last column $(-a_q, \ldots, -a_1)^{\mathsf{T}}$ and zeros elsewhere. Put $S_k = \operatorname{tr}(C_P^k)$ for $k \geq 0$, and define the *discriminant* $\operatorname{Disc}(P) = \det(S_{j+l})_{0\leq j,l\leq q-1}$, a polynomial with integer coefficients in $a_1, \ldots, a_q$. If the a_j are complex numbers and $\tau_1, \ldots, \tau_q$ are the roots of P, counted with multiplicity, then

$$\operatorname{Disc}(P) = \prod_{i<k} (\tau_i - \tau_k)^2; \tag{A.3}$$

in particular $\operatorname{Disc}(P) \neq 0$ if and only if the roots of P are distinct. Indeed, the characteristic polynomial of C_P is P, as one sees by expanding $\det(TI - C_P)$ along its last column, so after triangularizing C_P over $\mathbb{C}$ we find $S_k = \sum_i \tau_i^k$. Hence $(S_{j+l}) = \Lambda^{\mathsf{T}}\Lambda$ for the Vandermonde matrix $\Lambda = (\tau_i^j)_{1\leq i\leq q,\, 0\leq j<q}$, whose determinant is $\prod_{i<k}(\tau_k - \tau_i)$.

Lemma A.2.2 *Let $d \geq 0$, let $K = \operatorname{Frac}(\mathcal{O}_d)$, and let B be an integral domain containing $\mathcal{O}_d$ that is finitely generated as an $\mathcal{O}_d$-module and is generated as an $\mathcal{O}_d$-algebra by elements $b_1, \ldots, b_r$. Then the field of fractions L of B is a finite-dimensional vector space over K; let q be its dimension. Moreover:*

(1) *Let $x \in L$ be integral over $\mathcal{O}_d$. Then the minimal polynomial of x over K, that is, the monic polynomial of least degree in $K[T]$ vanishing at x, has degree at most q and coefficients in $\mathcal{O}_d$.*
(2) *For all $c \in \mathbb{C}^r$ outside finitely many proper linear subspaces, the powers $1, u, \ldots, u^{q-1}$ of $u = \sum_j c_j b_j$ form a basis of L over K. In particular, the minimal polynomial $W(T) = T^q + w_1 T^{q-1} + \cdots + w_q$ of u has degree q, and $w_1, \ldots, w_q \in \mathcal{O}_d$.*
(3) *For such u, the discriminant $\delta = \operatorname{Disc}(W) \in \mathcal{O}_d$ is not zero, and every $g \in L$ that is integral over $\mathcal{O}_d$ satisfies*

$$\delta g = \sum_{j=0}^{q-1} d_j u^j, \qquad d_j \in \mathcal{O}_d.$$

Proof We represent elements of L by multiplication matrices over K and evaluate these at points of $\mathbb{C}^d$; integrality bounds their eigenvalues, and Lemma A.1.7 turns bounded elements of K into elements of $\mathcal{O}_d$.

We first recall the determinant criterion for integrality. Let R be a subring of an integral domain D, let $b \in D$, and suppose that multiplication by b maps an R-submodule of D generated by elements $v_1, \dots, v_N$, not all zero, into itself: $bv_i = \sum_j a_{ij} v_j$ with $a_{ij} \in R$. Multiplying the matrix $bI - (a_{ij})$ by its adjugate shows that the monic polynomial $\det(bI - (a_{ij}))$ in b annihilates every v_i; since D is a domain, this determinant is zero, and b is integral over R. In particular, every element of B is integral over $\mathcal{O}_d$; take for the v_i generators of the $\mathcal{O}_d$-module B. Conversely, if $x_1, \dots, x_k \in D$ are integral over R, then $R[x_1, \dots, x_k]$ is a finitely generated R-module, since powers of each x_i can be reduced with its monic equation; by the criterion, sums and products of integral elements are integral.

The ring $KB = \{b/a : b \in B,\ a \in \mathcal{O}_d \setminus \{0\}\}$ is a domain, and it is spanned over K by finitely many module generators of B, so it is a finite-dimensional vector space over K. Multiplication by a nonzero element is an injective linear map of KB, hence surjective; so KB is a field, and it equals L. Choose a basis $e_1 = 1, e_2, \dots, e_q$ of L over K consisting of elements of B; it can be extracted from 1 and the module generators of B. For $x \in L$ let $[x] \in K^{q \times q}$ be the matrix of the K-linear map $y \mapsto xy$ in this basis. The map $x \mapsto [x]$ is an injective homomorphism of K-algebras, because the first column of $[x]$ is the coordinate vector of $xe_1 = x$. Let $\mathrm{Tr}(x) = \mathrm{tr}[x]$, which does not depend on the basis, and let $\chi_x(T) = \det(TI - [x]) \in K[T]$, a monic polynomial of degree q; by the Cayley–Hamilton theorem $[\chi_x(x)] = \chi_x([x]) = 0$, so $\chi_x(x) = 0$. The minimal polynomial μ_x of x exists, because $1, x, \dots, x^q$ are linearly dependent over K, and $\deg \mu_x \le q$. By Euclidean division in $K[T]$, μ_x divides every polynomial in $K[T]$ that vanishes at x.

Values of elements of K. Let U be a connected neighborhood of 0 in $\mathbb{C}^d$. An element of K is *defined* at $z' \in U$ if it can be written as a/h with a, h holomorphic on U and $h(z') \neq 0$; its value at z' is then $a(z')/h(z')$. This value does not depend on the representation, since two representations $a/h = a'/h'$ satisfy $ah' = a'h$ near 0, hence on U by the identity theorem. If finitely many elements of K are defined at z', so are their sums and products, and their quotients by those with nonzero value at z', with the corresponding values. Finitely many elements of K can be written with a common denominator $h \neq 0$ and numerators holomorphic on some connected neighborhood U of 0; they are then all defined at the points of $U \setminus \{h = 0\}$, a dense open subset of U. We shall use the following *boundedness criterion*: if the values of an element a/h of K at the points of $V \setminus \{h = 0\}$ are bounded, for some neighborhood $V \subset U$ of 0, then $a/h \in \mathcal{O}_d$. Indeed, $|a| \le C|h|$ on $V \setminus \{h = 0\}$, hence on V, since both sides are continuous functions on V and $V \setminus \{h = 0\}$ is dense in V; so $a/h \in \mathcal{O}_d$ by Lemma A.1.7. (For $d = 0$, the space $\mathbb{C}^0$ is a point and $K = \mathcal{O}_0 = \mathbb{C}$.)

Proof of (1). Let $x \in L$ be integral over $\mathcal{O}_d$, say $f(x) = 0$ with $f(T) = T^m + f_1 T^{m-1} + \cdots + f_m$ and $f_j \in \mathcal{O}_d$. Then $f([x]) = [f(x)] = 0$. Since $\chi_x(x) = 0$, we can write $\chi_x = \mu_x \rho$ with $\rho \in K[T]$. Write the entries of $[x]$ and the coefficients of μ_x and ρ with a common denominator h, so that they are all defined at the points of $U \setminus \{h = 0\}$ for a connected neighborhood U of 0; the coefficients of χ_x, polynomials in the entries of $[x]$, can then be written with the denominator h^q.

At the points $z' \in U \setminus \{h = 0\}$ near 0 the complex matrix $[x](z')$ satisfies $f(z', [x](z')) = 0$. Applying this to an eigenvector, we see that every eigenvalue λ of

$[x](z')$ satisfies $f(z',\lambda)=0$, so that $|\lambda|\le C_f$ by (A.1), where C_f is an upper bound of $2\max_j|f_j|^{1/j}$ near 0. The coefficients of $\chi_x(z',T)$, the characteristic polynomial of $[x](z')$, are up to sign the elementary symmetric functions of these eigenvalues, so they are bounded at the points of $V\setminus\{h=0\}$ for a neighborhood $V\subset U$ of 0, and $\chi_x\in\mathcal{O}_d[T]$ by the boundedness criterion; in particular $\mathrm{Tr}(x)\in\mathcal{O}_d$. The roots of $\mu_x(z',\cdot)$ are roots of $\chi_x(z',\cdot)=\mu_x(z',\cdot)\rho(z',\cdot)$, so they are also bounded by C_f; hence the coefficients of $\mu_x(z',\cdot)$, which are up to sign the elementary symmetric functions of these roots, are bounded at the points of $V\setminus\{h=0\}$. Therefore $\mu_x\in\mathcal{O}_d[T]$, again by the boundedness criterion, which proves (1).

Proof of (2). We evaluate the multiplication matrices at a point z_0', where they span, in a suitable basis, the algebra of all diagonal matrices; then a generic combination of the matrices $[b_k](z_0')$ has distinct eigenvalues. We first show that the trace form is nondegenerate. Let $\Theta=\det(\mathrm{Tr}(e_ie_j))_{i,j}\in K$. If $\Theta=0$, there are $x_1,\dots,x_q\in K$, not all zero, with $\sum_i x_i\,\mathrm{Tr}(e_ie_j)=0$ for all j; then $x=\sum_i x_ie_i\ne0$ satisfies $\mathrm{Tr}(xy)=0$ for all $y\in L$, by linearity, and $y=x^{-1}$ gives $\mathrm{Tr}(1)=q=0$, which is absurd. Write $e_ie_j=\sum_l\gamma_{ijl}e_l$ and $b_k=\sum_l\beta_{kl}e_l$ with $\gamma_{ijl},\beta_{kl}\in K$, and, since B is generated by the b_k as an $\mathcal{O}_d$-algebra, $e_l=G_l(b_1,\dots,b_r)$ for polynomials G_l with coefficients in $\mathcal{O}_d$. Choose a connected neighborhood U of 0 and a point $z_0'\in U$ at which the finitely many elements Θ, γ_{ijl}, β_{kl}, the entries of the matrices $[e_l]$ and $[b_k]$, and the coefficients of the G_l are defined, such that $\Theta(z_0')\ne0$.

Let $\mathcal{N}$ be the complex linear span of the matrices $N_l=[e_l](z_0')$, $1\le l\le q$. Since $[e_i][e_j]=[e_ie_j]=\sum_l\gamma_{ijl}[e_l]$, the set $\mathcal{N}$ is a commutative algebra of complex $q\times q$ matrices, and it contains $N_1=I$. The matrix $(\mathrm{tr}(N_iN_j))_{i,j}$ is the value of $(\mathrm{Tr}(e_ie_j))_{i,j}$ at z_0', which is invertible; hence the N_l are linearly independent, $\dim\mathcal{N}=q$, and the bilinear form $(N,N')\mapsto\mathrm{tr}(NN')$ on $\mathcal{N}$ is nondegenerate. If $N\in\mathcal{N}$ is nilpotent, then NN' is nilpotent for every $N'\in\mathcal{N}$, because N and N' commute; so $\mathrm{tr}(NN')=0$ for all N', and $N=0$. Consequently every $N\in\mathcal{N}$ is diagonalizable. Indeed, if $\lambda_1,\dots,\lambda_p$ are the distinct eigenvalues of N, then the minimal polynomial of N divides a power of $(T-\lambda_1)\cdots(T-\lambda_p)$, so the element $(N-\lambda_1I)\cdots(N-\lambda_pI)$ of $\mathcal{N}$ is nilpotent, hence zero, and the minimal polynomial of N has simple roots. Commuting diagonalizable matrices are simultaneously diagonalizable; so in a suitable basis of $\mathbb{C}^q$ all elements of $\mathcal{N}$ are diagonal, and since $\dim\mathcal{N}=q$, the algebra $\mathcal{N}$ consists of all diagonal matrices.

The matrices $[b_k](z_0')=\sum_l\beta_{kl}(z_0')N_l$ belong to $\mathcal{N}$; let $\lambda_k^{(i)}$ be the ith diagonal entry of $[b_k](z_0')$ in this basis, and $\lambda^{(i)}=(\lambda_1^{(i)},\dots,\lambda_r^{(i)})$. For $i\ne i'$ we have $\lambda^{(i)}\ne\lambda^{(i')}$: otherwise every matrix $N_l=G_l(z_0')([b_1](z_0'),\dots,[b_r](z_0'))$ would have equal ith and i'th diagonal entries, and so would every element of $\mathcal{N}$. Therefore, for c outside the finitely many proper linear subspaces $\{c:\sum_k c_k(\lambda_k^{(i)}-\lambda_k^{(i')})=0\}$, $i<i'$, the matrix $[u](z_0')=\sum_k c_k[b_k](z_0')$ is diagonal with q distinct diagonal entries $\tau_1,\dots,\tau_q$.

Fix such a c, and let $\Lambda\in K^{q\times q}$ be the matrix whose columns are the coordinate vectors of $1,u,\dots,u^{q-1}$; its entries are polynomials in the entries of the $[b_k]$, so they are defined at z_0'. The linear map $\mathcal{N}\to\mathbb{C}^q$ that takes a matrix to its first column, its product with the coordinate vector $(1,0,\dots,0)^{\mathsf T}$ of $e_1=1$, maps N_l

to the lth unit vector, the coordinate vector of $e_l e_1 = e_l$; so it is bijective. It maps $[u](z_0')^j = [u^j](z_0')$ to the jth column of $\Lambda(z_0')$, counting from $j = 0$. In the diagonal basis, $[u](z_0')^j = \operatorname{diag}(\tau_1^j, \dots, \tau_q^j)$, and these matrices are linearly independent for $0 \le j < q$, by the Vandermonde determinant, since the τ_i are distinct. Hence $\det \Lambda(z_0') \ne 0$, so $\det \Lambda \ne 0$, and $1, u, \dots, u^{q-1}$ is a basis of L. Consequently μ_u has degree q, and since $u \in B$ is integral over $\mathcal{O}_d$, its coefficients lie in $\mathcal{O}_d$ by (1). This proves (2).

Proof of (3). In the basis $1, u, \dots, u^{q-1}$, multiplication by u has the companion matrix C_W, because $u \cdot u^{q-1} = -\sum_j w_j u^{q-j}$. Hence $[u] = \Lambda C_W \Lambda^{-1}$, and $\operatorname{Tr}(u^k) = \operatorname{tr}(C_W^k) = S_k$ for $k \ge 0$, with S_k as in the definition of $\operatorname{Disc}(W)$. The w_j, being the coordinates of $-u^q$ in the basis $1, \dots, u^{q-1}$, are defined at z_0', because $\det \Lambda(z_0') \ne 0$; and $C_W(z_0') = \Lambda(z_0')^{-1}[u](z_0')\Lambda(z_0')$ has the q distinct eigenvalues $\tau_1, \dots, \tau_q$. Since the characteristic polynomial of $C_W(z_0')$ is $W(z_0', \cdot)$, these are its roots, and (A.3) shows that the value of δ at z_0' is not zero; so $\delta \ne 0$. Now let $g \in L$ be integral over $\mathcal{O}_d$, and write $g = \sum_{j<q} a_j u^j$ with $a_j \in K$. Multiplying by u^l and taking traces, we obtain

$$\sum_{j=0}^{q-1} S_{j+l}\, a_j = \operatorname{Tr}(g u^l), \qquad 0 \le l < q.$$

The elements gu^l are integral over $\mathcal{O}_d$, as products of integral elements, so $\operatorname{Tr}(gu^l) \in \mathcal{O}_d$ by the proof of (1). The matrix (S_{j+l}) has entries in $\mathcal{O}_d$ and determinant δ; multiplying the system by its adjugate gives $\delta a_j \in \mathcal{O}_d$, which is (3) with $d_j = \delta a_j$.□

Theorem A.2.3 (Local parametrization) *Let $(A, 0) \subset (\mathbb{C}^n, 0)$ be an irreducible analytic germ. There are integers $0 \le d \le n$ and $q \ge 1$, linear coordinates $z = (z', z'') \in \mathbb{C}^d \times \mathbb{C}^{n-d}$, which can be chosen arbitrarily close to any given linear coordinates, a representative A of the germ and arbitrarily small polydiscs $\Delta' \times \Delta''$ centered at 0 with the following properties. The projection*

$$\pi \colon A \cap (\Delta' \times \Delta'') \longrightarrow \Delta', \qquad \pi(z', z'') = z',$$

is proper; its fiber over 0 is $\{0\}$, and every fiber has at most q points. There is a holomorphic function $\delta \not\equiv 0$ on Δ' such that, over $\Delta' \setminus \{\delta = 0\}$, the map π is a connected q-sheeted covering whose sheets are graphs of holomorphic maps. Its total space $A_\delta = A \cap \pi^{-1}(\Delta' \setminus \{\delta = 0\})$, the covering locus*, is a complex submanifold of $(\Delta' \setminus \{\delta = 0\}) \times \Delta''$ of dimension d, open and dense in $A \cap (\Delta' \times \Delta'')$. Moreover, there are a linear form u on $\mathbb{C}^{n-d}$, called a* primitive linear form*, and a Weierstrass polynomial $W(z', T)$ in T of degree q with coefficients holomorphic on Δ', such that $\delta = \operatorname{Disc}(W)$ and, for every $z' \in \Delta' \setminus \{\delta = 0\}$, the numbers $u(z'')$, $z \in \pi^{-1}(z')$, are the q distinct roots of $W(z', \cdot)$.*

Proof We prove more generally the following statement for every prime ideal $J \subset \mathcal{O}_n$: the germ $V(J)$ has coordinates and polydiscs as in the theorem, and $I_{V(J),0} = J$. The theorem then follows by taking $J = I_{A,0}$, which is prime by Proposition A.2.1 and satisfies $V(J) = (A, 0)$. If $J = 0$, then $V(J)$ is a neighborhood of 0; we take $d = n$,

$q = 1$, $u = 0$ and $W(z', T) = T$, so that $\delta = \mathrm{Disc}(W) = 1$, and $I_{V(0),0} = 0$ by the identity theorem. Assume that $J \neq 0$. Steps 1–3 construct the covering and prove that $I_{V(J),0} = J$; Steps 4 and 5 prove the density and the connectedness of the covering locus.

Step 1: triangular equations. Since J is prime, it is a proper ideal, so its elements vanish at 0. Choose a nonzero element of J, of order $s_n \geq 1$, and linear coordinates in which it is distinguished in z_n. By the preparation theorem it is a unit times a Weierstrass polynomial $P_n \in J$ of degree s_n in z_n. Since a unit has order 0, the polynomial P_n also has order s_n, so its coefficient of $z_n^{s_n - j}$ has order at least j in $(z_1, \ldots, z_{n-1})$. The ideal $J \cap \mathcal{O}_{n-1}$ of germs in the first $n-1$ coordinates is again prime. If it is nonzero, we repeat the construction with a linear change of $z_1, \ldots, z_{n-1}$ only, which preserves the properties of P_n; we continue until $J \cap \mathcal{O}_d = 0$ for some d. This gives coordinates and Weierstrass polynomials

$$P_k = z_k^{s_k} + \sum_{j=1}^{s_k} a_{jk}(z_1, \ldots, z_{k-1}) z_k^{s_k - j} \in J, \qquad d < k \leq n,$$

with $a_{jk} = O(|(z_1, \ldots, z_{k-1})|^j)$. At each step only the zero set of a nonzero homogeneous polynomial is excluded in choosing the direction, so the coordinates can be chosen arbitrarily close to given ones. Fix generators of J, holomorphic on a neighborhood U of 0, and let $V(J)$ also denote their common zero set in U. Applying (A.1) successively to $P_{d+1}, \ldots, P_n$, we find a constant C with

$$|z''| \leq C|z'| \qquad \text{for } z \in V(J) \text{ near } 0.$$

Consequently, if Δ'' is a small polydisc and Δ' a polydisc whose radius is small compared with that of Δ'', the points of $V(J) \cap (\Delta' \times \Delta'')$ over each compact subset of Δ' form a compact set. Hence π is proper, its fiber over 0 is $\{0\}$, and its fibers are finite, since they lie in the sets of successive roots of $P_{d+1}, \ldots, P_n$.

Successive division by $P_n, \ldots, P_{d+1}$ (Proposition A.1.1) shows that every germ is congruent modulo J to a polynomial in $z_{d+1}, \ldots, z_n$ of degree less than s_k in z_k, with coefficients in $\mathcal{O}_d$. Since $J \cap \mathcal{O}_d = 0$, the ring $\mathcal{O}_d$ injects into

$$B = \mathcal{O}_n / J,$$

and B is a finitely generated $\mathcal{O}_d$-module, generated by the classes of the monomials $\prod_{k>d} z_k^{\alpha_k}$ with $0 \leq \alpha_k < s_k$.

Step 2: a primitive element and division by the discriminant. The ring B is a domain, because J is prime, and by Step 1 it is generated as an $\mathcal{O}_d$-algebra by the classes of $z_{d+1}, \ldots, z_n$. We apply Lemma A.2.2 to B and these generators, and let q be the dimension of the field of fractions of B over $K = \mathrm{Frac}(\mathcal{O}_d)$. Choose c as in Lemma A.2.2(2), and let $u = \sum_{k>d} c_k z_k$, which will be the primitive linear form of the statement; let $W(z', T) \in \mathcal{O}_d[T]$ be its minimal polynomial, of degree q, and let $\delta = \mathrm{Disc}(W) \in \mathcal{O}_d \setminus \{0\}$ be its discriminant. By Lemma A.2.2(3) there are polynomials $B_k(z', T) \in \mathcal{O}_d[T]$ of degree less than q with

$$\delta z_k \equiv B_k(z', u) \pmod{J}, \qquad d < k \le n.$$

The class of each z_k, $k > d$, lies in B and is therefore integral over $\mathcal{O}_d$, so by Lemma A.2.2(1) its minimal polynomial $W_k(z', T)$ over K is monic of degree at most q with coefficients in $\mathcal{O}_d$. These polynomials, and W, are Weierstrass polynomials in T. Indeed, the germ $W_k(z', z_k)$ belongs to J, so it vanishes at the origin, and 0 is a root of $W_k(0, \cdot)$ of some multiplicity $\mu_k \ge 1$. By Lemma A.1.3 in the variables (z', T) at the origin, $W_k = V_k Q_k$, where V_k is a Weierstrass polynomial in T of degree μ_k and Q_k is a polynomial in T with $Q_k(0, 0) \neq 0$. Substituting $T = z_k$, which vanishes at the origin, $Q_k(z', z_k)$ becomes a unit of $\mathcal{O}_n$, so $V_k(z', z_k) \in J$. By minimality $\deg V_k \ge \deg W_k$, hence $W_k = V_k$. The same argument applies to W and u.

Let $G \subset J$ be the ideal generated by $W(z', u)$ and the germs $\delta z_k - B_k(z', u)$. We shall use the following *divisibility property* several times: if $H \in \mathcal{O}_d[T]$ and $H(z', u) \in J$, then H is divisible by W in $\mathcal{O}_d[T]$; in particular, $H(z', u) \in G$, and $H = 0$ if $\deg H < q$. Indeed, H vanishes at the class of u in $\mathrm{Frac}(B)$, and so does the remainder of its Euclidean division by the monic polynomial W; this remainder has degree less than q, so it is zero, because $1, u, \dots, u^{q-1}$ are linearly independent over K.

We claim that, with $m = \max\{q, (n-d)(q-1)\}$, every germ $f \in \mathcal{O}_n$ satisfies

$$\delta^m f \equiv R_f(z', u) \pmod{G}, \qquad f \in J \Longrightarrow R_f = 0, \tag{A.4}$$

for a unique polynomial $R_f(z', T) \in \mathcal{O}_d[T]$ of degree less than q. To prove this, divide f successively by the Weierstrass polynomials $W_n(z', z_n), \dots, W_{d+1}(z', z_{d+1})$, whose coefficients depend only on z'. This writes f as a polynomial $\sum_\alpha e_\alpha(z') z''^\alpha$ with $\alpha_k \le q-1$ for all k, plus multiples of the $W_k(z', z_k)$. Multiplying by δ^m and substituting $\delta z_k \equiv B_k(z', u)$ modulo G, which is possible because $|\alpha| \le (n-d)(q-1) \le m$, we obtain a polynomial in u with coefficients in $\mathcal{O}_d$; dividing it by W gives R_f. The discarded multiples of $W_k(z', z_k)$ disappear modulo G, because $m \ge q$ and $\delta^q W_k(z', z_k) \in G$. Indeed, the polynomial

$$H_k(z', T) = \delta^q W_k\big(z', B_k(z', T)/\delta\big)$$

has coefficients in $\mathcal{O}_d$, since $\deg W_k \le q$. The same substitution gives $\delta^q W_k(z', z_k) \equiv H_k(z', u)$ modulo G; hence $H_k(z', u) \in J$, and by the divisibility property H_k is divisible by W and $H_k(z', u) \in G$. Two admissible polynomials R_f differ by a polynomial H of degree less than q with $H(z', u) \in G \subset J$, so they agree, by the divisibility property. If $f \in J$, then $R_f(z', u) \in J$, and $R_f = 0$ for the same reason.

Step 3: the covering off the discriminant. Choose polydiscs as in Step 1 so small that all the germs above and the generators of J are holomorphic near the closure of $\Delta' \times \Delta''$, and such that the following identities between germs hold there: (A.4) for the generators of J, the expressions of the generators of G as combinations of the generators of J, and the divisions of H_k and of $\sum_k c_k B_k(z', T) - \delta T$ by W used below. Let $z' \in \Delta'$ with $\delta(z') \neq 0$. Then $W(z', \cdot)$ has q distinct roots $\tau_1, \dots, \tau_q$, by (A.3), and the points of $V(G)$ over z' are exactly the q points

$$W(z',T)=0, \qquad z_k = B_k(z',T)/\delta(z'), \qquad T=\tau_1,\dots,\tau_q.$$

Indeed, a point of $V(G)$ satisfies these equations with $T=u(z'')$. Conversely, the polynomial $\sum_k c_k B_k(z',T)-\delta T$ takes at $T=u$ a value congruent to $\delta u-\delta u=0$ modulo J, so it is divisible by W, by the divisibility property of Step 2, and vanishes at each τ_i; hence the point defined by τ_i satisfies $u(z'')=\tau_i$ and lies in $V(G)$. These q points are distinct, because their values of u are distinct.

The simple roots τ_i, and hence the points, depend holomorphically on z' locally: if τ is a simple root of $W(z'_1,\cdot)$, the map $(z',T)\mapsto(z',W(z',T))$ has invertible derivative at (z'_1,τ), and its holomorphic local inverse (Proposition 7.1.2(1)) yields a holomorphic function $\tau(z')$ near z'_1 with $W(z',\tau(z'))=0$ and $\tau(z'_1)=\tau$. Since H_k is divisible by W, we have $\delta(z')^q W_k(z',B_k(z',\tau_i)/\delta(z'))=H_k(z',\tau_i)=0$, so the coordinate $z_k=B_k(z',\tau_i)/\delta(z')$ is a root of the Weierstrass polynomial $W_k(z',\cdot)$; by (A.1) it is small when z' is small. Shrinking Δ', we may therefore assume that these points lie in $\Delta'\times\Delta''$. They belong to $V(J)$, since $\delta^m f\in G$ for every generator f of J by (A.4), and $\delta(z')\neq 0$. As $G\subset J$, we conclude that over $\Delta'\setminus\{\delta=0\}$ the set $V(J)\cap(\Delta'\times\Delta'')$ is the union of the q disjoint local holomorphic graphs just described, on which u takes the values $\tau_1,\dots,\tau_q$; this gives the assertion about u and W, which persists when the polydiscs are shrunk later. Thus A_δ is a complex submanifold of dimension d, open in $V(J)\cap(\Delta'\times\Delta'')$, and a q-sheeted covering of $\Delta'\setminus\{\delta=0\}$; this base is connected by Lemma A.1.7.

Now let $f\in\mathcal{O}_n$ vanish on $V(J)$. By (A.4), the polynomial $R_f(z',\cdot)$ vanishes at the q distinct roots τ_i for every z' with $\delta(z')\neq 0$ near 0, because $\delta^m f$ and the elements of G vanish at the corresponding points of $V(J)$. Its degree is less than q, so $R_f=0$. Hence $\delta^m f\in G\subset J$, and since $\delta\notin J$ and J is prime, $f\in J$. This proves $I_{V(J),0}=J$.

Step 4: density of the covering locus and the bound on the fibers. For a complex linear form λ on $\mathbb{C}^{n-d}$ and $z'\in\Delta'\setminus\{\delta=0\}$, put

$$P_\lambda(z',T)=\prod_{z\in\pi^{-1}(z')\cap A_\delta}(T-\lambda(z'')).$$

Its coefficients are elementary symmetric functions of the values of λ on the sheets, hence holomorphic and bounded on $\Delta'\setminus\{\delta=0\}$; by Lemma A.1.7 they extend holomorphically to Δ'. The function $\delta(z')P_\lambda(z',\lambda(z''))$ vanishes on $V(J)\cap(\Delta'\times\Delta'')$: on A_δ the second factor vanishes, and elsewhere the first. By Step 3 its germ belongs to J, and since $\delta\notin J$, the germ of $P_\lambda(z',\lambda(z''))$ belongs to J; so this function vanishes on $V(J)$ near 0.

Let $z'_0\in\Delta'$ and let F be the fiber over z'_0 of the closure $\overline{A_\delta}$ of A_δ in $\Delta'\times\Delta''$. We claim that the roots of $P_\lambda(z'_0,\cdot)$ are exactly the values $\lambda(y'')$, $y\in F$. Indeed, every $y\in F$ is a limit of points $y_j\in A_\delta$ with $\pi(y_j)\to z'_0$, and $\lambda(y''_j)$ is a root of $P_\lambda(\pi(y_j),\cdot)$; since the coefficients of P_λ are continuous, $\lambda(y'')$ is a root of $P_\lambda(z'_0,\cdot)$. Conversely, choose $z'_j\to z'_0$ with $\delta(z'_j)\neq 0$. The q points of A_δ over z'_j lie in a compact subset of $\Delta'\times\Delta''$ by Step 1, so after passing to a subsequence they converge to points of F; their values of λ, which are the q roots of $P_\lambda(z'_j,\cdot)$, converge to the q

roots of $P_\lambda(z'_0,\cdot)$, counted with multiplicity. It follows that F has at most q points: a linear form taking distinct values at $q+1$ points of F would give $q+1$ distinct roots of a polynomial of degree q.

It remains to show that $V(J)$ is contained in $\overline{A_\delta}$ near 0; after shrinking the polydiscs once more, this gives the density of A_δ and the bound q for all fibers. Suppose not. Then there are points $y_j \in V(J)\setminus\overline{A_\delta}$ with $y_j \to 0$. For each j, the forms λ with $\lambda(y''_j)=\lambda(w'')$ for some point w of the finite fiber F_j of $\overline{A_\delta}$ over $\pi(y_j)$ form finitely many proper linear subspaces. We choose one form λ outside all these countably many subspaces: for instance, λ with coefficients $(1,t,\dots,t^{n-d-1})$, where t avoids the countably many roots of the nonzero polynomials $\sum_k v_k t^{k-1}$ attached to the vectors $v = y''_j - w''$. Then $\lambda(y''_j)$ is not a root of $P_\lambda(\pi(y_j),\cdot)$ for any j, by the previous paragraph. On the other hand, $P_\lambda(z',\lambda(z''))$ vanishes on $V(J)$ near 0, hence at y_j for all large j, a contradiction.

Step 5: connectedness of the covering locus. The covering A_δ of the connected manifold $\Delta'\setminus\{\delta=0\}$ has finitely many connected components, each of which is itself a covering with a positive number of sheets. Indeed, over a small ball $N\subset\Delta'\setminus\{\delta=0\}$ the set A_δ is the disjoint union of q connected holomorphic graphs, each of which lies in a single connected component of A_δ. Hence the number of points of a given component over z' is locally constant, hence constant on the connected set $\Delta'\setminus\{\delta=0\}$, and it is positive because the component is not empty.

Suppose that there are at least two components. For each component $A^{(i)}$ and each linear form λ, the product $P_\lambda^{(i)}$ of the factors $T-\lambda(z'')$ over the points of $A^{(i)}$ again has bounded holomorphic coefficients on $\Delta'\setminus\{\delta=0\}$, which extend to Δ', and $P_\lambda=\prod_i P_\lambda^{(i)}$. Choose points $z'_j\to 0$ with $\delta(z'_j)\neq 0$, and, by the countable choice of Step 4, a form λ that takes distinct values at the q points of A_δ over each z'_j. Since the germ of $P_\lambda(z',\lambda(z''))$ belongs to the prime ideal J, so does the germ of some factor $P_\lambda^{(i)}(z',\lambda(z''))$, which then vanishes on $V(J)$ near 0. Choose $k\neq i$. For large j the points of $A^{(k)}$ over z'_j are close to 0, by Step 1, so $P_\lambda^{(i)}(z'_j,\cdot)$ vanishes at their values of λ; but these values differ from the roots of $P_\lambda^{(i)}(z'_j,\cdot)$, which are the values of λ at the points of $A^{(i)}$ over z'_j. This contradiction shows that A_δ is connected, and completes the proof. □

We record a consequence that will be used repeatedly.

Corollary A.2.4 *Let $(A,0)\subset(\mathbb{C}^n,0)$ be an irreducible analytic germ, and let $(B,0)$ be an analytic subgerm different from $(A,0)$. Let $\Delta'\times\Delta''$ be a parametrization polydisc of $(A,0)$ as in Theorem A.2.3, with its representative A and covering locus A_δ, so small that $(B,0)$ has a representative B which is an analytic subset of $\Delta'\times\Delta''$. Then $A_\delta\setminus B$ is dense in $A\cap(\Delta'\times\Delta'')$. In particular, $(A\cap(\Delta'\times\Delta''))\setminus B$ is dense in $A\cap(\Delta'\times\Delta'')$, and so is the regular locus of $A\cap(\Delta'\times\Delta'')$, which contains A_δ.*

Proof The set $B\cap A_\delta$ is an analytic subset of the connected manifold A_δ. It is a proper subset, for otherwise the closed set B would contain the closure $A\cap(\Delta'\times\Delta'')$ of A_δ, and the two germs would agree. By Lemma A.1.6, $A_\delta\setminus B$ is dense in A_δ, and hence in $A\cap(\Delta'\times\Delta'')$, in which A_δ is dense. □

A first consequence is that the closure of the difference of two analytic sets is analytic. We shall use this for closures of graphs, in the constructions of normalizations and blowups.

Lemma A.2.5 *Let Z and Y be analytic subsets of a complex manifold M. Then the closure $\overline{Z \setminus Y}$ is analytic. More precisely, near each point $p \in Z$ it is the union of representatives of those branches of (Z, p) that are not contained in (Y, p).*

Proof Let $p \in Z$; near points outside Z there is nothing to prove, since $\overline{Z \setminus Y} \subset Z$. Choose representatives $Z_1, \ldots, Z_k$ of the branches of (Z, p) on an open neighborhood U of p with $Z \cap U = \bigcup_i Z_i$, so small that $Z_i \subset Y$ whenever $(Z_i, p) \subset (Y, p)$, and that, for each of the other branches, $Z_i \setminus Y$ is dense in Z_i; the latter is possible by Corollary A.2.4, applied to (Z_i, p) and its subgerm $(Z_i \cap Y, p)$, which differs from (Z_i, p). Then $(Z \setminus Y) \cap U$ is the union of the sets $Z_i \setminus Y$ over the branches not contained in (Y, p), and its closure in U is the union of the corresponding sets Z_i, which is analytic in U. □

The same holds for analytic subsets of a complex space, as we see by applying the lemma in local models (Appendix A.3).

Corollary A.2.6 (Analytic Nullstellensatz) *For every ideal $I \subset \mathcal{O}_n$, the ideal of germs vanishing on $V(I)$ is the radical $\sqrt{I} = \{f : f^k \in I \text{ for some } k \geq 1\}$.*

Proof Every element of $\sqrt{I}$ vanishes on $V(I)$. For the converse we first show that $\sqrt{I}$ is the intersection of the prime ideals containing I. A prime ideal containing I contains $\sqrt{I}$. Let $f \notin \sqrt{I}$, so that the ideals containing I and disjoint from $S = \{1, f, f^2, \ldots\}$ form a nonempty family; by Proposition A.1.2 it has a maximal element $\mathfrak{p}$, since otherwise there would be a strictly increasing infinite sequence of ideals. The ideal $\mathfrak{p}$ is prime: if $a, b \notin \mathfrak{p}$, then $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ meet S, say $f^k = p_1 + xa$ and $f^l = p_2 + yb$ with $p_1, p_2 \in \mathfrak{p}$; multiplying, we see that $ab \in \mathfrak{p}$ would give $f^{k+l} \in \mathfrak{p}$, which is impossible. Thus f lies outside some prime ideal containing I. Now let g vanish on $V(I)$. For every prime ideal $J \supset I$ we have $V(J) \subset V(I)$, so g vanishes on $V(J)$ and therefore $g \in J$, by the equality $I_{V(J),0} = J$ proved in Theorem A.2.3. Hence g lies in the intersection of all these prime ideals, which is $\sqrt{I}$. □

Proposition A.2.7 *Let $(A, 0)$ be an irreducible analytic germ. The regular points of every sufficiently small representative of $(A, 0)$ all have the same dimension d, equal to the base dimension in Theorem A.2.3; we call d the* dimension *of $(A, 0)$. Every sufficiently small representative is pure-dimensional: all its local irreducible branches, at all of its points, have dimension d. A proper analytic subgerm of an irreducible germ has strictly smaller dimension. For a reducible germ we define the dimension as the maximum of the dimensions of its branches.*

Proof Let A be the representative on a parametrization polydisc. Two local parametrizations of $(A, 0)$ have covering loci which are open and dense in A near 0; their intersection is therefore nonempty, and at its points A is a manifold whose dimension equals both base dimensions. Hence d is determined by the germ.

A regular point of A has a neighborhood in A which is a manifold; this neighborhood meets the dense covering locus, which has dimension d, so the regular point has dimension d as well. Now let $y \in A$, let $B_1, \dots, B_p$ be representatives of the branches of (A, y), and fix i. Since B_i is not contained in $\bigcup_{k \neq i} B_k$, Corollary A.2.4 gives points of the covering locus of a parametrization of B_i outside the other branches. Near such a point, A coincides with B_i and is a manifold of dimension $\dim B_i$; such points form a nonempty open subset of A, which meets the covering locus of A. Hence $\dim B_i = d$.

Let $(B, 0) \subset (A, 0)$ be germs with both irreducible, and let $J_A \subset J_B$ be their prime ideals. For linear coordinates z and $0 \leq k \leq n$, let $\mathcal{O}_k$ denote the ring of germs depending only on $z_1, \dots, z_k$. For a nonzero prime ideal J, Step 1 of the proof of Theorem A.2.3 chooses the coordinates downward from $k = n$. As long as $J \cap \mathcal{O}_k \neq 0$, it makes a linear change of $z_1, \dots, z_k$ in which a chosen nonzero element of $J \cap \mathcal{O}_k$ is distinguished in z_k; it stops at the largest d with $J \cap \mathcal{O}_d = 0$, which is the base dimension. The rest of that proof applies to every such choice. Since finitely many nonzero germs become distinguished in the last coordinate after one suitable linear change of coordinates, we may make these changes for J_B and, at the stages k with $J_A \cap \mathcal{O}_k \neq 0$, simultaneously for J_A. This gives parametrizations of both germs in common coordinates (if $J_A = 0$, the germ $(A, 0)$ is the germ of $\mathbb{C}^n$, with $d = n$). Since $J_A \subset J_B$, the ideal $J_B \cap \mathcal{O}_k$ is nonzero whenever $J_A \cap \mathcal{O}_k$ is, so $\dim B \leq \dim A$.

Suppose that the dimensions are equal, so that both germs are finite over the same base, and let δ_A, δ_B be the two discriminants. By Lemma A.1.7, the set A' of points of the connected covering locus of A lying over $\{\delta_B \neq 0\}$ is a connected manifold of dimension d, dense in A. The points of B over $\{\delta_A \delta_B \neq 0\}$ form a nonempty submanifold of A' of the same dimension, hence an open subset of A', since the inclusion is an injective immersion between manifolds of the same dimension and therefore open by the inverse function theorem. Thus $B \cap A'$ is an analytic subset of the connected manifold A' with nonempty interior; since proper analytic subsets of connected manifolds are nowhere dense (Lemma A.1.6), $B \supset A'$. Since B is closed and A' is dense in A, we get $(B, 0) = (A, 0)$. Hence a proper subgerm has strictly smaller dimension. For a general proper subgerm B of $(A, 0)$, each branch of B is a proper subgerm of $(A, 0)$, and the same conclusion follows. □

The dimension of an analytic set E at a point x is the dimension of its germ (E, x). Dimension is monotone: if $(E, x) \subset (F, x)$, every branch of (E, x) is contained in a branch of (F, x), so $\dim(E, x) \leq \dim(F, x)$. It is also upper semicontinuous: since small representatives of the branches of (E, x) are pure-dimensional, every branch of (E, y), for y near x, is contained in one of them and has dimension at most $\dim(E, x)$. We say that E is *pure-dimensional* if its dimension is the same at all points, and that $E \subset M$ has *codimension at least* k, respectively *codimension* k, if $\dim(E, x) \leq \dim_x M - k$, respectively $\dim(E, x) = \dim_x M - k$, for all $x \in E$.

Theorem A.2.8 (Cartan's coherence theorem) *For an analytic subset A of a complex manifold M, the sheaf $\mathcal{I}_A$ of holomorphic functions vanishing on A is coherent.*

Proof The assertion is local. Near a point of A, the set A is the union of representatives $A_1, \dots, A_p$ of the branches of its germ, and $\mathcal{I}_A = \bigcap_i \mathcal{I}_{A_i}$. The intersection of two coherent subsheaves $\mathcal{F}, \mathcal{G}$ of $\mathcal{O}_M$ is the kernel of the morphism $\mathcal{F} \oplus \mathcal{G} \to \mathcal{O}_M$, $(a, b) \mapsto a - b$, and is coherent by Theorem A.1.5 and Lemma A.1.4. It therefore suffices to prove that $\mathcal{I}_A$ is coherent near 0 when $(A, 0) \subset (\mathbb{C}^n, 0)$ is irreducible. We use the parametrization of Theorem A.2.3, with its primitive linear form u, its Weierstrass polynomial W and $\delta = \mathrm{Disc}(W)$, and we put $m = \max\{q, (n-d)(q-1)\}$. Since A may have several branches at points near 0, we describe all stalks near 0 at once: we construct holomorphic functions $G_1, \dots, G_N$ vanishing on A and a holomorphic function $h \not\equiv 0$ of z' such that, at every point x near 0, the stalk $\mathcal{I}_{A,x}$ consists of the germs f with $hf \in (G_{1,x}, \dots, G_{N,x})$; such a sheaf is coherent by Oka's theorem.

Step 1: dependence on the primitive form. For $c = (c_{d+1}, \dots, c_n) \in \mathbb{C}^{n-d}$ put $u_c = \sum_k c_k z_k$. For $z' \in \Delta' \setminus \{\delta = 0\}$ let $z^{(1)}, \dots, z^{(q)}$ be the points of A over z', and define

$$W_c(z', T) = \prod_{i=1}^{q} \big(T - u_c(z^{(i)})\big), \qquad \delta_c(z') = \prod_{i<l} \big(u_c(z^{(i)}) - u_c(z^{(l)})\big)^2,$$
$$B_{kc}(z', T) = \sum_{j<q} D_c(z') D_{kcj}(z') T^j,$$

where D_c is the Vandermonde determinant $\det(u_c(z^{(i)})^j)_{i,j}$ and D_{kcj} is obtained from it by replacing the jth column with the coordinates $z_k^{(i)}$. These expressions do not change under permutations of the points $z^{(i)}$, since D_c and D_{kcj} both change sign under a transposition. Hence their coefficients in T are polynomials in c whose coefficients are bounded holomorphic functions on $\Delta' \setminus \{\delta = 0\}$, and these extend holomorphically to Δ' by Lemma A.1.7. On A_δ we have $W_c(z', u_c(z'')) = 0$ and, by Cramer's rule in the form $\sum_j D_{kcj} u_c(z^{(i)})^j = D_c z_k^{(i)}$, also $\delta_c(z') z_k = B_{kc}(z', u_c(z''))$, for every c. Let $G_1, \dots, G_N$ be the coefficients, as polynomials in c, of the functions $W_c(z', u_c)$ and $\delta_c z_k - B_{kc}(z', u_c)$, $d < k \le n$. They are holomorphic on $\Delta' \times \Delta''$ and vanish on A_δ, hence on A, which is the closure of A_δ. For the coefficient vector c of the primitive linear form u, the values $u(z^{(i)})$ are the roots of $W(z', \cdot)$, by Theorem A.2.3, so $\delta_c = \mathrm{Disc}(W) = \delta$ by (A.3), and $\delta_c \not\equiv 0$ off $\{\delta = 0\}$; so the polynomial $\delta_c^m = \sum_\alpha h_\alpha(z') c^\alpha$ in c is not identically zero, and we choose a coefficient $h = h_\alpha \not\equiv 0$.

Step 2: a description of the stalks. We claim that on a small polydisc around 0,

$$\mathcal{I}_{A,x} = \{f \in \mathcal{O}_{M,x} : hf \in (G_{1,x}, \dots, G_{N,x})\} \tag{A.5}$$

for every point x. If $hf \in (G_{1,x}, \dots, G_{N,x})$, then hf vanishes on A near x, so f vanishes on $A \setminus \{h = 0\}$ near x. This set is dense in A: the covering locus A_δ is dense in A, and since π is a local biholomorphism on A_δ, the set where $h \circ \pi$ vanishes is nowhere dense in A_δ. Hence $f \in \mathcal{I}_{A,x}$, which proves the inclusion from right to left.

Step 3: the inclusion from left to right at points of A. Fix $x \in A$ and $f \in \mathcal{I}_{A,x}$. Call c admissible if $\delta_c \not\equiv 0$ and u_c takes distinct values at the points of the finite fiber of A over x'; the admissible c form a dense open subset of $\mathbb{C}^{n-d}$, the complement of an algebraic subset and finitely many proper linear subspaces. Fix an admissible c. Since $W_c(z', u_c(z''))$ vanishes on A by Step 1, the number $u_c(x'')$ is a root of $W_c(x', \cdot)$. By Lemma A.1.3, $W_c = W_c^x Q_c^x$ near $(x', u_c(x''))$, where W_c^x is a Weierstrass polynomial in $T - u_c(x'')$ and Q_c^x is a polynomial in T with $Q_c^x(x', u_c(x'')) \neq 0$.

We call z' *generic* if $\delta(z')\delta_c(z') \neq 0$. For generic z' near x', the roots of $W_c^x(z', \cdot)$ are distinct and are exactly the values of u_c at the points of A over z' that are close to x; let q_x be their number, which is the degree of W_c^x. Indeed, the roots of $W_c(z', \cdot)$ are the q distinct values of u_c at the points of A over z', and by properness these points are close to the points of the fiber over x'. The values of u_c at the points close to x are close to $u_c(x'')$, where $Q_c^x(z', \cdot)$ does not vanish, so they are roots of $W_c^x(z', \cdot)$. The values at the points close to the other points of the fiber are close to values of u_c different from $u_c(x'')$, since c is admissible, whereas the roots of the Weierstrass polynomial $W_c^x(z', \cdot)$ in $T - u_c(x'')$ tend to $u_c(x'')$ as $z' \to x'$, by (A.1).

Let $\mathfrak{g}_c \subset \mathcal{O}_{M,x}$ be the ideal generated by the germs at x of $W_c(z', u_c)$ and $\delta_c z_k - B_{kc}(z', u_c)$, $d < k \leq n$. These functions vanish on A, and they are complex linear combinations of $G_1, \ldots, G_N$, so $\mathfrak{g}_c \subset (G_{1,x}, \ldots, G_{N,x})$. The germ of $W_c^x(z', u_c)$ belongs to $\mathfrak{g}_c$, because it equals $Q_c^x(z', u_c)^{-1} W_c(z', u_c)$ and $Q_c^x(z', u_c)$ is a unit at x. We use the following divisibility property: *if $H(z', T)$ is a polynomial in T whose coefficients are germs at x', and $H(y', u_c(y'')) = 0$ at all points $y \in A$ close to x with y' generic, then the germ of $H(z', u_c)$ belongs to $\mathfrak{g}_c$.* Indeed, the remainder of the division of H by the monic polynomial W_c^x has degree less than q_x, and for generic z' near x' it vanishes at the q_x distinct roots of $W_c^x(z', \cdot)$; so it is zero, and $H(z', u_c)$ is a multiple of $W_c^x(z', u_c)$.

For $d < k \leq n$ and the unit vector e_k we have $u_{e_k} = z_k$, so by Step 1 the coordinate z_k of each point of A is a root of the monic polynomial $W_{e_k}(z', \cdot)$ of degree q. Let W_k^x be the Weierstrass polynomial in $T - x_k$ given by Lemma A.1.3 for W_{e_k} at the root x_k of $W_{e_k}(x', \cdot)$, so that $W_{e_k} = W_k^x Q_k^x$ near (x', x_k) with $Q_k^x(x', x_k) \neq 0$. Then $W_k^x(z', z_k)$ vanishes at the points of A close to x, and its degree q_k is at most q. Substituting $\delta_c z_k \equiv B_{kc}(z', u_c)$ modulo $\mathfrak{g}_c$, we get $\delta_c^{q_k} W_k^x(z', z_k) \equiv H_k(z', u_c)$ modulo $\mathfrak{g}_c$, where

$$H_k(z', T) = \delta_c^{q_k} W_k^x\big(z', B_{kc}(z', T)/\delta_c\big)$$

is a polynomial in T. At a point $y \in A$ close to x with y' generic, Step 1 gives $B_{kc}(y', u_c(y'')) = \delta_c(y') y_k$, so $H_k(y', u_c(y'')) = \delta_c(y')^{q_k} W_k^x(y', y_k) = 0$. By the divisibility property, $H_k(z', u_c) \in \mathfrak{g}_c$; hence $\delta_c^q W_k^x(z', z_k) \in \mathfrak{g}_c$.

Now divide f successively by the Weierstrass polynomials $W_k^x(z', z_k)$ at x, for $k = n, \ldots, d+1$ (Proposition A.1.1, in coordinates centered at x). This writes f as a polynomial $\sum_\alpha e_\alpha(z') z''^\alpha$, with germs e_α at x' and $\alpha_k < q_k \leq q$ for all k, plus multiples of the $W_k^x(z', z_k)$. After multiplication by δ_c^m, these multiples lie in $\mathfrak{g}_c$, because $m \geq q$. Substituting $\delta_c z_k \equiv B_{kc}(z', u_c)$, which is possible because $|\alpha| \leq (n-d)(q-1) \leq m$, we obtain $\delta_c^m f \equiv H(z', u_c)$ modulo $\mathfrak{g}_c$ for a polynomial

$H(z', T)$ whose coefficients are germs at x'. Since f and the elements of $\mathfrak{g}_c$ vanish on A near x, the divisibility property applies to H, and we conclude that

$$\delta_c^m f \in \mathfrak{g}_c \subset (G_{1,x}, \dots, G_{N,x}) \qquad \text{for every admissible } c.$$

The polynomials in c of degree at most $\deg_c \delta_c^m$ form a finite-dimensional space, and the evaluations at points of the dense open set of admissible c span its dual, since a polynomial vanishing on a nonempty open set is zero. Hence each coefficient h_α of δ_c^m is a complex linear combination of finitely many values $\delta^m_{c^{(l)}}$ with admissible $c^{(l)}$, and $h_\alpha f \in (G_{1,x}, \dots, G_{N,x})$. This proves (A.5) at points of A.

Step 4: points outside A, and conclusion. Let $x \notin A$. Choose c such that $u_c(x'')$ differs from the values of u_c at the finitely many points of A over x'. The roots of $W_c(x', \cdot)$ are values of u_c at points of A over x'. Indeed, choose $z'_j \to x'$ with $\delta(z'_j) \neq 0$; by properness, after passing to a subsequence, the q points $z_j^{(1)}, \dots, z_j^{(q)}$ of A over z'_j converge to points $y^{(1)}, \dots, y^{(q)}$ of A over x', so that $W_c(x', T) = \lim_j \prod_i (T - u_c(z_j^{(i)})) = \prod_i (T - u_c(y^{(i)}))$. Hence $W_c(x', u_c(x'')) \neq 0$, so one of the coefficients G_l does not vanish at x, and both sides of (A.5) are the unit ideal. The right side of (A.5) is the projection to the first coordinate of the kernel of the row matrix $(h, -G_1, \dots, -G_N)$, which is coherent by Theorem A.1.5 and Lemma A.1.4. Hence $\mathcal{I}_A$ is coherent. □

To study the singular locus, we need the ideal of a submanifold.

Lemma A.2.9 *Let M be a complex manifold, let (v, w), $w = (w_1, \dots, w_k)$ with $k \geq 0$, be holomorphic coordinates on an open set $U \subset M$, and let $N = \{w_1 = \cdots = w_k = 0\} \subset U$. Then for every $y \in U$ the ideal $I_{N,y}$ of germs vanishing on (N, y) is generated by the germs of $w_1, \dots, w_k$; it is the unit ideal if $y \notin N$. Moreover, for $y \in N$ the germ (N, y) is irreducible, and its dimension is $\dim_y M - k$, the dimension of N at y.*

Proof If $y \in U \setminus N$, then some w_l does not vanish at y, so it is a unit, and (N, y) is empty because N is closed in U; both ideals are the unit ideal. Let $y \in N$, and let g be holomorphic on a polydisc centered at y in the coordinates (v, w), vanishing on N. Then $g(v, 0) = 0$, and

$$g(v, w) = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} g(v, tw)\, \mathrm{d}t = \sum_l w_l \int_0^1 \partial_{w_l} g(v, tw)\, \mathrm{d}t,$$

where the integrals are holomorphic by differentiation under the integral sign. Hence the germ of g lies in the ideal generated by $w_1, \dots, w_k$; conversely, the w_l vanish on N. Consequently, the map $g \mapsto g(v, 0)$ identifies the quotient of $\mathcal{O}_{M,y}$ by this ideal with the ring of germs of holomorphic functions of v at $v(y)$, which is an integral domain (Appendix A.1); so $I_{N,y}$ is prime. Since $y \in N$, the germ (N, y) is nonempty, and it is irreducible by Proposition A.2.1. All points of N near y are regular points of N of dimension $\dim_y M - k$, so this is the dimension of (N, y), by Proposition A.2.7. □

Corollary A.2.10 *Let A be an analytic subset of a complex manifold M.*

(1) *The singular locus A_{sing} is an analytic subset, and at every point of A the intersection of A_{sing} with each irreducible branch of A has smaller dimension than that branch. More precisely, if $x \in A$ and $A_1, \ldots, A_p$ are sufficiently small representatives of the branches of (A, x), then near x*

$$A_{\text{sing}} = \bigcup_i (A_i)_{\text{sing}} \cup \bigcup_{i<j} (A_i \cap A_j);$$

thus a point near x is a regular point of A if and only if it lies on exactly one A_i and is a regular point of A_i.

(2) *Let A have dimension at most d at every point, and define $A^{(0)} = A$ and $A^{(j+1)} = (A^{(j)})_{\text{sing}}$. Then $A^{(d+1)} = \emptyset$, and A is the union of the sets $A^{(j)}_{\text{reg}}$, $0 \le j \le d$, where $A^{(j)}_{\text{reg}}$ is a complex submanifold of dimension at most $d - j$ of the open set $M \setminus A^{(j+1)}$.*

(3) *With d as in (2), the set A is locally a countable union of real submanifolds of dimensions at most $2d$, and also of images of compact cubes in $\mathbb{R}^r$, $r \le 2d$, under Lipschitz maps. The image of a compact cube in $\mathbb{R}^r$ under a Lipschitz map is $\mathcal{H}^s$-null, in the sense of Section 8.4, for every $s > r$. In particular, in every chart A is locally $\mathcal{H}^s$-null for every $s > 2d$.*

Proof (1) Let A first be a small representative of an irreducible germ of dimension e in $\mathbb{C}^n$, pure-dimensional by Proposition A.2.7, and choose generators $g_1, \ldots, g_N$ of $\mathcal{I}_A$ on a neighborhood, by Theorem A.2.8. At a regular point, these generate the ideal of the manifold A, which is generated by $n - e$ normal coordinates (Lemma A.2.9); differentiating, we see that the matrix of differentials of the g_j has rank $n - e$ there. Every minor of size $n - e + 1$ therefore vanishes on the dense covering locus, hence on A, so the rank is at most $n - e$ everywhere on A. Conversely, if it equals $n - e$ at $y \in A$, choose $n - e$ of the g_j with independent differentials at y. Together with e suitable coordinate functions they form holomorphic coordinates near y, by the holomorphic inverse function theorem (Proposition 7.1.2(2)), so their common zero set is an e-dimensional submanifold N near y, and it contains A. Every branch of (A, y) has dimension e and is contained in the germ (N, y), which is irreducible of dimension e by Lemma A.2.9, so it equals (N, y) by Proposition A.2.7; hence A is a manifold near y. Thus A_{sing} is the subset of A where all minors of size $n - e$ of this matrix vanish, an analytic subset.

For a general analytic set A near a point x, let $A_1, \ldots, A_p$ be small representatives of the branches of (A, x). A point y near x lying on only one A_i is regular for A if and only if it is regular for A_i. A point $y \in A_i \cap A_j$, $i \ne j$, is singular. Indeed, otherwise (A, y) would be irreducible, hence equal to some (A_k, y), and we may assume $k \ne i$; then $(A_i, y) \subset (A_i \cap A_k, y)$. But $\dim(A_i \cap A_k, x) < \dim A_i$ by Proposition A.2.7, so $\dim(A_i \cap A_k, y) < \dim A_i = \dim(A_i, y)$ by upper semicontinuity and purity, contradicting monotonicity. This proves the formula for A_{sing} near x, which is therefore analytic by the irreducible case. Finally, let $y \in A$ and let B be a branch of (A, y). By Corollary A.2.4, B contains points of the covering locus of a parametrization of B

outside the other branches of (A, y), arbitrarily close to y, and these are regular points of A. Hence the germ of $A_{\text{sing}} \cap B$ at y differs from B, and it has smaller dimension than B, by Proposition A.2.7.

(2) By (1), each $A^{(j)}$ is analytic, and $\dim(A^{(j+1)}, x) < \dim(A^{(j)}, x)$ at each point x of $A^{(j+1)}$, because every branch of $(A^{(j+1)}, x)$ lies in a branch of $(A^{(j)}, x)$ and in the singular locus. Hence $A^{(j)}$ has dimension at most $d - j$ at each of its points, and $A^{(d+1)} = \varnothing$, a set of dimension -1 being empty. Each $A^{(j)}_{\text{reg}} = A^{(j)} \setminus A^{(j+1)}$ is a complex submanifold of $M \setminus A^{(j+1)}$ of dimension at most $d - j$, and A is the union of these sets.

(3) Each of the submanifolds in (2) is locally a countable union of coordinate charts, so A is locally a countable union of real submanifolds of dimensions at most $2d$. Each such chart is a countable union of images of compact cubes in $\mathbb{R}^r$, $r \le 2d$, under Lipschitz maps, because the image of the chart in $\mathbb{R}^r$ is a countable union of compact cubes, on each of which the smooth inverse of the chart is Lipschitz, by the mean value inequality. Such an image is $\mathcal{H}^s$-null for $s > r$: covering the cube by $O(\delta^{-r})$ cubes of side δ, whose images have diameter $O(\delta)$, we cover the image by $O(\delta^{-r})$ balls of radii $O(\delta)$, and the sum of the sth powers of these radii is $O(\delta^{s-r})$, which tends to zero with δ. A countable union of $\mathcal{H}^s$-null sets is $\mathcal{H}^s$-null, which completes the proof. □

Hausdorff measures also bound the dimension of images of analytic sets under holomorphic maps (Lemma A.2.12), as needed for finite maps in Appendix A.4 and for Remmert's theorem (Theorem A.6.2). Such images are controlled by the rank of the differential, through the holomorphic constant rank theorem.

Lemma A.2.11 (Constant rank theorem) *Let F be a holomorphic map from an open subset of $\mathbb{C}^m$ to $\mathbb{C}^K$ whose differential has constant rank r near a point p. Then, after a permutation of the coordinates of $\mathbb{C}^K$, there are a biholomorphic map ϕ from a neighborhood of p onto a polydisc $Q' \times Q'' \subset \mathbb{C}^r \times \mathbb{C}^{m-r}$ and a biholomorphic map ψ of $Q' \times \mathbb{C}^{K-r}$ onto itself such that*

$$F \circ \phi^{-1}(y', y'') = \psi(y', 0) \qquad \text{for } (y', y'') \in Q' \times Q''.$$

In particular, the fibers of F near p are complex submanifolds of dimension $m - r$, and F maps a neighborhood of p onto a closed complex submanifold of dimension r of a neighborhood of $F(p)$.

Proof After permuting the coordinates of the source and of the target, we may assume that the matrix $(\partial F_i/\partial x_j(p))_{i,j\le r}$ is invertible. The map $\phi(x) = (F_1(x), \dots, F_r(x), x_{r+1}, \dots, x_m)$ has invertible derivative at p, so by the holomorphic inverse function theorem (Proposition 7.1.2(1)) it maps a neighborhood of p biholomorphically onto a polydisc $Q' \times Q''$ centered at $\phi(p)$, on which the rank of $G = F \circ \phi^{-1}$ is r. This map has the form $G(y) = (y', G''(y))$ with $y' = (y_1, \dots, y_r)$ and $G'' = (G_{r+1}, \dots, G_K)$. For $i, j > r$, the minor of the Jacobian matrix of G formed by the rows $1, \dots, r, i$ and the columns $1, \dots, r, j$ equals $\partial G_i/\partial y_j$, because the first r rows are $(I, 0)$; since the rank is r, these derivatives vanish. As

Q'' is convex, $G''(y', y'') = G''(y', c'')$, where c'' is the center of Q''. The map $\psi(w', w'') = (w', w'' + G''(w', c''))$ is a biholomorphism of $Q' \times \mathbb{C}^{K-r}$ onto itself, with inverse $(w', w'') \mapsto (w', w'' - G''(w', c''))$, and $G(y', y'') = \psi(y', 0)$. The fibers of G are the slices $\{y'\} \times Q''$, and its image is $\psi(Q' \times \{0\})$, a closed submanifold of $Q' \times \mathbb{C}^{K-r}$. □

Lemma A.2.12 *Let N be a complex manifold with a countable basis.*

(1) *If $F\colon N \to \mathbb{C}^K$ is a holomorphic map whose differential has rank at most k at every point, then $F(N)$ is $\mathcal{H}^s$-null for every $s > 2k$.*
(2) *If an analytic subset Y of an open subset of $\mathbb{C}^K$ is $\mathcal{H}^{2k+2}$-null, then $\dim(Y, y) \le k$ for every $y \in Y$.*
(3) *Let $A \subset N$ be an analytic subset with $\dim(A, x) \le d$ for all $x \in A$, and let $F\colon N \to M$ be a holomorphic map to a complex manifold such that $F(A)$ is an analytic subset of an open subset of M. Then $\dim(F(A), w) \le d$ for every $w \in F(A)$.*

Proof We use the properties of $\mathcal{H}^s$-null sets recalled in Section 8.4: subsets and countable unions of such sets, and their images under Lipschitz maps, are again $\mathcal{H}^s$-null, and an $\mathcal{H}^s$-null subset of $\mathbb{R}^b$ with $s \le b$ is Lebesgue null. Moreover, the image of a compact cube in $\mathbb{R}^r$ under a Lipschitz map is $\mathcal{H}^s$-null for $s > r$, by Corollary A.2.10(3), and a C^1 map is Lipschitz on every compact convex subset of its domain, by the mean value inequality.

(1) We argue by induction on the dimension of N. Since N has at most countably many connected components, each of them open with a countable basis, we may assume that N is connected, of dimension n. If $n = 0$, then $F(N)$ is a point. Let $n \ge 1$, and let $r \le k$ be the maximal rank of $\mathrm{d}F$ on N. The set N_1 of points where the rank is less than r is the common zero set of the minors of size r of the Jacobian matrix in local coordinates, so it is an analytic subset of N different from N. Hence it is nowhere dense, and its germs, being proper subgerms of the irreducible manifold germs of dimension n (Lemma A.2.9), have dimension at most $n - 1$ by Proposition A.2.7. On the open set $N \setminus N_1$ the rank is constantly r.

Let $p \in N \setminus N_1$, and choose ϕ, ψ and $Q' \times Q''$ as in Lemma A.2.11, in a chart around p. Let $C' \subset Q'$ be a compact cube, a product of closed squares, containing the first r coordinates of $\phi(p)$ in its interior, and let $V = \phi^{-1}(\operatorname{Int} C' \times Q'')$. Then $F(V) \subset \psi(C' \times \{0\})$, the image of a compact cube in $\mathbb{R}^{2r}$ under the restriction of ψ, which is Lipschitz there because $C' \times \{0\}$ is a compact convex subset of the convex domain of ψ. Hence $F(V)$ is $\mathcal{H}^s$-null for $s > 2r$, in particular for $s > 2k$. Countably many such sets V cover $N \setminus N_1$, because N has a countable basis. By Corollary A.2.10(2), N_1 is the union of finitely many complex submanifolds of open subsets of N, of dimensions at most $n - 1$; they have countable bases, and the restriction of F to each of them has rank at most k. By the induction hypothesis their images are $\mathcal{H}^s$-null for $s > 2k$, and therefore so is $F(N)$.

(2) Suppose that $\dim(Y, y) \ge k + 1$ for some $y \in Y$, and let Y_1 be a branch of (Y, y) of dimension $e \ge k + 1$. By Corollary A.2.4, Y_1 contains points of the covering locus of a parametrization of Y_1 outside the other branches of (Y, y); near such a

point p, the set Y is a complex submanifold of dimension e, so it contains a complex submanifold P of dimension $k+1$ through p, for instance a coordinate slice. The orthogonal projection Π of $\mathbb{C}^K$ onto the tangent space T_pP, a complex subspace of dimension $k+1$, is 1-Lipschitz, and its restriction to P has invertible derivative at p. By the inverse function theorem, $\Pi(P)$ contains a nonempty open subset of $T_pP \cong \mathbb{R}^{2k+2}$. On the other hand, $\Pi(P) \subset \Pi(Y)$ is $\mathcal{H}^{2k+2}$-null and hence Lebesgue null. This contradiction proves (2).

(3) By Corollary A.2.10(2), A is the union of finitely many complex submanifolds A_l of open subsets of N, of dimensions at most d. Let (V, χ) be a holomorphic chart of M around a point of $F(A)$, with values in $\mathbb{C}^K$. The restriction of $\chi \circ F$ to the open subset $A_l \cap F^{-1}(V)$ of A_l has rank at most $\dim A_l \le d$, so by (1) the set $\chi(F(A) \cap V) = \bigcup_l \chi \circ F(A_l \cap F^{-1}(V))$ is $\mathcal{H}^{2d+2}$-null. Near each of its points it is an analytic subset of an open subset of $\mathbb{C}^K$, so (2) shows that $\dim(F(A), w) \le d$ for $w \in F(A) \cap V$. □

An analytic subset Z of a complex manifold is *irreducible* if it is nonempty and is not the union of two analytic subsets different from Z.

Corollary A.2.13 *Let A be an analytic subset of a complex manifold M. The regular locus A_{reg} is dense in A. The closures of the connected components of A_{reg} are irreducible analytic subsets of A, called the* irreducible components *of A; they form a locally finite family with union A, and none of them is contained in the union of the others. Every locally finite family of irreducible analytic subsets with union A, none of which is contained in another, consists of these components. If M has a countable basis, there are at most countably many irreducible components. The regular locus of an irreducible component Z is connected, and a holomorphic function defined near Z that vanishes on a nonempty relatively open subset of Z_{reg} vanishes on Z.*

Proof Fix $x \in A$, let $A_1, \dots, A_p$ be representatives of the branches of (A, x), and for each i choose a parametrization polydisc P_i of (A_i, x) so small that $A \cap P_i = \bigcup_j A_j \cap P_i$. By Corollary A.2.10(1), $A_{\mathrm{sing}} \cap A_i$ is an analytic subset of A_i whose germ at x differs from (A_i, x). Hence A_{sing} meets the connected covering locus of A_i in a proper analytic subset, and by Lemma A.1.7 and Corollary A.2.4, the covering locus minus A_{sing} is connected and dense in $A_i \cap P_i$. The set $R_i = A_i \cap P_i \cap A_{\mathrm{reg}}$ lies between this connected set and its closure, so it is connected and dense in $A_i \cap P_i$. In particular A_{reg} is dense in A.

Let $\{S_\alpha\}$ be the connected components of the manifold A_{reg}, and $Z_\alpha = \overline{S_\alpha}$. Each R_i above is connected, so it lies in a single component $S_{\alpha(i)}$. Let $U = \bigcap_i P_i$. Then $S_\alpha \cap U$ is contained in the union of the A_i with $\alpha(i) = \alpha$, because points of $A_{\mathrm{reg}} \cap A_i \cap U$ lie in R_i; conversely $A_i \cap U \subset \overline{R_i} \subset Z_{\alpha(i)}$. Hence

$$Z_\alpha \cap U = \bigcup_{i:\, \alpha(i)=\alpha} A_i \cap U.$$

Thus each Z_α is analytic, only finitely many of them meet U, and their union is A. Since A_{reg} is open in A and its connected components are open in it, S_α is open in A.

So every point of S_α has a neighborhood that meets no other S_β, and therefore does not lie in $Z_\beta = \overline{S_\beta}$ for $\beta \neq \alpha$. Thus $S_\alpha \cap Z_\beta = \emptyset$ for $\beta \neq \alpha$, and no Z_α is contained in the union of the others. The same remark shows that $S_\alpha \subset (Z_\alpha)_{\mathrm{reg}}$: near a point of S_α, the set Z_α coincides with the manifold A. Hence $(Z_\alpha)_{\mathrm{reg}}$ lies between the connected set S_α and its closure, and is connected.

If Z_α were the union of two analytic subsets Z', Z'' different from it, then $S_\alpha \cap Z'$ and $S_\alpha \cap Z''$ would be analytic subsets of the connected manifold S_α different from S_α, since otherwise their closure would be Z_α. As proper analytic subsets of a connected manifold, they would be closed and nowhere dense in S_α (Lemma A.1.6), and so would their union, which is S_α. This contradiction shows that Z_α is irreducible.

If M has a countable basis, so does A_{reg}; each S_α is open in A_{reg} and contains a basic open set, and disjoint components contain different ones, so there are at most countably many components. Finally, a holomorphic function vanishing on a nonempty open subset of the connected manifold $(Z_\alpha)_{\mathrm{reg}}$ vanishes on all of it by the identity theorem, and hence on its closure Z_α.

It remains to prove uniqueness. Let $A = \bigcup_\beta Y_\beta$ be a locally finite decomposition into irreducible analytic subsets, none contained in another. A connected manifold is not a locally finite union of closed nowhere dense subsets, since such a union is closed and nowhere dense. Since proper analytic subsets of the connected manifold S_α are nowhere dense (Lemma A.1.6), each S_α lies in some Y_β, and then $Z_\alpha \subset Y_\beta$. Conversely, fix β. By what we have proved, applied to the analytic set Y_β, the set Y_β is the union of its irreducible components, and it has only one, for otherwise it would be the union of one of them and of the union of the others, two analytic subsets different from Y_β. Hence $(Y_\beta)_{\mathrm{reg}}$ is a connected manifold, dense in Y_β and covered by the locally finite family of closed analytic subsets $(Y_\beta)_{\mathrm{reg}} \cap Z_\alpha$; as before, $Y_\beta \subset Z_\alpha$ for some α. Choosing β' with $Z_\alpha \subset Y_{\beta'}$, we get $Y_\beta \subset Y_{\beta'}$, so $\beta' = \beta$ and $Y_\beta = Z_\alpha$. Thus every Y_β is an irreducible component of A; and every component Z_α, being contained in some $Y_\beta = Z_{\alpha'}$, equals it, because $S_\alpha \cap Z_{\alpha'} = \emptyset$ for $\alpha' \neq \alpha$. □

Corollary A.2.14 *Let Z be an irreducible analytic subset of a complex manifold M. Then Z_{reg} is a connected manifold, dense in Z, and:*

(1) *all branches of all germs (Z, x), $x \in Z$, have the dimension of Z_{reg}; in particular, Z is pure-dimensional;*
(2) *every analytic subset $Y \subset Z$ different from Z is nowhere dense in Z, and $\dim(Y, x) < \dim Z$ for every $x \in Y$;*
(3) *if $Z' \supset Z$ is an irreducible analytic subset of M with $\dim Z' = \dim Z$, then $Z' = Z$;*
(4) *if Z is contained in the union of a locally finite family of analytic subsets of M, then it is contained in one of them.*

Proof By the uniqueness in Corollary A.2.13, applied to the family consisting of Z alone, Z is its own only irreducible component; so Z_{reg} is connected and dense in Z, and its dimension is constant. (1) Let B be a branch of (Z, x). By Corollary A.2.4, B contains points of the covering locus of a parametrization of B outside the other

branches of (Z,x). Near such a point, Z coincides with B and is a manifold of dimension $\dim B$, so the point lies in Z_{reg}, and $\dim B = \dim Z_{\mathrm{reg}}$.

(2) The analytic subset $Y \cap Z_{\mathrm{reg}}$ of the connected manifold Z_{reg} has empty interior in Z_{reg}: otherwise it would be all of Z_{reg}, by Lemma A.1.6, and the closed set Y would contain the closure Z of Z_{reg}. Since Z_{reg} is open and dense in Z, it follows that the closed set Y has empty interior in Z. Now let $x \in Y$. Every branch Y_1 of (Y,x) is contained in a branch B of (Z,x), and it differs from B: otherwise Y would contain the points of B near x outside the other branches of (Z,x), which form a nonempty open subset of Z by Corollary A.2.4. By Proposition A.2.7 and (1), $\dim Y_1 < \dim B = \dim Z$.

(3) Let $e = \dim Z = \dim Z'$. By Corollary A.2.10(1) and by (1), Z'_{sing} has dimension less than e at each of its points, so by (2), applied to $Z'_{\mathrm{sing}} \cap Z \subset Z$, there is a point $p \in Z_{\mathrm{reg}} \setminus Z'_{\mathrm{sing}}$. Near p, the set Z is a submanifold of dimension e of the manifold Z' of dimension e; since the germ of a manifold is irreducible (Lemma A.2.9) and proper subgerms of irreducible germs have smaller dimension (Proposition A.2.7), Z and Z' coincide near p. Thus the analytic subset Z of Z' has interior points in Z', and $Z = Z'$ by (2).

(4) Let $Z \subset \bigcup_\beta Y_\beta$ with (Y_β) locally finite. The sets $Z_{\mathrm{reg}} \cap Y_\beta$ are closed analytic subsets of the connected manifold Z_{reg} with union Z_{reg}. A locally finite union of closed nowhere dense sets is closed and nowhere dense, so one of them has interior points and is therefore all of Z_{reg}. Then $Z \subset Y_\beta$, because Y_β is closed and Z_{reg} is dense in Z. □

Corollary A.2.15 *Let A be an analytic subset of codimension at least two of a complex manifold M. Then every plurisubharmonic function u on $M \setminus A$ is locally bounded above near A and extends uniquely to a plurisubharmonic function on M, whose value at $a \in A$ is $\overline{\lim}_{M\setminus A \ni y \to a} u(y)$.*

Proof The assertion is local, and extensions defined on overlapping charts agree, since their values on A are given by the same formula. Near a point where M has dimension at most one, the codimension condition forces $A = \varnothing$. Otherwise, let Ω be a connected coordinate chart of dimension $n \geq 2$. Then $\dim A \leq n-2$, and Corollary A.2.10(3) shows that $A \cap \Omega$ is locally $\mathcal{H}^{2n-3}$-null, since $2n-3 > 2(n-2)$. By Theorem 8.4.3, u extends uniquely to a plurisubharmonic function on Ω with the stated values on A; being upper semicontinuous, the extension is locally bounded above. □

For an irreducible analytic subset D of codimension one of a complex manifold M, we write $[D]$ for the current of integration over the complex hypersurface D_{reg} of $M \setminus D_{\mathrm{sing}}$ (Example 7.2.7), with coefficient measures extended by zero across D_{sing}. The following lemma shows that these coefficient measures have locally finite mass near D_{sing}, so that $[D]$ is a well-defined current on M, and that $[D]$ is closed and positive.

Lemma A.2.16 *Let D be an irreducible analytic subset of codimension one of a complex manifold M of dimension n. Every point $x \in D$ has an open neighborhood V*

with a holomorphic function f on V, a local equation *of D, such that $D \cap V = f^{-1}(0)$, the germ of f generates $\mathcal{I}_{D,y}$ for every $y \in V$, and f vanishes to order one along $D_{\mathrm{reg}} \cap V$: near each point of $D_{\mathrm{reg}} \cap V$ it is a unit times a holomorphic coordinate τ with $D = \{\tau = 0\}$. On V we have $[D] = \mathrm{dd}^{\mathrm{c}} \log |f|^2$. In particular, $[D]$ is a closed positive $(1,1)$-current on M with measure coefficients.*

Proof By Corollary A.2.14, D_{reg} is a connected manifold, dense in D, and all branches of D at all points have the dimension $n-1$ of D_{reg}. Let $x \in D$ and let $D_1, \dots, D_k$ be representatives of the branches of (D, x). We construct a generator f_i of the ideal of each D_i near x, and show that $f = f_1 \cdots f_k$ has the required properties.

Step 1: an equation of one branch. Fix i, and take a parametrization of (D_i, x) as in Theorem A.2.3, in coordinates centered at x. Its base dimension is $d = n-1$, by Proposition A.2.7, so $z'' = z_n$ is a single coordinate. For $z' \in \Delta' \setminus \{\delta = 0\}$, the q points of D_i over z' are distinct, and therefore so are their coordinates z_n; put $P(z', T) = \prod_{z \in \pi^{-1}(z')} (T - z_n)$. The coefficients of P are, up to sign, the elementary symmetric functions of these coordinates; they are holomorphic, because the sheets are graphs of holomorphic maps, and bounded, because Δ'' is bounded, so they extend holomorphically to Δ' by Lemma A.1.7. As $z' \to 0$, the points of D_i over z' tend to 0, because π is proper and $\pi^{-1}(0) = \{0\}$; hence the extended coefficients vanish at 0, and P is a Weierstrass polynomial of degree q in z_n. The function $P(z', z_n)$ vanishes on the covering locus, hence on $D_i \cap (\Delta' \times \Delta'')$, by density and continuity.

The germ of $P(z', z_n)$ generates the prime ideal $I_{D_i,x}$. Indeed, let $g \in I_{D_i,x}$, and let R be the remainder of the division of g by P (Proposition A.1.1), a polynomial in z_n of degree less than q with coefficients holomorphic near 0 in $\mathbb{C}^{n-1}$; it vanishes on D_i near 0, since g and $P(z', z_n)$ do. For z' near 0 with $\delta(z') \neq 0$, the q points of D_i over z' are close to 0, so $R(z', \cdot)$ vanishes at their q distinct coordinates z_n and is zero. The coefficients of R thus vanish on a dense subset of a neighborhood of 0, and $R = 0$.

Let f_i be a holomorphic function near x with the germ of $P(z', z_n)$. The sheaf $\mathcal{I}_{D_i}$ has finitely many generators near x by Theorem A.2.8, and each of them is a multiple of f_i at x, hence on a neighborhood; so f_i generates $\mathcal{I}_{D_i,y}$ for y near x. In particular, at every point near x outside D_i the ideal of D_i is the unit ideal, so f_i does not vanish there, and $f_i^{-1}(0) = D_i$ near x.

Step 2: the product $f = f_1 \cdots f_k$. Let $y \in D_{\mathrm{reg}}$ be close to x. By Corollary A.2.10(1), y lies on only one D_i, and $D = D_i = \{\tau = 0\}$ near y for a local coordinate τ; then f_i is a unit times τ, since both generate $\mathcal{I}_{D_i,y}$ (Lemma A.2.9), and $f_j(y) \neq 0$ for $j \neq i$, by Step 1. Thus f vanishes to order one along D_{reg} near x, and $f^{-1}(0) = D$ near x. Choose a neighborhood V of x on which all this holds and on which, in addition, $D_i \setminus \bigcup_{j \neq i} D_j$ is dense in D_i for every i; this is possible by Corollary A.2.4, since the branches are irredundant. Let $y \in V$ and $g \in \mathcal{I}_{D,y}$; we show by induction on l that $g \in (f_1 \cdots f_l)_y$. For $l = 0$ there is nothing to prove. If $g = f_1 \cdots f_l h$ with $l < k$, then h vanishes near y on $D_{l+1} \setminus (D_1 \cup \cdots \cup D_l)$, where the f_i, $i \leq l$, have no zeros; this set is dense in D_{l+1}, so h vanishes on D_{l+1} near y, and $h \in (f_{l+1})_y$. Hence f generates $\mathcal{I}_{D,y}$.

Finally, in the notation of Section 7.4, the regular part of the zero set of f is $D_{\mathrm{reg}} \cap V$, with multiplicity one. By Theorem 7.4.5, $[D] = \mathrm{dd}^{\mathrm{c}} \log |f|^2$ on V; hence $[D]$ is closed and positive, with measure coefficients of locally finite mass. □

Theorem A.2.17 *Let A be a nowhere dense analytic subset of a complex manifold M, and let R be a closed real $(1,1)$-current on M with measure coefficients and support in A. Then*

$$R = \sum_D c_D [D],$$

where D ranges over the irreducible components of A of codimension one, the real coefficients c_D are uniquely determined, and the sum is locally finite. If R is positive, then every $c_D \geq 0$. In particular $R = 0$ if A has codimension at least two.

Proof Working on each connected component of M separately, we may assume that M is connected, of dimension n.

Step 1: supports of high codimension. Suppose first that, near a point, the support of R lies in a real submanifold $\{g_1 = \cdots = g_k = 0\}$ with independent differentials $\mathrm{d}g_1, \ldots, \mathrm{d}g_k$, $k \geq 3$. Since R has measure coefficients and g_j vanishes on its support, $g_j R = 0$; since R is closed, $\mathrm{d}g_j \wedge R = \mathrm{d}(g_j R) = 0$. Take real coordinates x whose first three are g_1, g_2, g_3. For $l \leq 3$ and $l \notin \{a, b\}$, the coefficient of $\mathrm{d}x_l \wedge \mathrm{d}x_a \wedge \mathrm{d}x_b$ in $\mathrm{d}x_l \wedge R$ is, up to sign, the coefficient of $\mathrm{d}x_a \wedge \mathrm{d}x_b$ in R, so this coefficient vanishes. Since no pair $\{a, b\}$ contains all three indices $1, 2, 3$, every coefficient of R vanishes, and $R = 0$ near the point. Now let A have codimension at least two. Near a point of A_{reg}, the set A is a complex submanifold of real codimension at least four, so R vanishes there, and the support of R lies in A_{sing}. This is analytic of smaller dimension near each point by Corollary A.2.10(1), and again of codimension at least two; repeating the argument, we find after finitely many steps near each point that $R = 0$. This proves the last assertion.

Step 2: regular points of codimension one. Let $y \in A_{\mathrm{reg}}$ be a point where A is a complex hypersurface. Choose holomorphic coordinates (ζ, w) near y with $A = \{\zeta = 0\}$, write $\zeta = \xi + \mathrm{i}\eta$, and let $t \in \mathbb{R}^{2n-2}$ be the real coordinates of w. The identities $\mathrm{d}\xi \wedge R = \mathrm{d}\eta \wedge R = 0$ of Step 1 force $R = \mu\, \mathrm{d}\xi \wedge \mathrm{d}\eta$ for a real measure μ supported on $\{\xi = \eta = 0\}$, so $\mu = \delta_0(\xi, \eta) \otimes \sigma(t)$ for a real measure σ. Since R is closed, $\mathrm{d}R = \sum_l (\partial\mu/\partial t_l)\, \mathrm{d}t_l \wedge \mathrm{d}\xi \wedge \mathrm{d}\eta$ vanishes, the derivatives in ξ and η disappearing after the exterior product with $\mathrm{d}\xi \wedge \mathrm{d}\eta$; so $\partial\mu/\partial t_l = 0$ as distributions. Testing this against functions $\chi(\xi, \eta)\psi(t)$ with $\chi = 1$ near 0, we find that all distributional derivatives $\partial\sigma/\partial t_l$ vanish. On a smaller ball in the t-variables, the convolutions of σ with a smoothing kernel therefore have zero gradient and are constant; integrating them against a fixed test function of integral one, we see that these constants converge as the kernel shrinks, and that σ is a constant multiple c of Lebesgue measure. Hence $R = c[\{\zeta = 0\}]$ near y, with c real. (If $n = 1$, σ is a measure on a point and the same conclusion holds.)

Step 3: conclusion. Let $D = Z_\alpha$ be an irreducible component of codimension one, the closure of the connected component S_α of A_{reg}. By Step 2, near each point of S_α we have $R = c[D]$ for a real constant c, and this constant is locally constant on

the connected manifold S_α; we call it c_D. By Corollary A.2.13 the sum $\sum_D c_D[D]$ is locally finite, and by Lemma A.2.16 it is a closed real current with measure coefficients. The difference $R' = R - \sum_D c_D[D]$ vanishes near every point of A_{reg}: near a point of S_α with Z_α of codimension one by Step 2, since the other components do not meet S_α; and near a point of A_{reg} where A has codimension at least two by Step 1, since no component of codimension one passes through such a point. Thus the support of R' lies in A_{sing}.

We show that A_{sing} has codimension at least two; then $R' = 0$ by Step 1. A branch of A of dimension n at a point y would be a subgerm of the germ (M, y), which is irreducible of dimension n by Lemma A.2.9, hence equal to it by Proposition A.2.7; then A would contain a neighborhood of y, which is impossible because A is nowhere dense. So the branches of A have dimension at most $n - 1$, and A_{sing}, which has smaller dimension than each branch (Corollary A.2.10(1)), has codimension at least two. The coefficients are unique, because near a point of S_α the current R determines c_D; and if $R \geq 0$, then $c_D \geq 0$, because $[D]$ is positive and nonzero near such a point. □

A.3 Reduced spaces and normalization

The main result of this section is the existence and uniqueness of the normalization of a reduced complex space (Theorem A.3.6), which we construct locally from weakly holomorphic functions.

A *reduced complex space* is a Hausdorff topological space X with a sheaf $\mathcal{O}_X$ of continuous complex-valued functions, its structure sheaf, such that every point has an open neighborhood U and a homeomorphism ϕ of U onto an analytic subset A of a domain $\Omega \subset \mathbb{C}^N$ with the following property: a function g on an open set $V \subset U$ belongs to $\mathcal{O}_X(V)$ if and only if $g \circ \phi^{-1}$ is, near each point of $\phi(V)$, the restriction of a holomorphic function on an open subset of Ω. In other words, ϕ identifies $\mathcal{O}_X|_U$ with $\mathcal{O}_A = (\mathcal{O}_\Omega/\mathcal{I}_A)|_A$. We call (U, ϕ) a *local model*, or local embedding, of X; holomorphic functions on X extend locally to the ambient domains. As stated in the Conventions, all complex spaces are reduced, paracompact and Hausdorff; in this appendix we also assume that they have a countable basis of open sets. A compact space with such local models has a countable basis automatically, since finitely many local models cover it.

A continuous map $F\colon Y \to X$ of complex spaces is *holomorphic* if $g \circ F \in \mathcal{O}_Y$ for every local section g of $\mathcal{O}_X$, that is, if in local models of X the components of F belong to $\mathcal{O}_Y$. By the definition of $\mathcal{O}_Y$, these components extend near each point of Y to holomorphic functions on an open subset of the ambient space of a local model of Y; consequently, compositions of holomorphic maps are holomorphic. An isomorphism is a holomorphic bijection whose inverse is holomorphic. Two local models at the same point differ by an isomorphism of analytic sets which, together with its inverse, is locally the restriction of holomorphic maps between the ambient domains. Such an isomorphism maps analytic subsets to analytic subsets, and it maps

submanifolds to submanifolds, because a holomorphic map defined on a manifold that has a holomorphic left inverse is an injective immersion and a homeomorphism onto its image, so that its image is a submanifold by Lemma A.2.11. Hence regular and singular points, local branches and dimension are well defined on X, independently of the local model. An *analytic subset* of X is a closed subset that is analytic in every local model; with the restricted structure sheaf it is again a complex space.

The structure sheaf is coherent as a module over itself. Indeed, in a local model let $f_1, \dots, f_q$ be sections of $\mathcal{O}_X$ with holomorphic lifts $F_1, \dots, F_q$, and let $g_1, \dots, g_N$ generate $\mathcal{I}_A$ near a point, by Theorem A.2.8; the relations of the f_i over $\mathcal{O}_X$ are the images of the projections to the first q coordinates of the relations of $F_1, \dots, F_q, g_1, \dots, g_N$ over $\mathcal{O}_\Omega$, which are locally finitely generated by Theorem A.1.5. Thus Lemma A.1.4 applies to $\mathcal{O}_X$-modules. Each local ring $\mathcal{O}_{X,x}$ is a quotient of $\mathcal{O}_N$, hence Noetherian. The zero set $V(\mathfrak{a})$ of a coherent ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$ is an analytic subset; conversely, the ideal sheaf $\mathcal{I}_Y \subset \mathcal{O}_X$ of functions vanishing on an analytic subset $Y \subset X$ is locally generated by the images of generators of the ideal sheaf of Y in an ambient domain, so it is coherent by Theorem A.2.8 and Lemma A.1.4.

A holomorphic map $F\colon Y \to X$ is *proper* if inverse images of compact sets are compact, and *finite* if it is proper with finite fibers. It is *bimeromorphic* if there is a nowhere dense analytic subset $E \subset X$ such that $F^{-1}(E)$ is nowhere dense in Y and F restricts to an isomorphism $Y \setminus F^{-1}(E) \to X \setminus E$. A proper map is closed: if $C \subset Y$ is closed and $F(y_j) \to x$ with $y_j \in C$, the points y_j eventually lie in the compact inverse image of a compact neighborhood of x, so a subsequence converges to some $y \in C$, and $F(y) = x$.

Lemma A.3.1 *Let $F\colon Y \to X$ be a bimeromorphic holomorphic map of complex spaces. Then $F^{-1}(B)$ is nowhere dense in Y for every nowhere dense analytic subset $B \subset X$. Consequently, the set E in the definition may be enlarged by any nowhere dense analytic subset of X; if $U \subset X$ is the complement of a nowhere dense analytic subset, then $F^{-1}(U)$ is dense in Y; and every nonempty open subset of Y contains a nonempty open set that F maps isomorphically onto an open subset of X.*

Proof Let E be as in the definition, and let $G \subset F^{-1}(B)$ be open in Y. Then $G \setminus F^{-1}(E)$ is an open subset of $Y \setminus F^{-1}(E)$ contained in $F^{-1}(B \setminus E)$, which corresponds under the isomorphism $Y \setminus F^{-1}(E) \to X \setminus E$ to the nowhere dense subset $B \setminus E$ of $X \setminus E$; so $G \setminus F^{-1}(E)$ is empty, and G, an open subset of the nowhere dense set $F^{-1}(E)$, is empty as well. In particular F is also an isomorphism over $X \setminus (E \cup B)$, and $F^{-1}(E \cup B) = F^{-1}(E) \cup F^{-1}(B)$ is nowhere dense, as the union of two closed nowhere dense sets; so E may be replaced by $E \cup B$. If $U = X \setminus B$, then $F^{-1}(U)$ is the complement of the closed nowhere dense set $F^{-1}(B)$, hence dense. Finally, a nonempty open set $G \subset Y$ meets the dense open set $Y \setminus F^{-1}(E)$, and F maps the nonempty open set $G \setminus F^{-1}(E)$ isomorphically onto an open subset of $X \setminus E$. □

A *weakly holomorphic* function on an open set $U \subset X$ is a holomorphic function on $U \cap X_{\mathrm{reg}}$ that is locally bounded near $U \cap X_{\mathrm{sing}}$: every point of U has a neighborhood V such that the function is bounded on $V \cap X_{\mathrm{reg}}$. These functions form a sheaf of

rings $\widetilde{\mathcal{O}}_X$, which contains $\mathcal{O}_X$ because X_{reg} is dense in X (Corollary A.2.13). They can be computed branch by branch.

Lemma A.3.2 *Let x be a point of a complex space X, and let $X_1, \dots, X_p$ be sufficiently small representatives of the branches of (X, x) in a local model, on an open neighborhood U of x with $X \cap U = X_1 \cup \cdots \cup X_p$. Let $V \subset U$ be open. Then for each i the restriction of a weakly holomorphic function f on $X \cap V$ to $X_i \cap X_{\mathrm{reg}} \cap V$ extends uniquely to a weakly holomorphic function f_i on $X_i \cap V$, and $f \mapsto (f_1, \dots, f_p)$ is a bijection onto the p-tuples of weakly holomorphic functions on the sets $X_i \cap V$. In particular,*

$$\widetilde{\mathcal{O}}_{X,x} = \bigoplus_{i=1}^{p} \widetilde{\mathcal{O}}_{X_i,x}.$$

Proof By Corollary A.2.10(1), $X_{\mathrm{reg}} \cap U$ is the disjoint union of the sets $X_i \cap X_{\mathrm{reg}} = (X_i)_{\mathrm{reg}} \setminus \bigcup_{j \neq i} X_j$, each of which is open in X_{reg}. The analytic subset $(X_i)_{\mathrm{reg}} \cap \bigcup_{j \neq i} X_j$ of the manifold $(X_i)_{\mathrm{reg}}$ is nowhere dense in it, by Corollary A.2.4, applied to (X_i, x) and its subgerm $(X_i \cap \bigcup_{j \neq i} X_j, x)$, which differs from (X_i, x) because the branches are irredundant. Hence a bounded holomorphic function on $X_i \cap X_{\mathrm{reg}} \cap V$ extends uniquely across this analytic subset to a holomorphic function on $(X_i)_{\mathrm{reg}} \cap V$, by Lemma A.1.7 applied on connected coordinate balls of $(X_i)_{\mathrm{reg}}$, and the extension is bounded by the same bound. Applied near each point of $X_i \cap V$, this shows that the restriction of f has a unique extension f_i, which is weakly holomorphic on $X_i \cap V$. Conversely, if f_i is weakly holomorphic on $X_i \cap V$ for each i, the function equal to f_i on $X_i \cap X_{\mathrm{reg}} \cap V$ is holomorphic and locally bounded on $X_{\mathrm{reg}} \cap V$, hence weakly holomorphic on $X \cap V$; by the uniqueness of the extensions, the two constructions are inverse to each other. Passing to germs at x gives the last assertion. □

The total ring of fractions of the reduced local ring $\mathcal{O}_{X,x}$ is obtained by inverting its nonzerodivisors. These are exactly the germs that are not identically zero on any branch of (X, x). Indeed, if h is such a germ and $hg = 0$, then g vanishes on the complement of $\{h = 0\}$ in each branch, which is dense by Corollary A.2.4; so $g = 0$. If instead h vanishes identically on the branch X_1, choose for each $j \neq 1$ a germ g_j vanishing on X_j but not on X_1, which exists because $(X_1, x) \not\subset (X_j, x)$; the product $v = \prod_{j \neq 1} g_j$ does not vanish identically on X_1, because $I_{X_1,x}$ is prime, and $hv = 0$.

Proposition A.3.3 *Every point of a complex space X has a neighborhood U with a holomorphic function h on U whose germ at each $y \in U$ does not vanish identically on any branch of (X, y), such that*

$$h_y \widetilde{\mathcal{O}}_{X,y} \subset \mathcal{O}_{X,y} \qquad (y \in U).$$

For every $x \in X$, the weakly holomorphic germs at x are precisely the elements of the total ring of fractions of $\mathcal{O}_{X,x}$ that are integral over $\mathcal{O}_{X,x}$, and they form a finitely generated $\mathcal{O}_{X,x}$-module. If (X, x) is irreducible, every weakly holomorphic function near x has a finite limit at x along X_{reg}.

Proof *Step 1: an irreducible germ.* Let $(X,0)$ be irreducible, with a local parametrization as in Theorem A.2.3, its degree q, its primitive linear form u, its Weierstrass polynomial W and $\delta = \mathrm{Disc}(W)$. We show that $\delta_y \widetilde{\mathcal{O}}_{X,y} \subset \mathcal{O}_{X,y}$ for every y in the parametrization polydisc. Let f be weakly holomorphic near y and bounded on $V \cap X_{\mathrm{reg}}$ for a neighborhood V of y. Let $y = y^{(1)}, y^{(2)}, \dots, y^{(s)}$ be the points of X over y', choose disjoint neighborhoods $V_1 \subset V, V_2, \dots, V_s$ of them, and, using properness, a connected neighborhood N' of y' such that all points of X over N' lie in $\bigcup_l V_l$. Define F on $X_{\mathrm{reg}} \cap \pi^{-1}(N')$ by $F = f$ on V_1 and $F = 0$ on the other V_l.

For $z' \in N' \setminus \{\delta = 0\}$, let $z^{(1)}, \dots, z^{(q)}$ be the points of X over z'; the values $u(z^{(i)})$ are the distinct roots of $W(z', \cdot)$, by Theorem A.2.3, so there is a unique polynomial $\sum_{j<q} b_j(z')T^j$ taking the value $F(z^{(i)})$ at $T = u(z^{(i)})$ for every i. By Cramer's rule $\delta b_j = D \cdot D_j$, where D is the Vandermonde determinant of the $u(z^{(i)})$, so that $D^2 = \delta$ by (A.3), and D_j is obtained by replacing its jth column with the values $F(z^{(i)})$. The products $D \cdot D_j$ do not depend on the order of the points; they are holomorphic, because the points depend holomorphically on z' locally, and bounded, because u and F are bounded. By Lemma A.1.7 they extend holomorphically to N'. The holomorphic function $H(z) = \sum_j (\delta b_j)(z')u(z'')^j$ on $N' \times \Delta''$ equals δF on the covering locus over N', hence δf on $X_{\mathrm{reg}} \cap V_1 \cap \pi^{-1}(N')$, by density of the covering locus and continuity. Thus δf extends to the holomorphic germ of $H|_X$ at y.

Step 2: several branches. Fix $x \in X$, and let $X_1, \dots, X_p$ be sufficiently small representatives of the branches of (X,x), and let δ_j be a denominator for X_j as in Step 1, lifted to a holomorphic function of the ambient coordinates. Adding to δ_j a linear combination $\sum_l a_l e_{jl}$ of local generators e_{jl} of the ideal of X_j does not change it on X_j. For each $k \neq j$ some e_{jl} does not vanish identically on X_k, so the coefficients a for which $\delta_j + \sum_l a_l e_{jl}$ vanishes identically on X_k form a proper affine subspace or the empty set. Choosing a outside these finitely many subspaces, we obtain $\widetilde{\delta}_j$, equal to δ_j on X_j and not identically zero on any X_k. Then $\eta = \prod_j \widetilde{\delta}_j$ is not identically zero on any X_k, and it is a denominator for each X_j separately: if f is weakly holomorphic on X near a point y, the restriction of f to $X_j \cap X_{\mathrm{reg}}$ extends to a weakly holomorphic function f_j on X_j near y, by Lemma A.3.2, so by Step 1 $\eta f_j = g_j|_{X_j}$ for some holomorphic g_j near y.

For each j choose a product v_j of local equations of the branches X_k, $k \neq j$, that does not vanish identically on X_j, and put $h = \eta \sum_j v_j$. On X_j we have $h = \eta v_j$, which is not identically zero, and $hf = \sum_j v_j g_j$ holds on every $X_j \cap X_{\mathrm{reg}}$, hence on X_{reg} near y. Thus $h_y \widetilde{\mathcal{O}}_{X,y} \subset \mathcal{O}_{X,y}$ for all y near x. Finally, $\{h = 0\} \cap X_j$ has smaller dimension than X_j at x, hence at all points near x, by upper semicontinuity; every branch of (X,y) is a branch of some (X_j,y) and has dimension $\dim X_j$, by purity. So, after shrinking U, the germ of h at every $y \in U$ is not identically zero on any branch of (X,y).

Step 3: integrality, finiteness and limits. The map $f \mapsto (hf)/h$ embeds $\widetilde{\mathcal{O}}_{X,x}$ into the total ring of fractions of $\mathcal{O}_{X,x}$; it does not depend on the choice of h, and it is injective because $X_{\mathrm{reg}} \setminus \{h = 0\}$ is dense in X_{reg}. Multiplication by h identifies $\widetilde{\mathcal{O}}_{X,x}$ with an ideal of the Noetherian ring $\mathcal{O}_{X,x}$, so it is a finitely generated $\mathcal{O}_{X,x}$-module.

Every weakly holomorphic germ is integral. On an irreducible germ, in the notation of Step 1, the elementary symmetric functions of the values of f at the q points of X over $z' \in \Delta' \setminus \{\delta = 0\}$ are bounded holomorphic functions, which extend to Δ' by Lemma A.1.7. They are the coefficients of a monic polynomial $P(z', T)$ with $P(z', f) = 0$ on the covering locus, hence on $X_{\rm reg}$, and these coefficients belong to $\mathcal{O}_d \subset \mathcal{O}_{X,x}$. For several branches, the product of the polynomials of the branches, with coefficients lifted to holomorphic functions, is a monic equation for f. Conversely, let a/b be an element of the total ring of fractions, with b a nonzerodivisor, that is integral over $\mathcal{O}_{X,x}$. On $X_{\rm reg} \setminus \{b = 0\}$ near x the function a/b satisfies a monic equation with bounded coefficients, so it is bounded by (A.1). The set $\{b = 0\}$ meets each connected open subset of the manifold $X_{\rm reg}$ in a nowhere dense analytic subset, so by Lemma A.1.7 a/b extends to a holomorphic function on $X_{\rm reg}$ near x, which is weakly holomorphic and is mapped to a/b.

Let (X, x) be irreducible and f weakly holomorphic near x, satisfying the monic equation $P(z', f) = 0$ just constructed. As Q runs through the parametrization polydiscs around x, the sets $X_{\rm reg} \cap Q$ are the traces on $X_{\rm reg}$ of a basis of neighborhoods of x, and each of them is connected, because it lies between the connected covering locus and its closure. The cluster set of f at x is the intersection of the decreasing compact connected nonempty sets $\overline{f(X_{\rm reg} \cap Q)}$, hence it is nonempty, compact and connected. If w is a cluster value, then $w = \lim f(z_k)$ for points $z_k \to x$ of $X_{\rm reg}$, and $P(x', w) = \lim P(z'_k, f(z_k)) = 0$; so the cluster set is contained in the finite set of roots of $P(x', \cdot)$. Being connected, it is a single point, and f has a finite limit at x.□

A point $x \in X$ is *normal* if (X, x) is irreducible and $\mathcal{O}_{X,x}$ is integrally closed in its field of fractions. By Proposition A.3.3, this means that $\widetilde{\mathcal{O}}_{X,x} = \mathcal{O}_{X,x}$. Conversely, this equality forces (X, x) to be irreducible: otherwise the weakly holomorphic function equal to 1 on one branch and to 0 on the others would be holomorphic, hence continuous, at x, which is impossible because it takes both values at points arbitrarily close to x. Thus x is normal if and only if $\widetilde{\mathcal{O}}_{X,x} = \mathcal{O}_{X,x}$. The space X is *normal* if all its points are normal. Regular points are normal by Lemma A.1.7.

Lemma A.3.4 *The set of nonnormal points of a complex space X is an analytic subset of X. If X is normal and $E \subset X$ is a nowhere dense analytic subset, every holomorphic function on $X \setminus E$ that is locally bounded near E extends uniquely to a holomorphic function on X.*

Proof We show that the set of nonnormal points is the support of a coherent sheaf. We first record two facts about coherent sheaves. If $\mathcal{F}$ and $\mathcal{G}$ are coherent $\mathcal{O}_X$-modules, the sheaf $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ of module homomorphisms is coherent, and its stalk at x is $\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$. Indeed, $\mathcal{F}$ has local presentations $\mathcal{O}_X^a \to \mathcal{O}_X^b \to \mathcal{F} \to 0$, and $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ is the kernel of the induced morphism $\mathcal{G}^b \to \mathcal{G}^a$; we apply Lemma A.1.4.

Next, the support of a coherent module $\mathcal{F}$ is analytic. In a local presentation as above, given by a $b \times a$ matrix Φ of holomorphic functions, $\mathcal{F}_x = 0$ exactly when the matrix $\Phi(x)$ has rank b. If $\Phi(x)$ has rank b, then $\mathcal{F}_x = \mathfrak{m}_x \mathcal{F}_x$, where $\mathfrak{m}_x$ is the maximal ideal, and the following determinant argument gives $\mathcal{F}_x = 0$: writing each

of finitely many generators of $\mathcal{F}_x$ as a combination of them with coefficients in $\mathfrak{m}_x$, with coefficient matrix C, the adjugate identity shows that $\det(I - C)$ annihilates the generators, and this determinant is a unit. Conversely, if $\mathcal{F}_x = 0$, then Φ maps $\mathcal{O}_{X,x}^a$ onto $\mathcal{O}_{X,x}^b$, and reducing modulo $\mathfrak{m}_x$ shows that $\Phi(x)$ has rank b. Hence the support of $\mathcal{F}$ is the common zero set of the minors of size b of Φ.

Let h be a denominator as in Proposition A.3.3 on an open set U, let $\mathcal{I}$ be the coherent ideal sheaf of $\{h = 0\}$ in U, and put $\mathcal{F} = \mathcal{H}om(\mathcal{I}, \mathcal{I})$. Multiplication embeds $\mathcal{O}_U$ in $\mathcal{F}$, because $\mathcal{I}$ contains the nonzerodivisor h. Every $\varphi \in \mathcal{F}_y$ is multiplication by an element of the total ring of fractions: if $a = \varphi(h)$, then $h\varphi(g) = \varphi(hg) = ag$ for $g \in \mathcal{I}_y$, so φ is multiplication by a/h. Let $g_1, \dots, g_N$ generate $\mathcal{I}_y$, and write $(a/h)g_i = \varphi(g_i) = \sum_j c_{ij} g_j$ with $c_{ij} \in \mathcal{O}_{X,y}$. Multiplying the matrix $(a/h)I - (c_{ij})$ by its adjugate, we see that its determinant, a monic polynomial in a/h with coefficients in $\mathcal{O}_{X,y}$, annihilates every g_i, hence $\mathcal{I}_y$ and in particular h; since h is invertible in the total ring of fractions, this polynomial vanishes, and a/h is integral over $\mathcal{O}_{X,y}$. Hence, by Proposition A.3.3,

$$\mathcal{O}_U \subset \mathcal{F} \subset \widetilde{\mathcal{O}}_U.$$

We claim that the support of the coherent sheaf $\mathcal{F}/\mathcal{O}_U$ is the set of nonnormal points of U; this proves the first assertion. At a normal point $\widetilde{\mathcal{O}}_{X,y} = \mathcal{O}_{X,y}$, so $\mathcal{F}_y = \mathcal{O}_{X,y}$. Let y be a nonnormal point, so that $\widetilde{\mathcal{O}}_{X,y} \neq \mathcal{O}_{X,y}$. By the analytic Nullstellensatz (Corollary A.2.6), applied in a local model, every element of $\mathcal{I}_y$ has a power in $h\mathcal{O}_{X,y}$. Let $g_1, \dots, g_N$ generate $\mathcal{I}_y$, and choose e with $g_i^e \in h\mathcal{O}_{X,y}$ for all i. Every monomial of degree $\ell = N(e-1)+1$ in the g_i contains some g_i at least e times, so $\mathcal{I}_y^\ell \subset h\mathcal{O}_{X,y}$, and therefore $\mathcal{I}_y^\ell \widetilde{\mathcal{O}}_{X,y} \subset h\widetilde{\mathcal{O}}_{X,y} \subset \mathcal{O}_{X,y}$.

Let k be the least integer with $\mathcal{I}_y^k \widetilde{\mathcal{O}}_{X,y} \subset \mathcal{O}_{X,y}$; then $k \geq 1$, and we may choose $w \in \mathcal{I}_y^{k-1} \widetilde{\mathcal{O}}_{X,y}$ with $w \notin \mathcal{O}_{X,y}$. Then $w\mathcal{I}_y \subset \mathcal{O}_{X,y}$, and every product wg, $g \in \mathcal{I}_y$, vanishes on $\{h = 0\}$: at regular points because g vanishes there and w is holomorphic, and at singular points because w is locally bounded, g tends to zero, and wg is continuous. Thus $w\mathcal{I}_y \subset \mathcal{I}_y$, so multiplication by w belongs to $\mathcal{F}_y$; it is not multiplication by an element of $\mathcal{O}_{X,y}$, since $w \notin \mathcal{O}_{X,y}$ and h is a nonzerodivisor. This proves the claim.

For the last assertion, let f be a holomorphic function on $X \setminus E$ that is locally bounded near E. Since E is nowhere dense, an extension is unique. The set $E \cap X_{\mathrm{reg}}$ meets every connected open subset of the manifold X_{reg} in a nowhere dense analytic subset, so f extends holomorphically to X_{reg} by Lemma A.1.7. The extension is locally bounded near X_{sing}, because on a neighborhood of each point it is bounded by a bound of f on the dense subset $X_{\mathrm{reg}} \setminus E$; hence it is weakly holomorphic, and therefore holomorphic, because X is normal. It agrees with f on $X \setminus E$, since both functions are continuous there and agree on the dense subset $X_{\mathrm{reg}} \setminus E$. □

Normalizations are first constructed locally, as closures of graphs of weakly holomorphic functions.

Lemma A.3.5 *Let X be an analytic subset of a domain $\Omega \subset \mathbb{C}^N$, with coordinates z, let $x \in X$, and let $X_1, \dots, X_p$ be sufficiently small representatives of the branches*

of (X,x). There is an open neighborhood $W \subset \Omega$ of x with the following properties. For each i there are functions $f_{i1},\dots,f_{im_i}$, weakly holomorphic on $X_i \cap W$, and monic polynomials $P_{ij}(z,T)$ in T with coefficients holomorphic on W, such that $P_{ij}(z,f_{ij}) = 0$ on $(X_i)_{\mathrm{reg}} \cap W$. The closure Y_i in $W \times \mathbb{C}^{m_i}$ of the graph of $(f_{i1},\dots,f_{im_i})$ over $(X_i)_{\mathrm{reg}} \cap W$ is a normal analytic subset of $W \times \mathbb{C}^{m_i}$, on which $P_{ij}(z,w_j) = 0$ for all j, and the projection $\nu_i \colon Y_i \to X_i \cap W$, $(z,w) \mapsto z$, is finite and bimeromorphic, is an isomorphism over $(X_i)_{\mathrm{reg}} \cap W$, and has a single point over x. The map from the disjoint union of $Y_1,\dots,Y_p$ to $X \cap W$ given by the ν_i is finite and bimeromorphic, is an isomorphism over $X_{\mathrm{reg}} \cap W$, and has p points over x. All these properties persist when W is replaced by a smaller open neighborhood of x and each Y_i by its part over it.

Proof *Step 1: the construction for an irreducible germ.* Suppose first that $p = 1$, so that (X,x) is irreducible. Choose generators $f_1,\dots,f_m$ of the $\mathcal{O}_{X,x}$-module $\widetilde{\mathcal{O}}_{X,x}$ and a denominator h as in Proposition A.3.3, so that $hf_j = g_j$ with g_j holomorphic. Choose a neighborhood $W \subset \Omega$ of x on which h and the g_j are holomorphic, the f_j are weakly holomorphic, and each f_j satisfies a monic equation $P_j(z,f_j) = 0$ whose coefficients are holomorphic on W. On $(X \cap W) \setminus \{h = 0\}$ the functions g_j/h are holomorphic and agree with f_j on X_{reg}. Let Y be the closure in $W \times \mathbb{C}^m$ of the graph of $(g_1/h,\dots,g_m/h)$ over $(X \cap W) \setminus \{h = 0\}$. This graph is the part outside $\{h = 0\}$ of the analytic set

$$Z = \{(z,w) \in W \times \mathbb{C}^m : \ z \in X,\ h(z)w_j = g_j(z) \text{ for all } j\}.$$

By Lemma A.2.5, applied in $W \times \mathbb{C}^m$ to Z and the analytic set $\{h = 0\}$, the closure Y of $Z \setminus \{h = 0\}$ is analytic; near each point it is the union of the branches of Z on which h does not vanish identically.

The monic equations $P_j(z,g_j/h) = 0$ hold on $X_{\mathrm{reg}} \setminus \{h = 0\}$, which is dense in $X \setminus \{h = 0\}$, and their left-hand sides are continuous there; so they hold on the graph, hence on its closure Y: $P_j(z,w_j) = 0$. By (A.1) they bound w over compact subsets of $X \cap W$ and leave finitely many possible values of w over each point. Hence the projection $\nu \colon Y \to X \cap W$ is proper with finite fibers.

Over X_{reg}, the set Y is the graph of $f = (f_1,\dots,f_m)$: a point of Y over $z_0 \in X_{\mathrm{reg}}$ is a limit of graph points over points $z_k \to z_0$, and since the f_j are continuous at z_0, it equals $(z_0, f(z_0))$. Conversely, $(z_0, f(z_0))$ lies in Y, because z_0 is a limit of points of $X_{\mathrm{reg}} \setminus \{h = 0\}$: the germ of h at z_0 is not identically zero on the manifold germ (X, z_0), so $\{h = 0\}$ is nowhere dense in X_{reg} near z_0, by Lemma A.1.6. Hence ν is an isomorphism over $X_{\mathrm{reg}} \cap W$, with inverse $z \mapsto (z, f(z))$. This graph is dense in Y, because $X_{\mathrm{reg}} \setminus \{h = 0\}$ is dense in $X \setminus \{h = 0\}$ and $(g_1/h,\dots,g_m/h)$ is continuous there; so Y is the closure of the graph of f over $X_{\mathrm{reg}} \cap W$, the set $\nu^{-1}(X_{\mathrm{sing}})$ is nowhere dense, and ν is bimeromorphic. Every point of Y over x is a limit of points $(z_k, f(z_k))$ with $z_k \in X_{\mathrm{reg}}$ and $z_k \to x$; by the last assertion of Proposition A.3.3, the fiber over x is the single point $y = (x, \lim_{z \to x} f(z))$.

Step 2: normality of Y at y. Let a be a weakly holomorphic function on Y near y. Since $\nu^{-1}(X_{\mathrm{reg}})$ is a manifold, it is contained in Y_{reg}, and $b = a \circ (\nu|_{\nu^{-1}(X_{\mathrm{reg}})})^{-1}$ is holomorphic on X_{reg} near x. It is bounded near x: by properness and since the fiber

over x is $\{y\}$, the inverse image of a small neighborhood of x lies in a neighborhood of y on which a is bounded. Thus $b \in \widetilde{\mathcal{O}}_{X,x}$, and $b = \sum_j a_j f_j$ with $a_j \in \mathcal{O}_{X,x}$. Lift the a_j to holomorphic functions $\widetilde{a}_j$ near x in Ω. The function $\sum_j \widetilde{a}_j(z) w_j$ is holomorphic near y, and it equals a on $\nu^{-1}(X_{\mathrm{reg}})$ near y, which is dense in Y_{reg}; since both functions are continuous on Y_{reg}, they agree on Y_{reg} near y. Hence $\widetilde{\mathcal{O}}_{Y,y} = \mathcal{O}_{Y,y}$, and y is a normal point. By Lemma A.3.4, all points of Y near y are normal, and by properness the inverse image of a small neighborhood U of x lies in this set of normal points. Replacing W by U and Y by $\nu^{-1}(U)$, we obtain the assertions of the lemma for $p = 1$. They persist when W is replaced by a smaller open neighborhood of x, because the closure of the graph over the smaller set is the part of Y over it, and normality, finiteness, bimeromorphy and being an isomorphism over the regular locus are preserved under restriction over open subsets of X.

Step 3: several branches. In general, apply Steps 1 and 2 to each irreducible germ (X_i, x), with X_i regarded as an analytic subset of a ball around x in Ω, and let W be the intersection of the resulting neighborhoods; by the persistence just noted, the assertions about the individual Y_i hold over W. The disjoint union of the Y_i is normal, and its map to $X \cap W$ is finite, with p points over x. It is an isomorphism over $X_{\mathrm{reg}} \cap W$, because, by Corollary A.2.10(1), $X_{\mathrm{reg}} \cap W$ is the disjoint union of the open sets $X_i \cap X_{\mathrm{reg}} = (X_i)_{\mathrm{reg}} \setminus \bigcup_{j \neq i} X_j$, over which ν_i is an isomorphism. It is bimeromorphic, with the nowhere dense analytic subset $X_{\mathrm{sing}} \cap W$ of $X \cap W$. Indeed, by Corollary A.2.10(1), $X_{\mathrm{sing}} \cap X_i = (X_i)_{\mathrm{sing}} \cup (X_i \cap \bigcup_{j \neq i} X_j)$, which is a nowhere dense analytic subset of $X_i \cap W$ by Corollary A.2.4; so its inverse image under the bimeromorphic map ν_i is nowhere dense in Y_i, by Lemma A.3.1. These properties again persist for smaller neighborhoods of x. □

Theorem A.3.6 (Normalization) *Every complex space X has a* normalization*: a normal complex space $\widetilde{X}$ with a finite bimeromorphic holomorphic map $\nu\colon \widetilde{X} \to X$ that is an isomorphism over X_{reg}. It is unique up to a unique isomorphism over X. The fiber $\nu^{-1}(x)$ has one point for each irreducible branch of (X, x), and*

$$\nu_* \mathcal{O}_{\widetilde{X}} = \widetilde{\mathcal{O}}_X.$$

Proof *Step 1: local normalizations.* Let $x \in X$, and choose a local model of a neighborhood of x. By Lemma A.3.5, applied to sufficiently small representatives of the branches of (X, x), there is a neighborhood W of x such that $X \cap W$ has a normalization, namely the disjoint union of the normal spaces Y_i with the maps ν_i, whose fiber over x has one point for each branch of (X, x).

Step 2: uniqueness. Let $\nu_1\colon Y_1 \to X$ and $\nu_2\colon Y_2 \to X$ be finite bimeromorphic maps from normal complex spaces that are isomorphisms over X_{reg}. Over X_{reg} there is a unique isomorphism $\phi_0 = \nu_2^{-1} \circ \nu_1$ commuting with the maps to X; we show that it extends holomorphically. Let $p \in Y_1$ and $x = \nu_1(p)$. Choose disjoint open neighborhoods W_l of the finitely many points q_l of $\nu_2^{-1}(x)$, each relatively compact in the domain W_l' of a local model of Y_2, so that the coordinate functions of these models are bounded on W_l. By properness of ν_2, there is a neighborhood U of x

with $\nu_2^{-1}(U) \subset \bigcup_l W_l$. The analytic subset $E = \nu_1^{-1}(X_{\text{sing}})$ of Y_1 is nowhere dense, because ν_1 is bimeromorphic and X_{sing} is nowhere dense in X (Lemma A.3.1).

Since Y_1 is normal, the germ (Y_1, p) is irreducible, and a sufficiently small parametrization neighborhood V of p with $\nu_1(V) \subset U$ has the property that $V \setminus E$ is connected: its connected covering locus remains connected after removing the analytic subset E, which is nowhere dense in it, by Lemma A.1.7, and the result is dense in $V \setminus E$. Hence $\phi_0(V \setminus E)$ lies in a single W_l. The coordinate functions of W_l' composed with ϕ_0 are bounded holomorphic functions on $V \setminus E$, and they extend holomorphically to V by Lemma A.3.4. The extended map takes values in $\overline{W_l} \subset W_l'$, and it satisfies the equations of the local model, which hold on the dense set $V \setminus E$; so it is a holomorphic map $V \to Y_2$ extending ϕ_0. These local extensions are unique, by density of $V \setminus E$, and therefore glue to a holomorphic map $\phi\colon Y_1 \to Y_2$ with $\nu_2 \circ \phi = \nu_1$. Extending ϕ_0^{-1} in the same way, we obtain the inverse map, since the compositions are the identity on dense open sets. Any isomorphism over X agrees with ϕ_0 over X_{reg}, so it equals ϕ. This proves uniqueness.

Step 3: gluing. By the uniqueness just proved, applied over the overlaps of their domains, the local normalizations of Step 1 are canonically isomorphic there, and these isomorphisms are compatible on triple overlaps, because an isomorphism over X between two normalizations is unique; so they glue to a space $\widetilde{X}$ with a map $\nu\colon \widetilde{X} \to X$. The space $\widetilde{X}$ is Hausdorff, since points with different images are separated by inverse images of disjoint neighborhoods and points with the same image lie in one local normalization. Since X has a countable basis, countably many of the open sets over which local normalizations were constructed cover X, and each of these local normalizations, a finite disjoint union of analytic subsets of open subsets of the spaces $\mathbb{C}^{N+m_i}$, has a countable basis; hence so has $\widetilde{X}$. Being locally compact and Hausdorff with a countable basis, $\widetilde{X}$ is paracompact. Being finite, normal and bimeromorphic are local properties over X, so ν is a normalization. The description of the fibers follows from Step 1.

Step 4: the direct image. Let $U \subset X$ be open. If g is holomorphic on $\nu^{-1}(U)$, then $g \circ \nu^{-1}$ is holomorphic on $U \cap X_{\text{reg}}$ and locally bounded near $U \cap X_{\text{sing}}$, because ν is proper; so it is weakly holomorphic. Conversely, if f is weakly holomorphic on U, then $f \circ \nu$ is holomorphic on $\nu^{-1}(U \cap X_{\text{reg}})$ and locally bounded near the nowhere dense analytic subset $\nu^{-1}(U \cap X_{\text{sing}})$; by Lemma A.3.4, applied on the normal space $\nu^{-1}(U)$, it extends holomorphically to $\nu^{-1}(U)$. These two maps are inverse to each other, which proves $\nu_* \mathcal{O}_{\widetilde{X}} = \widetilde{\mathcal{O}}_X$. □

Lemma A.3.5 therefore describes the normalization locally: the normalization of each branch is the closure of the graph of finitely many weakly holomorphic functions satisfying monic equations.

Corollary A.3.7 *Let X be an analytic subset of a domain $\Omega \subset \mathbb{C}^N$, with coordinates z, let $\nu\colon \widetilde{X} \to X$ be its normalization, and let $x \in X$. Let W and $\nu_i\colon Y_i \to X_i \cap W$, $1 \le i \le p$, be as in Lemma A.3.5, for sufficiently small representatives $X_1, \dots, X_p$ of the branches of (X, x). Then each ν_i is a normalization of $X_i \cap W$ whose fiber over x is a single point, and there is an isomorphism of $\nu^{-1}(W)$ onto the disjoint union of $Y_1, \dots, Y_p$ under which ν corresponds to the maps ν_i. In particular, every*

holomorphic function on $\widetilde{X}$ near a point of $\nu^{-1}(x)$ corresponds, near that point, to the restriction to some Y_i of a holomorphic function of (z, w).

Proof By Lemma A.3.5, each Y_i is normal, and ν_i is finite, bimeromorphic and an isomorphism over $(X_i)_{\mathrm{reg}} \cap W$, with a single point over x; so ν_i is a normalization of $X_i \cap W$. Likewise, the disjoint union of the Y_i is a normalization of $X \cap W$. So is $\nu^{-1}(W) \to X \cap W$, because being finite, normal, bimeromorphic and an isomorphism over the regular locus are local properties over X. By the uniqueness in Theorem A.3.6, there is an isomorphism over $X \cap W$ between them. The last assertion holds because the holomorphic functions on the analytic set Y_i are, locally, restrictions of holomorphic functions of the ambient coordinates (z, w). □

Definition A.3.8 A complex space is *unibranch* if all its germs (X, x) are irreducible. Equivalently, each local ring $\mathcal{O}_{X,x}$ is a domain whose integral closure in its field of fractions is a local ring. Indeed, in a local model $\mathcal{O}_{X,x} = \mathcal{O}_N/I_{X,x}$, so by Proposition A.2.1 the germ (X, x) is irreducible if and only if $\mathcal{O}_{X,x}$ is a domain. In this case, by Proposition A.3.3, the integral closure is $\widetilde{\mathcal{O}}_{X,x}$, and taking the limit at x, which exists by the same proposition, is a ring homomorphism $\widetilde{\mathcal{O}}_{X,x} \to \mathbb{C}$. An element with nonzero limit is a unit, because its reciprocal is holomorphic and bounded on X_{reg} near x, hence weakly holomorphic; an element with limit zero is not a unit. So the non-units form the ideal of elements with limit zero, and $\widetilde{\mathcal{O}}_{X,x}$ is a local ring. Thus this definition agrees with the definition of unibranch local rings in commutative algebra.

Corollary A.3.9 *The normalization of a unibranch complex space is a homeomorphism. Every point x of a complex space with (X, x) irreducible, in particular every point of a unibranch space, has arbitrarily small neighborhoods U such that $U \setminus A$ is connected for every nowhere dense analytic subset A of U.*

Proof On a unibranch space the normalization has one-point fibers, so it is a continuous bijection; being proper, it is closed, hence a homeomorphism. For the second assertion, let U be the intersection of X with a parametrization polydisc of (X, x). Its covering locus is a connected manifold, dense in U; it remains connected after deleting A, by Lemma A.1.7, because A is nowhere dense, and the result is dense in $U \setminus A$. Hence $U \setminus A$ is connected. □

Example A.3.10 The curve $\{zw = 0\} \subset \mathbb{C}^2$ has two branches at the origin, the coordinate axes, whose germs are irreducible, being germs of submanifolds (Lemma A.2.9). The disjoint union of the two axes, mapped to the curve by the inclusions, is a manifold, hence normal, and the map is finite, bimeromorphic and an isomorphism over the regular locus $\{zw = 0\} \setminus \{0\}$; by the uniqueness in Theorem A.3.6 it is the normalization. Its fiber over the origin has two points, and the curve is not unibranch.

The cusp $C = \{w^2 = z^3\}$ is normalized by $\gamma(t) = (t^2, t^3)$. Indeed, γ is a holomorphic bijection of $\mathbb{C}$ onto C: for $(z, w) \in C$ with $z \neq 0$, the equations

$t^2 = z$, $t^3 = w$ force $t = w/z$, and $t = w/z$ satisfies them, since $(w/z)^2 = z$ and $(w/z)^3 = (w/z)z = w$; and $\gamma(t) = 0$ only for $t = 0$. The points of $C \setminus \{0\}$ are regular: at such a point $z \neq 0$, so the map $(z, w) \mapsto (w^2 - z^3, w)$ has the nonzero Jacobian determinant $-3z^2$ there, and by Proposition 7.1.2(1) the functions $w^2 - z^3$ and w form holomorphic coordinates near such a point, in which C is a coordinate line; so C is a submanifold there. On $C \setminus \{0\}$ the inverse of γ is the holomorphic function $t = w/z$. This function satisfies $t^2 = z$, so it tends to 0 at the origin; hence γ is a homeomorphism.

The function t is not the restriction of a holomorphic function $F(z, w)$ near the origin, because $F(t^2, t^3)$ has no term of degree one in t. Hence the origin is a singular point of C. Otherwise t, which is bounded near the origin, would extend holomorphically across it on the manifold C, by Lemma A.1.7, and would therefore be the restriction of a holomorphic function of (z, w): in coordinates in which a submanifold is a coordinate plane, a holomorphic function on it extends independently of the other coordinates. So t is weakly holomorphic on C, and integral, since $t^2 = z$, but it is not holomorphic at the origin, so C is not normal there. Thus γ is a finite bimeromorphic map from a manifold that is an isomorphism over $C_{\mathrm{reg}} = C \setminus \{0\}$, and by the uniqueness in Theorem A.3.6 it is the normalization. The cusp is therefore unibranch but not normal, and it distinguishes unibranch spaces from normal spaces.

A.4 Analytic curves and finite maps

Holomorphic curves through a point of a complex space reduce questions about functions near singular points to one complex variable: for instance, a holomorphic function whose modulus has a local maximum at a point is constant along every curve through that point. We parametrize irreducible germs of dimension one, construct curves through a given point that avoid a given analytic subgerm, and study finite maps onto balls, which control the dimension of zero sets of holomorphic functions.

Proposition A.4.1 *Let (C, x) be an irreducible analytic germ of dimension one. There are a disc $\Delta \subset \mathbb{C}$ centered at 0, a representative C of the germ and a holomorphic bijection $\gamma\colon \Delta \to C$ with $\gamma(0) = x$ that restricts to a biholomorphism of $\Delta \setminus \{0\}$ onto $C \setminus \{x\}$. In the coordinates of a local parametrization of (C, x), the projection of γ is $t \mapsto t^q$ for some integer $q \geq 1$. A normal complex space of dimension one is a complex manifold.*

Proof Embed C in a local model and apply Theorem A.2.3. The base is a disc $\Delta' = \{|z| < r\}$, and since the zeros of δ are isolated, we may take the parametrization polydisc so small that δ has no zero in $\Delta' \setminus \{0\}$. Let C be the representative in this polydisc and $C^* = C \setminus \{x\} = \pi^{-1}(\Delta' \setminus \{0\})$. Then C^* is a covering of the punctured disc $\Delta' \setminus \{0\}$ with q sheets, and it is connected: if $\delta(0) = 0$, it is the covering locus, and if $\delta(0) \neq 0$, then $q = 1$, because the fiber over 0 is $\{x\}$. We shall show that lifting a circle around the puncture permutes the q points of a fiber cyclically, and then obtain γ by lifting the exponential map to C^*.

We first recall two facts about coverings. A path in $\Delta' \setminus \{0\}$ lifts uniquely to C^* once its initial point is prescribed: subdivide the path into finitely many pieces lying in discs over which the covering is trivial, and lift piece by piece. If two paths with the same endpoints are homotopic with fixed endpoints, their lifts with the same initial point have the same endpoint. Indeed, let $h\colon [0,1]^2 \to \Delta' \setminus \{0\}$ be the homotopy, with $h(0,s)$ and $h(1,s)$ independent of s, and let $e(s)$ be the endpoint of the lift of $h(\cdot, s)$ with the prescribed initial point. By uniform continuity there is an integer $m \geq 1$ such that h maps each square $Q_{jl} = [(j-1)/m, j/m] \times [(l-1)/m, l/m]$ into a disc D_{jl} over which the covering is trivial. A lift of a path in D_{jl} stays in the sheet over D_{jl} that contains its initial point, because these sheets are open and disjoint.

Fix l and $s, s' \in [(l-1)/m, l/m]$. We show by induction on j that the lifts of $h(\cdot, s)$ and $h(\cdot, s')$ over $[(j-1)/m, j/m]$ lie in one sheet over D_{jl}. For $j = 1$ they start at the same point. If this holds for some $j < m$, the points of the two lifts at the parameter j/m are joined by the lift, inside this sheet, of the path $h(j/m, \cdot)$ between s and s'; this lift is a connected set lying over $D_{jl} \cap D_{j+1,l}$, so it lies in one sheet over $D_{j+1,l}$, which then contains both lifts over $[j/m, (j+1)/m]$. For $j = m$ we obtain two points in one sheet over D_{ml} lying over the same point $h(1,s) = h(1,s')$, so $e(s) = e(s')$. Hence e is constant on each interval $[(l-1)/m, l/m]$, and therefore on $[0,1]$.

Next, every loop based at a point $b \in \Delta' \setminus \{0\}$ is homotopic with fixed endpoints to the kth power of the circle around 0 through b, for some integer k. Indeed, lifting the argument continuously along the loop, we write the loop as exp of a path in the half-plane $H = \{\zeta : \operatorname{Re}\zeta < \log r\}$ from some ζ_0 to $\zeta_0 + 2\pi \mathrm{i} k$; the straight-line homotopy to the segment from ζ_0 to $\zeta_0 + 2\pi\mathrm{i}k$ stays in the convex set H. Let σ be the permutation of the fiber over b obtained by lifting the circle. Lifting any loop at b therefore induces a power of σ. Since C^* is connected, any two points of the fiber over b are joined by a path in C^*, whose image is a loop at b; hence the powers of σ act transitively on the q points of the fiber, and σ is a cycle of length q.

The map $\exp\colon H \to \Delta' \setminus \{0\}$ lifts to a map $\Phi\colon H \to C^*$ with $\pi \circ \Phi = \exp$: lift along paths from a base point ζ_0, starting at a chosen point of the fiber; since H is convex, the endpoint does not depend on the path. The map Φ is holomorphic, because locally it is exp followed by the inverse of π on a sheet. The vertical segment from ζ to $\zeta + 2\pi\mathrm{i}$ is mapped by exp onto the circle through e^{ζ}; so $\Phi(\zeta + 2\pi\mathrm{i}) = \sigma_\zeta(\Phi(\zeta))$, where σ_ζ is the permutation of the fiber over e^{ζ} obtained by lifting this circle. By the preceding paragraph, applied with $b = \mathrm{e}^{\zeta}$, it is a cycle of length q; hence $\Phi(\zeta + 2\pi\mathrm{i}q) = \Phi(\zeta)$. Therefore

$$\gamma(t) = \Phi(q \log t), \qquad 0 < |t| < r^{1/q},$$

does not depend on the branch of the logarithm; it is holomorphic, and $\pi(\gamma(t)) = t^q$. The coordinates of γ are bounded, because C lies in the bounded parametrization polydisc, so by the removable singularity theorem they extend holomorphically to $t = 0$. Since π is proper, $\gamma(t)$ stays in a compact subset of C as $t \to 0$; hence $\gamma(0) \in C$, and $\gamma(0) = x$ because $\gamma(0)$ lies over 0.

Let $\Delta = \{|t| < r^{1/q}\}$. The map $\gamma\colon \Delta \to C$ is surjective: the points of C over $z_0 = \mathrm{e}^{\zeta} \neq 0$ are $\Phi(\zeta + 2\pi\mathrm{i}k) = \gamma(\mathrm{e}^{(\zeta + 2\pi\mathrm{i}k)/q})$, $0 \leq k < q$, because σ_ζ is a q-cycle.

It is injective: if $\gamma(t_1) = \gamma(t_2)$ with $t_1, t_2 \neq 0$, then $t_1^q = t_2^q$, so $t_2 = t_1 e^{2\pi i k/q}$; writing $q \log t_1 = \zeta$, we get $\gamma(t_2) = \Phi(\zeta + 2\pi i k) = \sigma_\zeta^k(\gamma(t_1))$, which forces $k \equiv 0$ modulo q. On $\Delta \setminus \{0\}$ the map γ is a local biholomorphism into the manifold C^*, since $\pi \circ \gamma(t) = t^q$ has nonzero derivative; being bijective, it is a biholomorphism of $\Delta \setminus \{0\}$ onto $C \setminus \{x\}$.

Finally, let X be a normal complex space of dimension one, and let $x \in X$; if x is an isolated point, there is nothing to prove. Otherwise the germ (X, x) has dimension one, and it is irreducible by the definition of normal points; let $\gamma\colon \Delta \to C$ be the map just constructed for a representative C of this germ. Its inverse is holomorphic on the manifold $C \setminus \{x\}$, which is contained in the regular locus, and it is bounded; so it is weakly holomorphic and, by normality, extends to a holomorphic function τ near x. Then $\tau \circ \gamma = \mathrm{id}$ on $\Delta \setminus \{0\}$, hence on Δ, and $\gamma \circ \tau = \mathrm{id}$ on $C \setminus \{x\}$, hence on C, because these maps are continuous and the punctured sets are dense. Thus γ is an isomorphism of Δ onto a neighborhood of x, and x is a regular point. □

Proposition A.4.2 *Let (X, x) be an irreducible analytic germ of positive dimension, and let (B, x) be an analytic subgerm different from (X, x). There is a nonconstant holomorphic map γ from a disc Δ centered at 0 into X with $\gamma(0) = x$ and $\gamma(t) \in X_{\mathrm{reg}} \setminus B$ for $t \neq 0$.*

Proof The germ of $B \cup X_{\mathrm{sing}}$ is again different from (X, x), since (X, x) is irreducible and X_{sing} has smaller dimension by Corollary A.2.10(1). Since every analytic germ (A, x) satisfies $V(I_{A,x}) = (A, x)$, the ideals of these two germs differ, and we can choose a holomorphic germ f that vanishes on $B \cup X_{\mathrm{sing}}$ but not identically on X. Take a local parametrization of (X, x) over a polydisc $\Delta' \subset \mathbb{C}^d$, $d \geq 1$, with covering locus X_δ, and for $z' \in \Delta' \setminus \{\delta = 0\}$ let $N(z')$ be the product of the values of f at the q points of X over z'. This function is holomorphic and bounded, so it extends holomorphically to Δ' by Lemma A.1.7. It is not identically zero. Otherwise, choose a small ball in $\Delta' \setminus \{\delta = 0\}$ over which the q points are given by holomorphic sections $s_1, \ldots, s_q$; the product of the functions $f \circ s_i$ would vanish identically there, so one of these functions would vanish on an open set. Then f would vanish on a nonempty open subset of the connected manifold X_δ, hence on X_δ by the identity theorem, and on X by density, which is not the case.

The function $g = N\delta$ is holomorphic and not identically zero on Δ'. Choose a vector $v \in \mathbb{C}^d$ at which the first nonzero homogeneous term of g does not vanish, and $\rho > 0$ such that $g(tv) \neq 0$ for $0 < |t| < \rho$ and the ball $U = \{|z'| < \rho|v|\}$ lies in Δ'. Then $Y = X \cap \pi^{-1}(\{tv : |t| < \rho\})$ is an analytic subset of $X \cap \pi^{-1}(U)$. Over the punctured disc, Y consists of points of X_δ at which $f \neq 0$, and it is a covering of the punctured disc; over 0 it consists of the single point x. Thus $Y \setminus \{x\}$ is a complex manifold of dimension one, and by properness of π its points accumulate at x, so x is not an isolated point of Y. Let (C, x) be a branch of (Y, x) of positive dimension. Its regular points near x lie in the one-dimensional manifold $Y \setminus \{x\}$, so (C, x) has dimension one. Let $\gamma\colon \Delta \to C$ be the parametrization of Proposition A.4.1. For $t \neq 0$, the point $\gamma(t)$ lies in $C \setminus \{x\} \subset Y \setminus \{x\}$, hence in $X_\delta \subset X_{\mathrm{reg}}$ and outside $\{f = 0\} \supset B$. □

Corollary A.4.3 *Let f be a holomorphic function near a point x of a complex space X such that $|f|$ has a local maximum at x. Then f is constant on each irreducible branch of (X,x). Moreover, every compact analytic subset of an open subset of $\mathbb{C}^N$ is finite.*

Proof Let X_1 be a branch of (X,x). If X_1 has dimension zero, it is the point x. Otherwise, suppose that $f - f(x)$ does not vanish identically on X_1, and apply Proposition A.4.2 to X_1 and its subgerm $X_1 \cap \{f = f(x)\}$. The holomorphic function $f \circ \gamma$ on the disc has a local maximum of its modulus at 0, so it is constant by the maximum principle in one variable, and $f(\gamma(t)) = f(x)$ for all t, contradicting the choice of γ.

For the second assertion, let K be a compact analytic subset of an open set in $\mathbb{C}^N$, and let K_+ be the set of its non-isolated points. The isolated points form an open subset of K, so K_+ is compact. Suppose that $K_+ \neq \varnothing$, and let y be a point of K_+ where $|z|^2$ attains its maximum on K_+. Since y is not isolated, some branch of (K,y) has positive dimension, and its small representatives, being pure-dimensional, consist of non-isolated points of K. So Proposition A.4.2, applied to this branch and the subgerm $\{y\}$, gives a nonconstant holomorphic curve γ in K_+ with $\gamma(0) = y$. The function $|\gamma(t)|^2$ is subharmonic, with Laplacian $4\sum_j |\gamma_j'(t)|^2$, and it has a local maximum at 0; by the maximum principle (Proposition 3.1.4) it is constant near 0, so its Laplacian vanishes, all γ_j' vanish, and γ is constant, a contradiction. Hence $K_+ = \varnothing$, and K is compact and discrete, hence finite. □

The surjectivity of finite maps onto balls (Lemma A.4.5) rests on the following elementary fact.

Lemma A.4.4 *Let $B \subset \mathbb{R}^N$, $N \geq 2$, be an open ball, and let $E \subset B$ be relatively closed and $\mathcal{H}^{N-1}$-null. Then $B \setminus E$ is path connected.*

Proof Fix $a, b \in B \setminus E$. If $c \in B$ and the segment $[a,c]$ meets E at a point $e = a + s(c-a)$, then $0 < s \leq 1$, since $a \notin E$, so $c = a + t(e-a)$ with $t = 1/s \geq 1$. We show that the set of such c has N-dimensional Lebesgue measure zero. Exhaust E by the compact sets $E_k = E \cap \overline{B_k}$, where B_k are concentric balls increasing to B. Since $a \notin E_k$, the parameter t is bounded on $\{(e,t) : e \in E_k,\ a + t(e-a) \in B\}$, say by T_k, and the map $(e,t) \mapsto a + t(e-a)$ is Lipschitz on $E_k \times [1,T_k]$. Cover E_k by balls of radii r_l with $\sum_l r_l^{N-1}$ arbitrarily small. The image of each product of such a ball with $[1,T_k]$ is covered by at most C_k/r_l balls of radius $C_k r_l$, whose total volume is at most $C_k' r_l^{N-1}$. Hence the image of $E_k \times [1,T_k]$ is Lebesgue null, and so is the union over k. The same holds with b in place of a. Choose $c \in B$ outside both null sets. Then the path $[a,c] \cup [c,b]$ lies in the convex set B and does not meet E. □

Lemma A.4.5 *Let (X,x) be a pure d-dimensional analytic germ, and let $F\colon (X,x) \to (\mathbb{C}^d, 0)$ be a holomorphic map germ such that x is an isolated point of $F^{-1}(0)$. Then F has arbitrarily small representatives that map properly, with finite fibers, onto balls centered at 0: for every neighborhood V of x there are an open neighborhood $X' \subset V$ of x in a representative of (X,x) and a ball $B_\varepsilon \subset \mathbb{C}^d$ centered at 0 such that $F\colon X' \to B_\varepsilon$ is proper, has finite fibers and is surjective. In particular, the image under F of every neighborhood of x in X contains a neighborhood of 0.*

Proof For $d = 0$ the germ is the point x, and there is nothing to prove. Let $d \geq 1$, embed a representative X in a local model $X \subset \Omega \subset \mathbb{C}^N$ such that X is pure d-dimensional and F is the restriction of a holomorphic map on Ω, also denoted F, and choose a closed ball $\overline{B(x,\rho)} \subset \Omega$, with $X \cap B(x,\rho) \subset V$, whose intersection with $X \cap F^{-1}(0)$ is $\{x\}$. The compact set $X \cap \partial B(x,\rho)$ does not meet $F^{-1}(0)$, so $|F|$ has a positive minimum η on it (if it is empty, put $\eta = \infty$). Let B_ε be the ball of radius $\varepsilon < \eta$ about $0 \in \mathbb{C}^d$, and let $X' = X \cap B(x,\rho) \cap F^{-1}(B_\varepsilon)$. Then $F\colon X' \to B_\varepsilon$ is proper: the inverse image of a compact set $K \subset B_\varepsilon$ is $X \cap \overline{B(x,\rho)} \cap F^{-1}(K)$, which is compact and contains no point of the boundary sphere. Its fibers are compact analytic subsets of the open ball $B(x,\rho)$, hence finite by Corollary A.4.3.

It remains to prove surjectivity. On each connected component of the manifold X'_{reg}, which has dimension d, the differential of F has rank d at some point. Otherwise its maximal rank r on some component would be smaller; near a point where the rank is r, the rank would be constantly r, and by Lemma A.2.11, applied in a chart of X'_{reg}, the fibers near this point would be submanifolds of positive dimension $d - r$, contradicting their finiteness.

Let C be the union of X'_{sing} and of the set of regular points where the rank of F is less than d. In the local model, C is the union of X'_{sing} and of the common zero set in X' of the minors of size N of the matrix formed by the differentials of local generators of the ideal of X and the rows of $\mathrm{d}F$; so C is analytic. It contains no branch of X' at any point. Indeed, each branch contains a nonempty open subset of X'_{reg}, consisting of regular points of the branch outside the other branches (Corollary A.2.4), and on each component of X'_{reg} the points of rank less than d form a proper analytic subset, which is nowhere dense by Lemma A.1.6. Hence C has dimension at most $d - 1$ by Proposition A.2.7.

Its image $E = F(C)$ is closed in B_ε by properness, and it is $\mathcal{H}^{2d-1}$-null. Indeed, by Corollary A.2.10(2), C is a finite union of complex submanifolds of open subsets of Ω of dimensions at most $d - 1$, on each of which F has rank at most $d - 1$, so Lemma A.2.12(1) applies.

Over $B_\varepsilon \setminus E$, the map F is a local biholomorphism at every point of its source, by Proposition 7.1.2(1) applied in charts of X'_{reg}, and it is proper; so its image is open and closed in $B_\varepsilon \setminus E$. This image is nonempty: at a regular point of rank d, the image of a neighborhood is open, and it is not contained in E, which is Lebesgue null because it is $\mathcal{H}^{2d-1}$-null. Since $B_\varepsilon \setminus E$ is connected by Lemma A.4.4, the image contains $B_\varepsilon \setminus E$, which is dense in B_ε; being closed in B_ε, it equals B_ε. □

Proposition A.4.6 *Let X be a pure d-dimensional complex space, and let f be a holomorphic function on X that is not identically zero on any branch of (X, y), for every $y \in X$. Then the zero set of f is either empty or pure of dimension $d - 1$.*

Proof If $d = 0$, then f has no zeros. Let $d \geq 1$ and let x be a zero of f. We first show that $\dim(\{f = 0\}, x) = d - 1$. For each branch X_0 of (X, x), the germ $\{f = 0\} \cap X_0$ is a subgerm of X_0 different from it, so $\dim(\{f = 0\}, x) \leq d - 1$ by Proposition A.2.7. Fix a branch X_0, and choose linear forms $\ell_1, \ldots, \ell_{d-1}$ on the ambient space of a local model, vanishing at x, inductively such that ℓ_j does not vanish identically on any positive-dimensional branch of the germ at x of

$\{f = 0\} \cap X_0 \cap \{\ell_1 = \cdots = \ell_{j-1} = 0\}$. At each step this excludes finitely many proper linear subspaces, and, by Proposition A.2.7, it lowers the dimension of this germ by at least one as long as it is positive. Since the initial dimension is at most $d - 1$, the point x is isolated in the common zero set of $\Phi = (f, \ell_1, \ldots, \ell_{d-1})$ on X_0.

Suppose that $\dim(\{f = 0\}, x) < d - 1$; then $d \geq 2$, and by upper semicontinuity $\dim(\{f = 0\}, y) \leq d - 2$ for all zeros y of f near x. By Lemma A.4.5, applied to the germ (X_0, x), which is pure-dimensional, there is a neighborhood X_0' of x in a small representative of X_0, on which this bound holds, such that Φ maps X_0' onto a ball $B \subset \mathbb{C}^d$ centered at 0. Hence Φ maps $\{f = 0\} \cap X_0'$ onto the set $B \cap \{w_1 = 0\}$, which has positive $(2d - 2)$-dimensional Lebesgue measure; here w_1 is the first coordinate of $\mathbb{C}^d$. On the other hand, by Corollary A.2.10(2), $\{f = 0\} \cap X_0'$ is a finite union of complex submanifolds of dimensions at most $d - 2$ of open subsets of the ambient space, on each of which Φ has rank at most $d - 2$; by Lemma A.2.12(1), its image under Φ is $\mathcal{H}^{2d-2}$-null, hence Lebesgue null in $\{w_1 = 0\} \cong \mathbb{R}^{2d-2}$. This contradiction shows that $\dim(\{f = 0\}, x) = d - 1$.

Now let V_1 be a branch of $(\{f = 0\}, x)$. By Corollary A.2.4, there are regular points y of V_1 arbitrarily close to x that do not lie on the other branches. Near such a point, $\{f = 0\}$ coincides with V_1, which is a manifold of dimension $\dim V_1$ by purity. The hypotheses hold at y, so the first part gives $\dim(\{f = 0\}, y) = d - 1$, and therefore $\dim V_1 = d - 1$. □

A.5 Blowups and resolution

Blowing up an ideal replaces its zero set by the space of directions of its generators, and makes the pulled-back ideal locally principal. We construct blowups of complex spaces along coherent ideals, prove their universal property, and compute blowups of manifolds along submanifolds in explicit charts; these charts give the transformation rules for ideals of submanifolds needed in the next sections. At the end of the section we state the existence theorems for resolutions and principalizations and deduce that log resolutions exist.

Definition A.5.1 Let X be a complex space and let $\mathfrak{a} \subset \mathcal{O}_X$ be a coherent ideal sheaf that is not identically zero on any branch of (X, x), for every $x \in X$. On an open set where $\mathfrak{a}$ is generated by $f_0, \ldots, f_r$, the *blowup* $\mathrm{Bl}_{\mathfrak{a}} X$ is the closure, with its reduced structure, of the graph

$$\big\{\big(x, [f_0(x) : \cdots : f_r(x)]\big) : x \in X \setminus V(\mathfrak{a})\big\} \subset X \times \mathbb{P}^r.$$

Here $\mathbb{P}^r$ is the space of nonzero vectors in $\mathbb{C}^{r+1}$ modulo multiplication by nonzero scalars, with the charts $\{\xi_i \neq 0\}$ and the affine coordinates ξ_j/ξ_i. It is compact, because every point has a representative on the compact unit sphere and the quotient map is continuous. The projection to X is denoted by π.

An ideal sheaf on a complex space is *invertible* if it is locally generated by one nonzerodivisor. For a holomorphic map $g\colon Y \to X$ we write $\mathfrak{a}\mathcal{O}_Y$ for the ideal sheaf generated by the pullbacks $f \circ g$ of local sections f of $\mathfrak{a}$.

Proposition A.5.2 *The blowup is a complex space, independent of the chosen generators up to canonical isomorphism. Its projection π is proper and bimeromorphic, and $\mathfrak{a}\mathcal{O}_{\mathrm{Bl}_\mathfrak{a} X}$ is an invertible ideal. If $g\colon Y \to X$ is a holomorphic map from a complex space Y such that $\mathfrak{a}\mathcal{O}_Y$ is invertible, then g has a unique holomorphic lift $\widetilde{g}\colon Y \to \mathrm{Bl}_\mathfrak{a} X$, that is, $\pi \circ \widetilde{g} = g$.*

Proof In $X \times \mathbb{P}^r$ consider the analytic set Z defined by the equations of X and by $f_i\xi_j = f_j\xi_i$ for all i, j; in the chart $\{\xi_i \neq 0\}$ these are the holomorphic equations $f_j = f_i \cdot (\xi_j/\xi_i)$ and $f_k \cdot (\xi_j/\xi_i) = f_j \cdot (\xi_k/\xi_i)$. Over a point $x \notin V(\mathfrak{a})$, the only point of Z is $(x, [f(x)])$, because ξ must be proportional to the nonzero vector $f(x)$. Thus the graph is the part of Z outside the analytic set $\pi^{-1}(V(\mathfrak{a}))$, and by Lemma A.2.5 its closure is analytic; near each point it is the union of the branches of Z not contained in $\pi^{-1}(V(\mathfrak{a}))$. It is closed in $X \times \mathbb{P}^r$, and $\mathbb{P}^r$ is compact, so π is proper. Over $X \setminus V(\mathfrak{a})$ the blowup is the graph, and π is an isomorphism there, with inverse $x \mapsto (x, [f(x)])$, which is holomorphic because on $\{f_i \neq 0\}$ the affine coordinates f_j/f_i are holomorphic. The set $V(\mathfrak{a})$ is nowhere dense in X, by Corollary A.2.4, and the graph is dense in the blowup; so π is bimeromorphic.

On the chart $\{\xi_i \neq 0\}$ of the blowup, the relations $f_j \circ \pi = (f_i \circ \pi) \cdot (\xi_j/\xi_i)$ hold on the graph, hence everywhere, so the pulled-back ideal is generated by $f_i \circ \pi$ there. This generator is not identically zero on any branch of the blowup at any point of the chart, because the graph points of the chart, at which $f_i \circ \pi \neq 0$, are dense in every such branch. Hence it is a nonzerodivisor, and the pulled-back ideal is invertible.

Let $g\colon Y \to X$ be as in the statement. Near a point of Y, let d be a nonzerodivisor generating $\mathfrak{a}\mathcal{O}_Y$, and write $f_i \circ g = db_i$. Writing $d = \sum_i c_i\,(f_i \circ g) = d \sum_i c_i b_i$ and cancelling the nonzerodivisor d, we get $\sum_i c_i b_i = 1$; so the b_i have no common zero, and $y \mapsto (g(y), [b_0(y) : \cdots : b_r(y)])$ is a holomorphic map into $X \times \mathbb{P}^r$. Where $d \neq 0$, some $f_i(g(y))$ is nonzero, and this point is the graph point over $g(y)$; since $\{d = 0\}$ is nowhere dense, the whole image lies in the closure of the graph. Replacing d by ud for a unit u replaces b by $u^{-1}b$ and does not change the projective point, so these local lifts glue. A lift of g is uniquely determined over $Y \setminus g^{-1}(V(\mathfrak{a})) = Y \setminus \{d = 0\}$, where π is an isomorphism, hence everywhere by density.

Finally, the blowups defined by two families of generators both pull back $\mathfrak{a}$ to an invertible ideal. By the universal property, each maps to the other over X, and both compositions are the identity by uniqueness. These canonical isomorphisms also glue the local constructions on overlaps, compatibly on triple overlaps by the same uniqueness. The glued space is Hausdorff: points with different images in X are separated by the inverse images of disjoint neighborhoods, and points with the same image lie in one of the local constructions, which are Hausdorff. Since X has a countable basis, countably many of the open sets over which the local constructions are made cover X, and each local construction, an analytic subset of the product of

a local model with $\mathbb{P}^r$, has a countable basis; hence so has the glued space. Being locally compact and Hausdorff with a countable basis, it is paracompact. □

An *effective Cartier divisor* D on a complex space X is given by an invertible ideal sheaf, written $\mathcal{O}_X(-D)$; a local generator is a *local equation* of D, determined up to a unit. We write $\mathcal{O}_X(D)$ for the inverse invertible sheaf, generated locally by the reciprocal of a local equation. Addition of effective Cartier divisors corresponds to multiplication of their ideals. If $g\colon Y \to X$ is holomorphic and the local equations of D composed with g are nonzerodivisors, they define the pullback g^*D, with $\mathcal{O}_Y(-g^*D) = \mathcal{O}_X(-D)\mathcal{O}_Y$; restriction to an analytic subset is a special case.

On a complex manifold, an effective divisor has *simple normal crossings support*, and is then called a *simple normal crossings divisor*, if each irreducible component of its support is smooth and near each point there are holomorphic coordinates in which the support is $\{z_1 \cdots z_k = 0\}$. Its multiplicities may be arbitrary positive integers; the condition concerns only the support. A submanifold C has *normal crossings* with such a divisor if near each point of C there are coordinates in which the support of the divisor is a union of coordinate hyperplanes and C is a coordinate subspace.

For a proper bimeromorphic map between complex manifolds, an effective divisor is called *exceptional* if each irreducible component of its support has image of codimension at least two; these images are analytic by Theorem A.6.2, proved in the next section.

Proposition A.5.3 *Let C be a closed submanifold of a complex manifold M, of positive codimension at each of its points, and let $\pi\colon M' \to M$ be the blowup of M along $\mathcal{I}_C$. Then M' is a complex manifold, the inverse image $E = \pi^{-1}(C)$ is a smooth hypersurface with $\mathcal{I}_C\mathcal{O}_{M'} = \mathcal{O}_{M'}(-E)$, and the fiber of π over a point of C at which C has codimension r is $\mathbb{P}^{r-1}$. If C is a hypersurface, then π is the identity and $E = C$. Explicitly, let (z, w), $w = (w_1, \dots, w_r)$, be holomorphic coordinates on an open set $U \subset M$ with $C \cap U = \{w = 0\}$. Then $\pi^{-1}(U)$ is covered by r charts, one for each* pivot $i \in \{1, \dots, r\}$, *with coordinates (z, t, v), $v = (v_j)_{j\neq i}$, in which*

$$\pi(z, t, v) = (z, t\tilde{v}), \qquad \tilde{v}_i = 1, \quad \tilde{v}_j = v_j \ (j \neq i), \tag{A.6}$$

and $E = \{t = 0\}$; the Jacobian determinant of π in these coordinates is $\pm t^{r-1}$.

Let $W \subset M$ be a closed submanifold containing C, such that C has positive codimension in W at each of its points. Then the strict transform W' of W, the closure of $\pi^{-1}(W \setminus C)$, is the blowup $\mathrm{Bl}_C W$ *of W along $\mathcal{I}_C\mathcal{O}_W$, and*

$$\mathcal{I}_W\mathcal{O}_{M'} = \mathcal{O}_{M'}(-E)\mathcal{I}_{W'}. \tag{A.7}$$

In holomorphic coordinates (z, w), $w = (w_1, \dots, w_s)$, on an open set $U \subset M$ with $W \cap U = \{w = 0\}$ and $C \cap U = \{w = 0,\ z_1 = \cdots = z_k = 0\}$, which exist near every point of C, the center is generated by $z_1, \dots, z_k, w_1, \dots, w_s$, and the strict transform W' meets only the charts whose pivot is one of $z_1, \dots, z_k$. The chart with pivot z_i has coordinates (z'', t, u, v), where z'' consists of the coordinates z_l with $l > k$, and is given by $z_i = t$, $z_l = tu_l$ for $l \leq k$, $l \neq i$, and $w_j = tv_j$; in it, $W' = \{v = 0\}$. If instead

W is a closed submanifold disjoint from C, the strict transform $W' = \pi^{-1}(W)$ maps isomorphically onto W, and $\mathcal{I}_W \mathcal{O}_{M'} = \mathcal{I}_{W'}$.

Proof Let (z, w) be coordinates on an open set U as in the first part of the statement; such coordinates exist near every point of C. On U the ideal $\mathcal{I}_C$ is generated by $w_1, \dots, w_r$, by Lemma A.2.9. Let $Z \subset U \times \mathbb{P}^{r-1}$ be the analytic set defined by the equations $w_j \xi_i = w_i \xi_j$. Over a point of $U \setminus C$, its only point is $((z, w), [w])$, because ξ must be proportional to the nonzero vector w; so the part of Z over $U \setminus C$ is the graph of Definition A.5.1, whose closure is the blowup of U. In the chart $\{\xi_i \neq 0\}$, the equations $w_j \xi_i = w_i \xi_j$ say that $w_j = w_i v_j$ with $v_j = \xi_j / \xi_i$, $j \neq i$; hence this part of Z is the graph of the map $(z, t, v) \mapsto (z, w)$ with $w_i = t$ and $w_j = t v_j$, $j \neq i$. It is a manifold with coordinates (z, t, v), in which the graph part is $\{t \neq 0\}$, a dense subset. Hence these charts cover the blowup, M' is a manifold, and $\mathcal{I}_C \mathcal{O}_{M'}$ is generated by t in each chart. Thus $E = \{t = 0\}$ is a smooth hypersurface.

With w_i listed first among the coordinates w, the Jacobian matrix of (A.6) is block lower triangular with diagonal blocks I, 1 and $t I_{r-1}$, because $\partial w_i / \partial t = 1$, $\partial w_j / \partial t = v_j$ and $\partial w_j / \partial v_l = t \delta_{jl}$ for $j, l \neq i$, while z does not depend on (t, v) and w does not depend on z; so the Jacobian determinant of π is $\pm t^{r-1}$. The fiber over a point $(z, 0)$ is obtained by gluing the coordinate charts $v \in \mathbb{C}^{r-1}$, that is, it is $\mathbb{P}^{r-1}$. If $r = 1$, there is one chart, and the coordinates and the map are unchanged.

For the second part, choose coordinates (z, w) near a point of C with $W = \{w_1 = \cdots = w_s = 0\}$ and $C = \{w_1 = \cdots = w_s = z_1 = \cdots = z_k = 0\}$, where $k \geq 1$ is the codimension of C in W. Such coordinates exist: take coordinates (x, w) with $W = \{w = 0\}$, then holomorphic coordinates $z = \phi(x)$ on W near the point in which C is a coordinate plane, and use $(\phi(x), w)$, whose Jacobian matrix is block triangular and invertible, so that these are holomorphic coordinates near the point by Proposition 7.1.2(1). The center is now generated by $z_1, \dots, z_k, w_1, \dots, w_s$. In a chart whose pivot is z_i, we have $z_i = t$, $z_l = t u_l$ for $l \neq i$, $l \leq k$, and $w_j = t v_j$; here $\pi^{-1}(W \setminus C) = \{v = 0,\ t \neq 0\}$, so $W' = \{v = 0\}$ in this chart, and

$$\mathcal{I}_W \mathcal{O}_{M'} = (t v_1, \dots, t v_s) = (t)(v_1, \dots, v_s) = \mathcal{O}_{M'}(-E) \mathcal{I}_{W'}.$$

In a chart whose pivot is w_i, we have $w_i = t$, so $\pi^{-1}(W)$ lies in $\{t = 0\} = E$. Moreover, W' does not meet this chart, because the points of $\pi^{-1}(W \setminus C)$, and therefore those of its closure, have projective coordinates ξ whose components corresponding to $w_1, \dots, w_s$ vanish. The pulled-back ideal contains t and is therefore $(t) = \mathcal{O}_{M'}(-E)$, which proves (A.7) in this chart as well. In the charts of the first kind, W' has the coordinates (t, u) together with the remaining z-coordinates, and $\pi|_{W'}$ is given by the formulas of the blowup of W along C; so $W' = \mathrm{Bl}_C W$. Finally, if C is disjoint from W, then π is an isomorphism over the neighborhood $M \setminus C$ of W. □

We shall use the following two existence theorems, which we quote without proof. They are the analytic versions of [BM08, Theorems 1.1 and 1.3, pp. 611 and 614; Definitions 2.1–2.2, pp. 618–619]; that these results hold for complex analytic spaces is stated on p. 611 there. Further references are given in Appendix B.21.

Theorem A.5.4 (Resolution of singularities) *Every reduced complex space X has a proper bimeromorphic map $\pi\colon M \to X$ from a complex manifold M that is a composite of blowups with smooth centers and an isomorphism over X_{reg}. The divisor formed by the inverse images in M of the blowup centers has simple normal crossings. The sequence of blowups is finite when X is compact, and finite over each relatively compact open subset of X in general.*

Theorem A.5.5 (Principalization) *Let M be a complex manifold, let $\mathfrak{a} \subset \mathcal{O}_M$ be a coherent ideal that is not identically zero on any connected component of M, and let D be a simple normal crossings divisor on M, possibly empty. Then there is a sequence of blowups $\sigma_i\colon M_i \to M_{i-1}$, $i \geq 1$, with $M_0 = M$ and smooth centers $C_i \subset M_{i-1}$, finite over each relatively compact open subset of M, with the following properties. Put $\mathfrak{a}_0 = \mathfrak{a}$ and $E_i = \sigma_i^{-1}(C_i)$, and, for $i \geq 0$, call the union of the strict transforms of $D, E_1, \ldots, E_i$ in M_i the* accumulated divisor *on M_i. Each center C_i is contained in $V(\mathfrak{a}_{i-1})$ and has normal crossings with the accumulated divisor on M_{i-1}; the ideal*

$$\mathfrak{a}_i = \mathcal{O}_{M_i}(E_i)\, \mathfrak{a}_{i-1}\mathcal{O}_{M_i} \tag{A.8}$$

is holomorphic; and over each relatively compact open subset of M the last of these ideals is the unit ideal. The accumulated divisors have simple normal crossings, and the composite is an isomorphism off $V(\mathfrak{a})$.

In the notation of [BM08], Theorem A.5.5 is the theorem for marked ideals with mark 1 and $N = M$. The cosupport of $\mathfrak{a}_i$ is $V(\mathfrak{a}_i)$, and the transform (A.8) divides by a single exceptional factor at each step, even where $\mathfrak{a}_{i-1}$ vanishes to higher order along the center. If M is compact, the sequence is finite, say with t steps, and iterating (A.8) gives

$$\mathfrak{a}\mathcal{O}_{M_t} = \mathcal{O}_{M_t}(-H), \qquad H = \sum_{i=1}^{t} (\sigma_{i+1} \circ \cdots \circ \sigma_t)^* E_i,$$

where the empty composition for $i = t$ is the identity, so that the last term is E_t; the same holds over every relatively compact open subset of M.

The divisor H is effective, and its support has simple normal crossings. Indeed, for a blowup σ with center C and exceptional divisor E, and a closed set D, we have $\sigma^{-1}(D) \subset D' \cup E$, where D' is the strict transform, the closure of $\sigma^{-1}(D \setminus C)$. By induction, the inverse image in M_t of each E_i is contained in the accumulated divisor on M_t. The support of H is the union of these inverse images, the zero set of a local equation of H, which is not identically zero on any component of M_t; so it is empty or pure of codimension one, by Proposition A.4.6. By Corollary A.2.14(4) and (3), each of its irreducible components is contained in, and hence equal to, an irreducible component of the accumulated divisor. Thus the support of H is a union of some components of a divisor whose support has simple normal crossings.

Near a point, in coordinates in which the support of the accumulated divisor is $\{z_1 \cdots z_k = 0\}$, the germ of each of its components through the point is the germ of a connected submanifold of codimension one, hence irreducible by Lemma A.2.9. It is

contained in the union of the germs of the hyperplanes $\{z_l = 0\}$, so it is contained in one of them and, by Proposition A.2.7, equal to it. Hence the support of H is near this point the union of some of these hyperplanes, and it has simple normal crossings.

Let $\mathfrak{a}$ be a coherent ideal on a complex space X, not identically zero on any branch of any germ of X. A *log resolution* of $\mathfrak{a}$ is a proper bimeromorphic map $\pi\colon Y \to X$ from a complex manifold Y that is a composite of blowups with smooth centers, finite over each relatively compact open subset of X, such that $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-D)$ for an effective divisor D on Y whose support has simple normal crossings.

Log resolutions exist. Let $\rho\colon M \to X$ be a resolution given by Theorem A.5.4. The zero set of the pulled-back ideal $\mathfrak{a}\mathcal{O}_M$ is $\rho^{-1}(V(\mathfrak{a}))$, which is nowhere dense, because ρ is bimeromorphic and $V(\mathfrak{a})$ is nowhere dense in X (Lemma A.3.1); so $\mathfrak{a}\mathcal{O}_M$ is not identically zero on any component of M. We principalize it by Theorem A.5.5, taking as the given simple normal crossings divisor the exceptional divisor D_ρ of ρ, formed by the inverse images of its blowup centers, and obtain $\sigma\colon Y \to M$ with $\mathfrak{a}\mathcal{O}_Y = \mathcal{O}_Y(-D)$, $D = H$.

The argument given above for the support of H applies also to $\sigma^{-1}(D_\rho)$. This set lies in the accumulated divisor by the same induction, and it is the zero set of the pullback of a local equation of D_ρ. This pullback is not identically zero on any component of Y: the map σ is an isomorphism off the support of D, which is the nowhere dense set $\sigma^{-1}(V(\mathfrak{a}\mathcal{O}_M))$, so σ is bimeromorphic and $\sigma^{-1}(D_\rho)$ is nowhere dense by Lemma A.3.1. Hence the support of D together with $\sigma^{-1}(D_\rho)$ is a union of irreducible components of the accumulated divisor, and it has simple normal crossings. The composite $\pi = \rho \circ \sigma$ is proper, and it is bimeromorphic: it is an isomorphism over $X_{\mathrm{reg}} \setminus V(\mathfrak{a})$, the set $\sigma^{-1}(\rho^{-1}(V(\mathfrak{a})))$ is the support of D, a hypersurface, and $\sigma^{-1}(\rho^{-1}(X_{\mathrm{sing}}))$ is nowhere dense, because $\rho^{-1}(X_{\mathrm{sing}})$ is nowhere dense and σ is an isomorphism off the support of D.

When X is a manifold, the centers of a log resolution of $\mathfrak{a}$ are closed submanifolds of positive codimension at each of their points, because the ideal of a center must not vanish identically on any component (Definition A.5.1). No resolution step is needed then: Theorem A.5.5 with the empty divisor gives a log resolution that is an isomorphism off $V(\mathfrak{a})$. These are the log resolutions used in Chapters 12 and 13; in Section 13.1, a log resolution of the ideal in the type of a function with analytic singularities is also called a log resolution of that function.

A.6 Proper images and relative principalization

We now show that the image of a proper holomorphic map is analytic (Remmert's theorem). It is proved together with the Remmert–Stein theorem, which says that an analytic set whose branches all have large dimension remains analytic after adding its limit points on an analytic set of smaller dimension. In this book the mapping theorem is applied mainly to images of exceptional sets of compositions of blowups. We then prove a relative principalization theorem: one sequence of blowups principalizes

an ideal on a compact manifold and, at the same time, its restriction to a given submanifold, and the ideal of the submanifold transforms in a controlled way.

Theorem A.6.1 (Remmert–Stein) *Let X be a complex space, let $p \geq 0$ be an integer, and let $A \subset X$ be an analytic subset of dimension at most p. Let Z be an analytic subset of $X \setminus A$ all of whose local branches have dimension greater than p. Then the closure of Z in X is analytic.*

Theorem A.6.2 (Remmert) *The image of a proper holomorphic map between complex spaces is analytic.*

Proof We prove the two theorems together by induction on dimension. Let R_m denote the assertion of Theorem A.6.1 for sets Z of dimension at most m, and P_m the assertion of Theorem A.6.2 for maps whose source has dimension at most m. Both assertions are local on X. A proper map has compact inverse image over a compact neighborhood of a point of the target, and near this compact set its source has finite dimension; so, locally on the target, every proper map falls under some P_m. The assertion P_0 holds because a source of dimension zero is discrete, so the image of a proper map from it is closed and locally finite, hence analytic. The assertion R_0 holds trivially: a set Z of dimension zero has no branches of dimension greater than $p \geq 0$, so $Z = \varnothing$. We show that P_{m-1} implies R_m, and that R_m and P_{m-1} imply P_m.

Step 1: P_{m-1} implies R_m; reductions. Since the assertion is local, we may assume that X is an analytic subset of a domain $\Omega \subset \mathbb{C}^N$; then Z is an analytic subset of $\Omega \setminus A$, and its closure in X is its closure in Ω. We first reduce to the case where A is a submanifold of dimension p, by induction on p. Let A'' be the union of A_{sing} and of the irreducible components of A of dimension less than p; it is analytic of dimension at most $p-1$ (empty if $p = 0$), and $A \setminus A''$ is a closed submanifold of $\Omega \setminus A''$ of dimension p. Suppose that the closure Z_1 of Z in $\Omega \setminus A''$ is analytic. No branch B of a germ of Z_1 is contained in A: since Z is dense in Z_1, the branches not contained in A already have union Z_1 near the base point, and the decomposition into branches is irredundant. Hence $B \setminus A$ is dense in B, and at its regular points outside the other branches B coincides with Z, whose branches have dimension greater than p. Hence all branches of Z_1 have dimension greater than $p > p-1$, and by the induction hypothesis on p the closure of Z_1 in Ω, which is the closure of Z, is analytic.

Thus we may assume that A is a p-dimensional submanifold. Fix a point of A in the closure of Z; after a holomorphic change of coordinates it is the origin, and near it A is a linear subspace L of dimension p. Finally, the irreducible components of Z are pure-dimensional by Corollary A.2.14(1), and their dimensions exceed p, because each component contains regular points of Z near which Z coincides with it. The union of the components of a given dimension s is analytic in $\Omega \setminus A$, by Corollary A.2.13, and only the finitely many values $p < s \leq m$ occur; so we may assume that Z is pure of dimension s, with $p < s \leq m$. We shall present Z near the origin as a branched covering over an open subset of $\mathbb{C}^s$ and write down equations for its closure.

Choice of a projection. We choose linear forms $\xi_1, \ldots, \xi_s$ on $\mathbb{C}^N$ with the following properties: (i) for each $j < s$, the form ξ_{j+1} does not vanish identically

on any irreducible component of $Z \cap \{\xi_1 = \cdots = \xi_j = 0\}$; (ii) the restrictions of $\xi_1, \ldots, \xi_p$ to L are linearly independent; (iii) at a point z_α chosen in each connected component Z°_α of the manifold $Z_{\rm reg}$, the restrictions of $\xi_1, \ldots, \xi_s$ to the tangent space of Z are linearly independent.

We choose the forms one at a time. Suppose that $\xi_1, \ldots, \xi_j$, $j < s$, have been chosen and that $Z_j = Z \cap \{\xi_1 = \cdots = \xi_j = 0\}$ is pure of dimension $s - j$, which holds for $j = 0$. Since Ω has a countable basis, Z_j has countably many irreducible components and $Z_{\rm reg}$ has countably many connected components Z°_α, by Corollary A.2.13. The three properties exclude, for ξ_{j+1}, countably many proper linear subspaces of the dual space of $\mathbb{C}^N$: by (i), the forms vanishing on a given component of Z_j, which has positive dimension $s - j$; by (ii), if $j < p$, the forms whose restriction to L is a combination of those of $\xi_1, \ldots, \xi_j$; and by (iii), the forms whose restriction to the s-dimensional tangent space of Z at z_α is a combination of those of $\xi_1, \ldots, \xi_j$. A countable union of proper linear subspaces has Lebesgue measure zero, so ξ_{j+1} can be chosen. Then Z_{j+1} is empty or pure of dimension $s - j - 1$, by Proposition A.4.6; here we use that a holomorphic function vanishing identically on a local branch of Z_j vanishes on the irreducible component of Z_j containing this branch. Indeed, the branch contains regular points of Z_j outside the other branches, by Corollary A.2.4, and hence a nonempty open subset of the regular locus of this component, so Corollary A.2.13 applies.

Replacing ξ_j, $j > p$, by $\xi_j - \sum_{i \le p} a_{ji}\xi_i$ with suitable constants, we may assume that $\xi_{p+1}, \ldots, \xi_s$ vanish on L; this triangular change does not affect the zero sets $\{\xi_1 = \cdots = \xi_j = 0\}$ or the independence at the points z_α. Complete $\xi_1, \ldots, \xi_s$ by linear forms vanishing on L to linear coordinates $z = (z', z'')$ with $z' = (\xi_1, \ldots, \xi_s) \in \mathbb{C}^s$. Then $L = \mathbb{C}^p \times \{0\}$, and we let $\pi(z) = z'$.

A proper projection. By (i), the set $Z \cap \pi^{-1}(0)$ is discrete in $\Omega \setminus A$, and by (ii), $A \cap \pi^{-1}(0) = \{0\}$ near 0. Choose a ball B'' centered at 0 in the z''-space whose boundary sphere does not meet $Z \cap \pi^{-1}(0)$; this excludes countably many radii. The set $Z \cup A$ is closed near 0 and does not meet the compact set $\{0\} \times \partial B''$, so there is a ball $B' \subset \mathbb{C}^s$ centered at 0 with $(Z \cup A) \cap (\overline{B'} \times \partial B'') = \varnothing$. Let $L' = B' \cap (\mathbb{C}^p \times \{0\})$, so that $A \cap (B' \times B'')$ lies over L'. The projection of $Z \cap (B' \times B'')$ over $B' \setminus L'$ is proper, because over a compact set $K \subset B' \setminus L'$ it is the compact set $(Z \cup A) \cap (K \times \overline{B''})$, and its fibers are compact analytic subsets of B'', hence finite by Corollary A.4.3.

The exceptional sets in the base. Let S_1 be the set of singular points of Z over $B' \setminus L'$. It is analytic of dimension at most $s - 1 \le m - 1$, and its projection is proper, so $S'_1 = \pi(S_1)$ is analytic in $B' \setminus L'$ by P_{m-1}. Over $B' \setminus (L' \cup S'_1)$ the set Z is a manifold of dimension s; let S_2 be the set of its points where $\pi|_Z$ has rank less than s. It is analytic, given locally by a Jacobian determinant. Each connected component of this manifold is an open subset of some Z°_α, and on the connected manifold Z°_α the set where the rank of π is less than s is a proper analytic subset, since the rank is s at z_α by (iii); so it is nowhere dense, and S_2 has dimension at most $s - 1$. By P_{m-1}, $S'_2 = \pi(S_2)$ is analytic in $B' \setminus (L' \cup S'_1)$.

By Lemma A.2.12(3), the sets S'_1 and S'_2 have dimension at most $s - 1$, so they are nowhere dense in the open subsets of $\mathbb{C}^s$ in which they are analytic. Applying Lemma A.1.7 first to L', then to S'_1 and then to S'_2, we see that

$$G = B' \setminus (L' \cup S'_1 \cup S'_2)$$

is connected; applying it in the opposite order, we see that every bounded holomorphic function on G extends holomorphically to B'.

The set $Z \cap \pi^{-1}(G)$ is dense in $Z \cap (B' \times B'')$. Indeed, on each irreducible component $Z_\alpha = \overline{Z^\circ_\alpha}$, the functions $\xi_{p+1}, \dots, \xi_s$ do not all vanish identically, because $\pi|_Z$ is open near z_α and $s > p$; so $Z_\alpha \cap \pi^{-1}(L')$ is a proper analytic subset of Z_α, and it is nowhere dense by Corollary A.2.14(2). Moreover, the points at which $\pi|_Z$ has rank s are dense in Z, by the previous paragraph, and near such points π is open; hence $\pi^{-1}(S'_1)$ and $\pi^{-1}(S'_2)$, inverse images of nowhere dense sets, are nowhere dense in Z over $B' \setminus L'$ and over $B' \setminus (L' \cup S'_1)$, respectively.

Over G, the projection $Z \cap \pi^{-1}(G) \to G$ is proper and, by Proposition 7.1.2(1) applied in charts of the s-dimensional manifold Z over G, a local biholomorphism, so its image is open and closed in the connected set G; it is nonempty, because $Z \cap (B' \times B'')$ is nonempty and $Z \cap \pi^{-1}(G)$ is dense in it. Hence it is a covering of G with a fixed number $q \geq 1$ of sheets. Indeed, let $z' \in G$ and let $y_1, \dots, y_l$ be the points of the finite fiber over z'; choose disjoint neighborhoods V_i of the y_i mapped biholomorphically onto neighborhoods of z', and, by properness, a connected neighborhood N of z' in G, contained in all these images, such that every point of Z over N lies in $\bigcup_i V_i$. Then the part of Z over N is the disjoint union of l graphs over N, so the number of points in the fibers is locally constant, hence constant on G.

Equations for the closure. For a linear form λ on the z''-space and $z' \in G$ put

$$P_\lambda(z', T) = \prod_{z \in Z \cap \pi^{-1}(z')} (T - \lambda(z'')).$$

The coefficient of T^{q-k} is a homogeneous polynomial of degree k in the coefficients of λ, whose coefficients are symmetric polynomials in the coordinates of the q points of the fiber; these are bounded holomorphic functions on G, and they extend holomorphically to B'. Thus $P_\lambda(z', \lambda(z'')) = \sum_\beta g_\beta(z)\lambda^\beta$, where λ^β runs over the monomials in the coefficients of λ and the g_β are holomorphic on $B' \times B''$. Let $W = \{g_\beta = 0 \text{ for all } \beta\}$, the set of points where $P_\lambda(z', \lambda(z'')) = 0$ for every λ; it is analytic. Over G, the set W is Z: a point outside a finite fiber can be separated from all its points by a linear form. Every fiber of W over B' has at most q points, because a linear form taking distinct values at $q+1$ points of one fiber would give $q+1$ distinct roots of a polynomial of degree q.

The closure. Choose a smaller ball $B''_1 \subset B''$ whose boundary does not meet the finite set $W \cap \pi^{-1}(0)$, and then a ball $B'_1 \subset B'$ centered at 0 with $W \cap (\overline{B'_1} \times \partial B''_1) = \varnothing$. Then $W_1 = W \cap (B'_1 \times B''_1)$ is proper with finite fibers over B'_1. Every branch of W_1 has dimension at most s: at a regular point of a branch of dimension e, the projection restricted to the branch has finite fibers, so by Lemma A.2.11, applied near a point where its rank is maximal, its maximal rank is e, and $e \leq s$.

Let W_s be the union of the irreducible components of W_1 of dimension s, and $W_{<s}$ the union of the others. We claim that the closure of Z in $B'_1 \times B''_1$ is W_s, which is analytic; this proves R_m. Each component of W_s is pure of dimension s,

by Corollary A.2.14(1), and the image under π of every neighborhood of each of its points in it contains a neighborhood of the image point, by Lemma A.4.5; so its points over the dense set G are dense in it, and they belong to Z. Conversely, let $y \in Z \cap (B'_1 \times B''_1)$ lie over G. The germ $(W, y) = (Z, y)$ is a manifold germ of dimension s, hence irreducible by Lemma A.2.9, so it equals (W_s, y) or $(W_{<s}, y)$, and the latter has smaller dimension. Hence $y \in W_s$, and since $Z \cap \pi^{-1}(G)$ is dense in Z and W_s is closed, the closure of Z in $B'_1 \times B''_1$ is contained in W_s.

Step 2: R_m and P_{m-1} imply P_m. Let $F\colon Y \to X$ be proper, with $\dim Y \le m$. Let $\rho\colon M \to Y$ be a resolution given by Theorem A.5.4. It is proper, and it is surjective, because its image is closed and contains the dense set Y_{reg}; so replacing F by $F \circ \rho$ does not change the image. Moreover $\dim M \le m$: the set $\rho^{-1}(Y_{\mathrm{sing}})$ is nowhere dense, because ρ is bimeromorphic (Lemma A.3.1), so every connected component of M contains a nonempty open set mapped isomorphically onto an open subset of Y_{reg}. Fix a point of X, a relatively compact open neighborhood U' of it and an open neighborhood $U'' \Subset X$ of $\overline{U'}$ embedded in a domain of some $\mathbb{C}^N$. The compact set $(F \circ \rho)^{-1}(\overline{U'})$ meets only finitely many connected components $M_1, \dots, M_k$ of the manifold $(F \circ \rho)^{-1}(U'')$, and the restriction $F_j\colon M_j \to U''$ of $F \circ \rho$ is proper. It suffices to show that each $F_j(M_j)$ is analytic in U''.

Let r be the maximal rank of the differential of F_j on the connected manifold M_j. If $r = 0$, the components of F_j have zero differential and are constant, so the image is a point. Let $r \ge 1$. The critical set $C = \{\operatorname{rank} \mathrm{d}F_j < r\}$ is an analytic subset of M_j, defined by minors, and it is not all of M_j; so it has dimension at most $m - 1$, and $F_j(C)$ is analytic by P_{m-1}. It has dimension at most $r - 1$. Indeed, by Corollary A.2.10(2), C is a finite union of complex submanifolds of open subsets of M_j, and the restriction of F_j to each of them has rank at most $r - 1$; so $F_j(C)$ is $\mathcal{H}^{2r}$-null by Lemma A.2.12(1), and $\dim F_j(C) \le r - 1$ by Lemma A.2.12(2).

If $F_j(C) = F_j(M_j)$, we are done. Otherwise $F_j^{-1}(F_j(C))$ is an analytic subset of the connected manifold M_j different from M_j, and on its complement M'_j the rank of F_j is constantly r. The restriction $M'_j \to U'' \setminus F_j(C)$ is proper, and a proper map of constant rank has analytic image. Indeed, at a point w of the image, cover the compact fiber by finitely many open sets V_l such that, by Lemma A.2.11 applied in charts of M'_j, each $F_j(V_l)$ is a closed submanifold of dimension r of a neighborhood of w. By properness, the inverse image of a small neighborhood O of w lies in $\bigcup_l V_l$, so the image meets O in a finite union of closed submanifolds of dimension r. Thus $F_j(M_j) \setminus F_j(C) = F_j(M'_j)$ is an analytic subset of $U'' \setminus F_j(C)$, pure of dimension $r \le m$. Since $\dim F_j(C) < r$, the assertion R_m shows that its closure in U'' is analytic. This closure is $F_j(M_j)$, because M'_j is dense in M_j and $F_j(M_j)$ is closed by properness. This proves P_m and completes the induction. □

Corollary A.6.3 *The composite of two proper bimeromorphic holomorphic maps $g\colon Z \to Y$ and $f\colon Y \to X$ between complex spaces is proper and bimeromorphic.*

Proof The composite is proper. Let $E_X \subset X$ and $E_Y \subset Y$ be nowhere dense analytic subsets as in the definition of bimeromorphic maps for f and g. The image $f(E_Y)$ is analytic by Theorem A.6.2, and it is nowhere dense: an open set $G \subset f(E_Y)$

satisfies $G \setminus E_X \subset f(E_Y \setminus f^{-1}(E_X))$, which corresponds under the isomorphism $f\colon Y \setminus f^{-1}(E_X) \to X \setminus E_X$ to the nowhere dense set $E_Y \setminus f^{-1}(E_X)$; so $G \subset E_X$, and $G = \varnothing$. Hence $B = E_X \cup f(E_Y)$ is a nowhere dense analytic subset of X. The map f is an isomorphism over $X \setminus B$, and $f^{-1}(X \setminus B)$ does not meet E_Y, so $f \circ g$ is an isomorphism over $X \setminus B$. Finally, $f^{-1}(B)$ is a nowhere dense analytic subset of Y, and $(f \circ g)^{-1}(B) = g^{-1}(f^{-1}(B))$ is nowhere dense in Z, by two applications of Lemma A.3.1. □

Definition A.6.4 A composition of blowups with smooth centers on a complex manifold is *adapted* to a closed submanifold W if at each step the center is either contained in the current strict transform of W and of positive codimension in it at each of its points, or disjoint from it. The identity, the composition of no blowups, is allowed.

Theorem A.6.5 *Let M be a compact connected complex manifold, $W \subsetneq M$ a closed connected smooth submanifold, and $\mathfrak{a} \subset \mathcal{O}_M$ a coherent ideal whose restriction $\mathfrak{a}\mathcal{O}_W$ is nonzero. There is an adapted composition $f\colon M' \to M$, with strict transform W', and effective divisors H, H' on M' such that H is exceptional for f, H' is disjoint from W', and*

$$\mathcal{I}_W \mathcal{O}_{M'} = \mathcal{O}_{M'}(-H)\mathcal{I}_{W'}, \tag{A.9}$$

$$\mathfrak{a}\mathcal{O}_{M'} = \mathcal{O}_{M'}(-H - H'), \tag{A.10}$$

$$(\mathfrak{a}\mathcal{O}_W)\mathcal{O}_{W'} = \mathcal{O}_{W'}(-H|_{W'}). \tag{A.11}$$

The restriction $f|_{W'}\colon W' \to W$ is a composition of smooth-center blowups and an isomorphism outside a proper analytic subset of W.

Proof Let $\mathfrak{b} = \mathfrak{a}\mathcal{O}_W$. We first principalize $\mathfrak{b}$ by blowups of W, perform the same blowups on M, and then principalize the remaining ideal away from the strict transform of W. If $\dim W = 0$, then W is a point, $\mathfrak{b}$ is the unit ideal, the first sequence below is empty, and the argument applies with $H = 0$.

Step 1: principalizing the restriction. Apply Theorem A.5.5 to the ideal $\mathfrak{b}$ on the compact manifold W, with empty divisor. We obtain a finite sequence of blowups $W_i \to W_{i-1}$ with smooth centers $C_i \subset W_{i-1}$, $1 \le i \le t$, and ideals $\mathfrak{b}_i$ with $\mathfrak{b}_0 = \mathfrak{b}$, $\mathfrak{b}_i = \mathcal{O}_{W_i}(E_i^W)\mathfrak{b}_{i-1}\mathcal{O}_{W_i}$, where E_i^W is the inverse image of C_i, and $\mathfrak{b}_t = \mathcal{O}_{W_t}$. By induction on i, each W_{i-1} is connected, $\mathfrak{b}_{i-1}$ is nonzero, and C_i has positive codimension in W_{i-1} at each of its points; for $i = 1$, the first two assertions are hypotheses of the theorem. Assume them for some $i \le t$. A nonzero coherent ideal on a connected manifold has nonzero stalks at all points, since the points at which its stalk vanishes form an open set, and so, by the identity theorem, do the points at which it does not vanish; hence its zero set is nowhere dense. So $C_i \subset V(\mathfrak{b}_{i-1})$ is nowhere dense in W_{i-1}, that is, of positive codimension at each of its points. Then $W_i \setminus E_i^W \cong W_{i-1} \setminus C_i$ is connected by Lemma A.1.7 and dense in W_i, so W_i is connected; and $\mathfrak{b}_i$ is nonzero, because it agrees with $\mathfrak{b}_{i-1}$ on this set.

We perform the same blowups on the ambient manifolds: let $M_0 = M$ and let $\sigma_i \colon M_i \to M_{i-1}$ be the blowup along C_i, where $W_{i-1} \subset M_{i-1}$ is the strict transform of W. The center C_i is closed in M_{i-1}, because W_{i-1} is closed, and it has codimension at least two in M_{i-1}. By Proposition A.5.3, the strict transform of W_{i-1} in M_i is $\mathrm{Bl}_{C_i} W_{i-1} = W_i$, and $E_i \cap W_i = E_i^W$, where $E_i = \sigma_i^{-1}(C_i)$. More precisely, in the charts of that proposition whose pivot is one of the coordinates $z_1, \dots, z_k$ along W_{i-1}, which are the only ones meeting W_i, a local equation e_i of E_i restricts to a local equation of E_i^W. This sequence is adapted to W.

Step 2: the transformed ideals. Put $H_0 = 0$ and $\mathfrak{a}_0 = \mathfrak{a}$, and define inductively

$$H_i = \sigma_i^* H_{i-1} + E_i, \qquad \mathfrak{a}_i = \mathcal{O}_{M_i}(E_i)\mathfrak{a}_{i-1}\mathcal{O}_{M_i}.$$

We show by induction that $\mathfrak{a}_i$ is a holomorphic ideal and that $\mathfrak{a}_i \mathcal{O}_{W_i} = \mathfrak{b}_i$. Every local section h of $\mathfrak{a}_{i-1}$ vanishes on C_i, because its restriction to W_{i-1} belongs to $\mathfrak{b}_{i-1}$ and $C_i \subset V(\mathfrak{b}_{i-1})$. Hence $h \in \mathcal{I}_{C_i}$, and $h \circ \sigma_i$ belongs to $\mathcal{I}_{C_i}\mathcal{O}_{M_i} = \mathcal{O}_{M_i}(-E_i)$; that is, it is divisible by the local equation e_i of E_i, and $\mathfrak{a}_i$ is holomorphic. Restricting $(h \circ \sigma_i)/e_i$ to W_i divides $h|_{W_{i-1}} \circ \sigma_i|_{W_i}$ by the local equation $e_i|_{W_i}$ of E_i^W, which gives $\mathfrak{a}_i \mathcal{O}_{W_i} = \mathfrak{b}_i$; this holds also when C_i is a hypersurface in W_{i-1}. Iterating (A.7) and the definition of $\mathfrak{a}_i$, we obtain

$$\mathcal{I}_W \mathcal{O}_{M_t} = \mathcal{O}_{M_t}(-H_t)\mathcal{I}_{W_t}, \qquad \mathfrak{a}\mathcal{O}_{M_t} = \mathcal{O}_{M_t}(-H_t)\mathfrak{a}_t.$$

Since $\mathfrak{b}_t = \mathcal{O}_{W_t}$, the ideal $\mathfrak{a}_t$ is the unit ideal at each point of W_t: a local section whose restriction to W_t does not vanish at a point is a unit of the local ring of M_t there. Hence $V(\mathfrak{a}_t)$ is disjoint from W_t.

Step 3: principalizing away from W_t. The manifold M_t is compact, and it is connected by the argument given for the W_i in Step 1; the ideal $\mathfrak{a}_t$ is not zero, being the unit ideal near W_t. Apply Theorem A.5.5 to $\mathfrak{a}_t$ on M_t, with empty divisor. Each center lies in the zero set of the current transformed ideal; since the blowups are isomorphisms near the successive copies of W_t, which do not meet the centers, these transformed ideals remain the unit ideal near these copies, and all centers are disjoint from them. Let $g \colon M' \to M_t$ be the composite, and let H' be the effective divisor with $\mathfrak{a}_t \mathcal{O}_{M'} = \mathcal{O}_{M'}(-H')$ obtained by iterating (A.8); put $H = g^* H_t$ and $f = \sigma_1 \circ \cdots \circ \sigma_t \circ g$. The composition f is adapted to W, and the strict transform $W' = g^{-1}(W_t)$ is mapped isomorphically onto W_t. By Proposition A.5.3, $\mathcal{I}_{W_t}\mathcal{O}_{M'} = \mathcal{I}_{W'}$, and pulling back the two identities of Step 2 gives (A.9) and (A.10). The divisor H' is supported on the exceptional divisors of g, which are disjoint from W'.

Step 4: exceptional divisors and restriction. Every component of the support of H maps into the union of the images in M of the centers C_i; each such image lies in W, is analytic by Theorem A.6.2, and has dimension at most $\dim C_i < \dim W \le \dim M - 1$ by Lemma A.2.12(3), so it has codimension at least two in M. Hence H is exceptional for f, which is proper and bimeromorphic by Proposition A.5.2 and Corollary A.6.3.

No component of H contains W': by Step 1, the restriction to W_i of a local equation of E_i is a local equation of the hypersurface E_i^W of W_i, so by induction the restriction to W_t of a local equation of H_t is not identically zero, and g is an isomorphism near

W'. Hence $H|_{W'}$ is defined, and restricting (A.10) to W', where H' vanishes, gives (A.11). Finally, $f|_{W'}$ is the composition of the blowups $W_i \to W_{i-1}$ followed by the isomorphism $W' \to W_t$. It is an isomorphism outside the union of the images in W of the centers C_i that have codimension at least two in W_{i-1}; this union is a finite union of analytic subsets of dimension smaller than $\dim W$, hence a proper analytic subset of W. □

A.7 Connected fibers and resolution models

A proper bimeromorphic map can contract analytic sets, so its fibers may have positive dimension. Over a unibranch space they are nevertheless connected (Theorem A.7.1). Consequently a finite bimeromorphic map onto a normal space is an isomorphism, and a proper bimeromorphic map between manifolds is an isomorphism outside a set of codimension at least two. We then introduce smooth models and prime divisors over a complex space, and show that every prime divisor over a compact space is realized after finitely many blowups with smooth centers.

Theorem A.7.1 *Let $f\colon Y \to X$ be a proper bimeromorphic holomorphic map between complex spaces, and let $x \in X$ be a point at which the germ (X, x) is irreducible. Then the fiber $f^{-1}(x)$ is connected. In particular, if X is unibranch, every fiber of f is connected.*

Proof Choose a nowhere dense analytic subset $A \subset X$ such that $f^{-1}(A)$ is nowhere dense and f is an isomorphism over $X \setminus A$. Suppose that the compact fiber over x is disconnected, and write it as the union of two disjoint nonempty compact sets. Since Y is Hausdorff, they have disjoint open neighborhoods V_1 and V_2. Since f is proper, hence closed, $U = X \setminus f(Y \setminus (V_1 \cup V_2))$ is an open neighborhood of x with $f^{-1}(U) \subset V_1 \cup V_2$. As (X, x) is irreducible, Corollary A.3.9 allows us to shrink U so that $U \setminus A$ is connected. Each set $V_i \cap f^{-1}(U)$ is open and nonempty, so it meets the dense set $Y \setminus f^{-1}(A)$. Therefore V_1 and V_2 split $f^{-1}(U \setminus A)$ into two disjoint nonempty open sets. This contradicts the connectedness of $U \setminus A$, which is isomorphic to $f^{-1}(U \setminus A)$. □

Corollary A.7.2 *A finite bimeromorphic holomorphic map to a normal complex space is an isomorphism. Let $\phi\colon S \to X$ be a holomorphic map from a normal complex space to a complex space such that $\phi^{-1}(X_{\mathrm{sing}})$ is nowhere dense in S. Then ϕ factors uniquely through the normalization $\nu\colon \widetilde{X} \to X$: there is a unique holomorphic map $\psi\colon S \to \widetilde{X}$ with $\nu \circ \psi = \phi$.*

Proof Let $f\colon Y \to X$ be finite and bimeromorphic, with X normal, and let $A \subset X$ be a nowhere dense analytic subset such that f is an isomorphism over $X \setminus A$ and $f^{-1}(A)$ is nowhere dense. Normal spaces are unibranch, so the finite fibers of f are connected by Theorem A.7.1, hence consist of at most one point. The image of f is

closed, because f is proper, and contains the dense set $X \setminus A$; so f is a continuous closed bijection, that is, a homeomorphism. Let $x \in X$ and $y = f^{-1}(x)$. Choose a neighborhood V of y, relatively compact in the domain of a local model of Y, so that the coordinate functions of the model are bounded on V. Composed with f^{-1}, they are bounded holomorphic functions on $f(V) \setminus A$, where f^{-1} is holomorphic, and $f(V)$ is an open neighborhood of x. By Lemma A.3.4 they extend holomorphically to $f(V)$, and the extensions are the continuous functions (coordinates) $\circ f^{-1}$. Hence f^{-1} is holomorphic.

For the second assertion, over the dense open set $S \setminus \phi^{-1}(X_{\text{sing}})$ there is a unique lift $\psi_0 = \nu^{-1} \circ \phi$, because ν is an isomorphism over X_{reg}. We extend it by the argument used for the uniqueness in the proof of Theorem A.3.6. Let $s \in S$ and $x = \phi(s)$, choose disjoint neighborhoods W_l of the finitely many points of $\nu^{-1}(x)$, relatively compact in domains W_l' of local models of $\widetilde{X}$, and a neighborhood U of x with $\nu^{-1}(U) \subset \bigcup_l W_l$, which exists by properness. Since S is normal, hence unibranch, Corollary A.3.9 gives a neighborhood V of s with $\phi(V) \subset U$ such that $V \setminus \phi^{-1}(X_{\text{sing}})$ is connected. Then ψ_0 maps this set into a single W_l. The coordinate functions of W_l' composed with ψ_0 are bounded and extend holomorphically to V, by Lemma A.3.4; the extended map takes values in $\overline{W_l} \subset W_l'$, and it satisfies the equations of the local model, which hold on a dense subset of V. This gives a holomorphic lift on V, with $\nu \circ \psi = \phi$ by density. Lifts agree on the dense set where ψ_0 is defined, so these local lifts glue, and the lift is unique. □

Proposition A.7.3 *A proper bimeromorphic holomorphic map $f\colon M' \to M$ between complex manifolds is an isomorphism outside an analytic subset S of M of codimension at least two. In particular, every irreducible hypersurface D in M has a well-defined* strict transform *in M', namely the closure D' of $f^{-1}(D \setminus S)$, which does not depend on the choice of S; it is an irreducible hypersurface, $f(D') = D$, and f maps D' isomorphically onto D over the dense open subset $D \setminus S$ of D.*

Proof We take for S the image of the critical set of f; the main point is that a component of S of codimension one would contain a point over which f is finite, hence an isomorphism by Corollary A.7.2.

Step 1: the set S. We may work over each connected component of M, of dimension n say; every connected component of its inverse image has dimension n as well, because it contains a nonempty open set mapped isomorphically onto an open subset of M, by Lemma A.3.1. The map f is surjective, since its image is closed and dense. The critical set of f, where the Jacobian determinant vanishes in local coordinates, is analytic, and its image S is analytic by Theorem A.6.2. Over $M \setminus S$ the map f is proper and, by Proposition 7.1.2(1), a local biholomorphism, so its fibers are discrete and finite. They are connected by Theorem A.7.1, and nonempty; so they are single points, and f is a bijective local biholomorphism, hence an isomorphism, over $M \setminus S$. Since f is an isomorphism over a dense open subset of M, it has no critical points over this set, so S does not meet it and is nowhere dense.

Step 2: S has codimension at least two. Suppose that S has an irreducible component D_0 of codimension one. Choose a point of $(D_0)_{\text{reg}}$ outside the other components of

S, and a coordinate polydisc U centered at it in which $D_0 \cap U = S \cap U = \{h = 0\}$ for one of the coordinate functions h; then $D_0 \cap U$ is a connected manifold. Let $A \subset M$ be a nowhere dense analytic subset such that $f^{-1}(A)$ is nowhere dense and f is an isomorphism over $M \setminus A$. The function $h \circ f$ does not vanish identically on any connected component V of $f^{-1}(U)$: otherwise f would map the nonempty open set $V \setminus f^{-1}(A)$ isomorphically onto an open subset of M contained in the hypersurface D_0, which is impossible. By Proposition A.4.6, $E = f^{-1}(D_0 \cap U)$ is pure of dimension $n-1$.

Let $K \subset D_0 \cap U$ be a compact neighborhood of the chosen point in D_0. Only finitely many irreducible components $E_1, \dots, E_k$ of E meet the compact set $f^{-1}(K)$, and by Theorem A.6.2 each image $f(E_j)$ is an analytic subset of $D_0 \cap U$. If its dimension is $n-1$, then $f(E_j)$ contains an open subset of the connected manifold $D_0 \cap U$ and therefore equals it, since proper analytic subsets of connected manifolds are nowhere dense (Lemma A.1.6); we then call E_j dominating. Otherwise $f(E_j)$ has dimension at most $n-2$.

For a dominating E_j, the maximal rank of f restricted to the connected manifold $(E_j)_{\mathrm{reg}}$, as a map into the coordinate hyperplane $D_0 \cap U$, is $n-1$. Otherwise $f((E_j)_{\mathrm{reg}})$ would be $\mathcal{H}^{2n-2}$-null by Lemma A.2.12(1); so would $f((E_j)_{\mathrm{sing}})$, by the same lemma, since $(E_j)_{\mathrm{sing}}$ has dimension at most $n-2$ by Corollary A.2.10(1) and is therefore, by (2) of the same corollary, a finite union of complex submanifolds of dimension at most $n-2$; and $f(E_j)$ would have dimension at most $n-2$ by Lemma A.2.12(2).

Let R_j be the union of $(E_j)_{\mathrm{sing}}$ and of the regular points where this rank is less than $n-1$. It is analytic. Indeed, near a point of E_j, let e be a local equation of the irreducible hypersurface E_j as in Lemma A.2.16, and let f' be the map formed by the $n-1$ coordinates of f in the coordinate hyperplane $D_0 \cap U$. At a regular point p of E_j we have $\mathrm{d}e(p) \neq 0$ and $T_p E_j = \ker \mathrm{d}e(p)$, so the rank of $f'|_{E_j}$ at p is $n-1$ exactly when the Jacobian determinant of (e, f') does not vanish at p. Hence R_j is, near this point, the union of the analytic set $(E_j)_{\mathrm{sing}}$ and of the zero set in E_j of this Jacobian determinant. It is different from E_j, because the rank $n-1$ is attained on $(E_j)_{\mathrm{reg}}$; so it is nowhere dense in the irreducible set E_j and has dimension at most $n-2$, by Corollary A.2.14(2), and $f(R_j)$ is analytic of dimension at most $n-2$, by Theorem A.6.2 and Lemma A.2.12(3).

Choose a point a in the interior of K outside the finitely many sets $f(E_j)$, E_j not dominating, and $f(R_j)$, E_j dominating. Let $p \in f^{-1}(a)$. Since $f^{-1}(a) \subset E \cap f^{-1}(K)$, the point p lies on some of the components $E_1, \dots, E_k$. Each of them that contains p is dominating, and $p \notin R_j$, so $f|_{E_j}$ is a local biholomorphism at p onto an open subset of D_0, by Proposition 7.1.2(1) in charts, and p is isolated in $f^{-1}(a) \cap E_j$. The components of E not containing p form a locally finite family of closed sets, so near p the set $f^{-1}(a)$ lies in the union of the components containing p, and p is isolated in $f^{-1}(a)$. Thus $f^{-1}(a)$ is discrete, and being compact, it is finite.

Let $p_1, \dots, p_l$ be the points of $f^{-1}(a)$, and choose disjoint coordinate balls $B_1, \dots, B_l$ around them. By properness there is a neighborhood U_0 of a with $f^{-1}(U_0) \subset \bigcup_i B_i$. The fibers of f over U_0 are compact analytic subsets of these balls, hence finite by Corollary A.4.3. Thus f is finite and bimeromorphic over the manifold U_0, which is normal, and it is an isomorphism over U_0 by Corollary A.7.2.

In particular f has no critical points over U_0, which contradicts $a \in S$. Therefore S has codimension at least two.

Step 3: strict transforms. Let D be an irreducible hypersurface. Then $S \cap D$ has dimension at most $n-2$, so $D \setminus S$ is dense in D by Corollary A.2.14(2). The closure D' of $f^{-1}(D \setminus S) = f^{-1}(D) \setminus f^{-1}(S)$ is analytic by Lemma A.2.5, and f is an isomorphism from $D' \setminus f^{-1}(S) = f^{-1}(D \setminus S)$ onto $D \setminus S$. The set $f(D')$ is closed, because f is proper, it contains the dense subset $D \setminus S$ of D, and it is contained in the closed set D; so $f(D') = D$.

The set $Q = f^{-1}(D_{\mathrm{reg}} \setminus S)$ is a connected manifold of dimension $n-1$, because $D_{\mathrm{reg}} \setminus S$ is connected by Lemma A.1.7; it is open in D' and dense in D', since $D_{\mathrm{reg}} \setminus S$ is open and dense in $D \setminus S$. Hence Q consists of regular points of D' and lies in a single connected component of $(D')_{\mathrm{reg}}$, whose closure, an irreducible component of D' by Corollary A.2.13, contains $\overline{Q} = D'$; so D' is irreducible, of dimension $n-1$. Finally, if S_1 is another analytic subset of codimension at least two off which f is an isomorphism, then $D \setminus (S \cup S_1)$ is dense in $D \setminus S$ and in $D \setminus S_1$, by Corollary A.2.14(2), so $f^{-1}(D \setminus (S \cup S_1))$ is dense in both $f^{-1}(D \setminus S)$ and $f^{-1}(D \setminus S_1)$, and the two closures agree. □

A *smooth model* of a complex space X is a proper bimeromorphic holomorphic map $M \to X$ with M a complex manifold; resolutions (Theorem A.5.4) are smooth models. Two smooth models $f_1 \colon M_1 \to X$ and $f_2 \colon M_2 \to X$ have a common smooth model, that is, a smooth model M_3 with proper bimeromorphic maps $M_3 \to M_1$ and $M_3 \to M_2$ commuting with the maps to X.

To construct it, let $E_1, E_2 \subset X$ be nowhere dense analytic subsets as in the definition of bimeromorphic maps for f_1 and f_2, and put $U = X \setminus (E_1 \cup E_2)$. Both f_i are isomorphisms over U, and $f_i^{-1}(U)$ is dense in M_i by Lemma A.3.1. Let $\Gamma \subset M_1 \times_X M_2 = \{(p_1, p_2) : f_1(p_1) = f_2(p_2)\}$ be the closure of the graph of $f_2^{-1} \circ f_1$ over U. The fiber product is analytic in $M_1 \times M_2$, since the equality of the two images is expressed by coordinate equations in local embeddings of X. Over U the fiber product is the graph, so Γ is the closure of the complement in the fiber product of the analytic set of points lying over $E_1 \cup E_2$, and it is analytic by Lemma A.2.5.

The projections of Γ are proper: over a compact subset of M_1, the second coordinates lie in the compact inverse image under f_2 of its image in X, and Γ is closed. Their images are closed and contain the dense sets $f_i^{-1}(U)$, so they are surjective; they are isomorphisms over $f_i^{-1}(U)$, and the inverse images of the complements $f_i^{-1}(E_1 \cup E_2)$ are nowhere dense in Γ, because the graph over U is dense in Γ; hence they are bimeromorphic. Resolving Γ by Theorem A.5.4 and composing with the projections, we obtain the common smooth model; the composites are proper and bimeromorphic by Corollary A.6.3.

When X is compact and the two smooth models are themselves obtained from X by finite sequences of blowups with smooth centers, there is also a common refinement of the same kind. Let $X = X_0 \leftarrow X_1 \leftarrow \cdots \leftarrow X_k = M_1$ be the first sequence, where X_j is the blowup of X_{j-1} along the ideal of a smooth center $C_j \subset X_{j-1}$. We start from a resolution of X, which is such a sequence because X is compact, and lift it through the first sequence one center at a time.

Suppose that the current smooth model N, obtained from X by a finite sequence of blowups with smooth centers, has been constructed with a map to X_{j-1} over X. The maps $N \to X$ and $X_{j-1} \to X$ are proper and bimeromorphic by Corollary A.6.3; let $U \subset X$ be the complement of a nowhere dense analytic subset over which both are isomorphisms. The pullback of the ideal of C_j to N is not identically zero on any component of N. Indeed, its zero set is the inverse image of C_j, which is nowhere dense in X_{j-1} because the ideal of C_j is not identically zero on any branch of X_{j-1} (Definition A.5.1). Over U the map $N \to X_{j-1}$ is an isomorphism, so this inverse image meets the dense open part of N over U in a nowhere dense set, and its part over $X \setminus U$ is nowhere dense by Lemma A.3.1. Principalizing it by Theorem A.5.5, we obtain a sequence of blowups $N' \to N$ with smooth centers, finite because N is compact, on which the pulled-back ideal is invertible, and Proposition A.5.2 lifts $N' \to X_{j-1}$ to a map $N' \to X_j$.

Repeating this for the finitely many centers of the first sequence and then of the second, we obtain a finite sequence of blowups with smooth centers over X that dominates both models. All three models are proper and bimeromorphic over X, by Proposition A.5.2 and Corollary A.6.3; by Lemma A.3.1 there is a complement $U \subset X$ of a nowhere dense analytic subset over which all three are isomorphisms. The maps from the new model to the two given ones commute with the maps to X, so they are isomorphisms over the parts lying over U. The complements of these parts are nowhere dense analytic subsets of the three models, again by Lemma A.3.1; so the two maps are bimeromorphic, and they are proper because the source is compact. (For general proper bimeromorphic models we only have the common smooth model constructed above.)

Definition A.7.4 Let X be a complex space. A *prime divisor over* X is represented by a pair consisting of a smooth model $M \to X$ and an irreducible hypersurface $F \subset M$; two pairs are identified if the strict transforms of their hypersurfaces, in the sense of Proposition A.7.3, agree on a common smooth model. The prime divisor is *exceptional* if the image of F in X has codimension at least two.

Lemma A.7.5 *Let X be a complex space.*

(1) *The identification of pairs in Definition A.7.4 is an equivalence relation, and the image in X of the hypersurface F of a representative (M, F) depends only on the prime divisor.*

(2) *Let (M, F) represent a prime divisor over X, and let $g\colon M' \to M$ be the composite of the principalization of $\mathcal{I}_F$ given by Theorem A.5.5 with the empty divisor. Then g is proper and bimeromorphic, the strict transform F' of F is a smooth irreducible hypersurface of M', and (M', F') represents the same prime divisor.*

In particular, every prime divisor over X has a representative (M, F) with F smooth.

Proof We begin with two remarks used in the proof of (1). First, two maps from a smooth model M_3 to a smooth model M_1 that commute with the maps to X agree.

Indeed, let $U \subset X$ be the complement of a nowhere dense analytic subset over which both models are isomorphisms; the two maps agree over U, and the part of M_3 over U is dense, by Lemma A.3.1.

Second, strict transforms are compatible with composition. Let $g\colon P \to N$ and $h\colon N \to M$ be proper bimeromorphic maps between complex manifolds, so that $h \circ g$ is proper and bimeromorphic by Corollary A.6.3, and let $D \subset M$ be an irreducible hypersurface, with strict transform D^N in N. We claim that the strict transform of D^N under g is the strict transform D^P of D under $h \circ g$. Let $S_h, S_{hg} \subset M$ and $S_g \subset N$ be analytic subsets of codimension at least two as in Proposition A.7.3 for h, $h \circ g$ and g, and let $T = S_h \cup S_{hg} \cup h(S_g)$, which is analytic of codimension at least two by Theorem A.6.2 and Lemma A.2.12(3).

The set $O = D^N \cap h^{-1}(M \setminus T) = h^{-1}(D \setminus T)$ is open in D^N, nonempty, and does not meet S_g; its complement $D^N \cap h^{-1}(T)$ is an analytic subset of the irreducible set D^N different from D^N, hence nowhere dense by Corollary A.2.14(2). So O is dense in $D^N \setminus S_g$, and since g is an isomorphism over $N \setminus S_g$, the strict transform of D^N under g is the closure of $g^{-1}(O) = (h \circ g)^{-1}(D \setminus T)$. By the independence of the choice of S_{hg} in Proposition A.7.3, applied with T, this closure is also D^P.

Now let (M_i, F_i), $i = 1, 2, 3$, be pairs such that (M_1, F_1) and (M_2, F_2) are identified on a common smooth model N_{12}, and (M_2, F_2) and (M_3, F_3) on N_{23}. Let P be a common smooth model of N_{12} and N_{23}, and let $p_i\colon P \to M_i$ be the composite maps; by the first remark, the two composite maps $P \to M_2$, through N_{12} and through N_{23}, coincide. By the second remark, the strict transforms of F_1 and F_2 under p_1 and p_2 are the strict transforms in P of their strict transforms in N_{12}, which agree; similarly for F_2 and F_3. Hence F_1 and F_3 have the same strict transform in P, and the identification is transitive; it is reflexive, with M_1 itself as common smooth model, and symmetric by definition. Since p_i maps the strict transform of F_i onto F_i (Proposition A.7.3), the images of F_1 and F_3 in X both equal the image of their common strict transform in P; so the image of F in X depends only on the prime divisor.

We now prove (2). Let (M, F) represent a prime divisor. The ideal $\mathcal{I}_F$ is coherent by Theorem A.2.8, and it is not identically zero on any component of M, because F is nowhere dense. Let $g\colon M' \to M$ be as in (2), so that $\mathcal{I}_F \mathcal{O}_{M'} = \mathcal{O}_{M'}(-H)$ with H an effective divisor whose support has simple normal crossings; g is proper, it is an isomorphism off $g^{-1}(F)$, and $g^{-1}(F)$, the support of H, is nowhere dense, so g is bimeromorphic. The strict transform F' of F is an irreducible hypersurface contained in the support of H, so by Corollary A.2.14(4) and (3) it is one of its irreducible components, which are smooth. The pair (M', F') represents the same prime divisor, as we see on the common smooth model M', which is a smooth model of X by Corollary A.6.3. □

Theorem A.7.6 *Let X be a compact complex space. Every prime divisor over X is represented on a smooth model obtained from X by a finite sequence of blowups with smooth centers. Its representative may also be chosen smooth.*

Proof After reducing to the case of a manifold, we blow up the image of the divisor in X as long as it has codimension at least two; each step lowers the order of vanishing along the divisor of the Jacobian determinant of the model.

Step 1: reduction to a manifold. Let $\widehat{X} \to X$ be a resolution given by Theorem A.5.4, a finite sequence of blowups with smooth centers, since X is compact. Let the prime divisor be represented by (M_0, F_0), and let M be a common smooth model of M_0 and $\widehat{X}$, with $E \subset M$ the strict transform of F_0 (Proposition A.7.3); then (M, E) represents the same divisor, as we see on the common smooth model M. A finite sequence of blowups with smooth centers over $\widehat{X}$, composed with $\widehat{X} \to X$, is such a sequence over X, and smooth models of $\widehat{X}$ are smooth models of X, by Corollary A.6.3. Hence we may assume that X is a compact complex manifold, and we are given a proper bimeromorphic map $f\colon M \to X$ from a compact manifold and an irreducible hypersurface $E \subset M$.

Since E is irreducible, hence connected, it lies in a connected component M_E of M, and $f(M_E)$ lies in a connected component X_E of X. These components have the same dimension n, because M_E contains a nonempty open set mapped isomorphically onto an open subset of X (Lemma A.3.1); moreover $n \geq 1$ and $\dim E = n - 1$, since E is a nonempty hypersurface. If $n = 1$, the center $f(E)$ considered below is a point of X_E, hence a hypersurface.

Step 2: the order of the Jacobian. The Jacobian determinant J_f of f in local coordinates is holomorphic and determined up to multiplication by holomorphic units. It does not vanish identically near any point, because f is an isomorphism over a dense open subset of X whose inverse image is dense in M (Lemma A.3.1). Its order of vanishing $\operatorname{ord}_E J_f$ along E is the largest k such that J_f is divisible by the kth power of a local equation of E near points of E_{reg}; Definition 12.6.3 uses the same notion of order along a hypersurface. It is a finite nonnegative integer k, since a function divisible by all powers of a local coordinate y has vanishing Taylor coefficients in y and vanishes near the point. It does not depend on the point of E_{reg}, which is a connected manifold by Corollary A.2.14: for each integer j, the set of points of E_{reg} at which the order is at least j is open, and it is closed by the identity theorem on E_{reg}, since a holomorphic function vanishing on E near a regular point is divisible by a local equation of E.

For composable maps the chain rule gives

$$\operatorname{ord}_E J_{g\circ h} = \operatorname{ord}_E J_h + \operatorname{ord}_E (J_g \circ h), \tag{A.12}$$

because orders of vanishing along E add under multiplication. Replacing M by a common smooth model $h\colon M' \to M$ and E by its strict transform E' does not change k: by Proposition A.7.3, h is an isomorphism near generic points of E', so $\operatorname{ord}_{E'} J_h = 0$ and $\operatorname{ord}_{E'}(J_f \circ h) = k$.

Step 3: one principalization lowers the order. The center $Z = f(E)$ is an analytic subset of X by Theorem A.6.2, of dimension at most $\dim E = n - 1$ by Lemma A.2.12(3). It is irreducible: if $Z = Z_1 \cup Z_2$ with analytic subsets Z_1, Z_2, then the irreducible set E is the union of its analytic subsets $E \cap f^{-1}(Z_1)$ and $E \cap f^{-1}(Z_2)$, so it lies in one of them, and Z equals Z_1 or Z_2. The same argument shows that images of irreducible analytic sets under proper holomorphic maps are irreducible. Suppose first that Z is a hypersurface, and let $S \subset X$ be as in Proposition A.7.3 for f. The strict transform Z^M of Z in M is an irreducible hypersurface containing $f^{-1}(Z \setminus S)$.

The set $E \cap f^{-1}(Z \setminus S)$ is open in E, and it is nonempty, because $f(E) = Z$ is not contained in S, which has codimension at least two. So $E \cap Z^M$ is an analytic subset of E with interior points in E, and $E \subset Z^M$ by Corollary A.2.14(2); then $E = Z^M$ by Corollary A.2.14(3). Hence (X, Z) represents the given divisor, as we see on the common smooth model M.

Suppose now that Z has codimension at least two. Its ideal $\mathcal{I}_Z$ is coherent by Theorem A.2.8, and it is not identically zero on any component of X, since Z is nowhere dense. Let $\sigma = \sigma_1 \circ \cdots \circ \sigma_t \colon X' \to X$ be the finite sequence of blowups with centers $C_i \subset X_{i-1}$ and transformed ideals $\mathfrak{a}_i$, $\mathfrak{a}_0 = \mathcal{I}_Z$, given by Theorem A.5.5; it is proper and bimeromorphic by Corollary A.6.3, so X' is a smooth model of X. Let $h \colon M' \to M$ and $f' \colon M' \to X'$ be the maps from a common smooth model of M and X', so that $f \circ h = \sigma \circ f'$, let E' be the strict transform of E, let $\rho_i \colon M' \to X_i$ be the induced maps, and let $Z_i = \rho_i(E')$; thus $Z_0 = Z$.

Let $r \geq 2$ be the codimension of Z, which is pure by Corollary A.2.14(1). We claim that for some i the center C_i contains Z_{i-1}, and that the connected component of C_i containing Z_{i-1} has codimension r. The sets Z_j are irreducible, as images of the irreducible set E', and $\sigma_{j+1}(Z_{j+1}) = Z_j$. Put $\tau_j = \sigma_1 \circ \cdots \circ \sigma_j$. Suppose that $Z_{l-1} \not\subset C_l$ for all $l \leq j$. We show by induction on j that there are a nowhere dense analytic subset $A_j \subset Z$ and an open set $N_j \subset X \setminus A_j$ containing $Z_j^\circ = Z \setminus A_j$ such that τ_j is an isomorphism over N_j and, on $\tau_j^{-1}(N_j)$,

$$Z_j = \tau_j^{-1}(Z_j^\circ), \qquad \mathfrak{a}_j = \mathcal{I}_Z \mathcal{O}_{X_j}.$$

For $j = 0$ take $A_0 = \varnothing$ and $N_0 = X$.

Assume the assertion for some $j < t$, and suppose that $Z_j \not\subset C_{j+1}$. On $\tau_j^{-1}(N_j)$ we have $C_{j+1} \subset V(\mathfrak{a}_j) = \tau_j^{-1}(Z_j^\circ) \subset Z_j$. By Corollary A.2.14(2), $C_{j+1} \cap Z_j$ is nowhere dense in Z_j. Its image $\tau_j(C_{j+1} \cap Z_j) \subset Z$ is analytic by Theorem A.6.2, and it is nowhere dense in Z: its part in Z_j° corresponds under the isomorphism τ_j to the nowhere dense part of $C_{j+1} \cap Z_j$ in the open subset $\tau_j^{-1}(Z_j^\circ)$ of Z_j, and A_j is nowhere dense. Let $A_{j+1} = A_j \cup \tau_j(C_{j+1} \cap Z_j)$ and $N_{j+1} = N_j \setminus A_{j+1}$. Then $\tau_j^{-1}(N_{j+1})$ does not meet C_{j+1}, so σ_{j+1} is an isomorphism over it and does not change the ideal there: $\mathfrak{a}_{j+1} = \mathcal{O}_{X_{j+1}}(E_{j+1})\mathfrak{a}_j \mathcal{O}_{X_{j+1}} = \mathcal{I}_Z \mathcal{O}_{X_{j+1}}$ on $\tau_{j+1}^{-1}(N_{j+1})$. Moreover $Z_{j+1} \cap \tau_{j+1}^{-1}(N_{j+1}) = \sigma_{j+1}^{-1}(Z_j \cap \tau_j^{-1}(N_{j+1})) = \tau_{j+1}^{-1}(Z_{j+1}^\circ)$, because $\sigma_{j+1}(Z_{j+1}) = Z_j$ and σ_{j+1} is injective there. This completes the induction.

Since A_j is nowhere dense in Z, the set Z_j° is nonempty, and so is $\tau_j^{-1}(Z_j^\circ)$. If $Z_{l-1} \not\subset C_l$ held for all $l \leq t$, the induction would reach $j = t$, and $\mathfrak{a}_t$ would be the ideal of the nonempty set $\tau_t^{-1}(Z_t^\circ)$ on $\tau_t^{-1}(N_t)$, whereas $\mathfrak{a}_t$ is the unit ideal. So there is a least index i with $Z_{i-1} \subset C_i$, and the assertion holds for $j = i - 1$. On $\tau_{i-1}^{-1}(N_{i-1})$ we have $Z_{i-1} \subset C_i \subset V(\mathfrak{a}_{i-1}) = \tau_{i-1}^{-1}(Z_{i-1}^\circ) \subset Z_{i-1}$, so C_i coincides there with a copy of the open subset Z_{i-1}° of Z, which is pure of codimension r. The set Z_{i-1} is connected, being the closure of the connected manifold $(Z_{i-1})_{\mathrm{reg}}$, so it lies in one connected component of the manifold C_i, and this component has codimension r.

By (A.6) and Proposition A.5.3, near points over this component the Jacobian determinant of σ_i is, in any local coordinates, a unit times the $(r-1)$st power of

a local equation of the exceptional divisor $E_i = \sigma_i^{-1}(C_i)$. Since $Z_i = \rho_i(E')$ lies over $Z_{i-1} \subset C_i$, the function $J_{\sigma_i} \circ \rho_i$ vanishes on E'; it does not vanish identically near any point, because J_{σ_i} vanishes only on E_i, and $\rho_i^{-1}(E_i)$ is nowhere dense, ρ_i being proper and bimeromorphic as the composite of f' with $\sigma_{i+1} \circ \cdots \circ \sigma_t$ (Corollary A.6.3). Hence it has positive order along E'. The Jacobians of the other blowups composed with the corresponding maps ρ_j are holomorphic, so they have nonnegative order, and by the chain rule $J_\sigma \circ f' = \prod_j (J_{\sigma_j} \circ \rho_j)$. Thus

$$\operatorname{ord}_{E'}(J_\sigma \circ f') \geq 1.$$

Applying (A.12) to $f \circ h = \sigma \circ f'$ and using Step 2, we obtain

$$0 \leq \operatorname{ord}_{E'} J_{f'} = k - \operatorname{ord}_{E'}(J_\sigma \circ f') \leq k - 1.$$

Step 4: conclusion. The pair (M', E') represents the given prime divisor, as we see on the common smooth model M', and X' is a compact manifold. We repeat Step 3 with X', f' and E' in place of X, f and E. Each repetition in which the center has codimension at least two lowers the nonnegative integer k by at least one, so after at most k repetitions the center is a hypersurface on the current model, which is obtained from X by a finite sequence of blowups with smooth centers, and it represents the given prime divisor. Finally, by Lemma A.7.5(2), principalizing the ideal of this hypersurface by Theorem A.5.5 gives a further sequence of blowups with smooth centers, finite because the model is compact, on which the strict transform of the hypersurface is smooth and represents the same prime divisor. □

A.8 Plurisubharmonic functions on reduced spaces

At a singular point of a complex space there is no intrinsic complex Hessian, so we define plurisubharmonic functions on a complex space as local restrictions of plurisubharmonic functions on ambient domains. Their compositions with holomorphic discs are then subharmonic, and together with the curves of Appendix A.4 this gives a maximum principle on compact spaces and local upper bounds near analytic subsets of codimension at least two.

Definition A.8.1 Let X be a complex space. A function $u\colon X \to [-\infty, +\infty)$ is *locally plurisubharmonic* if every point of X has a local model, which identifies a neighborhood of the point with an analytic subset of a domain $\Omega \subset \mathbb{C}^N$, and a ball $V \subset \Omega$ around the point such that u is the restriction to $X \cap V$ of a function on V that is plurisubharmonic or identically $-\infty$. We write $u \in \mathrm{PSH}(X)$ if, in addition, u is not identically $-\infty$ on any irreducible component of X, the components being defined as in Corollary A.2.13. Locally plurisubharmonic functions are allowed to be identically $-\infty$ on some components; we use them in this generality when only local properties are needed.

The following lemma shows that this definition does not depend on the local models, and that it agrees with the definition of Section 8.2 on complex manifolds.

Lemma A.8.2 *Let X be a complex space.*

(1) *Let u be locally plurisubharmonic on X. For every local model of a neighborhood of a point of X, there is a ball V around the point in the ambient domain such that u is the restriction to $X \cap V$ of a function on V that is plurisubharmonic or identically $-\infty$.*
(2) *If u is locally plurisubharmonic on X and $F\colon Y \to X$ is a holomorphic map of complex spaces, then $u \circ F$ is locally plurisubharmonic on Y. In particular, if Y is a one-dimensional complex manifold, such as a disc, then $u \circ F$ is subharmonic, in the sense of Definition 3.1.1, in every local coordinate of Y, and on the whole disc if Y is a disc.*
(3) *If X is a complex manifold, then a function on X is locally plurisubharmonic if and only if, on each connected component of X, it is plurisubharmonic or identically $-\infty$. Moreover,* $\mathrm{PSH}(X)$ *agrees with the set defined in Section 8.2.*

Proof We prove (1) and the first assertion of (2) together. Let u be locally plurisubharmonic on X, let $F\colon Y \to X$ be holomorphic, and let $y \in Y$. Choose a local model of a neighborhood of $F(y)$, with ambient coordinates w and a ball V around $F(y)$ on which u is the restriction of a function v that is plurisubharmonic or identically $-\infty$, and an arbitrary local model of a neighborhood of y, with ambient coordinates z. By the definition of holomorphic maps and of $\mathcal{O}_Y$, the components of F in the w-model extend holomorphically to a neighborhood of y in the z-ambient space, and their joint extension G, being continuous with $G(y) = F(y)$, maps a smaller ball V' around y into V. By Proposition 8.2.3, $v \circ G$ is plurisubharmonic or identically $-\infty$ on V', and its restriction to $Y \cap V'$ is $u \circ F$. Hence $u \circ F$ is locally plurisubharmonic, with respect to every local model of Y. For $Y = X$ and F the identity, with two local models at the same point, this is (1).

If Y is a one-dimensional complex manifold, its holomorphic charts are local models. By what we have just proved, $u \circ F$ is therefore, in local coordinates near each point of Y, plurisubharmonic or identically $-\infty$, that is, subharmonic in the sense of Definition 3.1.1. Subharmonicity is a local property, so $u \circ F$ is subharmonic in every local coordinate, and on the whole disc if Y is a disc.

(3) Holomorphic charts are local models. So by (1), every point of X has a coordinate ball on which a locally plurisubharmonic function u is plurisubharmonic or identically $-\infty$, and by Lemma 8.2.4, u is plurisubharmonic or identically $-\infty$ on each connected component of X. Conversely, a function with this property is locally plurisubharmonic, with the charts as local models, since its expressions in charts with connected domains are plurisubharmonic or identically $-\infty$. The irreducible components of the manifold X are its connected components, by Corollary A.2.13. Hence $\mathrm{PSH}(X)$, in the sense of Definition A.8.1, consists of the functions that are plurisubharmonic on each connected component of X, which are the plurisubharmonic functions in the sense of Section 8.2, since plurisubharmonicity is a local property there. □

Proposition A.8.3 *A locally plurisubharmonic function on a compact connected complex space is constant, possibly equal to* $-\infty$.

Proof Let u be locally plurisubharmonic on the compact connected space X. It is upper semicontinuous, so it attains its maximum $M \in [-\infty, \infty)$. If $M = -\infty$, then $u \equiv -\infty$. Suppose that M is finite, and let x be a point with $u(x) = M$. We show that $u = M$ near x. If x is an isolated point, there is nothing to prove. Otherwise, let X_1 be a branch of (X, x) of positive dimension, and take a parametrization of (X_1, x), with its connected covering locus $X_{1,\delta}$, dense in the representative X_1. By Proposition A.4.2, applied to the subgerm consisting of the singular locus of X and the points of X_1 outside $X_{1,\delta}$, there is a nonconstant holomorphic curve γ with $\gamma(0) = x$ and $\gamma(t) \in X_{1,\delta} \cap X_{\text{reg}}$ for $t \neq 0$. The function $u \circ \gamma$ is subharmonic by Lemma A.8.2(2), and it attains its finite maximum M at 0; by the maximum principle (Proposition 3.1.4) it equals M near 0. Thus u attains the value M at points of $X_{1,\delta}$.

By Lemma A.8.2(2) and (3), applied to the inclusion of the connected manifold $X_{1,\delta}$, the restriction of u to $X_{1,\delta}$ is plurisubharmonic or identically $-\infty$, hence plurisubharmonic, since it takes the value M. On a coordinate ball around a point where $u = M$, it is subharmonic by Theorem 8.2.1, and it attains its maximum M at the center, so it equals M on this ball by Proposition 3.1.4. Thus the set where $u = M$ is open in $X_{1,\delta}$; it is closed by upper semicontinuity and nonempty, so $u = M$ on $X_{1,\delta}$. Since $X_{1,\delta}$ is dense in X_1, upper semicontinuity gives $u \geq M$, hence $u = M$, on all of X_1. Doing this for each branch, we find that $u = M$ on a neighborhood of x.

Hence the set $\{u = M\}$ is open. It is also closed, because $\{u < M\}$ is open for an upper semicontinuous function bounded above by M. Since X is connected, $u \equiv M$.□

Proposition A.8.4 *Let A be an analytic subset of a complex space X such that, at every point of A, the intersection of A with each irreducible branch of X has codimension at least two in that branch. Then every locally plurisubharmonic function on $X \setminus A$ is locally bounded above near A.*

Proof Let u be locally plurisubharmonic on $X \setminus A$, and let $x \in A$. We bound u over a circle that avoids the image of A under a finite projection, and carry the bound inward along one-dimensional slices by the maximum principle. It suffices to bound u from above near x on each branch of (X, x), and the restriction of u to a branch is locally plurisubharmonic by Lemma A.8.2(2); so we may work on the representative X_1 of a branch on a parametrization polydisc. Let $X_2, \ldots, X_p$ be representatives of the other branches.

The hypothesis is inherited by X_1 near x. Indeed, let $y \in X_1$ be close to x, and let Y be a branch of (X_1, y). It has dimension $\dim X_1$, by Proposition A.2.7, and it lies in a branch Y' of (X, y), which lies in some (X_j, y). If $j = 1$, then $Y' = Y$, because Y is a branch of (X_1, y). If $j \neq 1$, then $Y \subset (X_1 \cap X_j, y)$, which is impossible for y close to x, because $\dim(X_1 \cap X_j, y) \leq \dim(X_1 \cap X_j, x) < \dim X_1$, by upper semicontinuity and Proposition A.2.7. Hence every branch of (X_1, y) is a branch of (X, y). The dimension d of X_1 is at least two, since $A \cap X_1$ contains x and has codimension at least two in X_1. Now write X for X_1, and let $\pi \colon X \to \Delta' \subset \mathbb{C}^d$ be

the projection of the local parametrization (Theorem A.2.3); it is proper with finite fibers, and $\pi^{-1}(0) = \{x\}$.

Step 1: a cylinder avoiding $\pi(A)$. The image $\pi(A)$ is analytic by Theorem A.6.2, and its dimension is at most $d-2$ by Lemma A.2.12(3). In particular there is a holomorphic function $g \not\equiv 0$ near 0 vanishing on $\pi(A)$. Choose a complex line through 0 on which the first nonzero homogeneous term of g does not vanish, and write the base coordinates as (ζ, z'), with ζ in the direction of this line. Then $\pi(A)$ meets the line only at 0 near 0, and by compactness there are $r > 0$ and $\varepsilon > 0$ such that the cylinder

$$\{|\zeta| = r,\ |z'| \le \varepsilon\}$$

is compactly contained in Δ' and disjoint from $\pi(A)$. Its inverse image is compact and disjoint from A, so u has an upper bound $M < \infty$ there.

Step 2: slicing by curves. The projection to the z'-ball of $\pi(A) \cap \{|\zeta| < r,\ |z'| < \varepsilon\}$ is proper, because the closure of this set does not meet $\{|\zeta| = r\}$. By Theorem A.6.2, its image S is analytic in the ball $\{|z'| < \varepsilon\}$, and it has dimension at most $d-2$ by Lemma A.2.12(3), so it is nowhere dense in this $(d-1)$-dimensional ball. Fix z' with $|z'| < \varepsilon$ and $z' \notin S$. Then the compact set $\pi^{-1}(\{|\zeta| \le r\} \times \{z'\})$ is disjoint from A, and so is $C = \pi^{-1}(\{|\zeta| < r'\} \times \{z'\})$ for some $r' > r$.

The set C is a pure one-dimensional complex space, or empty. Indeed, it is cut out of X by the $d-1$ coordinate equations $z'_k = \text{const}$. At each stage, every branch of the current set at every point is mapped finitely, hence openly by Lemma A.4.5, to the space of the remaining base coordinates; so the next coordinate equation is not identically satisfied on it, and by Proposition A.4.6 the dimension drops by exactly one. Let $\nu\colon \widetilde{C} \to C$ be the normalization (Theorem A.3.6). By Proposition A.4.1, $\widetilde{C}$ is a one-dimensional complex manifold, and $\zeta \circ \pi \circ \nu\colon \widetilde{C} \to \{|\zeta| < r'\}$ is proper with finite fibers, hence open on each connected component by Lemma A.4.5.

By Lemma A.8.2(2), the function $v = u \circ \nu$ is subharmonic on the one-dimensional manifold $\widetilde{C}$. Let Q be a connected component of the open set $\{|\zeta \circ \pi \circ \nu| < r\} \subset \widetilde{C}$; its closure is compact, by properness. A point of $\overline{Q}$ over the open disc $\{|\zeta| < r\}$ has a connected neighborhood in $\{|\zeta \circ \pi \circ \nu| < r\}$ meeting Q, and therefore lies in Q; so the boundary of Q lies over the circle $\{|\zeta| = r\}$, where $v \le M$ by Step 1. This boundary is nonempty: otherwise Q would be a connected component of $\widetilde{C}$, and its image would be open and, by properness, closed in the disc $\{|\zeta| < r'\}$, hence equal to this disc, although it lies in $\{|\zeta| < r\}$.

The upper semicontinuous function v attains its maximum on the compact set $\overline{Q}$. If the maximum is finite and attained at a point of Q, the maximum principle (Proposition 3.1.4), applied on coordinate discs, shows that the set where v attains it is open in Q; this set is also closed, so v is constant on Q, and by upper semicontinuity at a boundary point the constant is at most M. Otherwise the maximum is $-\infty$ or is attained on the boundary of Q, so again it is at most M. Hence $v \le M$ on Q, and since ν is surjective, $u \le M$ on $\pi^{-1}(\{|\zeta| < r\} \times \{z'\})$.

Step 3: the remaining slices. We have shown that $u \le M$ on the part of $X \cap \pi^{-1}(\{|\zeta| < r,\ |z'| < \varepsilon\})$ lying over $z' \notin S$. Let $y \notin A$ be a point of this set lying over some $z' \in S$, and let Y_1 be a branch of (X, y). The set $T = \pi^{-1}(\{z' \in S\})$ does not

contain the germ Y_1: by purity, Y_1 has dimension d, so $\pi|_{Y_1}$ is open by Lemma A.4.5, while the image of T is nowhere dense. Since A is closed and $y \notin A$, Proposition A.4.2 gives a holomorphic curve γ in Y_1 with $\gamma(0) = y$ and $\gamma(t) \notin A \cup T$ for $t \neq 0$. Then $u \circ \gamma$ is subharmonic near 0, by Lemma A.8.2(2), and at most M on the punctured disc, so $u(y) \leq M$ by the submean inequality on small circles. Thus $u \leq M$ on $X \setminus A$ near x. □

Appendix B
Historical and bibliographical notes

These notes give the sources of the results in this book. For each chapter we indicate where the main results were first proved or where they can be found in the literature, compare our statements, proofs and normalizations with those of the sources, and point to alternative presentations. The sections follow the order of the book: each concerns one chapter, except that the notes on Chapter 17 occupy three sections, and the last section concerns Appendix A. A few ideas recur throughout the book: a function is recovered from its distributional derivatives, exceptional sets are controlled by capacities, nonlinear measures are constructed by approximation with bounded potentials, and singular potentials are compared through their envelopes. These ideas have different origins, and their formulation has changed as the theory moved from domains in Euclidean space to compact complex manifolds.

When the book is compared with another source, the conventions of that source matter as much as its hypotheses. We use $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/(2\pi)$; Lelong numbers are normalized so that $\log|z|^2$ has Lelong number 1 at the origin, and multiplier ideals are defined with the weight $\mathrm{e}^{-\varphi}$. Several of the sources cited below use the operator $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/\pi$, measure Lelong numbers against $\log|z|$ and use the weight $\mathrm{e}^{-2\varphi}$. A source with the last two conventions assigns to $\varphi = u/2$ the Lelong numbers and multiplier ideals that we assign to u. If the operator is multiplied by a constant $c > 0$, then at fixed potentials a product of p factors of the form $\mathrm{dd}^{\mathrm{c}}u$ is multiplied by c^p. We record these conversions where they matter.

Theorem and page numbers refer to the cited version of a work. When this version is a preprint rather than the published article, whose numbering or scope may differ, we say so. Two existence theorems are used without proof, namely the resolution and principalization theorems (Theorems A.5.4 and A.5.5); their source is given in Appendix B.21.

B.1 Measures and distributions

The representation of positive linear functionals by measures, Theorem 1.2.2, is the Riesz representation theorem. For its formulation on locally compact, σ-compact spaces compare [Tao09b, Theorem 24]; on open subsets of Euclidean space our proof constructs the measure from a metric outer measure and identifies the functional by slicing functions into level sets. The regularity and density statements in Propositions 1.1.2 and 1.2.3 correspond to [Tao09b, Theorem 21, Exercise 22 and Proposition 10]. For the measurability part of the outer-measure construction compare [Tao10b, Theorem 5]. The assertions on product measures and the interchange of integrals in Proposition 1.1.7 form the Tonelli–Fubini theorem for σ-finite measures; compare [Tao10b, Theorem 36 and its proof], where the monotone class lemma (Lemma 35 there) is applied to the Boolean algebra of finite disjoint unions of rectangles. We instead apply Dynkin's π–λ theorem, Lemma 1.1.6, to the rectangles themselves.

For the local order bound, regularization by mollifiers and the fact that positive distributions are measures (Definition 1.3.1, Proposition 1.4.3, and Theorem 1.4.5) compare [Tao09d, Section 1, Exercises 4–5; Section 2, Definition 1, Exercise 15 and Lemma 2]. That source writes the test function first and gives the space of distributions the conjugate complex structure. Our pairing $\langle u, \chi \rangle = \int u\chi \, \mathrm{d}x$ is complex linear in both variables, the convention used for currents later in the book; with this pairing the derivative of a distribution is defined with the same sign as in that source.

The selection theorem for sequences of measures, Theorem 1.5.5, is proved by a diagonal argument over countably many test functions. The compactness theorem for nets, Theorem 1.6.3, rests instead on Tychonoff's theorem for a product of intervals, which we prove with ultrafilters (Lemma 1.6.2); this is where the axiom of choice enters. Subnets are Willard subnets throughout the book, and Proposition 1.6.1 characterizes compactness in these terms. The Arzelà–Ascoli theorem, Proposition 1.6.4, is proved in the usual way, by showing that an equicontinuous and pointwise bounded family is uniformly bounded and totally bounded for the uniform metric.

For the differentiation of integrals compare [Tao10a, Section 2, Theorems 32 and 36, Exercise 38 and Lemma 39]. In Theorem 1.7.3 we use cubes instead of balls. The covering argument is the same: a finite disjoint subfamily of the cubes covers the original cubes after enlargement by the factor three, and the constant in the maximal inequality becomes 3^d. The Poincaré-type inequality Lemma 1.7.4 follows by integrating the fundamental theorem of calculus along segments parallel to the coordinate axes.

For the Radon–Nikodym theorem compare [Tao09a, Corollary 1, following Theorem 2]. The book uses only the case of a Borel measure that is dominated, at least locally, by a constant multiple of a Radon measure, Lemma 1.7.1. We prove it by an L^2 argument on half-open dyadic cubes, so that no assumption on the measure of cube boundaries is needed. This lemma provides, for instance, the densities of the coefficient measures in Lemma 9.13.1.

B.2 Harmonic functions and boundary values

Fundamental solutions, the regularity of distributional solutions of the Laplace equation, the Poisson formula and Harnack's inequality are treated in [Hör07, Theorems 3.1.2–3.1.7, pp. 117–123]. As there, the fundamental solution E of Proposition 2.2.1 satisfies $\Delta E = \delta_0$, whereas our Green functions are positive and satisfy $-\Delta G = \delta_x$. Our proof of Weyl's lemma, Theorem 2.1.4, uses radial averaging and convolution; Theorem 3.1.3 of that book contains in addition a statement on analytic continuation. The Poisson formula and Harnack's inequality (Theorem 2.3.1 and Proposition 2.3.2) are stated with the explicit Poisson kernel and its two-sided bounds, and they include the case of dimension one.

Perron's method, barriers and the solution of the Dirichlet problem for continuous boundary data are developed in [Hör07, Section 3.3, pp. 171–174, especially Theorem 3.3.3, Definition 3.3.4 and Theorem 3.3.5]. In Theorems 2.4.2 and 2.4.4 we work with continuous subsolutions. For the characterization of regular boundary points by barriers compare also [Hör07, Theorem 3.4.17, pp. 219–220]. Our local barrier criterion, Proposition 2.4.7, uses a positive continuous supersolution near the given boundary point whose continuous extension to the nearby boundary vanishes only at that point; the theorem in Hörmander's book allows more general exceptional sets. The Green function of Theorem 2.5.2 is constructed only on bounded regular domains; within this class the theorem also shows that Green functions increase with the domain and converge along increasing sequences of domains whose union is again bounded and regular.

B.3 Subharmonic functions and polar sets

The characterizations of subharmonic functions by comparison with harmonic functions and by mean values, their local integrability, the Riesz decomposition and the distributional characterization are developed in [Hör07, Theorem 3.2.3, Corollary 3.2.8 and Theorems 3.2.9–3.2.11, pp. 143–148]; see Theorems 3.1.3, 3.2.2 and 3.2.3 and Proposition 3.1.5. Since a subharmonic function is the limit of its means at every point, the canonical representative of a distribution with positive Laplacian is determined by the distribution at every point, not only almost everywhere. Our Riesz measure is Δu, without a factor $1/(2\pi)$; thus a logarithmic pole in the plane has Riesz mass 2π.

For the compactness theorem and Hartogs' lemma (Theorems 3.3.2 and 3.3.3) compare [Hör07, Theorems 3.2.12–3.2.13, pp. 149–151]. Theorem 3.2.13 there also contains convergence statements in Sobolev spaces and against measures; the latter are discussed in the next section. Choquet's lemma and the regularized upper envelope (Lemma 3.4.1 and Theorem 3.4.2) correspond to [Hör07, Proposition 3.4.4, pp. 205–206]; Theorem 3.4.2 also asserts that the pointwise supremum of an arbitrary family is Lebesgue measurable. For the local definition of polar sets and the removable singularity theorem (Definition 3.5.1 and Theorem 3.5.2) compare [Hör07, Definition

3.4.1 and Theorem 3.4.3, pp. 203–205]. Our proof works directly with the local subharmonic functions of the definition, and the connectedness of the complement of a closed polar set in a domain follows from the maximum principle.

B.4 Energy, capacity and exceptional sets

The continuity principle and the restriction of a measure to compact sets on which its potential is continuous (Theorem 4.2.2) follow the argument of [Hör07, Theorems 3.4.6–3.4.7, pp. 207–208]; the approximation lemma of Lusin type used there is Lemma 4.2.1. The criterion for polarity of compact sets is [Hör07, Theorem 3.4.5, pp. 206–208]. Our relative capacity of a compact set is the total mass of the Riesz measure Δu of its regularized relative extremal function u, and our Green functions are positive with $-\Delta G = \delta$; the numerical values of the capacity depend on these normalizations.

Strong subadditivity, outer regularity and capacitability are developed in [Hör07, Section 3.4, pp. 209–216, especially Theorems 3.4.9 and 3.4.12 and Proposition 3.4.13]. Choquet's capacitability theorem, Theorem 4.4.1, which underlies the capacitability assertion of that section, is proved here by approximating continuous images of countable intersections of countable unions of compact sets from inside by compact sets. The convergence of integrals against measures with continuous potentials in Lemma 4.4.3 is part of [Hör07, Theorem 3.2.13, pp. 150–151], and the fact that the exceptional sets of the limit theorems are polar, Theorem 4.4.4, corresponds to [Hör07, Theorem 3.4.14, pp. 216–217]. This polarity strengthens the equality almost everywhere of an upper envelope and its regularization in Theorem 3.4.2.

For the identity between Green energy and Dirichlet energy and for quasi-continuity compare [Hör07, Propositions 3.4.19 and 3.4.21, pp. 221–223]. In Proposition 4.5.1 we work with the explicit, symmetric Green function of a ball. The positivity of the Newtonian energy and of the logarithmic energy (Proposition 4.1.2) is proved by expressing the kernels as integrals of Gaussians. For the logarithmic kernel, positivity and the Cauchy–Schwarz inequality are proved, and used, only for signed measures of total mass zero. The equilibrium measure is obtained by minimizing the energy, and Proposition 4.4.5 describes its potential (Frostman's theorem) with our normalization of the kernels.

The example [Xia, Example 6.1.3] uses an atom-free measure on a polar Cantor set and refers to [Car83, p. 31] for the Cantor set. Such a measure μ is constructed explicitly in Proposition 4.2.4: the Cantor set K is built from a rapidly decreasing sequence of lengths, so that the subharmonic function $p(z) = \int_K \log|z-a|^2 \, \mathrm{d}\mu(a)$ equals $-\infty$ at every point of K. The remaining ingredients of that example, namely the potential on the projective line, its integrability and the computation of its mass in degree one, are Proposition 15.3.7, Corollary 12.3.2, and Example 15.3.8.

B.5 Fine topology and balayage

Fuglede attributes the classical fine topology to Cartan and the quasi-Lindelöf principle to Doob in [Fug71, Introduction, p. 227]. His preliminaries [Fug71, Section 1, pp. 228–230] are formulated in an axiomatic harmonic space satisfying the domination axiom and admitting a positive potential, and they assume the general duality theory of sweeping before fine connectedness is studied. In the Euclidean setting of this book the quasi-Lindelöf principle, Theorem 5.4.1, follows directly from Choquet's lemma and the polarity of the exceptional set of an upper envelope. By Example 5.1.7, the fine topology is not Lindelöf, so a countable subfamily can only be expected to cover the union of a family of finely open sets up to a polar set.

Reduction and balayage (Theorems 5.2.5, 5.3.1 and 5.3.2) are treated only on a ball, where the Green function is explicit and symmetric and Green energy is a Dirichlet integral by Proposition 5.2.2. In this setting the swept measure is determined by an orthogonality relation for the energy, which also gives additivity, and the general duality of sweeping recalled by Fuglede is not required. Example 5.3.3 expresses the swept potential of a Dirac measure through harmonic measure, for a compact set whose complement in the ball is a regular domain. Proposition 5.5.1 is proved directly from the definitions; it shows that for $n \geq 2$ the classical fine topology of $\mathbb{C}^n = \mathbb{R}^{2n}$ differs from the topology generated by the upper semicontinuous functions that are subharmonic or identically $-\infty$ on complex lines, which is the plurifine topology of Chapter 10 (Example 10.3.4).

B.6 Fourier analysis and complex-line incidence

For Fourier inversion on the Schwartz space and Plancherel's theorem (Lemma 6.1.1 and Theorem 6.1.2) compare [Tao09c, Exercises 38–42, formula (11), Theorem 43 and Exercise 44]. Our Fourier transform uses the character $e^{-2\pi i x\cdot\xi}$. Gaussian integration gives the exact constant in the Fourier expression of the Riesz energy, Theorem 6.1.4; the kernel there is $|x-y|^{-s}$, without the normalization of a fundamental solution.

For the dyadic decomposition compare [Tao06a, Proposition 4.3 and proof, pp. 23–24]; for the weak type $(1,1)$ estimate and its consequences by interpolation and duality compare [Tao06b, Lemma 2.7, Corollaries 2.9–2.10 and proofs, pp. 7–9]. We treat only the Hessian multipliers $-\xi_a\xi_b/|\xi|^2$ (Lemma 6.2.1), whose kernels are obtained by differentiating the heat kernel. Their weak type $(1,1)$ bound and their L^p bounds are Lemma 6.2.2 and Theorem 6.2.3; the exponents $p > 2$ are obtained by duality. The interpolation inequality Lemma 6.2.5 and Morrey's inequality Lemma 6.2.6 are used for metric Laplacians in Chapter 16.

The star-shaped annular sets, the projections onto complex hyperplanes and the incidence argument follow [PSHD, Lemmas 1.3–1.5, pp. 2–4]; this manuscript carries no author's name and no publication data. The Borel selection required by the argument is Lemma 6.4.1: the sets of high values on circles are closed and, by the submean inequality, have positive length, so a median with respect to arc length gives

a Borel choice, and no general measurable selection theorem for compact-valued maps is needed. The unitarily invariant measures are constructed from the Gaussian measure on matrices, and volumes in projective space are controlled by an explicit estimate for projective balls instead of volume charts. In Corollary 6.5.4 the function is assumed to be Borel; in the proof of Theorem 8.1.3 this measurability is established before the corollary is applied.

B.7 Differential forms, currents and holomorphic divisors

For positive forms and currents, the characterization of weak positivity by restriction to complex subspaces and the fact that positive currents have order zero, compare [Dem12, Chapter III, Section 1.A–B, pp. 129–134, especially (1.4), (1.6) and (1.14)]. In addition, Proposition 7.2.1 shows that β^p lies in the interior of the cone of strongly positive forms, which gives the trace bounds for positive currents; the proof averages over sign changes and permutations of the coordinates. Our currents are indexed by bidegree, whereas Demailly's subscripts there refer to bidimension. Pushforward and pullback of currents (Propositions 7.5.1 and 7.5.2) correspond to [Dem12, Chapter III, (1.17), p. 134]; for the pushforward the map need only be proper on the support of the current.

The mixed Hodge–Riemann theorem of linear algebra, due to Timorin, is recorded in [DN05, Proposition 2.1, p. 4] (second arXiv version). Lemma 7.2.3 is its case of real $(1,1)$-forms, written with mixed discriminants of Hermitian matrices; we prove it by induction on principal minors and deformation to identity matrices, where the form is given by an explicit trace formula.

For local potentials of closed positive $(1,1)$-currents compare [Dem12, Chapter III, Section 1.18 and (1.19), p. 135]. Our proof of Theorem 7.3.2 is different: we subtract the complex Hessian of the potential of the trace measure, and the remainder, which has trace zero and therefore harmonic coefficients, is dd^{c}-exact by the smooth local result Lemma 7.3.1. In Proposition 7.3.3 the composition with a holomorphic map is taken pointwise, for the canonical representative; it is proved by radial regularization.

Our proof of the Weierstrass preparation theorem, Theorem 7.4.2, is the proof by contour integrals in [Dem12, Chapter II, Section 2.A, pp. 79–80, Theorem (2.1)], where it is attributed to Siegel. The Poincaré–Lelong formula, Theorem 7.4.5, is [Dem12, Chapter III, (2.15), pp. 143–144]. The expression $(\mathrm{i}/\pi)\partial\bar\partial \log|f|$ appearing there equals our $\mathrm{dd}^{\mathrm{c}} \log|f|^2$, while the operator dd^{c} used subsequently in that book is twice ours. In our proof, slicing by coordinate lines and Lemma 7.4.4 show that $\mathrm{dd}^{\mathrm{c}} \log|f|^2$ puts no mass on the singular locus of the divisor, so that it equals the current of integration over the regular part, weighted by multiplicities. Example 7.4.6 gives the corresponding identity in L^1_{loc} for quotients of holomorphic functions.

B.8 Plurisubharmonic functions and invariant minima

The theorem that a function which is subharmonic or identically $-\infty$ on complex lines is automatically upper semicontinuous is proved in the unsigned manuscript [PSHD, Theorem 1.1 and its proof, pp. 1–6], and our proof of Theorem 8.1.3 completes the argument given there. In that manuscript, Lemma 1.2, pp. 1–2, proves measurability by induction on the dimension, using means over bounded fibers; Proposition 2.1, pp. 4–5, identifies a finite function that is subharmonic on lines with the canonical representative; and the final proof, p. 6, uses truncation. The incidence results needed for this argument are proved in Chapter 6, and Proposition 7.3.3 describes the restrictions of the canonical representative to complex lines without using the theorem. We keep the usual definition of plurisubharmonic functions (Definition 8.1.1), which requires upper semicontinuity; Theorem 8.1.3, proved shortly afterwards, shows that this condition is redundant.

The distributional characterization, upper envelopes and compactness of plurisubharmonic functions (Theorems 8.2.1, 8.3.2 and 8.3.4) are deduced from the corresponding results for subharmonic functions in Chapters 3 and 4. At this stage the set where an upper envelope differs from its regularization is only known to be real polar, that is, polar as a subset of $\mathbb{R}^{2n}$ in local charts; its pluripolarity is proved in Chapter 10. The extension theorem (Theorem 8.4.3) gives both local boundedness and extension across a relatively closed set that is locally $\mathcal{H}^{2n-3}$-null; to apply it to analytic sets of codimension at least two (Corollary A.2.15) one needs the local estimate for the Hausdorff measure of analytic sets in terms of their dimension.

For L^p integrability and convergence in L^p_{loc} compare [GZ17, Theorem 1.48, parts 1–2 and proof Steps 1–2, pp. 30–32]. As there, we pass from discs to polydiscs (Lemma 8.3.5, Proposition 8.3.6, and Theorem 8.3.7), but we use the Riesz decomposition with a compactly supported measure instead of the Poisson–Jensen formula on the boundary. Since our Riesz measure is Δv, its logarithmic potential carries the factor $1/(2\pi)$. The bound (8.4) gives uniform estimates on smaller polydiscs, and convergence in L^p_{loc} then follows from Hölder's inequality. The identity with the factor two in the truncation display of [GZ17, p. 32] is incorrect, as is also noted in an unofficial list of errata for that book.

The local exponential integrability estimates (Lemma 8.3.8 and Proposition 8.3.9) are proved with the same compactly supported Riesz decomposition and Jensen's inequality (Lemma 1.1.4), with an exponent that depends only on a local L^1 bound; a finite covering by charts then gives the uniform estimate for normalized quasi-plurisubharmonic functions on a compact manifold, Corollary 14.2.3.

The partial Legendre transform is due to Kiselman [Kis78]. For his minimum principle and the regularization argument in its proof compare [Dem12, Chapter I, Section 7.B, Theorem (7.5) and its proof, p. 57]. In Theorem 8.5.1 the existence of an invariant exhaustion is a hypothesis, and the smooth case is proved with a Schur complement. Such an exhaustion is constructed for product tubes over open convex sets in Corollary 8.5.2; the case of general pseudoconvex tubes, Corollary 11.2.6, uses in addition the plurisubharmonicity of $-\log$ of the boundary distance.

B.9 Bounded products and nonlinear capacity

The recursive definition of products of currents $\mathrm{dd^c} u$ with u locally bounded and plurisubharmonic, the Chern–Levine–Nirenberg estimates and continuity along decreasing sequences are developed in [Dem12, Chapter III, Section 3, (3.1)–(3.11), pp. 144–150]; see Theorems 9.1.3 and 9.2.5. In our proof of decreasing continuity, the integration by parts formula Lemma 9.2.1 for functions that coincide near the boundary of a ball is combined with an induction on the number of factors, which also gives the symmetry of the product at each stage. The weighted estimate Theorem 9.3.1 concerns only the measures $\mathrm{dd^c} u_1 \wedge \cdots \wedge \mathrm{dd^c} u_q \wedge \beta^{n-q}$; a version relative to a closed positive current, for one factor at a time, is Lemma 15.3.1.

For continuity along increasing sequences, the behavior of products under maxima and the comparison principle (Theorems 9.2.8 and 9.5.1 and Proposition 9.2.7) compare [GZ17, Theorem 3.23, pp. 89–91, and Theorems 3.27 and 3.29, pp. 93–94]. The domination principle Corollary 9.2.9 is [Xia, Proposition 1.2.10]; we prove it by a strictly plurisubharmonic quadratic perturbation and the comparison principle, applied on the given bounded open set itself.

The Monge–Ampère capacity, quasi-continuity and convergence in capacity (Definition 9.3.2 and Theorems 9.4.2 and 9.4.8) appear in [GZ17, Proposition 4.19, Theorem 4.20 and Lemma 4.21, pp. 110–112, and Theorem 4.26, pp. 114–115]. Our capacity is defined with competitors taking values between zero and one, whereas the competitors in [GZ17, Definition 4.16, p. 108] take values between -1 and 0; since adding a constant does not change Monge–Ampère measures, the two normalizations give the same capacity. Both Demailly's book and [GZ17] use the operator $\mathrm{dd^c} = (\mathrm{i}/\pi)\partial\bar\partial$, twice ours, so their Monge–Ampère measures and capacities are 2^n times ours.

The estimate of Proposition 9.3.5 is attributed to Błocki in [GZ17, Proposition 3.21, pp. 88–89]. The power M^{n+1} of the oscillation printed there should be M^n: the proof uses n integrations by parts, and the left-hand side is homogeneous of degree n in v. We prove the estimate with M^n, under the assumption that $u = h$ near the boundary of the ball, first for smooth functions and then, by regularization and Corollary 9.2.6, for bounded ones. Corollary 9.5.2 shows that the set where an upper envelope differs from its regularization has capacity zero; this is used in Chapter 10 to show that such sets are pluripolar.

The solution of the homogeneous Dirichlet problem on the ball, Theorem 9.6.3, goes back to [BT76]. Averaging over automorphisms that move the center exactly to a prescribed point is carried out for polydiscs in [Błoo2, Section 2.3, Theorem 2.3.2 and Proposition 2.3.3, pp. 50–52]; Lemma 9.6.1 provides the corresponding automorphisms of the ball. Compare also [GZ17, Theorems 5.13–5.14, pp. 143–149]. In Step 3 of the proof of Theorem 9.6.3, the determinant of the complex Hessian of the solution is shown to vanish almost everywhere by comparing the solution with its second-order Taylor polynomial at points where this expansion holds. For functions with bounded real Hessian, the expansion holds almost everywhere by Proposition 1.7.5, which is proved with the differentiation of integrals and the Poincaré inequality on cubes. The homogeneous balayage of Corollary 9.6.4, together with

the comparison principle, shows that a locally bounded plurisubharmonic function is maximal if and only if its Monge–Ampère measure vanishes (Theorem 9.6.6); in particular, maximality is a local property for locally bounded functions.

The characterization of the inequality $(\mathrm{dd}^{\mathrm{c}}u)^n \geq f\eta^n$, where $\eta = \mathrm{dd}^{\mathrm{c}}|z|^2$, by upper test functions and the linearization of the determinant (Proposition 9.7.3) are proved in [EGZ10, Proposition 1.5 and Lemma 1.7, pp. 5–7], the first arXiv version of the later article by Eyssidieux, Guedj and Zeriahi. The passage from upper test functions to distributional inequalities, which the proof in that preprint quotes from Hörmander, is Lemma 9.7.2. The trace matrices used there have determinant n^{-n}; multiplying them by n gives our matrices of determinant one.

The inequality for mixed products of smooth functions and its extension to bounded functions whose Monge–Ampère measures have densities with respect to Lebesgue measure are stated in [Din07, Theorems 1.1–1.2, pp. 1–2], the first arXiv version of Dinew's article; the extension is attributed there to Kołodziej and Błocki. We prove the case of continuous densities, Corollary 9.7.4, by linearization and decreasing convolution, and the case of a common measure in Corollary 9.11.2. At fixed potentials, the smooth measure displayed by Dinew is $(4\pi)^n$ times ours.

The Dirichlet problem on a ball with continuous density, Theorem 9.8.2, follows the argument of [GZ17, Theorems 5.13–5.14 and Corollary 5.15, pp. 143–149]. Since the automorphisms of Lemma 9.6.1 move the center exactly to a prescribed point, the preliminary Lipschitz comparison with translations used there is not needed. Under such an automorphism S the n-th root of the density transforms into $(f^{1/n} \circ S)|\det_{\mathbb{C}} DS|^{2/n}$; on p. 145 this factor is written first without the absolute value and later without the derivative. The symmetric error is quadratic, and the half-sum is taken only once; this corrects the formulas printed on pp. 145–146. The stability estimate (9.9) contains a term for the boundary data, which is omitted on p. 149, and its proof does not require monotone approximation of the boundary data and of the density. Densities are taken with respect to $\eta^n = (\mathrm{dd}^{\mathrm{c}}|z|^2)^n$; at fixed potentials, the operator of [GZ17] gives 2^n times our measures.

The extension to bounded densities (Lemma 9.9.1 and Theorem 9.9.2) corresponds to [GZ17, Theorem 5.17 and Lemma 5.18, pp. 150–151]. In Lemma 9.9.1 the comparison is made on strict sublevel sets, and only the shifted sublevel set is required to be relatively compact. In the proof of Theorem 9.9.2 the approximating densities have an explicit common bound. Convergence in capacity of the approximating solutions is proved without assuming that the solution for a bounded measurable density is continuous inside the ball, by applying Hartogs' lemma on compact sets outside the exceptional sets of quasi-continuity. Approximating the n-th roots of the densities by convolution extends the stability estimate (9.9) to bounded densities.

The class $\mathcal{E}_0(B)$ of Definition 9.10.1 is due to Cegrell: its definition on hyperconvex domains is recalled in [Ceg04, Section 3, p. 161], where the class is said to have been introduced in [Ceg98], and it is also recalled in [Din07, Section 2, p. 3]. The moment condition used in Dinew's article to approximate a measure by its averages over cubes is stated in Theorem 2.2, pp. 4–5, whose conclusions on approximation and stability are quoted from other sources. Theorem 9.10.2 establishes the moment condition for measures dominated by the Monge–Ampère measure of a bounded

plurisubharmonic function; it follows from Błocki's estimate Proposition 9.3.5 by integration over level sets, after truncating the function with zero boundary values and passing to a smaller ball. The constants depend explicitly on the mass of the measure and on the oscillation of the subsolution, and the estimate also holds in dimension one.

By Proposition 9.10.3, which combines this estimate with the local L^p bound Proposition 8.3.6, the cube averages satisfy the moment condition uniformly. Applied to the solutions with zero boundary values, this gives uniform bounds for their local energy moments, and applied to the measure η^n, a constant multiple of Lebesgue measure, it gives compactness in L^1 along subsequences. At fixed potentials, Dinew's operator multiplies measures and local energy moments by $(4\pi)^n$; because of the power of the mass on its right-hand side, both sides of (9.11) are multiplied by the same factor.

The convergence of the energy integrals of the approximating solutions is asserted in [Din07, Theorem 2.2(3), p. 5], where it is quoted from Cegrell. Theorem 9.10.5 proves it for compactly supported measures dominated by the Monge–Ampère measure of a bounded plurisubharmonic function, along every subsequence of solutions that converges in L^1. One ingredient is the continuity of integrals against a fixed measure that does not charge sets of capacity zero, Lemma 9.10.4.

Convergence in capacity, as a consequence of the convergence of the energy integrals, is quoted in [Din07, Theorem 2.3, p. 5]. In Theorem 9.10.9 we derive it from the comparison of capacities of strict sublevel sets (Lemma 9.9.1) and the bound on second moments. By Lemma 9.10.7, truncation preserves the total Monge–Ampère mass of a function in $\mathcal{E}_0(B)$; the lemma also bounds the change of the measure caused by truncation in terms of the energy, and shows that Monge–Ampère measures with bounded energy are uniformly absolutely continuous with respect to capacity.

The definition of the solution as an envelope of subsolutions is recalled in [Din07, Definition 2.4 and the following paragraph, p. 5]. We solve the Dirichlet problem with zero boundary values, and identify the solution with this envelope, for compactly supported measures on a ball that are dominated by the Monge–Ampère measure of a bounded plurisubharmonic function (Theorem 9.10.12).

The mixed energy inequalities used in the approximation argument are recorded in [Din07, Section 2, p. 4] and attributed there to Cegrell. For their proof by integration by parts and the Cauchy–Schwarz inequality compare [Ceg04, Lemma 5.4 and Theorem 5.5, pp. 168–170]; Theorem 5.5 there has a fixed weight and exponents $1/n$. Theorem 9.10.13 is the version with $n+1$ functions, with an explicit but weaker constant. If dd^c is multiplied by a constant $c > 0$, both sides of (9.26) are multiplied by c^n. Corollary 9.10.14 uses this bound to discard the sets where the functions are very negative before continuity in capacity is applied; the limit functions there are locally bounded.

Dinew's inequality for bounded functions and a common measure is [Din07, Theorem 3.3, p. 9]. Its proof by approximation (Proposition 3.1 and Lemma 3.2 there, pp. 7–9) relies on results on the local Dirichlet problem, energies and continuity in capacity that are quoted from other sources. Theorem 9.11.1 allows n different functions; its proof uses the same approximation by cube averages, with these

inputs proved in the preceding sections. The inequality for mixed discriminants, Lemma 9.7.5, follows from the reversed Cauchy–Schwarz inequality given by Lemma 7.2.3, and linearization of the determinant then gives the case of continuous densities, Corollary 9.7.6. In this bounded setting the common measure may charge pluripolar sets. The extension to unbounded functions in Cegrell's classes, [Din07, Theorem 3.4], is not treated here.

The computation on contact sets in [Xia, Lemma 2.4.7, Step 1] uses Stampacchia's theorem that weak derivatives vanish almost everywhere on level sets, but it also asserts that a function in $W^{2,1}$ has a classical real Hessian. A complete argument is given in [DNT19, Lemma 3.1 and Theorem 3.2, pp. 4–8] (first arXiv version). There $C^{1,\bar{1}}$ means that the Laplacian with respect to a metric is bounded, not that the full real Hessian is bounded, and elliptic $W^{2,p}$ regularity is used.

In this book the vanishing of weak derivatives on level sets is Lemma 9.12.1, proved with a smooth truncated chain rule, and functions with bounded Euclidean Laplacian have real $W^{2,2}$ derivatives by Lemma 9.12.2. With these lemmas we prove Propositions 9.12.3 and 9.12.4 for regular potentials and obstacles in $W^{2,1}_{\mathrm{loc}}$; the density on the contact set does not require the determinant of the obstacle to be integrable on the whole domain. The operator used in [DNT19] is $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar{\partial}$, which is 2π times ours.

B.10 Pluripolar sets and the plurifine topology

Josefson proved that locally pluripolar subsets of $\mathbb{C}^n$ are globally pluripolar [Jos78]; see Theorem 10.2.1. The approach through capacity is due to Bedford and Taylor [BT82]; for relative extremal functions compare [GZ17, Theorem 4.32, pp. 118–119, Theorem 4.34, pp. 119–121, and Theorem 4.40 and Corollary 4.41, pp. 124–125]. The identities for the masses of relative extremal functions of compact and of open sets (Propositions 10.1.2 and 10.1.3) are proved with the homogeneous balayage of Corollary 9.6.4; together with approximation by decreasing open sets they give the characterization of pluripolar sets by capacity, Theorem 10.1.4.

For the local base of plurifine neighborhoods in Proposition 10.3.2 compare [EMW06, arXiv version, Lemma 3.1, pp. 3–4]. The main theorem of that paper, the local connectedness of the plurifine topology, is not treated here; its proof also uses finely subharmonic functions. The plurifine quasi-Lindelöf principle, Theorem 10.4.1, follows from Choquet's lemma and the pluripolarity of the exceptional sets of upper envelopes (Corollary 10.2.2). The plurifine locality of Bedford–Taylor products, Theorem 10.4.3, is deduced from their behavior under maxima (Proposition 9.2.7), up to exceptional sets that products of bounded functions do not charge. In Section 10.5 the difference of two plurisubharmonic functions, and its quasi-extrema, are considered only outside their common pole set, where the difference is defined.

B.11 Weighted estimates and local extension

For unbounded operators between Hilbert spaces, their adjoints and complexes of such operators compare [Dem12, Chapter VIII, Section 1, pp. 363–365]; the facts used here are proved in Lemma 11.1.3 and Theorem 11.1.4. Hörmander's estimate on pseudoconvex domains, with the method of three weights, is presented in [Bło14, Lemma 7.2, Proposition 7.3 and Theorem 7.1, pp. 35–39], and our proof of Theorem 11.3.3 follows this presentation; in the density lemma (Lemma 11.3.1) we verify that multiplication by cutoff functions preserves the domain of the adjoint before smoothing. Theorem 11.3.4 allows a plurisubharmonic weight together with a smooth weight whose complex Hessian is positive semidefinite; it is proved by approximation with weights for which the right-hand sides are bounded pointwise.

For Richberg's regularization and the boundary distance of pseudoconvex domains (Lemma 11.2.3 and Theorem 11.2.4) compare [Dem12, Chapter I, (5.18)–(5.21), pp. 43–44, and (7.1)–(7.2), pp. 53–55]. These results give the invariant exhaustion needed in Corollary 11.2.6.

Our proof of the Ohsawa–Takegoshi extension theorem, Theorem 11.4.4, follows Chen's method with a bounded twisting function, as presented in [Bło14, Theorems 9.1–9.3 and Lemma 9.4, pp. 46–48]. The auxiliary weight of Lemma 11.4.3 is $-\log(s+\log s)$, where the function $s=-\log(|\zeta|^2+\varepsilon^2)$ is positive. The constant in Theorem 11.4.4 depends only on the radius in the normal direction, and iteration of the theorem gives the estimate with prescribed point value, Corollary 11.4.5.

B.12 Lelong numbers and multiplier ideals

For the Jensen–Lelong formula compare [Dem12, Chapter III, (6.1)–(6.11), pp. 162–166]. The book uses only its form on Euclidean balls, Theorem 12.1.3. For smooth functions the formula follows from the derivative of spherical means, Proposition 2.1.1, and the monotonicity of the density from the convexity of circle means on complex lines through the center; regularization, with lower bounds for spherical means given by the Poisson kernel, gives the general case. Demailly's operator there is twice ours, and he measures growth against $\log|z|$; the density in (12.3) is written with our dd^{c} and $\log|z|^2$, so that $\nu(\log|z|^2,0)=1$.

The coherence of multiplier ideals, Theorem 12.2.7, and its proof by generation from a weighted Hilbert space of holomorphic functions are presented in [Dem11, Proposition 5.7, pp. 36–37]; page numbers refer to the manuscript on Demailly's web page. The weight there is $\mathrm{e}^{-2\varphi}$, which becomes our $\mathrm{e}^{-\varphi}$ after doubling the potential. For the strong Noetherian property and the Artin–Rees lemma used in the proof compare [Dem12, Chapter II, (3.22), p. 90, and Chapter IX, (1.1)–(1.3), pp. 403–404]. Hilbert's basis theorem, the Artin–Rees lemma and Krull's intersection theorem are proved in Lemma 12.2.1. We prove the strong Noetherian property in Lemma 12.2.4, and the uniform generation by elements of the Hilbert space, with bounded coefficients, in Lemma 12.2.8, using the completeness of the Hilbert space.

Siu's theorem on the analyticity of the upper level sets of Lelong numbers, Theorem 12.3.4, goes back to [Siu74]. The proof through the orders of vanishing of multiplier ideals is Demailly's method; compare [Dem11, Corollary 14.3, p. 138] and [Bou23, Theorem 10.2 and Lemma 10.3, p. 42], the version of Boucksom's notes dated 11 September 2023. The proofs of the restriction formula and of subadditivity by products and the diagonal (Theorem 12.4.1), and the proof of the pullback formula by the graph (Corollary 12.4.2), are taken from [Dem11, (15.1)–(15.3), pp. 158–159]. In (12.12), the restriction of $\mathcal{I}(\varphi)$ to Z means its image ideal in $\mathcal{O}_Z$.

Strong openness, Corollary 12.5.4, is due to Guan and Zhou [GZ15]. Our proof of Theorem 12.5.3 and Corollary 12.5.4 follows their argument by analytic curves and extension in Sections 3.1–3.2, pp. 609–611, of the published article, and its monotone version in Section 3.3, pp. 611–612. The curve is constructed in Lemma 12.5.1 by blowing up an ideal, and the lower bound for the norms of extensions is Lemma 12.5.2. In the induction the slices are taken in a fixed polydisc, so that the constant of the extension theorem does not depend on the slice or on the weight.

That a weight with zero Lelong number at a point has trivial multiplier ideals there, Corollary 12.3.2, is a special case of Skoda's integrability theorem; compare [Dem11, Lemma 5.6(a), pp. 35–36]. Here it is deduced from the bounds for the orders of multiplier ideals in Lemma 12.3.1, which rely on the extension with prescribed point value of Corollary 11.4.5. In that source the potential $\varphi = u/2$ corresponds to our u, by the conversion described at the beginning of these notes.

For discrepancies, the change of variables formula and the valuative criterion for multiplier ideals compare [Bou23, Proposition 10.12, Theorem 10.13 and Corollary 10.18, pp. 44–46]; Boucksom's conventions $\mathrm{e}^{-2\varphi}$ and $\log|z|$ are converted as above. We state the criterion, Theorem 12.6.9, for a holomorphic germ, with one positive parameter ε that works for all divisors over a neighborhood of the point. For weights of the form $c\log\sum_j|g_j|^2+O(1)$, the strict threshold on a log resolution in Theorem 12.6.5 involves every component of the resolved divisor that meets the fiber over the point, including the strict transforms of hypersurfaces in X, not only the exceptional ones.

B.13 Analytic singularities and regularization

Approximation by Bergman kernels is Demailly's construction; compare [Dema, Proposition 3.1 and its proof, pp. 13–14] and [Dem11, Theorem 14.2, pp. 136–138]. Page numbers of the former refer to the manuscript on Demailly's web page. These sources use the weight $\mathrm{e}^{-2m\varphi}$ and the factor $(2m)^{-1}$, whereas Theorem 13.2.3 uses $\mathrm{e}^{-m\varphi}$ and m^{-1}, in accordance with our normalization of Lelong numbers. The comparison of the Bergman kernel with finitely many generators and its form on a log resolution (Lemma 13.2.2) are derived from the bounded generation of Lemma 13.2.1(3), a consequence of Lemma 12.2.8.

The computation for the logarithm of a sum of local Bergman kernels weighted by cutoff functions (Lemma 13.4.1) is [Dema, Lemma 3.5, p. 15], and the gluing

over overlapping balls in Lemma 3.6, pp. 16–17, of that paper is explained by the more general Lemma 4.6 and its proof, pp. 19–22. The comparison of the local kernels on overlaps and the gluing by maxima (Lemmas 13.3.1 and 13.3.2) follow the argument of Vu and Xia in the unpublished manuscript [VX, Lemmas `patching` and `regularization`]. Theorem 13.3.3 has exactly the two conclusions of the regularization lemma there, and its curvature constant depends only on the fixed geometry.

Our Definition 13.1.11 asks for the local form with smooth remainder on every log resolution of the ideal, that is, on every composite of blowups along closed submanifolds that transforms it into the ideal of a divisor with simple normal crossings support (Section 13.1). The definition in [Xia, Chapter on plurisubharmonic functions, definition of gentle analytic singularities] asks for such a form on one Kähler modification. Since log resolutions exist and are Kähler (Proposition 13.1.10), our requirement is the stronger one, and the approximations of Theorem 13.4.5 satisfy it.

For the comparison by extension from the diagonal, the dyadic correction and the passage from the integrability of a gap to inclusions of ideals compare [DPS01, arXiv version, Section 2.2, Steps 2–3, pp. 12–14]. Theorem 2.2.1 there, proved on pp. 9–14 of the same version, concerns equisingular approximation, and Remark 2.2.2 distinguishes the case of logarithmic poles. Theorem 13.4.5 is a different statement: the approximations have the prescribed gentle analytic types $(2^{-j}, \mathcal{I}(2^j\varphi))$, and their monotonicity and the inclusions of ideals between consecutive dyadic scales are derived from the kernels with cutoffs, the comparison on the diagonal and Lemma 13.4.3. To compare the approximations, the growth estimate Lemma 13.5.1, proved on finitely many local log resolutions, suffices, so that no division theorem is needed.

The ideal $\mathcal{I}_\infty$ of germs of bounded growth (Definition 13.6.1) is used in [Xia, Chapter on plurisubharmonic functions, the lemma following the definition of $\mathcal{I}_\infty$ and Lemma 1.6.3]. Its coherence for potentials with analytic singularities, Theorem 13.6.2, is proved with a cone of fractional powers, finite normalization and finitely many homogeneous generators; the direct image under the finite map is handled by division by monic polynomials, so the general theorem on direct images under proper maps is not needed. Corollary 13.6.3 derives from the discrepancies on log resolutions the inclusion for large multiples that is needed to compare approximations; the sharper estimate in the source, with a loss depending on the dimension, is not proved here.

B.14 Global potentials and positivity loci

For elliptic operators on compact manifolds and the Laplacian on forms compare [Dem12, Chapter VI, Section 2, pp. 289–290, and Sections 3.1–3.2, pp. 291–294]. The results on Sobolev spaces, compactness and regularity used here are proved directly in Lemmas 14.1.1 and 14.1.3, by Fourier truncation, difference quotients and duality. The proof of the dd^c-lemma, Theorem 14.1.7, uses the scalar trace equation

and the identity for the Hodge star of primitive $(1,1)$-forms; the factor $-1/(4\pi)$ in the trace identity (14.7) comes from our normalization of dd^{c} and from the sign convention that makes the Riemannian Laplacian nonnegative.

The method of primitive representatives for the mixed Hodge–Riemann relations is developed in [DN05, Propositions 2.3–2.4 and proof of Theorem A, pp. 5–8]. For real $(1,1)$-classes, Lemma 14.1.4 and Theorem 14.3.3 make a representative primitive at every point by solving a scalar equation in divergence form. The operator $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/\pi$ of that source is twice ours; doubling its correcting function gives the same representative. Corollary 14.3.4 derives the lower bound of mixed intersection numbers of nef classes by the geometric mean of their volumes from the concavity of the logarithms of mixed intersection numbers, first for Kähler classes and then, by perturbation, for nef classes.

For positive cones, potentials with minimal singularities and the divisorial Zariski decomposition compare [Bou04, Sections 2.1–2.3, pp. 47–49, Sections 2.7–2.8, pp. 51–53, and Section 3.1, pp. 54–55]. These locators refer to the published article, whose title and numbering differ from those of the arXiv version of 2002. Siu's decomposition, Theorem 14.3.7, including the uniqueness of the decomposition, is proved by subtracting divisors with the help of the uniform logarithmic growth of local potentials along a divisor (Proposition 12.3.6). The proof of Proposition 14.4.2 is the argument of Proposition 3.6 of the published article, by convex combination with a Kähler current. As in that article, minimal multiplicities are defined by means of currents $T \geq -\varepsilon\omega$; for big classes they coincide with the Lelong numbers of a potential with minimal singularities, and in general they are bounded above by these Lelong numbers.

That the non-Kähler locus of a big class is the pole set of a single Kähler current with analytic singularities is [Bou04, Definition 3.16 and Theorem 3.17, pp. 59–60]. Our proof of Theorem 14.4.4 follows Boucksom's argument, based on the stationarity of increasing sequences of coherent ideals on a compact manifold and on taking maxima, with the approximation results of Chapter 13. The theorem also contains the identity (14.15), with conventions that include classes that are not big or not pseudo-effective. The last part of Boucksom's Theorem 3.17, a criterion in terms of restrictions to arbitrary analytic subsets, is not treated here.

The conventions for psh metrics, written as quadratic forms, and for the Fubini–Study metrics defined by sections are those of [Xia, Chapter on plurisubharmonic functions, Definitions 1.8.3, 1.8.6 and 1.8.7]. In Proposition 14.5.8 the convex combinations have rational coefficients, and closure under these operations is proved by clearing denominators of the degrees and multiplying sections. The local weight of a metric in [Bou04, Section 2.4, pp. 49–50] is $\mathrm{e}^{-2\varphi}$; doubling the potential gives our weight $\mathrm{e}^{-\varphi}$ for quadratic metrics, and the curvature current is the same.

For the L^2 estimate for line bundles on Kähler manifolds, Lemma 14.6.5, compare [Dem12, Chapter VI, Section 6.1, pp. 304–306, and Chapter VII, Theorem 1.1, Theorem 1.2 and Corollary 1.3, pp. 329–330]. We prove the commutator identity, the density of smooth forms for the graph norm and the solvability in Hilbert space, and we show that for forms of top degree the curvature term in the estimate does not depend on the auxiliary complete metric.

Complete metrics and weights with logarithmic poles are also used in [His12, Theorem 5.1 and its proof, pp. 18–22] (second arXiv version), where Theorem 5.1 assumes a smooth metric and Theorem 1.4 assumes projectivity. Theorem 14.6.7 concerns extension from a point of a compact Kähler manifold. It is proved by applying the gentle approximations of Chapter 13 to the integrable error introduced by a fixed cutoff function. The weighted spaces of sections in Lemma 14.6.8 provide the finite-dimensional Hilbert spaces and the curvature of the background used in [Xia, Proposition 1.8.2].

For the twisted L^2 estimate and the treatment of the error from the initial gluing compare [Cao17, Section 2, pp. 5–9, and Appendix, pp. 14–15]; page numbers refer to the second arXiv version, posted on 3 October 2017 and dated 4 October in the manuscript. Theorem 1.3 there treats a submanifold defined by a global section of a vector bundle, under assumptions on curvature and approximation. Theorem 14.7.8 applies to an arbitrary closed connected submanifold Y. The pole along Y is given by the function R_Y of Lemma 14.7.4, a sum of local squared normal distances weighted by squares of cutoff functions.

The estimate (14.31) is first proved for weights with gentle analytic singularities. The local gain from the logarithmic factor is Lemma 14.7.3, the error of the initial gluing is controlled by strong openness and Hölder's inequality in Lemma 14.7.6, and Lemma 14.7.7 computes the limit over thin annular neighborhoods of Y for gentle weights. Approximation by decreasing gentle weights then gives the estimate for the original metric, with the same constant. Corollary 14.7.9 recovers [Xia, Theorem 1.8.1]; the threshold for the tensor power and the conversion between the volume forms, which involves factorials, are stated explicitly.

The extension of potentials from a submanifold, Theorem 14.7.12, follows [DRWN+26, arXiv version, Theorem 6.1 and its proof, Lemma 6.2, pp. 19–22]. That source uses the same operator $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/(2\pi)$ (p. 5) and the same notion of analytic singularities, with rational coefficients and bounded remainder (Definition 2.2, p. 6). The lemma on poles along the submanifold quoted there is replaced by Lemma 14.7.4. Strict positivity near the submanifold follows from an estimate for block matrices, and the local extensions are formed as logarithms of sums of exponentials (Lemma 14.7.11); the exponent of the normal pole is taken small enough to absorb the error coming from the Hölder continuity of the projection. Since the comparison of the local extensions on overlaps holds also at common poles, the glued maximum involves a fixed set of indices near each point of the submanifold.

Josefson's theorem on compact complex manifolds, Theorem 14.8.7, is due to Vu [Vu19, arXiv version, Theorem 1.1 and its proof, pp. 8–9]. The operator $\mathrm{i}\partial\bar\partial/\pi$ used there is twice ours. We follow Vu's argument, with a weighted sum of global poles and two extremal limits, but replace his comparison of capacities in exponential form by the estimate (14.46). Positivity of the mass is needed only for the one bounded limit function, not as a uniform lower bound over all bounded potentials.

For products with a background form that is not closed and the modified comparison principle compare [KN15, Section 1, pp. 3–4, and Theorem 2.3 and its proof, pp. 5–6], the second arXiv version of the article by Kołodziej and Nguyen, whose operator is again twice ours. The one-factor inequality on which the modified comparison

principle rests is Lemma 14.8.2; we prove it for smooth approximations with a convex cutoff and a partition of unity, and pass to the limit by the continuity of Bedford–Taylor products with polynomial weights (Corollary 9.2.6) and uniform approximation of the weights by polynomials. The modified comparison principle is proved here only with the fixed margin $\omega_v \geq \frac{1}{2}\omega$, Proposition 14.8.3.

In Proposition 14.8.6 the background form is frozen on small balls, so that the balayage of ordinary plurisubharmonic functions applies. In this way, the proof that the extremal function puts no Monge–Ampère mass outside the compact set avoids solving the homogeneous Dirichlet problem or smooth Monge–Ampère equations on Hermitian manifolds.

B.15 Non-pluripolar products

The definition of non-pluripolar products by truncation at finite levels, locally and on compact Kähler manifolds, is developed in [BEGZ10, Section 1.2, Proposition 1.4, Lemma 1.5 and Proposition 1.6, pp. 203–207]. Multilinearity is proved in Lemma 15.1.2 and Proposition 15.1.5 by truncating all factors at a common level. In Lemma 15.1.2 the convergence of the restricted products is stated for Borel sets E_k such that $\mathbf{1}_{E_k} \to 1$ almost everywhere for the trace measure of the product; this holds if the E_k increase and their union has pluripolar complement, whereas the condition on the union alone does not suffice. In Example 15.1.6 the computation of the Hessian shows that the function is plurisubharmonic with the sign opposite to the one displayed in [BEGZ10, Example 1.3, pp. 204–205].

That non-pluripolar products are closed, Theorem 15.2.1, is [BEGZ10, Theorem 1.8 and Lemma 1.9, pp. 207–210]. The gradient $\mathrm{d}v \wedge \Theta$ of the product of a bounded plurisubharmonic function v and a closed positive current Θ, and the corresponding chain rule, are established in Lemma 9.13.1 by decreasing approximation; the proof of Theorem 15.2.1 then estimates the error made by the cutoff functions. The gradient energy $\mathrm{d}v \wedge \mathrm{d}^{\mathrm{c}}v \wedge \Theta$, with d^{c} as in (7.6), is used only as a positive measure, and no closedness property of it is needed. On a compact Kähler manifold, Stokes' formula for bounded potentials (Proposition 15.4.1) and the mass bound with shifted backgrounds (Theorem 15.4.3) are proved before global truncations are used.

The lower bound for mixed products in terms of a common measure is [BEGZ10, Proposition 1.11 and its proof, p. 211]; it applies Dinew's inequality for bounded functions before passing to the restrictions at finite levels. Proposition 15.4.5 is the statement for n currents, deduced from Theorem 9.11.1; the case of two currents, Corollary 15.4.6, follows by repeating factors. Since non-pluripolar products put no mass on pluripolar sets, the domination hypotheses force the densities to vanish almost everywhere for the common measure on the union of the local pole sets; this allows us to remove the restrictions to finite levels for an arbitrary measure.

The compatibility of non-pluripolar products with Bedford–Taylor products of bounded factors, Proposition 15.3.3, is [Xia, Proposition 2.2.1(iii)]. Our proof uses the fact that Bedford–Taylor products with a current that does not charge pluripolar

sets again do not charge them, Lemma 15.3.2, which follows from the Chern–Levine–Nirenberg estimate relative to a current, Lemma 15.3.1; the products are then identified with the help of bounded plurisubharmonic cutoff functions.

For the identity in degree one, Proposition 15.3.6, compare [EK23, Proposition 3.3, pp. 3–4] (third arXiv version). That proposition concerns the domain of definition of the Monge–Ampère operator and refers to an earlier theorem for the equality. We prove the identity first in one variable, by approximating the Riesz measure from below by measures with continuous potentials, and then in several variables by integration over the fibers of coordinate projections. Strong positivity (Proposition 15.1.4) passes from products of smooth forms (Proposition 7.2.1) to Bedford–Taylor products and then, by Lemma 15.1.1, to their restrictions to Borel sets.

The fiber integration formula and its application to mixed products on compact manifolds follow [Xia, Chapter on non-pluripolar products, Lemma 2.1.1 and Corollary 2.2.1]. The measurability of the fiber measures that is needed to integrate indicator functions of level sets is proved in Lemma 15.3.4, the formula for Bedford–Taylor products of bounded functions is Lemma 15.3.5, and Corollary 15.5.1 passes to the limit along the increasing restricted products.

The comparison of masses in Lemma 15.7.1, for potentials that are locally bounded outside the closed pole set of a quasi-plurisubharmonic function, follows the argument with compact supports in [BEGZ10, Theorem 1.16 and its proof, pp. 217–218]. The auxiliary projective construction is due to Witt Nyström [WN19, arXiv version, Sections 3–4, pp. 4–9]. The projection onto the simplex and the integration over angles are carried out in Lemma 15.7.2. With the exterior powers used in this book, which are not divided by factorials, the formula of that lemma contains the factor $\binom{n+N}{n}$, and the Hessian contains $2\sum_i \mathrm{d}x_i \wedge \mathrm{d}^{\mathrm{c}}x_i$. Both factors follow by direct differentiation and expansion, and they are missing from the calculation displayed in the preprint; the common positive binomial factor cancels in the proof of monotonicity. Theorem 15.7.4 handles classes that need not be big by adding $\varepsilon\omega$, so it does not depend on the criterion for bigness by positive mass, Theorem 17.4.20.

The invariance of mixed masses under bounded perturbations, the lower bound for limits in capacity and the monotonicity of mixed masses are proved in [DDNL18, Proposition 2.1, Theorem 2.3 and Theorem 2.4, pp. 2054–2057]. Our Theorem 15.6.5 uses the local continuity of weighted products under convergence in capacity and truncations at fixed levels, and Corollary 15.7.5 obtains invariance from Theorem 15.7.4 by polarization and then monotonicity by decreasing maxima. Positive mixed intersection numbers and full mass are introduced in [BEGZ10, Definition 1.17, Proposition 1.20 and Definition 1.21, pp. 218–222]; Proposition 15.8.2 gives the bound for these masses that is used later. The comparison and domination principles, Proposition 15.8.5 and Corollary 15.8.6, follow [WN19, arXiv version, Corollaries 4.1–4.2, pp. 8–9]; they keep the comparison hypotheses on singularities, and pointwise domination requires a potential with minimal singularities, as stated there.

The version with varying weights, Corollary 15.6.8, is [DDNL25, arXiv version, Theorem 2.6 and its proof, pp. 10–12], the reference given in [Xia, Theorem 2.4.2]. We keep the argument with a cutoff at a finite level and complementary weights, but work locally in every mixed degree before gluing on the compact manifold;

the limit is taken with the theorem for bounded weights proved earlier. The survey uses $d^c = i(\bar\partial - \partial)/(2\pi)$, so its operator dd^c is twice ours. At fixed potentials and with doubled backgrounds, each product of p factors is 2^p times ours, and this common factor cancels in the inequalities for lower limits and for masses. Proposition 15.6.7 establishes the statement of [Xia, Proposition 2.3.1] for arbitrary monotone nets; the equivalence of the two notions of convergence in capacity is asserted for plurisubharmonic functions, not for arbitrary Borel functions with values in $[-\infty, \infty]$.

B.16 Envelopes and model potentials

Envelopes with obstacles, rooftop envelopes and envelopes of singularity types are used throughout [Xia]. The argument for lower semicontinuous obstacles and the two reductions for quasi-continuous obstacles are in [Xia, Lemma 2.4.3 and Theorem 2.4.4]. We follow them in Lemma 16.2.4 and Theorem 16.2.5, using the local homogeneous Dirichlet problem and the capacity theory of Chapter 9. Since potentials with minimal singularities are locally bounded outside the pole set of a reference potential with analytic singularities, the argument can be carried out in charts where they are bounded; it uses no differentiability of the envelope.

The concentration theorem of that source also allows pseudo-effective classes and quasi-continuous obstacles that are not bounded above. Obstacles that are not bounded above are already covered by Theorem 16.2.5, because capping the obstacle strictly above the envelope changes neither the envelope nor its contact set. Pseudo-effective classes are treated in Corollary 17.4.21, after the criterion for bigness: if the class is not big, every non-pluripolar measure of top degree vanishes.

For functions whose Laplacian with respect to a metric is bounded, and for the reduction to regular contact, compare [DNT19, p. 4 and Lemma 3.1, pp. 4–5]. Such functions are locally in $W^{2,2}$ by Lemma 16.2.8, which is derived from the elliptic theory of Lemma 14.1.3. Proposition 16.2.9 applies the vanishing of weak derivatives on level sets to obstacles in $W^{2,1}_{\mathrm{loc}}$, for instance continuous obstacles with bounded metric Laplacian; the computation on the contact set needs only this local regularity. The stronger elliptic $W^{2,p}$ estimate quoted in the source is proved separately in Proposition 16.2.11, by freezing the principal coefficients and testing the equation against a regularization of $u|u|^{p-2}$, and Corollary 16.2.12 then shows that the signed determinant of such obstacles is integrable on all of X.

Approximation of envelopes by Monge–Ampère equations with an exponential penalty, and the limit as the parameter tends to infinity, are due to Berman; the equations and estimates are presented in [Tos18, equations (2.1)–(2.2), pp. 282–283], where smooth solvability is referred to Aubin and Yau. We prove solvability for positive parameters in Theorem 16.2.21 by the continuity method, with a contraction argument in Sobolev spaces and a priori estimates along the path. The third-order estimate, Lemma 16.2.17, bounds the difference of two Chern connections. Its computation follows the tensor calculation in [SW12, Proposition 2.8 and Theorem

2.9, pp. 24–26], the first arXiv version of Song and Weinkove's lecture notes, where such estimates are traced back to a calculation of Calabi and the explicit formula in the parabolic case is attributed to Phong, Šešum and Sturm. Our calculation is elliptic: the penalized equation bounds the Ricci tensor and its covariant derivative in terms of the difference of connections.

The constants in the uniform trace bound and in the convergence to the envelope (Lemma 16.2.15 and Proposition 16.2.22) depend only on the fixed geometry, on a bound for the obstacle f and on an upper bound for the trace $\Lambda \mathrm{dd}^{\mathrm{c}} f = -Lf/(4\pi)$ of its complex Hessian, that is, on a lower bound for its metric Laplacian Lf. Tosatti writes forms with $\mathrm{i}\partial\bar\partial$; in the notation of Section 16.2.3, the coefficient matrices h and k represent $2\pi\omega$ and $2\pi\kappa$, so that $g_{i\bar j} = k_{i\bar j} + a_{i\bar j}$ is the coefficient matrix of $2\pi(\kappa + \mathrm{dd}^{\mathrm{c}} a)$.

That the envelope of the minimum of finitely many functions with bounded metric Laplacian again has bounded metric Laplacian is [DR14, Theorem 2.5(ii), p. 5] (first arXiv version). We prove it in Theorem 16.2.24 by smoothing the finite minima and applying the uniform trace estimate above; in the trace of the complex Hessian of the smoothed minimum, the variance of the gradients enters with a negative sign, so the upper bound for this trace is preserved. Convolution in charts and the elliptic estimates for all finite p (Proposition 16.2.11) control the commutator of the convolution with the operator $\Lambda \mathrm{dd}^{\mathrm{c}}$, whose coefficients vary.

The estimate for the full real Hessian (Lemma 16.2.25 and Theorem 16.2.26) follows Tosatti's argument with penalized equations, [Tos18, pp. 283–287]. For the absorption of the negative term with third derivatives, Tosatti refers to the calculation of Chu, Tosatti and Weinkove, which is given in full in [CTW17, arXiv version, Section 2, Lemmas 2.1–2.2, pp. 4–10]. In this book the formulas for derivatives of eigenvalues and every absorption step are proved directly. The perturbation by a projection subtracts one from the eigenvalues other than the largest one; keeping track of this shift gives the bound $2\lambda + 3$ in the comparison of positive forms. The real metric associated with $2\pi\kappa$ is $2\,\mathrm{Re}(k_{i\bar j}\,\mathrm{d}z_i \otimes \mathrm{d}\bar z_j)$, so that its Hessian trace is $2D_k a$. Since the estimate uses, besides bounds for the obstacle and its gradient, only an upper bound for its real Hessian, the same proof applies to minima of finitely many smooth functions, and then to rooftop envelopes of finitely many $C^{1,1}$ functions.

The regularity of envelopes in big classes on the ample locus is [DNT24, Theorem 1.1, pp. 1347–1349]; this locator refers to the statement in the published article. The complete proof and the extension to finitely many obstacles are given in [DNT24, arXiv version, Lemmas 3.1, 3.3–3.5, Section 4 and Theorem 4.2, pp. 3–19]. The smoothing by the flow of a Chern connection goes back to [Demb, Proposition 2.9, pp. 5–6, Proposition 3.8, pp. 8–12, and Theorem 4.1, pp. 12–16], a twenty-page manuscript cited with its own page numbers. In Lemma 16.2.27 we carry out the computation with jets of a fixed finite order instead of an extension to infinite order, and compute the integrals of the weights directly. Demailly's normalization $\mathrm{i}\partial\bar\partial/\pi$ is twice our dd^{c}, while the auxiliary operator in our kernel computation is $\mathcal{D} = 2\pi\mathrm{dd}^{\mathrm{c}}$.

Theorem 16.2.29 follows the two successive smoothing estimates of Di Nezza and Trapani and their use of high powers of the equations of the divisors on a resolution. Local holomorphic changes of the radius give the required uniform estimates without

introducing the total space of the line bundle used by them, and Lemma 16.2.28 handles the minimum over varying radii at the upper end point. We keep a fixed quadratic correction in the smoothing, keep the smooth background separate from the nonsmooth obstacle, and recover the varying metric Laplacian from the quadratic error by an expansion of the adjoint kernel. The final local estimate in real Sobolev spaces follows from the Euclidean Hessian multipliers of Chapter 6.

The formula for the Monge–Ampère measure on the contact set for arbitrary singular potentials is [DNT19, Theorem 3.2 and its proof, pp. 5–8], where the method is credited to earlier work of Witt Nyström. In Theorem 16.2.31 we simplify the passage to the limit in the weights at levels: the limiting inequality is restricted to the contact set, where the positive weight cancels, so that a single truncation level and a single positive parameter suffice. For a general obstacle, whose metric Laplacian need not control its full Hessian, we keep the regular intermediate envelope.

Corollary 16.2.32 gives the formula of the source for the density on the contact set of an ordinary obstacle in big classes. Corollary 17.4.22, proved after the criterion for bigness, also covers pseudo-effective classes that are not big, in which all non-pluripolar measures of top degree vanish. All measures are taken with our normalization; with the operator $\mathrm{i}\partial\bar\partial$ of the source, products of n factors, pure or mixed, are $(2\pi)^n$ times ours.

The argument showing that envelopes of singularity types preserve mixed masses is in [DDNL18, Section 3B, p. 2060]; the continuity along increasing sequences that it requires is provided by Corollary 15.7.6 and Proposition 16.3.2. The bounds for the Monge–Ampère measures of rooftop envelopes and of envelopes of singularity types are [DDNL18, Lemma 3.7 and Theorem 3.8, pp. 2060–2061]. In Lemma 16.2.2, Corollary 16.2.6, and Proposition 16.3.3 they are derived from convergence within a bounded singularity type and from the theorem on contact sets for obstacles. Here a measure is carried by the contact set if the complement of this Borel set has measure zero, which says nothing about the closed support of the measure.

The extrapolation argument and the domination principle (Theorems 16.4.4 and 16.4.6) follow [Xia, Lemma 2.4.4 and Theorem 2.4.5]. In Lemma 16.4.1, a fixed point that preserves the mass is obtained as the regularized limit of an increasing sequence that almost maximizes an integral height at every step; this replaces the transfinite iteration in the proof in the source. For Theorem 16.4.7 and Proposition 16.4.10 compare [DDNL18, Theorems 3.12 and 3.14, pp. 2063–2064]. There the domination principle is derived from the construction with prescribed densities in Corollary 3.9, p. 2062, whereas here, in a big class, it follows from extrapolation and the concentration on contact sets. The example with a $\log\log$ singularity, Example 16.4.8, follows [Xia, Example 3.1.1]; its mass is computed with the identity in degree one, Proposition 15.3.6.

The criterion for the existence of rooftop envelopes, the substitution of potentials of full mass and the mass inequality (Propositions 16.5.1 and 16.5.3 and Theorem 16.5.5) follow [Xia, Proposition 3.1.5, Lemma 3.1.2 and Theorem 3.1.3]. The bound for upper limits needed there is isolated in Lemma 16.5.4; it does not claim weak convergence of the measures along a decreasing sequence, which may lose mass.

Envelopes with an obstacle and a constraint on the singularity type (Definition 16.7.1 and Propositions 16.7.2 and 16.7.3) extend [Xia, Definition 3.1.4]. The bound on the contact set for bounded quasi-continuous obstacles follows from the theorems on relative envelopes and on concentration for obstacles; it does not identify a smooth density. Proposition 16.7.4 is [Xia, Proposition 3.1.15], with an explicit defect term for potentials that do not have full mass. The relative capacity of Definition 16.7.5, defined with competitors between $\phi - 1$ and ϕ, reduces to [Xia, Definition 2.3.1] when ϕ has minimal singularities. Theorem 16.8.1 gives a complete proof of [Xia, Theorem 6.1.1]; in big classes it provides the margin for rooftop envelopes used in [VX, Theorem 3.1.1].

Our definition of model potentials keeps the condition of positive mass from [Xia, Definition 3.1.3]. For the constraints by multiplier ideals used in the chapter on envelopes of that book, Proposition 16.6.1 combines the uniform valuative criterion Theorem 12.6.9 with Lemma 16.6.2, the formula $\nu(v, E) = \inf_j \nu(v_j, E)$ for the Lelong numbers along divisors of an upper envelope $v = (\sup_j v_j)^*$. In the proof of this formula, a local pullback argument shows that these Lelong numbers are preserved under the countable selection of Choquet's lemma. The proposition thus provides the closure property used in [Xia, Propositions 3.2.6 and 3.2.9]; the construction itself and its algebraic consequences are not treated here.

The comparison of capacities in Section 16.9 follows Lu [Lu21, arXiv version, Lemmas 3.1–3.4 and Theorem 3.5, pp. 8–13]. Integration by parts for bounded potentials on a compact manifold, Lemma 15.4.2, is derived from the calculus of gradients in Section 9.13, and, together with approximation by smooth bounded potentials, it gives the weighted estimate Lemma 16.9.2. The envelopes with increasing penalties of Lemma 16.9.3 provide the concentration on compact sets and the two growth bounds, which replace the properties of relative extremal functions quoted in the preprint. The comparison then proceeds in three steps: powers of model potentials in Kähler classes (Proposition 16.9.6), comparison downwards for potentials of full mass (Lemma 16.9.7), and comparison in both directions (Theorem 16.9.8). Finally, Proposition 16.9.9 compares, by a local extension construction, these capacities with the local capacities of Chapter 9; in particular, local convergence in capacity implies convergence in each relative capacity.

B.17 Relative energy and integration by parts

The identity for differences of bounded type, Lemma 17.1.2, follows [Lu21, arXiv version, Lemma 4.1, pp. 13–15]. Its proof reduces to bounded potentials by truncation, first for a strictly positive background; an expansion in polynomials then removes this restriction. The general integration by parts formula, Theorem 17.1.3, follows the argument by differentiation and polarization in [Lu21, arXiv version, proof of Theorem 1.2, pp. 15–16]. In his introduction, pp. 1–3, Lu credits the earlier integration by parts formula to Mingchen Xia. The general formula is stated in the introduction of [Xia19, Theorem 1.1 and discussion, pp. 1–2] (first arXiv version), which also

describes a proof by a projective construction; the proof followed here is Lu's later argument with capacities.

Relative energy classes and the relative energy are defined in [Xia, Definitions 3.1.5–3.1.6], and the cocycle formula for potentials of bounded type, the bounds by tangents, concavity and continuity along decreasing sequences are [Xia, Lemmas 3.1.3–3.1.5 and Proposition 3.1.16]. In Proposition 17.2.3 the first variation of the energy along a segment is computed directly from the integral formula for the energy and the integration by parts theorem, and the cocycle formula follows by integrating it. In Propositions 17.2.6 and 17.2.7 the energy is extended by means of the canonical truncations $\max\{u, \phi - k\}$, and continuity is proved for decreasing nets, not only for sequences, the limit being the pointwise limit.

For Theorem 17.2.8, Lemma 17.2.11, and Proposition 17.2.12 compare [Xia, Proposition 3.1.17, Lemma 3.1.6, Corollary 3.1.6 and Proposition 3.1.18]. In our proof, a uniform bound at finite levels for the integral of a potential against its own Monge–Ampère measure shows that finite energy by itself implies full mass.

Continuity of the energy along increasing nets, Proposition 17.3.1, follows [Xia, Proposition 3.1.20]; the uniform bound for the tails is (17.13). The proof of [Xia, Proposition 3.1.21] uses an exponential Monge–Ampère equation; in Proposition 17.3.6 we instead integrate the bound for the measures of rooftop envelopes on contact sets, Corollary 16.2.6. This closure of finite energy under rooftop envelopes guarantees that the slices of Perron segments have finite energy. The closure under upper envelopes of families and the envelopes of mutual singularities follow [Xia, Propositions 3.1.23–3.1.24].

The derivative of the energy along affine segments, Proposition 17.3.8, also appears in [Xia, Lemma 3.1.11]. Here it follows from the cocycle formula, which is polynomial in the parameter. The family of smooth potentials on $\mathbb{P}^1$ in Example 17.3.10 shows that convergence in L^1 does not imply convergence of the energy.

The characterization of solutions as maximizers of a functional and the derivative of the energy of projections (Theorem 17.4.4 and Proposition 17.4.3) follow the variational approach of [BBGZ13, Theorem 4.1 and Lemma 4.2, pp. 207–209], whose normalization of the operator agrees with ours. We prove the derivative of the energy of projections directly, with a model potential as reference, from the theorem on contact sets and the bounds of the energy by tangents.

For Monge–Ampère equations relative to a model potential, with prescribed measure and with an exponential factor, compare [DDNL21, arXiv version, Theorem 4.3, p. 20; Theorem 5.3 and Lemma 5.4, pp. 24–25]. That source uses $\mathrm{i}\partial\bar{\partial}$; its potential corresponding to our u is $u/(2\pi)$, and its exponential parameter is $2\pi\lambda$. We solve the prescribed-measure equation with bounded densities (Theorem 17.4.5) and the exponential equation for every finite non-pluripolar measure (Theorem 17.4.11) by compactness in the variational approach. The relative comparison and domination principles give Proposition 17.4.6 for every finite non-pluripolar measure, when the reference has positive mass.

For the continuity of integrals against a fixed non-pluripolar measure compare [BBGZ13, Theorem 3.10, pp. 203–205]. Lemma 17.4.9 proves the case of uniformly bounded functions. The upper limit is controlled outside a pluripolar set; for the lower

limit we use convex combinations of the tails with almost minimal L^2 norm for the given measure, which form a Cauchy sequence by the parallelogram identity. Applied to exponentials, the lemma allows the same variational construction for arbitrary finite non-pluripolar measures. The lower bound for limits in Lemma 17.4.12 follows [DDNL21, arXiv version, Lemma 2.8, pp. 12–13]; a fixed lower measure is restricted before weak limits are taken. The existence of solutions with prescribed measure, Theorem 17.4.13, is obtained instead by letting the exponential parameter tend to zero; for the existence statement compare Theorem 4.7, p. 23, of that preprint.

Uniqueness for the relative equations is proved in [DDNL18, Theorem 4.29 and its proof, pp. 2079–2080] by adapting the argument of Kołodziej and Dinew based on the concentration of mass; in that section the model reference has positive mass and small unbounded locus. Our Theorem 17.4.14 allows an arbitrary model reference of positive mass. It follows the same strategy, using the existence theorem, the comparison and domination principles for full mass and the lower bound for products with respect to a common measure (Corollary 15.4.7); the domination principle turns inequalities almost everywhere for the common measure into pointwise inequalities.

The iteration with capacities in Theorem 17.4.15 and Corollary 17.4.17 follows [DDNL21, arXiv version, Theorem 3.3 and its proof, pp. 14–15]. Here the measure is assumed to be dominated by the power $1 + \delta$ of the capacity, and an iteration with geometrically decreasing steps shows directly that the capacities of the relevant sublevel sets vanish beyond a finite level. The uniform bound for moments is Corollary 14.2.2.

The estimates of volume and of measures with L^p densities by capacity (Proposition 16.9.4 and Corollary 16.9.5) follow [DDNL18, Proposition 4.30, pp. 2081–2082], which assumes a small unbounded locus and uses uniform Skoda integrability; the formulation for arbitrary model potentials is [DDNL21, arXiv version, Lemma 3.9 and Proposition 3.10, p. 18]. Here the prerequisites are the uniform exponential bound Corollary 14.2.3 and the penalized envelopes of Lemma 16.9.3. At fixed potentials, the background $2\pi\theta$ of the source gives measures and capacities that are $(2\pi)^n$ times ours, while their ratios to the mass of the reference are unchanged.

The criterion that a class containing a current of positive non-pluripolar mass is big, which is used in [Xia, Proposition 2.4.1], is [BEGZ10, Proposition 1.22, p. 222]. There the criterion is attributed to Boucksom; in the published article [Bou02, Section 4.3, Theorem 4.7, pp. 1057–1061] the numbering differs from that citation, and the volume is defined through the determinant of the absolutely continuous part. Our proof of Theorem 17.4.20 uses instead the Monge–Ampère equations of Section 17.4, solved in the big classes obtained by adding $\varepsilon\{\omega\}$ to the given class, the mixed trace estimate Lemma 17.4.19 and Lamari's criterion. The proof also shows that a pseudo-effective class is big if and only if its volume is positive, without using an identity between the determinant volume and the non-pluripolar volume.

Lamari's duality criterion, Lemma 17.4.18, is [Lam99, Lemma 3.3 and its proof, pp. 275–276]; the cohomological statement on annihilators is on p. 267. We prove the Kähler case by extending a positive linear functional with the Hahn–Banach argument, together with the scalar Poisson equation and the identity for the trace and the Hodge star proved earlier.

The pointwise trace estimate for smooth forms and the integrated Cauchy–Schwarz argument are [Pop14, Lemmas 3.1–3.2 and their proofs, pp. 4–6] (first arXiv version). Lemma 17.4.19 states the estimate for an arbitrary strictly positive form τ of bidegree $(n-1, n-1)$, and extends it, by local decreasing convolution and linearization of the determinant, to currents whose local potentials are locally bounded outside a closed set of Lebesgue measure zero; the first factor on its left-hand side is the full mass of $S \wedge \tau$, including any mass on the poles. The preprint writes $\mathrm{i}\partial\bar\partial$ for potentials, whereas our proof uses the normalization of this book throughout.

The inequality between mixed masses and the geometric mean of the masses, with its consequences for the concavity of roots and logarithms of masses, is [DDNL21, arXiv version, Theorem 6.1 and its proof, p. 26]; it is the result referred to in [Xia, Theorem 2.4.1]. Theorem 17.4.23 follows the proof in the source by relative Monge–Ampère equations. Because this proof uses the criterion for bigness, the theorem comes after that criterion; the case of vanishing masses is treated separately. At fixed potentials, the operator $\mathrm{i}\partial\bar\partial$ of the source multiplies all measures of top degree by $(2\pi)^n$; by homogeneity this factor cancels in the inequality.

The identities for envelopes of arbitrary mass in Theorem 17.5.3 are [PSWZ26, Theorem 1.2, p. 4], an arXiv preprint; they include potentials of zero mass, to which Theorem 16.4.7 does not apply. Their proof there (Section 5, Theorem 5.1, pp. 37–42) uses a uniform bound for normalized tails and the recovery of volume for perturbations of positive mass; Proposition 17.5.1 follows this approach. Monotonicity gives the limit directly, and the solutions with constant density that are needed are provided by Theorem 17.4.5. Applying the limit to each member of the family $\mathcal{F}_u$ that defines the ceiling proves the identity $(\sup_{v\in\mathcal{F}_u} v)^* = P_\theta[u]$, without a separate formula for the regularization of the ceiling and without a second family of perturbations.

The source uses the operator $\mathrm{dd}^{\mathrm{c}} = \mathrm{i}\partial\bar\partial/\pi$ (p. 7). At fixed potentials, its backgrounds 2θ and 2ω give measures of top degree that are 2^n times ours, while the normalized tail ratios and the constant densities of the auxiliary equations are unchanged.

B.18 Weak geodesics

The definition of subgeodesics on a strip and their convexity and boundary behavior (Definition 17.6.1 and Proposition 17.6.4) follow [Xia, Definition 4.1.1 and Proposition 4.1.2], and Lemmas 17.6.5 and 17.6.6 and Theorem 17.6.8 follow the arguments of [Xia, Proposition 4.1.3, Lemma 4.1.1 and Proposition 4.2.1]. Rooftop envelopes and the identity for envelopes of mutual singularities show that the Perron segment is not identically $-\infty$ and has the prescribed endpoints. The formula of Proposition 17.6.10, which expresses the Perron segment through rooftop envelopes and a Legendre transform, follows by taking supporting affine functions of the convex function of time on each fiber where it is finite; the regularization in the space variable is retained.

That the relative energy is affine along weak geodesics, Theorem 17.7.3, follows the argument of [Xia, Proposition 4.3.3] on a compact extension of the strip. The proof

uses the fiber integration formula Corollary 15.5.1 and integration by parts for bounded differences, Theorem 17.1.3. The vanishing of the non-pluripolar Monge–Ampère measure of a maximal plurisubharmonic function, bounded or not, is Lemma 17.7.1; it is proved with the homogeneous balayages of the bounded truncations. The maximality of the complexified Perron segment, Lemma 17.7.2, is proved with its envelope under rotations of the annulus, or under imaginary translations of the strip. Approximating the endpoints by their decreasing truncations then extends the identity from potentials of bounded type to potentials of finite energy.

B.19 Extremal slopes and singularity cutoffs

The identity for the supremum of the difference along a weak geodesic is [Xia, Proposition 4.2.2], in the chapter on rays; for the infimum, [Xia, Proposition 4.2.3] refers to an analogous argument. Proposition 17.8.1 gives this argument by reversing the segment and using canonical representatives; it also identifies the essential extrema with the extrema over the set where the potentials are finite.

The assertion on bounded shifts that follows [Xia, Example 4.3.1] is justified there by comparing long segments with truncated endpoints. Proposition 17.8.2 makes this comparison quantitative: a bound C at the terminal endpoint contributes at most Cs/T at time s on a segment of length T. Passing to the regularized limits as $T \to \infty$ gives invariance under bounded perturbations and monotonicity with respect to the preorder $\preceq$ of singularities. The construction of the associated rays and the formula for their energy in terms of mixed masses are not treated here.

B.20 Convex duality and real Monge–Ampère measures

For the separation of convex sets and for the local regularity and subgradients of convex functions (Lemma 18.1.2 and Proposition 18.1.3) compare [Hör07, Theorem 2.1.10 and Corollaries 2.1.11–2.1.13, pp. 44–45; Theorem 2.1.22, pp. 55–56]. Closed convex functions and Legendre conjugation are treated in [Hör07, Propositions 2.2.1–2.2.2 and Theorem 2.2.4, pp. 66–68; Theorem 2.2.5, pp. 68–69; Theorem 2.2.8, p. 70]. The corresponding duality statements in this book are Theorem 18.2.2 and Proposition 18.2.3; in the formula for a sum in which the infimum is attained, both functions are assumed to be finite on the whole space.

The classes of convex functions, the formulas involving slope bodies and the real Monge–Ampère measures follow the appendix on convex analysis of [Xia]; as there, the factor $n!$ is kept, with Lebesgue measure normalized by the lattice. In Theorem 18.4.1, measurability with respect to the completed Lebesgue measure and countable additivity are proved using the fact that the inverse of the subgradient map is single-valued almost everywhere. Theorem 18.4.2 concerns vague convergence, and Example 18.4.3 shows that narrow convergence requires an additional condition that

prevents mass from escaping to infinity. The mixed measures of Proposition 18.4.4 are obtained from the positivity of matrices, convolution and polarization. We keep the convention of the source for full mass when a slope body has empty interior, and Example 18.3.5 shows that the hypotheses on slope bodies in the formula for sums cannot be omitted.

The lemma on the finiteness of rotation-invariant plurisubharmonic functions, [Xia, Lemma 1.2.1], is proved there by the logarithmic convexity of circle means, and it provides the finiteness step in [Xia, Lemma 5.2.1]. Proposition 18.1.5 and Corollary 18.1.6 give a direct proof, by smoothing and the fact that the complex Hessian of a function invariant under imaginary translations is one quarter of its real Hessian; this proof does not use Kiselman's minimum principle.

For the conjugate of the maximum of finitely many convex functions, the convex appendix of [Xia] uses [Roc70, Theorem 16.5]. Lemma 18.5.1 proves the case of functions whose domains lie in a compact set, which is the case of the finite suprema in the chapter on ample toric varieties, [Xia, Proposition 5.2.4]; compactness takes care of points whose weights tend to zero. The computation of the energy in [Xia, Proposition 5.2.9, Step 1] passes to the limit in integrals of a bounded function on a noncompact space against real Monge–Ampère measures; Lemma 18.5.2 justifies this passage to the limit by a cutoff and the convergence of total masses.

The asymptotic count of lattice points used in [Xia, Section 5.2] follows from the case of saturated cones in [KK12, Theorem 1.15, p. 939], where Kaveh and Khovanskii reduce the assertion to Riemann sums; their introduction states the more general theorem on the growth of semigroups on p. 929. Theorem 18.6.3 gives a complete proof for convex bodies by means of fundamental cells, allowing arbitrary translations, bodies of smaller dimension and sublinear errors at the boundary. The continuity of volume needed in its proof is Proposition 18.6.2, and Corollary 18.6.4 isolates the two-sided bound used in the toric computation of dimensions.

The chapter on ample toric varieties in [Xia] relies on the theory of toric varieties developed in [CLS11]. Here the smooth projective toric variety of a Delzant lattice polytope is constructed directly from its characters in Theorem 18.7.3; its fan and its boundary divisors are identified in Propositions 18.8.1 and 18.8.2, and the basis of global sections given by characters, Theorem 18.8.5, is obtained from Laurent expansions in the charts at the vertices. No general classification of toric varieties is needed.

The potential of Theorem 18.9.1, built from the function $G_0 = \sum_\rho \ell_\rho \log \ell_\rho$, is Guillemin's reference potential. For the Legendre computation compare [CDG03, Proposition 2, pp. 7–8; Theorem 1, pp. 9–10] (first arXiv version). That source uses $G = \frac{1}{2}\sum_\rho \ell_\rho \log \ell_\rho$ and holomorphic logarithmic coordinates $y + \mathrm{i}\theta$; with $x = 2y = \log|w|^2$ we have $G_0 = 2G$ and $F_0(x) = 2F(x/2)$, up to an additive constant. Its convention $\mathrm{d}^{\mathrm{c}} f = -\mathrm{d}f \circ J$ gives $\mathrm{dd}^{\mathrm{c}} = 2\mathrm{i}\partial\bar\partial$; with the curvature convention of this book the form is divided by 2π. Smooth extension and strict positivity in the monomial charts are proved by the inverse function theorem at every face, including the vertices.

B.21 Analytic spaces and modifications

The Weierstrass division theorem and the Noetherian property of the ring of germs of holomorphic functions (Propositions A.1.1 and A.1.2) follow [Dem12, Chapter II, (2.3)–(2.7), pp. 80–81]. For coherent sheaves and Oka's coherence theorem (Lemma A.1.4 and Theorem A.1.5) compare [Dem12, Chapter II, (3.8)–(3.22), pp. 87–90]. Local irreducible components, local parametrization, the Nullstellensatz, dimension and Cartan's theorem on the coherence of the ideal sheaf of an analytic set are treated in [Dem12, Chapter II, (4.1)–(4.31), pp. 91–100]. The algebraic facts on finite field extensions and discriminants used in these constructions are proved in Lemma A.2.2 by linear algebra over the field of fractions, with matrices of multiplication and the trace form, and they are applied in Theorem A.2.3. In our proof of Cartan's theorem, Theorem A.2.8, a single denominator works at all nearby stalks, which is more than finite generation of each stalk.

The Nullstellensatz and the strict decrease of dimension for proper analytic subgerms of an irreducible germ (Corollary A.2.6 and Proposition A.2.7) are deduced from the same local parametrization: division by the discriminant identifies each prime ideal with the ideal of its zero set, and the connectedness of the generic sheets gives the decrease of dimension. The proof of the Nullstellensatz includes the description of the radical as an intersection of prime ideals.

The support theorem for closed currents of order zero, Theorem A.2.17, is the case of bidegree $(1,1)$ of [Dem12, Chapter III, (2.10)–(2.14), pp. 141–143]. Our proof uses smooth normal coordinates, convolution in tangential directions and induction over the singular loci of the analytic set; the theorem is applied in Lemma 14.4.6. In the proof of Proposition 14.4.7, the pullback is corrected with the pushforward of a Kähler form, which gives the required strict positivity directly.

For universal denominators of weakly holomorphic functions and for normalization (Proposition A.3.3 and Theorem A.3.6) compare [Dem12, Chapter II, (7.1)–(7.15), pp. 110–116]. The cusp and the pair of crossing lines in Example A.3.10 illustrate the difference between being normal and being unibranch. The Remmert–Stein extension theorem and Remmert's proper mapping theorem are proved together in [Dem12, Chapter II, (8.7)–(8.8), pp. 118–121]. Our proof of Theorems A.6.1 and A.6.2 also treats them together, but it uses the resolution of singularities, Theorem A.5.4, to reduce to proper maps defined on manifolds, and it proves the statements on finite maps and on the dimension of hypersurfaces separately.

The resolution and principalization theorems (Theorems A.5.4 and A.5.5) are the analytic versions of [BM08, Theorems 1.1 and 1.3, pp. 611 and 614; Definitions 2.1–2.2, pp. 618–619]; that the results apply to complex analytic spaces is stated on p. 611 there. The relative principalization theorem, Theorem A.6.5, follows the lemma on relative principalization in the unpublished manuscript [VX, Lemma `relative-principalization`]. At each step of this construction, the centers in the ambient manifold divide the ideal by exactly one exceptional factor, as required for the controlled transform of the restricted ideal. In our statement the submanifold is closed and the ambient manifold is connected, so that the global blowups and the identities between ideals hold as stated.

Theorem A.7.1 states that a proper bimeromorphic map has connected fibers over the points at which the germ of the target is irreducible, in particular over unibranch spaces. Its proof follows the argument with complements of branches in [Xia, Appendix on unibranch spaces]; properness is used to place the inverse image of a small neighborhood inside neighborhoods of the pieces of the fiber. The local upper bound near analytic subsets of codimension at least two in a reduced space, Proposition A.8.4, is proved by slicing with curves over a finite projection and the maximum principle for subharmonic functions. It complements the extension theorem on manifolds, Corollary A.2.15, which follows from the criterion in terms of Hausdorff measure.

The corollary on the realization of divisors, in the appendix of [Xia] on unibranch spaces, underlies the comparison of multiplier ideals with Lelong numbers along divisors in Part I of that book. Theorem A.7.6 proves this corollary by principalizing the successive centers and decreasing the order of the Jacobian along a fixed divisor. The definition of plurisubharmonic functions on reduced spaces by local ambient extension, Definition A.8.1, suffices for pullback by holomorphic maps, for the maximum principle on compact fibers and for the upper bound obtained by slicing with curves. On the manifolds used in Part I of that book, the descent of quasi-plurisubharmonic functions under modifications, Proposition 14.2.8, is proved directly by extension across an analytic set of codimension at least two.

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